# Polyfold Theory, proof by proof

The formal proof of Hofer–Wysocki–Zehnder's Polyfold and Fredholm Theory, Parts I–III, Chapters 1–16, in words

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# Part I. The sc-calculus (HWZ Chapter 1, §§1.1–1.4)

Chapter 1 of Part I fixes the vocabulary of the whole theory: sc-Banach spaces, sc-operators, sc-subspaces and quotients, sc-Fredholm operators and their stability under $\mathrm{sc}^+$-perturbations, partial quadrants, $\mathrm{sc}^0$ and $\mathrm{sc}^1$ maps, the tangent construction and the inductive definitions of $\mathrm{sc}^k$ and sc-smooth, the relation between sc-differentiability and classical differentiability on the levels (Propositions 1.2.1–1.2.4 (preprint 1.16–1.19) and Corollary 1.2.5 (preprint 1.20), quoted from the 2010 paper), the chain rule, the degeneracy index of a point of a partial quadrant and boundary recognition. The formal proof establishes all of it, Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the text's own hypothesis ($C^{l+1}$ from $U_{m+l}$ to $F_m$ for $l \leq k$), by the 2010 paper's induction $(\mathrm{S}_k)$ on elementary maps (§I.17). So are the three items the text states without proof or example: Example 1.1.3 (preprint 1.2), the weighted Sobolev spaces (§I.22), the remark after Definition 1.1.15 (preprint 1.14) that $Df$ need not be norm continuous (§I.24), and Remark 1.1.17 (preprint 1.15) on partial cones (§I.25).

**What is proved, and what is not.** Every numbered statement of Chapter 1 is proved, and so are the three items the first revision of this part left open. The three gaps of the chapter are all closed. G.1, Example 1.1.3 (preprint 1.2), is proved with exactly the printed hypotheses, $\delta_0 = 0$ and $(\delta_m)$ strictly increasing, and level 0 is $L^2(\mathbb{R} \times S^1)$ (§I.22). G.2, Remark 1.1.17 (preprint 1.15), holds for closed convex partial cones with nonempty interior: relatively open subsets of such a cone are sc-domains, so §§1.2–1.3 apply to them, and the tangent cone is again a partial cone (§I.25). G.3, the remark after Definition 1.1.15 (preprint 1.14) that $x \mapsto Df(x)$ need not be continuous for the operator norm, is proved by a sequence-space example, since the text gives none (§I.24). No row of the chapter is narrower than the text, and no hypothesis is added to a numbered statement. The formalization goes beyond the text in two places: the index on every level of an sc-Fredholm operator (§§I.6, I.9), and the converse of Proposition 1.1.10 (preprint 1.9), which makes "Fredholm on level 0 and regularizing" a criterion for sc-Fredholm operators (§I.23). What is not done lies outside the statements of the text. The shift map of the 2010 paper, which the exposition's §6 uses on the weighted Sobolev spaces, is not built. And the sequence-space example of §I.24 is proved $\mathrm{sc}^1$, which is all the remark needs, not sc-smooth.

Two decisions shape everything. First, an sc-Banach space is an *indexed family* of Banach spaces $E_m$ with injective compact inclusions $E_{m+1} \to E_m$, not one space with a chain of subspaces (§I.1). "The point $x$ of $E_0$ lies in $E_m$" then means that $x$ is in the image of $E_m$, and every map has level maps $f_m : E_m \to F_m$ determined by injectivity. Second, the text's standing hypothesis "$U$ relatively open in a partial quadrant" enters the proofs of §§1.2–1.3 only through two facts about each level set $U_m$: it is the intersection of an open and a convex set, and derivatives within it are unique. These two facts make up the notion of an *sc-domain* (§I.10), which the tangent construction and the level shift preserve by a two-line argument, so

the inductions of §1.2 run on sc-domains, and the text's tuples $(U, C, E)$ are a special case.

Four proofs take a different road from the text's. The chain rule replaces the two integrals of §1.4.2 by the one-dimensional mean value inequality along a segment, and does not need the first integral at all (§I.12). The compactness argument behind Propositions 1.2.1–1.2.3 (preprint 1.16–1.18), the subsequence $h_n \to h$ in $E_m$ for unit vectors $h_n \in E_{m+1}$, is isolated once as the *compact-open lemma*, and its $C^k$ version replaces the multilinear maps $\Gamma(x)$ and the differentiation under the integral sign of the 2010 paper (§I.13). Theorem 1.3.5 (preprint 1.25), which the text quotes from another paper, is proved by the argument of §2.4 (preprint §2.3), for $\mathrm{sc}^1$ germs (§I.19). And the induction $(\mathrm{S}_k)$ behind Proposition 1.2.4 (preprint 1.19) runs on the order $s$ over all elementary maps $d^j f(\pi_0 z)(\pi_1 z, \dots, \pi_j z)$ at once, rather than on a description of the $2^{k+1}$ components of $T^{k+1} f$ (§I.17). Smaller departures are flagged where they occur: the complement in Proposition 1.1.11 (preprint 1.10) is built from a right inverse rather than a lifted basis; the regularizing step of §1.4.1 is a bootstrap; $k = k'$ in Lemma 1.3.4 (preprint 1.24) is derived from the lemma, not proved first. What Mathlib lacked, the Riesz–Schauder theorem, the stability of Fredholm operators under compact perturbations and the invariance of the Fredholm index, is proved in §I.7; the index statements of the exposition's Lemma 2.D and Proposition 1.1.11 (preprint 1.10) follow in §I.6 and §I.9. The lemmas that the exposition adds to make the text's remarks precise are proved where they belong: uniqueness of the sc-structure in finite dimensions (§I.1), sc-splittings as sc-projections (§I.3), the smooth points of a quotient (§I.5), the shift and tangent of a tuple (§I.10), compact-open continuity and $\mathrm{sc}^1 = C^1$ on quadrants in finite dimensions (§I.11).

**Map.**

- `Chapter1/ScBanach`: scales and sc-Banach spaces, the constant structure, sums and shifts; sc-operators, sc-isomorphisms, $\mathrm{sc}^+$-operators; sc-subspaces, sc-splittings, Proposition 1.1.7 (preprint 1.6); quotients, Proposition 1.1.8 (preprint 1.7).
- `Chapter1/FredholmPerturbation`: Fredholm operators between Banach spaces; Riesz–Schauder; compact perturbations.
- `Chapter1/FredholmIndex`: the index of a linear map, its additivity, its invariance under small and compact perturbations; the index on the levels of an sc-Fredholm operator; the index in Proposition 1.1.11 (preprint 1.10).
- `Chapter1/ExpositionLemmas`: the exposition's Lemmas 1.A(2)–(3), 2.B, 2.C, 3.F, and $\mathrm{sc}^1 = C^1$ in finite dimensions on quadrants.
- `Chapter1/ScFredholm`: Definition 1.1.9 (preprint 1.8), the index, the level-wise Fredholm property; Propositions 1.1.10 and 1.1.11 (preprint 1.9 and 1.10).
- `Chapter1/ScSmooth`: level sets, the tangent $TE$ and $TU$, level maps, $\mathrm{sc}^0$; $\mathrm{sc}^1$ with its derivative family, uniqueness of $Df$, $\mathrm{sc}^k$ and sc-smooth; partial quadrants, tuples, sc-domains.
- `Chapter1/ChainRule`: Theorem 1.3.1 (preprint 1.21) and its iterates.
- `Chapter1/Properties`: the compact-open continuity of $Df$; the base cases of Propositions 1.2.1, 1.2.4 (preprint 1.16, 1.19) and Corollary 1.2.5 (preprint 1.20).
- `Chapter1/Levels`: the compact-open lemma; the level operators $Df(x) : E_m \to F_m$; Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) on all levels; Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the stronger hypothesis "$C^{k+1}$ on every level".
- `Chapter1/LevelsWeak`: Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) as stated: the derivative tower, elementary maps, their $\mathrm{sc}^1$ step and tangent maps, the sum rule for $\mathrm{sc}^k$ maps, the induction $(\mathrm{S}_k)$.
- `Chapter1/Degeneracy`: Definition 1.3.3 (preprint 1.23) and Lemma 1.3.4 (preprint 1.24).
- `Chapter2/DegeneracyIndex`: Theorem 1.3.5 (preprint 1.25), by the argument of §2.4 (preprint §2.3).

- `Chapter1/ScFredholmCriterion`: regularizing operators; the converse of Proposition 1.1.10 (preprint 1.9); sc-operators from level-preserving bounded maps; restriction to sc-subspaces.
- `Chapter1/ScOneNotNormContinuous`: the sequence sc-Banach space $X$ and an $\mathrm{sc}^1$ map whose derivative is not norm continuous.
- `Chapter1/GapP1_Sobolev1–GapP1_Sobolev13`: Example 1.1.3 (preprint 1.2): weak derivatives on the cylinder, the spaces $H^{m,\delta}$, Fourier series on a box, Rellich's theorem, the compact inclusion, density of test functions.
- `Chapter1/GapP1_PartialCones`: Remark 1.1.17 (preprint 1.15).

## I.1 Scales and sc-Banach spaces (Definition 1.1.1 (preprint 1.1))

*The scale.* A **scale of Banach spaces** consists of real Banach spaces $E_m$, one for each $m \in \mathbb{N}$, together with continuous linear maps $\mathrm{incl}_m : E_{m+1} \to E_m$ that are injective and compact operators `Chapter1/ScBanach · ScScale`. Compactness is Mathlib's: the preimage of some neighbourhood of 0 is mapped into a compact set `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator`. The composite inclusion $\iota_m : E_m \to E_0$ is defined by recursion, $\iota_0 = \mathrm{id}$ and $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$ `Chapter1/ScBanach · ScScale.toBase`; it is injective `Chapter1/ScBanach · ScScale.toBase_injective`, and the composite $E_{m+k} \to E_m$ is defined likewise `Chapter1/ScBanach · ScScale.inclK` and is compatible with $\iota$ `Chapter1/ScBanach · ScScale.toBase_inclK`. The text's $E_m$ is here the image $\iota_m(E_m) \subseteq E_0$; the text's $E_m \supset E_{m+1}$ is the inclusion of images, and "$x \in E_m$" for $x \in E_0$ is "$x \in \mathrm{range}\,\iota_m$". The **smooth points** $E_\infty$ form the subspace of $E_0$ of points that lift to every level, $E_\infty = \bigcap_m \mathrm{range}\,\iota_m$ `Chapter1/ScBanach · ScScale.smooth` `Chapter1/ScBanach · ScScale.mem_smooth_iff`, and $E_\infty \cap E_m$ is its preimage in $E_m$ `Chapter1/ScBanach · ScScale.smoothLvl`; a smooth point of $E_m$ lifts to every $E_{m+k}$ `Chapter1/ScBanach · ScScale.exists_inclK_of_mem_smoothLvl`.

*The density axiom.* An **sc-Banach space** is a scale in which $E_\infty \cap E_m$ is dense in $E_m$ for every $m$ `Chapter1/ScBanach · ScBanach`. This is condition (2) of Definition 1.1.1 (preprint 1.1) exactly as the text states it. The weaker statement that $E_{m+k}$ has dense image in $E_m$ follows `Chapter1/ScBanach · ScBanach.denseRange_inclK` `Chapter1/ScBanach · ScBanach.denseRange_incl`; it does not obviously imply the axiom, which is why the strong form is the definition.

*Why a family.* Nothing in Chapter 1 uses more than the family: the text's identification of $E_{m+1}$ with a subset of $E_m$ is the injectivity of $\mathrm{incl}_m$, a field of the structure, and its "the inclusion is continuous" is part of compactness. The gain is that each level is a Banach space in its own right, with its own norm, and Mathlib's analysis applies on each level without carrying a filtration along.

*The constant structure.* For a finite-dimensional normed space $V$ the family $E_m = V$ with identity inclusions is a scale: the identity of a finite-dimensional space is a compact operator `Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · isCompactOperator_id_iff_finiteDimensional` `Chapter1/ScBanach · ScBanach.constScale`. Here $\iota_m$ is the identity `Chapter1/ScBanach · ScBanach.constScale_toBase`, every point is smooth `Chapter1/ScBanach · ScBanach.constScale_smoothLvl`, and density is trivial, so $V$ is an sc-Banach space `Chapter1/ScBanach · ScBanach.const` `Chapter1/ScBanach · ScBanach.const_toBase`.

*Only the constant structure (Lemma 1.A(2)–(3) of the exposition).* The text remarks that a finite-dimensional space has precisely one sc-structure. For families this reads: if $E$ is an sc-Banach space with $E_0$ finite-dimensional, the inclusions identify $E$ with the constant structure on $E_0$. First, every point is smooth: $E_\infty$ is a subspace of the finite-dimensional $E_0$, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional`, and dense by the axiom at level 0, hence everything `Chapter1/ExpositionLemmas · ScBanach.smooth_eq_top_of_finiteDimensional`. So every $x \in E_0$ lifts to ev-

ery level, and $\iota_m : E_m \to E_0$, injective by construction, is bijective `Chapter1/ExpositionLemmas · ScBanach.toBase_bijective_of_finiteDimensional`. A continuous linear bijection between Banach spaces is an isomorphism (open mapping theorem) `Mathlib/Analysis/Normed/Operator/Banach · ContinuousLinearEquiv.ofBijective`; the isomorphisms $\iota_m$ commute with the inclusions by the very definition $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$, so they form an sc-isomorphism of $E$ onto the constant structure on $E_0$ `Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional_apply`. In particular all the norms $|\cdot|_m$ are equivalent to $|\cdot|_0$. The second remark, that on an infinite-dimensional space the constant structure fails the compactness axiom, becomes: if an inclusion $\mathrm{incl}_m : E_{m+1} \to E_m$ of a scale is onto, then $E_m$ is finite-dimensional. Indeed $\mathrm{incl}_m$ is then a continuous linear bijection, hence an isomorphism $e$, and $\mathrm{id}_{E_m} = \mathrm{incl}_m \circ e^{-1}$ is compact as a compact operator composed with a bounded one `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm`; a Banach space with compact identity is finite-dimensional `Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id Chapter1/ExpositionLemmas · ScScale.finiteDimensional_of_surjective_incl`.

*Direct sums.* For scales $E$, $F$, the family $E_m \times F_m$ with the product inclusions is a scale: the product of two compact operators is compact, because the preimage of a product of neighbourhoods is mapped into the product of two compact sets `Chapter1/ScBanach · ScBanach.prodScale`. Its $\iota_m$ is the pair of the two `Chapter1/ScBanach · ScBanach.prodScale_toBase`, and its smooth points are the pairs of smooth points `Chapter1/ScBanach · ScBanach.prodScale_smoothLvl`; since a product of dense sets is dense, $E \oplus F$ is an sc-Banach space `Chapter1/ScBanach · ScBanach.prod Chapter1/ScBanach · ScBanach.prod_smoothLvl`.

*The level shift.* The family $E_{m+1}$ with the inclusions $\mathrm{incl}_{m+1}$ is a scale $E^1$ `Chapter1/ScBanach · ScBanach.shiftScale`. Its composite inclusion to its base $E_1$, followed by $\iota_1$, is $\iota_{m+1}$ `Chapter1/ScBanach · ScBanach.shiftScale_toBase`, and its smooth points at level $m$ are the smooth points of $E$ at level $m+1$ `Chapter1/ScBanach · ScBanach.shiftScale_smoothLvl`, so $E^1$ is an sc-Banach space `Chapter1/ScBanach · ScBanach.shift Chapter1/ScBanach · ScBanach.shift_toBase Chapter1/ScBanach · ScBanach.shift_smoothLvl`; $E^k$ is the $k$-fold iterate `Chapter1/ScBanach · ScBanach.shiftK`.

*Level sets.* For $U \subseteq E_0$ the level set $U_m \subseteq E_m$ is the preimage $\iota_m^{-1}(U)$ `Chapter1/ScSmooth · ScBanach.lvlSet`; $U_0 = U$ `Chapter1/ScSmooth · ScBanach.lvlSet_zero`, $U_{m+1} = \mathrm{incl}_m^{-1}(U_m)$ `Chapter1/ScSmooth · ScBanach.lvlSet_succ`, and level sets preserve inclusions, intersections, openness, closedness and convexity `Chapter1/ScSmooth · ScBanach.lvlSet_mono Chapter1/ScSmooth · ScBanach.lvlSet_inter Chapter1/ScSmooth · ScBanach.isOpen_lvlSet Chapter1/ScSmooth · ScBanach.isClosed_lvlSet Chapter1/ScSmooth · ScBanach.convex_lvlSet`. This is the text's filtration $A_m = A \cap E_m$ of a subset. The level sets of $U_1 \subseteq E_1$, taken in the shifted space $E^1$, are the $U_{m+1}$ `Chapter1/Levels · ScBanach.shift_lvlSet`, which is the text's $(A^1)_m = A_{1+m}$.

## I.2 Sc-operators, sc-isomorphisms, sc⁺-operators (Definitions 1.1.4–1.1.5 (preprint 1.3–1.4))

An **sc-operator** $T : E \to F$ is a family of continuous linear maps $T_m : E_m \to F_m$ commuting with the inclusions, $\mathrm{incl}_m^F \circ T_{m+1} = T_m \circ \mathrm{incl}_m^E$ `Chapter1/ScBanach · ScOperator`. Then $\iota_m T_m = T_0 \iota_m$ by induction `Chapter1/ScBanach · ScOperator.toBase_comm`, so the whole family is determined by the base map $T_0$ `Chapter1/ScBanach · ScOperator.ext_base`: this is the sense in which the text's single linear map $T$ "with $T(E_m) \subset F_m$ and $T : E_m \to F_m$ continuous" is the same thing. The text's formulation is also available as a predicate on a continuous linear $T_0 : E_0 \to F_0$, "for every $m$ there is a continuous linear $T_m$ with $\iota_m T_m = T_0 \iota_m$" `Chapter1/ScBanach · IsScOperator`, and the two notions pass into each other `Chapter1/ScBanach · ScOperator.isScOperator Chapter1/ScBanach · IsScOperator.toScOperator Chap-`

ter1/ScBanach · IsScOperator.toScOperator_zero. The identity is an sc-operator, sc-operators compose and add Chapter1/ScBanach · ScOperator.id Chapter1/ScBanach · ScOperator.comp Chapter1/ScBanach · ScOperator.add, and they map smooth points to smooth points, since a lift of $x$ to level $j$ is carried by $T_j$ to a lift of $T_0 x$ Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl.

An **sc-isomorphism** is a family of continuous linear equivalences $E_m \simeq F_m$ commuting with the inclusions Chapter1/ScBanach · ScIso; it has an underlying sc-operator Chapter1/ScBanach · ScIso.toScOperator, an inverse Chapter1/ScBanach · ScIso.symm, composites Chapter1/ScBanach · ScIso.trans and the identity Chapter1/ScBanach · ScIso.refl, and $\iota_m$ commutes with it and with its inverse Chapter1/ScBanach · ScIso.toBase_comm Chapter1/ScBanach · ScIso.symm_toBase_comm. The text defines an sc-isomorphism as a bijective sc-operator whose inverse is an sc-operator; the continuity of the inverse on each level, which there follows from the open mapping theorem, is here part of the data. The two are equivalent: a bijective sc-operator is an sc-isomorphism by the open mapping theorem on each level Chapter3/LinearizationChartsSc · ScIso.ofBijective. Likewise the continuity in "maps $E_m$ into $F_m$ continuously" is automatic for a bounded $T : E_0 \to F_0$, by the closed graph theorem Chapter1/ScFredholmCriterion · isScOperator_of_mapsTo (§I.23).

An **sc$^+$-operator** $E \to F$ is an sc-operator $E \to F^1$ into the shifted space, i.e. continuous linear maps $S_m : E_m \to F_{m+1}$ commuting with the inclusions Chapter1/ScBanach · ScPlus. The text's first condition, $S(E_m) \subset F_{m+1}$, is contained in the second and is dropped. The inclusion $F^1 \to F$, level by level $\mathrm{incl}_m : F_{m+1} \to F_m$, is itself an sc-operator Chapter1/ScBanach · ScBanach.shiftIncl, and composing gives the sc-operator $E \to F$ underlying $S$ Chapter1/ScBanach · ScPlus.toScOperator Chapter1/ScBanach · ScPlus.toScOperator_apply. Its level-$m$ map is the compact $\mathrm{incl}_m$ after the bounded $S_m$, hence compact Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm Chapter1/ScBanach · ScPlus.isCompactOperator: the text's remark that sc$^+$-operators are sc-compact. In terms of images, $S$ raises the level: if $x \in \operatorname{range} \iota_m$ then $S_0 x \in \operatorname{range} \iota_{m+1}$ Chapter1/ScFredholm · ScPlus.toScOperator_mem_range_toBase_succ.

## I.3 Sc-subspaces and sc-splittings (Definition 1.1.6 (preprint 1.5))

For a subspace $A \subseteq E_0$ its **level-$m$ part** $A_m \subseteq E_m$ is the preimage of $A$ under $\iota_m$ Chapter1/ScBanach · ScBanach.levelSub Chapter1/ScBanach · ScBanach.mem_levelSub; $A_0 = A$ Chapter1/ScBanach · ScBanach.levelSub_zero, the inclusions map $A_{m+1}$ into $A_m$ Chapter1/ScBanach · ScBanach.incl_mem_levelSub, and $A_m$ is closed when $A$ is, as the preimage of a closed set under a continuous map Chapter1/ScBanach · ScBanach.isClosed_levelSub.

$A$ is an **sc-subspace** if $A$ is closed and, for every $m$, $A_m$ lies in the closure of $A_m \cap E_\infty$ Chapter1/ScBanach · ScBanach.IsScSubspace Chapter1/ScBanach · ScBanach.IsScSubspace.isClosed. The text asks that $(A_m)$ be an sc-structure on $A$; of the conditions in that phrase, the closedness of $A_m$ in $E_m$ was just proved, and the compactness of the restricted inclusions $A_{m+1} \to A_m$ is automatic, since the restriction of a compact operator to a subspace, with values in a closed subspace, is compact. This is how the scale structure on a closed $A$ is built: levels $A_m$, inclusions the restrictions of $\mathrm{incl}_m$, injective because $\mathrm{incl}_m$ is, compact by the remark Chapter1/ScBanach · ScBanach.ofSubspaceScale. Its composite inclusion is that of $E$ on the underlying points Chapter1/ScBanach · ScBanach.ofSubspaceScale_coe_toBase, and its smooth points are the points of $A_m$ that are smooth in $E$ Chapter1/ScBanach · ScBanach.mem_ofSubspaceScale_smoothLvl Chapter1/ScBanach · ScBanach.ofSubspaceScale_smoothLvl. The density condition of `IsScSubspace` is then exactly the density axiom for this scale, so an sc-subspace is an sc-Banach space with the induced structure Chapter1/ScBanach · ScBanach.ofSubspace.

$E = F \oplus G$ is an **sc-splitting** if $F$ and $G$ are sc-subspaces and, for every $m$, $F_m$ and $G_m$ are complementary subspaces of $E_m$ `Chapter1/ScBanach · ScBanach.IsScSplitting`. The text asks for a *topological* direct sum $E_m = F_m \oplus G_m$. Since $F_m$ and $G_m$ are closed and complementary, the projection onto $F_m$ along $G_m$ is continuous by the closed graph theorem, which is Mathlib's "closed complemented from complementary closed subspaces" `Chapter1/ScBanach · ScBanach.IsScSplitting.closedComplemented`; so nothing is lost by asking only for complementarity. The splitting is symmetric in $F$ and $G$ `Chapter1/ScBanach · ScBanach.IsScSplitting.symm`.

*Sc-splittings are sc-projections (Lemma 2.B of the exposition).* The proof of Proposition 1.1.7 (preprint 1.6) in the text passes from "$P \circ P = P$, continuous on every level" to "a sc-projection" and an sc-complement $(1 - P)(E)$ without comment. Both directions are proved. (i) Given an sc-splitting $E = A \oplus B$, on each level $A_m$ and $B_m$ are complementary and closed, hence topological complements `Mathlib/Analysis/Normed/Module/Complemented · Submodule.IsCompl.isTopCompl_of_isClosed Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.isTopCompl`, so the projection $P_m$ of $E_m$ onto $A_m$ along $B_m$ is continuous `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL`. These projections commute with the inclusions: write $x = P_{m+1}x + (x - P_{m+1}x)$ with the first term in $A_{m+1}$ and the second in $B_{m+1}$ `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_sub_projectionL`; $\mathrm{incl}_m$ carries them into $A_m$ and $B_m$ (§I.3), where $P_m$ is the identity, resp. zero `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_iff Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_apply_eq_zero_iff`, so $P_m(\mathrm{incl}_m x) = \mathrm{incl}_m(P_{m+1}x)$. Hence the $P_m$ form an sc-operator $P$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_apply` with $P_m^2 = P_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_idem`, $P_m(E_m) = A_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.range_scProj` and $\ker P_m = B_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.ker_scProj`. (ii) Conversely let $P$ be an sc-operator with $P_0^2 = P_0$. Idempotence passes to every level by injectivity of $\iota_m$, since $\iota_m P_m P_m x = P_0 P_0 \iota_m x = P_0 \iota_m x = \iota_m P_m x$ `Chapter1/ExpositionLemmas · ScOperator.idem_lvl`. The level-$m$ part of $P_0(E_0)$ is $P_m(E_m)$: if $\iota_m x = P_0 y$ then $P_0 \iota_m x = \iota_m x$, i.e. $\iota_m P_m x = \iota_m x$, so $x = P_m x$ `Chapter1/ExpositionLemmas · ScOperator.levelSub_range_eq`; the level-$m$ part of $\ker P_0$ is $\ker P_m$ (§I.6, `Chapter1/ScFredholm · ScOperator.ker_lvl`). $P_0(E_0) = \{x : P_0 x = x\}$ is closed, and $\ker P_0$ is closed. Density of smooth points: $x \in P_m(E_m)$ satisfies $x = P_m x$ and is the limit of $P_m z_n$ for smooth $z_n \to x$, and the $P_m z_n$ are smooth points of $P_m(E_m)$ (§I.2); $x \in \ker P_m$ satisfies $x = x - P_m x$ and is the limit of $z_n - P_m z_n$, smooth points of $\ker P_m$. Finally $P_m(E_m)$ and $\ker P_m$ are complementary because $P_m$ is idempotent `Mathlib/LinearAlgebra/Projection · LinearMap.IsIdempotentElem.isCompl`. So $E = P_0(E_0) \oplus \ker P_0$ is an sc-splitting `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem`, and together: $E = A \oplus B$ is an sc-splitting iff an idempotent sc-operator has range $A$ and kernel $B$ `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`. The text's $(1-P)(E)$ is $\ker P$. The formal Proposition 1.1.7 (preprint 1.6) (§I.4) builds its complement directly and does not use this.

## I.4 Proposition 1.1.7 (preprint 1.6): finite-dimensional sc-subspaces

**Statement.** Let $F \subseteq E_0$ be finite-dimensional. Then $F$ is an sc-subspace if and only if $F \le E_\infty$ `Chapter1/ScBanach · ScBanach.proposition_1_1_7_iff`; and if $F \le E_\infty$ there is a $G$ with $E = F \oplus G$ an sc-splitting `Chapter1/ScBanach · ScBanach.proposition_1_1_7`.

**The equivalence.** If $F$ is an sc-subspace, the density condition at level 0 says $F \subseteq \overline{F \cap E_\infty}$. But $F \cap E_\infty$ is a subspace of the finite-dimensional $F$, hence finite-dimensional, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional`; so $F \subseteq F \cap E_\infty$, i.e. $F \le E_\infty$. Conversely, if $F \le E_\infty$ then $F$ is closed (finite-dimensional) and every point of $F_m$ is already a smooth point of $F_m$,

so the density condition holds trivially.

**The complement.** Mathlib provides a continuous linear projection $p : E_0 \to F$ with $p|_F = \mathrm{id}$, from the closed complementation of finite-dimensional subspaces (Hahn–Banach) `Mathlib/Analysis/LocallyConvex/HahnBanach · Submodule.ClosedComplemented.of_finiteDimensional`. This is the text's $P(x) = \sum \lambda_i(x) e_i$. Put $G := \ker p$. Since $F \le E_\infty$, the restriction of $\iota_m$ to $F_m$ is a bijection onto $F$ (injective by injectivity of $\iota_m$, surjective because every point of $F$ lifts to level $m$) `Chapter1/ScBanach · ScBanach.levelRestrict Chapter1/ScBanach · ScBanach.levelRestrict_bijective Chapter1/ScBanach · ScBanach.levelEquiv`; through this linear isomorphism $F_m \cong F$ the projection lifts to a continuous linear $P_m : E_m \to F_m$, $P_m = (\iota_m|_{F_m})^{-1} \circ p \circ \iota_m$ (continuous because $F_m$ is finite-dimensional) `Chapter1/ScBanach · ScBanach.finProj`. It fixes $F_m$ `Chapter1/ScBanach · ScBanach.finProj_apply_of_mem`, and $P_m x = 0$ iff $p(\iota_m x) = 0$ `Chapter1/ScBanach · ScBanach.finProj_eq_zero_iff`, so $G_m = \ker P_m$. Now $F$ is an sc-subspace by the first part, and $G$ is closed as a kernel. For the density of $G_m \cap E_\infty$ in $G_m$: given $g \in G_m$, let $Q := 1 - P_m$ on $E_m$; then $Qg = g$, $Q$ maps $E_m$ into $G_m$, and $Q$ maps smooth points to smooth points, since $P_m x$ is smooth ($F \le E_\infty$) and smooth points form a subspace `Chapter1/ScBanach · ScScale.smoothLvl_submodule_closed_sub`. Hence $g = Qg$ lies in the closure of $Q(E_\infty \cap E_m) \subseteq G_m \cap E_\infty$, because $E_\infty \cap E_m$ is dense and $Q$ is continuous. This is the text's $g_n = (1 - P) f_n \to g$. Finally $F_m$ and $G_m = \ker P_m$ are complementary because $P_m$ is a projection onto $F_m$ `Mathlib/LinearAlgebra/Projection · isCompl_of_proj`.

## I.5 Proposition 1.1.8 (preprint 1.7): quotients

**Statement.** Let $A \subseteq E_0$ be closed. The quotients $E_m/A_m$ form an sc-Banach space $E/A$ `Chapter1/ScBanach · ScBanach.quotient`, the quotient maps form an sc-operator $E \to E/A$ `Chapter1/ScBanach · ScBanach.quotientMap`, and, as a statement, there are an sc-Banach space $Q$ and an sc-operator $q : E \to Q$ with every $q_m$ surjective and $\ker q_m = A_m$ `Chapter1/ScBanach · ScBanach.proposition_1_1_8`. The text assumes $A$ is an sc-subspace; only closedness is used.

**The inclusions.** $A_m$ is closed in $E_m$ (§I.3), so $E_m/A_m$ is a Banach space. The map $E_{m+1}/A_{m+1} \to E_m/A_m$ is induced by $\mathrm{incl}_m$: it is well defined because $\mathrm{incl}_m$ maps $A_{m+1}$ into $A_m$ `Chapter1/ScBanach · ScBanach.quotIncl Chapter1/ScBanach · ScBanach.quotIncl_mk`. It is injective: if $\mathrm{incl}_m x \in A_m$ then $\iota_{m+1} x = \iota_m(\mathrm{incl}_m x) \in A$, so $x \in A_{m+1}$ `Chapter1/ScBanach · ScBanach.quotIncl_injective`. This injectivity is the text's identification of $E_m/A_m$ with the subspace $\{x + A : x \in E_m\}$ of $E/A$, and it holds precisely because $A_m$ is the *full* preimage of $A$. It is compact: the quotient map $E_m \to E_m/A_m$ after the compact $\mathrm{incl}_m$ is compact, and the induced map on $E_{m+1}/A_{m+1}$ sends the image of a neighbourhood of 0 (which is a neighbourhood, since quotient maps are open) into the same compact set `Chapter1/ScBanach · ScBanach.quotIncl_isCompactOperator`. So the quotients form a scale `Chapter1/ScBanach · ScBanach.quotientScale`, whose composite inclusion sends the class of $x$ to the class of $\iota_m x$ `Chapter1/ScBanach · ScBanach.quotientScale_toBase_mk`. The text's argument with a sequence $x_k + A_{m+1}$ of norm $\le 1$ and representatives of norm $\le 2$ is the sequential form of the same fact.

**Density.** The quotient map $E_m \to E_m/A_m$ is continuous and surjective, so the image of the dense set $E_\infty \cap E_m$ is dense. A class $[x]$ with $x$ smooth is a smooth point of the quotient scale: a lift $y$ of $x$ to level $j$ gives the lift $[y]$ of $[x]$. So the smooth points of $E_m/A_m$ are dense `Chapter1/ScBanach · ScBanach.quotient`. This is the inclusion "$\{x + A : x \in E_\infty\} \subseteq (E/A)_\infty$" of the text's proof, which is all its density claim needs.

**The smooth points of the quotient (Lemma 2.C of the exposition).** The text also asserts that the two sets are equal. Let $A$ be an sc-subspace and $x \in E_0$ with, for every $j$, a point $x_j \in E_j$ such that $\iota_j x_j - x \in A$.

We find a smooth $y$ with $y - x \in A$ Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset. *The differences.* $d_j := \mathrm{incl}_j x_{j+1} - x_j \in E_j$ has $\iota_j d_j = \iota_{j+1}x_{j+1} - \iota_j x_j = (\iota_{j+1}x_{j+1} - x) - (\iota_j x_j - x) \in A$, so $d_j \in A_j$. *A uniform bound for the inclusions.* Put $P_j := \prod_{i<j} \max(1, \|\mathrm{incl}_i\|) \geq 1$ Chapter1/ExpositionLemmas · ScBanach.inclBound Chapter1/ExpositionLemmas · ScBanach.one_le_inclBound. By induction, $\|\iota_j v\| \leq P_j \|v\|$ for $v \in E_j$ Chapter1/ExpositionLemmas · ScBanach.norm_toBase_le and $\|\mathrm{incl}_J^{J+k} v\| \leq P_{J+k}\|v\|$ for the composite inclusion $E_{J+k} \to E_J$ Chapter1/ExpositionLemmas · ScBanach.norm_inclK_le, each step multiplying by $\|\mathrm{incl}_i\| \leq \max(1, \|\mathrm{incl}_i\|)$ Chapter1/ExpositionLemmas · ScBanach.norm_incl_le. This replaces the exposition's $M_j = \max_{i \leq j} \|E_j \to E_i\|$, and avoids comparing levels $J + k$ with $j$. *Smooth corrections.* Since $A$ is an sc-subspace, $A_j \cap E_\infty$ is dense in $A_j$, so there are $c_j \in A_j \cap E_\infty$ with $\|d_j - c_j\| \leq 2^{-j}/P_j$ Mathlib/Topology/MetricSpace/Pseudo/Defs · Metric.mem_closure_iff. Then $b_j := d_j - c_j \in A_j$ has $P_j\|b_j\| \leq 2^{-j}$. *The series.* $\sum_j \iota_j b_j$ converges absolutely in $E_0$, its terms being bounded by $2^{-j}$ Mathlib/Analysis/Normed/Group/InfiniteSum · Summable.of_norm_bounded; let $S$ be its sum and $y := x_0 + S$. Its partial sums lie in $A$, which is closed, so $S \in A$ (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · HasProd.tendsto_prod_nat), and $y - x = (x_0 - x) + S \in A$. *$y$ is smooth.* Fix $J$. The tail $t := \sum_k \mathrm{incl}_J^{J+k} b_{J+k}$ converges in $E_J$ (terms bounded by $2^{-J}2^{-k}$), and $\iota_J t = \sum_k \iota_{J+k} b_{J+k}$ because $\iota_J$ is continuous and linear Mathlib/Topology/Algebra/InfiniteSum/Module · ContinuousLinearMap.map_tsum and $\iota_J \circ \mathrm{incl}_J^{J+k} = \iota_{J+k}$ (§I.1). The head telescopes: $\sum_{j<J} \iota_j d_j = \iota_J x_J - x_0$ (the additive form of Mathlib/Algebra/BigOperators/Group/Finset/Basic · Finset.prod_range_div), so $\sum_{j<J} \iota_j b_j = (\iota_J x_J - x_0) - \sum_{j<J} \iota_j c_j$, and the last sum is smooth, hence $\iota_J w$ for some $w \in E_J$. Splitting $S$ into head and tail (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · Multiplicable.prod_mul_tprod_nat_add) gives $y = \iota_J(x_J - w + t)$. As $J$ was arbitrary, $y \in E_\infty$. *As a statement about $E/A$.* The class of $x$ is a smooth point of $E/A$ iff for every $j$ it is the image of a class $[z]$, $z \in E_j$, i.e. iff $\iota_j z - x \in A$ for some $z$ (§I.5 and Mathlib/LinearAlgebra/Quotient/Defs · Submodule.Quotient.eq); by the above this happens iff $x + A$ contains a smooth point Chapter1/ExpositionLemmas · ScBanach.mk_mem_quotient_smooth_iff. This is the text's equality.

## I.6 Definition 1.1.9 (preprint 1.8): sc-Fredholm operators

**Level kernels and ranges.** For an sc-operator $T$, the kernel of $T_m$ is the level-$m$ part of the kernel of $T_0$ Chapter1/ScFredholm · ScOperator.ker_lvl, and the range of $T_m$ lies in the level-$m$ part of the range of $T_0$ Chapter1/ScFredholm · ScOperator.range_lvl_le; both from $\iota_m T_m = T_0 \iota_m$ and injectivity.

**The definition.** An **sc-Fredholm splitting** of $T : E \to F$ consists of subspaces $K, X \subseteq E_0$ and $C, Y \subseteq F_0$ such that $E = K \oplus X$ and $F = C \oplus Y$ are sc-splittings, $K = \ker T_0$, $Y = \mathrm{range}\, T_0$, $K$ and $C$ are finite-dimensional, and $T_m$ restricts to a bijection $X_m \to Y_m$ for every $m$ Chapter1/ScFredholm · ScFredholmSplitting; $T$ is **sc-Fredholm** if such a splitting exists Chapter1/ScFredholm · IsScFredholm. The text's condition (3), "$T : X \to Y$ is an sc-isomorphism", is here the bijectivity on each level; the continuity of the inverses $Y_m \to X_m$ is automatic by the open mapping theorem between the closed subspaces $X_m$ and $Y_m$, so nothing is lost.

**The index.** The index of a splitting is $\dim K - \dim C$ Chapter1/ScFredholm · ScFredholmSplitting.index, and the index of $T$ is $\dim \ker T_0 - \dim(F_0/\,\mathrm{range}\, T_0)$ Chapter1/ScFredholm · scIndex. They agree: $K = \ker T_0$, and $C$ is a complement of $Y = \mathrm{range}\, T_0$ in $F_0$, so $C \cong F_0/Y$ Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl Chapter1/ScFredholm · ScFredholmSplitting.index_eq. So the index does not depend on the splitting, which the text does not state.

**The remarks after the definition.** $K$ is a finite-dimensional sc-subspace, hence $K \leq E_\infty$ by Proposition 1.1.7 (preprint 1.6) Chapter1/ScFredholm · ScFredholmSplitting.K_le_smooth. $\ker T_m = K_m$ Chapter1/

ScFredholm · ScFredholmSplitting.ker_lvl and range $T_m = Y_m$: one inclusion from above, the other because $T_m$ maps $X_m$ onto $Y_m$ Chapter1/ScFredholm · ScFredholmSplitting.range_lvl. The text's "$F = C \oplus T(E_m)$" (read $F_m$) is the level-$m$ part of the splitting $F = C \oplus Y$ together with range $T_m = Y_m$. Finally, every $T_m$ is a Fredholm operator in Mathlib's sense: its kernel $K_m$ injects into the finite-dimensional $K$, its range $Y_m$ is closed, and $Y_m$ has the finite-dimensional complement $C_m$ (which injects into $C$), so its range has finite codimension Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_isCompl Chapter1/ScFredholm · ScFredholmSplitting.isFredholm, using the criterion of §I.7. This is the fact the text uses in the first line of §1.4.1.

**The index on every level (Lemma 2.D of the exposition).** $C$ is a finite-dimensional sc-subspace, so $C \leq F_\infty$ by Proposition 1.1.7 (preprint 1.6) Chapter1/FredholmIndex · ScFredholmSplitting.C_le_smooth. For any subspace $A \leq E_\infty$, $\iota_m$ restricts to a bijection $A_m \to A$ (injective as $\iota_m$ is, surjective because every point of $A$ lifts), so $\dim A_m = \dim A$ Chapter1/FredholmIndex · ScBanach.finrank_levelSub_of_le_smooth. Hence $\dim \ker T_m = \dim K_m = \dim K$, and $F_m / \operatorname{range} T_m = F_m / Y_m \cong C_m$ (complements, Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl) has dimension $\dim C$. So $T_m$ is Fredholm with index $\dim K - \dim C$ in the sense of §I.7 Chapter1/FredholmIndex · ScFredholmSplitting.fredholmIndex_lvl, which by the previous paragraph is the index of $T$ for every splitting Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl; on level 0 this is the definition, $\operatorname{ind}(T)$ being the Fredholm index of $T_0$ Chapter1/FredholmIndex · scIndex_eq_fredholmIndex. The text states the splittings on each level but not the index.

## I.7 Fredholm operators between Banach spaces: Riesz–Schauder and compact perturbations

Mathlib has a notion of Fredholm operator Mathlib/Analysis/Normed/Operator/Fredholm/Basic · IsFredholm: a continuous linear $u : E \to F$ that is a strict map, with closed range, finite-dimensional kernel, range of finite codimension Mathlib/RingTheory/Finiteness/Cofinite · CoFG and closed complemented kernel. It does not have the two theorems that §1.4.1 quotes as classical, that $1 +$ compact is Fredholm and that compact perturbations preserve the property. Both are proved here, for real Banach spaces.

**The criterion.** A continuous linear map with closed range between Banach spaces is a strict map Mathlib/Topology/Maps/Strict/Basic · IsStrictMap: by the open mapping theorem its corestriction to the range is an open quotient map Mathlib/Topology/Maps/Strict/Module · isStrictMap_iff_isOpenQuotientMap_rangeRestrict Chapter1/FredholmPerturbation · isStrictMap_of_isClosed_range. Hence a continuous linear map with finite-dimensional kernel, closed range and range of finite codimension is Fredholm, the kernel being closed complemented because it is finite-dimensional Chapter1/FredholmPerturbation · isFredholm_of_banach. A finite-rank operator is compact: the image of the unit ball is bounded in the finite-dimensional range, so its closure is compact Chapter1/FredholmPerturbation · isCompactOperator_of_finiteDimensional_range.

**Riesz–Schauder: the kernel.** Let $c : E \to E$ be compact and $N := \ker(1 + c)$. On $N$ the operator $c$ is $-\mathrm{id}$, and $c$ restricts to a compact operator of the closed subspace $N$ Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.restrict; so $\mathrm{id}_N$ is compact, and $N$ is finite-dimensional Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id Chapter1/FredholmPerturbation · finiteDimensional_ker_one_add.

**The range is closed.** Take a closed complement $G$ of $N$ (which exists since $N$ is finite-dimensional). $1 + c$ is bounded below on $G$: otherwise there are unit vectors $g_n \in G$ with $(1 + c)g_n \to 0$; the $cg_n$ lie in a compact set Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_

compact, so along a subsequence $cg_n \to y$ Mathlib/Topology/Sequences · IsCompact.tendsto_subseq, hence $g_n = (1+c)g_n - cg_n \to -y$; then $-y \in G$ (closed), $\|-y\| = 1$, and $(1+c)(-y) = 0$, so $-y \in G \cap N = 0$, a contradiction Chapter1/FredholmPerturbation · exists_bound_below_on_compl. A bounded-below operator from a Banach space has closed range (it is antilipschitz and uniformly continuous), and $(1+c)(G) = (1+c)(E)$ since $1+c$ kills $N$ Chapter1/FredholmPerturbation · isClosed_range_one_add.

**The range has finite codimension.** Suppose $R := \operatorname{range}(1+c)$ has infinite codimension. Pick, for every proper subspace, a vector outside it Chapter1/FredholmPerturbation · pick Chapter1/FredholmPerturbation · pick_notMem, and build the chain $M_0 = R$, $M_{j+1} = M_j + \mathbb{R}\operatorname{pick}(M_j)$ Chapter1/FredholmPerturbation · chain. Each $M_j$ is $R$ plus a finite-dimensional subspace Chapter1/FredholmPerturbation · chain_eq_sup, hence closed Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_sup_finiteDimensional Chapter1/FredholmPerturbation · chain_isClosed, and proper, since $R$ has infinite codimension Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_codisjoint Chapter1/FredholmPerturbation · chain_ne_top; the chain is increasing Chapter1/FredholmPerturbation · chain_mono. Riesz's lemma inside a subspace: for closed $M \subsetneq M'$ there is a unit vector of $M'$ at distance $\geq 1/2$ from $M$ (take $x \in M' \setminus M$, a point $y \in M$ with $\|x-y\| < 2\operatorname{dist}(x, M)$ Mathlib/Topology/MetricSpace/HausdorffDistance · infDist_lt_iff, and normalize $x-y$) Chapter1/FredholmPerturbation · exists_unit_far. This gives unit vectors $u_j \in M_{j+1}$ at distance $\geq 1/2$ from $M_j$. For $j < l$, $u_j + (1+c)u_l - (1+c)u_j \in M_l$, and $u_l - [u_j + (1+c)u_l - (1+c)u_j] = -(cu_l - cu_j)$, so $\|cu_l - cu_j\| \geq 1/2$. But the $cu_j$ lie in a compact set and so have a convergent subsequence, two of whose terms are within $1/2$ of each other: a contradiction Chapter1/FredholmPerturbation · coFG_range_one_add. Hence $1+c$ is Fredholm Chapter1/FredholmPerturbation · isFredholm_one_add.

**Compact perturbations.** Let $u : E \to F$ be Fredholm and $k$ compact. Mathlib gives a quasi-inverse $v : F \to E$ of $u$, with $vu - 1$ and $uv - 1$ of finite rank Mathlib/Analysis/Normed/Operator/Fredholm/Basic · isFredholm_iff_exists_isQuasiInverse; a finite-rank difference from the identity is compact Chapter1/FredholmPerturbation · isCompactOperator_sub_one_of_equiv. Then $v(u+k) = 1 + c_1$ and $(u+k)v = 1 + c_2$ with $c_1 = (vu - 1) + vk$ and $c_2 = (uv - 1) + kv$ compact. The kernel of $u+k$ lies in the finite-dimensional kernel of $1 + c_1$; the range of $u+k$ contains the range of $1 + c_2$, which is closed and of finite codimension, so $\operatorname{range}(u+k)$ has finite codimension Mathlib/RingTheory/Finiteness/Cofinite · CoFG.of_le and is closed, as a subspace containing a closed subspace of finite codimension Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_mono_of_finiteDimensional_quotient. By the criterion, $u+k$ is Fredholm Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator.

**The index.** Mathlib has no index; it is built here, first for linear maps over a field. For $u : E \to F$ put $\operatorname{ind}(u) := \dim\ker u - \dim(F/\operatorname{range} u) \in \mathbb{Z}$ Chapter1/FredholmIndex · linIndex, meaningful when $u$ has *finite index*, i.e. finite-dimensional kernel and cokernel Chapter1/FredholmIndex · HasFiniteIndex. *Additivity.* If $u : E \to F$ and $v : F \to G$ have finite index, so has $vu$, and $\operatorname{ind}(vu) = \operatorname{ind}(u) + \operatorname{ind}(v)$ Chapter1/FredholmIndex · HasFiniteIndex.comp_and_linIndex Chapter1/FredholmIndex · HasFiniteIndex.comp Chapter1/FredholmIndex · linIndex_comp. The proof is the dimension count along the exact sequence $0 \to \ker u \to \ker vu \to \operatorname{range} u \cap \ker v \to F/\operatorname{range} u \to G/\operatorname{range} vu \to G/\operatorname{range} v \to 0$, done as four applications of rank–nullity Mathlib/LinearAlgebra/FiniteDimensional/Lemmas · LinearMap.finrank_range_add_finrank_ker and two of the third isomorphism theorem Mathlib/LinearAlgebra/Isomorphisms · Submodule.quotientQuotientEquivQuotient: to $u$ restricted to $\ker vu$ (kernel $\ker u$, range $\operatorname{range} u \cap \ker v$), to the quotient map $F \to F/\operatorname{range} u$ restricted to $\ker v$, and to $v$ followed by $G \to G/\operatorname{range} vu$ (kernel $\operatorname{range} u + \ker v$); finiteness of the new kernel and cokernel comes out of the same maps Chapter1/FredholmIndex · finite_of_finite_ker_range Chapter1/FredholmIndex · finite_of_finite_quotient. Conversely, if $vu$ and $uv$ have finite index then so has $u$, since $\ker u \subseteq \ker vu$ and $\operatorname{range} uv \subseteq \operatorname{range} u$ Chapter1/FredholmIndex · finiteDimensional_ker_of_comp Chapter1/FredholmIndex · finiteDimensional_coker_of_comp Chap-

ter1/FredholmIndex · HasFiniteIndex.of_comp_of_comp. *Index zero.* A linear equivalence has index $0$ Chapter1/FredholmIndex · HasFiniteIndex.equiv Chapter1/FredholmIndex · linIndex_equiv. If $w - 1$ has finite-dimensional range $R$, then $\ker w \subseteq R$, $\operatorname{range} w + R = E$ and $R \cap \operatorname{range} w = w(R)$, so $\dim \ker w$ and $\dim(E/\operatorname{range} w)$ both equal $\dim R - \dim w(R)$ and $\operatorname{ind}(w) = 0$ Chapter1/FredholmIndex · HasFiniteIndex.of_finiteRange_sub_id_and_linIndex Chapter1/FredholmIndex · linIndex_eq_zero_of_finiteRange_sub_id. If $a$ is bijective and $r$ has finite rank, $a + r = a(1 + a^{-1}r)$ has index $0 + 0$ by additivity Chapter1/FredholmIndex · HasFiniteIndex.of_bijective_add_and_linIndex.

*Banach spaces.* For $u : E \to F$ continuous linear, the **Fredholm index** is $\operatorname{ind}(u)$ Chapter1/FredholmIndex · fredholmIndex; a Fredholm operator has finite index Chapter1/FredholmIndex · IsFredholm.hasFiniteIndex. *Small perturbations.* Let $v$ be a quasi-inverse of $u$ and $\|v\|\,\|s\| < 1$. Then $\|vs\|, \|sv\| < 1$ Mathlib/Analysis/Normed/Operator/Basic · ContinuousLinearMap.opNorm_comp_le, so $1 + vs$ and $1 + sv$ are invertible by the Neumann series Mathlib/Analysis/SpecificLimits/Normed · isUnit_one_sub_of_norm_lt_one Chapter1/FredholmIndex · bijective_one_add_of_norm_lt. Since $vu - 1$ and $uv - 1$ have finite rank Chapter1/FredholmIndex · finiteDimensional_range_sub_id_of_equiv, $v(u + s) = (1 + vs) + (vu - 1)$ and $(u + s)v = (1 + sv) + (uv - 1)$ are bijections plus finite-rank maps, hence of index $0$ Chapter1/FredholmIndex · hasFiniteIndex_comp_add_of_norm_lt; the same with $s = 0$ for $vu$ and $uv$. By the converse above, $u + s$, $u$ and $v$ have finite index, and additivity gives $\operatorname{ind}(u+s) + \operatorname{ind}(v) = 0 = \operatorname{ind}(u) + \operatorname{ind}(v)$, so $\operatorname{ind}(u+s) = \operatorname{ind}(u)$ Chapter1/FredholmIndex · fredholmIndex_add_of_norm_lt. *Compact perturbations.* Let $u$ be Fredholm and $k$ compact, and $\varphi(t) := \operatorname{ind}(u + tk)$ for $t \in \mathbb{R}$. At each $t_0$, $u + t_0 k$ is Fredholm (the previous paragraph, $t_0 k$ being compact Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.smul), with a quasi-inverse $v_0$; for $t$ so close to $t_0$ that $\|v_0\|\,\|(t - t_0)k\| < 1$, $\varphi(t) = \operatorname{ind}((u + t_0 k) + (t - t_0)k) = \varphi(t_0)$. So $\varphi$ is locally constant Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.iff_eventually_eq on the connected $\mathbb{R}$, hence constant Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.apply_eq_of_preconnectedSpace, and $\operatorname{ind}(u + k) = \varphi(1) = \varphi(0) = \operatorname{ind}(u)$ Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator. This replaces the citation of Kato IV.5.26 in the exposition's (F2).

## I.8 Proposition 1.1.10 (preprint 1.9): sc-Fredholm operators are regularizing

**Statement.** Let $T$ have an sc-Fredholm splitting, $e \in E_0$, and $T_0 e \in \operatorname{range} \iota_m$. Then $e \in \operatorname{range} \iota_m$ Chapter1/ScFredholm · proposition_1_1_10.

**Proof** (the text's). Write $T_0 e = \iota_m y$. Since $F_m = C_m \oplus Y_m$, $y = c + y'$ with $c \in C_m$ and $y' \in Y_m$, and $y' = T_m x'$ for some $x' \in X_m$ by the bijectivity of $T_m : X_m \to Y_m$. Then $T_0(e - \iota_m x') = \iota_m c$, so $\iota_m c$ lies in $C$ and in $Y = \operatorname{range} T_0$; these are complementary on level $0$, so $\iota_m c = 0$. Hence $e - \iota_m x' \in \ker T_0 = K \leq E_\infty$ (§I.6), so it lifts to some $z \in E_m$, and $e = \iota_m(z + x')$. The step "$c = 0$" is where $C \cap Y = 0$ enters, as the exposition notes. The converse, Fredholm on level $0$ and regularizing implies sc-Fredholm, is §I.23.

## I.9 Proposition 1.1.11 (preprint 1.10): compact perturbations of sc-Fredholm operators

**Statement.** If $T : E \to F$ is sc-Fredholm and $S : E \to F$ is an sc$^+$-operator, then $T + S$ is sc-Fredholm Chapter1/ScFredholm · proposition_1_1_11_preprint. Here $S$ is read as the sc-operator $E \to F$ underlying it

(§I.2), and $T + S$ is the sum of sc-operators.

**The bootstrap.** The text says: if $y = (T + S)e \in Y_m$ then, since $T$ is regularizing, $e \in E_m$. But $Te = y - Se$ with $Se$ only one level up. The formal proof iterates: if $(T + S)_0 e \in \operatorname{range} \iota_{m+1}$, then a fortiori $(T + S)_0 e \in \operatorname{range} \iota_m$ `Chapter1/ScFredholm · ScBanach.range_toBase_succ_le`, so $e \in \operatorname{range} \iota_m$ by induction, so $S_0 e \in \operatorname{range} \iota_{m+1}$ (§I.2), so $T_0 e = (T + S)_0 e - S_0 e \in \operatorname{range} \iota_{m+1}$, and Proposition 1.1.10 (preprint 1.9) gives $e \in \operatorname{range} \iota_{m+1}$ `Chapter1/ScFredholm · ScPlus.regularizing_add`. So $T + S$ is regularizing.

**The kernel.** Each $(T + S)_m = T_m + S_m$ is Fredholm, as a compact perturbation (§I.7) of the Fredholm operator $T_m$ (§I.6) `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`. Let $K' := \ker(T + S)_0$, finite-dimensional. Every $x \in K'$ is smooth: $(T + S)_0 x = 0 \in \operatorname{range} \iota_m$ for every $m$, so $x \in \operatorname{range} \iota_m$ by the bootstrap. By Proposition 1.1.7 (preprint 1.6) there is an sc-splitting $E = K' \oplus X'$ `Chapter1/ScBanach · ScBanach.proposition_1_1_7`. (The text's $K_m = K_{m+1}$ is the same bootstrap.)

**The range.** Let $Y' := \operatorname{range}(T + S)_0$, closed since $(T + S)_0$ is Fredholm. The range of $(T + S)_m$ is exactly $Y'_m$: one inclusion from §I.6; conversely if $\iota_m y = (T + S)_0 e$ then $e$ lifts to level $m$ by the bootstrap, and $y = (T + S)_m e'$ by injectivity of $\iota_m$. $Y'$ is an sc-subspace: for $y = (T + S)_m e \in Y'_m$, approximate $e$ by smooth points $e_n \in E_m$; then $(T+S)_m e_n \to y$ and the $(T+S)_m e_n$ are smooth points of $Y'_m$ (sc-operators preserve smooth points, §I.2). This is the text's argument "$y_n = (T + S)e_n \in Y_\infty$, $y_n \to y$".

**The complement.** $F_0/Y'$ is finite-dimensional. Let $p : F_0 \to F_0/Y'$ be the quotient map. The image $p(F_\infty)$ is a subspace of the finite-dimensional $F_0/Y'$, dense because $F_\infty$ is dense in $F_0$ and $p$ is a continuous surjection, hence closed, hence everything. Instead of lifting a basis, take a linear right inverse $g$ of $p|_{F_\infty}$ and let $\sigma := F_\infty \hookrightarrow F_0$ composed with $g$, so $p \circ \sigma = \mathrm{id}$; put $C' := \operatorname{range} \sigma$, a finite-dimensional subspace of $F_\infty$. On every level $Y'_m$ and $C'_m$ are complementary: disjoint because an element of $C'_m \cap Y'_m$ is $\sigma(q)$ with $p\sigma(q) = q = 0$; spanning because for $z \in F_m$ the point $c := \sigma(p(\iota_m z))$ is smooth, so it lifts to level $m$, and $z - c \in Y'_m$. So $F = C' \oplus Y'$ is an sc-splitting ($C'$ is an sc-subspace by Proposition 1.1.7 (preprint 1.6) `Chapter1/ScBanach · ScBanach.proposition_1_1_7_iff`).

**The bijections.** $(T + S)_m$ maps $X'_m$ into $Y'_m$; it is injective on $X'_m$ because its kernel on level $m$ is $K'_m$, which meets $X'_m$ in 0; and it is onto $Y'_m$ because every $e \in E_m$ splits as $k + x$ with $k \in K'_m$, $x \in X'_m$ and $(T + S)_m e = (T + S)_m x$. This is condition (3) of Definition 1.1.9 (preprint 1.8), which the text does not verify; the exposition supplies it with the open mapping theorem, the formalization needs only the bijections (§I.6). Hence $T + S$ is sc-Fredholm `Chapter1/ScFredholm · proposition_1_1_11_preprint`.

**The index.** The text does not say that the index is preserved; the exposition does. The index of $T+S$ is the Fredholm index of $(T+S)_0 = T_0+S_0$ (§I.6); $T_0$ is Fredholm and $S_0$ compact, so by the invariance under compact perturbations (§I.7) $\operatorname{ind}(T + S) = \operatorname{ind}(T)$ `Chapter1/FredholmIndex · proposition_1_1_11_index`. On every level, $T+S$ being sc-Fredholm, its level map $(T+S)_m$ is Fredholm of index $\operatorname{ind}(T+S) = \operatorname{ind}(T)$ by the index on every level (§I.6) `Chapter1/FredholmIndex · proposition_1_1_11_index_lvl`.

## I.10 Partial quadrants, tuples and sc-domains (Definition 1.1.12 (preprint 1.11))

**The quadrant.** In $\mathbb{R}^n \oplus W$, with $\mathbb{R}^n$ carrying its constant structure, the quadrant on level $m$ is the set of $(a, w)$ with $a_i \geq 0$ for all $i$ `Chapter1/ScSmooth · quadrant`; it is convex `Chapter1/ScSmooth · quadrant_convex`, closed `Chapter1/ScSmooth · quadrant_isClosed`, has non-empty interior (the point $(1, \dots, 1, 0)$) `Chapter1/`

ScSmooth · quadrant_interior_nonempty, and the level sets of the level-0 quadrant are the level quadrants Chapter1/ScSmooth · lvlSet_quadrant.

**Partial quadrants.** $C \subseteq E_0$ is a **partial quadrant** if $C = T_0^{-1}(\text{quadrant})$ for some $n$, some sc-Banach space $W$ and some sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ Chapter1/ScSmooth · IsPartialQuadrant. The text asks in addition that $C$ be closed and convex; both follow. Indeed the level set $C_m$ is $T_m^{-1}$ of the level-$m$ quadrant Chapter1/ScSmooth · IsPartialQuadrant.lvlSet_eq, hence convex Chapter1/ScSmooth · IsPartialQuadrant.convex_lvlSet, closed Chapter1/ScSmooth · IsPartialQuadrant.isClosed_lvlSet and with non-empty interior Chapter1/ScSmooth · IsPartialQuadrant.interior_lvlSet_nonempty; for $m = 0$ this is the text's "closed convex" Chapter1/ScSmooth · IsPartialQuadrant.convex Chapter1/ScSmooth · IsPartialQuadrant.isClosed. A convex set with non-empty interior has unique differentiability at every point of its closure Mathlib/Analysis/Calculus/TangentCone/Real · uniqueDiffWithinAt_convex, so $C_m$ has unique differentiability at each of its points Chapter1/ScSmooth · IsPartialQuadrant.uniqueDiffWithinAt_lvlSet; this is the exposition's Lemma 3.D, that derivatives on a quadrant are unique, in Mathlib's language. The whole space is a partial quadrant with $n = 0$ Chapter1/ScSmooth · isPartialQuadrant_univ.

**Tuples.** $(U, C, E)$ is a tuple, $U$ a **relatively open subset of the partial quadrant** $C$, if $C$ is a partial quadrant and $U = O \cap C$ for an open $O \subseteq E_0$ Chapter1/ScSmooth · IsRelOpenInQuadrant.

**Sc-domains.** $U \subseteq E_0$ is an **sc-domain** if for every $m$ the level set $U_m$ is the intersection of an open and a convex subset of $E_m$, and $U_m$ has unique differentiability at each of its points Chapter1/ScSmooth · IsScDomain. A relatively open subset of a partial quadrant is an sc-domain: $U_m = O_m \cap C_m$ with $O_m$ open and $C_m$ convex, and unique differentiability of $C_m$ at a point survives intersection with a neighbourhood Chapter1/ScSmooth · IsRelOpenInQuadrant.isScDomain. The property passes to the tangent: $(TU)_m = U_{m+1} \times E_m$ (§I.11) is $(O \times E_m) \cap (C \times E_m)$, and unique differentiability of a product is the product of the two Chapter1/ScSmooth · IsScDomain.tangentSet; and to the level shift, since the level sets of $U_1$ in $E^1$ are the $U_{m+1}$ Chapter1/Levels · IsScDomain.shift. This is what the proofs of §§1.2–1.3 use in place of the text's claim that $T(U, C, E)$ is again a tuple (Definition 1.1.14 (preprint 1.13)). Every theorem of §§1.2–1.3 below is stated for sc-domains; the tuples are a special case.

**The shift and the tangent of a tuple (Lemmas 3.A(2) and 3.B of the exposition).** The text's claim is nevertheless true, and is proved with Chapter 2, where retracts need it. If $T : E \to \mathbb{R}^n \oplus W$ presents $C$, its levels $T_{m+1}$ form an sc-isomorphism $T^1 : E^1 \to \mathbb{R}^n \oplus W^1$ Chapter2/LevelPolyfold · ScIso.shift, and $C_1 = T_1^{-1}(\text{quadrant})$ by the level description of a partial quadrant above, so $C_1$ is a partial quadrant in $E^1$ Chapter2/LevelPolyfold · IsPartialQuadrant.shift; with $U = O \cap C$, $U_1 = O_1 \cap C_1$ is relatively open in it Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift. For the tangent, $T_{m+1} \times T_m$ maps $(TE)_m = E_{m+1} \times E_m$ onto $(\mathbb{R}^n \times W_{m+1}) \times (\mathbb{R}^n \times W_m)$, and the reassociation $((a, w), q) \mapsto (a, (w, q))$ is an sc-isomorphism onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ Chapter2/TangentBundle · tangentQuadrantIso; the composite presents $TC = \{(x, h) : x \in C_1\}$, since the quadrant condition only looks at the $\mathbb{R}^n$-coordinates of $T_1 x$ Chapter2/TangentBundle · IsPartialQuadrant.tangentSet. And $TU = (O_1 \times E_0) \cap TC$ is relatively open in $TC$ Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet. The identity $T(U^1) = (TU)^1$ is §I.11's Chapter1/Levels · ScBanach.shift_tangentSet.

## I.11 Sc⁰ maps, the tangent, sc¹ maps, sc^k (Definitions 1.1.13–1.1.15 and the unnumbered definition after them (preprint Definitions 1.12–1.15))

**Level maps.** For $f : E_0 \to F_0$ and $x \in E_m$, the level map $f_m(x)$ is the inverse image under $\iota_m$ of $f(\iota_m x)$, when $f(\iota_m x) \in$ range $\iota_m$, and a junk value otherwise `Chapter1/ScSmooth · levelMap`; so $\iota_m f_m(x) = f(\iota_m x)$ whenever the right side lifts `Chapter1/ScSmooth · toBase_levelMap`, $f_0 = f$ `Chapter1/ScSmooth · levelMap_zero`, and $f_m(x)$ is characterized by $\iota_m y = f(\iota_m x)$ `Chapter1/ScSmooth · levelMap_eq_of_toBase_eq`. Maps are total functions on $E_0$; only the values on the domain $U$ will matter.

**Sc⁰.** $f$ is sc$^0$ **on** $U$ if for every $m$ and $x \in U_m$ the point $f(\iota_m x)$ lifts to $F_m$, and $f_m$ is continuous on $U_m$ `Chapter1/ScSmooth · IsSc0` `Chapter1/ScSmooth · IsSc0.toBase_levelMap`. Then $f$ is continuous on $U$ `Chapter1/ScSmooth · IsSc0.continuousOn_base`, $f_m$ maps $U_m$ into $V_m$ when $f$ maps $U$ into $V$ `Chapter1/ScSmooth · IsSc0.levelMap_mapsTo`, the property passes to subsets `Chapter1/ScSmooth · IsSc0.mono` and depends only on the values on $U$ `Chapter1/ScSmooth · IsSc0.congr`. Level maps compose: $(g \circ f)_m = g_m \circ f_m$ on $U_m$ `Chapter1/ScSmooth · IsSc0.levelMap_comp`, so compositions of sc$^0$ maps are sc$^0$ `Chapter1/ScSmooth · IsSc0.comp`. The identity is sc$^0$ `Chapter1/ScSmooth · isSc0_id`, and an sc-operator is sc$^0$ on all of $E_0$ with level maps $T_m$ `Chapter1/ScSmooth · ScOperator.isSc0` `Chapter1/ScSmooth · ScOperator.levelMap_eq`. The text's definition says "$f(U_m) \subset V_m$" with $V$ undefined; $V$ is $U'$, and the target tuple enters only as $f(U) \subseteq U'$.

**The tangent.** $TE := E^1 \oplus E$, with levels $E_{m+1} \times E_m$ `Chapter1/ScSmooth · ScBanach.tangent` `Chapter1/ScSmooth · ScBanach.tangent_toBase`; $TU \subseteq (TE)_0 = E_1 \times E_0$ is the set of $(x, h)$ with $x \in U_1$ `Chapter1/ScSmooth · ScBanach.tangentSet`, with level-$m$ part $U_{m+1} \times E_m$ `Chapter1/ScSmooth · ScBanach.mem_lvlSet_tangentSet` `Chapter1/ScSmooth · ScBanach.lvlSet_tangentSet`, monotone in $U$ `Chapter1/ScSmooth · ScBanach.tangentSet_mono`. The level-1 part of $TU$ is $T(U_1)$ in $T(E^1) = (TE)^1$, the identity $T(U^1) = (TU)^1$ that the inductions of §1.2 use `Chapter1/Levels · ScBanach.shift_tangentSet`.

**Sc¹.** Given a family $D : E_1 \to \mathcal{L}(E_0, F_0)$, the tangent map is $T_D f(x, h) = (f_1(x), D(x)h)$ `Chapter1/ScSmooth · tangentMapWith`. $f$ is sc$^1$ **on** $U$ **with derivatives** $D$ if (1) $f$ is sc$^0$ on $U$; (2) for every $x \in U_1$ the map $f \circ \iota_1 : E_1 \to F_0$ has derivative $D(x) \circ \iota_1$ at $x$ within $U_1$ `Mathlib/Analysis/Calculus/FDeriv/Defs · HasFDerivWithinAt`; (3) $T_D f$ is sc$^0$ on $TU$ `Chapter1/ScSmooth · IsSc1With`. Condition (2) is the text's limit $|f(x+h) - f(x) - Df(x)h|_0 / |h|_1 \to 0$ over $h \in E_1$ with $x + h \in U_1$, in Mathlib's words. $f$ is sc$^1$ **on** $U$ if some $D$ works `Chapter1/ScSmooth · IsSc1`; `scDeriv` is a chosen such $D$ `Chapter1/ScSmooth · scDeriv` `Chapter1/ScSmooth · IsSc1.isSc1With_scDeriv` and `tangentMap` the tangent map built from it `Chapter1/ScSmooth · tangentMap` `Chapter1/ScSmooth · IsSc1.tangentMap_isSc0`. If $f$ maps $U$ into $V$ then $T_D f$ maps $TU$ into $TV$ `Chapter1/ScSmooth · IsSc1With.tangentMapWith_mapsTo` `Chapter1/ScSmooth · IsSc1.tangentMap_mapsTo`; level 0 of (3) says $(x, h) \mapsto D(x)h$ is continuous on $U_1 \times E_0$ `Chapter1/ScSmooth · IsSc1With.continuousOn_deriv`; and both notions depend only on the values of $f$ on $U$ `Chapter1/ScSmooth · IsSc1With.congr` `Chapter1/ScSmooth · IsSc1.congr`.

**Uniqueness of $Df(x)$.** The text writes "the" operator $Df(x)$ and "the" tangent map. If $D$ and $D'$ both satisfy (2) at $x \in U_1$ and $U_1$ has unique differentiability at $x$, then $D(x) \circ \iota_1 = D'(x) \circ \iota_1$ `Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq`; since $\iota_1$ has dense range (§I.1) and both operators are continuous, $D(x) = D'(x)$ `Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on` `Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt`. Hence on a set $U$ whose $U_1$ has unique differentiability everywhere (an sc-domain), the chosen derivative agrees on $U_1$ with any admissible family `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn` and the chosen tangent map agrees on $TU$ with any $T_D f$ `Chapter1/ScSmooth · IsSc1.tangentMap_eqOn`. This is the exposition's Lemma 3.E, and the formal proofs invoke it wherever the text says "the tangent map". The uniform version of (3) at level 0: for $x \in U_1$, a compact

$K \subseteq E_0$ and $c > 0$ there is $\delta > 0$ such that $\|D(x')w - D(x)w\| \le c$ for all $x' \in U_1$ within $\delta$ of $x$ and all $w \in K$; this is the continuity of $(x', w) \mapsto D(x')w - D(x)w$ on $U_1 \times E_0$ turned uniform over the compact $K$ `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually` `Chapter1/Properties · IsSc1With.exists_delta_deriv`, and it is the text's remark that $x \mapsto Df(x)$ is continuous for the compact-open topology.

**Compact-open continuity as an equivalence (Lemma 3.F of the exposition).** For a metric space $X$, $s \subseteq X$ and $A : X \to \mathcal{L}(Y, Z)$: $(x, h) \mapsto A(x)h$ is continuous on $s \times Y$ if and only if for every $x \in s$, compact $K \subseteq Y$ and $\varepsilon > 0$, eventually for $x'$ near $x$ in $s$, $\|A(x')w - A(x)w\| < \varepsilon$ for all $w \in K$ `Chapter1/ExpositionLemmas · continuousOn_apply_iff_uniformOnCompact`. (□ ) As above, on the subtype $s$: for each $w \in K$, $(x', w') \mapsto A(x')w' - A(x)w'$ is continuous with value 0 at $(x, w)$, so it is small near $(x, w)$, and compactness of $K$ makes the neighbourhood of $x$ uniform in $w$ `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually`; neighbourhoods in the subtype are neighbourhoods within $s$ `Mathlib/Topology/NhdsWithin · eventually_nhds_subtype_iff`. (□ ) Continuity within $s \times Y$ at $(x, w)$ is tested on sequences `Mathlib/Order/Filter/AtTopBot/CountablyGenerated · tendsto_iff_seq_tendsto`: let $(x_n, w_n) \to (x, w)$ with $x_n \in s$. The set $K = \{w\} \cup \{w_n\}$ is compact `Mathlib/Topology/Compactness/Compact · Filter.Tendsto.isCompact_insert_range`, so $\|A(x_n)w_n - A(x)w_n\| \to 0$ by the hypothesis, and $A(x)w_n \to A(x)w$ by continuity of $A(x)$; adding, $A(x_n)w_n \to A(x)w$. For an sc$^1$ map, level 0 of condition (3) is the left side for $Df$ on $U_1$, so $x \mapsto Df(x)$ is compact-open continuous on $U_1$ `Chapter1/ExpositionLemmas · IsSc1With.deriv_uniformOnCompact`. The text's other half, that norm continuity can fail, is proved by a sequence-space example in §I.24 `Chapter1/ScOneNotNormContinuous · exists_isSc1_not_continuousOn_deriv`, not by the shift map of the exposition's §6.

**Sc^k and sc-smooth.** By recursion on $k$: sc$^0$ is sc$^0$, and $f$ is sc$^{k+1}$ on $U$ if it is sc$^1$ on $U$ and the chosen tangent map is sc$^k$ on $TU$ `Chapter1/ScSmooth · IsSck`; $f$ is **sc-smooth** on $U$ if it is sc$^k$ for every $k$ `Chapter1/ScSmooth · IsScSmooth`. An sc$^k$ map is sc$^0$ `Chapter1/ScSmooth · IsSck.isSc0`, sc$^{k+1}$ implies sc$^k$ `Chapter1/ScSmooth · IsSck.of_succ` `Chapter1/ScSmooth · IsSck.of_le`, and $f$ is sc-smooth iff it is sc$^1$ with sc-smooth tangent map `Chapter1/ScSmooth · isScSmooth_iff`. On an sc-domain the notion depends only on the values on $U$, by induction with the uniqueness of the tangent map `Chapter1/ChainRule · IsSc1.tangentMap_congr` `Chapter1/ChainRule · IsSck.congr`. **Finite dimensions.** The text's remark that sc$^1$ maps between finite-dimensional spaces are the $C^1$ maps holds on every sc-domain, in particular on relatively open subsets of quadrants, with boundary. Let $V$, $W$ carry their constant structures and $U \subseteq V$ be an sc-domain. All level sets of $U$ are $U$ itself `Chapter1/ExpositionLemmas · ScBanach.lvlSet_const` and all $f \circ \iota_m$ are $f$. If $f$ is sc$^1$, then $f \circ \iota_1 = f$ is $C^1$ within $U_1 = U$ by Proposition 1.2.1 (preprint 1.16) (§I.14, `Chapter1/Properties · IsSc1With.contDiffOn_base`); if $f$ is $C^1$ within $U$, it is $C^1$ on every level, so sc$^1$ by Corollary 1.2.5 (preprint 1.20) at $k = 0$ (§I.17, `Chapter1/Properties · corollary_1_2_5_zero`) `Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_const`. Mathlib's "$C^1$ within $U$" means, on a set with unique derivatives, differentiable within $U$ at every point with $x \mapsto$ fderivWithin $f\ U\ x$ continuous on $U$ `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_fderivWithin` `Chapter1/ExpositionLemmas · isSc1_iff_fderivWithin_const`: at a boundary point of a quadrant this is the one-sided derivative along $U$, unique because the quadrant has interior (§I.10). Likewise sc-smooth is $C^\infty$ within $U$, by Proposition 1.2.3 (preprint 1.18) at $m = 0$ (§I.16) and the sc-smooth case of Corollary 1.2.5 (preprint 1.20) `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`. For the text's tuples: `Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_of_isRelOpenInQuadrant` `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant`. The open case is also checked in the test file `Test · isSc1_iff_contDiffOn_of_finiteDimensional` `Test · isScSmooth_iff_contDiffOn_of_finiteDimensional`. By §I.1 every sc-structure on a finite-dimensional space is the constant one up to sc-isomorphism.

## I.12 Theorem 1.3.1 (preprint 1.21): the chain rule

**Statement.** Let $U \subseteq C \subseteq E$ and $V \subseteq D \subseteq F$ be relatively open subsets of partial quadrants, $f$ $\mathrm{sc}^1$ on $U$ with $f(U) \subseteq V$, and $g$ $\mathrm{sc}^1$ on $V$. Then $g \circ f$ is $\mathrm{sc}^1$ on $U$ and $T(g \circ f) = Tg \circ Tf$ on $TU$ `Chapter1/ChainRule · theorem_1_3_1`. The same holds for sc-domains `Chapter1/ChainRule · theorem_1_3_1'`, and the existence part needs still less: if $g$ is $\mathrm{sc}^1$ on $V$ with derivatives $Dg$, $f$ is $\mathrm{sc}^1$ on $U$ with derivatives $Df$, $f(U) \subseteq V$, and $V_1$ is open-in-convex, then $g \circ f$ is $\mathrm{sc}^1$ on $U$ with derivatives $x \mapsto Dg(f_1(x)) \circ Df(x)$, with no hypothesis on $U$ `Chapter1/ChainRule · IsSc1With.comp`. The text's third tuple $W \subset Q \subset G$ plays no role.

**The derivative.** Fix $x \in U_1$, let $y_0 := f_1(x) \in V_1$ and $K$ a compact set containing $Df(x)(\iota_1 \bar{B}_1)$, which exists because $\iota_1$ is compact `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_compact`. Write $V_1 = O \cap C$ with $O$ open and $C$ convex, and choose $\rho > 0$ with the $\rho$-ball around $y_0$ inside $O$. Given $c > 0$, two tolerances are fixed. First, by the uniform smallness of $[Dg(y) - Dg(y_0)](w + b)$ for $y \in V_1$ near $y_0$, $w \in K$ and $b$ small, which is the continuity of $(y, b, w) \mapsto [Dg(y) - Dg(y_0)](w + b)$ on $V_1 \times F_0 \times K$ made uniform over $K$ `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually Chapter1/ChainRule · IsSc1With.exists_delta`, there is $\delta$ with this expression of norm $\leq c/2$. Second, $\varepsilon := \min(\delta/2, (c/2)/(\|Dg(y_0)\| + 1))$. For $z \in U_1$ close to $x$ three things hold eventually: $\|f_1(z) - y_0\|_1 < \min(\rho, \delta)$ (continuity of $f_1$ on $U_1$), and $r := f(\iota z) - f(\iota x) - Df(x)\iota(z - x)$ has $\|r\| \leq \varepsilon \|z - x\|_1$ (condition (2) for $f$ `Mathlib/Analysis/Calculus/FDeriv/Defs · hasFDerivWithinAt_iff_isLittleO`). Let $k := f_1(z) - y_0 \in F_1$. The segment $y_0 + tk$, $t \in [0, 1]$, lies in $V_1$: inside the $\rho$-ball, hence in $O$, and in $C$ by convexity, since both endpoints are in $C$. The text states this inclusion without comment; it is the one place where the structure of $V$ enters.

*The mean value inequality.* The text's second integral is replaced by: for $\varphi(t) := g(\iota(y_0 + tk)) - t\, Dg(y_0)(\iota k)$ on $[0, 1]$, the derivative within $[0, 1]$ is $[Dg(y_0 + tk) - Dg(y_0)](\iota k)$ (condition (2) for $g$ along the segment, composed with the affine curve), and if its norm is $\leq C$ on $[0, 1]$ then $\|\varphi(1) - \varphi(0)\| \leq C$ `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le Chapter1/ChainRule · IsSc1With.norm_sub_le_of_segment`. Now write $\iota k = f(\iota z) - f(\iota x) = s(w_1 + b)$ with $s := \|z - x\|_1$, $w_1 := Df(x)\iota(u)$ for the unit vector $u := s^{-1}(z - x)$, so $w_1 \in K$, and $b := s^{-1} r$, so $\|b\| \leq \varepsilon \leq \delta/2 < \delta$. For every $t$, $y_0 + tk$ is within $\delta$ of $y_0$, so $\|[Dg(y_0 + tk) - Dg(y_0)](\iota k)\| \leq (c/2)\, s$, and the mean value inequality gives

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0)(\iota k)\| \leq \tfrac{c}{2} \|z - x\|_1.$$

(When $z = x$ both sides vanish.) The text's first integral is not needed: $\|Dg(y_0) r\| \leq \|Dg(y_0)\|\, \varepsilon \|z - x\|_1 \leq \tfrac{c}{2} \|z - x\|_1$ directly, because $Dg(y_0)$ is bounded on $F_0$. Adding the two estimates, since $\iota k = Df(x)\iota(z - x) + r$,

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0) Df(x) \iota(z - x)\| \leq c \|z - x\|_1$$

eventually, which is condition (2) for $g \circ f$ at $x$ with the operator $Dg(y_0) \circ Df(x)$ `Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp`. The text's subsequence argument for the uniform convergence of the second integrand is the compactness step above; the text's appeal to Proposition 1.2.3 (preprint 1.18) to make $g : V_1 \to G_0$ of class $C^1$ is not used.

**The tangent map.** $g \circ f$ is $\mathrm{sc}^0$ as a composite. The tangent map of $g \circ f$ with the composed derivatives equals $T_{Dg} g \circ T_D f$ on $TU$, since $(g \circ f)_1 = g_1 \circ f_1$ on $U_1$ `Chapter1/ChainRule · IsSc1With.tangentMapWith_comp`; the right side is a composite of $\mathrm{sc}^0$ maps ($T_D f$ maps $TU$ into $TV$), hence $\mathrm{sc}^0$: condition (3) `Chapter1/ChainRule · IsSc1With.comp`. For the chosen derivatives, $T(g \circ f) = Tg \circ Tf$ on $TU$ follows from the

uniqueness of the derivative on $U_1$ (§I.11), which is where the quadrant structure of $U$ enters `Chapter1/ChainRule · theorem_1_3_1' Chapter1/ChainRule · theorem_1_3_1`.

**Iterating.** If $U$, $V$ are sc-domains, $f$ is sc$^k$ on $U$ with $f(U) \subseteq V$ and $g$ is sc$^k$ on $V$, then $g \circ f$ is sc$^k$ on $U$: induction on $k$, the step being the theorem for the sc$^1$ parts, the induction hypothesis for $Tg \circ Tf$ on the sc-domains $TU$, $TV$, and the identity $T(g \circ f) = Tg \circ Tf$ with the independence of sc$^k$ from the values off $TU$ `Chapter1/ChainRule · IsSck.comp Chapter1/ChainRule · IsSck.comp_of_isRelOpenInQuadrant`. Sc-smooth maps compose to sc-smooth maps `Chapter1/ChainRule · IsScSmooth.comp`. This is the exposition's Corollary 4.A.

## I.13 The compact-open lemma and the level operators $Df(x) : E_m \to F_m$

The proofs of Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) in the 2010 paper all rest on one mechanism, which the formal proof isolates.

**The lemma, $k = 0$.** Let $A : X \to \mathcal{L}(Y, Z)$ with $(x, y) \mapsto A(x)y$ continuous on $s \times Y$, and $\iota : Y' \to Y$ a compact operator. Then $x \mapsto A(x) \circ \iota$ is continuous on $s$ into $\mathcal{L}(Y', Z)$ with the operator norm `Chapter1/Levels · continuousOn_clm_comp_of_isCompactOperator`. Proof: the image of the unit ball of $Y'$ under $\iota$ lies in a compact $K$; by uniform continuity on compact sets `Chapter1/Levels · exists_delta_of_continuousOn`, for $x' \in s$ near $x$ and all $w \in K$, $\|A(x')w - A(x)w\| \le \varepsilon/2$, so $\|(A(x') - A(x)) \circ \iota\| \le \varepsilon/2$ on unit vectors. This is the 2010 paper's subsequence argument (unit vectors $h_n \in E_{m+1}$ have a subsequence converging in $E_m$) in uniform form.

**The lemma, $C^k$.** Let $s$ be open-in-convex with unique differentiability at every point, and $(x, y) \mapsto A(x)y$ of class $C^k$ on $s \times Y$. Then $x \mapsto A(x) \circ \iota$ is $C^k$ on $s$ `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. Induction on $k$ over all $Z$ and $A$. For $k + 1$: the partial derivative in $x$, $B(x)y := D_x(A(x)y)$, is $C^k$ in $(x, y)$ `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn`, linear and continuous in $y$ for $x \in s$ (differentiate the linearity identities and use uniqueness of derivatives), so $B : s \to \mathcal{L}(Y, \mathcal{L}(X, Z))$; by the induction hypothesis $x \mapsto B(x) \circ \iota$ is $C^k$, and so is its flip $G'(x) \in \mathcal{L}(X, \mathcal{L}(Y', Z))$. $G'$ is the derivative of $x \mapsto A(x) \circ \iota$: for $z \in s$ near $x$ the segment from $x$ to $z$ lies in $s$ (open-in-convex), and for a unit vector $v \in Y'$ the mean value inequality applied to $\tau \mapsto A(x + \tau(z - x))(\iota v) - \tau B(x)(\iota v)(z - x)$ `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le`, with the uniform continuity of $B$ on $\{x\} \times \iota(\bar{B}_1)$, bounds $\|A(z)\iota v - A(x)\iota v - B(x)(\iota v)(z - x)\| \le c\|z - x\|$ uniformly in $v$. So $x \mapsto A(x) \circ \iota$ is differentiable within $s$ with a $C^k$ derivative, hence $C^{k+1}$ `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`. The 2010 paper's Proposition 2.3 carries the same information in the multilinear maps $\Gamma(x)(h_1, \dots, h_k, h)$ and a differentiation under the integral sign; the lemma replaces both.

**The level operators.** Let $f$ be sc$^1$ on $U$ with derivatives $D$, and $x \in E_{m+1}$. The **level-$m$ operator** $Df(x) : E_m \to F_m$ is the continuous linear $T$ with $\iota_m \circ T = D(\iota x) \circ \iota_m$, when such $T$ exists (it is unique by injectivity), and 0 otherwise `Chapter1/Levels · levelDeriv Chapter1/Levels · toBase_levelDeriv`; on level 0 it is $D(x)$ itself `Chapter1/Levels · levelDeriv_zero`. For $x \in U_{m+1}$ it exists: the level-$m$ tangent map sends $(x, v) \in U_{m+1} \times E_m$ to a pair whose second component $T_m(v)$ satisfies $\iota_m T_m(v) = D(\iota x)\iota_m v$; $T_m$ is linear (by injectivity of $\iota_m$) and continuous (condition (3) on level $m$) `Chapter1/Levels · IsSc1With.exists_levelDeriv Chapter1/Levels · IsSc1With.toBase_levelDeriv`. So the level-$m$ tangent map is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.levelMap_tangentMapWith`, and condition (3)

says exactly that $(x,h) \mapsto Df(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv`: this is condition (2) of Proposition 1.2.1 (preprint 1.16). By the compact-open lemma with $\iota = \mathrm{incl}_m : E_{m+1} \to E_m$, the map $x \mapsto Df(x) \circ \mathrm{incl}_m$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$ in the operator norm `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv_clm`: the continuity of $df$ in condition (1) of Proposition 1.2.1 (preprint 1.16). The base case, $x \mapsto D(x) \circ \iota_1$ continuous $U_1 \to \mathcal{L}(E_1, F_0)$, is also proved directly `Chapter1/Properties · IsSc1With.continuousOn_deriv_clm`, and with it $f \circ \iota_1 : U_1 \to F_0$ is $C^1$ `Chapter1/Properties · IsSc1With.contDiffOn_base`.

**Level-raising: differentiability of $f : U_{m+1} \to F_m$.** Let $U$ be an sc-domain. For $x \in U_{m+1}$, the map $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ has derivative $Df(x) \circ \mathrm{incl}_m$ at $x$ within $U_{m+1}$ `Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level`. This is the 2010 paper's integral representation $f(x+k) - f(x) = \int_0^1 Df(x+\tau k)k\,d\tau$. For $z \in U_{m+1}$ near $x$ and $k := z - x$, the segment $x + \tau k$ lies in $U_{m+1}$ (open-in-convex). The integrand $g(\tau) := Df(x+\tau k)(\mathrm{incl}_m k)$ is continuous into $F_m$ by condition (3) on level $m$, so its Bochner integral over $[0,1]$ exists in $F_m$. Its image under $\iota_m$ is the integral of $D(\iota(x+\tau k))\iota_{m+1}k$ in $F_0$, which by the fundamental theorem of calculus for the $C^1$ curve $\tau \mapsto f(\iota(x+\tau k))$ (condition (2) for $f$, composed with the affine curve) `Mathlib/MeasureTheory/Integral/IntervalIntegral/FundThmCalculus · integral_eq_sub_of_hasDerivAt_of_le` equals $f(\iota z) - f(\iota x) = \iota_m(f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x))$. By injectivity, $f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) = \int_0^1 g$ in $F_m$. Then

$$\left\| f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) - Df(x)\mathrm{incl}_m k \right\|_m \le \sup_\tau \|(Df(x+\tau k) - Df(x)) \circ \mathrm{incl}_m\|\, \|k\|_{m+1},$$

and the supremum tends to $0$ by the norm continuity just proved. The 2010 paper forms the integral in $F_0$ and remarks that the integrand is continuous into $F_m$; here it is formed in $F_m$ and pushed down. Consequently $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ is $C^1$ within $U_{m+1}$ `Chapter1/Levels · IsSc1With.contDiffOn_level`, which is condition (1) of Proposition 1.2.1 (preprint 1.16) in full.

**The level operator is the derivative of $f_m$.** If $f_m : U_m \to F_m$ happens to be differentiable at $\mathrm{incl}_m x$ within $U_m$ with derivative $L$, then $L = Df(x)$: both $\iota_m \circ L \circ \mathrm{incl}_m$ and $D(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at $x$ within $U_{m+1}$, hence equal by uniqueness; and since $\mathrm{incl}_m$ has dense range, $\iota_m \circ L = D(\iota x) \circ \iota_m$ on all of $E_m$ `Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on` `Chapter1/Levels · IsSc1With.levelDeriv_eq`. This is the exposition's Lemma 3.H and the key to Proposition 1.2.4 (preprint 1.19).

## I.14 Proposition 1.2.1 (preprint 1.16): sc¹ in terms of the levels

**The level conditions.** For a family $D_m(x) \in \mathcal{L}(E_m, F_m)$, $x \in E_{m+1}$, the **level conditions** are: (a) for every $m$ and $x \in U_{m+1}$, $f_m \circ \mathrm{incl}_m$ has derivative $D_m(x) \circ \mathrm{incl}_m$ at $x$ within $U_{m+1}$; (b) for every $m$, $(x,h) \mapsto D_m(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1Levels`. These are the text's (1) and (2) with $m-1$ renamed $m$ and the extensions $Df(x)$ made an explicit family. From them, $x \mapsto D_m(x) \circ \mathrm{incl}_m$ is norm continuous (compact-open lemma) `Chapter1/Levels · IsSc1Levels.continuousOn_clm`, so $f : U_{m+1} \to F_m$ is $C^1$ on an sc-domain `Chapter1/Levels · IsSc1Levels.contDiffOn`: the rest of the text's (1).

**Statement.** For an sc-domain $U$ and $f$ sc$^0$ on $U$: $f$ is sc$^1$ on $U$ iff some family $D_m$ satisfies the level conditions `Chapter1/Levels · proposition_1_2_1`.

**(□ ).** The level operators of §I.13 satisfy (a) (level-raising) and (b) (condition (3)) `Chapter1/Levels · IsSc1With.isSc1Levels`.

**(□ ), and what the text hides.** The 2010 paper calls this direction clear. It is, once one knows that the operators $D_m(x)$ for different $m$ are restrictions of one operator, which the text's notation $Df(x)$ assumes. Proof: for $x \in U_{m+1}$, both $\iota_m \circ D_m(x) \circ \mathrm{incl}_m$ and $D_0(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at $x$ within $U_{m+1}$ (compose (a) at level $m$ with $\iota_m$, and (a) at level 0 with the inclusion $E_{m+1} \to E_1$), so they agree by uniqueness, and since $\mathrm{incl}_m$ has dense range, $\iota_m \circ D_m(x) = D_0(\iota x) \circ \iota_m$ `Chapter1/Levels · IsSc1Levels.toBase_apply`. Then $f$ is sc$^1$ with derivatives $D_0$: condition (2) is (a) at level 0; condition (3) at level $m$ reads, after the compatibility, $(x, h) \mapsto (f_{m+1}(x), D_m(x)h)$ on $U_{m+1} \times E_m$, which is continuous by (b) and the continuity of $f_{m+1}$ `Chapter1/Levels · IsSc1Levels.isSc1With`.

**The "in particular".** If $f$ is sc$^1$ with derivatives $D$ and $x \in U_1$ is a smooth point, then $D(x) : E_0 \to F_0$ is an sc-operator: lift $x$ to $y \in U_{m+1}$; the level-$m$ tangent map at $(y, v)$ has second component $T_m(v)$ with $\iota_m T_m(v) = D(x)\iota_m v$, linear and continuous in $v$, which is the level-$m$ map required by `IsScOperator` `Chapter1/Properties · IsSc1With.scOperator_deriv`.

## I.15 Proposition 1.2.2 (preprint 1.17): lifting the index

**Statement.** If $U$ is an sc-domain in $E$ and $f$ is sc$^k$ on $U$, then $f_1 : E_1 \to F_1$ is sc$^k$ on $U_1$, as a map between the shifted spaces $E^1$ and $F^1$ `Chapter1/Levels · IsSck.shift`; sc-smooth maps lift to sc-smooth maps `Chapter1/Levels · IsScSmooth.shift`. The target $V$ of the text plays no role.

**Proof.** The level maps of $f_1$, computed in the shifted spaces, are the level maps of $f$ shifted by one `Chapter1/Levels · IsSc0.levelMap_shift`, so an sc$^0$ map lifts to an sc$^0$ map, with no hypothesis on $U$ `Chapter1/Levels · IsSc0.shift`. For sc$^1$ with derivatives $D$: the derivatives of $f_1$ are the level-1 operators $Df(x) : E_1 \to F_1$, $x \in U_2$; condition (2) for $f_1$ is the level-raising statement of §I.13 at $m = 1$, and condition (3) is the shift of condition (3) for $f$, since $T(U_1) = (TU)^1$ (§I.11) and the level tangent maps are $(x, h) \mapsto (f_{m+2}(x), Df(x)h)$ `Chapter1/Levels · IsSc1With.shift`. For sc$^{k+1}$: the tangent map of $f_1$ agrees on $T(U_1)$ with the level-1 map of $Tf$ (uniqueness of derivatives on the sc-domain $U_1$), which is sc$^k$ by the induction hypothesis applied to $Tf$ on $TU$ `Chapter1/Levels · IsSck.shift`. This is the 2010 paper's argument, with the compatibility $T(U^1) = (TU)^1$ and the restriction of $Df$ proved rather than assumed.

## I.16 Proposition 1.2.3 (preprint 1.18): sc^k maps are $C^k$ from $U_{m+k}$ to $F_m$

**Statement.** Let $U$ be an sc-domain and $f$ sc$^k$ on $U$. Then $f_m \circ (E_{m+k} \to E_m) : U_{m+k} \to F_m$ is $C^k$ within $U_{m+k}$ for every $m$ `Chapter1/Levels · IsSck.contDiffOn_levels`; $f : U_{m+l} \to F_m$ is $C^l$ for $l \le k$ `Chapter1/Levels · IsSck.contDiffOn_levels_of_le`; in particular $f \circ \iota_k : U_k \to F_0$ is $C^k$ `Chapter1/Levels · IsSck.contDiffOn_base`; and an sc-smooth map is $C^k$ from $U_{m+k}$ to $F_m$ for all $m$, $k$ `Chapter1/Levels · IsScSmooth.contDiffOn_levels`.

**Proof.** The 2010 paper proves the case $m = 0$ and reduces to it with Proposition 1.2.2 (preprint 1.17) applied $m$ times. The formal proof handles all levels at once, by induction on $k$ over all sc-domains, with a compatible level-lowering operator $\rho : E_n \to E_m$, $n = m + k$, $\iota_m \rho = \iota_n$, as a parameter `Chapter1/Levels · IsSck.contDiffOn_levels'`; this avoids identifying $(E^m)_k$ with $E_{m+k}$. For $k = 0$, $\rho$ is the identity and the claim is the continuity of $f_m$. For $k + 1$: write $\rho = \mathrm{incl}_m \circ \rho'$ with $\rho' : E_{m+k+1} \to E_{m+1}$ `Chapter1/Levels · ScBanach.incl_shift_inclK Chapter1/Levels · ScBanach.shift_inclK_mem_lvlSet`. By the level-raising statement of §I.13, $f_m \circ \rho$ has derivative $x \mapsto Df(\rho' x) \circ \rho$ on $U_{m+k+1}$, where $Df(\cdot)$ is the level-$m$ oper-

ator. It remains to see that this derivative is $C^k$ into $\mathcal{L}(E_{m+k+1}, F_m)$. The induction hypothesis for the $\mathrm{sc}^k$ map $Tf$ on the sc-domain $TU$, with the level-lowering operator of $TE$ (which is the pair of those of $E^1$ and $E$ `Chapter1/Levels · ScBanach.tangent_inclK`), says that the level-$m$ tangent map composed with it, $(x,h) \mapsto (f_{m+1}(\rho' x), Df(\rho' x)(E_{m+k} \to E_m)h)$, is $C^k$ on $U_{m+k+1} \times E_{m+k}$; its second component is $(x,h) \mapsto A(x)h$ with $A(x) := Df(\rho' x) \circ (E_{m+k} \to E_m)$. The compact-open lemma with $\iota = \mathrm{incl}_{m+k} : E_{m+k+1} \to E_{m+k}$ makes $x \mapsto A(x) \circ \mathrm{incl}_{m+k} = Df(\rho' x) \circ \rho$ of class $C^k$ `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. A map that is differentiable within an sc-domain with a $C^k$ derivative is $C^{k+1}$ `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`.

## I.17 Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20): sc^{k+1} from classical smoothness on the levels

**Statement.** Let $U$ be an sc-domain and $f$ $\mathrm{sc}^0$ on $U$. If for every $m \geq 0$ and $0 \leq l \leq k$ the map $f_m \circ (E_{m+l} \to E_m) : U_{m+l} \to F_m$ is $C^{l+1}$ within $U_{m+l}$, then $f$ is $\mathrm{sc}^{k+1}$ on $U$ `Chapter1/LevelsWeak · proposition_1_2_4_text`. This is the text's statement; its "$f$ induces a map $f : U_{m+l} \to F_m$" is the $\mathrm{sc}^0$ hypothesis (values of level $m$ on $U_m$; the continuity is then the case $l = 0$). The proof is the 2010 paper's (its Proposition 2.4) and lives in `Chapter1/LevelsWeak`; an earlier, shorter proof under a stronger hypothesis is kept and described at the end of this section.

**Why the hypothesis does not induct.** The obvious induction would show that $f$ is $\mathrm{sc}^1$ and that $Tf$ satisfies the hypothesis with $k-1$. It does not. The map $Tf : (TU)_{m+l} \to (TF)_m$ has the second component $(x,h) \mapsto Df(x)h$ on $U_{m+l+1} \times E_{m+l} \to F_m$, and $Df(x)$ on $E_{m+l}$ is the derivative of $f : U_{m+l} \to F_m$ (§I.13), a $C^{l+1}$ map; so this component is only $C^l$, one order short of the $C^{l+1}$ that the hypothesis for $Tf$ would ask. What survives the induction is a description of the components of the iterated tangent maps: they are built from the derivatives $d^j f(x)$ of $f$ itself, evaluated at points and vectors of various levels, and the regularity of $f$ on the coarser levels supplies the missing order. The 2010 paper's inductive statement $(\mathrm{S}_k)$ carries exactly this description. The formal proof organizes it in six steps: (1) the derivatives $d^j f(x)$ on all level pairs, with their compatibility, form a *derivative tower*; (2) an *elementary map* is $d^j f$ evaluated at sc-operators; (3) it is $\mathrm{sc}^0$, and $\mathrm{sc}^1$ with an explicit derivative, by the level criterion of Proposition 1.2.1 (preprint 1.16); (4) its tangent map is the pair of an elementary map and a sum of elementary maps; (5) by induction on $s$, every elementary map with $j + s \leq k+1$ is $\mathrm{sc}^s$; (6) $f$ is the elementary map with $j = 0$.

**The derivative tower.** A derivative tower of order $k+1$ for $f$ on $U$ `Chapter1/LevelsWeak · IsDerivTower` consists of base-level maps $\Phi_i(x)(h_1, \dots, h_i) \in F_0$ $(x, h_a \in E_0)$ and, for $x \in U_{m+l}$, continuous $i$-linear maps $\Psi_{i,m,l}(x)$ on $E_{m+l}$ with values in $F_m$, given for $i \leq l+1$ and $i \leq k+1$, such that: $\Phi_0 = f$; $\iota_m \Psi_{i,m,l}(x)(h_1, \dots, h_i) = \Phi_i(\iota_{m+l} x)(\iota_{m+l} h_1, \dots, \iota_{m+l} h_i)$, i.e. all the $\Psi$ are one map seen on different levels; $\Psi_{i,m,l}$ is continuous on $U_{m+l}$; and for $i \leq l$, $i \leq k$ it is differentiable within $U_{m+l}$ at every point, with derivative $\xi \mapsto \Psi_{i+1,m,l}(x)(\xi, \cdot)$. The ranges of $i$ are exactly what "$f : U_{m+l} \to F_m$ is $C^{l+1}$" provides; the compatibility is the text's use of one symbol $d^j f(x)$ on all levels. Two values with the same images in $E_0$ agree, by injectivity of $\iota_m$ `Chapter1/LevelsWeak · IsDerivTower.eq_of_toBase`.

*The tower of a map satisfying the hypothesis.* $\Psi_{i,m,l}$ is Mathlib's $i$-th derivative within $U_{m+l}$ of $f_m \circ (E_{m+l} \to E_m)$ `Chapter1/LevelsWeak · levelDerivTower` `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin`, and $\Phi_i$ is the $i$-th derivative of $f : U_{i-1} \to F_0$ at the lifts of its arguments to $E_{i-1}$ (arbitrary off the image of $E_{i-1}$) `Chapter1/LevelsWeak · baseDerivTower`. For $l > k$ the

hypothesis is used with order $k+1$: $f : U_{m+l} \to F_m$ is the $C^{k+1}$ map $f : U_{m+k} \to F_m$ restricted along the inclusion $E_{m+l} \to E_{m+k}$, and $U_{m+l}$ is the preimage of $U_{m+k}$ `Chapter1/LevelsWeak · ScBanach.inclLE` `Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet`; so $f : U_{m+l} \to F_m$ is $C^{\min(l,k)+1}$ for all $m$, $l$ `Chapter1/LevelsWeak · contDiffOn_levels_min`. Continuity of $\Psi_{i,m,l}$ for $i \le \min(l,k)+1$ is the continuity of the iterated derivatives of a $C^{\min(l,k)+1}$ map `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn.continuousOn_iteratedFDerivWithin`; for $i \le \min(l,k)$ the $i$-th derivative is $C^1$ `Mathlib/Analysis/Calculus/ContDiff/Comp · ContDiffWithinAt.iteratedFDerivWithin_right`, and its derivative is the $(i+1)$-st derivative with the new vector in front `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_succ_apply_left`. Both use the uniqueness of derivatives on the level sets of the sc-domain. The compatibility `Chapter1/LevelsWeak · levelDerivTower_toBase` is proved in three steps, for $i \le l+1$ (so that $i-1 \le m+l$): (i) composing on the left with the continuous linear $\iota_m$ commutes with iterated derivatives within a set `Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_left`; (ii) on $U_{m+l}$, $\iota_m \circ f_m \circ (E_{m+l} \to E_m)$ equals $(f : U_{i-1} \to F_0) \circ (E_{m+l} \to E_{i-1})$, since $f$ is sc$^0$ `Chapter1/ScSmooth · IsSc0.toBase_levelMap`, and a derivative within a set depends only on the values on the set `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_congr`; (iii) composing on the right with the continuous linear inclusion $E_{m+l} \to E_{i-1}$ commutes as well, the preimage of $U_{i-1}$ being $U_{m+l}$ `Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_right`. With the lifts in the definition of $\Phi_i$ `Chapter1/LevelsWeak · baseDerivTower_toBase` this gives the tower `Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels`.

**Shifted spaces.** $E^p$ is the sc-Banach space with $(E^p)_l = E_{p+l}$ by definition `Chapter1/LevelsWeak · ScBanach.shiftAdd` (the $p$-fold iterate of the shift of §I.1 has these levels only up to rewriting $p+l$). The identifications $E \cong E^0$ and $(E^p)^1 \cong E^{p+1}$ are sc-isomorphisms whose level maps are identities transported along $l = 0+l$, resp. $p+(l+1) = p+1+l$ `Chapter1/LevelsWeak · ScBanach.castCLE` `Chapter1/LevelsWeak · ScBanach.shiftAddZeroIso` `Chapter1/LevelsWeak · ScBanach.shiftAddShiftIso`; they are used only through their compatibility with the inclusions into the base level `Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm`. For $b \le a$ the inclusion $E_a \to E_b$ is compatible with the $\iota$'s `Chapter1/LevelsWeak · ScBanach.toBase_inclLE`.

**Elementary maps.** Elementary data on an sc-Banach space $X$ `Chapter1/LevelsWeak · ElemData` are an sc-operator $\pi_0 : X \to E^p$ (the base point), $i$ sc-operators $\pi_a : X \to E^{n_a}$ (the vectors) and a target shift $m$. The elementary map is

$$g(z) = d^i f(\pi_0 z)(\pi_1 z, \dots, \pi_i z) := \Phi_i(\pi_0 z)(\pi_1 z, \dots, \pi_i z),$$

lifted to $F_m = (F^m)_0$ `Chapter1/LevelsWeak · ElemData.map`, on the domain $W = \pi_0^{-1}(U_p)$ `Chapter1/LevelsWeak · ElemData.dom`, whose level-$l$ set is $\pi_0^{-1}(U_{p+l})$ `Chapter1/LevelsWeak · ElemData.mem_lvlSet_dom`. The exposition's map of type $(p, l; n_1, \dots, n_j)$ on $U^p \oplus E^{n_1} \oplus \cdots \oplus E^{n_j}$ is the case $X = E^p \oplus E^{n_1} \oplus \cdots$ with the projections as slots. The conditions are `Chapter1/LevelsWeak · ElemData.Good`

$$m+i \le p, \qquad m+i \le n_a+1 \ (1 \le a \le i), \qquad i \le k.$$

The second is the text's $j+l-1 \le n_i$. The first is $j+l \le k+1$ for the base point $x \in U_{k+1}$, which the text derives from $j+l-1 \le n_i \le k$ (and states separately, as $l \le k+1$, only for $j=0$ in the exposition). The third bounds the order of derivative that will be used. The text's upper bound $n_i \le k$ is not needed.

On level $l$, the base point and the vectors pushed down to a common level $q$ ($q \le p+l$, $q \le n_a+l$) are $\rho_0 = (E_{p+l} \to E_q) \circ \pi_0$ and $\rho_a = (E_{n_a+l} \to E_q) \circ \pi_a$ `Chapter1/LevelsWeak · ElemData.ρ₀` `Chapter1/LevelsWeak · ElemData.ρ`. Seen in $F_0$, $g$ at a point of level $l$ is $\Phi_i$ at the base-level images of $\pi_0 z$ and $\pi_a z$

Chapter1/LevelsWeak · ElemData.map_toBase, and its level-$l$ map is $z \mapsto \Psi_{i,m+l,\lambda}(\rho_0 z)(\rho_1 z, \dots, \rho_i z)$ for any $\lambda$ with $i \le \lambda+1$, $m+l+\lambda \le p+l$ and $m+l+\lambda \le n_a+l$ Chapter1/LevelsWeak · ElemData.levelMap_eq. The first two conditions make $\lambda = i-1$ admissible, and $i \le k$ gives $i \le k+1$, so the tower value exists. Hence $g$ is sc$^0$: its values at level-$l$ points lie in $F_{m+l}$, and its level-$l$ map is continuous on $W_l$ as the evaluation of the continuous $\Psi_{i,m+l,i-1} \circ \rho_0$ at the continuous $(\rho_1, \dots, \rho_i)$ Chapter1/LevelsWeak · ElemData.isSc0.

**The sc$^1$ step.** For $z \in W_{l+1}$ the candidate derivative $Dg(z) : X_l \to F_{m+l}$ is the derivative (3.5) of the exposition Chapter1/LevelsWeak · ElemData.deriv Chapter1/LevelsWeak · ElemData.deriv_apply:

$$Dg(z)\zeta = \Psi_{i+1,m+l,i}(\rho_0 z)(\rho_0\zeta, \rho_1 z, \dots, \rho_i z) + \sum_{a=1}^{i} \Psi_{i,m+l,i-1}(\rho_0 z)(\rho_1 z, \dots, \rho_a\zeta, \dots, \rho_i z),$$

the first term on the uniform level $m+l+i$, the sum on $m+l+i-1$. The conditions put every argument where it belongs: $\rho_0 z$ and $\rho_a z$ (with $z$ of level $l+1$) in $E_{m+l+i}$, because $m+i \le p+1$ and $m+i \le n_a+1$; $\rho_0\zeta$ (with $\zeta$ of level $l$) in $E_{m+l+i}$, because $m+i \le p$; $\rho_a\zeta$ in $E_{m+l+i-1}$, because $m+i-1 \le n_a$; and $\Psi_{i+1,m+l,i}$ exists because $i+1 \le k+1$, which is where $i \le k$ enters. On the base level, $Dg(z)\zeta$ is $\Phi_{i+1}(\pi_0 z)(\pi_0\zeta, \pi z) + \sum_a \Phi_i(\pi_0 z)(\pi z$ with $\pi_a z := \pi_a\zeta)$ Chapter1/LevelsWeak · ElemData.deriv_toBase. The level criterion of Proposition 1.2.1 (preprint 1.16) (§I.14) asks two things.

*Differentiability of $g : W_{l+1} \to F_{m+l}$ with derivative $Dg(z)$ restricted to $X_{l+1}$.* On $W_{l+1}$, the level-$l$ map of $g$ is $z' \mapsto \Psi_{i,m+l,i}(\rho_0 z')(\rho_1 z', \dots, \rho_i z')$ with the slots taken at level $l+1$ (the formula above with $\lambda = i$) Chapter1/LevelsWeak · ElemData.levelMap_incl_eq. In the tower, $\Psi_{i,m+l,i}$ is differentiable with derivative $\Psi_{i+1,m+l,i}$ ($i \le i$ and $i \le k$); the slots are continuous linear; and the evaluation $(A, h) \mapsto A(h_1, \dots, h_i)$ of continuous multilinear maps is differentiable with derivative $(B, \eta) \mapsto B(h) + \sum_a A(h$ with $h_a := \eta_a)$ Chapter1/LevelsWeak · hasFDerivAt_cmm_eval, proved from Mathlib's derivative of a multilinear map in its arguments Mathlib/Analysis/Calculus/FDeriv/Analytic · ContinuousMultilinearMap.hasFDerivAt and the linearity in $A$ (the evaluation written as a flipped multilinear map Chapter1/LevelsWeak · cmm_eval_eq). The chain rule gives a derivative whose first term is that of $Dg$ (the base point restricted from level $l+1$ to level $l$ is the same vector Chapter1/LevelsWeak · ElemData.ρ₀_incl) and whose sum terms are values of $\Psi_{i,m+l,i}$; these equal the $\Psi_{i,m+l,i-1}$ terms of $Dg$ because they have the same images in $E_0$ (the tower's compatibility) Chapter1/LevelsWeak · ElemData.hasFDerivWithinAt_level.

*Continuity of $(z, \zeta) \mapsto Dg(z)\zeta$ on $W_{l+1} \times X_l$:* $\Psi_{i+1,m+l,i}$ and $\Psi_{i,m+l,i-1}$ are continuous on their level sets, and the slots, the insertion of $\rho_a\zeta$ and the evaluation are continuous Chapter1/LevelsWeak · ElemData.continuousOn_deriv.

So $g$ satisfies the level conditions Chapter1/LevelsWeak · ElemData.isSc1Levels Chapter1/Levels · IsSc1Levels, and when $W$ is an sc-domain it is sc$^1$ with derivative $Dg$ at level 0 Chapter1/Levels · IsSc1Levels.isSc1With Chapter1/LevelsWeak · ElemData.isSc1With. This is the text's "$\Gamma$ is sc$^1$", with the level criterion in place of the direct check of Definition 1.1.15 (preprint 1.14) that the exposition carries out.

**The tangent map of an elementary map.** On $TX = X^1 \oplus X$ a slot $\pi : X \to E^n$ gives two slots: $\pi$ on the first factor, $TX \to E^{n+1}$ through $(E^n)^1 \cong E^{n+1}$ Chapter1/LevelsWeak · ScOperator.tangentX, and $\pi$ on the second factor, $TX \to E^n$ Chapter1/LevelsWeak · ScOperator.tangentH. Three kinds of elementary data on $TX$ arise from $g$. The first, $g'$, has all slots on the first factor, with levels $p+1$, $n_a+1$ and target shift $m+1$ Chapter1/LevelsWeak · ElemData.shiftData: it is $g$ one level up. The second, $g_0$, has the base point $\pi_0$ on the first factor (level $p+1$), $i+1$ vectors, namely $\pi_0$ on the second factor (level $p$) followed by the $\pi_a$ on the first (levels $n_a+1$), and target shift $m$ Chapter1/LevelsWeak · ElemData.derivData: it is the term $d^{i+1}f(x)(\delta x, h)$. The third, $g_a$ for $1 \le a \le i$, is like $g'$ but with $\pi_a$ moved to the second factor (level $n_a$) and target

shift $m$ Chapter1/LevelsWeak · ElemData.slotData: it is the term $d^i f(x)(h \text{ with } h_a := \delta h_a)$. All satisfy the conditions, which is the text's "one checks that the new indices satisfy the required inequalities": for $g'$, $m+1+i \le p+1$ and $m+1+i \le n_a+2$ Chapter1/LevelsWeak · ElemData.Good.shiftData; for $g_0$, $m+i+1 \le p+1$ (for the base point and for the new first vector, of level $p$) and $m+i+1 \le n_a+2$, and $i+1 \le k$, which has to be assumed Chapter1/LevelsWeak · ElemData.Good.derivData; for $g_a$, $m+i \le p+1$, $m+i \le n_a+1$ for the moved slot and $m+i \le n_b+2$ for the others Chapter1/LevelsWeak · ElemData.Good.slotData. The domain of all three (they share the base point $\pi_0$ on the first factor) is $TW$ Chapter1/LevelsWeak · ElemData.tangentSet_dom.

If $W$ is an sc-domain and $i+1 \le k$, the tangent map of $g$ agrees on $TW$ with the pair ($g'$ carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$) Chapter1/LevelsWeak · ElemData.tangentMap_eq0n. Proof: by the uniqueness of the derivative on the sc-domain $W$, the tangent map is ($g$ on level 1, $Dg$) Chapter1/ScSmooth · IsSc1.tangentMap_eq0n; the first components agree in $F_0$, both being $\Phi_i$ at the base-level images of the level-1 slots Chapter1/LevelsWeak · ElemData.map_toBase Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm; the second components agree in $F_0$ term by term, by the base-level formula for $Dg$ Chapter1/LevelsWeak · ElemData.deriv_toBase.

**Sums.** The constant map 0 is $\mathrm{sc}^k$, being the zero sc-operator Chapter1/LevelsWeak · ScOperator.zeroOp Chapter1/LevelsWeak · IsSck.zero. If $g_1$, $g_2$ are $\mathrm{sc}^k$ on an sc-domain, $g_1+g_2$ is the addition $Y \oplus Y \to Y$ Chapter1/LevelsWeak · ScOperator.addOp, an sc-operator and hence sc-smooth Chapter2/TangentBundle · ScOperator.isScSmooth, composed with the pairing $(g_1, g_2)$, which is $\mathrm{sc}^k$ Chapter2/StrongBundle · IsSck.scPair; so it is $\mathrm{sc}^k$ by the chain rule (§I.12, Chapter1/ChainRule · IsSck.comp) Chapter1/LevelsWeak · IsSck.add. Finite sums follow by induction over the index set Chapter1/LevelsWeak · IsSck.sum. Scalar multiples are not needed: the text's "linear combinations" are sums.

**The induction** $(\mathrm{S}_k)$**.** *Claim:* for every $s$, every sc-Banach space $X$ and all elementary data on $X$ satisfying the conditions, with sc-domain $W$ and $i+s \le k+1$, the elementary map is $\mathrm{sc}^s$ on $W$ Chapter1/LevelsWeak · ElemData.isSck. The induction is on $s$, for all $X$, $i$ and data at once. For $s=0$ this is the $\mathrm{sc}^0$ statement. For $s+1$: $g$ is $\mathrm{sc}^1$ by the $\mathrm{sc}^1$ step, and it remains to show that its tangent map is $\mathrm{sc}^s$. If $s=0$, the tangent map of an $\mathrm{sc}^1$ map is $\mathrm{sc}^0$ Chapter1/ScSmooth · IsSc1.tangentMap_isSc0. If $s \ge 1$, then $i+1 \le k$ (from $i+s+1 \le k+1$), so the tangent map agrees on $TW$ with ($g'$ carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$); $TW$ is an sc-domain Chapter1/ScSmooth · IsScDomain.tangentSet, equal to the domain of the three new data Chapter1/LevelsWeak · IsScDomain.of_eq Chapter1/LevelsWeak · IsSck.of_eq_set, and $\mathrm{sc}^s$ passes to maps that agree on the domain Chapter1/ChainRule · IsSck.congr. The pair is $\mathrm{sc}^s$ if its two components are Chapter2/StrongBundle · IsSck.scPair. Now $g'$ and the $g_a$ have $i$ vectors and $i+s \le k+1$, and $g_0$ has $i+1$ vectors and $(i+1)+s \le k+1$, so all are $\mathrm{sc}^s$ by the induction hypothesis (applied on $X' = TX$); $g'$ composed with the sc-isomorphism $(F^m)^1 \cong F^{m+1}$, an sc-operator Chapter1/ScBanach · ScIso.toScOperator, is $\mathrm{sc}^s$ by the chain rule; and $g_0 + \sum_a g_a$ by the sum rule.

This is the text's induction, run on $s$ over all elementary maps rather than on the description of $T^{k+1}f$. The 2010 paper proves that every term of $\pi \circ T^{k+1}f$ is $\mathrm{sc}^1$, hence $T^{k+1}f$ is, and that the terms of $T^{k+2}f$ are again elementary with admissible indices; the formal proof shows that an elementary map is $\mathrm{sc}^{s+1}$ because its tangent map is made of elementary maps that are $\mathrm{sc}^s$. The computations are the same — the derivative (3.5), the three kinds of new terms and their indices — and the formal version never names $T^{k+1}f$ or its $2^{k+1}$ components.

**Proposition 1.2.4 (preprint 1.19).** $f$ is the elementary map on $X = E$ with no vectors, $p = m = 0$ and $\pi_0 : E \cong E^0$ Chapter1/LevelsWeak · ElemData.base; its domain is $U$ Chapter1/LevelsWeak · ElemData.base_dom and the conditions hold trivially Chapter1/LevelsWeak · ElemData.base_good. With the tower of $f$ (step 1) and $s = k+1$, $0+(k+1) \le k+1$, this elementary map is $\mathrm{sc}^{k+1}$; composed with the sc-isomorphism

$F^0 \cong F$ it agrees with $f$ on $U$, since $\Phi_0 = f$; so $f$ is $\mathrm{sc}^{k+1}$ Chapter1/LevelsWeak · proposition_1_2_4_text. If the hypothesis holds for every $l$, then $f$ is $\mathrm{sc}^{k+1}$ for every $k$, i.e. sc-smooth Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_text.

**Corollary 1.2.5 (preprint 1.20).** For a finite-dimensional target $V$ with its constant structure, the level maps of $f : E_0 \to V$ are $f \circ \iota_m$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq Chapter1/ScBanach · ScBanach.const_toBase. Suppose $f \circ \iota_l$ is $C^{l+1}$ within $U_l$ for $0 \le l \le k$. Then $f$ is $\mathrm{sc}^0$ with no further hypothesis: $f$ is continuous on $U = U_0$ (the case $l = 0$), so each $f \circ \iota_m$ is continuous on $U_m$. And the hypothesis of Proposition 1.2.4 (preprint 1.19) holds: $f_m \circ (E_{m+l} \to E_m) = (f \circ \iota_l) \circ (E_{m+l} \to E_l)$, a $C^{l+1}$ map composed with a continuous linear one, on $U_{m+l}$, the preimage of $U_l$ Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet. So $f$ is $\mathrm{sc}^{k+1}$ Chapter1/LevelsWeak · corollary_1_2_5_text. If $f \circ \iota_l$ is $C^{l+1}$ within $U_l$ for every $l$, then $f$ is sc-smooth (for $k = 0$ the corollary gives $\mathrm{sc}^1$, hence $\mathrm{sc}^0$) Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_finiteDimensional_text; and if $f$ is $C^\infty$ on $U$, it is sc-smooth Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional (proved with the stronger version below, since $C^\infty$ on $U$ restricts to $C^{k+1}$ on every $U_m$). The two sc-smooth statements are the exposition's; the second is what the theory uses (bump functions, Chapter 5).

**The earlier version, under a stronger hypothesis.** `Chapter1/Levels` proves Proposition 1.2.4 (preprint 1.19) under the hypothesis that every level map $f_m : U_m \to F_m$ is $C^{k+1}$ within $U_m$ Chapter1/Levels · proposition_1_2_4. For $k \ge 1$ this asks more than the text ($C^{k+1}$ instead of $C^1$ on $U_m \to F_m$ itself), and in exchange the induction needs no $(\mathrm{S}_k)$; for $k = 0$ the hypotheses coincide.

$k = 0$. Suppose every $f_m$ is $C^1$ within $U_m$. Put $Df(x) :=$ the derivative of $f = f_0$ within $U$ at $\iota_1 x$, for $x \in E_1$ Mathlib/Analysis/Calculus/FDeriv/Defs · fderivWithin. Condition (2) holds because $f \circ \iota_1$ is the restriction of the differentiable $f$ along the continuous linear $\iota_1$. For condition (3) at level $m$: the derivative $df_m(x)$ of $f_m$ within $U_m$ at $x \in U_m$ is compatible with $Df$, $\iota_m \circ df_m(x) = Df(\iota_m x) \circ \iota_m$, because $\iota_m \circ f_m = f \circ \iota_m$ on $U_m$ and derivatives within $U_m$ are unique Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq; so the level-$m$ tangent map is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$, continuous on $U_{m+1} \times E_m$ because $f_m$ is $C^1$ (its derivative is continuous into $\mathcal{L}(E_m, F_m)$, and evaluation is continuous) Chapter1/Properties · proposition_1_2_4_zero. This is the text's proof of $(\mathrm{S}_0)$.

$k + 1$ *from* $k$. Suppose every $f_m$ is $C^{k+2}$. Then $f$ is $\mathrm{sc}^1$ by the case $k = 0$, with the derivative $Df$ above, and we must show $Tf$ is $\mathrm{sc}^{k+1}$ on $TU$. By the induction hypothesis (for the sc-domain $TU$ and the $\mathrm{sc}^0$ map $Tf$) it suffices that the level maps of $Tf$ are $C^{k+1}$. The level-$m$ map of $Tf$ is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ with the level operator (§I.13), and the level operator at $x \in U_{m+1}$ is the derivative of $f_m$ at $\mathrm{incl}_m x$ (Lemma 3.H, Chapter1/Levels · IsSc1With.levelDeriv_eq). So the level-$m$ map of $Tf$ is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$ on $U_{m+1} \times E_m$, whose components are $C^{k+1}$: the first as $f_{m+1}$ is, the second as the derivative of a $C^{k+2}$ map is $C^{k+1}$ and evaluation is bilinear Chapter1/Levels · proposition_1_2_4. This is where the stronger hypothesis is used, and where the text's hypothesis would fail (see "Why the hypothesis does not induct").

*Sc-smooth, and the corollary.* If $f$ maps each $U_m$ into $F_m$ and every $f_m$ is $C^\infty$, then $f$ is sc-smooth Chapter1/Levels · IsScSmooth.of_contDiffOn_levels. For a finite-dimensional target: $C^1$ on every $U_m$ gives $\mathrm{sc}^1$ Chapter1/Properties · corollary_1_2_5_zero, $C^{k+1}$ on every $U_m$ gives $\mathrm{sc}^{k+1}$ Chapter1/Levels · corollary_1_2_5, in particular $C^{k+1}$ on $U = U_0$ does, since restricting along $\iota_m$ preserves $C^{k+1}$ Chapter1/Levels · corollary_1_2_5_base.

## I.18 Definition 1.3.3 (preprint 1.23) and Lemma 1.3.4 (preprint 1.24): the degeneracy index

**Definitions.** For $p = (a, w) \in \mathbb{R}^n \oplus W$ let $\#(p)$ be the number of indices $i$ with $a_i = 0$ `Chapter1/Degeneracy · quadrantDeg`. For an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ presenting a partial quadrant $C$ and $x \in E_0$, let $d_T(x) := \#(T_0 x)$ `Chapter1/Degeneracy · degIndexWith`. The **degeneracy index** $d_C(x)$ is $d_T(x)$ for a presentation $T$ of $C$ chosen once and for all, and $0$ if $C$ is not a partial quadrant `Chapter1/Degeneracy · degIndex`; by Lemma 1.3.4 (preprint 1.24) below it equals $d_T(x)$ for every presentation and every $x \in C$ `Chapter1/Degeneracy · degIndex_eq`, and on the quadrant itself it is $\#$ `Chapter1/Degeneracy · degIndex_quadrant`. The text's interior points, boundary points and corners are the points with $d_C = 0$, $1$, $\geq 2$; Chapter 2 makes the boundary a definition.

**Lemma 1.3.4 (preprint 1.24), core form.** Let $S : \mathbb{R}^n \times W_0 \to \mathbb{R}^{n'} \times W_0'$ be a continuous linear equivalence with $q \in Q \iff Sq \in Q'$ for the two quadrants. Then $\#(p) = \#(Sp)$ for every $p \in Q$ `Chapter1/Degeneracy · quadrantDeg_eq_of_equiv`. Only level $0$ and linear algebra are used; the sc-structure plays no role. Proof. For $p \in Q$ let $E_p := \{(b, v) : b_i = 0 \text{ whenever } p_i = 0\}$, the kernel of the surjection $(b, v) \mapsto (b_i)_{i \in I(p)}$ `Chapter1/Degeneracy · coordMap` `Chapter1/Degeneracy · coordMap_surjective` `Chapter1/Degeneracy · mem_ker_coordMap`, so that $\dim(V/E_p) = \#(p)$ `Chapter1/Degeneracy · finrank_quotient_ker_coordMap`: this makes the text's "codimension of $E_x$" precise. For $u \in E_p$, $p + tu \in Q$ for all $t$ near $0$: coordinate by coordinate, if $p_i = 0$ then $u_i = 0$ and the coordinate stays $0$; if $p_i > 0$ it stays positive for small $t$ by continuity `Chapter1/Degeneracy · eventually_add_smul_mem_quadrant`. Hence $Sp + tSu \in Q'$ for $t$ near $0$, so for each $i$ with $(Sp)_i = 0$ we have $0 \leq t(Su)_i$ for all $t$ near $0$, of both signs, which forces $(Su)_i = 0$ `Chapter1/Degeneracy · eq_zero_of_eventually_nonneg_mul`; so $S(E_p) \subseteq E'_{Sp}$ `Chapter1/Degeneracy · map_ker_coordMap_le`. The same argument for $S^{-1}$ gives the reverse inclusion (the text concludes equality from one inclusion "since $S$ is an isomorphism", which for infinite-dimensional $W$ needs this second step), so $S(E_p) = E'_{Sp}$ and $S$ induces a linear isomorphism $V/E_p \cong V'/E'_{Sp}$; comparing dimensions, $\#(p) = \#(Sp)$.

**Lemma 1.3.4 (preprint 1.24).** If $T$ and $T'$ both present $C$, then $S := T_0' \circ T_0^{-1}$ carries the quadrant onto the quadrant (both are the image of $C$), and the core form at $p = T_0 x$ gives $d_T(x) = d_{T'}(x)$ for $x \in C$ `Chapter1/Degeneracy · lemma_1_3_4`. The text proves first that $k = k'$; here this is a consequence: $T_0^{-1}(0) \in C$ and $\#(0) = n$ in $\mathbb{R}^n \oplus W$ `Chapter1/Degeneracy · quadrantDeg_zero`, so the lemma at this point gives $n = n'$ `Chapter1/Degeneracy · lemma_1_3_4_dim`.

## I.19 Theorem 1.3.5 (preprint 1.25): boundary recognition

**Statement.** Let $U \subseteq C \subseteq E$ and $U' \subseteq C' \subseteq E'$ be relatively open subsets of partial quadrants, $f$ sc$^1$ on $U$ with $f(U) \subseteq U'$, $g$ sc$^1$ on $U'$ with $g(U') \subseteq U$, $g \circ f = \mathrm{id}$ on $U$ and $f \circ g = \mathrm{id}$ on $U'$. Then $d_C(x) = d_{C'}(f(x))$ for every $x \in U$ `Chapter2/DegeneracyIndex · theorem_1_3_5`. The text's "germ of a local sc-diffeomorphism" is undefined in Chapter 1 and is read this way, with $U$, $U'$ as small as one likes; the text has sc-smooth where sc$^1$ suffices, so the formal statement is stronger. The text gives no proof, referring to "[20], Theorem 1.19"; the formal proof is the argument of §2.4 (preprint §2.3) (Lemma 2.4.9 (preprint 2.29), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)) specialized to the identity retraction, and lives in the Chapter 2 file because it uses the retraction vocabulary. Smooth points are not needed; level-1 points suffice.

**The subspaces $E_x$.** For a presentation $T$ of $C$ and $x \in E_0$, let $E_x := \{v \in E_0 : (T_0 v)_i = 0 \text{ whenever } (T_0 x)_i = 0\}$, the kernel of a continuous linear map to a finite-dimensional space `Chapter2/`

DegeneracyIndex · minimalSubspaceWith; its codimension is $d_T(x)$ Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith. Two facts about the index: it is upper semicontinuous, $d_T(y) \le d_T(x)$ for $y$ near $x$ (a nonzero coordinate stays nonzero) Chapter2/DegeneracyIndex · eventually_degIndexWith_le; and every $x \in C$ is the limit of points of $C$ on level 1 with the same index, namely $T_0^{-1}(a, w_n)$ with $w_n \in W_1$ converging to $w$ in $W_0$, which exist by density Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith. These are the exposition's Lemma 5.A, with level 1 in place of smooth.

**The sign argument.** Let $f$ be sc$^1$ on $U \subseteq C$ with values in $C'$, $x \in U_1$, and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x + tv \in U_1$ for $t$ near 0 (the quadrant coordinates as in §I.18, and the openness of $U_1$) Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet, so $f(\iota(x + tv)) \in C'$ and, for each $j$ with $(T_0' f(x))_j = 0$, the function $t \mapsto (T_0' f(\iota(x + tv)))_j$ is $\ge 0$ with a zero at $t = 0$: a local minimum, so its derivative at 0, which is $(T_0' Df(x)\iota_1 v)_j$ by condition (2), vanishes Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1. So $Df(x)$ maps $E_x \cap \iota_1(E_1)$ into $E'_{f(x)}$. Since $f$ is differentiable only along $E_1$, the density of $E_x \cap \iota_1(E_1)$ in $E_x$ is needed to conclude $Df(x)(E_x) \subseteq E'_{f(x)}$; it holds because $E_x$ is the kernel of a continuous linear map to a finite-dimensional space and $\iota_1(E_1)$ is dense Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace. The text asserts the corresponding density ("the closure of $(T_x^R O)_\infty$ is $T_x^R O$") without proof.

**One inequality at level-1 points.** With $g$ the inverse, the chain rule gives $Dg(f_1(x)) \circ Df(x) = \mathrm{id}$ on $E_0$ at $x \in U_1$ Chapter2/DegeneracyIndex · scDeriv_comp_eq_self, so $Df(x)$ is injective and $E_x \cong Df(x)(E_x) \le E'_{f(x)}$ sits inside a subspace of codimension $d_{C'}(f(x))$, while $Dg(f_1(x))$ maps $E'_{f(x)}$ into $E_x$ by the sign argument for $g$; comparing codimensions, $d_C(x) \le d_{C'}(f(x))$ Chapter2/DegeneracyIndex · degIndexWith_le_of_germ. Applying this to $g$ at $f_1(x)$ gives the reverse inequality, so $d_C(x) = d_{C'}(f(x))$ for $x \in U_1$ Chapter2/DegeneracyIndex · theorem_1_3_5_lvl1.

**General points.** For $x \in U$, approximate $x$ by level-1 points $x_n \in U$ of $C$ with $d_C(x_n) = d_C(x)$; then $f(x_n) \to f(x)$ (continuity of $f$ on $U$), so $d_{C'}(f(x_n)) \le d_{C'}(f(x))$ eventually by semicontinuity, and $d_{C'}(f(x_n)) = d_C(x_n) = d_C(x)$ by the level-1 case; hence $d_C(x) \le d_{C'}(f(x))$. The symmetric argument for $g$ at $f(x)$ gives equality Chapter2/DegeneracyIndex · theorem_1_3_5.

## I.20 What the formalization taught us

- **The chain rule needs one integral less than the text writes, and one hypothesis more than it states.** The term $Dg(f(x))[f(x+h) - f(x) - Df(x)h]$ is $o(|h|_1)$ because $Dg(f(x))$ is bounded, with no integral Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp; the segment from $f(x)$ to $f(x+h)$ lies in $V_1$ only because $V_1$ is open-in-convex and $f : U_1 \to F_1$ is continuous, which the text does not say. The whole proof uses, of the quadrant structure of $U$, only the uniqueness of derivatives, and that only for $T(g \circ f) = Tg \circ Tf$ Chapter1/ChainRule · theorem_1_3_1'.
- **"The" derivative $Df(x)$ is a theorem, not a notation.** It is unique because $E_1$ is dense in $E_0$ and $U_1$ has unique differentiability Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt; without it the tangent map, hence sc$^2$, is undefined. The same density argument shows that the extensions $Df(x) : E_{m-1} \to F_{m-1}$ of Proposition 1.2.1 (preprint 1.16), written with one symbol in the text, are restrictions of one operator Chapter1/Levels · IsSc1Levels.toBase_apply, which is what makes the "clear" direction of that proposition work.
- **One compactness argument carries §1.2.** The compact-open lemma Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator turns joint continuity (or $C^k$-ness) of $(x, h) \mapsto Df(x)h$ into norm conti-

nuity (or $C^k$-ness) of $x \mapsto Df(x)|_{E_{m+1}}$; Propositions 1.2.1, 1.2.2 and 1.2.3 (preprint 1.16, 1.17 and 1.18) are this lemma plus bookkeeping of levels.

- **Proposition 1.2.4 (preprint 1.19) does not induct on its own hypothesis.** If $f$ satisfies it with $k$, then $Tf$ in general does not satisfy it with $k-1$: the second component $(x,h) \mapsto Df(x)h : U_{m+l+1} \times E_{m+l} \to F_m$ is one order short. The stronger hypothesis "$C^{k+1}$ on every level" makes the induction two lines `Chapter1/Levels · proposition_1_2_4`; the text's needs the 2010 paper's $(\mathrm{S}_k)$, which remembers $f$ through the elementary maps $d^j f$ on mixed levels `Chapter1/LevelsWeak · ElemData.isSck` `Chapter1/LevelsWeak · proposition_1_2_4_text`. Written out, $(\mathrm{S}_k)$ needs three things the text leaves implicit: one $d^j f(x)$ on all levels `Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels`, the base-point condition $j + l \le k+1$ in place of the bound $n_i \le k$ (which is not needed) `Chapter1/LevelsWeak · ElemData.Good`, and the derivative of the multilinear evaluation `Chapter1/LevelsWeak · hasFDerivAt_cmm_eval`.
- **Regularizing is a bootstrap.** In §1.4.1 the step "$e \in E_m$ by Proposition 1.9" needs $m$ applications, since $Se$ is only one level above $e$ `Chapter1/ScFredholm · ScPlus.regularizing_add`.
- **The index is an addition, and an easy one once additivity is there.** Neither Proposition 1.1.11 (preprint 1.10) nor the remark after Definition 1.1.9 (preprint 1.8) mentions the index of $T + S$ or of the level operators; with the additivity of the index `Chapter1/FredholmIndex · linIndex_comp` and a quasi-inverse, the index is locally constant along $u + tk$, hence invariant `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`, and the sc-statements follow on every level `Chapter1/FredholmIndex · proposition_1_1_11_index_lvl`.
- **The text's quick remarks are all true, and some need real arguments.** "Precisely one sc-structure" in finite dimensions `Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional`; "$(E/A)_\infty$ consists of the classes of $E_\infty$", a Mittag-Leffler series `Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset`; "it is a sc-projection" `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`; "$\mathrm{sc}^1$ is $C^1$ in finite dimensions", which at the boundary of a quadrant means derivatives within $U$ `Chapter1/ExpositionLemmas · isSc1_iff_fderivWithin_const`.
- **Mathlib had Fredholm operators but not Riesz–Schauder.** The theorem that $1 +$ compact is Fredholm, and the stability under compact perturbations, had to be proved `Chapter1/FredholmPerturbation · isFredholm_one_add` `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`; the finite codimension of the range comes from a chain of closed subspaces and Riesz's lemma `Chapter1/FredholmPerturbation · coFG_range_one_add`.
- **The degeneracy index knows nothing of the sc-structure.** Lemma 1.3.4 (preprint 1.24) holds for any linear equivalence of the base levels carrying one quadrant onto the other `Chapter1/Degeneracy · quadrantDeg_eq_of_equiv`, and $k = k'$ is its value at the origin `Chapter1/Degeneracy · lemma_1_3_4_dim`. Boundary recognition needs only $\mathrm{sc}^1$ and only level-1 points, not smooth ones `Chapter2/DegeneracyIndex · theorem_1_3_5`.
- **Example 1.1.3 (preprint 1.2) holds with exactly the printed hypotheses, and costs a small Sobolev theory.** $\delta_0 = 0$ and strict monotonicity are all that is used; $\delta_m \ge 0$ follows from them `Chapter1/GapP1_Sobolev13 · GapP1Sobolev.nonneg_of_strictMono`. Strict monotonicity is used once, for the tail estimate behind compactness `Chapter1/GapP1_Sobolev6 · GapP1Sobolev.tail_le`. Mathlib has no Sobolev spaces, so Sobolev's compact embedding is proved in the one case needed, on a bounded cylinder, by Fourier series on a box `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.rellich`.
- **The text's remark that $Df$ need not be norm continuous needs an example, and a sequence space suffices.** The factor $k$ in the derivative of $\cos(kt)x_k$ is absorbed by one level of the weighted $c_0$ scale `Chapter1/ScOneNotNormContinuous · ScOneExample.isSc1With`, while unit vectors at high frequency keep $Df$ from being norm continuous `Chapter1/ScOneNotNormContinuous · ScOneExample.two_le_norm_D_sub`. The shift map of the 2010 paper is not needed.

- **"Regularizing" characterizes sc-Fredholm operators among those Fredholm on level $0$.** The text does not state the converse of Proposition 1.1.10 (preprint 1.9); its §1.4.1 proves it in one case. As a criterion it is what the later chapters need for their unargued "is sc-Fredholm" `Chapter1/ScFredholmCriterion · isScFredholm_iff` `Chapter1/ScFredholmCriterion · IsScFredholm.restrict`. The continuity in Definition 1.1.4 (preprint 1.3) is automatic, by the closed graph theorem `Chapter1/ScFredholmCriterion · isScOperator_of_mapsTo`.
- **Remark 1.1.17 (preprint 1.15) needs less than it assumes.** For the calculus of §§1.2–1.3, a partial cone needs only convexity and nonempty interior; closedness and the cone property play no role `Chapter1/GapP1_PartialCones · IsPartialCone.isScDomain_of_convex`. What does not carry over is the degeneracy index and everything built on it, which use quadrant coordinates.

## I.21 The 2021 edition: Remark 1.1.2 and Proposition 1.1.11 with the index

The Springer edition (2021) changes Chapter 1 in two places. Remark 1.1.2 (p. 5) is new, and the statement of Proposition 1.1.11 (p. 9; preprint 1.10) now includes the index.

**Remark 1.1.2.** The remark makes "Banach space" mean a *Banachable* space, a vector space whose topology is induced by some complete norm, and says that all norms inducing the topology are equivalent "in view of the open mapping theorem". The formalization models one vector space with two norms as two normed spaces $E_1$, $E_2$ and a linear equivalence $e : E_1 \to E_2$, the identity of the underlying space. It records the remark's content in two readings. *(1) Same topology, equivalent norms* `Chapter1/Book2021 · remark_1_1_2`. If $e$ and $e^{-1}$ are continuous, they are bounded, $\|ex\| \le \|f\|\,\|x\|$ and $\|x\| = \|g(ex)\| \le \|g\|\,\|ex\|$ for the continuous linear maps $f$, $g$ they define, and $c = \max(\|f\|, \|g\|) + 1$ serves for both inequalities. No completeness enters. *(2) The open mapping theorem* `Chapter1/Book2021 · remark_1_1_2_openMapping`. If both norms are complete and $e$ is continuous, the bijective continuous linear map $e$ is a linear homeomorphism by the open mapping theorem `Mathlib/Analysis/Normed/Operator/Banach · ContinuousLinearEquiv.ofBijective`. So $e^{-1}$ is continuous, and (1) applies. As we read it, the sentence of the remark is true without the open mapping theorem. The theorem is what makes two complete norms equivalent when one of them dominates the other.

**Proposition 1.1.11 (preprint 1.10), 2021 statement.** Let $T : E \to F$ be sc-Fredholm and $S : E \to F$ an sc$^+$-operator. Then $T + S$ is sc-Fredholm, $\mathrm{ind}(T + S) = \mathrm{ind}(T)$, and every level map $(T + S)_m : E_m \to F_m$ is Fredholm of index $\mathrm{ind}(T)$ `Chapter1/Book2021 · proposition_1_1_11`. Nothing new has to be proved, because the formal statement is the conjunction of three earlier theorems: - the sc-Fredholm property of §I.9 `Chapter1/ScFredholm · proposition_1_1_11_preprint`; - the index on level $0$, from the invariance of the Fredholm index under compact perturbations `Chapter1/FredholmIndex · proposition_1_1_11_index`; - the index on every level `Chapter1/FredholmIndex · proposition_1_1_11_index_lvl`.

The 2021 proof in Appendix 1.4.1 is the earlier one and does not argue the index. The index statement, the exposition's addition in the earlier editions, is thus now the book's own. The level statement remains the formalization's.

## I.22 Example 1.1.3 (preprint 1.2): the weighted Sobolev spaces $H^{m,\delta_m}(\mathbb{R}\times S^1)$

**Statement.** Let $\delta_0 = 0 < \delta_1 < \delta_2 < \cdots$. The spaces $E_m = H^{m,\delta_m}(\mathbb{R}\times S^1)$ form an sc-Banach space whose level $m$ is $H^{m,\delta_m}$, whose inclusions are the identity on the underlying functions, and whose level 0 is $L^2(\mathbb{R}\times S^1)$ with its own norm `Chapter1/GapP1_Sobolev13 · GapP1Sobolev.example_1_1_3` `Chapter1/GapP1_Sobolev13 · GapP1Sobolev.example_1_1_3_lvl` `Chapter1/GapP1_Sobolev13 · GapP1Sobolev.level0_isometry` `Chapter1/GapP1_Sobolev13 · GapP1Sobolev.level0_surjective`. These are the text's hypotheses as printed. Non-negativity of the weights, which every estimate below uses, follows from $\delta_0 = 0$ and monotonicity `Chapter1/GapP1_Sobolev13 · GapP1Sobolev.nonneg_of_strictMono`. The text names the two ingredients, Sobolev's compact embedding on bounded domains and the strict monotonicity of $(\delta_m)$, and leaves the rest. Mathlib has neither Sobolev spaces nor the compact embedding, so the formal proof builds both; it takes thirteen files.

*The cylinder and weak derivatives.* $\mathbb{R}\times S^1$ is the strip $\mathbb{R}\times(0,1]$ with Lebesgue measure `Chapter1/GapP1_Sobolev1 · GapP1Sobolev.cyl`. The circle enters through the test functions: a test function is smooth, 1-periodic in $t$, and zero for $|s|$ large `Chapter1/GapP1_Sobolev1 · GapP1Sobolev.IsTest`. A function $g$ is the weak derivative $\partial_s^a\partial_t^b u$ if $\int u\,\partial_s^a\partial_t^b\varphi = (-1)^{a+b}\int g\,\varphi$ for every test function $\varphi$ `Chapter1/GapP1_Sobolev1 · GapP1Sobolev.HasWeakDeriv`. This is the distributional derivative on the cylinder, and it is unique almost everywhere `Chapter1/GapP1_Sobolev2 · GapP1Sobolev.HasWeakDeriv.ae_eq`, because an $L^2$ function orthogonal to all test functions vanishes `Chapter1/GapP1_Sobolev2 · GapP1Sobolev.ae_eq_zero_of_forall_test`; the proof of that uses the Fourier coefficients below.

*The spaces.* With the weight $e^{\delta|s|}$ `Chapter1/GapP1_Sobolev3 · GapP1Sobolev.wt`, $u \in H^{m,\delta}$ means that for all $a+b\le m$ the weak derivative $\partial_s^a\partial_t^b u$ exists and $e^{\delta|s|}\partial_s^a\partial_t^b u \in L^2$ `Chapter1/GapP1_Sobolev3 · GapP1Sobolev.MemH`. These $u$ form a subspace of $L^2$ `Chapter1/GapP1_Sobolev3 · GapP1Sobolev.Hsub`, and $H^{m,\delta}$ carries the norm

$$\|u\|_{m,\delta}^2 = \sum_{a+b\le m} \|e^{\delta|s|}\partial_s^a\partial_t^b u\|_{L^2}^2$$

`Chapter1/GapP1_Sobolev3 · GapP1Sobolev.H` `Chapter1/GapP1_Sobolev3 · GapP1Sobolev.norm_sq_H`. The text fixes no norm; this one is among the usual equivalent ones. By definition $u \mapsto (e^{\delta|s|}\partial^p u)_{|p|\le m}$ is an isometry into a finite product of copies of $L^2$ `Chapter1/GapP1_Sobolev3 · GapP1Sobolev.isometry_ΦH`. Its range is closed, because weak derivatives pass to $L^2$ limits: if $e^{\delta|s|}\partial^p u_n \to w_p$, then $e^{-\delta|s|}w_p$ is the weak derivative $\partial^p$ of $e^{-\delta|s|}w_0$, by continuity of the pairing with a test function `Chapter1/GapP1_Sobolev4 · GapP1Sobolev.isClosed_range_ΦH`. So $H^{m,\delta}$ is complete `Chapter1/GapP1_Sobolev4 · GapP1Sobolev.completeSpace_H`. For $m\le m'$ and $\delta\le\delta'$ the identity of $L^2$ maps $H^{m',\delta'}$ into $H^{m,\delta}$ with norm at most 1, since $e^{\delta|s|}\le e^{\delta'|s|}$ `Chapter1/GapP1_Sobolev4 · GapP1Sobolev.inclH` `Chapter1/GapP1_Sobolev4 · GapP1Sobolev.valCLM_inclH`; it is injective, because both sides are subspaces of $L^2$ `Chapter1/GapP1_Sobolev4 · GapP1Sobolev.inclH_injective`.

*Fourier series on a box.* On a box $(a,a+T]\times(0,1]$ the coefficients $\hat F_{jk} = T^{-1}\int e^{-2\pi i(js/T+kt)}F$ `Chapter1/GapP1_Sobolev2 · GapP1Sobolev.bc` satisfy Parseval's identity, $\sum_{j,k}|\hat F_{jk}|^2 = T^{-1}\int_{\text{box}}|F|^2$ `Chapter1/GapP1_Sobolev2 · GapP1Sobolev.hasSum_bc`. Mathlib has Parseval on one circle; the box is a product, and the identity follows by summing the $t$-Parseval identity of each slice $F(s,\cdot)$ and applying the $s$-Parseval identity to each $t$-coefficient. For a function vanishing near the ends of the box, the coefficients of a weak derivative are those of the function times $2\pi ij/T$, respectively $2\pi ik$ `Chapter1/GapP1_Sobolev2 · GapP1Sobolev.bc_ds` `Chapter1/GapP1_Sobolev2 · GapP1Sobolev.bc_dt`: test the weak derivative against a cut-off times the exponential.

*Rellich on a bounded cylinder.* For $R > 0$ and $C$, the set of truncations $1_{|s|\le R}v$ of the $v \in L^2$ with $\|v\|, \|\partial_s v\|, \|\partial_t v\| \le C$ is totally bounded in $L^2$ `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.rellich` `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.H1ball` `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.cutR`. This is the case of Sobolev's compact embedding that the example needs. *Proof.* Take a smooth cut-off $\chi$ equal to 1 on $[-R, R]$ and supported in a box of length $T = 2R + 4$ `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.exists_cutoff'`. Then $w = \chi v$ has weak derivatives bounded by a constant times $C$ (Leibniz), and by Parseval and the coefficient formulas, $\sum_{j,k}(j^2 + k^2)|\widehat{w}_{jk}|^2 \le M$ with $M$ depending only on $C$ and $R$ `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.summable_freq_bc`. The coefficients outside the square $|j|, |k| \le N$ therefore carry $L^2$ mass at most $M/N^2$ `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.tsum_sub_le` `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.exists_N_eta`. And $\|1_{|s|\le R}v\| \le \|w\|_{\mathrm{box}}$ `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.norm_cutR_sq_le`. So, up to $\varepsilon$, $1_{|s|\le R}v$ is determined by finitely many bounded complex coordinates, and a set approximated in this way is totally bounded `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.totallyBounded_image_of_coords` `Chapter1/GapP1_Sobolev5 · GapP1Sobolev.totallyBounded_of_approx`.

*The inclusion is compact.* For $\delta < \delta'$ the inclusion $H^{m+1,\delta'} \to H^{m,\delta}$ is a compact operator `Chapter1/GapP1_Sobolev6 · GapP1Sobolev.isCompactOperator_inclH`. *Proof.* Let $u$ lie in the unit ball of $H^{m+1,\delta'}$ and $|p| \le m$. The derivative $v = \partial^p u$ has weak derivatives $\partial_s v$, $\partial_t v$ of order at most $m+1$, and all three have $L^2$ norm at most $\|u\| < 1$, since $e^{\delta'|s|} \ge 1$ `Chapter1/GapP1_Sobolev6 · GapP1Sobolev.dvCLM_mem_H1ball`. Split the component $e^{\delta|s|}v$ of $u$ in $H^{m,\delta}$:

$$e^{\delta|s|}v = 1_{|s|\le R}\, e^{\delta|s|}\cdot 1_{|s|\le R}v \;+\; (1 - 1_{|s|\le R})\, e^{\delta|s|}e^{-\delta'|s|}\cdot e^{\delta'|s|}v.$$

The second term has norm at most $e^{-(\delta'-\delta)R}$, since the multiplier is bounded by that `Chapter1/GapP1_Sobolev6 · GapP1Sobolev.tail_le` and $\|e^{\delta'|s|}v\| < 1$. Here the strict inequality $\delta < \delta'$ enters, as the text says. Given $\varepsilon$, choose $R$ with $e^{-(\delta'-\delta)R} < \varepsilon$. The first term lies in the image of the totally bounded set of Rellich's theorem under multiplication by the bounded function $1_{|s|\le R}e^{\delta|s|}$ `Chapter1/GapP1_Sobolev6 · GapP1Sobolev.cutWt` `Chapter1/GapP1_Sobolev6 · GapP1Sobolev.abs_cutWt_le`. So the set of the $p$-th components is within $\varepsilon$ of a totally bounded set for every $\varepsilon$, hence totally bounded. The product of the closures over the finitely many $p$ is compact, and the isometry $u \mapsto (e^{\delta|s|}\partial^p u)_p$ pulls total boundedness back `Chapter1/GapP1_Sobolev6 · GapP1Sobolev.totallyBounded_of_isometry`.

*Test functions lie in every level.* A smooth function periodic in $t$ has its classical derivatives as weak derivatives, by integration by parts in $s$ (a test function has bounded support) and in $t$ (periodicity) `Chapter1/GapP1_Sobolev7 · GapP1Sobolev.SmPer.hasWeakDeriv`. A test function and all its derivatives are bounded and supported in a bounded strip, so they are in $L^2$ with every weight: $\varphi \in H^{m,\delta}$ for all $m$ and $\delta$ `Chapter1/GapP1_Sobolev7 · GapP1Sobolev.IsTest.mem_Hsub`.

*Test functions are dense in every level* `Chapter1/GapP1_Sobolev12 · GapP1Sobolev.dense_test`. This is the density of smooth points that Definition 1.1.1 (preprint 1.1) asks for. *Proof.* Two steps. - *Cut off in $s$.* For a smooth $\chi(s)$ whose derivatives up to order $m$ are bounded, $u \mapsto \chi u$ maps $H^{m,\delta}$ to itself `Chapter1/GapP1_Sobolev9 · GapP1Sobolev.mulH`, and its weak derivatives are given by the Leibniz rule in $s$ `Chapter1/GapP1_Sobolev8 · GapP1Sobolev.leibniz` `Chapter1/GapP1_Sobolev9 · GapP1Sobolev.dv_mulH`. With $\chi_n(s) = \rho(s/(n+1))$, $\rho$ a smooth bump equal to 1 on $[-1, 1]$ `Chapter1/GapP1_Sobolev10 · GapP1Sobolev.chi`, $\chi_n u \to u$ in $H^{m,\delta}$ `Chapter1/GapP1_Sobolev10 · GapP1Sobolev.tendsto_cut`. The term $(\chi_n - 1)e^{\delta|s|}\partial^p u$ tends to 0 by dominated convergence, and the terms with derivatives of $\chi_n$ carry a factor $(n+1)^{-j}$. - *Approximate a function of bounded support.* If the weak derivatives of $v \in H^{m,\delta}$ vanish for $|s| > \rho$, take a box containing $[-\rho, \rho] \times (0, 1]$, the partial Fourier sums $S_N v$ over $|j|, |k| \le N$ `Chapter1/GapP1_Sobolev11 · GapP1Sobolev.trig`, and a smooth window $\chi$ equal to 1 on $[-\rho, \rho]$ and zero near the ends of the box. Then $\chi \cdot \mathrm{Re}\, S_N v$ is a test function. Its derivatives

are $\chi$ times the partial Fourier sums of the derivatives of $v$, plus terms with derivatives of $\chi$ Chapter1/GapP1_Sobolev11 · GapP1Sobolev.pD_trig Chapter1/GapP1_Sobolev11 · GapP1Sobolev.bc_eexp. Each partial sum converges in $L^2$ of the box by Parseval Chapter1/GapP1_Sobolev11 · GapP1Sobolev.tendsto_box_sq, and the weight is bounded on the box. So $\chi \cdot \mathrm{Re}\, S_N v \to \chi v = v$ in $H^{m,\delta}$ Chapter1/GapP1_Sobolev12 · GapP1Sobolev.approx_supp.

Given $u$ and $\varepsilon$, the first step gives $\chi_n u$ within $\varepsilon/2$, of bounded support, and the second a test function within $\varepsilon/2$ of it.

*Assembly.* The levels $H^{m,\delta_m}$ with these inclusions form a scale Chapter1/GapP1_Sobolev13 · GapP1Sobolev.scaleH. A test function $\varphi$ lies in every level, and its images under the inclusions are the same function Chapter1/GapP1_Sobolev13 · GapP1Sobolev.toBase_toH, so it is a smooth point. As test functions are dense in each level, the scale is an sc-Banach space. For $\delta = 0$ the norm of $H^{0,0}$ is the $L^2$ norm, and every $L^2$ function lies in $H^{0,0}$, so level 0 is $L^2(\mathbb{R} \times S^1)$ itself Chapter1/GapP1_Sobolev13 · GapP1Sobolev.level0_isometry Chapter1/GapP1_Sobolev13 · GapP1Sobolev.level0_surjective.

## I.23 Definition 1.1.4 and Proposition 1.1.10 (preprint 1.3 and 1.9): the sc-Fredholm criterion

The text uses two facts about sc-operators without proving them. An operator that maps every level into the corresponding level is continuous on the levels. And an operator that is Fredholm on level 0 and regularizing is sc-Fredholm: this is what the text's "is sc-Fredholm" amounts to in several later places, among them §1.4.1 and Propositions 3.1.5 and 3.6.5 (preprint 3.5 and 3.52). Both are proved here.

**Sc-operators from level-preserving maps.** Let $T_0 : E_0 \to F_0$ be bounded and map the image of every $E_m$ into the image of $F_m$. Then $T_0$ is an sc-operator Chapter1/ScFredholmCriterion · isScOperator_of_mapsTo. *Proof.* By injectivity of $\iota_m$, the level map $g : E_m \to F_m$ is well defined and linear. Its graph is closed: if $x_n \to x$ in $E_m$ and $g x_n \to y$ in $F_m$, then applying the continuous maps $\iota_m$ and $T_0$ gives $\iota_m y = T_0 \iota_m x = \iota_m g x$, so $y = gx$. By the closed graph theorem $g$ is continuous. So "$T$ maps $E_m$ into $F_m$ continuously" in Definition 1.1.4 (preprint 1.3) is "$T$ maps $E_m$ into $F_m$". In the same way a bijective sc-operator is an sc-isomorphism: the open mapping theorem on each level makes the inverse continuous Chapter3/LinearizationChartsSc · ScIso.ofBijective.

**Regularizing operators.** An sc-operator $T$ is *regularizing* if $T_0 e \in F_m$ implies $e \in E_m$, for all $m$ Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing. Proposition 1.1.10 (preprint 1.9) says that sc-Fredholm operators are regularizing Chapter1/ScFredholmCriterion · IsScFredholm.isRegularizing. The kernel of a regularizing operator consists of smooth points, since $T_0 e = 0$ lies in every $F_m$ Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.ker_le_smooth. And if $T_0$ is onto, so is every $T_m$: given $y \in F_m$, choose $T_0 x = \iota_m y$, lift $x$ to $E_m$, and conclude by injectivity of $\iota_m$ Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.surjective.

**The criterion.** If $T_0$ is Fredholm and $T$ is regularizing, then $T$ is sc-Fredholm Chapter1/ScFredholmCriterion · isScFredholm_of_isRegularizing. *Proof.* This is the construction of §1.4.1 with the two properties used there as hypotheses. One builds the splittings of Definition 1.1.9 (preprint 1.8). - *The kernel and its complement.* $K = \ker T_0$ is finite-dimensional ($T_0$ is Fredholm) and consists of smooth points (regularizing). By Proposition 1.1.7 (preprint 1.6) (§I.4) it is an sc-subspace with an sc-complement $X$, $E = K \oplus X$. - *The range.* $Y = \mathrm{range}\, T_0$ is closed, since $T_0$ is Fredholm. On level $m$ it is $T(E_m)$: if $\iota_m y = T_0 e$, then $e \in E_m$ by regularity. Its smooth points are the images of smooth points, which are dense in $E_m$, so they are dense in $Y_m$, and $Y$ is an sc-subspace. - *The cokernel.* The quotient $F_0/Y$ is finite-dimensional. The smooth points of $F$ have dense image in it, so, since a

finite-dimensional subspace is closed, they map onto it. Choose a linear right inverse of $F_0 \to F_0/Y$ with values in the smooth points. Its range $C$ is a finite-dimensional space of smooth points, and $F_m = Y_m \oplus C_m$ on every level. - *Bijectivity.* $T_m$ maps $X_m$ onto $Y_m$, because every $y \in Y_m$ is $T_0 e$ with $e \in E_m$ and the $K$-component of $e$ is smooth. It is injective on $X_m$, because $X \cap K = 0$.

With the index of §I.6 on each level, $T_m$ is Fredholm of the index of $T_0$ for every $m$ `Chapter1/ScFredholmCriterion · isScFredholm_of_isRegularizing_index`. Together with Proposition 1.1.10 (preprint 1.9):

$$T \text{ sc-Fredholm} \iff T_0 \text{ Fredholm and } T \text{ regularizing}$$

`Chapter1/ScFredholmCriterion · isScFredholm_iff`. The text does not state the converse.

**Restriction to sc-subspaces.** If $T$ maps an sc-subspace $A$ into an sc-subspace $B$, it restricts to an sc-operator $A \to B$ `Chapter1/ScFredholmCriterion · ScOperator.restrict`. A point of $A$ lies on level $m$ of $A$ exactly when it lies on level $m$ of $E$ `Chapter1/ScFredholmCriterion · ScBanach.ofSubspace_mem_range_toBase_iff`, so the restriction of a regularizing operator is regularizing `Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.restrict`. Hence, by the criterion, the restriction of an sc-Fredholm operator is sc-Fredholm as soon as it is Fredholm on level 0, with the same index on every level `Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.isScFredholm_restrict` `Chapter1/ScFredholmCriterion · IsScFredholm.restrict`. Part III uses this for Propositions 3.1.5 and 3.6.5 (preprint 3.5 and 3.52).

## I.24 The remark after Definition 1.1.15 (preprint 1.14): an sc¹ map whose derivative is not norm continuous

**Statement.** There are sc-Banach spaces $E$, $F$, a relatively open $U$ in a partial quadrant of $E$, and an $\mathrm{sc}^1$ map $f : U \to F$ such that, for *every* derivative family $D$ of $f$, $x \mapsto D(x)$ is not continuous from $U_1$ into $\mathcal{L}(E_0, F_0)$ with the operator norm `Chapter1/ScOneNotNormContinuous · exists_isSc1_not_continuousOn_deriv`. With the compact-open continuity of §I.11 this is the whole remark. The text gives no example, and the exposition's §6 would use the shift map on the weighted Sobolev spaces; the example here uses sequences.

*The sequence space.* $c_0$ is the space of real sequences tending to $0$ with the sup norm `Chapter1/ScOneNotNormContinuous · ScOneExample.c0`. A bounded multiplier $c$ acts on it as a diagonal operator `Chapter1/ScOneNotNormContinuous · ScOneExample.diag`. The weight $w_k = 1/(k+1)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.w` gives the operator $\mathrm{diag}(w)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.inclOp`, which is injective and compact `Chapter1/ScOneNotNormContinuous · ScOneExample.inclOp_injective` `Chapter1/ScOneNotNormContinuous · ScOneExample.inclOp_isCompactOperator`: its truncations to the first $n$ coordinates have finite rank, hence are compact `Chapter1/ScOneNotNormContinuous · ScOneExample.isCompactOperator_diag_wTrunc`, and they converge to it in norm, at distance $\sup_{k \geq n} w_k = 1/(n+1)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.tendsto_diag_wTrunc`. The scale $X$ has every level equal to $c_0$, read as the coordinates $y_k = (k+1)^m x_k$ of a sequence $x$, so that $|x|_m = \sup_k (k+1)^m |x_k|$ and the inclusion $X_{m+1} \to X_m$ is $\mathrm{diag}(w)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.XScale`. Finitely supported sequences lift to every level `Chapter1/ScOneNotNormContinuous · ScOneExample.mem_smoothLvl_of_finSupp` and are dense in $c_0$, so $X$ is an sc-Banach space `Chapter1/ScOneNotNormContinuous · ScOneExample.X`. Then $E = \mathbb{R} \oplus X$, with the constant structure on $\mathbb{R}$ `Chapter1/ScOneNotNormContinuous · ScOneExample.E`, and $U = E_0$, the whole space, a partial quadrant.

*The map.* $f(t,x)_k = \cos(kt)\, x_k$ `Chapter1/ScOneNotNormContinuous · ScOneExample.f` `Chapter1/`

ScOneNotNormContinuous · ScOneExample.cosOp. It is the same formula on every level, since diagonal operators commute with the weights, and it is continuous there, so $f$ is $\mathrm{sc}^0$ Chapter1/ScOneNotNormContinuous · ScOneExample.isSc0_f. The candidate derivative at $(t, x) \in E_1$ is

$$D(t,x)(\tau,\xi)_k = -k\sin(kt)\,x_k\,\tau + \cos(kt)\,\xi_k$$

Chapter1/ScOneNotNormContinuous · ScOneExample.D. In level-1 coordinates $y_k = (k+1)x_k$ the first term is $-k\,w_k\sin(kt)\,y_k\,\tau$, a bounded multiplier ($kw_k \le 1$) applied to $y \in c_0$ Chapter1/ScOneNotNormContinuous · ScOneExample.sinMul Chapter1/ScOneNotNormContinuous · ScOneExample.natCast_mul_w_le. One level absorbs the factor $k$. $f : E_1 \to F_0$ has derivative $D(t,x)$ at every point Chapter1/ScOneNotNormContinuous · ScOneExample.hasFDerivAt_f, and the tangent map $(t,x,\tau,\xi) \mapsto (f(t,x), D(t,x)(\tau,\xi))$ is again given by the same formulas on every level of $TE$, hence $\mathrm{sc}^0$ Chapter1/ScOneNotNormContinuous · ScOneExample.continuous_tangentLevel. So $f$ is $\mathrm{sc}^1$ with derivative $D$ Chapter1/ScOneNotNormContinuous · ScOneExample.isSc1With Chapter1/ScOneNotNormContinuous · ScOneExample.isSc1.

*Discontinuity.* The points $(\pi/(n+1), 0)$ tend to 0 in $E_1$. Test $D(\pi/(n+1),0) - D(0,0)$ on the unit vector $(0, e_{n+1})$ of $E_0$: the $(n+1)$-st coordinate is $\cos\pi - \cos 0 = -2$. So $\|D(\pi/(n+1),0) - D(0,0)\| \ge 2$ for all $n$ Chapter1/ScOneNotNormContinuous · ScOneExample.two_le_norm_D_sub, and $D$ is not norm continuous at 0 Chapter1/ScOneNotNormContinuous · ScOneExample.not_continuousAt_D. Every derivative family of $f$ equals $D$ on $E_1$, because derivatives are unique there (§I.11), so no derivative family is norm continuous Chapter1/ScOneNotNormContinuous · ScOneExample.not_continuousOn_of_isSc1With. The family is compact-open continuous, as every $\mathrm{sc}^1$ derivative is (§I.11): on a compact set of $\xi$'s the high coordinates are uniformly small. The example is proved $\mathrm{sc}^1$ only; it is presumably sc-smooth, which the remark does not need.

## I.25 Remark 1.1.17 (preprint 1.15): partial cones

**Statement.** The remark proposes closed convex partial cones $P$ with nonempty interior in place of partial quadrants, and asks which arguments carry over. A *partial cone* is a closed convex $P \subseteq E_0$ with $\lambda P \subseteq P$ for $\lambda \ge 0$ and nonempty interior Chapter1/GapP1_PartialCones · IsPartialCone. For $O$ open, $O \cap P$ is an sc-domain Chapter1/GapP1_PartialCones · IsPartialCone.isScDomain, so every result of §§1.2–1.3, which are stated for sc-domains (§I.10), holds on relatively open subsets of partial cones. The tangent set $TP = P_1 \times E_0$ is again a partial cone in $TE$ Chapter1/GapP1_PartialCones · IsPartialCone.tangentSet, as the remark says.

*Proof.* Only convexity and nonempty interior are needed for the sc-domain property Chapter1/GapP1_PartialCones · IsPartialCone.isScDomain_of_convex. - *Open in convex.* On level $m$, $(O \cap P)_m = O_m \cap P_m$, with $O_m$ open and $P_m = \iota_m^{-1}(P)$ convex, as the preimage of a convex set under a linear map. - *Unique derivatives.* A convex set with nonempty interior has unique derivatives at each of its points (Mathlib's `uniqueDiffOn_convex`). The interior of $P_m$ is nonempty: the interior of $P$ is a nonempty open set of $E_0$, so it contains a smooth point, which lifts to $E_m$, and the preimage of the interior of $P$ is an open subset of $P_m$ Chapter1/GapP1_PartialCones · exists_toBase_mem_of_isOpen Chapter1/GapP1_PartialCones · interior_lvlSet_nonempty_of_interior. Uniqueness on the open-in-convex set $O_m \cap P_m$ follows.

Neither closedness nor the cone property enters. For the tangent cone: $P_1 \times E_0$ is closed and convex, invariant under nonnegative scaling, and its interior contains (interior of $P_1$) $\times E_0$.

A partial quadrant is a partial cone Chapter1/GapP1_PartialCones · IsPartialQuadrant.isPartialCone: the preimage under a linear isomorphism of $[0,\infty)^n \times W$ is an intersection of closed half-spaces, and its interior contains the preimage of $(0,\infty)^n \times W$. The chain rule on partial cones is an instance of Theorem 1.3.1

(preprint 1.21) on sc-domains `Chapter1/GapP1_PartialCones · remark_1_1_17_chainRule`. The degeneracy index, tameness and faces, from §1.3 on and in the later chapters, use coordinates of a quadrant, and stay with quadrants, as in the text.

# Part II. Differential geometry based on retracts (HWZ Chapter 2, §§2.1, 2.3–2.7 (preprint §§2.1–2.6))

Chapter 2 builds a differential geometry whose local models are not open sets of Banach spaces but images of sc-smooth retractions, sets whose "dimension" may jump. The formal proof establishes the chapter in its order: retractions and retracts and the independence of their tangent from the retraction (§§II.2–II.4), M-polyfolds with their filtration, metrizability and tangent bundle (§§II.5–II.9), sub-M-polyfolds (§II.10), the degeneracy index with the boundary recognition argument (§§II.11–II.13), tame retracts and the faces of tame M-polyfolds (§§II.14–II.16), products and fibred products (§II.17), and strong bundles with their pull-backs (§§II.18–II.20). The 2021 additions follow in §§II.22–II.37, and the items the first revision left open in §§II.38–II.48.

**What is proved, and what is not.** All fifteen gaps of the chapter are closed: G.4–G.15 from the 2014 text and G.76–G.78 from the 2021 additions. The items the first revision of this part left open are proved, in §§II.38–II.46: - the classical Proposition 2.1.2 (preprint 2.2), by conjugating $r$ to its derivative (G.4, §II.38); - the linear, Banach and sc-structure of $T_xX$ with Proposition 2.3.19 (3)–(4) (preprint 2.21) in $T_aX$ and the fibrewise linearity of $Tf$ (G.7, §II.39); - the openness of $p : TX \to X_1$ (G.8, §II.40); - Remark 2.4.8 (preprint 2.28) and the second statement of Theorem 2.4.12 (preprint 2.31) (G.9, G.11, §II.41); - Lemma 2.4.9 (preprint 2.29) in its formulation by sc-smooth paths (G.10, §II.42); - the extended faces of a partial quadrant from $C$ alone (G.12, §II.43); - Definition 2.5.8 (preprint 2.38) inside $T_xX$ (G.13, §II.44); - strong bundle isomorphisms and the category $\mathcal{SBR}$ (G.14, §II.45); - Michael's Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57) (G.15, §II.46).

G.5, G.6 and G.76–G.78 were closed in §§II.31–II.37. One thing remains unformalized, and it is not a statement: the categories $\mathcal{R}$ and $\mathcal{BR}$ of §2.1 are only named. Bundle retractions and bundle maps themselves are formalized (§II.32), and $\mathcal{SBR}$ is formalized, with forgetful functors that land in the category of sc-smooth retracts (§II.45).

*The narrower rows.* Four rows are narrower than the text, all in the 2021 appendix §2.7.4, and all for one reason. - (2.44) of Definition 2.7.11 does not hold as we read it for every neighbourhood $V$; it holds for $V$ centred at $x$, and the centred neighbourhoods are cofinal, which is what "taking $V$ sufficiently small" provides (V2.84; §II.28). - In Theorem 2.7.13 with Definition 2.7.14 (V2.89), and in Proposition 2.7.12 with the characterization after Theorem 2.7.13 (V2.103), "onto $\theta$" does not hold as we read it for every representative. In general the image of $\Phi_{(x,V,\theta)}$ is the open subset $\theta \cap V^x$ of $\theta$, which is $\theta$ for centred $V$. The Lean counterexample is the slab triple in

$[0,\infty)^3$: $\Phi$ is not onto $\theta$ there `Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective Chapter2/TameBoundaryNonCentred · MPolyfold.not_surjOn_phiTot`. - Proposition 2.7.12 itself holds as printed for every pair of representatives. Row V2.104 is narrower only in the characterization, where the image is $\theta \cap V^x$ (§II.37).

Three 2021 rows are now in book form, stated as one theorem each (§II.47): Remark 2.1.6 (V2.68), Definitions 2.5.18–2.5.19 with condition (3) in $T_yX$ (V2.78), and Proposition 2.7.12 (V2.87). Two statements are read rather than taken literally. In Proposition 2.1.2 (preprint 2.2) the printed normalization $\psi(0) = x$ is read as $\psi(x) = 0$ (§II.38). And the second sentence of Remark 2.4.8 (preprint 2.28), that $C_xO$ is in general not a partial quadrant, does not hold as we read it: at every smooth point of every sc-smooth retract, tame or not, $C_xO$ is a partial quadrant in $T_xO$ `Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant`. What can fail without tameness is only that its dimension equals $d_O(x)$ (§II.41).

Three choices shape the formal route. First, a retraction is a total map $r : E_0 \to E_0$ with an explicit domain $U$, and the tangent $TO$ of a retract is described as the *fixed-point set* of $Tr$ in $TU$ (§II.2); this description, which the text's proofs use without stating it, makes Propositions 2.1.11 and 2.1.13 (preprint 2.4 and 2.6) one-line consequences of the chain rule. Second, an M-polyfold is a space with *one* atlas, and every notion is defined through the maximal atlas of all charts compatible with it (§II.5); this needs the fact, not in the text, that compatibility is a local property, proved from the chain rule and the locality of sc-smoothness (§II.1). Third, where the text computes in the model $C = [0,\infty)^k \oplus W$, the formal statements carry a presentation $T : E \cong \mathbb{R}^n \oplus W$ of the partial quadrant and its coordinate functionals $\lambda_i = (T\,\cdot)_i$; the subspaces $E_x$ and cones $C_x$ are defined through $T$, and are shown not to depend on it (§II.14).

Six proofs take a different road from the text's (or the exposition's). Theorem 2.3.10 (preprint 2.16) (metrizability of the levels) is proved without Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57) (both proved since, §II.46), which need a regularity of $X_m$ the text does not check: a partition of unity of $X$ pulled back to $X_m$ gives an explicit metric, and the same argument proves Smirnov's theorem, which Mathlib lacks (§II.7). Proposition 2.4.10 (preprint 2.30) is proved from the sign arguments at points on level 1, not from Lemma 2.4.9 (preprint 2.29) at smooth points (§II.13); the lemma itself is §II.42. Theorem 2.4.12 (preprint 2.31) is proved for every chart of every M-polyfold by a direct isomorphism of quotients, without reducing to a chart that realizes the index (§II.13). Proposition 2.5.3 (preprint 2.35) uses only the second condition of tameness (§II.14). Proposition 2.5.6's (preprint 2.36) partial quadrant is proved for tame retracts without the equality case of Theorem 2.4.12 (preprint 2.31), the smooth vectors of its proof being written down explicitly (§II.14); the equality case itself, without its hypothesis, is §II.41. The second sentence of Proposition 2.5.16 (preprint 2.44) is proved by removing all coordinates of $\sigma$ at once rather than by the exposition's induction on $\#\sigma$ (§II.16).

**Map.**

- `Chapter2/Retracts`: restriction and locality of sc^k maps; retractions, retracts, Proposition 2.1.11 (preprint 2.4), Definitions 2.1.12, 2.1.14 (preprint 2.5, 2.7), Propositions 2.1.13, 2.3.1 (preprint 2.6, 2.10), Theorem 2.1.15 (preprint 2.8); local models.
- `Chapter2/TangentBundle`: the tangent of a retract as a retract, the tangent functor, locality of $Tf$; the tangent bundle $TX$, Proposition 2.3.15 (1)–(2) (preprint 2.19), $TX$ as an M-polyfold.
- `Chapter2/MPolyfold`: charts, atlases, M-polyfolds, Definition 2.3.8 (preprint 2.14), the levels $X_m$, Lemma 2.3.9 (preprint 2.15), sub-M-polyfolds (Definition 2.3.18 (preprint 2.20)), a retract as an M-polyfold.
- `Chapter2/LevelPolyfold`: $X^1$ as an M-polyfold; the locality of Definition 2.3.8 (preprint 2.14).
- `Chapter2/Metrizable`: Smirnov's theorem, Theorem 2.3.10 (preprint 2.16), Proposition 2.3.15 (3)

(preprint 2.19), Proposition 2.3.19 (3)–(4) (preprint 2.21).

- `Chapter2/TangentMap`: the tangent map $Tf : TX \to TY$ and the chain rule.
- `Chapter2/SubMPolyfold`: Proposition 2.3.19 (1)–(2) (preprint 2.21), Lemma 2.4.6 (preprint 2.26).
- `Chapter2/DegeneracyIndex`: $E_x$, the two-sided sign argument, the core of Theorem 2.4.12 (preprint 2.31), Theorem 1.3.5 (preprint 1.25), Definition 2.4.1 (preprint 2.22), Lemma 2.4.3 (preprint 2.23), Proposition 2.4.4 (preprint 2.24), Corollary 2.4.13 (preprint 2.32).
- `Chapter2/Boundary`: the boundary, the one-sided sign argument, Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31).
- `Chapter2/Tame`: the definitions of §2.5 (preprint §2.4) and of §2.6's (preprint §2.5) local objects.
- `Chapter2/TameRetracts`: Propositions 2.5.3, 2.5.6 (preprint 2.35, 2.36), the local geometry of faces, Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41), Propositions 2.5.14, 2.5.15 (preprint 2.42, 2.43) (the retraction onto a face).
- `Chapter2/TameFaces`: independence of the presentation; Proposition 2.5.15 (preprint 2.43) (faces are tame), Proposition 2.5.16 (preprint 2.44) (first sentence), Theorem 2.5.17 (preprint 2.45).
- `Chapter2/PartialCone`: Proposition 2.5.6 (preprint 2.36), $T^R_x O$ an sc-subspace and $C_x O$ a partial quadrant.
- `Chapter2/CornerGerms`: Proposition 2.5.16 (preprint 2.44) (second sentence), the corners $f_\sigma$.
- `Chapter2/FibredProduct`: products of M-polyfolds (the exposition's Lemma 4.D), Proposition 2.5.20 (preprint 2.46).
- `Chapter2/StrongBundle`, `Chapter2/StrongBundlePullback`: §2.6 (preprint §2.5).
- `Chapter2/GapP1_Prop212`: Proposition 2.1.2 (preprint 2.2).
- `Chapter2/TangentSpaceLinear`, `Chapter2/TangentMapLinear`, `Chapter2/SubMPolyfoldTangent`, `Chapter2/SubMPolyfoldTangentSc`: the linear, Banach and sc-structure of $T_x X$ (Definition 2.3.14 (preprint 2.18)), fibrewise linearity of $Tf$, Proposition 2.3.19 (3)–(4) (preprint 2.21) in $T_a X$.
- `Chapter2/TangentProjOpen`: Proposition 2.3.15 (2) (preprint 2.19), the openness of $p$.
- `Chapter2/PartialConeQuadrant`: Flor's theorem in the case needed, Remark 2.4.8 (preprint 2.28), the second statement of Theorem 2.4.12 (preprint 2.31).
- `Chapter2/GapP1_Lemma249`: Lemma 2.4.9 (preprint 2.29).
- `Chapter2/GapP1_ExtendedFaces`: the extended faces of a partial quadrant from $C$ alone.
- `Chapter2/ReducedTangentIntrinsic`, `Chapter2/GapP1_Def258`, `Chapter2/GapP1_Def258b`: Definition 2.5.8 (preprint 2.38) in $T_x X$.
- `Chapter2/StrongBundleRestrict`, `Chapter3/StrongBundleMaximalAtlas`, `Chapter3/StrongBundleIsoSBR`: transitivity of atlas equivalence, restriction of strong bundle charts, the maximal strong bundle atlas, strong bundle isomorphisms and $\mathcal{SBR}$.
- `Chapter2/GapP1_Michael`: Lemma 2.7.3 and Proposition 2.7.4 (preprint 2.56 and 2.57).
- `Chapter2/BookFormLocalCompleteness2021`, `Chapter2/BookFormTameSub2021`, `Chapter2/BookFormTameBoundary2021`: the 2021 statements in book form.
- `Chapter2/SmoothPaths`: smooth points of connected open sets are path-connected.
- `Chapter2/StrongBundleOne`: the M-polyfold $Y[1]$, $P[1]$, the chart independence of $Y_{m,k}$, sc$^+$-sections into $Y[1]$.

## II.1 Restriction and locality of sc^k maps

The text restricts sc-smooth maps to smaller sets ("$r : U \to U \cap U'$ is sc-smooth") and glues local sc-smoothness without comment. Both need an argument, because the tangent map of an sc$^1$ map is built

from a *chosen* derivative family (Part I, §I.11), and a chosen derivative on $U$ and one on $U' \subseteq U$ need not be the same objects.

*Restriction.* If $f$ is $\mathrm{sc}^1$ on $U$ with derivative family $D$ and $U' \subseteq U$, the same $D$ works on $U'$ `Chapter2/Retracts · IsSc1With.of_subset` `Chapter2/Retracts · IsSc1.of_subset`. When $U'$ is an sc-domain (unique differentiability on every level, Part I, §I.10), derivatives at points of $U'_1$ are unique, so the tangent maps built on $U'$ and on $U$ agree on $TU'$ `Chapter2/Retracts · IsSc1.tangentMap_eqOn_of_subset`. By induction on $k$, using that $TU'$ is again an sc-domain and that the tangent maps agree there, an $\mathrm{sc}^k$ map restricts to $\mathrm{sc}^k$ on every smaller sc-domain `Chapter2/Retracts · IsSck.of_subset`, and so does an sc-smooth map `Chapter2/Retracts · IsScSmooth.of_subset`; sc-smoothness depends only on the values on $U$ `Chapter2/Retracts · IsScSmooth.congr`. Relatively open subsets of quadrants, and their intersections with open sets, are sc-domains `Chapter2/Retracts · IsRelOpenInQuadrant.inter_isOpen` `Chapter2/Retracts · IsScDomain.inter_isOpen`, and $T(U \cap W) = TU \cap (W_1 \times E_0)$ `Chapter2/Retracts · ScBanach.tangentSet_inter_isOpen`.

*Locality.* $\mathrm{sc}^0$ is local: continuity on each level is a local property `Chapter2/Retracts · IsSc0.of_locally`. For $\mathrm{sc}^1$, suppose every $x \in U$ has an open $W_x \ni x$ with $f$ $\mathrm{sc}^1$ on $U \cap W_x$. Two such local structures give the same derivative at a common point, by uniqueness on the sc-domain `Chapter2/Retracts · IsSc1With.eq_of_eq_near`. Glue: at a point $x \in U_1$ take the derivative of the structure on $U \cap W_{\iota_1 x}$. This family $D$ agrees near every point with a local derivative family; the three conditions of Definition 1.1.15 (preprint 1.14) (the $\mathrm{sc}^0$ property, differentiability on $U_1$, and the $\mathrm{sc}^0$ property of the tangent map on $TU$) are local and hold `Chapter2/Retracts · IsSc1.of_locally`. The induction on $k$ applies the same argument to the tangent maps on $TU$ `Chapter2/Retracts · IsSck.of_locally`, and gives the sc-smooth case `Chapter2/Retracts · IsScSmooth.of_locally`. The identity is $\mathrm{sc}^1$ with derivative the identity `Chapter2/Retracts · isSc1With_id`, its tangent map is the identity on $TU$ `Chapter2/Retracts · tangentMap_id_eqOn`, and it is sc-smooth on every sc-domain `Chapter2/Retracts · isSck_id` `Chapter2/Retracts · isScSmooth_id`.

## II.2 Retractions, retracts and their tangent (Definitions 2.1.1, 2.1.3, 2.1.12 (preprint 2.1, 2.3, 2.5); Proposition 2.1.11 (preprint 2.4))

**Definition 2.1.1 (preprint 2.1).** An *sc-smooth retraction* on a partial quadrant $C \subseteq E_0$ is a set $U$ relatively open in $C$ and a map $r : E_0 \to E_0$ with $r(U) \subseteq U$, $r$ sc-smooth on $U$ and $r(r(x)) = r(x)$ for $x \in U$ `Chapter2/Retracts · ScRetraction`. Since $U$ is relatively open, it is an sc-domain and $r$ is $\mathrm{sc}^1$ `Chapter2/Retracts · ScRetraction.isScDomain` `Chapter2/Retracts · ScRetraction.isSc1`. The *retract* is $O = r(U)$ `Chapter2/Retracts · ScRetraction.retract`. It is contained in $U$ and is exactly the fixed-point set of $r$ in $U$: if $x = r(y)$ then $r(x) = r(r(y)) = x$ `Chapter2/Retracts · ScRetraction.retract_subset` `Chapter2/Retracts · ScRetraction.apply_eq_self_of_mem_retract` `Chapter2/Retracts · ScRetraction.mem_retract_iff`. On level 1, $r_1$ fixes the points of $O_1$ `Chapter2/Retracts · ScRetraction.levelMap_one_eq_self`, and $O_m \subseteq U_m$ `Chapter2/Retracts · ScRetraction.lvlSet_retract_subset`.

**Definition 2.1.3 (preprint 2.3).** $(O, C, E)$ is an *sc-smooth retract* if $O$ is the retract of some sc-smooth retraction on $C$ `Chapter2/Retracts · IsScRetract` `Chapter2/Retracts · ScRetraction.isScRetract`. Every relatively open $U \subseteq C$ is one, with $r = \mathrm{id}$ `Chapter2/Retracts · ScRetraction.id` `Chapter2/Retracts · IsRelOpenInQuadrant.isScRetract`. A *local model* is a triple $(E, C, O)$ with $C$ a partial quadrant and $O$ a retract in $C$ `Chapter2/Retracts · LocalModel`; these are the models of M-polyfold charts.

*The tangent of a retraction.* Put $Tr(TU) := \{Tr(p) : p \in TU\} \subseteq E_1 \times E_0$ `Chapter2/Retracts · ScRetraction.tangent`. The chain rule applied to $r \circ r = r$ (Part I, Theorem 1.3.1 (preprint 1.21)) gives $Tr \circ Tr =$

$Tr$ on $TU$ Chapter2/Retracts · ScRetraction.tangentMap_tangentMap Chapter1/ChainRule · theorem_1_3_1, since $Tr$ maps $TU$ into $TU$ Chapter2/Retracts · ScRetraction.tangentMap_mapsTo. Hence the **fixed-point description**: $p \in Tr(TU)$ iff $p \in TU$ and $Tr(p) = p$ Chapter2/Retracts · ScRetraction.mem_tangent_iff. Base points of $Tr(TU)$ lie in $O_1$ Chapter2/Retracts · ScRetraction.fst_mem_of_mem_tangent. For $x \in O_1$ the fixed-point condition reads $Dr(x)h = h$ Chapter2/Retracts · ScRetraction.mem_tangent_iff_of_mem, and the second component of $Tr \circ Tr = Tr$ at $(x, h)$ gives $Dr(x) \circ Dr(x) = Dr(x)$ Chapter2/Retracts · ScRetraction.scDeriv_comp_self — the remark after Definition 2.1.3 (preprint 2.3) that $Dr(x)$ is a projection. So for $x \in O_1$, $(x, h) \in Tr(TU)$ iff $h$ is in the range of $Dr(x)$ Chapter2/Retracts · ScRetraction.mem_tangent_iff_mem_range, and $Tr(TU) = \{(r(x), Dr(x)h) : x \in U_1, h \in E_0\}$ Chapter2/Retracts · ScRetraction.mem_tangent_iff_exists.

**Proposition 2.1.11 (preprint 2.4).** *If $r(U) \subseteq r'(U')$ then $r' \circ r = r$ on $U$*: for $y \in U$, $r(y) \in r'(U')$ is a fixed point of $r'$ Chapter2/Retracts · ScRetraction.comp_eq_self_of_retract_subset. *And then $Tr(TU) \subseteq Tr'(TU')$*: let $p \in Tr(TU)$, so $p \in TU$ and $Tr(p) = p$. Its base point lies in $O_1 \subseteq U'_1$, so $p \in TU'$. The map $r$ sends $U$ into $U'$ (into $O \subseteq U'$), so the chain rule applies to $r' \circ r$ on $U$: $T(r' \circ r)(p) = Tr'(Tr(p)) = Tr'(p)$; and $T(r' \circ r) = Tr$ on $TU$ because $r' \circ r = r$ on $U$ (tangent maps depend only on the values on $U$). So $Tr'(p) = Tr(p) = p$, which is the fixed-point condition for $r'$ Chapter2/Retracts · ScRetraction.tangent_subset_of_retract_subset. Applied in both directions, two retractions with the same retract have the same tangent image Chapter2/Retracts · proposition_2_1_11. This is the text's proof, with its step "$Tr'(x, h) = (x, h)$" made the definition of membership.

**Definition 2.1.12 (preprint 2.5).** $TO$ is $Tr(TU)$ for a retraction chosen (by choice) among those onto $O$, and the empty set if $O$ is not a retract Chapter2/Retracts · retractTangent; by Proposition 2.1.11 (preprint 2.4) it equals $Tr(TU)$ for *every* retraction onto $O$ Chapter2/Retracts · retractTangent_eq. The tangent space at $x \in O_1$ is the fibre $T_xO = \{h : (x, h) \in TO\}$ Chapter2/Retracts · tangentSpace, the range of $Dr(x)$ Chapter2/Retracts · tangentSpace_eq_range, a linear subspace Chapter2/Retracts · tangentSpace_eq_coe_submodule (closed, as the range of a continuous projection).

## II.3 Sc-smooth maps between retracts (Proposition 2.1.13 (preprint 2.6), Definition 2.1.14 (preprint 2.7), Theorem 2.1.15 (preprint 2.8), Proposition 2.3.1 (preprint 2.10))

**Proposition 2.1.13 (preprint 2.6).** Let $r$ on $U$ and $s$ on $V$ be retractions onto the same $O$. Since $s$ takes values in $O$, which $r$ fixes, $(f \circ r) \circ s = f \circ s$ on $V$ Chapter2/Retracts · ScRetraction.comp_comp_eqOn. (1) If $f \circ r$ is sc-smooth on $U$, then $f \circ r \circ s$ is sc-smooth on $V$ by the chain rule ($s$ maps $V$ into $O \subseteq U$), hence so is $f \circ s$ Chapter2/Retracts · proposition_2_1_13_one Chapter1/ChainRule · IsScSmooth.comp. (2) For $p \in TO$: $p \in TV$ and $Ts(p) = p$ (fixed-point description for $s$, after Proposition 2.1.11 (preprint 2.4)); so $T(f \circ r)(p) = T(f \circ r)(Ts(p)) = T((f \circ r) \circ s)(p) = T(f \circ s)(p)$ Chapter2/Retracts · proposition_2_1_13_two. (3) If $f$ maps $O$ into $O' = r'(U')$, then $r' \circ f \circ r = f \circ r$ on $U$; the chain rule gives $Tr'(T(f \circ r)(p)) = T(f \circ r)(p)$ for $p \in TU$, and $T(f \circ r)(p) \in TU'$, so $T(f \circ r)(p) \in TO'$ by the fixed-point description Chapter2/Retracts · proposition_2_1_13_three.

**Definition 2.1.14 (preprint 2.7).** $f$ is *sc-smooth from $(O, C, E)$ to $(O', C', E')$* if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth on $U$ for every retraction $r$ onto $O$ Chapter2/Retracts · IsScSmoothRetractMap. For a retract $O$ this is equivalent to "$f(O) \subseteq O'$ and $f \circ r$ sc-smooth for one $r$", by Proposition 2.1.13 (1) (preprint 2.6) Chapter2/Retracts · isScSmoothRetractMap_iff_exists. Quantifying over all retractions makes the definition in-

dependent of choices without further argument; the target triple enters only through $f(O) \subseteq O'$. The *tangent map* $Tf$ is $T(f \circ r)$ for the chosen retraction `Chapter2/Retracts · retractTangentMap`; on $TO$ it equals $T(f \circ r)$ for every $r$ onto $O$, by Proposition 2.1.13 (2) (preprint 2.6) `Chapter2/Retracts · retractTangentMap_eqOn`, it maps $TO$ into $TO'$ by (3) `Chapter2/Retracts · IsScSmoothRetractMap.tangentMap_mapsTo`, and $Tf(x,h) = (f(x), D(f\circ r)(x)h)$ `Chapter2/Retracts · retractTangentMap_apply`. An sc-smooth map between retracts is continuous on $O$ (level 0 of the $\mathrm{sc}^0$ property of $f\circ r$, and $f = f\circ r$ on $O$) `Chapter2/Retracts · IsScSmoothRetractMap.continuousOn` and maps points of $O$ on level $m$ to points on level $m$ `Chapter2/Retracts · IsScSmoothRetractMap.mem_range_toBase`.

**Theorem 2.1.15 (preprint 2.8).** Let $O = r(U)$, $O' = r'(U')$ be retracts, $f$ sc-smooth from $O$ to $O'$ and $g$ from $O'$ to $O''$. Since $f \circ r$ takes values in $O'$, which $r'$ fixes, $(g \circ r') \circ (f \circ r) = (g \circ f) \circ r$ on $U$ `Chapter2/Retracts · ScRetraction.comp_comp_comp_eqOn`. The left side is sc-smooth by the chain rule ($f \circ r$ maps $U$ into $O' \subseteq U'$, and $g \circ r'$ is sc-smooth on $U'$), so $g \circ f$ is sc-smooth from $O$ to $O''$ `Chapter2/Retracts · IsScSmoothRetractMap.comp`. For the tangent maps, at $p \in TO$: $T(g \circ f)(p) = T((g \circ f) \circ r)(p) = T(g \circ r')(T(f \circ r)(p)) = Tg(Tf(p))$, the last step because $Tf(p) \in TO'$ `Chapter2/Retracts · retractTangentMap_comp`. Together `Chapter2/Retracts · theorem_2_1_15`. The text gives no proof ("follows from Theorem 1.21"); this is the one it has in mind, with $r' \circ f \circ r = f \circ r$ made explicit. Further: the identity is sc-smooth `Chapter2/Retracts · isScSmoothRetractMap_id`, a retraction is sc-smooth as a map $(U,C,E) \to (O,C,E)$ and the inclusion $O \to U$ as a map $(O,C,E) \to (U,C,E)$ `Chapter2/Retracts · ScRetraction.isScSmoothRetractMap` `Chapter2/Retracts · ScRetraction.isScSmoothRetractMap_inclusion`, and sc-smoothness depends only on $f|_O$ `Chapter2/Retracts · IsScSmoothRetractMap.congr`.

**Proposition 2.3.1 (preprint 2.10).** (1) For $W \subseteq E_0$ open, $U \cap r^{-1}(W)$ is relatively open in $C$ because $r$ is continuous on $U$ `Chapter2/Retracts · ScRetraction.relOpen_inter_preimage`; the restriction of $r$ to it is a retraction `Chapter2/Retracts · ScRetraction.restrict` (sc-smooth by §II.1) with retract $O \cap W$ `Chapter2/Retracts · ScRetraction.restrict_retract`. So $(O \cap W, C, E)$ is a retract `Chapter2/Retracts · proposition_2_3_1_one` `Chapter2/Retracts · LocalModel.restrict`; "an open subset $O'$ of $O''$ is $O \cap W$ in the subspace topology. (2) If $V = r(U)$ and $s$ is sc-smooth from $(V,C,E)$ to itself with $s \circ s = s$ on $V$, then $\rho = s \circ r$ is sc-smooth on $U$ (that is what sc-smoothness of $s$ on the retract $V$ means), maps $U$ into $s(V) \subseteq V \subseteq U$, and is idempotent: $\rho(\rho(x)) = s(r(s(r(x)))) = s(s(r(x))) = s(r(x))$ because $r$ fixes $s(r(x)) \in V$. Its retract is $s(V)$ `Chapter2/Retracts · proposition_2_3_1_two`. As the text's proof.

*Restriction and locality between retracts.* An sc-smooth map between retracts restricts to $O \cap W$: a retraction onto $O \cap W$ composed with $r$ is handled by the chain rule `Chapter2/Retracts · IsScSmoothRetractMap.restrict`; the target may be shrunk to any set containing the image `Chapter2/Retracts · IsScSmoothRetractMap.mono_target`. Conversely, if every point of $O$ has an open $W$ with $f$ sc-smooth on $(O \cap W, C, E)$, then $f$ is sc-smooth on $(O,C,E)$: for a retraction $r$ onto $O$ and $y \in U$, the set $W$ around $r(y)$ pulls back under the continuous $r$ to an open $u \ni y$; on $U \cap u$ the map $f \circ r$ equals $(f \circ r') \circ r$ with $r'$ the restriction of $r$ onto $O \cap W$, which is sc-smooth, and the locality of §II.1 concludes `Chapter2/Retracts · IsScSmoothRetractMap.of_locally`. The text uses this silently every time it says "the definition does not depend on the choice of charts".

## II.4 The tangent functor on retracts

*The tangent of a partial quadrant.* If $C = T_0^{-1}(\text{quadrant})$ for $T : E \cong \mathbb{R}^n \oplus W$, then $TC = C_1 \times E_0$ is a partial quadrant in $TE = E^1 \oplus E$: compose the tangent of $T$, an sc-isomorphism $TE \cong T(\mathbb{R}^n \oplus W)$, with the re-association $((a,w),(b,v)) \mapsto (a,(w,(b,v)))$ onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ `Chapter2/TangentBundle · tangentQuadrantIso` `Chapter2/TangentBundle · IsPartialQuadrant.tangentSet`. $TU$ is relatively open

in $TC$ Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet (this is Lemma 3.B of the exposition of Chapter 1; Part I, V1.36). Sc-operators are sc-smooth on every sc-domain Chapter2/TangentBundle · ScOperator.isScSmooth.

*$Tr$ is a retraction.* $Tr$ is sc-smooth on $TU$ (Part I: the tangent map of an sc-smooth map is sc-smooth) Chapter2/Retracts · LocalModel.tangent_isScSmooth, maps $TU$ into itself and is idempotent; so it is an sc-smooth retraction on $TC$ Chapter2/TangentBundle · ScRetraction.tangentRetraction with retract $TO$ Chapter2/TangentBundle · ScRetraction.tangentRetraction_retract. Hence $(TO, TC, TE)$ is a retract Chapter2/TangentBundle · IsScRetract.tangentSet Chapter2/TangentBundle · LocalModel.tangent, base points of $TO$ lie in $O_1$ and $(x, 0) \in TO$ for $x \in O_1$ Chapter2/TangentBundle · retractTangent_fst_mem Chapter2/TangentBundle · mk_zero_mem_retractTangent.

*$Tf$ is sc-smooth.* If $f$ is sc-smooth from $O$ to $O'$ then $Tf$ is sc-smooth from $(TO, TC, TE)$ to $(TO', TC', TE')$: with the retraction $Tr$ onto $TO$, $Tf \circ Tr = T(f \circ r) \circ Tr = T(f \circ r \circ r) = T(f \circ r)$ on $TU$, which is sc-smooth Chapter2/TangentBundle · IsScSmoothRetractMap.tangent. $Tf$ depends only on $f|_O$ Chapter2/TangentBundle · retractTangentMap_congr, and a map equal to the identity on $O$ has tangent map the identity on $TO$ Chapter2/TangentBundle · retractTangentMap_id. With Theorem 2.1.15 (preprint 2.8) this is the tangent functor $T : \mathcal{R} \to \mathcal{R}$ of the text; the bundle structure of Definition 2.1.16 (preprint 2.9) is §II.32 ($Tr$ a bundle retraction, $Tf$ a bundle map), the category $\mathcal{BR}$ is not formalized.

*Locality of the tangent.* For $W$ open, $T(O \cap W) = TO \cap (W_1 \times E_0)$: with the restricted retraction, a fixed point of $Tr$ over a base point in $W_1$ is a fixed point of the restricted $Tr$, the two tangent maps agreeing on the smaller domain (§II.1) Chapter2/TangentBundle · retractTangent_inter_isOpen Chapter2/TangentBundle · retractTangent_mono_inter. And $Tf$ computed on $(O \cap W, C, E)$ agrees with $Tf$ computed on $(O, C, E)$ on $T(O \cap W)$ Chapter2/TangentBundle · retractTangentMap_inter_isOpen. These two facts are what Definitions 2.3.12–2.3.14 (preprint 2.17–2.18) use silently when the chain rule is applied on a triple overlap.

## II.5 Charts, atlases and M-polyfolds (Definitions 2.3.2–2.3.8 (preprint 2.11–2.14))

**Definition 2.3.2 (preprint 2.11).** A *chart* on $X$ is a local model $(O, C, E)$ and a partial equivalence $\varphi$ between $X$ and $E_0$ with open source $V$ and target $O$, continuous on $V$, with inverse continuous on $O$ Chapter2/MPolyfold · MChart. Images of open subsets of $V$ are open subsets $O \cap W$ of $O$ Chapter2/MPolyfold · MChart.exists_isOpen_image_eq, hence retracts (Proposition 2.3.1 (1) (preprint 2.10)) Chapter2/MPolyfold · MChart.isScRetract_image, and charts restrict to open subsets of their sources Chapter2/MPolyfold · MChart.restrict. The overlap $\varphi(V \cap V')$ Chapter2/MPolyfold · MChart.overlap is a retract Chapter2/MPolyfold · MChart.isScRetract_overlap, the transition map is $\psi \circ \varphi^{-1}$ Chapter2/MPolyfold · MChart.transition, mapping $\varphi(V \cap V')$ into $\psi(V \cap V')$ Chapter2/MPolyfold · MChart.transition_mapsTo, and two charts are *compatible* if both transition maps are sc-smooth between the overlap retracts Chapter2/MPolyfold · MChart.Compatible. Compatibility is symmetric, reflexive, and passes to restrictions Chapter2/MPolyfold · MChart.Compatible.symm Chapter2/MPolyfold · MChart.compatible_self Chapter2/MPolyfold · MChart.Compatible.restrict.

*Compatibility is local.* Let $\varphi$, $\psi$ be charts and $S$ a family of charts each compatible with $\varphi$ and with $\psi$, whose sources cover $V \cap V'$. Then $\varphi$ and $\psi$ are compatible Chapter2/MPolyfold · MChart.compatible_of_locally. By the locality between retracts (§II.3) it suffices to show that $\psi \circ \varphi^{-1}$ is sc-smooth near every point $\varphi(x)$ of the overlap. Take $\chi \in S$ around $x$ and the triple overlap $s = V \cap V' \cap V''$, with $\varphi(s) = \varphi(V \cap V') \cap W$

for an open $W$. On $\varphi(s)$, $\psi \circ \varphi^{-1} = (\psi \circ \chi^{-1}) \circ (\chi \circ \varphi^{-1})$ Chapter2/MPolyfold · MChart.transition_eq_comp_on; the two factors are sc-smooth on the restricted retracts $\varphi(s)$ and $\chi(s)$, and Theorem 2.1.15 (preprint 2.8) composes them. The reverse transition is the same argument with the roles exchanged. This is the step that makes "the equivalence class of atlases" a usable notion; the text does not discuss it.

**Definition 2.3.3 (preprint 2.12).** An *atlas* is a set of pairwise compatible charts covering $X$ Chapter2/MPolyfold · MAtlas; two atlases are *equivalent* if their union is an atlas Chapter2/MPolyfold · MAtlas.Equiv, i.e. if every chart of one is compatible with every chart of the other Chapter2/MPolyfold · MAtlas.equiv_iff. The relation is reflexive and symmetric Chapter2/MPolyfold · MAtlas.Equiv.refl Chapter2/MPolyfold · MAtlas.Equiv.symm, and transitive by the locality of compatibility (§II.45) Chapter2/StrongBundleRestrict · MAtlas.Equiv.trans.

**Definition 2.3.4 (preprint 2.13).** An *M-polyfold* structure on a Hausdorff paracompact space is an atlas Chapter2/MPolyfold · MPolyfold. Its *charts* are all charts compatible with every chart of the atlas Chapter2/MPolyfold · MPolyfold.IsChart Chapter2/MPolyfold · MPolyfold.isChart_of_mem. By locality of compatibility (with $S$ the atlas), any two of them are compatible Chapter2/MPolyfold · MPolyfold.IsChart.compatible; restrictions of charts are charts Chapter2/MPolyfold · MPolyfold.IsChart.restrict; every point lies in one Chapter2/MPolyfold · MPolyfold.exists_isChart_mem_source; and equivalent atlases have the same charts, again by locality Chapter2/MPolyfold · MPolyfold.isChart_congr. So the maximal atlas represents the equivalence class, and every later definition (levels, index, tangent bundle) quantifies over it.

**Definition 2.3.8 (preprint 2.14).** $f : X \to Y$ is *sc-smooth on the open set* $U$ if it is continuous on $U$ and for all charts $\varphi$ of $X$, $\psi$ of $Y$ and every open $V \subseteq U \cap \operatorname{dom}\varphi$ with $f(V) \subseteq \operatorname{dom}\psi$, the representation $\psi \circ f \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V), C, E)$ into $(O', C', E')$ Chapter2/MPolyfold · MPolyfold.IsScSmoothOn; *sc-smooth* means sc-smooth on $X$ Chapter2/MPolyfold · MPolyfold.IsScSmoothMap. Continuity is a field: the text's pairs of charts with $f(V) \subseteq V'$ exist around every point only for continuous $f$. The notion passes to open subsets and to maps agreeing on $U$ Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.mono Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.congr. *One pair of charts suffices*: if every $x \in U$ has charts $\varphi, \psi$ and an open $V \ni x$ with $\psi \circ f \circ \varphi^{-1}$ sc-smooth on $\varphi(V)$, then $f$ is sc-smooth on $U$ Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts. For arbitrary charts $\varphi'$, $\psi'$ and $V'$, locality reduces to a neighbourhood $\varphi'(V \cap V')$ of each point, where $\psi' \circ f \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1})$, three sc-smooth maps between retracts composed by Theorem 2.1.15 (preprint 2.8). This is the text's "the definition does not depend on the choice of charts". The identity is sc-smooth Chapter2/MPolyfold · MPolyfold.isScSmoothMap_id, and compositions are sc-smooth by the same three-factor argument with a chart of the middle space Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.comp Chapter2/MPolyfold · MPolyfold.IsScSmoothMap.comp.

## II.6 The filtration $X_m$ and the M-polyfold $X^1$ (Lemma 2.3.9 (preprint 2.15))

*Levels.* $x$ is on level $m$ if $\varphi(x) \in \iota_m(E_m)$ for some chart $\varphi$ around $x$ Chapter2/MPolyfold · MPolyfold.levelSet. *Independence of the chart*: if $\psi$ is another chart around $x$, the transition map $\varphi \circ \psi^{-1}$ is sc-smooth between the overlap retracts, hence maps points on level $m$ to points on level $m$ (§II.3), and $\varphi(x) = (\varphi \circ \psi^{-1})(\psi(x))$ Chapter2/MPolyfold · MPolyfold.mem_levelSet_iff. $X_0 = X$, $X_{m+1} \subseteq X_m$, and $X_\infty = \bigcap_m X_m$ Chapter2/MPolyfold · MPolyfold.levelSet_zero Chapter2/MPolyfold · MPolyfold.levelSet_succ_subset Chapter2/MPolyfold · MPolyfold.smoothSet. For $x \in V_m$ the lift of $\varphi(x)$ to $E_m$ Chapter2/MPolyfold · MPolyfold.levelLift satisfies $\iota_m(\text{lift}) = \varphi(x)$ and is compatible with the inclusions Chapter2/MPolyfold · MPolyfold.toBase_levelLift Chapter2/MPolyfold · MPolyfold.levelLift_eq_incl. $X_m$ carries the topology generated by the sets $\{x \in V : \text{lift}(x) \in W\}$, $W$ open in $E_m$ Chapter2/MPolyfold · MPolyfold.levelBasis Chapter2/

MPolyfold · MPolyfold.levelTopology; that this family is a basis is not needed.

**Lemma 2.3.9 (preprint 2.15).** The preimage under $X_{m+1} \to X_m$ Chapter2/MPolyfold · MPolyfold.levelIncl of a generating set $\{\text{lift}_m \in W\}$ is the generating set $\{\text{lift}_{m+1} \in \text{incl}_m^{-1}(W)\}$, by the compatibility of lifts with inclusions; so the inclusion is continuous Chapter2/MPolyfold · lemma_2_3_9. This is the text's proof.

*The space $X_m$.* As a type, $X_m$ is the subtype of points on level $m$ with this topology Chapter2/LevelPolyfold · MPolyfold.Level. The inclusion $X_m \to X$ is continuous: for $V$ open in $X$ and $x \in X_m$, a chart $\varphi$ around $x$ gives $\varphi(V \cap \operatorname{dom}\varphi) = O \cap W$, and the generating set $\{\text{lift}_m \in W_m\}$ is a neighbourhood of $x$ inside the preimage of $V$ Chapter2/LevelPolyfold · MPolyfold.continuous_levelVal Chapter2/LevelPolyfold · MPolyfold.isOpen_level_basic. Since this inclusion is injective and continuous into the Hausdorff $X$, $X_m$ is Hausdorff (an instance in `LevelPolyfold`, the text's argument with disjoint neighbourhoods).

*The level shift of the local data.* An sc-isomorphism shifts to $E^1 \to F^1$ Chapter2/LevelPolyfold · ScIso.shift, so $C_1$ is a partial quadrant in $E^1$ and $U_1$ relatively open in it Chapter2/LevelPolyfold · IsPartialQuadrant.shift Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift. A retraction $r$ induces the retraction $r_1$ on $U_1$ — sc-smooth by Proposition 1.2.2 (preprint 1.17) (Part I, §I.15), idempotent by the injectivity of $\iota_1$ Chapter2/LevelPolyfold · ScRetraction.shift — with retract $O_1$ Chapter2/LevelPolyfold · ScRetraction.shift_retract Chapter2/LevelPolyfold · IsScRetract.shift Chapter2/LevelPolyfold · LocalModel.shift. An sc-smooth map between retracts is $\text{sc}^0$ on $O$, maps $O_m$ into $O'_m$ and is continuous there Chapter2/LevelPolyfold · IsScSmoothRetractMap.isSc0 Chapter2/LevelPolyfold · IsScSmoothRetractMap.mapsTo_levelMap Chapter2/LevelPolyfold · IsScSmoothRetractMap.continuousOn_levelMap, and its level-1 map is sc-smooth from $O_1$ to $O'_1$: with the retraction $r_1$, $f_1 \circ r_1 = (f \circ r)_1$, sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter2/LevelPolyfold · IsScSmoothRetractMap.shift.

*The M-polyfold $X^1$.* For a chart $\varphi$ of $X$, $x \mapsto \text{lift}_1(x)$ on $V_1$ with model $(O_1, C_1, E^1)$ is a chart of $X_1$ Chapter2/LevelPolyfold · MPolyfold.levelChart (with a junk value for the inverse off $O_1$, since $X_1$ might be empty Chapter2/LevelPolyfold · MPolyfold.levelChartInv); its overlap with another level chart is the level-1 part of the overlap Chapter2/LevelPolyfold · MPolyfold.levelChart_overlap, its transition map the level-1 map of the transition map Chapter2/LevelPolyfold · MPolyfold.levelChart_transition, and so any two level charts are compatible by the level shift of sc-smooth maps Chapter2/LevelPolyfold · MPolyfold.levelChart_compatible. They form an atlas Chapter2/LevelPolyfold · MPolyfold.levelAtlas, and $X_1$ with it is the M-polyfold $X^1$ Chapter2/LevelPolyfold · MPolyfold.shift Chapter2/LevelPolyfold · MPolyfold.shift_isChart. Its paracompactness, an instance argument, is supplied by Theorem 2.3.10 (preprint 2.16) (§II.7). Only $m = 1$ is constructed; the levels of $X^1$ are those of $X$ with their topology and charts, so the iterates are the $X_m$ (§II.32; gap G.6, closed).

## II.7 Metrizability: Smirnov's theorem and Theorem 2.3.10 (preprint 2.16) (Theorem 2.7.1 (preprint 2.55))

The text proves Theorem 2.3.10 (preprint 2.16) with Smirnov's theorem (2.55), Michael's characterization of paracompactness (Lemma 2.7.3 (preprint 2.56)) and Proposition 2.7.4 (preprint 2.57) (a regular space with a locally finite closed cover by paracompact subspaces is paracompact). Proposition 2.7.4 (preprint 2.57) needs $X_m$ to be regular, which the text does not check. The formal proof needs neither 2.56 nor 2.57.

*The metrization lemma.* Let $Y$ have open subspaces $V_i$, each metrizable with a metric $d_i$, and contin-

uous $\rho_i : Y \to [0,\infty)$ with $\operatorname{supp}\rho_i \subseteq V_i$, the supports locally finite and $\rho_i(y) > 0$ for some $i$ at every $y$ Chapter2/Metrizable · metrizableSpace_of_partition. Put $\delta_i(x,y) := \min(d_i(x,y),1)$ for $x,y \in V_i$ and 1 otherwise Chapter2/Metrizable · truncDist — nonnegative, at most 1, symmetric, zero only on the diagonal, and satisfying the triangle inequality inside $V_i$ Chapter2/Metrizable · truncDist_nonneg Chapter2/Metrizable · truncDist_le_one Chapter2/Metrizable · truncDist_triangle Chapter2/Metrizable · truncDist_eq_zero — and the *cone metric*

$$e_i(x,y) := |\rho_i(x) - \rho_i(y)| + \min(\rho_i(x),\rho_i(y))\,\delta_i(x,y), \qquad D(x,y) := \sum_i e_i(x,y).$$

The sum is finite (only the $i$ with $x$ or $y$ in $\operatorname{supp}\rho_i$ contribute). $D(x,x) = 0$ and $D$ is symmetric. *Triangle inequality:* termwise, by an elementary inequality for $a,b,c \geq 0$ and $\delta$'s bounded by 1, which needs the triangle inequality for $\delta_i$ only when $\min(\rho_i(x),\rho_i(z)) > 0$ and $\rho_i(y) > 0$, i.e. when all three points lie in $V_i$ Chapter2/Metrizable · cone_triangle. *Separation:* if $D(x,y) = 0$, take $i$ with $\rho_i(x) > 0$; then $\rho_i(y) = \rho_i(x)$ and $\delta_i(x,y) = 0$, so $x = y$. *Topology:* $y \mapsto D(x,y)$ is continuous (locally a finite sum of continuous functions; the term $\min(\rho_i(x),\rho_i(y))\delta_i(x,y)$ is continuous where $y \in V_i$ and vanishes near points outside $\operatorname{supp}\rho_i$), so $D$-balls are open; conversely, if $s$ is open and $x \in s$, take $i$ with $\rho_i(x) > 0$ and a $d_i$-ball of radius $r \leq 1$ around $x$ inside $s$; then $D(x,y) < \frac{1}{2}\rho_i(x)r$ forces $\rho_i(y) > \frac{1}{2}\rho_i(x)$ (so $y \in V_i$) and $\frac{1}{2}\rho_i(x)\delta_i(x,y) < \frac{1}{2}\rho_i(x)r$, so $d_i(x,y) < r$ and $y \in s$. Mathlib then builds a metric space structure with this distance and the given topology Mathlib/Topology/MetricSpace/Defs · MetricSpace.ofDistTopology.

*Pulled back, and Smirnov.* If $\pi : Y \to X$ is continuous into a paracompact Hausdorff $X$ and every point of $X$ has an open $U$ with $\pi^{-1}(U)$ metrizable, then $Y$ is metrizable: a partition of unity of $X$ subordinate to these $U$ Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate, composed with $\pi$, satisfies the hypotheses of the lemma with $V_i = \pi^{-1}(U_i)$ Chapter2/Metrizable · metrizableSpace_of_pullback. With $\pi = \mathrm{id}$ this is **Smirnov's theorem** (Theorem 2.7.1 (preprint 2.55)) Chapter2/Metrizable · metrizableSpace_of_locally. Metrizable spaces are paracompact (Stone; in Mathlib through the instance for pseudo-emetric spaces) Chapter2/Metrizable · paracompactSpace_of_metrizableSpace.

**Theorem 2.3.10 (preprint 2.16).** *$X$ is metrizable*: chart domains are homeomorphic to subsets $O$ of the Banach space $E_0$ Chapter2/Metrizable · MChart.sourceHomeomorph Chapter2/Metrizable · MChart.metrizableSpace_source, and Smirnov applies Chapter2/Metrizable · MPolyfold.metrizableSpace. $V_m \cong O_m$: the lift $x \mapsto \mathrm{lift}_m(x)$ is a topological embedding of $V \cap X_m$ into $E_m$ Chapter2/Metrizable · MPolyfold.isEmbedding_levelLift. Its injectivity is that of $\varphi$; the induced topology is finer than that of $X_m$ because generating sets of $X_m$ inside $V$ are preimages of open sets; it is coarser because a generating set $\{\mathrm{lift}_m^{\psi} \in W\}$ of another chart, intersected with $V$, is the preimage under $\mathrm{lift}_m^{\varphi}$ of the open set $\{z \in \varphi(V \cap V')_m : (\psi\circ\varphi^{-1})_m(z) \in W\}$, open since the level-$m$ map of the transition is continuous on the level-$m$ part of the overlap retract. *$X_m$ is metrizable*: apply the pulled-back Smirnov to the continuous inclusion $X_m \to X$ and the chart domains $V$, whose preimages $V \cap X_m$ embed in $E_m$ Chapter2/Metrizable · MPolyfold.metrizableSpace_level; hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_level. $X_\infty$: the text does not define its topology; it is taken to be the initial topology of the inclusions $X_\infty \to X_m$ Chapter2/Metrizable · MPolyfold.SmoothLevel Chapter2/Metrizable · MPolyfold.smoothIncl Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq. It embeds in the countable product $\prod_m X_m$, which is metrizable Chapter2/Metrizable · metrizableSpace_pi_nat, so $X_\infty$ is metrizable Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel. Together Chapter2/Metrizable · theorem_2_3_10. *No hypothesis is added.* As a by-product every subset of an M-polyfold is metrizable, hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype, which discharges the paracompactness of sub-M-polyfolds (§II.10).

## II.8 The tangent bundle $TX$ (Definitions 2.3.12–2.3.14 (preprint 2.17–2.18), Proposition 2.3.15 (preprint 2.19))

*Tuples and transport.* A tangent tuple is $(x, \varphi, h)$ with $x \in X_1$, $\varphi$ a chart of the structure around $x$ and $(\mathrm{lift}_1 x, h) \in TO$ `Chapter2/TangentBundle · MPolyfold.TangentPre`; its vector is $(\mathrm{lift}_1 x, h)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.vec Chapter2/TangentBundle · MPolyfold.TangentPre.vec_mem`. For a vector of $TO$ over a point of an open $\varphi(s)$, membership in $TO$ and in $T(\varphi(s))$ agree (locality of the tangent, §II.4) `Chapter2/TangentBundle · MPolyfold.TangentPre.mem_retractTangent_image_iff`. The transport of the tuple to a chart $\psi$ around $x$ is $T(\psi \circ \varphi^{-1})(\mathrm{vec})$, the tangent map computed on the overlap retract `Chapter2/TangentBundle · MPolyfold.TangentPre.transport`. Transport to $\varphi$ itself does nothing, by $T(\mathrm{id}) = \mathrm{id}$ on the overlap $\varphi(V)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_self Chapter2/TangentBundle · MChart.transition_self_eqOn`; the transported vector lies in $TO'$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_mem` and has base point $\mathrm{lift}_1^{\psi}(x)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_fst`.

*The chain rule for transports.* Transporting to $\psi$ and then by $T(\chi \circ \psi^{-1})$ is transporting to $\chi$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport`. On the triple overlap $s = V \cap V' \cap V''$ the transition maps are sc-smooth from $\varphi(s)$ to $\psi(s)$ and from $\psi(s)$ to $\chi(s)$ `Chapter2/TangentBundle · MChart.Compatible.transition_image Chapter2/TangentBundle · MChart.image_eq_overlap_inter`, $\chi \circ \varphi^{-1} = (\chi \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1})$ there, and Theorem 2.1.15 (preprint 2.8) gives $T(\chi \circ \varphi^{-1}) = T(\chi \circ \psi^{-1}) \circ T(\psi \circ \varphi^{-1})$ on $T(\varphi(s))$. The three tangent maps here are computed on $\varphi(s)$ and $\psi(s)$, the transports on the full overlaps; the locality of $Tf$ (§II.4) identifies them. The text never checks this.

**Definition 2.3.12 (preprint 2.17).** Two tuples are *equivalent* if their points agree and the transport of the first to the chart of the second is the second `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel`. Equivalent tuples transport to the same vector in every chart (chain rule for transports) `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.transport`; hence the relation is reflexive, symmetric and transitive `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.refl Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.symm Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.trans`, an equivalence relation `Chapter2/TangentBundle · MPolyfold.TangentPre.setoid`.

**Definition 2.3.14 (preprint 2.18).** $TX$ is the quotient `Chapter2/TangentBundle · MPolyfold.TangentSpace Chapter2/TangentBundle · MPolyfold.TangentSpace.mk`, with $p : TX \to X^1$, $[(x, \varphi, h)] \mapsto x$ `Chapter2/TangentBundle · MPolyfold.tangentProj`, and $T_x X = p^{-1}(x)$ `Chapter2/TangentBundle · MPolyfold.tangentFiber`. For a chart $\varphi$: $TV = p^{-1}(V_1)$ `Chapter2/TangentBundle · MPolyfold.tangentChartSet` and $T\varphi : TV \to TO$ sends a class to the transport of a representative to $\varphi$, well defined by the previous paragraph `Chapter2/TangentBundle · MPolyfold.tangentChartMap Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk`; $T\varphi[(x, \varphi, h)] = (\varphi(x), h)$ `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_self`, $T\varphi$ lands in $TO$ over the lift of the base point `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mem Chapter2/TangentBundle · MPolyfold.tangentChartMap_fst`, and it is injective on $TV$: equal images have equal base points (injectivity of $\varphi$ and $\iota_1$) and then the representatives are equivalent `Chapter2/TangentBundle · MPolyfold.tangentChartMap_injOn`. Its inverse on $TO$ is $a \mapsto [(\varphi^{-1}(\iota_1 a_1), \varphi, a_2)]$ `Chapter2/TangentBundle · MPolyfold.TangentPre.ofVec Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec Chapter2/TangentBundle · MPolyfold.mk_ofVec_tangentChartMap`, and in another chart $T\psi \circ (T\varphi)^{-1} = T(\psi \circ \varphi^{-1})$ on $T(\varphi(V \cap V'))$ `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec_of_mem Chapter2/TangentBundle · MPolyfold.mk_ofVec_mem_tangentChartSet_iff`. The topology is generated by the sets $\{\alpha \in TV : T\varphi(\alpha) \in W\}$, $W$ open `Chapter2/TangentBundle · MPolyfold.tangentBasis Chapter2/TangentBundle · MPolyfold.isOpen_tangent_basic`; $TV$ itself is open `Chapter2/TangentBundle · MPolyfold.isOpen_tangentChartSet`. Here $T_x X$ is identified with $T_{\varphi(x)} O$ by $T\varphi$; its

own vector space structure, for which $T\varphi$ is a linear isomorphism in every chart, is §II.39 Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberLinearEquiv.

**Proposition 2.3.15 (preprint 2.19).** (2) *$p$ is continuous*: the preimage of a generating set $\{\mathrm{lift}_1^{\psi} \in W\}$ of $X_1$ is the generating set of $TX$ given by $\psi$ and $W \times E_0$ Chapter2/TangentBundle · MPolyfold.continuous_tangentProj; so is $TX \to X$ Chapter2/TangentBundle · MPolyfold.continuous_levelVal_tangentProj. Openness of $p$ is §II.40 Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj. (1) *$TX$ is Hausdorff* (the instance `T2Space` in `TangentBundle`): two classes over the same point lie in one $TV$ and have distinct images under the injective $T\varphi$, separated by open sets of $E_1 \times E_0$ whose preimages are basic; classes over distinct points are separated by preimages of disjoint open sets of $X$ under the continuous $TX \to X$. (3) *$TX$ is metrizable*: apply the pulled-back Smirnov theorem to $TX \to X$ and the chart domains $V$, whose preimages $TV$ are homeomorphic to subsets of $E_1 \times E_0$ through the charts of §II.8's next paragraph Chapter2/Metrizable · MPolyfold.metrizableSpace_tangentSpace Chapter2/Metrizable · MChart.metrizableSpace_source; hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_tangentSpace Chapter2/Metrizable · proposition_2_3_15_three.

*$TX$ as an M-polyfold.* For a chart $\varphi$ of $X$, $(TV, T\varphi, (TO, TC, TE))$ is a chart of $TX$ Chapter2/TangentBundle · MPolyfold.tangentChart Chapter2/TangentBundle · MPolyfold.tangentChartInv; its overlap with another is $T(\varphi(V \cap V'))$ Chapter2/TangentBundle · MPolyfold.tangentChart_overlap and its transition map is $T(\psi \circ \varphi^{-1})$ — formula (5) Chapter2/TangentBundle · MPolyfold.tangentChart_transition — which is sc-smooth between the tangent retracts by the tangent functor (§II.4); so the charts are compatible Chapter2/TangentBundle · MPolyfold.tangentChart_compatible, form an atlas Chapter2/TangentBundle · MPolyfold.tangentAtlas and make $TX$ an M-polyfold Chapter2/TangentBundle · MPolyfold.tangentPolyfold Chapter2/TangentBundle · MPolyfold.tangentPolyfold_isChart. *$p : TX \to X^1$ is sc-smooth*: in the charts $T\varphi$ and the level chart of $\varphi$ it is the projection $TO \to O_1$, $(a, h) \mapsto a$, which is sc-smooth between the retracts $(TO, TC, TE)$ and $(O_1, C_1, E^1)$ because, with the retraction $Tr$ onto $TO$, $\mathrm{pr}_1 \circ Tr = r_1 \circ \mathrm{pr}_1$ on $TU$, a composite of the sc-operator $\mathrm{pr}_1$ Chapter2/TangentBundle · ScBanach.tangentFst and the sc-smooth $r_1$ Chapter2/TangentBundle · MPolyfold.isScSmoothRetractMap_tangent_fst; one pair of charts per point suffices (§II.5) Chapter2/TangentBundle · MPolyfold.tangentProj_isScSmooth.

## II.9 The tangent map of an sc-smooth map between M-polyfolds

The text uses $Tf$ for maps between M-polyfolds from §2.4 (preprint §2.3) on without defining it. *The chain rule on chart images, with locality*: for $F$ sc-smooth on $\varphi(s_1)$ into a retract $O_1$ mapping $\varphi(t)$ into $O_1'$, $G$ sc-smooth on $O_1'$ and $H$ sc-smooth on $\varphi(s_2)$ with $H = G \circ F$ on $\varphi(t)$ ($t \subseteq s_1 \cap s_2$ open), $TH = TG \circ TF$ on $T(\varphi(t))$, the three tangent maps computed on their own domains Chapter2/TangentMap · MChart.retractTangentMap_comp_image — Theorem 2.1.15 (preprint 2.8) on $\varphi(t)$ plus the locality of $Tf$. *Levels*: an sc-smooth map maps $X_m$ into $Y_m$, because its coordinate representations do Chapter2/TangentMap · MPolyfold.IsScSmoothOn.mapsTo_levelSet Chapter2/TangentMap · MPolyfold.IsScSmoothMap.mapsTo_levelSet. *The image vector*: for a tuple $(x, \varphi, h)$ and a chart $\psi$ around $f(x)$, $T(\psi \circ f \circ \varphi^{-1})(\mathrm{vec})$, computed on $\varphi(V \cap f^{-1}(V'))$ Chapter2/TangentMap · MPolyfold.TangentPre.fdom Chapter2/TangentMap · MPolyfold.TangentPre.ftransport, lies in $TO'$ over $\psi(f(x))$ Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_mem Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_fst; changing $\psi$ changes it by $T$ of the transition map Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_transport, changing $\varphi$ by transport Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_eq_of_transport, and equivalent tuples give the same image Chapter2/TangentMap · MPolyfold.TangentPre.Rel.ftransport — all by the chain rule with locality. The image tuple $(f(x), \psi, \cdot)$ Chapter2/TangentMap · MPolyfold.TangentPre.fimageWith therefore has a class independent of $\psi$ and of the representative Chapter2/TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq Chapter2/

TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq_of_rel.

$Tf$. With a chosen $\psi$, $Tf[(x,\varphi,h)]$ is the class of the image tuple Chapter2/TangentMap · MPolyfold.tangentMap, computable in any chart around $f(x)$ Chapter2/TangentMap · MPolyfold.tangentMap_mk. It covers $f$ Chapter2/TangentMap · MPolyfold.levelVal_tangentProj_tangentMap, maps $T_xX$ into $T_{f(x)}Y$ Chapter2/TangentMap · MPolyfold.tangentMap_mapsTo_fiber, and in charts $T\psi \circ Tf = T(\psi \circ f \circ \varphi^{-1}) \circ T\varphi$ Chapter2/TangentMap · MPolyfold.tangentChartMap_tangentMap. $T(\mathrm{id}) = \mathrm{id}$ Chapter2/TangentMap · MPolyfold.tangentMap_id, and **the chain rule** $T(g \circ f) = Tg \circ Tf$ holds: with charts $\psi$ around $f(x)$ and $\chi$ around $g(f(x))$ and the open set $t = \mathrm{fdom}(\varphi, f, \psi) \cap f^{-1}(\mathrm{fdom}(\psi, g, \chi))$, the chain rule with locality gives $T(\chi \circ g \circ f \circ \varphi^{-1}) = T(\chi \circ g \circ \psi^{-1}) \circ T(\psi \circ f \circ \varphi^{-1})$ on $T(\varphi(t))$ Chapter2/TangentMap · MPolyfold.tangentMap_comp. $Tf$ is continuous: near a class in $TV$, the preimage of a generating set of $TY$ contains the generating set of $TX$ given by $\varphi$ and the preimage under the continuous $T(\psi \circ f \circ \varphi^{-1})$ Chapter2/TangentMap · MPolyfold.continuous_tangentMap; and $Tf$ is sc-smooth $TX \to TY$, its representation in the charts $T\varphi$, $T\psi$ being $T(\psi \circ f \circ \varphi^{-1})$, sc-smooth by the tangent functor Chapter2/TangentMap · MPolyfold.tangentMap_isScSmooth.

## II.10 Sub-M-polyfolds (Definition 2.3.18 (preprint 2.20), Proposition 2.3.19 (preprint 2.21))

**Definition 2.3.18 (preprint 2.20).** $A \subseteq X$ is a *sub-M-polyfold* if every $a \in A$ has an open $V \ni a$ and a map $r$ with $r(V) \subseteq V$, sc-smooth on $V$, idempotent on $V$, with $r(V) = A \cap V$ Chapter2/MPolyfold · MPolyfold.IsSubMPolyfold; $X$ is one Chapter2/MPolyfold · MPolyfold.isSubMPolyfold_univ. The data at a point are packaged Chapter2/SubMPolyfold · MPolyfold.SubData Chapter2/SubMPolyfold · MPolyfold.IsSubMPolyfold.data: $r$ maps $V$ into $A$, fixes $A \cap V$, and is continuous on $V$ Chapter2/SubMPolyfold · MPolyfold.SubData.apply_mem Chapter2/SubMPolyfold · MPolyfold.SubData.apply_eq_self Chapter2/SubMPolyfold · MPolyfold.SubData.continuousOn.

**Proposition 2.3.19 (preprint 2.21), the structure.** *Shrinking.* For a chart $\varphi$, put $W := (V \cap \mathrm{dom}\,\varphi) \cap r^{-1}(V \cap \mathrm{dom}\,\varphi)$ Chapter2/SubMPolyfold · MPolyfold.SubData.shrink: open (continuity of $r$) Chapter2/SubMPolyfold · MPolyfold.SubData.isOpen_shrink, containing every $a \in A \cap V \cap \mathrm{dom}\,\varphi$ Chapter2/SubMPolyfold · MPolyfold.SubData.mem_shrink, with $r(W) \subseteq W$ and $r(W) = W \cap A$ Chapter2/SubMPolyfold · MPolyfold.SubData.mapsTo_shrink Chapter2/SubMPolyfold · MPolyfold.SubData.image_shrink. *The retract.* $\rho := \varphi \circ r \circ \varphi^{-1}$ is sc-smooth from $\varphi(W)$ to itself — Definition 2.3.8 (preprint 2.14) for $r$ in the charts $\varphi$, $\varphi$ — idempotent, and $\rho(\varphi(W)) = \varphi(W \cap A)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image; so $\varphi(W \cap A)$ is a retract by Proposition 2.3.1 (2) (preprint 2.10) Chapter2/SubMPolyfold · MPolyfold.SubData.isScRetract_image. *The chart.* With the subspace topology on $A$, $\varphi$ restricted to $W \cap A$ is a chart of $A$ with model $(\varphi(W \cap A), C, E)$ Chapter2/SubMPolyfold · MPolyfold.subChart Chapter2/SubMPolyfold · MPolyfold.subChartInv. *Compatibility.* The overlap of two such charts is $\varphi((W \cap W') \cap A)$, a subset of $\varphi(V \cap V')$ Chapter2/SubMPolyfold · MPolyfold.subChart_overlap. An sc-smooth map between retracts is sc-smooth on every sub-retract $O_1 \subseteq O$: for a retraction $s$ onto $O_1$, $f \circ s = (f \circ r) \circ s$, sc-smooth by the chain rule Chapter2/SubMPolyfold · IsScSmoothRetractMap.mono_source — this is the text's "$(\psi' \circ \psi^{-1}) \circ s'' = (\varphi' \circ \varphi^{-1}) \circ s''$". The transition maps of the sub-charts are restrictions of those of $X$, hence sc-smooth Chapter2/SubMPolyfold · MPolyfold.subChart_compatible. They form an atlas Chapter2/SubMPolyfold · MPolyfold.subAtlas, and $A$ is an M-polyfold Chapter2/SubMPolyfold · MPolyfold.subPolyfold Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart. The structure requires $A$ to be paracompact (an instance argument); it is, as a subset of the metrizable $X$ (§II.7), and for closed $A$ also elementarily Chapter2/SubMPolyfold · MPolyfold.paracompactSpace_of_isClosed.

*(1) and (2).* The inclusion $A \to X$ is sc-smooth: in the sub-chart and $\varphi$ it is the identity of $\varphi(W \cap A)$ into $O$; one pair of charts per point suffices `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_inclusion_isScSmooth`; it is a homeomorphism onto its image by the choice of topology. For any data $(V, r)$, the map $i^{-1} \circ r$ (a total map with a junk value off $V$) `Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto` `Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto_val` is sc-smooth on $V$: near $x \in V$ take a chart $\varphi'$ around $x$, a chart $\varphi$ around $r(x)$ and $V_0 = V \cap \operatorname{dom} \varphi' \cap r^{-1}(\operatorname{dom} \varphi)$; on $\varphi'(V_0)$ the representation in $\varphi'$ and the sub-chart on $\varphi$ is $\varphi \circ r \circ \varphi'^{-1}$, sc-smooth by Definition 2.3.8 (preprint 2.14) for $r$ `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_retraction_isScSmooth`.

**Proposition 2.3.19 (3), (4) (preprint 2.21), in a chart.** Let $r$ be a retraction onto $O$, $s$ sc-smooth from $(O, C, E)$ to itself with $s \circ s = s$, and $t := s \circ r$, a retraction onto $s(O)$ (Proposition 2.3.1 (2) (preprint 2.10)) `Chapter2/Metrizable · ScRetraction.compRetraction` `Chapter2/Metrizable · ScRetraction.compRetraction_retract`. For $a \in s(O)_1$ put $P := Dr(a)$, $Q := Dt(a) = D(s \circ r)(a)$ — the text's "$Ds(a)$". Both are idempotent (§II.2). From $r \circ t = t$ and $t \circ r = t$ on $U$ ($s$ takes values in $O$, which $r$ fixes; and $r \circ r = r$), the chain rule at the point $a$ `Chapter2/DegeneracyIndex · scDeriv_comp_apply` gives $PQ = Q$ and $QP = Q$. If $a$ is smooth, $P$ and $Q$ are sc-operators (Part I, Proposition 1.2.1 (preprint 1.16)), and so is $P - Q$; identities between base maps of sc-operators hold on every level `Chapter2/Metrizable · ScOperator.lvl_comp_of_base` `Chapter2/Metrizable · ScOperator.lvl_comp_eq_zero_of_base` `Chapter2/Metrizable · ScOperator.lvl_eq_add_of_base`. Then $Q$ and $P - Q$ are commuting idempotent sc-operators with $Q(P - Q) = 0$, their ranges are sc-subspaces (Part I, §I.3: sc-splittings and sc-projections `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem`), and on every level the level parts of $QE$ and $(P - Q)E$ are disjoint with sum the level part of $PE$. Since $T_a O = PE$ and $T_a s(O) = QE$ (§II.2), this is the sc-splitting $T_a O = T_a A \oplus (Dr(a) - Dt(a))E$ `Chapter2/Metrizable · proposition_2_3_19_three`. For (4), $s$ is sc-smooth from $O$ onto $s(O)$ (shrinking the target), and $Ts(a, h) = (a, Qh)$ for $h \in T_a O$ because $Qh = QPh$; as $Q$ maps $PE$ onto $QE = Q(PE)$, $Ts(a)$ maps $T_a O$ onto $T_a s(O)$ `Chapter2/Metrizable · proposition_2_3_19_four`. The transport to $T_a X$ is §II.39 `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_3_linear` `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_4`.

## II.11 The degeneracy index of an M-polyfold (Definitions 2.4.1, 2.4.5 (preprint 2.22, 2.25); Lemmas 2.4.3, 2.4.6 (preprint 2.23, 2.26); Proposition 2.4.4 (preprint 2.24))

A retract is contained in its quadrant, and chart values lie in $C$ `Chapter2/DegeneracyIndex · LocalModel.O_subset_C` `Chapter2/DegeneracyIndex · MChart.apply_mem_C` `Chapter2/DegeneracyIndex · LocalModel.isPartialQuadrant`.
**Definition 2.4.1 (preprint 2.22).** $d(x, \varphi) := d_C(\varphi(x))$ `Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex`, and $d_X(x)$ is the least $d(x, \varphi)$ over the charts of the structure around $x$ `Chapter2/DegeneracyIndex · MPolyfold.degIndex`; $d_X(x) \le d(x, \varphi)$ for every such chart and the minimum is attained `Chapter2/DegeneracyIndex · MPolyfold.degIndex_le` `Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq`.
**Lemma 2.4.3 (preprint 2.23).** $d_C$ is upper semicontinuous: coordinates positive at $x$ stay positive nearby, so the vanishing set can only shrink `Chapter2/DegeneracyIndex · eventually_degIndexWith_le`; composed with the continuous chart, $d(\cdot, \varphi)$ is upper semicontinuous on $V$ `Chapter2/DegeneracyIndex · MPolyfold.eventually_chartDegIndex_le`, and with a chart realizing $d_X(x)$, $d_X(y) \le d(y, \varphi) \le d(x, \varphi) = d_X(x)$ near $x$ `Chapter2/DegeneracyIndex · lemma_2_4_3`. Hence $\{d_X < k\}$ is open and $\{d_X \ge k\}$ closed `Chapter2/Boundary · MPolyfold.isOpen_setOf_degIndex_lt` `Chapter2/Boundary · MPolyfold.isClosed_setOf_le_degIndex`.

**Proposition 2.4.4 (preprint 2.24).** An sc-diffeomorphism of open sets is $f$ sc-smooth on $U$, $g$ sc-smooth on

$V$, mapping into each other and mutually inverse Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn.symm. For a chart $\psi$ of $Y$, $\psi \circ f$ on $U \cap f^{-1}(\mathrm{dom}\,\psi)$, with inverse $g \circ \psi^{-1}$, is a chart of $X$ Chapter2/DegeneracyIndex · MPolyfold.transportChart; it is compatible with every chart $\varphi$ of the atlas of $X$, because its transition maps with $\varphi$ are $\psi \circ f \circ \varphi^{-1}$ and $\varphi \circ g \circ \psi^{-1}$, sc-smooth by Definition 2.3.8 (preprint 2.14) for $f$ and for $g$ Chapter2/DegeneracyIndex · MPolyfold.transportChart_isChart. Its index at $x$ is $d(f(x), \psi)$ Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex_transportChart. With $\psi$ realizing $d_Y(f(x))$: $d_X(x) \le d_Y(f(x))$ Chapter2/DegeneracyIndex · MPolyfold.degIndex_le_of_isScDiffeoOn, and with $g$ the reverse Chapter2/DegeneracyIndex · proposition_2_4_4. The text's "from the definitions one deduces immediately" is this construction.

**Definition 2.4.5 (preprint 2.25).** $\partial X = \{d_X \ge 1\}$ Chapter2/DegeneracyIndex · MPolyfold.boundary; without boundary means $d_X \equiv 0$, equivalently $\partial X = \emptyset$ Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary Chapter2/DegeneracyIndex · MPolyfold.isWithoutBoundary_iff. $\partial X$ and the corners $\{d_X \ge 2\}$ are closed, the interior $\{d_X = 0\}$ is open Chapter2/Boundary · MPolyfold.isClosed_boundary Chapter2/Boundary · MPolyfold.corners Chapter2/Boundary · MPolyfold.isClosed_corners Chapter2/Boundary · MPolyfold.isOpen_interior, and sc-diffeomorphisms preserve $\partial X$ by Proposition 2.4.4 (preprint 2.24) Chapter2/Boundary · MPolyfold.IsScDiffeoOn.mem_boundary_iff.

**Lemma 2.4.6 (preprint 2.26).** For a sub-M-polyfold $A$ and $a \in A$: take a chart $\varphi$ of $X$ realizing $d_X(a)$ and the data at $a$; the sub-chart built on $\varphi$ has the same quadrant $C$ and the same value $\varphi(a)$, so $d_A(a) \le d_C(\varphi(a)) = d_X(a)$ Chapter2/SubMPolyfold · lemma_2_4_6.

*A retract as an M-polyfold.* A nonempty retract $O$, as a subtype, has the single chart id Chapter2/MPolyfold · LocalModel.selfChart, compatible with itself, and is an M-polyfold Chapter2/MPolyfold · LocalModel.toMPolyfold (Hausdorff and paracompact as a subspace of $E_0$); so is a nonempty relatively open $U \subseteq C$ Chapter2/MPolyfold · IsRelOpenInQuadrant.toMPolyfold Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.selfChart_isChart — the text's $X_C$.

## II.12 The subspaces $E_x$, the cones $C_x$ and the sign arguments

*Coordinates.* For a presentation $T$ of $C$ and $x \in E_0$, the coordinates vanishing at $x$ form a continuous linear surjection $E_0 \to \mathbb{R}^{I(x)}$, $v \mapsto ((Tv)_i)_{i \in I(x)}$ Chapter2/DegeneracyIndex · coordProj Chapter2/DegeneracyIndex · coordMapL Chapter2/DegeneracyIndex · coordMapL_surjective. Its kernel is $E_x$ Chapter2/DegeneracyIndex · minimalSubspaceWith Chapter2/DegeneracyIndex · mem_minimalSubspaceWith, closed Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith, and by the first isomorphism theorem $E_0/E_x \cong \mathbb{R}^{I(x)}$, so $\dim(E_0/E_x) = d_C(x)$ Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith. The cone $C_x = \{v : (Tv)_i \ge 0 \text{ for } i \in I(x)\}$ Chapter2/Tame · partialQuadrantAtWith is convex and closed and contains $E_x$ Chapter2/Tame · convex_partialQuadrantAtWith Chapter2/Boundary · isClosed_partialQuadrantAtWith Chapter2/Tame · minimalSubspaceWith_subset_partialQuadrantAtWith. *Approximation*: if $Tx = (a, w)$, the points $T^{-1}(a, w')$ with $w' \in W_1$ (resp. $W_\infty$) converge to $x$ as $w' \to w$ and have the same vanishing coordinates, so $x$ is a limit of points on level 1 (resp. smooth points) of the same index, in $C$ iff $x$ is Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith Chapter2/TameRetracts · exists_tendsto_smooth_degIndexWith.

*Density on level* 1. For a continuous linear $L : E_0 \to V$, $V$ finite-dimensional, the level-1 points of $\ker L$ are dense in $\ker L$ Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase. The image $S$ of $E_1$ is dense in $E_0$; $L$ maps $S$ onto the finite-dimensional $L(S)$, which has a linear, hence continuous, right inverse $g$ with values in $S$. For $v \in \ker L$ take $u_k \in S$ with $u_k \to v$; then $L(u_k) \to 0$, $g(L(u_k)) \to 0$, and $u_k - g(L(u_k)) \in S \cap \ker L$ converges to $v$. This is the text's "the closure of $(T^R_x O)_\infty$ is $T^R_x O$" in the form

the proofs need; the level-1 points of $C_x$ are dense in $C_x$ likewise Chapter2/Boundary · partialQuadrantAtWith_subset_closure.

*The two-sided sign argument.* Let $f$ be sc¹ on $U$ relatively open in $C$, with values in $C'$, $x \in U_1$ and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x+tv \in U_1$ for $|t|$ small: the coordinates in $I(x)$ stay 0, the others stay positive, and $U_1$ is relatively open Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet. For $j$ with $(T'f(x))_j = 0$ the real function $g(t) = (T'f(\iota_1(x+tv)))_j$ is $\geq 0$ near 0 (values in $C'$), vanishes at 0, and has derivative $(T'Df(x)v)_j$ at 0 (Definition 1.1.15's (preprint 1.14) derivative along $E_1$, composed with the path); a local minimum has zero derivative Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero. So $Df(x)v \in E'_{f(x)}$ Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1, and by density and continuity of $Df(x)$ on $E_0$, $Df(x)$ maps $E_x$ into $E'_{f(x)}$ Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace.

*The one-sided sign argument.* For $v \in E_1$ with $\iota_1 v \in C_x$, $x + tv \in U_1$ for small $t \geq 0$ Chapter2/Boundary · eventually_add_smul_mem_lvlSet_of_cone, the same $g$ is $\geq 0$ on $[0, \varepsilon)$ with $g(0) = 0$, so its one-sided derivative is $\geq 0$ Chapter2/Boundary · nonneg_of_hasDerivWithinAt_Ici; hence $Df(x)v \in C'_{f(x)}$ Chapter2/Boundary · IsSc1With.deriv_mem_partialQuadrant_of_lvl1, and $Df(x)$ maps $C_x$ into $C'_{f(x)}$ Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant. These are the derivative-level content of Lemma 2.4.9 (1) and (2) (preprint 2.29); the lemma itself, about sc-smooth paths, is proved from them in §II.42 Chapter2/GapP1_Lemma249 · GapP1Lemma249.lemma_2_4_9_cone.

*The chain rule at a point.* For sc¹ maps $f$, $g$ on relatively open sets with $f(U) \subseteq U'$, $Dg(f(x))Df(x) = D(g \circ f)(x)$ at $x \in U_1$ Chapter2/DegeneracyIndex · scDeriv_comp_apply; if $g \circ f = \mathrm{id}$ on $U$ then $g_1(f_1(x)) = x$ and $Dg(f(x))Df(x) = \mathrm{id}$ Chapter2/DegeneracyIndex · levelMap_comp_eq_self Chapter2/DegeneracyIndex · scDeriv_comp_eq_self.

## II.13 Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)

**Definition 2.4.7 (preprint 2.27), Remark 2.4.8 (preprint 2.28).** For a retraction $r$, a presentation $T$ and $x \in O_1$: $T^R_x O := Dr(x)E_0 \cap E_x$ and $C_x O := Dr(x)E_0 \cap C_x$ Chapter2/Tame · reducedTangentSpaceWith Chapter2/Tame · partialConeWith, defined at every point on level 1; $T^R_x O \subseteq C_x O$ Chapter2/Tame · reducedTangentSpaceWith_subset_partialConeWith. The remark's further claim, that $C_x O$ is in general not a partial quadrant in $T_x O$, does not hold as we read it: at a smooth point it always is one (§II.41). The number $\dim(T_x O / T^R_x O)$ is $\dim(A/(A \cap B))$ for $A = T_x O$, $B = E_x$ Chapter2/Boundary · quotDim; since $A \to E_0/E_x$ has kernel $A \cap E_x$, it is at most $d_C(x)$ and finite Chapter2/Boundary · quotDim_inf_minimalSubspace_le Chapter2/Boundary · quotDim_reducedTangentSpaceWith_le.

*Germs between retracts.* A germ of sc-diffeomorphisms between $O = r(U)$ and $O' = r'(U')$ is a pair $F$, $G$ with $F \circ r$, $G \circ r'$ sc¹ (sc¹ suffices here; the text has sc-smooth), $F(O) \subseteq O'$, $G(O') \subseteq O$, inverse to each other on $O$, $O'$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm; an sc-smooth map between retracts with an sc-smooth inverse is one Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm, and two compatible charts give one between the overlap retracts, with the restricted retractions Chapter2/Boundary · MChart.Compatible.isDiffeoGerm. Since $(G \circ r') \circ (F \circ r) = r$ on $U$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.comp_eq, the chain rule at $x \in O_1$, $y = (F \circ r)_1(x) \in O'_1$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.image_mem Chapter2/

Boundary · ScRetraction.IsDiffeoGerm.image_image, gives

$$D(G\circ r')(y)\,D(F\circ r)(x)=Dr(x),\qquad D(F\circ r)(x)\,D(G\circ r')(y)=Dr'(y)$$

Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_deriv_inv. So $D:=D(F\circ r)(x)$ maps $T_xO$ into $T_yO'$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range, and by the sign arguments for $F\circ r$ (with values in $O'\subseteq C'$) it maps $E_x$ into $E'_y$ and $C_x$ into $C'_y$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_minimalSubspace Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_partialQuadrant.

**Proposition 2.4.10 (preprint 2.30).** $D(T^R_xO)=T^R_yO'$ and $D(C_xO)=C_yO'$, with $Tf(x)=D$ Chapter2/Boundary · proposition_2_4_10_reduced Chapter2/Boundary · proposition_2_4_10_cone. "$\subseteq$": $D$ maps $T_xO$ into $T_yO'$ and $E_x$ (resp. $C_x$) into $E'_y$ (resp. $C'_y$). "$\supseteq$": for $w\in T^R_yO'$, the vector $D'w$, $D'=D(G\circ r')(y)$, lies in $T_xO$ and in $E_x$ by the same facts for the inverse germ, and $D(D'w)=Dr'(y)w=w$ because $w$ is in the range of the projection $Dr'(y)$. This proof does not pass through Lemma 2.4.9 (preprint 2.29).

*The core of Theorem 2.4.12 (preprint 2.31).* Along a germ $F$, $\dim(T_xO/T^R_xO)\le d_{C'}(F(x))$ for $x\in O_1$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.quotDim_le. With $P=Dr(x)$, $P'=Dr'(y)$: $D$ maps $T_x=PE$ into $T_y=P'E'$ and $D'$ maps back, $D'D=P$ is the identity on $T_x$ and $DD'=P'$ the identity on $T_y$. The map $\Phi:T_x\to T_y/(T_y\cap E'_y)$, $v\mapsto[Dv]$, is onto ($[w]=\Phi(D'w)$) with kernel $T_x\cap E_x$ ($Dv\in E'_y$ implies $v=D'Dv\in E_x$). Hence $T_x/T^R_x\cong T_y/T^R_y$, which embeds in $E'_0/E'_y$, of dimension $d_{C'}(y)$. The same argument with $U_0$ relatively open in $C$ in place of the retract $O$ is the form used for Corollary 2.4.13 (preprint 2.32) and Theorem 1.3.5 (preprint 1.25) Chapter2/DegeneracyIndex · degIndexWith_le_of_germ: there $D'D=\mathrm{id}$, so $E_0/E_x\cong T_y/T^R_y\hookrightarrow E'_0/E'_y$ and $d_C(x)\le d_{C'}(F(x))$.

**Theorem 2.4.12 (preprint 2.31).** Let $\varphi$ be a chart of an M-polyfold around $x$ with $\varphi(x)=\iota_1 z$, $r$ a retraction onto its model and $T$ a presentation. For every other chart $\psi$ around $x$, the overlaps are $O\cap W$ and $O'\cap W'$, the transition maps form a germ between the restricted retractions, and the restricted retraction has the same derivative at $z$ Chapter2/Boundary · ScRetraction.scDeriv_restrict Chapter2/Boundary · reducedTangentSpaceWith_restrict; so the core gives $\dim(T_zO/T^R_zO)\le d(x,\psi)$ Chapter2/Boundary · MPolyfold.quotDim_le_chartDegIndex. Taking $\psi$ realizing $d_X(x)$: $\dim(T_zO/T^R_zO)\le d_X(x)$ Chapter2/Boundary · theorem_2_4_12. For a retract $O$ as an M-polyfold with the chart id this is the text's statement $\dim(T_xO/T^R_xO)\le d_O(x)$, at points on level 1 Chapter2/Boundary · theorem_2_4_12_retract. The text instead reduces to a chart realizing $d_O(x)$ and counts linearly independent $\mathbb{R}^d$-components; the quotient isomorphism makes the reduction unnecessary. The second statement (equality makes $C_xO$ a partial quadrant) is §II.41, proved without the hypothesis: at every smooth point of every retract, $C_xO$ is a partial quadrant of dimension $\dim(T_xO/T^R_xO)$ Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant.

**Corollary 2.4.13 (preprint 2.32).** Let $U$ be nonempty and relatively open in $C$, and $X_U$ its M-polyfold Chapter2/DegeneracyIndex · corollary_2_4_13. The chart id gives $d_{X_U}\le d_C$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le. At a point $x$ on level 1, a chart $\psi$ realizing $d_{X_U}(x)$ is compatible with id, so $F=\psi$ (on $U_0=\mathrm{dom}\,\psi$, relatively open in $C$) and $G=\psi^{-1}$ form a germ from $U_0$ to the retract $\psi(V)$, and the core gives $d_C(x)\le d_{C'}(\psi(x))=d_{X_U}(x)$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le_degIndex_of_lvl1. For a general $x$, approximate by level-1 points $\sigma_i\in U$ with $d_C(\sigma_i)=d_C(x)$ (§II.12); by Lemma 2.4.3 (preprint 2.23), $d_{X_U}(\sigma_i)\le d_{X_U}(x)$ eventually, so $d_C(x)=d_C(\sigma_i)\le d_{X_U}(\sigma_i)\le d_{X_U}(x)$. The text approximates by smooth points; level 1 suffices. **Theorem 1.3.5 (preprint 1.25)** (Part I, §I.19) is proved in the same file by the same argument for a germ between relatively open subsets of two quadrants Chapter2/DegeneracyIndex · theorem_1_3_5_lvl1

Chapter2/DegeneracyIndex · theorem_1_3_5.

## II.14 Tame retractions; Propositions 2.5.3 and 2.5.6 (preprint 2.35 and 2.36)

**Definition 2.5.1 (preprint 2.33).** $C_e$ and $E_e$ are the cone and subspace of §II.12 for the presentation $T$. They do not depend on it: for two presentations $T$, $T'$ of $C$ and $e \in C$, $v \in E_e$ (for $T$) gives $e+tv \in C$ for $|t|$ small Chapter2/TameFaces · eventually_add_smul_mem_of_minimalSubspace; for $j$ with $(T'e)_j = 0$, $(T'(e+tv))_j = t(T'v)_j \geq 0$ for $t = \pm\delta/2$, so $(T'v)_j = 0$ Chapter2/TameFaces · minimalSubspaceWith_le; by symmetry the two subspaces agree Chapter2/TameFaces · minimalSubspaceWith_eq. This is the argument of Lemma 1.3.4 (preprint 1.24).

**Definition 2.5.2 (preprint 2.34).** $r$ is *tame for* $T$ if (1) $d_C(r(x)) = d_C(x)$ on $U$ and (2) at every smooth $x \in O_1$ some $A \leq E_x$ makes $E = Dr(x)E_0 \oplus A$ an sc-splitting Chapter2/Tame · ScRetraction.IsTameWith; *tame* if tame for every presentation Chapter2/Tame · ScRetraction.IsTame, and by the independence of $E_x$ tameness for one presentation suffices Chapter2/TameFaces · ScRetraction.IsTameWith.isTame. A *tame retract* is the retract of a tame retraction Chapter2/Tame · IsTameRetract Chapter2/Tame · IsTameRetract.isScRetract. Identities of relatively open sets are tame ($T_xU = E$, $A = 0$) Chapter2/Tame · ScRetraction.id_isTame, and restrictions of tame retractions to $U \cap r^{-1}(V)$ are tame (same derivatives) Chapter2/TameFaces · ScRetraction.IsTameWith.restrict.

**Proposition 2.5.3 (preprint 2.35).** *$Dr(x)E_x \subseteq E_x$ for every retraction at every $x \in O_1$*: the two-sided sign argument for $r$, with $r(x) = x$ Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace. At a smooth $x$, $Dr(x)$ is an sc-operator Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv, so $E = Dr(x)E \oplus (1 - Dr(x))E$ is an sc-splitting Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv. *The proposition* Chapter2/TameRetracts · proposition_2_5_3: with $A \leq E_x$ from tameness (2), $E_0 = Dr(x)E_0 \oplus A$ algebraically (the level-0 part of the sc-splitting); write $v = t+a$; then $(1-Dr(x))v = (1-Dr(x))a = a - Dr(x)a \in E_x$, because $a \in E_x$ and $Dr(x)a \in E_x$. Only condition (2) is used; the text uses condition (1) for $Dr(x)E_x \subseteq E_x$. In coordinates: $\lambda_j(Dr(x)v) = \lambda_j(v)$ for $j \in I(x)$ Chapter2/TameRetracts · coord_scDeriv_eq.

**Proposition 2.5.6 (preprint 2.36).** *(13) at smooth points.* If $V = A' \oplus B$ and $B \leq K$, then $A'/(A' \cap K) \cong V/K$ Chapter2/TameRetracts · quotDim_eq_finrank_quotient_of_isCompl; with $A' = T_xO$, $B = A$ and $K = E_x$ this gives $\dim(T_xO/T_x^RO) = \dim(E_0/E_x) = d_C(x)$ Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq — the text's equations (15)–(16). *A tame chart realizes the index* Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith: let $\varphi$ be a chart of any M-polyfold whose model has a tame retraction $r$. At $y$ with $\varphi(y)$ smooth, $d(y,\varphi) = \dim(T/T^R) \leq d_X(y)$ by Theorem 2.4.12 (preprint 2.31), and $d_X(y) \leq d(y,\varphi)$ by definition. At a general $y$, with $\varphi(y) = o$, approximate $o$ by smooth points $\sigma_i$ of $C$ with the same vanishing coordinates; then $r(\sigma_i)$ are smooth points of $O$ (an $\mathrm{sc}^0$ map preserves smoothness Chapter2/TameRetracts · IsSc0.mem_smooth Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth) converging to $r(o) = o$, with $d_C(r(\sigma_i)) = d_C(\sigma_i) = d_C(o)$ by tameness (1); so the points $\varphi^{-1}(r(\sigma_i))$ converge to $y$ and have $d_X = d(\cdot,\varphi) = d_C(o)$, and Lemma 2.4.3 (preprint 2.23) gives $d_C(o) \leq d_X(y)$. For charts of the structure modelled on a tame retract Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract. Applied to the retract itself as an M-polyfold Chapter2/TameRetracts · ScRetraction.toLocalModel Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith: (14) $d_O = d_C$ on $O$ Chapter2/TameRetracts · proposition_2_5_6_degIndex and (13) $\dim(T_xO/T_x^RO) = d_O(x) = d_C(x)$ at smooth points Chapter2/TameRetracts · proposition_2_5_6. This is the text's proof, generalized from the retract to any chart; Proposition 2.5.14 (preprint 2.42) needs the

general form.

*Splitting off finitely many coordinates.* Let $B \le A$ be sc-subspaces of $E$, $\ell_1, \dots, \ell_d$ continuous functionals on $E_0$ with $B = A \cap \bigcap_j \ker \ell_j$, and $e_1, \dots, e_d \in A$ smooth with $\ell_i(e_j) = \delta_{ij}$. Each $e_j$ has a lift to every level $m$, compatible with the inclusions `Chapter2/PartialCone · smoothLift` `Chapter2/PartialCone · incl_smoothLift`. On level $m$ the maps $v \mapsto ((\ell_j v)_j,\ v - \sum_j \ell_j(v) e_j)$ and $(a,u) \mapsto u + \sum_j a_j e_j$ are continuous linear $A_m \to \mathbb{R}^d \times B_m$ and back `Chapter2/PartialCone · splitFwd` `Chapter2/PartialCone · splitBwd` — the bracket lies in $B$ because every $\ell_i$ kills it `Chapter2/PartialCone · sub_sum_mem` — and mutually inverse `Chapter2/PartialCone · splitBwd_splitFwd` `Chapter2/PartialCone · splitFwd_splitBwd`; they commute with the inclusions, so they form an sc-isomorphism $A \cong \mathbb{R}^d \oplus B$ `Chapter2/PartialCone · splitIso`, $A$ and $B$ with the sc-structures induced from $E$ `Chapter1/ScBanach · ScBanach.ofSubspace`. It carries $\{v \in A : \ell_j(v) \ge 0\ \forall j\}$ onto $[0,\infty)^d \oplus B$, so that set is a partial quadrant in $A$ `Chapter2/PartialCone · isPartialQuadrant_ofSubspace`. This is the map $S$ of the proof of Theorem 2.4.12 (preprint 2.31), with the $e_j$ smooth as the exposition requires.

**Proposition 2.5.6 (preprint 2.36), the remaining statements** `Chapter2/PartialCone · proposition_2_5_6_partialCone`. Let $r$ be tame for $T$, $x \in O_1$ smooth, $P = Dr(x)$ and $I(x) = \{i_1, \dots, i_d\}$. $T_xO = PE$ is an sc-subspace, the range of the sc-projection $P$ (Proposition 2.5.3 (preprint 2.35) above) `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. The vectors $\hat{e}_j := P(T^{-1}(e_{i_j}, 0))$ `Chapter2/TameRetracts · coordVec` lie in $T_xO$; they are smooth, because $T^{-1}(e_{i_j}, 0)$ is smooth `Chapter2/PartialCone · coordVec_mem_smooth` and $P$ is an sc-operator at the smooth point $x$ `Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv` `Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl`; and $\lambda_{i_k}(\hat{e}_j) = \lambda_{i_k}(T^{-1}(e_{i_j}, 0)) = \delta_{jk}$ by Proposition 2.5.3 (preprint 2.35) in coordinates `Chapter2/TameRetracts · coord_scDeriv_eq`. By definition $T_x^R O = T_xO \cap \bigcap_k \ker \lambda_{i_k}$ `Chapter2/TameRetracts · coordL`. $T_x^R O$ *is an sc-subspace.* It is closed, as an intersection of closed subspaces `Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith`. For the density of smooth points, let $u \in T_x^R O \cap E_m$ with $Tu = (a,w)$ on level $m$, and put $\phi(w') := P(T^{-1}(a,w'))$, a continuous map $W_m \to E_m$. For smooth $w'$ the point $\phi(w')$ is smooth, lies in $T_xO$, and lies in $E_x$, because $T^{-1}(a,w')$ has the same vanishing coordinates in $I(x)$ as $u$ and $PE_x \subseteq E_x$ `Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace`. And $\phi(w) = Pu = u$, since $u \in T_xO$ and $P$ is a projection `Chapter2/Retracts · ScRetraction.scDeriv_comp_self`. As $W_\infty$ is dense in $W_m$, $u$ lies in the closure of $\phi(W_\infty) \subseteq T_x^R O \cap E_\infty$ (this is the exposition's Lemma 3.A (1) in the case at hand). *The quadrant.* The splitting with $A = T_xO$, $B = T_x^R O$, $\ell_j = \lambda_{i_j}$ and $e_j = \hat{e}_j$ makes $\{v \in T_xO : \lambda_{i_j}(v) \ge 0\ \forall j\}$ a partial quadrant in $T_xO$, and this set is $C_xO = T_xO \cap C_x$. The text deduces the claim from the equality case of Theorem 2.4.12 (preprint 2.31), which needs smooth lifts $\hat{v}^j$ of a basis of $T_xO / T_x^R O$ (the exposition chooses them by a density argument); for a tame retraction they are the explicit $\hat{e}_j$, and the general equality case, proved in §II.41, is not needed here.

## II.15 Faces: local geometry, Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41), Proposition 2.5.14 (preprint 2.42)

**Definitions 2.5.7, 2.5.10 (preprint 2.37, 2.39).** $X$ is *tame* if an atlas equivalent to its atlas has only tame models `Chapter2/Tame · MPolyfold.IsTame`; charts of an equivalent atlas are charts of the structure `Chapter2/TameRetracts · MPolyfold.isChart_of_mem_equiv`, so every point lies in a chart of the structure with a tame model `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`. A *face* is the closure of the connected component in $\{d_X = 1\}$ of a point with $d_X = 1$ `Chapter2/Tame · MPolyfold.IsFace`, closed `Chapter2/Tame · MPolyfold.IsFace.isClosed` and inside $\partial X$ `Chapter2/Boundary · MPolyfold.IsFace.subset_boundary`; $X$ is *face-structured* if every $x$

lies in exactly $d_X(x)$ faces Chapter2/Tame · MPolyfold.IsFaceStructured.

*Coordinates of faces.* $I(z) = \{i : \lambda_i(z) = 0\}$ Chapter2/TameRetracts · zeroSet with $d_C = \#I$ Chapter2/TameRetracts · degIndexWith_eq_card_zeroSet; the extended face $f_i$ is the hyperplane $\{\lambda_i = 0\}$, closed Chapter2/TameRetracts · coordHyper Chapter2/TameRetracts · isClosed_coordHyper; $I(z) \subseteq I(e)$ for $z$ near $e$ Chapter2/TameRetracts · eventually_zeroSet_subset. *A connected piece of $\{d_C = 1\}$ lies in one face* (exposition Lemma 4.A): if $S \subseteq C$ is preconnected with $d_C = 1$ on $S$ and $\lambda_i(p) = 0$ at some $p \in S$, then $S \subseteq f_i$ Chapter2/TameRetracts · subset_coordHyper_of_isPreconnected. Points of $S$ have $I = \{i\}$ or $\lambda_i > 0$ Chapter2/TameRetracts · zeroSet_eq_singleton, so $S$ is covered by the open sets $u = \{\lambda_l > 0 \text{ for } l \neq i\}$ and $v = \{\lambda_i > 0\}$, which are disjoint on $S$ (a point in both would have $d_C = 0$); preconnectedness puts $S$ in one of them Mathlib/Topology/Connected/Clopen · isPreconnected_iff_subset_of_disjoint, and $p \in u$.

*Rays stay in their face* (exposition Lemma 4.B). If $f$ is sc¹, $x + tv \in U_1$ for small $t \geq 0$ and a functional $L$ vanishes on $f(x + tv)$ for $t \in [0, \varepsilon)$, then $L(Df(x)v) = 0$ Chapter2/TameRetracts · IsSc1With.apply_deriv_eq_zero_of_ray. Now let $r$ be tame, $x \in O_1$ smooth, $i \in I(x)$, $\lambda_i(y) = 0$ and $\lambda_l(y) > 0$ for the other $l \in I(x)$ Chapter2/TameRetracts · ray_zeroSet. For small $\tau > 0$, $x + \tau y \in U$ and $I(x + \tau y) = \{i\}$, so $d_C(r(x + \tau y)) = 1$ by tameness (1), and $I(r(x + \tau y)) \subseteq I(x)$ by continuity of $r$ at $x = r(x)$. The image of $(0, \tau_0]$ under $\tau \mapsto r(x + \tau y)$ is preconnected with $d_C = 1$, so it lies in one $f_j$ with $j \in I(x)$. Differentiating $\lambda_j(r(x + \tau y)) = 0$ at $\tau = 0$ gives $\lambda_j(Dr(x)y) = 0$, and $\lambda_j(Dr(x)y) = \lambda_j(y)$ by Proposition 2.5.3 (preprint 2.35) Chapter2/TameRetracts · coordL; as $\lambda_l(y) > 0$ for $l \in I(x) \setminus \{i\}$, $j = i$. This is the step of Lemma 2.5.12 (preprint 2.40) that the text omits, and it needs tameness (2).

*The local structure of a face* (exposition Lemma 4.C) Chapter2/TameRetracts · local_face_structure. Let $S = \{z \in O : d_C(z) = 1\}$ Chapter2/TameRetracts · ScRetraction.faceStratum and $e \in O$ with $\lambda_i(e) = 0$. (a) Choose $\varepsilon$ so that $B = B_\varepsilon(e) \cap C \subseteq U$ and $I(z), I(r(z)) \subseteq I(e)$ on $B$. (b) $A_i \cap B = \{z \in B : I(z) = \{i\}\}$ is convex Chapter2/TameRetracts · convex_ball_inter_zeroSet_eq, hence connected, and $r$ maps it into $S$; a smooth $u = r(T^{-1}(a, w')) \in O \cap B$ with $I(u) = I(e)$ and the ray of Lemma 4.B from $u$ in the direction $T^{-1}(b, 0)$, $b = 1$ on $I(e) \setminus \{i\}$ and 0 elsewhere Chapter2/TameRetracts · coordVec Chapter2/TameRetracts · rayDir Chapter2/TameRetracts · exists_lvl1_coordVec, produces a point $p \in S$ with $\lambda_i(p) = 0$ in $r(A_i \cap B)$; so $r(A_i \cap B)$ lies in the component of $S$ through $p$. (c) A point $e' \in f_i \cap B$ is the limit of points of $A_i \cap B$ (perturb the coordinates of $I(e') \setminus \{i\}$ upward), so $r(e')$ lies in the closure of that component.

**Lemma 2.5.12 (preprint 2.40).** A *face of the retract* is $\overline{\hat{F}} \cap O$ for a component $\hat{F}$ of $S$ Chapter2/TameRetracts · ScRetraction.IsRetractFace. A component of $S$ through a point with $\lambda_i = 0$ lies in $f_i$ Chapter2/TameRetracts · faceStratum_degIndexWith Chapter2/TameRetracts · connectedComponentIn_faceStratum_subset, and a component of $S$ in $f_i$ whose closure contains $e$ is the component of Lemma 4.C at $e$ Chapter2/TameRetracts · connectedComponentIn_eq_of_mem_closure. *The face through $x \in S$, $\lambda_i(x) = 0$, is the component of $f_i \cap O$ through $x$* Chapter2/TameRetracts · closure_connectedComponentIn_faceStratum: it is connected, lies in $f_i \cap O$, is closed in it, and is relatively open in it by Lemma 4.C(c); a preconnected subset meeting a relatively clopen set lies in it Chapter2/TameRetracts · subset_of_relClopen_of_isPreconnected. Conversely, the component of $f_i \cap O$ through any $e \in O$ with $\lambda_i(e) = 0$ is the face through the point $p$ of Lemma 4.C at $e$ Chapter2/TameRetracts · connectedComponentIn_coordHyper_eq. Together: the faces of the retract are the components of the sets $f_i \cap O$ Chapter2/TameRetracts · lemma_2_5_12. If the components of $f_i \cap O$ and $f_j \cap O$ through $x$ coincide, a point $z$ of the face in $S$ has $I(z) = \{i\}$ and is a limit of points with $I = \{j\}$, which near $z$ forces $j = i$ Chapter2/TameRetracts · lemma_2_5_12_injective. The faces of the M-polyfold $O$ (Definition 2.5.10 (preprint 2.39), with $d_O = d_C$ by Proposition 2.5.6 (preprint 2.36)) are exactly the faces of the retract, transferred along the subtype inclusion Chapter2/TameRetracts · image_val_connectedComponentIn Chapter2/TameRetracts · image_val_closure Chapter2/TameRetracts · LocalModel.isFace_toMPolyfold_iff.

**Corollary 2.5.13 (preprint 2.41).** The map $i \mapsto$ (component of $f_i \cap O$ through $x$) sends $I(x)$ onto the faces through $x$ (Lemma 2.5.12 (preprint 2.40)) and is injective (its second statement), so $x$ lies in exactly $\#I(x) = d_C(x)$ faces of the retract `Chapter2/TameRetracts · corollary_2_5_13_retract`, and the M-polyfold $O$ is face-structured `Chapter2/TameRetracts · LocalModel.isFaceStructured_toMPolyfold` `Chapter2/TameRetracts · corollary_2_5_13`.

**Proposition 2.5.14 (preprint 2.42).** Let $X$ be tame. *The global bound* `Chapter2/TameRetracts · MPolyfold.faces_encard_le`: in a tame chart $\varphi$ around $x$, $o = \varphi(x)$, Lemma 4.C gives for each $i \in I(o)$ a component $\hat{F}_i$ of $S$ containing all points of $O$ near $o$ with $I = \{i\}$. A face $F = \overline{G}$ through $x$ has points $y \in G$ near $x$; $d_C(\varphi(y)) = d_X(y) = 1$ (the tame chart realizes $d_X$) and $I(\varphi y) \subseteq I(o)$, so $I(\varphi y) = \{i\}$ for some $i \in I(o)$, and the preconnected set $\varphi^{-1}(\hat{F}_i) \subseteq \{d_X = 1\}$ meets $G$, hence lies in $G$. Distinct faces give distinct $i$; so at most $d_X(x)$ faces pass through $x$. *The local statement*: the tame chart realizes $d_X$ on its domain, the M-polyfold $O$ has $d_O = d_C$ (Proposition 2.5.6 (preprint 2.36)), and $O$ is face-structured (Corollary 2.5.13 (preprint 2.41)) `Chapter2/TameRetracts · proposition_2_5_14`. The neighbourhood $V$ of the text is represented by $O$.

## II.16 Faces are tame sub-M-polyfolds: Propositions 2.5.15, 2.5.16 (preprint 2.43, 2.44) and Theorem 2.5.17 (preprint 2.45)

*The projection onto a face.* $P_i := T^{-1} \circ (a \mapsto a \text{ with } a_i = 0) \circ T$ is an sc-operator `Chapter2/TameRetracts · killCoordL` `Chapter2/TameRetracts · killCoordModel` `Chapter2/TameRetracts · killCoord` which kills the $i$-th coordinate and keeps the others `Chapter2/TameRetracts · killCoord_coord` `Chapter2/TameRetracts · killCoord_coord_self`, fixes $f_i$ `Chapter2/TameRetracts · killCoord_eq_self` and maps $C$ into $C$ `Chapter2/TameRetracts · killCoord_mem`.

**Proposition 2.5.15 (preprint 2.43), the retraction.** Let $F$ be a face of the retract, through $x \in S$ with $I(x) = \{i\}$, so $F \subseteq f_i$ `Chapter2/TameRetracts · proposition_2_5_15`. For each $e \in F$, Lemma 4.C gives $\varepsilon_e$ with $B_e = B_{\varepsilon_e}(e) \cap C \subseteq U$ and $r(z) \in F$ for $z \in B_e \cap f_i$ (the component of Lemma 4.C at $e$ is the one of $F$). Put $U^* := \bigcup_{e \in F}(B_{\varepsilon_e}(e) \cap P_i^{-1}(B_{\varepsilon_e}(e))) \cap C$: relatively open in $C$, contains $F$ (as $P_i e = e$), and for $z \in U^*$ the point $P_i z$ lies in some $B_e \cap f_i$, so $r(P_i z) \in F$. Then $s := r \circ P_i$ is sc-smooth on $U^*$ (chain rule), maps $U^*$ into $F \subseteq U^*$, and fixes $F$; so it is a retraction onto $F$, and $U^* \subseteq U$. The preimages under $P_i$ are needed because $P_i$ need not be norm-decreasing for the norm of $E_0$; the exposition's "$|P_1 z - e|_0 \leq |z - e|_0$" holds only for special norms. *$F$ is a sub-M-polyfold of $O$*: in the chart id of $O$, $s$ restricted to $O \cap U^*$ is an sc-smooth retraction onto $F$ `Chapter2/TameRetracts · LocalModel.selfChart_symm_apply` `Chapter2/TameRetracts · LocalModel.isSubMPolyfold_of_isFace`.

*The face as a model space.* Insert and delete a coordinate: $\mathbb{R}^k \to \mathbb{R}^{k+1}$, $b \mapsto (b_1, \dots, 0_i, \dots, b_k)$, and back `Chapter2/TameFaces · insertCoordL` `Chapter2/TameFaces · removeCoordL`, as sc-operators on $\mathbb{R}^k \oplus W$ `Chapter2/TameFaces · insertCoordModel` `Chapter2/TameFaces · removeCoordModel`; composed with $T$ they give $\iota : \mathbb{R}^k \oplus W \to E$ onto $f_i$ and $\pi : E \to \mathbb{R}^k \oplus W$ `Chapter2/TameFaces · faceIncl` `Chapter2/TameFaces · faceProj`, with $\pi\iota = 1$, $\iota\pi = 1$ on $f_i$ `Chapter2/TameFaces · faceProj_faceIncl` `Chapter2/TameFaces · faceIncl_faceProj`, $\iota(C') \subseteq C$ and $\pi(C) \subseteq C'$ for $C' = [0, \infty)^k \oplus W$ `Chapter2/TameFaces · faceIncl_mem` `Chapter2/TameFaces · faceProj_mem`, and $d_C(y) = d_{C'}(\pi y) + 1$ on $f_i$ `Chapter2/TameFaces · degIndexWith_eq_quadrantDeg_faceProj_add_one`. The derivative of an sc-operator is itself `Chapter2/TameFaces · ScOperator.scDeriv_eq`, and the chain rule at a point holds between arbitrary sc-domains `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain`.

**Proposition 2.5.15 (preprint 2.43), the tame model** `Chapter2/TameFaces · proposition_2_5_15_tameModel`. With

$U^*$ and $s$ as above, put $U' := \iota^{-1}(U^*)$ (relatively open in $C'$) and $r' := \pi \circ r \circ \iota$ on $U'$. Since $r \circ \iota$ maps $U'$ into $F \subseteq f_i$, where $\iota\pi = 1$, $r'$ is an sc-smooth retraction with retract $\pi(F)$, and $\iota$ maps it back into $F$. *Tameness (1)*: for $z \in U'$, $d_{C'}(r'z) + 1 = d_C(r(\iota z)) = d_C(\iota z) = d_{C'}(z) + 1$. *Tameness (2)*: at a smooth $x \in \pi(F)$, $\iota x$ is a smooth point of $O$ and $Dr'(x)v = \pi(Dr(\iota x)(\iota v))$ by the chain rule; for $j$ with $x_j = 0$ the coordinate $j' = i.\mathrm{succAbove}(j)$ vanishes at $\iota x$, and Proposition 2.5.3 (preprint 2.35) for $r$ gives $\lambda_{j'}(Dr(\iota x)\iota v) = \lambda_{j'}(\iota v) = v_j$; so $\lambda'_j \circ Dr'(x) = \lambda'_j$, and $(1 - Dr'(x))E'$ is an sc-complement of $T_x\pi(F)$ inside $E'_x$. The exposition instead intersects the complement $A$ of $T_xO$ with $f_1$; that is not needed. And $d_{C'}(\pi y) + 1 = d_C(y)$ on $F$.

**Proposition 2.5.15 (preprint 2.43), faces are tame** `Chapter2/TameFaces · LocalModel.proposition_2_5_15_isTame`. For a face $F$ of a tame retract $O$ (paracompact, being closed), the chart $\pi : F \to \pi(F)$ `Chapter2/TameFaces · MPolyfold.faceChart` `Chapter2/TameFaces · MPolyfold.faceChartInv` is compatible with the sub-charts of $F$ built on the chart id of $O$, its transition maps being the sc-operators $\iota$ and $\pi$ `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`; a chart compatible with all sub-charts built on one chart of $X$ is a chart of the induced structure, by locality of compatibility `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible`. Its model is the tame retract $(\pi(F), C', E')$, so $F$ is tame (a space all of whose points lie in tame charts is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart`), and the tame chart realizes the index: $d_F(x) = d_{C'}(\pi x) = d_C(x) - 1 = d_O(x) - 1$ (equation (14) twice). The text calls this last statement obvious.

**Proposition 2.5.16 (preprint 2.44), first sentence** `Chapter2/TameFaces · MPolyfold.proposition_2_5_16`. In a tame chart $\varphi$ around $x$: $\varphi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold $O$ (Proposition 2.5.14 (preprint 2.42) with Corollary 2.5.13 (preprint 2.41)), and each face of $O$ is a tame sub-M-polyfold with $d_F + 1 = d_O$, where $d_O(\varphi x) = d_X(x)$.

*The corner as a model space.* For $\sigma \subseteq \{1, \dots, n\}$ list the coordinates outside $\sigma$ as $j \mapsto l_j$ `Chapter2/CornerGerms · cornerIdx` `Chapter2/CornerGerms · cornerIdx_injective` `Chapter2/CornerGerms · exists_cornerIdx`. Deleting the coordinates in $\sigma$ and inserting zeros there `Chapter2/CornerGerms · cornerRemoveL` `Chapter2/CornerGerms · cornerInsertL` are mutually inverse on $\{a : a_l = 0,\ l \in \sigma\}$ `Chapter2/CornerGerms · cornerRemoveL_cornerInsertL` `Chapter2/CornerGerms · cornerInsertL_cornerRemoveL`; as sc-operators on $\mathbb{R}^n \oplus W$ and $\mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerInsertModel` `Chapter2/CornerGerms · cornerRemoveModel`, composed with $T$, they give $\iota_\sigma : \mathbb{R}^{n-\#\sigma} \oplus W \to E$ onto the corner $f_\sigma = \bigcap_{i\in\sigma} f_i$ and $\pi_\sigma : E \to \mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl` `Chapter2/CornerGerms · cornerProj`, with $\pi_\sigma\iota_\sigma = 1$ and $\iota_\sigma\pi_\sigma = 1$ on $f_\sigma$ `Chapter2/CornerGerms · cornerProj_cornerIncl` `Chapter2/CornerGerms · cornerIncl_cornerProj`, $\iota_\sigma(C') \subseteq C$ and $\pi_\sigma(C) \subseteq C'$ for $C' = [0,\infty)^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl_mem` `Chapter2/CornerGerms · cornerProj_mem`, and $d_C(y) = d_{C'}(\pi_\sigma y) + \#\sigma$ on $f_\sigma$: the vanishing coordinates of $y$ are those of $\sigma$ together with the images $l_j$ of the vanishing coordinates of $\pi_\sigma y$, disjointly `Chapter2/CornerGerms · degIndexWith_eq_quadrantDeg_cornerProj_add_card`. The sc-projection $P_\sigma = \iota_\sigma \circ \pi_\sigma$ `Chapter2/CornerGerms · cornerKill` sets the coordinates in $\sigma$ to zero `Chapter2/CornerGerms · cornerKill_coord_mem`, fixes $f_\sigma$ `Chapter2/CornerGerms · cornerKill_eq_self` and maps $C$ into $C$ `Chapter2/CornerGerms · cornerKill_mem`. For $\sigma = \{i\}$ these are the $\iota$, $\pi$ and $P_i$ of Proposition 2.5.15 (preprint 2.43).

*The retraction onto a corner* `Chapter2/CornerGerms · corner_retraction`. Let $r$ be tame for $T$, $o \in O$ and $\sigma \subseteq I(o)$. For each $i \in \sigma$, Lemma 4.C at $o$ `Chapter2/TameRetracts · local_face_structure` gives a face $F_i$ of the retract with $o \in F_i \subseteq f_i$, containing a point with $d_C = 1$, and a ball around $o$ in which $r$ maps the points of $C \cap f_i$ into $F_i$. Distinct $i$ give distinct faces: a point $p \in F_i$ with $d_C(p) = 1$ has $I(p) = \{i\}$, so $F_i = F_j$ forces $i = j$ `Chapter2/TameRetracts · zeroSet_eq_singleton`. Take a ball $B$ around $o$ inside all these balls with $B \cap C \subseteq U$, an open $W_s$ with $W_s \cap U = r^{-1}(B) \cap U$ (continuity of $r$), and put $V := B \cap P_\sigma^{-1}(B \cap W_s)$,

an open set containing $o$ (since $P_\sigma o = o = r(o)$). On $V \cap C$ the map $s := r \circ P_\sigma$ is sc-smooth by the chain rule. For $z \in V \cap C$, $P_\sigma z \in C \cap f_\sigma \cap B$, so $r(P_\sigma z) \in F_i \subseteq f_i$ for every $i \in \sigma$: $s(z) \in O \cap f_\sigma$. Moreover $s(z) \in B$ (as $P_\sigma z \in W_s$), $P_\sigma s(z) = s(z)$, and $s(z) \in W_s$ (as $r(s(z)) = s(z) \in B$); so $s$ maps $V \cap C$ into itself. On $f_\sigma \cap O$, $P_\sigma$ and $r$ are the identity, so $s \circ s = s$: $s$ is an sc-smooth retraction with retract $O \cap V \cap f_\sigma$, and this set equals $O \cap V \cap \bigcap_{i\in\sigma} F_i$ (a point of $O \cap V \cap f_\sigma$ is fixed by $r$, which maps it into each $F_i$; conversely $F_i \subseteq f_i$). The exposition proceeds by induction on $\#\sigma$, identifying the germ of $F_1 \cap F_2$ with a face of $F_1$; the formal proof removes all coordinates of $\sigma$ at once.

*The tame model of a corner* `Chapter2/CornerGerms · corner_tameModel`. Let $s$ be an sc-smooth retraction on $U_s \subseteq U$, equal to $r \circ P_\sigma$ there, with retract $F \subseteq O \cap f_\sigma$; write $U_s = W_0 \cap C$ with $W_0$ open, and put $U' := \iota_\sigma^{-1}(W_0) \cap C'$ and $r' := \pi_\sigma \circ r \circ \iota_\sigma$ on $U'$. For $z \in U'$, $r(\iota_\sigma z) = s(\iota_\sigma z) \in F$ (as $P_\sigma \iota_\sigma = \iota_\sigma$), so $r'$ maps $U'$ into $\pi_\sigma(F) \subseteq U'$; it is sc-smooth by the chain rule and idempotent because $\iota_\sigma \pi_\sigma = 1$ on $F$ and $r = 1$ on $F \subseteq O$; its retract is $\pi_\sigma(F)$. *Tameness (1):* $d_{C'}(r'z) + \#\sigma = d_C(r(\iota_\sigma z)) = d_C(\iota_\sigma z) = d_{C'}(z) + \#\sigma$ (the index formula on $f_\sigma$ twice, and tameness (1) of $r$) `Chapter1/Degeneracy · degIndex_eq`. *Tameness (2):* at a smooth $x$ of the retract, $\iota_\sigma x$ is a smooth point of $O$, and $Dr'(x)v = \pi_\sigma(Dr(\iota_\sigma x)\, \iota_\sigma v)$ by the chain rule between sc-domains `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain` `Chapter2/TameFaces · ScOperator.scDeriv_eq`; for $j$ with $x_j = 0$ the coordinate $l_j$ vanishes at $\iota_\sigma x$ `Chapter2/CornerGerms · coord_cornerIncl_idx`, and Proposition 2.5.3 (preprint 2.35) gives $\lambda_{l_j}(Dr(\iota_\sigma x)\, \iota_\sigma v) = \lambda_{l_j}(\iota_\sigma v) = v_j$ `Chapter2/TameRetracts · coord_scDeriv_eq`; so $(1 - Dr'(x))E'$ lies in $E'_x$ and is an sc-complement of $T_x \pi_\sigma(F)$ `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. And $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$ for $y \in F$. This is the proof of the tame model of Proposition 2.5.15 (preprint 2.43) with $\sigma$ in place of $\{i\}$.

**Proposition 2.5.16 (preprint 2.44), second sentence** `Chapter2/CornerGerms · MPolyfold.proposition_2_5_16_corner`. Let $X$ be tame and $x \in X$, and take a chart $\varphi : V \to O$ of the structure around $x$ with a tame retraction $r$ onto $O$, tame for a presentation $T$ `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`. The face germs at $x$ are the faces $F_i$, $i \in I(\varphi x)$, of the retract (Corollary 2.5.13 (preprint 2.41): there are $d_X(x)$ of them), and $\sigma$ is a set of such indices. The retraction onto the corner gives the pairwise distinct faces $F_i \ni \varphi(x)$, $i \in \sigma$, an open $B \ni \varphi(x)$ and $s$ with retract $O \cap B \cap \bigcap_{i\in\sigma} F_i$. Put $A := \{y \in V : \varphi(y) \in B,\ \varphi(y) \in F_i\ \forall i \in \sigma\}$; its germ at $x$ is the germ $F^\sigma_x$. Since $\varphi(A)$ is the retract of $s$, $A$ is a sub-M-polyfold by the local criterion of Theorem 2.5.17 (preprint 2.45) `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local` (paracompact as every subset `Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype`). The chart $\pi_\sigma \circ \varphi$ of $A$ onto the retract $\pi_\sigma(\varphi(A))$ of $r'$ `Chapter2/TameFaces · MPolyfold.faceChart` is compatible with the sub-charts of $A$ built on $\varphi$, its transitions being $\iota_\sigma$ and $\pi_\sigma$ `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`, hence a chart of the induced structure `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible`; its model is tame `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame`, so $A$ is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart`. Both tame charts realize the index `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, so for $a \in A$: $d_A(a) + \#\sigma = d_{C'}(\pi_\sigma \varphi(a)) + \#\sigma = d_C(\varphi(a)) = d_X(a)$; in particular $d_A(x) = d_X(x) - \#\sigma$, and the codimension, the number of quadrant coordinates removed in the tame model, is $\#\sigma$. The text states the proposition without proof.

**Theorem 2.5.17 (preprint 2.45).** *The interior of a face.* $F \cap \{d_X = 1\} = G$ for $F = \overline{G}$ `Chapter2/TameRetracts · MPolyfold.theorem_2_5_17_interior`: at $y \in F$ with $d_X(y) = 1$, in a tame chart with $I(\varphi y) = \{i\}$, all points of $\{d_X = 1\}$ near $y$ lie in $\varphi^{-1}(\hat{F})$ for the single component $\hat{F}$ of Lemma 4.C at $\varphi(y)$, a connected subset of $\{d_X = 1\}$ containing $y$; it meets $G$ (as $y \in \overline{G}$), hence lies in $G$. *A local criterion for sub-M-polyfolds* `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local`: if near each $a \in A$, in a chart $\varphi$, $\varphi(A \cap V)$ agrees on an open $B \ni \varphi(a)$ with the retract of an sc-smooth retraction $s$ contained in $O$, then $A$ is a sub-M-polyfold, with the retraction $\varphi^{-1} \circ s \circ \varphi$ on $V' = \varphi^{-1}(B \cap W_0 \cap s^{-1}(B))$ — shrunk

so that $s$ maps it into itself, a step the exposition's "transport by $\varphi^{-1}$" omits. *$G$ is a sub-M-polyfold* Chapter2/TameFaces · MPolyfold.theorem_2_5_17_isSubMPolyfold_interior: at $y \in G$, the local face $\overline{\widehat{F}} \cap O$ agrees with $\varphi(G)$ on the ball of Lemma 4.C (both consist of the points of $S$ there), and Proposition 2.5.15 (preprint 2.43) gives its retraction. *The face-structured case.* Near $x \in F$, in a tame chart, $\varphi(F)$ agrees on a ball with a face of the retract Chapter2/TameFaces · MPolyfold.exists_localFace_of_isFaceStructured: the map $i \mapsto F_i$ (the closure of the component of $\varphi^{-1}(\widehat{F}_i)$) from $I(\varphi x)$ to the faces through $x$ is onto (as in Proposition 2.5.14 (preprint 2.42)), and since there are exactly $d_X(x) = \#I(\varphi x)$ faces through $x$, it is injective; so the points of $G$ near $x$ lie in $\varphi^{-1}(\widehat{F}_j)$ for the one $j$ with $F = F_j$. Hence $F$ is a sub-M-polyfold Chapter2/TameFaces · MPolyfold.theorem_2_5_17_isSubMPolyfold_face, and with the charts $\pi \circ \varphi$ onto the restrictions of the tame retractions $r'$ it is tame with $d_X(x) = d_F(x) + 1$ Chapter2/TameFaces · MPolyfold.theorem_2_5_17_isTame_face. The local and global face counts of the theorem are Proposition 2.5.14 (preprint 2.42) and the definition of face-structured.

## II.17 Products and fibred products: Lemma 4.D and Proposition 2.5.20 (preprint 2.46)

*Products of partial quadrants.* For presentations $T : E \cong \mathbb{R}^n \oplus W$ and $T' : E' \cong \mathbb{R}^{n'} \oplus W'$ of $C$ and $C'$, concatenating the coordinates ($\mathbb{R}^n \times \mathbb{R}^{n'} \cong \mathbb{R}^{n+n'}$ Chapter2/FibredProduct · finAppendL) gives a presentation $E \oplus E' \cong \mathbb{R}^{n+n'} \oplus (W \oplus W')$ Chapter2/FibredProduct · prodPresentation of $C \times C'$ Chapter2/FibredProduct · prod_eq_prodPresentation, so $C \times C'$ is a partial quadrant Chapter2/FibredProduct · IsPartialQuadrant.prod and $U \times U'$ is relatively open in it Chapter2/FibredProduct · IsRelOpenInQuadrant.prod. The vanishing coordinates of $(x, x')$ are those of $x$ followed by those of $x'$, so $d_{C\times C'}(x, x') = d_C(x) + d_{C'}(x')$ Chapter2/FibredProduct · degIndexWith_prodPresentation Chapter2/FibredProduct · degIndex_prod, and the minimal subspace is $E_x \oplus E'_{x'}$ Chapter2/FibredProduct · minimalSubspaceWith_prodPresentation.

*Products of retractions.* $f \times f'$ is sc-smooth when $f$ and $f'$ are Chapter2/FibredProduct · IsScSmooth.prodMap, with derivative $Df(x) \times Df'(x')$ Chapter2/FibredProduct · scDeriv_prodMap; so $r \times r'$ is an sc-smooth retraction Chapter2/FibredProduct · ScRetraction.prod with retract $O \times O'$ Chapter2/FibredProduct · ScRetraction.prod_retract Chapter2/FibredProduct · IsScRetract.prod, and products of sc-smooth maps between retracts are sc-smooth Chapter2/FibredProduct · IsScSmoothRetractMap.prodMap. Products of sc-subspaces and of sc-splittings are again such Chapter2/FibredProduct · IsScSubspace.prod Chapter2/FibredProduct · IsScSplitting.prod; hence if $r$, $r'$ are tame for $T$, $T'$, then $r \times r'$ is tame for the product presentation Chapter2/FibredProduct · ScRetraction.IsTameWith.prod: (1) is the additivity of the index, and for (2) the complement of $T_{(x,x')}(O \times O') = T_xO \oplus T_{x'}O'$ is $A \oplus A'$ inside $E_x \oplus E'_{x'}$.

**Lemma 4.D (the product M-polyfold).** $X \times Y$ is paracompact, being metrizable by Theorem 2.3.10 (preprint 2.16) Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod. The product charts Chapter2/FibredProduct · MChart.prod have the product retracts as models Chapter2/FibredProduct · LocalModel.prod, their overlaps are products Chapter2/FibredProduct · MChart.overlap_prod, and products of compatible charts are compatible Chapter2/FibredProduct · MChart.Compatible.prod; so they form an atlas of an M-polyfold structure on $X \times Y$ Chapter2/FibredProduct · MPolyfold.prod, in which products of charts of the factors are charts Chapter2/FibredProduct · MPolyfold.IsChart.prod. In a product chart the index is additive Chapter2/FibredProduct · MPolyfold.chartDegIndex_prod; choosing charts that realize $d_X(x)$ and $d_Y(y)$ gives $d_{X\times Y}(x, y) \le d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.degIndex_prod_le. If $X$ and $Y$ are tame, products of tame charts are tame charts, so $X \times Y$ is tame, and since a tame chart realizes the index (§II.14), $d_{X\times Y}(x, y) = d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.isTame_prod. The text's "$d_{X\times Y} = d_X + d_Y$"

is thus a statement about tame spaces. The projections are sc-smooth `Chapter2/FibredProduct · MPolyfold.isScSmoothMap_fst` `Chapter2/FibredProduct · MPolyfold.isScSmoothMap_snd` and so is the pairing $z \mapsto (f(z), g(z))$ of sc-smooth maps `Chapter2/FibredProduct · MPolyfold.IsScSmoothOn.prodMk`.

**Proposition 2.5.20 (preprint 2.46).** A *local sc-diffeomorphism* $f : X \to Z$ is read as: every point has an open neighbourhood $U$ that $f$ maps sc-diffeomorphically onto an open subset `Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo`; such an $f$ is continuous and sc-smooth `Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.continuous` `Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.isScSmoothMap`. The fibred product $X_f \times_g Y = \{(x, y) : f(x) = g(y)\}$ `Chapter2/FibredProduct · MPolyfold.fibredProduct` is closed in $X \times Y$ `Chapter2/FibredProduct · MPolyfold.isClosed_fibredProduct`, hence paracompact, which the text does not mention. *Sub-M-polyfold.* Near $(x, y)$, with $f|_U : U \to W$ an sc-diffeomorphism and $V = g^{-1}(W)$, the map $R(x', y') = ((f|_U)^{-1}(g(y')), y')$ on $U \times V$ is sc-smooth, idempotent, with image the fibred product near $(x, y)$; in a product chart it is an sc-smooth retraction `Chapter2/FibredProduct · MPolyfold.fibredSubDataL`, and so the fibred product is a sub-M-polyfold `Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct` (symmetrically when $g$ is the local sc-diffeomorphism `Chapter2/FibredProduct · MPolyfold.fibredSubDataR` `Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct'`). *$\pi_2$ is a local sc-diffeomorphism.* On the fibred product near $(x, y)$, $\pi_2$ is sc-smooth with inverse $y' \mapsto ((f|_U)^{-1}(g(y')), y')$, a pairing of sc-smooth maps with values in the fibred product, hence sc-smooth into the sub-M-polyfold by Proposition 2.3.19 (2) (preprint 2.21) `Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct`; symmetrically for $\pi_1$ when $g$ is a local sc-diffeomorphism `Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_fst_fibredProduct`. The exposition factors this inverse through $y' \mapsto (x, y')$, which is not sc-smooth unless $x$ is a smooth point (a constant map is sc$^0$ only when its value lies in $X_\infty$); the formula above has the same values and avoids it. *The index.* Proposition 2.4.4 (preprint 2.24) along $\pi_2$ gives $d_{X_f \times_g Y}(x, y) = d_Y(y)$ for arbitrary $Y$ `Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_snd`, and along $\pi_1$, when $g$ is a local sc-diffeomorphism, $= d_X(x)$ `Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_fst`. *Tameness.* If $Y$ is tame, the charts $\psi \circ \pi_2$, $\psi$ a tame chart of $Y$ restricted to $g^{-1}(W)$, are charts of the fibred product with tame models `Chapter2/FibredProduct · MPolyfold.isTame_fibredProduct`. Together: Proposition 2.5.20 (preprint 2.46) `Chapter2/FibredProduct · MPolyfold.proposition_2_5_20`, and in its last statement $d = d_X = d_Y$ for arbitrary $X$, $Y$ and, for tame $X$, $Y$, $2d = d_{X \times Y}$ by Lemma 4.D `Chapter2/FibredProduct · MPolyfold.proposition_2_5_20_of_isLocalScDiffeo`.

## II.18 Strong bundle retractions (Definitions 2.6.1–2.6.3 (preprint 2.47–2.49))

$U \lhd F$. The double filtration is $U_m \times F_k$ `Chapter2/Tame · doubleLevel`. $C \times F_0$ is a partial quadrant in $E \oplus F$, through $(\mathbb{R}^n \oplus W) \oplus F \cong \mathbb{R}^n \oplus (W \oplus F)$ `Chapter2/StrongBundle · prodQuadrantIso` `Chapter2/StrongBundle · IsPartialQuadrant.prod_univ`, and $U \times F_0$ is relatively open in it `Chapter2/StrongBundle · IsRelOpenInQuadrant.prod_univ` — so "sc-smooth on $(U \lhd F)[i] = U \oplus F^{i}$" makes sense (exposition Lemma 1.B of Chapter 1). Pairings of sc$^k$ maps are sc$^k$, with the tangent map rearranged by an sc-operator `Chapter2/StrongBundle · scPair` `Chapter2/StrongBundle · tangentProdSwap` `Chapter2/StrongBundle · IsSck.scPair` `Chapter2/StrongBundle · IsScSmooth.scPair`, and the projections and inclusions of products are sc-operators `Chapter2/StrongBundle · ScBanach.prodFstOp` `Chapter2/StrongBundle · ScBanach.prodSndOp` `Chapter2/StrongBundle · ScBanach.prodInlOp` `Chapter2/StrongBundle · ScBanach.prodShiftInclOp`.

**Definition 2.6.1 (preprint 2.47).** A strong bundle map maps $U \times F_0$ into $U' \times F_0'$, has first component independent of $h$ and second linear in $h$, maps $U_m \times F_k$ into $U'_m \times F'_k$ for $k \leq m + 1$,

is sc-smooth on $U \times F_0$, and its fibre-lifted map on $U \times F_1$ Chapter2/Tame · shiftFiber is sc-smooth for $E \oplus F^1$ Chapter2/Tame · IsStrongBundleMap; a strong bundle retraction is an idempotent one Chapter2/Tame · IsStrongBundleRetraction. *Lemma 5.A.* The base $r(u) = R(u,0)_1$ Chapter2/Tame · IsStrongBundleMap.base Chapter2/StrongBundle · IsStrongBundleRetraction.apply_fst is an sc-smooth retraction on $U$: sc-smooth as $R$ composed with $u \mapsto (u,0)$ and the first projection, idempotent by the first component of $R \circ R = R$ Chapter2/StrongBundle · IsStrongBundleRetraction.baseRetraction. For $u \in O$, $R(u,h) = (u, \Gamma(u,h))$, so the second component of $R \circ R = R$ reads $\Gamma(u,\Gamma(u,h)) = \Gamma(u,h)$: $\Gamma(u,\cdot)$ is a projection Chapter2/StrongBundle · IsStrongBundleRetraction.fiber_idem. $R$ is an sc-smooth retraction on $U \times F_0$ in $C \times F_0$ Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetraction and, through the lift of the fibre component to $F_1$ (possible since $R$ preserves the double filtration at $(0,1)$ Chapter2/StrongBundle · IsStrongBundleRetraction.snd_mem_range Chapter2/StrongBundle · IsStrongBundleRetraction.toBase_shiftFiber_snd), on $U \times F_1$ for $E \oplus F^1$ Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne.

**Definition 2.6.2 (preprint 2.48).** $K$ is a local strong bundle retract if $K = R(U \times F_0)$ Chapter2/Tame · IsStrongBundleRetract, with $K_{m,k}$, $K[i]_m$ and $K[1] \subseteq E_0 \times F_1$ Chapter2/Tame · strongLevel Chapter2/Tame · strongLevelShift Chapter2/Tame · strongShiftOne and the projection $p$ Chapter2/Tame · strongProj. $K = \{(u,h) : u \in O, \Gamma(u,h) = h\}$ Chapter2/StrongBundle · IsStrongBundleRetraction.mem_image_iff. *Lemma 5.B.* $K[1]$ is the image of $R$ on $U \times F_1$ Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne_retract, so $K$ and $K[1]$ are retracts Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract_one; the base $p(K)$ is $O$ Chapter2/StrongBundle · IsStrongBundleRetract.base Chapter2/StrongBundle · IsStrongBundleRetract.base_eq; and $p : K[i] \to O$ is sc-smooth, because $p \circ R = r \circ \mathrm{pr}_1$ is Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one, as is the inclusion $K[1] \to K[0]$ Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl.

**Definition 2.6.3 (preprint 2.49).** An sc-smooth section is sc-smooth from $(O,C,E)$ into $(K, C \times F_0, E \oplus F)$ with $f(x)_1 = x$ Chapter2/Tame · IsScSmoothSection; an sc⁺-section is sc-smooth into $K[1]$ Chapter2/Tame · IsScPlusSection Chapter2/Tame · IsScPlusSection.mem. Composed with the inclusion $K[1] \to K[0]$ (Theorem 2.1.15 (preprint 2.8)), an sc⁺-section is an sc-smooth section Chapter2/StrongBundle · IsScPlusSection.isScSmoothSection.

## II.19 Strong bundles over M-polyfolds (Definitions 2.6.4–2.6.7 (preprint 2.50–2.53))

**Definition 2.6.4 (preprint 2.50).** For $P : Y \to X$ with linear structures on the fibres, a strong bundle chart is a chart $\varphi$ of $X$ (of the structure — the text says only "a homeomorphism onto $O$"), a strong bundle retraction $R$ on $U \subseteq C$ with base retract $O$, and a homeomorphism $\Phi : P^{-1}(V) \to K$ with $\Phi(y)_1 = \varphi(P(y))$, linear on fibres Chapter2/StrongBundle · StrongBundleChart Chapter2/StrongBundle · StrongBundleChart.K. $K$ is a local strong bundle retract and a retract on $[0]$ and $[1]$ Chapter2/StrongBundle · StrongBundleChart.isStrongBundleRetract Chapter2/StrongBundle · StrongBundleChart.isScRetract Chapter2/StrongBundle · StrongBundleChart.isScRetract_one Chapter2/StrongBundle · StrongBundleChart.model, the source $P^{-1}(V)$ is open Chapter2/StrongBundle · StrongBundleChart.isOpen_source, $\Phi(P^{-1}(V)) = K$ and $p(K) = O$ Chapter2/StrongBundle · StrongBundleChart.image_source Chapter2/StrongBundle · StrongBundleChart.base_K. "Bounded on the fibres" needs no hypothesis: the restriction of the continuous $\Phi$ to a fibre is continuous Chapter2/StrongBundle · StrongBundleChart.continuous_fiber. Two charts are compatible if the transitions $\Phi' \circ \Phi^{-1}$ Chapter2/StrongBundle · StrongBundleChart.overlap Chapter2/

StrongBundle · StrongBundleChart.transition and back are sc-smooth between the overlaps on $[0]$ and, fibre-lifted, on $[1]$, and carry fibre components in $F_1$ to $F_1'$ Chapter2/StrongBundle · StrongBundleChart.Compatible Chapter2/StrongBundle · StrongBundleChart.Compatible.symm; the last condition, implicit in the text, is what makes $(\Phi' \circ \Phi^{-1})[1]$ the restriction of $\Phi' \circ \Phi^{-1}$. Atlases and their equivalence Chapter2/StrongBundle · StrongBundleAtlas Chapter2/StrongBundle · StrongBundleAtlas.Equiv, an equivalence relation (§II.45) Chapter2/StrongBundleRestrict · StrongBundleAtlas.equivalence; Banachable fibres Chapter2/StrongBundle · IsBanachable.

**Definition 2.6.5 (preprint 2.51).** A strong bundle is a continuous surjection with Banachable fibres and a strong bundle atlas, $Y$ Hausdorff paracompact Chapter2/StrongBundle · StrongBundle. Each $\Phi$ is a chart of $Y$ with model $(K, C \times F_0, E \oplus F)$ Chapter2/StrongBundle · StrongBundleChart.toMChart, compatibility of strong bundle charts on $[0]$ is compatibility of these charts Chapter2/StrongBundle · StrongBundleChart.Compatible.toMChart, and they form the atlas of the M-polyfold $Y[0]$ Chapter2/StrongBundle · StrongBundle.atlas0 Chapter2/StrongBundle · StrongBundle.toMPolyfold0 Chapter2/StrongBundle · StrongBundle.isChart_toMChart. $P : Y[0] \to X$ is sc-smooth: in the charts $\Phi$ and $\varphi$ it is $p : K \to O$ (Lemma 5.B), and one pair of charts per point suffices Chapter2/StrongBundle · StrongBundle.isScSmoothMap_proj. The double filtration $Y_{m,k}$ is defined through all the charts of the atlas around a point Chapter2/StrongBundle · StrongBundle.doubleLevel.

*$Y_{m,k}$ does not depend on the chart* Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff. Let $y$ lie in the domain of a chart $\Phi$ of the atlas with $\Phi(y) \in E_m \times F_k$, $k \le m+1$, and let $\Phi'$ be another chart around $y$; then $\Phi'(y) = (\Phi' \circ \Phi^{-1})(\Phi(y))$ Chapter2/StrongBundlePullback · StrongBundleChart.transition_apply. *The base component:* $\Phi'(y)_1 = \varphi'(\varphi^{-1}(\Phi(y)_1))$, and the base transition, sc-smooth and so $\mathrm{sc}^0$, maps points of level $m$ to points of level $m$ Chapter2/MPolyfold · MChart.transition_apply. *The fibre component, $k \le m$:* $\Phi(y)$ lies on level $k$ of $E \oplus F$ (levels decrease Chapter2/StrongBundleOne · StrongBundle.range_toBase_mono), and the $[0]$-transition, $\mathrm{sc}^0$ on the overlap retract Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap, keeps it on level $k$. *The case $k = m+1$:* the fibre component of $\Phi(y)$ lies in $F_1$, so its lift $\Phi(y)[1]$ to $E_0 \times F_1$ Chapter2/StrongBundleOne · StrongBundleChart.liftOne Chapter2/StrongBundleOne · StrongBundleChart.toBase_liftOne is a point of level $m$ of $E \oplus F^1$ (level $m$ of $F^1$ is $F_{m+1}$ Chapter1/ScBanach · ScBanach.shiftScale_toBase); the $[1]$-transition, $\mathrm{sc}^0$ for $E \oplus F^1$ on the overlap Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one, keeps it on level $m$, and its value is the lift $\Phi'(y)[1]$, because the $F_1$-condition of compatibility puts the fibre component of $\Phi'(y)$ in $F_1'$. So $\Phi'(y) \in E_m' \times F_{m+1}'$. The case $k \le m$ uses the $[0]$-transition only; the case $k = m+1$ uses exactly the $F_1$-condition made explicit in Definition 2.6.4 (preprint 2.50).

*The set $Y[1]$ and its topology.* $Y_{0,1}$ is the set of points whose fibre component lies in $F_1$ in every chart of the atlas around them Chapter2/StrongBundleOne · StrongBundle.oneSet; by the $F_1$-condition one chart suffices Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff. $Y[1]$ is $Y_{0,1}$ as a type Chapter2/StrongBundleOne · StrongBundle.One with the topology generated by the sets $\{y \in \mathrm{dom}\,\Phi : \Phi(y)[1] \in W\}$, $W$ open in $E_0 \times F_1$, as for $X_m$ in §2.3 (preprint §2.2) Chapter2/StrongBundleOne · StrongBundle.isOpen_one_basic. The inclusion $Y[1] \to Y$ is continuous Chapter2/StrongBundleOne · StrongBundle.continuous_oneVal: inside the domain of a chart $\Phi$, an open subset of $Y$ is the preimage under $\Phi$ of an open $W \subseteq E_0 \times F_0$ ($\Phi$ is a homeomorphism onto $K$ Chapter2/MPolyfold · MChart.exists_isOpen_image_eq), and its points in $Y[1]$ form the basic set of $(1 \times \iota)^{-1}(W)$. A continuous injection into the Hausdorff space $Y$ makes $Y[1]$ Hausdorff Mathlib/Topology/Separation/Hausdorff · T2Space.of_injective_continuous.

*The charts $\Phi[1]$* Chapter2/StrongBundleOne · StrongBundle.oneChart. $y \mapsto \Phi(y)[1]$ maps $P^{-1}(V) \cap Y[1]$ into $K[1]$ Chapter2/StrongBundleOne · StrongBundleChart.liftOne_mem, with inverse $z \mapsto \Phi^{-1}(z_1, \iota z_2)$, a point of $Y_{0,1}$ Chapter2/StrongBundleOne · StrongBundle.symm_mem_oneSet Chapter2/StrongBundleOne · StrongBundle.oneChartInv Chapter2/StrongBundleOne · StrongBundle.liftOne_symm. It is continuous by the definition of the topology. Its

inverse is continuous: a basic open set $\{\Phi'(y)[1] \in W\}$ pulls back to the set of $z \in K[1]$ in the $[1]$-part of the overlap with $(\Phi' \circ \Phi^{-1})[1](z) \in W$; the overlap is $K \cap W_0$ for an open $W_0$ `Chapter2/StrongBundlePullback · StrongBundleChart.overlap_eq_inter` and the $[1]$-transition is continuous on it, so this set is open in $K[1]$. The overlap of $\Phi[1]$ and $\Phi'[1]$ is the $[1]$-part of the overlap of $\Phi$ and $\Phi'$ `Chapter2/StrongBundleOne · StrongBundle.oneChart_overlap`, and their transition map is the $[1]$-transition $(\Phi' \circ \Phi^{-1})[1]$ `Chapter2/StrongBundleOne · StrongBundle.oneChart_transition`, sc-smooth in both directions by Definition 2.6.4 (preprint 2.50); so the charts $\Phi[1]$ are compatible `Chapter2/StrongBundleOne · StrongBundle.oneChart_compatible` and form an atlas of $Y[1]$ `Chapter2/StrongBundleOne · StrongBundle.atlas1`. $Y[1]$ is metrizable: $P$ restricted to $Y[1]$ is continuous into the paracompact Hausdorff $X$, and over the domain $V$ of a chart of $X$ its preimage is the source of a chart $\Phi[1]$, homeomorphic to a retract and so metrizable `Chapter2/Metrizable · MChart.metrizableSpace_source`; the pull-back form of Smirnov's theorem applies `Chapter2/Metrizable · metrizableSpace_of_pullback` `Chapter2/StrongBundleOne · StrongBundle.metrizableSpace_one`. Hence $Y[1]$ is paracompact `Chapter2/StrongBundleOne · StrongBundle.paracompactSpace_one`, which an M-polyfold must be and the text does not mention, and $Y[1]$ **is an M-polyfold** `Chapter2/StrongBundleOne · StrongBundle.toMPolyfold1` whose charts include the $\Phi[1]$ `Chapter2/StrongBundleOne · StrongBundle.isChart_oneChart`.

*$P[1]$ and the inclusion $Y[1] \to Y[0]$.* In the charts $\Phi[1]$ and $\varphi$, $P[1]$ is the projection $K[1] \to O$ of Lemma 5.B `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one`, and one pair of charts per point suffices `Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts`; so $P[1] : Y[1] \to X$ **is sc-smooth** `Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_proj1`. In the charts $\Phi[1]$ and $\Phi$ the inclusion $Y[1] \to Y[0]$ is $(u, h) \mapsto (u, \iota h)$ `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl`, so it is sc-smooth `Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_oneVal` (not stated in the text; it gives that an sc⁺-section is an sc-smooth section).

**Definition 2.6.6 (preprint 2.52).** A section has $P \circ f = \mathrm{id}$ `Chapter2/StrongBundle · IsSection`; it is sc-smooth if sc-smooth into $Y[0]$ `Chapter2/StrongBundle · StrongBundle.IsScSmoothSection`, and an sc⁺-section if in every chart its fibre values lie in $F_1$ and its fibre-lifted representative `Chapter2/StrongBundle · StrongBundle.localRepOne` is sc-smooth into $K[1]$ `Chapter2/StrongBundle · StrongBundle.IsScPlusSection`. *This is the text's definition* `Chapter2/StrongBundleOne · StrongBundle.isScPlusSection_iff`: $f$ is an sc⁺-section iff it is a section and $f = \iota \circ g$ for an sc-smooth $g : X \to Y[1]$, $\iota$ the inclusion. Forward: the values of $f$ lie in $Y_{0,1}$ (one chart suffices), which defines $g$; on the domain of a chart $\varphi$ of $X$ under a strong bundle chart $\Phi$, $g = \Phi[1]^{-1} \circ (\text{the local representative}) \circ \varphi$, which gives the continuity of $g$, and the representative of $g$ in the charts $\varphi$, $\Phi[1]$ is the chart-wise representative, sc-smooth by hypothesis. Backward: the fibre values lie in $F_1$ because $g$ takes values in $Y_{0,1}$, and the chart-wise representative is the representative of $g$ in $\varphi$, $\Phi[1]$, sc-smooth by Definition 2.3.8 (preprint 2.14) (a chart with empty domain is handled separately).

**Definition 2.6.7 (preprint 2.53).** $f^*Y = \{(z, y) : P(y) = f(z)\}$ `Chapter2/StrongBundle · pullbackSpace` with the projections `Chapter2/StrongBundle · pullbackProj` `Chapter2/StrongBundle · pullbackProj'`, $P \circ P' = f \circ P_f$ `Chapter2/StrongBundle · pullback_comm`; $P_f$ is continuous and $f^*Y$ closed in $Z \times Y$ `Chapter2/StrongBundle · continuous_pullbackProj` `Chapter2/StrongBundle · isClosed_pullbackSpace`.

## II.20 Proposition 2.6.8 (preprint 2.54): the pull-back strong bundle

*The pulled-back retraction.* For a retraction $r'$ on $U'$ and $g : O' \to O$ sc-smooth between retracts, $R_g(w, h) := (r'(w), \Gamma(g(r'(w)), h))$ `Chapter2/StrongBundle · pullbackRetraction`, where $g(r'(w)) \in O \subseteq U$ `Chapter2/StrongBundle · pullback_mem_O` `Chapter2/StrongBundle · pullback_mem_U`. It is a strong bundle retraction `Chapter2/StrongBundle · isStrongBundleRetraction_pullback`: it maps $U' \times F_0$ into itself; its first component does

not depend on $h$ and its second is linear in $h$; it preserves the double filtration ($r'$ and $g \circ r'$ preserve levels, and $R$ preserves the double filtration); on $[0]$ it is the pairing of $r' \circ \mathrm{pr}_1$ with the second component of $R \circ ((g \circ r') \circ \mathrm{pr}_1, \mathrm{pr}_2)$, sc-smooth by the chain rule and the pairing lemma, and likewise on $[1]$; and $R_g \circ R_g = R_g$ because $r' \circ r' = r'$ and $\Gamma(o, \cdot)$ is a projection for $o = g(r'(w)) \in O$ (Lemma 5.A). Its image is $K_g = \{(w,h) : w \in O', (g(w), h) \in K\}$ `Chapter2/StrongBundle · pullback_image` `Chapter2/StrongBundle · isStrongBundleRetract_pullback`.

*The fibres.* $P_f^{-1}(z) \to P^{-1}(f(z))$, $(z,y) \mapsto y$, is a bijection `Chapter2/StrongBundlePullback · pullbackFiberEquiv`; transporting the linear structure along it `Chapter2/StrongBundlePullback · pullbackFiber_addCommGroup` `Chapter2/StrongBundlePullback · pullbackFiber_module` makes it a linear homeomorphism `Chapter2/StrongBundlePullback · pullbackFiberCLE`, so the fibres of $f^*Y$ are Banachable.

*The charts.* The data: a strong bundle chart $c$ of $Y$, a chart $\psi$ of $Z$ with $f(\mathrm{dom}\,\psi) \subseteq \mathrm{dom}\,\varphi$, a retraction $r_Z$ onto the model of $\psi$, and the sc-smoothness of $g = \varphi \circ f \circ \psi^{-1}$ `Chapter2/StrongBundlePullback · PullbackChartData` `Chapter2/StrongBundlePullback · PullbackChartData.g` `Chapter2/StrongBundlePullback · PullbackChartData.g_smooth`. The chart $\Phi_f(z,y) = (\psi(z), \mathrm{pr}_F \Phi(y))$ maps $P_f^{-1}(\mathrm{dom}\,\psi)$ into $K_g$ — because $\Phi(y) = (g(\psi(z)), \cdot) \in K$ `Chapter2/StrongBundlePullback · PullbackChartData.g_apply` `Chapter2/StrongBundlePullback · PullbackChartData.mem_Kg` — with inverse $(w,h) \mapsto (\psi^{-1}(w), \Phi^{-1}(g(w), h))$ `Chapter2/StrongBundlePullback · PullbackChartData.symm_mem_pullback` `Chapter2/StrongBundlePullback · PullbackChartData.invFun`; both are continuous, it covers $\psi$ and is linear on fibres, so it is a strong bundle chart of $f^*Y$ `Chapter2/StrongBundlePullback · PullbackChartData.toChart` `Chapter2/StrongBundlePullback · PullbackChartData.image_eq_Kg`.

*Compatibility.* Sc-operators and pairings are sc-smooth between retracts `Chapter2/StrongBundlePullback · ScOperator.isScSmoothRetractMap` `Chapter2/StrongBundlePullback · IsScSmoothRetractMap.scPair` `Chapter2/StrongBundlePullback · isScRetract_univ`, and overlaps of strong bundle charts are retracts on $[0]$ and $[1]$ `Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap` `Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one`. On the overlap of two pulled-back charts the transition map is

$$(w,h) \mapsto (\psi_2 \psi_1^{-1}(w),\ \mathrm{pr}_F(\Phi_2 \Phi_1^{-1})(g_1(w), h))$$

`Chapter2/StrongBundlePullback · PullbackChartData.transitionFormula` `Chapter2/StrongBundlePullback · PullbackChartData.transition_eqOn`: its first component is the transition of $Z$ composed with the projection, its second the fibre component of the transition of $Y$ composed with $(w,h) \mapsto (g_1(w), h)$ `Chapter2/StrongBundlePullback · PullbackChartData.gPair`, which maps the overlap into the overlap of $Y$'s charts `Chapter2/StrongBundlePullback · PullbackChartData.gPair_mem`. Each factor is sc-smooth between retracts, and Theorem 2.1.15 (preprint 2.8) and the pairing lemma give the sc-smoothness on $[0]$ `Chapter2/StrongBundlePullback · PullbackChartData.transition_isScSmooth`; the same with $F^1$ on $[1]$ `Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_eqOn` `Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_isScSmooth`; fibre components in $F_1$ go to $F_1$ because the transitions of $Y$ have this property `Chapter2/StrongBundlePullback · PullbackChartData.transition_snd_mem_range`. So any two pulled-back charts are compatible `Chapter2/StrongBundlePullback · PullbackChartData.compatible_toChart`.

**Proposition 2.6.8 (preprint 2.54).** $f^*Y$ is paracompact: $Z$ and $Y[0]$ are metrizable (Theorem 2.3.10 (preprint 2.16)), so $Z \times Y$ and its subspace $f^*Y$ are metrizable, hence paracompact `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` — a requirement of Definition 2.6.5 (preprint 2.51) that the text does not mention. The pulled-back charts, built around every point from a chart of $Y$'s atlas and a chart of $Z$ restricted to $f^{-1}(\mathrm{dom}\,\varphi)$, cover $f^*Y$ and form an atlas `Chapter2/StrongBundlePullback · pullbackAtlas`. With the continuous surjection $P_f$ and its Banachable fibres this is a

strong bundle structure on $f^*Y \to Z$ Chapter2/StrongBundlePullback · proposition_2_6_8 ("the easy proof is left to the reader").

## II.21 What the formalization taught us

- **Theorem 2.3.10 (preprint 2.16) needs no added hypothesis, and not the text's route.** The text's proof applies Proposition 2.7.4 (preprint 2.57) to $X_m$ without checking that $X_m$ is regular. A partition of unity of $X$, pulled back along the continuous inclusion, metrizes $X_m$ directly Chapter2/Metrizable · metrizableSpace_of_pullback, and the same lemma proves Smirnov's theorem Chapter2/Metrizable · metrizableSpace_of_locally, Proposition 2.3.15 (3) (preprint 2.19) Chapter2/Metrizable · proposition_2_3_15_three and the paracompactness of $f^*Y$ Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace. With it every paracompactness hypothesis carried by the constructions is discharged Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype.
- **Compatibility is local, and the text needs it.** That the charts compatible with an atlas are compatible with each other is not in the text; it is the chain rule plus the locality of sc-smoothness Chapter2/MPolyfold · MChart.compatible_of_locally, and every chart-independence claim of §2.3 (preprint §2.2) rests on it.
- $TO$ **is a fixed-point set.** Proposition 2.1.11 (preprint 2.4) and Proposition 2.1.13 (preprint 2.6) are one-line consequences of $p \in Tr(TU) \iff Tr(p) = p$ Chapter2/Retracts · ScRetraction.mem_tangent_iff, which the text's proofs use without stating.
- **The tangent bundle needs the locality of** $Tf$**.** That Definition 2.3.12's (preprint 2.17) relation is an equivalence relation needs the chain rule on the triple overlap and the agreement of $Tf$ computed on an open subset with $Tf$ computed on the whole retract Chapter2/TangentBundle · retractTangentMap_inter_isOpen Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport.
- **Level** $1$ **suffices for §2.4 (preprint §2.3).** Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31) and Corollary 2.4.13 (preprint 2.32) hold at points on level 1 and for sc[1] germs Chapter2/Boundary · proposition_2_4_10_reduced Chapter2/Boundary · theorem_2_4_12 Chapter2/DegeneracyIndex · corollary_2_4_13; the text's "closure of $(T^R_x O)_\infty$" is the density of level-1 points in the kernel of a finite-rank map Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase.
- **Proposition 2.5.3 (preprint 2.35) needs only tameness (2), and Lemma 2.5.12 (preprint 2.40) needs (2) through it.** $Dr(x)E_x \subseteq E_x$ holds for every retraction Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace; that the retraction keeps points of a face in that face is the ray lemma Chapter2/TameRetracts · ray_zeroSet, which uses Proposition 2.5.3 (preprint 2.35).
- **Three slips in the exposition.** "$|P_1 z - e|_0 \le |z - e|_0$" in the exposition's proof of Proposition 2.5.15 (preprint 2.43) is false for a general norm on $E_0$; the formal neighbourhood intersects balls with their $P_i$-preimages Chapter2/TameRetracts · proposition_2_5_15. The "transport by $\varphi^{-1}$" in Theorem 2.5.17 (preprint 2.45) needs a shrinking so that the retraction maps its domain into itself Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local. In Proposition 2.5.20 (preprint 2.46) the inverse of $\pi_2$ cannot be factored through the constant-in-$x$ map $y' \mapsto (x, y')$, which is not sc-smooth at a non-smooth $x$ Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct.
- **Tameness is needed only for the additivity of the index under products.** $d_{X\times Y} \le d_X + d_Y$ in general Chapter2/FibredProduct · MPolyfold.degIndex_prod_le, with equality for tame factors Chapter2/FibredProduct · MPolyfold.isTame_prod; the identities $d = d_X = d_Y$ of Proposition 2.5.20 (preprint 2.46) hold without tameness Chapter2/FibredProduct · MPolyfold.proposition_2_5_20_of_isLocalScDiffeo.

- **Definition 2.6.4 (preprint 2.50) needs two explicit conditions.** The base homeomorphism of a strong bundle chart must be a chart of $X$, and the $[1]$-transition must preserve fibre components in $F_1$ `Chapter2/StrongBundle · StrongBundleChart.Compatible`. The second condition is exactly what makes $Y_{0,1}$, and so the M-polyfold $Y[1]$, independent of the chart `Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff` `Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff`.
- **Proposition 2.5.6 (preprint 2.36) does not need the equality case of Theorem 2.4.12 (preprint 2.31).** For a tame retraction the smooth vectors of the splitting are $Dr(x)T^{-1}(e_{i_j}, 0)$, dual to the coordinates by Proposition 2.5.3 (preprint 2.35), and $T^R_x O$ is an sc-subspace because $Dr(x)$ preserves $E_x$ `Chapter2/PartialCone · proposition_2_5_6_partialCone`.
- **Corners at once, not by induction.** The intersections $F^\sigma_x$ of Proposition 2.5.16 (preprint 2.44) are retracts of $r \circ P_\sigma$, by Lemma 4.C for each $i \in \sigma$; no identification of the germ of $F_1 \cap F_2$ with a face of $F_1$ is needed `Chapter2/CornerGerms · corner_retraction`.
- **$C_x O$ is always a partial quadrant.** The second sentence of Remark 2.4.8 (preprint 2.28) does not hold as we read it. At every smooth point of every sc-smooth retract, $C_x O$ is a partial quadrant in $T_x O$ `Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant`. The reason is finite-dimensional: $Dr(x)$ induces a nonnegative idempotent matrix on $E/E_x$, and the cone of such a matrix is simplicial (Flor) `Chapter2/PartialConeQuadrant · flor_nonnegRange_simplicial`. What tameness adds is only that the quadrant has the full dimension $d_C(x)$. The second statement of Theorem 2.4.12 (preprint 2.31) needs no hypothesis.
- **The linear structure of $T_x X$ is chart independent because the fibre transitions are restrictions of derivatives.** Definition 2.3.14 (preprint 2.18) adds in one chart and does not check independence. It holds because $T\psi \circ (T\varphi)^{-1}$ is the restriction of $D(\psi \circ \varphi^{-1} \circ r)(\varphi(x))$ `Chapter2/TangentSpaceLinear · MPolyfold.exists_transition_clm`. At a smooth point this restriction is an sc-isomorphism of the models, which makes the sc-structure of $T_x X$ chart independent too `Chapter2/GapP1_Def258b · MPolyfold.tangentFiberScIso`.
- **The openness of $\mathrm{pr}_1 : TO \to O_1$ has a one-line proof.** Appendix 2.7.3 states it without proof. The section $o' \mapsto Tr(o', h)$ through a given tangent vector is continuous because $Tr$ is $\mathrm{sc}^0$ `Chapter2/TangentProjOpen · ScRetraction.exists_isOpen_tangent_section`.
- **Michael's lemma needs a shrinking.** The closures of a locally finite open refinement lie in members of the cover only after the cover is shrunk by regularity `Chapter2/GapP1_Michael · GapP1Michael.exists_closed_refinement`. The converse direction needs no separation at all `Chapter2/GapP1_Michael · GapP1Michael.paracompactSpace_of_closed_refinement`.
- **Proposition 2.1.2 (preprint 2.2) is a conjugation.** $\varphi(y) = P(r(y) - x) + (1 - P)(y - r(y))$ has derivative $1$ at $x$ and satisfies $\varphi \circ r = P \circ \varphi$, so the inverse function theorem replaces the remainder estimates of Appendix 2.7.1 `Chapter2/GapP1_Prop212 · GapP1Prop212.retractChart_comp`.
- **"An equivalence class of atlases" needs transitivity, which rests on the locality of compatibility.** The text does not check this, for M-polyfolds or for strong bundles `Chapter2/StrongBundleRestrict · MAtlas.Equiv.trans` `Chapter2/StrongBundleRestrict · StrongBundleAtlas.Equiv.trans`.

## II.22 The 2021 edition: Proposition 2.1.13 and local completeness (§2.1)

The Springer edition (2021) adds to §2.1 a theory of *local completeness*: Proposition 2.1.4, Definition 2.1.5, Remark 2.1.6, Corollary 2.1.7, Proposition 2.1.8, Lemma 2.1.9 and Proposition 2.1.10, pp. 22–25. It also changes one statement, Proposition 2.1.13.

**Proposition 2.1.13 (preprint 2.6), the 2021 statement.** For sc-smooth retractions $r$, $s$ onto the same retract

$O$, a retraction $\rho$ onto $O'$, and $f$ with $f(O) \subseteq O'$, three things hold Chapter2/Book2021 · proposition_2_1_13: - (1) $f \circ r$ is sc-smooth iff $f \circ s$ is; - (2) if $f \circ r$ is sc-smooth, $T(f \circ r) = T(f \circ s)$ on $TO$; - (3) if $f \circ r$ is sc-smooth, $T(f \circ r)$ maps $TO$ into $TO'$.

The 2021 statement of (3) carries the hypothesis "$f \circ r$ sc-smooth", without which $T(f \circ r)$ is not defined. The formal statement of §II.3 had read (3) this way from the start Chapter2/Retracts · proposition_2_1_13_three, so the 2021 theorem is the conjunction of the three earlier ones. The first is used in both directions Chapter2/Retracts · proposition_2_1_13_one Chapter2/Retracts · proposition_2_1_13_two.

**Complete metrics on a subspace.** "$W$ admits a metric $d$ defining its topology for which $K$ is complete" is a metric structure on $W$ whose topology is the given one, with $K$ complete for it Chapter2/LocalTopology · AdmitsCompleteMetricOn. A metric space admits its own metric Chapter2/LocalTopology · admitsCompleteMetricOn_of_isComplete, and such a $W$ is metrizable Chapter2/LocalTopology · AdmitsCompleteMetricOn.metrizableSpace. The property passes to smaller sets Chapter2/LocalTopology · AdmitsCompleteMetricOn.mono and along embeddings: an embedding $j : T' \to T$ pulls the metric back, and a closed subset of a complete set is complete Chapter2/LocalTopology · isComplete_inter_of_isClosed Chapter2/LocalTopology · AdmitsCompleteMetricOn.of_isEmbedding.

**Definition 2.1.5.** $X$ is *locally complete* if every $x$ has open $V \ni x$ and $W$ with $\mathrm{cl}_X(V) \subseteq W$ such that $W$ admits a metric defining its topology for which $\mathrm{cl}_X(V)$ is complete Chapter2/LocalTopology · IsLocallyComplete. The text's $V(x) \subseteq W(x)$ follows from $\mathrm{cl}_X(V) \subseteq W$. The text also assumes $X$ Hausdorff; the formal definition does not, because it follows. - *$W$ arbitrarily small* Chapter2/LocalTopology · IsLocallyComplete.exists_subset. Given a neighbourhood $N$ of $x$, take a metric ball $V'$ around $x$ in $W$ whose closure in $W$ lies in $W \cap N$. Replace $W$ by the open $W \cap \operatorname{int} N$ and $V$ by $V \cap V'$; the metric restricts. - *Regular* Chapter2/LocalTopology · IsLocallyComplete.regularSpace. A neighbourhood $N$ of $x$ contains the closed neighbourhood $\mathrm{cl}_X(V)$ for a $W \subseteq N$; this is where the global condition $\mathrm{cl}_X(V) \subseteq W$ enters. - $T_1$ Chapter2/LocalTopology · IsLocallyComplete.t1Space. Points are closed in the metrizable $W$. For $y \neq x$, a neighbourhood of $x$ missing $y$ comes from $W$ when $y \in W$; otherwise $W$ itself serves. - *Hausdorff* Chapter2/LocalTopology · IsLocallyComplete.t2Space: regular and $T_1$.

Witnesses: a complete metric space ($W = V = X$) Chapter2/LocalTopology · isLocallyComplete_of_completeSpace, $\mathbb{R}$ Chapter2/LocalTopology · isLocallyComplete_real, and a metric space whose points have open $V \subseteq W$ with $\mathrm{cl}(V) \subseteq W$ and $\mathrm{cl}(V)$ complete for the given metric Chapter2/LocalTopology · isLocallyComplete_of_metric.

**Remark 2.1.6.** If $\mathrm{cl}_X(V) \subseteq W$, the closure of $V$ in the subspace $W$ is $\mathrm{cl}_X(V)$ Chapter2/LocalTopology · remark_2_1_6. The modified notion asks only $V \subseteq W$, with completeness of $\mathrm{cl}_W(V)$ Chapter2/LocalTopology · IsLocallyCompleteRel. Local completeness implies it, by the identity just proved Chapter2/LocalTopology · IsLocallyComplete.isLocallyCompleteRel. The remark's "weaker" is formalized as this implication; that the modified notion is *strictly* weaker is §II.31 (gap G.76, closed).

**Proposition 2.1.4.** Let $O = r(U)$ be a retract with the metric of $E_0$. *Closed pieces of $O$:* for $\varepsilon$ small, $O \cap \bar{B}_\varepsilon(x)$ is closed in $E_0$ Chapter2/LocalTopology · ScRetraction.exists_isClosed_retract_inter_closedBall. Choose $\delta$ with $\bar{B}_\delta(x) \cap C \subseteq U$ ($U$ is relatively open in the closed $C$). A limit $v$ of points $v_k \in O \cap \bar{B}_\varepsilon(x)$ lies in $C \cap \bar{B}_\varepsilon(x) \subseteq U$, and $v_k = r(v_k) \to r(v)$ by continuity of $r$ on $U$, so $v = r(v) \in O$. *The neighbourhoods* Chapter2/LocalTopology · IsScRetract.exists_isComplete_closure: $V$ the open $\varepsilon/2$-ball and $W$ the open $\varepsilon$-ball in $O$, inside any given neighbourhood. The closure of $V$ in $O$ lies in $O \cap \bar{B}_{\varepsilon/2}(x)$, closed in the Banach space $E_0$ and therefore complete. This proves the proposition Chapter2/LocalTopology · proposition_2_1_4. The text's set $V'$ with $r(V') \subseteq V'$ is not needed, because the closed balls already give closed pieces of

$O$.

**Corollary 2.1.7.** By Proposition 2.1.4, with the metric $d_O$ restricted to $W$ `Chapter2/LocalTopology · isLocallyComplete_of_metric`, a retract is locally complete `Chapter2/LocalTopology · corollary_2_1_7`. So is $E_0$, the retract of the identity `Chapter2/LocalTopology · isLocallyComplete_univ_retract`.

**Proposition 2.1.8.** *Embeddings with locally closed image* `Chapter2/LocalTopology · IsLocallyComplete.of_isEmbedding`. Let $j : Y \to X$ be an embedding with $j(Y) = U \cap Z$, $U$ open and $Z$ closed. For $y \in Y$, take $W$, $V$ around $j(y)$ with $W \subseteq U$. The preimages $j^{-1}(W)$ and $j^{-1}(V)$ serve for $y$, for two reasons. First, the closure of $j^{-1}(V)$ corresponds to $\mathrm{cl}_X(V) \cap j(Y)$, and $j(Y) \cap W = Z \cap W$ is closed in $W$. Second, the embedding $j^{-1}(W) \to W$ has closed image, so the metric of $W$ pulls back with the completeness of the closure. Parts (1)–(3) follow: - (1) a locally closed subspace `Chapter2/LocalTopology · proposition_2_1_8_1`; - (2) a homeomorphic space, the image being everything `Chapter2/LocalTopology · proposition_2_1_8_2`; - (3) an open subspace `Chapter2/LocalTopology · proposition_2_1_8_3`.

*The open-cover statement* `Chapter2/LocalTopology · proposition_2_1_8_4`. Let $y \in Y_i$, with $U \ni y$ open, $U \subseteq Y_i$ and $\mathrm{cl}_Y(U) \subseteq Y_i$ (condition (b)), and $W'$, $V'$ for $y$ in the locally complete $Y_i$. Put $V = V' \cap U$ and $W = W'$, both open in $Y$ because $Y_i$ is. Then $\mathrm{cl}_Y(V) \subseteq \mathrm{cl}_Y(U) \subseteq Y_i$ by (b), so $\mathrm{cl}_Y(V)$ is the closure of $V$ in $Y_i$ and lies in $W$. The metric of $W' \subseteq Y_i$ is transported to $W \subseteq Y$, the identity being a homeomorphism. No Hausdorff hypothesis is used anywhere.

**Lemma 2.1.9 and Proposition 2.1.10.** *First countable* `Chapter2/LocalTopology · lemma_2_1_9`: the neighbourhood filter of $x$ in $X$ is the image of that of $x$ in the open, metrizable $W$, and the latter is countably generated. *Completely regular* `Chapter2/LocalTopology · IsLocallyComplete.completelyRegularSpace`. For $x$ outside a closed $A$, take $W \subseteq X \setminus A$ (arbitrarily small $W$). In the metrizable $W$, a continuous $g : W \to [0,1]$ has $g(x) = 0$ and $g = 1$ off $V$. Extended by 1 on $X \setminus \mathrm{cl}_X(V)$, it is continuous on the two open sets $W$ and $X \setminus \mathrm{cl}_X(V)$, which cover $X$ because $\mathrm{cl}_X(V) \subseteq W$. Mathlib's complete regularity uses this normalization ($f(x) = 0$, $f = 1$ on $A$); the text's $\beta$ is $1 - f$. With Hausdorff, the space is $T_{3\frac{1}{2}}$ `Chapter2/LocalTopology · proposition_2_1_10`. The text argues continuity with sequences; the formal proof works with filters and open covers, so first countability is not needed for it.

## II.23 The 2021 edition: local path-connectedness and local contractibility (§2.2)

§2.2 of the 2021 edition, pp. 30–33, is new.

**Lemma 2.2.1.** Let $U$ be relatively open in a partial quadrant $C$ and connected. For $k \in \mathbb{N}$, $U_k = U \cap E_k$ carries the topology of $E_k$ `Chapter1/ScSmooth · ScBanach.lvlSet`. For $k = \infty$ the text does not say which topology $U_\infty$ carries. The formalization takes the topology of all the norms: $U_\infty$ is the set of compatible sequences $(v_m) \in \prod_m E_m$ with $v_0 \in U$ `Chapter2/LocalTopology · ScBanach.lvlSetInfty`, with the product topology. A smooth point $x$ gives its sequence of lifts `Chapter2/LocalTopology · ScBanach.smoothSeq` `Chapter2/LocalTopology · ScBanach.toBase_smoothSeq` `Chapter2/LocalTopology · ScBanach.smoothSeq_mem` `Chapter2/LocalTopology · ScBanach.smoothSeq_zero`. The image of $U_\infty$ in $E_0$ is $U \cap E_\infty$ `Chapter2/LocalTopology · ScBanach.image_lvlSetInfty`.

The proof uses three facts: - two points of $U_\infty$ whose base points span a segment inside $U$ are joined in $U_\infty$ by the segment, which is continuous into every level `Chapter2/LocalTopology · ScBanach.joinedIn_lvlSetInfty_of_segment`; - smooth points are dense in every level $C_m$

Chapter2/LocalTopology · IsPartialQuadrant.exists_smooth_near, the text's "$U_\infty$ is dense in every $U_k$"; - every $z \in U$ has a ball with $B_\varepsilon(z) \cap C \subseteq U$, a convex set Chapter2/LocalTopology · IsRelOpenInQuadrant.exists_ball_inter_subset.

*The case $k = \infty$* Chapter2/LocalTopology · lemma_2_2_1_infty. Fix a point $a$ of $U_\infty$, and say that $z \in U$ is good if near $z$ every point of $U_\infty$ is joined to $a$. Joining propagates inside a convex $B_\varepsilon(z) \cap C \subseteq U$ by segments, and smooth points are dense, so the set of good points is open and closed in $U$. It is nonempty, so it is all of $U$ because $U$ is connected. *The case $k \in \mathbb{N}$* Chapter2/LocalTopology · lemma_2_2_1: project the path-connected $U_\infty$ to $U_k$ and join points of $U_k$ to nearby smooth points by segments. The text's proof replaces a path in $U_0$ by a polygon through smooth points, using compactness of the path; the clopen argument avoids both the path and the polygon.

**Lemma 2.2.2.** For $x \in O$ and $V = V' \cap U$ with $V' \ni x$ open in $E_0$, the set $W = V \cap r^{-1}(V)$ has these properties Chapter2/LocalTopology · lemma_2_2_2: - it is relatively open in $C$, by continuity of $r$ on $U$; - it contains $x$, since $r(x) = x$; - $W \cap O = V \cap O$, because points of $O$ are fixed; - $r(W) \subseteq W$, since $r(r(y)) = r(y) \in V$; - $r(W) = O \cap V$.

**Definition 2.2.3 and Theorem 2.2.4.** Mathlib's local path-connectedness is the text's: every open neighbourhood contains a path-connected open neighbourhood Chapter2/LocalTopology · definition_2_2_3. A retract is locally path-connected Chapter2/LocalTopology · theorem_2_2_4. This was proved earlier for Chapter 7 (2017, p. 183, recalls it without proof) Chapter7/LocConnected · IsScRetract.locallyPathConnectedSpace: points $o, p$ of $O$ near each other are joined in $O$ by $\tau \mapsto r(o + \tau(p - o))$, the segment lying in a convex piece of $U$. The text goes through Lemma 2.2.2 and the component of $W(x)$ containing $x$.

**Definition 2.2.5 and Proposition 2.2.6.** *Almost locally contractible* Chapter2/LocalTopology · IsAlmostLocallyContractible: for $z$ and open $V \ni z$, an open $W$ with $z \in W \subseteq V$ and a continuous $\Phi : [0,1] \times W \to Z$ with values in $V$, $\Phi(0, \cdot) = \mathrm{id}$, $\Phi(t, z) = z$ and $\Phi(1, \cdot) \equiv z$. For a retract Chapter2/LocalTopology · proposition_2_2_6, choose $\varepsilon$ with $B_\varepsilon(z) \cap C \subseteq U$ and $r(B_\varepsilon(z) \cap C) \subseteq V$, by continuity of $r$ at $z = r(z)$. Then $\Phi(t, y) = r((1-t)y + tz)$ on $W = B_\varepsilon(z) \cap O$ is the text's homotopy: the segment stays in the convex $B_\varepsilon(z) \cap C$, and $r$ fixes $y$ and $z$. The model $E_0$ itself is a witness Chapter2/LocalTopology · isAlmostLocallyContractible_univ.

## II.24 The 2021 edition: paracompactness and M-polyfolds (§2.3, Appendix 2.7.2)

**The footnote to Definition 2.3.4 (preprint 2.13).** The 2021 footnote says that paracompactness does not include the Hausdorff property; it asks only that every open cover have a locally finite open refinement. In the formalization, this paracompactness* is defined with covers as sets of open sets Chapter2/LocalTopology · IsLocallyFiniteSetFamily Chapter2/LocalTopology · IsParacompactStar and coincides with Mathlib's paracompactness, which does not include the Hausdorff property either Chapter2/LocalTopology · isParacompactStar_iff. So "Hausdorff paracompact" *is exactly the hypothesis of the M-polyfold structure (§II.5)* Chapter2/LocalTopology · definition_2_3_4, *and $\mathbb{R}$ is paracompact* Chapter2/LocalTopology · isParacompactStar_real. The footnote changes nothing for M-polyfolds, whose Hausdorff property is required separately.

*The convention of Appendix 2.7.2.* The appendix (p. 72; 2017 Appendix 2.6.2, p. 51) states the convention with a locally finite open *subcovering* Chapter2/LocalTopology · IsParacompactBySubcover. Read literally, it excludes $\mathbb{R}$ Chapter2/LocalTopology · not_isParacompactBySubcover_real. The cover by the intervals $(-n-1, n+1)$

has no locally finite subcover: a subcover is infinite, since $\mathbb{R}$ is unbounded, and the infinitely many intervals of it all contain $0$. So the wording does not hold as we read it, and we read it as "refinement", as in the footnote (possible misprint).

**Lemmas 2.7.2 and 2.3.5.** In a regular space every open $U \ni x$ contains an open $V \ni x$ with $\mathrm{cl}(V) \subseteq U$, because closed neighbourhoods form a basis `Chapter2/LocalTopology · lemma_2_7_2`. The text states this for regular Hausdorff spaces. A Hausdorff paracompact space is normal, hence regular, and the shrinking can be done inside any given neighbourhood `Chapter2/LocalTopology · exists_isOpen_closure_subset_of_paracompact`. This is Lemma 2.3.5 for M-polyfolds, $V(x)$ "arbitrarily small" `Chapter2/LocalTopology · lemma_2_3_5`.

**Proposition 2.3.6.** A chart domain is homeomorphic to a retract, hence locally complete by Corollary 2.1.7 and Proposition 2.1.8 (2) `Chapter2/LocalTopology · MChart.isLocallyComplete_source`. The chart domains form an open cover of the M-polyfold $X$, and condition (b) of the open-cover part of Proposition 2.1.8 holds by Lemma 2.3.5. So $X$ is locally complete `Chapter2/LocalTopology · MPolyfold.isLocallyComplete`, with $W(x)$ arbitrarily small `Chapter2/LocalTopology · proposition_2_3_6`. The text argues directly in a chart, with Proposition 2.1.4 and Lemma 2.3.5. "The topology on level 0" in the text is the topology of $X$.

**Proposition 2.3.7.** Every M-polyfold is locally path-connected `Chapter2/LocalTopology · proposition_2_3_7`. As for Theorem 2.2.4, this was already proved for Chapter 7 `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace`, by transporting the local path-connectedness of retracts through charts.

## II.25 The 2021 edition: tame sub-M-polyfolds (Definitions 2.5.18, 2.5.19)

**Definition 2.5.18.** The structure induced on a sub-M-polyfold $Y$ (§II.10) is *tame* if the M-polyfold $Y$, paracompact as a subspace, is tame `Chapter2/TameBoundary · MPolyfold.HasTameInducedStructure`.

**Definition 2.5.19.** Condition (3) asks for an sc-complement of $T_yY$ in $T_yX$ inside the reduced tangent space $T^R_yX$. Here (3) is stated in a chart; with the linear structure of $T_yX$ (gap G.7, closed, §II.39) it is restated in $T_yX$ in §II.47. For retracts $Q \subseteq O$ in $C$, a presentation $T$ of $C$, and a level-1 point $z$ of $Q$, there must be an sc-subspace $A$ of $E$ with $A \subseteq T_zO \cap E_z$ and $(T_zO)_m = (T_zQ)_m \oplus A_m$ on every level `Chapter2/TameBoundary · MPolyfold.HasReducedComplementAt`. ($E_z$ is the subspace of §II.12 and $T_zO \cap E_z$ the reduced tangent space of Definition 2.4.7 (preprint 2.27).) This is literally Chapter 4's general-position condition, restated so that Chapter 2 does not import Chapter 4. $A = 0$ serves for $Q = O$ `Chapter2/TameBoundary · MPolyfold.hasReducedComplementAt_self`.

The data at a point consist of sub-M-polyfold data $(V, r)$, so that $r(V) = Y \cap V$, together with two conditions `Chapter2/TameBoundary · MPolyfold.TameSubData`: - $d_X(r(x)) = d_X(x)$ on $V$; - at every smooth $y \in Y \cap V$, the chart form of (3) for some chart around $y$, with $Q$ the chart image of $Y$ near $y$.

$Y$ is a tame sub-M-polyfold of the tame $X$ if every point of $Y$ has such data `Chapter2/TameBoundary · MPolyfold.IsTameSubMPolyfold`; it is then a sub-M-polyfold `Chapter2/TameBoundary · MPolyfold.IsTameSubMPolyfold.isSubMPolyfold`. Satisfiability: in a tame $X$, $Y = X$ with $r = \mathrm{id}$ has tame induced structure, the sub-charts built on tame charts having the same tame models `Chapter2/TameBoundary · MPolyfold.hasTameInducedStructure_univ`. It is a tame sub-M-polyfold, with $Q = O$ and $A = 0$ in every chart `Chapter2/TameBoundary · MPolyfold.isTameSubMPolyfold_univ`. The independence of (3) of the chart, the remark that a tame sub-M-polyfold is tame, and the two examples in $[0, \infty)^2 \times \mathbb{R}$ are §II.35 (gap G.77, closed).

## II.26 The 2021 edition: faces of open sets and the corner axioms (§2.7.4, Propositions 2.7.5, 2.7.7)

Appendix §2.7.4 is new in the 2021 edition. It builds from a tame M-polyfold $X$ a space $bX$ of "points with a chosen face" and a map $B : bX \to X$. The formalization proves all of it for an abstract $d : X \to \mathbb{N}$ satisfying *corner axioms*, then specializes to the degeneracy index of a tame M-polyfold. The same proofs serve for the object spaces of tame ep-groupoids in 2021 §7.6.5.

**Faces of open sets.** For an open $V$ and $p \in V$ with $d(p) = 1$, the face of $V$ through $p$ is the closure in $V$ of the connected component of $V \cap \{d = 1\}$ through $p$ `Chapter2/TameBoundary · TameBoundary.faceAt` `Chapter2/TameBoundary · TameBoundary.IsFaceIn`. $F_X(y, V)$ is the set of faces of $V$ through $y$ `Chapter2/TameBoundary · TameBoundary.facesThrough`. For an M-polyfold the faces of $X$ itself are those of Definition 2.5.10 (preprint 2.39) `Chapter2/TameBoundary · MPolyfold.isFaceIn_univ_iff`. The text's faces of $V$ are those of $V$ with its induced structure, whose degeneracy index is $d_X|V$ (2021 Proposition 2.4.4), so defining them from $d_X$ is the same thing.

**The corner axioms** `Chapter2/TameBoundary · TameBoundary.IsCornerIndex`. First, $d$ is upper semicontinuous (Lemma 2.4.3 (preprint 2.23)). Second, every $x$ has an open $D$ such that every $y \in D$ has *local pieces* in arbitrarily small neighbourhoods $N$: an open $B \ni y$ in $N$ and $d(y)$ preconnected sets $\ell_i \subseteq N \cap \{d = 1\}$ accumulating at $y$ and covering $B \cap \{d = 1\}$ `Chapter2/TameBoundary · TameBoundary.LocalPieces`. Third, the pieces are separated in $D$: no preconnected subset of $D \cap \{d = 1\}$ meets two of them. In a tame chart the pieces are the images under $r$ of the sets $\{z \in C : |z - \varphi(y)| < \varepsilon,\ I(z) = \{i\}\}$, $i \in I(\varphi(y))$, from the local face structure of §II.15 `Chapter2/TameBoundary · MPolyfold.exists_localPieces_chart`. So the degeneracy index of a tame M-polyfold satisfies the axioms `Chapter2/TameBoundary · MPolyfold.isCornerIndex_degIndex`, and $d \equiv 0$ does trivially `Chapter2/TameBoundary · TameBoundary.isCornerIndex_zero`. Faces of nested opens are compatible: a face of $V$ meets an open $U \subseteq V$ in a union of faces of $U$ `Chapter2/TameBoundary · TameBoundary.faceAt_inter_eq`. A face $\theta$ of $U$ lies in a unique face of $V$, the union $\theta'$ of the faces of $V$ containing it `Chapter2/TameBoundary · TameBoundary.faceExt` `Chapter2/TameBoundary · TameBoundary.faceExt_unique`. Every face of $V$ through a point of $U$ comes from a face of $U$ `Chapter2/TameBoundary · TameBoundary.exists_faceExt_eq`.

**Proposition 2.7.5.** *Face-structured neighbourhoods exist* `Chapter2/TameBoundary · TameBoundary.exists_nbhd_forall_isFaceStructuredOpen`. Take $D$ from the axioms. For an open $V \subseteq D$ and $y \in V$, the local pieces of $y$ inside $V$ lie on faces of $V$, and every face through $y$ contains a piece. Distinct pieces lie on distinct faces: a face through two pieces would give a preconnected subset of $D \cap \{d = 1\}$ meeting both, which separation forbids. So $y$ lies on exactly $d(y)$ faces of $V$. Then $V = D \cap \{d \le d(x)\}$, open by semicontinuity, gives (1) and (2). For (3), on any open $V \ni x$, take the set $B$ of the local pieces of $x$ inside $V$: a face of $V$ through a point of $B$ contains one of the pieces of $x$, which accumulate at $x$, and faces are closed in $V$ `Chapter2/TameBoundary · TameBoundary.exists_nbhd_faces_mem`. Together these prove the proposition `Chapter2/TameBoundary · TameBoundary.proposition_2_7_5` `Chapter2/TameBoundary · MPolyfold.proposition_2_7_5`.

**Definition 2.7.6 and Proposition 2.7.7.** The definitions: - an open $V$ is *face-structured* if every $y \in V$ lies on exactly $d(y)$ faces of $V$ `Chapter2/TameBoundary · TameBoundary.IsFaceStructuredOpen` `Chapter2/TameBoundary · TameBoundary.faceStructuredOpens`; - $N_X$ is the set of the nonempty face-structured $V$ with $d|V$ bounded `Chapter2/TameBoundary · TameBoundary.faceStructuredNbhds`; - $N_{X,x}$ is the set of those $V \in N_X$ with $x \in V$ and $d \le d(x)$ on $V$ `Chapter2/TameBoundary · TameBoundary.faceStructuredNbhdsAt`.

$N_X = \bigcup_x N_{X,x}$, because a $V \in N_X$ contains a point where the bounded $d|V$ is maximal `Chapter2/TameBoundary · TameBoundary.faceStructuredNbhds_eq_iUnion`. An open subset of a face-structured set is face-

structured: its faces through a point are the traces of those of the larger set, by the local pieces Chapter2/TameBoundary · TameBoundary.IsFaceStructuredOpen.mono. Hence Proposition 2.7.7: - $N_{X,x}$ is nonempty (Proposition 2.7.5) Chapter2/TameBoundary · TameBoundary.faceStructuredNbhdsAt_nonempty, so $N_X$ covers $X$ Chapter2/TameBoundary · TameBoundary.sUnion_faceStructuredNbhds; - nonempty open subsets of members of $N_X$ are in $N_X$ Chapter2/TameBoundary · TameBoundary.mem_faceStructuredNbhds_of_subset; - nonempty intersections of members of $N_X$ are in $N_X$ Chapter2/TameBoundary · TameBoundary.inter_mem_faceStructuredNbhds; - $N_{X,x}$ is closed under intersections Chapter2/TameBoundary · TameBoundary.inter_mem_faceStructuredNbhdsAt.

For tame M-polyfolds this is Chapter2/TameBoundary · MPolyfold.proposition_2_7_7.

*Quadrants.* In $[0,\infty)^n \oplus W$, $d$ counts the vanishing half-line coordinates Chapter2/TameBoundaryExample · TameBoundaryExample.dq_eq. $\{d = 1\}$ is the union of the strata $\{p_k = 0,\ p_l > 0\ (l \neq k)\}$ Chapter2/TameBoundaryExample · TameBoundaryExample.part Chapter2/TameBoundaryExample · TameBoundaryExample.dq_eq_one_iff. A preconnected subset of $\{d = 1\}$ lies in one stratum Chapter2/TameBoundaryExample · TameBoundaryExample.subset_part, so a face through a point of the $k$-th stratum lies in $\{p_k = 0\}$ Chapter2/TameBoundaryExample · TameBoundaryExample.faceAt_subset_coord_eq_zero. A point of an open $V$ with $y_k = 0$ lies on the face of $V$ through a nearby point of the $k$-th stratum Chapter2/TameBoundaryExample · TameBoundaryExample.exists_part_mem_faceAt. Hence every open subset of a quadrant is face-structured Chapter2/TameBoundaryExample · TameBoundaryExample.isFaceStructuredOpen_quadrant.

## II.27 The 2021 edition: $L_X$, $F_X$, $bX$ and $U(x,V,\theta)$ (Definitions 2.7.8–2.7.11)

**Definition 2.7.8.** There is a morphism $(x,V) \to (y,W)$ iff $V \subseteq W$, $x$ lies on $d(x)$ faces of $W$, and each of them contains $y$ Chapter2/TameBoundary · TameBoundary.IsMor. The objects are the pairs $(x,V)$ with $d(x) \geq 1$ and $V \in N_{X,x}$ Chapter2/TameBoundary · TameBoundary.LObj. There is at most one morphism between two objects, so $L_X$ is a preorder, a thin category. *Composition* Chapter2/TameBoundary · TameBoundary.IsMor.trans follows the text's argument. Given $(x,V) \to (y,W) \to (z,Q)$, every face $\gamma$ of $Q$ through $x$ extends a face $\theta$ of $W$ through $x$ Chapter2/TameBoundary · TameBoundary.exists_faceExt_eq. That face contains $y$, so $y \in \gamma$, and then $z \in \gamma$.

**Definition 2.7.9.** The functor sends $(x,V)$ to $F_X(x,V)$ and a morphism to $\theta \mapsto \theta'$, the face of $W$ containing $\theta$ Chapter2/TameBoundary · TameBoundary.boundaryFunctor Chapter2/TameBoundary · MPolyfold.LocalBoundaryCategory Chapter2/TameBoundary · MPolyfold.boundaryStructureFunctor. It is the text's "face $\theta'$ of $W$ through $x$ and $y$ with $\theta \cap \theta'$ open in $\theta$" Chapter2/TameBoundary · TameBoundary.faceExt_unique. Identities and composition hold because extending twice is extending once. $F_X(\varphi)$ is injective, the text's injection (2.43) Chapter2/TameBoundary · TameBoundary.boundaryFunctor_map_injective. It is bijective when the point does not change, since both sets have $d(x)$ elements Chapter2/TameBoundary · TameBoundary.boundaryFunctor_map_bijective.

**Definition 2.7.10.** An admissible triple is an object $(x,V)$ with a face $\theta \in F_X(x,V)$ Chapter2/TameBoundary · TameBoundary.BTriple. $(x,V,\theta) \sim (x',V',\theta')$ iff $x = x'$ and some $V'' \in N_{X,x}$, $V'' \subseteq V \cap V'$, has a face through $x$ sent to $\theta$ and to $\theta'$ Chapter2/TameBoundary · TameBoundary.BRel. Reflexivity, symmetry and transitivity hold Chapter2/TameBoundary · TameBoundary.BRel.refl Chapter2/TameBoundary · TameBoundary.BRel.symm Chapter2/TameBoundary · TameBoundary.BRel.trans; for transitivity, intersect the two $V''$ and use the bijectivity at a fixed point. $bX$ is the quotient Chapter2/TameBoundary · TameBoundary.bSetoid Chapter2/TameBoundary · TameBoundary.BSpace Chapter2/TameBoundary · MPolyfold.TameBoundarySpace, and $B[x,V,\theta] = x$

Chapter2/TameBoundary · TameBoundary.bMap Chapter2/TameBoundary · MPolyfold.tameBoundaryMap. The image of $B$ is $\{d \geq 1\} = \partial X$ Chapter2/TameBoundary · TameBoundary.range_bMap Chapter2/TameBoundary · MPolyfold.range_tameBoundaryMap. *Fibres.* $B^{-1}(x)$ has exactly $d(x)$ points Chapter2/TameBoundary · TameBoundary.encard_bMap_preimage Chapter2/TameBoundary · MPolyfold.encard_tameBoundaryMap_preimage: for a fixed $V \in N_{X,x}$, the classes over $x$ correspond to the $d(x)$ faces of $V$ through $x$, because every triple over $x$ is equivalent to one with this $V$ and distinct faces give distinct classes. This is the set-level content of Theorem 2.7.13 (4).

**Definition 2.7.11.** $U(x, V, \theta)$ is the set of classes with a representative $(y, V', \theta')$ such that $y \in \theta$, there is a morphism $(y, V') \to (x, V)$, and it sends $\theta'$ to $\theta$ Chapter2/TameBoundary · TameBoundary.uSet. The formalization reads membership as "for some representative". $\Phi_{(x,V,\theta)}$ is $B$ restricted to it, with values in $\theta$ Chapter2/TameBoundary · TameBoundary.phi. It is injective: two classes over the same $y$ in $U(x, V, \theta)$ are sent to the same face $\theta$ of $V$, and $F_X$ of the morphism to $(x, V)$ is injective Chapter2/TameBoundary · TameBoundary.phi_injective.

*The image.* Put $V^x = \{y \in V : \text{every face of } V \text{ through } y \text{ contains } x\}$ Chapter2/TameBoundary · TameBoundary.reach, an open set Chapter2/TameBoundary · TameBoundary.isOpen_reach. For $y \in V$ there is a morphism $(y, V') \to (x, V)$ exactly when $y \in V^x$. So the image of $\Phi_{(x,V,\theta)}$ is $\theta \cap V^x$ Chapter2/TameBoundary · TameBoundary.bMap_image_uSet. If $V$ is centred at $x$, i.e. every face of $V$ contains $x$ Chapter2/TameBoundary · TameBoundary.IsCenteredAt, then $V^x = V$ and $\Phi$ is a bijection onto $\theta$ Chapter2/TameBoundary · TameBoundary.phi_bijective_of_isCenteredAt. $V^x$ is itself centred at $x$ Chapter2/TameBoundary · TameBoundary.isCenteredAt_reach and lies in $N_{X,x}$ Chapter2/TameBoundary · TameBoundary.reach_mem_faceStructuredNbhdsAt. So the centred neighbourhoods are cofinal, which is what the text's "take $V$ sufficiently small" (p. 85) provides. The text calls $\Phi_{(x,V,\theta)}$ a canonical bijection for every triple (2.44); §II.28 shows that some such restriction is needed.

## II.28 The 2021 edition: (2.44) and an example in $[0, \infty)^3$

Let $X = [0, \infty)^3 \oplus W$, $W$ any sc-Banach space; the trivial $W$ gives the setting $[0, \infty)^3$. Its degeneracy index satisfies the corner axioms Chapter2/TameBoundaryExample · TameBoundaryExample.dq Chapter2/TameBoundaryExample · TameBoundaryExample.isCornerIndex_dq. Take: - $x = (0, 0, 1)$, a corner point with $d(x) = 2$; - $V = \{p_0 < 1/2,\ 1/2 < p_1 + p_2 < 3/2\}$, a convex open slab around the segment from $x$ to $(0, 1, 0)$ Chapter2/TameBoundaryExample · TameBoundaryExample.slab.

On $V$, $d \leq 2$. $V$ is face-structured, like every open subset of a quadrant (§II.26), so $V \in N_{X,x}$ Chapter2/TameBoundaryExample · TameBoundaryExample.slab_mem_faceStructuredNbhdsAt. The open segment $\{(0, t, 1 - t)\}$ lies in $\{d = 1\}$, and the face $\theta$ of $V$ through it contains its endpoints $x$ and $(0, 1, 0)$ Chapter2/TameBoundaryExample · TameBoundaryExample.slabFace. $(x, V, \theta)$ is an admissible triple Chapter2/TameBoundaryExample · TameBoundaryExample.counterTriple. The point $(0, 1, 0)$ also lies on the face of $V$ through the nearby points of the stratum $\{p_2 = 0\}$. That face lies in $\{p_2 = 0\}$, so it misses $x$ Chapter2/TameBoundaryExample · TameBoundaryExample.exists_face_not_mem. Hence $(0, 1, 0) \notin V^x$, and $(0, 1, 0) \in \theta$ is not in the image $\theta \cap V^x$ of $\Phi_{(x,V,\theta)}$ Chapter2/TameBoundaryExample · TameBoundaryExample.pt_010_not_mem_image. So $\Phi_{(x,V,\theta)}$ is not onto $\theta$ Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective, and not every $\Phi$ is a bijection Chapter2/TameBoundaryExample · TameBoundaryExample.not_forall_bijective_phi. As we read it, (2.44) does not hold for every $V \in N_{X,x}$. It holds for centred $V$ (§II.27). The example does not depend on reading membership in $U(x, V, \theta)$ as "for some representative", because $\Phi$ misses $(0, 1, 0)$ already on the level of points.

## II.29 The 2021 edition: the topology of $bX$ (Lemma 2.7.15, Propositions 2.7.12, 2.7.16, Theorem 2.7.13 (1))

**The topology.** $\mathcal{U} \subseteq bX$ is in $T_{bX}$ if every class $p \in \mathcal{U}$ has a representative $(x, V, \theta)$ and a set $O \ni p$, $O \subseteq \mathcal{U} \cap U(x, V, \theta)$, with $B(O)$ open in $\theta$ `Chapter2/TameBoundaryTopology · TameBoundary.IsBOpen`. We read $O$ as containing $p$: otherwise $O = \emptyset$ would make every set open. Two tools: - a triple can be shrunk into any open $N \ni x$, keeping its class and shrinking its $U$-set `Chapter2/TameBoundaryTopology · TameBoundary.BTriple.exists_shrink`; - equivalent triples have a common refinement whose $U$-set lies in both `Chapter2/TameBoundaryTopology · TameBoundary.BRel.exists_common`.

If two triples with the same $V$ have meeting $U$-sets, their faces agree `Chapter2/TameBoundaryTopology · TameBoundary.eq_of_mem_uSet`. These give the alternative characterization: $\mathcal{U} \in T_{bX}$ iff every $p \in \mathcal{U}$ has a representative with $U(x, V, \theta) \subseteq \mathcal{U}$ `Chapter2/TameBoundaryTopology · TameBoundary.isBOpen_iff`. In that form the axioms of a topology are immediate. For intersections, two representatives of $p$ have a common refinement with $U$-set inside both `Chapter2/TameBoundaryTopology · TameBoundary.bTopology` `Chapter2/TameBoundaryTopology · TameBoundary.isOpen_iff_isBOpen` `Chapter2/TameBoundaryTopology · TameBoundary.isOpen_iff_forall_uSet`. This proves Lemma 2.7.15 `Chapter2/TameBoundaryTopology · TameBoundary.lemma_2_7_15` `Chapter2/TameBoundaryTopology · MPolyfold.lemma_2_7_15`. The text's proof of the intersection axiom uses the sc-smoothness of transition maps; this one uses only faces.

**Proposition 2.7.12, topological part.** The sets $U(x, V, \theta)$ are open `Chapter2/TameBoundaryTopology · TameBoundary.isOpen_uSet` and form a basis `Chapter2/TameBoundaryTopology · TameBoundary.isTopologicalBasis_uSet`, and $B$ is continuous `Chapter2/TameBoundaryTopology · TameBoundary.continuous_bMap`. *The germ of a face:* if $\theta$ is a face of $U \subseteq V$ through $y$, then near $y$ the extended face $\theta'$ of $V$ coincides with $\theta$ `Chapter2/TameBoundaryTopology · TameBoundary.exists_nbhd_faceExt_inter_subset`. Hence $\Phi$ is open: $B(\mathcal{U} \cap U(x, V, \theta))$ is open in $\theta$ for open $\mathcal{U}$ `Chapter2/TameBoundaryTopology · TameBoundary.exists_isOpen_bMap_image`. With injectivity and continuity, every $\Phi_{(x,V,\theta)}$ is an open embedding into $\theta$ `Chapter2/TameBoundaryTopology · TameBoundary.isOpenEmbedding_phi`, and a homeomorphism onto $\theta$ for $V$ centred at $x$ `Chapter2/TameBoundaryTopology · TameBoundary.phiHomeomorph`. For two triples, the image of $U(x, V, \theta) \cap U(x', V', \theta')$ is open in $\theta$ and in $\theta'$, and the transition map is the identity on points of $X$, both maps being restrictions of $B$ `Chapter2/TameBoundaryTopology · TameBoundary.proposition_2_7_12` `Chapter2/TameBoundaryTopology · MPolyfold.proposition_2_7_12`. Not part of this section: the sc-diffeomorphism half of Proposition 2.7.12. For centred representatives it is proved with the M-polyfold structure of $bX$ (§II.36), for every representative in §II.37 (gap G.78, closed).

**Proposition 2.7.16.** *Hausdorff.* Two classes over different points of the Hausdorff $X$ are separated by preimages under the continuous $B$. Two classes over the same $x$ have representatives with a common centred $V$ and different faces, and their $U$-sets are disjoint (Hausdorff instance in `Chapter2/TameBoundaryTopology.lean`). *Metrizable* `Chapter2/TameBoundaryTopology · TameBoundary.proposition_2_7_16`. For $x \in \partial X$ and a centred $W \in N_{X,x}$, write $(x, W, \theta)$ for the triple of a face $\theta$ of $W$ through $x$ `Chapter2/TameBoundaryTopology · TameBoundary.tripleOf`. The sets $U(x, W, \theta)$ cover $B^{-1}(W)$ `Chapter2/TameBoundaryTopology · TameBoundary.exists_mem_uSet_tripleOf`; they are open, pairwise disjoint and metrizable through $\Phi$. So $B^{-1}(W)$ embeds into $X \times \mathbb{R}^{F_X(x,W)}$ by $B$ and the indicators of these clopen pieces, and is metrizable `Chapter2/TameBoundaryTopology · TameBoundary.metrizableSpace_preimage`. The pull-back form of Smirnov's theorem along $B$ (§II.7) `Chapter2/Metrizable · metrizableSpace_of_pullback` then metrizes $bX$. For tame M-polyfolds, $bX$ is Hausdorff and metrizable `Chapter2/TameBoundaryTopology · MPolyfold.proposition_2_7_16`. The text's route instead builds a locally finite refinement from one of the cover $(V_x)$ of $\partial X$, proves

paracompactness, and applies Nagata–Smirnov.

**Theorem 2.7.13 (1) and the topological characterization.** A topology in which every $U(x, V, \theta)$ is open, and every $\Phi$ maps open sets to relatively open subsets of $\theta$, is $T_{bX}$ `Chapter2/TameBoundaryTopology · TameBoundary.eq_of_isOpen_uSet`. With the results above this gives the topological part of the theorem `Chapter2/TameBoundaryTopology · TameBoundary.theorem_2_7_13_topology` `Chapter2/TameBoundaryTopology · MPolyfold.theorem_2_7_13`: - $bX$ is metrizable; - the $U$-sets are open and the $\Phi$ are open embeddings; - $T_{bX}$ is unique with these properties; - $B$ is continuous, with image $\partial X$ and fibres of $d_X(x)$ points.

The tame boundary of Definition 2.7.14, as a continuous map, is $B$ `Chapter2/TameBoundaryTopology · MPolyfold.tameBoundary`. Parts (2)–(4) (the tame M-polyfold structure, $B$ sc-smooth, $d_{bX}$) and the iterated boundaries are §II.36. *Witness.* The quadrant $[0, \infty)^n \oplus W$ is a tame M-polyfold `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold` `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_isTame`. Its corner has exactly $n$ preimages in $bX$ `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_encard_preimage`, so $bX \neq \emptyset$ for $n \geq 1$ `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_tameBoundarySpace_nonempty`. It is a tame sub-M-polyfold of itself `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_univ`.

## II.30 What the 2021 additions taught us

- **Local completeness needs no Hausdorff hypothesis.** The global condition $\mathrm{cl}_X(V) \subseteq W$ gives regularity, and the metrizable $W$ gives $T_1$ `Chapter2/LocalTopology · IsLocallyComplete.t2Space`. Proposition 2.1.8, Lemma 2.1.9 and complete regularity hold without it.
- **One permanence lemma.** Parts (1)–(3) of Proposition 2.1.8 are one statement about embeddings with locally closed image `Chapter2/LocalTopology · IsLocallyComplete.of_isEmbedding`. Proposition 2.3.6 is then the open-cover part applied to chart domains.
- **"Subcovering" in Appendix 2.7.2 does not hold as we read it.** With it, $\mathbb{R}$ would not be paracompact `Chapter2/LocalTopology · not_isParacompactBySubcover_real`. The footnote to Definition 2.3.4 has "refinement", which is Mathlib's notion `Chapter2/LocalTopology · isParacompactStar_iff` (possible misprint).
- **$U_\infty$ needs a topology.** Lemma 2.2.1 for $k = \infty$ is proved for the topology of all the norms `Chapter2/LocalTopology · lemma_2_2_1_infty`, by a clopen argument rather than polygons.
- **§2.7.4 is combinatorics of $d$.** Every statement of the appendix about faces, $L_X$, $bX$ and its topology follows from the upper semicontinuity of $d$ and separated local pieces `Chapter2/TameBoundary · TameBoundary.IsCornerIndex`, which is why it applies to ep-groupoids as well.
- **(2.44) needs $V$ centred.** $\Phi_{(x,V,\theta)}$ has image $\theta \cap V^x$ `Chapter2/TameBoundary · TameBoundary.bMap_image_uSet`, and in $[0, \infty)^3$ it is not onto for a thin slab `Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective`. Centred neighbourhoods are cofinal `Chapter2/TameBoundary · TameBoundary.reach_mem_faceStructuredNbhdsAt`.
- **The topology of $bX$ needs no sc-calculus.** Lemma 2.7.15 and Proposition 2.7.16 are proved from faces and the pull-back form of Smirnov's theorem `Chapter2/TameBoundaryTopology · TameBoundary.metrizableSpace_preimage`. The text uses the sc-smoothness of transition maps and a paracompactness argument.

## II.31 The 2021 edition, continued: Remark 2.1.6 is strict (the K-topology)

§II.22 proved the implication of Remark 2.1.6. That the modified notion is *strictly* weaker needs an example, which the text does not give.

**A criterion.** If every point has an open neighbourhood $W$ that is completely metrizable, the modified definition holds with $V = W$, the closure of $W$ in itself being $W$ `Chapter2/LocalTopologyExample · isLocallyCompleteRel_of_forall_isCompletelyMetrizable`. A space embedded in $\mathbb{R}$ onto the complement of a sequence $(a_n)$ is completely metrizable: $y \mapsto (y, ((y - a_n)^{-1})_n)$ is a closed embedding into $\mathbb{R} \times \mathbb{R}^{\mathbb{N}}$, with image $\{(u, z) : z_n(u - a_n) = 1 \text{ for all } n\}$ `Chapter2/LocalTopologyExample · isCompletelyMetrizableSpace_of_isEmbedding_compl_range`. This replaces the theorem that $G_\delta$ subsets of complete metric spaces are completely metrizable, which Mathlib lacks.

**The K-topology.** Let $K = \{1/(n + 1)\}$ `Chapter2/LocalTopologyExample · kSeq` and give $\mathbb{R}$ the topology generated by the usual open sets and $\mathbb{R} \setminus K$ `Chapter2/LocalTopologyExample · kTop` `Chapter2/LocalTopologyExample · RealKTop`. It is finer than the usual topology, so Hausdorff. The open sets $\mathbb{R} \setminus K$ and $\mathbb{R} \setminus \{0\}$ cover it, carry the usual subspace topology, and are complements of sequences; by the criterion the modified definition holds `Chapter2/LocalTopologyExample · RealKTop.isLocallyCompleteRel`. But $0$ and the closed set $K$ cannot be separated: every closed neighbourhood of $0$ contains irrational points arbitrarily close to some $1/(n + 1)$, hence meets $K$ `Chapter2/LocalTopologyExample · RealKTop.not_regularSpace`. A locally complete space is regular (§II.22), so this one is not locally complete `Chapter2/LocalTopologyExample · RealKTop.not_isLocallyComplete`.

**Remark 2.1.6, strictness.** Some Hausdorff space satisfies the modified definition and not Definition 2.1.5 `Chapter2/LocalTopologyExample · remark_2_1_6_strict`. So the remark holds as we read it, and the example shows that the global condition $\mathrm{cl}_X(V) \subseteq W$ is what makes a locally complete space regular.

## II.32 The 2021 edition: bundle retractions and the levels of $X^1$ (Definition 2.1.16, after Definition 2.3.8)

**Definition 2.1.16 (preprint 2.9).** A retraction $R$ of $E \oplus F$ is a bundle retraction covering the retraction $r$ of $E$ if its domain is $U \oplus F$, $U$ the domain of $r$, and $R(u, v) = (r(u), \rho(u)v)$ there, with $\rho(u)$ a continuous linear map `Chapter2/TangentLift · ScRetraction.IsBundleRetractionOver`. A map $\Phi$ is a bundle map on $B$ covering $a$ if $\Phi(u, v) = (a(u), \phi(u)v)$ on $B$ `Chapter2/TangentLift · IsBundleMapOn`; then the first component of $\Phi$ is $a$ of the first component `Chapter2/TangentLift · IsBundleMapOn.fst_eq`. The text writes "$p' \circ \Phi = p$", which we read as $p' \circ \Phi = a \circ p$ (possible misprint). Sc-smoothness and $R \circ R = R$ are part of the data, as for every retraction (§II.2). The tangent retraction is a bundle retraction covering the level-1 retraction $r^1$: $Tr(u, h) = (r(u), Dr(u)h)$ `Chapter2/TangentLift · ScRetraction.tangentRetraction_isBundleRetractionOver`. The tangent map of an sc-smooth map between retracts is a bundle map covering $f^1$ `Chapter2/TangentLift · IsScSmoothRetractMap.retractTangentMap_isBundleMapOn`. The categories $\mathcal{R}$ and $\mathcal{BR}$ and their forgetful functors only name these objects and maps; they are not formalized. The category $\mathcal{SBR}$ of §2.6 is (§II.45).

**The levels of $X^1$ (gap G.6).** §II.6 built the M-polyfold $X^1$ on $X_1$; its levels are those of $X$ shifted by one as sets. Their topologies agree too. In a chart, the level-$m$ lift of a point is moved to another chart by the level-$m$ map of the transition map `Chapter2/TangentLift · MPolyfold.levelLift_transition`, and the

level-$m$ lift of the level chart $\varphi^1$ of $X^1$ is the level-$(m+1)$ lift of $\varphi$, since $(E^1)_m = E_{m+1}$ Chapter2/TangentLift · MPolyfold.levelLift_levelChart. The identity of points is therefore continuous from $(X^1)_m$ to $X_{m+1}$ and back, chart by chart, for an arbitrary chart of $X^1$ through level maps of transition maps Chapter2/TangentLift · MPolyfold.shiftLevelMap Chapter2/TangentLift · MPolyfold.continuous_shiftLevelMap Chapter2/TangentLift · MPolyfold.shiftLevelInv Chapter2/TangentLift · MPolyfold.continuous_shiftLevelInv. So $(X^1)_m \cong X_{m+1}$ Chapter2/TangentLift · MPolyfold.shiftLevelHomeomorph, with the charts of $X_{m+1}$ Chapter2/TangentLift · MPolyfold.shift_level_eq. Composing these homeomorphisms, each iterate $(X^1)^1 \cdots$ is $X_m$ with its topology and charts, which is the text's statement after Definition 2.3.8.

## II.33 The 2021 edition: Theorem 2.3.16 and Proposition 2.3.17, $T(X^i) \equiv (TX)^i$

**In a local model.** For an sc¹ map $f$, the tangent map of the level-1 map $f^1$ is the level-1 map of $Tf$ on $T(U^1) = (TU)^1$ Chapter2/TangentLift · tangentMap_levelMap_one Chapter2/TangentLift · toBase_tangentMap_levelMap_one. For a retract, $T(O^1) = (TO)^1$ Chapter2/TangentLift · retractTangent_shift_of_isScRetract, and for sc-smooth maps between retracts $T(f^1)$ is again the level-1 map of $Tf$ Chapter2/TangentLift · toBase_retractTangentMap_shift.

**Theorem 2.3.16, bundle form.** For an M-polyfold $X$, the tangent charts belong to the structure of $TX$ (§II.8), $p : TX \to X^1$ is sc-smooth, and in a tangent chart $p$ is the first projection onto the level chart $\varphi^1$. Every model $(TO, TC, TE)$ is a bundle retract, $Tr$ covering $r^1$, and the transition map of two tangent charts is a bundle map covering the level-1 map of the transition map of the charts Chapter2/TangentLift · theorem_2_3_16. "M-polyfold bundle" is read through the models and transition maps; the linear structure of the fibres $T_xX$ and the linearity of $Tf$ on them are §II.39 (gap G.7, closed).

**Proposition 2.3.17 for $i = 1$.** A tangent vector $q$ of $X^1$ at $x \in (X^1)_1 = X_2$, read in the level chart $\varphi^1$ of a chosen chart $\varphi$ of $X$ around $x$ Chapter2/TangentLift · MPolyfold.someChart, is $(\varphi^1(x), h')$ with $h' \in E_1$; it gives the tuple $(x, \varphi, h')$ of $TX$ Chapter2/TangentLift · MPolyfold.TangentPre.lower. Read in any chart $\psi$ of $X$, this tuple is the inclusion of $q$ read in $\psi^1$, because $T(\psi \circ \varphi^{-1})$ on $(TO)^1$ is the tangent map of $(\psi \circ \varphi^{-1})^1$ Chapter2/TangentLift · MPolyfold.TangentPre.transport_lower. So it defines an inclusion $\iota : T(X^1) \to TX$ Chapter2/TangentLift · MPolyfold.tangentLower. Its values lie on level 1 of $TX$, giving $\Phi : T(X^1) \to (TX)^1$ Chapter2/TangentLift · MPolyfold.tangentShiftMap, a bijection Chapter2/TangentLift · MPolyfold.tangentShiftEquiv and a homeomorphism Chapter2/TangentLift · MPolyfold.tangentShiftHomeomorph. In every chart $\psi$ around the base point, $T\psi(\Phi(\alpha))$ is the inclusion of $T(\psi^1)(\alpha)$ Chapter2/TangentLift · MPolyfold.tangentChartMap_tangentLower, so $\Phi$ does not depend on the chosen chart; in the charts $T(\varphi^1)$ and $(T\varphi)^1$ it is the identity of $T(O^1) = (TO)^1$ Chapter2/TangentLift · MPolyfold.levelChart_tangentShiftMap. Hence $\Phi$ is an sc-diffeomorphism Chapter2/TangentLift · MPolyfold.tangentShiftEquiv_isScDiffeoOn, it covers the identity of $(X^1)^1$, and the level-1 map $p^1 : (TX)^1 \to (X^1)^1$ is sc-smooth Chapter2/TangentLift · MPolyfold.tangentProjShift Chapter2/TangentLift · proposition_2_3_17.

**$i = 2$.** A bijection of M-polyfolds sc-smooth in both directions raises to an sc-diffeomorphism of the lifts Chapter2/TangentLift · MPolyfold.levelOneEquiv Chapter2/TangentLift · MPolyfold.levelOneEquiv_isScDiffeoOn. Composing $\Phi$ for $X^1$ with the raise of $\Phi^{-1}$ gives $T(X^2) \cong (TX)^2$ Chapter2/TangentLift · MPolyfold.tangentShiftEquivTwo, covering the identity of $(X^1)^2 = (X^2)^1$ Chapter2/TangentLift · MPolyfold.tangentProjShiftTwo Chapter2/TangentLift · proposition_2_3_17_two.

**Every $i$.** To speak of $X^i$ as a function of $i$, an M-polyfold is bundled with its space Chapter2/

TangentLiftAll · BundledMPolyfold; then $X^0 = X$, $X^{i+1} = (X^i)^1$ Chapter2/TangentLiftAll · BundledMPolyfold.lift, and $TX$ is again bundled Chapter2/TangentLiftAll · BundledMPolyfold.tangent. Sc-diffeomorphisms and sc-smooth maps are packaged with their index raises Chapter2/TangentLiftAll · MPolyfold.ScEquiv Chapter2/TangentLiftAll · MPolyfold.ScEquiv.shift Chapter2/TangentLiftAll · MPolyfold.ScMap, with $\Phi$ and $p$ as instances Chapter2/TangentLiftAll · MPolyfold.tangentShiftScEquiv Chapter2/TangentLiftAll · MPolyfold.tangentProjScMap. By recursion, $\Phi_0 = \mathrm{id}$ and $\Phi_{i+1} = (\Phi_i)^1 \circ \Phi$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv; likewise $J_i : (X^1)^i \to (X^i)^1$ Chapter2/TangentLiftAll · BundledMPolyfold.shiftLiftEquiv and $p^i$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentProjLift. The inclusions $X^i \to X$ Chapter2/TangentLiftAll · BundledMPolyfold.incl are injective, and $T(X^i) \to TX$ is the composite of the inclusions $\iota$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentIncl. With these, $\Phi_i$ and $J_i$ are sc-diffeomorphisms, the projections are sc-smooth, $J_i \circ p^i \circ \Phi_i = p$, $\Phi_i$ commutes with the inclusions into $TX$ and with those of the two towers $T(X^{i+1}) \to T(X^i)$ and $(TX)^{i+1} \to (TX)^i$, $J_i$ is the identity on points, and these properties determine $\Phi_i$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_unique Chapter2/TangentLiftAll · proposition_2_3_17_all. $\Phi_1$ is the $\Phi$ above Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_one. Because $\iota \circ T(f^1) = Tf \circ \iota$ for every sc-smooth $f$ Chapter2/TangentLiftAll · MPolyfold.tangentLower_tangentMap, $\Phi_i$ is natural in $X$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_natural. A bundled M-polyfold exists, the half-line Chapter2/TangentLiftAll · BundledMPolyfold.nonempty. Linearity of $\Phi_i$ on the fibres is not stated; in charts $\Phi_i$ is a composite of index raises of identities. The linear structure of $T_xX$ in which it could be stated is §II.39. Lemma 4.4.4 for every $i$ uses this identification (Chapter 4).

## II.34 The 2021 edition: Remark 2.4.2, the diagonal half-line

The text's example is $\Delta = \{(s,s,s) : s \geq 0\}$ in $C = [0,\infty)^3$; here $C = [0,\infty)^3 \oplus W$ for any sc-Banach space $W$, with $W = 0$ the text's case. The constant linear maps between $\mathbb{R}$ and $\mathbb{R}^3$ are sc-operators Chapter2/TangentLift · ScOperator.ofConst, and replacing the three coordinates by their average Chapter2/TangentLift · diagonalAverage is an sc-smooth retraction of $C$ onto $\Delta \oplus W$ Chapter2/TangentLift · diagonalRetraction Chapter2/TangentLift · mem_diagonalRetraction_retract_iff, the orthogonal projection onto the diagonal; the text gives no retraction. Its retract is a local model Chapter2/TangentLift · diagonalModel and an M-polyfold Chapter2/TangentLift · diagonalPolyfold.

$(s,w) \mapsto ((s,s,s),w)$ Chapter2/TangentLift · diagonalEmbed, with inverse the first coordinate Chapter2/TangentLift · diagonalProject, is an sc-diffeomorphism from the half-line M-polyfold $[0,\infty) \oplus W$ onto $\Delta \oplus W$ Chapter2/TangentLift · diagonalEmbed_isScDiffeoOn. In $C$, $d_C((s,s,s),w)$ is 3 at $s = 0$ and 0 otherwise Chapter2/TangentLift · degIndex_diagonal_quadrant; on the M-polyfold, $d_\Delta$ is 1 at $s = 0$ and 0 otherwise, by the invariance of the index under sc-diffeomorphisms (Proposition 2.4.4 (preprint 2.24), §II.11) applied to the half-line Chapter2/TangentLift · diagonalPolyfold_degIndex. One chart, defined on all of $\Delta \oplus W$, has the model $[0,\infty) \oplus W$, and in it the chart index is $d_\Delta$ Chapter2/TangentLift · diagonalHalfLineChart Chapter2/TangentLift · diagonalHalfLineChart_spec Chapter2/TangentLift · remark_2_4_2. So the defining chart realizes $d_C$, the more economical one $d_\Delta$, as the remark says.

## II.35 The 2021 edition: tame sub-M-polyfolds, continued (Definition 2.5.19 and the text after it)

§II.25 stated condition (3) of Definition 2.5.19 in one chart around each smooth point, with one presentation of its partial quadrant.

**Chart independence.** A point is smooth iff its image in one chart is a smooth point of the model Chapter2/TameSub · MPolyfold.mem_smoothSet_iff_of_isChart. Condition (3) in a chart is the general-position condition of Chapter 4 for the retraction of the model and the chart image of $Y$, word for word. Chapter 4's chart change for that condition (MPolyfold.isGeneralPositionRetractAt_chart_change, Chapter 4) moves it to every chart $\varphi \ni y$, every presentation of the partial quadrant of $\varphi$, every level-1 representative, and the chart image of any other sub-M-polyfold retraction onto $Y$ near $y$ Chapter2/TameSub · MPolyfold.TameSubData.hasReducedComplementAt_of_isChart, in particular of the same one Chapter2/TameSub · MPolyfold.TameSubData.complement_forall_charts. (3) is a statement in charts here; its form in $T_yX$, equivalent to it, is §II.47.

**A tame sub-M-polyfold is tame.** Sc-splittings compose: if $E = M \oplus B$ is an sc-splitting and $M_m = N_m \oplus A_m$ on every level for sc-subspaces $N, A \subseteq M$, then $E = N \oplus (A \oplus B)$ is one Chapter2/TameSub · ScBanach.levelSub_sup_of_le_of_isScSplitting Chapter2/TameSub · ScBanach.isScSplitting_sup_of_complementIn. In a chart $\varphi$ with a tame model, whose tame retraction is $t$, the chart image $\varphi(W \cap Y)$ is the retract of $(\varphi r \varphi^{-1}) \circ t$. That retraction is tame: the index condition comes from (2) of Definition 2.5.19 and $d_X = d_C \circ \varphi$ in a tame chart; the sc-complement of the tangent space of $\varphi(W \cap Y)$ inside $E_x$ is $A \oplus B$, with $A$ from (3) in the chart $\varphi$ (chart independence) and $B$ the complement of $T_xO$ from the tameness of $t$ Chapter2/TameSub · MPolyfold.TameSubData.isTameRetract_image. So the sub-charts of $Y$ built on tame charts of $X$ have tame models, and $Y$ has a tame induced structure (Definition 2.5.18) Chapter2/TameSub · MPolyfold.IsTameSubMPolyfold.hasTameInducedStructure. The text calls this not difficult to see and gives no proof.

**The examples.** In $X = [0,\infty)^n \oplus W$, the fixed set of an idempotent sc-operator $L$ mapping the quadrant into itself Chapter2/TameSub · TameSubExample.fixSet is a sub-M-polyfold, $L$ restricted to the quadrant being a retraction Chapter2/TameSub · TameSubExample.isSubMPolyfold_fixSet; if $L$ keeps the $\mathbb{R}^n$-component, $L$ preserves $d_X$ (condition (2)) and (3) holds in the identity chart, so the fixed set is a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.isTameSubMPolyfold_fixSet. For $n = 2$, $A = \{(x, y, 0)\}$ Chapter2/TameSub · TameSubExample.setA is the case $L(a, w) = (a, 0)$: a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.isTameSubMPolyfold_setA with tame induced structure Chapter2/TameSub · TameSubExample.hasTameInducedStructure_setA. $B = \{(x, x, 0)\}$ Chapter2/TameSub · TameSubExample.setB is a sub-M-polyfold Chapter2/TameSub · TameSubExample.isSubMPolyfold_setB. No retraction onto $B$ with (1) and (2) is defined near the corner: for small $t > 0$ the point $(t, 0, 0)$ has $d_X = 1$, while $d_X$ takes only the values 0 and 2 on $B$ Chapter2/TameSub · TameSubExample.not_mem_tameSubData_setB. So $B$ is not a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.not_isTameSubMPolyfold_setB, and (3) plays no role. Yet $t \mapsto (t, t, 0)$ is an sc-diffeomorphism from the half-line onto $B$ with its induced structure Chapter2/TameSub · TameSubExample.halfLineToB_isScDiffeoOn, which is therefore tame Chapter2/TameSub · TameSubExample.hasTameInducedStructure_setB. For $W = \mathbb{R}$ these are the text's statements Chapter2/TameSub · TameSubExample.examples_A_B.

## II.36 The 2021 edition: the tame M-polyfold $bX$ (Theorem 2.7.13 (2)–(4), Definition 2.7.14, (2.45)–(2.47))

**Local faces in a tame chart.** Every point of a tame M-polyfold has tame charts with domain inside a given neighbourhood Chapter2/TameBoundaryStructure · MPolyfold.exists_tame_chart_subset, and near a point of a face $\theta$ of a face-structured open $V$ the face is, in such a chart, the zero set of one coordinate, as in Theorem 2.5.17 (preprint 2.45) (§II.16) Chapter2/TameBoundaryStructure · MPolyfold.exists_localFace_of_faceIn.

**Chart data.** A chart datum consists of a triple $(x, V, \theta)$ with $V$ centred at $x$ (§II.27), a chart $\varphi$ of $X$, an open $V_1$ inside its domain, an sc-isomorphism $T$ of its model space with $\mathbb{R}^{k+1} \oplus W$ and a coor-

dinate $i$ vanishing on $T\varphi(\theta \cap V_1)$, such that the face projection $\pi$ of Proposition 2.5.15 (preprint 2.43) maps $\varphi(\theta \cap V_1)$ onto a tame retract of $[0,\infty)^k \oplus W$, and $d_X = d \circ \pi \circ \varphi + 1$ on $\theta \cap V_1$ Chapter2/TameBoundaryStructure · MPolyfold.BChartData. Its chart is $p \mapsto \pi(\varphi(B(p)))$ on $U(x,V,\theta) \cap B^{-1}(V_1)$ Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart. Two such charts have the transition map $\pi' \circ (\varphi' \circ \varphi^{-1}) \circ \iota$, $\iota$ the face inclusion, which is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart_transition_isScSmooth, so they are compatible Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart_compatible. By the local faces, every point of $\theta$ lies in the $V_1$ of a datum with the triple $(x,V,\theta)$, and every class has a centred representative Chapter2/TameBoundaryStructure · TameBoundary.BSpace.exists_centred_rep; so the charts cover $bX$ Chapter2/TameBoundaryStructure · MPolyfold.exists_bChartData and form an atlas Chapter2/TameBoundaryStructure · MPolyfold.bAtlas.

**Theorem 2.7.13 (2)–(4).** $T_{bX}$ is metrizable (§II.29), hence paracompact Chapter2/TameBoundaryStructure · MPolyfold.paracompactSpace_tameBoundarySpace, and with the atlas $bX$ is an M-polyfold Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold, tame because the models are tame retracts Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_isTame (2). In these charts $B$ is the face inclusion $\iota$ followed by $\varphi^{-1}$, so $B$ is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryMap_isScSmooth (3). The index of the face model is one less than that of the quadrant: $d_X(B(p)) = d_{bX}(p) + 1$ Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap, which is (4) as printed Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap_sub_one.

**The characterization, centred representatives.** A face $\theta$ of a face-structured open $V$ is a sub-M-polyfold of $X$ Chapter2/TameBoundaryStructure · MPolyfold.isSubMPolyfold_of_isFaceIn, which we take as "the structure induced from $V$" Chapter2/TameBoundaryStructure · MPolyfold.facePolyfold. For $V$ centred at $x$, $\Phi_{(x,V,\theta)}$ Chapter2/TameBoundaryStructure · MPolyfold.phiTot is an sc-diffeomorphism of $U(x,V,\theta)$ onto $\theta$, because in the charts above it is the face inclusion Chapter2/TameBoundaryStructure · MPolyfold.phi_isScDiffeoOn. Every M-polyfold structure on $bX$ with the topology $T_{bX}$ for which all these $\Phi$ are sc-diffeomorphisms is equivalent to this one Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_unique, which has the property Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_satisfies. The sc-half of Proposition 2.7.12 for centred representatives follows Chapter2/TameBoundaryStructure · MPolyfold.proposition_2_7_12_scDiffeo, and one statement collects (2)–(4), the characterization and its uniqueness Chapter2/TameBoundaryStructure · MPolyfold.theorem_2_7_13_structure.

**Definition 2.7.14 and the iterated boundaries.** The tame boundary is $bX$ with this structure and the sc-smooth $B$ Chapter2/TameBoundaryStructure · MPolyfold.definition_2_7_14. To iterate $b$, a tame M-polyfold is bundled with its space Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle, so that $bX$ is again such a bundle Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.b. Then $b^0X = X$, $b^{k+1}X = b^k(bX)$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iter, and $B^0 = \mathrm{id}$, $B^{k+1} = B_X \circ B^k_{bX}$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iterMap, which unwinds to the text's $B^k = B_X \circ B_{bX} \circ \cdots \circ B_{b^{k-1}X}$. By (3) and (4) and induction, $B^k$ is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iterMap_isScSmooth, $d_X(B^k(p)) = d_{b^kX}(p) + k$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.degIndex_iterMap, and the image of $B^k$ is $\{d_X \geq k\}$, by induction from the image $\partial X$ of $B$ (§II.29) and the index formula Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.range_iterMap.

*Witness.* For the tame quadrant $[0,\infty)^n \oplus W$, $n \geq 1$, chart data exist Chapter2/TameBoundaryStructure · MPolyfold.quadrantPolyfold_bChartData_nonempty, and the $n$ points of $bX$ over the corner have $d_{bX} = n - 1$ Chapter2/TameBoundaryStructure · MPolyfold.quadrantPolyfold_degIndex_tameBoundary.

## II.37 The 2021 edition: Proposition 2.7.12 and the characterization for every representative

**Centred refinements.** Every triple $s$ has an equivalent triple $s'$ centred at its point with $U(s') \subseteq U(s)$ `Chapter2/TameBoundaryNonCentred · TameBoundary.BTriple.exists_centred_subset`: replace $V$ by $V^x$ (§II.27). On the image $\theta \cap V^x$ of $\Phi_{(x,V,\theta)}$ `Chapter2/TameBoundaryNonCentred · MPolyfold.reachSet`, an open subset of $\theta$ `Chapter2/TameBoundaryNonCentred · MPolyfold.isOpen_reachSet` equal to $\theta$ for centred $V$ `Chapter2/TameBoundaryNonCentred · MPolyfold.reachSet_eq_univ_of_isCenteredAt`, the inverse is a total map `Chapter2/TameBoundaryNonCentred · TameBoundary.phiInv` `Chapter2/TameBoundaryNonCentred · TameBoundary.phiInv_spec`.

**Sc-smoothness.** $\Phi$ is sc-smooth on $U(x, V, \theta)$, as $B$ followed by the inclusion into the sub-M-polyfold $\theta$ `Chapter2/TameBoundaryNonCentred · MPolyfold.phiTot_isScSmoothOn`. Near each point $y$ of $\theta \cap V^x$, $\Phi^{-1}$ is $\Phi_{s'}^{-1}$ for a centred refinement $s'$ at $y$, composed with the inclusion of an open piece of $\theta$ into the face of $s'$; that inclusion is sc-smooth because a map into a sub-M-polyfold is sc-smooth as soon as it is sc-smooth into $X$. This replaces the open sc-embeddings of faces in the text's argument `Chapter2/TameBoundaryNonCentred · MPolyfold.phiInv_isScSmoothOn`. So $\Phi$ is an sc-diffeomorphism of $U(x, V, \theta)$ onto the open set $\theta \cap V^x$ `Chapter2/TameBoundaryNonCentred · MPolyfold.phi_isScDiffeoOn_nonCentred`.

**Proposition 2.7.12 as printed.** For every pair of representatives, the images of $U \cap U'$ lie in $\theta \cap V^x$ and $\theta' \cap V'^{x'}$ and are open there, and the transition map is the composite of an inverse and a forward map, hence an sc-diffeomorphism, with inverse the reverse transition map and the identity on points of $X$ `Chapter2/TameBoundaryNonCentred · MPolyfold.proposition_2_7_12_scDiffeo_nonCentred`.

**The characterization.** An M-polyfold structure on $bX$ in which every $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto $\theta \cap V^x$ is equivalent to that of §II.36 `Chapter2/TameBoundaryNonCentred · MPolyfold.tameBoundaryPolyfold_unique_allReps`, which has the property `Chapter2/TameBoundaryNonCentred · MPolyfold.tameBoundaryPolyfold_satisfies_allReps`. With "onto $\theta$" it does not hold as we read it for every representative: for the slab triple of §II.28, $\Phi$ does not map $U(x, V, \theta)$ onto $\theta$ `Chapter2/TameBoundaryNonCentred · MPolyfold.not_surjOn_phiTot`. The strongest version that holds is the one onto $\theta \cap V^x$, which for centred $V$ is the printed statement (§II.36).

**What these additions taught us.** - **"Weaker" is strict**, through a classical non-regular space `Chapter2/LocalTopologyExample · remark_2_1_6_strict`; the global condition of Definition 2.1.5 is what gives regularity. - **The levels of $X^1$ are those of $X$**, chart by chart `Chapter2/TangentLift · MPolyfold.shiftLevelHomeomorph`, so the iterates realize the $X_m$ of the text. - **$T(X^i) \equiv (TX)^i$ is one statement for all $i$** once M-polyfolds are bundled with their spaces `Chapter2/TangentLiftAll · proposition_2_3_17_all`; fibre linearity of $Tf$ is §II.39. - **Condition (3) of Definition 2.5.19 is Chapter 4's general-position condition**, and its chart change is Chapter 4's `Chapter2/TameSub · MPolyfold.TameSubData.hasReducedComplementAt_of_isChart`. - **$bX$ is built from face charts of centred triples** `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold`; non-centred representatives need the image $\theta \cap V^x$ `Chapter2/TameBoundaryNonCentred · MPolyfold.phi_isScDiffeoOn_nonCentred`.

## II.38 Proposition 2.1.2 (preprint 2.2): classical smooth retractions have submanifold images

**Statement.** Let $r : U \to U$ be a $C^\infty$ retraction of an open subset $U$ of a Banach space $E$, $O = r(U)$ and $x \in O$. With $P = Dr(x)$, $R = \operatorname{range} P$ and $N = \ker P$, the space splits as $E = R \oplus N$ into closed subspaces. There are open $V \ni x$, $W \ni 0$ and a $C^\infty$ diffeomorphism $\psi : V \to W$ with $\psi(x) = 0$ and

$\psi(O \cap V) = R \cap W$ Chapter2/GapP1_Prop212 · GapP1Prop212.proposition_2_1_2. The text prints the normalization as $\psi(0) = x$ for a map $\psi : V \to W$; we read it as $\psi(x) = 0$.

*Proof.* The text estimates remainders in Appendix 2.7.1. The formal proof conjugates $r$ to the linear projection $P$ instead. - *$P$ is a projection.* From $r \circ r = r$ near $x$ and $r(x) = x$, the chain rule gives $P^2 = P$ Chapter2/GapP1_Prop212 · GapP1Prop212.isIdempotentElem_fderiv. So $R$ and $N$ are closed and complementary. - *The conjugating map.* Put

$$\varphi(y) = P(r(y) - x) + (1 - P)(y - r(y))$$

Chapter2/GapP1_Prop212 · GapP1Prop212.retractChart. It is $C^\infty$ on $U$ Chapter2/GapP1_Prop212 · GapP1Prop212.contDiffOn_retractChart, with $\varphi(x) = 0$. Its derivative at $x$ is $P \cdot P + (1 - P)(1 - P) = P + (1 - P) = 1$ Chapter2/GapP1_Prop212 · GapP1Prop212.hasFDerivAt_retractChart. - *Conjugation.* $\varphi(r(y)) = P(r(y) - x)$, because $r(r(y)) = r(y)$, and $P\varphi(y) = P(r(y) - x)$, because $P(1 - P) = 0$. So $\varphi \circ r = P \circ \varphi$ on $U$ Chapter2/GapP1_Prop212 · GapP1Prop212.retractChart_comp. - *Inverse function theorem.* Mathlib's inverse function theorem makes $\varphi$ a $C^\infty$ diffeomorphism from an open $V_0 \ni x$ onto an open set. Shrink $V_0$ to the open set $V$ of points $y \in V_0$ with $r(y) \in V_0$ at which $D\varphi(y)$ is invertible, and let $\psi = \varphi|_V$ and $W = \psi(V)$. - *The image of $O$.* If $y \in O \cap V$, then $\psi(y) = \varphi(r(y)) = P\varphi(y) \in R$. Conversely, if $\psi(y) \in R$, then $\varphi(r(y)) = P\varphi(y) = \varphi(y)$; both $y$ and $r(y)$ lie in $V_0$, where $\varphi$ is injective, so $r(y) = y$ and $y \in O$.

## II.39 Definition 2.3.14 (preprint 2.18), Proposition 2.3.19 (3)–(4) (preprint 2.21) and Theorem 2.3.16: the linear structure of $T_xX$

Definition 2.3.14 (preprint 2.18) makes $T_xX$ a vector space by adding representatives in one chart, and the text says after it that $T_xX$ inherits the Banach structure of $T_{\varphi(x)}O$, and the sc-structure at a smooth $x$. That the result does not depend on the chart is not argued. It holds because the fibre transitions are linear.

**The models.** For a retract $O$ and $a \in O_1$, the tangent space $T_aO = Dr(a)E_0$ is a closed subspace, the range of a continuous projection Chapter2/TangentSpaceLinear · tangentSubmodule Chapter2/TangentSpaceLinear · isClosed_tangentSubmodule. For a chart $\varphi$ around $x \in X_1$, the model of the fibre is $T_{\varphi(x)}O$ Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberModel, and $T\varphi$ restricts to a bijection $T_xX \to T_{\varphi(x)}O$ with inverse $h \mapsto [(x, \varphi, h)]$ Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberEquiv.

**The transitions are linear.** For two charts $\varphi$, $\psi$ around $x$, the fibre transition $T\psi \circ (T\varphi)^{-1}$ is the restriction to $T_{\varphi(x)}O$ of the bounded operator $D(\psi \circ \varphi^{-1} \circ r)(\varphi(x))$ Chapter2/TangentSpaceLinear · MPolyfold.exists_transition_clm Chapter2/TangentSpaceLinear · MPolyfold.transitionCLM. This comes from the description of the tangent map of a map between retracts, $(a, h) \mapsto (g(a), Dg(a)h)$ (Theorem 2.1.15 (preprint 2.8), §II.3). The transitions in the two directions are inverse to each other on the models Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberTransition_transition.

**The vector space.** Transport the vector space structure of $T_{\varphi_0(x)}O$ along $T\varphi_0$ for one chosen chart $\varphi_0$ around $x$ Chapter2/TangentSpaceLinear · MPolyfold.fiberChart. For every other chart $\varphi$, $T\varphi = (\text{transition}) \circ T\varphi_0$ is then linear, so $T\varphi : T_xX \to T_{\varphi(x)}O$ is a linear isomorphism for every chart Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberLinearEquiv. This is the chart independence of the definition. The text's formula $\lambda[(x, \varphi, h_1)] + \mu[(x, \varphi, h_2)] = [(x, \varphi, \lambda h_1 + \mu h_2)]$ holds in every chart Chapter2/TangentSpaceLinear · MPolyfold.definition_2_3_14.

**The Banach space.** Give $T_xX$ the subspace topology of $TX$. Then $T\varphi$ is a homeomorphism onto

$T_{\varphi(x)}O$, since the tangent charts are homeomorphisms (§II.8) `Chapter2/TangentSpaceLinear · MPolyfold.continuous_tangentFiberEquiv` `Chapter2/TangentSpaceLinear · MPolyfold.continuous_tangentFiberEquiv_symm`. So it is a topological linear isomorphism onto a Banach space `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberContinuousLinearEquiv`. The norm pulled back through a chart induces the topology and is complete `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberNormedAddCommGroup` `Chapter2/TangentSpaceLinear · MPolyfold.completeSpace_tangentFiber`. Two charts give equivalent norms, because the transition is an isomorphism of Banach spaces `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberTransitionEquiv`.

**The sc-structure at a smooth point.** If $x$ is smooth, so is $\varphi(x)$, and $Dr(\varphi(x))$ is an sc-projection. Its range $T_{\varphi(x)}O$ is therefore an sc-subspace of $E$ `Chapter2/TangentSpaceLinear · MPolyfold.isScSubspace_tangentFiberModel`. The transition operator is an sc-operator at a smooth point `Chapter2/TangentSpaceLinear · MPolyfold.isScOperator_transitionCLM` and maps level $m$ of one model into level $m$ of the other `Chapter2/TangentSpaceLinear · MPolyfold.transitionCLM_mem_range_toBase`. With its inverse, it restricts to an sc-isomorphism of the models `Chapter2/GapP1_Def258b · MPolyfold.tangentFiberScIso` (§II.44). So the sc-structures through two charts agree.

**Tangent maps are linear on the fibres (Theorem 2.3.16).** For $f$ sc-smooth near $x$, $Tf$ restricted to $T_xX \to T_{f(x)}Y$ is, in charts $\varphi$ and $\psi$, the bounded operator $D(\psi \circ f \circ \varphi^{-1} \circ r)(\varphi(x))$, and this operator is an sc-operator if $x$ is smooth `Chapter2/TangentMapLinear · MPolyfold.exists_clm_tangentMapOnFiber`. Hence $Tf$ is continuous and linear on the fibres `Chapter2/TangentMapLinear · MPolyfold.tangentMapOnCLM` `Chapter2/TangentMapLinear · MPolyfold.tangentMapOnCLM_apply` `Chapter2/TangentMapLinear · MPolyfold.tangentMapCLM`. Together with §II.33 this is Theorem 2.3.16 with fibre linearity `Chapter2/TangentMapLinear · MPolyfold.theorem_2_3_16_linear`. The text leaves this implicit in "$Tf$ is a bundle map".

**Proposition 2.3.19 (3)–(4) (preprint 2.21) in $T_aX$.** Let $A \subseteq X$ be a sub-M-polyfold, $a \in A_1$, and $r : V \to V$ an sc-smooth retraction with $r(V) = A \cap V$. The tangent of the inclusion, $Ti : T_aA \to T_aX$ `Chapter2/SubMPolyfoldTangent · MPolyfold.subInclusionCLM`, and the tangent of $i^{-1} \circ r$, $T_aX \to T_aA$ `Chapter2/SubMPolyfoldTangent · MPolyfold.subRetractionCLM`, are continuous linear maps. Since $i^{-1} \circ r \circ i = \mathrm{id}$ near $a$, the chain rule gives $T(i^{-1} \circ r) \circ Ti = \mathrm{id}$ `Chapter2/SubMPolyfoldTangent · MPolyfold.subRetractionCLM_subInclusionCLM`. So $Ti$ is injective, $Ti \circ T(i^{-1} \circ r)$ is a continuous projection of $T_aX$, and

$$T_aX = Ti(T_aA) \oplus \ker T(i^{-1} \circ r)$$

is a splitting into closed subspaces `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_3_linear`. The projection is $Tr(a)$, so its range is $Tr(a)T_aX = Ti(T_aA)$, and $T(i^{-1} \circ r)$ is onto: this is (4) `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_4`. At a smooth $a$ the splitting is an sc-splitting. In a chart, the images of the two summands are the ranges of the idempotent sc-operators $L \circ Dr(a)$ and $Dr(a) - L \circ Dr(a)$, where $L$ is the chart representative of $Tr(a)$. Both are sc-subspaces, complementary on every level of $T_{\varphi(a)}O$ `Chapter2/SubMPolyfoldTangentSc · MPolyfold.proposition_2_3_19_3_sc`. This is the text's proof, which writes $Dt(a)$ for $L \circ Dr(a)$, carried to $T_aX$. The chart statements of §II.10 remain.

## II.40 Proposition 2.3.15 (2) (preprint 2.19): $p : TX \to X_1$ is open

**Statement.** For every M-polyfold, the projection $p : TX \to X^1$ is open `Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj`. With the continuity of §II.8 this is (2) in full `Chapter2/TangentProjOpen · proposition_2_3_15_two`. Appendix 2.7.3 reduces it to the openness of $\mathrm{pr}_1 : TO \to O_1$

and states that without proof.

*The local step* Chapter2/TangentProjOpen · ScRetraction.exists_isOpen_tangent_section. Let $O = r(U)$, $(o,h) \in TO$, and $W$ open in $E_1 \times E_0$ with $(o,h) \in W$. The map $o' \mapsto Tr(o',h) = (r(o'), Dr(o')h)$ is continuous on $U_1$, because $Tr$ is $\mathrm{sc}^0$. At a point $o' \in O_1$ it equals $(o', Dr(o')h)$, a point of $TO$, because $r(o') = o'$ and $Dr(o')h \in T_{o'}O$. At $o$ it equals $(o,h)$, because $Dr(o)$ is the identity on $T_oO$. By continuity there is an open $N \ni o$ in $E_1$ such that every $o' \in N \cap O_1$ carries the vector $Dr(o')h$ with $(o', Dr(o')h) \in TO \cap W$. So $\mathrm{pr}_1(TO \cap W) \supseteq N \cap O_1$.

*The global step.* An open $S \subseteq TX$ around a class $\alpha$ is, in the tangent chart $T\varphi$ of a representative of $\alpha$, a set $TO \cap W$. The open set $N$ of the local step gives the basic open set $\{y \in V : \varphi(y) \in N\}$ of $X^1$ (§II.6), which contains $p(\alpha)$ and lies in $p(S)$. This makes the hypothesis "$p$ open" of Chapter 7's smooth regularity, and of the results of §§7.4 and 8.1 that assume it, a theorem.

## II.41 Remark 2.4.8 (preprint 2.28) and Theorem 2.4.12 (preprint 2.31), second statement: $C_xO$ is a partial quadrant

**Statement.** Let $r$ be an sc-smooth retraction with retract $O$ in a partial quadrant $C = T^{-1}([0,\infty)^n \oplus W)$, and $x \in O$ smooth. Then $T_xO$ and $T_x^RO$ are sc-subspaces of $E$, and $C_xO$ is a partial quadrant in the sc-Banach space $T_xO$ Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant Chapter2/PartialConeQuadrant · remark_2_4_8_partialConeWith. Its dimension is $\dim(T_xO/T_x^RO) \le d_C(x)$. No tameness is assumed. The remark's second sentence, that in general $C_xO$ is not a partial quadrant, therefore does not hold as we read it; the text gives no example, and by this theorem none exists. Under the hypothesis of Theorem 2.4.12 (preprint 2.31), $\dim(T_xO/T_x^RO) = d_O(x)$, and the theorem's second statement follows, without the hypothesis being needed for the quadrant. What tameness adds is only the equality of the dimension with $d_C(x)$ (§II.14).

*Notation.* Let $I(x) = \{i : \lambda_i(x) = 0\}$, $E_x$ and $C_x$ as in §II.12, and $P = Dr(x)$, so that $T_xO = PE$. Let $\Lambda : E \to \mathbb{R}^{I(x)}$, $\Lambda v = (\lambda_i v)_{i \in I(x)}$, with kernel $E_x$, and $\sigma : \mathbb{R}^{I(x)} \to E$, $\sigma a = T^{-1}(\tilde{a}, 0)$ with $\tilde{a}$ the extension by zero, a right inverse of $\Lambda$.

*Step 1: $C_xO = P(C_x)$.* The two sign arguments of §II.12, applied to $r$ itself at $x = r(x)$, give $PE_x \subseteq E_x$ and $PC_x \subseteq C_x$ Chapter2/PartialConeQuadrant · ScRetraction.scDeriv_mapsTo_partialQuadrant. As $P$ is the identity on $T_xO$, $C_xO = T_xO \cap C_x = P(C_x)$ Chapter2/PartialConeQuadrant · partialConeWith_eq_image_scDeriv.

*Step 2: a nonnegative idempotent matrix.* Let $M = \Lambda P \sigma$ on $\mathbb{R}^{I(x)}$ Chapter2/PartialConeQuadrant · coneMatrix. Since $v - \sigma\Lambda v \in E_x$ and $P$ maps $E_x$ into $\ker\Lambda$, $\Lambda P = M\Lambda$ Chapter2/PartialConeQuadrant · coordMapL_scDeriv. Hence $M^2 = \Lambda P \sigma \Lambda P \sigma = \Lambda P P \sigma = M$ Chapter2/PartialConeQuadrant · coneMatrix_idem. And $M$ maps the orthant into itself, since $\sigma$ maps it into $C_x$ and $PC_x \subseteq C_x$ Chapter2/PartialConeQuadrant · coneMatrix_mapsTo_orthant. Moreover $\Lambda(T_xO) = \mathrm{range}\, M$, and a $v \in T_xO$ lies in $C_x$ exactly when $\Lambda v \ge 0$.

*Step 3: Flor's theorem, in the case needed.* For a nonnegative idempotent matrix $M$, the cone $K = \mathrm{range}\, M \cap [0,\infty)^\iota$, which equals $M([0,\infty)^\iota)$ Chapter2/PartialConeQuadrant · image_orthant_eq_nonnegRange, is simplicial: it is the set of nonnegative combinations of linearly independent columns $c_1, \dots, c_s$ of $M$ Chapter2/PartialConeQuadrant · flor_nonnegRange_simplicial. Write $B(j)$ for the support of column $j$. - From $M^2 = M \ge 0$, $M_{im} \ge M_{ij}M_{jm}$ Chapter2/PartialConeQuadrant · matEntry_mul_le. So $m \in B(j)$ implies $B(m) \subseteq B(j)$ Chapter2/PartialConeQuadrant · colSupp_subset. - Call $j$ *minimal* if $B(j)$ is nonempty and minimal among the nonempty column supports. A minimal column has a private index, one that lies in

no other minimal support except an equal one `Chapter2/PartialConeQuadrant · IsMinCol.exists_private`. Pick $i \in B(j)$. Then $M_{ij} = \sum_m M_{im} M_{mj} > 0$ gives $m \in B(j)$ with $i \in B(m) \subseteq B(j)$, so $B(m) = B(j)$ by minimality, and $m$ is private. - *Generation.* Every $y \in K$ is a nonnegative combination of minimal columns, one per minimal support. The proof is by induction on the size of the support of $y$. From $y = My$, a coordinate $y_i > 0$ yields $m$ with $y_m > 0$, $i \in B(m)$ and $B(m) \subseteq \operatorname{supp} y$, and then a minimal $j$ with $B(j) \subseteq B(m)$. With $t = \min_{l \in B(j)} y_l / M_{lj}$, the vector $y - t\, c_j$ is nonnegative, lies in range $M$, and has smaller support `Chapter2/PartialConeQuadrant · exists_nonneg_comb_coneGen`. - *Independence.* Evaluating a vanishing combination at the private indices shows that the chosen columns are linearly independent `Chapter2/PartialConeQuadrant · linearIndependent_coneGen`.

So $\Lambda(C_x O) \cong C_x O / T_x^R O$ is a simplicial cone, at every $x \in O_1$ `Chapter2/PartialConeQuadrant · remark_2_4_8_quotient_simplicial`.

*Step 4: the quadrant.* At a smooth $x$, $T_x^R O$ is an sc-subspace, because the sc-projection $P$ preserves $E_x$ `Chapter2/PartialConeQuadrant · ScRetraction.isScSubspace_reducedTangentSpaceWith`. This is the argument of Proposition 2.5.6 (preprint 2.36) in §II.14, which did not use tameness. The vectors $\hat{e}_s = P\sigma(c_s) \in T_x O$ are smooth, with $\Lambda \hat{e}_s = c_s$. With a linear left inverse $\rho$ of $t \mapsto \sum_s t_s c_s$, the functionals $\ell_s = (\rho \circ \Lambda)_s$ satisfy $\ell_s(\hat{e}_t) = \delta_{st}$, and $T_x^R O = T_x O \cap \bigcap_s \ker \ell_s$. For $v \in T_x O$: $v \in C_x \iff \Lambda v \in K \iff \ell_s(v) \geq 0$ for all $s$. The sc-isomorphism $T_x O \cong \mathbb{R}^s \oplus T_x^R O$ of §II.14 `Chapter2/PartialCone · splitIso` therefore carries $C_x O$ onto $[0,\infty)^s \oplus T_x^R O$ `Chapter2/PartialCone · isPartialQuadrant_ofSubspace`.

The quadrant has dimension $s = \operatorname{rank} M = \dim(T_x O / T_x^R O)$. For a tame retraction $M$ is the identity (Proposition 2.5.3 (preprint 2.35)), so $s = d_C(x)$.

## II.42 Lemma 2.4.9 (preprint 2.29): $T_x^R O$ and $C_x O$ through sc-smooth paths

**Statement.** Let $x \in O$ be smooth. (1) $T_x^R O$ is the closure of the set of derivatives $\dot{\alpha}(0)$ of sc-smooth paths $\alpha : (-\varepsilon, \varepsilon) \to O$ with $\alpha(0) = x$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.lemma_2_4_9_reduced Chapter2/GapP1_Lemma249 · GapP1Lemma249.twoSidedPathDerivs`. (2) $C_x O$ is the closure of the set of derivatives of sc-smooth paths $\alpha : [0, \varepsilon) \to O$ with $\alpha(0) = x$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.lemma_2_4_9_cone Chapter2/GapP1_Lemma249 · GapP1Lemma249.oneSidedPathDerivs`. (3) $T_x O = C_x O - C_x O$, at every $x \in O_1$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.lemma_2_4_9_tangent`. The proof is the text's.

*Paths.* The intervals $(-\varepsilon, \varepsilon)$ and $[0, \varepsilon)$ are sc-domains in $\mathbb{R}$ with the constant structure `Chapter2/GapP1_Lemma249 · GapP1Lemma249.isScDomain_Ioo Chapter2/GapP1_Lemma249 · GapP1Lemma249.isScDomain_Ico`. The derivative $\dot{\alpha}(0)$ is $D\alpha(0)1$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.pathDeriv`, the classical two-sided or one-sided derivative `Chapter2/GapP1_Lemma249 · GapP1Lemma249.hasDerivWithinAt_pathDeriv`.

*Derivatives of paths lie in the cones.* From $r \circ \alpha = \alpha$ the chain rule gives $Dr(x)\dot{\alpha}(0) = \dot{\alpha}(0)$, so $\dot{\alpha}(0) \in T_x O$. For $i \in I(x)$ the coordinate $\lambda_i \alpha(t) \geq 0$ vanishes at $t = 0$. For a two-sided path it has a local minimum there, so its derivative is 0, and $\dot{\alpha}(0) \in E_x$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.pathDeriv_mem_minimalSubspaceWith`. For a one-sided path the derivative from the right is $\geq 0$, and $\dot{\alpha}(0) \in C_x$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.pathDeriv_mem_partialQuadrantAtWith`. Both cones are closed, so the closures lie in them.

*Paths through every smooth vector.* For a smooth $v$, the path $\alpha(t) = r(x + tv)$ is sc-smooth for $t$ small, lies in $O$, starts at $x$, and has $\dot{\alpha}(0) = Dr(x)v$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.path_retract_segment`. If

$v \in T_x^R O$ is smooth, $x+tv$ stays in $C$ for small $|t|$, since $v \in E_x$, and $Dr(x)v = v$. If $v \in C_x O$ is smooth, $x + tv \in C$ for small $t \geq 0$. So every smooth vector of $T_x^R O$, respectively $C_x O$, is a path derivative.

*Density.* The smooth vectors of $C_x O$ are dense in $C_x O$ `Chapter2/GapP1_Lemma249 · GapP1Lemma249.partialConeWith_subset_closure`. Write $v = Pu \in C_x O$ and let $a$ be the $\mathbb{R}^n$-component of $T(Pu)$. The map $w' \mapsto P\,T^{-1}(a, w')$ on $W_0$ is continuous. It sends smooth points of $W$ to smooth points of $C_x O$, because $P$ is an sc-operator at the smooth point $x$ and $T^{-1}(a, w')$ lies in $C_x$. It sends the $W$-component of $T(Pu)$ to $P^2 u = v$. Smooth points of $W$ are dense, so $v$ is a limit of smooth points of $C_x O$. For $T_x^R O$ the density is that of the smooth points of an sc-subspace, since $T_x^R O$ is one (§II.41) `Chapter2/PartialConeQuadrant · ScRetraction.isScSubspace_reducedTangentSpaceWith`. With the two previous steps this proves (1) and (2).

*(3).* Let $Pk \in T_x O$, and let $a$ be the $\mathbb{R}^n$-component of $Tk$. Put $k^- = T^{-1}(\max(-a, 0), 0)$, coordinatewise. Then $k + k^-$ and $k^-$ have all coordinates $\lambda_i \geq 0$, so $P(k + k^-)$ and $Pk^-$ lie in $C_x O$ by the one-sided sign argument, and $Pk = P(k + k^-) - Pk^-$. The converse inclusion holds because $C_x O \subseteq T_x O$. Smoothness of $x$ is not used.

## II.43 The extended faces of a partial quadrant, intrinsically (§2.5 before Definition 2.5.1 (preprint §2.4 before Definition 2.33))

The text defines the extended faces of a partial quadrant $C \cong [0, \infty)^k \oplus W$ from $C$ alone, through the components of $\{d_C = 1\}$. The library works with a presentation $T$, $C = T^{-1}([0, \infty)^n \oplus W)$, and the coordinate hyperplanes $f_i = \{\lambda_i = 0\}$. This subsection proves that the two agree, so the presentation enters the proofs only.

**The components of $\{d_C = 1\}$.** The set $\{e \in C : d_C(e) = 1\}$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.stratumOne` is the disjoint union of the sets $A_i = \{\lambda_i = 0,\ \lambda_j > 0 \text{ for } j \neq i\}$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.faceOpen`. Each $A_i$ is convex, hence connected `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.convex_faceOpen`, and relatively open in $\{d_C = 1\}$, since the conditions $\lambda_j > 0$ are open. So the connected component of a point of $A_i$ is $A_i$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.connectedComponentIn_eq`, and the components are exactly the $A_i$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.faceComponents_eq`. The $A_i$ are distinct and nonempty: the point with $\lambda_i = 0$, the other coordinates equal to 1 and $W$-component 0 lies in $A_i$ only `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.facePoint_mem`. So there are exactly $n$ components `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.ncard_faceComponents`.

**Extended faces and half-spaces.** The span of $A_i$ is $f_i$, since $A_i$ is a nonempty relatively open subset of $f_i$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.span_faceOpen`. So the extended faces are the $f_i$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.extendedFaces_eq`, and there are exactly $n$ of them `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.ncard_extendedFaces`. The closed half-space containing $C$ with boundary $f_i$ is $C + f_i = \{\lambda_i \geq 0\}$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.add_coordFace Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.isClosed_add_coordFace`. For $e \in C$, the faces containing $e$ are the $f_i$ with $\lambda_i(e) = 0$, so $d_C(e) = \#\mathcal{F}(e)$ `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.ncard_facesAt`. And $\bigcap_{f \in \mathcal{F}(e)} (C + f) = C_e$, the partial quadrant of Definition 2.5.1 (preprint 2.33) as the library defines it `Chapter2/GapP1_ExtendedFaces · GapP1ExtendedFaces.iInter_facesAt`.

## II.44 Definition 2.5.8 (preprint 2.38): $T^R_x X$ and $C_x X$ inside $T_x X$

**Statement.** For a chart $\varphi$ around $x \in X_1$ and a presentation of its quadrant, $T^R_x X$ and $C_x X$ are the preimages of $T^R_{\varphi(x)} O$ and $C_{\varphi(x)} O$ under $T\varphi : T_x X \to T_{\varphi(x)} O$ `Chapter2/ReducedTangentIntrinsic · MPolyfold.fiberCoord` `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber` `Chapter2/ReducedTangentIntrinsic · MPolyfold.partialConeFiber`. They do not depend on the chart or the presentation `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber_eq` `Chapter2/ReducedTangentIntrinsic · MPolyfold.partialConeFiber_eq`. At a smooth $x$, $C_x X$ is a partial quadrant in $T_x X$, and $d_{C_x X}(0_x) = \dim(T_x X / T^R_x X)$. In a tame chart both equal $d_X(x)$ `Chapter2/GapP1_Def258 · MPolyfold.definition_2_5_8_degIndex`. The sc-structure of $T_x X$ and the quadrant do not depend on the chart `Chapter2/GapP1_Def258b · MPolyfold.definition_2_5_8_scStructure`.

*Chart independence.* For two charts, the transition maps form a germ of sc-diffeomorphisms between the overlap retracts `Chapter2/ReducedTangentIntrinsic · MPolyfold.exists_isDiffeoGerm_transition`. Its derivative at $\varphi(x)$ is the fibre transition of §II.39. Proposition 2.4.10 (preprint 2.30) (§II.13) says that this derivative maps $T^R$ onto $T^R$ and $C$ onto $C$. So the preimages agree. This holds at every point on level 1, not only at smooth points. The quotient dimension in a tame chart at a smooth point is $d_X(x)$ by Proposition 2.5.6 (preprint 2.36) `Chapter2/ReducedTangentIntrinsic · MPolyfold.finrank_quotient_reducedTangentFiber_eq_degIndex`.

*The quadrant and its degree.* For a partial quadrant $C$ in an sc-Banach space $F$, $d_C(0) = \dim(F_0/(C \cap -C))$ `Chapter2/GapP1_Def258 · degIndex_zero_eq_finrank_quotient`. The degree at the apex therefore depends only on the set. Inside $T_x X$, $T^R_x X = C_x X \cap -C_x X$ `Chapter2/GapP1_Def258 · MPolyfold.mem_reducedTangentFiber_iff_lineality`, because $E_x = C_x \cap -C_x$ `Chapter2/GapP1_Def258 · mem_minimalSubspaceWith_iff_lineality`. At a smooth $x$, in any chart, tame or not, the image of $C_x X$ in $T_{\varphi(x)} O$ `Chapter2/GapP1_Def258 · MPolyfold.partialConeFiberModel` is a partial quadrant, by §II.41 `Chapter2/GapP1_Def258 · MPolyfold.isPartialQuadrant_partialConeFiberModel`. Its degree at $0$ is $\dim(T_x X / T^R_x X)$ `Chapter2/GapP1_Def258 · MPolyfold.degIndex_partialConeFiberModel_eq_finrank`.

*The sc-structure through two charts.* An sc-operator with an sc-operator inverse between sc-subspaces restricts to an sc-isomorphism of them `Chapter2/GapP1_Def258b · ScBanach.subspaceIso`. Applied to the fibre transition and its inverse, which preserve levels at a smooth point (§II.39), this gives an sc-isomorphism $T_{\varphi(x)} O \cong T_{\psi(x)} O'$ `Chapter2/GapP1_Def258b · MPolyfold.tangentFiberScIso`. It carries $T\varphi(\alpha)$ to $T\psi(\alpha)$ `Chapter2/GapP1_Def258b · MPolyfold.tangentFiberScIso_comp` and the model of $C_x X$ in $\varphi$ onto the model in $\psi$ `Chapter2/GapP1_Def258b · MPolyfold.tangentFiberScIso_image_partialConeFiberModel`. So the identity of $T_x X$ is an sc-isomorphism between the two transported structures, and it preserves $C_x X$.

## II.45 Strong bundles: atlas equivalence, restriction, the maximal atlas, isomorphisms and $\mathcal{SBR}$ (Definitions 2.3.3, 2.6.1, 2.6.3–2.6.5 (preprint 2.12, 2.47, 2.49–2.51))

**Atlas equivalence is an equivalence relation.** The text speaks of "an equivalence class" of atlases, for M-polyfolds (Definition 2.3.3 (preprint 2.12)) and for strong bundles (Definition 2.6.5 (preprint 2.51)), without checking transitivity. It holds by the locality of compatibility (§II.1). If $\mathcal{A} \sim \mathcal{A}'$ and $\mathcal{A}' \sim \mathcal{A}''$, a chart of $\mathcal{A}$ and a chart of $\mathcal{A}''$ are compatible near each point of their overlap, through a chart of $\mathcal{A}'$ there, and so they are compatible `Chapter2/StrongBundleRestrict · MAtlas.Equiv.trans` `Chapter2/StrongBundleRestrict · MAtlas.equivalence`. For strong bundles the same argument runs on

$[0]$ and on $[1]$. The condition that the $[1]$-transition preserves fibre components in $F_1$ (§II.21) passes through the factorization `Chapter2/StrongBundleRestrict · StrongBundleChart.compatible_of_locally` `Chapter2/StrongBundleRestrict · StrongBundleAtlas.equiv_iff` `Chapter2/StrongBundleRestrict · StrongBundleAtlas.Equiv.trans` `Chapter2/StrongBundleRestrict · StrongBundleAtlas.equivalence`.

**Restriction of a strong bundle chart.** Let $s$ be open in the domain of the base chart $\varphi$, with $\varphi(s) = O \cap W$ for an open $W$. The chart restricts over $\varphi|_s$, with the strong bundle retraction restricted to $(U \cap r^{-1}(W)) \lhd F$ `Chapter2/StrongBundleRestrict · IsStrongBundleRetraction.restrict` `Chapter2/StrongBundleRestrict · StrongBundleChart.restrict`. Its retract is $\{(u,h) \in K : u \in \varphi(s)\}$ `Chapter2/StrongBundleRestrict · StrongBundleChart.restrict_K`, and its domain $\{u \in U : r(u) \in \varphi(s)\}$ does not depend on $W$ `Chapter2/StrongBundleRestrict · StrongBundleChart.restrict_U`. Compatibility passes to restrictions `Chapter2/StrongBundleRestrict · StrongBundleChart.Compatible.restrict`. The overlaps are $\Phi(P^{-1}(t)) = K \cap (W \times F)$ `Chapter2/StrongBundleRestrict · StrongBundleChart.image_preimage_eq`, open in $K$ through the base chart, so continuity of $P$ is never used.

**The maximal atlas.** The later chapters speak of "a strong bundle chart of $P$". Two such charts are compatible `Chapter3/StrongBundleMaximalAtlas · StrongBundle.IsChart.compatible`. All of them form an atlas equivalent to the given one `Chapter3/StrongBundleMaximalAtlas · StrongBundle.maximalAtlas` `Chapter3/StrongBundleMaximalAtlas · StrongBundle.maximalAtlas_equiv`. A chart of $P$ is a chart of some atlas of the class `Chapter3/StrongBundleMaximalAtlas · StrongBundle.isChart_iff_exists_equiv`, and the charts depend only on the class `Chapter3/StrongBundleMaximalAtlas · StrongBundle.isChart_congr`. Restrictions of charts of $P$ are charts of $P$ `Chapter3/StrongBundleMaximalAtlas · StrongBundle.IsChart.restrict`.

**Strong bundle isomorphisms and $\mathcal{SBR}$ (Definition 2.6.1 (preprint 2.47), pp. 66–67).** Strong bundle isomorphisms between the $U \lhd F$ contain the identity and are closed under inverse and composition `Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.refl` `Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.symm` `Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.trans`. In the text's wording, they are the invertible strong bundle maps with a strong bundle inverse `Chapter3/StrongBundleIsoSBR · exists_isStrongBundleIso_iff`. The category $\mathcal{SBR}$ `Chapter3/StrongBundleIsoSBR · SBR` has as objects the local strong bundle retracts $(K, C \lhd F, E \lhd F)$. Its morphisms are the maps $K \to K'$ that extend to a map $\Phi : E \lhd F \to E' \lhd F'$ which is linear on the fibres, preserves the double filtration, and has $\Phi[0]$ and $\Phi[1]$ sc-smooth between the retracts `Chapter3/StrongBundleIsoSBR · SBR.IsHom` `Chapter3/StrongBundleIsoSBR · SBR.Hom`. These are closed under identity and composition `Chapter3/StrongBundleIsoSBR · SBR.IsHom.id` `Chapter3/StrongBundleIsoSBR · SBR.IsHom.comp`. Equality of morphisms is equality on $K$, so the category laws are those of maps. A strong bundle map of Definition 2.6.1 is a morphism `Chapter3/StrongBundleIsoSBR · IsStrongBundleMap.isHom`, and a strong bundle isomorphism is an isomorphism `Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.toIso` `Chapter3/StrongBundleIsoSBR · SBR.isoOfInverse`. The forgetful functors forget$[i] : K \mapsto K[i]$, $\Phi \mapsto \Phi[i]$ `Chapter3/StrongBundleIsoSBR · SBR.forget0` `Chapter3/StrongBundleIsoSBR · SBR.forget1` land in the category of sc-smooth retracts `Chapter3/StrongBundleIsoSBR · ScRetr`. The text has them land in $\mathcal{BR}$, which is only named (§II.32), so $K[i]$ is recorded without its bundle structure.

## II.46 Lemma 2.7.3 and Proposition 2.7.4 (preprint 2.56 and 2.57): closed locally finite refinements

**Statement.** A regular Hausdorff space is paracompact if and only if every open cover has a locally finite refinement by closed sets `Chapter2/GapP1_Michael · GapP1Michael.lemma_2_7_3`. A regular Hausdorff space covered by a locally finite family of closed paracompact subspaces is paracompact `Chapter2/GapP1_`

`Michael · GapP1Michael.proposition_2_7_4`. Both hold as printed, with Mathlib's paracompactness (locally finite open refinements).

*Paracompact implies closed refinements* `Chapter2/GapP1_Michael · GapP1Michael.exists_closed_refinement`. The text takes the closures of a locally finite open refinement. Such a closure need not lie in a member of the cover, so first shrink: by regularity each $x$ has an open $W_x$ with $\overline{W_x}$ inside a member $U_{i(x)}$. Take a locally finite open refinement $(V_j)$ of $(W_x)$. Its closures are locally finite and each lies in some $\overline{W_x} \subseteq U_{i(x)}$.

*Closed refinements imply paracompact* `Chapter2/GapP1_Michael · GapP1Michael.paracompactSpace_of_closed_refinement`. This direction holds in every topological space, without separation. Let $(U_i)$ be an open cover. Take a locally finite closed refinement $(A_b)$, with $A_b \subseteq U_{i(b)}$. For each $x$ let $B(x)$ be the complement of the union of the $A_b$ not containing $x$ (open, by local finiteness), intersected with the $U_{i(b)}$ for the finitely many $A_b \ni x$. Take a locally finite closed refinement $(C_k)$ of $(B(x))_x$, and put $A'_b = X \setminus \bigcup\{C_k : C_k \cap A_b = \emptyset\}$. Then $A'_b$ is open, contains $A_b$, and $A'_b \cap U_{i(b)}$ is an open refinement. It is locally finite: a neighbourhood of $x$ meets only finitely many $C_k$, and each $C_k \subseteq B(x_k)$ meets only the $A_b$ with $x_k \in A_b$, finitely many.

*Proposition 2.7.4.* Given the closed paracompact $Y_i$ and an open cover, refine the cover on each $Y_i$ by a locally finite closed family $(C^i_b)$, using the lemma in the subspace $Y_i$. The family of all $C^i_b$, indexed by pairs $(i, b)$ instead of the text's unions $\bigcup_i C^i_b$, is a locally finite closed refinement of the cover, and the lemma applies.

The library's proof of Theorem 2.3.10 (preprint 2.16) still goes around these results (§II.7), since the text applies Proposition 2.7.4 to $X_m$ without checking that $X_m$ is regular.

## II.47 The 2021 statements in book form: Remark 2.1.6, Definitions 2.5.18–2.5.19, Proposition 2.7.12, Theorem 2.7.13

For four 2021 statements, the earlier subsections proved the parts separately. Each is now one theorem in the book's shape.

**Remark 2.1.6.** The closure identity, the implication from local completeness to the modified notion, and its strictness (the K-topology, §II.31) together `Chapter2/BookFormLocalCompleteness2021 · remark_2_1_6_book`. The text does not say whether "weaker" means strictly weaker, so both readings are included.

**Definitions 2.5.18–2.5.19 with (3) in $T_yX$.** With the linear structure of §II.39 and $T^R_yX$ of §II.44, condition (3) can be stated as the text states it. $T_yY \subseteq T_yX$ is read as the image of $Ti$ `Chapter2/BookFormTameSub2021 · MPolyfold.subTangentRange`. In a chart $\varphi$ of $X$ and the sub-chart of $Y$ built on it, $T\varphi \circ Ti$ is the chart map of the sub-chart `Chapter2/BookFormTameSub2021 · MPolyfold.fiberCoord_subInclusionCLM`. So $T\varphi$ maps $T_yY$ onto the tangent space of the chart image of $Y$ `Chapter2/BookFormTameSub2021 · MPolyfold.coe_map_range_subInclusionCLM`. Condition (3) in $T_yX$ asks for a subspace $A \subseteq T^R_yX$ with $T_yX = T_yY \oplus A$ whose chart image is an sc-subspace complementary on every level `Chapter2/BookFormTameSub2021 · MPolyfold.HasReducedScComplement`. It is equivalent to the chart form of §II.25 `Chapter2/BookFormTameSub2021 · MPolyfold.hasReducedScComplement_iff`. Definition 2.5.19 with this (3) `Chapter2/BookFormTameSub2021 · MPolyfold.TameSubData2021` `Chapter2/BookFormTameSub2021 · MPolyfold.IsTameSubMPolyfold2021` is therefore the chart definition `Chapter2/BookFormTameSub2021 · MPolyfold.isTameSubMPolyfold2021_iff`. It implies the tame induced structure of Definition 2.5.18, and $Y = X$ satisfies it `Chapter2/BookFormTameSub2021 · MPolyfold.definition_2_5_19_book`. Through the equivalence, the two examples of §II.35 apply to it.

**Proposition 2.7.12 and Theorem 2.7.13.** Proposition 2.7.12 as printed, for every pair of representatives `Chapter2/BookFormTameBoundary2021 · MPolyfold.proposition_2_7_12_book`. Theorem 2.7.13 (1)–(4) with the characterization, the uniqueness of topology and structure, and range $B = \partial X$ of Definition 2.7.14, in one statement `Chapter2/BookFormTameBoundary2021 · MPolyfold.theorem_2_7_13_book`. In the characterization the image of $\Phi_{(x,V,\theta)}$ is $\theta \cap V^x$, which is $\theta$ for centred $V$, as described in the opening of this part.

## II.48 §2.2 in M-polyfolds: smooth points of connected open sets are path-connected

Lemma 2.2.1 proves that the levels $U_k$, $k \leq \infty$, of a connected relatively open $U \subseteq C$ are path-connected. Chapters 6 and 7 use the M-polyfold version without proof: for $U \subseteq X$ open and connected, $U \cap X_\infty$ is connected. It is proved here in two steps. - *Local.* Every $u \in U$ has an open neighbourhood $N \subseteq U$ in which any two smooth points are joined in $U \cap X_\infty$ by a path continuous for the topology of $X_\infty$ `Chapter2/SmoothPaths · MPolyfold.exists_nhds_joinedIn_smooth`. The path is the retracted segment $t \mapsto \varphi^{-1}(r(\varphi(x) + \rho(t)(\varphi(y) - \varphi(x))))$ in a small ball, which is an sc-smooth path. - *Global.* For $U$ open and connected, $U \cap X_\infty$ is path-connected in $X_\infty$ `Chapter2/SmoothPaths · MPolyfold.isPathConnected_smoothVal_preimage`. Fix a smooth $x_0 \in U$. The set of points of $U$ all of whose nearby smooth points are joined to $x_0$ is open. By the local step and the density of $X_\infty$, it is also closed in $U$, so it is $U$.

# Part III. Sc-Fredholm theory and the implicit function theorem (HWZ Chapter 3, §§3.1–3.8)

Chapter 3 is the reason for the first two chapters: it defines the sections of strong bundles to which an implicit function theorem applies, the *sc-Fredholm sections*, and proves that theorem. The engine is a fixed point problem with finitely many parameters, $w = B(a, w)$, $a \in \widetilde{C} \subseteq \mathbb{R}^n$, $w \in W$, where $B(a, \cdot)$ contracts with an arbitrarily small constant near 0 *on every level* $W_m$ of the scale. Its solution $w = \delta(a)$ is the *solution germ*; the germ implicit function theorem (Theorem 3.3.6 (preprint 3.33)) says that $\delta$ is as regular as $B$. Around it the chapter builds: germs and their calculus (§III.1), sections of strong bundles in charts (§III.2), fillers (§§III.3–III.4), basic germs and their linearization (§III.5), sc-Fredholm germs and sections with the stability theorem 3.15 (§III.6); the germ implicit function theorem itself (§§III.7–III.11); local regularity and compactness, Theorems 3.5.1 and 3.1.13 (preprint 3.40 and 3.12) (§§III.12–III.13); the finite-dimensional geometry of good position and the inverse and implicit function theorems in partial quadrants, which the formalization proved first and the rest uses (§§III.14–III.17); the basic class and local solvability (§§III.18–III.20); the linearization of an sc-Fredholm section at a zero, good position, and the graph description of the solution set of Theorem 3.6.8 (preprint 3.54) (§§III.21–III.23); the tangent germ property, its invariance, and the manifold structure of solution sets (§III.24); and Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61) of the basic germ criterion (§III.25). Sections III.27–III.39 were added after the part was first assembled, for the Lean written since: tameness under sc-diffeomorphisms (§III.27), Corollary 3.4.3 (preprint 3.39) (§III.28), good complements whose projection maps the quadrant into itself and local solvability with such a complement (§III.29), the index in Proposition 3.6.5 (preprint 3.52) (§III.30), the linearization at a zero in charts and its chart independence (§III.31), Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros (§§III.32–III.33), Theorem 3.6.8 (2) (preprint 3.54) (§§III.34–III.36), the implicit function theorems 3.54, 3.23, 3.19, 3.24, 3.56, 3.57 assembled (§III.37), the basic germ criterion, Theorem 3.7.1 (preprint 3.58) (§III.38), and what this addendum taught us (§III.39). Sections III.40–III.46 were added in a second addendum: Theorem 3.1.21 (preprint 3.18) (§III.40); Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, by an implicit function theorem at the nearby zero (§§III.41–III.43); the dimension of the solution set as the Fredholm index and its local constancy (§III.44); the kernel as tangent space, good position at interior zeros, and the filler condition of §III.33 as a theorem (§III.45); and what the second addendum taught us (§III.46). Section III.47 repairs a tameness hypothesis, and §III.48 records the switch of Definition 3.1.11 (preprint 3.11) to the text's germ reading; the passages above that use the filling or the sc$^+$-section have been rewritten for it. Section III.49 proves Definition 6.2.1 and Proposition 6.2.2 (preprint 6.1 and 6.2), and §§III.50–III.55 compare the part with the 2021 edition. Sections III.56–III.66

were added with the waves of November 2026, which closed the remaining gaps of the chapter: Theorem 3.2.3 (preprint 3.27) under the tangent germ property alone (§III.56); the implicit function theorems with the 2021 definitions (§III.57); Remark 3.3.8 (preprint 3.36) (§III.58); the extreme-ray lemmas of the appendix (§III.59); Definition 3.8.14, Lemmas 3.8.20 and 3.8.21 (preprint 3.71, 3.76, 3.77) (§III.60); Lemma 3.8.13 (preprint 3.70) in a general presentation and $d_M = d_X$ (§III.61); the intrinsic linearization and $\ker f'(y) = T_yS$ (§III.62); Theorem 3.7.1 (preprint 3.58) for a general sc-splitting (§III.63); an sc-smooth family of projections whose rank jumps (§III.64); Theorem 3.6.8 (preprint 3.54) with untamed 2021 germ data, a counterexample (§III.65); and what is left and what was learnt (§III.66).

**What is proved, and what is not.** All of Chapter 3 is now formalized. The sixteen gaps the comparison listed for the chapter (G.16–G.30 and G.79) are closed, G.79 by a counterexample, and no hypothesis remains that the text neither states nor takes in its proof.

*The germ implicit function theorem.* It is proved with the text's hypotheses (Theorems 3.3.2–3.3.7 (preprint 3.29–3.34) and Theorem 3.35 (preprint; not in the 2021 edition)), and in the quantitative form of Remark 3.3.8 (preprint 3.36) (§III.58, G.19). What the formalization adds is precision about *where* the solution germ is regular. A germ is formalized as a fixed sequence of sets, and $\delta$ is $\mathrm{sc}^k$ on the germ shifted by $s_k = 2^k - 1$ levels. So "if $f$ is sc-smooth so is its solution germ" holds for every $k$ on a germ depending on $k$ (§III.11).

*Germs, sections and their linearization.* - Definition 3.1.11 (preprint 3.11) is formalized as the text writes it, with the filling and the $\mathrm{sc}^+$-section sc-smooth *germs* (§§III.6, III.48). - Proposition 3.1.10 (preprint 3.10) holds on every level (§III.5, G.21). - Proposition 3.6.5 (preprint 3.52) gives an sc-Fredholm operator in every chart (§§III.30, III.57, G.22). - Corollary 3.4.3 (preprint 3.39) is §III.28 (G.20), and Theorem 3.1.21 (preprint 3.18) is §III.40 (G.16).

*Theorem 3.2.3 (preprint 3.27)* is proved under the tangent germ property alone (§III.56). The two regularity conditions with which it was first proved (§III.24) are theorems: re-graphing over a projecting good complement gives the one, re-centring the other. The closedness of the good complement in Definition 3.2.1 (preprint 3.25) and Lemma 3.6.9 (preprint 3.55), once listed as an added hypothesis, follows from the text's sc-complement. Two steps of the text's proof do not hold as we read it, each with a Lean counterexample (§III.56).

*The implicit function theorems.* Theorem 3.6.8 (preprint 3.54) is proved with the text's hypotheses (§§III.41–III.43, G.23). With it come Theorems 3.1.22 and 3.1.26 (preprint 3.19 and 3.23), Corollary 3.1.27 (preprint 3.24) and Theorems 3.6.10 and 3.6.11 (preprint 3.56 and 3.57) (§§III.37, III.43, G.17, G.18). The dimension of the solution set is the Fredholm index (§III.44), and $\ker f'(y) = T_yS$ holds in $T_yX$ for the intrinsic linearization (§III.62, G.30). All of these are also stated in the book's form with the 2021 definitions (§III.57).

"*Tame strong bundle*". The phrase is read as germ data and good position in a tame chart (§III.47). This is the wording of the 2017 text and the preprint, and the chart the 2021 proof takes on p. 137; it is not derived from the 2021 text as printed. The reading is needed: with the 2021 germ data as printed, in any compatible chart, Theorem 3.6.8 does not hold as we read it. The witness is a tame strong bundle over $\mathbb{R}$ whose solution set near a zero in good position is a closed half-line (§III.65, G.79). The reading is satisfiable: the graph section of a smooth map of a trivial bundle has tame germ data (§III.47).

*The basic germ criterion*, Theorem 3.7.1 (preprint 3.58), is proved as printed, for a general sc-splitting (§III.63, G.24). The proof reduces it to the coordinates of the text's proof (§III.38).

*The appendix.* - Lemmas 3.8.1–3.8.3 and 3.8.9–3.8.11 (preprint 3.62–3.64 and 3.66–3.68) are §III.59 (G.25, G.26). - Definition 3.8.14 (preprint 3.71) agrees with the $C^k$ the formalization uses (§III.60, G.27). - Lemma 3.8.20 (preprint 3.76) is proved with Lemma 3.8.21 (preprint 3.77) in the general form its use needs (§III.60, G.28). - Lemma 3.8.13 (preprint 3.70) and Proposition 3.8.12 (preprint 3.69) hold for a general presentation of the partial quadrant, which gives $d_M = d_X$ in Theorem 3.2.3 (§III.61, G.29).

*From Chapter 6.* Definition 6.2.1 and Proposition 6.2.2 (preprint 6.1 and 6.2) are proved with the text's hypotheses for paths on $\mathbb{R}$ (§III.49).

*The four narrower rows.* Four comparison rows are narrower than the text. In each, the printed statement does not hold as we read it, and a Lean counterexample shows it: - V3.73: the re-centring step in the proof of Theorem 3.2.3, for every pair of data. The lemma is proved with the first graph map $C^1$ on its whole domain (§§III.35, III.56). - V3.120: Lemma 3.8.6 (2) at $N = 0$ (§III.54). - V3.123: Lemma 3.8.1 at $P = \{0\}$ (§III.59). - V3.130: Theorem 3.6.8 with the 2021 germ data as printed (§III.65).

**Seven structural choices.** (1) Germs are fixed sequences of sets, and the calculus the text uses on them silently is proved (§III.1). (2) Contraction germs carry explicit contraction regions instead of the text's balls, because the regions of the iterated germ $f^{(1)}$ must be unbounded in the tangent directions (§III.7). (3) "A strong bundle chart" in the definitions of §3.1 is a chart *compatible* with the atlas (§III.2). (4) Tameness, which the text assumes throughout, is used only in Theorem 3.6.8 (preprint 3.54), where it cannot be dropped (§III.65), and only for the chart of the germ data at the zero: through Proposition 2.5.3 (preprint 2.35), $(1 - Dr(0))E \subseteq E_0$, and through the tameness of the model of the chart $\psi \circ \varphi$ (§III.27). The theorems assume germ data and good position in tame charts, as Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) do; the first formalization assumed every compatible chart tame, which cannot be met (§III.47). (5) The coordinate change of Theorem 3.4.2 (preprint 3.38) is linear, which makes the transport of solution sets in §3.5 linear algebra. (6) Classical regularity in the appendix is Mathlib's $C^k$ within a set, and the iterations of the text's Lemmas 3.8.20 and 3.8.26 (preprint 3.76 and 3.82) are replaced by bootstrapping a derivative formula; the text's Definition 3.8.14 (preprint 3.71) is shown to be this notion, and Lemma 3.8.20 and Lemma 3.8.21 in the general form its use needs are proved as well (§III.60). (7) In Definition 3.1.11 (preprint 3.11) the filling $g$ and the sc$^+$-section $s$ are germs on germs of neighbourhoods of their own, and the basic germ is shrunk to the germ $G''$ of the points that $\psi^{-1}$ sends into both, so that the nearby zeros produced by the implicit function theorem are points where $g$ and $s$ are sc$^1$ (§§III.6, III.48).

**Map.**

- `Chapter3/Germ`, `Chapter3/GermLevels`: Definitions 3.1.1, 3.1.2 (preprint 3.1, 3.2); the calculus of germs; Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs.
- `Chapter3/StrongBundleSections`: the section algebra, compatible charts, Definitions 3.1.3, 3.1.6, 3.1.11, 3.1.16 (preprint 3.3, 3.6, 3.11, 3.14), Theorem 3.1.18 (preprint 3.15).
- `Chapter3/Filler`, `Chapter3/FillerLinearAlgebra`, `Chapter3/FillerLinearization`: Definition 3.1.4 (preprint 3.4), Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17), Proposition 3.1.5 (preprint 3.5), Remark 3.6.2 (preprint 3.49).
- `Chapter3/BasicGermLinearization`: Lemma 3.1.9 (preprint 3.9) at 0, Proposition 3.1.10 (preprint 3.10).
- `Chapter3/ContractionGerm`, `Chapter3/GermIFT`: Definition 3.3.1 (preprint 3.28), Theorems 3.3.2, 3.3.3 (preprint 3.29, 3.30), Lemma 3.3.4 (preprint 3.31), the induction (Lemma 3.3.5 (preprint 3.32), Theorem 3.3.6 (preprint 3.33)), Corollary 3.3.7 (preprint 3.34), Theorem 3.35 (preprint; not in the 2021 edition).

- `Chapter3/LocalCompactness`, `Chapter3/SectionCompactness`: the parametrized fixed point theorem, Definition 3.1.7 (preprint 3.7), Theorem 3.5.1 (preprint 3.40), Theorem 3.1.13 (preprint 3.12), Corollary 3.1.15 (preprint 3.13).
- `Chapter3/GoodPosition`, `Chapter3/GoodPositionGraph`: Definition 3.1.24 (preprint 3.21), Proposition 3.1.25 (preprint 3.22), Lemmas 3.5.9, 3.8.8 (preprint 3.45, 3.65), Proposition 3.8.12 (preprint 3.69), Lemma 3.8.13 (preprint 3.70).
- `Chapter3/QuadrantInverse`, `Chapter3/QuadrantIFT`: Theorems 3.8.15, 3.8.22 (preprint 3.72, 3.78) with their lemmas.
- `Chapter3/BasicClass`, `Chapter3/SolutionLinearization`, `Chapter3/LocalSolvability`: Theorem 3.4.2 (preprint 3.38), Corollary 3.5.3 (preprint 3.41), Lemmas 3.5.5, 3.5.8, 3.5.10 (preprint 3.42, 3.44, 3.46), Theorem 3.5.6 (preprint 3.43), Theorem 3.5.11 (preprint Corollary 3.47).
- `Chapter3/FredholmGermLinearization`, `Chapter3/GoodPositionGerm`, `Chapter3/GoodPositionOpen`, `Chapter3/GlobalIFT`: Proposition 3.6.5 (preprint 3.52), Definition 3.6.6 (preprint 3.53), Lemma 3.6.9 (preprint 3.55), Theorem 3.6.8 (preprint 3.54) (the linear part and the graph description), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57) from the tangent germ property.
- `Chapter3/TangentGerm`: Definition 3.2.1 (preprint 3.25), Proposition 3.2.2 (preprint 3.26), Theorem 3.2.3 (preprint 3.27).
- `Chapter3/BasicGermCriterion`, `Chapter3/InverseFamily`: Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) (abstract form), Lemma 3.7.4 (preprint 3.61).
- Added later (§§III.27–III.39): `Chapter2/TameInvariance` (tameness under sc-diffeomorphisms); `Chapter3/BasicClassComposition` (Corollary 3.4.3 (preprint 3.39), the calculus of strong bundle isomorphisms); `Chapter3/GoodComplementProjection`, `Chapter3/LocalSolvabilityClosed` (good complements with a projection, Theorems 3.5.6, 3.5.11 (preprint 3.43, 3.47) with a closed projecting complement); `Chapter3/FredholmGermIndex` (Proposition 3.6.5 (preprint 3.52) with its index); `Chapter3/LinearizationCharts` (the linearization at a zero in charts, chart independence, Definition 3.6.6 (preprint 3.53) from a non-centred chart); `Chapter3/GlobalIFTFull`, `Chapter3/GlobalIFTNearbyGraph`, `Chapter3/GlobalIFTNearbyKernel`, `Chapter3/GlobalIFTNearbySurj`, `Chapter3/GlobalIFTNearbyPerturb` (Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros); `Chapter3/GlobalIFTCenter`, `Chapter3/SolutionGermRegularity` (Theorem 3.6.8 (2) (preprint 3.54) at the centre); `Chapter3/SolutionGermRecentre` (the re-centring, regular data, Proposition 3.8.17 (preprint 3.73) / Lemma 3.8.21 (preprint 3.77) with weak hypotheses); `Chapter3/GlobalIFTTheorems` (Theorem 3.2.3 (preprint 3.27) with locally constant dimension, Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19), Corollary 3.1.27 (preprint 3.24), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)); `Chapter3/BasicGermCriterionProof`, `Chapter3/BasicGermCriterionIndex` (Theorem 3.7.1 (preprint 3.58)).
- Added in the second addendum (§§III.40–III.46): `Chapter3/ParamPerturbation` (Theorem 3.1.21 (preprint 3.18), the pull-back bundle, inert parameters, the linearization of the perturbed section); `Chapter3/GlobalIFTNearbyOntoLemmas`, `Chapter3/GlobalIFTNearbyOntoGerm`, `Chapter3/GlobalIFTNearbyOntoXi`, `Chapter3/GlobalIFTNearbyOntoCurve`, `Chapter3/GlobalIFTNearbyOnto` (Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19) with them); `Chapter3/GlobalIFTIndex` (the index of the linearization in every chart, dimension = index, the index locally constant, the kernel as tangent space, good position at interior zeros, the filler condition as a theorem).
- Added with the repair of the tameness hypothesis (§III.47): `Chapter3/TameGerms` (tame charts, Definitions 3.6.3, 3.6.4, 3.6.6 (preprint 3.50, 3.51, 3.53) in a tame chart, good position read in a tame chart, Theorem 3.1.18 (preprint 3.15) for tame germ data); `Chapter3/TameVacuity` (a non-tame retraction

onto a product with a point).

- Added with the germ reading of Definition 3.1.11 (preprint 3.11) (§III.48): `Chapter3/GermVsChart` (a germ sc-smooth on no neighbourhood); `Chapter3/GermFilling`, `Chapter3/GermFillingChart` (Definitions 3.1.6 and 3.1.11 (preprint 3.6 and 3.11) on germs, the bridges to the whole-chart notions); `Chapter3/GermCalculus`, `Chapter3/GermCalculusParam`, `Chapter3/GermPushforward`, `Chapter3/GermFillerLinearization` (sums, chain rules with whole-chart maps, the fill defect, push-forwards and products of germs, Proposition 3.1.5 (preprint 3.5) for filling germs); `Chapter3/BasicGermShrink`, `Chapter3/BasicGermShrinkIFT` (the shrunk basic germ $G''$ and Corollary 3.5.3 (preprint 3.41) on it); and the files `Chapter3/…G` (`FredholmSectionsG`, `TameGermsG`, `GoodPositionGermG`, `FredholmGermLinearizationG`, `GlobalIFTNearbyGraphG`, `GlobalIFTTheoremsG`, …), where the canonical names of Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4, 3.6.6 (preprint 3.11, 3.14, 3.50, 3.51, 3.53) and of the theorems built on them now live. Their original files (`StrongBundleSections`, `GlobalIFT`, …) keep the whole-chart versions, named `…WholeChart` / `…_wholeChart`.
- Added in the fourth addendum (§§III.47, III.49): `Chapter3/TameGermData` (condition (3) of Definition 3.1.4 (preprint 3.4) as a condition on the chart; a tame atlas whose chart carries no germ data); `Chapter3/IndexAlongPaths`, `Chapter3/IndexAlongPathsDevice` (Definition 6.2.1 (preprint 6.1), bump functions on $\mathbb{R}^n \times X$, the section along a path, the slice of a section of the pull-back, Proposition 6.2.2 (preprint 6.2)).
- Added in the fifth addendum (§III.49): `Chapter3/RestrictOpen` (the restriction of M-polyfolds and strong bundles to open sets); `Chapter3/RestrictOpenGerms` (germ data and the index on the restriction); `Chapter3/IndexLocallyConstantLocal` (sc$^+$-sections through the fibre points from a chart over the whole base, the index along the zero set without bump functions); `Chapter3/IndexAlongPathsLocal` (Definition 6.2.1 (preprint 6.1) with local sections, Proposition 6.2.2 (preprint 6.2) with the text's hypotheses); `Chapter3/IndexAlongPathsLocalBump` (the two readings of Definition 6.2.1 (preprint 6.1) agree under bump functions).
- Added with the 2021 edition (§§III.50–III.55): `Chapter3/Book2021Germs` (the 2021 filling and germs, the corner filling, the step on p. 138, Theorem 3.5.1 (preprint 3.40) with the sequential conclusion); `Chapter3/Book2021Items` (cones, Lemmas 3.8.4–3.8.6, Corollary 3.8.7); `Chapter3/Counterexamples2021` (the set $\Sigma$ of Theorem 3.2.3 (preprint 3.27), Lemma 3.8.6 (2) at $N = 0$).
- Added in the November 2026 waves (§§III.56–III.65): `Chapter3/TangentGermProjecting`, `Chapter3/GoodComplementClosed`, `Chapter3/RecentreCounterexample2021` (Theorem 3.2.3 (preprint 3.27) under the tangent germ property alone, the closed good complement, the re-centring step); `Chapter3/BasicGermLinearizationLevel` (Proposition 3.1.10 (preprint 3.10) on every level); `Chapter3/MainTheorems2021`, `Chapter3/BookFormIFT2021` (the implicit function theorems with the 2021 definitions); `Chapter3/GapP1_Remark338`; `Chapter3/GapP1_ExtremeRays`, `Chapter3/GapP1_Lemma389`, `Chapter3/GapP1_Lemma389Sc` (Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11); `Chapter3/GapP1_Definition3814`, `Chapter3/GapP1_Lemma3821`, `Chapter3/GapP1_Lemma3820`; `Chapter3/GapP1_Lemma3813`, `Chapter3/GapP1_Theorem323Deg` (general presentations, $d_M = d_X$); `Chapter3/GapP1_LinKey`, `Chapter3/GapP1_LinIntrinsic`, `Chapter3/GapP1_Theorem3122Ker` (the intrinsic linearization, $\ker f'(y) = T_y S$); `Chapter3/GapP1_ScIsoConj`, `Chapter3/GapP1_Theorem371Split`, `Chapter3/GapP1_Theorem371Transport`, `Chapter3/GapP1_Theorem371General` (Theorem 3.7.1 (preprint 3.58) as printed); `Chapter3/TameGermTransfer` (tame germ data at interior points); `Chapter3/GapP1_UntamedJump`, `Chapter3/GapP1_UntamedCounterexample`, `Chapter3/GapP1_UntamedBundle`, `Chapter3/GapP1_UntamedBundleFrame`, `Chapter3/GapP1_UntamedBundleTame`, `Chapter3/GapP1_UntamedBundleTheorem` (the jumping projections and the counterexample for untamed germ data).

Throughout, $P$ and $Q$ are finite-dimensional normed spaces with the constant sc-structure (the text's $\mathbb{R}^n$ and $\mathbb{R}^N$), $\widetilde{C} \subseteq P$ a set containing $0$, convex, with nonempty interior (the text's $[0,\infty)^k \oplus \mathbb{R}^{n-k}$), $W$ an sc-Banach space, $E = P \oplus W$ with levels $P \times W_m$ `Chapter3/ContractionGerm · ScBanach.paramProd`. Because all levels of $P$ coincide, the condition "$\widetilde{C} \times W$ is quadrant-like" is just "$0 \in \widetilde{C}$, convex, nonempty interior" `Chapter3/ContractionGerm · isQuadrantLike_const_iff` `Chapter3/ContractionGerm · IsQuadrantLike.paramProd`.

## III.1 Germs and their calculus (Definitions 3.1.1, 3.1.2 (preprint 3.1, 3.2))

**Definition 3.1.1 (preprint 3.1).** An *sc-germ of neighbourhoods* around $0 \in C$ is a family of sets $U_m \subseteq E_m$, each of the form $O \cap C_m$ with $O$ open in $E_m$, containing $0$, and nested along the inclusions: $\iota U_{m+1} \subseteq U_m$ `Chapter3/Germ · ScNbhdGerm`. Its *tangent germ* has levels $U_{m+1} \times E_m$ `Chapter3/Germ · ScNbhdGerm.tangent`. The text's example, a relatively open $U \ni 0$ with $U_m = U \cap E_m$, is `ofSet` `Chapter3/Germ · ScNbhdGerm.ofSet`. The germ calculus needs one property of $C$: every level $C_m$ is convex with nonempty interior and $0 \in C$ ("quadrant-like") `Chapter3/Germ · IsQuadrantLike`. Partial quadrants have it `Chapter3/Germ · IsPartialQuadrant.isQuadrantLike`, and it passes to $TC$ `Chapter3/Germ · IsQuadrantLike.tangentSet`. It makes every $U_m$ a set of unique differentiability `Chapter3/Germ · ScNbhdGerm.uniqueDiffWithinAt`, the intersection of an open and a convex set `Chapter3/Germ · ScNbhdGerm.relOpenConvex`.

A germ is a *fixed* sequence of sets. The text says "the size of the sets $U_m$ does not matter", i.e. it reads germs up to shrinking; the formalization cannot, and must say on which sequence each statement holds. One trivial remark becomes important in §III.9: the re-indexed sequence $(U_{m+1})_m$ is in general not a germ, since $U_{m+1}$ is relatively open in $C_{m+1}$ and not in $C_m$; it is one when all levels coincide, i.e. for germs in the finite-dimensional parameter space $P$.

**Definition 3.1.2 (preprint 3.2).** $f$ is an *sc⁰-germ* on $G = (U_m)$ if $f(U_m) \subseteq F_m$ and the level map $f_m : U_m \to F_m$ is continuous for every $m$ `Chapter3/Germ · IsSc0Germ`. It is an *sc¹-germ with derivative family* $D$ if it is sc⁰, differentiable at every $x \in U_1$ within $U_1$ in the sense of Definition 1.1.15 (preprint 1.14) (increments in $E_1$, remainder measured in $F_0$) with derivative $D(x)$, and its tangent map $(x,h) \mapsto (f(x), D(x)h)$ is an sc⁰-germ on the tangent germ `Chapter3/Germ · IsSc1GermWith`; an sc¹-germ if some such $D$ exists `Chapter3/Germ · IsSc1Germ`, in which case a derivative family is chosen `Chapter3/Germ · scDerivGerm` and gives the tangent germ `Chapter3/Germ · tangentMapGerm`. On quadrant-like $C$ the derivative is unique on $U_1$ `Chapter3/Germ · IsSc1GermWith.eq_of_uniqueDiffWithinAt`, so the choice does not matter there `Chapter3/Germ · IsSc1Germ.tangentMapGerm_eqOn`. Then sc^k is defined by recursion — sc⁰ for $k = 0$, and sc¹ with sc^k tangent germ for $k+1$ `Chapter3/Germ · IsSckGerm` — and sc-smooth means sc^k for all $k$ `Chapter3/Germ · IsScSmoothGerm`. These are the notions of Part I, §I.11, with $U \cap E_m$ replaced by $U_m$, and the bridge is exact: an sc^k map on a relatively open $U \ni 0$ is an sc^k-germ on `ofSet U` `Chapter3/Germ · IsSck.isSckGerm` `Chapter3/Germ · IsScSmooth.isScSmoothGerm`.

**The calculus of germs.** The text restricts, composes and pairs germs without comment. All of it is proved, with the proofs of Part I, §§I.11–I.12. (1) sc^k-germs depend only on the values on $U_0$ `Chapter3/Germ · IsSckGerm.congr` and restrict to smaller germs `Chapter3/Germ · IsSckGerm.mono` — by induction on $k$, using that the tangent germs of the smaller germ and of the larger one agree on the smaller tangent germ `Chapter3/Germ · IsSc1Germ.tangentMapGerm_mono`. (2) *The chain rule.* A composition $g \circ f$ makes sense level by level only if $f$ maps $U_m$ into the domain $V_m$ of $g$ for every $m$ `Chapter3/Germ · ScNbhdGerm.MapsToGerm`; then sc⁰-germs compose `Chapter3/Germ · IsSc0Germ.comp`, the derivative of $g \circ f$ at $x \in U_1$ is $Dg(f(x)) \circ Df(x)$ — proved as in §I.12 with the mean value inequality along segments in $V_1$ and the uniform smallness of $Dg(y) - Dg(y_0)$ on compact sets `Chapter3/Germ · IsSc1GermWith.hasFDerivWithinAt_comp` — so $g \circ f$ is sc¹ with

$T(g\circ f)=Tg\circ Tf$ on the tangent germ `Chapter3/Germ · IsSc1GermWith.comp` `Chapter3/Germ · IsSc1Germ.comp`, and by induction on $k$ it is sc^k `Chapter3/Germ · IsSckGerm.comp` `Chapter3/Germ · IsScSmoothGerm.comp`. (3) An sc-operator $T$ is an sc¹-germ with derivative $T$ itself `Chapter3/Germ · ScOperator.isSc1GermWith`, its tangent map is the sc-operator $(T,T)$ `Chapter3/Germ · ScOperator.tangentOp`, and so it is an sc-smooth germ on every germ `Chapter3/Germ · ScOperator.isSckGerm` `Chapter3/Germ · ScOperator.isScSmoothGerm`; composing with one on the outside preserves sc^k `Chapter3/Germ · IsSckGerm.operator_comp`. (4) The pairing $(f,g)$ of two sc^k-germs is an sc^k-germ `Chapter3/Germ · pairMap` `Chapter3/Germ · IsSckGerm.pair`, through the reshuffle $TF\oplus TG\cong T(F\oplus G)$ `Chapter3/Germ · tangentProdComm`. (5) Where the text says "near $0$" it shrinks germs: for an sc⁰-germ $f$ with $f(0)=0$ and a germ $G'$ the sets $\{x\in U_m : f_m(x)\in U'_m\}$ form the largest sub-germ on which $f$ maps into $G'$ `Chapter3/GermFilling · ScNbhdGerm.preimage` `Chapter3/GermFilling · ScNbhdGerm.preimage_mapsToGerm`.

**Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs.** The text applies these to germs ("recalling Proposition 1.17" in Lemma 3.1.9 (preprint 3.9); Lemma 3.3.4 (preprint 3.31); Corollary 3.3.7 (preprint 3.34)) but states them for relatively open sets. For an sc¹-germ with derivative family $D$ and $x\in U_{m+1}$, $D(x)$ restricts to a bounded level operator $E_m\to F_m$ `Chapter3/GermLevels · IsSc1GermWith.exists_levelDeriv` `Chapter3/GermLevels · IsSc1GermWith.toBase_levelDeriv`, the level-$m$ tangent map is $(x,h)\mapsto(f_{m+1}(x),D_m(x)h)$ `Chapter3/GermLevels · IsSc1GermWith.levelMap_tangentMapWith`, jointly continuous `Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv` and continuous in $x$ in the operator norm (the compact-open lemma of §I.13) `Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv_clm`; and $f : U_{m+1}\to F_m$ is differentiable within $U_{m+1}$ with derivative $D_m(x)$, by the integral representation $f(x+k)-f(x)=\int_0^1 D(x+\tau k)k\,d\tau$ in $F_m$ (Proposition 1.2.1 (preprint 1.16)) `Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level`. The shifted sequence $(U_{m+1})_m$ is a germ in $E^1$ around $0\in C^1$ `Chapter3/GermLevels · ScNbhdGerm.shift`, and an sc^k-germ on $G$ is an sc^k-germ $G^1\to F^1$ (Proposition 1.2.2 (preprint 1.17)) `Chapter3/GermLevels · IsSc0Germ.shift` `Chapter3/GermLevels · IsSc1GermWith.shift` `Chapter3/GermLevels · IsSckGerm.shift`. An sc^k-germ is $C^k$ from $U_{m+k}$ to $F_m$ (Proposition 1.2.3 (preprint 1.18)), by the proof of §I.16 with $U_m$ for $U\cap E_m$ `Chapter3/GermLevels · IsSckGerm.contDiffOn_levels'` `Chapter3/GermLevels · IsSckGerm.contDiffOn_levels`. sc^k depends on a germ only through its sets `Chapter3/GermLevels · IsSckGerm.of_U_eq`.

## III.2 Sections of strong bundles in charts (Definition 3.1.3 (preprint 3.3))

Let $P : Y\to X$ be a strong bundle over an M-polyfold (Part II, §II.19). A strong bundle chart $(\Phi,P^{-1}(V),K,U\lhd F)$ has a model $K=R(U\lhd F)$ for a strong bundle retraction $R$. Its base and fibre parts are read off: $r(u)=(R(u,0))_1$ `Chapter3/Filler · StrongRetr.r`, $\rho(u)h=(R(u,h))_2$ `Chapter3/Filler · StrongRetr.ρ`, and $R(u,h)=(r(u),\rho(u)h)$ because the base component of a strong bundle map does not depend on the fibre variable `Chapter3/Filler · StrongRetr.apply_eq`. Comparing components of $R\circ R=R$ gives $r\circ r=r$ `Chapter3/Filler · StrongRetr.r_r` and $\rho(r(u))\rho(u)=\rho(u)$ `Chapter3/Filler · StrongRetr.ρ_r_ρ`; for $x\in O=r(U)$, $r(x)=x$ `Chapter3/Filler · StrongRetr.r_eq_self` and $\rho(x)$ is a projection `Chapter3/Filler · StrongRetr.ρ_ρ`; $R$ fixes the points of $K$ `Chapter3/Filler · StrongRetr.apply_eq_self_of_mem_K`, so $\rho(x)h=h$ for $(x,h)\in K$ `Chapter3/Filler · StrongRetr.ρ_eq_self_of_mem_K`. $\rho(u)$ is linear by Definition 2.6.1 (preprint 2.47) and continuous because $R$ is sc⁰ on level $0$ `Chapter3/Filler · StrongRetr.ρ_continuous`; it is a bounded operator `Chapter3/Filler · StrongRetr.ρCLM` — the text's "$\rho(u):F\to F$ is a bounded linear operator", without comment.

*Principal parts and the section algebra.* For a section $f$ (a map with $P\circ f=\mathrm{id}$), $\Phi\circ f\circ\varphi^{-1}(z)=(z,\mathbf{f}(z))$; $\mathbf{f}$ is its *principal part* `Chapter3/StrongBundleSections · StrongBundleChart.principal` `Chapter3/`

StrongBundleSections · StrongBundleChart.apply_section. The fibrewise sum $f + s$ is defined with the vector space structure of the fibres $P^{-1}(x)$ Chapter3/StrongBundleSections · sectionAdd and is a section Chapter3/StrongBundleSections · IsSection.sectionAdd. Since $\Phi$ is linear on the fibres Chapter3/StrongBundleSections · StrongBundleChart.snd_add and maps the zero of $P^{-1}(x)$ to $(\varphi(x), 0)$ Chapter3/StrongBundleSections · StrongBundleChart.snd_fiberZero, the principal part of $f + s$ is $\mathbf{f} + \mathbf{s}$ Chapter3/StrongBundleSections · StrongBundleChart.principal_sectionAdd and $f(x) = 0$ iff $\mathbf{f}(\varphi(x)) = 0$ Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff Chapter3/StrongBundleSections · IsZeroAt. The fibres of $K$ over $O$ are linear: $(u, a), (u, b) \in K$ give $(u, a + b) \in K$, because membership is $\rho(u)a = a$ Chapter3/StrongBundleSections · StrongBundleChart.add_mem_K. *The sum rule*, used silently in "$f + s$ is an sc-smooth section", is pairing (Part I) followed by the addition sc-operator $F \oplus F \to F$ Chapter3/StrongBundleSections · ScBanach.addOp Chapter3/StrongBundleSections · IsScSmooth.add Chapter3/StrongBundleSections · IsScSmooth.sub Chapter3/StrongBundleSections · IsScSmooth.neg; a map sc-smooth into $F^1$ is sc-smooth into $F$ (compose with the inclusion) Chapter3/StrongBundleSections · IsScSmooth.toBase_shift.

*Which charts.* "A strong bundle chart" in Definitions 3.1.3, 3.1.6 and 3.1.11 (preprint 3.3, 3.6 and 3.11) must mean a chart *compatible* with the atlas of $P$ Chapter3/StrongBundleSections · StrongBundle.IsChart Chapter3/StrongBundleSections · StrongBundle.isChart_of_mem: a chart of the atlas need not be centred at the point in question, and the notions must not depend on the atlas within its class. A compatible chart is a chart of the M-polyfold $Y[0]$ Chapter3/StrongBundleSections · StrongBundle.IsChart.toMChart. The price is that representatives of sections must be sc-smooth in every compatible chart. For an sc-smooth section, $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$ by Definition 2.3.8 (preprint 2.14) as it stands Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothRetractMap. For an sc⁺-section, sc-smooth into $Y[1]$, the principal part takes values in $F_1$ in every compatible chart Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.principal_mem_range, and $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$: in a chart $\Phi'$ of the atlas this is the definition, and in the given chart it is the composition with the change of chart $(\Phi \circ \Phi'^{-1})[1]$ on $K[1]$, glued over the atlas by locality Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.isScSmoothRetractMap_one. This is precisely the compatibility on $[1]$ of Definition 2.6.4 (preprint 2.50), used here for the first time.

**Definition 3.1.3 (preprint 3.3).** $(f, x_0)$ is an *sc-smooth section germ* if $x_0$ is smooth and in some compatible strong bundle chart with $\varphi(x_0) = 0$ the map $\mathbf{f} \circ r$ is an sc-smooth germ on some sc-germ of neighbourhoods $G$ of $0$ with $G_0 \subseteq U$, whose levels may shrink with $m$ Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSectionGerm — the text's germ reading, as for Definition 3.1.11 (preprint 3.11) (§III.48); until the fourth addendum the germ was the chart's $U \cap E_m$, which is stricter (§III.48). The text's "continuous section on a neighbourhood $V$" is the section on the chart domain; near $x_0$ continuity is a consequence. An sc-smooth section is an sc-smooth section germ at every smooth point with a centred chart: its representative composed with $r$ is sc-smooth on the relatively open $U$, and §III.1's bridge applies on the germ $U \cap E_m$ Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothSectionGerm. No theorem assumes the definition.

## III.3 Fillers (Definitions 3.1.4, 3.1.6 (preprint 3.4, 3.6); Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17); Remark 3.6.2 (preprint 3.49))

**Definition 3.1.4 (preprint 3.4).** For the principal parts $f, g : E_0 \to F_0$ (with $O = r(U)$, $K = R(U \lhd F)$, $0 \in U$), *$g$ fills $f$* if (1) $f = g$ on $O$ near 0; (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in $O$; (3) the map $y \mapsto [1 - \rho(r(y))]g(y)$ is sc¹ on $U$, and its derivative at 0 restricts to a topological linear

isomorphism $\ker Dr(0) \to \ker \rho(0)$ `Chapter3/Filler · IsFillerOf`. The text's condition (3) presupposes the linearization; requiring the map to be sc¹ loses nothing, since it is sc-smooth when $g$ and $R$ are. The sc-smoothness of $g$ itself is added in Definition 3.1.6 (preprint 3.6). **Definition 3.1.6 (preprint 3.6)**: a *filled version* of **f** is a germ $(g, 0)$, and it is formalized as one, a *filling germ* `Chapter3/GermFilling · IsFillingGerm`: $g$ is an sc-smooth germ (§III.1) on a germ $G$ with $G_0 \subseteq U$, whose levels may shrink with $m$; (1) and (2) hold near 0; and in (3) the fill defect $k(y) = [1 - \rho(r(y))]g(y)$ `Chapter3/GermFilling · fillDefect` is an sc¹-germ on $G$ with a derivative family $Dk$ whose value $Dk(0)$ — unique (§III.1) — restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$. The first formalization took a filler sc-smooth on the whole $U$ of the chart `Chapter3/StrongBundleSections · IsFilledVersion`, on the ground that a chart can always be restricted to $U \cap W$. That is the special case \$G = \$ `ofSet U` `Chapter3/GermFillingChart · IsFilledVersion.isFillingGerm` `Chapter3/GermFillingChart · IsFillingGerm.isFilledVersion`, and it is narrower: restricting a chart on level 0 never makes the levels shrink, and there are sc-smooth germs with shrinking levels that are sc-smooth on no neighbourhood of 0 `Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth` (§III.48). The lemmas below about fillers on $U$ are the whole-chart ones; their germ forms, which the canonical notions use, are cited next to them.

**Lemma 3.1.19 (preprint 3.16).** For a section $s$ of $K \to O$ (that is, $(x, s(x)) \in K$ for $x \in O$), the extension $\tilde{s} = s \circ r$ satisfies $\tilde{s} = s$ on $O$ (as $r(x) = x$ there), and $R(r(u), s(r(u))) = (r(u), s(r(u)))$, i.e. $\rho(r(u))\tilde{s}(u) = \tilde{s}(u)$, because $s(r(u))$ is a point of $K$ and $R$ fixes $K$ `Chapter3/Filler · lemma_3_1_19`. The text's computation uses $s(x) \in K$ without saying so; it is the hypothesis that $s$ is a section. **Lemma 3.1.20 (preprint 3.17).** If $g$ fills $f$ and $s$ is a section of $K \to O$, then $g + s \circ r$ fills $f + s$ `Chapter3/Filler · lemma_3_1_20`: (1) on $O$, $g + s \circ r = f + s$; (2) by Lemma 3.1.19 (preprint 3.16), $(g + \tilde{s})(y) = \rho(r(y))(g + \tilde{s})(y)$ is equivalent to $g(y) = \rho(r(y))g(y)$; (3) by the same identity $[1 - \rho(r(y))](g + \tilde{s})(y) = [1 - \rho(r(y))]g(y)$ on $U$, so the two maps of (3) coincide on $U$ and have the same derivative at 0 (derivatives on $U_1$ are unique). *For a filling germ* $g$ on $G$ and $s$ sc-smooth on $U$ the proof is the same: the identity of (3) holds on $U$, hence on $G_0$, so the derivative family $Dk$ of the fill defect of $g$ is one of $g + s \circ r$, a filling germ on the same $G$ `Chapter3/GermCalculus · IsFillingGerm.add_section`; for $s$ itself a germ on $G_s$ the filling lives on $G \cap r^{-1}(G_s)$ `Chapter3/GermCalculus · IsFillingGerm.add_section_germ`. The regularity statements of both lemmas — $\tilde{s}$ sc⁺, $g + \tilde{s}$ sc-smooth — are proved where they are used, in Theorem 3.1.18 (preprint 3.15) (§III.6). Condition (1) only involves $f$ on $O$ `Chapter3/StrongBundleSections · IsFillerOf.congr_left`.

**Remark 3.6.2 (preprint 3.49).** On a smaller neighbourhood $W$ of 0 condition (1) holds on all of $O \cap W$ and (2) for all $y \in U \cap W$: intersect the two neighbourhoods given by (1) and (2) `Chapter3/FillerLinearization · IsFillerOf.remark_3_6_2`.

## III.4 The linearization of a filler (Proposition 3.1.5 (preprint 3.5))

The text derives, "by linearization at $y = 0$", the matrix of $Dg(0)$ in the splittings $E = Dr(0)E \oplus \ker Dr(0)$, $F = \rho(0)F \oplus \ker \rho(0)$, and from it Proposition 3.1.5 (preprint 3.5). The formal proof has two halves: the derivative identities, and the linear algebra.

*The derivative identities.* $r$ is sc-smooth on $U$ `Chapter3/FillerLinearization · StrongRetr.r_isScSmooth` with $r(0) = 0$ since $0 \in O$ `Chapter3/FillerLinearization · StrongRetr.r_zero`, and the chain rule applied to $r \circ r = r$ makes $Dr(0)$ idempotent `Chapter3/FillerLinearization · StrongRetr.scDeriv_r_idem`; $\rho(0)$ is idempotent `Chapter3/FillerLinearization · StrongRetr.ρCLM_idem`. The fibre part $(u, k) \mapsto \rho(u)k$ is sc¹ on $U \oplus F$ `Chapter3/FillerLinearization · StrongRetr.fiber_isSc1`, and its derivative at $(0, 0)$ is $(h, k) \mapsto \rho(0)k$: in the fibre direction because $k \mapsto \rho(0)k$ is linear, in the base direction because $\rho(u)0 = 0$;

both directional derivatives are computed on the dense level 1 and extended by continuity Chapter3/FillerLinearization · StrongRetr.scDeriv_fiber_zero. By the chain rule for $(u,k) \mapsto \rho(u)k$ composed with $y \mapsto (r(y), g(y))$ at the point $(r(0), g(0)) = (0,0)$ — this is where $g(0) = 0$ enters — the derivative of $y \mapsto [1-\rho(r(y))]g(y)$ at 0 is $(1-\rho(0))Dg(0)$ Chapter3/FillerLinearization · IsFillerOf.scDeriv_sub_eq Chapter3/FillerLinearization · scDeriv_comp_apply'. Next, $[1-\rho(r(y))]g(r(y)) = 0$ near 0, from $g = f$ on $O$ and $\rho(x)f(x) = f(x)$ on $O$ ($f$ a section of $K$) and the continuity of $r$; linearizing gives $(1-\rho(0))Dg(0)Dr(0) = 0$ Chapter3/FillerLinearization · IsFillerOf.deriv_comp_r_eq_zero, i.e. $\rho(0)Dg(0)Dr(0) = Dg(0)Dr(0)$ Chapter3/FillerLinearization · IsFillerOf.ρ_deriv_comp_r. Condition (3) becomes: $(1-\rho(0))Dg(0)$ restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$ Chapter3/FillerLinearization · IsFillerOf.exists_kerEquiv. Finally $f \circ r = g \circ r$ near 0, so $D(f\circ r)(0) = Dg(0)Dr(0)$ — the text's "$Tg(0) = Tf(0)$ on $T_0O$" Chapter3/FillerLinearization · IsFillerOf.scDeriv_section_comp_r.

*The linear algebra.* Let $p$, $q$ be bounded idempotents on Banach spaces $E$, $F$ and $L : E \to F$ bounded with $(1-q)Lp = 0$ and $(1-q)L : \ker p \to \ker q$ a topological isomorphism $C$. Then $A := L|_{\operatorname{range} p} : \operatorname{range} p \to \operatorname{range} q$ is well defined Chapter3/FillerLinearAlgebra · fillerRestrict. Two facts carry everything: if $(1-q)Lw = 0$ then $w \in \operatorname{range} p$ (injectivity of $C$, applied to the $\ker p$-component of $w$) Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_of; and $(1-q)Lz = y$ is solvable in $\ker p$ for $y \in \ker q$ (surjectivity of $C$) Chapter3/FillerLinearAlgebra · fillerRestrict_exists_of. From them: $A$ is onto iff $L$ is Chapter3/FillerLinearAlgebra · fillerRestrict_surjective_iff; the inclusion $\ker A \to \ker L$ is a bijection — the text's $\ker Dg(0) = \ker Tf(0) \oplus \{0\}$ Chapter3/FillerLinearAlgebra · fillerRestrict_kerEquiv; $\operatorname{range} A = \operatorname{range} L \cap \operatorname{range} q$ Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_iff and $\operatorname{range} q/\operatorname{range} A \cong F/\operatorname{range} L$ Chapter3/FillerLinearAlgebra · fillerRestrict_cokerEquiv. So the indices are equal as $\dim\ker - \dim\operatorname{coker}$ with no finiteness assumption Chapter3/FillerLinearAlgebra · fredholmIndex_fillerRestrict. Closedness of the ranges transfers both ways: $\operatorname{range} A$ is closed in $\operatorname{range} q$ when $\operatorname{range} L$ is Chapter3/FillerLinearAlgebra · fillerRestrict_isClosed_range_of, and conversely $\operatorname{range} L$ is the preimage of $\operatorname{range} A$ under the continuous $M(y) = y - LC^{-1}(1-q)y$ Chapter3/FillerLinearAlgebra · isClosed_range_of_fillerRestrict — this is where "topological" in (3) is used. Hence $A$ is Fredholm iff $L$ is Chapter3/FillerLinearAlgebra · fillerRestrict_isFredholm_iff, and the three statements together Chapter3/FillerLinearAlgebra · proposition_3_1_5_preprint. The text's proof factorizes $\begin{pmatrix} A & B \\ 0 & C \end{pmatrix} = \begin{pmatrix} 1 & BC^{-1} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$ and uses the additivity of the index; the isomorphisms of kernels and cokernels avoid both.

**Proposition 3.1.5 (preprint 3.5).** For a filler $g$ of the section $f$ of $K \to O$, $g$ sc-smooth with $g(0) = 0$: with $p = Dr(0)$, $q = \rho(0)$, $L = Dg(0)$ the hypotheses of the linear algebra hold by the identities above, and $Tf(0) := Dg(0)|_{Dr(0)E} : Dr(0)E \to \rho(0)F$ is onto iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, with equal index Chapter3/FillerLinearization · IsFillerOf.proposition_3_1_5. (The text writes the target of the lower right entry as "$(1-D\rho(0))F$"; it is $(1-\rho(0))F$.) *For a filling germ* (§III.3) the proof is the same with any derivative family $Dg$ of $g$ on any germ inside $U$: the fill defect is an sc-smooth germ on the germ of $g$, by the chain rule with the whole-chart outer map $R$ (§III.48) Chapter3/GermCalculus · fillDefect_isScSmoothGerm, and the value at 0 of its derivative family is $(1-\rho(0))Dg(0)$ Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho, which replaces `scDeriv_sub_eq`; so $Tf(0) = Dg(0)|_{Dr(0)E}$ is onto, Fredholm, of the index of $Dg(0)$ iff $Dg(0)$ is Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_1_5. $Dg(0)$ does not depend on the family or the germ Chapter3/GoodPositionGermG · IsSc1GermWith.deriv_zero_eq.

## III.5 Basic germs, Lemma 3.1.9 (preprint 3.9) and Proposition 3.1.10 (preprint 3.10)

**Definition 3.1.7 (preprint 3.7) (and 3.37).** $f : P \times W_0 \to Q \times W_0$ is a *basic germ* on a germ $G$ around $0$ in $\widetilde{C} \oplus W$, with contraction map $B$, if $f$ is an sc-smooth germ with $f(0) = 0$ whose $W$-component is $w - B(a, w)$ on $U_0$, and $B$ is an sc$^0$-contraction germ in the sense of Definition 3.3.1 (preprint 3.28) (§III.7) `Chapter3/LocalCompactness · IsBasicGerm`. ($B = w - f_W$ is then itself an sc-smooth germ `Chapter3/LocalSolvability · IsBasicGerm.B_isScSmoothGerm`.)

**Lemma 3.1.9 (preprint 3.9).** The text: for every $\varepsilon > 0$ and $m$, $|D_2B(a,w)\zeta|_m \le \varepsilon|\zeta|_m$ when $(a, w) \in E_{m+1}$ is close to $0$ in $E_m$; in particular $D_2B(0,0) = 0$. Its proof: for $\zeta \in W_{m+1}$ the difference quotient of $t \mapsto B(a, w + t\zeta)$ is bounded by the contraction estimate, and the derivative is the limit (this needs $B : U_{m+1} \to W_m$ differentiable, Proposition 1.2.1 (preprint 1.16) for germs — "recalling Proposition 1.17" in the text); then density of $W_{m+1}$ in $W_m$. Only sc$^1$ is used. The formalization proves the three forms the chapter needs. (a) On level $m$ in the working form of §III.7, with the constant $\frac{1}{2}$: at $(a, w) \in U_{m+1}$ with $a \in N(m+1) \cap \widetilde{C}$, $w \in O(m+1)$ and $\iota w \in O(m)$, $\|D_2B(a,w)\|_{L(W_m)} \le \frac{1}{2}$ `Chapter3/ContractionGerm · ContractionGerm.norm_levelDeriv_snd_le`; a linear map of norm $\le \frac{1}{2}$ then has no nonzero fixed point `Chapter3/ContractionGerm · ContractionGerm.eq_zero_of_norm_le_half`. (b) $D_2B(0,0) = 0$, from the difference quotients along $t \mapsto (0, t\zeta)$, bounded by $\varepsilon|\zeta|$ for every $\varepsilon$, then density `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`. (c) On level $0$ with any $\varepsilon$: $|D_2B(x)\zeta|_0 \le \varepsilon|\zeta|_0$ for $x \in U_1$ small *on level* $1$ `Chapter3/SolutionLinearization · IsSc0ContractionGerm.exists_norm_deriv_snd_le` — the curve $t \mapsto x + t(0, \zeta)$, $\zeta \in W_1$, must stay in $U_1$, where $B$ is differentiable.

**Proposition 3.1.10 (preprint 3.10).** Let $\widetilde{C}$ be a partial quadrant, $h$ a basic germ, $s$ an sc$^1$-germ into $(Q \oplus W)^1$ (an sc$^+$-germ) with derivative family $Ds$, and $f = h + \iota s$ sc$^1$ with derivative family $Df$. Then $Df(0)$ is an sc-operator, sc-Fredholm of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · proposition_3_1_10`. *Proof.* The derivative of an sc$^1$-germ at the smooth point $0$ is an sc-operator, its level-$m$ map being the fibre component of the level-$m$ tangent map at $(0, \cdot)$ (the remark after Proposition 1.2.1 (preprint 1.16)) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`. The operator (32) of the text, $T_0(\delta a, \delta w) = (0, \delta w)$ `Chapter3/BasicGermLinearization · ScBanach.projW0p`, is sc-Fredholm with kernel $P \oplus 0$ and cokernel $Q \oplus 0$ `Chapter3/BasicGermLinearization · ScBanach.projWSplitting` `Chapter3/BasicGermLinearization · ScBanach.isScFredholm_projW0p`, of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · ScBanach.scIndex_projW0p`. The text says "$Df(0)$ is an sc$^+$-perturbation of (32)"; the formal proof builds the sc$^+$-operator $S = Df(0) - T_0$ level by level `Chapter3/BasicGermLinearization · proposition_3_1_10_scPlus`: its $Q$-component is sc$^+$ because $Q$ has all levels equal; $\mathrm{pr}_W Dh(0)(\delta a, 0) = -D_1B(0)\delta a = -DB(0)(\delta a, 0)$ has values in $W_\infty$ ($(\delta a, 0)$ is smooth and $DB(0)$ an sc-operator) and its level maps $E_m \to W_{m+1}$ factor through $DB(0)_{m+1}$; $\iota Ds(0)$ is sc$^+$ because $Ds(0)$ is an sc-operator into $(Q \oplus W)^1$; and $\mathrm{pr}_W Dh(0)(0, \delta w) = \delta w$ because $D_2B(0) = 0$ (Lemma 3.1.9 (preprint 3.9) (b)). The identities $Df(0) = Dh(0) + \iota Ds(0)$ and $\mathrm{pr}_W Dh(0) = \mathrm{pr}_W - DB(0)$ are derivative identities within $U_1$, extended to $E_0$ because two bounded operators agreeing on the dense $E_1$ are equal `Chapter3/BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one`. Proposition 1.1.11 (preprint 1.10) with its index statement concludes `Chapter1/ScFredholm · proposition_1_1_11_preprint` `Chapter1/FredholmIndex · proposition_1_1_11_index`. For a basic germ alone ($s = 0$) `Chapter3/FredholmGermLinearization · ScOperator.zero` `Chapter3/FredholmGermLinearization · proposition_3_1_10_basic`, with its level-0 consequence: $Dh(0)$ is Fredholm of index $\dim P - \dim Q$ `Chapter3/FredholmGermLinearization · proposition_3_1_10_basic_level_zero`.

The second statement of Proposition 3.1.10 (preprint 3.10) ($Df(a,w)$ Fredholm on every level $m$ for $(a,w)$ small) is proved on level 0 for the germs of Theorem 3.4.2 (preprint 3.38), at points of $U_1$ small on level 1 (§III.19). On every level $m$ it is proved by the text's argument run on level $m$ (gap G.21) `Chapter3/BasicGermLinearizationLevel · proposition_3_1_10_level`. At $x \in U_{m+1}$ small on level $m$ the level-$m$ operator splits as $Df(x)_m = T_0 + S(x) + K(x)$ `Chapter3/BasicGermLinearizationLevel · IsBasicGerm.levelDeriv_add_eq`. Here $T_0(\delta a, \delta w) = (0, \delta w)$ is Fredholm. $S(x) = (0, -D_2B(x)_m)$ is small by Lemma 3.1.9 (preprint 3.9) on level $m$. $K(x)$, the finite-rank part plus the sc$^+$-part through $F_{m+1} \to F_m$, is compact `Chapter3/BasicGermLinearizationLevel · IsBasicGerm.isCompactOperator_levelK`. The index is stable under the small perturbation $S(x)$. Both statements of the proposition together are `Chapter3/BasicGermLinearizationLevel · proposition_3_1_10_full`.

## III.6 Sc-Fredholm germs and sections; Theorem 3.1.18 (preprint 3.15) (Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4 (preprint 3.11, 3.14, 3.50, 3.51))

**Definition 3.1.11 (preprint 3.11).** A *strong bundle isomorphism* $\Psi : U \lhd F \to U' \lhd F'$ is a pair of strong bundle maps (Definition 2.6.1 (preprint 2.47)) inverse to each other on $U \oplus F$ and $U' \oplus F'$ `Chapter3/StrongBundleSections · IsStrongBundleIso`; its base map is $\psi$. A principal part $k : U \to F$ is *conjugated to a basic germ* if there are a strong bundle isomorphism with $\psi(0) = 0$ onto $U'$ relatively open in $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant of a finite-dimensional $P$), a basic germ $h$ on a germ $G$ with $G_0 \subseteq U'$, and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0 `Chapter3/StrongBundleSections · IsConjugateToBasicGerm`; this depends only on $k$ on $U$ `Chapter3/StrongBundleSections · IsConjugateToBasicGerm.congr`. $(f, x_0)$ is an *sc-Fredholm germ* if in a compatible chart centred at $x_0$ the principal part has a filling germ $g$ (§III.3), and there is a local sc$^+$-section $s$ — an sc-smooth germ into $F^1$ on a germ $G_s$ with $(G_s)_0 \subseteq U$ `Chapter3/GermFilling · IsScPlusGerm` — with $\iota s(0) = g(0)$ such that $g - \iota s$ is conjugated to a basic germ `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm`. This is the text's reading: $g$ and $s$ are germs, on germs of neighbourhoods of their own whose levels may shrink with the level, while $\Psi$ is an isomorphism of the whole local bundles and the conjugation identity is asked near 0 only, so it depends only on the germ of $g - \iota s$ `Chapter3/GermFillingChart · IsConjugateToBasicGerm.congr_nhds`. The first formalization asked $g$ and $s$ to be sc-smooth on $U \cap E_m$ for every $m$ `Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart`. That notion implies the germ reading `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGermWholeChart.isScFredholmGerm` and is exactly the germ reading with $g$ and $s$ sc-smooth on $U$ `Chapter3/GermFillingChart · StrongBundle.isScFredholmGerm_iff_germG_wholeChart`; whether every germ datum can be traded for a whole-chart one is open (§III.48). Definitions 3.1.16, 3.6.3, 3.6.4, 3.6.6 (preprint 3.14, 3.50, 3.51, 3.53) and every theorem of this part built on them are stated for the germ reading; the whole-chart notions and theorems are kept under the names `…WholeChart`, `…_wholeChart`. The text's $C' = T\psi(0)C$, used in the proof of Theorem 3.6.8 (preprint 3.54), is not part of the data; it is a theorem (§III.23).

**Definition 3.1.16 (preprint 3.14).** $f$ is an *sc-Fredholm section* if it is sc-smooth, regularizing ($x \in X_m$ and $f(x) \in Y_{m,m+1}$ imply $x \in X_{m+1}$) and an sc-Fredholm germ at every smooth point `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection`. The text assumes a tame bundle; nothing in Definitions 3.1.3–3.1.16 (preprint 3.3–3.14) or in Theorem 3.1.18 (preprint 3.15) uses tameness, so it is not assumed. (Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51) restate 3.11, 3.14.) A zero of a regularizing section is a smooth point: the zero section lies in $Y_{m,m+1}$ over $X_m$ `Chapter3/SectionCompactness · StrongBundle.fiberZero_mem_doubleLevel`, so a zero on level $m$ is on level $m+1$, and by induction on every level `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`.

**Theorem 3.1.18 (preprint 3.15).** If $f$ is an sc-Fredholm section and $s$ an sc$^+$-section, then $f+s$ is an sc-Fredholm section `Chapter3/FredholmSectionsG · theorem_3_1_18_preprint`. *Proof.* The text says "$f+s$ is an sc-smooth section which is also regularizing" and checks the germ property; each clause is a proof.

*Sc-smooth.* In a chart the representative of $f+s$ is $z \mapsto (z, \mathbf{f}(z)+\mathbf{s}(z))$. The map $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$, $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$ (§III.2), and composing the latter with $K[1] \subseteq K$ and adding fibre components (sum rule) gives an sc-smooth map with values in $K$, since its fibres are linear `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.add_isScSmoothRetractMap`; continuity of $f+s$ on each chart domain is read off the same representative `Chapter3/StrongBundleSections · StrongBundle.continuousOn_sectionAdd` `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.continuousOn_source`. So $f+s$ is sc-smooth `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.sectionAdd`.

*Regularizing.* For $x \in X_m$, the representative of $s$ maps $O_m$ into $K[1]_m$, so $\mathbf{s}(\varphi(x)) \in F_{m+1}$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.snd_mem_range`. If $(f+s)(x) \in Y_{m,m+1}$ then $\mathbf{f}(\varphi(x)) = (\mathbf{f}+\mathbf{s})(\varphi(x)) - \mathbf{s}(\varphi(x)) \in F_{m+1}$, i.e. $f(x) \in Y_{m,m+1}$ ($\Phi$ is linear on fibres `Chapter3/StrongBundleSections · StrongBundleChart.snd_sub`), and $x \in X_{m+1}$ `Chapter3/FredholmSectionsG · StrongBundle.regularizing_sectionAdd`.

*The germ property at a smooth point $x$* `Chapter3/FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd`. Take the chart, filling germ $g$ on $G_g$, sc$^+$-germ $t_0$ on $G_{t_0}$ with $\iota t_0(0) = g(0)$ and conjugation $\Psi$ for $(f,x)$. Let $t = \mathbf{s}$ be the principal part of $s$ in this chart; it is a section of $K \to O$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.mem_K` with lift $t_1$ to $F_1$, $\iota t_1 = t$ on $O$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.toBase_principalLift`. Then: $\tilde{t} = t \circ r$ is sc-smooth on the whole $U$, and $t_1 \circ r$ is sc-smooth into $F^1$ on $U$ (sc-smoothness on the retract $O$, Definition 2.1.14 (preprint 2.7)) — $s$ is a section of the bundle, not a germ; $g + \tilde{t}$ is a filling germ of $\mathbf{f}+\mathbf{s}$ on the same $G_g$ (Lemma 3.1.20 (preprint 3.17) for germs, §III.3 `Chapter3/GermCalculus · IsFillingGerm.add_section`, with the filler condition (1) transported along $\mathbf{f}+\mathbf{s}$ on $O$ `Chapter3/FredholmSectionsG · IsFillingGerm.congr`); $t_1 \circ r + t_0$ is an sc$^+$-germ on the same $G_{t_0}$, a germ plus a whole-chart map (§III.48), with value $(g+\tilde{t})(0)$ at $0$ `Chapter3/FredholmSectionsG · StrongBundle.sectionAdd_chartData_germ`; and $(g+\tilde{t}) - \iota(t_1 \circ r + t_0) = g - \iota t_0$ *on* $U$ (where $\iota t_1(r(u)) = t(r(u))$ because $r(u) \in O$), which is conjugated to a basic germ by the same $\Psi$. This last identity holds on $U$ only, which is why the conjugation condition is stated for $u \in U$. The germs of the data do not change, so nothing of $g$ or $t_0$ away from $G_g$, $G_{t_0}$ is used; the whole-chart proof `Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart.sectionAdd` is the same with the sum rule on $U$ in place of the sum of a germ and a whole-chart map. □

# III.7 Contraction germs: Definition 3.3.1 (preprint 3.28) and its working form

**Definition 3.3.1 (preprint 3.28).** For a germ $G$ around $0$ in $\widetilde{C} \oplus W$, $B : P \times W_0 \to W_0$ is an *sc$^0$-contraction germ* (i.e. $f(a,w) = w - B(a,w)$ is one `Chapter3/ContractionGerm · contractionMap`) if $B$ is an sc$^0$-germ on $G$, $B(0,0) = 0$, and for every $\varepsilon > 0$ and $m$ there is $r > 0$ such that every $(a,w)$ with $a \in \widetilde{C}$, $|a| < r$, $|w|_m < r$ lies in $U_m$ and $|B_m(a,w) - B_m(a,w')|_m \le \varepsilon |w - w'|_m$ for such points `Chapter3/ContractionGerm · IsSc0ContractionGerm`. (The text: "for all $(a,w)$, $(a,w')$ on level $m$ sufficiently close to $(0,0)$"; that these points are in the domain $U_m$ is presupposed.) $B_m$ is the level-$m$ map `Chapter3/ContractionGerm · levelB`, with $B_m(0) = 0$ `Chapter3/ContractionGerm · levelB_zero`.

*The working form.* The induction of §III.10 builds from $f$ a contraction germ $f^{(1)}$ whose fibre is $TW =$

$W^1 \oplus W$ and whose map is affine in the second factor; its natural contraction region is unbounded there, which balls cannot express. So the proofs work with explicit regions: a *contraction germ in working form* consists of $B$ and $G$ as above with open neighbourhoods $N(0) \supseteq N(1) \supseteq \cdots$ of 0 in $P$, open $O(m) \ni 0$ and closed $K(m) \subseteq O(m)$ in $W_m$ with $\iota K(m+1) \subseteq K(m)$, such that for $a \in N(m) \cap \widetilde{C}$: $\{a\} \times O(m) \subseteq U_m$, $B_m(a,\cdot)$ maps $K(m)$ into $K(m)$ and is a $\frac{1}{2}$-contraction on $O(m)$ `Chapter3/ContractionGerm · ContractionGerm`.

*Lemma: every sc⁰-contraction germ has a working form* `Chapter3/GermIFT · IsSc0ContractionGerm.toContractionGerm`. Apply Definition 3.3.1 (preprint 3.28) with $\varepsilon = \frac{1}{2}$ to get radii $r_m$ `Chapter3/GermIFT · IsSc0ContractionGerm.rad` `Chapter3/GermIFT · IsSc0ContractionGerm.rad_spec`. Make them decrease fast enough for the inclusions: $r'_0 = r_0$, $r'_{m+1} = \min(r_{m+1}, r'_m/(\|\iota_m\| + 1))$ `Chapter3/GermIFT · IsSc0ContractionGerm.rad'` `Chapter3/GermIFT · IsSc0ContractionGerm.norm_incl_le_of_le`. Put $O(m) = \{|w|_m < r'_m\}$, $K(m) = \{|w|_m \le r'_m/2\}$; then $|\iota w|_m \le \|\iota_m\| r'_{m+1}/2 \le r'_m/2$ for $w \in K(m+1)$. Since $B_m$ is continuous at 0 with $B_m(0) = 0$ `Chapter3/GermIFT · IsSc0ContractionGerm.levelB_zero_eq`, there is $\rho^0_m > 0$ with $|B_m(x)|_m < r'_m/4$ for $x \in U_m$, $|x|_m < \rho^0_m$ `Chapter3/GermIFT · IsSc0ContractionGerm.exists_cont_rad` `Chapter3/GermIFT · IsSc0ContractionGerm.contRad`. Let $\rho_m$ decrease with $\rho_m \le \min(\rho^0_m, r'_m)$ `Chapter3/GermIFT · IsSc0ContractionGerm.prad` `Chapter3/GermIFT · IsSc0ContractionGerm.prad_anti` and $N(m) = \{|a| < \rho_m\}$. For $a \in N(m) \cap \widetilde{C}$ and $w \in K(m)$, $|B_m(a,w)|_m \le |B_m(a,w) - B_m(a,0)|_m + |B_m(a,0)|_m \le \frac{1}{2} \cdot \frac{r'_m}{2} + \frac{r'_m}{4} = \frac{r'_m}{2}$.

## III.8 Existence of the solution germ: the parametrized fixed point theorem and Theorem 3.3.2 (preprint 3.29)

**The parametrized Banach fixed point theorem.** The text invokes "the parametrized version of Banach's fixed point theorem" in Theorems 3.3.2 and 3.5.1 (preprint 3.29 and 3.40) without stating it; Mathlib has the theorem without the parameter. Let $X$ be a complete metric space, $K \subseteq X$ closed and nonempty, and $g_l : K \to K$ ($l \in \Lambda'$, a subset of a topological space) $\frac{1}{2}$-contractions such that $l \mapsto g_l(x)$ is continuous for every $x \in K$. Then the fixed point $\phi(l) \in K$ is unique in $K$ and continuous in $l$: from $d(\phi(l), \phi(l_0)) \le \frac{1}{2} d(\phi(l), \phi(l_0)) + d(g_l\phi(l_0), g_{l_0}\phi(l_0))$ one gets $d(\phi(l), \phi(l_0)) \le 2d(g_l(\phi(l_0)), g_{l_0}(\phi(l_0)))$, which tends to 0 by continuity at the single point $\phi(l_0)$ `Chapter3/LocalCompactness · exists_fixedPoint_continuousOn`.

**Theorem 3.3.2 (preprint 3.29).** In the working form, for $a \in N(m) \cap \widetilde{C}$ the map $B_m(a,\cdot) : K(m) \to K(m)$ is a $\frac{1}{2}$-contraction of a closed subset of $W_m$, with a fixed point $\delta_m(a) \in K(m)$ `Chapter3/ContractionGerm · ContractionGerm.exists_fixed` `Chapter3/ContractionGerm · ContractionGerm.fixLvl`, which is the only fixed point in the larger set $O(m)$ (two fixed points in $O(m)$ satisfy $|w - w'|_m \le \frac{1}{2}|w - w'|_m$) `Chapter3/ContractionGerm · ContractionGerm.fixed_unique` `Chapter3/ContractionGerm · ContractionGerm.eq_fixLvl`. For $a \in N(m+1) \cap \widetilde{C}$, $\iota\delta_{m+1}(a) \in K(m) \subseteq O(m)$ is a fixed point of $B_m(a,\cdot)$ because $B$ commutes with the inclusions; hence $\iota\delta_{m+1}(a) = \delta_m(a)$ `Chapter3/ContractionGerm · ContractionGerm.incl_fixLvl`. So $\delta := \delta_0$ `Chapter3/ContractionGerm · ContractionGerm.sol` maps $V_m := N(m) \cap \widetilde{C}$ into $W_m$ with level-$m$ map $\delta_m$ `Chapter3/ContractionGerm · ContractionGerm.levelMap_sol`. Each $\delta_m$ is continuous on $V_m$ by the parametrized fixed point theorem ($g_a = B_m(a,\cdot)$ on $K(m)$, continuous in $a$ because $B_m$ is) `Chapter3/ContractionGerm · ContractionGerm.continuousOn_fixLvl`. Write $\mathrm{germ}(s) = (N(m+s) \cap \widetilde{C})_m$ `Chapter3/ContractionGerm · ContractionGerm.germ` — legitimately a germ because $P$ is finite-dimensional — and call $\delta$ a *solution germ* on a germ $V$ if it is an sc⁰-germ there with $\delta(0) = 0$, $(a, \delta_m(a)) \in U_m$ on $V_m$ and $B(a, \delta(a)) = \delta(a)$ on $V_0$ `Chapter3/ContractionGerm · ContractionGerm.IsSolutionGerm`. Then: $\delta$ is a solution germ on $\mathrm{germ}(0)$, and every $w \in O(0)$ with $B(a,w) = w$, $a \in N(0) \cap \widetilde{C}$, equals $\delta(a)$ `Chapter3/ContractionGerm · theorem_3_3_2` `Chapter3/ContractionGerm · ContractionGerm.eq_sol`. The property passes to every

germ$(s)$ Chapter3/ContractionGerm · ContractionGerm.isSolutionGerm_germ. In the text's form, from Definition 3.3.1 (preprint 3.28) with balls: the working form of §III.7 gives the solution germ, unique among solutions $w$ with $|w|_0 < r_0'$ Chapter3/GermIFT · theorem_3_3_2_of_isSc0ContractionGerm. The text's "uniquely determined" leaves this region implicit (Remark 3.3.8 (preprint 3.36) names it, $|w|_0 \le \tau_0$).

## III.9 The solution germ is sc¹: Theorem 3.3.3 (preprint 3.30)

The text cites [21, Theorem 2.3], "the hard part of the regularity theory", and gives no proof. The formal proof shows that $\delta$ is sc¹ on the germ shifted by one level, $V_m = N(m+1) \cap \widetilde{C}$, and the shift is forced by the argument.

*The operators.* Let $D$ be the derivative family of $B$. For $a \in N(m+1) \cap \widetilde{C}$ the point $(a, \delta_{m+1}(a))$ lies in $U_{m+1}$, and the level-$m$ operator of §III.1 there is $L_m(a) := DB(a, \delta_{m+1}(a)) : P \times W_m \to W_m$ Chapter3/ContractionGerm · ContractionGerm.Lop, with fibre part $T_m(a) = D_2B(a, \delta_{m+1}(a))$ Chapter3/ContractionGerm · ContractionGerm.Top and $L_m(a)(h,\zeta) = L_m(a)(h,0) + T_m(a)\zeta$ Chapter3/ContractionGerm · ContractionGerm.Lop_apply. By Lemma 3.1.9 (preprint 3.9) (a), $\|T_m(a)\| \le \frac{1}{2}$ Chapter3/ContractionGerm · ContractionGerm.norm_Top_le, so $1 - T_m(a)$ is invertible (Neumann series) Chapter3/ContractionGerm · ContractionGerm.isUnit_one_sub_Top, and the candidate derivative is $A_m(a) := (1 - T_m(a))^{-1} L_m(a)(\cdot, 0) \in L(P, W_m)$ Chapter3/ContractionGerm · ContractionGerm.Aop. It satisfies the *key identity* $L_m(a)(h, A_m(a)h) = A_m(a)h$ Chapter3/ContractionGerm · ContractionGerm.Lop_Aop, and is the unique solution $\zeta$ of $L_m(a)(h,\zeta) = \zeta$ Chapter3/ContractionGerm · ContractionGerm.eq_Aop. Because the level operators commute with the inclusions Chapter3/ContractionGerm · ContractionGerm.toBase_Lop, so do the $A_m$: $\iota A_m(a)h = A_0(a)h$ Chapter3/ContractionGerm · ContractionGerm.toBase_Aop.

*Differentiability at $a_0 \in N(2) \cap \widetilde{C}$* Chapter3/ContractionGerm · ContractionGerm.hasFDerivWithinAt_sol. Let $x_1 = (a_0, \delta_1(a_0)) \in U_1$, $A_1 = A_1(a_0)$, $A_0 = A_0(a_0)$. For $a$ near $a_0$ put $h = a - a_0$ and $k = (h, A_1 h) \in E_1$, so $|k|_1 \le (1 + \|A_1\|)|h|$. *This is where $a_0 \in N(2)$ is needed*: $A_1$ exists only because $(a_0, \delta_2(a_0)) \in U_2$, and without it the increment $A_0 h$ need not lie in $W_1$, so the sc¹-differentiability of $B$ at $x_1$, which controls only increments in $E_1$, could not be used. For $h$ small, $x_1 + k \in U_1$ and $y := \delta(a_0) + A_0 h \in O(0)$. By the sc¹ property, $B(a,y) = B(a_0, \delta(a_0)) + D(x_1)\iota k + \text{rem} = \delta(a_0) + L_0(a_0)(h, A_0 h) + \text{rem} = y + \text{rem}$ with $|\text{rem}|_0 = o(|h|)$, using the key identity on level 0. So $y$ is an approximate fixed point of $B(a,\cdot)$, and the $\frac{1}{2}$-contraction on $O(0)$ gives $|\delta(a) - y|_0 \le \frac{1}{2}|\delta(a) - y|_0 + |\text{rem}|_0$, i.e. $|\delta(a) - \delta(a_0) - A_0 h|_0 \le 2|\text{rem}|_0 = o(|h|)$.

*The tangent map is sc⁰.* On level $m$ of the tangent of germ$(1)$, i.e. on $(N(m+2) \cap \widetilde{C}) \times P$, the tangent map is $(a,h) \mapsto (\delta_{m+1}(a), A_m(a)h)$ Chapter3/ContractionGerm · ContractionGerm.tangent_toBase_sol. Its first component is continuous (§III.8); the second because $(x,v) \mapsto D_m(x)v$ is jointly continuous and $\zeta(a) = A_m(a)h$ satisfies $\zeta(a) = L_m(a)(h,0) + T_m(a)\zeta(a)$, whence $|\zeta(a) - \zeta(a')|_m \le 2|L_m(a)(h,0) - L_m(a')(h,0)|_m + 2|(T_m(a) - T_m(a'))\zeta(a')|_m$ Chapter3/ContractionGerm · ContractionGerm.continuousOn_Aop. With the derivative family $a \mapsto A_0(a)$ on germ$(1)$ Chapter3/ContractionGerm · ContractionGerm.solDeriv, $\delta$ is an sc¹-germ there Chapter3/ContractionGerm · ContractionGerm.isSc1GermWith_sol **— Theorem 3.3.3 (preprint 3.30)**, on the shifted germ Chapter3/ContractionGerm · theorem_3_3_3. At parameters $a_0 \in N(1) \setminus N(2)$ the argument does not apply, and whether $\delta$ is differentiable there is neither needed nor addressed.

## III.10 The contraction germ $f^{(1)}$: Lemma 3.3.4 (preprint 3.31)

**Lemma 3.3.4 (preprint 3.31).** The text: if $f$ is of class sc^k, $k \geq 1$, and $\delta$ of class sc^j, then $f^{(1)}(v,b,u,w) = (u - B(v,u), w - DB(v,\delta(v))(b,w))$ is an sc⁰-contraction germ of class sc^{min(k−1, j)} whose solution germ is $T\delta$ — "as one verifies by comparing $Tf$ with $f^{(1)}$", and "from the uniqueness". The formal version: if $B$ is sc^{k+1} with derivative family $D$ and $\delta$ is sc^j on germ$(s)$, then $f^{(1)}$ is a contraction germ in working form of class sc^{min(k, j)} with solution germ $T\delta$ on $(N(2+s) \cap \widetilde{C}) \times P$ `Chapter3/GermIFT · ContractionGerm.lemma_3_3_4`. The parameters are $(v,b) \in \widetilde{C} \times P \subseteq P \times P$ (quadrant-like `Chapter3/GermIFT · ContractionGerm.isQuadrantLike_next`), the unknowns $(u,w) \in TW$, with levels $W_{m+1} \times W_m$.

*Step 1: the germ and the regions.* The germ of $f^{(1)}$ is $U_m^{(1)} = \{((v,b),(u,w)) : (v,u) \in U_{m+1},\ v \in N(m+1+s)\}$ `Chapter3/GermIFT · ContractionGerm.nextG`, and the regions are $N^{(1)}(m) = N(m+2+s) \times P$, $O^{(1)}(m) = O(m+1) \times W_m$, $K^{(1)}(m) = K(m+1) \times W_m$ `Chapter3/GermIFT · ContractionGerm.next`. They are unbounded in $b$ and $w$, necessarily: the solution must be $(\delta(v), D\delta(v)b)$ for all $b \in P$.

*Step 2: the map.* With the reshuffling sc-operator $\sigma((v,b),(u,w)) = ((v,u),(b,w))$ `Chapter3/GermIFT · tangentShuffle` and $\Gamma((v,b),(u,w)) = ((v,\delta(v)),(b,w))$, where $v \mapsto \delta(v)$ is regarded as the *shifted* germ into $W^1$ `Chapter3/GermIFT · ContractionGerm.nextΓ`, formula (38) is $B^{(1)} = (\mathrm{pr}_1 \circ TB \circ \sigma, \mathrm{pr}_2 \circ TB \circ \Gamma)$ `Chapter3/GermIFT · ContractionGerm.nextB` `Chapter3/GermIFT · ContractionGerm.nextB_apply`. On level $m$, for $v \in N(m+1) \cap \widetilde{C}$, $B_m^{(1)}((v,b),(u,w)) = (B_{m+1}(v,u), L_m(v)(b,w))$ `Chapter3/GermIFT · ContractionGerm.levelB_next`: the point $(v, \delta_{m+1}(v))$ lies in $U_{m+1}$, which is where the parameter region $N(m+1+\cdot)$ comes from.

*Step 3: contraction and invariance.* For $(v,b) \in N^{(1)}(m)$ and $(u,w),(u',w') \in O^{(1)}(m)$, the two components contract by $\frac{1}{2}$: $B_{m+1}(v,\cdot)$ on $O(m+1)$, and $T_m(v)$ with $\|T_m(v)\| \leq \frac{1}{2}$ (Lemma 3.1.9 (preprint 3.9) (a)); $K(m+1) \times W_m$ is mapped into itself; $B^{(1)}(0) = 0$ `Chapter3/GermIFT · ContractionGerm.nextB_zero`.

*Step 4: the class of $B^{(1)}$.* $TB$ is an sc^k-germ on the tangent germ; $\sigma$, the projections $(v,b,u,w) \mapsto v$ into $P^1$ and $\mapsto (b,w)$ `Chapter3/GermIFT · projV` `Chapter3/GermIFT · projBW` and the identification $P^1 \oplus W^1 = (P \oplus W)^1$ `Chapter3/GermIFT · shiftParamOp` are sc-operators; $\Gamma$ is the pairing of $(b,w)$ with $(v,\delta(v))$, which is sc^j because $\delta$ is and so is its shift into $W^1$ (Proposition 1.2.2 (preprint 1.17) for germs) `Chapter3/GermIFT · ContractionGerm.nextΓ_isSckGerm`; and $\Gamma$ maps $U_m^{(1)}$ into $U_{m+1} \times E_m$ level by level `Chapter3/GermIFT · ContractionGerm.levelMap_nextΓ` `Chapter3/GermIFT · ContractionGerm.nextΓ_mapsToGerm`, as does $\sigma$ `Chapter3/GermIFT · ContractionGerm.tangentShuffle_mapsToGerm`. The chain rule and the pairing rule of §III.1 give $B^{(1)}$ sc^j `Chapter3/GermIFT · ContractionGerm.nextB_isSckGerm`.

*Step 5: the solution germ is $T\delta$.* For $v \in N(2+s) \cap \widetilde{C}$ and $b \in P$, the point $(\delta_1(v), A_0(v)b) \in O(1) \times W_0$ is a fixed point of $B^{(1)}(v,b,\cdot)$ — the first component by $B_1(v,\delta_1(v)) = \delta_1(v)$, the second by the key identity of §III.9 — so by the uniqueness of Theorem 3.3.2 (preprint 3.29) *in* $O^{(1)}(0)$ it is the solution germ of $f^{(1)}$ at $(v,b)$ `Chapter3/GermIFT · ContractionGerm.next_sol`, and since $A_0(v) = D\delta(v)$ (Theorem 3.3.3 (preprint 3.30)) this is $T\delta(v,b)$ `Chapter3/GermIFT · ContractionGerm.next_sol_eq_tangentMapWith`. The text's "from the uniqueness" is true because the regions of Step 1 contain $T\delta(v,b)$ for every $b$.

## III.11 The induction: Lemma 3.3.5 (preprint 3.32), Theorem 3.3.6 (preprint 3.33), Corollary 3.3.7 (preprint 3.34), Theorem 3.35 (preprint)

The text proves Lemma 3.3.5 (preprint 3.32) by iterating $f \mapsto f^{(1)}$ and Theorem 3.3.6 (preprint 3.33) by contradiction (if $\delta$ were sc^j but not sc^{j+1}, $f^{(j)}$ would be sc¹, so $T^j\delta$ would be sc¹ by Theorem 3.3.3 (preprint 3.30)). The formal proof is the same argument as one induction on $j$, with the statement quantified over *all* contraction germs, since $f^{(1)}$ has parameter space $P \times P$ and fibre $TW$ and the induction hypothesis must apply to it. It also keeps track of the germ.

*Claim ($S_j$): for every contraction germ in working form, over any finite-dimensional parameter space, of class sc^k with $k \geq j$, the solution germ is an sc^j-germ on* $\mathrm{germ}(s)$ *for some* $s$ `Chapter3/GermIFT · ContractionGerm.sol_isSckGerm`. For $j = 0$ take $s = 0$ (Theorem 3.3.2 (preprint 3.29)). *Step* $j \to j+1$ `Chapter3/GermIFT · ContractionGerm.isSckGerm_succ_of_next`: let $B$ be sc^k, $k = k' + 1 \geq j + 1$. (1) By ($S_j$) for $f$, $\delta$ is sc^j on $\mathrm{germ}(s)$. (2) By Lemma 3.3.4 (preprint 3.31) with this $s$, $f^{(1)}$ is a contraction germ of class sc^{min(k', j)} = sc^j. (3) By ($S_j$) for $f^{(1)}$, its solution germ is sc^j on its $\mathrm{germ}(s')$, whose level $m$ is $(N(m + s' + 2 + s) \cap \widetilde{C}) \times P$. (4) By Theorem 3.3.3 (preprint 3.30) $\delta$ is sc¹ on $\mathrm{germ}(1)$, hence on $\mathrm{germ}(s + s' + 1)$ (restriction), whose tangent germ has level $m$ equal to the set in (3). (5) On that set the solution germ of $f^{(1)}$ is $T\delta$ (Lemma 3.3.4 (preprint 3.31), Step 5); the identification of $T\widetilde{C} = \widetilde{C} \times P$ inside $TP$ with $\widetilde{C} \times P$ inside $P \times P$ is the identity on every level, an sc-isomorphism `Chapter3/GermIFT · constTangentOp'` through which the chain rule transports the germ. (6) So $\delta$ is sc¹ with sc^j tangent germ: sc^{j+1} on $\mathrm{germ}(s + s' + 1)$. Since the proof does the same for every contraction germ, $s$ and $s'$ are the same function of $j$, so $s_0 = 0$, $s_{j+1} = 2s_j + 1$: $s_j = 2^j - 1$.

**Theorem 3.3.6 (preprint 3.33)** follows from ($S_k$): there is $s$ such that $\delta$ is a solution germ and an sc^k-germ on $\mathrm{germ}(s)$ `Chapter3/GermIFT · theorem_3_3_6`; if $B$ is sc-smooth this holds for every $k$ `Chapter3/GermIFT · theorem_3_3_6_smooth`; and from Definition 3.3.1 (preprint 3.28) with balls through the working form `Chapter3/GermIFT · theorem_3_3_6_of_isSc0ContractionGerm`. Lemma 3.3.5 (preprint 3.32) is contained in the induction: its step from $j$ to $j+1$ is items (2)–(3), without the iterated germs $f^{(j)}$. **Corollary 3.3.7 (preprint 3.34)**: if $B$ is sc-smooth, for $m$ and $k$ take $s$ for $k$; by Proposition 1.2.3 (preprint 1.18) for germs, $\delta$ maps $V_{m,k} := N(m + k + s) \cap \widetilde{C}$ into $W_m$ by a $C^k$ map `Chapter3/GermIFT · corollary_3_3_7` `Chapter3/GermLevels · IsSckGerm.contDiffOn_levels`, the inclusions of the constant structure being the identity `Chapter3/GermIFT · ScBanach.const_inclK`. **Theorem 3.35 (preprint)**: for $\widetilde{C}$ a partial quadrant (which is quadrant-like), the same `Chapter3/GermIFT · theorem_3_35_preprint` `Chapter3/GermIFT · theorem_3_35_smooth_preprint`.

**What the shrinking means.** Definition 3.1.2 (preprint 3.2) calls $\delta$ sc-smooth on a germ if it is sc^k *on that germ* for every $k$. The proof gives, for each $k$, the germ $\mathrm{germ}(2^k - 1)$; these shrink with $k$, their intersection is in general not a neighbourhood of 0, and no diagonal choice gives one germ, because being sc^k on a germ involves every level of it. So Theorem 3.3.6 (preprint 3.33) and the first sentence of Theorem 3.35 (preprint) hold for germs read up to shrinking, and the last sentences of Corollary 3.3.7 (preprint 3.34) and Theorem 3.35 (preprint) hold in the sense of Corollary 3.3.7 (preprint 3.34) — for every $k$ on a germ depending on $k$ — and are proved neither by the text nor here as `IsScSmoothGerm` on one germ. In the text's reading of germs the two senses coincide. Whether the shrinking is a property of contraction germs or only of the method is not decided: in the toy model of the exposition (Chapter3.md §3) the solution germ is sc-smooth on the original germ. Remark 3.3.8 (preprint 3.36), the quantitative form with radii $\varepsilon_i$ shrinking with the level, is §III.58.

## III.12 Local regularity and compactness: Theorem 3.5.1 (preprint 3.40)

**Theorem 3.5.1 (preprint 3.40)** (text): for $f = h + s$ on a relatively open $U \ni 0$ in $C$, $h$ a basic germ, $s$ an sc$^+$-germ with $s(0) = 0$, and $S = \{f = 0\}$, there are relatively open neighbourhoods $U \supseteq O(0) \supseteq O(1) \supseteq \cdots$ of 0 on level 0 with $\mathrm{cl}(S \cap O(m)) \subseteq C_m$. Of $f$ only the $W$-component matters: $S$ lies in the set of solutions of $w = B(a, w) - \mathbf{s}(a, w)$, $\mathbf{s} = P_W \circ s$. The formal statement uses only that $B$ is an sc$^0$-contraction germ (not sc-smooth), that $\mathbf{s}$ is an sc$^0$-germ into $W^1$ with $\mathbf{s}(0) = 0$ (the sc$^+$ property level by level), and that $\widetilde{C}$ is closed (true for partial quadrants; the theorem is proved for convex $\widetilde{C}$, and closedness makes the solution set closed in $E_0$, where the text takes closures in $C_0$). It is *stronger* than the text's in the way Theorem 3.1.13 (preprint 3.12) needs: for any set $S$ of solutions in $U_0$ there are nested open $O(i) \ni 0$ in $E_0$ and compact $K_i \subseteq E_i$ with $\mathrm{cl}_{E_0}(S \cap O(i)) = \iota_i(K_i)$ `Chapter3/LocalCompactness · theorem_3_5_1_compactness`; in particular $\mathrm{cl}(S \cap O(i)) \subseteq E_i$ `Chapter3/LocalCompactness · theorem_3_5_1_regularity`.

*Step 0: the level-$m$ contractions with a parameter.* Let $r_m$ be the radii of Definition 3.3.1 (preprint 3.28) for $\varepsilon = \frac{1}{2}$ and $c_m = \min(r_m, r_0/(\|\iota_m\| + 1))$, $\iota_m : W_m \to W_0$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.crad`, so that $|\iota_m y|_0 < r_0$ whenever $|y|_m \le c_m/2$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.norm_toBase_lt_rad` (the text's $\tau_1' < \tau_0/c_1$). Points with $a \in \widetilde{C}$, $|a|, |w|_m < r_m$ lie in $U_m$ and satisfy the contraction estimate `Chapter3/LocalCompactness · IsSc0ContractionGerm.mem_G_of_lt` `Chapter3/LocalCompactness · IsSc0ContractionGerm.contr_of_lt`. Choose $\rho_m \in (0, c_m]$ with $|B_m(a, 0)|_m \le c_m/8$ for $|a| < \rho_m$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_cprad` `Chapter3/LocalCompactness · IsSc0ContractionGerm.cprad`. On $\Sigma_m = \{(a, z) : a \in \widetilde{C}, |a| < \rho_m, |z|_m \le c_m/8\}$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.Sigma` the maps $w \mapsto B_m(a, w) - z$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap` send the closed ball $\bar{B}_m(c_m/2)$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.cBall` into itself ($\frac{1}{2} \cdot \frac{c_m}{2} + \frac{c_m}{8} + \frac{c_m}{8} = \frac{c_m}{2}$) `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_mapsTo`, are $\frac{1}{2}$-contractions `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_contr` and depend continuously on $(a, z)$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_continuousOn`. The parametrized fixed point theorem gives a continuous $\delta_m : \Sigma_m \to \bar{B}_m(c_m/2)$ with $\delta_m(a, z) = B_m(a, \delta_m(a, z)) - z$, unique in the ball `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_fixed`.

*Step 1: the lifts — the induction the text leaves to the reader.* Let $\mathrm{Sol}(\tau)$ be the solutions $x = (a, w)$ with $a \in \widetilde{C}$, $|a| \le \tau$, $|w|_0 \le \tau$ `Chapter3/LocalCompactness · solSet`. *Claim:* for every $m$ there are $\tau_m > 0$ and a map $L_m : E_0 \to E_m$ such that for every $x \in \mathrm{Sol}(\tau_m)$: $\iota_m L_m x = x$, $L_m x \in U_m$, $|(L_m x)_W|_m \le c_m/2$, and $L_m$ is continuous on $\mathrm{Sol}(\tau_m)$ *for the level-*0 *topology of the source* `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_lift`. For $m = 0$, $L_0 = \mathrm{id}$. Given $L_m$: by continuity of $\mathbf{s}_m : U_m \to W_{m+1}$ at 0 (where $\mathbf{s}_m(0) = 0$ `Chapter3/LocalCompactness · levelMap_s_zero`) there is $\eta$ with $|\mathbf{s}_m(y)|_{m+1} < c_{m+1}/8$ for $|y|_m < \eta$; by continuity of $L_m$ at 0 there is $\theta$ with $|L_m x|_m < \eta$ for $x \in \mathrm{Sol}(\tau_m)$, $|x|_0 < \theta$. Put $\tau_{m+1} = \min(\tau_m, \rho_{m+1}/2, \theta/2)$ and, for $x = (a, w) \in \mathrm{Sol}(\tau_{m+1})$, $z = \mathbf{s}_m(L_m x)$, so $(a, z) \in \Sigma_{m+1}$, and $L_{m+1} x := (a, \delta_{m+1}(a, z))$. *It lifts $x$:* $w' = \iota_{m+1}\delta_{m+1}(a, z)$ solves $w' = B(a, w') - \mathbf{s}(x)$ (apply $\iota_{m+1}$, which commutes with $B$, and use $\iota_m L_m x = x$); so does $w$; both are smaller than $r_0$ on level 0 by the choice of $c_{m+1}$, so the level-0 contraction gives $w' = w$ — the text's (43), on every level. *Continuity:* $x \mapsto (a, \mathbf{s}_m(L_m x))$ is continuous from the level-0 topology into $\Sigma_{m+1}$, and $\delta_{m+1}$ is continuous. The induction step needs the continuity of $L_m$: the bound $|\mathbf{s}_m(L_m x)|_{m+1} \le c_{m+1}/8$ requires $L_m x$ small on level $m$, whereas $x$ is small only on level 0. In the text's level-1 step $L_0 = \mathrm{id}$ and this does not show; existence, bound and continuity of the lifts have to be proved together.

*Step 2: compactness.* $\mathrm{Sol}(\tau)$ is closed in $E_0$ ($\widetilde{C}$ closed, $B$ and $\mathbf{s}$ continuous on level 0) `Chapter3/`

`LocalCompactness · IsSc0ContractionGerm.isClosed_solSet`. For $x \in \mathrm{Sol}(\tau_{m+1})$ the $W$-component is $\iota_{m+1}$ of a vector of norm $\le c_{m+1}/2$, and $\iota_{m+1} : W_{m+1} \to W_0$ is compact `Chapter3/LocalCompactness · ScBanach.toBase_succ_isCompactOperator`; with $|a| \le \tau_{m+1}$ in the finite-dimensional $P$, a sequence of lifts has a subsequence converging in $E_{m+1}$ (parameters in a compact ball, $W_0$-parts in a compact image, and the continuity of the lift for the level-0 topology) `Chapter3/LocalCompactness · IsSc0ContractionGerm.isCompact_lift_image`.

*Step 3: the neighbourhoods.* $O(i)$ are the balls of radius $\varrho_i$, $\varrho_0 = \tau_1$, $\varrho_{i+1} = \min(\varrho_i, \tau_{i+2})$ — one level ahead of the text's $O(m)$, because compactness in $E_i$ comes from boundedness in $E_{i+1}$. Then $\mathrm{cl}(S \cap O(i)) \subseteq \mathrm{Sol}(\tau_{i+1})$, every point of it is $\iota_i$ of the image under $E_{i+1} \to E_i$ of its lift $L_{i+1}x$, and $K_i$ is the compact set of these images intersected with the closed preimage of $\mathrm{cl}(S \cap O(i))$.

For a basic germ plus an sc$^+$-germ ($S$ the zero set of $h + \iota s$; its $Q$-component only shrinks $S$) this is the **local Theorem 3.1.13 (preprint 3.12)** `Chapter3/LocalCompactness · theorem_3_1_13_local`, and **Corollary 3.1.15 (preprint 3.13) in local form**: solutions $x_k \to 0$ in $E_0$ lie in $\mathrm{Sol}(\tau_m)$ for $k$ large, and $L_m x_k \to L_m 0 = 0$ in $E_m$ `Chapter3/LocalCompactness · corollary_3_1_15_local`.

# III.13 Theorem 3.1.13 (preprint 3.12) and Corollary 3.1.15 (preprint 3.13) for sections

**Theorem 3.1.13 (preprint 3.12).** Let $f$ be a section of a strong bundle with $(f, x_0)$ an sc-Fredholm germ and $f(x_0) = 0$. Then there are nested open $O(i) \ni x_0$ in $X$ such that $\mathrm{cl}_X\{x \in O(i) \mid f(x) = 0\}$ is the image of a compact subset of $X_i$ (with the level topology of Part II, §II.6) under the inclusion $X_i \to X$ `Chapter3/SectionCompactnessG · theorem_3_1_13_preprint`. Of "$f$ an sc-smooth section of a tame strong bundle" only $P \circ f = \mathrm{id}$ is used; the regularity enters through the germ data, and of the filling germ $g$ only its continuity on level $0$ near $0$. The text calls the theorem "an immediate consequence of Theorem 3.40"; the proof makes explicit six steps.

(i) *Normalizations.* $f(x_0) = 0$ gives $\mathbf{f}(0) = 0$, so $g(0) = 0$ (filler (1)) and $s(0) = 0$ ($\iota s(0) = g(0)$). (ii) *The zeros correspond.* $f(x) = 0 \iff \mathbf{f}(\varphi(x)) = 0$ (§III.2) $\iff g(\varphi(x)) = 0$ (filler (1)) near $x_0$; zeros of $g$ in $U$ near $0$ lie in $O$ (filler (2)); and $g(u) = 0 \iff (h + t)(\psi(u)) = 0$, $t(\psi(u)) := (\Psi(u, s(u)))_2$, because $\Psi(u, k) = (\psi(u), \Gamma(u, k))$ `Chapter3/SectionCompactness · IsStrongBundleMap.apply_eq` with $\Gamma(u, \cdot)$ linear `Chapter3/SectionCompactness · IsStrongBundleMap.snd_sub` and injective `Chapter3/SectionCompactness · IsStrongBundleIso.snd_eq_zero_iff`. (iii) *$t$ is an sc$^+$-germ with $t(0) = 0$:* the base map of a strong bundle map is sc-smooth `Chapter3/SectionCompactness · IsStrongBundleMap.base_isScSmooth`, $\psi'(0) = 0$ `Chapter3/SectionCompactness · IsStrongBundleIso.base_zero'`, and the fibre component of $\Psi[1]$ lifts that of $\Psi$ to $F_1'$ `Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd`. Theorem 3.5.1 (preprint 3.40) needs $t$ only as an sc$^0$-germ into $F'^1$ on the germ of the basic germ. With $s$ a germ on $G_s$, $t$ is one on the germ of the points of the germ of $h$ that $\psi^{-1}$ maps into $G_s$ level by level `Chapter3/GermFilling · ScNbhdGerm.preimage` (germ composition with the whole-chart $\psi^{-1}$ and $\Psi[1]$, §III.48), and the basic germ restricts to that germ `Chapter3/BasicGermShrink · IsBasicGerm.mono` (a form of the shrinking of §III.48). (iv) *Local compactness for a subset.* The local theorem for an arbitrary subset of the solution set of $h + \iota t$ `Chapter3/SectionCompactness · theorem_3_1_13_local_subset` is applied on the restricted basic germ to $\psi\varphi$ of the zeros near $x_0$; its $O'(i)$ are intersected with a small ball, so that $E_0$-closures stay inside $\psi(U)$ and the region of (ii) (the closure of a subset of the solution set is still compact). (v) *Back to $X$.* The $E_0$-closure consists of zeros of $h + \iota t$, a limit of zeros in the domain of continuity being a zero `Chapter3/SectionCompactness · eq_of_mem_closure_of_continuousOn`, hence of zeros of $g$ in $U$, which lie in $O$. Here $g$ must be continuous at the limits $\psi^{-1}(u')$: the

level-0 neighbourhood of the filler and conjugation identities is chosen inside the open set $O_g$ with $(G_g)_0 = O_g \cap C$, so the limits lie in $(G_g)_0$, where the sc⁰-germ $g$ is continuous. The whole-chart theorem `Chapter3/SectionCompactness · theorem_3_1_13_wholeChart` took $t$ sc-smooth on all of $U'$ and $g$ continuous on $U$; nothing of either away from a neighbourhood of 0 was used. An $X$-closure of a subset of the chart domain could a priori leave it; compactness in $E_0$ shows it does not. (vi) *Levels.* A compact subset of $E_i$ is carried to a compact subset of $X_i$ because $\psi^{-1}$ is sc⁰ and the inverse chart $O_i \to X_i$ `Chapter3/SectionCompactness · MPolyfold.levelChartSymm` is continuous for the level topology of $X_i$, the transition maps being sc⁰ `Chapter3/SectionCompactness · MPolyfold.continuousOn_levelChartSymm` — the paragraph after Definition 2.3.8 (preprint 2.14), for every level.

**Corollary 3.1.15 (preprint 3.13).** If $f(x_k) = 0$ and $x_k \to x_0$, fix $m$ and take $O(m)$ and the compact $K_m \subseteq X_m$; for $k$ large $x_k \in O(m)$, so $x_k = \iota y_k$ with $y_k \in K_m$, and $x_0 = \iota y_0$. $X_m$ is not known to be first countable here (its metrizability, Theorem 2.3.10 (preprint 2.16), is not used), so subsequences are avoided: a sequence in a compact set whose images under a continuous injection converge, converges `Chapter3/SectionCompactness · tendsto_of_isCompact_of_tendsto_map`. So $y_k \to y_0$ in $X_m$ `Chapter3/SectionCompactnessG · corollary_3_1_15_preprint`. The text states the corollary for an arbitrary limit $x$ while the hypothesis concerns the germ at $x_0$; for an sc-Fredholm section and any limit $x$ of zeros, $x$ is a zero (the principal part is continuous in a chart `Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto`), hence a smooth point (regularizing), hence the germ property holds at $x$ and the first part applies `Chapter3/Chapter5BridgeG · corollary_3_1_15_section`.

## III.14 Good position (Definition 3.1.24 (preprint 3.21), Lemma 3.8.8 (preprint 3.65), Proposition 3.1.25 (preprint 3.22), Lemma 3.5.9 (preprint 3.45), Proposition 3.8.12 (preprint 3.69))

**Definition 3.1.24 (preprint 3.21).** $P$ is a *good complement* of the finite-dimensional $N$ with respect to $C$ if $E_0 = N \oplus P$ algebraically and for some $\varepsilon > 0$, $|p|_0 \le \varepsilon |n|_0$ implies $(n \in C \iff n + p \in C)$ `Chapter3/GoodPosition · IsGoodComplement`; $N$ is *in good position* if $N \cap C$ has nonempty interior in $N$ and $N$ has a good complement `Chapter3/GoodPosition · IsInGoodPosition`. The text asks for an sc-complement (or, for non-smooth $N$, a topological one); none of Proposition 3.1.25 (preprint 3.22), Lemmas 3.5.9, 3.8.8 (preprint 3.45, 3.65) or Proposition 3.8.12 (preprint 3.69) uses more than the algebraic splitting, so only that is required. Where closedness is needed (Lemma 3.6.9 (preprint 3.55), Definition 3.2.1 (preprint 3.25)) it is asked there; it is not an added hypothesis, since the text's sc-complement is closed (§III.56).

*Set-up in coordinates.* After a presentation $E \cong \mathbb{R}^n \oplus W$, $C = \{\ell_i \ge 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$ `Chapter3/GoodPosition · coordFun` `Chapter3/GoodPosition · coordFun_surj`. Inside $N$, $Q := N \cap C$ is the cone $\{\ell_i \ge 0\}$ `Chapter3/GoodPosition · quadCone`, closed `Chapter3/GoodPosition · quadCone_isClosed`. Let $\Sigma$ be the indices $i$ with $\ell_i(a) = 0$ for some nonzero $a \in Q$ `Chapter3/GoodPosition · vanIdx`; by definition, for $i \notin \Sigma$ and $x \in Q \setminus 0$, $\ell_i(x) > 0$ `Chapter3/GoodPosition · pos_of_not_mem_vanIdx`. Let $N \cap W = \{x \in N : \ell_i(x) = 0 \ \forall i\}$ `Chapter3/GoodPosition · nullSpace` and $Q_\Sigma = (\ell_i)_{i \in \Sigma} : N \to \mathbb{R}^\Sigma$ `Chapter3/GoodPosition · sigmaCoord`.

**Lemma 3.8.8 (preprint 3.65).** A good complement satisfies $\ell_i(p) = 0$ for $p \in P$, $i \in \Sigma$: take $a \in Q \setminus 0$ with $\ell_i(a) = 0$, scale $p$ so that $|p| \le \varepsilon |a|$; $a \in C$ gives $a + p \in C$, so $\ell_i(p) = \ell_i(a + p) \ge 0$; the same for $-p$ `Chapter3/GoodPosition · lemma_3_8_8`. *Consequences.* $Q_\Sigma$ is onto: for $k \in \mathbb{R}^\Sigma$ pick $v$ with $\ell_i(v) = k_i$ and split $v = n + p$; then $Q_\Sigma(n) = k$ (the text's (55), preprint p. 125) `Chapter3/GoodPosition · sigmaCoord_surjective`.

If an interior point $y$ of $Q$ has $\ell_i(y) = 0$, then $\ell_i = 0$ on $N$ ($y \pm tu \in Q$) `Chapter3/GoodPosition · restrFun_eq_zero_of_interior`.

**Proposition 3.1.25 (preprint 3.22), the normal form.** There are $k$, a subspace $N_0$ and a linear isomorphism $e : N \to \mathbb{R}^k \times N_0$ with $Q = e^{-1}([0,\infty)^k \times N_0)$, $k = \dim N - \dim(N \cap W)$, the $\mathbb{R}^k$-component of $e$ vanishing exactly on $N \cap W$, and the vanishing coordinates of $e(x)$ as many as the vanishing $\ell_i(x)$ for $x \in Q \setminus N \cap W$ `Chapter3/GoodPosition · QuadrantNormalForm` `Chapter3/GoodPosition · exists_quadrantNormalForm`. Three cases. *Case A*, $Q = N$: every $\ell_i$ is $\geq 0$ on a vector space, hence 0; $k = 0$ `Chapter3/GoodPosition · quadrantNormalForm_of_eq_univ` (a case the text's case distinction does not single out). *Case B*, $Q \neq N$, $\Sigma = \emptyset$: an interior point $y$ is nonzero with all $\ell_i(y) > 0$; for $u \in N$ the set $\{t : u + ty \in Q\}$ is closed, contains all large $t$ and is bounded below; at its minimum $t^*$ some $\ell_i(u + t^*y) = 0$, so $u + t^*y = 0$: $N$ is the line $\mathbb{R}y$ and $Q$ a half-line `Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx_empty` (the text's Lemma 3.8.11 (preprint 3.68), second case). *Case C*, $\Sigma \neq \emptyset$: first $Q = \{x \in N : \ell_i(x) \geq 0 \text{ for } i \in \Sigma\}$ — move from such an $x$ towards an interior point $y$ along $x + sy$; at the first parameter $s^*$ with $x + s^*y \in Q$, if $s^* > 0$ the point $z = x + s^*y$ is nonzero (else $\ell_i(x) < 0$ for $i \in \Sigma$), so it has $\ell_i(z) > 0$ for all $i$ ($\ell_i(z) \geq s^*\ell_i(y) > 0$ in $\Sigma$, by definition of $\Sigma$ outside), and finitely many strict inequalities survive a small step back, contradicting minimality `Chapter3/GoodPosition · mem_quadCone_of_vanIdx`. This replaces the text's extreme rays (Lemmas 3.8.1–3.8.3 (preprint 3.62–3.64), cited from [21], and 3.66–3.68); those lemmas are now proved as well, and are still not used (§III.59). Then a linear right inverse $g$ of $Q_\Sigma$ gives $e(x) = (Q_\Sigma x, x - gQ_\Sigma x) \in \mathbb{R}^\Sigma \times \ker Q_\Sigma$ `Chapter3/GoodPosition · splitEquiv`, and $\ker Q_\Sigma = N \cap W$ `Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx`. For a partial quadrant in good position this applies in a presentation `Chapter3/GoodPosition · IsInGoodPosition.quadrantNormalForm`, and choosing a basis of $N_0$ makes $(N, N \cap C)$ isomorphic to $(\mathbb{R}^{\dim N}, [0,\infty)^k \oplus \mathbb{R}^{\dim N - k})$ through an sc-isomorphism of constant structures `Chapter3/GoodPosition · scIsoOfLinearEquiv` `Chapter3/GoodPosition · preimage_eq_of_normalForm`: $N \cap C$ **is a partial quadrant in** $N$ `Chapter3/GoodPosition · proposition_3_1_25`.

**Lemma 3.5.9 (preprint 3.45) (linear form).** A continuous linear isomorphism $S : E_0 \to E_0'$ with $x \in C \iff Sx \in C'$ carries a good complement $P$ of $N$ to one of $S(N)$, with $\varepsilon' = \varepsilon/(\|S\|\|S^{-1}\| + 1)$: for $|Sp| \leq \varepsilon'|Sn|$, $|p| \leq \|S^{-1}\|\varepsilon'\|S\||n| \leq \varepsilon|n|$ `Chapter3/GoodPosition · IsGoodComplement.map`; and the interior condition is transported by the homeomorphism $S|_N$ `Chapter3/GoodPosition · IsInGoodPosition.map`. The text states it for $T\varphi(0)$; since the coordinate change of Theorem 3.4.2 (preprint 3.38) is linear, this is the form §§III.18–III.23 need.

**Proposition 3.8.12 (preprint 3.69).** In the model $\mathbb{R}^n \oplus W$, for $x \in N \cap C$: $d_{N\cap C}(x) = d_C(x)$ if $x \notin N \cap W$, and $d_{N\cap C}(x) + \dim(N \cap W) = \dim N$ if $x \in N \cap W$ `Chapter3/GoodPosition · proposition_3_8_12` — read off the normal form ($d_{N\cap C}$ does not depend on the presentation, Lemma 1.3.4 (preprint 1.24)).

## III.15 The degeneracy index along a graph: Lemma 3.8.13 (preprint 3.70)

**Lemma 3.8.13 (preprint 3.70).** In the model, for $N$ in good position with good complement $Y$ and $\tau : N \to E$ with values in $Y$, $C^1$ on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$, and $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $v \notin N \cap W$ `Chapter3/GoodPositionGraph · lemma_3_8_13`. The text distinguishes $\dim N = 1$ (a contradiction argument with $\tau(v) = o(|v|)$) and $\dim N \geq 2$ (the sign of a directional derivative). The formal proof unifies them.

*Ray decomposition.* From the normal form, $u_i = e^{-1}(\mathbf{e}_i, 0)$ are finitely many nonzero points of $Q$,

and every $v \in Q \setminus N \cap W$ has some $i$ and $s > 0$ with $v - su_i \in Q$ and $s|u_i| \le c|v|$ (take $s = \lambda_i > 0$ for $e(v) = (\lambda, n_0)$) `Chapter3/GoodPositionGraph · exists_rayDecomp` — the text's basis $e^j$ of $\widetilde{N}$. *Smallness.* $D\tau(0) = 0$ and continuity of $D\tau$ on the convex $Q$ make $\tau$ $\eta$-Lipschitz on $Q \cap B(\rho)$ for every $\eta$ `Chapter3/GoodPositionGraph · exists_lipschitz_small` `Chapter3/GoodPositionGraph · quadCone_convex`. *The margin.* Choose $\eta \le \varepsilon$ with $\eta\|\ell_j\||u_i| < \ell_j(u_i)$ for all $i$ and $j \notin \Sigma$ — a positive lower bound of finitely many positive numbers `Chapter3/GoodPositionGraph · exists_pos_le_of_forall_pos`. *Conclusion* `Chapter3/GoodPositionGraph · lemma_3_8_13_core`. For $|v| < \rho/(1+c)$: $|\tau(v)| \le \varepsilon|v|$, so $v + \tau(v) \in C$ (good complement). For $j \in \Sigma$, $\ell_j(\tau(v)) = 0$ (Lemma 3.8.8 (preprint 3.65)). For $v \notin N \cap W$, $w = v - su_i \in Q$, the segment $[w, v]$ lies in $Q \cap B(\rho)$ — which is why the radius is $\rho/(1+c)$, the text's "close to 0" — and for $j \notin \Sigma$,

$$\ell_j(v + \tau(v)) = \ell_j(w + \tau(w)) + s\ell_j(u_i) + \ell_j(\tau(v) - \tau(w)) \ge 0 + s\ell_j(u_i) - \|\ell_j\|\eta s|u_i| > 0.$$

So $v$ and $v + \tau(v)$ have the same vanishing coordinates, and Proposition 3.8.12 (1) (preprint 3.69) gives the lemma in the model, with the coordinate functionals `Chapter3/GoodPositionGraph · coordCLM`. (The level $Y_1$ of the text's $\tau$ plays no role.) For a partial quadrant with an arbitrary presentation the same proof runs with the coordinate functionals of the presentation, with "$v \notin N \cap W$" read as $-v \notin C$ (§III.61).

## III.16 The quadrant inverse function theorem (Theorem 3.8.15 (preprint 3.72), Proposition 3.8.17 (preprint 3.73), Lemmas 3.8.18, 3.8.21 (preprint 3.74, 3.77), Corollary 3.8.19 (preprint 3.75))

*Interior points.* For the model quadrant, $x$ is interior iff all its half-line coordinates are positive `Chapter3/QuadrantInverse · interior_quadrant`, so $d_C(x) = 0$ exactly at interior points `Chapter3/QuadrantInverse · degIndex_eq_zero_iff`. A linear isomorphism $S$ with $S(C) = C$ preserves $d_C$ (Lemma 1.3.4 (preprint 1.24) for the presentation $T \circ S$) `Chapter3/QuadrantInverse · ScIso.ofCLE` `Chapter3/QuadrantInverse · degIndex_map_eq`. A linear map carrying a set with nonempty interior onto itself is onto `Chapter3/QuadrantInverse · surjective_of_image_eq` — the text's "$\mathbb{R}^n = C - C$".

**Proposition 3.8.17 (preprint 3.73).** The hypotheses (`InvData`): $U$ relatively open and convex in the partial quadrant $C$ of a finite-dimensional $V$; $f : U \to C$ of class $C^j$, $j \ge 1$; $\|Df(x) - 1\| \le \frac{1}{2}$; and $x$ interior iff $f(x)$ interior — the only consequence of $d_C(x) = d_C(f(x))$ the proof uses `Chapter3/QuadrantInverse · InvData`. The text's $0 \in U$, $f(0) = 0$ are not needed. Conclusion: $f(U)$ is relatively open in $C$, $f$ is injective on $U$, with an inverse of class $C^j$ on $f(U)$ `Chapter3/QuadrantInverse · proposition_3_8_17`.

**Lemma 3.8.18 (preprint 3.74).** $x \mapsto x - f(x)$ is a $\frac{1}{2}$-contraction on the convex $U$ (mean value inequality) `Chapter3/QuadrantInverse · InvData.norm_sub_sub_le`, which gives (70): $\frac{1}{2}|x - x'| \le |f(x) - f(x')| \le \frac{3}{2}|x - x'|$ `Chapter3/QuadrantInverse · InvData.half_norm_le` `Chapter3/QuadrantInverse · InvData.norm_sub_le`, and injectivity `Chapter3/QuadrantInverse · InvData.injOn`. *Openness*: if $B(x', 3r) \cap C \subseteq U$ then $B(f(x'), r) \cap C \subseteq f(U)$ `Chapter3/QuadrantInverse · InvData.ball_inter_subset_image`. The set $A = f(\bar{B}(x', 2r) \cap C)$ is compact ($\bar{B}(x', 2r) \cap C \subseteq U$ is compact in finite dimensions). Interior points first: for an interior $y_1 \in B(y', r)$ choose an interior $x_0 \in U$ with $f(x_0) \in B(y', r)$ — the text takes any interior $x_0 \in B(x', 3r) \cap C$, which does not suffice — and follow the segment $y_t$ from $f(x_0)$ to $y_1$, which stays interior. $\{t : y_t \in A\}$ is closed and contains 0; at its maximum $t^* < 1$, $y_{t^*} = f(x^*)$ with $x^*$ interior and $|x^* - x'| < 2r$, and at the interior point $x^*$, $Df(x^*)$ is a unit `Chapter3/QuadrantInverse · InvData.isUnit_deriv` and $U$ a neighbourhood `Chapter3/QuadrantInverse · InvData.U_mem_nhds`, so the classical inverse function theorem maps neighbourhoods onto

neighbourhoods `Chapter3/QuadrantInverse · InvData.map_nhds_eq`, pushing $t^*$ further — a contradiction. The compactness of $A$ replaces the text's Cauchy sequence and also gives the remaining points by density. So $f(U)$ is relatively open `Chapter3/QuadrantInverse · InvData.image_relOpen`. **Corollary 3.8.19 (preprint 3.75)**: $f$ and its inverse `Chapter3/QuadrantInverse · InvData.inv` (2-Lipschitz by (70) `Chapter3/QuadrantInverse · InvData.norm_inv_sub_le`) are continuous and inverse to each other `Chapter3/QuadrantInverse · corollary_3_8_19`.

**Lemma 3.8.21 (preprint 3.77).** The inverse $g$ is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$ `Chapter3/QuadrantInverse · InvData.hasFDerivWithinAt_inv`: with $x_0 = g(y_0)$, $x_0 + \delta = g(y_0 + h)$, the quotient $|g(y_0 + h) - g(y_0) - Df(x_0)^{-1}h|/|h|$ is $|\delta|/|h|$ (bounded by (70)) times the remainder quotient of $f$ at $x_0$. The lemma's extra hypotheses (relatively open image, homeomorphism, invertible derivative) are consequences of those of Proposition 3.8.17 (preprint 3.73). *Regularity (the text: Lemma 3.8.20 (preprint 3.76) on the iterated tangent maps $T^k f$).* $x \mapsto [Df(x)]^{-1}$ is $C^{j-1}$ (inversion is smooth on units) `Chapter3/QuadrantInverse · InvData.contDiffOn_inverse_deriv`, so $Dg = [Df]^{-1} \circ g$ and $g \in C^i$, $i < j$, give $g \in C^{i+1}$: $g$ is $C^j$ `Chapter3/QuadrantInverse · InvData.contDiffOn_inv`. Lemma 3.8.20 (preprint 3.76) is not needed here; it is proved for the comparison, with Lemma 3.8.21 (preprint 3.77) in the general form the text's iteration needs (§III.60).

**Theorem 3.8.15 (preprint 3.72).** $U$ a relatively open neighbourhood of 0 in $C$, $f$ of class $C^j$ with $f(U) \subseteq C$, $f(0) = 0$, $Df(0)C = C$, $d_C(x) = d_C(f(x))$ on $U$: then $f$ is a $C^j$-diffeomorphism between relatively open neighbourhoods $U'$, $V'$ of 0 `Chapter3/QuadrantInverse · theorem_3_8_15`. *Proof.* $Df(0)$ is onto, hence invertible, with $Df(0)^{-1}C = C$; $\tilde{f} = Df(0)^{-1}f$ maps into $C$, $D\tilde{f}(0) = 1$, and preserves $d_C$ ($d_C \circ Df(0)^{-1} = d_C$), hence interior points. On $U' = B(0,\rho) \cap C$ with $\|D\tilde{f} - 1\| \le \frac{1}{2}$, Proposition 3.8.17 (preprint 3.73) applies, and $V' = Df(0)\tilde{f}(U')$ is relatively open since $Df(0)$ is a linear homeomorphism preserving $C$. The text's hypothesis (2) is used only through interior points; some such hypothesis is needed ($f(x,y) = (x, y + x^2)$ on $[0,\infty)^2$ satisfies (1) but $f(C)$ is not a neighbourhood of 0 in $C$).

## III.17 The implicit function theorem in partial quadrants (Theorem 3.8.22 (preprint 3.78), Lemmas 3.8.23–3.8.26 (preprint 3.79–3.82))

**Theorem 3.8.22 (preprint 3.78).** $U$ a relatively open neighbourhood of 0 in the partial quadrant $C$ of a finite-dimensional $V$, $f : U \to F$ of class $C^j$, $j \ge 1$, with $f(0) = 0$, $Df(0)$ onto, $K = \ker Df(0)$ in good position with good complement $Y$. Then there are a relatively open $V' \ni 0$ in $K \cap C$, $\tau : K \to Y$ of class $C^j$ on $V'$ and $\varepsilon, \sigma > 0$ with (1) $\tau(0) = 0$, $D\tau(0) = 0$, and $k + \tau(k) \in U$, $f(k + \tau(k)) = 0$ for $k \in V'$; (2) $|\tau(k)| \le \sigma$ for $|k| \le \varepsilon$; (3) if $k \in V'$, $|k| \le \varepsilon$, $y \in Y$, $|y| \le \sigma$, $k + y \in U$ and $f(k + y) = 0$, then $y = \tau(k)$ `Chapter3/QuadrantIFT · theorem_3_8_22`. The two bold clauses are not in the text; the first is the content of the theorem (its proof has $\tau(k) = B(k, \tau(k))$), the second is needed for $f(k + y)$ to make sense.

*Normalization (71)–(72).* With $L = (Df(0)|_Y)^{-1}$, in coordinates $x = k + y \in K \oplus Y$, $Lf(k + y) = y - B(k, y)$ with $B(k, y) := y - Lf(k + y)$, $B(0,0) = 0$, $DB(0,0) = 0$. The data after normalization: on $K \times Y$ a convex set $C_p$ with nonempty interior, a relatively open $U_p \ni 0$, $B$ of class $C^j$ on $U_p$ with $B(0) = 0$, $DB(0) = 0$, the good-position estimate and the interior condition `Chapter3/QuadrantIFT · ContractionData`. Only convexity and interior are used; the quadrant enters through good position.

*Lemmas 3.8.23, 3.8.24 (preprint 3.79, 3.80): the constants* `Chapter3/QuadrantIFT · Radii` `Chapter3/QuadrantIFT · ContractionData.exists_radii`. With $\varepsilon$ the good-position constant and $c_0 = \varepsilon/(2(1 + \varepsilon)) \le \frac{1}{2}$ `Chapter3/QuadrantIFT · Radii.c₀`, continuity of $DB$ at 0 gives $a, b$ with $\varepsilon a \le b$, the region

$R = \{|k| \le a, |y| \le b\} \cap C_p \subseteq U_p$ Chapter3/QuadrantIFT · Radii.R Chapter3/QuadrantIFT · Radii.R_sub_U and $\|DB\| \le c_0$ on $R$. $R$ is convex Chapter3/QuadrantIFT · Radii.R_convex, so the mean value inequality gives $|B(q) - B(p)| \le c_0|q - p|$ on $R$ Chapter3/QuadrantIFT · Radii.norm_sub_le Chapter3/QuadrantIFT · Radii.norm_B_sub_le_snd and $|B(p)| \le c_0|p|$ Chapter3/QuadrantIFT · Radii.norm_B_le. For $k \in V_0 = \{k \in K \cap C : |k| < a\}$ Chapter3/QuadrantIFT · Radii.V₀ and $|y| \le \varepsilon|k|$: good position puts $(k, y) \in C_p$, $|y| \le \varepsilon a \le b$, so $(k, y) \in R$ and $|B(k, y)| \le c_0(1 + \varepsilon)|k| = \frac{\varepsilon}{2}|k|$. So $B(k, \cdot)$ maps $X_k = \bar{B}_Y(\varepsilon|k|)$ into itself Chapter3/QuadrantIFT · Radii.mapsTo_ball and has a fixed point $\tau(k)$ there Chapter3/QuadrantIFT · Radii.exists_fixed Chapter3/QuadrantIFT · Radii.τ; any fixed point $y$ with $(k, y) \in R$ equals $\tau(k)$ Chapter3/QuadrantIFT · Radii.eq_τ. (In Lemma 3.8.24 (preprint 3.80) the maximum is over $B((1 + \varepsilon)a) \cap C$, where $DB$ is defined.)

*Lemma 3.8.25 (preprint 3.81): continuity.* $|\tau(k)| \le \frac{\varepsilon}{2}|k|$ Chapter3/QuadrantIFT · Radii.norm_τ_le. If $|k_0| \le 2|k|$, then $(k, \tau(k_0)) \in R$ and $|\tau(k) - \tau(k_0)| \le \frac{1}{2}|\tau(k) - \tau(k_0)| + c_0|k - k_0|$, so $|\tau(k) - \tau(k_0)| \le \varepsilon|k - k_0|$ — the text's estimate, whose argument needs $(k, \tau(k_0))$ in the domain. If $|k_0| > 2|k|$, then $|k - k_0| \ge \frac{1}{2}|k_0|$ and $|\tau(k) - \tau(k_0)| \le \frac{\varepsilon}{2}(|k| + |k_0|) \le \frac{3\varepsilon}{2}|k - k_0|$. In all cases the constant is $1 + 2\varepsilon$ Chapter3/QuadrantIFT · Radii.norm_τ_sub_le, and $\tau$ is continuous Chapter3/QuadrantIFT · Radii.continuousOn_τ. Since $X_k$ depends on $k$, the parametrized fixed point theorem of §III.8 does not apply, and continuity is proved by hand, as in the text.

*Lemma 3.8.26 (preprint 3.82): regularity.* With $D_1B = DB \circ \mathrm{inl}$, $D_2B = DB \circ \mathrm{inr}$ Chapter3/QuadrantIFT · Radii.D₁ Chapter3/QuadrantIFT · Radii.D₂, $\|D_2B\| \le \frac{1}{2}$ on $R$ Chapter3/QuadrantIFT · Radii.norm_D₂_le, $M = (1 - D_2B)^{-1}$ (Neumann) Chapter3/QuadrantIFT · Radii.M and $\Phi = MD_1B$ Chapter3/QuadrantIFT · Radii.Φ: from the fixed-point identities and the little-o composition of $B$ with the Lipschitz $k' \mapsto (k', \tau(k'))$, $\tau$ is differentiable within $V_0$ with $D\tau(k) = \Phi(k, \tau(k))$ Chapter3/QuadrantIFT · Radii.hasFDerivWithinAt_τ Chapter3/QuadrantIFT · Radii.fderivWithin_τ. $\Phi$ is $C^{j-1}$ on $R$ Chapter3/QuadrantIFT · Radii.contDiffOn_Φ, so $\tau \in C^i$, $i < j$, gives $D\tau = \Phi \circ (\mathrm{id}, \tau) \in C^i$ and $\tau \in C^{i+1}$: $\tau$ is $C^j$ on all of $V_0$ Chapter3/QuadrantIFT · Radii.contDiffOn_τ. The text iterates $f \mapsto f^{(1)}$ (the finite-dimensional model of Lemma 3.3.4 (preprint 3.31)) and concludes "on a possibly smaller neighbourhood"; in finite dimensions there are no levels to shift and no shrinking is needed. *Assembly* Chapter3/QuadrantIFT · ContractionData.exists_solution: $V' = V_0$, $\sigma = b$, $\varepsilon$ below $a$; (1) $Lf(k + \tau(k)) = \tau(k) - B(k, \tau(k)) = 0$ and $L$ is injective; (3) $f(k + y) = 0$ means $B(k, y) = y$ with $(k, y) \in R$.

## III.18 Weak stability of basic germs: Theorem 3.4.2 (preprint 3.38)

**Theorem 3.4.2 (preprint 3.38)** (text): for a basic germ $f$ and the principal part $s$ of an sc⁺-section with $s(0) = 0$ there is a strong bundle isomorphism $\Phi$ covering an sc-diffeomorphism $\varphi$ such that $(\Phi_*(f + s))^1$, with the levels raised by one, is a basic germ, and $n - N = n' - N'$. The text's proof: with $A = P_W \circ D_2 s(0)$, $1 + A$ is sc-Fredholm of index 0, with sc-splittings $W = C \oplus X = R \oplus Z$, $C = \ker(1 + A)$, $R = \mathrm{range}(1 + A)$; $P_W \circ s(a, w) = Aw + S(a, w)$ with $S$ a small contraction in $w$ on every level $m \ge 1$; and with $L = (1 + A)|_X : X \to R$ the equation becomes $x - L^{-1}P_2\bar{B}(a, c + x) = 0$ in the new parameters $(a, c)$ and the new fibre variable $x$. The formal proof follows it with two observations: the coordinate change is *linear*, and the index raise is a *change of sc-structure*.

*The linear part of $s$.* For an sc¹-germ $s$ into $W^1$, $A = D_2 s(0)$ is an sc⁺-operator whose level-$m$ map is the level operator of Proposition 1.2.1 (preprint 1.16) at $0 \in U_{m+1}$ Chapter3/BasicClass · scPlusOfDeriv. The remainder $S(a, w) = s(a, w) - Aw$ on level $m + 1$ Chapter3/BasicClass · scPlusRemainder is, for $a$ and $w$ small depending on $m$ and $\varepsilon$, an $\varepsilon$-contraction in $w$: $D_2S(0) = 0$ on level $m + 1$, the derivative of $s : U_{m+1} \to W_{m+1}$ is continuous in the operator norm (§III.1), and the mean value inequality on a ball

for fixed $a$ concludes Chapter3/BasicClass · scPlusRemainder_contraction. The partial derivative of $S$ in $a$ does not vanish and plays no role. The text gives this one sentence ("$S$ is … an arbitrary small contraction on every level $m \geq 1$").

*The data and the splitting.* The data of the theorem are the germ $G$, the $W$-component $s$ of the sc$^+$-germ with $s(0) = 0$, and an sc-Fredholm splitting $W = C \oplus X = R \oplus Z$ of $T = 1 + A$ Chapter3/BasicClass · StabilityData. Such data exist: the identity is sc-Fredholm of index 0 (the trivial splitting) Chapter3/BasicClass · isScFredholm_id Chapter3/BasicClass · scIndex_id, so $1 + A$ is sc-Fredholm by Proposition 1.1.11 (preprint 1.10) Chapter3/BasicClass · StabilityData.exists_of_scPlus. With the projections $P_1 : W_m \to X_m$ along $C_m$, $P_2 : W_m \to R_m$ along $Z_m$ Chapter3/BasicClass · StabilityData.projX Chapter3/BasicClass · StabilityData.projR, $L = T|_X : X_m \to R_m$ is bijective Chapter3/BasicClass · StabilityData.Lm_bijective, a topological isomorphism by the open mapping theorem Chapter3/BasicClass · StabilityData.Lequiv, and $L^{-1}P_2$ is an sc-operator $W \to X$ Chapter3/BasicClass · StabilityData.LinvPOp. The finite-dimensional $C$, $Z \subseteq W_\infty$ lift to every level Chapter3/BasicClass · ScBanach.finLift.

*The coordinate changes.* On the index-raised spaces, the new source is $E' = (P \oplus C) \oplus X^1$ Chapter3/BasicClass · StabilityData.E' and the new target $F' = (Q \oplus Z) \oplus X^1$ Chapter3/BasicClass · StabilityData.F' — the text's $\tau : Z \to C$, making the new finite-dimensional target $\mathbb{R}^N \oplus C$, is not applied. The base change $\psi(a, w) = ((a, (1 - P_1)w), P_1 w)$ Chapter3/BasicClass · StabilityData.split and its inverse $\psi^{-1}((a, c), x) = (a, c + x)$ Chapter3/BasicClass · StabilityData.glue are sc-operators inverse to each other Chapter3/BasicClass · StabilityData.glue_split Chapter3/BasicClass · StabilityData.split_glue; so are the fibre change $\Psi(q, w) = ((q, (1 - P_2)w), L^{-1}P_2 w)$ Chapter3/BasicClass · StabilityData.fiber and $\Psi^{-1}((q, z), x) = (q, z + (1 + A)x)$ Chapter3/BasicClass · StabilityData.fiberInv Chapter3/BasicClass · StabilityData.fiber_fiberInv Chapter3/BasicClass · StabilityData.fiberInv_fiber, and $\Psi$ vanishes only at 0 Chapter3/BasicClass · StabilityData.fiber_eq_zero_iff. No general strong bundle isomorphism is needed.

*The push-forward.* The perturbed germ is $f + \iota s$ with the inclusion $W^1 \to Q \oplus W$ Chapter3/BasicClass · ScBanach.inrOp Chapter3/BasicClass · StabilityData.fsum, sc-smooth by the pairing and operator rules Chapter3/BasicClass · StabilityData.fsum_isScSmoothGerm. Its germ of neighbourhoods on $E'$ is $\psi(U_{m+1})$, the preimage of $U_{m+1}$ under $\psi^{-1}$ Chapter3/BasicClass · StabilityData.pushGerm Chapter3/BasicClass · StabilityData.glue_mapsToGerm, and the push-forward is $(\Phi_*(f + s))^1 = \Psi \circ (f + \iota s)^1 \circ \psi^{-1}$ with the level-shifted operators Chapter3/BasicClass · ScOperator.shiftOp Chapter3/BasicClass · StabilityData.pushMap. It is sc-smooth: $(f + \iota s)^1$ is sc-smooth on the shifted germ (Proposition 1.2.2 (preprint 1.17) for germs) and $\Psi$, $\psi^{-1}$ are sc-operators Chapter3/BasicClass · StabilityData.pushMap_isScSmoothGerm. With $\bar{B} = B - S$ Chapter3/BasicClass · StabilityData.Bbar, the $W$-component of $(f + \iota s)^1$ at a level-1 point is $(1 + A)w - \bar{B}(a, w)$ Chapter3/BasicClass · StabilityData.snd_levelMap_fsum, so the $X$-component of the push-forward is $x - \hat{B}((a, c), x)$ with $\hat{B}((a, c), x) = L^{-1}P_2\bar{B}(a, c + x)$ Chapter3/BasicClass · StabilityData.pushB Chapter3/BasicClass · StabilityData.snd_pushMap — the text's "desired formula". $\hat{B}$ is an sc$^0$-contraction germ on the index-raised structure: on level $m$ of $E'$, i.e. level $m + 1$ of $W$, $B$ contracts by Definition 3.1.7 (preprint 3.7) and $S$ by the remainder estimate, and $L^{-1}P_2$ is bounded Chapter3/BasicClass · StabilityData.levelMap_pushB Chapter3/BasicClass · StabilityData.pushB_isSc0ContractionGerm. So **the push-forward is a basic germ with contraction map** $\hat{B}$ Chapter3/BasicClass · theorem_3_4_2. Its zeros correspond to those of $f + \iota s$ under $\psi$, since $\Psi$ is a linear isomorphism Chapter3/BasicClass · StabilityData.pushMap_eq_zero_iff. *The index*: $\mathrm{ind}(1 + A) = \mathrm{ind}(1) = 0$ (Proposition 1.1.11 (preprint 1.10) with its index), i.e. $\dim C = \dim Z$, so $n' - N' = (n + \dim C) - (N + \dim Z) = n - N$ Chapter3/BasicClass · StabilityData.finrank_Cs_eq_finrank_Zs.

*The same on level* 0. For §§III.19–III.20 the coordinate changes are also needed before the index raise: $\psi$, $\psi^{-1}$, $\Psi$, $\Psi^{-1}$ on every level `Chapter3/BasicClass · StabilityData.split₀` `Chapter3/BasicClass · StabilityData.glue₀` `Chapter3/BasicClass · StabilityData.fiber₀`, the germ $\psi(U_m)$ `Chapter3/BasicClass · StabilityData.pushGerm₀`, and $g = \Phi_*(f+s) = \Psi \circ (f + \iota s) \circ \psi^{-1}$ on level 0 `Chapter3/BasicClass · StabilityData.pushMap₀`, sc-smooth `Chapter3/BasicClass · StabilityData.pushMap₀_isScSmoothGerm`, whose level-1 map is the index-raised push-forward `Chapter3/BasicClass · StabilityData.pushMap_eq_levelMap_pushMap₀`.

## III.19 The linearization at nearby solutions: Corollary 3.5.3 (preprint 3.41)

**Corollary 3.5.3 (preprint 3.41).** For the data of Theorem 3.4.2 (preprint 3.38) with $f := h$ basic and $D(h+\iota s)(0)$ surjective, there is $r > 0$ such that every zero $x$ of $h + \iota s$ in $U_0$ with $|x|_0 < r$ is $\iota(x_1)$ with $x_1 \in U_1$, and $D(h+\iota s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$ `Chapter3/SolutionLinearization · StabilityData.corollary_3_5_3`. The text's sc⁺-map $s$ has an $\mathbb{R}^N$-component; it can be absorbed into the basic germ (as in §III.23), so the data carry only the $W$-component.

*The derivative on* $U_1$. For $x \in U_1$, $D(h+\iota s)(x) = Dh(x) + \iota Ds(x)$ — uniqueness of the derivative within $U_1$, then density of $E_1$ `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq` — and $\mathrm{pr}_W Dh(x) = \mathrm{pr}_W - DB(x)$, from $\mathrm{pr}_W h = \mathrm{pr}_W - B$ on $U_0$ `Chapter3/SolutionLinearization · StabilityData.snd_Df_eq`; $D_2B(0) = 0$ `Chapter3/SolutionLinearization · StabilityData.DB_zero_snd`. *The decomposition.* $D(h+\iota s)(x) = D(h+\iota s)(0) + S(x) + K(x)$ with $S(x)(\alpha,\zeta) = (0, -D_2B(x)\zeta)$ `Chapter3/SolutionLinearization · StabilityData.Sop` `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq_add` of norm $\le \varepsilon$ for $x \in U_1$ small on level 1, by Lemma 3.1.9 (preprint 3.9) (c) `Chapter3/SolutionLinearization · StabilityData.exists_norm_Sop_le`, and $K(x)$ `Chapter3/SolutionLinearization · StabilityData.Kop` compact: the sum of finite-rank operators through $Q$ and through $P$ and of $\iota(Ds(x) - Ds(0))$ through the compact inclusion $W_1 \to W_0$ `Chapter3/SolutionLinearization · StabilityData.Kop_isCompactOperator`.

*Fredholm with index* $n - N$ *at every small* $x \in U_1$. $D(h+\iota s)(0)$ is Fredholm of index $n - N$: $Dh(0)$ is (Proposition 3.1.10 (preprint 3.10) for the basic germ, §III.5) and $\iota Ds(0)$ is compact `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`. A small perturbation of a Fredholm operator is Fredholm: if $v$ is a quasi-inverse of $u$ and $\|v\|\|s\| < 1$, then $(1+vs)^{-1}v$ is a quasi-inverse of $u+s$ (Neumann series) `Chapter3/SolutionLinearization · isFredholm_add_of_norm_lt`, with the same index `Chapter3/SolutionLinearization · isFredholm_fredholmIndex_add_of_norm_lt`; a compact perturbation does not change the index `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`. So $D(h+\iota s)(x)$ is Fredholm of index $n-N$ for *every* $x \in U_1$ small on level 1, not only at solutions `Chapter3/SolutionLinearization · StabilityData.exists_isFredholm_Dfs'`.

*Surjectivity: the compactness argument.* Let $N$ be a closed complement of $\ker D(h+\iota s)(0)$. Then $\ker D(h+\iota s)(x) \cap N = 0$ for $x \in U_1$ small on level 1 `Chapter3/SolutionLinearization · StabilityData.exists_ker_Dfs'_inf_eq_bot`. Otherwise there are $x_k \to 0$ in $E_1$ and unit vectors $(\alpha_k, \zeta_k) \in N$ in the kernels; the $W$-component of the equation reads $\zeta_k = D_1B(x_k)\alpha_k + D_2B(x_k)\zeta_k - \iota Ds(x_k)(\alpha_k,\zeta_k)$. $\alpha_k$ has a convergent subsequence ($P$ finite-dimensional); $Ds(x_k)(\alpha_k,\zeta_k)$ is bounded in $W_1$ (the derivatives near 0 are uniformly bounded `Chapter3/SolutionLinearization · IsSc1GermWith.exists_norm_deriv_le`), so its image in $W_0$ has a convergent subsequence; $D_1B(x_k)\alpha_k \to D_1B(0)\alpha$ by the joint continuity of the tangent map `Chapter3/SolutionLinearization · IsSc1GermWith.tendsto_deriv`; and $\|D_2B(x_k)\| \to 0$. So $\zeta_k$ converges, $(\alpha_k,\zeta_k) \to (\alpha, z_0) \in N$ of norm 1 in $\ker D(h+\iota s)(0)$ — a contradiction. (The text uses the Neumann series $\zeta_k = \sum (D_2B(x_k))^l z_k$; here $|\zeta_k - z_k| \le \varepsilon_k \to 0$ suffices.) Then

$\dim\ker D(h+\iota s)(x) \le \dim\ker D(h+\iota s)(0) = n - N = \mathrm{ind}$, and an operator of finite index with $\dim\ker \le \mathrm{ind}$ is onto `Chapter3/SolutionLinearization · HasFiniteIndex.surjective_of_finrank_ker_le` — instead of the text's "$Df(x)|_N$ has index 0 and is injective". *Zeros are on level* 1 *and small there*: by the lifts of Theorem 3.5.1 (preprint 3.40), continuous at 0 (§III.12); the text's detour through "bounded on level 2, hence convergent on level 1" is not needed.

## III.20 Local solvability: Lemma 3.5.10 (preprint 3.46), Theorem 3.5.6 (preprint 3.43), Lemmas 3.5.5, 3.5.8 (preprint 3.42, 3.44), Theorem 3.5.11 (preprint Corollary 3.47)

**Lemma 3.5.10 (preprint 3.46)** (Appendix 3.8.3 (preprint 3.8.2)). If $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$, the image of the graph map `Chapter3/LocalSolvability · graphMap` (with $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ iff $\alpha \in \widetilde{C}$ `Chapter3/LocalSolvability · graphMap_mem_prod_iff`), has a good complement $Y'$ with respect to $\widetilde{C} \oplus W$, then $\widetilde{K}$ has a good complement $Z$ with respect to $\widetilde{C}$ such that $Z \times W$ is a good complement of $K'$ `Chapter3/LocalSolvability · lemma_3_5_10`. *Proof.* $Z' := \mathrm{pr}_P(Y')$ satisfies $P = \widetilde{K} + Z'$; choose $Z \subseteq Z'$ with $P = \widetilde{K} \oplus Z$ and a linear right inverse $\rho : Z' \to Y'$ of the projection (bounded, finite dimensions); for $|z| \le \varepsilon'|\alpha|$, $\varepsilon' = \varepsilon/(\|\rho\| + 1)$, $y = \rho(z) = (z, w) \in Y'$ has $|y| \le \varepsilon|(\alpha, L\alpha)|$, so $(\alpha, L\alpha) \in C'$ iff $(\alpha, L\alpha) + y \in C'$, i.e. $\alpha \in \widetilde{C}$ iff $\alpha + z \in \widetilde{C}$. The text's intermediate complement $Y_0$ is not needed. Good position of $K'$ gives good position of $\widetilde{K}$ `Chapter3/LocalSolvability · lemma_3_5_10_isInGoodPosition`.

**Projecting solutions into the quadrant.** A gap in the text's proof of Theorem 3.5.6 (3) (preprint 3.43): a solution $a = k + z$ ($k \in \widetilde{K}$, $z \in Z$) of $H(a) = 0$ is matched with the solution of Theorem 3.8.22 (preprint 3.78) by its uniqueness clause, which needs $k \in \widetilde{C}$ — not verified in the text. It holds because $z$ is small compared with $k$: from $H(a) = 0 = H(0)$ and $DH(0)k = 0$, $|DH(0)z| = o(|a|)$; $DH(0)$ is injective on $Z$, hence bounded below there `Chapter3/LocalSolvability · exists_norm_le_of_injOn`; so $|z| \le \varepsilon|k|$ near 0, and the good-complement property gives $k \in \widetilde{C}$ from $k + z = a \in \widetilde{C}$ `Chapter3/LocalSolvability · exists_proj_mem_of_eq_zero`.

**The data of Theorem 3.5.6 (preprint 3.43).** The text states it for $g = \Phi_*(f + s)$; the formal statement is for any sc-smooth germ $g$ on $G$ with $g(0) = 0$ whose index-raised germ $g^1$ on the shifted germ is a basic germ with contraction map $B$ `Chapter3/LocalSolvability · ShiftedBasicData`. Its solution germ $\delta$ (of the working form of $B$) takes values in $W_1$ `Chapter3/LocalSolvability · ShiftedBasicData.δ`, with $\mathrm{pr}_W g(a, \delta(a)) = 0$ on level 0 `Chapter3/LocalSolvability · ShiftedBasicData.snd_g_sol` and uniqueness in the contraction region `Chapter3/LocalSolvability · ShiftedBasicData.eq_δ`. Put $H(a) = \mathrm{pr}_Q\, g(a, \delta(a))$ `Chapter3/LocalSolvability · ShiftedBasicData.H`. *Regularity of* $H$ (the text: "$H$ restricted to $U_i'$ is of class $C^{i-1}$"): for every $j \ge 1$ there is a relatively open $V \ni 0$ in $\widetilde{C}$ on which $H$ is $C^j$ — by Corollary 3.3.7 (preprint 3.34) for $g^1$, $\delta$ lifts to a $C^j$ map into $W_{2j}$, and $g : E_{2j} \to F_j$ is $C^j$ by Proposition 1.2.3 (preprint 1.18) for germs `Chapter3/LocalSolvability · ShiftedBasicData.H_contDiffOn`. A single $C^1$ neighbourhood $V_1$ carries the derivative data `Chapter3/LocalSolvability · ShiftedBasicData.Deriv` `Chapter3/LocalSolvability · ShiftedBasicData.exists_deriv`: $D\delta(0) : P \to W_1$, $L = \iota D\delta(0)$, and $DH(0) = \mathrm{pr}_Q Dg(0)(\alpha, L\alpha)$ `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀` `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_H`, by the chain rule for $a \mapsto g(a, \delta(a))$ `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_comp`.

**Lemma 3.5.8 (preprint 3.44).** "Recalling from Lemma 3.9 that $D_2B(0) = 0$": Lemma 3.1.9 (preprint 3.9) gives this for $g^1$, on level 1; the kernel of $Dg(0)$ is computed on level 0. So first the $W$-component

of $Dg(0)(0,\zeta)$ is $\zeta$ for $\zeta \in W_1$, by Lemma 3.1.9 (preprint 3.9) (b) for $g^1$, and then for all $\zeta \in W_0$ by density Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_inr. Differentiating $\mathrm{pr}_W g(a,\delta(a)) = 0$ at 0 gives $\mathrm{pr}_W Dg(0)(\alpha, L\alpha) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_graph. Hence $Dg(0)(\alpha,\zeta) = 0$ iff $\zeta = L\alpha$ and $DH(0)\alpha = 0$: $\ker Dg(0) = \{(\alpha, D\delta(0)\alpha) : \alpha \in \ker DH(0)\}$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq, and $DH(0)$ is onto when $Dg(0)$ is Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀_surjective.

**Theorem 3.5.6 (preprint 3.43)** Chapter3/LocalSolvability · theorem_3_5_6. Hypotheses: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position to $\widetilde{C} \oplus W$, and *the regularity of Lemma 3.5.5 (preprint 3.42) as an explicit hypothesis*: for every $\varepsilon > 0$ there is $r > 0$ such that every zero $x$ of $g$ in $U_0$ with $|x|_0 < r$ has $W$-component $\iota(w_1)$, $|w_1|_1 < \varepsilon$. *The choices* Chapter3/LocalSolvability · ShiftedBasicData.Sol Chapter3/LocalSolvability · ShiftedBasicData.exists_sol: $\widetilde{K} = \ker DH(0)$ with $K'$ its graph (Lemma 3.5.8 (preprint 3.44)); the good complement $Z$ of $\widetilde{K}$ with $Z \oplus W$ good for $K'$ (Lemma 3.5.10 (preprint 3.46)); and the solution $\sigma : V' \to Z$ of $H(k+\sigma(k)) = 0$ with its uniqueness constants, from Theorem 3.8.22 (preprint 3.78) applied to $H$ (§III.17). Then $\tau(y) = (\sigma(v), \delta(v+\sigma(v)) - Lv)$ for $y = (v, Lv)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ on $V = \pi^{-1}(V')$, relatively open in $K' \cap C'$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.V Chapter3/LocalSolvability · ShiftedBasicData.Sol.V_relOpen, with values in $Z \oplus W$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_mem_Y, so that $y + \tau(y) = (a, \delta(a))$ with $a = v + \sigma(v)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.coe_add_τ. (3), first half: $y + \tau(y)$ lies in $U_0$ and is a zero, since $\mathrm{pr}_W g(a,\delta(a)) = 0$ and $\mathrm{pr}_Q g(a,\delta(a)) = H(a) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.sol_mem_and_eq_zero. Regularity: $\tau$ is $C^1$ on $V$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_contDiffOn with $W$-component on level 1 Chapter3/LocalSolvability · ShiftedBasicData.Sol.snd_τ_mem_range, and $D\tau(0)\eta = -D\delta(0)\pi(\eta) + D\delta(0)\pi(\eta) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.hasFDerivWithinAt_τ Chapter3/LocalSolvability · ShiftedBasicData.Sol.fderivWithin_τ — (2). (3), second half Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ: a zero $x = (a, w)$ near 0 has, by the regularity hypothesis, $w$ on level 1 and small there, so $w = \delta(a)$ by the uniqueness of the solution germ on level 1, so $H(a) = 0$; write $a = k + z$ with the projection along $Z$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.projK; then $k \in \widetilde{K} \cap \widetilde{C}$ (the gap above) and $z = \sigma(k)$ by the uniqueness clause of Theorem 3.8.22 (preprint 3.78); so $x = y + \tau(y)$ for $y = (k, Lk)$. *Clause (1), "$\tau$ is an sc-smooth germ"*, is proved level-wise: for every $k \geq 1$ there are a relatively open $V_k \subseteq V$ containing 0 and a $C^k$ map $\tau_k : V_k \to E_k = P \oplus W_k$ with $\iota_k \tau_k = \tau$ on $V_k$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level — the reading of Corollary 3.3.7 (preprint 3.34) and Definition 3.2.1 (ii) (preprint 3.25). The proof applies Theorem 3.8.22 (preprint 3.78) to $H$ at regularity $k$ on the neighbourhood of its regularity lemma, and identifies the $C^k$ solution $\sigma_k$ with $\sigma$ near 0 through the uniqueness clause (3) of Theorem 3.8.22 (preprint 3.78) — which needs $\sigma$ continuous at 0 with $\sigma(0) = 0$, so that $|\sigma(y)|$ is below the uniqueness constant for $y$ small; the text says nothing about why the solutions produced for the different $k$ coincide. The $W$-component $\iota\delta(a) - L\pi(y)$, $a = \pi y + \sigma(\pi y)$, lifts to $W_k$ through the $C^k$ lift of $\delta$ into $W_{2k}$, with $L$ the derivative of that lift at 0 composed with the inclusion (uniqueness of the derivative of $\iota\delta$ at 0). The full statement is Chapter3/LocalSolvability · theorem_3_5_6_sc_smooth; the statement without (1) is kept for Theorem 3.6.8 (preprint 3.54).

**Lemma 3.5.5 (preprint 3.42).** For the push-forward $g = \Phi_*(f+s)$ of Theorem 3.4.2 (preprint 3.38) — packaged as the data of Theorem 3.5.6 (preprint 3.43), $g$ on level 0 with index-raised basic germ $(\Phi_*(f+s))^1$ Chapter3/LocalSolvability · StabilityData.toShiftedBasicData — the regularity hypothesis holds: a zero $x'$ of $g$ in $\psi(U_0)$ with $|x'|_0$ small has $\psi^{-1}(x')$ a zero of $f + \iota s$, which by the lifts of Theorem 3.5.1 (preprint 3.40), continuous at 0, lies on level 1 and is small there; $\psi$ is linear on level 1 Chapter3/

LocalSolvability · StabilityData.lemma_3_5_5.

**Theorem 3.5.11 (preprint Corollary 3.47)** Chapter3/LocalSolvability · theorem_3_5_11_sc_smooth Chapter3/LocalSolvability · theorem_3_5_11. For a basic germ $f$ and the data of Theorem 3.4.2 (preprint 3.38) with $D(f+\iota s)(0)$ surjective and kernel $K$ in good position to $C = \widetilde{C} \oplus W$: a good complement $Y$, $V$ relatively open in $K \cap C$, $\sigma : V \to Y$ of class $C^1$ with values on level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$, the graph consisting of zeros of $f + \iota s$, an open $U'' \ni 0$ in which every zero lies on the graph, and for every $k$ a $C^k$ lift of $\sigma$ into $E_k$ near 0. *Proof.* Theorem 3.5.6 (preprint 3.43) for $g = \Phi_*(f+s)$, with Lemma 3.5.5 (preprint 3.42) for its regularity hypothesis. Since the coordinate change is linear, $Dg(0) = \Psi \circ D(f+\iota s)(0) \circ \psi^{-1}$ (chain rule for sc-operators) Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀, so surjectivity and the kernels correspond, $\ker Dg(0) = \psi(K)$; good position of $\psi(K)$ follows by Lemma 3.5.9 (preprint 3.45) (finite dimensionality of $K$ from Lemma 3.5.8 (preprint 3.44)); and the conclusion is transported back by $\psi^{-1}$: $Y = \psi^{-1}(Y')$, $V = \psi^{-1}(V')$, $\sigma = \psi^{-1}\tau\psi$. The text transports the solution set back with the nonlinear $\varphi$ and justifies the good position with the degeneracy index (Lemma 3.8.13 (preprint 3.70), Theorem 3.8.15 (preprint 3.72)); with $\psi$ linear neither is needed. The transport uses that $\widetilde{C} \oplus C$ is a partial quadrant for finite-dimensional $C$ Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ.

# III.21 The linearization at a zero (Proposition 3.6.5 (preprint 3.52)) and good position of a germ (Definition 3.6.6 (preprint 3.53))

**Proposition 3.6.5 (preprint 3.52)** (text): if $f$ is an sc-Fredholm section of a tame strong bundle and $f(x) = 0$, then $Tf(x) : T_xX \to Y_x$ is an sc-Fredholm operator. Formalized in the chart of Definition 3.1.11 (preprint 3.11) centred at a zero $x_0$, with the level-0 Fredholm property Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint Chapter3/FredholmGermLinearizationG · proposition_3_6_5_local: $Dg(0) : E_0 \to F_0$ and $Tf(x_0) = Dg(0)|_{T_0O} : T_0O \to K_0$ are Fredholm with $\operatorname{ind} Tf(x_0) = \operatorname{ind} Dg(0)$. Here $Dg(0)$ is the value at 0 of a derivative family $Dg$ of the filling germ on a germ $G$ with $G_0 \subseteq U$, the same for every family and every germ (§III.4); the statement exposes $G$ and $Dg$, where the whole-chart statement Chapter3/FredholmGermLinearization · proposition_3_6_5_wholeChart has the derivative on $U$, a junk value for a germ. (The sc-Fredholm property of $Tf(x)$ would need $T_0O = Dr(0)E$ and $K_0 = \rho(0)F$ as sc-Banach spaces, not set up. The statements quantify the index as "$n - N$ for some natural numbers $n$, $N$", which every integer satisfies; the identification with $\dim P - \dim Q$ of the conjugating basic germ is made inside the proof, by `proposition_3_1_10_basic_level_zero`, and not exported.)

*Step (a): the derivative of the conjugation* Chapter3/FredholmGermLinearizationG · IsConjugateToBasicGerm.scDeriv_isFredholm. Let $k = g - \iota s$, an sc¹-germ on $G_g \cap G_s$ (§III.48) with $k(0) = 0$ ($s(0) = g(0)$ in Definition 3.1.11 (preprint 3.11), $g(0) = \mathbf{f}(0) = 0$ by the filler) and $(\Psi(u, k(u)))_2 = h(\psi(u))$ near 0. Linearizing needs: the chain rule for the germ $h \circ \psi$ on a germ small enough that $\psi$ maps it into the germ of $h$ (§III.1, `ScNbhdGerm.preimage`); the derivative of $u \mapsto \Gamma(u)k(u)$ at 0, where $\Gamma(u) = (\Psi(u,\cdot))_2$ is a bounded operator Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM and $D(\mathrm{pr}_{F'} \circ \Psi)(0,0)(h,k) = \Gamma(0)k$ (as in §III.4) Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_zero, so the derivative is $\Gamma(0)Dk(0)$ because $k(0) = 0$ Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp — for the germ $k$ by the chain rule with the whole-chart outer map $\mathrm{pr}_{F'} \circ \Psi$, the two derivative families agreeing at 0 on $G_k \cap \psi^{-1}(G)$ Chapter3/FredholmGermLinearizationG · IsStrongBundleIso.fiberCLM_comp_deriv_zero_eq_germ, so no shrinking of the basic germ is needed at 0; and the invertibility of

$D\psi(0)$ and $\Gamma(0)$ (inverses $D\psi'(0)$, $\Gamma'(0)$, chain rule for $\psi' \circ \psi = \mathrm{id}$). Then $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, so $Dk(0) = \Gamma(0)^{-1}Dh(0)D\psi(0)$ is Fredholm with the index of $Dh(0)$: Fredholm operators and their index are stable under composition with isomorphisms on both sides `Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv`, and $Dh(0)$ is Fredholm by Proposition 3.1.10 (preprint 3.10) for the basic germ `Chapter3/FredholmGermLinearization · proposition_3_1_10_basic_level_zero`. *Step (b)*: $Dg(0) = Dk(0) + \iota Ds(0)$ — for germs this is algebra at 0: $g - \iota s$ has the derivative family $Dg - \iota Ds$ on $G_g \cap G_s$ `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus` (on $U$ it was the uniqueness of the derivative within $U_1$ and density) — a compact perturbation since $\iota Ds(0)$ is the level-0 map of an sc$^+$-operator ($s$ is an sc$^1$-germ into $F^1$ and 0 is smooth); the index is unchanged (Part I, §I.9). *Step (c)*: Proposition 3.1.5 (preprint 3.5) (§III.4) transfers both to $Tf(x_0)$; it needs $g(0) = 0$.

**Definition 3.6.6 (preprint 3.53)** (text): $(f, x)$ with $f(x) = 0$ is in good position if (1) $f'(x) : T_xX \to Y_x$ is onto and (2) if $d_X(x) \geq 1$, $\ker f'(x)$ is in good position to the partial cone $C_xX$ in $T_xX$. In the chart: Definition 3.1.24 (preprint 3.21) *inside a subspace* $T \subseteq E_0$ — $N \subseteq T$, interior of $N \cap C$ in $N$ nonempty, and a good complement $P \subseteq T$ of $N$ in $T$ `Chapter3/GoodPositionGerm · IsInGoodPositionIn` (for $T = E_0$ it is Definition 3.1.24 (preprint 3.21) `Chapter3/GoodPositionGerm · isInGoodPositionIn_top_iff`); and the germ is in good position if $Tf(0) = Dg(0)|_{T_0O}$ is onto and $\ker Dg(0)$ is in good position to $T_0O \cap C$ inside $T_0O = Dr(0)E$ `Chapter3/GoodPositionGerm · IsGoodPositionAt Chapter3/GoodPositionGerm · IsGoodPositionAt.surjective_fillerRestrict`. Here $\ker Dg(0) \subseteq T_0O$, because $(1 - \rho(0))Dg(0)$ is injective on $\ker Dr(0)$ (filler (3)) and $(1-\rho(0))Dg(0)Dr(0) = 0$ `Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range`, and it is the kernel of $Tf(0)$ `Chapter3/GoodPositionGerm · IsFillerOf.map_ker_fillerRestrict` — the text's "$\ker Tf(0) \oplus \{0\} = \ker Dg(0)$ by the properties of the filler". The case distinction $d_X(x) \geq 1$ is dropped: at an interior point the partial cone is the whole tangent space and (2) holds for every subspace. For a filling germ the condition is one on the operator $Dg(0)$ (the text's $f'(x)$ is defined through the filler at the centre only), which the chart notion takes as a parameter `Chapter3/GoodPositionGermG · IsGoodPositionAtGerm` — the notion above is the case of the derivative on $U$ `Chapter3/GoodPositionGermG · isGoodPositionAt_iff_germ` — and the two kernel facts hold with the same proofs `Chapter3/GoodPositionGermG · IsFillingGerm.ker_deriv_le_range Chapter3/GoodPositionGermG · IsFillingGerm.map_ker_fillerRestrict`. For a section, $(f, x_0)$ is in good position if it has the data of Definition 3.1.11 (preprint 3.11) at $x_0$ and a derivative family $Dg$ of the filling germ on a germ inside $U$, with the germ in good position at $Dg(0)$ in that chart `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm`; as $Dg(0)$ does not depend on the family, "for some family" and "for every family" agree `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.forall_isGoodPositionAtGerm`; it is then an sc-Fredholm germ `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.isScFredholmGerm`. The definition is in a chart, not in $T_xX$; the transport of $C_xX$ to $T_xX$ (Definition 2.5.8 (preprint 2.38)) is not formalized.

## III.22 Good position along a graph: Lemma 3.6.9 (preprint 3.55)

**Lemma 3.6.9 (preprint 3.55)** (text, Appendix 3.8.4 (preprint 3.8.3)): for $N$ in good position to $C$ with good complement $Y$, $V \subseteq N \cap C$ relatively open and $\tau : V \to Y_1$ of class $C^1$ with $\tau(0) = 0$, $D\tau(0) = 0$, for $v$ near 0: (1) $v + \tau(v) \in C_1$; (2) $Y$ is a topological complement of $N_v = \{n + D\tau(v)n\}$; (3) for some $\gamma_v > 0$, $n \in C_x \iff n + y \in C_x$ for $n \in N_v$, $y \in Y$, $|y|_0 \leq \gamma_v |n|_0$, $x = v + \tau(v)$. *Formalized* `Chapter3/GoodPositionOpen · lemma_3_6_9_preprint` in the model $\mathbb{R}^n \oplus W$, with the core in coordinates for any $C$ cut out by functionals `Chapter3/GoodPositionOpen · lemma_3_6_9_core`: for $N$ in good position with good complement $Y$, $Y$ **closed**, and $\tau : N \to E$ with values in $Y$, $C^1$ on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$ and $Y$ is a good complement of $N_v$ (the subspace $\{n + Ln\}$

for $L = D\tau(v)$ `Chapter3/GoodPositionOpen · graphSpace`) with respect to the partial quadrant $C_{v+\tau(v)}$.

*The added hypothesis.* The step "$D\tau(v)n \in Y$ for all $n \in N$" needs $Y$ closed: the derivative within $N \cap C$ at a point of unique differentiability of a map with values in a closed subspace takes values there, because the tangent cone of a subset of a closed subspace lies in it `Chapter3/GoodPositionOpen · tangentConeAt_subset_of_subset_closed Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo`; for a non-closed algebraic complement it fails. The text's $Y = Y_0$ is a topological complement, hence closed. With $Y$ closed, (2) is linear algebra: $E_0 = N_v \oplus Y$ whenever $E_0 = N \oplus Y$ and $L$ takes values in $Y$ `Chapter3/GoodPositionOpen · graphSpace_isCompl`, and $N_v$ is closed (finite-dimensional) `Chapter3/GoodPositionOpen · graphSpace_isClosed`.

*The estimate.* (1) is Lemma 3.8.13's (preprint 3.70) smallness: $|\tau(v)| \le \varepsilon|v|$ near 0 and good position. For (3), take $\varepsilon_0 = \min(\varepsilon, 1)$ and the neighbourhood where $\|D\tau(v)\| \le \varepsilon_0/4$; then $\gamma = \varepsilon_0/4$ works *for every* $v$, since $\gamma(1+\varepsilon_0/4) + \varepsilon_0/4 \le \varepsilon_0$ (the text: $\gamma_v(1+\varepsilon) + \varepsilon < \gamma$, with $\gamma_v$ chosen after $v$). If $v \in N \cap W$, $C_x = C$ and the estimate transfers the good-complement property from $N$ to $N_v$. If $v \notin N \cap W$, no normal form is needed: if $\ell_i(v) = 0$ then $i \in \Sigma$, so $\ell_i$ vanishes on $Y$ (Lemma 3.8.8 (preprint 3.65)), and $\ell_i(v + \tau(v)) = 0$ iff $\ell_i(v) = 0$ (Lemma 3.8.13 (preprint 3.70)); so $\ell_i(h + y) = \ell_i(h)$ for every coordinate $i$ that defines $C_x$, and (3) holds for every $\gamma$. The text's cases $\dim N = 1$, $\dim N > 1$ are not needed, and the sc-level of the values of $\tau$ plays no role. The formal $\tau$ is given on all of $N \cap C$ rather than on a neighbourhood $V$; the conclusion is local.

## III.23 Theorem 3.6.8 (preprint 3.54): the linear part and the graph description of the solution set

**Theorem 3.6.8 (preprint 3.54)** (text): for an sc-Fredholm section $f$ of a tame strong bundle with $f(x) = 0$ and $(f, x)$ in good position there is an open $V \ni x$ such that $S = \{y \in V : f(y) = 0\}$ satisfies (1) $(f, y)$ is in good position at every $y \in S$, and (2) $S$ is a sub-M-polyfold with the structure of a manifold with boundary with corners. The proof: the linear part (surjectivity and good position transported from $T_xX$ to the chart and the push-forward); (3), the graph description of $S$ from Theorem 3.5.11 (preprint Corollary 3.47) for $\Psi_*(g) = h + t$; then Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) for (1), and Theorem 3.2.3 (preprint 3.27) for (2). Formalized: the linear part and (3).

*The linear part* `Chapter3/GoodPositionGermG · theorem_3_6_8_linear`. In the chart, with the germ in good position and $(1 - Dr(0))E \subseteq E_0$ (the minimal subspace at the corner 0; Proposition 2.5.3 (preprint 2.35) for the tame retraction $r$ `Chapter3/GoodPositionGermG · theorem_3_6_8_linear_of_isTameWith Chapter2/TameRetracts · proposition_2_5_3`): $Dg(0)$ is onto (Proposition 3.1.5 (preprint 3.5) for the filling germ, §III.4, from the surjectivity of $Tf(0)$), and $\ker Dg(0)$ is in good position to $C$ in $E$; only the filler at 0 enters, and $r$ is whole-chart, so $Dr(0)$ is a genuine derivative. The latter is the *transport lemma*: for a bounded idempotent $\pi$ with $(1-\pi)E_0$ inside the minimal subspace at 0, good position inside $\pi E_0$ gives good position in $E_0$, with the good complement $P \oplus (1-\pi)E_0$ and the constant $\varepsilon/(\|\pi\| + 1)$ `Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition`. Two points the text passes over: for $p = z + k$ with $|p| \le \varepsilon'|n|$ the $T_0O$-component is $z = Dr(0)p$, so $|z| \le \|Dr(0)\|\varepsilon'|n|$ — the splitting must be topological; and $\ker Dr(0) \subseteq E_0$ is used only as "the $\mathbb{R}^n$-coordinates of $k$ vanish, so $n+z+k \in C \iff n+z \in C$". *This inclusion is the only place where tameness enters.*

$C' = T\psi(0)C$. Not part of the data of a conjugation (§III.6); it is a theorem: for an sc-diffeomorphism $\psi$ between relatively open subsets of partial quadrants with $\psi(0) = 0$, $x \in C \iff D\psi(0)x \in C'$, by the one-sided sign argument for $\psi$ and $\psi^{-1}$ (Part II, §II.12) `Chapter2/Boundary · IsSc1With.mapsTo_`

partialQuadrant, since $C_0 = C$ at the corner Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero Chapter3/GoodPositionGerm · scDeriv_mem_iff_of_inverse.

*The push-forward.* $\Psi_*(g)(u') = (\Psi(\psi^{-1}(u'), g(\psi^{-1}(u'))))_2$ Chapter3/GermPushforward · pushforward is an sc-smooth germ on the germ of the points that $\psi^{-1}$ maps into the germ of $g$ Chapter3/GermPushforward · IsStrongBundleIso.pullGerm Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isScSmoothGerm (chain rules with the whole-chart $\Psi$ and $\psi^{-1}$, §III.48), with $\Psi_*(g)(0) = 0$ and $D(\Psi_*g)(0) = \Gamma(0)Dg(0)D\psi(0)^{-1}$ for its explicit derivative family Chapter3/GermPushforward · IsStrongBundleIso.pushforwardDeriv_zero ($Dk(0) = \Gamma(0)Dg(0)$ for $k(u) = (\Psi(u, g(u)))_2$) Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization; so it is onto Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_surjective, its kernel $D\psi(0)(\ker Dg(0))$ is in good position to $C'$ (Lemma 3.5.9 (preprint 3.45)) Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_ker_isInGoodPosition, and it is Fredholm (transported from $Dg(0)$, Proposition 3.6.5 (preprint 3.52)) — the hypotheses of Theorem 3.5.11 (preprint Corollary 3.47), for *any* conjugating isomorphism Chapter3/GoodPositionGermG · theorem_3_6_8_pushforward, and in bundle form Chapter3/GoodPositionGermG · theorem_3_6_8_pushforward_section. The text instead obtains the Fredholm property of $D(h+t)(0)$ from Propositions 3.1.10 and 1.1.11 (preprint 3.10 and 1.10).

**The graph description** Chapter3/GlobalIFTG · theorem_3_6_8_graph. Let the charts of the bundle have tame base retractions (`htame`, the hypothesis of the first formalization; it cannot be met, and it is replaced by good position in a tame chart, §III.47), $f(x_0) = 0$, $(f, x_0)$ in good position. The proof, in the order of the Lean.

1. *The chart and the identities.* Take the chart of Definition 3.6.6 (preprint 3.53) centred at $x_0$, filling germ $g$ on $G_g$, sc⁺-germ $t_0$ on $G_s$ and conjugation $\Psi$ covering $\psi$, with inverse $\Psi'$ covering $\psi'$. *Shrink the basic germ* to the germ $G''$ of the points of $G$ that $\psi'$ sends into $G_g \cap G_s$ level by level: same $\Psi$, $h$, $B$ and conjugation identity Chapter3/BasicGermShrink · ConjugationData.shrink (§III.48); below, $G$ denotes $G''$. There is an open $N_1 \ni 0$ on which $\mathbf{f} = g$ on $O$, zeros of $g$ in $U$ lie in $O$, and $\Psi(u, (g - \iota t_0)(u))_2 = h(\psi(u))$ (filler (1), (2) and the conjugation).
2. *The sc⁺-germ $t$.* $t(u') = \Psi[1](\psi'(u'), t_0(\psi'(u')))_2$ is an sc-smooth germ into $(Q \oplus W)^1$ on $G$: $t_0 \circ \psi'$ is a germ on the preimage of $G_s$, which contains $G = G''$, and the whole-chart $\Psi[1]$ is composed on the outside (§III.48). (With $t_0$ sc-smooth on $U$, $t$ was sc-smooth on all of $U'$.) On $U \cap N_1$, $\Psi_*(g)(\psi(u)) = h(\psi(u)) + \iota t(\psi(u))$, by the linearity of the fibre maps.
3. *Absorbing the $\mathbb{R}^N$-component.* The data of Theorem 3.4.2 (preprint 3.38) carry only the $W$-component of an sc⁺-germ. Let $t_Q$, $t_W$ be the two components of $t$, through the sc-operators $(q, w) \mapsto (q, 0)$ and $(q, w) \mapsto w$ Chapter3/GlobalIFT · ScBanach.qPartOp Chapter3/GlobalIFT · ScBanach.wPartOp. Then $h' = h + (t_Q, 0)$ is again a basic germ with the same contraction $B$ (Definition 3.1.7 (preprint 3.7) constrains only the $W$-component), and with the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) for $s = t_W$ (they exist, §III.18) one has $\mathcal{S}.\mathrm{fsum}\, h' = h' + \iota t_W = h + \iota t$.
4. *The hypotheses of Theorem 3.5.11 (preprint Corollary 3.47).* Tameness gives $(1 - Dr(0))E \subseteq E_0$ (Proposition 2.5.3 (preprint 2.35)), so the push-forward step applies: $D(\Psi_*g)(0)$ is surjective with kernel in good position. Since $\mathcal{S}.\mathrm{fsum}\, h' = \Psi_*(g)$ only on a neighbourhood $W'$ of 0, the two derivatives at 0 are identified by the uniqueness of the derivative within $G_1''$ and density of $E_1$ Chapter3/GlobalIFT · scDeriv_eq_of_eqOn_nhds; $\Psi_*(g)$ is a germ on the preimage of $G_g$, which contains $G''$.
5. *Theorem 3.5.11 (preprint Corollary 3.47) for $h + t$* gives $Y$, $V_0$, $\sigma$ (good complement, $C^1$, level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$), the graph consisting of zeros, and $U''$ in which every zero lies on the graph.
6. *The neighbourhoods* (the text does not say which $V$). Let $\Omega = U'' \cap W'$ and $V' = V_0 \cap (1 + \sigma)^{-1}(\Omega)$,

relatively open in $K \cap C'$ because $v \mapsto v + \sigma(v)$ is continuous. Let $V_x = \{y \in \operatorname{dom}\varphi : \varphi(y) \in N_1,\ \psi\varphi(y) \in \Omega \cap G_0\}$, open and containing $x_0$ by continuity of $\psi \circ \varphi$. $D\sigma(0) = 0$ persists on the smaller $V'$ (unique differentiability at 0 of $V'$).

7. *The conclusions.* $\psi \circ \varphi$ is injective on $V_x$ ($\varphi$ is, and $\psi' \circ \psi = \mathrm{id}$ on $U$). For $y \in V_x$: $f(y) = 0 \iff \mathbf{f}(\varphi y) = 0 \iff g(\varphi y) = 0 \iff (h + \iota t)(\psi\varphi y) = 0$ (items 1–2). If $f(y) = 0$, then $\psi\varphi(y)$ is a zero in $U'' \cap G_0$, hence $v + \sigma(v)$ with $v \in V_0$, and $v \in V'$ since $v + \sigma(v) \in \Omega$. Conversely, for $v \in V'$, $u = \psi'(v + \sigma(v))$ lies in $U \cap N_1$, $g(u) = 0$, so $u \in O$ and $u = \varphi(y)$ for some $y$, which lies in $V_x$ and is a zero.

So $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$. When this section was written, (1) and (2) were not formalized; they are now: the kernel and good position at the nearby zeros in §§III.32–III.33 (the surjectivity of $f'(y)$ under an added hypothesis), the data of Definition 3.2.1 (preprint 3.25) at every zero in §§III.34–III.36, and the theorem in §III.37. The graph description also records, since then, that $Y$ is closed, that the kernel is a smooth subspace and the level-wise lifts of $\sigma$ (Theorem 3.5.11 (1) (preprint Corollary 3.47)), and it is a corollary of the version with a good complement whose projection maps $C'$ into itself (§III.29).

**Theorems 3.6.10 and 3.6.11 (preprint 3.56 and 3.57), the step from the tangent germ property.** With $S = \{f = 0\}$ `Chapter3/GlobalIFT · solutionSet`: if $S \neq \emptyset$ has at every point regular data of Definition 3.2.1 (preprint 3.25) (§III.24), then $S$ is a sub-M-polyfold whose induced structure is tame and an $n$-dimensional manifold with boundary with corners (Theorem 3.2.3 (preprint 3.27)) `Chapter3/GlobalIFT · theorem_3_6_11_of_tangentGerm`; if $X$ has no boundary, the induced structure is without boundary, since $d_S \leq d_X = 0$ (Lemma 2.4.6 (preprint 2.26)) `Chapter2/SubMPolyfold · lemma_2_4_6` `Chapter3/GlobalIFT · theorem_3_6_10_of_tangentGerm`. The first step of the text's proof — good position at every zero gives the tangent germ property — was not formalized when this section was written; the theorems with the text's hypotheses are in §III.37.

## III.24 The tangent germ property: Definition 3.2.1 (preprint 3.25) and Theorem 3.2.3 (preprint 3.27)

**Definition 3.2.1 (preprint 3.25).** The data at $x \in M$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart`: a chart $\varphi : (U, x) \to (O, 0)$ onto a tame retract $(O, C, E)$; an $n$-dimensional smooth $N \subseteq E$ in good position to $C$ with a good complement $Y$, $Y$ **closed**; a relatively open $V \ni 0$ in $N \cap C$; $\delta : V \to E$ continuous with values in $Y$; (i) $\varphi(M \cap U) = \{v + \delta(v) : v \in V\}$; (ii) $\delta(0) = 0$, $D\delta(0) = 0$ and $\delta$ an sc^∞-germ, read as: for every $m$, $k$ a relatively open $V_{m,k} \ni 0$ in $N \cap C$ with $\delta(V_{m,k}) \subseteq E_m$ and $\delta : V_{m,k} \to E_m$ of class $C^k$ `Chapter3/TangentGerm · IsScInftyGermAt`. This $k$-dependent reading is what the germ implicit function theorem delivers (§III.11); the reading "sc-smooth on one fixed germ" implies it by Proposition 1.2.3 (preprint 1.18) for germs `Chapter3/TangentGerm · IsScSmoothGerm.isScInftyGermAt`, as does sc-smoothness on $V$ as a set `Chapter3/TangentGerm · IsScSmooth.isScInftyGermAt`, so the theorem covers all three. $M$ has the property if $M \subseteq X_\infty$ and every point has such data `Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty`. With these data: $N \cap C$ is a partial quadrant in $N$ (Proposition 3.1.25 (preprint 3.22)) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isPartialQuadrant`, and $N$, $Y$ are topological complements — this is where $Y$ closed is used — so the projection $\pi : E \to N$ along $Y$ is continuous `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isTopCompl` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π`; with $\Gamma(v) = v + \delta(v)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ`, $\pi\Gamma = \mathrm{id}$ and $\varphi(y) = \Gamma(\pi\varphi(y))$ for $y \in M \cap U$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`. "$Y$ closed"

is not an added hypothesis: the text's good complement is an sc-complement, hence closed, and the two notions of datum agree (§III.56).

**Proposition 3.2.2 (preprint 3.26)** Chapter3/TangentGerm · proposition_3_2_2. For an sc-diffeomorphism $\psi : X \to X'$ (sc-smooth on all of $X$ with sc-smooth inverse $\psi'$ on all of $X'$) and $\psi(M) = M'$, $M'$ has the $n$-dimensional tangent germ property iff $M$ has it. One direction suffices, applied to $\psi$ and $\psi'$ Chapter3/TangentGerm · proposition_3_2_2_image: $\psi(M) \subseteq X'_\infty$ because sc-smooth maps preserve the levels, and the data at $\psi(x)$ are the transported chart $\varphi \circ \psi^{-1}$ Chapter2/DegeneracyIndex · transportChart, whose model is that of $\varphi$ Chapter3/TangentGerm · MPolyfold.transportChart_model_0, with the same $N$, $Y$, $V$, $\delta$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.transport: condition (i) is transported because $\varphi\psi^{-1}(\psi(M) \cap \psi(U)) = \varphi(M \cap U)$. As in the text; tameness of $X$ plays no role.

**Two gaps in the text's proof of Theorem 3.2.3 (preprint 3.27), and the conditions that close them** Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular. (a) The text asserts that $\delta$ is an sc^∞-germ near every $v_0 \in V$, by linearizing at $v_0$ and claiming that a certain map $N' \to N$ is an isomorphism. That needs $\delta$ differentiable at $v_0$, known only near 0, and fails: in $X = \mathbb{R}^2$ take $M$ the graph of $g$ with $g(s) = s^2$ near 0 and $g(s) = 4 + (s-2)^{1/3}$ near 2, a smooth embedded curve with a vertical tangent at $(2, 4)$; with $\varphi = \mathrm{id}$, $N$ the horizontal axis, $Y$ the vertical one and $\delta = g$, the data at 0 satisfy Definition 3.2.1 (preprint 3.25), $M$ has the tangent germ property (at $(2, 4)$ use a rotated chart), but $\delta$ is not differentiable at 2 and the chart $\pi\varphi$ on all of $M \cap U$ is not smoothly compatible with the chart at $(2, 4)$. So condition (a): $\delta$ is sc-smooth on $V$ as a map on a set. (b) The text's $\Sigma = \{v + y \in C : v \in V, y \in Y\}$ is claimed relatively open in $C$, so that $t(v + w) = v + \delta(v)$ is defined near 0 in $O$. This does not hold as we read it: $E = \mathbb{R}^3$, $C = [0, \infty)^3$, $N = \mathbb{R}(1, 1, 1)$, $Y = \{x + y - z = 0\}$ is a good complement, and $\pi(x, y, z) = (x + y - z)(1, 1, 1)$, so $\pi z \in C$ iff $x + y \geq z$; $\Sigma$ is not open at $(1, 0, 1)$ and points $(\varepsilon, 0, \varepsilon + \varepsilon^2) \notin \Sigma$ approach 0. So condition (b): $\pi(O) \subseteq C$; then $\Sigma \cap O = \{z \in O : \pi z \in V\}$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig is relatively open in $O$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig_eq. (A good complement with $\pi'(C) \subseteq N \cap C$ always exists, via the normal form, and $M \cap U$ can be rewritten as a graph along it with the quadrant inverse function theorem; with the re-centring of §III.36 this makes (a) and (b) theorems rather than hypotheses, §III.56, where both examples are also Lean counterexamples.)

**Theorem 3.2.3 (preprint 3.27)** Chapter3/TangentGerm · theorem_3_2_3_of_isRegular. If $M \neq \emptyset$ has at every point data satisfying (a), (b), then (since §III.56 the conditions are not needed: such data exist at every point):

*$M$ is a sub-M-polyfold* Chapter3/TangentGerm · theorem_3_2_3_isSubMPolyfold, with the text's retraction. $t(v + w) = v + \delta(v)$, i.e. $t = \Gamma \circ \pi$ on $\Sigma \cap O$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.t, is idempotent with image $\Gamma(V)$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_t Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_eq_self_of_mem. It is sc-smooth: $\pi$ is a continuous linear map into a finite-dimensional space, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)) Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmooth; $\Gamma = \iota + \delta$ is sc-smooth on $V$ by (a) and because the inclusion of a smooth finite-dimensional subspace is sc-smooth, its level maps being the linear lifts $N \to E_m$ Chapter3/TangentGerm · finSmoothLift Chapter3/TangentGerm · isScSmooth_subtype_val Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmooth; and for a retraction $R$ onto $\Sigma \cap O$, $t \circ R = \Gamma \circ \pi \circ R$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_isScSmoothRetractMap. The text says nothing about either. On $U_0 = \varphi^{-1}(\Sigma \cap O)$, open Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀, the map $s = \varphi^{-1} \circ t \circ \varphi$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.s is an sc-smooth retraction Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn onto $M \cap U_0$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s — Definition 2.3.18 (preprint 2.20) at $x$.

*The charts $\pi \circ \varphi$.* On $M \cap U_0$, $y \mapsto \pi\varphi(y)$ is a chart with model $(V, N \cap C, N)$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.model, a tame retract (the identity retraction) Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract, with inverse $v \mapsto \varphi^{-1}(v + \delta(v))$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.chartInv Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart, a finite-dimensional corner chart of dimension $n$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner containing $x$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source. It is compatible with every chart $\varphi'|_{W\cap M}$ of the induced structure of Proposition 2.3.19 (preprint 2.21): the transition maps are $\varphi' \circ \varphi^{-1} \circ \Gamma$ and $\pi \circ \varphi \circ \varphi'^{-1}$, compositions of sc-smooth maps between retracts Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmoothRetractMap Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmoothRetractMap Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart. So it is a chart of the induced structure — **Theorem 3.2.3 (2) (preprint 3.27)** Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_isChart Chapter2/SubMPolyfold · subPolyfold; these charts are pairwise compatible by locality Chapter3/TangentGerm · MPolyfold.tangentCharts_compatible and form an atlas Chapter3/TangentGerm · MPolyfold.germAtlas equivalent to the induced one Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_atlas_equiv, with tame models: **the induced structure is tame** Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_isTame, and **sc-smoothly equivalent to an $n$-dimensional manifold with boundary with corners** Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_isManifoldWithCorners. "Manifold with boundary with corners" belongs to Chapter 4; here it is an equivalent atlas of charts on relatively open subsets of $n$-dimensional partial quadrants Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners, whose transition maps are classically smooth because sc-smooth maps between finite-dimensional constant structures are $C^\infty$ Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const. Tameness of $X$ is not used: the text's "since $O$ is a tame retract, there exists … a tame sc-smooth retraction $r$" is Proposition 2.3.1 (1) (preprint 2.10) for any retract. *The degeneracy index*: since $\pi\varphi$ has a tame model, $d_M(y) = d_{N\cap C}(\pi\varphi(y))$ (Proposition 2.5.6 (preprint 2.36)) Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_degIndex; Lemma 3.8.13 (preprint 3.70) in a general presentation turns this into $d_M(y) = d_X(y)$ near the centre, off the lineality space (§III.61).

## III.25 Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61): towards the basic germ criterion

**Lemmas 3.7.2 and 3.7.3 (preprint 3.59 and 3.60), abstract form.** In the proof of Theorem 3.7.1 (preprint 3.58) the family $L(b) = P_Y \circ Df(b)|_X : X \to Y$ is shown to be injective for $b$ near 0 (Lemma 3.7.2 (preprint 3.59)) and to satisfy uniform lower bounds $|L(b)h|_m \geq c_m|h|_m$ (Lemma 3.7.3 (preprint 3.60)) — the hypotheses of Lemma 3.7.4 (preprint 3.61). Both are sequence arguments, and all they use of the compactness condition (2) of Theorem 3.7.1 (preprint 3.58) is a property of the family itself: if $b_j \to b$, $(h_j)$ is bounded in $X_m$ and $L(b_j)h_j \to 0$ in $Y_m$, then $(h_j)$ has a convergent subsequence in $X_m$ Chapter3/BasicGermCriterion · FamilyCompactness. (For $L = P_Y \circ Df|_X$ it follows from (2), since $Df(b_j)h_j - L(b_j)h_j = (1 - P_Y)Df(b_j)h_j$ lies in the finite-dimensional smooth complement $A$ and is bounded on every level.) The two lemmas are formalized for an abstract family of sc-operators with jointly continuous level maps. *Lemma 3.7.2 (preprint 3.59)*: if $L(0)$ is injective, so is $L(b)$ for $b$ near 0 — otherwise $b_j \to 0$ and unit vectors $h_j$ with $L(b_j)h_j = 0$ give, by the compactness property, a limit $h$ with $|h| = 1$ and $L(0)h = 0$ Chapter3/BasicGermCriterion · lemma_3_7_2. *Lemma 3.7.3 (preprint 3.60)*: on a compact parameter set $K$ on which every $L(b)_m$ is injective there is $c_m > 0$ with $c_m|h|_m \leq |L(b)h|_m$ — otherwise $b_j \in K$ and unit $h_j$ with $|L(b_j)h_j|_m \to 0$ give, after subsequences, $b_j \to b \in K$, $h_j \to h$, $|h|_m = 1$, $L(b)h = 0$ Chapter3/BasicGermCriterion · lemma_3_7_3. The text's step "$P_Y \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism" is needed only to feed Lemma 3.7.4 (preprint 3.61). The setting of Theorem

3.7.1 (preprint 3.58) (the splitting as coordinates, the complement $A$, the index computation), the strong bundle map $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ and the contraction estimate from condition (3) were not formalized when this section was written; they are now, in the coordinates of the text's proof, and there the hypotheses of the two lemmas are derived from those of the theorem (§III.38).

**Lemma 3.7.4 (preprint 3.61)** (quoted by the text from [25, Proposition 4.8] without proof). Let $V$ be a germ of relatively open neighbourhoods of 0 in a partial quadrant $\widetilde{C}$ of a finite-dimensional $G$, $V \oplus E$ the product germ with levels $V_m \times E_m$ `Chapter3/InverseFamily · ScNbhdGerm.prodFull`, and $L(v) : E \to F$ sc-isomorphisms such that $\Phi(v,h) = L(v)h$ `Chapter3/InverseFamily · famMap` is an sc-smooth germ on $V \oplus E$ and $c_m|h|_m \le |L(v)h|_m$ for $v \in V_0$. Then $\Lambda(v,k) = L(v)^{-1}k$ `Chapter3/InverseFamily · famInv` is an sc-smooth germ on $V \oplus F$ `Chapter3/InverseFamily · lemma_3_7_4`. Operator-norm continuity of $v \mapsto L(v)$ is not assumed.

*sc⁰.* On level $m$, with $h_0 = L(v_0)^{-1}k_0$, $|L(v)^{-1}k - L(v_0)^{-1}k_0|_m \le c_m^{-1}(|k-k_0|_m + |(L(v_0)-L(v))h_0|_m)$, and $v \mapsto L(v)h_0$ is continuous `Chapter3/InverseFamily · famInv_isSc0Germ`.

*Differentiability* `Chapter3/InverseFamily · famInv_hasFDerivWithinAt`. The candidate is $D\Lambda(v,k)(\delta v,\delta k) = L(v)^{-1}(\delta k - D_1\Phi(v,L(v)^{-1}k)\delta v)$ `Chapter3/InverseFamily · famInvDeriv`. With $h = \Lambda(v,k)$ and $w = D\Lambda(v,k)(\delta v,\delta k)$, $\Lambda(v',k') - \Lambda(v,k) - w = L(v')^{-1}[(L(v)-L(v'))w] - L(v')^{-1}r$, where $r = \Phi(v',h) - \Phi(v,h) - D_1\Phi(v,h)\delta v = o(|\delta v|)$ is the remainder of $\Phi$ along $(\delta v, 0)$ with $h$ fixed, so that only the finite-dimensional variable moves. The term $(L(v)-L(v'))w$ is $o(|(\delta v,\delta k)|_1)$ *because the increment is measured on level* 1: $w/|(\delta v,\delta k)|_1$ lies in the compact set $D\Lambda(v,k)(\iota(B_1))$ ($\iota : F_1 \to F_0$ compact), and $(L(v)-L(v'))u \to 0$ uniformly for $u$ in a compact set — a tube lemma for jointly continuous maps vanishing on $\{x_0\} \times K$ `Chapter3/InverseFamily · exists_forall_norm_le_of_isCompact`. Strong continuity plus the compactness of $F_1 \to F_0$ replace the operator-norm continuity; the differentiability of $\Lambda$ as a map $(V \oplus F)_0 \to E_0$ would need the latter and is not claimed.

*The tangent map and the induction.* $T\Lambda((v,k),(\delta v,\delta k)) = (\Lambda(v,k), \Lambda(v,\delta k - (T\Phi((v,\Lambda(v,k)),(\delta v,0)))_2))$, a composition of $\Lambda$, $T\Phi$ and sc-operators (the projections `Chapter3/InverseFamily · tangentBaseOp` `Chapter3/InverseFamily · tangentInlOp` `Chapter3/InverseFamily · tangentFiberOp` and subtraction `Chapter3/InverseFamily · ScBanach.subOp`); it is sc⁰ `Chapter3/InverseFamily · famInv_tangent_isSc0`, so $\Lambda$ is sc¹ `Chapter3/InverseFamily · famInv_isSc1GermWith`; and if $\Phi$ is sc^{k+1}, then $\Lambda$ (sc^k by induction) and $T\Phi$ (sc^k) make $T\Lambda$ sc^k, so $\Lambda$ is sc^{k+1} `Chapter3/InverseFamily · famInv_isSckGerm`. The sc¹ step uses no quadrant hypothesis; the induction uses the composition rules for germs.

## III.26 What the formalization taught us

1. **The germ implicit function theorem holds with the text's hypotheses**, but on shrinking germs: the solution germ is sc^k on the germ shifted by $2^k - 1$ levels. In the text's reading of germs (up to shrinking) this is the theorem; read with fixed germs, "if $f$ is sc-smooth so is its solution germ" holds for every $k$ on a germ depending on $k$. Every later "sc-smooth germ" (Definition 3.2.1 (ii) (preprint 3.25), Theorem 3.5.6 (1) (preprint 3.43), Theorem 3.5.11 (1) (preprint Corollary 3.47)) must be read the same way. Whether the shrinking is intrinsic is open.
2. **Theorem 3.3.3 (preprint 3.30) has a short proof**, which the text cites from [21]: an approximate fixed point and the $\frac{1}{2}$-contraction on level 0, at parameters where the level-1 operators exist.
3. **The contraction regions must be explicit** for "from the uniqueness we conclude $\delta^{(1)} = T\delta$" (Lemma 3.3.4 (preprint 3.31)) to be true.

4. **Several "immediate consequences" are proofs of their own**: the compactness of Theorem 3.1.13 (preprint 3.12) (not in Theorem 3.5.1 (preprint 3.40)), the induction of Theorem 3.5.1 (preprint 3.40) (lifts, bounds and continuity together), the sc-smoothness and regularizing property in Theorem 3.1.18 (preprint 3.15), the $\mathrm{sc}^+$ property in Proposition 3.1.10 (preprint 3.10), the projection of solutions into the quadrant in Theorem 3.5.6 (preprint 3.43).
5. **Three added hypotheses**: $Y$ closed in Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25); and the two regularity conditions of Theorem 3.2.3 (preprint 3.27), each with a counterexample to the step of the text's proof it replaces. (None of the three remains: §III.56.)
6. **Tameness is nearly unused**: not in §3.1 at all, and in §3.6 only through $(1-Dr(0))E \subseteq E_0$ and, for the tangent germ property of the solution set, the tameness of the model of the chart $\psi \circ \varphi$ (§III.27).
7. **Linear coordinate changes** in Theorem 3.4.2 (preprint 3.38) make the text's degeneracy-index argument for Theorem 3.5.11 (preprint Corollary 3.47) unnecessary.
8. **The extreme-ray lemmas and the iterated tangent maps are not needed**: Proposition 3.1.25 (preprint 3.22) by a segment argument, the regularity in Theorems 3.8.15 and 3.8.22 (preprint 3.72 and 3.78) by bootstrapping, without shrinking. (They are proved as well since, §§III.59–III.60.)
9. **What remained of Theorem 3.6.8 (preprint 3.54)** when this list was written — the re-centring of good position at the nearby zeros, and the level-wise regularity of the solution map in the data of Definition 3.2.1 (preprint 3.25) at every zero — is done in §§III.31–III.37, with one added hypothesis for Theorem 3.6.8 (1) (preprint 3.54); see §III.39.

## III.27 Tameness is carried by sc-diffeomorphisms, and by open subsets

The text never states that tameness is invariant under sc-diffeomorphisms, or that an open piece of a tame retract is a tame retract, but it uses both. In the proof of Theorem 3.6.8 (preprint 3.54) the chart $\psi \circ \varphi$ onto $\psi(O)$ must be a chart of the tame M-polyfold $X$ for Definition 3.2.1 (preprint 3.25) to apply; in the proof of Proposition 3.2.2 (preprint 3.26) it writes "$X' = \psi(X)$ is tame"; and wherever a chart is restricted to a smaller domain (the shrinking of the data of Definition 3.2.1 (preprint 3.25) in §§III.36–III.37) the model of the restricted chart must again be a tame retract. Both facts are proved here, with Definition 2.5.2 (preprint 2.34) as the formalization states it (Part II).

**The objects.** An sc-diffeomorphism between relatively open subsets $U \subseteq C$, $U' \subseteq C'$ of partial quadrants is a pair $\psi : U \to U'$, $\psi' : U' \to U$ of sc-smooth maps, each mapping its domain into the other's, inverse to each other on $U$ and on $U'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant` (its inverse is again one `Chapter2/TameInvariance · IsScDiffeoQuadrant.symm`). A retraction $r : U \to U$ is *tame* for a presentation $T$ of $C$ if (1) $d_C(r(x)) = d_C(x)$ for $x \in U$ and (2) at every smooth point $x$ of $O = r(U)$ on level 1 the tangent space $T_xO = Dr(x)E$ has an sc-complement $A$ contained in the minimal subspace $E_x$ `Chapter2/Tame · ScRetraction.IsTameWith`; it is tame if it is tame for every presentation `Chapter2/Tame · ScRetraction.IsTame`, and $(O, C, E)$ is a tame retract if $O$ is the image of a tame retraction `Chapter2/Tame · IsTameRetract`.

**The transported retraction.** For an sc-smooth retraction $r$ with domain $U$, put $r' = \psi \circ r \circ \psi' : U' \to U'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport`. It maps $U'$ into $U'$ because $\psi'(U') \subseteq U$, $r(U) \subseteq U$ and $\psi(U) \subseteq U'$; it is sc-smooth by the chain rule applied twice; and it is idempotent, $r'(r'(u')) = \psi(r(\psi'\psi(r(\psi'u')))) = \psi(r(r(\psi'u'))) = r'(u')$, since $\psi'\psi = \mathrm{id}$ on $U$ and $r$ is idempotent. Its image is $\psi(O)$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport_retract`: $r'(u') = \psi(r(\psi'u'))$ with $r(\psi'u') \in O$; conversely, for $z \in O$, $r'(\psi z) = \psi(r(z)) = \psi(z)$.

*Condition (1).* The degeneracy index is preserved by $\psi$, $d_{C'}(\psi(u)) = d_C(u)$ for $u \in U$: this is Theorem

1.3.5 (preprint 1.25) `Chapter2/TameInvariance · IsScDiffeoQuadrant.degIndex_eq` `Chapter2/DegeneracyIndex · theorem_1_3_5`. Then for $u' \in U'$: $d_{C'}(r'(u')) = d_C(r(\psi' u')) = d_C(\psi' u') = d_{C'}(\psi\psi' u') = d_{C'}(u')$, the middle step being condition (1) for $r$.

*The derivative of $r'$.* On level 0, $D\psi'(\psi(x)) \circ D\psi(x) = 1$ for $x \in U_1$ and $D\psi(\psi'(y)) \circ D\psi'(y) = 1$ for $y \in U_1'$, by the chain rule for $\psi' \circ \psi = \mathrm{id}$ and $\psi \circ \psi' = \mathrm{id}$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_inv_deriv` `Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_deriv_inv` `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`. If $y \in U_1'$ and $\psi'(y) \in O$, then with $x = \psi_1'(y)$,

$$Dr'(y) = D\psi(x) \circ Dr(x) \circ D\psi'(y)$$

`Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport`. The chain rule is applied to $\psi \circ (r \circ \psi')$ and then to $r \circ \psi'$ `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain`; the outer derivative of $\psi$ is taken at the level-1 point of $r \circ \psi'$ over $y$, and this point is $x$ because $r$ fixes $\psi'(y) \in O$. Without $\psi'(y) \in O$ the formula would involve $Dr$ at $\psi'(y)$ and $D\psi$ at the different point $r(\psi'(y))$.

*Sc-isomorphisms carry sc-splittings.* Let $D : E \to E'$ be an sc-operator with an sc-operator inverse $D'$. Its level maps are topological linear isomorphisms $E_m \cong E'_m$ `Chapter2/TameInvariance · scIsoLevel` (inverse to each other by the injectivity of the inclusions into level 0); they carry $A_m = A \cap E_m$ onto $(DA)_m$ `Chapter2/TameInvariance · levelSub_map_eq` and smooth points to smooth points `Chapter2/TameInvariance · mem_smoothLvl_of_isScOperator`. Hence an sc-subspace $A$ (closed, with the smooth points of $A_m$ dense in $A_m$ for every $m$) is carried to the sc-subspace $DA$: $DA = (D')^{-1}(A)$ is closed, and the image of the closure of the smooth points of $A_m$ lies in the closure of the image `Chapter2/TameInvariance · IsScSubspace.map`. An sc-splitting $E = F \oplus A$ (sc-subspaces, complementary on every level) is carried to the sc-splitting $E' = DF \oplus DA$, since an isomorphism of each level carries complements to complements `Chapter2/TameInvariance · IsScSplitting.map`.

*Condition (2)* `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameWith_transport`. Let $T$, $T'$ be presentations of $C$, $C'$, $r$ tame for $T$, and $y$ a smooth point of $\psi(O)$ on level 1. Then $x = \psi_1'(y)$ lies on level 1 of $O$ (if $y = \psi(z)$ with $z \in O$, then $\psi'(y) = z$) and is smooth, because sc-smooth maps carry smooth points to smooth points. Tameness of $r$ at $x$ gives an sc-splitting $E = T_xO \oplus A$ with $A \subseteq E_x$. Because $x$ and $y$ are smooth points, $D\psi(x)$ and $D\psi'(y)$ are sc-operators (the remark after Proposition 1.2.1 (preprint 1.16)) `Chapter1/Properties · IsSc1With.scOperator_deriv` — at a point that is only on level 1 they need not be — and they are inverse to each other. From the formula for $Dr'(y)$ and $D\psi'(y)D\psi(x) = 1$: $T_y\psi(O) = \operatorname{range} Dr'(y) = D\psi(x)(\operatorname{range} Dr(x)) = D\psi(x)(T_xO)$. So $E' = T_y\psi(O) \oplus D\psi(x)A$ is an sc-splitting. Finally $D\psi(x)A \subseteq E'_y$: the derivative of an sc¹-map between relatively open subsets of partial quadrants maps the minimal subspace $E_x$ into $E'_{\psi(x)}$ (the one-sided sign argument, and density) `Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace`, which needs $\psi(U) \subseteq C'$, and $\psi(x) = y$. Since $T'$ was arbitrary and the presentation $T$ of $C$ is available, the transported retraction of a tame retraction is tame `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport`, and **the image $(\psi(O), C', E')$ of a tame retract under an sc-diffeomorphism is a tame retract**, with retraction $\psi \circ r \circ \psi^{-1}$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image`.

**Open subsets.** For an open $W \subseteq E_0$, the restriction of $r$ to $U \cap r^{-1}(W)$ is an sc-smooth retraction onto $O \cap W$ (Proposition 2.3.1 (1) (preprint 2.10)) `Chapter2/Retracts · ScRetraction.restrict`. It is tame if $r$ is `Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame`: condition (1) is inherited pointwise; for (2), at a smooth point $x$ of $O \cap W$ on level 1 the derivative of the restricted retraction is $Dr(x)$, because the two maps agree on the smaller relatively open domain, which contains $x$ since $r(x) = x \in W$, and the

derivative at a level-1 point depends only on the map near the point `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`; so the complement $A$ of $r$ at $x$ serves. Hence $O \cap W$ is a tame retract whenever $O$ is `Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen`. For a chart $\varphi$ and an open $s$ inside its source, $\varphi(s) = O \cap W$ for an open $W$ `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq`, so the model of the restricted chart is a tame retract when that of $\varphi$ is `Chapter3/SolutionGermRecentre · MChart.isTameRetract_image`.

These are used in §III.34 (the chart $\psi \circ \varphi$ of Theorem 3.6.8 (preprint 3.54) has the tame model $\psi(O)$) and in §§III.36–III.37 (the shrunk data of Definition 3.2.1 (preprint 3.25) keep tame models). The text's sentence "$X' = \psi(X)$ is tame" in Proposition 3.2.2 (preprint 3.26) is not needed there (§III.24): the tameness Definition 3.2.1 (preprint 3.25) asks for is that of the chart models, which the transport of charts leaves unchanged.

## III.28 Corollary 3.4.3 (preprint 3.39): composition with a given conjugation

**Corollary 3.4.3 (preprint 3.39)** (text): if the sc-germ $g$ is equivalent, by a strong bundle isomorphism $\Phi$, to a basic germ and $s$ is an sc$^+$-germ, there is a strong bundle map $\Psi$ such that $((\Psi \circ \Phi)_*(g+s))^1$ is a basic germ. The text's proof: $\Phi_*(g) = h$ is basic, $t = \Phi_*(s)$ is an sc$^+$-germ with $t(0) = 0$, $\Phi_*(g+s) = h + t$, and Theorem 3.4.2 (preprint 3.38) gives $\Psi$ with $(\Psi_*(h+t))^1$ basic; take $\Psi \circ \Phi$. Four things are silent: that Theorem 3.4.2's (preprint 3.38) coordinate change is a strong bundle *isomorphism*; that the index raise of a strong bundle isomorphism is one between the shifted structures; that strong bundle isomorphisms and push-forwards compose; and that $\Phi_*(g+s) = h + t$ holds only where the conjugation identity holds. The formalization proves all four and states the corollary with the composite explicit.

*Linear strong bundle isomorphisms.* Two sc-operators $S : E \to E'$, $T : F \to F'$ give the sc-operator $S \oplus T$ `Chapter3/BasicClassComposition · ScOperator.prodMap`, and if $S(U) \subseteq U'$ the map $(u,k) \mapsto (Su, Tk)$ is a strong bundle map $U \lhd F \to U' \lhd F'$ (Definition 2.6.1 (preprint 2.47) as formalized `Chapter2/Tame · IsStrongBundleMap`): its base component does not depend on $k$, it is linear in $k$, it preserves the double filtration because sc-operators preserve the levels, it is the sc-operator $S \oplus T$ on $(U \lhd F)[0]$, and on $(U \lhd F)[1]$ it is $S \oplus T^1$ with $T^1$ the shifted operator `Chapter3/BasicClassComposition · ScOperator.isStrongBundleMap_prodMap`. With inverse sc-operators $S'$ (inverse to $S$ on $U$, $U'$) and $T'$ it is a strong bundle isomorphism `Chapter3/BasicClassComposition · ScOperator.isStrongBundleIso_prodMap`.

*Theorem 3.4.2's (preprint 3.38) coordinate change as an isomorphism.* In §III.18 the change of coordinates of Theorem 3.4.2 (preprint 3.38) is the pair of linear sc-isomorphisms `$□ = $ split`, `$□ = $ fiber` (inverses `glue`, `fiberInv`) on the index-raised spaces `Chapter3/BasicClass · StabilityData.split` `Chapter3/BasicClass · StabilityData.fiber`. Put $\Lambda((a,w),(q,w')) = (\psi(a,w), \Psi(q,w'))$ `Chapter3/BasicClassComposition · StabilityData.linIso`. For every $U$ relatively open in the shifted quadrant $(\widetilde{C} \oplus W)^1$, $\psi(U) = \mathrm{glue}^{-1}(U)$ is relatively open in $(\widetilde{C} \oplus C) \oplus X^1$ (`glue` is continuous and preserves the quadrant coordinates) `Chapter3/BasicClassComposition · StabilityData.isRelOpen_glue_preimage`, and $\Lambda : U \lhd (\mathbb{R}^N \oplus W)^1 \to \psi(U) \lhd ((\mathbb{R}^N \oplus Z) \oplus X^1)$ is a strong bundle isomorphism by the preceding paragraph `Chapter3/BasicClassComposition · StabilityData.linIso_isStrongBundleIso`. Before the index raise, the same construction with the level-0 operators `split₀`, `fiber₀` gives $\Lambda_0$ `Chapter3/BasicClassComposition · StabilityData.linIso₀`, a strong bundle isomorphism $U \lhd (\mathbb{R}^N \oplus W) \to \psi(U) \lhd ((\mathbb{R}^N \oplus Z) \oplus X)$ — the text's "strong bundle map covering $\psi$" `Chapter3/BasicClassComposition · StabilityData.linIso₀_isStrongBundleIso` — and its index raise is $\Lambda$, because `split` and `fiber` are `split₀` and `fiber₀` on the levels $m+1$ `Chapter3/BasicClassComposition · StabilityData.strongShift_`

linIso$_0$.

*The index raise.* For $\Psi : E_0 \times F_0 \to E_0' \times F_0'$ let $\Psi^1$ be its level-1 map on $E_1 \times F_1$ Chapter3/BasicClassComposition · strongShift. For a strong bundle map it exists on $U_1 \oplus F_1$ and satisfies $\iota\Psi^1(p) = \Psi(\iota p)$, because $\Psi$ maps $U_1 \oplus F_1$ into $U_1' \oplus F_1'$ (the filtration with $k = 1 \le m + 1$) Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift; its base map is the level-1 map of the base map of $\Psi$ Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_base_strongShift. **$\Psi^1$ is a strong bundle map** $U^1 \lhd F^1 \to U'^1 \lhd F'^1$ of the shifted structures Chapter3/BasicClassComposition · IsStrongBundleMap.shift: independence of the base component and fibre linearity are read through the injective inclusions; the double filtration $(U^1 \lhd F^1)_{m,k} = U_{m+1} \oplus F_{k+1}$ is preserved because $k + 1 \le (m + 1) + 1$ whenever $k \le m + 1$, so the filtration property of $\Psi$ applies; $(U^1 \lhd F^1)[0]$ is the shift of $(U \lhd F)[0]$, so $\Psi^1$ is sc-smooth there by Proposition 1.2.2 (preprint 1.17) Chapter1/Levels · IsScSmooth.shift; and $(U^1 \lhd F^1)[1]$, with levels $E_{m+1} \oplus F_{m+2}$, is the shift of $(U \lhd F)[1]$, on which $(\Psi^1)[1]$ agrees with $(\Psi[1])^1$ (the lift of the fibre component to $F_2'$ is the same element, by the injectivity of $F_2' \to F_1' \to F_0'$), so Proposition 1.2.2 (preprint 1.17) applies again. For a strong bundle isomorphism with inverse $\Psi'$, $\Psi'^1$ is the inverse of $\Psi^1$, again checked after the inclusion into level 0 Chapter3/BasicClassComposition · IsStrongBundleIso.shift. The text uses this whenever it writes $((\Psi \circ \Phi)_*(\cdot))^1$.

*Composition.* The base map of $\Lambda \circ \Phi$ is the composite of the base maps on $U$ Chapter3/BasicClassComposition · IsStrongBundleMap.base_comp, and $(\Lambda \circ \Phi)[1] = \Lambda[1] \circ \Phi[1]$ on $U \oplus F_1$ Chapter3/BasicClassComposition · IsStrongBundleMap.shiftFiber_comp; so the composite is a strong bundle map — fibre linearity because over $u$ it is the fibre map of $\Lambda$ over $\varphi(u)$ after that of $\Phi$ over $u$, sc-smoothness on $[0]$ and $[1]$ by the chain rule Chapter3/BasicClassComposition · IsStrongBundleMap.comp — and the composite of strong bundle isomorphisms is one, with the inverses composed in the opposite order Chapter3/BasicClassComposition · IsStrongBundleIso.comp. Push-forwards compose: $(\Lambda \circ \Phi)_* g = \Lambda_*(\Phi_* g)$ on the target of $\Lambda$, since the base of $(\Lambda \circ \Phi)^{-1}$ is $\varphi^{-1} \circ \lambda^{-1}$ and $\Phi(\varphi^{-1}(w), \cdot)$ lies over $w$ Chapter3/BasicClassComposition · IsStrongBundleIso.pushforward_comp Chapter3/GermPushforward · pushforward. The index raise commutes with composition, $(\Lambda \circ \Phi)^1 = \Lambda^1 \circ \Phi^1$ on $U_1 \oplus F_1$ Chapter3/BasicClassComposition · IsStrongBundleMap.strongShift_comp.

**Corollary 3.4.3 (preprint 3.39) as formalized** Chapter3/BasicClassComposition · corollary_3_4_3. Let $U$ be relatively open in a partial quadrant $C$ of $E$ with $0 \in U$, $g : U \to F$ sc-smooth (the text's "sc-germ"; sc-smoothness is what makes $(g + s)^1$ take values on level 1), $d$ conjugation data of $g$ (§III.30: a strong bundle isomorphism $\Phi : U \lhd F \to U' \lhd (\mathbb{R}^N \oplus W)$ covering $\varphi$ with $\varphi(0) = 0$ and a basic germ $h$ on a germ $G$ inside $U'$, with $(\Phi(u, g(u)))_2 = h(\varphi(u))$ for $u \in U$ near 0; the text's "equivalent by a strong bundle isomorphism to the basic germ $\Phi_* g$"), and $s : U \to F^1$ sc-smooth into $F^1$ with $s(0) = 0$ (the text's sc$^+$-germ $s : O(C, 0) \to (F, 0)$). The proof, in the order of the Lean:

1. *The sc$^+$-germ $t = \Phi[1]_* s$.* Put $t(u') = (\Phi[1](\varphi^{-1}u', s(\varphi^{-1}u')))_2$ Chapter3/BasicClassComposition · ConjugationData.pushPlus. It is sc-smooth from $U'$ into $(\mathbb{R}^N \oplus W)^1$: the pairing of $\varphi^{-1}$ with $s \circ \varphi^{-1}$ maps $U'$ into $U \oplus F^1$, and $\Phi[1]$ is sc-smooth there by Definition 2.6.1 (preprint 2.47); restricted to the germ $G \subseteq U'$ it is an sc-smooth germ. And $t(0) = 0$, from $\varphi^{-1}(0) = 0$, $s(0) = 0$ and $\Phi(0, 0)_2 = 0$ (fibre linearity).
2. *Absorbing the $\mathbb{R}^N$-component.* As in Theorem 3.6.8 (preprint 3.54) (§III.23, step 3), $h' = h + (t_{\mathbb{R}^N}, 0)$ Chapter3/BasicClassComposition · ConjugationData.absorb Chapter3/GlobalIFT · ScBanach.qPartOp is a basic germ with the same contraction map: it is an sc-smooth germ (pairing and the sum operator), $h'(0) = h(0) + 0 = 0$, and its $W$-component is that of $h$. The $W$-component $t_W$ is an sc-smooth germ into $W^1$ vanishing at 0, so the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) exist for $s = t_W$ on

the germ $G$ of $h$ Chapter3/BasicClass · StabilityData.exists_of_scPlus, and $\mathcal{S}.\mathrm{fsum}\, h' = h + \iota t$ Chapter3/BasicClass · StabilityData.fsum.

3. $\Phi_*(g+s) = h + t$ *near* $0$. For $u \in U$ at which the conjugation identity holds, $\Phi_*(g+\iota s)(\varphi(u)) = (\Phi(u, g(u)+\iota s(u)))_2$ because $\varphi^{-1}\varphi(u) = u$; by fibre linearity this is $(\Phi(u, g(u)))_2 + (\Phi(u, \iota s(u)))_2 = h(\varphi(u)) + \iota t(\varphi(u))$, where $\iota(\Phi[1](u, s(u)))_2 = (\Phi(u, \iota s(u)))_2$ Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd. So $\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ at the points $\varphi(u)$, $u \in U$ near $0$ — not everywhere, as the text's "clearly" suggests.
4. *The composite.* $\Gamma = \Lambda \circ \Phi^1$ Chapter3/BasicClassComposition · ConjugationData.compIso, with inverse $\Phi'^1 \circ \Lambda^{-1}$ Chapter3/BasicClassComposition · ConjugationData.compIsoInv, is a strong bundle isomorphism $U^1 \lhd F^1 \to \psi(U'^1) \lhd ((\mathbb{R}^N \oplus Z) \oplus X^1)$ Chapter3/BasicClassComposition · ConjugationData.compDomain: the index raise of $\Phi$ (relatively open $U_1'$ in the shifted quadrant) followed by $\Lambda$, composed by the composition lemma. Its base map is $u \mapsto \psi(\varphi_1(u))$, which vanishes at 0; $\Gamma_* = \Lambda_* \circ \Phi^1_*$ (composition of push-forwards); and the germ of $\mathcal{S}.\mathrm{pushMap}\, h'$ lies in the target, because `glue` maps the level-1 part of $G$ into $U_1'$.
5. *The basic germ.* $\mathcal{S}.\mathrm{pushMap}\, h' = \mathrm{fiber} \circ (\mathcal{S}.\mathrm{fsum}\, h')^1 \circ \mathrm{glue}$ Chapter3/BasicClass · StabilityData.pushMap is a basic germ by Theorem 3.4.2 (preprint 3.38) (§III.18) Chapter3/BasicClass · theorem_3_4_2.
6. *The conjugation identity on level* $1$. For $u \in U^1$ near $0$: $(\Gamma(u, (g+s)^1(u)))_2 = \mathcal{S}.\mathrm{pushMap}\, h'(\gamma(u))$. Since $\gamma(u) = \mathrm{split}(\varphi_1(u))$ and $\mathrm{glue} \circ \mathrm{split} = \mathrm{id}$, both sides are `fiber` applied to elements of $(\mathbb{R}^N \oplus W)_1$, which are compared after the injective inclusion into level 0: $\iota(\Phi^1(u, (g+s)^1(u)))_2 = (\Phi(\iota u, (g+s)(\iota u)))_2$ Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift_snd, which is $\mathcal{S}.\mathrm{fsum}\, h'(\varphi(\iota u))$ by step 3, and this is $\iota$ of $(\mathcal{S}.\mathrm{fsum}\, h')^1(\varphi_1(u))$. "Near 0" means: $\iota u$ lies in the neighbourhood where the conjugation identity holds (continuity of $\iota : E_1 \to E_0$), and $\varphi_1(u)$ lies in the level-1 part of the germ $G$, a relatively open condition at 0 by the continuity of $\varphi$ on level 1.

The conclusion of `corollary_3_4_3` collects these: $\mathcal{S}$ with $\mathcal{S}.G = G$ and $\mathcal{S}.s = t_W$, $t(0) = 0$, $h'$ basic, $\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ near $0$, $\Gamma$ a strong bundle isomorphism onto a relatively open subset of $(\widetilde{C} \oplus C) \oplus X^1$ with $\gamma(0) = 0$ and $\Gamma_* = \Lambda_* \circ \Phi^1_*$, the basic germ $\mathcal{S}.\mathrm{pushMap}\, h'$ inside the target, and the identity of step 6. In the form of Definition 3.1.11 (preprint 3.11): $(g+s)^1 : U^1 \to F^1$ **is conjugated to a basic germ** on the shifted structures $E^1$, $F^1$ Chapter3/BasicClassComposition · corollary_3_4_3_isConjugate, with parameter space $\mathbb{R}^n \oplus C$, target $\mathbb{R}^N \oplus Z$ and fibre $X^1$ (the text's $\tau : Z \to C$ is not applied, as in §III.18). The text's form "$\Gamma = \Psi \circ \Phi$, then raise the index" is the same map: $\Lambda_0 \circ \Phi$ is a strong bundle isomorphism on level 0 and $(\Lambda_0 \circ \Phi)^1 = \Lambda_0^1 \circ \Phi^1 = \Lambda \circ \Phi^1$ on $U_1 \oplus F_1$ Chapter3/BasicClassComposition · ConjugationData.compIso_eq_strongShift. No hypothesis is added beyond the two readings ("sc-germ" as sc-smooth on $U$; "equivalent by a strong bundle isomorphism" as conjugation data). The first reading is the whole-chart one of §III.6, narrower than the text's germs; no theorem of the germ reading uses Corollary 3.4.3 (preprint 3.39) (the facts at the nearby zeros come from the chart at $x_0$ through the linear coordinate change of Theorem 3.4.2 (preprint 3.38), §III.43).

## III.29 Good complements with a projection into the quadrant; Theorems 3.5.6, 3.5.11 and 3.6.8 (3) (preprint 3.43, 3.47 and 3.54) with a closed projecting complement

Two steps of the text depend on the choice of a good complement without saying so. In the proof of Theorem 3.2.3 (preprint 3.27) the retraction $t(v+w) = v + \delta(v)$ is defined on $\Sigma = \{v + y \in C : v \in V,\ y \in Y\}$, which the text calls relatively open in $C$; §III.24 shows that this fails for the good

complement $Y = \{x + y - z = 0\}$ of $N = \mathbb{R}(1,1,1)$ in $C = [0,\infty)^3$, and makes the remedy a hypothesis: condition (b) of the regularity conditions, $\pi(O) \subseteq C$ for the projection $\pi$ onto $N$ along $Y$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`. In the proof of Theorem 3.5.6 (preprint 3.43) (Lemma 3.5.10 (preprint 3.46)) a good complement $Z$ of $\widetilde{K} = \ker DH(0)$ is produced by projecting a good complement of $K'$. This section shows that for solution sets the complement can be chosen once and for all, at the bottom of the chain Theorem 3.8.22 (preprint 3.78) □ Theorem 3.5.6 (preprint 3.43) □ Theorem 3.5.11 (preprint Corollary 3.47) □ Theorem 3.6.8 (3) (preprint 3.54), so that its projection maps the partial quadrant into itself; condition (b) then holds for the chart of §III.34 without any hypothesis (Findings, Chapter 3, items 130–133). Along the way the closedness of the complement, which Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25) need (items 49, 81), is recorded in the statements (item 93).

**The projection property.** For subspaces $N$, $Y$ of a vector space and a set $C$, *$N$ projects $C$ into itself along $Y$* if $z \in C$, $k \in N$ and $z - k \in Y$ imply $k \in C$ `Chapter3/GoodComplementProjection · ProjectsInto`. The definition avoids naming the projection; when $E = N \oplus Y$ it says exactly that the projection $\pi$ onto $N$ along $Y$ maps $C$ into $C$ `Chapter3/GoodComplementProjection · ProjectsInto.projectionOnto_mem` `Chapter3/GoodComplementProjection · projectsInto_of_projectionOnto_mem` ($z - \pi z \in Y$, and conversely $z - k \in Y$, $k \in N$ force $\pi z = k$). It is carried by a linear isomorphism $S$ with $x \in C \iff Sx \in C'$: given $z' \in C'$, $k \in N$ and $z' - Sk = Sy$ with $y \in Y$, the point $S^{-1}z'$ lies in $C$ and $S^{-1}z' - k = y$, so $k \in C$ and $Sk \in C'$ `Chapter3/GoodComplementProjection · ProjectsInto.map`, in particular for continuous linear isomorphisms `Chapter3/GoodComplementProjection · ProjectsInto.mapL` — the analogue of Lemma 3.5.9 (preprint 3.45) (§III.14).

**The finite-dimensional lemma.** *If the finite-dimensional subspace $N$ is in good position to the partial quadrant $C$ of the sc-Banach space $E$, then $N$ has a closed good complement $Y$ such that the projection onto $N$ along $Y$ maps $C$ into $N \cap C$* `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto`. The text has no such statement; Definition 3.1.24 (preprint 3.21) and Proposition 3.1.25 (preprint 3.22) do not need it. The proof works in the coordinates of §III.14: after a presentation, $C = \{\ell_i \geq 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$, the cone $Q = N \cap C$ is $\{x \in N : \ell_i(x) \geq 0\}$, $\Sigma$ is the set of indices vanishing at some nonzero point of $Q$, and for a set $S$ of indices $Q_S = (\ell_i)_{i \in S} : N \to \mathbb{R}^S$ `Chapter3/GoodComplementProjection · coordOn` (for $S = \Sigma$ it is the map $Q_\Sigma$ of Proposition 3.1.25 (preprint 3.22) `Chapter3/GoodComplementProjection · sigmaCoord_eq_coordOn`).

*Step 1: the choice of coordinates* `Chapter3/GoodComplementProjection · exists_coordSet`. Wanted is a set $S$ of indices with (1) $Q_S$ onto, (2) $Q = \{x \in N : \ell_i(x) \geq 0 \text{ for } i \in S\}$, (3) every $\ell_i$, $i \notin S$, positive on $Q \setminus 0$. Property (3) holds for $S = \Sigma$ by the definition of $\Sigma$ `Chapter3/GoodPosition · pos_of_not_mem_vanIdx`, and (1) for $S = \Sigma$ is the consequence of Lemma 3.8.8 (preprint 3.65) proved in §III.14 `Chapter3/GoodPosition · sigmaCoord_surjective`. Three cases, those of the normal form. If $Q = N$, take $S = \Sigma$; (2) is trivial. If $\Sigma \neq \emptyset$, take $S = \Sigma$; (2) is the segment argument of Proposition 3.1.25 (preprint 3.22) `Chapter3/GoodPosition · mem_quadCone_of_vanIdx`. In the remaining case, $\Sigma = \emptyset$ and $Q \neq N$, the choice $S = \Sigma$ fails, and here the sketch of Finding 83, which takes $S = \Sigma$ throughout, misses a case (item 131). With $S = \emptyset$, (2) would say $Q = N$; and the formula of Step 4 below would reduce to an arbitrary projection onto $N$, which need not map $C$ into $C$: for $C = [0,\infty)^2$ and $N = \mathbb{R}(1,1)$ every nonzero point of $Q$ has both coordinates positive, so $\Sigma = \emptyset$; the line $\mathbb{R}(1,2)$ is a good complement of $N$ (for $|s| \leq \varepsilon|t|$ the coordinates of $t(1,1) + s(1,2)$ have the sign of $t$), and the projection along it maps $(0,1) = -(1,1) + (1,2)$ to $-(1,1) \notin C$. The formal proof keeps one coordinate. Some $x_1 \in N$ lies outside $C$, so there is at least one coordinate, $j_0$ say, and $S = \{j_0\}$. An interior point $y$ of $Q$ is nonzero — if $0$ were interior, $Q$ would contain a ball of $N$ around $0$ and, being closed under positive scaling, all of $N$ `Chapter3/GoodComplementProjection · mem_quadCone_of_zero_mem_interior` `Chapter3/GoodPosition · smul_mem_quadCone` — so $\ell_{j_0}(y) > 0$ by (3), and $Q_{\{j_0\}}$ is

onto: $k \mapsto (k/\ell_{j_0}(y))\, y$ is a preimage. Property (2) for $S = \{j_0\}$ is the next step.

*Step 2: the segment argument for any $S$* `Chapter3/GoodComplementProjection · mem_quadCone_of_coordSet`. Suppose (3) holds for $S$, some $i_0 \in S$ is positive at a point $y \in Q$, and $x \in N$ has $\ell_i(x) \geq 0$ for $i \in S$. The set $T = \{s \geq 0 : x + sy \in Q\}$ is nonempty (for $i \in S$, $\ell_i(x) + s\ell_i(y) \geq 0$ for all $s \geq 0$; for $i \notin S$, $\ell_i(y) > 0$, so large $s$ work), closed and bounded below; let $s^* = \min T$. If $s^* > 0$, the point $z = x + s^* y$ is nonzero, since $\ell_{i_0}(z) \geq s^* \ell_{i_0}(y) > 0$; so by (3) all $\ell_i$, $i \notin S$, are positive at $z$ and, by continuity, at $x + sy$ for $s$ near $s^*$; the coordinates in $S$ stay $\geq 0$ for all $s \geq 0$; so a point $s_1 = \max(0, s^* - \delta/2) < s^*$ lies in $T$, contradicting minimality. Hence $s^* = 0$ and $x \in Q$. This is the argument of Proposition 3.1.25 (preprint 3.22), Case C (§III.14), with the positivity outside $S$ as the only input; with $S = \{j_0\}$ and $y$ interior it gives (2) in the third case of Step 1.

*Step 3: a lower bound* `Chapter3/GoodComplementProjection · exists_coord_ge`. A coordinate positive on $Q \setminus 0$ is bounded below by $c|x|$ on $Q$, $c > 0$: its minimum on the compact set $Q \cap \{|x| = 1\}$ is positive, and $Q$ is a cone.

*Step 4: the projection and the estimate* `Chapter3/GoodComplementProjection · exists_proj_quadCone`. Take $S$ from Step 1, a continuous projection $p_N : E_0 \to N$ (Hahn–Banach, $N$ finite-dimensional), a linear right inverse $g$ of $Q_S$, and

$$\pi(z) = p_N(z) + g\big((\ell_i(z - p_N z))_{i \in S}\big).$$

Then $\pi$ is continuous, $\pi = \mathrm{id}$ on $N$ (the bracket vanishes), and $\ell_i(\pi z) = \ell_i(p_N z) + \ell_i(z - p_N z) = \ell_i(z)$ for $i \in S$, so for $z \in C$ the coordinates of $\pi z$ in $S$ are $\geq 0$ and $\pi z \in Q$ by (2): $\pi(C) \subseteq N \cap C$. The kernel of $\pi$ is a complement of $N$, and it is a good complement. For $p \in \ker \pi$ the coordinates in $S$ vanish ($\ell_i(p) = \ell_i(\pi p) = 0$). Let $c_i$ be the constants of Step 3 for $i \notin S$ and $\varepsilon' = \min_i c_i/(\|\ell_i\| + 1)$ `Chapter3/GoodComplementProjection · exists_pos_le_of_finite`. Let $x \in N$, $p \in \ker \pi$, $|p| \leq \varepsilon'|x|$. If $x \in C$: for $i \in S$, $\ell_i(x + p) = \ell_i(x) \geq 0$; for $i \notin S$, $\ell_i(x + p) \geq c_i|x| - \|\ell_i\|\,|p| \geq c_i|x| - \|\ell_i\|\varepsilon'|x| \geq 0$. If $x + p \in C$: the coordinates of $x$ in $S$ are those of $x + p$, hence $\geq 0$, and $x \in Q$ by (2). The reverse implication needs no estimate.

*Step 5: partial quadrants.* For a partial quadrant presented by an sc-isomorphism $T$ the functionals are the coordinate functions of $T$, continuous `Chapter3/GoodComplementProjection · continuous_coordFun` and jointly onto `Chapter3/GoodPosition · coordFun_surj`; good position supplies a good complement $P$ with its constant and the interior point. Step 4 then gives a continuous projection $\pi : E_0 \to N$ with $\pi(C) \subseteq C$ whose kernel is a good complement `Chapter3/GoodComplementProjection · exists_proj_goodComplement` ($\ker \pi$ is a topological complement of $N$, as the kernel of a continuous projection). With $Y = \ker \pi$, closed as the kernel of a continuous map, the projection property is immediate: $z \in C$, $k \in N$, $z - k \in Y$ give $k = \pi k = \pi z \in C$ `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto`.

**Theorem 3.5.6 (preprint 3.43) with a prescribed good complement.** In the text's proof the complement $Z$ of $\widetilde{K} = \ker DH(0)$ in $\mathbb{R}^{n'}$ comes from a good complement of $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$ through Lemma 3.5.10 (preprint 3.46) (§III.20). Nothing later uses that origin. *For every good complement $Z$ of $\widetilde{K}$ (constant $\varepsilon$), $Z \oplus W$ is a good complement of $K'$ with respect to $\widetilde{C} \oplus W$, with constant $\varepsilon/(\|L\| + 1)$* `Chapter3/GoodComplementProjection · IsGoodComplement.prod_top_graph`. The splitting: $K' \cap (Z \oplus W) = 0$ because $\widetilde{K} \cap Z = 0$, and $(a, w) = (\alpha, L\alpha) + (z, w - L\alpha)$ for $a = \alpha + z$. The estimate: with the maximum norm, $|(\alpha, L\alpha)| \leq (\|L\| + 1)|\alpha|$, so $|(z, w)| \leq \varepsilon/(\|L\| + 1) \cdot |(\alpha, L\alpha)|$ gives $|z| \leq \varepsilon|\alpha|$, and $(\alpha, L\alpha) + (z, w) \in \widetilde{C} \oplus W$ iff $\alpha + z \in \widetilde{C}$ iff $\alpha \in \widetilde{C}$ iff $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ `Chapter3/LocalSolvability · graphMap_mem_prod_iff`. This is the second half of the proof of Lemma 3.5.10 (preprint 3.46) for an arbitrary $Z$; Lemma 3.5.10 (preprint 3.46) itself is needed only for the good position of $\widetilde{K}$. The projection property passes the same way: if

$z \in \widetilde{C} \oplus W$ and $z - (\alpha, L\alpha) \in Z \oplus W$, then $z_1 - \alpha \in Z$, so $\alpha \in \widetilde{C}$ when $(\widetilde{K}, Z)$ has the property, and $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ `Chapter3/GoodComplementProjection · ProjectsInto.prod_top_graph`; the projection along $Z \oplus W$ is $(a, w) \mapsto (\pi_Z a, L\pi_Z a)$, and $\widetilde{C} \oplus W$ constrains only the first component.

So the data of the proof of Theorem 3.5.6 (preprint 3.43) (§III.20: the derivative data, the graph description $K' = \operatorname{graph}(L|_{\widetilde{K}})$ of Lemma 3.5.8 (preprint 3.44), a complement $Z$, and the solution $\sigma$ of $H(k + \sigma(k)) = 0$ from Theorem 3.8.22 (preprint 3.78) with its uniqueness constants `Chapter3/LocalSolvability · ShiftedBasicData.Sol`) exist with any prescribed good complement $Z$ of $\widetilde{K}$ `Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_of_goodComplement`: $K'$ is the graph `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq`, so $\widetilde{K}$ is in good position (Lemma 3.5.10 (preprint 3.46)) `Chapter3/LocalSolvability · lemma_3_5_10_isInGoodPosition`, $Z \oplus W$ is a good complement of $K'$ by the lemma above, and Theorem 3.8.22 (preprint 3.78) is applied to $H$ with $Z$ `Chapter3/QuadrantIFT · theorem_3_8_22`. Choosing $Z$ by the finite-dimensional lemma in $\mathbb{R}^{n'}$ (constant sc-structure, partial quadrant $\widetilde{C}$) gives data whose $Z$ projects $\widetilde{C}$ into $\widetilde{K} \cap \widetilde{C}$ `Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_proj`.

**Theorem 3.5.6 (preprint 3.43), closed and projecting.** With the hypotheses of Theorem 3.5.6 (preprint 3.43) (§III.20: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position, the regularity of Lemma 3.5.5 (preprint 3.42) as a hypothesis), the conclusions of the level-wise form `Chapter3/LocalSolvability · theorem_3_5_6_sc_smooth` hold with a good complement $Y = Z \oplus W$ that is **closed** `Chapter3/LocalSolvabilityClosed · theorem_3_5_6_closed` — $Z$ is finite-dimensional, hence closed, and $Z \times W_0$ is closed — and, with the data of the last paragraph, that has the **projection property** for $\widetilde{C} \oplus W$ `Chapter3/GoodComplementProjection · theorem_3_5_6_proj` (rewrite $K'$ as the graph and apply the passage of the property to $Z \oplus W$). The other conclusions ($V$, $\tau$ with values in $Y$ and $W$-components on level 1, $C^1$, $\tau(0) = 0$, $D\tau(0) = 0$, the graph of zeros, the $C^k$ lifts $\tau_k$, the neighbourhood $U''$ in which every zero lies on the graph) are those of §III.20, read off the same data `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level` `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ`.

**Theorem 3.5.11 (preprint Corollary 3.47), closed and projecting** `Chapter3/LocalSolvabilityClosed · theorem_3_5_11_proj`. For a basic germ $f$, the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) with $D(f + \iota s)(0)$ onto and kernel $K$ in good position to $\widetilde{C} \oplus W$: the conclusions of the level-wise Theorem 3.5.11 (preprint Corollary 3.47) `Chapter3/LocalSolvability · theorem_3_5_11_sc_smooth` hold with $Y$ closed and $K$ projecting $\widetilde{C} \oplus W$ into itself along $Y$. The proof is that of §III.20 with the two new properties carried along. Let $S$ be the linear coordinate change $\psi(a, w) = ((a, (1 - P_1)w), P_1 w)$ of Theorem 3.4.2 (preprint 3.38) on level 0, a continuous linear isomorphism with inverse the gluing map. The push-forward $g = \Phi_*(f + s)$, packaged as data of Theorem 3.5.6 (preprint 3.43) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`, has $Dg(0) = \Psi \circ D(f + \iota s)(0) \circ S^{-1}$ `Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀`, so $\ker Dg(0) = S(K)$, $Dg(0)$ is onto, and $K$ is finite-dimensional (Lemma 3.5.8 (preprint 3.44) for $g$ and the injectivity of $S^{-1}$). $S$ carries $\widetilde{C} \oplus W$ onto $(\widetilde{C} \oplus C_s) \oplus X$, where $C_s$ is the finite-dimensional space of the splitting of Theorem 3.4.2 (preprint 3.38) (§III.18) and $\widetilde{C} \oplus C_s$ is a partial quadrant of $\mathbb{R}^n \oplus C_s$ `Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ`; so $S(K)$ is in good position (Lemma 3.5.9 (preprint 3.45), §III.14). Theorem 3.5.6 (preprint 3.43), closed and projecting, for $g$, with Lemma 3.5.5 (preprint 3.42) for its regularity hypothesis `Chapter3/LocalSolvability · StabilityData.lemma_3_5_5`, gives $Y'$, $V'$, $\tau$. Transported back: $Y = S^{-1}(Y')$ is a good complement of $K = S^{-1}(S(K))$ (Lemma 3.5.9 (preprint 3.45)) `Chapter3/GoodPosition · IsGoodComplement.map`, closed as the image of a closed set under a linear homeomorphism, and projecting, by the transport of the projection property `Chapter3/GoodComplementProjection · ProjectsInto.mapL`; $V = (S|_K)^{-1}(V')$ and $\sigma = S^{-1} \circ \tau \circ S|_K$. The $W$-component of $\sigma(y)$ is $c + w$ with $c \in C_s \subseteq W_\infty$ `Chapter3/BasicClass · StabilityData.Cs_le_smooth` and $w$ the level-1 $X$-component of $\tau$, so it lies on level 1;

$D\sigma(0) = S^{-1} \circ D\tau(0) \circ S|_K = 0$, by the chain rule within $V$, which has unique differentiability at 0 as a relatively open subset of the convex cone $K \cap C$ with interior; the graph points are zeros because the linear fibre map of Theorem 3.4.2 (preprint 3.38) is injective; the lifts are $\psi^{-1}$ on level $k$ applied to $\tau_k$; and $U'' := S^{-1}(U'')$. The form with only the closedness is a corollary `Chapter3/LocalSolvabilityClosed · theorem_3_5_11_closed`.

**Theorem 3.6.8 (3) (preprint 3.54), closed and projecting** `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj`. The graph description of §III.23 — for a section $f$ of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and $(f, x_0)$ in good position: the chart $\varphi$ centred at $x_0$, the conjugating sc-diffeomorphism $\psi$, the germ $h + t$ with surjective linearization and kernel $K$ in good position, $Y$, $V'$, $\sigma$, and the open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$ — is proved with Theorem 3.5.11 (preprint Corollary 3.47) closed and projecting in place of the plain Theorem 3.5.11 (preprint Corollary 3.47), and the statement now records five more facts. (i) The inverse $\psi'$ of $\psi$, sc-smooth on $U'$, with $\psi'\psi = \mathrm{id}$ on $U$ and $\psi\psi' = \mathrm{id}$ on $U'$ (the base maps of the conjugating strong bundle isomorphism and its inverse); §III.34 needs it to make $\psi \circ \varphi$ a chart. (ii) $Y$ is closed, and (iii) $K$ projects $C' = \widetilde{C} \oplus W$ into itself along $Y$, both from Theorem 3.5.11 (preprint Corollary 3.47) — the shrinking of $V'$ in step 6 of §III.23 ($V' = V_0 \cap O_1$ with $O_1$ open, chosen so that $v + \sigma(v)$ stays in $\Omega = U'' \cap W'$) does not touch $Y$. (iv) $K$ is a smooth subspace `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth` (proved in §III.34). (v) Every $v + \sigma(v)$, $v \in V'$, is a zero of $h + t$ in the germ, and for every $k \geq 1$ the $C^k$ lift $\sigma_k$ of Theorem 3.5.11 (1) (preprint Corollary 3.47) survives the shrinking: on $V_k \cap O_1$, relatively open in $K \cap C'$, containing 0 and contained in $V'$, it is still $C^k$ and still lifts $\sigma$. The theorem of §III.23, with the same statement except the projection property, is a corollary `Chapter3/GlobalIFTG · theorem_3_6_8_graph`.

*Where the formal route departs from the text.* The text never chooses the good complement; Definition 3.2.1 (preprint 3.25) lets it be any good complement, and for an arbitrary subset with the tangent germ property the repair of the claim "$\Sigma$ is relatively open" would need a re-graphing over a better complement, which is not done ($\pi(O) \subseteq C$ remains a hypothesis of Theorem 3.2.3 (preprint 3.27), §III.24). For solution sets the complement is not given but produced, at Theorem 3.8.22 (preprint 3.78), and every link of the chain accepts a prescribed complement; so the repair is a choice made at the bottom of the chain, not a change in the proof of Theorem 3.2.3 (preprint 3.27) (item 130). The closedness of $Y$ is part of the text's sc-complement and is recorded because Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25) use it at the nearby zeros (items 49, 81, 93).

## III.30 Proposition 3.6.5 (preprint 3.52) with its index

**Proposition 3.6.5 (preprint 3.52)** (text): the linearization $Tf(x) : T_xX \to Y_x$ of an sc-Fredholm section at a zero is an sc-Fredholm operator; in the proof (and in Theorem 3.6.8 (preprint 3.54)) its index is the index $n - N$ of the basic germ $h : O(([0,\infty)^k \oplus \mathbb{R}^{n-k}) \oplus W, 0) \to (\mathbb{R}^N \oplus W, 0)$ to which $g - s$ is conjugated. The statements of §III.21 state the index as "$n - N$ for some natural numbers $n$, $N$", which every integer is. The reason is that the conjugation of Definition 3.1.11 (preprint 3.11) is an existential proposition `Chapter3/StrongBundleSections · IsConjugateToBasicGerm`, so the dimensions of the basic germ cannot be named outside it. Here the witnesses become data.

*Conjugation data* `Chapter3/FredholmGermIndex · ConjugationData`: finite-dimensional normed spaces $P_n$ (the text's $\mathbb{R}^n$) and $Q$ (the text's $\mathbb{R}^N$), an sc-Banach space $W$, a partial quadrant $\widetilde{C} \subseteq P_n$, a relatively open $U' \subseteq \widetilde{C} \oplus W$, a strong bundle isomorphism $\Psi : U \lhd F \to U' \lhd (Q \oplus W)$ with inverse $\Psi'$ and $\psi(0) = 0$ for its base map, a germ $G$ with $G_0 \subseteq U'$, a basic germ $h$ on $G$ with contraction map $B$, and

$(\Psi(u,k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0. A map is conjugated to a basic germ iff such data exist Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty.

**The derivative of a conjugated map** Chapter3/FredholmGermIndexG · ConjugationData.scDeriv_isFredholm_index. Let $U$ be relatively open in the partial quadrant $C$, $0 \in U$, $k$ an sc¹-germ with derivative family $Dk$ on a germ $G_k$ with $(G_k)_0 \subseteq U$, $k(0) = 0$, and $d$ conjugation data of $k$. Then $Dk(0) : E_0 \to F_0$ is Fredholm of index $\dim P_n - \dim Q$; for $k$ sc-smooth on $U$ this is the whole-chart form Chapter3/FredholmGermIndex · ConjugationData.scDeriv_isFredholm_index_wholeChart. The proof is that of §III.21, step (a), with the dimensions kept:

*(a)* $Dh(0)$ is Fredholm of index $\dim P_n - \dim Q$ (Proposition 3.1.10 (preprint 3.10) for the basic germ on level 0) Chapter3/FredholmGermLinearization · proposition_3_1_10_basic_level_zero.

*(b)* $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, with $\Gamma(0) = (\Psi(0,\cdot))_2$ the fibre operator of $\Psi$ at 0 Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM. The left side is the derivative at 0 of $u \mapsto (\Psi(u,k(u)))_2$, because $k(0) = 0$ Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp. For the right side, let $N_1$ be an open neighbourhood of 0 on which the conjugation identity holds; the germ $\{U \cap N_1\}$ has a subgerm $G_1$ mapped by $\psi$ into $G$ level by level Chapter3/GermFilling · ScNbhdGerm.preimage; on $G_1$ the chain rule for germs gives the derivative family $Dh(\psi_1(x))D\psi(x)$ of $h \circ \psi$ Chapter3/Germ · IsSc1GermWith.comp; and $u \mapsto (\Psi(u,k(u)))_2$, an sc¹-germ on $G_k$ by the chain rule with the whole-chart outer map (§III.48), agrees with $h \circ \psi$ on $G_1 \cap G_k$, so the two derivative families agree at 0 (uniqueness of the derivative of a germ at points of $U_1$) Chapter3/Germ · IsSc1Germ.scDerivGerm_eqOn, with $\psi_1(0) = 0$.

*(c)* $D\psi(0)$ and $D\psi'(0)$ are inverse to each other (chain rule for $\psi' \circ \psi = \mathrm{id}$ on $U$ and $\psi \circ \psi' = \mathrm{id}$ on $U'$) Chapter2/DegeneracyIndex · scDeriv_comp_eq_self, and so are $\Gamma(0)$ and the fibre operator $\Gamma'(0)$ of $\Psi'$ at 0: $\Psi'(\Psi(0,y)) = (0,y)$ with $\Psi(0,y) = (0,\Gamma(0)y)$ because $\psi(0) = 0$, and symmetrically.

*(d)* Hence $Dk(0) = \Gamma(0)^{-1} \circ Dh(0) \circ D\psi(0)$, a Fredholm operator composed with isomorphisms on both sides, of the same index Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv.

Consequently **the index $n - N$ does not depend on the conjugation**: two conjugation data $d$, $d'$ of the same $k$ give $\dim d.P_n - \dim d.Q = \operatorname{ind} Dk(0) = \dim d'.P_n - \dim d'.Q$ Chapter3/FredholmGermIndexG · ConjugationData.index_eq. This is the text's remark after Theorem 3.4.2 (preprint 3.38) that "the Fredholm index is invariant under strong bundle isomorphisms", for the conjugations of one germ.

**Proposition 3.6.5 (preprint 3.52) in the chart, with the index** Chapter3/FredholmGermIndexG · proposition_3_6_5_local_index. Hypotheses: $U$ relatively open in $C$, $R$ a strong bundle retraction on $U \lhd F$ with base retraction $r$, $0 \in U$ and $0 \in O = r(U)$; $g$ a filling germ of the principal part $\hat{f}$ (§III.3) and $Dg$ a derivative family of $g$ on a germ $G$ with $G_0 \subseteq U$; $\hat{f}(0) = 0$ and $(x, \hat{f}(x)) \in K$ for $x \in O$; $s$ an sc⁺-germ with derivative family $Ds$ and $\iota s(0) = g(0)$; and conjugation data $d$ of $k = g - \iota s$. Conclusion: $Dg(0) : E_0 \to F_0$ and $Tf(0) = Dg(0)|_{T_0O} : T_0O \to K_0$ Chapter3/FillerLinearAlgebra · fillerRestrict are Fredholm with index $\dim d.P_n - \dim d.Q$. Proof: $g(0) = \hat{f}(0) = 0$ (filler condition (1) at 0), so $k$ is an sc¹-germ on $G \cap G_s$ with the family $Dg - \iota Ds$ and $k(0) = 0$ Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift, and $Dk(0)$ is Fredholm of index $\dim d.P_n - \dim d.Q$ by the preceding theorem. Since 0 is a smooth point, $Ds(0) : E \to F^1$ is an sc-operator (the remark after Proposition 1.2.1 (preprint 1.16), for germs) Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero, so $\iota Ds(0)$ is the level-0 map of an sc⁺-operator and compact. $Dg(0) = Dk(0) + \iota Ds(0)$ is then the value at 0 of the identity $Dg = (Dg - \iota Ds) + \iota Ds$ Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus; on $U$ it was the uniqueness of derivatives within $U_1$ and the density of $E_1$ Chapter3/

BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one. A compact perturbation of a Fredholm operator is Fredholm of the same index Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator. Proposition 3.1.5 (preprint 3.5) (§III.4) transfers the Fredholm property and the index to $Tf(0)$ Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_1_5.

**Proposition 3.6.5 (preprint 3.52), with the index** Chapter3/FredholmGermIndexG · proposition_3_6_5_index. Let $f$ be a section of the strong bundle $P : Y \to X$ with $(f, x_0)$ an sc-Fredholm germ and $f(x_0) = 0$. In the strong bundle chart of Definition 3.1.11 (preprint 3.11) centred at $x_0$, with the filling germ $g$ of $\widehat{f}$, its germ $G$ and a derivative family $Dg$, and the local sc$^+$-germ $s$, conjugation data of $g - s$ exist, $Dg(0)$ and $Tf(x_0)$ are Fredholm, and **for every** conjugation data $d$ of $g - s$ their index is $\dim d.P_n - \dim d.Q$. Proof: unpack Definition 3.1.11 (preprint 3.11); $0 \in O$ because the chart is centred at $x_0$; $\widehat{f}(0) = 0$ from $f(x_0) = 0$; $(x, \widehat{f}(x)) \in K$ because the principal part of a section lies in the retract $K$ of the chart; data $d_0$ exist by the equivalence above and give the Fredholm property; and the local theorem applied to an arbitrary $d$ gives the index. No hypothesis is added. As in §III.21, only the level-0 Fredholm property is proved ($T_0O$ and $K_0$ are not set up as sc-Banach spaces). The statements of §III.21 stand next to these, with the same proofs and without the dimensions; the whole-chart forms are kept as well Chapter3/FredholmGermIndex · proposition_3_6_5_index_wholeChart.

## III.31 The linearization at a zero in charts, its chart independence, and Definition 3.6.6 (preprint 3.53) from a non-centred chart

Definition 3.6.6 (preprint 3.53) is stated in $T_xX$, with the linearization $f'(x) : T_xX \to Y_x$ and the partial cone $C_xX$ of Definition 2.5.8 (preprint 2.38). The text defines $f'(x)$ only at the *centre* of a chart, before Definition 3.1.4 (preprint 3.4), as the restriction of $D(\mathbf{f} \circ r)(0)$ to $T_0O$, identifies it with $Dg(0)|_{T_0O}$ for a filler $g$, and appeals to Proposition 2.4.10 (preprint 2.30) for the independence of $T^R_xX$ and $C_xX$ of the chart. That $f'(x)$ itself, and the two conditions of Definition 3.6.6 (preprint 3.53), do not depend on the chart is used — the proof of Theorem 3.6.8 (preprint 3.54) repeats "as above" the argument at $x_0$ in a chart centred at a nearby zero $y$ — but not proved (Finding 110). The formalization (§III.21) states Definition 3.6.6 (preprint 3.53) in the chart of Definition 3.1.11 (preprint 3.11) centred at the zero; this section proves the local form of the missing independence at a zero, in charts, and the assembly step that Theorem 3.6.8 (1) (preprint 3.54) needs: Definition 3.6.6 (preprint 3.53) at a zero $y$ from data in a chart which is *not* centred at $y$ (the chart at $x_0$). Throughout, $z$ is a point on level 1 (the text has smooth points), and tameness is not used.

**Linear algebra of conjugate operators.** Let $L : T \to K$ and $L' : T' \to K'$ be bounded operators between normed spaces and $A : T \cong T'$, $\Gamma : K \cong K'$ topological linear isomorphisms with $L'A = \Gamma L$. Then $L' = \Gamma L A^{-1}$ Chapter3/LinearizationCharts · eq_comp_of_conj, so $L$ is onto iff $L'$ is Chapter3/LinearizationCharts · surjective_iff_of_conj; $\ker L' = A(\ker L)$ Chapter3/LinearizationCharts · ker_eq_map_of_conj; for complete spaces $L$ is Fredholm iff $L'$ is, with the same index, because Fredholm operators and their index are stable under composition with isomorphisms on both sides Chapter3/LinearizationCharts · isFredholm_iff_of_conj Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv; and if $D \subseteq K$ is dense and contained in the range of $L$, then $\Gamma(D)$ is dense in $K'$ and contained in the range of $L'$ Chapter3/LinearizationCharts · dense_image_of_conj. The isomorphisms that occur are restrictions of bounded operators $a : E_0 \to E_0'$, $a' : E_0' \to E_0$ which map a subspace $T$ into $T'$ and $T'$ into $T$ and are inverse to each other there Chapter3/LinearizationCharts · restrictEquiv. A subspace $N \subseteq E_0$ is trans-

ported by $A : T \cong T'$ to $A(N) = \{Ax : x \in T \cap N\} \subseteq T'$ `Chapter3/LinearizationCharts · transportSub`, with $y \in A(N) \iff A^{-1}y \in N$ for $y \in T'$ `Chapter3/LinearizationCharts · mem_transportSub_iff` and $A^{-1}(A(N)) = N$ for $N \subseteq T$ `Chapter3/LinearizationCharts · transportSub_symm`.

**Good position inside a subspace is transported** `Chapter3/LinearizationCharts · IsInGoodPositionIn.transport`. Good position inside $T$ (§III.21, `Chapter3/GoodPositionGerm · IsInGoodPositionIn`) asks: $N \subseteq T$, the interior of $N \cap C$ in $N$ nonempty, and a complement $P \subseteq T$ of $N$ *in* $T$ with the $\varepsilon$-condition of Definition 3.1.24 (preprint 3.21). If $A : T \cong T'$ respects the cones ($x \in C \iff Ax \in C'$ for $x \in T$) and $N$ is in good position to $C$ inside $T$, then $A(N)$ is in good position to $C'$ inside $T'$. *The interior*: $A$ restricts to a homeomorphism $N \cong A(N)$ which carries $N \cap C$ onto $A(N) \cap C'$, so it carries interior points to interior points. *The complement*: $A(P)$; $A(N) \cap A(P) = 0$ and $A(N) + A(P) = T'$ by pulling back with $A^{-1}$. *The constant*: with $a = \|A\|$, $b = \|A^{-1}\|$, take $\varepsilon' = \varepsilon/(ab+1)$; for $y = Ax$, $q = Ap$ with $|q| \le \varepsilon'|y|$, $|p| \le b|q| \le b\varepsilon' a|x| \le \varepsilon|x|$, so $x \in C \iff x + p \in C$, i.e. $y \in C' \iff y + q \in C'$ — the constant of Lemma 3.5.9 (preprint 3.45), no finite dimensionality needed (Finding 115). Applied to $A$ and $A^{-1}$ this gives an equivalence `Chapter3/LinearizationCharts · isInGoodPositionIn_transport_iff`.

**Two derivative tools at a level-1 point.** (i) *Uniqueness*: if $f_1 = f_2$ on a relatively open $V \subseteq U_1 \cap U_2$ containing $z$, then $Df_1(z) = Df_2(z)$: both are derivatives within $V_1$ (level 1), $V$ is an sc-domain with unique differentiability, and two operators $E_0 \to F_0$ that agree after composition with the dense inclusion $E_1 \to E_0$ are equal `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`. (ii) *The fibre derivative at a point of the zero section* `Chapter3/LinearizationCharts · scDeriv_fiber_of_vanishing`: let $\Phi$ be sc¹ on a set $D \subseteq E \oplus F$ containing $\{(u,0) : u \in S\}$ ($S$ relatively open, $z \in S_1$) and the whole fibre $\{z\} \times F$, with $\Phi(u,0) = 0$ on $S$ and $\Phi(z,k) = \Gamma k$ for a bounded $\Gamma$. Then $D\Phi(z,0)(h,k) = \Gamma k$. Proof: along $k_1 \mapsto (z,k_1)$, $k_1 \in F_1$, the composite is $\Gamma \circ \iota$, so $D\Phi(z,0)$ restricted to the fibre direction is $\Gamma$ after composition with the inclusion, hence equal to $\Gamma$ by density; along $u_1 \mapsto (u_1,0)$, $u_1 \in S_1$, the composite vanishes, and unique differentiability of $S_1$ at $z$ gives zero in the base direction. This is the generalization from 0 to $z$ of the identities of §III.4 and §III.21; it is the reason why only the fibre part survives at a zero (Finding 112).

**The linearization at a point of the retract, without a filler.** For the principal part $\mathbf{f}$ of a section in a strong bundle chart with retraction $R(u,h) = (r(u), \rho(u)h)$ and a level-1 point $z$ of $O = r(U)$, put

$$\mathrm{Lin}(z) = D(\mathbf{f} \circ r)(z)|_{T_zO}, \qquad T_zO = Dr(z)E_0$$

`Chapter3/LinearizationCharts · linAt`; $Dr(z)$ is idempotent `Chapter3/LinearizationCharts · scDeriv_r_idem_at`. Three facts make it the right object.

*Its values lie in $K_z = \rho(z)F$ at a zero* `Chapter3/LinearizationCharts · rho_scDeriv_comp_r` (the text's "$\rho(0)Tf(0) = Tf(0)$", now at any zero). Since $(r(u), \mathbf{f}(r(u))) \in K$, $\rho(r(u))\mathbf{f}(r(u)) = \mathbf{f}(r(u))$ on $U$ `Chapter3/Filler · ρ_eq_self_of_mem_K`; so $\mathbf{f} \circ r$ equals $\Psi_2(u) = (R(r(u), \mathbf{f}(r(u))))_2$, and their derivatives at $z$ agree by (i). The chain rule for $\Psi_2 = (\text{fibre part of } R) \circ (r, \mathbf{f} \circ r)$ at $z$, where $(r, \mathbf{f} \circ r)$ passes through $(z,0)$ (because $r(z) = z$ and $\mathbf{f}(z) = 0$), and (ii) for the fibre map $(u,k) \mapsto \rho(u)k$, which vanishes on the zero section and is $\rho(z)$ on the fibre over $z$, give $D(\mathbf{f} \circ r)(z)v = \rho(z)D(\mathbf{f} \circ r)(z)v$. The fibre $K_z$ is the fibre of $K$ over $z$ `Chapter3/LinearizationCharts · mem_rangeS_ρCLM_iff`.

*The bridge to fillers* `Chapter3/LinearizationCharts · scDeriv_comp_r_eq_of_eventuallyEq` (the text's "$Tg(0) = Tf(0)$ on $T_0O$", now at any point): if $g$ is sc¹ and agrees with $\mathbf{f}$ on $O$ near $z$, say on $O \cap W$ with $W$ open, then $\mathbf{f} \circ r = g \circ r$ on the relatively open $U \cap r^{-1}(W)$, so by (i) and the chain rule ($r$ maps $z$ to itself on level 1) $D(\mathbf{f} \circ r)(z) = Dg(z)Dr(z)$. Hence $\mathrm{Lin}(z) = Dg(z)$ on $T_zO$ `Chapter3/LinearizationCharts · linAt_`

eq_scDeriv, by idempotence of $Dr(z)$. At a centre, for a filled version, $\mathrm{Lin}(0)$ is the $Tf(0) = Dg(0)|_{T_0O}$ of Proposition 3.1.5 (preprint 3.5) Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt Chapter3/FillerLinearAlgebra · fillerRestrict. At $z \neq 0$ the full $Dg(z) : E \to F$ is not related to $\mathrm{Lin}(z)$ (condition (3) of the filler holds at 0 only), so the relevant kernel is $\ker Dg(z) \cap T_zO = \ker \mathrm{Lin}(z)$ (Finding 111). *For a filling germ $g$* on $G$ the bridge needs two things the whole-chart one did not: $Dg(z)$ exists only for $z$ on level 1 of $G$, and the chain rule for $g \circ r$ is the inner one on the germ $r^{-1}(G)$, which is defined when $r(0) = 0$, i.e. $0 \in O$ Chapter3/LinearizationChartsG · scDeriv_comp_r_eq_germ. Under these two conditions $\mathrm{Lin}(z) = Dg(z)$ on $T_zO$ Chapter3/LinearizationChartsG · linAt_eq_germDeriv Chapter3/LinearizationChartsG · StrongBundle.linAt_eq_germDeriv_chart, and at the centre, which lies in every germ, $\mathrm{Lin}(0) = Tf(0)$ Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt. Both hold wherever the bridge is used: the chart is centred at a zero, and the points $z \neq 0$ are the level-1 lifts of nearby zeros, which lie in the germ of $g$ by the shrinking of the basic germ (§§III.32, III.48).

**Fibres of strong bundle charts and of transitions.** For a strong bundle chart $\Phi : P^{-1}(V) \to K$ covering $\varphi$: points of $K$ lie over $O$ Chapter3/LinearizationCharts · StrongBundleChart.fst_mem_0; $\Phi^{-1}(x, \cdot)$ is additive and homogeneous on the fibre of $K$ over $x$, since $\Phi$ is a fibrewise linear bijection of $P^{-1}(\varphi^{-1}(x))$ onto that fibre Chapter3/LinearizationCharts · StrongBundleChart.symm_add Chapter3/LinearizationCharts · StrongBundleChart.symm_smul; the fibre of $K$ is closed under scalars and contains $(u, \rho(u)k)$ Chapter3/LinearizationCharts · StrongBundleChart.mem_K_ρ. For a second chart $\Phi'$ the transition $\Theta = \Phi' \circ \Phi^{-1}$ covers the base transition $\tau = \varphi' \circ \varphi^{-1}$ Chapter3/LinearizationCharts · StrongBundleChart.transition_fst, is linear on the fibres Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_add Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_smul, maps the zero section to the zero section Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_zero and $K$ into $K'$ Chapter3/LinearizationCharts · StrongBundleChart.transition_mem_K, and is inverted by the reverse transition Chapter3/LinearizationCharts · StrongBundleChart.transition_transition. Over a point $x$ of the base overlap, $\Gamma(x)k = (\Theta(x, \rho(x)k))_2$ is a bounded operator $F \to F'$ — linear by the fibre linearity, continuous as a composition of $\rho(x)$, $\Phi^{-1}$ and $\Phi'$ Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM — which on $K_x$ is the transition itself Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM_apply_of_mem. The text's "bounded linear on the fibres" (Definition 2.6.4 (preprint 2.50)) is about $\Phi$ only; the linearity of $\Phi^{-1}$ on the fibres of $K$ is this lemma (Finding 113). Finally, if $\varphi(V \cap V') = O \cap W$ with $W$ open, the overlap of $K$ is $K \cap (W \times F)$ Chapter3/LinearizationCharts · StrongBundleChart.overlap_eq_inter_prod.

**The chart change at a zero** Chapter3/LinearizationCharts · linAt_chart_change. *Statement.* Let $c$, $c'$ be strong bundle charts whose transition $\Theta$ is sc-smooth on $[0]$ (as a map between the overlap retracts of $K$ and $K'$), $f$ a section whose representatives $z \mapsto (z, \mathbf{f}(z))$ are sc-smooth $O \to K$ and $O' \to K'$, $y$ a zero of $f$ in both chart domains, and $z$, $z'$ level-1 points over $\varphi(y)$, $\varphi'(y)$. Then there are topological isomorphisms $A : T_zO \cong T_{z'}O'$ and $\Gamma : K_z \cong K'_{z'}$, with $\Gamma \circ \mathrm{Lin}(z) = \mathrm{Lin}'(z') \circ A$, and $x \in C_z \iff Ax \in C'_{z'}$, for $x \in T_zO$, where $C_z$ is the partial quadrant at $\varphi(y)$ of a presentation of $C$ Chapter2/Tame · partialQuadrantAtWith. The proof, in the order of the Lean:

1. *The base transition as a germ of sc-diffeomorphisms.* The base charts are compatible (both are charts of $X$, Chapter2/MPolyfold · IsChart.compatible). Choose open $W$, $W'$ with $\varphi(V \cap V') = O \cap W$ and $\varphi'(V \cap V') = O' \cap W'$ Chapter2/MPolyfold · exists_isOpen_image_eq and restrict the base retractions to $U \cap r^{-1}(W)$, $U' \cap r'^{-1}(W')$ (Proposition 2.3.1 (1) (preprint 2.10)); their retracts are the overlaps. Then $\tau$, $\tau' = \varphi \circ \varphi'^{-1}$ form a germ of sc-diffeomorphisms between them Chapter2/Boundary · ScRetraction.IsDiffeoGerm, and $\tau \circ r_W$ maps $z$ to $z'$ on level 1. The derivatives of the restricted retraction and of $\mathbf{f} \circ r$ at $z$ are those on the full domains, by (i) (Finding 114).

2. *The transition near the zero section.* Restrict the strong bundle retraction to $W \times F$; its retract is the overlap of $K$, so $\Theta \circ R_W$ is sc-smooth and its fibre part $\Phi$ is sc¹. $\Phi(z,k) = \Gamma(z)k$ because $R(z,k) = (z, \rho(z)k)$; $\Phi(u,0) = 0$ because $R(u,0) = (r(u),0)$ and $\Theta$ preserves the zero section.
3. *The section in the two charts.* For $u$ in the restricted domain, $r(u) = \varphi(y_1)$ for some $y_1 \in V \cap V'$, and both $\mathbf{f}' \circ r' \circ \tau \circ r_W$ and $\Phi \circ (r, \mathbf{f} \circ r)$ take at $u$ the value $(\Phi'(f(y_1)))_2$ (the principal parts are the fibre parts of $\Phi \circ f \circ \varphi^{-1}$ and $\Phi' \circ f \circ \varphi'^{-1}$, `Chapter3/StrongBundleSections · StrongBundleChart.apply_section'`). By (i) their derivatives at $z$ agree.
4. *The key identity.* The chain rule on both sides, with $(r, \mathbf{f} \circ r)$ passing through $(z,0)$ since $\mathbf{f}(z) = 0$, and (ii) for $\Phi$ give
$$D(\mathbf{f}' \circ r')(z')\, D(\tau \circ r_W)(z)v = \Gamma(z)\, D(\mathbf{f} \circ r)(z)v.$$
The derivative of $x \mapsto \Gamma(x)\mathbf{f}(z)$ in the base direction has dropped out *because* $\mathbf{f}(z) = 0$; away from zeros the linearization is not chart independent (Finding 112).
5. *The isomorphism $A$.* $A_0 = D(\tau \circ r_W)(z)$ maps $T_zO$ into $T_{z'}O'$ and $A_0' = D(\tau' \circ r'_{W'})(z')$ maps back `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range`; on these subspaces they are inverse `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv` (with the idempotence of $Dr(z)$); $A$ is the induced isomorphism (`restrictEquiv`).
6. *The isomorphism $\Gamma$.* $\Gamma(z)$ maps $K_z$ into $K'_{z'}$ (the transition maps $K$ into $K'$ and covers $\tau$), the fibre operator $\Gamma'(z')$ of the reverse transition maps back, and they are inverse on these fibres (the reverse transition inverts the transition).
7. *Conclusions.* For $x \in T_zO$, $\mathrm{Lin}(z)x = D(\mathbf{f} \circ r)(z)x$ and $\mathrm{Lin}'(z')(Ax) = D(\mathbf{f}' \circ r')(z')A_0x$, so step 4 is $\Gamma\,\mathrm{Lin}(z) = \mathrm{Lin}'(z')A$. The cones: Proposition 2.4.10 (preprint 2.30) for the partial cone, applied to the germ of step 1 `Chapter2/Boundary · proposition_2_4_10_cone`, gives $A_0(T_zO \cap C_z) = T_{z'}O' \cap C'_{z'}$ for the formalized partial cone `Chapter2/Tame · partialConeWith`, which is the cone condition on $T_zO$.

*Hypotheses.* Of the compatibility of Definition 2.6.4 (preprint 2.50) only the sc-smoothness of $\Theta$ on $[0]$ is used — not on $[1]$, not the reverse transition (Finding 116). For two strong bundle charts of $P$ it holds because both are charts of the M-polyfold $Y[0]$, hence compatible `Chapter3/LinearizationCharts · StrongBundle.IsChart.transition_isScSmoothRetractMap`; and for an sc-smooth section the representatives are sc-smooth in every strong bundle chart of $P$, so no hypothesis is added.

**Chart independence** `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. For two strong bundle charts of $P$ around a zero $y$ of an sc-smooth section, with level-1 points $z$, $z'$ over $\varphi(y)$, $\varphi'(y)$: the linearizations $L = \mathrm{Lin}(z) : T_zO \to K_z$ and $L' = \mathrm{Lin}'(z') : T_{z'}O' \to K'_{z'}$ (corestricted to the fibres by the first fact above) are simultaneously surjective, simultaneously Fredholm with the same index, and $\ker L$ is in good position to $C_z$ inside $T_zO$ iff $\ker L'$ is in good position to $C'_{z'}$ inside $T_{z'}O'$. The four spaces are complete, being ranges of the idempotents $Dr(z)$, $\rho(z)$ and their primed versions `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`; the rest is the conjugation algebra, with $\ker L' = A(\ker L)$ and the transport of good position.

**Definition 3.6.6 (preprint 3.53) at a zero, from a non-centred chart** `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`. *Statement.* Let $f$ be an sc-smooth section, $y$ a zero with $(f,y)$ an sc-Fredholm germ (Definition 3.1.11 (preprint 3.11)), $c$ a strong bundle chart of $P$ around $y$ (not necessarily centred at $y$), $z$ on level 1 over $\varphi(y)$, and $g$ sc¹ on $U$ agreeing with $\mathbf{f}$ on $O$ near $\varphi(y)$ — the case in view was the chart at $x_0$ with a whole-chart filled version. Assume (a) the range of $Dg(z)|_{T_zO}$ contains $\rho(z)\hat{k}$ for every smooth $\hat{k} \in F_\infty$, and (b) $\ker Dg(z) \cap T_zO$ is in good position to $C_{\varphi(y)}$ inside $T_zO$. Then $(f,y)$ is in good position in the whole-chart sense `Chapter3/GoodPositionGerm · StrongBundle.IsGoodPositionGermWholeChart`, in the chart $c'$ centred at $y$ given by Definition

3.1.11 (preprint 3.11), with its filled version $g'$. *Proof.* The chart change between $c$ at $z$ and $c'$ at 0 gives $A$, $\Gamma$ with $\Gamma\,\mathrm{Lin}(z) = \mathrm{Lin}'(0)A$; the bridges give $\mathrm{Lin}(z) = Dg(z)$ on $T_zO$ and $\mathrm{Lin}'(0) = Dg'(0)$ on $T_0O'$ (filler condition (1)). *Surjectivity of $Tf(y) = Dg'(0)|_{T_0O'}$ onto $K_0'$.* By Proposition 3.6.5 (preprint 3.52) in the centred chart $c'$ `Chapter3/FredholmGermLinearizationG · proposition_3_6_5_local` it is Fredholm, so its range $S$ is closed in the closed subspace $K_0'$, hence closed in $F'$. The map $k \mapsto \Gamma(\rho(z)k)$ is continuous $F \to F'$; by (a) and the conjugation it maps $F_\infty$ into $S$; it maps $F$ onto $K_0'$ (preimages by $\Gamma^{-1}$, which lands in $K_z$ where $\rho(z)$ is the identity); since $F_\infty$ is dense in $F$, every point of $K_0'$ is a limit of points of $S$, hence in $S$. So only a dense subset of $K_z$ in the range is needed (Finding 117 (b)). *The kernel.* (b) is transported by $A$ to good position inside $T_0O'$ with respect to $C_0' = C'$ (the partial quadrant at the corner is $C'$ itself `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero`); and the transported subspace is $\ker Dg'(0)$: $\ker Dg'(0) \subseteq T_0O'$ by condition (3) of the filler $g'$ *at its centre* `Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range`, and for $w \in T_0O'$, $Dg'(0)w = 0 \iff \Gamma(Dg(z)A^{-1}w) = 0 \iff A^{-1}w \in \ker Dg(z)$. This is where the hypothesis is on $\ker Dg(z) \cap T_zO$ and not on $\ker Dg(z)$: at $z \neq 0$ condition (3) is not available and $\ker Dg(z)$ need not lie in $T_zO$ (Finding 117 (a)). Variants: with all of $K_z$ in the range `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_chart`, and with the hypotheses on $\mathrm{Lin}(z)$ itself, taking $g = \mathbf{f} \circ r$, which agrees with $\mathbf{f}$ on $O$ because $r$ is the identity there `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_linAt` `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj`.

*In the germ reading* the assembly is the same, with $g$ a germ with derivative family $Dg$, $z$ on level 1 of its germ, and the bridge for germs at $z$ and at 0; the conclusion is good position at $Dg'(0)$ for the filling germ $g'$ at $y$. The dense form needs the closed range of $Dg'(0)|_{T_0O'}$, which is now a hypothesis at chart level `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart_dense`, supplied at section level by Proposition 3.6.5 (preprint 3.52) for germs in a chart centred at $y$ `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. The forms with all of $K_z$ in the range need no Proposition 3.6.5 (preprint 3.52) at all: $\mathrm{Lin}'(0) = \Gamma\,\mathrm{Lin}(z)A^{-1}$ is then onto $K_0'$ directly `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart` (the whole-chart variant had gone through the dense one).

What remains is to supply (a) and (b) in the chart at $x_0$, at the zeros $z_0 = \varphi(y)$ near 0: (b) with the text's hypotheses (§III.32), (a) first under an added hypothesis (§III.33) and then with the text's hypotheses (§III.43).

## III.32 Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros: Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) on the graph, and the kernel

The text proves property (4) of Theorem 3.6.8 (preprint 3.54) — at every zero $y \in V$, with $\psi\varphi(y) = v + \sigma(v)$, "ker $f'(y)$ is in good position to $C_yX$ and $f'(y)$ is surjective" — in one sentence: it follows from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55). Unpacked, three steps are needed beyond the graph description of §III.23. First, Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) have to be applied at the graph points, Lemma 3.6.9 (preprint 3.55) on the relatively open $V'$ of Theorem 3.5.11 (preprint Corollary 3.47) and for a general presentation of $C'$ (Finding 92). Second, Lemma 3.6.9 (preprint 3.55) is about the tangent space $N_v = \{n + D\sigma(v)n\}$ of the graph, not about a kernel; the identification $N_v = \ker D(h+t)(v+\sigma(v))$ is not commented on (Finding 91 (a), 101). Third, the statements live in the push-forward chart $\psi \circ \varphi$, about $D(h+t)$ on $E'$, while Definition 3.6.6 (preprint 3.53) at $y$ (via §III.31)

asks for statements about $Dg(z_0)$ on $T_{z_0}O$ in the chart at $x_0$ (Findings 102–104). This section proves all of it for the kernel; the surjectivity of $f'(y)$ is §III.33. No hypothesis beyond those of the graph description is used.

**Lemma 3.6.9 (preprint 3.55) for a relatively open domain and a general presentation** `Chapter3/GlobalIFTFull · lemma_3_6_9_relOpen`. *Statement.* Let $C = \{x : \ell_i(x) \geq 0\}$ be a partial quadrant presented by an sc-isomorphism $T$, with the coordinate functionals $\ell_i(x) = (Tx)_i$ `Chapter3/GlobalIFTFull · coordCLMWith`, $N$ finite-dimensional in good position to $C$ with a good complement $Y$, $Y$ **closed**, $V$ relatively open in $N \cap C$ with $0 \in V$, and $\sigma : N \to E_0$ with values in $Y$, $C^1$ on $V$, $\sigma(0) = 0$, $D\sigma(0) = 0$ (within $V$). Then there is $\rho > 0$ such that for $v \in V$ with $|v| < \rho$ and $z = v + \sigma(v)$: $z \in C$; $Y$ is a good complement of $N_v = \{n + D\sigma(v)n\}$ `Chapter3/GoodPositionOpen · graphSpace` with respect to the partial quadrant $C_z$; and $N_v$ is in good position to $C_z$. *Proof.* $D = N \cap C$ is convex with nonempty interior in $N$ (good position), so $D$ and $V = D \cap O$ have unique differentiability. *The extension.* The text's Lemma 3.6.9 (preprint 3.55) (§III.22, `Chapter3/GoodPositionOpen · lemma_3_6_9_core`) wants a map on the whole cone. Take a ball $B(0,r) \subseteq O$ and a smooth bump $b$ equal to 1 on $\bar{B}(0,r/2)$ and 0 outside $B(0,r)$, and $\Phi(v) = b(v)v$. $\Phi$ maps $D$ into $V$: $D$ is closed under nonnegative scaling, and $|b(v)v| < r$. So $\tau = \sigma \circ \Phi$ is $C^1$ on the cone, agrees with $\sigma$ on $B(0,r/2)$, and there its derivatives within the cone are those of $\sigma$ within $V$ (a neighbourhood of the point inside $O$ does not change derivatives within a set); in particular $\tau(0) = 0$, $D\tau(0) = 0$. *Lemma 3.6.9 (preprint 3.55) in coordinates* for $\tau$, with the good-complement estimate of $(N, Y)$ rewritten in the $\ell_i$ and the joint surjectivity of the coordinates ($T^{-1}(k,0)$ has coordinates $k$), gives $\rho_1$, the membership $z \in C$, and the good complement $Y$ of $N_v$ with respect to $C_z$. *The interior*, which Lemma 3.6.9 (preprint 3.55) does not state but good position requires: by the continuity of $D\sigma$ at 0 (a $C^1$ map on a set of unique differentiability) there is $\rho_2$ with $\|D\sigma(v)\| < \varepsilon$, the good-complement constant of $(N, Y)$. $L = D\sigma(v)$ takes values in $Y$ — this is where $Y$ closed is used `Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo` — so $n \mapsto n + Ln$ is injective ($N \cap Y = 0$), a linear homeomorphism $N \cong N_v$ in finite dimensions. It carries the interior of $D$ to an open subset of $N_v$ contained in $C_z$: for $w$ in the interior of $D$, $w \in C$ and $|Lw| \leq \varepsilon|w|$, so $w + Lw \in C$ by the good-complement property, and $C \subseteq C_z$. Take $\rho = \min(\rho_1, r/2, \rho_2)$.

**Corollary 3.5.3 (preprint 3.41) at the graph points** `Chapter3/GlobalIFTFull · StabilityData.exists_surjective_Dfs'_graph`. For the data of Theorem 3.4.2 (preprint 3.38) with $D(h+s)(0)$ surjective and a continuous $\sigma$ on $V \ni 0$, $\sigma(0) = 0$, whose graph points $v + \sigma(v)$ are zeros in the germ: there is $\rho > 0$ such that for $v \in V$, $|v| < \rho$, the zero $v + \sigma(v)$ lies on level 1, at a point $x_1$ of the level-1 germ, and $D(h+s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$. Corollary 3.5.3 (preprint 3.41) `Chapter3/SolutionLinearization · StabilityData.corollary_3_5_3` gives a radius $r$ for the zeros, and continuity of $v \mapsto v + \sigma(v)$ at 0 gives $\rho$.

**Property (4) in the push-forward chart** `Chapter3/GlobalIFTFullG · theorem_3_6_8_nearby`. *Statement.* For a section $f$ of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and $(f, x_0)$ in good position: the data of the graph description (§III.23, `Chapter3/GlobalIFTG · theorem_3_6_8_graph`: chart, $\psi$, $h+t$, $K$, closed $Y$, $V'$, $\sigma$, $V_x$), a presentation $T$ of $C' = \widetilde{C} \oplus W$, and an open $V_{x,1} \subseteq V_x$ containing $x_0$ such that for every zero $y \in V_{x,1}$, with $\psi\varphi(y) = v + \sigma(v)$, $v \in V'$: the point lies on level 1 and $D(h+t)$ there is surjective Fredholm of index $n - N$; $v + \sigma(v) \in C'$; $Y$ is a good complement of $N_v$ with respect to $C'_{v+\sigma(v)}$, and $N_v$ is in good position to it. Only $f$ a section is assumed, not an sc-Fredholm section. *Proof.* Take the radii $\rho_1$ (Corollary 3.5.3 (preprint 3.41) at the graph points) and $\rho_2$ (Lemma 3.6.9 (preprint 3.55), relatively open form), $\rho = \min(\rho_1, \rho_2)$, and the projection $\pi$ onto $K$ along $Y$, continuous because $Y$ is closed and $K$ finite-dimensional. Put $V_{x,1} = V_x \cap (\psi\varphi)^{-1}B(0, \rho/(\|\pi\| + 1))$, open since $\psi \circ \varphi$ is continuous on $V_x$. For a zero $y \in V_{x,1}$, $\psi\varphi(y) = v + \sigma(v)$ by the graph description, $v = \pi(\psi\varphi(y))$ because $\sigma(v) \in Y$, so

$|v| \le \|\pi\|\,|\psi\varphi(y)| < \rho$, and both lemmas apply at $v$.

**Derivative identities at a nearby zero.** Four lemmas move these statements from $E'$ and $N_v$ to $E$ and $\ker Dg(z_0)$.

*Uniqueness at a point of $C_1$* `Chapter3/GlobalIFTNearbyGraph · scDeriv_eq_of_eqOn_nhds_at`: two maps with sc¹-derivatives $D_1$, $D_2$ at $x \in C_1$ that agree on $W_1 \cap C_1$ for an open $W_1 \ni x$ of $E_1$ have $D_1 = D_2$ — uniqueness of the derivative within the quadrant on level 1, then density of $E_1$. The text writes $\Psi_*(g) = h + t$ as an identity; formally they agree only on $U' \cap W'$, and this lemma identifies their derivatives at every nearby point (Finding 104).

*The fibre derivative of a strong bundle map at $(u_0, 0)$* `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_at`: for a strong bundle map $\Psi$ with fibre operators $\Gamma(u) = (\Psi(u,\cdot))_2$ `Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM` and $u_0 \in U_1$, $D(\mathrm{pr}_{F'} \circ \Psi)(u_0,0)(h,k) = \Gamma(u_0)k$; hence the derivative of $u \mapsto \Gamma(u)k(u)$ at a zero $u_0$ of $k$ is $\Gamma(u_0)Dk(u_0)$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_comp_at` — §III.21's identity at 0, moved to $u_0$.

*The chain rule for the push-forward at a zero* `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_pushforward_at`: for a strong bundle isomorphism $\Psi$ covering $\psi$ with inverse $\Psi'$ covering $\psi'$, and a zero $u_0 \in U_1$ of the sc-smooth $g$, $\Psi_*(g) = \kappa \circ \psi'$ with $\kappa(u) = (\Psi(u, g(u)))_2$, so

$$D(\Psi_* g)(\psi(u_0)) = \Gamma(u_0) \circ Dg(u_0) \circ D\psi'(\psi(u_0)).$$

$D\psi'(\psi(u_0))$ and $D\psi(u_0)$ are inverse on level 0 `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_base_inv_base`, and $\Gamma(u_0)$ is injective because $\Psi'$ inverts $\Psi$ on the fibres `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.fiberCLM_injective`; hence $\ker Dg(u_0) = D\psi'(\psi(u_0))(\ker D(\Psi_* g)(\psi(u_0)))$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.ker_scDeriv_eq_map`. *For a germ $g$* with derivative family $Dg$ the formula is an identity of the explicit derivative family of $\Psi_*(g)$ (§III.23), valid at every level-1 zero $u_0$ and for every family `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.pushforwardDeriv_at`, and the kernel formula follows as before `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.ker_deriv_eq_map_germ`; that $g$ is sc¹ at $u_0$ then means that $u_0$ lies on level 1 of the germ of $g$.

*Corresponding partial quadrants* `Chapter3/GlobalIFTNearbyGraph · scDeriv_mem_partialQuadrantAtWith_iff`: for an sc-diffeomorphism $\psi : U \to U'$ between relatively open subsets of partial quadrants with inverse $\psi'$ and $z \in U_1'$, $x \in C_z' \iff D\psi'(z)x \in C_{\psi'(z)}$: the one-sided sign argument `Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant` for $\psi'$ at $z$ gives one direction, for $\psi$ at $\psi'(z)$ the other, and $D\psi(\psi'(z))D\psi'(z) = 1$. At the centre this was "$C' = T\psi(0)C$" (§III.23).

*Tangent vectors of a family of points of a retract* `Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed`: let $\rho$ be sc¹ on $U$ and $\Theta_1 : N \to E_1$ differentiable at $v$ within $V$ (unique differentiability at $v$), with values in $U_1$ and $\rho(\iota\Theta_1(w)) = \iota\Theta_1(w)$ for $w \in V$. Then $D\rho(\Theta_1(v))$ fixes $\iota(D\Theta_1(v)n)$ for every $n$: differentiate $\rho \circ \iota \circ \Theta_1 = \iota \circ \Theta_1$ at $v$. For $\rho$ a retraction onto $O$: the tangent vectors of a $C^1$ family of points of $O$ lie in $T_{z_0}O$.

$N_v = \ker D(h+s)(v+\sigma(v))$ (Finding 101). *The inclusion* `Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_le_ker_Dfs'`. For the solution graph $\sigma$ of Theorem 3.5.11 (preprint Corollary 3.47) on $V$, a $C^1$-lift $\sigma_1 : V_1 \to E_1$ of $\sigma$ on a relatively open $V_1 \subseteq V$ (Theorem 3.5.11 (1) (preprint Corollary 3.47), §III.20), $v \in V_1$ and the level-1 point $x_1$ over $v + \sigma(v)$ in the germ: $N_v \subseteq \ker D(h+s)(x_1)$. *Proof.* $h + s$ is differentiable on level 1 only, so the graph is lifted: the kernel $K$ is a smooth sub-

space Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth, hence lifts linearly to $E_1$ Chapter3/BasicClass · finLift, and $\Gamma_1 = \text{lift} + \sigma_1$ is a $C^1$ map $V_1 \to E_1$ with $\iota\Gamma_1(w) = w + \sigma(w)$ and $\Gamma_1(v) = x_1$. $V_1 = V \cap O_1$ Chapter3/GlobalIFTNearbyGraph · relOpen_cone_inter has unique differentiability at its points Chapter3/GlobalIFTNearbyGraph · uniqueDiffWithinAt_relOpen_cone, and $D\sigma(v) = \iota \circ D\sigma_1(v)$ Chapter3/GlobalIFTNearbyGraph · fderivWithin_eq_toBase_comp. The composite $(h+s)\circ\iota\circ\Gamma_1$ vanishes on $V_1$, and near $v$ (on $V_1 \cap \Gamma_1^{-1}$ of the open set defining the level-1 germ) $\Gamma_1$ maps into the level-1 germ, where $h+s$ has the derivative $D(h+s)(x_1)$ composed with $\iota$ Chapter3/SolutionLinearization · StabilityData.isSc1GermWith_Dfs'. The chain rule and uniqueness give $D(h+s)(x_1)(n + D\sigma(v)n) = 0$ for $n \in K$. *Equality* Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_eq_ker_Dfs' under the conclusions of Corollary 3.5.3 (preprint 3.41) at $x_1$ (Fredholm, index $\dim P - \dim Q$, surjective) and with the closed good complement $Y$: $D\sigma(v)$ takes values in $Y$, so $\dim N_v = \dim K$ Chapter3/GlobalIFTNearbyGraph · finrank_graphSpace_of_compl; for a surjective Fredholm operator $\dim\ker = \text{ind}$ Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective, so $\dim K = \text{ind}\, D(h+s)(0) = \dim P - \dim Q = \text{ind}\, D(h+s)(x_1) = \dim\ker D(h+s)(x_1)$, and an inclusion of finite-dimensional spaces of the same dimension is an equality.

**Good position restricts to a subspace containing the kernel** Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le: if $N$ is in good position to $C$ in $E_0$ and $N \subseteq T$, a good complement $P$ restricts to the good complement $P \cap T$ of $N$ in $T$, with the same constant ($n + p \in T$ and $n \in T$ give $p \in T$).

**The graph description with the filler data exposed** Chapter3/GlobalIFTNearbyGraphG · theorem_3_6_8_chart_graph. This is the body of the graph description (§III.23) stated in the chart, with the filling germ $g$ of $\mathbf{f}$ on $G_g$ and a derivative family $Dg$, the sc⁺-germ $t_0$ on $G_s$, the conjugation $\Psi$ (base $\psi$, inverse $\psi'$), the basic germ on a germ $G$ that $\psi'$ maps into $G_g \cap G_s$ level by level ($G \subseteq G''$, §III.48), and the tameness consequence $(1 - Dr(0))E \subseteq E_0$ as hypotheses, so that the later steps can use $g$ at the points $z_0$. Its conclusions: the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) on $G$ itself and the basic germ $h' = h + (t_{\mathbb{R}^N}, 0)$ with $\mathcal{S}.\text{fsum}\, h' = \Psi_*(g)$ on $U' \cap W'$; $D(h+t)(0)$ surjective with kernel $K$ in good position; $Y$ (closed), $V'$, $\sigma$ with its $C^k$-lifts; an open $N_1 \ni 0$ on which the filler identities (1), (2) and the conjugation identity hold, with $\psi'(U' \cap W') \subseteq N_1$; and an open $N_2 \subseteq N_1$ with $\psi(U \cap N_2)$ inside the germ and $U' \cap W'$, every zero of $g$ in $U \cap N_2$ in $O$, $g(u) = 0 \iff (h+t)(\psi(u)) = 0$ on $O \cap N_2$, and $\psi(\{u \in O \cap N_2 : g(u) = 0\}) = \{v + \sigma(v) : v \in V'\}$. The proof is that of §III.23 (items 1–7), in the chart; $D(h+t)(0) = D(\Psi_* g)(0)$ for the explicit family is part of the conclusion.

**The kernel at the nearby zeros, in the chart at $x_0$** Chapter3/GlobalIFTNearbyKernelG · theorem_3_6_8_chart_kernel. *Statement.* In the data of the preceding theorem there is an open $N_3 \subseteq N_2$ containing $0$ such that for every zero $z_0 \in O \cap N_3$ of $g$ there are $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point $z_{0,1}$ over $z_0$, **on level 1 of the germ $G_g$ of the filling**, and $x_1 = \psi_1(z_{0,1})$ in the level-1 germ over $v + \sigma(v)$, with ($Dg(z_{0,1})$ the value of the derivative family): (i) $D(h+t)(x_1)$ surjective Fredholm of index $n - N$; (a) $N_v = \ker D(h+t)(x_1)$; (b) $D(\Psi_* g)(x_1) = D(h+t)(x_1)$ and $\ker Dg(z_{0,1}) = D\psi'(x_1)(\ker D(h+t)(x_1))$, and $Dg(z_{0,1})$ is surjective; (c) $\ker Dg(z_{0,1}) \subseteq T_{z_0}O = Dr(z_{0,1})E$; (d) $\ker Dg(z_{0,1})$ is in good position to $C_{z_0}$ inside $T_{z_0}O$. *Proof.*

1. *The neighbourhood.* Take the $C^1$-lift $\sigma_1$ on $V_1 = V' \cap O_1$ (Theorem 3.5.11 (1) (preprint Corollary 3.47) with $k = 1$), the radii $\rho_1$ (Corollary 3.5.3 (preprint 3.41) at the graph points), $\rho_2$ (Lemma 3.6.9 (preprint 3.55) relatively open, for the presentation of $C' = \widetilde{C} \oplus W$), $\rho_3$ with $B(0, \rho_3) \subseteq O_1$, and $\rho = \min(\rho_1, \rho_2, \rho_3)$. With the projection $\pi$ onto $K$ along $Y$, let $N_3 = N_2 \cap O_3$ where $O_3 \cap U = \psi^{-1}(B(0, \rho/(\|\pi\| + 1))) \cap U$ (continuity of $\psi$ on $U$). For a zero $z_0 \in O \cap N_3$ of $g$, $\psi(z_0) = v + \sigma(v)$ with $v \in V'$, and as in `theorem_3_6_8_nearby` $|v| < \rho$, so $v \in V_1$ and both lemmas apply.
2. *(i), and the level-1 points.* Corollary 3.5.3 (preprint 3.41) at $v$ gives $x_1$ on level 1 of $\mathcal{S}.G =$

$G \subseteq G''$ (§III.48, `Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_5_3_shrink`); $x_1$ lies in $U_1'$, so $z_{0,1} = \psi_1'(x_1)$ is a level-1 point over $\psi'(\psi(z_0)) = z_0$, and $\psi_1(z_{0,1}) = x_1$ `Chapter2/DegeneracyIndex · levelMap_comp_eq_self`. Since $\psi_1'$ maps $G_1''$ into $(G_g)_1$, $z_{0,1}$ lies in $(G_g)_1$ `Chapter3/BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem`: this is the step that needs the shrinking of the basic germ, since $g$ is differentiable only there. (With $g$ sc-smooth on $U$, any level-1 point would do.)

3. *(a)* is $N_v = \ker D(h+t)(x_1)$ above, fed with the lift and (i).
4. $D(\Psi_* g)(x_1) = D(h+t)(x_1)$: the two maps agree on $U' \cap W'$, so the uniqueness lemma at $x_1$ applies on the open set of $E_1'$ cut out by $W' \cap U_o$ (with $U' = U_o \cap C'$) and the open set defining the level-1 germ $G_1$; $\Psi_*(g)$ is an sc¹-germ on the preimage of $G_g$, which contains $G$ `Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isSc1GermWith` (whole-chart: sc-smooth on $U'$ `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization`).
5. *(b)*: the kernel formula is the chain rule for the push-forward at the zero $z_{0,1}$ of $g$, in the germ form above. *Surjectivity of $Dg(z_{0,1})$*: given $k$, solve $D(h+t)(x_1)w = \Gamma(z_0)k$ by (i); then $\Gamma(z_0)Dg(z_{0,1})D\psi'(x_1)w = \Gamma(z_0)k$, and $\Gamma(z_0)$ is injective. This is the surjectivity of the *filled* $Dg(z_0) : E \to F$, not of $Tf(z_0) = Dg(z_0)|_{T_{z_0}O} : T_{z_0}O \to K_{z_0}$ (§III.33).
6. *(c), the tangency to $O$* (Finding 102). Condition (3) of Definition 3.1.4 (preprint 3.4), which gave $\ker Dg(0) \subseteq T_0 O$ at the centre, is not available at $z_0$. Instead: the transported retraction $r' = \psi \circ r \circ \psi'$ on $U'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport` fixes every graph point, because each $w + \sigma(w)$, $w \in V'$, is $\psi(u)$ for a zero $u \in O$ of $g$ (the graph description), and $r(u) = u$. The lifted graph map $\Gamma_1 = \text{lift} + \sigma_1$ maps $V_1$ into $U_1'$, so the tangent-vector lemma for $r'$ gives $Dr'(x_1)$ fixing $n + D\sigma(v)n$ for every $n \in K$, i.e. $Dr'(x_1)$ is the identity on $N_v$. The derivative of the transported retraction is $Dr'(x_1) = D\psi(z_{0,1})Dr(z_{0,1})D\psi'(x_1)$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport` (§III.27); applying $D\psi'(x_1)$ and using that it inverts $D\psi(z_{0,1})$, $Dr(z_{0,1})$ fixes $D\psi'(x_1)m$ for every $m \in N_v$. By (a) and (b), these are exactly the elements of $\ker Dg(z_{0,1})$, so $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$.
7. *(d)* (Finding 103). Lemma 3.6.9 (preprint 3.55) gives $N_v$ in good position to $C'_{v+\sigma(v)}$ in $E'$. The isomorphism $S = D\psi'(x_1) : E_0' \cong E_0$ (inverse $D\psi(z_{0,1})$, `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.derivInvEquiv`) carries $C'_{v+\sigma(v)}$ onto $C_{z_0}$ by the corresponding-quadrants lemma, so Lemma 3.5.9 (preprint 3.45) `Chapter3/GoodPosition · IsInGoodPosition.map` gives $S(N_v) = \ker Dg(z_{0,1})$ (by (a), (b)) in good position to $C_{z_0}$ in $E$; by (c) and the restriction lemma it is in good position inside $T_{z_0}O$.

**Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros, the kernel, for the section** `Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_nearby_kernel`. *Statement.* For a section $f$ of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and $(f, x_0)$ in good position: the data of the graph description, with the chart $c$ centred at $x_0$, its filling germ $g$ with a derivative family and the sc⁺-germ $t_0$ exposed (and $K$ smooth, $Y$ closed, the lifts), and an open $V_{x,2} \subseteq V_x$ containing $x_0$, such that for every zero $y \in V_{x,2}$: $\mathbf{f} = g$ on $O$ near $\varphi(y)$, $g(\varphi(y)) = 0$, and with $z_0 = \varphi(y)$ the conclusions (i), (a)–(d) of the chart statement hold, with $z_{0,1}$ on level 1 of the germ of $g$. *Proof.* Tameness gives $(1 - Dr(0))E \subseteq E_0$ (Proposition 2.5.3 (preprint 2.35), `Chapter2/TameRetracts · proposition_2_5_3`, at the smooth corner point 0 of the tame base retraction; §III.23). The conjugation data are first shrunk to $G''$ (§III.48), so that the chart statements apply; they give $N_1$, $N_2$, $N_3$, with $\psi(U \cap N_2) \subseteq G_0''$; put $V_x = \varphi^{-1}(N_2)$ and $V_{x,2} = \varphi^{-1}(N_3)$ within the chart domain, both open by continuity of $\varphi$. On $V_x$, $\mathbf{f}(\varphi(y)) = g(\varphi(y))$ (filler (1) on $N_1$), so $f(y) = 0 \iff g(\varphi(y)) = 0 \iff (h+t)(\psi\varphi(y)) = 0$ `Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff`; $\psi \circ \varphi$ is injective on $V_x$; and the image of the zero

set is the graph. For a zero $y \in V_{x,2}$, $\varphi(y) \in O \cap N_3$ is a zero of $g$, and the kernel statement applies. The hypotheses are those of the graph description: **the kernel part of Theorem 3.6.8 (1) (preprint 3.54) needs no added hypothesis**, and of $f$ only that it is a section (the regularity enters through the filling germ $g$, at 0 and at the lifts $z_{0,1}$ of the nearby zeros, which lie in its germ).

## III.33 Theorem 3.6.8 (1) (preprint 3.54) under the filler condition at the nearby zeros, and the $\lambda$-extension

At a zero $y$ near $x_0$, with $z_0 = \varphi(y)$ in the chart at $x_0$, §III.32 gives the kernel condition (b) of the assembly step of §III.31 and the surjectivity of the *filled* $Dg(z_0) : E \to F$. Condition (a) asks for more: that $Tf(z_0) = Dg(z_0)|_{T_{z_0}O}$ maps onto $K_{z_0} = \rho(z_0)F$ (or onto a dense part of it). The text obtains it from Corollary 3.5.3 (preprint 3.41). This section shows exactly what is missing, proves Theorem 3.6.8 (1) (preprint 3.54) under the **added hypothesis** that closes the gap, and records the formalized part of an attempt to close it without that hypothesis (Findings 100, 105–107).

**The block form at a zero.** With respect to $E = T_{z_0}O \oplus \ker Dr(z_0)$ and $F = K_{z_0} \oplus \ker \rho(z_0)$ ($Dr(z_0)$ and $\rho(z_0)$ are idempotent),

$$Dg(z_0) = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}, \qquad A = Tf(z_0), \quad C = (1-\rho(z_0))Dg(z_0)|_{\ker Dr(z_0)} : \ker Dr(z_0) \to \ker \rho(z_0).$$

The lower left block vanishes at a zero: $Dg(z_0)Dr(z_0) = D(\mathbf{f} \circ r)(z_0)$ (the bridge, §III.31) takes values in $K_{z_0}$ (the values of the linearization at a zero, §III.31). If $Dg(z_0)$ is onto, $C$ is onto, and (Finding 100) $b \mapsto [Bb]$ induces an isomorphism $\ker C \cong \operatorname{coker} A$: so $Tf(z_0)$ **is onto if and only if** $C$ **is injective** — condition (3) of Definition 3.1.4 (preprint 3.4) at $z_0$. Definition 3.1.4 (preprint 3.4) imposes it at 0 only, and it does not propagate: $u \mapsto Dg(u)$ is continuous for the strong operator topology, not in norm, and a strongly continuous family of surjections may pass from an isomorphism to a non-injective operator. In index terms: Corollary 3.5.3 (preprint 3.41) gives $\operatorname{ind} Dg(z_0) = \dim K$ at all nearby zeros, $\operatorname{ind} Dg(z_0) = \operatorname{ind} A + \operatorname{ind} C = \dim\ker A$ when $C$ is onto, and $\operatorname{ind}(f,y) = \operatorname{ind} A = \dim K - \dim\operatorname{coker} A$; so $\operatorname{coker} Tf(z_0) = 0$ is the local constancy $\operatorname{ind}(f,y) = \operatorname{ind}(f,x_0)$ along the solution set — the index stability of Proposition 6.2.2 (preprint 6.2) of the text, whose proof uses the same step ("$F'$ surjective" for the intrinsic linearization, from the implicit function theorem for the filled one). The text's "(4) follows from Corollary 3.41 and Lemma 3.55" therefore has a gap for the surjectivity of $f'(y)$; the same sentence stands in the book's arXiv version (Theorem 3.13 there, with Definition 3.13, the filling, still imposing (3) at 0 only; Finding 157). Formalized of this paragraph is the direction used:

**Proposition 3.1.5 (1) (preprint 3.5) at a zero of a filler** `Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective`. *Statement.* Let $U$ be relatively open in a partial quadrant, $R$ a strong bundle retraction on $U \lhd F$, $\mathbf{f}$ with $\mathbf{f} \circ r$ sc¹ and $(x, \mathbf{f}(x)) \in K$ on $O$, $g$ sc¹, $z$ on level 1 with $\iota z \in O$, $\mathbf{f}(\iota z) = 0$, $\mathbf{f} = g$ on $O$ near $\iota z$. If $Dg(z)$ is surjective and $(1-\rho(z))Dg(z)$ is injective on $\ker Dr(z)$ (in the form: $Dr(z)b = 0$ and $\rho(z)Dg(z)b = Dg(z)b$ imply $b = 0$), then for every $k \in F$, $\rho(z)k = Dg(z)x$ for some $x \in T_zO$. For a germ $g$, with $Dg(z)$ the value of a derivative family at a level-1 point of its germ, the proof is the same with the bridge for germs (§III.31) `Chapter3/GlobalIFTNearbySurjG · exists_range_deriv_eq_ρ_of_normal_injective_germ`. *Proof.* Write $Dg(z)e = \rho(z)k$, $a = Dr(z)e$, $b = e - a \in \ker Dr(z)$ (idempotence). $Dg(z)a = D(\mathbf{f}\circ r)(z)e$ lies in $K_z$ (bridge and values at a zero), so $\rho(z)Dg(z)a = Dg(z)a$; then $\rho(z)Dg(z)b = \rho(z)(\rho(z)k - Dg(z)a) = \rho(z)k - Dg(z)a = Dg(z)b$ ($\rho(z)$ is linear and idempotent

on the fibre). So $b = 0$, $e = a \in T_z O$. The converse direction and the index identity above are not formal statements; they explain why the hypothesis is the right one.

**The added hypothesis** `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`. $(f, x_0)$ is in good position *with a filler satisfying condition (3) at the nearby zeros* if $x_0$ is smooth and there are the data of Definition 3.6.6 (preprint 3.53) in the chart form (a strong bundle chart centred at $x_0$, tame since §III.47, a filling germ $g$ of $\mathbf{f}$ with a derivative family $Dg$ on a germ $G$ inside $U$, a local sc$^+$-germ $s$ with $s(0) = g(0)$ and $g - s$ conjugated to a basic germ, the germ in good position at $Dg(0)$) together with an open $N \ni 0$ such that at every level-1 point $z_1$ *of* $G$ with $\iota z_1 \in O \cap N$ and $g(\iota z_1) = 0$, $(1 - \rho(\iota z_1))Dg(z_1)$ is injective on $\ker Dr(z_1)$. The condition is imposed only where $Dg$ is defined, which is where the lifts of the nearby zeros lie (§III.32); the whole-chart form imposed it at every level-1 point over $O \cap N$ and implies it `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilledWholeChart.isGoodPositionGermFilled`. It implies Definition 3.6.6 (preprint 3.53) `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.isGoodPositionGerm` (forget $N$). According to the docstring, the fillers of the applications satisfy (3) at every point; Definition 3.1.4 (preprint 3.4) as stated does not ask for it. It is recorded as an added hypothesis in the docstrings and in the Findings (items 100, 153).

**Theorem 3.6.8 (1) (preprint 3.54)** `Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one`. *Statement.* Let $f$ be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$, and $(f, x_0)$ in good position with a filler satisfying condition (3) at the nearby zeros. Then there is an open $V_x \ni x_0$ such that $(f, y)$ is in good position (Definition 3.6.6 (preprint 3.53), `IsGoodPositionGerm`) at every zero $y \in V_x$. *Proof.*

1. In the chart $c$ at $x_0$ with the filling germ $g$: $(1 - Dr(0))E \subseteq E_0$ from tameness (Proposition 2.5.3 (preprint 2.35)); the germ $G$ of the condition is intersected with the germ of $g$, and the basic germ is shrunk to the points that $\psi'$ maps into this intersection and into the germ of $s$ (§III.48); then the graph description in the chart and the kernel statement of §III.32 give $N_1$, $N_2$, $N_3$. Put $V_x = \varphi^{-1}(N_3 \cap N)$ in the chart domain, with $N$ the neighbourhood of the added hypothesis.
2. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O \cap N_3$, $\mathbf{f}(z_0) = 0$, and $g(z_0) = \mathbf{f}(z_0) = 0$ by filler (1) on $N_1$; $\mathbf{f} = g$ on $O$ near $z_0$. The kernel statement gives the level-1 point $z_{0,1}$, on level 1 of $G$, $Dg(z_{0,1})$ surjective, $\ker Dg(z_{0,1}) \subseteq T_{z_0} O$, and $\ker Dg(z_{0,1})$ in good position to $C_{z_0}$ inside $T_{z_0} O$.
3. $y$ is a smooth point, because $f$ is regularizing `Chapter3/TameGermsG · IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so $(f, y)$ is an sc-Fredholm germ (Definition 3.1.16 (preprint 3.14)); and $\mathbf{f} \circ r$ is sc$^1$ for an sc-smooth section `Chapter3/GlobalIFTNearbySurj · StrongBundle.IsScSmoothSection.isSc1_principal_comp_r`.
4. Apply the assembly step of §III.31 in the chart $c$ at $z_{0,1}$ with $g$, in its germ form, the closed range in the tame chart at $y$ coming from Proposition 3.6.5 (preprint 3.52) for germs `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. *Condition (a)*: the added hypothesis at $z_{0,1}$ (with $\iota z_{0,1} = z_0 \in O \cap N$, $g(z_0) = 0$) is the injectivity of $C$, and Proposition 3.1.5 (1) (preprint 3.5) at $z_0$ turns the surjectivity of $Dg(z_{0,1})$ into $\rho(z_0)k \in Dg(z_{0,1})(T_{z_0} O)$ for every $k$, smooth or not. *Condition (b)*: $\ker Dg(z_{0,1}) \cap T_{z_0} O = \ker Dg(z_{0,1})$ by the tangency (c), and the good position is (d).

So Theorem 3.6.8 (1) (preprint 3.54) holds under condition (3) of the filler at the nearby zeros; the kernel half holds with the text's hypotheses (§III.32). When this section was written, **the surjectivity of $f'(y)$ at the nearby zeros with Definition 3.1.4 (preprint 3.4) as stated was open in the formalization**; it is proved in §§III.41–III.43 by another argument, and the condition (3) at the nearby zeros turns out to be a consequence of good position (§III.45). The global theorems are not affected: they assume good position at every zero, which supplies at each $x_0$ the nearby good position that the regularity argument needs

(§III.36–§III.37, Finding 156).

**The λ-extension: what the perturbation argument of Proposition 6.2.2 (preprint 6.2) gives.** The text's device for index stability (Proposition 6.2.2 (preprint 6.2)) is to add finitely many parameters $\lambda$ multiplying sc⁺-sections, so that the extended problem is surjective, and to apply the implicit function theorem to it. In the chart at $x_0$ it can be set up without new bundle isomorphisms: for a finite-dimensional $\Lambda_0 \subseteq F_\infty$, the sc⁺-germ $\tilde{e}(\lambda, u) = \rho(r(u))(\lambda_1(\pi_K \psi(u)) + \hat{\jmath}(\lambda_2))$ (Finding 105). Two points are not addressed by the text, and the first is formalized.

*The obstruction at the corner* `Chapter3/GlobalIFTNearbyPerturb · mem_lineality_of_goodComplement_prod`. Theorem 3.5.11 (preprint Corollary 3.47) for the extended problem needs its kernel in good position to the extended partial quadrant at the corner. Let $C \subseteq V$ contain 0 and be closed under nonnegative scaling, $N \subseteq V$ a subspace, and $P \subseteq \mathbb{R} \times V$ a subspace with the $\varepsilon$-condition of Definition 3.1.24 (preprint 3.21) for $\mathbb{R} \times N$ `Chapter3/GlobalIFTNearbyPerturb · prodSub` with respect to $\mathbb{R} \times C$ `Chapter3/GlobalIFTNearbyPerturb · prodCone`. Then every $p = (\mu, y) \in P$ has $y \in C$ and $-y \in C$. *Proof.* $n = (1, 0)$ lies in $\mathbb{R} \times N$ and in $\mathbb{R} \times C$, $|n| = 1$; for $t = \pm\varepsilon/|p|$, $|tp| \le \varepsilon|n|$, so $n + tp \in \mathbb{R} \times C$, i.e. $ty \in C$, and scaling by $1/|t|$ gives $\pm y \in C$. Hence a good complement projects into the lineality space $C \cap (-C)$, and $\mathbb{R} \times N$ is in good position only if every $v \in V$ is $n + y$ with $n \in N$, $y \in C \cap (-C)$ `Chapter3/GlobalIFTNearbyPerturb · sup_lineality_eq_top_of_goodComplement_prod`. For the naive extension (parameters multiplying smooth vectors with a factor vanishing at 0) the kernel is $\mathbb{R} \times K$, so good position fails unless $K$ maps onto all boundary coordinates — the text's "has a complement contained in $T^R$" is exactly the statement that the complement lies in the lineality space.

*The fix: a tilted kernel* `Chapter3/GlobalIFTNearbyPerturb · isInGoodPosition_extendedKernel`. In $(\Lambda \times P_n) \oplus W$ with the partial quadrant $(\Lambda \times \widetilde{C}) \oplus W$ (the parameters $\lambda$ unconstrained), let $K$, $Y'$ be subspaces of $P_n \oplus W$ and $j : \Lambda \to P_n \oplus W$ linear with $K \cap j(\Lambda) = 0$, $(K + j(\Lambda)) \oplus Y' = P_n \oplus W$, $Y'$ in the lineality space of $\widetilde{C} \oplus W$ ($v + y \in \widetilde{C} \oplus W \iff v \in \widetilde{C} \oplus W$ for $y \in Y'$), and $K + j(\Lambda)$ containing an interior point of $\widetilde{C} \oplus W$. Then the kernel $\widetilde{N} = \{((\lambda, a), w) : (a, w) + j(\lambda) \in K\}$ of $((\lambda, a), w) \mapsto D(h + t)(0)((a, w) + j(\lambda))$ `Chapter3/GlobalIFTNearbyPerturb · extendedKernel Chapter3/GlobalIFTNearbyPerturb · extProj Chapter3/GlobalIFTNearbyPerturb · extParam` is in good position. *Proof.* Choose a complement $\Lambda_c$ of $\ker j$; every element of $j(\Lambda)$ is $j(\mu)$ with $\mu \in \Lambda_c$. *Interior*: if $k_0 + j(\mu_0)$ is interior, $n_0 = ((-\mu_0, k_0 + j(\mu_0)))$ lies in $\widetilde{N}$ and its $(P_n \oplus W)$-part is interior, an open condition. *The complement* $\Lambda_c \times Y'$ `Chapter3/GlobalIFTNearbyPerturb · extendedCompl`: an element of both has $(a, w) \in (K + j(\Lambda)) \cap Y' = 0$, then $j(\lambda) \in K \cap j(\Lambda) = 0$, $\lambda \in \ker j \cap \Lambda_c = 0$; and every $x = ((\lambda, a), w)$ splits by decomposing $(a, w) + j(\lambda) = k + j(\mu) + y$. *The constant*: since $Y'$ lies in the lineality space, adding an element of $\Lambda_c \times Y'$ never changes membership in the quadrant, so the $\varepsilon$-condition holds for every $\varepsilon$ (the proof takes $\varepsilon = 1$). The directions $j(\Lambda) \subseteq T_0(\psi(O)) \cap E'_\infty$ with $K + j(\Lambda) + E'_0 = E'$ exist by tameness ($T_0 O + E_0 = E$) and would be realized by the sc⁺-perturbations $\rho(r(u)) Dg(0) D\psi^{-1}(0) j(\lambda)$.

*Why the argument does not close the gap* (Findings 105–106). Carried through (not formalized: the extended basic germ and its sc⁺-perturbation, its derivative at the zeros, Theorem 3.5.11 (preprint Corollary 3.47) for it, and the tangency argument), the argument gives for each finite-dimensional $\Lambda_0 \subseteq F_\infty$ a neighbourhood $V(\Lambda_0)$ of $x_0$ with $\rho(z_0)\Lambda_0 \subseteq \operatorname{range} Tf(z_0)$ at every zero in $V(\Lambda_0)$. The assembly step at a fixed zero $y$ needs $\rho(z_0)\hat{k}$ in the range for *all* smooth $\hat{k}$, and $V(\Lambda_0)$ comes from Theorem 3.5.11 (preprint Corollary 3.47) for the $\Lambda_0$-extended problem with no control in $\Lambda_0$: the radii of Theorems 3.5.1, 3.5.6, 3.5.11 (preprint 3.40, 3.43, 3.47) would have to be uniform in the finite-dimensional parameter space, which the statements do not provide; $\operatorname{coker} Tf(z_0)$ is finite-dimensional and spanned by smooth vectors, but the spanning set depends on $y$. The formalization of this route was stopped there; §§III.41–III.43 take another one.

## III.34 Theorem 3.6.8 (2) (preprint 3.54) at the centre: the data of Definition 3.2.1 (preprint 3.25) for the solution set at $x_0$

The text ends the proof of Theorem 3.6.8 (preprint 3.54) with $\psi\circ\varphi(\{y \in V : f(y) = 0\}) = \{v+\sigma(v) : v \in V'\}$ and concludes that the solution set has the $d$-dimensional tangent germ property, $d = \dim\ker f'(x)$, so that Theorem 3.2.3 (preprint 3.27) applies. Definition 3.2.1 (preprint 3.25) asks for much more than the graph identity (§III.24): a chart of $X$ centred at the point onto a *tame* retract, a *smooth* finite-dimensional subspace $N$ in good position with a *closed* good complement $Y$, a relatively open $V \ni 0$ in $N \cap C$, and $\delta : V \to Y$ continuous with the graph identity, $\delta(0) = 0$, $D\delta(0) = 0$ within $N \cap C$, and the sc$^\infty$-germ property in the $k$-dependent reading `Chapter3/TangentGerm · MPolyfold.TangentGermChart Chapter3/TangentGerm · IsScInftyGermAt`. This section assembles all of it at the centre $x_0$ from the graph description of §III.29, adding the three facts the text leaves silent: the chart is $\psi\circ\varphi$ and it is a chart of $X$; its model $\psi(O)$ is a tame retract; and $K = \ker D(h+t)(0)$ consists of smooth points. With the projecting complement of §III.29 the data also satisfy condition (b) of the regularity conditions. The data at the other zeros and condition (a) come in §§III.36–III.37.

**The sc$^\infty$-germ property from the level-wise lifts.** Theorem 3.5.11 (1) (preprint Corollary 3.47) gives, for every $k \geq 1$, a relatively open $V_k \ni 0$ in the cone and a $C^k$ map $\sigma_k : V_k \to E_k$ with $\iota_k\sigma_k = \sigma$; Definition 3.2.1 (ii) (preprint 3.25) asks, for every level $m$ and every $k$, for a relatively open $V_{m,k} \ni 0$ on which $\sigma$ takes values in $E_m$ and is $C^k$ into $E_m$. Given $m$ and $k$, take $j = \max(m,k)+1$, $V_{m,k} = V_j$, and compose $\sigma_j$ with the inclusion $E_j \to E_m$, a continuous linear map compatible with the inclusions into $E_0$ `Chapter3/SolutionGermRegularity · exists_clm_toBase_eq` (an iterated composition of the level inclusions): it is a lift of $\sigma$ into $E_m$, of class $C^j$, hence $C^k$, and it coincides with the level map of $\sigma$ on $V_j$ by the injectivity of the inclusions `Chapter3/SolutionGermRegularity · isScInftyGermAt_of_lifts`. So the level-wise reading of Theorem 3.5.11 (1) (preprint Corollary 3.47) (§III.20) is exactly what Definition 3.2.1 (ii) (preprint 3.25) needs (item 94 (a)).

**The kernel is smooth** `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth`. For a basic germ $f$ with contraction map $B$ and the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) (sc$^+$-germ $s$ into $W^1$), every $x = (a,w) \in \ker D(f+\iota s)(0)$ is a smooth point of $\mathbb{R}^n \oplus W$. The text takes the kernel of an sc-Fredholm operator to be smooth without comment. *The kernel equation.* $D(f+\iota s)(0) = Df(0) + \iota Ds(0)$ `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq`, and the $W$-component of $Df(0)x$ is $w - DB(0)x$ `Chapter3/SolutionLinearization · StabilityData.snd_Df_eq`. Lemma 3.1.9 (preprint 3.9) at $(0,0)$ gives $DB(0)(0,w) = 0$ `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`, so $DB(0)x = DB(0)(a,0)$, and the $W$-component of the kernel equation reads

$$w = DB(0)(a,0) - \iota Ds(0)x.$$

*The bootstrap.* $DB(0)$ is an sc-operator (the derivative of an sc$^1$-germ at the smooth point 0) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`, and so is $Ds(0) : E \to W^1$, which raises the level by one. By induction on $m$, $w \in W_m$: if $w = \iota y$ with $y \in W_m$, then $(a,0)$ lies on level $m+1$ (the parameter space has all levels equal), so $DB(0)(a,0) \in W_{m+1}$, and $(a,y) \in E_m$ gives $Ds(0)x \in W_{m+1}$; hence $w \in W_{m+1}$. So $x \in E_\infty$.

**The chart $\psi\circ\varphi$.** An sc-smooth map $g$ on a relatively open $U$ is an sc-smooth map between retracts $(O,C,E) \to (O',C',E')$ for every retract $O \subseteq U$ mapped into $O'$: for every sc-smooth retraction $R$ onto $O$, $g \circ R$ is sc-smooth by the chain rule `Chapter3/GlobalIFTCenter · IsScSmooth.isScSmoothRetractMap_of_subset`. For a chart $\varphi : V \to O$ of $X$ and an sc-diffeomorphism $\psi : U \to U'$ between relatively open

subsets of partial quadrants with inverse $\psi'$ (§III.27) `Chapter2/TameInvariance · IsScDiffeoQuadrant` and $O \subseteq U$, given that $\psi(O)$ is a retract of $C'$, the map $\psi \circ \varphi$ is a chart onto the model $(\psi(O), C', E')$, with inverse $\varphi^{-1} \circ \psi'$ and the source of $\varphi$ `Chapter3/GlobalIFTCenter · MChart.compDiffeo` (continuity of both maps: $\psi$, $\psi'$ are $\mathrm{sc}^0$, hence continuous on level 0). *It is a chart of the M-polyfold structure* `Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart`: for a chart $\varphi'$ of the atlas, the overlap of $\psi\varphi$ with $\varphi'$ is $\psi$ of the overlap of $\varphi$ with $\varphi'$, and the two transition maps are $(\varphi' \circ \varphi^{-1}) \circ \psi'$ and $\psi \circ (\varphi \circ \varphi'^{-1})$; both are compositions of sc-smooth maps between retracts — the transitions of $\varphi$ and $\varphi'$, compatible because both are charts of $X$, and $\psi'$ resp. $\psi$ restricted to the overlap retracts by the lemma above.

**The tame model.** For the base retraction $r$ of the centred strong bundle chart, tame by the hypothesis on the charts, and the sc-diffeomorphism $\psi$ of the conjugation, $(\psi(O), C', E')$ is a tame retract with the retraction $\psi \circ r \circ \psi'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image` (§III.27); the text takes this for granted when it calls $\psi \circ \varphi$ a chart of the tame M-polyfold (item 95).

**Condition (b) from the choice of complement.** For data of Definition 3.2.1 (preprint 3.25) whose complement $Y$ has the projection property for $C$, $\pi(O) \subseteq C$: the retract lies in $C$, and $\pi$ maps $C$ into $C$ `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.π_mem_of_projectsInto`. So the regularity conditions reduce to condition (a), the sc-smoothness of $\delta$ on $V$ in the set sense `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.isRegular_of_projectsInto`.

**Theorem 3.6.8 (2) (preprint 3.54) at the centre** `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart_proj`. *Let $f$ be a section of a strong bundle whose charts compatible with the atlas have tame base retractions, $f(x_0) = 0$ and $(f, x_0)$ in good position. Then there are an open $V_x \ni x_0$ and a dimension $d$ such that $S_{V_x} = \{y \in V_x : f(y) = 0\}$ has at $x_0$ the data of Definition 3.2.1 (preprint 3.25) of dimension $d$, with $\pi(O) \subseteq C$ for their projection and retract.* The proof fills the fields of the data one by one from the graph description, closed and projecting (§III.29) `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj`:

1. *The chart.* $\psi$ and $\psi'$ form an sc-diffeomorphism $U \to U'$ (item (i) of §III.29); $O = r(U) \subseteq U$ since the retraction maps $U$ into $U$ `Chapter3/Filler · StrongRetr.r_mem`; $\psi(O)$ is a tame retract (above, with the retract of the base retraction identified with $O = r(U)$ `Chapter3/StrongBundleSections · StrongBundleChart.r_image`); the chart is $\chi = \psi \circ \varphi$, a chart of $X$, containing $x_0$, with $\chi(x_0) = \psi(0) = 0$.
2. *The subspace.* $N = K = \ker D(h + t)(0)$, finite-dimensional because $D(h + t)(0)$ is Fredholm `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`, smooth (above), in good position to $C'$ (the hypothesis of Theorem 3.5.11 (preprint Corollary 3.47) established in §III.23); $d := \dim K$.
3. *The complement and the domain.* $Y$ closed, a good complement; $V = V'$, relatively open in $K \cap C'$, containing 0; $\delta = \sigma$, continuous (it is $C^1$), with values in $Y$.
4. *The graph identity.* $\chi(S_{V_x} \cap \text{source}) = \psi\varphi(S_{V_x})$, since $V_x$ lies in the source of $\varphi$; this is the graph identity of Theorem 3.6.8 (3) (preprint 3.54).
5. *Regularity at* 0. The $\mathrm{sc}^\infty$-germ property from the lifts (above); $\sigma(0) = 0$; and $D\sigma(0) = 0$ within the whole cone $K \cap C'$, from $D\sigma(0) = 0$ within $V' = O \cap (K \cap C')$ with $O$ open, since derivatives within a set do not change when the set is intersected with a neighbourhood of the point.
6. *Condition (b).* From the projection property of $Y$, by the lemma above.

Without the last item this is the plain form `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart`. *What the statement gives about $d$.* The dimension is existentially quantified in the statement; the proof takes $d = \dim\ker D(h + t)(0)$, the dimension of the kernel of the linearization of the push-forward, and the statement does not identify $d$ with $\dim\ker f'(x_0)$ (the docstring's "$n = \dim\ker f'(x_0)$" is not part of the statement). The local constancy of §III.37 makes $d$ intrinsic in another way: it is the dimension of

every datum of Definition 3.2.1 (preprint 3.25) for the solution set at points of the chart domain.

*What is missing at this stage, and where it is supplied.* Condition (a), the sc-smoothness of $\sigma$ on $V'$ in the set sense, is not provided by Theorem 3.5.11 (1) (preprint Corollary 3.47), which gives the sc$^\infty$-germ property at 0 only (item 82); it is obtained in §III.36 by re-centring the data at the nearby zeros, which needs the data of this section at every zero near $x_0$, hence good position there (Theorem 3.6.8 (1) (preprint 3.54), §§III.32–III.33). The solution set here is the one in $V_x$; §III.37 restricts the chart so that it describes the global solution set.

## III.35 The re-centring lemma

The text's proof of Theorem 3.2.3 (preprint 3.27) begins (preprint p. 84) by showing that the map $\delta$ of Definition 3.2.1 (preprint 3.25) is an sc$^\infty$-germ near *every* point $v_0 \in V$, not only near 0: take the point $x_0 = \varphi^{-1}(v_0 + \delta(v_0))$ of $M$, the data $(\psi, N', Y', V', \tau)$ of Definition 3.2.1 (preprint 3.25) at $x_0$, and consider $w \mapsto \pi \circ \varphi \circ \psi^{-1}(w + \tau(w))$; its linearization at 0 is claimed to be an isomorphism $N' \to N$, so it is a local diffeomorphism near 0, and $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) = (1-\pi)\varphi\psi^{-1}(1+\tau)$ shows that $\delta$ is regular near $v_0$. For an arbitrary set with the tangent germ property the claim does not hold as we read it (§III.56 has the Lean counterexample): $\delta$ need not even be differentiable at $v_0$ (the curve with a vertical tangent of §III.24), and that is why condition (a) of `IsRegular` had to be added to Theorem 3.2.3 (preprint 3.27). What the argument really needs is that $\delta$ be $C^1$ at $v_0$. The formal *re-centring lemma* is the text's argument under that hypothesis, for two graph descriptions of the same set: one in a chart $\chi_1$ by a map $\sigma_1$ of class $C^1$ on all of its domain, one in a chart $\chi_2$ centred at the point over $v_0$ by a map $\sigma_2$ with the level-wise $C^k$-lifts of Theorem 3.5.11 (1) (preprint Corollary 3.47) at 0. Its conclusion is that $\sigma_1$ is $C^k$ into every level $E_{1,m}$ near $v_0$, and that the two subspaces have the same dimension. Section III.36 applies it to the shrunk data of Definition 3.2.1 (preprint 3.25); for the solution set of an sc-Fredholm section, §III.37 supplies the second description at every nearby zero. Four ingredients are proved first: the inverse function machinery of Proposition 3.8.17 (preprint 3.73) with the hypotheses its proofs use, a local inverse theorem for $C^k$ homeomorphisms between relatively open subsets of convex sets, Sard's lemma in fixed dimension, and two facts about relatively open subsets of convex sets.

**The inverse function machinery with the hypotheses its proofs use** `Chapter3/SolutionGermRecentre · InvDataW`. On a finite-dimensional space $V$, let $U \subseteq V$ be convex with unique differentiability at each of its points, $f : V \to V$ of class $C^j$ on $U$ with $j \geq 1$, $\|Df(x) - 1\| \leq \frac{1}{2}$ for $x \in U$ (derivatives within $U$), and let the image $f(U)$ have unique differentiability at each of its points. These are the data of Proposition 3.8.17 (preprint 3.73) as formalized in §III.16 (`InvData`, `Chapter3/QuadrantInverse · InvData`) with the partial quadrant removed: `InvData` asks for a partial quadrant $C$, $U$ relatively open in $C$ and convex, $f(U) \subseteq C$, and that $x$ be an interior point of $C$ iff $f(x)$ is; from the last hypothesis it derives that $f(U)$ is relatively open in $C$ `Chapter3/QuadrantInverse · InvData.image_relOpen`, and the only use of that openness in the regularity proof is the unique differentiability of $f(U)$. The weak data keep convexity and the derivative bound and replace the quadrant hypotheses by unique differentiability of $U$ and of $f(U)$. The proofs are those of §III.16, retold here because they now stand on their own.

*The bi-Lipschitz estimate* (Lemma 3.8.18 (preprint 3.74), (70)). $f - \mathrm{id}$ has derivative $Df - 1$ of norm $\leq \frac{1}{2}$ on the convex $U$, so by the mean value inequality it is a $\frac{1}{2}$-contraction there: $|(f(x') - x') - (f(x) - x)| \leq \frac{1}{2}|x' - x|$ `Chapter3/SolutionGermRecentre · InvDataW.norm_sub_sub_le`. Writing $x' - x = (f(x') - f(x)) - ((f(x') - x') - (f(x) - x))$ gives $\frac{1}{2}|x' - x| \leq |f(x') - f(x)|$ `Chapter3/SolutionGermRecentre · InvDataW.`

half_norm_le, hence $f$ is injective on $U$ Chapter3/SolutionGermRecentre · InvDataW.injOn. *The derivative is invertible*: $Df(x) = 1 - (1 - Df(x))$ with $\|1 - Df(x)\| \le \frac{1}{2} < 1$ is a unit by the Neumann series Chapter3/SolutionGermRecentre · InvDataW.derivUnit Chapter3/SolutionGermRecentre · InvDataW.isUnit_deriv. *The inverse* $g$ on $f(U)$ is the inverse of $f$ on $U$ (a choice of preimage, unique by injectivity) Chapter3/SolutionGermRecentre · InvDataW.inv Chapter3/SolutionGermRecentre · InvDataW.inv_apply; the estimate gives $|g(y') - g(y)| \le 2|y' - y|$ Chapter3/SolutionGermRecentre · InvDataW.norm_inv_sub_le, so $g$ is Lipschitz, hence continuous on $f(U)$ Chapter3/SolutionGermRecentre · InvDataW.continuousOn_inv.

*Lemma 3.8.21 (preprint 3.77)* Chapter3/SolutionGermRecentre · InvDataW.hasFDerivWithinAt_inv: $g$ is differentiable within $f(U)$ at every $y \in f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$. With $x = g(y)$: as $y' \to y$ in $f(U)$, $g(y') \to x$ in $U$ (continuity), so $f(g(y')) - f(x) - Df(x)(g(y') - x) = o(|g(y') - x|)$, and $|g(y') - x| \le 2|y' - y|$ turns this into $o(|y' - y|)$; applying the bounded operator $Df(x)^{-1}$ and using $f(g(y')) = y'$ gives $g(y') - g(y) - Df(x)^{-1}(y' - y) = o(|y' - y|)$. No hypothesis on $f(U)$ enters here.

*Proposition 3.8.17 (preprint 3.73), the regularity of the inverse* Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv: $g$ is $C^j$ on $f(U)$. First, $x \mapsto [Df(x)]^{-1}$ is $C^{j-1}$ on $U$: the derivative within $U$ of a $C^j$ map is $C^{j-1}$ there because $U$ has unique differentiability, and inversion is smooth near the units Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inverse_deriv. Then by induction on $i \le j$: $g$ is $C^0$ (continuous); if $g$ is $C^i$ with $i < j$, then $g$ has the derivative $y \mapsto [Df(g(y))]^{-1}$ at every point of $f(U)$, which is $C^i$ as the composite of the $C^{j-1}$ map $x \mapsto [Df(x)]^{-1}$ with $g$, and a map with a $C^i$ derivative on a set of unique differentiability is $C^{i+1}$ there Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn. This last step is the one place where the unique differentiability of $f(U)$ is used. It is the bootstrapping of §III.16 in place of the text's iterated tangent maps (Lemma 3.8.20 (preprint 3.76)).

**Relatively open subsets of convex sets.** Let $D \subseteq V_1$ be convex with nonempty interior and $A = O \cap D$ with $O$ open. Then $A$ has unique differentiability at each of its points (a convex set with interior has it, and intersecting with an open set keeps it) Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex; and if $A$ is nonempty, it has nonempty interior: for $a \in A$ and an interior point $w$ of $D$, the points $a + t(w - a)$ with $0 < t$ small lie in $O$ by continuity and in the interior of $D$, because the open segment from a point of the closure of a convex set to an interior point lies in the interior Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex.

**Sard's lemma in fixed dimension** Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty. If $f : V_1 \to V_2$ between finite-dimensional spaces is differentiable on $s$ and $f(s)$ has nonempty interior, then $\dim V_2 \le \dim V_1$. Suppose $\dim V_1 < \dim V_2$. There is a linear surjection $p : V_2 \to V_1$ (send the first $\dim V_1$ vectors of a basis of $V_2$ to a basis of $V_1$ and the rest to 0) Chapter3/SolutionGermRecentre · exists_linearMap_surjective. The map $f \circ p$ on $p^{-1}(s) \subseteq V_2$ has the derivative $Df(p(x)) \circ p$, an endomorphism of $V_2$ whose range lies in the range of $Df(p(x))$, of dimension $\le \dim V_1 < \dim V_2$; so it is not injective and its determinant vanishes. By Mathlib's change-of-variables estimate, a set on which a map is differentiable with vanishing Jacobian determinant has an image of Haar measure zero Mathlib/MeasureTheory/Function/Jacobian · MeasureTheory.addHaar_image_eq_zero_of_det_fderivWithin_eq_zero. But $f(s) \subseteq (f \circ p)(p^{-1}(s))$ since $p$ is onto, and the nonempty open interior of $f(s)$ has positive Haar measure: a contradiction. The lemma uses only differentiability (not continuity of the derivative) and only the finite dimension of both spaces.

**A $C^k$ homeomorphism with invertible derivative has a $C^k$ inverse** Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse. Let $D_1 \subseteq V_1$, $D_2 \subseteq V_2$ be convex with nonempty interior, $A$ relatively open in $D_1$, $\beta$ of class $C^k$ on $A$ with $k \ge 1$, $\beta(A)$ relatively open in $D_2$, $\alpha$ continuous on $\beta(A)$ with $\alpha(\beta(w)) = w$

for $w \in A$, and $D\beta(w_0) = L$ (derivative within $A$) a linear isomorphism $V_1 \to V_2$ at some $w_0 \in A$. Then there is a relatively open $B' \subseteq \beta(A)$ in $D_2$ containing $\beta(w_0)$ on which $\alpha$ is $C^k$. *Proof.* Normalize: $\hat\beta = L^{-1} \circ \beta$ is $C^k$ on $A$ with $D\hat\beta(w_0) = 1$. Its derivative is continuous within $A$ ($C^k$, $k \geq 1$, on a set of unique differentiability), so $\|D\hat\beta(w) - 1\| \leq \frac{1}{2}$ for $w \in A$ in a ball $B(w_0, r_1)$; take $r \leq r_1$ so small that $B(w_0, r) \subseteq O$, where $A = O \cap D_1$. The set $U = D_1 \cap B(w_0, r)$ is convex, lies in $A$, is relatively open in $D_1$ (hence has unique differentiability), and the derivatives of $\hat\beta$ within $U$ and within $A$ agree there (locality of the derivative within a set), so $\|D\hat\beta - 1\| \leq \frac{1}{2}$ on $U$. Its image $\beta(U) = \beta(A) \cap \alpha^{-1}(B(w_0, r))$ — because $\alpha$ inverts $\beta$ on $A$ — is relatively open in $D_2$ by the continuity of $\alpha$, so $\beta(U)$, and with it $\hat\beta(U) = L^{-1}(\beta(U))$, has unique differentiability. So $(U, \hat\beta, k)$ are weak inverse data, and $\hat\beta^{-1}$ is $C^k$ on $\hat\beta(U)$ (Proposition 3.8.17 (preprint 3.73) above). On $B' = \beta(U)$, $\alpha = \hat\beta^{-1} \circ L^{-1}$, since $\alpha(\beta(w)) = w = \hat\beta^{-1}(\hat\beta(w))$; hence $\alpha$ is $C^k$ on $B'$. The relative openness of $\beta(U)$ comes from the continuity of the given inverse $\alpha$, not from a degeneracy index; that is the point of the weak data.

**The re-centring lemma** `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`. Let $\chi_1$, $\chi_2$ be charts of the M-polyfold $X$ `Chapter2/MPolyfold · MPolyfold.IsChart.compatible`, with models $(O_i, C_i, E_i)$, and $S \subseteq X$. Suppose:

- (in $\chi_1$) $K_1 \subseteq E_1$ is a finite-dimensional smooth subspace in good position to $C_1$, $Y_1$ a closed algebraic complement of $K_1$, $V_1$ relatively open in $K_1 \cap C_1$, $\sigma_1 : K_1 \to E_{1,0}$ with values in $Y_1$ and of class $C^1$ on $V_1$; $V_{x,1}$ is open in $X$ inside the domain of $\chi_1$ and $\chi_1(S \cap V_{x,1}) = \{v + \sigma_1(v) \mid v \in V_1\}$;
- (in $\chi_2$) the same kind of data $K_2$, $Y_2$, $V_2 \ni 0$, $\sigma_2$ with values in $Y_2$, continuous on $V_2$, $\sigma_2(0) = 0$, $D\sigma_2(0) = 0$ within $V_2$, and for every $k \geq 1$ a relatively open $V_{2,k} \subseteq V_2$ containing $0$ and a map $\sigma_{2,k} : K_2 \to E_{2,k}$ of class $C^k$ on $V_{2,k}$ with $\iota_k \sigma_{2,k} = \sigma_2$ there (the lifts of Theorem 3.5.11 (1) (preprint Corollary 3.47)); $V_{x,2}$ open inside the domain of $\chi_2$ with $\chi_2(S \cap V_{x,2}) = \{w + \sigma_2(w) \mid w \in V_2\}$;
- $y \in V_{x,1} \cap V_{x,2}$ with $\chi_2(y) = 0$, and $v_0 \in V_1$ with $\chi_1(y) = v_0 + \sigma_1(v_0)$.

Then $\dim K_1 = \dim K_2$, and for every level $m$ and every $k$ there is a relatively open $B' \subseteq V_1$ in $K_1 \cap C_1$ containing $v_0$ on which $\sigma_1$ takes values in $E_{1,m}$ and its level map $\sigma_1 : B' \to E_{1,m}$ is of class $C^k$.

The hypotheses are exactly what each step uses: the closedness of $Y_i$ makes the projection $\pi_i$ onto $K_i$ along $Y_i$ continuous; good position gives $K_i \cap C_i$ a nonempty interior in $K_i$ (for unique differentiability and for Sard); smoothness of $K_i$ gives the linear lifts of $K_i$ to every level `Chapter3/BasicClass · ScBanach.finLift`. Note that $V_1$ need not contain $0$ and $\sigma_1$ need not vanish anywhere: $\chi_1$ is an arbitrary chart in which $S$ is a $C^1$ graph.

*Step 1: notation and the points of $S$.* Write $\Gamma_1(v) = v + \sigma_1(v)$, $\Gamma_2(w) = w + \sigma_2(w)$, and $\pi_1$, $\pi_2$ for the continuous projections along $Y_1$, $Y_2$, so that $\pi_i \Gamma_i = \mathrm{id}$. `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` For $v \in V_1$ the point $\chi_1^{-1}(\Gamma_1(v))$ lies in $S \cap V_{x,1}$ and $\chi_1$ maps it back to $\Gamma_1(v)$, which therefore lies in $O_1$; likewise for $w \in V_2$. Since $\sigma_2(0) = 0$, $\Gamma_2(0) = 0$, and both $\chi_1^{-1}(\Gamma_1(v_0))$ and $\chi_2^{-1}(\Gamma_2(0))$ are $y$.

*Step 2: the transition and the retractions onto the overlaps.* The charts are compatible, so the transition $T = \chi_1 \circ \chi_2^{-1}$ is an sc-smooth map from the overlap retract $\chi_2(V_{\chi_2} \cap V_{\chi_1})$ to $\chi_1(V_{\chi_1} \cap V_{\chi_2})$, and $T' = \chi_2 \circ \chi_1^{-1}$ in the other direction `Chapter2/MPolyfold · MChart.transition` `Chapter2/MPolyfold · MChart.overlap`. The overlaps are sc-retracts `Chapter2/MPolyfold · MChart.isScRetract_overlap`: choose sc-smooth retractions $r_2$, $r_1$ onto them, with domains $U_{r_2}$, $U_{r_1}$. Then $T \circ r_2$ is sc-smooth on $U_{r_2}$, $T' \circ r_1$ on $U_{r_1}$, $T(\chi_2(q)) = \chi_1(q)$ on the domain of $\chi_2$, and $T' \circ T = \mathrm{id}$ on the overlap. Here the formal proof departs from the text, which treats $\varphi \circ \psi^{-1}$ as a map one can differentiate along the graph; a transition is only an sc-smooth map between retracts,

and every derivative below is taken of $T \circ r_2$ or $T' \circ r_1$, on relatively open subsets of partial quadrants.

*Step 3: the parameter sets and the maps $\beta$, $\alpha$.* Let $A = \{w \in V_2 \mid \chi_2^{-1}(\Gamma_2(w)) \in V_{x,1}\}$ and $B_s = \{v \in V_1 \mid \chi_1^{-1}(\Gamma_1(v)) \in V_{x,2}\}$: the parameters whose points of $S$ lie in both chart neighbourhoods. By continuity of $w \mapsto \chi_2^{-1}(\Gamma_2(w))$ on $V_2$ (and of its counterpart), $A$ is relatively open in $K_2 \cap C_2$, $B_s$ in $K_1 \cap C_1$; $0 \in A$ and $v_0 \in B_s$. Define

$$\beta(w) = \pi_1(T(\Gamma_2(w))), \qquad \alpha(v) = \pi_2(T'(\Gamma_1(v))).$$

For $w \in A$, with $q = \chi_2^{-1}(\Gamma_2(w)) \in S \cap V_{x,1} \cap V_{x,2}$: $T(\Gamma_2(w)) = \chi_1(q)$ lies on the graph of $\sigma_1$, say $= \Gamma_1(v)$ with $v \in V_1$; applying $\pi_1$ gives $\beta(w) = v$, so $T(\Gamma_2(w)) = \Gamma_1(\beta(w))$, $\beta(w) \in B_s$, $\alpha(\beta(w)) = w$, and $\Gamma_2(w)$ lies in the overlap. Symmetrically, for $v \in B_s$: $T'(\Gamma_1(v)) = \Gamma_2(\alpha(v))$, $\alpha(v) \in A$, $\beta(\alpha(v)) = v$. Moreover $\beta(0) = v_0$, and $\alpha$ is continuous on $B_s$ ($\pi_2 \circ \chi_2 \circ \chi_1^{-1} \circ \Gamma_1$). Consequently, for every $A' \subseteq A$, $\beta(A') = \{v \in B_s \mid \alpha(v) \in A'\}$, which is relatively open in $K_1 \cap C_1$ whenever $A'$ is relatively open in $K_2 \cap C_2$. This is the text's "local diffeomorphism" in its topological half, obtained without any derivative. `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`

*Step 4: the lifted graph maps.* For $j \geq 1$ let $A_j = A \cap V_{2,j}$ and $\Gamma_{2,j}(w) = \lambda_j(w) + \sigma_{2,j}(w) \in E_{2,j}$, where $\lambda_j : K_2 \to E_{2,j}$ is the linear lift `Chapter3/BasicClass · ScBanach.finLift`. Then $A_j$ is relatively open and contains $0$, $\Gamma_{2,j}$ is $C^j$ on $A_j$ with $\iota_j \Gamma_{2,j} = \Gamma_2$ there, $\Gamma_{2,j}$ maps $A_j$ into the level-$j$ part of the domain of $r_2$, and $r_2(\Gamma_2(w)) = \Gamma_2(w)$ for $w \in A_j$ (the graph points lie in the overlap, where $r_2$ is the identity). Its derivative within $A_j$ at $0$ is $\lambda_j$: indeed $\iota_j \circ D\sigma_{2,j}(0) = D\sigma_2(0)$ within $A_j$, which equals the derivative within $V_2$ (the two sets agree near $0$) and so vanishes; $\iota_j$ is injective, so $D\sigma_{2,j}(0) = 0$. Here the unique differentiability of $A_j$ at $0$ is that of a relatively open subset of the convex cone $K_2 \cap C_2$ with interior `Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex`.

*Step 5: $\beta$ is $C^j$ on $A_j$.* On $A_j$, $\beta = \pi_1 \circ ((T \circ r_2) \circ \iota_j) \circ \Gamma_{2,j}$, because $r_2$ fixes $\Gamma_2(w) = \iota_j \Gamma_{2,j}(w)$. The map $(T \circ r_2) \circ \iota_j$ is $C^j$ on the level-$j$ part of $U_{r_2}$ — Proposition 1.2.3 (preprint 1.18): an sc$^j$ map is $C^j$ from level $j$ to level $0$ `Chapter1/Levels · IsSck.contDiffOn_base` — and $\pi_1$ is continuous linear; so $\beta$ is $C^j$ on $A_j$. This is where the lifts are indispensable: $\sigma_2$ itself is only continuous into level $0$, and $T \circ r_2$ is not differentiable on level $0$.

*Step 6: the chain rule at $0$.* Take $j = 2$; then $\beta$ is $C^2$, in particular $C^1$, on $A_2$, with derivative $D\beta = D\beta(0)$ within $A_2$ at $0$. Read the lift on level $1$: $\Gamma_{2,1} = \iota_{1,2} \circ \Gamma_{2,2}$, with $\iota_1 \Gamma_{2,1} = \Gamma_2$ on $A_2$, values in the level-$1$ part of $U_{r_2}$, derivative $\iota_{1,2}\lambda_2$ at $0$, and $\iota_1 \iota_{1,2} \lambda_2(n) = n$ for $n \in K_2$. Let $\mathcal{D} = D(T \circ r_2)(\Gamma_{2,1}(0))$, the sc-derivative at the level-$1$ point over $0$. The map $w \mapsto T(r_2(\iota_1 \Gamma_{2,1}(w)))$ has, by the chain rule for the sc$^1$ map $T \circ r_2$ (differentiable from level $1$ to level $0$) and the differentiable $\Gamma_{2,1}$, the derivative $n \mapsto \mathcal{D}n$ at $0$ within $A_2$. On $A_2$ this map equals $\Gamma_1 \circ \beta$ (Step 3 and $r_2 \circ \Gamma_2 = \Gamma_2$), and $\Gamma_1 \circ \beta$ has, at $0$, the derivative $n \mapsto D\beta\, n + D\sigma_1(v_0) D\beta\, n$ by the chain rule — here the $C^1$-property of $\sigma_1$ *at* $v_0$, within $V_1$, is used, together with $\beta(A_2) \subseteq V_1$ and $\beta(0) = v_0$. By uniqueness of the derivative within $A_2$ at $0$,

$$\mathcal{D}n = D\beta\, n + D\sigma_1(v_0)\, D\beta\, n \qquad (n \in K_2).$$

*Step 7: $D\beta(0)$ is injective.* Let $D\beta\, n = 0$; then $\mathcal{D}n = 0$. The composite $(T' \circ r_1) \circ (T \circ r_2)$ is sc-smooth on $U_{r_2}$ ($T \circ r_2$ maps $U_{r_2}$ into the overlap retract of $\chi_1$, hence into $U_{r_1}$), and it equals $r_2$ there: $r_2(u)$ lies in the overlap, $T(r_2(u))$ in the other overlap where $r_1$ is the identity, and $T' \circ T = \mathrm{id}$ on the overlap. By the chain rule for sc$^1$ maps between relatively open subsets of partial quadrants `Chapter2/DegeneracyIndex · scDeriv_comp_apply`, $D((T' \circ r_1) \circ (T \circ r_2))(\Gamma_{2,1}(0))n = D(T' \circ r_1)(\cdot)\, \mathcal{D}n = 0$.

Since the composite agrees with $r_2$ on the relatively open $U_{r_2}$, its derivative at the level-1 point is $Dr_2(\Gamma_{2,1}(0))$ Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen. And $Dr_2(\Gamma_{2,1}(0))n = n$: the family $w \mapsto \iota_1\Gamma_{2,1}(w)$ consists of fixed points of $r_2$, so the derivative of $r_2$ fixes its tangent vectors at 0, which are the vectors $\iota_1\iota_{1,2}\lambda_2(n) = n$ Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed — the tangent space $K_2$ of the graph lies in the tangent space of the retract. So $n = 0$. Hence $D\beta(0)$ is injective and $\dim K_2 \le \dim K_1$.

*Step 8: surjectivity by counting dimensions.* $\beta$ is differentiable on $A_2$, and $\beta(A_2)$ is a nonempty relatively open subset of the convex cone $K_1 \cap C_1$ (Step 3; it contains $\beta(0) = v_0$), which has nonempty interior in $K_1$ by good position; so $\beta(A_2)$ has nonempty interior Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex, and Sard's lemma in fixed dimension gives $\dim K_1 \le \dim K_2$ Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty. So $\dim K_1 = \dim K_2$, the injective $D\beta(0)$ is bijective, and $L = D\beta(0) : K_2 \to K_1$ is a linear isomorphism. This is the first conclusion of the lemma, and also the reason why the dimension of a set with two such descriptions cannot jump.

*Step 9: the inverse, and the regularity of* $\sigma_1$. Fix $m$ and $k$ and take $j = m + k + 2$. By Step 5, $\beta$ is $C^j$, hence $C^{\max(k,1)}$, on $A_j$, $\beta(A_j) \subseteq B_s$, and its derivative within $A_j$ at 0 is again $L$ (the derivatives within $A_j$ and within $A_2$ agree at 0, both sets being relatively open near 0 in the same cone). Now apply the local inverse theorem above with $D_1 = K_2 \cap C_2$, $D_2 = K_1 \cap C_1$ (convex with interior), $A = A_j$, the continuous inverse $\alpha$ on $\beta(A_j)$ and $w_0 = 0$ Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse: there is a relatively open $B' \subseteq \beta(A_j)$ in $K_1 \cap C_1$ containing $v_0$ on which $\alpha$ is $C^{\max(k,1)}$, with $\alpha(B') \subseteq A_j$. Next the level-$m$ map

$$\Psi(w) = (T \circ r_2)_m(\iota_{m,k+2}\,\Gamma_{2,j}(w)) - \lambda_m(\beta(w)) \in E_{1,m}, \qquad w \in A_j,$$

where $\iota_{m,k+2} : E_{2,m+k+2} \to E_{2,m}$ and $\lambda_m$ is the lift of $K_1$ to level $m$. It is $C^k$ on $A_j$: the $\mathrm{sc}^{k+2}$ map $T \circ r_2$ is $C^{k+2}$ from level $m+k+2$ to level $m$ (Proposition 1.2.3 (preprint 1.18) Chapter1/Levels · IsSck.contDiffOn_levels), $\Gamma_{2,j}$ is $C^j$ into level $j = m+k+2$, and $\lambda_m \circ \beta$ is $C^j$. Including into level 0, $\iota_m\Psi(w) = T(r_2(\Gamma_2(w))) - \beta(w) = \Gamma_1(\beta(w)) - \beta(w) = \sigma_1(\beta(w))$. Hence for $v \in B'$, $v = \beta(\alpha(v))$ and $\sigma_1(v) = \iota_m\Psi(\alpha(v))$: $\sigma_1$ takes values in $E_{1,m}$ on $B'$, its level-$m$ map is $\Psi \circ \alpha$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq, which is $C^k$ on $B'$ as a composite of $C^k$ maps, and $B' \subseteq B_s \subseteq V_1$. This is the text's identity $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) = (1-\pi)\varphi\psi^{-1}(1+\tau)$, written on level $m$ through the lifts.

*Comparison with the text.* Four things in the formal argument are not in the sketch of preprint p. 84. (a) The isomorphism claim needs $\sigma_1$ differentiable at $v_0$, which for an arbitrary set with the tangent germ property fails; here $\sigma_1$ is assumed $C^1$ on all of $V_1$, and §III.36 arranges this by shrinking $V_1$ to the neighbourhood of 0 on which the germ property makes $\delta$ of class $C^1$. (b) Only the *injectivity* of the linearization comes from the chain rule, and it uses that the tangent space $K_2$ of the second graph lies in the tangent space of the overlap retract (Step 7). The symmetric argument for surjectivity would differentiate $T' \circ \Gamma_1 = \Gamma_2 \circ \alpha$ at $v_0$; but $T' \circ r_1$ is differentiable only from level 1 to level 0, so this would need $\sigma_1$ differentiable *into level* 1 at $v_0$, which is known near the centre of $\chi_1$ (Theorem 3.5.11 (1) (preprint Corollary 3.47)) and not at $v_0$. Surjectivity therefore comes from the dimension count by Sard's lemma (Step 8), which at the same time shows that the dimension is locally constant. (c) The regularity of $\beta^{-1} = \alpha$ is not the quadrant inverse function theorem 3.72 (§III.16): that theorem asks for $d_C(x) = d_C(\beta(x))$ (formally: interior points to interior points), which here would be a statement about the intrinsic degeneracy index of $S$ — not available. It is Lemma 3.8.21 (preprint 3.77) and the bootstrapping of Proposition 3.8.17 (preprint 3.73) under the weak data, with the relative openness of $\beta(U)$ supplied by the continuity of the inverse $\alpha$. (d) The neighbourhood $B'$ depends on $(m,k)$ through the lift $\Gamma_{2,m+k+2}$; this is harmless,

because sc-smoothness in the set sense is a local property of the level maps (§III.36).

## III.36 Regular data by re-centring, and Theorem 3.2.3 (preprint 3.27) with locally constant dimension

This section turns the re-centring lemma of §III.35 into condition (a) of `IsRegular` (§III.24) and adapts Theorem 3.2.3 (preprint 3.27) to data whose dimension varies from point to point. The tool is the shrinking of the data of Definition 3.2.1 (2) (preprint 3.25): the chart is restricted to an open subset of its domain and the parameter set $V$ to a smaller relatively open set, with the same $N$, $Y$ and $\delta$. The data at a point $x$ are shrunk into the neighbourhood of $0$ where $\delta$ is $C^1$; then, at every point of the shrunk graph, the data at that point (which a set with the tangent germ property has everywhere) provide the second graph description that the re-centring lemma needs.

*Open subsets of tame retracts* (told in §III.27). The restriction of a tame retraction $r : U \to U$ to $U \cap r^{-1}(W)$, $W$ open (Proposition 2.3.1 (1) (preprint 2.10)) `Chapter2/Retracts · ScRetraction.restrict`, is tame: the degeneracy-index condition of Definition 2.5.2 (preprint 2.34) is inherited pointwise, and at a smooth point $x$ of $O \cap W$ the sc-complement of $T_xO$ of the original retraction serves, because the derivative of the restricted retraction at $x$ is $Dr(x)$ — the two maps agree on a relatively open neighbourhood `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen` `Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame`. So $O \cap W$ is a tame retract `Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen`, and the model $\varphi(s)$ of a chart restricted to an open $s$ of its domain is tame when the model of $\varphi$ is, since $\varphi(s) = O \cap W$ for some open $W$ `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq` `Chapter3/SolutionGermRecentre · MChart.isTameRetract_image`. The text uses this silently whenever it shrinks a chart of a tame M-polyfold.

**Shrinking the data** `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink`. Let $D = (\varphi, N, Y, V, \delta)$ be data of Definition 3.2.1 (2) (preprint 3.25) for $M$ at $x$ of dimension $n$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart`, $s$ an open subset of the domain of $\varphi$ containing $x$, $V' \subseteq V$ relatively open in $N \cap C$ containing $0$, and $M'$ a set with $\varphi(M' \cap s) = \{v + \delta(v) \mid v \in V'\}$. Then the chart $\varphi$ restricted to $s$ `Chapter2/MPolyfold · MChart.restrict` — a chart of the structure `Chapter2/MPolyfold · MPolyfold.IsChart.restrict` with a tame model by the paragraph above — together with the same $N$, $Y$, $\delta$ and the parameter set $V'$, are data of Definition 3.2.1 (2) (preprint 3.25) for $M'$ at $x$ of dimension $n$. Every field is inherited: good position, the closed good complement, $\delta(0) = 0$, $D\delta(0) = 0$ and the germ property of $\delta$ concern $N$, $Y$, $\delta$ and the cone $N \cap C$ only, which are unchanged, and continuity on $V'$ follows from continuity on $V$. The set $M'$ may differ from $M$; this is how §III.37 passes between the solution set in a neighbourhood and the global solution set. The shrunk data have the source $s$, the model $\varphi(s)$, the same chart values, the same $N$, $\delta$ and projection $\pi$, and the parameter set $V'$ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_source` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_model_0` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_apply` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_N` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_V` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_δ` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π`. They are regular `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular` as soon as $\delta$ is sc-smooth on $V'$ in the set sense and the original data satisfy condition (b), $\pi(O) \subseteq C$: condition (b) passes to the smaller retract $\varphi(s) \subseteq O$ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`.

**The shrinking data** `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_shrinkData`. For an open

$W \ni x$ in $X$ and an open $O'' \ni 0$ in $N$, let

$$s = \{y \in \operatorname{dom}\varphi \cap W \mid \pi\varphi(y) \in O''\}, \qquad V' = \{v \in V \cap O'' \mid \varphi^{-1}(v + \delta(v)) \in W\}.$$

Then $s$ is open (the preimage of $O''$ under the continuous $\pi \circ \varphi$ on the open $\operatorname{dom}\varphi \cap W$) and contains $x$ ($\pi\varphi(x) = \pi(0) = 0$); $V'$ is relatively open in $N \cap C$ because $v \mapsto \varphi^{-1}(v + \delta(v))$ is continuous on $V$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_continuousOn`, and contains $0$ ($\varphi^{-1}(0) = x \in W$); $s \subseteq W$, $V' \subseteq V \cap O''$; and $\varphi(M \cap s) = \{v + \delta(v) \mid v \in V'\}$. For the last identity: a point $y \in M \cap s$ has $\varphi(y) = \Gamma(v)$ with $v = \pi\varphi(y) \in V$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`, $v \in O''$ because $y \in s$, and $\varphi^{-1}(\Gamma(v)) = y \in W$; conversely for $v \in V'$ the point $\varphi^{-1}(\Gamma(v))$ lies in $M$ and in the domain `Chapter3/TangentGerm · MPolyfold.TangentGermChart.symm_Γ_mem`, in $W$ by definition of $V'$, and $\pi\varphi$ of it is $v \in O''$ since $\pi\Gamma = \mathrm{id}$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_Γ`.

**The lifts from the germ property.** Definition 3.2.1 (ii) (preprint 3.25), in the reading of §III.24 `Chapter3/TangentGerm · IsScInftyGermAt`, gives for every $m$, $k$ a relatively open $V_{m,k} \ni 0$ on which $\delta$ takes values in $E_m$ and is $C^k$ into $E_m$. With $m = k$: for every $k \geq 1$ the level map of $\delta$ on $V_{k,k} \cap V$ is a $C^k$-lift $\delta_k$ of $\delta$ into $E_k$ — the hypothesis of the re-centring lemma on the centred side `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_lift`. With $(m, k) = (0, 1)$: $\delta$ is $C^1$ on a relatively open neighbourhood of $0$ in $N \cap C$ `Chapter3/TangentGerm · IsScInftyGermAt.contDiffOn_one`. And $D\delta(0) = 0$ within $V$, since the data give it within $N \cap C$ and $V$ agrees with $N \cap C$ near $0$ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.fderivWithin_V_zero`.

**Condition (a) of `IsRegular` by re-centring** `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`. Let $X$ be an M-polyfold (Hausdorff, paracompact), $M \subseteq X$, $D$ data of Definition 3.2.1 (2) (preprint 3.25) for $M$ at $x$ of dimension $n$ satisfying condition (b), $\pi(O) \subseteq C$, and $U \ni x$ open such that every point of $M \cap U$ has data of Definition 3.2.1 (2) (preprint 3.25) for $M$ of *some* dimension. Then there are data $D'$ for $M$ at $x$ of dimension $n$ which are regular (`IsRegular`: (a) and (b)), whose chart domain lies in $U$, and such that every datum of Definition 3.2.1 (2) (preprint 3.25) for $M$ at a point of $M$ in the chart domain of $D'$ has dimension $n$. (The proof constructs $D'$ as the shrinking of $D$, with the same $N$, $Y$, $\delta$; the statement records only the regularity, the domain and the dimension.)

*Step 1: shrinking into the $C^1$ neighbourhood.* By the germ property, $\delta$ is $C^1$ on $V_1 = O_1 \cap (N \cap C)$ with $O_1$ open and $0 \in O_1$. The shrinking data for $W = U$ and $O'' = O_1$ give $s \subseteq U$ and $V' \subseteq V \cap O_1$, so $V' \subseteq V_1$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_subset`, with $\varphi(M \cap s) = \Gamma(V')$; let $D_1$ be the shrunk data for $M$ itself. Then $\delta$ is $C^1$ on all of $D_1$'s parameter set $V'$: this is hypothesis (a) of the re-centring lemma on the side of $D_1$, and the reason for the shrinking.

*Step 2: the re-centring at every point.* Let $y \in M \cap s$ and let $D''$ be any data of Definition 3.2.1 (2) (preprint 3.25) for $M$ at $y$, of dimension $n'$. Put $v_0 = \pi\varphi(y)$; then $v_0 \in V'$ and $\varphi(y) = \Gamma(v_0)$. The re-centring lemma applies with $\$\square_1 = \$$ the chart of $D_1$ (graph of $\delta$ over $V'$ on its whole domain $s$, $\delta$ of class $C^1$ on $V'$) and $\$\square_2 = \$$ the chart of $D''$, centred at $y$ (graph of $\delta''$ over $V''$ on its whole domain, $\delta''(0) = 0$, $D\delta''(0) = 0$ within $V''$, the lifts from the germ property of $D''$); all the remaining hypotheses — smooth finite-dimensional $N$, closed good complement, good position, relatively open parameter sets — are fields of the two data `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`. It gives $n = \dim N = \dim N'' = n'$, and for every $m$, $k$ a relatively open $B' \subseteq V'$ in $N \cap C$ containing $v_0$ on which $\delta$ takes values in $E_m$ and is $C^k$ into $E_m$.

*Step 3: $\delta$ is sc-smooth on $V'$ in the set sense.* $V'$ is a relatively open subset of the partial quadrant $N \cap C$ of $N$ with the constant sc-structure `Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_isRelOpenInQuadrant`, all

of whose levels are $V'$. By Proposition 1.2.4 (preprint 1.19) in the form "maps every level into the corresponding level, and the level maps are $C^\infty$" Chapter1/Levels · IsScSmooth.of_contDiffOn_levels, it suffices that for every $m$: (i) $\delta(V') \subseteq E_m$, and (ii) the level map $\delta : V' \to E_m$ is $C^k$ for every $k$. For $v \in V'$ the point $y = \varphi^{-1}(\Gamma(v))$ lies in $M \cap s \subseteq M \cap U$, so by hypothesis there are data at $y$ of some dimension, and Step 2 applies with $v_0 = \pi\varphi(y) = v$. For (i), the set $B'$ for $(m, 0)$ contains $v$, and $\delta$ maps it into $E_m$. For (ii), fix $k$; around every $v \in V'$ the set $B' = O_B \cap (N \cap C)$ for $(m, k)$ contains $v$, and the level map is $C^k$ on $O_B \cap V' \subseteq B'$; a map that is $C^k$ near every point of a set is $C^k$ on the set Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_of_locally_contDiffOn. Here the dependence of $B'$ on $(m, k)$ (§III.35, Step 9) does no harm: the property is checked one $(m, k)$ at a time, and locally.

*Step 4: conclusion.* With (a) from Step 3 and (b) inherited from $D$, the shrunk data $D_1$ are regular Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular; their chart domain is $s \subseteq U$; and by Step 2 every datum at a point of $M \cap s$ has dimension $n$.

*What the statement gives for an arbitrary subset.* The hypotheses of the theorem are about $M$ and $D$ only; no solution set, section or Fredholm property enters. If $M$ has the $n$-dimensional tangent germ property Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty, every point of $M$ has data of dimension $n$, so the hypothesis "every point of $M \cap U$ has data of some dimension" holds with $U = X$. Hence, read off the statement (this is not a separate Lean theorem): *for any $M$ with the tangent germ property, any data at $x$ that satisfy condition (b) can be shrunk to regular data at $x$, of the same dimension.* Condition (a) of `IsRegular` is therefore not an independent hypothesis of Theorem 3.2.3 (preprint 3.27): if at every point of $M$ some $n$-dimensional data satisfy (b), the regular data that Theorem 3.2.3 (preprint 3.27) asks for exist at every point Chapter3/TangentGerm · theorem_3_2_3_of_isRegular. (The inclusion $M \subseteq X_\infty$ of Definition 3.2.1 (1) (preprint 3.25) is not used.) The counterexample to the text's step "$\delta$ is an sc$^\infty$-germ near every $v_0$" (§III.24) is not contradicted: there the data are regular only after shrinking $V$ away from the point with a vertical tangent, which is what Step 1 does. What remains a genuine restriction is condition (b), which fails for some good complements (§III.24) and is obtained for solution sets by the choice of the complement (§III.29, §III.34).

*The dimension is locally constant.* The last clause of the theorem says more than the text: around every point where regular data exist, all data of Definition 3.2.1 (2) (preprint 3.25) at nearby points of $M$ have the same dimension. The dimension of a set with the tangent germ property at varying dimensions is therefore locally constant along $M$ — but not necessarily globally constant, and Theorem 3.2.3 (preprint 3.27) as formalized fixes $n$ (§III.24). For the solution set of an sc-Fredholm section, whose dimension $\dim \ker f'(x)$ can change between components, the fixed-dimension theorem would need $S$ split first; instead Theorem 3.2.3 (preprint 3.27) is restated for locally constant dimension.

**Manifolds with corners of locally constant dimension** Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc. An M-polyfold structure is that of a manifold with boundary with corners of locally constant dimension if it has an equivalent atlas all of whose charts have models $(O, C, E)$ with $O$ relatively open in the partial quadrant $C$ and $E_0$ finite-dimensional — the definition of §III.24 Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners without the requirement that every chart have dimension $n$. A manifold with corners of dimension $n$ is one of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCorners.isManifoldWithCornersLoc.

**Theorem 3.2.3 (preprint 3.27) with locally constant dimension** Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular. Let $X$ be an M-polyfold, $M \subseteq X$ nonempty, and suppose every $x \in M$ has regular data of Definition 3.2.1 (2) (preprint 3.25) for $M$ of some dimension $n(x)$. Then $M$ is a sub-M-polyfold of $X$; its induced structure (Proposition 2.3.19 (preprint 2.21)) is tame and that of a manifold with boundary

with corners of locally constant dimension; and for every $x \in M$, all regular data $d$ at $x$ of any dimension and every base point $a_0 \in M$ (the value of the chart inverse off its domain), the chart $y \mapsto \pi\varphi(y)$ of $d$ is a chart of the induced structure whose domain contains $x$. The proof is that of Theorem 3.2.3 (preprint 3.27) (§III.24) with the dimension quantified pointwise; every step there is either pointwise or does not use $n$.

*Sub-M-polyfold* Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isSubMPolyfold. Definition 2.3.18 (preprint 2.20) is pointwise: at $a \in M$ take regular data $d$ (of whatever dimension); the open set $U_0 = \varphi^{-1}(\Sigma \cap O)$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_U₀ and the sc-smooth retraction $s = \varphi^{-1} \circ t \circ \varphi$ onto $M \cap U_0$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.s Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s are those of §III.24, exactly as in the fixed-dimension proof Chapter3/TangentGerm · theorem_3_2_3_isSubMPolyfold.

*The charts and the atlas.* Let the charts be all maps $\pi\varphi$ of all regular data at all points of $M$, of all dimensions, with all base points Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart. Each is a chart of the induced structure Chapter2/SubMPolyfold · subPolyfold: it is compatible with every chart $\varphi'|_{W\cap M}$ of the induced atlas, by the lemma of §III.24, whose proof never compares dimensions Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isChart. Any two of them are compatible, by the locality of compatibility: both are compatible with every chart of the induced atlas, whose domains cover $M$ Chapter2/SubMPolyfold · MPolyfold.subAtlas Chapter2/MPolyfold · MChart.compatible_of_locally Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc_compatible. The argument never mentions the dimension of either chart. They form an atlas: every $y \in M$ has regular data, and the chart of these data, with the base point a fixed element of $M$ (here $M \neq \emptyset$ is used), contains $y$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source Chapter3/GlobalIFTTheorems · MPolyfold.germAtlasLoc. This atlas is equivalent to the induced atlas of Proposition 2.3.19 (preprint 2.21), since each of its charts is a chart of the induced structure Chapter2/MPolyfold · MAtlas.equiv_iff Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_atlas_equiv.

*Tame, and a manifold with corners.* The model of each chart $\pi\varphi$ is $(V, N\cap C, N)$ with the identity retraction, a tame retract Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract; so the induced structure is tame, witnessed by the equivalent atlas just built Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isTame. Each model is relatively open in the partial quadrant $N \cap C$ of the finite-dimensional $N$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner; so the induced structure is that of a manifold with boundary with corners of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isManifoldWithCornersLoc. The assembled statement collects these, with the chart property of every $\pi\varphi$ Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular.

As for Theorem 3.2.3 (preprint 3.27), the hypotheses are the regularity conditions at every point; by the observation above, condition (a) there can be replaced by condition (b) plus shrinking. Theorem 3.6.11 (preprint 3.57) (§III.37) applies this version to the solution set, where regular data exist at every zero by §§III.34–III.36 and the dimension $\dim\ker f'(x)$ is locally constant by Step 2 of the theorem above; the fixed-dimension statement for each part $S_n$ of the solution set is obtained from Theorem 3.2.3 (preprint 3.27) itself.

# III.37 The implicit function theorems assembled: Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19), Corollary 3.1.27 (preprint 3.24), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)

The text derives all its implicit function theorems from Theorem 3.6.8 (preprint 3.54): (1) good position at every zero near $x$, (2) the sub-M-polyfold structure of the solution set near $x$, through the tangent germ property and Theorem 3.2.3 (preprint 3.27). In the formalization the pieces are: the data of Definition 3.2.1 (preprint 3.25) at the centre, with condition (b) (§III.34); good position at the nearby zeros, under an added hypothesis (§III.33); the re-centring, which turns data at all nearby points into regular data and shows that the dimension is locally constant (§§III.35–III.36); and Theorem 3.2.3 (preprint 3.27), with fixed dimension (§III.24) or with locally constant dimension `Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular` (§III.36). This section assembles them. Throughout, $f$ is a section of a strong bundle $P : Y \to X$ over an M-polyfold, "the charts have tame base retractions" means that the base retraction of every strong bundle chart compatible with the atlas is tame (the formal reading of "tame strong bundle", §III.23), $S = \{x \in X : f(x) = 0\}$ is the global solution set `Chapter3/GlobalIFT · solutionSet`, and good position is Definition 3.6.6 (preprint 3.53) in the chart form `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm` (§III.21).

**The data for the global solution set** `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_tangentGermChart_solutionSet`. *If the charts have tame base retractions, $f(x_0) = 0$ and $(f, x_0)$ is in good position, then $S$ has at $x_0$ data of Definition 3.2.1 (preprint 3.25) of some dimension $d$ with $\pi(O) \subseteq C$.* §III.34 gives data $D$ for $S_{V_x} = \{y \in V_x : f(y) = 0\}$. Restrict its chart to the open set $s = \text{source} \cap V_x \ni x_0$, keeping $N$, $Y$, $V$, $\delta$ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink` `Chapter2/MPolyfold · MChart.restrict`: the graph identity holds for the global $S$ because $S \cap s = S_{V_x} \cap \text{source}$. The restricted model $\varphi(s)$ is a tame retract (an open subset of a tame retract, §III.36), and condition (b) passes to it, since $\varphi(s) \subseteq O$ `Chapter2/MPolyfold · MChart.image_subset_model` and the projection is unchanged `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π`.

**Theorem 3.6.8 (2) (preprint 3.54), from the nearby good position** `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_two`. *Let the charts have tame base retractions, $f(x_0) = 0$, $(f, x_0)$ in good position, and let every zero of $f$ in an open $V_{x_0} \ni x_0$ be in good position. Then there is a dimension $d$ and regular data of Definition 3.2.1 (preprint 3.25) of dimension $d$ for $S$ at $x_0$ (condition (a): $\delta$ sc-smooth on $V$ in the set sense; condition (b): $\pi(O) \subseteq C$) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`, on a chart whose source lies in $V_{x_0}$; and every datum of Definition 3.2.1 (preprint 3.25) for $S$ at a point of $S$ in that source has dimension $d$.* Only that $f$ is a section is assumed; the regularity of $f$ enters through the good position at $x_0$ and at the nearby zeros, which carry the sc-Fredholm germ data. *Proof.* Take the data $D$ at $x_0$ above. Every $y \in S \cap V_{x_0}$ is a zero in good position, so the same theorem at $y$ gives data of Definition 3.2.1 (preprint 3.25) for $S$ at $y$, of some dimension. These are the hypotheses of the re-centring theorem of §III.36 `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally` for $D$ (which satisfies (b)) and the open set $V_{x_0}$: it shrinks $D$ to regular data $D'$ with source in $V_{x_0}$, of the same dimension $d$, along which every datum has dimension $d$. This is the text's "proceeding as above we can construct a map $\sigma$" at every zero, made into a proof: the regularity condition (a), an added hypothesis of Theorem 3.2.3 (preprint 3.27) for general sets (§III.24), is a theorem for solution sets (items 150–151).

**Theorem 3.6.8 (1) and (2) (preprint 3.54)** `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8`. *Let $f$ be an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and let $(f, x_0)$ be in good position* ***with a filler that satisfies condition (3) of Definition 3.1.4 (preprint 3.4) at the zeros of the filler near*** $0$ `Chapter3/`

GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled. *Then there is an open $V_x \ni x_0$ such that (1) $(f, y)$ is in good position at every zero $y \in V_x$, and (2) the conclusions of Theorem 3.6.8 (2) (preprint 3.54) above hold with $V_{x_0} = V_x$.* (1) is §III.33 Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one; (2) is the previous theorem, fed with (1), with the strengthened good position weakened to good position Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.isGoodPositionGerm and the section property of the sc-Fredholm section. The filler condition was an **added hypothesis** when this form was proved: Definition 3.1.4 (preprint 3.4) imposes condition (3) at 0 only, and the surjectivity of $f'(y)$ at the nearby zeros is equivalent to it there (§III.33; items 100, 105–106); the kernel half of (1) needs no added hypothesis (§III.32). Since the second addendum the condition is a consequence of good position (§III.45), and Theorem 3.6.8 (preprint 3.54) is proved with the text's hypotheses (§III.43). Part (3) of the text's proof, the graph description, is §III.29 Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj, and part (4), the kernel at the nearby zeros in the chart at $x_0$, is §III.32 Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_nearby_kernel. The text's neighbourhood $V$ is the $V_x$ of (3), shrunk in (2) to the chart source of the re-centred data.

**Theorem 3.1.26 (preprint 3.23), boundary case, from the nearby good position** Chapter3/GlobalIFTTheoremsG · theorem_3_1_26_of_nearby. *Let $f$ be an sc-Fredholm section, the charts with tame base retractions, $f(x) = 0$, $(f, x)$ in good position ($f'(x)$ onto with kernel in good position, Definition 3.6.6 (preprint 3.53)), and every zero in an open $V_{x_0} \ni x$ in good position. Then there are an open $U$ with $x \in U \subseteq V_{x_0}$ and a dimension $d$ such that every zero in $U$ is a smooth point, $S(f, U) = \{y \in U : f(y) = 0\}$ is a sub-M-polyfold of $X$, and its induced M-polyfold structure (Proposition 2.3.19 (preprint 2.21)) Chapter2/SubMPolyfold · subPolyfold is tame and equivalent to that of a $d$-dimensional manifold with boundary with corners* Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners. *Proof.* Theorem 3.6.8 (2) (preprint 3.54) at $x$ gives regular data $D$ of dimension $d$ with source $U \subseteq V_{x_0}$ and the local constancy of the dimension; let $S' = S(f, U)$. At every $y \in S'$, Theorem 3.6.8 (2) (preprint 3.54) at $y$, with the open neighbourhood $U \cap V_{x_0}$ in place of $V_{x_0}$, gives regular data $D_y$ for $S$ with source in $U \cap V_{x_0}$, of a dimension $d_y$; since $y \in S \cap U$, the local constancy along $D$ gives $d_y = d$. Because the source of $D_y$ lies in $U$, $S' \cap \text{source} = S \cap \text{source}$, so $D_y$ restricted to its own source is a datum for $S'$ Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink, still regular Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular. So every point of $S'$ has regular data of dimension $d$, $S' \ni x$ is nonempty, and Theorem 3.2.3 (preprint 3.27) with fixed dimension (§III.24) Chapter3/TangentGerm · theorem_3_2_3_of_isRegular gives the sub-M-polyfold with its tame induced structure equivalent to a $d$-dimensional manifold with boundary with corners. The zeros are smooth points because an sc-Fredholm section is regularizing Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt. The text's "a uniquely determined structure … the same as the one defined in Proposition 2.21" is the induced structure itself, with the equivalent atlas of the charts $\pi \circ \varphi$; uniqueness is not stated separately.

**Theorem 3.1.26 (preprint 3.23)** Chapter3/GlobalIFTTheoremsG · theorem_3_1_26. *The same under the added filler hypothesis at $x$ instead of the nearby good position, with in addition: every zero in $U$ is in good position.* Theorem 3.6.8 (1) (preprint 3.54) (§III.33) supplies the open $V_x$ of nearby good position, and the previous theorem applies with $V_{x_0} = V_x$.

**Theorem 3.1.22 (preprint 3.19), interior case.** The first clause, "$f'(x_0)$ is sc-Fredholm", is Proposition 3.6.5 (preprint 3.52) in the chart at $x_0$ Chapter3/GlobalIFTTheoremsG · theorem_3_1_22_linearization Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint, applied to the sc-Fredholm germ contained in the good position: $Dg(0)$, for a derivative family $Dg$ of the filling germ (§III.21), and $Tf(x_0) = Dg(0)|_{T_0O}$ are Fredholm on level 0 with the same index. The statement inherits the index clause of that proposition, "$= n - N$ for some natural numbers $n$, $N$", which carries no information; the index as the $n - N$ of the conjugating basic germ is §III.30 Chapter3/FredholmGermIndexG · proposition_3_6_5_index. *The theorem* Chap–

ter3/GlobalIFTTheoremsG · theorem_3_1_22: *let $d_X \equiv 0$* Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary*, $f$ an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and $(f, x_0)$ in good position with the filler condition at the nearby zeros. Then there are an open $U \ni x_0$ and $d$ such that every zero in $U$ is in good position (in particular $f'(y)$ is onto) and a smooth point, and $S(f, U)$ is a sub-M-polyfold whose induced structure is equivalent to a $d$-dimensional manifold with boundary with corners and has degeneracy index $0$ everywhere. Proof.* Theorem 3.1.26 (preprint 3.23) gives everything but the last clause; for that, Lemma 2.4.6 (preprint 2.26), $d_S(y) \le d_X(y)$ Chapter2/SubMPolyfold · lemma_2_4_6, and $d_X = 0$. So the interior case is the boundary case plus Lemma 2.4.6 (preprint 2.26); $d_X \equiv 0$ is used for nothing else. *Two readings and two omissions* (item 154), all resolved in the second addendum. (a) The text's hypothesis is "$f'(x_0)$ surjective"; good position adds condition (2) of Definition 3.6.6 (preprint 3.53), which the text calls an empty condition when $d_X(x) = 0$. Formally condition (2) is stated in the chart for the chart's partial quadrant, so this form takes good position as in the boundary case; that the condition is void at $d_X = 0$ in a tame chart, and Theorem 3.1.22 (preprint 3.19) with "$f'(x_0)$ surjective", are §III.45. The filler condition at the nearby zeros was the added hypothesis of Theorem 3.6.8 (1) (preprint 3.54); the form with the text's hypotheses is §III.43. (b) "The dimension agrees with the Fredholm index": this statement quantifies $d$ existentially; the proof takes $d = \dim\ker D(h+t)(0)$ (§III.34), and $\operatorname{ind} D(h+t)(0) = \dim P - \dim Q = \operatorname{ind} f'(x_0)$ holds for the conjugation data of the proof (§§III.18, III.30); the graph description with the conjugation exposed, and $\dim\ker = \operatorname{ind}$ for the surjective $D(h+t)(0)$, give the dimension as the index in §III.44. (c) "$\ker f'(y) = T_y S(f, U)$": in the chart at $x_0$, $\ker D(h+t)(z) = N_v$, the tangent space of the graph (§III.32); the chart form of the statement is §III.45, and the identification with $T_y S$ as a subspace of $T_y X$, with the intrinsic linearization, is §III.62.

**Theorem 3.6.11 (preprint 3.57), with the text's hypotheses** Chapter3/GlobalIFTTheoremsG · theorem_3_6_11. *Let $f$ be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with $(f, x)$ in good position at every zero $x$, and let $S \neq \emptyset$. Then $S$ is a sub-M-polyfold of $X$, its induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension* Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc *(an equivalent atlas of charts on relatively open subsets of partial quadrants of finite-dimensional spaces), and at every $x \in S$ there are regular data of Definition 3.2.1 (preprint 3.25) of a dimension $d(x)$ such that every datum at a point of $S$ in their chart source has dimension $d(x)$. Proof.* At every $x \in S$, Theorem 3.6.8 (2) (preprint 3.54) with $V_{x_0} = X$ — the hypothesis "every zero near $x$ in good position" is part of the global hypothesis — gives the regular data and the local constancy. Theorem 3.2.3 (preprint 3.27) with locally constant dimension (§III.36) Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular then gives the sub-M-polyfold, the tameness and the corner atlas. **No hypothesis is added**: good position at every zero is the text's hypothesis, and it supplies at every point the nearby good position that the re-centring needs, so the gap of Theorem 3.6.8 (1) (preprint 3.54) does not enter (item 156). $S \neq \emptyset$ is needed for the formal charts, whose inverses need a point of $S$ as a value off the chart (for $S = \emptyset$ the statement says nothing). "Tame strong bundle" is read as above. The text states no dimension; the theorem says that the dimension is locally constant, which the text's tangent germ property of a fixed dimension $d$ near each point implies but does not state (item 152).

**Corollary 3.1.27 (preprint 3.24)** Chapter3/GlobalIFTTheoremsG · corollary_3_1_27 is Theorem 3.6.11 (preprint 3.57) without the last clause: if at every zero $f'(x)$ is onto with kernel in good position, the solution set is a sub-M-polyfold with tame induced structure equivalent to a manifold with boundary with corners of locally constant dimension.

**Theorem 3.6.10 (preprint 3.56), $\partial X = \emptyset$, with the text's hypotheses** Chapter3/GlobalIFTTheoremsG · theorem_3_6_10. *Under the hypotheses of Theorem 3.6.11 (preprint 3.57) and $d_X \equiv 0$, $S$ is a sub-M-polyfold whose induced structure is equivalent to a manifold with corners of locally constant dimension and has degeneracy index $0$ every-*

*where*: Theorem 3.6.11 (preprint 3.57) and Lemma 2.4.6 (preprint 2.26), as for Theorem 3.1.22 (preprint 3.19). (The statement keeps only these two conclusions; the tameness of the induced structure and the regular data are those of Theorem 3.6.11 (preprint 3.57).) The text's hypothesis "$f'(x)$ surjective at every zero" is read as good position, which is the text's Definition 3.6.6 (preprint 3.53) for $d_X(x) = 0$; that condition (2) is void in a tame chart at $d_X = 0$, and Theorem 3.6.10 (preprint 3.56) with "$f'(x)$ surjective" and the tameness of the induced structure, are §III.45.

**The parts of fixed dimension** `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11_dim`. Under the hypotheses of Theorem 3.6.11 (preprint 3.57) (without $S \neq \emptyset$), let $S_n$ be the set of points of $S$ at which $S$ has data of Definition 3.2.1 (preprint 3.25) of dimension $n$ `Chapter3/GlobalIFTTheorems · solutionSetDim`, and $U_n$ the union of the sources of the regular data of dimension $n$ along which the dimension is constantly $n$ `Chapter3/GlobalIFTTheorems · solutionSetDimOpen`, an open set `Chapter3/GlobalIFTTheorems · isOpen_solutionSetDimOpen`. *Then $S$ is the union of the $S_n$, the $S_n$ are pairwise disjoint, $S_n = S \cap U_n$ and $S_n = S \cap F_n$ for a closed $F_n$ (so $S_n$ is open and closed in $S$), and each nonempty $S_n$ is a sub-M-polyfold whose induced structure is tame and equivalent to an $n$-dimensional manifold with boundary with corners. Proof.* At every $x \in S$, Theorem 3.6.8 (2) (preprint 3.54) with $V_{x_0} = X$ gives regular data $D_x$ of a dimension $d(x)$ with the local constancy. *The dimension at a point is well defined*: two data at $x$ of dimensions $n$, $m$ both have the dimension $d(x)$ by the local constancy along $D_x$, since $x$ lies in its source; so the $S_n$ cover $S$ (by $D_x$) and are pairwise disjoint. $S_n = S \cap U_n$: for $x \in S_n$, $d(x) = n$ and $D_x$ is one of the data whose sources make up $U_n$; conversely, if $x \in S$ lies in the source of regular data of dimension $n$ along which the dimension is $n$, then $d(x) = n$ and $D_x$ witnesses $x \in S_n$. *$S_n$ is closed in $S$*: $S_n = S \setminus \bigcup_{m \neq n} U_m$, since a point of $S$ lies in exactly one $S_m$; take $F_n$ the complement of that open union. *Each nonempty $S_n$*: at $x \in S_n$, restrict $D_x$ (of dimension $n$) to a datum for $S_n$ — $S_n \cap$ source $= S \cap$ source because every point of $S$ in the source has dimension $n$ — still regular `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`, and apply Theorem 3.2.3 (preprint 3.27) with fixed dimension $n$ `Chapter3/TangentGerm · theorem_3_2_3_of_isRegular`.

*Where the formal route departs from the text.* Two steps of the text's proof are replaced (items 150, 157): the regularity of the solution germ away from the centre, which the proof of Theorem 3.2.3 (preprint 3.27) asserts for any set and which fails for general sets (§III.24), is proved for solution sets by re-centring with the $C^1$-property of $\sigma$ on $V'$ and the level-wise lifts at the nearby zero (§§III.35–III.36); and the relative openness of $\Sigma$, which does not hold as we read it for an arbitrary good complement, is obtained by choosing the complement (§III.29). With these, the global theorems 3.56, 3.57 and Corollary 3.1.27 (preprint 3.24) are proved with the text's hypotheses, and the local Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) under the hypothesis that the zeros near the point are in good position, which is Theorem 3.6.8 (1) (preprint 3.54); this section proves that one under the filler condition at the nearby zeros, which Definition 3.1.4 (preprint 3.4) does not impose, and §III.43 with the text's hypotheses. Not stated here: the uniqueness of the manifold structure separately from the equivalence of atlases; the dimension $d$ of these statements is existentially quantified (only as the common dimension of all data of Definition 3.2.1 (preprint 3.25) for $S$ near the point) — its identification with $\dim \ker f'(x) = \operatorname{ind} f'(x)$ is §III.44, and $\ker f'(y) = T_y S$ is §III.45 in the chart and §III.62 in $T_y X$.

## III.38 The basic germ criterion: Theorem 3.7.1 (preprint 3.58)

**Theorem 3.7.1 (preprint 3.58)** (text): let $U$ be a relatively open neighbourhood of $0$ in a partial quadrant $C \subseteq E$ and $f : U \to F$ sc-smooth, with (1) $Df(x)$ sc-Fredholm at the smooth points, of constant index; (2) an sc-splitting $E = B \oplus X$, $B$ finite-dimensional containing $\ker Df(0)$, with a compactness property: for $b_j \to b$ in $B \cap U$ small and $(\eta_j)$ bounded in $X_m$ with $Df(b_j)\eta_j = y_j + z_j$, $y_j \to 0$ in $F_m$, $(z_j)$

bounded in $F_{m+1}$, $(\eta_j)$ has a convergent subsequence in $X_m$; (3) for every $m$ and $\varepsilon > 0$, $|[D_2 f(b,0) - D_2 f(b,x)]h|_m \le \varepsilon |h|_m$ for $b$ small and $x \in X_{m+1}$ small on level $m$. Then $f - f(0)$ is conjugated near 0 to a basic germ. The proof: $Y = Df(0)X$, $F = A \oplus Y$, $P$ the projection onto $Y$; Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) and an index argument make $L(b) = P \circ Df(b)|_X$ sc-isomorphisms with uniform lower bounds; Lemma 3.7.4 (preprint 3.61) makes $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ a strong bundle isomorphism; and $H(b,x) = x - L(b)^{-1}Ph(b,x)$, $h = f - f(0)$, is a contraction by (3).

**The coordinates.** The formalization works in the coordinates of the text's proof: $E = \mathbb{R}^n \oplus X$ with $\mathbb{R}^n$ a finite-dimensional normed space $P$ (constant sc-structure), $C = \widetilde{C} \oplus X$ with $\widetilde{C}$ a partial quadrant of $P$, $F = \mathbb{R}^N \oplus Y$ with $\mathbb{R}^N = Q$ finite-dimensional, and "$P\circ$" the $Y$-component. Condition (3) becomes `Chapter3/BasicGermCriterionProof · DerivCondition`: for every $m$ and $\varepsilon > 0$ there is $r > 0$ with $|Df(b,0)(0,h) - Df(b,x)(0,h)|_m \le \varepsilon |h|_m$ for $h \in X_m$, $b \in \widetilde{C}$, $|b| < r$, $x \in X_{m+1}$, $|x|_m < r$, $(b,0)$ and $(b,x)$ in $U_{m+1}$ (the level operators of Proposition 1.2.1 (preprint 1.16)). Condition (2) becomes `Chapter3/BasicGermCriterionIndex · CompactnessCondition`: there is $r > 0$ such that for every $m$, every $b_j \to b_0$ in $\widetilde{C}$ with $(b_j,0) \in U$ and $|b_0| < r$, every bounded $(\eta_j) \subseteq X_m$ with $Df(b_j,0)(0,\eta_j) = y_j + \iota z_j$, $y_j \to 0$ in $F_m$ and $(z_j)$ bounded in $F_{m+1}$, $(\eta_j)$ has a subsequence converging in $X_m$. (The smallness of $b$ is read as uniform in $m$, as the order of the text's quantifiers suggests; Lemma 3.7.3 (preprint 3.60) needs the property on every level on one neighbourhood.)

**Theorem 3.7.1 (preprint 3.58) as formalized** `Chapter3/BasicGermCriterionIndex · theorem_3_7_1`. Let $U$ be relatively open in $\widetilde{C} \oplus X$ with $0 \in U$ and $f : U \to Q \oplus Y$ sc-smooth such that (1) for every $b \in \widetilde{C}$ with $(b,0) \in U$ the linearization $Df(b,0)$, an sc-operator `Chapter3/BasicGermCriterionIndex · derivParamOp`, is sc-Fredholm with the sc-index of $Df(0)$; (2) $h \mapsto (Df(0)(0,h))_2$ is bijective $X_0 \to Y_0$, and the compactness condition holds; (3) the derivative condition holds. Then there are an open $V_o \ni 0$ in $P$ and sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ for $b \in V_o \cap \widetilde{C}$, such that the push-forward $k(b,x) = ((f - f(0))_1(b,x), L(b)^{-1}(f - f(0))_2(b,x))$ is a basic germ on $U \cap (V_o \oplus X)$ with contraction map $H(b,x) = x - L(b)^{-1}(f(b,x) - f(0))_2$, $\Phi$ is a strong bundle isomorphism covering the identity, and $f - f(0)$ is conjugated to a basic germ (Definition 3.1.11 (preprint 3.11)). The hypotheses differ from the text's in three places: (1) is asked only at the points $(b,0)$, $b \in \widetilde{C}$, which is all the proof uses; "constant index" is equality with the index at 0; and the text's "$B \supseteq \ker Df(0)$, $Y = Df(0)X$" is replaced by its consequence, the bijectivity of $P \circ Df(0)|_X$ on level 0. The reduction of a general sc-splitting $E = B \oplus X$, $F = A \oplus Y$ to these coordinates is §III.63.

The proof has two halves, as in the text.

**First half: Theorem 3.7.1 (preprint 3.58) from the family of sc-isomorphisms.**

**Theorem 3.7.1 (preprint 3.58), given the family** `Chapter3/BasicGermCriterionProof · theorem_3_7_1_of_family`. In addition to the coordinates, $f$ sc-smooth on $U$ and (3), suppose: $V_o$ open with $0 \in V_o$ and $(b,0) \in U$ for $b \in V_o \cap \widetilde{C}$; sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ on level 0 for $b \in V_o \cap \widetilde{C}$; and for every $m$ a constant $c_m > 0$ with $c_m |h|_m \le |L(b)h|_m$ for $b \in V_o \cap \widetilde{C}$, $h \in X_m$. Then, with $U' = U \cap (V_o \oplus X)$, the conclusions above hold.

*Step 1: $(b,h) \mapsto L(b)h$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus X$* `Chapter3/BasicGermCriterionProof · famMap_isScSmooth_of_deriv` (the text: "since $B$ is a finite-dimensional sc-smooth space, the map $(b,x) \mapsto P \circ Df(b)x$ is sc-smooth"). It is $\mathrm{pr}_Y \circ Tf \circ J$ with the sc-operator $J(b,h) = ((b,0),(0,h))$ into the tangent space `Chapter3/BasicGermCriterionProof · tangentInjOp`; $J$ maps into the tangent set because $(b,0) \in U$ lies on every level; and the tangent map of the sc-smooth $f$ is sc-smooth.

*Step 2: $(b,y) \mapsto L(b)^{-1}y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$* `Chapter3/BasicGermCriterionProof · famInv_isScSmooth_`

of. This is Lemma 3.7.4 (preprint 3.61) (§III.25) Chapter3/InverseFamily · lemma_3_7_4 for the germ $\{V_o \cap \widetilde{C}\}$ with the uniform lower bounds, translated between maps on sets and germs on the special germ of a set: an sc$^k$-germ on $\{U\}$ is an sc$^k$-map on $U$ Chapter3/BasicGermCriterionProof · IsSckGerm.isSck_ofSet Chapter3/BasicGermCriterionProof · IsScSmoothGerm.isScSmooth_ofSet, by induction on $k$: the sc$^0$ case is a matter of definitions Chapter3/BasicGermCriterionProof · IsSc0Germ.isSc0_ofSet; for the step, the tangent germ of $\{U\}$ is the germ of the tangent set, and the derivative family of the germ agrees with the derivative of the map on the tangent set; the converse is `IsSck.isSckGerm` (§III.1).

*Step 3: the strong bundle isomorphism* $\Phi$. For a family $M(b) : Y \to X$ of sc-isomorphisms put $\Phi_M((b,z),(c,y)) = ((b,z),(c,M(b)y))$ Chapter3/BasicGermCriterionProof · fiberIsoMap. If $(b,y) \mapsto M(b)y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$ and the parameters of $U'$ lie in $V_o \cap \widetilde{C}$, $\Phi_M$ is a strong bundle map $U' \lhd (Q \oplus Y) \to U' \lhd (Q \oplus X)$ covering the identity Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleMap: it is linear on the fibres; it preserves the filtration because $M(b)$ maps $Y_k$ to $X_k$; on $[0]$ it is sc-smooth as the pairing of the projection with $(c, M(b)y)$, the latter the composite of the sc-operator $((b,z),(c,y)) \mapsto (b,y)$ Chapter3/BasicGermCriterionProof · paramFiberOp with $(b,y) \mapsto M(b)y$ Chapter3/BasicGermCriterionProof · fiberIsoMap_isScSmooth; on $[1]$, $\Phi_M[1]$ is $\Phi_{M^1}$ for the shifted family $M^1(b) = M(b) : Y^1 \to X^1$ Chapter3/BasicGermCriterionProof · shiftFam Chapter3/BasicGermCriterionProof · shiftFiber_fiberIsoMap, and $(b,y) \mapsto M^1(b)y$ is sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter3/BasicGermCriterionProof · famMap_shiftFam_isScSmooth — the text's "if we raise the index by one, the map is also sc-smooth by Proposition 1.17". With $M = L^{-1}$ (Step 2) and $M = L$ (Step 1) the two maps are inverse to each other, so $\Phi = \Phi_{L^{-1}}$ is a strong bundle isomorphism Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleIso.

*Step 4: the push-forward and the contraction map.* With $h = f - f(0)$ Chapter3/BasicGermCriterionProof · shiftedGerm, $\Phi_* h = k$, $k(b,x) = (h_1(b,x), L(b)^{-1}h_2(b,x))$ Chapter3/BasicGermCriterionProof · pushGerm358, and the $X$-component of $k$ is $x - H(b,x)$ with $H(b,x) = x - L(b)^{-1}h_2(b,x)$ Chapter3/BasicGermCriterionProof · contrMap358 Chapter3/BasicGermCriterionProof · pushGerm358_snd. Both are sc-smooth on $U'$: sums, pairings and the composite of $(b,y) \mapsto L(b)^{-1}y$ with $(b,x) \mapsto (b, h_2(b,x))$. On level $m$, $H_m(b,w) = w - L(b)^{-1}_m((f_m(b,w))_2 - (f_m(0))_2)$ Chapter3/BasicGermCriterionProof · levelMap_contrMap358, and $|L(b)^{-1}_m| \le c_m^{-1}$ Chapter3/BasicGermCriterionProof · norm_symm_le.

*Step 5: the contraction estimate* Chapter3/BasicGermCriterionProof · contrMap358_contraction. For every $m$ and $\varepsilon > 0$ there is $r > 0$ such that $(a,w) \in U'_m$ and $|H(a,w) - H(a,w')|_m \le \varepsilon |w - w'|_m$ for $a \in \widetilde{C}$, $|a| < r$, $w, w' \in X_m$ small on level $m$. Choose $r$ below the radius $r_3$ of (3) for $\varepsilon c_m$, below the radius of a ball around 0 in $V_o$, and so small that $(a,z) \in U_m$ whenever $|a| < r$ and $|z|_m < r$ ($U = V \cap C$ with $V$ open; the level-$m$ norm of $z$ controls its level-0 norm through the norm of the inclusion) Chapter3/BasicGermCriterionProof · mem_lvlSet_of_small. First let $w_1, w_1' \in X_{m+1}$ with $|w_1|_m, |w_1'|_m < r$. On the convex set $S = \{z \in X_{m+1} : |z|_m < r\}$ the map $g(z) = f_m(a, \iota z)$ has the derivative $Df(a,z)(0,\iota\,\cdot)$ (Proposition 1.2.1 (preprint 1.16) on level $m$) Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level, and by (3) $|Df(a,z)(0,v) - Df(a,0)(0,v)|_m \le \varepsilon c_m |v|_m$. **The mean value inequality with the derivative bound in the weaker norm** Chapter3/BasicGermCriterionProof · norm_sub_sub_le_of_hasFDerivWithinAt_segment — the one-dimensional mean value inequality for $t \mapsto g(w_1' + t(w_1 - w_1')) - T\iota(w_1' + t(w_1 - w_1'))$ on $[0,1]$, which measures the increment in the norm of $X_m$ rather than of $X_{m+1}$, as condition (3) requires — gives

$$|g(w_1) - g(w_1') - T\iota(w_1 - w_1')|_m \le \varepsilon c_m |\iota(w_1 - w_1')|_m, \qquad T = Df(a,0)(0,\cdot).$$

The $Y$-component of $T$ is $L(a)_m$ Chapter3/BasicGermCriterionProof · snd_levelDeriv_eq_L (the level-$m$ operator of Proposition 1.2.1 (preprint 1.16) restricts the level-0 one). Then $H(a,w_1) - H(a,w_1') = L(a)^{-1}[L(a)(w_1 - w_1') - (f(a,w_1) - f(a,w_1'))_2]$, whose norm is at most $c_m^{-1}\varepsilon c_m|w_1 - w_1'|_m$. This is the text's integral $\int_0^1 D_2 f(b, tx + (1-t)x')(x - x')\,dt$, made precise. For general $w, w' \in X_m$: approximate them by sequences from $X_{m+1}$ (dense in $X_m$), small on level $m$ for large indices, and pass to the limit with the continuity of $H_m$ on $U_m'$ — the text's "by the density of $X_{m+1}$ in $X_m$".

*Step 6: the basic germ and the conjugation.* $k$ is a basic germ on the germ $\{U'\}$ with contraction map $H$ (Definition 3.1.7 (preprint 3.7) as formalized Chapter3/LocalCompactness · IsBasicGerm): it is an sc-smooth germ (Step 4), $k(0) = 0$, its $X$-component is $x - H(b,x)$, and $H$ is an sc$^0$-contraction germ — sc$^0$, $H(0) = 0$ since $L(0)^{-1}0 = 0$, and the contraction estimate on every level with every $\varepsilon$ (Step 5). The conjugation identity $(\Phi(u, h(u)))_2 = k(u)$ holds on all of $U'$, because $\Phi$ covers the identity; with $\Phi$ from Step 3 these are the data of Definition 3.1.11 (preprint 3.11) for $h = f - f(0)$.

**Second half: the family from conditions (1) and (2).**

*The family.* At the smooth point $(b,0)$ the linearization $Df(b,0)$ is an sc-operator Chapter3/BasicGermCriterionIndex · derivParamOp Chapter1/Properties · IsSc1With.scOperator_deriv whose level maps are the level operators of Proposition 1.2.1 (preprint 1.16) Chapter3/BasicGermCriterionIndex · derivParamOp_lvl; $L(b) = \mathrm{pr}_Y \circ Df(b,0) \circ \mathrm{inr}_X$ is an sc-operator $X \to Y$ (set to 0 where $(b,0) \notin U$) Chapter3/BasicGermCriterionIndex · derivFamOp; and $(b,h) \mapsto L(b)h$ is jointly continuous on $O \times X_m$ for parameters with $(b,0) \in U$, by the continuity of the level operators (Proposition 1.2.1 (2) (preprint 1.16)) Chapter3/BasicGermCriterionIndex · continuousOn_levelDeriv_param Chapter3/BasicGermCriterionIndex · continuousOn_derivFamOp.

*The compactness property of the family* Chapter3/BasicGermCriterionIndex · familyCompactness_derivFamOp. Let $O$ be a set of parameters in $\widetilde{C}$ with $(b,0) \in U$ and $|b| < r$ ($r$ from (2)), $b_j \to b_0$ in $O$, $(h_j)$ bounded in $X_m$, $L(b_j)h_j \to 0$. Write $Df(b_j,0)(0,h_j) = (0, L(b_j)h_j) + ((Df(b_j,0)(0,h_j))_Q, 0)$. The first summand tends to 0 in $F_m$. The second lies in $Q \oplus 0$, where all levels coincide, and is bounded: a jointly continuous family of operators is uniformly bounded near a parameter Chapter3/BasicGermCriterionIndex · exists_bound_of_continuousOn_clm (continuity at $(b_0,0)$, then scaling), so $|Df(b_j,0)(0,\cdot)|$ is bounded for $j$ large. Condition (2) applied to the tails of the sequences gives a convergent subsequence: this is the compactness property of the abstract Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) (§III.25) Chapter3/BasicGermCriterion · FamilyCompactness. The text's "$z_j = Df(b_j)h_j \in A$ is bounded on every level since $A$ is a smooth subspace" omits the uniform operator bound.

*Lemma 3.7.2 (preprint 3.59).* Let $V' = \{b : (b,0) \in V\}$ ($U = V \cap C$) and $O = V' \cap B(0,r) \cap \widetilde{C}$, which contains 0. $L(0)$ is injective on $X_0$ by (2), so Lemma 3.7.2 (preprint 3.59) Chapter3/BasicGermCriterion · lemma_3_7_2 gives $r_1 > 0$ with $L(b)$ injective on $X_0$ for $b \in O$, $|b| < r_1$, hence on every level (the inclusions are injective and commute with $L(b)$) Chapter3/BasicGermCriterionIndex · ScOperator.injective_lvl.

*Lemma 3.7.3 (preprint 3.60).* Take a closed ball $\bar{B}(0, r_1')$ inside $V' \cap B(0,r) \cap B(0,r_1)$ and $K = \bar{B}(0,r_1') \cap \widetilde{C}$, compact (since $\widetilde{C}$ is closed) and inside $O$. On $K$ every $L(b)_m$ is injective, the family is jointly continuous and has the compactness property (restricted from $O$ Chapter3/BasicGermCriterionIndex · FamilyCompactness.mono), so Lemma 3.7.3 (preprint 3.60) Chapter3/BasicGermCriterion · lemma_3_7_3 gives $c_m > 0$ with $c_m|h|_m \le |L(b)h|_m$ for $b \in K$.

*The index argument.* The text: "$P \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism". The formal proof uses a block computation instead. **The index of a block operator** Chapter3/BasicGermCriterionIndex · fredholmIndex_block: for a Fredholm $T : P \oplus X \to Q \oplus Y$ with $P$, $Q$

finite-dimensional and the block $L = \mathrm{pr}_Y \circ T \circ \mathrm{inr}_X$ of finite index, $\operatorname{ind} T = \dim P + \operatorname{ind} L - \dim Q$. Indeed $T - \mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is $(p,x) \mapsto T(p,0) + ((T(0,x))_Q, 0)$, of finite rank, hence compact, so $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is Fredholm with the index of $T$ `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator` `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`; and the indices of $\mathrm{pr}_X$ and $\mathrm{inr}_Y$ are $\dim P$ and $-\dim Q$ `Chapter3/BasicGermCriterionIndex · hasFiniteIndex_snd_and_linIndex` `Chapter3/BasicGermCriterionIndex · hasFiniteIndex_inr_and_linIndex`, which add under composition `Chapter1/FredholmIndex · linIndex_comp`. At $b = 0$, $L(0)$ is bijective on level $0$, of index $0$, so $\operatorname{ind} Df(0) = \dim P - \dim Q$ on level $0$, which is the sc-index (an sc-Fredholm operator has the same index on every level) `Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl`. **Surjectivity from the index** `Chapter3/BasicGermCriterionIndex · surjective_of_fredholmIndex_block`: if $\operatorname{ind} T = \dim P - \dim Q$ and the block $L$ is injective, then $L$ has finite index — its cokernel is a quotient of the cokernel of the Fredholm operator $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ — and index $0$ by the block formula, so it is surjective. For $b \in K$ and every $m$, $Df(b,0)_m$ is Fredholm of index $\operatorname{ind} Df(0) = \dim P - \dim Q$ by (1) and the level-wise index, and $L(b)_m$ is injective; so $L(b)_m$ is bijective, a topological isomorphism by the open mapping theorem, and the $L(b)$, $b \in K$, are sc-isomorphisms. The sc-Fredholm property of the projection $P$ and the composition rule for sc-Fredholm operators are not needed.

*Conclusion.* Extend $b \mapsto L(b)$ outside $K$ by $L(0)$, put $V_o = B(0, r_1')$ (so $V_o \cap \widetilde{C} \subseteq K$), and apply the first half with the lower bounds of Lemma 3.7.3 (preprint 3.60). The neighbourhood replacement $U' = U \cap (V_o \oplus X)$ is the text's "replacing, if necessary, $U$ by a smaller neighbourhood".

## III.39 What the addendum taught us

1. **The global implicit function theorems hold with the text's hypotheses.** Theorem 3.6.11 (preprint 3.57), Corollary 3.1.27 (preprint 3.24) and Theorem 3.6.10 (preprint 3.56) are proved for an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with good position at every zero `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11` `Chapter3/GlobalIFTTheoremsG · corollary_3_1_27` `Chapter3/GlobalIFTTheoremsG · theorem_3_6_10` (§III.37). Theorem 3.6.10 (preprint 3.56) reads "$f'(x)$ surjective" as good position, which the text's Definition 3.6.6 (preprint 3.53) makes the same thing at points with $d_X(x) = 0$; that the kernel condition is void in the chart form was not formalized when this was written; it is now (§III.45).
2. **Two steps of the text's proof had to be replaced, and both repairs are specific to solution sets.** The regularity of the solution germ away from the centre, which the proof of Theorem 3.2.3 (preprint 3.27) asserts for any set and which fails in general (§III.24), follows for two graph descriptions of one set from the $C^1$-property of the first and the level-wise lifts of the second, by the re-centring lemma `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` (§III.35). The relative openness of $\Sigma$, which does not hold as we read it for an arbitrary good complement, holds for a complement whose projection maps the partial quadrant into itself, and the proof of Theorem 3.6.8 (preprint 3.54) can produce such a complement from the start `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto` (§III.29). Read off the statement of the regularity theorem `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`, the first repair works for every set with the tangent germ property whose data satisfy the second regularity condition (§III.36).
3. **The proof of the local Theorem 3.6.8 (1) (preprint 3.54) has a gap, and it is the same gap as in the book's arXiv version.** At a nearby zero, Corollary 3.5.3 (preprint 3.41) gives the surjectivity of the filled linearization; the surjectivity of $f'(y)$ needs condition (3) of Definition 3.1.4 (preprint 3.4) at that zero, which the definition imposes at the centre only `Chapter3/GlobalIFTNearbySurj · exists_range_`

scDeriv_eq_ρ_of_normal_injective. The local theorems were therefore first proved under that condition at the nearby zeros Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one (§III.33); the kernel half needs nothing (§III.32). The global theorems do not need it, because their hypothesis already gives good position at every zero. The second addendum proves the surjectivity with Definition 3.1.4 (preprint 3.4) as stated (§§III.41–III.43).

4. **Definition 3.6.6 (preprint 3.53) presupposes a chart independence the text does not prove.** It holds at zeros, because only the fibre part of a chart transition survives there Chapter3/LinearizationCharts · linAt_chart_change (§III.31), and it is what lets the good position found in the chart at $x_0$ be transported to the chart centred at a nearby zero.
5. **Three silent steps are theorems of their own**: tameness is carried by sc-diffeomorphisms and by open subsets Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image (§III.27); the kernel of the linearization of a basic germ plus an $\mathrm{sc}^+$-germ consists of smooth points Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth (§III.34); and the index raise of a strong bundle isomorphism is one Chapter3/BasicClassComposition · IsStrongBundleIso.shift (§III.28).
6. **The dimension of the solution set is locally constant, not constant**, and the dimension count that proves the re-centring is also what shows it (Sard's lemma in fixed dimension Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty, §III.35). Theorem 3.2.3 (preprint 3.27) is therefore also proved with a dimension that may vary Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular (§III.36), and the solution set of Theorem 3.6.11 (preprint 3.57) splits into open and closed parts of fixed dimension Chapter3/GlobalIFTTheoremsG · theorem_3_6_11_dim.
7. **Proposition 3.6.5's (preprint 3.52) index needs the conjugation as data.** With the witnesses of the conjugation named, the index of $Tf(x_0)$ is the $n - N$ of every conjugating basic germ, and that number does not depend on the conjugation Chapter3/FredholmGermIndexG · proposition_3_6_5_index Chapter3/FredholmGermIndexG · ConjugationData.index_eq (§III.30).
8. **The quadrant inverse function theorem is not the right tool away from the centre**: it needs the degeneracy index to be preserved, which for the solution set is unknown; Lemma 3.8.21's (preprint 3.77) bootstrapping with the hypotheses its proof uses suffices Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv (§III.35).
9. **The basic germ criterion is proved in the coordinates of its proof**, with condition (1) needed only along the finite-dimensional directions and the text's index argument replaced by a block index formula Chapter3/BasicGermCriterionIndex · fredholmIndex_block Chapter3/BasicGermCriterionIndex · theorem_3_7_1 (§III.38); the form as printed reduces to it (§III.63).
10. **What was left of Chapter 3** after this addendum: the surjectivity in Theorem 3.6.8 (1) (preprint 3.54) with Definition 3.1.4 (preprint 3.4) as stated; in Theorem 3.1.22 (preprint 3.19), the dimension as the Fredholm index and $\ker f'(y) = T_y S$; Proposition 3.6.5 (preprint 3.52) as an sc-Fredholm operator on $T_x X$; Theorem 3.7.1 (preprint 3.58) for a general sc-splitting; Theorem 3.1.21 (preprint 3.18); Lemma 3.8.13 (preprint 3.70) for a general presentation of the partial quadrant; and Theorem 3.2.3 (preprint 3.27) for sets whose data violate the second regularity condition. The second addendum (§§III.40–III.46) settles the first three items, the third in charts, and Theorem 3.1.21 (preprint 3.18). The rest is closed since: §§III.56, III.57, III.61, III.62, III.63.

## III.40 Theorem 3.1.21 (preprint 3.18): parametrized perturbations

The text states Theorem 3.1.21 (preprint 3.18) after a five-line argument: the section $\tilde{f}(r,x) = ((r,x), f(x))$ of the pull-back bundle $\pi^*P \to \mathbb{R}^n \times X$, $\pi(r,x) = x$, "is a sc-Fredholm section as is readily verified", $\tilde{s}(r,x) = ((r,x), \sum_i r_i s_i(x))$ is an sc$^+$-section, and Theorem 3.1.18 (preprint 3.15) adds them. The formalization follows this route. What it has to build first is everything the five lines take for granted: $\mathbb{R}^n$ as an M-polyfold with enough charts, the pull-back bundle as a strong bundle, the sc-calculus for a scalar factor, the product chart with its filler, and the extension of a basic germ by $n$ inert parameters (Findings 160–167). Tameness is not used anywhere, as for Theorem 3.1.18 (preprint 3.15).

**$\mathbb{R}^n$ and $\mathbb{R}^n \times X$.** $\mathbb{R}^n$ carries the constant sc-structure `Chapter3/ParamPerturbation · rnSc` and the local model $(\mathbb{R}^n, \mathbb{R}^n, \mathbb{R}^n)$ `Chapter3/ParamPerturbation · rnModel`. For every $a$ the translation $r \mapsto r - a$ is a chart `Chapter3/ParamPerturbation · rnChart`; two translation charts are compatible, because their transition maps are translations and a $C^\infty$ map between finite-dimensional spaces is sc-smooth for the constant structures `Chapter3/ParamPerturbation · isScSmoothRetractMap_const_of_contDiff Chapter3/ParamPerturbation · rnChart_compatible`. The M-polyfold $\mathbb{R}^n$ has the identity chart as its atlas `Chapter3/ParamPerturbation · rnPolyfold`, and every translation chart is a chart of it `Chapter3/ParamPerturbation · isChart_rnChart`. These charts are needed: Definition 3.1.11 (preprint 3.11) asks for a chart *centred* at the smooth point $(r_0, x)$, and the product of the identity of $\mathbb{R}^n$ with a chart of $X$ is centred only at $r_0 = 0$. $\mathbb{R}^n$ is tame `Chapter3/ParamPerturbation · rnPolyfold_isTame` and every point lies on every level `Chapter3/ParamPerturbation · rnPolyfold_levelSet`; so $(r,x)$ is on level $m$ of the product iff $x$ is `Chapter3/ParamPerturbation · prod_rn_mem_levelSet_iff`, smooth iff $x$ is `Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff`, and $\mathbb{R}^n \times X$ is tame when $X$ is `Chapter3/ParamPerturbation · prod_rn_isTame` — which the text needs, since Definition 3.1.16 (preprint 3.14) is stated for tame bundles.

**Two rules of the sc-calculus.** Scalar multiplication $\mathbb{R} \oplus F \to F$ is continuous and bilinear on every level, hence sc-smooth by Proposition 1.2.4 (preprint 1.19) `Chapter3/ParamPerturbation · isScSmooth_smul`; composed with a pairing, $x \mapsto a(x)v(x)$ is sc-smooth for sc-smooth $a : U \to \mathbb{R}$ and $v : U \to F$ `Chapter3/ParamPerturbation · IsScSmooth.smul`; finite sums of sc-smooth maps are sc-smooth `Chapter3/ParamPerturbation · IsScSmooth.finsetSum`. The coordinate $r \mapsto r_i$ is an sc-operator `Chapter3/ParamPerturbation · rnCoordOp`. Chapter 1 states neither rule.

**The pull-back bundle.** Proposition 2.6.8 (preprint 2.54) is formalized as an existence statement `Chapter2/StrongBundlePullback · proposition_2_6_8`; here it is made a structure, the strong bundle $\pi^*P$ with the atlas of all pulled-back charts `Chapter3/ParamPerturbation · StrongBundle.pullback`, whose total space is paracompact by Theorem 2.3.10 (preprint 2.16) `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace`. A pulled-back chart is made from a chart $\psi$ of the base $Z$ around $z$ and a strong bundle chart $c$ of $P$ around $\pi(z)$, with $\pi$ mapping the domain of $\psi$ into that of $c$ `Chapter3/ParamPerturbation · exists_pullbackChartData`; such a chart is a chart of $\pi^*P$ as soon as $c$ is merely compatible with the atlas of $P$ `Chapter3/ParamPerturbation · PullbackChartData.isChart_toChart`. A map $\tau : Z \to Y$ over $\pi$ is a section $z \mapsto (z, \tau(z))$ of $\pi^*Y$ `Chapter3/ParamPerturbation · pullbackLift Chapter3/ParamPerturbation · isSection_pullbackLift`, whose value in a pulled-back chart is $(\psi(z), \mathrm{pr}_F \Phi(\tau(z)))$ `Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift`. For $\tau = \sigma \circ \pi$ the representative is $w \mapsto (w, \hat{\sigma}(g(w)))$ with $g = \varphi \circ \pi \circ \psi^{-1}$, which maps the retract of $\psi$ into that of $\varphi$ `Chapter3/ParamPerturbation · PullbackChartData.g_mem_0 Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift_comp`; so the pull-back of an sc-smooth section along an sc-smooth map is sc-smooth `Chapter3/ParamPerturbation · StrongBundle.IsScSmoothSection.pullback` (Finding 167).

**The three sections.** $F(r,x) = f(x) + \sum_i r_i s_i(x)$ is computed in the fibre $P^{-1}(x)$, from the

fibre combination $\sum_i r_i s_i(x)$ Chapter3/ParamPerturbation · fiberComb, and read in the fibre of $\pi^*Y$ over $(r,x)$ Chapter3/ParamPerturbation · paramSection; $\tilde{f}$ Chapter3/ParamPerturbation · paramLift and $\tilde{s}$ Chapter3/ParamPerturbation · paramComb are the text's sections, and $F = \tilde{f} + \tilde{s}$ in the fibres of $\pi^*Y$ Chapter3/ParamPerturbation · paramSection_eq_sectionAdd.

**$\tilde{s}$ is an sc⁺-section** Chapter3/ParamPerturbation · StrongBundle.isScPlusSection_paramComb. The atlas of $\pi^*P$ contains the charts pulled back through *every* chart $\psi$ of $\mathbb{R}^n \times X$, not only product charts, so the check must work for an arbitrary $\psi$. In a pulled-back chart, by the linearity of $\Phi$ on the fibre over $x = \pi(z)$, $\tilde{s}(z)$ has the value $(\psi(z), \sum_i r_i \hat{s}_i(x))$ Chapter3/ParamPerturbation · PullbackChartData.toChart_paramComb, so the principal part is $w \mapsto \sum_i \lambda_i(w)\hat{s}_i(g(w))$ with $\lambda(w) = (\psi^{-1}(w))_1$. The map $\lambda$ is sc-smooth on the retract of $\psi$: it is the projection $\mathbb{R}^n \times X \to \mathbb{R}^n$ (an sc-smooth map of M-polyfolds) read in $\psi$ and in the identity chart of $\mathbb{R}^n$ Chapter3/ParamPerturbation · isScSmooth_chart_coord. Each $\hat{s}_i \circ g$ is sc-smooth into $F^1$, $s_i$ being sc⁺. The product rule and the sum rule conclude.

**The product chart and its filler.** For a strong bundle chart $c$ of $P$ and $r_0 \in \mathbb{R}^n$, the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd is $\psi = (r \mapsto r - r_0) \times \varphi$ on $\mathbb{R}^n \times V$, with the retraction $1 \times r$ of $\mathbb{R}^n \times U$ onto $\mathbb{R}^n \times O$; there $g = \varphi \circ \pi \circ \psi^{-1}$ is the projection $(a,u) \mapsto u$. *Filler conditions.* If $g$ fills $\hat{f}$ in $c$, then $(a,u) \mapsto g(u)$ fills $(a,u) \mapsto \hat{f}(u)$ in the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd_isFillerOf. Conditions (1) and (2) are those of $g$, because the fibre projection of the product retraction at $(a,u)$ is $\rho(r(u))$ — which uses $\varphi(\varphi^{-1}(r(u))) = r(u)$ on $O$. Condition (3): $D(1 \times r)(0) = 1 \times Dr(0)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_scDeriv_r, so its kernel is $0 \times \ker Dr(0)$; the fibre projection at 0 is $\rho(0)$ (from $r(0) = 0$); and $[1 - \rho(r(u))]g(u)$ does not depend on $a$, so its derivative at 0 on $0 \times \ker Dr(0)$ is that of $g$. *For a filling germ $g$* on $G$ the same holds on the product germ $\mathbb{R}^n \times G$, with levels $\mathbb{R}^n \times G_m$, unbounded in the parameter direction, which Definition 3.1.1 (preprint 3.1) allows Chapter3/GermCalculusParam · ScNbhdGerm.univProd: $(a,u) \mapsto g(u)$ is a germ on it with the derivative family $Dg(u) \circ \mathrm{pr}_2$ Chapter3/GermCalculusParam · IsSc1GermWith.univProd, and the fill defect in the product chart is the pull-back along $\mathrm{pr}_2$ of that of $g$, so condition (3) is the kernel identification just made, without a uniqueness argument Chapter3/GermCalculusParam · PullbackChartData.ofProd_isFillingGerm. *Principal parts* ($a = r - r_0$ the chart coordinate): $\tilde{s}$ has $(a,u) \mapsto \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramComb and $F$ has $(a,u) \mapsto \hat{f}(u) + \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramSection, the text's $(r,u) \mapsto \hat{f}(u) + \sum_i r_i \hat{s}_i(u)$.

**Inert parameters.** Let $\Lambda$ be finite-dimensional. The product $1_\Lambda \times \Phi$, $((a,u),k) \mapsto ((a,\varphi(u)),\Gamma(u,k))$, of a map $\Phi$ of $U \lhd F$ with the identity of $\Lambda$ Chapter3/ParamPerturbation · paramMap is a strong bundle map when $\Phi$ is: on $[0]$ and $[1]$ it is $1 \times \Phi$ up to the reassociations of the factors, which are sc-operators Chapter3/ParamPerturbation · ScBanach.assocOp Chapter3/ParamPerturbation · ScBanach.assocInvOp Chapter3/ParamPerturbation · IsStrongBundleMap.paramMap; and a strong bundle isomorphism, with inverse $1_\Lambda \times \Phi'$, when $\Phi$ is Chapter3/ParamPerturbation · IsStrongBundleIso.paramMap. The $\Lambda$-extension of a germ of neighbourhoods is $U''_m = \{((a,\lambda),w) \mid (a,w) \in U_m\}$ Chapter3/ParamPerturbation · ScNbhdGerm.paramExtend, unbounded in $\lambda$, which Definition 3.1.1 (preprint 3.1) allows. *The $\Lambda$-extension of a basic germ is basic* Chapter3/ParamPerturbation · IsBasicGerm.paramExtend: if $h(a,w) = (h_1(a,w), w - B(a,w))$ is basic on $\widetilde{C} \oplus W$, then $h'((a,\lambda),w) = h(a,w)$ is basic on $(\widetilde{C} \times \Lambda) \oplus W$, with $B'((a,\lambda),w) = B(a,w)$: it is the composite of $h$ with the projection forgetting $\lambda$ Chapter3/ParamPerturbation · paramForgetOp, and the contraction estimate of Definition 3.3.1 (preprint 3.28) holds with the same radius since $|a| \le |(a,\lambda)|$. *The $\Lambda$-extension of a conjugation* Chapter3/ParamPerturbation · IsConjugateToBasicGerm.paramExtend: if $k$ is conjugated to $h$ by $\Psi$, then $(\lambda,u) \mapsto k(u)$ is conjugated to $h'$ by $\Psi''((\lambda,u),k) = ((((\psi(u))_1,\lambda),(\psi(u))_2),\Gamma(u,k))$,

the composite Chapter3/BasicClassComposition · IsStrongBundleIso.comp of $1_\Lambda \times \Psi$ with the linear reassociation $(\lambda,(a,w)) \mapsto ((a,\lambda),w)$ Chapter3/ParamPerturbation · paramSwapOp Chapter3/ParamPerturbation · paramSwapInvOp, a strong bundle isomorphism as a pair of sc-isomorphisms Chapter3/BasicClassComposition · ScOperator.isStrongBundleIso_prodMap. The parameter quadrant is $\widetilde{C} \times \Lambda$ in $P \times \Lambda$, put second so as to reuse the product of a partial quadrant with a whole space; the order is immaterial (Finding 163).

**$\tilde{f}$ is an sc-Fredholm section** Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSection.param. *Sc-smooth*: it is the pull-back of $f$ along the projection. *Regularizing* Chapter3/ParamPerturbationG · StrongBundle.regularizing_paramLift: if $(r,x)$ is on level $m$ and $\tilde{f}(r,x)$ on level $(m,m+1)$ of $\pi^*Y$, then $x \in X_m$ and $f(x) \in Y_{m,m+1}$ — tested in the pulled-back charts, whose fibre parts are those of $P$ — so $x \in X_{m+1}$ because $f$ is regularizing, and $(r,x)$ is on level $m+1$. *The germ at a smooth point* $(r_0,x)$ Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmGerm.param: $x$ is smooth, so Definition 3.1.11 (preprint 3.11) at $x$ gives a centred chart $c$, a filling germ $g$ on $G_g$, an sc$^+$-germ $t_0$ on $G_{t_0}$ with $t_0(0) = g(0)$ and a conjugation of $g - t_0$ to a basic germ. In the product chart at $(r_0,x)$, which is centred there, the principal part of $\tilde{f}$ is $(a,u) \mapsto \hat{f}(u)$, its filling germ $(a,u) \mapsto g(u)$ on $\mathbb{R}^n \times G_g$ (the filler conditions above), the sc$^+$-germ to subtract $(a,u) \mapsto t_0(u)$ on $\mathbb{R}^n \times G_{t_0}$ Chapter3/GermCalculusParam · IsScSmoothGerm.univProd, and $(a,u) \mapsto (g-t_0)(u)$ is conjugated to the $\mathbb{R}^n$-extension of the basic germ; the conjugation concerns $g-t_0$ near 0 in $U$, and $1 \times \Psi$ and the extended basic germ are whole-chart objects, so no germ calculus enters it. This is the "readily verified".

**Theorem 3.1.21 (preprint 3.18)** Chapter3/ParamPerturbationG · theorem_3_1_21_preprint. *Statement.* For a strong bundle $P : Y \to X$, an sc-Fredholm section $f$ and sc$^+$-sections $s_1, \dots, s_n$, the section $F$ is an sc-Fredholm section of $\pi^*P$. *Proof.* $F = \tilde{f} + \tilde{s}$; $\tilde{f}$ is sc-Fredholm and $\tilde{s}$ is sc$^+$; Theorem 3.1.18 (preprint 3.15) Chapter3/FredholmSectionsG · theorem_3_1_18_preprint. The paracompactness of $\mathbb{R}^n \times X$ and of $\pi^*Y$ (Definitions 2.3.4, 2.6.5 (preprint 2.13, 2.51)) enters as instance arguments; the version without them supplies both by Theorem 2.3.10 (preprint 2.16) Chapter3/ParamPerturbationG · theorem_3_1_21' Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod. The text's sketch of the germ of $F$ — filled version $(r,u) \mapsto g(u) + \sum_i r_i \hat{s}_i(r(u))$, sc$^+$-part $t_0(u) + \sum_i r_i \hat{s}_i(r(u))$, conjugation $\Psi'(r,u,k) = (r - r_0, \Psi(u,k))$ — is exactly what Theorem 3.1.18 (preprint 3.15) produces from the germ data of $\tilde{f}$ and the principal part of $\tilde{s}$ Chapter3/FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd; the translation $r - r_0$ sits in the chart rather than in the conjugation (Finding 164).

**The linearization** (after Theorem 3.1.22 (preprint 3.19) in the text, used for the finite-dimensional reduction and for transversality). In the product chart at $(r_0,x)$, for a chart $c$ with $0 \in O$, the derivative of $\hat{F} \circ (1 \times r)$ at 0 is

$$(\delta a, \delta u) \mapsto D(\hat{G} \circ r)(0)\delta u + \sum_i \delta a_i \hat{s}_i(0), \qquad \hat{G} = \hat{f} + \sum_i r_{0,i} \hat{s}_i$$

Chapter3/ParamPerturbationG · scDeriv_paramSection: the product rule at $a = 0$, where the terms $\sum_i a_i D(\hat{s}_i \circ r)$ vanish. The tangent space of the product retract at 0 lies in $\mathbb{R}^n \times T_0O$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_mem_tangent, and in the chart form of the linearization (§III.31) $\mathrm{Lin}_F(0)(\delta a, \delta u) = \mathrm{Lin}_{f+\sum r_{0,i}s_i}(0)\delta u + \sum_i \delta a_i \hat{s}_i(0)$ Chapter3/ParamPerturbationG · linAt_paramSection. At $r_0 = 0$ this is the text's formula $\tilde{f}'(0,x)(h,u) = f'(x)u + \sum_i h_i e_i$ Chapter3/ParamPerturbationG · linAt_paramSection_zero. The formula is the derivative at the centre of a chart and does not use $F(r_0,x) = 0$; at a general $(r_0,x)$ its first term is the linearization of $f + \sum r_{0,i}s_i$, not $f'(x)$ (Finding 165). The text's "sc$^1$-sections $s_i$" (p. 101 (preprint p. 75)) must be sc$^+$ for Theorem 3.1.21 (preprint 3.18) to apply (Finding 166).

# III.41 Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses: the plan and the elementary lemmas

§III.33 showed what the text's argument for property (4) of Theorem 3.6.8 (preprint 3.54) leaves open: at a zero $z_0 = \varphi(y)$ near 0, in the chart at $x_0$, the filled operator $Dg(z_0)$ is onto (Corollary 3.5.3 (preprint 3.41)), but $f'(y) = Dg(z_0)|_{T_{z_0}O}$ is onto only if condition (3) of the filler holds at $z_0$, which Definition 3.1.4 (preprint 3.4) does not ask; and the perturbation argument of the text's Proposition 6.2.2 (preprint 6.2) gives $\rho(z_0)\Lambda_0 \subseteq$ range $f'(y)$ only on a neighbourhood depending on the finite-dimensional $\Lambda_0$. This section and the next two prove the surjectivity anyway, with the text's hypotheses, by a new argument found while formalizing (Findings 180–189). The text's statement is true; its proof, and that of the book's arXiv version, is incomplete.

**The plan.** By the assembly step of §III.31 it suffices to show, in the chart at $x_0$ and for every zero $z_0$ of $g$ in $O$ near 0, that $\rho(z_0)k \in Dg(z_0)(T_{z_0}O)$ for every smooth $k \in F_\infty$; the closed range of $f'(y)$ (Proposition 3.6.5 (preprint 3.52) at $y$) does the rest. Fix $z_0$ and $k$. Suppose $u(\mu)$, $0 \le \mu < \mu_0$, is a curve through $u(0) = z_0$ of zeros of the *perturbed filler* $g + \mu\rho(r(\cdot))k$, right differentiable at 0 into $E_1$ with velocity $\xi$. The curve lies in $O$ (the first lemma below), so $r(u(\mu)) = u(\mu)$ and $Dr(z_0)\xi = \xi$, i.e. $\xi \in T_{z_0}O$; differentiating $g(u(\mu)) = -\mu\rho(r(u(\mu)))k$ at $\mu = 0$ gives $Dg(z_0)\xi = -\rho(z_0)k$. So one curve for each pair $(z_0, k)$ suffices. It is produced by an implicit function theorem *at the nearby zero*, in the normal form of Theorem 3.4.2 (preprint 3.38): its infinite-dimensional part is Theorem 3.3.3 (preprint 3.30) for a contraction germ extended by the parameter $\mu$, applied at the centre, whose regions in the parameter $a$ do not depend on $k$ (§III.42); its finite-dimensional part is a one-sided implicit function theorem at the image of $z_0$, in the direction of an interior vector of the partial quadrant that good position provides (this section and §III.43). The set of zeros to which the argument applies is cut out by conditions on the data at $x_0$ alone. This section proves the elementary lemmas, with no sc-structure involved.

**Zeros of the perturbed filler lie in $O$** `Chapter3/GlobalIFTNearbyOntoLemmas · StrongRetr.mem_O_of_perturbed_zero`. *Statement.* Let $R$ be a strong bundle retraction on $U \lhd F$, $u \in U$, $\mu \in \mathbb{R}$, $k \in F$, and let condition (2) of Definition 3.1.4 (preprint 3.4) hold at $u$ ($g(u) = \rho(r(u))g(u)$ implies $u \in O$). If $g(u) + \mu\rho(r(u))k = 0$, then $u \in O$. *Proof.* $r(u) \in U$ `Chapter3/Filler · StrongRetr.r_mem` lies in $O$, $\rho(r(u))$ is linear `Chapter3/Filler · StrongRetr.ρ_isLinear` and idempotent on the fibre over a point of $O$ `Chapter3/Filler · StrongRetr.ρ_ρ`; so $\rho(r(u))g(u) = -\mu\rho(r(u))\rho(r(u))k = -\mu\rho(r(u))k = g(u)$, and condition (2) applies. □

**An inward direction from good position** `Chapter3/GlobalIFTNearbyOntoLemmas · exists_inward_of_isInGoodPosition`. *Statement.* Let $C = \{x : \ell_i(x) \ge 0\}$ be a partial quadrant presented by an sc-isomorphism $T$, with coordinate functionals $\ell_i = (T\cdot)_i$, $z \in C$, and $N \ne 0$ a subspace in good position to the partial quadrant $C_z = \{v : \ell_i(v) \ge 0$ for the $i$ with $\ell_i(z) = 0\}$. Then there are $v \in N$ and $\eta, t_0 > 0$ with $z + t(v + \zeta) \in C$ for all $t \in [0, t_0]$ and $|\zeta| < \eta$. *Proof.* Let $v$ be an interior point of $N \cap C_z$ in $N$ (good position). *For an active index $i$,* $\ell_i(v) > 0$. Otherwise $\ell_i(v) = 0$; for $u \in N$ and small $t$, $v \pm tu \in C_z$, so $\pm t\ell_i(u) \ge 0$ and $\ell_i$ vanishes on $N$. Let $P$ be a good complement with constant $\varepsilon$ and $p \in P$; choose $n_0 \ne 0$ in $N \cap C_z$ (it has interior and $N \ne 0$) and scale it so that $|p| \le \varepsilon|n_0|$. Good position gives $n_0 \pm p \in C_z$, so $\ell_i(\pm p) = \ell_i(n_0 \pm p) \ge 0$ and $\ell_i(p) = 0$. Then $\ell_i$ vanishes on $N \oplus P = E$, which contradicts the surjectivity of the coordinates of a presentation. *The estimates.* A finite family of positive numbers has a positive lower bound `Chapter3/GlobalIFTNearbyOntoLemmas · exists_pos_le_of_forall_pos_fin`: $\ell_i(v) \ge m$ for the active $i$ and $\ell_i(z) \ge m$ for the others. For the active $i$, $\ell_i(z + t(v + \zeta)) = t(\ell_i(v) + \ell_i(\zeta)) \ge t(m - \|\ell_i\|\,|\zeta|) \ge 0$ when $|\zeta|$ is small; for the others, $\ell_i(z + t(v + \zeta)) \ge m - t\|\ell_i\|(|v| + |\zeta|) \ge 0$ when $t$ is small. □ The lemma needs $N \ne 0$: for the zero

subspace good position is an empty condition, and the case $\dim K = 0$ of Theorem 3.6.8 (1) (preprint 3.54) is treated separately (§III.43). The note behind the formalization phrased it as "$N$ meets the interior of $C_z$"; the cone condition is what transports along the linear coordinate change (Finding 185).

**Inward directions absorb any vector after rescaling** `Chapter3/GlobalIFTNearbyOntoLemmas · inward_add_smul`. If $a_0 + t(\alpha+\beta) \in \widetilde{C}$ for $t \in [0,t_0]$ and $|\beta| < \eta$, then for every $a_1$ and $s \geq (|a_1|+1)/\eta$ the vector $\alpha_0 = a_1 + s\alpha$ satisfies $a_0 + t(\alpha_0+\beta) \in \widetilde{C}$ for $t \in [0, t_0/s]$ and $|\beta| < 1$: indeed $a_0 + t(\alpha_0+\beta) = a_0 + ts(\alpha + (a_1+\beta)/s)$ with $ts \leq t_0$ and $|(a_1+\beta)/s| < (|a_1|+1)/s \leq \eta$.

**A one-sided curve of zeros** `Chapter3/GlobalIFTNearbyOntoLemmas · exists_curve_of_inward` (the finite-dimensional implicit function theorem at a boundary point, in an inward direction). *Statement.* Let $\widetilde{C} \subseteq P$ be convex, $\Omega \subseteq \mathbb{R}\times P$ open, $D = \Omega \cap (\mathbb{R}\times\widetilde{C})$, $a_0 \in \widetilde{C}$ with $(0,a_0) \in \Omega$, $H : \mathbb{R}\times P \to Q$ of class $C^1$ on $D$, $H(0,a_0) = 0$, $D_aH(0,a_0)$ onto (the derivative within $D$, restricted to $0\times P$), and $\alpha_0$ with $DH(0,a_0)(1,\alpha_0) = 0$ and $a_0 + t(\alpha_0+\beta) \in \widetilde{C}$ for $t \in [0,t_0]$, $|\beta| < \eta$. Then there are $\mu_0 > 0$ and $a : \mathbb{R} \to P$ with $a(0) = a_0$, right derivative $\alpha_0$ at 0, and $(\mu, a(\mu)) \in D$, $H(\mu,a(\mu)) = 0$ for $0 \leq \mu < \mu_0$. *Proof. Unique derivatives.* The inward property puts a ball around $a_0 + t_0\alpha_0$ into $\widetilde{C}$, so $\mathbb{R}\times\widetilde{C}$ is convex with nonempty interior and $D$ has unique derivatives; derivatives within $D$ are well defined. *The contraction.* Let $S$ be a right inverse of $M_0 = D_aH(0,a_0)$ and $Z$ its range; $SM_0$ is the identity on $Z$. For $\mu \geq 0$ and $\zeta \in Z$ with $|\zeta| \leq c\mu$, $c = \eta/2$, the point $(\mu, a_0 + \mu\alpha_0 + \zeta)$ lies in $D$ and close to $(0,a_0)$ `Chapter3/GlobalIFTNearbyOntoLemmas · dist_curve_le`, and $a_0 + \mu\alpha_0 + \zeta = a_0 + \mu(\alpha_0 + \zeta/\mu)$ is in $\widetilde{C}$ by the inward property ($|\zeta/\mu| \leq c < \eta$). Put $T_\mu(\zeta) = \zeta - SH(\mu, a_0+\mu\alpha_0+\zeta)$. Its derivative in $\zeta$ is $1 - SD_aH(\mu, a_0+\mu\alpha_0+\zeta)|_Z$, which tends to $1 - SM_0|_Z = 0$ as $(\mu,\zeta) \to 0$ by the continuity of $DH$ at $(0,a_0)$; the mean value inequality on the convex ball makes $T_\mu$ a $\frac{1}{2}$-contraction for small $\mu$. The differentiability of $H$ at $(0,a_0)$ in the direction $(1,\alpha_0)$, with $DH(0,a_0)(1,\alpha_0) = 0$, gives $|T_\mu(0)| = |SH(\mu, a_0+\mu\alpha_0)| = o(\mu)$; so $|T_\mu(0)| \leq c\mu/2$ for small $\mu$ and $T_\mu$ maps the ball of radius $c\mu$ into itself. *The curve.* The fixed point $\zeta(\mu)$ has $SH(\mu, a_0+\mu\alpha_0+\zeta(\mu)) = 0$, hence $H = 0$ there ($S$ is injective), and $|\zeta(\mu)| \leq 2|T_\mu(0)| = o(\mu)$; so $a(\mu) = a_0 + \mu\alpha_0 + \zeta(\mu)$ has right derivative $\alpha_0$ at 0. □

**Two facts on right derivatives.** If $k$ is continuous at 0 from the right, $\mu \mapsto \mu k(\mu)$ has the right derivative $k(0)$ at 0 `Chapter3/GlobalIFTNearbyOntoLemmas · hasDerivWithinAt_smul_right_of_continuousWithinAt`; and right derivatives at 0 on $[0,\infty)$ are unique `Chapter3/GlobalIFTNearbyOntoLemmas · eq_of_hasDerivWithinAt_Ici`.

## III.42 The $\lambda$-extended contraction germ and the perturbation in the normal form

Throughout, the normal form is the data of Theorem 3.4.2 (preprint 3.38) from §III.20 `Chapter3/LocalSolvability · ShiftedBasicData`: a germ $g : O(\widetilde{C}\oplus W, 0) \to Q\oplus W$, sc-smooth, whose index raise $g^1 = (G_1, w - B(a,w))$ is a basic germ on the shifted structures; at the chart at $x_0$ it is obtained from the stability data of the graph description by Theorem 3.4.2 (preprint 3.38) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`. (In this section $g$ and $e$ denote the normal form and its perturbation; the filler of the chart at $x_0$ returns in §III.43.) The norms of $w$ below are those of $W_{m+1}$, the shifted level $m$.

**The perturbation on the shifted structure.** For an sc$^+$-perturbation $e : O(\widetilde{C}\oplus W, 0) \to (Q\oplus W)^1$, its $W$-component on level 1 `Chapter3/GlobalIFTNearbyOntoGerm · perturbEW` is sc-smooth on the index-raised germ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_isScSmoothGerm`, by Proposition 1.2.2 (preprint 1.17) for germs applied to the level-raised map `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelMap_e_isScSmoothGerm` `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_eq_on`. Its level maps are those of $e$ read

on the shifted levels Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_eW.

**Contraction regions that do not depend on the perturbation** Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii. A system of regions consists of radii $\nu_m$ in $P$ and $\rho_m$ in $W_{m+1}$ such that for $a \in \widetilde{C}$, $|a| < \nu_m$ and $|w|, |w'| < \rho_m$: $(a, w)$ lies in the domain of $g^1$ on level $m$, $|B(a,w) - B(a,w')| \le \frac{1}{4}|w - w'|$, and $|B(a,0)| \le \rho_m/8$; $\nu$ decreasing Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii.ν_anti_le and the $\rho_m$ compatible with the inclusions. *They exist below any prescribed radii* $\delta_m$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii: on each level, Definition 3.3.1 (preprint 3.28) with $\varepsilon = \frac{1}{4}$ gives a radius on which $B(a,\cdot)$ is a $\frac{1}{4}$-contraction, and since $B(0,0) = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_B_zero and $B$ is continuous at $0$ on each level, $|B(a,0)| \le \rho_m/8$ for $|a|$ small; the radii are then shrunk level by level to be monotone and below $\delta_m$. Nothing here involves a perturbation.

**The λ-extended germ.** On $(\mathbb{R} \times \widetilde{C}) \oplus W^1$ take the germ of neighbourhoods $\{((\mu, a), w) \mid (a, w) \in U^1_m\}$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbGerm, with the projections forgetting $\mu$ and keeping $\mu$ as sc-operators Chapter3/GlobalIFTNearbyOntoGerm · perturbProj Chapter3/GlobalIFTNearbyOntoGerm · perturbMuProj and $\mathbb{R} \times \widetilde{C}$ quadrant-like Chapter3/GlobalIFTNearbyOntoGerm · isQuadrantLike_perturb, and put

$$B_\mu(a, w) = B(a, w) - \mu\, Pe(a, w)$$

Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB, $P$ the projection onto $W$, with $B_\mu(0,0) = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_zero. It is sc-smooth on the germ: $B$ composed with the projection, minus the scalar product of the $\mu$-coordinate with $Pe$ composed with the projection, by the product rule with a scalar factor of §III.40 Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_isScSmoothGerm Chapter3/ParamPerturbation · IsScSmooth.smul; its level maps are $B_m - \mu(Pe)_m$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_perturbB. Given constants $c_m \ge 0$ with $|Pe(a,w)| \le c_m$ and $|Pe(a,w) - Pe(a,w')| \le c_m|w - w'|$ on the regions, define the $\mu$-radii $\mu_m = \min_{j \le m} \min(\frac{1}{4}, \rho_j/8)/(c_j + 1)$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu, positive and decreasing Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_pos Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_anti, with $\mu_m c_m \le \frac{1}{4}$ and $\mu_m c_m \le \rho_m/8$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_mul_le.

**Lemma: $B_\mu$ is a contraction germ in the working form** Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction, with the parameter regions $(-\mu_m, \mu_m) \times \{|a| < \nu_m\}$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.mem_perturbContraction_N_iff, the open balls $\{|w| < \rho_m\}$ and the closed balls $\{|w| \le \rho_m/2\}$ (Definition 3.A, §III.7). *The contraction*: $|B_\mu(a,w) - B_\mu(a,w')| \le (\frac{1}{4} + |\mu|c_m)|w - w'| \le \frac{1}{2}|w - w'|$. *The closed balls are invariant*: for $|w| \le \rho_m/2$, $|B_\mu(a,w)| \le |B(a,w) - B(a,0)| + |B(a,0)| + |\mu||Pe(a,w)| \le \frac{1}{4} \cdot \frac{\rho_m}{2} + \frac{\rho_m}{8} + |\mu|c_m \le \frac{\rho_m}{2}$. The regions in $a$ and $w$ are those of $B$ and do not depend on $e$; only the $\mu$-intervals do. The general construction of a working form from Definition 3.3.1 (preprint 3.28) Chapter3/GermIFT · toContractionGerm is *not* used: its radii in $a$ would depend on $e$, and the finite-dimensional step needs the nearby zero inside the level-3 region for every perturbation (Finding 182).

**Theorem 3.3.3 (preprint 3.30) for $B_\mu$** Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_isSc1GermWith. The solution germ $\delta_\mu(\mu, a)$ is sc¹ on the germ re-indexed by one level, $((-\mu_{m+1}, \mu_{m+1}) \times \{|a| < \nu_{m+1}\}) \cap (\mathbb{R} \times \widetilde{C})$, with the derivative family of the proof of Theorem 3.3.3 (preprint 3.30) Chapter3/ContractionGerm · theorem_3_3_3. *At* $\mu = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_sol_zero: a fixed point $w \in W_1$ of $B(a,\cdot)$ with $|w| < \rho_0$, $|a| < \nu_0$, $a \in \widetilde{C}$, is $\delta_\mu(0, a)$,

by the uniqueness of the fixed point in the open ball of level 0.

**The perturbation in the normal form.** Now the chart at $x_0$ returns: the strong bundle retraction $R$ on $U \lhd F$, the conjugation $\Psi$ covering $\psi$ with inverse $\Psi'$, and the linear maps glue, split, fiber of the coordinate change of Theorem 3.4.2 (preprint 3.38) (§III.18). The perturbation $u \mapsto \rho(r(u))k$ of the filler, pushed to the normal form, is $\hat{e}_k(y) = \Xi(y,k)$ with the *joint* map

$$\Xi(y,\kappa) = \operatorname{fiber}\Big(\Psi[1](\psi^{-1}(\operatorname{glue} y),\ (R[1](r(\psi^{-1}(\operatorname{glue} y)),\kappa))_F)\Big)_{F'}$$

`Chapter3/GlobalIFTNearbyOntoXi · xiMap` on $\psi(U) \oplus F^1$ `Chapter3/GlobalIFTNearbyOntoXi · xiDom` `Chapter3/GlobalIFTNearbyOntoXi · mem_xiDom`, a relatively open set (glue$^{-1}(U')$ is relatively open in the quadrant of the normal form) `Chapter3/GlobalIFTNearbyOntoXi · glue₀_preimage_relOpen` `Chapter3/GlobalIFTNearbyOntoXi · xiDom_relOpen`. $\Xi$ *is sc-smooth into the index-raised target* `Chapter3/GlobalIFTNearbyOntoXi · xiMap_isScSmooth`: it is the composite of the linear glue, $\psi^{-1} \times 1$, the map $(u,\kappa) \mapsto (u,(R[1](r(u),\kappa))_F)$, the strong bundle map $\Psi[1]$ on the index-raised bundle (Definition 2.6.1 (preprint 2.47)), a projection and the linear fiber. $\Xi$ *is linear in* $\kappa$ `Chapter3/GlobalIFTNearbyOntoXi · xiMap_smul` `Chapter3/GlobalIFTNearbyOntoXi · xiMap_zero`, also on every level `Chapter3/GlobalIFTNearbyOntoXi · levelMap_xiMap_smul`, since $R$ and $\Psi$ are linear on the fibres. For a smooth $k$ with level-1 representative $k_1$, $\hat{e}_k = \Xi(\cdot,k_1)$ `Chapter3/GlobalIFTNearbyOntoXi · ehat` is sc-smooth into the index-raised target `Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmooth` (a constant map with smooth value is sc-smooth `Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth`), an sc$^+$-perturbation on the germ of Theorem 3.4.2 (preprint 3.38) `Chapter3/GlobalIFTNearbyOntoXi · pushGerm₀_U_subset` `Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmoothGerm`, with level maps $(\hat{e}_k)_m(y) = \Xi_m(y,k_{m+1})$ `Chapter3/GlobalIFTNearbyOntoXi · levelMap_ehat`.

**Lemma: bounds uniform in** $k$ `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_bounds`. *For every level $m$ there are $\delta > 0$ and $M \geq 0$ such that for $|a| < \delta$, $|\kappa|_{m+2} < \delta$ and a convex set $S$ of $w \in W_{m+1}$ with $|w| < \delta$ on $S$ (inside the domain of $\Xi$ on level $m+1$): $|\Xi_m((a,w),\kappa)| \leq 1$ and $|\Xi_m((a,w),\kappa) - \Xi_m((a,w'),\kappa)| \leq M|w-w'|$ on $S$. Proof.* By Proposition 1.2.3 (preprint 1.18) `Chapter1/Levels · IsScSmooth.contDiffOn_levels`, $\Xi$ is $C^1$ from level $m+1$ to level $m$ of its index-raised target, which is the shifted level $m$ of the normal form: on the shifted structure the perturbation is $C^1$ on every level, which is what contraction estimates on the shifted levels need. *This is where the index raise of Theorem 3.4.2 (preprint 3.38) is used* (Finding 183). $\Xi_m$ vanishes at $\kappa = 0$, so by continuity of $\Xi_m$ and of its derivative at 0 both are small on a small ball, and the mean value inequality on the convex $S$ gives the Lipschitz bound. □ *The constants for* $\hat{e}_k$ `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds`: there are radii $\delta_m > 0$, depending on $\Xi$ only, such that for every system of regions with $\nu_m, \rho_m \leq \delta_m$ and every smooth $k$, the constants $c_m = (2|k|_{m+2}/\delta_m)(M_m+1)$ bound $(P\hat{e}_k)_m$ and its Lipschitz constant in $w$ on the regions. The proof is the *scaling* $\Xi_m(y,\kappa) = (2|\kappa|/\delta_m)\,\Xi_m(y,(\delta_m/2|\kappa|)\kappa)$, which needs only the linearity of $\Xi$ in $\kappa$; the note behind the formalization used the linearity of $D_y\Xi$ in $\kappa$ instead.

## III.43 The curve of perturbed zeros, the surjectivity in the chart at $x_0$, and Theorem 3.6.8 (1) (preprint 3.54)

**The reduction at a nearby zero** (in the normal form, with a perturbation $e$, regions $R$ and constants $c$ as in §III.42). Let $z = (a_0,w_0)$ be a level-1 zero of $g$ in the domain of the germ with $a_0 \in \widetilde{C}$, $|a_0| < \nu_3$ and $|w_0|_1 < \rho_0$. Then $w_0$ is a fixed point of $B(a_0,\cdot)$ `Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.B_eq_of_zero` (because $g^1 = (G_1, w - B)$), so

$w_0 = \delta_\mu(0,a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.sol_zero_eq. On the domain $D_3 = (-\mu_3,\mu_3)\times(\{|a| < \nu_3\}\cap\widetilde{C})$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDom, which has unique derivatives Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.uniqueDiffOn_curveDom, put $(\mu,a)\mapsto(a,\delta_\mu(\mu,a))\in E_1$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift_mem and

$$\widetilde{G}(\mu,a) = (g+\mu\,\iota e)(a,\delta_\mu(\mu,a))$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap. *Its $W$-component vanishes* Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap_snd: $\delta_\mu(\mu,a)$ is the fixed point of $B_\mu(a,\cdot)$, and the $W$-component of $g+\mu\iota e$ is $w-B_\mu(a,w)$ (read on level 1, Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.snd_g_toBase). So the whole equation is carried by $H = (\widetilde{G})_Q$. *$\widetilde{G}$ is $C^1$* on $D_3$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.contDiffOn_curveMap: $\delta_\mu$ is $C^1$ into $W_2$ by Theorem 3.3.3 (preprint 3.30) on the re-indexed germ (Proposition 1.2.1 (preprint 1.16) on level 1), and $g$, $e$ are $C^1$ from level 1 to level 0. *Its derivative at $(0,a_0)$* Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_curveMap is

$$D\widetilde{G}(0,a_0)(\dot\mu,\alpha) = Dg(z)(\alpha,\ \iota D\delta_\mu(0,a_0)(\dot\mu,\alpha)) + \dot\mu\,\iota e(z)$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solD Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv: the chain rule, with the derivative of the solution germ on level 1 Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level, the derivative of $g$ at $z$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_Dg', and the product rule for $\mu\,\iota e$, whose derivative at $\mu = 0$ in the base directions vanishes. Its $W$-component vanishes too Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_snd.

**Lemma: $D_aH(0,a_0)$ is onto, by a dimension count** Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq. *Statement.* Let moreover $\ker Dg(z)$ be finite-dimensional of dimension $\dim P - \dim Q$. Then $M_0 = D_aH(0,a_0) : P\to Q$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.DH is onto, and $\ker Dg(z)$ is the graph of $L = \iota D_a\delta_\mu(0,a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solL over $\ker M_0$. *Proof.* At $\dot\mu = 0$ the derivative formula reads $Dg(z)(\alpha,L\alpha) = (M_0\alpha,0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_inr Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.Dg'_graph, so the graph of $L$ over $\ker M_0$ lies in $\ker Dg(z)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.map_graph_le_ker. The graph map is injective, so $\dim\ker M_0 \le \dim\ker Dg(z) = \dim P - \dim Q$; rank–nullity gives $\operatorname{rank} M_0 \ge \dim Q$, so $M_0$ is onto, and then $\dim\ker M_0 = \dim\ker Dg(z)$ and the inclusion is an equality. □ The obvious route, eliminating the $W$-block with $(1-D_2B(z))^{-1}$, needs $\|D_2B(z)\|\le\frac12$ on the level where $Dg(z)$ is known to be onto, i.e. the surjectivity on the shifted structure; the dimension count uses only the kernel dimension on level 0 (Finding 184).

**Proposition: the curve of perturbed zeros** Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.exists_perturbed_curve. *Statement.* Under the hypotheses of the lemma, let $v\in\ker Dg(z)$ with $a_0+t(v_1+\beta)\in\widetilde{C}$ for $t\in[0,t_0]$, $|\beta|<\eta$ ($v_1$ the $P$-component). Then there are $\mu_0>0$ and a curve $u'(\mu) = (a(\mu),\delta_\mu(\mu,a(\mu)))$, $0\le\mu<\mu_0$, on level 2, in the domain of the germ with $a(\mu)\in\widetilde{C}$, with $u'(0)=z$ (on level 1), right differentiable at 0 as a map into $E_2$, and $(g+\mu\iota e)(u'(\mu)) = 0$. *Proof.* (i) By the lemma, $M_0$ is onto and $v_1\in\ker M_0$ (as $v=(v_1,Lv_1)$). (ii) Choose $a_1$ with $M_0a_1 = -D_\mu H(0,a_0)$, and $\alpha_0 = a_1 + sv_1$ with $s$ from the rescaling lemma of §III.41: $\alpha_0$ is inward at $a_0$ (with $|\beta|<1$) and

$DH(0,a_0)(1,\alpha_0) = D_\mu H(0,a_0) + M_0 a_1 + sM_0 v_1 = 0$. (iii) $H$ is $C^1$ on $D_3$ with $H(0,a_0) = 0$, and $\widetilde{C}$ is convex; the one-sided curve lemma (§III.41) on $\$\square = \$$ the open region of level 3 gives $a(\mu)$ with right derivative $\alpha_0$ and $H(\mu, a(\mu)) = 0$. (iv) The level-1 map $(\mu, a) \mapsto \delta_\mu(\mu, a) \in W_2$ is differentiable at $(0, a_0)$ within $D_3$ (Theorem 3.3.3 (preprint 3.30), `Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level`), so by the chain rule for $\mu \mapsto (\mu, a(\mu))$ the curve $u'$ is right differentiable at 0 into $E_2$. (v) At $\mu = 0$, $\delta_\mu(0, a_0) = w_0$; and $(g + \mu\iota e)(u'(\mu)) = \widetilde{G}(\mu, a(\mu)) = (H(\mu, a(\mu)), 0) = 0$. □

**The normal form at a nearby point.** For the chart at $x_0$: at every level-1 point $y$ of $\psi(U_1)$ in the germ, $Dg(y) = \text{fiber} \circ D(h + t)(\text{glue}\, y) \circ \text{glue}$ for the normal form $g$ of $h + t$ `Chapter3/GlobalIFTNearbyOnto · StabilityData.scDerivGerm_pushMap₀_at` (the coordinate change is linear), and the kernel of fiber $\circ\, T \circ$ glue is $\text{split}(\ker T)$ `Chapter3/GlobalIFTNearbyOnto · StabilityData.ker_fiber₀_comp_glue₀`, split being injective `Chapter3/GlobalIFTNearbyOnto · StabilityData.split₀_injective`. Corollary 3.4.3 (preprint 3.39) is not used: the facts at the nearby zero come from the chart at $x_0$ through these linear maps (Finding 187).

**Theorem A: the surjectivity in the chart at $x_0$** `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_chart_onto`. *Statement.* Take the data of the graph description in the chart at $x_0$ (§III.29, `Chapter3/GlobalIFTNearbyGraphG · theorem_3_6_8_chart_graph`): the strong bundle retraction $R = (r, \rho)$ on $U$, a filling germ $g$ of $\mathbf{f}$ on $G_g$ with a derivative family $Dg$, $\mathbf{f}(0) = 0$, the conjugation $\Psi$ covering $\psi$, $\psi(0) = 0$, the stability data on a germ that $\psi'$ maps into $G_g$ level by level (inside $G''$, §III.48) with the basic germ $h'$, $D(h' + \iota t)(0)$ onto with kernel $K$ in good position, a closed good complement $Y$ of $K$, the $C^1$ graph $\sigma : V' \to Y$ with $\sigma(0) = 0$, $D\sigma(0) = 0$ and its $C^k$-lifts, an open $W_o$ on which $h' + \iota t$ is the push-forward of $g$, an open $N_2 \ni 0$ on which $\psi$ maps into the germ and into $U' \cap W_o$, and on which $\psi$ maps the zeros of $g$ in $O$ onto the graph. Assume condition (2) of Definition 3.1.4 (preprint 3.4) on $U \cap N_2$ and $K \neq 0$. Then there is an open $N_4 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_4$ of $g$, every level-1 point $z_{0,1}$ over $z_0$ and every smooth $k$, there is $x \in T_{z_0}O = Dr(z_{0,1})E_0$ with $Dg(z_{0,1})x = \rho(z_0)k$. *Proof*, in the order of the Lean:

1. *Data at $x_0$, independent of $k$.* Lemma 3.6.9 (preprint 3.55) on the graph with its radius $\rho_2$ `Chapter3/GlobalIFTFull · lemma_3_6_9_relOpen` (§III.32); the kernel at the nearby zeros on an open $N_3 \subseteq N_2$ `Chapter3/GlobalIFTNearbyKernelG · theorem_3_6_8_chart_kernel` (§III.32); the normal form $\mathcal{D}$ of Theorem 3.4.2 (preprint 3.38) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`; the radii $\delta_m$ of the bounds of $\Xi$ `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds` and contraction regions below them `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii`; and the radius $r_{42}$ of Lemma 3.5.5 (preprint 3.42) for the bound $\rho_0$ `Chapter3/LocalSolvability · StabilityData.lemma_3_5_5`: zeros of the normal form on level 0 smaller than $r_{42}$ have their $W$-component on level 1, smaller than $\rho_0$ there.
2. *The neighbourhood.* Let $\pi$ be the projection onto $K$ along $Y$ (a topological complement: $Y$ closed, $K$ finite-dimensional), $c_s = \|\text{split}\| + 1$, and $r' = \min(\rho_2/(\|\pi\| + 1), r_{42}/c_s, \nu_3/c_s)$. By the continuity of $\psi$ on $U$, $\psi^{-1}(B(0, r')) \cap U = O_4 \cap U$ for an open $O_4 \ni 0$; put $N_4 = N_3 \cap O_4$. Nothing here depends on $k$.
3. *At a zero $z_0 \in O \cap N_4$.* The kernel theorem at $z_0$ gives $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point $x_1$ over it, $D(h' + \iota t)(x_1)$ onto and Fredholm of index $\dim P - \dim Q$, its kernel $N_v$, and $\ker Dg(z_{0,1}) = D\psi^{-1}(x_1)N_v \subseteq T_{z_0}O$. Since $|v| = |\pi(\psi(z_0))| \leq \|\pi\| r' < \rho_2$, Lemma 3.6.9 (preprint 3.55) at $v$ gives $x_1 \in C'$ and $N_v$ in good position to $C'_{x_1}$.
4. *The zero of the normal form.* $z = \text{split}(x_1)$ lies in the germ on level 1, with $a_0 = (x_1)_P \in \widetilde{C}$; $|\text{split}\, \psi(z_0)| < c_s r'$, so $|a_0| < \nu_3$, and Lemma 3.5.5 (preprint 3.42) gives $|w_0|_1 < \rho_0$; $g$ vanishes at $z$ because $h' + \iota t$ vanishes at the graph point.

5. *Its kernel.* $Dg(z) = \text{fiber} \circ D(h' + \iota t)(x_1) \circ \text{glue}$, so $\ker Dg(z) = \text{split}(N_v)$, finite-dimensional of dimension $\dim\ker D(h' + \iota t)(x_1) = \dim P - \dim Q$ (for a surjective Fredholm operator the kernel dimension is the index, `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`).
6. *The inward direction.* $N_v \neq 0$: its dimension is $\dim P - \dim Q$, which is also $\dim K$ ($D(h' + \iota t)(0)$ is onto and Fredholm of that index, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`), and $K \neq 0$. The inward lemma at $x_1$ (§III.41) gives $v_v \in N_v$ with $x_1 + t(v_v + \zeta) \in C'$; then split $v_v \in \ker Dg(z)$, and its $P$-component is inward at $a_0$ in the quadrant of the normal form for $|\beta| < \eta/(\|\text{glue}\| + 1)$, since $\text{glue}(\beta, 0)$ has norm at most $\|\text{glue}\|\,|\beta|$.
7. *The perturbation and the curve.* Take the level-1 representative $k_1$ of $k$; $\hat{e}_k$ is an sc$^+$-perturbation (§III.42) with constants $c_m$ on the regions; the proposition above gives $\mu_0$ and the curve $u'$ of zeros of $g + \mu\iota\hat{e}_k$ through $z$.
8. *Back in the chart.* $u(\mu) = \psi^{-1}(\text{glue}\, u'(\mu))$ on level 1: $\psi^{-1}$ is $C^1$ from level 2 to level 1 (Proposition 1.2.3 (preprint 1.18), `Chapter1/Levels · IsScSmooth.contDiffOn_levels`), so $u$ is right differentiable at 0 with some velocity $\xi_1$, $u(0) = z_{0,1}$, and its base points $u_\mu = \psi^{-1}(\text{glue}\, \iota u'(\mu))$ lie in $U$.
9. *The perturbed equation.* By continuity at $\mu = 0$, $\text{glue}\, \iota u'(\mu) \in W_o$ and $u_\mu \in N_2$ for $0 \leq \mu < \mu_2$ `Chapter3/GlobalIFTNearbyOnto · exists_Ico_of_eventually_nhdsWithin_Ici`. There the push-forward identity of $g$, the description of $\hat{e}_k$ as the $\Psi$-fibre image of $\rho(r(u_\mu))k$ transported by the linear fiber, and the injectivity of the fibre maps of $\Psi$ turn $(g + \mu\iota\hat{e}_k)(u'(\mu)) = 0$ into $g(u_\mu) + \mu\rho(r(u_\mu))k = 0$.
10. *Tangency.* By the first lemma of §III.41 and condition (2) on $N_2$, $u_\mu \in O$.
11. *Differentiation.* $r$ is sc$^1$ on $U$, and $g$ is an sc$^1$-germ on $G_g$; the curve $u(\mu)$ stays on level 1 of $G_g$, because $\text{glue}\, \iota u'(\mu)$ lies on level 1 of the germ of the stability data and $\psi_1'$ maps that into $(G_g)_1$ `Chapter3/BasicGermShrink · ConjugationData.levelMap_inclK_mem_of_subset_shrinkGerm`. So both are differentiable from level 1 to level 0 at $z_{0,1}$ along the curve, $g$ by the chain rule within $(G_g)_1$ with derivative $Dg(z_{0,1})$ (with $g$ sc-smooth on $U$ no such condition was needed). Along the curve, $r(u_\mu) = u_\mu$, so the right derivatives at 0 agree, $Dr(z_{0,1})\xi_0 = \xi_0$ ($\xi_0 = \iota\xi_1$; uniqueness of right derivatives, §III.41): $\xi_0 \in T_{z_0}O$. The map $\mu \mapsto \rho(r(u_\mu))k$ is continuous at 0 ($R$ is continuous on level 0), so $\mu \mapsto \mu\rho(r(u_\mu))k$ has the right derivative $\rho(r(z_0))k = \rho(z_0)k$; differentiating $g(u_\mu) = -\mu\rho(r(u_\mu))k$ gives $Dg(z_{0,1})\xi_0 = -\rho(z_0)k$. Take $x = -\xi_0$. □

*Where the hypotheses enter.* Tameness and good position at $x_0$ through the graph description, Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) (steps 1, 3) and the inward direction at boundary zeros (step 6); condition (2) of the filler through the tangency (step 10); $K \neq 0$ for the inward direction. The sc-Fredholm property at $y$ is not used here; it enters only through the closed range in the assembly step. Condition (3) of the filler at the nearby zeros is not used. *Why there is no uniformity problem.* The only $k$-dependent quantities are the constants $c_m$ and the $\mu$-intervals; $N_4$ is fixed in step 2 before $k$ is chosen. The perturbation argument of §III.33 applied the implicit function theorem at $x_0$ to the $\Lambda_0$-extended problem, whose neighbourhood depends on $\Lambda_0$; here it is applied at the nearby zero (Finding 181).

**Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses** `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'`. *Statement.* Let $f$ be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$, and $(f, x_0)$ in good position. Then there is an open $V_x \ni x_0$ such that $(f, y)$ is in good position at every zero $y \in V_x$. *Proof.* Destructure the good position at $x_0$: the centred chart $c$, the filling germ $g$ with its derivative family, the sc$^+$-germ, the conjugation, shrunk to $G''$ (§III.48), and the good position in the chart. Tameness gives $(1 - Dr(0))E \subseteq E_0$ by Proposition 2.5.3 (preprint 2.35) `Chapter2/TameRetracts · proposition_2_5_3`; the graph description in the chart `Chapter3/GlobalIFTNearbyGraphG · theorem_3_6_8_chart_graph` and the kernel at the nearby zeros `Chapter3/GlobalIFTNearbyKernelG · theorem_3_6_8_chart_kernel` give $N_1 \supseteq N_2 \supseteq N_3$. A zero $y$ of $f$ in the chart domain with $\varphi(y) \in N_2$ is a zero of $g$ ($\mathbf{f} = g$ on $O \cap N_1$).

*Case* $K = 0$. Put $V_x = \varphi^{-1}(N_2)$ in the chart domain. A zero $y \in V_x$ maps by $\psi\varphi$ onto a graph point $v + \sigma(v)$ with $v \in K = 0$, so $\psi(\varphi(y)) = \sigma(0) = 0 = \psi(0)$, $\varphi(y) = 0$ (as $\psi^{-1}\psi = 1$ on $U$), and $y = x_0$ by the injectivity of the chart: the only zero is $x_0$, in good position by hypothesis. *Case* $K \neq 0$. Theorem A gives $N_4$ (condition (2) on $U \cap N_2$ is the filler's, §III.29); put $V_x = \varphi^{-1}(N_3 \cap N_4)$ in the chart domain. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O$ is a zero of $g$; the kernel theorem gives the level-1 point $z_{0,1}$, $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$ and its good position inside $T_{z_0}O$; $y$ is smooth `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so $(f, y)$ is an sc-Fredholm germ; $\mathbf{f} = g$ on $O$ near $z_0$. The assembly step in its germ form `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense` (§III.31; whole-chart `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`) applies at $z_{0,1} \in (G_g)_1$ with (a) from Theorem A — $\rho(z_0)\hat{k}$ in the range of $Dg(z_{0,1})|_{T_{z_0}O}$ for every smooth $\hat{k}$ — and (b) from the kernel theorem, $\ker Dg(z_{0,1}) \cap T_{z_0}O = \ker Dg(z_{0,1})$. □ **No hypothesis is added**: the statement has exactly the hypotheses of the kernel theorem of §III.32 plus the sc-Fredholm property (Finding 189).

**Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) with the text's hypotheses.** Theorem 3.6.8 (1) and (2) (preprint 3.54) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8'`: (1) as above on $V_x$, and (2), the regular data of Definition 3.2.1 (preprint 3.25) at $x_0$ of a dimension $d$ that every nearby datum shares, by §III.37 `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_two` fed with (1). Theorem 3.1.26 (preprint 3.23) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_26'`: (1) supplies the open set of nearby good position that the boundary case from the nearby good position needs `Chapter3/GlobalIFTTheoremsG · theorem_3_1_26_of_nearby`, and every zero of the resulting $U$ is in good position. Theorem 3.1.22 (preprint 3.19) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_22'`: Theorem 3.1.26 (preprint 3.23) and Lemma 2.4.6 (preprint 2.26), $d_S \leq d_X = 0$ `Chapter2/SubMPolyfold · lemma_2_4_6`. The first forms under the filler condition at the nearby zeros (§§III.33, III.37) are kept; that condition is now a theorem (§III.45).

## III.44 The dimension of the solution set is the Fredholm index, and the index is locally constant

Theorem 3.1.22 (preprint 3.19) says that the solution set is a manifold "whose dimension agrees with the Fredholm index", and the proof of Theorem 3.6.8 (preprint 3.54) takes $d = \dim\ker f'(x)$. In the statements of §§III.34–III.37 the dimension was existential. This section identifies it, at every zero and in every chart, and deduces the local constancy of the index along the solution set (Findings 170–171).

**"$f'(y)$ is onto, Fredholm, of index $d$", chart-intrinsically** `Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex`: in every strong bundle chart around $y$ compatible with the atlas, at the level-1 point $z$ over $\varphi(y)$, every corestriction $L : T_zO \to K_z$ of the linearization $\mathrm{Lin}(z)$ (§III.31) is onto and Fredholm with $\mathrm{ind}\, L = \dim\ker L = d$. "$f'(y)$ is onto" alone `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt`: in *some* chart, $\mathrm{Lin}(z)$ maps onto the fibre $\{k \mid (\varphi(y), k) \in K\}$; by the chart independence it is then onto in every chart.

**Proposition 3.6.5 (preprint 3.52) in every chart** `Chapter3/GlobalIFTIndexG · StrongBundle.linAt_index_of_conjugationData`. *Statement.* Let $f$ be an sc-smooth section, $x_0$ a zero, $c$ a strong bundle chart centred at $x_0$ with a filling germ $g$ of $\mathbf{f}$ and a derivative family $Dg$ on a germ inside $U$, an sc⁺-germ $s$ with $s(0) = g(0)$, and conjugation data $d_c$ of $g - s$ to a basic germ `Chapter3/FredholmGermIndex · ConjugationData` with finite-dimensional spaces $P_n$, $Q$. Then in every strong bundle chart $c'$ around $x_0$, at the level-1 point $z'$ over $\varphi'(x_0)$, every corestriction $L'$ of $\mathrm{Lin}'(z')$ is Fredholm of index $\dim P_n - \dim Q$,

and onto if the germ is in good position at $Dg(0)$ in $c$. *Proof.* Proposition 3.6.5 (preprint 3.52) in $c$ (§III.30, `Chapter3/FredholmGermIndexG · proposition_3_6_5_local_index`): $Tf(0)$ is Fredholm of index $\dim P_n - \dim Q$. $Tf(0) = Dg(0)|_{T_0O}$ is $\mathrm{Lin}(0)$ by the bridge for germs at the centre, which lies in every germ `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt` (whole-chart: `Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt`), up to the identification of the fibres $\rho(0)F$ and $\rho(\iota 0)F$, which are equal subspaces but different types in Lean `Chapter3/GlobalIFTIndex · exists_rangeS_pCLM_equiv`. The chart independence (§III.31, `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`) carries the Fredholm property, the index and the surjectivity from $c$ at $0$ to $c'$ at $z'$; the four spaces are complete as ranges of idempotents `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`. □ At a zero in good position this gives $\mathrm{IsSurjLinOfIndex}(f, x_0, \dim P_n - \dim Q)$ `Chapter3/GlobalIFTIndexG · StrongBundle.isSurjLinOfIndex_of_conjugationData`, because for a surjective Fredholm operator the kernel dimension is the index `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`; and the $d$ with $\mathrm{IsSurjLinOfIndex}(f, x_0, d)$ equals $\dim P_n - \dim Q$ for every conjugation in the chart of Definition 3.6.6 (preprint 3.53) `Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.exists_isSurjLinOfIndex` (a conjugation exists, `Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty`). The first clause of Theorem 3.1.22 (preprint 3.19) with the index `Chapter3/GlobalIFTIndexG · theorem_3_1_22_linearization_index`: for an sc-Fredholm section and a zero in good position, in the chart of Definition 3.6.6 (preprint 3.53), $Dg(0)$ and $Tf(x_0)$ are Fredholm, $Tf(x_0)$ is onto, and $\mathrm{ind}\, Dg(0) = \mathrm{ind}\, Tf(x_0) = \dim\ker Tf(x_0) = \dim d.P_n - \dim d.Q$ for every conjugation $d$ — the old formulation's "$n - N$ for some natural numbers" made precise.

**The conjugation exposed.** The graph description and the tangent germ data at the centre are built from the chart of Definition 3.6.6 (preprint 3.53) by destructuring the existential good position, so their statements did not record which conjugation they used. Their proofs now prove versions that expose it `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj_conj` `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart_proj_conj`: the filling germ with its germ and derivative family $Dg$, the sc⁺-germ, the good position at $Dg(0)$ in the chart and conjugation data $d_c$, with $\dim K = \dim d_c.P_n - \dim d_c.Q$ for $K = \ker D(h+t)(0)$, because $D(h+t)(0)$ is onto and Fredholm of index $n - N$ (§III.18, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`). The earlier statements are corollaries.

**Dimension = index.** *At the centre* `Chapter3/GlobalIFTIndexG · theorem_3_6_8_tangentGermChart_index`: for an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero $x_0$ in good position, the data of Definition 3.2.1 (preprint 3.25) at $x_0$ for $\{y \in V_x : f(y) = 0\}$ (with $\pi(O) \subseteq C$) have a dimension $d$ with $\mathrm{IsSurjLinOfIndex}(f, x_0, d)$: the exposed $d = \dim K = \dim d_c.P_n - \dim d_c.Q$ and the previous paragraph. The same for the global solution set `Chapter3/GlobalIFTIndexG · theorem_3_6_8_tangentGermChart_solutionSet_index` (restriction of the chart, §III.37) and for the regular data of Theorem 3.6.8 (2) (preprint 3.54) `Chapter3/GlobalIFTIndexG · theorem_3_6_8_two_index` (the re-centring keeps the dimension, `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`). *At a nearby zero* `Chapter3/GlobalIFTIndexG · isSurjLinOfIndex_of_forall_dim`: if $y$ is a zero in good position at which every datum of Definition 3.2.1 (preprint 3.25) for the solution set has dimension $d$, then $\mathrm{IsSurjLinOfIndex}(f, y, d)$ — the data at $y$ from the centre theorem at $y$ have dimension $\mathrm{ind}\, f'(y)$, and that dimension is $d$.

**The index is locally constant along the solution set** `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index_locally_constant`. *Statement.* For an sc-Fredholm section of a strong bundle whose charts have tame base retractions, a zero $x_0$ in good position: there are an open $V_x \ni x_0$ and $d$ with $\mathrm{IsSurjLinOfIndex}(f, y, d)$ at every zero $y \in V_x$; in particular $\mathrm{ind}\, f'(y) = \mathrm{ind}\, f'(x_0)$. *Proof.* Theorem 3.6.8 (1) (preprint 3.54) (§III.43) gives an open $V_{x,1}$ of nearby good position; Theorem 3.6.8 (2) (preprint 3.54) with the index on it gives regular data $D$ of dimension $d$ with source in $V_{x,1}$ along which every datum has dimension $d$; on the

source of $D$, every zero is in good position with all its data of dimension $d$, and the previous paragraph applies. □ The proof does not repeat the kernel computation of the matrix argument (Finding 188): "the dimension is locally constant" (from the re-centring and Sard's lemma in fixed dimension, §III.35) and "the dimension is the index" give "the index is locally constant". Along the solution set this is the index stability of the text's Proposition 6.2.2 (preprint 6.2), whose proof needs exactly the surjectivity of §III.43. *Theorem 3.6.8 (preprint 3.54) with the index* `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index`: on an open $V_x \ni x_0$ every zero is in good position with IsSurjLinOfIndex$(f, y, d)$, and the solution set has at $x_0$ regular data of dimension $d$ whose chart domain *is* $V_x$, every datum at a point of the solution set in $V_x$ having dimension $d$.

**Theorems 3.1.26, 3.1.22 and 3.6.11 (preprint 3.23, 3.19 and 3.57) with the index.** Theorem 3.1.26 (preprint 3.23) `Chapter3/GlobalIFTIndexG · theorem_3_1_26_index`: for an sc-Fredholm section with tame base retractions and a zero $x$ in good position, an open $U \ni x$ and $d$ such that every zero $y \in U$ is in good position with IsSurjLinOfIndex$(f, y, d)$, the zeros in $U$ are smooth, and $S(f, U)$ is a sub-M-polyfold with tame induced structure equivalent to a $d$-dimensional manifold with boundary with corners — the proof of the boundary case from the nearby good position (§III.37) with Theorem 3.6.8 (2) (preprint 3.54) with the index. Theorem 3.1.22 (preprint 3.19) `Chapter3/GlobalIFTIndexG · theorem_3_1_22_index`: for $d_X \equiv 0$ the same, with degeneracy index 0 everywhere (Lemma 2.4.6 (preprint 2.26)): "a smooth manifold whose dimension agrees with the Fredholm index", now with the tameness of the induced structure. Theorem 3.6.11 (preprint 3.57) `Chapter3/GlobalIFTIndexG · theorem_3_6_11_index`: under the hypotheses of §III.37 (good position at every zero, $S \neq \emptyset$), at every $x \in S$ there are regular data of a dimension $d$, shared by every datum at the points of $S$ in their chart domain, with IsSurjLinOfIndex$(f, y, d)$ at every zero $y$ of that domain.

## III.45 The kernel as tangent space, good position at interior zeros, and the filler condition as a theorem

**"$\ker f'(y) = T_y S(f, U)$", in a chart** `Chapter3/GlobalIFTIndexG · theorem_3_6_8_ker_eq_tangent`. *Statement.* For an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero $x_0$ in good position, there are the strong bundle chart $c$ centred at $x_0$ of Definition 3.6.6 (preprint 3.53), the conjugating sc-diffeomorphism $\psi : U \to U' \subseteq C'$ with inverse $\psi'$ and $\psi(0) = 0$, the finite-dimensional $K = \ker D(h + t)(0)$, the solution graph $\sigma : V' \to E'$ ($C^1$, $\sigma(0) = 0$, $D\sigma(0) = 0$, $V'$ relatively open in $K \cap C'$), an open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$, and an open $V_{x,2} \subseteq V_x$ containing $x_0$, such that at every zero $y \in V_{x,2}$, with $\psi(\varphi(y)) = v + \sigma(v)$, and every level-1 point $z$ over $\varphi(y)$:

$$D\psi(z)(\ker \mathrm{Lin}(z)) = N_v = \{n + D\sigma(v)n \mid n \in K\},$$

the tangent space at $v$ of the graph, which is the solution set in the chart $\psi \circ \varphi$ of its tangent germ data. *Proof.* From the kernel statement at the nearby zeros (§III.32, `Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_nearby_kernel`): $z$ lies on level 1 of the germ of $g$, $\ker Dg(z) = D\psi'(\psi z)N_v$, $\ker Dg(z) \subseteq T_z O$, and $\mathrm{Lin}(z) = Dg(z)$ on $T_z O$ (the bridge for germs, which needs exactly $z$ in the germ and $0 \in O$, `Chapter3/LinearizationChartsG · linAt_eq_germDeriv`; whole-chart `Chapter3/LinearizationCharts · linAt_eq_scDeriv`), so $\ker \mathrm{Lin}(z) = \ker Dg(z)$; and $D\psi(z)D\psi'(\psi z) = 1$ on level 0 by the chain rule for $\psi \circ \psi' = 1$ on $U'$. □ The intrinsic statement, with $T_y S$ as a subspace of $T_y X$ (Definition 2.5.8 (preprint 2.38)), is not formalized: $T_y X$ has no linear structure in the formalization (Chapter 2 gap G.13); the chart form is the statement (Finding 172).

**Good position at a zero with $d_X = 0$ is surjectivity.** *Linear algebra* Chapter3/GlobalIFTIndex · isInGoodPositionIn_of_subset: a subspace $N$ of $T \subseteq E_0$ is in good position inside $T$ to every cone $C$ containing $T$ — $N \cap C = N$ has interior in $N$, and any algebraic complement of $N$ in $T$ is a good complement with any constant, both sides of the condition being true. *The statement* Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_linAt_surj_of_degIndex_eq_zero: let $f$ be an sc-smooth section, $y$ a zero with $(f, y)$ an sc-Fredholm germ and $d_X(y) = 0$, $c$ a strong bundle chart around $y$ with tame base retraction, and $\mathrm{Lin}(z)$ onto $K_z$ at the level-1 point $z$ over $\varphi(y)$; then $(f, y)$ is in good position. *Proof.* The model of $c$ is a tame retract, so $d_C(\varphi(y)) = d_X(y) = 0$ by the strong form of Proposition 2.5.6 (preprint 2.36) Chapter2/TameRetracts · degIndex_eq_chartDegIndex_of_isTameRetract; no coordinate of the presentation vanishes at $\varphi(y)$, so the partial quadrant at $\varphi(y)$ is the whole space, and the kernel condition of the assembly step holds by the linear algebra. The assembly step with all of $K_z$ in the range Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj concludes. □ For an sc-Fredholm section with tame base retractions, "$f'(y)$ onto in some chart" therefore gives good position at a zero with $d_X(y) = 0$ Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt ($y$ is smooth, so the germ is sc-Fredholm); conversely good position gives "$f'(y)$ onto" in the chart of Definition 3.6.6 (preprint 3.53) Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.isLinSurjAt; so at such a zero the two are equivalent Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt — the text's remark after Theorem 3.6.8 (preprint 3.54). In a non-tame chart $d_C(\varphi(y))$ can exceed $d_X(y)$, and the cone would have to be transported from a chart realizing $d_X(y)$; the standing tameness hypothesis makes that unnecessary (Finding 173).

**Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) with "$f'(x)$ surjective".** Theorem 3.1.22 (preprint 3.19) Chapter3/GlobalIFTIndexG · theorem_3_1_22_of_surjective: for $d_X \equiv 0$, an sc-Fredholm section with tame base retractions, $f(x_0) = 0$ and $f'(x_0)$ onto in some chart: the equivalence gives good position at $x_0$, and Theorem 3.1.22 (preprint 3.19) with the index (§III.44) applies. Theorem 3.6.10 (preprint 3.56) with the tameness and the regular data Chapter3/GlobalIFTIndexG · theorem_3_6_10_tame: for $d_X \equiv 0$ and good position at every zero, $S \neq \emptyset$: Theorem 3.6.11 (preprint 3.57) with the index (§III.44) and Lemma 2.4.6 (preprint 2.26) — the solution set is a sub-M-polyfold with tame induced structure, a manifold of locally constant dimension without boundary, with regular data at every zero of a dimension equal to the index at the zeros of their chart domain. With "$f'(x)$ surjective at every zero" Chapter3/GlobalIFTIndexG · theorem_3_6_10_of_surjective: the equivalence at every zero, then the previous statement.

**The converse of the matrix argument, and the filler condition as a theorem.** *The converse* Chapter3/GlobalIFTIndex · normal_injective_of_exists_range_scDeriv_eq_ρ. Let $z$ be a level-1 point over a point $z_0$ of the retract with $\ker Dg(z) \subseteq T_{z_0}O$, and suppose $\rho(z_0)k \in Dg(z)(T_{z_0}O)$ for every $k \in F$. Then $Dr(z)b = 0$ and $\rho(z_0)Dg(z)b = Dg(z)b$ imply $b = 0$: choose $a \in T_{z_0}O$ with $Dg(z)a = \rho(z_0)Dg(z)b = Dg(z)b$; then $a - b \in \ker Dg(z) \subseteq T_{z_0}O$, so $b \in T_{z_0}O$; writing $b = Dr(z)w$, idempotence gives $b = Dr(z)b = 0$ Chapter3/LinearizationCharts · scDeriv_r_idem_at. The surjectivity of $Dg(z)$ is not needed for this direction. With the direction of §III.33 Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective, under $Dg(z)$ onto and $\ker Dg(z) \subseteq T_{z_0}O$ (and $g$ agreeing with **f** on $O$ near a zero $z_0$ of **f**): $\rho(z_0)F \subseteq Dg(z)(T_{z_0}O)$ iff condition (3) of Definition 3.1.4 (preprint 3.4) holds at $z_0$ Chapter3/GlobalIFTIndex · exists_range_scDeriv_eq_ρ_iff_normal_injective — the "iff" of Finding 100, both directions formalized. Both are linear algebra in the operator $Dg(z)$ and hold for a germ at the points of its germ Chapter3/GlobalIFTIndexG · normal_injective_of_exists_range_deriv_eq_ρ_germ Chapter3/GlobalIFTIndexG · exists_range_deriv_eq_ρ_iff_normal_injective_germ. *The filler condition* Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm: for an sc-Fredholm section with tame base retractions and a zero $x_0$ in good position, $(f, x_0)$ is in good position with the filler condition at

the nearby zeros `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`. *Proof.* Take the chart and filler of the kernel statement (§III.32) and the open $V_{x,1}$ of nearby good position (§III.43); let $N$ be an open neighbourhood of $0$ on which $\mathbf{f} = g$ on $O$ and $O \cap N = \varphi(V_{x,2} \cap V_{x,1})$ (`Chapter2/MPolyfold · exists_isOpen_image_eq`). At a level-1 zero $z_1$ of $g$ over a point of $O \cap N$, on level $1$ of the germ of $g$ (the condition asks no more, §III.33): $z_0 = \varphi(y)$ for a zero $y$ of $f$ in $V_{x,2} \cap V_{x,1}$; $y$ is in good position, so $\mathrm{Lin}(z_1)$ is onto $K_{z_0}$ in the chart $c$ (§III.44); the kernel statement gives $\ker Dg(z_1) \subseteq T_{z_0}O$; and the converse gives condition (3) at $z_0$. □ So the added hypothesis of the first forms of Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19) (§§III.33, III.37) holds whenever their other hypotheses do: it was never a restriction, but proving that needed the surjectivity of §III.43 (Finding 174).

## III.46 What the second addendum taught us

1. **The text's Theorem 3.6.8 (1) (preprint 3.54) is true, and the gap of its proof can be closed.** The surjectivity of $f'(y)$ at the zeros near a zero in good position does not follow from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55), as the text (and the book's arXiv version) says; it follows from an implicit function theorem *at the nearby zero* for the perturbed filler $g + \mu\rho(r(\cdot))k$, one smooth direction $k$ at a time, whose curve of zeros is tangent to the retract by condition (2) of the filler `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_chart_onto` `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'` (§§III.41–III.43). With it, Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) hold with the text's hypotheses.
2. **The uniformity problem disappears when the contraction regions are written down.** Theorem 3.3.3 (preprint 3.30) is applied at the centre to the $k$-extended contraction germ, whose regions in $a$ are those of the unperturbed germ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction`; only the $\mu$-interval depends on $k$, and the bounds of the perturbation scale linearly in $k$ `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds`. The black-box use of Theorem 3.5.11 (preprint Corollary 3.47) for an extended problem gives neighbourhoods depending on the extension.
3. **The index raise of Theorem 3.4.2 (preprint 3.38) is what makes the perturbation $C^1$ on the shifted levels** (§III.42), and the surjectivity of the reduced finite-dimensional problem comes from a dimension count on level $0$ `Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq`, not from inverting $1 - D_2B$.
4. **The index is locally constant along the solution set** `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index_locally_constant`, the special case of the text's Proposition 6.2.2 (preprint 6.2), and the dimension of the solution set is the Fredholm index at every zero, in every chart `Chapter3/GlobalIFTIndexG · theorem_3_1_22_index` (§III.44); $\ker f'(y)$ is the tangent space of the solution set in the chart of the proof `Chapter3/GlobalIFTIndexG · theorem_3_6_8_ker_eq_tangent` (§III.45).
5. **The filler condition at the nearby zeros was never a restriction** `Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm`, and good position at a zero with $d_X = 0$ is surjectivity for a tame strong bundle `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt` (§III.45), so Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) hold with "$f'(x)$ surjective".
6. **Theorem 3.1.21's (preprint 3.18) "readily verified" is a construction**: translation charts of $\mathbb{R}^n$, the pull-back bundle as a structure, the product rule for a scalar factor, and the extension of a basic germ by inert parameters `Chapter3/ParamPerturbation · IsBasicGerm.paramExtend` `Chapter3/ParamPerturbationG · theorem_3_1_21_preprint` (§III.40).

7. **What is left of Chapter 3**: Proposition 3.6.5 (preprint 3.52) as an sc-Fredholm operator on $T_xX$; $\ker f'(y) = T_yS$ with $T_yS \subseteq T_yX$; Theorem 3.7.1 (preprint 3.58) for a general sc-splitting; Lemma 3.8.13 (preprint 3.70) for a general presentation of the partial quadrant; Theorem 3.2.3 (preprint 3.27) for sets whose data violate the second regularity condition; and Proposition 6.2.2 (preprint 6.2) beyond solution sets (since proved, with the text's hypotheses for paths on $\mathbb{R}$, §III.49). The only added hypotheses are the closedness of good complements and the regularity conditions of Theorem 3.2.3 (preprint 3.27) for general subsets. All of this is closed since, and neither hypothesis remains (§§III.56–III.57, III.61–III.63).

## III.47 The tameness hypothesis repaired

The implicit function theorems of §§III.23 and III.32–III.45 were first formalized with the hypothesis, `htame` in the Lean, that the base retraction of **every** strong bundle chart compatible with the atlas is tame; the phrase "a strong bundle whose charts have tame base retractions" in those sections describes it. No strong bundle over a nonempty base satisfies it, so the statements that assumed it were vacuously true (Finding 190). This section records why, and what replaced it. The proofs told above are unchanged; the hypotheses and the charts of the conclusions changed, and the names of the theorems did not.

**Why the hypothesis fails.** Compatibility of strong bundle charts (Definition 2.6.4 (preprint 2.50)) constrains the transition maps between the retracts $K$, not the retraction whose image $K$ is. From a compatible chart with base retraction $r$ on $U \subseteq C$, the chart $(1, \varphi)$ onto $\{1\} \times O$, with model $\mathbb{R} \oplus E$, quadrant $[0,\infty) \times C$ and base retraction $r'(a,u) = (1, r(u))$ on $[0,\infty) \times U$, has the transition maps of the first chart composed with affine maps, so it is compatible with every chart the first one is; but $r'$ lowers the degeneracy index at $(0,u)$. The last point is machine-checked. The constant retraction $(a,w) \mapsto (1,0)$ of $[0,\infty) \oplus W$ `Chapter3/TameVacuity · ScRetraction.halfLineConst` has retract $\{(1,0)\}$ `Chapter3/TameVacuity · ScRetraction.halfLineConst_retract`, and its product with any sc-smooth retraction $r$ of a nonempty $U$ is not tame `Chapter3/TameVacuity · ScRetraction.halfLineConst_prod_not_isTame`: the degeneracy index of a product is the sum of the indices `Chapter2/FibredProduct · degIndex_prod`, so it is $1 + d_C(u)$ at $((0,0),u)$, $u \in r(U)$, and $d_C(u)$ at the image $((1,0),u)$, against condition (1) of Definition 2.5.2 (preprint 2.34). That the chart $(1,\varphi)$ is compatible is not formalized.

**The new hypotheses.** The text asks for the tameness of particular charts: the chart of an sc-Fredholm germ has "a tame strong local bundle" $K \to O$ (Definitions 3.1.6 and 3.6.3 (preprint 3.6 and 3.50)), and the proof of Proposition 3.6.5 (preprint 3.52) starts from "a tame strong bundle chart". A *tame chart* is a compatible strong bundle chart whose base retraction is tame `Chapter3/TameGerms · StrongBundle.IsTameChart`. *Tame germ data* at $x$ are the data of Definition 3.1.11 (preprint 3.11) (§III.6) in a tame chart `Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame`; *good position in a tame chart* adds Definition 3.6.6 (preprint 3.53) in that chart (§III.21) `Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame`; and an sc-Fredholm section *with tame germ data* has them at every smooth point `Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionTame`. Each implies the notion without tameness `Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame.isScFredholmGerm` `Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame.isGoodPositionGerm`. They can be met: tameness depends only on the domain of a retraction and its values there `Chapter3/TameGerms · ScRetraction.IsTame.congr`, and the identity retraction is tame `Chapter2/Tame · ScRetraction.id_isTame`, so a compatible chart whose base retraction is the identity of $U$ is tame `Chapter3/TameGerms · StrongBundle.isTameChart_of_eqOn_id`.

**The restated theorems.** The proofs used the tameness for three charts only (Finding 191): the chart

of the germ data at $x_0$, for $(1 - Dr(0))E \subseteq E_0$ in the linear part (§III.23) `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj_conj`; the chart $\psi \circ \varphi$ built from it, whose model is tame because $r$ is (§§III.27, III.34) `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart_proj_conj`; and the chart in which $f'(y)$ is onto at a zero with $d_X(y) = 0$ (§III.45). So each theorem now takes $(f, x_0)$ in good position in a tame chart in place of good position and `htame`; where it concludes something at the other zeros it takes an sc-Fredholm section with tame germ data; its conclusions are good position in tame charts, and the charts it produces are tame. This applies to Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'`, to its first form, whose filler condition is now stated in a tame chart `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one`, to Theorems 3.1.26 and 3.1.22 (preprint 3.23 and 3.19) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_26' Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_22'`, to the global theorems, with good position in a tame chart at every zero `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11 Chapter3/GlobalIFTTheoremsG · theorem_3_6_10 Chapter3/GlobalIFTTheoremsG · corollary_3_1_27`, to the forms with the index `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index`, and to the filler condition as a theorem `Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm`.

**Reading the conclusion in the tame chart.** At a nearby zero $y$ the good position is proved in the chart at $x_0$ and carried to a chart centred at $y$ (§III.31). That chart was the one Definition 3.1.11 (preprint 3.11) supplies at $y$, of which nothing says that it is tame. The chart change now gives the good position in **every** chart carrying germ data at $y$ `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionAt_of_chart_dense`, the statement of §III.31 being its corollary `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`. A good position given in any chart therefore holds in the tame chart of the germ data `Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame`: in the chart of the given good position $Tf(0)$ maps onto $K_0 \supseteq \rho(0)F_\infty$, $\ker Dg(0) \subseteq T_0O$ and $C_0 = C$, which are the hypotheses of the chart change at $z = 0$; in the germ reading the point $z = 0$ lies in every germ, $Tf(0)$ is onto all of $K_0$, and the form of the chart change with all of $K_0$ in the range needs no Proposition 3.6.5 (preprint 3.52) (§III.31) `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart`. The same step gives good position from "$f'(y)$ surjective" at a zero with $d_X(y) = 0$: the surjectivity, given in one chart, holds in every chart `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective`, in particular in the tame chart of the germ data at $y$, where $d_C = d_X = 0$ `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`.

**Theorems 3.1.18 and 3.1.21 (preprint 3.15 and 3.18) keep tame germ data.** The proof of Theorem 3.1.18 (preprint 3.15) (§III.6) does not change the chart, so $f + s$ has germ data in the tame charts of $f$ `Chapter3/TameGermsAddG · StrongBundle.IsScFredholmGermTame.sectionAdd Chapter3/TameGermsAddG · theorem_3_1_18_tame`. For Theorem 3.1.21 (preprint 3.18) (§III.40) the germ of $\tilde{f}$ at $(r_0, x)$ is taken in the product chart $(r \mapsto r - r_0) \times \varphi$, whose base retraction $1 \times r$ on $\mathbb{R}^n \times U$ is tame when $r$ is: the identity retraction of $\mathbb{R}^n$ is tame, so $d(a, u) = d_C(u)$, and the minimal subspaces and tangent spaces of the product are products `Chapter2/FibredProduct · ScRetraction.IsTameWith.prod Chapter3/ParamPerturbation · PullbackChartData.ofProd_isTame`. Hence $\tilde{f}$ has tame germ data `Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmGermTame.param Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSectionTame.param`, and so has $F(r, x) = f(x) + \sum_i r_i s_i(x)$ `Chapter3/ParamPerturbationG · theorem_3_1_21_tame`, which is what the implicit function theorems need when applied to $F$.

**Not done** (Finding 199): the compatibility of the chart $(1, \varphi)$, that is, the vacuity of `htame` at the level of charts; and the passage from an atlas of tame charts to tame germ data (Definition 3.1.11 (preprint 3.11) □ Definition 3.6.3 (preprint 3.50)). The text builds the tame chart into Definition 3.6.3 (preprint 3.50), and so do the theorems. The passage does not hold in general (§III.65).

**Germ data and the charts of a tame atlas** (Findings 350–353, fourth addendum). The passage was to be made by re-centring a tame chart of the atlas at the point and transporting the filler and the conjugation to it. That strategy fails, because a filling is bound to its chart. *Condition (3) of Definition 3.1.4 (preprint 3.4) is a condition on the chart*: a filling germ in a chart with strong bundle retraction $R = (r, \rho)$ gives a topological isomorphism $\ker Dr(0) \cong \ker \rho(0)$, the linearization of the fill defect $y \mapsto [1 - \rho(r(y))]g(y)$ on $\ker Dr(0)$ `Chapter3/TameGermData · IsFillingGerm.nonempty_kerEquiv`; so in a chart with $\ker Dr(0) \neq 0$ and $\rho(0)$ injective no map fills anything `Chapter3/TameGermData · not_isFillingGerm_of_ker`. *A tame chart with no filling.* For the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ take the model $\mathbb{R} \oplus \mathbb{R}^n$ without boundary, the base retraction $r(a, u) = (0, u)$, the product of the constant retraction of $\mathbb{R}$ onto $0$ with the identity `Chapter3/TameGermData · TrivialBundle.Reven`, the chart $x \mapsto (0, x)$, and $R((a, u), h) = ((0, u), h)$ `Chapter3/TameGermData · TrivialBundle.evenBundleChart`. It is compatible with the translation charts `Chapter3/TameGermData · TrivialBundle.isChart_evenBundleChart` and tame — the degeneracy index of $\mathbb{R} \oplus \mathbb{R}^n$ is $0$ everywhere and $\mathbb{R} \oplus 0$ is an sc-complement of $T_x O = 0 \oplus \mathbb{R}^n$ `Chapter3/TameGermData · TrivialBundle.Reven_isTame` `Chapter3/TameGermData · TrivialBundle.isTameChart_evenBundleChart` — but $\ker Dr(0) = \mathbb{R} \oplus 0$ while $\rho(0) = 1$, so no map fills the principal part of any section in it `Chapter3/TameGermData · TrivialBundle.not_isFillingGerm_evenBundleChart`. *A tame atlas without germ data in its chart.* With the one-chart atlas $\{\text{this chart}\}$ `Chapter3/TameGermData · TrivialBundle.bundleEven` every chart of the atlas is tame `Chapter3/TameGermData · TrivialBundle.bundleEven_atlas_isTameChart`, the graph section of a $C^\infty$ map has tame germ data — in the compatible translation charts `Chapter3/TameGermData · TrivialBundle.isScFredholmSectionTame_graphSection_even` — and in the chart of the atlas no section has a filling `Chapter3/TameGermData · TrivialBundle.tame_atlas_no_germData_in_atlas_chart`. So Definition 3.6.3 (preprint 3.50) must be the existential over all compatible tame charts, as formalized; read as "in a chart of the tame atlas" it would make every section of this bundle non-Fredholm. Whether germ data in some compatible chart give germ data in some compatible tame chart, for a tame atlas, was left open here (Finding 353, gap G.79): the transition between two charts is defined on the retracts only, while the filling lives off them, so it cannot simply be pushed forward. The answer is no, even for a tame strong bundle over a tame M-polyfold without boundary (§III.65); it is yes when the chart of the data is centred at an interior point of its partial quadrant (§III.65).

## III.48 Definition 3.1.11 (preprint 3.11) on germs

The text states Definitions 3.1.4, 3.1.6 and 3.1.11 (preprint 3.4, 3.6 and 3.11) for germs: the filled version is "a germ $(g, 0)$", the sc$^+$-section a local one, and a germ of neighbourhoods (Definition 3.1.1 (preprint 3.1)) is a sequence $U_m$ that may shrink with $m$. The first formalization asked $g$ and $s$ to be sc-smooth on the whole chart domain $U$, on the ground that a chart can be restricted. This section records why that was narrower, and the two devices on which the proofs in the text's germ reading rest: the calculus of germs with whole-chart maps, and the shrinking of the basic germ (Findings 260–334).

**Germs are not restrictions.** On $\mathbb{R}$ with the constant structure, $f(x) = \sum_n c_n (x - a_n)^{n+1} |x - a_n|$, $a_n = 1/(n+2)$, $c_n = (4(n+2))^{-(n+2)}$, is $C^k$ on $U_k = (-1/(k+2), 1/(k+2))$ — the terms $n \leq k$ have their singular points outside, the tail is $C^{k+2}$ — and $C^{n+2}$ on no open set containing $a_n$. So $f$ is an sc-smooth germ on $(U_m)$ and sc-smooth on no open $U \ni 0$ `Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth`. Restricting a chart on level $0$ never makes the levels shrink, so the whole-chart reading is a restriction of the class of sections; for a section sc-smooth on $X$ the two readings of Definition 3.1.11 (preprint 3.11) differ exactly in the regularity of $g$ and $s$ away from their germs (§III.6). Whether every germ datum can be traded for a whole-chart one is open: replacing the sc$^+$-germ by a whole-chart sc$^+$-section costs one level (the index raise of Theorem 3.4.2 (preprint 3.38)), and on level $0$ no argument

was found in either direction (Finding 264). No theorem depends on the answer, since the theorems are proved in the germ reading.

**The calculus with whole-chart maps.** The text forms $g + s$, $g - s$, $[1 - \rho(r(y))]g(y)$ and $\Psi_*(g)$ for germs without comment. Sums, differences, scalar multiples and compositions with sc-operators of sc¹-germs are sc¹-germs with the explicit derivative families, on $G_1 \cap G_2$ for germs on two germs `Chapter3/GermCalculus · IsSc1GermWith.add` `Chapter3/GermCalculus · IsScSmoothGerm.add_inter`. The chain rule of §III.1 composes two germs *at* 0, but $R$ and $\Psi$ are applied at $(r(y), g(y))$, which is $(0, 0)$ only when $g(0) = 0$ and $0 \in O$; so the chain rule is proved again with the outer map sc¹ on a set $V$ and the inner one a germ mapping $G_0$ into $V$, with no condition at a base point `Chapter3/GermCalculus · IsSc1With.comp_germ` `Chapter3/GermCalculus · IsScSmooth.comp_germ`, and with a whole-chart inner map $\psi$, $\psi(0) = 0$, on the preimage germ `Chapter3/GermCalculus · IsScSmoothGerm.comp_preimage`. Hence the fill defect of a germ is a germ on the same germ (§III.4); the push-forward $\Psi_*(g)$ is a germ on the germ of the points that $\psi^{-1}$ maps into the germ of $g$, with $D(\Psi_* g)(0) = \Gamma(0) Dg(0) D\psi(0)^{-1}$ (§III.23); a germ plus a whole-chart map is a germ on the same germ (Theorem 3.1.18 (preprint 3.15), §III.6); and $(a, u) \mapsto g(u)$ is a germ on $\mathbb{R}^n \times G$ (Theorem 3.1.21 (preprint 3.18), §III.40). The strong bundle isomorphism $\Psi$ and the basic germ $h$ remain what they were.

**A basic germ restricts.** Definitions 3.1.7 and 3.3.1 (preprint 3.7 and 3.28) pass to every smaller germ with the same $h$ and $B$: the contraction field after shrinking its radius, since every level of a germ is a relatively open neighbourhood of 0, and the sc-smoothness by the uniqueness of derivatives in the quadrant `Chapter3/BasicGermShrink · IsBasicGerm.mono`.

**The shrunk basic germ $G''$.** The text does not relate the germ $G$ of the basic germ to the germs $G_g$, $G_s$ of $g$ and $s$, and nothing forces $\psi^{-1}(G_m) \subseteq (G_g)_m$. For conjugation data with base map $\psi$, inverse $\psi'$ and basic germ on $G$, let $G''_m = \{x \in G_m \mid \psi'_m(x) \in (G_g)_m \cap (G_s)_m\}$, the preimage germ along the sc⁰-germ $\psi'|_G$ ($\psi'$ is sc-smooth on all of $U'$ and $\psi'(0) = 0$) `Chapter3/BasicGermShrink · ConjugationData.shrinkGerm`. The conjugation data shrunk to $G''$ have the same $\Psi$, $h$, $B$ and conjugation identity `Chapter3/BasicGermShrink · ConjugationData.shrink` `Chapter3/BasicGermShrink · ConjugationData.basic_shrinkGerm`; a point of $G''_m$ is sent by $\psi'_m$ into $(G_g)_m \cap (G_s)_m$ `Chapter3/BasicGermShrink · ConjugationData.levelMap_mem_inter_of_mem_shrinkGerm`, and read on the $E$-side, $u \in U_m$ with $\psi_m(u) \in G''_m$ lies in both germs `Chapter3/BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem`. *Corollary 3.5.3 (preprint 3.41) on $G''$*: every zero $x$ of the normal form with $|x|_0$ small is $\iota x_1$ with $x_1$ on level 1 of $G''$ — it is small on level 1 (Theorem 3.5.1 (preprint 3.40)), and the contraction field puts a level-1 ball around 0 inside $G''_1$ — so $\psi'_1(x_1)$ lies on level 1 of $G_g$ and $G_s$ `Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_5_3_shrink`. This is the device of every argument at the nearby zeros: the level-1 lift $z_{0,1}$ of a zero near $x_0$ lies in the germ of $g$, where $Dg(z_{0,1})$ is defined (§§III.32, III.33, III.43, III.45). It relies on the lifts being small on level 1; a point small on level 0 only need not lie in $G''_1$. At the centre no shrinking is needed: 0 lies in every germ, and $Dg(0)$ is the same for all germs and families (§III.4).

**What the proofs use of $g$ and $s$.** At 0: $Dg(0)$ and $Ds(0)$ (Propositions 3.1.5 and 3.6.5 (preprint 3.5 and 3.52), Definition 3.6.6 (preprint 3.53), the linear part of Theorem 3.6.8 (preprint 3.54); §§III.4, III.21, III.23, III.30). On level 0 near 0: continuity of $g$ and of the sc⁺-part (Theorem 3.1.13 (preprint 3.12), §III.13; the local step of Theorem 5.2.2 (preprint 5.5), §V.6). At the level-1 lifts of the nearby zeros: $Dg$, through $G''$ (Theorem 3.6.8 (1) (preprint 3.54), §§III.32–III.33, III.43, III.45). Nowhere $g$ or $s$ on $U \cap E_m$ away from their germs. So every theorem of Chapter 3 holds in the germ reading with the text's hypotheses and its former conclusions, and no hypothesis is added; the statements that named the derivative of $g$ on $U$ at 0 (Proposition 3.6.5 (preprint 3.52), the linearization statements of Theorems 3.1.22 and 3.6.8 (preprint 3.19

and 3.54)) now name the germ, a derivative family $Dg$ and $Dg(0)$, and Corollary 5.2.5 (preprint 5.8) in a chart has an sc⁺-germ on the shifted germ for its sc⁺-part (§V.7).

**The switch.** The canonical names now denote the germ reading: Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4 and 3.6.6 (preprint 3.11, 3.14, 3.50, 3.51 and 3.53) with the tame, filled and sc variants of good position `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm` `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection` `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm`, and the theorems built on them. The whole-chart notions and theorems keep their statements under the names `…WholeChart` and `…_wholeChart`, and each whole-chart notion implies its germ twin `Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionWholeChart.isScFredholmSection` `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGermWholeChart.isGoodPositionGerm`. The non-vacuity witnesses are whole-chart data, hence germ data. Chapter 5 is stated with the germ notions: the local step of Theorem 5.2.2 (preprint 5.5) and the spanning sections changed locally, and Corollary 5.2.5 (preprint 5.8) in a chart changed its statement (§§V.6, V.12, V.7).

## III.49 Definition 6.2.1 (preprint 6.1) and Proposition 6.2.2 (preprint 6.2): the index along paths

The text's Chapter 6 opens with the index of an sc-Fredholm germ at a smooth point which need not be a zero (Definition 6.2.1 (preprint 6.1), pp. 237–240 (preprint pp. 200–201)) and proves that it is constant along sc-smooth paths of smooth points (Proposition 6.2.2 (preprint 6.2), p. 240 (preprint p. 201)). Its proof is the device that §III.44 used along the zero set: stabilize by finitely many sc⁺-sections and apply the implicit function theorem. This section retells both: first with sc⁺-sections on all of $X$, which needs bump functions (Findings 340–348), then as the text does it, with locally defined sections on the restriction of the bundle to a chart domain and without bump functions (Findings 349, 360–367).

**Definition 6.2.1 (preprint 6.1)** `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt`. *$f$ has index $i$ at $x$* if there is an sc⁺-section $s$ of $P$ with $(f+s)(x)=0$ and $(f+s)'(x)$ Fredholm of index $i$ in every strong bundle chart around $x$ (the chart-intrinsic statement of §III.44). The text's $f-s$ with $s(x)=f(x)$ is $f+(-s)$; the text's section is local, this one global; the reading with local sections, and the agreement of the two readings under bump functions, close this section. *Well defined.* At a zero of an sc-Fredholm section the index in `IsLinOfIndex` is unique, because in a chart $\mathrm{Lin}(z)$ is a bounded operator into $K_z$ (§III.44), so the quantification over corestrictions is not vacuous `Chapter3/IndexAlongPaths · StrongBundle.IsLinOfIndex.unique`. Adding an sc⁺-section $s$ with $s(x)=0$ at a zero $x$ of $f$ does not change the index: the germ datum of $f$ at $x$, with conjugation data $dc$, gives the index $\dim P_n - \dim Q$ of $f'(x)$ (Proposition 3.6.5 (preprint 3.52)), and Theorem 3.1.18 (preprint 3.15) gives germ data of $f+s$ in the same chart whose conjugation data are $dc$ up to their values on $U$, since $(g+\hat{s}\circ r)-(\hat{s}_1\circ r+t_0)=g-t_0$ `Chapter3/IndexAlongPaths · StrongBundle.isLinOfIndex_sectionAdd_of_isZeroAt`. For two sections $s, s'$ of the definition, $s'-s$ is an sc⁺-section vanishing at $x$ `Chapter3/IndexAlongPaths · isZeroAt_sectionDiff_of_isZeroAt_sectionAdd` and $(f+s)+(s'-s)=f+s'$ `Chapter3/IndexAlongPaths · sectionAdd_sectionAdd_sectionDiff`, so the indices agree `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.unique`. The text argues instead that $(t-s)'(x)$ is an sc⁺-operator, hence compact on every level, and invokes Proposition 1.1.11 (preprint 1.10); the project has no lemma that the linearization of an sc⁺-section is compact, and the germ data make it unnecessary. At a zero the index of the germ is that of $f'(x)$ `Chapter3/IndexAlongPaths · StrongBundle.isIndexAt_of_isZeroAt` `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.isLinOfIndex_of_isZeroAt`. *Existence.* If $X$ admits bump functions, every smooth point has an index: $f(x)$ is a smooth point of $Y$, Lemma 5.3.3 (preprint 5.11) gives an sc⁺-section $s_0$ with $s_0(x)=f(x)$, and for $s=-s_0$ the germ data of the sc-Fredholm section $f+s$

at its zero $x$ (Theorem 3.1.18 (preprint 3.15)) give the index Chapter3/IndexAlongPaths · StrongBundle.exists_isIndexAt.

**Along the zero set, without good position** (Findings 240–242). Let $X$ admit bump functions, $f$ be an sc-Fredholm section with tame germ data and $f(x_0) = 0$; neither good position nor surjectivity at $x_0$ is assumed. In a chart centred at $x_0$, finitely many smooth vectors span a complement of the range of $\mathrm{Lin}(0)$, closed of finite codimension by Proposition 3.6.5 (preprint 3.52), and Lemma 5.3.3 (preprint 5.11) realizes them as allowable sc⁺-sections $s_1, \dots, s_k$ Chapter3/IndexLocallyConstantG · exists_spanning_sections_germ; with further sections that tilt the kernel into good position at a corner (the conditions (i)–(ii) of §V.19), the parametrized section $F(\lambda, x) = f(x) + \sum_i \lambda_i s_i(x)$ of Theorem 3.1.21 (preprint 3.18) is in good position at $(0, x_0)$ in a tame chart Chapter3/IndexLocallyConstantG · exists_sections_isParamGP_germ Chapter3/IndexLocallyConstantG · isGoodPositionGermTame_paramSection_of_isParamGP. So the index of $F'$ is one integer $d$ at the zeros of $F$ near $(0, x_0)$ (§III.44), among them $(0, y)$ for every zero $y$ of $f$ near $x_0$. At $(0, y)$, in the product chart, $\mathrm{Lin}_F(0)(\delta\lambda, \delta u) = \mathrm{Lin}_f(0)\delta u + \sum_i \delta\lambda_i \widehat{s}_i(0)$, a finite-rank extension, so $d = k + \mathrm{ind}\, f'(y)$ Chapter3/IndexLocallyConstant · fredholmIndex_eq_of_param Chapter3/IndexLocallyConstantG · StrongBundle.isLinOfIndex_of_paramSection_isLinOfIndex. Hence $f'(y)$ has index $d - k$ in every chart at every zero $y$ near $x_0$ Chapter3/IndexLocallyConstantG · index_locally_constant_of_admitsBumpFunctions, also as an sc-Fredholm operator on every level Chapter3/IndexLocallyConstantG · theorem_3_6_8_index_locally_constant_general. This is the text's device for Proposition 6.2.2 (preprint 6.2) in the case of a zero.

**The local form.** For $X$ with bump functions, $f$ with tame germ data and an sc⁺-section $s$ with $(f + s)(x_0) = 0$, the index of the germ is one integer at the zeros of $f + s$ near $x_0$: $f + s$ has tame germ data (§III.47) and the previous paragraph applies to it Chapter3/IndexAlongPaths · isIndexAt_locally_constant_of_isZeroAt_sectionAdd. Hence along a path continuous at $t_0$ on which $f + s$ vanishes near $t_0$ the index is constant near $t_0$ Chapter3/IndexAlongPaths · isIndexAt_eventually_const_of_path, and on a preconnected parameter space a locally constant index is constant: by uniqueness the index is a locally constant function of $t$ Chapter3/IndexAlongPaths · isIndexAt_const_of_forall_eventually. What remains is a section $s$, depending on $t_0$, that vanishes along the path — the text's $s(t, x)$ on $[0, 1] \times X$ — and the passage from the product back to $X$.

**Bump functions on $\mathbb{R}^n \times X$.** The device works on the product of Theorem 3.1.21 (preprint 3.18) (§III.40), which must admit bump functions when $X$ does. A $C^\infty$ function on $\mathbb{R}^n$ is sc-smooth on the M-polyfold $\mathbb{R}^n$ Chapter3/IndexAlongPaths · isScSmoothMap_rn_real_of_contDiff, products of sc-smooth functions are sc-smooth by the product rule in every chart Chapter3/IndexAlongPaths · MPolyfold.IsScSmoothMap.mul, and $(r, x) \mapsto \beta_1(r)\beta_2(x)$ has support in $\mathrm{supp}\,\beta_1 \times \mathrm{supp}\,\beta_2$ Chapter3/IndexAlongPaths · tsupport_prod_mul_subset; so $\mathbb{R}^n \times X$ admits bump functions Chapter3/IndexAlongPaths · MPolyfold.admitsBumpFunctions_prod_rn. Then for the pull-back $\tilde{f} = f \circ \pi$ and an sc⁺-section $S$ of $\pi^* P$ with $\tilde{f} + S$ vanishing along the graph $t \mapsto (t, \varphi(t))$ near $t_0$, the index of $(\tilde{f} + S)'(t, \varphi(t))$ in every chart is one integer for $t$ near $t_0$: $\tilde{f} + S$ has tame germ data (§§III.40, III.47), and the local constancy along its zero set applies Chapter3/IndexAlongPaths · isLinOfIndex_paramLift_sectionAdd_eventually_const. The proposition thereby reduces to two steps, the section along the path (E) and the index count on a slice (S) Chapter3/IndexAlongPaths · proposition_6_2_2_of_sections_of_slice.

**(E): the text's section along the path** Chapter3/IndexAlongPathsDevice · exists_isScPlusSection_pullback_of_path. For an sc-smooth section $f$ (sc-Fredholm is not needed), an sc-smooth path $\varphi : \mathbb{R}^n \to X$ and $t_0$, there is an sc⁺-section $S$ of $\pi^* P$ with $\tilde{f} + S = 0$ along the graph near $t_0$. Take a strong bundle chart $c$ around $\varphi(t_0)$. First the path is reparametrized by a $C^\infty$ map $\sigma$ of $\mathbb{R}^n$ equal to the identity near $t_0$ with values in $\varphi^{-1}(\mathrm{dom}\,\varphi_c)$, $\sigma(t) = t_0 + b(t - t_0)(t - t_0)$ for a smooth bump $b$ Chapter3/IndexAlongPathsDevice · exists_contDiff_eq_id_near_mapsTo; $C^\infty$ self-maps of $\mathbb{R}^n$ are sc-smooth Chapter3/IndexAlongPathsDevice · isScSmoothMap_

rn_of_contDiff, so $\psi = \varphi \circ \sigma$ is an sc-smooth path that stays in the chart. The representative $t \mapsto \Phi(f(\psi(t)))$ is sc-smooth from the constant structure $\mathbb{R}^n$ into $E \oplus F$; Proposition 1.2.2 (preprint 1.17) makes it sc-smooth from $(\mathbb{R}^n)^1 = \mathbb{R}^n$ into $(E \oplus F)^1 = E^1 \oplus F^1$, and its $F^1$-component $\hat{e}_1$ is the curve of smooth points that the text's $s(t,x) = R(x, f(\varphi(t)))$ needs in $F^1$ — no new analysis is required. In the product chart $(r \mapsto r - t_0) \times \varphi_c$ of $\pi^*P$ the local section is $\tau(a,u) = -R[1](u, \hat{e}_1(a + t_0))_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is cut off by a bump of $X$ equal to 1 near $\varphi(t_0)$ with support in the chart domain (Proposition 5.5.3 (preprint 5.30)) and extended by 0 (§V.9). Along the graph $\Phi(S(t,\varphi(t)))_2 = -R(\varphi_c(\varphi(t)), \Phi(f(\varphi(t)))_2)_2 = -\Phi(f(\varphi(t)))_2$, since $\Phi(f(\varphi(t))) \in K$ is fixed by $R$, so $\tilde{f} + S$ vanishes there for $t$ near $t_0$, where $\sigma(t) = t$ and the bump is 1. The bump function and the reparametrization make the text's local section global; the text needs neither.

**(S): the slice.** The slice $S(t,\cdot)$ of a section of $\pi^*P$, read in $Y$ through the fibres $(\pi^*Y)_{(t,x)} = P^{-1}(x)$, is a section of $P$ Chapter3/IndexAlongPathsDevice · pullbackSlice Chapter3/IndexAlongPathsDevice · IsSection.pullbackSlice; $G(t,x) = 0$ iff the slice vanishes at $x$ Chapter3/IndexAlongPathsDevice · isZeroAt_pullback_iff_pullbackSlice, and the slice of $\tilde{f} + S$ is $f + S(t,\cdot)$ Chapter3/IndexAlongPathsDevice · pullbackSlice_sectionAdd_paramLift. *The slice of an sc⁺-section is sc⁺* Chapter3/IndexAlongPathsDevice · StrongBundle.IsScPlusSection.pullbackSlice: for a chart $c$ of the atlas of $P$, the product chart $(r \mapsto r - t) \times \varphi_c$ is a chart of the atlas of $\pi^*P$, and the local representative of $S(t,\cdot)$ is that of $S$ composed with the sc-operator $u \mapsto (0,u)$ Chapter3/IndexAlongPathsDevice · ScBanach.prodSndLeftOp and projected. *The linearization on the slice direction*: in the product chart at $(t,x)$, whose base retraction is $(a,u) \mapsto (a, r(u))$ Chapter3/IndexAlongPathsDevice · PullbackChartData.r_ofProd, $\mathrm{Lin}_G(0)(0,\delta u) = \mathrm{Lin}_{G(t,\cdot)}(0)\delta u$, by the chain rule for $u \mapsto \widehat{G}(0, r(u))$ Chapter3/IndexAlongPathsDevice · StrongBundle.linAt_ofProd_slice. *The index count*: if $G$ is an sc-Fredholm section of $\pi^*P$ whose slice $g = G(t,\cdot)$ is an sc-Fredholm section with $g(x) = 0$, and $G'(t,x)$ is Fredholm of index $i$ in every chart, then $g'(x)$ is Fredholm of index $i - n$ in every chart: $\mathrm{Lin}_G(0)(\delta a, \delta u) = \mathrm{Lin}_g(0)\delta u + \mathrm{Lin}_G(0)(\delta a, 0)$ by linearity and the slice identity, the last term has rank at most $n$, and the parameter count of the zero-set case applies, the chart independence coming from the germ datum of $g$ at $x$ Chapter3/IndexAlongPathsDevice · StrongBundle.isLinOfIndex_pullbackSlice_of_isLinOfIndex. For $G = \tilde{f} + S$ ($n = 1$) this is $\mathrm{ind}(f,x) = i - 1$, with $S(t,\cdot)$ as the section of Definition 6.2.1 (preprint 6.1) Chapter3/IndexAlongPathsDevice · StrongBundle.isIndexAt_of_pullback_isLinOfIndex.

**Proposition 6.2.2 (preprint 6.2)** Chapter3/IndexAlongPathsDevice · proposition_6_2_2. *Let $X$ admit sc-smooth bump functions, $f$ be an sc-Fredholm section with tame germ data and $\varphi : \mathbb{R} \to X$ an sc-smooth path. Then* $\mathrm{ind}(f, \varphi(t))$ *is one integer for all $t$*; in particular $\mathrm{ind}(f,\varphi(0)) = \mathrm{ind}(f,\varphi(1))$. For each $t_0$, (E) gives $S$; the index of $\tilde{f} + S$ is constant near $t_0$ along the graph; (S) turns it into $\mathrm{ind}(f,\varphi(t)) + 1$; and $\mathbb{R}$ is connected. The paracompactness of $\mathbb{R} \times X$ and of $\pi^*Y$ is supplied by Theorem 2.3.10 (preprint 2.16) Chapter3/IndexAlongPathsDevice · proposition_6_2_2'. *Against the text.* (i) The path is defined on $\mathbb{R}$, not on $[0,1]$: $\mathbb{R} \times X$ is the product of Theorem 3.1.21 (preprint 3.18) and has no boundary, and the text's "tame M-polyfold $[0,1] \times X$" is not used; a path on $[0,1]$ that extends sc-smoothly to $\mathbb{R}$ is covered, and by the reparametrization in (E) only a neighbourhood of each $t_0$ matters. (ii) The hypotheses are tame germ data, the reading of "tame strong bundle" of §III.47, and bump functions, which the text does not assume ("we do not need a sc-smooth partition of unity"): they are used for the stabilizing sections (Lemma 5.3.3 (preprint 5.11)) and for the cut-off of the section along the path, both globalizations of the text's locally defined sections. Without them the device runs on the restriction of $P$ to a chart domain, as the text intends; that is the rest of this section. (iii) Definition 6.2.1 (preprint 6.1) is proved well defined without Proposition 1.1.11 (preprint 1.10). (iv) The two points the text's proof does not address — the tilting sections for the kernel condition at a corner and the uniformity in the zero — are those of the zero-set case above.

**Without bump functions: the restriction to an open set** (Findings 349, 360). The text restricts the bundle to a chart domain without comment ("in a chart, $s(y) = R(y, f(x))$"; "we do not need a sc-smooth partition of unity", preprint p. 202). For an open $V \subseteq X$, a chart of $X$ whose domain lies in $V$ is transported to a chart of the subtype $V$ with the same model `Chapter3/RestrictOpen · MChart.restrictTo`, compatibility is transported `Chapter3/RestrictOpen · MChart.Compatible.restrictTo`, and the charts of $X$ inside $V$ form the atlas of the M-polyfold $V$ `Chapter3/RestrictOpen · MPolyfold.restrictOpen`, whose levels are the traces of those of $X$ `Chapter3/RestrictOpen · MPolyfold.mem_levelSet_restrictOpen_iff`; an sc-smooth map with values in $V$ is sc-smooth into $V$ `Chapter3/RestrictOpen · MPolyfold.IsScSmoothMap.codRestrict_restrictOpen`. The restriction $P|V$ lives on the pull-back $\iota^*Y = \{(v, y) \mid P(y) = v\}$ of Definition 2.6.7 (preprint 2.53), as the restricted sections of Definition 5.3.13 (preprint 5.20) do (§V.23), and its fibres are those of $P$. A strong bundle chart of $P$ with base domain inside $V$ is transported with the same $F$, $U$, $R$, $K$ `Chapter3/RestrictOpen · StrongBundleChart.restrictTo` `Chapter3/RestrictOpen · StrongBundleChart.Compatible.restrictTo`, and these charts form the atlas of $P|V$ `Chapter3/RestrictOpen · StrongBundle.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.restrictTo_mem_restrictOpen_atlas`; every chart of $P$ restricts to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpen · StrongBundleChart.restrictInter`. The restriction $f|_V$ of an sc-smooth, sc$^+$ or regularizing section is again one `Chapter3/RestrictOpen · StrongBundle.IsScSmoothSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.IsScPlusSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.regularizing_restrictOpen`. Two choices of the formalization: the paracompactness of $V$ and of $\iota^*Y$ is supplied by Theorem 2.3.10 (preprint 2.16), and $V$ is nonempty, for the junk values of the transported inverse charts.

**Germ data and the index on $P|V$** (Findings 361, 362). The data of Definition 3.1.11 (preprint 3.11) on a relatively open $U$ restrict to every relatively open $U_0 \subseteq U$ containing 0: the germs of the filling and of the sc$^+$-part are intersected with $U_0$, and $Dr(0)$ and $\rho(0)$ do not change, so the kernel isomorphism of condition (3) is the same `Chapter3/RestrictOpenGerms · IsFillingGerm.of_subset` `Chapter3/RestrictOpenGerms · IsScPlusGerm.of_subset`; the strong bundle isomorphism $\Psi$ of the conjugation is restricted to $U_0 \lhd F \to \psi(U_0) \lhd F'$, where $\psi(U_0) = U' \cap \psi'^{-1}(U_0)$ is relatively open by the continuity of $\psi'$, and the basic germ is shrunk to $G \cap \psi'^{-1}(U_0)$ as for $G''$ (§III.48), with the same $P_n$, $Q$, $h$, $B$ `Chapter3/RestrictOpenGerms · ConjugationData.ofSubset` `Chapter3/RestrictOpenGerms · IsConjugateToBasicGerm.of_subset`. So a germ datum at $x \in V$ survives in its chart restricted to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpenGerms · StrongBundleChart.germData_restrictInter`, which is tame when the chart is `Chapter3/RestrictOpenGerms · StrongBundle.IsTameChart.restrictInter`, and an sc-Fredholm section with tame germ data restricts to one on $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsScFredholmSectionTame.restrictOpen`. *The index.* At a zero $x \in V$, $f'(x)$ has index $i$ in every chart of $P$ iff $(f|_V)'(x)$ has index $i$ in every chart of $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.restrictOpen` `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.of_restrictOpen`. The charts are not compared — a chart of $P|V$ need not come from one of $P$ —: in either bundle the index is $\dim P_n - \dim Q$ for the germ datum (§III.44), whose restriction keeps $P_n$ and $Q$ `Chapter3/RestrictOpenGerms · StrongBundle.isLinOfIndex_restrictOpen_of_conjugationData`, and uniqueness identifies it with $i$. The same argument passes from $P|V$ to $P|W$ for an open $W \subseteq V$, where the chart restricted to $\operatorname{dom}\varphi \cap W$ is common to both and the two linearizations agree in it `Chapter3/IndexAlongPathsLocal · sectionRestrict` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsScPlusSection.restrictOpen_of_subset` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsLinOfIndex.restrictOpen_of_subset`; Definition 6.2.1 (preprint 6.1) restricts accordingly `Chapter3/RestrictOpenGerms · StrongBundle.IsIndexAt.restrictOpen` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.restrictOpen_of_subset`.

**The stabilizing sections of a chart, without cut-off** (Finding 363). In the zero-set argument above, bump functions enter only through Lemma 5.3.3 (preprint 5.11), as sc$^+$-sections through given smooth points of the fibres. That property — through every smooth point $e$ of the fibre over a smooth point $x$ there

is an sc⁺-section $s$ with $s(x) = e$ — is isolated `Chapter3/IndexLocallyConstantLocal · StrongBundle.HasScPlusSectionsThrough`. Bump functions give it by Lemma 5.3.3 (preprint 5.11) `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`, and so does a chart of the atlas over the whole base, with no cut-off `Chapter3/IndexLocallyConstantLocal · StrongBundle.hasScPlusSectionsThrough_of_chart`: $\hat{e} = \Phi(e)_2$ is a smooth point of $F$, with a lift $\hat{e}_1 \in F^1$, and the section $y \mapsto \Phi^{-1}(\varphi(y), R(\varphi(y), \hat{e}))$ — the text's $R(\cdot, e)$ — has the local representative $z \mapsto R[1](z, \hat{e}_1)_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is the cut-off section of Lemma 5.3.3 (preprint 5.11) with $\beta = 1$ (§V.9), and its value at $x$ is $e$ since $\Phi(e) \in K$ is fixed by $R$. The restriction $P|V$ to a chart domain $V$ has such a chart: the chart itself, transported to $V$. The three steps of the zero-set argument — the spanning sections, the tilting sections that put the parametrized section in good position, the local constancy — are repeated with this property in place of bump functions `Chapter3/IndexLocallyConstantLocal · exists_spanning_sections_of_sections` `Chapter3/IndexLocallyConstantLocal · exists_sections_isParamGP_of_sections` `Chapter3/IndexLocallyConstantLocal · index_locally_constant_of_sections`; the support and norm bounds of Lemma 5.3.3 (preprint 5.11), which the index argument does not use, are dropped. *The index along the zero set with the text's hypotheses* `Chapter3/IndexLocallyConstantLocal · index_locally_constant_tame`: *let $f$ be an sc-Fredholm section with tame germ data of any strong bundle and $x_0$ a zero; then there are an open $V \ni x_0$ and $i \in \mathbb{Z}$ with $f'(y)$ Fredholm of index $i$ in every chart at every zero $y \in V$.* Take a chart around $x_0$ with domain $V$: $f|_V$ has tame germ data on $P|V$, which has sc⁺-sections through every smooth fibre point, so the index of $f|_V$ is one integer at its zeros near $x_0$, and at each of them it is the index on $P$. In sc form, with the sc-Fredholm property of $\mathrm{Lin}(z)$ at every zero in every chart `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm`, $\mathrm{Lin}(z)$ is sc-Fredholm of index $i$ on every level `Chapter3/IndexLocallyConstantLocal · theorem_3_6_8_index_locally_constant_general_tame`. This is the localization that Finding 343 described; the gap ind $C(z) = 0$ of Finding 243 remains only without tame germ data.

**Definition 6.2.1 (preprint 6.1) with local sections** (Finding 364) `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal`. *$f$ has index $i$ at $x$* if there are an open $V \ni x$ and an sc⁺-section $s$ of $P|V$ with $(f|_V + s)(x) = 0$ and $(f|_V + s)'(x)$ Fredholm of index $i$ in every chart of $P|V$: the global definition at the beginning of this section, on $P|V$. *Well defined.* Two local sections, on $V$ and on $V'$, both restrict to $V \cap V'$ (the restriction of Definition 6.2.1 (preprint 6.1) to a smaller open set, above), where the uniqueness on $P|V \cap V'$ applies `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal.unique`. A global section gives a local one, with $V = X$ `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.isIndexAtLocal`, and at a zero the index is that of $f'(x)$ `Chapter3/IndexAlongPathsLocal · StrongBundle.isIndexAtLocal_of_isZeroAt`. *Existence, without bump functions.* On the domain $V$ of a chart around $x$, the section through $f(x)$ of the previous paragraph, negated, is an sc⁺-section $s$ of $P|V$ with $(f|_V + s)(x) = 0$, and Theorem 3.1.18 (preprint 3.15) gives the germ datum of $f|_V + s$ at $x$, hence an index `Chapter3/IndexAlongPathsLocal · StrongBundle.exists_isIndexAtLocal`. As for the global reading, a locally constant local index along a map from a preconnected space is constant `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_const_of_forall_eventually`.

**Proposition 6.2.2 (preprint 6.2) with the text's hypotheses** (Finding 365) `Chapter3/IndexAlongPathsLocal · proposition_6_2_2_local`. *Let $f$ be an sc-Fredholm section with tame germ data of a strong bundle $P : Y \to X$ — no bump functions — and $\varphi : \mathbb{R} \to X$ an sc-smooth path. Then* $\mathrm{ind}(f, \varphi(t))$*, read with local sections, is one integer for all $t$;* any two values of the index at $\varphi(0)$ and at $\varphi(1)$ agree `Chapter3/IndexAlongPathsLocal · proposition_6_2_2_local'`. The proof is that of (E), (S) and Proposition 6.2.2 (preprint 6.2) above, run near each $t_0$ on $P|V$, $V$ the domain of a chart $c$ around $\varphi(t_0)$ `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_eventually_const_of_path`. The path is reparametrized into $V$ as in

(E), and is then an sc-smooth path of the M-polyfold $V$. The section along the path is the one of (E) with the cut-off as a datum Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_bump, taken equal to 1, which is possible because the chart $c|_V$ of $P|V$ covers its base Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_chart. The product chart of $c|_V$ covers $\mathbb{R} \times V$, so the pull-back has sc$^+$-sections through every smooth fibre point without cut-off Chapter3/IndexAlongPathsLocal · hasScPlusSectionsThrough_pullback_rn_of_chart, and the index of $\tilde{f} + S$ is constant along the graph near $t_0$ Chapter3/IndexAlongPathsLocal · isLinOfIndex_paramLift_sectionAdd_eventually_const_of_sections. The slice relation (S), applied to $P|V$ unchanged, turns it into $\mathrm{ind}(f|_V, \varphi(t)) = i - 1$ near $t_0$, a local index of $f$, and the connectedness of $\mathbb{R}$ ends the proof. *Against the text*, what remains is the domain of the path: it is $\mathbb{R}$, and a path sc-smooth on $[0,1]$ only is not covered, the project having no M-polyfold $[0,1]$. The hypotheses on $f$ are the text's, tame germ data being the reading of "tame strong bundle" of §III.47.

**The two readings agree under bump functions** (Finding 367) Chapter3/IndexAlongPathsLocalBump · StrongBundle.isIndexAtLocal_iff_isIndexAt. If $X$ admits bump functions, a local section of Definition 6.2.1 (preprint 6.1) gives a global one with the same index Chapter3/IndexAlongPathsLocalBump · StrongBundle.IsIndexAtLocal.isIndexAt: the local section $s$ on $V$ is restricted to $W = \mathrm{dom}\,\varphi \cap V$ for a chart $\varphi$ around $x$, read in that chart as $\tau : O \to F^1$, and cut off by a bump $\beta$ supported in $W$ and equal to 1 near $x$; the cut-off of Lemma 5.3.3 (preprint 5.11) is made here in a chart of $P$ that need not belong to the atlas Chapter3/IndexAlongPathsLocalBump · StrongBundle.cutoffSection_mem_oneSet_of_isChart Chapter3/IndexAlongPathsLocalBump · StrongBundle.isScPlusSection_cutoffSection_of_isChart. Then $f + \beta\tau$ vanishes at $x$, and its linearization there has the index of $(f|_W + s)'(x)$: the two sections agree where $\beta = 1$, so the two linearizations coincide in the chart restricted to that set, and the germ datum of $f + \beta\tau$ at $x$ carries the index to every chart. So the global definition of the beginning of this section and the text's local one are the same notion whenever the hypotheses of Chapter 5 hold, and only the text's is available without them.

## III.50 The 2021 edition: the filling without tameness (Definitions 3.1.4, 3.6.1 (preprint 3.4, 3.48)) and a corner example

**The 2021 filling.** Definition 3.1.4 (p. 91) and its repetition Definition 3.6.1 (p. 134) of the Springer edition ask the strong bundle retraction $R(u,h) = (r(u), \rho(u)h)$ to cover an *sc-smooth* retraction $r$ with $r(0) = 0$. The 2017 text and the preprint ask for a *tame* one. Otherwise the conditions are those of §III.3: - (1) $\hat{f} = g$ on $O$ near 0; - (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in $O$; - (3) the linearization at 0 of $k(y) = [1 - \rho(r(y))]g(y)$, restricted to $\ker Dr(0)$, is a topological linear isomorphism onto $\ker \rho(0)$.

The formal definition is a structure with $r(0) = 0$ as a field and (1)–(3) in the germ reading of §III.48 Chapter3/Book2021Germs · IsFilling2021 Chapter3/Book2021Germs · HasFilling2021. The 2017 filling adds the tameness of $r$ Chapter3/Book2021Germs · IsFilling2017 Chapter3/Book2021Germs · isFilling2017_iff.

**The project's filling is the 2021 one.** The germ filling of §III.3 never asked $r$ to be tame Chapter3/GermFilling · IsFillingGerm. A 2021 filling is exactly a germ filling with $r(0) = 0$ Chapter3/Book2021Germs · isFilling2021_iff Chapter3/Book2021Germs · IsFilling2021.isFillingGerm. Conversely, a germ filling is a 2021 filling as soon as $0 \in O$, because points of $O$ are fixed by $r$ Chapter3/Book2021Germs · IsFillingGerm.isFilling2021; so is a whole-chart filled version Chapter3/Book2021Germs · IsFilledVersion.isFilling2021. In a chart centred at a point of $X$, $0 = \varphi(x_0)$ lies in $O$, so the text's standing assumption $r(0) = 0$ costs nothing there. A 2017 filling is a 2021 filling Chapter3/

Book2021Germs · IsFilling2017.isFilling2021. Both notions are satisfiable: over the identity strong bundle retraction, which is tame, every sc-smooth $g$ fills itself Chapter3/Book2021Germs · isFilling2017_id, for instance $g = 0$ on $\mathbb{R}$ Chapter3/Book2021Germs · isFilling2017_rn.

**A 2021 filling that is not a 2017 filling.** Take the following data: - $E = \mathbb{R} \oplus \mathbb{R}$, $C = U = [0, \infty) \times \mathbb{R}$, $F = \mathbb{R}$; - $R((a, b), h) = ((0, b), 0)$ Chapter3/Book2021Germs · CornerFilling.Rc, a strong bundle retraction Chapter3/Book2021Germs · CornerFilling.isStrongBundleRetraction_Rc, so $r(a, b) = (0, b)$, $O = \{0\} \times \mathbb{R}$ and $\rho \equiv 0$ Chapter3/Book2021Germs · CornerFilling.ρCLM_Rc; - the zero section $\hat{f} = 0$ of $K \to O$, and $g(a, b) = a$ Chapter3/Book2021Germs · CornerFilling.gc Chapter3/Book2021Germs · CornerFilling.isScSmooth_gc.

The derivatives at 0 are $Dr(0)(a, b) = (0, b)$ and $Dg(0)(a, b) = a$ Chapter3/Book2021Germs · CornerFilling.scDeriv_r Chapter3/Book2021Germs · CornerFilling.scDeriv_g. Since $\rho \equiv 0$, the fill defect $k$ is $g$ itself Chapter3/Book2021Germs · CornerFilling.fillDefect_gc. Now check the three conditions: - (1) $g = 0$ on $O$; - (2) $g(y) = \rho(r(y))g(y) = 0$ forces $y \in O$; - (3) $\ker Dr(0) = \mathbb{R} \oplus 0$ Chapter3/Book2021Germs · CornerFilling.mem_ker_scDeriv_r, and $Dk(0)$ maps it by $(a, 0) \mapsto a$ onto $\ker \rho(0) = \mathbb{R}$ Chapter3/Book2021Germs · CornerFilling.kerMap Chapter3/Book2021Germs · CornerFilling.kerMap_bijective.

So $g$ is a filled version Chapter3/Book2021Germs · CornerFilling.isFilledVersion_gc. With $0 \in O$ Chapter3/Book2021Germs · CornerFilling.zero_mem_O, it is a 2021 filling Chapter3/Book2021Germs · CornerFilling.isFilling2021_gc. But $r$ is not tame: $d_C(1, 0) = 0$, while $d_C(r(1, 0)) = d_C(0, 0) = 1$, against Definition 2.5.2 (1) (preprint 2.33 (1)) Chapter3/Book2021Germs · CornerFilling.not_isTame_r. So $g$ is not a 2017 filling Chapter3/Book2021Germs · CornerFilling.not_isFilling2017_gc, and the 2021 notion is strictly more general Chapter3/Book2021Germs · exists_isFilling2021_not_isFilling2017.

**Sc-Fredholm germs and sections.** With the 2021 filling in place of the filling of §III.6: - the 2021 sc-Fredholm germ (Definition 3.1.11 = 3.6.3) Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2021; - the 2021 sc-Fredholm section (Definition 3.1.16 = 3.6.4) Chapter3/Book2021Germs · StrongBundle.IsScFredholmSection2021; - the 2017 germ, with the 2017 filling Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2017.

In a chart centred at $x_0$, $0 \in O$ Chapter3/Book2021Germs · StrongBundle.zero_mem_O_of_chart. So the 2021 germ is the project's germ Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2021_iff Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm, and the 2021 section the project's section Chapter3/Book2021Germs · StrongBundle.isScFredholmSection2021_iff Chapter3/TameGermsG · StrongBundle.IsScFredholmSection. The 2017 germ is the germ with tame germ data of §III.47 Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2017_iff Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame, and 2017 implies 2021 Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2017.isScFredholmGerm2021. The graph section of a $C^\infty$ map $\mathbb{R}^n \to \mathbb{R}^k$ satisfies both Chapter3/Book2021Germs · isScFredholmSection2021_graphSection Chapter3/Book2021Germs · isScFredholmGerm2017_graphSection. Whether every 2021 germ is a 2017 germ is the question of Finding 353, the passage from germ data in some compatible chart to germ data in a tame one (gap G.79). It is not: §III.65 gives a 2021 germ of a section of a tame strong bundle that is not a 2017 germ Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.isScFredholmGerm2021_not_2017.

## III.51 The 2021 edition: the theorems whose proofs do not use tameness

The project's proofs of the following results never used the tameness of $r$; each 2021 statement is the earlier theorem transported through the equivalences of §III.50: - **Proposition 3.1.5 (preprint 3.5)** (§III.4).

If $g$ is a 2021 filling of $\hat{f}$ with $\hat{f}(0) = 0$, then $0 \in O$ by $r(0) = 0$, and $g(0) = \hat{f}(0) = 0$ by (1). So the germ form of the proposition applies Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_1_5. $Dg(0)$ maps $T_0O = \operatorname{range} Dr(0)$ into $K_0 = \operatorname{range} \rho(0)$. The restriction $T\hat{f}(0) : T_0O \to K_0$ is onto iff $Dg(0)$ is, and is Fredholm iff $Dg(0)$ is, with the same index Chapter3/Book2021Germs · proposition_3_1_5. - **Theorem 3.1.13 and Corollary 3.1.15 (preprint 3.12 and 3.13)** (§III.13) Chapter3/SectionCompactnessG · theorem_3_1_13_preprint Chapter3/SectionCompactnessG · corollary_3_1_15_preprint Chapter3/Book2021Germs · theorem_3_1_13 Chapter3/Book2021Germs · corollary_3_1_15. - **Theorem 3.1.18 (preprint 3.15)** (§III.6) Chapter3/FredholmSectionsG · theorem_3_1_18_preprint Chapter3/Book2021Germs · theorem_3_1_18. - **Theorem 3.1.21 (preprint 3.18)** (§III.40) Chapter3/ParamPerturbationG · theorem_3_1_21_preprint Chapter3/Book2021Germs · theorem_3_1_21. - **Proposition 3.6.5 (preprint 3.52)** (§§III.21, III.30) Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint Chapter3/Book2021Germs · proposition_3_6_5.

So the 2021 change of the filling does not affect these statements, and the 2021 text's version of them is verified as printed.

## III.52 The 2021 edition: Theorem 3.6.8 (preprint 3.54), the step on p. 138

The proof of Theorem 3.6.8 (pp. 137–138) starts from a germ in good position (Definition 3.6.6, §III.21). There $\ker T\tilde{f}(0)$ is in good position to the partial cone $C_0O = T_0O \cap C$ in $T_0O$. The proof passes to the good position of $\ker Dg(0)$ to $C$ in $E$, the hypothesis of the implicit function theorem in the quadrant (§III.17). The passage uses an sc-complement of $T_0O$ inside the subspace $E_0$ of $C$ at $0$, namely $\ker Dr(0)$. Proposition 2.5.3 (preprint 2.35) supplies it for a tame retract. In the linear part of §III.23 this is the hypothesis $(1 - Dr(0))E \subseteq E_0$ Chapter3/GoodPositionGermG · theorem_3_6_8_linear. The 2021 text keeps the sentence that the retract is tame by assumption, while its Definition 3.6.1 no longer assumes it.

**The step does not go through without tameness.** In the example of §III.50, every other hypothesis of the linear part holds: - $g$ is a 2021 filling of $\hat{f} = 0$, $0 \in O$, $\hat{f}(0) = 0$, and $\hat{f}$ is a section of $K$; - $g$ is sc$^1$ with derivative $Dg(0)$, and $Dg(0)$ is onto; - the germ is in good position in the chart Chapter3/GoodPositionGermG · IsGoodPositionAtGerm Chapter3/Book2021Germs · CornerFilling.isGoodPositionAtGerm_gc: $T\tilde{f}(0) : T_0O \to K_0 = 0$ is onto, and its kernel $T_0O = 0 \oplus \mathbb{R}$ is all of $T_0O \cap C$. Here $\ker Dg(0) = \operatorname{range} Dr(0)$ Chapter3/Book2021Germs · CornerFilling.ker_Lg_eq_range_Lr.

But $(1 - Dr(0))(1, 0) = (1, 0)$ is not in $E_0 = 0 \oplus \mathbb{R}$ Chapter3/Book2021Germs · CornerFilling.not_range_le_minimalSubspace. And $\ker Dg(0) = 0 \oplus \mathbb{R}$ is not in good position to $C$ in $E$ Chapter3/Book2021Germs · CornerFilling.not_isInGoodPosition_gc, for two reasons: - a complement $P$ contains a point $p$ with first coordinate 1, from writing $(1, 0) = n + p$; - the first coordinate vanishes at $(0, 1) \in \ker Dg(0) \cap C$, so by Lemma 3.8.8 (preprint 3.65) every good complement lies in $\{a = 0\}$ Chapter3/GoodPosition · lemma_3_8_8.

The whole is one statement Chapter3/Book2021Germs · theorem_3_6_8_tame_step. So this step of the proof does not hold as we read it for the 2021 definitions. The conclusion of the theorem still holds in this example: its solution set $O \cong \mathbb{R}$ is a manifold. Since $X = \mathbb{R}$ there has no boundary, even the proof of Theorem 3.6.10 (preprint 3.56), where $\partial X = \emptyset$, passes through this step.

The project states Theorem 3.6.8 and what rests on it (Theorems 3.1.22, 3.1.26, Corollary 3.1.27, Theorems 3.6.10 and 3.6.11 (preprint 3.19, 3.23, 3.24, 3.56 and 3.57)) with tame germ data, the 2017 notion (§III.47). For the untamed 2021 data, read with the definitions as printed, Theorem 3.6.8 does not hold as we read it (gap G.79, settled): §III.65 builds on the corner of §III.50 a tame strong bundle over $\mathbb{R}$ with a 2021

sc-Fredholm section whose solution set near a zero in good position is a closed half-line `Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not`. In that example the step on p. 138 is exactly where the proof needs the tame chart.

## III.53 The 2021 edition: Theorem 3.5.1 (preprint 3.40) with the sequential conclusion

The 2021 statement of Theorem 3.5.1 (p. 118) adds a second conclusion. If points $(a_k, w_k)$ of the solution set $S$ converge to 0 in $E_0$, then for every $m$ they eventually lie in $C_m$ and converge to 0 there. The formal statement has both conclusions `Chapter3/Book2021Germs · theorem_3_5_1`, in the following setting: - $h$ a basic germ on an sc-germ $G$ of neighbourhoods of 0 in $C = \widetilde{C} \times W$, with $\widetilde{C}$ closed; - $U = G_0$, $s$ an sc$^+$-germ on $G$ with $s(0) = 0$, and $S = \{x \in U : h(x) + s(x) = 0\}$.

*The neighbourhoods.* Take the open sets $O'(m)$ of the local Theorem 3.1.13 of §III.12 `Chapter3/LocalCompactness · theorem_3_1_13_local` and put $O(m) = O'(m) \cap G_0$. These are nested, relatively open in $C$, contain 0 and lie in $U$. The closure of $S \cap O(m)$ is taken in $E_0$. It lies in $C$, which is closed and contains $G_0$, and it lies in the image of $E_m$, because it is the image of the compact set $K_m$ of §III.12.

*The sequences.* A zero of $h + s$ solves $w = B(a, w) - \mathbf{s}(a, w)$ with $\mathbf{s}$ the $W$-component of $s$ `Chapter3/Book2021Germs · IsBasicGerm.solves`. $\mathbf{s}$ is an sc$^0$-germ into $W^1$ with $\mathbf{s}(0) = 0$. So the local Corollary 3.1.15 of §III.12 applies `Chapter3/LocalCompactness · corollary_3_1_15_local`. For every $m$ there are $y_k \in E_m$ with $\iota_m(y_k) = x_k$ for $k$ large and $y_k \to 0$ in $E_m$; these are the lifts $L_m x_k$. "Converges on level $m$" is thus stated with level-$m$ lifts, as in Corollary 3.1.15 (preprint 3.13).

## III.54 The 2021 edition: cones and Lemmas 3.8.4–3.8.6, Corollary 3.8.7 (Appendix 3.8.1.3)

The appendix on "cones and partial quadrants in infinite dimensions" (pp. 145–149) is new in the 2021 edition. Its notions: - a *closed convex cone* $K$ in a finite-dimensional space is a closed convex set with $K \cap (-K) = \{0\}$ and $tK \subseteq K$ for $t \geq 0$ `Chapter3/Book2021Items · IsClosedConvexCone`; - *$a$ generates an extreme ray* if $a \in K \setminus \{0\}$, and $y \in K$, $a - y \in K$ imply $y \in \mathbb{R}_+ a$ `Chapter3/Book2021Items · IsExtremeRayGen`.

A closed convex cone is pointed `Chapter3/Book2021Items · IsClosedConvexCone.eq_zero`. $[0, \infty)$ is one, generated by the extreme ray of 1 `Chapter3/Book2021Items · isClosedConvexCone_Ici` `Chapter3/Book2021Items · isExtremeRayGen_one`.

The setting is the good position of §III.14: $C = T^{-1}([0, \infty)^n \oplus W)$ for a presentation $T$ (the text's case is $T = \mathrm{id}$), and $N$ a finite-dimensional subspace in good position. Within it: - $N \cap C$ is the cone of $N$ `Chapter3/GoodPosition · quadCone`; - $\Gamma$, the union of the sets $\sigma_a = \{i : a_i = 0\}$ over $0 \neq a \in N \cap C$, is the vanishing index set `Chapter3/GoodPosition · vanIdx`; - $P : N \to \mathbb{R}^\Gamma$ is the coordinate projection `Chapter3/GoodPosition · sigmaCoord`, whose kernel is $N \cap (\mathbb{R}^{\Gamma^c} \oplus W)$.

**The quadrant form of a pointed cone.** If $N \cap C$ is pointed, a linear isomorphism $e : N \cong \mathbb{R}^{\dim N}$ carries $N \cap C$ onto $[0, \infty)^{\dim N}$ `Chapter3/Book2021Items · exists_quadrantEquiv_of_pointed`. It comes from the normal form of §III.14, $N \cap C \cong [0, \infty)^k \times N_0$ `Chapter3/GoodPosition · IsInGoodPosition.quadrantNormalForm`, in which pointedness forces $N_0 = 0$. The vectors $e^{-1}(e_j)$ generate extreme rays `Chapter3/Book2021Items · isExtremeRayGen_symm_single`. A nonzero $a \in N \cap C$ has a coordinate $e(a)_j > 0$

with $a - e(a)_j e^{-1}(e_j) \in N \cap C$ Chapter3/Book2021Items · exists_sub_single_mem. As in §III.14, this replaces the Krein–Milman-type Lemmas 3.8.1–3.8.3 (preprint 3.62–3.64): the cone is a quadrant at once. Those lemmas are now proved too, by Krein–Milman (§III.59).

**Lemma 3.8.4** Chapter3/Book2021Items · lemma_3_8_4. Let $N^\perp$ be a good complement and $N \cap C$ a closed convex cone. Parts (1) and (3) say that $y_i = 0$ for $y \in N^\perp$ and $i \in \sigma_a$, $a$ a generator, respectively $i \in \Gamma$. They hold without the cone hypothesis, because they are Lemma 3.8.8 (preprint 3.65) (§III.14) Chapter3/GoodPosition · lemma_3_8_8. For (2), $\Gamma = \bigcup_{a \text{ generator}} \sigma_a$, the inclusion $\supseteq$ is the definition. For $\subseteq$, let $0 \neq a \in N \cap C$ with $a_i = 0$, and take $j$ and $b = e^{-1}(e_j)$ as above. Then $0 \leq (a - e(a)_j b)_i = -e(a)_j b_i \leq 0$ forces $b_i = 0$, so $i \in \sigma_b$. The text writes $a$ as a sum of extreme-ray generators (Lemma 3.8.1); here only pointedness is used.

**Lemma 3.8.5** Chapter3/Book2021Items · lemma_3_8_5. The text's statement is: either $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, in which case $\dim N = 1$ and $\Gamma = \emptyset$, or $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$. The proof assumes $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) \neq \{0\}$, and the formal statement is the dichotomy it shows. $P$ is onto Chapter3/GoodPosition · sigmaCoord_surjective. - $\Gamma \neq \emptyset$. A point $x$ of $N$ with $x_i = 0$ for $i \in \Gamma$ lies in $N \cap C$, and so does $-x$ Chapter3/GoodPosition · mem_quadCone_of_vanIdx, so $x = 0$. - $\Gamma = \emptyset$ *and* $N \neq 0$. The kernel is all of $N$. If $\dim N \geq 2$, the generators $b_0, b_1$ have all coordinates positive (no index vanishes on a nonzero point of the cone). Then $b_0 - t b_1 \in N \cap C$ for small $t > 0$, but its $e$-coordinate $j_1$ is $-t < 0$. - $N = 0$. The kernel is $\{0\}$, and the first alternative holds.

*Read literally, the first alternative does not hold as we read it for $N = 0$* Chapter3/Book2021Items · lemma_3_8_5_not_literal. In $E = \mathbb{R}$, $C = [0, \infty)$, the subspace $N = 0$ is in good position, $N \cap C = \{0\}$ is a closed convex cone, and $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$; but $\dim N = 0$ and $N \ncong \mathbb{R}$. The dichotomy holds there through its other alternative.

**Lemma 3.8.6.** (1) If $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$, then $P$ is a linear isomorphism carrying $N \cap C$ onto $[0, \infty)^\Gamma$ Chapter3/Book2021Items · lemma_3_8_6. This needs no cone hypothesis: $P$ is injective by hypothesis and onto. A point whose $\Gamma$-coordinates are $\geq 0$ is in $N \cap C$ by the segment argument of §III.14 when $\Gamma \neq \emptyset$; when $\Gamma = \emptyset$, $N = 0$. (2) If $N \cap C$ is a closed convex cone, $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$ and $N \neq 0$, then $\dim N = 1$ by Lemma 3.8.5. A linear isomorphism $N \cong \mathbb{R}$ then carries $N \cap C$ onto $[0, \infty)$ Chapter3/Book2021Items · lemma_3_8_6_two. As printed, (2) does not hold as we read it for $N = 0$, by the same example Chapter3/Counterexamples2021 · Counterexamples2021.not_lemma_3_8_6_two (the narrower row V3.120). With $N \neq 0$ its hypotheses are met by $N = E = \mathbb{R}$, $C = [0, \infty)$ Chapter3/Book2021Items · halfLine_witness.

**Corollary 3.8.7.** If $N \cap C$ is a closed convex cone, it is a partial quadrant in $N$, and $(N, N \cap C) \cong (\mathbb{R}^{\dim N}, [0, \infty)^{\dim N})$ Chapter3/Book2021Items · corollary_3_8_7. The first part is Proposition 3.1.25 (preprint 3.22) (§III.14), which needs no cone hypothesis Chapter3/GoodPosition · proposition_3_1_25. The second is the quadrant form above, which the text also derives on p. 149.

## III.55 What the 2021 additions taught us

- **The 2021 filling is the formalization's.** The germ filling never asked for tameness. The 2021 text drops it, so the formal definitions and the printed ones now agree Chapter3/Book2021Germs · isFilling2021_iff Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2021_iff. The 2017 notion is strictly narrower Chapter3/Book2021Germs · exists_isFilling2021_not_isFilling2017.
- **One proof step needs the dropped hypothesis.** All the theorems of Chapter 3 that the project proved without tameness hold for the 2021 definitions as printed. The passage of Theorem 3.6.8 from good position in $T_0 O$ to good position in $E$ (p. 138) does not hold as we read it without a tame

chart `Chapter3/Book2021Germs · theorem_3_6_8_tame_step`. The implicit function theorems remain verified with tame germ data; for untamed data Theorem 3.6.8 does not hold as we read it (§III.65; G.79 settled).

- **Krein–Milman is not needed.** A pointed cone in good position is a quadrant `Chapter3/Book2021Items · exists_quadrantEquiv_of_pointed`. Lemma 3.8.4 (2) follows by one subtraction, and (1) and (3) need no cone hypothesis. (Lemmas 3.8.1–3.8.3, which use it, are proved in §III.59 for the comparison.)
- $N = 0$. Lemma 3.8.5, read as an implication, and Lemma 3.8.6 (2) need $N \neq 0$ `Chapter3/Book2021Items · lemma_3_8_5_not_literal`. The dichotomy the proof shows holds for all $N$.

## III.56 Theorem 3.2.3 (preprint 3.27) under the tangent germ property alone; the good complement is closed

§III.24 proved Theorem 3.2.3 (preprint 3.27) for data with two regularity conditions, (a) $\delta$ sc-smooth on $V$ as a map on a set and (b) $\pi(O) \subseteq C$, and with the good complement $Y$ closed. None of the three is a hypothesis any more.

**The good complement of Definition 3.2.1 (preprint 3.25) is closed.** Definition 3.2.1 as written `Chapter3/GoodComplementClosed · MPolyfold.TangentGermChart2021` asks for $N$ in good position in the sense of Definition 3.1.24 (preprint 3.21), whose good complement is an *sc-complement*; it defines $\delta$ on $V$ only, and reads "sc$^\infty$-germ" with $k$-dependent neighbourhoods, the weakest reading (§III.24). An sc-complement is closed `Chapter3/GoodPositionSc · IsScGoodComplement.isClosed` and makes $N$ smooth by Proposition 1.1.7 (preprint 1.6) `Chapter3/GoodPositionSc · IsScGoodComplement.le_smooth`. So the book's datum is the project's datum, with $\delta$ extended by 0 off $V$ `Chapter3/GoodComplementClosed · MPolyfold.TangentGermChart2021.toTangentGermChart`; conversely a closed good complement of a smooth $N$ is an sc-complement `Chapter3/GoodComplementClosed · MPolyfold.TangentGermChart.toTangentGermChart2021`. The two tangent germ properties coincide `Chapter3/GoodComplementClosed · MPolyfold.hasTangentGermProperty2021_iff`, and Lemma 3.6.9 (preprint 3.55) with the book's good complement is the lemma of §III.22, whose closedness hypothesis the sc-complement supplies `Chapter3/GoodComplementClosed · lemma_3_6_9`. The "$Y$ closed" of §§III.22, III.24 and III.32 is therefore the text's own hypothesis.

**Condition (b) is a choice of the complement.** Let $(\varphi, N, Y, V, \delta)$ be a datum at $x$. In a presentation of $C$ by functionals $\ell_i$, let $\Sigma$ be the indices that vanish on some nonzero point of $N \cap C$ (§III.14). The interior of $N \cap C$ in $N$ is $\{\ell_i > 0,\ i \in \Sigma\}$ `Chapter3/TangentGermProjecting · interior_quadCone`, and the $\ell_i$, $i \in \Sigma$, vanish on every good complement (Lemma 3.8.8 (preprint 3.65)) `Chapter3/TangentGermProjecting · IsGoodComplement.coordFun_eq_zero`.

1. *A projecting complement.* There is a continuous projection $\pi' : E \to N$ with $\pi'(C) \subseteq N \cap C$ whose kernel $Y'$ is a closed good complement `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_projData` `Chapter3/GoodComplementProjection · exists_proj_goodComplement` (§III.29).
2. *The re-graphing map* $g(v) = \pi'(v + \delta(v))$, $V \to N \cap C$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraphMap`. It has $g(0) = 0$ and $Dg(0) = 1$, because $D\delta(0) = 0$ and $\pi' = 1$ on $N$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.hasFDerivWithinAt_regraphMap_zero`. It preserves the coordinates $\ell_i$, $i \in \Sigma$, since these vanish on $Y$ and on $Y'$; hence $g(v)$ is interior iff $v$ is `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraphMap_interior_iff`.
3. *The quadrant inverse function theorem applies.* Its hypothesis "$d_C(x) = d_C(g(x))$" is used only as "interior iff interior" (§III.16), which step 2 gives. On a small convex ball $B(0,r) \cap N \cap C$, $\|Dg - 1\| \leq \frac{1}{2}$,

so $g$ satisfies the hypotheses of Proposition 3.8.17 (preprint 3.73), of class $C^k$ on a ball depending on $k$, since $\delta$ is $C^k$ into $E_m$ only near 0 on neighbourhoods depending on $(m,k)$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_invData_ball`. The inverse is the same map on all these balls.

4. *The re-graphed datum.* Shrink the chart domain to $\{y : \pi\varphi(y) \in B(0,r)\}$. Keep the chart and $N$, take the complement $Y' = \ker\pi'$, $V' = g(V)$ and $\delta' = (1-\pi')\circ(1+\delta)\circ g^{-1}$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraph`. It describes the same set, $\varphi(M\cap U) = \{w+\delta'(w) : w \in V'\}$, because $g(u)+\delta'(g(u)) = u+\delta(u)$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraphδ_regraphMap`, and its projection is $\pi'$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.π_eq_of_ker`.

So every datum at $x$ can be replaced, on a smaller chart domain, by one whose projection maps $O$ into $C$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_projecting` — condition (b).

**Condition (a) follows.** Given (b), the re-centring theorem of §III.36 shrinks a datum to one where $\delta$ is sc-smooth on $V$, provided every nearby point of $M$ has a datum `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally` — which the tangent germ property gives. Hence every datum can be shrunk to regular data `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_isRegular`, and under the tangent germ property every point has regular data `Chapter3/TangentGermProjecting · MPolyfold.HasTangentGermProperty.exists_isRegular`.

**Theorem 3.2.3 (preprint 3.27) as stated** `Chapter3/TangentGermProjecting · theorem_3_2_3`. *If $M \neq \emptyset$ has the $n$-dimensional tangent germ property, then $M$ is a sub-M-polyfold, its induced structure is tame and sc-smoothly equivalent to an $n$-dimensional manifold with boundary with corners, and the charts $\pi\circ\varphi$ of regular data are charts of it.* This is §III.24 fed with the regular data. The same holds with the dimension depending on the point `Chapter3/TangentGermProjecting · theorem_3_2_3_loc`, and with the data of Definition 3.2.1 as written `Chapter3/TangentGermProjecting · theorem_3_2_3_2021def`. Statement (2), "$\pi\circ\varphi$ is a chart", is stated for regular data, which exist at every point; for an arbitrary datum it would need the step (a′) below.

**The two steps of the text's proof, as Lean counterexamples.** Each of them does not hold as we read it; the examples of §III.24 are now machine-checked.

- (a′) *"$\delta$ is an sc$^\infty$-germ near every $v_0 \in V$."* The text re-centres at $x_0 = \varphi^{-1}(v_0+\delta(v_0))$ and claims that $k(w) = \pi\varphi\psi^{-1}(w+\tau(w))$ has an invertible linearization at 0. In $X = \mathbb{R}^2$ let $M = \{(t^3,t)\}$, a smooth embedded curve with a vertical tangent at the origin. It has the 1-dimensional tangent germ property `Chapter3/RecentreCounterexample2021 · Recentre2021.CE.hasTangentGermPropertyLit_M`. At $(1,1)$ take $N = \mathbb{R}(3,1)$, $Y = \mathbb{R}(0,1)$, $V = N$ and $\delta(v) = (\sqrt[3]{1+v_1} - 1 - v_1/3)(0,1)$ `Chapter3/RecentreCounterexample2021 · Recentre2021.CE.germData_bad`. This $\delta$ is smooth near 0 with $D\delta(0) = 0$, but not differentiable at $v_0 = -\frac{1}{3}(3,1)$, the point over the origin `Chapter3/RecentreCounterexample2021 · Recentre2021.CE.not_germNear`. So the claim does not hold as we read it `Chapter3/RecentreCounterexample2021 · Recentre2021.not_theorem_3_2_3_germNearEveryPoint`, and neither does the re-centring step for every pair of data `Chapter3/RecentreCounterexample2021 · Recentre2021.not_theorem_3_2_3_recentreStep`. This is why row V3.73 is narrower: the re-centring lemma is proved with the first graph map $C^1$ on its whole domain (§III.35).
- (b′) *"$\Sigma = \{v+y \in C : v \in V, y \in Y\}$ is relatively open in $C$."* In $\mathbb{R}^3$ with $C = [0,\infty)^3$, $N = \mathbb{R}(1,1,1)$ has the good sc-complement $Y = \{x+y-z = 0\}$ `Chapter3/Counterexamples2021 · Counterexamples2021.isScGoodComplement_cube`. On $\Sigma$, $x+y-z \geq 0$, so the points $(a,0,a+a^2) \in C$, $a>0$, lie outside $\Sigma$ and tend to 0: for no $V \subseteq N\cap C$ is $\Sigma$ a relatively open neighbourhood of 0 `Chapter3/Counterexamples2021 · Counterexamples2021.theorem_3_2_3_sigma_not_relOpen`.

The theorem itself holds; only the route changes. Step (b′) is replaced by the choice of the projecting complement and the re-graphing, and step (a′) by the re-centring along solution data, which needs the nearby data that the tangent germ property supplies.

## III.57 Theorems 3.6.8 and 3.1.22 (preprint 3.54 and 3.19) in the book's form, with the 2021 definitions

§§III.37 and III.43–III.45 prove the implicit function theorems with the project's notions. Two files restate them with the 2021 definitions literally, so that a reader checks only those.

**The dictionary** `Chapter3/MainTheorems2021 · StrongBundle.IsScFredholmGermTame2021`. The 2021 germ data (Definition 3.6.3 = 3.1.11, with the 2021 filling of §III.50) in a *tame* strong bundle chart are the project's tame germ data. In a tame chart the 2021 and the 2017 fillings agree, so these are also the 2017 germ data `Chapter3/MainTheorems2021 · StrongBundle.isScFredholmGermTame2021_iff_2017`. The section with such data at every smooth point `Chapter3/MainTheorems2021 · StrongBundle.IsScFredholmSectionTame2021` and good position in a tame chart `Chapter3/MainTheorems2021 · StrongBundle.IsGoodPositionGermTame2021` complete the dictionary. Reading "tame strong bundle" as germ data in a tame chart is the reading of §III.47 (row V3.103); §III.65 shows that it cannot be dropped.

**The theorems.** Each is a corollary of the project's form by the dictionary, using the primed theorems of §III.43, which carry no added filler hypothesis: Theorem 3.6.8 (1), (2) `Chapter3/MainTheorems2021 · theorem_3_6_8_2021def`, Theorem 3.1.26 `Chapter3/MainTheorems2021 · theorem_3_1_26_2021def`, Theorem 3.1.22 `Chapter3/MainTheorems2021 · theorem_3_1_22_2021def`, Corollary 3.1.27 `Chapter3/MainTheorems2021 · corollary_3_1_27_2021def`, Theorems 3.6.10 and 3.6.11 `Chapter3/MainTheorems2021 · theorem_3_6_10_2021def` `Chapter3/MainTheorems2021 · theorem_3_6_11_2021def`.

**Theorem 3.6.8 (preprint 3.54) with property (4) of its proof** `Chapter3/BookFormIFT2021 · theorem_3_6_8_book`. *Let $f$ be an sc-Fredholm section with tame germ data, $f(x) = 0$, and $(f, x)$ in good position in a tame chart. There are an open $V \ni x$ and $d$ such that (1) every zero in $V$ is in good position; (4) at every zero $y \in V$, $f'(y)$ is onto in every strong bundle chart, with index $d$; (2) $S = \{y \in V : f(y) = 0\}$ is a sub-M-polyfold whose induced structure is that of a $d$-dimensional manifold with boundary with corners.* The proof is the index form of Theorem 3.1.26 (§III.44) `Chapter3/GlobalIFTIndexG · theorem_3_1_26_index`. **Its last step** `Chapter3/BookFormIFT2021 · theorem_3_6_8_tangentGermProperty`: there are an open $U \ni x$ and $d = \operatorname{ind} f'(x) = \dim \ker f'(x)$ such that $S(f, U)$ has the $d$-dimensional tangent germ property *as written in Definition 3.2.1*, and Theorem 3.2.3 applies to it. *Proof.* Theorem 3.6.8 (1) gives an open $V_x$ of nearby good position `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'`. Theorem 3.6.8 (2) with the index gives data of Definition 3.2.1 of dimension $d$ at every zero in a smaller open set `Chapter3/GlobalIFTIndexG · theorem_3_6_8_two_index`. These are the project's data, which are the book's by §III.56 `Chapter3/GoodComplementClosed · MPolyfold.hasTangentGermProperty2021_iff`. Theorem 3.2.3 for regular data (§III.24) `Chapter3/TangentGerm · theorem_3_2_3_of_isRegular` closes the argument.

**Theorem 3.1.22 (preprint 3.19)** `Chapter3/BookFormIFT2021 · theorem_3_1_22_book`. *Let $d_X \equiv 0$, $f$ an sc-Fredholm section with tame germ data, $f(x_0) = 0$. Then $f'(x_0)$ is sc-Fredholm in every strong bundle chart around $x_0$* `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm`. *If it is onto, there are $U \ni x_0$ and $d$ with $f'(y)$ onto of index $d = \dim \ker f'(y)$ at every zero $y \in U$, and $S(f, U)$ a sub-M-polyfold with the structure of a $d$-dimensional manifold without boundary* `Chapter3/GlobalIFTIndexG · theorem_3_1_22_of_surjective`. The sc-Fredholm property in every chart is Proposition 3.6.5 (preprint 3.52) in sc form, carried between charts by the chart change at a zero (§§III.30–III.31, gap G.22) `Chapter3/FredholmGermLinearizationScG · proposition_`

3_6_5_sc. The last clause, $\ker f'(y) = T_y S(f, U)$, is §III.62.

## III.58 Remark 3.3.8 (preprint 3.36): the quantitative solution germ

**Remark 3.3.8 (preprint 3.36)** Chapter3/GapP1_Remark338 · remark_3_3_8. *Let $f(a,w) = w - B(a,w)$ be an sc$^0$-contraction germ which is an sc-smooth germ with $f(0) = 0$, on a quadrant-like $\widetilde{C}$. There are non-increasing sequences $\varepsilon_i, \tau_i > 0$ such that (1) the solution germ $\delta$ is continuous on $\{a \in \widetilde{C} : |a| \le \varepsilon_0\}$ with values in $\{|w|_0 \le \tau_0\}$, $\delta(0) = 0$; (2) every solution of $f(a,w) = 0$ with $|a| \le \varepsilon_0$, $|w|_0 \le \tau_0$ is $w = \delta(a)$; (3) $|a| \le \varepsilon_i$ implies $\delta(a) \in W_i$ and $|\delta(a)|_i \le \tau_i$; (4) $\delta : \{|a| \le \varepsilon_i\} \to W_i$ is of class $C^i$.* "Decreasing" is read as non-increasing.

*Proof.* 1. *Level* 0. The contraction region of Theorem 3.3.2 (preprint 3.29) at 0 gives $\tau_0'$ and the solution germ $\delta$, continuous, with uniqueness in the region Chapter3/GapP1_Remark338 · ContractionGerm.exists_radius. Put $\tau_i = \tau_0'/2^{i+1}$; uniqueness gives (2) Chapter3/ContractionGerm · ContractionGerm.eq_sol. 2. *Level $i$.* Corollary 3.3.7 (preprint 3.34) with $m = k = i$ (§III.11) gives a relatively open $V_i \ni 0$ on which $\delta$ maps into $W_i$ by a $C^i$ map Chapter3/GermIFT · corollary_3_3_7. The level-$i$ fixed points are continuous, so a radius $e_i$ with $\bar{B}(0,e_i) \cap \widetilde{C} \subseteq V_i$ and $|\delta(a)|_i \le \tau_i$ for $|a| \le e_i$ exists. At $i = 0$ the level-0 map is $\delta$ itself Chapter3/GapP1_Remark338 · ContractionGerm.fixLvl_zero_eq. 3. *Monotone radii.* $\varepsilon_i = \min(e_0, \dots, e_i)$, the running minima Chapter3/GapP1_Remark338 · exists_antitone_le.

Like Corollary 3.3.7, the remark lives on balls that shrink with $i$, the shrinking of §III.11 made explicit.

## III.59 Extreme rays: Lemmas 3.8.1–3.8.3 and 3.8.9–3.8.11 (preprint 3.62–3.64 and 3.66–3.68)

The formalization proved Proposition 3.1.25 (preprint 3.22) and Lemmas 3.8.4–3.8.6 without extreme rays (§§III.14, III.54). The text's lemmas on extreme rays are now proved as well, for the comparison; nothing else uses them.

**Definitions.** A closed convex cone $P$ in a finite-dimensional $N$ is closed, convex, with $P \cap (-P) = \{0\}$ and $\mathbb{R}_+ P = P$ Chapter3/Book2021Items · IsClosedConvexCone. The ray of $x$ is $\mathbb{R}_+ x$ Chapter3/GapP1_ExtremeRays · rayOf; it is an extreme ray if $x \in P \setminus 0$ and $y, x - y \in P$ imply $y \in \mathbb{R}_+ x$ Chapter3/GapP1_ExtremeRays · IsExtremeRay Chapter3/GapP1_ExtremeRays · extremeRays. This is the generator notion of §III.54 Chapter3/GapP1_ExtremeRays · IsClosedConvexCone.isExtremeRay_rayOf_iff. A quadrant is a cone carried onto $[0,\infty)^n$ by a linear isomorphism $N \cong \mathbb{R}^n$ Chapter3/GapP1_ExtremeRays · IsQuadrantIn.

**A strictly positive functional** Chapter3/GapP1_ExtremeRays · IsClosedConvexCone.exists_pos_functional. *There are $\varphi \in N^*$ and $c > 0$ with $\varphi(x) \ge c|x|$ on $P$.* 1. For a unit vector $z \in P$, $-z \notin P$, and Hahn–Banach separation of $-z$ from the closed convex $P$ gives a functional positive at $z$ and $\ge 0$ on $P$ Chapter3/GapP1_ExtremeRays · IsClosedConvexCone.exists_functional_pos. 2. Each is positive on an open neighbourhood of $z$. Finitely many neighbourhoods cover the compact unit sphere of $P$. The sum $\varphi$ of their functionals is positive on that sphere, and its minimum there is the constant $c$.

So the base $P \cap \{\varphi = 1\}$ is compact and convex.

**Lemma 3.8.1 (preprint 3.62)** Chapter3/GapP1_ExtremeRays · lemma_3_8_1. *If $P \neq \{0\}$, $P$ is the closed convex hull of its extreme rays.* 1. Mathlib's Krein–Milman theorem makes the compact convex base the closed convex hull of its extreme points. 2. An extreme point of the base spans an extreme ray of $P$ Chapter3/GapP1_

ExtremeRays · IsClosedConvexCone.rayOf_mem_extremeRays. 3. A nonzero $x \in P$ is $\varphi(x)$ times the point $x/\varphi(x)$ of the base, so it lies in the closed convex hull of the rays.

*For $P = \{0\}$ the statement does not hold as we read it* Chapter3/GapP1_ExtremeRays · not_lemma_3_8_1_zero: $\{0\}$ has no extreme ray, and the closed convex hull of the empty family is empty. With 0 adjoined the lemma holds for every closed convex cone Chapter3/GapP1_ExtremeRays · lemma_3_8_1_zero_union. This is the narrower row V3.123; Lemma 3.8.11 below uses the form with 0 adjoined.

**Lemma 3.8.2 (preprint 3.63)** Chapter3/GapP1_ExtremeRays · lemma_3_8_2. *A closed convex cone with nonempty interior is a quadrant iff it has exactly* dim $N$ *extreme rays.* - (□ ) The extreme rays of $[0,\infty)^n$ are the $\mathbb{R}_+ e_i$ Chapter3/GapP1_ExtremeRays · extremeRays_nonnegOrthant, and the count is transported by the isomorphism. - (□ ) Pick generators $a_1, \dots, a_n$. By Lemma 3.8.1 and the nonempty interior they span $N$; by the count they are a basis. The cone is the preimage of the orthant under the dual coordinates.

**Lemma 3.8.3 (preprint 3.64)**, in the model $C = [0,\infty)^n \oplus W$. *If $N \cap C$ is a closed convex cone and $a$ generates an extreme ray, then* dim $N - 1 \le \#\sigma_a$, $\sigma_a = \{i : a_i = 0\}$, *with equality if $N$ is in good position.* 1. Let $\Pi : N \to \mathbb{R}^{\sigma_a}$ be the coordinate projection, the text's map (3.34) Chapter3/GapP1_ExtremeRays · zeroIdxProj. 2. Its kernel lies in $\mathbb{R}a$ (the text's (3.33)) Chapter3/GapP1_ExtremeRays · ker_zeroIdxProj_le. For $z$ in the kernel, $la + z$ and $la - z$ lie in $N \cap C$ when $l = \sum_{i \notin \sigma_a} |z_i|/a_i + 1$. They add up to $2la$, so extremality puts $z$ on the line $\mathbb{R}a$. 3. Rank–nullity gives the inequality Chapter3/GapP1_ExtremeRays · lemma_3_8_3. 4. Under good position, the coordinates in $\sigma_a$ vanish on the good complement (Lemma 3.8.8 (preprint 3.65)), so $\Pi$ is onto and equality holds Chapter3/GapP1_ExtremeRays · lemma_3_8_3_goodPosition. Only the algebraic splitting is used.

**Lemmas 3.8.9–3.8.11 (preprint 3.66–3.68).** Let $W$ be $\{0\} \oplus W$ Chapter3/GapP1_Lemma389 · wPart, $\widetilde{N}$ an algebraic complement of $N \cap W$ in $N$, $\Sigma$ the union of the $\sigma_a$ over nonzero $a \in N \cap C$ Chapter3/GapP1_Lemma389 · bigSigma, and $\mathbb{R}^{\Sigma^c} \oplus W$ the subspace where the coordinates in $\Sigma$ vanish Chapter3/GapP1_Lemma389 · sigmaCompl. - *Lemma 3.8.9: $\widetilde{N}$ is in good position, with good complement $(N \cap W) \oplus N^\perp$.* The proof uses only that $N^\perp$ is closed Chapter3/GapP1_Lemma389 · lemma_3_8_9_of_isClosed: 1. $E_0 = \widetilde{N} \oplus ((N \cap W) \oplus N^\perp)$ by linear algebra. 2. The good-complement constant is $\varepsilon/(1 + \|\mathrm{pr}\|)$, with pr the continuous projection onto $N$ along $N^\perp$ — the text's $c_1$. 3. The interior of $\widetilde{N} \cap C$ comes from that of $N \cap C$ by translating along $N \cap W$, which does not change membership in $C$.

So the lemma holds for the text's sc-complement Chapter3/GapP1_Lemma389 · lemma_3_8_9. With the text's notions on both sides, $(N \cap W) \oplus N^\perp$ is again an sc-complement Chapter3/GapP1_Lemma389Sc · lemma_3_8_9_sc: a finite-dimensional smooth $A$ plus an sc-subspace is an sc-subspace Chapter3/GapP1_Lemma389Sc · ScBanach.isScSubspace_sup_of_le_smooth, and the splitting holds on every level Chapter3/GapP1_Lemma389Sc · ScBanach.isCompl_levelSub_of_isScSplitting. - *Lemma 3.8.10: $C \cap \widetilde{N}$ is a closed convex cone, and a generator $a$ of an extreme ray has* dim $\widetilde{N} - 1 = \#\sigma_a = d_C(a)$ Chapter3/GapP1_Lemma389 · lemma_3_8_10. The cone property needs only $\widetilde{N} \cap W = 0$ Chapter3/GapP1_Lemma389 · isClosedConvexCone_of_disjoint. The formula is Lemma 3.8.3 for $\widetilde{N}$. - *Lemma 3.8.11: either $\widetilde{N} \cap (\mathbb{R}^{\Sigma^c} \oplus W) = 0$, or $\widetilde{N} \subseteq \mathbb{R}^{\Sigma^c} \oplus W$,* dim $\widetilde{N} = 1$ *and $\Sigma = \emptyset$* Chapter3/GapP1_Lemma389 · lemma_3_8_11. By the text's argument: 1. Let $0 \neq x$ lie in the intersection, and let $b$ generate an extreme ray. The coordinates of $b$ outside $\Sigma$ are positive and those of $x$ in $\Sigma$ vanish, so with $\lambda = \sum_{i \notin \Sigma} |x_i|/b_i + 1$ both $\lambda b + x$ and $\lambda b - x$ lie in the cone Chapter3/GapP1_Lemma389 · large_smul_add_mem. They add up to $2\lambda b$, so extremality puts $x$ on $\mathbb{R}b$ Chapter3/GapP1_Lemma389 · exists_eq_smul_of_isExtremeRayGen. 2. So every extreme ray lies on $\mathbb{R}x$. By Lemma 3.8.1 with 0 adjoined, $C \cap \widetilde{N} \subseteq \mathbb{R}x$. 3. The nonempty interior gives $\widetilde{N} = \mathbb{R}x$, of dimension 1. 4. An extreme ray exists, and Lemma 3.8.10 gives $\#\sigma = 0$, so $\Sigma = \emptyset$.

# III.60 Definition 3.8.14 (preprint 3.71), Lemma 3.8.21 (preprint 3.77) in general form, and Lemma 3.8.20 (preprint 3.76)

**Definition 3.8.14 (preprint 3.71): class $C^k$ on partial quadrants.** - $f : U \to \mathbb{R}^m$ on a relatively open $U \subseteq C$ is $C^1$ if it has a continuous family of one-sided derivatives `Chapter3/GapP1_Definition3814 · IsClassC1On`. The derivative condition is the text's limit over $x + h \in U$ `Chapter3/GapP1_Definition3814 · isClassC1On_iff_tendsto`. - The tangent map is $Tf(x,h) = (f(x), Df(x)h)$ on $U \oplus \mathbb{R}^n$ `Chapter3/GapP1_Definition3814 · bookTangentMap`. $f$ is $C^k$ if $f$ and $Tf$ are $C^{k-1}$ `Chapter3/GapP1_Definition3814 · IsClassCOn`, and smooth if it is $C^k$ for all $k$ `Chapter3/GapP1_Definition3814 · IsClassSmoothOn`.

*Agreement.* On a set of unique differentiability every derivative family agrees with the derivative within the set `Chapter3/GapP1_Definition3814 · IsClassC1On.eqOn_fderivWithin`, so $Tf$ is well defined. 1. $C^1$ is Mathlib's $C^1$ within $U$ `Chapter3/GapP1_Definition3814 · isClassC1On_iff_contDiffOn`. 2. $C^{k+2}$ is $C^{k+1}$ of $Tf$ `Chapter3/GapP1_Definition3814 · contDiffOn_add_two_iff_tangentMap`, so by induction $C^k$ is Mathlib's $C^k$ within $U$ `Chapter3/GapP1_Definition3814 · isClassCOn_iff_contDiffOn Chapter3/GapP1_Definition3814 · isClassSmoothOn_iff_contDiffOn`. This holds in particular on relatively open subsets of partial quadrants `Chapter3/GapP1_Definition3814 · isClassCOn_iff_contDiffOn_of_isRelOpenInQuadrant`. 3. The text's $C^1$ and smooth are sc¹ and sc-smooth for the constant structure `Chapter3/GapP1_Definition3814 · isSc1_iff_isClassC1On Chapter3/GapP1_Definition3814 · isScSmooth_iff_isClassSmoothOn`.

So the $C^k$ used in §§III.16–III.17 is the text's notion (row V3.126).

**Lemma 3.8.21 (preprint 3.77), general form** `Chapter3/GapP1_Lemma3821 · lemma_3_8_21`. *Let $U$ be relatively open in a partial quadrant $C \subseteq \mathbb{R}^n$, $f : U \to C$ of class $C^1$, $V = f(U)$ relatively open, $f : U \to V$ a homeomorphism and $Df(x)$ invertible on $U$. Then $g = f^{-1}$ is differentiable within $V$ with $Dg(y) = [Df(g(y))]^{-1}$, and is $C^1$ on $V$.* The text adds "satisfying the assumptions of Proposition 3.8.17"; its proof of that proposition applies the lemma to the iterated tangent maps $T^k f$, which do not satisfy $\|D(T^k f) - 1\| \le \frac{1}{2}$, so the form used is this general one. 1. *Derivative.* Mathlib's derivative of a local left inverse needs only the invertibility of $Df(g(y))$ and the continuity of $g$; the text's bi-Lipschitz estimate is not needed. Relatively open subsets of a quadrant have unique derivatives `Chapter3/GapP1_Lemma3821 · IsRelOpenInQuadrant.uniqueDiffOn_const Chapter3/GapP1_Lemma3821 · hasFDerivWithinAt_inverse_of_isUnit`. 2. *Bootstrap.* $x \mapsto [Df(x)]^{-1}$ is $C^{j-1}$ `Chapter3/GapP1_Lemma3821 · contDiffOn_inverse_deriv_comp`, and $Dg = [Df]^{-1} \circ g$, so $f \in C^j$ gives $g \in C^j$ `Chapter3/GapP1_Lemma3821 · lemma_3_8_21_contDiffOn` and $f$ smooth gives $g$ smooth `Chapter3/GapP1_Lemma3821 · lemma_3_8_21_smooth`.

The printed form is a special case `Chapter3/GapP1_Lemma3821 · lemma_3_8_21_printed` (row V3.62, now stronger).

**Lemma 3.8.20 (preprint 3.76)** `Chapter3/GapP1_Lemma3820 · lemma_3_8_20`. *Under the hypotheses of Proposition 3.8.17, every $T^k f : T^k U \to T^k V$ is a smooth homeomorphism of relatively open subsets of the partial quadrant $T^k C$, with smooth inverse.* 1. *The data.* The induction runs over bundled data, a space, a quadrant, a set and a map `Chapter3/GapP1_Lemma3820 · QuadMapData`. Their tangent is $(V \times V, C \times V, U \times V, Tf)$ `Chapter3/GapP1_Lemma3820 · QuadMapData.tangent`; the product of a partial quadrant with a whole space is one `Chapter3/GapP1_Lemma3820 · constProdIso`. 2. *The property inherited.* The hypotheses of the general Lemma 3.8.21 — partial quadrant, relatively open domain and image, smooth, continuous inverse, invertible derivative `Chapter3/GapP1_Lemma3820 · QuadMapData.IsSmoothHomeo` — pass from $f$ to $Tf$ `Chapter3/GapP1_Lemma3820 · QuadMapData.IsSmoothHomeo.tangent`. - The derivative of $Tf$ at $(x,h)$ is $(a,b) \mapsto (Df(x)a, D^2 f(x)(a,h) + Df(x)b)$ `Chapter3/GapP1_Lemma3820 · hasFDerivWithinAt_tangentMap`. It is injective because $Df(x)$ is, hence invertible in finite dimensions `Chapter3/GapP1_Lemma3820 · isUnit_fderivWithin_tangentMap`. - The inverse is the text's

$\Phi(y,l) = (g(y), [Df(g(y))]^{-1}l)$. 3. *Iteration.* By induction on $k$, $T^k f$ is $(\text{tangent}^{[k]}d).f$ Chapter3/GapP1_Lemma3820 · QuadMapData.iterTangent, and Lemma 3.8.21 applies at each stage.

The lemma as printed, with the hypotheses of Proposition 3.8.17 for every order, is a special case Chapter3/GapP1_Lemma3820 · lemma_3_8_20_printed. §III.16 does not need the lemma: there the bootstrap of step 2 above replaces the iteration.

## III.61 Lemma 3.8.13 and Proposition 3.8.12 (preprint 3.70 and 3.69) in a general presentation; $d_M = d_X$ in Theorem 3.2.3 (preprint 3.27)

§§III.14–III.15 proved Proposition 3.8.12 and Lemma 3.8.13 in the model $C = [0,\infty)^n \oplus W$. Their uses are in charts, whose partial quadrant $C$ has an arbitrary presentation $T : E \cong \mathbb{R}^n \oplus W$.

**The reading of "$v \notin N \cap W$".** In the model, $\{0\} \oplus W$ is the lineality space $C \cap (-C)$, so for $v \in C$ the condition reads $-v \notin C$, which makes sense for every partial quadrant. In a presentation, $v \in C \cap (-C)$ iff all coordinates of $Tv$ vanish Chapter3/GapP1_Lemma3813 · mem_lineality_iff Chapter3/GapP1_Lemma3813 · mem_nullSpace_coordFun_iff.

**Proposition 3.8.12 (preprint 3.69), general** Chapter3/GapP1_Lemma3813 · proposition_3_8_12_general. For $N$ in good position to any partial quadrant $C$: 1. $d_{N\cap C}(x) = d_C(x)$ if $-x \notin C$; 2. $d_{N\cap C}(x) + \dim(N \cap C \cap (-C)) = \dim N$ otherwise.

**Lemma 3.8.13 (preprint 3.70), general** Chapter3/GapP1_Lemma3813 · lemma_3_8_13_general. Under the hypotheses of §III.15 for any partial quadrant: 1. near 0, $v + \tau(v) \in C$; 2. $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $-v \notin C$.

Both proofs are the model proofs, run with the coordinate functionals $x \mapsto (Tx)_i$ Chapter3/GapP1_Lemma3813 · coordCLMOf in place of the model coordinates. The cores were already stated for jointly onto functionals (§III.15). A local form, for $\tau$ given only on a relatively open $V \ni 0$, extends $\tau$ by a radial cut-off Chapter3/GapP1_Lemma3813 · lemma_3_8_13_local.

$d_M = d_X$ Chapter3/GapP1_Theorem323Deg · MPolyfold.theorem_3_2_3_degIndex_eq. *Let $M$ have regular data at every point and $(\varphi, N, Y, V, \delta)$ regular data at $x$. There is $\rho > 0$ such that $d_M(y) = d_X(y)$ for every $y \in M$ in the chart domain with $v = \pi\varphi(y)$ of norm $< \rho$ and $-v \notin C$.* 1. The chart $\pi \circ \varphi$ gives $d_M(y) = d_{N\cap C}(v)$ (§III.24) Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_degIndex. 2. The tame chart $\varphi$ gives $d_X(y) = d_C(\varphi(y))$ (Proposition 2.5.6 (preprint 2.36)), and $\varphi(y) = v + \delta(v)$. 3. The local Lemma 3.8.13 equates the two.

Some restriction is needed. For $X = [0,\infty)^2$ and $M$ the diagonal, $d_M(0) = 1$ and $d_X(0) = 2$; there $v = 0$ lies in the lineality space $\{0\}$.

With Lemma 3.8.13 in general form, Lemma 3.6.9 (preprint 3.55) holds in every presentation as well (row V3.80); its proof in coordinates (§III.22, §III.32) was already written for functionals cut out by a presentation.

## III.62 The intrinsic linearization, and $\ker f'(y) = T_yS$ (Theorem 3.1.22 (preprint 3.19), last clause)

§§III.31 and III.45 stated the linearization at a zero and "$\ker f'(y) = T_yS$" in a chart. With the linear structure of $T_yX$ from Chapter 2 (Definition 2.5.19 in book form), both are now intrinsic.

**The fibre map of a chart.** For a strong bundle chart $c$ and $y$ in its domain, the fibre $Y_y$ maps linearly into $F$ by the second component of $c$ `Chapter3/GapP1_LinIntrinsic · StrongBundleChart.fiberLin`. This map is injective `Chapter3/GapP1_LinIntrinsic · StrongBundleChart.fiberLin_injective`, its range is the fibre of $K$ over $\varphi(y)$ `Chapter3/GapP1_LinIntrinsic · StrongBundleChart.mem_range_fiberLin`, and between two charts it changes by the fibre part of the transition `Chapter3/GapP1_LinIntrinsic · StrongBundleChart.transitionFiberCLM_fiberLin`.

**The intrinsic linearization.** In the chart, the linearization at a zero on level 1 is $Tf(z)$ composed with the tangent of the chart `Chapter3/GapP1_LinIntrinsic · StrongBundleChart.linChart`. At a zero its values lie in the fibre of $K$, the range of the fibre map `Chapter3/GapP1_LinIntrinsic · StrongBundleChart.linChart_mem`. So it pulls back to $f'(y) : T_yX \to Y_y$ `Chapter3/GapP1_LinIntrinsic · StrongBundle.linIntr`, with $fibre map □ f'(y) = $ the chart linearization `Chapter3/GapP1_LinIntrinsic · StrongBundle.fiberLin_linIntr`.

*Chart independence* `Chapter3/GapP1_LinIntrinsic · StrongBundle.linIntr_chart_independent`. At a zero only the fibre part of a chart transition survives in the linearization `Chapter3/GapP1_LinKey · linAt_chart_change_key` (§III.31), and it matches the change of the fibre maps.

ker $f'(y)$ **in the chart** `Chapter3/GapP1_Theorem3122Ker · StrongBundle.coe_map_ker_linIntr`. The tangent of the chart carries ker $f'(y)$ onto the kernel of the chart linearization $T_zO \to F$, because the fibre map is injective.

ker $f'(y) = T_yS$ **in** $T_yX$ `Chapter3/GapP1_Theorem3122Ker · StrongBundle.ker_linIntr_eq_subTangentRange`. *Let $f$ have tame germ data, let every zero in an open $U$ be in good position in a tame chart, and let $S = S(f,U)$ be a sub-M-polyfold. Then at every $y \in S$ on level* 1, ker $f'(y)$ *is the image of $T_yS$ in $T_yX$* `Chapter2/BookFormTameSub2021 · MPolyfold.subTangentRange`. *Proof.* 1. In the chart of the tangent germ data at $y$, $T_yS$ is the kernel of the chart linearization (§III.45) `Chapter5/GeneralPosition · exists_tangentSpace_eq_linKer`. 2. The inclusion $S \subseteq X$ has tangent map with that image `Chapter2/BookFormTameSub2021 · MPolyfold.coe_map_range_subInclusionCLM`. 3. The previous paragraph identifies the kernels.

**Theorem 3.1.22 (preprint 3.19), with its last clause** `Chapter3/GapP1_Theorem3122Ker · theorem_3_1_22_book_ker`. *Theorem 3.1.22 as in §III.57, and in addition* ker $f'(y) = T_yS(f,U)$ *in $T_yX$ at every $y \in S(f,U)$ on level* 1, *for the intrinsic $f'(y)$ through any strong bundle chart.* Gap G.30 is closed with it.

## III.63 Theorem 3.7.1 (preprint 3.58) for a general sc-splitting

§III.38 proved the basic germ criterion in the coordinates of the text's proof, $E = \mathbb{R}^n \oplus X$, $F = \mathbb{R}^N \oplus Y$. The theorem as printed has a partial quadrant $C \subseteq E$ and an sc-splitting $E = B \oplus X$ with $X \subseteq C$.

**The statement** `Chapter3/GapP1_Theorem371General · theorem_3_7_1_general`. *Let $U$ be a relatively open neighbourhood of* 0 *in a partial quadrant $C \subseteq E$ and $f : U \to F$ sc-smooth, with* 1. *at every smooth point $x \in U$, $Df(x)$ sc-Fredholm* `Chapter3/GapP1_Theorem371General · derivOpAt`, *with index independent of $x$;* 2. *an sc-splitting $E = B \oplus X$, $B$ finite-dimensional containing* ker $Df(0)$, *$X \subseteq C$, with the compactness property* `Chapter3/GapP1_Theorem371General · CompactnessCondition371`; 3. *the estimate for the derivative along $X$* `Chapter3/GapP1_Theorem371General · DerivCondition371`.

*Then $f - f(0)$ is conjugated near* 0 *to a basic germ.*

Three readings, none of them a restriction (row V3.89, restated): - (3) is printed for $h \in E_m$; it is read for $h \in X_m$, since $D_2f$ is the derivative along $X$. - In (2) one radius serves all levels, the text's order of quantifiers; $B \subseteq E_\infty$ is finite-dimensional. - A smooth point is a point of $U \cap E_\infty$.

**The proof is a reduction to §III.38.** 1. *The target splitting* Chapter3/GapP1_Theorem371General · exists_targetSplitting371. As in the text, $Y = Df(0)X$ has the finite-dimensional complement $A = C_0 \oplus Df(0)B \subseteq F_\infty$, with $C_0$ a smooth complement of the range of the sc-Fredholm $Df(0)$. $A$ has an sc-complement $Y'$ by Proposition 1.1.7 (preprint 1.6). 2. *Coordinates.* The sc-splittings give sc-isomorphisms $S : E \cong B \oplus X$ and $T : F \cong A \oplus Y'$ Chapter3/GapP1_Theorem371Split · ScBanach.splitScIso. Since $X \subseteq C$, $S$ carries $C$ to $(B \cap C) \oplus X$ Chapter3/GapP1_Theorem371Split · splitScIso_preimage, and $B \cap C$ is a partial quadrant of $B$. Indeed $X$, a subspace inside $C$, lies in the lineality space $C \cap (-C)$, and adding such a point does not change membership in $C$ Chapter3/GapP1_Theorem371Split · IsPartialQuadrant.add_mem_iff_of_mem; so $X$ is a good complement of $B$ with any $\varepsilon$, and $B$ is in good position Chapter3/GapP1_Theorem371Split · isInGoodPosition_of_isScSplitting Chapter3/GapP1_Theorem371Split · isPartialQuadrant_of_isScSplitting Chapter3/GapP1_Theorem371Split · isPartialQuadrant_split. The preimage of $U$ is relatively open Chapter3/GapP1_Theorem371General · isRelOpenInQuadrant_symm_preimage. 3. *The conjugate* $g = T \circ f \circ S^{-1}$ Chapter3/GapP1_ScIsoConj · scIsoConjMap is sc-smooth Chapter3/GapP1_ScIsoConj · isScSmooth_conjMap, with $Dg = T\,Df\,S^{-1}$ on level $0$ and on every level Chapter3/GapP1_ScIsoConj · scDeriv_conjMap Chapter3/GapP1_ScIsoConj · levelDeriv_conjMap. So condition (1) passes to $g$ Chapter3/GapP1_ScIsoConj · isScFredholm_of_conj. - Condition (2): the compactness property passes with the radius $\min(r, \delta)$ Chapter3/GapP1_Theorem371General · compactnessCondition_conj. - Condition (3): the estimate passes with $\varepsilon$ rescaled by $\|T_m\| + 1$ Chapter3/GapP1_Theorem371General · derivCondition_conj. - The block $h \mapsto (Dg(0)(0,h))_{Y'}$ is bijective on level $0$ Chapter3/GapP1_Theorem371General · bijective_splitScIso_snd. 4. *Conclusion.* The coordinate theorem of §III.38 conjugates $g - g(0)$ to a basic germ Chapter3/BasicGermCriterionIndex · theorem_3_7_1. The conjugation moves back through $S$ and $T$ Chapter3/GapP1_Theorem371Transport · IsConjugateToBasicGerm.of_scIso.

Gap G.24 is closed with it.

## III.64 An sc-smooth family of projections whose rank jumps

The counterexample of §III.65 needs a strong bundle whose fibre dimension jumps: an sc-smooth family of projections $\pi_t$ of an sc-Banach space $W$, with $\pi_t = 0$ for $t \le 0$ and $\pi_t$ of rank one for $0 < t < 1$. Such a family is sc-smooth but not norm-continuous; the sc-smooth maps of the earlier sections all came from Proposition 1.2.4 (preprint 1.19), which needs $C^1$ on equal levels and cannot produce one.

**A criterion** Chapter3/GapP1_UntamedJump · Jump.Family. A *jump family* on $V$ consists of: - operators $A_m(t) : V_m \to V_m$ and $A'_m(t) : V_{m+1} \to V_m$, compatible with the inclusions; - $(t,h) \mapsto A_m(t)h$ continuous on each level; - $t \mapsto A_m(t)|_{V_{m+1}}$ norm-differentiable with derivative $A'_m(t)$; - $(s,t,h) \mapsto sA'(t)h$ sc-smooth from $\mathbb{R}^2 \oplus V^1$ to $V$.

*Then $(t,h) \mapsto A(t)h$ is sc-smooth* Chapter3/GapP1_UntamedJump · Jump.Family.isScSmooth_map. *Proof.* By induction on $k$, over all jump families at once Chapter3/GapP1_UntamedJump · Jump.Family.isSck_map. The tangent map is $T\pi(t,h;\delta t,\delta h) = (A(t)h, \delta t\, A'(t)h + A(t)\delta h)$. Its second summand is the given sc-smooth map; its third is again a jump family, on the shifted space Chapter3/GapP1_UntamedJump · Jump.Family.shift.

**The moving rank-one projection** Chapter3/GapP1_UntamedJump · Jump.family. 1. *The scale.* Weighted $c_0$, with $|x|_m = \sup_k (k+1)^m |x_k|$, and positions $K_i = 16^i - 1$ Chapter3/GapP1_UntamedJump · Jump.pos, where the weights are powers of 16. 2. *The profile.* $\Lambda$, built from Mathlib's smooth transition and supported in $[\frac12, 2]$ Chapter3/GapP1_UntamedJump · Jump.prof. Put $e(t) = \sum_i \Lambda(2^i t) e_{K_i}$, $\ell_t = \sum_i \Lambda(2^i t)\delta_{K_i}$ and $\pi_t = \ell_t \otimes e(t)$. For $0 < t < 1$ exactly two consecutive coefficients are nonzero, equal to $(\cos\theta, \sin\theta)$. 3. *The derivatives on levels.* From source level $b+g$ to target level $b$, the $j$-th $t$-derivative is a tridiagonal operator Chapter3/GapP1_

UntamedJump · Jump.T, with coefficients $2^{ij}16^{-ig}$ times derivatives of products of $\Lambda$. It is norm-differentiable in $t$ when $j+2 \le 4g$ Chapter3/GapP1_UntamedJump · Jump.hasDerivAt_T and continuous at $g=0$ Chapter3/GapP1_UntamedJump · Jump.continuous_T_zero: one level pays for one derivative. 4. *The criterion's last hypothesis* holds by Proposition 1.2.4 Chapter3/GapP1_UntamedJump · Jump.isScSmooth_deriv.

**The projections** Chapter3/GapP1_UntamedJump · Jump.proj have the following properties: - $\pi_t = 0$ for $t \le 0$ Chapter3/GapP1_UntamedJump · Jump.proj_of_nonpos; - $\pi_t$ is of rank one for $0 < t < 1$ Chapter3/GapP1_UntamedJump · Jump.exists_proj_eq Chapter3/GapP1_UntamedJump · Jump.proj_ne_zero; - $\pi_t^2 = \pi_t$ for $t < 1$ Chapter3/GapP1_UntamedJump · Jump.proj_proj; - a vector fixed by $\pi_t$ for $t \le 0$ is $0$ Chapter3/GapP1_UntamedJump · Jump.eq_zero_of_proj_eq.

## III.65 Theorem 3.6.8 (preprint 3.54) with untamed 2021 germ data: the counterexample

The 2021 filling (§III.50) drops the tameness of the retraction, and Definition 3.6.3 asks for germ data in *some* strong bundle chart. §III.52 showed that the step on p. 138 needs a tame chart. The question left there (Finding 353, gap G.79) was whether germ data in an arbitrary compatible chart can be moved to a tame chart, or else whether the theorem fails. It fails. *Read with the 2021 definitions as printed, Theorem 3.6.8 does not hold as we read it* Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not.

**The local model** (namespace `UntamedCorner`). Let $\pi_t$ be the projections of §III.64, $v$ a smooth vector with $\pi_t v \neq 0$ for $0 < t < 1$, and $\nu$ a functional with $\nu \neq 0$ on $\operatorname{ran}\pi_t \setminus 0$. - $E = \mathbb{R}^2 \oplus W \ni ((a,t),w)$, $C = [0,\infty)\times\mathbb{R}\times W$, $U = C \cap \{t < 1\}$ and $F = \mathbb{R}\oplus W$. - $R(((a,t),w),(x,h)) = (((0,t),0),(0,\pi_t h))$ is a strong bundle retraction Chapter3/GapP1_UntamedCounterexample · UntamedCorner.isStrongBundleRetraction_R. It covers $r(a,t,w) = (0,t,0)$, so $O \cong \mathbb{R}$, and $r$ is not tame Chapter3/GapP1_UntamedCounterexample · UntamedCorner.not_isTame_baseRetr: $d_C = 1$ on $O$, but $0$ at $(1,t,0)$. The fibre of $K \to O$ is $0$ for $t \le 0$ and a line for $0 < t < 1$. - The filling is $g((a,t),w) = (a+\nu(w)^2, w+\pi_t v)$.

*It is a 2021 filling* Chapter3/GapP1_UntamedCounterexample · UntamedCorner.isFilling2021_g: 1. $g$ is the section on $O$. 2. If $g(u) = \rho(r(u))g(u)$, then $a+\nu(w)^2 = 0$ and $\pi_t w = w$. Since $a \ge 0$, this forces $a=0$, $\nu(w)=0$, $w \in \operatorname{ran}\pi_t$, so $w=0$ and $u \in O$. **Here the boundary $a \ge 0$ of the untamed chart is used.** 3. The fill defect has $Dk(0)((a,t),w) = (a,w)$, an isomorphism $\ker Dr(0) = \{t=0\} \to \ker\rho(0) = F$ Chapter3/GapP1_UntamedCounterexample · UntamedCorner.kerEquiv.

*The rest of the germ data.* - $g$ is a basic germ, with $B = -\pi_t v$ constant in $w$, and $s = 0$ Chapter3/GapP1_UntamedCounterexample · UntamedCorner.isBasicGerm_g Chapter3/GapP1_UntamedCounterexample · UntamedCorner.isConjugateToBasicGerm_g. - $Dg(0)((a,t),w) = (a,w)$ is onto Chapter3/GapP1_UntamedCounterexample · UntamedCorner.surjective_scDeriv_g, with kernel the $t$-axis, $T_0O$. So the germ is in good position in its chart Chapter3/GapP1_UntamedCounterexample · UntamedCorner.isGoodPositionAtGerm_g.

*The solution set.* $g(u) = 0$ iff $w = -\pi_t v$ and $a = -\nu(\pi_t v)^2$. In $C$ this forces $a=0$ and $\pi_t v = 0$, so $S = \{((0,t),0) : t \le 0\}$ Chapter3/GapP1_UntamedCounterexample · UntamedCorner.solutionSet_eq, a closed half-line in $O \cong \mathbb{R}$ ending at the zero.

*Why $S$ is not a sub-M-polyfold.* A smooth retraction $\varrho$ of $(-\varepsilon,\varepsilon)$ onto $(-\varepsilon,0]$ would have $\varrho'(0) = 1$, so $\varrho(t) > 0$ for small $t > 0$ Chapter3/GapP1_UntamedCounterexample · UntamedCorner.not_exists_retraction. The

retract $O$ with its one chart is a tame M-polyfold Chapter3/GapP1_UntamedCounterexample · UntamedCorner.poly0 Chapter3/GapP1_UntamedCounterexample · UntamedCorner.isTame_poly0, with $d_X \equiv 0$ Chapter3/GapP1_UntamedCounterexample · UntamedCorner.degIndex_poly0, and $S$ is not a sub-M-polyfold of it Chapter3/GapP1_UntamedCounterexample · UntamedCorner.not_isSubMPolyfold_solutionSet. The local statements are collected in Chapter3/GapP1_UntamedCounterexample · UntamedCorner.theorem_3_6_8_untamed_local.

**The strong bundle.** $K \to O$ is made a strong bundle over $O$ Chapter3/GapP1_UntamedBundle · UntamedCorner.bundle, with the chart $R$ above Chapter3/GapP1_UntamedBundle · UntamedCorner.chart0 and the section $x \mapsto (x, g(x))$ Chapter3/GapP1_UntamedBundle · UntamedCorner.sec, sc-smooth Chapter3/GapP1_UntamedBundle · UntamedCorner.isScSmoothSection_sec, whose zero set is $S$ Chapter3/GapP1_UntamedBundle · UntamedCorner.solutionSet_sec. Each printed hypothesis of Theorem 3.6.8 holds: - *The bundle is tame* in both readings of the phrase "tame strong bundle", which the text does not define separately: - the base is tame; - a tame chart covers all of $O$, namely $R'(t,(x,h)) = (t,(0,\pi_t h))$, which covers the identity Chapter3/GapP1_UntamedBundleTame · UntamedCorner.isStrongBundleRetraction_RT Chapter3/GapP1_UntamedBundleTame · UntamedCorner.chartTame, is compatible with the chart of $R$ Chapter3/GapP1_UntamedBundleTame · UntamedCorner.compatible_chart0_chartTame, and is tame Chapter3/GapP1_UntamedBundleTame · UntamedCorner.isTameChart_chartTame; - so the bundle is tame Chapter3/GapP1_UntamedBundleTheorem · StrongBundle.IsTameBundle Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.isTameBundle_bundle. - *At the zero $x_0$ ($t = 0$),* the section is a 2021 sc-Fredholm germ in the chart of $R$ Chapter3/GapP1_UntamedBundle · UntamedCorner.isScFredholmGerm2021_sec_zero and is in good position Chapter3/GapP1_UntamedBundle · UntamedCorner.isGoodPositionGerm_sec_zero. Here $d_X(x_0) = 0$ Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.isWithoutBoundary_poly0 and the fibre over $x_0$ is $0$ Chapter3/GapP1_UntamedBundle · UntamedCorner.subsingleton_fiber_zero, so every reading of Definition 3.6.6 holds. - *At the other points* there are germ data in tame charts, from frames in which the section is constant Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.Frame Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.Frame.chart Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.Frame.compatible_chart0 Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.Frame.isScFredholmGermInterior_sec: - the zero frame over $t < 0$ Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.frame0; - over $0 < t < 1$ the frame $s \mapsto s\pi_t v$ with coframe $h \mapsto \nu(h)/\nu(\pi_t v)$ Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.frame1.

The re-centred charts are centred at interior points of their quadrants, hence tame Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.isScFredholmGermTame_sec_of_ne. So the section is sc-Fredholm in the sense of 2021 Definitions 3.1.16 = 3.6.4 Chapter3/GapP1_UntamedBundleFrame · UntamedCorner.isScFredholmSection2021_sec.

**The conclusion fails.** No neighbourhood of $x_0$ meets the zero set in a sub-M-polyfold Chapter3/GapP1_UntamedBundle · UntamedCorner.not_isSubMPolyfold_zeroSet. There are no tangent germ data at $x_0$ either Chapter3/GapP1_UntamedBundle · UntamedCorner.isEmpty_tangentGermChart: in a tame chart $d_C(0) = d_X(x_0) = 0$, so the data would be a graph over a neighbourhood of $0$ in $N$, and a continuous injective function on an interval has no interior maximum. - The facts above are collected in one statement Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_example. - The conclusion of Theorem 3.6.8 fails for this section, even with (1) weakened to good position in any chart Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.not_theorem_3_6_8_conclusion. - As a universal statement over tame strong bundles with 2021 sc-Fredholm sections, the theorem does not hold as we read it Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not, and neither do Theorems 3.1.22 and 3.1.26, whose hypothesis $d_X \equiv 0$ the example meets Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.theorem_3_1_22_untamed_not.

This is the narrower row V3.130.

**Untamed germ data cannot be moved to a tame chart.** - The germ at $x_0$ has no tame germ data `Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.not_isScFredholmGermTame_sec_zero`. Otherwise the section would have tame germ data everywhere, and the good position would hold in the tame chart (§III.47) `Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame`. Theorem 3.6.8 with tame data (§III.43) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8'` would then give tangent germ data at $x_0$, which do not exist. - So "every 2021 sc-Fredholm germ of a tame strong bundle has tame germ data" does not hold `Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.isScFredholmGerm2021_not_tame`, and a 2021 germ need not be a 2017 germ `Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.isScFredholmGerm2021_not_2017` (row V3.131).

This answers the question of §§III.47 and III.50 in the negative.

**The interior case, and the reading that is needed.** Germ data in a compatible chart centred at a point *interior* to its partial quadrant, $d_C(\varphi(x_0)) = 0$, do give tame germ data: the chart shrunk into the interior is tame and carries the restricted data `Chapter3/TameGermTransfer · StrongBundle.isScFredholmGermTame_of_chart_interior`. A section with such data at every point is a section with tame germ data `Chapter3/TameGermTransfer · StrongBundle.isScFredholmSectionTame_of_interior`. The corner filling of §III.50 is centred on the boundary `Chapter3/TameGermTransfer · CornerFilling.zero_not_mem_interior`, as is the chart of $R$ here, where $d_X(x_0) = 0 < 1 = d_C(\varphi(x_0))$. So the interior result is sharp.

Remark 3.6.7's "if $d_X(x) = 0$ we do not need additional assumptions" presumably means $d_C(\varphi(x)) = 0$ in the chart of the germ data. Theorem 3.6.8 holds in the reading of §III.47 — germ data and good position in a tame chart, the wording of the 2017 text and the preprint, and the chart the 2021 proof itself takes on p. 137 (§§III.43, III.57; rows V3.85, V3.94, V3.103). This reading is not derived from the 2021 text as printed; it is satisfiable, since the graph section of a smooth map has tame germ data (§III.47).

## III.66 What is not formalized, and what the November 2026 waves taught us

**What is not formalized.** Nothing of Chapter 3. The sixteen gaps of the chapter are closed: - G.16 (Theorem 3.1.21, §III.40); - G.17, G.18 (Theorems 3.1.22, 3.1.26, Corollary 3.1.27, §§III.37, III.43, III.57); - G.19 (Remark 3.3.8, §III.58); - G.20 (Corollary 3.4.3, §III.28); - G.21 (Proposition 3.1.10 on every level, §III.5); - G.22 (Proposition 3.6.5 as an sc-Fredholm operator, §§III.30, III.57); - G.23 (Theorem 3.6.8 (1) with the text's hypotheses, §III.43); - G.24 (Theorem 3.7.1, §III.63); - G.25, G.26 (Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11, §III.59); - G.27, G.28 (Definition 3.8.14, Lemma 3.8.20, §III.60); - G.29 (Lemma 3.8.13 in general presentations, $d_M = d_X$, §III.61); - G.30 (dimension = index, $\ker f'(y) = T_y S$, §§III.44, III.62); - G.79 (untamed germ data, settled by the counterexample, §III.65).

Four comparison rows remain narrower than the text, each because the printed statement does not hold as we read it, with a Lean counterexample: - V3.73, the re-centring step for every pair of data (§III.56, `Recentre2021.not_theorem_3_2_3_recentreStep`); - V3.120, Lemma 3.8.6 (2) at $N = 0$ (§III.54, `Counterexamples2021.not_lemma_3_8_6_two`); - V3.123, Lemma 3.8.1 at $P = \{0\}$ (§III.59, `not_lemma_3_8_1_zero`); - V3.130, Theorem 3.6.8 with the 2021 germ data as printed (§III.65, `UntamedCorner.theorem_3_6_8_untamed_not`).

Side questions that are not items of the comparison stay as they were: whether the shrinking of solution germs is intrinsic (§III.11), whether every germ datum can be traded for a whole-chart one (§III.48), and the compatibility of the chart $(1, \varphi)$ of §III.47.

**What the waves taught us.** 1. **No hypothesis of Chapter 3 had to be added after all.** The closedness of good complements is the text's sc-complement (§III.56). The two regularity conditions of Theorem 3.2.3 are theorems: condition (b) is a choice of complement, reached by re-graphing with the quadrant inverse function theorem, whose degeneracy hypothesis is needed only at interior points and holds there because the good complements share the vanishing coordinates; condition (a) follows by re-centring along the data at nearby points (§III.56). The two proof steps they replace do not hold as we read it, and both are now Lean counterexamples. 2. **"Tame strong bundle" must mean germ data in a tame chart.** The 2021 edition moved tameness from the filling to the phrase "tame strong bundle". In the weaker reading — a tame base and a tame atlas, with germ data anywhere — Theorem 3.6.8 does not hold as we read it (§III.65). The mechanism is a fibre that jumps: an sc-smooth but not norm-continuous family of projections (§III.64), which no finite-dimensional or norm-continuous example can show. The boundary of the untamed chart is used in condition (2) of the filling; at points interior to the chart's quadrant the untamed data are tame (§III.65). 3. **Proposition 3.8.12 and Lemma 3.8.13 are intrinsic** once "$v \notin N \cap W$" is read as "$-v \notin C$" (§III.61). Then $d_M = d_X$ near the centre of a datum, off the lineality space, and not at it (the diagonal of the quarter plane). 4. **The extreme-ray lemmas hold, with one degenerate case.** Lemma 3.8.1 needs $P \neq \{0\}$ or 0 adjoined (§III.59); Lemma 3.8.6 (2) needs $N \neq 0$ (§III.54). The formal route through good position (§III.14) remains shorter. 5. **Lemma 3.8.21 must be used in a form more general than its statement**, because the iteration of Lemma 3.8.20 leaves the hypotheses of Proposition 3.8.17 (§III.60). The text's Definition 3.8.14 is Mathlib's $C^k$ within a set. 6. **The basic germ criterion reduces to its coordinates** by an sc-isomorphic change of both spaces; the text's hypothesis $X \subseteq C$ puts $X$ in the lineality space of $C$, which makes $B \cap C$ a partial quadrant of $B$ (§III.63). 7. $f'(y)$ **is intrinsic.** With the fibre maps of the charts, the linearization at a zero is a map $T_y X \to Y_y$ independent of the chart, and its kernel is $T_y S$ (§III.62).

# Part IV. Manifolds with boundary with corners, and sc-differential forms (HWZ Chapter 4, §§4.1–4.5)

Chapter 4 asks when an M-polyfold — a space glued from images of sc-smooth retractions — is secretly an ordinary manifold with boundary with corners, and why the fibres of a family of such manifolds are good for almost every parameter. The pivot is the *sc⁺ condition*: a retraction that gains a level. At a smooth point its derivative is compact, so its retract has finite-dimensional tangent spaces (Lemma 4.1.7 (preprint 4.7)), and near a point where the tangent space is in good position to the partial quadrant the retract is the graph of a map over that tangent space (Proposition 4.1.9 (preprint 4.9)). Theorem 4.1.2 (preprint 4.2) characterizes the manifolds with corners among M-polyfolds by these two properties; Theorem 4.2.7 (preprint 4.18) applies the characterization to submanifolds in good position; Theorem 4.2.11 (preprint 4.22) describes a submanifold near a point of general position; Theorem 4.3.1 (preprint 4.23) combines that description with Sard's theorem on the strata of the corners. The last section, rewritten in the 2021 edition, builds sc-differential forms, their exterior derivative, Stokes' theorem and a Poincaré lemma, with vector fields and their bracket in Appendix 4.5. This Part retells: finite-dimensional sc-spaces and corner charts (§IV.1); sc⁺-maps (§IV.2); sc⁺-retractions, Lemmas 4.1.5 and 4.1.7 (preprint 4.5 and 4.7) (§IV.3); Proposition 4.1.8 (preprint 4.8) with its chart form and Remark 4.1.3 (preprint 4.3) (§IV.4); charts centred at a smooth point, which the text uses and never constructs (§IV.5); Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), by a direct proof (§§IV.6–IV.9); Theorem 4.1.2 (preprint 4.2) (§IV.10); smooth finite-dimensional submanifolds and Remark 4.2.2 (preprint 4.14) (§IV.11); Proposition 4.2.3 (preprint 4.15) (§IV.12); good and general position in charts and their chart independence (§IV.13); Lemma 4.2.8 (preprint 4.19) and Theorem 4.2.7 (preprint 4.18) (§IV.14); Remark 4.2.9 (preprint 4.20) (§IV.15); the linear algebra at the corner of a centred chart (§IV.16); Theorem 4.2.11 (preprint 4.22) (§§IV.17–IV.18) and Remark 4.3.2 (preprint 4.24) (§IV.19); Sard's theorem (§IV.20); the finite-dimensional core of Theorem 4.3.1 (preprint 4.23) (§§IV.21–IV.22); the fibre over a regular value (§IV.23); Theorem 4.3.1 (preprint 4.23) (§IV.24); what is not formalized (§IV.25); what the formalization taught us (§IV.26); Remark 4.1.3's (preprint 4.3) example (§IV.27); and the 2021 edition's Theorems 4.2.11 and 4.3.1, vector fields and the Lie bracket (§§IV.28–IV.29). The later subsections close the remaining gaps of §§4.1–4.3: Lemma 4.1.11 (preprint 4.10) (§IV.30); Remark 4.2.2 (preprint 4.14) in both wordings and Question 4.2.4 (§IV.31); the local faces of Theorem 4.2.11 (1) (preprint 4.22) (§IV.32); the chart independence of stratified regularity (§IV.33); and Lemma 4.5.1 between open subsets with the pull-back bundle $TX|X^2$ (§IV.34). They then retell §4.4 and Appendix 4.5 of the 2021 edition: the Whitney sum and Definition 4.4.1 (preprint 4.25) (§IV.35); the exterior derivative, $d^2 = 0$ and Definition 4.4.6 (preprint 4.27) (§§IV.36–IV.37); the for-

mula of p. 197 (§IV.38); pull-back, wedge product and the graded differential algebra (§IV.39); forms on $\partial X$ and the relative complex (§IV.40); integration and Stokes' theorem on quadrants and on Mathlib manifolds with corners (§§IV.41–IV.42); Theorem 4.4.7 (preprint 4.28) for sc-forms (§IV.43); and the Poincaré Lemmas 4.5.2 and 4.4.8 (§§IV.44–IV.45). Last, the wedge product on the forms on $\partial X$ and on the relative complex (§IV.46).

**What is proved, and what is not.** Theorems 4.1.2, 4.2.7, 4.2.11 and 4.3.1 (preprint 4.2, 4.18, 4.22 and 4.23) are proved with the text's hypotheses, and Theorem 4.1.2 (preprint 4.2) needs no tameness at all.

*§§4.1–4.3.* - Proposition 4.1.9 (preprint 4.9) is proved with two hypotheses the text leaves implicit: the good complement $Y$ is closed, and $p$ is the projection onto $T_0O$ *along* $Y$ rather than $Dt(0)$. The proof is a direct argument (§§IV.6–IV.9). Where the proposition is applied (Theorems 4.1.2, 4.2.7, 4.2.11 (preprint 4.2, 4.18, 4.22), Remark 4.2.9 (preprint 4.20)), the closed complement is chosen, so nothing is added there. - The text's own route goes through Lemma 4.1.11 (preprint 4.10). That lemma is now proved as well, as an sc-Fredholm section in the sense of 2021 Definition 3.1.16 on all of $U$ (gap G.32 closed, §IV.30). Proposition 4.1.9 does not use it. - One step in the proof of Lemma 4.2.8 (preprint 4.19) does not hold as we read it. It is repaired, and the lemma holds (§IV.14). - Theorem 4.2.11 (preprint 4.22): part (4), which the text leaves to "the previous arguments", is proved (§IV.18). Part (1) is proved in the text's sense, with the local faces of the neighbourhood $U$ rather than the faces of the chart model (gap G.35 closed, §IV.32). - Theorem 4.3.1 (preprint 4.23): "regular value" has to be read as the stratified notion of its proof. This notion is now shown not to depend on the product chart (gap G.36 closed, §IV.33). The finite cover of the proof is replaced by a countable one (§IV.24). - Remark 4.2.2 (preprint 4.14) is proved in the preprint wording ($d_X = 0$ gives a manifold) and in the 2021 wording ($d_A = 0$ gives a manifold, $d_A \leq 1$ a manifold with boundary), without tameness (gap G.33 closed, §IV.31). - Question 4.2.4 (preprint: the open question after Proposition 4.15) is a question, so nothing in it is to be formalized (G.34). Its settled last sentence is proved: the "T" is not homeomorphic to an sc$^+$-retract (§IV.31). - Remark 4.1.3's (preprint 4.3) "$X^1 \neq X$ is possible", for which the preprint and 2017 texts give no example, is proved by one: an M-polyfold whose retraction loses a level in its first derivative, with a new criterion for sc-smoothness (gap G.31 closed, §IV.27). The example the 2021 edition adds holds as we read it and is not formalized. - Proved although the text does not construct them: charts and strong bundle charts centred at a smooth point (§IV.5), an sc$^+$-retraction onto the fibre $A_\lambda$ (§IV.23), Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form (§IV.21), and Sard's theorem in all dimensions (§IV.20).

*§4.4 and Appendix 4.5 (2021 edition).* - Vector fields, Lemma 4.4.4, Proposition 4.4.5 and Lemma 4.5.1 are proved (§§IV.28–IV.29). They now also cover local sc-diffeomorphisms between open subsets, and the bracket is a section of the pull-back bundle $TX|X^2 \to X^2$ (gap G.92 closed, §IV.34). - Gaps G.37–G.40 and G.80 are closed. The formalized parts are: - sc-differential forms on the Whitney sum (Definition 4.4.1 (preprint 4.25)), the system (4.4) and $\Omega^*_\infty(X)$ (§IV.35); - the exterior derivative, chart independent, with $d^2 = 0$, commuting with (4.4), and $H^*_{sc}$ (Definition 4.4.6 (preprint 4.27)) (§§IV.36–IV.37); - pull-back, the wedge product (classical normalization, since the text does not define it) and the graded differential algebra (§IV.39); - forms on $\partial X$, read through the tame boundary, with $j^*$ and the relative complex (§IV.40), and the wedge product on both, on the relative complex the product of the mapping cone of $j^*$ (§IV.46); - Theorem 4.4.7 (preprint 4.28), Stokes' theorem for sc-forms with the sum over the faces (§§IV.41–IV.43). It has one added hypothesis, not derived from the text: the boundary being "a union of smooth faces" is read as face-structured. It is satisfied, for instance, without boundary and by the quadrants. - the Poincaré Lemmas 4.5.2 and 4.4.8 (§§IV.44–IV.45). - One row is narrower. The formula for $d\omega$ on p. 197, with the printed sign $(-1)^{\ell+k}$, does not hold as we read it (the counterexample `Chapter4/ScExteriorDerivative2021 · invariantFormula_printedSign_counterexample`). It holds with $(-1)^{\ell+j+1}$ for the text's

bracket (V4.83, §IV.38).

*Not formalized* (§IV.25): - the pull-back of forms on $\partial X$ along a general sc-smooth map beyond the faces of finite-dimensional sub-M-polyfolds, which no statement of Parts I–II uses (gap G.80 is closed); - the argument that an example for Remark 4.1.3 needs a retraction that loses a level, and the 2021 example; - the classical manifold structure behind Definition 4.1.1 (preprint 4.1) beyond the $C^\infty$ transitions.

**Six structural choices.** (1) *sc⁺ is a property of maps.* The text defines it for retractions and operators and uses it for maps; here a map is sc⁺ when its lift into $F^1$ is sc-smooth, and the calculus of such maps is proved first (§IV.2). (2) *Tangent spaces of $X$ are read in charts.* $T_xX$ carries no linear structure in the formalization (gap G.7 of Chapter 2), so $T_aA$, good position, general position and stratified regularity are stated in a chart, and proved independent of the chart (§IV.13; for stratified regularity §IV.33). (3) *Centred charts are re-centred charts.* A chart with $\psi(x) = 0$ is the chart $\varphi - \varphi(x)$ on the set where the partial quadrant $C$ agrees with $C_{\varphi(x)}$, which exists only at smooth points (§IV.5). (4) *Proposition 4.1.9 (preprint 4.9) on fixed sets.* The sc-diffeomorphism $p : O \cap W \to V$ is proved with $\Gamma = p^{-1}$ sc-smooth on a *fixed* relatively open $V$, by contraction arguments on level 1, a frame formula for $D\Gamma$ and a $C^k$-bootstrap along the finite-dimensional tangent space. (5) *Stratified Sard.* Sard's theorem is applied stratum by stratum in the chart of Theorem 4.2.11 (preprint 4.22), and the null sets of critical values of countably many charts are collected. (6) *Forms are read in charts, and integrals on Mathlib manifolds.* The Whitney sum $\oplus_k TX$ is never built as an M-polyfold: a form is a family of alternating forms on the tangent spaces whose chart representatives on $\oplus_k TO$ are sc-smooth (§IV.35). Forms on $\partial X$ are forms on the tame boundary $bX$ that agree over $B : bX \to X$ (§IV.40). Integration happens on a finite-dimensional sub-M-polyfold made a Mathlib manifold with corners by its corner structure, with signed charts (§§IV.41–IV.43).

**Map.**

- `Chapter4/ScPlus`: sc⁺-maps, Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7), Proposition 4.1.8 (preprint 4.8) and its chart form, Remark 4.1.3 (preprint 4.3), finite-dimensional sc-spaces.
- `Chapter4/Recentre`: the partial quadrant $C_e$ and its neighbourhood $G(e)$, translations, re-centred retracts, charts and strong bundle charts.
- `Chapter4/ScPlusGraph`: the mean value estimate for sc⁺-retractions, Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), the frame formula and the bootstrap.
- `Chapter4/Characterization`: Definition 4.1.1 (preprint 4.1), corner charts, Theorem 4.1.2 (preprint 4.2).
- `Chapter4/Submanifolds`: Definitions 4.2.1, 4.2.5, 4.2.6, 4.2.10 (preprint 4.13, 4.16, 4.17, 4.21), Remark 4.2.2 (preprint 4.14), Proposition 4.2.3 (preprint 4.15), Theorem 4.2.7 (preprint 4.18), Lemma 4.2.8 (preprint 4.19), Remark 4.2.9 (preprint 4.20), the chart independence of good and general position.
- `Chapter4/LocalStructure`: Theorem 4.2.11 (1)–(4) (preprint 4.22), Remark 4.3.2 (preprint 4.24), the linear algebra of a centred chart, Lemmas 7.A and 7.B of the exposition.
- `Vendor/MilnorSard/…`: Sard's theorem, copied from our Milnor project (`math_faithful`) with the namespace renamed.
- `Chapter4/FamiliesLocal`, `Chapter4/Families`: the strata and Sard on them, Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form, the fibre retraction, the sc⁺-retraction onto $A_\lambda$ and its general position, Theorem 4.3.1 (preprint 4.23).
- Added in the fourth addendum (§IV.27): `Chapter4/RemarkFourThreeExample` (parameter families of level operators with one level of loss, their sc-smoothness, the reduction of Remark 4.1.3's

(preprint 4.3) example), `Chapter4/RemarkFourThreeFamily` (the family of moving projections, the M-polyfold of a retract, the example).

- The 2021 edition (§§IV.28–IV.29, IV.34): `Chapter4/Book2021`, `Chapter4/VectorFields`, `Chapter4/BracketLevel`, `Chapter4/VectorFieldsGlobal`, `Chapter4/BookFormLieBracket2021`, `Chapter4/GapP1_Lemma451` (open subsets), `Chapter4/GapP1_Lemma451Bundle` (the pull-back bundle $TX|X^2$).
- Gaps of §§4.1–4.3: `Chapter4/GapP1_Lemma4111`, `Chapter4/GapP1_Lemma4111Split`, `Chapter4/GapP1_LocalStrongBundle` (Lemma 4.1.11, §IV.30); `Chapter4/RemarkLocalForm`, `Chapter4/GapP1_Remark422`, `Chapter4/GapP1_Remark422One`, `Chapter4/GapP1_Question424` (Remark 4.2.2, Question 4.2.4, §IV.31); `Chapter4/GapP1_LocalFaces`, `Chapter4/GapP1_Theorem4211` (§IV.32); `Chapter4/GapP1_Theorem431` (§IV.33).
- sc-differential forms (§§IV.35–IV.40): `Chapter4/ScForms2021` (Whitney sum, Definition 4.4.1, the system (4.4)), `Chapter4/ScExteriorDerivative2021` (the local $d$, the formula of p. 197), `Chapter4/ScFormsNatural2021`, `Chapter4/ScFormsGlobal2021`, `Chapter4/ScFormsSquare2021`, `Chapter4/ScFormsLower2021` ($d$ on an M-polyfold, $d^2 = 0$, $H_{sc}$), `Chapter4/ScFormsPullback2021`, `Chapter4/ScFormsWedge2021`, `Chapter4/ScFormsWedge2021b` (pull-back, wedge, graded algebra), `Chapter4/BoundaryForms2021`, `Chapter4/A3BoundaryForms2021`, `Chapter4/A3BoundaryForms2021b` (forms on $\partial X$, the relative complex), `Chapter4/BoundaryFormsPullback2021` (forms on $\partial X$ pulled back to faces of a sub-M-polyfold).
- Integration and Stokes (§§IV.41–IV.43): `Chapter4/EuclideanForms2021`, `Chapter4/EuclideanStokes2021`, `Chapter4/ManifoldForms2021`, `Chapter4/ManifoldIntegral2021`, `Chapter4/ManifoldBoundary2021`, `Chapter4/ManifoldStokes2021`, `Chapter4/ManifoldFaceForms2021`, `Chapter4/GapP1_StokesFaces`, `Chapter4/GapP1_StokesFaceCharts`, `Chapter4/GapP1_StokesFaceOrientation`, `Chapter4/GapP1_StokesFaceIntegral`, and in Chapter 9 `Chapter9/MathlibFormsBridge2021`, `Chapter9/GapP1_Stokes2021`.
- The Poincaré lemma (§§IV.44–IV.45): `Chapter4/Poincare2021` (parametric integrals), `Chapter4/Poincare2021b` (the homotopy operator, Lemmas 4.5.2 and 4.4.8), `Chapter4/GapP1_OpenForms`, `Chapter4/Poincare448Shift`, `Chapter4/Poincare448`.
- The wedge product on $\partial X$ and on the relative complex (§IV.46): `Chapter4/GapP3_K4_BoundaryWedge`, `Chapter4/GapP3_K4_RelWedge`.

Throughout, $C = T^{-1}([0,\infty)^n \oplus W)$ is a partial quadrant presented by an sc-isomorphism $T : E \cong \mathbb{R}^n \oplus W$, with coordinates $\lambda_1, \dots, \lambda_n$; $I(e) = \{i : \lambda_i(e) = 0\}$, $d_C(e) = \#I(e)$, $C_e = \{\lambda_i \geq 0, i \in I(e)\}$ and $E_e = \{\lambda_i = 0, i \in I(e)\}$ (Definition 2.5.1 (preprint 2.33), §II.12). For a retract $O = r(U)$ and a level-1 point $z$, $T_zO = Dr(z)E_0$ is the tangent space of Definition 2.3.12 (preprint 2.17) and $T_z^R O = T_zO \cap E_z$ the reduced tangent space.

# IV.1 Finite-dimensional sc-spaces and corner charts (Definition 4.1.1 (preprint 4.1))

**Finite-dimensional sc-spaces.** Definition 4.1.1 (preprint 4.1) models its charts on finite-dimensional spaces "with the constant sc-structure"; the text remarks after Definition 1.1.1 (preprint 1.1) that a finite-dimensional space has only that one. This is proved. If $E_0$ is finite-dimensional, every level $E_m$ is, since it injects into $E_0$ `Chapter4/ScPlus · ScBanach.finiteDimensional_lvl`. The range of $E_m \to E_0$ is then a finite-dimensional, hence closed, subspace; it contains the smooth points, which are dense (Definition 1.1.1 (2) (preprint 1.1)); so it is all of $E_0$ `Chapter4/ScPlus · ScBanach.toBase_surjective`. The inclusions are therefore bijective, and they form an sc-isomorphism of $E$ onto $E_0$ with the constant structure `Chapter4/`

ScPlus · ScBanach.scIsoConst; in particular every point of $E_0$ is smooth Chapter4/Characterization · ScBanach.mem_smooth_of_finiteDimensional. The density axiom is what makes this work: $E_0 = \mathbb{R}$, $E_m = 0$ for $m \geq 1$ has compact injective inclusions and levels that differ (Findings, Chapter 4, item 6).

**Definition 4.1.1 (preprint 4.1).** A chart of $X$ is a *corner chart* if its local model is $(U, C, V)$ with $V$ a finite-dimensional real normed space carrying the constant sc-structure, $C$ a partial quadrant in $V$, $U$ relatively open in $C$, and the identity retraction Chapter4/Characterization · MChart.IsCornerChart. Its model is then relatively open in a partial quadrant Chapter4/Characterization · MChart.IsCornerChart.isRelOpenInQuadrant, finite-dimensional Chapter4/Characterization · MChart.IsCornerChart.finiteDimensional, a tame retract — the identity retraction is tame Chapter2/Tame · ScRetraction.id_isTame Chapter4/Characterization · MChart.IsCornerChart.isTameRetract — and an sc$^+$-retract by Remark 4.1.3 (preprint 4.3) (§IV.4) Chapter4/Characterization · MChart.IsCornerChart.isScPlusRetract; every point of its model space is smooth Chapter4/Characterization · MChart.IsCornerChart.mem_smooth. *$X$ has a corner structure* if it has an atlas of corner charts equivalent to its own atlas Chapter4/Characterization · MPolyfold.HasCornerStructure. The dimension of $V$ may vary from chart to chart, as in the text. The text's "sc-smoothly compatible charts", "$\varphi$ an sc-diffeomorphism", is read as compatibility with the atlas of $X$, as for tameness (Definition 2.5.7 (preprint 2.37)).

*The definition is local* Chapter4/Characterization · MPolyfold.hasCornerStructure_iff: $X$ has a corner structure iff every point lies in a corner chart that is a chart of the M-polyfold structure. One direction takes a chart of the equivalent atlas through the point. For the other, the corner charts that are charts of $X$ form an atlas: they cover $X$ by hypothesis, any two charts of $X$ are compatible Chapter2/MPolyfold · MPolyfold.IsChart.compatible, and the atlas is equivalent to that of $X$ because each of its members is a chart of $X$. *A corner structure makes $X$ tame* Chapter4/Characterization · MPolyfold.HasCornerStructure.isTame: the same atlas is an atlas of tame charts.

**The transition maps are $C^\infty$.** For a chart whose model is relatively open in a partial quadrant, every overlap $\chi(V \cap V')$ is relatively open Chapter4/Characterization · MChart.isRelOpenInQuadrant_overlap. For compatible corner charts, the transition map is sc-smooth between the overlap retracts; composed with the identity retraction of the overlap it is sc-smooth on a relatively open subset of a partial quadrant in a finite-dimensional space with the constant structure, and such maps are exactly the $C^\infty$ maps with derivatives within the set Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const Chapter4/Characterization · MChart.IsCornerChart.contDiffOn_transition. That is the text's "classically smooth"; "the atlas defines the structure of a smooth manifold with boundary with corners" is not formalized beyond it, because Mathlib's manifolds with corners have a single model space and here the partial quadrants vary from chart to chart.

**Arbitrary finite-dimensional models.** The constant structure is no restriction. A partial quadrant of a finite-dimensional $E$ is one of $E_0$ with the constant structure Chapter4/Characterization · IsPartialQuadrant.toConst Chapter4/Characterization · IsRelOpenInQuadrant.toConst, and the identity of $E_0$ is an sc-diffeomorphism between the two models Chapter4/Characterization · IsRelOpenInQuadrant.isScDiffeoQuadrant_toConst. Composing a chart with it (the construction `compDiffeo` of Chapter 3) Chapter3/GlobalIFTCenter · MChart.compDiffeo gives, with the same map and domain, a corner chart of the same dimension Chapter4/Characterization · MChart.exists_isCornerChart_of_finiteDimensional. Hence $X$ has a corner structure iff it has an equivalent atlas of charts onto relatively open subsets of partial quadrants in sc-Banach spaces with finite-dimensional $E_0$ Chapter4/Characterization · MPolyfold.hasCornerStructure_iff_finiteDimensional — verbatim the definition of the manifolds with corners of locally constant dimension of Theorem 3.2.3 (preprint 3.27) (§III.36) — and the manifolds with corners of dimension $n$ of Theorem 3.2.3 (preprint 3.27) are the corner structures all of whose charts have dimen-

sion $n$ Chapter4/Characterization · MPolyfold.isManifoldWithCorners_iff Chapter4/Characterization · MPolyfold.IsManifoldWithCorners.hasCornerStructure.

## IV.2 Sc⁺-maps

The text uses "sc⁺-map" in the proof of Lemma 4.1.7 (preprint 4.7) and composes sc⁺-retractions with sc-smooth maps in Lemma 4.1.5 (preprint 4.5), but defines sc⁺ only for retractions (Definition 4.1.4 (preprint 4.4)) and operators (Definition 1.1.5 (preprint 1.4)). The formalization defines it for maps on any set, as the neighbouring notions of the project are (sc⁺-operators are sc-operators into $F^1$, sc⁺-sections sc-smooth maps into $K[1]$).

**The lift.** For $x \in E_1 \subseteq E_0$, $1_0^1 x$ is the point of $E_1$ over $x$, with a junk value off $E_1$ Chapter4/ScPlus · ScBanach.liftOne Chapter4/ScPlus · ScBanach.toBase_liftOne Chapter4/ScPlus · ScBanach.liftOne_toBase.

**Definition.** $f$ is *sc⁺ on* $U$ if $f(U) \subseteq F_1$ and the lift $x \mapsto 1_0^1 f(x)$, a map $U \to F^1$, is sc-smooth Chapter4/ScPlus · IsScPlus. Then $f(U_m) \subseteq F_{m+1}$: the lift is sc⁰, so it maps $U_m$ into $(F^1)_m = F_{m+1}$ Chapter4/ScPlus · IsScPlus.mem_range_succ. On an sc-domain the canonical lift can be replaced by any sc-smooth $g : U \to F^1$ with $\iota \circ g = f$, which is how the text phrases it Chapter4/ScPlus · IsScPlus.iff_exists. The calculus: an sc⁺-map is sc-smooth, being $\iota\circ$ (lift) with the sc-operator $\iota : F^1 \to F$ Chapter4/ScPlus · IsScPlus.isScSmooth; the property depends only on the values on $U$ Chapter4/ScPlus · IsScPlus.congr and passes to smaller sc-domains Chapter4/ScPlus · IsScPlus.mono; *sc⁺ ◻ sc-smooth is sc⁺*, the lift of $g \circ f$ being (lift of $g$) $\circ f$, sc-smooth by the chain rule Chapter1/ChainRule · IsScSmooth.comp Chapter4/ScPlus · IsScPlus.comp_isScSmooth; *sc-smooth ◻ sc⁺ is sc⁺*, the lift of $g \circ f$ being $g^1\circ$ (lift of $f$) with $g^1 : V^1 \to G^1$ the map $g$ induces on level 1, sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter1/Levels · IsScSmooth.shift Chapter4/ScPlus · IsScSmooth.comp_isScPlus; and an sc⁺-operator is an sc⁺-map Chapter4/ScPlus · ScPlus.isScPlus.

**The derivative at a smooth point** Chapter4/ScPlus · IsScPlus.exists_scPlus_deriv. Let $f$ be sc⁺ on an sc-domain $U$ and $x \in U_1$ smooth. The lift $g$ is sc-smooth into $F^1$, so its derivative at the smooth point $x$ is an sc-operator $E \to F^1$ (the remark after Proposition 1.2.1 (preprint 1.16)), that is an sc⁺-operator $S : E \to F$. Since $f = \iota \circ g$ on $U$, the chain rule and the uniqueness of the derivative on an sc-domain give $Df(x) = \iota\circ Dg(x)$, the level-0 operator of $S$. This is the text's step "$Dr(o)$ is a sc⁺-operator, since $r$ is a sc⁺-map". Consequently $Df(x)$ is compact, and so is every level operator of $S$ Chapter4/ScPlus · IsScPlus.isCompactOperator_deriv.

**A compact projection has finite-dimensional range** Chapter4/ScPlus · finiteDimensional_range_of_isCompactOperator_of_idem: if $P = P \circ P$ is compact, its range is the closed set $\{v : Pv = v\}$, $P$ restricts to a compact operator of the range which is the identity, and a normed space with compact identity is finite-dimensional (Riesz). The text takes this for granted.

## IV.3 Sc⁺-retractions and sc⁺-retracts (Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7))

**Definition 4.1.4 (preprint 4.4).** An sc-smooth retraction $r : U \to U$ is an *sc⁺-retraction* if it is an sc⁺-map on $U$ Chapter4/ScPlus · ScRetraction.IsScPlus. The text's condition $r(U_m) \subseteq U_{m+1}$ is then a consequence, not a hypothesis Chapter4/ScPlus · ScRetraction.IsScPlus.mem_lvlSet_succ. On an open subset $V$ of an M-polyfold, an sc⁺-retraction is a map $s$, sc-smooth on $V$, with $s(V) \subseteq V$, $s\circ s = s$ on $V$, $s(V) \subseteq X_1$, and $1_0^1 \circ s : V \to$

$X^1$ sc-smooth Chapter4/ScPlus · MPolyfold.IsScPlusRetractionOn; the junk value of $1_0^1$ off $X_1$ is quantified over and plays no role.

**Lemma 4.1.5 (preprint 4.5)** Chapter4/ScPlus · lemma_4_1_5. *If $t$ is an sc⁺-retraction onto $O$ and $s$ an sc-smooth retraction (of a possibly different relatively open set) onto $O$, then $s$ is sc⁺.* The text's proof: $s$ maps its domain into $O$, which lies in the domain of $t$, and $t$ fixes $O$, so $s = t \circ s$, which is sc⁺ □ sc-smooth. (The text writes "$s(V) = 0$" for $s(V) = O$.)

**Definition 4.1.6 (preprint 4.6).** $(O, C, E)$ is an *sc⁺-retract* if it is the retract of an sc⁺-retraction Chapter4/ScPlus · IsScPlusRetract; it is then a retract Chapter4/ScPlus · IsScPlusRetract.isScRetract, every sc-smooth retraction onto it is sc⁺ by Lemma 4.1.5 (preprint 4.5) Chapter4/ScPlus · IsScPlusRetract.isScPlus, and so the property is intrinsic Chapter4/ScPlus · isScPlusRetract_iff_forall.

**Lemma 4.1.7 (preprint 4.7)** (i) Chapter4/ScPlus · lemma_4_1_7_i. *The retract of an sc⁺-retraction consists of smooth points.* By induction on $m$: a point $y \in O$ lies in $E_m$; then $y \in U_m$, and $y = r(y) \in E_{m+1}$ by the level gain. (ii) Chapter4/ScPlus · lemma_4_1_7_ii. *At a level-1 point $o$ over a point of $O$, $Dr(o)E_0$ is finite-dimensional.* $o$ is smooth by (i), so $Dr(o)$ is compact (§IV.2); it is a projection because $r \circ r = r$ gives $Dr(o) \circ Dr(o) = Dr(o)$ at points of $O$ Chapter2/Retracts · ScRetraction.scDeriv_comp_self; and a compact projection has finite-dimensional range. In the form with the tangent space of Definition 2.3.12 (preprint 2.17) Chapter4/ScPlus · lemma_4_1_7_ii_tangentSpace: every $o \in O$ has a smooth lift $o_1 \in E_1$ Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth, and $T_{o_1}O$ is the range of $Dr(o_1)$ Chapter2/Retracts · tangentSpace_eq_coe_submodule, finite-dimensional. Tangent spaces are indexed by level-1 points, which is why the statement mentions the lift.

# IV.4 Proposition 4.1.8 (preprint 4.8), its chart form, and Remark 4.1.3 (preprint 4.3)

**Proposition 4.1.8 (preprint 4.8)** Chapter4/ScPlus · proposition_4_1_8. *A retract $(O, C, E)$ is an sc⁺-retract iff $O \subseteq E_\infty$ and the lift $1_0^1$ is sc-smooth from $(O, C, E)$ to $(O_1, C_1, E^1)$* in the sense of Definition 2.1.14 (preprint 2.7). (□) $O \subseteq E_\infty$ is Lemma 4.1.7 (i) (preprint 4.7); and by the criterion of Definition 2.1.14 (preprint 2.7) through one retraction Chapter2/Retracts · isScSmoothRetractMap_iff_exists, the lift is sc-smooth between the retracts because $1_0^1 \circ r$ is the lift of the sc⁺-map $r$. (□) For a retraction $r$ onto $O$, the composite $U \to O \to O_1 \to E^1$ is sc-smooth (Definition 2.1.14 (preprint 2.7), the field `comp_isScSmooth` of Chapter2/Retracts · IsScSmoothRetractMap) and is the lift of $r$, whose values lie in $E_1$ because $O \subseteq E_\infty$. The text's proof is complete; it ends "the lemma is proved".

**The chart form.** Theorem 4.1.2 (preprint 4.2) needs the statement for the charts of an M-polyfold, which the text does not state. The map $1_0^1 : X \to X^1$ is defined on $X_1$, with a junk value $x_0$ off it Chapter4/ScPlus · MPolyfold.toLevelOne Chapter4/ScPlus · MPolyfold.levelVal_toLevelOne Chapter4/ScPlus · MPolyfold.toLevelOne_levelVal. In the chart $\varphi$ of $X$ and the level chart $(V_1, \varphi, (O_1, C_1, E^1))$ of $X^1$ Chapter2/LevelPolyfold · MPolyfold.levelChart it is the lift $O \to O_1$ Chapter4/ScPlus · MPolyfold.levelChart_toLevelOne. A point of a chart domain is smooth iff its image is Chapter4/ScPlus · MPolyfold.mem_smoothSet_iff. So *the model of a chart $\varphi : V \to O$ is an sc⁺-retract iff $V \subseteq X_\infty$ and $1_0^1$ is sc-smooth on $V$* Chapter4/ScPlus · MPolyfold.proposition_4_1_8_chart: (□) continuity on $V$ comes from the lift read in the charts, and sc-smoothness is checked in the pair of charts $\varphi$, `levelChart` $\varphi$ Chapter2/LevelPolyfold · IsScSmoothOn.of_exists_charts, where it is Proposition 4.1.8 (preprint 4.8); (□) the same pair of charts turns the sc-smoothness of $1_0^1$ on $V$ into that of the lift on $O$. Hence "$X_0 = X_\infty$ and $1_0^1$ sc-smooth" holds iff every chart of $X$ is modelled on an

sc⁺-retract Chapter4/ScPlus · MPolyfold.isScSmoothMap_toLevelOne_iff, and already if some cover of $X$ by such charts exists Chapter4/Characterization · MPolyfold.isScSmoothMap_toLevelOne_of_cover — sc-smoothness being local.

**Remark 4.1.3 (preprint 4.3).** The inclusion $\iota : (O_1, C_1, E^1) \to (O, C, E)$ is sc-smooth between retracts Chapter4/ScPlus · isScSmoothRetractMap_toBase_one, so $1^0_1 : X^1 \to X$ is always sc-smooth, being $\iota$ in the charts `levelChart` $\varphi$ and $\varphi$ Chapter4/ScPlus · MPolyfold.remark_4_1_3_levelVal. If every point lies in $X_1$ and $1^1_0$ is sc-smooth, it is an sc-diffeomorphism with inverse $1^0_1$ Chapter4/ScPlus · MPolyfold.remark_4_1_3_isScDiffeo. For a finite-dimensional $V$ with the constant structure, $V^1 = V$ definitionally, the lift is the identity, and every retract in $V$ is an sc⁺-retract, its retraction being its own lift Chapter4/ScPlus · remark_4_1_3_const. The remark's "it is possible that $X^1 \neq X$ as M-polyfolds although the sets agree" is given without an example; one is constructed in §IV.27.

## IV.5 Charts centred at a smooth point

Theorem 4.1.2 (ii) (preprint 4.2) asks for charts with $\psi(x) = 0$, the proofs of Theorems 4.2.7 and 4.2.11 (preprint 4.18 and 4.22) and of Lemma 5.3.3 (preprint 5.11) use them, and the text never says where they come from. Translating a chart $\varphi : V \to O \subseteq C$ by $e = \varphi(x)$ fails twice: translation by $e$ preserves every level only if $e$ is smooth, and $C - e$ is not a partial quadrant unless $e \in C \cap (-C)$. The construction (Findings, items 1, 2) uses the partial quadrant $C_e$ and the neighbourhood on which it agrees with $C$.

**The partial quadrant $C_e$ and the neighbourhood $G(e)$.** Let $G(e) = \{v : \lambda_i(v) > 0$ whenever $\lambda_i(e) \neq 0\}$ Chapter4/Recentre · recentreNbhd, open Chapter4/Recentre · isOpen_recentreNbhd and containing $e$ when $e \in C$ Chapter4/Recentre · mem_recentreNbhd. On it $C$ and $C_e$ agree: $G(e) \cap C = G(e) \cap C_e$, because on $G(e)$ the coordinates not vanishing at $e$ are positive anyway Chapter4/Recentre · recentreNbhd_inter_eq. $C_e$ is invariant under translation by $e$ Chapter4/Recentre · add_mem_partialQuadrantAtWith_iff, so $C_e - e = C_e$ Chapter4/Recentre · image_sub_partialQuadrantAtWith. *$C_e$ is a partial quadrant* — Definition 2.5.1 (preprint 2.33) asserts it without proof: the vanishing indices $I(e)$ Chapter4/Recentre · zeroIdx, $d = \#I(e)$ of them Chapter4/Recentre · recentreDim, are put first, $\mathbb{R}^n \cong \mathbb{R}^{I(e)} \times \mathbb{R}^{\text{rest}}$ Chapter4/Recentre · splitCoordsL, which gives an sc-isomorphism $\mathbb{R}^n \oplus W \cong \mathbb{R}^{I(e)} \oplus (\mathbb{R}^{\text{rest}} \oplus W)$ Chapter4/Recentre · splitCoordsIso and the presentation $T_e$ of $C_e$ Chapter4/Recentre · recentreIso Chapter4/Recentre · partialQuadrantAtWith_eq_preimage Chapter4/Recentre · isPartialQuadrant_partialQuadrantAtWith: the coordinates not vanishing at $e$ join the linear factor. *Re-centring keeps the tame data near $e$.* For $v \in G(e)$ the vanishing coordinates of $v - e$ for $T_e$ are those of $v$ for $T$ Chapter4/Recentre · recentreIso_fst_sub Chapter4/Recentre · zero_of_mem_recentreNbhd, so $d_{C_e}(v-e) = d_C(v)$ Chapter4/Recentre · degIndexWith_recentreIso Chapter4/Recentre · degIndex_sub_recentre, and the minimal subspaces and partial quadrants agree: $E^{T_e}_{v-e} = E^T_v$ Chapter4/Recentre · minimalSubspaceWith_recentreIso, $(C_e)_{v-e} = C_v$ Chapter4/Recentre · partialQuadrantAtWith_recentreIso.

**Translations.** Translation by a smooth vector $a$ is sc-smooth on every sc-domain: its level maps are $x \mapsto x + a_m$, $a_m$ the lift of $a$ to $E_m$, and the criterion of Proposition 1.2.4 (preprint 1.19) applies Chapter1/Levels · IsScSmooth.of_contDiffOn_levels Chapter4/Recentre · isScSmooth_add_const Chapter4/Recentre · isScSmooth_sub_const. For $U$ relatively open in $C$ with $U \subseteq G(e)$, the translate $U - e$ is relatively open in $C_e$ Chapter4/Recentre · relOpen_translate and, for smooth $e$, $v \mapsto v - e$ is an sc-diffeomorphism $U \to U - e$ Chapter4/Recentre · isScDiffeoQuadrant_translate.

**Re-centred retracts.** A retraction $r$ restricted to $U \cap r^{-1}(W) \cap W$, $W$ open, is a retraction whose domain lies in $W$ Chapter4/Recentre · ScRetraction.restrictIn, onto $r(U) \cap W$ Chapter4/Recentre · ScRetraction.restrictIn_retract, with the same derivative Chapter4/Recentre · ScRetraction.scDeriv_restrictIn, and tame

if $r$ is Chapter4/Recentre · ScRetraction.IsTameWith.restrictIn Chapter4/Recentre · ScRetraction.restrictIn_isTame. Transporting the restriction to $G(e)$ along the translation Chapter2/TameInvariance · IsScDiffeoQuadrant.transport shows that $((O \cap G(e)) - e, C_e, E)$ is a retract, with retraction $u \mapsto r(u+e) - e$ Chapter4/Recentre · isScRetract_recentre, and tame if $(O, C, E)$ is Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport Chapter4/Recentre · isTameRetract_recentre — the transport of tameness uses exactly the invariance of $d$ and of the minimal subspaces above.

**Re-centred charts.** For a chart $\varphi$, a presentation of its partial quadrant and a smooth $e$, the domain $V \cap \varphi^{-1}(G(e))$ Chapter4/Recentre · MChart.recentreSource is open, and the chart $\varphi_e = \varphi - e$ on it is $\varphi$ restricted Chapter2/MPolyfold · MChart.restrict and composed with the translation Chapter4/Recentre · MChart.recentre, modelled on $((O \cap G(e)) - e, C_e, E)$. It is a chart of the M-polyfold, its transition maps being those of $\varphi$ composed with translations Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart Chapter4/Recentre · MChart.recentre_isChart; $\varphi_e(x) = 0$ when $\varphi(x) = e$ Chapter4/Recentre · MChart.recentre_apply_self; its model is tame when that of $\varphi$ is Chapter4/Recentre · MChart.isTameRetract_recentre; and it computes the same degeneracy index Chapter4/Recentre · MChart.chartDegIndex_recentre. So *every smooth point $x$ of a chart domain is the centre of a chart of $X$* inside that domain, tame if the original is Chapter4/Recentre · MPolyfold.exists_centredChart, and *in a tame M-polyfold every smooth point is the centre of a tame chart* Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart Chapter4/Recentre · MPolyfold.IsTame.exists_centredChart. At the centre every coordinate of $C_e$ vanishes, so the minimal subspace there is $\{\lambda = 0\}$, the text's "$\{0\} \oplus W$", which is what §§IV.14, IV.16 and IV.17 use.

**Re-centred strong bundle charts** (for Lemma 5.3.3 (preprint 5.11)). Only the base is translated: $(u, h) \mapsto (u \mp e, h)$ Chapter4/Recentre · subFst Chapter4/Recentre · addFst is sc-smooth on $E \oplus F$ and on $E \oplus F^1$ because $(e, 0)$ is smooth in both Chapter4/Recentre · mem_smooth_prod_zero Chapter4/Recentre · isScSmooth_addFst Chapter4/Recentre · isScSmooth_subFst. A strong bundle retraction restricts to $(U \cap r^{-1}(G) \cap G) \lhd F$ Chapter4/Recentre · IsStrongBundleRetraction.restrictIn and translates to $R_e(u, h) = (r(u+e) - e, \Gamma(u+e, h))$ Chapter4/Recentre · IsStrongBundleRetraction.translate; with $\Phi_e(y) = (\Phi(y)_1 - e, \Phi(y)_2)$ this is the re-centred chart Chapter4/Recentre · StrongBundleChart.recentre. Its overlaps are translates of open parts of the old ones Chapter4/Recentre · StrongBundleChart.recentre_overlap Chapter4/Recentre · StrongBundleChart.overlap_recentre_eq, its transition maps on $[0]$ and $[1]$ are the old ones composed with translations by $(e, 0)$, so compatibility passes to it Chapter4/Recentre · StrongBundleChart.Compatible.recentre, and it is a strong bundle chart Chapter4/Recentre · StrongBundle.IsChart.recentre. Every smooth point of the base is the centre of one Chapter4/Recentre · StrongBundle.IsChart.exists_centred. No hypothesis beyond the smoothness of the point is needed.

## IV.6 Proposition 4.1.9 (preprint 4.9): the data, and the mean value estimate

**The text.** Let $t : U \to U$ be an sc$^+$-retraction onto $O \ni 0$ with $T_0O$ in good position to $C, Y$ a good complement and $p = Dt(0)$ "the sc-projection". Then $p$ maps a neighbourhood of $0$ in $O$ sc-diffeomorphically onto one in $T_0O \cap C$. The proof: the section $\mathbf{f}(u) = (1-p)(u - t(u))$ of $U \lhd Y$ is sc-Fredholm (Lemma 4.1.11 (preprint 4.10)), Theorem 3.5.11 (preprint Corollary 3.47) describes its zeros near $0$ as a graph $\{a + \delta(a)\}$, Lemma 4.1.12 (preprint 4.11) says that every small zero is on the graph, Lemma 4.1.13 (preprint 4.12) that the graph lies in $O$.

**What changes, and why.** (a) $Dt(0)$ is a projection onto $T_0O$ along $\ker Dt(0)$, which nothing relates to $Y$; Lemma 4.1.13 (preprint 4.12) writes $t(a + \delta(a)) = p\,t(a + \delta(a)) + \delta(a)$ with $\delta(a) \in Y$, which needs the

projection *along* $Y$. (b) $Y$ must be closed, or that projection is not continuous. (c) Theorem 3.5.11 (preprint Corollary 3.47) gives $\delta$ over a complement of its own choosing and sc-smooth only as a germ — $C^k$ into $E_m$ on neighbourhoods shrinking with $m$ and $k$ (§III.20) — while the claim is an sc-diffeomorphism between fixed open sets. The formalization therefore takes $p$ to be the projection along a closed $Y$ and proves the proposition directly, with the sc$^+$ property doing the work of the Fredholm theory (Findings, items 20–28).

**The data** `Chapter4/ScPlusGraph · GraphData`: the sc$^+$-retraction $t$ with $0 \in O$; $N$ with $N = T_0O$, which equals $Dt(0)E_0$ `Chapter4/ScPlusGraph · GraphData.N_eq_range` and is finite-dimensional by Lemma 4.1.7 (preprint 4.7) `Chapter4/ScPlusGraph · GraphData.finiteDimensional`; $N$ in good position; a good complement $Y$, closed; and $p : E_0 \to N$ continuous linear with $p = 1$ on $N$, $p = 0$ on $Y$. The tangent space consists of smooth points: $Dt(0)$ is an sc$^+$-operator and $N$ its fixed vectors `Chapter4/ScPlusGraph · GraphData.smooth`. $Y = \ker p$ `Chapter4/ScPlusGraph · GraphData.mem_Y_iff`. Because $N \subseteq E_\infty$ is finite-dimensional, it lifts continuously to every level `Chapter3/TangentGerm · finSmoothLift` `Chapter4/ScPlusGraph · GraphData.liftN`, and $p$, $q = 1 - p$ act on every level $E_m$ `Chapter4/ScPlusGraph · GraphData.pL` `Chapter4/ScPlusGraph · GraphData.qL`. The identities the text uses hold for this $p$ on every level: $(1 - p)Dt(0) = 0$, because the range of $Dt(0)$ is $N$ `Chapter4/ScPlusGraph · GraphData.qL_lvlDeriv_zero`, and $Dt(0) = 1$ on $N$ `Chapter4/ScPlusGraph · GraphData.lvlDeriv_zero_of_mem_N`. The good complement has a constant $\varepsilon$: for $n \in N$, $y \in Y$, $|y|_0 \le \varepsilon|n|_0$, $n \in C \iff n + y \in C$ `Chapter4/ScPlusGraph · GraphData.eps` `Chapter4/ScPlusGraph · GraphData.eps_spec`. Its level-1 version is $\varepsilon' = \varepsilon/(\|\iota_1\|\|\mathrm{lift}_N\| + 1)$ `Chapter4/ScPlusGraph · GraphData.eps'`: $|y|_1 \le \varepsilon'|a|_1$ implies $|y|_0 \le \varepsilon|a|_0$ for $a \in N$ `Chapter4/ScPlusGraph · GraphData.norm_toBase_le_eps_mul`.

**The level data of an sc$^+$-retraction.** The lift $t^1 : U \to E^1$ `Chapter4/ScPlusGraph · ScRetraction.liftOne` is sc-smooth `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isScSmooth_liftOne`, sc$^1$ with derivative family $Dt^1$ `Chapter4/ScPlusGraph · ScRetraction.liftDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isSc1With_liftOne`. By Proposition 1.2.1 (preprint 1.16) for $t^1$, the map $t : U_{k+1} \to E_{k+1}$ `Chapter4/ScPlusGraph · ScRetraction.lvlMap` is differentiable within $U_{k+1}$ with a derivative $Dt(x) \in \mathcal{L}(E_{k+1})$ `Chapter4/ScPlusGraph · ScRetraction.lvlDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.hasFDerivWithinAt_lvlMap` that depends continuously on $x$ in the operator norm `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlDeriv`; $t$ is continuous on $U_{k+1}$ `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlMap`, fixes $O_{k+1}$ `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlMap_eq_self`, is $t$ itself through the inclusions `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlMap`, and its level derivatives are restrictions of $Dt$ `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_liftDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlDeriv`, projections at points of $O$ `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlDeriv_lvlDeriv`.

**The mean value estimate** (the text's (76)–(77)). The abstract form `Chapter4/ScPlusGraph · exists_norm_sub_sub_le_of_hasFDerivWithinAt`: if $s$ is differentiable within a convex set $S$ with derivative $D$ continuous at $u_0 \in S$, then for every $\theta > 0$ there is $r > 0$ with $|s(z') - s(z) - D(u_0)(z' - z)| \le \theta|z' - z|$ for $z, z' \in S \cap B(u_0, r)$ — the mean value inequality for $s - D(u_0)$. For an sc$^+$-retraction `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.exists_mvt`: at $u_0 \in U_{k+1}$ a ball meets $C_{k+1}$ inside $U_{k+1}$ `Chapter4/ScPlusGraph · ScRetraction.exists_ball_inter_subset` `Chapter4/ScPlusGraph · ScRetraction.lvlSet_U_subset_lvlSet_C`, the set $C_{k+1} \cap B(u_0, r_0)$ is convex, and the abstract form and the continuity of $Dt$ give $r$ with $C_{k+1} \cap B(u_0, r) \subseteq U_{k+1}$, $\|Dt(z) - Dt(u_0)\| \le \theta$ and the estimate with constant $\theta$ there. The text integrates along segments in $U_{k+1}$, which need not be convex; the convex set $C_{k+1} \cap B$ is the repair.

A shortcut does not work: inverting $p \circ t$ on the finite-dimensional quadrant $N \cap C$ by Proposition 3.8.17 (preprint 3.73). In $\mathbb{R}^3$ with $C = [0, \infty)^2 \times \mathbb{R}$, $\Gamma(a) = (a_1, a_2, a_2^2)$ and $t(x) = \Gamma(x_1 + (x_3 - x_2^2)^2, x_2)$, $t$ is a retraction with $N = \mathbb{R}^2 \times 0$, but $p\,t(x_1, x_2, 0) = (x_1 + x_2^4, x_2)$ misses $(0, y_2)$ for $y_2 > 0$.

## IV.7 Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12): uniqueness and existence on level 1

**Lemma 4.1.12 (preprint 4.11), the contraction estimate** `Chapter4/ScPlusGraph · GraphData.exists_contraction_radius`. For $\theta > 0$ the mean value estimate at $u_0 = 0$ gives $r > 0$ with the estimates of §IV.6 on $C_{k+1} \cap B(0,r)$ and, for $u, u' \in O_{k+1}$ with norms below $r$, $|(1-p)(u'-u)|_{k+1} \le \|1-p\|\theta|u'-u|_{k+1}$: since $u = t(u)$, $u' = t(u')$ and $(1-p)Dt(0) = 0$, $(1-p)(u'-u) = (1-p)[t(u') - t(u) - Dt(0)(u'-u)]$.
**Uniqueness** `Chapter4/ScPlusGraph · GraphData.eq_of_p_eq`: if $p(u) = p(u')$ and $|(1-p)(u'-u)| \le c|u'-u|$ with $c < 1$, then $u' - u \in Y$, so $(1-p)(u'-u) = u'-u$ and $u' = u$. This is what the text's Lemma 4.1.12 (preprint 4.11) is used for; its statement, "$y = \delta(a)$", presupposes the $\delta$ of Theorem 3.5.11 (preprint Corollary 3.47).

**Lemma 4.1.13 (preprint 4.12), existence by a contraction** `Chapter4/ScPlusGraph · GraphData.exists_graph_point`. Suppose the estimates hold on $C_1 \cap B(0,r)$ with $\theta < 1$, $\|1-p\|\theta < 1$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$. As $t(0) = 0$ and $t : U_1 \to E_1$ is continuous at 0, there is $\delta$ with $|t(u)|_1 < r$ for $|u|_1 < \delta$; put $r' = \min(r, \delta)$ and $\rho = r'/(1+\varepsilon')$. For $a \in N \cap C$ with $|a|_1 < \rho$:

1. The set $B_a = \{y \in Y_1 : |y|_1 \le \varepsilon'|a|_1\}$ is closed, hence complete, because $Y$ is closed — the first use of that hypothesis.
2. For $y \in B_a$, $a + y \in C_1$ by the good complement property, $|y|_0 \le \varepsilon|a|_0$ coming from $|y|_1 \le \varepsilon'|a|_1$; and $|a+y|_1 \le (1+\varepsilon')|a|_1 < r'$. So $a + y$ lies in $U_1$, where $t$ is defined — a point the text does not check.
3. $\Phi_a(y) = (1-p)t(a+y)$ maps $B_a$ into itself: $\Phi_a(y) = (1-p)[t(a+y) - t(0) - Dt(0)(a+y)]$, of norm at most $\|1-p\|\theta(1+\varepsilon')|a|_1 \le \varepsilon'|a|_1$, and it lies in $Y$.
4. $\Phi_a$ is a contraction with constant $\|1-p\|\theta < 1$, by the estimate between $a+y$ and $a+y'$.
5. Its fixed point $y$ (Banach's theorem on the complete $B_a$) gives $u = a + y$ with $(1-p)(t(u) - u) = 0$, i.e. $w - u \in N$ for $w = t(u)$. Now $w \in O$, $|w|_1 < r$ by the choice of $\delta$, $t(w) = w$, and $Dt(0)(w-u) = w - u$. The estimate between $u$ and $w$ reads $|t(w) - t(u) - Dt(0)(w-u)|_1 = |w-u|_1 \le \theta|w-u|_1$, so $w = u$: $u = t(u) \in O$ — the text's argument "$a_1 = a$".

So $u \in O_1$, $p(u) = a$, $u - a \in Y$, $|u-a|_1 \le \varepsilon'|a|_1$ and $|u|_1 < r$. The text's "$|a(\tau)|_1 \le \varepsilon_1$ if $|a|_0 < \varepsilon$" is the level-1 continuity of $t$ at 0 used in step 5.

## IV.8 The step $p(O \cap W) \subseteq C$, and the neighbourhoods

**$p$ maps points of $O$ near 0 into $C$** `Chapter4/ScPlusGraph · GraphData.p_mem_C`, which the conclusion "$p : U_0 \to V \subseteq T_0O \cap C$" needs and the text never checks. If $u \in O_1$ has $|(1-p)u|_1 \le \kappa|u|_1$ with $\kappa \le \frac{1}{2}$, then $|u|_1 \le |pu|_1 + \frac{1}{2}|u|_1$, so $|u|_1 \le 2|pu|_1 \le 2\|\mathrm{lift}_N\||pu|_0$, and $|(1-p)u|_0 \le \|\iota_1\|\kappa|u|_1 \le 2\|\iota_1\|\kappa\|\mathrm{lift}_N\||pu|_0$; if that constant is at most $\varepsilon$, the good complement property gives $p(u) \in C \iff p(u) + (1-p)u = u \in C$, and $u \in O \subseteq C$. The hypothesis on $u$ is the contraction estimate between 0 and $u$.

**The neighbourhoods** `Chapter4/ScPlusGraph · GraphData.Nbhd` `Chapter4/ScPlusGraph · GraphData.exists_nbhd`. For every $\theta_0 > 0$ the proof chooses, in this order:

1. $\theta = \min(\theta_0, \theta_1, \theta_2)$ with $\theta_1 = \varepsilon'/(2(\|1-p\|+1)(1+\varepsilon'))$ (so that $\|1-p\|\theta \le \frac{1}{2}$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$, for Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12)) and $\theta_2 = \varepsilon/(2\|\iota_1\|\|\mathrm{lift}_N\|(\|1-p\|+1)+1)$ (for $p(O \cap W) \subseteq C$);

2. $r$ from the contraction estimate on level 1, and $\rho$ and the graph points $u(a)$ from Lemma 4.1.13 (preprint 4.12);
3. $\delta_0$ from the continuity of the lift $t^1 : U \to E_1$ at 0 on level 0: $|t(u)|_1 < r$ for $u \in U$, $|u|_0 < \delta_0$;
4. $\rho' = \min(\rho, r/(1+\varepsilon'), \delta_0/(\|\iota_1\|(1+\varepsilon')+1))$, the open $V' = \{a \in N : |a|_1 < \rho'\}$ of $N$, $V = V' \cap C$, and $W = B_0(0,\delta_0) \cap p^{-1}(V')$, open in $E_0$;
5. $\Gamma(a) = u(a)$ for $a \in V$.

Then $\Gamma(a) \in O \cap W$ (on level 0, $|\Gamma(a)|_0 \le \|\iota_1\|(1+\varepsilon')|a|_1 < \delta_0$), $p\Gamma(a) = a$, $\Gamma(a) - a \in Y$, $|\Gamma(a)|_1 < r$. A point $u_0 \in O \cap W$ satisfies $u_0 = t(u_0)$, hence $|u_0|_1 < r$ by step 3 — *this is how a neighbourhood open on level* 0 *is controlled on level* 1, the sc⁺ property a second time; the contraction estimate between 0 and $u_0$ and the first constraint on $\theta$ then give $p(u_0) \in C$, so $p(u_0) \in V$, and uniqueness gives $\Gamma(p(u_0)) = u_0$. Also $\|Dt(z) - Dt(0)\|_{\mathcal{L}(E_1)} \le \theta_0$ on $C_1 \cap B(0,r)$, and $|(1-p)(u'-u)|_1 \le \frac{1}{2}|u'-u|_1$ on $O_1 \cap B_1(0,r)$. Consequently $\Gamma(V) = O \cap W$ Chapter4/ScPlusGraph · GraphData.Nbhd.image_V, $\Gamma(0) = 0$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_zero, and $p$ is injective on $O \cap W$ Chapter4/ScPlusGraph · GraphData.Nbhd.injOn_p.

## IV.9 The regularity of $\Gamma$ on the fixed $V$: the frame formula and the bootstrap

**Continuity into every level.** $\Gamma(V) \subseteq O \subseteq E_\infty$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_mem_smooth, so $\Gamma$ has lifts $\Gamma_m : V \to E_m$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl with values in $O_m$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_mem_retract. $\Gamma_1$ is Lipschitz: with $u = \Gamma(a)$, $u' = \Gamma(a')$, $|u'-u|_1 \le |p(u'-u)|_1 + |(1-p)(u'-u)|_1 \le \|\mathrm{lift}_N\||a'-a| + \frac{1}{2}|u'-u|_1$, so $|u'-u|_1 \le 2\|\mathrm{lift}_N\||a'-a|$ Chapter4/ScPlusGraph · GraphData.Nbhd.norm_Γ_lvl_one_sub_le Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl_one. And $\Gamma_{k+1} = t^1 \circ \Gamma_k$ on level $k$, because $\Gamma(a) = t(\Gamma(a))$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_succ; the lift $t^1$ is continuous from $U_k$ into $E_{k+1}$, so by induction $\Gamma$ is continuous into every level Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ.

**The frame at a point $u \in O$ near 0.** Let $A(u) = p \circ Dt(u)|_N : N \to N$ Chapter4/ScPlusGraph · GraphData.A, $M(u) = A(u)^{-1}$ Chapter4/ScPlusGraph · GraphData.M and $B_u = Dt(u) \circ M(u)$, the *frame* Chapter4/ScPlusGraph · GraphData.frame, with its level-$(k+1)$ version Chapter4/ScPlusGraph · GraphData.frameL. For $n \in N$, $Dt(u)n - n = \iota_1(Dt_1(u) - Dt_1(0))\mathrm{lift}(n)$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_eq. Choose $\theta_0$ with $\|p\|\|\iota_1\|\|\mathrm{lift}_N\|\theta_0 \le \frac{1}{2}$ and $\|1-p\|\theta_0 < 1$ Chapter4/ScPlusGraph · GraphData.FrameSmall Chapter4/ScPlusGraph · GraphData.exists_frameSmall, and let $\|Dt_1(u) - Dt_1(0)\| \le \theta_0$. Then (L1) $|A(u)n - n| \le \frac{1}{2}|n|$, so $A(u)$ is injective Chapter4/ScPlusGraph · GraphData.A_injective, hence an isomorphism of the finite-dimensional $N$ Chapter4/ScPlusGraph · GraphData.Aeq, with $A(u)M(u) = 1$ Chapter4/ScPlusGraph · GraphData.A_M and $p \circ B_u = 1$ Chapter4/ScPlusGraph · GraphData.p_frame. (L2) $T_uO \cap Y = 0$: a vector $v = Dt(u)v$ in $Y$ satisfies $v = (1-p)v = (1-p)(Dt(u) - Dt(0))v$ on level 1, so $|v|_1 \le \|1-p\|\theta_0|v|_1$ Chapter4/ScPlusGraph · GraphData.eq_zero_of_mem_Y_of_scDeriv_eq. Hence for $u \in O_1$: $Dt(u)x - B_u(p\,Dt(u)x)$ lies in $T_uO \cap Y$ and vanishes, $(1 - B_up)Dt(u) = 0$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_frame; $Dt(u)B_u = B_u$ Chapter4/ScPlusGraph · GraphData.scDeriv_frame; and both hold on every level Chapter4/ScPlusGraph · GraphData.p_toBase_frameL Chapter4/ScPlusGraph · GraphData.lvlDeriv_sub_frameL. On the points of $\Gamma(V)$ the derivative bound holds by the choice of $r$ Chapter4/ScPlusGraph · GraphData.Nbhd.deriv_bound_Γ.

**The abstract step.** On a set $S$ of a normed space, let $s$ satisfy the mean value estimate with derivative $D_0$ at $u_0$ and constant $\theta$, let $q$, $B$ be continuous linear with $q \circ B = 1$ and $(1-Bq)D_0 = 0$, and $\theta\|1-Bq\| \le \frac{1}{2}$. For fixed points $u, u'$ of $s$ in $S$, $(1-Bq)(u'-u) = (1-Bq)[s(u') - s(u) - D_0(u'-u)]$, of norm at

most $\frac{1}{2}|u'-u|$, so $|u'-u| \le 2\|B\||q(u'-u)|$ Chapter4/ScPlusGraph · norm_sub_le_of_projection. And *every continuous right inverse $\Gamma$ of $q$ with values in the fixed points of $s$ in $S$, through $u_0$, is differentiable at $z_0 = q(u_0)$ with derivative $B$* Chapter4/ScPlusGraph · hasFDerivWithinAt_of_projection: $\Gamma(z) - \Gamma(z_0) - B(z-z_0) = (1 - Bq)(\Gamma(z) - u_0) = (1 - Bq)[(\Gamma(z) - u_0) - D_0(\Gamma(z) - u_0)]$, which is $o(|\Gamma(z) - u_0|)$ by the differentiability of $s$ at $u_0$, and $|\Gamma(z) - u_0| \le 2\|B\||z - z_0|$ by the Lipschitz estimate.

**The frame formula** Chapter4/ScPlusGraph · GraphData.Nbhd.hasFDerivWithinAt_Γ_lvl. At $a_0 \in V$ and on level $k+1$, apply the abstract step with $s = t$, $u_0 = \Gamma_{k+1}(a_0)$, $q = p \circ \iota_{k+1}$, $B$ the level-$(k+1)$ frame at $u_0$, $\theta = 1/(2(\|1 - Bq\| + 1))$, and $S = C_{k+1} \cap B(u_0, r)$ from the mean value estimate *at* $u_0$; $\Gamma_{k+1}$ stays in $B(u_0, r)$ near $a_0$ by continuity. So

$$D\Gamma(a) = Dt(\Gamma(a))|_N \circ \left(p \circ Dt(\Gamma(a))|_N\right)^{-1} \quad \text{into every level } E_{k+1}.$$

The estimate is used at $\Gamma(a)$ with the projection $1 - B_{\Gamma(a)}p$, not at 0 with $1 - p$: this is why the regularity holds at every point of the fixed $V$.

**The $C^k$-bootstrap.** By Proposition 1.2.3 (preprint 1.18) for the tangent map of the lift $t^1$, $(x, w) \mapsto Dt_{j+1}(x)w$ is $C^k$ on $U_{j+k+1} \times E_{j+k+1}$, read through the inclusions into level $j+1$ Chapter4/ScPlusGraph · GraphData.contDiffOn_lvlDeriv_apply. If $\Gamma_{j+k+1}$ and $w : V \to N$ are $C^k$, then so is $a \mapsto Dt_j(\Gamma_{j+1}(a))\,\mathrm{lift}(w(a))$ Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_lvlDeriv_Γ; in particular $a \mapsto A(\Gamma(a))$ is $C^k$ when $\Gamma_{k+1}$ is Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_A_Γ, and so is its inverse, inversion being smooth on the invertible operators Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_M_Γ. Now induct on $k$, for all levels at once Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl: $\Gamma_{j+1}$ is $C^0$ (continuity); if every $\Gamma_{j'+1}$ is $C^k$, the derivative $a \mapsto \mathrm{frame}_j(\Gamma_{j+1}(a))$ applied to $n$ is $Dt_j(\Gamma_{j+1}(a))\mathrm{lift}(M(\Gamma_1(a))n)$, which is $C^k$ because $\Gamma_{j+k+1}$ is (the hypothesis at $j' = j+k$) and $M(\Gamma_1(\cdot))n$ is (the hypothesis at $j' = k$, through $A$); so $\Gamma_{j+1}$ is $C^{k+1}$. Derivatives of $t$ are only ever applied to the smooth vectors of the finite-dimensional $N$, so the loss of levels in $Tt$ is absorbed by $\Gamma$ being $C^k$ into *every* level and does not accumulate. Hence $\Gamma$ is $C^\infty$ into every level Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_succ_infty Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_infty and sc-smooth on $V$ by the criterion of Proposition 1.2.4 (preprint 1.19) for a finite-dimensional source Chapter4/ScPlusGraph · GraphData.Nbhd.isScSmooth_Γ.

**The sc-diffeomorphism.** $(O \cap W, C, E)$ is a retract (Proposition 2.3.1 (1) (preprint 2.10)) Chapter2/Retracts · proposition_2_3_1_one Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_inter_W; $(V, N \cap C, N)$ is a local model, $N \cap C$ being a partial quadrant of $N$ (Proposition 3.1.25 (preprint 3.22)) and $V$ relatively open in it Chapter4/ScPlusGraph · GraphData.Nbhd.V_isRelOpenInQuadrant Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_V. $p$ is sc-smooth $E \to N$ as a continuous linear map into a finite-dimensional space (Corollary 1.2.5 (preprint 1.20)) Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional Chapter4/ScPlusGraph · GraphData.p_isScSmooth, so $p$ and $\Gamma$ are sc-smooth between the retracts Chapter4/ScPlusGraph · GraphData.Nbhd.p_isScSmoothRetractMap Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_isScSmoothRetractMap.

**Proposition 4.1.9 (preprint 4.9)** Chapter4/ScPlusGraph · proposition_4_1_9. *Statement.* Let $t$ be an sc$^+$-retraction onto $O \ni 0$, $N = T_0O$ in good position to $C$, $Y$ a closed good complement of $N$ and $p : E_0 \to N$ the projection along $Y$. There are an open $W \ni 0$ in $E_0$, a relatively open $V \ni 0$ of $N \cap C$ and $\Gamma : N \to E_0$, sc-smooth on $V$, with $p \circ \Gamma = 1$ on $V$, $\Gamma(a) - a \in Y$, $\Gamma(V) = O \cap W$, $\Gamma \circ p = 1$ on $O \cap W$, $\Gamma(0) = 0$, and $p$, $\Gamma$ sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *Proof.* $\theta_0$ from the frame condition, the neighbourhoods of §IV.8 for it, and the regularity above. The finite dimension of $N$, an instance argument needed to form $N$ as an sc-Banach space, is Lemma 4.1.7 (preprint 4.7) Chapter4/

ScPlusGraph · finiteDimensional_tangentSpace_zero.

## IV.10 Theorem 4.1.2 (preprint 4.2): the characterization

**Statement** Chapter4/Characterization · MPolyfold.theorem_4_1_2_general. For every M-polyfold $X$ and every junk value $x_0 \in X_1$ of $1^1_0$, the following are equivalent. (i) $X$ has a corner structure. (ii) Every point of $X$ is smooth, $1^1_0 : X \to X^1$ is sc-smooth, and every $x$ lies in a chart $\psi$ of $X$ with $\psi(x) = 0$ whose model is a tame retract and whose tangent space $N = T_0O$ is finite-dimensional and in good position to $C$. The text's form, for tame $X$, is the same statement with an unused hypothesis Chapter4/Characterization · MPolyfold.theorem_4_1_2. Tameness is used neither in the text's proof nor here, and (i) implies it (§IV.1).

**Charts composed with an sc-diffeomorphism between retracts.** The proof of (ii) □ (i) composes a chart with $p$, which is an sc-diffeomorphism between retracts, not between relatively open sets, so the construction of Chapter 3 does not apply. For a chart $\varphi$ onto $(O, C, E)$ and mutually inverse $f$, $g$ sc-smooth between $(O, C, E)$ and $(O', C', E')$, $f \circ \varphi$ is a chart onto $(O', C', E')$ Chapter4/Characterization · MChart.compRetract, and a chart of the M-polyfold when $\varphi$ is Chapter4/Characterization · MChart.compRetract_isChart: for a chart $\varphi'$ of $X$ the transition maps are $(\varphi' \circ \varphi^{-1}) \circ g$ and $f \circ (\varphi \circ \varphi'^{-1})$, compositions of maps sc-smooth between retracts (Theorem 2.1.15 (preprint 2.8)) Chapter2/Retracts · IsScSmoothRetractMap.comp. This is the compatibility the text's "we obtain a smoothly compatible atlas" skips (Findings, item 55).

**(i) □ (ii).** Corner charts are modelled on sc⁺-retracts, so every point is smooth and $1^1_0$ is sc-smooth by Proposition 4.1.8 (preprint 4.8) on a cover (§IV.4). The text stops here; (ii) also asks for a centred tame chart with $T_0O$ finite-dimensional and in good position (Findings, item 53). Let $x$ lie in a corner chart $\chi$. Its image is smooth (§IV.1), so the re-centred chart $\psi = \chi - \chi(x)$ exists; its model is a relatively open subset of $C_{\chi(x)}$ in the same finite-dimensional space Chapter4/Characterization · image_sub_const_eq_preimage_add' Chapter4/Characterization · MChart.IsCornerChart.exists_centred. With the identity retraction it is a tame retract; the tangent space of a relatively open subset of a partial quadrant is the whole space, because the derivative of the identity retraction is the identity Chapter4/Characterization · tangentSpace_relOpen; and the whole space is in good position to every partial quadrant, with the complement $0$, the interior of a partial quadrant being nonempty Chapter4/Characterization · isInGoodPosition_top Chapter4/Characterization · MChart.IsCornerChart.exists_centred_tame.

**(ii) □ (i).** *The data of the corner chart* Chapter4/Characterization · CornerChartData for a chart $\psi$ and a finite-dimensional $N$: a closed good complement $Y$, the projection $p$ along it, $W$, $V$, $\Gamma$ with $p$, $\Gamma$ mutually inverse and sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *They exist* when the model of $\psi$ is an sc⁺-retract containing $0$ and $N = T_0O$ is in good position Chapter4/Characterization · exists_cornerChartData: here no complement is prescribed, so one can be chosen — the normal form of Proposition 3.1.25 (preprint 3.22) gives a continuous projection onto $N$ whose kernel is a good complement, closed, even mapping $C$ into $C$ Chapter3/GoodComplementProjection · exists_proj_goodComplement — and Proposition 4.1.9 (preprint 4.9) is applied to an sc⁺-retraction onto $O$ (Definition 4.1.6 (preprint 4.6)). The issue "$p = Dt(0)$" of §IV.6 does not arise, and nothing is added to Theorem 4.1.2 (preprint 4.2) (Findings, item 54). *The chart* $p \circ \psi$ on $\psi^{-1}(W)$ Chapter4/Characterization · CornerChartData.chart is $\psi$ restricted to $\psi^{-1}(W)$ composed with $p$ by `compRetract`, modelled on $(V, N \cap C, N)$ Chapter4/Characterization · CornerChartData.model; it is a corner chart Chapter4/Characterization · CornerChartData.chart_isCornerChart, a chart of $X$ Chapter4/Characterization · CornerChartData.chart_isChart, contains $x$ when $\psi(x) = 0$ Chapter4/Characterization · CornerChartData.mem_chart_source and sends it to $0$ Chapter4/

Characterization · CornerChartData.chart_apply_eq_zero. *At every point* Chapter4/Characterization · MPolyfold.theorem_4_1_2_chart: if all points are smooth and $1^1_0$ is sc-smooth, the model of every chart is an sc⁺-retract (§IV.4), so for a chart $\psi$ with $\psi(x) = 0$ and $N = T_0O$ finite-dimensional and in good position the corner chart $p \circ \psi$ exists. With such a chart at every point, $X$ has a corner structure by the local form of Definition 4.1.1 (preprint 4.1) Chapter4/Characterization · MPolyfold.hasCornerStructure_of_centred — whether or not the charts $\psi$ are tame.

## IV.11 Smooth finite-dimensional submanifolds; the chart retraction; Remark 4.2.2 (preprint 4.14)

**Definition 4.2.1 (preprint 4.13).** $A \subseteq X$ is a *smooth finite-dimensional submanifold* if every $a \in A$ has an open $V \ni a$ and an sc⁺-retraction $s$ on $V$ with $s(V) = A \cap V$ Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold; the data at a point are a structure Chapter4/Submanifolds · MPolyfold.SubmanifoldData Chapter4/Submanifolds · MPolyfold.isSmoothSubmanifold_iff. An sc⁺-retraction is an sc-smooth retraction, so the data are those of Definition 2.3.18 (preprint 2.20) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.toSubData, $A$ is a sub-M-polyfold Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.isSubMPolyfold and carries the induced structure of Proposition 2.3.19 (preprint 2.21) Chapter2/SubMPolyfold · MPolyfold.subPolyfold. Its points lie on level 1, as $a = s(a) \in X_1$ Chapter4/Submanifolds · MPolyfold.SubmanifoldData.mem_levelSet_one.

**The sc⁺ condition read in a chart** (Lemma 6.A of the exposition). The text never reads the sc⁺ condition in charts, and all of §§4.2–4.3 rests on it (Findings, item 60). Let $\varphi$ be a chart of $X$ and $W = V \cap \operatorname{dom}\varphi \cap s^{-1}(V \cap \operatorname{dom}\varphi)$, the open set of the proof of Proposition 2.3.19 (preprint 2.21) Chapter2/SubMPolyfold · MPolyfold.SubData.shrink. (a) The lift of $\varphi \circ s \circ \varphi^{-1}$ into $E^1$ is sc-smooth from $(\varphi(W), C, E)$ to $(O_1, C_1, E^1)$ Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScSmoothRetractMap_lift: it is the sc-smoothness of $1^1_0 \circ s$ read in the charts $\varphi$ of $X$ and `levelChart` $\varphi$ of $X^1$, where $1^1_0$ is the lift. (b) For every retraction $r$ onto $O$: $\varphi(W) = O \cap G$ for an open $G$; $\rho = \varphi \circ s \circ \varphi^{-1}$ is sc-smooth on the retract $\varphi(W)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth and idempotent there Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem, with image $\varphi(W \cap A)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image; so $t = \rho \circ r$ on $U \cap r^{-1}(G)$ — the restriction of $r$ Chapter2/Retracts · ScRetraction.restrict composed with $\rho$ Chapter2/Metrizable · ScRetraction.compRetraction — is a retraction onto $\varphi(W \cap A)$, and it is sc⁺: its values $\varphi(s(x))$ lie in $E_1$ because $s(x) \in X_1$, and its lift is the map of (a) composed with $r$ Chapter4/Submanifolds · MPolyfold.SubmanifoldData.exists_chartRetraction. This is the text's $t = \varphi \circ s \circ \varphi^{-1} \circ r$ of the proof of Theorem 4.2.7 (preprint 4.18), with $W$ in place of "$V = \varphi^{-1}(O)$". Hence the model $\varphi(W \cap A)$ of the chart of $A$ induced by $\varphi$ is an sc⁺-retract Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScPlusRetract_image, and $\varphi(a)$ is smooth for $a \in A \cap V \cap \operatorname{dom}\varphi$ (Lemma 4.1.7 (i) (preprint 4.7)) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.apply_mem_smooth.

**Tangent spaces of sub-retracts.** The tangent space of a retract is a germ: for an open $K$, $T_z(Q \cap K) = T_zQ$ Chapter4/Submanifolds · tangentSpace_inter_isOpen. For retracts $Q \subseteq O = r(U)$, $T_zQ \subseteq T_zO$: a retraction $s$ onto $Q$ satisfies $s = r \circ s$, so $Ds(z) = Dr(z)Ds(z)$ Chapter4/Submanifolds · tangentSpace_subset_of_subset. And if $F \circ r$ is sc-smooth and $F$ maps the sub-retract $Q$ into a retract $Q'$, then $D(F \circ r)(z)$ maps $T_zQ$ into $T_{F(z)}Q'$ (Proposition 2.1.13 (3) (preprint 2.6) for $F|_Q$) Chapter4/Submanifolds · scDeriv_mem_tangentSpace_of_subset.

**Remark 4.2.2 (preprint 4.14) in a chart** Chapter4/Submanifolds · MPolyfold.SubmanifoldData.remark_4_2_2. The text: $T_aA = Ts(a)(T_aX)$ is the image of the sc⁺-operator $Ts(a)$, hence finite-dimensional. In a chart, at $a \in A \cap V \cap \operatorname{dom}\varphi$ and the level-1 point $z$ over $\varphi(a)$: $z$ is smooth (Lemma 4.1.7 (i) (preprint 4.7) for the

chart retraction $t$); the tangent map of $\rho$ at $z$ maps $T_zO = T_z(\varphi(W))$ onto $T_z(\varphi(W \cap A))$ (Proposition 2.3.19 (4) (preprint 2.21) for $\rho$) `Chapter2/Metrizable · proposition_2_3_19_four`; $T_z(\varphi(W \cap A)) \subseteq T_zO$; and it is the range of $Dt(z)$, finite-dimensional by Lemma 4.1.7 (ii) (preprint 4.7). The sc⁺-operator is $Dt(z)$, not $Ts(a)$.

**Points with $d_X = 0$.** The remark adds that near points with $d_X(a) = 0$, $A$ is a classical manifold. Every subspace is in good position to the whole space, any algebraic complement being good `Chapter4/Submanifolds · isInGoodPosition_univ`, and at a point with $d_C(e) = 0$ the partial quadrant $C_e$ is the whole space `Chapter4/Submanifolds · partialQuadrantAtWith_eq_univ_of_degIndex_eq_zero`. So at $a \in A$ with $d_X(a) = 0$, in a chart realizing $d_X(a) = 0$ `Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq`, $A$ is in good position (Definition 4.2.5 (preprint 4.16), §IV.13) `Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_of_degIndex_eq_zero`; and if $d_X = 0$ on all of $A$, re-centring at each point gives charts modelled in $C_{\varphi(a)} = E$, in which $T_0(\psi(W \cap A))$ is in good position, so the induced structure has a corner structure by the core of Theorem 4.2.7 (preprint 4.18) (§IV.14) — without tameness of $X$ `Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_degIndex_eq_zero`. The local statement for the open piece $A \cap \{d_X = 0\}$ of an arbitrary $A$, and the 2021 wording with $d_A$ in place of $d_X$, are §IV.31.

**The open question** after Proposition 4.2.3 (preprint 4.15), Question 4.2.4 in the 2021 edition — is a sub-M-polyfold with $A = A_\infty$ and finite, locally constant $\dim T_aA$ a smooth finite-dimensional submanifold; is the "T" $\{(x,0) : |x| < 1\} \cup \{(0,y) : 0 \le y < 1\}$ homeomorphic to a retract $O = O_\infty$ — is a question, and nothing in it is to be formalized (gap G.34). Its last sentence, that this is impossible for sc⁺-retracts, is proved in §IV.31: an sc⁺-retract without boundary is locally a graph over its tangent space, and the "T" is not a topological manifold at its branch point.

## IV.12 Proposition 4.2.3 (preprint 4.15)

**(1)** $A \subseteq X_\infty$ `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.subset_smoothSet`: in a chart at $a$, $\varphi(a)$ lies in the retract of the sc⁺ chart retraction (Lemma 4.1.7 (i) (preprint 4.7)). The text does not distinguish this from "$A = A_\infty$ for the induced structure", which holds because the levels of the induced structure are the traces of those of $X$, computed in the induced charts `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart` `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.mem_levelSet_subPolyfold_iff` `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_one`.

**(2)** *$A$ is locally closed* `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_two`. The text's definition is garbled — "a point $b \in V(a)$ belongs to $A$ if $U \cap A \neq \emptyset$ for all open neighbourhoods $U$ of $b$", with $U$ also the letter of the domain — and is read as $V(a) \cap \overline{A} \subseteq A$, one of the equivalent forms of Mathlib's `IsLocallyClosed` (the intersection of an open and a closed set). Proof: for $b \in V \cap \overline{A}$, $b$ is in the closure of $A \cap V$ ($V$ open), so the filter of neighbourhoods of $b$ within $A \cap V$ is nontrivial; $s$ is continuous at $b$ and equal to the identity on $A \cap V$, so $s(b)$ and $b$ are both limits of the identity along that filter, and $s(b) = b$ ($X$ Hausdorff); hence $b \in s(V) \subseteq A$. A filter replaces the text's sequence $a_k \to b$, so no metrizability is needed.

**(3)** $\partial A = \{d_A \ge 1\} \subseteq \partial X$ `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_three`, by $d_A \le d_X$ (Lemma 2.4.6 (preprint 2.26)) `Chapter2/SubMPolyfold · lemma_2_4_6`. The text's argument for $d_X(a) = 0$ is the proof of that lemma; it needs the chart re-centred at $a$, since only $C_{\varphi(a)}$, not $C$, is the whole space.

**(4)** $1^1_0 : A \to A^1$ *is sc-smooth* `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_four`. The

text says only that "the postulated sc⁺-retraction is the identity on the retract $A$". The proof: every point of $A$ lies in an induced chart `Chapter2/SubMPolyfold · MPolyfold.subChart` whose model $\varphi(W \cap A)$ is an sc⁺-retract (§IV.11), and Proposition 4.1.8 (preprint 4.8) on a cover applies (§IV.4).

## IV.13 Good and general position in charts, and their chart independence (Definitions 4.2.5, 4.2.6, 4.2.10 (preprint 4.16, 4.17, 4.21))

**Definition 4.2.5 (preprint 4.16) in a chart.** For retracts $Q \subseteq O$, a presentation $T$ and a level-1 point $z$: $T_zQ$ is *in good position to $C_z$ inside $T_zO$* `Chapter4/Submanifolds · IsGoodPositionRetractAt` — in the sense of Definition 3.1.24 (preprint 3.21) inside a subspace `Chapter3/GoodPositionGerm · IsInGoodPositionIn`: $T_zQ \subseteq T_zO$, $T_zQ \cap C_z$ has interior points in $T_zQ$, and $T_zQ$ has a complement inside $T_zO$ with a constant $\varepsilon$. The condition depends only on the two tangent spaces `Chapter4/Submanifolds · IsGoodPositionRetractAt.congr`. For the data $d$ of Definition 4.2.1 (preprint 4.13), a chart $\varphi$ and $Q = \varphi(W \cap A)$ it is the chart condition `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGoodPositionChart`, and $A$ *is in good position at $a$* if it holds for some data at $a$, some chart at $a$, some presentation and the level-1 point over $\varphi(a)$ `Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt`. Explicitly, for retractions $r$ onto $O$ and $t$ onto $\varphi(W \cap A)$, the condition says that $Dt(z)E$ is in good position to $C_z$ inside $Dr(z)E$ `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isGoodPositionChart_iff`. The text states Definition 4.2.5 (preprint 4.16) for tame $X$ so that $C_aX$ is a partial quadrant (Proposition 2.5.6 (preprint 2.36)); the chart form does not need it.

**Definition 4.2.10 (preprint 4.21) in a chart.** *$T_zQ$ has in $T_zO$ an sc-complement contained in $T^R_zO$* `Chapter4/Submanifolds · IsGeneralPositionRetractAt`: an sc-subspace $B \subseteq T_zO \cap E_z$ of $E$ with $(T_zO)_m = (T_zQ)_m \oplus B_m$ on every level, the sc-structure of $T_zO$ being the one induced from $E$; again it depends only on the tangent spaces `Chapter4/Submanifolds · IsGeneralPositionRetractAt.congr`. The chart condition `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGeneralPositionChart` and *general position at $a$* `Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt` are defined as for good position. **Definition 4.2.6 (preprint 4.17)**: $A$ is *tame* if its induced structure is `Chapter4/Submanifolds · MPolyfold.IsTameSubmanifold`.

**Transport by a germ of sc-diffeomorphisms.** Let $F$ be a germ of sc-diffeomorphisms between retracts $O = r(U)$ and $O' = r'(U')$ with inverse $G$ `Chapter2/Boundary · IsDiffeoGerm`, mapping sub-retracts $Q$, $Q'$ into each other. At $z \in Q_1$, $\mathcal{A} = D(F \circ r)(z)$ restricts to an isomorphism $T_zO \cong T_{F(z)}O'$ with inverse $D(G \circ r')(F(z))$, carrying the partial cones onto each other (Proposition 2.4.10 (preprint 2.30)) `Chapter2/Boundary · proposition_2_4_10_cone` and $T_zQ$ onto $T_{F(z)}Q'$ (§IV.11 in both directions). Good position inside a subspace is transported by such an isomorphism (Lemma 3.5.9 (preprint 3.45)) `Chapter3/LinearizationCharts · IsInGoodPositionIn.transport`, so *good position of $Q$ at $z$ gives good position of $Q'$ at $F(z)$* `Chapter4/Submanifolds · IsGoodPositionRetractAt.transport`. For general position $z$ must be smooth: then $\mathcal{A}$ is an sc-operator, it carries $T^R_zO$ onto $T^R_{F(z)}O'$ (Proposition 2.4.10 (preprint 2.30)) `Chapter2/Boundary · proposition_2_4_10_reduced`, and the image of the sc-complement $B$ is an sc-complement inside $T^R_{F(z)}O'$, level by level `Chapter4/Submanifolds · IsGeneralPositionRetractAt.transport`.

**The chart change** `Chapter4/Submanifolds · MPolyfold.chart_change_of_transport`. Let $\Phi$ be a property of $(Q \subseteq O, T, z)$ that depends only on the tangent spaces and is carried by germs of sc-diffeomorphisms at smooth points. If $\Phi$ holds for $\varphi(G \cap A) \subseteq O$ at $z$ (smooth), it holds for $\varphi'(G' \cap A) \subseteq O'$ at $z'$, for any presentations: on $H = G \cap G'$ the transition $\varphi' \circ \varphi^{-1}$ is a germ of sc-diffeomorphisms between the overlap retracts `Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm` carrying $\varphi(H \cap A)$ onto $\varphi'(H \cap A)$ `Chapter4/Submanifolds · MChart.image_inter_eq_inter`, and the tangent spaces at $z$ are those of the open pieces (the germ property). Applied to the two properties `Chapter4/Submanifolds · MPolyfold.isGoodPositionRetractAt_chart_`

change Chapter4/Submanifolds · MPolyfold.isGeneralPositionRetractAt_chart_change, with $G$, $G'$ the open sets of two submanifold charts: *if $A$ is in good position at $a$, the chart condition holds for every data, chart, presentation and lift* Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isGoodPositionChart, and good position is equivalent to the condition for all of them Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_iff_forall; likewise for general position Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart Chapter4/Submanifolds · MPolyfold.isGeneralPositionAt_iff_forall, the smoothness of the point holding on $A$. The text takes this independence for granted (Findings, item 66).

## IV.14 Lemma 4.2.8 (preprint 4.19) and Theorem 4.2.7 (preprint 4.18)

**Lemma 4.2.8 (preprint 4.19), the complement kept explicit** Chapter4/Submanifolds · isGoodComplement_sup_range. Let $\pi$ be a bounded idempotent with $(1-\pi)E_0$ inside the minimal subspace at 0, $E_0^{\min} = \{\lambda = 0\}$, and $Y \subseteq \pi E_0$ a complement of $N$ inside $\pi E_0$ with constant $\varepsilon$. Then $Y \oplus (1-\pi)E_0$ is a good complement of $N$ in $E_0$ with constant $\varepsilon/(\|\pi\|+1)$: for $y \in Y$, $k \in (1-\pi)E_0$ with $|y+k| \le \frac{\varepsilon}{\|\pi\|+1}|n|$, $y = \pi(y+k)$, so $|y| \le \varepsilon|n|$ and $n \in C \iff n+y \in C$; and adding $k \in E_0^{\min}$ changes no coordinate, so $n+y \in C \iff n+y+k \in C$. The splitting: $N \oplus Y = \pi E_0$ and $\pi E_0 \oplus (1-\pi)E_0 = E_0$.
**Lemma 4.2.8 (preprint 4.19)** Chapter4/Submanifolds · lemma_4_2_8: *in the situation of the lemma, if $N$ is in good position to $C$ inside $\pi E_0$, then $N$ is in good position to $C$ in $E_0$, with good complements $Y \oplus (1-\pi)E_0$.* The good-position half is the transport lemma of the proof of Theorem 3.6.8 (preprint 3.54) Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition (§III.23).

*One step of the text's proof does not hold as we read it.* It concludes $n+y \in C$ from $n+y+z \in C$ "since $z \in Z \subset C$". This needs $-z \in C$: in $C = [0,\infty)$, $z = 1$ and $n+y = -\frac{1}{2}$ give $n+y+z \in C$, $n+y \notin C$. What holds, and what is used, is $Z \subseteq E_0^{\min} \subseteq C \cap (-C)$ — Proposition 2.5.3 (preprint 2.35) at the corner of the centred chart. The text's norm $|y|_0 + |z|_0$ on $Y \oplus Z$ is not the norm of $E$; the constant $\varepsilon/(\|\pi\|+1)$ compares them (Findings, item 69).

**Good position read in a tame centred chart** Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isInGoodPosition_of_tame. If $A$ is in good position at $a$ and $\psi$ is a chart of $X$ with $\psi(a) = 0$ modelled on a tame retract $O = r(U)$, then $N = T_0(\psi(W \cap A))$ is in good position to $C$ in $E$: by chart independence the chart condition holds in $\psi$, i.e. $N$ is in good position to $C_0$ inside $T_0O = \operatorname{range} Dr(0)$; at the corner of the centred chart $C_0 = C$ Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero; Proposition 2.5.3 (preprint 2.35) for the tame $r$ at the smooth point 0 gives $(1 - Dr(0))E_0 \subseteq E_0^{\min}$ Chapter2/TameRetracts · proposition_2_5_3; and $Dr(0)$ is idempotent. Lemma 4.2.8 (preprint 4.19) with $\pi = Dr(0)$ concludes. Tameness of $X$ enters Theorem 4.2.7 (preprint 4.18) here and nowhere else.

**The corner charts of $A$** Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_centred_isInGoodPosition. If every $a \in A$ lies in a chart $\psi$ of $X$ with $\psi(a) = 0$ in which $T_0(\psi(W \cap A))$ is in good position to $C$ in $E$, the induced structure has a corner structure. Proof (for nonempty $A$): every point of $A$ is smooth and $1_0^1 : A \to A^1$ is sc-smooth (Proposition 4.2.3 (4) (preprint 4.15)); the chart of $A$ induced by $\psi$ is a chart of the induced structure, centred at $a$, modelled on $\psi(W \cap A)$; its tangent space at 0 is the range of $Dt(0)$ for the sc⁺ chart retraction, finite-dimensional (Lemma 4.1.7 (preprint 4.7)), and in good position by hypothesis; Theorem 4.1.2 (ii) (preprint 4.2) □ (i) for the M-polyfold $A$ (§IV.10) concludes.

**Theorem 4.2.7 (preprint 4.18)** Chapter4/Submanifolds · MPolyfold.theorem_4_2_7. *Statement.* Let $X$ be tame and $A$ a smooth finite-dimensional submanifold in good position at every point. Then the induced structure on $A$ has a corner structure, and $A$ is tame. *Proof* (the text's route through Theorem 4.1.2 (preprint 4.2)): at $a \in A$ take a tame chart centred at $a$ (§IV.5) — the "sc-smooth tame chart $\varphi : (V, a) \to (O, 0)$" the text

takes for granted; good position read in it (above) gives the hypothesis of the previous paragraph; and a corner structure makes $A$ tame (§IV.1). What the text skips, besides the step of Lemma 4.2.8 (preprint 4.19) repaired above: the centred chart; the set $W$ in place of "$V = \varphi^{-1}(O)$" (§IV.11); the sc$^+$ condition in charts; the closed complement and the projection along it in place of "$p = Dt(0)$" (§IV.10); and the compatibility of the charts $p \circ \varphi$ (§IV.10). The text's "$V(a) = \varphi^{-1}(V_0) \cap A$" should be $(p \circ \varphi)^{-1}(V_0) \cap A$; its "Proposition 4.2" is Theorem 4.1.2 (preprint 4.2) (Findings, item 68). No hypothesis is added.

## IV.15 Remark 4.2.9 (preprint 4.20): good position is an open condition

**Statement** `Chapter4/Submanifolds · MPolyfold.remark_4_2_9`. For tame $X$, if $A$ is in good position at $a$, there is an open $U \ni a$ with $A$ in good position at every point of $U \cap A$. The text omits "tame" (Definition 4.2.5 (preprint 4.16) is for tame $X$, and Lemma 4.2.8 (preprint 4.19) is used) and argues "since the tangent spaces of $A$ move only slowly, Lemma 3.55 implies…". The formal proof makes this precise.

**The tangent spaces along a graph** `Chapter4/Submanifolds · tangentSpace_eq_range_graph`. If the retract $Q$ is the image of $\Gamma : (V, N \cap C, N) \to (Q, C, E)$, sc-smooth between retracts with inverse $p$, then at $v \in V$ and the level-1 point $y$ over $\Gamma(v)$: $\Gamma$ is differentiable within $V$ with derivative its sc-derivative `Chapter4/Submanifolds · hasFDerivWithinAt_scDeriv_const`, $T_yQ = D\Gamma(v)N$, and $p \circ D\Gamma(v) = 1$ — the tangent maps of the germ of sc-diffeomorphisms $\Gamma$, $p$.

**Proof of Remark 4.2.9 (preprint 4.20).** Take a tame chart $\psi$ centred at $a$ with tame retraction $r$, the sc$^+$ chart retraction $t$ onto $\psi(W \cap A)$, and $N = T_0$, in good position to $C$ in $E$ (§IV.14). The normal form of Proposition 3.1.25 (preprint 3.22) gives a continuous projection $\pi$ onto $N$ with closed kernel a good complement `Chapter3/GoodComplementProjection · exists_proj_goodComplement`, and Proposition 4.1.9 (preprint 4.9) gives $W_1$, $V$, $\Gamma$ over $N \cap C$. Write $\Gamma(v) = v + \sigma(v)$ with $\sigma(v) \in \ker \pi$. Then $\sigma$ is $C^1$ on $V$, $\sigma(0) = 0$, and $D\sigma(0) = 0$: $D\Gamma(0)m$ lies in $T_0 = N$, so $D\Gamma(0)m = \pi D\Gamma(0)m = m$. The tangent space at $\Gamma(v)$ is the graph $\{n + D\sigma(v)n\}$ `Chapter3/GoodPositionOpen · graphSpace`. Lemma 3.6.9 (preprint 3.55), with the closed complement $\ker \pi$ `Chapter3/GlobalIFTFull · lemma_3_6_9_relOpen`, gives $\rho > 0$ such that for $|v| < \rho$ this graph is in good position to $C_{\Gamma(v)}$ in $E$. It lies in $T_{\Gamma(v)}O$ (Remark 4.2.2 (preprint 4.14) in the chart), and good position in $E$ of a subspace of $T_{\Gamma(v)}O$ gives good position inside $T_{\Gamma(v)}O$, with the complement cut down by the modular law `Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le`. That is Definition 4.2.5 (preprint 4.16) at $b = \psi^{-1}(\Gamma(v))$, read in $\psi$. The neighbourhood is $U = W \cap \psi^{-1}(W_1 \cap \{|\pi u| < \rho\})$.

## IV.16 Linear algebra at the corner of a centred chart; Lemmas 7.A and 7.B of the exposition

In a chart centred at $0$, every coordinate vanishes at $0$, so the minimal subspace there is $E_0^{\min} = \{\lambda = 0\}$, the text's "$\{0\} \oplus W$" `Chapter4/LocalStructure · mem_minimalSubspaceWith_zero`, and it lies in every $E_x$ `Chapter4/LocalStructure · minimalSubspaceWith_zero_le`. Adding a vector of $E_0^{\min}$ changes no coordinate `Chapter4/LocalStructure · coord_add_of_mem_zero`, hence neither the degeneracy index `Chapter4/LocalStructure · degIndexWith_add_of_mem_zero` nor membership in $C$ `Chapter4/LocalStructure · add_mem_iff_of_mem_zero`.

**Closed complements.** If $N$ is finite-dimensional, $K$ closed and $N + K$ the whole space, $N$ has a closed complement $Y \subseteq K$: take a continuous projection $q$ onto the finite-dimensional $N \cap K$ and $Y = K \cap \ker q$

Chapter4/LocalStructure · exists_isClosed_isCompl_le. The projection along a closed complement of a finite-dimensional subspace is continuous, with kernel exactly the complement Chapter4/LocalStructure · exists_proj_of_isCompl.

**Good position from $N+E_0^{\min}=E_0$.** Let $N$ be finite-dimensional with $N+E_0^{\min}=E_0$. The coordinates restricted to $N$ Chapter4/LocalStructure · coordsOn map $N$ onto $\mathbb{R}^n$ Chapter4/LocalStructure · coordsOn_surjective, so $N\cap C$ has interior points, those with all coordinates positive Chapter4/LocalStructure · interior_nonempty_of_sup_eq_top. A complement inside $E_0^{\min}$ is good with every $\varepsilon$ Chapter4/LocalStructure · isGoodComplement_of_le_zero. Hence $N$ is in good position, with a closed good complement $Y\subseteq E_0^{\min}$ Chapter4/LocalStructure · exists_goodComplement_le_zero. **The degeneracy index on the slice** Chapter4/LocalStructure · degIndex_slice: $N\cap C$ is the partial quadrant of $N$ presented by $v\mapsto(\lambda(v),v-s(\lambda(v)))$, $s$ a linear section of $\lambda|_N$, and $d_{N\cap C}(v)=d_C(v)$ — both count the vanishing coordinates. This replaces the text's splitting $T_0A=N\oplus(T_0A\cap W)$, $E=N\oplus W$ (Findings, item 83).

**Lemma 7.A: an sc-complement inside an sc-subspace** Chapter4/LocalStructure · exists_scComplement_le. Let $N\subseteq E_\infty$ be finite-dimensional, $R$ an sc-subspace and $M=N+R$. Then $N$ has in $M$ an sc-complement $B\subseteq R$: $K=N\cap R$ is finite-dimensional and smooth, so it has an sc-complement $G$ in $E$ (Proposition 1.1.7 (preprint 1.6)) Chapter1/ScBanach · ScBanach.proposition_1_1_7; put $B=R\cap G$. Then $R=K\oplus B$, $N+R=N+B$, $N\cap B\subseteq K\cap G=0$, and $M_m=N\oplus B_m$ on every level because $N\subseteq E_\infty$. $B$ is closed with $B_m=(1-P)R_m$, $P$ the sc-projection onto $K$, so the smooth points of $B$ are dense in every level: for $x\in B_m$ and smooth $u\in R_m$ near $x$, $u-Pu\in B$ is smooth and near $x-Px=x$.

**General position from $T_yQ+E_0^{\min}=E_0$** Chapter4/LocalStructure · isGeneralPositionRetractAt_of_sup_eq_top. Let $r$ be tame, $Q\subseteq O$ a retract, $y\in Q_1$ smooth, and $T_yQ$ finite-dimensional, smooth, with $T_yQ+E_0^{\min}=E_0$. As $E_0^{\min}\subseteq E_y$, intersecting with $T_yO\supseteq T_yQ$ gives $T_yO=T_yQ+(T_yO\cap E_0^{\min})\subseteq T_yQ+T_y^RO$ (modular law). $T_y^RO$ is an sc-subspace by Proposition 2.5.6 (preprint 2.36) for the tame $r$ Chapter2/PartialCone · proposition_2_5_6_partialCone, and Lemma 7.A gives the sc-complement inside it. This is the step of Theorem 4.2.11 (3) (preprint 4.22) "implying that $T_aA$ has in $T_aO$ a sc-complement contained in $T_a^RO$" (Findings, item 85).

**Lemma 7.B: general position implies good position** Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt. For $r$ tame and a smooth $z\in O_1$: if $T_zQ$ has in $T_zO$ an sc-complement $B\subseteq T_z^RO$, then $T_zQ$ is in good position to $C_z$ inside $T_zO$. $B\subseteq E_z$ changes none of the coordinates $\lambda_i$, $i\in I(z)$, so it is a good complement with every $\varepsilon$. The coordinates $\lambda_{I(z)}$ restricted to $T_zO$ have kernel $T_z^RO$, whose codimension is $d_C(z)=\#I(z)$ (Proposition 2.5.6 (preprint 2.36)) Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq, so they map $T_zO$ onto $\mathbb{R}^{I(z)}$; they kill $B$, so they map $T_zQ$ onto $\mathbb{R}^{I(z)}$, and the preimage of the open positive orthant gives interior points of $T_zQ\cap C_z$. The text never states this step, which Theorem 4.2.11 (4) (preprint 4.22) and Theorem 4.3.1 (preprint 4.23) need for tameness and corner structures.

# IV.17 Theorem 4.2.11 (1)–(3) (preprint 4.22): local structure in general position

**$T_0Q+E_0^{\min}=E_0$ from general position at the centre** Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top. If $T_0Q$ has in $T_0O$ an sc-complement $B\subseteq T_0^RO\subseteq E_0^{\min}$ and $r$ is tame, then $E_0=T_0O+(1-Dr(0))E_0$ with $(1-Dr(0))E_0\subseteq E_0^{\min}$ (Proposition 2.5.3 (preprint 2.35)), so $E_0=T_0Q+B+(1-Dr(0))E_0\subseteq T_0Q+E_0^{\min}$. (The text: "$E=T_0O\oplus Z=T_0A\oplus\widetilde{A}\oplus Z$, so $T_0A$ has in $E$ an sc-complement

$Y$ contained in $W$"; only $T_0A + W = E$ is used afterwards.)

$T_yQ + E_0^{\min} = E_0$ **along the graph** `Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top_graph`. If $\Gamma : V \to Q \cap W$ is the graph map of Proposition 4.1.9 (preprint 4.9) for a projection $p$ onto $N$ whose kernel lies in $Y \subseteq E_0^{\min}$, and $N + E_0^{\min} = E_0$, then at $y$ over $\Gamma(v)$, $T_yQ = D\Gamma(v)N$ (§IV.15) and $D\Gamma(v)n - n \in \ker p \subseteq E_0^{\min}$ because $p \circ D\Gamma(v) = 1$; so $T_yQ + E_0^{\min} = N + E_0^{\min} = E_0$. This is the text's "by means of the particular form of the map $\alpha : h \mapsto h + D\delta(v)h$, also $T_aA + W = E$", which needs $D\delta(v)h \in W$.

**Theorem 4.2.11 (preprint 4.22) in a tame centred chart** `Chapter4/LocalStructure · theorem_4_2_11_chart`. Let $r$ be tame onto $O \ni 0$, $t$ an sc⁺-retraction onto $Q \subseteq O$ with $0 \in Q$, $N = T_0Q$, and suppose $Q$ is in general position at $0$. Then: $N + E_0^{\min} = E_0$ (above); $N$ is in good position with a closed good complement $Y \subseteq E_0^{\min}$ (§IV.16) and the projection $p$ along it, with kernel exactly $Y$; Proposition 4.1.9 (preprint 4.9) for this $Y$ gives $W$, $V$, $\Gamma$; (2) $d_{N\cap C}(v) = d_C(\Gamma(v))$ for $v \in V$, because $d_{N\cap C}(v) = d_C(v)$ on the slice and $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate; (3) at every level-1 point $y$ over $Q \cap W$, $Q$ is in general position: $y$ is smooth (Lemma 4.1.7 (i) (preprint 4.7)), $T_yQ$ is the range of $Dt(y)$, finite-dimensional (Lemma 4.1.7 (ii) (preprint 4.7)) and smooth — the range of $Dt(y)$ is an sc-subspace at a smooth point `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`, finite-dimensional sc-subspaces are smooth `Chapter1/ScBanach · ScBanach.proposition_1_1_7_iff` — and $T_yQ + E_0^{\min} = E_0$ along the graph gives general position (§IV.16).

**Theorem 4.2.11 (preprint 4.22)** `Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint`. *Statement.* Let $X$ be tame, $A$ a smooth finite-dimensional submanifold, in general position at $x \in A$. There are a chart $\psi$ of $X$ with $\psi(x) = 0$, a tame retraction $r$ onto its model for a presentation with $n$ half-line coordinates, and an open $U \ni x$ inside the domain of $\psi$, such that: (1) $d_X(x) = n$, and $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold $O$; (2) $d_X(a) = d_A(a)$ for $a \in A \cap U$; (3) $A$ is in general position at every point of $A \cap U$; (4) for every $\sigma \subseteq \{1, \dots, n\}$ the intersection of local faces $F_\sigma$ is a tame sub-M-polyfold inside $U$ with $d_{F_\sigma} + \#\sigma = d_X$, and $A \cap F_\sigma$ is a tame smooth finite-dimensional submanifold of $F_\sigma$, in general position in $F_\sigma$ at each of its points. *Proof.* $x$ is smooth (Proposition 4.2.3 (1) (preprint 4.15)), so there is a tame chart $\psi$ centred at $x$ (§IV.5), with a tame retraction $r$ and a presentation $T$; the chart retraction $t$ of $A$ in $\psi$ is sc⁺ (§IV.11), and general position at $x$ read in $\psi$ (chart independence, §IV.13) is general position of $t(U)$ at $0$. Theorem 4.2.11 (preprint 4.22) in the chart gives $Y$, $p$, $W_1$, $V$, $\Gamma$; put $U = W \cap \psi^{-1}(W_1)$, $W$ the open set of the submanifold chart.

1. *(1) is a statement about the chart.* At the centre all $n$ coordinates vanish, so $d_C(0) = n$; the tame chart realizes $d_X$ `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, so $d_X(x) = n$. The face count is Corollary 2.5.13 (preprint 2.41) for the M-polyfold $O$ `Chapter2/TameRetracts · isFaceStructured_toMPolyfold` `Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith`. "Precisely $d$ local faces contained in $U$" is read here for the faces $O \cap \{\lambda_i = 0\}$ of the chart model; their identification with the local faces of the neighbourhood $U$ itself (Definition 2.5.10 for $U$), and (1) in that sense, are §IV.32 (Findings, item 86).
2. *(2).* The text: "$p : A \to (T_0A \cap C) \cap V$ is a sc-diffeomorphism … by Proposition 2.24, $d_A(a) = d_{T_0A\cap C}(p(a))$". The sc-diffeomorphism is the corner chart $p \circ \psi$ of $A$: the data of §IV.10 for the chart of $A$ induced by $\psi$, with the complement $Y \subseteq E_0^{\min}$, give a chart of the induced structure modelled on $(V, N \cap C, N)$ with the identity retraction. That model is tame, so the chart realizes $d_A$ `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract`; $\psi$ realizes $d_X$; and $d_{N\cap C}(p\psi(a)) = d_C(\Gamma(p\psi(a))) = d_C(\psi(a))$ by (2) of the chart form (Findings, item 84).
3. *(3)* is (3) of the chart form, read back through the chart: at $b \in U \cap A$ the level-1 point over $\psi(b)$ is

smooth, and the chart condition in $\psi$ is general position at $b$.

4. *(4)* is §IV.18.

The text's further claims "$\delta(0) = 0$ and $D\delta(0) = 0$" hold (§IV.15) but are not used.

## IV.18 Theorem 4.2.11 (4) (preprint 4.22): the faces, and $A \cap F_\sigma$

The text: "in view of the arguments in the proof of Proposition 2.43, the sets $F_\sigma$ are tame M-polyfolds. Moreover, $A \cap F_\sigma$ are smooth submanifolds of $F_\sigma$. The previous arguments, but now applied to $A \cap F_\sigma$ conclude the proof." Four things are missing (Findings, item 89): Proposition 2.5.15 (preprint 2.43) is about one face, and intersections need Proposition 2.5.16 (preprint 2.44) (second sentence); "smooth submanifold of $F_\sigma$" needs an sc⁺-retraction *in the M-polyfold* $F_\sigma$; general position in $F_\sigma$ needs an argument; and tameness of $A \cap F_\sigma$ needs Theorem 4.2.7 (preprint 4.18) in $F_\sigma$ and good position.

**Killing coordinates.** For $\sigma \subseteq \{1, \dots, n\}$, $a \mapsto a \cdot 1_\sigma$ keeps the coordinates in $\sigma$ `Chapter4/LocalStructure · maskL`, and for a linear section $s$ of $\lambda|_N$, $\kappa_\sigma(v) = v - s(\lambda(v)1_\sigma)$ `Chapter4/LocalStructure · killOn` has $\lambda_i(\kappa_\sigma v) = 0$ for $i \in \sigma$ and $\lambda_j(\kappa_\sigma v) = \lambda_j(v)$ otherwise `Chapter4/LocalStructure · coordsOn_killOn`, fixes the vectors of the stratum $f_\sigma = \{\lambda_i = 0, i \in \sigma\}$ `Chapter4/LocalStructure · killOn_eq_self`, is idempotent `Chapter4/LocalStructure · killOn_idem` and maps $N \cap C$ into itself `Chapter4/LocalStructure · killOn_mem`. A continuous linear map into a finite-dimensional space is an sc-operator for the constant structure `Chapter4/LocalStructure · opToConst`.

**The sc⁺-retraction onto $A \cap F_\sigma$ in the chart** `Chapter4/LocalStructure · exists_scPlus_retraction_corner`. In the situation of the chart form of Theorem 4.2.11 (preprint 4.22), for every open $B \ni 0$ there are an open $G \ni 0$, $G \subseteq W \cap B$, and an sc⁺-retraction $t_\sigma = \Gamma \circ \kappa_\sigma \circ p \circ t$ on $U \cap t^{-1}(G)$ onto $Q_\sigma = Q \cap G \cap f_\sigma$: (a) it is sc⁺, as sc-smooth □ sc⁺ (§IV.2), $p$, $\kappa_\sigma$ and $\Gamma$ being sc-smooth; (b) it is a retraction, because $t\Gamma = \Gamma$, $p\Gamma = 1$ and $\kappa_\sigma^2 = \kappa_\sigma$; (c) its image is $Q \cap G \cap f_\sigma$, because $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate, so $\lambda(\Gamma(v)) = \lambda(v)$; (d) at a point $y$ of $Q_\sigma$ the tangent vectors of $Q$ lying in $f_\sigma$ are tangent to $Q_\sigma$: for such $h$, $k = ph$ has $\lambda(k) = \lambda(h)$, so $\kappa_\sigma k = k$ and $Dt_\sigma(y)h = D\Gamma(w)\kappa_\sigma pDt(y)h = D\Gamma(w)k = h$, $w = p(y)$, by the chain rule and $T_yQ = D\Gamma(w)N$.

**The corner model** `Chapter4/LocalStructure · exists_cornerModel_retraction`. For a retraction $S$ with retract in $f_\sigma$, $\pi_\sigma \circ S \circ \iota_\sigma$ on $\iota_\sigma^{-1}(S.U)$, with the corner inclusion and projection of Chapter 2 `Chapter2/CornerGerms · cornerIncl` `Chapter2/CornerGerms · cornerProj`, is a retraction of $[0,\infty)^{n-\#\sigma} \oplus W$ onto $\pi_\sigma(S(U))$, sc⁺ if $S$ is.

**Theorem 4.2.11 (4) (preprint 4.22) in the chart** `Chapter4/LocalStructure · theorem_4_2_11_four_chart`. For every $\sigma$ and every open $B_0 \ni 0$ there are faces $F_i \ni 0$ of $O$ ($i \in \sigma$) and an open $G \ni 0$, $G \subseteq W \cap B_0$, such that: *$F_\sigma$ near 0 is a tame retract* — $O \cap G \cap f_\sigma$ is the retract of the face retraction of Proposition 2.5.16 (preprint 2.44) `Chapter2/CornerGerms · corner_retraction` restricted to $G$, whose corner model is the retract of a tame retraction $r'$ with $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$ `Chapter2/CornerGerms · corner_tameModel`; *$A \cap F_\sigma$ near 0 is an sc⁺-retract* — $Q \cap G \cap f_\sigma$ is the retract of $t_\sigma$, and its corner model that of the sc⁺-retraction $t'$, inside the retract of $r'$; *general position in $F_\sigma$* — at a level-1 point $y'$ of $\pi_\sigma(Q \cap G \cap f_\sigma)$, with $y = \iota_\sigma(y')$: $T_yQ + E_0^{\min} = E_0$ (§IV.17) gives for every $b \in \mathbb{R}^{n-\#\sigma}$ a $v \in T_yQ$ with $\lambda(v) = \iota(b)$, zero on $\sigma$; $v \in f_\sigma$, so $v$ is tangent to $Q_\sigma$, and $\pi_\sigma v$ is tangent to $\pi_\sigma(Q_\sigma)$ at $y'$ (§IV.11), with coordinates $b$; so the tangent space of the corner model plus its minimal subspace at 0 is everything, and general position follows (§IV.16), in the corner model with the identity presentation.

**sc⁺-retractions in a chart are sc⁺-retractions on open subsets** `Chapter4/LocalStructure · MPolyfold.`

`isScPlusRetractionOn_of_chart` — the converse of §IV.11, which the text does not state. For a chart $\chi$ and an sc⁺-retraction $\tau$ of its model with $\tau(U) \subseteq O$, $s = \chi^{-1} \circ \tau \circ \chi$ is an sc⁺-retraction on the open set $\chi^{-1}(\tau.U)$, with image $\chi^{-1}(\tau(U))$: sc-smoothness of $s$ and of $1^1_0 \circ s$ are read in the charts $\chi$ of $X$ and `levelChart` $\chi$ of $X^1$, where they are $\tau$ and its lift; the values lie in $X_1$ since $\tau(U) \subseteq E_\infty$ (Lemma 4.1.7 (i) (preprint 4.7)).

**(4) on the M-polyfold, for one $\sigma$** `Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_four_of_chart`. In the tame chart $\psi$ centred at $x$ with the chart data of §IV.17: the faces $F_i$ and $G$ of the chart form, with $G$ inside the domain of the submanifold chart, give $F_\sigma = \psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ `Chapter4/LocalStructure · MChart.cornerPreimage`, which contains $x$ and lies in $W$. It is a sub-M-polyfold, a local property `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local`; its face chart $\pi_\sigma \circ \psi$, compatible with the induced charts `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`, is a chart of the induced structure `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible` modelled on the tame corner model, so $F_\sigma$ is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart` with $d_{F_\sigma} + \#\sigma = d_X$. $A \cap F_\sigma$ is, in the face chart, the retract of the sc⁺-retraction $t'$, which the previous paragraph turns into an sc⁺-retraction on an open subset of $F_\sigma$: it is a smooth finite-dimensional submanifold of $F_\sigma$. It is in general position at each of its points by the chart form read in the face chart. Its tameness is Theorem 4.2.7 (preprint 4.18) in the tame M-polyfold $F_\sigma$, good position following from general position (Lemma 7.B). $F_\sigma$ is a germ, chosen inside $U$ — the text's "contained in $U$" — and $G$ depends on $\sigma$. Paracompactness of $F_\sigma$ and $A \cap F_\sigma$, instance arguments of the induced structures, holds for every subset `Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype`.

The proof of Theorem 4.3.1 (preprint 4.23) uses $\dim(A \cap F_\sigma) = \dim A - \#\sigma$, which Theorem 4.2.11 (preprint 4.22) does not state; it is proved in the model in §IV.21.

## IV.19 Remark 4.3.2 (preprint 4.24)

If $T_zO = T_zQ + B$ with $B \subseteq K$, the image of $T_zQ$ spans $T_zO/K$, so $\dim(T_zO/K) \leq \dim T_zQ$ `Chapter4/LocalStructure · quotDim_le_finrank_of_eq_sup`. **In a chart** `Chapter4/LocalStructure · remark_4_3_2_chart`: at a smooth point $z$ of a tame retract, $\dim(T_zO/T^R_zO) = d_C(z)$ (Proposition 2.5.6 (preprint 2.36)), and if $T_zQ$ has in $T_zO$ an sc-complement inside $T^R_zO$, then $d_C(z) \leq \dim T_zQ$. **On the M-polyfold**: for tame $X$ and smooth $x$, in a tame chart at $x$ the reduced tangent space has codimension $d_X(x)$ `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart` `Chapter4/LocalStructure · MPolyfold.remark_4_3_2_codim` — the remark's first sentence, which needs $X$ tame and $x$ smooth; if $A$ is in general position at $a$, then $d_X(a) \leq \dim T_aA$ in any tame chart at $a$ `Chapter4/LocalStructure · MPolyfold.remark_4_3_2`; and a submanifold in general position with $T_aA = 0$ does not meet $\partial X$ at $a$ `Chapter4/LocalStructure · MPolyfold.remark_4_3_2_not_mem_boundary`. The remark's "$x \in A_\lambda$ can only belong to a corner if $d_X(x) \leq \dim A_\lambda$" holds for every submanifold in general position, so it applies to the $A_\lambda$ of Theorem 4.3.1 (preprint 4.23) once their general position is proved (§IV.24).

## IV.20 Sard's theorem, vendored

The proof of Theorem 4.3.1 (preprint 4.23) applies "Sard's theorem" to maps $A \cap F^\circ_\sigma \to \mathbb{R}^m$ between spaces of arbitrary dimensions. Mathlib has the cases $\dim E \leq \dim F$ only. The Milnor project of this repository proves the general theorem by Milnor's induction; its import closure is copied into `Polyfold/Vendor/MilnorSard/` with the namespace renamed and no statement or proof changed (Findings, item

100). The proof, in Milnor's steps:

- *The objects.* Crit $= \{x \in U : Df(x) \text{ not onto}\}$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet`, and $C_k$, the points where all derivatives of orders $1, \dots, k$ vanish `Vendor/MilnorSard/Basic · MilnorSard.Sard.vanishSet`, with Crit $\subseteq$ (Crit $\setminus C_1$) $\cup \bigcup_{1 \le i < k} (C_i \setminus C_{i+1}) \cup C_k$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet_subset_union`. The statement in dimension $n$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.SardStatement`.
- *Straightening.* A smooth $w$ with $Dw(\bar{x}) \neq 0$ is straightened by $H(x) = (w(x), Px)$, a local diffeomorphism by the inverse function theorem `Vendor/MilnorSard/InverseFunction · MilnorSard.isLocalDiffeomorphAt_of_isInvertible_mfderiv` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.Straightening` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.exists_straightening`.
- *Step 1,* $f(\text{Crit} \setminus C_1)$ null, locally `Vendor/MilnorSard/Step1 · MilnorSard.Sard.step1_local`: in straightened coordinates $g = f \circ H^{-1}$ maps hyperplanes to hyperplanes, the critical points of $g$ on a hyperplane are those of its restriction, the induction hypothesis in dimension $n-1$ applies to the restrictions, and Fubini concludes.
- *Step 2,* $f(C_k \setminus C_{k+1})$ null, locally `Vendor/MilnorSard/Step2 · MilnorSard.Sard.step2_local`: a $k$-th derivative $w$ vanishes on $C_k$ with $Dw \neq 0$; after straightening, $C_k$ lies in a hyperplane, on which every point of $C_k$ is critical, and the induction hypothesis applies. Local null sets glue along a countable cover `Vendor/MilnorSard/Step2 · MilnorSard.Sard.measure_image_eq_zero_of_locally`.
- *Step 3,* $f(C_k \cap K)$ null for $(k+1)\dim F > \dim E$ and $K$ compact convex `Vendor/MilnorSard/Step3 · MilnorSard.Sard.measure_image_vanishSet_inter_eq_zero`: Taylor's bound $|f(y) - f(x)| \le c|y - x|^{k+1}$ for $x \in C_k$ `Vendor/MilnorSard/Step3 · MilnorSard.Sard.norm_sub_le_of_iteratedFDeriv_eq_zero` makes $f$ Hölder of exponent $k+1$ there, so $f(C_k \cap K)$ has Hausdorff dimension at most $\dim E/(k+1) <$ $\dim F$, hence measure zero `Vendor/MilnorSard/Basic · MilnorSard.Sard.addHaar_eq_zero_of_dimH_lt_finrank` — Milnor's cube count is the standard proof of that estimate.
- *The induction* `Vendor/MilnorSard/Sard · MilnorSard.Sard.sardStatement`, with the zero-dimensional target treated separately (no critical points) `Vendor/MilnorSard/Sard · MilnorSard.Sard.critSet_eq_empty_of_subsingleton`.

**Sard's theorem** `Vendor/MilnorSard/Sard · MilnorSard.sard`. *For $E$, $F$ finite-dimensional real normed spaces in the same universe, a Haar measure $\mu$ on $F$, $U \subseteq E$ open and $f$ of class $C^\infty$ on $U$, $\mu(f(\text{Crit})) = 0$;* and the complement of the critical values is dense `Vendor/MilnorSard/Sard · MilnorSard.dense_compl_image_critSet`.

# IV.21 The strata, Sard on the strata, and Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form

The finite-dimensional model of the proof of Theorem 4.3.1 (preprint 4.23): a finite-dimensional $V$ (in the application $T_0A$ of the chart of Theorem 4.2.11 (preprint 4.22)) with linear coordinates $\ell : V \to \mathbb{R}^n$ (those of the tame chart, onto), the partial quadrant $C = \{\ell \ge 0\}$ `Chapter4/FamiliesLocal · quadC`, a relatively open $U = O \cap C$ and a map $f$ of class $C^\infty$ on $U$ (in the application the projection to $\mathbb{R}^m$ composed with the graph map $\Gamma$).

**The strata.** The stratum subspace of $\sigma$ is $\{\ell_i = 0, i \in \sigma\}$ `Chapter4/FamiliesLocal · stratSub` `Chapter4/FamiliesLocal · mem_stratSub`, the tangent space of $A \cap F_\sigma$; $\sigma(v) = \{i : \ell_i(v) = 0\}$ is the set of faces through $v$ `Chapter4/FamiliesLocal · zeroSetOf`, and $v$ lies in the stratum subspace of $\sigma(v)$ `Chapter4/FamiliesLocal · mem_stratSub_zeroSetOf`. **The dimension count** `Chapter4/FamiliesLocal · finrank_stratSub`: if $\ell$ is onto, the stratum subspace of $\sigma$ has dimension $\dim V - \#\sigma$ — it is the kernel of $v \mapsto (\ell_i(v))_{i \in \sigma}$, which is onto $\mathbb{R}^\sigma$. This is

the count "$A \cap F_\sigma^\circ$ is a smooth manifold … having dimension $\dim(A) - \#\sigma$", which the text attributes to "Theorem 4.23" (it means Theorem 4.2.11 (preprint 4.22), which does not state it).

**Stratified critical points** `Chapter4/FamiliesLocal · stratCritSet`: the $v \in U$ at which $Df(v)$ restricted to the stratum subspace of $\sigma(v)$ is not onto — the critical points of the text's $p_\sigma : A \cap F_\sigma^\circ \to \mathbb{R}^m$ for $\sigma = \sigma(v)$. **Sard on the strata** `Chapter4/FamiliesLocal · measure_image_stratCritSet`: *the stratified critical values form a null set for every Haar measure.* The open stratum of $\sigma$, $\{k \in O : \ell_i(k) = 0\ (i \in \sigma),\ \ell_j(k) > 0\ (j \notin \sigma)\}$, is an open subset of the stratum subspace and lies in $U$; $f$ restricted to it is $C^\infty$, with derivative $Df(v)$ restricted to the stratum subspace; so a stratified critical point $v$ with $\sigma = \sigma(v)$ is a critical point of $f$ on the open stratum of $\sigma$, and Sard's theorem applies to each of the finitely many strata. The union of the open strata is $U$, and a point is stratified-regular iff it is a regular point of the $p_\sigma$ for its own $\sigma$; so the stratified critical values are the union of the critical values of all the $p_\sigma$.

**Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form.** Theorem 3.8.22 (preprint 3.78) (§III.17) is stated for one $j$, with $j$-dependent constants in its proof; the fibre retraction below needs one solution that is $C^\infty$. In the normalized form `Chapter4/FamiliesLocal · ContractionData.exists_solution_smooth`: if the contraction data of the proof hold for every $j \geq 1$, the radii of Lemmas 3.8.23–3.8.24 (preprint 3.79–3.80) `Chapter3/QuadrantIFT · ContractionData.exists_radii` and the fixed point $\tau(k)$ are those for $j = 1$, independent of $j$, and only the regularity statement depends on $j$ `Chapter3/QuadrantIFT · Radii.contDiffOn_τ`; so one $\tau$ is $C^j$ for every $j$. **Theorem 3.8.22 (preprint 3.78), $C^\infty$** `Chapter4/FamiliesLocal · theorem_3_8_22_smooth`: for $U$ relatively open in a partial quadrant $C$ of a finite-dimensional $V$, $0 \in U$, $f$ of class $C^\infty$ on $U$ with $f(0) = 0$ and $Df(0)$ onto, $K = \ker Df(0)$ in good position to $C$ with good complement $Y$: there are a relatively open $V' \ni 0$ of $K \cap C$ and $\tau : K \to Y$, $C^\infty$ on $V'$, with $\tau(0) = 0$, $D\tau(0) = 0$, $k + \tau(k) \in U$ and $f(k + \tau(k)) = 0$, and the uniqueness of $\tau(k)$ among small $y$ — the proof of Theorem 3.8.22 (preprint 3.78) `Chapter3/QuadrantIFT · theorem_3_8_22` (the splitting $V \cong K \times Y$, $L = Df(0)$ an isomorphism $Y \to F$, $B(k, y) = y - L^{-1} f(k + y)$) with the $C^\infty$ normalized form in place of the $C^j$ one.

# IV.22 The fibre over a stratified regular point

**Partial quadrants of the model.** $\{\ell_i \geq 0, i \in \sigma\}$ `Chapter4/FamiliesLocal · quadAt` is, for $\sigma = \sigma(v_0)$, the partial quadrant $C_{v_0}$ of Definition 2.5.1 (preprint 2.33). With a linear section $s$ of $\ell$, $v \mapsto (\ell(v), v - s(\ell(v)))$ is a linear isomorphism $V \cong \mathbb{R}^n \times \ker \ell$ `Chapter4/FamiliesLocal · sliceEquiv`, which presents both as partial quadrants `Chapter4/FamiliesLocal · isPartialQuadrant_quadC Chapter4/FamiliesLocal · isPartialQuadrant_quadAt`; $C$ is convex `Chapter4/FamiliesLocal · convex_quadC` with nonempty interior `Chapter4/FamiliesLocal · interior_quadC_nonempty`, so derivatives within $O \cap C$ are unique `Chapter4/FamiliesLocal · uniqueDiffWithinAt_quadC`.

**The kernel at a stratified regular point is in good position** `Chapter4/FamiliesLocal · exists_goodComplement_stratSub` — the argument of Lemma 5.26 (preprint; not in the 2021 edition), which the text needs here and does not give. Let $K + \{\ell_i = 0, i \in \sigma\} = V$ (for $K = \ker Df(v_0)$, $\sigma = \sigma(v_0)$, this is stratified regularity). The stratum subspace is closed, so $K$ has a closed complement $Y$ inside it (§IV.16). Adding a vector of $Y$ changes none of the coordinates $\ell_i$, $i \in \sigma$, so $Y$ is a good complement with $\varepsilon = 1$. For the interior points: choose $x$ with $\ell(x) = (1, \dots, 1)$ and write $x = k + y$; then $\ell_i(k) = 1$ for $i \in \sigma$, and the open set $\{k' \in K : \ell_i(k') > 0, i \in \sigma\}$ contains $k$ and lies in $K \cap \{\ell_i \geq 0, i \in \sigma\}$.

**The fibre retraction** `Chapter4/FamiliesLocal · exists_fibre_retraction`. *Let $\ell$ be onto, $f$ $C^\infty$ on $O \cap C$ and $v_0 \in O \cap C$ not stratified-critical. There are an open $O' \ni v_0$, $O' \subseteq O$, and $\rho$, $C^\infty$ on $O' \cap C$, mapping $O' \cap C$ into itself and into $f^{-1}(f(v_0))$, equal to the identity on the fibre there, with $D\rho(v_0)$ of range* $\ker Df(v_0)$ *and equal to the identity on it.* Proof:

1. Stratified regularity gives $Df(v_0)$ onto (its restriction to the stratum already is) and $K+\text{stratum} = V$ for $K = \ker Df(v_0)$; so $K$ is in good position to $C_{v_0}$ with a closed good complement $Y$ in the stratum subspace.
2. Near $0$, $v_0 + C_{v_0}$ and $C$ agree: on $G_0 = \{w : \ell_j(v_0+w) > 0, j \notin \sigma\}$, a point $v_0+w$ is in $C$ iff $w \in C_{v_0}$, since $\ell_i(v_0) = 0$ for $i \in \sigma$ — the re-centring of §IV.5 in the model. So $U_1 = (O - v_0) \cap G_0 \cap C_{v_0}$ is relatively open in $C_{v_0}$, contains $0$, and $w \mapsto v_0 + w$ maps it into $U$.
3. $g(w) = f(v_0+w) - f(v_0)$ is $C^\infty$ on $U_1$, $g(0) = 0$, $Dg(0) = Df(v_0)$ onto; Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form gives the fibre near $v_0$ as the graph $\{v_0 + k + \tau(k)\}$ of a $C^\infty$ map $\tau : V' \to Y$, $V' \subseteq K \cap C_{v_0}$, unique for small $k$ and $y$.
4. With $q(v)$ the projection of $v - v_0$ onto $K$ along $Y$ `Chapter4/LocalStructure · exists_proj_of_isCompl`, put $\rho(v) = v_0 + q(v) + \tau(q(v))$ on $O' \cap C$, $O'$ a small ball around $v_0$ chosen so that $q$ maps it into the uniqueness region and $\rho(O' \cap C) \subseteq O'$. The projection along $Y$ keeps the coordinates in $\sigma$, so $q$ maps $C$ near $v_0$ into $C_{v_0}$; $\rho$ lands on the fibre; and on the fibre $\rho(v) = v$ by uniqueness.
5. $D\tau(0) = 0$ gives $D\rho(v_0) = q$, whose range is $K$ and which is the identity on $K$.

# IV.23 Stratified regularity, the sc⁺-retraction onto $A_\lambda$, and its general position

**The family.** For tame $X$, $\mathbb{R}^m \times X$ with the product charts is tame `Chapter3/ParamPerturbation · prod_rn_isTame` (§III.40 for $\mathbb{R}^m$ as an M-polyfold `Chapter3/ParamPerturbation · rnPolyfold` with its translation charts `Chapter3/ParamPerturbation · rnChart`). The fibre of $A \subseteq \mathbb{R}^m \times X$ over $\lambda$ is $A_\lambda = \{x : (\lambda, x) \in A\}$ `Chapter4/Families · fibreSet`. $\mathbb{R}^m$ is presented as a partial quadrant without half-line coordinates `Chapter4/Families · rnPres` `Chapter4/Families · rnPres_eq`, and the product chart $\chi = (r \mapsto r - a) \times \varphi$ has the product presentation `Chapter2/FibredProduct · prodPresentation`, whose minimal subspace at $z$ is $E_z = \mathbb{R}^m \times (E_X)_{\varphi(z_2)}$.

**Stratified regular points** `Chapter4/Families · IsStratRegularPoint`. $z \in A$ is a stratified regular point if, for some data of Definition 4.2.1 (preprint 4.13) at $z$, some $a \in \mathbb{R}^m$, some chart $\varphi$ of $X$ at $z_2$, some presentation of its partial quadrant and the level-1 point $y$ over $\chi(z)$, every $\mu \in \mathbb{R}^m$ is the first component of a tangent vector $v$ of the chart model of $A$ at $y$ with $v_2 \in E_{\varphi(z_2)}$: the projection maps $T_zA \cap E_z$ — the tangent space of the stratum through $z$ — onto $\mathbb{R}^m$. $\lambda$ is a *stratified regular value* if every point of $A$ over $\lambda$ is a stratified regular point `Chapter4/Families · IsStratRegularValue`. This is the text's "regular value of every $p_\sigma$". The condition does not depend on these choices (§IV.33). *The naive notion does not suffice*: in $\mathbb{R} \times [0,\infty)$, $A = \{(x,x) : 0 \le x < 1\} \cup (-1,1) \times \{5\}$ and $\varepsilon = \frac{1}{2}$ satisfy the hypotheses of Theorem 4.3.1 (preprint 4.23) (at $(0,0)$, $T_zA = \mathbb{R}(1,1)$ has the complement $\mathbb{R} \times 0 = T_z^R Z$), $Dp$ is onto everywhere, but $A_0 = \{0, 5\}$ is zero-dimensional and meets $\partial X$, so it is not in general position (Remark 4.3.2 (preprint 4.24)); $(0,0)$ is stratified-critical, $T_zA \cap E_z = 0$.

**The slice** $e \mapsto (c, e)$: the inclusion $h \mapsto (0,h)$ `Chapter4/Families · ScBanach.prodInrOp` followed by the translation by the smooth vector $(c,0)$ `Chapter4/Families · mem_smooth_rn_zero` is sc-smooth `Chapter4/Families · isScSmooth_slice` with derivative $h \mapsto (0,h)$ `Chapter4/Families · scDeriv_slice`. On a finite-dimensional source the sc-derivative is the classical one `Chapter4/Families · hasFDerivWithinAt_scDeriv_const'`.

**The sc⁺-retraction onto $A_\lambda$** `Chapter4/Families · exists_slice_retraction` — Definition 4.2.1 (preprint 4.13) requires it and the text does not construct it. In the model $\mathbb{R}^m \oplus E$ of a product chart, let $t$ be an sc⁺-retraction (the chart model of $A$), described near a point as the graph $\Gamma : V \to t(U) \cap W$ with inverse $p$ (Proposition 4.1.9 (preprint 4.9)), and $\rho$ a $C^\infty$ retraction of $O' \cap (\mathbb{R}^m \times C) \subseteq V$ onto the points $v$ with $\Gamma(v)_1 = c$

(the fibre retraction of §IV.22 for $f = (\Gamma)_1$). Then

$$\tau(e) = \big(\Gamma(\rho(p(t(c,e))))\big)_2$$

is an sc⁺-retraction of $E$ (partial quadrant $C$) onto $\{e : (c,e) \in t(U) \cap W,\ p(c,e) \in O'\}$: it is sc⁺ because $t$ is sc⁺ and the slice, $p$, $\rho$ (sc-smooth as a $C^\infty$ map on a finite-dimensional set, Corollary 1.2.5 (preprint 1.20)), $\Gamma$ and the second projection `Chapter2/StrongBundle · ScBanach.prodSndOp` are sc-smooth (§IV.2); and it is a retraction, because for $e' = \tau(e)$, $\Gamma(w) = (c, e')$ with $w = \rho(\cdot)$ on the fibre, so $t(c,e') = \Gamma(w)$, $p\Gamma(w) = w = \rho(w)$ and $\tau(e') = e'$. *Its tangent space*: if $D\rho(v_0) = 1$ on the $h$ with $(D\Gamma(v_0)h)_1 = 0$, then at the level-1 point over $\Gamma(v_0)_2$ the tangent space of the retract of $\tau$ contains $(D\Gamma(v_0)h)_2$ for all such $h$, by the chain rule $D\tau(y)w = (D\Gamma(v_0)D\rho(v_0)p\,Dt(c,y)(0,w))_2$, $Dt$ fixing $T(t(U)) = D\Gamma(v_0)N$ and $p \circ D\Gamma(v_0) = 1$.

**General position of the fibre** `Chapter4/Families · isGeneralPositionRetractAt_slice`. Let $r_X$ be tame, the chart model of $A$ in general position at $z = (c,y)$ with respect to id $\times\, r_X$ and the product presentation, $z$ regular on its stratum (every $\mu$ is the first component of some $a \in T_zA \cap E_z$), and $\tau$ an sc⁺-retraction onto $Q \subseteq O$ whose tangent space at the smooth $y$ contains the second components of the tangent vectors of $A$ at $z$ with first component $0$. Then $Q$ is in general position at $y$. For $w \in T_yO$ write $(0,w) = a + b$ with $a \in T_zA$ and $b \in T^R_zZ = \mathbb{R}^m \times T^R_yO$ (general position of $A$ at $z$); correct $a$ by $a_2 \in T_zA \cap E_z$ with the same first component as $a$ (regularity on the stratum); then $(a - a_2)_2 \in T_yQ$ and $(b + a_2)_2 \in T^R_yO$, $a_2$ lying in $E_z$, so $w \in T_yQ + T^R_yO$. Proposition 2.5.6 (preprint 2.36) makes $T^R_yO$ an sc-subspace and Lemma 7.A gives the sc-complement (§IV.16). This is the text's "$T_{z'}(A \cap F^\circ_\sigma) = T_{z'}((\{\lambda\} \times A_\lambda) \cap F^\circ_\sigma) \oplus \xi_{z'}$ … in view of the assumption (2)", made precise: regularity on the stratum is what puts $a_2$ in $E_z$.

## IV.24 Theorem 4.3.1 (preprint 4.23): families and Sard's theorem

**The conclusions at a point** `Chapter4/Families · FibreLocal`: $z = (\lambda, x)$ is a stratified regular point, $A_\lambda$ is near $x$ the image of an sc⁺-retraction on an open subset of $X$, and $A_\lambda$ is in general and in good position at $x$.

**The local step** `Chapter4/Families · exists_local_family`. *For $X$ tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold in general position at every point, every $z_0 \in A$ has an open neighbourhood $U$ and a Lebesgue-null* CV $\subseteq \mathbb{R}^m$ *such that the conclusions hold at every $z \in A \cap U$ with $z_1 \notin$ CV.* Proof:

1. *The chart.* $z_{0,2}$ is smooth `Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff`, so there is a tame chart $\varphi$ of $X$ centred at it (§IV.5); the product chart $\chi = (r \mapsto r - \lambda_0) \times \varphi$ is a chart of $\mathbb{R}^m \times X$ `Chapter3/ParamPerturbation · isChart_rnChart` `Chapter2/FibredProduct · MPolyfold.IsChart.prod` centred at $z_0$, with the tame retraction id $\times\, r_X$ `Chapter2/FibredProduct · ScRetraction.prod` `Chapter2/FibredProduct · ScRetraction.IsTameWith.prod` and coordinates those of $\varphi$ ($\mathbb{R}^m$ has none).
2. *Theorem 4.2.11 (preprint 4.22) in the chart.* The chart retraction $t$ of $A$ in $\chi$ is sc⁺ (§IV.11), and general position at $z_0$ read in $\chi$ is general position at $0$; Theorem 4.2.11 (preprint 4.22) in the chart (§IV.17) gives $N = T_0(\chi A)$ with $N + E^{\min}_0 = E_0$, so the coordinates $\ell$ are onto on $N$, a closed good complement $Y \subseteq E^{\min}_0$, the projection $p$, and $\Gamma : V \to \chi(A)$ over $V = O_V \cap \{\ell \geq 0\} \subseteq N$, with $\ell \circ \Gamma = \ell$.
3. *Sard.* $f(v) = \Gamma(v)_1 + \lambda_0$, the projection $P$ read in the chart, is sc-smooth on $V$ as the first projection `Chapter2/StrongBundle · ScBanach.prodFstOp` composed with $\Gamma$, hence $C^\infty$ (§IV.1); CV $= f($stratified critical points of $f)$ is null (§IV.21). $U = W_A \cap \chi^{-1}(W)$, with $W_A$ the open set of the submanifold chart (§IV.11) and $W$ that of Proposition 4.1.9 (preprint 4.9).

4. *A point $z \in A \cap U$ with $z_1 \notin$ CV.* $\chi(z) \in t(U) \cap W$, so $v_0 = p(\chi z) \in V$ with $\Gamma(v_0) = \chi(z)$ and $f(v_0) = z_1$; hence $v_0$ is not stratified-critical. Since $\ell(D\Gamma(v)h) = \ell(h)$, $D\Gamma(v_0)$ maps the stratum subspace of $\sigma(v_0)$ onto $T_yA \cap E_y$, $y = \Gamma(v_0)$; so $z$ is a stratified regular point, in the chart $\chi$.
5. *The fibre.* The fibre retraction $\rho$ of $f$ at $v_0$ (§IV.22), the sc$^+$-retraction $\tau$ onto the chart image of $A_{z_1}$ (§IV.23), the sc$^+$-retraction $\varphi^{-1} \circ \tau \circ \varphi$ on an open subset of $X$ (§IV.18), whose image is $A_{z_1}$ near $z_2$; the tangent space of $\tau$'s retract contains the vertical tangent vectors of $A$, since $D\rho(v_0) = 1$ on $\ker Df(v_0)$; general position of $A_{z_1}$ at $z_2$ (§IV.23); good position from general position (Lemma 7.B).

**The cover** `Chapter4/Families · theorem_4_3_1_of_subset`. *For $X$ tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with compact closure, in general position at every point, and any set $S \subseteq \mathbb{R}^m$ of parameters, there is $\Sigma \subseteq S$ with $S \setminus \Sigma$ null, consisting of stratified regular values, with the conclusions at every point of every $A_\lambda$, $\lambda \in \Sigma$.* The text covers $\{z \in A : |p(z)| \le \varepsilon\}$ "with the finitely many such neighborhoods $U(z_1), \dots, U(z_k)$". That set need not be compact: in $\mathbb{R} \times \mathbb{R}$, $A = (-2\varepsilon, 2\varepsilon) \times \{2\} \cup \{(\varepsilon(1-x), x) : 0 < x < 1\}$ has compact closure and compact non-empty fibres over $|\lambda| \le \varepsilon$, but $\{|p| \le \varepsilon\}$ contains the open segment, which accumulates at $(\varepsilon, 0) \notin A$, and neighbourhoods of radius less than half the distance to $(\varepsilon, 0)$ have no finite subfamily covering it. A countable subcover suffices: $\mathbb{R}^m \times X$ is metrizable (Theorem 2.3.10 (preprint 2.16)) `Chapter2/Metrizable · MPolyfold.metrizableSpace`, so the compact $\overline{A}$, and with it $A$, is separable, hence second countable; countably many of the neighbourhoods of the local step cover $A$, the countable union of their null sets is null, and $\Sigma = S \setminus \bigcup_i \mathrm{CV}(z_i)$ (Findings, item 103). Some countability is necessary: an uncountable discrete $A$ can have critical values of positive measure; here it comes from the compactness of $\overline{A}$, which the text assumes.

**Theorem 4.3.1 (preprint 4.23)** `Chapter4/Families · theorem_4_3_1_preprint`. *Statement.* Let $X$ be tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, and (1) for $|\lambda| \le \varepsilon$ ($|\lambda|^2 = \sum \lambda_i^2$) the set $A \cap \{z_1 = \lambda\}$ is compact and non-empty, (2) $A$ is in general position at every point. Then there is $\Sigma \subseteq \{|\lambda| < \varepsilon\}$ with null complement in $\{|\lambda| < \varepsilon\}$, consisting of stratified regular values, such that for $\lambda \in \Sigma$ the fibre $A_\lambda$ is compact, a smooth finite-dimensional submanifold of $X$, in general position at each of its points, with a corner structure (Definition 4.1.1 (preprint 4.1)), and tame. *Proof.* The cover for $S = \{|\lambda| < \varepsilon\}$. $A_\lambda$ is the image of the compact $A \cap \{z_1 = \lambda\}$ under the projection to $X$, hence compact. It is a smooth finite-dimensional submanifold and in general position by the conclusions at its points; it is in good position at every point (Lemma 7.B), so Theorem 4.2.7 (preprint 4.18) gives the corner structure and tameness. Hypothesis (2), as printed ("there exists a sc-complement $T_zA$ in $T_zZ$ which is contained in $T_z^RZ$"), is read as Definition 4.2.10 (preprint 4.21), as the proof uses it. Not used: the tameness of the induced structure of $A$, the non-emptiness in (1), $\varepsilon > 0$ (Findings, item 101). Every stratified regular value is a regular value — $T_zA \cap E_z \subseteq T_zA$ — so $\Sigma$ consists of regular values in the text's sense; but the conclusion is not true at every ordinary regular value (§IV.23).

# IV.25 What is not formalized

- **The pull-back of forms on $\partial X$ along a general sc-smooth map** of tame M-polyfolds. The text pulls back forms on $\partial X$ along a map only in the proof of Theorem 6.7.1, for $p_1$ on the faces of the solution set, and that case is formalized: the pull-back to the faces of a finite-dimensional sub-M-polyfold $S$ of a second tame M-polyfold, along an sc-smooth $G : S \to X$ with $d_S = d_X \circ G$ `Chapter4/BoundaryFormsPullback2021 · MPolyfold.isChartFaceForm_faceAlongData` `Chapter4/BoundaryFormsPullback2021 · MPolyfold.faceAlongData_formsD` (Part VI). No statement of Parts I–II uses the

general case, which is estimated at 600–1,500 lines. Gap G.80 is closed: the wedge product on the relative complex is §IV.46 (Findings, Chapter 4, items 158, 167–173, 177, 178).

- **Remark 4.1.3 (preprint 4.3).** The example of §IV.27 closes gap G.31. Not formalized: the argument that an example cannot come from a retraction whose level maps are $C^1$, and the example the 2021 edition adds (the space of its Subsection 1.4.4), which holds as we read it (Findings, Chapter 4, item 159).
- **Question 4.2.4 (preprint: the open question after Proposition 4.15)** is a question, so nothing is to be formalized (G.34). Its settled last sentence is §IV.31.
- The classical manifold structure behind Definition 4.1.1 (preprint 4.1) beyond the $C^\infty$ transitions (§IV.1).

Closed since the first version of this Part: - Lemma 4.1.11 (preprint 4.10) (G.32, §IV.30); - the local form of Remark 4.2.2 (preprint 4.14) (G.33, §IV.31); - the local faces in Theorem 4.2.11 (1) (preprint 4.22) (G.35, §IV.32); - the chart independence of stratified regularity (G.36, §IV.33); - §4.4 apart from the item above (G.37–G.40, §§IV.35–IV.45); - Lemma 4.5.1 between open subsets and the pull-back bundle $TX|X^2$ (G.92, §IV.34); - the wedge product on the forms on $\partial X$ and on the relative complex (G.80, §IV.46).

## IV.26 What the formalization taught us

1. *Centred charts are a construction.* They exist only at smooth points and live in the partial quadrant $C_{\varphi(x)}$, on the set where it agrees with $C$; four theorems of the chapter and Lemma 5.3.3 (preprint 5.11) rely on them, and the text never builds them (§IV.5).
2. *Proposition 4.1.9 (preprint 4.9) is a statement about fixed sets, and its proof is not the implicit function theorem.* The text's route, as we read it, gives a germ over a different complement from the one needed. The direct proof uses only the sc$^+$ property — a mean value estimate on every level and the level-1 continuity of $t$ — and its regularity argument rests on two observations: the estimate is applied at $\Gamma(a)$ with the frame $B_{\Gamma(a)}$, not at 0, and derivatives of $t$ only meet the finite-dimensional smooth tangent space, so the loss of levels does not accumulate (§§IV.6–IV.9). The projection must be the one along the closed good complement, not $Dt(0)$.
3. *Theorem 4.1.2 (preprint 4.2) does not need tameness*, and its compatibility claim needs charts composed with sc-diffeomorphisms between retracts (§IV.10).
4. *The sc$^+$ condition has to be read in charts*, in both directions: a submanifold's chart retraction is sc$^+$ (§IV.11), and an sc$^+$-retraction in a chart is one on an open set (§IV.18). Every statement of §§4.2–4.3 passes through these two facts.
5. *One step in the proof of Lemma 4.2.8 (preprint 4.19) does not hold as we read it* ($C = [0,\infty)$, $z = 1$, $n + y = -\frac{1}{2}$), and the repair is the observation that the complement of $T_0O$ lies in $C \cap (-C)$ at the corner of a centred chart (§IV.14).
6. *General position implies good position* (§IV.16), a step the text uses in Theorem 4.2.11 (4) (preprint 4.22) and Theorem 4.3.1 (preprint 4.23) without stating it.
7. *Theorem 4.3.1 (preprint 4.23) needs stratified regular values and a countable cover*, and the fibre's manifold structure needs the implicit function theorem at corner points with the kernel in good position — the argument of Lemma 5.26 (preprint) — and Theorem 3.8.22 (preprint 3.78) in $C^\infty$ form (§§IV.21–IV.24). Sard's theorem had to be supplied in all dimensions (§IV.20).
8. *In a tame M-polyfold the good position of Theorem 4.1.2 (2) (preprint 4.2) is automatic, and for $d \leq 1$ tameness is not needed.* The case that would block good position, a boundary coordinate vanishing on the tangent space, is excluded by the graph argument of 2021 Lemma 9.2.13. This makes the

sentence of Remark 4.2.2 (preprint 4.14) on $d_A = 1$ a theorem without hypotheses (§IV.31).

9. *The formula for $d\omega$ on p. 197 does not hold as we read it with the printed sign.* The text's bracket is the negative of the classical one, so the classical sign fails too; $(-1)^{\ell+j+1}$ is the sign that works (§IV.38). The formal $d$ is defined by the local alternating sum of base derivatives, and it is chart independent by the symmetry of second derivatives (§§IV.36–IV.37).
10. *Forms on $\partial X$ need a reading, and the reading closes.* $\partial X$ is not an M-polyfold; forms on it are forms on $bX$ that agree over $B$. That $d$ preserves agreement needs no corner strata as M-polyfolds, only the map $r \circ P_i \circ P_j$ into two faces at once (§IV.40).
11. *Stokes' theorem needs the faces as manifolds, signed charts and an added reading.* "A union of smooth faces" is read as face-structured. $[0,1]$ with the quadrant model has no oriented atlas, so integrals use signed charts. The face orientation is the outward vector first (§§IV.41–IV.43).
12. *The Poincaré lemma needs a smooth centre other than 0.* In the proof of Lemma 4.4.8 the convex set $B_\delta(\varphi x) \cap C$ need not contain 0. The homotopy about a smooth centre works with the same proof, and its sc-smoothness is a theorem on parametric integrals of its own (§§IV.44–IV.45).
13. *The text uses $\wedge$ without defining it.* With the classical normalization the Leibniz sign $(-1)^k$ holds as we read it, and associativity and graded commutativity are identities between permutation sums that need no alternation (§IV.39).

## IV.27 Remark 4.1.3's (preprint 4.3) example: a retraction that loses a level

Remark 4.1.3 (preprint 4.3) says that "$X^1 \neq X$" is possible as M-polyfolds although the underlying sets agree, and gives no example. By Proposition 4.1.8 (preprint 4.8) in charts (§IV.4) an example is an sc-retract $O = r(U)$ with $O = O_\infty$ that is not an sc$^+$-retract. It cannot come from a retraction whose level maps $U_m \to E_m$ are $C^1$ — the maps that Proposition 1.2.4 (preprint 1.19), Corollary 1.2.5 (preprint 1.20) and their composites produce: then $T_oO = Dr(o)E_0$ is finite-dimensional by a Baire category argument ($O$ is a countable union of the compact sets $\mathrm{cl}_0\{x \in O : |x|_1 \leq n\}$), and the topologies of $O$ in $E_0$ and $E_1$ agree near $o$ (Findings 141; an argument, not formalized). So the retraction must lose a level in its first derivative, as the splicings of HWZ do, and its sc-smoothness needs a criterion the project did not have (Findings 140–146).

**A criterion with loss.** Let $V$ be finite-dimensional and $L(p)_m : Z_m \to W_m$, $p \in V$, be bounded operators compatible with the inclusions, such that $(p,z) \mapsto L(p)_m z$ is continuous on $V \times Z_m$ — strong continuity, not continuity of $p \mapsto L(p)_m$ in the operator norm — and $p \mapsto A_m(p) = L(p)_m \circ \iota$ is $C^\infty$ as a map into the operators $Z_{m+1} \to W_m$: one level of loss buys all the $p$-derivatives `Chapter4/RemarkFourThreeExample · IsLossFamily` `Chapter4/RemarkFourThreeExample · famLoss` `Chapter4/RemarkFourThreeExample · famDeriv`. *Then $(p,z) \mapsto L(p)z$ is sc-smooth on $V \oplus Z$* `Chapter4/RemarkFourThreeExample · lossFamMap` `Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth`. The map is sc$^0$ by the strong continuity `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc0`, and sc$^1$ with derivative $(\pi,\zeta) \mapsto \partial_p A_0(p)[\pi]z + L(p)\zeta$ at a level-1 point $(p,z)$ `Chapter4/RemarkFourThreeExample · IsLossFamily.differentiable_famLoss` `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc1With`: the increment in $p$ is taken in the operator norm of $A_0$, which needs $z \in Z_1$, and the derivative family is continuous on the tangent levels by the $C^\infty$ dependence of $\partial_p A_m$ `Chapter4/RemarkFourThreeExample · IsLossFamily.continuous_famDeriv`. The tangent map is again a map of the same kind: after the sc-isomorphism that reorders $(p,z,\pi,\zeta)$ as $((p,\pi),(z,\zeta))$ `Chapter4/RemarkFourThreeExample · famSwap` it is the map of the *tangent family* $\tilde{L}(p,\pi)(z,\zeta) = (L(p)z, \partial_p A(p)[\pi]z + L(p)\zeta)$ on $V \times V$ `Chapter4/`

RemarkFourThreeExample · famTangent Chapter4/RemarkFourThreeExample · tangentMapWith_lossFamMap_eq, and $\tilde{L}$ is a loss family Chapter4/RemarkFourThreeExample · IsLossFamily.tangent. Induction on $k$ gives $\mathrm{sc}^k$ for every $k$ Chapter4/RemarkFourThreeExample · isSck_lossFamMap. Only one level of loss is allowed: maps whose $j$-th derivative costs $j$ levels, such as the map $(t,x) \mapsto (\cos(kt)x_k)_k$ of the example after Definition 1.1.15 (preprint 1.14), are not covered.

**The reduction.** For a loss family of projections ($L(p)_0^2 = L(p)_0$), $r(p,z) = (p, L(p)z)$ is an sc-smooth retraction of $V \oplus Z$ Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth_famRetr Chapter4/RemarkFourThreeExample · IsLossFamily.retraction; if the ranges of the $L(p)_0$ are smooth points, $O = O_\infty$ Chapter4/RemarkFourThreeExample · IsLossFamily.retract_subset_smooth; and if the lift $O \to E_1$ is discontinuous, Proposition 4.1.8 (preprint 4.8) says that $O$ is not an $\mathrm{sc}^+$-retract Chapter4/RemarkFourThreeExample · IsLossFamily.not_isScPlusRetract. These are the data of the example Chapter4/RemarkFourThreeExample · RemarkFourThreeData Chapter4/RemarkFourThreeExample · exists_retract_smooth_not_scPlus_of_data.

**The family.** On the weighted sequence space $X$ of the example after Definition 1.1.15 (preprint 1.14) ($|x|_m = \sup_k (k+1)^m |x_k|$; Chapter1/ScOneNotNormContinuous · ScOneExample.X) the moving vector must pass between coordinates of comparable weight, or the level-$m$ operators are unbounded, so the transitions must be fast and the gain per step geometric: the indices are $n_k = 2^k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.nk. With nodes $1/(k+1)$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.node and angles $\theta_k(t)$ turning from $0$ (for $t \geq 1/(k+1)$) to $\pi/2$ (for $t \leq 1/(k+2)$) through Mathlib's smooth transition, flat at both ends Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition, the coefficients $b_k = \sin\theta_{k-1}\cos\theta_k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb vanish except $b_k = \cos\theta_k$ and $b_{k+1} = \sin\theta_k$ at the active block $k = \lfloor 1/t \rfloor - 1$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_eq_zero_of_pos Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt_succ, and all vanish for $t \leq 0$. $L(t) = \sum_k$ (block $k$) Chapter4/RemarkFourThreeFamily · RemarkFourThree.blk Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam is a sum with two non-zero terms, which on level $0$ is the projection onto $b_k e_{n_k} + b_{k+1} e_{n_{k+1}}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_of_pos; on level $m$ its entries carry the weight ratios $((a+1)/(b+1))^m \leq 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.elemL. It is a loss family Chapter4/RemarkFourThreeFamily · RemarkFourThree.isLossFamily_fam: compatible with the inclusions Chapter4/RemarkFourThreeFamily · RemarkFourThree.toBase_fam; uniformly bounded, $\|L(t)_m\| \leq 3 + 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.norm_fam_le; strongly continuous, from the bound and the norm continuity after the dense inclusion by an $\varepsilon/3$ argument Chapter4/RemarkFourThreeFamily · RemarkFourThree.continuous_apply_of_bound; and $t \mapsto L(t)_m \circ \iota$ is $C^\infty$ in the operator norm, by Mathlib's differentiation of series Mathlib/Analysis/Calculus/SmoothSeries · contDiff_tsum: $j$ derivatives of block $k$ are bounded by a polynomial in $k$ of degree $2j$ (Leibniz's rule and the derivatives of the flat transition), while one level of loss gains a factor $(2 + 2^m)2^{-k}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.contDiff_fam_comp. Every $L(t)_0$ is a projection Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_idem with finitely supported, hence smooth, range Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_mem_smooth. The lift is discontinuous: the points $o_k = (1/(k+1), (1/(k+1))e_{n_k})$ lie in $O$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_node_smul_6v and tend to $0$ in $E_0$, while their level-1 coordinates $(2^k+1)/(k+1)\, e_{n_k}$ have norm at least $1$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.not_continuousOn_liftOne. So the data exist Chapter4/RemarkFourThreeFamily · RemarkFourThree.remarkFourThreeData, and there is an sc-retract $O = O_\infty$ that is not an $\mathrm{sc}^+$-retract Chapter4/RemarkFourThreeFamily · remark_4_1_3_retract_example.

**The M-polyfold.** A retract with its single chart $O \to O$ is an M-polyfold Chapter4/

RemarkFourThreeFamily · retractChart Chapter4/RemarkFourThreeFamily · retractPolyfold; if $O \subseteq E_\infty$ and $O$ is not an sc⁺-retract, every point of it is smooth and $1^1_0 : X \to X^1$ is not sc-smooth, for every junk value (Proposition 4.1.8 (preprint 4.8) in charts, §IV.4) Chapter4/RemarkFourThreeFamily · retractPolyfold_smooth_not_toLevelOne. *Remark 4.1.3's (preprint 4.3) example* Chapter4/RemarkFourThreeFamily · remark_4_1_3_example: there is an M-polyfold with $X_0 = X_\infty$ for which $1^1_0$ is not sc-smooth, hence (by the remark's first part, §IV.4) not an sc-diffeomorphism: $X^1 \neq X$ as M-polyfolds. In this example $X$ and $X^1$ even carry different topologies on the same set, since $o_k \to 0$ in $X$ and not in $X^1$. "$X^1 \neq X$" is read as "$1^1_0$ is not sc-smooth"; the M-polyfold is a single retract, which is all the remark needs; its parameter space is $\mathbb{R}$ and its partial quadrant all of $\mathbb{R} \oplus X$, so there is no boundary.

## IV.28 The Springer 2021 edition: Theorems 4.2.11 and 4.3.1, vector fields and the Lie bracket

The 2021 edition changes two statements of §§4.2–4.3 and rewrites §4.4, adding sc-smooth vector fields (Definition 4.4.3, Lemma 4.4.4), a proof of Proposition 4.4.5 (preprint 4.26) in a new Appendix 4.5, and a Poincaré lemma. The files are `Chapter4/Book2021` (the two theorems), `Chapter4/VectorFields` (vector fields and the bracket) and `Chapter4/BracketLevel` (an example on the level of the bracket). The forms of §4.4 and the Poincaré lemma are §§IV.35–IV.45 (Findings, Chapter 4, items 150–159).

**Theorem 4.2.11 (preprint 4.22), 2021 form** Chapter4/Book2021 · MPolyfold.theorem_4_2_11. *Statement.* As in §IV.17–§IV.18, with (1) strengthened: the $d = d_X(x)$ local faces lie in $U$ *and contain* $x$. Formally: a tame chart $\psi$ centred at $x$ with tame retraction $R$ and presentation $T : E \cong \mathbb{R}^n \oplus W$, and an open $U \ni x$ in its domain, such that $d_X(x) = n$, $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold $O$, and for every $i < n$ there are a face $F_i \ni 0$ of the retract in the $i$-th coordinate hyperplane and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap F_i) \subseteq U$; (2)–(4) unchanged. *Proof.* Everything except the new clause is Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint, which already counts the faces *through* $\psi(x)$ — the 2021 reading of (1). For the new clause apply (4) to $\sigma = \{i\}$: it gives a family $(F_j)_{j\in\{i\}}$ of faces through 0 in the coordinate hyperplanes and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap \bigcap_j F_j) \subseteq U$; since the index set has one element, the intersection over it is $F_i$ itself (an extensionality step on the preimage set). The local faces here are those of the chart model; the faces of the neighbourhood $U$ itself are §IV.32.

**Theorem 4.3.1 (preprint 4.23), 2021 form** Chapter4/Book2021 · theorem_4_3_1. *Statement.* $X$ tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, (1) $A \cap \{z_1 = \lambda\}$ compact for $|\lambda| \leq \varepsilon$ — no longer "non-empty" — and (2) general position at every point. Then there is a *measurable* $\Sigma \subseteq B = \{|\lambda| < \varepsilon\}$ with $B \setminus \Sigma$ null, of stratified regular values Chapter4/Families · IsStratRegularValue, such that each $A_\lambda$, $\lambda \in \Sigma$, is compact, a smooth finite-dimensional submanifold in general position at its points, with a corner structure, and tame. *Proof.* The cover of §IV.24 for $S = B$ Chapter4/Families · theorem_4_3_1_of_subset gives $\Sigma_0 \subseteq B$ with $B \setminus \Sigma_0$ null and the local conclusions at every point of every fibre. $B$ is measurable (a strict sublevel set of a continuous function). Let $N$ be a measurable null set containing $B \setminus \Sigma_0$ (Mathlib's measurable hull Mathlib/MeasureTheory/Measure/MeasureSpaceDef · MeasureTheory.toMeasurable) and $\Sigma = B \setminus N$: measurable, inside $\Sigma_0$, with $B \setminus \Sigma \subseteq N$ null. For $\lambda \in \Sigma$, the fibre $A_\lambda$ is the image of the compact set $A \cap \{z_1 = \lambda\}$ (hypothesis (1), $|\lambda| < \varepsilon$) under the second projection, hence compact; it is a submanifold and in general and good position by the local conclusions; Theorem 4.2.7 (preprint 4.18) Chapter4/Submanifolds · MPolyfold.theorem_4_2_7 gives the corner structure and tameness. Neither non-emptiness, nor the tameness of $A$, nor $\varepsilon > 0$ is used.

**Vector fields in a local model (Definition 4.4.3).** For a local model $(O, C, E)$ a map $A : E_1 \to E_0$ is

a *vector field* if $x \mapsto (x, A(x))$ is an sc-smooth map between retracts from $(O^1, C^1, E^1)$ to the tangent $(TO, TC, TE)$ `Chapter4/VectorFields · IsRetractVectorField` — the text's sc-smooth section of $TX \to X^1$, read in a chart: its domain is $X^1$. The zero map is one on every sc-smooth retract: $(x, 0)$ lies in $TO$ `Chapter2/TangentBundle · mk_zero_mem_retractTangent` and $x \mapsto (x, 0)$ is the pair of the retraction and the zero operator `Chapter4/VectorFields · isRetractVectorField_zero`. Consequences: $(x, A(x)) \in TO$ for $x \in O_1$ `Chapter4/VectorFields · IsRetractVectorField.mem_retractTangent`; for a retraction $r$ onto $O$, $\tilde{A} = A \circ r^1$ is sc-smooth on $U^1$, as the second component of the section composed with the shifted retraction `Chapter4/VectorFields · IsRetractVectorField.isScSmooth_comp`; and $Dr(x)A(x) = A(x)$ for $x \in O_1$, formula (4.8), because $T_xO$ is the fixed space of $Dr(x)$ `Chapter4/VectorFields · IsRetractVectorField.scDeriv_apply`.

**Vector fields on an M-polyfold (Definition 4.4.3).** $A : X^1 \to TX$ is an sc-smooth vector field if it is sc-smooth for the M-polyfold structures of $X^1$ and of $TX$ and $p(A(x)) = x$ `Chapter4/VectorFields · MPolyfold.IsScVectorField`. The zero section takes at $x$ the zero tangent vector in a chart chosen around $x$ `Chapter4/VectorFields · MPolyfold.zeroTangentPre` `Chapter4/VectorFields · MPolyfold.zeroVectorField`; the tangent transition maps are linear in the fibre, so in every chart $\psi$ around $x$ it reads $(\psi(x), 0)$ `Chapter4/VectorFields · MPolyfold.zeroTangentPre_transport` `Chapter4/VectorFields · MPolyfold.tangentChartMap_zeroVectorField`. Hence it is continuous (the preimage of a basic open set of $TX$ is a basic open set of $X^1$) `Chapter4/VectorFields · MPolyfold.continuous_zeroVectorField` and, in each chart, the sc-smooth map $z \mapsto (z, 0)$: the zero section is an sc-smooth vector field `Chapter4/VectorFields · MPolyfold.isScVectorField_zero`.

**Raising the index (Lemma 4.4.4, local form, $i = 1$).** *For a retraction $r$ onto $O$, $T(O^1) = (TO)^1$* `Chapter4/VectorFields · retractTangent_shift`: a point $(x, h)$ of $E_2 \oplus E_1$ lies in the tangent of the shifted retract iff $Tr^1(x, h) = (x, h)$, and the base point of $Tr^1(x, h)$ in $E_1 \oplus E_0$ is $Tr(x, h)$, because the derivative of the shifted retraction is the level-1 restriction of $Dr$ `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn`; the inclusion of levels being injective, the two fixed-point conditions agree. *If $A$ is a vector field on $(O, C, E)$, its level map $A_1 : E_2 \to E_1$ is a vector field on $(O^1, C^1, E^1)$* `Chapter4/VectorFields · IsRetractVectorField.shift`: shifting the sc-smooth map $x \mapsto (x, A(x))$ gives an sc-smooth map from $O^2$ into $(TO)^1$ `Chapter2/LevelPolyfold · IsScSmoothRetractMap.shift`, which by the identity just proved is $T(O^1)$, and its level-1 values are $(x, A_1(x))$ `Chapter1/ScSmooth · toBase_levelMap`. Larger $i$ by iteration; the M-polyfold form, through the identification $T(X^1) \cong (TX)^1$ of Proposition 2.3.17 (gap G.6, closed), is in §IV.29, for $i = 1$ and for every $i$.

**The second derivative and its symmetry.** For $f$ sc² on $U$, $z \in U_2$ and $a, b \in E_1$, $D^2f(z)(a, b)$ is defined as the second component of $D(Tf)(z, a)(b, 0)$ — the derivative of $z \mapsto Df(z)a$ in the direction $b$ `Chapter4/VectorFields · scSecondDeriv`. (The text writes $D^2r(x)(A(x), B(x))$ with $A(x), B(x) \in E_1$, where $r : U_2 \to E_0$ has a second derivative only in directions of $E_2$.) The vertical part of $D(Tf)$ is $D(Tf)(z, a)(0, l) = (0, Df(z)l)$ `Chapter4/VectorFields · scDeriv_tangentMap_inr`. *Symmetry* `Chapter4/VectorFields · scSecondDeriv_symm`: for $U$ relatively open in a partial quadrant, $D^2f(z)(a, b) = D^2f(z)(b, a)$. (i) On $E_2$: $f \circ \iota : U_2 \to F_0$ is $C^2$ within $U_2$ (sc² gives $C^2$ on level 2), every point of $U_2$ lies in the closure of the interior of $U_2$ `Chapter4/VectorFields · IsRelOpenInQuadrant.mem_closure_interior_lvlSet`, and Mathlib's symmetry of second derivatives within a set applies `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffWithinAt.isSymmSndFDerivWithinAt`; for $v, w \in E_2$ the iterated derivative within $U_2$ is $D^2f(z)(\iota v, \iota w)$ `Chapter4/VectorFields · fderivWithin_fderivWithin_eq`, by the derivative of $Df$ along level 2 `Chapter4/VectorFields · hasFDerivWithinAt_two` `Chapter4/VectorFields · hasFDerivWithinAt_scDeriv_apply`. (ii) On $E_1$: $(a, b) \mapsto D^2f(z)(a, b)$ is continuous ($D(Tf)$ is continuous on level 0), $E_2 \times E_2$ is dense in $E_1 \times E_1$, and two continuous functions equal on a dense set are equal.

**The bracket in a local model (§4.5.1).** For a retraction $r$ onto $O$ and $x \in E_2$, $[A, B](x) = D\tilde{A}(x)B_1(x) -$

$D\tilde{B}(x)A_1(x)$ with $\tilde{A} = A \circ r^1$, $\tilde{B} = B \circ r^1$ Chapter4/VectorFields · retractLieBracket; $[A,B] = -[B,A]$ Chapter4/VectorFields · retractLieBracket_antisymm. On $O^2$, $(A \circ r^1)_1 = A_1$, since $r^1$ fixes $O^1$ Chapter4/VectorFields · levelMap_comp_shift_eq. *The differentiated identity* Chapter4/VectorFields · IsRetractVectorField.diff_identity: for $x \in O_2$ and $b$ with $Dr^1(x)b = b$, $D^2r(x)(A(x),b) + Dr(x)D\tilde{A}(x)b = D\tilde{A}(x)b$. The map $\alpha : y \mapsto (r^1(y), \tilde{A}(y))$ is sc-smooth on $U^1$ with values in $TO$ (as $r^1(y) \in O^1$), so $Tr \circ \alpha = \alpha$ on all of $U^1$, $TO$ being the fixed set of $Tr$. Differentiate at $x$ in the direction $b$ by the chain rule Chapter1/ChainRule · theorem_1_3_1'; the derivative of $\alpha$ is $(Dr^1(x)b, D\tilde{A}(x)b) = (b, D\tilde{A}(x)b)$, and $\alpha_1(x) = (x, A_1(x))$ since $r^1$ fixes $x$. Split $(b, D\tilde{A}(x)b) = (b,0) + (0, D\tilde{A}(x)b)$: the first part gives $D^2r(x)(A(x),b)$ by definition, the second the vertical part $Dr(x)D\tilde{A}(x)b$; compare second components. *Tangency* Chapter4/VectorFields · retractLieBracket_mem_tangent: by Lemma 4.4.4, $B_1(x)$ and $A_1(x)$ are fixed by $Dr^1(x)$; apply the identity to $\tilde{A}$ in the direction $B_1(x)$ and to $\tilde{B}$ in the direction $A_1(x)$, subtract, and cancel the second derivatives by symmetry: $Dr(x)[A,B](x) = [A,B](x)$.

**Proposition 4.4.5 (preprint 4.26), local form** Chapter4/VectorFields · proposition_4_4_5_local. *Statement.* For vector fields $A$, $B$ on $(O,C,E)$ with retraction $r$: $(x,[A,B](x)) \in TO$ for $x \in O_2$, and $x \mapsto (x,[A,B](x))$ is an sc-smooth map from $(O^2,C^2,E^2)$ to $(TO,TC,TE)$. *Proof.* The first clause is tangency, $T_xO$ being the fixed space of $Dr(x)$ at $x \in O_1$. For sc-smoothness between retracts it suffices to give an sc-smooth extension to $U^2$ that agrees on $O^2$ Chapter2/Retracts · isScSmoothRetractMap_iff_exists. First component: the inclusion $E^2 \to E^1$ composed with $r^2$. Second component: $y \mapsto D\tilde{A}(r^2y)(\tilde{B})_1(y) - D\tilde{B}(r^2y)(\tilde{A})_1(y)$. Each term is the second component of the tangent map $T\tilde{A}$, which is sc-smooth because $\tilde{A}$ is, composed with the sc-smooth pair $(r^2,(\tilde{B})_1)$ Chapter4/VectorFields · isScSmooth_scDeriv_apply; and $B_1 \circ r^2 = (\tilde{B})_1$ on $U^2$ Chapter4/VectorFields · levelMap_shift_shift_eq, so the extension is $[A,B] \circ r^2$. The statement on an M-polyfold, with Lemma 4.5.1, whose proof the text leaves to the reader, is in §IV.29 (gap G.38, closed).

**The level of the bracket: an example** Chapter4/BracketLevel · lieBracket_level_counterexample. The text concludes from tangency that the bracket lies in "the fiber $(T_xO)_1$ whenever $x \in O_2$" (p. 200) and that for $x \in O_2$ it "belongs to the level 1" (p. 201). Read as $[A,B](x) \in E_1$, this does not hold as we read it. *Generalities.* For $O = E$ (no boundary, identity retraction Chapter4/BracketLevel · idRetraction), $TO = TE$ Chapter4/BracketLevel · retractTangent_univ; every sc-operator $S : E^1 \to E$ is a vector field Chapter4/BracketLevel · isRetractVectorField_scOperator, and the bracket of two is the commutator $S_0T_1 - T_0S_1$, the derivative of a linear map being itself Chapter4/BracketLevel · retractLieBracket_scOperator. *The space.* $E$ is the weighted sequence space with $|x|_m = \sup_k (k+1)^m |x_k|$ Chapter1/ScOneNotNormContinuous · ScOneExample.X, level $m$ being $c_0$ in the coordinates $(k+1)^m x_k$. *The operators.* $(Sx)_k = (-1)^k(k+1)x_k$ Chapter4/BracketLevel · BracketLevel.opS, on every level the diagonal $\mathrm{diag}((-1)^k)$ in the weighted coordinates; $(Tx)_k = (k+1)x_{k+1}$ Chapter4/BracketLevel · BracketLevel.opT, on level $m$ the weighted shift with weights $((k+1)/(k+2))^{m+1} \le 1$ Chapter4/BracketLevel · BracketLevel.wshift Chapter4/BracketLevel · BracketLevel.tw; both are vector fields Chapter4/BracketLevel · BracketLevel.isRetractVectorField_opS Chapter4/BracketLevel · BracketLevel.isRetractVectorField_opT. *The computation.* $[S,T]x = ((-1)^k(k+1)(2k+3)x_{k+1})_k$ — the sign $(-1)^k$ makes the two terms add instead of cancelling; with polynomial weights the commutator of a non-oscillating diagonal operator and a shift would lose only one level. For $x_k = (k+1)^{-3}$, a point of $E_2$ Chapter4/BracketLevel · BracketLevel.xPt, the level-1 coordinates $(k+1)[S,T](x)_k$ have absolute value $P^3 + P^2 \ge 1/4$, $P = (k+1)/(k+2) \ge 1/2$, so they do not tend to 0 and $[S,T]x \notin E_1$ Chapter4/BracketLevel · BracketLevel.bracket_not_mem_level_one. The universal statement therefore fails Chapter4/BracketLevel · not_forall_lieBracket_mem_level_one. Here $T_xO = E_0$, so $(T_xO)_1 = E_1$ and the sentence of p. 200 is covered too. What holds, and what Proposition 4.4.5 states, is the shifted statement $x \in O_{m+2} \Rightarrow [A,B](x) \in E_m$: $[A,B]$ is a tangent vector at the point $x$ of level 1 of $X^1$ — the

local form above. A first example on a weighted $\ell^2$ space ($E_m = \ell^2$ with weights $4^{mk}$, $S = \mathrm{diag}(2^k)$, $(Tx)_k = 2^k x_{k+1}$, $x_j = 4^{-j}/j$) is sketched in the header of `Chapter4/VectorFields` and not formalized, the project having no weighted $\ell^2$ sc-structure.

**What remains.** Nothing of this subsection: Lemma 4.4.4 and Proposition 4.4.5 on an M-polyfold and Lemma 4.5.1 are §IV.29 and §IV.34; Definitions 4.4.1–4.4.2, $d$ and $d \circ d = 0$, Definition 4.4.6, Theorem 4.4.7 and Lemmas 4.4.8 and 4.5.2 are §§IV.35–IV.45. The example the 2021 edition adds to Remark 4.1.3 (preprint 4.3), the space of its Subsection 1.4.4, for which $1^1_0$ is not sc-smooth because Theorem 4.1.2 (preprint 4.2) would make it a manifold of constant local dimension, holds as we read it and is not formalized (Findings, Chapter 4, item 159); gap G.31 is closed by the example of §IV.27.

## IV.29 The Springer 2021 edition: vector fields and the bracket on an M-polyfold (Lemmas 4.4.4, 4.5.1, Proposition 4.4.5)

The local forms of §IV.28 are carried to M-polyfolds in `Chapter4/VectorFieldsGlobal`, on top of the identification $\Phi : T(X^1) \cong (TX)^1$ of Part I's Proposition 2.3.17 `Chapter2/TangentLift · MPolyfold.tangentShiftEquiv Chapter2/TangentLift · proposition_2_3_17`. The plan: Lemma 4.5.1 in a local model, by the computation of §4.5.2; principal parts in charts and the independence of the chart; the bracket $X^2 \to TX$; Lemma 4.5.1 on M-polyfolds; Lemma 4.4.4 for $i = 1$ and for every $i$ (gap G.38 closed; Findings, Chapter 4, items 160–164).

**Lemma 4.5.1, local form (the computation of §4.5.2)** `Chapter4/VectorFieldsGlobal · retractLieBracket_natural`. *Statement.* Let $r$, $r'$ be retractions onto $O$, $O'$, $f$ an sc-smooth map from the retract $O$ to the retract $O'$, $g = f \circ r$, and $A$, $B$ vector fields on $O$, $A'$, $B'$ on $O'$ with $Dg(x)A(x) = A'(f(x))$ and $Dg(x)B(x) = B'(f(x))$ for $x \in O_1$. Then $Dg(x)[A,B](x) = [A',B'](f(x))$ for $x \in O_2$. The text states it for a local sc-diffeomorphism $\varphi : X \to X$ with the relations at points of $X_2$, and leaves the proof to the reader (p. 201); §4.5.2 is the computation, and it differentiates the relations, so they are assumed on the open set $O_1$ of level 1. *Proof.* (i) *The relation differentiated* `Chapter4/VectorFieldsGlobal · IsRetractVectorField.diff_related`: the map $y \mapsto Dg(r^1 y)\tilde{A}(y) - \tilde{A}'(g_1(y))$, $\tilde{A} = A \circ r^1$, $\tilde{A}' = A' \circ r'^1$, vanishes on $U^1$ (since $r^1 y \in O_1$ and $g \circ r = g$); differentiating at $x \in O_2$ in a direction $b$ with $Dr^1(x)b = b$, as in §IV.28's differentiated identity, gives $D^2 g(x)(A(x),b) + Dg(x)D\tilde{A}(x)b = D\tilde{A}'(g_2(x))Dg^1(x)b$. (ii) *The relation on level* 1 `Chapter4/VectorFieldsGlobal · IsRetractVectorField.related_shift`: the derivative of the shifted map $g^1$ at $x \in O_2$ is the level-1 restriction of $Dg(x)$ `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn`, so $Dg^1(x)B_1(x) = B'_1(g_2(x))$; and $g_2 = f_2$ on $O_2$ `Chapter4/VectorFieldsGlobal · levelMap_levelMap_comp_eq`. (iii) Apply (i) to $A$ with $b = B_1(x)$ and to $B$ with $b = A_1(x)$ (both fixed by $Dr^1(x)$, §IV.28), rewrite the right sides by (ii), subtract, and cancel $D^2 g(x)(A(x),B(x)) - D^2 g(x)(B(x),A(x))$ by symmetry `Chapter4/VectorFields · scSecondDeriv_symm`: $Dg(x)[A,B](x) = D\tilde{A}'(f_2 x)B'_1(f_2 x) - D\tilde{B}'(f_2 x)A'_1(f_2 x) = [A',B'](f(x))$. §4.5.2 ends with "The proof of Proposition 4.5.1 is complete"; there is no Proposition 4.5.1, presumably Lemma 4.5.1 is meant.

*On an open part.* A transition map of charts is defined only on an open part $O \cap W$ of a model. A vector field on $O$ is one on $O \cap W = r'(U')$ for the restricted retraction $r'$ `Chapter4/VectorFieldsGlobal · IsRetractVectorField.restrict`, and at points of $(O \cap W)_2$ the two brackets agree, since $r'$ is $r$ on the smaller open domain $U'$ and a derivative on an open subdomain is the restriction of the derivative `Chapter4/VectorFieldsGlobal · retractLieBracket_restrict`. So for $f$ sc-smooth on $O \cap W$ with $Tf(z,A(z)) = (f(z),A'(f(z)))$ and $Tf(z,B(z)) = (f(z),B'(f(z)))$ on $(O \cap W)_1$, also $Tf(x,[A,B](x)) = (f(x),[A',B'](f(x)))$ on $(O \cap W)_2$ `Chapter4/VectorFieldsGlobal · retractLieBracket_natural_inter`.

**Principal parts and the independence of the chart.** Tangent charts change by $T\psi = T(\psi \circ \varphi^{-1}) \circ T\varphi$ over the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_change` (formula (2.9)). Choose a retraction onto the model of each chart `Chapter4/VectorFieldsGlobal · MPolyfold.chartRetraction Chapter4/VectorFieldsGlobal · MPolyfold.chartRetraction_retract`. The *principal part* of a vector field $A$ in $\varphi$ is the second component of $T\varphi(A(\varphi^{-1}z))$ `Chapter4/VectorFieldsGlobal · MPolyfold.principalPart`, so $T\varphi(A(y)) = (\varphi(y), A_\varphi(\varphi(y)))$ for $y$ of level 1 in the domain of $\varphi$ `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_eq_principalPart`; read through the charts $\varphi^1$ of $X^1$ and $T\varphi$ of $TX$, the sc-smoothness of $A$ says that $A_\varphi$ is a vector field on the model `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.isRetractVectorField_principalPart`. The principal parts in two charts are related by the transition map on level 1 of the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.principalPart_related`. The bracket of the principal parts `Chapter4/VectorFieldsGlobal · MPolyfold.chartLieBracket` satisfies the local Proposition 4.4.5 `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.chartLieBracket_spec`, and — the use of Lemma 4.5.1 that p. 201 names — the brackets in $\varphi$ and $\psi$ are related by $T(\psi\varphi^{-1})$ on level 2 of the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.chartLieBracket_change`: the overlap is $O \cap W$ for an open $W$, and the local form on an open part applies.

**Proposition 4.4.5 (preprint 4.26) on an M-polyfold** `Chapter4/VectorFieldsGlobal · proposition_4_4_5`. *Statement.* For sc-smooth vector fields $A$, $B$ on $X$, the bracket $[A,B] : X^2 \to TX$ is sc-smooth and $p([A,B](x)) = x$. The text's "sc-smooth section of $TX|X^2 \to X^{2}$" is read as an sc-smooth map $X^2 \to TX$ covering the inclusion $X^2 \to X^1$; the pull-back bundle itself, and the bracket as its section in the text's form, are §IV.34. *Construction.* At $x \in X^2$ take the chart $\varphi$ chosen around $x$; the level lift of $x$ in the chart $(\varphi^1)^1$ of $X^2$ lies in $O_2$ and is $\varphi(x)$ on level 1 `Chapter4/VectorFieldsGlobal · MPolyfold.levelLift_levelChart_spec`, so $(\varphi(x), [A_\varphi, B_\varphi](\varphi(x)))$ lies in $TO$ and defines a tangent vector at $x$ `Chapter4/VectorFieldsGlobal · MPolyfold.lieBracketPre Chapter4/VectorFieldsGlobal · MPolyfold.lieBracket` over $x$ `Chapter4/VectorFieldsGlobal · MPolyfold.tangentProj_lieBracket`. *Independence of the chart* `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_lieBracket`: in every chart $\psi$ around $x$, $T\psi([A,B](x)) = (\psi(x), [A_\psi, B_\psi](\psi(x)))$, by the change of tangent charts and the relation of the brackets in $\varphi$ and $\psi$. *Proof of the statement.* On the domain of $(\psi^1)^1$, $T\psi \circ [A,B]$ is the section $z \mapsto (z, [A_\psi, B_\psi](z))$ composed with the chart `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_lieBracket_eq`. That section is continuous and sc-smooth by the local proposition, so $[A,B]$ is continuous (the preimage of a basic open set of $TX$ is open in each chart domain of $X^2$) `Chapter4/VectorFieldsGlobal · MPolyfold.continuous_lieBracket`, and sc-smooth in the charts $(\psi^1)^1$ of $X^2$ and $T\psi$ of $TX$.

**Lemma 4.5.1 on M-polyfolds** `Chapter4/VectorFieldsGlobal · lemma_4_5_1`. *Statement.* Let $f : X \to Y$ be sc-smooth, with level maps $f_1 : X^1 \to Y^1$ `Chapter4/VectorFieldsGlobal · MPolyfold.liftLevelOneMap` and $f_2 : X^2 \to Y^2$ `Chapter4/VectorFieldsGlobal · MPolyfold.levelTwoMap`, and $A$, $B$ on $X$, $A'$, $B'$ on $Y$ sc-smooth vector fields with $Tf \circ A = A' \circ f_1$ and $Tf \circ B = B' \circ f_1$ on $X^1$. Then $Tf \circ [A,B] = [A',B'] \circ f_2$ on $X^2$. This is more general in the map than the text's local sc-diffeomorphism $X \to X$; one between open subsets of $X$ is §IV.34. *Proof.* Fix $x \in X^2$, a chart $\varphi$ around $x$ and $\psi$ around $f(x)$. Both sides lie over $f(x)$; in $T\psi$ the right side is $(\psi f(x), [A'_\psi, B'_\psi](\psi f(x)))$ by the independence of the chart, and on level 1, $(\psi f \varphi^{-1})_1(\varphi(x)) = \psi(f(x))$ `Chapter4/VectorFieldsGlobal · MPolyfold.levelMap_coord_levelLift`. The left side, through $\varphi$, is $T(\psi f \varphi^{-1})(\varphi(x), [A_\varphi, B_\varphi](\varphi(x)))$. The map $\psi f \varphi^{-1}$ is sc-smooth on the open part of the model of $\varphi$ that $\varphi(f^{-1}(\text{domain of } \psi))$ cuts out, and the hypotheses read in the charts say that it relates $A_\varphi$, $B_\varphi$ to $A'_\psi$, $B'_\psi$ on level 1 there; the local form on an open part gives the equality.

**Lemma 4.4.4 on an M-polyfold,** $i = 1$ `Chapter4/VectorFieldsGlobal · lemma_4_4_4`. *Statement.* For an sc-smooth vector field $A$, $A^{(1)} = \Phi^{-1} \circ A_1 : X^2 \to T(X^1)$ `Chapter4/VectorFieldsGlobal · MPolyfold.raiseIndex`, $A_1 : X^2 \to (TX)^1$ the level-1 map of $A$ `Chapter4/VectorFieldsGlobal · MPolyfold.vectorFieldLevelOne`, is an

sc-smooth vector field on the M-polyfold $X^1$, and $\Phi(A^{(1)}(x)) = A(x)$ in $(TX)^1$. *Proof.* $A_1$ is sc-smooth, as the level-1 map of the sc-smooth $A$; $\Phi^{-1}$ is sc-smooth (Proposition 2.3.17), so $A^{(1)}$ is; $\Phi$ commutes with the projections to $X^2$, so $p(A^{(1)}(x)) = x$ follows from $p(A(x)) = x$; and $\Phi \circ \Phi^{-1}$ is the identity. Every $i$ follows by applying the lemma to $X^1$, $X^2$, …, as the next paragraph does. A witness for the hypotheses of the section: the zero section `Chapter4/VectorFields · MPolyfold.isScVectorField_zero`.

**Lemma 4.4.4 on an M-polyfold, every $i$** `Chapter2/TangentLiftAll · lemma_4_4_4_all`. To state the lemma once for all $i$, the M-polyfolds are bundled with their spaces `Chapter2/TangentLiftAll · BundledMPolyfold`, so that the lifts $X^0 = X$, $X^{i+1} = (X^i)^1$ form a family `Chapter2/TangentLiftAll · BundledMPolyfold.lift`. Part I's Proposition 2.3.17 for every $i$ `Chapter2/TangentLiftAll · proposition_2_3_17_all` gives the sc-diffeomorphisms $\Phi_i : T(X^i) \to (TX)^i$ `Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv` and $J_i : (X^1)^i \to (X^i)^1$ `Chapter2/TangentLiftAll · BundledMPolyfold.shiftLiftEquiv`, and $\Phi_i$ commutes with the inclusions into $TX$. *Statement.* With $A^{(0)} = A$ and $A^{(i+1)} = (A^{(i)})^{(1)}$, the case $i = 1$ on $X^i$ `Chapter2/TangentLiftAll · BundledMPolyfold.raiseIndexAll`: for every $i$, $A^{(i)}$ is an sc-smooth vector field on $X^i$; its image in $TX$ at $x \in X^{i+1}$ is $A(x)$ `Chapter2/TangentLiftAll · BundledMPolyfold.tangentIncl_raiseIndexAll`; and $\Phi_i \circ A^{(i)} \circ J_i$ is the $i$-th index raise of $A$, viewed as an sc-smooth map $X^1 \to TX$ `Chapter2/TangentLiftAll · BundledMPolyfold.vectorFieldScMap` `Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_raiseIndexAll` — the diagram printed before the lemma. *Proof.* The first claim is the case $i = 1$, applied $i$ times. The second by induction: the case $i = 1$ for $X^i$ says that $A^{(i+1)}(x)$ and $A^{(i)}(x)$ agree after $T(X^{i+1}) \to T(X^i)$. For the third, both sides are compared after the inclusion $(TX)^i \to TX$, which is injective: on the left it is the inclusion $T(X^i) \to TX$ after $\Phi_i$, so the second claim applies; on the right it is $A$ on the underlying point. *Witness:* the zero vector field, for every $i$ `Chapter2/TangentLiftAll · BundledMPolyfold.lemma_4_4_4_all_zero` (Findings, Chapter 4, item 164).

**What remains.** Nothing: Lemma 4.5.1 between open subsets and the pull-back bundle $TX|X^2 \to X^2$ are §IV.34 (gap G.92 closed), the forms of §4.4 §§IV.35–IV.45.

## IV.30 Lemma 4.1.11 (preprint 4.10): the section $u \mapsto (1-p)(u - t(u))$ is sc-Fredholm

The text proves Proposition 4.1.9 (preprint 4.9) through this lemma and the implicit function theorem. The formalization proves the proposition directly (§§IV.6–IV.9), so the lemma is not used there. It is proved here on its own, in the 2021 notion of an sc-Fredholm section.

**Statement** `Chapter4/GapP1_Lemma4111 · lemma_4_1_11`. Take the data of Proposition 4.1.9 (preprint 4.9): an sc$^+$-retraction $t : U \to U$ of a relatively open $U \subseteq C$ onto $O = t(U) \ni 0$, $N = T_0O$ in good position, a good complement $Y$ with $E = N \oplus Y$ an sc-splitting, and $p$ the projection onto $N$ along $Y$. Then the section $u \mapsto (u, (1-p)(u - t(u)))$ of the local strong bundle $U \lhd Y \to U$ is an sc-Fredholm section in the sense of 2021 Definition 3.1.16: it is sc-smooth, it is regularizing, and it is an sc-Fredholm germ at every smooth point. It holds on all of $U$, not only near $0$. That $E = N \oplus Y$ is an sc-splitting is the text's "good sc-complement" (Definition 3.1.24 asks for an sc-complement), not an added hypothesis. $N$ is smooth and finite-dimensional by Lemma 4.1.7 (preprint 4.7) (§IV.3).

**The bundle and the section.** The local strong bundle $U \lhd Y$ lives over the M-polyfold $U$, which has one chart and the identity retraction. It is a strong bundle with one strong bundle chart `Chapter4/GapP1_LocalStrongBundle · LocalBundle.bundle`. The section `Chapter4/GapP1_LocalStrongBundle · LocalBundle.sec` is sc-smooth `Chapter4/GapP1_LocalStrongBundle · LocalBundle.isScSmoothSection_sec`: $t$ is sc-smooth, and $1 - p$ is

an sc-operator $E \to Y$ Chapter4/GapP1_Lemma4111Split · qOp.

**Regularizing.** Let $u \in U_m$ with $(1-p)(u - t(u)) \in Y_{m+1}$. Write

$$u = (1-p)(u - t(u)) + t(u) + p(u - t(u)).$$

Here $t(u) \in E_{m+1}$ because $t$ is sc$^+$, and $p(u - t(u)) \in N \subseteq E_\infty$. So $u \in E_{m+1}$.

**The germ at a smooth point $e$.** Go into the strong bundle chart centred at $e$ Chapter4/GapP1_LocalStrongBundle · LocalBundle.centred. Its retraction is pointwise the identity, so the principal part fills itself and no filler has to be built Chapter4/GapP1_LocalStrongBundle · isFilling2021_of_R_eq_self. In this chart the principal part is $g(v) = (1-p)(v + e - t(v+e))$ Chapter4/GapP1_LocalStrongBundle · LocalBundle.principal_centred. The text's sc$^+$-section is $s(v) = (1-p)(e - t(v+e))$. It is sc$^+$ because $t$ is, read as a map into $Y^1$. Then $g - s = (1-p)v$, and it remains to conjugate this germ to a basic germ.

**The splitting $E = (N \oplus B) \oplus W$.** Let $\lambda$ be the coordinates of the partial quadrant. Their values on the smooth points of $Y$ already fill $\lambda(Y)$: the smooth points are dense and $\lambda(Y)$ is finite-dimensional Chapter4/GapP1_Lemma4111Split · map_le_map_smooth. So there are smooth $e_1, \dots, e_d \in Y$ and functionals $\ell_j$ with $\ell_i(e_j) = \delta_{ij}$ such that, on $Y$, the $\ell_j$ vanish exactly where $\lambda$ does Chapter4/GapP1_Lemma4111Split · exists_splitData. Put $B = \operatorname{span}(e_j)$, a smooth subspace inside $Y_\infty$, and $W = Y \cap \bigcap_j \ker \ell_j$ Chapter4/GapP1_Lemma4111Split · splitW. $W$ is an sc-subspace Chapter4/GapP1_Lemma4111Split · isScSubspace_splitW, and $Y = B \oplus W$. Adding a vector of $W$ changes no coordinate. So near 0 the partial quadrant is $((N \oplus B) \cap C) \oplus W$, and the quadrant on $N \oplus B$ comes from the corner coordinates, not from good position Chapter4/GapP1_Lemma4111Split · isPartialQuadrant_preimage_quadrant. In the coordinates $(a+b, w)$, $a \in N$, $b \in B$, $w \in W$, the germ $(1-p)v$ becomes $(a+b, w) \mapsto (b, w)$. This is a basic germ with contraction part 0 Chapter4/GapP1_Lemma4111Split · isConjugateToBasicGerm_qOp.

The proof follows the text. We read it differently from the printed text in a few places, all recorded in Findings, Chapter 4, item 57: the sign of $D\mathbf{f}(u)$; $Dt(u)$, not $D\mathbf{f}(u)$, is the sc$^+$-operator; "a perturbation of the identity" is said of an operator into $Y$; $L(Y)$ is meant where $L(T_0O)$ is printed; and the text omits the interior points of $(T_0O \oplus B) \cap C$.

## IV.31 Remark 4.2.2 (preprint 4.14) in both wordings, and Question 4.2.4 (preprint: the question after Proposition 4.15)

**The preprint wording, $d_X(a) = 0$.** Let $A$ be a smooth finite-dimensional submanifold. The open piece $A \cap \{d_X = 0\}$ is again a smooth finite-dimensional submanifold. Its induced structure has a corner structure with degeneracy index 0 everywhere: a manifold without boundary Chapter4/RemarkLocalForm · MPolyfold.remark_4_2_2_interior. The pointwise form is Chapter4/RemarkLocalForm · MPolyfold.remark_4_2_2_near. The piece is open because $d_X$ is upper semicontinuous. The corner structure is the theorem of §IV.11 for $d_X = 0$ Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_degIndex_eq_zero, applied to this piece.

**The 2021 wording, $d_A(a) = 0$** Chapter4/GapP1_Remark422 · MPolyfold.remark_4_2_2_dA_zero Chapter4/GapP1_Remark422 · MPolyfold.remark_4_2_2_dA_zero_near. The 2021 text speaks of $d_A$, not $d_X$. Since $d_A \le d_X$ (Lemma 2.4.6 (preprint 2.26)), this covers more points. For example, the parabola $\{(y^2, y)\}$ in $[0,\infty) \times \mathbb{R}$ has $d_A \equiv 0$, while $d_X(0,0) = 1$ (Findings, Chapter 4, items 165, 176). The proof is one step. $A$ with its induced structure is a smooth finite-dimensional submanifold of itself Chapter4/GapP1_Remark422 · MPolyfold.

IsSmoothSubmanifold.isSmoothSubmanifold_univ_subPolyfold: the retraction is the identity, and it is sc$^+$ because $A = A_\infty$ and $1^1_0 : A \to A^1$ is sc-smooth (Proposition 4.2.3 (4) (preprint 4.15), §IV.12). The preprint form, applied to $A$ inside $A$, has $d_A$ in place of $d_X$. No tameness of $X$ or $A$ is used.

**The sentence on $d_A(a) = 1$.** The text says, without proof, that near such points $A$ is a manifold with boundary. The key step is a general statement Chapter4/GapP1_Remark422One · MPolyfold.hasCornerStructure_of_degIndex_le_one: *an M-polyfold with $X_0 = X_\infty$, $1^1_0$ sc-smooth and $d_X \le 1$ everywhere has a corner structure*, with no tameness. By Theorem 4.1.2 (preprint 4.2) in its centred form Chapter4/Characterization · MPolyfold.hasCornerStructure_of_centred, it suffices to find at every $x$ a chart centred at $x$ with $N = T_0O$ in good position. Take a chart realizing $d_X(x)$ and re-centre it at $e = \varphi(x)$ (§IV.5). The new partial quadrant $C_e$ is cut out by the at most one coordinate that vanishes at $e$.

- If $d_X(x) = 0$, then $C_e = E$, and every subspace is in good position Chapter4/Submanifolds · isInGoodPosition_univ.
- If $d_X(x) = 1$, then $C_e = \{\lambda \ge 0\}$. If $\lambda$ does not vanish on $N$, then $N$ is in good position with a complement inside $\ker \lambda$ Chapter4/GapP1_Remark422One · isInGoodPosition_halfSpace.
- If $\lambda$ vanishes on $N$, then $T_0O$ lies in the reduced tangent space. The model is then near $0$ a graph over an open subset of $N$, a chart without boundary, so $d_X(x) = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.degIndex_eq_zero_of_isReducedTangentAt. This is the graph argument of 2021 Lemma 9.2.13, and it contradicts $d_X(x) = 1$.

The same holds for the induced structure of a submanifold with $d_A \le 1$ Chapter4/GapP1_Remark422One · MPolyfold.IsSmoothSubmanifold.hasCornerStructure_of_degIndex_le_one. The set $\{d_A \le 1\}$ is open Chapter4/GapP1_Remark422One · MPolyfold.isOpen_degIndex_le. So it is a smooth submanifold of $A$ whose induced structure is a manifold with boundary Chapter4/GapP1_Remark422One · MPolyfold.remark_4_2_2_dA_one, and the same holds near one point with $d_A(a) = 1$ Chapter4/GapP1_Remark422One · MPolyfold.remark_4_2_2_dA_one_near.

**In a tame M-polyfold, good position is automatic.** At the centre of a tame chart, $T_0O \cap C$ has interior points in $T_0O$ Chapter4/GapP1_Remark422 · ScRetraction.interior_rangeS_scDeriv_zero_nonempty. With Lemma 4.2.8 (preprint 4.19), $T_0O$ is then in good position Chapter4/GapP1_Remark422 · ScRetraction.isInGoodPosition_rangeS_of_isTameWith. So *a tame M-polyfold with $X_0 = X_\infty$ and $1^1_0$ sc-smooth has a corner structure* Chapter4/GapP1_Remark422 · MPolyfold.hasCornerStructure_of_isTame: the good-position hypothesis of Theorem 4.1.2 (2) (preprint 4.2) holds by itself. This gives a second route to the $d_A = 1$ sentence when $A$ is tame Chapter4/GapP1_Remark422 · MPolyfold.remark_4_2_2_dA_one_of_isTame.

**Question 4.2.4 (preprint: the open question after Proposition 4.15).** It asks two things. Is a sub-M-polyfold with $A = A_\infty$ and finite, locally constant $\dim T_aA$ a smooth finite-dimensional submanifold? Is the "T" $T = \{(x,0) : |x| < 1\} \cup \{(0,y) : 0 \le y < 1\}$ Chapter4/GapP1_Question424 · Question424.tSet homeomorphic to an sc-retract $O = O_\infty$ without boundary? These are questions, and nothing in them is to be formalized (gap G.34). The text adds that this is impossible for sc$^+$-retracts. That sentence is proved Chapter4/GapP1_Question424 · Question424.question_4_2_4_scPlus in two steps.

1. *An sc$^+$-retract $(O, E, E)$ is locally a graph over its finite-dimensional tangent space* Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_flat. This is the argument of 2021 Lemma 9.2.13 with $C = E$, so that $T_zO \subseteq E_z = E$.
2. *$T$ is not a topological manifold at the branch point* Chapter4/GapP1_Question424 · Question424.not_homeomorph_open. No relatively open neighbourhood of $(0,0)$ in $T$ is homeomorphic to an open subset of a finite-dimensional space $\mathbb{R}^k$. The proof splits by $k$:

- For $k = 0$: a point is not a neighbourhood in $T$.
- For $k = 1$: an injective arc leaves its starting point to one side `Chapter4/GapP1_Question424 · Question424.arc_side`, so two of the three arms go to the same side of the image of $(0, 0)$, and their images overlap `Chapter4/GapP1_Question424 · Question424.arc_overlap`.
- For $k \geq 2$: a punctured ball is connected, but a punctured neighbourhood of $(0, 0)$ in $T$ is not.

## IV.32 Theorem 4.2.11 (1) (preprint 4.22): the local faces of the neighbourhood

In the statement of Theorem 4.2.11, the local faces at $x$ are the faces of a small open $U \ni x$: the closures in $U$ of the components of $U \cap \{d_X = 1\}$, as recalled before Proposition 2.5.16 (preprint 2.44) (Definition 2.5.10 for $U$). In the proof, the text uses the chart faces $O \cap \{\lambda_i = 0\}$ instead. §IV.17 proved (1) for the chart faces. This subsection identifies the two.

**The faces of a neighbourhood in a tame centred chart** `Chapter4/GapP1_LocalFaces · MPolyfold.exists_faces_chart`. Let $\psi$ be a chart centred at $x$, modelled on a tame retract $O = r(U_r)$ with $n$ half-line coordinates, and let $U \ni x$ be open inside its domain. Then there are an open $U' \ni x$ inside $U$ and pairwise distinct faces $\theta_0, \dots, \theta_{n-1}$ of $U$ such that $\theta_i \cap U' = U' \cap \psi^{-1}\{\lambda_i = 0\}$. Moreover, every face $F \ni 0$ of the retract with $F \subseteq \{\lambda_i = 0\}$ satisfies $U' \cap \psi^{-1}(F) = U' \cap \psi^{-1}\{\lambda_i = 0\}$. If $d_X(x) = n$, the faces of $U$ through $x$ are exactly the $\theta_i$.

*Proof.* Fix $i$. 1. For small $\delta$, the set $\{z \in C : |z| < \delta,\ I(z) = \{i\}\}$ is convex, so it is connected. 2. Its image under $r$ lies in $\{d = 1\}$, so it lies in one component of $U \cap \{d_X = 1\}$. Let $\theta_i$ be the closure of that component, a face of $U$. 3. Near 0, every point of $O \cap \{\lambda_i = 0\}$ is a limit of such images. This is Lemma 4.C of the exposition `Chapter2/TameRetracts · local_face_structure`. So $\theta_i$ contains the chart face near $x$. 4. Conversely, a connected subset of $\{d = 1\}$ near 0 stays inside one coordinate hyperplane. This is Lemma 4.A `Chapter2/TameRetracts · subset_coordHyper_of_isPreconnected`, and it gives the reverse inclusion. 5. The $\theta_i$ are distinct because they lie in distinct hyperplanes.

**(1) and (4) with the faces of $U$** `Chapter4/GapP1_LocalFaces · MPolyfold.theorem_4_2_11_faces`. This is Theorem 4.2.11 (preprint 4.22) with the local faces $\theta_i$ of $U$. Near $x$, the text's $F_\sigma = \bigcap_{i \in \sigma} \theta_i$ agrees with the set $\psi^{-1}(G \cap \bigcap_{i \in \sigma} F_i)$ of the formal (4) (§IV.18), for every $\sigma$. So (4) holds for the germ of $F_\sigma$ at $x$.

**(1) as printed** `Chapter4/GapP1_Theorem4211 · MPolyfold.theorem_4_2_11_one`. There is an open $U \ni x$ such that: - the faces of $U$ through $x$ number exactly $d_X(x)$, and each is contained in $U$; - (2) and (3) hold on $U$; - $U$ is face-structured: every $y \in U$ lies in exactly $d_X(y)$ faces of $U$.

*Proof.* Take the neighbourhood of §IV.17 and shrink it. A tame $X$ has at $x$ an open $D \ni x$ all of whose small open subsets are face-structured `Chapter2/TameBoundary · TameBoundary.exists_nbhd_forall_isFaceStructuredOpen`. Intersect with $D$ and apply the identification above.

## IV.33 Theorem 4.3.1 (preprint 4.23): stratified regularity does not depend on the chart

The definition of a stratified regular point (§IV.23) asks for the condition in *some* product chart $\chi = (r \mapsto r - a) \times \varphi$ of $\mathbb{R}^m \times X$, with some data of Definition 4.2.1 (preprint 4.13), some presentation and some lift. In that chart the condition is that the projection maps $T_z A \cap (\mathbb{R}^m \times E_{z_2})$ onto $\mathbb{R}^m$ `Chapter4/GapP1_`

Theorem431 · StratRegCond. Like the conditions of §IV.13, it depends only on the tangent spaces Chapter4/GapP1_Theorem431 · StratRegCond.congr. The text treats the notion as intrinsic to $A$ and its strata (Findings, Chapter 4, items 102, 108).

**Transport.** Let $F$ be a germ of sc-diffeomorphisms whose $\mathbb{R}^m$-component is a translation. Then $F$ carries the condition Chapter4/GapP1_Theorem431 · StratRegCond.transport, for three reasons: - its derivative maps $T_zQ$ onto $T_{F(z)}Q'$ (§IV.11); - by the sign argument of Proposition 2.4.10 (preprint 2.30), it maps $E_z$ into $E'_{F(z)}$; - it keeps the $\mathbb{R}^m$-component of every vector. If the first components of two sc¹ maps differ by a constant, so do the first components of their derivatives Chapter4/GapP1_Theorem431 · scDeriv_fst_eq_of_fst_eq.

**The chart change** Chapter4/GapP1_Theorem431 · stratRegCond_chart_change. The transition of two product charts is $w \mapsto (w_1 + a - a', \varphi' \circ \varphi^{-1}(w_2))$ Chapter4/GapP1_Theorem431 · transition_rnChart_prod_fst. This is a germ of the kind just treated, so the chart-change argument of §IV.13 applies.

So the condition holds in every product chart once it holds in one Chapter4/GapP1_Theorem431 · IsStratRegularPoint.stratRegCond. The chart-free forms are Chapter4/GapP1_Theorem431 · isStratRegularPoint_iff_forall Chapter4/GapP1_Theorem431 · isStratRegularValue_iff_forall. Since $T_zA$ has no chart-free linear structure here (gap G.7), "intrinsic" is read as "independent of every choice in the product charts".

## IV.34 Lemma 4.5.1 between open subsets; the pull-back bundle $TX|X^2 \to X^2$ and Proposition 4.4.5 as printed

**Open subsets.** An open $U \subseteq X$ is an M-polyfold by Part I's restriction of the atlas Chapter3/RestrictOpen · MPolyfold.restrictOpen. Its inclusion is sc-smooth, and $TU$ is the part of $TX$ over $U$. A vector field $A$ on $X$ restricts to $U$ as $A|_U = (T\iota)^{-1} \circ A \circ \iota_1$ Chapter4/GapP1_Lemma451 · MPolyfold.restrictVectorField. This restriction is an sc-smooth vector field of $U$ Chapter4/GapP1_Lemma451 · MPolyfold.IsScVectorField.restrictOpen with $T\iota(A|_U(x)) = A(x)$ Chapter4/GapP1_Lemma451 · MPolyfold.tangentMap_restrictVectorField.

The bracket is local: $T\iota[A|_U, B|_U](x) = [A, B](x)$ for $x \in U^2$ Chapter4/GapP1_Lemma451 · MPolyfold.tangentMap_lieBracket_restrictOpen. In other words, the bracket of the restrictions is the restriction of the bracket Chapter4/GapP1_Lemma451 · MPolyfold.lieBracket_restrictOpen. The reason is that both are computed in the same charts by the local formula, and Lemma 4.5.1 (§IV.29) applies to the inclusion.

**Lemma 4.5.1 for maps between open subsets** Chapter4/GapP1_Lemma451 · lemma_4_5_1_open. Let $U, U' \subseteq X$ be open and $\varphi : U \to U'$ sc-smooth; a local sc-diffeomorphism between open subsets is one such map. Suppose $T\varphi \circ A|_U = A|_{U'} \circ \varphi_1$ and $T\varphi \circ B|_U = B|_{U'} \circ \varphi_1$ on $U^1$. Then $T\varphi \circ [A|_U, B|_U] = [A|_{U'}, B|_{U'}] \circ \varphi_2$ on $U^2$. This is §IV.29 for the M-polyfolds $U$ and $U'$. Read in $TX$, for $g = \iota' \circ \varphi$, it says $Tg((T\iota)^{-1}[A, B](x)) = [A, B](g(x))$ Chapter4/GapP1_Lemma451 · lemma_4_5_1_open_ambient. No hypothesis is added. The relations are asked on $U^1$, as in §IV.29.

**The pull-back bundle.** The text says that $[A, B]$ is an sc-smooth section of $TX|X^2 \to X^2$ and gives this bundle no structure. We use the sub-M-polyfold

$$TX|X^2 = \{(x, \alpha) \in X^2 \times TX \mid p(\alpha) = x\}$$

Chapter4/GapP1_Lemma451Bundle · MPolyfold.tangentPullback. Take a chart $\varphi$ of $X$. The product chart of $X^2 \times TX$ is the lifted chart times $T\varphi$ Chapter4/GapP1_Lemma451Bundle · MPolyfold.tangentPullbackChart. In it, $p(\alpha) = x$ says $y = \iota z$ for $(z, (y, v))$ Chapter4/GapP1_Lemma451Bundle · MPolyfold.mem_tangentPullback_iff, where $\iota : E_2 \to$

$E_1$ is the inclusion. The local retraction is

$$(z,(y,v)) \mapsto (z, Tr(\iota z, v))$$

`Chapter4/GapP1_Lemma451Bundle · tangentPullbackLocal`, for $r$ a retraction onto the model of $\varphi$. It is the composite of the linear sc-operator $(z,(y,v)) \mapsto (z,(\iota z,v))$ with id $\times\, Tr$, hence sc-smooth `Chapter4/GapP1_Lemma451Bundle · isScSmoothRetractMap_tangentPullbackLocal`.

So $TX|X^2$ is a sub-M-polyfold in the sense of Definition 2.3.18 (preprint 2.20) `Chapter4/GapP1_Lemma451Bundle · MPolyfold.isSubMPolyfold_tangentPullback`, and an M-polyfold `Chapter4/GapP1_Lemma451Bundle · MPolyfold.tangentPullbackPolyfold`. Its projection to $X^2$ is sc-smooth `Chapter4/GapP1_Lemma451Bundle · MPolyfold.isScSmoothMap_tangentPullbackProj`, and its fibre over $x$ is $T_xX$ `Chapter4/GapP1_Lemma451Bundle · MPolyfold.tangentPullbackFiberEquiv`.

**Proposition 4.4.5 (preprint 4.26) as printed** `Chapter4/GapP1_Lemma451Bundle · proposition_4_4_5_section`. The map $x \mapsto (x,[A,B](x))$ `Chapter4/GapP1_Lemma451Bundle · MPolyfold.lieBracketSection` is an sc-smooth section of $TX|X^2 \to X^2$. A map into the sub-M-polyfold is sc-smooth when its two components are `Chapter4/GapP1_Lemma451Bundle · MPolyfold.isScSmoothMap_tangentPullback_mk`, and both components are sc-smooth by §IV.29. The form "in every chart, the usual formula" is collected in one statement `Chapter4/BookFormLieBracket2021 · MPolyfold.proposition_4_4_5_book`.

**Lemma 4.5.1 on sections** `Chapter4/GapP1_Lemma451Bundle · lemma_4_5_1_section`. For $f : X \to Y$ sc-smooth, the bundle map $(f_2, Tf) : TX|X^2 \to TY|Y^2$ `Chapter4/GapP1_Lemma451Bundle · MPolyfold.tangentPullbackMap` is sc-smooth `Chapter4/GapP1_Lemma451Bundle · MPolyfold.isScSmoothMap_tangentPullbackMap`. Under the hypotheses of §IV.29, $(f_2, Tf) \circ [A,B] = [A',B'] \circ f_2$ (Findings, Chapter 4, item 163).

## IV.35 The Whitney sum $\oplus_k TX$, Definition 4.4.1 (preprint 4.25), and the system (4.4)

The 2021 text defines a $k$-form on $X$ as an sc-smooth map $\oplus_k TX \to \mathbb{R}$ on the Whitney sum, linear and alternating in the fibres. The Whitney sum is not constructed in the text, and $T_xX$ has no linear structure in the formalization (gap G.7). So $\oplus_k TX$ is read in its charts $\oplus_k T\varphi$.

**The Whitney sum of a local model** `Chapter4/ScForms2021 · ScBanach.whitney`. For an sc-Banach space $E$, the space $E^1 \oplus E^{\oplus k}$ has the fibre power $E^{\oplus k}$ `Chapter4/ScForms2021 · ScBanach.fiberPow`. For a retraction $r$ onto $O$, the map

$$r_k(x,h_1,\dots,h_k) = (r(x), Dr(x)h_1, \dots, Dr(x)h_k)$$

is a retraction `Chapter4/ScForms2021 · ScRetraction.whitney` `Chapter4/ScForms2021 · ScRetraction.whitney_retract`. Its image $\oplus_k TO$ is a retract `Chapter4/ScForms2021 · IsScRetract.whitney` `Chapter4/ScForms2021 · retractWhitney`. An sc-smooth map $f$ between retracts induces the sc-smooth $\oplus_k Tf$ `Chapter4/ScForms2021 · retractWhitneyMap` `Chapter4/ScForms2021 · IsScSmoothRetractMap.whitney`. This is the chain rule of Proposition 2.1.13 (preprint 2.6), applied in each fibre.

**Definition 4.4.1 (preprint 4.25) in charts.** A family $\omega$ of alternating $k$-forms $\omega(x)$ on the tangent spaces $T_xX$, $x \in X^1$, has in a chart $\varphi$ the representative

$$\omega_\varphi(y)(h_1,\dots,h_k) = \omega(x)(T\varphi^{-1}h_1, \dots)$$

Chapter4/ScForms2021 · MPolyfold.chartForm Chapter4/ScForms2021 · MPolyfold.chartForm_eq. $\omega$ is an *sc-form* if every representative is sc-smooth on $\oplus_k TO$ Chapter4/ScForms2021 · MPolyfold.IsScForm Chapter4/ScForms2021 · IsRetractScForm. Two representatives differ by $\oplus_k T$ of the transition map Chapter4/ScForms2021 · MPolyfold.chartForm_transition. So it suffices to check the condition on a cover by charts Chapter4/ScForms2021 · MPolyfold.isScForm_of_cover, and the condition does not quantify over all charts in a way that could be vacuous. The sc-forms of degree $k$ form a subspace $\Omega^k(X)$ Chapter4/ScForms2021 · MPolyfold.scForms. It contains the constant functions in degree $0$ Chapter4/ScForms2021 · MPolyfold.constForm_mem_scForms.

**The system (4.4) and Definition 4.4.2.** A form on $X^i$ restricts to a form on $X^{i+1}$ along the inclusion. On tangent fibres the inclusion is a continuous linear map Chapter4/ScForms2021 · MPolyfold.lowerFiberCLM, and the restriction is Chapter4/ScForms2021 · MPolyfold.formLower. It maps sc-forms to sc-forms Chapter4/ScForms2021 · MPolyfold.formLower_mem_scForms. The M-polyfolds $X^i$ are the lifts of the bundled M-polyfold (§IV.29). They give the spaces $\Omega^k(X^i)$ Chapter4/ScForms2021 · BundledMPolyfold.forms and the maps $\Omega^k(X^i) \to \Omega^k(X^j)$, $i \le j$ Chapter4/ScForms2021 · BundledMPolyfold.formsMap. These compose Chapter4/ScForms2021 · BundledMPolyfold.formsMap_comp, so they form a directed system Chapter4/ScForms2021 · BundledMPolyfold.directedSystem_forms. $\Omega^k_\infty(X)$ is its direct limit Chapter4/ScForms2021 · BundledMPolyfold.formsInfty, with the canonical maps Chapter4/ScForms2021 · BundledMPolyfold.formsInftyOf Chapter4/ScForms2021 · BundledMPolyfold.formsInftyOf_formsMap. $\Omega^*_\infty(X) = \bigoplus_k \Omega^k_\infty(X)$ is the graded space Chapter4/ScForms2021 · BundledMPolyfold.formsInftyGraded.

## IV.36 The exterior derivative in a local model, and its naturality

**The local $d$.** Let $\Omega$ be an sc-smooth map on $\oplus_k TU$, alternating in the fibres. At a point $(y, h_0, \dots, h_k)$ on level 1 define

$$d\Omega(y)(h_0, \dots, h_k) = \sum_{j=0}^{k} (-1)^j \, D\Omega(y, h_0, \dots, \hat{h}_j, \dots, h_k)(h_j, 0)$$

Chapter4/ScExteriorDerivative2021 · ambientD Chapter4/ScExteriorDerivative2021 · ambientD_apply. Each term is the derivative of $\Omega$ in the base direction $h_j$ with the fibre fixed. The sc-derivative of $\Omega$ in the direction $(h, 0)$ is determined by the level-1 derivative and the density of $E_2$ in $E_1$ Chapter4/ScExteriorDerivative2021 · scDeriv_relation. So $d\Omega(y)$ is the alternation of the base derivative Chapter4/ScExteriorDerivative2021 · ambientD_whitneyMk, an alternating $(k+1)$-form. $d\Omega$ is again sc-smooth Chapter4/ScExteriorDerivative2021 · IsScSmooth.ambientD Chapter4/ScExteriorDerivative2021 · IsAmbientScForm.ambientD. On a retract, $d$ of $\Omega$ is $d$ of $\Omega \circ r_k$, restricted to the retract Chapter4/ScExteriorDerivative2021 · retractD Chapter4/ScExteriorDerivative2021 · IsRetractScForm.retractD.

**Naturality** Chapter4/ScFormsNatural2021 · ambientD_comp_whitneyMap. For $f$ sc-smooth, $d(\Omega \circ \oplus_k Tf) = d\Omega \circ \oplus_{k+1} Tf^1$. The derivative of $\Omega(f(y), Df(y)h_\bullet)$ in the direction $h_j$ has two parts: - a term $D\Omega(\dots)(Df(y)h_j, 0)$; - terms with the second derivative $D^2 f(y)(h_j, h_i)$.

The second-derivative terms are symmetric in $(i, j)$ (§IV.28) Chapter4/VectorFields · scSecondDeriv_symm. They cancel in the alternating sum Chapter4/ScFormsNatural2021 · sum_update_symm_eq_zero. The derivatives involved are computed by the chain rule for sc-derivatives Chapter4/ScFormsNatural2021 · scDeriv_comp and in the fibres of the Whitney sum Chapter4/ScFormsNatural2021 · scDeriv_whitney_fiber Chapter4/ScFormsNatural2021 · scDeriv_whitneyMap_whitneyMk. On retracts, $d$ commutes with $\oplus_k Tf$ Chapter4/ScFormsNatural2021 · retractD_comp_retractWhitneyMap, and it is local: it does not change when the retract is shrunk to an open part Chapter4/ScFormsNatural2021 · retractD_eq_of_subset.

## IV.37 $d$ on an M-polyfold, $d \circ d = 0$, and Definition 4.4.6 (preprint 4.27)

**The global $d$.** In a chart $\varphi$, $d$ of the representative is `Chapter4/ScFormsGlobal2021 · MPolyfold.chartD`. By naturality, the representatives in two charts are related by $\oplus_{k+1} T$ of the transition map `Chapter4/ScFormsGlobal2021 · MPolyfold.chartD_transition`. So

$$d\omega(x)(\alpha_0, \dots, \alpha_k) = d(\omega_\varphi)(\varphi(x))(T\varphi^1 \alpha_0, \dots)$$

for one chosen chart at $x$ defines a form on $X^1$ `Chapter4/ScFormsGlobal2021 · MPolyfold.extD` `Chapter4/ScFormsGlobal2021 · MPolyfold.extD_apply`. Its representative in every chart is $d$ of the representative `Chapter4/ScFormsGlobal2021 · MPolyfold.chartForm_extD`. It maps $\Omega^k(X)$ to $\Omega^{k+1}(X^1)$ `Chapter4/ScFormsGlobal2021 · MPolyfold.extD_mem_scForms`, and it is linear `Chapter4/ScFormsGlobal2021 · MPolyfold.extD_add` `Chapter4/ScFormsGlobal2021 · MPolyfold.extD_smul` `Chapter4/ScFormsGlobal2021 · MPolyfold.extD`. The 2021 text writes $d : \Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$. The preprint's $\Omega^k(X^{i+1}) \to \Omega^{k+1}(X^i)$ presumably means the same map, with the indices exchanged (Findings, Chapter 4, item 120).

**$d \circ d = 0$.** In a local model, $dd\Omega(y)(h_0, \dots, h_{k+1})$ is a double alternating sum of second derivatives of $\Omega$ in base directions `Chapter4/ScFormsSquare2021 · scDeriv_ambientD_whitneyMk`. These are symmetric (§IV.28), so the sum cancels in pairs `Chapter4/ScFormsSquare2021 · sum_removeNth_symm_eq_zero` `Chapter4/ScFormsSquare2021 · ambientD_ambientD`. The same holds on retracts `Chapter4/ScFormsSquare2021 · retractD_shift_retractD` and on an M-polyfold `Chapter4/ScFormsSquare2021 · MPolyfold.extD_extD`. The text asserts $d^2 = 0$ (Findings, Chapter 4, item 121).

**Passing to $\Omega^*_\infty$.** $d$ commutes with the restriction to $X^{i+1}$ `Chapter4/ScFormsLower2021 · MPolyfold.extD_formLower`. It gives maps $\Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$ `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsD` that commute with the system (4.4) `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsD_formsStep` `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsD_formsMap`. Hence $d$ is defined on the limit `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyD` `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyD_of` and on the graded space `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyGradedD`, with $d \circ d = 0$ there as well `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyD_comp_formsInftyD` `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyGradedD_comp_self`.

**Definition 4.4.6 (preprint 4.27).** The image of $d$ lies in its kernel `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyDRange` `Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyDRange_le_ker`. $H^k_{sc}(X)$ is the quotient `Chapter4/ScFormsLower2021 · BundledMPolyfold.scDeRham`, a real vector space, with classes `Chapter4/ScFormsLower2021 · BundledMPolyfold.scDeRhamMk`. Two closed forms have the same class iff they differ by an exact form `Chapter4/ScFormsLower2021 · BundledMPolyfold.scDeRhamMk_eq_iff`. The relative version is §IV.40.

## IV.38 The formula of p. 197 for $d\omega$, and its sign

The text gives $d\omega(A^{(1)}_0, \dots, A^{(1)}_k)$ by "the familiar formula". It is the sum over $j$ of $(-1)^j D(\omega(A_0, \dots, \widehat{A}_j, \dots, A_k)) \cdot A^{(1)}_j$, with the summation index printed as $i$, plus the sum over $\ell < j$ of $(-1)^{\ell+k}\omega([A_\ell, A_j], \dots)$. Here $[A, B] = DA \cdot B - DB \cdot A$ (p. 200). The formula is checked in the classical setting, a real normed space, to which it reduces in a chart.

**The text's bracket** `Chapter4/ScExteriorDerivative2021 · bookBracket` is the negative of Mathlib's Lie bracket `Chapter4/ScExteriorDerivative2021 · bookBracket_eq_neg_lieBracket`. Write the formula with a sign function on the pairs $\ell < j$ `Chapter4/ScExteriorDerivative2021 · invariantFormula`. With the sign $(-1)^{\ell+j+1}$ `Chapter4/`

ScExteriorDerivative2021 · correctSign, it is Mathlib's exterior derivative for vector fields differentiable at the point Chapter4/ScExteriorDerivative2021 · extDeriv_eq_invariantFormula.

**With the printed sign the formula does not hold as we read it** Chapter4/ScExteriorDerivative2021 · invariantFormula_printedSign_counterexample. Take $k = 1$, $\omega = dy$ on $\mathbb{R}^2$, $A_0 = (1,0)$ and $A_1(p) = (0,p_1)$: - with the printed sign $(-1)^{\ell+k}$ Chapter4/ScExteriorDerivative2021 · printedSign, the right side at $0$ is $2$, while $d\omega = 0$; - the field $A_1' = 0$ agrees with $A_1$ at $0$, and with it the right side is $0$, so the printed formula is not tensorial; - the classical sign $(-1)^{\ell+j}$ Chapter4/ScExteriorDerivative2021 · item158Sign, proposed earlier in Findings, gives $2$ on the same example, because the text's bracket is minus the classical one.

The printed sign may go with another bracket convention. The index $i$ of the first sum is read as $j$ (Findings, Chapter 4, items 158, 167). This is the one narrower row of the chapter (V4.83). The formal $d$ of §§IV.36–IV.37 is defined by the local formula, so the sign does not enter it.

## IV.39 The pull-back $f^*$, the wedge product, and the graded differential algebra

**The pull-back** (p. 198; the 2017 text and the preprint state only that $f$ induces $f^*$). Let $f : X \to Y$ be sc-smooth. On tangent fibres, $Tf$ is a continuous linear map Chapter4/ScFormsPullback2021 · MPolyfold.tangentFiberMap. Define $f^*\omega(x)(\alpha_\bullet) = \omega(f(x))(Tf\alpha_\bullet)$ Chapter4/ScFormsPullback2021 · MPolyfold.formPullback. In charts it is $\omega_\psi \circ \oplus_k T(\psi f \varphi^{-1})$ Chapter4/ScFormsPullback2021 · MPolyfold.chartForm_formPullback, so it maps $\Omega^k(Y)$ to $\Omega^k(X)$ Chapter4/ScFormsPullback2021 · MPolyfold.formPullback_mem_scForms. It satisfies $df^*\omega = (f^1)^* d\omega$ Chapter4/ScFormsPullback2021 · MPolyfold.extD_formPullback, by naturality of the local $d$ (§IV.36). The text leaves this identity to the reader; it holds as we read it. Pull-back is functorial Chapter4/ScFormsPullback2021 · MPolyfold.formPullback_comp Chapter4/ScFormsPullback2021 · MPolyfold.formPullback_id and commutes with (4.4) Chapter4/ScFormsPullback2021 · MPolyfold.formLower_formPullback. On the lifts, the maps $(f^i)^*$ Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsPullback pass to the limit Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsInftyPullback. There $df^* = f^* d$ Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsInftyD_comp_formsInftyPullback, pull-back is functorial Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsInftyPullback_comp, and it induces a map on $H^k_{sc}$ Chapter4/ScFormsPullback2021 · BundledMPolyfold.scDeRhamPullback.

**The wedge product.** The text uses $\wedge$ through "$(\Omega^*_\infty(X), d)$ is a graded differential algebra" but never defines it. We take the classical normalization:

$$(a \wedge b)(v) = (k!\,l!)^{-1} \sum_{\sigma \in S_{k+l}} \operatorname{sign}\sigma \; a(v_{\sigma(1)}, \dots, v_{\sigma(k)})\, b(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)})$$

Chapter4/ScFormsWedge2021 · wedgeFun Chapter4/ScFormsWedge2021 · altWedge Chapter4/ScFormsWedge2021 · coe_altWedge. Write $\delta A(v) = \sum_j (-1)^j A(v_j)(v_0, \dots, \hat{v}_j, \dots)$ for the alternating sum that defines $d$. Two identities hold:

$$\delta(A \wedge b) = \delta A \wedge b, \qquad \delta(a \wedge B) = (-1)^k a \wedge \delta B$$

Chapter4/ScFormsWedge2021 · deltaFun_wedgeFun_left Chapter4/ScFormsWedge2021 · deltaFun_wedgeFun_right. They are identities between sums over permutations. They use the decomposition $S_{n+1} \cong \{0, \dots, n\} \times S_n$ by insertion, and they need no alternation. With the product rule, they give the local Leibniz rule Chapter4/ScFormsWedge2021 · ambientD_lwedge.

On an M-polyfold, the pointwise wedge Chapter4/ScFormsWedge2021 · MPolyfold.formWedge maps $\Omega^k(X) \times \Omega^l(X)$ into $\Omega^{k+l}(X)$ Chapter4/ScFormsWedge2021 · MPolyfold.formWedge_mem_scForms. It satisfies

$$d(\omega \wedge \tau) = d\omega \wedge \iota^*\tau + (-1)^k \iota^*\omega \wedge d\tau$$

Chapter4/ScFormsWedge2021 · MPolyfold.extD_formWedge and $f^*(\omega \wedge \tau) = f^*\omega \wedge f^*\tau$ Chapter4/ScFormsWedge2021 · MPolyfold.formPullback_formWedge. It passes to $\Omega^*_\infty(X)$ Chapter4/ScFormsWedge2021 · BundledMPolyfold.formsInftyWedge Chapter4/ScFormsWedge2021 · BundledMPolyfold.formsInftyD_formsInftyWedge.

Associativity and graded commutativity are again identities between permutation sums Chapter4/ScFormsWedge2021b · wedgeFun_assoc Chapter4/ScFormsWedge2021b · wedgeFun_comm. The sign $(-1)^{kl}$ is that of a rotation of $k + l$ places by $k$. The unit is the constant $1$ Chapter4/ScFormsWedge2021b · MPolyfold.formOne_mem_scForms. On $\Omega^*_\infty(X)$ Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyWedge_assoc Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyWedge_comm, they make $\bigoplus_k \Omega^k_\infty(X)$ a graded $\mathbb{R}$-algebra Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyGAlgebra that is graded commutative Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyGraded_of_mul_of_comm. $d$ is a derivation of degree $+1$ Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyGradedD_of_mul_of with $d^2 = 0$ (§IV.37). This is the graded differential algebra of p. 198. $f^*$ is a morphism of graded algebras Chapter4/ScFormsWedge2021 · BundledMPolyfold.formsInftyPullback_formsInftyWedge Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyPullback_formsInftyOne. The Leibniz sign $(-1)^k$ holds as we read it.

## IV.40 Forms on $\partial X$, $j^*$, and the relative complex (pp. 198–199)

For a tame $X$ the text says that the inclusion $\partial X \to X$ is sc-smooth on local faces, so differential forms on $\partial X$ "make sense". It gives no definition. $\partial X$ is not an M-polyfold (Findings, Chapter 4, item 122).

**The reading.** Forms on $\partial X$ are read through the tame boundary $B : bX \to X$ of Theorem 2.7.13, an sc-smooth map from an M-polyfold that is locally the inclusion of a local face. Let $f : X \to Y$ be sc-smooth. A family $\tau$ on $X$ *agrees over* $f$ Chapter4/BoundaryForms2021 · MPolyfold.IsAgreeingOver if $\tau_p(v) = \tau_{p'}(v')$ whenever $f(p) = f(p')$ and $Tf\, v_j = Tf\, v'_j$ for all $j$. The base-point condition matters in degree $0$. A form on $\partial X^i$ is an sc-form on $bX^i$ that agrees over $B^i$ Chapter4/BoundaryForms2021 · MPolyfold.IsFaceCompatible. So on two local faces through a point, the form coincides on vectors with the same image in $TX$ (reading R-a; Findings, Chapter 4, item 173).

The definition Chapter4/BoundaryForms2021 · MPolyfold.bForms also lists the agreement of $d\tau$, which makes $\Omega^*(\partial X)$ the largest $d$-stable family of agreeing forms Chapter4/BoundaryForms2021 · MPolyfold.le_bForms. With it, $d$ maps $\Omega^k(\partial X^i)$ to $\Omega^{k+1}(\partial X^{i+1})$ Chapter4/BoundaryForms2021 · MPolyfold.extD_mem_bForms.

**The agreement of $d\tau$ follows from that of $\tau$** Chapter4/A3BoundaryForms2021b · MPolyfold.isFaceCompatible_extD. So the extra condition is automatic, and $\Omega^k(\partial X^i)$ is the set of agreeing forms Chapter4/A3BoundaryForms2021b · MPolyfold.bForms_eq_agreeForms Chapter4/A3BoundaryForms2021b · MPolyfold.mem_bForms_iff. The general statement is for any sc-smooth $f$ with two properties Chapter4/A3BoundaryForms2021 · BundledMPolyfold.isAgreeingOver_extD: - $Tf$ is injective on each tangent space Chapter4/A3BoundaryForms2021 · BundledMPolyfold.IsTangentInjective; - $f$ has a *corner cover* Chapter4/A3BoundaryForms2021 · BundledMPolyfold.HasCornerCover. For $p \neq p'$ with $f(p) = f(p')$, there are sc-smooth $g, g' : Z \to X$ with $f \circ g = f \circ g'$ and $g(z) = p$, $g'(z) = p'$ such that every pair of tangent vectors at $p$, $p'$ with the same image is $(Tg\, v, Tg'\, v)$.

Given these, $g^*\tau = g'^*\tau$, hence $g^*d\tau = g'^*d\tau$ by $df^* = f^*d$ Chapter4/A3BoundaryForms2021 · MPolyfold.formPullback_extD_eq_of_isAgreeingOver. The two properties are stated on level 0 and transported to every $X^i$.

For $B$, $TB$ is the face inclusion in tangent charts Chapter4/A3BoundaryForms2021b · MPolyfold.tangentChartMap_bdryMap, so it is injective Chapter4/A3BoundaryForms2021b · MPolyfold.isTangentInjective_bdryMap. The corner cover is built as follows Chapter4/A3BoundaryForms2021b · MPolyfold.hasCornerCover_bdryMap: 1. At two faces through $x$, take a tame chart with retraction $r$ and coordinates $\lambda_i$, $\lambda_j$ vanishing on them Chapter4/A3BoundaryForms2021b · MPolyfold.CornerChart. 2. The map $r \circ P_i \circ P_j$, where $P_i$ kills $\lambda_i$, sends a relatively open piece of the partial quadrant into both faces. 3. Its two lifts to $bX$ have the same composite with $B$. 4. Every vector of $T_xF \cap T_xF'$ is in the image of its tangent map Chapter4/A3BoundaryForms2021b · MPolyfold.CornerChart.exists_tangent, since $D(rP_iP_j)(\varphi x)h = Dr(\varphi x)h = h$ for $h$ with $\lambda_i h = \lambda_j h = 0$.

No corner strata are built as M-polyfolds, and the point $x$ need not be smooth. The text writes $d\tau$ without comment; this holds as we read it.

**$j^*$.** The pull-back along $B$, by §IV.39, maps $\Omega^k(X)$ into $\Omega^k(\partial X)$ Chapter4/BoundaryForms2021 · MPolyfold.formsPullback_mem_bForms. It passes to the limits Chapter4/BoundaryForms2021 · MPolyfold.bFormsInfty Chapter4/BoundaryForms2021 · MPolyfold.jStar, with $dj^* = j^*d$ Chapter4/BoundaryForms2021 · MPolyfold.jStar_comp_D. The text states this identity without proof (Findings, Chapter 4, item 121).

**The relative complex.** In degree 0 it is $\Omega^0_\infty(X)$, and in degree $k+1$ it is $\Omega^{k+1}_\infty(X) \times \Omega^k_\infty(\partial X)$ Chapter4/BoundaryForms2021 · MPolyfold.RelForms. The differential is $d(\omega, \tau) = (d\omega, j^*\omega - d\tau)$ Chapter4/BoundaryForms2021 · MPolyfold.relD. It squares to 0 Chapter4/BoundaryForms2021 · MPolyfold.relD_comp_relD, because $d^2 = 0$ and $dj^* = j^*d$. $H^k_{dR}(X, \partial X)$ is its cohomology Chapter4/BoundaryForms2021 · MPolyfold.relDeRham Chapter4/BoundaryForms2021 · MPolyfold.relDeRhamMk_eq_iff. If $\partial X = \emptyset$, then $bX$ is empty and $H^k_{dR}(X, \partial X) \cong H^k_{sc}(X)$ Chapter4/BoundaryForms2021 · MPolyfold.relDeRhamEquivOfNoBoundary. The text calls the relative complex a differential algebra. The wedge product on it is §IV.46.

## IV.41 Integration and Stokes' theorem on quadrants

Theorem 4.4.7 (preprint 4.28) integrates over finite-dimensional sub-M-polyfolds with corners. The integration is built from the bottom: first in $\mathbb{R}^{n+1}$, then on Mathlib manifolds with corners (§IV.42), then for sc-forms (§IV.43).

**Top forms in $\mathbb{R}^{n+1}$.** For a top-degree form $\omega$, define $\int_s \omega = \int_s \omega(x)(e_0, \dots, e_n)\, dx$ on the standard frame Chapter4/EuclideanForms2021 · stdFrame Chapter4/EuclideanForms2021 · topFormIntegral. Pull-back Chapter4/EuclideanForms2021 · formPullback multiplies the frame value by the determinant Chapter4/EuclideanForms2021 · alternatingMap_comp_eq_det_mul. So the change of variables formula holds with the sign of the determinant Chapter4/EuclideanForms2021 · topFormIntegral_image.

**Faces and their orientation.** The quadrant is $Q_S = \{x_i \geq 0, i \in S\}$. Its boundary differs from that of the open quadrant by a null set Chapter4/EuclideanForms2021 · openQuadrant_ae_eq_quadrant, and the corners are null Chapter4/EuclideanForms2021 · image_faceEmbedding_openQuadrant. The face $\{x_i = 0\}$ is parametrized by $\iota_i : \mathbb{R}^n \to \mathbb{R}^{n+1}$, inserting a zero at place $i$ Chapter4/EuclideanForms2021 · faceEmbedding. It carries the pulled-back form Chapter4/EuclideanForms2021 · facePullback and the sign $(-1)^{i+1}$ Chapter4/EuclideanForms2021 · faceSign. This is the sign of the frame $(-e_i, \text{face frame})$ Chapter4/EuclideanForms2021 · det_cons_neg_stdFrame_faceFrame, the induced orientation with the outward vector $-e_i$ first (2021 Definition 9.3.8).

**Stokes on a quadrant** Chapter4/EuclideanStokes2021 · stokes_quadrant. For a $C^1$ form $\omega$ of degree $n$ vanishing off a compact set,

$$\int_{Q_S} d\omega = \sum_{i \in S} (-1)^{i+1} \int_{\mathbb{R}^n_{\geq 0}} \iota_i^* \omega.$$

The engine is Mathlib's divergence theorem on a box Chapter4/EuclideanStokes2021 · integral_divergence_piQuadrant. Take the box $[a, b]$ with $a_i = 0$ for $i \in S$, $a_i = -R$ otherwise, and $b_i = R$. The faces $x_i = -R$ and $x_i = R$ contribute nothing, because $\omega$ vanishes there. A form defined only on an open $U$ is extended by $0$ Chapter4/EuclideanStokes2021 · stokes_quadrant_inter_open.

# IV.42 Forms, integrals and Stokes' theorem on Mathlib manifolds with corners

**Forms and $d$.** A $k$-form on a manifold $M$ modelled on $[0, \infty)^{n+1}$ is a family of alternating forms on the tangent spaces Chapter4/ManifoldForms2021 · ManifoldForm. It has chart representatives Chapter4/ManifoldForms2021 · chartForm, which are pull-backs of one another Chapter4/ManifoldForms2021 · chartForm_eq_formPullback, and a smoothness class Chapter4/ManifoldForms2021 · ContMDiffForm. $d$ is the derivative of the chart representative within the range of the model Chapter4/ManifoldForms2021 · ManifoldForm.extDeriv. It commutes with pull-back by chart changes Chapter4/ManifoldForms2021 · extDerivWithin_formPullback, so it has the right representative in every chart Chapter4/ManifoldForms2021 · chartForm_extDeriv.

**Integration.** The interval $[0, 1]$, with the quadrant model, has no oriented atlas: a chart at 1 reverses the orientation. So the charts are signed Chapter4/ManifoldIntegral2021 · signedOr Chapter4/ManifoldIntegral2021 · orSign. The integral in one chart carries the sign of the orientation there Chapter4/ManifoldIntegral2021 · chartIntegral Chapter4/ManifoldIntegral2021 · chartIntegral_eq_integral_sign. *Integration data* Chapter4/ManifoldIntegral2021 · IntegrationData consist of finitely many charts, regions on which the orientation has a constant sign, and $C^\infty$ functions $\rho_j$ summing to 1 on the compact support. They exist Chapter4/ManifoldIntegral2021 · exists_integrationData, by a subordinate smooth partition of unity. Any two give the same sum Chapter4/ManifoldIntegral2021 · IntegrationData.integral_eq, by the change of variables in the overlaps. This defines $\int_M \omega$ Chapter4/ManifoldIntegral2021 · ManifoldForm.integral Chapter4/ManifoldIntegral2021 · ManifoldForm.integral_eq_integrationData. The integral is additive, and it changes sign with the orientation Chapter4/ManifoldIntegral2021 · ManifoldForm.integral_neg_orientation.

**The boundary integral.** In one chart, the boundary integral is $\sum_i (-1)^{i+1} \int \iota_i^* \omega_e$ Chapter4/ManifoldStokes2021 · chartBoundaryIntegral. Through integration data this defines $\int_{\partial M} \omega$ Chapter4/ManifoldStokes2021 · ManifoldForm.boundaryIntegral. It does not depend on the data Chapter4/ManifoldStokes2021 · IntegrationData.boundaryIntegral_eq. The proof of independence goes through Stokes: both sums equal $\int_M d\omega$ for an extension. The orientation of the faces has to be compatible across charts. The chart change maps a face to a face Chapter4/ManifoldBoundary2021 · exists_face_chartChange, inward vectors to inward vectors, with $(De_i)_j > 0$. The sign identity faceSign$(i)$ det $A = (Ae_i)_j$ faceSign$(j)$ det $\bar{A}$, where $\bar{A}$ is the face restriction Chapter4/ManifoldStokes2021 · faceSign_mul_det_eq, gives the compatibility Chapter4/ManifoldBoundary2021 · orientation_face_chartChange.

**Stokes' theorem** Chapter4/ManifoldStokes2021 · stokes_of_compactSpace. For $M$ compact with a continuous orientation and $\omega$ of class $C^1$, $\int_M d\omega = \int_{\partial M} \omega$. The proof has two steps: 1. Stokes in one chart is §IV.41 Chapter4/ManifoldStokes2021 · chartIntegral_extDeriv_eq. 2. With integration data, $d\omega = \sum_j d(\rho_j \omega)$, so the general case is the sum of the chart cases Chapter4/ManifoldStokes2021 · IntegrationData.stokes Chapter4/

ManifoldStokes2021 · integral_extDeriv_eq_boundaryIntegral.

**Forms on the faces of $M$.** These are face data in the charts Chapter4/ManifoldFaceForms2021 · IsChartFaceForm Chapter4/ManifoldFaceForms2021 · ChartFaceForm. On each chart face there is a smooth form, compatible under chart changes at points of degeneracy index 1, and agreeing at corners on vectors tangent to both faces. Every smooth form restricts to such data Chapter4/ManifoldFaceForms2021 · ChartFaceForm.ofForm, so they are satisfiable. Compactly supported face data extend to a form on $M$ Chapter4/ManifoldFaceForms2021 · exists_extension, by an inclusion–exclusion sum over the faces of box charts. This defines the integral of face data as the boundary integral of an extension Chapter4/ManifoldFaceForms2021 · ChartFaceForm.integral. It also gives $\int_{\partial M} d\tau = 0$ Chapter4/ManifoldFaceForms2021 · ChartFaceForm.integral_d_eq_zero, by Stokes and $d^2 = 0$ Chapter4/ManifoldFaceForms2021 · ManifoldForm.extDeriv_extDeriv.

## IV.43 Theorem 4.4.7 (preprint 4.28): Stokes' theorem for sc-forms, with the sum over the faces

**Statement** Chapter9/GapP1_Stokes2021 · MPolyfold.theorem_4_4_7_hyp. Let $X$ be an M-polyfold and $S \subseteq X$ a compact sub-M-polyfold. Assume: - $S$ has a corner structure of dimension $n+1$ (Definition 4.1.1 (preprint 4.1)); - $S$ is face-structured; - $S$ has a continuous tangent orientation $o$.

Then for $\omega \in \Omega^n_\infty(X)$,

$$\int_S (d\omega)|_S = \sum_F \int_F (\omega|_S)|_F,$$

the sum over the finitely many faces $F$ of $S$. Each face is a manifold modelled on $[0,\infty)^n$ with the induced orientation, outward vector first.

The text's "boundary $\partial N$, a union of smooth faces $F$" is read as: $S$ is face-structured (Definition 2.5.10), so every point lies on exactly $d_S(x)$ faces. This hypothesis is added; it is not derived from the text. It is satisfiable. It holds when $S$ has no boundary Chapter9/GapP1_Stokes2021 · MPolyfold.isFaceStructured_of_forall_degIndex_eq_zero, and then the theorem says $\int_S d\omega = 0$. It also holds for the quadrants Chapter11/EpEquivalenceGeometryFaces · isFaceStructured_quadrant. The corner structure and the continuity of $o$ are the meaning of "a finite-dimensional oriented manifold with boundary with corners" in the text. The 2021 text writes $\int_N$, the preprint and 2017 texts $\int_M$ (Findings, Chapter 4, items 123, 171, 172).

**Step 1: the restriction of sc-forms to $S$.** The corner structure makes $S$ a Mathlib manifold, and $S \subseteq X_\infty$ Chapter9/MathlibFormsBridge2021 · MPolyfold.subset_smoothSet_of_hasCornerStructureOfDim. An sc-form $\omega$ on $X^i$ restricts to a classical form on $S$ Chapter9/MathlibFormsBridge2021 · MPolyfold.scFormRestrict, through the pull-back of §IV.39 along the inclusion lifted to $S \to X^i$. This works because $S = S^1 = S^i$ as sets. In a chart of $X$ the restricted form is $\omega_\varphi$ evaluated on the model derivatives of the corner chart Chapter9/MathlibFormsBridge2021 · MPolyfold.scFormRestrict_apply_modelDeriv. It commutes with (4.4) Chapter9/MathlibFormsBridge2021 · MPolyfold.scFormRestrict_formLower and passes to $\Omega_\infty$ Chapter9/MathlibFormsBridge2021 · MPolyfold.formsInftyRestrict. It is $C^\infty$ Chapter9/MathlibFormsBridge2021 · MPolyfold.contMDiffForm_formsInftyRestrict, and $d(\omega|_S) = (d\omega)|_S$ Chapter9/MathlibFormsBridge2021 · MPolyfold.extDeriv_formsInftyRestrict. With §IV.42 this gives $\int_S (d\omega)|_S = \int_{\partial S} \omega|_S$ Chapter9/MathlibFormsBridge2021 · MPolyfold.stokes_formsInftyRestrict. Such $S$ exist for every tame $X$ with a submanifold in good position Chapter9/MathlibFormsBridge2021 · exists_hasCornerStructureOfDim_of_tame.

**Step 2: the faces of $S$.** A compact tame M-polyfold has finitely many faces Chapter4/GapP1_

StokesFaces · MPolyfold.finite_setOf_isFace. In a face-structured one, near each of its points and in any chart modelled on a tame retract, a face is exactly a coordinate face Chapter4/GapP1_StokesFaces · MPolyfold.exists_face_eq_coordHyper. In corner coordinates of $S$ this gives, at $x \in F$, a coordinate $i$ and an open set on which $F = \{y_i = 0\}$ Chapter4/GapP1_StokesFaceCharts · MPolyfold.CornerCoords.exists_faceData. Removing the coordinate $i$ gives a corner chart of $F$. So *every face has a corner structure of dimension* $n$ Chapter4/GapP1_StokesFaceCharts · MPolyfold.hasCornerStructureOfDim_face, as the sub-M-polyfold of Theorem 2.5.17.

**Step 3: the induced orientation.** Take a chart of a face and the chart of $S$ it comes from. The chart change of two such pairs has derivative the face restriction of the chart change of $S$ Chapter4/GapP1_StokesFaceOrientation · FaceChartPair.chartChangeDeriv_face. This derivative maps the face to the face and inward vectors to inward vectors, at every point, corners included Chapter4/GapP1_StokesFaceOrientation · FaceChartPair.inward. The orientation induced on the face reads, in every chart pair, as the sign of $o$ times $(-1)^{i+1}$ Chapter4/GapP1_StokesFaceOrientation · IsInducedOrientation. It exists, is continuous and is unique Chapter4/GapP1_StokesFaceOrientation · inducedOrientation Chapter4/GapP1_StokesFaceOrientation · isContinuous_inducedOrientation Chapter4/GapP1_StokesFaceOrientation · IsInducedOrientation.unique. On $S$ it is Chapter9/GapP1_Stokes2021 · MPolyfold.faceOrientation.

**Step 4: the face sum is the boundary integral** Chapter4/GapP1_StokesFaceIntegral · sum_integral_restrictForm_eq_boundaryIntegral. The form $\omega|_F$ is the pull-back along $F \hookrightarrow S$ Chapter4/GapP1_StokesFaceIntegral · restrictForm. Its representative in a face chart is $\iota_i^*$ of that of $\omega$ Chapter4/GapP1_StokesFaceIntegral · FaceChartPair.chartForm_restrictForm. Locally, on a *good region*, the face sum equals the chart boundary integral Chapter4/GapP1_StokesFaceIntegral · GoodRegion.sum_integral_eq. A good region is a signed region inside the face chart domains and the neighbourhood of Proposition 2.7.5 (3) Chapter9/GapP1_Stokes2021 · MPolyfold.goodRegion. Inside it, distinct faces through a point get distinct coordinates, because a point of degeneracy index 1 lies on exactly one face Chapter9/GapP1_Stokes2021 · MPolyfold.face_eq_of_degIndex_eq_one. Integration data subordinate to good regions Chapter4/GapP1_StokesFaceIntegral · ManifoldForm.exists_integrationData_of_regions sum the local identities. With Step 1 the theorem follows.

## IV.44 Parametric integrals, and Lemma 4.5.2: the Poincaré lemma on a convex piece of a partial quadrant

**Parametric integrals** Chapter4/Poincare2021 · isSck_intervalIntegral Chapter4/Poincare2021 · isScSmooth_intervalIntegral. If $g$ is $\mathrm{sc}^k$ (sc-smooth) on $[0,1] \times U$, with $[0,1]$ carrying the constant sc-structure, then $x \mapsto \int_0^1 g(t,x)\,dt$ is $\mathrm{sc}^k$ (sc-smooth) on $U$. Its derivative is the integral of the partial derivative Chapter4/Poincare2021 · scDeriv_intervalIntegral. The proof is an induction on $k$ through the tangent map, $T\int g = \int Tg$. The derivative is differentiation under the integral sign, uniform by the compactness of $[0,1]$, within the relatively open $U$. The level-wise $C^{l+1}$ criteria lose one level too many and are not used.

**The homotopy operator** (2021 p. 203) Chapter4/Poincare2021b · homotopyForm. For a closed $(k+1)$-form $\omega$ on a convex $D$ and a centre $c$, put

$$\tau(y)(v_1,\dots,v_k) = \int_0^1 t^k\,\omega(c + t(y-c))(y-c, v_1,\dots,v_k)\,dt.$$

$\tau$ is sc-smooth if $c$ is smooth `Chapter4/Poincare2021b · isScSmooth_homotopyForm`: the integrand is sc-smooth on $[0,1]\times D$ because $y \mapsto c+t(y-c)$ is, on every level. Then $d\tau = \omega$ on $D^1$ `Chapter4/Poincare2021b · ambientD_homotopyForm`. This is the classical computation: 1. Write $d\tau(y)(v_0,\dots,v_k)$ with the alternating sum $\delta$ of §IV.39. 2. Use $d\omega = 0$ to trade the alternating sum of base derivatives for the radial derivative of $\omega$. 3. The integrand becomes $\frac{d}{dt}(t^{k+1}\omega(c+t(y-c))(v_0,\dots,v_k))$, whose integral is $\omega(y)(v_\bullet)$.

The vectors are taken on level 1 at points of $D^1$, so no density argument is needed.

**Lemma 4.5.2 (2021, p. 202).** In local form `Chapter4/Poincare2021b · lemma_4_5_2`: let $D$ be relatively open in a partial quadrant and convex, and $c \in D$ smooth. Every closed sc-form of degree $k+1$ on $D$ has a primitive of degree $k$ on $D^1$. On the M-polyfold $D$, with forms as in §IV.35 `Chapter4/Poincare2021b · hasPoincare_relOpen`, the bridge is that on a relatively open $D$ the local $d$ of §IV.36 does not depend on the retraction `Chapter4/Poincare2021b · retractD_eq_ambientD`. The printed statement, with $0 \in U$ and centre $0$, is the case $c = 0$ `Chapter4/Poincare2021b · hasPoincare_relOpen_of_zero_mem`. The hypotheses are satisfiable: take $D = C = E$ and $c = 0$.

Two readings, neither an added hypothesis. The text's degrees $k \geq 1$, $k-1$ are $k+1$, $k$ here. "$d\tau = \omega$ on $U^1$" is $d\tau = \iota^*\omega$ with $\iota : U^1 \to U$. Any smooth centre is allowed. Lemma 4.4.8 needs this generality (§IV.45); the text's remark after Lemma 4.5.2, with centre $(0, w_0)$, does not cover the case needed (Findings, Chapter 4, item 177).

## IV.45 Lemma 4.4.8 (2021, p. 199): the Poincaré lemma on an M-polyfold

**Forms on open subsets.** An open $U \subseteq X$ is an M-polyfold (§IV.34). Restriction maps $\Omega^k(X)$ to $\Omega^k(U)$ `Chapter4/GapP1_OpenForms · MPolyfold.formRestrict` and commutes with $d$ `Chapter4/GapP1_OpenForms · MPolyfold.extD_formRestrict`. On the limits, for nonempty $U$ as a bundled M-polyfold `Chapter4/GapP1_OpenForms · BundledMPolyfold.openSub`, there are restrictions $\Omega^k_\infty(X) \to \Omega^k_\infty(U)$ and $\Omega^k_\infty(U) \to \Omega^k_\infty(V)$ for $V \subseteq U$ `Chapter4/GapP1_OpenForms · BundledMPolyfold.formsInftyResOpen` `Chapter4/GapP1_OpenForms · BundledMPolyfold.formsInftyResOpenOpen`. The text's "$[\omega] = [\tau]$ on $V$" of (4.7) `Chapter4/GapP1_OpenForms · BundledMPolyfold.FormsInftyEqOn` means: for representatives on one level $i$, the restrictions agree on $V^\ell$ for some $\ell \geq i$ `Chapter4/GapP1_OpenForms · BundledMPolyfold.formsInftyEqOn_of_iff`.

**Statement** `Chapter4/Poincare2021b · lemma_4_4_8`. Every $x \in X$ has open sets $V(x) \subseteq U(x)$ with the following property. For every $k$, every closed $a \in \Omega^{k+1}_\infty(U(x))$ has $b \in \Omega^k_\infty(V(x))$ with $db = a|_{V(x)}$. The text's $k \geq 1$ is $k+1$ here.

*Proof* `Chapter4/Poincare448 · lemma_4_4_8_of_local`. Lemma 4.5.2 with a smooth centre `Chapter4/Poincare448 · LocalPoincare` is §IV.44 `Chapter4/Poincare2021b · localPoincare`. 1. **The chart and the radii** `Chapter4/Poincare448 · exists_poincareChart`. At $x$, take a chart $\varphi$ with model retraction $r$, and $\varepsilon, \delta > 0$ such that $B_\varepsilon(\varphi x) \cap C$ lies in the domain of $r$ and $r(B_\delta(\varphi x) \cap C) \subseteq B_\varepsilon(\varphi x)$. Put $U(x) = \varphi^{-1}(B_\varepsilon(\varphi x))$ and $V(x) = \varphi^{-1}(B_\delta(\varphi x))$. Since $r$ is the identity on the model, this is the text's $r(B_\delta \cap r^{-1}(B_\delta))$. 2. **The two maps.** $D = B_\delta(\varphi x) \cap C$ is convex and relatively open, with a smooth point `Chapter4/Poincare448 · exists_smooth_mem_relOpen`. The map $\rho = \varphi^{-1} \circ r : D \to U(x)$ is sc-smooth `Chapter4/Poincare448 · isScSmoothMap_retractChart`, and $j = \varphi : V(x) \to D$ is sc-smooth. Their composite $\rho \circ j$ is the inclusion $V(x) \to U(x)$. 3. **Every level.** A representative of $a$ lives on some $U(x)^m$, so the lemma is needed on the $m$-th lift of $D$. That lift is the M-polyfold of $D^m \subseteq E^m$ `Chapter4/Poincare448Shift · LocalModel.liftEquiv`, the text's "working with $C^i \subset E^i$". The Poincaré property is invariant under sc-diffeomorphisms `Chapter4/Poincare448Shift · MPolyfold.HasPoincare`, so it holds on every lift `Chapter4/Poincare448 · hasPoincare_lift`. 4.

**The primitive** Chapter4/Poincare448 · exists_primitive_of_factor. $\rho^* a$ is closed on $D$, so it has a primitive $\tau$. Then $b = j^*\tau$ satisfies $db = j^*\rho^* a = (\rho \circ j)^* a = a|_{V(x)}$, by §IV.39 on the direct limits.

Here $\varphi x$ is only a level-0 point of a general partial quadrant, so $0 \notin D$ in general. This is why Lemma 4.5.2 is needed with a smooth centre other than 0 (§IV.44).

## IV.46 The wedge product on the forms on $\partial X$ and on the relative complex (gap G.80)

The text (p. 199) calls $\Omega^*_\infty(X, \partial X)$ a differential algebra and gives no product. Two products are formalized.

**On the forms on $\partial X$.** A form on $\partial X$ is an sc-form on $bX$ that agrees over $B$ (§IV.40). The wedge product of two forms that agree over an sc-smooth $f$ agrees over $f$, because at a point it is computed from the values of the two forms on the same tangent vectors Chapter4/GapP3_K4_BoundaryWedge · MPolyfold.IsAgreeingOver.formWedge Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsWedge. It passes to the limits Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge_of_of and, up to the casts of degrees Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyCast, it satisfies the Leibniz rule Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyD_agreeDFormsInftyWedge, is associative Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge_assoc and has the constant 1 as a unit on both sides Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyOne Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge_agreeDFormsInftyOne Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyCast_agreeDFormsInftyOne_wedge. The inclusion into $\Omega^*_\infty(bX)$ is multiplicative Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeDFormsInftyIncl_agreeDFormsInftyWedge, and so is the pull-back Chapter4/GapP3_K4_BoundaryWedge · BundledMPolyfold.agreeInftyPullback_formsInftyWedge. For $f = B$ this is the wedge product on $\Omega^*_\infty(\partial X)$, and $j^*(\omega \wedge \omega') = j^*\omega \wedge j^*\omega'$ Chapter4/GapP3_K4_BoundaryWedge · MPolyfold.jStar_formsInftyWedge.

**On the relative complex: the product of the mapping cone of $j^*$.** For $(\omega, \tau)$ of degree $k$ and $(\omega', \tau')$ of degree $l$,

$$(\omega, \tau)(\omega', \tau') = (\omega \wedge \omega', (-1)^k\, j^*\omega \wedge \tau')$$

Chapter4/GapP3_K4_RelWedge · MPolyfold.relWedge Chapter4/GapP3_K4_RelWedge · MPolyfold.relFst_relWedge Chapter4/GapP3_K4_RelWedge · MPolyfold.relSnd_relWedge_succ Chapter4/GapP3_K4_RelWedge · MPolyfold.relSnd_relWedge_zero. This choice is ours: the text gives no product, and this is the usual product on the mapping cone of a multiplicative map of complexes. - *$d$ is a derivation of degree $+1$:* $d(xy) = (dx)y + (-1)^k x\, dy$ Chapter4/GapP3_K4_RelWedge · MPolyfold.relD_relWedge. With $d(\omega, \tau) = (d\omega, j^*\omega - d\tau)$ (§IV.40), both sides have first component $d\omega \wedge \omega' + (-1)^k \omega \wedge d\omega'$ and second component $j^*\omega \wedge j^*\omega' - (-1)^k j^* d\omega \wedge \tau' - j^*\omega \wedge d\tau'$, by the Leibniz rules on $X$ and on $\partial X$ and the multiplicativity of $j^*$. - *Associativity* Chapter4/GapP3_K4_RelWedge · MPolyfold.relWedge_assoc, from the associativity on $X$ and on $\partial X$ and $j^*(\omega \wedge \omega') = j^*\omega \wedge j^*\omega'$. - *The unit.* $(1, 0)$ is a unit on the left Chapter4/GapP3_K4_RelWedge · MPolyfold.relOne Chapter4/GapP3_K4_RelWedge · MPolyfold.relCast_relWedge_relOne. On the right, $(\omega, \tau)(1, 0) = (\omega, 0)$ Chapter4/GapP3_K4_RelWedge · MPolyfold.relWedge_relOne, so it is not a unit there; relative cohomology has no unit in general either. The product is not graded commutative on forms. - *The graded ring.* With Mathlib's graded multiplication Chapter4/GapP3_K4_RelWedge · MPolyfold.relGMul Chapter4/GapP3_K4_RelWedge · MPolyfold.relGNonUnitalNonAssocSemiring Chapter4/GapP3_K4_RelWedge · MPolyfold.relGraded_of_mul_of and associativity

`Chapter4/GapP3_K4_RelWedge · MPolyfold.relGraded_mul_assoc`, $\bigoplus_k \Omega^k_\infty(X, \partial X)$ is an associative ring without unit `Chapter4/GapP3_K4_RelWedge · MPolyfold.relGradedNonUnitalRing`. The differential on it `Chapter4/GapP3_K4_RelWedge · MPolyfold.relGradedD` squares to $0$ `Chapter4/GapP3_K4_RelWedge · MPolyfold.relGradedD_comp_self` and is a derivation `Chapter4/GapP3_K4_RelWedge · MPolyfold.relGradedD_of_mul_of`. The projection to $\Omega^*_\infty(X)$ is multiplicative.

So the relative complex is read as a differential graded algebra without a right unit (V4.91). The pull-back of forms on $\partial X$ along a general sc-smooth map is not formalized; the one use the text makes of it is (§IV.25). Gap G.80 is closed (Findings, Chapter 4, item 178).

# Part V. The Fredholm package for M-polyfolds (HWZ Chapter 5, §§5.1–5.5 (preprint §§5.1–5.4))

Chapter 5 turns the implicit function theorems of Chapter 3 into the working tools of a Fredholm theory: a way to measure perturbations (auxiliary norms, §5.1), the compactness that keeps the solution sets of small perturbations under control (§5.2), a supply of perturbations built from sc-smooth bump functions (the start of §5.3), and the perturbation theorems themselves — transversality when there is no boundary (Theorem 5.3.5 (preprint 5.13)), general position (Theorem 5.3.10 (preprint 5.18)), good position relative to the boundary (Theorem 5.3.12 (preprint 5.19) and Theorem 5.25 (preprint; not in the 2021 edition)), and the Morse-type Theorem 5.3.15 (preprint 5.22). §5.2 (preprint Appendix 5.4.1) proves Theorem 5.2.2 (preprint 5.5); §5.5 (preprint Appendix 5.4.2) relates sc-smooth bump functions and partitions of unity to the geometry of Banach spaces. This Part retells §5.1 (§§V.1–V.3); §5.2, with the proof that the preprint gives in Appendix 5.4.1 (§§V.4–V.7), the longest retelling being Theorem 5.2.2 (preprint 5.5) (§§V.5–V.6); the start of §5.3 — bump functions, the calculus of sections, Lemma 5.3.3 (preprint 5.11), allowable sections (§§V.8–V.10); Theorem 5.3.5 (preprint 5.13) (§§V.11–V.15); general position, Lemma 5.26 (preprint; not in the 2021 edition) and Theorem 5.3.10 (preprint 5.18) (§§V.16–V.20); Theorem 5.3.12 (preprint 5.19) and Theorem 5.25 (preprint) (§§V.21–V.22); the homotopy lemmas (§V.23); §5.5 (preprint Appendix 5.4.2) (§§V.24–V.26); what is not formalized (§V.27); and what the formalization taught us (§V.28). A fourth addendum adds Corollary 5.2.5 (preprint 5.8) on the shifted strong bundle (§V.29), the metric space of the allowable sections (§V.30) and weak against pointwise bump functions (§V.31); §V.32 compares with the Springer 2021 edition. A fifth addendum adds Remark 5.3.8 (preprint 5.16) with Theorem 5.3.10 (preprint 5.18) relative to a compact part (§V.33), the dense form of Theorem 5.3.12 (preprint 5.19) (§V.34), Remark 5.3.14 (preprint 5.21) (§V.35), Theorem 5.3.15 (preprint 5.22) (§§V.36–V.41), Theorem 5.5.10 (preprint 5.37) for every chart (§V.42), and the theorems that §5.5 cites: Theorem 5.5.5 (preprint 5.32) with the Sobolev remark (§§V.43–V.44), Theorem 5.5.13 (preprint 5.40) (§V.45) and Theorem 5.5.16 (preprint 5.43) (§§V.46–V.47). A sixth addendum (Part III, wave 1) adds the items of the 2021 edition that the preprint does not have, numbered by the 2021 edition with the 2017 numbers in parentheses: weak sequential compactness in reflexive spaces (§V.48); reflexive $(0,1)$-fibres, Lemma 5.1.5, mixed convergence, reflexive auxiliary norms and Proposition 5.1.8 (§§V.49–V.51); the reflexive local compactness property with Proposition 5.2.7, Theorem 5.2.8 and Remarks 5.2.9, 5.2.10 (§§V.52–V.53); and §5.4, Definition 5.4.1 and the $\partial$-extension Theorem 5.4.2 (§§V.54–V.55).

**What is proved, and what is not.** Theorems 5.2.2, 5.2.3, 5.2.4, 5.3.5, 5.3.10, 5.3.12, 5.3.15, 5.5.5, 5.5.7,

5.5.10, 5.5.13, 5.5.16 (preprint 5.5, 5.6, 5.7, 5.13, 5.18, 5.19, 5.22, 5.32, 5.34, 5.37, 5.40, 5.43) and Theorem 5.25 (preprint) are proved: Theorems 5.2.2–5.2.4 (preprint 5.5–5.7) without tameness, Theorem 5.3.12 (preprint 5.19) also in its dense form (§V.34; gap G.45), Theorem 5.5.10 (preprint 5.37) for every chart (§V.42). So are Lemmas 5.1.2, 5.3.3, 5.3.16, 5.3.17 (preprint 5.2, 5.11, 5.23, 5.24) and Lemma 5.26 (preprint), Propositions 5.1.3, 5.3.2, 5.5.3, 5.5.8, 5.5.11, 5.5.15 (preprint 5.3, 5.10, 5.30, 5.35, 5.38, 5.42), Remarks 5.3.4, 5.3.8, 5.3.14 (preprint 5.12, 5.16, 5.21) (§§V.10, V.33, V.35; gaps G.44, G.47), and Corollary 5.5.17 (preprint 5.44), now without hypothesis (§V.46). Since 2026-11-13 the items of the 2021 edition that the preprint does not have are proved as well (§§V.48–V.55): Definitions 5.1.4, 5.1.6, 5.1.7, 5.2.6 and 5.4.1, Lemma 5.1.5, Proposition 5.1.8, whose proof the text leaves to the reader, Proposition 5.2.7 and the remark after it, in a stronger form that does not use the reflexive local compactness, Theorem 5.2.8, in a stronger form for every continuous section (tameness and the sc-Fredholm property are not used), and the $\partial$-extension Theorem 5.4.2 of §5.4, which the text only sketches (2017: Definitions 5.2, 5.3, 5.4, 5.6, 5.11, Lemma 5.2, Propositions 5.2, 5.3, Theorems 5.4, 5.9). Remark 5.2.9 restates Theorem 5.2.8. The weak sequential compactness of bounded sets in reflexive spaces, which the text uses without comment, is proved (§V.48). Every definition of the chapter is formalized. Theorem 5.3.15 (preprint 5.22) (gap G.48) is proved with no added hypothesis, Definition 5.3.13 (preprint 5.20) being read in its 2021 form, whose condition (3) supplies the hypothesis $dt(x) \neq 0$ that the preprint's proof uses (§§V.23, V.32); "Morse-type" is read through smooth parametrizations of the solution set and shown independent of the parametrization (§§V.36–V.41). The cited Theorems 5.5.5, 5.5.13, 5.5.16 (preprint 5.32, 5.40, 5.43) (gaps G.50, G.52, G.53) are proved, the last for complete spaces, the text's "Banach" (§§V.43, V.45–V.47). The Sobolev remark (gap G.51) is proved for every domain on level 0; the sc-structure $E_m = W^{1+m,p}(\Omega)$ is built for $\Omega = (0,1)$ and every $p > 1$, and on other domains its compactness and density are hypotheses — the one narrower row of the chapter, V5.76 (§V.44). The step of Theorem 5.3.12 (preprint 5.19) at the boundary zeros that the formal proof avoids (Finding 88, gap G.46) is shown not to hold as we read it by a Lean counterexample (§V.22). Corollary 5.2.5 (preprint 5.8) is proved on the shifted strong bundle $P^1 : Y^1 \to X^1$, which the text uses without defining it, and by iteration for every fixed $i$, with the filler condition on level 1 that its "it is a priori clear" hides (§V.29; gap G.41). The perturbation theorems take $f$ with *tame germ data* (Definition 3.6.4 (preprint 3.51)), the reading of the tame strong bundle with which §5.3 opens and the hypothesis under which the §3.6 implicit function theorems are proved since their repair (Part III, Findings of Chapter 3, items 190–199); Theorem 5.3.10 (preprint 5.18) also assumes $X$ tame, which its proof uses through Theorems 4.2.11 and 4.3.1 (preprint 4.22 and 4.23). The metric space $(\Gamma^{+,1}_U(P), \rho)$ is built, and the text's "not complete" holds for the trivial bundle but does not hold as we read it without a hypothesis (§V.30; gap G.42). Nothing to formalize: Remark 5.3.6 (preprint 5.14), a comment on earlier work (gap G.43), Remark 5.2.10 (2017: 5.2), a comment on inductive perturbation arguments, and Theorem 5.27 (preprint), which is not in the 2021 edition (gap G.49; its general-position analogue is in §V.33). Not formalized (§V.27): Corollary 5.2.5 (preprint 5.8) as one statement for a family indexed by $i$, and its tame form; non-completeness of $(\Gamma^{+,1}_U(P), \rho)$ in the text's setting; weak against pointwise bump functions at a non-smooth point; the comparison of the parametrizations of §V.36 with the Mathlib charts of the solution set; the Sobolev sc-structure on domains other than $(0,1)$.

**Four structural choices.** (1) *The linearization is read in charts.* $T_xX$ carries no linear structure in the formalization (gap G.7 of Chapter 2), so $f'(x)$ is the operator $\mathrm{Lin}(z) : T_zO \to K_z$ of a strong bundle chart (Part III), transversality is surjectivity of Lin in a chart, and general position is stated in a tame chart; its chart independence is proved where it matters (§V.18). (2) *"Generic" is "off a null set".* Every perturbation theorem is proved first in a generic form — for $\lambda$ outside a Lebesgue null set of a small ball, $s_0 + \sum \lambda_i t_i$ does the job — and the text's dense-set form is derived from it. (3) *The parametrized solution*

*set is not built as a manifold.* The Sard step is local: at each zero of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ in good position, Theorem 3.6.8 (2) (preprint 3.54) gives a regular tangent germ chart, Sard's theorem is applied on its finite-dimensional model, and finitely many null sets are collected (§V.14). Where Theorem 4.3.1 (preprint 4.23) needs a submanifold, the *bridge lemma* supplies it (§V.17). (4) *Density is pointwise.* "Within $\varepsilon$" is $N(s(x) - s'(x)) \le \varepsilon$ for all $x$; this is $\rho(s, s') \le \varepsilon$ for the metric $\rho(s, s') = \sup_x N(s(x) - s'(x))$, which the perturbation theorems never needed and which was built afterwards (§V.30).

**Map.**

- `Chapter5/AuxNorm`: the charts $\Phi[1]$, the fibre $(Y_{0,1})_x$ and its chart norms, Definition 5.1.1 (preprint 5.1), Lemma 5.1.2 (preprint 5.2), Proposition 5.1.3 (preprint 5.3).
- `Chapter5/Compactness`: Definition 5.2.1 (preprint 5.4), Theorems 5.2.2–5.2.4 (preprint 5.5–5.7), Corollary 5.2.5 (preprint 5.8) in a chart, Remark 5.3.4 (preprint 5.12).
- `Chapter5/BumpFunctions`: $\mathbb{R}$ as an M-polyfold, Definitions 5.3.1, 5.5.2, 5.5.6 (preprint 5.9, 5.29, 5.33), Propositions 5.3.2, 5.5.3, 5.5.8 (preprint 5.10, 5.30, 5.35), Theorem 5.5.7 (preprint 5.34), the calculus of sections, Lemma 5.3.3 (preprint 5.11), $\Gamma^{+,1}_U(P)$.
- `Chapter5/Transversality`: Definitions 5.3.7, 5.3.9 (preprint 5.15, 5.17), Lemma 5.26 (preprint), the parametrized section, the Sard step, Theorem 5.3.5 (preprint 5.13).
- `Chapter5/GeneralPosition`: the bridge lemma, $T_y S = \ker f'(y)$, chart independence and openness of general position, properties (i)–(ii), Theorems 5.3.10, 5.3.12 (preprint 5.18, 5.19) and Theorem 5.25 (preprint).
- `Chapter5/Homotopy`: Definition 5.3.13 (preprint 5.20), Lemmas 5.3.16, 5.3.17 (preprint 5.23, 5.24).
- `Chapter5/PartitionsOfUnity`: the algebra of sc-smooth functions, Definitions 5.5.1, 5.5.4, 5.5.9, 5.5.12, 5.5.14 (preprint 5.28, 5.31, 5.36, 5.39, 5.41), Theorem 5.5.10 (preprint 5.37), Propositions 5.5.11, 5.5.15 (preprint 5.38, 5.42), Corollary 5.5.17 (preprint 5.44), the statements of Theorems 5.5.13, 5.5.16 (preprint 5.40, 5.43).
- `Chapter5/Book2021`: the 2021 forms of Definitions 5.2.1, 5.3.13 and Theorems 5.2.2, 5.3.5, the witness of Definition 5.3.13 (§V.32).
- Added in the fourth addendum (§§V.29–V.31): `Chapter5/StrongBundleShift` (the shifted strong bundle $P^1$, raising the index of sc-smooth maps and sections), `Chapter5/Corollary58` (the filler condition on every level, Corollary 5.2.5 (preprint 5.8)), `Chapter5/AllowableSectionsMetric` (the metric $\rho$, completeness and its failure), `Chapter5/BumpWeakPointwise` (bump points, an sc-Banach space as an M-polyfold).
- Added in the fifth addendum (§§V.33–V.47): `Chapter5/CylinderPerturbation2021` (the chart criterion of general position, Theorem 5.3.10 relative to a compact part, the cobordism between admissible sections), `Chapter5/GapP1_Remark538` (Remark 5.3.8), `Chapter5/GapP1_DenseGoodPosition` (the dense form of Theorem 5.3.12), `Chapter5/GapP1_Remark5314` (Remark 5.3.14); for Theorem 5.3.15 `Chapter5/Theorem5315_Ends`, `_GeneralPosition` (the ends and conclusion (1)), `_ParametricMorse`, `_MorseLagrange`, `_MorseFamily` (the parametrized Sard step, Lagrange multipliers), `_MorseParam`, `_MorseTransfer`, `_Morse` (smooth parametrizations, the families $a_i\sigma$, the covering argument), `_LocalParam`, `_LocalTangent`, `_LocalData` (the local data), `_LocalInvariance`, `_LocalInvarianceClause` (independence of the parametrization), `_Final`; `Chapter5/GapP1_ApproxEveryChart` (Theorem 5.5.10 for every chart); `Chapter5/GapP1_LpWeightedProduct`, `Chapter5/GapP1_BonicFrampton` (Theorem 5.5.5); `Chapter5/GapP1_SobolevRemark1–3`, `Chapter5/GapP1_SobolevCube1–5` (the Sobolev remark); `Chapter5/GapP1_Torunczyk` (Theorem 5.5.13); `Chapter5/GapP1_SeparablePOU`, `Chapter5/GapP1_HilbertPOU` (separable and Hilbert spaces, Corollary 5.5.17 without hypothesis); `Chapter5/GapP1_C0Base`,

`Chapter5/GapP1_LindenstraussPair`, `Chapter5/GapP1_WCGPRI`, `Chapter5/GapP1_GTWZ` (Theorem 5.5.16).

- Added in the sixth addendum (§§V.48–V.55): `Chapter5/P3R_WeakSeqCompact` (weak convergence, weak sequential compactness in reflexive spaces), `Chapter5/P3R_Reflexive2021` (Definitions 5.1.4, 5.1.6, 5.1.7, Lemma 5.1.5, Proposition 5.1.8), `Chapter5/P3R_ReflexiveCompactness2021` (Definition 5.2.6, Proposition 5.2.7, Theorem 5.2.8, the witnesses on the trivial bundle), `Chapter5/P3B_BoundaryExtension2021` (Definition 5.4.1, Theorem 5.4.2); the local extension it uses is `Chapter12/P3B_LocalBoundaryExtension` (Part XII, §XII.8).

Throughout, $P : Y \to X$ is a strong bundle over an M-polyfold (both paracompact and Hausdorff), a section is a map $s : X \to Y$ with $P \circ s = \mathrm{id}$, $0_x$ is the zero of the fibre over $x$, and $\operatorname{supp} s$ is the closure of $\{s \neq 0\}$ `Chapter5/BumpFunctions · sectionSupport`. A strong bundle chart $\Phi$ has base chart $\varphi : V \to O$, $O = r(U)$ a retract, and local bundle $K \to O$; the *principal part* of $f$ is $\widehat{f}(u) = \Phi(f(\varphi^{-1}(u)))_2$, and $\mathrm{Lin}(z)$ is its linearization at a level-1 point $z$ (Part III). $|\lambda|$ is the maximum norm on $\mathbb{R}^m$.

## V.1 The fibre $Y_{0,1}$ and the charts $\Phi[1]$

§5.1 works on the part $Y_{0,1}$ of biregularity $(0,1)$, the M-polyfold $Y[1]$ of §2.6 (preprint §2.5) (Part II), and assumes two things the project did not have: charts $\Phi[1]$ for *every* strong bundle chart, and a Banach structure on the fibres of $Y_{0,1}$.

**$Y_{0,1}$ in any chart.** For a strong bundle chart $\Phi$ compatible with the atlas — the text's "strong bundle isomorphism $\Phi : Y|V \to K$" — a point $y$ over $\operatorname{dom}\varphi$ lies in $Y_{0,1}$ iff $\Phi(y)_2 \in F_1$: compare with a chart of the atlas through the transition map, which is a strong bundle map and so preserves biregularity `Chapter5/AuxNorm · StrongBundle.mem_oneSet_iff_of_isChart`. On $P^{-1}(V) \cap Y_{0,1}$ the map $w \mapsto \Phi(w)$ lifted to $K[1]$ is then a partial homeomorphism `Chapter5/AuxNorm · StrongBundle.oneChartOf`, compatible with every chart $\Phi'[1]$ of the atlas because the transition is the $[1]$-transition of Definition 2.6.4 (preprint 2.50) `Chapter5/AuxNorm · StrongBundle.oneChartOf_transition` `Chapter5/AuxNorm · StrongBundle.oneChartOf_compatible`, hence a chart of the M-polyfold $Y[1]$ `Chapter5/AuxNorm · StrongBundle.isChart_oneChartOf`. For a chart of the atlas it is the chart of Part II `Chapter5/AuxNorm · StrongBundle.oneChart_eq_oneChartOf`.

**The fibre.** The fibres $P^{-1}(x)$ carry linear structures; the points of $P^{-1}(x)$ in $Y_{0,1}$ form a linear subspace $(Y_{0,1})_x$, because the charts are linear on the fibres and $F_1 \subseteq F_0$ is a subspace `Chapter5/AuxNorm · StrongBundle.oneFiber`. Its points are points of $Y[1]$ and conversely `Chapter5/AuxNorm · StrongBundle.toOne` `Chapter5/AuxNorm · StrongBundle.ofOne`; its zero is the point $0_x$ of $Y[1]$ `Chapter5/AuxNorm · StrongBundle.zeroOne` `Chapter5/AuxNorm · StrongBundle.fiberZero_mem_oneSet`.

**The chart norm $|h|_1$.** In a chart around $x$, the fibre component of $\Phi[1]$ is a linear map $(Y_{0,1})_x \to F_1$ `Chapter5/AuxNorm · StrongBundle.liftOneFiber`, injective because $\Phi$ is injective on the fibre `Chapter5/AuxNorm · StrongBundle.liftOneFiber_injective`. Its image is the slice $\{h \in F_1 \mid (\varphi(x), h) \in K[1]\}$ `Chapter5/AuxNorm · StrongBundle.mem_range_liftOneFiber`, the fixed point set of the continuous map $h \mapsto R[1](\varphi(x), h)$, hence closed in $F_1$ `Chapter5/AuxNorm · StrongBundle.isClosed_slice`. So a sequence of the fibre that is Cauchy for $|\Phi[1](\cdot)_2|_1$ converges: its images converge in $F_1$, the limit stays in the slice, and it is the image of a point of the fibre `Chapter5/AuxNorm · StrongBundle.complete_liftOneFiber`. This is the text's "the fibers $Y_x$ have the structure of Banach spaces". The norm depends on the chart, but only up to equivalence, and locally uniformly: by Lemma 5.1.2's (preprint 5.2) upper bound (§V.2) applied to $g = |\Phi_1[1](\cdot)_2|_1$, the norms of two charts are comparable with one constant on the fibres over a neighbourhood of $x$ `Chapter5/AuxNorm · StrongBundle.exists_norm_liftOne_le_mul`, and so are finitely many charts and one more `Chapter5/`

AuxNorm · StrongBundle.exists_uniform_comparison.

## V.2 Auxiliary norms and Lemma 5.1.2 (preprint 5.2)

**Definition 5.1.1 (preprint 5.1).** A function $N : Y[1] \to \mathbb{R}$ is a *complete norm on the fibre over* $x$ if $N(v) = 0$ only for $v = 0$, $N(tv) = |t|N(v)$, $N(v + w) \le N(v) + N(w)$, and every $N$-Cauchy sequence of $(Y_{0,1})_x$ converges Chapter5/AuxNorm · StrongBundle.IsCompleteFiberNorm. An *auxiliary norm* is a continuous such $N$, complete on every fibre, with condition (2): if $P(w_k) \to x$ and $N(w_k) \to 0$, then $w_k \to 0_x$ in $Y[1]$ Chapter5/AuxNorm · StrongBundle.IsAuxNorm. It is non-negative Chapter5/AuxNorm · StrongBundle.IsAuxNorm.nonneg and homogeneous on the fibres Chapter5/AuxNorm · StrongBundle.IsAuxNorm.smul.

**Lemma 5.1.2 (preprint 5.2), the upper bound.** Let $g : Y[1] \to \mathbb{R}$ be continuous at $0_x$ and positively homogeneous on the fibres near $x$, and $\Phi$ a chart around $x$. By continuity at $0_x$ — read through the chart $\Phi[1]$, whose neighbourhoods of $(\varphi(x), 0)$ are the sets $|p - \varphi(x)|_0 + |h|_1 < \varepsilon$ — $g < 1$ where $|p - \varphi(x)|_0 < \varepsilon/2$ and $|h|_1 \le \varepsilon/2$; homogeneity scales every vector of a nearby fibre to $|h|_1 = \varepsilon/2$, which gives $g(w) \le (2/\varepsilon)|h|_1$ on the fibres over a neighbourhood $V$ of $x$ Chapter5/AuxNorm · StrongBundle.exists_le_mul_norm_liftOne. This is the text's first half of "(i) □ (ii)".

**Lemma 5.1.2 (preprint 5.2), the lower bound.** For an auxiliary norm, $|h|_1 \le A\,N(w)$ near $x$ Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le. Suppose not: then for every $k$ there is $w_k$ over $\varphi^{-1}(B(\varphi(x), 1/(k+1)))$ with $|h_k|_1 > (k+1)N(w_k)$; rescale to $|h_k|_1 = 1$. Then $N(w_k) \to 0$ and $P(w_k) \to x$ — the points of the shrinking balls converge to $x$ because $\varphi^{-1}$ is continuous on $O$, so no metrizability of $X$ is needed — hence $w_k \to 0_x$ by condition (2), and $\Phi[1](w_k) \to (\varphi(x), 0)$ by the continuity of $\Phi[1]$, contradicting $|h_k|_1 = 1$. Together: **in every chart around** $x$ there are a neighbourhood $V$ and $0 < a < A$ with $aN(w) \le |h|_1 \le AN(w)$ over $V$ Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_bounds.

**Lemma 5.1.2 (preprint 5.2).** For $N$ continuous and a complete norm on every fibre, $N$ is an auxiliary norm iff every $x$ has a chart and such two-sided bounds near $x$ Chapter5/AuxNorm · lemma_5_1_2. "(i) □ (ii)" is the previous paragraph. "(ii) □ (i)", "obvious" in the text: if $P(w_k) \to x$ and $N(w_k) \to 0$, the bound gives $|h_k|_1 \to 0$, so $\Phi[1](w_k) \to (\varphi(x), 0)$, and $w_k \to 0_x$ because $\Phi[1]^{-1}$ is continuous on $K[1]$ (§V.1). In the statement $V$ is a neighbourhood of $x$ on which the bounds hold; the text's $V$ is the domain of $\Phi$, but its own proof gives them "for $P(w)$ close to $x$".

## V.3 Proposition 5.1.3 (preprint 5.3)

**Existence.** For the charts $\Phi_i$ of the atlas and a partition of unity $(\rho_i)$ subordinate to their domains — it exists since $X$ is paracompact and Hausdorff, hence normal Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate — the *glued norm* is $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)_2|_1$ Chapter5/AuxNorm · StrongBundle.gluedNorm Chapter5/AuxNorm · StrongBundle.gluedNorm_eq_sum. It is continuous on $Y[1]$: each term vanishes off the closed set $P^{-1}(\operatorname{supp}\rho_i)$, which lies in the open domain of $\Phi_i[1]$, where the term is continuous, and the sum is locally finite Chapter5/AuxNorm · StrongBundle.continuous_gluedNorm. Near a point $x$ only finitely many $\rho_i$ are non-zero, all with $x$ in the domain of $\Phi_i$ and summing to $1$ Chapter5/AuxNorm · StrongBundle.sum_partition_eq_one; their chart norms are uniformly comparable with the norm of any one chart $\Phi'$ around $x$ (§V.1), so $N$ is two-sided comparable with $|\Phi'[1](\cdot)_2|_1$ near $x$ Chapter5/AuxNorm · StrongBundle.gluedNorm_bounds. On each fibre $N$ is a convex combination of norms, hence a norm, and comparable with a complete norm, hence complete

Chapter5/AuxNorm · StrongBundle.complete_of_bounds Chapter5/AuxNorm · StrongBundle.isCompleteFiberNorm_gluedNorm. Lemma 5.1.2 (preprint 5.2) "(ii) □ (i)" concludes Chapter5/AuxNorm · proposition_5_1_3_exists. The text gives this as one sentence.

**Comparison.** For auxiliary norms $N_0$, $N_1$, Lemma 5.1.2 (preprint 5.2) "(i) □ (ii)" for both in the *same* chart around $y$ — which is why "(i) □ (ii)" was proved in every chart — gives $r_y > 0$ with $r_y N_0 \le N_1$ and $r_y N_1 \le N_0$ over a neighbourhood of $y$. At each point the admissible constants $r$ form a convex set ($r$ admissible and $0 < r' \le r$ give $r'$ admissible), so a partition of unity glues the local constants into a continuous $f > 0$ with values in these sets Mathlib/Analysis/Convex/PartitionOfUnity · exists_continuous_forall_mem_convex_of_local_const, and $f(x)N_0 \le N_1 \le f(x)^{-1}N_0$ on the fibre over $x$ Chapter5/AuxNorm · proposition_5_1_3_compare. Tameness plays no role in §5.1.

## V.4 Definition 5.2.1 (preprint 5.4) and proper sections

"$N(y) \le r$" for a point $y$ of $Y$ is read with the text's convention $N(y) = \infty$ off $Y_{0,1}$: it means $y \in Y_{0,1}$ and $N(y) \le r$ Chapter5/Compactness · StrongBundle.AuxLe, a property of the point only Chapter5/Compactness · StrongBundle.AuxLe.congr. A section $f$ is *weakly proper* if $f^{-1}(0)$ is compact and for every auxiliary norm $N$ there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section $s$ with support in $U$ and $N(s) \le 1$ Chapter5/Compactness · StrongBundle.IsWeaklyProper; *proper* if instead $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le 1\}$ is compact Chapter5/Compactness · StrongBundle.IsProper.

*Proper implies weakly proper* for a continuous $f$ Chapter5/Compactness · StrongBundle.IsProper.isWeaklyProper. Take the $U$ of properness. A solution of $f(x) = s(x)$ outside $U$ has $s(x) = 0$, so $f(x) = 0$ and $x \in f^{-1}(0) \subseteq U$ after all; inside $U$, $f(x) = s(x)$ gives $N(f(x)) = N(s(x)) \le 1$. So $\{f = s\}$ lies in the compact closure, and it is closed because $f$ and $s$ are continuous and $Y$ is Hausdorff. Weakly proper sections have compact solution sets by definition Chapter5/Compactness · StrongBundle.IsWeaklyProper.isCompact_solutionSet. The text's "obviously" hides the continuity of $f$.

## V.5 Theorem 5.2.2 (preprint 5.5): the local form for basic germs

Theorem 5.2.2 (preprint 5.5) says that for sc-Fredholm sections a compact solution set already forces properness. The appendix proves it in the coordinates of a basic germ, and that local statement is the heart of the matter.

**The setting.** A basic germ (Definition 3.1.7 (preprint 3.7), Part III) is $h : \tilde{C} \oplus W \supseteq U_0 \to \mathbb{R}^N \oplus W$ whose $W$-component is $w - B(a, w)$, where $B$ is an sc$^0$-contraction germ: on small balls, $w \mapsto B(a, w)$ contracts with constant $\le 1/2$ on level 0, uniformly in $a$, and $B(0, 0) = 0$. Part III built the *parametrized fixed point* $\delta(a, z)$, the unique solution $w$ in a closed ball $\bar{B}(r)$ of $w = B(a, w) - z$ for $(a, z)$ in a set $\Sigma$ of small parameters, continuous in $(a, z)$ on level 0 Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_continuousOn Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero.

**The local form** Chapter5/Compactness · IsSc0ContractionGerm.exists_isCompact_bdd. Assume $\tilde{C}$ closed. There is an open $O \ni 0$, *independent of $c$*, such that for every $c$ the points $x = (a, w) \in U_0 \cap O$ with $w - B(a, w) \in W_1$ and $|w - B(a, w)|_1 \le c$ lie in a compact set. The proof:

1. *A small ball.* $B(a, w) - w$ is continuous at 0 within $U_0$ and vanishes there, so for $|(a, w)|_0 < \rho_1$ it has norm $< r/8$. Let $\rho = \min(\rho_1, \rho'/2, r/2)$, with $\rho'$ the radius of the parameter set, and $O = B(0, \rho)$.

2. *The compact set of parameters.* The inclusion $W_1 \to W_0$ is compact, so the image of the closed level-1 ball of radius $c$ lies in a compact $K_c \subseteq W_0$. The set $L = (\tilde{C} \cap \bar{B}(\rho)) \times (K_c \cap \bar{B}(r/8))$ is compact (here $\tilde{C}$ closed is used) and lies in $\Sigma$.
3. *The candidate.* The compact set is the image of $L$ under the continuous map $(a, z) \mapsto (a, \delta(a, z))$.
4. *Every bounded point is hit.* Let $x = (a, w) \in U_0 \cap O$ with $w - B(a, w) = \iota(z_1)$, $|z_1|_1 \le c$. Put $p = (a, B(a, w) - w)$. Then $a \in \tilde{C}$ with $|a| < \rho$; $B(a, w) - w = \iota(-z_1)$ lies in $K_c$ and has norm $< r/8$ by step 1; so $p \in L$. And $w$ lies in the ball $\bar{B}(r)$ and solves $w = B(a, w) - (B(a, w) - w)$, so $w = \delta(p)$ by uniqueness `Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero`, i.e. $x$ is the image of $p$.

Only the level-0 contraction is used. For a basic germ $f$ itself the $W$-component of $f(x)$ is $w - B(a, w)$, so the same $O$ works with the level-1 bound $|f(x)|_1 \le c$ `Chapter5/Compactness · IsBasicGerm.exists_isCompact_bdd`. This is the appendix's (82)–(90), where the text writes "$a \in \mathbb{R}^{n}$" for $a \in \tilde{C}$ ($B$ is defined on $\tilde{C} \oplus W$ only), "$\mathbb{R}^n \oplus Y_1$" for $\mathbb{R}^N \oplus W_1$, and in (89) mixes the $\mathbb{R}^N$- and $W$-components; the $t$ of its "$h + t$" and its estimates (83), (85) disappear in the next step.

## V.6 Theorem 5.2.2 (preprint 5.5): the local step and the covering

**Uniform fibre bounds.** A strong bundle map $\Psi$ (Definition 2.6.1 (preprint 2.47)) is linear on the fibres and $\Psi[1]$ is continuous; at $(u_0, 0)$, where $\Psi[1](u_0, 0)_2 = 0$ `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_zero`, continuity gives $|\Psi[1](u, k)_2|_1 \le 1$ for $|u - u_0|_0 < \delta$, $|k|_1 \le \eta$, and fibre linearity `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_smul` turns this into $|\Psi[1](u, k)_2|_1 \le \eta^{-1}|k|_1$ for all $k \in F_1$ `Chapter5/Compactness · IsStrongBundleMap.exists_fiber_bound_one`. The text uses this bound silently when it replaces $N(f(x)) \le 1$ by a bound in basic germ coordinates.

**The local step** `Chapter5/Compactness · exists_nbhd_isCompact_closure_bdd`. Let $(f, x_0)$ be an sc-Fredholm germ (Definition 3.1.11 (preprint 3.11)) — $x_0$ need not be a zero. Its data, in the text's germ reading (§III.6): a strong bundle chart $\Phi$ centred at $x_0$, a filling germ $g$ of the principal part $\hat{f}$ on a germ inside the domain $U$ of the retraction, an sc$^+$-germ $t_0$ on a germ $G_s$, and a strong bundle isomorphism $\Psi$ with base map $\psi$ conjugating $g - t_0$ to a basic germ $h$: $h(\psi(u)) = \Psi(u, g(u) - t_0(u))_2$ for $u \in U$ near 0. The claim is an open $V \ni x_0$, independent of $C$, such that for every $C$ the closure in $X$ of $Z_C = \{x \in V \mid |\Phi(f(x))_2|_1 \le C\}$ is compact.

1. *The neighbourhood.* Near 0 the filler agrees with the principal part on $O$ and the conjugation identity holds (a set $N_1$); $t_0$ is continuous into $F_1$ on level 0 of its germ, $(G_s)_0 = O_s \cap C$ with $O_s$ open — not on $U$ — so the neighbourhood is also cut with $O_s$, and $|t_0(u)|_1 \le |t_0(0)|_1 + 1$ for $u \in U \cap O_s$, $|u|_0 < \delta_2$ (with $t_0$ sc-smooth on $U$ the cut was not needed); the fibre bound of $\Psi$ holds for $|u|_0 < \delta_3$ with constant $M$. On the side of the basic germ, the local form (§V.5) gives an open $O'$; $\psi$ maps a neighbourhood of 0 into a closed ball $\bar{B}(r_0) \cap C'$ inside the domain $U'$ of $\psi^{-1}$ (relative openness of $U'$). $V$ is the set of points of the chart domain whose image lies in all of these.
2. *The bound transported.* For $x \in Z_C$, $u = \varphi(x) \in O$, $g(u) = \hat{f}(u)$ lies in $F_1$ with $|g(u)|_1 \le C$, so $|g(u) - t_0(u)|_1 \le C + |t_0(0)|_1 + 1$, and $|h(\psi(u))|_1 = |\Psi[1](u, g(u) - t_0(u))_2|_1 \le M(C + |t_0(0)|_1 + 1) =: c'$. *Subtracting the sc$^+$-part before the conjugation* is what makes the text's $t$ and its estimates unnecessary, and the neighbourhood independent of $C$.
3. *Compactness on the side of the basic germ.* So $\psi(\varphi(Z_C))$ lies in the compact set $K_{c'}$ of the local form, intersected with the closed ball $\bar{B}(r_0) \cap C'$; its closure $A$ is compact and inside $U'$.
4. *Back to $U$.* $\psi^{-1}$ is continuous on $U'$, so $T = \psi^{-1}(A)$ is compact, contains $\varphi(Z_C)$ and lies in $U$.

5. *Back to $O$.* The closure of $\varphi(Z_C)$ lies in $T \subseteq U$ and in the closure of $O$; since $O$ is the fixed point set of the continuous retraction $r$ on $U$, it is closed in $U$, so the closure lies in $O$.
6. *Back to $X$.* $\varphi^{-1}$ is continuous on $O$, so $\varphi^{-1}$ of that closure is compact and contains $Z_C$, hence its closure.

The text's "it suffices to find in local coordinates an open neighborhood" leaves steps 4–6 implicit: its set $\{(a,w) \in U \mid |(h+t)(a,w)|_1 \le c\}$ lives in the chart domain $U$ of the filling, which contains points not in the retract.

**Theorem 5.2.2 (preprint 5.5)** `Chapter5/Compactness · theorem_5_2_2_preprint`. Let $f$ be sc-Fredholm with compact solution set, and $N$ an auxiliary norm. At each zero $x$ the germ $(f,x)$ is sc-Fredholm, so the local step gives a chart $\Phi$ and an open $V(x)$; Lemma 5.1.2's (preprint 5.2) lower bound in that chart `Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le` gives $V_1(x)$ and $A$ with $|\Phi(w)_2|_1 \le A\,N(w)$ over $V_1(x)$, so on $V(x) \cap V_1(x)$ the condition $N(f(y)) \le 1$ implies $|\Phi(f(y))_2|_1 \le A$, and $\mathrm{cl}\{y \in V(x) \cap V_1(x) \mid N(f(y)) \le 1\}$ is a closed subset of a compact set. Finitely many of these neighbourhoods cover the compact $f^{-1}(0)$; their union is $U$, and the closure of $\{y \in U \mid N(f(y)) \le 1\}$ lies in the finite union of the compact closures. That is the text's last sentence, "since the solution set of $f$ is compact, the proof is complete". Tameness is not used, as in the appendix's statement of the theorem.

**The equivalence.** For an sc-Fredholm section, which is continuous, proper □ weakly proper (§V.4) □ compact solution set □ proper (Theorem 5.2.2 (preprint 5.5)) `Chapter5/Compactness · theorem_5_2_2_iff`.

## V.7 Theorems 5.2.3, 5.2.4 (preprint 5.6, 5.7) and Corollary 5.2.5 (preprint 5.8)

**Theorem 5.2.3 (preprint 5.6)** is Theorem 3.1.13 (preprint 3.12) at a zero: a zero of an sc-Fredholm section is a smooth point (regularizing), where $(f,x)$ is an sc-Fredholm germ, and Theorem 3.1.13 (preprint 3.12) gives nested open $U(i) \ni x$ with $\mathrm{cl}\{y \in U(i) \mid f(y) = 0\}$ the image of a compact subset of $X_i$ `Chapter3/SectionCompactnessG · theorem_3_1_13_preprint` `Chapter5/Compactness · theorem_5_2_3`.

**The space $X_\infty$.** The text says "$X_\infty$ is a metric space" but never gives it a topology. It is the set of smooth points with the initial topology of the inclusions $X_\infty \to X_m$ `Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq`, metrizable by Theorem 2.3.10 (preprint 2.16) `Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel`, and a sequence converges in it iff it converges on every level `Chapter5/Compactness · MPolyfold.tendsto_smoothLevel_iff`; the inclusions into $X$ and into each $X_i$ are continuous `Chapter5/Compactness · MPolyfold.continuous_smoothVal` `Chapter5/Compactness · MPolyfold.continuous_smoothIncl`.

**Theorem 5.2.4 (preprint 5.7)** `Chapter5/Compactness · theorem_5_2_4`. $S = f^{-1}(0)$ lies in $X_\infty$ because $f$ is regularizing. Take a sequence $(x_k)$ in $S$; a subsequence converges in $X$ to some $x$. The limit is a zero, since a limit of zeros of a continuous section is a zero `Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto`, so it is smooth and $(f,x)$ is an sc-Fredholm germ. Now *Corollary 3.1.15 (preprint 3.13) at the point $x$* gives the convergence $x_k \to x$ on every level `Chapter3/Chapter5BridgeG · corollary_3_1_15_section`, hence in $X_\infty$. The text cites Theorem 5.2.3 (preprint 5.6), which gives compactness in $X_i$, not the convergence of a given sequence. Tameness is not used.

**Corollary 5.2.5 (preprint 5.8), the compact solution set.** The zeros in $X_i$ are the image of the compact subset of $X_\infty$ of Theorem 5.2.4 (preprint 5.7) under the continuous inclusion $X_\infty \to X_i$ `Chapter5/Compactness · isCompact_solutionSet_level`.

**Corollary 5.2.5 (preprint 5.8) in a chart.** "It is a priori clear that $f^i$ is sc-Fredholm" is not clear, and $Y^i \to X^i$ is not defined in the text. What is formalized is the shift of the germ data. A basic germ $f$ on $\tilde{C} \oplus W$ shifts to a basic germ $f^1$ on $\tilde{C} \oplus W^1$ with contraction germ $B^1$: the shifted neighbourhood germ `Chapter5/Compactness · ScNbhdGerm.shiftParam`, the contraction estimates on the shifted levels `Chapter5/Compactness · IsSc0ContractionGerm.shift`, and sc-smoothness of $f^1$ by Proposition 1.2.2 (preprint 1.17) for germs, which needs $\tilde{C}$ a partial quadrant `Chapter5/Compactness · IsBasicGerm.shift`. If $k$ is conjugated to $h$ by $\Psi$, then $k^1$ is conjugated to $h^1$ by the index raise $\Psi^1$ `Chapter3/BasicClassComposition · IsStrongBundleIso.shift Chapter5/Compactness · IsConjugateToBasicGerm.shift`. Hence, for an sc-Fredholm section with compact solution set, the zeros in $X_1$ are compact, and at every smooth point, in the chart of the germ, $(g - s)^1$ is conjugated to a basic germ and $s^1$ is an sc$^+$-germ on the shifted germ $G^1_s$ (Proposition 1.2.2 (preprint 1.17) for germs, `Chapter3/GermLevels · IsSckGerm.shift`), the text's setting `Chapter5/Compactness · corollary_5_2_5_chart`. In the germ reading of Definition 3.1.11 (preprint 3.11) (§III.48) $g$ is a filling germ and $g - s$ only an sc$^0$-germ on the intersection of the two germs; the level shift of the conjugation needs no more, since the conjugation identity is used near $0$ only `Chapter5/Compactness · IsConjugateToBasicGerm.shift_germ`. The statement changed with the switch: in the whole-chart reading $s^1$ was an sc$^+$-section of all of $U^1 \lhd F^1$. The filler of $g^1$ was the gap: its condition (3) on level $1$ does not follow from level $0$. It is closed, together with the shifted strong bundle and the whole corollary, in §V.29.

## V.8 $\mathbb{R}$ as an M-polyfold, and bump functions

**$\mathbb{R}$.** "An sc-smooth map $f : X \to \mathbb{R}$" needs $\mathbb{R}$ as an M-polyfold: the constant sc-structure `Chapter5/BumpFunctions · realSc` and the single chart id `Chapter5/BumpFunctions · realChart Chapter5/BumpFunctions · realPolyfold Chapter5/BumpFunctions · isChart_realChart`. Sc-smoothness of a map between M-polyfolds can be checked in one pair of charts around each point, continuity included, since the local representatives determine the map near the point `Chapter5/BumpFunctions · MPolyfold.isScSmoothMap_of_local Chapter5/BumpFunctions · isScSmoothMap_real_of_local`; an sc-smooth function read in any chart is sc-smooth on the retract `Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.real_chart`; constants are sc-smooth `Chapter5/BumpFunctions · isScSmoothMap_real_const`; a $C^\infty$ function $\sigma : \mathbb{R} \to \mathbb{R}$ is an sc-smooth self-map of $\mathbb{R}$ (Corollary 1.2.5 (preprint 1.20) in the constant structure) `Chapter5/BumpFunctions · isScSmoothMap_real_of_contDiff`, so $\sigma \circ f$ is sc-smooth `Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.comp_contDiff`.

**Definitions 5.3.1 and 5.5.2 (preprint 5.9 and 5.29).** *$X$ admits sc-smooth bump functions* if every $x$ and open $U \ni x$ have an sc-smooth $f$ with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions`. The text has two definitions: Definition 5.3.1 (preprint 5.9) asks "$f \neq 0$", Definition 5.5.2 (preprint 5.29) "not identically zero". The proof of Proposition 5.5.3 (preprint 5.30) starts from "a bump function $g$ with $g(0) = 1$", i.e. $g(x) \neq 0$ at the given point; on an M-polyfold there are no translations to move a bump to $x$ (even on an sc-Banach space, translation by a non-smooth vector does not preserve the levels). So Definition 5.3.1 (preprint 5.9) is read with $f(x) \neq 0$; Definition 5.5.2 (preprint 5.29) as written `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctionsWeak` is implied `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions.weak`. How far apart the two readings are is §V.31.

**Propositions 5.3.2 and 5.5.3 (preprint 5.10 and 5.30).** Given $x$ and $U$, take a bump $g$ with $g(x) \neq 0$ and let $f = \tau(3g/g(x) - 1)$, $\tau$ the smooth transition from $0$ on $(-\infty, 0]$ to $1$ on $[1, \infty)$ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`. Then $f$ is sc-smooth, $0 \leq f \leq 1$, $f(x) = \tau(2) = 1$, $f \equiv 1$ on the open set $\{g/g(x) > 2/3\} \ni x$, and $f \neq 0$ only where $g/g(x) > 1/3$, so $\operatorname{supp} f \subseteq \operatorname{supp} g \subseteq U$ `Chapter5/BumpFunctions · proposition_5_5_3 Chapter5/BumpFunctions · proposition_5_3_2`. The text's $\sigma$ ("0 for $s \leq 1$ and 1 for $s \geq 1$") is not a function.

## V.9 The calculus of sections, and Lemma 5.3.3 (preprint 5.11)

**$\Gamma^+(P)$ and its algebra.** $\Gamma^+(P)$ is the set of sc⁺-sections (Definition 2.6.6 (preprint 2.52)) `Chapter5/BumpFunctions · StrongBundle.scPlusSections`; their values lie in $Y_{0,1}$, because in a chart their principal parts take values in $F_1$ `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.mem_oneSet`. The text uses, and does not state, the rules that make $\Gamma^+(P)$ a module over the sc-smooth functions. The zero section is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_fiberZero`. *The product rule:* for $f$ sc-smooth and $s$ sc⁺, the fibrewise product $f \cdot s$ `Chapter5/BumpFunctions · sectionSmul` is sc⁺: in a chart its principal part on level 1 is $(f \circ \varphi^{-1}) \cdot \hat{s}$, the product of an sc-smooth function and an sc-smooth map into $F^1$, sc-smooth by the product rule of the sc-calculus `Chapter3/ParamPerturbation · IsScSmooth.smul` `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.smul_left`. Sums are sc⁺, since principal parts add in a chart `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.add`, and by induction finite sums `Chapter5/BumpFunctions · sectionSum` `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_sectionSum`.

**Cut-offs.** Lemma 5.3.3 (preprint 5.11) builds a section in one chart and "extends it by 0". For a chart $\Phi$ of the atlas, $\tau : O \to F_1$ with $z \mapsto (z, \tau(z))$ sc-smooth into $K[1]$, and $\beta$ sc-smooth with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$, the section $y \mapsto \Phi^{-1}(\varphi(y), \beta(y)\tau(\varphi(y)))$ on $\operatorname{dom}\varphi$, $0_y$ outside `Chapter5/BumpFunctions · StrongBundle.cutoffSection`, is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_cutoffSection`: on $\operatorname{dom}\varphi$ its representative in $\Phi[1]$ is $z \mapsto (z, \beta(\varphi^{-1}(z))\tau(z))$, sc-smooth by the product rule; on the open set $X \setminus \operatorname{supp}\beta$ it is the zero section; the two open sets cover $X$.

**Lemma 5.3.3 (preprint 5.11)** `Chapter5/BumpFunctions · lemma_5_3_3`. Let $X$ admit bump functions, $N$ be an auxiliary norm, $x$ smooth, $e$ a smooth point of $Y$ over $x$, $\varepsilon > 0$ and $U \ni x$ open. Take any chart $\Phi$ of the atlas around $x$ — neither a centred chart nor tameness is needed — and $\hat{e} = \Phi(e)_2$, a smooth point of $F$ because $e$ is smooth. The constant $\hat{e}$ is sc-smooth into $F^1$ `Chapter5/BumpFunctions · isScSmooth_const_of_mem_smooth`, and $R[1]$ is sc-smooth, so $t(z) = R(z, \hat{e})$ gives an sc⁺-section near $x$ with $t(x) = e$. Since $N$ is continuous on $Y[1]$ and $t$ is continuous into $Y[1]$, the set $\{y \mid N(t(y)) < N(e) + \varepsilon\}$ is an open neighbourhood of $x$. Proposition 5.5.3 (preprint 5.30) gives $\beta : X \to [0,1]$ with $\beta(x) = 1$ and support in that set, in $U$ and in the chart domain, and the cut-off $s = \beta \cdot t$ is sc⁺ with $s(x) = e$ and $\operatorname{supp} s \subseteq U$. Finally, at every $y$, $N(s(y)) = \beta(y)N(t(y)) \le N(t(y)) < N(e) + \varepsilon$, because $N$ is a norm on the fibre and $0 \le \beta \le 1$; off the support, $N(s(y)) = N(0_y) = 0$. The text states the bound only at $x$, where it is trivial; its proof gives, and §5.3 needs, the bound everywhere.

## V.10 Allowable sections and Remark 5.3.4 (preprint 5.12)

**$\Gamma^{+,1}_U(P)$.** For an auxiliary norm $N$ and an open $U$, the allowable sections are the sc⁺-sections with support in $U$ and $N(s(x)) \le 1$ at every point `Chapter5/BumpFunctions · StrongBundle.allowableSections` — a condition at every point because sc⁺-sections take values in $Y_{0,1}$. The text's metric $\rho(s, s') = \sup_x N(s(x) - s'(x))$ is used through its pointwise form: $N(s(x) - s'(x))$, the difference taken in the fibre $(Y_{0,1})_x$ `Chapter5/Transversality · StrongBundle.sectionDistAt`. The metric space itself is §V.30.

**Remark 5.3.4 (preprint 5.12)** `Chapter5/Compactness · remark_5_3_4`. Let $S_f \subseteq U$ with $\operatorname{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact (the $U$ of properness), and let $s$ vanish outside $U$ with $N(s(x)) \le 1$ everywhere — properties that pass to uniform limits of allowable sections, which is all the remark uses of the completion. At a zero $x$ of $f + s$: if $x \notin U$, then $s(x) = 0$ and $f(x) = 0$, so $x \in S_f \subseteq U$, a contradiction; so $x \in U$, and $f(x) = -s(x)$ gives $N(f(x)) = N(s(x)) \le 1$ `Chapter5/Compactness · StrongBundle.IsAuxNorm.auxLe_of_sectionAdd`. So $S_{f+s} \subseteq \{x \in U \mid N(f(x)) \le 1\}$, whose closure is compact. The remark

claims the solution set compact, which needs it closed; for an allowable $s$ this holds because $f+s$ is sc-Fredholm (Theorem 3.1.18 (preprint 3.15)) `Chapter3/FredholmSectionsG · theorem_3_1_18_preprint` and solution sets of continuous sections are closed `Chapter5/Transversality · isClosed_solutionSet`. This is how part (1) of Theorems 5.3.5, 5.3.10, 5.3.12 (preprint 5.13, 5.18, 5.19) and Theorem 5.25 (preprint) is proved: $U$ from Theorem 5.2.2 (preprint 5.5), then Remark 5.3.4 (preprint 5.12) and closedness.

## V.11 Transversality and the parametrized section

**Definition 5.3.7 (preprint 5.15).** $g$ is transversal if at every zero its linearization is onto in a chart `Chapter5/Transversality · StrongBundle.IsTransversal`, hence in every chart `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. The kernel of $\mathrm{Lin}(z)$ in a chart is a subspace of $E_0$ inside $T_zO$ `Chapter5/Transversality · StrongBundleChart.linKer`.

**The parametrized section.** For a section $g$ and sections $t_1,\dots,t_m$, Theorem 3.1.21 (preprint 3.18) (Part III) makes $F(\lambda,y)=g(y)+\sum\lambda_i t_i(y)$ a section of the pull-back of $P$ to $\mathbb{R}^m\times X$, sc-Fredholm when $g$ is and the $t_i$ are sc$^+$, and with tame germ data when $g$ has them `Chapter3/ParamPerturbationG · theorem_3_1_21_tame`. Its *slice* at $\lambda$ is the section $g+\sum\lambda_i t_i$ of $P$ `Chapter5/Transversality · paramSlice Chapter5/Transversality · paramSlice_eq_sectionAdd`, and $(\lambda,y)$ is a zero of $F$ iff $y$ is a zero of the slice `Chapter5/Transversality · isZeroAt_paramSection_iff`. The slices are sc-Fredholm, since $\sum\lambda_i t_i$ is sc$^+$ (§V.9, Theorem 3.1.18 (preprint 3.15)) `Chapter5/Transversality · isScFredholmSection_paramSlice`; slices of sc$^+$-sections are sc$^+$ `Chapter5/Transversality · isScPlusSection_paramSlice` with support in the union of the supports `Chapter5/Transversality · sectionSupport_paramSlice_subset` and vanish off it `Chapter5/Transversality · isZeroAt_paramSlice_of_not_mem_support`. *Allowability:* if $N(s_0)\le 1-\varepsilon$ and $N(t_i)\le 1$, the triangle inequality on the fibre gives $N(s_0+\sum\lambda_i t_i)\le 1-\varepsilon+m|\lambda|$ `Chapter5/Transversality · StrongBundle.IsAuxNorm.toOne_sum_le`, which is $\le 1$ when $m|\lambda|\le\varepsilon$ `Chapter5/Transversality · auxLe_paramSlice` — the text's "$N(\sum\lambda_j t_j(y))<\varepsilon/2$", with $|\lambda|$ the maximum norm.

**The linear algebra of "regular value ⇒ transversal".** Let $M(a,u)=Lu+\sum a_i e_i:\mathbb{R}^m\oplus T\to K$ be onto. If the projection $\ker M\to\mathbb{R}^m$ is onto, $L$ is onto: given $k=Lu+\sum a_i e_i$, choose $u'$ with $(a,u')\in\ker M$, so $k=L(u-u')$ `Chapter5/Transversality · surjective_of_forall_exists_ker_fst`. Conversely, if $L$ is onto, every $a$ lifts to $\ker M$ `Chapter5/Transversality · forall_exists_ker_fst_of_surjective`. In the product chart $(r\mapsto r-\lambda)\times\Phi$ of a chart $\Phi$ centred at $y$, the linearization of $F$ at $(\lambda,y)$ is $\mathrm{Lin}_F(0)(a,u)=\mathrm{Lin}_{g_\lambda}(0)u+\sum a_i\hat{t}_i(0)$ `Chapter3/ParamPerturbationG · linAt_paramSection`, the fibre of the product chart over its centre being the fibre $K_0$ of $\Phi$ `Chapter5/Transversality · mem_K_ofProd_zero_iff`. So if $\mathrm{Lin}_F$ is onto at $(\lambda,y)$ and its kernel projects onto $\mathbb{R}^m$, the slice $g_\lambda$ is transversal at $y$ `Chapter5/Transversality · isLinSurjAt_paramSlice_of_ker_fst`. In the other direction, if in a chart centred at a zero $y$ of $g$ the range of $\mathrm{Lin}_g(0)$ and the $\hat{t}_i(0)$ span $K_0$, then $F$ is transversal at $(0,y)$ `Chapter3/ParamPerturbationG · linAt_paramSection_zero Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_spanning`; and adding sections keeps this, because a combination over a subfamily is one over the family `Chapter5/Transversality · exists_sum_comp_eq Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_comp`. Only "regular value ⇒ transversal" is needed in Theorem 5.3.5 (preprint 5.13).

## V.12 Spanning sections

"For every $x\in S_{f+s_0}$ we find finitely many allowable sc$^+$-sections such that the range of $(f+s_0)'(x)$ together with the sections span $Y_x$": the text does not say how. Two ingredients.

**Linear algebra** `Chapter5/Transversality · exists_fin_mem_of_dense_of_closedComplemented`. Let $R \subseteq K$ be closed, complemented and of finite codimension, and $D \subseteq K$ a dense subspace. With $\pi$ a continuous projection onto $R$, $Q = \ker \pi$ is finite-dimensional, and $(1-\pi)(D)$ is a dense subspace of $Q$, hence closed, hence all of $Q$; a basis of $Q$ lifts to finitely many $d_i \in D$, and $K = R + \operatorname{span}\{d_i\}$.

**The sections** `Chapter5/Transversality · exists_spanning_sections`. Let $g$ be sc-Fredholm, $g(y) = 0, U \ni y$ open. In the chart $\Phi$ of the germ, centred at $y$, $\mathrm{Lin}(0) : T_0O \to K_0$ is $Dg(0)|_{T_0O}$ for a derivative family $Dg$ of the filling germ on its germ (the bridge for germs at the centre, `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt`), hence Fredholm (Proposition 3.6.5 (preprint 3.52) for germs, which exposes the germ and $Dg$) `Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint`, so its range $R$ is closed, complemented and of finite codimension in $K_0 = \rho(0)F$. The smooth vectors $\rho(0)F_\infty$ are dense in $K_0$ — $\rho(0)$ is continuous and preserves the levels at the smooth point 0, and $F_\infty$ is dense — so finitely many smooth $k_i$ span a complement. Each $k_i$ is $\Phi(e_i)_2$ for a smooth point $e_i$ of the fibre over $y$ `Chapter5/Transversality · mem_smoothSet_toMPolyfold0_of_chart`; Lemma 5.3.3 (preprint 5.11) gives $s_i$ with $s_i(y) = e_i$, support in $U$ and $N(s_i) < N(e_i) + 1$, and $s_i/(N(e_i)+1)$ is allowable, with support unchanged `Chapter5/Transversality · sectionSupport_sectionSmul_const_subset`, $N$ scaled `Chapter5/Transversality · StrongBundle.IsAuxNorm.oneMk_sectionSmul_const` and principal part scaled `Chapter5/Transversality · principal_sectionSmul_const`. The density of the smooth vectors is the point the text omits: Lemma 5.3.3 (preprint 5.11) only reaches smooth points of the fibre.

## V.13 Theorem 5.3.5 (preprint 5.13): openness, and the omitted compactness step

**No boundary on $\mathbb{R}^m \times X$.** "The interior case, Theorem 3.19" is applied to $F$ on $\mathbb{R}^m \times X$, so $\partial(\mathbb{R}^m \times X) = \emptyset$ is needed: $d_{\mathbb{R}^m} \equiv 0$ `Chapter5/Transversality · rnPolyfold_degIndex` and $d_{\mathbb{R}^m \times X} \le d_{\mathbb{R}^m} + d_X$ `Chapter2/FibredProduct · MPolyfold.degIndex_prod_le Chapter5/Transversality · isWithoutBoundary_prod_rn`. At a point with $d = 0$ good position is surjectivity `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`.

**Transversality spreads.** If $g$ has tame germ data and $F$ is transversal at $(0,x)$, $x$ a zero of $g$, then $(F,(0,x))$ is in good position, and Theorem 3.6.8 (1) (preprint 3.54) gives an open $V \ni (0,x)$ all of whose zeros are in good position, hence transversal; $U(x) = \{y \mid (0,y) \in V\}$ is the text's neighbourhood `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one' Chapter5/Transversality · exists_isOpen_isLinSurjAt_zero_near`. Likewise, if $F$ is transversal at every $(0,y)$, $y \in S_g$, the union of these sets is an open $V \supseteq \{0\} \times S_g$ all of whose zeros of $F$ are in good position in a tame chart `Chapter5/Transversality · exists_isOpen_goodPosition_of_isLinSurjAt`.

**The omitted step.** The text passes from the neighbourhood $U$ of $\{0\} \times S_{f+s_0}$ to "$U \subset \{|\lambda| < \delta_0\} \oplus X$" without argument; what is needed is that *every* zero of $F$ with $|\lambda|$ small lies in $V$. The topological core: if $Z \subseteq \mathbb{R}^m \times X$ is compact and the open $V$ contains $Z \cap \{\lambda = 0\}$, then $Z \setminus V$ is compact, its projection to $\mathbb{R}^m$ is compact and misses 0, and any $\delta$ below its distance to 0 works `Chapter5/Transversality · exists_pos_forall_mem_of_isCompact`. The compact $Z$ is the set of zeros of $F$ with $|\lambda| \le \delta_1$: for such $\lambda$ the perturbation $s_0 + \sum \lambda_i t_i$ is allowable (§V.11), so by Remark 5.3.4 (preprint 5.12) its zeros lie in the compact $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ of Theorem 5.2.2 (preprint 5.5) — $Z$ is a closed subset of $\bar{B}(\delta_1)\times$ (that compact set), closed because $F$ is continuous.

## V.14 Theorem 5.3.5 (preprint 5.13): the Sard step

"$\tilde{S} := F^{-1}(0)$ is a smooth manifold"; "taking a regular value $\lambda_0$ for the projection $\tilde{S} \to \mathbb{R}^m$, it is easily verified that $f + s_0 + s(\lambda_0, \cdot)$ is transversal". The formalization does not build $\tilde{S}$; it proves the needed statement near each zero `Chapter5/Transversality · exists_null_isLinSurjAt_paramSlice_near`.

*Claim.* Let $g$ have tame germ data, $t_i$ be sc$^+$, and $z_0$ a zero of $F$ in good position in a tame chart. There are an open $W \ni z_0$ and a null set $CV \subseteq \mathbb{R}^m$ such that for every zero $(\lambda, y) \in W$ with $\lambda \notin CV$ the slice $g_\lambda$ is transversal at $y$.

1. *The chart of $\tilde{S}$.* Theorem 3.6.8 (2) (preprint 3.54) gives at $z_0$ a regular tangent germ chart $D$ of the solution set of $F$ (Definition 3.2.1 (2) (preprint 3.25)): a chart $D.\varphi$ of $\mathbb{R}^m \times X$ with domain $W$, a finite-dimensional $N$, a relatively open $V \subseteq N \cap C$ and an sc-smooth graph map $\Gamma$ with $D.\varphi^{-1} \circ \Gamma$ parametrizing $\tilde{S} \cap W$ `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_two`.
2. *The projection.* $q = p \circ D.\varphi^{-1} \circ \Gamma : V \to \mathbb{R}^m$, $p$ the projection to the parameter, is sc-smooth on a finite-dimensional space with the constant structure, hence $C^\infty$ `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`.
3. *Sard on the strata.* $V$ is relatively open in a partial quadrant of $N$; Sard's theorem on each stratum of the quadrant (Part IV) makes the set $CV$ of stratified critical values of $q$ null `Chapter4/FamiliesLocal · measure_image_stratCritSet`. Since $\partial X = \emptyset$ the points over the zeros are interior and classical Sard would do, but proving that needs the degeneracy index of the induced structure; the stratified form makes the question unnecessary.
4. *The point.* A zero $(\lambda, y) \in W$ is $D.\varphi^{-1}(\Gamma(v))$ for some $v \in V$ with $q(v) = \lambda$. Let $\Phi$ be a chart centred at $y$ and $\psi = (r \mapsto r - \lambda) \times \varphi$ the product chart. On a relatively open $V' \subseteq V$ around $v$, gmap $= \psi \circ D.\varphi^{-1} \circ \Gamma$ is sc-smooth (the transition between the two charts of $\mathbb{R}^m \times X$ restricted to the overlap), and $q = \mathrm{pr}_1 \circ \mathrm{gmap} + \lambda$ there.
5. *Kernel vectors.* If $\lambda \notin CV$, $Dq(v)$ maps the stratum through $v$ onto $\mathbb{R}^m$. The values of $D\,\mathrm{gmap}(v)$ lie in the tangent space $T_0 O$ of the product chart, and in $\ker \mathrm{Lin}_F(0)$: the principal part of $F$ vanishes along $r \circ \mathrm{gmap}$, and the chain rule gives $\mathrm{Lin}_F(0) \circ D\,\mathrm{gmap}(v) = 0$. So for every $a \in \mathbb{R}^m$ there is a kernel vector with first component $a$.
6. *Conclusion.* $F$ is transversal at $(\lambda, y)$ (good position near $z_0$), and §V.11 gives the transversality of the slice at $y$.

Only range $D\,\mathrm{gmap}(v) \subseteq \ker \mathrm{Lin}_F$ is used, not "$T\tilde{S} = \ker F'$".

## V.15 Theorem 5.3.5 (preprint 5.13): the assembly

**The generic form** `Chapter5/Transversality · theorem_5_3_5_generic`. Let $\partial X = \emptyset$, $X$ admit bump functions, $f$ have tame germ data and compact solution set, $N$ be an auxiliary norm.

1. *$U$ and part (1).* Theorem 5.2.2 (preprint 5.5) gives $U \supseteq S_f$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact `Chapter5/Compactness · theorem_5_2_2_preprint`; for allowable $s$, Remark 5.3.4 (preprint 5.12) and closedness give $S_{f+s}$ compact (§V.10).
2. *The section $g = f + s_0$.* For $s_0$ sc$^+$ with $\mathrm{supp}\, s_0 \subseteq U$ and $N(s_0) \le 1 - \varepsilon$, $g$ has tame germ data `Chapter3/TameGermsAddG · theorem_3_1_18_tame`, a compact solution set (part (1)), and $S_g \subseteq U$ (Remark 5.3.4 (preprint 5.12)).
3. *Local families.* At every zero $x$ of $g$, spanning sections $s_1^x, \dots, s_{k_x}^x$ supported in $U$ (§V.12) make $F_x$ transversal at $(0, x)$, and at $(0, y)$ for the zeros $y$ in an open $U(x)$ (§V.13).

4. *One family.* Finitely many $U(x_j)$ cover the compact $S_g$; the families are united through an equivalence $\Sigma_j \mathrm{Fin}(k_{x_j}) \simeq \mathrm{Fin}(m)$ into $t_1, \dots, t_m$, allowable and supported in $U$; adding sections keeps transversality (§V.11), so $F$, whose slice at $\lambda = 0$ is $g$ `Chapter5/Transversality · paramSlice_zero`, is transversal at every $(0, y)$, $y \in S_g$.
5. *Good position near the zero slice.* An open $V \supseteq \{0\} \times S_g$ of zeros of $F$ in good position (§V.13).
6. *Allowability and compactness.* For $|\lambda| \le \delta_1 = \varepsilon/(m+1)$ the slices $s_0 + \sum \lambda_i t_i$ are allowable, and the zeros of $F$ over $|\lambda| \le \delta_1$ form a compact $Z$ (§V.13).
7. *The omitted step.* $Z \cap \{|\lambda| < \delta_0\} \subseteq V$ for some $\delta_0 > 0$.
8. *Sard.* With $\delta = \min(\delta_0, \delta_1)$, the zeros $Z_0$ of $F$ over $|\lambda| \le \delta/2$ are compact and lie in $V$; the Sard step (§V.14) at each point of $Z_0$ gives an open $W_z$ and a null $CV_z$; finitely many $W_z$ cover $Z_0$, and $CV = \bigcup CV_z$ is null.
9. *The conclusions.* For $|\lambda| < \delta/2$, $\lambda \notin CV$: $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable; every zero $y$ of $f + s_\lambda$ gives a zero $(\lambda, y) \in Z_0$, inside some $W_z$, so $f + s_\lambda$ is transversal at $y$; $S_{f+s_\lambda}$ is compact by part (1); and if it is non-empty, Theorem 3.6.10 (preprint 3.56) for a transversal section over $X$ without boundary, applied to $f + s_\lambda$ with tame germ data, makes it a sub-M-polyfold whose induced structure is tame, a manifold without boundary of locally constant dimension, with regular tangent germ data of dimension the Fredholm index at every zero `Chapter3/GlobalIFTIndexG · theorem_3_6_10_of_surjective` `Chapter5/Transversality · StrongBundle.SolutionSetIsManifold`.

**The dense form** `Chapter5/Transversality · theorem_5_3_5_preprint`. Given an allowable $s'$ and $\varepsilon > 0$, let $\varepsilon' = \min(\varepsilon, 1)$ and $s_0 = (1 - \varepsilon'/2)s'$, so $N(s_0) \le 1 - \varepsilon'/2$. The generic form gives $t$, $\delta$, $CV$; since $CV$ is null and the ball $\{|\lambda| < \min(\delta, \varepsilon'/(2m+2))\}$ has positive measure, some $\lambda$ in it avoids $CV$. Then $s = s_0 + \sum \lambda_i t_i$ has $N(s(x) - s'(x)) \le (\varepsilon'/2) N(s'(x)) + \sum |\lambda_i| N(t_i(x)) \le \varepsilon'$. The text's "$N(s_0(y) + s(\lambda_0, y)) \le 1 - \varepsilon/2$" is presumably meant as $\le 1$, its "$N(f(x)) \le 1$" in the statement is $N(s(x)) \le 1$, and its "the germ $(f, x)$ is in good position" is $(f + s, x)$.

## V.16 Definition 5.3.9 (preprint 5.17) and Lemma 5.26 (preprint)

**Definition 5.3.9 (preprint 5.17), in a tame chart** `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm`. $(f, y)$, $f(y) = 0$, is in general position if in some tame strong bundle chart around $y$ (Definition 3.6.3 (preprint 3.50)) `Chapter3/TameGerms · StrongBundle.IsTameChart`, at the level-1 point $z$ over $\varphi(y)$, $\mathrm{Lin}(z)$ is onto $K_z$, and for every presentation $T$ of the partial quadrant the kernel has an sc-complement $B$ in $T_z O$ inside the reduced tangent space $T_z^R O = T_z O \cap E_z$ `Chapter2/Tame · reducedTangentSpaceWith`, on every level: $(T_z O)_m = (\ker \mathrm{Lin}(z))_m \oplus B_m$. The chart is tame because $T_x^R X$ is the reduced tangent space of a tame M-polyfold and the dimension count below needs it. General position includes transversality `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isLinSurjAt`.

**The linear algebra of Lemma 5.26 (preprint)** `Chapter5/Transversality · isInGoodPositionIn_of_scComplement_le_reduced`. Let $r$ be a tame retraction, $z$ a smooth point of the retract, and $N, B \subseteq T_z O$ with $T_z O = N \oplus B$ and $B \subseteq T_z^R O$. Then $N$ is in good position to the cone $C_z$ inside $T_z O$ (Definition 3.1.24 (preprint 3.21)): $B$ is a good complement, because adding a vector of $E_z$ does not change membership in $C_z$ (the coordinates $\lambda_i$, $i \in I(z)$, vanish on $E_z$); and $N \cap C_z$ has non-empty interior in $N$, because the coordinates $\lambda_i$, $i \in I(z)$, map $T_z O$ onto $\mathbb{R}^{I(z)}$ — $\dim T_z O / T_z^R O = d_C(z)$ in a tame retract (Proposition 2.5.6 (preprint 2.36)) — and vanish on $B$, so they map $N$ onto $\mathbb{R}^{I(z)}$, and a vector of $N$ with all these coordinates equal to 1 is interior. The text omits the interior condition; this is the argument of Part IV for tangent spaces `Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt`, for a subspace not given as one.

**Lemma 5.26 (preprint)** `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGerm`. For $f$ sc-Fredholm and $(f, y)$ in general position: in the chart of Definition 5.3.9 (preprint 5.17) $\mathrm{Lin}(z)$ is onto and its kernel is in good position by the linear algebra; Definition 3.6.6 (preprint 3.53) then follows `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj`. For $f$ with tame germ data, good position holds in the tame chart of the germ data `Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGermTame` — the form the §3.6 theorems take. $X$ itself need not be tame. (The text places the lemma after Theorem 5.25 (preprint) and calls it "clear" after Theorem 5.3.10 (preprint 5.18); it is used in Theorem 5.3.10 (preprint 5.18) and Theorem 5.25 (preprint) here.)

## V.17 The bridge lemma, and $T_yS = \ker f'(y)$

Theorem 5.3.10 (preprint 5.18) applies Theorem 4.3.1 (preprint 4.23) to the parametrized solution set $\tilde{S}$, which needs $\tilde{S}$ to be a smooth submanifold in the sense of Definition 4.2.1 (preprint 4.13) — locally the image of an sc⁺-retraction. The implicit function theorem gives tangent germ data (Definition 3.2.1 (preprint 3.25)), and the text never connects the two.

**sc⁺ on a finite-dimensional domain** `Chapter5/GeneralPosition · isScPlus_of_isScSmooth_const`. On a finite-dimensional space with the constant structure all levels of the domain coincide, so an sc-smooth $f$ is $C^\infty$ into every $F_m$, and its lift into $F^1$ is sc-smooth: $f$ is sc⁺.

**The bridge lemma.** Let $D$ be regular tangent germ data for $M$ at $x$ (Definition 3.2.1 (2) (preprint 3.25)), with graph map $\Gamma : V \to E$ over the finite-dimensional $N$, projection $\pi$ onto $N$, and $r$ a retraction onto the chart model. Then $\Gamma$ is sc⁺ on $V$ `Chapter5/GeneralPosition · MPolyfold.TangentGermChart.Γ_isScPlus`, so $\tau = \Gamma \circ \pi \circ r$, on the set of points whose image under $\pi \circ r$ lies in $V$, is an sc-smooth retraction (as in the proof of Theorem 3.2.3 (preprint 3.27)) `Chapter3/TangentGerm · theorem_3_2_3_of_isRegular` which is sc⁺ (sc⁺ ⊡ sc-smooth), with $\tau \circ \Gamma = \Gamma$ on $V$ and retract the chart image of $M$ `Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_scPlusRetraction`. Transported to $X$ it is an sc⁺-retraction on an open set with image $M \cap V$ `Chapter4/LocalStructure · MPolyfold.isScPlusRetractionOn_of_chart Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_isScPlusRetractionOn`. Hence a set with regular tangent germ data at each point is a smooth finite-dimensional submanifold `Chapter5/GeneralPosition · MPolyfold.isSmoothSubmanifold_of_isRegular`. For solution sets: if all zeros of $f$ in an open $W$ are in good position in a tame chart, Theorem 3.6.8 (2) (preprint 3.54) gives regular data at each, on charts inside $W$, and $S_f \cap W$ is a smooth submanifold `Chapter5/GeneralPosition · StrongBundle.isSmoothSubmanifold_solutionSet_inter`.

$T_yS \subseteq \ker f'(y)$ `Chapter5/GeneralPosition · tangentSpace_subset_linKer`. In a chart, let $Q \subseteq O$ be a retract on which the principal part $\hat{f}$ vanishes, $S$ a retraction onto $Q$, $z$ a level-1 point of $Q$. Then $T_zQ = \mathrm{range}\, DS(z) \subseteq T_zO$, and $\hat{f} \circ r \circ S = 0$ gives $D(\hat{f} \circ r)(z) \circ DS(z) = 0$ by the chain rule, i.e. $T_zQ \subseteq \ker \mathrm{Lin}(z)$ `Chapter5/GeneralPosition · StrongBundleChart.mem_linKer_iff`.

**Equality, by counting** `Chapter5/GeneralPosition · exists_tangentSpace_eq_linKer`. Let $f$ have tame germ data and $(f, x_0)$ be a zero in good position in a tame chart. The index form of Theorem 3.6.8 (preprint 3.54) gives an open $V \ni x_0$ whose zeros are in good position, and regular data $D$ of a fixed dimension $d$ at each zero, with $\dim\ker \mathrm{Lin}(z) = d$ in *every* chart `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex`. On the other side, the tangent space of the bridge retraction $\tau$ at the point over $\Gamma(v)$ contains the $d$ vectors $D\Gamma(v)b_i$, $b_i$ a basis of $N$: $D\tau \circ D\Gamma(v) = D(\tau \circ \Gamma)(v) = D\Gamma(v)$, and $\pi \circ D\Gamma(v) = 1$ makes them independent `Chapter5/GeneralPosition · MPolyfold.`

TangentGermChart.exists_linearIndependent_tangentSpace. "Having $d$ independent tangent vectors" Chapter5/GeneralPosition · HasIndepTangent is carried by germs of sc-diffeomorphisms, since $D(F \circ r)(z)$ maps $T_zQ$ into $T_{F(z)}Q'$ and is injective on $T_zO$ Chapter5/GeneralPosition · HasIndepTangent.transport, so it passes from the chart of $D$ to any chart $\Phi$ through the transition maps Chapter4/Submanifolds · MPolyfold.chart_change_of_transport. A $d$-dimensional subspace of the $d$-dimensional $\ker \mathrm{Lin}(z)$ — tangent spaces of retracts are germs, so the retract $\varphi(G \cap S_f \cap V)$ of any small open $G$ has the same tangent space — is all of it.

## V.18 General position: chart independence and openness

**Definition 5.3.9 (preprint 5.17) is Definition 4.2.10 (preprint 4.21)** Chapter5/GeneralPosition · exists_isGeneralPositionGerm_iff. Near a zero $x_0$ in good position (with $f$ of tame germ data) there is an open $V$ such that every zero in $V$ is in good position, $S = S_f \cap V$ is a smooth submanifold (§V.17), and at every zero $y \in V$: $(f, y)$ is in general position iff $S$ is in general position at $y$ (Definition 4.2.10 (preprint 4.21), Part IV). Since $T_zS = \ker \mathrm{Lin}(z)$ in every chart, an sc-complement of $\ker \mathrm{Lin}(z)$ inside $T^R_zO$ *is* an sc-complement of $T_zS$ inside $T^R_zO$. Definition 4.2.10 (preprint 4.21) is chart independent (Part IV) Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart, so Definition 5.3.9 (preprint 5.17) is too, at these zeros.

**From submanifold general position to good position** Chapter5/GeneralPosition · isGoodPositionGermTame_of_isGeneralPositionAt. If $f'(y)$ is onto and a smooth submanifold $A \subseteq S_f$ is in general position at $y$, then in the tame chart of the germ data $T_zA \subseteq \ker \mathrm{Lin}(z)$ (§V.17) Chapter5/GeneralPosition · StrongBundleChart.linKer_le and $T_zA + T^R_zO = T_zO$; so $\ker \mathrm{Lin}(z) + T^R_zO = T_zO$, the kernel has an algebraic complement inside $T^R_zO$ Chapter5/GeneralPosition · exists_compl_le_of_sup_eq, and §V.16 gives good position.

**Openness** Chapter5/GeneralPosition · exists_isOpen_isGeneralPositionGerm. Let $X$ be tame, $f$ with tame germ data, $(f, x_0)$ in general position. By Lemma 5.26 (preprint) it is in good position; the previous paragraph turns its general position into general position of the submanifold $S_f \cap V$ at $x_0$; Theorem 4.2.11 (3) (preprint 4.22) for tame $X$ (Part IV) makes that open along the submanifold Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint; Definition 4.2.10 (preprint 4.21) depends only on the germ of the submanifold, since tangent spaces of chart models agree on open sets Chapter5/GeneralPosition · MPolyfold.IsGeneralPositionAt.congr_nhds; and the equivalence turns it back into general position of $(f, y)$ at the nearby zeros. The text uses this without proof ("they will also hold at $(0, y)$ for $y \in S_f$ close to $x$"; "the properties (i)–(ii) hold for all $(\lambda, y) \in \widetilde{S}$").

## V.19 Properties (i)–(ii)

In a chart centred at $y$, with a presentation $T$, the properties of the proof of Theorem 5.3.10 (preprint 5.18) are Chapter5/GeneralPosition · StrongBundleChart.IsParamGP: (i) $\mathrm{Lin}_g(0)(T_0O) + \mathrm{span}\{\hat{t}_i(0)\} = K_0$ — $F'(0, y)$ onto; (ii) every $w \in T_0O$ is $u + e$ with $e \in E_0$ and $\mathrm{Lin}_g(0)u + \sum a_i\hat{t}_i(0) = 0$ for some $a$ — $\ker F'(0, y)$ transversal to $\mathbb{R}^m \oplus T^R$. Adding sections keeps both, by linear algebra Chapter5/GeneralPosition · StrongBundleChart.IsParamGP.comp.

**(i)–(ii) give general position** Chapter5/GeneralPosition · isGeneralPositionGerm_paramSection_of_isParamGP. In the tame product chart $(r \mapsto r) \times \Phi$ of a tame chart centred at $y$ Chapter5/GeneralPosition · pc0, whose tangent space at the centre is $\mathbb{R}^m \times T_0O$ Chapter5/GeneralPosition · ofProd_mem_rangeS and whose reduced tangent space is $\mathbb{R}^m \times T^R_0O$, the minimal subspace of $\mathbb{R}^m$ being everything Chapter5/GeneralPosition · minimalSubspaceWith_rnPres: (i) is the surjectivity of $F'(0, y)$; (ii) says $\ker F'(0, y) + T^R =$

$T(\mathbb{R}^m \times O)$ on level 0, so the kernel has an algebraic complement inside $T^R$, and §V.16 gives good position of $(F,(0,y))$. *Then* $\ker F'(0,y) = T_{(0,y)}\tilde{S}$ (§V.17), which is finite-dimensional and smooth; a finite-dimensional smooth subspace has an sc-complement inside any sc-subspace it complements algebraically `Chapter4/LocalStructure · exists_scComplement_le`, which gives Definition 5.3.9 (preprint 5.17) for $F$. The text's "at this point the combined system of sections already satisfies (i) and (ii)" compresses these two steps.

**General position gives (i)–(ii)** in every tame chart centred at $y$ `Chapter5/GeneralPosition · isParamGP_of_isGeneralPositionGerm`: (i) is surjectivity read in the product chart; for (ii), $\tilde{S}$ is in general position at $(0,y)$ (§V.18), and in the product chart its tangent space is $\ker F'(0,y)$, which therefore has a complement inside $\mathbb{R}^m \times T^R$.

**Achieving (i)–(ii)** `Chapter5/GeneralPosition · exists_sections_isParamGP`. Spanning sections give (i) (§V.12) `Chapter5/GeneralPosition · spanning_of_isLinSurjAt`. For (ii), "take a linear subspace $L$ complementing $T^R_x X$ in $T_x X$ and add sections which at $x$ span $f'(x)L$": the sections must come from Lemma 5.3.3 (preprint 5.11), which needs *smooth* points of the fibre, so $L$ must consist of smooth vectors. The coordinates $\lambda_{I(0)}$ map $T_0O$ onto $\mathbb{R}^{I(0)}$ with kernel $T^R_0O$ (Proposition 2.5.6 (preprint 2.36)), and already map the dense smooth vectors of $T_0O$ onto it (a dense subspace of a finite-dimensional image is all of it); so there are smooth $l_i \in T_0O$ with $\lambda(l_i) = e_i$. $\mathrm{Lin}_g(0)l_i$ is smooth (Lin is an sc-operator at the smooth point), Lemma 5.3.3 (preprint 5.11) gives allowable sections through $\Phi^{-1}(0, \mathrm{Lin}_g(0)l_i)$ supported in $U$, and with these added every $w \in T_0O$ is $u+e$: subtract $\sum \lambda_i(w) l_i$, whose image under $\mathrm{Lin}_g(0)$ is cancelled by the new sections.

## V.20 Theorem 5.3.10 (preprint 5.18)

**From a family** `Chapter5/GeneralPosition · theorem_5_3_10_of_family`. Let $X$ be tame, $f$ with tame germ data and compact solution set, $N$ auxiliary, $U$ from Theorem 5.2.2 (preprint 5.5). Fix $s_0$ with $\operatorname{supp} s_0 \subseteq U$, $N(s_0) \le 1-\varepsilon$, and allowable $t_1, \dots, t_m$ with $F$ in general position at every $(0,x)$, $x \in S_g$, $g = f + s_0$.

1. *General position near the zero slice.* $\mathbb{R}^m \times X$ is tame `Chapter3/ParamPerturbation · prod_rn_isTame` and $F$ has tame germ data `Chapter3/ParamPerturbationG · theorem_3_1_21_tame`, so openness (§V.18) gives, around each $(0,x)$, an open set on which the zeros of $F$ are in good and general position; their union $W \supseteq \{0\} \times S_g$.
2. *Compactness and the omitted step* as in §V.13: the zeros $Z$ of $F$ over $|\lambda| \le \delta_1$ are compact, and $Z \cap \{|\lambda| < \delta_0\} \subseteq W$.
3. *The submanifold.* $A = \tilde{S} \cap W \cap \{|\lambda| < \delta\}$ is a smooth finite-dimensional submanifold of $\mathbb{R}^m \times X$ (the bridge lemma, §V.17), in general position at each of its points (Definition 4.2.10 (preprint 4.21), by §V.18 and locality), with compact closure (inside $Z$).
4. *Theorem 4.3.1 (preprint 4.23).* In its countable-cover form (Part IV) `Chapter4/Families · theorem_4_3_1_of_subset`, it gives a set $\Sigma$ of parameters with null complement over which the fibre $A_\lambda$ is a smooth submanifold of $X$ in general position.
5. *Transversality.* The Sard step of §V.14 on the compact zeros over $|\lambda| \le \delta/2$ gives a further null set off which the slices are transversal at their zeros.
6. *The conclusions for $\lambda$.* For $|\lambda| < \delta/2$ off both null sets, with $h = f + s_\lambda$: the fibre of $A$ over $\lambda$ is $S_h$; $h$ is transversal; $S_h$ is a submanifold in general position, so $(h,y)$ is in good position (§V.18), and then in general position by the equivalence of §V.18 applied at $y$ together with the locality of Definition 4.2.10 (preprint 4.21); Theorem 4.2.7 (preprint 4.18) gives the corner structure and tameness of $S_h$

Chapter4/Submanifolds · MPolyfold.theorem_4_2_7, and Theorem 4.2.11 (2) (preprint 4.22) gives $d_{S_h} = d_X$ on $S_h$ Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint. These are the conclusions collected in Chapter5/GeneralPosition · StrongBundle.SolutionSetInGeneralPosition.

**The generic form** Chapter5/GeneralPosition · theorem_5_3_10_param. At every zero $x$ of $g$ choose a tame chart centred at $x$ with a presentation; sections with (i)–(ii) there (§V.19) make $F_x$ general at $(0, x)$, hence (openness, §V.18) at $(0, y)$ for the zeros $y$ in an open $U(x)$. Finitely many $U(x_j)$ cover $S_g$; the united family $t$ keeps (i)–(ii) at every $(0, y)$ — read (i)–(ii) off general position in the chart of $x_j$, add sections, go back — so $F$ is in general position at every $(0, y)$, and the previous theorem applies. "Replacing $f + s_0$ by $f$ and $N$ by $cN$" is not needed; the proof keeps $s_0$.

**The dense form** Chapter5/GeneralPosition · theorem_5_3_10_dense is derived exactly as for Theorem 5.3.5 (preprint 5.13) (§V.15).

The text's $\tilde{S}$ omits the equation $F = 0$, and its "$p^{-1}(\{|\lambda| \le \varepsilon\})$ is compact" is the omitted step of §V.13.

## V.21 Theorem 5.25 (preprint)

Chapter5/GeneralPosition · theorem_5_25_preprint. Let $X$ be tame with bump functions, $f$ proper with tame germ data. Properness gives a compact solution set; Theorem 5.3.10 (preprint 5.18) with $s_0 = 0$ gives $U$, $t$, $\delta$ and a null $CV$ Chapter5/GeneralPosition · StrongBundle.IsAuxNorm.fiberZero_eq Chapter5/GeneralPosition · sectionAdd_fiberZero; the ball $\{|\lambda| < \delta\}$ has positive measure, so some $\lambda$ avoids $CV$, and $s = \sum \lambda_i t_i$ is allowable with $f + s$ in general position at every zero and all the conclusions of Theorem 5.3.10 (preprint 5.18); Lemma 5.26 (preprint) turns general into good position in the tame chart of the germ data (§V.16). The text's proof says "in view of Theorem 3.54 … in general position", which is the openness of §V.18 (Theorem 3.6.8 (preprint 3.54) is about good position), and writes $x \in S_f$ for $x \in S_{f+s_0}$. Here $U$ is the neighbourhood Theorem 5.3.10 (preprint 5.18) produces; the text lets the reader choose "an associated" one.

## V.22 Theorem 5.3.12 (preprint 5.19)

Chapter5/GeneralPosition · theorem_5_3_12_param. Let $X$ admit bump functions, $f$ have tame germ data and compact solution set, and $(f, x)$ be in good position at the zeros $x \in \partial X$. The text's "as before" applies the implicit function theorem to $F$ at the boundary zeros $(0, x)$, where all $t_i$ vanish and $\ker F'(0, x) = \mathbb{R}^m \times \ker f'(x)$; this is in good position to $\mathbb{R}^m \times C_x$ only if $\ker f'(x) + E_x \supseteq T_x X$, a general position condition that good position does not imply (Finding 88, below). The formal proof avoids those points:

1. *Near the boundary zeros.* Theorem 3.6.8 (1) (preprint 3.54) at each zero of $f$ on $\partial X$ gives an open set of zeros in good position Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'; their union $V_{bd}$, and an open $V_1$ with $S_f \cap \partial X \subseteq V_1$ and $\operatorname{cl} V_1 \subseteq V_{bd}$.
2. *Away from them.* The zeros of $f$ outside $V_1$ are not on $\partial X$, so interior ($d = 0$). Spanning sections at these zeros are taken with supports in $U \setminus (\partial X \cup \operatorname{cl} V_1)$, an open set containing them; finitely many families cover the compact $S_f \setminus V_1$, and $F$ is onto at every $(0, y)$, $y \in S_f$ outside $V_1$.
3. *Good position of $F$ off the boundary.* At the interior points good position is surjectivity Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt; Theorem 3.6.8 (1) (preprint 3.54) for $F$ gives an open $W \supseteq \{0\} \times (S_f \setminus V_1)$ of zeros in good position, off $\mathbb{R}^m \times \partial X$.
4. *Compactness and Sard* as in §V.15 for the zeros of $F$ over small $\lambda$ outside $\mathbb{R}^m \times V_1$.

5. *The conclusions.* For small $\lambda$ off the null set, $h = f + s_\lambda$. On $V_1$ all $t_i$ vanish, so $h = f$ there, and the good position of $(f, y)$ at the zeros in $V_1$ passes to $(h, y)$ because good position depends only on the germ of the section `Chapter3/Chapter5BridgeG · StrongBundle.IsGoodPositionGermTame.congr_nhds`. Off $V_1$ the zeros of $h$ are interior, $h$ is transversal there by the Sard step, hence in good position. Theorem 3.6.11 (preprint 3.57) makes $S_h$, if non-empty, a sub-M-polyfold whose induced structure is tame and a manifold with boundary with corners `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11`.

$s_\lambda = \sum \lambda_i t_i$ has support disjoint from $\partial X$, so it vanishes on the open neighbourhood $X \setminus \operatorname{supp} s_\lambda$ of $\partial X$, and it is "arbitrarily small" as $\lambda \to 0$. The text's tameness of $X$ is not used, and its "$(f, x)$ in good position for every $x \in S_{f+s}$" is $(f + s, x)$. The dense form is §V.34.

**Finding 88 in Lean** (gap G.46). The smallest instance `Chapter3/Counterexamples2021 · Counterexamples2021.theorem_5_3_12_goodPosition_not_stable`: the line $\mathbb{R}(1, 1)$ is in good position to the corner $[0, \infty)^2$, with good complement $\mathbb{R}(1, -1)$ `Chapter3/Counterexamples2021 · Counterexamples2021.isGoodComplement_diag_antidiag` `Chapter3/Counterexamples2021 · Counterexamples2021.isInGoodPosition_diag`, also with the sc-complement of the definition as written `Chapter3/Counterexamples2021 · Counterexamples2021.isInGoodPositionSc_diag`; but $\mathbb{R} \times \mathbb{R}(1, 1)$ is not in good position to $\mathbb{R} \times [0, \infty)^2$, not even with an algebraic complement `Chapter3/Counterexamples2021 · Counterexamples2021.not_isInGoodPosition_prod_diag`. A good complement $Y$ would have $(a, 0) + y \in \mathbb{R} \times [0, \infty)^2$ for $y \in Y$ and $|a|$ large, so the second components of $Y$ lie in $[0, \infty)^2 \cap -[0, \infty)^2 = 0$, $Y \subseteq \mathbb{R} \times 0$, and $(0, (1, 0))$ is not in $\mathbb{R} \times \mathbb{R}(1, 1) + Y$. So the step "as before" — the implicit function theorem for good position applied to $F$ at the boundary zeros — does not hold as we read it; the theorem does, by the route above.

## V.23 Definition 5.3.13 (preprint 5.20), and Lemmas 5.3.16 and 5.3.17 (preprint 5.23 and 5.24)

**Definition 5.3.13 (preprint 5.20)** `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy`. $X$ tame, $f$ with tame germ data and compact solution set, $t : X \to \mathbb{R}$ sc-smooth with $t(X) = [0, 1]$, and for every $\tau \in [0, 1]$: $X_\tau = t^{-1}(\tau)$ a sub-M-polyfold (Definition 2.5.7 (2017: 2.19)) with tame induced structure, on which the restriction $f|X_\tau$ — the section $a \mapsto (a, f(a))$ of the pull-back of $P$ along the inclusion `Chapter5/Homotopy · restrictSection` `Chapter3/ParamPerturbation · StrongBundle.pullback`, the inclusion being sc-smooth (Proposition 2.3.19 (preprint 2.21)) — has tame germ data. The text never says what $Y|X_t$ is; the pull-back of Proposition 2.6.8 (preprint 2.54) is the strong bundle structure available. The end problems are the restrictions to $t^{-1}(0)$ and $t^{-1}(1)$ (or problems isomorphic to them, Remark 5.3.14 (preprint 5.21), §V.35); their solution sets are $S_f \cap X_\tau$ `Chapter5/Homotopy · isZeroAt_restrictSection_iff`, closed in the compact $S_f$, hence compact `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy.isCompact_solutionSet_restrict`.

**Lemma 5.3.16 (preprint 5.23) and the hidden hypothesis.** The proof of Theorem 5.3.15 (preprint 5.22) fixes a critical point $x$ of $t|S$ and says "since $x$ is a smooth point, we find a one-dimensional smooth linear sc-subspace $Z \subset T_x X$ such that $T_x X = Z \oplus \ker(dt(x))$". For $dt(x) = 0$, $\ker dt(x) = T_x X$ and the only complement is 0: a non-zero complement exists only if $\ell \neq 0$ `Chapter5/Homotopy · ne_zero_of_isCompl_ker`. Theorem 5.3.15 (preprint 5.22) does not assume that $t$ is a submersion at the critical points of $t|S$, so this is a hidden hypothesis of the preprint. The 2021 condition (3) supplies it (§V.32), and Theorem 5.3.15 is proved in §§V.36–V.41. When $\ell \neq 0$, some vector $e$ of any spanning set (e.g. the dense smooth vectors) has $\ell(e) \neq 0$, and $\mathbb{R}e$ complements $\ker \ell$ `Chapter5/Homotopy · exists_isCompl_ker_of_ne_zero`. Lemma 5.3.16 (preprint 5.23) is then linear algebra: $\ell|_Z$ is an isomorphism onto $\mathbb{R}$, so $b = (\ell|_Z)^{-1} \circ \tau$ is the unique

$b : S \to Z$ with $\ell \circ b = \tau$, and adding $a$ with values in $\ker \ell$ does not change $\ell \circ b$ `Chapter5/Homotopy · lemma_5_3_16_preprint`.

**The calculus.** At a level-1 point $z$, $D(a \cdot w)(z)v = Da(z)v \cdot w(z) + a(z)Dw(z)v$ for sc-smooth $a : U \to \mathbb{R}$, $w : U \to F$ `Chapter5/Homotopy · scDeriv_smul_apply`, and $D(\ell(\rho(\cdot) - p))(z) = \ell \circ D\rho(z)$ `Chapter5/Homotopy · scDeriv_clm_sub_apply`. The function $g = \beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ on $\operatorname{dom}\varphi$, 0 outside `Chapter5/Homotopy · chartLinCutoff`, is sc-smooth when $\beta$ is with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$ `Chapter5/Homotopy · isScSmoothMap_chartLinCutoff`.

**Lemma 5.3.17 (preprint 5.24).** *The construction* `Chapter5/Homotopy · exists_scPlusSection_linAt`: for a chart $\Phi$ around the smooth point $x$, a smooth point $e$ over $x$, a continuous linear $\ell$ on the model space and $U \ni x$, let $s_e$ be the sc⁺-section through $e$ of Lemma 5.3.3 (preprint 5.11), supported in $U \cap \operatorname{dom}\varphi$, $\beta$ a bump with $\beta \equiv 1$ near $x$ and support in the chart domain, and $s = g \cdot s_e$ with $g$ as above. Then $s$ is sc⁺ (§V.9), $s(x) = g(x)e = 0$ `Chapter5/Homotopy · principal_sectionSmul`, $\operatorname{supp} s \subseteq U$ `Chapter5/Homotopy · sectionSupport_sectionSmul_subset`, and in the chart, by the product rule, $\mathrm{Lin}_s(z)v = D(g \circ \varphi^{-1} \circ r)(z)v \cdot \hat{s}_e(\varphi(x)) + g(x)\mathrm{Lin}_{s_e}(z)v = \ell(Dr(z)v)\Phi(e)_2 = \ell(v)\Phi(e)_2$ for $v \in T_zO$, since $g \circ \varphi^{-1} = \ell(\cdot - \varphi(x))$ near $\varphi(x)$ `Chapter5/Homotopy · chartLinCutoff_of_mem` and $Dr(z) = 1$ on $T_zO$. *The lemma* `Chapter5/Homotopy · lemma_5_3_17`: for a finite-dimensional $K \subseteq T_zO$ (the text's $T_xS = \ker f'(x)$) and a linear $b : K \to T_zO$ with $\mathrm{Lin}_f(z)(bv) = \lambda(v)\Phi(e)_2$, extend $-\lambda$ (linear on the finite-dimensional $K$) to a continuous linear functional $\ell$ on the model space (Hahn–Banach) and apply the construction: $\mathrm{Lin}_f(z)(bv) + \mathrm{Lin}_s(z)v = \lambda(v)\Phi(e)_2 - \lambda(v)\Phi(e)_2 = 0$. The text's graph of $S$ over $T_xS$, its complement $L$ with $j : L \to \mathbb{R}$, and its case "if this operator is the zero operator we can take $s = 0$" are not needed. That $f'(x) \circ b_\tau$ has its image in the line of a *smooth* point is a hypothesis; the text's justification ("the image of any vector in $T_xS$ belongs to $Y_\infty$") is not formalized; the proof of Theorem 5.3.15 in §V.41 does not need it, because it takes the vector of the section's value at a smooth point directly.

## V.24 Bump functions in local models: Theorem 5.5.7 (preprint 5.34) and Proposition 5.5.8 (preprint 5.35)

**Definition 5.5.6 (preprint 5.33)** `Chapter5/BumpFunctions · LocalModel.HasBumpProperty`. For $x \in O$ and $W$ open in $E_0$ with $\mathrm{cl}(O \cap W) \subseteq O$, an sc-smooth $f$ on $O$ with $f(x) \neq 0$ (as in Definition 5.3.1 (preprint 5.9)) and support in $O$ inside $W$.

**A retract is closed near its points** `Chapter5/BumpFunctions · closure_inter_ball_subset`. If $B(z,r) \subseteq W'$ and $U = C \cap W'$ is the domain of a retraction $\rho$ onto $O$, then $O \cap \mathrm{cl}\,B(z, r/2) \subseteq O$: a limit of points of $O$ in the closed half ball lies in $C$ ($C$ closed) and in $B(z,r)$, hence in $U$, where $\rho$ is continuous, and it is a fixed point of $\rho$.

**Theorem 5.5.7 (preprint 5.34)** `Chapter5/BumpFunctions · theorem_5_5_7`. "□ ": every chart's model has the property — given $x$, $W$, a bump of $X$ at $\varphi^{-1}(x)$ supported in $\varphi^{-1}(O \cap W)$, read in the chart. "□ ", "clearly" in the text, hides a support condition: a local bump $f_0$ with support in $U(x)$, $\mathrm{cl}_E(U(x)) \subseteq O$, gives $f = f_0 \circ \varphi$ on the chart domain, but $\mathrm{cl}_X\{f \neq 0\}$ may leave the domain. For $X = E$ a Hilbert space, the chart $\varphi(u) = u/(1 - |u|_0^2)$ from the open unit ball onto $O = E$, $U(x) = O$ and $f_0 \equiv 1$, the extension by 0 is the indicator of the ball. The proof chooses $U_1 \ni x$ with $\mathrm{cl}_X U_1 \subseteq U \cap \operatorname{dom}\varphi$ (regularity of $X$) and $r$ with $B(\varphi(x), r) \cap O \subseteq \varphi(U_1)$, takes the local bump in $W = B(\varphi(x), r/2)$ — admissible by the previous paragraph — and extends $f_0 \circ \varphi$ by 0: its support lies in $\mathrm{cl}_X U_1 \subseteq U$, and it is sc-smooth because it vanishes on the open complement of its support and is sc-smooth in the chart (§V.8).

**Proposition 5.5.8 (preprint 5.35).** If $\kappa\|v\|^2 \le b(v,v)$ for a continuous bilinear $b$ on $E_0$, then $f(v) = \tau(1 - b(v-x, v-x)/\rho)$ is $C^\infty$ on $E_0$, equal to 1 at $x$ and supported in a small ball; it is sc-smooth on $E$ by Corollary 1.2.5 (preprint 1.20) from the base level `Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional`, and so is its composition with any retraction onto $O$ `Chapter5/BumpFunctions · LocalModel.hasBumpProperty_of_quadratic`. "$E_0$ is a Hilbert space" is stated as $\|v\|^2 = b(v,v)$ `Chapter5/BumpFunctions · proposition_5_5_8`: an inner product structure on $E_0$ would be a second normed structure next to that of the sc-scale. Translating by a non-smooth $x$ is legitimate here because $f$ is $C^\infty$ on $E_0$ itself; for a merely sc-smooth function it would not be. With Theorem 5.5.7 (preprint 5.34): an M-polyfold covered by charts on sc-Hilbert spaces admits sc-smooth bump functions, the remark before Definition 5.3.1 (preprint 5.9) `Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert`.

## V.25 Partitions of unity: Theorem 5.5.10 (preprint 5.37) and Proposition 5.5.11 (preprint 5.38)

**The algebra of sc-smooth functions.** Sums, differences and products of sc-smooth functions are sc-smooth, on retracts `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_add` `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_mul` and on $X$ `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_add` `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_sub` `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_mul`, and so are finite sums and products `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetSum` `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetProd`. Sc-smoothness is local `Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_eventuallyEq`, so a locally finite sum, which near each point is a finite sum, is sc-smooth `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsum`; and a function supported in a chart domain with sc-smooth representative is sc-smooth on $X$ `Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_chart`.

**Definitions 5.5.1 and 5.5.9 (preprint 5.28 and 5.36).** A partition of unity is Mathlib's (continuous $\rho_i \ge 0$, locally finite supports, $\sum \rho_i = 1$), sc-smooth if every $\rho_i$ is `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothPartitionOfUnity`; $X$ admits sc-smooth partitions of unity if every open cover has a subordinate one `Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity`. The approximation property: for $V = O \cap W$ with $\mathrm{cl}\, V \subseteq O$, continuous $f : O \to [0,1]$ supported in $V$ and $\varepsilon > 0$, an sc-smooth $g : O \to [0,1]$ supported in $V$ with $|f - g| \le \varepsilon$ `Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty`. Partitions of unity give bump functions: the function attached to $U$ of a partition subordinate to $\{U, X \setminus \{x\}\}$ is 1 at $x$ `Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity.admitsBumpFunctions` — the easy half of the open problem the text states after Definition 5.5.2 (preprint 5.29).

**"(i) ⇒ (ii)"** `Chapter5/PartitionsOfUnity · hasApproxProperty_restrict`. The text calls $f \circ \varphi$ "sc-smooth" ($f$ is only continuous) and asserts $\mathrm{cl}_X(\varphi^{-1}(V)) \subseteq U$; with the chart of the unit ball above, $V = O$ and $f \equiv 1$, the extension by 0 is not continuous. Since (ii) asks only for *some* atlas, the proof restricts each chart to an open $s$ with $\mathrm{cl}_X(s) \subseteq \mathrm{dom}\, \varphi$. Then for closed $A \subseteq \varphi(s)$ the set $\varphi^{-1}(A)$ is closed in $X$ `Chapter5/PartitionsOfUnity · MChart.isClosed_inter_preimage`, so $F = f \circ \varphi$ extended by 0 is continuous. Cover $X$ by $\{F < \varepsilon/2\}$ and the sets $U_x = \{|F - F(x)| < \varepsilon/2\} \cap \varphi^{-1}(V)$, take a subordinate sc-smooth partition of unity $(\beta_0, \beta_x)$, and let $\hat{g} = \sum_x \beta_x F(x)$: a locally finite sum of sc-smooth functions with values in $[0,1]$ `Chapter5/PartitionsOfUnity · PartitionOfUnity.finsum_mul_mem_Icc`, supported in $\varphi^{-1}(V)$ `Chapter5/PartitionsOfUnity · PartitionOfUnity.tsupport_finsum_mul_subset`, and within $\varepsilon/2$ of $F$, since every $F(x)$ with $\beta_x(y) \ne 0$ is `Chapter5/PartitionsOfUnity · PartitionOfUnity.abs_sub_finsum_mul_le`. Every chart has the property, by a different route (§V.42).

**"(ii) □ (i)".** Every point of a chart domain has small open neighbourhoods $N$ with $\mathrm{cl}_X N \subseteq \mathrm{dom}\,\varphi$ and $\mathrm{cl}\,\varphi(N) \subseteq O$ — half a ball inside the domain of a retraction, closed by §V.24 `Chapter5/PartitionsOfUnity · MChart.exists_small_nhds` — and the approximation property transports to continuous functions on $X$ supported in such $N$ `Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty.exists_approx`. *Sc-smooth Urysohn functions* `Chapter5/PartitionsOfUnity · exists_isScSmoothMap_urysohn`: for disjoint closed $s, t$, cover $t$ by small neighbourhoods $W_x$ inside $X \setminus s$, take continuous $f_x = 1$ near $x$ supported in $W_x$, a locally finite refinement, approximations $\hat{g}_x$ within $1/2$ — so $\hat{g}_x \geq 1/2$ where $f_x = 1$ — and $F = \tau(4\sum \hat{g}_x - 1)$, sc-smooth, $0$ on $s$ and $1$ on $t$. Then Mathlib's bump coverings produce a covering indexed by the given cover with sc-smooth functions `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate_of_prop`, and $\rho_i = f_i \prod_{j<i}(1 - f_j)$, locally a finite product, is the partition of unity `Mathlib/Topology/PartitionOfUnity · BumpCovering.toPartitionOfUnity` `Chapter5/PartitionsOfUnity · admitsScPartitionsOfUnity_of_approx`. This replaces the text's refinement "with the same index set", which does not exist in general (the cover $\{X\}$); metrizability is not needed.

**Theorem 5.5.10 (preprint 5.37)** `Chapter5/PartitionsOfUnity · theorem_5_5_10` combines the two directions, "atlas" read as a family of charts covering $X$.

**Proposition 5.5.11 (preprint 5.38)** `Chapter5/PartitionsOfUnity · proposition_5_5_11` is not "an immediate consequence of Theorem 5.37": that theorem produces some atlas, not the given model, and applied to $O$ would need partitions of unity on an open subset of $C$. The proof repeats "(i) □ (ii)" in $E_0$: $A = \{x \in O \mid f(x) \geq \varepsilon/2\}$ is closed in $E_0$ because $\mathrm{cl}\,V \subseteq O$; $E_0 \setminus A$ and balls $B(x, r_x) \subseteq W$, $x \in V$, on which $f$ varies by less than $\varepsilon/2$, cover $E_0$; a $C^\infty$ partition of unity of $E_0$ gives $g = \sum \beta_x f(x)$, $C^\infty$ on $E_0$, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)), composed with a retraction onto $O$.

## V.26 Banach space geometry: Definitions 5.5.4, 5.5.12, 5.5.14 (preprint 5.31, 5.39, 5.41), Proposition 5.5.15 (preprint 5.42), Corollary 5.5.17 (preprint 5.44)

A Banach space admits a $C^k$-bump function if some $C^k$ function, not identically zero, has bounded support `Chapter5/PartitionsOfUnity · HasContDiffBump`, and $C^k$ partitions of unity if every open cover has a subordinate $C^k$ one `Chapter5/PartitionsOfUnity · AdmitsContDiffPartitionsOfUnity`. $c_0(\Gamma)$ is the subspace of $\ell^\infty(\Gamma)$ of functions tending to $0$ along the cofinite filter `Chapter5/PartitionsOfUnity · cZero`, equivalently with $\{|f(\gamma)| > \varepsilon\}$ finite for every $\varepsilon$ `Chapter5/PartitionsOfUnity · mem_cZero_iff`, normed by the supremum `Chapter5/PartitionsOfUnity · norm_cZero`; coordinate-wise $C^k$ embeddings are as in the text `Chapter5/PartitionsOfUnity · IsCoordinatewiseContDiffEmbedding`, and Toruńczyk's Theorem 5.5.13 (preprint 5.40) is recorded as a statement `Chapter5/PartitionsOfUnity · TorunczykStatement`, proved in §V.45. WCG means a weakly compact set with dense span `Chapter5/PartitionsOfUnity · IsWCG`; reflexive, used in Proposition 5.5.15 (preprint 5.42) and not defined in the text, means $V \to V''$ onto `Chapter5/PartitionsOfUnity · IsReflexive`.

**Proposition 5.5.15 (preprint 5.42).** (i) For reflexive $V$ the weak closure of the closed unit ball is weakly compact — Kakutani's theorem, in Mathlib as Banach–Alaoglu in $V''$ through the embedding of $(V, \text{weak})$ into $(V'', \text{weak-*})$ `Mathlib/Analysis/Normed/Module/DoubleDual · NormedSpace.isCompact_closure_of_isBounded` — and its span contains the ball, hence is $V$ `Chapter5/PartitionsOfUnity · span_closedBall_eq_top` `Chapter5/PartitionsOfUnity · proposition_5_5_15_reflexive`. (ii) For separable $V$, a dense sequence $(y_n)$ scaled by $1/((n+1)(|y_n|+1))$ is a null sequence; with $0$ it is norm compact, hence weakly compact, and its span contains every $y_n$ `Chapter5/PartitionsOfUnity · proposition_5_5_15_separable`. The text's "$\{0\} \cup \{x_n/n\}$" has

its index range missing.

**Corollary 5.5.17 (preprint 5.44)** `Chapter5/PartitionsOfUnity · corollary_5_5_17`. Let $\|v\|^2 = b(v,v)$ on $E_0$, and assume Theorem 5.5.16 (preprint 5.43) for $E_0$ and $k = \infty$, recorded as a statement `Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement`. A Hilbert space is reflexive: the parallelogram law holds, so the norm comes from an inner product `Mathlib/Analysis/InnerProductSpace/OfNorm · InnerProductSpace.ofNorm`, and every $\phi \in V''$ is evaluation at the Riesz representative of $x \mapsto \phi(\langle x, \cdot\rangle)$ `Chapter5/PartitionsOfUnity · isReflexive_of_hilbert`; hence WCG by Proposition 5.5.15 (i) (preprint 5.42). It has the $C^\infty$-bump $\tau(1 - b(v,v))$ `Chapter5/PartitionsOfUnity · hasContDiffBump_of_hilbert`. Theorem 5.5.16 (preprint 5.43) gives $C^\infty$ partitions of unity of $E_0$, and Proposition 5.5.11 (preprint 5.38) — not Theorem 5.5.10 (preprint 5.37), as the text says — the approximation property. With Theorem 5.5.10 (preprint 5.37), an M-polyfold on such charts admits sc-smooth partitions of unity `Chapter5/PartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert`. The hypothesis is removed in §V.46: $E_0$ is separable, and Theorem 5.5.16 (preprint 5.43) is proved for separable spaces there and for every WCG space in §V.47.

## V.27 What is not formalized

Of the chapter's gaps, G.41, G.42, G.44–G.48 and G.50–G.53 are closed, G.41 for each fixed $i$ and G.42 with a correction; G.43 and G.49 have nothing to formalize. What remains:

- **Corollary 5.2.5 (preprint 5.8) as one statement for all $i$.** It is proved for each fixed $i$ (§V.29; gap G.41 closed in the fourth addendum). Not formalized: a family $P^i$ indexed by $i$, and the tame form of the conclusion, which needs the tameness of the index raises of tame charts. The identification of the iterated base $(X^1)^1$ with $X_2$ as a topological space is proved since (§II.32; gap G.6, closed). Condition (3) of a filling on level 1 does not follow from level 0 alone — on $E_m = \{x \mid \sum e^{2mk} x_k^2 < \infty\}$ the right shift $T$ makes $1 - T/2$ invertible on $E_0$ while $(1-T/2)^{-1} e_0 = \sum 2^{-n} e_n \notin E_1$ — so the corollary's proof needs the regularity argument of §V.29, through Lemma 3.1.20 (preprint 3.17) `Chapter3/Filler · lemma_3_1_20`.
- **The non-completeness of $(\Gamma^{+,1}_U(P), \rho)$ in the text's setting**, beyond the trivial bundle, and the identification of the completion with the sections of Remark 5.3.4 (preprint 5.12) (§V.30; gap G.42).
- **Weak against pointwise bump functions at a non-smooth point** (§V.31).
- **Theorem 5.3.15 (preprint 5.22)**, proved (§§V.36–V.41; gap G.48), leaves the comparison of the smooth parametrizations through which "Morse-type" is read with the charts of the Mathlib manifold structure on the solution set. That the complement $Z$ of Lemma 5.3.16 (preprint 5.23) can be chosen smooth is not formalized; the formal proof does not use Lemmas 5.3.16 and 5.3.17 (preprint 5.23 and 5.24), proved in §§V.23 and V.32.
- **Remark 5.3.8 (preprint 5.16)** is formalized on $\mathbb{R}^1 \times X$ with the time window $[0,1]$, not on $[0,1] \times X$, which would need a tame structure on the product with an interval (§V.33).
- **Theorem 5.27 (preprint; not in the 2021 edition)** (gap G.49, nothing to formalize in the 2021 edition; the preprint leaves the proof to the reader). Its general-position analogue is §V.33. A good-position form at the corner points $\{i\} \times \partial X$ would need the step that Finding 88 shows not to hold as we read it `Chapter3/Counterexamples2021 · Counterexamples2021.not_isInGoodPosition_prod_diag`; no counterexample to the theorem itself is built (Findings, Chapter 5, items 104, 186).
- **The Sobolev sc-structure on other domains** (row V5.76, narrower): compactness of the inclusions $W^{k+1,p}(\Omega) \to W^{k,p}(\Omega)$ and density of $E_\infty$ are proved for $\Omega = (0,1)$ and every $p > 1$; for cubes in $\mathbb{R}^n$, $n \geq 2$, and other domains they are hypotheses (§V.44). The bump claims hold for every $\Omega$.

- **The 2021 remark that the notions of Definition 5.2.1 differ for general sc-smooth sections** (§V.32).
- **Remark 5.3.6 (preprint 5.14)** (gap G.43): nothing to formalize; the formalization assumes neither separable sc-Hilbert fibres nor further properties of auxiliary norms.

## V.28 What the formalization taught us

**The tameness hypothesis.** The first formal versions of Theorem 5.3.5 (preprint 5.13) and Lemma 5.26 (preprint) assumed, as the §3.6 theorems then did, that the base retraction of *every* compatible strong bundle chart — and of every chart of the pull-backs to $\mathbb{R}^k \times X$ — is tame. No strong bundle over a nonempty base satisfies this (a compatible chart $(1, \varphi)$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ is never tame), so those statements were vacuous. The repair is the text's own Definitions 3.6.3, 3.6.4, 3.6.6 (preprint 3.50, 3.51, 3.53) in a tame chart: the perturbation theorems take $f$ with tame germ data, the parametrized sections inherit them through the tame product chart, and Definition 5.3.9 (preprint 5.17) lives in a tame chart, where Lemma 5.26's (preprint) count $\dim T_zO/T_z^RO = d_C(z)$ holds.

**What "it is easily verified" costs.** The perturbation theorems of §5.3 are sketches. Their formal proofs needed the density of smooth vectors (§V.12), the compactness of the zeros over a small parameter ball (§V.13), a local, stratified Sard step (§V.14), the bridge from tangent germ data to Definition 4.2.1 (preprint 4.13) (§V.17), $T_yS = \ker f'(y)$ by a dimension count, which gives the chart independence of Definition 5.3.9 (preprint 5.17) and the openness of general position (§§V.17–V.18), and, in Theorem 5.3.12 (preprint 5.19), a route around the implicit function theorem at boundary points where its hypothesis fails (§V.22).

**Theorem 5.3.15 without a trivialization.** The text's proof moves the solution set $S$ along a trivialization $\Phi_\lambda : S \to S_\lambda$ and differentiates the equation twice. The formal proof takes the perturbations of the form $a_i\sigma_{g(i)}$, so that every $S_\lambda$ is a level set inside one fixed solution manifold of $f + \sum \mu_j\sigma_j$, and replaces the trivialization by Lagrange multipliers and the bordered Hessian (§§V.38–V.39). The "parameterized version of Sard's theorem" is the inverse function theorem plus Sard's lemma in equal dimensions (§V.38). When the families of several critical points are united, the transversality obtained at one of them must survive the others; the bounded coordinates $\arctan(q_i)/2$ and the constant 1 make it do so, because a multiplier of the larger family restricts to a non-zero multiplier of the smaller one (§V.39; Findings, Chapter 5, item 183).

**The corner case is no case.** The boundary of the solution set might have corners where $t|S$ is not regular. It cannot: at a zero on $\partial X$, $t$ takes the value 0 or 1 and is an extremum, so $dt(x)$ vanishes on the two-sided directions $E_x \supseteq T_x^RX$; general position and $dt(x) \neq 0$ then make $dt(x)$ non-zero on $\ker f'(x)$, at every degeneracy index (§V.41).

**The hidden hypothesis is the 2021 condition.** The hypothesis $dt(x) \neq 0$ that §V.23 found missing from the preprint is Definition 5.3.13 (3) of the 2021 edition, and with it Theorem 5.3.15 needs nothing beyond the text (§§V.32, V.36).

**Finding 88 is machine-checked.** Good position is not stable under $\mathbb{R} \times \cdot$ (§V.22), so the implicit function theorem cannot be applied "as before" at the boundary zeros of the parametrized section; Theorem 5.3.12, its dense form and the cobordisms of §V.33 are proved on routes that avoid those points.

**$E_0$ is always separable.** The inclusion $E_1 \to E_0$ of an sc-Banach space is compact with dense range, so $E_0$ is separable (§V.46). In the sc-calculus, Theorem 5.5.16 (preprint 5.43) is needed only for separable spaces, where it is elementary, and Corollary 5.5.17 (preprint 5.44) holds without it. The general

WCG case was proved anyway, without Krein's theorem, the Eberlein–Šmulian theorem or a transfinite recursion: Mazur's theorem and finite character of a closure operator suffice (§V.47).

**Every chart, by shells.** The text's derivation of the approximation property in a chart uses $\mathrm{cl}_X(W) \subseteq U$, which does not follow from $\mathrm{cl}_C(V) \subseteq O$ (§V.25; Findings, Chapter 5, item 111). Exhausting the chart domain by shells $\{1/(n+2) \le d \le 1/(n+1)\}$ and summing local approximations gives the property for every chart (§V.42).

**The Sobolev remark lives on level 0.** Its bump claims use only $E_0 = W^{1,4}(\Omega)$ or $W^{1,3/2}(\Omega)$: a $C^k$ bump on $E_0$ is $\mathrm{sc}^k$ on any sc-structure. The sc-structure $E_m = W^{1+m,p}(\Omega)$ itself — Rellich compactness and density of smooth functions — is a separate matter, proved here for $(0,1)$ (§V.44).

## V.29 Corollary 5.2.5 (preprint 5.8) on the shifted strong bundle

Corollary 5.2.5 (preprint 5.8) says that, for an sc-Fredholm section $f$ with compact solution set, $f^i$ is an sc-Fredholm section of $P^i : Y^i \to X^i$ with compact solution set, and that "it is a priori clear that $f^i$ is sc-Fredholm". The text defines $Y[i]$ for $i = 0, 1$ only, never $Y^i$, and "a priori clear" hides the filler condition on level 1 (§V.7). Both are supplied here (Findings, Chapter 5, items 140–146).

**The reading of $Y^1$.** $Y^1 = Y_{1,1}$ with $(Y^1)_{m,k} = Y_{m+1,k+1}$ — the convention of $E^1$ and of $(U \lhd F)^1 = U^1 \lhd F^1$ — not $Y[1] = Y_{0,1}$ and not $Y_{1,2}$. It is forced by "$f^i$ is obtained by raising the index": an sc-smooth section maps $X_1$ into $Y_{1,1}$, and in general not into $Y_{1,2}$.

**Fibres of a strong bundle.** Through a chart, $P^{-1}(x)$ is linearly homeomorphic to $K_{\varphi(x)} = \{h \mid \Gamma(\varphi(x), h) = h\}$ `Chapter5/StrongBundleShift · StrongBundleChart.fiberK` `Chapter5/StrongBundleShift · StrongBundleChart.fiberCLE`, the fixed points of a continuous projection, a closed subspace of the fibre model `Chapter5/StrongBundleShift · StrongBundleChart.isClosed_fiberK`; so every fibre is a Banach space `Chapter5/StrongBundleShift · StrongBundleChart.isBanachable_fiber`, which Definition 2.6.5 (preprint 2.51) asks of a strong bundle.

**The levels of $X^1$, and raising the index of maps.** $(X^1)_m = X_{m+1}$ and $(X^1)_\infty = X_\infty$ as sets `Chapter5/StrongBundleShift · MPolyfold.mem_shift_levelSet_iff` `Chapter5/StrongBundleShift · MPolyfold.mem_shift_smoothSet_iff` (their topologies agree too, §II.32; gap G.6, closed). An sc-smooth map $f : X \to Z$ between M-polyfolds induces an sc-smooth $f_1 : X^1 \to Z^1$: continuity, and in the level charts the local representatives are the level-1 maps of those of $f$, sc-smooth by Proposition 1.2.2 (preprint 1.17) for retracts `Chapter5/StrongBundleShift · MPolyfold.levelChart_apply_levelOne` `Chapter5/StrongBundleShift · MPolyfold.IsScSmoothMap.levelOne`.

**The shifted strong bundle** `Chapter5/StrongBundleShift · StrongBundle.shift`. Its total space is $Y_{1,1}$, the level 1 of the M-polyfold $Y[0]$ with its level topology, Hausdorff and paracompact by Theorem 2.3.10 (preprint 2.16) `Chapter5/StrongBundleShift · StrongBundle.ShiftSpace`; the projection is $P^1 = P|_{Y_{1,1}}$ `Chapter5/StrongBundleShift · StrongBundle.shiftProj`, continuous and onto `Chapter5/StrongBundleShift · StrongBundle.continuous_shiftProj` `Chapter5/StrongBundleShift · StrongBundle.surjective_shiftProj`; the fibre $P^{-1}(x) \cap Y_{1,1}$ is a linear subspace — in a chart it is the preimage of $F_1$ under the linear fibre map `Chapter5/StrongBundleShift · StrongBundle.shiftFiberSubmodule` — carries the topology of $F_1$, not that of $P^{-1}(x)$, and is a Banach space, a closed subspace of $F_1$ through a chart `Chapter5/StrongBundleShift · StrongBundle.banachable_shift`. The atlas consists of the *index raises* of the strong bundle charts `Chapter5/StrongBundleShift · StrongBundleChart.shift` `Chapter5/StrongBundleShift · StrongBundle.shiftAtlas`: base chart the level chart $\varphi^1 : V_1 \to O_1$, fibre model $F^1$, retraction $R$ on $U^1 \lhd F^1$

Chapter5/StrongBundleShift · IsStrongBundleRetraction.shift, bundle chart the level chart of $\Phi$ in $Y[0]$. Compatible charts have compatible index raises Chapter5/StrongBundleShift · StrongBundleChart.Compatible.shift: on $[0]$ the transitions are level charts of compatible charts, on $[1]$ the transition $(\Phi'^1 \circ (\Phi^1)^{-1})[1]$ is the level-1 map of $(\Phi' \circ \Phi^{-1})[1]$ (Proposition 1.2.2 (preprint 1.17) for retracts) Chapter5/StrongBundleShift · StrongBundleChart.shift_transition_one, and the $F_1$-condition of Definition 2.6.4 (preprint 2.50) holds because the $[1]$-transition is sc$^0$ Chapter5/StrongBundleShift · StrongBundleChart.shift_compatible_range. Checks: $(Y^1)_{m,k} = Y_{m+1,k+1}$ Chapter5/StrongBundleShift · StrongBundle.mem_shift_doubleLevel_iff, the M-polyfold $Y^1[0]$ is $Y[0]^1$ Chapter5/StrongBundleShift · StrongBundle.isChart_shift_toMPolyfold0_iff, and the index raise of every chart of $P$ is a chart of $P^1$ Chapter5/StrongBundleShift · StrongBundle.isChart_shift.

**The section $f^1$.** $f^1 = f|_{X_1}$ Chapter5/StrongBundleShift · StrongBundle.sectionShift is a section of $P^1$ Chapter5/StrongBundleShift · StrongBundle.isSection_sectionShift, sc-smooth by the raising of the index of maps Chapter5/StrongBundleShift · StrongBundle.IsScSmoothSection.shift, with the zeros of $f$ in $X_1$ as its zeros Chapter5/StrongBundleShift · StrongBundle.solutionSet_sectionShift, and regularizing, from $(X^1)_m = X_{m+1}$ and $(Y^1)_{m,m+1} = Y_{m+1,m+2}$ Chapter5/StrongBundleShift · StrongBundle.regularizing_sectionShift. In the index raise of a chart the principal part of $f^1$ is the level-1 map of that of $f$ Chapter5/Corollary58 · StrongBundleChart.principal_sectionShift, and the raised chart is centred where the chart is Chapter5/Corollary58 · StrongBundleChart.shift_φ_eq_zero.

**Condition (3) on every level, at every centre** Chapter5/Corollary58 · IsFillingGerm.mem_range_toBase_of_fillDefect. *Let $(g, s)$ be a datum of Definition 3.1.11 (preprint 3.11) in a chart — $g$ a filling germ of $\hat{f}$ with fill defect $k = [1 - \rho(r)]g$ and derivative family $Dk$, $s$ an sc$^+$-germ with $s(0) = g(0)$, $g - s$ conjugated to a basic germ. If $v \in \ker Dr(0)$ and $Dk(0)v \in F_m$, then $v \in E_m$* — at the centre of the datum, which need not be a zero. At a zero this is the regularity of §III.45 Chapter3/LinearizationChartsScG · IsFillingGerm.mem_range_toBase_of_scDeriv_r_eq_zero. The reduction to a zero needs neither a global sc$^+$-section nor a bump function: $g(0) = \iota s(0)$ is a smooth point of $F$ Chapter5/Corollary58 · IsScSmoothGerm.toBase_zero_mem_smooth; $\hat{s}(z) = \Gamma(z, -g(0))$ is sc-smooth on $U$ Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth and a section of $K \to O$, so by Lemma 3.1.20 (preprint 3.17) $g' = g + \hat{s} \circ r$ fills $\hat{f} + \hat{s}$ with the *same* fill defect Chapter3/GermCalculus · IsFillingGerm.add_section, and $g'(0) = g(0) - \Gamma(0, g(0)) = 0$. With the sc$^+$-germ $s' = s + \Gamma^1(r(\cdot), -s(0))$, $g' - s' = g - s$ on $U$, so Proposition 3.6.5 (preprint 3.52) in sc form makes $Dg'(0)$ sc-Fredholm Chapter3/FredholmGermLinearizationScG · proposition_3_6_5_local_sc; and $Dk(0) = (1 - \rho(0))Dg'(0)$ Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho, so the case of a zero applies to $g'$.

**The filling on level** $1$ Chapter5/Corollary58 · IsFillingGerm.germ_shift. The fill defect of $g^1$ is the level-1 map of $k$, an sc$^1$-germ with derivative family $(Dk)_1$ by Proposition 1.2.2 (preprint 1.17) for germs Chapter3/GermLevels · IsSc1GermWith.shift; $\iota\, Dr^1(0) = Dr(0)\, \iota$, by the uniqueness of the derivative of $r^1 = r_1$ on the shifted germ Chapter5/Corollary58 · toBase_scDeriv_r_shift, and $\iota\, \rho^1(0) = \rho(0)\, \iota$ Chapter5/Corollary58 · toBase_ρCLM_shift. So $Dk^1(0) = Dk(0)|_{E_1}$ maps $\ker Dr^1(0)$ to $\ker \rho^1(0)$; it is injective by condition (3) on level $0$, onto by the regularity just proved with $m = 1$, and a topological isomorphism by the open mapping theorem. Conditions (1) and (2) of the filling shift directly Chapter5/Corollary58 · StrongBundleChart.shift_filling_eq_on Chapter5/Corollary58 · StrongBundleChart.shift_filling_mem_of_fixed, and $g^1$ is a filling germ of the principal part of $f^1$ in $\Phi^1$ Chapter5/Corollary58 · StrongBundleChart.shift_isFillingGerm.

**Corollary 5.2.5 (preprint 5.8).** At every smooth point of $X^1$, the datum $(g, s)$ of $f$ in a chart $\Phi$ gives the datum $(g^1, s^1)$ of $f^1$ in $\Phi^1$: $g^1$ fills, $s^1$ is an sc$^+$-germ on the shifted germ, $s^1(0) = g^1(0)$, and $g^1 - s^1 = (g - s)_1$ near $0$ is conjugated to a basic germ by the index raise of the conjugation (§V.7) Chapter5/Corollary58 · isScFredholmGerm_sectionShift. *For an sc-Fredholm section $f$, $f^1$ is an sc-Fredholm section*

*of* $P^1$ Chapter5/Corollary58 · corollary_5_2_5_isScFredholmSection; *if* $f$ *has compact solution set, so has* $f^1$, by the compactness on level 1 (§V.7) Chapter5/Corollary58 · corollary_5_2_5. Since this holds for every strong bundle, it applies to $P^1$ and $f^1$: $f^2 = (f^1)^1$ is an sc-Fredholm section of $P^2 = (P^1)^1$ with compact solution set Chapter5/Corollary58 · corollary_5_2_5_two, and so on for every fixed $i$. The unpacked data in the shifted chart are kept for reference Chapter5/Corollary58 · corollary_5_2_5_data. No hypothesis beyond the text's is added, and tameness is not used. *Not done*: a family $P^i$ indexed by $i$ (a dependent type of strong bundles) and the tame form of the conclusion — germ data of $f^1$ in tame charts of $P^1$ — which needs the presentation independence of the degeneracy index for $C^1$ and that $A \cap E_1$ is an sc-complement of $T_xO^1$ when $A$ is one of $T_xO$; there is no counterexample either.

## V.30 The metric $\rho$ on $\Gamma_U^{+,1}(P)$

The text: $\Gamma_U^{+,1}(P)$ "becomes a metric space" for $\rho(s,s') = \sup_x N(s(x) - s'(x))$, and "is not complete" (p. 217 (preprint p. 174)). The first is proved here; the second turns out to need a hypothesis (Findings, Chapter 5, items 150–153).

**The metric.** $\rho(s,s')$ is the supremum of the pointwise distances of §V.10 Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist. For an auxiliary norm the fibre norm is symmetric and subadditive (Definition 5.1.1 (1) (preprint 5.1)) Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.neg Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_le Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_triangle, so for allowable sections $N(s(x) - s'(x)) \le N(s(x)) + N(s'(x)) \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.sectionDistAt_le_two: the supremum is finite and $\rho \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_two. $\rho(s,s) = 0$, symmetry and the triangle inequality come from the fibre norms Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_self Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_comm Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_triangle, and $\rho(s,s') = 0$ forces $s = s'$ because $N$ vanishes only at $0_x$ Chapter5/AllowableSectionsMetric · StrongBundle.eq_of_allowableDist_eq_zero. So $\rho$ is a metric Chapter5/AllowableSectionsMetric · StrongBundle.allowableMetric, carried by a type synonym of $\Gamma_U^{+,1}(P)$, since the sections $X \to Y$ carry the product topology Chapter5/AllowableSectionsMetric · StrongBundle.AllowableSpace. Only Definition 5.1.1 (1) (preprint 5.1) is used. The density statements of Theorems 5.3.5 and 5.3.10 (preprint 5.13 and 5.18) (§§V.15, V.20) are $\rho(s,s') \le \varepsilon$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_iff.

**Not complete, by an example** Chapter5/AllowableSectionsMetric · TrivialBundle.not_completeSpace_allowableSections. Take the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ of the non-vacuity witnesses of Chapter 3, the fibre norm $N(x,v) = |v|$ — an auxiliary norm, including the condition on the topology of $Y[1]$ Chapter5/AllowableSectionsMetric · TrivialBundle.fibNorm Chapter5/AllowableSectionsMetric · TrivialBundle.isAuxNorm_fibNorm — and $U = \mathbb{R}$. Graphs of $C^\infty$ maps are sc⁺-sections Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_graphSection, so $s_j(x) = (x, \sqrt{x^2 + 1/(j+1)^2}/(2\sqrt{1+x^2}))$ is allowable, with $N \le 1/2$ Chapter5/AllowableSectionsMetric · TrivialBundle.approx Chapter5/AllowableSectionsMetric · TrivialBundle.graphSection_approx_mem, and $\rho(s_i,s_j) \le (\varepsilon_i + \varepsilon_j)/2$ with $\varepsilon_j = 1/(j+1)$, by $|\sqrt{x^2+a^2} - |x|| \le a$ Chapter5/AllowableSectionsMetric · TrivialBundle.abs_sqrt_sub_abs_le Chapter5/AllowableSectionsMetric · TrivialBundle.abs_approx_sub_limitFn. A $\rho$-limit $s$ would have, by pointwise convergence, the fibre part $|x|/(2\sqrt{1+x^2})$ Chapter5/AllowableSectionsMetric · TrivialBundle.limitFn, which is not differentiable at $0$ Chapter5/AllowableSectionsMetric · TrivialBundle.not_differentiableAt_limit; but the fibre part of an sc⁺-section of the trivial bundle is $C^1$, by Proposition 1.2.3 (preprint 1.18) in the chart $\Phi = \mathrm{id}$ Chapter5/

AllowableSectionsMetric · TrivialBundle.differentiable_snd_of_isScPlusSection.

**Complete, in two cases.** For $U = \emptyset$ every allowable section is the zero section, so the space is a point, for every bundle and auxiliary norm Chapter5/AllowableSectionsMetric · StrongBundle.subsingleton_allowableSections_empty Chapter5/AllowableSectionsMetric · StrongBundle.completeSpace_allowableSections_empty. Over the point $\mathbb{R}^0$ with fibre $\mathbb{R}^k$ every section is the graph of a constant, hence sc$^+$ Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_of_isSection_point, and the space is the closed unit ball of $\mathbb{R}^k$ (or $\{0\}$), complete for every $U$ Chapter5/AllowableSectionsMetric · TrivialBundle.completeSpace_allowableSections_point. So the text's "not complete" does not hold as we read it without a hypothesis and is read as "not complete in general". In the text's setting — $U$ a neighbourhood of the solution set of an sc-Fredholm section on an M-polyfold with bump functions — one expects non-completeness as soon as $U$ contains a non-isolated point over which $Y_{0,1}$ is nonzero, by the same mechanism (bump functions times a non-smooth uniform limit); this is not formalized, and neither is the identification of the metric completion with the sections of Remark 5.3.4 (preprint 5.12) (§V.10).

## V.31 Weak and pointwise bump functions

§V.8 reads Definition 5.3.1 (preprint 5.9) pointwise, $f(x) \neq 0$, because Proposition 5.5.3 (preprint 5.30) needs it, and keeps Definition 5.5.2's (preprint 5.29) "not identically zero" as the weak reading. How far apart are they? (Findings, Chapter 5, items 160–163.)

**Bump points.** $x$ is a *bump point* if every open $U \ni x$ carries an sc-smooth $f$ with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ Chapter5/BumpWeakPointwise · MPolyfold.HasBumpAt. The pointwise reading says that every point is a bump point Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctions_iff_forall_hasBumpAt, and the weak reading holds as soon as the bump points are dense: a bump at a bump point inside $U$ is not identically zero and is supported in $U$ Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctionsWeak_of_dense. (The converse, weak $\Box$ dense bump points, is a nested-neighbourhoods argument with Cantor's theorem in a chart; not formalized.)

**On an sc-Banach space.** An sc-Banach space $E$ is an M-polyfold with the identity chart and model $(E, E, E)$ Chapter5/BumpWeakPointwise · scChart Chapter5/BumpWeakPointwise · scPolyfold, whose sc-smooth functions are the sc-smooth maps $E \to \mathbb{R}$ Chapter5/BumpWeakPointwise · isScSmoothMap_scPolyfold_iff. *There the weak reading gives a bump point at every smooth point* Chapter5/BumpWeakPointwise · MPolyfold.AdmitsBumpFunctionsWeak.hasBumpAt_of_mem_smooth: given $x \in E_\infty$ and $B(x, \varepsilon) \subseteq U$, a weak bump supported in $B(x, \varepsilon/2)$ is non-zero on a non-empty open set, which contains a smooth point $y$ because $E_\infty$ is dense; $a = y - x$ is smooth, so $v \mapsto f(v + a)$ is sc-smooth Chapter4/Recentre · isScSmooth_add_const, non-zero at $x$, with support in $B(x, \varepsilon) \subseteq U$. The argument stops at $E_\infty$: translation by a vector $a \notin E_m$ is not even sc$^0$, since it sends $0 \in E_m$ out of $E_m$ Chapter5/BumpWeakPointwise · not_isSc0_add_const. On M-polyfolds built on sc-Hilbert spaces both readings hold at every point Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert. *Open*: whether the weak reading gives a bump function non-zero at a non-smooth point, on an sc-Banach space or an M-polyfold; a counterexample would need an $E_0$ without $C^1$ bump functions.

# V.32 The Springer 2021 edition: Definition 5.2.1, Theorems 5.2.2 and 5.3.5, Definition 5.3.13

The 2021 edition adds two compactness properties to Definition 5.2.1 (preprint 5.4) and extends Theorem 5.2.2 (preprint 5.5) to them, concludes in Theorem 5.3.5 (preprint 5.13) that the perturbed solution sets are compact sub-M-polyfolds, and changes Definition 5.3.13 (preprint 5.20): $t$ need not be onto, the fibres $X_\tau$ are no longer asked to be Fredholm problems, and $dt(x)$ is asked onto at every smooth point. Chapter 5 has no item that is new in the 2021 edition. The file is `Chapter5/Book2021.lean` (Findings, Chapter 5, items 170–179).

**Definition 5.2.1, 2021 form** `Chapter5/Book2021 · StrongBundle.IsWeaklyProperOnU` `Chapter5/Book2021 · StrongBundle.IsWeaklyProperLtInfty` `Chapter5/Book2021 · StrongBundle.IsProperLtInfty`. *Statement.* (2) weakly proper: compact solution set, and for every auxiliary norm $N$ an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section $s$ supported in $U$ with $N(s(y)) \le 1$ for $y \in U$; (4) weakly proper$_{<\infty}$: the same for every such $s$ with $N(s) \le C$ on $U$ for some constant $C$; (5) proper$_{<\infty}$: compact solution set, and for every $N$ an open $U \supseteq f^{-1}(0)$ with $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le C\}$ compact for every $C > 0$, where $N(f(x)) \le C$ includes $f(x) \in Y_{0,1}$ `Chapter5/Compactness · StrongBundle.AuxLe`. *The bound on $U$ is the old bound everywhere* `Chapter5/Book2021 · StrongBundle.isWeaklyProperOnU_iff`: a section supported in $U$ vanishes off $U$ `Chapter5/Book2021 · StrongBundle.isZeroAt_of_not_mem_of_support_subset`, and the zero of a fibre has $N \le r$ for every $r \ge 0$ `Chapter5/Book2021 · StrongBundle.IsAuxNorm.auxLe_fiberZero`; so (2) is the `IsWeaklyProper` of §V.4 `Chapter5/Compactness · StrongBundle.IsWeaklyProper`, and (1), (3) are unchanged `Chapter5/Compactness · StrongBundle.IsProper`.

**The a priori dependencies (p. 209).** Proper$_{<\infty}$ gives proper with $C = 1$ `Chapter5/Book2021 · StrongBundle.IsProperLtInfty.isProper`; weakly proper$_{<\infty}$ gives weakly proper, since the bound 1 is a constant `Chapter5/Book2021 · StrongBundle.IsWeaklyProperLtInfty.isWeaklyProperOnU`; weakly proper includes a compact solution set `Chapter5/Book2021 · StrongBundle.IsWeaklyProperOnU.isCompact_solutionSet`. For a continuous $f$, proper gives weakly proper by §V.4 `Chapter5/Book2021 · StrongBundle.IsProper.isWeaklyProperOnU`, and proper$_{<\infty}$ gives weakly proper$_{<\infty}$ `Chapter5/Book2021 · StrongBundle.IsProperLtInfty.isWeaklyProperLtInfty`: let $N(s) \le C$ on $U$. A solution $x$ of $f = s$ lies in $U$ — otherwise $s(x) = 0$, so $x \in f^{-1}(0) \subseteq U$ — and there $N(f(x)) = N(s(x)) \le \max(C, 1)$; so $\{f = s\}$, closed because $f$ and $s$ are continuous, lies in the compact $\mathrm{cl}\{x \in U \mid N(f(x)) \le \max(C, 1)\}$ `Chapter5/Book2021 · StrongBundle.AuxLe.mono`. The text's remark that the notions differ for general sc-smooth sections is not formalized.

**Theorem 5.2.2 (preprint 5.5), 2021 form** `Chapter5/Book2021 · theorem_5_2_2`. *Statement.* An sc-Fredholm section with compact solution set is proper$_{<\infty}$. *Proof.* As for §V.6 `Chapter5/Compactness · theorem_5_2_2_preprint`. At a zero $x$, the local step gives a chart $c$, an open $V \ni x$ and for every bound $b$ a compact set containing the closure of the points of $V$ whose fibre part has level-1 norm $\le b$ `Chapter5/Compactness · exists_nbhd_isCompact_closure_bdd`; near $x$ the chart norm is at most $A \cdot N$ `Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le`. So on a smaller $V$ the closure of $\{N(f) \le C\}$ lies in the compact set for $b = A \cdot C$, and $V$ does not depend on $C$. Finitely many such $V$ cover $f^{-1}(0)$, and the closure of the finite union lies in the finite union of the compact closures. □ The text's first display writes $N(f(x)) \le d$ for the points $z$ of the set, presumably meant as $N(f(z)) \le d$; the later display has $y$ throughout. *The five properties are equivalent* for sc-Fredholm sections `Chapter5/Book2021 · theorem_5_2_2_tfae`: the cycle (5) □ (3) □ (2) □ (1) □ (5) and (5) □ (4) □ (2), with $f$ continuous because it is sc-smooth.

**Theorem 5.3.5 (preprint 5.13), 2021 form** `Chapter5/Book2021 · theorem_5_3_5`. *Statement.* The conclusions of §V.15 `Chapter5/Transversality · theorem_5_3_5_preprint`, and moreover $S_{f+s}$ is a compact sub-M-polyfold

for the transversal perturbations $s$. *Proof.* `theorem_5_3_5_preprint` gives $S_{f+s}$ compact and, if non-empty, a manifold, a sub-M-polyfold in particular; the empty set is a sub-M-polyfold `Chapter5/Book2021 · isSubMPolyfold_solutionSet_of_manifold`. □ The perturbations are the allowable sections, bounded by $N(s) \leq 1$ `Chapter5/BumpFunctions · StrongBundle.allowableSections`; the printed "$N(f(x)) \leq 1$ for all $x$" is read as $N(s(x)) \leq 1$, presumably what is meant, as in §V.15.

**Definition 5.3.13 (preprint 5.20), 2021 form** `Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy`. *Statement.* $X$ tame; $f$ an sc-Fredholm section with tame germ data and compact solution set; $t : X \to \mathbb{R}$ sc-smooth with values in $[0, 1]$; for $i = 0, 1$ the end $t^{-1}(i)$ a sub-M-polyfold with tame induced structure over which the restriction of $f$ to the pull-back bundle has tame germ data `Chapter5/Book2021 · StrongBundle.IsCompactFredholmRestriction` `Chapter5/Homotopy · restrictSection`; and (3) $dt(x)$ onto at every smooth $x$. Since $T_xX$ has no linear structure in the project, (3) is read in a chart $\varphi$ around $x$ with a retraction $R$ onto its model and the level-1 point $z$ over $\varphi(x)$: every $a \in \mathbb{R}$ is the derivative of $t \circ \varphi^{-1} \circ R$ at $z$ on some $DR(z)v \in T_zO$ `Chapter5/Book2021 · MPolyfold.HasSurjDerivAt`. *Relation to §V.23.* The data of the earlier reading `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy` together with (3) give the new data `Chapter5/Book2021 · StrongBundle.IsGenCompactHomotopy.isSubmersiveGenCompactHomotopy` ($t$ onto $[0, 1]$ gives values in $[0, 1]$, and the fibre conditions at $\tau = 0, 1$ are the end conditions). Every fibre $t^{-1}(\tau)$ still has a compact solution set: it is closed, and the solution set of the restriction is the trace of $f^{-1}(0)$ `Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy.isCompact_solutionSet_restrict`.

**A witness, and the converse.** Over an empty end the end conditions hold: $\emptyset$ is a tame sub-M-polyfold whose atlas has no charts, and every section over it is tame sc-Fredholm `Chapter5/Book2021 · StrongBundle.isCompactFredholmRestriction_of_eq_empty` `Chapter5/Book2021 · isScSmoothRetractMap_of_domain_empty`. On the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ take the graph section of the identity, with solution set $\{0\}$ `Chapter5/Book2021 · GenCompactHomotopyWitness.solutionSet_graphSection_id` and tame germ data `Chapter3/NonVacuityG · TrivialBundle.isScFredholmSectionTame_graphSection`, and $t(x) = 1/2 + \arctan(x)/\pi$ `Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime`, a $C^\infty$ function with values in $(0, 1)$ `Chapter5/Book2021 · GenCompactHomotopyWitness.contDiff_arctanTime` `Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime_mem_Ioo`. Both ends are empty `Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime_preimage_eq_empty`, and $t'(x) = 1/(\pi(1+x^2)) \neq 0$ `Chapter5/Book2021 · GenCompactHomotopyWitness.hasFDerivAt_arctanTime`, so $dt(x)$ is onto in the identity chart `Chapter5/Book2021 · GenCompactHomotopyWitness.hasSurjDerivAt_arctanTime`. So the 2021 definition is satisfiable `Chapter5/Book2021 · GenCompactHomotopyWitness.isSubmersiveGenCompactHomotopy_witness`; the same data are not a generalized compact homotopy in the earlier sense, as $0$ is not a value of $t$ `Chapter5/Book2021 · GenCompactHomotopyWitness.not_isGenCompactHomotopy_witness`.

**Lemma 5.3.16 (preprint 5.23) under (3)** `Chapter5/Book2021 · lemma_5_3_16`. *Statement.* If $dt(x)$ is onto, then in a chart the derivative $\ell$ on $T_zO$ is non-zero, $\ker \ell$ has a one-dimensional complement $Z$, and the conclusions of Lemma 5.3.16 hold. *Proof.* Take $v$ with $\ell(DR(z)v) = 1$; so $\ell \neq 0$. A vector on which $\ell$ is non-zero spans a complement of $\ker \ell$ `Chapter5/Homotopy · exists_isCompl_ker_of_ne_zero`, and §V.23 applies `Chapter5/Homotopy · lemma_5_3_16_preprint`. □ This is the hypothesis $dt(x) \neq 0$ that §V.23 found missing from Theorem 5.3.15 (preprint 5.22): the 2021 condition (3) supplies it. That $Z$ can be chosen smooth is not formalized; Theorem 5.3.15 itself is proved without it (§§V.36–V.41; gap G.48).

**Theorem 5.5.5 (preprint 5.32).** The 2017 and 2021 statements print the exponent $p$ that the preprint omits in (1) and (2); the theorem is proved as printed there, with the exponent (§V.43; gap G.50).

# V.33 Remark 5.3.8 (preprint 5.16): Theorem 5.3.10 relative to a compact part, and the cobordism

Remark 5.3.8 joins two transversal sections $f_0$, $f_1$ by an sc-Fredholm arc $(t,x)\mapsto f_t(x)$ with compact solution set and says that a small perturbation vanishing near $t=0,1$ makes the arc transversal, its solution set a compact cobordism; "the reader will be able to carry out this construction". Two things are needed: Theorem 5.3.10 (preprint 5.18) for a section that is already in general position off a compact part of its zeros, and the arc itself. The arc lives on $[0,1]\times X$ in the text; it is read on $\mathbb{R}^1\times X$ with the time window $[0,1]$ (reading D2 of Part VI), so that no tame structure on $[0,1]\times X$ is needed (Findings, Chapter 5, item 186; gap G.44).

**A chart criterion** `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_of_sup_eq`. If in a tame chart centred at a zero $y$ the linearization $\mathrm{Lin}_f(0)$ is onto and $\ker\mathrm{Lin}_f(0)+T_0^RO=T_0O$ on level 0, then $(f,y)$ is in general position. This is the second half of the argument of §V.19, now for any section: the kernel has an algebraic complement inside $T_0^RO$ `Chapter5/GeneralPosition · exists_compl_le_of_sup_eq`, §V.16 gives good position, the kernel is then the tangent space of the solution set, finite-dimensional and smooth (§V.17), and it has an sc-complement inside $T_0^RO$ `Chapter4/LocalStructure · exists_scComplement_le`. Conversely general position gives $\ker\mathrm{Lin}_f(0)+T_0^RO=T_0O$ in every tame chart centred at $y$ `Chapter5/CylinderPerturbation2021 · isParamGP_elim0_of_isGeneralPositionGerm`. Two consequences: if $(g,y)$ is in general position, the family $F(\lambda,x)=g(x)+\sum\lambda_it_i(x)$ is in general position at $(0,y)$ whatever sections are added `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramSection_of_isGeneralPositionGerm`, and the lift $\tilde g(r,x)=g(x)$ to $\mathbb{R}^m\times X$ is in general position at every $(r_0,y)$, since in the product chart $\mathrm{Lin}_{\tilde g}(0)(a,u)=\mathrm{Lin}_g(0)u$ `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramLift`. General position, like good position, depends only on the germ of the section `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_congr_nhds`.

**The perturbation step relative to an open set** `Chapter5/CylinderPerturbation2021 · exists_null_isGeneralPositionGerm_paramSlice`. Let $G$ have tame germ data, $N$ be an auxiliary norm, $W$ open with $\mathrm{cl}\{x\in W\mid N(G(x))\le 1\}$ compact, and $t_1,\dots,t_m$ sc$^+$-sections with $N(t_i)\le 1$ such that $F$ is in general position at $(0,y)$ for every zero $y$ of $G$. Then there are $\delta>0$ and a null set $CV$ such that for $|\lambda|<\delta$, $\lambda\notin CV$, $N(\sum\lambda_it_i)\le 1$ and $G+\sum\lambda_it_i$ is in general position at every zero in $W$. This is the mechanism of §V.20 with $s_0=0$ and only the zeros in $W$ controlled: openness of general position near the zero slice, compactness of the zeros over a small parameter ball inside the margin, Theorem 4.3.1 (preprint 4.23) and the Sard step.

**Theorem 5.3.10 relative to a compact part** `Chapter5/CylinderPerturbation2021 · theorem_5_3_10_relative_hyp`. Let $X$ be tame with sc-smooth bump functions, $G$ with tame germ data, $N$ auxiliary, $K\subseteq W$ with $K$ compact and $W$ open, such that $(G,x)$ is in general position at every zero $x\notin K$ and $\mathrm{cl}\{x\in W\mid N(G(x))\le 1\}$ is compact. Then for every $\varepsilon>0$ there is an sc$^+$-section $p$ supported in $W$ with $N(p)\le\varepsilon$ and $(G+p,x)$ in general position at every zero. *Proof.* Sections supported in $W$ make $F$ general at $(0,y)$ for the zeros $y$ near the compact $K\cap S_G$, by (i)–(ii) of §V.19 at finitely many centres `Chapter5/CylinderPerturbation2021 · exists_sections_isGeneralPositionGerm_near`; at the other zeros $F$ is general because $G$ is (the chart criterion). The relative step, for the norm $N/\eta$ with $\eta=\min(\varepsilon,1)$, gives $p=\sum\lambda_it_i$ with the zeros in $W$ in general position. A zero outside $W$ has a neighbourhood on which $p$ vanishes, since $\operatorname{supp}p\subseteq W$ is closed, so $G+p=G$ there; such a zero is a zero of $G$ not in $K$, in general position, and germ locality concludes. □ The text's theorem is the case $K=S_G$, $W=U$; here $S_G$ need not be compact.

**The margin** Chapter5/GapP1_Remark538 · exists_margin_of_isCompact. For a compact set $K$ of zeros of an sc-Fredholm section $f$ and an open $V \supseteq K$, some open $W$ with $K \subseteq W \subseteq V$ has cl$\{x \in W \mid N(f(x)) \leq 1\}$ compact: the local step of Theorem 5.2.2 (§V.6) at the points of $K$, finitely many of them.

**Remark 5.3.8 relative to a time function** Chapter5/GapP1_Remark538 · remark_5_3_8_relative. Let $X$ be tame with sc-smooth bump functions, $F$ with tame germ data, $N$ auxiliary, $\tau : X \to \mathbb{R}$ continuous, $c < a$, $b < d$. Suppose $(F, x)$ is in general position at every zero with $\tau(x) \notin (a, b)$, and the zeros with $\tau \in [c, d]$ form a compact set. Then for every $\varepsilon > 0$ there are $c < c' < a$, $b < d' < d$ and an sc$^+$-section $p$ with $N(p) \leq \varepsilon$, vanishing where $\tau \notin (c', d')$, such that $F + p$ is in general position at every zero and its zeros with $\tau \in [c, d]$ are compact. *Proof.* The zeros over $[a, b]$ are compact; the margin gives $W$ around them inside $\tau^{-1}(c', d')$, and Theorem 5.3.10 relative to that compact set gives $p$ supported in $W$. □

**The arc** Chapter5/GapP1_Remark538 · remark_5_3_8_arc. For $F(t, x) = f_t(x)$ a section of the pull-back of $P$ to $\mathbb{R}^1 \times X$ with tame germ data, in general position at its zeros with $t \notin (a, b)$, $0 < a$, $b < 1$, and with compact zeros over $[0, 1]$, the instance $\tau = t$, $[c, d] = [0, 1]$ gives an sc$^+$-section $s$ with $N(s) \leq \varepsilon$, $s(t, \cdot) = 0$ for $t \notin (c', d')$, and $F + s$ in general position at every zero with compact zeros over $[0, 1]$. "$f_0$, $f_1$ transversal" is read as general position of $F$ at the zeros near $t = 0, 1$; for $X$ without boundary and $F$ stationary near the ends it is the transversality of $f_0$, $f_1$. Its orientation, Remark 6.6.19 (preprint 6.42), is formalized in Part VI (gap G.66).

**The cobordism between admissible sections** Chapter5/CylinderPerturbation2021 · exists_cobordism, the form Theorem 6.7.1 uses (Part VI). Let $U \supseteq S_f$ be open with cl$\{x \in U \mid N(f(x)) \leq 2\}$ compact Chapter5/CylinderPerturbation2021 · exists_adapted_margin, and $s_0, s_1$ allowable with $f + s_i$ in general position at every zero. There is an sc$^+$-section $S$ of the pull-back to $\mathbb{R}^1 \times X$ with $S(t, \cdot) = s_0$ for $t \leq 1/8$, $S(t, \cdot) = s_1$ for $t \geq 7/8$, $S = 0$ over $X \setminus U$, $N(S) \leq 2$, $\tilde{f} + S$ in general position at every zero, and compact zeros over every $[a, b]$. *Proof.* $S = \hat{s} + p$, where $\hat{s} = (1 - \chi)\tilde{s}_0 + \chi\tilde{s}_1$ with the $C^\infty$ cut-off $\chi$, 0 for $t \leq 3/16$ and 1 for $t \geq 13/16$ Chapter5/CylinderPerturbation2021 · cylCut, an sc$^+$-section Chapter5/CylinderPerturbation2021 · isScPlusSection_cylInterp; near the ends $\tilde{f} + \hat{s}$ is the lift of $f + s_i$, general by the lift lemma; $p$ comes from the relative theorem for $G = \tilde{f} + \hat{s}$, the pulled-back auxiliary norm Chapter5/CylinderPerturbation2021 · StrongBundle.IsAuxNorm.pullback, $W = (1/8, 7/8) \times U$ and $K$ the zeros over $[3/16, 13/16]$. □ The homotopy version is Chapter6/ProperHomotopy2021 · exists_homotopy_cobordism.

## V.34 Theorem 5.3.12 (preprint 5.19): the dense form

Theorems 5.3.5 and 5.3.10 have dense forms (§§V.15, V.20); Theorem 5.3.12 is printed with "arbitrarily small", and §V.22 proves that form. The dense form is proved too (gap G.45).

**The generic form around a given section** Chapter5/GapP1_DenseGoodPosition · theorem_5_3_12_param_rel. Let $f$ have tame germ data, $N$ be auxiliary, $U \supseteq S_f$ open with cl$\{x \in U \mid N(f(x)) \leq 1\}$ compact, $(f, x)$ in good position at the zeros $x \in \partial X$, and $s_0$ an sc$^+$-section supported in $U$, with $N(s_0) \leq 1 - \varepsilon$ and support disjoint from $\partial X$. Then there are allowable $t_1, \ldots, t_m$ supported in $U \setminus \partial X$, $\delta > 0$ and a null $CV$ such that for $|\lambda| < \delta$, $\lambda \notin CV$, the section $s_0 + \sum \lambda_i t_i$ is allowable, its support misses $\partial X$, and $f + s_0 + \sum \lambda_i t_i$ is transversal and in good position at every zero. *Proof.* The proof of §V.22 with $g = f + s_0$ in place of $f$. Off the support of $s_0$, $g = f$ Chapter5/GapP1_DenseGoodPosition · sectionAdd_eq_of_not_mem_support, so near $\partial X$ the zeros of $g$ are those of $f$, in good position, and germ locality Chapter3/Chapter5BridgeG · StrongBundle.IsGoodPositionGermTame.congr_nhds carries this to $g$; $g$ has tame germ data (§V.15, step 2), and its zeros lie in the compact margin (Remark 5.3.4 (preprint 5.12), §V.10). Steps 1–5 of §V.22 then apply to $g$; on the slices the value of a sum at a zero is read off Chapter5/GapP1_DenseGoodPosition · paramSlice_apply_of_isZeroAt.

□ For a section, "vanishing near $\partial X$" and "support disjoint from $\partial X$" are the same `Chapter5/GapP1_DenseGoodPosition · eventually_isZeroAt_nhdsSet_iff`.

**The dense form** `Chapter5/GapP1_DenseGoodPosition · theorem_5_3_12_dense`. Let $X$ admit bump functions, $f$ have tame germ data and compact solution set, $N$ be auxiliary, and $(f, x)$ be in good position at the zeros on $\partial X$. There is an open $U \supseteq S_f$ with $S_{f+s}$ compact for every $s \in \Gamma_U^{+,1}(P)$, and for every $s' \in \Gamma_U^{+,1}(P)$ vanishing near $\partial X$ and every $\varepsilon > 0$ an $s \in \Gamma_U^{+,1}(P)$ vanishing near $\partial X$ with $N(s(x) - s'(x)) \le \varepsilon$ for all $x$, $f + s$ transversal and in good position at every zero, and $S_{f+s}$ compact and, if non-empty, a tame sub-M-polyfold with the structure of a manifold with boundary with corners. *Proof.* $U$ from Theorem 5.2.2 (preprint 5.5), compactness by Remark 5.3.4 (§V.10). With $\varepsilon' = \min(\varepsilon, 1)$ and $s_0 = (1 - \varepsilon'/2)s'$, $N(s_0) \le 1 - \varepsilon'/2$ and $\operatorname{supp} s_0 \subseteq \operatorname{supp} s'$ misses $\partial X$; the generic form gives $t$, $\delta$, $CV$; the ball of radius $\min(\delta, \varepsilon'/(2m+2))$ is not null, so it contains a $\lambda \notin CV$, and $s = s_0 + \sum \lambda_i t_i$ has $N(s - s') \le \varepsilon'/2 + m|\lambda| \le \varepsilon'$, exactly as in §V.15. □

## V.35 Remark 5.3.14 (preprint 5.21): isomorphic end problems

Definition 5.3.13 asks $(f|X_i, P|Y_i) = (f_i, P_i)$; Remark 5.3.14 proposes to read the equalities up to isomorphism and to make the isomorphisms part of the data. Formalized as follows (gap G.47; Findings, Chapter 5, item 100).

**Isomorphic problems** `Chapter5/GapP1_Remark5314 · StrongBundle.IsProblemIso`. $(P, f)$ and $(P', f')$ are isomorphic if there is a strong bundle isomorphism $(h, \Phi)$, with inverse $(g, \Psi)$, carrying $f$ to $f'$: $\Phi \circ f = f' \circ h$. Strong bundle isomorphisms are those of the formalization of Chapter 8 `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleIso` (the file imports them, a forward import that builds). The relation is reflexive, symmetric and transitive `Chapter5/GapP1_Remark5314 · StrongBundle.IsProblemIso.refl` `Chapter5/GapP1_Remark5314 · StrongBundle.IsProblemIso.symm` `Chapter5/GapP1_Remark5314 · StrongBundle.IsProblemIso.trans`. A strong bundle map sends the zero of the fibre over $x$ to the zero over $h(x)$ `Chapter5/GapP1_Remark5314 · StrongBundle.IsStrongBundleMap.map_fiberZero`, so the base map of the inverse carries $S_{f'}$ onto $S_f$ `Chapter5/GapP1_Remark5314 · StrongBundle.IsProblemIso.image_solutionSet`, and compactness of the solution sets is invariant `Chapter5/GapP1_Remark5314 · StrongBundle.IsProblemIso.isCompact_solutionSet_iff`.

**Ends up to isomorphism.** For a sub-M-polyfold $A \subseteq X$, the restricted problem — $f|A$ in the pull-back bundle, as in §V.23 — is isomorphic to a given problem `Chapter5/GapP1_Remark5314 · StrongBundle.EndIso`; end isomorphisms compose with isomorphisms of the given problem `Chapter5/GapP1_Remark5314 · StrongBundle.EndIso.trans` and give compact end solution sets `Chapter5/GapP1_Remark5314 · StrongBundle.EndIso.isCompact_solutionSet`. A *generalized compact homotopy between* $(P_0, f_0)$ *and* $(P_1, f_1)$ is a generalized compact homotopy in the 2021 sense (§V.32) with end isomorphisms at $t^{-1}(0)$ to $(P_0, f_0)$ and at $t^{-1}(1)$ to $(P_1, f_1)$ `Chapter5/GapP1_Remark5314 · StrongBundle.IsGenCompactHomotopyBetween`. It is invariant under isomorphism of the end problems `Chapter5/GapP1_Remark5314 · StrongBundle.IsGenCompactHomotopyBetween.congr_left` `Chapter5/GapP1_Remark5314 · StrongBundle.IsGenCompactHomotopyBetween.congr_right`, its end solution sets are compact `Chapter5/GapP1_Remark5314 · StrongBundle.IsGenCompactHomotopyBetween.isCompact_solutionSet_left` `Chapter5/GapP1_Remark5314 · StrongBundle.IsGenCompactHomotopyBetween.isCompact_solutionSet_right`, and every generalized compact homotopy is one between its own restrictions, with identity isomorphisms `Chapter5/GapP1_Remark5314 · StrongBundle.IsSubmersiveGenCompactHomotopy.isGenCompactHomotopyBetween` — so the notion is satisfiable whenever Definition 5.3.13 is (§V.32).

## V.36 Theorem 5.3.15 (preprint 5.22): the statement, and "Morse-type"

Theorem 5.3.15 says: let $P$ be a strong bundle over a tame M-polyfold with sc-smooth bump functions, $(f,t)$ a generalized compact homotopy (Definition 5.3.13) whose ends $X_i = t^{-1}(i)$ have no boundary and carry problems in general position, with $\partial X = X_0 \sqcup X_1$, and $N$ an auxiliary norm. Then there is an open $U \supseteq S_f$ with $S_{f+s}$ compact for $s \in \Gamma_U^{+,1}(P)$, and for every $\varepsilon > 0$ an $s_0 \in \Gamma_U^{+,1}(P)$ with $N(s_0) \le \varepsilon$, vanishing near $\partial X$, such that (1) $(f + s_0, x)$ is in general position at every zero and (2) $t : S_{f+s_0} \to [0,1]$ has only Morse-type critical points. The proof (pp. 223–228) finds that $d(t|S)$ has no zeros near $\partial S$, builds at each critical point $x$ of $t|S$ sections with Lemmas 5.3.16 and 5.3.17 that make $(\lambda, y) \mapsto d(t\circ\Phi_\lambda)(y)$ transversal to the zero section near $(0,x)$, through a trivialization $\Phi_\lambda : S \to S_\lambda$ and the second derivative of the equation, unites the sections of finitely many critical points, and concludes with "a parameterized version of Sard's theorem". The formal proof keeps this outline and changes one device: Lagrange multipliers on level sets of the solution manifold of $f + \sum \mu_j \sigma_j$ replace $\Phi_\lambda$ (§§V.38–V.40). §§V.36–V.41 retell it (gap G.48; Findings, Chapter 5, items 101, 178, 183).

**The ends** `Chapter5/Theorem5315_Ends · StrongBundle.IsGeneralPositionAtEnds`. For $i = 0, 1$ and every proof that $t^{-1}(i)$ is a sub-M-polyfold: the induced M-polyfold has no boundary, and the restriction of $f$ to the pull-back bundle (§V.23) is in general position at each of its zeros. The text's $(f_i, P_i)$ are these restrictions (Definition 5.3.13 (2) as read in §V.32), or problems isomorphic to them (§V.35).

**Smooth parametrizations** `Chapter5/Theorem5315_MorseParam · MPolyfold.IsSmoothParam`. "Morse-type critical points" is not defined in the text, and $S_{f+s_0}$ is a manifold with boundary with corners. It is read through parametrizations. A *smooth parametrization* of $A \cap G$, $G \subseteq X$ open, by a set $\Omega$ of a normed space $E$ consists of $p$, continuous on $\Omega$ with $p(\Omega) = A \cap G$, and $q : X \to E$ with sc-smooth components and $q \circ p = \mathrm{id}$ on $\Omega$, such that $h \circ p$ is $C^\infty$ on $\Omega$ for every sc-smooth $h : X \to \mathbb{R}$. A point $x$ of $A$ is a *regular point or a Morse-type critical point of* $t|A$ `Chapter5/Theorem5315_MorseParam · MPolyfold.IsMorseOrRegularPoint` if some parametrization by a set $\Omega \subseteq \mathbb{R}^n$ with unique derivatives, $x = p(u_0)$, has the property: if $d(t\circ p)(u_0) = 0$, then $\Omega$ is a neighbourhood of $u_0$ and the Hessian of $t\circ p$ at $u_0$ is injective. So critical points are interior and non-degenerate, and boundary points must be regular. *$t$ has only Morse-type critical points on $A$* if every point of $A$ is such a point `Chapter5/Theorem5315_MorseParam · MPolyfold.HasOnlyMorseCriticalPoints`. Any finite-dimensional parameter space and open $\Omega$ will do `Chapter5/Theorem5315_MorseParam · MPolyfold.isMorseOrRegularPoint_of_param Chapter5/Theorem5315_MorseParam · MPolyfold.IsSmoothParam.comp_equiv`. In a chart the notion is the classical one `Chapter5/Theorem5315_ParametricMorse · HasOnlyMorseCriticalPointsOn`.

**Independence of the parametrization.** For two parametrizations $p$, $p'$ through one point, the transition $\theta = q' \circ p$ is $C^\infty$ on $\Omega$, $\theta(u) = u'$ and $p' \circ \theta = p$ near $u$ `Chapter5/Theorem5315_LocalInvariance · MPolyfold.IsSmoothParam.transition`; the chain rule within the parameter sets `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.hasFDerivWithinAt_transition` gives $D(q \circ p')(u') \circ D(q' \circ p)(u) = 1$ `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.fderivWithin_transition_comp`. Hence $d(t \circ p')(u') \neq 0$ implies $d(t \circ p)(u) \neq 0$, at boundary points too `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.fderivWithin_ne_zero_congr`; $\Omega'$ a neighbourhood of $u'$ makes $\Omega$ one of $u$, by the inverse function theorem for the transition `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.mem_nhds_congr`; and at an interior critical point the second-order chain rule `Chapter5/Theorem5315_MorseLagrange · fderiv_fderiv_comp_apply`, whose first-order term vanishes there, conjugates the two Hessians by the invertible $D\theta(u)$ `Chapter5/Theorem5315_LocalInvariance · MPolyfold.IsSmoothParam.hessian_injective_congr`. So the whole clause transfers `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.morseClause_congr`: if $x$ is a regular or Morse-type critical point of $t|A$, then in *every* smooth parametrization of $A$ around $x$ by a set with unique derivatives

a critical point is interior with injective Hessian `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsMorseOrRegularPoint.morseClause`. The comparison with the charts of the Mathlib manifold structure that the project builds on solution sets is not formalized.

**Theorem 5.3.15** `Chapter5/Theorem5315_Final · theorem_5_3_15`. *Statement.* $X$ with sc-smooth bump functions, $(f,t)$ a generalized compact homotopy in the 2021 sense `Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy` — so $X$ tame, $f$ with tame germ data and compact solution set, $t$ sc-smooth with values in $[0,1]$ and $dt(x)$ onto at every smooth $x$ — with the end hypotheses above, $\partial X = t^{-1}(0) \cup t^{-1}(1)$, and $N$ auxiliary. Then: an open $U \supseteq S_f$ with $S_{f+s}$ compact for $s \in \Gamma^{+,1}_U(P)$, and for every $\varepsilon > 0$ an $s_0 \in \Gamma^{+,1}_U(P)$ with $N(s_0(x)) \le \varepsilon$ for all $x$, $s_0 = 0$ near $\partial X$, (1) $(f + s_0, x)$ in general position at every $x \in S_{f+s_0}$ and (2) $t$ with only Morse-type critical points on $S_{f+s_0}$. No hypothesis is added; the bump functions on the $X_i$ are not assumed. The hypotheses are satisfiable: the witness of §V.32 has empty ends and $\partial\mathbb{R} = \emptyset = X_0 \cup X_1$ `Chapter5/Book2021 · GenCompactHomotopyWitness.isSubmersiveGenCompactHomotopy_witness` `Chapter5/Theorem5315_Ends · theorem_5_3_15_generalPosition_witness`. *Proof.* $s_0 = s_1 + s_2$: $s_1$ gives (1) (§V.37); $s_2$ comes from the covering argument for $f + s_1$ (§§V.38–V.40) `Chapter5/Theorem5315_Morse · theorem_5_3_15_hyp`, whose local data are derived from the implicit function theorem and Definition 5.3.13 (3) (§V.41) `Chapter5/Theorem5315_LocalData · StrongBundle.hasMorseLocalData`. □ The hidden hypothesis $dt(x) \neq 0$ of the preprint (§V.23) is the 2021 condition (3) (§V.32); of $\partial X = X_0 \sqcup X_1$ the proof uses $\partial X \subseteq X_0 \cup X_1$ for (1) and $t(\partial X) \subseteq \{0,1\}$ for (2).

## V.37 Theorem 5.3.15 (preprint 5.22): compactness and general position, from the ends

"In view of the assumption that $(f_i, X_i)$ are already in general position" the text perturbs away from $\partial X$. What this needs is general position of $(f,x)$ at the zeros $x \in \partial X$.

**From the restriction to $X$** `Chapter5/Theorem5315_Ends · isGeneralPositionGerm_of_restrict`. Let $f$ have tame germ data, $A \subseteq X$ be a sub-M-polyfold without boundary with $f|A$ sc-smooth in the pull-back bundle, and $a \in A$ a zero with $(f|A)'(a)$ onto. Then $(f,a)$ is in general position in $X$. *Proof.* Take the tame chart $c$ of the germ data, centred at $a$. The sub-chart of $A$ built from $c$ `Chapter5/Theorem5315_Ends · subChartOf` and the chart of the pull-back bundle over it `Chapter5/Theorem5315_Ends · subPullbackData` make $\varphi\circ\iota\circ\psi^{-1}$ the inclusion $\varphi(W \cap A) \subseteq O$, and the principal part of $f|A$ is $\widehat{f}$ on $\varphi(W \cap A)$ `Chapter5/Theorem5315_Ends · principal_restrictSection_subPullback`. Surjectivity of $(f|A)'(a)$ is chart independent, and the derivative of a retraction is the identity on the tangent space of a smaller retract `Chapter5/Theorem5315_Ends · scDeriv_apply_eq_self_of_retract_subset`, so every $k$ in the fibre is $D(\widehat{f}\circ r)(z)v$ for some $v$ tangent to $\varphi(W \cap A)$ `Chapter5/Theorem5315_Ends · exists_scDeriv_eq_of_isLinSurjAt_restrict`. Since $A$ has no boundary, these tangent vectors lie in $E_z$: the reduced tangent space is chart independent `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber_eq`, and in a chart of $A$ at a point of index 0 the point is interior, so $T^R_a A = T_a A$ `Chapter5/Theorem5315_Ends · mem_minimalSubspaceWith_of_isWithoutBoundary`. So $\mathrm{Lin}_f(0)$ is onto already on a subspace of $T^R_0 O$, $\ker \mathrm{Lin}_f(0) + T^R_0 O = T_0 O$, and the chart criterion (§V.33) concludes. □ Of the end hypotheses only the surjectivity of $f_i'(x)$ and $d_{X_i} \equiv 0$ are used; the kernel complement in $X_i$ and the degeneracy index of $x$ in $X$ are not.

**At the boundary** `Chapter5/Theorem5315_Ends · StrongBundle.IsSubmersiveGenCompactHomotopy.isGeneralPositionGerm_of_mem_boundary`. If $\partial X \subseteq t^{-1}(0) \cup t^{-1}(1)$, a zero $x \in \partial X$ lies in an end, a sub-M-polyfold without boundary on which the restriction is in general position, hence onto; the previous lemma gives $(f,x)$ in general position.

**Compactness and conclusion (1)** `Chapter5/Theorem5315_Ends · theorem_5_3_15_generalPosition`, the hypotheses of §V.36. *Proof* `Chapter5/Theorem5315_GeneralPosition · theorem_5_3_15_generalPosition_hyp`. $U$ from Theorem 5.2.2 (preprint 5.5); compactness of $S_{f+s}$ for $s \in \Gamma^{+,1}_U(P)$ by Remark 5.3.4 (§V.10). General position at the zeros on $\partial X$ holds at the zeros of an open $W_b \supseteq S_f \cap \partial X$ (openness, §V.18) `Chapter5/Theorem5315_GeneralPosition · exists_isOpen_isGeneralPositionGerm_of_forall`; Theorem 5.3.10 relative to the compact part $K = S_f \setminus W_b$ (§V.33), with the perturbation supported in $U \setminus \partial X$, gives $s_0$ small, vanishing near $\partial X$, with $(f + s_0, x)$ in general position at every zero. □

## V.38 Theorem 5.3.15 (preprint 5.22): the parametrized Sard step, and Lagrange multipliers

The text's last step is "a parameterized version of Sard's theorem" (p. 228). Here it is proved in a chart from the inverse function theorem and Sard's lemma in equal dimensions, which Mathlib has.

**Parametric transversality in equal dimensions.** A surjective $L : \Lambda \times E \to F$ with $\dim E = \dim F$ is completed by a linear $T : \Lambda \times E \to \Lambda$, a projection onto $\ker L \cong \Lambda$, to an isomorphism $(L, T)$ `Chapter5/Theorem5315_ParametricMorse · exists_clm_prod_equiv`. *The local step* `Chapter5/Theorem5315_ParametricMorse · exists_isOpen_measure_image_fst_eq_zero`: near a point where $G'(p_0)$ is onto, $(G, T)$ is a local diffeomorphism by the inverse function theorem; in its coordinates the zeros of $G$ form a graph over $\Lambda$, a parameter $\lambda$ of a zero at which $\partial_z G$ is not injective is a critical value of a map between spaces of the same dimension $\dim \Lambda$, and Sard's lemma in equal dimensions makes these parameters null. *The theorem* `Chapter5/Theorem5315_ParametricMorse · ae_forall_injective_of_surjective`: if $G : \Lambda \times E \to F$ is $C^1$ on an open $W$, $\dim E = \dim F$, and $G'(p)$ is onto at every zero $p \in W$, then for almost every $\lambda$, at every zero $(\lambda, z) \in W$, $\partial_z G(\lambda, z)$ is injective. Countably many local steps cover $W$.

**Morse functions for almost every parameter** `Chapter5/Theorem5315_ParametricMorse · ae_hasOnlyMorseCriticalPointsOn`. If $g : \Lambda \times E \to \mathbb{R}$ is $C^2$ on the open $W$ and $(\lambda, z) \mapsto \partial_z g(\lambda, z) \in E^*$ — the text's $(\lambda, y) \mapsto d(t \circ \Phi_\lambda)(y)$ in a chart — has a surjective derivative at its zeros, then for almost every $\lambda$, $g(\lambda, \cdot)$ has only Morse-type critical points on its slice: apply the previous theorem to $G = \partial_z g$, with $\dim E^* = \dim E$ `Chapter5/Theorem5315_ParametricMorse · finrank_clm_real` and $\partial_z G$ the Hessian `Chapter5/Theorem5315_ParametricMorse · fderiv_slice_eq`. The hypothesis is satisfiable: for a $C^2$ function $h$ and almost every linear form $\ell$, $h + \ell$ has only Morse-type critical points, since $D(\partial_z g)(\ell, z)(\delta\ell, 0) = \delta\ell$ `Chapter5/Theorem5315_ParametricMorse · ae_hasOnlyMorseCriticalPointsOn_add_linear`.

**Lagrange multipliers.** For a constraint $h : V \to F_c$, $F_c$ finite-dimensional, and $T : V \to \mathbb{R}$, the Lagrangian is $L(v, c) = T(v) - c(h(v))$, $c \in F_c^*$ `Chapter5/Theorem5315_MorseLagrange · lagrangian`. $(v, c)$ is critical iff $h(v) = 0$ and $dT(v) = c \circ dh(v)$ `Chapter5/Theorem5315_MorseLagrange · fderiv_lagrangian_eq_zero_iff`; when $dh(v)$ is onto, a functional vanishing on $\ker dh(v)$ factors through $dh(v)$ `Chapter5/Theorem5315_MorseLagrange · exists_clm_comp_eq_of_ker_le`, so a critical point of $T$ on $\{h = 0\}$ has a multiplier. Near a regular point the level set is the image of a $C^\infty$ map $\gamma$ on an open set of $\ker dh(v)$, with an affine left inverse and $\operatorname{im} d\gamma(u) = \ker dh(\gamma(u))$ at every point `Chapter5/Theorem5315_MorseLagrange · exists_levelSet_param` (the classical implicit function theorem). The multiplier is *non-degenerate* if the Hessian of $L$ at $(v, c)$ is injective `Chapter5/Theorem5315_MorseLagrange · IsLagrangeNondegenerate`. *The bordered Hessian* `Chapter5/Theorem5315_MorseLagrange · injective_hessian_comp_of_lagrange`: if $\gamma$ parametrizes $\{h = 0\}$ near $\gamma(u)$, $c$ is a multiplier there and the Lagrangian is non-degenerate, the Hessian of $T \circ \gamma$ at $u$ is injective. By the second-order chain rule and $dh \circ D^2\gamma = -D^2h(d\gamma, d\gamma)$ (from $h \circ \gamma = 0$), the Hessian of $T \circ \gamma$ is the $v$-Hessian of $L$ on $\ker dh$; a vector $a$ in its kernel gives the kernel vector $(d\gamma(u)a, \delta c)$ of the Hessian of

$L$, with $\delta c$ from the factorization through $dh$, so $d\gamma(u)a = 0$ and $a = 0$.

**The family of level sets.** For $\mu,\ B_1, \dots, B_d : V \to F_c$, the constraints $h_\lambda = \mu - \sum \lambda_i B_i$ `Chapter5/Theorem5315_MorseFamily · famConstraint` and the gradient of the Lagrangian of the family, $(\lambda, v, c) \mapsto D_{(v,c)}(T - c(h_\lambda))$ `Chapter5/Theorem5315_MorseFamily · famGradient`. *The transversality (5.13)–(5.14) in Lagrange form* `Chapter5/Theorem5315_MorseFamily · surjective_fderiv_famGradient`: if $d\mu(v_0)$ is onto and the forms $\sum \delta_i\, c(dB_i(v_0)\,\cdot)$, over the $\delta$ with $\sum \delta_i B_i(v_0) = 0$, restrict onto the dual of $\ker d\mu(v_0)$, then the derivative of the gradient at $(0, v_0, c)$ is onto. The gradient has a component in $V^*$ and one in $F_c$. The direction $\delta v$ reaches every value of the $F_c$-component, $d\mu(v_0)$ being onto `Chapter5/Theorem5315_MorseFamily · fderiv_famGradient_point`; the direction $\delta c$ gives $-\delta c \circ d\mu(v_0)$ in $V^*$, all functionals vanishing on $\ker d\mu(v_0)$ `Chapter5/Theorem5315_MorseFamily · fderiv_famGradient_mult`; and the direction $\delta\lambda$ with $\sum \delta_i B_i(v_0) = 0$ gives the forms of the hypothesis, onto the dual of $\ker d\mu(v_0)$ `Chapter5/Theorem5315_MorseFamily · fderiv_famGradient_param`. Only first derivatives of $\mu$ and $B_i$ at $v_0$ enter; the text differentiates the equation $f(\Phi_\lambda(z)) + \sum \lambda_i s_i(\Phi_\lambda(z)) = 0$ twice.

**The Sard step for the family** `Chapter5/Theorem5315_MorseFamily · ae_isLagrangeNondegenerate_family`. Let $T$, $\mu$, $B_i$ be $C^\infty$ on the open $\Omega$, $d\mu(v_0)$ onto, and the hypothesis above hold for every multiplier $c$ at $v_0$ for $\lambda = 0$. Then there are $\varepsilon > 0$ and an open $\Omega' \ni v_0$ such that for almost every $\lambda$ with $|\lambda| < \varepsilon$, at every $v \in \Omega'$ with $h_\lambda(v) = 0$, $dh_\lambda(v)$ is onto and every multiplier at $v$ is non-degenerate. Surjective operators into a finite-dimensional space form an open set `Chapter5/Theorem5315_MorseFamily · isOpen_setOf_surjective_clm` and the multiplier of a regular level set depends continuously on the point, so the transversality holds near every $(0, v_0, c)$; the parametric Sard theorem above, applied to the Lagrangian of the family on $\Lambda \times (V \times F_c^*)$, gives the rest. This is the text's open neighbourhood $U(x)$ of p. 227.

## V.39 Theorem 5.3.15 (preprint 5.22): the families $a_i\sigma$, and transversality at one centre

**The solution manifold of a family.** For sc⁺-sections $\sigma_1, \dots, \sigma_k$ let $M = \{(\mu, y) \mid f(y) + \sum \mu_j \sigma_j(y) = 0\} \subseteq \mathbb{R}^k \times X$ `Chapter5/Theorem5315_Morse · famZero`. The sections are taken of the form $s_i = a_i \cdot \sigma_{g(i)}$, $a_i$ sc-smooth functions and $g : \{1, \dots, d\} \to \{1, \dots, k\}$ — the form $g \cdot s_e$ of the section in the proof of Lemma 5.3.17 (§V.23). Then $y$ is a zero of $f + \sum \lambda_i a_i \sigma_{g(i)}$ iff $(\Lambda(\lambda, y), y) \in M$, $\Lambda(\lambda, y)_j = \sum_{g(i)=j} \lambda_i a_i(y)$ `Chapter5/Theorem5315_MorseParam · famShift` `Chapter5/Theorem5315_Morse · fiberComb_smul_eq` `Chapter5/Theorem5315_Morse · isZeroAt_paramSlice_smul_iff`. So $S_\lambda$ lies in the fixed $M$, and $\lambda$ enters through the sc-smooth map $\Lambda$ `Chapter5/Theorem5315_MorseParam · isScSmoothMap_famShift`, sc-smooth into $\mathbb{R}^k$ because its components are `Chapter5/Theorem5315_MorseParam · isScSmoothMap_rn_of_components`. No trivialization $\Phi_\lambda$ and no second derivative of $f$ are needed.

**The local statement** `Chapter5/Theorem5315_MorseParam · ae_isMorseOrRegularPoint_famSlice`. Let $M$ be smoothly parametrized by $p : \Omega \to M \cap G$, $\Omega$ open in a finite-dimensional space, $p(v_0) = (0, y_0)$; let $t$ and $a_1, \dots, a_d$ be sc-smooth and $S_\lambda = \{y \mid (\Lambda(\lambda, y), y) \in M\}$. Suppose $\mu \circ p$ has a surjective derivative at $v_0$ ($\mu$ the projection to $\mathbb{R}^k$), and for every $c$ with $d(t\circ p)(v_0) = c \circ d(\mu\circ p)(v_0)$ the forms $\sum \delta_i c_{g(i)}\, d(a_i \circ p)(v_0)$, over the $\delta$ with $\Lambda(\delta, y_0) = 0$, restrict onto the dual of $\ker d(\mu \circ p)(v_0)$. Then there are $\varepsilon > 0$ and an open $O \ni y_0$ such that for almost every $\lambda$ with $|\lambda| < \varepsilon$ every point of $S_\lambda \cap O$ is a regular point or a Morse-type critical point of $t|S_\lambda$. *Proof.* In the parameter space of $p$, with $T = t \circ \mathrm{pr}_2 \circ p$ and $B_i = (a_i \circ \mathrm{pr}_2 \circ p) e_{g(i)}$, $S_\lambda$ near $y_0$ is the level set $\{h_\lambda = 0\}$; the Sard step for the family (§V.38) applies. At a point of the level set, the level set parametrization composed with $p$ is a smooth parametrization of $S_\lambda$ `Chapter5/Theorem5315_MorseParam · MPolyfold.IsSmoothParam.of_levelSet`; a critical point of $t$ on it has a multiplier, non-degenerate,

so the bordered Hessian is injective, and the point is interior because $\Omega'$ is open. □

**Multipliers and critical points.** A Lagrange multiplier of $t$ on $M$ at $(0, y)$ makes $y$ a critical point of $t|S_f$ `Chapter5/Theorem5315_MorseTransfer · MPolyfold.IsSmoothParam.fderivWithin_eq_zero_of_multiplier Chapter5/Theorem5315_Morse · not_isRegularPointOn_of_multiplier`. So at a regular point $y$ of $t|S_f$ there is no multiplier, and the hypothesis of the local statement holds vacuously. Regular points form an open subset `Chapter5/Theorem5315_MorseTransfer · MPolyfold.IsSmoothParam.exists_isOpen_regular` and satisfy the clause of §V.36, also for a set that coincides with $A$ near the point `Chapter5/Theorem5315_MorseTransfer · MPolyfold.IsSmoothParam.isMorseOrRegularPoint_of_regular_of_eq Chapter5/Theorem5315_Morse · MPolyfold.IsRegularPointOn`.

**The functions.** At a critical point $x$, with one section $\sigma_x$ and the parametrization $(p_x, q_x)$ of $M_x = \{f+\mu\sigma_x = 0\} \subseteq \mathbb{R}\times X$, the functions are $a_0 = 1$ and $a_i = \arctan(q_{x,i}(0,\cdot))/2$, the bounded coordinates of the parametrization: $|a_i| \le 1$ `Chapter5/Theorem5315_Morse · abs_arctan_div_two_le_one`, and $\arctan(\cdot)/2$ is $C^\infty$ `Chapter5/Theorem5315_Morse · contDiff_arctan_div_two` with non-vanishing derivative.

**The transversality survives in the larger family** `Chapter5/Theorem5315_MorseTransfer · exists_spanning_of_group`. Let $M_1 \subseteq \mathbb{R} \times X$ be the $j$-th one-parameter slice of $M \subseteq \mathbb{R}^k \times X$ `Chapter5/Theorem5315_MorseTransfer · inclJ`, both smoothly parametrized near $(0, y)$, the functions $a_0 = 1$, $a_i = \arctan(q_{1,i}(0,\cdot))/2$ built from the left inverse $q_1$ of the parametrization of $M_1$, and every multiplier of $t$ on $M_1$ at $(0, y)$ non-zero. Then for every multiplier $c$ of $t$ on $M$ at $(0, y)$, $c_j \neq 0$, and the forms $\sum \delta_i c_j\, d(a_i \circ p)$, $\sum \delta_i a_i(y) = 0$, restrict onto the dual of $\ker d(\mu \circ p)$: the hypothesis of the local statement. Restricted to the slice, $c$ gives the multiplier $c_j$ on $M_1$, non-zero; the differentials of the coordinates $q_{1,i}$, and so of the $a_i$, restrict onto the dual of the tangent space of the solution set at $y$; and the constant $a_0 = 1$, whose differential vanishes, adjusts any $\delta$ to meet $\sum \delta_i a_i(y) = 0$. The multipliers on $M_x$ at $(0, y)$ are non-zero exactly when $(0, y)$ is not critical for $t$ on $M_x$, an open condition (above) that defines $U(x)$. This is the text's "by construction, the smooth map has, at every point $(0, y)$ satisfying $d(t|S)(y) = 0$, a linearization which is surjective" (p. 227); when the families of several centres are united it needs this argument (Findings, Chapter 5, item 183).

## V.40 Theorem 5.3.15 (preprint 5.22): the covering argument

**The local data.** For a section $g$ `Chapter5/Theorem5315_Morse · StrongBundle.HasMorseLocalData`: (A1) for all sc⁺-sections $\sigma_1, \dots, \sigma_k$ and every zero $y \notin \partial X$, the solution set of $g + \sum \mu_j\sigma_j$ has a smooth parametrization around $(0, y)$ by an open set on which $\mu$ is a submersion `Chapter5/Theorem5315_Morse · HasFamilyParam`; (A2) every zero $x \notin \partial X$ has, inside every open $W \ni x$ and with arbitrarily small $N$, an sc⁺-section $\sigma$ for which $(0, x)$ is a regular point of $t$ on the solution set of $g + \mu\sigma$ `Chapter5/Theorem5315_Morse · HasRegularFamilyParam`; (A3) the zeros on $\partial X$ are regular points of $t|S_g$.

**The covering argument** `Chapter5/Theorem5315_Morse · exists_morse_perturbation_hyp`. Let $X$ be tame, $f$ with tame germ data and compact $S_f$, in general position at every zero, $N$ auxiliary, $t$ sc-smooth, $V \supseteq S_f$ open, and (A1)–(A3) hold for $f$. Then there is an open $U$, $S_f \subseteq U \subseteq V$, with $S_{f+s}$ compact for $s \in \Gamma^{+,1}_U(P)$, and for every $\varepsilon > 0$ an $s_0 \in \Gamma^{+,1}_U(P)$ with $N(s_0) \le \varepsilon$, vanishing near $\partial X$, $(f + s_0, x)$ in general position at every zero, and only Morse-type critical points of $t$ on $S_{f+s_0}$. *Proof.*

1. *The critical set.* The critical points $C$ of $t|S_f$ form a closed, hence compact, subset of $S_f$ (regular points form an open set, §V.39), and $C \cap \partial X = \emptyset$ by (A3).
2. *The centres.* At $x \in C$, (A2) gives a small $\sigma_x$ supported in $V \setminus \partial X$ near $x$ and a parametrization of $M_x$ with $(0, x)$ regular for $t$; the regular points form an open $U(x)$. Finitely many $U(x_1), \dots, U(x_k)$

cover $C$.

3. *The family.* $\sigma_j = \sigma_{x_j}$, the functions 1 and $\arctan(q_{x_j,i}(0,\cdot))/2$ of §V.39, and the sections $s_i = a_i\sigma_{g(i)}$, allowable and supported in $V \setminus \partial X$.
4. *General position.* $f$ is in general position at every zero, so the family is in general position at every $(0, y)$ (§V.33), and the relative step gives a null set off which the small slices are in general position at every zero `Chapter5/CylinderPerturbation2021 · exists_null_isGeneralPositionGerm_paramSlice` — Theorem 5.3.10's mechanism.
5. *The local statements.* Every zero $y$ of $f$ has an open $O_y$ and a null set. For $y \in \partial X$ the perturbations vanish near $y$, so $S_\lambda = S_f$ there, and the regularity of (A3) persists on an open set. For $y \notin \partial X$, (A1) parametrizes $M$ at $(0, y)$ with $\mu$ a submersion; if $y$ lies in some $U(x_j)$, the hypothesis of §V.39's local statement comes from the survival of the transversality, and otherwise $y$ is regular and the hypothesis is vacuous. The local statement gives $O_y$ and the null set.
6. *Size.* For small $|\lambda|$, $N(\sum \lambda_i s_i) \le \varepsilon$ and the sum is allowable.
7. *Compactness.* Finitely many $O_y$ cover $S_f$, and the zeros of the slices for small $|\lambda|$ lie in their union: the compactness argument of §V.13 in the margin of Theorem 5.2.2.
8. *The parameter.* Off the finitely many null sets a small $\lambda$ exists; $s_0 = \sum \lambda_i s_i$. Every zero of $f + s_0$ lies in some $O_y$, where it is regular or a Morse-type critical point, and $(f + s_0, \cdot)$ is in general position at every zero. □

**With the ends** `Chapter5/Theorem5315_Morse · theorem_5_3_15_hyp`: the hypotheses of §V.36 and (A1)–(A3) for every perturbed section $f+s$ ($s$ sc$^+$, vanishing near $\partial X$, with compact solution set and general position at every zero). Let $\eta = \min(\varepsilon, 1)/2$. §V.37 gives $U$ and $s_1$ with $N(s_1) \le \eta$, vanishing near $\partial X$, $f_1 = f + s_1$ in general position at every zero; $S_{f_1}$ is compact and inside $U$. The covering argument for $f_1$ inside $V = U$ gives $s_2$ with $N(s_2) \le \eta$. Then $f + (s_1 + s_2) = f_1 + s_2$ `Chapter5/Theorem5315_Morse · sectionAdd_sectionAdd_eq`, $N(s_1 + s_2) \le \min(\varepsilon, 1)$ `Chapter5/CylinderPerturbation2021 · StrongBundle.IsAuxNorm.auxLe_sectionAdd`, its support lies in $U$, and it vanishes near $\partial X$ `Chapter5/Theorem5315_Morse · isZeroAt_sectionAdd_of_isZeroAt`. □

## V.41 Theorem 5.3.15 (preprint 5.22): the local data from the implicit function theorem

(A1)–(A3) hold for every section $g$ with tame germ data in general position at every zero, when $X$ has bump functions and $t$ is as in Definition 5.3.13 `Chapter5/Theorem5315_LocalData · StrongBundle.hasMorseLocalData`: $t$ sc-smooth with values in $[0, 1]$, $dt(x)$ onto at every smooth $x$ (the 2021 condition (3), read in a chart `Chapter5/Book2021 · MPolyfold.HasSurjDerivAt`), and $\partial X \subseteq t^{-1}(0) \cup t^{-1}(1)$.

**Parametrizations from tangent germ data** `Chapter5/Theorem5315_LocalParam · MPolyfold.TangentGermChart.exists_isSmoothParam`. For regular tangent germ data $D$ of $M$ at $x$ (Definition 3.2.1 (2) (preprint 3.25)), $p = D.\varphi^{-1} \circ \Gamma$ on $\Omega = V \cap O_E$, $O_E$ open in the finite-dimensional $N$, is a smooth parametrization of $M$ with $p(0) = x$ and $q = \beta \cdot (\pi \circ D.\varphi)$, $\beta$ a bump function equal to 1 near $x$ and supported in the chart domain, the cut-off of §V.23 `Chapter5/Homotopy · chartLinCutoff`; $h \circ p$ is $C^\infty$ because $\Gamma$ is sc-smooth on a finite-dimensional domain (§V.14, step 2). *At a zero in general position* `Chapter5/Theorem5315_LocalData · StrongBundle.exists_isSmoothParam_solutionSet`, the index form of Theorem 3.6.8 (§V.17) gives $D$; the parametrization is by a set of $\mathbb{R}^n$ with unique derivatives ($V$ is relatively open in a partial quadrant), and by an open set when $d_X(y) = 0$, since the degeneracy index of the tame chart is $d_X(y)$ and at index 0 the partial quadrant is the whole space `Chapter5/Theorem5315_LocalData · MPolyfold.IsSmoothParam.exists_isOpen_of_mem_nhds`.

**Kernel vectors are tangent vectors.** For a smooth parametrization $p$ of $A$ and a chart $\varphi$ with retraction $R$ around $p(u_0)$, let $S$ be a retraction whose retract lies in $\varphi(A)$ and passes through $z = \varphi(p(u_0))$. Every $w \in T_z$ of that retract has a $v$, namely $D(q \circ \varphi^{-1} \circ S)(z)w$, with $d(g \circ p)(u_0)v = D(g \circ \varphi^{-1} \circ R)(z)w$ for all sc-smooth $g$, at interior and boundary points alike `Chapter5/Theorem5315_LocalTangent · MPolyfold.IsSmoothParam.exists_fderivWithin_eq_scDeriv`; the chosen sc-derivative is any bounded operator that restricts to the level-1 derivative `Chapter5/Theorem5315_LocalTangent · scDeriv_eq_of_hasFDerivWithinAt`. At a zero in general position, every strong bundle chart carries a retraction $S$ with retract in $\varphi(S_f)$ through $\varphi(y)$ and range $DS(z) = \ker \mathrm{Lin}(z)$ `Chapter5/Theorem5315_LocalTangent · StrongBundle.exists_retract_range_eq_linKer`, by the equivalence of §V.18 `Chapter5/GeneralPosition · exists_isGeneralPositionGerm_iff`. Together: every $w \in \ker \mathrm{Lin}(z)$ is tangent to every smooth parametrization of $S_f$ through $y$ `Chapter5/Theorem5315_LocalTangent · StrongBundle.exists_fderivWithin_eq_scDeriv_of_mem_linKer`. Since condition (3) is stated in an existential chart, one change of chart for the derivative of a function is needed `Chapter5/Theorem5315_LocalTangent · MPolyfold.exists_scDeriv_eq_chart_change`.

**(A1)** `Chapter5/Theorem5315_LocalData · StrongBundle.hasFamilyParam`. The family $F(\mu, y) = g(y) + \sum \mu_j \sigma_j(y)$ is in general position at $(0, y)$ (§V.33), so its solution set has a parametrization around $(0, y)$ — open for $y \notin \partial X$ — whose tangent vectors include $\ker F'(0, y)$ in the product chart `Chapter5/Theorem5315_LocalData · StrongBundle.exists_isSmoothParam_famZero`. In the product chart the derivative of $\mu_j$ is $(\delta a, \delta u) \mapsto \delta a_j$ `Chapter5/Theorem5315_LocalData · scDeriv_pc0_fst_apply`. Since $g'(y)$ is onto, every $\delta\mu$ has a $w$ with $g'(y)w = -\sum \delta\mu_j \hat{\sigma}_j(y)$, i.e. $(\delta\mu, w) \in \ker F'(0, y)$; so $d(\mu \circ p)(v_0)$ is onto.

**(A2)** `Chapter5/Theorem5315_LocalData · StrongBundle.exists_hasRegularFamilyParam`. Let $x \notin \partial X$ be a zero, $W \ni x$ open, $\eta > 0$. In a tame chart centred at $x$, $dt(x) \neq 0$ and $E_\infty$ is dense in $E_0$, so there is a smooth $v_1$ with $dt(x)(Dr(0)v_1) \neq 0$. $w_1 = Dr(0)v_1$ and $k = \mathrm{Lin}(0)w_1$ are smooth points, because $Dr(0)$ and $D(\hat{g} \circ r)(0)$ are sc-operators at the smooth point 0 and sc-operators map smooth points to smooth points `Chapter5/Theorem5315_LocalData · IsScOperator.apply_mem_smooth`. Lemma 5.3.3 (preprint 5.11) gives an sc⁺-section through $\Phi^{-1}(0, k)$ supported in $W$, rescaled by $\theta = \eta/(N(e) + 1)$ to $\sigma$; then $(-1/\theta, w_1) \in \ker F'(0, x)$, and the derivative of $t \circ \mathrm{pr}_2$ in the product chart `Chapter5/Theorem5315_LocalData · scDeriv_pc0_snd_apply` is $dt(x)w_1 \neq 0$ on it: $(0, x)$ is a regular point of $t$ on the solution set of $g + \mu\sigma$. Neither the open mapping theorem nor density in the fibre is needed. This is where the text's $b(e_i)$, in an algebraic complement $Z$ (Lemma 5.3.16), is replaced: Lemma 5.3.3 reaches only smooth points of the fibre, and $-c\,g'(x)w_1$ is one because $w_1$ is smooth.

**(A3) at every corner** `Chapter5/Theorem5315_LocalData · StrongBundle.isRegularPointOn_of_extremum`. *The extremum lemma* `Chapter5/Theorem5315_LocalTangent · scDeriv_eq_zero_of_extremum`: if $U$ is relatively open in a partial quadrant $C = T^{-1}([0, \infty)^n \oplus W)$ and an sc¹ function $\tau$ on $U$ has a minimum or a maximum at a level-1 point $z$, then $D\tau(z)e = 0$ for every $e \in E_z$. A direction $e \in E_z$ keeps at 0 the coordinates that vanish at $z$, so $z + se \in C$ for small $|s|$ of both signs; for $e$ of level 1, $s \mapsto \tau(z + se)$ is differentiable with an extremum at 0; the level-1 vectors are dense in $E_z$ `Chapter5/Theorem5315_LocalTangent · mem_closure_levelOne_minimalSubspaceWith` and $D\tau(z)$ is continuous on $E_0$. At a zero $x \in \partial X$, $t(x) \in \{0, 1\}$ and $0 \leq t \leq 1$, so $t$ has an extremum, and $dt(x)$ vanishes on $E_x \supseteq T^R_x X$, whatever the codimension of $T^R_x X$. General position gives $T_x X = \ker g'(x) + T^R_x X$ `Chapter5/CylinderPerturbation2021 · isParamGP_elim0_of_isGeneralPositionGerm` and $dt(x) \neq 0$, so $dt(x)$ does not vanish on $\ker g'(x)$, the tangent space of the parametrization: $x$ is a regular point of $t|S_g$. No hypothesis on the degeneracy index $d_X(x)$ is needed; of $\partial X = X_0 \sqcup X_1$ only $t(\partial X) \subseteq \{0, 1\}$ enters here, and "the $X_i$ have no boundary" only in §V.37.

**The theorem.** With (A1)–(A3) for every perturbed section, `theorem_5_3_15_hyp` (§V.40) is the theorem without hypothesis (§V.36). Theorem 5.3.15 holds as printed, with Definition 5.3.13 read in its 2021 form.

## V.42 Theorem 5.5.10 (preprint 5.37): every chart has the approximation property

§V.25 proved "(i) □ (ii)" for restricted charts, because the text's $\mathrm{cl}_X(W) \subseteq U$ does not follow from $\mathrm{cl}_C(V) \subseteq O$ (Findings, Chapter 5, item 111). Every chart has the property, by a different route `Chapter5/GapP1_ApproxEveryChart · hasApproxProperty_of_isChart`, and Theorem 5.5.10 holds with "every chart" `Chapter5/GapP1_ApproxEveryChart · theorem_5_5_10_every_chart`.

**Tools on $X$.** With sc-smooth partitions of unity, disjoint closed sets have sc-smooth Urysohn functions `Chapter5/GapP1_ApproxEveryChart · MPolyfold.AdmitsScPartitionsOfUnity.exists_urysohn`, and a continuous function with values in $[0,1]$ and support closed in an open set is approximated within $\varepsilon$ by sc-smooth functions with closed support in that set `Chapter5/GapP1_ApproxEveryChart · MPolyfold.AdmitsScPartitionsOfUnity.exists_approx` — the argument of §V.25 on $X$ itself.

**The shells.** Let $f : O \to [0,1]$ be continuous with support in $V = O \cap W$, $\mathrm{cl}\, V \subseteq O$, and $F = f \circ \varphi$ on the chart domain; $F$ extended by 0 need not be continuous (§V.25). Write the chart domain as $\{d > 0\}$ with $d$ continuous `Chapter5/GapP1_ApproxEveryChart · exists_continuous_pos_iff`. Take sc-smooth $\kappa_n$, 1 on $\{d \geq 1/(n+1)\}$ and 0 on $\{d \leq 1/(n+2)\}$, and the weights $\lambda_n = \kappa_n \prod_{j<n}(1-\kappa_j)$, which telescope to 1 on the chart domain and are locally finite. Where $\lambda_n \neq 0$, $\kappa_{n+1} = 1$, so $F = F\kappa_{n+1}$ there; $F\kappa_{n+1}$ has support closed in $X$, and its approximation $\hat{g}_n$ has closed support in the chart domain. $g = \sum_n \lambda_n \hat{g}_n$ is finite near each point, hence sc-smooth, with values in $[0,1]$ `Chapter5/GapP1_ApproxEveryChart · sum_mul_mem_Icc`, and within $\varepsilon$ of $F$ as a convex combination of $\varepsilon$-approximations `Chapter5/GapP1_ApproxEveryChart · abs_sub_sum_mul_le`. Read in the chart, $g$ gives the approximation on $O$. □ The restricted charts of §V.25 are kept.

## V.43 Theorem 5.5.5 (preprint 5.32): Bonic–Frampton

**The statement** `Chapter5/GapP1_BonicFrampton · BonicFrampton.theorem_5_5_5`. On the real $L^p(\mu)$ of any measure space, $p \geq 1$, $F(f) = \|f\|^p$ is (1) $C^\infty$ for $p$ an even integer, (2) $C^{p-1}$ for $p$ an odd integer, (3) $C^{[p]}$ for $p$ not an integer — as printed (gap G.50; Findings, Chapter 5, items 117, 120, 179).

**Weighted products** `Chapter5/GapP1_LpWeightedProduct · BonicFrampton.wMul`. For $w \in L^q$ with $q^{-1} + m/p = 1$, $(h_1, \dots, h_m) \mapsto \int w \prod h_i$ is a continuous $m$-linear form of norm $\leq \|w\|_q$, by the generalized Hölder inequality `Chapter5/GapP1_LpWeightedProduct · BonicFrampton.eLpNorm_prod_le` `Chapter5/GapP1_LpWeightedProduct · BonicFrampton.norm_wMul_le`. *The even case* `Chapter5/GapP1_LpWeightedProduct · BonicFrampton.theorem_5_5_5_even`: $\|f\|^{2k} = \int f^{2k}$ `Chapter5/GapP1_LpWeightedProduct · BonicFrampton.norm_rpow_eq_integral` is the diagonal of the $2k$-linear form with weight $1 \in L^\infty$, hence $C^\infty$.

**The general case.** Let $g_j$ be the $j$-th derivative of $|t|^p$, of the form $a_j \max(t,0)^{p-j} + b_j \max(-t,0)^{p-j}$ `Chapter5/GapP1_BonicFrampton · BonicFrampton.gK` `Chapter5/GapP1_BonicFrampton · BonicFrampton.hasDerivAt_gK`. Two one-variable estimates for such homogeneous kernels carry the analysis: $|g(s+t) - g(s)| \leq \varepsilon|s|^r + C_\varepsilon|t|^r$ for $r > 0$, from the continuity of $z^r$ at 1 and homogeneity `Chapter5/GapP1_BonicFrampton · BonicFrampton.kern_sub_le`, and the first-order remainder bound $|t|(\varepsilon|s|^{r-1} + C_\varepsilon|t|^{r-1})$ by the mean value theorem `Chapter5/GapP1_BonicFrampton · BonicFrampton.taylor1_le` `Chapter5/GapP1_BonicFrampton · BonicFrampton.gK_taylor_le`. For $j < p$, $g_j(f) \in L^{p/(p-j)}$ `Chapter5/GapP1_BonicFrampton · BonicFrampton.memLp_gK`, and the exponents form Hölder triples `Chapter5/GapP1_BonicFrampton · BonicFrampton.holderTriple_qE`. The forms $T_j(f) = \int g_j(f) \prod h_i$ `Chapter5/GapP1_BonicFrampton · BonicFrampton.Tj` are continuous in $f$ (the first estimate, integrated) `Chapter5/`

GapP1_BonicFrampton · BonicFrampton.continuous_Tj, and the derivative of $T_j$ is $T_{j+1}$ curried (the second estimate with Hölder $L^p \times L^{p/(p-j-1)} \to L^{p/(p-j)}$) Chapter5/GapP1_BonicFrampton · BonicFrampton.hasFDerivAt_Tj. So $(T_j)$ is a Taylor series of $\|\cdot\|^p$ up to every integer $n < p$ Chapter5/GapP1_BonicFrampton · BonicFrampton.hasFTaylorSeriesUpTo_normRpow, and $\|\cdot\|^p$ is $C^n$ Chapter5/GapP1_BonicFrampton · BonicFrampton.contDiff_normRpow_of_lt. Case (2) is $n = 2k < 2k+1$, case (3) is $n = [p] < p$. No calculus of Nemytskii operators is needed beyond these two estimates.

## V.44 The Sobolev remark after Theorem 5.5.5

After Theorem 5.5.5 the text says that the usual norms of $W^{k,p}(\Omega)$ have the differentiability of $\|\cdot\|_{L^p}$, that $\beta(\|x\|^p)$ is a bump function of $L^p$, that $W^{1,4}(\Omega)$ with $E_m = W^{1+m,4}(\Omega)$ admits sc-smooth bump functions, while on $W^{1,3/2}(\Omega)$ the straightforward bump is only sc¹, and it asks whether an sc-smooth one exists (pp. 230–231). Formalized, with the sc-structure built on one domain (row V5.76, narrower; gap G.51; Findings, Chapter 5, items 118, 185).

$W^{k,p}(\Omega)$ Chapter5/GapP1_SobolevRemark2 · SobolevRemark.Wsub. For $\Omega \subseteq \mathbb{R}^n$ and $p \geq 1$: the tuples $(v_\alpha)_{|\alpha|\leq k}$ in the $\ell^p$-product of copies of $L^p(\Omega)$ with $\int_\Omega v_0\, \partial^\alpha \varphi = (-1)^{|\alpha|} \int_\Omega v_\alpha \varphi$ for all test functions $\varphi$ on $\Omega$. Each identity is a continuous functional Chapter5/GapP1_SobolevRemark2 · SobolevRemark.wk, so the subspace is closed Chapter5/GapP1_SobolevRemark2 · SobolevRemark.isClosed_Wsub, a Banach space with $\|u\|^p = \sum_\alpha \|D^\alpha u\|^p_{L^p}$ Chapter5/GapP1_SobolevRemark2 · SobolevRemark.norm_rpow_W; weak derivatives are unique on open $\Omega$ Chapter5/GapP1_SobolevRemark2 · SobolevRemark.Dop_zero_injective.

**The norms and the bumps.** Any norm with $\|u\|^p = \sum_i \|T_i u\|^p_{L^p}$, $T_i$ continuous linear, has the differentiability of Theorem 5.5.5 Chapter5/GapP1_SobolevRemark1 · SobolevRemark.theorem_5_5_5_sum, which is the first claim Chapter5/GapP1_SobolevRemark2 · SobolevRemark.remark_5_5_5_sobolev. For $\beta$ smooth with compact support, $\beta(\|x\|^p)$ has bounded support Chapter5/GapP1_SobolevRemark1 · SobolevRemark.bump_comp_rpow Chapter5/GapP1_SobolevRemark1 · SobolevRemark.hasContDiffBump_of_rpow; "non-vanishing $\beta$" is read as $\beta(0) \neq 0$, as in the Hilbert example of p. 229. This gives bumps of $L^p$ Chapter5/GapP1_SobolevRemark1 · SobolevRemark.hasContDiffBump_Lp and of $W^{k,p}(\Omega)$ Chapter5/GapP1_SobolevRemark2 · SobolevRemark.remark_5_5_5_sobolev_bump, $C^\infty$, $C^{p-1}$ or $C^{[p]}$ in the three cases.

**From level 0 to the sc-structure.** A $C^\infty$ bump on $E_0$ gives the sc-smooth bump property of every local model on $E$: rescale and translate it to $x$ as in Proposition 5.5.8 (§V.24); it is $C^\infty$ on $E_0$, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)), and composed with the retraction Chapter5/GapP1_SobolevRemark1 · LocalModel.hasBumpProperty_of_hasContDiffBump Chapter5/GapP1_SobolevRemark1 · MPolyfold.admitsBumpFunctions_of_hasContDiffBump. Hence every sc-Banach space with $E_0 \cong W^{1,4}(\Omega)$ admits sc-smooth bump functions, as do its local models and the M-polyfolds modelled on it Chapter5/GapP1_SobolevRemark3 · SobolevRemark.remark_5_5_5_W14; on $W^{1,3/2}(\Omega)$ the bump $\beta(\|x\|^{3/2})$ is $C^1 = C^{[3/2]}$ on $E_0$, hence sc¹ Chapter5/GapP1_SobolevRemark1 · exists_isSck_bump Chapter5/GapP1_SobolevRemark3 · SobolevRemark.remark_5_5_5_W1_3_2. These use only the level 0: they hold for every sc-structure on these spaces and every $\Omega$. The closing question is a question, not a claim.

**The sc-structure** $E_m = W^{1+m,p}(\Omega)$ Chapter5/GapP1_SobolevRemark3 · SobolevRemark.sobolevSc, with the inclusions $W^{k+1,p} \to W^{k,p}$ Chapter5/GapP1_SobolevRemark2 · SobolevRemark.inclW, injective on open $\Omega$ Chapter5/GapP1_SobolevRemark2 · SobolevRemark.inclW_injective, takes the compactness of the inclusions and the density of $E_\infty$ as hypotheses; $\Omega = \emptyset$ is the trivial instance Chapter5/GapP1_SobolevRemark3 · SobolevRemark.sobolevSc_hyps_empty. Both are proved for $\Omega = (0,1) \subset \mathbb{R}^1$ and every $p > 1$ Chapter5/GapP1_SobolevCube5 · SobolevCube.cubeSc:

1. *The Volterra operator* $Vg(x) = \int_0^x g$ is compact on $L^p(0,1)$ Chapter5/GapP1_SobolevCube1 · SobolevCube.isCompactOperator_V: by Hölder $|Vg(x) - Vg(y)| \le \|g\|\,|x-y|^{1-1/p}$ Chapter5/GapP1_SobolevCube1 · SobolevCube.abs_Vf_sub_le, so the image of the unit ball is uniformly close to step functions on a grid of $N$ intervals Chapter5/GapP1_SobolevCube1 · SobolevCube.dist_V_le, hence totally bounded.
2. *Rellich.* A function with weak derivative $0$ is constant Chapter5/GapP1_SobolevCube2 · SobolevCube.eq_cst_of_weakDeriv_zero, so every component of $u \in W^{k,p}$ is a constant plus $V$ of the next one Chapter5/GapP1_SobolevCube2 · SobolevCube.eq_cst_add_V Chapter5/GapP1_SobolevCube3 · SobolevCube.comp_eq_cst_add_V, with the constant bounded by the norm Chapter5/GapP1_SobolevCube4 · SobolevCube.abs_cst_le; the inclusion $W^{k+1,p} \to W^{k,p}$ is compact Chapter5/GapP1_SobolevCube4 · SobolevCube.isCompactOperator_inclW_cube.
3. *Density.* Polynomials are dense Chapter5/GapP1_SobolevCube2 · SobolevCube.denseRange_poly; approximating the top derivative and integrating gives smooth approximations on every level Chapter5/GapP1_SobolevCube5 · SobolevCube.exists_smooth_approx, so $E_\infty$ is dense Chapter5/GapP1_SobolevCube5 · SobolevCube.dense_smoothLvl_cube.

On this structure the claims hold for the text's sc-structure Chapter5/GapP1_SobolevCube5 · SobolevCube.remark_5_5_5_W14_cube Chapter5/GapP1_SobolevCube5 · SobolevCube.remark_5_5_5_W14_instance Chapter5/GapP1_SobolevCube5 · SobolevCube.remark_5_5_5_W1_3_2_cube. Cubes in $\mathbb{R}^n$, $n \ge 2$, and other bounded domains — where, as we read it, a regular boundary is presupposed — are not done; hence the row is narrower.

## V.45 Theorem 5.5.13 (preprint 5.40): Toruńczyk

The statement recorded in §V.26 Chapter5/PartitionsOfUnity · TorunczykStatement — $C^k$ partitions of unity iff a coordinate-wise $C^k$ homeomorphic embedding into some $c_0(\Gamma)$ — holds for every real Banach space and every $k \in \mathbb{N} \cup \{\infty\}$ Chapter5/GapP1_Torunczyk · theorem_5_5_13 (gap G.52). The coordinates of $c_0(\Gamma)$ are 1-Lipschitz Chapter5/GapP1_Torunczyk · abs_cz_sub_le and only finitely many exceed a given $\varepsilon$ Chapter5/GapP1_Torunczyk · finite_cz_gt.

**(□ )** Chapter5/GapP1_Torunczyk · torunczyk_forward. For each $n$ take a $C^k$ partition of unity $(\rho_{n,c})_{c\in V}$ subordinate to the balls $B(c, 1/(n+1))$, and $h(x) = (\rho_{n,c}(x)/(n+1))_{(n,c)}$. It lies in $c_0(\mathbb{N}\times V)$, since the partitions are locally finite and the weights tend to $0$; it is continuous, because up to $\varepsilon$ only finitely many $n$ matter; its coordinates are $C^k$; and it is an embedding: if $h(y)$ is close to $h(x)$, some coordinate $(n,c)$ large at $x$ is still non-zero at $y$, so $y$ and $x$ share a ball of radius $1/(n+1)$.

**(□ )** Chapter5/GapP1_Torunczyk · torunczyk_backward, through $C^k$ Urysohn functions Chapter5/GapP1_Torunczyk · exists_contDiff_urysohn_of_embedding and Mathlib's bump coverings Chapter5/GapP1_Torunczyk · admitsContDiffPartitionsOfUnity_of_urysohn. With a $C^\infty$ bump $\theta$, $1$ on $[-1/2,1/2]$ and $0$ outside $(-3/5,3/5)$ Chapter5/GapP1_Torunczyk · thetaB, the *lattice bumps* are $b_{n,\sigma}(x) = \prod_\gamma \theta((n+1)h(x)(\gamma) - \sigma(\gamma))$ for finitely supported $\sigma : \Gamma \to \mathbb{Z}$ Chapter5/GapP1_Torunczyk · latticeBump. Near a point all but finitely many factors are $1$, so they are $C^k$ Chapter5/GapP1_Torunczyk · contDiff_latticeBump (locally finite sums and products of $C^k$ functions are $C^k$ Chapter5/GapP1_Torunczyk · contDiff_finsum_of_locallyFinite); for fixed $n$ they are locally finite in $\sigma$ Chapter5/GapP1_Torunczyk · locallyFinite_latticeBump; at $x$, the bump of the rounding $\sigma$ of $(n+1)h(x)$ equals $1$ Chapter5/GapP1_Torunczyk · exists_latticeBump_eq_one, and its support lies in $\{y \mid \|h(y) - h(x)\| \le 2/(n+1)\}$ Chapter5/GapP1_Torunczyk · tsupport_latticeBump_subset_ball. For disjoint closed $s$, $t$, the bumps whose support misses $s$ or misses $t$ cover $V$, $h$ being an embedding; Michael's products, which multiply the bumps of level $n$ by $\eta$ Chapter5/GapP1_Torunczyk · etaB of the sums

of the levels $m < n$, make them locally finite across $n$, and the quotient of the sum over the bumps on the side of $t$ by the sum of all is the Urysohn function.

## V.46 Theorem 5.5.16 (preprint 5.43) for separable spaces, and Corollary 5.5.17 (preprint 5.44) without hypothesis

**$E_0$ is separable** `Chapter5/GapP1_SeparablePOU · ScBanach.separableSpace_zero`, for every sc-Banach space: the inclusion $E_1 \to E_0$ is a compact operator with dense range; the range of a compact operator lies in $\bigcup_n nK$ with $K$ compact, so it is separable `Chapter5/GapP1_SeparablePOU · IsCompactOperator.isSeparable_range`, and a space with a dense separable subset is separable. So in the sc-calculus Theorem 5.5.16 is needed only for separable spaces (Findings, Chapter 5, item 184).

**The separable case.** From a $C^k$ bump, a $C^k$ function $\psi : V \to [0,1]$, $1$ on $B(0,1)$, supported in $B(0,R)$ `Chapter5/GapP1_SeparablePOU · HasContDiffBump.exists_one_on_ball`. For disjoint closed $s$, $t$: a countable cover of $t$ by balls $B(x_n, r_n)$ whose $R$-fold enlargements miss $s$ (Lindelöf), $\psi_n = \psi((\cdot - x_n)/r_n)$, and the telescoping products $f_n = \psi_n \prod_{j<n}(1-\psi_j)$ with $\sum_{n<N} f_n = 1 - \prod_{n<N}(1-\psi_n)$ `Chapter5/GapP1_SeparablePOU · sum_range_mul_prod_one_sub` give a $C^k$ Urysohn function `Chapter5/GapP1_SeparablePOU · exists_contDiff_urysohn_of_separable`, and bump coverings give $C^k$ partitions of unity `Chapter5/GapP1_SeparablePOU · admitsContDiffPartitionsOfUnity_of_separable`: Theorem 5.5.16 for separable spaces, the Bonic–Frampton construction `Chapter5/GapP1_SeparablePOU · godefroyTroyanskiWhitfieldZizlerStatement_of_separable`.

**Corollary 5.5.17 (preprint 5.44)** `Chapter5/GapP1_SeparablePOU · corollary_5_5_17_unconditional`. If $\|v\|^2 = b(v,v)$ on $E_0$, the local model has the sc-smooth approximation property, with no further hypothesis: $E_0$ is separable, has the $C^\infty$ bump $\tau(1 - b(v,v))$ (§V.26), hence $C^\infty$ partitions of unity, and Proposition 5.5.11 (preprint 5.38) applies. The M-polyfold form `Chapter5/GapP1_SeparablePOU · MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional` drops the hypothesis of §V.26.

**Hilbert spaces of any density** `Chapter5/GapP1_HilbertPOU · admitsContDiffPartitionsOfUnity_of_hilbertSpace`. For a Hilbert basis $(b_\gamma)$, $x \mapsto (\|x\|^2, (\langle b_\gamma, x\rangle)_\gamma)$ maps into $c_0$ with $C^\infty$ coordinates, and it is a homeomorphic embedding by a Kadec–Klee estimate in $\ell^2$ `Chapter5/GapP1_HilbertPOU · l2_inverse_estimate`: if $|v_\gamma - u_\gamma| \le \delta$ for all $\gamma$ and $|\|v\|^2 - \|u\|^2| \le \delta$, then $\|v-u\|^2 \le n\delta^2 + 2\delta + 2n\delta(2\|u\| + \delta) + 4\,\mathrm{tail}_F(u)$, with $F$ a set of $n$ coordinates carrying most of $u$. Toruńczyk (□) concludes `Chapter5/GapP1_HilbertPOU · godefroyTroyanskiWhitfieldZizlerStatement_of_hilbertSpace`.

## V.47 Theorem 5.5.16 (preprint 5.43): every WCG Banach space

`Chapter5/GapP1_GTWZ · theorem_5_5_16`. A weakly compactly generated Banach space with a $C^k$ bump admits $C^k$ partitions of unity, for every $k$ (gap G.53). The recorded statement `Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement` leaves completeness out; the theorem assumes it, the text's "Banach". The route follows Deville–Godefroy–Zizler, reorganized so that neither Krein's theorem nor the Eberlein–Šmulian theorem nor a transfinite recursion is needed.

**Bases** `Chapter5/GapP1_C0Base · IsRelC0Base`. A *base relative to $Q$* is a locally $c_0$-finite family of $C^k$ functions with values in $[-1,1]$ whose cozero sets are $Q$-small. A base gives partitions of unity: $x \mapsto (f_i(x))_i$ is a coordinate-wise $C^k$ embedding into $c_0(I)$, and Toruńczyk applies `Chapter5/GapP1_C0Base · admitsContDiffPartitionsOfUnity_of_isRelC0Base`; conversely $f_{(c,n)} = \rho_{n,c}/(n+1)$ `Chapter5/GapP1_`

C0Base · exists_isRelC0Base_of_admits. Bases transfer along operators Chapter5/GapP1_C0Base · IsRelC0Base.comp, and a bump gives a cut-off Chapter5/GapP1_C0Base · HasContDiffBump.exists_cutoff. *The gluing lemma* Chapter5/GapP1_C0Base · isRelC0Base_glue: if $X$ has a $C^k$ bump, $\Delta_i$, $Q_i$ are operators with $\|\Delta_i\| \le C$, $(\|\Delta_i x\|)_i \in c_0$ for every $x$, every $x \neq 0$ is close to some $Q_i x$ with $\Delta_i x \neq 0$, and each $Q_i$ carries a base relative to $Q_i$, then $X$ has a base. The new functions are $(m{+}1)^{-1}(N{+}1)^{-1}\psi((m{+}1)\Delta_i y)\,\chi((N{+}1)(y{-}Q_i y))\,g_{ij}(y)$ and $(N+1)^{-1}\chi((N+1)y)$, with $\psi = 1-\chi$.

**Projections from Lindenstrauss pairs** Chapter5/GapP1_LindenstraussPair · IsLPair. For $K$ weakly compact with dense span, a *Lindenstrauss pair* $(A, D)$: $A$ a $\mathbb{Q}$-subspace, $D$ a set of functionals norming $A$, and $A \cap K$ dense in $K$ for the topology of $D$. It gives a norm-one projection onto $\operatorname{cl} A$ along $D_\perp$ Chapter5/GapP1_LindenstraussPair · IsLPair.exists_proj, by weak compactness (the finite intersection property) and Mazur's theorem; the convexity of $K$ (Krein) is not used. A monotone closure operator $S \mapsto (A(S), D(S))$, an $\omega$-iteration of finitary operations with global choices of norming functionals and witnesses Chapter5/GapP1_LindenstraussPair · lclA Chapter5/GapP1_LindenstraussPair · lclD, yields a pair Chapter5/GapP1_LindenstraussPair · isLPair_lcl of finite character Chapter5/GapP1_LindenstraussPair · lclA_finite and of cardinality at most $\max(\#S, \aleph_0)$ Chapter5/GapP1_LindenstraussPair · card_lcl.

**The resolution of the identity** Chapter5/GapP1_WCGPRI · priP. With $e$ enumerating a dense set by the ordinals below $\mu$, $P_\alpha$ is the projection of the closure of $\{e_\beta \mid \beta < \alpha\}$. The projections commute, $P_\mu = 1$ Chapter5/GapP1_WCGPRI · priP_top, and the family is continuous at limit ordinals Chapter5/GapP1_WCGPRI · PRICont Chapter5/GapP1_WCGPRI · priP_cont because the closure has finite character — so no recursion over the ordinals is needed. For such a family the increments of $x$ are in $c_0$ Chapter5/GapP1_WCGPRI · pri_finite, and $x$ is approximated by some $P_{\alpha+1}x$ with a non-zero increment, through the least $\beta$ with $P_\beta x = x$ Chapter5/GapP1_WCGPRI · pri_approx.

**The induction** Chapter5/GapP1_GTWZ · admitsContDiffPartitionsOfUnity_of_wcg_card, strong induction on a cardinal $\kappa$ bounding a dense set. For $\kappa \le \aleph_0$ the space is separable (§V.46). Otherwise the pieces $P_{\alpha+1}X$ are WCG Chapter5/GapP1_GTWZ · IsWCG.of_proj, carry a $C^k$ bump Chapter5/GapP1_GTWZ · HasContDiffBump.submodule, and have dense sets of smaller cardinality Chapter5/GapP1_GTWZ · exists_dense_card_le, so by induction they admit partitions of unity, hence bases; with $\Delta_\alpha = P_{\alpha+1} - P_\alpha$ and $Q_\alpha = P_{\alpha+1}$ the gluing lemma gives a base of $X$, and partitions of unity.

## V.48 Weak sequential compactness in reflexive spaces

The text uses without comment that a bounded sequence in a reflexive Banach space has a weakly convergent subsequence (proof of Proposition 5.2.7, "after perhaps taking a subsequence"), and that the compact inclusion $F_1 \to F_0$ maps weakly convergent sequences to norm convergent ones (proof of Lemma 5.1.5). Mathlib has the sequential Banach–Alaoglu theorem only for the dual of a separable space, so both are proved here. A space is *reflexive* if the canonical map $F \to F^{**}$ is onto Chapter5/PartitionsOfUnity · IsReflexive (the definition of §V.26); finite-dimensional spaces are reflexive Chapter5/P3R_WeakSeqCompact · isReflexive_of_finiteDimensional. *Weak convergence* $u_k \rightharpoonup h$ is $g(u_k) \to g(h)$ for every $g \in F^*$ Chapter5/P3R_WeakSeqCompact · WeakTendsto. Weakly convergent sequences are bounded, by uniform boundedness Chapter5/P3R_WeakSeqCompact · WeakTendsto.exists_bound, and the norm is weakly lower semicontinuous, $\|h\| - \varepsilon < \|u_k\|$ eventually Chapter5/P3R_WeakSeqCompact · WeakTendsto.eventually_norm_gt.

**Weak sequential compactness** Chapter5/P3R_WeakSeqCompact · exists_subseq_weakTendsto: in a reflexive $F$ every bounded sequence has a weakly convergent subsequence; also with a bound that holds only eventually Chapter5/P3R_WeakSeqCompact · exists_subseq_weakTendsto_of_eventually. *Step 1.* The closed span $M$ of the se-

quence is separable, so countably many norm-one functionals $g_d$ norm the points of $M$. *Step 2.* The values $(g_d(u_k))_d$ lie in a cube $\prod_d [-C, C]$, compact and metrizable; a subsequence makes every $g_d(u_k)$ converge. *Step 3.* The images $J(u_k)$ in the bidual lie in a weak-*compact ball (Banach–Alaoglu). Every weak* cluster point is $J(p)$ for some $p \in M$, by reflexivity and Hahn–Banach, and the $g_d$ determine $p$; so the subsequence has one cluster point, and a sequence in a compact set with one cluster point converges. **Two consequences.** A compact operator maps weakly convergent sequences to norm convergent ones `Chapter5/P3R_WeakSeqCompact · WeakTendsto.tendsto_of_isCompactOperator`. In a reflexive space, a bounded sequence whose image under an injective bounded operator converges to the image of $h$ converges weakly to $h$ `Chapter5/P3R_WeakSeqCompact · weakTendsto_of_bounded_of_tendsto`: every subsequence has a weakly convergent subsequence, whose limit is $h$ by injectivity.

## V.49 Definition 5.1.4 (2017: 5.2; not in the preprint) and Lemma 5.1.5 (2017: 5.2; not in the preprint)

**Definition 5.1.4** (restated) `Chapter5/P3R_Reflexive2021 · StrongBundle.HasReflexiveFibers`: $P : Y \to X$ has reflexive $(0,1)$-fibres if every $x \in X$ lies in the domain of a strong bundle chart compatible with the atlas whose model has $F_1$ reflexive. The text asks for an equivalent atlas of such charts; this is read as "the reflexive compatible charts cover $X$", which is the same, since chart domains are of the form $P^{-1}(V)$. It holds if every chart of the given atlas is reflexive `Chapter5/P3R_Reflexive2021 · StrongBundle.hasReflexiveFibers_of_atlas`. In a chart $c$ the fibre part of a point $y \in Y_{0,1}$ is $h \in F_1$ `Chapter5/P3R_Reflexive2021 · StrongBundleChart.liftOneSnd`.

**Lemma 5.1.5** `Chapter5/P3R_Reflexive2021 · lemma_5_1_5`: let $y_k \in Y_{0,1}$ with $P(y_k) \to x$ on level 0, and $\Psi$, $\Psi'$ charts around $x$ with $F_1$, $F_1'$ reflexive; then the fibre parts $h_k$ converge weakly in $F_1$ iff the $h_k'$ converge weakly in $F_1'$. The strong form `Chapter5/P3R_Reflexive2021 · lemma_5_1_5_limit` identifies the limits, which the text's proof uses implicitly: if $h_k \rightharpoonup h$ (and only $F_1'$ needs to be reflexive), there is one $y \in Y_{0,1}$ over $x$ with $\Psi$-part $h$, $y_k \to y$ in $Y$, and $h_k'$ converges weakly to the $\Psi'$-part of $y$ `Chapter5/P3R_Reflexive2021 · StrongBundle.weakTendsto_transfer`. *Step 1, the limit point.* $h_k \rightharpoonup h$ in $F_1$ gives $h_k \to h$ in $F_0$, through the compact inclusion (§V.48). The limit $(a, h)$ lies in the local model $K$, because the retraction is continuous on $U \lhd F$ and fixes $K$ `Chapter5/P3R_Reflexive2021 · StrongBundleChart.mem_K_of_tendsto`; it is a point $y$ of $Y_{0,1}$ with $y_k \to y$ `Chapter5/P3R_Reflexive2021 · StrongBundle.exists_limit_of_weakTendsto`. *Step 2, the other chart.* The $h_k'$ are bounded in $F_1'$, by the local comparison of the norms $|\cdot|_1$ of two charts (Lemma 5.1.2), and converge in $F_0'$ to the fibre part of $y$, because $y_k \to y$ in $Y$ and $\Psi'$ is continuous; so they converge weakly in the reflexive $F_1'$ (§V.48) `Chapter5/P3R_Reflexive2021 · StrongBundle.weakTendsto_liftOneSnd_of_tendsto`. Fibre parts are read in a chart only for large $k$.

## V.50 Definitions 5.1.6 and 5.1.7 (2017: 5.3 and 5.4; not in the preprint): mixed convergence and reflexive auxiliary norms

**Definition 5.1.6** `Chapter5/P3R_Reflexive2021 · StrongBundle.MixedTendsto`: $y_k \in Y_{0,1}$ is mixed convergent to $y \in Y_{0,1}$ if $P(y_k) \to P(y)$ in $X_0$ and, in some chart around $P(y)$ with $F_1$ reflexive, the fibre parts converge weakly. The text's "in view of this lemma we can define" is proved: the fibre parts then converge weakly in every compatible reflexive chart around $P(y)$ `Chapter5/P3R_Reflexive2021 · StrongBundle.MixedTendsto.weakTendsto_of_isChart`, and $y_k \to y$ in $Y$ `Chapter5/P3R_Reflexive2021 · StrongBundle.MixedTendsto.tendsto_oneVal`, both by Lemma 5.1.5. *Mixed convergent subsequences* `Chapter5/P3R_Reflexive2021 · StrongBundle.`

exists_mixedTendsto_subseq: if $P(y_k) \to x$ and $N(y_k) \le r$ for an auxiliary norm $N$, a subsequence is mixed convergent; the fibre parts in a reflexive chart are bounded by Lemma 5.1.2, and §V.48 applies.

**Definition 5.1.7** Chapter5/P3R_Reflexive2021 · StrongBundle.IsReflexiveAuxNorm: a reflexive auxiliary norm is an auxiliary norm with $N(y) \le \liminf_k N(y_k)$ whenever $y_k$ is mixed convergent to $y$. The text prints $\liminf_{k\to 0}$; presumably $k \to \infty$. The lim inf is taken in the extended reals, so that an unbounded sequence has $\liminf = \infty$. Equivalent forms: if $N(y_k) \le r$ infinitely often then $N(y) \le r$ Chapter5/P3R_Reflexive2021 · StrongBundle.IsReflexiveAuxNorm.le_of_frequently_le, and a criterion through eventual bounds Chapter5/P3R_Reflexive2021 · StrongBundle.le_liminf_coe_of_eventually.

## V.51 Proposition 5.1.8 (2017: 5.2; not in the preprint): reflexive auxiliary norms exist

**Proposition 5.1.8** Chapter5/P3R_Reflexive2021 · proposition_5_1_8: with reflexive $(0,1)$-fibres a reflexive auxiliary norm exists Chapter5/P3R_Reflexive2021 · proposition_5_1_8_exists, and for an auxiliary norm $N_0$ and a reflexive one $N_1$ there is a continuous $f : X \to (0,\infty)$ with $f(x)N_0 \le N_1 \le f(x)^{-1}N_0$ on the fibre over $x$ Chapter5/P3R_Reflexive2021 · proposition_5_1_8_compare. The text leaves the proof to the reader. *Existence.* The norm of Proposition 5.1.3 (preprint 5.3) built over the reflexive compatible charts, $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)|_1$ for a partition of unity $(\rho_i)$ subordinate to their domains, is an auxiliary norm by Lemma 5.1.2 (preprint 5.2). *Lower semicontinuity* Chapter5/P3R_Reflexive2021 · StrongBundle.le_liminf_gluedNorm: near $P(y)$ only finitely many terms are nonzero; $\rho_i(P(y_k)) \to \rho_i(P(y))$; in each chart the fibre parts converge weakly, by the chart independence of §V.50, and the norm is weakly lower semicontinuous (§V.48). So $N$ is a reflexive auxiliary norm Chapter5/P3R_Reflexive2021 · StrongBundle.isReflexiveAuxNorm_gluedNorm. *Comparison.* A reflexive auxiliary norm is an auxiliary norm, and Proposition 5.1.3 (preprint 5.3) compares any two. Two tools for later chapters: $(g \circ P)\cdot N$ is again a (reflexive) auxiliary norm for continuous $g > 0$ Chapter5/P3R_Reflexive2021 · StrongBundle.IsReflexiveAuxNorm.comp_mul, and with reflexive fibres every auxiliary norm lies between two reflexive ones Chapter5/P3R_Reflexive2021 · StrongBundle.exists_reflexiveAuxNorm_sandwich.

## V.52 Definition 5.2.6 and Proposition 5.2.7 (2017: 5.6 and 5.3; not in the preprint)

**Definition 5.2.6** Chapter5/P3R_ReflexiveCompactness2021 · StrongBundle.HasReflexiveLocalCompactness: $f$ has the reflexive local compactness property if for every reflexive auxiliary norm $N$ and every $x$ there is an open $U(x) \ni x$ with $\mathrm{cl}_X\{q \in U(x) \mid N(f(q)) \le 1\}$ compact. The text prints $N(f(x)) \le 1$ inside the set; presumably $N(f(q))$. $N(f(q)) \le 1$ includes $f(q) \in Y_{0,1}$, as in §V.32. The standing assumptions of the text (reflexive fibres, $f$ sc-smooth and regularizing) are hypotheses of the statements that use it.

**Proposition 5.2.7** (stronger) Chapter5/P3R_ReflexiveCompactness2021 · proposition_5_2_7. With the text's hypotheses: (1) every $x \in X_0 \setminus X_1$ has a neighbourhood on which $N(f(q)) > 1$; (2) so has every $x$ with $N(f(x)) > 1$. *(2) for every continuous section* Chapter5/P3R_ReflexiveCompactness2021 · StrongBundle.exists_nhds_not_auxLe. If not, $x$ is a limit of points $x_k$ with $N(f(x_k)) \le 1$; $X$ is metrizable (Theorem 2.3.10), so a sequence suffices. A subsequence of $f(x_k)$ is mixed convergent to some $y$ (§V.50), and $f(x_k) \to y$ in $Y$, so $y = f(x)$ by continuity of $f$; lower semicontinuity gives $N(f(x)) \le 1$. *(1)*: for a regularizing $f$, $f(x) \notin Y_{0,1}$ at $x \notin X_1$, so $N(f(x)) = \infty$ Chapter5/P3R_ReflexiveCompactness2021 · StrongBundle.not_auxLe_of_not_mem_

`levelSet`, and (2) applies. The reflexive local compactness property is not used, and (2) needs only continuity of $f$. In other words, $\{x \mid N(f(x)) \le 1\}$ is closed `Chapter5/P3R_ReflexiveCompactness2021 · StrongBundle.isClosed_setOf_auxLe`.

**The remark after Proposition 5.2.7** (stronger) `Chapter5/P3R_ReflexiveCompactness2021 · StrongBundle.closure_auxLe_subset`: the closure of $\{x \in U \mid N(f(x)) \le 1\}$ lies in $\{x \in \mathrm{cl}_X U \mid N(f(x)) \le 1\}$, for every $U$ and no compactness assumption, since the second set is closed. The text prints $\{x \in X \mid \dots\}$ in the closure; presumably $x \in U$, which is what the argument above it uses.

## V.53 Theorem 5.2.8 (2017: 5.4; not in the preprint) and Remarks 5.2.9, 5.2.10 (2017: 5.1, 5.2)

**Theorem 5.2.8** (stronger) `Chapter5/P3R_ReflexiveCompactness2021 · theorem_5_2_8`. Hypotheses as in the text: $X$ tame, reflexive $(0,1)$-fibres, $f$ sc-Fredholm with compact solution set $S$ and the reflexive local compactness property, $N$ a reflexive auxiliary norm, and $U_\partial$ an open neighbourhood of $S_\partial = S \cap \partial X$ in $\partial X$ with $\mathrm{cl}\{x \in U_\partial \mid N(f(x)) \le 1\}$ compact; "open in $\partial X$" is read as $U_\partial = W \cap \partial X$ with $W$ open. Conclusion: an open $U \supseteq S$ with $U \cap \partial X = U_\partial$ and $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact. Tameness and the sc-Fredholm property are not used; the theorem holds for every continuous section of a strong bundle with reflexive fibres over any M-polyfold `Chapter5/P3R_ReflexiveCompactness2021 · theorem_5_2_8_of_continuous`. *Step 1, the boundary part.* Finitely many $U(\bar{x}_1), \dots, U(\bar{x}_\ell)$ of Definition 5.2.6 cover the compact $\mathrm{cl}\{x \in U_\partial \mid N(f(x)) \le 1\}$; with the open set $\{N \circ f > 1\}$ (§V.52) they form $\tilde{U}$, which contains $U_\partial$. *Step 2, the interior part.* $S \setminus \tilde{U}$ is compact and misses $\partial X$; finitely many $U(x) \setminus \partial X$ cover it, and their union with $\tilde{U}$ is an open $U'$. *Step 3, compactness.* $\{x \in U' \mid N(f(x)) \le 1\}$ lies in the finite union of the compact sets $\mathrm{cl}\{q \in U(x_i) \mid N(f(q)) \le 1\}$, because on $\{N \circ f > 1\}$ there is nothing to add. *Step 4, the boundary.* $U = U' \cap (W \cup (X \setminus \partial X))$ contains $S$, has $U \cap \partial X = U_\partial$, and its controlled set is a closed subset of the compact union. Witness for all hypotheses at once: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ with the section id (solution set $\{0\}$, $U_\partial = \emptyset$) `Chapter5/P3R_ReflexiveCompactness2021 · TrivialBundle.theorem_5_2_8_witness`.

**Remark 5.2.9** restates Theorem 5.2.8 in words: if $(U_\partial, N)$ controls compactness on the boundary, some $(U, N)$ with $U \cap \partial X = U_\partial$ controls it on $X$. **Remark 5.2.10**, that the theorem is crucial for inductive perturbation arguments, has no mathematical content to formalize.

## V.54 Definition 5.4.1 (2017: 5.11; not in the preprint): sc⁺-sections over the boundary

**Definition 5.4.1** (restated) `Chapter5/P3B_BoundaryExtension2021 · StrongBundle.IsBoundaryScPlusSection`: a section $s$ of $W|\partial X \to \partial X$ is an sc⁺-section over $\partial X$ if its restrictions to the local faces at every $x \in \partial X$ are sc⁺-sections. Read chart-wise: on $\partial X$, $s$ is a section with values in $W_{0,1}$, and for every tame strong bundle chart $\Phi$ around $x$, every tame presentation $T$ of the corner of its model and every $i \in I(\varphi(x))$, the local representative $\Phi[1] \circ s \circ \varphi^{-1}$ restricted to the face germ $O \cap \{t_i = 0\}$ at $\varphi(x)$ is an sc⁺-section of $K \to O$ near $\varphi(x)$ (Part XII, §XII.8). That $s(x) \in W_{0,1}$ is stated outright, so that $N(s(x))$ in Theorem 5.4.2 is defined at every boundary point. The definition quantifies over all tame charts and presentations; its witness is the zero section, for every strong bundle `Chapter5/P3B_BoundaryExtension2021 · StrongBundle.isBoundaryScPlusSection_fiberZero`. The text's $\mathrm{supp}(s)$ is the closure of $\{x \in \partial X \mid s(x) \neq 0_x\}$ `Chapter5/`

# V.55 Theorem 5.4.2 (2017: 5.9; not in the preprint): the ∂-extension theorem

**Theorem 5.4.2** Chapter5/P3B_BoundaryExtension2021 · theorem_5_4_2. Let $P : W \to X$ be a strong bundle over a tame M-polyfold admitting sc-smooth partitions of unity, $s$ an sc⁺-section over $\partial X$, $N$ an auxiliary norm, $\tilde{U}$ an open neighbourhood of $\mathrm{supp}(s)$, and $f : X \to [0,\infty)$ continuous, supported in $\tilde{U}$, with $N(s(x)) < f(x)$ on $\mathrm{supp}(s)$. Then there is an sc⁺-section $\tilde{s}$ with (1) $N(\tilde{s}) \le f$, (2) $\mathrm{supp}(\tilde{s}) \subset \tilde{U}$, (3) $\tilde{s}|\partial X = s$. The text gives a sketch: Proposition 12.3.7 locally, glued by a partition of unity. "Strong bundle over a tame M-polyfold" is the project's tame strong bundle, whose base is tame and every point of which lies in a tame strong bundle chart; this fixes the spaces in the lowest universe, and the core below takes only the chart hypothesis. *Step 1, the local step* Chapter5/P3B_BoundaryExtension2021 · StrongBundle.exists_local_extension. At $x \in \partial X$, in a tame chart, Proposition 12.3.7 (Part XII, §XII.8) extends the local representative of $s$ near $\varphi(x)$; a boundary point of $X$ is a boundary point of the model Chapter5/P3B_BoundaryExtension2021 · StrongBundle.degIndex_ne_zero_of_mem_boundary. Cut off by a bump function equal to $1$ near $x$ (Proposition 5.5.3 (preprint 5.30)), the extension becomes a global sc⁺-section agreeing with $s$ on $\partial X$ near $x$, by the locality of sc⁺ Chapter5/P3B_BoundaryExtension2021 · StrongBundle.isScPlusSection_of_locally_p3b. *Step 2, local bounds* Chapter5/P3B_BoundaryExtension2021 · StrongBundle.exists_local_bounded. Every $x$ has an open $U \ni x$ and an sc⁺-section $\sigma$ with $N(\sigma) \le f$ on $U$ and $\sigma = s$ on $\partial X \cap U$. At $x \in \mathrm{supp}(s)$, $\sigma$ is the local extension, and $N(\sigma) < f$ near $x$ by continuity of $N \circ \sigma$ and of $f$. Elsewhere $\sigma = 0$ Chapter5/P3B_BoundaryExtension2021 · StrongBundle.aux_fiberZero: off $\partial X$, and on $\partial X \setminus \mathrm{supp}(s)$, where $s$ vanishes on a neighbourhood. *Step 3, gluing* Chapter5/P3B_BoundaryExtension2021 · StrongBundle.exists_boundary_extension. With an sc-smooth partition of unity $(\rho_i)$ subordinate to these $U_i$, $\tilde{s} = \sum_i \rho_i \sigma_i$ is sc⁺ by locality; $N(\tilde{s}) \le \sum_i \rho_i N(\sigma_i) \le f$ by the triangle inequality and homogeneity of $N$; on $\partial X$, $\tilde{s} = \sum_i \rho_i s = s$ since $\sum_i \rho_i = 1$. (2) follows from (1): where $f = 0$, $N(\tilde{s}) = 0$, so $\mathrm{supp}(\tilde{s}) \subset \mathrm{supp}(f) \subset \tilde{U}$. That $\tilde{U}$ is open and contains $\mathrm{supp}(s)$ is kept from the text and not used.

# Part VI. Linearizations, orientations and invariants (HWZ Chapter 6, §§6.2–6.8 (preprint §§6.1–6.7))

Chapter 6 builds the orientation theory of sc-Fredholm sections and, at its end, the invariant of oriented proper sections. At a smooth point $x$ it collects the linearizations of $f$ into a convex set $\mathrm{Lin}(f,x)$ of Fredholm operators (§6.2 (preprint §6.1)); it fixes the conventions for determinant lines and for the isomorphism attached to a four-term exact sequence (§6.3 (preprint §6.2)); it defines the determinant $\det(T)=\lambda(\ker T)\otimes\lambda^*(\mathrm{coker}\,T)$ of a Fredholm operator, compares it with the determinants of $PT$ for good left-projections $P$ and of the stabilizations $T_\varphi$, and builds the determinant line bundle $\mathrm{DET}(E,F)$ over the Fredholm operators (§§6.4–6.5 (preprint §§6.3–6.4)); it then restricts that bundle to $\mathrm{Lin}(f,x)$, propagates its orientations along sc-smooth paths of $X$, first in small neighbourhoods (Theorem 6.6.11 (preprint 6.35)) and then globally, as a functor on the path category (Theorem 6.6.13 (preprint 6.37)), puts a topology on the orientation space (Theorem 6.6.17, new in 2021), and defines orientability and orientations of $f$, orients the solution set (Theorem 6.6.18 (preprint 6.41)) and the sections $f+s$ over $X\times[0,1]$ (Remark 6.6.19 (preprint 6.42)) (§6.6 (preprint §6.5)); §6.7 (preprint §6.6) defines the admissible perturbations and the invariant $\Psi_{(f,o)}:H^*_{dR}(X,\partial X)\to\mathbb{R}$ and proves that it is well defined and invariant under proper oriented homotopies (Theorem 6.7.1 (preprint 6.43)); and the appendix §6.8 (preprint §6.7) proves Lemma 6.3.3 (preprint 6.5) and Proposition 6.4.11 (preprint 6.12). This Part retells §6.2 (preprint §6.1) (§VI.1); §6.3 (preprint §6.2) — determinant lines, Definition 6.3.1 (preprint 6.3) and Proposition 6.3.2 (preprint 6.4), the isomorphism $\Phi_E$, Lemma 6.3.3 (preprint 6.5) and Proposition 6.3.5 (preprint 6.6) (§§VI.2–VI.3); §6.4 (preprint §6.3) — Definitions 6.4.1–6.4.3 (preprint 6.7–6.9), Lemma 6.10 (preprint; not in the 2021 edition), the isomorphisms $\gamma_T^P$, the stabilization and the direct sums (§§VI.4–VI.6); §6.5 (preprint §6.4) — the local determinant bundles and $\mathrm{DET}(E,F)$ (§§VI.7–VI.8); §6.6 (preprint §6.5) — orientations over $\mathrm{Lin}(f,x)$ (§VI.9), Proposition 6.6.3 (preprint 6.28) (§VI.10), Lemmas 6.6.4–6.6.6 (preprint 6.29–6.31) (§§VI.11–VI.13), the push-forward (§§VI.14–VI.15), Lemma 6.6.8 (preprint 6.33), Lemma 6.6.10 (preprint 6.34) and Proposition 6.6.7 (preprint 6.32) (§§VI.16–VI.17), Theorem 6.6.11 (preprint 6.35) and Definition 6.6.12 (preprint 6.36) (§§VI.18–VI.19), the path category (§VI.20), the comparison of charts (§VI.21), the transport and Theorem 6.6.13 (preprint 6.37) (§§VI.22–VI.23), Definitions 6.6.14–6.6.16 (preprint 6.38–6.40) (§VI.24), Theorem 6.6.18 (preprint 6.41) (§VI.25) and Remark 6.6.19 (preprint 6.42) (§VI.26); §6.7 (preprint §6.6), the admissible sections and Theorem 6.7.1 (preprint 6.43) with its full proof (§VI.27); the appendix §6.8 (preprint §6.7) (§§VI.28–VI.29); what is not formalized (§VI.30); what the formalization taught us (§VI.31); the 2021 additions — Lemma 6.4.4, Proposition 6.4.13, Lemmas 6.6.8 and 6.6.9, Theorem 6.6.17 (§§VI.32–VI.33); and the material formalized since the

first edition of this Part: the two conventions of §§6.3–6.4 and the remark that $\mathrm{DET}(E,F)$ is not orientable (§VI.34), Lemma 6.5.3 (preprint 6.19) for general vector bundles (§VI.35), Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds (§VI.36), paths and families on $[0,1]$ and the facts on paths before Proposition 6.2.2 (preprint 6.2) (§VI.37), and the orientation bundle after Theorem 6.6.17 (§VI.38). Proposition 6.2.2 (preprint 6.2) was formalized with Chapter 3 and is retold in §III.49. Each item carries the number of its comparison item (V6.1–V6.82) in `Exposition/verification/ch06.md`.

**What is proved, and what is not.** *Theorem 6.7.1 (preprint 6.43) is proved, with the hypotheses of the text only* — $X$ tame with sc-smooth bump functions — and so is the case $\partial X = \emptyset$ after it (§VI.27; rows V6.76, V6.77 restated; gaps G.54, G.55 closed). The proof follows the text's: one admissible perturbation defines $\Psi$, Stokes' theorem makes it a map on cohomology, and a cobordism $S_F$ over $\mathbb{R}^1 \times X$ compares two perturbations and two ends of a homotopy; the decomposition of $\partial S_F$ is replaced by a cut-off in time. Theorems 6.6.11, 6.6.13, 6.6.17 and 6.6.18 (preprint 6.35, 6.37, —, 6.41) are proved, the last three with the text's hypotheses (§§VI.18, VI.23, VI.33, VI.25); Theorem 6.6.18 (preprint 6.41) in the book's shape, the solution set a compact Mathlib manifold with boundary and corners carrying a continuous tangent orientation which is the Fredholm one, read compatibly in any two charts, and the paragraph after it on perturbed solution sets (gaps G.67–G.69 closed). Remark 6.6.19 (preprint 6.42) gives an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40), the 2021 convention $e$ first, with no sign (§VI.26; gap G.66 closed). Proved besides: Propositions 6.3.2, 6.3.5, 6.4.10, 6.4.11, 6.4.13, 6.4.15, 6.5.4, 6.5.7, 6.5.8, 6.6.1, 6.6.7 (preprint 6.4, 6.6, Lemma 6.11, 6.12, —, 6.15, 6.20, 6.23, 6.24, 6.26, 6.32), Lemmas 6.3.3, 6.4.4, 6.4.14, 6.5.1, 6.5.2, 6.5.5, 6.5.9, 6.6.6, 6.6.8–6.6.9, 6.6.10 (preprint 6.5, —, 6.14, 6.17, 6.18, 6.21, 6.25, 6.31, 6.33, 6.34) and Lemma 6.10 (preprint), Corollary 6.4.12 (preprint 6.13), Lemma 6.5.3 (preprint 6.19) for general topological vector bundles (§VI.35; G.59 closed), Lemma 6.6.4 (preprint 6.29) with property (P) for the slices and Lemma 6.6.5 (preprint 6.30) as a statement about manifolds, with $\pi$ a submersion and $L$ a smooth vector bundle (§VI.36; G.62, G.63 closed), Proposition 6.6.3 (preprint 6.28) with (1) as printed in 2021 and with (1′), on $\mathbb{R}$ and on $[0,1]$ (§VI.10; G.61 closed), the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), the sc-Fredholm property of $\mathrm{Lin}(f,x)$ on every level (§VI.1; G.64 closed), the facts on paths recalled before Proposition 6.2.2 (preprint 6.2) (§VI.37; G.56 closed), the alternative determinant and the identification (6.25) (preprint (95)) (§VI.34; G.57, G.58 closed), Proposition 6.4.13, last sentence (§VI.33; G.81 closed), property (2) of Theorem 6.6.17 without further hypotheses (§VI.33; G.90 closed), the orientation bundle, the $\mathbb{Z}_2$-action and $\det_f$ after it (§VI.38; G.91 closed), and the classical determinant bundle at the opening of §6.6 (§VI.34). Every definition of the chapter is formalized. **Gaps:** G.54–G.64, G.66–G.69, G.81, G.90, G.91 closed (G.61 since 2026-11-13, with Proposition 6.6.3 (1) in its 2021 wording); G.65 (paths and families on $[0,1]$) in part; no gap is open. **Narrower rows**, with their reasons: V6.48, Theorem 6.6.11 (preprint 6.35) — the statement does not introduce the section $f$ (presumably an omission; it is read for an sc-Fredholm section with tame germ data), and $U$ is not shown contractible in itself, the preprint's (1); the 2021 form of (1), a smaller $U'$ contractible in $U$, is proved, and (1′) is what the proof uses; V6.80, the remark that $\mathrm{DET}(E,F)$ is not orientable — it does not hold as we read it for finite-dimensional $E$, $F$ (Lean counterexample `exists_two_lineOrientation_det_of_finiteDimensional`) and is proved for infinite-dimensional Hilbert spaces and under complex structures, not for all Banach spaces; V6.81, the classical case — $C^1$ maps in place of smooth ones, and a topological, not smooth, line bundle. Several places needed more than a reading. The projection of the proof of Lemma 6.10 (preprint) does not lie below the two given projections as we read it, and the lemma is proved by another construction (§VI.4). The isomorphism $\Phi_E$ of formula (94) does not vary continuously in the topology of DET where $\dim\operatorname{coker} T$ changes parity, as we read it, so the propagation uses $(-1)^{\dim\operatorname{coker} T}\Phi_E$ (§VI.14). Proposition 6.6.3 (1) is proved in its 2021 wording, a smaller $U'$ contractible in $U$; the preprint's "$U$ is

sc-smoothly contractible" is not established by its proof as we read it, and the proofs use (1′) (§§VI.10, VI.18). The form $\bar{\omega} = p_1^*\omega \wedge p_2^* dt$ of the proof of Theorem 6.7.1 (preprint 6.43) has one degree too many for the comparison, as we read it, and $p_1^*\omega$ is used (§VI.27). Seven proofs the text omits are supplied: the orientability of a line bundle over a convex set (§VI.9), the general position of Lemma 6.6.4 (preprint 6.29) (§VI.11), the "simple computation" of Lemma 6.6.8 (preprint 6.33) (§VI.16), the covering argument of Theorem 6.6.13 (preprint 6.37) (§VI.22), the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) (§VI.24), the construction of the topology of Theorem 6.6.17 (§VI.33), and the orientation of the cobordism on its ends in Theorem 6.7.1 (preprint 6.43) (§VI.27).

**Six structural choices.** (1) *Chart points.* $T_xX$ carries no linear structure in the formalization (gaps G.6, G.7, G.13 of Chapter 2), so $\mathrm{Lin}(f,x)$, $\mathrm{DET}(f,x)$ and $\mathrm{Or}_x^f$ are made at a *chart point* $(c,z)$ of $x$, with operators $T_zO = Dr(z)E \to K_z = \rho(z)F$ on level 0, and they are identified between charts by explicit transition maps (§VI.21); the text's $f-s$ with $s(x) = f(x)$ is $f+s$ with $(f+s)(x) = 0$. (2) *Global sections, from bump functions.* The text assumes sc-smooth partitions of unity from §6.3 (preprint §6.2) on, "to simplify the presentation" (p. 241 (preprint p. 203)). The formalization assumes sc-smooth bump functions, which partitions of unity give (§V.25), and uses them only to make global sc$^+$-sections (Findings, Chapter 6, item 43); Theorem 6.7.1 (preprint 6.43) assumes them in its statement. (3) *Families on $\mathbb{R}^n$, and on $[0,1]^n$.* Paths, homotopies and families are first sc-smooth maps on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ whose conditions are imposed on $[0,1]$, $[0,1]^2$, $[0,1]^n$, and $X \times [0,1]$ is read as $\mathbb{R}^n \times X$ with the pull-back $\pi^*P$ of the bundle along the projection, the parameter first (decision D2, as for Proposition 6.2.2 (preprint 6.2) in §III.49; Findings, Chapter 6, item 102). Most of these statements now also hold as printed, on the M-polyfolds $[0,1]$, $[0,1]^2$, $[0,1]^n \times X$ (§VI.37); what is still read on $\mathbb{R}$ is listed in gap G.65 (§VI.30). The cobordism of Theorem 6.7.1 (preprint 6.43) lives over $\mathbb{R}^1 \times X$. (4) *The sign $\alpha(T) = (-1)^{\dim \operatorname{coker} T}$* at both end points of every propagation (§VI.14). (5) *(1′), and orientable neighbourhoods as pairs $V \subseteq U$*, since what the proofs use of Proposition 6.6.3 (preprint 6.28) is (1′) rather than contractibility (§§VI.10, VI.19). (6) *Dyadic subdivisions* in place of the finite coverings of the proof of Theorem 6.6.13 (preprint 6.37) (§VI.22). An orientation of a line is a ray (Mathlib's `Module.Ray`); an orientation of a topological line bundle is a choice of rays given near every point by a continuous nowhere-zero section; an orientation of a Mathlib manifold is a continuous tangent orientation (§VI.25).

**Map.**

- `Chapter6/LinSet`: chart points, $T_zO$, $K_z$, sc$^+$-operators, Definition 6.2.1 (preprint 6.1), the index and convexity of $\mathrm{Lin}(f,x)$.
- `Chapter6/DetLine`: $\lambda(E)$, $\lambda^*(E)$, wedges, $v^*$, $\lambda(\Phi)$, Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4).
- `Chapter6/ExactSequence`: four-term exact sequences, the choices of (91), $\Phi_E$, Lemma 6.3.3 (preprint 6.5) with the proof of §6.8.1 (preprint §6.7.1), (92), Proposition 6.3.5 (preprint 6.6).
- `Chapter6/FredholmDet`: Definitions 6.4.1, 6.4.2 (preprint 6.7, 6.8).
- `Chapter6/GoodProjections`: Definition 6.4.3 (preprint 6.9), the order on $\Pi_T$, the facts after it, Lemma 6.10 (preprint) and the example on $\mathbb{R}^2$.
- `Chapter6/ProjectionSequence`: Proposition 6.4.10 (preprint Lemma 6.11), $\gamma_T^P$, Proposition 6.4.11 (preprint 6.12) with the proof of §6.8.2 (preprint §6.7.2), Corollary 6.4.12 (preprint 6.13).
- `Chapter6/Stabilization`: Lemma 6.4.14 (preprint 6.14), formula (94), Proposition 6.4.15 (preprint 6.15).
- `Chapter6/DirectSumDet`: Definition 6.4.16 (preprint 6.16).
- `Chapter6/LocalDetBundle`: Lemmas 6.5.1, 6.5.2 (preprint 6.17, 6.18), the frames of Proposition 6.5.4 (preprint 6.20), Lemma 6.5.5 (preprint 6.21).

- `Chapter6/DetBundle`: $\mathrm{DET}(E,F)$ as a vector prebundle, Definition 6.5.6 (preprint 6.22), Propositions 6.5.7, 6.5.8 (preprint 6.23, 6.24), Lemma 6.5.9 (preprint 6.25), Proposition 6.6.1 (preprint 6.26).
- `Chapter6/LineBundleOrientation`: orientations of line bundles, the orientation double cover.
- `Chapter6/LinOrientation`: Definition 6.6.2 (preprint 6.27).
- `Chapter6/OrientableNbhd`: sc-smooth homotopies, Proposition 6.6.3 (preprint 6.28).
- `Chapter6/Lemma629`: Lemma 6.6.4 (preprint 6.29).
- `Chapter6/KernelFamily`: the kernels of Lemma 6.6.5 (preprint 6.30).
- `Chapter6/SubspaceBundle`: continuous families of subspaces, their determinant lines, continuation over $[0,1]$.
- `Chapter6/StabilizationContinuity`: families of kernels and their images, the stabilization sign, the example without the sign.
- `Chapter6/ParamKernel`: the stabilized section in a product chart, Lemma 6.6.6 (preprint 6.31).
- `Chapter6/Continuation`: continuation over a simply connected base, the propagation $\Phi_{E_1}^{-1}\circ$ continuation $\circ\,\Phi_{E_0}$.
- `Chapter6/Propagation`: stabilized kernels and the push-forward $\varphi_*$; Lemmas 6.6.8 and 6.6.10 (preprint 6.33 and 6.34) for it.
- `Chapter6/StabKernelsExist`: the stabilized kernels exist.
- `Chapter6/PushData`: push-forwards along paths, their existence, halves and reparametrizations.
- `Chapter6/AddSections`: the linear algebra of Lemma 6.6.8 (preprint 6.33).
- `Chapter6/FamilySections`: sections over parameter families.
- `Chapter6/PropagationWellDefined`: Proposition 6.6.7 (preprint 6.32), the homotopy argument, Theorem 6.6.11 (preprint 6.35).
- `Chapter6/ScPaths`: sc-smooth paths in a chart, concatenation.
- `Chapter6/ContinuousOrientation`: Definition 6.6.12 (preprint 6.36), continuous orientations.
- `Chapter6/ScHomotopy`: reparametrization homotopies.
- `Chapter6/PathCategory`: homotopy classes and the path category $PX$.
- `Chapter6/ChartTransfer`: conjugation of determinants, the transfer between chart points.
- `Chapter6/PushTransfer`: the push-forward does not depend on the chart.
- `Chapter6/SectionOrientation`: orientable neighbourhoods as structures, $\mathrm{Or}_x^f$ in a chosen chart, Definitions 6.6.15, 6.6.16 (preprint 6.39, 6.40).
- `Chapter6/PathTransport`: transport along paths from local push-forwards.
- `Chapter6/OrientationFunctor`: Theorem 6.6.13 (preprint 6.37), Definitions 6.6.14–6.6.16 (preprint 6.38–6.40) and their equivalences.
- `Chapter6/SolutionOrientation`: Theorem 6.6.18 (preprint 6.41).
- `Chapter6/ProductOrientation`: Remark 6.6.19 (preprint 6.42).
- `Chapter6/Book2021`, `Chapter6/Book2021Continuity`: Lemma 6.4.4, Proposition 6.4.13, Lemmas 6.6.8 and 6.6.9 of the 2021 edition (§§VI.32–VI.33).
- `Chapter6/OrientationSpace`: the orientation space $O_f$ and Theorem 6.6.17 (§VI.33).
- `Chapter6/LinSetFredholm`: the members of $\mathrm{Lin}(f,x)$ are sc-Fredholm on every level (§VI.1).
- `Chapter6/GapP1_DetConventions`, `Chapter6/GapP1_DetNonOrientable`, `…Ell2`, `…Transport`, `Chapter6/GapP1_ClassicalDetBundle`: the alternative determinant, (6.25), the remark that $\mathrm{DET}(E,F)$ is not orientable, the classical case (§VI.34).
- `Chapter6/DetVectorBundles2021`: Lemma 6.5.3 (preprint 6.19) (§VI.35).
- `Chapter6/SolutionManifoldFamily2021`, `Chapter6/GapP1_SubmersionNearFamily`, `Chapter6/GapP1_SlicePropertyP`, `Chapter6/GapP1_SmoothL1–3`, `Chapter6/GapP1_Cube∗`: Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds, on $\mathbb{R}^n$ and on $[0,1]^n$ (§VI.36).

- `Chapter6/SmoothPathComponents2021`, `Chapter6/PathsInterval`, `Chapter6/GapP1_PathCategoryInterval`, `Chapter6/GapP1_UnitIntervalSections`: paths on $[0,1]$, the facts before Proposition 6.2.2 (preprint 6.2), Proposition 6.6.3 (3), (4) on $[0,1]\times X$ (§VI.37).
- `Chapter6/OrientationMainClean`, `Chapter6/OrientationConvention2021`: Theorems 6.6.13, 6.6.17, 6.6.18 with the text's hypotheses; Remark 6.6.19 in the 2021 convention.
- `Chapter6/ManifoldOrientation`, `Chapter6/SolutionSetOrientation`, `Chapter6/SolutionOrientationCharts`, `Chapter6/PerturbedOrientation`, `Chapter6/SolutionSetOpen2021`: orientations of Mathlib manifolds; the solution set as an oriented manifold; two charts; perturbed solution sets (§VI.25).
- `Chapter6/ProductSectionOrientation2021`, `…2021b–f`: the orientation $e\wedge o$ of $\hat f$ over $\mathbb{R}^1\times X$ (§VI.26).
- `Chapter6/GapP1_OrientationBundle`: the orientation bundle, the $\mathbb{Z}_2$-action, $\det_f$ (§VI.38).
- `Chapter5/CylinderPerturbation2021`: margins, admissible sections, the relative form of Theorem 5.3.10 (preprint 5.18), the cobordism data of part (1) (§VI.27).
- `Chapter6/ProperHomotopy2021`: proper oriented homotopies and the cobordism data of part (2) (§VI.27).
- `Chapter6/InvariantSingle2021`: the index parts $S_k$, the integrals, $\Psi_s$ for one admissible $s$ (§VI.27).
- `Chapter6/InvariantProductEnds2021`, `…2021b`: product ends of the cobordism, Fubini on them (§VI.27).
- `Chapter6/InvariantCobordism2021`: the cut-off identity and $S_F$ (§VI.27).
- `Chapter6/CobordismRelativeForms2021`, `…2021b`, `Chapter4/BoundaryFormsPullback2021`: the relative form on the cobordism (§VI.27).
- `Chapter6/CobordismOrientation2021`, `Chapter6/CobordismOrientationEnds2021`, `…2021b–d`: the orientation of the cobordism and its value on the ends (§VI.27).
- `Chapter6/Theorem671_2021`, `Chapter6/Theorem671Assembly2021`, `Chapter6/Theorem671Final2021`: the assembly of Theorem 6.7.1 (preprint 6.43) and the case $\partial X=\emptyset$ (§VI.27).
- `Chapter6/GapP3_K4_Contractible`, `Chapter6/GapP3_K4_IntervalHomotopy`: Proposition 6.6.3 (1) as printed in 2021, and (1′) on $[0,1]$ (§VI.10); `Chapter6/GapP3_K4_SmoothLInterval`: $L$ of Lemma 6.6.5 over $[0,1]$ (§VI.30).

Throughout, $E$ and $F$ are real Banach spaces, $T:E\to F$ is bounded, $R(T)$ is its range and $\operatorname{coker}T=F/R(T)$; $T$ is Fredholm if $\ker T$ and $\operatorname{coker}T$ are finite-dimensional, with index $\dim\ker T-\dim\operatorname{coker}T$. A *line* is a one-dimensional real space. $P:Y\to X$ is a strong bundle over an M-polyfold, both paracompact and Hausdorff, and from §VI.10 on $X$ is tame with sc-smooth bump functions and $f$ is an sc-Fredholm section with tame germ data (Definition 3.6.4 (preprint 3.51)), the reading of Chapter 5 (Part V). A strong bundle chart $c$ has base chart $\varphi$, retraction $r$ on an open set $U$ of a partial quadrant $C$, retract $O=r(U)$, and fibre projections $\rho(u)$; the *principal part* of a section is its fibre coordinate in the chart. $\mathbb{R}^1$ is the M-polyfold of the parameter $t$, with $t$ the point of $\mathbb{R}^1$ `Chapter6/OrientableNbhd · pt1`, and $(\tau,t)$ the point of $\mathbb{R}^2$ `Chapter6/OrientableNbhd · pt2`; $\kappa$ is Mathlib's smooth transition function, $0$ on $(-\infty,0]$, $1$ on $[1,\infty)$.

## VI.1 Definition 6.2.1 (preprint 6.1): the space $\operatorname{Lin}(f,x)$

§6.2 (preprint §6.1) (pp. 237–240 (preprint pp. 200–201)) works at a smooth point $x$. Since $f(x)$ is a smooth point of the fibre, there is an sc⁺-section $s$ with $s(x)=f(x)$ — in a chart, $s(y)=R(y,f(x))$ —

and $(f-s)'(x) : T_xX \to Y_x$ is sc-Fredholm. Definition 6.2.1 (preprint 6.1) collects these linearizations, perturbed by sc⁺-operators, into $\mathrm{Lin}(f,x)$; the index is the index of any member, and Proposition 6.2.2 (preprint 6.2) says that it is constant along sc-smooth paths.

**The chart point** (V6.1). A chart point of $x$ is a strong bundle chart $c$ of the atlas whose base chart contains $x$, with the level-1 point $z$ of the model over $\varphi(x)$ `Chapter6/LinSet · ChartPoint`. In it $T_xX$ is $T_zO = Dr(z)E$, the range of the derivative of the retraction `Chapter6/LinSet · ChartPoint.T`, and $Y_x$ is $K_z = \rho(z)F$, the range of the fibre projection over $z$ `Chapter6/LinSet · ChartPoint.K`; both are closed, as ranges of continuous projections, hence Banach spaces. An sc⁺-operator $T_xX \to Y_x$ is level 0 of an sc⁺-operator between these sc-subspaces `Chapter6/LinSet · ChartPoint.IsScPlusOp`.

**Definition 6.2.1 (preprint 6.1)** (V6.1). $\mathrm{Lin}(f,x)$, at the chart point $p$, is the set of the operators $(f+s)'(x)+a : T_zO \to K_z$ for an sc⁺-section $s$ with $(f+s)(x)=0$ and an sc⁺-operator $a$ `Chapter6/LinSet · ChartPoint.linSet` `Chapter6/LinSet · ChartPoint.mem_linSet_iff`. The linearization of $f+s$ at its zero $x$ is the derivative at $z$ of the principal part composed with $r$, restricted to $T_zO$ — the $\mathrm{Lin}(z)$ of Part III. Replacing the text's $s$ by $-s$ turns $f-s$ with $s(x)=f(x)$ into $f+s$ with $(f+s)(x)=0$.

**Two sections differ by an sc⁺-operator.** For two sections $s$, $t$ with $(f+s)(x)=(f+t)(x)=0$, the principal parts of $f+s$ and $f+t$ differ by that of the sc⁺-section $s-t$, whose derivative at $z$, followed by $\rho(z)$, is level 0 of an sc⁺-operator: the derivative of an sc⁺-map is sc⁺ on the shifted levels. So $\mathrm{Lin}(f,x)$ is $(f+s)'(x)$ plus all the sc⁺-operators, for any one $s$ `Chapter6/LinSet · ChartPoint.exists_isScPlusOp_linAt_eq`, and two members differ by an sc⁺-operator `Chapter6/LinSet · ChartPoint.isScPlusOp_sub_of_mem_linSet`. The paragraph before the definition puts this as "$(f-s)'(x)$ is an sc-Fredholm operator if and only if $(f-s)'(x)$ is"; it is read with $(f-t)'(x)$ in the second place (possible misprint; Findings, Chapter 6, item 41).

**Not empty.** The text makes its section locally, in a chart. The formalization uses global sc⁺-sections: $\mathrm{Lin}(f,x)$ is not empty when $P$ has sc⁺-sections through the smooth points of its fibres, one through $-f(x)$ giving a member `Chapter6/LinSet · ChartPoint.linSet_nonempty`, and bump functions give such sections `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`. This is the hidden hypothesis of the definition: an sc⁺-section through $f(x)$.

**Fredholm, of one index.** Every member is Fredholm on level 0, with the index $\mathrm{ind}(f,x)$ of the germ at $x$ (§III.49) `Chapter6/LinSet · ChartPoint.isFredholm_isIndexAt_of_mem_linSet` `Chapter6/LinSet · ChartPoint.exists_index_linSet` `Chapter6/LinSet · ChartPoint.fredholmIndex_of_mem_linSet`. The proof: $f+s$ has germ data at $x$ (an sc-Fredholm germ plus an sc⁺-germ is one), so its linearization at the zero $x$ is Fredholm of the index of the germ, through the conjugation to a basic germ (Proposition 3.6.5 (preprint 3.52), §III.30); adding an sc⁺-operator, compact on level 0, keeps the Fredholm property and the index; and the index of the germ does not depend on the choices. *The members are sc-Fredholm on every level*, as the text says, all of one index `Chapter6/LinSetFredholm · ChartPoint.exists_isScFredholm_of_mem_linSet` `Chapter6/LinSetFredholm · ChartPoint.definition_6_2_1_linSet_isScFredholm`: for $L=(f+s)'(x)+a$, the section $f+s$ is sc-Fredholm with a zero at $x$, so its linearization $T_zOoK_z$ is sc-Fredholm, between these sc-subspaces with their induced sc-structures (row V3.96); $a$ is level 0 of an sc⁺-operator; and an sc-Fredholm operator plus an sc⁺-operator is sc-Fredholm of the same index on every level (Proposition 1.1.11). The level-0 operator is $L$, so the indices agree. The determinant lines use level 0 only (gap G.64 closed; Findings, Chapter 6, items 40, 122).

**Convex** `Chapter6/LinSet · ChartPoint.convex_linSet`. $(1-\tau)(L+a)+\tau(L+b)=L+((1-\tau)a+\tau b)$, and the sc⁺-operators form a vector space. The text notes this on preprint p. 219, where Proposition 6.6.1

(preprint 6.26) needs it.

**Proposition 6.2.2 (preprint 6.2)** (V3.98). The constancy of the index along sc-smooth paths was formalized with Chapter 3 and is retold in §III.49: for paths on $\mathbb{R}$, with the text's hypotheses — tame germ data, no bump functions, the locally defined sc$^+$-sections of Definition 6.2.1 (preprint 6.1). It is not repeated here. The facts on paths recalled before it on preprint p. 201 — an M-polyfold is locally path connected, and two smooth points in one path component are joined by an sc$^+$-smooth path — are formalized, with paths on $[0, 1]$ (§VI.37; gap G.56 closed; Findings, Chapter 6, items 42, 124).

## VI.2 Determinant lines, Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4)

**$\lambda(E)$ and $\lambda^*(E)$** (V6.2). For a finite-dimensional real space $E$, $\lambda(E) = \Lambda^{\max}E$ is the top exterior power `Chapter6/DetLine · maxPow` and $\lambda^*(E)$ its dual `Chapter6/DetLine · maxPowDual`; both are lines `Chapter6/DetLine · finrank_maxPow`, $\lambda(\{0\}) = \mathbb{R}$ being the zeroth exterior power. A family $a_1, \dots, a_n$ with $n = \dim E$ has its wedge $a_1 \wedge \cdots \wedge a_n$ `Chapter6/DetLine · wedge`. Two rules carry every computation of the chapter. A change of basis multiplies the wedge by its determinant `Chapter6/DetLine · wedge_eq_det_smul`, also when only one block of a two-block wedge $a \wedge b$ is changed `Chapter6/DetLine · wedge_append_basis_left` `Chapter6/DetLine · wedge_append_basis_right`; and adding to the vectors of one block vectors of the span of the other does not change the wedge `Chapter6/DetLine · wedge_append_add_left` `Chapter6/DetLine · wedge_append_add_right`. For $v \neq 0$ in a line, $v^*$ is the functional with $v^*(v) = 1$ `Chapter6/DetLine · dualVec` `Chapter6/DetLine · dualVec_self`, and that value determines it `Chapter6/DetLine · eq_dualVec_of_apply_eq_one`. For $\dim E = \dim F$ and $\Phi : E \to F$, $\lambda(\Phi)(a_1 \wedge \cdots \wedge a_n) = \Phi a_1 \wedge \cdots \wedge \Phi a_n$ `Chapter6/DetLine · maxPowMap` `Chapter6/DetLine · maxPowMap_wedge`, with $\lambda(\Psi\Phi) = \lambda(\Psi)\lambda(\Phi)$ `Chapter6/DetLine · maxPowMap_comp`, and $\lambda(\Phi) \neq 0$ iff $\Phi$ is bijective `Chapter6/DetLine · maxPowMap_ne_zero_iff`; $\lambda^*(\Phi)$ is its dual `Chapter6/DetLine · maxPowDualMap` (pp. 241–243 (preprint pp. 203–204)).

**Definition 6.3.1 (preprint 6.3)** (V6.3). $\iota : \lambda(E^*) \to \lambda^*(E)$ is $\iota(e_1^* \wedge \cdots \wedge e_n^*)(a_1 \wedge \cdots \wedge a_n) = \det(e_i^*(a_j))$ `Chapter6/DetLine · iotaDual` `Chapter6/DetLine · iotaDual_wedge_wedge`. The formula is alternating and multilinear in both families, so it descends to the exterior powers; that is the text's well-definedness. $\iota$ is a nonzero map of lines `Chapter6/DetLine · iotaDual_ne_zero`, and for a basis with its dual basis $\iota(e_1^* \wedge \cdots \wedge e_n^*) = (e_1 \wedge \cdots \wedge e_n)^*$ `Chapter6/DetLine · iotaDual_wedge_dualBasis`, since the matrix is the identity. The text's matrix $(e_i^*(a_j))$ is the transpose of the one of Mathlib's pairing, with the same determinant (preprint p. 204).

**Proposition 6.3.2 (preprint 6.4)** (V6.4) `Chapter6/DetLine · proposition_6_3_2`. For an isomorphism $\Phi : E \to F$, $\lambda^*(\Phi) \circ \iota = \iota \circ \lambda(\Phi^*)$. Both sides, evaluated at $f_1^* \wedge \cdots \wedge f_n^*$ and $a_1 \wedge \cdots \wedge a_n$, give $\det(f_i^*(\Phi a_j))$: the left side by the definition of $\lambda^*(\Phi)$, the right side because $(\Phi^* f_i^*)(a_j) = f_i^*(\Phi a_j)$ (p. 243 (preprint p. 205)).

## VI.3 Four-term exact sequences: $\Phi_E$, Lemma 6.3.3 (preprint 6.5), Proposition 6.3.5 (preprint 6.6)

**The data of (91)** (V6.5). Let $\mathbf{E} : 0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \xrightarrow{\gamma} D \to 0$ be an exact sequence of finite-dimensional spaces `Chapter6/ExactSequence · FourTermExact`. A *choice* `Chapter6/ExactSequence · FourTermExact.Choice` consists of a complement $Z$ of $\alpha(A)$ in $B$, a complement $V$ of $\beta(B)$ in $C$, bases $a$ of $A$, $d$ of $D$, $b'$ of $Z$, and the

$\gamma$-preimages $c$ in $V$ of the basis $d$. Then $\alpha(a), b'$ is a basis of $B$, and $c, \beta(b')$ is one of $C$, because $\beta$ maps $Z$ isomorphically onto $\beta(B)$ and $\gamma$ maps $V$ isomorphically onto $D$. Formula (91) assigns to the choice the value $(\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}) \otimes (c_1 \wedge \cdots \wedge c_l \wedge \beta(b'_1) \wedge \cdots \wedge \beta(b'_{m-n}))^*$ as the image of $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, with the text's two ordering conventions `Chapter6/ExactSequence · FourTermExact.Choice.value`. $\Phi_E : \lambda(A) \otimes \lambda^*(D) \to \lambda(B) \otimes \lambda^*(C)$ is the linear map of lines defined by one fixed choice `Chapter6/ExactSequence · FourTermExact.canonicalChoice` `Chapter6/ExactSequence · FourTermExact.phi`; it is nonzero `Chapter6/ExactSequence · FourTermExact.phi_ne_zero`, hence an isomorphism `Chapter6/ExactSequence · FourTermExact.phiEquiv` (preprint p. 206).

**Lemma 6.3.3 (preprint 6.5)** (V6.6) `Chapter6/ExactSequence · FourTermExact.lemma_6_3_3`. The text: $\Phi_E$ does not depend on the choices. The formalization proves the strongest form: $\Phi_E$, defined from one choice, maps the source element $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$ of *any other* choice to the right-hand side of (91) computed with that choice. This is the form every later computation uses — $\gamma_T^P$ is always evaluated in bases adapted to the problem at hand (Findings, Chapter 6, item 1). The proof is §6.8.1 (preprint §6.7.1) (§VI.28).

**Proposition 6.3.5 (preprint 6.6)** (V6.7). With the contraction $\bar{\iota}(v \otimes b \otimes \varphi) = \varphi(v)\, b$ `Chapter6/ExactSequence · contractDual`, (92) defines $\Psi_E(c \otimes h) = \bar{\iota}(c \otimes \Phi_E(h))$ `Chapter6/ExactSequence · FourTermExact.psi`. The proposition `Chapter6/ExactSequence · FourTermExact.proposition_6_3_5`: $\Psi_E$ maps the wedge of the basis $c, \beta(b')$ of $C$, tensored with $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, to $\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}$. By Lemma 6.3.3 (preprint 6.5) in its strong form, $\Phi_E$ of the source is the value of (91) for this choice, and the dual wedge evaluated on the wedge is 1. The wedge in $C$ is taken in the order of (91); the statement exchanges the names of its two blocks (possible misprint; Findings, Chapter 6, item 2) (preprint p. 207).

## VI.4 Determinants of Fredholm operators, good left-projections, and Lemma 6.10 (preprint)

**Definition 6.4.1 (preprint 6.7)** (V6.8). $\det(T) = \lambda(\ker T) \otimes \lambda^*(F/R(T))$ `Chapter6/FredholmDet · linDet` `Chapter6/FredholmDet · fredholmDet`, a line for $T$ Fredholm `Chapter6/FredholmDet · finrank_fredholmDet`. Equal operators have identified determinants `Chapter6/FredholmDet · linDetEquivOfEq`: $PQT$ and $PT$ are the same operator when $P \le Q$, but not the same expression, and Proposition 6.4.11 (preprint 6.12) compares their determinants. The alternative definition $\lambda(\ker T) \otimes \lambda(\mathrm{coker}(T)^*)$ mentioned after the definition is naturally isomorphic to $\det(T)$ through $\iota$ (§VI.34; gap G.57 closed) (preprint p. 208).

**Definition 6.4.2 (preprint 6.8)** (V6.9). An orientation of $T$ is a ray of $\det(T)$ `Chapter6/FredholmDet · FredholmOrientation`; there is one `Chapter6/FredholmDet · nonempty_fredholmOrientation`, and exactly two `Chapter6/FredholmDet · exists_two_fredholmOrientation` (preprint p. 208).

**Definition 6.4.3 (preprint 6.9) and the facts after it** (V6.10). A *good left-projection* for $T$ is a bounded projection $P : F \to F$ with $\dim F/R(P) < \infty$ and $R(PT) = R(P)$ `Chapter6/GoodProjections · IsGoodLeftProj`; $\Pi_T$ is their set `Chapter6/GoodProjections · goodLeftProjs`. $P \le Q$ iff $P = PQ = QP$ `Chapter6/GoodProjections · ProjLE`, a partial order `Chapter6/GoodProjections · ProjLE.refl` `Chapter6/GoodProjections · ProjLE.antisymm` `Chapter6/GoodProjections · ProjLE.trans`. A good left-projection is Fredholm of index 0 — $\ker P$ is a complement of $R(P)$, of dimension $\dim F/R(P)$ `Chapter6/GoodProjections · IsGoodLeftProj.isFredholm` `Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_eq_zero` — so $PT$ is Fredholm with $\mathrm{ind}(PT) = \mathrm{ind}(T)$ when $T$ is, by the index of a composite `Chapter6/GoodProjections · IsGoodLeftProj.isFredholm_comp` `Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_comp`. For $P, P' \in \Pi_T$, $R(P) \cap R(P')$ has finite codimen-

sion Chapter6/GoodProjections · IsGoodLeftProj.finiteDimensional_quotient_inf. And $\Pi_T \neq \emptyset$ for $T$ Fredholm Chapter6/GoodProjections · exists_isGoodLeftProj — the projection onto $R(T)$ along a finite-dimensional complement is one — which the text uses from Proposition 6.4.10 (preprint Lemma 6.11) on without stating it (Findings, Chapter 6, item 3) (preprint p. 208).

**Lemma 6.10 (preprint)** (V6.11) Chapter6/GoodProjections · lemma_6_10_preprint. *For $P, P' \in \Pi_T$ there is $P'' \in \Pi_T$ with $P'' \leq P$ and $P'' \leq P'$*, for any bounded $T$. The text's $P''$ projects onto $R(P) \cap R(P')$ along a complement assembled from $\ker P$, $\ker P'$ and a further piece, and $P'' = P''P$ needs $\ker P \subseteq \ker P''$. As we read it, this does not hold in general: on $\mathbb{R}^2$ with $T = 1$, let $P$ project onto the $x$-axis along the $y$-axis and $P'$ along the diagonal; $R(P) \cap R(P')$ is the $x$-axis, and a projection onto it below both would equal both Chapter6/GoodProjections · lemma_6_10_text_reading_preprint. The lemma holds with a smaller range. *Step 1.* $K = \ker P + \ker P'$ is finite-dimensional, each kernel having the dimension of a finite-dimensional quotient. *Step 2.* $K$ has a closed complement $N$ (finite-dimensional subspaces are complemented, by Hahn–Banach), and $R'' = R(P) \cap R(P') \cap N$ is closed, of finite codimension, and meets $K$ only in $0$. *Step 3.* $R''$ has a complement $C \supseteq K$, topological because the quotient by $R''$ is finite-dimensional; $P''$ is the projection onto $R''$ along $C$. *Step 4.* $\ker P \subseteq C = \ker P''$ gives $P''P = P''$, and $R'' \subseteq R(P)$ gives $PP'' = P''$; the same for $P'$. *Step 5.* $R(P''T) = P''(R(PT)) = P''(R(P)) = R''$, since $P''$ is the identity on $R'' \subseteq R(P)$. Nothing uses that $T$ is Fredholm. The 2017 version drops the lemma and remarks that a common lower bound need not exist; with the order of Definition 6.4.3 (preprint 6.9), used by both versions, it always exists, as we read it, so the preprint's route to Lemma 6.5.9 (preprint 6.25) is sound and is the one formalized (Findings, Chapter 6, item 4) (preprint pp. 208–209).

## VI.5 Proposition 6.4.10 (preprint Lemma 6.11), $\gamma_T^P$, Proposition 6.4.11 (preprint 6.12) and Corollary 6.4.12 (preprint 6.13)

**Proposition 6.4.10 (preprint Lemma 6.11)** (V6.12). For $P \in \Pi_T$, the sequence $\mathbf{E}(T,P) : 0 \to \ker T \to \ker(PT) \to F/R(P) \to \operatorname{coker} T \to 0$, with the inclusion Chapter6/ProjectionSequence · projSeqJ, $x \mapsto Tx + R(P)$ Chapter6/ProjectionSequence · projSeqPhi and $y + R(P) \mapsto (I-P)y + R(T)$ Chapter6/ProjectionSequence · projSeqPi, is exact Chapter6/ProjectionSequence · proposition_6_4_10, a four-term exact sequence of §VI.3 Chapter6/ProjectionSequence · projSeq. The four checks use only $P^2 = P$ and $R(PT) = R(P)$. At $\ker(PT)$: if $PTx = 0$ and $Tx \in R(P)$, then $Tx \in R(P) \cap \ker P = 0$. At $F/R(P)$: the last map is well defined because $(I-P)R(P) = 0$; it kills $[Tx]$ for $x \in \ker(PT)$, since $(I-P)Tx = Tx$; and if $(I-P)y = Tx$, then $PTx = 0$ and $y - Tx = Py \in R(P)$. Onto: $Py = PTe$ for some $e$, and then $(I-P)(y-Te) = y - Te$, which has the class of $y$ in $\operatorname{coker} T$ (preprint p. 209).

**The isomorphism $\gamma_T^P$** (V6.13). $\gamma_T^P : \det(T) \to \det(PT)$ is $\Phi_{\mathbf{E}(T,P)}$ followed by the identification of $\lambda^*(F/R(P))$ with $\lambda^*(F/R(PT))$ Chapter6/ProjectionSequence · gammaProj Chapter6/ProjectionSequence · gammaProj_tmul — the text's "recalling $R(P) = R(PT)$" (Findings, Chapter 6, item 5). By Lemma 6.3.3 (preprint 6.5) it is given on wedges by (91) for any choice Chapter6/ProjectionSequence · gammaProj_wedge_tmul (preprint p. 210).

**Proposition 6.4.11 (preprint 6.12)** (V6.14). *For $P \leq Q$ in $\Pi_T$, $QT$ is Fredholm, $P \in \Pi_{QT}$, and $\gamma_{QT}^P \circ \gamma_T^Q = \gamma_T^P$*, up to the identification of $\det(P(QT))$ with $\det(PT)$ Chapter6/ProjectionSequence · proposition_6_4_11. $P \in \Pi_{QT}$ because $R(PQT) = R(PT) = R(P)$ Chapter6/ProjectionSequence · IsGoodLeftProj.of_projLE. The identity is a computation in adapted bases Chapter6/ProjectionSequence · comp_comp_eq_of_projLE, the proof of §6.8.2 (preprint §6.7.2), retold in §VI.29 (preprint p. 211).

**Corollary 6.4.12 (preprint 6.13)** (V6.15) `Chapter6/ProjectionSequence · corollary_6_4_12`. For $P \le S$ and $P \le R$ in $\Pi_T$, $(\gamma^P_{RT})^{-1} \circ \gamma^P_{ST} = \gamma^R_T \circ (\gamma^S_T)^{-1}$. Proposition 6.4.11 (preprint 6.12) twice gives $\gamma^P_{ST}\gamma^S_T = \gamma^P_T = \gamma^P_{RT}\gamma^R_T$; rearranging gives the claim. So the left side, the comparison of two local frames through a common lower bound, does not depend on the lower bound (preprint p. 211).

## VI.6 The stabilization: Lemma 6.4.14 (preprint 6.14), formula (94), Proposition 6.4.15 (preprint 6.15), Definition 6.4.16 (preprint 6.16)

**Lemma 6.4.14 (preprint 6.14)** (V6.16). For $T$ Fredholm and $\varphi : \mathbb{R}^n \to F$ with $T_\varphi(e,r) = Te + \varphi r$ onto, the sequence $0 \to \ker T \xrightarrow{j} \ker T_\varphi \xrightarrow{\pi} \mathbb{R}^n \xrightarrow{[\varphi]} F/R(T) \to 0$, with $j(e) = (e,0)$ `Chapter6/Stabilization · stabJ`, $\pi(e,r) = r$ `Chapter6/Stabilization · stabPi` and $[\varphi](r) = \varphi(r) + R(T)$ `Chapter6/Stabilization · stabClass`, is exact `Chapter6/Stabilization · lemma_6_4_14` `Chapter6/Stabilization · stabSeq`. $j$ is injective; $\ker\pi \cap \ker T_\varphi$ is $\{(e,0) \mid Te = 0\} = j(\ker T)$; for $(e,r) \in \ker T_\varphi$, $\varphi r = -Te \in R(T)$, and conversely $\varphi r = -Te$ puts $(e,r)$ in $\ker T_\varphi$; $[\varphi]$ is onto because $T_\varphi$ is. A stabilizing $\varphi$ exists for every Fredholm $T$ `Chapter6/Stabilization · exists_surjective_coprod`: send a basis of $\mathbb{R}^n$ to representatives of a basis of the cokernel. The proof writes $j(e) = (0,e)$, read as $(e,0)$ (possible misprint; Findings, Chapter 6, item 8) (pp. 251–256 (preprint pp. 211–212)).

**Formula (94)** (V6.17). $\psi : \lambda^*(\mathbb{R}^n) \to \lambda^*(\{0\})$ maps $(e_1 \wedge \cdots \wedge e_n)^*$ to $1^*$ for the standard basis `Chapter6/Stabilization · stdWedge` `Chapter6/Stabilization · stabOne` `Chapter6/Stabilization · stabPsi`. Then $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c_1,\dots,c_n]^{-1}\, 1^*$ for every basis `Chapter6/Stabilization · stabPsi_dualVec`: $c_1 \wedge \cdots \wedge c_n = \det[c]\, e_1 \wedge \cdots \wedge e_n$, and the dual vectors scale inversely (p. 256 (preprint p. 212)).

**Proposition 6.4.15 (preprint 6.15)** (V6.18). Since $T_\varphi$ is onto, $\det(T_\varphi) = \lambda(\ker T_\varphi) \otimes \lambda^*(\{0\})$, and $\iota_\varphi = (1 \otimes \psi) \circ \Phi_{\mathbf{E}}$, for the sequence $\mathbf{E}$ of Lemma 6.4.14 (preprint 6.14), is an isomorphism $\det(T) \to \det(T_\varphi)$ `Chapter6/Stabilization · stabIota` `Chapter6/Stabilization · stabIota_bijective`. The formula `Chapter6/Stabilization · proposition_6_4_15`: for a basis $a$ of $\ker T$, vectors $\bar c_1,\dots,\bar c_m$ with $[\varphi](\bar c_i)$ a basis of $F/R(T)$, and $\bar b_i = (b_i, r_i)$ completing $(a,0)$ to a basis of $\ker T_\varphi$, $\iota_\varphi(h) = \det(\bar c, r)^{-1}\,((a_1,0) \wedge \cdots \wedge (a_k,0) \wedge \bar b_1 \wedge \cdots \wedge \bar b_l) \otimes 1^*$. These data are a choice of (91) for $\mathbf{E}$ — $Z$ the span of the $\bar b_i$, $V$ the span of the $\bar c_j$ — so Lemma 6.3.3 (preprint 6.5) gives $\Phi_{\mathbf{E}}(h) = ((a,0) \wedge \bar b) \otimes (\bar c \wedge r)^*$, and (94) turns the second factor into $\det(\bar c, r)^{-1} 1^*$. To form $\det(\bar c, r)$ one needs $m + l = n$: the statement takes it as a hypothesis, and it is proved to hold always `Chapter6/Stabilization · stab_card_eq`; that the $r_i$ and $\bar c_j$ form a basis of $\mathbb{R}^n$, which the text notices, is proved — the $r_i$ span $\pi(\ker T_\varphi) = \ker[\varphi]$, and the $\bar c_j$ a complement of it `Chapter6/Stabilization · isCompl_range_stabPi_span`. The statement calls $\psi$ "$\varphi$" (possible misprint). The identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after the proposition are formalized separately (§VI.34; gap G.58 closed); the statements here use only (94) (Findings, Chapter 6, items 9, 10) (preprint p. 213).

**Definition 6.4.16 (preprint 6.16)** (V6.19). For Fredholm $T$, $S$, the operator $T \oplus S$ has $\ker(T \oplus S) \cong \ker T \times \ker S$ and $\operatorname{coker}(T \oplus S) \cong \operatorname{coker} T \times \operatorname{coker} S$, through the inclusions of the factors `Chapter6/DirectSumDet · kerInl` `Chapter6/DirectSumDet · kerInr` `Chapter6/DirectSumDet · cokerInl` `Chapter6/DirectSumDet · cokerInr` `Chapter6/DirectSumDet · kerProdEquiv` `Chapter6/DirectSumDet · cokerProdEquiv`; the dimensions add `Chapter6/DirectSumDet · finrank_ker_prodMap` `Chapter6/DirectSumDet · finrank_coker_prodMap`, so $T \oplus S$ is Fredholm `Chapter6/DirectSumDet · isFredholm_prodMap`. The map $\det(T) \otimes \det(S) \to \det(T \oplus S)$, $h \otimes h' \mapsto (a \wedge a') \otimes (d \wedge d')^*$, is defined by the formula in one choice of bases `Chapter6/DirectSumDet · directSumVec` `Chapter6/DirectSumDet · directSumDet`, satisfies it for every choice `Chapter6/`

DirectSumDet · directSumVec_eq_smul Chapter6/DirectSumDet · definition_6_4_16 — a change of bases multiplies source and target by the same product of determinants, the block wedges by the block determinants — and is an isomorphism Chapter6/DirectSumDet · directSumDet_bijective. "Natural" is read as the formula for every choice, as in Lemma 6.3.3 (preprint 6.5) (Findings, Chapter 6, item 11) (p. 258 (preprint p. 214)).

## VI.7 Local determinant bundles: Lemmas 6.5.1–6.5.3 (preprint 6.17–6.19), Proposition 6.5.4 (preprint 6.20), Lemma 6.5.5 (preprint 6.21)

**Lemma 6.5.1 (preprint 6.17)** (V6.20). The Fredholm operators form a subspace of $L(E,F)$ Chapter6/LocalDetBundle · FredholmOp. *For a bounded $T$, $P \in \Pi_T$ and a topological complement $Y$ of* $\ker(PT)$*, there is $\varepsilon > 0$ such that for $\|S-T\| < \varepsilon$, $P \in \Pi_S$ and $PS|_Y : Y \to R(P)$ is a topological isomorphism* Chapter6/LocalDetBundle · lemma_6_5_1. *Step 1.* $PT|_Y$ is injective, since $Y \cap \ker(PT) = 0$, and onto $R(P)$, since $PT(E) = PT(Y)$ and $R(PT) = R(P)$; as a bijection between the Banach spaces $Y$ and $R(P)$ (both closed) it is an isomorphism by the open mapping theorem. *Step 2.* $\|PS|_Y - PT|_Y\| \le \|P\|\,\|S-T\|$ Chapter6/LocalDetBundle · projRestrict, and the invertible operators form an open set. *Step 3.* $R(PS) \supseteq PS(Y) = R(P)$, so $P \in \Pi_S$. That $T$ is Fredholm is not used (Findings, Chapter 6, item 20); a topological complement is an algebraic complement with a continuous projection (p. 258 (preprint p. 214)).

**Lemma 6.5.2 (preprint 6.18)** (V6.21). A *local determinant chart* is the data $(T,P,Y,\varepsilon)$ of Lemma 6.5.1 (preprint 6.17) with its conclusions Chapter6/LocalDetBundle · LocalDetChart. For $k \in \ker(PT)$, $y(k,S) = -(PS|_Y)^{-1}PSk$ Chapter6/LocalDetBundle · LocalDetChart.ySol is continuous in $(k,S)$, inversion being continuous, and $k + y(k,S) \in \ker(PS)$. The map $(S,k) \mapsto (S, k+y(k,S))$ Chapter6/LocalDetBundle · LocalDetChart.kerMap is onto $\bigcup_{\|S-T\|<\varepsilon}\{S\} \times \ker(PS)$, inverted by the continuous projection onto $\ker(PT)$ along $Y$ Chapter6/LocalDetBundle · LocalDetChart.kerProj: for $w \in \ker(PS)$ write $w = k + y$ with $y \in Y$; then $PSy = -PSk$, so $y = y(k,S)$ Chapter6/LocalDetBundle · LocalDetChart.lemma_6_5_2. "Is a trivial vector bundle" is stated as this trivialization and its inverse (Findings, Chapter 6, item 21) (preprint p. 215).

**Lemma 6.5.3 (preprint 6.19) and Proposition 6.5.4 (preprint 6.20)** (V6.22). For a basis $k$ of $\ker(PT)$, $k_i(S) = k_i + y(k_i,S)$ is a continuous frame of the kernels Chapter6/LocalDetBundle · LocalDetChart.kerFrame, and with a constant basis $e$ of $F/R(P) = F/R(PS)$, $\omega(S) = (k_1(S) \wedge \cdots \wedge k_n(S)) \otimes (e_1 \wedge \cdots \wedge e_l)^*$ Chapter6/LocalDetBundle · LocalDetChart.frame is a nowhere-zero frame Chapter6/LocalDetBundle · LocalDetChart.frame_ne_zero of $\mathrm{DET}(T,P,\varepsilon) = \bigcup_S \det(PS)$. The frames of two charts with the same $P$ differ by a continuous nowhere-zero factor Chapter6/LocalDetBundle · proposition_6_5_4: the determinant of the matrix expressing one kernel frame in the other, continuous because both frames are continuous in $S$, times the constant change of the bases of $F/R(P)$. Lemma 6.5.3 (preprint 6.19) — exterior powers, duals, sums and tensor products of topological vector bundles — is applied here only to trivial bundles, and Proposition 6.5.4 (preprint 6.20) is proved with the explicit frames; the lemma in general is §VI.35 (gap G.59 closed); "in a natural way" is read as this relation between frames (Findings, Chapter 6, item 22) (preprint p. 216).

**Lemma 6.5.5 (preprint 6.21)** (V6.23). *For $Q \le P$, $\gamma^Q_{PS}$ maps the $P$-frame at $S$ to a continuous nowhere-zero multiple of the $Q$-frame, on the common ball of any two local determinant charts* Chapter6/LocalDetBundle · LocalDetChart.gammaProj_frame_eq Chapter6/LocalDetBundle · lemma_6_5_5. The text's sketch — extend the kernel sections of $PS$ to sections spanning $\ker(QS)$ — is the proof. For a basis $f_1, \ldots$ of $R(P) \cap \ker Q$, $z_m(S) = (PS|_Y)^{-1} f_m$ is continuous and $PSz_m(S) = f_m$; the $k_i(S)$ and $z_m(S)$ span $\ker(QPS) = \ker(QS)$; and in the choice of (91) for $\mathbf{E}(PS,Q)$ these vectors go with the *constant* classes

$[f_m]$. So $\gamma^Q_{PS}(\omega_P(S))$ is $(k(S) \wedge z(S)) \otimes (e \wedge [f])^*$, continuous in $S$, and its ratio to the $Q$-frame of the other chart is the determinant of a continuous change of basis. The lemma is stated for two charts with different centres and radii, the form Lemma 6.5.9 (preprint 6.25) uses (Findings, Chapter 6, item 23) (pp. 259–261 (preprint pp. 216–217)).

## VI.8 $\mathrm{DET}(E, F)$: Definition 6.5.6 (preprint 6.22), Propositions 6.5.7 and 6.5.8 (preprint 6.23 and 6.24), Lemma 6.5.9 (preprint 6.25)

**The prebundle** (V6.24). The fibre over a Fredholm $S$ is $\det(S)$ `Chapter6/DetBundle · DetFiber`. A local determinant chart gives the pretrivialization $(S, h) \mapsto (S, \omega(S)^*(\gamma^P_S h))$ over its ball, with model fibre $\mathbb{R}$ `Chapter6/DetBundle · LocalDetChart.pretriv Chapter6/DetBundle · LocalDetChart.coord` — the text's $\gamma_{(T,P,\varepsilon)}$ read in the frame of Proposition 6.5.4 (preprint 6.20) (Findings, Chapter 6, item 24). Every Fredholm operator is the centre of one `Chapter6/DetBundle · exists_localDetChart`: $\Pi_T \neq \emptyset$ (§VI.4) and the finite-dimensional $\ker(PT)$ has a topological complement. The charts form a Mathlib vector prebundle `Chapter6/DetBundle · detPrebundle`, which needs exactly that the coordinate changes be continuous, Lemma 6.5.9 (preprint 6.25) below; its topology and bundle structure are those of $\mathrm{DET}(E, F)$ (preprint p. 217).

**Lemma 6.5.9 (preprint 6.25)** (V6.28), proved first. *The transition between two charts is $(S, t) \mapsto (S, g(S)t)$ with $g$ continuous and nowhere zero on the common ball* `Chapter6/DetBundle · lemma_6_5_9_local Chapter6/DetBundle · lemma_6_5_9`. The proof is the text's: a common lower bound $Q \leq P, P'$ (Lemma 6.10 (preprint), §VI.4); Corollary 6.4.12 (preprint 6.13) to pass from $\gamma^P$ to $\gamma^{P'}$ through $Q$; and Lemma 6.5.5 (preprint 6.21) for each chart, which turns both frames into multiples of $Q$-frames (preprint p. 218).

**Definition 6.5.6 (preprint 6.22), Propositions 6.5.7 and 6.5.8 (preprint 6.23 and 6.24).** $\mathcal{B}$ is the collection of sets lying over the ball of some chart whose image in ball $\times \mathbb{R}$ is open (V6.25) `Chapter6/DetBundle · detBasis`. It is a basis of the topology of the prebundle (V6.26) `Chapter6/DetBundle · proposition_6_5_7`: its members are open, the charts being homeomorphisms onto ball $\times \mathbb{R}$, and every open set is the union of its intersections with chart domains, which are members; so the topology $\mathcal{B}$ generates is that one. $\mathrm{DET}(E, F)$ is a topological vector bundle with model fibre $\mathbb{R}$, and every $\gamma_{(T,P,\varepsilon)}$, read in its frame, is a trivialization (V6.27) `Chapter6/DetBundle · proposition_6_5_8` (preprint p. 218). The remark that $\mathrm{DET}(E, F)$ is not orientable, which does not hold as we read it for finite-dimensional $E$, $F$, and the classical case at the opening of §6.6 (preprint §6.5) are §VI.34 (gap G.60 closed; Findings, Chapter 6, items 26, 135).

## VI.9 Orientations of line bundles, Proposition 6.6.1 (preprint 6.26), Definition 6.6.2 (preprint 6.27)

**Orientations of line bundles** (V6.29). An orientation of a topological line bundle is a choice of rays of the fibres given near every point by a continuous nowhere-zero section `Chapter6/LineBundleOrientation · IsLineOrientation Chapter6/LineBundleOrientation · LineOrientation`. Two orientations that agree on one fibre agree over a preconnected base `Chapter6/LineBundleOrientation · LineOrientation.eq_of_apply_eq`: the set where they agree is open and closed. The rays of the fibres form a two-sheeted covering `Chapter6/LineBundleOrientation · OrCover Chapter6/LineBundleOrientation · isCoveringMap_orCover`, and an orientation is a continuous section of it; over a simply connected, locally path-connected base the lifting criterion extends every orientation of one fibre `Chapter6/LineBundleOrientation · exists_lineOrientation_apply_eq`, and over a nonempty such base there are exactly two orientations `Chapter6/`

`LineBundleOrientation · exists_two_lineOrientation`. The text gives "since the base space is contractible" as the reason, without the theorem it needs (Findings, Chapter 6, item 25) (p. 263 (preprint p. 219)).

**Proposition 6.6.1 (preprint 6.26)** (V6.30). Over a nonempty convex set $C$ of Fredholm operators `Chapter6/DetBundle · fredholmIncl`, the restriction $\mathrm{DET}(E,F)|_C$ `Chapter6/DetBundle · DetOn` is a topological line bundle with exactly two orientations `Chapter6/DetBundle · proposition_6_6_1`: a nonempty convex set of a normed space is contractible, hence simply connected, and locally path-connected. The proposition is stated for every nonempty convex $C$; $\mathrm{Lin}(f,x)$ is one (§VI.1) (preprint p. 219).

**Definition 6.6.2 (preprint 6.27)** (V6.31). At a chart point the members of $\mathrm{Lin}(f,x)$ are Fredholm `Chapter6/LinOrientation · ChartPoint.isFredholm_of_mem_linSet`; $\mathrm{DET}(f,x)$ is the pull-back of $\mathrm{DET}(T_zO,K_z)$ to $\mathrm{Lin}(f,x)$ `Chapter6/LinOrientation · ChartPoint.detLin`, and $\mathrm{Or}_x^f$ is the set of its orientations `Chapter6/LinOrientation · ChartPoint.Or`, with exactly two elements `Chapter6/LinOrientation · ChartPoint.exists_two_or`. For each $L \in \mathrm{Lin}(f,x)$, $o \mapsto o(L)$ is a bijection onto the two orientations of $\det(L)$ `Chapter6/LinOrientation · ChartPoint.orEquivRay`: injective because an orientation is determined by one value over the connected base, onto because each ray extends. The category $\mathcal{O}_f$ is read through its objects, the two-element sets, with bijections as morphisms; between two two-element sets there are bijections `Chapter6/OrientationFunctor · nonempty_equiv_of_two`, each determined by one value `Chapter6/OrientationFunctor · equiv_eq_of_apply_eq_of_two`. No negation is set up on the tensor products, so $\{o_x, -o_x\}$ is stated as "exactly two elements" (Findings, Chapter 6, item 44) (preprint pp. 219–220). The comparison of $\mathrm{Or}_x^f$ in two charts is §VI.21.

## VI.10 Proposition 6.6.3 (preprint 6.28): the neighbourhoods of the propagation

The text: every smooth point $x$ has an open neighbourhood $U$ which (1) is sc-smoothly contractible, (2) has $\{y \in \mathrm{cl}(U) \mid f(y) = 0\}$ compact, (3) carries, for every sc-smooth path $\varphi : [0,1] \to U$, an sc⁺-section $s : U \times [0,1] \to Y$ with $s(\varphi(t),t) = f(\varphi(t))$, and (4) carries sc⁺-sections $q : U \times [0,1] \to Y$ through every smooth point of every fibre over $U_\infty$, at every $t_0$. The proof takes local strong bundle coordinates, $P : K \to O$ with $x = 0$. This is the preprint's wording of (1). The 2021 text (p. 263) prints (1) with a second neighbourhood: $U' \subset U$ is sc-smoothly contractible in $U$ to the point $x$; that form is proved at the end of this section.

**The statement formalized** (V6.32) `Chapter6/OrientableNbhd · proposition_6_6_3`. *With sc-smooth bump functions, every smooth $x$ has open $U \subseteq W$ in the domain of one chart of the atlas with* (1′) *any two sc-smooth paths in $U$ with the same end points are sc-smoothly homotopic in $W$ with the end points fixed* `Chapter6/OrientableNbhd · IsScHomotopyRel`; (2) $\{y \in \mathrm{cl}(U) \mid f(y) = 0\}$ *is compact*; (3) *for every sc-smooth family $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n) \subseteq W$ there is an sc⁺-section $S$ of $\pi^*P \to \mathbb{R}^n \times X$ with $(f+S)(q,\Phi(q)) = 0$ for $q$ near $[0,1]^n$, depending on $q \in [0,1]^n$ only through $\Phi(q)$*; (4) *$\pi^*P$ has sc⁺-sections through all smooth fibre points*. A path "in $U$" is one with $\gamma([0,1]) \subseteq U$.

**The neighbourhoods.** Take a chart $c$ of the atlas at $x$, $u_0 = \varphi(x)$, the retraction $r$ on $U_c \subseteq C$ with $O = r(U_c)$. Choose $\delta$ with $B_\delta \cap C \subseteq U_c$ (the domain of the retraction is relatively open); $\delta_2$ with $r(B_{\delta_2} \cap U_c) \subseteq B_\delta$ (continuity of $r$ at $u_0$ on level 0, $r(u_0) = u_0$); a neighbourhood $N$ of $x$ whose closure meets the zeros of $f$ in a compact set `Chapter6/OrientableNbhd · exists_nhds_isCompact_zeros` — at a zero by the local compactness of Theorem 5.2.3 (preprint 5.6) (§V.7), elsewhere because the zero set is closed and misses a neighbourhood of $x$; a closed neighbourhood $T \subseteq N$ of $x$ and $\delta_3$ with $\varphi^{-1}(B_{\delta_3} \cap O) \subseteq T$.

With $\delta' = \min(\delta, \delta_2, \delta_3)$, $W = \varphi^{-1}(B(u_0, \delta) \cap O)$ and $U = \varphi^{-1}(B(u_0, \delta') \cap O)$.

**(1′).** Let $\gamma_0, \gamma_1$ be sc-smooth paths in $U$ with the same end points. Since paths are maps on $\mathbb{R}$ that lie in $U$ only for $t \in [0,1]$, first take a $C^\infty$ map $\sigma : \mathbb{R} \to \mathbb{R}$ equal to the identity near $[0,1]$ with values in the open set $\gamma_0^{-1}(U) \cap \gamma_1^{-1}(U)$ `Chapter6/OrientableNbhd · exists_contDiff_eq_id_near_Icc_mapsTo`. Then $\Phi(\tau, t) = \varphi^{-1}(r((1-\kappa(\tau))\,\varphi(\gamma_0(\sigma(t))) + \kappa(\tau)\,\varphi(\gamma_1(\sigma(t)))))$. The convex combination lies in the convex set $B_{\delta'} \cap C$, hence in $U_c$; $r$ maps it into $B_\delta \cap O$, so $\Phi$ takes values in $W$; it is sc-smooth as a composite of sc-smooth maps in the chart; at $\tau = 0, 1$ it is $\gamma_0, \gamma_1$ on $[0,1]$, because $\sigma$ is the identity there and $r$ fixes $O$; and the end points stay fixed, because $\gamma_0$ and $\gamma_1$ agree there. The text's contraction $\widehat{r}(t, y) = r(ty)$ contracts $r(V)$ for a convex $V$; for $U = r(V')$ it needs $r(V') \subseteq V'$, which is not stated, and a convex domain of the retraction is not available in general. As we read it, the preprint's (1) is not established for $U$ (Findings, Chapter 6, item 45); the 2021 wording of (1) is proved below. What Theorem 6.6.11 (preprint 6.35) uses is (1′).

**(2).** $\mathrm{cl}(U) \subseteq T \subseteq \mathrm{cl}(N)$, since $U \subseteq \varphi^{-1}(B_{\delta_3} \cap O) \subseteq T$ and $T$ is closed; so the zeros in $\mathrm{cl}(U)$ are a closed subset of a compact set.

**(3), sections along families** `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_family`. This is the text's $s(y) = R(y, f(x))$ made to depend on the parameter. An sc-smooth map on the finite-dimensional $\mathbb{R}^n$ takes values in the smooth points, and the principal part $q \mapsto \widehat{f}(\Phi(q))$ lifts to a $C^\infty$ map into the level-1 fibre model. In the chart, the section over $(q, y)$ whose fibre coordinate is the projection onto $K_{\varphi(y)}$ of that lift of $-\widehat{f}(\Phi(q))$ — after a reparametrization of the parameter, equal to the identity near $[0,1]^n$, that keeps the family in the chart domain — is an sc$^+$-section of the pull-back over the chart domain, and at $y = \Phi(q)$ it is $-f(\Phi(q))$. Cut off by an sc-smooth function on $X$ supported in the chart domain and equal to 1 near a compact set containing $\Phi([0,1]^n)$ `Chapter6/OrientableNbhd · exists_isScSmoothMap_eq_one_nhdsSet`, it becomes a global sc$^+$-section, with $f + S = 0$ at the points $(q, \Phi(q))$ where the cut-off is 1 `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_mapsTo_of_bump`. For $q \in [0,1]^n$ it is the projection of $-\widehat{f}(\Phi(q))$, so it depends on $q$ only through $\Phi(q)$. The text's (3) is for paths; it is stated for families because the homotopies of Theorem 6.6.11 (preprint 6.35) need it (Findings, Chapter 6, item 46).

**(4)** `Chapter6/OrientableNbhd · hasScPlusSectionsThrough_pullback_rn_of_bump`. Products of bump functions on $\mathbb{R}^n$ and on $X$ are bump functions on $\mathbb{R}^n \times X$, and they give the sections. $\mathbb{R}^n \times X$ `Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod` and the total space of $\pi^*P$ `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` are paracompact, as the project's M-polyfolds and strong bundles must be. *Hidden hypotheses:* bump functions, for the global sections of (3) and (4), where the text assumes partitions of unity from §6.3 (preprint §6.2) on. The construction uses neither that $x$ is smooth nor that $X$ is tame (pp. 263–264 (preprint pp. 220–221)). (3) and (4) also hold as printed, with sections over the M-polyfolds $[0,1] \times X$ and $[0,1]^n \times X$ (§VI.37).

**(1), as printed in 2021** `Chapter6/GapP3_K4_Contractible · proposition_6_6_3_one`. *For every smooth $x$ and every open $U \ni x$ there are an open $U' \ni x$ in $U$ and an sc-smooth $H$ on the product M-polyfold $[0,1] \times X$ with values in $U$ on $[0,1] \times U'$, $H(0,\cdot) = x$, $H(1,\cdot) = <built-in function id>$ on $U'$, and $H(t,x) = x$* `Chapter6/GapP3_K4_Contractible · IsScContraction` `Chapter6/GapP3_K4_Contractible · exists_isScContraction`. In a chart $\varphi$ with retraction $r$ and $u_0 = \varphi(x)$, take $H(t,y) = \varphi^{-}1r(u_0 + t(\varphi(y) - u_0))$ and $U' = \varphi^{-}1(B_\varepsilon(u_0))$, the ball chosen so that $B_\varepsilon \cap C$ lies in the domain of $r$ and $\varphi^{-}1r(B_\varepsilon \cap C) \subseteq U$. The segment stays in the convex set $B_\varepsilon \cap C$; at $t = 1$ it is $\varphi(y)$, which $r$ fixes; at $t = 0$ it is $u_0$. The smoothness of $x$ is used: the constant $u_0$ must be a smooth point of $E$ for $H$ to be sc-smooth. The text's $\widehat{r}(t,y) = r(ty)$ is this map for $x = 0$. The form with parameter $\mathbb{R}$ is `Chapter6/GapP3_K4_Contractible · exists_isScContractionRn`. `proposition_6_6_3_one` is the proposition with (1) in this form and with (1′), (2), (3), (4), for $U' \subseteq U \subseteq W$ in

one chart (V6.32, restated; gap G.61, closed; Findings, Chapter 6, item 137).

**(1′) on $[0,1]$** `Chapter6/GapP3_K4_IntervalHomotopy · exists_isScHomotopyI`. For paths on the M-polyfold $[0,1]$ no reparametrization $\sigma$ is needed. For sc-smooth $\gamma_0, \gamma_1 : [0,1] \to U$ with the same end points, the segment $H(\tau,t) = \varphi^{-}1r(\varphi\gamma_0(t) + \tau(\varphi\gamma_1(t) - \varphi\gamma_0(t)))$ is sc-smooth on the M-polyfold $[0,1]^2$: in the charts $t$ and $1-t$ of $[0,1]$ its representative is $(v,a,b) \mapsto r(a + s(v)(b-a))$ with $s(v) = v$ or $1-v$ `Chapter6/GapP3_K4_IntervalHomotopy · chartSegment` `Chapter6/GapP3_K4_IntervalHomotopy · isScSmoothOn_chartSegment`. The parameter has to stay in $[0,1]$, since outside it the segment leaves the partial quadrant on which $r$ is defined. The edge conditions hold exactly, and they are the hypotheses of the invariance of the transport along $[0,1]$-paths `Chapter6/PathsInterval · orTransportI_eq_of_homotopy` (§VI.37). Together, (1) with parameter $[0,1]$, (1′) for $[0,1]$-paths, (2) and (3), for one $U$: `Chapter6/GapP3_K4_IntervalHomotopy · proposition_6_6_3_interval` (Findings, Chapter 6, item 138).

## VI.11 Lemma 6.6.4 (preprint 6.29): general position along a family

The text (preprint pp. 221–222): along a path $\varphi$ with a section $s(\cdot,t)$, the section $F(t,y,\lambda) = f(y) - s(y,t) + \sum \lambda_i s_i(y,t)$ of the bundle over $[0,1] \times \mathbb{R}^k \times X$; Lemma 6.6.4 (preprint 6.29) asserts that there are $s_1, \dots, s_k$ with property (P): for every $t$, the slice $F_t$ is in general position to the boundary of $X \times \mathbb{R}^k$ at $(\varphi(t), 0)$. The proof is the sentence "using the results from the transversality theory".

**The statement formalized** (V6.33) `Chapter6/Lemma629 · lemma_6_6_4`. *Over a tame $X$ with bump functions, for $f$ with tame germ data, an sc-smooth $\Phi : \mathbb{R}^n \to X$ and an sc⁺-section $S$ of $\pi^*P$ with $(f+S)(q,\Phi(q)) = 0$ near $[0,1]^n$, there are finitely many sc⁺-sections $s_1, \dots, s_k$ of $\pi^*P$ such that $F(\lambda,(q,y)) = f(y) + S(q,y) + \sum \lambda_i s_i(q,y)$ is in general position at $(0,(q,\Phi(q)))$ for all $q$ near $[0,1]^n$.*

**The proof** is that of Theorem 5.3.10 (preprint 5.18) (§V.20). The points $(0,(q,\Phi(q)))$ are zeros of $F$ `Chapter6/Lemma629 · isZeroAt_paramSection_zero`, and for $q$ in a compact neighbourhood of $[0,1]^n$ they form a compact set. At each of them finitely many sections make the parametrized section satisfy properties (i)–(ii) of §V.19; general position is open (§V.18); finitely many neighbourhoods cover the compact set, and the sections of all of them together do the job `Chapter6/Lemma629 · exists_sections_isGeneralPositionGerm_of_isCompact`. *Hidden hypotheses:* $X$ tame, bump functions, tame germ data, the standing hypotheses of §6.6 (preprint §6.5) and of Theorem 5.3.10 (preprint 5.18). The conclusion is the general position of $F$ at the points $(t,\varphi(t),0)$, which the text derives from (P) and which Lemmas 6.6.5 and 6.6.6 (preprint 6.30 and 6.31) use. (P) itself, the general position of each slice $F_t$, follows from it by a chart computation — the tangent vectors of the family of zeros lie in $\ker F'$ and $\mathbb{R}^n$ has no boundary — and is §VI.36 (gap G.63 closed; Findings, Chapter 6, item 47).

## VI.12 Lemma 6.6.5 (preprint 6.30): kernels as continuous families of subspaces

The text (preprint p. 222): near $(t,\varphi(t),0)$ the solution set of $F$ is a manifold with boundary with corners, $\pi(t,y,\lambda) = t$ restricted to it is a submersion, and the kernels $L_t$ form a smooth vector bundle $L$ over $[0,1]$. What the propagation needs is less: the kernels form a continuous family, so that an orientation of one of them can be continued to the others.

**The kernels** (V6.34). At a zero with surjective linearization and germ data of index $d$, $\dim\ker = d$ `Chapter6/KernelFamily · StrongBundle.IsSurjLinOfIndex.finrank_linKer`. Near a zero in good position the kernel

is the tangent space of the solution set (Theorem 3.6.8 (preprint 3.54), §III.45), the range of the derivative of an sc-retraction onto the solution set read in the chart. So along a continuous family $y_q$ of zeros in good position, read in one chart at continuously varying level-1 points $z_q$, near every $q_0$ the vectors $DS(z_q)b_i$, for a basis $b$ of the kernel at $q_0$ and that retraction $S$, are a continuous frame of the kernels `Chapter6/KernelFamily · exists_local_frame_linKer` — continuous because $(z, v) \mapsto DS(z)v$ is, independent near $q_0$ because independence is open, and spanning because the dimension is constant. The kernels form a continuous family of subspaces of the model space, of dimension the index `Chapter6/KernelFamily · exists_subspaceFamily_linKer`. For an sc-smooth family of zeros $\mathbb{R}^n \to X$ read in a chart `Chapter6/KernelFamily · hasFDerivAt_scDeriv_rn`, differentiating $\widehat{f}(\Phi(q)) = 0$ shows that the derivative lies in the kernel of the linearization `Chapter6/KernelFamily · fderiv_mem_linKer`. For the propagation the bundle $L$ is used as a continuous family of subspaces; the solution set as a manifold with boundary with corners near $(t, \varphi(t), 0)$, the submersion $\pi$ and the smooth structure of $L$ are §VI.36 (gap G.62 closed; Findings, Chapter 6, items 48, 126).

**Continuous families of subspaces** (V6.35). A family of $n$-dimensional subspaces $L_b$ of a normed space, spanned near every point by a continuous frame `Chapter6/SubspaceBundle · SubspaceFamily`, has the line bundle $b \mapsto \lambda(L_b)$ `Chapter6/SubspaceBundle · SubspaceFamily.det`, with the wedges of the frames as trivializations `Chapter6/SubspaceBundle · SubspaceFamily.Frame.pretriv`; the coordinate change between two frames is the determinant of the matrix expressing one in the other, continuous because that matrix solves a linear system with continuous coefficients `Chapter6/SubspaceBundle · SubspaceFamily.Frame.coordChange_continuousOn`. Families and orientations restrict along continuous maps `Chapter6/SubspaceBundle · SubspaceFamily.comp` `Chapter6/SubspaceBundle · LineOrientation.comp`. Over $[0, 1]$ an orientation of $\lambda(L_0)$ extends uniquely to an orientation of $\lambda(L)$ (§VI.9), and its value at 1 is the continuation `Chapter6/SubspaceBundle · SubspaceFamily.transport` `Chapter6/SubspaceBundle · SubspaceFamily.transport_eq`, injective `Chapter6/SubspaceBundle · SubspaceFamily.transport_injective` and compatible with restriction `Chapter6/SubspaceBundle · SubspaceFamily.transport_comp`; over a simply connected, locally path-connected base the continuation along a path depends only on its end points. This is the text's "an orientation of the lines … determines by continuation an orientation of all the other lines" (pp. 264–266 (preprint pp. 222–223)), and the bundles $\tilde{L}$ of pp. 266–273 (preprint pp. 224–227). "Vector bundle" is read as a family of subspaces of one model space with continuous local frames.

**Families of kernels and their images** (V6.36). The kernels of a norm-continuous family of surjective Fredholm operators over a connected base form a continuous family `Chapter6/StabilizationContinuity · exists_subspaceFamily_ker`: their dimension is the index, and near $q_0$ the kernel of $A_q$ is the graph over $\ker A_{q_0}$ given by Lemma 6.5.2 (preprint 6.18) with $P = 1$. The images of a continuous family under a jointly continuous family of linear maps injective on its members form one `Chapter6/StabilizationContinuity · SubspaceFamily.exists_map`, and continuous orientations are carried along `Chapter6/StabilizationContinuity · SubspaceFamily.isLineOrientation_map`. Cutting a continuous family by the kernel of a fixed $\ell$ that is onto on every member gives one `Chapter6/StabilizationContinuity · SubspaceFamily.exists_inf_ker`. "As $\tau$ varies, the bundle varies continuously" (pp. 265–272 (preprint pp. 223–226)) is read in the operator norm, and for maps into a model space as joint continuity.

## VI.13 The stabilized section in a product chart, and Lemma 6.6.6 (preprint 6.31)

The text (pp. 264–266 (preprint pp. 222–223)): with $T = (f - s(\cdot,t))'(\varphi(t))$, the kernel $L_t$ of $F_t'(\varphi(t),0)$ fits into the sequence (96) $0 \to \ker T \to L_t \to \mathbb{R}^k \to \operatorname{coker} T \to 0$, and Lemma 6.6.6 (preprint 6.31) says that it is exact.

**The product chart** (V6.37). The parametrized section $F$ lives on $\mathbb{R}^k \times (\mathbb{R}^n \times X)$; it is read in the product chart $(\lambda \mapsto \lambda) \times ((q \mapsto q) \times \varphi)$ at $(0,(q,z))$, whose tangent space and fibre are products `Chapter6/ParamKernel · PullbackChartData.ofProd_scDeriv_r_apply` `Chapter6/ParamKernel · PullbackChartData.ofProd_mem_tangent_iff` `Chapter6/ParamKernel · PullbackChartData.mem_K_ofProd_iff`. There the linearization of $F$ is $(\lambda,(\sigma,v)) \mapsto D(\hat{g}_q \circ r)(z)v + \sum \lambda_i \hat{s}_i(q,\cdot)(z) + D_q\sigma$, with $g_q = f + S(q,\cdot)$ and $D_q$ the derivative in the parameter `Chapter6/ParamKernel · StrongBundle.scDeriv_ofProd_slice_apply` `Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_apply` `Chapter6/ParamKernel · StrongBundleChart.paramDerivAt` `Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_ofProd_ofProd`. With the projection to $\sigma$ and the inclusion of the slice $\sigma = 0$ `Chapter6/ParamKernel · sliceProj` `Chapter6/ParamKernel · sliceIncl`, the kernel cut by $\sigma = 0$ is the image of $\ker(T \oplus \varphi)$ `Chapter6/ParamKernel · linKer_inf_ker_sliceProj_eq_map` `Chapter6/ParamKernel · exists_sliceOps`, where $T = g_q'(x)$ lies in $\mathrm{Lin}(f,x)$ `Chapter6/ParamKernel · mem_linSet_sectionAdd_pullbackSlice` and $\varphi(\lambda) = \sum \lambda_i s_i(q,x)$. *$T \oplus \varphi$ is onto when $F$ is in general position at* $(0,(q,\Phi(q)))$ `Chapter6/ParamKernel · surjective_coprod_of_isLinSurjAt`: given $w$, the surjectivity of $F'$ gives $(\lambda,\sigma,v)$ with image $w$; the derivative of the curve of zeros, $(0,\sigma,D\Phi\,\sigma)$, lies in $\ker F'$ (§VI.12); subtracting it gives $(\lambda,0,v - D\Phi\,\sigma)$ with the same image, a preimage under $T \oplus \varphi$. The kernel is then a closed subspace of the product with the right inclusion `Chapter6/ParamKernel · exists_sliceKernel`. $\ker F_t'(\varphi(t),0)$ is read as the kernel of the linearization of the whole $F$ cut by $\sigma = 0$ (Findings, Chapter 6, item 49) (pp. 263–266 (preprint pp. 221–223)).

**Lemma 6.6.6 (preprint 6.31)** (V6.38) `Chapter6/ParamKernel · lemma_6_6_6`. With these operators, (96) $0 \to \ker T \to \ker(T \oplus \varphi) \to \mathbb{R}^k \to K_z/R(T) \to 0$ is the stabilization sequence of $T$ and $\varphi$ (Lemma 6.4.14 (preprint 6.14), §VI.6), hence exact. The text has $c$ onto by (P); here it is onto by the general position of $F$ (§VI.11) and the derivative of the family of zeros, which absorbs the $\mathbb{R}^n$-directions (preprint p. 223).

## VI.14 Continuation, the sign, and the push-forward

The text (pp. 265–269 (preprint pp. 223–224)): $\Phi_{E_t}$ of (94) identifies $\det(T_t)$ with $\lambda(L_t)$; an orientation $o_0$ of $\det(T_0)$ gives one of $\lambda(L_0)$, continuation carries it to $\lambda(L_1)$, and $\Phi_{E_1}^{-1}$ carries it to $\det(T_1)$: this defines $o_1 = \varphi_* o_0$.

**Continuation** (V6.39). Over a simply connected, locally path-connected base the continuation of a ray of $\lambda(L_{q_0})$ to $\lambda(L_{q_1})$ is the value at $q_1$ of the unique orientation extending it `Chapter6/Continuation · SubspaceFamily.cont` `Chapter6/Continuation · SubspaceFamily.cont_eq`. It is the identity from a point to itself `Chapter6/Continuation · SubspaceFamily.cont_self`, composes `Chapter6/Continuation · SubspaceFamily.cont_trans`, is a bijection of rays `Chapter6/Continuation · SubspaceFamily.contEquiv`, is compatible with pull-back `Chapter6/Continuation · SubspaceFamily.cont_comp`, and along $[0,1]$ it is the continuation of §VI.12 `Chapter6/Continuation · SubspaceFamily.cont_eq_transport`. For a stabilization $T \oplus \varphi$ that is onto, $\det(T \oplus \varphi) \cong \lambda(\ker(T \oplus \varphi)) \otimes \lambda^*(\{0\}) \cong \lambda(\ker(T \oplus \varphi))$ `Chapter6/Continuation · detEquivOfSurjective`, and through $\iota_\varphi$ (§VI.6) and an identification of the kernel with a member of $L$ the rays of $\det(T)$ and of $\lambda(L)$ correspond `Chapter6/Continuation · stabRayEquiv`. The propagation is $\Phi_{E_0}$, then continuation,

then $\Phi_{E_1}^{-1}$ on rays Chapter6/Continuation · propagate, with the identity on a constant family Chapter6/Continuation · propagate_self, composition Chapter6/Continuation · propagate_trans and pull-back Chapter6/Continuation · propagate_comp. It is stated over any simply connected, locally path-connected base, so that one statement serves the paths and the squares; the fixed identification $\lambda^*(\mathbb{R}^k) \cong \lambda^*(\{0\})$ of (94) occurs at both ends.

**The sign** (V6.40). In a local determinant chart with projection $P$, the chart coordinate of $\iota_\psi(x) \in \det(S \oplus \psi)$ is $(-1)^{k \dim F/R(P) + \dim \operatorname{coker} S} D(S,\psi)$ times that of $x$ Chapter6/StabilizationContinuity · coord_stabIota, with $D$ Chapter6/StabilizationContinuity · stabD continuous Chapter6/StabilizationContinuity · continuousOn_stabD and nowhere zero Chapter6/StabilizationContinuity · stabD_ne_zero near the centre. The first summand of the exponent is constant on a chart, the second is not. So where $\dim \operatorname{coker} T$ changes parity along a family, the rays carried by $\Phi_E$ jump. With $\alpha(T) = (-1)^{\dim \operatorname{coker} T}$ Chapter6/StabilizationContinuity · stabSign: *for a continuous orientation $o$ of* DET*, a norm-continuous family $T_q$ and a continuous stabilization $\varphi_q$ with $T_q \oplus \varphi_q$ onto, the rays $\alpha(T_q)\,\Phi(o(T_q))$ of $\lambda(\ker(T_q \oplus \varphi_q))$ form an orientation* Chapter6/StabilizationContinuity · isLineOrientation_stabRay. Without the sign they need not Chapter6/StabilizationContinuity · not_isLineOrientation_stabRay_uncorrected: $E = F = \mathbb{R}$, $k = 1$, $T_q = q/(1+|q|)$ and $\varphi(r) = r_0$, where $\dim \operatorname{coker} T_q$ is 1 at $q = 0$ and 0 elsewhere. As we read it, the orientations carried by $\Phi_E$ of (94) along a continuous family are not continuous where $\dim \operatorname{coker} T$ changes parity, and Lemmas 6.6.8, 6.6.10 (preprint 6.33, 6.34) and the homotopy argument before Theorem 6.6.11 (preprint 6.35) use that they are. The propagation uses $\alpha(T)\Phi_E$, which on each operator carries the text's orientation or its opposite (Findings, Chapter 6, item 50) (p. 256 (preprint p. 212), as used on pp. 265–272 (preprint pp. 223–226)).

**The push-forward** (V6.41). At a chart point $p$ of a point $y$ Chapter6/Propagation · chartPt, a section $S$ with $(f+S)(y) = 0$ and sections $s_1, \dots, s_k$ give $T = (f+S)'(y) \in \operatorname{Lin}(f, y)$ Chapter6/Propagation · ChartPoint.stabT Chapter6/Propagation · ChartPoint.stabT_mem_linSet and $\varphi(\lambda) = \sum \lambda_i \hat{s}_i(z)$ Chapter6/Propagation · ChartPoint.stabPhi, with appended sections giving the appended stabilization Chapter6/Propagation · ChartPoint.stabPhi_finAppend; $T \oplus \varphi$ is Fredholm Chapter6/Propagation · isFredholm_stabCoprod, and the sign is an involution Chapter6/Propagation · stabSign_inv Chapter6/Propagation · toPerm_stabSign_symm. A family of *stabilized kernels* Chapter6/Propagation · IsStabKernels over a parameter space $Q$ consists of a global sc$^+$-section $S$ of the pull-back with $f + S = 0$ at the points $(q, y(q))$, sections $s_1, \dots, s_k$, a continuous lift of $q \mapsto \varphi(y(q))$ to level 1 of the chart, and a continuous family $\Lambda$ of subspaces of $E_0 \times \mathbb{R}^k$ such that at every chart point of $y(q)$ in the chart $T_q \oplus \varphi_q$ is onto and $\Lambda_q$ is the image of its kernel. It restricts along continuous maps Chapter6/Propagation · IsStabKernels.comp. Its *push-forward* between two parameters $a, b$ Chapter6/Propagation · IsStabKernels.push: $o \in \operatorname{Or}^f_{y(a)}$ gives the ray $o(T_a)$, then $\alpha(T_a)\Phi_{E_a}(o(T_a))$, a ray of $\lambda(\Lambda_a)$; continuation carries it to $\lambda(\Lambda_b)$; $\alpha(T_b)\Phi_{E_b}^{-1}$ carries it to $\det(T_b)$; and $\operatorname{Or}^f_{y(b)} \cong \{\text{rays of } \det(T_b)\}$ (§VI.9) gives the image. It depends only on the data Chapter6/Propagation · IsStabKernels.push_congr. With $\operatorname{orRay}_q(o)$, the ray of $\lambda(\Lambda_q)$ read from $o$ at $q$ Chapter6/Propagation · IsStabKernels.orRay, $\varphi_* = \operatorname{orRay}_b^{-1} \circ \operatorname{cont}_{ab} \circ \operatorname{orRay}_a$ Chapter6/Propagation · IsStabKernels.push_eq_orRay. *Along a path with continuously varying data and a fixed point, continuation carries the rays read at one parameter to those read at the next* Chapter6/Propagation · IsStabKernels.cont_orRay_path: the operators $T_q$, $\varphi_q$ vary continuously; their kernels form a continuous family (§VI.12); the sign makes the rays $\alpha\Phi(o(T_q))$ an orientation; and the inclusion into $E_0 \times \mathbb{R}^k$ carries that orientation to $\lambda(\Lambda)$. The push-forward is compatible with pull-back Chapter6/Propagation · IsStabKernels.push_comp and with concatenation Chapter6/Propagation · IsStabKernels.push_trans. For given end-point operators the result differs from the text's by $\alpha(T_a)\alpha(T_b)$; the sections enter pointwise, and the continuity in the parameter is that of $\Lambda$ (Findings, Chapter 6, item 50) (preprint p. 224).

## VI.15 The stabilized kernels exist; push-forwards along paths

**Lemma 6.6.5 (preprint 6.30) for the push-forward** (V6.42) `Chapter6/StabKernelsExist · exists_isStabKernels`. *For an sc-smooth $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n)$ in the domain of a chart, an sc$^+$-section $S$ with $f + S = 0$ along $\Phi$ near $[0,1]^n$, and sections $s_1, \dots, s_k$ as in Lemma 6.6.4 (preprint 6.29), the stabilized kernels of the slices form a continuous family over $[0,1]^n$.* The family $q \mapsto \varphi(\Phi(q))$ has a continuous lift to level 1 `Chapter6/StabKernelsExist · exists_continuousOn_lift_chart`, the values of an sc-smooth map on $\mathbb{R}^n$ being smooth points. The zeros $(0,(q,\Phi(q)))$ of $F$ are in general position, hence in good position; their kernels form a continuous family (§VI.12); cutting it by $\{d\sigma = 0\}$, a fixed functional onto on every member because the derivative of the family of zeros lies in the kernel, keeps it continuous (§VI.12); and §VI.13 identifies the cut with $\ker(T_q \oplus \varphi_q)$. The kernels also form a smooth vector bundle (§VI.36); the continuation uses only their continuity.

**Push-forwards along paths** (V6.43). The parameter points of $[0,1]$ `Chapter6/PushData · pt1_mem_Icc`, level-1 lifts at smooth points `Chapter6/PushData · exists_lift_of_smooth` and the chart points they give `Chapter6/PushData · chartPtOf` are the bookkeeping. *$e : \mathrm{Or}^f_{\gamma(0)} \to \mathrm{Or}^f_{\gamma(1)}$ is a push-forward along an sc-smooth path $\gamma$ in a chart* `Chapter6/PushData · IsPush` if there are global sc$^+$-sections $S, s_1, \dots, s_k$ of $\pi^*P \to \mathbb{R} \times X$ with $(f+S)(t,\gamma(t)) = 0$ for $t$ near $[0,1]$, and a family of stabilized kernels of their slices along $\gamma$ whose push-forward is $e$. Equal data give equal push-forwards `Chapter6/PushData · IsStabKernels.push_eq_of_eq`. Pulled back along an affine reparametrization the sections still vanish along the reparametrized path `Chapter6/PushData · isZeroAt_sectionAdd_reparamSection` and give a push-forward along it `Chapter6/PushData · isPush_reparam`; so a push-forward along $\gamma$ is the composite of push-forwards along its two halves, each reparametrized to $[0,1]$ `Chapter6/PushData · IsPush.split`. *Along every sc-smooth path in a set with property (3) of Proposition 6.6.3 (preprint 6.28) there is a push-forward* `Chapter6/PushData · exists_isPush`: (3) gives $S$, Lemma 6.6.4 (preprint 6.29) gives $s_1, \dots, s_k$, and the stabilized kernels exist. Global sections of the pull-back stand in place of the text's sections over $U \times [0,1]$, and (P) is used only to construct a push-forward (preprint p. 224).

## VI.16 Lemma 6.6.8 (preprint 6.33) and sections over families

**Lemma 6.6.8 (preprint 6.33)** (V6.45) `Chapter6/Propagation · IsStabKernels.push_append`. *Appending sc$^+$-sections $\bar{s}_1, \dots, \bar{s}_d$ after $s_1, \dots, s_k$ does not change the push-forward.* The text proves it with a homotopy $\widetilde{F}(\tau,\cdot)$ and "one verifies by a simple computation". The computation is this. For $\ell_1, \dots, \ell_d \in \ker(T \oplus (\varphi,\psi))$ whose $\psi$-coordinates are the standard basis of $\mathbb{R}^d$, the map $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$ `Chapter6/AddSections · wedgeExt` `Chapter6/AddSections · wedgeExt_wedge` does not depend on the $\ell_j$ — two choices differ by vectors of the image of $\ker(T \oplus \varphi)$, the span of the first block `Chapter6/AddSections · wedgeExt_congr` — and is bijective `Chapter6/AddSections · wedgeExt_bijective`. With the appended stabilization `Chapter6/AddSections · appendStab` and the kernel inclusion $(v,\lambda) \mapsto (v,\lambda,0)$ `Chapter6/AddSections · kerAppend`, *the stabilization isomorphism of $(\varphi,\psi)$ is that of $\varphi$ followed by $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$, exactly, with no sign* `Chapter6/AddSections · detEquiv_stabIota_append`: a choice of Proposition 6.4.15 (preprint 6.15) for $(\varphi,\psi)$ is one for $\varphi$ with the $\ell_j$ appended to the kernel basis and the standard basis vectors to the $r_i$, and with the appended sections last the matrix $(\bar{c}, r)$ becomes block triangular with an identity block, so the determinant in (94) is unchanged. Along a family the $\ell_j$ are chosen continuously `Chapter6/AddSections · SubspaceFamily.exists_continuous_lifts` — the larger kernels project onto $\mathbb{R}^d$ because $T \oplus \varphi$ is already onto — and the wedge extension carries continuous orientations along `Chapter6/AddSections · isLineOrientation_wedgeExt`, through the kernel inclusion `Chapter6/AddSections · kerEquivMap`. So the stabilized rays `Chapter6/AddSections · stabRayEquiv_append` and the propaga-

tions Chapter6/AddSections · propagate_append agree, and with them the push-forwards. The homotopy of the text's proof is not needed (Findings, Chapter 6, item 51) (pp. 266–271 (preprint pp. 224–225)).

**Sections over families** (V6.46). For $g$ affine, $(q, y) \mapsto S(g(q), y)$ Chapter6/FamilySections · reparamSection is again an sc⁺-section of the pull-back Chapter6/FamilySections · StrongBundle.IsScPlusSection.reparam, through the sc-smooth product map $g \times 1$ Chapter6/FamilySections · rnProdMapOp, with the expected slices Chapter6/FamilySections · pullbackSlice_reparamSection. Sc⁺-sections may be checked in one chart around each point Chapter6/FamilySections · StrongBundle.isScPlusSection_of_local. Adding sections keeps the general position of $F$ at $(0, y)$ Chapter6/FamilySections · StrongBundle.isGeneralPositionGerm_paramSection_of_comp: the linearization only gains columns, and the properties of Definition 5.3.9 (preprint 5.17) survive. These are the sections $s(y, \tau, t)$ over a square and the text's "adding more sections, the properties will still hold" (pp. 266–272 (preprint pp. 224–226)).

## VI.17 Lemma 6.6.10 (preprint 6.34) and Proposition 6.6.7 (preprint 6.32)

**Lemma 6.6.10 (preprint 6.34)** (V6.47) Chapter6/Propagation · IsStabKernels.push_square. *For a family of stabilized kernels over the square $[0,1]^2$ with the end points $x_0$, $x_1$ fixed and the end-point operators varying continuously, the push-forwards along the edges $\tau = 0$ and $\tau = 1$ agree.* The edges are the restrictions to affine segments of the square Chapter6/PropagationWellDefined · sqEdge Chapter6/PropagationWellDefined · sqPath. On the edge $t = 0$ the point is $x_0$ and the operators vary continuously, so by §VI.14 continuation along it carries $\mathrm{orRay}_{(0,0)}(o)$ to $\mathrm{orRay}_{(1,0)}(o)$; the same on $t = 1$. The square is simply connected, so continuation from $(0,0)$ to $(1,1)$ is the same through $(0,1)$ and through $(1,0)$. Writing both push-forwards as $\mathrm{orRay}^{-1} \circ \mathrm{cont} \circ \mathrm{orRay}$ gives the claim. The text's convex homotopy $s^\tau$ is part of the square of Proposition 6.6.7 (preprint 6.32) below, and the continuity of the end-point orientations along $\tau$ is that of §VI.14, with the sign (preprint p. 226).

**Proposition 6.6.7 (preprint 6.32)** (V6.44) Chapter6/PropagationWellDefined · proposition_6_6_7. *Two push-forwards along the same sc-smooth path in the domain of a chart agree.* Let the data be $(S^0; s^0_1, \dots, s^0_{k^0})$ and $(S^1; s^1_1, \dots, s^1_{k^1})$. The text combines Lemma 6.6.8 (preprint 6.33) (more sections) and Lemma 6.6.10 (preprint 6.34) (a homotopy of the data); the formalization does both over one square (Findings, Chapter 6, item 52). *Step 1, the square.* $\Phi(\tau, t) = \gamma(t)$, constant in $\tau$, and $S^\tau = (1-\tau)S^0 + \tau S^1$, a section of the pull-back over $\mathbb{R}^2 \times X$ by §VI.16; $f + S^\tau$ vanishes along $\Phi$ since both $f + S^0$ and $f + S^1$ do and the fibres are linear Chapter6/PropagationWellDefined · isZeroAt_sectionAdd_combo. *Step 2, the sections.* With $\hat{s}^i_j = s^i_j$ for $j < k^i$ and 0 otherwise, the sections $(1-\tau)\hat{s}^0_j + \tau \hat{s}^1_j$, followed by the sections that Lemma 6.6.4 (preprint 6.29) gives on the square. The parametrized section is in general position along the square — adding sections keeps it (§VI.16) — so the stabilized kernels exist over $[0,1]^2$ (§VI.15). *Step 3, the edges $\tau = 0, 1$.* At $\tau = 0$ the sections $s^0$ are the first $k^0$ of the family and at $\tau = 1$ the sections $s^1$ are the first $k^1$, so Lemma 6.6.8 (preprint 6.33) compares each datum with the edge of the square without reordering Chapter6/PropagationWellDefined · IsStabKernels.push_eq_of_prefix, the edges being data of the same kind Chapter6/PropagationWellDefined · IsStabKernels.congr_data Chapter6/PropagationWellDefined · IsStabKernels.push_congr_data. *Step 4, the edges $t = 0, 1$.* There the point is fixed and the operators are affine in $\tau$ Chapter6/PropagationWellDefined · linAt_principal_combo, hence continuous Chapter6/PropagationWellDefined · continuous_principal_pullbackSlice. *Step 5.* Lemma 6.6.10 (preprint 6.34) equates the push-forwards along the edges $\tau = 0$ and $\tau = 1$, and Step 3 identifies them with the two given ones. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) ($X$ tame, bump functions, tame germ data). "Does not depend on the choices" is read as the agreement of any two data (pp. 266–272 (preprint pp. 224–226)).

## VI.18 Theorem 6.6.11 (preprint 6.35), and paths in a chart

**Theorem 6.6.11 (preprint 6.35)** (V6.48). $U$ has the *local propagation property* in a chart $c$ `Chapter6/PropagationWellDefined · HasLocalPropagation` if $U$ lies in the domain of $c$, along every sc-smooth path in $U$ there is a push-forward, and any two push-forwards along paths in $U$ with the same end points agree. `Chapter6/PropagationWellDefined · theorem_6_6_11`: *over a tame $X$ with bump functions, for $f$ with tame germ data, every smooth point $x$ has an open $U \ni x$ and a chart $c$ with the local propagation property.* The statement names the bundle and the point but not the section (possible omission); it is read for an sc-Fredholm section with tame germ data (Findings, Chapter 6, item 53). *Step 1.* $U \subseteq W$ and $c$ from Proposition 6.6.3 (preprint 6.28); push-forwards exist along paths in $W$, a fortiori in $U$, by (3) (§VI.15). *Step 2, homotopy invariance* `Chapter6/PropagationWellDefined · push_eq_of_homotopy`. For an sc-smooth $\Phi : \mathbb{R}^2 \to X$ with $\Phi([0,1]^2) \subseteq W$ and the end points fixed, (3) for $n = 2$ gives $S$, Lemma 6.6.4 (preprint 6.29) gives sections, and the stabilized kernels exist over the square. The edge $t \mapsto (\tau, t)$ is an affine map, so the restriction to it is a push-forward datum along $\Phi(\tau, \cdot)$ (§§VI.14–VI.15). Along $t = 0, 1$ the point is constant and so is the section $S(q, \cdot)$, because it depends on $q$ only through $\Phi(q)$; the end-point operators are then constant, and Lemma 6.6.10 (preprint 6.34) applies. With Proposition 6.6.7 (preprint 6.32), any push-forward along $\Phi(0, \cdot)$ equals any along $\Phi(1, \cdot)$. *Step 3, the reparametrization.* The homotopy of (1′) is $\gamma_0$ and $\gamma_1$ at its edges only for $t \in [0,1]$, while a push-forward datum asks for zeros of $f + S$ for $t$ near $[0,1]$. The proof therefore passes through $\gamma_i \circ \kappa$, which maps all of $\mathbb{R}$ into $\gamma_i([0,1])$, with three homotopies whose edges are exact for all $t$: $H(\sigma, t) = \gamma_0((1-\sigma)t + \sigma\kappa(t))$ from $\gamma_0$ to $\gamma_0 \circ \kappa$; the homotopy $(\tau, t) \mapsto \Phi(\tau, \kappa(t))$ from $\gamma_0 \circ \kappa$ to $\gamma_1 \circ \kappa$; and the analogue of $H$ for $\gamma_1$. Step 2 applied three times gives the agreement. $U$ is not shown to be sc-smoothly contractible (gap G.61); what the proof uses is (1′) (pp. 271–273 (preprint pp. 226–227)). For paths on the M-polyfold $[0,1]$ the theorem is §VI.37.

**Paths in a chart** (V6.49). Sc-smooth maps that agree near every point of a set are sc-smooth together `Chapter6/ScPaths · MPolyfold.isScSmoothMap_of_eventuallyEq`; $C^\infty$ maps of the parameter compose `Chapter6/ScPaths · pt1_apply` `Chapter6/ScPaths · contDiff_pt1_comp`. *For an open $U$ in the domain of a chart and $x \in U$ there is an open $V \ni x$ in $U$ whose smooth points are pairwise joined by sc-smooth paths in $U$* `Chapter6/ScPaths · exists_nhds_scPath`: in the chart, $t \mapsto \varphi^{-1}(r(u_1 + \kappa(t)(u_2 - u_1)))$, for $u_1, u_2$ in a small ball around $\varphi(x)$, as in (1′). The concatenation `Chapter6/ScPaths · concatPath` is $\gamma_1(\kappa(3s))$ for $s \le 1/2$ and $\gamma_2(\kappa(3s-2))$ for $s > 1/2$ `Chapter6/ScPaths · concatPath_of_le` `Chapter6/ScPaths · concatPath_of_gt`. It is constant on $[1/3, 2/3]$, with value $\gamma_1(1) = \gamma_2(0)$, so the two pieces glue sc-smoothly `Chapter6/ScPaths · isScSmoothMap_concatPath`; it has the right end points `Chapter6/ScPaths · concatPath_zero` `Chapter6/ScPaths · concatPath_one` and halves `Chapter6/ScPaths · concatPath_half₁` `Chapter6/ScPaths · concatPath_half₂`. Reparametrizing by a $C^\infty$ map keeps sc-smoothness `Chapter6/ScPaths · isScSmoothMap_reparam`, and push-forwards compose along concatenations `Chapter6/ContinuousOrientation · HasLocalPropagation.eq_trans`. Paths are joined here locally, inside a chart; the global facts recalled on preprint p. 201 are §VI.37 (gap G.56 closed) (pp. 272–274 (preprint pp. 227–228)).

## VI.19 Definition 6.6.12 (preprint 6.36): orientable neighbourhoods and continuous orientations

**Definition 6.6.12 (preprint 6.36)** (V6.50). The text: an orientable neighbourhood of $(f, x)$ is an open, sc-smoothly contractible $U$ with the local propagation property, and a continuous orientation over it is a choice $y \mapsto o_y$ carried into itself by the propagation. Since Proposition 6.6.3 (preprint 6.28) gives (1′), an orientable neighbourhood is a pair $V \subseteq U$ `Chapter6/SectionOrientation · IsOrientableNbhd`: $U$ open with the local propagation property in a chart, $V \ni x$ open with any two smooth points joined by an

sc-smooth path in $U$. A *continuous orientation* is a choice $y \mapsto o_y$ on $V_\infty$, read in the chart, carried into itself by the push-forwards along the paths in $U$ Chapter6/ContinuousOrientation · IsContinuousOrientation. *There are exactly two* Chapter6/ContinuousOrientation · exists_two_isContinuousOrientation: given $o_x$, set $o_y$ to be the push-forward of $o_x$ along any path in $U$ from $x$ to $y$, well defined by the local propagation property; push-forwards compose along concatenations, so this is a continuous orientation; the two values of $o_x$ give two different ones; and a continuous orientation is determined by its value at $x$. *Every smooth point has an orientable neighbourhood* Chapter6/ContinuousOrientation · exists_orientableNbhd Chapter6/SectionOrientation · exists_isOrientableNbhd, by Theorem 6.6.11 (preprint 6.35) and §VI.18. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) (Findings, Chapter 6, item 54) (preprint p. 227).

## VI.20 The path category $PX$

**Reparametrization homotopies** (V6.51). The values of an sc-smooth path at parameters are smooth points Chapter6/ScHomotopy · mem_smoothSet_of_isScSmoothMap, and $C^\infty$ maps of $\mathbb{R}^2$ compose Chapter6/ScHomotopy · contDiff_pt2_comp. *For a $C^\infty$ $\theta$ with $\theta(0) = 0$, $\theta(1) = 1$, $\gamma \circ \theta$ is sc-smoothly homotopic to $\gamma$ with the end points fixed* Chapter6/ScHomotopy · isScHomotopyRel_reparam, by $H(\tau, t) = \gamma((1-\tau)t + \tau\theta(t))$, and inside every set containing $\gamma([0,1])$ when $\theta([0,1]) \subseteq [0,1]$ Chapter6/ScHomotopy · reparam_mem. With the halves of a path Chapter6/ScHomotopy · leftHalf Chapter6/ScHomotopy · rightHalf, the concatenation of the two halves of $\gamma$ is such a reparametrization Chapter6/ScHomotopy · halvesReparam Chapter6/ScHomotopy · concatPath_halves (preprint p. 228).

**The category** (V6.52). The tools are the value $\kappa(1/2)$ Chapter6/PathCategory · smoothTransition_half, constant maps at smooth points Chapter6/PathCategory · isScSmoothMap_const_of_mem_smoothSet, maps defined by cases on closed pieces that agree near the seams Chapter6/PathCategory · isScSmoothMap_ite Chapter6/PathCategory · isScSmoothMap_ite₃ and the middle third of a concatenation Chapter6/PathCategory · concatPath_of_le_two_thirds Chapter6/PathCategory · concatPath_of_one_third_le. Sc-smooth homotopy with the end points fixed Chapter6/PathCategory · ScHomotopic is an equivalence relation on the sc-smooth paths from $x$ to $y$ Chapter6/PathCategory · ScHomotopic.refl Chapter6/PathCategory · ScHomotopic.symm Chapter6/PathCategory · ScHomotopic.trans. Transitivity needs care: a homotopy meets its edge conditions only on $[0,1]$ (decision D2), and two homotopies glued in $\tau$ need not agree off $[0,1]$, so all paths and homotopies are first reparametrized by $\kappa$ in $t$ (Findings, Chapter 6, item 71). The paths with smooth end points Chapter6/PathCategory · ScPath Chapter6/PathCategory · ScPath.source_mem_smoothSet Chapter6/PathCategory · ScPath.target_mem_smoothSet have the constant paths Chapter6/PathCategory · ScPath.const, concatenation Chapter6/PathCategory · ScPath.concat and reversal Chapter6/PathCategory · ScPath.reverse. Concatenation is well defined on classes Chapter6/PathCategory · ScPath.concat_congr, has the constant paths as units Chapter6/PathCategory · ScPath.const_concat Chapter6/PathCategory · ScPath.concat_const, is associative Chapter6/PathCategory · ScPath.concat_assoc — the two bracketings differ by an explicit reparametrization Chapter6/PathCategory · assocReparam, joined by a three-piece homotopy Chapter6/PathCategory · isScHomotopyRel_triHomotopy — and the reversed paths are inverses Chapter6/PathCategory · ScPath.concat_reverse Chapter6/PathCategory · ScPath.reverse_concat. The classes Chapter6/PathCategory · ScPathClass with their composition Chapter6/PathCategory · ScPathClass.comp make the smooth points a category $PX$ Chapter6/PathCategory · PathCat Chapter6/PathCategory · PathCat.category, indeed a groupoid Chapter6/PathCategory · PathCat.groupoid, with the morphism of a path Chapter6/PathCategory · PathCat.homMk, composition and identities on representatives Chapter6/PathCategory · PathCat.comp_mk Chapter6/PathCategory · PathCat.id_eq_mk, equality of morphisms as homotopy Chapter6/PathCategory · PathCat.mk_eq_mk_iff, every morphism represented Chapter6/PathCategory · PathCat.exists_eq_mk and inverses by reversal Chapter6/PathCategory · PathCat.inv_mk. The text does not name the

identities; they are the classes of the constant paths. The second piece $\psi(\beta(s))$ of the composition on preprint p. 228 leaves $[0,1]$, since $\beta([1/2,1]) = [1,2]$; it is read as $\psi(\beta(s)-1)$ (possible misprint; Findings, Chapter 6, item 55). Mathlib's $[\varphi] \gg [\psi]$ is the text's $[\psi] * [\varphi]$ (pp. 272–274 (preprint pp. 227–228)). The category with paths on $[0,1]$, as printed, is equivalent to this one (§VI.37).

## VI.21 Comparing charts: the transfer, and $\mathrm{Or}_x^f$ in a chosen chart

The proof of Theorem 6.6.13 (preprint 6.37) carries an orientation along orientable neighbourhoods of different points of a path, which may lie in different charts. The text takes $T_xX$, $Y_x$, $\mathrm{Lin}(f,x)$ and $\mathrm{Or}_x^f$ to be the same whichever chart is used (preprint p. 219); here they are made at chart points (§VI.1), so the charts have to be compared (Findings, Chapter 6, item 70).

**Conjugation of determinants** (V6.53). For isomorphisms $A$, $\Gamma$ with $S'A = \Gamma S$, $A$ maps $\ker S$ onto $\ker S'$ `Chapter6/ChartTransfer · conjKerEquiv` and $\Gamma$ induces $\operatorname{coker} S \cong \operatorname{coker} S'$ `Chapter6/ChartTransfer · conjCokerEquiv`; together they give $\det(S) \cong \det(S')$ `Chapter6/ChartTransfer · conjDet`, on wedges $(a \wedge \dots) \otimes (d \wedge \dots)^* \mapsto (Aa \wedge \dots) \otimes ([\Gamma d] \wedge \dots)^*$ `Chapter6/ChartTransfer · conjDet_wedge_tmul`, compatible with composition, equal data and the identity `Chapter6/ChartTransfer · conjDet_trans` `Chapter6/ChartTransfer · conjDet_congr` `Chapter6/ChartTransfer · conjDet_refl`. Good left-projections are conjugated, $\Gamma P \Gamma^{-1} \in \Pi_{S'}$ `Chapter6/ChartTransfer · IsGoodLeftProj.conj`, and the $\gamma^P$ commute with the conjugation `Chapter6/ChartTransfer · gammaProj_conjDet`; read in local determinant charts the conjugation has a continuous nowhere-zero coordinate `Chapter6/ChartTransfer · coord_conjDet_coordInv`, so continuous orientations of DET are carried along `Chapter6/ChartTransfer · isLineOrientation_conj` `Chapter6/ChartTransfer · LineOrientation.conj` `Chapter6/ChartTransfer · LineOrientation.conj_apply`, with the same rules `Chapter6/ChartTransfer · LineOrientation.conj_refl` `Chapter6/ChartTransfer · LineOrientation.conj_trans` `Chapter6/ChartTransfer · LineOrientation.conj_congr`.

**The transfer between two chart points** (V6.53). For chart points $p$, $p'$ of $x$, $A$ is the derivative at $z$ of the transition of the base charts `Chapter6/ChartTransfer · ChartPoint.baseDeriv`, with the cocycle rules `Chapter6/ChartTransfer · ChartPoint.baseDeriv_self` `Chapter6/ChartTransfer · ChartPoint.baseDeriv_trans`, an sc-operator `Chapter6/ChartTransfer · ChartPoint.isScOperator_baseDeriv`; $\Gamma$ is the fibre part of the transition of the strong bundle charts `Chapter6/ChartTransfer · ChartPoint.fiberMap` `Chapter6/ChartTransfer · ChartPoint.fiberMap_self` `Chapter6/ChartTransfer · ChartPoint.fiberMap_trans` `Chapter6/ChartTransfer · ChartPoint.isScOperator_fiberMap`. They restrict to isomorphisms $T_zO \cong T_{z'}O'$ and $K_z \cong K_{z'}$ `Chapter6/ChartTransfer · ChartPoint.transferT` `Chapter6/ChartTransfer · ChartPoint.transferK`, again with the cocycle rules `Chapter6/ChartTransfer · ChartPoint.transferT_self_apply` `Chapter6/ChartTransfer · ChartPoint.transferT_trans_apply`. At a zero the linearizations in the two charts are related by them, the chain rule applied to the principal parts `Chapter6/ChartTransfer · ChartPoint.scDeriv_principal_baseDeriv`, and sc⁺-operators conjugate to sc⁺-operators `Chapter6/ChartTransfer · ChartPoint.IsScPlusOp.conj`; so $L \in \mathrm{Lin}_p(f,x)$ iff $\Gamma L A^{-1} \in \mathrm{Lin}_{p'}(f,x)$ `Chapter6/ChartTransfer · ChartPoint.mem_linSet_transfer_iff`. Conjugation of orientations over this bijection of the bases is the transfer $\mathrm{Or}_x^f(p) \cong \mathrm{Or}_x^f(p')$ `Chapter6/ChartTransfer · ChartPoint.orTransfer` `Chapter6/ChartTransfer · ChartPoint.orTransfer_apply`, with the cocycle rules `Chapter6/ChartTransfer · ChartPoint.orTransfer_self` `Chapter6/ChartTransfer · ChartPoint.orTransfer_trans`.

**The push-forward does not depend on the chart** (V6.54). The sign is invariant, conjugation preserving $\dim\operatorname{coker}$ `Chapter6/PushTransfer · stabSign_conj`; the stabilization is natural — $(\Gamma S A^{-1}) \oplus \Gamma\varphi$ is the conjugate of $S \oplus \varphi$, with the conjugated kernel, onto iff $S \oplus \varphi$ is `Chapter6/PushTransfer · coprod_conj` `Chapter6/PushTransfer · ker_coprod_conj` `Chapter6/PushTransfer · surjective_coprod_conj` — and so is its isomor-

phism on rays Chapter6/PushTransfer · conj_stab_key Chapter6/PushTransfer · stabRayEquiv_conjDet; continuation is natural under jointly continuous families of injective linear maps Chapter6/PushTransfer · SubspaceFamily.cont_map. The principal parts of the sections transform by $\Gamma$ Chapter6/PushTransfer · ChartPoint.fiberMap_principal, so the operators $T$ and $\varphi$ of a datum in one chart are the conjugates of those in the other Chapter6/PushTransfer · ChartPoint.conj_of_linAt Chapter6/PushTransfer · ChartPoint.conj_of_principal, and $A$ varies continuously along a path Chapter6/PushTransfer · continuous_baseDeriv. A family of stabilized kernels in one chart is therefore carried by $A \times 1$ to one in the second chart, with the same sections Chapter6/PushTransfer · IsStabKernels.exists_transfer, the rays read from an orientation corresponding under the transfer Chapter6/PushTransfer · IsStabKernels.orRay_transfer; a push-forward in one chart, conjugated by the transfers at the end points, is a push-forward in the other Chapter6/PushTransfer · IsPush.transfer, and the local propagations of two charts agree Chapter6/PushTransfer · HasLocalPropagation.push_compat. The text does not compare charts (pp. 264–274 (preprint pp. 222–228)).

$\mathrm{Or}_x^f$ **in a chosen chart** (V6.55). A chart of the atlas at $x$ is chosen for every $x$ Chapter6/SectionOrientation · orChart Chapter6/SectionOrientation · isChart_orChart Chapter6/SectionOrientation · mem_orChart, and $\mathrm{Or}_x^f$ is the set of the orientations read there Chapter6/SectionOrientation · Orx, with two elements Chapter6/OrientationFunctor · exists_two_orx. It is identified with the orientations read in any other chart at $x$ by the transfer Chapter6/SectionOrientation · toChartOr Chapter6/SectionOrientation · toChartOr_orChart, compatibly with composition Chapter6/OrientationFunctor · toChartOr_trans, and a map $y \mapsto o_y$ is read in the chart of an orientable neighbourhood through it Chapter6/SectionOrientation · restrictOr. A chosen chart stands for the text's chart-free $\mathrm{Or}_x^f$ (preprint pp. 220, 228).

## VI.22 Transport along paths from local push-forwards

The proof of Theorem 6.6.13 (preprint 6.37) (p. 273 (preprint p. 228)): for a path $\varphi$, "using the compactness of $[0,1]$ and a finite covering" by orientable neighbourhoods, $\varphi_*$ is the composite of the local propagations; "by a covering argument" it depends only on $[\varphi]$; and "it is obvious that $\Gamma_f$ is unique". The formalization separates this argument from orientations: it is proved for any family of types over $X_\infty$ (Findings, Chapter 6, item 72).

**Local transport data** (V6.56) Chapter6/PathTransport · LocalTransport: a family of *good* open sets covering $X_\infty$, all fibres in bijection, and for every good set $U$ a push-forward along every sc-smooth path in $U$, such that push-forwards in one good set compose and do not depend on the good set containing the path. Then the push-forward along a loop in a good set is the identity Chapter6/PathTransport · LocalTransport.push_eq_refl, in a good set it depends only on the end points Chapter6/PathTransport · LocalTransport.push_eq_push, and only on the path on $[0,1]$ Chapter6/PathTransport · LocalTransport.push_congr.

**The dyadic transport.** At level 0 it is the push-forward in a chosen good set containing $\gamma([0,1])$ (an arbitrary bijection if there is none); at level $n+1$ it is the composite of the transports at level $n$ along the two halves Chapter6/PathTransport · LocalTransport.transportN. On a path in a good set every level gives the push-forward Chapter6/PathTransport · LocalTransport.transportN_eq_push, by the composition rule. The pieces of level $N$ lie in good sets Chapter6/PathTransport · LocalTransport.IsFine for some $N$: the preimages of the good sets cover the compact $[0,1]$, so there is a Lebesgue number $\delta$ Chapter6/PathTransport · LocalTransport.IsLebesgueNumber Chapter6/PathTransport · LocalTransport.exists_isLebesgueNumber, and $2^N\delta > 1$ suffices Chapter6/PathTransport · LocalTransport.isFine_of_isLebesgueNumber. From such an $N$ on the levels agree Chapter6/PathTransport · LocalTransport.transportN_eq_of_isFine, since at the next level each piece is split inside its good set. The *transport* is the value at such a level Chapter6/PathTransport · LocalTransport.transport.

**Its properties.** It is the push-forward on a path in a good set `Chapter6/PathTransport · LocalTransport.transport_eq_push` — property (□) — depends only on the path on $[0,1]$ `Chapter6/PathTransport · LocalTransport.transport_congr`, is the identity on paths constant on $[0,1]$ `Chapter6/PathTransport · LocalTransport.transport_const`, and is the composite along the two halves `Chapter6/PathTransport · LocalTransport.transport_halves`. *Homotopy invariance* `Chapter6/PathTransport · LocalTransport.transport_eq_of_isScHomotopyRel` is the covering argument. For a homotopy $\Phi : \mathbb{R}^2 \to X$ the preimages of the good sets cover $[0,1]^2$, with a Lebesgue number $\delta$ `Chapter6/PathTransport · LocalTransport.exists_isSqLebesgueNumber`. *The square lemma* `Chapter6/PathTransport · LocalTransport.transportN_square`: for a rectangle of $[0,1]^2$ with sides $< 2^n\delta$, the transports at level $n$ along its vertical, horizontal and diagonal segments `Chapter6/PathTransport · vSeg` `Chapter6/PathTransport · hSeg` `Chapter6/PathTransport · dSeg` commute around its boundary — left side then top equals bottom then right side. By induction on $n$: for $n = 0$ the rectangle lies in one good set, and both composites are the push-forward along the diagonal; for $n + 1$ the rectangle is cut into four quadrants and the four commutative squares are pasted, the interior edges cancelling `Chapter6/PathTransport · equiv_paste_squares`. Applied to $[0,1]^2$ at a level with $2^n\delta > 1$, with the sides $\tau \mapsto \Phi(\tau,0)$, $\tau \mapsto \Phi(\tau,1)$ constant on $[0,1]$ and of trivial transport, it gives the invariance. The transport is invariant under reparametrization `Chapter6/PathTransport · LocalTransport.transport_reparam` (§VI.20) and takes concatenations to composites `Chapter6/PathTransport · LocalTransport.transport_concatPath`, the halves of a concatenation being reparametrizations of the two paths. *Uniqueness* `Chapter6/PathTransport · LocalTransport.transport_unique`: a family of bijections along the sc-smooth paths that is homotopy invariant, takes concatenations to composites and is the push-forward on paths in good sets, is the transport — $G(\gamma) = G(\text{left half}) \circ G(\text{right half})$ through the reparametrization, and induction on the dyadic level. Dyadic subdivisions of $[0,1]$ and of $[0,1]^2$ stand in place of arbitrary finite coverings (preprint p. 228).

## VI.23 Theorem 6.6.13 (preprint 6.37): the functor $\Gamma_f$

The text: *for a strong bundle over a tame M-polyfold and an sc-Fredholm section $f$ there is a uniquely determined functor $\Gamma_f : PX \to \mathcal{O}_f$, $x \mapsto \mathrm{Or}_x^f$, with (□): for every smooth $x$, every sc-smoothly contractible open $U \ni x$ on which $f$ has the local propagation property, and every sc-smooth path $\varphi : [0,1] \to U$, the propagation $\varphi_*$ and $\Gamma_f([\varphi])$ coincide.*

**The local data of $f$** (V6.57). $U$ is a propagation neighbourhood if it is open with the local propagation property in some chart `Chapter6/OrientationFunctor · IsPropagationNbhd`. Along a path in such a $U$ the push-forward, read in the chosen charts of the end points through the transfer (§VI.21), is orPush `Chapter6/OrientationFunctor · orPush`; it equals every push-forward along the path in every chart in which the path lies in a set with the local propagation property `Chapter6/OrientationFunctor · orPush_eq` — by the chart independence of §VI.21 and the cocycle rule of the transfer — it does not depend on the neighbourhood `Chapter6/OrientationFunctor · orPush_compat`, and it composes `Chapter6/OrientationFunctor · orPush_trans`. With Theorem 6.6.11 (preprint 6.35), which makes the propagation neighbourhoods cover $X_\infty$, and §VI.9, which puts every $\mathrm{Or}_x^f$ in bijection with every other, these are local transport data `Chapter6/OrientationFunctor · orLocalTransport`, and their transport satisfies (□) `Chapter6/OrientationFunctor · orLocalTransport_transport_eq`.

**The functor.** $\Gamma_f([\varphi])$ is the transport along $\varphi$, well defined on classes by homotopy invariance `Chapter6/OrientationFunctor · orFunctorMap` `Chapter6/OrientationFunctor · orFunctorMap_mk`; it takes identities to identities `Chapter6/OrientationFunctor · orFunctorMap_id` and compositions to composites `Chapter6/OrientationFunctor · orFunctorMap_comp`, by the constancy and concatenation rules of §VI.22; and it is a func-

tor $PX \to$ Type `Chapter6/OrientationFunctor · orFunctor`. Property (□ ) for a family $G$ on the morphisms of $PX$, on the members of a family of open sets `Chapter6/OrientationFunctor · IsStarPropertyOn` and on all of them `Chapter6/OrientationFunctor · IsStarProperty`, reads $G([\varphi]) = \mathrm{toChartOr}^{-1}_{\varphi(1)} \circ e \circ \mathrm{toChartOr}_{\varphi(0)}$ for every chart $c$, every open $U$ with the local propagation property in $c$, every sc-smooth path $\varphi$ in $U$ with smooth end points and every push-forward $e$ along $\varphi$ in $c$. $\Gamma_f$ has it `Chapter6/OrientationFunctor · orFunctorMap_isStarProperty`, so the property is satisfiable. *Uniqueness*: a family with the rules for identities and composites and with (□ ) on the members of a family of such open sets covering $X_\infty$ is $\Gamma_f$ `Chapter6/OrientationFunctor · orFunctorMap_unique_of_cover` — restrict the local data to the cover `Chapter6/OrientationFunctor · LocalTransport.restrict` and apply the uniqueness of the transport — in particular with (□ ) everywhere `Chapter6/OrientationFunctor · orFunctorMap_unique`.

**Theorem 6.6.13 (preprint 6.37)** `Chapter6/OrientationFunctor · theorem_6_6_13_hyp`. *Over a tame $X$ with bump functions, for $f$ with tame germ data and $P$ with sc⁺-sections through the smooth fibre points, there is exactly one family of bijections $\Gamma_f([\varphi]) : \mathrm{Or}^f_x \to \mathrm{Or}^f_y$ on the morphisms of $PX$ taking identities to identities and compositions to composites, with property (□ ). Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5), and sc⁺-sections through every smooth fibre point, so that every $\mathrm{Or}^f_x$ has two elements (bump functions give them). (□ ) is asked for every open set with the local propagation property, not only for sc-smoothly contractible ones (the preprint's (1), §VI.10), and the uniqueness needs it only on a covering family — for instance on the sc-smoothly contractible ones, wherever they cover $X_\infty$. $\mathcal{O}_f$ is read as types and bijections, and "uniquely determined" for the maps on morphisms: on objects there is no choice (preprint p. 228). With the text's hypotheses only — $X$ tame with bump functions, $f$ with tame germ data — the sc⁺-sections through the smooth fibre points come from the bump functions (§VI.1) and the paracompactness of $\mathbb{R}^n \times X$ and of the pull-backs is proved, so the theorem holds without the further hypothesis `Chapter6/OrientationMainClean · theorem_6_6_13` (Findings, Chapter 6, item 122).

## VI.24 Definitions 6.6.14–6.6.16 (preprint 6.38–6.40): orientability and orientations

**Definition 6.6.14 (preprint 6.38)** (V6.58). *$f$ is orientable if for every pair of smooth points $x$, $y$ in the same path component, $\Gamma_f([\varphi])$ does not depend on the class $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · IsOrientableSection`. "In the same path component" is read as "joined by an sc-smooth path" `Chapter6/OrientationFunctor · scPathSetoid`; for points not so joined there is nothing to check (Findings, Chapter 6, item 73). The reading is justified in §VI.37: for smooth points, one connected component, one path component and one sc-smooth path component are the same. *Equivalently there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for every $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · isOrientableSection_iff_exists`: from left to right, choose a base point in each class of points joined by sc-smooth paths and set $o_y = \Gamma_f([\varphi])o_{\mathrm{base}}$ for any $[\varphi]$, which does not depend on $[\varphi]$; from right to left, $\Gamma_f([\varphi])$ and $\Gamma_f([\psi])$ are bijections between two-element sets with the same value at $o_x$, hence equal (§VI.9) (preprint p. 229).

**Definition 6.6.15 (preprint 6.39)** (V6.59). *$f$ is orientable if there is $y \mapsto o_y$ on $X_\infty$ such that every smooth point has an orientable neighbourhood over which it is one of the two continuous orientations* `Chapter6/SectionOrientation · IsOrientableSection'`. The definition writes "$y \in U_\infty$" before any $U$ is introduced; it is read as $y \in X_\infty$ (possible misprint; Findings, Chapter 6, item 74). *This holds iff there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi]$* `Chapter6/OrientationFunctor · isOrientableSection'_iff_exists`. From right to left: on an orientable neighbourhood $V \subseteq U$ the push-forwards along paths in $U$ are values of $\Gamma_f$, by (□ ), so the restriction is a continuous orientation. From left to right: elements carried into each other by the

transport along the paths in the members of an open covering are carried into each other along every path `Chapter6/OrientationFunctor · LocalTransport.transport_apply_eq_of_local` — the transport is computed on dyadic pieces inside orientable neighbourhoods. *So Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) are equivalent* `Chapter6/OrientationFunctor · isOrientableSection_iff`. The text calls them equivalent without proof (Findings, Chapter 6, item 73) (preprint p. 229).

**Definition 6.6.16 (preprint 6.40)** (V6.60). *An orientation of $f$ is a map $y \mapsto o_y$ on $X_\infty$ whose restriction to every orientable neighbourhood of every smooth point is a continuous orientation* `Chapter6/SectionOrientation · IsSectionOrientation`. Equivalently $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi] : x \to y$ `Chapter6/OrientationFunctor · isSectionOrientation_iff`, by the argument of Definition 6.6.15 (preprint 6.39); so $f$ has an orientation iff it is orientable `Chapter6/OrientationFunctor · exists_isSectionOrientation_iff`, two orientations that agree at a point agree on the smooth points joined to it by sc-smooth paths `Chapter6/OrientationFunctor · IsSectionOrientation.apply_eq_iff`, and there are exactly two on each such class `Chapter6/OrientationFunctor · IsSectionOrientation.exists_two`. An orientable neighbourhood is the pair of §VI.19 (Findings, Chapter 6, item 75) (preprint p. 229).

## VI.25 Theorem 6.6.18 (preprint 6.41): the natural orientation of the solution set

The text: *over a tame $X$, let $f$ be a proper sc-Fredholm section in general position to the boundary at every zero, and $o^f$ an orientation of $f$; then $S = f^{-1}(0)$ is a smooth compact manifold with boundary with corners with a natural orientation.* The proof: $S$ is such a manifold "from the transversality discussion"; $\ker f'(x) = T_xS$ and, $f'(x)$ being onto, $T_xS$ is oriented by $o^f_x$; and "one verifies that the differential geometric local prolongation of the orientation on $S$ is the same as the Fredholm one".

**The first assertion** (V6.61) `Chapter6/SolutionOrientation · theorem_6_6_18_solutionSet`. *For $X$ tame and a proper $f$ with tame germ data in general position (Definition 5.3.9 (preprint 5.17)) at every zero, $S$ is compact, a smooth submanifold in general position with the induced structure of a tame manifold with boundary with corners* — the conclusions of Theorem 5.3.10 (preprint 5.18) (§V.20), with good position from general position by Lemma 4.2.8 (preprint 4.19) (§IV.14) `Chapter6/SolutionOrientation · MPolyfold.IsGeneralPositionAt.isGoodPositionAt`. In a chart, $T_xS = \ker f'(x)$ `Chapter6/SolutionOrientation · tangentSpace_solutionSet_eq_linKer` (§V.17). "Proper" is the text's hypothesis; the proof uses only the compactness of $S$, which for sc-Fredholm sections is equivalent to it (Theorem 5.2.2 (preprint 5.5), §V.6), so nothing of the hypothesis is lost (Findings, Chapter 6, item 90).

**The tangent orientation.** At a zero, $f'(x)$ — the linearization of $f + s$ with $s = 0$ `Chapter6/SolutionOrientation · ChartPoint.linZero` — lies in $\mathrm{Lin}(f, x)$ `Chapter6/SolutionOrientation · ChartPoint.linZero_mem_linSet` and is onto `Chapter6/SolutionOrientation · ChartPoint.linZero_surjective`, with kernel $T_xS$ in the chart `Chapter6/SolutionOrientation · ChartPoint.map_ker_linZero`. So an element $o_x$ of $\mathrm{Or}^f_x$ gives a ray of $\det f'(x) \cong \lambda(\ker f'(x)) = \lambda(T_xS)$, the *tangent orientation* `Chapter6/SolutionOrientation · ChartPoint.tangentOr`, read in a chart at a zero `Chapter6/SolutionOrientation · tangentOrAt`. The sign of a surjective operator is $1$ `Chapter6/SolutionOrientation · stabSign_of_surjective`, and the stabilization by $\mathbb{R}^0$ of a surjective operator is $\det(T) \cong \lambda(\ker T)$ on rays `Chapter6/SolutionOrientation · stabRayEquiv_zero`, so the tangent orientation is the stabilized ray of §VI.14 for the empty family of sections `Chapter6/SolutionOrientation · ChartPoint.stabRayEquiv_eq_tangentOr`.

**"The local prolongation is the Fredholm one."** Along an sc-smooth path $\gamma$ of zeros in the domain of a

chart, the datum $S = 0$, $k = 0$ is a family of stabilized kernels Chapter6/SolutionOrientation · isStabKernels_zero, its kernels being those of $f'(\gamma_q)$ Chapter6/SolutionOrientation · SubspaceFamily.congrL; they form a continuous family of subspaces, by Lemma 6.6.5 (preprint 6.30) (§VI.12) Chapter6/SolutionOrientation · exists_tangentFamily Chapter6/SolutionOrientation · tangentFamily Chapter6/SolutionOrientation · tangentFamily_L, and the rays read from an orientation are the tangent orientations Chapter6/SolutionOrientation · IsStabKernels.orRay_eq_tangentOr. So along a path of zeros the push-forward is continuation between tangent orientations Chapter6/SolutionOrientation · cont_tangentOr, and *the tangent orientations given by an orientation of* DET *along the path form an orientation of the line bundle* $\lambda(\Lambda)$ Chapter6/SolutionOrientation · isLineOrientation_tangentOr. Intermediate parameters are reached by the paths $s \mapsto \gamma(t\kappa(s))$, which stay in $\gamma([0,t])$ for all $s \in \mathbb{R}$. For an orientation $o$ of $f$ and an sc-smooth path of zeros in an orientable neighbourhood $V \subseteq U$, the tangent orientations of $o$ along it form an orientation of $\lambda(\Lambda)$ Chapter6/SolutionOrientation · isLineOrientation_tangentOr_of_isSectionOrientation, and together this is the theorem Chapter6/SolutionOrientation · theorem_6_6_18_hyp (Findings, Chapter 6, item 91).

**Two charts** (V6.61). The tangent orientations are read at chart points, so the text's "natural" needs that they do not depend on the chart. *For $o \in \mathrm{Or}_x^f$ at a zero with $f'(x)$ onto and two charts $c$, $c'$ at $x$, the orientation of* ker $f'(x)$ *read in $c'$ is the image of the one read in $c$ under the tangent map $D(\tau \circ r)(z)$ of the transition* Chapter6/SolutionOrientationCharts · tangentOrAt_toChartOr. The tangent map restricts to an isomorphism of the two kernels Chapter6/SolutionOrientationCharts · ChartPoint.tangentKerEquiv, it conjugates the two linearizations, and the transfer of §VI.21, which is the conjugation of determinants, maps the stabilized ray for the empty family to the stabilized ray; on a surjective operator the determinant is $\lambda(\ker)$ and the conjugation is $\lambda$ of the tangent map. (Gap G.68 closed.)

**Theorem 6.6.18 (preprint 6.41) in the book's shape** Chapter6/OrientationMainClean · theorem_6_6_18. *Over a tame $X$, for a proper $f$ with tame germ data in general position at every zero and an orientation $o$ of $f$ (Definition 6.6.16 (preprint 6.40)), $S$ is a compact manifold with boundary with corners in general position with the natural orientation given by $o$* Chapter6/OrientationMainClean · IsNaturalSolutionOrientation: at every zero $o_x$ orients $T_xS = \ker f'(x)$; the orientations read in two charts correspond under the tangent map of the transition; and along every sc-smooth path of zeros in an orientable neighbourhood they form an orientation of the kernel line bundle. The three clauses are the three paragraphs above.

**$S$ as an oriented manifold** (V6.61). Mathlib has manifolds with corners but no orientations of them. A *tangent orientation* of a Mathlib manifold $M$ is an orientation of every tangent space, and it is *continuous* if, read in the charts of the maximal atlas, it is locally constant Chapter6/ManifoldOrientation · TangentOrientation.IsContinuous; the tangent maps of chart changes Chapter6/ManifoldOrientation · chartChangeDeriv have continuous, nowhere-zero Jacobian determinant, so this is well posed. An *oriented atlas* is a covering family of charts whose changes have positive Jacobian determinant Chapter6/ManifoldOrientation · IsOrientedAtlas; a continuous tangent orientation gives one whenever the positive charts cover $M$ Chapter6/ManifoldOrientation · isOrientedAtlas_positiveCharts. The solution set is a compact Mathlib manifold with corners, modelled on the quadrant of $\mathbb{R}^d$, $d$ the index Chapter9/MathlibSolutionSets · theorem_6_6_18_isManifold, from the corner structure of Theorem 3.1.26 (Part III). *Theorem* Chapter6/SolutionSetOrientation · theorem_6_6_18_oriented: *with the bump functions of §6.6 (preprint §6.5), $S$ carries a continuous tangent orientation $o_S$ which at every zero $x$, in every chart $c$ of $X$ at $x$, is $o_x$: the tangent map $\iota$ of $S \hookrightarrow X$ at $x$ is an isomorphism $T_xS \cong \ker f'(x)$, and a basis $b$ of $T_xS$ is positive for $o_S$ iff $\iota b_1 \wedge \cdots \wedge \iota b_d$ is positive for the tangent orientation of $o_x$. Step 1, the isomorphism.* The tangent map read in the corner coordinates $\kappa$ of $S$ and the chart $c$ is injective with values in $\ker f'(x)$, and both have dimension $d$ Chapter6/SolutionSetOrientation · solKerEquiv. *Step 2, the orientation at a point.* $o_S(x)$ is the pull-back of the tangent orientation along it, read in the corner chart of $S$ at $x$ and the chosen chart of $X$ at $x$ Chapter6/

SolutionSetOrientation · solOr Chapter6/SolutionSetOrientation · solOr_spec. *Step 3, the chart of $X$ plays no role*: two charts of $X$ change the isomorphism by the tangent map of the transition (chain rule), which also relates the two tangent orientations (two charts, above) Chapter6/SolutionSetOrientation · solReadOr_chartChange. *Step 4, continuity.* Along a path of $S$ in corner coordinates the wedges $\iota_q b_1 \wedge \cdots \wedge \iota_q b_d$ of a fixed basis are a continuous section of the kernel line bundle — the derivative of the inverse corner chart is continuous Chapter6/SolutionSetOrientation · isLineOrientation_solKer — and the tangent orientations of $o$ along the path are an orientation of the same bundle (the prolongation paragraph); two orientations of a line bundle over an interval that agree at one point agree everywhere, so $o_S$ is locally constant in the corner charts Chapter6/SolutionSetOrientation · isContinuous_solOr. (Gap G.67 closed.) *Hidden hypothesis:* the bump functions, which give the orientable neighbourhoods on which $o$ is continuous; the text assumes partitions of unity from §6.3 (preprint §6.2) on, which give them, and they hold on Hilbert models Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert. "A natural orientation" of the manifold is read as a continuous tangent orientation, not as an oriented atlas: for the quadrant model the two differ — $[0,1]$ has no atlas of quadrant charts with positive changes, since a chart at 0 increases and one at 1 decreases, and in dimension 0 every atlas is oriented while a tangent orientation is a sign at every point (Findings, Chapter 6, item 127). The same holds for an open part of the solution set, without properness: if every zero in an open $V$ is in general position with index $d$, $S \cap V$ is a Hausdorff manifold with corners with the continuous tangent orientation of $o$ Chapter6/SolutionSetOpen2021 · theorem_6_6_18_oriented_open; Theorem 6.7.1 (preprint 6.43) uses it for the index parts of $S_{f+s}$ and for the cobordism $S_F$ (§VI.27).

**The paragraph after the theorem** (V6.61). For $f$ not in general position the text perturbs by an sc$^+$-section $s$ supported near $S$ and orients $S_{f+s}$ from $o^f$, using $\mathrm{Lin}(f+s,x) = \mathrm{Lin}(f,x)$. This identity holds at every chart point Chapter6/PerturbedOrientation · ChartPoint.linSet_eq_of_sectionAdd: $(f+s)+t$ ranges over the same sections as $f+(s+t)$, and the sc$^+$-sections form a group. So $\mathrm{DET}(f+s,x) = \mathrm{DET}(f,x)$, the push-forwards of $f+s$ are those of $f$ (the sections $S$ of the push-forward data absorb $s$), and an orientation of $f$ is one of $f+s$ Chapter6/PerturbedOrientation · isSectionOrientation_sectionAdd Chapter6/PerturbedOrientation · SectionOrientation.sectionAdd; for $f+s$ proper and in general position, $S_{f+s}$ has the natural orientation coming from $o^f$ Chapter6/PerturbedOrientation · theorem_6_6_18_sectionAdd, and this applies to the perturbations of Theorem 5.3.10 (preprint 5.18) in its pointwise reading Chapter6/PerturbedOrientation · theorem_6_6_18_perturbation. (Gap G.69 closed.) *Readings:* tame germ data, the standing hypothesis of Chapter 5; the orientation $o$ exists iff $f$ is orientable (§VI.24) (Findings, Chapter 6, items 92, 93, 123, 127) (pp. 274–276 (preprint pp. 229–230)).

## VI.26 Remark 6.6.19 (preprint 6.42): the orientation of $\hat{f}$

The text: for an oriented $(f,o)$ and an sc$^+$-section $s$ of $Y \to X \times [0,1]$, $\hat{f}(x,t) = f(x) + s(x,t)$ has a natural orientation from $o$ and the standard orientation of $[0,1]$: for $L = (f-t)'(x)$ oriented by $h = (a_1 \wedge \cdots \wedge a_n) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, orient $\tilde{L}(a,b) = La$ by $\tilde{h} = (a_1 \wedge \cdots \wedge a_n \wedge e) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, $e$ the standard orientation of $[0,1]$ (the preprint's order; the 2021 edition puts $e$ first); $\tilde{L}$ lies in the linearization space of $\hat{f}$ at $(x,t)$ (p. 276 (preprint p. 230)).

**The linear algebra** (V6.62). For Fredholm $L$, $\tilde{L} = L \oplus 0 : E \times \mathbb{R} \to F$ Chapter6/ProductOrientation · extOp is Fredholm Chapter6/ProductOrientation · isFredholm_extOp with $\ker \tilde{L} = \ker L \times \mathbb{R}$ — one dimension more Chapter6/ProductOrientation · finrank_ker_extOp, the new direction $e = (0,1)$ Chapter6/ProductOrientation · extE — and the same range Chapter6/ProductOrientation · range_extOp and cokernel Chapter6/ProductOrientation · extCoker. The map $h \mapsto \tilde{h}$, $(a_1 \wedge \cdots \wedge a_n) \otimes d^* \mapsto ((a_1,0) \wedge \cdots \wedge (a_n,0) \wedge e) \otimes d^*$,

is an isomorphism $\det(L) \cong \det(\tilde{L})$ Chapter6/ProductOrientation · extDet Chapter6/ProductOrientation · extDet_wedge_tmul, the wedge extension of §VI.16 with one vector. In local determinant charts its coordinate is $(-1)^{\dim F/R(P)+\dim \operatorname{coker} L}$ times a continuous nowhere-zero function Chapter6/ProductOrientation · coord_extDet. Over the convex $\mathrm{Lin}(f,x)$, where $\dim \operatorname{coker} L$ can change parity, the orientation defined by $\tilde{h}$ at one $L$ could therefore differ from the one defined at another, as we read it. With the sign of §VI.14, $(-1)^{\dim \operatorname{coker} L}\, h \mapsto \tilde{h}$ Chapter6/ProductOrientation · extDetN has a coordinate continuous up to a sign constant on a chart Chapter6/ProductOrientation · coord_extDetN, agrees with the text's map when $L$ is onto Chapter6/ProductOrientation · extDetN_eq_of_surjective, and pulls continuous orientations back along $L \mapsto \tilde{L}$ Chapter6/ProductOrientation · isLineOrientation_extPull Chapter6/ProductOrientation · LineOrientation.extPull (Findings, Chapter 6, item 95). Transported along isomorphisms $A$, $\Gamma$, $\hat{L} = \Gamma^{-1}(L \oplus 0)A$ Chapter6/ProductOrientation · extConjOp gets the isomorphism $h \mapsto \tilde{h}$ followed by the conjugation of §VI.21 Chapter6/ProductOrientation · extConjDet; and for a preconnected $C$ and a simply connected, locally path-connected $\hat{C}$ containing the $\hat{L}$, an orientation $o$ of DET over a nonempty $C$ gives exactly one orientation $\hat{o}$ over $\hat{C}$ with $\hat{o}(\hat{L})$ the image of $o(L)$ for every $L \in C$ Chapter6/ProductOrientation · existsUnique_extConjOr.

**In charts.** $X \times [0,1]$ is read as $\mathbb{R} \times X$ (decision D2), with the parameter first; $s$ is an sc⁺-section $S$ of $\pi^*P$, and $\hat{f} = \tilde{f} + S$ with $\tilde{f}(t,y) = f(y)$. At the product chart point $(t,z)$ Chapter6/ProductOrientation · ChartPoint.prodPt the tangent space is $\mathbb{R} \times T_zO$ Chapter6/ProductOrientation · ChartPoint.mem_prodPt_T_iff and the fibre $K_z$ Chapter6/ProductOrientation · ChartPoint.mem_prodPt_K_iff. The text's $\tilde{L}(a,b) = La$ puts the $\mathbb{R}$-factor second, so $\hat{L} = \Gamma^{-1}(L \oplus 0)A$ with $A$ the swap of the factors and $\Gamma = 1$ Chapter6/ProductOrientation · ChartPoint.prodT Chapter6/ProductOrientation · ChartPoint.prodK Chapter6/ProductOrientation · ChartPoint.prodLin. Sc⁺-operators lift to the product Chapter6/ProductOrientation · ChartPoint.IsScPlusOp.prodPt, sections of $P$ lift to $t$-constant sections of $\pi^*P$ Chapter6/ProductOrientation · constLift, and so $\hat{L} \in \mathrm{Lin}(\hat{f},(t,x))$ for every $L \in \mathrm{Lin}(f,x)$ Chapter6/ProductOrientation · ChartPoint.prodLin_mem_linSet — the derivative in $t$ of a $t$-constant lift vanishes, and $S$ only contributes an sc⁺-operator. *An orientation $o$ of* $\mathrm{DET}(f,x)$ *gives exactly one orientation $\hat{o}$ of* $\mathrm{DET}(\hat{f},(t,x))$ *with $\hat{o}(\hat{L})$ the image of $o(L)$ for every $L$* Chapter6/ProductOrientation · ChartPoint.existsUnique_prodOr Chapter6/ProductOrientation · ChartPoint.prodOr Chapter6/ProductOrientation · ChartPoint.prodOr_apply, and $o \mapsto \hat{o}$ is injective Chapter6/ProductOrientation · ChartPoint.prodOr_injective (Findings, Chapter 6, item 96). *Hidden hypotheses:* $x$ smooth and $\mathrm{Lin}(f,x)$ nonempty (bump functions give it, §VI.1); only the orientation of $\mathrm{DET}(f,x)$ at the one point is used, not an orientation of $f$. That these pointwise orientations, in the 2021 convention, form an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40) is the last paragraph below.

**The 2021 convention,** $e$ **first** (V6.62). The 2021 edition prints $\tilde{h} = (e \wedge a_1 \wedge \cdots \wedge a_n) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, with $e$ first; the preprint puts it last. Moving $e$ across $n$ vectors costs $(-1)^n$ Chapter6/OrientationConvention2021 · wedge_snoc_eq_smul_wedge_cons. The 2021 map $h \mapsto \tilde{h}$ Chapter6/OrientationConvention2021 · extDetFirst Chapter6/OrientationConvention2021 · extDetFirst_wedge_tmul is $(-1)^{\mathrm{ind}\,L}$ times the sign-normalized map of the previous paragraph Chapter6/OrientationConvention2021 · extDetFirst_eq_indSign_smul_extDetN, because $(-1)^{\dim \ker L} = (-1)^{\mathrm{ind}\,L}(-1)^{\dim \operatorname{coker} L}$ Chapter6/OrientationConvention2021 · kerSign_eq_indSign_mul_stabSign. The index is locally constant on the Fredholm operators Chapter6/OrientationConvention2021 · eventually_fredholmIndex_eq Chapter6/OrientationConvention2021 · eventually_indSign_eq, and a line orientation times a locally constant sign is a line orientation, so *the text's own map pulls continuous orientations back, over any set of Fredholm operators* Chapter6/OrientationConvention2021 · isLineOrientation_extPullFirst. With the 2021 convention no sign has to be changed: the factor $(-1)^{\dim \operatorname{coker} L}$ of the preprint's map and the $(-1)^{\dim \ker L}$ of moving $e$ to the front combine to a locally constant sign. In charts, *an orientation $o$ of* $\mathrm{DET}(f,x)$

*gives exactly one orientation $\hat{o}$ of* $\mathrm{DET}(\hat{f},(t,x))$ *with* $\hat{o}(\hat{L}) = e \wedge o(L)$ *for every* $L \in \mathrm{Lin}(f,x)$ Chapter6/OrientationConvention2021 · ChartPoint.existsUnique_prodOrFirst Chapter6/OrientationConvention2021 · ChartPoint.prodOrFirst, and on $\mathrm{Lin}(f,x)$ the 2021 map is the preprint's times the one constant sign $(-1)^{\mathrm{ind}(f,x)}$ Chapter6/OrientationConvention2021 · ChartPoint.exists_extConjDetFirst_eq_sign_smul (Findings, Chapter 6, item 122).

**The orientation of $\hat{f}$** (V6.62) Chapter6/ProductSectionOrientation2021f · SectionOrientation.prodFirst. *Over a tame $X$ with bump functions, for $f$ with tame germ data, an orientation $o$ of $f$ (Definition 6.6.16 (preprint 6.40)) and an sc⁺-section $S$ of $\pi^*P \to \mathbb{R}^1 \times X$, the map $(t,x) \mapsto e \wedge o_x$ is an orientation of $\hat{f} = \tilde{f} + S$ in the sense of Definition 6.6.16 (preprint 6.40).* The text's "natural orientation", with its remark that the choices made in the construction are compatible, is this statement. *Step 1, the map.* $(t,x) \mapsto e \wedge o_x$ is defined in the product chart point over any chart point of $x$ Chapter6/ProductSectionOrientation2021f · prodFirstFun Chapter6/ProductSectionOrientation2021e · prodOrOf, and its reading in any chart of $\mathbb{R}^1 \times X$ is computed through the product chart Chapter6/ProductSectionOrientation2021f · toChartOr_prodFirstFun; the chart of $X$ plays no role, since the transfer of §VI.21 between two product chart points is the transfer between the chart points of $X$ with $e$ put in front Chapter6/ProductSectionOrientation2021b · ChartPoint.orTransfer_prodPt_prodOrFirst Chapter6/ProductSectionOrientation2021f · prodOrOf_orTransfer. *Step 2, the lifted data.* A push-forward datum of $f$ along a path $\gamma$ — sections $S$, $s_1,\dots,s_k$ and stabilized kernels $\Lambda$ (§§VI.14–VI.15) — lifts to one of $\tilde{f}$ along $q \mapsto (\tau(q),\gamma(q))$: the sections are lifted constant in $t$ Chapter6/ProductSectionOrientation2021c · liftProd Chapter6/ProductSectionOrientation2021c · isScPlusSection_liftProd, the linearization of a $t$-constant lift is $(\sigma,v) \mapsto g'(x)v$, and the stabilized kernels become $\mathbb{R} \times \Lambda$ Chapter6/ProductSectionOrientation2021c · SubspaceFamily.prodLine Chapter6/ProductSectionOrientation2021c · IsStabKernels.prodLift. *Step 3, no sign.* The stabilization isomorphism of $\hat{L}$ maps $e \wedge h$ to $\hat{e} \wedge J(\cdot)$ of the stabilization of $h$, $J$ the inclusion $v \mapsto (0,v)$ of the old kernel Chapter6/ProductSectionOrientation2021c · stabRayEquiv_extConjDetFirst Chapter6/ProductSectionOrientation2021 · detEquiv_stabIota_extDetFirst: in adapted bases, the appended vector $e$ is first on both sides of formula (91), so the computation is that of Proposition 6.4.15 (preprint 6.15) with one more kernel vector. Continuation commutes with $\hat{e} \wedge J(\cdot)$, proved directly in frames Chapter6/ProductSectionOrientation2021c · SubspaceFamily.cont_prodLine, and the rays read from the product orientation in $\lambda((\mathbb{R} \times \Lambda)_q)$ are $\hat{e} \wedge J$ of those read from $o$ Chapter6/ProductSectionOrientation2021d · ChartPoint.prodOrFirst₀ Chapter6/ProductSectionOrientation2021e · IsStabKernels.orRay_prodLift. Hence *the lifted push-forward maps $e \wedge o$ to $e\wedge$ (the push-forward of $o$)* Chapter6/ProductSectionOrientation2021e · IsPush.prodLift. *Step 4.* An orientation of $f$ is carried by every push-forward along paths in orientable neighbourhoods (§VI.24); by Step 3 so is $e \wedge o$ along the lifted paths, and every path of $\mathbb{R}^1 \times X$ in a product chart is such a lift, so $e \wedge o$ is an orientation of $\tilde{f}$ Chapter6/ProductSectionOrientation2021f · isSectionOrientation_prodFirstFun Chapter6/ProductSectionOrientation2021f · SectionOrientation.prodFirst₀, and of $\tilde{f} + S$ by the paragraph after Theorem 6.6.18 (preprint 6.41) (§VI.25). (Gap G.66 closed; Findings, Chapter 6, items 96, 97, 122.) *Readings:* $X \times [0,1]$ is read as $\mathbb{R}^1 \times X$ with the parameter first (decision D2; gap G.65 in part), and the hypotheses — $X$ tame with sc-smooth bump functions, tame germ data — are those of Theorem 6.6.18 (preprint 6.41) in §VI.25. On a product end of the solution set of $\hat{f}$ this orientation is the product orientation, the time vector first (§VI.27).

# VI.27 §6.7 (preprint §6.6) and Theorem 6.7.1 (preprint 6.43): the invariant $\Psi_{(f,o)}$

§6.7 (preprint §6.6) (p. 276 (preprint p. 230)) recalls, for a proper oriented $(f,o)$ over a tame $X$ with sc-smooth bump functions and an auxiliary norm $N$, the neighbourhood $U$ of $S_f$ of Theorem 5.3.10 (preprint 5.18) and the sc$^+$-perturbations $s$ supported in $U$ with $N(s) \le 1$ for which $f+s$ is in general position at its zeros; it calls them *admissible for the pair* $(N,U)$. Theorem 6.7.1 (preprint 6.43) (p. 277 (preprint p. 231)): *there is a well-defined map assigning to a proper oriented $(f,o)$ a linear map $\Psi_{(f,o)} : H^*_{dR}(X,\partial X) \to \mathbb{R}$ such that* (1) *for every auxiliary norm $N$ and every $s$ admissible for $(N,U)$, $S_{f+s}$ is an oriented compact manifold with boundary with corners and $\Psi_{(f,o)}([\omega,\tau]) = \int_{S_{f+s}} \omega - \int_{\partial S_{f+s}} \tau$, the integrals being zero when the dimensions do not agree;* (2) $\Psi_{(f_0,o_0)} = \Psi_{(f_1,o_1)}$ *for a proper oriented homotopy between $(f_0,o_0)$ and $(f_1,o_1)$.* The proof (pp. 277–279 (preprint pp. 231–233)) defines $\Psi$ from one admissible $s_0$, joins two admissible sections by a homotopy $s_t$ whose solution set $S_F$ over $X\times[0,1]$ is an oriented compact manifold with boundary with corners, decomposes $\partial S_F = S_0 \cup S_1 \cup \tilde{S}$, and compares the two sides by Stokes' theorem applied to an extension $\bar\omega$ of $\omega$ and to $\tau$; part (2) is "the same homotopy argument". For $\partial X = \emptyset$ (p. 279 (preprint p. 233)), $\Psi([\omega]) = \int_{S_{f+s}} \omega$ for $[\omega] \in H^n_{dR}(X)$, $n$ the index, and zero in the other degrees.

**The theorem formalized** (V6.76) `Chapter6/Theorem671Final2021 · theorem_6_7_1`. *Let $P : Y \to X$ be a strong bundle over a tame M-polyfold with sc-smooth bump functions. There is $\Psi$ assigning to every proper sc-Fredholm section $f$ with tame germ data and orientation $o$ (Definition 6.6.16 (preprint 6.40)) linear maps $\Psi_{(f,o)} : H^k_{dR}(X,\partial X) \to \mathbb{R}$, $k \ge 0$, such that* (1) *for every auxiliary norm $N$ there are open $U \supseteq S_f$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \le C\}$ compact for every $C>0$; for every such $U$, $S_{f+s}$ is compact for every allowable $s$, admissible sections exist, and for every admissible $s$, $S_{f+s}$ is compact and the union of its index parts $S_k$, each $S_k$ a compact manifold with boundary and corners of dimension $k$ oriented by $o$, and $\Psi_{(f,o)}([\omega,\tau]) = \int_{S_k}\omega - \int_{\partial S_k}\tau$ in degree $k$;* (2) $\Psi_{(f_0,o_0)} = \Psi_{(f_1,o_1)}$ *in every degree for a proper oriented homotopy.* No hypothesis beyond the text's is used: the proof below assumes nothing of its own steps. The readings (Findings, Chapter 6, items 128–133, 136):

- *R-a, the cohomology.* $H^k_{dR}(X,\partial X)$ is the relative cohomology of Part IV: pairs $(\omega,\tau)$ of an sc-form on $X$ and a form on the faces of $X$, with $d(\alpha,\beta) = (d\alpha, j^*\alpha - d\beta)$ `Chapter4/BoundaryForms2021 · MPolyfold.RelForms` `Chapter4/BoundaryForms2021 · MPolyfold.relD`. The integrals are integrals of Mathlib forms over the Mathlib manifolds $S_k$ and their faces (Part IX).
- *R-b, the neighbourhood $U$ with a margin.* The text takes $U$ from Theorem 5.3.10 (preprint 5.18), for one $N$. It is read existentially and for every $N$, with the margin "$\mathrm{cl}\{x \in U \mid N(f(x)) \le C\}$ compact for every $C>0$", which the cobordism below needs for $C=2$ (item 128).
- *R-c, "zero if the dimensions do not agree".* The index of $f$ need not be constant. Degree $k$ is paired with the part $S_k$ of $S_{f+s}$ where the index is $k$ (item 129).
- *R-d, the proper oriented homotopy*, which the text does not define. It is read through Remark 6.6.19 (preprint 6.42): an sc$^+$-section $S$ of $\pi^*P \to \mathbb{R}^1 \times X$ with $S(0,\cdot)=0$, $f_1 = f_0 + S(1,\cdot)$ and $o_1$ the orientation $o_0$ carried to $f_1$, compact zeros of $\tilde{f_0}+S$ over $t\in[0,1]$, and the properness clause of Definition 5.2.1 (3) on that slab `Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy` `Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopyBetween`; then $f_0$, $f_1$ are proper `Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy.isProper_zero` `Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy.isProper_one`, and $S=0$ is one `Chapter6/ProperHomotopy2021 · witness_isProperOrientedHomotopyBetween` (item 130).

- *R-e, the extension of $\omega$.* The proof's $\bar\omega = p_1^*\omega \wedge p_2^*dt$ has degree $k+1$, one more than the dimension $k$ of $\partial S_F$, while the comparison needs $\int_{S_0}\omega$ and $\int_{S_1}\omega$; as we read it, the argument goes through with $p_1^*\omega$ and $p_1^*\tau$ in their place, and that is what is formalized (item 131).
- *R-f, signs.* The boundary orientation is outward vector first, which gives Stokes' theorem with the sign $+$, as the text uses it; $S_F$ is oriented by $e \wedge o$, time first (item 132). The cut-off route below does not need the signs of the pieces of $\partial S_F$.

**Step 1: the neighbourhood and the admissible sections** (V6.76). A proper $f$ has compact $S_f$ (Theorem 5.2.2 (preprint 5.5), §V.6). *For an auxiliary norm $N$ and every open $V \supseteq S_f$ there is an open $S_f \subseteq U \subseteq V$ with* $\mathrm{cl}\{x \in U \mid N(f(x)) \le C\}$ *compact for every $C > 0$* `Chapter5/CylinderPerturbation2021 · exists_adapted_margin`: for compact $S_f$, Theorem 5.2.2 (preprint 5.5) gives an open $U_0 \supseteq S_f$ with this property for every $C > 0$ (§V.6), and $U = U_0 \cap V$ keeps it. With such a $U$, *$S_{f+s}$ is compact for every $s$ allowable for $(N, U)$* `Chapter6/Theorem671_2021 · isCompact_solutionSet_sectionAdd_of_margin`: off $U$ the zeros of $f+s$ are zeros of $f$, which lie in $U$; in $U$, $f(x) = -s(x)$ gives $N(f(x)) \le 1$; so $S_{f+s}$ is a closed part of a compact set (Remark 5.3.4 (preprint 5.12)). *Admissible* is the text's notion: $s$ is allowable for $(N, U)$ — supported in $U$, $N(s) \le 1$ — and $f+s$ is in general position (Definition 5.3.9 (preprint 5.17)) at every zero `Chapter6/InvariantSingle2021 · StrongBundle.IsAdmissible` (gap G.55 closed). They exist by a relative form of Theorem 5.3.10 (preprint 5.18) `Chapter5/CylinderPerturbation2021 · theorem_5_3_10_relative_hyp`: *for $G$ with tame germ data, $N$ an auxiliary norm, $K \subseteq W$ with $K$ compact and $W$ open, $G$ in general position at every zero outside $K$ and* $\mathrm{cl}\{x \in W \mid N(G(x)) \le 1\}$ *compact, there is for every $\varepsilon > 0$ an sc$^+$-section $p$ supported in $W$ with $N(p) \le \varepsilon$ and $G + p$ in general position at every zero.* Its proof is that of §V.20: finitely many sections make $G(x) + \sum \lambda_i s_i(x)$ general at the points $(0, y)$ near the compact $K \cap S_G$ `Chapter5/CylinderPerturbation2021 · exists_sections_isGeneralPositionGerm_near`, and at the other zeros, where $G$ is already general `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramSection_of_isGeneralPositionGerm`; Sard's theorem and Theorem 4.3.1 give a regular value $\lambda$ of norm as small as needed `Chapter5/CylinderPerturbation2021 · exists_null_isGeneralPositionGerm_paramSlice`; off $W$ the perturbation vanishes near the zero, and general position there is that of $G$ `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_congr_nhds`. The text's theorem is the case $K = S_G$, $W = U$; here $S_G$ need not be compact, which the cobordisms need. With $K = S_f$ it gives an admissible section for $(N, V)$ inside any $V$ `Chapter5/CylinderPerturbation2021 · exists_admissible` `Chapter6/InvariantSingle2021 · StrongBundle.exists_isAdmissible_nbhd`, and, applied to the maximum of finitely many norms `Chapter5/CylinderPerturbation2021 · IsAuxNorm.max` `Chapter5/CylinderPerturbation2021 · allowableSections_max_subset`, one section admissible for two pairs $(N_0, U_0)$, $(N_1, U_1)$ at once `Chapter5/CylinderPerturbation2021 · exists_admissible_subset` `Chapter5/CylinderPerturbation2021 · exists_admissible_two` — the step that compares two norms.

**Step 2: the index parts as oriented manifolds** (V6.76). For admissible $s$, $S_k$ is the set of zeros of $f+s$ at which $f$ has index $k$ `Chapter6/InvariantSingle2021 · StrongBundle.solPart`. The index is locally constant (Proposition 6.2.2 (preprint 6.2), §III.49), so $S_k$ is open and closed in $S_{f+s}$ `Chapter6/InvariantSingle2021 · StrongBundle.isClosed_solPart`, compact `Chapter6/InvariantSingle2021 · StrongBundle.isCompact_solPart`, and $S_{f+s} = \bigcup_k S_k$ `Chapter6/InvariantSingle2021 · StrongBundle.solutionSet_eq_iUnion_solPart`. Theorem 6.6.18 (preprint 6.41) for an open part (§VI.25), applied to $f+s$ with the orientation $o$ carried to it (§VI.25, the paragraph after the theorem), makes $S_k$ a compact Mathlib manifold of dimension $k$ `Chapter6/InvariantSingle2021 · isManifold_solPart` with a continuous tangent orientation which is the Fredholm one at every point `Chapter6/InvariantSingle2021 · solPartOr` `Chapter6/InvariantSingle2021 · solPartIsOrientedManifold`. For a constant index $n$, $S_n = S_{f+s}$ and $S_k = \emptyset$ for $k \ne n$ `Chapter6/InvariantSingle2021 · StrongBundle.solPart_eq_solutionSet` `Chapter6/InvariantSingle2021 · StrongBundle.solPart_eq_empty_of_ne`: the text's clause "zero if the dimensions do not agree".

**Step 3: $\Psi_s$ for one admissible section** (V6.76). On relative forms of degree $k$, $\Psi_s(\omega,\tau)=\int_{S_k}\omega-\int_{\partial S_k}\tau$ Chapter6/InvariantSingle2021 · solIntegral Chapter6/InvariantSingle2021 · solBoundaryIntegral Chapter6/InvariantSingle2021 · psiForms: $\omega$ restricted to $S_k$ is a Mathlib form Chapter9/MathlibFormsBridge2021 · MPolyfold.formsInftyRestrict, and $\tau$ is read on the faces of $S_k$, which lie in faces of $X$ (Theorem 4.2.11, §IV) Chapter9/SolutionFaceForms2021 · MPolyfold.solFaceForm. *$\Psi_s$ vanishes on exact relative forms* Chapter6/InvariantSingle2021 · psiForms_relD: for $d(\alpha,\beta)=(d\alpha, j^*\alpha-d\beta)$,

$$\int_S d\alpha-\int_{\partial S} j^*\alpha+\int_{\partial S} d\beta=0,$$

by Stokes' theorem on the compact manifold with corners $S_k$ Chapter9/MathlibFormsBridge2021 · MPolyfold.stokes_formsInftyRestrict, the face form of $j^*\alpha$ being $\alpha$ restricted to the faces Chapter9/SolutionFaceForms2021 · MPolyfold.solFaceForm_jStar_integral, and $\int_{\partial S} d\beta = 0$ since the face form of $d\beta$ is $d$ of the face form Chapter9/SolutionFaceForms2021 · MPolyfold.solFaceForm_d and exact forms on the compact faces integrate to zero Chapter4/ManifoldFaceForms2021 · ChartFaceForm.integral_d_eq_zero_of_compactSpace. So $\Psi_s$ descends to $H^k_{dR}(X,\partial X)$ Chapter6/InvariantSingle2021 · psiAt Chapter6/InvariantSingle2021 · psiAt_mk. It depends on $s$ only, not on $(N,U)$ Chapter6/InvariantSingle2021 · psiAt_congr_admissible. In degree 0 it is a signed count, $\Psi_s([\omega]) = \sum_{x\in S_0}\varepsilon(x)\,\omega(x)$ Chapter6/InvariantSingle2021 · psiAt_mk_zero_eq_sum Chapter6/InvariantSingle2021 · ManifoldForm.integral_eq_sum_of_dim_zero, and it is 0 when $S_{f+s}=\emptyset$ Chapter6/InvariantSingle2021 · psiAt_of_eq_empty. *An integrand.* A form $a$ on $S_k$ with $a=\omega|_{S_k}+d\gamma$, $\gamma$ a compactly supported $C^\infty$ form extending $-\tau|_{\partial S}$ ($a=\omega|_{S_0}$ in degree 0), is an *integrand* of $(\omega,\tau)$ Chapter6/InvariantCobordism2021 · IsPsiIntegrand; integrands exist, by extension of face data on the compact $S_k$ Chapter6/InvariantCobordism2021 · exists_isPsiIntegrand, and $\Psi_s([\omega,\tau]) = \int_{S_k} a$, since $\int_S d\gamma = \int_{\partial S} j^*\gamma = -\int_{\partial S}\tau$ Chapter6/InvariantCobordism2021 · IsPsiIntegrand.psiAt_eq. This is the form $\omega-d\beta$ with $j^*\beta=\tau$ of the text's computation, written on $S_k$.

**Step 4: the cobordism data of part (1)** (V6.76). The text joins $s_0$ and $s_1$ by an sc$^+$-homotopy $s_t$ (Remark 5.3.8 (preprint 5.16)) with $F(x,t)=f(x)+s_t(x)$ in general position. Here $X\times[0,1]$ is $\mathbb{R}^1\times X$ with the parameter first (decision D2), and the homotopy is stationary near the ends. *Cobordism data* for $f$, $\sigma_0$, $\sigma_1$ are an sc$^+$-section $\widehat{S}$ of $\pi^*P$ with $\widehat{S}(t,\cdot)=\sigma_0$ for $t\le 1/8$ and $\sigma_1$ for $t\ge 7/8$, $f+\sigma_i$ in general position at every zero, $F=\tilde{f}+\hat{S}$ in general position at every zero, and the zeros of $F$ over every $[a,b]$ compact Chapter6/ProperHomotopy2021 · IsCobordismData. *For $U$ with the margin and $s_0$, $s_1$ admissible for $(N,U)$, cobordism data for $f$, $s_0$, $s_1$ exist, with $\widehat{S}=0$ off $U$ and $N(\widehat{S})\le 2$* Chapter5/CylinderPerturbation2021 · exists_cobordism Chapter6/ProperHomotopy2021 · exists_isCobordismData. *Proof.* Let $\chi$ be a smooth cut-off, 0 for $t\le 3/16$ and 1 for $t\ge 13/16$ Chapter5/CylinderPerturbation2021 · cylCut, and $\hat{s}=(1-\chi)\tilde{s}_0+\chi\tilde{s}_1$ Chapter5/CylinderPerturbation2021 · cylInterp Chapter5/CylinderPerturbation2021 · isScPlusSection_cylInterp, with $N(\hat{s})\le 1$. $G=\tilde{f}+\hat{s}$ is $(f+s_0)\tilde{}$ for $t<3/16$ and $(f+s_1)\tilde{}$ for $t>13/16$, where it is in general position at its zeros, since the lift of a section in general position is Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramLift Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramLift_add_of_eq. Let $K$ be the zeros of $G$ over $[3/16,13/16]$, compact because they lie in $[3/16,13/16]\times \mathrm{cl}\{x\in U \mid N(f(x))\le 1\}$, and $W=(1/8,7/8)\times U$. Where $N(G)\le 1$ on $W$, $N(f)\le N(G)+N(\hat{s})\le 2$; this is where the margin $C=2$ is used. The relative Theorem 5.3.10 (preprint 5.18) (Step 1) for $G$, the pulled-back norm, $K$, $W$ and $\varepsilon=1$ gives $p$, and $\widehat{S}=\hat{s}+p$: it equals $s_0$ for $t\le 1/8$ and $s_1$ for $t\ge 7/8$, vanishes off $U$, has $N(\widehat{S})\le 2$, and $\tilde{f}+\widehat{S}$ is in general position at every zero; its zeros over $[a,b]$ lie in $[a,b]\times\mathrm{cl}\{x\in U\mid N(f(x))\le 2\}$, hence are compact Chapter5/CylinderPerturbation2021 · isCompact_slab_paramLift_add. □

**Step 5: the cobordism data of part (2)** (V6.76) Chapter6/ProperHomotopy2021 · exists_homotopy_cobordism.

*For a proper homotopy $S$ from $f_0$ to $f_1 = f_0 + S(1,\cdot)$, an auxiliary norm $N$, a neighbourhood $W$ of the zeros over the slab with* $\mathrm{cl}\{w \in W \mid t \in [0,1],\ N(F(w)) \le 2\}$ *compact* Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy.exists_slab_nbhd, *$s_0$ admissible for $f_0$ and $(N,U_0)$ and $s_1$ for $f_1$ and $(N,U_1)$ with $\{0\}\times U_0$, $\{1\}\times U_1 \subseteq W$, there are cobordism data for $f_0$, $s_0$ and $S(1,\cdot)+s_1$.* So $\tilde f_0 + \hat S$ is $f_0+s_0$ for $t \le 1/8$ and $f_1+s_1$ for $t \ge 7/8$. *Proof.* $\hat S = S\circ(\chi\times 1) + w_0\tilde s_0 + w_1\tilde s_1 + p$, with a time change $\chi$, 0 for $t \le 3/8$ and 1 for $t \ge 5/8$ Chapter6/ProperHomotopy2021 · homTime, a weight $w_0$ that switches $s_0$ off over $[3/16, 5/16]$, before the homotopy moves, and a weight $w_1$ that switches $s_1$ on over $[11/16, 13/16]$, after it has stopped Chapter6/ProperHomotopy2021 · homWeightZero Chapter6/ProperHomotopy2021 · homWeightOne Chapter6/ProperHomotopy2021 · homInterp, and $p$ from the relative Theorem 5.3.10 (preprint 5.18) on $(1/8, 7/8)\times \chi^{-1}(W)$, as in Step 4. The zeros lie over the projection of $\mathrm{cl}\{w \in W \mid t \in [0,1],\ N(F(w)) \le 2\}$, which is compact. □ The slab neighbourhood and the supports of $s_0$, $s_1$ in it are hypotheses of this step only; they can be met inside any given neighbourhoods of $S_{f_0}$ and $S_{f_1}$ Chapter6/ProperHomotopy2021 · exists_homotopy_cobordism_data, so nothing is added to the theorem.

**Step 6: the cobordism and its ends** (V6.76). Let $F = \tilde f_0 + \hat S$ for cobordism data. The index-$(k+1)$ part $S_F^{k+1}$ of its solution set Chapter6/InvariantCobordism2021 · cobPart is, by Theorem 6.6.18 (preprint 6.41) for an open part over the tame $\mathbb{R}^1 \times X$, a $(k+1)$-dimensional submanifold with a corner structure Chapter6/InvariantCobordism2021 · isSubmanifoldOfDim_cobPart Chapter6/InvariantCobordism2021 · hasCornerStructureOfDim_cobPart, σ-compact Chapter6/InvariantCobordism2021 · sigmaCompactSpace_cobPart, with the smooth time function $t$ Chapter6/InvariantCobordism2021 · cobTime whose slabs $\{t \in [c,d]\}$ are compact Chapter6/InvariantCobordism2021 · isCompact_cobSlab. *The ends.* Over $t \le 1/8$ the solution set of $F$ is $\{t \le 1/8\}\times S_{f+\sigma_0}$, and over $t \ge 7/8$ it is $\{t \ge 7/8\} \times S_{f+\sigma_1}$ Chapter6/InvariantProductEnds2021 · solutionSet_left_eq_prod Chapter6/InvariantProductEnds2021 · solutionSet_right_eq_prod, since $F$ is the $t$-constant lift there. *The index on the ends*: at a zero $(t,x)$ where $F$ is the lift of $g = f + \sigma_i$ and in general position, $F'(t,x)$ has index $k+1$ exactly when $g'(x)$ has index $k$ Chapter6/InvariantProductEnds2021b · isLinOfIndex_end_iff: the lifted linearization is $(\sigma, v) \mapsto g'(x)v$, with one more kernel dimension, and the index of $F'$ is unique. So $S_F^{k+1}$ over $(0,1/8)$ is $(0,1/8)\times S_k(f+\sigma_0)$, and over $(7/8,1)$ it is $(7/8,1)\times S_k(f+\sigma_1)$ Chapter6/InvariantCobordism2021 · cobPart_end_left Chapter6/InvariantCobordism2021 · cobPart_end_right Chapter6/InvariantCobordism2021 · cobEndLeft Chapter6/InvariantCobordism2021 · cobEndRight. *As manifolds*: a *product end* of a $(k+1)$-manifold $M$ over a $k$-manifold $N$ is an open part identified with $(a,b)\times N$ such that the product charts $(t,y) \mapsto (t-a, e(y))$ belong to the maximal atlas of $M$ Chapter6/InvariantProductEnds2021 · ProductEnd. The corner coordinates of $(a,b)\times A$ are the products of those of $A$ with the time coordinate Chapter6/InvariantProductEnds2021b · MPolyfold.CornerCoords.prodRn Chapter6/InvariantProductEnds2021b · isChart_prodEndChart, and they agree with the charts of the identification Chapter6/InvariantProductEnds2021b · endChart_eqOnSource, so the two ends are product ends Chapter6/InvariantProductEnds2021b · productEndOfSub Chapter6/InvariantProductEnds2021b · solProductEnd. *Fubini on an end*: if $M$ carries the product orientation $e \wedge o_N$ on the end, then for $h$ supported in $(a,b)$, $\int_M h(t)\,dt \wedge p_1^*\alpha = (\int_a^b h)\int_N \alpha$ Chapter6/InvariantProductEnds2021 · ProductEnd.IsEndProductOrientation Chapter6/InvariantProductEnds2021 · ProductEnd.integral_end_fubini.

**Step 7: the orientation of $S_F$** (V6.76). The text orients $S_F$ "from the orientation $o$ of $f$ and the standard orientation of $[0,1]$". This is Remark 6.6.19 (preprint 6.42) as an orientation of $F$ (§VI.26): $e \wedge o_0$ is an orientation of $F = \tilde f_0 + \hat S$, and Theorem 6.6.18 (preprint 6.41) for an open part turns it into a continuous tangent orientation of $S_F^{k+1}$ Chapter6/CobordismOrientation2021 · cobOr Chapter6/CobordismOrientation2021 · isContinuous_cobOr. What the comparison needs is its value on the ends: *on $(0,1/8)\times S_k(f+\sigma_0)$ and $(7/8,1)\times S_k(f+\sigma_1)$ it is the product orientation $e \wedge o_S$, time first, with no sign* Chapter6/CobordismOrientationEnds2021d · isEndProductOrientation_cobOr — the step the text leaves to

"taking the orientations into account". *Proof.* Let $(t,x)$ be a point of an end, $g = f + \sigma_i$. (a) *The linearization.* $F$ and the $t$-constant lift $\tilde{g}$ agree near $(t,x)$, and two sc-Fredholm sections that agree near a common zero have the same linearization there Chapter6/CobordismOrientationEnds2021c · ChartPoint.linZero_congr_of_eventuallyEq; that of $\tilde{g}$ in the product chart point is $\hat{L}(\sigma,v) = g'(x)v$ Chapter6/CobordismOrientationEnds2021c · ChartPoint.linZero_prodPt_paramLift, so $\ker F'(t,x) = \mathbb{R} \times \ker g'(x)$. (b) *The orientation of the kernel.* For a surjective $T$ and $\hat{T} = \Gamma^{-1}(T \oplus 0)A$, the kernel ray of the 2021 image $e \wedge r$ of a ray $r$ of $\det T$ is $\hat{e} \wedge J(\cdot)$ of the kernel ray of $r$ Chapter6/CobordismOrientationEnds2021b · surjRay_extConjDetFirst (the stabilization by $\mathbb{R}^0$ of §VI.25 with one more vector in front); in coordinates $\hat{\iota}(s,w) = (s,\iota w)$ of the kernels, the pulled-back orientation is the standard one of $\mathbb{R}^{k+1}$ iff the one of $\ker T$ is the standard one of $\mathbb{R}^k$ Chapter6/CobordismOrientationEnds2021b · pullOr_surjRay_extConjDetFirst_iff Chapter6/CobordismOrientationEnds2021 · pullOr_wedgeExtLeft_eq_std_iff, the time coordinate being first on both sides. In chart points this is Chapter6/CobordismOrientationEnds2021c · ChartPoint.pullOr_tangentOrOf_prodOrFirst₀_iff, and the tangent orientation of $e \wedge o$ in a product chart is that of the product orientation of $o$ read in the chart of $X$ Chapter6/CobordismOrientationEnds2021c · tangentOr_toChartOr_prodFirst. (c) *The tangent map.* In the corner coordinates of the product end, the tangent map of $S_F \hookrightarrow \mathbb{R}^1 \times X$ is $(s,w) \mapsto (s, T\iota(w))$, $T\iota$ that of $S \hookrightarrow X$ Chapter6/CobordismOrientationEnds2021 · MPolyfold.derivIn_prodRn Chapter6/CobordismOrientationEnds2021c · modelDeriv_prodRn. (d) By (a)–(c) the readings of the orientation of $S_F$ in the product coordinates and of $o_S$ in the coordinates of $S$ correspond Chapter6/CobordismOrientationEnds2021d · solReadOrOpen_prodFirst_iff, which is the product form on the end. □ The orientation on the end is that of $S_k(g)$ for the orientation of $g = f_0 + \sigma_i$ carried from $o_0$. *Matching the ends.* In part (1), $o_1 = o_0$ and both ends carry $o_0$ carried to $f + s_i$. In part (2) the right end carries $o_0$ carried to $f_0 + S(1,\cdot) + s_1 = f_1 + s_1$, while $\Psi_{s_1}(f_1, o_1)$ uses $o_1 = o_0$ carried to $f_1$, then carried to $f_1 + s_1$; the two have the same values at every point of every chart Chapter6/CobordismOrientationEnds2021d · IsInducedSectionOrientation Chapter6/CobordismOrientationEnds2021d · IsInducedSectionOrientation.sectionAdd, because $\mathrm{Lin}(g+s,x) = \mathrm{Lin}(g,x)$ (§VI.25). Hence the orientation of the cobordism is of product form on both ends, for both parts Chapter6/CobordismOrientationEnds2021d · cobordismOrientation_of_induced Chapter6/CobordismOrientation2021 · cobordismOrientation_of_ends Chapter6/CobordismOrientationEnds2021d · cobordismOrientationHyp (Findings, Chapter 6, item 134).

**Step 8: the relative form on the cobordism** (V6.76). Fix a closed relative form $(\omega,\tau)$ of degree $k$ and integrands $a_i = \omega|_{S_i} + d\gamma_i$ on the ends $S_i = S_k(f+\sigma_i)$ (Step 3). The text's form on $S_F$ is $p_1^*\omega - d\beta$ with $j^*\beta = p_1^*\tau$ on $\partial S_F$. *Why $p_1^*\tau$ lives on the faces of $S_F$*, which the text does not say: the degeneracy index of $S_F$ is that of $\mathbb{R}^1 \times X$ (Theorem 4.2.11 (2)) Chapter6/CobordismRelativeForms2021b · degIndex_cobPart, and that of $\mathbb{R}^1 \times X$ at $(t,x)$ is $d_X(x)$ Chapter4/BoundaryFormsPullback2021 · MPolyfold.CornerCoords.zero_coord_face_along, so $p_1$ maps the faces of $S_F$ into faces of $X$ and $\tau$ pulls back along it Chapter4/BoundaryFormsPullback2021 · MPolyfold.faceAlongData Chapter6/CobordismRelativeForms2021b · MPolyfold.pullOver Chapter6/CobordismRelativeForms2021b · MPolyfold.faceRestriction_pullOver (item 136 (1)). *The form* Chapter6/CobordismRelativeForms2021b · exists_closed_form_of_ends: *on a part $A'$ of $\mathbb{R}^1 \times X$ which is $T_0 \times A_0$ over the open time set $T_0$ and $T_1 \times A_1$ over $T_1$, with compact slabs, and cut-offs $\rho_0$, $\rho_1$ in time supported over $T_0$, $T_1$ with $\rho_0 + \rho_1 = 1$ outside a compact interval, there is a closed $C^\infty$ form $A = p_1^*\omega|_{A'} + d\gamma$ with vanishing face restriction which reads as the pull-back of $\omega|_{A_i} + d\gamma_i$ where $\rho_i = 1$.* Here $\gamma$ extends $-p_1^*\tau$ on the faces and is $p_1^*\gamma_i$ where $\rho_i = 1$: $\gamma = \rho_0\, p_1^*\gamma_0 + \rho_1\, p_1^*\gamma_1 + (1-\rho_0-\rho_1)\gamma_{\mathrm{mid}}$, $\gamma_{\mathrm{mid}}$ an extension of the face data cut off in time Chapter6/CobordismRelativeForms2021 · exists_glued_extension. $A$ is closed since $dp_1^*\omega = p_1^*d\omega = 0$, and its face restriction is $p_1^*j^*\omega + d(-p_1^*\tau) = p_1^*(j^*\omega - d\tau) = 0$. $p_1^*\omega$ and $p_1^*\gamma_i$ read as pull-backs on the product ends Chapter6/InvariantProductEnds2021b · MPolyfold.isEndPullback_

formsInftyRestrict_snd Chapter6/CobordismRelativeForms2021 · ProductEnd.isEndPullback_of_pullAt, and so does $dp_1^*\gamma_i$ Chapter6/CobordismRelativeForms2021 · ProductEnd.IsEndPullback.extDeriv. *The inner ends* (item 136 (2)). The gluing needs room in time on both sides of the region where $\rho_i$ falls from 1 to 0, inside the product end. The cobordism data fix the slices only for $t \le 1/8$ and $t \ge 7/8$, so $\rho_0 = 1 - \mathrm{cutRise}(1/16, 1/8)$ and $\rho_1 = \mathrm{cutRise}(7/8, 15/16)$ are used, and $A$ reads as $p_1^*a_i$ on the *inner ends* $(0, 1/16) \times S_0$ and $(15/16, 1) \times S_1$ Chapter6/CobordismRelativeForms2021b · cobordismRelativeFormsInner_succ Chapter6/CobordismRelativeForms2021b · cobEndLeftIn Chapter6/CobordismRelativeForms2021b · cobEndRightIn; the orientation is still of product form there Chapter6/CobordismRelativeForms2021 · ProductEnd.IsEndProductOrientation.of_subEnd. In degree 0, $A = p_1^*\omega|_{S_F}$, a closed function on the 1-dimensional $S_F^1$ vanishing at its boundary points Chapter6/CobordismRelativeForms2021b · cobordismRelativeFormsInner_zero. So the relative form exists for all cobordism data Chapter6/CobordismRelativeForms2021b · cobordismRelativeFormsInner. The statement on the full ends Chapter6/InvariantCobordism2021 · CobordismRelativeForms would need a smooth extension across $t = 1/8$, and is not used.

**Step 9: the cut-off identity** (V6.76) Chapter6/InvariantCobordism2021 · integral_eq_of_cutCobordism. In place of the decomposition $\partial S_F = S_0 \cup S_1 \cup \tilde{S}$, which would need face integrals over $\tilde{S} \subseteq \partial X \times (0,1)$ and a Fubini theorem for face forms, the proof cuts off in time (item 133). *Let the $(k+1)$-manifold $M$, with a continuous orientation and a smooth time function $t$ with compact slabs, have cut ends over $N_0$ on $t^{-1}(a_0, b_0)$ and over $N_1$ on $t^{-1}(a_1, b_1)$ — product ends with the product orientation, on which the closed form $A$ with vanishing face restriction reads as $p_1^*\alpha_i$* Chapter6/InvariantCobordism2021 · ProductEnd.IsCutEnd Chapter6/InvariantCobordism2021 · HasCutEnd *— or an empty $N_i$ and no points over $(a_i, b_i)$. Then $\int_{N_0} \alpha_0 = \int_{N_1} \alpha_1$. Proof.* Let $g_i(t) = \mathrm{cutRise}(a_i, b_i)$, 0 before the first third of $(a_i, b_i)$ and 1 after the second, so that $g_i'$ is supported in $(a_i, b_i)$ with $\int g_i' = 1$ Chapter6/InvariantCobordism2021 · cutRise Chapter6/InvariantCobordism2021 · integral_deriv_cutRise, and $g = g_0 - g_1$. $gA$ is supported over the compact slab $[a_0, b_1]$, so Stokes' theorem applies: $\int_M d(gA) = \int_{\partial M} gA = 0$, the face restriction of $A$ being 0 Chapter6/InvariantCobordism2021 · integral_extDeriv_smulFun_eq_zero_of_faces. Since $dA = 0$, $d(g_iA) = g_i'(t)\,dt \wedge A$ is supported in the $i$-th end Chapter6/InvariantCobordism2021 · tsupport_extDeriv_smulFun_subset, where it is $g_i'(t)\,dt \wedge p_1^*\alpha_i$, and by Fubini on the end $\int_M d(g_iA) = \int_{N_i} \alpha_i$ Chapter6/InvariantCobordism2021 · integral_extDeriv_smulFun_cutRise. Subtracting, $\int_{N_0} \alpha_0 - \int_{N_1} \alpha_1 = \int_M d(gA) = 0$. □ *Applied to $S_F^{k+1}$* with the inner ends Chapter6/CobordismRelativeForms2021b · hasCutEnd_leftIn, the form $A$ of Step 8 and $\alpha_i = a_i$: by Step 3,

$$\Psi_{s_0}(f_0, o_0)([\omega, \tau]) = \int_{S_0} a_0 = \int_{S_1} a_1 = \Psi_{s_1}(f_1, o_1)([\omega, \tau])$$

Chapter6/CobordismRelativeForms2021b · psiAt_eq_of_cobordism_of_orientation Chapter6/InvariantCobordism2021 · psiAt_eq_of_cobordism. This is the text's identity $\int_{S_0} \omega - \int_{\partial S_0} \tau = \int_{S_1} \omega - \int_{\partial S_1} \tau$.

**Step 10: the assembly** (V6.76). (A) *For $U$ with the margin and $s_0$, $s_1$ admissible for $(N, U)$, $\Psi_{s_0} = \Psi_{s_1}$* Chapter6/Theorem671_2021 · PsiIndepAdmissible Chapter6/CobordismRelativeForms2021b · psiAt_eq_of_admissible_of_orientation: Steps 4 and 9 with $o_1 = o_0$. (B) *For a proper oriented homotopy and $s_i$ admissible for $f_i$, $\Psi_{s_0}(f_0, o_0) = \Psi_{s_1}(f_1, o_1)$* Chapter6/Theorem671_2021 · PsiIndepHomotopy Chapter6/CobordismRelativeForms2021b · psiAt_eq_of_homotopy_of_orientation: Steps 5 and 9. *The map.* For each $(f, o)$ choose a norm, a neighbourhood with the margin and an admissible section Chapter6/Theorem671_2021 · AdmissibleData Chapter6/Theorem671_2021 · nonempty_admissibleData, and let $\Psi_{(f,o)}$ be $\Psi_s$ for the chosen $s$ Chapter6/Theorem671_2021 · psiInvariant. *(1)*: for any $N$, $U$ with the margin and $s$ admissible for $(N, U)$, take $s_2$ admissible for both $(N, U)$ and the chosen pair (Step 1); (A) for the chosen pair, $\Psi_{s_2}$ not depending on the pair, and (A) for $(N, U)$ give $\Psi_{(f,o)} = \Psi_s$ Chapter6/Theorem671_2021 · psiInvariant_eq_

psiAt Chapter6/Theorem671_2021 · psiInvariant_spec. *(2)*: by (1) at both ends and (B) for sections admissible inside the neighbourhoods of Step 5 Chapter6/Theorem671_2021 · psiInvariant_homotopy. So the theorem follows from (A) and (B) alone Chapter6/Theorem671_2021 · theorem_6_7_1_of_indep, (A) and (B) follow from the orientation of the cobordism Chapter6/CobordismRelativeForms2021b · theorem_6_7_1_of_orientation, and that is Step 7 Chapter6/Theorem671Final2021 · theorem_6_7_1. The earlier form of the theorem with the two cobordism statements as hypotheses Chapter6/Theorem671Assembly2021 · theorem_6_7_1_hyp Chapter6/Theorem671Assembly2021 · CobordismOrientationHyp is kept; both hypotheses are now discharged, the second by its inner-end form. (Gap G.54 closed.)

**The case $\partial X = \emptyset$** (V6.77) Chapter6/Theorem671Final2021 · theorem_6_7_1_noBoundary. *For $X$ without boundary there is $\Phi$ assigning to every proper oriented $(f, o)$ linear maps $\Phi_{(f,o)} : H^k_{dR}(X) \to \mathbb{R}$ such that, for every admissible $s$ and $n$ the index if it is constant, $S_{f+s} = S_n$, $\Phi([\omega]) = \int_{S_{f+s}} \omega$ in degree $n$ and $\Phi = 0$ in the other degrees, invariant along proper oriented homotopies.* Without boundary the face forms vanish and $H^k_{dR}(X, \partial X) \cong H^k_{dR}(X)$ Chapter4/BoundaryForms2021 · MPolyfold.relDeRhamEquivOfNoBoundary, the boundary integral is $0$ Chapter6/InvariantSingle2021 · psiAt_mk_of_isWithoutBoundary, and $\Phi$ is $\Psi$ composed with this isomorphism Chapter6/Theorem671_2021 · psiInvariantNoBoundary Chapter6/Theorem671_2021 · psiInvariantNoBoundary_mk; off degree $n$, $S_k = \emptyset$ and $\Phi = 0$ Chapter6/Theorem671_2021 · psiInvariantNoBoundary_of_ne Chapter6/Theorem671_2021 · theorem_6_7_1_noBoundary_of_indep. $H^n_{dR}(X)$ is read as the cohomology of sc-forms of Part IV (Chapter 4, item 173).

**Not vacuous** Chapter6/Theorem671_2021 · theorem_6_7_1_witness. On the point $X = \mathbb{R}^0$ with the zero bundle and $f = 0$ the hypotheses hold, $f$ is a proper tame sc-Fredholm section with an orientation, and on the class of the constant 0-form 1, $\Psi_s([1]) = \pm 1$ for every admissible $s$: $S_{f+s}$ is the point, of index 0, counted with its sign. Every $\Psi$ with property (1) agrees with $\Psi_s$, so it is not $0$ either.

**Readings and what is left.** The hypotheses are the text's: $X$ tame with sc-smooth bump functions (the first sentence of the theorem); "sc-Fredholm" is the tame form of Definition 3.6.4 (preprint 3.51); "proper" is Definition 5.2.1 (3); "oriented" is Definition 6.6.16 (preprint 6.40). The readings are R-a to R-f above. Nothing of the proof is assumed. What the formalization adds to the text: the margin of $U$ and its use for $C = 2$; the relative Theorem 5.3.10 (preprint 5.18), for sets of zeros that are not compact; the index parts; the cut-off in place of the decomposition of $\partial S_F$; the inner ends; the reason why $p_1$ maps faces to faces; and the computation that $e \wedge o$ is the product orientation on the ends, with no sign (Findings, Chapter 6, items 100, 101, 103, 128–134, 136) (pp. 276–279 (preprint pp. 230–233)).

## VI.28 Appendix §6.8.1 (preprint §6.7.1): the proof of Lemma 6.3.3 (preprint 6.5)

**The text's proof** (preprint pp. 234–235) has three steps: a change of the basis of the complement of $\alpha(A)$ multiplies both wedges of (91) by its determinant; a change of the bases of $A$ and $D$ multiplies the source by $\det(\tau)/\det(\varepsilon)$; and a change of complements adds to the basis of the complement of $\alpha(A)$ vectors of $\alpha(A)$, and to the $\gamma$-preimages vectors of $\beta(B)$, which leaves the wedges unchanged.

**The formalized proof** (V6.63) does the three steps at once. For two choices $\chi$, $\chi'$ there is a scalar $\mu$ with $\text{source}(\chi') = \mu\,\text{source}(\chi)$ and $\text{value}(\chi') = \mu\,\text{value}(\chi)$ Chapter6/ExactSequence · FourTermExact.Choice.exists_smul_source_value. *Step 1.* Project the basis $b''$ of the new complement $Z'$ onto $Z$ along $\alpha(A)$; the projections form a basis of $Z$, related to $b'$ by some $\sigma$, and since $\beta \circ \alpha = 0$ they have the same $\beta$-images as $b''$. *Step 2.* In $B$, $\alpha(a') \wedge b'' = \alpha(a') \wedge p(b'')$, adding vectors of the span of the first block; with $a' = \tau a$ this is $\det\tau\ \det\sigma\ \ \alpha(a) \wedge b'$. *Step 3.* In $C$, the $\gamma$-preimages $c'$ of $d' = \varepsilon d$ in $V'$ differ from

the $\gamma$-preimages in $V$ by vectors of $\ker\gamma = \beta(B)$, the span of the second block, so the wedge of the basis of $C$ `Chapter6/ExactSequence · FourTermExact.Choice.wedgeC` is $\det\varepsilon\,\det\sigma$ times that of $\chi$. *Step 4.* So the value scales by $\det\tau\,\det\sigma/(\det\varepsilon\,\det\sigma) = \det\tau/\det\varepsilon$, as the source does, and Lemma 6.3.3 (preprint 6.5) follows from the fixed choice that defines $\Phi_E$. The step "independent of the choices of the basis if and only if $\det(\tau) = \det(\varepsilon)$" is read as this homogeneity of (91) in the bases of $A$ and $D$, $\tau$ being the $\sigma$ just introduced; the corrections $p_i$ lie in $\beta(B)$, written $\beta(C)$, and are combinations of the $\beta$-images, whose names are exchanged with those of the $\gamma$-preimages in the last step (possible misprints; Findings, Chapter 6, item 1).

## VI.29 Appendix §6.8.2 (preprint §6.7.2): the proof of Proposition 6.4.11 (preprint 6.12)

(V6.64.) Let $P \le Q$ in $\Pi_T$. *Step 1, adapted bases.* Take a basis $a$ of $\ker T$ and classes $[d_i]$ forming a basis of $\operatorname{coker} T$ with $d_i \in \ker Q$, possible because $R(QT) = R(Q)$; complete $a$ by $b$ to a basis of $\ker(QT)$, and $(a,b)$ by $\tilde b$ to a basis of $\ker(PQT) = \ker(PT)$. *Step 2, the three values.* By (91) with these choices (Lemma 6.3.3 (preprint 6.5), §VI.3), $\gamma_T^Q(h_0) = (a\wedge b)\otimes([d]_Q\wedge[Tb]_Q)^*$; $\gamma_{QT}^P\gamma_T^Q(h_0) = (a\wedge b\wedge\tilde b)\otimes([d]_P\wedge[Tb]_P\wedge[QT\tilde b]_P)^*$, the classes $[d]_P$, $[Tb]_P$ being preimages of $[d]_Q$, $[Tb]_Q$ because $Pd_i = PQd_i = 0$ and $PTb\in R(P)\subseteq R(Q)$; and $\gamma_T^P(h_0) = (a\wedge b\wedge\tilde b)\otimes([d]_P\wedge[Tb]_P\wedge[T\tilde b]_P)^*$ `Chapter6/ProjectionSequence · gammaProj_wedge_tmul`. *Step 3, the spanning.* $[T\tilde b]_P - [QT\tilde b]_P = [(I-Q)T\tilde b]_P$, and $(I-Q)T\tilde b\in\ker Q$. If $Qg = 0$, then $g = \sum c_i d_i + Tx$ with $x\in\ker(QT)$ — write $[g] = \sum c_i[d_i]$ in $\operatorname{coker} T$ and use $Qd_i = 0$ — so the classes of the $d_i$ and of the $Tb_j$ span what the text calls $\ker\tilde Q$ `Chapter6/ProjectionSequence · mem_span_append_of_ker`. *Step 4.* The last block of the third value differs from that of the second by vectors of the span of the first two blocks, which does not change the wedge (§VI.2), and with the identification of $\det(P(QT))$ with $\det(PT)$ the two values agree `Chapter6/ProjectionSequence · comp_comp_eq_of_projLE`. The spanning is proved directly in $F$; the name $\alpha_q$ in the last part is read as $\alpha^P$ (possible misprint; Findings, Chapter 6, items 6, 7) (pp. 280–286 (preprint pp. 235–237)).

## VI.30 What is not formalized

Theorem 6.7.1 (preprint 6.43), the admissible sections, the orientation of the solution set and of $\hat f$, Lemma 6.5.3 (preprint 6.19), Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds, the remarks of §§6.3–6.6 and after Theorem 6.6.17, and the facts on paths are now formalized, and so is Proposition 6.6.3 (1) in its 2021 wording (gaps G.54–G.64, G.66–G.69, G.81, G.90, G.91 closed). What remains:

- **Proposition 6.6.3 (1) in the preprint's wording** (gap G.61 is closed): "$U$ is sc-smoothly contractible". As we read it, the proof does not establish it for its $U$: it needs $r(V')\subseteq V'$, which is not stated. No counterexample is known to us. The 2021 wording, a smaller $U'$ contractible in $U$, is proved, and the proofs use (1′) (§§VI.10, VI.18, VI.19, VI.23).
- **Paths and families on** $[0,1]$ (gap G.65, in part): most statements hold on $[0,1]$, $[0,1]^2$, $[0,1]^n\times X$ as printed (§§VI.36, VI.37), and so does Proposition 6.6.3 (1′) for paths on $[0,1]$ (§VI.10). For $n = 1$ the bundle $L$ of Lemma 6.6.5 is a smooth vector bundle over the manifold with boundary $[0,1]$ itself, the pull-back along the inclusion `Chapter6/GapP3_K4_SmoothLInterval · lemma_6_6_5_smoothBundle_interval` `Chapter6/GapP3_K4_SmoothLInterval · SmoothSubspaceFamily.comp` `Chapter6/GapP3_K4_SmoothLInterval · contMDiff_unitIntervalToOpens`. Still read on $\mathbb{R}$ (decision D2): $L$ over a neighbourhood

of $[0,1]^n$ for $n \geq 2$ (V6.34; Mathlib has no manifold structure on the cube, estimated at 300–600 lines), the stabilized kernels and their continuation for families with sections over $[0,1]^n \times X$ (V6.42, 400–700 lines), and Remark 6.6.19 (preprint 6.42) over $\mathbb{R}^1 \times X$ (V6.62, 1,500–2,500 lines), with the cobordisms of Theorem 6.7.1 (preprint 6.43).

- **The narrower rows.** V6.48, Theorem 6.6.11 (preprint 6.35): the section $f$, which the statement does not introduce, is supplied (presumably an omission), and the neighbourhood is not shown contractible in itself (the preprint's (1)). V6.80, the remark that $\mathrm{DET}(E,F)$ is not orientable: it does not hold as we read it for finite-dimensional $E$, $F$ (`exists_two_lineOrientation_det_of_finiteDimensional`), and it is proved only for spaces isomorphic to an infinite-dimensional Hilbert space or to $H \times H$ with $H \cong \mathbb{R} \times H$, and under complex structures with an odd-dimensional kernel; infinite-dimensional Banach spaces without such structures are not covered (§VI.34). V6.81, the classical case: $C^1$ in place of smooth, and the smoothness of the line bundle is not formalized (§VI.34). V6.32, Proposition 6.6.3, is restated since 2026-11-13.
- **The chart-free objects** $T_x X$, $Y_x$, $\mathrm{Lin}(f,x)$, $\mathrm{Or}_x^f$ (gaps G.6, G.7, G.13 of Chapter 2): they are made in chart points and identified between charts (§VI.21).

## VI.31 What the formalization taught us

**The sign of (94).** The main change of substance is a sign. The coordinate of $\Phi_E$ in a local determinant chart carries the factor $(-1)^{\dim \operatorname{coker} S}$, which jumps where the parity of the cokernel dimension changes; the orientations it carries along a continuous family are then not continuous, as we read it, and an example on $\mathbb{R}$ shows it (§VI.14). The continuity statements of pp. 265–272 (preprint pp. 223–226) are not written in coordinates in the text, and it is in coordinates that the sign appears. The normalized $\alpha(T)\Phi_E$ at both ends of every propagation changes nothing on surjective operators — the tangent orientations of Theorem 6.6.18 (preprint 6.41) are the text's — and the same sign is used in Remark 6.6.19 (preprint 6.42) (§VI.26).

**Lemma 6.10 (preprint) and the 2017 remark.** The common lower bound of two good left-projections exists, but not with the range $R(P) \cap R(P')$: a projection onto a smaller, finite-codimensional piece of it, along a complement containing both kernels, is below both (§VI.4). This keeps the preprint's construction of $\mathrm{DET}(E,F)$ — one lower bound, Corollary 6.4.12 (preprint 6.13), Lemma 6.5.5 (preprint 6.21) — intact.

**Contractibility was never needed.** Proposition 6.6.3 (1) (preprint 6.28) in the preprint's wording is not established for the neighbourhood its proof builds, as we read it; the 2021 wording, with a smaller $U'$, holds (§VI.10). Every later use of contractibility is a use of (1′): two paths in $U$ with the same end points are homotopic in a larger region where sections along families exist. Orientable neighbourhoods become pairs $V \subseteq U$, and property ( ) of Theorem 6.6.13 (preprint 6.37) holds on every set with the local propagation property (§§VI.10, VI.19, VI.23).

**Paths on $\mathbb{R}$ cost reparametrizations.** With paths and homotopies on $\mathbb{R}$ whose conditions hold on $[0,1]$ (decision D2), the homotopy of (1′) has the right edges only on $[0,1]$ while push-forward data need zeros near $[0,1]$; transitivity of homotopy, associativity of concatenation and Theorem 6.6.11 (preprint 6.35) all pass through the smooth transition function $\kappa$ (§§VI.18, VI.20).

**Charts must be compared.** The text's chart-free $\mathrm{Or}_x^f$ becomes a set read in a chosen chart, and the invariance of the local propagation under a change of chart — the naturality of the stabilization isomorphism, of the sign and of continuation — is a file of its own (§VI.21).

**"By a covering argument" and "one verifies".** The covering argument of Theorem 6.6.13 (preprint 6.37) is a dyadic transport with a square lemma, proved for any family of types (§VI.22); the "simple computation" of Lemma 6.6.8 (preprint 6.33) is an exact identity of stabilization isomorphisms, with no sign, once the appended sections come last (§VI.16); and the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), stated without proof, needs the transport computed on dyadic pieces inside orientable neighbourhoods (§VI.24).

**Theorem 6.7.1 is proved as the text proves it, with two changes of route.** The invariant is defined from one admissible perturbation, made a map on cohomology by Stokes' theorem, and compared along a cobordism, as in the text. The decomposition $\partial S_F = S_0 \cup S_1 \cup \tilde{S}$ is replaced by a cut-off in time: $\int d(gA) = 0$ for a closed form $A$ with vanishing face restriction, while $d(g_i A)$ lives on the product ends, where Fubini gives the integrals over $S_0$ and $S_1$. This avoids face integrals over $\tilde{S} \subseteq \partial X \times (0,1)$ and the signs of the pieces of $\partial S_F$. And the relative form is built on inner ends, because gluing $p_1^*\gamma_i$ to an extension needs room in time inside the product end (§VI.27).

**The extension $\bar{\omega}$ has one degree too many, as we read it.** The proof extends $\omega$ to $X \times [0,1]$ by $p_1^*\omega \wedge p_2^* dt$, a form of degree $k+1$ on the $(k+1)$-dimensional $S_F$, whose integrals over the $k$-dimensional pieces of $\partial S_F$ vanish. The comparison goes through with $p_1^*\omega$ and $p_1^*\tau$ (§VI.27, reading R-e).

**The orientation of the cobordism is a computation, and has no sign.** "Taking the orientations into account" is: the linearization of $F$ on an end is that of a $t$-constant lift, its kernel is $\mathbb{R} \times \ker$, the 2021 map puts $e$ in front of the kernel orientation, and the tangent map of the end is $(s,w) \mapsto (s, T\iota(w))$; with the time vector first on both sides, $e \wedge o$ is the product orientation $e \wedge o_{S_i}$ (§VI.27). The 2021 convention $\tilde{h} = e \wedge h$ also removes the sign that the preprint's $h \mapsto \tilde{h}$ needed: it differs from the continuous map by $(-1)^{\mathrm{ind}L}$, which is locally constant (§VI.26).

**Hidden data of the theorem.** The neighbourhood $U$ needs a margin, $\mathrm{cl}\{N(f) \leq 2\}$ compact, for the cobordism between two admissible sections; Theorem 5.3.10 (preprint 5.18) is needed relative to a compact part of a zero set that need not be compact; the degree of a form is paired with the part of $S_{f+s}$ of that index, since the index need not be constant; the proper oriented homotopy, which the text does not define, is read through Remark 6.6.19 (preprint 6.42); and $p_1$ maps the faces of $S_F$ to faces of $X$ because the degeneracy index of $\mathbb{R}^1 \times X$ is that of $X$ (§VI.27).

$\mathrm{DET}(E,F)$ **and dimension.** The remark that $\mathrm{DET}(E,F)$ is not orientable does not hold as we read it for finite-dimensional $E$, $F$: there $L(E,F)$ is convex and Proposition 6.6.1 (preprint 6.26) gives two orientations (`exists_two_lineOrientation_det_of_finiteDimensional`). It holds for infinite-dimensional Hilbert spaces, through a loop of surjective operators whose kernels form a Möbius band; in finite dimension the cokernels form a second Möbius band and the two cancel in det (§VI.34).

**Orientations of manifolds with corners.** Mathlib has none. An oriented atlas, the usual definition, does not fit the quadrant model: $[0,1]$ has no atlas of quadrant charts with positive changes, and in dimension $0$ every atlas is oriented. A continuous tangent orientation does, and it is the form in which Theorem 6.6.18 (preprint 6.41), Stokes' theorem and the cobordism argument use orientations (§§VI.25, VI.27).

**(P) follows from the general position of $F$.** The text derives the general position of $F$ along the family from property (P) of the slices. The formalization goes the other way: general position of $F$ is proved by the method of Theorem 5.3.10 (preprint 5.18), and (P) follows from it by a chart computation, since the tangent vectors of the family of zeros lie in $\ker F'$ and have reduced $X$-components; on the cube $[0,1]^n$ the two are equivalent (§VI.36).

# VI.32 The Springer 2021 edition: Lemma 6.4.4, Proposition 6.4.13, Lemmas 6.6.8 and 6.6.9

The 2021 edition adds index equalities to Lemma 6.4.4 (2017: 6.3), makes the hypothesis of Proposition 6.4.13 (2017: 6.6) explicit, and splits preprint Lemma 6.33 (2017: 6.15) into Lemmas 6.6.8 and 6.6.9. Neither Lemma 6.4.4 nor Proposition 6.4.13 has a preprint counterpart. The file is `Chapter6/Book2021.lean` (Findings, Chapter 6, items 110–115).

**Lemma 6.4.4 (2017: 6.3)** `Chapter6/Book2021 · lemma_6_4_4`. *Statement.* Let $E$, $F$ be Banach spaces, $T_0 : E \to F$ Fredholm and $P_0 \in \Pi_{T_0}$ `Chapter6/GoodProjections · IsGoodLeftProj`. There is $\varepsilon > 0$ such that for every $T$ and every projection $P$ with $\|T - T_0\| < \varepsilon$, $\|P - P_0\| < \varepsilon$: $T$ is Fredholm, $P \in \Pi_T$, and $\operatorname{ind}(P \circ T) = \operatorname{ind}(T) = \operatorname{ind}(T_0) = \operatorname{ind}(P_0 \circ T_0)$, all indices of maps $E \to F$ `Chapter1/FredholmIndex · fredholmIndex`. *Proof.* (a) With a quasi-inverse $v$ of $T_0$, every $T$ with $\|T - T_0\| < 1/(\|v\| + 1)$ is Fredholm of the index of $T_0$ `Chapter3/SolutionLinearization · isFredholm_add_of_norm_lt` `Chapter1/FredholmIndex · fredholmIndex_add_of_norm_lt`. (b) Instead of the text's map $\Phi_{(P,T)}(x,h) = PTx + (I - P)h$, take the intertwiner $J(P) = PP_0 + (1 - P)(1 - P_0)$ `Chapter6/Book2021 · projIntertwiner`. It is continuous in $P$ `Chapter6/Book2021 · continuous_projIntertwiner`, $J(P_0) = 1$ `Chapter6/Book2021 · projIntertwiner_self`, so $J(P)$ is a unit for $P$ near $P_0$; and $J(P)P_0 = PJ(P)$ for every projection $P$ `Chapter6/Book2021 · projIntertwiner_mul`, so $P = JP_0J^{-1}$. (c) $J^{-1}T$ is near $T_0$, so by Lemma 6.5.1 (preprint 6.17) $P_0 \in \Pi_{J^{-1}T}$ `Chapter6/LocalDetBundle · lemma_6_5_1`. Then $R(P) = JR(P_0)$ has finite codimension, $(1 - P)F$ complements it, and $R(PT) = JR(P_0J^{-1}T) = JR(P_0) = R(P)$: $P \in \Pi_T$. (d) The indices of $P \circ T$ and $P_0 \circ T_0$ are those of $T$ and $T_0$ `Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_comp`. □ As a map onto $R(P)$ the operator $P \circ T$ would have index $\dim\ker(PT)$; the text's equalities hold for the map $E \to F$. $E$ and $F$ are taken in the universe `Type`, that of (a).

**A common complement** `Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq`. *Statement.* Two finite-dimensional subspaces $K_1$, $K_2$ of the same dimension of a real vector space have a common algebraic complement. *Proof.* Induction on $n = \dim(K_1 + K_2) - \dim K_1$ `Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq_aux`. For $n = 0$, $K_1 = K_2$. Otherwise neither contains the other; take $a \in K_2 \setminus K_1$, $b \in K_1 \setminus K_2$. Then $a+b$ lies in neither, $K_i + \mathbb{R}(a+b)$ has dimension $\dim K_i + 1$ `Chapter6/Book2021 · finrank_sup_span_singleton_of_notMem` and their sum is still $K_1 + K_2$, so the induction gives a common complement $X'$ of $K_1 + \mathbb{R}(a + b)$ and $K_2 + \mathbb{R}(a + b)$, and $X' + \mathbb{R}(a + b)$ complements $K_1$ and $K_2$ `Chapter6/Book2021 · isCompl_sup_span_singleton`. □ The text takes $X$ closed; that is not needed for the algebraic part below, and a closed one is §VI.33.

**Proposition 6.4.13 (2017: 6.6), the algebraic part** `Chapter6/Book2021 · proposition_6_4_13`. *Statement.* Let $T$ be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$ (the 2021 hypothesis $\Pi_T \cap \mathcal{P}_H \neq \emptyset$ is the existence of such projections). There are linear isomorphisms $A_T : \ker(PT) \to \ker(QT)$, $B_T : F/R(P) \to F/R(Q)$ with $A_T \circ j_P = j_Q$, $\Phi_Q \circ A_T = B_T \circ \Phi_P$ and $\pi_Q \circ B_T = \pi_P$, for the maps of $E(T,P)$, $E(T,Q)$ (§VI.5) `Chapter6/ProjectionSequence · projSeqJ` `Chapter6/ProjectionSequence · projSeqPhi` `Chapter6/ProjectionSequence · projSeqPi`. *Proof (the text's).* $PT$ and $QT$ are Fredholm `Chapter6/GoodProjections · IsGoodLeftProj.isFredholm_comp` of index $\operatorname{ind}(T)$, and $F/R(P) = F/R(Q)$, so $\dim\ker(PT) = \dim\ker(QT)$; let $X$ be a common complement. $QT$ maps $X$ isomorphically onto $R(Q)$; for $e \in \ker(PT)$ let $x(e) \in X$ solve $QTx = QTe$ and put $A_T(e) = e - x(e) \in \ker(QT)$. It is injective ($A_T(e) = 0$ puts $e$ in $X \cap \ker(PT) = 0$), the identity on $\ker T$, and onto by the dimensions. For $f \in F$ write $(1 - P)f = Tx + \tilde{f}$ with $x \in X$, $\tilde{f} \in (1 - Q)F$ (solve $QTx = Q(1 - P)f$), and put $B_T(f + H) = \tilde{f} + H$. The three identities are direct computations; $B_T$ is injective by a direct com-

putation, hence an isomorphism, the two quotients having the same finite dimension (the text uses the five lemma). □ The last sentence, that $A_T$ and $B_T$ depend continuously on $T$, is §VI.33, with a closed $X$ (gap G.81, closed).

**Lemmas 6.6.8 and 6.6.9** `Chapter6/Book2021 · lemma_6_6_8` `Chapter6/Book2021 · lemma_6_6_9`. *Statement.* For stabilized kernels of the data $(f, s_1, \dots, s_k)$ along a family `Chapter6/Propagation · IsStabKernels` and of the data with $d$ sections appended, the push-forwards of orientations agree: for zero sections (Lemma 6.6.8, the parameters $\mu$ as dummy variables) and for arbitrary sections $\bar{s}_1, \dots, \bar{s}_d$ (Lemma 6.6.9). *Proof.* Both are §VI.16 `Chapter6/Propagation · IsStabKernels.push_append`, Lemma 6.6.8 with $\bar{s}_j = 0$. □ The statement of preprint Lemma 6.33 (2017: 6.15) is Lemma 6.6.9, and the first step of its proof is Lemma 6.6.8; the formalization of §VI.16 never needed that step.

**Already covered.** The other 2021 changes of the chapter concern statements verified before: Proposition 6.4.15 with its hypotheses made explicit (§VI.6) `Chapter6/Stabilization · proposition_6_4_15`, Proposition 6.6.3 for a tame $X$ and sections valued in $Y \times [0,1]$ (§VI.10, without tameness) `Chapter6/OrientableNbhd · proposition_6_6_3`, and Lemma 6.6.5 with the solution set near $(t, \varphi(t), 0)$ in the points $(t, y, \lambda)$, whose kernel family (§VI.12) is unaffected `Chapter6/KernelFamily · exists_subspaceFamily_linKer`.

## VI.33 The Springer 2021 edition, continued: Proposition 6.4.13 (last sentence) and Theorem 6.6.17

The last sentence of Proposition 6.4.13 (2017: 6.6), the continuity of $A_T$ and $B_T$ in $T$, is proved with a closed common complement. Theorem 6.6.17 is new in the 2021 edition: a topology on the orientation space $O_f$ of pp. 274–275, whose proof the text leaves to the reader. It is proved with properties (1)–(3); property (2) uses a hypothesis on $X$, connected smooth neighbourhoods, which every M-polyfold satisfies (gap G.90, closed). The files are `Chapter6/Book2021Continuity.lean` and `Chapter6/OrientationSpace.lean` (Findings, Chapter 6, items 116–121).

**A common closed complement** `Chapter6/Book2021Continuity · exists_isClosed_isTopCompl_isTopCompl_of_finrank_eq`. *Statement.* Two finite-dimensional subspaces $K_1$, $K_2$ of the same dimension of a real normed space $E$ have a common closed complement $X$: $E = K_1 \oplus X = K_2 \oplus X$ topologically. *Proof.* Let $X_0$ be a common algebraic complement (§VI.32) `Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq` and $Z$ a closed complement of the finite-dimensional $K_1 + K_2$. Then $X = Z + X_0 \cap (K_1 + K_2)$ complements $K_i$: inside $K_1 + K_2$, $X_0 \cap (K_1 + K_2)$ complements $K_i$, and $Z$ complements $K_1 + K_2$ `Chapter6/Book2021Continuity · isCompl_sup_inf_of_isCompl`. $X$ is closed, a closed subspace plus a finite-dimensional one, and an algebraic complement of a finite-dimensional subspace that is closed is a topological one. □

**Proposition 6.4.13 (2017: 6.6), the last sentence** `Chapter6/Book2021Continuity · proposition_6_4_13_continuity`. *Statement.* Let $T$ be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$. There are a closed $X \subseteq E$ and $\varepsilon > 0$ such that on $\|S - T\| < \varepsilon$ the maps $S \mapsto A_S = 1 - \xi_S \circ S$ `Chapter6/Book2021Continuity · kerTransition` and $S \mapsto B_S = (1-P) - S\xi_S(1-P)$ `Chapter6/Book2021Continuity · cokerTransition`, where $\xi_S = (QS|_X)^{-1} \circ Q$ `Chapter6/Book2021Continuity · solveOp` `Chapter6/Book2021Continuity · projCod`, are continuous into $L(E)$, $L(F)$ in the operator norm, and jointly with the vector; and for every such $S$: $P, Q \in \Pi_S$, $E = \ker(PS) \oplus X = \ker(QS) \oplus X$ topologically, $e - A_S e \in X$ and $QSA_S e = 0$ (6.15)–(6.16) `Chapter6/Book2021Continuity · kerTransition_apply`, $QB_S f = 0$ and $Sx + B_S f = (1-P)f$ for some $x \in X$ (6.18)–(6.19) `Chapter6/Book2021Continuity · cokerTransition_apply` `Chapter6/Book2021Continuity · solveOp_mem`, and $A_S$, $B_S$ induce isomorphisms $\ker(PS) \to \ker(QS)$, $F/R(P) \to F/R(Q)$ filling the diagram of

$E(S,P)$, $E(S,Q)$ Chapter6/Book2021Continuity · diagram_of_solveOp. *Proof.* Take $X$ from the previous statement for $\ker(PT)$, $\ker(QT)$, which have the same dimension (§VI.32). (a) By Lemma 6.5.1 (preprint 6.17) Chapter6/LocalDetBundle · lemma_6_5_1 there is $\varepsilon$ with $P,Q \in \Pi_S$ and $QS|_X : X \to R(Q)$ an isomorphism for $\|S-T\| < \varepsilon$ Chapter6/LocalDetBundle · projRestrict. (b) $S \mapsto QS|_X$ is continuous Chapter6/LocalDetBundle · continuous_projRestrict and inversion is continuous at isomorphisms, $X$ being complete; so $\xi_S$ is continuous in $S$ Chapter6/Book2021Continuity · continuousAt_solveOp, and so are $A_S$, $B_S$, which are built from $\xi_S$, $S$ and $P$ by composition. (c) $QS\xi_S = Q$ Chapter6/Book2021Continuity · apply_solveOp_of_projRestrict_eq gives (6.15)–(6.19). (d) $\ker(QS)\cap X = 0$ because $QS|_X$ is injective, and $\ker(QS)+X = E$ because $e - \xi_S S e \in \ker(QS)$ Chapter6/Book2021Continuity · disjoint_ker_comp_of_projRestrict_eq; the projection $1-\xi_S S$ onto $\ker(QS)$ along $X$ is continuous Chapter6/Book2021Continuity · isTopCompl_ker_comp_of_projRestrict_eq; the same for $P$, $R(P)=R(Q)$. (e) The diagram is §VI.32 for $S$ with this $X$. □ *Satisfiable:* $T=1$ on $\mathbb{R}$ and $P=Q$ any element of $\Pi_T$ Chapter6/Book2021Continuity · proposition_6_4_13_continuity_hyps. "Depend continuously" is read as the continuity of $A_S$ on $E$ and $B_S$ on $F$; these determine the maps between the varying kernels $\ker(PS)$, $\ker(QS)$ and on $F/H$, and the joint continuity is continuity on the kernel bundle $\bigcup_S \{S\}\times\ker(PS)$ of Remark 6.4.5 (gap G.81, closed).

**The orientation space** Chapter6/OrientationSpace · OrSpace Chapter6/OrientationSpace · orProj. *Definition.* $O_f$ is the set of pairs $(x,o)$ with $x \in X_\infty$ and $o \in \mathrm{Or}^f_x$, read in the chosen chart (§VI.21) Chapter6/SectionOrientation · Orx, and $\pi_f(x,o) = x$; $X_\infty$ carries the topology of Theorem 2.3.10 (§II.7) Chapter2/Metrizable · MPolyfold.SmoothLevel. For an sc-smooth path $\varphi$ (a map $\mathbb{R}\to X$, decision D2) the pull-back $\varphi^*O_f$ is the subspace of $[0,1]\times O_f$ of the $(t,p)$ with $\varphi(t)=\pi_f(p)$ Chapter6/OrientationSpace · OrPullback, and a continuous section over an open $U$ is a continuous $\sigma : U_\infty \to O_f$ with $\pi_f\circ\sigma = \mathrm{id}$ Chapter6/OrientationSpace · IsContSectionOver. The three properties of the theorem: (1) $\pi_f$ is a local homeomorphism; (2) every $x \in X_\infty$ has an open $U$ in $X$ over whose smooth part there are exactly two continuous sections Chapter6/OrientationSpace · HasTwoContSections; (3) for every sc-smooth path $\varphi$ and continuous section $t \mapsto (\varphi(t), o_t)$ of $\varphi^*O_f$, for every $t_0$ and $t$ near $t_0$, $o_t$ is the push-forward of $o_{t_0}$ along the segment $\varphi|[t_0,t]$ Chapter6/OrientationSpace · segPath Chapter6/OrientationSpace · isScSmoothMap_segPath in a set with the local propagation property Chapter6/OrientationFunctor · IsPropagationNbhd Chapter6/OrientationFunctor · orPush, which does not depend on the set Chapter6/OrientationFunctor · orPush_compat Chapter6/OrientationSpace · IsPropagationContinuous. An *orientation topology* is a Hausdorff topology with (1)–(3) Chapter6/OrientationSpace · IsOrientationTopology. This is our reading of the text's "local propagation of $o_{t_0}$".

**Sc-smooth paths in $X_\infty$** Chapter6/OrientationSpace · continuous_smoothMk_comp Chapter6/OrientationSpace · eventually_exists_scPath_smoothLevel. *Statement.* (a) An sc-smooth $\Gamma : \mathbb{R}^n \to X$ is continuous into $X_\infty$. (b) If $N$ is a neighbourhood of $x$ in $X_\infty$, every $y \in X_\infty$ near $x$ is joined to $x$ by an sc-smooth path with all its points in $N$. *Proof.* (a) Its values are smooth points Chapter6/ScHomotopy · mem_smoothSet_of_isScSmoothMap; in a chart, $\Gamma$ is continuous into every level $m$ Chapter6/OrientationSpace · continuousOn_levelLift_comp Chapter6/OrientationSpace · rnChart_toBase, which is continuity into $X_\infty$. (b) In a chart at $x$, take the retracted segment $t \mapsto r((1-t)u_0 + tu_1)$ Chapter6/OrientationSpace · chartSegPath, sc-smooth Chapter6/OrientationSpace · isScSmoothMap_chartSegPath; for $y$ near $x$ in $X_\infty$ its points are near $x$ in every level, hence in $N$. □ The neighbourhood version for open sets of $X$ is Chapter6/OrientationSpace · exists_nhds_scPath_of_isOpen, from §VI.18 Chapter6/ScPaths · exists_nhds_scPath.

**The topology** Chapter6/OrientationSpace · orTopology. *Definition.* Fix local transport data $L$ (§VI.22) Chapter6/PathTransport · LocalTransport. A *continuous orientation* over an open $U$ is a family $y \mapsto o_y$ on $U_\infty$ carried along by the transport of $L$ along every sc-smooth path in $U$ Chapter6/OrientationSpace · IsContOrOn; its sheet is $\{(y,o_y) : y \in U_\infty\}$ Chapter6/OrientationSpace · orSec Chapter6/OrientationSpace · orBasic. A set

$W \subseteq O_f$ is open if $\{y \in U_\infty : (y, o_y) \in W\}$ is open in $X_\infty$ for every continuous orientation $o$ over an open $U$. *Properties, for every $L$.* (a) Two continuous orientations that agree at a point agree near it `Chapter6/OrientationSpace · IsContOrOn.eventually_iff`; so the sheets are open `Chapter6/OrientationSpace · isOpen_orBasic` and $\pi_f$ is continuous `Chapter6/OrientationSpace · continuous_orProj`. (b) Every point of $O_f$ lies on a sheet `Chapter6/OrientationSpace · exists_isContOrOn`: push the orientation from $x$ along the paths of §VI.18 in a good set of $L$. (c) On a sheet $\pi_f$ is a homeomorphism onto $U_\infty$, with inverse $y \mapsto (y, o_y)$ `Chapter6/OrientationSpace · continuousOn_orSec` `Chapter6/OrientationSpace · eq_orSec_of_mem_orBasic` `Chapter6/OrientationSpace · orSec_mem_orBasic` `Chapter6/OrientationSpace · orChartHomeo`; with (b), property (1) `Chapter6/OrientationSpace · isLocalHomeomorph_orProj`. (d) Hausdorff `Chapter6/OrientationSpace · t2Space_orTopology`: points over different $x$ are separated by $\pi_f$, $X_\infty$ being metrizable `Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel`; the two points over one $x$ lie on two sheets which, by (a), are disjoint over a neighbourhood of $x$. □ The text gives no construction ("in view of Proposition 6.6.3 and Theorem 6.6.11").

**Property (2), under connected smooth neighbourhoods** `Chapter6/OrientationSpace · hasTwoContSections_orTopology`. *Statement.* Suppose every smooth point has, inside every open $W \ni x$, an open $U \ni x$ with $U_\infty$ preconnected in $X_\infty$ `Chapter6/OrientationSpace · MPolyfold.HasConnectedSmoothNbhds`, and every $\mathrm{Or}_y^f$ has two elements. Then the topology has property (2) for every $L$, the two sections being $y \mapsto (y, \pm o_y)$. *Proof.* Take a sheet of $o$ over $V \ni x$ (property (b)) and such a $U \subseteq V$. Every point over $y$ is $(y, o_y)$ or $(y, -o_y)$ `Chapter6/OrientationSpace · eq_orSec_or_eq_orSec`, and both sheets are open; so for a continuous section the sets where it equals $(y, o_y)$ and where it equals $(y, -o_y)$ are open and complementary in $U_\infty$, which is preconnected. □ The hypothesis holds for $\mathbb{R}^n$ (convex balls) `Chapter6/OrientationSpace · rnPolyfold_hasConnectedSmoothNbhds` and for every M-polyfold on a subsingleton `Chapter6/OrientationSpace · MPolyfold.hasConnectedSmoothNbhds_of_subsingleton`. It holds on every M-polyfold (below; gap G.90 closed).

**Theorem 6.6.17** `Chapter6/OrientationSpace · theorem_6_6_17_hyp`. *Statement.* Let $X$ be tame with bump functions and $f$ an sc-Fredholm section with tame germ data and sc⁺-sections through the smooth fibre points, the hypotheses under which $\Gamma_f$ is built (§VI.23) `Chapter6/OrientationFunctor · theorem_6_6_13_hyp` `Chapter6/OrientationFunctor · orLocalTransport`. If $X$ has connected smooth neighbourhoods, $O_f$ carries exactly one Hausdorff topology with (1), (2), (3). Without that hypothesis, $O_f$ carries exactly one Hausdorff topology with (1) and (3) `Chapter6/OrientationSpace · existsUnique_orientationTopology`. *Proof.* Existence: the topology for the transport data of $\Gamma_f$. It has (1) and is Hausdorff (above), has (2) by the previous statement, every $\mathrm{Or}_y^f$ having two elements `Chapter6/OrientationFunctor · exists_two_orx`, and has (3) `Chapter6/OrientationSpace · isPropagationContinuous_orTopology`: near $t_0$ a continuous section of $\varphi^* O_f$ stays in one sheet `Chapter6/OrientationSpace · eq_orSec_iff` `Chapter6/OrientationSpace · eventually_segPath_mem`, and a sheet is carried along the segments of $\varphi$ by the transport. Uniqueness: let $t$ be any topology with (1) and (3) `Chapter6/OrientationSpace · eq_orTopology_of`. (i) A $t$-continuous local section $s$ over an open $N \subseteq X_\infty$ that agrees with a continuous orientation $o$ at $y_0$ agrees with it near $y_0$ `Chapter6/OrientationSpace · eventually_eq_orSec_of_continuousOn`: join $y$ near $y_0$ to $y_0$ by a path in $N$ (statement (b) above), which is continuous into $X_\infty$ (statement (a)); then $s$ along the path is a continuous section of the pull-back, and by (3) the set of parameters where it equals $o$ is open and closed in $[0, 1]$. (ii) Through every point a local inverse of $\pi_f$ for $t$ is such a section. By (i) the $t$-open sets and the open sets of the topology coincide. □ Neither Hausdorffness nor (2) is used in the uniqueness. The text's "uniquely defined" is read as "exactly one".

**Property (2) without the hypothesis** (V6.74). *Every M-polyfold has connected smooth neighbourhoods* `Chapter6/OrientationMainClean · MPolyfold.hasConnectedSmoothNbhds`: inside an open $W \ni x$, the connected component $U$ of $x$ in $W$ is open, M-polyfolds being locally connected (Proposition 2.3.7), and $U \cap X_\infty$

is path connected in $X_\infty$ Chapter2/SmoothPaths · MPolyfold.isPathConnected_smoothVal_preimage, since nearby smooth points are joined by paths in $X_\infty$ Chapter2/SmoothPaths · MPolyfold.exists_nhds_joinedIn_smooth and $U$ is connected. So property (2) holds without the hypothesis, and *Theorem 6.6.17 holds with (1), (2), (3) under the text's hypotheses* — $X$ tame with bump functions, $f$ with tame germ data Chapter6/OrientationMainClean · theorem_6_6_17 (gap G.90 closed; Findings, Chapter 6, items 120, 122). The remarks after the theorem — the orientation bundle, orientability as a global continuous section, the $\mathbb{Z}_2$-action and $\det_f$ — are §VI.38 (gap G.91 closed).

## VI.34 §§6.4–6.6 (preprint §§6.3–6.5), remarks: the alternative determinant, the identification (6.25) (preprint (95)), the remark that $\mathrm{DET}(E,F)$ is not orientable, and the classical case

Four remarks of the text, without proofs, are formalized here; the chapter's statements do not use them (Findings, Chapter 6, items 10, 26, 135). The files are `Chapter6/GapP1_DetConventions.lean`, `Chapter6/GapP1_DetNonOrientable.lean`, `…Ell2.lean`, `…Transport.lean` and `Chapter6/GapP1_ClassicalDetBundle.lean`.

**The alternative determinant** (V6.78; after Definition 6.4.1 (preprint 6.7), p. 246 (preprint p. 208)). *Some authors use* $\lambda(\ker T)\otimes\lambda(\mathrm{coker}(T)^*)$*; it is naturally isomorphic to* $\det(T)$. The map $\iota:\lambda(A^*)\to\lambda^*(A)$ of Definition 6.3.1 (preprint 6.3) is a nonzero map of lines (§VI.2), hence an isomorphism Chapter6/GapP1_DetConventions · iotaDualEquiv, natural by Proposition 6.3.2 (preprint 6.4). With $A=\mathrm{coker}\,T$, $\mathrm{id}\otimes\iota$ is the isomorphism Chapter6/GapP1_DetConventions · altLinDetEquiv Chapter6/GapP1_DetConventions · altFredholmDetEquiv, and on a cokernel basis $d$ with its dual basis it is $(a_1\wedge\dots)\otimes(d_1^*\wedge\cdots\wedge d_m^*)\mapsto(a_1\wedge\dots)\otimes(d_1\wedge\cdots\wedge d_m)^*$ Chapter6/GapP1_DetConventions · altLinDetEquiv_tmul_dualBasis, since the matrix of the pairing is the identity. (Gap G.57 closed.)

**The identification (6.25) (preprint (95)) and the maps after Proposition 6.4.15 (preprint 6.15)** (V6.79). $\mathbb{R}\cong\lambda(\mathbb{R}^n)$, $1\mapsto e_1\wedge\cdots\wedge e_n$, has inverse $c_1\wedge\cdots\wedge c_n\mapsto\det[c_1,\dots,c_n]$, the matrix with columns $c_j$ Chapter6/GapP1_DetConventions · stdLine Chapter6/GapP1_DetConventions · stdLine_symm_wedge, by the change-of-basis rule of §VI.2; its transpose $\lambda^*(\mathbb{R}^n)\cong\mathbb{R}^*$ maps $(e_1\wedge\cdots\wedge e_n)^*$ to $1^*$ Chapter6/GapP1_DetConventions · stdLineDual_dualVec, the isomorphism (94) of Proposition 6.4.15 (preprint 6.15). The maps $V\otimes\lambda(\mathbb{R}^n)\to V$ and $V\otimes\lambda^*(\mathbb{R}^n)\to V$, $v\otimes(e_1\wedge\cdots\wedge e_n)\mapsto v$ and $v\otimes(e_1\wedge\cdots\wedge e_n)^*\mapsto v$, are isomorphisms Chapter6/GapP1_DetConventions · tensorStdLine_tmul_stdWedge Chapter6/GapP1_DetConventions · tensorStdLineDual_tmul_dualVec for every vector space $V$; the text asks $V$ finite-dimensional, which is not needed (stronger). (Gap G.58 closed.)

**The remark that** $\mathrm{DET}(E,F)$ **is not orientable** (V6.80; end of §6.5, p. 262 (preprint p. 218)). The text: $\mathrm{DET}(E,F)$ is known to be non-orientable, but orientable over certain subsets (Proposition 6.6.1 (preprint 6.26)); no hypothesis on $E,F$, and no proof or reference. "Not orientable" is read as: no orientation in the sense of §VI.9. *For finite-dimensional $E,F$ the remark does not hold as we read it*: every operator is Fredholm, the base $L(E,F)$ is convex, and $\mathrm{DET}(E,F)$ has exactly two orientations by Proposition 6.6.1 (preprint 6.26) Chapter6/GapP1_DetNonOrientable · exists_two_lineOrientation_det_of_finiteDimensional — the Lean counterexample. *It holds under hypotheses.* (a) *A loop criterion*: a line bundle with a continuous nowhere-zero section along a map from a preconnected space that takes the values $v$ and $-v$ in one fibre has no orientation Chapter6/GapP1_DetNonOrientable · isEmpty_lineOrientation_of_loop — an orientation would have to agree with the section everywhere or nowhere. (b) *A section of* DET *from kernel bases*: for surjective Fredholm $A_x$,

norm-continuous in $x$, and continuous bases $v_x$ of the kernels, $(v_{x,1}\wedge\cdots\wedge v_{x,n})\otimes 1^*$ is a continuous section of $\mathrm{DET}(E,F)$ along $x\mapsto A_x$ Chapter6/GapP1_DetNonOrientable · continuous_kerDetMk: in a local determinant chart with $P=1$ its coordinate is the determinant of the projections of the $v_{x,i}$ to the kernel at the centre. (c) Hence, *if such a family has $A_{x_1}=A_{x_0}$, $v_{x_1}=-v_{x_0}$ and odd $n$,* $\mathrm{DET}(E,F)$ *has no orientation* Chapter6/GapP1_DetNonOrientable · isEmpty_lineOrientation_det_of_kernel_loop, since $(-v_1)\wedge\cdots\wedge(-v_n)=-(v_1\wedge\cdots\wedge v_n)$. (d) *If $E$ and $F$ carry bounded $J$, $J'$ with $J^2=-1$, $J'^2=-1$ and there is a surjective $T:E\to F$ with kernel of odd finite dimension,* $\mathrm{DET}(E,F)$ *has no orientation* Chapter6/GapP1_DetNonOrientable · isEmpty_lineOrientation_det_of_complexStructure: with $R_\theta=\cos\theta+\sin\theta\,J$ and $R'_\theta$ likewise, the loop $A_\theta=R'_\theta T R_{-\theta}$, $\theta\in[0,\pi]$, consists of surjective operators with kernels $R_\theta(\ker T)$ and bases $R_\theta v$; at $\theta=\pi$, $R_\pi=-1$, so $A_\pi=T$ and $R_\pi v=-v$, and (c) applies. The kernels form a Möbius band. (e) *Consequences*: $\mathrm{DET}(H\times H, H\times H)$ has no orientation when $H$ has a surjective operator with kernel of odd finite dimension Chapter6/GapP1_DetNonOrientable · isEmpty_lineOrientation_det_prod, for instance $H=\ell^2$ Chapter6/GapP1_DetNonOrientableEll2 · isEmpty_lineOrientation_det_ellTwo; orientability depends only on the isomorphism classes of $E$ and $F$ Chapter6/GapP1_DetNonOrientableTransport · nonempty_lineOrientation_det_iff_of_equiv; so $\mathrm{DET}(H,H)$ has no orientation for every infinite-dimensional real Hilbert space $H$ Chapter6/GapP1_DetNonOrientableTransport · isEmpty_lineOrientation_det_hilbert, nor $\mathrm{DET}(E,F)$ for $E$, $F$ isomorphic to such an $H$ or to $H\times H$ with $H\cong\mathbb{R}\times H$ Chapter6/GapP1_DetNonOrientableTransport · isEmpty_lineOrientation_det_of_equiv_hilbert Chapter6/GapP1_DetNonOrientableTransport · isEmpty_lineOrientation_det_of_equiv_prod. Infinite dimension enters through the surjective operator with a kernel; on a finite-dimensional space the same loop also has a Möbius band of cokernels, and the two cancel in det. *Not claimed*: infinite-dimensional Banach spaces without such structures. The remark presumably intends infinite-dimensional $E$, $F$ of the kind met in the applications (Findings, Chapter 6, item 135). (Gap G.60 closed, with this corrected form; row V6.80 narrower.)

**The classical case** (V6.81; opening of §6.6, p. 262). The text: for a smooth $f:U\to F$ on an open $U\subseteq E$ with Fredholm derivatives, the determinant lines of the $Df(x)$ form a continuous, even smooth, line bundle, with two orientations when $U$ is contractible. For $f$ of class $C^1$, $x\mapsto Df(x)$ is a continuous map into the Fredholm operators Chapter6/GapP1_ClassicalDetBundle · derivFredholm, the pull-back of $\mathrm{DET}(E,F)$ along it is a topological line bundle over $U$ Chapter6/GapP1_ClassicalDetBundle · ClassicalDet Chapter6/GapP1_ClassicalDetBundle · classicalDet_vectorBundle, and for contractible $U$ it has exactly two orientations, every orientation of one line extending to exactly one of the bundle Chapter6/GapP1_ClassicalDetBundle · classicalDet_two_orientations: $U$ is then simply connected, and locally path connected as an open subset of a Banach space (§VI.9). *Narrower* (V6.81): $C^1$ in place of smooth, which is weaker, and the bundle is topological; its smoothness, which would need smooth frames of the local determinant charts, is not formalized.

## VI.35 Lemma 6.5.3 (preprint 6.19): determinant bundles of vector bundles

The text (p. 260 (preprint p. 216)): *for finite-dimensional topological vector bundles $E$, $F$ over $X$, the bundles $\lambda(E)$, $\lambda^*(F)$, $E\oplus F$ and $E\otimes F$ have in a natural way the structure of topological vector bundles*, a standard lemma whose proof is left to the reader. §VI.7 applied it only to trivial bundles (V6.22). The file is `Chapter6/DetVectorBundles2021.lean` (Findings, Chapter 6, item 125).

**The statement formalized** (V6.22) Chapter6/DetVectorBundles2021 · DetBundle.lemma_6_5_3. *For Mathlib topological vector bundles $E$, $F$ with finite-dimensional normed model fibres over a topological space, the fibrewise $\lambda(E_b)$,*

$\lambda^*(E_b)$, $E_b \times F_b$ *and* $E_b \otimes F_b$ *are Mathlib topological vector bundles, with trivializations induced by those of* $E$ *and* $F$. *Proof,* $\lambda(E)$. A trivialization $e$ of $E$ and a basis $f_1, \dots, f_n$ of the model fibre give the frame $b \mapsto e^{-1}(b)f_1, \dots, e^{-1}(b)f_n$ over the base set of $e$, whose wedge $w_e(b)$ is a continuous nowhere-zero section of $\lambda(E)$; the trivialization is $x \mapsto w_e(b)^*(x)$ `Chapter6/DetVectorBundles2021 · DetBundle.lamPrebundle`. By the change-of-basis rule of §VI.2 the coordinate changes are the determinants of those of $E$ `Chapter6/DetVectorBundles2021 · DetBundle.coord_wedgeAt_eq_det`, continuous because the coordinate changes of $E$ are; this is what a Mathlib vector prebundle asks, and it gives the topology and the bundle structure. $\lambda^*(E)$: $\xi \mapsto \xi(w_e(b))$, with the inverse determinants `Chapter6/DetVectorBundles2021 · DetBundle.lamDualPrebundle`. $E \oplus F$ is Mathlib's fibrewise product with its vector bundle structure. $E \otimes F$: a pair of trivializations gives $u \otimes w \mapsto (i \mapsto (eu)_i\, e_1 w)$ into $F_1^n$, $n = \dim F_0$ `Chapter6/DetVectorBundles2021 · DetBundle.tenPrebundle`, with the matrices of $A \otimes C$ as coordinate changes for those $A$, $C$ of $E$, $F$ `Chapter6/DetVectorBundles2021 · DetBundle.tenMap_tenInv_eq`. □ *Readings.* "In a natural way" is read as: the trivializations of the new bundles are induced by those of $E$ and $F$, so that local frames give local frames. The model fibre of $E \otimes F$ is $F_1^n$, since Mathlib puts no norm on $F_0 \otimes F_1$; this does not change the bundle. Proposition 6.5.4 (preprint 6.20) is still proved with the explicit frames of §VI.7. (Gap G.59 closed; row V6.22 restated.)

## VI.36 Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds: property (P), the solution set near the family, $\pi$ a submersion, $L$ a smooth bundle

§§VI.11–VI.12 proved what the propagation uses: the general position of $F$ along the family, and the kernels as a continuous family of subspaces. The text says more: property (P) for every slice $F_t$; the solution set of $F$ near $(t, \varphi(t), 0)$ is a manifold with boundary with corners on which $\pi(t, y, \lambda) = t$ is a submersion; its fibres $S_t$ are manifolds with tangent spaces $L_t$; and $L$ is a smooth vector bundle (pp. 264–265 (preprint p. 222)). All of it is now formalized, on $\mathbb{R}^n$ (decision D2) and on the cube $[0,1]^n$ itself. The setting is that of §VI.11: $G = \tilde{f} + S$ with $G(q, \Phi(q)) = 0$ near $[0,1]^n$, and $F(\lambda, (q, y)) = G(q, y) + \sum \lambda_i s_i(q, y)$ (Findings, Chapter 6, items 47, 48, 126).

**Property (P) for the slices** (V6.33) `Chapter6/GapP1_SlicePropertyP · lemma_6_6_4_propertyP`. *Under the hypotheses of Lemma 6.6.4 (preprint 6.29) there are finitely many sc⁺-sections* $s_1, \dots, s_k$ *such that for every* $q$ *near* $[0,1]^n$ *the slice* $F_q(\lambda, y) = G(q, y) + \sum \lambda_i s_i(q, y)$ *is in general position at* $(0, \Phi(q))$ *— property (P) — and* $F$ *is in general position at* $(0, (q, \Phi(q)))$. The second clause is §VI.11. The first is derived from it, the reverse of the text's order `Chapter6/GapP1_SlicePropertyP · isGeneralPositionGerm_slice`. *The chart computation* `Chapter6/GapP1_SlicePropertyP · isParamGP_slice_of_isParamGP`: in a tame chart centred at $\Phi(q)$ and the product chart $(r \mapsto r - q) \times c$, properties (i)–(ii) of §V.19 for $G$ and the $s_i$ at $(q, \Phi(q))$ give (i)–(ii) for the slice at $\Phi(q)$. The $\mathbb{R}^n$-directions are absorbed by the derivative of the curve of zeros $q' \mapsto (q', \Phi(q'))$, which lies in $\ker G'$ (§VI.12); and its $X$-component lies in the reduced tangent space, because every corner coordinate of $\varphi \circ \Phi$ is $\geq 0$ near $q$ and vanishes at $q$, so its derivative vanishes `Chapter6/GapP1_SlicePropertyP · fderiv_mem_minimalSubspaceWith_of_centre` — $\mathbb{R}^n$ has no boundary. *The fibres*: the zeros of $F_q$ are the fibre $S_q$ of the solution set of $F$ over $q$; the slice is tame sc-Fredholm `Chapter6/GapP1_SlicePropertyP · isScFredholmSectionTame_slice`, and given (P), $S_q$ is near $(0, \Phi(q))$ a Mathlib manifold with boundary and corners `Chapter6/GapP1_SlicePropertyP · lemma_6_6_5_fibre_isManifold`, by Theorem 3.1.26 (Part III) for $F_q$. (Gap G.63 closed; row V6.33 restated.)

**The solution set near the family, and** $\pi$ (V6.34, V6.42). *For* $q$ *near* $[0,1]^n$, $(0, (q, \Phi(q)))$ *has an open neighbourhood in which the solution set of* $F$ *is a Mathlib manifold with boundary and corners* `Chap-`

ter6/SolutionManifoldFamily2021 · lemma_6_6_5_isManifold, *a smooth submanifold of dimension the index* Chapter6/SolutionManifoldFamily2021 · lemma_6_6_5_isSubmanifoldOfDim (Theorems 3.1.26 and 3.6.8 applied to $F$). *At the points of the family $d\pi$ maps the tangent space onto $\mathbb{R}^n$* Chapter6/SolutionManifoldFamily2021 · lemma_6_6_5_submersion: in the product chart the tangent space is $\ker F'$, $d\pi$ is the projection to the parameter, and the derivative of the curve of zeros lies in $\ker F'$ with parameter component the identity. *Off the family* Chapter6/GapP1_SubmersionNearFamily · lemma_6_6_5_submersion_nhds: surjectivity of $d\pi$ on the kernels is open along the zeros Chapter6/GapP1_SubmersionNearFamily · exists_nhds_surjective_linKer, because the zeros near a point in good position are the image of an sc$^+$-retraction, so the level-1 chart points vary continuously along them, the kernels have a continuous frame (§VI.12), and a determinant that is 1 at one point is nonzero nearby. So $\pi$ is a submersion on a neighbourhood of every point of the family. (Gap G.62 closed; rows V6.34, V6.42 restated.)

**$L$ is a smooth vector bundle** (V6.34) Chapter6/GapP1_SmoothL3 · lemma_6_6_5_smoothBundle. *In the setting of §VI.15 (with (P)), there are an open connected $V \supseteq [0,1]^n$ and a smooth family of subspaces over the manifold $V$ whose fibre at $q$ is $L_q = \ker F'(0,(q,z_q)) \cap \{dq = 0\}$, a Mathlib $C^\infty$ vector bundle. Step 1.* An sc-smooth map on $\mathbb{R}^n$ read in a chart is $C^\infty$ into every level on the open sets it maps into the chart Chapter6/GapP1_SmoothL1 · IsScSmooth.contDiffOn_levelMap_const. *Step 2.* The kernel vectors of an sc-Fredholm linearization are smooth points Chapter6/GapP1_SmoothL1 · IsSc1With.apply_mem_smooth_of_finiteDimensional. *Step 3, a smooth frame* Chapter6/GapP1_SmoothL3 · exists_local_smooth_frame_linKer: with an sc-retraction $S$ onto the solution set (Theorem 3.6.8) and a basis $b$ of one kernel, $e_i(q) = DS(z_q)b_i$ is $C^\infty$ into every level, since $q \mapsto (z_q, b_i)$ is sc-smooth into the tangent bundle and the tangent map of $S$ is sc-smooth; it is a frame near the base point, as in §VI.12. *Step 4.* A family of subspaces of a normed space with smooth local frames is a smooth vector bundle Chapter6/GapP1_SmoothL2 · SmoothSubspaceFamily.contMDiffVectorBundle, and cutting it by $\{dq = 0\}$, onto which the projection is surjective, keeps it smooth Chapter6/GapP1_SmoothL2 · SmoothSubspaceFamily.infKer. The continuation of orientations still uses only continuity. (Rows V6.34, V6.42 restated.)

**On the cube itself** (V6.33, V6.34). The same statements hold for families on $[0,1]^n$, boundary and corners included, with sections over the M-polyfold with corners $[0,1]^n \times X$ Chapter6/GapP1_CubeSections · cubePolyfold, whose degeneracy index counts the coordinates in $\{0,1\}$ Chapter6/GapP1_CubeSections · cubePolyfold_degIndex. The pull-back of a tame sc-Fredholm section to $[0,1]^n \times X$ is tame sc-Fredholm Chapter6/GapP1_CubeFredholm · StrongBundle.IsScFredholmSectionTame.paramCube, with charts of $[0,1]^n$ centred at every point Chapter6/GapP1_CubeCharts · UnitCube.cChart and basic germs extended by the mixed quadrants Chapter6/GapP1_CubeGerms · IsConjugateToBasicGerm.quadExtend. Lemma 6.6.4 (preprint 6.29) holds there Chapter6/GapP1_CubeLemmas · lemma_6_6_4_cube Chapter6/GapP1_CubeLemmas · lemma_6_6_4_cube_family, and near the family the solution set is a manifold with boundary and corners and a submanifold of dimension the index Chapter6/GapP1_CubeLemmas · lemma_6_6_5_isManifold_cube Chapter6/GapP1_CubeLemmas · lemma_6_6_5_isSubmanifoldOfDim_cube. On the cube, *(P) at $q$ is equivalent to the general position of $F$ at $(0,(q,\Phi(q)))$*, boundary and corners included Chapter6/GapP1_CubeBoundaryP · isGeneralPositionGerm_cubeSlice_iff Chapter6/GapP1_CubeBoundaryP · lemma_6_6_4_propertyP_cube: the tangent vectors of the family of zeros in reduced directions of $[0,1]^n$ have reduced $X$-components, by a one-sided minimum argument Chapter6/GapP1_CubeBoundarySlice · mem_minimalSubspaceWith_of_hasFDerivWithinAt. At these points $\pi$ is a submersion in the corner sense: $d\pi$ maps $\ker F'$ onto $\mathbb{R}^n$ and its reduced part onto the reduced tangent space of $[0,1]^n$ Chapter6/GapP1_CubeBoundarySubmersion · lemma_6_6_5_submersion_cube.

# VI.37 Paths and families on $[0,1]$; the facts on paths before Proposition 6.2.2 (preprint 6.2)

Decision D2 reads the paths, homotopies and families of the chapter on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$. Most statements now also hold as printed, for maps on the M-polyfolds $[0,1]$, $[0,1]^2$, $[0,1]^n$ (gap G.65, in part; Findings, Chapter 6, items 102, 122, 124).

**$[0,1]$, $[0,1]^2$ and $X \times [0,1]$.** $[0,1]$ is a tame M-polyfold with boundary `Chapter3/UnitIntervalPolyfold · unitIntervalPolyfold`, and every sc-smooth path on it is the reparametrization by $\sin^2(\pi r/2)$ of an sc-smooth path on $\mathbb{R}$ `Chapter3/UnitIntervalPolyfold · MPolyfold.IsScSmoothMap.exists_rn_reparam`; this transfers the statements on $\mathbb{R}$. $X \times [0,1]$ is tame for tame $X$, with $d(x,t) = d_X(x) + d_{[0,1]}(t)$ `Chapter6/PathsInterval · isTame_prod_unitInterval`, and $[0,1]^2$ is an M-polyfold with corners `Chapter6/PathsInterval · unitSquarePolyfold`.

**Theorems 6.6.11, 6.6.13 and 6.6.17 for paths on $[0,1]$** (V6.48, V6.57, V6.71). The push-forward along $\varphi : [0,1] \to U$ is read along the path $\varphi \circ \sin^2(\pi r/2)$ on $\mathbb{R}$, independently of the reparametrization `Chapter6/PathsInterval · theorem_6_6_11_interval`; the transport of Theorem 6.6.13 (preprint 6.37) is invariant under sc-smooth homotopies $[0,1]^2 \to X$ with fixed end points `Chapter6/PathsInterval · orTransportI_eq_of_homotopy`, and Theorem 6.6.17 holds with property (3) for paths on $[0,1]$ `Chapter6/PathsInterval · theorem_6_6_17_interval`.

**The path category on $[0,1]$** (V6.52). Sc-smooth homotopies $[0,1]^2 \to X$ with fixed end points `Chapter6/GapP1_PathCategoryInterval · ScHomotopicI` are an equivalence relation, and two paths on $[0,1]$ are homotopic in this sense iff their reparametrized paths on $\mathbb{R}$ are homotopic in the sense of §VI.20 `Chapter6/GapP1_PathCategoryInterval · scHomotopicI_iff`. The composition is the text's $\gamma(s) = \varphi(\beta(s))$ on $[0,1/2]$ and $\psi(\beta(s)-1)$ on $[1/2,1]$, with $\beta(s) = \kappa(3s) + \kappa(3s-2)$ `Chapter6/GapP1_PathCategoryInterval · betaConcat` `Chapter6/GapP1_PathCategoryInterval · ScPathI.concat` `Chapter6/GapP1_PathCategoryInterval · ScPathI.concat_toFun_of_le` `Chapter6/GapP1_PathCategoryInterval · ScPathI.concat_toFun_of_ge`. The classes form a category `Chapter6/GapP1_PathCategoryInterval · PathCatI` `Chapter6/GapP1_PathCategoryInterval · PathCatI.category`, a groupoid `Chapter6/GapP1_PathCategoryInterval · PathCatI.groupoid`, and $[\varphi] \mapsto [\varphi \circ \sin^2(\pi r/2)]$ is a full and faithful functor to the category of §VI.20, bijective on objects `Chapter6/GapP1_PathCategoryInterval · PathCatI.toPathCat_isEquivalence`, so the two readings give isomorphic categories.

**Proposition 6.6.3 (3) and (4) on $[0,1] \times X$ and $[0,1]^n \times X$** (V6.32). *With bump functions, for an sc-smooth path $\varphi : [0,1] \to X$ in the domain of a strong bundle chart there is an sc$^+$-section $S$ of the pull-back of $P$ to the M-polyfold $[0,1] \times X$ whose fibre part is the text's $-\beta(y)R(y, f(\varphi(t)))$, $\beta$ an sc-smooth cut-off equal to $1$ near $\varphi([0,1])$, with $(f+S)(t,\varphi(t)) = 0$ for every $t \in [0,1]$* `Chapter6/GapP1_UnitIntervalSections · proposition_6_6_3_three_interval`. The construction works in any product chart $\chi \times \varphi_c$ of $Q \times X$ `Chapter6/GapP1_UnitIntervalSections · PullbackChartData.ofProdChart` `Chapter6/GapP1_UnitIntervalSections · exists_isScPlusSection_prodChart`; on $[0,1]$ the principal part lifts to level 1, the parameter space being finite-dimensional, and the two charts $t$, $1-t$ of $[0,1]$ are glued with a partition of unity in $t$. $[0,1] \times X$ admits sc-smooth bump functions when $X$ does, which gives (4) `Chapter6/GapP1_UnitIntervalSections · proposition_6_6_3_four_interval`. The same holds for families on $[0,1]^n$, with sections over $[0,1]^n \times X$ `Chapter6/GapP1_CubeSections · proposition_6_6_3_three_cube` `Chapter6/GapP1_CubeSections · proposition_6_6_3_four_cube` `Chapter6/GapP1_CubeSections · admitsBumpFunctions_prod_cube`. The sign is that of $f + s$ (structural choice (1)).

**The facts on paths before Proposition 6.2.2 (preprint 6.2)** (V6.49) `Chapter6/`

SmoothPathComponents2021 · MPolyfold.paths_before_proposition_6_2_2. The text recalls (p. 240 (preprint p. 201)): *an M-polyfold is locally path connected, and two smooth points in one connected component (preprint: path component) are joined by an sc⁺-smooth path $\varphi : [0,1] \to X$. Step 1, the local step* Chapter6/SmoothPathComponents2021 · MPolyfold.exists_nhds_scJoinedIn: every $u$ in an open $U$ has an open $N \ni u$ in $U$ whose smooth points are pairwise joined by sc-smooth paths in $U$ — the retracted segment of a chart of $X$, without a strong bundle chart. Local path connectedness is proved from the charts, which are retracts of open sets of partial quadrants Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace. *Step 2, the global step* Chapter6/SmoothPathComponents2021 · MPolyfold.scJoinedIn_of_isConnected: in a connected open $U$ any two smooth points are joined by an sc-smooth path in $U$, by a chain argument over the neighbourhoods of Step 1, with the concatenation of paths (§VI.18), using that $U$ is connected. *Step 3, sc⁺* Chapter6/SmoothPathComponents2021 · MPolyfold.IsScSmoothMap.exists_scPlus_lift: $[0,1]$ is modelled on sc⁺-retracts, so the inclusion $[0,1] \to [0,1]^1$ is sc-smooth, and an sc-smooth path on $[0,1]$ lifts to an sc-smooth path into $X^1$. For smooth points, being in one connected component, in one path component and being joined by an sc-smooth path on $[0,1]$ are equivalent Chapter6/SmoothPathComponents2021 · mem_connectedComponent_iff_exists_scPath Chapter6/SmoothPathComponents2021 · MPolyfold.mem_pathComponent_iff_exists_scPath, which justifies reading "in the same path component" in Definitions 6.6.2 and 6.6.14 (preprint 6.27 and 6.38) as "joined by an sc-smooth path" (§VI.24); and two smooth points of one connected component have the same index Chapter6/SmoothPathComponents2021 · proposition_6_2_2_connectedComponent_2021def. "Sc⁺-smooth path" is read as an sc-smooth path that lifts to $X^1$; every sc-smooth path on $[0,1]$ does. (Gap G.56 closed.)

**What is still read on** $\mathbb{R}$ (gap G.65, in part; Proposition 6.6.3 (1′) holds on $[0,1]$ since 2026-11-13, §VI.10): the smooth bundle $L$ of Lemma 6.6.5 (preprint 6.30) over a neighbourhood of $[0,1]^n$ in $\mathbb{R}^n$ (V6.34), the stabilized kernels and their continuation (V6.42), and Remark 6.6.19 (preprint 6.42) over $\mathbb{R}^1 \times X$ (V6.62), with the cobordisms of Theorem 6.7.1 (preprint 6.43) (§VI.27).

## VI.38 The remarks after Theorem 6.6.17: the orientation bundle, orientability, $\mathbb{Z}_2$ and $\det_f$

The text (p. 275, new in the 2021 edition): with the topology of Theorem 6.6.17, $\pi_f : O_f \to X_\infty$ is the orientation bundle of $f$; $f$ is orientable iff $\pi_f$ has a global continuous section; $\mathbb{Z}_2$ acts continuously on $O_f$ by $(x, o_x) \mapsto (x, -o_x)$; and $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f \to X_\infty$ is the determinant bundle of $f$. The file is `Chapter6/GapP1_OrientationBundle.lean` (V6.82; Findings, Chapter 6, item 121).

**Sections** (V6.82). For local transport data whose fibres have two elements, with the topology of §VI.33: *$y \mapsto (y, o_y)$ is continuous iff the transport along every sc-smooth path carries $o$ along* Chapter6/GapP1_OrientationBundle · continuous_orSec_iff — near each point a continuous section stays in one sheet, and a sheet is carried along by the transport (§VI.33) — and every section of $\pi_f$ has this form Chapter6/GapP1_OrientationBundle · exists_eq_orSec. For the transport of $\Gamma_f$: *$y \mapsto o_y$ is an orientation of $f$ (Definition 6.6.16 (preprint 6.40)) iff $y \mapsto (y, o_y)$ is continuous* Chapter6/GapP1_OrientationBundle · isSectionOrientation_iff_continuous, by §VI.24, and so *$f$ is orientable iff $\pi_f$ has a global continuous section* Chapter6/GapP1_OrientationBundle · isOrientableSection_iff_exists_continuous_section.

**The $\mathbb{Z}_2$-action.** $(x, o) \mapsto (x, -o)$, $-o$ the other element of $\mathrm{Or}_x^f$, maps each sheet onto the sheet of the opposite continuous orientation, so it is a homeomorphism Chapter6/GapP1_OrientationBundle · orNegHomeomorph, an involution without fixed points over $\pi_f$; it is the action of $\mathbb{Z}^\times \cong \mathbb{Z}_2$ Chapter6/GapP1_OrientationBundle · orMulAction, continuous Chapter6/GapP1_OrientationBundle · continuous_orUnitsAct.

$\det_f$. A Mathlib vector bundle core with fibre $\mathbb{R}$ over $X_\infty$, with a trivialization for every continuous orientation $(U, o)$ and coordinate changes $\pm 1$ according as two continuous orientations agree — locally constant by §VI.33 (a) `Chapter6/GapP1_OrientationBundle · detCore` `Chapter6/GapP1_OrientationBundle · detBundleCore`. Its fibre over $x$ is $\mathbb{R} \otimes_{\mathbb{Z}_2} \mathrm{Or}_x^f$: $r \otimes a$ and $r' \otimes a'$ have the same class iff $(r', a') = (r, a)$ or $(-r, -a)$ `Chapter6/GapP1_OrientationBundle · detMk_eq_iff`, and $r \otimes a$ has coordinate $r$ in the trivialization of $(U, o)$ if $a = o_x$, and $-r$ if not `Chapter6/GapP1_OrientationBundle · localTriv_detMk`. (Gap G.91 closed; row V6.82 restated.)

# Part VII. Ep-groupoids (HWZ Chapter 7, §§7.1–7.3, 7.5, 7.6 (2017 §§7.1–7.5))

Chapter 7 opens Part II of the book. Part II has no 2014 preprint, so its numbers and pages are those of the book (Springer 2021, pp. 287–356), with the 2017 text's in parentheses (2017 pp. 225–256; see `BookConcordance.md`). The chapter introduces ep-groupoids, the sc-smooth version of étale proper Lie groupoids. §7.1 defines groupoids, isotropy groups and ep-groupoids (Definitions 7.1.1–7.1.3 (2017: 7.1–7.3)), the local sc-diffeomorphism $\widehat{\varphi}$ of a morphism and its tangent (Definition 7.1.4 (2017: 7.4)), the levels and the degeneracy index of the orbit space (Definition 7.1.5 (2017: 7.5)), tame ep-groupoids and the category $\mathcal{EP}$ (Definitions 7.1.6, 7.1.7 (2017: 7.6, 7.7)). It proves that $\pi$ is open, that $|X|$ has countable bases of neighbourhoods, the local closure relation and the finiteness of the isotropy groups (Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1)). It states the natural representation of $G_x$ (Theorem 7.1.13 (2017: 7.1)), discusses the translation groupoid $G_x \ltimes U(x)$ and the action $\widehat{\Phi}$ (pp. 293–297 (2017 pp. 231–233)), extends the natural representation to finite sets of isomorphic objects (Proposition 7.1.15 (2017: 7.5), Definition 7.1.18 (2017: 7.8)), identifies $|U(x)|$ with $G_x \backslash U(x)$ (Proposition 7.1.19 (2017: 7.6)) and defines local uniformizers and ep-subgroupoids (Definitions 7.1.20, 7.1.21 (2017: 7.9, 7.10), Proposition 7.1.22 (2017: 7.7)). §7.2 treats effective ep-groupoids, the subgroups $T_x$, the reduced morphism space $\mathbf{X}^R$ and the reduced ep-groupoid $X_R$ (Definitions 7.2.1–7.2.15 (2017: 7.11–7.13), Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Propositions 7.2.4, 7.2.10 (2017: 7.8, 7.9) and Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14)). §7.3 proves that $|X|$ is locally metrizable, regular and Hausdorff (Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2), Definition 7.3.7 (2017: 7.14)) and lifts ep-groupoids to level 1 (Proposition 7.3.9 (2017: 7.11)). §7.5 (2017 §7.4) defines paracompact orbit spaces and sc-smooth partitions of unity (Definitions 7.5.1, 7.5.5 (2017: 7.15, 7.16), Remarks 7.5.2, 7.5.6 (2017: 7.1, 7.2)), proves Proposition 7.5.3 (2017: 7.12) and Theorems 7.5.4, 7.5.7 (2017: 7.3, 7.4), and characterizes the closed subsets of $|X|$ (Definition 7.5.8 (2017: 7.17), Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14)). The appendix §7.6 (2017 §7.5) proves Theorem 7.1.13 (2017: 7.1) as Theorem 7.6.1 (2017: 7.5) (Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9)) and Theorem 7.5.7 (2017: 7.4) as Theorem 7.6.4 (2017: 7.6) (Lemma 7.6.5 (2017: 7.10)), and ends with Corollary 7.6.6 (2017: 7.3). This Part retells §7.1 in §§VII.1–VII.12 — Definitions 7.1.1, 7.1.2 (2017: 7.1, 7.2) (§VII.1), Definition 7.1.3 (2017: 7.3) and properness (§VII.2), Definitions 7.1.4–7.1.6 (2017: 7.4–7.6) (§VII.3), Definition 7.1.7 (2017: 7.7) and the remark after it (§VII.4), Propositions 7.1.8–7.1.10 (2017: 7.1–7.3) (§VII.5), Lemma 7.1.11 (2017: 7.1) and Proposition 7.1.12 (2017: 7.4) (§VII.6), Theorem 7.1.13 (2017: 7.1) (§VII.7), pp. 293–297 (2017 pp. 231–233) (§VII.8), the uniqueness of natural representations (§VII.9), Proposition 7.1.15 (2017: 7.5) and Definition 7.1.18 (2017: 7.8) (§VII.10), Proposition 7.1.19 (2017: 7.6) and Definition 7.1.20 (2017: 7.9) (§VII.11), Definition 7.1.21 (2017: 7.10) and Proposition 7.1.22 (2017: 7.7) (§VII.12); §7.2 in §§VII.13–VII.17 — Definition 7.2.1 (2017: 7.11) and Lemmas 7.2.2, 7.2.3 (2017:

7.2, 7.3) (§VII.13), Proposition 7.2.4 (2017: 7.8) (§VII.14), $T_x$ and Definition 7.2.7 (2017: 7.12) (§VII.15), Proposition 7.2.10 (2017: 7.9) (§VII.16), $X_R$, Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14) and Definition 7.2.15 (2017: 7.13) (§VII.17); §7.3 in §§VII.18–VII.22 — Theorem 7.3.1 (2017: 7.2) and Lemmas 7.3.2, 7.3.3 (2017: 7.4, 7.5) (§VII.18), Lemmas 7.3.4, 7.3.5 (2017: 7.6, 7.7) and Corollary 7.3.6 (2017: 7.1) (§VII.19), Michael's theorem of Part I (§VII.20), Definition 7.3.7 (2017: 7.14) and Corollary 7.3.8 (2017: 7.2) (§VII.21), Proposition 7.3.9 (2017: 7.11) (§VII.22); §7.5 (2017 §7.4) in §§VII.23–VII.25 — Definition 7.5.1 (2017: 7.15), Remark 7.5.2 (2017: 7.1) and Proposition 7.5.3 (2017: 7.12) (§VII.23), Theorem 7.5.4 (2017: 7.3), Definition 7.5.5 (2017: 7.16), Remark 7.5.6 (2017: 7.2) and the statement of Theorem 7.5.7 (2017: 7.4) (§VII.24), Definition 7.5.8 (2017: 7.17) and Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14) (§VII.25); the appendix in §§VII.26–VII.29 — the proof of Theorem 7.1.13 (2017: 7.1) (§VII.26), Lemma 7.6.5 (2017: 7.10) (§VII.27), the proof of Theorem 7.5.7 (2017: 7.4) (§VII.28) and Corollary 7.6.6 (2017: 7.3) (§VII.29); the witnesses and counterexamples (§VII.30); what is not formalized (§VII.31); and what the formalization taught us (§VII.32). Each item carries the number of its comparison item (V7.1–V7.68) in `Exposition/verification/ch07.md`. The additions of the 2021 edition (V7.69–V7.104) are retold in §§VII.33–VII.43: §7.1 (§VII.33), §7.2 (§§VII.34–VII.35), Appendices 7.6.3 and 7.6.4 (§VII.36), §7.4, new in the 2021 edition (§§VII.37–VII.39), Appendix 7.6.5 (§VII.40), what is not formalized and what they taught us (§VII.41), Theorem 7.6.9 in full, the boundary ep-groupoid (§VII.42), and §7.4 on every lift $X^i$ (§VII.43). The work that closed the remaining gaps is retold in §§VII.44–VII.50: $T\varphi$ as an sc-isomorphism, the facts (3), (4), $|X_\infty|$ and the density of $X_\infty$ (§VII.44), the Part I facts of §7.4 as theorems (§VII.45), Theorem 7.6.7 and Corollary 7.6.8 for every $i$ (§VII.46), Zhou's retract of Appendix 7.6.3 (§VII.47), the 2021 forms of Propositions 7.2.10, 7.2.16, Theorem 7.2.14 and Corollary 7.6.6 as printed (§VII.48), an ep-groupoid whose orbit space is not paracompact (§VII.49), and Theorem 7.6.9 for every representative and in the book's form (§VII.50).

**What is proved, and what is not.** Every statement of the chapter is formalized, and every gap of the chapter is closed: G.70 ($T\varphi$, §VII.44), G.71 (the M-polyfold $\mathbf{X}^R$, §§VII.34, VII.48), G.72 ($|X_\infty|$, §VII.44), G.73 (the density of $X_\infty$, §VII.44), G.82 (Theorem 7.6.7 for every $i$, §VII.46), G.83 (Zhou's retract, §VII.47), G.84 (§7.4 on every lift, §VII.43) and G.85 (the Part I facts of §7.4, §VII.45). Theorems 7.1.13–7.6.4 (2017: 7.1–7.6) are proved, and so are Propositions 7.1.8–7.2.4 and 7.3.9–7.5.10 (2017: 7.1–7.8 and 7.11–7.14), Lemmas 7.1.11–7.6.5 (2017: 7.1–7.10) and Corollaries 7.3.6–7.6.6 (2017: 7.1–7.3); Corollary 7.6.6 (2017: 7.3) as printed, Part I's Theorem 5.5.16 (2017: 5.14) of Godefroy–Troyanski–Whitfield–Zizler being now a theorem (§§VII.29, VII.48). Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) are proved in their 2021 form: the natural charts of $\mathbf{X}^R$ with no hypothesis (Proposition 7.2.10), paracompactness for every ep-groupoid (Proposition 7.2.12), and the reduced ep-groupoid under the 2021 text's hypothesis that $T_R$ is Hausdorff (Theorem 7.2.14, Proposition 7.2.16) (§§VII.34, VII.48); for every ep-groupoid, as the 2017 text states them, they do not hold as we read it, and $\mathbb{Z}/2$ acting on a dimension-jumping retract is the counterexample `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor` (§VII.16). The 2021 additions to §§7.1, 7.2, the new §7.4 and Appendices 7.6.3–7.6.5 are proved (§§VII.33–VII.50). Remark 7.5.6 (2017: 7.2) is an assessment, with nothing to formalize (row V7.57). One statement is formalized in a narrower form: in Theorem 7.6.9 (row V7.98), "$\Phi$ is a homeomorphism onto $\theta$" holds as we read it only for $V$ centred at $x$; for the other representatives $\Phi$ is an sc-diffeomorphism onto the open set $\theta \cap V^x$, and both are proved (§§VII.40, VII.50). The open question of p. 340 (2017 p. 249), whether property (5) follows from (1)–(4), is settled negatively: the Čech groupoid of a Prüfer-type surface is an ep-groupoid whose orbit space is not paracompact `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_epGroupoid_not_hasParacompactOrbitSpace` (§VII.49). Five places of the 2017 text needed more than a reading. The remark after Definition 7.1.7 (2017: 7.7), that a functor sc-smooth on objects is sc-

smooth on morphisms, does not hold as we read it, and a counterexample on the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$ shows it; it holds for functors continuous on morphisms, which is what its argument uses and what the 2021 text asks (§§VII.4, VII.33). The proof of Proposition 7.2.10 (2017: 7.9) gives the charts of $\mathbf{X}^R$ and their sc-smooth transitions but, as we read it, neither Hausdorffness nor paracompactness; Hausdorffness is the hypothesis of the 2021 Theorem 7.2.14, and paracompactness is proved (§§VII.16, VII.34). The natural representation of Theorem 7.1.13 (2017: 7.1) is taken on a connected neighbourhood, which the text chooses on 2017 p. 231 but does not put in the statement; connectedness makes $\Phi$ a homomorphism and $\Gamma$ a functor, and the local path connectedness of M-polyfolds that supplies it is proved (§VII.7). The tameness of $X^1$ in Proposition 7.3.9 (2017: 7.11) needs that the index lift of a tame retraction is tame, which Part I left open; it is proved (§VII.22). And several omitted steps are supplied: the uniqueness of natural representations behind "adapting restrictions" (§VII.9), the conjugation $\varphi T_x \varphi^{-1} = T_y$ that the composition of $X_R$ needs (§VII.15), the passage from compact to component compact sets in Corollary 7.3.8 (2017: 7.2) (§VII.21), the support bound that makes the sum in the proof of Theorem 7.5.7 (2017: 7.4) locally finite (§§VII.27–VII.28) and Michael's theorem on unions of closed paracompact sets (§VII.20). Proposition 7.1.15 (2017: 7.5) is proved by a construction without induction (§VII.10), Proposition 7.5.3 (2017: 7.12) by Smirnov's theorem pulled back to $|X^1|$ (§VII.23), Proposition 7.5.10 (2) (2017: 7.14) without Lemma 2.7.3 (2017: 2.7) (§VII.25), Lemma 7.3.4 (2017: 7.6) from the closedness of points and regularity (§VII.19), and the normalization of Theorem 7.5.7 (2017: 7.4) by bump coverings (§VII.28). Nothing of the chapter is left unformalized (§VII.31).

**Seven structural choices.** (1) *A groupoid is a structure of maps.* Objects and morphisms are types; $s, t, u, \iota$ are maps and the composition $m$ is a total map whose value on a non-composable pair is a junk value never used; the groupoid laws are required on composable pairs. A bridge to Mathlib's groupoids makes "full" and "faithful" Mathlib's. (2) *Properness is Mathlib's.* Definition 7.1.3 (3) (2017: 7.3) is the properness, in Mathlib's sense, of $t$ restricted to $s^{-1}(\mathrm{cl}_X V)$, for some open $V \ni x$, per point; on metrizable spaces it is equivalent to the sequential form the text uses (§VII.2). (3) *Condition (4) in two forms.* It is stated literally, on the fibred product of Part I's Proposition 2.5.20 (2017: 2.15; preprint 2.46, §II.17), and used in the equivalent *sheet form*: for every local inverse $\sigma$ of $s$, the map $\psi \mapsto m(\sigma(t\psi), \psi)$ is sc-smooth. (4) *Natural representations are data.* "An open neighbourhood $U(x)$ equipped with the natural $G_x$-action" is always a datum $(U, \Phi, \Gamma)$ with $U$ open, connected and containing $x$, never a hypothesis "for every such $U(x)$". (5) *$G \times U$ is a product with a discrete M-polyfold*, the finite disjoint union of copies of $U$, on which sc-smoothness is sc-smoothness on each slice (§VII.8). (6) *The reduced morphisms* $\mathbf{X}^R$ are a quotient whose M-polyfold structure is built under the hypothesis that it is Hausdorff, the 2021 text's hypothesis on $T_R$; its paracompactness is proved, for every ep-groupoid (§§VII.16, VII.34). (7) *Michael's theorem without Hausdorff.* Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17) (gap G.15 there) are proved for regular spaces, and Mathlib's paracompactness, like Definition 7.5.1 (2017: 7.15), does not include Hausdorff (§VII.20).

**Map.**

- `Chapter7/Groupoid`: Definitions 7.1.1, 7.1.2 (2017: 7.1, 7.2); the orbit space, saturations, the conjugation of isotropy groups, regular objects, full subcategories, translation groupoids, functors and the bridge to Mathlib.
- `Chapter7/EpGroupoid`: Definition 7.1.3 (2017: 7.3) with the fibred-product and sheet forms of (4), Definition 7.1.4 (2017: 7.4), the levels, Definitions 7.1.5–7.5.8 (2017: 7.5–7.17) as structures and predicates, the trivial ep-groupoid and its witnesses.
- `Chapter7/ProperMaps`: proper maps between metrizable spaces, the sequential criterion, proper restrictions $t : s^{-1}(Q) \to X$.

- `Chapter7/LocalDiffeo`: local sc-diffeomorphisms, their levels, degeneracy index and tameness; Part I's partitions of unity on $\mathbb{R}^n$.
- `Chapter7/Basic`: the remark after Definition 7.1.7 (2017: 7.7), Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1), the conjugation $G_x \cong G_y$.
- `Chapter7/Discrete`: finiteness from proper locally injective maps, discrete M-polyfolds, $G \times X$ and its sheets.
- `Chapter7/LocConnected`: local path connectedness of retracts and M-polyfolds, uniqueness of sections on connected sets.
- `Chapter7/NaturalRep`: Theorem 7.1.13 (2017: 7.1) = Theorem 7.6.1 (2017: 7.5) with Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9).
- `Chapter7/NaturalAction`: the consequences of Theorem 7.1.13 (2017: 7.1) on pp. 293–297 (2017 pp. 231–233) and the uniqueness of natural representations.
- `Chapter7/NaturalRepFinite`: Proposition 7.1.15 (2017: 7.5) by the star construction, Definition 7.1.18 (2017: 7.8).
- `Chapter7/Uniformizer`: Proposition 7.1.19 (2017: 7.6), Definition 7.1.20 (2017: 7.9) and the claims after it.
- `Chapter7/SubGroupoid`: Definition 7.1.21 (2017: 7.10), the M-polyfold $\mathbf{A}$, Proposition 7.1.22 (2017: 7.7).
- `Chapter7/Effective`: Definition 7.2.1's (2017: 7.11) consequences, Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Proposition 7.2.4 (2017: 7.8), $T_x$.
- `Chapter7/Reduced`: Definition 7.2.7 (2017: 7.12) made precise, Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), Definition 7.2.15 (2017: 7.13).
- `Chapter7/OrbitTopology`: Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2).
- `Chapter7/Michael`: Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17), unions of closed paracompact sets.
- `Chapter7/Lift`: Proposition 7.3.9 (2017: 7.11), the map $|X^1| \to |X|$, Definition 7.5.1's (2017: 7.15) consequences, Remark 7.5.2 (2017: 7.1), Proposition 7.5.3 (2017: 7.12).
- `Chapter7/ProperSubsets`: Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14).
- `Chapter7/PartitionsOfUnity`: Theorem 7.5.4 (2017: 7.3), Lemma 7.6.5 (2017: 7.10), Theorem 7.5.7 (2017: 7.4) = Theorem 7.6.4 (2017: 7.6), Corollary 7.6.6 (2017: 7.3).
- `Chapter7/NonVacuity`: the translation groupoid of a finite group action, $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection, the trivial $\mathbb{Z}/3$-action, the counterexamples.
- `Chapter7/Book2021Basic`: the 2021 forms of the remark after Definition 7.1.7 and of Proposition 7.1.15, the data (7.3), Proposition 7.1.14, Lemma 7.1.16, Corollary 7.1.17 (§VII.33).
- `Chapter7/ReducedCriterion`: the 2021 §7.2 from Definition 7.2.5 on, LUC, Theorem 7.2.22, Appendices 7.6.3 and 7.6.4 (§§VII.34–VII.36).
- `Chapter7/Zhou`: §7.4, the index lifts, the Part I facts as hypotheses, (ZHOU), (SLUC), (VSLUC), (LR), (SLR) (§§VII.37–VII.39).
- `Chapter7/TameBoundaryEp`: Appendix 7.6.5, the boundary groupoid of a tame ep-groupoid, Theorem 7.6.9 (topological part) (§VII.40).
- `Chapter7/TameBoundaryEpStructure`: Theorem 7.6.9 in full, on Part I's Theorem 2.7.13 (2)–(4): the boundary ep-groupoid and the ep-functor $B$ (§VII.42).
- `Chapter7/ZhouLevels`: §7.4 on every lift $X^i$: Propositions 7.4.9, 7.4.16–7.4.18, and Corollary 7.4.11, Theorems 7.4.13, 7.4.24 for every $X^i$ (§VII.43).
- `Chapter7/TangentMorphism`, `Chapter7/TangentMorphismLinear`: Definition 7.1.4, $T\varphi$ as a bijec-

tion, a bounded linear isomorphism and an sc-isomorphism (§VII.44).

- `Chapter7/SubGroupoidTangent`, `Chapter7/SubGroupoidTangentSc`: the facts (3), (4) on $A$ (§VII.44).
- `Chapter7/SmoothPointsDense`: the density of $X_\infty$ in every $X_m$, the filtration of $|X|$ (§VII.44).
- `Chapter7/ZhouFacts`, `Chapter7/SmoothConnectedReps`, `Chapter7/SmoothRegularAssembly`: the Part I facts of §7.4 for every M-polyfold, and §7.4, Appendix 7.6.4 without them (§VII.45).
- `Chapter7/GapP2_ARaise`: Theorem 7.6.7 and Corollary 7.6.8 for every $i$, as isomorphisms of ep-groupoids (§VII.46).
- `Chapter7/GapP2_A2Line`, `GapP2_A2Shift`, `GapP2_A2Flat`, `GapP2_A2Family`, `GapP2_A2ZhouGroupoid`, `GapP2_A2Sigma`: Zhou's retract of Appendix 7.6.3: the scale $H^{m,\delta_m}(\mathbb{R})$, the sc-smooth retraction, the ep-groupoid and its non-effective part (§VII.47); `Chapter7/GapP2_AZhouRetract`: the same computation for the analogue (§VII.36).
- `Chapter7/GapP2_AReduced`, `Chapter7/GapP2_ACor766`: Proposition 7.2.10, Theorem 7.2.14, Proposition 7.2.16 in the 2021 form; Corollary 7.6.6 as printed (§VII.48).
- `Chapter7/PropertyFiveResearch`: an ep-groupoid with Hausdorff, non-paracompact orbit space (§VII.49).
- `Chapter7/TameBoundaryFull`, `Chapter7/BookFormTameBoundaryEp2021`: Theorem 7.6.9 for every representative, and in the book's form (§VII.50).

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid: $X$ is the object M-polyfold and $\mathbf{X}$ the morphism M-polyfold, both Hausdorff and paracompact, as every M-polyfold of the project (Definition 2.3.4 (2017: 2.8), §II.5); $s, t : \mathbf{X} \to X$ are source and target, $m(\varphi, \psi) = \varphi \circ \psi$ the composition, $u(x) = 1_x$ the unit and $\iota(\varphi) = \varphi^{-1}$ the inverse. $|X|$ is the orbit space with the quotient topology and $\pi : X \to |X|$ the projection; $G_x$ is the isotropy group of $x$, $\mathrm{cl}_X$ the closure in $X$, and $\mathbf{U}(x)$ the set of morphisms with source and target in $U(x)$. For a natural representation $(U, \Phi, \Gamma)$ at $x$, $g * y = \Phi(g)(y)$. A *sheet* is an open set $\mathbf{W} \subseteq \mathbf{X}$ on which $s$ is an sc-diffeomorphism onto an open set, its inverse a *local inverse* of $s$. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses; the Lean names carry the 2021 numbers (Lemma 2.7.3 and Proposition 2.7.4, 2017 Lemma 2.7 and Proposition 2.17, are `lemma_2_7_3` and `proposition_2_7_4`; Theorem 5.5.10, 2017 Theorem 5.12, is `theorem_5_5_10`), and the Lean names of Chapter 7 carry the 2021 numbers too, with `_2017` for the statements that exist only in 2017. Findings are those of `Exposition/Findings.md`, "Chapter 7".

## VII.1 Definitions 7.1.1 and 7.1.2 (2017: 7.1 and 7.2): groupoids, the orbit space, isotropy

**Definition 7.1.1 (2017: 7.1)** (V7.1). A groupoid is a structure of maps $s, t, m, u, \iota$ on a type of objects and a type of morphisms `Chapter7/Groupoid · GroupoidStr`, with associativity, units and inverses required on composable pairs; the morphisms $x \to y$ are a set `Chapter7/Groupoid · GroupoidStr.mor`, and so are the morphisms between points of a set $A$ `Chapter7/Groupoid · GroupoidStr.morOn`. The composition of a non-composable pair is a junk value; no statement uses it. The text's list of consequences — $(\varphi^{-1})^{-1} = \varphi$, $1_x^{-1} = 1_x$, $(\varphi \circ \psi)^{-1} = \psi^{-1} \circ \varphi^{-1}$, uniqueness of inverses — is derived from the laws `Chapter7/Groupoid · GroupoidStr.inv_inv` `Chapter7/Groupoid · GroupoidStr.inv_unit` `Chapter7/Groupoid · GroupoidStr.inv_comp_rev` `Chapter7/Groupoid · GroupoidStr.eq_inv_of_comp_eq_unit`. Objects are isomorphic if there is a morphism between them `Chapter7/Groupoid · GroupoidStr.Isomorphic`, an equivalence relation `Chapter7/Groupoid · GroupoidStr.orbitSetoid` whose quotient is $|X|$ `Chapter7/Groupoid · GroupoidStr.OrbitSpace` with projection $\pi$ `Chapter7/Groupoid · GroupoidStr.π`; $\pi(x) = \pi(y)$ iff $x \cong y$ `Chapter7/Groupoid · GroupoidStr.π_eq_π_iff`,

$\pi(s\varphi) = \pi(t\varphi)$ Chapter7/Groupoid · GroupoidStr.π_src_eq_π_tgt and $\pi$ is onto Chapter7/Groupoid · GroupoidStr.surjective_π. The saturation $\pi^{-1}(\pi(A))$, used from Definition 7.1.21 (2017: 7.10) on, is defined here Chapter7/Groupoid · GroupoidStr.saturation, with saturated sets Chapter7/Groupoid · GroupoidStr.IsSaturated Chapter7/Groupoid · GroupoidStr.isSaturated_iff. The bridge to Mathlib's groupoids is a type synonym of the objects Chapter7/Groupoid · GroupoidStr.Ob with the groupoid instance Chapter7/Groupoid · GroupoidStr.instGroupoid, whose isomorphism classes are those of the orbit relation Chapter7/Groupoid · GroupoidStr.orbitSetoid_eq (Findings, Chapter 7, item 1) (pp. 287–288 (2017 pp. 225–226)).

**Definition 7.1.2 (2017: 7.2)** (V7.2). $G_x$, the morphisms $x \to x$ Chapter7/Groupoid · GroupoidStr.Isotropy, is a group under composition with $gh = g \circ h$ Chapter7/Groupoid · GroupoidStr.Isotropy.val_mul Chapter7/Groupoid · GroupoidStr.Isotropy.val_one Chapter7/Groupoid · GroupoidStr.Isotropy.val_inv, the text's order. Mathlib's vertex group multiplies in the other order; it is identified with the opposite group Chapter7/Groupoid · GroupoidStr.isotropyEquivVertexGroup, and $G_x$ with the automorphisms of $x$ in the bridge Chapter7/Groupoid · GroupoidStr.isotropyEquivAut (Findings, Chapter 7, item 2) (p. 288 (2017 p. 226)).

## VII.2 Definition 7.1.3 (2017: 7.3), and properness

**Definition 7.1.3 (2017: 7.3)** (V7.3) Chapter7/EpGroupoid · EpGroupoid. An ep-groupoid is a groupoid with M-polyfold structures on the objects and on the morphisms such that (1) $s$ and $t$ are local sc-diffeomorphisms, (2) $u$ and $\iota$ are sc-smooth, (3) every $x$ has an open neighbourhood $V(x)$ with $t : s^{-1}(\mathrm{cl}_X V(x)) \to X$ proper, and (4) $m$ is sc-smooth on $\mathbf{X}\,{}_s\times_t \mathbf{X}$. "In particular all structure maps are sc-smooth": $s$ and $t$ are Chapter7/EpGroupoid · EpGroupoid.isScSmoothMap_src Chapter7/EpGroupoid · EpGroupoid.isScSmoothMap_tgt, and continuous Chapter7/EpGroupoid · EpGroupoid.continuous_src Chapter7/EpGroupoid · EpGroupoid.continuous_tgt Chapter7/EpGroupoid · EpGroupoid.continuous_unit Chapter7/EpGroupoid · EpGroupoid.continuous_inv (pp. 288–290 (2017 pp. 226–227)).

*Condition (3).* It is existential, one open $V(x)$ per point, as the text writes it. The universal reading, "for every open $V \ni x$", fails: in the pair groupoid of $\mathbb{Z}$ (objects $\mathbb{Z}$, one morphism between any two objects) $t$ is proper over $s^{-1}(\{x\})$ but not over $s^{-1}(\mathbb{Z})$ Chapter7/NonVacuity · Examples.pair_isProperMap Chapter7/NonVacuity · Examples.pair_not_isProperMap_univ Chapter7/NonVacuity · Examples.pair_not_forall_isProperMap; every smaller open neighbourhood of $x$ works as soon as one does (§VII.25) (Findings, Chapter 7, items 3, 103).

*Condition (4).* The text cites "Lemma 2.8 in [38]" for the M-polyfold structure of the fibred product; this is an external reference, and the statement used is Part I's Proposition 2.5.20 (2017: 2.15) (§II.17): since $s$ is a local sc-diffeomorphism, $\mathbf{X}\,{}_s\times_t \mathbf{X}$ is a sub-M-polyfold of $\mathbf{X} \times \mathbf{X}$ Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct, and (4) is stated on it Chapter7/EpGroupoid · IsScSmoothComp. The proofs use an equivalent *sheet form* Chapter7/EpGroupoid · IsScSmoothCompSheet: for every sc-diffeomorphism $s : \mathbf{W} \to V$ with inverse $\sigma$, the map $\psi \mapsto m(\sigma(t\psi), \psi)$ is sc-smooth on $t^{-1}(V)$. *Step 1.* The graph of $\sigma \circ t$ over $t^{-1}(V)$ is an open piece of the fibred product, and the second projection is an sc-diffeomorphism from it onto $t^{-1}(V)$, with inverse $\psi \mapsto (\sigma(t\psi), \psi)$. *Step 2.* So the literal (4) gives the sheet form by composition, and conversely the fibred product is covered by such graphs Chapter7/EpGroupoid · isScSmoothComp_iff_sheet. Every ep-groupoid satisfies the sheet form Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet. The "for every $\sigma$" of the sheet form carries no vacuity risk, being equivalent to the literal condition (Findings, Chapter 7, item 4).

**Properness in sequential form** (V7.4). The text uses properness through sequences: in Proposition 7.1.10 (2017: 7.3), Lemma 7.1.11 (2017: 7.1), Proposition 7.1.22 (2017: 7.7), Proposition 7.3.9 (2017: 7.11), Proposition 7.5.9 (2017: 7.13) and Lemma 7.6.2 (2017: 7.8). For a continuous map between metrizable spaces,

properness is equivalent to: every sequence whose images converge has a convergent subsequence `Chapter7/ProperMaps · isProperMap_iff_exists_tendsto_subseq`. *Step 1, from properness.* The images and their limit form a compact set, whose preimage is compact, hence sequentially compact `Chapter7/ProperMaps · exists_tendsto_subseq_of_isProperMap`. *Step 2, to properness.* Preimages of compact sets are sequentially compact, hence compact in a metrizable space, and a continuous map with compact fibres of compacta into a compactly generated space is proper (Mathlib's characterization) `Chapter7/ProperMaps · isProperMap_of_exists_tendsto_subseq`. For the restriction of $t$ to $s^{-1}(Q)$ the criterion reads: every sequence with $s(\varphi_k) \in Q$ and $t(\varphi_k)$ convergent has a subsequence converging to a morphism over $Q$ `Chapter7/ProperMaps · isProperMap_domRestrict_iff`, over $\mathrm{cl}\,Q$ when $Q$ is closed `Chapter7/ProperMaps · isProperMap_domRestrict_iff_of_isClosed` `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff`, and it is the compactness of $s^{-1}(Q) \cap t^{-1}(K)$ for compact $K$ `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff_isCompact_preimage`. Three consequences recur. Properness passes to smaller closures, so any open $V' \subseteq V(x)$ around $x$ will do `Chapter7/ProperMaps · isProperMap_domRestrict_closure_mono`; the image $t(s^{-1}(Q))$ is closed for closed $Q$, a proper map being closed `Chapter7/ProperMaps · isClosed_image_preimage_of_isProperMap_domRestrict`; and the fibres over points are finite when $t$ is locally injective `Chapter7/ProperMaps · finite_setOf_of_isProperMap_domRestrict`. If $t$ is proper over $s^{-1}(Q)$ and the units exist, $Q$ is closed: a limit $x$ of $x_n \in Q$ is the source of a limit of a subsequence of the units $1_{x_n}$ `Chapter7/ProperMaps · isClosed_of_isProperMap_domRestrict`. The hidden hypothesis is metrizability, which M-polyfolds have (Theorem 2.3.10 (2017: 2.2), §II.7) (Findings, Chapter 7, item 3).

## VII.3 Definition 7.1.4 (2017: 7.4), the levels, Definitions 7.1.5 and 7.1.6 (2017: 7.5 and 7.6)

**Definition 7.1.4 (2017: 7.4)** (V7.5). For a morphism $\varphi : x \to y$ the text takes open $\mathbf{U}(\varphi)$, $U(x)$, $U(y)$ with $s : \mathbf{U}(\varphi) \to U(x)$ and $t : \mathbf{U}(\varphi) \to U(y)$ sc-diffeomorphisms and sets $\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} : U(x) \to U(y)$. Such a sheet exists at every morphism `Chapter7/EpGroupoid · EpGroupoid.Sheet` `Chapter7/EpGroupoid · EpGroupoid.exists_sheet`: (1) gives open neighbourhoods of $\varphi$ on which $s$, resp. $t$, is an sc-diffeomorphism onto an open set, and on their intersection both are, an sc-diffeomorphism restricting to one on every open subset. A sheet is chosen at every morphism `Chapter7/EpGroupoid · EpGroupoid.sheetAt` `Chapter7/EpGroupoid · EpGroupoid.mem_sheetAt`, and $\widehat{\varphi}$ is defined from it `Chapter7/EpGroupoid · EpGroupoid.Sheet.hat` `Chapter7/EpGroupoid · EpGroupoid.hat`: an sc-diffeomorphism `Chapter7/EpGroupoid · EpGroupoid.Sheet.isScDiffeoOn_hat` `Chapter7/EpGroupoid · EpGroupoid.isScDiffeoOn_hat`, a composite of sc-diffeomorphisms, with $\widehat{\varphi}(x) = y$ `Chapter7/EpGroupoid · EpGroupoid.hat_src`. It depends on the sheet only as a germ. $T\varphi = T\widehat{\varphi}(x)$ is read through the chain rule as $Tt \circ (Ts|_{T_\varphi \mathbf{X}})^{-1}$ with Part I's tangent maps `Chapter7/EpGroupoid · EpGroupoid.tangentMor`; that it equals $T\widehat{\varphi}(s(\varphi))$, is a bounded linear isomorphism on level 0 for $\varphi$ on level 1, and an sc-isomorphism for smooth $\varphi$, is proved in §VII.44, with the linear structure of $T_xX$ that Part I now provides (gaps G.7, G.70, closed). The display "$\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} \circ U(x) \to U(y)$" is read with : for the last $\circ$, and "$T\varphi : T_{s(\varphi)}X \to T_{t(\varphi)}$" with the missing $X$ (possible misprints; Findings, Chapter 7, items 5, 110) (2017 p. 227).

**The levels and the degeneracy index** (V7.6). A local sc-diffeomorphism preserves the levels and the degeneracy index: in a sheet it is an sc-diffeomorphism, and Part I's Proposition 2.4.4 (2017: 2.7) (§II.11) applies `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.mem_levelSet_iff` `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.degIndex_eq`. Hence $\varphi \in \mathbf{X}_m$ iff $s(\varphi) \in X_m$ iff $t(\varphi) \in X_m$ `Chapter7/EpGroupoid · EpGroupoid.mem_levelSet_src_iff` `Chapter7/EpGroupoid · EpGroupoid.mem_levelSet_tgt_iff`, and $d_X(s\varphi) = d_{\mathbf{X}}(\varphi) = d_X(t\varphi)$ `Chapter7/EpGroupoid · EpGroupoid.degIndex_src` `Chapter7/EpGroupoid · EpGroupoid.degIndex_tgt`. The text's $ml(x)$,

the largest $m$ with $x \in X_m$, is the maximal level `Chapter7/LocalDiffeo · MPolyfold.maxLevel`, preserved too `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.maxLevel_eq`. $|X_k| = \pi(X_k)$ is defined `Chapter7/EpGroupoid · EpGroupoid.orbitLevel`; the set $|X_\infty|$, the identity $|X_\infty| = \bigcap_k |X_k|$ and the density of $X_\infty$ in $X_k$ are §VII.44 (gaps G.72, G.73, closed; Findings, Chapter 7, items 6, 112, 116, 153) (pp. 289–291 (2017 pp. 227–228)).

**Definition 7.1.5 (2017: 7.5)** (V7.7). $d_{|X|}(|x|) = d_X(x)$ is well defined by the previous paragraph `Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit` `Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit_π` (2017 p. 228).

**Definition 7.1.6 (2017: 7.6)** (V7.8). $X$ is tame if the object M-polyfold is tame `Chapter7/EpGroupoid · EpGroupoid.IsTame`, "tame" being the existence of an equivalent tame atlas (Definition 2.5.7 (2017: 2.19), §II.14) `Chapter2/Tame · MPolyfold.IsTame` — not "every chart is tame", which would be unsatisfiable. *Then* **X** *is tame.* At a morphism, a sheet $s : \mathbf{W} \to V$ composed with a tame chart of $X$ around $s(\varphi)$ is a tame chart of **X**, since tameness is carried by sc-diffeomorphisms (§III.27) `Chapter7/LocalDiffeo · MPolyfold.IsScDiffeoOn.exists_tame_chart`; these charts are compatible with the atlas of **X** and form a tame atlas `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame`; conversely a surjective local sc-diffeomorphism transfers tameness down `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame_iff_of_surjective`. The witnesses are the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn` and the reflection groupoid of §VII.30 (Findings, Chapter 7, items 7, 106) (2017 p. 228).

## VII.4 Definition 7.1.7 (2017: 7.7), and the remark after it

**Definition 7.1.7 (2017: 7.7)** (V7.9). A functor between groupoids is a pair of maps commuting with $s$, $t$, $m$ on composable pairs and $u$ `Chapter7/Groupoid · GroupoidFunctor`; it commutes with $\iota$, by the uniqueness of inverses `Chapter7/Groupoid · GroupoidFunctor.mor_inv`, has identities and composites `Chapter7/Groupoid · GroupoidFunctor.id` `Chapter7/Groupoid · GroupoidFunctor.comp`, a Mathlib functor `Chapter7/Groupoid · GroupoidFunctor.toFunctor` and a map of orbit spaces `Chapter7/Groupoid · GroupoidFunctor.orbitMap`. An ep-functor is sc-smooth on objects and on morphisms `Chapter7/EpGroupoid · EpFunctor` `Chapter7/EpGroupoid · EpFunctor.id` `Chapter7/EpGroupoid · EpFunctor.comp` `Chapter7/EpGroupoid · EpFunctor.ext`, and $\mathcal{EP}$ is the category of bundled ep-groupoids and ep-functors `Chapter7/EpGroupoid · EP` (Findings, Chapter 7, item 8) (2017 p. 228).

**The remark after Definition 7.1.7 (2017: 7.7)** (V7.10). The text says that it suffices to ask sc-smoothness between the object spaces, because near $\varphi$ one has $f(\gamma) = (s|\mathbf{U}(\psi))^{-1}(f(s(\gamma)))$ for $\gamma$ in a neighbourhood $\mathbf{U}(\varphi)$ with $f(\mathbf{U}(\varphi)) \subseteq \mathbf{U}(\psi)$, $\psi = f(\varphi)$. Choosing that neighbourhood uses the continuity of $f$ on morphisms, and without it the claim does not hold as we read it. *The counterexample.* On the trivial action of $\mathbb{Z}/3$ on $\mathbb{R}$ (objects $\mathbb{R}$, morphisms $\mathbb{Z}/3 \times \mathbb{R}$, $s = t$ the projection), let $f$ be the identity on objects and $(g, y) \mapsto (g, y)$ for $y \le 0$, $(g, y) \mapsto (g^2, y)$ for $y > 0$ `Chapter7/NonVacuity · Examples.remarkFunctor`. Since $g \mapsto g^2$ is an automorphism of $\mathbb{Z}/3$, this is a functor; it is sc-smooth on objects `Chapter7/NonVacuity · Examples.isScSmoothMap_remarkFunctor_obj` but not continuous on morphisms, jumping at $(g, 0)$ for $g \neq 1$ `Chapter7/NonVacuity · Examples.not_continuous_remarkFunctor_mor`, hence not sc-smooth there `Chapter7/NonVacuity · Examples.not_isScSmoothMap_remarkFunctor_mor`, and no ep-functor has these components `Chapter7/NonVacuity · Examples.not_exists_epFunctor_remarkFunctor` `Chapter7/NonVacuity · Examples.remark_definition_7_1_7_counterexample`. *The remark for functors continuous on morphisms.* Take a sheet $\mathbf{U}(\psi)$ of $s$ at $\psi = f(\varphi)$, with inverse $\sigma$, and $\mathbf{U}(\varphi) = f^{-1}(\mathbf{U}(\psi))$, open by continuity. For $\gamma \in \mathbf{U}(\varphi)$, $f(\gamma)$ lies in $\mathbf{U}(\psi)$ and has source $f(s\gamma)$, so $f(\gamma) = \sigma(f(s\gamma))$ `Chapter7/Basic · EpGroupoid.mor_eq_srcInv_obj_src`, a composite of sc-smooth maps `Chapter7/Basic · EpGroupoid.isScSmoothMap_mor_of_continuous`. The sheet at $\varphi$

and the neighbourhoods $U(s\varphi)$, $U(s\psi)$ of the text are not used. Since every ep-functor is continuous on morphisms `Chapter7/Basic · EpFunctor.continuous_mor`, the remark becomes an equivalence for functors sc-smooth on objects `Chapter7/Basic · EpGroupoid.isScSmoothMap_mor_iff_continuous`, and a functor sc-smooth on objects and continuous on morphisms is an ep-functor `Chapter7/Basic · EpFunctor.ofContinuous`. *The 2021 sentence* (p. 291) asks continuity on level 0 between the morphism spaces, which is the hypothesis of the form just proved, so the 2021 remark holds as we read it (§VII.33); the counterexample concerns the 2017 sentence only `Chapter7/NonVacuity · EpGroupoid.Examples.remark_definition_7_1_7_counterexample` (Findings, Chapter 7, items 9, 104) (pp. 290–292 (2017 pp. 228–229)).

## VII.5 Propositions 7.1.8–7.1.10 (2017: 7.1–7.3): the topology of $|X|$

**Proposition 7.1.8 (2017: 7.1)** (V7.11) `Chapter7/Basic · EpGroupoid.proposition_7_1_8`. *$\pi$ is open.* $|X|$ carries the quotient topology, so $\pi$ is continuous `Chapter7/EpGroupoid · EpGroupoid.continuous_π`, and $\pi(U)$ is open iff its saturation $\pi^{-1}(\pi(U))$ is. *Step 1.* The saturation of $U$ is $t(s^{-1}(U))$: $y \cong x \in U$ iff some $\varphi$ has $s\varphi \in U$ and $t\varphi = y$ `Chapter7/Basic · EpGroupoid.preimage_π_image_eq`. *Step 2.* $s^{-1}(U)$ is open since $s$ is continuous, and $t$ is an open map, being a local sc-diffeomorphism `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isOpenMap`. The text argues with the sheets $V(\varphi)$, which is the openness of $t$ unpacked. So $\pi$ is an open quotient map `Chapter7/Basic · EpGroupoid.proposition_7_1_8_isOpenQuotientMap` and carries neighbourhood filters to neighbourhood filters `Chapter7/Basic · EpGroupoid.proposition_7_1_8_map_nhds` (Findings, Chapter 7, item 10) (2017 p. 229).

**Proposition 7.1.9 (2017: 7.2)** (V7.12) `Chapter7/Basic · EpGroupoid.proposition_7_1_9`. *Every $z \in |X|$ has a countable basis of open neighbourhoods, chosen monotone.* $X$ is metrizable (Theorem 2.3.10 (2017: 2.2)); for $z = \pi(x)$ the images of the balls $B(x, 1/(k+1))$ are open by Proposition 7.1.8 (2017: 7.1) and decreasing, and they form a basis because $\pi$ maps the neighbourhood filter of $x$ onto that of $z$ `Chapter7/Basic · EpGroupoid.proposition_7_1_9_of_hasAntitoneBasis`. The statement is Mathlib's antitone basis indexed by $\mathbb{N}$ (radii $1/(k+1)$, the index starting at 0), and $|X|$ is first countable `Chapter7/Basic · EpGroupoid.proposition_7_1_9_firstCountableTopology` (Findings, Chapter 7, item 10) (2017 p. 229).

**Proposition 7.1.10 (2017: 7.3)** (V7.13) `Chapter7/Basic · EpGroupoid.proposition_7_1_10`. *Every $x$ has an open $U$ with $\pi(\mathrm{cl}_X A) = \mathrm{cl}_{|X|}\pi(A)$ for every $A \subseteq U$.* The text proves "$\supseteq$" with a sequence $\varphi_k : a_k \to b_k$, $b_k \to b$, $a_k \in A$, of which properness extracts a convergent subsequence. The formalization proves more, for every $U$ with $t$ proper over $s^{-1}(\mathrm{cl}_X U)$ and every $A \subseteq \mathrm{cl}_X U$, and without sequences `Chapter7/Basic · EpGroupoid.proposition_7_1_10_of_isProperMap`. *Step 1.* Since $\pi$ is open and continuous, $\pi^{-1}(\mathrm{cl}\,\pi(A)) = \mathrm{cl}\,\pi^{-1}(\pi(A)) = \mathrm{cl}\,t(s^{-1}(A))$. *Step 2.* $t(s^{-1}(\mathrm{cl}A))$ is closed, as the image of the closed $s^{-1}(\mathrm{cl}A) \subseteq s^{-1}(\mathrm{cl}U)$ under a proper, hence closed, map (§VII.2), and contains $t(s^{-1}(A))$; so $\mathrm{cl}\,\pi(A) \subseteq \pi(t(s^{-1}(\mathrm{cl}A))) = \pi(\mathrm{cl}A)$. *Step 3.* "$\subseteq$" is the continuity of $\pi$. "For every $z$ and every representative $x$" is "for every $x$" (Findings, Chapter 7, item 11) (pp. 291–293 (2017 pp. 229–230)).

## VII.6 Lemma 7.1.11 (2017: 7.1), Proposition 7.1.12 (2017: 7.4), and the conjugation $G_x \cong G_y$

**Lemma 7.1.11 (2017: 7.1)** (V7.14) `Chapter7/Basic · EpGroupoid.lemma_7_1_11`. *If $t : s^{-1}(\mathrm{cl}_X U(x_0)) \to X$ is proper, only finitely many $\psi$ have $s(\psi) = x$ and $t(\psi) \in U(x_0)$.* **The text's proof** argues by contradiction with distinct $\psi_n : s(\psi_n) \to x$, $s(\psi_n) \in U(x_0)$; properness gives a convergent subsequence, $t$ is a local

sc-diffeomorphism near the limit, and $t(\psi_n) = t(\psi_m) = x$ forces $\psi_n = \psi_m$. The proof thus treats the inverses of the morphisms of the statement, with $s$ and $t$ exchanged, and writes "$t : V(\psi) \to V(y)$" for $V(x)$ and "$s(\psi) \in s^{-1}(\mathrm{cl}_X(U(x_0)))$"; parentheses are unbalanced in the statement and the proof (possible misprints; Findings, Chapter 7, items 12, 96). **The formalized proof** does the proof's form first and has no sequences. *Step 1.* The fibre of the proper restriction of $t$ over $x$ is compact. *Step 2.* It is discrete, $t$ being locally injective as a local sc-diffeomorphism `Chapter7/Discrete · MPolyfold.IsLocalScDiffeo.isLocallyInjective` `Chapter7/Discrete · IsLocallyInjective.isDiscrete_preimage_singleton`. *Step 3.* A compact discrete set is finite `Chapter7/Discrete · IsProperMap.finite_preimage_singleton_of_isLocallyInjective` `Chapter7/Discrete · finite_setOf_mem_of_isProperMap_domRestrict` `Chapter7/Discrete · MPolyfold.finite_setOf_mem_of_isLocalScDiffeo`. This gives the finiteness of $\{\psi \mid s\psi \in \mathrm{cl}_X U(x_0),\ t\psi = x\}$ `Chapter7/Basic · EpGroupoid.lemma_7_1_11_swap_of_isProperMap` `Chapter7/Basic · EpGroupoid.lemma_7_1_11_swap`. *Step 4.* The statement's form follows by $\iota$, which exchanges $s$ and $t$ and is injective, and holds even for targets in $\mathrm{cl}_X U(x_0)$ `Chapter7/Basic · EpGroupoid.lemma_7_1_11_of_isProperMap`. The openness of $U(x_0)$ and $x_0 \in U(x_0)$ are kept as hypotheses but not used (2017 p. 230).

**Proposition 7.1.12 (2017: 7.4)** (V7.15) `Chapter7/Basic · EpGroupoid.proposition_7_1_12`. *$G_x$ is finite.* Lemma 7.1.11 (2017: 7.1) with $x_0 = x$ and $U(x_0) = V(x)$ of Definition 7.1.3 (3) (2017: 7.3): $G_x$ is contained in the finite set of morphisms with source $x$ and target in $V(x)$ `Chapter7/Discrete · MPolyfold.finite_isotropy_of_isLocalScDiffeo`. The proposition is an instance, so the cardinality of $G_x$ is its order as a finite group (Findings, Chapter 7, item 13) (2017 p. 230).

**The conjugation** (V7.16). For $\varphi : x \to y$, $g \mapsto \varphi \circ g \circ \varphi^{-1}$ is an isomorphism $G_x \to G_y$ `Chapter7/Groupoid · GroupoidStr.isotropyConj` `Chapter7/Groupoid · GroupoidStr.isotropyConj_val` `Chapter7/Basic · EpGroupoid.conjIsotropy` `Chapter7/Basic · EpGroupoid.conjIsotropy_val`, so isomorphic objects have isomorphic isotropy groups of the same order `Chapter7/Basic · EpGroupoid.nonempty_isotropy_mulEquiv` `Chapter7/Basic · EpGroupoid.nat_card_isotropy_eq`. "There is no canonical identification" is given its content: two choices $\varphi, \varphi'$ give isomorphisms that differ by conjugation with $\varphi'\varphi^{-1} \in G_y$, an inner automorphism `Chapter7/Basic · EpGroupoid.isotropyConj_eq_isotropyConj_conj` (Findings, Chapter 7, item 13) (2017 p. 230).

## VII.7 Theorem 7.1.13 (2017: 7.1), and local path connectedness

**Theorem 7.1.13 (2017: 7.1)** (V7.17) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13`. *For every object $x$ and every open $V \ni x$ there are an open neighbourhood $U \subseteq V$ of $x$, a homomorphism $\Phi : G_x \to \mathrm{Diff}_{sc}(U)$ and an sc-smooth $\Gamma : G_x \times U \to \mathbf{X}$ with (1) $\Gamma(g, x) = g$, (2) $s(\Gamma(g, y)) = y$ and $t(\Gamma(g, y)) = \Phi(g)(y)$, (3) every morphism $h : y \to z$ between points of $U$ is $\Gamma(g, y)$ for a unique $g$.* The data form a *natural representation* at $x$ `Chapter7/EpGroupoid · EpGroupoid.NaturalRep`: $U$ open, connected and containing $x$; $\Phi(g)$ an sc-diffeomorphism of $U$ with inverse $\Phi(g^{-1})$ and $\Phi(gh) = \Phi(g) \circ \Phi(h)$ on $U$; $\Gamma$ sc-smooth on each slice $\{g\} \times U$, which is sc-smoothness on $G_x \times U$ (§VII.8), with (1)–(3). The first consequences: $\Phi(g)$ maps $U$ into $U$ `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.mapsTo_Φ`, $\Phi(1)$ is the identity on $U$ `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Φ_one`, every $\Phi(g)$ fixes $x$ `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Φ_self`, $\Gamma(g, y)$ is a morphism between points of $U$ `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Γ_mem_morOn`, $\Phi(g^{-1})\Phi(g) = 1$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_inv_Φ`, and (3) is a bijection $G_x \times U \to \mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.bijOn_Γ`. The action $g * y = \Phi(g)(y)$ is a group action on $U$ `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.smul_val`. The proof is the text's Theorem 7.6.1 (2017: 7.5) in the appendix (§VII.26). *Hidden hypothesis: $U$* connected. The text chooses "the connected open neighborhood $U(x)$" on p. 293 (2017 p. 231) but does not put it into the statement; as we read it, the homomorphism property of $\Phi$, the functoriality of $\Gamma$ and the uniqueness of $(\Phi, \Gamma)$ all need it (Findings, Chapter 7, item 20) (2017 p. 231).

**Local path connectedness** (V7.18). Connected neighbourhoods exist because M-polyfolds are locally path connected, which the text uses without statement (2017 p. 231, Lemma 7.6.3 (2017: 7.9), Corollary 7.3.8 (2017: 7.2)). *Step 1, a retract.* For $o \in O = r(U)$, $U$ open in a partial quadrant $C$, and a neighbourhood $W$ of $o$, there is $\varepsilon > 0$ such that every $p \in O$ with $\|p - o\| < \varepsilon$ is joined to $o$ inside $O \cap W$ by the path $\tau \mapsto r(o + \tau(p - o))$: the segment lies in the convex set $B_\varepsilon(o) \cap C \subseteq U$, $r$ is continuous on level $0$ and fixes $o$ and $p$ `Chapter7/LocConnected · ScRetraction.exists_ball_joinedIn`. So an sc-smooth retract is locally path connected `Chapter7/LocConnected · IsScRetract.locallyPathConnectedSpace`. *Step 2, an M-polyfold.* Through any chart around a point, path components of open sets are neighbourhoods `Chapter7/LocConnected · MChart.pathComponentIn_mem_nhds`, and $X$ is locally path connected `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace`, hence locally connected `Chapter7/LocConnected · MPolyfold.locallyConnectedSpace`. *Step 3, consequences.* Open connected neighbourhoods form a basis `Chapter7/LocConnected · MPolyfold.hasBasis_isOpen_isConnected` `Chapter7/LocConnected · MPolyfold.hasBasis_isOpen_isPathConnected` `Chapter7/LocConnected · MPolyfold.exists_isOpen_isConnected_subset`; components, also of open sets, are open `Chapter7/LocConnected · MPolyfold.isOpen_connectedComponent` `Chapter7/LocConnected · MPolyfold.isOpen_connectedComponentIn`; an open set is connected iff path connected `Chapter7/LocConnected · MPolyfold.isConnected_iff_isPathConnected`; every quotient of $X$, in particular $|X|$, is locally path connected `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace_quotient`; and finitely many points of a Hausdorff locally connected space have pairwise disjoint connected open neighbourhoods inside a given open set `Chapter7/LocConnected · exists_pairwiseDisjoint_isOpen_isConnected`. *Step 4, the tool for connectedness.* Two continuous sections of a locally injective map into a Hausdorff space that agree at one point of a preconnected set agree on all of it `Chapter7/LocConnected · eqOn_sections_of_isPreconnected`, by Mathlib's statement for separated maps `Mathlib/Topology/SeparatedMap · eqOn_of_comp_eqOn`. For $s$ this says that a local inverse of $s$ on a connected $U$ is determined by one value; it is what makes $\Phi$ a homomorphism (§VII.26) and $\Gamma$ a functor (§VII.8). The paths are continuous on level $0$; sc-smooth paths between smooth points are Chapter 6's (§VI.18). This is the first fact of gap G.56 of Part I, now formalized (Findings, Chapter 7, items 21, 114).

## VII.8 Pages 231–233: the sheets $\mathbf{U}(g)$, $G_x \ltimes U(x)$, the functor $\Gamma$, and $\hat{\Phi}$

Everything on these pages is derived from one natural representation $(U, \Phi, \Gamma)$; the text's "using the proof of Theorem 7.1 one verifies" is not needed. No possible misprint was found on pp. 293–297 (2017 pp. 231–233) (Findings, Chapter 7, item 22).

**The splitting $\mathbf{U}(x) = \bigsqcup_g \mathbf{U}(g)$** (V7.19). Put $\mathbf{U}(g) = \Gamma(g, U)$. *Step 1, open.* The image of a continuous section of a local homeomorphism over an open set is open `Chapter7/NaturalAction · isOpen_image_of_section`; $\Gamma(g, \cdot)$ is a section of $s$ by (2) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_image_Γ`, containing $\Gamma(g, y)$ for every $y \in U$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_mem_image`. *Step 2, connected, disjoint, covering.* $\mathbf{U}(g)$ is connected as a continuous image of $U$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isConnected_image_Γ`; two sheets are disjoint by the uniqueness in (3) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_inj` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.pairwise_disjoint_image_Γ`; they cover $\mathbf{U}(x)$ by the existence in (3) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.iUnion_image_Γ`, which is therefore open `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_morOn`. *Step 3, components.* Open, connected, disjoint and covering, they are the connected components of $\mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.connectedComponentIn_morOn`, and there are $\#G_x$ of them `Chapter7/NaturalAction · EpGroupoid.NaturalRep.injective_image_Γ`. *Step 4.* $s, t : \mathbf{U}(g) \to U$ are sc-diffeomorphisms, with inverses $\Gamma(g, \cdot)$ and $\Gamma(g, \cdot) \circ \Phi(g)^{-1}$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_src_image` `Chapter7/`

NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_tgt_image. Every $\varphi \in \mathbf{U}(x)$ has a sheet index $h$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.index with $\varphi = \Gamma(h, s\varphi)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_index and $t\varphi = \Phi(h)(s\varphi)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.tgt_eq_Φ_index, locally constant on $\mathbf{U}(x)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.eventually_index_eq. Once $U$ is connected, the text's "choose $U(x)$ so small that $\mathbf{U}(x)$ splits into $\#G_x$-many components" needs no shrinking. "$\mathbf{U}(x)$ fits into an arbitrarily small open neighbourhood of $G_x$" belongs to the construction: Lemma 7.6.2 (2017: 7.8) holds for arbitrary open $N_g \ni g$ (§VII.26) Chapter7/NaturalRep · EpGroupoid.lemma_7_6_2 (2017 p. 231).

**The M-polyfold** $G \times U$ (V7.20). "A natural M-polyfold structure as the finite disjoint union of copies of $U$" is the product with a discrete M-polyfold. *Step 1.* A discrete space is an M-polyfold with the charts $\{g\} \to \mathbb{R}^0$ Chapter7/Discrete · discreteChart Chapter7/Discrete · discretePolyfold Chapter7/Discrete · isChart_discreteChart, pairwise compatible because their transitions are the identity of the one-point space or the empty map Chapter7/Discrete · discreteChart_compatible; every point is smooth Chapter7/Discrete · discretePolyfold_smoothSet, and a map into it is sc-smooth iff continuous Chapter7/Discrete · isScSmoothMap_discrete_iff. *Step 2.* In $G \times X$ the slices $\{g\} \times X$ are open Chapter7/Discrete · isOpen_sheet, $x \mapsto (g, x)$ is sc-smooth Chapter7/Discrete · isScSmoothMap_sheet and the projection of a slice is an sc-diffeomorphism Chapter7/Discrete · isScDiffeoOn_snd_sheet, with $d_{G\times X}(g, x) = d_X(x)$ Chapter7/Discrete · degIndex_prod_discrete. *Step 3.* Hence a map on $G \times U$ is sc-smooth iff it is on each slice Chapter7/Discrete · isScSmoothOn_prod_discrete_iff Chapter7/Discrete · isScSmoothOn_univ_prod_discrete_iff Chapter7/Discrete · isScSmoothMap_prod_discrete_iff, the text's sheet-by-sheet reading of "$\Gamma$ sc-smooth on $G \times U$". For a natural representation the objects are the open-subset M-polyfold $U$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.objPolyfold, the morphisms $G_x \times U$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.morPolyfold, and $(g, y) \mapsto g*y$ is sc-smooth Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScSmoothMap_smul (Findings, Chapter 7, item 23) (2017 p. 232).

**The translation groupoid** $G_x \ltimes U(x)$ **is an ep-groupoid** (V7.21). The groupoid has objects $U$, morphisms $G\times U$, $s(g, y) = y$, $t(g, y) = g*y$, $(g, h*z)\circ(h, z) = (gh, z)$ Chapter7/Groupoid · GroupoidStr.translation Chapter7/EpGroupoid · EpGroupoid.NaturalRep.translationGroupoid. *(1).* $s$ is the projection, a local sc-diffeomorphism by Step 2 above Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isLocalScDiffeo_src; $t$ is the projection after the shear $(g, y) \mapsto (g, g * y)$, an sc-diffeomorphism with inverse $(g, y) \mapsto (g, g^{-1} * y)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.shearHomeomorph Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isLocalScDiffeo_tgt. *(3), for every* $V$. $t$ is the shear followed by the projection $G \times U \to U$, proper because $G_x$ is finite Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isProperMap. *(4), sheet form.* For a local inverse $\sigma$ of $s$, $\sigma(t\psi) \circ \psi = (k(\psi)\psi_1, \psi_2)$ with $k$ locally constant, sc-smooth on each slice Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isScSmoothCompSheet. So $G_x \ltimes U(x)$ is an ep-groupoid Chapter7/NaturalAction · EpGroupoid.NaturalRep.translationEpGroupoid, built from $(\Phi, \Gamma)$ alone (2017 p. 232).

**The functor** $\Gamma : G_x \ltimes U(x) \to X$ (V7.22). *Step 1, composition.* $y \mapsto \Gamma(g, h*y)\circ\Gamma(h, y)$ and $y \mapsto \Gamma(gh, y)$ are continuous sections of $s$ over $U$ — the first by (4) in the sheet form, for the sheet $\mathbf{U}(g)$ — with the same value $g \circ h$ at $x$ by (1); on the connected $U$ they agree (§VII.7) Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_mul. *Step 2.* In the same way $\Gamma(1, y) = 1_y$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_one, and $\Gamma(g^{-1}, g*y) = \Gamma(g, y)^{-1}$ by the uniqueness of inverses Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_inv. *Step 3.* So $\Gamma$, the inclusion on objects, is a functor Chapter7/NaturalAction · EpGroupoid.NaturalRep.functor, full and faithful by (3) Chapter7/NaturalAction · EpGroupoid.NaturalRep.full Chapter7/NaturalAction · EpGroupoid.NaturalRep.faithful, onto the full subcategory on $U$ Chapter7/Groupoid · GroupoidStr.fullSub, and an ep-functor Chapter7/NaturalAction · EpGroupoid.NaturalRep.epFunctor. *Hidden hypothesis:* $U$ connected, without which the sheet index of $\Gamma(g, h * y) \circ \Gamma(h, y)$ is only locally constant (2017 p. 232).

**The action $\hat{\Phi}$ on $\mathbf{U}(x)$** (V7.23). The text defines $\hat{\Phi}(g)(\varphi) = \Gamma(ghg^{-1}, g * s(\varphi))$ for $\varphi \in \mathbf{U}(h)$. The formalization defines the total map $\hat{\Phi}(g)(\varphi) = \Gamma(g, t\varphi) \circ \varphi \circ \Gamma(g, s\varphi)^{-1}$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat` and recovers the text's formula from $\Gamma_{\text{mul}}$ and $\Gamma_{\text{inv}}$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_Γ`, together with $\Phi(g)\Phi(h)\Phi(g)^{-1} = \Phi(ghg^{-1})$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_conj`. Then $\hat{\Phi}(g)$ maps $\mathbf{U}(h)$ onto $\mathbf{U}(ghg^{-1})$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_mem_image` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.image_Φhat` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.index_Φhat`; the two diagrams are $s \circ \hat{\Phi}(g) = \Phi(g) \circ s$ and $t \circ \hat{\Phi}(g) = \Phi(g) \circ t$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.src_Φhat` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.tgt_Φhat`; $\hat{\Phi}(1) = 1$ and $\hat{\Phi}(gh) = \hat{\Phi}(g)\hat{\Phi}(h)$ on $\mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_one` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_mul`; and $\hat{\Phi}(g)$ is an sc-diffeomorphism of $\mathbf{U}(x)$ with inverse $\hat{\Phi}(g^{-1})$, sc-smooth by (4) in the sheet form and the sc-smoothness of $\iota$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_Φhat` (pp. 294–297 (2017 pp. 232–233)).

## VII.9 Uniqueness and restriction of natural representations

(V7.24.) The text uses without statement that natural representations can be shrunk and compared: "adapting the corresponding restrictions" (Lemma 7.2.3 (2017: 7.3)), "$U(x)$ can be chosen so small" ($T_x$, 2017 p. 241). Let $R = (U, \Phi, \Gamma)$ and $R' = (U', \Phi', \Gamma')$ be natural representations at $x$ and $C$ the connected component of $x$ in $U \cap U'$, open and connected (§VII.7) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_connectedComponentIn_inter` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isConnected_connectedComponentIn_inter`. *Step 1.* $\Gamma(g, \cdot)$ and $\Gamma'(g, \cdot)$ are continuous sections of $s$ over $C$ with the value $g$ at $x$, so they agree on $C$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_eqOn`; applying $t$, $\Phi(g) = \Phi'(g)$ on $C$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_eqOn`. *Step 2.* $C$ is invariant: $\Phi(g)(C)$ is connected, contains $x$ and lies in $U \cap U'$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.mapsTo_Φ_connectedComponentIn`. *Step 3.* So a natural representation restricts to every invariant connected open neighbourhood of $x$ inside $U$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrict` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.restrict` `Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_restrict`, in particular to $C$, where the restrictions of $R$ and $R'$ coincide `Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrictInter` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrictInter_eq`, and natural representations exist inside every open $V \ni x$ `Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_shrink`. *Step 4.* If $\Phi(g) = 1$ for one natural representation, then $\Phi'(g) = 1$ near $x$ for every one `Chapter7/NaturalAction · EpGroupoid.NaturalRep.eventually_Φ_eq_of_actsTrivially`. Uniqueness holds on the component $C$; $U \cap U'$ itself need not be connected (Findings, Chapter 7, item 24).

## VII.10 Proposition 7.1.15 (2017: 7.5) and Definition 7.1.18 (2017: 7.8): the star construction

**Proposition 7.1.15 (2017: 7.5)** (V7.25) `Chapter7/NaturalRepFinite · EpGroupoid.proposition_7_1_15_2017`. *Let $Y$ be a finite set of mutually isomorphic objects and $V$ a neighbourhood of $Y$. There is a functor $f$ assigning to every $y \in Y$ a connected open $U(y) \subseteq V$, mutually disjoint, and to every $\psi : y \to y'$ between points of $Y$ an sc-diffeomorphism $f(\psi) : U(y) \to U(y')$, such that the morphisms between points of $\bigcup U(y)$ split into components $\mathbf{U}(\psi)$ with (1) the natural action of $G_y$ on $U(y)$, (2) $s : \mathbf{U}(\psi) \to U(y)$, $t : \mathbf{U}(\psi) \to U(y')$ sc-diffeomorphisms, (3) $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$.* The datum is a structure `Chapter7/EpGroupoid · EpGroupoid.NaturalRepFinite` (Definition 7.1.18 (2017: 7.8), V7.26), "decomposes into the connected components" being

read as: the $\mathbf{U}(\psi)$ are open, connected, pairwise disjoint and cover; they are then the components Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.connectedComponentIn_morOn (pp. 295–305 (2017 pp. 233–236)).

**The text's proof** (pp. 297–305 (2017 pp. 234–236)) is an induction on $\#Y$. $\#Y = 1$ is Theorem 7.1.13 (2017: 7.1). For $Y = \{y_1, y_2\}$ it fixes $\gamma_1 : y_1 \to y_2$ with a sheet and $\hat{\gamma}_1 = t \circ (s|\cdot)^{-1}$, takes a $G_{y_1}$-invariant connected $U(y_1)$ with $\hat{\gamma}_1(U(y_1))$ inside a $G_{y_2}$-invariant $W(y_2)$, puts $U(y_2) = \hat{\gamma}_1(U(y_1))$ and shows it invariant; the other morphisms $y_1 \to y_2$ are $g \circ \gamma_1$, $g \in G_{y_2}$, with components $\Phi_g(\mathbf{U}(\gamma_1))$, $\Phi_g(\varphi) = \Gamma(g, t\varphi) \circ \varphi$, and the morphisms $y_2 \to y_1$ get the inverse components. For $Y = Y' \cup \{y_{k+1}\}$ it fixes $\delta_j : y_{k+1} \to y_j$, shrinks as in the case of two objects, and adjusts the components of the morphisms of $Y'$ to $(s|\mathbf{U}'(\gamma))^{-1}(U(y_i))$, showing $t(\mathbf{U}(\gamma)) = U(y_j)$.

**The formalized proof** (V7.25) is a *star*: everything is transported from one base object, and nothing is shrunk afterwards. *Step 1, transport along a sheet.* Let $R = (U, \Phi, \Gamma)$ be a natural representation at $x$ and $\mathbf{S}_1, \mathbf{S}_2$ sheets of morphisms out of $x$ with local inverses $a_i = (s|\mathbf{S}_i)^{-1}$ and $U \subseteq s(\mathbf{S}_i)$, $\hat{\gamma}_i = t \circ a_i$. The images $\hat{\gamma}_i(U)$ are open and connected Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isOpen_transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isConnected_transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_hat_transportSet, $\hat{\gamma}_i^{-1}$ is the inverse there Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.back, and the conjugates $\hat{\gamma}_2 \circ \Phi(g) \circ \hat{\gamma}_1^{-1}$ are sc-diffeomorphisms compatible with composition Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap_conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap_one. The transported morphisms are $\Theta_g(z) = a_2(\Phi(g)u) \circ \Gamma(g, u) \circ a_1(u)^{-1}$, $u = \hat{\gamma}_1^{-1}(z)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport, with source $z$ and target $\hat{\gamma}_2(\Phi(g)u)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.src_transport Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.tgt_transport, equal to $\gamma_2 \circ g \circ \gamma_1^{-1}$, $\gamma_i = a_i(x)$, at the point $\hat{\gamma}_1(x)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport_hat_self, sc-smooth by (4) in the sheet form, used twice, and the sc-smoothness of $\iota$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScSmoothOn_transport. *Step 2, the bijection.* A morphism $k$ from $\hat{\gamma}_1(U)$ to $\hat{\gamma}_2(U)$ pulls back to the morphism $a_2(\cdot)^{-1} \circ k \circ a_1(\cdot)$ between points of $U$, which is $\Gamma(g, u)$ for a unique $g$ by Theorem 7.1.13 (3) (2017: 7.1); so $k = \Theta_g(sk)$ for a unique $g$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.existsUnique_transport Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport_injective. *Step 3, a natural representation at $y$.* For $\gamma : x \to y$, $h \mapsto \gamma^{-1} h \gamma$ is an isomorphism $G_y \to G_x$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportConj, and $U(y) = \hat{\gamma}(U)$, $\Phi_y(h) = \hat{\gamma} \circ \Phi(\gamma^{-1} h \gamma) \circ \hat{\gamma}^{-1}$, $\Gamma_y(h, \cdot) = \Theta_{\gamma^{-1} h \gamma}$ is a natural representation at $y$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_U Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_Φ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_Γ, after shrinking $R$ (same $\Phi$, $\Gamma$) into $s$ of a sheet of $\gamma$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.exists_spoke. *Step 4, the star.* Fix one $x$ isomorphic to all of $Y$, morphisms $\gamma_y : x \to y$ with sheets $\mathbf{K}_y$, and by Hausdorff separation of the finite $Y$ pairwise disjoint open $W_y \ni y$; take $U_0 = \bigcap_y (s(\mathbf{K}_y) \cap \hat{\gamma}_y^{-1}(W_y \cap V))$ and a natural representation inside $U_0$ (Theorem 7.1.13 (2017: 7.1)) Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.exists_star. Then $U(y) = \hat{\gamma}_y(U(x)) \subseteq W_y \cap V$ are pairwise disjoint. *Step 5, the functor.* $\psi \mapsto g_\psi = \gamma_{y'}^{-1} \circ \psi \circ \gamma_y \in G_x$ is a bijection from the morphisms $y \to y'$ onto $G_x$ compatible with composition, units and inverses Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_comp Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_unit Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_inv Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_isotropy; $f(\psi) = \hat{\gamma}_{y'} \circ \Phi(g_\psi) \circ \hat{\gamma}_y^{-1}$ Chapter7/NaturalRepFinite · EpGroupoid.

NatRepFinite.Star.F is an sc-diffeomorphism Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.isScDiffeoOn_F and a functor, since $\psi \mapsto g_\psi$ is one and $\Phi$ is a homomorphism Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.F_comp Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.F_unit. *Step 6, the components.* $\mathbf{U}(\psi) = \Theta_{g_\psi}(U(y))$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.0 Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.sheet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.mem_sheet are pairwise disjoint Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.pairwiseDisjoint_sheet and cover the morphisms between points of $\bigcup U(y)$, by Step 2 Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.morOn_eq; $s$ and $t$ are sc-diffeomorphisms on them, as images of sections Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_src_image_of_section Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_tgt_image_of_section, and $t \circ (s|\mathbf{U}(\psi))^{-1} = f(\psi)$. *Step 7, (1).* The natural representation of $G_y$ on $U(y)$ is the transported one of Step 3, whose $\Phi_y$ is $f$ on $G_y$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.toNaturalRepFinite; by §VII.9 it agrees near $y$ with every other one. The text instead chooses $U(y)$ invariant under a natural representation of $G_y$ given beforehand. The text's case of two objects already has this form ($U(y_2) = \hat{\gamma}_1(U(y_1))$); the adjustment of the components of $Y'$ in the inductive step has no counterpart. "A neighbourhood $V$ of $Y$" is a neighbourhood of the set Chapter7/NaturalRepFinite · EpGroupoid.proposition_7_1_15_of_mem_nhdsSet, reduced to an open one; $Y = \emptyset$ is allowed Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.emptyNaturalRepFinite, and enlarging $V$ keeps a datum Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.mono. "Mutually isomorphic" is used only through one $x$ isomorphic to all of $Y$. Among the possible misprints of the proof (Findings, Chapter 7, item 26): "$\Phi_g$" for $g \in G_{y_1}$ where $t(\varphi) \in U(y_2)$ requires $G_{y_2}$; "$\mathbf{U}(\gamma_g) := \Psi_g(\mathbf{U}(\gamma_1))$" for $\Phi_g$; the induction step "let $k \geq 3$", which must start at $k \geq 2$ to reach $\#Y = 3$; "$\varphi = \Gamma(g, s(\varphi)) \circ \psi$" for $\Gamma(g, t(\psi)) \circ \psi$; "$y : \mathbf{U}(\delta_j) \to U(y_j)$" for $t$; and "$g \in G_{y_i}$" for $G_{y_j}$ on 2017 p. 236 (Findings, Chapter 7, items 25, 26).

## VII.11 Proposition 7.1.19 (2017: 7.6) and Definition 7.1.20 (2017: 7.9): $|U(x)| = G_x\backslash U(x)$, local uniformizers

**Proposition 7.1.19 (2017: 7.6)** (V7.27) Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19. *For a natural representation at $x$, $\tau : G_x\backslash U(x) \to |U(x)| = \pi(U(x))$, $G_x * y \mapsto |y|$, is a homeomorphism.* $G_x\backslash U(x)$ is the orbit space of the action $g * y$ with the quotient topology. **The text's proof**: $\tau$ is well defined and onto; it is injective because a morphism $\varphi : y \to y'$ is $\Gamma(g, y)$ by Theorem 7.1.13 (3) (2017: 7.1), so $y' = g * y$; $|U(x)|$ is open by Proposition 7.1.8 (2017: 7.1); and $\tau \circ \sigma = \pi|U(x)$ for the open quotient map $\sigma$, so $\tau$ is continuous and open. **The formalized proof.** *Step 1.* $|y| = |y'|$ iff $y' = g * y$ for some $g$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.π_eq_π_iff_exists_smul. *Step 2.* $\tau$ as a map into $|X|$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.orbitQuotientMap is continuous by the quotient topology and injective by Step 1. *Step 3, open.* The image of an open $Q$ is $\pi(\sigma^{-1}(Q))$, and $\sigma^{-1}(Q)$ is open in the open $U(x)$, so Proposition 7.1.8 (2017: 7.1) applies; the openness of $\sigma$ is not needed. So $\tau$ is an open embedding into $|X|$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_orbitQuotientMap with range $\pi(U(x))$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_orbitQuotientMap Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19_apply_mk. *The second form.* The functor $G_x \ltimes U(x) \to X$ of §VII.8 is fully faithful Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19_fullyFaithful, and the same argument for the orbit space of the translation groupoid, whose classes are the same orbits, gives an open embedding with range $\pi(U(x))$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_orbitMap Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_functor_orbitMap Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19_orbitMap, with $|U(x)|_{G_x \ltimes U(x)} \cong G_x\backslash U(x)$ Chapter7/

Uniformizer · EpGroupoid.translationOrbitHomeomorph (Findings, Chapter 7, item 27) (pp. 299–306 (2017 pp. 236–237)).

**Definition 7.1.20 (2017: 7.9)** (V7.28). A local uniformizer around $x$ is a fully faithful functor $\Psi_x : G_x \ltimes U(x) \to X$, the inclusion on objects, inducing a homeomorphism of $|U(x)|_{G_x \ltimes U(x)}$ onto the open $\pi(U(x))$ Chapter7/EpGroupoid · EpGroupoid.LocalUniformizer. "Embedding" is not a separate field: the injectivity on objects and morphisms, which the text notes after the definition, follows from the inclusion on objects and faithfulness Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.injective_obj Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.injective_mor; for the uniformizers built from natural representations it is also topological, $\Psi_x$ being an open embedding on objects and on morphisms Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_obj Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_mor Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_functor_mor. "There exists a local uniformizer around every object": every natural representation is one Chapter7/Uniformizer · EpGroupoid.NaturalRep.toLocalUniformizer, and one exists inside every open $V \ni x$ Chapter7/Uniformizer · EpGroupoid.exists_localUniformizer; its orbit map is a homeomorphism onto the open $|U(x)|$ Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.isOpen_image Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.orbitHomeomorph. "$\Psi_x \circ G_x \ltimes U(x) \to X$" is read with :, and "a a local uniformizer" has a doubled article (possible misprints; Findings, Chapter 7, item 28) (2017 p. 237).

## VII.12 Definition 7.1.21 (2017: 7.10), the M-polyfold A, and Proposition 7.1.22 (2017: 7.7)

**Definition 7.1.21 (2017: 7.10)** (V7.29). An ep-subgroupoid is the full subcategory on a set $A$ of objects which is a sub-M-polyfold (Definition 2.3.18 (2017: 2.12), §II.10) and saturated Chapter7/EpGroupoid · EpGroupoid.IsEpSubgroupoid. Witnesses: $X$ itself Chapter7/EpGroupoid · EpGroupoid.isEpSubgroupoid_univ, every saturated open set Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_of_isOpen, the saturation of every open set, open by Proposition 7.1.8 (2017: 7.1) Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_saturation, and every sub-M-polyfold of a trivial ep-groupoid Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_trivial (Findings, Chapter 7, items 40, 105) (2017 p. 237).

**The facts (1)–(4) on** $A$ (V7.30). By Part I's Proposition 2.3.19 (2017: 2.6) (§II.10), $A$ has an M-polyfold structure Chapter2/SubMPolyfold · MPolyfold.subPolyfold Chapter7/SubGroupoid · EpGroupoid.subObjPolyfold with (1) the inclusion sc-smooth and a homeomorphism onto its image Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_val_subObj and (2) $i^{-1} \circ r$ sc-smooth for every sc-smooth retraction $r$ onto $A \cap V$ Chapter7/SubGroupoid · EpGroupoid.isScSmoothOn_retractInto_subObj. The tangent facts (3), (4) — $T_aA$ a closed, sc-complemented subspace of $T_aX$, and $T_aA = Tr(a)T_aX$ — are proved in §VII.44 with the linear structure of tangent spaces Chapter7/SubGroupoidTangent · EpGroupoid.fact_3_subObj Chapter7/SubGroupoidTangent · EpGroupoid.fact_4_subObj; in a chart they are Part I's Chapter2/Metrizable · proposition_2_3_19_three, and nothing below uses them (gap G.70, closed; Findings, Chapter 7, items 40, 154) (2017 pp. 237–238).

**A is a sub-M-polyfold** (V7.31). The text takes, for $\varphi \in \mathbf{A}$ with $a = s(\varphi)$, a sheet $s : \mathbf{U}(\varphi) \to U(a)$ and a retraction $r$ of $U(a)$ onto $U(a) \cap A$, and shows that $\mathbf{r} = (s|\mathbf{U}(\varphi))^{-1} \circ r \circ s$ retracts $\mathbf{U}(\varphi)$ onto $\mathbf{U}(\varphi) \cap \mathbf{A}$. *Step 1, saturation.* For saturated $A$, $t(\varphi) \in A$ iff $s(\varphi) \in A$ Chapter7/SubGroupoid · EpGroupoid.src_mem_of_tgt_mem_of_isSaturated, so $\mathbf{A} = s^{-1}(A) = t^{-1}(A)$ Chapter7/SubGroupoid · EpGroupoid.morOn_eq_preimage_src Chapter7/SubGroupoid · EpGroupoid.morOn_eq_preimage_tgt. *Step 2, a more general statement.* The preimage of a sub-M-polyfold under a local sc-diffeomorphism is one: the retraction is pulled back

along a sheet, after shrinking $U(a)$ to $(V \cap V_0) \cap r^{-1}(V \cap V_0)$ so that $r$ maps it into itself Chapter7/SubGroupoid · isSubMPolyfold_preimage_of_isLocalScDiffeo Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_preimage_src Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_preimage_tgt Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_morOn. So $\mathbf{A}$ is an M-polyfold Chapter7/SubGroupoid · EpGroupoid.subMorPolyfold with facts (1), (2) Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_val_subMor Chapter7/SubGroupoid · EpGroupoid.isScSmoothOn_retractInto_subMor; its paracompactness, which the project's M-polyfolds need, is that of a subspace of a metrizable space. The text's "$s(\mathbf{r}(\psi)) = r(s(\psi)) \in A$ and $t(\mathbf{r}(\psi)) \in U(a)$" is not used: the second claim does not hold in general as we read it, $t(\mathbf{r}(\psi))$ lying near $t(\varphi)$ rather than near $a$, and $s(\mathbf{r}(\psi)) \in A$ with saturation gives $t(\mathbf{r}(\psi)) \in A$ (possible misprint or extraction artefact for $s(\mathbf{r}(\psi)) \in U(a)$; Findings, Chapter 7, item 41) (2017 p. 238).

**Proposition 7.1.22 (2017: 7.7)** (V7.32) Chapter7/SubGroupoid · EpGroupoid.proposition_7_1_22. *For a saturated sub-M-polyfold $A$, the full subcategory on $A$ is an ep-groupoid; $A$ and $\mathbf{A}$ are sub-M-polyfolds, and all structure maps are the restrictions of those of $X$.* The text checks the conditions of Definition 7.1.3 (2017: 7.3) in a few lines; the formalization supplies the unstated step that *a map into a sub-M-polyfold which is sc-smooth into the ambient M-polyfold is sc-smooth into the sub-M-polyfold*: near a point it is $(i^{-1} \circ r) \circ i \circ f$, by fact (2) Chapter7/SubGroupoid · isScSmoothOn_subPolyfold_of_val. *(1).* A sheet of $s$ maps $\mathbf{U}(\varphi) \cap \mathbf{A}$ onto $U(a) \cap A$; with the unstated step, $s$ and $t$ restrict to local sc-diffeomorphisms Chapter7/SubGroupoid · EpGroupoid.isLocalScDiffeo_fullSub_src Chapter7/SubGroupoid · EpGroupoid.isLocalScDiffeo_fullSub_tgt. *(2).* $u$ and $\iota$ are restrictions Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_fullSub_unit Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_fullSub_inv. *(3).* The text uses a $U(a)$ with the natural $G_a$-action and $t$ proper over $s^{-1}(\mathrm{cl}_X U(a))$ and notes that $U(a) \cap A$ is invariant; the invariance is not used, and any $V(a)$ of Definition 7.1.3 (3) (2017: 7.3) works with $V(a) \cap A$: a sequence with sources in $\mathrm{cl}_A(V(a) \cap A) \subseteq \mathrm{cl}_X V(a)$ and convergent targets in $A$ has a subsequence converging in $\mathbf{X}$ (§VII.2), whose source lies in $A$ by saturation and in $\mathrm{cl}_A(V(a) \cap A)$ since this set is closed in $A$ Chapter7/SubGroupoid · EpGroupoid.exists_isProperMap_fullSub. *(4), literally.* The inclusion $\mathbf{A}\,{}_s\!\times_t \mathbf{A} \to \mathbf{X}\,{}_s\!\times_t \mathbf{X}$ is sc-smooth between the fibred products of Proposition 2.5.20 (2017: 2.15), by the unstated step applied twice, and $m$ of $A$ is $m$ of $X$ composed with it Chapter7/SubGroupoid · EpGroupoid.fullSub_comp_val Chapter7/SubGroupoid · EpGroupoid.isScSmoothComp_fullSub. So the full subcategory is an ep-groupoid Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid whose structure maps are the restrictions Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid_src_val Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid_comp_val, and the inclusion is a full and faithful ep-functor Chapter7/SubGroupoid · EpGroupoid.subEpFunctor Chapter7/SubGroupoid · EpGroupoid.subEpFunctor_full Chapter7/SubGroupoid · EpGroupoid.subEpFunctor_faithful, which the text does not state. *Hidden hypotheses:* none; the paracompactness of $A$ and $\mathbf{A}$ is proved (Findings, Chapter 7, item 42) (pp. 306–308 (2017 pp. 238–239)).

## VII.13 Definition 7.2.1 (2017: 7.11), Lemmas 7.2.2 and 7.2.3 (2017: 7.2 and 7.3)

**Definition 7.2.1 (2017: 7.11)** (V7.33). $X$ is *effective* if for every $x$, every $g \in G_x \setminus \{1_x\}$ and every open neighbourhood $\mathbf{U}(g)$ there is $\psi \in \mathbf{U}(g)$ with $s(\psi) \neq t(\psi)$ Chapter7/EpGroupoid · EpGroupoid.IsEffective; $x$ is *regular* if $G_x = \{1_x\}$ Chapter7/Groupoid · GroupoidStr.IsRegular, and $X_{reg}$ is the set of regular objects Chapter7/Groupoid · GroupoidStr.regularSet. "If $s(\varphi)$ is regular, so is $t(\varphi)$": conjugation with $\varphi$ is an isomorphism of isotropy groups (§VII.6) Chapter7/Groupoid · GroupoidStr.isRegular_tgt_iff Chapter7/Effective · EpGroupoid.isRegular_tgt_of_isRegular_src, so $X_{reg}$ is saturated Chapter7/Effective · EpGroupoid.isSaturated_regularSet. The definition is satisfiable non-vacuously, by $\mathbb{Z}/2$ acting on $\mathbb{R}$ by reflection (§VII.30) Chapter7/NonVacuity · Examples.isEffective_reflection_nonvacuous (Findings, Chapter 7, items 50, 101)

(2017 p. 239).

**Lemma 7.2.2 (2017: 7.2)** (V7.34) `Chapter7/Effective · EpGroupoid.lemma_7_2_2`. $X_{reg}$ *is open.* For regular $x$ and a natural representation at $x$, every morphism $\varphi : a \to b$ between points of $U(x)$ is $\Gamma(1_x, a)$ by Theorem 7.1.13 (3) (2017: 7.1), and $\Gamma(1_x, a) = 1_a$ (§VII.8); so $b = a$, $\varphi = 1_a$, and $G_a$ is trivial. The formalization proves the stronger statement that $U(x) \subseteq X_{reg}$ for *every* natural representation at a regular $x$ `Chapter7/Effective · EpGroupoid.NaturalRep.subset_regularSet`. *Hidden hypothesis:* $\Gamma(1, a) = 1_a$, which needs $U(x)$ connected. "Examples show that the regular set can be empty" is the trivial $\mathbb{Z}/3$-action (§VII.30) (Findings, Chapter 7, item 51) (p. 308 (2017 p. 239)).

**Lemma 7.2.3 (2017: 7.3)** (V7.35) `Chapter7/Effective · EpGroupoid.lemma_7_2_3`. *Equivalent: (1) $X$ is effective; (2) every open $V(x)$ contains a natural representation with $\Phi : G_x \to \mathrm{Diff}_{sc}(U(x))$ injective.* Injectivity is read as "$\Phi(g) = \mathrm{id}$ on $U$ only for $g = 1$" `Chapter7/Effective · EpGroupoid.NaturalRep.injective_Φ_iff`, the triviality of the kernel of $G_x \to \mathrm{Perm}(U)$ `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ` `Chapter7/Effective · EpGroupoid.NaturalRep.mem_kerΦ` `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ_eq_bot_iff` `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ_eq_bot_iff_injective`. **The text's proof.** (1) □ (2) by contradiction: if $\Phi(h) = 1$ with $h \neq 1$, effectiveness gives $\varphi_k \to h$ with $s(\varphi_k) \neq t(\varphi_k)$; $\varphi_k = \Gamma(h_k, s\varphi_k)$ with $h_k \neq h$, after a subsequence $h_k = g$, and in the limit $h = \Gamma(g, x) = g$. (2) □ (1): $\Phi(g) \neq 1$ gives $a$ with $\Phi(g)(a) \neq a$, and $\Gamma(g, a)$ has $s \neq t$; "by taking smaller and smaller neighborhoods $U(x)$ and adapting the corresponding restrictions of $\Gamma$" one finds such morphisms converging to $g$. **The formalized proof.** *(1) □ (2), for every natural representation, without sequences* `Chapter7/Effective · EpGroupoid.IsEffective.eq_one`. If $\Phi(h) = 1$ on $U$, the sheet $\Gamma(h, U)$ is an open neighbourhood of $h$ on which $s = t$, since $t(\Gamma(h, y)) = \Phi(h)(y) = y$; effectiveness forces $h = 1$. So effectiveness is the injectivity of every natural representation `Chapter7/Effective · EpGroupoid.isEffective_iff_forall_naturalRep`, and natural representations exist inside every $V(x)$ (§VII.9). *(2) □ (1).* Fix one natural representation $R_0 = (U_0, \Phi_0, \Gamma_0)$, $g \neq 1$ and an open $\mathbf{U}(g)$. *Step 1.* Apply (2) once, to $V = \{y \in U_0 \mid \Gamma_0(g, y) \in \mathbf{U}(g)\}$, open and containing $x$. *Step 2.* The natural representation $R$ that (2) gives inside $V$ has $\Phi(g) \neq 1$, so some $a \in U$ has $\Phi(g)(a) \neq a$. *Step 3.* $R$ need not be a restriction of $R_0$, as we read "adapting the corresponding restrictions"; it agrees with $R_0$ on the component of $x$ in $U \cap U_0 = U$, i.e. on $U$ (§VII.9) `Chapter7/Effective · EpGroupoid.NaturalRep.Γ_eqOn_of_subset` `Chapter7/Effective · EpGroupoid.NaturalRep.Φ_eqOn_of_subset`. *Step 4.* So $\Gamma_0(g, a) = \Gamma(g, a) \in \mathbf{U}(g)$ has source $a$ and target $\Phi(g)(a) \neq a$. One application of (2) suffices (Findings, Chapter 7, item 52) (pp. 307–309 (2017 pp. 239–240)).

## VII.14 Proposition 7.2.4 (2017: 7.8)

**Proposition 7.2.4 (2017: 7.8)** (V7.36) `Chapter7/Effective · EpGroupoid.proposition_7_2_4`. *If $X$ is effective, $X_{reg}$ is open and dense.* **The text's proof**: in a $U(x_0)$ take $x$ with $\#G_x$ minimal; if $\#G_x > 1$, Lemma 7.2.3 (2017: 7.3) gives $U(x) \subseteq U(x_0)$ with $\Phi$ injective and, for $g \neq 1$, a point $a$ with $\Phi(g)(a) \neq a$; $G_a$ is isomorphic to $H = \{h \in G_x \mid \Phi(h)(a) = a\}$, so $\#G_x \leq \#G_a = \#H \leq \#G_x$, $H = G_x$, a contradiction. **The formalized proof.** *Step 1.* For a natural representation at $x$ and $a \in U$, $h \mapsto \Gamma(h, a)$ is an isomorphism from the stabilizer of $a$ onto $G_a$, by Theorem 7.1.13 (3) (2017: 7.1) and the functoriality of $\Gamma$ `Chapter7/Effective · EpGroupoid.NaturalRep.stabilizerEquivIsotropy`. *Step 2.* Given a nonempty open $O$, take $x \in O$ minimizing $\#G_x$ over $O$ itself, so the text's $U(x_0)$ is not needed. *Step 3.* If $G_x \neq 1$, take a natural representation inside $O$ (§VII.9), injective by Lemma 7.2.3 (2017: 7.3), $g \neq 1$ and $a \in U$ with $\Phi(g)(a) \neq a$; then $\#G_a = \#\mathrm{Stab}(a) < \#G_x$, contradicting minimality. So $O$ meets $X_{reg}$; openness is Lemma 7.2.2 (2017: 7.2). Only $\#G_a < \#G_x$ is used. *The converse* holds too, and is not in the text `Chapter7/Effective · EpGroupoid.isEffective_of_dense_regularSet` `Chapter7/Effective · EpGroupoid.isEffective_`

iff_dense_regularSet: if $X_{reg}$ is dense and $g \neq 1$, a regular $a$ near $x$ with $\Gamma(g,a)$ in a given $\mathbf{U}(g)$ has $\Gamma(g,a) \neq 1_a = \Gamma(1,a)$, so $s \neq t$ there (Findings, Chapter 7, item 53) (pp. 308–311 (2017 pp. 240–241)).

## VII.15 The subgroups $T_x$, and Definition 7.2.7 (2017: 7.12)

$T_x$ (V7.37). $g \in G_x$ *acts trivially* if $\Phi(g) = 1$ for some natural representation Chapter7/EpGroupoid · EpGroupoid.ActsTrivially, and $T_x$ is the set of such $g$ Chapter7/EpGroupoid · EpGroupoid.trivialSet Chapter7/Groupoid · EpGroupoid.one_mem_trivialSet. The text: since $G_x$ is finite, $U(x)$ can be chosen so small that the kernel of $\Phi$ is $T_x$, a normal subgroup as a kernel. Since sc-smooth maps are not analytic, $\Phi(g) = \mathrm{id}$ near $x$ does not give $\Phi(g) = \mathrm{id}$ on a connected $U(x)$, so the kernel can depend on $U(x)$; the text's "small" is made precise in three statements. *Step 1, for every representation.* $g \in T_x$ iff $\Phi(g) = \mathrm{id}$ near $x$, for any given natural representation (§VII.9, Step 4) Chapter7/Effective · EpGroupoid.NaturalRep.actsTrivially_iff. *Step 2, intrinsically.* $g \in T_x$ iff $s = t$ on a neighbourhood of $g$ in $\mathbf{X}$ Chapter7/Effective · EpGroupoid.mem_trivialSet_iff_eventually: one direction is the sheet $\Gamma(g,U)$; for the other, a continuous section of $s$ through $g$ along which $s = t$ near $x$ gives $\Phi(g) = \mathrm{id}$ near $x$ Chapter7/Effective · EpGroupoid.mem_trivialSet_of_section. *Step 3, small neighbourhoods.* For one $R_0$, the common fixed set in $U_0$ of the finitely many elements of $T_x$ is a neighbourhood $N$ of $x$, and $\ker\Phi = T_x$ for *every* natural representation inside $N$ Chapter7/Effective · EpGroupoid.exists_mem_nhds_trivialSet_iff Chapter7/Effective · EpGroupoid.exists_naturalRep_trivialSet_iff Chapter7/Effective · EpGroupoid.exists_naturalRep_kerΦ_eq. *Step 4.* So $T_x$ is a kernel, a normal subgroup Chapter7/Effective · EpGroupoid.trivialSubgroup Chapter7/Effective · EpGroupoid.mem_trivialSubgroup Chapter7/Effective · EpGroupoid.trivialSubgroup_normal, its own normal closure Chapter7/Effective · EpGroupoid.normalClosure_trivialSet, and by Step 2 Definition 7.2.1 (2017: 7.11) says exactly $T_x = \{1_x\}$ for all $x$ Chapter7/Effective · EpGroupoid.isEffective_iff_trivialSubgroup_eq_bot. *At nearby points.* For $g \in \ker\Phi$, $\Gamma(g,y) \in T_y$ for $y \in U$, which the proof of Proposition 7.2.10 (2017: 7.9) uses Chapter7/Effective · EpGroupoid.NaturalRep.mem_trivialSet_of_val_eq_Γ. *Along a morphism, not stated in the text.* For $\varphi : x \to y$ and $g \in T_x$, $z \mapsto \theta(z) \circ \Gamma(g, s\theta(z)) \circ \theta(z)^{-1}$, $\theta = (t|\mathbf{U}(\varphi))^{-1}$, is a continuous section of $s$ through $\varphi g \varphi^{-1}$ — by (4) in the sheet form, twice — along which $s = t$ near $y$ Chapter7/Effective · EpGroupoid.exists_conj_section; by Step 2, $\varphi g \varphi^{-1} \in T_y$ Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet_iff, and $\varphi T_x \varphi^{-1} = T_y$ Chapter7/Effective · EpGroupoid.map_isotropyConj_trivialSubgroup. This is what the descent of the composition to $\mathbf{X}^R$ needs. "Diff$_{sc}(U(x)$" lacks a parenthesis (possible misprint or extraction artefact). Witnesses: $T_x = 1$ for the trivial ep-groupoid Chapter7/Effective · EpGroupoid.trivialSubgroup_trivial and the reflection, $T_x = G_x \cong \mathbb{Z}/3$ for the trivial $\mathbb{Z}/3$-action (§VII.30) (Findings, Chapter 7, item 54) (2017 p. 241).

**Definition 7.2.7 (2017: 7.12)** (V7.38). $\mathbf{X}^T = \bigcup_x T_x$ Chapter7/EpGroupoid · EpGroupoid.nonEffectivePart; $\varphi \sim \varphi'$ iff $s(\varphi) = s(\varphi') = x$ and $\varphi = \varphi' \circ g$ with $g \in T_x$ Chapter7/EpGroupoid · EpGroupoid.ReducedRel; $\mathbf{X}^R$ is the set of classes Chapter7/EpGroupoid · EpGroupoid.ReducedMor. *Step 1, $\mathbf{X}^T$ intrinsically.* By §VII.15 Step 2, $\mathbf{X}^T$ is the interior of $\{\varphi \mid s\varphi = t\varphi\}$ Chapter7/Reduced · EpGroupoid.mem_nonEffectivePart_iff Chapter7/Reduced · EpGroupoid.nonEffectivePart_eq_interior Chapter7/Reduced · EpGroupoid.val_mem_nonEffectivePart_iff, open Chapter7/Reduced · EpGroupoid.isOpen_nonEffectivePart and closed under units, inverses, composition and conjugation Chapter7/Reduced · EpGroupoid.unit_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.inv_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.comp_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.conj_mem_nonEffectivePart. So $\varphi \sim \varphi'$ iff $\varphi = \varphi' \circ k$ with $k \in \mathbf{X}^T$ Chapter7/Reduced · EpGroupoid.reducedRel_iff, a form without dependent types. *Step 2, an equivalence relation,* because $T_x$ is a subgroup Chapter7/Reduced · EpGroupoid.reducedRel_refl Chapter7/Reduced · EpGroupoid.ReducedRel.symm Chapter7/Reduced · EpGroupoid.ReducedRel.trans Chapter7/Reduced · EpGroupoid.reducedSetoid Chapter7/Reduced · EpGroupoid.reducedMor_eq_quotient. *Step 3, the left*

*form.* By the conjugation $\varphi T_x \varphi^{-1} = T_y$, $\varphi \sim \psi$ iff $\varphi = k \circ \psi$ with $k \in T_{t\psi}$ `Chapter7/Reduced · EpGroupoid.reducedRel_comp_left` `Chapter7/Reduced · EpGroupoid.ReducedRel.exists_comp_left`; the relation respects $s$, $t$, $\iota$ and composition on both sides `Chapter7/Reduced · EpGroupoid.ReducedRel.src_eq` `Chapter7/Reduced · EpGroupoid.ReducedRel.tgt_eq` `Chapter7/Reduced · EpGroupoid.ReducedRel.inv` `Chapter7/Reduced · EpGroupoid.ReducedRel.comp_left` `Chapter7/Reduced · EpGroupoid.ReducedRel.comp_right`. "The subset $\mathbf{X}^R$ of reduced morphisms" is a quotient, as the next sentence of the text says (possible misprint; Findings, Chapter 7, item 55) (2017 p. 241).

## VII.16 Proposition 7.2.10 (2017: 7.9): the M-polyfold $\mathbf{X}^R$

**Proposition 7.2.10 (2017: 7.9)** (V7.39) `Chapter7/Reduced · EpGroupoid.proposition_7_2_10`. *$\mathbf{X}^R$ has a natural M-polyfold structure.* **The text's proof**: $\mathbf{X}^R$ carries the quotient topology; on a sheet $s : U(\varphi_0) \to U(s\varphi_0)$ the projection $\pi'$ is continuous and injective; its image $U([\varphi_0])$ is open, by the map $\delta(\psi') = \psi' \circ \Gamma(g, s\psi')$ near an equivalent $\psi$, with $U(s\psi)$ chosen with $\ker \Phi = T_{s\psi}$; the same argument shows $\Psi_{\varphi_0} : U(\varphi_0) \to U([\varphi_0])$ open; and the transitions are, near $\varphi$ with $\psi = \varphi \circ g$, the sc-smooth $\delta$. **The formalized proof.** *Step 1, the topology.* The projection $\pi^R : \mathbf{X} \to \mathbf{X}^R$ `Chapter7/Reduced · EpGroupoid.reducedπ` `Chapter7/Reduced · EpGroupoid.reducedπ_eq_iff` `Chapter7/Reduced · EpGroupoid.surjective_reducedπ` is a quotient map `Chapter7/Reduced · EpGroupoid.isQuotientMap_reducedπ` `Chapter7/Reduced · EpGroupoid.continuous_reducedπ`. *Step 2, $\pi^R$ is open* `Chapter7/Reduced · EpGroupoid.isOpenMap_reducedπ`. The text's $\delta$ composes on the right with a section of $s$; its continuity holds by (4) applied to inverses. The left form is used instead: for $\varphi \sim \psi$, $\varphi = \Gamma(g, t\psi) \circ \psi$ near $\psi$ with $g \in T_{t\psi}$ and a natural representation with $\ker \Phi = T_{t\psi}$ (§VII.15); $\delta(\psi') = \Gamma(g, t\psi') \circ \psi'$ is sc-smooth by (4) in the sheet form `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScSmoothOn_comp_Γ_tgt` and $\psi' \sim \delta(\psi')$ since $\Gamma(g, t\psi') \in T_{t\psi'}$. So the saturation of an open set is open. *Step 3, sheets.* On every sheet $s : \mathbf{W} \to V$, $\pi^R$ is injective — equivalent morphisms have the same source — and hence an open embedding `Chapter7/Reduced · EpGroupoid.injOn_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ_image_Γ`; "small" sheets are not needed. *Step 4, charts.* For a chart $\psi$ of $\mathbf{X}$ with domain in a sheet $\mathbf{W}$, $\sigma = (s|\mathbf{W})^{-1}$, the chart of $\mathbf{X}^R$ is $[\varphi'] \mapsto \psi(\sigma(s[\varphi']))$ on $\pi^R(\operatorname{dom}\psi)$ `Chapter7/Reduced · EpGroupoid.reducedChart`; every point lies in one `Chapter7/Reduced · EpGroupoid.exists_chart_sheet`. *Step 5, transitions.* On an overlap, an element of the second sheet equivalent to $\varphi'$ has the source $s(\varphi')$, so the text's $\delta$ equals $\sigma_2 \circ s$ and the transition is $\psi_2 \circ \sigma_2 \circ s \circ \psi_1^{-1}$ `Chapter7/Reduced · EpGroupoid.reducedChart_transition`, sc-smooth; what needs the text's argument is that the overlap is open, i.e. Step 2 `Chapter7/Reduced · EpGroupoid.overlap_reducedChart` `Chapter7/Reduced · EpGroupoid.reducedChart_compatible`. The charts form an atlas `Chapter7/Reduced · EpGroupoid.reducedCharts` `Chapter7/Reduced · EpGroupoid.reducedAtlas`, all without hypotheses. *Step 6, Hausdorff and paracompact.* An M-polyfold is a Hausdorff paracompact space (Definition 2.3.4 (2017: 2.8)). The proof gives neither as we read it, and $\mathbf{X}^R$ need not be Hausdorff (the example at the end of this paragraph, formalized); so the M-polyfold $\mathbf{X}^R$ is built under the hypothesis that $\mathbf{X}^R$ is Hausdorff, the 2021 text's hypothesis on $T_R$ (§VII.34); paracompactness, a second hypothesis in the first reading, is now an instance, proved for every ep-groupoid as Proposition 7.2.12 `Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_reducedMor` `Chapter7/Reduced · EpGroupoid.reducedPolyfold` `Chapter7/Reduced · EpGroupoid.isChart_reducedChart`, and then $\pi^R$ is a local sc-diffeomorphism, with inverse $\sigma \circ s$ on $\pi^R(\mathbf{W})$ `Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedπ` `Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedπ` `Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedπ` (gap G.71, closed, §§VII.34, VII.48). Steps 1–5 are the 2021 Proposition 7.2.10, with no hypothesis `Chapter7/GapP2_AReduced · EpGroupoid.proposition_7_2_10_natural_charts`. *The hypotheses are satisfiable.* For effective $X$, $\sim$ is equality `Chapter7/Reduced · EpGroupoid.reducedRel_iff_eq_of_isEffective`, $\pi^R$ a homeomorphism `Chapter7/Reduced · EpGroupoid.reducedHomeomorph`, and $\mathbf{X}^R$ is Haus-

dorff, paracompact and regular `Chapter7/Reduced · EpGroupoid.t2Space_reducedMor_of_isEffective` `Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_of_isEffective` `Chapter7/Reduced · EpGroupoid.t3Space_reducedMor_of_isEffective`; this covers the trivial ep-groupoid of every M-polyfold `Chapter7/Reduced · EpGroupoid.t2Space_reducedMor_trivial` `Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_trivial`. Paracompactness also follows from regularity of $\mathbf{X}^R$ when $\mathbf{X}$ is second countable `Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_of_t3`; these partial results of the first reading are superseded by Proposition 7.2.12, which needs no hypothesis. Four possible misprints of the proof: "$\psi' = (s|U(\psi))^{-1}(s(\psi))$" for $s(\psi')$, "$U(\varphi) \subset \Psi_{\varphi_0}(U([\varphi_0]) \cap U([\psi_0]))$" for $\Psi_{\varphi_0}^{-1}$, "$T_x$ agrees with the kernel of the $G_{s(\varphi)}$-action" for $T_{s(\varphi)}$, and a garbled sentence on the smoothness of the source and target maps (Findings, Chapter 7, items 56, 57, 60, 111) (2017 pp. 242–243). *The Hausdorff hypothesis cannot be dropped* `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor`. Let $\mathbb{Z}/2$ act by $(t, z) \mapsto (t, \pm z)$ on the retract $O$ of Part I's Remark 4.2.2 (2017: 4.3) example, a point over each $t \leq 0$ and a line over each $t > 0$; the action is the restriction of an sc-operator `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.isScSmoothMap_smul`. Over $t < 0$ both elements act trivially, so $(-1, (t, 0)) \in \mathbf{X}^T$ `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.mem_nonEffectivePart`; at $(0, 0)$ points of $O$ with $t > 0$ tend to $0$ and are moved by $-1$, so $(-1, (0, 0)) \notin \mathbf{X}^T$ `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_mem_nonEffectivePart`. The classes of $(\pm 1, (-1/(n+1), 0))$ coincide and converge to both $[(-1, 0)] \neq [(1, 0)]$ `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_t2Space_reducedMor`. So Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), for every ep-groupoid, do not hold as we read them; the 2021 text takes the Hausdorff hypothesis (Findings, Chapter 7, item 107; found on paper while reading, proved in the round-trip audit).

## VII.17 The reduced ep-groupoid $X_R$, Proposition 7.10 (2017), Definition 7.2.15 (2017: 7.13)

$X_R$ (V7.40). Objects $X$, morphisms $\mathbf{X}^R$, $s([\varphi]) = s(\varphi), t([\varphi]) = t(\varphi)$, with unit, inverse and composition descended by §VII.15 `Chapter7/Reduced · EpGroupoid.reducedSrc` `Chapter7/Reduced · EpGroupoid.reducedTgt` `Chapter7/Reduced · EpGroupoid.reducedUnit` `Chapter7/Reduced · EpGroupoid.reducedInv` `Chapter7/Reduced · EpGroupoid.reducedComp` `Chapter7/Reduced · EpGroupoid.reducedComp_π`, $s$ and $t$ onto `Chapter7/Reduced · EpGroupoid.surjective_reducedSrc` `Chapter7/Reduced · EpGroupoid.surjective_reducedTgt`, a groupoid `Chapter7/Reduced · EpGroupoid.reducedGroupoidStr`; the composition of a non-composable pair is a junk value, as in §VII.1 (2017 p. 243).

**Proposition 7.10 (2017)** (V7.40) `Chapter7/Reduced · EpGroupoid.proposition_7_10_2017`. *Under the hypotheses of §VII.16, $X_R$ is an ep-groupoid, and it is effective. (1).* For a sheet $\mathbf{W}$ of Definition 7.1.4 (2017: 7.4), $s, t : \pi^R(\mathbf{W}) \to X$ are $s \circ (\pi^R|\mathbf{W})^{-1}$ and $t \circ (\pi^R|\mathbf{W})^{-1}$, the text's $s(\Psi_\varphi(\varphi')) = s(\varphi')$ `Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedSrc` `Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedTgt` `Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedSrc` `Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedTgt`. *(2).* $u = \pi^R \circ u_X$ and $\iota = \pi^R \circ \iota_X \circ (\pi^R|\mathbf{W})^{-1}$ on $\pi^R(\mathbf{W})$ `Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedUnit` `Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedInv`. *(3), "proved similarly".* For the same $V(x)$, $t \circ \pi^R = t_X$ on $s^{-1}(\mathrm{cl}V(x))$, with $\pi^R$ a continuous surjection onto the corresponding set of $\mathbf{X}^R$; a map whose composite with a continuous surjection is proper is proper (Mathlib) `Chapter7/Reduced · EpGroupoid.exists_isProperMap_reduced`. *(4), "reduced by our local charts",* in the sheet form: for an sc-diffeomorphism $s : \mathbf{W}' \to V'$ of $\mathbf{X}^R$ with inverse $\sigma'$, $\kappa = \sigma_2 \circ s \circ \sigma'$ is the inverse of $s$ on an open set of $\mathbf{X}$ — a step the text leaves implicit — and $m(\sigma'(t\psi), \psi) = \pi^R(m(\kappa(t\psi), \psi))$ `Chapter7/Reduced · EpGroupoid.isScSmoothCompSheet_reduced`. So $X_R$ is an ep-groupoid `Chapter7/Reduced · EpGroupoid.reducedEpGroupoid` and $X \to X_R$ an ep-functor `Chapter7/Reduced · EpGroupoid.reductionFunctor`. *Effective.* The text's point $a$ with

$\Phi(g)(a) \neq a$ is replaced by the intrinsic form of $T_x$: if $s = t$ near $[g]$ in $\mathbf{X}^R$, then $s = t$ near $g$ in $\mathbf{X}$, $\pi^R$ being continuous, so $g \in T_x$ and $[g] = [1_x]$ Chapter7/Reduced · EpGroupoid.isEffective_reducedEpGroupoid. For effective $X$, $X_R \cong X$ in $\mathcal{EP}$ Chapter7/Reduced · EpGroupoid.reductionInv Chapter7/Reduced · EpGroupoid.reductionIso. In the 2021 form, Theorem 7.2.14 with Proposition 7.2.16, only the Hausdorff hypothesis remains Chapter7/GapP2_AReduced · EpGroupoid.theorem_7_2_14_reduced (§VII.48) (Findings, Chapter 7, items 58, 60) (2017 p. 244).

**The isotropy of $X_R$, and Definition 7.2.15 (2017: 7.13)** (V7.41). $g \mapsto [g]$ is a surjective homomorphism from $G_x$ onto the isotropy of $x$ in $X_R$ Chapter7/Reduced · EpGroupoid.reducedIsotropyHom Chapter7/Reduced · EpGroupoid.surjective_reducedIsotropyHom with kernel $T_x$ — the text's "$[1_x] = T_x$" Chapter7/Reduced · EpGroupoid.ker_reducedIsotropyHom — so that isotropy is $G_x/T_x$ Chapter7/Reduced · EpGroupoid.quotientTrivialSubgroupEquiv. The effective isotropy group $G_x^{\text{eff}} = G_x/T_x$ Chapter7/EpGroupoid · EpGroupoid.EffIsotropy is defined as the quotient by the normal closure of $T_x$, which is $G_x/T_x$ by §VII.15 Chapter7/Effective · EpGroupoid.effIsotropyEquiv, and it is the isotropy group of $X_R$ Chapter7/Reduced · EpGroupoid.effIsotropyEquivReduced. "For and object" (possible misprint; Findings, Chapter 7, item 59) (2017 p. 244).

## VII.18 Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2 and 7.3.3 (2017: 7.4 and 7.5)

**Theorem 7.3.1 (2017: 7.2)** (V7.42) Chapter7/OrbitTopology · EpGroupoid.theorem_7_3_1. *$|X|$ is locally metrizable, regular and Hausdorff; if $|X|$ is paracompact, it is metrizable.* "Locally metrizable" is read as: every point has an open neighbourhood which is metrizable in the subspace topology. The three properties are Lemmas 7.3.2, 7.3.5 and 7.3.4 (2017: 7.4, 7.7 and 7.6). The metrizability Chapter7/OrbitTopology · EpGroupoid.theorem_7_3_1_metrizable is the text's "Nagata/Smirnov" in Part I's form (§II.7): a paracompact Hausdorff locally metrizable space is metrizable Chapter2/Metrizable · metrizableSpace_of_locally. The hypothesis "$|X|$ paracompact" is satisfiable, by the trivial ep-groupoid Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial (Findings, Chapter 7, item 70) (2017 p. 244).

**Lemma 7.3.2 (2017: 7.4)** (V7.43) Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_2. *Every $z \in |X|$ has an open metrizable neighbourhood; one can take $|U(x)|$ for a natural representation at $x$.* **The text's proof**: $U$ is metrizable with a metric $d$; $\rho(y, y') = \max_g d(g * y, g * y')$ is an equivalent $G_x$-invariant metric; $\hat{\rho}([y], [y']) = \min\{\rho(g * y, h * y') \mid g, h \in G_x\}$ defines the quotient topology on $G_x\backslash U$; and Proposition 7.1.19 (2017: 7.6) carries it to $|U|$. **The formalized proof** follows it, for any finite group acting by homeomorphisms on a metrizable space Chapter7/OrbitTopology · metrizableSpace_orbitRel_quotient. *Step 1.* $\rho$ is the maximum over the finite $G_x$ Chapter7/OrbitTopology · maxDist, invariant Chapter7/OrbitTopology · maxDist_smul, and $d \leq \rho$. *Step 2.* $\hat{\rho}$ is computed as $\min_h \rho(y, h * y')$, the same number by invariance Chapter7/OrbitTopology · quotDist Chapter7/OrbitTopology · quotDist_mk; it is symmetric and satisfies the triangle inequality by invariance, and $\hat{\rho}([y], [y']) = 0$ forces $y' \in G_x * y$, orbits of a finite group being closed. *Step 3, the topology.* A set of $G_x\backslash U$ is open in the quotient topology iff it contains a $\hat{\rho}$-ball around each of its points: $\{y' \mid \rho(y, y') < \varepsilon\} = \bigcap_g g^{-1}(B_d(g * y, \varepsilon))$ is open, and $d \leq \rho$ gives the converse. This is Mathlib's construction of a metric space with a prescribed topology; $\rho$ itself is never installed as a metric on $U$. So $G_x\backslash U(x)$ is metrizable Chapter7/OrbitTopology · EpGroupoid.metrizableSpace_orbitRel_quotient_naturalRep and so is $|U(x)|$, its homeomorphic image (§VII.11) Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_2_naturalRep (Findings, Chapter 7, item 71) (pp. 319–325 (2017 pp. 244–245)).

**Lemma 7.3.3 (2017: 7.5)** (V7.44) Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_3. *Every point of $|X|$ has a monotone countable basis of open neighbourhoods.* This is Proposition 7.1.9 (2017: 7.2) (§VII.5), proved there from

the metrizability of $X$ and the openness of $\pi$; the text derives it from Lemma 7.3.2 (2017: 7.4) (Findings, Chapter 7, item 72) (2017 p. 245).

## VII.19 Lemmas 7.3.4 and 7.3.5 (2017: 7.6 and 7.7), Corollary 7.3.6 (2017: 7.1), subsets of $|X|$

The text proves Lemma 7.3.4 (2017: 7.6) before Lemma 7.3.5 (2017: 7.7); the formalization proves Lemma 7.3.5 (2017: 7.7) first and derives Lemma 7.3.4 (2017: 7.6) from it (Findings, Chapter 7, item 72).

**Lemma 7.3.5 (2017: 7.7)** (V7.46) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_5`. $|X|$ *is regular.* **The text's proof**: for closed $A \not\ni z = \pi(x)$ choose $O \ni x$ with $\pi(O) \cap A = \emptyset$ and $t$ proper over $s^{-1}(\mathrm{cl}O)$, and $x \in O' \subseteq \mathrm{cl}O' \subseteq O$; every $y \in \pi^{-1}(A)$ has a $V(y)$ which no morphism from $\mathrm{cl}O'$ reaches, else properness gives $\varphi$ with $s\varphi \in \mathrm{cl}O'$ and $t\varphi = y$; then $\pi(\bigcup V(y)) \supseteq A$ and $\pi(O')$ are disjoint. **The formalized proof** uses a closed neighbourhood basis `Mathlib/Topology/Separation/Regular · RegularSpace.of_exists_mem_nhds_isClosed_subset`. *Step 1.* For a neighbourhood $N$ of $\pi(x)$, take a closed neighbourhood $O'$ of $x$ inside $\pi^{-1}(N) \cap V(x)$, $X$ being regular and $V(x)$ from Definition 7.1.3 (3) (2017: 7.3). *Step 2.* $\pi(O')$ is closed by Proposition 7.1.10 (2017: 7.3), $O' \subseteq \mathrm{cl}_X V(x)$ `Chapter7/Basic · EpGroupoid.proposition_7_1_10_of_isProperMap`, a neighbourhood of $\pi(x)$ by Proposition 7.1.8 (2017: 7.1), and inside $N$. The same properness replaces the text's sets $V(y)$ (2017 p. 246).

**Lemma 7.3.4 (2017: 7.6)** (V7.45) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_4`. $|X|$ *is Hausdorff.* **The text's proof** is by contradiction: $x_k \to x$, $x'_k \to x'$ with morphisms $\varphi_k : x_k \to x'_k$, of which properness extracts a subsequence converging to a morphism $x \to x'$. **The formalized proof** is two lines. *Step 1, points are closed* `Chapter7/OrbitTopology · EpGroupoid.t1Space_orbitSpace`: the point $\{x\}$ is compact, hence a proper subset `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_of_isCompact`, and $\pi(\{x\})$ is closed by Proposition 7.5.10 (1) (2017: 7.14) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_10_one` (§VII.25), which uses the same properness as the text's sequence. *Step 2.* A regular $T_1$ space is Hausdorff, with Lemma 7.3.5 (2017: 7.7) (pp. 324–326 (2017 pp. 245–246)).

**Corollary 7.3.6 (2017: 7.1)** (V7.47) `Chapter7/OrbitTopology · EpGroupoid.corollary_7_3_6`. *A second countable* $|X|$ *is metrizable*: Urysohn's metrization theorem, an instance of Mathlib, $|X|$ being regular and Hausdorff by Lemmas 7.3.4, 7.3.5 (2017: 7.6, 7.7) (Findings, Chapter 7, item 73) (2017 p. 247).

**Subsets of** $|X|$ (V7.48). "The induced topology on a subset of $|X|$ is again locally metrizable, regular and Hausdorff": local metrizability passes to subspaces `Chapter7/OrbitTopology · EpGroupoid.exists_metrizable_nhds_subtype`, regularity and Hausdorffness are Mathlib's subspace instances, and so a paracompact subset of $|X|$ is metrizable, by Smirnov's theorem in the subspace `Chapter7/OrbitTopology · EpGroupoid.metrizableSpace_of_paracompactSpace_subtype` (Findings, Chapter 7, item 73) (2017 p. 247).

## VII.20 Michael's theorem: Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17)

(V7.50.) Corollary 7.3.8 (2017: 7.2) needs that a finite union of closed paracompact subsets of the regular $|X|$ is paracompact, and the text's route to Proposition 7.5.3 (2017: 7.12) uses Lemma 2.7.3 (2017: 2.7). Part I's Lemma 2.7.3 (2017: 2.7) (Michael: a regular Hausdorff space is paracompact iff every open cover has a locally finite closed refinement) and Proposition 2.7.4 (2017: 2.17) (a regular Haus-

dorff space that is the union of a locally finite family of closed paracompact subspaces is paracompact), 2017 p. 51, were gap G.15 of Part I; they are formalized here, for regular spaces and without Hausdorff. Michael's Lemma 1 has four conditions on a regular space: (a) paracompact; (b) every open cover has a σ-locally finite open refinement; (c) every open cover has a locally finite refinement by arbitrary sets; (d) every open cover has a locally finite closed refinement `Chapter7/Michael · michael_tfae`. *(d) □ (a), in every topological space* `Chapter7/Michael · paracompactSpace_of_closed_refinement`. Let $(A_b)$ be a closed locally finite refinement of the cover $(U_i)$, with $A_b \subseteq U_{f(b)}$; cover the space by open sets each meeting finitely many $A_b$ and take a closed locally finite refinement $(B_c)$ of that cover; then $V_b = (\bigcup\{B_c \mid B_c \cap A_b = \emptyset\})^c \cap U_{f(b)}$ is open (a locally finite union of closed sets is closed), contains $A_b$, and each $B_c$ meets only the finitely many $V_b$ with $B_c \cap A_b \neq \emptyset$; so $(V_b)$ is a locally finite open refinement. *(c) □ (d) in a regular space* `Chapter7/Michael · closed_refinement_of_refinement`: refine first a cover by open sets whose closures lie in the cover, and take closures. *(b) □ (c), in every space* `Chapter7/Michael · refinement_of_sigma_refinement`: $t_{n,b} \setminus \bigcup_{k<n} \bigcup_{b'} t_{k,b'}$. Hence (c) □ (a) and (b) □ (a) in a regular space `Chapter7/Michael · paracompactSpace_of_refinement` `Chapter7/Michael · paracompactSpace_of_sigma_refinement`, and (a) □ (d) by Mathlib's shrinking lemma, with the same index and $v_i \subseteq u_i$ `Chapter7/Michael · exists_closed_precise_refinement` `Chapter7/Michael · exists_closed_precise_refinement_set`. **Lemma 2.7.3 (2017: 2.7)** is the equivalence of (a) and (d) `Chapter7/Michael · lemma_2_7_3`. **Proposition 2.7.4 (2017: 2.17)** `Chapter7/Michael · proposition_2_7_4` follows the text: closed locally finite refinements $(C^i_j)_j$ of the traces $(Y_i \cap U_j)_j$ by Lemma 2.7.3 (2017: 2.7) in each $Y_i$, which is regular because $Y$ is; $C_j = \bigcup_i C^i_j$ is locally finite in $Y$ because $(Y_i)$ is and a locally finite family in a closed subspace is locally finite in the space `Chapter7/Michael · locallyFinite_image_val`; and (d) □ (a) in $Y$. The subspace forms are what Chapter 7 uses: a locally finite family of closed paracompact subsets has a paracompact union `Chapter7/Michael · paracompactSpace_iUnion`, so do finite families and pairs `Chapter7/Michael · paracompactSpace_iUnion_of_finite` `Chapter7/Michael · paracompactSpace_biUnion_of_finite` `Chapter7/Michael · paracompactSpace_union`, and a closed subset of a paracompact subspace is paracompact `Chapter7/Michael · paracompactSpace_of_isClosed_subset`. *The Hausdorff property is never used*; regularity enters through (c) □ (d) and (a) □ (d). Part I's 2017 p. 51 says "subcovering" in the definition of paracompactness, read as "refinement", as on 2017 p. 249 (Findings, Chapter 7, items 75, 114).

## VII.21 Definition 7.3.7 (2017: 7.14) and Corollary 7.3.8 (2017: 7.2)

**Definition 7.3.7 (2017: 7.14)** (V7.49). $K \subseteq |X|$ is component compact if it meets every connected component of $|X|$ in a compact set `Chapter7/EpGroupoid · EpGroupoid.IsComponentCompact` (2017 p. 247).

**Corollary 7.3.8 (2017: 7.2)** (V7.51) `Chapter7/OrbitTopology · EpGroupoid.corollary_7_3_8`. *A component compact $K$ has an open neighbourhood $U$ with* $\mathrm{cl}_{|X|}U$ *paracompact; then* $\mathrm{cl}U$ *and $U$ are metrizable.* **The text's proof**: work in a connected component of $K$, so that $K$ may be assumed compact; every $z \in K$ has an open $U(z)$ with metrizable, hence paracompact, closure; finitely many $U(z_i)$ cover $K$; $U = \bigcup \mathrm{cl}U(z_i)$ is paracompact, Hausdorff, regular and locally metrizable, hence metrizable, and so is $U = \bigcup U(z_i)$. **The formalized proof.** *Step 1, the compact case, inside a given open $W \supseteq K$* `Chapter7/OrbitTopology · EpGroupoid.corollary_7_3_8_of_isCompact`. Every $z$ has an open $U(z)$ with $\mathrm{cl}U(z) \subseteq W$ metrizable, by Lemma 7.3.2 (2017: 7.4) and the regularity of $|X|$ `Chapter7/OrbitTopology · EpGroupoid.exists_isOpen_metrizableSpace_closure`; finitely many cover $K$; $U = \bigcup U(z_i)$ has $\mathrm{cl}U = \bigcup \mathrm{cl}U(z_i)$, paracompact by Michael (§VII.20), which the text does not cite, and metrizable by §VII.19. *Step 2, the components are open.* "We work in a connected component of $K$" is read in a component $C$ of $|X|$, where $K \cap C$ is compact. This needs the components of $|X|$ to be open, which the text does not state: $|X|$ is a quotient of the locally path connected $X$ (§VII.7), hence locally connected `Chapter7/OrbitTopology · EpGroupoid.locallyPathConnectedSpace_orbitSpace`. *Step 3, back*

*to $K$, not in the text.* Step 1 in each component $C$, with $W = C$, gives $U_C \supseteq K \cap C$ with $\mathrm{cl}U_C \subseteq C$ paracompact. The components are open and pairwise disjoint, so $(\mathrm{cl}U_C)_C$ is locally finite, $\mathrm{cl}(\bigcup U_C) = \bigcup \mathrm{cl}U_C$, and this locally finite union of closed paracompact sets is paracompact (Proposition 2.7.4 (2017: 2.17), §VII.20) `Chapter7/Michael · paracompactSpace_iUnion`. The text's $U$ names both $\bigcup \mathrm{cl}U(z_i)$ and $\bigcup U(z_i)$ (possible misprint); here $U$ is the open one (Findings, Chapter 7, item 74) (2017 p. 247).

## VII.22 Proposition 7.3.9 (2017: 7.11): index lifting

**Proposition 7.3.9 (2017: 7.11)** (V7.52) `Chapter7/Lift · EpGroupoid.proposition_7_3_9`. *If $X = (X, \mathbf{X})$ is an ep-groupoid, so is $X^1 = (X^1, \mathbf{X}^1)$; if $X$ is tame, so is $X^1$.* $X^1$ is the M-polyfold of Part I with the levels shifted by one (§II.6). **The structure maps of $X^1$.** The text restricts $s, t, u, \iota, m$ to level 1 without comment. Sc-smooth maps preserve the levels, so $s, t, u, \iota$ restrict `Chapter7/LocalDiffeo · MPolyfold.levelOneMap Chapter7/Lift · EpGroupoid.liftSrc Chapter7/Lift · EpGroupoid.liftTgt Chapter7/Lift · EpGroupoid.liftUnit Chapter7/Lift · EpGroupoid.liftInv`; a composite of composable morphisms on level 1 is on level 1, since $ml(t(\varphi \circ \psi)) = ml(t\varphi) = ml(\varphi)$ (§VII.3) `Chapter7/Lift · EpGroupoid.comp_mem_levelSet Chapter7/Lift · EpGroupoid.liftComp`; the groupoid is `Chapter7/Lift · EpGroupoid.liftGroupoidStr`. **The text's proof** says that (1), (2) and (4) hold trivially, and proves (3) by sequences: for $x \in X_1$ and $V(x)$ with $t$ proper over $s^{-1}(\mathrm{cl}_X V(x))$, $W(x) = V(x) \cap X_1$ is open in $X_1$ and $\mathrm{cl}_{X_1} W(x) \subseteq \mathrm{cl}_X V(x)$; for $(\varphi_k)$ with $(t\varphi_k)$ in a compact subset of $X_1$, after a subsequence (a) $\varphi_k \to \varphi$ in $\mathbf{X}_0$, (b) $s, t$ converge in $X_0$, (c) $t(\varphi_k) \to t(\varphi)$ in $X_1$, so (d) $\varphi \in \mathbf{X}_1$, and $\varphi_k = (t|\mathbf{U}(\varphi))^{-1}(t\varphi_k) \to \varphi$ in $\mathbf{X}^1$. **The formalized proof.** *(1).* The level-1 lift of a local sc-diffeomorphism is one: an sc-diffeomorphism on an open set lifts with its inverse `Chapter7/LocalDiffeo · MPolyfold.IsScDiffeoOn.levelOne Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.levelOne Chapter7/Lift · EpGroupoid.isLocalScDiffeo_liftSrc Chapter7/Lift · EpGroupoid.isLocalScDiffeo_liftTgt`. *(2).* Part I's lifting of the index for sc-smooth maps, Proposition 1.2.2 (preprint 1.17) in charts (§I.15) `Chapter5/StrongBundleShift · MPolyfold.IsScSmoothMap.levelOne Chapter7/Lift · EpGroupoid.isScSmoothMap_liftUnit Chapter7/Lift · EpGroupoid.isScSmoothMap_liftInv`. *(4).* The text's "trivially", as we read it, uses $(\mathbf{X}\,{}_s\!\times_t \mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t \mathbf{X}^1$ as M-polyfolds, which is not proved here; it is avoided through the sheet form (§VII.2). A local inverse $\sigma_1$ of $s$ on level 1 agrees near each point with the level-1 lift of a local inverse $\sigma$ of $s$, both inverting $s$ near that point, and $\psi \mapsto m(\sigma(t\psi), \psi)$ lifts by the lifting of sc-smooth maps on open sets `Chapter7/LocalDiffeo · MPolyfold.IsScSmoothOn.levelOne Chapter7/Lift · EpGroupoid.isScSmoothCompSheet_lift Chapter7/Lift · EpGroupoid.isScSmoothComp_lift`. *(3), the text's argument* `Chapter7/Lift · EpGroupoid.exists_isProperMap_lift`. "$(t\varphi_k)$ in a compact subset of $X_1$" is replaced by "$t\varphi_k$ converges in $X_1$", the sequential criterion for a closed set (§VII.2) `Chapter7/ProperMaps · isProperMap_domRestrict_iff_of_isClosed`, $X^1$ and $\mathbf{X}^1$ being metrizable (Theorem 2.3.10 (2017: 2.2)) `Chapter2/Metrizable · MPolyfold.metrizableSpace_level`; $\mathrm{cl}_{X_1} W(x) \subseteq \mathrm{cl}_X V(x)$ is the continuity of $X_1 \to X$; (a) is properness on level 0, (c) the hypothesis, (d) that $t$ is a local sc-diffeomorphism and preserves levels; step (b) is not needed; and "$(t|\mathbf{U}(\varphi))^{-1}$ is $\mathrm{sc}^0$" is used as the continuity on level 1 of its level-1 lift. So $X^1$ is an ep-groupoid `Chapter7/Lift · EpGroupoid.liftOne`, paracompact on both levels `Chapter2/Metrizable · MPolyfold.paracompactSpace_level`. The display (7.5), "$s^{-1}(\mathrm{cl}_{X_1}(W(x)) \to X_1$", has an unbalanced parenthesis (possible misprint; Findings, Chapter 7, items 80, 96).

**Tameness of $X^1$** `Chapter7/Lift · EpGroupoid.isTame_liftOne`. The text calls it trivial. As we read it, it needs that the index lift of a tame retraction is tame, which Part I left open (Findings, Chapter 5, item 145); it is proved `Chapter7/Lift · MPolyfold.IsTame.levelOne`. *Step 1, the shifted presentation.* For a tame $r$ with a presentation $T : E \cong \mathbb{R}^n \oplus W$ of the partial quadrant $C$, the presentation $T^1 : E^1 \cong \mathbb{R}^n \oplus W^1$ of $C^1$ has the same $\mathbb{R}^n$-coordinates `Chapter7/Lift · shiftPres Chapter7/Lift · shift_presentation`, so

$d_{C^1} = d_C$ on $C^1$ `Chapter7/Lift · degIndex_shift` and the subspace $E_x$ for $T^1$ is $E_x \cap E_1$ `Chapter7/Lift · mem_minimalSubspaceWith_shift`. *Step 2, the tangent.* $Dr^1(x) = Dr(x)|_{E_1}$ (Proposition 1.2.2 (preprint 1.17)), whose range is $T_xO \cap E_1$ since $Dr(x)$ is a projection. *Step 3, the complement.* An sc-complement $A$ of $T_xO$ gives the sc-complement $A \cap E_1$ of $T_xO^1$, an sc-splitting shifting to one `Chapter7/Lift · isScSplitting_levelSub_one` `Chapter7/Lift · isScSubspace_levelSub_one`. *Step 4.* So $r^1$ is tame for $T^1$ `Chapter7/Lift · isTameWith_shift` `Chapter7/Lift · isTameRetract_shift`, hence tame by Part I's independence of the presentation `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame`, and the level charts of tame charts form a tame atlas of $X^1$ (Findings, Chapter 7, item 81) (pp. 326–328 (2017 pp. 247–248)).

## VII.23 Definition 7.5.1 (2017: 7.15), Remark 7.5.2 (2017: 7.1), and Proposition 7.5.3 (2017: 7.12)

**Definition 7.5.1 (2017: 7.15)** (V7.53). An ep-groupoid has paracompact orbit space if in addition (5) $|X|$ is paracompact `Chapter7/EpGroupoid · EpGroupoid.HasParacompactOrbitSpace`; the text stipulates that paracompact does not include Hausdorff, and Mathlib's notion does not. Since $|X|$ is Hausdorff and locally metrizable (Theorem 7.3.1 (2017: 7.2)), (5) is equivalent to the metrizability of $|X|$ `Chapter7/Lift · EpGroupoid.hasParacompactOrbitSpace_iff_metrizableSpace`, and it holds when $|X|$ is second countable (Corollary 7.3.6 (2017: 7.1)) `Chapter7/Lift · EpGroupoid.hasParacompactOrbitSpace_of_secondCountableTopology`. *The sentence before Remark 7.5.2 (2017: 7.1)*, "in case $X$ has a second countable topology, (1)–(4) imply (5)", is read with $X$ the object space `Chapter7/Lift · EpGroupoid.remark_7_5_2`: $|X|$ is then second countable, $\pi$ being open and onto (Proposition 7.1.8 (2017: 7.1)), hence metrizable by Corollary 7.3.6 (2017: 7.1), hence paracompact. Remark 7.5.2 (2017: 7.1) itself, on the second countability required in [38], has no formal content. The open question the text states here, whether (5) is a consequence of (1)–(4), is settled negatively in §VII.49: the Čech groupoid of a Prüfer-type surface is an ep-groupoid with Hausdorff, non-paracompact orbit space `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_epGroupoid_not_hasParacompactOrbitSpace` (Findings, Chapter 7, items 82, 113, 152) (p. 340 (2017 p. 249)).

**The map $|X^1| \to |X|$.** The text writes $|X_1|$ for the orbit space of $X^1$ and treats it as a subset of $|X|$ ("$A_x \cap |X_1|$", 2017 p. 250). Here $|X^1|$ keeps its own quotient topology: the inclusion $X^1 \to X$ is a functor `Chapter7/Lift · EpGroupoid.liftIncl` and induces $|X^1| \to |X|$ `Chapter7/Lift · EpGroupoid.orbitIncl` `Chapter7/Lift · EpGroupoid.orbitIncl_π`, continuous `Chapter7/Lift · EpGroupoid.continuous_orbitIncl` and injective, because a morphism between points of $X_1$ lies in $\mathbf{X}_1$ ($ml(s\varphi) = ml(t\varphi)$, §VII.3) `Chapter7/Lift · EpGroupoid.orbitIncl_injective`, with range $|X_1| = \pi(X_1)$ `Chapter7/Lift · EpGroupoid.range_orbitIncl`; the preimage of $\pi(U)$ is $\pi_1(U \cap X_1)$ `Chapter7/Lift · EpGroupoid.preimage_orbitIncl_image`.

**Proposition 7.5.3 (2017: 7.12)** (V7.54) `Chapter7/Lift · EpGroupoid.proposition_7_5_3`. *If $X$ has paracompact orbit space, so has $X^1$.* **The text's proof**: by Lemma 2.7.3 (2017: 2.7) the cover $(|U(x)|)$ of $|X|$ has a locally finite closed refinement $(A_x)$, $A_x \subseteq |U(x)|$; $A_{x,1} = A_x \cap |X_1|$ is a locally finite closed cover of $|X_1|$; $G_x$ acts on $U(x)_1$ and, as in Lemma 7.3.2 (2017: 7.4), an invariant metric passes to $|U(x)|_1 = |U(x)_1|$, so $A_{x,1}$ is metrizable, hence paracompact, and $|X_1|$ is a locally finite union of closed paracompact sets (Proposition 2.7.4 (2017: 2.17)). **The formalized proof** proves more, that $|X^1|$ is metrizable `Chapter7/Lift · EpGroupoid.metrizableSpace_liftOne_orbitSpace`, by Smirnov's theorem pulled back along $|X^1| \to |X|$ (Part I): a continuous injection into a paracompact Hausdorff space whose points have open neighbourhoods with metrizable preimages has a metrizable source `Chapter2/Metrizable · metrizableSpace_of_pullback`. *Step 1, the restricted action.* For a natural representation at $x$, $G_x$ acts on $U(x)_1 = U(x) \cap X_1$ by the level-1 lifts of the $\Phi(g)$, homeomorphisms of $U(x)_1$

Chapter7/Lift · EpGroupoid.NaturalRep.liftU Chapter7/Lift · EpGroupoid.NaturalRep.liftΦ Chapter7/Lift · EpGroupoid.NaturalRep.liftMulAction Chapter7/Lift · EpGroupoid.NaturalRep.continuous_liftSmul. "$U(x)_1$ has the natural $G_x$-action" is read as this restricted action only: $U(x)_1$ need not be connected in $X^1$, so it is not a natural representation of $X^1$. *Step 2, the fibres.* Points of $U(x)_1$ have the same image in $|X^1|$ iff they are in one $G_x$-orbit, by Theorem 7.1.13 (3) (2017: 7.1) and §VII.3 Chapter7/Lift · EpGroupoid.NaturalRep.liftOne_π_eq_iff. *Step 3, the metric.* $G_x\backslash U(x)_1$ is metrizable by the text's $\hat{\rho}$, the argument of Lemma 7.3.2 (2017: 7.4) repeated for a finite group acting by homeomorphisms on a metrizable space Chapter7/Lift · lvMaxDist Chapter7/Lift · lvQuotDist Chapter7/Lift · metrizableSpace_orbitRel_quotient_of_finite. *Step 4, Proposition 7.1.19 (2017: 7.6) on level* 1. $G_x\backslash U(x)_1 \to |X^1|$ is an open embedding Chapter7/Lift · EpGroupoid.NaturalRep.liftOrbitQuotientMap Chapter7/Lift · EpGroupoid.NaturalRep.isOpenEmbedding_liftOrbitQuotientMap onto the preimage of $|U(x)|$ Chapter7/Lift · EpGroupoid.NaturalRep.range_liftOrbitQuotientMap, so that preimage is metrizable Chapter7/Lift · EpGroupoid.NaturalRep.metrizableSpace_preimage_orbitIncl. *Step 5.* $|X|$ is paracompact by hypothesis and Hausdorff by Lemma 7.3.4 (2017: 7.6), and every point has the open neighbourhood $|U(x)|$; the pulled-back Smirnov theorem gives $|X^1|$ metrizable, hence paracompact (Stone). The regularity of $|X^1|$ and Michael's theorem are not needed on this route, though both are available (§§VII.19, VII.20) (Findings, Chapter 7, item 83) (pp. 340–342 (2017 pp. 249–250)).

## VII.24 Theorem 7.5.4 (2017: 7.3), Definition 7.5.5 (2017: 7.16), Remark 7.5.6 (2017: 7.2), and the statement of Theorem 7.5.7 (2017: 7.4)

**Theorem 7.5.4 (2017: 7.3)** (V7.55) Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_4. *If $|X|$ is paracompact, every open cover $(O_\lambda)$ of $|X|$ has a subordinate partition of unity $(\sigma_\lambda)$, and every cover $(U_\lambda)$ of $X$ by saturated open sets has a subordinate continuous partition of unity $(\beta_\lambda)$ taking equal values on isomorphic objects. Step 1.* The first statement is Mathlib's existence of subordinate partitions of unity on a paracompact normal space Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate. *Hidden hypothesis:* normality, which needs $|X|$ Hausdorff — Lemma 7.3.4 (2017: 7.6) — besides paracompact. *Step 2.* For the second, $O_\lambda = \pi(U_\lambda)$ is open (Proposition 7.1.8 (2017: 7.1)), and $\beta_\lambda = \sigma_\lambda \circ \pi$ Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_apply has $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_src. *Step 3, supports.* $\operatorname{supp}(\sigma_\lambda \circ \pi) \subseteq \pi^{-1}(\operatorname{supp}\sigma_\lambda) \subseteq \pi^{-1}(\pi(U_\lambda)) = U_\lambda$, since $U_\lambda$ is saturated Chapter7/PartitionsOfUnity · EpGroupoid.tsupport_orbitPullback_subset Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_isSubordinate. *Step 4, local finiteness.* If an open $V \ni \pi(x)$ meets finitely many supports of the $\sigma_\lambda$, the saturated open $\pi^{-1}(V)$ meets finitely many supports of the $\beta_\lambda$ Chapter7/PartitionsOfUnity · EpGroupoid.exists_isSaturated_nhds_finite_orbitPullback. "Subordinate" is Mathlib's, $\operatorname{supp}\sigma_\lambda \subseteq O_\lambda$; "some of the $\sigma_\lambda \equiv 0$" says only that the index set is $\Lambda$ (Findings, Chapter 7, item 90) (2017 p. 250).

**Definition 7.5.5 (2017: 7.16)** (V7.56). $X$ has sc-smooth partitions of unity if every cover of $X$ by saturated open sets has a subordinate sc-smooth partition of unity $(\beta_\lambda)$ with (1) every point having a saturated open neighbourhood meeting finitely many supports and (2) $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ Chapter7/EpGroupoid · EpGroupoid.HasScPartitionsOfUnity, the partitions being Part I's Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothPartitionOfUnity with supports the closed supports and index types in the universe of $X$. The hypothesis is satisfiable: for the trivial ep-groupoid it reduces to Part I's sc-smooth partitions of unity Chapter7/EpGroupoid · EpGroupoid.hasScPartitionsOfUnity_trivial, which $\mathbb{R}^n$ has Chapter7/LocalDiffeo · rnPolyfold_admitsScPartitionsOfUnity (Findings, Chapter 7, item 91) (pp. 341–343 (2017 pp. 250–251)).

**Remark 7.5.6 (2017: 7.2)** (V7.57). Apart from sc-Hilbert spaces, the existence of sc-smooth partitions of unity is "a zoo"; the remark is an assessment, with nothing to formalize, and the sc-Hilbert case is Corollary 7.6.6, now as printed (§§VII.29, VII.48) (2017 p. 251).

**Theorem 7.5.7 (2017: 7.4)** (V7.58) `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_7`. *If $|X|$ is paracompact and the object M-polyfold admits sc-smooth partitions of unity, $X$ has sc-smooth partitions of unity.* The proof is the text's Theorem 7.6.4 (2017: 7.6) (§§VII.27–VII.28). The hypotheses are satisfiable, by the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_7_hypotheses_rn` `Chapter7/PartitionsOfUnity · EpGroupoid.hasScPartitionsOfUnity_trivial_rn_of_theorem_7_5_7` (Findings, Chapter 7, items 94, 105) (2017 p. 251).

## VII.25 Definition 7.5.8 (2017: 7.17), Propositions 7.5.9 and 7.5.10 (2017: 7.13 and 7.14)

**Definition 7.5.8 (2017: 7.17)** (V7.59). $Q \subseteq X$ is *proper* if $t : s^{-1}(Q) \to X$ is proper `Chapter7/EpGroupoid · EpGroupoid.IsProperSubset`, and an open $U$ has the *properness property* if $\mathrm{cl}_X U$ is proper `Chapter7/EpGroupoid · EpGroupoid.HasPropernessProperty`; every $x$ has such a $U$, by Definition 7.1.3 (3) (2017: 7.3) `Chapter7/EpGroupoid · EpGroupoid.exists_hasPropernessProperty`. In metrizable spaces $Q$ is proper iff $s^{-1}(Q) \cap t^{-1}(K)$ is compact for every compact $K$ `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iff_isCompact`, and for closed $Q$ iff the sequential criterion holds `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iff_exists_tendsto_subseq`. Basic facts: a closed subset of a proper set is proper `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.of_subset`; $\emptyset$, finite unions and compact sets are proper `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_empty` `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.union` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_biUnion` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iUnion` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_of_isCompact`; an open set inside one with the properness property has it `Chapter7/ProperSubsets · EpGroupoid.HasPropernessProperty.isProperSubset` `Chapter7/ProperSubsets · EpGroupoid.HasPropernessProperty.mono`, and every neighbourhood of $x$ contains an open, also an open connected, neighbourhood of $x$ with the properness property `Chapter7/ProperSubsets · EpGroupoid.exists_hasPropernessProperty_subset` `Chapter7/ProperSubsets · EpGroupoid.exists_hasPropernessProperty_isConnected_subset` (Findings, Chapter 7, item 92) (2017 p. 251).

**Proposition 7.5.9 (2017: 7.13)** (V7.60) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_9`. *A proper subset is closed.* The text: for $x_n \to x$ in $Q$, the units $1_{x_n}$ lie in the compact $(t|s^{-1}(Q))^{-1}(\{x_n\} \cup \{x\})$, a subsequence converges to some $\varphi$ with $s(\varphi) \in Q$, and $s(\varphi) = x$. The formalization is this argument with the compact preimage (§VII.2) `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.isClosed` (2017 p. 251).

**Proposition 7.5.10 (2017: 7.14)** (V7.61). *(1) If $Q$ is proper, $\pi(Q)$ is closed. (2) If $|X|$ is paracompact, $A \subseteq |X|$ is closed iff $A = \pi(Q)$ for a proper $Q$.* **(1)** `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_10_one`. The text uses sequences $\varphi_k : x_k \to y_k$ with $x_k \in Q$, $y_k \to y$. The formalization uses the closed-map property: a set of $|X|$ is closed iff its preimage is `Chapter7/ProperSubsets · EpGroupoid.isClosed_iff_preimage_π`, and $\pi^{-1}(\pi(Q)) = t(s^{-1}(Q))$ is the image of the closed $s^{-1}(Q)$ (Proposition 7.5.9 (2017: 7.13)) under the proper, hence closed, $t$ `Chapter7/ProperSubsets · EpGroupoid.isClosed_saturation_of_isProperSubset`. The proof's "$x = s(\varphi) \in Q$" reuses the name $x$ (possible misprint).

**(2)** `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_10_two`. One direction is (1). **The text's proof** of the other chooses for $z \in A$ a representative $x_z$ and a neighbourhood $U(x_z)$ with the natural action whose closure $Q_z$ is proper; by Lemma 2.7.3 (2017: 2.7) the closed, hence paracompact, $A$ has a locally finite

cover by closed $A_i \subseteq \pi(Q_{z_i})$; $P_i = (\pi|\mathrm{cl}_X U(x_{z_i}))^{-1}(A_i)$ and $Q = \bigcup P_i$ has $\pi(Q) = A$; it is proper because only finitely many $A_i$ meet a neighbourhood of $\pi(y)$, so the sources lie in finitely many $P_i$. Lemma 2.7.3 (2017: 2.7) assumes a regular Hausdorff space. **The formalized proof** needs only the paracompactness of $|X|$. *Step 1.* Representatives $x_z$ are chosen by the quotient, and $V(x_z)$ is any open neighbourhood with the properness property; the natural action is not used. *Step 2, an open refinement.* The open sets $\pi(V(x_z))$ cover the closed $A$; Mathlib gives an open locally finite precise refinement $(v_z)$ of this cover of $A$, from paracompactness alone `Mathlib/Topology/Compactness/Paracompact · precise_refinement_set`. *Step 3, closures.* $P_z = \mathrm{cl}_X V(x_z) \cap \pi^{-1}(\mathrm{cl}(v_z) \cap A)$ is a closed subset of the proper $\mathrm{cl}_X V(x_z)$, hence proper, with $\pi(P_z) \subseteq \mathrm{cl}(v_z) \cap A$; and $\pi(P_z) \supseteq v_z \cap A$, since $v_z \subseteq \pi(V(x_z))$. As the $v_z$ cover $A$, $\pi(\bigcup P_z) = A$. *Step 4, properness of $Q = \bigcup P_z$.* The closures $\mathrm{cl}(v_z)$ of a locally finite family are locally finite, and a union of proper sets whose images form a locally finite family is proper: a compact $K$ has compact $\pi(K)$, which meets only finitely many $\mathrm{cl}(v_z)$ `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iUnion_of_locallyFinite`. Neither Lemma 2.7.3 (2017: 2.7) nor the regularity of $|X|$ is used, and no version under a regularity hypothesis is needed; the closed refinement of Michael's theorem would give the text's route as well (§VII.20). The text writes $\mathrm{cl}_X(U_{x_{z_i}})$ for $\mathrm{cl}_X(U(x_{z_i}))$ (possible misprint; Findings, Chapter 7, item 92) (pp. 342–344 (2017 pp. 251–252)).

## VII.26 Appendix §7.6.1 (2017 §7.5.1): the proof of Theorem 7.1.13 (2017: 7.1) (Theorem 7.6.1 (2017: 7.5), Lemmas 7.6.2 and 7.6.3 (2017: 7.8 and 7.9))

**Theorem 7.6.1 (2017: 7.5)** (V7.62) `Chapter7/NaturalRep · EpGroupoid.theorem_7_6_1` is Theorem 7.1.13 (2017: 7.1) with $\Phi(g) = t_g \circ s_g^{-1}$; it is stated as the same theorem. **The text's proof** (pp. 344–346 (2017 pp. 253–254)): for every $g \in G_{x_0}$ choose contractible open $N_g^t$, $N_g^s$ on which $t$ and $s$ are sc-diffeomorphisms onto an open $U_0 \ni x_0$, disjoint for distinct $g$ since $G_{x_0}$ is finite; $N_g = N_g^t \cap N_g^s$, $s_g = s|N_g$, $t_g = t|N_g$; Lemmas 7.6.2 and 7.6.3 (2017: 7.8 and 7.9) give $U$; $\Phi(g) = t_g \circ s_g^{-1}$, a homomorphism "since the neighborhoods $N_g$ are disjoint and since the structure maps are continuous"; $\Gamma(g, y) = s_g^{-1}(y)$. **The formalized proof.** *Step 1, the sheets* `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.exists_sheets`. $G_{x_0}$ is finite (Proposition 7.1.12 (2017: 7.4)) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.finite_isotropy`; Hausdorff separation of the finite set $G_{x_0} \subseteq \mathbf{X}$ gives pairwise disjoint open $N_g \ni g$, shrunk so that $s$ is an sc-diffeomorphism from $N_g$ onto its open image with inverse $\sigma_g = s_g^{-1}$; $t$ is one too, which is recorded but not used. Only connectedness will be needed, not contractibility. *Step 2, $U \subseteq V$.* The proof does not mention $V$; it starts at $U_0 = V \cap \bigcap_g s(N_g)$. *Step 3, Lemma 7.6.2 (2017: 7.8), then Lemma 7.6.3 (2017: 7.9)* (below). *Step 4, the homomorphism* `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isScSmoothOn_hat`. On the connected $U$, $y \mapsto \sigma_{gh}(y)$ and $y \mapsto \sigma_g(\Phi(h)y) \circ \sigma_h(y)$ are continuous sections of $s$ — the second by (4) in the sheet form — agreeing at $x_0$, so they agree on $U$ (§VII.7) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eqOn_σ_mul` and $\Phi(gh) = \Phi(g)\Phi(h)$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat_mul`. In the same way $\sigma_1$ is the unit section $u$ on $U$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eqOn_σ_one` and $\Phi(1) = \mathrm{id}$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat_one`, which the text does not mention and which makes $\Phi(g)$ an sc-diffeomorphism with inverse $\Phi(g^{-1})$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isScDiffeoOn_hat`. As we read it, "since the $N_g$ are disjoint and the structure maps are continuous" shows only that the sheet index is locally constant; connectedness of $U$ makes

it constant. *Step 5, (3).* A morphism $h$ between points of $U$ lies in some $N_{g'}$ by Lemma 7.6.2 (2017: 7.8), and $s$ is injective there, so $h = \sigma_{g'}(sh)$; disjointness of the $N_g$ gives uniqueness `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eq_of_mem_N`. *Step 6.* The natural representation is built on every open connected invariant $U \subseteq \bigcap_g s(N_g)$ with the conclusion of Lemma 7.6.2 (2017: 7.8) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.mkRep`, and Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9) provide one `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.exists_naturalRep_sheets`. "$\Phi : G_x \to \mathrm{Diff}_{sc}(U)$" is read with $G_{x_0}$ (possible misprint; Findings, Chapter 7, item 29).

**Lemma 7.6.2 (2017: 7.8)** (V7.63) `Chapter7/NaturalRep · EpGroupoid.lemma_7_6_2`. *There is an open $U_1 \subseteq U_0$ of $x_0$ such that every morphism with source and target in $U_1$ lies in some $N_g$* — for any open $N_g \ni g$ and any neighbourhood $U_0$; the sheet property is not used. **The text's proof**: otherwise there are $h_k \notin \bigcup N_g$ with $s(h_k), t(h_k) \to x_0$; by properness $h_{k_l} \to h$, necessarily $h \in G_{x_0}$, so $h \in N_g$ for some $g$, a contradiction. **The formalized proof** follows it. *Step 1.* The sequence is taken along a countable decreasing basis of $x_0$, $X$ being metrizable. *Step 2.* "By the properness assumption" is Definition 7.1.3 (3) (2017: 7.3) at $x_0$, which applies to the $h_k$ with $s(h_k) \in \mathrm{cl}V(x_0)$, i.e. to all large $k$ (§VII.2). *Step 3.* "Necessarily $h \in G_{x_0}$": $t(h) = x_0$ comes with the properness, $s(h) = x_0$ from the continuity of $s$ and the uniqueness of limits. *Step 4.* $N_g$ is an open neighbourhood of the limit, so eventually $h_{k_l} \in N_g$. "The properness assumption of ep-polyfolds" is read as that of ep-groupoids (possible misprint; Findings, Chapter 7, item 30) (pp. 344–346 (2017 pp. 253–254)).

**Lemma 7.6.3 (2017: 7.9)** (V7.64) `Chapter7/NaturalRep · EpGroupoid.lemma_7_6_3`. *There is an open $U_2 \subseteq U_1$ of $x_0$ such that $U = \bigcup_g t_g s_g^{-1}(U_2)$ lies in $U_1$ and is invariant under all $t_g s_g^{-1}$.* **The text's proof**: choose $U_2$ so small that $U$ and $t_g s_g^{-1}(U)$ lie in $U_1$; for $x = t_h s_h^{-1}(u)$, the point $v = t_g s_g^{-1} t_h s_h^{-1}(u) \in U_1$ is the target of a morphism $u \to v$, which by Lemma 7.6.2 (2017: 7.8) is $s_{g'}^{-1}(u)$, so $v = t_{g'} s_{g'}^{-1}(u) \in U$. **The formalized proof.** *Step 1.* Each $t_g s_g^{-1}$ is continuous at its fixed point $x_0$ and $G_{x_0}$ is finite, so a $U_2$ with $t_h s_h^{-1}(U_2) \subseteq U_1$ and $t_g s_g^{-1} t_h s_h^{-1}(U_2) \subseteq U_1$ for all $g, h$ exists; it is taken *connected* (§VII.7). *Step 2.* $U$ is open, $t$ being open and $s_g^{-1}(U_2) = N_g \cap s^{-1}(U_2)$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.unionHat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isOpen_image_hat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isOpen_unionHat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.mem_unionHat_iff`, and connected, a union of connected sets through $x_0 = t_g s_g^{-1}(x_0)$, so no passage to a component is needed `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isConnected_unionHat`. *Step 3, invariance.* The morphism $u \to v$ is $\sigma_g(x) \circ \sigma_h(u)$, composable because $x \in U_1 \subseteq s(N_g)$; it lies in some $N_{g'}$ by Lemma 7.6.2 (2017: 7.8), and "it has necessarily the form $v = t_{g'} s_{g'}^{-1}(u)$" uses that $s$ is injective on $N_{g'}$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.image_σ` (Findings, Chapter 7, item 31) (2017 p. 254).

## VII.27 Appendix §7.6.2 (2017 §7.5.2): Lemma 7.6.5 (2017: 7.10)

**Theorem 7.6.4 (2017: 7.6)** (V7.65) `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_6_4` is Theorem 7.5.7 (2017: 7.4), stated again (Findings, Chapter 7, item 95) (2017 p. 255).

**Lemma 7.6.5 (2017: 7.10)** (V7.66) `Chapter7/PartitionsOfUnity · EpGroupoid.lemma_7_6_5`. *Let $|X|$ be paracompact and $X$ admit sc-smooth partitions of unity; let $U(x) \subseteq \mathrm{cl}_X U(x) \subseteq V(x)$ be open, $V(x)$ with the natural $G_x$-action and $U(x)$ invariant. A continuous $\beta : X \to [0,1]$ supported in $U(x)$ and invariant on $V(x)$ lies below an sc-smooth $f \geq 0$ supported in $V(x)$ and invariant on $V(x)$.* The formalization adds the conclusion $\mathrm{supp} f \subseteq \mathrm{cl}_X U(x)$, which the proof gives and Theorem 7.5.7 (2017: 7.4) needs, and omits the hypothesis $x \in U(x)$, which the proof does not use and Theorem 7.5.7 (2017: 7.4) cannot supply; the paracompactness of $|X|$

and the continuity of $\beta$ are kept, unused. **The text's proof**: open $W(z)$, $z \in V(x)$, with numbers $a_z$ — $\mathrm{cl}W(z) \subseteq U(x)$ and $\beta < a_z$ on $W(z)$ for $z \in U(x)$, otherwise $W(z) \subseteq V(x)$, $\beta|W(z) = 0$, $a_z = 0$; an sc-smooth partition of unity $(\sigma_z)$ of $V(x)$ subordinate to them; $\gamma = \sum \sigma_z a_z \geq \beta$, supported in $\mathrm{cl}U(x)$; and $f(y) = |G_x|^{-1} \sum_g \gamma(g * y)$. **The formalized proof.** *Step 1, a cover of $X$.* The hypothesis is that $X$, not the open subset $V(x)$, admits sc-smooth partitions of unity, so the cover is completed by $X \setminus \mathrm{cl}U(x)$, where $\beta = 0$ and $a = 0$; and it is coarsened to two sets, $U(x)$ with $a = 1$ (as $\beta \leq 1$) and $X \setminus \mathrm{supp}\,\beta$ with $a = 0$. The text's $\mathrm{cl}W(z) \subseteq U(x)$ and $\beta < a_z$ are not needed. *Step 2, $\gamma$.* $\gamma$ is the function attached to $U(x)$ of a partition of unity subordinate to $\{U(x), X \setminus \mathrm{supp}\,\beta\}$: sc-smooth, $\geq 0$, supported in $U(x)$, and $\gamma = 1 \geq \beta$ on $\mathrm{supp}\,\beta$, where the other function vanishes. *Step 3, the average.* $f = |G_x|^{-1} \sum_g \gamma \circ \Phi(g)$ on $V(x)$ and 0 elsewhere. Each $\gamma \circ \Phi(g)$, extended by 0, is supported in $\Phi(g)^{-1}(\mathrm{cl}U(x)) = \mathrm{cl}U(x) \subseteq V(x)$ by the invariance of $U(x)$, hence sc-smooth on $X$ `Chapter7/PartitionsOfUnity · isScSmoothMap_real_of_tsupport_subset` `Chapter7/PartitionsOfUnity · isScSmoothMap_real_of_isScSmoothOn_nhds`. *Step 4.* $\beta \leq f$ by the invariance of $\beta$: $\beta(y) = \beta(g*y) \leq \gamma(g*y)$ for every $g$. $f$ is invariant by the reindexing $g \mapsto gh$, and $\mathrm{supp} f \subseteq \mathrm{cl}U(x)$ (Findings, Chapter 7, item 93) (2017 p. 255).

## VII.28 Appendix §7.6.2 (2017 §7.5.2): the proof of Theorem 7.5.7 (2017: 7.4)

**The text's proof** (pp. 346–347 (2017 pp. 255–256)). For every $x$ choose an open $V(x)$ with (1) an open $W(x)$ with the natural $G_x$-action, $V(x) \subseteq \mathrm{cl}V(x) \subseteq W(x)$ and $V(x)$ invariant; (2) $t : s^{-1}(\mathrm{cl}V(x)) \to X$ proper; (3) $V(x) \subseteq U_\lambda$ for some $\lambda$; (4) $(\pi^{-1}(|V(x)|) \setminus V(x)) \cap W(x) = \emptyset$. Take a locally finite refinement $(V_i')$ of $(|V(x)|)$ with $V_i' \subseteq |V(x_i)|$, open $V_i'' \subseteq \mathrm{cl}V_i'' \subseteq V_i'$ covering $|X|$, $U_i'' = V(x_i) \cap \pi^{-1}(V_i'')$ and a partition of unity $(\beta_i)$ subordinate to $(V_i'')$. "Approximate $(\beta_i \circ \pi)|V(x_i)$ using Lemma 7.10" gives $f_i$ supported in $V(x_i)$; averaging, extending to the saturation and by 0 gives $h_i$; $h = \sum h_i > 0$ is locally finite and $(h_i/h)$ is the partition of unity.

**The formalized proof** (V7.67) builds the same functions $h_i$, but for a Urysohn lemma, and normalizes with Mathlib's bump coverings, as Part I's Theorem 5.5.10 (2017: 5.12) does (§V.25) `Chapter5/PartitionsOfUnity · admitsScPartitionsOfUnity_of_approx`. The four points where the text needed more than a reading are marked (a)–(d).

*Step 1, properties (1)–(3)* `Chapter7/PartitionsOfUnity · EpGroupoid.exists_invariant_nhds_of_isOpen`. Given $x$ and an open $O \ni x$, take $V_0 \subseteq O$ with the properness property (§VII.25), $W(x) = U$ of a natural representation inside $V_0$ (Theorem 7.1.13 (2017: 7.1)), an open $N \ni x$ with $\mathrm{cl}N \subseteq W(x)$, $X$ being regular, and $V(x) = \bigcap_g \Phi(g)^{-1}(N)$: open, invariant, $x \in V(x)$, $\mathrm{cl}V(x) \subseteq \mathrm{cl}N \subseteq W(x)$, and proper over its closure by §VII.25. Elements of one orbit have one image in $|X|$ `Chapter7/PartitionsOfUnity · EpGroupoid.π_Φ_eq`.

*Step 2, (4) follows from (1)* and is not a further choice. The morphisms between points of $W(x)$ are the $\Gamma(g, \cdot)$ (Theorem 7.1.13 (3) (2017: 7.1)), so a point of $W(x)$ isomorphic to a point of the invariant $V(x)$ is $g * v$ for some $v \in V(x)$, hence lies in $V(x)$ `Chapter7/PartitionsOfUnity · EpGroupoid.mem_of_π_mem_image_of_invariant` `Chapter7/PartitionsOfUnity · EpGroupoid.saturation_diff_inter_eq_empty`.

*Step 3, extension to the saturation.* A $G_x$-invariant $f$ on $W(x)$ defines $F$ on $|X|$ by $F(|y|) = f(y)$ for $y \in W(x)$ and 0 elsewhere, well defined because points of $W(x)$ in one orbit differ by the natural action (§VII.11) `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension_π` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension_nonneg` `Chapter7/PartitionsOfUnity · EpGroupoid.support_orbitExtension_subset`. (d) *$F \circ \pi$ is sc-smooth* when $f$ is

sc-smooth with $\operatorname{supp} f \subseteq \operatorname{cl} V(x)$ `Chapter7/PartitionsOfUnity · EpGroupoid.isScSmoothMap_orbitExtension`: $F \circ \pi$ vanishes off the saturation of $\operatorname{supp} f$, which is closed because $\operatorname{supp} f \subseteq \operatorname{cl} V(x)$ is proper by (2) `Chapter7/ProperSubsets · EpGroupoid.isClosed_saturation_of_isProperSubset`; near a point $y \cong z$ of that saturation, with $\varphi : z \to y$, $F \circ \pi = f \circ s \circ (t|\mathbf{U}(\varphi))^{-1}$ (Definition 7.1.4 (2017: 7.4)). The text does not mention the closedness.

*Step 4, the invariant Urysohn lemma* `Chapter7/PartitionsOfUnity · EpGroupoid.exists_isScSmoothMap_orbit_urysohn`: *for disjoint closed $s', t' \subseteq |X|$ there is a continuous $F : |X| \to [0,1]$ with $F \circ \pi$ sc-smooth, $F = 0$ on $s'$ and $F = 1$ on $t'$.* For every $x \in \pi^{-1}(t')$ Step 1 gives $V(x) \subseteq \operatorname{cl} V(x) \subseteq W(x) \subseteq \pi^{-1}(|X| \setminus s')$. The open sets $|V(x)|$ cover the closed $t'$; take an open locally finite precise refinement $(V'_x)$ of this cover of $t'$ `Mathlib/Topology/Compactness/Paracompact · precise_refinement_set` and a bump covering $(b_x)$ of $t'$ subordinate to it `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate`, whose $\operatorname{supp} b_x \subseteq V'_x$ takes the place of the text's $V''_x$; put $U''_x = V(x) \cap \pi^{-1}(V'_x)$. (b) *The cut-off.* $c_x = b_x \circ \pi$ on $V(x)$, $0$ elsewhere, is continuous and supported in $U''_x$: a point of $\operatorname{cl} V(x) \cap \pi^{-1}(\operatorname{supp} b_x)$ lies in $W(x)$ and is isomorphic to a point of $V(x)$, as $\operatorname{supp} b_x \subseteq |V(x)|$, hence lies in $V(x)$ by (4). (a) *Lemma 7.6.5 (2017: 7.10)* is applied with $W(x)$ in the role of $V(x)$ and $U''_x$ in the role of $U(x)$: its hypothesis $\operatorname{cl} U''_x \subseteq W(x)$, with the natural action on $W(x)$, holds by (1), and $U''_x$ is invariant. The text's "approximate $(\beta_i \circ \pi)|V(x_i)$ using Lemma 7.10" reads as $V(x_i)$ in the role of $V(x)$ (possible misprint). It gives an invariant sc-smooth $f_x \geq c_x$ with $\operatorname{supp} f_x \subseteq \operatorname{cl} U''_x$, and $F_x$ its extension (Step 3), using $\operatorname{cl} U''_x \subseteq \operatorname{cl} V(x)$. (c) *Local finiteness.* $\operatorname{supp}(F_x \circ \pi) \subseteq \pi^{-1}(\operatorname{cl} V'_x)$ by the support bound of Lemma 7.6.5 (2017: 7.10), and $(\operatorname{cl} V'_x)$ is locally finite, so $G = \sum_x F_x$ is a locally finite sum. The text does not state this bound: Lemma 7.6.5 (2017: 7.10) as stated gives only $\operatorname{supp} f_x \subseteq W(x)$, and the saturations of the $W(x)$ need not be locally finite. *The function.* $G \circ \pi$ is sc-smooth, $G = 0$ on $s'$ since every $W(x)$ misses $\pi^{-1}(s')$, and $G \geq 1$ on $t'$, since at every point of $t'$ some $b_x$ equals 1, and then $f_x \geq c_x = 1$ over it; $F = \tau \circ G$ with Mathlib's smooth transition function $\tau$ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`, $0$ on $(-\infty, 0]$ and $1$ on $[1, \infty)$.

*Step 5, the normalization* `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_7`. Given a saturated open cover $(U_\lambda)$, the $\pi(U_\lambda)$ cover $|X|$ `Chapter7/PartitionsOfUnity · EpGroupoid.iUnion_image_π_eq_univ`. Mathlib's existence of bump coverings subordinate to an open cover with all functions satisfying a predicate for which Urysohn's lemma holds `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate_of_prop`, for the class of the $g$ with $g \circ \pi$ sc-smooth and Step 4 as its input, gives $(b_\lambda)$ on the paracompact normal $|X|$, subordinate to $(\pi(U_\lambda))$, each $b_\lambda \circ \pi$ sc-smooth. The partition $\rho_\lambda = b_\lambda \prod_{\mu<\lambda}(1 - b_\mu)$ `Mathlib/Topology/PartitionOfUnity · BumpCovering.toPartitionOfUnity` is locally a finite product of such functions, so $\rho_\lambda \circ \pi$ is sc-smooth, and $\beta_\lambda = \rho_\lambda \circ \pi$, Theorem 7.5.4's (2017: 7.3) pull-back (§VII.24), has the properties (1), (2) of Definition 7.5.5 (2017: 7.16). This replaces the text's $(h_i/h)$ and the passage from the index set of the refinement back to $\Lambda$, which the text leaves implicit. "We average using the $G_{x_i}$-action": $f_x$ is already invariant by Lemma 7.6.5 (2017: 7.10), and no second average is taken. "$t : s^{-1}(\mathrm{cl}_X(V(x)) \to X$" has an unbalanced parenthesis (possible misprint; Findings, Chapter 7, items 94, 96) (pp. 346–347 (2017 pp. 255–256)).

## VII.29 Corollary 7.6.6 (2017: 7.3)

**Corollary 7.6.6 (2017: 7.3)** (V7.68) `Chapter7/PartitionsOfUnity · EpGroupoid.corollary_7_6_6`. *If $|X|$ is paracompact and the object M-polyfold has an atlas whose local models have Hilbert spaces $E_0$, every saturated open cover of $X$ has a subordinate sc-smooth partition of unity of $X$.* The text cites Part I's Proposition 5.5.8 (2017: 5.6), Theorem 5.5.10 (2017: 5.12) and Corollary 5.5.17 (2017: 5.2), the ingredients of Part

I's theorem that such an M-polyfold admits sc-smooth partitions of unity (§§V.25–V.26) Chapter5/PartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert, and Theorem 7.5.7 (2017: 7.4) applies. "An atlas consisting of charts whose local models have Hilbert spaces" is existential, one chart per point; "every compatible chart has a Hilbert $E_0$" would be unsatisfiable. In the first reading, as in Part I, the theorem of Godefroy–Troyanski–Whitfield–Zizler (Part I's Theorem 5.5.16 (2017: 5.14)) was an explicit hypothesis Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement (gap G.53), which holds for finite-dimensional spaces Chapter7/LocalDiffeo · godefroyTroyanskiWhitfieldZizlerStatement_of_finiteDimensional Chapter7/LocalDiffeo · admitsContDiffPartitionsOfUnity_of_finiteDimensional. The hypotheses are satisfiable on $\mathbb{R}^1$ Chapter7/LocalDiffeo · rnPolyfold_one_hilbert_hypothesis Chapter7/LocalDiffeo · rnPolyfold_one_admitsScPartitionsOfUnity_of_hilbert Chapter7/PartitionsOfUnity · EpGroupoid.corollary_7_6_6_hypotheses_rn. *The corollary as printed.* Part I now proves Theorem 5.5.16 Chapter5/GapP1_GTWZ · theorem_5_5_16; the case used here is that of a separable space Chapter5/GapP1_SeparablePOU · godefroyTroyanskiWhitfieldZizlerStatement_of_separable, $E_0$ being separable for every sc-Banach space, so an M-polyfold with Hilbert $E_0$ admits sc-smooth partitions of unity with no hypothesis Chapter5/GapP1_SeparablePOU · MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional, and Corollary 7.6.6 holds as printed Chapter7/GapP2_ACor766 · EpGroupoid.corollary_7_6_6_unconditional, with the same witness on $\mathbb{R}^1$ Chapter7/GapP2_ACor766 · EpGroupoid.corollary_7_6_6_hypotheses_rn_unconditional (gap G.53, closed; §VII.48) (Findings, Chapter 7, items 95, 106) (p. 347 (2017 p. 256)).

## VII.30 Witnesses and counterexamples

Every hypothesis of the chapter that is a predicate on ep-groupoids or on their data is shown satisfiable, and for the readings that do not hold a counterexample is given (Findings, Chapter 7, items 100–107, 152, 156).

**The trivial ep-groupoid** of an M-polyfold $Z$, with only units, exists on every M-polyfold Chapter7/EpGroupoid · EpGroupoid.trivial. It is tame for tame $Z$ and for $\mathbb{R}^n$ Chapter7/EpGroupoid · EpGroupoid.isTame_trivial Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn, effective with $X_{reg} = Z$ Chapter7/EpGroupoid · EpGroupoid.isEffective_trivial Chapter7/EpGroupoid · EpGroupoid.regularSet_trivial, every closed set is proper and every open set has the properness property Chapter7/EpGroupoid · EpGroupoid.isProperSubset_trivial Chapter7/EpGroupoid · EpGroupoid.hasPropernessProperty_trivial, $|X| \cong Z$ Chapter7/EpGroupoid · EpGroupoid.trivialOrbitHomeomorph is paracompact Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial, and it has natural representations on every connected open neighbourhood Chapter7/EpGroupoid · EpGroupoid.trivialNaturalRep Chapter7/EpGroupoid · EpGroupoid.trivialNaturalRepFinite. Its witnesses are vacuous for the isotropy, which the next ones are not.

**The translation groupoid** $K \ltimes U$ of a finite discrete group acting on an M-polyfold by sc-smooth maps, with morphisms the product with the discrete M-polyfold Chapter7/NonVacuity · Examples.actionMorPolyfold, is an ep-groupoid Chapter7/NonVacuity · Examples.actionEpGroupoid: $s$ is a projection, $t$ a projection after the shear Chapter7/NonVacuity · Examples.actionShear Chapter7/NonVacuity · Examples.action_isLocalScDiffeo_src Chapter7/NonVacuity · Examples.action_isLocalScDiffeo_tgt, (4) holds in the sheet form Chapter7/NonVacuity · Examples.action_isScSmoothCompSheet and $t$ is proper on all of $\mathbf{X}$ Chapter7/NonVacuity · Examples.action_isProperMap_tgt. Its isotropy is the stabilizer Chapter7/NonVacuity · Examples.isotropyEquivStabilizer Chapter7/NonVacuity · Examples.isRegular_action_iff Chapter7/NonVacuity · Examples.regularSet_action, its orbit space $U/K$ Chapter7/NonVacuity · Examples.action_π_eq_π_iff Chapter7/NonVacuity · Examples.actionOrbitHomeomorph, its proper sets the closed sets Chapter7/NonVacuity · Examples.isProperSubset_action Chapter7/NonVacuity · Examples.isProperSubset_action_iff Chapter7/NonVacuity · Examples.hasPropernessProperty_action, and its orbit space is paracompact for

second countable $U$ Chapter7/NonVacuity · Examples.hasParacompactOrbitSpace_action.

$\mathbb{Z}/2 \ltimes \mathbb{R}$ **by reflection** Chapter7/NonVacuity · Examples.isScSmoothMap_units_smul Chapter7/NonVacuity · Examples.reflectionEpGroupoid is effective Chapter7/NonVacuity · Examples.isEffective_reflection with a non-trivial isotropy element $(-1, 0)$ Chapter7/NonVacuity · Examples.reflectionIsotropyNegOne, so Definition 7.2.1 (2017: 7.11) is met non-vacuously Chapter7/NonVacuity · Examples.isEffective_reflection_nonvacuous; $X_{reg} = \mathbb{R} \setminus \{0\}$ Chapter7/NonVacuity · Examples.regularSet_reflection, $G_0 \cong \mathbb{Z}/2$ Chapter7/NonVacuity · Examples.reflectionIsotropyZeroEquiv, $T_x = 1$ Chapter7/NonVacuity · Examples.trivialSubgroup_reflection; it is tame and has paracompact orbit space Chapter7/NonVacuity · Examples.isTame_reflection Chapter7/NonVacuity · Examples.hasParacompactOrbitSpace_reflection.

**The trivial action of** $\mathbb{Z}/3$ **on** $\mathbb{R}$ Chapter7/NonVacuity · Examples.trivialZ3EpGroupoid has $G_x \cong \mathbb{Z}/3$ Chapter7/NonVacuity · Examples.trivialZ3IsotropyEquiv, $T_x = G_x$ Chapter7/NonVacuity · Examples.trivialSubgroup_trivialZ3 Chapter7/NonVacuity · Examples.trivialZ3TrivialSubgroupEquiv, is not effective Chapter7/NonVacuity · Examples.not_isEffective_trivialZ3, and its regular set is empty Chapter7/NonVacuity · Examples.regularSet_trivialZ3: the text's "examples show that the regular set can be empty" (p. 308 (2017 p. 239)). It carries the counterexample to the remark after Definition 7.1.7 (2017: 7.7) (§VII.4).

**The pair groupoid of** $\mathbb{Z}$ Chapter7/NonVacuity · Examples.pairGroupoidStr Chapter7/NonVacuity · Examples.pairEpGroupoid is an ep-groupoid in which properness fails over all of $\mathbb{Z}$ Chapter7/NonVacuity · Examples.pair_exists_not_isProperMap, refuting the universal reading of Definition 7.1.3 (3) (2017: 7.3) (§VII.2). The abstract finiteness statements of §VII.6 are met by the trivial action of a finite discrete group, whose isotropy set has $|G|$ elements Chapter7/Discrete · isotropy_snd_witness.

$\mathbb{Z}/2$ **acting on a dimension-jumping retract** Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.X, by $(t, z) \mapsto (t, \pm z)$ on the retract of Part I's Remark 4.2.2 (2017: 4.3) example, is a non-effective ep-groupoid whose reduced morphism space $\mathbf{X}^R$ is not Hausdorff Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor: the hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) cannot be dropped (§VII.16; Findings, Chapter 7, item 107). Its non-effective part is computed exactly: $\mathbf{X}^T = (\{1\} \times O) \cup (\{-1\} \times \{t < 0\})$, with $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$ one point Chapter7/GapP2_AZhouRetract · EpGroupoid.Examples.JumpingRetract.appendix_7_6_3 (§VII.36).

**Zhou's retract** Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.X, $\mathbb{Z}/2$ acting by $(t, f) \mapsto (t, -f)$ on the sc-smooth retract of $\mathbb{R} \oplus L^2(\mathbb{R})$ of Appendix 7.6.3, for every choice of the data Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data, of which one exists Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data.std: an ep-groupoid in which $T_R$ is not Hausdorff and LUC and (ZHOU) fail (§VII.47; Findings, Chapter 7, item 156).

**The Čech groupoid of a Prüfer-type surface** Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.epGroupoid: an ep-groupoid whose orbit space is Hausdorff and not paracompact, so property (5) of Definition 7.5.1 (2017: 7.15) is not a consequence of (1)–(4) (§VII.49; Findings, Chapter 7, item 152).

## VII.31 What is not formalized

Nothing of the chapter: every gap is closed, and the one narrower row is a reading (§VII.50). In detail:

- **Closed since the first reading.** $T\varphi$ as a bounded linear isomorphism and an sc-isomorphism (gap G.70), with the tangent facts (3), (4) on $A$ (§VII.44); $|X_\infty|$ and $|X_\infty| = \bigcap_k |X_k|$ (gap G.72) and the density of $X_\infty$ in $X_k$ (gap G.73) (§VII.44); the M-polyfold $\mathbf{X}^R$ (gap G.71): paracompact for every

ep-groupoid, Hausdorff exactly under LUC, which is the 2021 text's hypothesis (§§VII.34, VII.35, VII.48); Corollary 7.6.6 (2017: 7.3) as printed, Part I's gap G.53 being closed (§§VII.29, VII.48).

- **The 2017 statements that do not hold as we read them.** The remark after Definition 7.1.7 (2017: 7.7) for functors that are only sc-smooth on objects (§VII.4; the 2021 remark holds), and Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) for every ep-groupoid (§VII.16; the 2021 forms hold). Each has a counterexample in Lean.
- **The open question of p. 340 (2017 p. 249)**, whether (5) follows from (1)–(4): settled negatively (§VII.49). Remark 7.5.2 (2017: 7.1) has no formal content, and Remark 7.5.6 (2017: 7.2) is an assessment, with nothing to formalize (§§VII.23, VII.24; Findings, Chapter 7, items 82, 113, 152).
- **Not needed.** The contractibility of $N_g^t$, $N_g^s$ in the proof of Theorem 7.6.1 (2017: 7.5) (p. 344 (2017 p. 253)): connectedness suffices (§VII.26; Findings, Chapter 7, item 29). The identification $(\mathbf{X}\,{}_s\!\times_t \mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t \mathbf{X}^1$, which the sheet form avoids (§VII.22). The comparison with the classical properness of $(s,t)$ on 2017 p. 227 has no formal content.
- **Gaps of Part I taken up**: G.15 (Lemma 2.7.3 (2017: 2.7), Proposition 2.7.4 (2017: 2.17)) is formalized (§VII.20); the local path connectedness of M-polyfolds (§VII.7) and, now, the path connectedness of $U \cap X_\infty$ (§VII.45); the tameness of $X^1$, left open in Part I, is proved (§VII.22) (Findings, Chapter 7, item 114).

## VII.32 What the formalization taught us

**Continuity is the content of the remark after Definition 7.1.7 (2017: 7.7).** The formula $f(\gamma) = (s|\mathbf{U}(\psi))^{-1}(f(s\gamma))$ is right wherever it applies; what it needs is a neighbourhood of $\varphi$ mapped into the sheet at $f(\varphi)$, and for that the functor must be continuous on morphisms. A functor of the trivial $\mathbb{Z}/3$-action that swaps two isotropy elements on one side of 0 shows that sc-smoothness on objects alone does not give it, as we read the remark (§VII.4).

**Connectedness is the hidden hypothesis of the natural representation.** Every proof that $\Phi$ is a homomorphism, that $\Gamma$ is a functor, that $\Gamma(1,a) = 1_a$ (Lemma 7.2.2 (2017: 7.2)), and that two natural representations agree compares two continuous sections of $s$ that agree at $x$; without connectedness they agree only near $x$. The text chooses a connected $U(x)$ on 2017 p. 231 and never states it again. Making $U$ connected a field of the datum, and proving that M-polyfolds are locally path connected, settles all of these at once (§§VII.7–VII.9, VII.26).

**"Small" neighbourhoods and the kernel of $\Phi$.** Since sc-smooth maps are not analytic, the kernel of $\Phi$ can depend on $U(x)$, and the text's "for $U(x)$ small, $\ker \Phi = T_x$" needs a precise form: a neighbourhood $N$ of $x$ such that every natural representation inside $N$ has kernel $T_x$, and the intrinsic description $g \in T_x$ iff $s = t$ near $g$. That description turns Definition 7.2.1 (2017: 7.11) into $T_x = \{1_x\}$, proves the effectiveness of $X_R$ in one line, and gives the conjugation $\varphi T_x \varphi^{-1} = T_y$ that the composition of $X_R$ needs and the text does not state (§§VII.15, VII.17).

**The reduced charts do not give Hausdorffness.** The proof of Proposition 7.2.10 (2017: 7.9) builds charts and sc-smooth transitions for $\mathbf{X}^R$ without any hypothesis; an M-polyfold also has to be Hausdorff and paracompact, and these do not follow from the charts as we read it. The formalization separates what is proved unconditionally from the two hypotheses, and shows them satisfiable (§VII.16).

**A star instead of an induction.** Proposition 7.1.15 (2017: 7.5) becomes a construction transported from one object along chosen morphisms, with nothing shrunk afterwards; the text's inductive adjustment of components, and most of its possible misprints, have no counterpart (§VII.10).

**Properness does more than sequences.** Where the text extracts convergent subsequences, the formalization mostly uses that a proper map is closed or that preimages of compact sets are compact: Proposition 7.1.10 (2017: 7.3), Proposition 7.5.10 (1) (2017: 7.14), the closedness of points behind Lemma 7.3.4 (2017: 7.6), and the closed saturations behind the sc-smoothness in Theorem 7.5.7 (2017: 7.4). With compact sets in place of sequences, Proposition 7.5.10 (2) (2017: 7.14) needs only paracompactness, no regularity (§§VII.5, VII.19, VII.25, VII.28).

**Paracompact without Hausdorff.** Definition 7.5.1 (2017: 7.15) stipulates that paracompact does not include Hausdorff, and Michael's theorem, which the text quotes for regular Hausdorff spaces, holds for regular spaces; one direction holds in every space. This closes gap G.15 of Part I and gives Corollary 7.3.8 (2017: 7.2) its finite and locally finite unions (§§VII.20, VII.21). For Proposition 7.5.3 (2017: 7.12) a shorter route proves more: $|X^1|$ is metrizable (§VII.23).

**Lemma 7.6.5 (2017: 7.10) needs its support bound.** As stated, Lemma 7.6.5 (2017: 7.10) bounds the support of $f$ by $V(x)$; the proof of Theorem 7.5.7 (2017: 7.4) needs the bound $\operatorname{cl} U(x)$, which the lemma's own proof gives, for the sum $\sum h_i$ to be locally finite, and it applies the lemma to sets $U_i''$ that need not contain $x_i$. Stating the lemma with the stronger conclusion and the weaker hypothesis makes the proof of Theorem 7.5.7 (2017: 7.4) go through as written, with $W(x_i)$ in the role of $V(x)$ (§§VII.27–VII.28).

**Paracompactness of the orbit space is a genuine condition.** The text leaves open whether (5) follows from (1)–(4). It does not: the orbit space of an ep-groupoid is locally a finite quotient of an M-polyfold, so every Hausdorff, locally Euclidean space whose atlas has a Čech groupoid with proper target maps is one, and a Prüfer-type surface is such a space without being paracompact. The object space of the example is not second countable, as it must be by Remark 7.5.2 (2017: 7.1) (§VII.49).

**What the tangent of a morphism needs is the linear structure of $T_xX$.** Once Part I gives $T_xX$ its linear structure through charts, Definition 7.1.4 (2017: 7.4) is the chain rule and the fact that the derivative at a smooth point is an sc-operator; nothing in Chapter 7 itself was missing (§VII.44).

## VII.33 The 2021 additions to §7.1: the remark after Definition 7.1.7, Propositions 7.1.14, 7.1.15, Lemma 7.1.16, Corollary 7.1.17

The Springer edition (2021) changes the remark after Definition 7.1.7 and Proposition 7.1.15 and adds Proposition 7.1.14, Lemma 7.1.16 and Corollary 7.1.17; Sections VII.33–VII.41 retell the additions of the 2021 edition, with its numbers and pages (V7.69–V7.98).

**The remark after Definition 7.1.7, 2021 form** (V7.69) `Chapter7/Book2021Basic · EpGroupoid.remark_definition_7_1_7`. *A functor sc-smooth on objects and continuous on morphisms is sc-smooth on morphisms, through $f(\gamma) = (s|U(\psi))^{-1}(f(s(\gamma)))$ on a neighbourhood $U(\varphi)$ mapped into a sheet $U(\psi)$ at $\psi = f(\varphi)$.* The 2021 text adds the continuity on morphisms, the hypothesis under which §VII.4 proved the 2017 remark; the continuity gives the neighbourhood $U(\varphi)$, the sheets give the formula, and sc-smoothness follows as in §VII.4. The functor of the counterexample of §VII.4 is sc-smooth on objects and not continuous on morphisms, so the 2021 hypothesis excludes it `Chapter7/Book2021Basic · EpGroupoid.remark_definition_7_1_7_remarkFunctor` (Findings, Chapter 7, item 117) (p. 291).

**The data (7.3) and Proposition 7.1.14** (V7.70) `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_14`. *For data (7.3) — an open $W(x)$ and sheets $U^s(g)$, $U^t(g)$ of $s$ and $t$ at each $g \in G_x$ onto $W(x)$ — and a representation $(\Phi, \Gamma)$ with the properties of Theorem 7.1.13, there is a $\Phi$-invariant open $V(x) \subseteq W(x) \cap$*

$U(x)$ *on which* $\Gamma(g,v) = s_g^{-1}(v)$ *and* $\Phi(g)(v) = t(s_g^{-1}(v))$. The data are a structure `Chapter7/Book2021Basic · EpGroupoid.IsotropySheetData` and exist at every object `Chapter7/Book2021Basic · EpGroupoid.exists_isotropySheetData`; "neighbourhoods $U^s(g)$, $U^t(g)$ for $g$" is read as neighbourhoods of $g$, without which $\Gamma(g,x) = g = s_g^{-1}(x)$ fails. *The proof.* On the open set $N$ of the $v \in W(x) \cap U(x)$ with $\Gamma(g,v) \in U^s(g) \cap U^t(g)$ for all $g$, $\Gamma(g,v) = s_g^{-1}(s(\Gamma(g,v))) = s_g^{-1}(v)$; applying $t$ gives the formula for $\Phi(g)$, and $\Gamma(g,v) = t_g^{-1}(\Phi(g)(v))$, the identity (7.4) of the proof, is part of the conclusion. $V(x) = \bigcap_g \Phi(g)^{-1}(N)$, a finite intersection, is open and invariant. Property (3) of Theorem 7.1.13 is a hypothesis as printed and is not used. The same holds for every natural representation `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_14_of_naturalRep` (Findings, Chapter 7, item 118) (p. 296).

**Proposition 7.1.15, 2021 form** (V7.71) `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_15`. *For a finite set $Y$ of mutually isomorphic objects and $V \in \mathcal{N}(Y)$ there is a natural representation of $Y$ in $V$ whose sheets $\mathbf{U}(\psi) \ni \psi$ are the components of the morphisms between points of $\bigcup_y U(y)$, with $s : \mathbf{U}(\psi) \to U(s(\psi))$, $t : \mathbf{U}(\psi) \to U(t(\psi))$ sc-diffeomorphisms, $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$ and $t(\psi') = f(\psi)(s(\psi'))$ on $\mathbf{U}(\psi)$.* This is the star construction of §VII.10, whose structure already had the 2021 form of (2). *The 2017 form of (2).* As printed in 2017, $s, t : \mathbf{U}(\psi) \to U(y)$ with one $y$ for both maps; with $y \in U(y)$, disjoint $U(y)$ and $\psi \in \mathbf{U}(\psi)$ this forces $s(\psi) = t(\psi)$ `Chapter7/Book2021Basic · EpGroupoid.src_eq_tgt_of_printed_2017`, and for $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection with $Y = \{1, -1\}$ the hypotheses hold, a natural representation of the 2021 form exists, and no data of the printed 2017 form exist `Chapter7/Book2021Basic · EpGroupoid.not_printed_2017_reflection`. So the 2017 form does not hold as we read it, and the 2021 form is the one proved (Findings, Chapter 7, items 25, 119) (p. 297).

**Lemma 7.1.16** (V7.72) `Chapter7/Book2021Basic · EpGroupoid.lemma_7_1_16`. *If $\psi'_k \in \mathbf{U}(\psi)$ converge to $\psi'$ and $s(\psi'_k)$ or $t(\psi'_k)$ converges to a point of $U(s(\psi)) \cup U(t(\psi))$, then $\psi' \in \mathbf{U}(\psi)$ and the sources and targets converge inside $U(s(\psi))$, $U(t(\psi))$.* The four cases reduce to two, since a limit lying in $U(b)$ of points of $U(a)$ lies in $U(a)$, the $U(y)$ being open and disjoint `Chapter7/Book2021Basic · EpGroupoid.NaturalRepFinite.mem_U_of_tendsto`; then $s$ on $\mathbf{U}(\psi)$ is a homeomorphism onto $U(s(\psi))$ and the limits agree (Findings, Chapter 7, item 120) (p. 300).

**Corollary 7.1.17 (4)–(7)** (V7.73). *(4) For $x \in Y$, the natural representation of $G_x$ on $U(x)$ has $\Gamma(g,z) = (s|\mathbf{U}(g))^{-1}(z)$ and $f(g) = \Phi(g)$* `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_4` — by a topological argument, the image of the connected $U(x)$ under $\Gamma(g,\cdot)$ lying in one component, since $f(\psi) = f(\psi')$ does not force $\psi = \psi'$ in a non-effective ep-groupoid. *(5)* $f(\varphi)$ is equivariant along the conjugation $g \mapsto \varphi \circ g \circ \varphi^{-1}$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_5`. *(6)* The multiplication is an sc-diffeomorphism from $\mathbf{U}(\psi) \,{}_s\!\times_t \mathbf{U}(\varphi)$, an open subset of the fibred-product sub-M-polyfold of Part I's Proposition 2.5.20 (2017: 2.15), onto $\mathbf{U}(\psi \circ \varphi)$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_6`: it is $(s|\mathbf{U}(\psi \circ \varphi))^{-1} \circ s \circ \mathrm{pr}_2$ once its image is known to lie in $\mathbf{U}(\psi \circ \varphi)$, which follows, as in the text, from the connectedness of the domain and the continuity of the multiplication, without sequences. *(7)* The $U(y)$ can be taken with the properness property `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_7` (Findings, Chapter 7, item 120) (p. 301).

## VII.34 The 2021 §7.2: $T_x$, the topology $T_R$, Lemma 7.2.11, Propositions 7.2.12, 7.2.13 and Theorem 7.2.14

**Definition 7.2.5 and Lemma 7.2.6** (V7.74) `Chapter7/ReducedCriterion · EpGroupoid.definition_7_2_5` `Chapter7/ReducedCriterion · EpGroupoid.lemma_7_2_6`. *$T_x$ is the maximal subgroup of $G_x$ acting trivially on some small $W(x)$; conjugation by $\varphi : x \to y$ maps $T_x$ onto $T_y$.* For a natural representation, $T_x^V$ is the group

of the $g$ with $\Phi(g) = \mathrm{id}$ on $V \cap U(x)$ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.mem_kerOn; it grows as $V$ shrinks Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_anti, is normal for invariant $V$ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_normal and lies in the intrinsic $T_x$ of §VII.15 Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_le_trivialSubgroup. "Maximal" is read as two statements: $T_x \supseteq T_x^W$ for every open $W \ni x$, with equality for every small $W$, by §VII.15 Step 3. Lemma 7.2.6 is the conjugation $\varphi T_x \varphi^{-1} = T_y$ that §VII.15 had proved for the composition of $X_R$, now an isomorphism Chapter7/ReducedCriterion · EpGroupoid.trivialSubgroupEquiv Chapter7/ReducedCriterion · EpGroupoid.trivialSubgroupEquiv_apply (Findings, Chapter 7, item 121) (p. 310).

**Remark 7.2.8 and Lemma 7.2.9** (V7.75) Chapter7/ReducedCriterion · EpGroupoid.remark_7_2_8. *$T_R$ is the finest topology on $\mathbf{X}^R$ for which $\pi$ is continuous; in general $(X, \mathbf{X}^R)$ cannot be made an ep-groupoid, but it can when $T_R$ is Hausdorff.* The finest such topology is the quotient topology of §VII.16, and $\pi$ is an open quotient map Chapter7/ReducedCriterion · EpGroupoid.isOpenQuotientMap_reducedπ; Lemma 7.2.9 is §VII.16 Steps 2 and 3 Chapter7/Reduced · EpGroupoid.isOpenMap_reducedπ Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ. In Definition 7.2.7 the target map is printed "$s([\varphi]) = t_X(\varphi)$", a possible misprint for $t([\varphi])$ Chapter7/Reduced · EpGroupoid.reducedTgt. The remark's "in general not" is what the counterexample of §VII.16 shows (Findings, Chapter 7, items 122, 123) (p. 311).

**Lemma 7.2.11** (V7.76). *If $\varphi_1, \dots, \varphi_k$ are the other members of the class of $\varphi_0$, there are neighbourhoods $U(\varphi_i)$ with disjoint closures, each mapped by $\pi$ homeomorphically onto one $U([\varphi_0])$, which contain every $\psi$ with $[\psi] \in U([\varphi_0])$.* The class is finite Chapter7/ReducedCriterion · EpGroupoid.finite_reducedπ_preimage, by properness. The conclusion is a predicate Chapter7/ReducedCriterion · EpGroupoid.IsReducedEvenlyCovered, with "homeomorphism onto $U([\varphi_0])$" read as: $\pi(U(\varphi_i)) = U([\varphi_0])$ and $\pi$ an open embedding on $U(\varphi_i)$. *As we read it, the lemma needs $T_R$ Hausdorff.* The text's proof of (3) takes a sequence $\psi_k$ with $[\psi_k] \to [\varphi_0]$ and uses that the limit of $[\psi_k]$ is unique. Without that, (1)–(3) at every unit already make $\mathbf{X}^T$ closed Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_isReducedEvenlyCovered, and (1)–(3) at every morphism are equivalent to $T_R$ Hausdorff Chapter7/ReducedCriterion · EpGroupoid.isReducedEvenlyCovered_iff_t2Space; in the ep-groupoid of §VII.16 they fail at $(1, o_0)$, whose class contains only $(1, o_0)$ while the morphisms $(-1, (-1/(n+1), 0))$ have classes converging to it Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.reducedπ_preimage_one Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isReducedEvenlyCovered. *Under $T_R$ Hausdorff*, the setting of Theorem 7.2.14 where the lemma is used, every open $N \supseteq \pi^{-1}([\varphi_0])$ contains $\pi^{-1}(O)$ for an open $O \ni [\varphi_0]$, which gives (3) for sheets at the finitely many members of the class, chosen with disjoint closures Chapter7/ReducedCriterion · EpGroupoid.exists_preimage_reducedπ_subset_of_t2Space Chapter7/ReducedCriterion · EpGroupoid.isReducedEvenlyCovered_of_t2Space, and the lemma holds Chapter7/ReducedCriterion · EpGroupoid.lemma_7_2_11 (Findings, Chapter 7, item 124) (p. 315).

**Proposition 7.2.12** (V7.77) Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_12. *$(\mathbf{X}^R, T_R)$ is paracompact, for every ep-groupoid.* **The text's proof** refines an open cover through the neighbourhoods of Lemma 7.2.11, which as we read it need $T_R$ Hausdorff. **The formalized proof** needs no hypothesis. *Step 1, descent.* Paracompactness descends along an open continuous surjection $p$ if every point $q$ has a compact $K$ such that every open $N \supseteq K$ contains $p^{-1}(O)$ for some open $O \ni q$ Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_of_isOpenMap: the images of a locally finite precise refinement upstairs form one downstairs, with no separation axiom. *Step 2, the compact set.* For $q = [\varphi]$ and $V$ the open set of Definition 7.1.3 (3) at $s(\varphi)$, $K = \{\psi \in s^{-1}(\mathrm{cl}\, V) \mid (s,t)(\psi) = (s,t)(\varphi)\}$ is compact by properness and contains the class of $\varphi$; $(s,t)$ is proper, hence closed, on $s^{-1}(\mathrm{cl}\, V)$, so an open $N \supseteq K$ contains $\pi^{-1}(O)$ for an open $O \ni q$ Chapter7/ReducedCriterion · EpGroupoid.exists_isCompact_preimage_reducedπ_subset. So `ParacompactSpace X.ReducedMor` is an instance Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_reducedMor,

and of the two hypotheses of §VII.16 only the Hausdorff one remains, the 2021 text's own (gap G.71, closed) (Findings, Chapter 7, items 111, 125) (p. 317).

**Proposition 7.2.13 and Theorem 7.2.14** (V7.78) `Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_13` `Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_14`. *The natural charts make $(\mathbf{X}^R, T_R)$ an M-polyfold iff $T_R$ is Hausdorff; if $T_R$ is Hausdorff, $(X, \mathbf{X}^R)$ is naturally an ep-groupoid.* An M-polyfold is Hausdorff and paracompact by definition, so Proposition 7.2.13 is read as the equivalence of "an M-polyfold structure with the atlas of §VII.16 exists" and $T_R$ Hausdorff; one direction is the definition, the other is §VII.16 with Proposition 7.2.12. Theorem 7.2.14 is Proposition 7.10 (2017) of §VII.17 under exactly the hypothesis the formalization had taken: $X_R$ `Chapter7/Reduced · EpGroupoid.reducedEpGroupoid`, with the M-polyfold $\mathbf{X}^R$ `Chapter7/Reduced · EpGroupoid.reducedPolyfold`, the groupoid structure of §VII.17 and the ep-functor $X \to X_R$; it is effective (Proposition 7.2.16) `Chapter7/Reduced · EpGroupoid.isEffective_reducedEpGroupoid`. The 2021 text thus agrees with the reading of §§VII.16–VII.17 (gap G.71, closed in the 2021 form); the 2021 statements, with the Hausdorff hypothesis alone, are §VII.48 `Chapter7/GapP2_AReduced · EpGroupoid.proposition_7_2_16` (Findings, Chapter 7, items 56, 107, 111, 126, 143) (p. 318).

## VII.35 Proposition 7.2.17, the local unique continuation property, and Theorem 7.2.22

**Proposition 7.2.17** (V7.79) `Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_17`. *$T_R$ is Hausdorff iff $\mathbf{X}^T$ is closed in $\mathbf{X}$. Step 1, the relation.* $[\varphi] = [\psi]$ iff $t\varphi = t\psi$ and $\psi^{-1} \circ \varphi \in \mathbf{X}^T$ `Chapter7/ReducedCriterion · EpGroupoid.reducedπ_eq_iff_comp_inv`; in particular $\varphi \in \mathbf{X}^T$ iff $[\varphi]$ is the class of a unit `Chapter7/ReducedCriterion · EpGroupoid.mem_nonEffectivePart_iff_reducedπ`. *Step 2, closed relation.* $(\varphi, \psi) \mapsto \psi^{-1} \circ \varphi$ is continuous on $\{s\varphi = s\psi\}$ `Chapter7/ReducedCriterion · EpGroupoid.continuous_comp_fibredProduct` `Chapter7/ReducedCriterion · EpGroupoid.continuousOn_comp_inv`, so the relation is closed when $\mathbf{X}^T$ is `Chapter7/ReducedCriterion · EpGroupoid.isClosed_reducedπ_eq_of_isClosed`, and a quotient by an open quotient map is Hausdorff iff its relation is closed (Mathlib). *Step 3, conversely,* if $T_R$ is Hausdorff, $\mathbf{X}^T = \{\varphi \mid [\varphi] = [1_{s(\varphi)}]\}$ is closed, the set where two continuous maps into a Hausdorff space agree `Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_t2Space`. Properness is not needed. Always, $\mathrm{cl}\,\mathbf{X}^T \subseteq \{s = t\}$ `Chapter7/ReducedCriterion · EpGroupoid.isClosed_src_eq_tgt` `Chapter7/ReducedCriterion · EpGroupoid.closure_nonEffectivePart_subset` (Findings, Chapter 7, item 127) (p. 320).

**Definition 7.2.18 (= 7.4.1), Remarks 7.2.19 and 7.2.20** (V7.80). *(LUC): every object has a natural representation on which $\hat{g} = \mathrm{id}$ on a nonempty open subset forces $\hat{g} = \mathrm{id}$.* `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.HasLUC` `Chapter7/ReducedCriterion · EpGroupoid.IsLUC`. The choice of $U(x)$ does not matter: if $\mathbf{X}^T$ is closed, every natural representation has the property `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.hasLUC_of_isClosed`, so LUC holds iff it holds for all of them `Chapter7/ReducedCriterion · EpGroupoid.isLUC_iff_forall_naturalRep`, and Remark 7.2.20 (inside every $O(x)$) follows `Chapter7/ReducedCriterion · EpGroupoid.remark_7_2_20`. Remark 7.2.19, some ep-groupoid is not LUC, is §VII.36 `Chapter7/ReducedCriterion · EpGroupoid.Examples.remark_7_2_19`. Effective ep-groupoids are LUC, $\mathbf{X}^T$ being the closed set of units `Chapter7/ReducedCriterion · EpGroupoid.isClosed_range_unit` `Chapter7/ReducedCriterion · EpGroupoid.nonEffectivePart_eq_range_unit_of_isEffective` `Chapter7/ReducedCriterion · EpGroupoid.isLUC_of_isEffective`, and so are the trivial ep-groupoid, the reflection and the non-effective trivial $\mathbb{Z}/3$-action `Chapter7/ReducedCriterion · EpGroupoid.isLUC_trivial` `Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_reflection` `Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_trivialZ3` (Findings, Chapter 7, items 128, 144) (p. 321).

**Proposition 7.2.21, Theorem 7.2.22 and Proposition 7.2.23** (V7.81) `Chapter7/ReducedCriterion · EpGroupoid.`

theorem_7_2_22. *$T_R$ Hausdorff, LUC and $\mathbf{X}^T$ closed are equivalent; under LUC, $X_R$ is an ep-groupoid. Proposition 7.2.21, by connectedness.* In a natural representation, $\Gamma(g,y) \in \mathbf{X}^T$ iff $\Phi(g) = \mathrm{id}$ near $y$ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.eventually_Φ_eq_of_Γ_mem Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.Γ_mem_of_eventually. So the set of $y \in U(x)$ near which $\Phi(g) = \mathrm{id}$ is open, and closed in $U(x)$ when $\mathbf{X}^T$ is closed, hence empty or all of the connected $U(x)$: LUC. Conversely, under LUC, a morphism $\varphi \in \mathrm{cl}\,\mathbf{X}^T$ has $s\varphi = t\varphi = x$, so $g = \varphi \in G_x$; the open sheet $\Gamma(g, U(x)) \ni \varphi$ meets $\mathbf{X}^T$ in some $\Gamma(g,y)$, so $\Phi(g) = \mathrm{id}$ near $y$, LUC gives $\Phi(g) = \mathrm{id}$ on $U(x)$, and $\varphi \in T_x$ Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_isLUC Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_21. With Proposition 7.2.17 this is Theorem 7.2.22, and Proposition 7.2.23 is Theorem 7.2.14 under LUC Chapter7/ReducedCriterion · EpGroupoid.t2Space_reducedMor_of_isLUC Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_23. For the trivial $\mathbb{Z}/3$-action, $\mathbf{X}^T = \mathbf{X}$ is closed and $T_R$ is Hausdorff Chapter7/ReducedCriterion · EpGroupoid.Examples.nonEffectivePart_trivialZ3 Chapter7/ReducedCriterion · EpGroupoid.Examples.t2Space_reducedMor_trivialZ3, the case the first reading had left open (Findings, Chapter 7, items 60, 128, 129) (pp. 321–323).

## VII.36 Appendices 7.6.3 and 7.6.4: Zhou's example, and the reduced groupoid of $X^1$

**Appendix 7.6.3** (V7.82) Chapter7/ReducedCriterion · EpGroupoid.Examples.remark_7_2_19. *Zhou's example: $\mathbb{Z}/2$ acting on the retract of Part I's Subsection 1.4.4 by $(s,f) \mapsto (s,-f)$; $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$ is one point, so $T_R$ is not Hausdorff and LUC fails.* The formalization uses its own example of §VII.16, $\mathbb{Z}/2$ acting by $(t,z) \mapsto (t, \pm z)$ on the retract of Part I's Remark 4.2.2 (2017: 4.3), which has the same features: the action is trivial over $t < 0$ and moves points arbitrarily close to the origin over $t > 0$. The point $(-1, o_0)$ is a limit of $(-1,(t,0))$, $t<0$, which lie in $\mathbf{X}^T$, and is not in $\mathbf{X}^T$ Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.mem_closure_nonEffectivePart Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isClosed_nonEffectivePart, so by Theorem 7.2.22 LUC fails Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isLUC and $T_R$ is not Hausdorff, as §VII.16 had shown directly Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_t2Space_reducedMor. *$\mathbf{X}^T$ exactly.* Since then the non-effective part of the analogue is computed: $\mathbf{X}^T = (\{1\} \times O) \cup (\{-1\} \times \{(t,z) \in O \mid t < 0\})$ — the element 1 acts trivially everywhere, $-1$ near exactly the points over $t < 0$, since near a point over $t \geq 0$ there are points of $O$ with $z \neq 0$ — and $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T = \{(-1, o_0)\}$, one point, as the text says of its example Chapter7/GapP2_AZhouRetract · EpGroupoid.Examples.JumpingRetract.nonEffectivePart_eq Chapter7/GapP2_AZhouRetract · EpGroupoid.Examples.JumpingRetract.closure_nonEffectivePart_diff Chapter7/GapP2_AZhouRetract · EpGroupoid.Examples.JumpingRetract.appendix_7_6_3. Zhou's example itself is §VII.47 Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.appendix_7_6_3 (gap G.83, closed). The counterexample found in the first reading is thus consistent with the 2021 text: it is of the kind Remark 7.2.8 announces and Appendix 7.6.3 gives, and it fails LUC as Theorem 7.2.22 requires (Findings, Chapter 7, items 107, 129, 130, 156) (p. 348).

**Appendix 7.6.4, Theorem 7.6.7 and Corollary 7.6.8** (V7.83) Chapter7/ReducedCriterion · EpGroupoid.theorem_7_6_7. *Under (SLUC), $(\mathbf{X}^T)^1 = (\mathbf{X}^1)^T$, and the identification of $(X^1)_R$ with $(X_R)^1$ is an sc-diffeomorphism; likewise for every $i$. (7.24).* A morphism of level 1 in $\mathbf{X}^T$ lies in $(\mathbf{X}^1)^T$, since $s = t$ near it in $\mathbf{X}$ gives $s = t$ near it in $\mathbf{X}^1$ Chapter7/ReducedCriterion · EpGroupoid.mem_nonEffectivePart_liftOne_of_levelVal. *The converse* follows the text: for $g$ acting trivially near $x$ in $X^1$, $\hat{g} = \mathrm{id}$ on $U(x) \cap X_1$ by LUC for $X^1$, through the level-one form of $\Gamma$ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.levelOneΓ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.continuousOn_levelOneΓ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.eventually_of_levelOneΓ_mem Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.levelOneΓ_mem_of_eventually, and on $U(x)$ by density

Chapter7/ReducedCriterion · EpGroupoid.levelVal_mem_nonEffectivePart_of_liftOne. The text uses that $U(x) \cap X_1$ is connected in $X^1$ and dense in $U(x)$; both were a hypothesis Chapter7/ReducedCriterion · EpGroupoid.HasConnectedDenseLevelOneReps, the density being that of $X_1$ in $X$ (gap G.73), and both now hold for every ep-groupoid Chapter7/SmoothConnectedReps · EpGroupoid.hasConnectedDenseLevelOneReps (§VII.45), so that $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ needs only LUC for $X^1$ Chapter7/SmoothConnectedReps · EpGroupoid.nonEffectivePart_liftOne_eq_of_isClosed. Then $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ Chapter7/ReducedCriterion · EpGroupoid.nonEffectivePart_liftOne_eq, the two relations agree on level 1 Chapter7/ReducedCriterion · EpGroupoid.liftOne_reducedπ_eq_iff, and $[\varphi]_1 \mapsto [\varphi]$ Chapter7/ReducedCriterion · EpGroupoid.reducedLevelOneMap is well defined, injective and continuous with range $\pi(\mathbf{X}_1)$. Of (SLUC) only LUC for $X^1$ is used. Corollary 7.6.8 takes "(ZHOU) implies LUC for $X^1$" as a hypothesis Chapter7/ReducedCriterion · EpGroupoid.corollary_7_6_8, which §VII.38 discharges Chapter7/Zhou · EpGroupoid.corollary_7_6_8_of_isZhou. All hypotheses hold for the trivial ep-groupoid of a point Chapter7/ReducedCriterion · EpGroupoid.Examples.pointEpGroupoid Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_liftOne_point Chapter7/ReducedCriterion · EpGroupoid.Examples.hasConnectedDenseLevelOneReps_point Chapter7/ReducedCriterion · EpGroupoid.Examples.theorem_7_6_7_point Chapter7/Zhou · EpGroupoid.Examples.corollary_7_6_8_of_isZhou_point. The sc-diffeomorphism, which needs the M-polyfold $(\mathbf{X}^R)^1$, and $i \geq 2$ are §VII.46 (gap G.82, closed) (Findings, Chapter 7, item 131) (pp. 348–351).

## VII.37 §7.4 (new in the 2021 edition): index lifts, Lemma 7.4.4, the Part I facts, and $Tg$

§7.4 has no 2017 counterpart. It introduces the Zhou condition and the unique continuation and regularity properties that the reduced groupoid and the index lifts need.

**Index lifts, Proposition 7.4.2 and Remark 7.4.3** (V7.84). The lifts $X^i$ are bundled ep-groupoids Chapter7/Zhou · EP.liftOne Chapter7/Zhou · EP.lift, with $X^0 = X$, $X^{i+1} = (X^i)^1 = (X^1)^i$ Chapter7/Zhou · EP.lift_zero Chapter7/Zhou · EP.lift_succ Chapter7/Zhou · EP.lift_succ'. *Under LUC, the property passes from a connected invariant $U(x)$ to a larger $W(x)$, so $U(x)$ may be taken small or large.* By §VII.35 every natural representation has the property under LUC, so the hypotheses on $U(x)$ are not used Chapter7/Zhou · EpGroupoid.proposition_7_4_2 Chapter7/Zhou · EpGroupoid.remark_7_4_3 (Findings, Chapter 7, item 132) (pp. 328–329).

**Lemma 7.4.4** (V7.85) Chapter7/Zhou · EpGroupoid.lemma_7_4_4. *If $X^i$ has LUC, so has every $X^k$, $k \leq i$.* For $i = 1$ the text shows that the points of $U(x) \cap X_1$ near which $\hat{g} = \mathrm{id}$ in $X^1$ form an open, closed and nonempty subset of the connected $U(x) \cap X_1$, and concludes by density Chapter7/Zhou · EpGroupoid.NaturalRep.eqOn_of_isClosed_liftOne Chapter7/Zhou · EpGroupoid.isLUC_of_isLUC_liftOne. The connectedness of $U(x) \cap X_1$ in $X^1$ is used without proof, and the density is that of $X_1$ in $X$ (gap G.73); both are a hypothesis on each $X^k$, $k < i$ Chapter7/Zhou · EpGroupoid.HasLevelOneConnectedDenseReps (gap G.85, closed: the hypothesis holds for every ep-groupoid, §VII.45; the lemma is not restated without it), satisfied on subsingletons and by every lift of the point Chapter7/Zhou · EpGroupoid.hasLevelOneConnectedDenseReps_of_subsingleton Chapter7/Zhou · EpGroupoid.Examples.hasLevelOneConnectedDenseReps_lift_pointEP Chapter7/Zhou · EpGroupoid.Examples.lemma_7_4_4_pointEP. In the proof, "$s : W(g) \to W(y)$" for $W(h)$ (possible misprint) (Findings, Chapter 7, item 133) (p. 329).

**The Part I facts** (V7.86). The proofs of §7.4 use, without stating them, facts about an M-polyfold that Part I did not state; in the first reading they were hypotheses, and §VII.45 proves them for every M-polyfold (gap G.85, closed): $X_\infty$ dense in every $X_m$ (gap G.73) Chapter7/Zhou · MPolyfold.DenseSmooth, $U \cap X_\infty$ connected for connected open $U$ Chapter7/Zhou · MPolyfold.SmoothConnected, maps agreeing near a smooth point in $X_\infty$ have the same tangent there Chapter7/Zhou · MPolyfold.TangentDeterminedSmooth Chapter7/Zhou · MPolyfold.TangentDeterminedSmooth.eq_self, and tangent vectors are approximated by tangent

vectors at nearby smooth points, which follows from the openness of $p : TX \to X^1$ (gap G.8) Chapter7/Zhou · MPolyfold.eqOn_tangentFiber_of_mem_closure. They are bundled as one hypothesis Chapter7/Zhou · MPolyfold.SmoothRegular, with the density of $(T_xX)_\infty$ in $T_xX$ separate Chapter7/Zhou · MPolyfold.DenseSmoothTangent, and consequences Chapter7/Zhou · MPolyfold.DenseSmooth.dense Chapter7/Zhou · MPolyfold.DenseSmooth.subset_closure_inter Chapter7/Zhou · MPolyfold.DenseSmooth.exists_smooth. They hold on subsingletons with all points smooth and on the discrete M-polyfold of a point Chapter7/Zhou · MPolyfold.smoothRegular_of_subsingleton Chapter7/Zhou · MPolyfold.smoothRegular_discrete_unit Chapter7/Zhou · MPolyfold.denseSmoothTangent_discrete_unit (gap G.85) (Findings, Chapter 7, item 134) (pp. 331–340).

**The tangent** $Tg$ (V7.87). For a smooth $x$ and $g \in G_x$, Chapter 7's tangent of a morphism (Definition 7.1.4, $Tt \circ (Ts|T_g\mathbf{X})^{-1}$, without charts, §VII.3) is, on $T_xX$, the tangent $T\Phi(g)$ of $\hat{g}$, for every natural representation Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_Γ Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_eq_tΦ Chapter7/Zhou · EpGroupoid.NaturalRep.tΦ_eq_tΦ: $t = \Phi(g) \circ s$ on the sheet $\Gamma(g, \cdot)$. At a fixed point $z$ of $\Phi(g)$, $\Gamma(g, z) \in G_z$ acts near $z$ as $g$ Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ_val Chapter7/Zhou · EpGroupoid.NaturalRep.eventually_Φ_eq_Γ Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_isoΓ, and lies in $T_z$ iff $\Phi(g) = \mathrm{id}$ near $z$ Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ_mem_trivialSubgroup_iff. $Tg$ is a set map on the fibre; (ZHOU) asks only that it be the identity, so the linear structure of $T_xX$ (gap G.7) is not needed (Findings, Chapter 7, item 135) (p. 331).

## VII.38 §7.4: (ZHOU), (ZHOU'), unique continuation, (SLUC) and (VSLUC)

The forms below take the Part I facts of §VII.37 as hypotheses; since these hold for every M-polyfold, every result also holds without them, and the restatements are listed in §VII.45.

**Definitions 7.4.5 and 7.4.7** (V7.88). *(ZHOU): for every smooth $x$ and $g \in G_x$, $Tg = \mathrm{id}$ on $T_xX$ implies $g \in T_x$* Chapter7/Zhou · EpGroupoid.IsZhou. $T_x^\infty$, the $g$ acting trivially near $x$ in $X_\infty$ Chapter7/Zhou · EpGroupoid.trivialSetInf Chapter7/Zhou · EpGroupoid.trivialSubgroupInf, testable in any natural representation Chapter7/Zhou · EpGroupoid.mem_trivialSetInf_iff Chapter7/Zhou · EpGroupoid.mem_trivialSubgroupInf_iff, contains $T_x$ Chapter7/Zhou · EpGroupoid.trivialSubgroup_le_trivialSubgroupInf. *(ZHOU'): $Tg = \mathrm{id}$ on $(T_xX)_\infty$ implies $g \in T_x^\infty$* Chapter7/Zhou · EpGroupoid.IsZhou'. Both hold when every $T_x = G_x$ Chapter7/Zhou · EpGroupoid.isZhou_of_trivialSubgroup_eq_top Chapter7/Zhou · EpGroupoid.isZhou'_of_trivialSubgroup_eq_top, on subsingletons Chapter7/Zhou · EpGroupoid.isZhou_of_subsingleton Chapter7/Zhou · EpGroupoid.isZhou'_of_subsingleton, for the trivial epgroupoid Chapter7/Zhou · EpGroupoid.isZhou_trivial Chapter7/Zhou · EpGroupoid.isZhou'_trivial and for the trivial $\mathbb{Z}/3$-action, where $T_x \neq 1$ Chapter7/Zhou · EpGroupoid.Examples.isZhou_trivialZ3 Chapter7/Zhou · EpGroupoid.Examples.isZhou'_trivialZ3 (Findings, Chapter 7, items 135, 144) (p. 331).

**Lemma 7.4.6 and Proposition 7.4.8** (V7.89) Chapter7/Zhou · EpGroupoid.proposition_7_4_8. *(ZHOU) and (ZHOU') are equivalent. (ZHOU) □ (ZHOU')*: if $Tg = \mathrm{id}$ on the dense $(T_xX)_\infty$, then on $T_xX$, and (ZHOU) gives $g \in T_x \subseteq T_x^\infty$ Chapter7/Zhou · EpGroupoid.lemma_7_4_6. *(ZHOU') □ (ZHOU)*: the text uses sequences; the formalization shows that the smooth points $y$ of $U(x)$ that $\Phi(g)$ fixes with tangent the identity form an open and closed subset of the connected $U(x)_\infty$ Chapter7/Zhou · EpGroupoid.NaturalRep.mem_tangentU_of_fiber Chapter7/Zhou · EpGroupoid.NaturalRep.tΦ_eq_self_of_eqOn Chapter7/Zhou · EpGroupoid.NaturalRep.fix_and_tangent_of_mem_closure Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_smooth_clopen Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_smooth_tangent; continuity and density give $\Phi(g) = \mathrm{id}$ near $x$ Chapter7/Zhou · eq_of_mem_closure_of_continuousAt Chapter7/Zhou · EpGroupoid.isZhou_of_isZhou'. The Part I facts of §VII.37 are hypotheses, satisfied by the point Chapter7/Zhou · EpGroupoid.Examples.proposition_7_4_8_point.

"$z_k \to Y$" for $z_k \to y$ in the proof (possible misprint) (Findings, Chapter 7, item 136) (pp. 331–332).

**Proposition 7.4.10 and Corollary 7.4.11** (V7.90) `Chapter7/Zhou · EpGroupoid.proposition_7_4_10`. *(ZHOU) implies LUC, and then every $X^i$ has LUC.* The text shows that the smooth points near which $\hat{g} = \mathrm{id}$ form an open and closed subset of $U(x)_\infty$, closedness by approximating tangent vectors at a limit point by tangent vectors at nearby smooth points. The formalization isolates the conclusion as unique continuation through $X_\infty$ `Chapter7/Zhou · EpGroupoid.IsSmoothUC` `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_isZhou`, which gives LUC for $X$, by the density of $X_\infty$, and for $X^1$, through Proposition 7.2.21, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ being closed in $\mathbf{X}^1$ `Chapter7/Zhou · EpGroupoid.isLUC_of_isSmoothUC` `Chapter7/Zhou · EpGroupoid.isLUC_liftOne_of_isSmoothUC` `Chapter7/Zhou · EpGroupoid.corollary_7_4_11`. This form for every $X^i$ takes (ZHOU) for every $X^i$, the conclusion of Proposition 7.4.9, as a hypothesis `Chapter7/Zhou · EpGroupoid.corollary_7_4_11_lift`; with (ZHOU) for $X$ only it is §VII.43 (gap G.84, closed) (Findings, Chapter 7, items 136, 137) (pp. 333–334).

**Definition 7.4.12 and Theorem 7.4.13** (V7.91) `Chapter7/Zhou · EpGroupoid.theorem_7_4_13`. *(SLUC), every $X^i$ has LUC* `Chapter7/Zhou · EP.IsSLUC`, *follows from (ZHOU).* Proposition 7.4.9, (ZHOU) for $X$ gives (ZHOU) for every $X^i$, needs the identification of $(X^i)_\infty$ with $X_\infty$ and of the tangent spaces; this form of the theorem takes (ZHOU) and the Part I facts for every $X^i$, and Proposition 7.4.9 itself, with the theorem for (ZHOU) on $X$ only, is §VII.43 (gap G.84, closed). They hold for every lift of the point `Chapter7/Zhou · EpGroupoid.Examples.pointEP` `Chapter7/Zhou · EpGroupoid.Examples.lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.smoothRegular_lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.isZhou_lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.isSLUC_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.theorem_7_4_13_pointEP` (Findings, Chapter 7, item 137) (p. 334).

**Definition 7.4.14, Lemma 7.4.15, Propositions 7.4.16–7.4.18** (V7.92). *(VSLUC): LUC for $X_\infty$, on $G_x$-invariant open $V(x) \subseteq X_\infty$* `Chapter7/Zhou · EpGroupoid.IsVSLUC`, read with $V(x)$ open in $X_\infty$, inside a natural representation and invariant. It is equivalent to unique continuation through $X_\infty$, given the Part I facts `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_isVSLUC` `Chapter7/Zhou · EpGroupoid.isVSLUC_of_isSmoothUC`, so it gives LUC (Lemma 7.4.15) `Chapter7/Zhou · EpGroupoid.lemma_7_4_15` for $X$ and $X^1$ `Chapter7/Zhou · EpGroupoid.proposition_7_4_16` by the argument of Proposition 7.4.10, with connectedness of $U(x)_\infty$ in place of the text's maximality argument; (ZHOU) implies it `Chapter7/Zhou · EpGroupoid.isVSLUC_of_isZhou`. This form of Proposition 7.4.17 takes (VSLUC) for every $X^i$ `Chapter7/Zhou · EpGroupoid.proposition_7_4_17`; Proposition 7.4.16 for every $X^i$, Proposition 7.4.17 for (VSLUC) on $X$ only and Proposition 7.4.18, which needs natural representations of $X^N$ in terms of those of $X$, are §VII.43 (gap G.84, closed). Witnesses: `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_subsingleton` `Chapter7/Zhou · EpGroupoid.isVSLUC_of_subsingleton` `Chapter7/Zhou · EpGroupoid.isLUC_of_subsingleton_mor` `Chapter7/Zhou · EpGroupoid.isVSLUC_trivial` `Chapter7/Zhou · EpGroupoid.isSmoothUC_trivial` (Findings, Chapter 7, items 136, 137) (pp. 335–336).

## VII.39 §7.4: (LR), (SLR), and the example of Appendix 7.6.3

As in §VII.38, the Part I facts taken as hypotheses below are theorems (§VII.45).

**Definitions 7.4.19, 7.4.21 and Lemma 7.4.20** (V7.93). *(LR): inside every neighbourhood of $x$ a natural representation such that every sc-diffeomorphism $\beta$ of $U(x)$ with $|\beta(y)| = |y|$ equals some $\hat{g}$* `Chapter7/Zhou · EpGroupoid.IsLR`; *(LR'): the same for $\beta$ onto an open image with $\beta(x) = x$* `Chapter7/Zhou · EpGroupoid.IsLR'`. "Arbitrarily small" is "inside every open $O \ni x$", and $|\beta(y)| = |y|$ is $\pi(\beta(y)) = \pi(y)$. *Lemma 7.4.20: such a $\beta$ maps $U(x)$ onto $U(x)$* `Chapter7/Zhou · EpGroupoid.lemma_7_4_20`. Two points of $U(x)$ in one orbit are related by the natural action, so where $\beta(y) \in U(x)$, $\beta(y) = \hat{g}(y)$ for some $g$ `Chapter7/Zhou · EpGroupoid.NaturalRep.exists_Φ_eq_of_π_eq`; so $U(x) \cap \beta^{-1}(U(x))$ is covered by the finitely many closed sets $\{\beta = \hat{g}\}$, and connected-

ness of $U(x)$, instead of the text's paths, gives $\beta(U(x)) \subseteq U(x)$ and then equality; the only point of $U(x)$ in the orbit of $x$ is $x$ Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_π_eq. Hence (LR) and (LR') are equivalent Chapter7/Zhou · EpGroupoid.isLR_iff_isLR' (Findings, Chapter 7, item 138) (pp. 337–338).

**Proposition 7.4.22, Definition 7.4.23 and Theorem 7.4.24** (V7.94) Chapter7/Zhou · EpGroupoid.proposition_7_4_22. *(ZHOU) implies (LR), and (SLR), every $X^i$ has (LR)* Chapter7/Zhou · EP.IsSLR. The text passes to the reduced ep-groupoid and uses the density of the regular points. The formalization does neither: $U(x)_\infty$ is covered by the closed sets $\{\beta = \hat{g}\}$; the union of their $X_\infty$-interiors is open and dense, and two pieces whose closures meet have $T\hat{g} = T\hat{g}'$ at a common smooth point, so $\hat{g} = \hat{g}'$ by (ZHOU) and LUC Chapter7/Zhou · EpGroupoid.NaturalRep.eqOn_Φ_of_tangent; connectedness of $U(x)_\infty$ and density finish Chapter7/Zhou · EpGroupoid.NaturalRep.exists_eq_Φ_of_isZhou. This form of Theorem 7.4.24 takes (ZHOU) for every $X^i$, and the form for (ZHOU) on $X$ only is §VII.43 (gap G.84, closed) Chapter7/Zhou · EpGroupoid.theorem_7_4_24 Chapter7/Zhou · EpGroupoid.Examples.isSLR_pointEP (Findings, Chapter 7, item 138) (pp. 338–340).

**The example of Appendix 7.6.3 and (ZHOU)** (V7.95) Chapter7/Zhou · EpGroupoid.Examples.JumpingRetract.not_isZhou. *The example fails the Zhou condition (p. 331).* For the analogue of §VII.36: it is not LUC, and (ZHOU) implies LUC by Proposition 7.4.10, so it fails (ZHOU). In the first reading this was conditional on the Part I facts for its object M-polyfold; they hold for every M-polyfold (§VII.45), so the analogue fails (ZHOU) with no hypothesis Chapter7/SmoothRegularAssembly · EpGroupoid.Examples.JumpingRetract.not_isZhou_unconditional, and Zhou's example itself fails it by the same two steps Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_isZhou (§VII.47; gap G.83, closed) (Findings, Chapter 7, items 139, 156) (p. 331).

## VII.40 Appendix 7.6.5: the boundary of a tame ep-groupoid, Theorem 7.6.9 (topological part)

**Transport along index-preserving local homeomorphisms** (V7.96). The maps (7.25)–(7.28) are defined through sc-diffeomorphisms of sheets; what they need of them is topological. A map that is a homeomorphism of an open $W$ onto an open set and preserves the corner index Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn Chapter7/TameBoundaryEp · TameBoundary.IsDLocalHomeo Chapter7/TameBoundaryEp · TameBoundary.isDHomeoOn_id carries faces, face-structured neighbourhoods, the morphisms of Chapter 2's $L_X$ and $F_X$ and their extensions to the corresponding objects Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.isFaceIn_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.mem_faceStructuredNbhdsAt_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.isMor_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.image_faceExt, so it maps triples $(y, V, \theta)$ to triples Chapter7/TameBoundaryEp · TameBoundary.BTriple.map with a class independent of $W$ Chapter7/TameBoundaryEp · TameBoundary.map_cls_eq. For an index-preserving local homeomorphism $f$, $bf[y, V, \theta] = [f(y), f(V), f(\theta)]$ Chapter7/TameBoundaryEp · TameBoundary.bLift Chapter7/TameBoundaryEp · TameBoundary.bLift_cls is the unique lift through $B$ of this kind Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf.unique Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf.comp Chapter7/TameBoundaryEp · TameBoundary.isLiftOf_id, continuous, open and a local homeomorphism Chapter7/TameBoundaryEp · TameBoundary.continuous_bLift Chapter7/TameBoundaryEp · TameBoundary.isOpenMap_bLift Chapter7/TameBoundaryEp · TameBoundary.isLocalHomeomorph_bLift; $B$ has compact preimages of compact sets Chapter7/TameBoundaryEp · TameBoundary.isCompact_preimage_bMap, and representatives can be centred Chapter7/TameBoundaryEp · TameBoundary.exists_rep_centered. Sc-diffeomorphisms preserve the degeneracy index Chapter7/TameBoundaryEp · MPolyfold.IsScDiffeoOn.isDHomeoOn Chapter7/TameBoundaryEp · MPolyfold.IsLocalScDiffeo.isDLocalHomeo (Findings, Chapter 7, item 140)

(pp. 353–355).

**(7.25)–(7.29), the boundary groupoid** (V7.97) Chapter7/TameBoundaryEp · EpGroupoid.boundaryGroupoid. For a tame ep-groupoid, $\mathbf{X}$ is tame Chapter7/TameBoundaryEp · EpGroupoid.isTame_PMor, and $bX$, $b\mathbf{X}$ are its tame boundaries Chapter7/TameBoundaryEp · EpGroupoid.cornerObj Chapter7/TameBoundaryEp · EpGroupoid.cornerMor Chapter7/TameBoundaryEp · EpGroupoid.bObj Chapter7/TameBoundaryEp · EpGroupoid.bMor. $s$, $t$, $\iota$, $u$ are index-preserving local homeomorphisms Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_src Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_tgt Chapter7/TameBoundaryEp · EpGroupoid.isScDiffeoOn_inv Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_inv Chapter7/TameBoundaryEp · EpGroupoid.isDHomeoOn_unit Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_unit — $u$ as a continuous section of the local homeomorphism $s$, onto the open set $u(X)$ — so (7.25), (7.27), (7.28) are their lifts Chapter7/TameBoundaryEp · EpGroupoid.bSrc Chapter7/TameBoundaryEp · EpGroupoid.bTgt Chapter7/TameBoundaryEp · EpGroupoid.bInv Chapter7/TameBoundaryEp · EpGroupoid.bUnit, with the printed formulas Chapter7/TameBoundaryEp · EpGroupoid.bSrc_cls Chapter7/TameBoundaryEp · EpGroupoid.bTgt_cls Chapter7/TameBoundaryEp · EpGroupoid.bInv_cls Chapter7/TameBoundaryEp · EpGroupoid.bUnit_cls. *(7.29).* A class of $b\mathbf{X}$ is determined by its image under $B$ and its source Chapter7/TameBoundaryEp · EpGroupoid.eq_of_bMap_eq_of_bSrc_eq, so $m(a,b)$ is the unique class over $\psi \circ \varphi$ with source $s(b)$ Chapter7/TameBoundaryEp · EpGroupoid.exists_bComp Chapter7/TameBoundaryEp · EpGroupoid.bComp Chapter7/TameBoundaryEp · EpGroupoid.bComp_spec; near $\varphi$, $\gamma \mapsto (s|\mathbf{W})^{-1}(t\gamma) \circ \gamma$ agrees with a sheet inverse of $s$ composed with $s$ Chapter7/TameBoundaryEp · EpGroupoid.eventuallyEq_comp_sheet, so the printed class $[\psi \circ \varphi, \mathbf{W}\,{}_s\!\circ_t \mathbf{V}, \sigma\,{}_s\!\circ_t \theta]$ is this one Chapter7/TameBoundaryEp · EpGroupoid.bComp_eq. Off the composable pairs $m$ takes a junk value, as in §VII.1. The groupoid laws hold and $B$ is a functor Chapter7/TameBoundaryEp · EpGroupoid.boundaryFunctor (Findings, Chapter 7, item 140) (pp. 354–355).

**Theorem 7.6.9, topological part** (V7.98) Chapter7/TameBoundaryEp · EpGroupoid.theorem_7_6_9. *For a tame ep-groupoid, (1) $bX$ and $b\mathbf{X}$ are metrizable, (3) both maps $B$ are continuous (and sc-smooth), (4) a point of index $d$ has $d$ preimages, (5) $(bX, b\mathbf{X})$ is a tame ep-groupoid; the topology is characterized by the openness of the sets $U(x, V, \theta)$ and the maps $\Phi$.* Formalized: metrizability Chapter7/TameBoundaryEp · EpGroupoid.metrizableSpace_bObj Chapter7/TameBoundaryEp · EpGroupoid.metrizableSpace_bMor from Chapter 2's tame boundary; the characterization; continuity of $B$ and the fibre count; $s$, $t$ local homeomorphisms Chapter7/TameBoundaryEp · EpGroupoid.isLocalHomeomorph_bSrc Chapter7/TameBoundaryEp · EpGroupoid.isLocalHomeomorph_bTgt, $u$, $\iota$ continuous Chapter7/TameBoundaryEp · EpGroupoid.continuous_bSrc Chapter7/TameBoundaryEp · EpGroupoid.continuous_bTgt Chapter7/TameBoundaryEp · EpGroupoid.continuous_bInv Chapter7/TameBoundaryEp · EpGroupoid.continuous_bUnit, $m$ continuous on composable pairs Chapter7/TameBoundaryEp · EpGroupoid.continuousOn_bComp, and the properness property of Definition 7.1.3 (3), by the compact preimages under $B$ Chapter7/TameBoundaryEp · EpGroupoid.exists_isProperMap_bTgt. "$\Phi$ is a homeomorphism onto $\theta$" holds, as we read it, only for $V$ centred at $x$, as for (2.44) in Chapter 2; in general $\Phi$ is an open embedding, the form used. Witness: the trivial ep-groupoid of $[0,\infty)^n \oplus W$, $n \geq 1$, with $n$ classes over the corner Chapter7/TameBoundaryEp · EpGroupoid.bSrc_trivial Chapter7/TameBoundaryEp · EpGroupoid.trivial_quadrant_witness. The rest — (2), the sc-smoothness in (3) and (5), $d_{bX}$ in (4) — depends on 2021 Theorem 2.7.13 (2) and is §VII.42; the characterization for every representative and the book's form of the theorem are §VII.50 (Findings, Chapter 7, items 140, 141, 145–147, 151, 155) (p. 355).

## VII.41 The 2021 additions: what is not formalized, and what they taught us

Every gap of the 2021 additions is closed:

- **Theorem 7.6.7 for every** $i$ (gap G.82, closed): the identification $(X^1)_R \cong (X_R)^1$ is an sc-diffeomorphism and an isomorphism of ep-groupoids, and $(X^i)_R \cong (X_R)^i$ for every $i$ under (SLUC) (§VII.46; Findings, Chapter 7, item 131).
- **Zhou's retract** (gap G.83, closed): the example itself, with the sc-smoothness the text cites from [60], and its non-effective part computed as printed; the analogue's non-effective part is computed exactly as well (§§VII.36, VII.47; Findings, Chapter 7, items 130, 139, 156).
- $X^i$**,** $i \geq 2$**, in §7.4** (gap G.84, closed): Propositions 7.4.9 and 7.4.16–7.4.18 and the results for every $X^i$ are §VII.43; of "(ZHOU') is the same for $X$ and $X^i$" only the direction from $X$ to $X^i$, the one the proof uses, is formalized (Findings, Chapter 7, items 137, 148–150).
- **The Part I facts** (gap G.85, closed): they hold for every M-polyfold, and every natural representation has $U(x) \cap X_m$ connected and $U(x) \cap X_1$ dense, so §7.4 and Appendix 7.6.4 hold with no added hypothesis (§VII.45; Findings, Chapter 7, item 153). Lemma 7.4.4 is not restated without its hypothesis, which now holds.
- **Theorem 7.6.9** (row V7.98, narrower): "$\Phi$ is a homeomorphism onto $\theta$" holds as we read it only for $V$ centred at $x$; the image is $\theta \cap V^x$ in general, and the theorem is proved for every representative in that form (§VII.50; Findings, Chapter 7, items 141, 151, 155).

**The counterexample and the 2021 text agree.** The first reading found that $\mathbf{X}^R$ need not be Hausdorff, on a dimension-jumping retract, and took Hausdorffness and paracompactness as hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017). The 2021 text rewrites §7.2 to the same effect: Remark 7.2.8 says the structure fails in general, Theorem 7.2.14 assumes $T_R$ Hausdorff, Theorem 7.2.22 characterizes it by LUC and the closedness of $\mathbf{X}^T$, and Appendix 7.6.3 gives Zhou's example of the same kind. Paracompactness, which the formalization had left open, holds always (§§VII.34–VII.36; gap G.71, closed).

**Hausdorffness sits inside Lemma 7.2.11.** Its conclusion at every morphism is equivalent to $T_R$ Hausdorff, so it cannot serve, as we read it, as a step towards Hausdorffness; the paracompactness of Proposition 7.2.12 is proved instead from the openness of $\pi$ and the properness of $(s,t)$ (§VII.34).

**Connectedness replaces sequences.** Propositions 7.2.21, 7.4.8, 7.4.10, 7.4.22, Lemmas 7.4.4, 7.4.15 and 7.4.20 are proved by showing a set open and closed in a connected $U(x)$, $U(x)_\infty$ or $U(x) \cap X_1$; what this needs of Part I — density and connectedness of the smooth points, locality of tangents, the openness of $p : TX \to X^1$ — becomes visible as explicit hypotheses (§§VII.35–VII.39).

$Tg$ **needs no linear structure.** (ZHOU) asks that a map be the identity; Chapter 7's chart-free tangent of a morphism suffices (§VII.37).

## VII.42 Appendix 7.6.5: Theorem 7.6.9 in full (the boundary ep-groupoid)

Theorem 7.6.9 (2)–(5) rest on Part I's Theorem 2.7.13 (2)–(4): for a tame M-polyfold $A$, the tame boundary $bA$ carries a tame M-polyfold structure `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold` `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_isTame` whose charts are $p \mapsto \pi(\varphi(Bp))$, $\pi$ a coordinate projection of a tame chart $\varphi$ and the face read off the triple; $B : bA \to A$ is sc-smooth `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryMap_isScSmooth` and $d_A(Bp) = d_{bA}(p) + 1$ `Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap` `Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap_sub_one`; the structure is characterized by the charts $\Phi$ of centred triples, and

unique with that property Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_unique Chapter2/TameBoundaryStructure · MPolyfold.theorem_2_7_13_structure. The text leaves the verification of Theorem 7.6.9 to the reader, with a sketch: for $[\varphi, \mathbf{V}, \theta]$ choose $\mathbf{V}$ with $s : \mathbf{V} \to V$ an sc-diffeomorphism; in the charts $\Phi_{(\varphi,\mathbf{V},\theta)}$ and $\Phi_{(x,V,s(\theta))}$ the map $s$ becomes $s : \theta \to s(\theta)$, an sc-diffeomorphism of faces. The formalization replaces this sc-diffeomorphism by a computation in the charts of $bA$, which needs only that the map below is sc-smooth (Findings, Chapter 7, items 145–147).

**Maps of tame boundaries** (V7.99) Chapter7/TameBoundaryEpStructure · MPolyfold.isScSmoothOn_of_bMap. *Let $f : A \to B$ be sc-smooth on an open $U$ and $g : bA \to bB$ continuous on an open $O \subseteq B^{-1}(U)$ with $B \circ g = f \circ B$ on $O$. Then $g$ is sc-smooth on $O$. Proof.* Sc-smoothness is local. Near $p \in O$ take the chart $\chi = \pi \circ \varphi \circ B$ of $bA$ at $p$ and $\chi' = \pi' \circ \varphi' \circ B$ of $bB$ at $g(p)$, with $\varphi, \varphi'$ tame charts at $Bp$ and $f(Bp)$. On a neighbourhood of $\chi(p)$ in the face, $\chi' \circ g \circ \chi^{-1} = \pi' \circ (\varphi' \circ f \circ \varphi^{-1}) \circ \iota$, with $\iota$ the inclusion of the face into the model of $\varphi$: indeed $B \circ \chi^{-1} = \varphi^{-1} \circ \iota$ near $\chi(p)$ and $B \circ g = f \circ B$. The right side is a composition of sc-smooth maps, $\pi'$ and $\iota$ being linear. *The lifts.* The lift $bf$ of an sc-smooth index-preserving local homeomorphism (§VII.40) covers $f$ and is continuous, so it is sc-smooth Chapter7/TameBoundaryEpStructure · MPolyfold.isScSmoothMap_bLift. For a local sc-diffeomorphism $f$ and a sheet $f : W \to V$ with inverse $\sigma$, a lift of $\sigma$ takes $bf(c)$ back to $c$ for $Bc \in W$, by the uniqueness of lifts Chapter7/TameBoundaryEpStructure · MPolyfold.isLiftOf_bLift_sheet; so $bf$ is injective on $B^{-1}(W)$, maps it onto $B^{-1}(V)$, and its inverse, covering $\sigma$ and continuous, is sc-smooth by the same statement: $bf$ is a local sc-diffeomorphism Chapter7/TameBoundaryEpStructure · MPolyfold.isLocalScDiffeo_bLift (Findings, Chapter 7, item 146).

**The boundary ep-groupoid** (V7.100) Chapter7/TameBoundaryEpStructure · EpGroupoid.boundaryEpGroupoid. For a tame ep-groupoid, $bX$ and $b\mathbf{X}$ are paracompact Chapter7/TameBoundaryEpStructure · EpGroupoid.paracompactSpace_bObj Chapter7/TameBoundaryEpStructure · EpGroupoid.paracompactSpace_bMor and carry the structures of Theorem 2.7.13 (2) for the object and the morphism M-polyfold Chapter7/TameBoundaryEpStructure · EpGroupoid.bPObj Chapter7/TameBoundaryEpStructure · EpGroupoid.bPMor ($\mathbf{X}$ is tame, §VII.40). Since $s$, $t$ are local sc-diffeomorphisms of $(X, \mathbf{X})$, their lifts (7.25) are local sc-diffeomorphisms Chapter7/TameBoundaryEpStructure · EpGroupoid.isLocalScDiffeo_bSrc Chapter7/TameBoundaryEpStructure · EpGroupoid.isLocalScDiffeo_bTgt; $u$ and $\iota$ are sc-smooth index-preserving local homeomorphisms, so their lifts (7.28), (7.27) are sc-smooth Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothMap_bUnit Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothMap_bInv. *The multiplication (7.29).* Condition (4) of Definition 7.1.3 (2017: 7.3) is proved in its sheet form (§VII.1, V7.3): for every sheet $\sigma$ of $s : b\mathbf{X} \to bX$ over $V$, $b \mapsto m(\sigma(tb), b)$ is sc-smooth on $t^{-1}(V)$ Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothCompSheet_bComp. Near $b$ take a sheet $\sigma_1$ of $s : \mathbf{X} \to X$ through $B(\sigma(tb))$; then $b' \mapsto m(\sigma(tb'), b')$ is continuous and covers $\gamma \mapsto m(\sigma_1(t\gamma), \gamma)$, which is sc-smooth by (4) for $(X, \mathbf{X})$ in sheet form Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet, so the statement on maps of tame boundaries applies; the sheet form gives (4) Chapter7/EpGroupoid · isScSmoothComp_iff_sheet Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothComp_bComp. With the groupoid and the properness of §VII.40 this is an ep-groupoid, tame by Theorem 2.7.13 (2) Chapter7/TameBoundaryEpStructure · EpGroupoid.isTame_boundaryEpGroupoid. *$B$ is an ep-functor* Chapter7/TameBoundaryEpStructure · EpGroupoid.boundaryEpFunctor: a functor (§VII.40), sc-smooth on objects and morphisms, so a morphism of $\mathcal{EP}$ (Definition 7.1.7 (2017: 7.7)) Chapter7/EpGroupoid · EpFunctor. *The diagram of the sketch* Chapter7/TameBoundaryEpStructure · EpGroupoid.phiTot_bSrc: for $(\varphi, \mathbf{V}, \theta)$ with $\mathbf{V}$ in a sheet of $s$, $s$ maps $U(\varphi, \mathbf{V}, \theta)$ into $U(x, s(\mathbf{V}), s(\theta))$ and $\Phi_{(x,s(\mathbf{V}),s(\theta))} \circ s = s \circ \Phi_{(\varphi,\mathbf{V},\theta)}$ there (Findings, Chapter 7, items 145, 146) (p. 355).

**Theorem 7.6.9** (V7.101) Chapter7/TameBoundaryEpStructure · EpGroupoid.theorem_7_6_9_full. *For a tame ep-groupoid: (1) $bX$, $b\mathbf{X}$ are metrizable; (2) they carry the tame M-polyfold structures above, characterized as in Theorem 2.7.13 — for a triple with $V$ centred at $x$, $U(x, V, \theta)$ is open and $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto*

*the face $\theta$, inverse to the homeomorphism of §VII.40 — and every structure with this property is equivalent to it; (3) both maps $B$ are sc-smooth; (4) $d_X(B(p)) - 1 = d_{bX}(p)$ and $d_{\mathbf{X}}(B(a)) - 1 = d_{b\mathbf{X}}(a)$; (5) the boundary ep-groupoid is tame, with the structure maps (7.25)–(7.29), $s$, $t$ local sc-diffeomorphisms, $u$, $\iota$ sc-smooth, $m$ sc-smooth on the fibred product; and $B$ is an ep-functor. Proof.* (1) is §VII.40; (2)–(4) are Theorem 2.7.13 (2)–(4) for the object and the morphism M-polyfold; (5) and the ep-functor are the previous paragraph. *Readings.* "Natural" in (2) is read through the characterization and its uniqueness, for representatives with $V$ centred at $x$ (in Part I every representative is covered since, with image $\theta \cap V^x$, §II.37; gap G.78, closed; for the boundary ep-groupoid, §VII.50); the hypotheses of the two uniqueness statements are satisfied by the structures of (2) themselves. (4) is stated as printed, with the subtraction of natural numbers ($B$ takes values in $\partial X$, where $d_X \geq 1$); the additive form is Part I's. *Witness* `Chapter7/TameBoundaryEpStructure · EpGroupoid.trivial_quadrant_boundaryEp_witness`: the trivial ep-groupoid of the tame quadrant $[0,\infty)^n \oplus W$, $n \geq 1$, is tame, its boundary ep-groupoid is tame with nonempty $bX$, $b\mathbf{X}$, and the classes over the corner have $d_{bX} = n-1$ (Findings, Chapter 7, items 145, 147) (p. 355).

## VII.43 §7.4 on every lift $X^i$: Propositions 7.4.9, 7.4.16–7.4.18

The per-level hypotheses below — the Part I facts on every $X^i$, the density of $X_\infty$ and level-connected natural representations — hold for every ep-groupoid (§VII.45), where the forms without them are listed.

**The smooth points of $X^1$** (V7.102) `Chapter7/ZhouLevels · MPolyfold.continuous_levelDown`. The inclusion $X_{m+1} \to X_m$ is continuous; so the identity $(X^1)_\infty \to X_\infty$ and its inverse are continuous `Chapter7/ZhouLevels · MPolyfold.shiftSmoothMap` `Chapter7/ZhouLevels · MPolyfold.shiftSmoothInv` `Chapter7/ZhouLevels · MPolyfold.continuous_shiftSmoothMap` `Chapter7/ZhouLevels · MPolyfold.continuous_shiftSmoothInv`, a homeomorphism `Chapter7/ZhouLevels · MPolyfold.shiftSmoothHomeomorph`. This is the text's "$X_\infty = X^i_\infty$" in Propositions 7.4.9 and 7.4.16, as topological spaces.

**Proposition 7.4.9** (V7.102) `Chapter7/ZhouLevels · EP.proposition_7_4_9`. *If $X$ satisfies (ZHOU), so does every $X^i$.* The text says (ZHOU') is the same for $X$ and $X^i$, by $X_\infty = X^i_\infty$ and Proposition 7.4.8. *(ZHOU') passes from $X$ to $X^1$.* An isotropy element of $x \in X_1$ in $X^1$ is one in $X$ `Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal`; a natural representation of $X^1$ at $x$ with $U'(x) \subseteq U(x) \cap X_1$ acts, on $U'(x)$, by the natural action of $X$ `Chapter7/ZhouLevels · EpGroupoid.levelVal_Φ_liftOne`, and the tangent of a smooth morphism of $\mathbf{X}^1$ is, under $T(X^1) \to TX$ (Part I's Proposition 2.3.17 for $i = 1$), that of $\mathbf{X}$ `Chapter7/ZhouLevels · EpGroupoid.tangentLower_tangentMor_liftOne`. So $Tg = \mathrm{id}$ on $(T_xX^1)_\infty$ gives $Tg = \mathrm{id}$ on $(T_xX)_\infty$, (ZHOU') for $X$ gives $\hat{g} = \mathrm{id}$ near $x$ in $X_\infty$, hence near $x$ in $(X^1)_\infty$ `Chapter7/ZhouLevels · EpGroupoid.isZhou'_liftOne`, and by induction on $i$ for every $X^i$ `Chapter7/ZhouLevels · EP.isZhou'_lift`. The converse, from $X^i$ to $X$, is not used and not formalized. *The proposition.* (ZHOU) for $X$ gives (ZHOU') for $X$ by Lemma 7.4.6, which uses $(T_xX)_\infty$ dense in $T_xX$; then (ZHOU') for $X^i$, and (ZHOU) for $X^i$ by Proposition 7.4.8 on $X^i$, which uses the Part I facts on $X^i$ (§VII.37). Witness `Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_9_pointEP` (Findings, Chapter 7, item 148) (p. 333).

**Corollary 7.4.11, Theorems 7.4.13 and 7.4.24 for every $X^i$** (V7.102). With Proposition 7.4.9, the forms of §§VII.38–VII.39 that took (ZHOU) for every $X^i$ hold for (ZHOU) on $X$ only: every $X^i$ has LUC `Chapter7/ZhouLevels · EP.corollary_7_4_11_all`, $X$ has (SLUC) `Chapter7/ZhouLevels · EP.theorem_7_4_13_all` and (SLR) `Chapter7/ZhouLevels · EP.theorem_7_4_24_all`. Witnesses `Chapter7/ZhouLevels · EpGroupoid.Examples.theorem_7_4_13_all_pointEP` `Chapter7/ZhouLevels · EpGroupoid.Examples.theorem_7_4_24_all_pointEP` (pp. 334, 340).

**Propositions 7.4.16 and 7.4.17** (V7.103) `Chapter7/ZhouLevels · EP.proposition_7_4_16_all`. *If $X$ has (VSLUC), so does every $X^i$, and in particular every $X^i$ has LUC; so (VSLUC) implies (SLUC).* For $x \in X_1$ smooth,

the text's $V(x)$ is taken inside a natural representation $U'(x)$ of $X^1$ at $x$, with $U'(x) \subseteq U(x)$: $V'(x) = V(x) \cap U'(x)$, an open invariant subset of $(X^1)_\infty = X_\infty$, and $\hat{g}$ is the same map on it `Chapter7/ZhouLevels · EpGroupoid.isVSLUC_liftOne` `Chapter7/ZhouLevels · EP.isVSLUC_lift`. The density of $X_\infty$ in every level passes to $X^1$ `Chapter7/ZhouLevels · MPolyfold.DenseSmooth.shift` and so to every $X^i$ `Chapter7/ZhouLevels · EP.denseSmooth_lift`. Lemma 7.4.15 on every $X^i$ (§VII.38), with `SmoothConnected` on every $X^i$, gives LUC; and (SLUC) follows `Chapter7/ZhouLevels · EP.proposition_7_4_17_all`. Witnesses `Chapter7/ZhouLevels · EpGroupoid.Examples.isVSLUC_pointEP` `Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_17_all_pointEP` (Findings, Chapter 7, item 149) (p. 336).

**Proposition 7.4.18** (V7.104) `Chapter7/ZhouLevels · EP.proposition_7_4_18`. *(VSLUC) and (SLUC) are equivalent.* One direction is Proposition 7.4.17. *(SLUC)* □ *(VSLUC)*, the text's proof: take $V(x) = U(x) \cap X_\infty$; an open subset of it on which $\hat{g} = \mathrm{id}$ contains $W' \cap X_\infty$ for a nonempty open $W'$ of some $X_N$; by density and continuity $\hat{g} = \mathrm{id}$ on $W'$, and LUC for $X^N$ gives $\hat{g} = \mathrm{id}$ on $U(x)$, $U(x) \cap X_N$ being connected "by construction". *The neighbourhood basis.* The inclusion $X_N \to X_0$ through the levels `Chapter7/ZhouLevels · MPolyfold.levelDownTo` `Chapter7/ZhouLevels · MPolyfold.continuous_levelDownTo` is continuous, and the sets $O \cap X_\infty$, $O$ open in some $X_{N+1}$, form a neighbourhood basis of every smooth point `Chapter7/ZhouLevels · MPolyfold.exists_level_nhds_smooth`. *Natural representations one level up.* An isotropy element of $x \in X_1$ in $X$ is one in $X^1$ `Chapter7/ZhouLevels · EpGroupoid.isotropyLevelMk` `Chapter7/ZhouLevels · EpGroupoid.isotropyLevelMk_levelVal` `Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal_mul` `Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal_inv`. If $U(x) \cap X_1$ is connected in $X^1$, the natural action of $X$ restricts to $U(x) \cap X_1$ `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.liftOneΦ` `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.levelVal_liftOneΦ` `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.mapsTo_liftOneΦ` `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.isScDiffeoOn_liftOneΦ` and is a natural representation of $X^1$ at $x$ `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.liftOneRep`, with the same $\hat{g}$ `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.eqOn_of_liftOneRep`. The text's "by construction" is a hypothesis: every $U(x) \cap X_m$ connected in $X_m$ `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.LevelConnected`, which passes to the restriction `Chapter7/ZhouLevels · EpGroupoid.NaturalRep.levelConnected_liftOneRep`, so LUC for $X^N$ applies to the restriction of $U(x)$ to $X^N$, built one level at a time `Chapter7/ZhouLevels · EP.eqOn_of_level_smooth`. Hence, with the density of $X_\infty$ and level-connected natural representations at every smooth point, (SLUC) gives (VSLUC) `Chapter7/ZhouLevels · EP.isVSLUC_of_isSLUC`. Witnesses `Chapter7/ZhouLevels · EpGroupoid.Examples.levelConnected_pointEP` `Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_18_pointEP` (gaps G.73, G.85) (Findings, Chapter 7, item 150) (p. 336).

## VII.44 Definition 7.1.4 (2017: 7.4): $T\varphi$ as an sc-isomorphism; the facts (3), (4); $|X_\infty|$ and the density of $X_\infty$

The first reading left these statements of §7.1 as definitions, for want of the linear structure of $T_xX$ and of the density of smooth points in an M-polyfold. Part I now gives $T_xX$ the linear structure of Definition 2.3.14, transported from the tangent space $T_oO = Dr(o)E$ of a retract through any chart (gap G.7, closed), and the remaining steps are the chain rule and a density argument.

**$T\varphi$ as a map** (V7.5) `Chapter7/TangentMorphism · EpGroupoid.definition_7_1_4`. *For $\varphi \in \mathbf{X}_1$: $\hat{\varphi}(s(\varphi)) = t(\varphi)$; $T\varphi = T\hat{\varphi}(s(\varphi))$ on $T_{s(\varphi)}X$; $T\varphi$ is a bijection $T_{s(\varphi)}X \to T_{t(\varphi)}X$ with inverse $T(\varphi^{-1})$; $T(\psi\circ\varphi) = T\psi\circ T\varphi$; and in charts it is a bounded linear isomorphism. Step 1, $T\varphi = T\hat{\varphi}$.* On the sheet of Definition 7.1.4, $t = \hat{\varphi}\circ s$, and the chain rule for Part I's tangent maps gives $Tt = T\hat{\varphi}\circ Ts$ on $T_\varphi\mathbf{X}$; so $Tt \circ (Ts|_{T_\varphi\mathbf{X}})^{-1}$, the definition of §VII.3, is $T\hat{\varphi}$ at $s(\varphi)$ `Chapter9/Tangent · EpGroupoid.tangentMor_eq_tangentMapOn_hat`. *Step 2, the inverse.* The inverse map $\iota$ exchanges $s$ and $t$, so $T(\varphi^{-1})$ inverts $T\varphi$ on the fibres, and $T\varphi$ is a bijection be-

tween them Chapter7/TangentMorphism · EpGroupoid.bijOn_tangentMor Chapter7/TangentMorphism · EpGroupoid.invOn_tangentMor. *Step 3, in charts.* For charts $\chi$ around $s(\varphi)$ and $\psi$ around $t(\varphi)$, $T\varphi$ is represented on the fibres by $L = D(\psi \circ \widehat{\varphi} \circ \chi^{-1} \circ r)(\chi(s\varphi))$, a bounded operator Chapter7/TangentMorphism · EpGroupoid.tangentMor_chart, and $L$ and the operator $L'$ of $\varphi^{-1}$ restrict to mutually inverse maps $T_{\chi(s\varphi)}O \rightleftarrows T_{\psi(t\varphi)}O'$, by Step 2 Chapter7/TangentMorphism · EpGroupoid.tangentMor_chart_iso.

**Linearity** (V7.5) Chapter7/TangentMorphismLinear · EpGroupoid.definition_7_1_4_linear. *For $\varphi \in \mathbf{X}_1$, $T\varphi : T_{s(\varphi)}X \to T_{t(\varphi)}X$ is additive, homogeneous, continuous and bijective, with inverse $T(\varphi^{-1})$.* The linear structure of $T_xX$ is the one transported by a chart, so in the charts of Step 3 the fibre map $T\varphi$ is the restriction of the bounded linear $L$ to $T_{\chi(s\varphi)}O$; additivity, homogeneity and continuity are those of $L$, and bijectivity is Step 2. So $T\varphi$ is a continuous linear equivalence of the fibres Chapter7/TangentMorphismLinear · EpGroupoid.tangentMorCLE.

**An sc-isomorphism for smooth $\varphi$** (V7.5) Chapter7/TangentMorphismLinear · EpGroupoid.definition_7_1_4_sc. *If $\varphi$ is smooth, $T\varphi$ is an sc-isomorphism.* For smooth $\varphi$ the point $\chi(s\varphi)$ is smooth, and the derivative at a smooth point of an sc-smooth map is an sc-operator (Part I), so $L$ and $L'$ are sc-operators Chapter7/TangentMorphismLinear · EpGroupoid.tangentMor_chart_sc; they restrict to mutually inverse maps of the sc-subspaces $T_{\chi(s\varphi)}O$ and $T_{\psi(t\varphi)}O'$, which are $T_{s(\varphi)}X$ and $T_{t(\varphi)}X$ as sc-Banach spaces. This closes gap G.70; the paragraph's "bounded linear isomorphism" on level 0 for $\varphi \in \mathbf{X}_1$ and "sc-isomorphism" for smooth $\varphi$ hold as we read them (Findings, Chapter 7, items 5, 110, 154) (p. 289 (2017 p. 227)).

**The facts (3), (4) on $A$** (V7.30). For a saturated sub-M-polyfold $A$ of the object M-polyfold, $a \in A_1 \cap V$ and an sc-smooth retraction $r$ of $V$ onto $A \cap V$, the tangent of the inclusion $Ti : T_aA \to T_aX$ Chapter7/SubGroupoidTangent · EpGroupoid.subObjInclusionCLM and that of $i^{-1} \circ r$, $T_aX \to T_aA$ Chapter7/SubGroupoidTangent · EpGroupoid.subObjRetractionCLM, are bounded linear. *(3)* $Ti(T_aA)$ is a closed subspace of $T_aX$ with the closed complement $\ker T(i^{-1} \circ r)$ Chapter7/SubGroupoidTangent · EpGroupoid.fact_3_subObj, and at a smooth $a$ the complement is an sc-complement, read in a chart around $a$ Chapter7/SubGroupoidTangentSc · EpGroupoid.fact_3_subObj_sc. *(4)* $T(i^{-1} \circ r)$ is onto, $Ti \circ T(i^{-1} \circ r) = Tr(a)$, and $Tr(a)T_aX = Ti(T_aA)$, i.e. $T_aA = Tr(a)T_aX$ Chapter7/SubGroupoidTangent · EpGroupoid.fact_4_subObj. *Proof.* Both are Part I's Proposition 2.3.19 (3) in its linear form, applied to the object M-polyfold Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_3_linear: since $r \circ r = r$ near $a$, $Tr(a)$ is a bounded projection of $T_aX$ with range $Ti(T_aA)$, and $i^{-1} \circ r \circ i = \mathrm{id}$ gives the surjectivity (Findings, Chapter 7, items 40, 154) (pp. 305–307 (2017 pp. 237–238)).

**The density of $X_\infty$** (V7.6) Chapter7/SmoothPointsDense · MPolyfold.denseSmooth. *For every M-polyfold and every $m$, $X_\infty$ is dense in $X_m$, with the topology of $X_m$.* The text states this on p. 289 without proof; Part I had it for sc-Banach spaces only. *Step 1, a retract.* For an sc-smooth retraction $r$ of an open subset $U$ of a partial quadrant $C$ with retract $O$, the points of $O_\infty$ are dense in $O_m$ Chapter7/SmoothPointsDense · IsScRetract.lvlSet_subset_closure_smooth: for a partial quadrant $C$, $C_m \cap E_\infty$ is dense in $C_m$, since the smooth points of $E$ are dense in $E_m$ and $C$ is defined by finitely many continuous linear inequalities; the same holds for $U$ relatively open in $C$, the fact Part I uses in the proof of Lemma 2.2.1; and a point $o \in O_m$ is the limit in $E_m$ of smooth points $e_n$ of $U$, $r$ maps smooth points to smooth points and is continuous on level $m$, so $r(e_n) \to r(o) = o$ in $O_m$. *Step 2, the M-polyfold.* The level lift of a chart is an embedding of the chart domain in $X_m$ onto the level $m$ of its retract, so density passes to $X_m$. For an ep-groupoid, $X_\infty$ is dense in every $X_k$ and $\mathbf{X}_\infty$ in every $\mathbf{X}_k$ Chapter7/SmoothPointsDense · EpGroupoid.p289_dense; with the density of $X_\infty$ in $X$, Chapter 7's hypothesis of the first reading becomes a theorem (gap G.73, closed; Findings, Chapter 7, items 116, 153).

**The filtration of $|X|$** (V7.6) Chapter7/SmoothPointsDense · EpGroupoid.p290_orbit_filtration. $|X| = |X_0| \supset$

$|X_1| \supset \cdots \supset |X_\infty| = \bigcap_k |X_k|$. $|X_\infty| = \pi(X_\infty)$ Chapter7/SmoothPointsDense · EpGroupoid.orbitSmooth. Since morphisms preserve the levels (§VII.3), $X_k$ is saturated and $\pi^{-1}(|X_k|) = X_k$ Chapter7/SmoothPointsDense · EpGroupoid.preimage_orbitLevel; so the $|X_k|$ decrease Chapter7/SmoothPointsDense · EpGroupoid.orbitLevel_antitone and their intersection is $\pi$ of $\bigcap_k X_k = X_\infty$ Chapter7/SmoothPointsDense · EpGroupoid.orbitSmooth_eq_iInter. The text does not state that $|X_\infty|$ is dense in $|X|$; it follows from the density of $X_\infty$, $\pi$ being continuous and onto Chapter7/SmoothPointsDense · EpGroupoid.dense_orbitSmooth. For M-polyfolds the density is also recorded in the form Chapter 9 uses Chapter7/SmoothPointsDense · MPolyfold.denseSmooth (gap G.72, closed; Findings, Chapter 7, items 6, 112) (pp. 289–290 (2017 pp. 227–228)).

## VII.45 §7.4: the Part I facts as theorems, and §7.4 without them

§VII.37 listed the facts about M-polyfolds that the proofs of §7.4 use without statement; the first reading took them as one hypothesis, satisfiable on a point. They hold for every M-polyfold Chapter7/SmoothRegularAssembly · MPolyfold.smoothRegular, and every result of §7.4, Appendix 7.6.4 and Proposition 8.1.17 is restated without them (gap G.85, closed; Findings, Chapter 7, item 153).

**The four facts** (V7.86). *(a) Density.* $X_\infty$ is dense in every $X_m$ (§VII.44). *(b) Connectedness.* For a connected open $U$, $U \cap X_\infty$ is connected in $X_\infty$ Chapter7/SmoothConnectedReps · MPolyfold.smoothConnected. The local step: every point $u$ of an open $U$ has a neighbourhood $N \subseteq U$ in which any two smooth points are joined inside $U \cap X_\infty$, for the topology of $X_\infty$, by the retracted segment between their images in a chart, Lemma 2.2.1 for one convex piece followed by the retraction. The global step: for a fixed smooth point $x_0$, the points of $U$ having a neighbourhood whose smooth points are joined to $x_0$ form an open set, and by the local step so does its complement in $U$; by density it is nonempty, so it is all of the connected $U$, and $U \cap X_\infty$ is path connected Chapter2/SmoothPaths · MPolyfold.isPathConnected_smoothVal_preimage. *(c) Openness of* $p : TX \to X^1$ Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj, Part I's gap G.8, closed. *(d) Locality of tangents.* Two maps sc-smooth on open sets that agree near a smooth point $y$ in $X_\infty$ have the same tangent on $T_yX$ Chapter7/ZhouFacts · MPolyfold.tangentDeterminedSmooth: the sets $O \cap X_\infty$, $O$ open in some $X_{N+1}$, form a neighbourhood basis of $y$ in $X_\infty$ (§VII.43), so the maps agree on $O \cap X_\infty$, by density and continuity on the open set $O$ of $X_{N+1}$, and in a chart the derivative at $y$ of an $\mathrm{sc}^1$ map is determined by its values on a level-1 neighbourhood Chapter7/ZhouFacts · IsSc1.scDeriv_eq_of_eqOn_level. The fifth fact, kept apart in §VII.37: *(e)* $(T_xX)_\infty$, read as $T_xX \cap (TX)_\infty$, is dense in $T_xX$ for smooth $x$ Chapter7/ZhouFacts · MPolyfold.denseSmoothTangent: in a chart the tangent vectors at $o$ are $(o, Dr(o)k)$, $Dr(o)$ is an sc-operator at the smooth point $o$, so $k \in E_\infty$ gives smooth tangent vectors, and they are dense since $E_\infty$ is dense in $E$ and $Dr(o)$ maps $E$ continuously onto $T_oO$.

**Natural representations** (V7.83, V7.85, V7.104). *Every natural representation $U(x)$ has $U(x) \cap X_m$ connected in $X_m$ for every $m$, and $U(x) \cap X_1$ dense in $U(x)$.* The paths of (b) lie in $X_\infty$ with its topology, finer than that of every $X_m$, so $U(x) \cap X_\infty$ is connected in $X_m$; it is dense in $U(x) \cap X_m$ by (a), and the closure of a connected set is connected Chapter7/SmoothConnectedReps · EpGroupoid.NaturalRep.levelConnected Chapter7/SmoothConnectedReps · EpGroupoid.NaturalRep.isPreconnected_liftU; the density is that of $X_\infty \subseteq X_1$ in $X$ Chapter7/SmoothConnectedReps · EpGroupoid.NaturalRep.subset_closure_inter_levelSet. So the hypotheses of Appendix 7.6.4 and of Proposition 7.4.18 hold for every ep-groupoid Chapter7/SmoothConnectedReps · EpGroupoid.hasConnectedDenseLevelOneReps Chapter7/SmoothConnectedReps · EpGroupoid.exists_naturalRep_levelConnected; the text's choice of $U(x)$ "by construction" is automatic.

**The restatements.** With (a)–(e) and the natural representations, the results of §§VII.36–VII.39 and VII.43 hold with only the hypotheses the text states. *(ZHOU).* Lemma 7.4.6 Chapter7/

ZhouFacts · EpGroupoid.lemma_7_4_6_of_isZhou and Proposition 7.4.8 Chapter7/SmoothRegularAssembly · EpGroupoid.proposition_7_4_8_unconditional; Proposition 7.4.10 and the unique continuation through $X_\infty$ Chapter7/SmoothRegularAssembly · EpGroupoid.proposition_7_4_10_of_isZhou Chapter7/SmoothRegularAssembly · EpGroupoid.isSmoothUC_of_isZhou'; Corollary 7.4.11 for $X^1$ and for every $X^i$ Chapter7/SmoothRegularAssembly · EpGroupoid.corollary_7_4_11_of_isZhou Chapter7/SmoothRegularAssembly · EP.corollary_7_4_11_all_of_isZhou; Proposition 7.4.9 Chapter7/SmoothRegularAssembly · EP.proposition_7_4_9_of_isZhou; Theorem 7.4.13 Chapter7/SmoothRegularAssembly · EP.theorem_7_4_13_all_of_isZhou; Proposition 7.4.22 and Theorem 7.4.24 Chapter7/SmoothRegularAssembly · EpGroupoid.proposition_7_4_22_of_isZhou Chapter7/SmoothRegularAssembly · EP.theorem_7_4_24_all_of_isZhou, all under (ZHOU) for $X$ alone. *(VSLUC).* Lemma 7.4.15 and its unique continuation Chapter7/SmoothConnectedReps · EpGroupoid.isSmoothUC_of_isVSLUC' Chapter7/SmoothConnectedReps · EpGroupoid.lemma_7_4_15_of_isVSLUC; Propositions 7.4.16 and 7.4.17, for $X^1$ and for every $X^i$ Chapter7/SmoothConnectedReps · EpGroupoid.proposition_7_4_16_of_isVSLUC Chapter7/SmoothConnectedReps · EP.proposition_7_4_16_all_of_isVSLUC Chapter7/SmoothConnectedReps · EP.proposition_7_4_17_of_isVSLUC Chapter7/SmoothConnectedReps · EP.proposition_7_4_17_all_of_isVSLUC, under (VSLUC) for $X$ alone; and Proposition 7.4.18, (VSLUC) $\Leftrightarrow$ (SLUC), with no hypothesis at all Chapter7/SmoothConnectedReps · EP.isVSLUC_of_isSLUC' Chapter7/SmoothConnectedReps · EP.proposition_7_4_18_unconditional. *Appendix 7.6.4.* Theorem 7.6.7 for $i = 1$ as sets under LUC for $X^1$, and Corollary 7.6.8 under (ZHOU) Chapter7/SmoothConnectedReps · EpGroupoid.theorem_7_6_7_of_isLUC Chapter7/SmoothConnectedReps · EpGroupoid.corollary_7_6_8_of_imp Chapter7/SmoothRegularAssembly · EpGroupoid.corollary_7_6_8_of_isZhou'; the full theorem is §VII.46. *The example.* The analogue of Zhou's example fails (ZHOU) with no hypothesis (§VII.39). Lemma 7.4.4 is not restated; its hypothesis now holds (§VII.37). The forms with the hypotheses are kept, since the hypotheses are defined in files the new proofs import (Findings, Chapter 7, items 133, 134, 136–139, 148–150, 153) (pp. 329–340, 349).

## VII.46 Appendix 7.6.4: Theorem 7.6.7 and Corollary 7.6.8 for every $i$

**Theorem 7.6.7 for $i = 1$** (V7.83) Chapter7/GapP2_ARaise · EpGroupoid.theorem_7_6_7_one. *If $T_R$ is Hausdorff for $X$ and for $X^1$ (equivalently, by Theorem 7.2.22, $X$ and $X^1$ have LUC), the natural identification of $(X^1)_R$ with $(X_R)^1$ — the identity on the common object M-polyfold $X^1$ and $[\varphi]_1 \mapsto [\varphi]$ on morphisms — is an sc-diffeomorphism on objects and on morphisms, and the two functors are inverse isomorphisms of ep-groupoids.* §VII.36 had the identification as a continuous injection of sets. *Step 1, the objects.* $(X^1)_R$ and $(X_R)^1$ both have the object M-polyfold $X^1$, by definition. *Step 2, the projection on level* 1. The projection $\pi^R : \mathbf{X} \to \mathbf{X}^R$ is a local sc-diffeomorphism (§VII.16), so its lift $\pi^{R,1} : \mathbf{X}^1 \to (\mathbf{X}^R)^1$ is one Chapter7/GapP2_ARaise · EpGroupoid.reducedπOne Chapter7/GapP2_ARaise · EpGroupoid.isLocalScDiffeo_reducedπOne. *Step 3, the map.* $[\varphi]_1 \mapsto [\varphi]$ Chapter7/GapP2_ARaise · EpGroupoid.reducedRaise is well defined and injective because the two relations agree on level 1: $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ under the Hausdorff hypothesis on $X^1$ Chapter7/GapP2_ARaise · EpGroupoid.nonEffectivePart_liftOne_eq_of_t2 (§VII.36, (7.24) and its converse); it is onto, since the class of a level-1 point of $\mathbf{X}^R$ consists of level-1 morphisms (levels are preserved, §VII.3) Chapter7/GapP2_ARaise · EpGroupoid.bijective_reducedRaise. *Step 4, sc-smoothness.* Near $[\varphi]_1$ the map is $\pi^{R,1} \circ (\pi_1^R|U)^{-1}$, with $\pi_1^R : \mathbf{X}^1 \to (\mathbf{X}^1)^R$ the projection of $X^1$, a local sc-diffeomorphism by §VII.16 for $X^1$, and $U$ a sheet; its inverse is locally $\pi_1^R \circ (\pi^{R,1}|U)^{-1}$. Both are sc-smooth Chapter7/GapP2_ARaise · EpGroupoid.isScSmoothMap_reducedRaise Chapter7/GapP2_ARaise · EpGroupoid.isScSmoothMap_reducedRaiseInv, so the map is an sc-diffeomorphism Chapter7/GapP2_ARaise · EpGroupoid.isScDiffeoOn_reducedRaise. *Step 5, the functors.* The map commutes with sources and composition Chapter7/GapP2_ARaise · EpGroupoid.src_reducedRaise Chapter7/GapP2_ARaise · EpGroupoid.reducedRaise_comp, so with the identity on objects it is an ep-functor

$(X^1)_R \to (X_R)^1$ Chapter7/GapP2_ARaise · EpGroupoid.raiseReducedFunctor, and its inverse another Chapter7/GapP2_ARaise · EpGroupoid.raiseReducedFunctorInv, the two composites being identities. The hypotheses hold on the point Chapter7/GapP2_ARaise · EpGroupoid.Examples.theorem_7_6_7_pointEP.

**Theorem 7.6.7 for every $i$, and Corollary 7.6.8** (V7.83) Chapter7/GapP2_ARaise · EP.theorem_7_6_7. *Under (SLUC), $(X^i)_R \cong (X_R)^i$ in $\mathcal{EP}$ for every $i$; under (ZHOU), the same.* Under LUC, $T_R$ is Hausdorff (Theorem 7.2.22), so $X_R$ is an object of $\mathcal{EP}$ Chapter7/GapP2_ARaise · EP.reducedOfLUC, and the case $i = 1$ is an isomorphism $(X^1)_R \cong (X_R)^1$ Chapter7/GapP2_ARaise · EP.raiseReducedIso. Lifting to level one carries isomorphisms of $\mathcal{EP}$ to isomorphisms Chapter7/GapP2_ARaise · EP.liftOneIso. The isomorphism for $i$ is built by recursion Chapter7/GapP2_ARaise · EP.raiseReducedIsoLift: the identity for $i = 0$, and for $i + 1$ the composite $(X^{i+1})_R = ((X^i)^1)_R \cong ((X^i)_R)^1 \cong ((X_R)^i)^1 = (X_R)^{i+1}$ of the case $i = 1$ for $X^i$, which uses LUC for $X^i$ and $X^{i+1}$, and the lift of the isomorphism for $i$. This is the composite of identifications of the text's proof. Corollary 7.6.8 follows, since (ZHOU) implies (SLUC) (Theorem 7.4.13, §VII.45) Chapter7/GapP2_ARaise · EP.corollary_7_6_8. *Reading.* "The identity on objects" is stated for $i = 1$ only: for variable $i$ the object types of $(X^i)_R$ and $(X_R)^i$ are equal but not by definition, and the statement is an isomorphism in $\mathcal{EP}$ (gap G.82, closed; Findings, Chapter 7, item 131) (pp. 349–351).

## VII.47 Appendix 7.6.3: Zhou's retract

*Appendix 7.6.3 (p. 348), paraphrased.* On $E = L^2(\mathbb{R})$ with the levels $E_m = H^{m,\delta_m}(\mathbb{R})$, $0 = \delta_0 < \delta_1 < \cdots$, and a smooth compactly supported $\beta \geq 0$ with $\int \beta^2 = 1$, the map $r(t, f) = (t, \pi_t f)$ on $\mathbb{R} \oplus E$, with $\pi_t = 0$ for $t \leq 0$ and $\pi_t f = \langle f, \beta_c \rangle \beta_c$, $\beta_c = \beta(\cdot + c)$, $c = e^{1/t}$, for $t > 0$, is an sc-smooth retraction (the text cites [60] and Part I's Subsection 1.4.4). Its retract $Y$ is a point over each $t \leq 0$ and a line over each $t > 0$. $\mathbb{Z}/2$ acts on $Y$ by $(t, f) \mapsto (t, -f)$; the text computes the non-effective part on $\Sigma \cong Y$ and finds one point in $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$. The first reading used an analogue (§VII.36); the example itself is formalized as follows.

**The sc-Hilbert space** (V7.82) Chapter7/GapP2_A2Line · ZhouLine.lineSc. *For $0 = \delta_0 < \delta_1 < \cdots$, $E_m = H^{m,\delta_m}(\mathbb{R})$ is an sc-structure on $L^2(\mathbb{R})$, with level 0 equal to $L^2(\mathbb{R})$* Chapter7/GapP2_A2Line · ZhouLine.level0_line. *Step 1, the levels.* $H^{m,\delta}(\mathbb{R})$ is taken inside $L^2(\mathbb{R})$ as the $f$ whose lift $f \circ \mathrm{pr}_1$ to the cylinder $\mathbb{R} \times S^1$ lies in Part I's weighted Sobolev space $H^{m,\delta}(\mathbb{R} \times S^1)$, normed by the lift Chapter7/GapP2_A2Line · ZhouLine.H1 Chapter7/GapP2_A2Line · ZhouLine.liftH; a function of $s$ alone has no $t$-derivatives, so the norm is $\sum_{a \leq m} \|e^{\delta|s|} \partial_s^a f\|^2_{L^2}$. Completeness holds because the range of the lift is closed. *Step 2, compactness.* The inclusions $E_{m+1} \to E_m$ are restrictions of Part I's compact inclusions on the cylinder Chapter7/GapP2_A2Line · ZhouLine.isCompactOperator_incl1. *Step 3, density.* Averaging over $S^1$ Chapter7/GapP2_A2Line · ZhouLine.avgH, of norm at most 1 on every level and the adjoint of the lift, maps the dense test functions of the cylinder to test functions of $\mathbb{R}$, which are therefore dense in every level Chapter7/GapP2_A2Line · ZhouLine.dense_ofTest1. *Reading.* The level is defined through the lift; the equivalence with the one-dimensional definition by weak derivatives is not formalized, and the norms are the same.

**Translates** (V7.82). The translate $\tau_c \chi = \chi(\cdot + c)$ of a test function lies in every level Chapter7/GapP2_A2Shift · ZhouLine.trH, with $\|\tau_c \chi\|_{H^{m,\delta}} \leq C e^{\delta(|c|+R)}$ for $\chi$ supported in $[-R, R]$ Chapter7/GapP2_A2Shift · ZhouLine.norm_trH_le, while the pairing with a far translate is small, $\|\langle \tau_c \chi, \cdot \rangle\|_{(H^{m,\delta})'} \leq C e^{-\delta(c-R)}$, since $e^{-\delta|s|} \leq e^{-\delta(c-R)}$ on the support of $\tau_c \chi$ Chapter7/GapP2_A2Shift · ZhouLine.norm_ellL_tr0_le. Both depend smoothly on $c$, with derivative given by $\chi'$ Chapter7/GapP2_A2Shift · ZhouLine.hasDerivAt_trH Chapter7/GapP2_A2Shift · ZhouLine.hasDerivAt_ellT.

**Flatness at $t = 0$** (V7.82). *If $g = 0$ for $t \leq 0$, $g$ is $C^\infty$ on $t > 0$ and every derivative tends to 0 at $0^+$, then*

*$g$ is $C^\infty$* Chapter7/GapP2_A2Flat · ZhouFlat.contDiff_of_flat. *If $Q$ is $C^\infty$ with every derivative bounded by $Ke^{-\varepsilon c}$, $\varepsilon > 0$, then $t \mapsto Q(e^{1/t})$ for $t > 0$, $0$ for $t \le 0$, is $C^\infty$* Chapter7/GapP2_A2Flat · ZhouFlat.contDiff_flat_comp: by Faà di Bruno's formula the derivatives of $Q(e^{1/t})$ are sums of derivatives of $Q$ at $e^{1/t}$ times polynomials in $1/t$ and $e^{1/t}$, and the decay $e^{-\varepsilon e^{1/t}}$ beats them as $t \to 0^+$.

**The retraction** (V7.82) Chapter7/GapP2_A2Family · ZhouLine.zhouRetraction. *$r(t, f) = (t, \pi_t f)$ is an sc-smooth retraction of $\mathbb{R} \oplus E$.* The operators $\pi_t$ on each level Chapter7/GapP2_A2Family · ZhouLine.zhouFam form a loss family in the sense of Part I's criterion of Remark 4.1.3 Chapter7/GapP2_A2Family · ZhouLine.isLossFamily_zhouFam, which gives the sc-smoothness of $r$: *compatibility*, $\pi_t$ commutes with the inclusions of the levels; *continuity*, $(t, f) \mapsto \pi_t f$ is continuous on every level, from a bound on $\|\pi_t\|$ uniform in $t$ and the continuity on the dense test functions; *loss of one level*, $t \mapsto \pi_t \circ \iota$, $\iota : E_{m+1} \to E_m$, is $C^\infty$ into the bounded operators: for $t > 0$ it is $Q(e^{1/t})$ with $Q(c) = \langle \cdot, \beta_c \rangle \beta_c$ as an operator $E_{m+1} \to E_m$, whose derivatives satisfy $\|D^n Q(c)\| \le Ke^{-(\delta_{m+1} - \delta_m)c}$ by the two bounds on translates Chapter7/GapP2_A2Family · ZhouLine.norm_iteratedFDeriv_zhouQ_le, and flatness applies with $\varepsilon = \delta_{m+1} - \delta_m > 0$; *idempotence*, $\pi_t \pi_t = \pi_t$ since $\langle \beta_c, \beta_c \rangle = \int \beta^2 = 1$. So the sc-smoothness the text cites from [60] is proved, and only one level of loss is used. *The data* Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data are the weights and a test function $\beta$ with $\int \beta^2 = 1$; the text's $\beta \ge 0$ is not needed. A $\beta$ as the book requires exists Chapter7/GapP2_A2Family · ZhouLine.exists_beta, and $\delta_m = m$ with a normalized bump is a witness of the data Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data.std.

**The ep-groupoid and its non-effective part** (V7.82) Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.appendix_7_6_3. *For $\mathbb{Z}/2$ acting on $Y$ by $(t, f) \mapsto (t, -f)$: $\mathbf{X}^T = (\{1\} \times Y) \cup (\{-1\} \times \{t < 0\})$, $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T = \{(-1, (0, 0))\}$, $T_R$ is not Hausdorff, and LUC and (ZHOU) fail.* The action is the restriction of the sc-operator $(t, f) \mapsto (t, \pm f)$, which commutes with $r$ Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.isScSmoothMap_smul, and the translation groupoid of §VII.30 is an ep-groupoid Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.X. *Step 1, the retract as printed.* $Y$ is $\{(t, 0)\}$ over $t \le 0$ and the line $\mathbb{R}\beta_{e^{1/t}}$ over each $t > 0$ Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.mem_0_iff'. *Step 2, $\mathbf{X}^T$.* $1$ acts trivially; $-1$ acts trivially near a point over $t < 0$, where $Y$ is the half-line of points $(t, 0)$, and near no point over $t \ge 0$, where points with $f \ne 0$ accumulate Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.nonEffectivePart_eq. *Step 3, the closure.* $(-1, (0, 0))$ is the limit of $(-1, (t, 0))$, $t \to 0^-$, and the only point of the closure not in $\mathbf{X}^T$ Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.closure_nonEffectivePart_diff. *Step 4, the consequences.* $\mathbf{X}^T$ is not closed, so $T_R$ is not Hausdorff Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_t2Space_reducedMor and LUC fails Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_isLUC (Theorem 7.2.22); (ZHOU) implies LUC (Proposition 7.4.10, whose Part I facts hold, §VII.45), so (ZHOU) fails Chapter7/GapP2_A2ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_isZhou.

**$\Sigma$ as printed** (V7.82) Chapter7/GapP2_A2Sigma · EpGroupoid.Examples.ZhouRetract.appendix_7_6_3_sigma. The text works on $\Sigma = \{(s, t) \in \mathbb{R}^2 \mid t = 0 \text{ if } s \le 0\}$, with $\mathbb{Z}/2$ written $\{0, 1\}$. $(s, t) \mapsto (s, t\beta(\cdot + e^{1/s}))$ (and $(s, 0)$ for $s \le 0$) is a homeomorphism $\Sigma \cong Y$, with inverse $(s, f) \mapsto (s, \langle f, \beta_{e^{1/s}} \rangle)$, using $\|\beta_c\| = 1$ Chapter7/GapP2_A2Sigma · EpGroupoid.Examples.ZhouRetract.sigmaHomeo; pulled back to $\{0, 1\} \times \Sigma$ (with $0 \mapsto 1$, $1 \mapsto -1$), the non-effective part and $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$, the single point $(1, (0, 0))$ in the text's notation, are the printed sets Chapter7/GapP2_A2Sigma · EpGroupoid.Examples.ZhouRetract.nonEffectivePart_sigma Chapter7/GapP2_A2Sigma · EpGroupoid.Examples.ZhouRetract.closure_nonEffectivePart_sigma. Every claim of the appendix holds as we read it (gap G.83, closed; rows V7.82, V7.95; Findings, Chapter 7, items 130, 139, 156) (pp. 331, 348).

## VII.48 Propositions 7.2.10, 7.2.16 and Theorem 7.2.14 in the 2021 form; Corollary 7.6.6 as printed

**Proposition 7.2.10, 2021 form** (V7.39) `Chapter7/GapP2_AReduced · EpGroupoid.proposition_7_2_10_natural_charts`. *$\mathbf{X}^R$ with the quotient topology $T_R$ has the natural charts $\Psi_\varphi = \psi \circ \sigma \circ s$; their domains cover $\mathbf{X}^R$ and any two are sc-smoothly compatible.* This is Steps 1–5 of §VII.16, which need no hypothesis; the 2021 text states the proposition as the charts, and leaves Hausdorffness to Theorem 7.2.14, as the formalization had. The charts form the atlas of §VII.16 `Chapter7/Reduced · EpGroupoid.reducedAtlas` (p. 313).

**Theorem 7.2.14 and Proposition 7.2.16** (V7.40) `Chapter7/GapP2_AReduced · EpGroupoid.theorem_7_2_14_reduced` `Chapter7/GapP2_AReduced · EpGroupoid.proposition_7_2_16`. *If $T_R$ is Hausdorff, $X_R = (X, \mathbf{X}^R)$ is an ep-groupoid with the structure maps of $X$ descended to $\mathbf{X}^R$, the object M-polyfold of $X$ and the natural M-polyfold structure on $\mathbf{X}^R$; it is effective; its isotropy group at $x$ is $G_x/T_x$; and $\pi$ is an ep-functor $X \to X_R$.* The proof is §VII.17 — Proposition 7.10 (2017) — with the Hausdorff hypothesis only: paracompactness is Proposition 7.2.12 (§VII.34), an instance, so the M-polyfold $\mathbf{X}^R$ of §VII.16 is built from Hausdorffness alone. Each clause is a component already proved: the groupoid and the structures are those of §§VII.16–VII.17 by definition, effectiveness is §VII.17, the isotropy is §VII.17's $G_x/T_x$, and the functor is the reduction functor. The hypothesis is the text's own, satisfied by every effective ep-groupoid and by every ep-groupoid with LUC (§VII.35) (rows V7.39, V7.40, now same; Findings, Chapter 7, items 56, 60, 107, 111, 125, 126) (pp. 318–319).

**Corollary 7.6.6 as printed** (V7.68) `Chapter7/GapP2_ACor766 · EpGroupoid.corollary_7_6_6_unconditional`. *If $|X|$ is paracompact and the object M-polyfold has an atlas whose local models have Hilbert spaces $E_0$, $X$ has sc-smooth partitions of unity.* The text's proof: Part I's Corollary 5.5.17 and Theorem 5.5.10 give sc-smooth partitions of unity on the object M-polyfold, and Theorem 7.5.7 applies. With Part I's Theorem 5.5.16 now a theorem (§VII.29), the first step needs no hypothesis `Chapter5/GapP1_SeparablePOU · MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional` and the corollary is the text's two lines. The hypotheses are satisfiable on $\mathbb{R}^1$ `Chapter7/GapP2_ACor766 · EpGroupoid.corollary_7_6_6_hypotheses_rn_unconditional` (row V7.68, now same; Findings, Chapter 7, item 95) (p. 347). Remark 7.5.6, which points to this case, is an assessment with nothing to formalize (row V7.57; §VII.24).

## VII.49 Property (5) does not follow from (1)–(4): an ep-groupoid with a non-paracompact orbit space

*The text (p. 340 (2017 p. 249)).* After Definition 7.5.1 the text recalls that $|X|$ is always Hausdorff (Lemma 7.3.4), says that it is presently not known whether (5) is a consequence of (1)–(4), and that it is when $X$ is second countable. **The answer is no** `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_epGroupoid_not_hasParacompactOrbitSpace`. The orbit space of an ep-groupoid is Hausdorff and locally the quotient of an M-polyfold by a finite group; a non-metrizable surface is such a space, as the orbit space of the Čech groupoid of an atlas. The example is the Čech groupoid of the atlas of a Prüfer-type surface (after Prüfer; Calabi–Rosenlicht 1953).

**The surface.** One chart $\mathbb{R}^2$ for each $a$ in an uncountable discrete index set $\iota$, injected into $\mathbb{R}$ by $c$; the chart $a$ is glued to the chart $b$ along the open upper half-planes by $(u, y) \mapsto (u+(a-b)/y, y)$. In the canonical coordinates $(X, l)$, $y = e^l$, $X = u + ae^{-l}$, the gluing is the identity; the coordinate change $\mathrm{chartPt}_a(X, l) = (X - ae^{-l}, e^l)$ `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.chartPt` is a diffeomorphism of $\mathbb{R}^2$

onto the open upper half-plane, with inverse $\mathrm{canon}_a$ Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.canon. So the surface is the open upper half-plane with one closed lower half-plane attached along its own copy of the real axis for each $a$.

**The ep-groupoid** Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.epGroupoid. Objects $\iota \times \mathbb{R}^2$ and morphisms $(\iota \sqcup \{(a,b) \mid a \neq b\}) \times \mathbb{R}^2$, disjoint unions of copies of $\mathbb{R}^2$ over discrete sets, hence M-polyfolds, metrizable and paracompact. The sheet of $a$ carries the units of the chart $a$, and the sheet of $(a,b)$ the morphisms from $(a, \mathrm{chartPt}_{c(a)}(p))$ to $(b, \mathrm{chartPt}_{c(b)}(p))$, $p$ the canonical coordinates. There is at most one morphism between two objects, so composition, units and inverses are determined by sources and targets, and the groupoid laws reduce to that uniqueness. *(1)* On each sheet, $s$ and $t$ are the identity or chartPt, diffeomorphisms onto open sets; a $C^\infty$ map on an open subset of $\mathbb{R}^m$ is sc-smooth there Chapter7/PropertyFiveResearch · isScSmoothOn_rn_rn_of_contDiffOn. *(2), (4)* are checked sheet by sheet, (4) in the sheet form Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.isScSmoothCompSheet. *(3)* With $V(x)$ a unit ball in one chart: a morphism with source over $\mathrm{cl}\, V(x)$ and target in a compact $K$ lies in one of finitely many sheets, and on the sheet $(a,b)$, $a \neq b$, both $e^l$ and $(a-b)e^{-l}$ are bounded, so the canonical coordinates are bounded Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_isProperMap.

**Not (5)** Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.not_paracompactSpace_orbitSpace. *Step 1.* The point $d_a = |(a,0)|$ is the only point of its orbit in the charts, since the axis $y = 0$ is not glued. The open sets $W_a = |X| \setminus \{d_b \mid b \neq a\}$ cover $|X|$, and $d_a \in W_b$ only for $b = a$; so a locally finite open refinement has, for every $a$, a member containing $d_a$, and these members are distinct for distinct $a$. *Step 2, the density step.* Every open set containing $d_a$ contains $|(a,(0,y))|$ for small $y > 0$, which is a point $|(a_0,q)|$ of a fixed chart $a_0$, and so a point $|(a_0,r)|$ with $r \in \mathbb{Q}^2$ Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_rat_mem_of_mem. *Step 3.* Each of the countably many points $|(a_0,r)|$ lies in finitely many members of the refinement, so only countably many members contain some $d_a$, and $\iota$ would be countable. For $\iota = \mathbb{R}$ with the discrete topology Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.pruefer this is absurd: the orbit space is not paracompact Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.pruefer_not_hasParacompactOrbitSpace, though Hausdorff, as Lemma 7.3.4 says Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.t2Space_orbitSpace_pruefer. *Reading.* "(1)–(4)" are the four conditions of Definition 7.1.3 as formalized, including the existential reading of (3) and the sheet form of (4); no second countability is assumed. The object space has uncountably many components and is not second countable, in line with the text's remark that second countability gives (5) (§VII.23) (Findings, Chapter 7, items 82, 113, 152) (p. 340 (2017 p. 249)).

## VII.50 Appendix 7.6.5: Theorem 7.6.9 for every representative, and in the book's form

**Theorem 7.6.9 for every representative** (V7.101) Chapter7/TameBoundaryFull · EpGroupoid.theorem_7_6_9_allReps. *For a tame ep-groupoid: (1)–(5) and "$B$ is an ep-functor" as in §VII.42, and, for every representative $(x,V,\theta)$ of $bX$ and of $b\mathbf{X}$, centred or not: $U(x,V,\theta)$ and $\theta \cap V^x$ are open, $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism of $U(x,V,\theta)$ onto $\theta \cap V^x$, which is all of $\theta$ for $V$ centred at $x$; and every M-polyfold structure with this property for every representative is equivalent to the one of (2).* §VII.42 characterized the structure through centred representatives. The proof applies Part I's Theorem 2.7.13 for every representative — the chart $\Phi$ of a non-centred triple is an sc-diffeomorphism onto $\theta \cap V^x$, and the uniqueness holds for that family — to the object and to the morphism M-polyfold (§II.37; gap G.78, closed). No hypothesis is added; tameness is met by the trivial

ep-groupoid of a quadrant (§VII.42) (Findings, Chapter 7, item 151).

**Theorem 7.6.9 in the book's form** (V7.98) `Chapter7/BookFormTameBoundaryEp2021 · EpGroupoid.theorem_7_6_9_book`. *For a tame ep-groupoid: (1) $bX$, $b\mathbf{X}$ are metrizable, with the only topologies for which every $U(x, V, \theta)$ is open and every $\Phi_{(x,V,\theta)}$ is open into $\theta$; (2) the structures are tame, every $\Phi$ is an open embedding and an sc-diffeomorphism onto the open set $\theta \cap V^x$, onto $\theta$ for centred $V$, and the structures are unique up to equivalence; (3) both maps $B$ are sc-smooth; (4) the index formulas hold; (5) the boundary ep-groupoid is tame with the structure maps (7.25)–(7.29); and $B$ is an ep-functor.* The statement assembles the topological half of §VII.40 and the previous paragraph into the order of the text. *Reading.* After the theorem the text says that $\Phi$ is a homeomorphism onto $\theta$. That holds as we read it for $V$ centred at $x$ only; for other representatives the image is $\theta \cap V^x$, a proper subset in general, as in Chapter 2's (2.44); the counterexample is a non-centred triple of the quadrant $[0, \infty)^3 \oplus W$, whose $\Phi$ is not onto $\theta$ `Chapter2/TameBoundaryNonCentred · MPolyfold.not_surjOn_phiTot`, and every class has a centred representative `Chapter7/TameBoundaryEp · TameBoundary.exists_rep_centered`. This is the one row of the chapter formalized in a narrower form (row V7.98; Findings, Chapter 7, items 141, 155) (p. 355).

# Part VIII. Bundles and covering functors (HWZ Chapter 8, §§8.1–8.3, 8.5, 8.6 (2017 §§8.1–8.5))

Chapter 8 continues Part II; as for Chapter 7, its numbers and pages are those of the book (Springer 2021, pp. 357–400), with the 2017 text's in parentheses (2017 pp. 257–289). It treats bundles over ep-groupoids and covering functors. §8.1 equips the tangent $T(X,\mathbf{X}) = (TX, T\mathbf{X})$ of an ep-groupoid with the structure maps (8.1)–(8.9) and proves that it is an ep-groupoid, tame if $X$ is, with an sc-smooth projection functor to $X^1$ (Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8)); it describes the morphisms of $T\mathbf{X}$ by pairs $(T\varphi, a)$ (Remark 8.1.10 (2017: 8.1)), proves that the tangent of an sc-smooth functor is an sc-smooth functor and that $T : \mathcal{EP} \to \mathcal{EP}$ is a functor (Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3)), defines natural transformations and natural equivalence (Definition 8.1.13 (2017: 8.1)), proves that the tangent of a natural transformation is one (Proposition 8.1.14 (2017: 8.1)), and that the paracompactness of $|X|$ passes to $|TX|$ (Theorem 8.1.15 (2017: 8.4)). §8.2 defines sc-differential forms on ep-groupoids and the sc-de Rham complex (Definitions 8.2.1, 8.2.2 (2017: 8.2, 8.3)). §8.3 defines strong bundles over ep-groupoids (Definition 8.3.1 (2017: 8.4)) with their functorial picture (pp. 375–377 (2017 pp. 271–272)), strong bundle maps and strong bundle functors (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) and section functors (Definition 8.3.4 (2017: 8.7)). §8.5 (2017 §8.4) defines proper covering functors (Definition 8.5.1 (2017: 8.8)), proves that they compose (Proposition 8.5.2 (2017: 8.2)), describes them near a fibre (Theorem 8.5.3 (2017: 8.5), Remark 8.5.4 (2017: 8.2)), defines covering functors of finite groupoids and geometric lifts (Definition 8.5.5 (2017: 8.9)), constructs the proper covering functor of a geometric lift (Proposition 8.5.7 (2017: 8.3)), shows that every proper covering functor is locally of this form (Theorem 8.5.8 (2017: 8.6)), and defines proper strong bundle covering functors (Definition 8.5.9 (2017: 8.10)). The appendix counts the fibres of a proper covering functor (§8.6.1 (2017 §8.5.1): Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11), Proposition 8.6.4 (2017: 8.4)) and describes strong bundle coverings fibrewise (§8.6.2 (2017 §8.5.2)). This Part retells §8.1 in §§VIII.1–VIII.9 — the tangent data (8.1)–(8.9) and the tangents of maps defined on open sets (§VIII.1), Theorem 8.1.1 (2017: 8.1) and Lemma 8.1.2 (2017: 8.1) (§VIII.2), Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5) (§VIII.3), Lemmas 8.1.7 and 8.1.8 (2017: 8.6 and 8.7) (§VIII.4), Lemma 8.1.9 (2017: 8.8) (§VIII.5), Remark 8.1.10 (2017: 8.1) (§VIII.6), Theorems 8.1.11 and 8.1.12 (2017: 8.2 and 8.3) (§VIII.7), Definition 8.1.13 (2017: 8.1), the equivalence relation $\simeq$ and Proposition 8.1.14 (2017: 8.1) (§VIII.8), Theorem 8.1.15 (2017: 8.4) (§VIII.9); §8.3 in §§VIII.10–VIII.14 — strong bundle maps and the pull-back along $s$ (§VIII.10), Definition 8.3.1 (2017: 8.4) and the fibre maps (§VIII.11), the functorial picture (§VIII.12), Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6) and their equivalence (§VIII.13), Definition 8.3.4 (2017: 8.7) (§VIII.14); §8.5 (2017 §8.4) in §§VIII.15–VIII.22 — Definition 8.5.1 (2017: 8.8) and the degeneracy remark (§VIII.15),

Proposition 8.5.2 (2017: 8.2) (§VIII.16), Theorem 8.5.3 (2017: 8.5) and Remark 8.5.4 (2017: 8.2) (§VIII.17), $\mathrm{Diff}_{sc}$, Definition 8.5.5 (2017: 8.9) and the discussion after it (§VIII.18), disjoint unions of M-polyfolds (§VIII.19), Proposition 8.5.7 (2017: 8.3) (§VIII.20), Theorem 8.5.8 (2017: 8.6) (§VIII.21), Definition 8.5.9 (2017: 8.10) (§VIII.22); the appendix in §§VIII.23–VIII.25 — §8.6.1 (2017 §8.5.1) and Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11) (§VIII.23), Proposition 8.6.4 (2017: 8.4) (§VIII.24), §8.6.2 (2017 §8.5.2) (§VIII.25); the witnesses and the counterexample (§VIII.26); what is not formalized, among it §8.2 (§VIII.27); and what the formalization taught us (§VIII.28). Each item carries the number of its comparison item (V8.1–V8.43) in `Exposition/verification/ch08.md`. The additions of the 2021 edition (V8.44–V8.52) are retold in §§VIII.29–VIII.33: §8.1 (§VIII.29), §§8.3–8.4 (§VIII.30), §8.5 (§VIII.31), what is not formalized and what they taught us (§VIII.32), and Proposition 8.1.17 (2) for every $(TX)^i$ (§VIII.33). The work that closed the remaining gaps is retold in §§VIII.34–VIII.40: Lemma 8.1.2 as printed and the M-polyfold of pairs of Remark 8.1.10 (§VIII.34), Definition 8.2.1 (§VIII.35), Definition 8.2.2 and the pull-back of forms (§VIII.36), the Poincaré lemma, Proposition 8.2.3 (§VIII.37), Proposition 8.1.16 without added hypothesis (§VIII.38), Propositions 8.4.1 and 8.4.3 (§VIII.39), and Lemma 8.5.10, Proposition 8.5.7 for $D = \emptyset$, Theorem 8.5.8 in every universe and Proposition 8.1.17 without added hypotheses (§VIII.40).

**What is proved, and what is not.** Every statement and every definition of the chapter is formalized, and every gap of the chapter is closed: G.74 (Definitions 8.2.1, 8.2.2, §§VIII.35–VIII.36), G.75 (Lemma 8.1.2 as printed, §VIII.34), G.86 (Proposition 8.4.1 (3), §VIII.39), G.87 (the Poincaré lemma, Proposition 8.2.3, §VIII.37), G.88 (Proposition 8.1.16 without added hypothesis, §VIII.38) and G.89 (Lemma 8.5.10, §VIII.40). Theorems 8.1.1–8.5.8 (2017: 8.1–8.6) are proved, Theorem 8.5.8 (2017: 8.6) for ep-groupoids in any one universe, and Theorem 8.5.3 (2017: 8.5) with (5) read with $s$ on both vertical arrows; so are Propositions 8.1.14, 8.5.2 and 8.6.4 (2017: 8.1, 8.2 and 8.4), the 2021 Propositions 8.1.16, 8.1.17, 8.2.3, 8.4.1 and 8.4.3, Proposition 8.5.7 (2017: 8.3) for nonempty $D$, Lemmas 8.1.2–8.6.3 (2017: 8.1–8.11), Lemma 8.1.2 (2017: 8.1) also as printed, Lemma 8.5.10, Remark 8.1.10 (2017: 8.1) with its M-polyfold of pairs and its triples, and Remark 8.5.4 (2017: 8.2). Three rows are formalized in a narrower form. In Remark 8.1.10 (2017: 8.1) (row V8.12) the map $G(\varphi, h) = (T\varphi, Ts(h))$ is a bijection with $T\varphi$ labelled by $\varphi$; read with the bare linear map $T\varphi$, it does not hold as we read it, and the trivial $\mathbb{Z}/3$-action is the counterexample `Chapter8/TangentEpNotInjective · EpGroupoid.Examples.remark_8_1_10_not_injective` (§VIII.6). The degeneracy remark after Definition 8.5.1 (2017: 8.8) (row V8.28) is proved for tame $X, Y$, a hypothesis the text does not state, and as an inequality in general; equality without tameness is open in the project (§VIII.15). Proposition 8.5.7 (2017: 8.3) (row V8.35) is proved for nonempty $D$; for $D = \emptyset$ it does not hold as we read it, and a covering functor of the empty groupoid with a geometric lift on $\mathbb{R}$ is the counterexample `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty` (§§VIII.20, VIII.40). Three places of the 2017 text needed more than a reading. Theorem 8.5.3 (5) (2017: 8.5), as printed with $t$ on the right vertical arrow, does not hold as we read it: the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ violates it in every chart at $0$, and the square holds with $s$ (§VIII.17). Definition 8.5.5 (2017: 8.9) allows $D = \emptyset$, and then the functor of Proposition 8.5.7 (2017: 8.3) is surjective on objects only if $U(x) = \emptyset$; Proposition 8.5.7 (2017: 8.3) is proved for nonempty $D$, which holds where Theorem 8.5.8 (2017: 8.6) uses it (§VIII.20). The degeneracy remark after Definition 8.5.1 (2017: 8.8) uses the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, which Part I has for tame M-polyfolds; it is proved for tame $X, Y$, and as an inequality in general (§VIII.15). Several omitted steps are supplied: the tangents of maps defined on open sets, which the proofs of §8.1 differentiate without a definition (§VIII.1); the tameness of $Tr$ in Lemma 8.1.9 (2017: 8.8), asserted without argument (§VIII.5); that $\simeq$ is an equivalence relation (§VIII.8); the open embedding $G_x \backslash TU(x) \to |TX|$ in the proof of Theorem 8.1.15 (2017: 8.4) (§VIII.9); the ep-groupoid of the functorial picture (§VIII.12); the equivalence of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6) (§VIII.13); and the "one verifies" of Proposition 8.5.7 (2017:

8.3) and of Theorem 8.5.8 (2017: 8.6) (§§VIII.20, VIII.21). Several proofs take another route: Lemmas 8.1.4–8.1.6 (2017: 8.3–8.5), Theorem 8.1.11 (2017: 8.2) and Proposition 8.1.14 (2017: 8.1) by fibre injectivity instead of differentiating identities (§§VIII.3, VIII.7, VIII.8); condition (4) of Definition 7.1.3 (2017: 7.3) for $T\mathbf{X}$ in Chapter 7′s sheet form (§VIII.4); Theorem 8.1.15 (2017: 8.4) by metrizability instead of a closed paracompact cover (§VIII.9); Theorem 8.5.3 (2017: 8.5) through Definition 8.5.1 (3) (2017: 8.8) instead of sheets and natural representations in $Y$ (§VIII.17). Not formalized: a few remarks that nothing uses, and equality in the degeneracy remark without tameness (§VIII.27).

**Seven structural choices.** (1) *The tangent is Part I's.* $TX$ and $T\mathbf{X}$ are Part I's tangent M-polyfolds and $Ts$, $Tt$, $Tu$, $T\iota$ Part I's tangent maps (§§II.8, II.9); for $X$ in `Type u` they live in `Type (max u 1)`, and so does the tangent functor of Theorem 8.1.12 (2017: 8.3). The fibres $T_xX$ carry Part I's linear structure, transported through charts (Definition 2.3.14; gap G.7, closed since the first reading), which gives $T\varphi$ as a linear isomorphism (§VII.44) and the forms of §8.2 (§VIII.35); the proofs of §8.1 do not use it. (2) *The composition of $T\mathbf{X}$ is defined by choice*: total, with a junk value off composable pairs, as Chapter 7′s composition, and unique on composable pairs. (3) *Fibre injectivity.* $s$ being a local sc-diffeomorphism, two vectors of $T\mathbf{X}$ with the same base point and the same image under $Ts$ are equal; most identities of §8.1 are proved this way instead of by differentiating identities. (4) *Tangents of maps defined on open sets* are built from the charts, as Part I builds $Tf$, with a junk value off $p^{-1}(U)$. (5) *Condition (4) in sheet form*, as in Chapter 7 (§VII.2): for $T\mathbf{X}$, for the functorial picture of a strong bundle and for the ep-groupoid of a geometric lift; Lemma 8.1.2 (2017: 8.1) is stated in this form. (6) *Strong bundle maps between strong bundles over M-polyfolds are a reading*: Part I defines strong bundle maps only between local models, and the global notion is a map covering a map of the bases, linear on the fibres, sc-smooth on $[0]$ and on $[1]$ (§VIII.10). (7) *Data, not hypotheses.* "A connected and sufficiently small $U(x)$ admitting the natural $G_x$-action" in Theorems 8.5.3 and 8.5.8 (2017: 8.5 and 8.6) is a datum, a chart of Theorem 8.5.3 (2017: 8.5), as natural representations are in Chapter 7; Definition 8.5.5 (2017: 8.9) is stated with Mathlib's groupoids, $C$ being `SingleObj G`, and its finiteness enters as instance arguments of the results that use it.

**Map.**

- `Chapter8/Basic`: the tangent data (8.1)–(8.9), Definitions 8.1.13, 8.5.1, 8.5.5 (2017: 8.1, 8.8, 8.9) and the category $\mathrm{Diff}_{sc}$.
- `Chapter8/TangentLocal`: tangents of maps sc-smooth on open sets, of sc-diffeomorphisms between open sets and of local sc-diffeomorphisms.
- `Chapter8/TangentEp`: Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.8 (2017: 8.1–8.7), Remark 8.1.10 (2017: 8.1), Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3), Proposition 8.1.14 (2017: 8.1).
- `Chapter8/TangentTame`: Lemma 8.1.9 (2017: 8.8), the tangent of a tame retraction, the tameness clauses of Theorems 8.1.1 and 8.1.12 (2017: 8.1 and 8.3).
- `Chapter8/NatTrans`: natural equivalence is an equivalence relation; whiskering.
- `Chapter8/TangentOrbit`: the action of $G_x$ on $TU(x)$, the orbit space of $TX$, Theorem 8.1.15 (2017: 8.4).
- `Chapter8/StrongBundleMap`: strong bundle maps between strong bundles over M-polyfolds, the pull-back along $s$.
- `Chapter8/EpStrongBundle`: Definitions 8.3.1–8.3.4 (2017: 8.4–8.7), the functorial picture, Definition 8.5.9 (2017: 8.10), §8.6.2 (2017 §8.5.2).
- `Chapter8/ProperCovering`: the degeneracy remark after Definition 8.5.1 (2017: 8.8), Proposition 8.5.2 (2017: 8.2), §8.6.1 (2017 §8.5.1) with Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11) and Proposition 8.6.4 (2017: 8.4).
- `Chapter8/ProperCoveringLocal`: Theorem 8.5.3 (2017: 8.5) and Remark 8.5.4 (2017: 8.2).

- `Chapter8/SigmaPolyfold`: disjoint unions of M-polyfolds.
- `Chapter8/GeometricLift`: the discussion after Definition 8.5.5 (2017: 8.9), Proposition 8.5.7 (2017: 8.3), Theorem 8.5.8 (2017: 8.6).
- `Chapter8/NonVacuity`: covering functors and a geometric lift; the counterexample to Theorem 8.5.3 (5) (2017: 8.5) as printed.
- `Chapter8/NonVacuityBundle`: witnesses for Definitions 8.3.1–8.5.1 and 8.5.9 (2017: 8.4–8.8 and 8.10).
- `Chapter8/TangentZhou`: Propositions 8.1.16, 8.1.17 and Definition 8.4.2 (§§VIII.29–VIII.30).
- `Chapter8/Book2021`: Definition 8.3.5, Proposition 8.4.1, Example 8.5.6, Definition 8.5.9 in the 2021 form, Lemma 8.5.10 (§§VIII.30–VIII.31).
- `Chapter8/TangentZhouLevels`: Proposition 8.1.17 (2) for every $(TX)^i$, through $T(X^i) \cong (TX)^i$ as ep-groupoids (§VIII.33).
- `Chapter8/GapP2_B_TangentFibred`: $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ and Lemma 8.1.2 as printed (§VIII.34).
- `Chapter8/GapP2_B_RemarkPairs`, `Chapter8/TangentEpNotInjective`: the M-polyfold of pairs and the triples of Remark 8.1.10, and the bare-map reading (§§VIII.6, VIII.34).
- `Chapter8/GapP2_B_EpForms`, `GapP2_B_EpFormsObj`, `GapP2_B_EpFormsFunctor`: Definitions 8.2.1, 8.2.2, the direct limit, the sc-de Rham complex, pull-backs by functors (§§VIII.35–VIII.36).
- `Chapter8/GapP2_B_EpPoincare` and `Chapter11/GapP2_B2_Averaging`, `GapP2_B2_TranslationAverage`, `GapP2_B2_SubGroupoidModel`, `GapP2_B2_Poincare`: Proposition 8.2.3 (§VIII.37); the proof uses the push-forward of §11.2, hence the files of Chapter 11.
- `Chapter8/TangentVertical2021`, `Chapter8/TangentZhouReflect2021`: vertical vectors of $T(TX)$ and Proposition 8.1.16 (§VIII.38).
- `Chapter8/BundleOrbitParacompact2021`, `GapP2_CProp841W1`, `BundleLocalRegularity2021`: Propositions 8.4.1 and 8.4.3 (§VIII.39).
- `Chapter8/PullbackStrongBundle2021`, `Chapter8/GeometricLiftEmpty2021`: Lemma 8.5.10 and Proposition 8.5.7 for $D = \emptyset$ (§VIII.40); `Chapter7/SmoothRegularAssembly`, `Chapter7/SmoothConnectedReps`: Proposition 8.1.17 without added hypotheses (§VIII.40). `Chapter8/ProperCoveringUniverse` is an earlier copy of Proposition 8.5.7 and Theorem 8.5.8 for one universe, made unnecessary when Definition 8.5.5 itself was stated in any universe.

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid as in Part VII, with source and target $s, t$, composition $m(\psi, \varphi) = \psi \circ \varphi$, unit $u(x) = 1_x$ and inverse $\iota(\varphi) = \varphi^{-1}$. $TX$ and $T\mathbf{X}$ are Part I's tangent M-polyfolds (§II.8), $p$ the projection of a tangent vector to its base point on level 1, and $Tf$ Part I's tangent map of an sc-smooth $f$ (§II.9). For a vector $v \in T\mathbf{X}$ we write $p(v) \in \mathbf{X}_1$ for its base point, so $p(Ts(v)) = s(p(v))$. A *sheet* of $s$ is an open set $\mathbf{W} \subseteq \mathbf{X}$ on which $s$ is an sc-diffeomorphism onto an open $V$, its inverse $\sigma$ a *local inverse* of $s$; for open $U \subseteq X$, $TU = p^{-1}(U)$. $G_x$ is the isotropy group of $x$, and a natural representation $(U(x), \Phi, \Gamma)$ is Chapter 7's (§VII.7), with $U(x)$ open and connected. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses (Definition 2.5.2, tame retractions, is 2017 Definition 2.17, §II.14; Proposition 2.5.20, fibred products, is 2017 Proposition 2.15, §II.17; Proposition 2.6.8, the pull-back strong bundle, is 2017 Proposition 2.16, §II.20; Proposition 2.4.4, 2017 Proposition 2.7, is `proposition_2_4_4`, §II.11; Lemma 1.3.4 is 2017 Lemma 1.1, §I.18); the Lean names of Parts I and II carry the 2021 numbers. Findings are those of `Exposition/Findings.md`, "Chapter 8".

## VIII.1 The tangent data (8.1)–(8.9), and tangents of maps on open sets

**The tangent data** (V8.1). The objects of $T(X, \mathbf{X})$ are the points of $TX$ Chapter2/TangentBundle · MPolyfold.TangentSpace Chapter2/TangentBundle · MPolyfold.tangentPolyfold, the morphisms those of $T\mathbf{X}$; a vector $(\varphi, h) \in T_\varphi\mathbf{X}$ is viewed as a morphism $Ts(\varphi)(h) \to Tt(\varphi)(h)$ (8.2), so that $s_T = Ts$ and $t_T = Tt$ (8.3) Chapter8/Basic · EpGroupoid.tSrc Chapter8/Basic · EpGroupoid.tTgt, Part I's tangent maps Chapter2/TangentMap · MPolyfold.tangentMap. The unit is $u_T = Tu$ and the inverse $\iota_T = T\iota$ (8.9) Chapter8/Basic · EpGroupoid.tUnit Chapter8/Basic · EpGroupoid.tInv, and the base point $p : T\mathbf{X} \to \mathbf{X}^1$ is Part I's projection Chapter8/Basic · EpGroupoid.tBase Chapter2/TangentBundle · MPolyfold.tangentProj. *The composition* (8.5), $(\psi, k) \circ (\varphi, h) = (\psi \circ \varphi, Ts(\psi \circ \varphi)^{-1}(Ts(\varphi)(h)))$ for $Ts(\psi)(k) = Tt(\varphi)(h)$ (8.4), needs the inverse of $Ts$ on one fibre. *Step 1, fibre bijectivity.* Since $s$ is a local sc-diffeomorphism, $Ts$ maps every fibre $T_\varphi\mathbf{X}$ bijectively onto $T_{s(\varphi)}X$ (Step 4 of the next paragraph). So two vectors of $T\mathbf{X}$ with the same base point and the same image under $Ts$ are equal Chapter8/TangentEp · EpGroupoid.tangent_ext, and every vector over $s(\varphi)$ is $Ts(v)$ for a $v$ over $\varphi$ Chapter8/TangentEp · EpGroupoid.exists_tSrc_eq. *Step 2, the choice.* For a composable pair, $s(p\,k) = t(p\,h)$ Chapter8/TangentEp · EpGroupoid.src_tBase_eq, and $k \circ h$ is chosen as a vector over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$ Chapter8/Basic · EpGroupoid.tComp Chapter8/TangentEp · EpGroupoid.tComp_spec Chapter8/TangentEp · EpGroupoid.tBase_tComp Chapter8/TangentEp · EpGroupoid.tSrc_tComp; off composable pairs it is the junk value $h$, which no statement uses. By Step 1 the choice is the only possible one, which is (8.5) in this form Chapter8/TangentEp · EpGroupoid.tComp_eq. *Step 3, the units.* $u_T = Tu$ is (8.8), $1_a = (1_x, Ts(1_x)^{-1}a)$: $Tu(a)$ lies over $1_x$, and $Ts(Tu(a)) = a$ by the chain rule for $s \circ u = \mathrm{id}$, so Step 1 identifies the two Chapter8/TangentEp · EpGroupoid.tUnit_eq. (8.5) writes $T_{\varphi\circ\psi}\mathbf{X}$ for $T_{\psi\circ\varphi}\mathbf{X}$ (possible misprint) (Findings, Chapter 8, items 1, 2) (pp. 357–359 (2017 pp. 257–258)).

**Tangents of maps on open sets** (V8.2). The proofs of §8.1 differentiate maps that are defined only on open sets, such as a local inverse $(s|\mathbf{U}(\psi))^{-1}$ and the map $\alpha$ of Lemma 8.1.3 (2017: 8.2), and write $TU$ for the part of $TX$ over an open $U$; Part I defines $Tf$ only for maps sc-smooth on the whole M-polyfold. Here $TU$ is the open set $p^{-1}(U)$ Chapter8/TangentLocal · MPolyfold.tangentPreimage Chapter8/TangentLocal · MPolyfold.isOpen_tangentPreimage, and it is the tangent of the open sub-M-polyfold $U$ when $U \neq \emptyset$ Chapter8/TangentLocal · MPolyfold.isScDiffeoOn_tangentMap_subtypeVal Chapter8/TangentLocal · MPolyfold.isOpenEmbedding_tangentMap_subtypeVal. *Step 1, the construction.* For $f$ sc-smooth on $U$, $Tf$ on $p^{-1}(U)$ is built from the charts as Part I builds $Tf$ (§II.9), with $U \cap V \cap f^{-1}(V')$ in place of $V \cap f^{-1}(V')$ Chapter8/TangentLocal · MPolyfold.TangentPre.fimageOn Chapter8/TangentLocal · MPolyfold.tangentMapOn; off $p^{-1}(U)$ it takes a junk value, which exists because a nonempty M-polyfold has points on level 1 ($E_1$ is dense in $E_0$ and a nonempty retract meets it) Chapter8/TangentLocal · MPolyfold.exists_mem_levelSet_one Chapter8/TangentLocal · MPolyfold.nonempty_tangentSpace. *Step 2, the calculus.* $Tf$ lies over $f$ Chapter8/TangentLocal · MPolyfold.levelVal_tangentProj_tangentMapOn, agrees with Part I's tangent when $f$ is sc-smooth everywhere Chapter8/TangentLocal · MPolyfold.tangentMapOn_eq_tangentMap, satisfies the chain rule Chapter8/TangentLocal · MPolyfold.tangentMapOn_comp and is sc-smooth on $p^{-1}(U)$ Chapter8/TangentLocal · MPolyfold.tangentMapOn_isScSmoothOn. *Step 3, sc-diffeomorphisms between open sets.* The tangent of an sc-diffeomorphism $U \to V$ is an sc-diffeomorphism $p^{-1}(U) \to p^{-1}(V)$ with the tangent of the inverse as its inverse, by the chain rule Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.tangent Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.tangentMap, and it is bijective on every fibre Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.bijOn_tangentFiber; this is the step "$Ts : T\mathbf{U}(\varphi) \to TU(x)$ is an sc-diffeomorphism" of Lemma 8.1.8 (2017: 8.7). *Step 4, local sc-diffeomorphisms.* A sheet of $Tf$ around $v$ is $p^{-1}(U)$ for a sheet $U$ of $f$ around $p(v)$, so $Tf$ is a local sc-diffeomorphism Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.tangentMap, bijective on the fibres Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.bijOn_tangentFiber and surjective when $f$ is Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.surjective_tangentMap. Step 4 is Lemma 8.1.7's (2017: 8.6) "by con-

struction". The chain rule with a globally sc-smooth outer map, $TF \circ Tf = T(F \circ f)$ on $p^{-1}(U)$, is used throughout `Chapter8/TangentEp · MPolyfold.tangentMap_tangentMapOn` (Findings, Chapter 8, item 3) (used on pp. 358–364 (2017 pp. 258–263)).

## VIII.2 Theorem 8.1.1 (2017: 8.1), and Lemma 8.1.2 (2017: 8.1)

**Theorem 8.1.1 (2017: 8.1)** (V8.3) `Chapter8/TangentEp · EpGroupoid.theorem_8_1_1`. $T(X, \mathbf{X}) = (TX, T\mathbf{X})$ *with the structure maps (8.3)–(8.9) is an ep-groupoid; it is tame if $X$ is; and the projections form an sc-smooth functor $(p, \mathbf{p}) : T(X, \mathbf{X}) \to (X, \mathbf{X})^1$.* The text proves it on pp. 363–365 (2017 pp. 263–264) by collecting Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8), and the formalization does the same. The groupoid laws are Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5) (§VIII.3) `Chapter8/TangentEp · EpGroupoid.tangentGroupoidStr`; conditions (1), (2) and (4) of Definition 7.1.3 (2017: 7.3) are Lemma 8.1.7 (2017: 8.6) and condition (3) is Lemma 8.1.8 (2017: 8.7) (§VIII.4); together they make the ep-groupoid `Chapter8/TangentEp · EpGroupoid.tangentEpGroupoid`, and `theorem_8_1_1` records that it has the M-polyfolds $TX$, $T\mathbf{X}$, the structure maps of the text, and the base point and source of a composite. The tameness sentence is Lemma 8.1.9 (2017: 8.8) (§VIII.5) `Chapter8/TangentTame · EpGroupoid.theorem_8_1_1_tame`. *The projection functor.* $(X, \mathbf{X})^1$ is Chapter 7's lift to level 1 (§VII.22) `Chapter7/Lift · EpGroupoid.liftOne`, and $(p, \mathbf{p})$ is a functor into it `Chapter8/TangentEp · EpGroupoid.tangentProjFunctor`: it commutes with sources and targets because $p \circ Ts = s \circ p$ and $p \circ Tt = t \circ p$, with the composition because $p(k \circ h) = p(k) \circ p(h)$ (Step 2 of §VIII.1, the text's "This is obvious"), and with the units because $Tu(a)$ lies over $1_x$; the two projections are sc-smooth by Part I (Findings, Chapter 8, items 2, 9) (2017 pp. 258, 263–264).

**Lemma 8.1.2 (2017: 8.1)** (V8.4) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_2`. *As printed*: with $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X})$ identified with $T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$, the composition (8.5) is $(\psi, k) \circ (\varphi, h) = Tm(\psi, \varphi)(k, h)$. **The text's proof** varies $\varphi'$ near $\varphi$, writes the partner as $\psi' = F(\varphi') = (s|\mathbf{U}(\psi))^{-1}(t(\varphi'))$, applies $s$ to $m(F(\varphi'), \varphi')$, which gives $s(\varphi')$, and differentiates this identity at $\varphi$; since $s$ is a local sc-diffeomorphism, $Tm(k, h) = Ts(\psi \circ \varphi)^{-1} Ts(\varphi)(h)$. **The formalized statement** is the form the proof uses; the identification of the tangent of the fibred product with the fibred product of the tangents, and $Tm$ on it, the lemma as printed, are §VIII.34 (gap G.75, closed). Let $s : \mathbf{W} \to V$ be a sheet with inverse $\sigma$ (the text's $(s|\mathbf{U}(\psi))^{-1}$), and $k, h$ composable with $p(k) \in \mathbf{W}$. *Step 1.* $g(\varphi') = m(\sigma(t\varphi'), \varphi')$, the text's $m(F(\varphi'), \varphi')$, is sc-smooth on $t^{-1}(V)$: this is Chapter 7's sheet form of condition (4) (§VII.2) `Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet`. *Step 2,* $k = T\sigma(Tt(h))$. $Tt(h) = Ts(k)$, and $T\sigma$ inverts $Ts$ on $p^{-1}(\mathbf{W})$ (Step 3 of §VIII.1) `Chapter8/TangentEp · EpGroupoid.tangentMapOn_srcInv_tSrc`; in particular $h$ lies over $t^{-1}(V)$ `Chapter8/TangentEp · EpGroupoid.mem_tangentPreimage_tgt`. *Step 3,* $k \circ h = Tg(h)$. $Tg(h)$ lies over $g(p\,h) = m(\sigma(s(p\,k)), p\,h) = p(k) \circ p(h)$, and $Ts(Tg(h)) = Ts(h)$ by the chain rule for $s \circ g = s$ on $t^{-1}(V)$, which is the text's identity; by the uniqueness of (8.5) the two vectors agree `Chapter8/TangentEp · EpGroupoid.tComp_eq_tangentMapOn`. The lemma also restates that $k \circ h$ lies over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$. The proof writes $s|\mathbf{U}(\varphi \circ \psi)$ for $s|\mathbf{U}(\psi \circ \varphi)$ (possible misprint) (Findings, Chapter 8, items 5, 111) (2017 p. 259).

## VIII.3 Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5): the groupoid laws of $T(X, \mathbf{X})$

**Lemma 8.1.3 (2017: 8.2)** (V8.5) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_3`. $s_T(\Psi \circ \Phi) = s_T(\Phi)$ *and* $t_T(\Psi \circ \Phi) = t_T(\Psi)$. The first is (8.5) itself (Step 2 of §VIII.1). **The text's proof** of the second takes the map $\alpha = t \circ (s|\mathbf{U}(\psi \circ \varphi))^{-1}$, notes $\alpha \circ s = t \circ (s|\mathbf{U}(\psi))^{-1} \circ t$ near $\varphi$, and differentiates both sides at $\varphi$ in the direction $h$. **The formalized proof** does the same with the tangents of §VIII.1, on a sheet $s : \mathbf{W} \to V$ with inverse $\sigma$ around $p(k)$. *Step 1.* By Lemma 8.1.2 (2017: 8.1), $k \circ h = Tg(h)$ with $g(\varphi') = m(\sigma(t\varphi'), \varphi')$. *Step 2.*

On $t^{-1}(V)$, $t\circ g = t\circ\sigma\circ t$ (the text's identity), so $Tt(Tg(h)) = T(t\circ\sigma\circ t)(h) = Tt(T\sigma(Tt(h))) = Tt(k)$, the last step by Step 2 of Lemma 8.1.2 (2017: 8.1) `Chapter8/TangentEp · EpGroupoid.tTgt_tangentMapOn`. This is the only identity of §8.1 that is genuinely differentiated; the others follow from fibre injectivity. The proof has "==", ": $\mathbf{U}(\psi\circ\varphi) \to U(x)$" without the map $s$, and "$Tt)\varphi)(h)$" (possible misprints) (Findings, Chapter 8, items 4, 6) (pp. 359–361 (2017 pp. 259–260)).

**Lemma 8.1.4 (2017: 8.3)** (V8.6) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_4`. *The composition of $T\mathbf{X}$ is associative.* **The text's proof** computes both sides of $(\sigma, l)\circ((\psi, k)\circ(\varphi, h))$ with (8.5) and cancels $Ts(\psi\circ\varphi)(Ts(\psi\circ\varphi))^{-1}$. **The formalized proof** compares the two sides by fibre injectivity (§VIII.1): both lie over $p(l)\circ p(k)\circ p(h)$, by the associativity of $\mathbf{X}$, and both have $Ts$ equal to $Ts(h)$, by Lemma 8.1.3 (2017: 8.2) applied twice; composability of the outer pairs is Lemma 8.1.3's (2017: 8.2) target formula (Findings, Chapter 8, item 4) (pp. 360–362 (2017 pp. 260–261)).

**Lemma 8.1.5 (2017: 8.4)** (V8.7) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_5`. $1_q = (1_x, Ts(1_x)^{-1}q)$ *is a morphism $q \to q$ and a two-sided unit.* $1_q = Tu(q)$ (Step 3 of §VIII.1). *Step 1,* $1_q : q \to q$. $Ts(Tu(q)) = q$ and $Tt(Tu(q)) = q$ are the chain rule for $s\circ u = \mathrm{id}$ and $t\circ u = \mathrm{id}$ `Chapter8/TangentEp · EpGroupoid.tSrc_tUnit` `Chapter8/TangentEp · EpGroupoid.tTgt_tUnit`. The text argues the second through $t\circ(s|\mathbf{U}(1_x))^{-1} = \mathrm{id}$ near $x$; the identity $t\circ u = \mathrm{id}$ holds everywhere and needs no local inverse. *Step 2, the unit laws.* $k\circ 1_{Ts(k)}$ and $k$ lie over $p(k)\circ 1 = p(k)$ and have the same $Ts$, and likewise on the left; the uniqueness of (8.5) gives both `Chapter8/TangentEp · EpGroupoid.tComp_tUnit` `Chapter8/TangentEp · EpGroupoid.tUnit_tComp` (Findings, Chapter 8, items 4, 6) (2017 p. 261).

**Lemma 8.1.6 (2017: 8.5)** (V8.8) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_6`. $\iota_T(\varphi, k) = (\varphi^{-1}, T\iota(\varphi)k)$ *is a two-sided inverse.* As in the text, the chain rule for $s\circ\iota = t$ and $t\circ\iota = s$ makes $\iota_T(v)$ a morphism $t_T(v) \to s_T(v)$ `Chapter8/TangentEp · EpGroupoid.tSrc_tInv` `Chapter8/TangentEp · EpGroupoid.tTgt_tInv`. The text then computes $\iota_T(\varphi, k)\circ(\varphi, k)$ with (8.5); here the uniqueness of (8.5) does it: $\iota_T(v)\circ v$ lies over $\varphi^{-1}\circ\varphi = 1_x$ and has $Ts$ equal to $Ts(v)$, which characterizes $1_{Ts(v)}$, and the other side is the same `Chapter8/TangentEp · EpGroupoid.tInv_tComp` `Chapter8/TangentEp · EpGroupoid.tComp_tInv` (Findings, Chapter 8, item 4) (pp. 361–363 (2017 pp. 261–262)).

## VIII.4 Lemmas 8.1.7 and 8.1.8 (2017: 8.6 and 8.7): the differential conditions

**Lemma 8.1.7 (2017: 8.6)** (V8.9) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_7`. $s_T$, $t_T$ *are surjective local sc-diffeomorphisms and all structure maps of $T(X, \mathbf{X})$ are sc-smooth. Step 1, $s_T$ and $t_T$.* "By construction" is Step 4 of the tangents on open sets (§VIII.1): the tangent of a surjective local sc-diffeomorphism is one `Chapter8/TangentEp · EpGroupoid.isLocalScDiffeo_tSrc` `Chapter8/TangentEp · EpGroupoid.isLocalScDiffeo_tTgt` `Chapter8/TangentEp · EpGroupoid.surjective_tSrc` `Chapter8/TangentEp · EpGroupoid.surjective_tTgt`. *Step 2, $u_T$ and $\iota_T$* are Part I's tangents of the sc-smooth $u$ and $\iota$ `Chapter8/TangentEp · EpGroupoid.isScSmoothMap_tUnit` `Chapter8/TangentEp · EpGroupoid.isScSmoothMap_tInv`. *Step 3, the multiplication.* **The text** says it is sc-smooth "by inspection", or because $T\mathbf{X}_{s_T}\times_{t_T} T\mathbf{X}$ is the tangent of $\mathbf{X}_s\times_t\mathbf{X}$ and $m_{T\mathbf{X}}$ "is precisely $Tm$". **The formalized proof** uses Chapter 7's sheet form of condition (4) (§VII.2) `Chapter7/EpGroupoid · IsScSmoothCompSheet`: for a sheet of $s_T$ with inverse $\sigma_T$, the map $\Psi \mapsto \sigma_T(t_T\Psi)\circ\Psi$ is sc-smooth on $t_T^{-1}(V_T)$. Near $\Psi_0$, take a sheet $s : \mathbf{W} \to V$ around the base point of $\sigma_T(t_T\Psi_0)$; on the open set of $\Psi$ with $p(\sigma_T(t_T\Psi)) \in \mathbf{W}$, the map is $Tg$ for the $g$ of Lemma 8.1.2 (2017: 8.1), by Step 3 of Lemma 8.1.2 (2017: 8.1), and $Tg$ is sc-smooth `Chapter8/TangentEp · EpGroupoid.isScSmoothCompSheet_tangent`; sc-smoothness is local `Chapter7/Discrete · MPolyfold.IsScSmoothOn.of_locally`. Chapter 7's equivalence of the two forms gives

(4) on the fibred product itself `Chapter7/EpGroupoid · isScSmoothComp_iff_sheet Chapter8/TangentEp · EpGroupoid.isScSmoothComp_tangent`, without $T(\mathbf{X}_s\times_t\mathbf{X}) = T\mathbf{X}_{Ts}\times_{Tt}T\mathbf{X}$ (gap G.75) (Findings, Chapter 8, item 7) (2017 p. 262).

**Lemma 8.1.8 (2017: 8.7)** (V8.10) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_8`. *Every $a \in TX$ has an open $V' \ni a$ with $t_T : s_T^{-1}(\operatorname{cl} V') \to TX$ proper*, namely $V' = p^{-1}(V)$ for an open $V$ around $p(a)$ as in Definition 7.1.3 (3) (2017: 7.3) for $X$. **The text's proof** takes $(\varphi_k, h_k)$ with sources in $\operatorname{cl} TV$ and convergent targets, gets $\varphi_k \to \varphi$ in $\mathbf{X}$ from the properness of $X$, then in $\mathbf{X}^1$, and concludes that $(\varphi_k, h_k)$ converges since $Ts : T\mathbf{U}(\varphi) \to TU(x)$ is an sc-diffeomorphism. **The formalized proof** checks Chapter 7's sequential criterion for properness on metrizable spaces (§VII.2) `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff_of_isClosed`. *Step 1.* Let $\Phi_k$ have $s_T(\Phi_k) \in \operatorname{cl} V'$ and $t_T(\Phi_k) \to b$. The base points have $s(p\,\Phi_k) \in \operatorname{cl} V$, $p$ being continuous, and $t(p\,\Phi_k) \to p(b)$; properness of $X$ gives a subsequence with $p(\Phi_{k_j}) \to \varphi_0$ in $\mathbf{X}$ `Chapter7/ProperMaps · MPolyfold.exists_tendsto_subseq_of_isProperMap_domRestrict`. *Step 2.* Take a sheet $t : \mathbf{W} \to V''$ around $\varphi_0$. Its tangent is an sc-diffeomorphism $p^{-1}(\mathbf{W}) \to p^{-1}(V'')$ with continuous inverse $(Tt)^{-1}$ (Step 3 of §VIII.1), and $b \in p^{-1}(V'')$; eventually $p(\Phi_{k_j}) \in \mathbf{W}$, so $\Phi_{k_j} = (Tt)^{-1}(t_T\Phi_{k_j}) \to (Tt)^{-1}(b)$. The text's step "$\varphi_k \to \varphi$ in $\mathbf{X}^1$" is not needed: convergence in $\mathbf{X}$ already puts the $\varphi_k$ in the sheet, and $TX$, $T\mathbf{X}$ are metrizable, so the sequential criterion applies. The proof writes $t(\varphi_k, h_k) = Ts(\varphi_k)(h_k)$ for $Tt(\varphi_k)(h_k)$ (possible misprint) (Findings, Chapter 8, item 7) (pp. 362–364 (2017 pp. 262–263)).

## VIII.5 Lemma 8.1.9 (2017: 8.8): the tangent of a tame ep-groupoid

**Lemma 8.1.9 (2017: 8.8)** (V8.11) `Chapter8/TangentTame · EpGroupoid.lemma_8_1_9`. *If $X$ is tame, so is $TX$*; tame means that the object M-polyfold has an equivalent tame atlas (Definition 7.1.6 (2017: 7.6), §VII.3) `Chapter7/EpGroupoid · EpGroupoid.IsTame`. **The text's proof** is one sentence: for a tame chart onto a tame retract $O = r(U)$, "$Tr : TU \to TU$ is also a tame retraction". Part I does not state this, and the text gives no argument; it is proved here from Part I's Definition 2.5.2 (2017: 2.17) (§II.14) `Chapter2/Tame · ScRetraction.IsTame`. The tangent of the retract $(O, C, E)$ is $(TO, TC, TE)$, $TC = C^1 \oplus E$, with $Tr(a,b) = (r(a), Dr(a)b)$ `Chapter2/TangentBundle · ScRetraction.tangentRetraction Chapter2/TangentBundle · LocalModel.tangent`. *Step 1, a presentation of $TC$.* An sc-isomorphism of $TE$ with $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$, built from a presentation $T$ of $C$, presents $TC$ as a partial quadrant whose $\mathbb{R}^n$-coordinates at $(a,b)$ are those of $a$ `Chapter8/TangentTame · ScIso.tangentPres Chapter8/TangentTame · tangentSet_eq_tangentPres`; so $d_{TC}(a,b) = d_C(a)$ `Chapter8/TangentTame · degIndex_tangentSet`, and the minimal subspace at $(a,b)$ is $(E_a \cap E_1) \oplus E$ `Chapter8/TangentTame · mem_minimalSubspaceWith_tangentPres`. *Step 2, condition (1)* of Definition 2.5.2 (2017: 2.17) for $Tr$ at $(a,b)$ is condition (1) for $r$ at $a$, by Step 1 and $Tr(a,b) = (r(a), Dr(a)b)$. *Step 3, condition (2)* at a smooth point $(a,b) \in TO$. With $A$ the sc-complement of $T_aO = Dr(a)E$ in $E_a$ that the tameness of $r$ gives, the complement for $Tr$ is $A' = (A \cap E_1) \oplus A$. With $L = D(Tr)(a,b)$, three facts suffice: $L$ is a projection, since $Tr \circ Tr = Tr$ `Chapter2/Retracts · ScRetraction.scDeriv_comp_self`; its first row is $(Dr(a), 0)$ `Chapter8/TangentTame · ScRetraction.scDeriv_tangentRetraction_fst`; and $L(0,l) = (0, Dr(a)l)$ `Chapter8/TangentTame · ScRetraction.scDeriv_tangentRetraction_inr`. Then $LTE \cap A' = 0$ (if $(k,l) = L(k,l) \in A'$, then $k \in T_aO \cap A = 0$ and $l = Dr(a)l \in T_aO \cap A = 0$) and $LTE + A' = TE$ (split $k$, then the remainder of $l$), on every level. The second derivative of $r$, which enters $L$, is never needed `Chapter8/TangentTame · ScRetraction.IsTameWith.tangent`. *Step 4, all presentations.* Tameness quantifies over all presentations of $TC$; Part I's Lemma 1.3.4 (2017: 1.1), the presentation independence of $d_C$ (§I.18), passes from one to all `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame Chapter8/TangentTame · ScRetraction.IsTame.tangentRetraction Chapter8/TangentTame · IsTameRetract.tangent`. *Step 5, from charts to $TX$.* Around

$\alpha \in TX$ the tangent chart of a chart onto a tame retract around $p(\alpha)$ is modelled on the tame retract $T(O, C, E)$ `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`, so these charts form an equivalent tame atlas `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart Chapter8/TangentTame · MPolyfold.IsTame.tangent`. The tameness clauses of Theorem 8.1.1 (2017: 8.1) and Theorem 8.1.12 (2017: 8.3) follow `Chapter8/TangentTame · EpGroupoid.theorem_8_1_1_tame Chapter8/TangentTame · theorem_8_1_12_tame`, and the hypothesis is satisfiable: the tangent of the trivial ep-groupoid of $\mathbb{R}^n$ is tame `Chapter8/TangentTame · EpGroupoid.isTame_tangentEpGroupoid_trivial_rn` (Findings, Chapter 8, items 8, 104) (2017 p. 263).

## VIII.6 Remark 8.1.10 (2017: 8.1): the morphisms as pairs $(T\varphi, a)$

**Remark 8.1.10 (2017: 8.1)** (V8.12) `Chapter8/TangentEp · EpGroupoid.remark_8_1_10`. The text describes $T\mathbf{X}$ a second way: a morphism $\varphi : x \to y$ on level 1 has a linear isomorphism $T\varphi : T_xX \to T_yX$, the tangent of $t \circ (s|\mathbf{U}(\varphi))^{-1}$ at $x$; the pairs $(T\varphi, a)$, $a \in T_{s(\varphi)}X$, composed by $(T\psi, h) \circ (T\varphi, l) = (T(\psi \circ \varphi), l)$, form a groupoid, and $G(\varphi, h) = (T\varphi, Ts(h))$ is a bijection compatible with the compositions; moreover $t_T \circ (s_T|T_\varphi\mathbf{X})^{-1} = T\varphi$. $T\varphi$ is Chapter 7's $Tt \circ (Ts|T_\varphi\mathbf{X})^{-1}$ (§VII.3) `Chapter7/EpGroupoid · EpGroupoid.tangentMor`. *Step 1, well defined.* $T\varphi(Ts(h)) = Tt(h)$ for every $h \in T_\varphi\mathbf{X}$, which is the last sentence of the remark `Chapter8/TangentEp · EpGroupoid.tangentMor_tSrc`. *Step 2, functorial.* $T(\psi \circ \varphi) = T\psi \circ T\varphi$ on $T_{s(\varphi)}X$: given $a$, take $h$ over $\varphi$ with $Ts(h) = a$ (fibre surjectivity) and $k$ over $\psi$ with $Ts(k) = Tt(h)$; then $k \circ h$ lies over $\psi \circ \varphi$ with $Ts = a$, and $Tt(k \circ h) = Tt(k)$ by Lemma 8.1.3 (2017: 8.2) `Chapter8/TangentEp · EpGroupoid.tangentMor_comp`. *Step 3, the map G.* As a bare map $(\varphi, h) \mapsto (T\varphi, Ts(h))$ need not be injective: distinct morphisms can have the same $T\varphi$, as the three morphisms at a point of the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$ (§VII.30), all with $T\varphi = \mathrm{id}$. So the first entry is read as $T\varphi$ labelled by $\varphi$ `Chapter8/TangentEp · EpGroupoid.remarkG`; then $G$ is injective by fibre injectivity, its range is the set of pairs $(\varphi, a)$ with $\varphi \in \mathbf{X}_1$ and $a$ over $s(\varphi)$ by fibre surjectivity, composability of $(k, h)$ is $Ts(k) = T\varphi(Ts(h))$ with $\varphi = p(h)$, and $G(k \circ h) = (p(k) \circ p(h), Ts(h))$, the text's computation. *The bare-map reading.* Read with the bare map $T\varphi$ as first entry, $G$ is not injective, so the remark does not hold as we read it in that reading: on the trivial $\mathbb{Z}/3$-action $T\varphi = \mathrm{id}$ for every morphism `Chapter8/TangentEpNotInjective · EpGroupoid.Examples.tangentMor_trivialZ3`, and lifting one tangent vector along $Ts$ over $(1, x)$ and over $(g, x)$, $g \neq 1$, gives two vectors of $T\mathbf{X}$ with the same image `Chapter8/TangentEpNotInjective · EpGroupoid.Examples.remark_8_1_10_not_injective` (row V8.12, narrower). *Since the first reading.* That $T\varphi$ is a linear isomorphism, and the tangent of $\hat{\varphi}$ at $x$, is §VII.44 `Chapter7/TangentMorphism · EpGroupoid.definition_7_1_4 Chapter7/TangentMorphismLinear · EpGroupoid.definition_7_1_4_linear Chapter7/TangentMorphismLinear · EpGroupoid.tangentMorCLE` (gap G.70, closed); the M-polyfold structure on the pairs, for which $G$ is an sc-diffeomorphism, and the triples $(a, T\varphi, b)$ are §VIII.34. The remark writes "$T\varphi : T_xX \to T_yY$" for $T_yX$, and calls $T\varphi$ the linearization "at $\varphi$" on 2017 p. 265 where 2017 p. 264 has "at $x$" (possible misprints) (Findings, Chapter 8, items 10, 112, 113, 131) (pp. 364–366 (2017 pp. 264–265)).

## VIII.7 Theorems 8.1.11 and 8.1.12 (2017: 8.2 and 8.3): the tangent functor

**Theorem 8.1.11 (2017: 8.2)** (V8.13) `Chapter8/TangentEp · EpFunctor.theorem_8_1_11`. *For an sc-smooth functor $(\Phi, \Phi) : X \to Y$, $(T\Phi, T\Phi)$ is an sc-smooth functor $TX \to TY$* `Chapter8/TangentEp · EpFunctor.tangent`. **The text's proof** differentiates $\Phi(m(\psi, \varphi)) = m(\Phi\psi, \Phi\varphi)$ on the fibred product, using Lemma 8.1.2 (2017: 8.1) as printed, and differentiates $s \circ \Phi = \Phi \circ s$ and $t \circ \Phi = \Phi \circ t$; it leaves the units to the reader. **The formalized proof.** *Step 1.* The tangent maps are sc-smooth by Part I, and compatibility

with sources and targets is the chain rule for $s \circ \Phi = \Phi \circ s$, $t \circ \Phi = \Phi \circ t$, as in the text. *Step 2, the composition, by fibre injectivity in $TY$.* $T\Phi(k \circ h)$ lies over $\Phi(p(k) \circ p(h)) = \Phi(p\,k) \circ \Phi(p\,h)$, and its $Ts$ is $T\Phi(Ts(k \circ h)) = T\Phi(Ts(h)) = Ts(T\Phi(h))$; the composite $T\Phi(k) \circ T\Phi(h)$ has the same base point and the same $Ts$. No tangent of the fibred product is needed. *Step 3, the units*, not checked in the text, follow from the chain rule for $\Phi \circ u = u \circ \Phi$ (Findings, Chapter 8, items 4, 11) (pp. 365–367 (2017 pp. 265–266)).

**Theorem 8.1.12 (2017: 8.3)** (V8.14) `Chapter8/TangentEp · theorem_8_1_12`. *$T$ is a functor $\mathcal{EP} \to \mathcal{EP}$, $X \mapsto TX$, $\Phi \mapsto T\Phi$, and $TX$ is tame if $X$ is.* On objects it is Theorem 8.1.1 (2017: 8.1) `Chapter8/TangentEp · EP.tangent`, on morphisms Theorem 8.1.11 (2017: 8.2), and the functor laws are Part I's chain rule and the tangent of the identity `Chapter2/TangentMap · MPolyfold.tangentMap_comp` `Chapter2/TangentMap · MPolyfold.tangentMap_id` `Chapter8/TangentEp · tangentFunctor`; $\mathcal{EP}$ is Chapter 7's category (§VII.4) `Chapter7/EpGroupoid · EP`. Since $TX$ lives in `Type (max u 1)`, the functor goes from $\mathcal{EP}$ in that universe to itself, an artifact of Part I's `TangentSpace`. The tameness sentence is Lemma 8.1.9 (2017: 8.8) (§VIII.5) `Chapter8/TangentTame · theorem_8_1_12_tame` (Findings, Chapter 8, items 2, 11) (2017 p. 266).

## VIII.8 Definition 8.1.13 (2017: 8.1), the equivalence relation $\simeq$, and Proposition 8.1.14 (2017: 8.1)

**Definition 8.1.13 (2017: 8.1)** (V8.15). A natural transformation $\tau : F \to G$ between sc-smooth functors $X \to Y$ is an sc-smooth map $\tau : X \to \mathbf{Y}$ with $\tau(x) : F(x) \to G(x)$ and $\tau(x') \circ F(h) = G(h) \circ \tau(x)$ for every $h : x \to x'$ `Chapter8/Basic · EpNatTrans`; $F \simeq G$ if one exists `Chapter8/Basic · EpFunctor.NaturallyEquivalent` (Findings, Chapter 8, item 12) (2017 p. 266).

**The equivalence relation** (V8.16) `Chapter8/NatTrans · EpFunctor.naturallyEquivalent_equivalence`. The text states without proof that $\simeq$ is an equivalence relation. Reflexivity is $x \mapsto 1_{F(x)}$ `Chapter8/NatTrans · EpNatTrans.refl`, symmetry $x \mapsto \tau(x)^{-1}$, sc-smooth since $\iota$ is, natural by inverting the square of $h^{-1}$ `Chapter8/NatTrans · EpNatTrans.symm`, and transitivity $x \mapsto \tau'(x) \circ \tau(x)$ `Chapter8/NatTrans · EpNatTrans.trans` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.refl` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.symm` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.trans`. *The one analytic point* is the sc-smoothness of $x \mapsto \tau'(x) \circ \tau(x)$ `Chapter8/NatTrans · EpNatTrans.isScSmoothMap_comp_app`. Near $x_0$, take a sheet $\mathbf{W} \ni \tau'(x_0)$ of $s$ with inverse $\sigma$; for $x \in \tau'^{-1}(\mathbf{W})$, $\tau'(x) = \sigma(s\,\tau'(x)) = \sigma(G(x)) = \sigma(t\,\tau(x))$, so $\tau'(x) \circ \tau(x) = m(\sigma(t\psi), \psi)$ at $\psi = \tau(x)$, which is sc-smooth by Chapter 7's sheet form of condition (4), composed with the sc-smooth $\tau$. Whiskering, not in the text, is added `Chapter8/NatTrans · EpNatTrans.whiskerLeft` `Chapter8/NatTrans · EpNatTrans.whiskerRight` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_left` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right` (Findings, Chapter 8, item 12) (2017 p. 267).

**Proposition 8.1.14 (2017: 8.1)** (V8.17) `Chapter8/TangentEp · proposition_8_1_14`. *If $\tau : \Phi \to \Psi$ is a natural transformation, $T\tau : TX \to T\mathbf{Y}$ is a natural transformation $T\Phi \to T\Psi$; hence $\Phi \simeq \Psi$ implies $T\Phi \simeq T\Psi$* `Chapter8/TangentEp · EpNatTrans.tangent`. *Step 1, as in the text.* $T\tau$ is sc-smooth by Part I, and the chain rule for $s \circ \tau = \Phi$ and $t \circ \tau = \Psi$ makes $T\tau(a)$ a morphism $T\Phi(a) \to T\Psi(a)$ of $TY$. *Step 2, naturality.* **The text** differentiates the naturality square $\tau(t\alpha) \circ \Phi(\alpha) = \Psi(\alpha) \circ \tau(s\alpha)$ at $\alpha_0 \in \mathbf{X}^1$, again through Lemma 8.1.2 (2017: 8.1) as printed. **The formalized proof** compares $T\tau(Tt(v)) \circ T\Phi(v)$ and $T\Psi(v) \circ T\tau(Ts(v))$ by fibre injectivity in $TY$: they lie over the two sides of the naturality square at $p(v)$, which are equal, and both have $Ts$ equal to $T\Phi(Ts(v))$, by Lemma 8.1.3 (2017: 8.2) and Step 1. The proof has "$T\tau(x_0(h))$", and labels the lower arrow of the last diagram $T\tau(x_0)h$ for $T\tau(x_0')h'$ (possible misprints, the second perhaps an artefact of the text extraction) (Findings, Chapter 8, items 4, 13) (pp. 367–372 (2017 pp. 267–268)).

## VIII.9 Theorem 8.1.15 (2017: 8.4): the orbit space of $TX$

**Theorem 8.1.15 (2017: 8.4)** (V8.18) `Chapter8/TangentOrbit · EpGroupoid.theorem_8_1_15`. *If $|X|$ is paracompact, so is $|TX|$*; paracompact orbit space is Chapter 7's Definition 7.5.1 (2017: 7.15) (§VII.23) `Chapter7/EpGroupoid · EpGroupoid.HasParacompactOrbitSpace`. **The text's proof**: $|X^1|$ is paracompact (Proposition 7.5.3 (2017: 7.12)); for $z \in |X^1|$ take $x_z \in X_1$ and $U(x_z)$ with the natural $G_{x_z}$-action such that $TU(x_z)$ is, as a bundle over $U(x_z)^1$, sc-isomorphic to a local bundle model $TO \to O^1$ with an sc-smooth bundle action, so that $|TU(x_z)|$ and $|U(x_z)^1|$ are metrizable; a locally finite closed refinement $(A_z)$ of the cover $(|U(x_z)^1|)$ gives closed paracompact sets $\tilde{A}_z \subseteq |TX|$ covering it locally finitely, and $|TX|$ is regular Hausdorff, so Part I's Proposition 2.7.4 (2017: 2.17) applies. **The formalized proof** proves more, that $|TX|$ is metrizable `Chapter8/TangentOrbit · EpGroupoid.metrizableSpace_tangent_orbitSpace`, along the route of Proposition 7.5.3 (2017: 7.12) (§VII.23): Smirnov's theorem pulled back along the continuous map $|p| : |TX| \to |X^1|$ (§II.7) `Chapter2/Metrizable · metrizableSpace_of_pullback`. *Step 1, the map $|p|$.* The projection functor of Theorem 8.1.1 (2017: 8.1) induces $|p|$ `Chapter8/TangentOrbit · EpGroupoid.tangentOrbitProj Chapter8/TangentOrbit · EpGroupoid.continuous_tangentOrbitProj`, and $|p|^{-1}(\pi_1(U \cap X_1)) = \pi_T(p^{-1}(U))$ for every $U \subseteq X$, by fibre surjectivity of $Ts$: a vector over a point isomorphic in $X^1$ to a point of $U \cap X_1$ is isomorphic in $TX$ to a vector over it `Chapter8/TangentOrbit · EpGroupoid.preimage_tangentOrbitProj_image`. *Step 2, the action of $G_x$ on $TU(x)$.* For a natural representation at $x \in X_1$, $TU(x) = p^{-1}(U(x))$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangentU` carries the action $g \mapsto T\Phi(g)$ by the tangents of the natural action, sc-diffeomorphisms of $TU(x)$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tΦ Chapter8/TangentOrbit · EpGroupoid.NaturalRep.isScDiffeoOn_tΦ Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangentMulAction`, with the tangent $T\Gamma(g)$ of the sheet `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tΓ`. Its linearity on the fibres is not formalized (gap G.7), and the local bundle model $TO \to O^1$ is not used. *Step 3, the open embedding.* Two vectors of $TU(x)$ have the same image in $|TX|$ iff they are in one $G_x$-orbit `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangent_π_eq_iff`: a morphism $k : v \to w$ of $TX$ lies over a morphism between points of $U(x)$, so $p(k) = \Gamma(g, s(p\,k))$ by Theorem 7.1.13 (3) (2017: 7.1), and $k = T\Gamma(g)(v)$ by fibre injectivity. So $G_x\backslash TU(x) \to |TX|$ is an open embedding onto $|TU(x)|$, the argument of Proposition 7.1.19 (2017: 7.6) for $TX$, which the text leaves implicit `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.isOpenEmbedding_tangentOrbitQuotientMap Chapter8/TangentOrbit · EpGroupoid.NaturalRep.range_tangentOrbitQuotientMap`. *Step 4, metrizability.* $TU(x)$ is metrizable as a subspace of $TX$ (Part I's Proposition 2.3.15 (3) (2017: 2.5)) `Chapter2/Metrizable · MPolyfold.metrizableSpace_tangentSpace`, and $G_x\backslash TU(x)$ by the metric $\hat{\rho}$ of Lemma 7.3.2 (2017: 7.4), repeated for a finite group acting by homeomorphisms on a metrizable space; hence $|TU(x)|$ is metrizable `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.metrizableSpace_image_tangentU`, and by Step 1 it is $|p|^{-1}(|U(x)_1|)$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.image_tangentU_eq_preimage`, $|U(x)_1|$ being open in $|X^1|$. These are the text's two metrizability claims `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.metrizableSpace_image_liftU`. *Step 5.* $|X^1|$ is metrizable by Proposition 7.5.3 (2017: 7.12) `Chapter7/Lift · EpGroupoid.metrizableSpace_liftOne_orbitSpace`, hence paracompact and Hausdorff; every point of it lies in some $|U(x)_1|$, for a natural representation at a representative (Theorem 7.1.13 (2017: 7.1)) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13`, whose preimage is metrizable by Step 4. The pulled-back Smirnov theorem makes $|TX|$ metrizable, hence paracompact (Stone). The closed sets $\tilde{A}_z$, the regularity of $|TX|$ and Proposition 2.7.4 (2017: 2.17) are not used; the step shared with the text is the metrizability of $|TU(x_z)|$. The theorem is not vacuous `Chapter8/TangentOrbit · EpGroupoid.hasParacompactOrbitSpace_tangent_trivial`. "parcompact" (possible misprint) (Findings, Chapter 8, items 20, 21, 104) (2017 p. 268).

## VIII.10 Strong bundle maps, and the pull-back along $s$

**Strong bundle maps and the pull-back** (V8.19). §8.3 works with strong bundle maps between strong bundles over M-polyfolds, the structure map $\mu : \mathbf{X}_s\times_P W \to W$ first of all, and with the pull-back of $P : W \to X$ along $s$; it defines neither notion beyond Part I. *The pull-back* is Part I's Proposition 2.6.8 (2017: 2.16) (§II.20) `Chapter2/StrongBundle · pullbackSpace` `Chapter3/ParamPerturbation · StrongBundle.pullback`, and its paracompactness, which Part I's definition of a strong bundle requires, is proved `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` `Chapter8/StrongBundleMap · StrongBundle.paracompactSpace_pullback`. Added here: its atlas, built from pairs of charts `Chapter8/StrongBundleMap · StrongBundle.pullback_atlas` `Chapter8/StrongBundleMap · PullbackChartData.exists_mem_atlas`; the two projections are sc-smooth `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackProj` `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackProj'`; its universal property on $[0]$, a map into the pull-back being sc-smooth iff its two components are `Chapter8/StrongBundleMap · StrongBundle.isScSmoothOn_to_pullback_iff` `Chapter8/StrongBundleMap · pullbackPair` `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackPair`, sc-smoothness being checked chart by chart `Chapter8/StrongBundleMap · MPolyfold.isScSmoothMap_of_forall_charts`; and over a sheet $s : \mathbf{W} \to V$ with inverse $\sigma$ the projection $(\varphi, w) \mapsto w$ is an sc-diffeomorphism onto $P^{-1}(V)$, with inverse $w \mapsto (\sigma(Pw), w)$ `Chapter8/StrongBundleMap · pullbackSheetInv` `Chapter8/StrongBundleMap · StrongBundle.isScDiffeoOn_pullbackProj'`, so a local sc-diffeomorphism `Chapter8/StrongBundleMap · StrongBundle.isLocalScDiffeo_pullbackProj'`. *Strong bundle maps.* Part I defines them only between local models (Definition 2.6.1 (2017: 2.22), Definition 2.47 of the preprint, §II.18), and asks the transition maps of a strong bundle atlas to be sc-smooth on $[0]$ and $[1]$. The global notion is a reading `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap`: $\Phi : Y \to Y'$ with $P' \circ \Phi = f \circ P$, linear on the fibres `Chapter8/StrongBundleMap · fiberRestrict`, sc-smooth $Y[0] \to Y'[0]$, mapping $Y_{0,1}$ into $Y'_{0,1}$ and sc-smooth $Y[1] \to Y'[1]$. In the charts the last two conditions say that the local representatives $K[i] \to K'[i]$ are sc-smooth, as Part I asks of transition maps. The preservation of the double filtration, part of Definition 2.6.1 (2017: 2.22), is not assumed but follows `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.mapsTo_doubleLevel`, and so do the continuity of $\Phi$, the boundedness of its fibre maps and the sc-smoothness of the base map $f = P' \circ \Phi \circ 0$ `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.continuous` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.fiberMap` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.isScSmoothMap_base`. Strong bundle maps compose and identities are strong bundle maps `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.comp` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.id`; an isomorphism is a pair of inverse strong bundle maps `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleIso`; and the projection of the pull-back to $W$ is a strong bundle map covering $s$, the identity on each fibre `Chapter8/StrongBundleMap · StrongBundle.isStrongBundleMap_pullbackProj'` `Chapter8/StrongBundleMap · StrongBundle.fiberMap_pullbackProj'`. Strong bundle maps over open subsets of the base are not formalized (§VIII.27) (Findings, Chapter 8, items 30, 31) (used on pp. 373–396 (2017 pp. 270–285)).

## VIII.11 Definition 8.3.1 (2017: 8.4): strong bundles over ep-groupoids, and the fibre maps

**Definition 8.3.1 (2017: 8.4)** (V8.20) `Chapter8/EpStrongBundle · EpStrongBundle`. *A strong bundle over $X$ is a pair $(P, \mu)$: a strong bundle $P : W \to X$ over the object M-polyfold and a strong bundle map $\mu : \mathbf{X}_s\times_P W \to W$ covering $t$, with (1) $\mu$ a surjective local sc-diffeomorphism, linear on the fibres; (2) $\mu(1_x, w) = w$; (3) $\mu(\psi \circ \gamma, w) = \mu(\psi, \mu(\gamma, w))$.* The formalization takes the definition as printed, on Part I's pull-back and with

the strong bundle maps of §VIII.10; the conditions $s(\psi) = P(w)$ are carried by the points of the pull-back, the pairs $(1_{Pw}, w)$ are points of it `Chapter8/EpStrongBundle · EpGroupoid.unitPair`, and the composite $P \circ \mu$ is $t \circ \pi_1$ `Chapter8/EpStrongBundle · EpStrongBundle.P_μ`. The surjectivity in (1) follows from (2) and is kept, as in the text; $\mu$ is continuous `Chapter8/EpStrongBundle · EpStrongBundle.continuous_μ`. The definition is met by every strong bundle over an M-polyfold, as a strong bundle over the trivial ep-groupoid with $\mu(z, w) = w$ `Chapter8/NonVacuityBundle · EpStrongBundle.trivial` (Findings, Chapter 8, items 32, 102) (2017 p. 271).

**The fibre maps** $\mu(\psi, \cdot)$ (V8.21) `Chapter8/EpStrongBundle · EpStrongBundle.fiberMorEquiv`. *For $\psi : x \to y$, $\mu(\psi, \cdot) : W_x \to W_y$ is a linear isomorphism*, stated after the definition without proof. It is proved in the stronger form of a topological linear isomorphism `Chapter8/EpStrongBundle · EpStrongBundle.fiberMor`, whose value at $w$ is $\mu(\psi, w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fiberMor_apply`. *Step 1, linear and bounded.* $\mu$ is linear on the fibres of $\mathbf{X}_s\times_P W \to \mathbf{X}$, so $\mu(\psi, \cdot)$ is additive and homogeneous `Chapter8/EpStrongBundle · EpStrongBundle.μ_add` `Chapter8/EpStrongBundle · EpStrongBundle.μ_smul`, and bounded, the fibre maps of a strong bundle map being bounded (§VIII.10). *Step 2, the inverse* is $\mu(\psi^{-1}, \cdot)$: by (3) and (2), $\mu(\psi^{-1}, \mu(\psi, w)) = \mu(1_x, w) = w$, and symmetrically `Chapter8/EpStrongBundle · EpStrongBundle.μ_inv_μ` `Chapter8/EpStrongBundle · EpStrongBundle.μ_μ_inv` (Findings, Chapter 8, item 32) (2017 p. 271).

## VIII.12 The functorial picture: the ep-groupoid $(W, \mathbf{W})$

**The functorial picture** (V8.22) `Chapter8/EpStrongBundle · EpStrongBundle.toEpGroupoid`. *With $\mathbf{W} = \mathbf{X}_s\times_P W$, $s(\psi, w) = w$, $t(\psi, w) = \mu(\psi, w)$, units $(1_x, w)$, inverses $(\psi^{-1}, \mu(\psi, w))$ and composition $(\psi, w') \circ (\gamma, w) = (\psi \circ \gamma, w)$, $(W, \mathbf{W})$ is an ep-groupoid, and $(P, \pi_1) : (W, \mathbf{W}) \to (X, \mathbf{X})$ is an sc-smooth functor.* **The text** says "we can define the ep-groupoid" and gives no proof. **The formalized proof** checks Definition 7.1.3 (2017: 7.3). *Step 1, the groupoid.* The composition is total, with the junk value $(\gamma, w)$ off composable pairs, and the composite of a composable pair has target $\mu(\psi \circ \gamma, w) = \mu(\psi, w')$ by Definition 8.3.1 (3) (2017: 8.4) `Chapter8/EpStrongBundle · EpStrongBundle.morComp` `Chapter8/EpStrongBundle · EpStrongBundle.morComp_val` `Chapter8/EpStrongBundle · EpStrongBundle.μ_morComp`; the inverse is a morphism $\mu(\psi, w) \to w$ by Definition 8.3.1 (2), (3) (2017: 8.4) `Chapter8/EpStrongBundle · EpStrongBundle.morInv`, and the groupoid laws are those of $\mathbf{X}$ in the first entry `Chapter8/EpStrongBundle · EpStrongBundle.toGroupoidStr`. *Step 2, (1) and (2).* $s_{\mathbf{W}}$ is the projection of the pull-back along the local sc-diffeomorphism $s$, a surjective local sc-diffeomorphism by §VIII.10, and $t_{\mathbf{W}} = \mu$ is one by Definition 8.3.1 (1) (2017: 8.4); the units and inverses are sc-smooth by the universal property of the pull-back. *Step 3, (3), by sequences.* For $w \in W$ take $V(Pw)$ as in Definition 7.1.3 (3) (2017: 7.3) for $X$ and $V = P^{-1}(V(Pw))$. If $(\psi_k, w_k)$ has $w_k \in \operatorname{cl} V$ and $\mu(\psi_k, w_k) \to w'$, then $s(\psi_k) = P(w_k) \in \operatorname{cl} V(Pw)$ and $t(\psi_k) = P(\mu(\psi_k, w_k)) \to P(w')$, so a subsequence has $\psi_k \to \psi$ by the properness of $X$, and $w_k = \mu(\psi_k^{-1}, \mu(\psi_k, w_k)) \to \mu(\psi^{-1}, w')$ by the continuity of $\mu$ and $\iota$ `Chapter8/EpStrongBundle · EpStrongBundle.exists_isProperMap_tgt`. *Step 4, (4)* is proved in Chapter 7's sheet form, through a sheet of $s$ in $\mathbf{X}$ and condition (4) for $X$ `Chapter8/EpStrongBundle · EpStrongBundle.isScSmoothCompSheet_toGroupoidStr`. *Step 5, the functor.* $s_{\mathbf{W}}$, $t_{\mathbf{W}}$ are strong bundle maps covering $s$, $t$ `Chapter8/EpStrongBundle · EpStrongBundle.isStrongBundleMap_src` `Chapter8/EpStrongBundle · EpStrongBundle.isStrongBundleMap_tgt`, $(P, \pi_1)$ is an sc-smooth functor `Chapter8/EpStrongBundle · EpStrongBundle.projFunctor`, and $|P| : |W| \to |X|$ is continuous and surjective `Chapter8/EpStrongBundle · EpStrongBundle.continuous_orbitMap_projFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.surjective_orbitMap_projFunctor`, since a point isomorphic to $P(w)$ is the image of a point isomorphic to $w$, namely $\mu(\psi, w)$ `Chapter8/EpStrongBundle · EpStrongBundle.exists_isomorphic_of_isomorphic`. An `EpGroupoid` has its objects and morphisms in one universe, so from here on $W$ lives in the universe of the objects and morphisms of $X$ (Findings, Chapter 8, items 2, 33) (pp. 375–377 (2017

pp. 271–272)).

## VIII.13 Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6), and "these are identical notions"

**Definition 8.3.2 (2017: 8.5)** (V8.23) `Chapter8/EpStrongBundle · EpStrongBundle.IsMap`. *A strong bundle map $(P,\mu) \to (P',\mu')$ is a strong bundle map $\Phi : W \to W'$ covering the object map of an sc-smooth functor $\phi : X \to X'$, with $\Phi(\mu(\gamma,w)) = \mu'(\phi(\gamma),\Phi(w))$.* As printed, with the strong bundle maps of §VIII.10; the identity is one `Chapter8/NonVacuityBundle · EpStrongBundle.isMap_id` (Findings, Chapter 8, item 34) (2017 p. 273).

**Definition 8.3.3 (2017: 8.6)** (V8.24) `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor`. *A strong bundle functor is a functor $\Phi : (W,\mathbf{W}) \to (W',\mathbf{W}')$, linear on the fibres, covering an sc-smooth functor $\phi$ and inducing strong bundle maps $W \to W'$ and $\mathbf{W} \to \mathbf{W}'$.* "Linear on the fibres", "covers $\phi$" and "induces strong bundle maps" are read together as two strong bundle maps in the sense of §VIII.10: on objects, for $P$ and $P'$, covering $\phi$ on objects, which contains $P' \circ \Phi = \phi \circ P$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.projFunctor_comp`; and on morphisms, for the pull-backs $\mathbf{W} \to \mathbf{X}$ and $\mathbf{W}' \to \mathbf{X}'$, covering $\phi$ on morphisms. The second condition forces the morphism map into the form $(\gamma,w) \mapsto (\phi(\gamma),\Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.mor_val`. The identity is a strong bundle functor `Chapter8/NonVacuityBundle · EpStrongBundle.isStrongBundleFunctor_id` (Findings, Chapter 8, item 34) (2017 p. 274).

**The equivalence** (V8.25) `Chapter8/EpStrongBundle · EpStrongBundle.isMap_iff_strongBundleFunctor`. *$\Phi : W \to W'$ is a strong bundle map of Definition 8.3.2 (2017: 8.5) iff it is the object map of a strong bundle functor of Definition 8.3.3 (2017: 8.6), and the functor, $\Phi(\gamma,w) = (\phi(\gamma),\Phi(w))$, is unique.* The text states that "these are identical notions" and that the functor is recovered from the map. *Step 1, from the functor to the map.* The object map of a strong bundle functor is a strong bundle map by definition, and its compatibility with $\mu$ is the functor's compatibility with targets, read through the form of the morphism map: $\Phi(\mu(\gamma,w)) = \Phi(t(\gamma,w)) = t'(\phi(\gamma),\Phi(w)) = \mu'(\phi(\gamma),\Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.isMap`. *Step 2, from the map to the functor.* $(\gamma,w) \mapsto (\phi(\gamma),\Phi(w))$ lands in the pull-back, since $s(\phi(\gamma)) = \phi(Pw) = P'(\Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.liftMor`; it is a functor by the functor laws of $\phi$ and the compatibility with $\mu$ `Chapter8/EpStrongBundle · EpStrongBundle.IsMap.liftFunctor`. It is a strong bundle map of the pull-backs: linear on the fibres `Chapter8/EpStrongBundle · isLinearMap_fiberRestrict_pullbackMap`, sc-smooth on $[0]$ by the universal property of §VIII.10, and on $[1]$ by the universal property of the pull-back on level $[1]$, which is proved here `Chapter8/EpStrongBundle · StrongBundle.pullbackProj'One` `Chapter8/EpStrongBundle · StrongBundle.isScSmoothMap_to_pullback_one` `Chapter8/EpStrongBundle · EpStrongBundle.IsMap.isStrongBundleFunctor_liftFunctor`. *Step 3, uniqueness.* Every strong bundle functor with object map $\Phi$ has this morphism map, by the form of Definition 8.3.3's (2017: 8.6) morphism maps `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.eq_liftFunctor` (Findings, Chapter 8, item 34) (pp. 377–379 (2017 pp. 273–274)).

## VIII.14 Definition 8.3.4 (2017: 8.7): section functors

**Definition 8.3.4 (2017: 8.7)** (V8.26). *(1) An sc-smooth section functor of $(P,\mu)$ is an sc-smooth section $f$ of $P$ with $f(y) = \mu(\varphi,f(x))$ for every $\varphi : x \to y$; (2) an sc$^+$-section functor takes values in $W_{0,1}$ and is sc-smooth into $W[1]$; (3) an sc-Fredholm section functor is an sc-smooth section functor that is*

*an sc-Fredholm section of $P$. The sc-smooth section functors form a vector space $\Gamma(P,\mu)$, the sc$^+$-section functors a subspace $\Gamma^+(P,\mu)$.* The three notions `Chapter8/EpStrongBundle · EpStrongBundle.IsFunctorialSection` `Chapter8/EpStrongBundle · EpStrongBundle.IsScSmoothSectionFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.IsScPlusSectionFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.IsScFredholmSectionFunctor` and their sets `Chapter8/EpStrongBundle · EpStrongBundle.sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.fredholmSectionFunctors` are as printed; (3) uses Part I's Definition 3.1.16 (2017: 3.8) in its germ reading (§III.48) `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection`, and Part I's sections `Chapter2/StrongBundle · StrongBundle.IsScSmoothSection`. *"Vector space"* is read as closure under Part I's fibrewise zero section, sums and constant multiples `Chapter3/StrongBundleSections · fiberZero` `Chapter3/StrongBundleSections · sectionAdd`; no `Module` structure is built. *Step 1, the analytic point.* Part I does not state that the sum of two sc-smooth sections, or a constant multiple of one, is sc-smooth. It is proved in the charts: a section is sc-smooth as soon as its principal parts are, as maps into the local strong bundle models `Chapter8/EpStrongBundle · StrongBundle.isScSmoothSection_of_principal`, and the principal part of a sum is the sum of the principal parts `Chapter8/EpStrongBundle · StrongBundle.IsScSmoothSection.sectionAdd_of_isScSmoothSection` `Chapter8/EpStrongBundle · StrongBundle.IsScSmoothSection.sectionSmul_const` `Chapter8/EpStrongBundle · StrongBundle.isScSmoothSection_fiberZero`. *Step 2, functoriality* is preserved because each $\mu(\varphi,\cdot)$ is linear (§VIII.11). So $\Gamma$ contains the zero section and is closed under sums and constant multiples `Chapter8/EpStrongBundle · EpStrongBundle.fiberZero_mem_sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionAdd_mem_sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionSmul_mem_sectionFunctors`, and so is $\Gamma^+ \subseteq \Gamma$, the values in $W_{0,1}$ and the sc-smoothness into $W[1]$ being preserved in the same way `Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors_subset` `Chapter8/EpStrongBundle · EpStrongBundle.fiberZero_mem_plusSectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionAdd_mem_plusSectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionSmul_mem_plusSectionFunctors`; the sc-Fredholm section functors lie in $\Gamma$ `Chapter8/EpStrongBundle · EpStrongBundle.fredholmSectionFunctors_subset`. All three sets are inhabited by non-zero sections: over a trivial ep-groupoid every section is functorial, and the graph section of a smooth map $\mathbb{R}^n \to \mathbb{R}^k$ lies in $\Gamma$ and is sc-Fredholm (§VIII.26) (Findings, Chapter 8, items 35, 102, 103) (pp. 378–384 (2017 pp. 274–275)).

## VIII.15 Definition 8.5.1 (2017: 8.8), and the degeneracy remark after it

**Definition 8.5.1 (2017: 8.8)** (V8.27) `Chapter8/Basic · EpFunctor.IsProperCovering`. *An sc-smooth functor $F : Y \to X$ is a proper covering functor if (1) $F$ is a surjective local sc-diffeomorphism on objects; (2) every $F^{-1}(x)$ is finite, and $x$ has a neighbourhood $U(x)$ with $F^{-1}(U(x))$ the disjoint union of neighbourhoods $U(y)$, $y \in F^{-1}(x)$, each mapped sc-diffeomorphically onto $U(x)$; (3) $L_F : \varphi \mapsto (F(\varphi), s(\varphi))$ is an sc-diffeomorphism $\mathbf{Y} \to \mathbf{X}\,{}_s\!\times_F Y$.* In (2) $U(x)$ is open and the local inverses $\sigma(y)$ of $F$ on the $U(y)$ are part of the data; (3) is stated on Part I's fibred-product sub-M-polyfold of $\mathbf{X} \times Y$ (Proposition 2.5.20 (2017: 2.15), Proposition 2.46 of the preprint, §II.17) `Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct` `Chapter8/Basic · EpFunctor.fibredMap`, as Chapter 7 states Definition 7.1.3 (4) (2017: 7.3). The identity functor of every ep-groupoid is a proper covering functor `Chapter8/NonVacuityBundle · EpFunctor.isProperCovering_id` (Findings, Chapter 8, items 50, 102) (2017 p. 275).

**The degeneracy remark** (V8.28). *For $\varphi \in \mathbf{Y}$, $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}\,{}_s\!\times_F Y}(F(\varphi), s(\varphi)) = \frac{1}{2}\, d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi))$.* **The text** argues that $L_F$ is an sc-diffeomorphism and that $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, with $d_{\mathbf{X}}(F(\varphi)) = d_Y(s(\varphi)) = d_{\mathbf{Y}}(\varphi)$, $F$ and $s$ being local sc-diffeomorphisms. **The formalized proof** splits the claims. *Step 1, in general.* $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}\,{}_s\!\times_F Y}(L_F(\varphi))$ and $d_{\mathbf{X}}(F(\varphi)) = d_{\mathbf{Y}}(\varphi)$, by Part I's Proposition 2.4.4 (2017: 2.7), the invariance of $d$ under sc-diffeomorphisms (§II.11) `Chapter2/DegeneracyIndex · proposition_2_`

4_4 Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_fibredMap Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_mor. *Step 2, the product formula.* Part I's formalization has $d_{X\times Y} = d_X + d_Y$ for tame factors Chapter2/FibredProduct · MPolyfold.isTame_prod and in general only $d_{X\times Y} \le d_X + d_Y$ Chapter2/FibredProduct · MPolyfold.degIndex_prod_le, and Part I's Proposition 2.5.20 (2017: 2.15) states its formula $d = \frac{1}{2} d_{X\times Y}$ for tame $X$, $Y$. So $d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi)) = 2\, d_{\mathbf{Y}}(\varphi)$, and with it the remark's second equality, is proved for tame $X$ and $Y$ ($\mathbf{X}$ is then tame, $s$ being a local sc-diffeomorphism) Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_prod_fibredMap, and in general as the inequality $\le$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_prod_fibredMap_le. As we read it, the remark needs $X$ and $Y$ tame, which the text does not state; whether the equality holds without tameness is not settled here (Findings, Chapter 8, item 51) (pp. 379–385 (2017 pp. 275–276)).

## VIII.16 Proposition 8.5.2 (2017: 8.2): composites of proper covering functors

**Proposition 8.5.2 (2017: 8.2)** (V8.29) Chapter8/ProperCovering · EpFunctor.IsProperCovering.proposition_8_5_2. *If $F : Y \to X$ and $H : Z \to Y$ are proper covering functors, so is $F \circ H$.* **The text's proof** says that (1) and (2) "follow easily", and for (3) writes $L_{F\circ H} = \tilde{K} \circ L_H$ with $\tilde{K}(\psi, z) = (F(\psi), z)$, whose inverse is $\tilde{L}(\varphi, z) = (L_F^{-1}(\varphi, H(z)), z)$. **The formalized proof.** *Step 1, (1)* holds since composites of surjective local sc-diffeomorphisms are such, and the fibre of $F \circ H$ over $x$ is the finite union of the fibres of $H$ over the points of $F^{-1}(x)$. *Step 2, (2).* The even covering of $F$ at $x$ has to be shrunk: take $U(x)$ for $F$ and, for each $y \in F^{-1}(x)$, the evenly covered neighbourhood $U'(y)$ of $H$; the set $U_0 = \bigcap_y F(U(y) \cap U'(y))$, a finite intersection, is an open neighbourhood of $x$ with $(F|U(y))^{-1}(U_0) \subseteq U'(y)$, and over $U_0$ the sheets of $F \circ H$ are those of $H$ over the $U'(y)$, cut down to the preimages of the sets $(F|U(y))^{-1}(U_0)$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.evenlyCovered_comp. *Step 3, (3)*, through the text's $\tilde{K}$ and $\tilde{L}$, which are inverse to each other Chapter8/ProperCovering · EpFunctor.IsProperCovering.isScDiffeo_fibredMap_comp. Their sc-smoothness, as maps between sub-M-polyfolds of products, uses Part I's Proposition 2.3.19 (2) (2017: 2.6) (a map into a sub-M-polyfold is sc-smooth as soon as it is sc-smooth into the ambient M-polyfold), proved again privately, as in Chapter 7. The proof has "fo" for "for" (possible misprint) (Findings, Chapter 8, item 52) (pp. 384–386 (2017 pp. 276–277)).

## VIII.17 Theorem 8.5.3 (2017: 8.5) (Proper Covering I), and Remark 8.5.4 (2017: 8.2)

**Theorem 8.5.3 (2017: 8.5)** (V8.30) Chapter8/ProperCoveringLocal · EpFunctor.theorem_8_5_3. *Let $F : Y \to X$ be a proper covering functor and $x \in X$. For a connected, sufficiently small open $U(x)$ admitting the natural $G_x$-action: (0) $F^{-1}(U(x))$ is the disjoint union of connected open $U(y) \ni y$, $y \in F^{-1}(x)$, with $F : U(y) \to U(x)$ sc-diffeomorphisms; (1) $\mathbf{U}(x) = \{\varphi \mid s(\varphi), t(\varphi) \in U(x)\}$ is the disjoint union of the components $\mathbf{U}(g)$, $g \in G_x$, on which $s, t$ are sc-diffeomorphisms onto $U(x)$; (2) the morphisms from $U(y)$ to $U(y')$ form the disjoint union of connected open $\mathbf{U}(\psi_0) \ni \psi_0$, $\psi_0 : y \to y'$, with (3) $s : \mathbf{U}(\psi_0) \to U(y)$ and $t : \mathbf{U}(\psi_0) \to U(y')$ sc-diffeomorphisms, (4) $F : \mathbf{U}(\psi_0) \to \mathbf{U}(F(\psi_0))$ an sc-diffeomorphism, and (5) a commutative square of sc-diffeomorphisms with top arrow $F$, left arrow $s$, right arrow $t$ and bottom arrow $F : U(s(\psi_0)) \to U(t(F(\psi_0)))$.* The data form a chart Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart: a natural representation $R$ of $G_x$ on $U(x)$ (Chapter 7's datum, §VII.7, with $U(x)$ open and connected), the sets $U(y)$ and $\mathbf{U}(\psi_0)$, and (0)–(5) as fields; $\mathbf{U}(g)$ is read as the connected component of $\mathbf{U}(x)$ at $g$, which contains $F(\psi_0)$ by Re-

mark 8.5.4 (2017: 8.2) (a). "Sufficiently small" is proved in two forms: charts with $U(x)$ inside every open $V \ni x$ (`theorem_8_5_3`), and one open $W \ni x$ such that every natural representation on a connected $U(x) \subseteq W$ is the $R$ of a chart `Chapter8/ProperCoveringLocal · EpFunctor.theorem_8_5_3_of_naturalRep`.

**The text's proof** starts from $U'(x)$ and the $U'(y)$ of Definition 8.5.1 (2) (2017: 8.8), takes connected $V(y) \subseteq U'(y)$ with natural representations of the $G_y$ and, for each $\psi : y \to y'$, sheets $V(\psi)$ with $\widehat{\psi} = t \circ (s|V(\psi))^{-1} : V(y) \to V(y')$ (8.15), so that the morphisms between points of $\bigcup V(y)$ form the disjoint union of the $V(\psi)$. It shrinks to a $G_x$-invariant connected $U(x)$, puts $U(y) = (F|V(y))^{-1}(U(x))$, shows that $\widehat{\psi}$ maps $U(y)$ onto $U(y')$, and sets $\mathbf{U}(\psi) = (s|V(\psi))^{-1}(U(y))$; (4) comes from $F = (s|\mathbf{U}(g))^{-1} \circ F \circ s$ on $\mathbf{U}(\psi_0)$, and (5) is called obvious. **The formalized proof** uses Definition 8.5.1 (3) (2017: 8.8) instead of natural representations in $Y$: the sheets $\mathbf{U}(\psi_0)$ are the images of continuous sections of $s$ given by $L_F^{-1}$. *Step 1, the input* `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data`: $U'(x)$ with the $U'(y)$ and local inverses $\sigma(y)$ of $F$ (Definition 8.5.1 (2) (2017: 8.8)), the continuous inverse $L_F^{-1}$ of the sc-diffeomorphism $L_F$ (Definition 8.5.1 (3) (2017: 8.8)), and a natural representation $R$ with $U(x) \subseteq U'(x)$; one $W = U'(x)$ serves for every such $R$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.exists_data`. *Step 2, (0).* $U(y) = U'(y) \cap F^{-1}(U(x)) = \sigma(y)(U(x))$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.U` is open, connected as the image of the connected $U(x)$, and mapped sc-diffeomorphically onto $U(x)$. *Step 3, (1)* involves only $R$ and is Chapter 7's description of $\mathbf{U}(x)$ by the sheets $\Gamma(g, U(x))$ (§VII.8) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.iUnion_image_Γ` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.connectedComponentIn_morOn`. *Step 4, the sheets.* For $\psi_0 : y \to y'$ with $g = F(\psi_0) \in G_x$, put $\tau(z) = L_F^{-1}(\Gamma(g, F(z)), z)$ for $z \in U(y)$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.τ` and $\mathbf{U}(\psi_0) = \tau(U(y))$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.sheet`. $\tau$ is a continuous section of $s$ with $\tau(y) = \psi_0$, and a continuous section of a local sc-diffeomorphism over an open set is sc-smooth, being locally a local inverse `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.isScSmoothOn_of_section` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScSmoothOn_τ`; this gives $s$ in (3). *Step 5, $t$ in (3).* $F(t(\tau z)) = t(\Gamma(g, Fz)) = \Phi(g)(Fz) \in U(x)$, so $t \circ \tau$ maps $U(y)$ into $F^{-1}(U(x))$, the disjoint union of the open $U(y'')$; since $U(y)$ is connected and $t(\tau(y)) = y'$, it maps into $U(y')$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.mapsTo_of_isPreconnected` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.tgt_τ_mem`. This is the counterpart of the text's claim that $\widehat{\psi}$ maps $U(y)$ onto $U(y')$; there $t \circ \tau = \sigma(y') \circ \Phi(g) \circ F$, an sc-diffeomorphism with inverse $\sigma(y) \circ \Phi(g)^{-1} \circ F$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.isScDiffeoOn_congr_left` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScDiffeoOn_hat`. *Step 6, (2).* A morphism $\psi$ from $U(y)$ to $U(y')$ has $F(\psi) \in \mathbf{U}(x)$, so $F(\psi) = \Gamma(g, F(s\psi))$ for one $g \in G_x$ by (1), and $\psi = L_F^{-1}(\Gamma(g, F(s\psi)), s\psi)$ lies in the sheet of the morphism $\psi_0$ out of $y$ with $F(\psi_0) = g$; distinct $\psi_0$ give disjoint sheets by the injectivity of $L_F$, and the sheets are the connected components `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.connectedComponentIn_eq_of_iUnion`. *Step 7, (4)* is the text's formula: $F = \Gamma(g, \cdot) \circ F \circ s$ on $\mathbf{U}(\psi_0)$, an sc-diffeomorphism onto $\Gamma(g, U(x)) = \mathbf{U}(g)$ with inverse $\tau \circ \sigma(y) \circ s$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScDiffeoOn_mor_sheet`. *Step 8, (5)* is the functoriality $F(s\psi) = s(F\psi)$ (the field `comm_src`), together with the companion square $F(t\psi) = t(F\psi)$ (the field `comm_tgt`); the arrows are sc-diffeomorphisms by (0), (1), (3), (4). The chart is assembled `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.toChart`, and charts exist inside every $V$ because natural representations do (Theorem 7.1.13 (2017: 7.1)) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13`. The finiteness of $F^{-1}(x)$ and Definition 8.5.1 (1) (2017: 8.8) are not used.

*(5) as printed.* With $t$ on the right arrow and the corner $U(t(F(\psi_0)))$, the square says $F(s(\psi)) = t(F(\psi))$ on $\mathbf{U}(\psi_0)$. A functor gives $F(s(\psi)) = s(F(\psi))$, and $F(\psi) = \Gamma(g, F(s\psi))$ has $t(F(\psi)) = \Phi(g)(F(s\psi))$,

which differs from $F(s\psi)$ wherever $\Phi(g)$ does not fix $F(s\psi)$. So (5) as printed does not hold as we read it: for the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$, a proper covering functor, every chart at $0$ contains, in the sheet of $\psi_0 = (-1, 0)$ `Chapter8/NonVacuity · CoveringExamples.reflectionψ₀`, a $\psi = (-1, z)$ with $F(s\psi) = z \neq -z = t(F\psi)$ `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed`, and such charts exist `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed_nonvacuous`. The square holds with $s$ on both vertical arrows (the corner $U(s(F(\psi_0))) = U(x)$ is the same set), which suggests $t$ as a possible misprint for $s$. Possible misprints in the proof: "$g \in G_y$" and "$G_y$-invariant" for $G_x$, "$F(z') = s(F(\varphi))$" for $t(F(\varphi))$, "admitting admitting", and "morphismsm" in (2) (Findings, Chapter 8, items 53, 54, 55) (pp. 385–388 (2017 pp. 277–279)).

**Remark 8.5.4 (2017: 8.2)** (V8.31). *(a) A morphism between points of $F^{-1}(x)$ is mapped into $G_x$; (b) everything in Theorem 8.5.3 (2017: 8.5) is determined by the choice of $U(x)$.* (a) is functoriality `Chapter8/ProperCoveringLocal · EpFunctor.remark_8_5_4_a`, and defines the element of $G_x$ `Chapter8/ProperCoveringLocal · EpFunctor.isotropyMor`. (b) is read as: two charts with the same $U(x)$ have the same $\Phi$ and $\Gamma$ on $U(x)$, the same $U(y)$ and the same $\mathbf{U}(\psi_0)$ `Chapter8/ProperCoveringLocal · EpFunctor.remark_8_5_4_b`. *Step 1.* On a connected $U(x)$, $\Gamma(g, \cdot)$ is the continuous section of $s$ through $g$, and two such sections agree (§VII.9) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_eqOn`; $\Phi(g) = t \circ \Gamma(g, \cdot)$ follows. *Step 2.* The $U(y)$ and the $\mathbf{U}(\psi_0)$ are the connected components of $F^{-1}(U(x))$ at $y$ and of the morphisms from $U(y)$ to $U(y')$ at $\psi_0$, so they are determined by $U(x)$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart.connectedComponentIn_U` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart.connectedComponentIn_sheet`. The remarks on the naturality of Theorem 7.1.13's (2017: 7.1) construction have no further formal content (Findings, Chapter 8, item 56) (pp. 385–387 (2017 pp. 277–278)).

## VIII.18 $\mathrm{Diff}_{sc}$, Definition 8.5.5 (2017: 8.9), and the discussion after it

**$\mathrm{Diff}_{sc}$ and Definition 8.5.5 (2017: 8.9)** (V8.32). *$\mathrm{Diff}_{sc}$ has the M-polyfolds as objects and the sc-diffeomorphisms as morphisms. For finite groupoids $C$, with one object, and $D$, a functor $f : D \to C$ is a covering functor if $\psi \mapsto (f(\psi), s(\psi))$ is a bijection $\mathbf{D} \to C \,{}_s\!\times_f D$; a geometric lift of $f$ is a pair of functors $L_D : D \to \mathrm{Diff}_{sc}$, $L_C : C \to \mathrm{Diff}_{sc}$ with a natural transformation $\tau : L_D \to L_C \circ f$.* $\mathrm{Diff}_{sc}$ is the category of bundled M-polyfolds of one universe `Chapter8/Basic · MPolyfoldCat` with Part I's sc-diffeomorphisms, bundled with their inverses `Chapter8/Basic · ScDiffeo` `Chapter8/Basic · ScDiffeo.id` `Chapter8/Basic · ScDiffeo.comp` `Chapter8/Basic · ScDiffeo.ext`. $C$ is Mathlib's `SingleObj G` for a group $G$, its automorphism group, and $D$ a Mathlib groupoid, both in `Type`; then $C \,{}_s\!\times_f D = G \times D$ `Chapter8/Basic · coveringMap` `Chapter8/Basic · IsCoveringFunctor` `Chapter8/Basic · GeometricLift`. The covering property says that a $g \in G$ lifts uniquely at every $y$ `Chapter8/GeometricLift · IsCoveringFunctor.existsUnique_lift` `Chapter8/GeometricLift · IsCoveringFunctor.lift`. The finiteness ("finitely many objects and morphisms") is not part of the two structures; it is carried as instance arguments by the results that use it, Proposition 8.5.7 (2017: 8.3) and Theorem 8.5.8 (2017: 8.6). $D = \emptyset$ is allowed, as in the text `Chapter8/GeometricLift · isCoveringFunctor_of_isEmpty` (§VIII.20). The definition is not vacuous in either direction: the identity of `SingleObj G` is a covering functor, and for non-trivial $G$ the functor sending everything to 1 is not `Chapter8/NonVacuity · CoveringExamples.isCoveringFunctor_id` `Chapter8/NonVacuity · CoveringExamples.not_isCoveringFunctor_one` (Findings, Chapter 8, items 57, 100) (pp. 387–389 (2017 pp. 279–280)).

**The discussion after Definition 8.5.5 (2017: 8.9)** (V8.33). *With $U(x) = L_C(\star)$, $U(y) = L_D(y)$, the maps $\tau(y) : U(y) \to U(x)$ and $\varphi_{yy'}(\psi) = L_D(\psi)$ are sc-diffeomorphisms, $\varphi_{yy''}(\psi \circ \gamma) = \varphi_{y'y''}(\psi) \circ \varphi_{yy'}(\gamma)$, $G_y$ acts on $U(y)$ and $G_x$ on $U(x)$, and $\tau(y)(h * z) = f(h) * \tau(y)(z)$.* The text states these as consequences of the definition; they are proved. $U(x)$ `Chapter8/GeometricLift · GeometricLift.Ux` and

$\tau(y)$ Chapter8/GeometricLift · GeometricLift.τy Chapter8/GeometricLift · GeometricLift.isScDiffeoOn_τy; $\varphi_{yy'}$ Chapter8/GeometricLift · GeometricLift.φ Chapter8/GeometricLift · GeometricLift.isScDiffeoOn_φ is functorial because $L_D$ is Chapter8/GeometricLift · GeometricLift.φ_comp Chapter8/GeometricLift · GeometricLift.φ_id Chapter8/GeometricLift · GeometricLift.φ_inv_φ. The actions are $h * z = \varphi_{yy}(h)(z)$ and $g * z = L_C(g)(z)$ Chapter8/GeometricLift · GeometricLift.smul_def Chapter8/GeometricLift · GeometricLift.smul_def', sc-smooth Chapter8/GeometricLift · GeometricLift.isScSmoothMap_smul; the action of $G_x$ is a left action through the composition of `SingleObj G`, in which $g \gg h = hg$. The naturality of $\tau$ at a morphism, $\tau(y')(\varphi_{yy'}(\psi)(z)) = f(\psi) * \tau(y)(z)$ Chapter8/GeometricLift · GeometricLift.τy_φ, specializes to the equivariance Chapter8/GeometricLift · GeometricLift.τ_equivariant. Functoriality is stated with Mathlib's $\gamma \gg \psi$ for the text's $\psi \circ \gamma$, and $G_y$ is Mathlib's monoid `End y` (Findings, Chapter 8, item 57) (pp. 388–390 (2017 pp. 280–281)).

## VIII.19 Disjoint unions of M-polyfolds

**Disjoint unions** (V8.34). *In the proof of Proposition 8.5.7 (2017: 8.3), "$Y = \bigsqcup U(y)$ possesses a natural M-polyfold structure".* The text does not describe it. It is read as the structure whose charts are those of the summands, transported along the inclusions Chapter8/SigmaPolyfold · MChart.pushforward Chapter8/SigmaPolyfold · MPolyfold.sigmaChart Chapter8/SigmaPolyfold · MPolyfold.sigma, for any index type. *Step 1, compatibility.* Charts of different summands have disjoint domains, and Part I's compatibility then asks for the sc-smoothness of maps between empty retracts, which holds, so nothing is added Chapter8/SigmaPolyfold · MChart.compatible_of_disjoint; the charts of the union are exactly the transported charts Chapter8/SigmaPolyfold · MPolyfold.isChart_sigmaChart_iff. A transported chart is extended by 0 off its image, so no summand needs to be nonempty. *Step 2, paracompactness* is proved summand by summand, without metrizability Chapter8/SigmaPolyfold · paracompactSpace_sigma. *Step 3, summand-wise properties.* The summands are open and sc-diffeomorphic to the $X_i$ Chapter8/SigmaPolyfold · MPolyfold.isScDiffeoOn_sigmaMk; sc-smoothness, local sc-diffeomorphisms, the degeneracy index and tameness are checked summand by summand Chapter8/SigmaPolyfold · MPolyfold.isScSmoothMap_sigma_iff Chapter8/SigmaPolyfold · MPolyfold.isLocalScDiffeo_sigma_iff Chapter8/SigmaPolyfold · MPolyfold.degIndex_sigma Chapter8/SigmaPolyfold · MPolyfold.isTame_sigma_iff. *Step 4, products.* Products distribute over the union Chapter8/SigmaPolyfold · MPolyfold.isScDiffeoOn_sigmaProdDistrib, and $G \times X \cong \bigsqcup_G X$ for discrete $G$ Chapter8/SigmaPolyfold · isScDiffeoOn_prod_discrete_sigma, which identifies the text's $\mathrm{mor}(y, y') \times U(y)$ with the model of §VIII.20 (Findings, Chapter 8, item 58) (2017 p. 281).

## VIII.20 Proposition 8.5.7 (2017: 8.3): the proper covering functor of a geometric lift

**Proposition 8.5.7 (2017: 8.3)** (V8.35) Chapter8/GeometricLift · proposition_8_5_7. *A covering functor $f : D \to C$ with a geometric lift gives a proper covering functor $F : Y \to X$, with $X = G_x \ltimes U(x)$, $Y = \bigsqcup_y U(y)$, $\mathbf{Y} = \bigsqcup_{y,y'} \mathrm{mor}(y, y') \times U(y)$, $F|U(y) = \tau(y)$ and $F(\psi, z) = (f(\psi), \tau(y)(z))$.* **The text's proof** defines $X$, $Y$ and $F$, says that "one verifies" that $Y$ is an ep-groupoid and $F$ a functor, and shows that (8.16), $L_F$, is injective, surjective and a local sc-diffeomorphism. **The formalized proof** is for nonempty $D$. *Step 1, $X$* is Chapter 7's translation groupoid (§VII.30) Chapter7/NonVacuity · EpGroupoid.Examples.actionEpGroupoid Chapter8/GeometricLift · geometricLiftTarget, with $G$ discrete and lifted by `ULift` to the universe of $U(x)$; a finite $G$ makes it an ep-groupoid. *Step 2, $Y$.* Objects and morphisms are disjoint unions (§VIII.19) Chapter8/GeometricLift · GeometricLift.

YObj Chapter8/GeometricLift · GeometricLift.YMor Chapter8/GeometricLift · GeometricLift.yObjPolyfold Chapter8/GeometricLift · GeometricLift.yMorPolyfold, $\mathrm{mor}(y,y') \times U(y)$ being modelled as $\bigsqcup_{\psi: y \to y'} U(y)$. The composition $(\psi, z') \circ (\gamma, z) = (\psi \circ \gamma, z)$ is total, with a junk value off composable pairs Chapter8/GeometricLift · homComp Chapter8/GeometricLift · GeometricLift.sourceStr. On the summand of $\psi$, $s$ is the identity of $U(y)$ and $t$ is $\varphi_{yy'}(\psi)$, so both are local sc-diffeomorphisms Chapter8/GeometricLift · GeometricLift.isLocalScDiffeo_ySrc Chapter8/GeometricLift · GeometricLift.isLocalScDiffeo_yTgt. "One verifies" is carried out: $t$ is proper, finitely many summands each being mapped homeomorphically onto a summand, so Definition 7.1.3 (3) (2017: 7.3) holds with $V = Y$ Chapter8/GeometricLift · GeometricLift.isProperMap_yTgt; and (4) holds in the sheet form, since near $(\psi_0, z_0)$ the map $\psi \mapsto \sigma(t\psi) \circ \psi$ is $(\psi_0, z) \mapsto (\gamma_0 \circ \psi_0, z)$, the summand index $\gamma_0$ of $\sigma(t\psi)$ being locally constant Chapter8/GeometricLift · GeometricLift.isScSmoothCompSheet_y Chapter8/GeometricLift · geometricLiftSource. *Step 3,* $F$ is a functor by the naturality of $\tau$ (§VIII.18) Chapter8/GeometricLift · geometricLiftFunctor. *Step 4, Definition 8.5.1 (1), (2) (2017: 8.8).* $F$ is a local sc-diffeomorphism, each $\tau(y)$ being one on an open summand Chapter8/GeometricLift · geometricLiftFunctor_isLocalScDiffeo, and surjective since $D \neq \emptyset$ Chapter8/GeometricLift · geometricLiftFunctor_surjective; its fibres have one point in each summand and are finite Chapter8/GeometricLift · geometricLiftFunctor_finite_fiber, and (2) holds with $U(x)$ the whole object space of $X$, the sheets being the summands Chapter8/GeometricLift · geometricLiftFunctor_evenlyCovered. *Step 5, (8.16).* $L_F(\psi, z) = ((f(\psi), \tau(y)z), (y, z))$ is bijective because $f$ is a covering functor. The text deduces the sc-diffeomorphism from "bijection and local sc-diffeomorphism"; here the inverse is shown sc-smooth directly: near $((g_0, q_0), (y_0, z_0))$ it is $((g_0, q), (y_0, z)) \mapsto (\psi_0, z)$, $\psi_0$ the lift of $g_0$ at $y_0$ Chapter8/GeometricLift · GeometricLift.fibredInv Chapter8/GeometricLift · GeometricLift.isScSmoothMap_fibredInv Chapter8/GeometricLift · geometricLiftFunctor_isScDiffeo_fibredMap. *$D$ nonempty.* Definition 8.5.5 (2017: 8.9) allows $D = \emptyset$, and then $Y = \emptyset$ while $U(x)$ is arbitrary: $F$ is surjective on objects iff $D \neq \emptyset$ or $U(x) = \emptyset$ Chapter8/GeometricLift · geometricLiftFunctor_surjective_iff. The text does not say so; the hypothesis is used for the surjectivity only, and holds in Theorem 8.5.8 (2017: 8.6). For $D = \emptyset$ the proposition does not hold as we read it: the empty groupoid has a covering functor to $\mathbb{Z}/2$ with a geometric lift on $\mathbb{R}$, whose functor is not surjective Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty (§VIII.40). The proposition is not vacuous: $\mathbb{Z}/2$ acting on $\mathbb{R}$ by reflection is a geometric lift of the identity of `SingleObj` $\mathbb{Z}/2$ (§VIII.26). The proof's "$F : X \to Y$" is read as $F : Y \to X$ (possible misprint) (Findings, Chapter 8, items 59, 101, 130) (pp. 389–391 (2017 pp. 281–282)).

## VIII.21 Theorem 8.5.8 (2017: 8.6) (Proper Covering II)

**Theorem 8.5.8 (2017: 8.6)** (V8.36) Chapter8/GeometricLift · EpFunctor.theorem_8_5_8. *Let $F : Y \to X$ be a proper covering functor, $x \in X$, $C$ the full subcategory on $x$ and $D$ the full subcategory on $F^{-1}(x)$. Then $f = F|D : D \to C$ is a covering functor, and for $U(x)$ as in Theorem 8.5.3 (2017: 8.5) there are a geometric lift $A : K \to L$ of $f$ and fully faithful open embeddings $\Phi : L \to X$, $\Psi : K \to Y$ whose images are the full subcategories on $U(x)$ and $F^{-1}(U(x))$, with $F \circ \Psi = \Phi \circ A$.* **The text's proof** puts $L_C(g) = \Phi(g)$, $L_D(y) = U(y)$, $L_D(\psi) = t \circ (s|\mathbf{U}(\psi_0))^{-1}$, $\tau(y) = F|U(y)$, says "one verifies" that this is a geometric lift, takes $A$ from Proposition 8.5.7 (2017: 8.3) and puts $\Phi(g, z) = \Gamma(g, z)$, $\Psi(\psi, z) = (s|\mathbf{U}(\psi_0))^{-1}(z)$. **The formalized proof** follows it, and carries out the "one verifies" through $F$ being injective on each $U(y)$. *Step 1, $f$.* $C$ is `SingleObj` $G_x$ and $D$ the full subcategory on $F^{-1}(x)$ Chapter8/GeometricLift · GroupoidStr.FullOb Chapter8/GeometricLift · EpFunctor.FiberCat; $F$ sends its morphisms into $G_x$ Chapter8/GeometricLift · EpFunctor.fiberIsotropy Chapter8/GeometricLift · EpFunctor.restrictFiber. $f$ is a covering functor because $L_F$ is a bijection and a morphism $\psi$ with $s(\psi) \in F^{-1}(x)$ and $F(\psi) \in G_x$

has $t(\psi) \in F^{-1}(x)$ `Chapter8/GeometricLift · EpFunctor.covering_restrict`. $D$ is finite by Definition 8.5.1 (2) (2017: 8.8) and Proposition 7.1.12 (2017: 7.4) `Chapter8/GeometricLift · EpFunctor.fintypeFiberCat`, and nonempty by the surjectivity of $F$ `Chapter8/GeometricLift · EpFunctor.nonempty_fiberCat`. *Step 2, the lift.* For a chart of Theorem 8.5.3 (2017: 8.5), let $\sigma_\psi$ be the inverse of $s$ on $\mathbf{U}(\psi)$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.σSrc`. By (1) and (4) of Theorem 8.5.3 (2017: 8.5), $F(\sigma_\psi(z)) = \Gamma(f(\psi), F(z))$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.mor_σSrc`, so $\tau(y') \circ L_D(\psi) = \Phi(f(\psi)) \circ \tau(y)$, the naturality of $\tau$. The functoriality of $L_D$ follows: $t\sigma_{\gamma \gg \psi}(z)$ and $t\sigma_\psi(t\sigma_\gamma(z))$ lie in the same $U(y'')$ and have the same image $\Phi(f(\psi))\Phi(f(\gamma))F(z)$ under $F$, which is injective on $U(y'')$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.hat_comp`. The sets $U(x)$, $U(y)$ become objects of $\mathrm{Diff}_{sc}$ as open sub-M-polyfolds `Chapter8/GeometricLift · MPolyfoldCat.ofOpen` `Chapter8/GeometricLift · ScDiffeo.ofOpen`, and the geometric lift is assembled `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.geometricLift`; $A$ is its functor of Proposition 8.5.7 (2017: 8.3), a proper covering functor since $D \neq \emptyset$. *Step 3, $\Phi$ and $\Psi$.* $\Phi$ is the inclusion on objects and $\Gamma$ on morphisms `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.Φ`; $\Psi$ is the inclusion on objects and $\Psi(\psi, z) = \sigma_\psi(z)$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.Ψ`, which preserves compositions and units by the unique lifting of Definition 8.5.1 (3) (2017: 8.8): $\Psi(\psi, z') \circ \Psi(\gamma, z)$ and $\Psi(\gamma \gg \psi, z)$ have the same source and the same image under $F$. "Fully faithful open embedding" is read as full and faithful on the underlying groupoids, and an injective local sc-diffeomorphism on objects and on morphisms `Chapter8/GeometricLift · EpFunctor.IsFullyFaithfulOpenEmbedding` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Φ` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Ψ`; the images are the full subcategories, on objects and on morphisms `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Φ_obj` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Φ_mor` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Ψ_obj` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Ψ_mor`, the last two by Theorem 8.5.3 (0) and (2) (2017: 8.5). *Step 4, the square.* $F(\Psi(\psi, z)) = F(\sigma_\psi(z)) = \Gamma(f(\psi), F(z)) = \Phi(A(\psi, z))$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.comp_Ψ`. *Readings.* The theorem is stated for ep-groupoids $X, Y$ whose objects and morphisms lie in any one universe, Definition 8.5.5 (2017: 8.9) taking $D$, $G$ and $\mathrm{Diff}_{sc}$ in that universe; the first reading had it in `Type` only, and no proof changed (§VIII.40) `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_8_id`; "a sufficiently small $U(x)$ as in Theorem 8.5" is any chart of Theorem 8.5.3 (2017: 8.5), and such charts exist inside every open $V \ni x$, which `theorem_8_5_8` restates; "a geometric lift $A : K \to L$" is a geometric lift with its functor $A$ of Proposition 8.5.7 (2017: 8.3); and since the morphisms of $D$ are the $\psi_0$ themselves, "$\psi \in \mathbf{U}(\psi_0)$" in the definition of $L_D$ is read as $\psi = \psi_0$. Possible misprints: "there exists exists", "$\tau(t)$" for $\tau(y)$, and "$A(\psi, z) = (f(z), F(z))$" for $(f(\psi), F(z))$, as in Proposition 8.5.7 (2017: 8.3) (Findings, Chapter 8, items 2, 60, 125) (2017 pp. 283–284).

## VIII.22 Definition 8.5.9 (2017: 8.10): proper strong bundle covering functors

**Definition 8.5.9 (2017: 8.10)** (V8.37) `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering`. *Let $(P, \mu)$ be a strong bundle over $Y$, $(Q, \nu)$ one over $X$, and $A : W \to V$ a strong bundle map with $Q \circ A = F \circ P$. $A$ is a proper strong bundle covering functor if (1) $A$ is surjective and a local strong bundle isomorphism covering a proper covering functor $F$; (2) over the even coverings of Definition 8.5.1 (2) (2017: 8.8), $A : W|U(y) \to V|U(x)$ are strong bundle isomorphisms; (3) $(\psi, w) \mapsto ((F(\psi), A(w)), w)$ is a strong bundle isomorphism $\mathbf{W} \to \mathbf{V}\,{}_s\!\times_A W$ covering $L_F : \mathbf{Y} \to \mathbf{X}\,{}_s\!\times_F Y$. Readings.* $A$ is taken in the functorial picture, as a strong bundle functor covering $F$ (Definition 8.3.3 (2017: 8.6); equivalently its object map is a strong bundle map of Definition 8.3.2 (2017: 8.5), §VIII.13). "$A : W|U \to V|U'$ is a strong

bundle isomorphism" means that $F : U \to U'$ is an sc-diffeomorphism and $A : P^{-1}(U) \to Q^{-1}(U')$ an sc-diffeomorphism on $[0]$ and on $[1]$ `Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn`; its inverse is then linear on the fibres, so the fibre maps are topological linear isomorphisms `Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn.fiberEquiv`. The finiteness of $F^{-1}(x)$ in (2) is that of Definition 8.5.1 (2) (2017: 8.8) and is not repeated. *The strong bundle of (3).* The text does not say which strong bundle structure $\mathbf{V}\,{}_s\!\times_A W \to \mathbf{X}\,{}_s\!\times_F Y$ carries. It is read as the pull-back of $P : W \to Y$ along the sc-smooth projection $\mathrm{pr}_2 : \mathbf{X}\,{}_s\!\times_F Y \to Y$ `Chapter8/EpStrongBundle · EpFunctor.fibredSnd` `Chapter8/EpStrongBundle · EpFunctor.isScSmoothMap_fibredSnd`, on Part I's fibred-product sub-M-polyfold. Its points $((\varphi, y), w)$ with $P(w) = y$ correspond to the text's $((\varphi, v), w)$ with $v = A(w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fibredBundleEquiv`; under the correspondence the projection is the text's $((\varphi, v), w) \mapsto (\varphi, P(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.pullbackProj_fibredBundleEquiv`, and the map of (3) is $(\psi, w) \mapsto ((F(\psi), s(\psi)), w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fibredLift`, which the correspondence turns into the text's $((F(\psi), A(w)), w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fibredBundleEquiv_symm_fibredLift` and which covers $L_F$ `Chapter8/EpStrongBundle · EpStrongBundle.pullbackProj_fibredLift`. The definition is satisfiable: the identity of every strong bundle over an ep-groupoid is a proper strong bundle covering functor, with (3) on this reading (§VIII.26). The closing sentence, "all structural statements about proper covering functors lift to the strong bundle situation", is not formalized beyond §8.6.2 (2017 §8.5.2) (Findings, Chapter 8, items 70, 102, 103) (2017 p. 285).

## VIII.23 Appendix §8.6.1 (2017 §8.5.1): the morphisms over a fibre, and Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11)

**The morphisms over a fibre** (V8.38). *For a proper covering functor $F : Y \to X$ and $x \in X$, $\psi \mapsto (F(\psi), s(\psi))$ is a bijection $\mathrm{mor}(F^{-1}(x)) \to G_x \times F^{-1}(x)$, so $\#\mathrm{mor}(F^{-1}(x)) = \#G_x \cdot \#F^{-1}(x)$; $F^{-1}(x)$ splits into isomorphism classes $\Theta_1, \dots, \Theta_{q'}$ and points of one class have isomorphic isotropy groups.* $\mathrm{mor}(F^{-1}(x))$ is the set of morphisms between points of $F^{-1}(x)$. *Step 1, the bijection.* A morphism between points of $F^{-1}(x)$ is mapped into $G_x$ (Remark 8.5.4 (2017: 8.2) (a)) `Chapter8/ProperCovering · EpFunctor.isotropyOfMorOn`, and by Definition 8.5.1 (3) (2017: 8.8) every $(g, y)$ has exactly one lift `Chapter8/ProperCovering · EpFunctor.IsProperCovering.bijective_fibredMap` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.existsUnique_lift`, whose target lies again in $F^{-1}(x)$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.morOnFiberEquiv`. *Step 2, the count.* $\#$ is the cardinality of finite sets: the fibre is finite by Definition 8.5.1 (2) (2017: 8.8), the isotropy groups by Proposition 7.1.12 (2017: 7.4) (§VII.6) `Chapter7/Basic · EpGroupoid.proposition_7_1_12`, hence so is $\mathrm{mor}(F^{-1}(x))$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.finite_morOn_fiber` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_morOn_fiber`. *Step 3, the classes* are indexed by their images in $|Y|$: $\Theta_c = F^{-1}(x) \cap \pi^{-1}(c)$ `Chapter8/ProperCovering · EpFunctor.fiberClass` `Chapter8/ProperCovering · EpFunctor.fiberIndex` `Chapter8/ProperCovering · EpFunctor.fiberClasses` `Chapter8/ProperCovering · EpFunctor.mem_fiberClass_iff`; they cover the fibre and are pairwise disjoint `Chapter8/ProperCovering · EpFunctor.iUnion_fiberClass` `Chapter8/ProperCovering · EpFunctor.pairwiseDisjoint_fiberClass`, with $\Theta_c = \emptyset$ for $c \notin \pi(F^{-1}(x))$, and isomorphic points have conjugate, hence equinumerous, isotropy groups `Chapter8/ProperCovering · EpFunctor.nat_card_isotropy_eq_of_mem_fiberClass`. With this indexing the correspondence $\Theta_j \leftrightarrow \Theta'_j$ of Lemma 8.6.2 (2017: 8.10) is "the same index" (Findings, Chapter 8, item 71) (2017 p. 286).

**Lemma 8.6.1 (2017: 8.9)** (V8.39) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_1`. *$\#G_x \cdot \#\Theta_j = \#\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} = \#G_{y_0} \cdot (\#\Theta_j)^2$, and $\#G_x = \#G_{y_0} \cdot \#\Theta_j$ for $y_0 \in \Theta_j$.* **The text's proof** restricts the bijection to (8.17), $\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} \to G_x \times \Theta_j$, and, choosing

$\psi_y : y_0 \to y$, uses the bijections $\mathrm{mor}(y_0, y_0) \to \mathrm{mor}(y, y')$, $\psi \mapsto \psi_{y'} \circ \psi \circ \psi_y^{-1}$. **The formalized proof** does the same. *Step 1.* The set of the lemma is $\mathrm{mor}(\Theta_j)$ `Chapter8/ProperCovering · EpFunctor.morOn_fiberClass_eq`, and (8.17) is a bijection because $t(\psi) \in \Theta_j$ is automatic for $s(\psi) \in \Theta_j$ and $t(\psi) \in F^{-1}(x)$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.morOnClassEquiv`. *Step 2.* The text's bijections assemble into one, $\mathrm{mor}(\Theta_j) \cong (\Theta_j \times \Theta_j) \times G_{y_0}$ `Chapter8/ProperCovering · EpFunctor.nonempty_morOnClass_equiv_isotropy`. *Step 3.* The third claim divides by $\#\Theta_j \geq 1$. (8.17) writes "$\psi \to (F\psi), s(\psi))$" for $\psi \mapsto (F(\psi), s(\psi))$ (possible misprint) (Findings, Chapter 8, items 71, 72) (pp. 396–398 (2017 pp. 286–287)).

**Lemma 8.6.2 (2017: 8.10)** (V8.40) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_2`. *For isomorphic $x, x'$ the isomorphism classes of $F^{-1}(x)$ and $F^{-1}(x')$ correspond one to one, and isomorphic $y \in F^{-1}(x)$, $y' \in F^{-1}(x')$ lie in corresponding classes $\Theta_j$, $\Theta'_j$ for exactly one $j$.* **The text** argues before the lemma by lifting a morphism $x \to x'$ through Definition 8.5.1 (3) (2017: 8.8). **The formalized proof** does the same: every $y \in F^{-1}(x)$ is the source of a lift of $\varphi : x \to x'$, whose target lies in $F^{-1}(x')$ and is isomorphic to $y$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.exists_isomorphic_of_isomorphic`. So the index sets in $|Y|$ agree `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberIndex_eq`, and with the classes indexed by their images in $|Y|$ the unique common index of $y$ and $y'$ is their common image. As a bijection between the sets of classes the correspondence is `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_2_equiv` (Findings, Chapter 8, items 71, 73) (2017 p. 287).

**Lemma 8.6.3 (2017: 8.11)** (V8.41) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_3`. *For isomorphic $x, x'$, $\#\Theta_j = \#\Theta'_j$ and $\#F^{-1}(x) = \#F^{-1}(x')$.* As in the text: $\#\Theta_j = \#G_x / \#G_{y_0}$ by Lemma 8.6.1 (2017: 8.9), $\#G_x = \#G_{x'}$, and $\#G_{y_0} = \#G_{y'_0}$ for $y_0 \in \Theta_j$, $y'_0 \in \Theta'_j$, isomorphic points having conjugate isotropy groups; the division is by $\#G_{y_0} \geq 1$. For an index outside both index sets both classes are empty. The fibre is the disjoint union of its classes `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_fiber_eq_sum` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_fiber_eq_of_isomorphic`. Possible misprints: after Lemma 8.6.2 (2017: 8.10), "the third claim of Lemma 8.10" for that of Lemma 8.6.1 (2017: 8.9), which has the claim $\#G_x = \#G_{y_0} \cdot \#\Theta_j$; and in the lemma, $\sum_{i=1}^{q} \#\Theta_j$ sums over $i$ with summands indexed by $j$ (Findings, Chapter 8, item 73) (2017 p. 287).

## VIII.24 Proposition 8.6.4 (2017: 8.4): the number of sheets

**Proposition 8.6.4 (2017: 8.4)** (V8.42) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.proposition_8_6_4`. *$\#F^{-1}(x)$ is constant on the preimage of every connected component of $|X|$, and $x \mapsto \#F^{-1}(x)$ is an sc-smooth functor $X \to \mathbb{N}$.* **The text** argues that under small perturbations of $x$ "the structure of isomorphism classes stays the same". **The formalized proof** uses only the local constancy of $\#F^{-1}(x)$. *Step 1.* For $x' \in U(x)$, Definition 8.5.1 (2) (2017: 8.8) gives the bijection $y \mapsto (F|U(y))^{-1}(x')$ from $F^{-1}(x)$ onto $F^{-1}(x')$, the $U(y)$ being disjoint and covering $F^{-1}(U(x))$; so $\#F^{-1}$ is locally constant on $X$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.eventually_ncard_fiber_eq` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.isLocallyConstant_ncard_fiber`. *Step 2.* By Lemma 8.6.3 (2017: 8.11) it descends to $|X|$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberCardOrbit`, where it is locally constant in the quotient topology, a level set upstairs being open `Chapter8/ProperCovering · EpFunctor.IsProperCovering.isLocallyConstant_fiberCardOrbit`, hence constant on connected components. *Step 3, the functor.* $\mathbb{N}$ is Lean's, with $0$, as the trivial ep-groupoid of the discrete M-polyfold `Chapter7/EpGroupoid · EpGroupoid.trivial` `Chapter7/Discrete · discretePolyfold`; a morphism $\varphi$ goes to the unit of $\#F^{-1}(s\varphi) = \#F^{-1}(t\varphi)$, and the object map is sc-smooth because it is locally constant `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberCardFunctor`. The values are $\geq 1$, $F$ being surjective, which matches the text's $\mathbb{N} = \{1, 2, \dots\}$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.one_le_ncard_fiber` (Findings, Chapter 8, item 74) (2017

p. 288).

## VIII.25 Appendix §8.6.2 (2017 §8.5.2): the structure of strong bundle coverings

**§8.6.2 (2017 §8.5.2)** (V8.43) `Chapter8/EpStrongBundle · EpStrongBundle.section_8_5_2`. *For a proper strong bundle covering functor $A$ over $F$, $A(\varphi, w) = (F(\varphi), A_{s(\varphi)}(w))$ with topological linear isomorphisms $A_y : W_y \to V_{F(y)}$, and $\mu_X(F(\varphi)) \circ A_{s(\varphi)} = A_{t(\varphi)} \circ \mu_Y(\varphi)$: $y \mapsto A_y$ is a natural transformation $\mu_Y \to \mu_X \circ F$ of functors into* Ban. **The text** writes the morphism map in this form and computes $t(A(\varphi, w)) = A(t(\varphi, w))$. **The formalized proof** does the same. *Step 1.* The morphism map has the form $(\varphi, w) \mapsto (F(\varphi), A(w))$, as for every strong bundle functor (§VIII.13) `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering.mor_val`, and the fibre maps $A_y$ are topological linear isomorphisms by Definition 8.5.9 (1) (2017: 8.10), the local strong bundle isomorphisms having linear inverses `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering.fiberEquiv`. *Step 2.* A strong bundle functor commutes with the targets, $\nu(F(\varphi), A(w)) = A(\mu(\varphi, w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.μ_comm`, which is the text's computation. The natural transformation is stated as its naturality identity; the fibres $Q^{-1}(F(s\varphi))$ and $Q^{-1}(s(F\varphi))$ are identified through their values in $V$, and the category Ban is not formalized. Only the strong bundle functor part of Definition 8.5.9 (2017: 8.10) enters the identity (Findings, Chapter 8, item 75) (2017 pp. 288–289).

## VIII.26 Witnesses and counterexamples

Every hypothesis of the chapter is shown satisfiable, and each statement that does not hold as we read it has a counterexample (Findings, Chapter 8, items 100–105, 125, 130).

**Covering functors** (Definition 8.5.5 (2017: 8.9)). The identity of `SingleObj G` is a covering functor for every group $G$ `Chapter8/NonVacuity · CoveringExamples.isCoveringFunctor_id`; for non-trivial $G$ the functor sending every morphism to $1$ is not `Chapter8/NonVacuity · CoveringExamples.not_isCoveringFunctor_one`; the empty groupoid gives a covering functor, with a geometric lift for which Proposition 8.5.7 (2017: 8.3) does not hold as we read it `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty` (§§VIII.20, VIII.40). Theorem 8.5.8 (2017: 8.6) applies to the identity functor of every ep-groupoid in every universe `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_8_id`.

**A geometric lift, and Proposition 8.5.7 (2017: 8.3) applied.** $\mathbb{Z}/2 = \mathbb{Z}^\times$ acting on $\mathbb{R}$ `Chapter8/NonVacuity · CoveringExamples.rCat` by $y \mapsto \pm y$ `Chapter8/NonVacuity · CoveringExamples.reflectionDiffeo` `Chapter8/NonVacuity · CoveringExamples.reflectionHom` is a geometric lift of the identity of `SingleObj` $\mathbb{Z}^\times$, with $L_D = L_C$ and $\tau = 1$ `Chapter8/NonVacuity · CoveringExamples.reflectionLift`. Proposition 8.5.7 (2017: 8.3) makes it a proper covering functor onto the translation groupoid $\mathbb{Z}/2 \ltimes \mathbb{R}$, with non-trivial isotropy at $0$ `Chapter8/NonVacuity · CoveringExamples.isProperCovering_reflectionLift`.

**Identities and trivial strong bundles.** The identity functor of every ep-groupoid is a proper covering functor `Chapter8/NonVacuityBundle · EpFunctor.isScDiffeoOn_fibredMap_id` `Chapter8/NonVacuityBundle · EpFunctor.isProperCovering_id`. Every strong bundle over an M-polyfold is a strong bundle over its trivial ep-groupoid with $\mu(z, w) = w$ (Definition 8.3.1 (2017: 8.4)) `Chapter8/NonVacuityBundle · EpStrongBundle.trivial`; the identity is a strong bundle map and a strong bundle functor (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) `Chapter8/NonVacuityBundle · EpStrongBundle.isMap_id` `Chapter8/NonVacuityBundle · EpStrongBundle.`

isStrongBundleFunctor_id, and the identity of every strong bundle over an ep-groupoid is a proper strong bundle covering functor Chapter8/NonVacuityBundle · StrongBundle.isStrongBundleIsoOn_id Chapter8/NonVacuityBundle · EpStrongBundle.isStrongBundleIso_fibredLift_id Chapter8/NonVacuityBundle · EpStrongBundle.isProperStrongBundleCovering_id. Over a trivial ep-groupoid every section is functorial Chapter8/NonVacuityBundle · EpStrongBundle.isFunctorialSection_trivial, so $\Gamma$, $\Gamma^+$ and the sc-Fredholm section functors are Part I's sc-smooth, sc$^+$- and sc-Fredholm sections Chapter8/NonVacuityBundle · EpStrongBundle.mem_sectionFunctors_trivial Chapter8/NonVacuityBundle · EpStrongBundle.mem_plusSectionFunctors_trivial Chapter8/NonVacuityBundle · EpStrongBundle.mem_fredholmSectionFunctors_trivial. The witnesses use that maps of pull-backs $(z, y) \mapsto (g(z), \Phi(y))$ are strong bundle maps Chapter8/NonVacuityBundle · pullbackMap Chapter8/NonVacuityBundle · StrongBundle.isStrongBundleMap_pullbackMap.

**A strong bundle over an ep-groupoid in Type.** Part I's trivial strong bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ Chapter3/NonVacuity · TrivialBundle.bundle over the trivial ep-groupoid of $\mathbb{R}^n$ Chapter8/NonVacuityBundle · TrivialBundle.epBundle has an ep-groupoid as functorial picture Chapter8/NonVacuityBundle · TrivialBundle.epBundleEpGroupoid, its identity is a proper strong bundle covering functor Chapter8/NonVacuityBundle · TrivialBundle.epBundle_isMap_id Chapter8/NonVacuityBundle · TrivialBundle.epBundle_isProperStrongBundleCovering, $0 \in \Gamma^+$ Chapter8/NonVacuityBundle · TrivialBundle.fiberZero_mem_plusSectionFunctors_epBundle, and for a smooth $\phi : \mathbb{R}^n \to \mathbb{R}^k$ the graph section lies in $\Gamma$ and is sc-Fredholm Chapter8/NonVacuityBundle · TrivialBundle.graphSection_mem_sectionFunctors_epBundle Chapter8/NonVacuityBundle · TrivialBundle.graphSection_mem_fredholmSectionFunctors_epBundle, from Part I's results on graph sections Chapter3/NonVacuity · TrivialBundle.isScSmoothSection_graphSection.

**Tangent witnesses.** The tangent of the trivial ep-groupoid of $\mathbb{R}^n$ is tame Chapter8/TangentTame · EpGroupoid.isTame_tangentEpGroupoid_trivial_rn (Lemma 8.1.9 (2017: 8.8); the trivial ep-groupoid itself is tame by Chapter 7 Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn), and the tangent of the trivial ep-groupoid of any M-polyfold has a paracompact orbit space Chapter8/TangentOrbit · EpGroupoid.hasParacompactOrbitSpace_tangent_trivial (Theorem 8.1.15 (2017: 8.4); its hypothesis is Chapter 7's Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial).

**Theorem 8.5.3 (5) (2017: 8.5) as printed.** For the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§VII.30) Chapter7/NonVacuity · EpGroupoid.Examples.reflectionEpGroupoid, every chart of Theorem 8.5.3 (2017: 8.5) at $0$ has, in the sheet of $\psi_0 = (-1, 0)$, a morphism $\psi$ with $F(s(\psi)) \neq t(F(\psi))$ Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed, and such charts exist Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed_nonvacuous (§VIII.17).

**Remark 8.1.10 (2017: 8.1), the bare-map reading.** On the trivial $\mathbb{Z}/3$-action every $T\varphi$ is the identity, and $(\varphi, h) \mapsto (T\varphi, Ts(h))$ is not injective Chapter8/TangentEpNotInjective · EpGroupoid.Examples.remark_8_1_10_not_injective (§VIII.6).

**Forms.** The constant 0-forms are forms on every ep-groupoid Chapter8/GapP2_B_EpForms · EpGroupoid.constForm_mem_epForms, and on an ep-groupoid with $s = t$ every form of the object M-polyfold is one Chapter11/GapP2_B2_Poincare · EP.epFormsInftyToObj_surjective_of_src_eq_tgt (§§VIII.35, VIII.37).

*What the witnesses do not exercise.* "A sufficiently small $U(x)$" is a chart datum, and charts exist inside every open $V \ni x$, so part (c) of Theorem 8.5.8 (2017: 8.6), which quantifies over all charts, is not vacuous; no hypothesis of the form "for all compatible charts" is made in the chapter. But every proper covering functor constructed has one-point fibres $F^{-1}(x)$ (identities, and the functor of the reflection lift, whose $D$ has one object), so the even coverings of Definitions 8.5.1 (2) and 8.5.9 (2) (2017: 8.8 and 8.10) are met with one sheet and the counting of §8.6.1 (2017 §8.5.1) only with $q = 1$, $\#\Theta_1 = 1$; no proper strong

bundle covering functor other than identities is constructed, and $\Gamma^+$ is witnessed by the zero section only (Findings, Chapter 8, item 105).

## VIII.27 What is not formalized

Every gap of the chapter is closed (G.74, G.75, G.86–G.89, §§VIII.34–VIII.40). What remains:

- **The degeneracy remark for non-tame $X, Y$** (pp. 379–385 (2017 pp. 275–276); row V8.28, narrower): only $d_{\mathbf{X}\times Y} \leq 2\, d_{\mathbf{Y}}$ along the image of $L_F$; equality without tameness is open in the project, Part I proving the product formula only for tame M-polyfolds (§VIII.15; Findings, Chapter 8, item 51).
- **The statements that do not hold as we read them**, each with a counterexample: Theorem 8.5.3 (5) (2017: 8.5) as printed, where the square with $s$ on both vertical arrows and the companion square with $t$ on both are proved (§VIII.17; Findings, Chapter 8, item 54); Proposition 8.5.7 (2017: 8.3) for $D = \emptyset$ (§§VIII.20, VIII.40; Findings, Chapter 8, items 59, 130); and Remark 8.1.10 (2017: 8.1) in the bare-map reading (§VIII.6; Findings, Chapter 8, item 10).
- **The local bundle model in the proof of Theorem 8.1.15 (2017: 8.4)** (2017 p. 268): the sc-isomorphism of $TU(x_z)$ with $TO \to O^1$ and the linearity of the bundle action of $G_{x_z}$; not needed by the proof here (§VIII.9; Findings, Chapter 8, items 21, 113).
- $W[1]$ **as an ep-groupoid**: Proposition 8.4.1 (3) needs only its orbit space, which is built directly (§VIII.39; Findings, Chapter 8, item 133).
- **Remarks without formal content**: the naturality remarks of Remark 8.5.4 (2017: 8.2) (b), strong bundle maps over open subsets of the base, the closing sentence of Definition 8.5.9 (2017: 8.10), "the structure of isomorphism classes stays the same" before Proposition 8.6.4 (2017: 8.4), and the category Ban of §8.6.2 (2017 §8.5.2); none is used elsewhere in the chapter, so none is given a gap (Findings, Chapter 8, item 113).

## VIII.28 What the formalization taught us

**Fibre injectivity replaces calculus on fibred products.** The text proves the groupoid laws of $T(X, \mathbf{X})$, Theorem 8.1.11 (2017: 8.2) and Proposition 8.1.14 (2017: 8.1) by differentiating identities on $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$, which needs the tangent of a fibred product. Since $s$ is a local sc-diffeomorphism, a vector of $T\mathbf{X}$ is determined by its base point and its image under $Ts$; base points are computed in $\mathbf{X}$ and images under $Ts$ by Lemma 8.1.3 (2017: 8.2). Only one identity of §8.1, the target formula of Lemma 8.1.3 (2017: 8.2), is genuinely differentiated, and the identification $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ is never needed (§§VIII.3, VIII.7, VIII.8).

**The tangent calculus of §8.1 lives on open sets.** Every proof of §8.1 differentiates a map defined only near a point: a local inverse of $s$, the map $\alpha$ of Lemma 8.1.3 (2017: 8.2), $\widehat{\psi}$. Part I differentiates only globally sc-smooth maps. Tangents of maps on open sets, with the chain rule and the tangents of local sc-diffeomorphisms, are the infrastructure the chapter rests on, and Lemma 8.1.7's (2017: 8.6) "by construction" is exactly their last property (§VIII.1).

**Definition 8.5.1 (3) (2017: 8.8) carries Theorem 8.5.3 (2017: 8.5).** The text builds the sheets $\mathbf{U}(\psi_0)$ from natural representations of the groups $G_y$ in $Y$ and never invokes condition (3). With (3), the sheets are images of the continuous sections $z \mapsto L_F^{-1}(\Gamma(g, Fz), z)$ of $s$, and connectedness of $U(y)$ does the rest; neither the finiteness of the fibre nor condition (1) is used. The same condition counts the morphisms

over a fibre in §8.6.1 (2017 §8.5.1) and gives the unique lifting behind Theorem 8.5.8 (2017: 8.6) (§§VIII.17, VIII.21, VIII.23).

**A functor commutes with $s$, not with "$s$ on one side and $t$ on the other".** Theorem 8.5.3 (5) (2017: 8.5) as printed equates $F(s(\psi))$ with $t(F(\psi))$; a non-trivial element of $G_x$ moves points, so the square fails for the reflection of $\mathbb{R}$ in every chart at 0. The two squares that functoriality gives, with $s$ on both sides and with $t$ on both sides, are what the proofs downstream use (§VIII.17).

**Empty groupoids and non-tame products.** Two statements needed a hypothesis that the text does not state: Proposition 8.5.7 (2017: 8.3) needs $D \neq \emptyset$, because Definition 8.5.5 (2017: 8.9) admits the empty groupoid, over which the functor of Proposition 8.5.7 (2017: 8.3) is not surjective; and the degeneracy remark after Definition 8.5.1 (2017: 8.8) needs tameness, the only form of the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$ that Part I proves. In both cases the hypothesis holds where the text uses the statement (§§VIII.15, VIII.20).

**Metrizability is the shorter road to paracompactness, again.** As for Proposition 7.5.3 (2017: 7.12), Theorem 8.1.15 (2017: 8.4) follows from Smirnov's theorem pulled back along $|TX| \to |X^1|$: $|TX|$ is metrizable, which is stronger than paracompact, and the closed paracompact sets $\tilde{A}_z$, the regularity of $|TX|$ and Proposition 2.7.4 (2017: 2.17) drop out. What remains of the text's argument is the metrizability of $|TU(x_z)|$, and for it the open embedding $G_x \backslash TU(x) \to |TX|$, which the text leaves implicit (§VIII.9).

**Strong bundles over ep-groupoids need more of Part I than Part I states.** A global notion of strong bundle map, the atlas and universal property of the pull-back on both levels $[0]$ and $[1]$, the sum of sc-smooth sections, and the strong bundle structure on $\mathbf{V}_{s}\times_{A} W$ are all used in §8.3 and Definition 8.5.9 (2017: 8.10) without being defined or proved in Part I. With them in place, the claims of §8.3 that the text states without proof, the functorial picture and the identity of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6), are routine (§§VIII.10–VIII.14, VIII.22).

**Compatibility with morphisms is $s^*\omega = t^*\omega$.** Definition 8.2.1 asks $(T\varphi)^*\omega_y = \omega_x$ for every morphism. Since $T\varphi = Tt \circ (Ts|T_{\varphi}\mathbf{X})^{-1}$, this is the equality of the pull-backs of $\omega$ by the two global sc-smooth maps $s, t : \mathbf{X} \to X$. In that form every closure property of §8.2 — under $d$, the wedge product, restriction to $X^1$, pull-back by functors — is one line, from the naturality of pull-backs in Part I; the text's local maps $t \circ (s|\mathbf{U}(\varphi_0))^{-1}$ are not needed (§VIII.35).

**The Poincaré lemma on an ep-groupoid is a statement about equivalences.** The text averages a primitive over $G_x$ and extends it to the saturation by morphisms. Both steps are instances of general facts: averaging works for any finite family of self-functors covering the morphisms of a groupoid, and the extension is the push-forward of forms along the equivalence $G_x \ltimes U(x) \to X|\pi^{-1}(\pi(U(x)))$ of §11.2. So the proof of Proposition 8.2.3 rests on Chapter 11 (§VIII.37).

**Vertical vectors are the derivative the text takes.** Proposition 8.1.16 differentiates (8.16) and uses that $T\tau_X$ is onto. On the vertical vectors of $T(TX)$ the tangent of an isotropy element of $TX$ acts as $Tg$, and that suffices; no surjectivity is used (§VIII.38).

## VIII.29 The 2021 additions to §8.1: Propositions 8.1.16 and 8.1.17, the Zhou condition and LUC for $TX$

The Springer edition (2021) adds Propositions 8.1.16, 8.1.17, Definition 8.3.5, §8.4, Example 8.5.6, condition (0) of Definition 8.5.9 and Lemma 8.5.10; Sections VIII.29–VIII.33 retell them, with the 2021 numbers

and pages (V8.44–V8.52). The conditions of Chapter 7's §7.4 are those of §§VII.35–VII.39.

**Proposition 8.1.16** (V8.44) `Chapter8/TangentZhou · EpGroupoid.proposition_8_1_16`. *If $X$ satisfies (ZHOU), so does $TX$.* **The text's proof**: an isotropy element of a smooth $b \in TX$ over $x$ is a pair $(g,h)$, $g \in G_x$; differentiating (8.16) and using that $T\tau$ is onto, $T(g,h) = \mathrm{id}$ gives $Tg = \mathrm{id}$, so $g \in T_x$ by (ZHOU), and then $(g,h) \in T_b$. **The formalized proof.** *Step 1.* A smooth point of $TX$ lies over a smooth point of $X$ `Chapter8/TangentZhou · MPolyfold.levelVal_tangentProj_mem_smoothSet`, and the base $p(k)$ of an isotropy element $k$ of $b$ is an isotropy element of $p(b)$. *Step 2.* The step "$T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$" needs the second tangent bundle $T(TX)$ and $T\tau$, which are not formalized; it was the hypothesis `Chapter8/TangentZhou · EpGroupoid.TangentZhouReflect` in the first reading, and holds for every ep-groupoid by §VIII.38 (gap G.88, closed). *Step 3.* (ZHOU) for $X$ puts $p(k)$ in $\mathbf{X}^T$, and if $s = t$ near $p(v)$ then $Ts = Tt$ near $v$, so $k \in (T\mathbf{X})^T$ `Chapter8/TangentZhou · EpGroupoid.mem_nonEffectivePart_tangent`. As in Chapter 7, $Tg$ is a set map on the fibre, no linear structure is used (§VII.37). For every trivial ep-groupoid the tangent of a morphism is the identity and the hypothesis holds `Chapter8/TangentZhou · EpGroupoid.tangentMor_trivial Chapter8/TangentZhou · EpGroupoid.tangentZhouReflect_trivial Chapter8/TangentZhou · EpGroupoid.Examples.proposition_8_1_16_point` (Findings, Chapter 8, item 115) (p. 369).

**Proposition 8.1.17** (V8.45) `Chapter8/TangentZhou · EpGroupoid.proposition_8_1_17_one`. *(1) If $X^1$ has LUC, so has $TX$, with $(T\mathbf{X})^T = p^{-1}((\mathbf{X}^1)^T)$. (2) (SLUC) for $X$ gives (SLUC) for $TX$. (1).* If $v \in (T\mathbf{X})^T$, then $Ts = Tt$ on a neighbourhood $O$ of $v$, and $s = t$ on the open set $p(O)$, $p$ being open: $p(v) \in (\mathbf{X}^1)^T$. Conversely, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ by Appendix 7.6.4 (§VII.36), and Step 3 of Proposition 8.1.16 applies `Chapter8/TangentZhou · EpGroupoid.nonEffectivePart_tangent_eq`. So $(T\mathbf{X})^T$ is closed as a preimage of the closed $(\mathbf{X}^1)^T$ (Proposition 7.2.21 for $X^1$), and $TX$ has LUC by Proposition 7.2.21 for $TX$. The hypotheses are those of Appendix 7.6.4 — $U(x) \cap X_1$ connected in $X^1$ and dense in $U(x)$ (the density is gap G.73) — and the openness of $p : T\mathbf{X} \to \mathbf{X}^1$ (gap G.8); both now hold for every ep-groupoid (§VII.45), and the proposition without them is §VIII.40. They are satisfied by the point `Chapter8/TangentZhou · EpGroupoid.isOpenMap_tangentProj_of_subsingleton Chapter8/TangentZhou · EpGroupoid.Examples.proposition_8_1_17_one_point`. *(2)* for $(TX)^0 = TX$ is (1) `Chapter8/TangentZhou · EpGroupoid.proposition_8_1_17_two`; for every $(TX)^i$ it needs $(TX)^i = T(X^i)$, which is §VIII.33 (gap G.88, closed) (Findings, Chapter 8, items 116, 124, 126) (p. 371).

## VIII.30 Definition 8.3.5, Proposition 8.4.1 and Definition 8.4.2: auxiliary norms, paracompactness, local regularity

**Definition 8.3.5** (V8.46) `Chapter8/Book2021 · EpStrongBundle.IsAuxNorm`. *An auxiliary norm for a strong bundle over an ep-groupoid is a continuous $N : W_{0,1} \to [0,\infty)$, a complete norm on every fibre, with $h_k \to 0_x$ whenever $P(h_k) \to x$ and $N(h_k) \to 0$, and invariant: $N(h) = N(k)$ if $h = \mu(\varphi, k)$.* The first three conditions are Part I's auxiliary norm (Definition 5.1.1) on the underlying strong bundle, so $N$ lives on its level-one space; values in $[0,\infty)$ follow from (1). In (3), $h \in W_{0,1}$ is given and $h = \mu(\varphi,k)$ is an equation in $W$; it is not vacuous, since $\mu(\varphi,k) \in W_{0,1}$ for $k \in W_{0,1}$ `Chapter8/Book2021 · EpStrongBundle.exists_oneVal_eq_μ`. Over the trivial ep-groupoid every auxiliary norm of Part I is one `Chapter8/Book2021 · EpStrongBundle.exists_isAuxNorm_trivial`, and $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ has one `Chapter8/Book2021 · TrivialBundle.exists_isAuxNorm_epBundle` (Findings, Chapter 8, item 117) (p. 379).

**Proposition 8.4.1** (V8.47) `Chapter8/Book2021 · EpStrongBundle.proposition_8_4_1_two_imp_one`. *$|X|$, $|W[0]|$ and $|W[1]|$ are paracompact together. (2) □ (1).* $\mu(\varphi, 0_{s(\varphi)}) = 0_{t(\varphi)}$ `Chapter8/Book2021 · EpStrongBundle.μ_fiberZero`, so the zero section induces $|0| : |X| \to |W[0]|$ `Chapter8/Book2021 · EpStrongBundle.zeroOrbitMap`, continuous `Chapter8/Book2021 · EpStrongBundle.continuous_zeroOrbitMap`, with left inverse $|P|$

Chapter8/Book2021 · EpStrongBundle.leftInverse_zeroOrbitMap, hence a topological embedding Chapter8/Book2021 · EpStrongBundle.isEmbedding_zeroOrbitMap. The text concludes as a closed subset of a paracompact space; the formalized proof goes through metrizability: $|W[0]|$ paracompact is metrizable by Theorem 7.3.1 and Smirnov's theorem (§VII.23), so is its subspace $|X|$, hence paracompact. (1) □ (2) and the statements on $|W[1]|$ are §VIII.39, without bundle charts (gap G.86, closed) (Findings, Chapter 8, items 118, 127, 133) (p. 380).

**Definition 8.4.2** (V8.48) Chapter8/TangentZhou · EpStrongBundle.IsLR. *The strong bundle has the local regularity property if every $x$ has arbitrarily small $U(x)$ such that every strong bundle isomorphism $B$ of $W|U(x)$ with $|B(w)| = |w|$ is $w \mapsto g * w$ for some $g \in G_x$,* the natural lift $g * w = \mu(\Gamma(g, P(w)), w)$ Chapter8/TangentZhou · EpStrongBundle.natLift. As for Definition 8.5.9 (§VIII.22), the strong bundle isomorphism is read as a strong bundle map $W \to W$ that is Part I's strong bundle isomorphism on $W|U(x)$, "arbitrarily small" as "inside every open $O \ni x$", and $|B(w)| = |w|$ as equality of orbits in $(W, \mathbf{W})$, as for (LR) (§VII.39). Every strong bundle over a trivial ep-groupoid has it Chapter8/TangentZhou · EpStrongBundle.isLR_trivial. Proposition 8.4.3, effective $X$ with (LR) gives the property, is §VIII.39, with the openness of $P$ from Chapter 10 (Findings, Chapter 8, items 119, 128) (p. 382).

## VIII.31 Example 8.5.6, Definition 8.5.9 in the 2021 form, and Lemma 8.5.10

**The remark after Definition 8.5.5 and Example 8.5.6** (V8.49) Chapter8/Book2021 · example_8_5_6. *For a covering functor $f : D \to C$, $f : G_d \to G_{f(d)}$ is injective* Chapter8/Book2021 · IsCoveringFunctor.injective_map_end — by the bijectivity required in Definition 8.5.5 — *but need not be surjective*: $D$ with two objects and one morphism between any two Chapter8/Book2021 · Example856D, mapped to $\mathbb{Z}^\times$ as a one-object category with $f(\varphi) = f(\varphi^{-1}) = -1$ Chapter8/Book2021 · example856Functor, is a covering functor whose $G_a$, $G_b$ are trivial while $G_x \cong \mathbb{Z}/2$ (Findings, Chapter 8, item 120) (p. 388).

**Definition 8.5.9, 2021 form** (V8.50) Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021. *A strong bundle map $A : W \to V$ over an ep-functor $F$ with $Q \circ A = F \circ P$ is a proper strong bundle covering functor if (0) $\tau(F(\psi), A(w)) = A(\mu(\psi, w))$ and (1)–(3) of the 2017 definition hold.* It is stated for the object map, (0) being the strong bundle map property of Definition 8.3.2 (2017: 8.5); (1)–(3) are those of §VIII.22. The reading of §VIII.22 had taken 2017's "strong bundle map $A$" as a strong bundle functor, which contains (0); the two are equivalent Chapter8/Book2021 · EpStrongBundle.isProperStrongBundleCovering2021_iff Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering.isProperStrongBundleCovering2021: the functor is determined by its object map, on morphisms $(\psi, w) \mapsto (F(\psi), A(w))$. Identities are witnesses Chapter8/Book2021 · EpStrongBundle.isProperStrongBundleCovering2021_id Chapter8/Book2021 · TrivialBundle.epBundle_isProperStrongBundleCovering2021 (Findings, Chapter 8, items 70, 121) (p. 394).

**Lemma 8.5.10** (V8.51) Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.lemma_8_5_10. *$B : W \to F^*V$, $w \mapsto (P(w), A(w))$, is a strong bundle isomorphism over $Y$ for $\mu_{F^*V}(\varphi, (y, v)) = (t(\varphi), \mu_X(F(\varphi), v))$, with (8.23) $B \circ \mu_Y = \mu_{F^*V} \circ B$, and $A = \bar{A} \circ B$.* $F^*V$ is Part I's pull-back, whose paracompactness always holds, and $\mu_{F^*V}$ is its formula Chapter8/Book2021 · EpStrongBundle.pullbackμ. *Step 1.* $B$ lands in $F^*V$ Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_val, and (8.23) is (0) Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_μ. *Step 2, bijective.* On the fibre over $y$, $A$ is the linear isomorphism $A_y : W_y \to V_{F(y)}$ of a local strong bundle isomorphism of Definition 8.5.9 (1) at $y$, so $B$ is bijective Chapter8/

Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bijective_bundleB. *Step 3, the inverse.* Near every point, $B^{-1}(y,v) = \Psi(v)$ for the inverse $\Psi$ of a local strong bundle isomorphism of Definition 8.5.9 (1), whose base map is injective Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.eq_of_isStrongBundleIsoOn Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv_eq_of, so $B$ and $B^{-1}$ are strong bundle maps Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleB Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.mapsTo_oneSet_bundleBInv Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleBInv. "Strong bundle isomorphism over the ep-groupoid $Y$" is read as a strong bundle isomorphism $W \to F^*V$ covering the identity, with (8.23); that $\mu_{F^*V}$ makes $F^*V$ a strong bundle over the ep-groupoid $Y$, and $B$ an isomorphism of strong bundles over $Y$, is §VIII.40 (gap G.89, closed) (Findings, Chapter 8, items 122, 129) (p. 395).

## VIII.32 The 2021 additions: what is not formalized, and what they taught us

Every gap of the 2021 additions is closed:

- **Proposition 8.1.16 without `TangentZhouReflect`** (gap G.88, closed): through the vertical vectors of $T(TX)$ (§VIII.38; Findings, Chapter 8, items 115, 126). Proposition 8.1.17 (2) for every $(TX)^i$ is §VIII.33; of $(TX)^i = T(X^i)$ the compatibility with unit, inverse and composition is not formalized, since LUC does not use it (Findings, Chapter 8, item 124).
- **Proposition 8.2.3** (p. 373; gap G.87, closed): the Poincaré lemma, as printed, with the forms of §8.2 (gap G.74, closed) (§§VIII.35–VIII.37; Findings, Chapter 8, items 110, 123, 132).
- **Proposition 8.4.1 and Proposition 8.4.3** (gap G.86, closed): (1) □ (2) □ (3), with $|W[1]|$ built as an orbit space, and the local regularity of strong bundles (§VIII.39; Findings, Chapter 8, items 118, 119, 127, 128, 133).
- **$F^*V$ as a strong bundle over the ep-groupoid $Y$** in Lemma 8.5.10 (gap G.89, closed) (§VIII.40; Findings, Chapter 8, items 122, 129).

**The tangent inherits the Chapter 7 conditions through the base.** Both propositions of §8.1 reduce to Chapter 7: a vector of $T\mathbf{X}$ is in $(T\mathbf{X})^T$ as soon as its base is in $(\mathbf{X}^1)^T$, and the openness of $p$ gives the converse. The passage from $T(g,h)$ to $Tg$ needs the second tangent bundle; it goes through the vertical vectors of $T(TX)$ (§VIII.38).

**Condition (0) was already in the reading.** The 2021 Definition 8.5.9 adds (0); the first reading had read 2017's "strong bundle map" as a strong bundle functor, which contains it, and the two forms are equivalent (§VIII.31).

## VIII.33 Proposition 8.1.17 (2) for every $(TX)^i$

**Equivalences of ep-groupoids** (V8.52) Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv. The text's "naturally $(TX)^i = T(X^i)$" is used only to carry LUC from $T(X^i)$ to $(TX)^i$. For that it suffices to have sc-diffeomorphisms of the object and of the morphism M-polyfolds commuting with source and target; an sc-diffeomorphism is a homeomorphism Chapter8/TangentZhouLevels · MPolyfold.ScEquiv.toHomeomorph. Such equivalences compose and contain the identities Chapter8/TangentZhouLevels · EpGroupoid.

EpEquiv.refl Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.trans, pass to the index raises Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.shift, map $\mathbf{X}^T$ onto $\mathbf{X}'^T$, since $s = t$ near a morphism is preserved Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.nonEffectivePart_eq, and so transport LUC, by the characterization of Proposition 7.2.21 ($\mathbf{X}^T$ closed) Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.isLUC. The compatibility with unit, inverse and composition is not formalized, since LUC does not use it.

$T(X^i) \cong (TX)^i$ (V8.52) Chapter8/TangentZhouLevels · EP.tangentLiftEpEquiv. Part I's Proposition 2.3.17 for $i = 1$, on the object and the morphism M-polyfold, gives an equivalence $T(X^1) \cong (TX)^1$, commuting with $s$ and $t$ by the naturality of the inclusion $T(X^1) \to TX$ Chapter8/TangentZhouLevels · EpGroupoid.tangentShiftEpEquiv. By recursion, composing with the index raise of the previous equivalence, $T(X^i) \cong (TX)^i$ for every $i$, for the bundled tangent ep-groupoid Chapter8/TangentZhouLevels · EP.tangentEP, as Part I's $\Phi_i$ for M-polyfolds Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv Chapter2/TangentLiftAll · proposition_2_3_17_all.

**Proposition 8.1.17 (2)** (V8.52) Chapter8/TangentZhouLevels · EP.proposition_8_1_17_two_all. *If $X$ has (SLUC), so has $TX$.* **The text's proof**: (SLUC) gives LUC for every $X^{i+1} = (X^i)^1$; by (1) every $T(X^i)$ has LUC; and $(TX)^i = T(X^i)$. **The formalized proof** is the same: (1) applied to $X^i$ Chapter8/TangentZhouLevels · EP.isLUC_tangent_lift, then the equivalence. The hypotheses of (1) — natural representations with $U(x) \cap (X^i)_1$ connected in $(X^i)^1$ and dense in $U(x)$ (gap G.73), and the openness of $p : T(\mathbf{X}^i) \to (\mathbf{X}^i)^1$ (gap G.8) — are taken on every $X^i$; both hold for every ep-groupoid (§VII.45), and the proposition without them is §VIII.40 Chapter7/SmoothRegularAssembly · EP.isLUC_tangent_lift_of_isSLUC. They hold for the point ep-groupoid and all its lifts Chapter8/TangentZhouLevels · EpGroupoid.hasConnectedDenseLevelOneReps_of_subsingleton Chapter8/TangentZhouLevels · EpGroupoid.Examples.proposition_8_1_17_two_all_pointEP (gap G.88, closed for (2)) (Findings, Chapter 8, item 124) (p. 371).

## VIII.34 Lemma 8.1.2 (2017: 8.1) as printed, and the M-polyfold of pairs of Remark 8.1.10 (2017: 8.1)

§VIII.2 proved Lemma 8.1.2 in the form the construction of $T\mathbf{X}$ uses: the composition (8.5) is defined through the local sc-diffeomorphisms $Ts$, $Tt$, and $T\mathbf{X}$ is an ep-groupoid. The text states more: under the identification of $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X})$ with $T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$, the multiplication of $T\mathbf{X}$ is $Tm$. This is now formalized as printed (V8.4; gap G.75, closed).

**The M-polyfold $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$** Chapter8/GapP2_B_TangentFibred · EpGroupoid.compPolyfold is the sub-M-polyfold of $\mathbf{X} \times \mathbf{X}$ of Definition 7.1.3 (4); its second projection $\pi_2$ is a local sc-diffeomorphism, since $s$ is one Chapter8/GapP2_B_TangentFibred · EpGroupoid.isLocalScDiffeo_compSnd, and the multiplication $m$ is sc-smooth by condition (4) Chapter8/GapP2_B_TangentFibred · EpGroupoid.compMul.

**The identification** Chapter8/GapP2_B_TangentFibred · EpGroupoid.tangentCompPairs is $v \mapsto (T\pi_1 v, T\pi_2 v)$. *Step 1.* As $\pi_2$ is a local sc-diffeomorphism, $T\pi_2$ is bijective on the fibres, and a vector $v$ of $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X})$ is determined by its base point and $T\pi_2 v$. *Step 2.* $T\pi_1(v)$ is the vector over $\pi_1$ with $Ts(T\pi_1 v) = Tt(T\pi_2 v)$, by the chain rule for $s \circ \pi_1 = t \circ \pi_2$. So the identification is injective with range the pairs $(k, h)$ with $Ts(k) = Tt(h)$ Chapter8/GapP2_B_TangentFibred · EpGroupoid.range_tangentCompPairs. *Step 3, $Tm$ is the composition.* $Tm(v)$ lies over $m(\pi_1 v, \pi_2 v)$ and $Ts(Tm\, v) = Ts(T\pi_2 v)$, by the chain rule for $s \circ m = s \circ \pi_2$; since $Ts$ is injective on the fibres of $\mathbf{X}$, this characterizes the composition (8.5): $Tm(v) = T\pi_1(v) \circ T\pi_2(v)$ Chapter8/GapP2_B_TangentFibred · EpGroupoid.lemma_8_1_2_tangentMul. In the text's words: for $Ts(k) = Tt(h)$ there is exactly one $v$ over $(k, h)$, and $k \circ h = Tm(v)$, a vector over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$

Chapter8/GapP2_B_TangentFibred · EpGroupoid.lemma_8_1_2_book; the multiplication of the tangent ep-groupoid on its fibred product is $Tm$ transported by the identification Chapter8/GapP2_B_TangentFibred · EpGroupoid.lemma_8_1_2_compMul. *Step 4, as M-polyfolds.* The identification is an sc-diffeomorphism onto the sub-M-polyfold $T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ of $T\mathbf{X}\times T\mathbf{X}$ Chapter8/GapP2_B_TangentFibred · EpGroupoid.isScDiffeoOn_tangentCompPairs: it is a bijection which, composed with the local sc-diffeomorphism $T\pi_2$ on the right side, gives the local sc-diffeomorphism $T\pi_2$ on the left, and such a map is a local sc-diffeomorphism Chapter8/GapP2_B_TangentFibred · MPolyfold.isLocalScDiffeo_of_comp_eq. **A reading.** The text derives $Ts(k\circ h) = Ts(h)$ by differentiating $s(m(\psi,\varphi)) = s(\varphi)$; the proof above uses the same identity, in the form $s\circ m = s\circ\pi_2$ on the M-polyfold of pairs (p. 359).

**Remark 8.1.10, the M-polyfold of pairs** (V8.12). §VIII.6 formalized the bijection of the remark between morphisms of $T\mathbf{X}$ and pairs $(T\varphi, a)$ with the label $\varphi$ in place of the bare map $T\varphi$ (as a bare map the assignment is not injective, §VIII.6). Here $T\varphi$, defined as $Tt\circ(Ts|T_\varphi\mathbf{X})^{-1}$, is the tangent at $s(\varphi)$ of the local sc-diffeomorphism $\hat\varphi = t\circ(s|\mathbf{U}(\varphi))^{-1}$, as the text has it Chapter9/Tangent · EpGroupoid.tangentMor_eq_tangentMapOn_hat, a bijection $T_{s(\varphi)}X\to T_{t(\varphi)}X$ with inverse $T(\varphi^{-1})$ Chapter7/TangentMorphism · EpGroupoid.bijOn_tangentMor, represented in charts by mutually inverse bounded linear maps Chapter7/TangentMorphism · EpGroupoid.tangentMor_chart_iso. The remark also says that the pairs form an M-polyfold for which the bijection is an sc-diffeomorphism. *The pairs* $(\varphi,a)$, $\varphi\in\mathbf{X}^1$, $a\in T_{s(\varphi)}X$, form the fibred product $\mathbf{X}^1\,{}_s\!\times_p TX$, a sub-M-polyfold of $\mathbf{X}^1\times TX$ by Proposition 2.5.20, $s:\mathbf{X}^1\to X^1$ being a local sc-diffeomorphism Chapter8/GapP2_B_RemarkPairs · EpGroupoid.tangentPairsPolyfold. *The map* $G(v) = (p(v), Ts(v))$ Chapter8/GapP2_B_RemarkPairs · EpGroupoid.remarkGInto is an sc-diffeomorphism $T\mathbf{X}\to\mathbf{X}^1\,{}_s\!\times_p TX$ Chapter8/GapP2_B_RemarkPairs · EpGroupoid.remark_8_1_10_scDiffeo, by the argument of Step 4: it is bijective and commutes with the local sc-diffeomorphisms $Ts: T\mathbf{X}\to TX$ and $\pi_2:\mathbf{X}^1\,{}_s\!\times_p TX\to TX$. *The triples.* $v\mapsto(Ts\,v, p(v), Tt\,v)$ Chapter8/GapP2_B_RemarkPairs · EpGroupoid.remarkTriple is injective, with range the triples $(a,\varphi,b)$ with $a\in T_{s(\varphi)}X$ and $b = T\varphi(a)$ Chapter8/GapP2_B_RemarkPairs · EpGroupoid.remark_8_1_10_triples, the text's "$(a, T\varphi, b)$" with $\varphi$ recorded (pp. 364–365). The row stays narrower only for the reading of $T\varphi$ as a label (Findings, Chapter 8, item 131).

## VIII.35 Definition 8.2.1 (2017: 8.2): sc-differential forms on ep-groupoids

§VIII.27 of the first reading recorded §8.2 as not formalized, since Part I's forms (§4.4) were not yet available. With them (Part IV), Definition 8.2.1 is formalized as printed (V8.53; gap G.74, closed).

**The condition** Chapter8/GapP2_B_EpForms · EpGroupoid.IsMorCompatible. *An sc-differential $k$-form on the ep-groupoid $(X,\mathbf{X})$ is an sc-differential $k$-form $\omega$ on the object M-polyfold, Definition 4.4.1, with $(T\varphi)^*\omega_y = \omega_x$ for every morphism $\varphi: x\to y$,* $T\varphi$ being the bounded linear isomorphism $T_xX\to T_yX$ of Definition 7.1.4 (§VIII.6). The forms satisfying it form the vector space $\Omega^k_{ep}(X)$ Chapter8/GapP2_B_EpForms · EpGroupoid.epForms. **The key reformulation**: $\omega$ is compatible with the morphisms iff $s^*\omega = t^*\omega$ on the morphism M-polyfold $\mathbf{X}$, the pull-backs being Part I's pull-backs by the sc-smooth maps $s,t:\mathbf{X}\to X$ Chapter8/GapP2_B_EpForms · EpGroupoid.isMorCompatible_iff_formPullback; for $T\varphi = Tt\circ(Ts)^{-1}$ on the fibre over $\varphi$ is the definition of §VIII.6. Every closure property follows from this form. *(a)* A form on $X$ is a form on $X^1$ Chapter8/GapP2_B_EpForms · EpGroupoid.formLower_mem_epForms. *(b)* $d\omega$ is compatible with the morphisms: $s^*d\omega = d\,s^*\omega = d\,t^*\omega = t^*d\omega$, by the naturality of Part I's $d$; so $d:\Omega^k_{ep}(X)\to\Omega^{k+1}_{ep}(X^1)$ Chapter8/GapP2_B_EpForms · EpGroupoid.extD_mem_epForms. The text argues with the local sc-diffeomorphisms $f = t\circ(s|\mathbf{U}(\varphi_0))^{-1}$ and $f^*d = df^*$; the global maps $s$, $t$ give the same in one line. *(c)* Wedge products

Chapter8/GapP2_B_EpForms · EpGroupoid.formWedge_mem_epForms. *(d)* For an sc-smooth functor $F : X \to Y$, $F^*$ maps $\Omega^k_{ep}(Y)$ to $\Omega^k_{ep}(X)$, as $s_Y \circ F = F \circ s_X$ Chapter8/GapP2_B_EpForms · EpGroupoid.formPullback_mem_epForms, and naturally equivalent functors (Definition 8.1.13) have the same pull-back: for $\theta : F \to F'$, $F = s \circ \theta$ and $F' = t \circ \theta$, so $F^*\omega = \theta^* s^* \omega = \theta^* t^* \omega = F'^* \omega$ Chapter8/GapP2_B_EpForms · EpGroupoid.formPullback_eq_of_natTrans. The constant 0-forms are a witness (§VIII.26).

**The direct limit.** For the index lifts $X^i$ (an ep-groupoid by Proposition 7.3.9), $\Omega^k_{ep}(X^i)$ Chapter8/GapP2_B_EpForms · EP.epForms, with the pull-backs by the inclusions as the maps of a directed system Chapter8/GapP2_B_EpForms · EP.epFormsMap, and $\Omega^k_{ep,\infty}(X)$ is its direct limit Chapter8/GapP2_B_EpForms · EP.epFormsInfty, Mathlib's direct limit of modules. Every class has a representative on some $X^i$ Chapter8/GapP2_B_EpForms · EP.exists_epFormsInftyOf_eq, and two forms on $X^i$, $X^j$ represent the same class iff their pull-backs to some $X^l$, $l \geq i, j$, agree Chapter8/GapP2_B_EpForms · EP.epFormsInftyOf_eq_iff, as the text says. *The exterior derivative* $d[\omega] = [d\omega]$ Chapter8/GapP2_B_EpForms · EP.epFormsInftyD Chapter8/GapP2_B_EpForms · EP.epFormsInftyD_of is well defined since $d$ commutes with the steps, and $d \circ d = 0$ from Part I Chapter8/GapP2_B_EpForms · EP.epFormsInftyD_comp_epFormsInftyD. **Readings.** "An sc-smooth map $\oplus_k TX \to \mathbb{R}$, linear in each argument and skew-symmetric" is Part I's Definition 4.4.1, as the text says ("already introduced in Section 4.4"), so a form is a family of alternating maps on the fibres; compatibility on level $i$ is with the morphisms of $X^i$ on level one, that is of $\mathbf{X}_{i+1}$; and $T_yY$ in the definition is read as $T_yX$ (Findings, Chapter 8, items 110, 132) (pp. 372–373).

## VIII.36 Definition 8.2.2 (2017: 8.3): the sc-de Rham complex, and pull-backs by functors

**The complex** (V8.54). $\Omega^*_{ep,\infty}(X) = \oplus_k \Omega^k_{ep,\infty}(X)$ Chapter8/GapP2_B_EpForms · EP.epFormsInftyGraded with $d$ Chapter8/GapP2_B_EpForms · EP.epFormsInftyGradedD and $d \circ d = 0$ Chapter8/GapP2_B_EpForms · EP.epFormsInftyGradedD_comp_self is the sc-de Rham complex; its cohomology $\ker d / \operatorname{im} d$ Chapter8/GapP2_B_EpForms · EP.epDeRham is that of the same complex as a cochain complex of real vector spaces Chapter8/GapP2_B_EpFormsObj · EP.epDeRhamComplex. *It is a subcomplex of Part I's complex of the object M-polyfold.* The text defines $d$ as "the exterior derivative of an sc-differential form on the object M-polyfold". The inclusion $\Omega^k_{ep,\infty}(X) \to \Omega^k_\infty(\operatorname{Ob} X)$, the object M-polyfold of $X^i$ being the lift $(\operatorname{Ob} X)^i$ of Part I, is injective Chapter8/GapP2_B_EpFormsObj · EP.epFormsInftyToObj Chapter8/GapP2_B_EpFormsObj · EP.epFormsInftyToObj_injective and commutes with $d$ Chapter8/GapP2_B_EpFormsObj · EP.formsInftyD_comp_epFormsInftyToObj (p. 373).

**Pull-backs by sc-smooth functors.** Used by Proposition 8.2.3 and by §11.2 (Part XI), and stated in Theorem 11.2.1 (1), (2): for $F : X \to Y$ with index raises $F^i : X^i \to Y^i$, $F^*[\omega] = [(F^i)^* \omega]$ Chapter8/GapP2_B_EpFormsFunctor · EP.epFormsInftyPullback, well defined since the $(F^i)^*$ commute with the steps; it commutes with $d$ Chapter8/GapP2_B_EpFormsFunctor · EP.epFormsInftyD_comp_epFormsInftyPullback, and naturally equivalent functors give the same map, by §VIII.35 (d) on every level, applied to the index raises of the natural transformation Chapter8/GapP2_B_EpFormsFunctor · EP.epFormsInftyPullback_eq_of_natTrans. The inclusion into the forms of the object M-polyfold commutes with these pull-backs Chapter11/GapP2_B2_Poincare · EP.epFormsInftyToObj_epFormsInftyPullback.

# VIII.37 Proposition 8.2.3 (new in the 2021 edition; the 2017 text promised it for later): the Poincaré lemma

**Proposition 8.2.3** (V8.55; gap G.87, closed) `Chapter11/GapP2_B2_Poincare · EP.proposition_8_2_3`. *Every object $x$ of an ep-groupoid $X$ has saturated open neighbourhoods $V \subseteq U$ such that every closed $[\omega] \in \Omega^k_{ep,\infty}(X|U)$, $k \geq 1$, is $d[\sigma]$ on $V$ for some $[\sigma] \in \Omega^{k-1}_{ep,\infty}(X|V)$.* Here $X|U$ is the restriction of Proposition 7.1.22 `Chapter11/GapP2_B2_SubGroupoidModel · EP.sub`, $k \geq 1$ is written $k+1$, and "on $V$" is the pull-back by the inclusion functor $X|V \to X|U$ `Chapter11/GapP2_B2_SubGroupoidModel · EP.subIncl`. The proposition holds also inside every saturated open neighbourhood $U$ given in advance `Chapter11/GapP2_B2_Poincare · EP.proposition_8_2_3_of_isOpen`; the statement above takes $U = X$. **The text's proof**: Lemma 4.4.8 on the object M-polyfold gives a primitive $\tau$ on a neighbourhood $V'(x)$ invariant under $G_x$; the average $\tau' = \frac{1}{\#G_x}\sum_g g^*\tau$ is invariant with $d\tau' = \omega$, and is extended to the saturation by $\sigma(y) = \psi^*\tau'(x')$ for a morphism $\psi : x' \to y$. **The formalized proof** follows these steps; the extension is the push-forward along an equivalence.

*Step 1, the object M-polyfold.* Lemma 4.4.8 `Chapter4/Poincare2021b · lemma_4_4_8`, applied to the image of $a$ in the forms of the object M-polyfold of $X|U$, gives an open $V_0 \ni x$ and a primitive among the forms of the object M-polyfold on $V_0$ `Chapter8/GapP2_B_EpPoincare · EP.proposition_8_2_3_object`.

*Step 2, the translation groupoid.* Theorem 7.1.13 `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13` gives a natural representation of $G_x$ on an open $U(x) \subseteq V_0$, and with it the translation ep-groupoid $G_x \ltimes U(x)$, the M-polyfold $U(x)$ with only the unit morphisms, the functor $J : U(x) \to G_x \ltimes U(x)$, the identity on objects `Chapter11/GapP2_B2_TranslationAverage · EpGroupoid.NaturalRep.unitsFunctor`, the functors $\Phi_g : y \mapsto g*y$ of $U(x)$ `Chapter11/GapP2_B2_TranslationAverage · EpGroupoid.NaturalRep.smulFunctor` and the natural transformations $\theta_g : J \to J \circ \Phi_g$, $y \mapsto (g, y)$ `Chapter11/GapP2_B2_TranslationAverage · EpGroupoid.NaturalRep.smulNatTrans`. In an ep-groupoid with $s = t$, such as $U(x)$ with its units, every form of the object M-polyfold is a form of the ep-groupoid `Chapter11/GapP2_B2_Poincare · EP.epFormsInftyToObj_surjective_of_src_eq_tgt`, so the primitive of Step 1, restricted to $U(x)$, is a $\beta \in \Omega^k_{ep,\infty}(U(x))$ with $d\beta = J^*c$, $c$ the pull-back of $a$ to $G_x \ltimes U(x)$; $\beta$ need not be invariant.

*Step 3, averaging.* This is done for any $J : W \to T$ which is a bijective local sc-diffeomorphism on objects, with a finite family $\Phi_g$ closed under composition up to permutations $\sigma_g$ of the index set ($\Phi_h \circ \Phi_g = \Phi_{\sigma_g(h)}$), and $\theta_g : J \to J \circ \Phi_g$ whose values exhaust the morphisms of $T$ `Chapter11/GapP2_B2_Averaging · EP.exists_primitive_of_average`. On every level $i$ the sum $\sum_g (\Phi^i_g)^*\beta$ is invariant under every $(\Phi^i_h)^*$, by the permutations, and is therefore $(J^i)^*\tau$ for a form $\tau$ of the ep-groupoid $T^i$: compatibility with a morphism $\theta_g(y)$ of $T$ is invariance under $\Phi_g$ `Chapter11/GapP2_B2_Averaging · EP.exists_epForms_sum_pullback`. Dividing by $\#G_x$, $J^*(d\tau) = dJ^*\tau = J^*c$, and $J^*$ is injective since $J$ is a surjective local sc-diffeomorphism on objects `Chapter11/GapP2_B2_Averaging · EP.injective_epFormsInftyPullback_of_surjective`; so $c = d\tau$ with $\tau$ the text's $\tau'$ `Chapter11/GapP2_B2_TranslationAverage · EpGroupoid.NaturalRep.exists_primitive_translation`.

*Step 4, extension to the saturation.* The functor $G_x \ltimes U(x) \to X|\pi^{-1}\pi(U(x))$ `Chapter11/GapP2_B2_SubGroupoidModel · EP.modelFunctor` is an equivalence `Chapter11/GapP2_B2_SubGroupoidModel · EP.isEquivalence_modelFunctor`: full and faithful by the uniqueness property of $\Gamma$ (Theorem 7.1.13), essentially surjective by the definition of the saturation, and a local sc-diffeomorphism on objects. If $E : T \to Z$ is an equivalence and $E^*a = d\tau$, then $a = d(E_*\tau)$, with $E_*$ the push-forward of Definition 11.2.3 `Chapter11/GapP2_B2_EquivFormsInfty · EP.epFormsInftyPush` `Chapter11/GapP2_B2_Averaging · EP.exists_primitive_of_equivalence`; $E_*\tau$ at $y$ is $\psi^*\tau(x')$ for a morphism $\psi : x' \to y$, the text's $\sigma$. So $V = \pi^{-1}\pi(U(x))$, saturated, open and contained in $U$. No hypothesis is added (Findings, Chapter 8, items 123, 132) (pp. 373–374).

## VIII.38 Proposition 8.1.16 without added hypothesis: vertical vectors

§VIII.29 proved Proposition 8.1.16 with the step "$T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$" as the hypothesis `TangentZhouReflect`. The step holds for every ep-groupoid `Chapter8/TangentZhouReflect2021 · EpGroupoid.tangentZhouReflect`, so *(ZHOU) for $X$ gives (ZHOU) for $TX$* with the text's hypothesis only `Chapter8/TangentZhouReflect2021 · EpGroupoid.proposition_8_1_16_2021` (V8.44; gap G.88, closed).

**Vertical vectors.** For a smooth $\alpha \in T_xX$ (level one of $TX$) and $w \in T_xX$, the vertical vector $\mathrm{vert}_\alpha w \in T_\alpha(TX)$ `Chapter8/TangentVertical2021 · MPolyfold.vert` is the vector with chart representative $(T\varphi(\alpha), (0,h))$, $h$ the representative of $w$, in a chart of $TX$ at $\alpha$ `Chapter8/TangentVertical2021 · MPolyfold.vertChart`; it does not depend on the chart, its representative in every chart has the same form `Chapter8/TangentVertical2021 · MPolyfold.tangentChartMap_vert`. $w \mapsto \mathrm{vert}_\alpha w$ is injective `Chapter8/TangentVertical2021 · MPolyfold.vert_injective` and natural: $T(Tf)(\mathrm{vert}_\alpha w) = \mathrm{vert}_{Tf\alpha}(Tf\,w)$ for sc-smooth $f$ `Chapter8/TangentVertical2021 · MPolyfold.tangentMap_tangentMap_vert`. The local form is $D(Tf)(x,h)(0,v) = (0, Df(x)v)$, from the linearity of $Tf$ in the fibre `Chapter8/TangentVertical2021 · IsSc1.scDeriv_tangentMap_vert`.

**The proof.** Let $k = (g,h)$ be in the isotropy group of a smooth $b \in TX$ over $x$, with $Tk = \mathrm{id}$, and let $v \in T_xX$. Choose $u \in T_g\mathbf{X}$ with $Ts(u) = v$ ($Ts$ is bijective on the fibres) and put $V = \mathrm{vert}_k u \in T_k(T\mathbf{X})$. By naturality, $T(Ts)V = \mathrm{vert}_b v$ and $T(Tt)V = \mathrm{vert}_b(Tt\,u)$. $Tk$ sends $T(Ts)V$ to $T(Tt)V$, so $Tk = \mathrm{id}$ gives $\mathrm{vert}_b(Tt\,u) = \mathrm{vert}_b v$, hence $Tt(u) = v$ by injectivity, that is $Tg(v) = Tt(u) = v$. This is the text's "differentiate (8.16)" restricted to the vertical directions of $T_b(TX)$, on which $Tk$ acts as $Tg$; the surjectivity of $T\tau$ is not used. Steps 1 and 3 of §VIII.29 are unchanged (Findings, Chapter 8, items 115, 126) (p. 369).

## VIII.39 Propositions 8.4.1 and 8.4.3

**Proposition 8.4.1** (V8.47; gap G.86, closed) `Chapter8/GapP2_CProp841W1 · EpStrongBundle.proposition_8_4_1`. *For a strong bundle $P : W \to X$ over an ep-groupoid, the following are equivalent: (1) $|X|$ is paracompact; (2) $|W[0]|$ is paracompact; (3) $|W[1]|$ is paracompact.* (2) □ (1) is §VIII.30.

*(1) □ (2)* `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.proposition_8_4_1_one_imp_two`. *Step 1, the local action.* For a natural representation of $G_x$ on $U(x)$, $G_x$ acts on $W|U(x) = P^{-1}(U(x))$ by $g * w = \mu(\Gamma(g, P(w)), w)$, the text's lift on p. 382, by homeomorphisms `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.bundleMulAction`. *Step 2.* Two points of $W|U(x)$ have the same orbit in $|W|$ iff they are $G_x$-related `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.π_eq_π_iff_exists_smul`, so $G_x\backslash(W|U(x)) \to |W|$ is an open embedding `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.isOpenEmbedding_bundleOrbitQuotientMap` onto $|P|^{-1}(|U(x)|)$ `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.range_bundleOrbitQuotientMap`. *Step 3.* The quotient of a metrizable space by a finite group of homeomorphisms is metrizable, by the metric $\hat\rho$ of Lemma 7.3.2 `Chapter8/BundleOrbitParacompact2021 · metrizableSpace_orbitRel_quotient_finite`; $W$ is metrizable as an M-polyfold, so every $|P|^{-1}(|U(x)|)$ is metrizable. *Step 4.* $|X|$ paracompact is metrizable (Theorem 7.3.1), and Smirnov's theorem, pulled back along $|P|$ as for $TX$ in §VII.23, makes $|W[0]|$ metrizable `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.metrizableSpace_orbitSpace`, hence paracompact `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.proposition_8_4_1_one_iff_two`. This replaces the text's closed cover and Proposition 2.7.4 by the partition-of-unity form of Smirnov's theorem; no bundle charts are needed, only the continuity of $\mu$.

*(1) □ (3).* $W[1]$ is not built as an ep-groupoid in the project; $|W[1]|$ is taken as the orbit space of the groupoid $(W[1], \mathbf{X}_s \times_P W[1])$, the quotient of the M-polyfold $W[1]$ by $w \sim \mu(\varphi, w)$ `Chapter8/GapP2_CProp841W1 · EpStrongBundle.OrbitSpaceOne` `Chapter8/GapP2_CProp841W1 · EpStrongBundle.πOne_eq_πOne_iff`, its ep-groupoid structure being unused by the statement. The quotient map is open, as in Proposition 7.1.8 `Chapter8/GapP2_CProp841W1 · EpStrongBundle.isOpenMap_πOne`. The action of $G_x$ on $W[1]|U(x)$ is continuous for the level-one topology, since $\mu : \mathbf{X}_s \times_P W[1] \to W[1]$ is sc-smooth (part of "strong bundle map") `Chapter8/GapP2_CProp841W1 · EpStrongBundle.continuous_bundleOneSmul`, and Steps 2–4 repeat `Chapter8/GapP2_CProp841W1 · EpStrongBundle.isOpenEmbedding_bundleOneOrbitQuotientMap` `Chapter8/GapP2_CProp841W1 · EpStrongBundle.metrizableSpace_orbitSpaceOne`: (1) □ (3) `Chapter8/GapP2_CProp841W1 · EpStrongBundle.proposition_8_4_1_one_imp_three`. Conversely the zero section is an sc$^+$-section, so $|0| : |X| \to |W[1]|$ is continuous with continuous left inverse $|P[1]|$ and closed image `Chapter8/GapP2_CProp841W1 · EpStrongBundle.isClosedEmbedding_zeroOrbitMapOne`, and $|X|$ is paracompact as a closed subspace: (3) □ (1) `Chapter8/GapP2_CProp841W1 · EpStrongBundle.proposition_8_4_1_three_imp_one` `Chapter8/GapP2_CProp841W1 · EpStrongBundle.proposition_8_4_1_one_iff_three`. No hypothesis beyond the text's (Findings, Chapter 8, items 118, 127, 133) (p. 380).

**Proposition 8.4.3** (V8.48) `Chapter8/BundleLocalRegularity2021 · EpStrongBundle.proposition_8_4_3`. *If $X$ is effective and has (LR), the strong bundle has the local regularity property of Definition 8.4.2.* Take $U(x)$ from (LR) for $X$, and a strong bundle isomorphism $\Phi$ of $W|U(x)$ with $|\Phi(w)| = |w|$, over $\beta = g * \cdot$ by (LR). *Step 1.* $|\Phi(w)| = |w|$ gives a morphism $(\psi, w) : w \to \Phi(w)$, $\psi : P(w) \to \beta(P(w))$ `Chapter8/BundleLocalRegularity2021 · EpStrongBundle.exists_mor_of_π_eq`. *Step 2.* At a regular object two morphisms with the same source and target are equal `Chapter8/BundleLocalRegularity2021 · EpGroupoid.eq_of_isRegular_src`, so over a regular $P(w)$, $\psi = \Gamma(g, P(w))$ and $\Phi(w) = \mu(\psi, w) = g * w$ directly (the text's $C(w) = w$). *Step 3.* The regular points are open and dense (Proposition 7.2.4), and $P$ is an open map `Chapter10/LocalizationLREffectiveBundle · StrongBundle.isOpenMap_proj`, so $P^{-1}(X_{reg})$ is dense in $W|U(x)$, and the closed set where $\Phi(w) = g * w$ is all of $W|U(x)$. The uniqueness of $g$, which the text takes from effectiveness, is not needed (Findings, Chapter 8, items 119, 128) (p. 382).

## VIII.40 Lemma 8.5.10, Proposition 8.5.7 for $D = \emptyset$, Theorem 8.5.8 in every universe, Proposition 8.1.17 without added hypotheses

**Lemma 8.5.10, $F^*V$ over the ep-groupoid $Y$** (V8.51; gap G.89, closed). §VIII.31 proved the lemma with $F^*V$ as Part I's pull-back and $\mu_{F^*V}$ as a formula. *The strong bundle.* For every sc-smooth functor $F : Y \to X$ and strong bundle $Q : V \to X$, $(F^*Q, \mu_{F^*V})$ is a strong bundle over the ep-groupoid $Y$ (Definition 8.3.1: $\mu_{F^*V}$ is a surjective strong bundle map covering $t$, a local sc-diffeomorphism, with the unit and composition laws) `Chapter8/PullbackStrongBundle2021 · EpStrongBundle.pullbackBundle2021`: it is Chapter 10's pull-back along $F$ (Part X), and its structure map is the text's formula $\mu_{F^*V}(\varphi, (y, v)) = (t(\varphi), \mu_X(F(\varphi), v))$ `Chapter8/PullbackStrongBundle2021 · EpStrongBundle.pullbackBundle2021_μ`, so the text's sentence "defines the strong bundle" holds `Chapter8/PullbackStrongBundle2021 · EpStrongBundle.pullbackBundle2021_isStrongBundle`. *The isomorphism.* For a proper strong bundle covering functor, $B(w) = (P(w), A(w))$ and $B^{-1}$ are strong bundle maps of strong bundles over ep-groupoids (Definition 8.3.2) covering the identity functor of $Y$ `Chapter8/PullbackStrongBundle2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isMap_bundleB` `Chapter8/PullbackStrongBundle2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isMap_bundleBInv`, the compatibility with $\mu$ being (8.23); so $B$ is a strong bundle isomorphism over the ep-groupoid $Y$ `Chapter8/PullbackStrongBundle2021 · EpStrongBundle.IsProperStrongBundleCovering2021.lemma_8_5_10_full` (Findings,

Chapter 8, items 122, 129) (p. 395).

**Proposition 8.5.7 for $D = \emptyset$** (V8.35). Definition 8.5.5 allows $D = \emptyset$, and then Proposition 8.5.7 does not hold as we read it `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty`. The functor from the empty category to $\mathbb{Z}^\times$, as a one-object category, is a covering functor, both sides of the bijection of Definition 8.5.5 being empty `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.emptyCover`; with $L_C(\star) = \mathbb{R}$, $\mathbb{Z}/2$ acting by reflection, it has a geometric lift `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.emptyLift`. The induced $F : \emptyset \to \mathbb{Z}/2 \ltimes \mathbb{R}$ is not surjective on objects, hence not a proper covering functor. The project's Proposition 8.5.7 carries $D \neq \emptyset$, which the text presumably intends (§VIII.20; Findings, Chapter 8, item 130) (p. 390).

**Theorem 8.5.8 in every universe** (V8.36). The first reading proved Theorem 8.5.8 for ep-groupoids in `Type` only. Read with Definition 8.5.5 in one universe — $D$, $G$ and the category $\mathrm{Diff}_{sc}$ in which the lift takes values — no proof changes, and $X$, $Y$ share the universe because $\mathrm{Diff}_{sc}$ is one category (§VIII.21). The identity of a one-object category, with its geometric lift, is a witness `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_8_id` (Findings, Chapter 8, item 125) (p. 392).

**Proposition 8.1.17 without added hypotheses** (V8.45, V8.52). The hypotheses of §VIII.29 and §VIII.33 — natural representations with $U(x) \cap X_1$ connected in $X^1$ and dense in $U(x)$ (gap G.73), and the openness of $p : T\mathbf{X} \to \mathbf{X}^1$ (gap G.8) — hold for every ep-groupoid (§VII.45): the first `Chapter7/SmoothConnectedReps · EpGroupoid.hasConnectedDenseLevelOneReps`, the second by Part I `Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj`. With only the second still taken as an assumption the proposition is `Chapter7/SmoothConnectedReps · EpGroupoid.proposition_8_1_17_one_of_isOpenMap`; with none: *(1) if $X^1$ has LUC, so has $TX$, and $(T\mathbf{X})^T = p^{-1}((\mathbf{X}^1)^T)$* `Chapter7/SmoothRegularAssembly · EpGroupoid.proposition_8_1_17_one_of_isLUC` `Chapter7/SmoothRegularAssembly · EpGroupoid.nonEffectivePart_tangent_eq_of_isLUC`; *(2) if $X$ has (SLUC), $TX$ has LUC* `Chapter7/SmoothRegularAssembly · EP.proposition_8_1_17_two_of_isSLUC`, *every $T(X^i)$ has LUC* `Chapter7/SmoothRegularAssembly · EP.isLUC_tangent_lift_of_isSLUC`, *and $TX$ has (SLUC)* `Chapter7/SmoothRegularAssembly · EP.proposition_8_1_17_two_all_of_isSLUC`, through $(TX)^i \cong T(X^i)$ of §VIII.33 (Findings, Chapter 8, items 116, 124, 126) (p. 371).

# Part IX. Branched ep⁺-subgroupoids (HWZ Chapter 9, §§9.1–9.5 and §9.6.1 (2017 §§9.1–9.5))

Chapter 9 continues Part II; its numbers and pages are those of the book (Springer 2021, pp. 401–458), with the 2017 text's in parentheses (2017 pp. 291–332). It introduces the objects that will carry the solution sets of perturbed sc-Fredholm sections with symmetries: functors $\Theta : X \to \mathbb{Q}^+$ on an ep-groupoid that are, near every point of their support, finite weighted sums of indicator functions of properly embedded smooth finite-dimensional submanifolds. §9.1 gives the definition (Definition 9.1.2 (2017: 9.1)), the local closedness of the orbit support (Lemma 9.1.5 (2017: 9.1)), symmetric local branching structures (Lemma 9.1.7 (2017: 9.2)) and the dimensional decomposition (Propositions 9.1.11 (2017: 9.2) and 9.1.12, the latter new in 2021 with its proof in Appendix §9.6.1). §9.2 attaches to $\Theta$ a section $T_\Theta$ of formal sums of tangent spaces (Theorem 9.2.1 (2017: 9.1)), proved through good and bad points and a graph representation of the sheets near a bad point (Proposition 9.2.8, Lemmas 9.2.9–9.2.11 (2017: 9.3, 9.6–9.8)); it defines the tangent $T\Theta$ (Definition 9.2.12 (2017: 9.6)) and, through the position of a submanifold relative to $\partial X$ (Lemma 9.2.13 (2017: 9.9)), the boundary $\partial\Theta$ (Proposition 9.2.15, Definition 9.2.16 (2017: 9.4, 9.7)). §9.3 orients $\Theta$ (Definition 9.3.2 (2017: 9.8)), compares orientations at good points (Lemma 9.3.4 (2017: 9.10)), gives an example (Example 9.3.5 (2017: 9.1)) and the boundary orientation (Definitions 9.3.6–9.3.10). §9.4 studies essential points and sets of measure zero of the sheets and decomposes an oriented structure (Propositions 9.4.9, 9.4.10 (2017: 9.5, 9.6)). §9.5 integrates sc-differential forms and proves Stokes' theorem: the sets of measure zero in $|\mathrm{supp}(\Theta)|$ and the $\sigma$-algebras $L(S)$, $L(\partial S)$ (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16)), the branched integrals (Theorems 9.5.4, 9.5.5 (2017: 9.2, 9.3), Definition 9.5.6, Lemmas 9.5.7, 9.5.9–9.5.11 (2017: 9.17–9.19), Proposition 9.5.13), the boundary measures (Theorems 9.5.15, 9.5.16 (2017: 9.4, 9.5)) and Stokes' theorem (Theorems 9.5.17, 9.5.18 (2017: 9.6, 9.7)). Since the first version of this part every gap of §§9.1–9.4 has been closed and §9.5 has been formalized with the text's hypotheses, except part of Remark 9.5.20; the new material is in §§IX.26–IX.39, and the subsections IX.1–IX.25 are brought up to date in place. The state described is that of wave K (2026-11-13), on top of commit 8e5afeb.

**What is proved, and what is not.** All definitions of §§9.1–9.5 and §9.6.1 are formalized. Of the eighteen gaps of the chapter, seventeen are closed (G.93–G.109) and one is formalized in part (G.110, for Remark 9.5.20 only); none is open and none has nothing to formalize. *§§9.1–9.4.* The facts of level-0 sc-calculus that the first version of this part took as hypotheses on a general M-polyfold (the local constancy of the dimension of Remark 9.1.1, the graph representation of Lemmas 9.2.9–9.2.11, invariance of domain, the level-1 continuity of frames) are theorems on every M-polyfold (§IX.27). So Proposition 9.1.12, Theo-

rem 9.2.1 (2017: 9.1), Lemma 9.2.7 (2), Lemma 9.6.4 and Proposition 9.6.5 hold for every branched ep⁺-subgroupoid of every ep-groupoid, Lemma 9.3.4 (2017: 9.10) for every $n$-dimensional one, and $T\Theta$ is a branched ep⁺-subgroupoid of $TX$ when $|X|$ is paracompact (§IX.11). Also proved: Remark 9.1.4, Proposition 9.1.9 (2017: 9.1) and Remark 9.1.10 (§IX.26); the global decomposition of Proposition 9.1.11 (§IX.4); the footnote to Definition 9.1.2 (§IX.1); the classification of the orientations of Example 9.3.5 (§IX.29); a tame sheet as a Mathlib manifold with corners, the boundary operation built from the sheet, the continuity of $o_{\partial M}$, the boundary stratum and Figure 9.1 (§IX.30); Lemma 9.4.3 (§IX.18); Lemmas 9.4.4 and 9.4.6 for the sheets of an M-polyfold, with no hypothesis (§IX.28). Lemmas 9.1.5, 9.1.7, 9.2.3, 9.2.5, 9.2.7 (1), 9.3.7, 9.4.2, 9.4.8, 9.6.1, 9.6.2, Propositions 9.2.15, 9.3.9, 9.4.9, 9.4.10 and the remarks with formal content are proved, several in a stronger form. *§9.5.* The Lebesgue null sets of the sheets of an M-polyfold are built (Hausdorff measure in the charts), and Lemma 9.5.2 holds for $n$-dimensional $\Theta$ (§IX.28); Theorems 9.5.4 and 9.5.5 hold as printed for the integral of sc-differential forms, with "oriented, tame" read as: near every point of the support an oriented tame structure describes $\hat{T}_\Theta$ (§§IX.31–IX.32); (9.21), (9.22), Lemmas 9.5.7, 9.5.9–9.5.11 and Proposition 9.5.13 hold as identities of measures (§§IX.31, IX.33); $L(\partial S)$ is built with $(n-1)$-dimensional null sets (§IX.34). Theorems 9.5.15–9.5.18 are proved as printed, with the text's hypotheses only, for every $n \geq 1$ (G.109 closed, V9.67 and V9.68 restated): the class conditions `hcov` and `hcovS` of the first proofs (§§IX.34–IX.35) are derived (§IX.36), the local formula of Theorems 9.5.15 and 9.5.16 holds for every oriented tame structure (§IX.37), and for $n = 1$ the 0-forms are integrated over the finite sets of oriented boundary points (§IX.38). Witnesses with nonempty boundary, with the local unique continuation property, are the closed upper hemisphere ($n = 2$) and the closed half-circle ($n = 1$) (§§IX.35, IX.38). **Formalized in part** (G.110): Remark 9.5.20. Its M-polyfold $X$, $\partial X$, $\mathbb{Z}_3 \ltimes X$ and the effective isotropy at $\partial X$, three read in $X$ and one read in $\partial X$, are proved; $\Theta$, $\oint_\Theta df$ and the two boundary integrals are not (§§IX.24, IX.39).

**Narrower than printed, and why.** V9.2: the dimension is measured in a chart. V9.9 (Proposition 9.1.11): the uniqueness among all decompositions does not hold as we read it; it holds among local decompositions, and local finiteness off the support needs a bound on the dimensions (§IX.4). V9.27 (Definition 9.2.16): the uniqueness does not hold as we read it; the characterization is unique with (2) asked on the larger support (§IX.13). V9.32 (Lemma 9.3.4): $\Theta$ is $n$-dimensional; for mixed dimensions the lemma does not hold as we read it (§IX.15). V9.33 (Example 9.3.5): "otherwise different" does not hold as we read it, and "four possibilities" holds for each structure but not for $\Theta$, which has infinitely many orientations, six of them with integer weights (§§IX.15, IX.29). V9.42 (Lemma 9.4.4): the sheets have one dimension (§IX.17). V9.65 (Remark 9.1.10): one dimension $n$, since a Mathlib manifold has one model space; the composition of the groupoid is sc-smooth only, Mathlib having no orbifolds (§IX.26). V9.67 and V9.68 (Theorems 9.5.15–9.5.18) are restated since 2026-11-13 (§§IX.36–IX.38). **Statements that do not hold as we read them**, each with a Lean counterexample: the uniqueness in Proposition 9.1.11 (§IX.4); Lemma 9.2.13 (2) □ (1) at interior points (§IX.12); the uniqueness in Definition 9.2.16 (§IX.13); Lemma 9.3.4 for mixed dimensions and two claims of Example 9.3.5 (§§IX.15, IX.29); the footnote to Definition 9.1.2 for $\Theta = 0$ (§IX.1); Lemma 9.4.4 and the transfer of null sets in Lemma 9.5.2 across dimensions (§§IX.17, IX.19, IX.28).

**Seven structural choices.** (1) *Tangent sets, linear algebra in charts.* When this part was first written, Part I's $T_xX$ carried no linear structure (gap G.7, since closed). The tangent space $T_yM$ of a submanifold is the set of tangent vectors fixed by the tangent of the retraction that presents $M$, and $\mathrm{Gr}(x)$ is generated by such sets; in a chart the image of $T_yM$ is a linear subspace of the right dimension, and every argument that needs linear algebra is carried out there (§IX.6). (2) *Functors as functions.* $\Theta$ is a function on objects with $\Theta(s(\varphi)) = \Theta(t(\varphi))$, valued in $\mathbb{Q}$, nonnegativity a field. (3) *Dimension labellings.* The

splitting by dimension is done with labellings $\dim : I \to \mathbb{N}$, unique when they exist (§IX.1); since the dimension is locally constant on every M-polyfold, a labelled structure exists at every point of the support (§IX.27). (4) *Slices instead of sequences.* Proposition 9.2.8 is used in the text to count the weights of a class of tangent sheets by the values of $\Theta$ at good points; the formalization counts them on one finite slice transverse to the class (§IX.9). The same device proves Proposition 9.6.5 (§IX.22). (5) *Hypotheses first, then theorems.* The facts of the text's §9.2 that need level-0 sc-calculus inside a partial quadrant were first named hypotheses (`HasGraphChartAt`, `HasLocallyConstantDimAt`, `HasDomainInvarianceAt`, `HasContinuousLevelOneOn`), proved on $\mathbb{R}^m$; they are now theorems on every M-polyfold, with the graph representation in the weaker form of one cone per tangent space, which is all the slice arguments use (§IX.27). The statements under the hypotheses are kept. (6) *Sheet geometry and null sets as structures.* §9.4 is formalized for Mathlib manifolds with corners and, through the structure `SheetGeometry`, for sheets in an M-polyfold (§IX.17); the null sets of §9.5 are a structure `SheetNullSets` (§IX.19). Both are now instantiated on every M-polyfold by the $n$-dimensional Hausdorff measure of the chart images (§IX.28). (7) *Integration as a structure.* The integral of a form over a measurable part of an oriented sheet enters §9.5 through a structure `SheetIntegration` with the properties the text uses (linearity, countable additivity, vanishing on null sets, the comparison of two sheets, invariance under morphisms); Theorems 9.5.4 and 9.5.5 and the lemmas before them are proved for every such structure (§IX.31), and the structure is then built for sc-differential forms by integrating in graph charts (§IX.32).

**Map.**

- `Chapter9/Basic`: functors, supports, Definition 9.1.2 (2017: 9.1), Lemma 9.1.5 (2017: 9.1), Definitions 9.1.6, 9.1.8 (2017: 9.2, 9.3), the indicator of a point.
- `Chapter9/Symmetric`: the restriction remark and Lemma 9.1.7 (2017: 9.2).
- `Chapter9/Propositions`: Remark 9.1.1, $n$-dimensional $\Theta$, Remark 9.1.3 (b), the statements and the local construction of Propositions 9.1.11 and 9.1.12, the counterexample to the uniqueness of 9.1.11, submanifolds of $\mathbb{R}^n$.
- `Chapter9/GoodPoints`: Definitions 9.2.2, 9.2.6, Lemmas 9.2.3, 9.2.5, 9.2.7 (1) (openness), Remark 9.2.4.
- `Chapter9/Tangent`: $T_yM$, $\mathrm{Gr}(X)$, Theorem 9.2.1 under the independence, Definition 9.2.12.
- `Chapter9/TangentSlicesBasic`, `TangentSlicesRn`, `TangentSlices`: the chart form of $T_yM$, the graph representation and tangent slices, Lemma 9.2.7 (1) (density) and (2), the case of $\mathbb{R}^m$.
- `Chapter9/AppendixSlices`: one graph chart for all dimensions, Proposition 9.6.5, Proposition 9.1.12 and the independence of Theorem 9.2.1 on $\mathbb{R}^m$ without hypotheses.
- `Chapter9/Appendix`: Lemmas 9.6.1, 9.6.2, 9.6.4, Definition 9.6.3, Proposition 9.1.12 under `TopMultiplicity`.
- `Chapter9/BoundaryPosition`, `BoundaryPositionConverse`: Lemma 9.2.13, Remark 9.2.14, Definition 9.3.6, Lemma 9.3.7.
- `Chapter9/Boundary`: Proposition 9.2.15, Definition 9.2.16, Remark 9.2.17.
- `Chapter9/OrientationTangent`, `OrientationBasic`, `OrientationExample`, `OrientationUniqueness`, `OrientationBoundary`, `OrientationRn` (index `Chapter9/Orientation`): §9.3.
- `Chapter9/BranchingGeometryMeasure`, `BranchingGeometryManifold`, `BranchingGeometry`: §9.4.
- `Chapter9/Measure`: Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2, the remark after Definition 9.5.3, witnesses.
- `Chapter9/MeasureEffective`: the isotropy and set parts of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13.
- `Chapter9/GeneralLevelZero`, `GeneralCorollaries`, `GeneralBoundaryPoints`: the level-0 facts on

every M-polyfold and their consequences (§IX.27).

- `Chapter9/DimDecompositionExists`, `DimDecompositionUnique`: the global decomposition of Proposition 9.1.11 (§IX.4).
- `Chapter9/LocallyEquidimensional2021`: the footnote to Definition 9.1.2 and "locally equidimensional" (§IX.1).
- `Chapter9/TangentSubmanifoldSmooth`: $TM_i$ smooth in $TX$ (§IX.11).
- `Chapter9/MathlibManifoldBridge`: tame sheets as Mathlib manifolds with corners (§IX.16).
- `Chapter9/BoundaryOperation`, `BoundaryOrientationContinuity2021`, `GapP2_DBoundaryStratum`, `GapP2_E6_Stratum*`, `GapP2_E6_BoundaryOrientation`, `GapP2_D3FaceOrientation`: Definitions 9.3.8, 9.3.10 and Figure 9.1 (§IX.30).
- `Chapter9/PropertyTShrinking2021`: Lemma 9.4.3 (§IX.18).
- `Chapter9/SheetEssentialChart2021`, `GapP2_DSheet*`: Lemmas 9.4.4, 9.4.6 and the null sets for M-polyfold sheets, Lemma 9.5.2 (§IX.28).
- `Chapter9/GapP2_CRemark914`, `GapP2_CProp919*`, `GapP2_CDescentPolyfold`, `GapP2_C2*`: Remark 9.1.4, Proposition 9.1.9, Remark 9.1.10 (§IX.26).
- `Chapter9/GapP2_DExample935`, `GapP2_D2Example935*`, `GapP2_DOrientationDim`, `GapP2_DBoundaryChar`: Example 9.3.5, Lemma 9.3.4, Definition 9.2.16 (§§IX.13, IX.15, IX.29).
- `Chapter9/GapP2_E1_*`, `GapP2_E2MeasureGlue`: branched integration for an abstract sheet integration (§IX.31).
- `Chapter9/GapP2_E3_*`, `GapP2_E4_*`: the integral of sc-forms, Theorems 9.5.4, 9.5.5, (9.22), Proposition 9.5.13 (2), the circle (§§IX.32–IX.33).
- `Chapter9/GapP2_E2BoundaryNull`, `GapP2_E5_BoundaryNull*`, `GapP2_E6_BoundarySheets`, `GapP2_E7_*`: $L(\partial S)$ and Theorems 9.5.15, 9.5.16 (§IX.34).
- `Chapter9/GapP2_E2Stokes`, `GapP2_E5_*`, `GapP2_E6_StokesDim1*`, `GapP2_E8F_*`, `GapP2_E8_*`, `GapP2_E8W_*`: Theorems 9.5.17, 9.5.18 and the hemisphere (§IX.35).
- `Chapter9/GapP3_K1_CornerFinite`, `GapP3_K1_BoundaryCover`, `GapP3_K1_Theorems`, `GapP3_K1_Hemisphere`: the class conditions derived, Theorems 9.5.15–9.5.18 with the text's hypotheses for $n \geq 2$ (§IX.36); `GapP3_K1_AllStructures`: the local formula for every structure (§IX.37).
- `Chapter9/GapP3_K2_*`: $n = 1$, 0-forms, finite sets of points, the half-circle, and the local unique continuation property of the witnesses (§IX.38).
- `Chapter9/GapP3_K3_Retract`, `GapP3_K3_MPolyfold`, `GapP3_K3_Groupoid`: Remark 9.5.20, the M-polyfold, the groupoid and the effective isotropy (§IX.39).

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid as in Part VII, $s, t$ its source and target, $\pi : X \to |X|$ the orbit map, $X^\infty$ the full subcategory of smooth objects, $G_x$ the isotropy group of $x$ with a natural representation $(\Phi, \Gamma)$ on a neighbourhood $U(x)$ (§VII.4). A *submanifold* is a smooth finite-dimensional submanifold in the sense of Part I (§IV.2): the image, near each of its points, of an sc$^+$-retraction. For $\Theta : X \to \mathbb{Q}$, $\mathrm{supp}(\Theta) = \{\Theta > 0\}$ and $|\mathrm{supp}(\Theta)| = \pi(\mathrm{supp}(\Theta))$. A local branching structure at $x$ is $(U(x), (M_i)_{i\in I}, (\sigma_i)_{i\in I})$ as in Definition 9.1.2; we write $I_y = \{i \mid y \in M_i\}$ and $M_U = \bigcup_i M_i$.

# IX.1 Definition 9.1.2 (2017: 9.1), and the dimension of a submanifold (Remark 9.1.1 (2017: 9.1))

**Definition 9.1.2 (2017: 9.1)** (V9.1) `Chapter9/Basic · EpGroupoid.IsBranchedEpSubgroupoid` `Chapter9/Basic · EpGroupoid.LocalBranchingStructure`. A branched ep$^+$-subgroupoid is a functor $\Theta \geq 0$ `Chapter9/Basic · EpGroupoid.IsFunctorial` with $\mathrm{supp}(\Theta) \subseteq X_\infty$ `Chapter9/Basic · EpGroupoid.supp`, $|\mathrm{supp}(\Theta)|$

paracompact Chapter9/Basic · EpGroupoid.orbitSupp, and a local branching structure at every point of the support. The formalization reads the paracompactness of the induced topology as a `ParacompactSpace` instance on the subtype $|\mathrm{supp}(\Theta)|$, and "the inclusion $M_i \to U(x)$ is proper" as `IsProperMap` of the inclusion of subtypes. For a closed $M_i$ in $U(x)$ the latter is automatic Chapter9/Symmetric · isProperMap_inclusion_iff. Witnesses: the zero functor Chapter9/Basic · EpGroupoid.isBranchedEpSubgroupoid_zero and the indicator of a smooth point of $\mathbb{R}^n$, a 0-dimensional submanifold Chapter9/Basic · rnPolyfold_isSmoothSubmanifold_singleton Chapter9/Basic · EpGroupoid.localBranchingStructure_indicator Chapter9/Basic · EpGroupoid.isBranchedEpSubgroupoid_indicator. The 2017 text refers in (4) to "(ii) and (iii)", the 2021 text to (2) and (3).

**The dimension** (V9.2). Remark 9.1.1 says that $m \mapsto \dim T_m M$ is locally constant. With $T_m M$ a set of tangent vectors, the dimension is measured in a chart: $M$ has dimension $k$ at $a$ if the tangent space at the chart image of $a$ of the sub-retract presenting $M$ is $k$-dimensional Chapter9/Propositions · MPolyfold.HasDimAt. *Step 1, chart independence.* A germ of sc-diffeomorphisms between retracts restricts to a linear isomorphism of the tangent spaces of sub-retracts Chapter9/Propositions · finrank_span_tangentSpace_eq, so the value is the same for every chart and every presentation Chapter9/Propositions · MPolyfold.finrank_tangentSpace_chart_change Chapter9/Propositions · MPolyfold.HasDimAt.finrank_eq; it is unique Chapter9/Propositions · MPolyfold.HasDimAt.unique and exists at every point Chapter9/Propositions · MPolyfold.IsSmoothSubmanifold.exists_hasDimAt. *Step 2, invariance.* It is unchanged by intersecting with an open set and by sc-diffeomorphisms Chapter9/Propositions · MPolyfold.hasDimAt_inter_isOpen_iff Chapter9/Propositions · MPolyfold.hasDimAt_image_iff. *Step 3, local constancy* Chapter9/AppendixSlices · MPolyfold.HasLocallyConstantDimAt holds on every M-polyfold Chapter9/GeneralLevelZero · MPolyfold.hasLocallyConstantDimAt: in a chart the dimension is the rank of $Dt$ for the chart sc$^+$-retraction $t$ presenting the submanifold Chapter9/GeneralLevelZero · hasDimAt_finrank, and this rank is locally constant along the retract Chapter9/GeneralLevelZero · eventually_finrank_eq (§IX.27). On $\mathbb{R}^m$ it was proved first: $T_y M$ is the range of the projection $Ds(y)$, whose rank is its trace, continuous in $y$ and integer valued Chapter9/AppendixSlices · MPolyfold.hasLocallyConstantDimAt_rn. The dimensions of the sheets of one structure enter as labellings; all sheets of one structure pass through $x$, so a labelling is unique Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dim_unique, and by local constancy every point of the support has a labelled structure Chapter9/GeneralCorollaries · hasLabelledBranchingAt_general.

**$n$-dimensional $\Theta$.** The 2021 text says the sheets "can be taken" $n$-dimensional (2017: are $n$-dimensional), and this is the formal reading: some local branching structure at each point of the support has only $n$-dimensional sheets Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.IsOfDim Chapter9/Propositions · EpGroupoid.IsNDimensional. The zero functor is $n$-dimensional for every $n$ Chapter9/Propositions · EpGroupoid.isNDimensional_zero. **The footnote**, that an $n$-dimensional $\Theta$ is not $m$-dimensional for $m \neq n$, is proved for $\Theta$ with nonempty support Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote_of_ne_zero. *Step 1.* At a point of the support two structures, one with $n$-dimensional and one with $m$-dimensional sheets, give the same tangent sum (Theorem 9.2.1, now unconditional, §IX.10), so some sheet of the first has the tangent space of some sheet of the second Chapter9/LocallyEquidimensional2021 · exists_submanifoldTangent_eq. *Step 2.* Equal tangent spaces have equal dimensions, so $m = n$ Chapter9/LocallyEquidimensional2021 · isOfDim_unique. For $\Theta = 0$ the footnote does not hold as we read it, the zero functor being $n$-dimensional for every $n$ Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote_zero. "Locally equidimensional" is formalized in both readings, which agree: at every point of the support some structure has sheets of one dimension Chapter9/LocallyEquidimensional2021 · IsLocallyEquidimensional, iff at every such point there is an $n$ such that every structure there, restricted to a small enough neighbourhood, has only $n$-dimensional sheets Chapter9/LocallyEquidimensional2021 · isLocallyEquidimensional_iff, the book's "suffi-

ciently small"; nearby points of the support then have structures of the same dimension Chapter9/LocallyEquidimensional2021 · IsLocallyEquidimensional.exists_isOpen_isNDimensionalAt (gap G.93, closed).

**Submanifolds of $\mathbb{R}^n$** (V9.3). The witnesses of the chapter are built from closed submanifolds of $\mathbb{R}^n$. The image of a smooth idempotent $s$ is one, of dimension rank $Ds(a)$ Chapter9/Propositions · rnPolyfold_isSmoothSubmanifold_range Chapter9/Propositions · rnPolyfold_hasDimAt_range. The graph of a smooth $\beta : \mathbb{R} \to \mathbb{R}$ is the image of $(s,t) \mapsto (s, \beta(s))$ Chapter9/Propositions · graphRetraction Chapter9/Propositions · rnPolyfold_isSubmanifoldOfDim_graph2. Finite weighted sums of closed submanifolds, and indicators of locally closed ones, are branched ep$^+$-subgroupoids of the trivial ep-groupoid Chapter9/Propositions · EpGroupoid.isBranchedEpSubgroupoid_sum_rn Chapter9/Propositions · EpGroupoid.isNDimensional_sum_rn Chapter9/Propositions · EpGroupoid.isBranchedEpSubgroupoid_indicator_rn, whose orbit space is $\mathbb{R}^n$, so the paracompactness is free Chapter9/Propositions · EpGroupoid.paracompactSpace_orbitSupp_trivial_rn.

## IX.2 Remark 9.1.3 (b) (2017: 9.2 (b)), Lemma 9.1.5 (2017: 9.1), Definitions 9.1.6 and 9.1.8 (2017: 9.2 and 9.3)

**Remark 9.1.3 (b) (2017: 9.2 (b))** (V9.4) Chapter9/Propositions · EpGroupoid.remark_9_1_3_b. The text takes $\beta_1$, $\beta_2$ with prescribed signs, the graphs $G_1$, $G_2$ with weight 1, and $\Theta(p) = \#\{i \mid p \in G_i\}$ Chapter9/Propositions · EpGroupoid.thetaGraphs Chapter9/Propositions · EpGroupoid.thetaGraphs_eq, and says that the graphs of $\gamma_1 = 0$ and of some $\gamma_2$ positive off 0 describe $\Theta$ too. *Step 1.* The sign conditions are the structure `Remark913Hyp` Chapter9/Propositions · EpGroupoid.Remark913Hyp, satisfied by $\beta_1(s) = e(s)$, $\beta_2(s) = e(-s)$ with Mathlib's `expNegInvGlue` $e$ Chapter9/Propositions · EpGroupoid.remark913Hyp_expNegInvGlue. *Step 2.* For the second structure to describe the same $\Theta$, $\gamma_2$ must be $\beta_1 + \beta_2$ Chapter9/Propositions · EpGroupoid.Remark913Hyp.eq_add_of_thetaGraphs_eq, which is smooth and positive off 0 Chapter9/Propositions · EpGroupoid.Remark913Hyp.γ₂_pos. The two structures are Chapter9/Propositions · EpGroupoid.Remark913Hyp.structureβ and Chapter9/Propositions · EpGroupoid.Remark913Hyp.structureγ. *Step 3.* "Not unique, even close to $x$" is read as: no sheet of the first structure agrees with a sheet of the second on any neighbourhood of the origin, since each of $\beta_1$, $\beta_2$ differs from 0 and from $\beta_1 + \beta_2$ at points arbitrarily close to 0.

**Lemma 9.1.5 (2017: 9.1)** (V9.5) Chapter9/Basic · EpGroupoid.lemma_9_1_5. *$|\mathrm{supp}(\Theta)|$ is locally closed in $|X|$.* **The text's proof** takes a neighbourhood with the natural $G_x$-action, the properness property and a local branching structure, and shows that the image of $\bigcup_i M_i$ is closed in the image of $U(x)$. **The formalized proof.** *Step 1.* For a functor the support is saturated, $\pi^{-1}(|\mathrm{supp}(\Theta)|) = \mathrm{supp}(\Theta)$ Chapter9/Basic · EpGroupoid.preimage_π_orbitSupp. *Step 2.* On $U(x)$ the support is $\bigcup_i M_i$ Chapter9/Basic · EpGroupoid.LocalBranchingStructure.mem_supp_iff, a finite union of sets closed in $U(x)$ Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.isClosed_M Chapter9/Basic · EpGroupoid.LocalBranchingStructure.isClosed_preimage_val_supp, so $\mathrm{supp}(\Theta)$ is locally closed in $X$ Chapter9/Basic · EpGroupoid.isLocallyClosed_supp. *Step 3.* $\pi$ is open (Chapter 7), and the image of the locally closed saturated set is locally closed. The properness property of the text is not used.

**Definitions 9.1.6 and 9.1.8 (2017: 9.2 and 9.3)** (V9.6, V9.8) are read literally Chapter9/Basic · EpGroupoid.IsClosedBranched Chapter9/Basic · EpGroupoid.IsCompactBranched Chapter9/Basic · EpGroupoid.IsOrbifoldType Chapter9/Basic · EpGroupoid.IsManifoldType; the indicator of a point is compact and of manifold type Chapter9/Basic · EpGroupoid.isCompactBranched_indicator Chapter9/Basic · EpGroupoid.isManifoldType_indicator. Proposition 9.1.9 and Remark 9.1.10 (natural manifold and orbifold structures on $|\mathrm{supp}(\Theta)|$) and Remark 9.1.4 (metrizability, the compactness criterion) are proved in §IX.26 (gaps G.94, G.95, closed).

## IX.3 The restriction remark, and Lemma 9.1.7 (2017: 9.2)

**The restriction** (V9.7). Before Lemma 9.1.7 the text replaces $M_i$ by $N_i = M_i \cap V(x_0)$ for $V(x_0) \subseteq U(x)$ and notes that some $N_i$ may be empty. Definition 9.1.2 asks $x \in M_i$, which the $N_i$ need not satisfy at $x_0$; the remark is formalized for an open $V \subseteq U(x)$ containing $x$ itself, which is the case the symmetrization uses `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.restrict`. The $N_i$ are submanifolds `Chapter9/Symmetric · MPolyfold.IsSmoothSubmanifold.inter_isOpen`, and they are properly embedded in $V$ because they are closed in $U(x)$.

**Lemma 9.1.7 (2017: 9.2)** (V9.7) `Chapter9/Symmetric · EpGroupoid.lemma_9_1_7` `Chapter9/Symmetric · EpGroupoid.lemma_9_1_7'`. *At $x \in \operatorname{supp}(\Theta)$ there is a local branching structure on the domain of a natural representation of $G_x$, with a $G_x$-action on the index set, $g * M_i = M_{g(i)}$ and $\sigma_{g(i)} = \sigma_i$; it can be found inside every open $V \ni x$.* **The text's proof** sets $J = I \times G_x$, $M_{(i,g)} = g * M_i$, $\sigma_{(i,g)} = \sigma_i/|G_x|$ and computes $\sum_{y \in g*M_i} \sigma_{(i,g)} = \frac{1}{|G_x|}\sum_g \Theta(g^{-1} * y) = \Theta(y)$ by functoriality. **The formalized proof** follows it `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.symmetrize` `Chapter9/Symmetric · EpGroupoid.SymmetricBranchingStructure`. *Step 1.* Restrict the given structure to the domain $U \subseteq V \cap U(x)$ of a natural representation (Theorem 7.1.13, §VII.4). *Step 2.* $\Phi(g)(M_i \cap U)$ is a submanifold properly embedded in $U$ through $x$, since $\Phi(g)$ is an sc-diffeomorphism of $U$ fixing $x$ `Chapter9/Symmetric · MPolyfold.IsSmoothSubmanifold.image_of_isScDiffeoOn`; the text uses this without saying so. *Step 3.* The weights: $\Theta(\Phi(g)(y)) = \Theta(y)$ `Chapter9/Symmetric · EpGroupoid.IsFunctorial.apply_Φ` and $y \in \Phi(g)(A) \iff \Phi(g)^{-1}(y) \in A$ `Chapter9/Symmetric · EpGroupoid.NaturalRep.mem_image_Φ_iff` turn the text's computation into a sum over $G_x$ with $|G_x| =$ `Nat.card` of the finite group. *Step 4.* $G_x$ acts on $J$ by left multiplication on the second factor `Chapter9/Symmetric · EpGroupoid.mulActionProdLeft`.

## IX.4 Proposition 9.1.11 (2017: 9.2): the local construction, and the uniqueness

**Proposition 9.1.11 (2017: 9.2)** (V9.9) `Chapter9/Propositions · EpGroupoid.Proposition_9_1_11`. *A branched ep⁺-subgroupoid determines a unique sequence $\Theta_0, \Theta_1, \ldots$ of branched ep⁺-subgroupoids with $\Theta_k$ $k$-dimensional and $\Theta = \sum_k \Theta_k$, locally finite.* The formal statement: for every branched $\Theta$ there is exactly one family $(\Theta_k)$ with `IsDimDecomposition` `Chapter9/Propositions · EpGroupoid.IsDimDecomposition`, every $\Theta_k$ branched and $k$-dimensional, at each object finitely many nonzero and $\Theta$ their sum, and every such family is locally finite `Chapter9/Propositions · EpGroupoid.IsLocallyFiniteFamily`.

**The local construction.** The text splits the indices of a symmetric structure by the dimension of $M_i$ and puts $\Theta_k(y) = \sum_{i \in I_k, y \in M_i} \sigma_i$. Formally, for a labelling dim, $\Theta_k$ `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart` has the $k$-dimensional local branching structure $(M_i)_{\dim i = k}$ `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartStructure` `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartStructure_isOfDim` and $\Theta = \sum_k \Theta_k$ on $U(x)$ `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.eq_sum_dimPart`; $\Theta_k = 0$ if no sheet has dimension $k$ `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart_eq_zero`, and $\Theta_n = \Theta$ for an $n$-dimensional structure `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart_eq_of_isOfDim`. For a symmetric structure $G_x$ preserves dimension, since $\Phi(g)$ is an sc-diffeomorphism `Chapter9/Propositions · EpGroupoid.SymmetricBranchingStructure.dim_smul`, so $\Theta_k$ is $G_x$-invariant `Chapter9/Propositions · EpGroupoid.SymmetricBranchingStructure.dimPart_Φ`. That the local $\Theta_k$ does not depend on the structure is Proposition 9.1.12 (§IX.5).

**The uniqueness, as we read it, does not hold** Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn. On the trivial ep-groupoid of $\mathbb{R}^2$, let $L$ be the $x$-axis Chapter9/Propositions · EpGroupoid.axis2 and $\Theta = 1_L$. *Family A:* $\Theta_0 = 1_{\{0\}}$, $\Theta_1 = 1_{L\setminus\{0\}}$ Chapter9/Propositions · EpGroupoid.decompA Chapter9/Propositions · EpGroupoid.decompA_isDimDecomposition. *Family B:* $\Theta_0 = 0$, $\Theta_1 = 1_L$ Chapter9/Propositions · EpGroupoid.decompB Chapter9/Propositions · EpGroupoid.decompB_isDimDecomposition. In family A, $L \setminus \{0\}$ is a 1-dimensional submanifold properly embedded in the open set $\mathbb{R}^2 \setminus \{0\}$, which is all Definition 9.1.2 asks at its points, and $1_{\{0\}}$ is 0-dimensional; both families satisfy (1) and (2). The proof of the text constructs family B.

**The repaired uniqueness.** Call a decomposition *local* if near every point of $\mathrm{supp}(\Theta)$ its $\Theta_k$ is the $k$-dimensional part of a local branching structure of $\Theta$, which is what the text's proof produces Chapter9/DimDecompositionUnique · IsLocalDimDecomposition. Family B is local and family A is not Chapter9/DimDecompositionUnique · decompB_isLocalDimDecomposition Chapter9/DimDecompositionUnique · decompA_not_isLocalDimDecomposition. Two local decompositions are equal: near a point of the support both are dimensional parts of structures there, which agree near the point by Proposition 9.1.12, and off the support both vanish Chapter9/DimDecompositionUnique · isDimDecomposition_unique. Proposition 9.1.12 now holds on every ep-groupoid (§IX.5), so the uniqueness among local decompositions has no hypothesis Chapter9/GeneralCorollaries · proposition_9_1_11_unique.

**The global decomposition** (gap G.96, closed). *Step 1, the definition.* $\Theta_k(y)$ is the sum of the weights of the $k$-dimensional sheets of a chosen labelled structure at $y$, and 0 if there is none Chapter9/DimDecompositionExists · HasLabelledBranchingAt Chapter9/DimDecompositionExists · dimComponent. Every sheet passes through its base point, so this is the value at $y$ of the local $\Theta_k$ at $y$. *Step 2, independence and locality.* By Proposition 9.1.12, $\Theta_k(x)$ is the same for every labelled structure at $x$ Chapter9/DimDecompositionExists · dimComponent_eq_self, and on all of $U(x)$ it is the $k$-dimensional part of any labelled structure at $x$, restricted to $y$ Chapter9/DimDecompositionExists · dimComponent_eq_of_mem Chapter9/DimDecompositionExists · dimPart_restrictAt. *Step 3, functoriality.* Along a morphism $\varphi$, a labelled structure at $s(\varphi)$ is transported to $t(\varphi)$, preserving dimensions Chapter9/DimDecompositionExists · isSubmanifoldOfDim_transport, and Step 2 at the target gives $\Theta_k(s(\varphi)) = \Theta_k(t(\varphi))$ Chapter9/DimDecompositionExists · isFunctorial_dimComponent; the text's symmetric structures are not needed. *Step 4, paracompactness*, which the text does not discuss: $|\mathrm{supp}(\Theta_k)|$ is closed in $|\mathrm{supp}(\Theta)|$ Chapter9/DimDecompositionExists · isClosed_orbitSupp_dimComponent Chapter9/DimDecompositionExists · paracompactSpace_orbitSupp_dimComponent. *Step 5.* So each $\Theta_k$ is a $k$-dimensional branched ep⁺-subgroupoid Chapter9/DimDecompositionExists · isBranchedEpSubgroupoid_dimComponent Chapter9/DimDecompositionExists · isNDimensional_dimComponent, the family is a local decomposition Chapter9/GeneralCorollaries · isLocalDimDecomposition_dimComponent_general, and for every ep-groupoid there is exactly one local decomposition Chapter9/GeneralCorollaries · existsUnique_localDimDecomposition_general. *Step 6, local finiteness.* At the points of the support only finitely many $\Theta_k$ are nonzero nearby, those of the sheets of one structure Chapter9/GeneralCorollaries · dimComponent_locallyFinite_of_mem_general. At points of the closure of the support outside it, the formalization needs a bound on the dimensions of the submanifolds of $X$ Chapter9/GeneralCorollaries · proposition_9_1_11_local_of_le, which holds on $\mathbb{R}^m$ Chapter9/DimDecompositionExists · rn_hasDimAt_le Chapter9/DimDecompositionExists · proposition_9_1_11_local_rn. For the line $L$ the construction gives family B Chapter9/DimDecompositionExists · dimComponent_axis2. Row V9.9 is narrower for these two reasons: "uniquely" is read among local decompositions, and local finiteness off the support uses the bound.

## IX.5 Proposition 9.1.12: the statement and its cases

**Proposition 9.1.12** (V9.10) `Chapter9/Propositions · EpGroupoid.Proposition_9_1_12`, new in the 2021 edition. *For two local branching structures of $\Theta$ on $U(x)$ there is an open $G_x$-invariant $V(x) \subseteq U(x)$ on which the dimensional decompositions agree.* The formal conclusion is `DimPartsAgreeNear`: an open $V \ni x$ in both domains on which $\Theta_k = \Theta'_k$ for every $k$ `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear`. "$G_x$-invariant, with the usual properties" is obtained by shrinking $V$ to the domain of a natural representation `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear.exists_naturalRep`. *The equidimensional case* is immediate, since then $\Theta_n = \Theta$ for both `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_isOfDim`. The general case is proved in Appendix §9.6.1 (§§IX.21–IX.22): under the hypothesis `TopMultiplicity` `Chapter9/Appendix · EpGroupoid.proposition_9_1_12_of_topMultiplicity`, under the graph representation `Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_of_hasGraphChartAt`, without hypothesis on every ep-groupoid over $\mathbb{R}^m$ `Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_rn`, and now without hypothesis on every ep-groupoid `Chapter9/GeneralCorollaries · proposition_9_1_12`: `TopMultiplicity` holds on every M-polyfold, by the slice argument of §IX.22 run with one cone per tangent space (§IX.27) `Chapter9/GeneralCorollaries · topMultiplicity`. Mixed structures exist: $\Theta = 1 + \delta_0$ on $\mathbb{R}^1$ has sheets of dimensions $0$ and $1$ `Chapter9/Appendix · exists_mixed_rn_one`, and on $\mathbb{R}^2$ two labelled structures of one $\Theta$ with sheets of dimensions $1$ and $0$ have different sheets and agreeing dimensional parts `Chapter9/AppendixSlices · EpGroupoid.Remark913Hyp.exists_mixed_rn_two`.

## IX.6 The category $\mathrm{Gr}(X)$, and the tangent spaces $T_yM$

$T_yM$ (V9.12). Part I's Remark 4.2.2 (preprint: 4.14) defines $T_yM = Tr(y)(T_yX)$ for an sc$^+$-retraction $r$ with image $M$ near $y$. Here $T_yM$ is the set of tangent vectors at $y$ fixed by the tangent of $r$ `Chapter9/Tangent · MPolyfold.submanifoldTangent`. *Step 1.* It does not depend on the retraction `Chapter9/Tangent · MPolyfold.mem_submanifoldTangent_iff` `Chapter9/Tangent · MPolyfold.submanifoldTangent_congr`, lies in the fibre over $y$ `Chapter9/Tangent · MPolyfold.submanifoldTangent_subset_tangentFiber`, is local `Chapter9/Tangent · MPolyfold.submanifoldTangent_inter_isOpen`, and is mapped onto $T_{f(y)}f(M)$ by an sc-diffeomorphism $f$ `Chapter9/Tangent · MPolyfold.submanifoldTangent_image`. *Step 2, the chart form.* In a chart the second components of the images of $T_yM$ form a set `Chapter9/TangentSlicesBasic · MPolyfold.chartTangent` equal to the tangent space of the chart model of $M$ `Chapter9/TangentSlicesBasic · MPolyfold.chartTangent_eq`, a linear subspace `Chapter9/TangentSlicesBasic · MPolyfold.exists_submodule_chartTangent` of dimension $\dim_y M$ `Chapter9/TangentSlicesBasic · MPolyfold.finrank_chartTangent`; a chart change acts on it by a continuous linear map `Chapter9/TangentSlicesBasic · MPolyfold.exists_clm_tangentChartMap_snd` `Chapter9/TangentSlicesBasic · MPolyfold.chartTangent_chart_change`, a linear isomorphism `Chapter9/TangentSlicesBasic · MPolyfold.exists_linearEquiv_chartTangent`. So the generators $T_xM_i$ of $T_\Theta(x)$ are linear subspaces in every chart `Chapter9/TangentSlicesBasic · EpGroupoid.LocalBranchingStructure.exists_submodule_of_mem_support`, which is what the text means by "finite-dimensional linear subspaces of $T_xX$".

$\mathrm{Gr}(X)$ (V9.11). $\mathrm{Gr}(x)$ is the additive group of finitely supported $\mathbb{Q}$-valued functions on subsets of $T_xX$ `Chapter9/Tangent · EpGroupoid.Gr`, a morphism $\varphi : x \to y$ acts by the image under its tangent $T\varphi$ (Chapter 7) `Chapter9/Tangent · EpGroupoid.grPush`, with $T1_x = \mathrm{id}$ `Chapter9/Tangent · EpGroupoid.tangentMor_unit_of_eq` and $T\varphi$ injective on the fibre `Chapter9/Tangent · EpGroupoid.tangentMor_injOn`; the push-forward is functorial `Chapter9/Tangent · EpGroupoid.grPush_comp` `Chapter9/Tangent · EpGroupoid.grPush_unit`. The objects $(x, a)$ `Chapter9/Tangent · EpGroupoid.GrObj` form a category over the full subcategory $X^\infty$ `Chapter9/Tangent · EpGroupoid.SmoothCat` with the projection $\pi$ `Chapter9/Tangent · EpGroupoid.grProj`.

## IX.7 Good points: Definitions 9.2.2 and 9.2.6 (2017: 9.4 and 9.5), Lemmas 9.2.3 and 9.2.5 (2017: 9.3 and 9.4), Remark 9.2.4 (2017: 9.4)

**Definition 9.2.2 (2017: 9.4)** (V9.15) `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.IsGoodPoint`. $y \in M_U$ `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.MU` is good if there is an open $V(y) \subseteq U(x)$ on which all sheets through $y$ coincide. The text states this for equidimensional $\Theta$; no proof below uses equidimensionality, so the definition and Lemmas 9.2.3 and 9.2.5 are stated for every $\Theta$ with a local branching structure.

**The intrinsic form** `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.isGoodPoint_iff`. *For $y \in U(x)$: $y$ is good iff $y \in \operatorname{supp}(\Theta)$ and $\Theta$ is constant on $\operatorname{supp}(\Theta)$ near $y$. Step 1.* Near $y$ take $V$ in the complement of the sheets not through $y$ `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.nbhdAt` `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.isOpen_nbhdAt`; there $\Theta(z) \leq \Theta(y)$ `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.le_of_mem_nbhdAt`, with equality iff $z$ lies on all sheets through $y$ `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.eq_iff_of_mem_nbhdAt`. *Step 2.* If all sheets through $y$ coincide on $V$, $\Theta$ is $\Theta(y)$ on $\operatorname{supp}(\Theta) \cap V$; conversely, if $\Theta$ is constant on the support near $y$, every point of a sheet through $y$ near $y$ lies on all of them, by Step 1.

**Lemma 9.2.3 (2017: 9.3)** (V9.16) `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.lemma_9_2_3` follows at once, for any two structures, at possibly different points, whose domains contain $y$: the intrinsic form does not mention the structure. This is the text's argument, in which $Q(z) = V(y) \cap W(y)$ is presumably $Q(y)$.

**Remark 9.2.4 (2017: 9.4)** (V9.17). (a) `Chapter9/GoodPoints · EpGroupoid.remark_9_2_4_a` is Lemma 9.2.3 with the good set of Definition 9.2.6 `Chapter9/GoodPoints · EpGroupoid.mem_goodSet_iff`. (b) `Chapter9/GoodPoints · EpGroupoid.remark_9_2_4_b`: near a good $y$, $\Theta = \Theta(y) \cdot 1_M$ with $M = M_i \cap V(y)$ for any $i \in I_y$, properly embedded in $V(y)$ `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.restrictAt`.

**Lemma 9.2.5 (2017: 9.4)** (V9.18) `Chapter9/GoodPoints · EpGroupoid.lemma_9_2_5`. *$s(\varphi)$ is good iff $t(\varphi)$ is good.* The text transports local branching structures along morphisms. The formal proof uses the intrinsic form and the sheet of $\varphi$: on a sheet $\mathbf{W}$ of $s$ around $\varphi$, $\widehat{\varphi} = t \circ (s|\mathbf{W})^{-1}$ is an sc-diffeomorphism between open neighbourhoods of $s(\varphi)$ and $t(\varphi)$ with $\Theta \circ \widehat{\varphi} = \Theta$ `Chapter9/GoodPoints · EpGroupoid.Sheet.apply_hat`, so "$\Theta$ constant on the support near the point" passes from one to the other `Chapter9/GoodPoints · EpGroupoid.src_mem_goodSet_of_tgt`.

**Definition 9.2.6 (2017: 9.5)** (V9.19) `Chapter9/GoodPoints · EpGroupoid.goodSet` `Chapter9/GoodPoints · EpGroupoid.badSet`. Both sets are saturated `Chapter9/GoodPoints · EpGroupoid.preimage_π_image_goodSet` `Chapter9/GoodPoints · EpGroupoid.preimage_π_image_badSet`, by Lemma 9.2.5. The text cites "Remark 9.2.4 (c)" here; the remark has (a) and (b), and Lemma 9.2.5 is what the descent to $|X|$ uses.

## IX.8 Lemma 9.2.7 (2017: 9.5): good points are dense and accumulate on every sheet

**Lemma 9.2.7 (1) (2017: 9.5 (1))** (V9.20). The openness of the good set in $\operatorname{supp}(\Theta)$ is the definition `Chapter9/GoodPoints · EpGroupoid.lemma_9_2_7_1`. **The text** derives the density from (2). **The formalized proof** needs neither (2) nor equidimensionality `Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_1_dense`. *Step 1.* Given an open $W \subseteq U(x)$ meeting the support, shrink $W$ one sheet at a time so that $M_U \cap W$ either lies in $M_j$ or

misses it Chapter9/TangentSlices · exists_isOpen_subset_inter_eq_empty_or_subset: if $M_U \cap W \not\subseteq M_j$, a point of $M_U \cap W$ off the closed set $M_j$ has a neighbourhood in $W$ missing $M_j$, and the sheets already decided stay decided. *Step 2.* On the final $W$ every sheet through a point of $M_U \cap W$ contains $M_U \cap W$, so all sheets through it agree on $W$ and it is good. So the good set is dense Chapter9/TangentSlices · EpGroupoid.dense_goodSet and the bad set nowhere dense Chapter9/TangentSlices · EpGroupoid.isNowhereDense_badSet.

**Lemma 9.2.7 (2) (2017: 9.5 (2))** (V9.21) Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_2. *For an $n$-dimensional structure at $x$ and every $i$, good points of $M_i$ converge to $x$.* **The text's proof** fixes $i_0$, sets $\Sigma_i = M_i \cap M_{i_0} \cap D_k(x)$ and either finds a point of $M_{i_0} \cap D_k(x)$ off all $\Sigma_i$ or moves a point off the boundaries of those $\Sigma_i$ with interior; such a point is "clearly" good. **The formalized proof** does this with Step 1 above applied inside $M_{i_0}$. The point found is good only if $M_{i_0} \subseteq M_i$ near it forces $M_{i_0} = M_i$ near it, which is invariance of domain for submanifolds of equal dimension. It was first the hypothesis `HasDomainInvarianceAt` Chapter9/TangentSlices · MPolyfold.HasDomainInvarianceAt, proved on $\mathbb{R}^m$ by the inverse function theorem Chapter9/TangentSlices · MPolyfold.hasDomainInvarianceAt_rn Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_2_rn. It now holds on every M-polyfold Chapter9/GeneralLevelZero · hasDomainInvarianceAt: in a chart both submanifolds are the ranges of chart sc⁺-retractions, and two such retracts with the same tangent space at the point, one inside the other, agree near it Chapter9/GeneralLevelZero · eq_of_range_eq (§IX.27). So Lemma 9.2.7 (2) holds for every ep-groupoid Chapter9/GeneralCorollaries · lemma_9_2_7_2_general (gap G.98, closed). The statement holds at every $x$, not only at bad points. In Remark 9.1.3 (b) the origin is bad Chapter9/TangentSlices · EpGroupoid.Remark913Hyp.zero_mem_badSet and is approached by good points on each graph Chapter9/TangentSlices · EpGroupoid.Remark913Hyp.exists_seq_goodSet_structureβ.

## IX.9 Proposition 9.2.8 and Lemmas 9.2.9–9.2.11 (2017: 9.3 and 9.6–9.8): graph charts and tangent slices

**The text.** At a bad point $x$ the indices with the same $T_xM_i = L$ form a class $\Xi$. Locally, $x = 0$ in a partial quadrant $C$; $L \cap C$ has interior in $L$ (Lemma 9.2.9); the $M_i$, $i \in \Xi$, are graphs $a + B_i(a)$ over a cone around a unit $h_0$ in that interior, with an sc-complement $Z$ of $L$ (Lemma 9.2.10), through the inversion of a map $\Phi$ on the cone (Lemma 9.2.11). Proposition 9.2.8 then splits $\Xi = I^1 \sqcup \cdots \sqcup I^p$ with sequences of good points $x_k^e \to x$ lying on $M_i$ exactly for $i \in I^e$, and the proof of Theorem 9.2.1 counts $\sum_{i\in\Xi} \sigma_i = \sum_e \Theta(x_k^e)$.

**What the count needs.** One finite set $F$ near $x$ that meets each $M_i$, $i \in \Xi$, exactly once and misses the other sheets, so that $\sum_{i\in\Xi} \sigma_i = \sum_{y\in F} \Theta(y)$: a *tangent slice* Chapter9/Tangent · MPolyfold.HasTangentSlicesAt. The sequences of Proposition 9.2.8 are one way to produce such sets; a slice needs only one good direction $h_0 \in L$, off the finitely many other tangent spaces. So Proposition 9.2.8 is formalized in this slice form (V9.22).

**The graph representation, first as a hypothesis.** Lemmas 9.2.9–9.2.11 rest on sc-calculus at level 0 inside a partial quadrant (sc-complements, the mean value formula for sc⁺-retractions, an inverse function theorem on a cone). In the first version of this part their content in a chart was the hypothesis `IsGraphChartAt`, which is now proved on every M-polyfold (§IX.27; gap G.97, closed): every $n$-dimensional submanifold through $x$ is, along each admissible ray $th$ of its tangent space, met by the affine fibre through $th$, with secants uniformly approximated by $T_xM$ Chapter9/TangentSlicesBasic · MPolyfold.IsGraphChartAt Chapter9/TangentSlicesBasic · MPolyfold.HasGraphChartsAt. Rays replace the cones of Lemma 9.2.10, and the admissible directions play the role of $C$. *From graphs to slices* Chapter9/TangentSlicesBasic · MPolyfold.hasTangentSlicesAt_of_hasGraphChartsAt: with $Q$ a projection onto $L$, the slice is $\{y \mid Q(\varphi y - \varphi x) = th_0,\ \|\varphi y - \varphi x - th_0\| < \varepsilon t\}$

for small $t$; it meets each sheet of the class once, by the graph property, and misses the other sheets, since their tangent spaces do not contain $h_0$ Chapter9/TangentSlicesBasic · exists_mem_ball_forall_not_mem Chapter9/TangentSlicesBasic · eq_zero_of_proj_eq_zero. *On* $\mathbb{R}^m$ the graph representation holds at every point and in every dimension Chapter9/TangentSlicesRn · MPolyfold.isGraphChartAt_rn Chapter9/TangentSlicesRn · MPolyfold.hasGraphChartsAt_rn Chapter9/TangentSlicesRn · MPolyfold.hasTangentSlicesAt_rn: a submanifold is near $x$ the image of a smooth retraction $s$, and the inverse function theorem for $Q \circ s$ restricted to $L$ gives the graph Chapter9/TangentSlicesRn · exists_mem_graph_of_hasStrictFDerivAt Chapter9/TangentSlicesRn · MPolyfold.rn_exists_retraction.

**One chart for all dimensions.** For Proposition 9.6.5 and the independence of Theorem 9.2.1 without equidimensionality, the sheets of all dimensions through $x$ must be compared in one chart: `HasGraphChartAt` asks one chart and one set of admissible directions for all of them Chapter9/AppendixSlices · MPolyfold.HasGraphChartAt Chapter9/AppendixSlices · MPolyfold.HasGraphChartAt.hasGraphChartsAt, which holds on $\mathbb{R}^m$ Chapter9/AppendixSlices · MPolyfold.hasGraphChartAt_rn. In it, every class has slices inside every neighbourhood Chapter9/AppendixSlices · MPolyfold.exists_slice_of_isGraphChartAt, which can be chosen to miss finitely many given submanifolds of smaller dimension Chapter9/AppendixSlices · MPolyfold.exists_generic_slice. For Lemma 9.3.4 good points must be found near the slice, which a ray does not provide; the ball form `HasGraphBallAt` asks the graph property over a ball of foot points Chapter9/OrientationRn · MPolyfold.HasGraphBallAt Chapter9/OrientationRn · MPolyfold.HasGraphBallAt.hasGraphChartsAt, again true on $\mathbb{R}^m$ Chapter9/OrientationRn · hasGraphBallAt_rn. On a general M-polyfold, one cone of admissible directions for all submanifolds through a boundary point is more than the slice arguments use: they need a cone for each tangent space separately, `HasClassGraphChartAt` Chapter9/GeneralLevelZero · HasClassGraphChartAt, which holds at every point Chapter9/GeneralLevelZero · hasClassGraphChartAt; the stronger `HasGraphChartAt` and `HasGraphBallAt` hold at interior points Chapter9/GeneralLevelZero · hasGraphChartAt_of_mem_interior Chapter9/GeneralLevelZero · hasGraphBallAt_of_mem_interior (§IX.27).

## IX.10 Theorem 9.2.1 (2017: 9.1): the section $T_\Theta$

**Theorem 9.2.1 (2017: 9.1)** (V9.13) Chapter9/Tangent · EpGroupoid.theorem_9_2_1. *There is a section functor* $T_\Theta : X^\infty \to \mathrm{Gr}(X)$, $\pi \circ T_\Theta = \mathrm{Id}$, *with* $T_\Theta(x) = \sum_i \sigma_i T_x M_i$ *for every local branching structure at* $x \in \mathrm{supp}(\Theta)$, *the zero sum off the support, and* $T_\Theta(\varphi) = (T_\Theta(s(\varphi)), \varphi)$. The theorem is proved in two parts. *Part 1, the section, given the independence.* Write $\sum_i \sigma_i T_x M_i$ as an element of $\mathrm{Gr}(x)$ Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum. If any two structures at $x$ give the same sum Chapter9/Tangent · EpGroupoid.HasIndependentTangentSums, $T_\Theta(x)$ is that sum for a chosen structure, and $0$ off the support Chapter9/Tangent · EpGroupoid.tangentValue Chapter9/Tangent · EpGroupoid.tangentValue_eq. For a morphism $\varphi$, a structure at $s(\varphi)$ is transported to $t(\varphi)$ by $\hat{\varphi}$ Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.transport, which maps the sheets' tangent sets by $T\varphi$ Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_transport Chapter9/Tangent · EpGroupoid.tangentMor_eq_tangentMapOn_hat; by the independence $T\varphi_*(T_\Theta(s(\varphi))) = T_\Theta(t(\varphi))$ Chapter9/Tangent · EpGroupoid.grPush_tangentValue, which makes $T_\Theta$ a functor Chapter9/Tangent · EpGroupoid.tangentSection.

**Part 2, the independence** (V9.14). *At a good point* there is no hypothesis: near it all sheets through it coincide, so the sum is $\Theta(y) \cdot T_y M$ for every structure Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isGoodPoint Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_subset_goodSet. *Equidimensional structures* Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_isOfDim: group the sheets of both structures through $x$ by tangent set; for each class $L$, a tangent slice $F$ (§IX.9) meeting each

sheet of either structure with tangent space $L$ once gives $\sum_{T_xM_i=L}\sigma_i=\sum_{y\in F}\Theta(y)=\sum_{T_xM'_j=L}\sigma'_j$. With sheets of one dimension a slice for one structure is one for the other. *In general* the text reduces to the equidimensional case through the dimensional decomposition, and so does the formalization: if the dimensional parts of two structures agree near $x$ (Proposition 9.1.12), their tangent sums agree dimension by dimension `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_sum_dimPartStructure` `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_dimPartsAgreeNear` `Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_dimPartsAgreeNear`. With one graph chart for all dimensions and the local constancy of the dimension, which supplies the labellings, the independence follows `Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_of_hasGraphChartAt_of_hasLocallyConstantDimAt`; restricting a structure to a neighbourhood where the dimensions are constant is `Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.exists_restrict_labelled`. On every ep-groupoid over $\mathbb{R}^m$ both hypotheses hold, so $T_\Theta$ exists for every branched $\Theta$ `Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn`. Now Proposition 9.1.12 holds on every ep-groupoid and labelled structures exist at every point (§IX.27), so the independence, and with it Theorem 9.2.1, holds for every branched ep⁺-subgroupoid of every ep-groupoid `Chapter9/GeneralCorollaries · hasIndependentTangentSums`; the tangent slices of the equidimensional case exist at every point too `Chapter9/GeneralCorollaries · hasTangentSlicesAt`. In Remark 9.1.3 (b) the two structures have the same tangent sum at the bad origin `Chapter9/TangentSlicesRn · EpGroupoid.Remark913Hyp.tangentSum_structureβ_eq_structurey`. The text writes "it suffices to prove the proposition" (p. 410) for the theorem and closes with "This completes the proof of Theorem 9.2.8" (p. 418), presumably Theorem 9.2.1; in $\sum_{i\in I|y\in M_j}\sigma_iT_yM_i$ (p. 418) the index is presumably $i$.

## IX.11 Definition 9.2.12 (2017: 9.6): the tangent $T\Theta$

**Definition 9.2.12 (2017: 9.6)** (V9.23) `Chapter9/Tangent · EpGroupoid.tangentTheta`. $T\Theta(h)=0$ over the complement of the support and $T\Theta(h)=\sum_{h\in TM_i}\sigma_i$ over $U(x)$, with $h\in TM_i$ read as $h\in T_{p(h)}M_i$ `Chapter9/Tangent · EpGroupoid.tangentSubmanifold`. The formal definition evaluates $T_\Theta(p(h))$ at the tangent sets containing $h$ `Chapter9/Tangent · EpGroupoid.grEval`, so it does not depend on a structure by construction; that it agrees with the text's local formula for every structure `Chapter9/Tangent · EpGroupoid.tangentTheta_eq` is the text's "the local definitions fit together", and comes from the independence through the identity $\sum_{h\in T_yM_i}\sigma_i=\sum_{h\in T_yM'_j}\sigma'_j$ `Chapter9/Tangent · EpGroupoid.sum_tangent_eq_sum_tangent`. $T\Theta\geq 0$ `Chapter9/Tangent · EpGroupoid.tangentTheta_nonneg` and it is a functor on the tangent ep-groupoid of Chapter 8 `Chapter9/Tangent · EpGroupoid.isFunctorial_tangentTheta`.

**Is $T\Theta$ a branched ep⁺-subgroupoid?** The text says so without discussion. It needs that the $TM_i$ are smooth submanifolds of $TX$ of smooth points, closed in $TU(x)$, and that $|\mathrm{supp}(T\Theta)|$ is paracompact; these are the fields of `TangentBranchingHyp` `Chapter9/Tangent · EpGroupoid.TangentBranchingHyp`, under which the theorem holds `Chapter9/Tangent · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta` `Chapter9/Tangent · EpGroupoid.tangentBranchingStructure`. *Closedness* is proved `Chapter9/TangentSlices · EpGroupoid.isClosed_tangentSubmanifold`. *Paracompactness* follows from that of $|X|$ `Chapter9/TangentSlices · EpGroupoid.paracompactSpace_orbitSupp_tangentTheta`. So only the smoothness of the $TM_i$ remains `Chapter9/TangentSlices · EpGroupoid.tangentBranchingHyp_of_isSmoothSubmanifold` `Chapter9/TangentSlices · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_of_isSmoothSubmanifold`. $TM_i$ is a sub-M-polyfold of $TX$, the image of the retraction $Ts$ `Chapter9/AppendixSlices · MPolyfold.IsSmoothSubmanifold.isSubMPolyfold_tangent` `Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.isSubMPolyfold_tangentSubmanifold`. *That $Ts$ is an sc⁺-retraction of $TX$* (gap G.99, closed): for the sc⁺-retraction $s$ presenting a smooth finite-dimensional submanifold, $g=1^1_0\circ s$ is sc-smooth into $X^1$ and $Ts=T\iota\circ Tg$ for

the inclusion $\iota : X^1 \to X$. The tangent of $\iota$ is the natural inclusion $T(X^1) \to TX$ of Proposition 2.3.17, in a chart the tangent of the sc-operator $E^1 \to E$, which is itself `Chapter9/TangentSubmanifoldSmooth · retractTangentMap_shiftIncl` `Chapter9/TangentSubmanifoldSmooth · tangentLower_tangentMapOn_one`. Hence $Ts$ maps into $(TX)_1$ `Chapter9/TangentSubmanifoldSmooth · tmap_mem_levelSet`, and $1^1_0 \circ Ts$ is $Tg$ followed by the sc-diffeomorphism $T(X^1) \cong (TX)^1$, so $Ts$ is an sc⁺-retraction of $TX$ `Chapter9/TangentSubmanifoldSmooth · isScPlusRetractionOn_tmap`. So $TA$ is a smooth submanifold of $TX$ `Chapter9/TangentSubmanifoldSmooth · IsSmoothSubmanifold.isSmoothSubmanifold_tangent`, the field holds for every local branching structure `Chapter9/TangentSubmanifoldSmooth · LocalBranchingStructure.isSmoothSubmanifold_tangentSubmanifold`, and for $|X|$ paracompact $T\Theta$ is a branched ep⁺-subgroupoid of $TX$ on every ep-groupoid `Chapter9/TangentSubmanifoldSmooth · tangentBranchingHyp` `Chapter9/TangentSubmanifoldSmooth · isBranchedEpSubgroupoid_tangentTheta_general`. On $\mathbb{R}^m$ this was proved first with the field as the only hypothesis `Chapter9/AppendixSlices · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_rn`; the tangent of a line in $\mathbb{R}^2$ is a sub-M-polyfold of $T\mathbb{R}^2$ `Chapter9/AppendixSlices · rnPolyfold_isSubMPolyfold_tangent_graph2`. The full hypothesis has the zero functor as witness `Chapter9/Tangent · EpGroupoid.tangentBranchingHyp_zero` `Chapter9/Tangent · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_zero`. The text's introduction (p. 408) calls $T\Theta$ a "branched ep-subgroupoid", Definition 9.2.12 an ep⁺-subgroupoid.

## IX.12 Lemma 9.2.13 (2017: 9.9) and Remark 9.2.14 (2017: 9.5): the position of $M$ relative to $\partial X$

**Reduced tangent position in a chart** (V9.24). $T^R_x X$ is not a subspace of a vector space in the project; in a chart $\varphi$ with $\varphi(x) = z$ in a partial quadrant, "$T_xM \subseteq T^R_x X$" is the inclusion of the chart image of $T_xM$ in the reduced tangent space $E_z$ `Chapter9/BoundaryPosition · IsReducedTangentAt` `Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_reduced`, equivalently in the partial quadrant or the partial cone at $z$ `Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_partialQuadrant` `Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_partialCone`. It does not depend on the chart `Chapter9/BoundaryPosition · MPolyfold.isReducedTangentAt_chart_change` `Chapter9/BoundaryPosition · MPolyfold.isTangentReducedAt_iff_forall`, and defines `IsTangentReducedAt` `Chapter9/BoundaryPosition · MPolyfold.IsTangentReducedAt`. $d_M(x)$ is the degeneracy index of the induced M-polyfold structure `Chapter9/BoundaryPosition · MPolyfold.subDegIndex`, at most $d_X(x)$ `Chapter9/BoundaryPosition · MPolyfold.subDegIndex_le`.

**Lemma 9.2.13 (2017: 9.9).** *For $x \in M$: (1) $x \in \partial X$ and $d_M(x) = 0$ iff (2) $T_xM \subseteq T^R_x X$.* **As printed (2) □ (1) does not hold at interior points**: for a point $p$ of $\mathbb{R}^n$, $\{p\}$ is in reduced tangent position at $p$ and $p \notin \partial\mathbb{R}^n$ `Chapter9/BoundaryPosition · MPolyfold.not_lemma_9_2_13_two_one`. The lemma is about $x \in M \cap \partial X$, where the text uses it, and is formalized there. *(1) □ (2)* `Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13` `Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13_chart`: if $d_M(x) = 0$, Part I's Theorem 2.5.? (preprint: 2.31) gives a chart of $M$ without boundary near $x$, whose tangent space lies in the reduced tangent space of $X$; the boundary hypothesis is not used, and the implication holds off $\partial X$ too `Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13_of_not_mem_boundary`. *(2) □ (1)* `Chapter9/BoundaryPositionConverse · MPolyfold.lemma_9_2_13_two_one`. **The text's proof** builds a complement $Y$ and a map $F$ and asserts that the image is $M$ near 0. **The formalized proof.** *Step 1.* In a chart, $M$ is the image of an sc⁺-retraction $s$ near $z$, with $T_zM \subseteq E_z$. *Step 2,* "the image is $M$ near 0", which the text does not spell out: the projection $p$ onto $T_zM$ along a complement is injective on $M$ near $z$, by a level-1 contraction with the mean value inequality `Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_injOn_proj`. *Step 3.* By Mathlib's inverse function theorem for $p \circ s$ on $T_zM$, $\chi = p(\cdot - z)$ is a chart of $M \cap G$ onto an open subset of $T_zM \subseteq E_z$,

with inverse $s \circ (z + \cdot) \circ \Psi^{-1}$, a model without boundary Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_flat. *Step 4.* So $d_M(x) = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.subDegIndex_eq_zero_of_isTangentReducedAt Chapter9/BoundaryPositionConverse · MPolyfold.degIndex_eq_zero_of_isReducedTangentAt. The complement $Y$ and the map $F$ are not needed. Together: the equivalence at $x \in \partial X$ Chapter9/BoundaryPositionConverse · MPolyfold.lemma_9_2_13_iff, and at every $x \in M$: $d_M(x) = 0 \iff T_xM \subseteq T^R_xX$ Chapter9/BoundaryPositionConverse · MPolyfold.subDegIndex_eq_zero_iff. The witness is a point of $\mathbb{R}^n$; a witness at a boundary point needs a submanifold API for quadrants, not yet available.

**Remark 9.2.14 (2017: 9.5)** (V9.25). $\partial M$ is the boundary of the induced M-polyfold Chapter9/BoundaryPosition · MPolyfold.mem_boundary_subPolyfold_iff. *$x \in \partial M$ iff $x \in \partial X$ and $T_xM \not\subseteq T^R_xX$* Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_iff: □ with $d_M(x) \geq 1$ from Lemma 9.2.13 Chapter9/BoundaryPosition · MPolyfold.remark_9_2_14; □ since $\partial M \subseteq \partial X$ Chapter9/BoundaryPosition · MPolyfold.remark_9_2_14_mem_boundary and a point of $M$ in reduced position has $d_M = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_not_isTangentReducedAt Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_not_mem_boundary_iff.

## IX.13 Proposition 9.2.15 (2017: 9.4), Definition 9.2.16 (2017: 9.7) and Remark 9.2.17 (2017: 9.6): the boundary $\partial\Theta$

**The boundary of a sheet.** $y \in \partial M$ is characterized through $T_yM$ (Remark 9.2.14), so the set of $y$ with $y \in \partial M_i$ Chapter9/Boundary · MPolyfold.submanifoldBoundary Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_iff is read off the tangent sets: $y \in \partial M$ iff $y \in \partial X$ and $T_yM$ is not in reduced position Chapter9/Boundary · MPolyfold.IsReducedTangentSet Chapter9/Boundary · MPolyfold.isReducedTangentSet_iff. It is local and invariant under sc-diffeomorphisms Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_inter_isOpen_iff Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_image_iff.

**Proposition 9.2.15 (2017: 9.4)** (V9.26). *(1)* The boundary weight $\sum_{y \in \partial M_i} \sigma_i$ is the part of $T_\Theta(y)$ on tangent sets not in reduced position Chapter9/Boundary · EpGroupoid.boundaryPart Chapter9/Boundary · EpGroupoid.filter_tangentValue_eq; so, under the independence of Theorem 9.2.1, it is the same for any two structures whose domains contain $y$ Chapter9/Boundary · EpGroupoid.proposition_9_2_15_1. *(2)* If $d_X(y) = 0$, no tangent set is out of reduced position, so the weight and the formal sum vanish for every structure, with no independence Chapter9/Boundary · EpGroupoid.proposition_9_2_15_2. *(3)* $\partial\Theta$, 0 off $\partial X$ and the boundary part of $T_\Theta$ on it Chapter9/Boundary · EpGroupoid.boundaryTheta, is a functor Chapter9/Boundary · EpGroupoid.isFunctorial_boundaryTheta, nonnegative, supported in $\partial X$ and given by the local formula on every structure Chapter9/Boundary · EpGroupoid.boundaryTheta_eq Chapter9/Boundary · EpGroupoid.proposition_9_2_15_3. (1) and (2) need neither $x \in \operatorname{supp}(\Theta) \cap \partial X$ nor that $\Theta$ be branched.

**Definition 9.2.16 (2017: 9.7)** (V9.27). *$\partial\Theta$ is uniquely characterized by (1) $\operatorname{supp}(\partial\Theta) \subseteq \partial X$ and (2) the local formula at points of $\operatorname{supp}(\partial\Theta)$* Chapter9/Boundary · EpGroupoid.IsBoundaryCharacterized. $\partial\Theta$ has (1) and (2) Chapter9/Boundary · EpGroupoid.isBoundaryCharacterized_boundaryTheta. **The uniqueness, as we read it, does not hold**: (2) is asked only on $\operatorname{supp}(F)$, so $F = 0$ has (1) and (2) for every $\Theta$ Chapter9/Boundary · EpGroupoid.isBoundaryCharacterized_zero, and for $\Theta = 1$ on $[0, \infty)$, where $\partial\Theta(0) = 1$ Chapter9/Boundary · boundaryTheta_halfLine_one, both 0 and $\partial\Theta$ do Chapter9/Boundary · not_unique_isBoundaryCharacterized_halfLine. With (2) asked at the points of $\operatorname{supp}(\Theta)$ and $F = 0$ off $\operatorname{supp}(\Theta)$, $F = \partial\Theta$ Chapter9/Boundary · EpGroupoid.eq_boundaryTheta. **A reading with the printed (2)** Chapter9/GapP2_DBoundaryChar · definition_9_2_16_existsUnique: among the $F \geq 0$ vanishing off $\operatorname{supp}(\Theta)$ with (1) and (2), there is exactly one whose support contains the supports of all the others, and it is $\partial\Theta$. *Step 1.* Such an $F$ agrees with $\partial\Theta$ on its own support, by (2) and the local

formula for $\partial\Theta$ Chapter9/GapP2_DBoundaryChar · IsBoundaryCharacterized.eq_boundaryTheta_on, so its support lies in $\operatorname{supp}(\partial\Theta)$ Chapter9/GapP2_DBoundaryChar · IsBoundaryCharacterized.supp_subset. *Step 2.* Hence $F = \partial\Theta$ iff its support contains $\operatorname{supp}(\partial\Theta)$ Chapter9/GapP2_DBoundaryChar · IsBoundaryCharacterized.eq_boundaryTheta_iff, and $\partial\Theta$ itself has (1) and (2). Row V9.27 is narrower for this reason. Before the definition the text writes "In view of Proposition 9.2.13" (2017: "Proposition 9.9", the number of Lemma 9.2.13 there), presumably Proposition 9.2.15, and in it "branched ep⁺-groupoid".

**Remark 9.2.17 (2017: 9.6)** (V9.28). $\partial\Theta \le \Theta$ Chapter9/Boundary · EpGroupoid.boundaryTheta_le, and $\partial\Theta$ is not $\Theta$ restricted to $\partial X$: for the indicator of the corner $0$ of $[0,\infty)$, $\Theta(0) = 1$ and $\partial\Theta(0) = 0$ Chapter9/Boundary · boundaryTheta_halfLineCorner, while for $\Theta = 1$ on $[0,\infty)$, whose sheet is all of $[0,\infty)$ Chapter9/Boundary · MPolyfold.HasCornerStructure.isSmoothSubmanifold_univ Chapter9/Boundary · EpGroupoid.isBranchedEpSubgroupoid_one, $\partial\Theta(0) = 1$ Chapter9/Boundary · EpGroupoid.boundaryTheta_one. For the indicator of an interior point $\partial\Theta = 0$ Chapter9/Boundary · EpGroupoid.boundaryTheta_indicator_of_mem. The witnesses satisfy the independence Chapter9/Boundary · EpGroupoid.hasIndependentTangentSums_indicator_of_mem. The remark's "In Section 9.5, Definition 9.3.6" refers to §9.3.

## IX.14 Orientations: $\widehat{\mathrm{Gr}}(X)$, Definition 9.3.2 (2017: 9.8) and Remark 9.3.3 (2017: 9.8)

**Oriented tangent sets** (V9.29). An orientation of a finite-dimensional vector space is a class of bases; for a tangent set it is a set of frames, lists of vectors of the set, whose chart images Chapter9/OrientationTangent · MPolyfold.chartVec Chapter9/OrientationTangent · MPolyfold.chartVec_injOn are the positive bases of an orientation of the chart image Chapter9/OrientationTangent · MPolyfold.OrTangent Chapter9/OrientationTangent · MPolyfold.chartFrames Chapter9/OrientationTangent · MPolyfold.IsChartOrientation Chapter9/OrientationTangent · MPolyfold.IsOrientedAt. A chart change acts linearly Chapter9/OrientationTangent · MPolyfold.exists_clm_chartVec, so this does not depend on the chart Chapter9/OrientationTangent · MPolyfold.IsOrientedAt.isChartOrientation Chapter9/OrientationTangent · MPolyfold.isOrientedAt_iff_forall; every tangent set with linear chart image has exactly two orientations Chapter9/OrientationTangent · MPolyfold.exists_two_orientations, and sc-diffeomorphisms and morphisms push them forward Chapter9/OrientationTangent · MPolyfold.OrTangent.push Chapter9/OrientationTangent · MPolyfold.IsOrientedAt.push_tangentMapOn Chapter9/OrientationTangent · EpGroupoid.isOrientedAt_push_tangentMor. $\widehat{\mathrm{Gr}}(x)$ is generated by oriented tangent sets Chapter9/OrientationTangent · EpGroupoid.GrHat Chapter9/OrientationTangent · EpGroupoid.grHatPush Chapter9/OrientationTangent · EpGroupoid.GrHatObj Chapter9/OrientationTangent · EpGroupoid.grHatProj, and forgetting the orientation is a functor over $X^\infty$ Chapter9/OrientationTangent · EpGroupoid.grForget Chapter9/OrientationTangent · EpGroupoid.grForgetFunctor Chapter9/OrientationTangent · EpGroupoid.grForgetFunctor_comp_grProj, adding the two orientations of a set to the set Chapter9/OrientationTangent · EpGroupoid.grForget_single_add_single. Remark 9.3.1 (2017: 9.7), on an earlier notion of orientation, has no formal content.

**Definition 9.3.2 (2017: 9.8)** (V9.30) Chapter9/OrientationBasic · EpGroupoid.IsOrientation. An orientation of a submanifold is a continuous choice of orientations of its tangent sets Chapter9/OrientationBasic · MPolyfold.SubmanifoldOrientation, continuity being local constancy of the chart frames; an oriented local branching structure carries one per sheet Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure, with oriented sum $\sum_{y\in M_i} \sigma_i T_y(M_i, o_i)$ Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure.orientedSum forgetting to the tangent sum Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure.grForget_orientedSum. An orientation of $\Theta$ assigns to each smooth object an element of $\widehat{\mathrm{Gr}}$, naturally

in morphisms, forgetting to $T_\Theta$, and locally an oriented sum; the "lifts $T_\Theta$" clause uses $T_\Theta$, hence the independence of Theorem 9.2.1 Chapter9/OrientationBasic · EpGroupoid.IsOrientation.mk', which on $\mathbb{R}^m$ is free Chapter9/OrientationRn · EpGroupoid.isOrientation_sum_trivial_rn. It defines a section functor Chapter9/OrientationBasic · EpGroupoid.orientedSection Chapter9/OrientationBasic · EpGroupoid.orientedSection_comp_grHatProj Chapter9/OrientationBasic · EpGroupoid.orientedSection_comp_grForgetFunctor_obj. Witnesses: $\Theta = 0$ Chapter9/OrientationBasic · EpGroupoid.isOrientation_zero, and finite weighted sums of oriented closed submanifolds of $\mathbb{R}^n$ Chapter9/OrientationBasic · EpGroupoid.isOrientation_sum_rn.

**Remark 9.3.3 (2017: 9.8)** (V9.31) Chapter9/OrientationBasic · EpGroupoid.remark_9_3_3. For the indicator of a point $p$, $T_q = q \cdot (\{p\}, +) + (1-q) \cdot (\{p\}, -)$, $q \in (0,1) \cap \mathbb{Q}$, are pairwise distinct orientations: a 0-dimensional tangent set has two orientations, and any split of the weight 1 between them is an orientation.

## IX.15 Lemma 9.3.4 (2017: 9.10) and Example 9.3.5 (2017: 9.1)

**Lemma 9.3.4 (2017: 9.10)** (V9.32). *Two orientations of $\Theta$ that agree at good points are the same.* **Without equidimensionality, as we read it, the lemma does not hold** Chapter9/OrientationRn · lemma_9_3_4_mixed_dim_rn Chapter9/OrientationUniqueness · lemma_9_3_4_mixed_dim: on $\mathbb{R}^2$ let $\Theta = 1_{\mathbb{R}\times 0} + 1_{(0,0)}$ Chapter9/OrientationUniqueness · mixedTheta Chapter9/OrientationUniqueness · mixedTheta_isBranched. The origin is bad Chapter9/OrientationUniqueness · zero_not_mem_goodSet_mixed, the good points are the other points of the line, and two orientations that orient the line the same way and the point $\{0\}$ oppositely agree at all good points and differ at the origin. The lemma is formalized for $n$-dimensional $\Theta$ Chapter9/OrientationUniqueness · EpGroupoid.IsOrientationOfDim.

**The text's proof** takes, at a bad $x$, the sequences of good points of Proposition 9.2.8 for the first structure and compares the orientations of the two structures along them. As we read it, this needs that the same sequences serve the second structure, which the text does not show, and that frames of different sheets can be compared along a sequence. **The formalized proof** Chapter9/OrientationUniqueness · EpGroupoid.lemma_9_3_4. *Step 1.* Its hypothesis `HasOrientedGoodPointsAt` Chapter9/OrientationUniqueness · MPolyfold.HasOrientedGoodPointsAt asks, for each class of sheets of both structures with one tangent set at $x$, good points near $x$ on every sheet of the class, on which the orientations of the sheets can be compared; this is Proposition 9.2.8 applied to both families together. Under it, the oriented sums at $x$ agree class by class, which is the lemma. *Step 2,* from the ball graph representation (§IX.9) Chapter9/OrientationRn · MPolyfold.hasOrientedGoodPointsAt_of_hasGraphBallAt: over a ball of foot points in the class tangent space each sheet of the class is a graph, the good points are dense in it (Lemma 9.2.7 (1)), and the relative orientations of the sheets are locally constant. *Step 3,* local constancy: the relative orientation of two frame fields is the sign of a determinant Chapter9/OrientationRn · frameDet Chapter9/OrientationRn · orientation_eq_iff_frameDet Chapter9/OrientationRn · MPolyfold.frames_eq_iff_frameDet, continuous if the chart frames are continuous on level 1 Chapter9/OrientationRn · MPolyfold.continuousOn_chartVec_of_continuousOn Chapter9/OrientationRn · MPolyfold.SubmanifoldOrientation.exists_isOpen_tangent_eq_iff. This continuity was first the hypothesis `HasContinuousLevelOneOn` Chapter9/OrientationRn · MPolyfold.HasContinuousLevelOneOn, true on $\mathbb{R}^m$ Chapter9/OrientationRn · hasContinuousLevelOneOn_rn, and Chapter9/OrientationRn · EpGroupoid.lemma_9_3_4_of_hasGraphBallAt holds under the two hypotheses, on every ep-groupoid over $\mathbb{R}^m$ without them Chapter9/OrientationRn · EpGroupoid.lemma_9_3_4_rn Chapter9/OrientationRn · hasOrientedGoodPointsAt_rn; witness in dimension 0: Chapter9/OrientationUniqueness · hasOrientedGoodPointsAt_rn_zero. *Step 4, no hypothesis* (gap G.100, closed). The level-1 continuity holds along every smooth finite-dimensional submanifold of every M-polyfold Chapter9/GeneralLevelZero · hasContinuousLevelOneOn: on the subman-

ifold the level-1 lift of the chart is $1_0^1 \circ t$ composed with the chart, for the chart sc⁺-retraction $t$ Chapter9/GeneralLevelZero · levelLift_eq_plusLift, continuous since $t$ is sc⁺. The ball graph representation holds at interior points, which gave the lemma for $\Theta$ supported where $d_X = 0$ Chapter9/GeneralCorollaries · lemma_9_3_4_of_hasGraphBallAt' Chapter9/GeneralCorollaries · lemma_9_3_4_of_degIndex_eq_zero. At boundary points a ball of foot points is not available, but it is not needed: the good direction $h_0$ off the other tangent spaces is chosen near an admissible direction $h_1$, so that every $h$ near $h_0$ is admissible with one radius and one factor Chapter9/GeneralBoundaryPoints · exists_small_cone, and the tangent sheets are graphs over a small ball $\{u \in T \mid \|u - th_0\| < \tau t\}$ for $t$ below a uniform bound Chapter9/GeneralBoundaryPoints · exists_graph_unif_chart. So the good-point input holds at every point Chapter9/GeneralBoundaryPoints · hasOrientedGoodPointsAt, and Lemma 9.3.4 holds for every $n$-dimensional $\Theta$ of every ep-groupoid Chapter9/GeneralBoundaryPoints · lemma_9_3_4_general, every orientation of an $n$-dimensional $\Theta$ being $n$-dimensional Chapter9/GapP2_DOrientationDim · IsOrientation.isOrientationOfDim Chapter9/GapP2_DOrientationDim · lemma_9_3_4_of_isNDimensional. Row V9.32 stays narrower only because of the mixed-dimensional counterexample above.

**Example 9.3.5 (2017: 9.1)** (V9.33) Chapter9/OrientationExample · example_9_3_5 Chapter9/OrientationRn · example_9_3_5_rn. Take $\beta^{++}(s) = e(s) + e(-s)$ with $e =$ `expNegInvGlue`, $\beta^{--} = -\beta^{++}$, $\beta^{+-}$ equal to $\beta^{++}$ for $s \le 0$ and to $\beta^{--}$ for $s > 0$, and $\beta^{-+}$ the other way Chapter9/OrientationExample · betaPP Chapter9/OrientationExample · betaMM Chapter9/OrientationExample · betaPM Chapter9/OrientationExample · betaMP. The two pairs of graphs cover each point equally often Chapter9/OrientationExample · example_9_3_5_count, which defines $\Theta$ Chapter9/OrientationExample · example935Theta Chapter9/OrientationExample · example935Theta_isBranched. Orienting the graphs by $\pm(1, \beta')$ Chapter9/OrientationExample · graphOrientation Chapter9/OrientationExample · chartVec_image_graph2 gives $\hat{T}^1$ from $G^{++}, G^{--}$ and $\hat{T}^2$ from $G^{+-}, G^{-+}$ Chapter9/OrientationExample · exampleT1 Chapter9/OrientationExample · exampleT2. Verified as stated: they coincide at the origin Chapter9/OrientationExample · example_9_3_5_eq_origin, differ at $(s, \beta^{++}(s))$ for $s > 0$ Chapter9/OrientationExample · example_9_3_5_ne_of_pos, and no signs for the second structure reproduce $\hat{T}^1$ near the origin Chapter9/OrientationExample · example_9_3_5_cannot_coincide. **Two claims do not hold as we read them.** "Otherwise different": they also coincide wherever $x < 0$, where $G^{+-} = G^{++}$ and $G^{-+} = G^{--}$ with the same orientations Chapter9/OrientationExample · example_9_3_5_eq_of_neg. "Four possibilities to orient": the four sign choices for the first structure give four different orientations Chapter9/OrientationExample · orientedLift1_injective, and $\hat{T}^2$ is none of them Chapter9/OrientationExample · exampleT2_ne_orientedLift1. Counting both structures, the sign choices give exactly six orientations, the two structures sharing the two constant choices Chapter9/GapP2_DExample935 · orientedLift1_eq_orientedLift2_iff Chapter9/GapP2_DExample935 · orientedLift2_injective Chapter9/GapP2_DExample935 · ncard_signOrientations Chapter9/GapP2_DExample935 · example_9_3_5_six, and with fractional weights there are infinitely many, as in Remark 9.3.3. So "four possibilities" holds for each structure and, for $\Theta$, does not hold as we read it; the full classification is §IX.29 (gap G.101, closed). The good points distinguish $\hat{T}^1$ from $\hat{T}^2$, in line with Lemma 9.3.4 Chapter9/OrientationRn · exists_goodSet_exampleT1_ne_exampleT2.

## IX.16 Definition 9.3.6 (2017: 9.10), Lemma 9.3.7, and the boundary orientation ((9.11), Definition 9.3.8, Proposition 9.3.9, Definition 9.3.10)

**Definition 9.3.6 (2017: 9.10)** (V9.34) Chapter9/BoundaryPosition · EpGroupoid.TameBranchingStructure Chapter9/BoundaryPosition · EpGroupoid.IsTameBranched. A tame structure has a natural representation on its domain, the properness property, and sheets with tame induced structures in the sense of Part I's Definition 2.5.18. Witnesses: $\Theta = 0$ Chapter9/BoundaryPosition · EpGroupoid.isTameBranched_

zero and a point of $\mathbb{R}^n$ Chapter9/BoundaryPosition · EpGroupoid.isTameBranched_indicator Chapter9/BoundaryPosition · MPolyfold.hasTameInducedStructure_of_degIndex_eq_zero. "In particular every $M_i$ has an equivalent structure of a manifold with boundary with corners" (gap G.102, closed): for a tame structure with $n$-dimensional sheets, each $M_i$ with the subspace topology is a Mathlib manifold with corners modelled on $[0,\infty)^n$ Chapter9/MathlibManifoldBridge · TameBranchingStructure.hasCornerStructureOfDim Chapter9/MathlibManifoldBridge · TameBranchingStructure.isManifold. *Step 1.* A tame induced structure gives corner coordinates near each point, charts onto relatively open subsets of partial quadrants of $\mathbb{R}^n$ with smooth transitions Chapter9/MathlibManifoldBridge · CornerCoords Chapter9/MathlibManifoldBridge · IsSubmanifoldOfDim.hasCornerStructureOfDim_of_hasTameInducedStructure. *Step 2.* A Mathlib manifold has one model, so each partial quadrant is moved into the full quadrant by a linear isomorphism Chapter9/MathlibManifoldBridge · exists_cle_partialQuadrant, which gives a charted space Chapter9/MathlibManifoldBridge · cornerChartedSpace with smooth transitions Chapter9/MathlibManifoldBridge · isManifold_cornerChartedSpace. The same bridge serves Part I's solution sets.

**Lemma 9.3.7** (V9.35) Chapter9/BoundaryPosition · MPolyfold.lemma_9_3_7, new in the 2021 edition. *For tame $Y$ and submanifolds $M_1$, $M_2$ of one dimension with tame induced structures, $T_yM_1 = T_yM_2$ implies $d_{M_1}(y) = d_{M_2}(y)$.* **The text's proof** describes $T_yM$ in a tame chart of $M$ by the cone $C_yM$ and shows that $T_0O \cap C$ is the reduced tangent space. **The formalized proof.** For a submanifold with tame induced structure, $d_M(y)$ is the codimension of the reduced part of the chart image of $T_yM$, which depends only on that linear space and on the partial quadrant of the chart of $Y$ Chapter9/BoundaryPosition · MPolyfold.subDegIndex_eq_quotDim_of_hasTameInducedStructure (Remark 2.5.9 in every chart). So equal tangent spaces give equal indices Chapter9/BoundaryPosition · MPolyfold.subDegIndex_eq_of_tangentSpace_eq Chapter9/OrientationBoundary · MPolyfold.subDegIndex_eq_of_submanifoldTangent_eq, with neither "$Y$ tame" nor "of the same dimension". The text's "$T_0O \cap C = T_0^R O$" (a cone against a subspace) is read through the linearity of the chart image of $T_0M$.

**Definition 9.3.8 (2017: 9.9)** (V9.36) Chapter9/OrientationBoundary · MPolyfold.boundaryFrames. *For an oriented $(M, o)$, $y \in \partial M$ with $d_M(y) = 1$: a basis $(a_1, \dots, a_{n-1})$ of $T_y\partial M$ is positive iff $(e, a_1, \dots, a_{n-1})$ is positive, $e$ outward.* Linear algebra first: the convention orients a hyperplane $K$ Chapter9/OrientationBoundary · exists_boundary_orientation, by the determinant criterion Chapter9/OrientationBoundary · exists_basis_orientation_iff_det_pos Chapter9/OrientationBoundary · det_cons_smul_add. In a tangent set: for $K \subseteq T_yM$ with $(m-1)$-dimensional chart image and $e \notin K$, the frames orient $K$ Chapter9/OrientationBoundary · MPolyfold.isOrientedAt_boundaryFrames, and $e \mapsto ce + k$, $c > 0$, $k \in K$, does not change them Chapter9/OrientationBoundary · MPolyfold.boundaryFrames_eq; both orientations occur, on a graph Chapter9/OrientationBoundary · exists_boundaryFrames_graph. $T_y\partial M$ and $e$ are data in this definition. That $T_y\partial M = T_y^R M$ and the outward half-space are determined by $M$ through the cones $C_yM$, and that the resulting orientation is continuous along the stratum $\{d_M = 1\}$, is proved in §IX.30 (gap G.103, closed); row V9.36 is now explicit.

**(9.11) and Proposition 9.3.9** (V9.37), new in the 2021 edition. Split the oriented sum at $y$ by $\ell = d_{M_i}(y)$ Chapter9/OrientationBoundary · EpGroupoid.degreePart Chapter9/OrientationBoundary · EpGroupoid.degreePart_orientedSum; by Lemma 9.3.7 tangent sets of different degree never coincide Chapter9/OrientationBoundary · MPolyfold.HasSubDegree Chapter9/OrientationBoundary · MPolyfold.SubmanifoldOrientation.hasSubDegree_tangent_iff, so two oriented structures with the same oriented sum at $y$ have the same degree-$\ell$ parts for every $\ell$ Chapter9/OrientationBoundary · EpGroupoid.eq_9_11. Applying a boundary operation $T_y(M, o) \mapsto T_y(\partial M, o_{\partial M})$, a function `bd` of the oriented tangent set, to the degree-1 part gives (9.12), which therefore does not depend on the structure Chapter9/OrientationBoundary · EpGroupoid.proposition_9_3_9. **Definition 9.3.10 (2017: 9.11)** (V9.38) is (9.12) with 0 off the support Chapter9/OrientationBoundary · EpGroupoid.boundaryLift; the condition $y \in \partial M_i$ is implied by $d_{M_i}(y) = 1$.

The operation `bd` is now built from the sheet, and (9.12) and Proposition 9.3.9 hold with it Chapter9/BoundaryOperation · EpGroupoid.proposition_9_3_9_boundaryOp Chapter9/BoundaryOperation · EpGroupoid.boundaryOrientation; the continuity of $o_{\partial M}$ along $\partial M$ and its extension to the faces (Figure 9.1) are §IX.30. Remark 9.3.11, on the non-tame case, has no formal content.

## IX.17 §9.4: essential points and sets of measure zero (Definitions 9.4.1, 9.4.5, 9.4.7, Lemmas 9.4.2–9.4.8 (2017: 9.12–9.14, 9.11–9.15))

**For Mathlib manifolds.** Definitions 9.4.1 and 9.4.5 are about a smooth finite-dimensional manifold $M$ with boundary and corners, formalized for Mathlib manifolds with corners. *Definition 9.4.1 (1)* (V9.39): $T_z^K M$ is the span of the unit limits $\varphi(z_k)/|\varphi(z_k)|$, $z_k \to z$ in $K \setminus \{z\}$, in a chart $\varphi$ centred at $z$. In a chart it is the span of Mathlib's tangent cone of the chart image of $K$ Chapter9/BranchingGeometryManifold · chartImage Chapter9/BranchingGeometryManifold · span_tangentConeAt_chartImage_eq_map Chapter9/BranchingGeometryMeasure · span_unitDirections Chapter9/BranchingGeometryMeasure · unitDirections Chapter9/BranchingGeometryManifold · relTangent, and a chart change maps it by the derivative Chapter9/BranchingGeometryManifold · relTangent_eq_map, which is the text's "independent of the choice of $\varphi$". *Definition 9.4.1 (2)*: $z$ is essential if it is a limit of points of $K$ where $T^K M$ is everything Chapter9/BranchingGeometryManifold · relTangent_eq_top_iff Chapter9/BranchingGeometryManifold · fullSet Chapter9/BranchingGeometryManifold · essentialSet Chapter9/BranchingGeometryManifold · essentialSet_eq. For two crossing lines in $\mathbb{R}^2$ the essential set is $\{0\}$ Chapter9/BranchingGeometryManifold · crossLines Chapter9/BranchingGeometryManifold · essentialSet_crossLines Chapter9/BranchingGeometryManifold · nonEssential_crossLines. *Lemma 9.4.2 (2017: 9.11)* (V9.40), "obvious from the definition": $K^e$ is a closure taken in $K$ Chapter9/BranchingGeometryManifold · lemma_9_4_2 Chapter9/BranchingGeometryManifold · lemma_9_4_2_open. *Definition 9.4.5 (2017: 9.13)* (V9.43) Chapter9/BranchingGeometryManifold · IsMeasureZero: null in every chart; countable unions Chapter9/BranchingGeometryManifold · isMeasureZero_iUnion, a chart cover suffices Chapter9/BranchingGeometryManifold · isMeasureZero_of_forall_mem, and $\mathbb{R}^2$ is not null Chapter9/BranchingGeometryManifold · not_isMeasureZero_univ. *Lemma 9.4.6 (2017: 9.14)* (V9.44) Chapter9/BranchingGeometryManifold · lemma_9_4_6, "proved similarly as Lemma 2.8 in [57]": *Step 1.* At a point where $T_z^K M \neq T_z M$ there is a unit direction $v$ off the span, and points of $K$ near $z$ lie outside a cone around $v$ Chapter9/BranchingGeometryMeasure · eventually_abs_le_of_tangentConeAt. *Step 2.* A cone has positive density at its vertex Chapter9/BranchingGeometryMeasure · exists_pos_measure_cone_lt, so $z$ is not a Lebesgue density point of $K$; by Lebesgue's density theorem the set of such $z$ is null Chapter9/BranchingGeometryMeasure · addHaar_setOf_span_tangentConeAt_ne_top Chapter9/BranchingGeometryManifold · isMeasureZero_setOf_relTangent_ne_top. *Step 3.* A non-essential point has a neighbourhood in $K$ of such points, by Lemma 9.4.2. The statement holds for every $K$, not only for $K(i, i')$.

**For sheets in an M-polyfold.** The sheets $M_i$ are submanifolds of an M-polyfold, which in general carry no Mathlib manifold structure (a tame sheet does, §IX.16). So the essential points and null sets of the sheets are the data of a `SheetGeometry` Chapter9/BranchingGeometry · MPolyfold.SheetGeometry, with Lemma 9.4.4 (`ess_comm`), Lemma 9.4.6 (`isNull_diff_ess`) and the note of Definition 9.4.7 (`submanifoldTangent_eq`) as fields; it was first witnessed by pieces of graphs in $\mathbb{R}^2$, with the essential sets and null sets of the Mathlib part read in the chart $z \mapsto z_0$ Chapter9/BranchingGeometry · graphSheets Chapter9/BranchingGeometry · graphSheetGeometry. It is now built on the $n$-dimensional submanifolds of every M-polyfold with no hypothesis, Definitions 9.4.1 and 9.4.5 being read in the M-polyfold charts as in the text's proof of Lemma 9.4.4 Chapter9/GapP2_DSheetEssential · sheetGeometry (§IX.28; gap G.105, closed). So Lemma 9.4.8 and Propositions 9.4.9, 9.4.10 hold for every $n$-dimensional $\Theta$ with no

sheet-geometry hypothesis Chapter9/GapP2_DSheetGeometryCor · lemma_9_4_8_of_isNDimensional Chapter9/GapP2_DSheetGeometryCor · proposition_9_4_9_of_isNDimensional Chapter9/GapP2_DSheetGeometryCor · proposition_9_4_10_of_isNDimensional.

**Lemma 9.4.4 (2017: 9.13)** (V9.42). *Essential for $K(i,i') \subseteq M_i$ iff for $K(i,i') \subseteq M_{i'}$*, "left to the reader". **As we read it, it needs sheets of one dimension** Chapter9/BranchingGeometry · lemma_9_4_4_mixed_dim: $\mathbb{R} \times 0$ as a subset of $\mathbb{R}$ is essential at every point, as a subset of $\mathbb{R}^2$ nowhere, since its relative tangent space is a line. For an $n$-dimensional $\Theta$ the text's reduction to $T_0^D N$ in a chart applies, and is now proved for $n$-dimensional submanifolds of every M-polyfold Chapter9/SheetEssentialChart2021 · lemma_9_4_4_chart and for the sheets of every structure of an $n$-dimensional $\Theta$ Chapter9/GapP2_DSheetMeasure · lemma_9_4_4_of_isNDimensional (§IX.28).

**Definition 9.4.7 (2017: 9.14)** (V9.45). $i \sim_y i'$ through a chain of links $y \in K(i_{p-1}, i_p)^e$ Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.IsLink Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.YEquiv is an equivalence relation Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.equivalence_yEquiv Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.ySetoid; equivalent sheets have the same tangent set at $y$ Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.submanifoldTangent_eq_of_yEquiv; $\#P_y$ is Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.numClasses.

**Lemma 9.4.8 (2017: 9.15)** (V9.46) Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.lemma_9_4_8. $\{y \in M_U \mid \#P_y \geq 2\}$ *is null.* "Null in $M_U$" is null in every sheet through the point Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.IsNullIn. The text decomposes $M_U$ by $(I_y, P_y)$ and concludes that the bad part lies in $\bigcup_{j,j'} K(j,j')^{ne}$. As we read it, $K(j,j')^{ne}$ is null only in $M_j$ and $M_{j'}$; inside a sheet $M_i$ through $y$ the proof uses $K(i,j)^{ne}$ for $j$ not $y$-equivalent to $i$: if $y$ had two classes, the class of $i$ omits some $j \in I_y$, so $y \in K(i,j)$ is not essential. Hence $M_U \setminus M_U^*$ is null Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.MStar Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.isNullIn_diff_MStar.

## IX.18 Property (T), Propositions 9.4.9 and 9.4.10 (2017: 9.5 and 9.6)

**Property (T)** (V9.41) Chapter9/BranchingGeometry · EpGroupoid.OrientedBranchingStructure.PropertyT: the orientation relation of two sheets with the same tangent space is the same at every common point with the same tangent space; the first structure of Example 9.3.5 has it Chapter9/BranchingGeometry · propertyT_example935B1. Under (T) the sign map $K(i,i')^e \to \{\pm 1\}$ is constant Chapter9/BranchingGeometry · EpGroupoid.tangent_eq_iff_of_mem_ess. **Lemma 9.4.3** (V9.41) Chapter9/PropertyTShrinking2021 · lemma_9_4_3 Chapter9/PropertyTShrinking2021 · lemma_9_4_3_propertyT. *Every oriented local branching structure has Property (T) after restriction to a smaller open $U'(x)$* (gap G.104, closed). **The text's proof** argues along sequences of points converging to $x$. **The formalized proof** finds, for each of the finitely many pairs $(i,i')$, an open neighbourhood of $x$ on which the pair satisfies (T), and intersects Chapter9/PropertyTShrinking2021 · SubmanifoldOrientation.exists_isOpen_pairT Chapter9/PropertyTShrinking2021 · restrict Chapter9/PropertyTShrinking2021 · propertyT_restrict_iff. *Case 1, different dimensions:* two submanifolds with the same tangent set at a point have the same dimension there Chapter9/PropertyTShrinking2021 · HasDimAt.eq_of_submanifoldTangent_eq, so (T) asks nothing of such a pair. *Case 2, dimension 0:* a point where a submanifold is 0-dimensional is isolated in it, by the chord estimate with vanishing derivative Chapter9/PropertyTShrinking2021 · HasDimAt.exists_isOpen_eq_of_mem. *Case 3, equal tangent spaces at $x$:* where the tangent spaces agree, the relative orientation is the sign of a frame determinant, locally constant by the level-1 continuity (§IX.15) Chapter9/OrientationRn · MPolyfold.SubmanifoldOrientation.exists_isOpen_tangent_eq_iff.

*Case 4, different tangent spaces at $x$:* they stay different nearby, since a continuous frame field of one, written in the chart, has a nonzero minor against the other `Chapter9/PropertyTShrinking2021 · exists_chartTangent_field` `Chapter9/PropertyTShrinking2021 · exists_frameDet_ne_zero` `Chapter9/PropertyTShrinking2021 · exists_isOpen_submanifoldTangent_ne`, so after shrinking (T) is empty for the pair.

**Proposition 9.4.9 (2017: 9.5)** (V9.47) `Chapter9/BranchingGeometry · EpGroupoid.proposition_9_4_9`. *Off a null set $M_U = M_U^* = \bigsqcup_J M_U^J$, and on $M_U^J$: $\Theta = \sigma_J$, $T_\Theta = \sigma_J T^J$, $\hat{T}_\Theta = a_J(T^J, o_J^a) + b_J(T^J, o_J^b)$, $a_J + b_J = \sigma_J$, $o_J^a = -o_J^b$. Step 1.* The null set is Lemma 9.4.8; the pieces are the level sets of $y \mapsto I_y$ `Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.MStarPart`. *Step 2.* On $M_U^J$ all sheets of $J$ are $y$-equivalent, so they have one tangent set $T^J$ and $T_\Theta(y) = \sigma_J T^J$ `Chapter9/BranchingGeometry · EpGroupoid.LocalBranchingStructure.tangentValue_eq_sum`, which uses the independence of Theorem 9.2.1. *Step 3.* By (T) the orientations of the sheets of $J$ at $y$ fall into two opposite classes; $o_J^a = -o_J^b$ is "$o_J^a \neq o_J^b$ on the same tangent set", since it has exactly two orientations `Chapter9/BranchingGeometry · MPolyfold.OrTangent.eq_or_eq_or_eq_of_isOrientedAt` `Chapter9/BranchingGeometry · MPolyfold.exists_ne_isOrientedAt`; if one class is empty its orientation is the other one. The footnote's "measure zero" in $M_U$ refers forward to Definition 9.5.1 and Lemma 9.5.2 (p. 437), consistently as we read it.

**Proposition 9.4.10 (2017: 9.6)** (V9.48) `Chapter9/BranchingGeometry · EpGroupoid.proposition_9_4_10`. *Two oriented structures on $U(x)$ with (a) the same weight sums and (b) the same oriented sums agree off a null set, piece by piece, as in (1)–(5).* **The text** takes the union $B''$ of the two structures, which describes $2\Theta$, applies Proposition 9.4.9 to $B$, $B'$, $B''$ and intersects. **The formalized proof** takes $M_U^\dagger = M_U''^*$ for the union only, not intersected with $M_U^*$ and $M_U'^*$, which (1)–(5) do not need. The pieces $A_p$ are cut out by the finitely many values of $y \mapsto (I_y, I'_y, J_+(y), J'_+(y))$, with $J_\pm(y)$ the two orientation classes at $y$, so Property (T) is not used. The two structures of Example 9.3.5, all signs $+$, satisfy (a) and (b) `Chapter9/BranchingGeometry · example935B1` `Chapter9/BranchingGeometry · example935B2` `Chapter9/BranchingGeometry · example935_hypotheses_9_4_10` `Chapter9/BranchingGeometry · example935_MU_eq`; at the origin $G^{++}$ and $G^{--}$ are tangent but not 0-equivalent, so the set of Lemma 9.4.8 is not empty `Chapter9/BranchingGeometry · example935_two_le_numClasses`.

## IX.19 §9.5: sets of measure zero and the σ-algebras $L(S)$, $L(\partial S)$ (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16))

**The setting of §9.5.** $\Theta$ is oriented, tame, compact and $n$-dimensional, and $S = |\mathrm{supp}(\Theta)|$. Measure zero in a sheet $M_i$ is that of Definition 9.4.5, which the project first had for Mathlib manifolds only (§IX.17). So, as in §9.4, the null sets of the sheets are data, now instantiated on every M-polyfold by the $n$-dimensional Hausdorff measure of the chart images (§IX.28; gap G.106, closed): a system `SheetNullSets` on a class $\mathcal{S}$ of sheets `Chapter9/Measure · SheetNullSets`, with null sets closed under subsets, countable unions `Chapter9/Measure · SheetNullSets.isNull_iUnion_countable` and finite unions `Chapter9/Measure · SheetNullSets.isNull_union`, local, induced alike by two sheets of $\mathcal{S}$ on their overlap (`isNull_transfer`), and a nonempty sheet not null in itself. A §9.4 sheet geometry with these properties is a `SheetMeasureGeometry` `Chapter9/Measure · MPolyfold.SheetMeasureGeometry` `Chapter9/Measure · MPolyfold.SheetMeasureGeometry.toSheetNullSets`, and then null preimages are §9.4's "null in $M_U$" `Chapter9/Measure · EpGroupoid.LocalBranchingStructure.hasNullPreimages_iff_isNullIn`. *The covering criterion* `Chapter9/Measure · SheetNullSets.isNull_of_cover`: a subset $C$ of a sheet $N \in \mathcal{S}$ is null in $N$ if near each of its points it is covered by finitely many sheets of $\mathcal{S}$ in which it is null; by locality and the transfer property.

**Definition 9.5.1 (2017: 9.15)** (V9.49) Chapter9/Measure · EpGroupoid.IsMeasureZeroSupp. $A \subseteq |X|$ has measure zero if every $x \in \operatorname{supp}(\Theta)$ has a tame local branching structure (Definition 9.3.6, the text's "suitable $U(x)$") with sheets in $\mathcal{S}$ whose preimages $M_i \cap \pi^{-1}(A)$ are null in $M_i$ Chapter9/Measure · EpGroupoid.LocalBranchingStructure.HasNullPreimages. The variants ask this of some local branching structure Chapter9/Measure · EpGroupoid.IsMeasureZeroSuppAny, or of every structure with sheets in $\mathcal{S}$ Chapter9/Measure · EpGroupoid.IsMeasureZeroSuppAll; the standing assumption is that tame structures with sheets in $\mathcal{S}$ exist Chapter9/Measure · EpGroupoid.HasTameSheetsIn. Compactness and orientation are not used.

**Lemma 9.5.2 (2017: 9.16)** (V9.50) Chapter9/Measure · EpGroupoid.lemma_9_5_2. *The notion does not depend on the choice of the local branching structures.* **The text** cites the literature, without proof. **The formalized proof.** *Step 1, the core* Chapter9/Measure · EpGroupoid.isMeasureZeroSuppAll_of_any. Suppose some structure at every point has null preimages of $A$, and let $B'$ be any structure with sheets in $\mathcal{S}$ and $M'_j$ one of its sheets. Each $z \in M'_j \cap \pi^{-1}(A)$ lies in the support, so a structure $B$ at $z$ with null preimages exists. Near $z$, $M'_j \subseteq \operatorname{supp}(\Theta) \cap U_B = \bigcup_i M_i$, so $M'_j \cap \pi^{-1}(A)$ is covered near $z$ by the finitely many $M_i$, in which it is null. The covering criterion gives that it is null in $M'_j$. *Step 2.* A tame structure is a structure Chapter9/Measure · EpGroupoid.isMeasureZeroSuppAny_of_isMeasureZeroSupp, and under the standing assumption every point has a tame one with sheets in $\mathcal{S}$; the three notions agree. *Step 3.* The sets of measure zero form a σ-ideal Chapter9/Measure · EpGroupoid.isSigmaIdeal_isMeasureZeroSuppAll Chapter9/Measure · EpGroupoid.isSigmaIdeal_isMeasureZeroSupp, and $S \neq \emptyset$ is not null, since a sheet through a point of the support is not null in itself Chapter9/Measure · EpGroupoid.not_isMeasureZeroSuppAny_orbitSupp Chapter9/Measure · EpGroupoid.not_isMeasureZeroSupp_orbitSupp.

**The transfer property, as we read it, needs one dimension** Chapter9/Measure · isNull_transfer_mixed_dim. Step 1 uses that a set null in the sheets $M_i$ is null in $M'_j$ on their overlap. For sheets of different dimensions this does not hold: the axis $\mathbb{R} \times 0$ has Lebesgue measure zero in the sheet $\mathbb{R}^2$ and is not null in itself, a 1-dimensional sheet; so no single system contains the planar null sets and those of graphs. The lemma is therefore proved for structures whose sheets lie in one class, as do the $n$-dimensional structures of an $n$-dimensional $\Theta$. That *every* structure of an $n$-dimensional $\Theta$ has $n$-dimensional sheets is now proved on every ep-groupoid, from the independence of Theorem 9.2.1 Chapter9/GapP2_DSheetMeasure · isOfDim_of_isNDimensional, so for $n$-dimensional $\Theta$ Lemma 9.5.2 holds as printed (§IX.28; row V9.50 explicit). With the Hausdorff null sets the transfer across dimensions does not hold as we read it either: the axis of $\mathbb{R}^2$ has 2-dimensional and not 1-dimensional measure zero Chapter9/GapP2_DSheetMeasure · isHNull_mixed_dim.

**Definition 9.5.3 (2017: 9.16) and the remark after it** (V9.51). *Step 1, a general fact.* For a family $\mathcal{N}$ of subsets, the σ-algebra generated by the Borel sets and $\mathcal{N}$ Chapter9/Measure · borelNullSigma Chapter9/Measure · borel_le_borelNullSigma Chapter9/Measure · measurableSet_borelNullSigma_of_mem is, if $\mathcal{N}$ is a σ-ideal Chapter9/Measure · IsSigmaIdeal, the family $\{A \mid A \,\Delta\, B \in \mathcal{N} \text{ for a Borel } B\}$ Chapter9/Measure · borelNullCompletion Chapter9/Measure · measurableSet_borelNullSigma_iff: that family is a σ-algebra, by $\bigcup_k A_k \,\Delta\, \bigcup_k B_k \subseteq \bigcup_k (A_k \,\Delta\, B_k)$ Chapter9/Measure · symmDiff_iUnion_subset, contains the Borel sets and $\mathcal{N}$, and lies in every σ-algebra containing both. *Step 2.* $L(S)$ is this σ-algebra on the subspace $S$ for the sets of measure zero Chapter9/Measure · EpGroupoid.measureZeroSets Chapter9/Measure · EpGroupoid.lebesgueSigma, a σ-ideal under the standing assumption Chapter9/Measure · EpGroupoid.isSigmaIdeal_measureZeroSets. *Step 3.* The text's remark that $L(S)$ is obtained by adding the sets of measure zero is read as the completion: $K \in L(S)$ iff $K \,\Delta\, B$ has measure zero for a Borel $B$ Chapter9/Measure · EpGroupoid.remark_9_5_3; sets of measure zero belong to $L(S)$ Chapter9/Measure · EpGroupoid.measurableSet_lebesgueSigma_of_isMeasureZeroSupp. The form $K = B \cup N$ would need Borel null hulls of null sets, which an abstract system does not provide.

**Definition 9.5.14 (2017: 9.17) and the paragraph before it** (V9.52). $\partial S = |\operatorname{supp}(\partial\Theta)| \subseteq S$ Chapter9/

Measure · EpGroupoid.orbitSupp_boundaryTheta_subset. $(n-1)$-dimensional measure zero in the boundaries $\partial M_i$ Chapter9/Boundary · MPolyfold.submanifoldBoundary is a second system on a class of boundaries, and $K$ has measure zero in $\partial S$ if at every point of $\mathrm{supp}(\partial\Theta)$ a tame structure has boundary preimages null in the $\partial M_i$ Chapter9/Measure · EpGroupoid.IsMeasureZeroBoundary Chapter9/Measure · EpGroupoid.IsMeasureZeroBoundaryAll. *"A well-defined notion"* Chapter9/Measure · EpGroupoid.isMeasureZeroBoundaryAll_of: the proof of Lemma 9.5.2 applies to the boundaries. By the local formula for $\partial\Theta$ (Proposition 9.2.15), which holds for every structure given the independence of Theorem 9.2.1, a point of $\partial M'_j$ lies in $\mathrm{supp}(\partial\Theta)$, and a point of $\mathrm{supp}(\partial\Theta)$ lies in some $\partial M_i$ of any structure at it, a positive sum having a nonzero term Chapter9/Measure · EpGroupoid.exists_mem_of_sum_pos; so the boundary preimage in $\partial M'_j$ is covered near each point by the $\partial M_i$ of a structure with null boundary preimages. $L(\partial S)$ is the σ-algebra generated by the Borel sets of $\partial S$ and these sets Chapter9/Measure · EpGroupoid.boundaryMeasureZeroSets Chapter9/Measure · EpGroupoid.lebesgueSigmaBoundary. The $(n-1)$-dimensional null sets themselves, a witness with $\partial S \neq \emptyset$ and the σ-algebra $L(\partial S)$ built with them are §IX.34 (gap G.108, closed).

**Witnesses** (V9.53). *Tools.* Measure zero of Definition 9.4.5 is local Chapter9/Measure · isMeasureZero_image_of_local, holds for subsingletons Chapter9/Measure · isMeasureZero_of_subsingleton, and on $\mathbb{R}^2$ is Lebesgue measure zero Chapter9/Measure · isMeasureZero_rn2_iff. On the trivial ep-groupoid of $\mathbb{R}^n$ every local branching structure restricts to a tame one, on the domain of a natural representation Chapter9/Measure · exists_tameBranchingStructure_trivial_rn Chapter9/Measure · hasTameSheetsIn_trivial_rn, and $\pi$ is injective Chapter9/Measure · injective_π_trivial_rn. *Example 9.3.5.* The graph sheets of §9.4 carry a `SheetMeasureGeometry` Chapter9/Measure · graphSheetMeasureGeometry Chapter9/Measure · graphNull; for the $\Theta$ of the two graphs $G^{++}$, $G^{--}$, which touch at $0$ Chapter9/Measure · theta935 Chapter9/Measure · isBranchedEpSubgroupoid_theta935 Chapter9/Measure · hasTameSheetsIn_theta935 Chapter9/Measure · zero_mem_supp_theta935, the point $|0|$ has measure zero Chapter9/Measure · isMeasureZeroSupp_theta935_zero, $S$ does not Chapter9/Measure · not_isMeasureZeroSupp_theta935, and both are in $L(S)$ Chapter9/Measure · lebesgueSigma_theta935. *The plane.* For $\Theta = 1$ on $\mathbb{R}^2$, with the open subsets of $\mathbb{R}^2$ as sheets and Lebesgue null sets Chapter9/Measure · planeSheets Chapter9/Measure · planeNull Chapter9/Measure · planeSheet Chapter9/Measure · isSmoothSubmanifold_univ_rn2 Chapter9/Measure · thetaPlane Chapter9/Measure · isBranchedEpSubgroupoid_thetaPlane Chapter9/Measure · mem_supp_thetaPlane Chapter9/Measure · hasTameSheetsIn_thetaPlane, some member of $L(S)$ is not Borel Chapter9/Measure · exists_mem_lebesgueSigma_not_borel_thetaPlane: every subset of the null axis is in $L(S)$, there are $2^{\mathfrak{c}}$ of them, and only $\mathfrak{c}$ Borel sets.

## IX.20 §9.5: the isotropy factors (Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 (2017: Lemmas 9.18, 9.19))

The measures $\mu_\omega$ of Theorems 9.5.4 and 9.5.5 are local weighted integrals of a form over the sheets, divided by $\#G_x$ or $\#G_x^{\mathrm{eff}}$. This subsection gives what the proofs of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 rest on, the orders of the isotropy groups and the decompositions of the preimages; the measure identities themselves are stated and proved in §IX.31 (Lemmas 9.5.7, 9.5.9–9.5.11, Proposition 9.5.13 (1)) and §IX.33 (Proposition 9.5.13 (2)), and rows V9.54–V9.56 are explicit.

**Lemma 9.5.9 (2017: 9.18)** (V9.54). **The text's proof** identifies $H = \{g \mid g * y = y\}$ with $G_y$, writes $G = \bigcup_{r\in R} rH$, chooses $V(y)$ invariant under $H$ and disjoint from its other translates, notes that $\tilde{M}_j = V \cap M_j$ is a branching structure on $V$, decomposes the preimage of $K$ as $\bigcup_r \varphi_r(\tilde{K})$, and uses $\#R = \#G/\#H$. **The formalized part.** *Step 1,* $\#R \cdot \#G_y = \#G_x$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_isotropy: the stabilizer $H$ of $y$ for the action of $G_x$ on $U(x)$ is isomorphic to $G_y$

by the natural representation (Chapter 7) Chapter9/MeasureEffective · EpGroupoid.NaturalRep.mem_stabilizer_iff_Φ, and $R$ is `Quotient.out` on $G_x/H$. *Step 2, the slice* Chapter9/MeasureEffective · EpGroupoid.NaturalRep.exists_slice: inside any neighbourhood of $y$ there is an open $V$ with $\varphi_g(V) = V$ for $g \in H$ and $\varphi_g(V) \cap V = \emptyset$ for $g \notin H$: separate $y$ from $\varphi_g(y) \neq y$ for the finitely many $g \notin H$, and make the neighbourhood $H$-invariant. *Step 3, the preimage.* $|\varphi_g(z)| = |z|$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.π_Φ, so $\varphi_g$ maps $M_U = \mathrm{supp}(\Theta) \cap U$ to itself, $\Theta$ being a functor Chapter9/MeasureEffective · EpGroupoid.NaturalRep.mapsTo_Φ_MU; for $K \subseteq |M_U \cap V|$, $M_U \cap \pi^{-1}(K) = \bigcup_g \varphi_g(M_U \cap V \cap \pi^{-1}(K))$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion, the same with $g$ running over the representatives $r$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion_quotient, and the union over $R$ is disjoint Chapter9/MeasureEffective · EpGroupoid.NaturalRep.pairwise_disjoint_image_Φ_out. The restriction $\tilde{M}_j$ is §IX.3's `restrictAt`: the project's structures pass through their base point, so at $y$ the sheets through $y$ are kept.

**Lemma 9.5.10** (V9.55), new in the 2021 edition. *Step 1.* $\#T_x \cdot \#G_x^{\mathrm{eff}} = \#G_x$ for the subgroup $T_x$ acting trivially near $x$ Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_mul_card_trivialSubgroup. *Step 2.* Under LUC for the given $U(x)$ (Definition 7.2.18, `NaturalRep.HasLUC`), $T_x$ is the kernel of the representation Chapter9/MeasureEffective · EpGroupoid.NaturalRep.kerΦ_eq_trivialSubgroup, hence lies in every stabilizer Chapter9/MeasureEffective · EpGroupoid.NaturalRep.kerΦ_le_stabilizer, and corresponds to $T_y$ under $H \cong G_y$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.stabilizerEquivIsotropy_mem_trivialSubgroup_iff, so $\#T_y = \#T_x$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_trivialSubgroup_eq. *Step 3, (9.23).* Dividing Step 1 of Lemma 9.5.9 by $\#T_x = \#T_y$ gives $\#R \cdot \#G_y^{\mathrm{eff}} = \#G_x^{\mathrm{eff}}$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_effIsotropy.

**Lemma 9.5.11 (2017: 9.19)** (V9.56). The orders $\#G_x$ and $\#G_x^{\mathrm{eff}}$ are invariant along morphisms, by conjugation with $\varphi$ (Lemma 7.2.6) Chapter9/MeasureEffective · EpGroupoid.card_isotropy_eq_of_mor Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_of_mor. A functor is constant on orbits Chapter9/MeasureEffective · EpGroupoid.IsFunctorial.eq_of_π_eq, so for structures on $U$, $U'$ and any map $\hat{\varphi}$ of $U$ onto $U'$ with $|\hat{\varphi}(z)| = |z|$, $|M_U| = |M_{U'}|$ Chapter9/MeasureEffective · EpGroupoid.image_π_MU_eq; the maps $\varphi_g$ of a natural representation are such maps Chapter9/MeasureEffective · EpGroupoid.NaturalRep.Φ_hyps. The "change of variables" in the integrals is the invariance of the sheet integration under the local sc-diffeomorphism $\hat{\varphi}$, and gives Lemma 9.5.11 as a measure identity in §IX.31.

**Proposition 9.5.13** (V9.57), new in the 2021 edition. *(1)* For effective $X$, $T_x = \{1\}$, so $\#G_x^{\mathrm{eff}} = \#G_x$ Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_of_isEffective and the normalizing factors of the two integrals agree. *(2)* The reduced ep-groupoid $X_R$ (Chapter 7) has the object M-polyfold of $X$, with Hausdorff and paracompact morphism space as instances (Hausdorff under LUC); functors Chapter9/MeasureEffective · EpGroupoid.isFunctorial_reduced_iff and local branching structures Chapter9/MeasureEffective · EpGroupoid.LocalBranchingStructure.toReduced of $X$ and $X_R$ coincide, and $\#G_x^{\mathrm{eff}} = \#G_x^R = \#(G_x^R)^{\mathrm{eff}}$ Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_reduced. The identity of the integrals, $\oint_\Theta \omega = \int_{\hat{\Theta}_R} \omega$, is §IX.33 (row V9.57 restated).

# IX.21 Appendix §9.6.1: Lemmas 9.6.1 and 9.6.2, Definition 9.6.3 and Lemma 9.6.4

**In place of "measure zero."** The text's proofs of Lemmas 9.6.1, 9.6.2 and 9.6.4 remove sets of smaller dimension or not containing a sheet near the point, "of measure zero", and move a point slightly. The formalization uses a topological covering lemma instead, with no Lebesgue measure and no Baire cate-

gory. *Step 1.* If $A \cap W \subseteq N$ for an open $W \ni a$, then $\dim_a A \le \dim_a N$, by the inclusion of the chart tangent spaces `Chapter9/Appendix · MPolyfold.HasDimAt.le_of_inter_subset`. *Step 2*, the covering lemma `Chapter9/Appendix · MPolyfold.exists_mem_forall_notMem`: a nonempty open piece of a $k$-dimensional submanifold is not covered by finitely many submanifolds of smaller dimension closed in an open set, by induction on the family: removing one closed set of smaller dimension leaves a nonempty open piece, by Step 1. *Step 3*, "moving $z$ slightly": a point of the piece off all the given sets `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsGeneric` `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.exists_isGeneric`.

**Lemma 9.6.1** (V9.58) `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_1`. *Two structures at $x$ have the same top dimension* `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.topDim`. If the first has top dimension $k$ and the second $\ell < k$, a generic point of a $k$-dimensional sheet of the first has $\Theta > 0$ by the first structure and $\Theta = 0$ by the second. So $\dim_\Theta(x)$ is well defined `Chapter9/Appendix · EpGroupoid.localTopDim` `Chapter9/Appendix · EpGroupoid.localTopDim_eq`, $-\infty$ off the support `Chapter9/Appendix · EpGroupoid.localTopDim_eq_bot`.

**Lemma 9.6.2** (V9.59) `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_2`. *The unions $M^k_U$ of the top-dimensional sheets agree on $U \cap U'$* `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.topUnion` `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.mem_topUnion_iff_of_mem`: a generic point of $M^k_U \setminus M'^k_{U'}$ would have $\Theta > 0$ by the first structure and, off the lower-dimensional sheets of the second, $\Theta = 0$.

**Definition 9.6.3** (V9.60) `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsTopGoodPoint` `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsTopBadPoint`, as in the text. At points top-good for both structures the top parts agree, with no hypothesis `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_isTopGoodPoint`: near such a point the top part is $\sigma \cdot 1_M$ for both, and $\sigma$ is read off $\Theta$ at a generic point of $M$.

**Lemma 9.6.4** (V9.61). *Top-goodness does not depend on the structure.* **The text's proof**: near a top-good $z$, $\Theta = \sigma \cdot \delta_M + \Theta_0 + \cdots + \Theta_{k-1}$; a point of $M$ near $z$ off the lower-dimensional sets has $\Theta = \sigma$, and "similarly as good points", $z$ is top-good for the second structure. **As we read it, this needs that the top weight of the second structure at $y$ is $\sigma$**, an argument the text does not give: the second structure could a priori have, through $y$, several top-dimensional sheets with tangent spaces that differ. **The formalized proof** makes this step the hypothesis `TopMultiplicity` `Chapter9/Appendix · MPolyfold.TopMultiplicity`: at a point where $\Theta$ equals $\sigma$ on generic points of a $k$-dimensional sheet, the top weight of any structure is $\sigma$. Under it the lemma holds `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_4` `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.isTopBadPoint_iff`, the top parts agree on $U \cap U'$ `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_topMultiplicity`, and Proposition 9.1.12 follows by the descending induction of p. 454: subtract the top parts `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.subOn` and repeat `Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_topMultiplicity` `Chapter9/Appendix · EpGroupoid.proposition_9_1_12_of_topMultiplicity`. Witnesses: $k = 0$ on every M-polyfold `Chapter9/Appendix · MPolyfold.topMultiplicity_zero`, and $k \ge n$ on $\mathbb{R}^n$, where an $n$-dimensional submanifold is a neighbourhood of each of its points `Chapter9/Appendix · rnPolyfold_mem_nhds_of_hasDimAt` `Chapter9/Appendix · topMultiplicity_rn_of_le`; so Proposition 9.1.12 holds on $\mathbb{R}^0$ and $\mathbb{R}^1$ unconditionally `Chapter9/Appendix · proposition_9_1_12_rn_zero` `Chapter9/Appendix · proposition_9_1_12_rn_one`. The step itself is proved from the graph representation in §IX.22 `Chapter9/AppendixSlices · MPolyfold.topMultiplicityAt_of_hasGraphChartAt` `Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.lemma_9_6_4_of_hasGraphChartAt`, and now on every M-polyfold `Chapter9/GeneralCorollaries · topMultiplicityAt` `Chapter9/GeneralCorollaries · topMultiplicity`, so Lemma 9.6.4 holds for every ep-groupoid `Chapter9/GeneralCorollaries · lemma_9_6_4_general` `Chapter9/`

GeneralCorollaries · isTopBadPoint_iff_general. In the proof the top-good point is called $z$ and then $y$.

## IX.22 Proposition 9.6.5, the formula on p. 457, and the proof of Proposition 9.1.12

**Proposition 9.6.5** (V9.62) Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.proposition_9_6_5. *At a top-bad $x$, a class $\Xi$ of top-dimensional indices with one tangent space splits as $I^1 \sqcup \cdots \sqcup I^p$ with sequences of top-good points $x_j^e \to x$ lying on $M_i$ exactly for $i \in I^e$.* In (1) the text writes "$I_k = I^1 \sqcup \cdots \sqcup I^p$"; the $I^e$ being subsets of $\Xi$, it reads as $\Xi = I^1 \sqcup \cdots \sqcup I^p$. **The slice form.** Under `HasGraphChartAt` at $x$, for every class and every neighbourhood $W$ of $x$ there is a finite top slice in $W$ Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.IsTopSlice: it meets each sheet of the class once, misses the other top-dimensional sheets and all lower-dimensional ones, using a generic slice (§IX.9). Top slices in every $W$ replace the top-good sequences.

**The formula on p. 457** Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_sum_slices. $\Theta_k(x) = \sum_\Xi \sum_{y \in \text{slice}} \Theta(y)$; for one class the slice gives $\sum_{i \in \Xi} \sigma_i$ Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.sum_class_eq_sum_of_isTopSlice. The text's $\Theta_k(x_e^k)$ uses points $x_e^k$ not introduced, presumably points $x_j^e$ of the sequences. **The independence of the right-hand side**, which the text asserts while $\Xi$, $I^e$ and $x^e$ come from the first structure, is an argument the text does not give. It is proved by choosing one slice for the sheets of both structures, off all lower-dimensional sets of both, so that both class weights are the same sum of values of $\Theta$ Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.sum_class_eq_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_hasGraphChartAt. The same grouping proves `TopMultiplicityAt` Chapter9/AppendixSlices · MPolyfold.TopMultiplicityAt Chapter9/AppendixSlices · MPolyfold.topMultiplicity_of_hasGraphChartAt, on $\mathbb{R}^m$ in all dimensions Chapter9/AppendixSlices · MPolyfold.topMultiplicity_rn. A per-dimension graph representation is not enough here: the lower-dimensional sheets through $y$ must be avoided in the chart in which the top sheets are graphs, which is why one chart for all dimensions was asked. One cone per tangent space in that chart suffices, since the slice for a class only uses the cone of that class; with this change the slices exist at every point of every M-polyfold Chapter9/GeneralCorollaries · exists_slice_of_classGraphChart Chapter9/GeneralCorollaries · exists_generic_slice_general, and Proposition 9.6.5 and the formula on p. 457 hold for every ep-groupoid Chapter9/GeneralCorollaries · proposition_9_6_5_general Chapter9/GeneralCorollaries · sum_class_eq Chapter9/GeneralCorollaries · dimPart_topDim_eq.

**The proof of Proposition 9.1.12.** With the top parts agreeing near $x$, subtract them from both structures and repeat in lower dimensions Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_of_hasGraphChartAt; on $\mathbb{R}^m$ without hypothesis Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_rn, and on every ep-groupoid Chapter9/GeneralCorollaries · LocalBranchingStructure.dimPartsAgreeNear Chapter9/GeneralCorollaries · proposition_9_1_12. "This completes the proof of Proposition 9.6.5" (p. 457) refers to Proposition 9.1.12, Proposition 9.6.5 being stated without proof.

## IX.23 Witnesses, counterexamples, and the hypotheses on $\mathbb{R}^m$

**Hypotheses first taken on a general M-polyfold, now theorems.** Each was proved on $\mathbb{R}^m$ first, and each now holds on every M-polyfold or every ep-groupoid; the statements under

the hypotheses are kept. The independence of Theorem 9.2.1, `HasIndependentTangentSums`: first every $\Theta$ over $\mathbb{R}^m$ `Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn` and the indicator of a point `Chapter9/Tangent · hasIndependentTangentSums_indicator`, now every $\Theta$ `Chapter9/GeneralCorollaries · hasIndependentTangentSums`. Tangent slices `HasTangentSlicesAt`: $\mathbb{R}^m$ `Chapter9/TangentSlicesRn · MPolyfold.hasTangentSlicesAt_rn`, $\mathbb{R}^0$ `Chapter9/Tangent · hasTangentSlicesAt_rn_zero`, every M-polyfold `Chapter9/GeneralCorollaries · hasTangentSlicesAt`. Graph charts: `IsGraphChartAt`, `HasGraphChartsAt`, `HasGraphChartAt`, `HasGraphBallAt` at every point of $\mathbb{R}^m$ (§IX.9); on every M-polyfold the class form at every point `Chapter9/GeneralLevelZero · hasClassGraphChartAt` and the one-cone and ball forms at interior points `Chapter9/GeneralLevelZero · hasGraphChartAt_of_mem_interior`. The local constancy of the dimension `Chapter9/GeneralLevelZero · MPolyfold.hasLocallyConstantDimAt`, invariance of domain `Chapter9/GeneralLevelZero · hasDomainInvarianceAt` and the level-1 continuity `Chapter9/GeneralLevelZero · hasContinuousLevelOneOn`: every M-polyfold. `TopMultiplicity`: every M-polyfold `Chapter9/GeneralCorollaries · topMultiplicity`. `HasOrientedGoodPointsAt`: every point `Chapter9/GeneralBoundaryPoints · hasOrientedGoodPointsAt`. `TangentBranchingHyp`: every $\Theta$ with $|X|$ paracompact `Chapter9/TangentSubmanifoldSmooth · tangentBranchingHyp`. `SheetGeometry`: the $n$-dimensional submanifolds of every M-polyfold `Chapter9/GapP2_DSheetEssential · sheetGeometry`, first graphs in $\mathbb{R}^2$ `Chapter9/BranchingGeometry · graphSheetGeometry`. Property (T): after shrinking, every oriented structure `Chapter9/PropertyTShrinking2021 · lemma_9_4_3`, first Example 9.3.5 `Chapter9/BranchingGeometry · propertyT_example935B1`. `SheetNullSets`: the Hausdorff null sets of every M-polyfold `Chapter9/GapP2_DSheetNull · sheetNullSets`, first graphs and open subsets of $\mathbb{R}^2$ `Chapter9/Measure · graphSheetMeasureGeometry` `Chapter9/Measure · planeNull`. `HasTameSheetsIn`: every structure on $\mathbb{R}^n$ `Chapter9/Measure · hasTameSheetsIn_trivial_rn` and every tame $n$-dimensional $\Theta$ for the $n$-dimensional sheets `Chapter9/GapP2_DSheetMeasure · hasTameSheetsIn_of_isNDimensional`. `Remark913Hyp`: `expNegInvGlue` `Chapter9/Propositions · EpGroupoid.remark913Hyp_expNegInvGlue`.

**Hypotheses and readings that remain, with their witnesses.** The class hypotheses (a) and (b) of the first forms of Theorems 9.5.15–9.5.18 are derived from the text's hypotheses since 2026-11-13 (§§IX.36–IX.38); the forms with them are kept. (a) `hcov` (Theorems 9.5.15, 9.5.16): at every point of $\mathrm{supp}(\partial\Theta)$ an oriented tame structure describing $\hat{T}_\Theta$ whose nonempty boundary sheets lie in the class of the integration; witnesses: $\Theta = 0$ `Chapter9/GapP2_E7_BoundaryMeasure · hcov_zero`, compact boundary sheets `Chapter9/GapP2_E7_BoundaryMeasure · hcov_msheets_of_isCompact`, and the closed upper hemisphere `Chapter9/GapP2_E8W_Witness · e8w_hcov` (§IX.34). (b) `hcovS` (Theorems 9.5.17, 9.5.18): at every point of $\mathrm{supp}(\Theta)$ such a structure whose sheets lie in one graph chart, for 9.5.18 with the local unique continuation property; witness: the closed upper hemisphere, $m = 1$ `Chapter9/GapP2_E8W_Witness · e8w_stokes_hypotheses` `Chapter9/GapP2_E8_Hemisphere · E8W.e8_hemisphere_stokes`, and for the LUC part the same hemisphere (§IX.38). Not derived from the text: (c) the bound on the dimensions of submanifolds for the local finiteness in Proposition 9.1.11 off the support; witness $\mathbb{R}^m$ `Chapter9/DimDecompositionExists · rn_hasDimAt_le` (§IX.4). Readings: (d) "oriented, tame" in §9.5 as `hloc`, near every point of the support an oriented tame structure describing $\hat{T}_\Theta$; derived when $d_X = 0$ on the support `Chapter9/GapP2_E4_Hloc · hloc_of_degIndex_eq_zero`, witnessed by the circle `Chapter9/GapP2_E4_Circle · circle_witness` (§IX.32). (e) (9.22) for $\tau = f \cdot \omega$ read through the evaluation on frames, `IsMul`, witnessed by constants (§IX.33).

**Unconditional.** On every ep-groupoid: Propositions 9.1.11 (read among local decompositions, local finiteness on the support) and 9.1.12, Theorem 9.2.1, Lemma 9.2.7 (2), Lemma 9.3.4 for $n$-dimensional $\Theta$, Lemma 9.4.3, Lemmas 9.4.4, 9.4.6 for $n$-dimensional $\Theta$, Lemma 9.5.2 for $n$-dimensional $\Theta$, Lemma 9.6.4, Proposition 9.6.5, Remark 9.1.4, Proposition 9.1.9; Definition 9.2.12 with $|X|$ paracompact; Theorems 9.5.4, 9.5.5 and 9.5.15–9.5.18 under the reading (d).

**Witnesses for §9.5.** The unit circle in $\mathbb{R}^2$, counter-clockwise: Theorems 9.5.4, 9.5.5 with compact

nonempty support Chapter9/GapP2_E4_FiniteCircle · circle_theorem_9_5_4_full, the branched integral as the integral over $S^1$ Chapter9/GapP2_E4_CircleMulti · circle_multiBranchedIntegral, Proposition 9.5.13 (2) Chapter9/GapP2_E4_ReducedCircle · circle_proposition_9_5_13_2. The half-line $[0,\infty)$ with $\Theta = 1$: $\partial S = \{|0|\} \neq \emptyset$ and $L(\partial S)$ Chapter9/GapP2_E2BoundaryNull · halfLine_lebesgueSigmaBoundary. The closed upper hemisphere: Theorems 9.5.15–9.5.18 with $\partial S \neq \emptyset$ (§§IX.35, IX.36, IX.38). The closed half-circle in $[0,\infty) \times \mathbb{R}$: the same for $n = 1$ (§IX.38). The groupoid $\mathbb{Z}_3 \ltimes X$ of Remark 9.5.20 (§IX.39). The $x$-axis of $\mathbb{R}^2$: Remark 9.1.10 Chapter9/GapP2_C2Witness · remark_9_1_10_hypotheses_axis and the one-chart class Chapter9/GapP2_E3_TameOpen · hcov_axis.

**Counterexamples** (each to a statement as we read it): the uniqueness in Proposition 9.1.11 Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn; the footnote to Definition 9.1.2 for $\Theta = 0$ Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote_zero; Lemma 9.2.13 (2) $\Rightarrow$ (1) off $\partial X$ Chapter9/BoundaryPosition · MPolyfold.not_lemma_9_2_13_two_one; the uniqueness in Definition 9.2.16 Chapter9/Boundary · not_unique_isBoundaryCharacterized_halfLine; Lemma 9.3.4 without equidimensionality Chapter9/OrientationRn · lemma_9_3_4_mixed_dim_rn; "otherwise different" and "four possibilities" in Example 9.3.5 Chapter9/OrientationExample · example_9_3_5_eq_of_neg Chapter9/GapP2_DExample935 · example_9_3_5_six Chapter9/GapP2_D2Example935Class · example_9_3_5_infinite; Lemma 9.4.4 for sheets of different dimensions Chapter9/BranchingGeometry · lemma_9_4_4_mixed_dim; the transfer of null sets in Lemma 9.5.2 across dimensions Chapter9/Measure · isNull_transfer_mixed_dim Chapter9/GapP2_DSheetMeasure · isHNull_mixed_dim.

## IX.24 What is not formalized

As of wave K (2026-11-13), on top of commit 8e5afeb, gaps G.93–G.109 are closed, G.110 is formalized in part, for Remark 9.5.20 only, and no gap of the chapter is open.

- **G.110 (in part), Remark 9.5.20.** Formalized: the M-polyfold $X \subseteq \mathbb{R} \times \mathbb{C}$, $\partial X = \{(\pm 1, 0)\}$, $\mathbb{Z}_3 \ltimes X$ with the local unique continuation property, $\partial X$ as a sub-ep-groupoid, and the effective isotropy at a point of $\partial X$, three read in $X$ and one read in $\partial X$ (§IX.39). Not formalized: $\Theta(t, z) = \delta_0(z)$ with its orientation, $\oint_\Theta df = \frac{1}{3}(\phi(1) - \phi(-1))$ and the two boundary integrals, estimated at 2,340–3,790 lines; the main risks are sc-smooth partitions of unity on the $c_0$ scale of the example and the tameness of $M$ at the infinite-dimensional ends. Theorems 9.5.17 and 9.5.18 themselves are proved as printed for every $n \geq 1$ (§§IX.36–IX.38).
- **Closed in wave K: G.109**, Theorems 9.5.15 and 9.5.16 as printed for every $n \geq 1$ (§§IX.36–IX.38), and with it the Theorems 9.5.17, 9.5.18 part of G.110.
- **Left inside closed gaps.** Local finiteness of the decomposition of Proposition 9.1.11 off the support without a bound on the dimensions (G.96). The form $f \cdot \omega$ for a non-constant sc-smooth $f$, read through the evaluation (G.107); in dimension 0 the integration is the abstract one Chapter9/GapP2_E1_PointIntegration · pointIntegration. The orientations of the faces of the faces, the last sentence of the caption of Figure 9.1 (G.103). The comparison of the face-by-face null sets with those taken in the ambient charts (G.108), and of the essential points in M-polyfold charts with those in corner charts (G.105), which no statement needs. A witness of Lemma 9.2.13 at a boundary point.
- **Nothing to formalize.** Remarks 9.3.1, 9.3.11 and 9.5.8 (the last has no content beyond Lemma 9.5.11); Remarks 9.5.12 and 9.5.19 and §§9.6.2–9.6.3 discuss open questions.

## IX.25 What the formalization taught us

The definition of a branched ep⁺-subgroupoid is local and says nothing about how the local pictures fit together, and three statements that read as uniqueness claims, Proposition 9.1.11, Definition 9.2.16 and Lemma 9.3.4, turned out to depend on exactly that: an indicator can be split into a point and a punctured line, a characterization asked only on its own support is met by zero, and a lower-dimensional sheet through a bad point escapes the good points. In each case the repair is a condition that ties the object to the local branching structures of $\Theta$ (local decompositions, the largest support, equidimensionality). The analytic heart of §9.2, the graph representation of tangent sheets near a bad point, is a statement about one chart in which sheets of all dimensions are graphs; separated from the counting it serves, it became a single hypothesis, true on $\mathbb{R}^m$ by the inverse function theorem, from which Proposition 9.1.12, Theorem 9.2.1, Lemma 9.6.4 and Proposition 9.6.5 follow. Two steps the text leaves to "similarly" (Lemma 9.6.4) or asserts (the independence on p. 457) needed that both structures be compared on one common slice. Finally, "measure zero" in §9.6.1 is not needed: the arguments only use that a piece of a $k$-dimensional submanifold is not covered by finitely many thinner ones, which is topology. In §9.5 the opposite happens: the measure-theoretic part needs only that the null sets of the sheets form a local σ-ideal compatible on overlaps, and that compatibility is exactly where dimensions must agree; and of the isotropy computations that normalize the integrals, everything except the integrals themselves is finite group theory.

The second round of work taught more. *The level-0 hypotheses were one fact.* Every level-0 statement of §9.2 (the cone, the graphs, invariance of domain, the local constancy of rank, the level-1 continuity of frames) follows from one observation: the lift $1_0^1 \circ r$ of an sc⁺-retraction to level 1 is $C^1$ on the retract, so that Mathlib's inverse function theorem applies on level 1 and the conclusions descend through the inclusion. The project's first hypothesis, one cone for all submanifolds through a boundary point, was more than the text uses; one cone per tangent space is enough, and holds everywhere (§IX.27). *A tacit finiteness in §9.5.* The text integrates $\omega$ over measurable subsets of the sheets, which presupposes $\int |\omega| < \infty$. For the sheets themselves this follows from compactness and the properness of $\operatorname{cl} U(x)$ (§IX.32); for the boundary sheets $\partial_1 M_i$ it was first an added hypothesis of Theorems 9.5.15–9.5.18 (§§IX.34–IX.35), and it follows from the continuity of Part I's chart forms up to the corners and, toward the frontier of $U(x)$, from a comparison of tangent spaces with other structures, in which $\{d = 1\}$ depends on the tangent space alone (§§IX.36–IX.37). The same proof needs facts the text does not state: that the sheets are Lindelöf, so that countably many neighbourhoods suffice in Theorems 9.5.4, 9.5.5; that $M_i \cap \operatorname{supp} \beta_k$ is compact in the proof of Stokes' theorem; that chart changes are differentiable along a finite-dimensional submanifold for the level-0 norms, which makes Definitions 9.4.1 and 9.4.5 chart independent; and, for Proposition 9.5.13 (2), a natural action of $G_x^R$ built from that of $G_x$ under the local unique continuation property. *Some hypotheses are not used.* Proposition 9.1.9 does not use the paracompactness of $|X|$, the continuity of $o_{\partial M}$ needs no tameness, only reduced codimension 1, and the effective clause of Lemma 9.5.11 needs no local unique continuation. *Examples are more rigid than they look.* In Example 9.3.5 every one-dimensional submanifold inside the support through the origin has tangent $\mathbb{R}(1,0)$ and meets every vertical line, and this rigidity determines all orientations: a three-parameter family, of which six have integer weights (§IX.29). *Paracompactness through metrizability.* In Remark 9.1.4 the step from a paracompact neighbourhood to a component compact set goes through metrizability, since a subspace of a paracompact space need not be paracompact (§IX.26).

*Remark 9.5.20 needs infinite dimensions.* In finite dimensions a finite group that fixes a boundary point and acts trivially on the face acts trivially on the inward normal as well, so the effective isotropy of a

boundary point is the same read in $X$ and in $\partial X$. The remark's example has to be an M-polyfold whose local dimension jumps at its boundary, $0 \to 3 \to 0$ (§IX.39).

## IX.26 Remark 9.1.4, Proposition 9.1.9 (2017: 9.1) and Remark 9.1.10 (2017: 9.2)

**Remark 9.1.4** (V9.63), new in the 2021 edition `Chapter9/GapP2_CRemark914 · EpGroupoid.remark_9_1_4`. *A paracompact $L \subseteq |X|$ is metrizable, and if $|\mathrm{supp}(\Theta)|$ is component compact, the paracompactness in Definition 9.1.2 (1) is automatic.* **The text's argument**, followed exactly. *Step 1.* $|X|$ is locally metrizable, regular and Hausdorff (Theorem 7.3.1), so a paracompact subspace is metrizable by Smirnov's theorem `Chapter9/GapP2_CRemark914 · EpGroupoid.remark_9_1_4_metrizable`. *Step 2.* A component compact $K$ has an open paracompact neighbourhood $U$ (Corollary 7.3.8), metrizable by Step 1 `Chapter9/GapP2_CRemark914 · EpGroupoid.remark_9_1_4_exists_nhds`. *Step 3*, which the text does not spell out: $K$ is paracompact. A subspace of a paracompact space need not be paracompact, but a subspace of the metrizable $U$ is metrizable, hence paracompact `Chapter9/GapP2_CRemark914 · EpGroupoid.metrizableSpace_of_isComponentCompact` `Chapter9/GapP2_CRemark914 · EpGroupoid.remark_9_1_4_paracompactSpace`. *Step 4.* So a functor with the other properties of Definition 9.1.2 and component compact $|\mathrm{supp}(\Theta)|$ is a branched ep$^+$-subgroupoid `Chapter9/GapP2_CRemark914 · EpGroupoid.isBranchedEpSubgroupoid_of_isComponentCompact`, compact supports in particular `Chapter9/GapP2_CRemark914 · EpGroupoid.paracompactSpace_orbitSupp_of_isCompact`.

**Proposition 9.1.9 (2)** (V9.64), *for $\Theta$ of orbifold type, $X$ induces a natural orbifold structure on $Z = |\mathrm{supp}(\Theta)|$.* **The text's proof.** Near $x \in \mathrm{supp}(\Theta)$ all sheets coincide, $\Theta = 1_{M_x}$; the groupoid with objects $\bigsqcup_x M_x$ and morphisms $\bigsqcup M(x,x')$ is étale and proper, the functor $\beta : M \to Z$ induces a homeomorphism $|\beta|$, and other choices give an equivalent structure. **The formalized proof.** *Step 1, one sheet.* $\Theta$ takes only the values 0 and 1, so a sheet missing a point of the support near $x$ would make $0 < \Theta < 1$ there; hence $\mathrm{supp}(\Theta) \cap U(x) = M_i$ for every $i$ `Chapter9/GapP2_CProp919 · EpGroupoid.supp_inter_eq_of_isOrbifoldType`. *Step 2.* So $\mathrm{supp}(\Theta)$ is a smooth finite-dimensional submanifold of $X$ `Chapter9/GapP2_CProp919 · EpGroupoid.isSmoothSubmanifold_supp_of_isOrbifoldType`, saturated, hence an ep-subgroupoid, whose full subcategory is an ep-groupoid (Proposition 7.1.22) `Chapter9/GapP2_CProp919 · EpGroupoid.isEpSubgroupoid_supp_of_isOrbifoldType`. *Step 3.* For a saturated $A$, the map $|A| \to \pi(A)$ induced by the inclusion is a homeomorphism `Chapter9/GapP2_CProp919 · EpGroupoid.orbitSuppHomeomorph`. This gives a choice-free form `Chapter9/GapP2_CProp919 · EpGroupoid.proposition_9_1_9_two`. *Step 4, the text's groupoid.* For a choice of local branching structures at all points of the support `Chapter9/GapP2_C2Prop919Text · SheetChoice`, the sets $M_x = \mathrm{supp}(\Theta) \cap U(x)$ form an open cover `Chapter9/GapP2_C2Prop919Text · sheetCover`, and the text's $M$ is the refinement of Proposition 10.1.3 of the full ep-subgroupoid along it `Chapter9/GapP2_C2Prop919Text · textGroupoid`: objects $(x,y)$ with $y \in M_x$, morphisms $((x,x'),\varphi)$ with $s(\varphi) \in M_x$, $t(\varphi) \in M_{x'}$. It is an ep-groupoid, $|\beta|$ is a homeomorphism `Chapter9/GapP2_C2Prop919Text · textOrbitHomeomorph`, and $M \to \mathrm{supp}(\Theta)$ is an equivalence over $Z$ `Chapter9/GapP2_C2Prop919Text · proposition_9_1_9_two_text`. *Step 5, independence.* A refinement of covers gives an equivalence of refinements `Chapter9/GapP2_C2Prop919Text · OpenCover.isEquivalence_refineFunctor`; for two choices $c$, $c'$ the common refinement by the $M_x \cap M'_{x'}$ gives $M_c \leftarrow M_{c,c'} \to M_{c'}$, a generalized isomorphism of Chapter 10 (Theorem 10.3.10) compatible with the homeomorphisms to $Z$ `Chapter9/GapP2_C2Prop919Text · proposition_9_1_9_two_independent`. "Equivalent orbifold structure" is read as this generalized isomorphism. The construction assumes $\mathrm{supp}(\Theta) \neq \emptyset$; for $\Theta = 0$ there is nothing to construct.

**Proposition 9.1.9 (1)**, *for $\Theta$ of manifold type, a natural manifold structure on $Z$. Step 1.* Between two objects of the support there is at most one morphism, so the isotropy of the ep-subgroupoid is trivial `Chapter9/`

GapP2_CProp919Manifold · EpGroupoid.subsingleton_isotropy_of_isManifoldType, and $\pi$ is injective on the domain of a natural representation Chapter9/GapP2_CProp919Manifold · EpGroupoid.injOn_π_of_subsingleton_isotropy, the text's "$\Psi_x$ is injective". *Step 2, descent.* A general lemma: a continuous, open, surjective, locally injective map $p$ from an M-polyfold to a Hausdorff paracompact space, such that $p(a) = p(b)$ is realized near $a$ by an sc-smooth $f$ with $f(a) = b$ and $p \circ f = p$ Chapter9/GapP2_CDescentPolyfold · MPolyfold.IsDescentMap, pushes the charts forward Chapter9/GapP2_CDescentPolyfold · MPolyfold.descChart to compatible charts, an M-polyfold structure on the target Chapter9/GapP2_CDescentPolyfold · MPolyfold.descend for which $p$ is a local sc-diffeomorphism Chapter9/GapP2_CDescentPolyfold · MPolyfold.isLocalScDiffeo_descend. *Step 3.* The orbit map $\mathrm{supp}(\Theta) \to Z$ is such a map, with $f = t \circ (s|U(\varphi))^{-1}$ Chapter9/GapP2_CProp919Manifold · EpGroupoid.isDescentMap_orbitSupp, which gives the structure Chapter9/GapP2_CProp919Manifold · EpGroupoid.proposition_9_1_9_one. *Step 4, "natural"* is read as uniqueness: any two M-polyfold structures on $Z$ for which the orbit map is a local sc-diffeomorphism agree Chapter9/GapP2_C2Prop919Text · EpGroupoid.proposition_9_1_9_one_unique. The paracompactness of $|X|$, a hypothesis of the proposition, is not used.

**Remark 9.1.10** (V9.65, narrower). *For $\Theta$ $n$-dimensional with $d_{|X|} = 0$ on $Z$, the structures are classical. Step 1.* The support is an $n$-dimensional submanifold of degeneracy index $0$ Chapter9/GapP2_C2Remark9110 · EpGroupoid.isSubmanifoldOfDim_supp Chapter9/GapP2_C2Remark9110 · EpGroupoid.degIndex_eq_zero_of_mem_supp, so it has a corner structure of dimension $n$ (§IX.16), which a surjective local sc-diffeomorphism carries to $Z$ Chapter9/GapP2_C2Remark9110 · MPolyfold.HasCornerStructureOfDim.of_isLocalScDiffeo_surjective. *Step 2.* In case (1) $Z$ is a Mathlib manifold, `IsManifold (□ n) ∞` Chapter9/GapP2_C2Remark9110 · EpGroupoid.remark_9_1_10_manifold, and the orbit map is a Mathlib local diffeomorphism Chapter9/GapP2_C2Remark9110Etale · EpGroupoid.remark_9_1_10_manifold_proj. *Step 3.* In case (2), for every choice, the object and morphism spaces of $M$ are Mathlib $n$-manifolds Chapter9/GapP2_C2Remark9110 · EpGroupoid.remark_9_1_10_orbifold, and source, target, unit, inverse and the equivalence to the support are Mathlib local diffeomorphisms Chapter9/GapP2_C2Remark9110Etale · EpGroupoid.remark_9_1_10_orbifold_etale, since a local sc-diffeomorphism between such corner structures of degeneracy index $0$ is a Mathlib local diffeomorphism Chapter9/GapP2_C2Remark9110Etale · MPolyfold.isLocalDiffeomorph_of_isLocalScDiffeo. The row is narrower for two reasons: the dimension $n$ is fixed, a Mathlib manifold having one model space while the text's $Z$ may have components of different dimensions; and the composition of $M$ is sc-smooth, as for every ep-groupoid, but not stated as Mathlib-smooth, Mathlib having no notion of an orbifold. Witness: the $x$-axis of $\mathbb{R}^2$ Chapter9/GapP2_C2Witness · EpGroupoid.remark_9_1_10_hypotheses_axis.

## IX.27 The level-0 facts on every M-polyfold: Remark 9.1.1 (2017: 9.1), Lemmas 9.2.9–9.2.11 (2017: 9.6–9.8), invariance of domain, and the level-1 continuity

These are the facts the first version of this part took as hypotheses (gaps G.93, G.97, G.98, G.100, all closed). **The text** works at level 0 in a chart, with $x = 0$ in a partial quadrant $C$ and $M$ the image of an sc$^+$-retraction: $L = T_0M$ meets $C$ in a set with interior in $L$ (Lemma 9.2.9); $M$ is a graph $a + B(a)$ over a cone in $L \cap C$ with $DB(0) = 0$ (Lemma 9.2.10), by inverting $\Phi(a) = a + PB(a)$ (Lemma 9.2.11); $\dim T_mM$ is locally constant (Remark 9.1.1); a point in the interior of the sets it meets is "clearly" good (Lemma 9.2.7 (2)). The text treats $B$ as $C^1$ at level 0, while the retraction is only sc-smooth.

**The level-1 calculus of an sc$^+$-retraction.** *Step 1.* For an sc$^+$-retraction $r$ the lift $1^1_0 \circ r$ into level 1 Chapter9/GeneralLevelZero · plusLift is $C^1$ on level 1 Chapter9/GeneralLevelZero · hasFDerivWithinAt_plusLift Chapter9/GeneralLevelZero · continuousOn_plusDeriv, strictly differentiable Chapter9/GeneralLevelZero · exists_strict_

plusLift, and its derivative brought down by the inclusion $\iota$ is $Dr$ Chapter9/GeneralLevelZero · toBase_plusDeriv. All estimates are proved on level 1 and brought down by $\iota$; on the finite-dimensional $T = \operatorname{range} Dr(z)$ the two norms are equivalent. *Step 2, the chord estimate* ($DB(0) = 0$): chords of the retract are close to $T$ Chapter9/GeneralLevelZero · exists_approx Chapter9/GeneralLevelZero · exists_strict_range. *Step 3, Lemma 9.2.9:* $Dr$ maps into the cone of the quadrant Chapter9/GeneralLevelZero · nonneg_scDeriv, so a closed ball of $T$ around an admissible direction $h_1$, scaled by small $s$, stays in $C$ Chapter9/GeneralLevelZero · ScRetraction.exists_cone. *Step 4, Lemmas 9.2.10 and 9.2.11:* the projection of the retract onto $T$ approximates the identity of $T$ on small balls, so Mathlib's surjectivity theorem for maps approximating a linear map (`ApproximatesLinearOn`) inverts it over a cone, uniformly in the foot point Chapter9/GeneralLevelZero · exists_graph_unif Chapter9/GeneralLevelZero · exists_graph, and over a ball of $T$ at interior points Chapter9/GeneralLevelZero · exists_graph_ball. *Step 5, invariance of domain:* if two sc⁺-retractions $r_1, r_2$ have the same tangent space at $z$, every point $b$ of the first retract near $z$ with $r_2(b)$ in the first retract is fixed by $r_2$. On level 1, with $u$ the difference of the lifts of $r_2(b)$ and $b$, the chord estimate for both retractions gives $\|u\|_1 \le c\eta\|u\|_1$ with $c\eta < 1$, so $u = 0$ Chapter9/GeneralLevelZero · eq_of_range_eq. *Step 6, local constancy of the rank:* the idempotents $Dr(z)$, read on level 1, depend continuously on the point in the operator norm, and an idempotent at distance $< 1$ from another has rank at most that of the other Chapter9/GeneralLevelZero · finrank_range_le_of_norm_sub_lt; applied both ways, the rank is locally constant along the retract Chapter9/GeneralLevelZero · eventually_finrank_eq.

**In an M-polyfold.** *Step 7.* A submanifold $A$ is, in a chart, the range of a chart sc⁺-retraction $t$ Chapter9/GeneralLevelZero · exists_plusRetraction, whose derivative has range the chart tangent space Chapter9/GeneralLevelZero · chartTangent_eq_range, and on $A$ the level-1 lift of the chart is $1_0^1 \circ t$ Chapter9/GeneralLevelZero · levelLift_eq_plusLift; so the dimension is the rank of $Dt$ Chapter9/GeneralLevelZero · hasDimAt_finrank. *Step 8, consequences.* Remark 9.1.1 Chapter9/GeneralLevelZero · MPolyfold.hasLocallyConstantDimAt (Step 6), invariance of domain Chapter9/GeneralLevelZero · hasDomainInvarianceAt (Step 5), the level-1 continuity along submanifolds Chapter9/GeneralLevelZero · hasContinuousLevelOneOn (Step 7). *Step 9, the graph representation.* Each tangent space $T_xA$ has its admissible cone of directions Chapter9/GeneralLevelZero · goodDirections, and every submanifold through $x$ is a graph over it Chapter9/GeneralLevelZero · isGraphChartAt_goodDirections. This is the class form `HasClassGraphChartAt`, one cone per tangent space, at every point Chapter9/GeneralLevelZero · hasClassGraphChartAt; at interior points one cone and the ball form serve all submanifolds Chapter9/GeneralLevelZero · exists_chart_mem_interior_of_degIndex_eq_zero Chapter9/GeneralLevelZero · isGraphBallAt_of_mem_interior Chapter9/GeneralLevelZero · hasGraphChartAt_of_mem_interior.

**What follows** Chapter9/GeneralCorollaries · exists_slice_of_classGraphChart. The slice arguments of §§IX.9–IX.10 and IX.21–IX.22 only use the cone of the class at hand, so they run with the class form: tangent slices at every point Chapter9/GeneralCorollaries · hasTangentSlicesAt, generic slices missing finitely many thinner sets Chapter9/GeneralCorollaries · exists_generic_slice_general, `TopMultiplicity` Chapter9/GeneralCorollaries · topMultiplicity, Proposition 9.1.12 Chapter9/GeneralCorollaries · proposition_9_1_12, Lemma 9.6.4 Chapter9/GeneralCorollaries · lemma_9_6_4_general, Proposition 9.6.5 Chapter9/GeneralCorollaries · proposition_9_6_5_general, the independence of Theorem 9.2.1 Chapter9/GeneralCorollaries · hasIndependentTangentSums, Lemma 9.2.7 (2) Chapter9/GeneralCorollaries · lemma_9_2_7_2_general, Proposition 9.1.11 in the reading of §IX.4 Chapter9/GeneralCorollaries · existsUnique_localDimDecomposition_general, and, with the boundary argument of §IX.15, Lemma 9.3.4 Chapter9/GeneralBoundaryPoints · lemma_9_3_4_general. Proposition 9.2.8 itself stays in slice form (V9.22).

# IX.28 Lemmas 9.4.4 and 9.4.6 (2017: 9.13, 9.14) for the sheets of an M-polyfold, and Lemma 9.5.2 (2017: 9.16) with Lebesgue null sets

The sheets of an M-polyfold carry no manifold charts of their own, so Definitions 9.4.1 and 9.4.5 are read in the M-polyfold charts $\psi$ of the ambient space, as the text itself does in the proof of Lemma 9.4.4 (gaps G.105, G.106, closed).

**Definition 9.4.1 in M-polyfold charts.** $T^K_z M$ is the span of Mathlib's tangent cone of $\psi(K)$ at $\psi(z)$ in $E_0$ `Chapter9/SheetEssentialChart2021 · chartRelTangent`, and $z$ is essential if it is a limit of points where $T^K M = TM$ in every chart `Chapter9/SheetEssentialChart2021 · IsRelTangentFull` `Chapter9/SheetEssentialChart2021 · chartEssentialSet`. *Step 1.* For $K \subseteq M$, $T^K_z M \subseteq T_z M$, since chords of the chart retract are approximated by its tangent space (§IX.27, Step 2) `Chapter9/SheetEssentialChart2021 · chartRelTangent_le`. *Step 2.* Conversely $T_z M \subseteq T^M_z M$: for level-1 directions $p$ into the quadrant, $Dr(z)p$ is a limit direction of the retract, and these span the range of $Dr(z)$ since $E_1$ is dense in $E_0$ `Chapter9/SheetEssentialChart2021 · ScRetraction.range_scDeriv_subset_span_tangentConeAt` `Chapter9/SheetEssentialChart2021 · chartTangent_subset_chartRelTangent`. So every point of $M$ is essential for $M \subseteq M$ and the reading is not degenerate `Chapter9/SheetEssentialChart2021 · chartEssentialSet_self`. *Step 3, chart independence.* On level 0 a chart change is only continuous, but along a finite-dimensional submanifold it is differentiable for the level-0 norms: the points of the chart image lie on level 1 with level-1 distance controlled by level-0 distance, because the chart retraction is $\mathrm{sc}^+$ `Chapter9/GapP2_DSheetChartChange · ScRetraction.exists_lift_lipschitz`, and an $\mathrm{sc}^1$ map is differentiable from level 1 to level 0 (Proposition 1.2.1) `Chapter9/GapP2_DSheetChartChange · ScRetraction.hasFDerivWithinAt_of_isSc1` `Chapter9/GapP2_DSheetChartChange · exists_hasFDerivWithinAt_transition`. The derivative maps tangent cones to tangent cones `Chapter9/GapP2_DSheetChartChange · exists_clm_tangentCone_chart_change`, so "$T^K_z M = T_z M$" in one chart gives it in all `Chapter9/GapP2_DSheetChartChange · isRelTangentFull_of_chart`.

**Lemma 9.4.4** (V9.42, narrower) `Chapter9/SheetEssentialChart2021 · lemma_9_4_4_chart`. *For $n$-dimensional $M$, $M'$, $K = M \cap M'$ has the same essential points in $M$ and in $M'$.* The text's reduction to $T^D_0 N$ is the pointwise statement, which holds in each chart separately: $T^K_z M = T_z M$ iff $T^K_z M' = T_z M'$, both tangent spaces being $n$-dimensional and containing $T^K_z$ `Chapter9/SheetEssentialChart2021 · isRelTangentFull_iff`. At essential points $T_z M = T_z M'$, the note in Definition 9.4.7 `Chapter9/SheetEssentialChart2021 · submanifoldTangent_eq_of_mem_chartEssentialSet`. For different dimensions the lemma does not hold as we read it (§IX.17).

**Definition 9.4.5 in M-polyfold charts.** *Step 1.* `HNull d s`: the $d$-dimensional Hausdorff measure of $s$ vanishes `Chapter9/GapP2_DSheetNull · HNull`. A subset $C$ of a sheet has $n$-dimensional measure zero if its separable pieces have $n$-null chart images in every chart `Chapter9/GapP2_DSheetNull · IsHNull`; $E_0$ need not be separable, and a non-separable set has infinite Hausdorff measure, but the sheets are locally separable, so nothing is lost. It is a local σ-ideal `Chapter9/GapP2_DSheetNull · isHNull_iUnion` `Chapter9/GapP2_DSheetNull · isHNull_of_local`. *Step 2, why these are Lebesgue null sets.* Near a point of a sheet the tangent projection $p$ onto $T_x M$ satisfies $\|(\psi a - \psi b) - p(\psi a - \psi b)\| \le \frac{1}{2}\|\psi a - \psi b\|$ `Chapter9/GapP2_DSheetNull · exists_half_approx`, so it is bi-Lipschitz there, and on $\mathbb{R}^n$ the $n$-dimensional Hausdorff measure is Lebesgue measure `Chapter9/GapP2_DSheetNullBridge · hNull_iff_volume_euclidean`. On a tame sheet these are exactly the null sets of Definition 9.4.5 in the smooth charts of the manifold with corners of §IX.16 `Chapter9/GapP2_DSheetNullBridge · isHNull_iff_isMeasureZero_cornerChartedSpace` `Chapter9/GapP2_DSheetNullBridge · isHNull_iff_isMeasureZero_sheet`. *Step 3, chart independence:* a map Lipschitz at each point sends separable null sets to null sets `Chapter9/GapP2_DSheetChartChange · hNull_image_of_pointwise_lipschitz`, which with Step 3 above gives `Chapter9/GapP2_`

DSheetChartChange · isHNull_of_forall_exists_chart. *Step 4, positivity:* an $n$-dimensional sheet is not null in itself, its tangent projection containing a ball of $T_xM$ (§IX.27, the graph over a cone) Chapter9/GapP2_DSheetNull · not_isHNull_of_hasDimAt. *Step 5.* So the null sets form a `SheetNullSets` on the $n$-dimensional submanifolds with no hypothesis Chapter9/GapP2_DSheetNull · sheetNullSets; the transfer property is immediate, since the definition does not mention the sheet.

**Lemma 9.4.6** (V9.44) Chapter9/GapP2_DSheetEssential · lemma_9_4_6_chart. *For $K \subseteq M$, $M$ $n$-dimensional, $K \setminus K^e$ has measure zero.* The proof of §IX.17 transported through the tangent projection: every limit direction of $p(\psi(A))$ is the projection of a limit direction of $\psi(A)$, chords of $M$ being close to the finite-dimensional tangent space Chapter9/GapP2_DSheetEssential · exists_mem_tangentConeAt_proj; the points of $K$ where $T^K \neq T$ project to points of $p(\psi(K))$ where the tangent cone does not span, a Lebesgue null set by the density theorem Chapter9/GapP2_DSheetEssential · exists_hNull_chart_not_isRelTangentFull Chapter9/GapP2_DSheetEssential · isHNull_setOf_not_isRelTangentFull. The $n$-dimensional submanifolds carry the `SheetGeometry` and `SheetMeasureGeometry` of §§IX.17, IX.19 with no hypothesis Chapter9/GapP2_DSheetEssential · sheetGeometry Chapter9/GapP2_DSheetEssential · sheetMeasureGeometry.

**For $n$-dimensional $\Theta$, and Lemma 9.5.2** (V9.50, explicit). *Step 1.* Every local branching structure of an $n$-dimensional $\Theta$ has $n$-dimensional sheets Chapter9/GapP2_DSheetMeasure · isOfDim_of_isNDimensional: at a point of the support, a structure with $n$-dimensional sheets and any other structure have the same tangent sum (Theorem 9.2.1), so each sheet of the second has the tangent space of a sheet of the first. *Step 2.* So Lemmas 9.4.4 and 9.4.6 hold for the sheets of every structure of $\Theta$ Chapter9/GapP2_DSheetMeasure · lemma_9_4_4_of_isNDimensional Chapter9/GapP2_DSheetMeasure · lemma_9_4_6_of_isNDimensional, and Lemma 9.4.8, Propositions 9.4.9, 9.4.10 follow (§IX.17). *Step 3.* Definition 9.5.1 with these null sets Chapter9/GapP2_DSheetMeasure · IsMeasureZeroH, and Lemma 9.5.2 as printed: null preimages for some choice of structures iff for every structure Chapter9/GapP2_DSheetMeasure · lemma_9_5_2_of_isNDimensional, and for tame $\Theta$ iff for a choice of tame structures Chapter9/GapP2_DSheetMeasure · lemma_9_5_2_tame; the proof is that of §IX.19 with one class of sheets. *Step 4.* The sets of measure zero form a $\sigma$-ideal Chapter9/GapP2_DSheetMeasure · isSigmaIdeal_isMeasureZeroH, $S$ is not one of them Chapter9/GapP2_DSheetMeasure · not_isMeasureZeroH_orbitSupp, and $L(S)$ is the completion of §IX.19 Chapter9/GapP2_DSheetMeasure · remark_9_5_3_H Chapter9/GapP2_DSheetMeasure · lebesgueSigmaH_eq. Across dimensions the transfer does not hold as we read it Chapter9/GapP2_DSheetMeasure · isHNull_mixed_dim.

## IX.29 Example 9.3.5 (2017: 9.1): all orientations of $\Theta$

**The text** says that $\Theta$ has "four different possibilities to orient". With the first claim of §IX.15 settled, the question is what all orientations of $\Theta$ are (gap G.101, closed; V9.33, narrower). The support is $G^{++} \cup G^{--}$ Chapter9/GapP2_D2Example935Sheets · Ex935.supp_example935Theta, and $\operatorname{supp}(\Theta) \setminus \{0\}$ consists of four open arcs, $G^{++}$ and $G^{--}$ over $s > 0$ and over $s < 0$; $\Theta = 1$ on the arcs and $\Theta(0) = 2$ Chapter9/GapP2_D2Example935Sheets · Ex935.example935Theta_eq_one Chapter9/GapP2_D2Example935Sheets · Ex935.example935Theta_zero.

**The sheets inside the support.** Every sheet of a local branching structure of $\Theta$ lies in the support and is 1-dimensional (§IX.28). *Step 1, velocities.* A curve in the support through a point $z$ of $G_b$ has velocity in $\mathbb{R}(1, \beta_b'(z_0))$: off the origin it runs in $G_b$; at the origin $|\gamma_1| = \beta^{++}(\gamma_0)$ is $o(t)$ since $\beta^{++}$ is flat, so the velocity is horizontal Chapter9/GapP2_D2Example935Sheets · Ex935.hasDerivAt_snd_eq_of_mem_Z. Hence every 1-dimensional submanifold inside the support has, at $z \in G_b$, the tangent space of $G_b$, and at the origin the tangent $\mathbb{R}(1, 0)$ Chapter9/GapP2_D2Example935Sheets · Ex935.submanifoldTangent_eq_of_subset_Z. *Step 2, signs.* Its orientation is one of $\pm(1, \beta')$ at every point Chapter9/GapP2_D2Example935Sheets · Ex935.tangent_eq_orTan_or,

with a locally constant sign Chapter9/GapP2_D2Example935Sheets · Ex935.exists_isOpen_tangent_eq_orTan_iff. *Step 3, vertical lines.* A sheet through the origin meets every vertical line $\{s\} \times \mathbb{R}$ for small $|s|$, by the ball graph representation over $T_0M = \mathbb{R} \times 0$ Chapter9/GapP2_D2Example935Sheets · Ex935.exists_mem_fst_eq.

**The classification** Chapter9/GapP2_D2Example935Class · example_9_3_5_classification. *Step 1, normal form.* At a point $y$ of the support an orientation is $c \cdot (1, \beta') + (\Theta(y) - c) \cdot (-(1, \beta'))$ Chapter9/GapP2_D2Example935Class · Ex935.eq_single_add_single. *Step 2, constancy along arcs.* Near a point of an arc every sheet of a structure is the arc with a locally constant sign (Step 2 above), so the coefficient $a$ is constant along each arc, $0 \le a \le 1$ Chapter9/GapP2_D2Example935Class · Ex935.coef_pt_eq_of_pos Chapter9/GapP2_D2Example935Class · Ex935.coef_pt_eq_of_neg. *Step 3, the origin.* Every sheet of a structure at $0$ meets each vertical line near $0$ (Step 3 above), and the count $\Theta(s, \beta^{++}(s)) + \Theta(s, \beta^{--}(s)) = 2 = \Theta(0)$ forces it to meet it exactly once. With the constancy of the signs, the weight $A$ of $(1, 0)$ at the origin is $a(+, G^{++}) + a(+, G^{--})$, and also $a(-, G^{++}) + a(-, G^{--})$ Chapter9/GapP2_D2Example935Class · Ex935.coef_zero_eq. *Step 4, the family.* So every orientation is a member of the family classOrient$(a)$, $a$ indexed by side and branch, with the two sums equal Chapter9/GapP2_D2Example935Class · Ex935.classOrient Chapter9/GapP2_D2Example935Class · Ex935.eq_classOrient, and different $a$ give different orientations Chapter9/GapP2_D2Example935Class · Ex935.classOrient_injective. *Step 5, the converse.* Each member is an orientation: it is the oriented sum of the four graphs $G^{++}, G^{--}, G^{+-}, G^{-+}$, each with both orientations, with weights solving a $2 \times 2$ transport problem between the sides $s > 0$ and $s < 0$ Chapter9/GapP2_D2Example935Class · Ex935.exists_transport Chapter9/GapP2_D2Example935Class · Ex935.isOrientation_classOrient.

**Consequences.** $\Theta$ has infinitely many orientations, a three-parameter family Chapter9/GapP2_D2Example935Class · example_9_3_5_infinite; those with integer weights are exactly the six sign choices of §IX.15 Chapter9/GapP2_D2Example935Class · example_9_3_5_integral Chapter9/GapP2_D2Example935Class · example_9_3_5_integral_ncard. So "four different possibilities" holds for each of the two structures, and for $\Theta$ it does not hold as we read it.

## IX.30 The boundary operation, the continuity of $o_{\partial M}$, the stratum $\{d_M = 1\}$ and Figure 9.1 (Definitions 9.3.8 and 9.3.10 (2017: 9.9 and 9.11))

**The text.** For a tame $M$ the points with $d_M(y) = 1$ form an open dense part of $\partial M$ with a manifold structure; for $(M, o)$ oriented, a basis $(a_1, \dots, a_{n-1})$ of $T_y\partial M$ is positive iff $(e, a_1, \dots, a_{n-1})$ is positive, $e$ outward; these orientations "fit continuously together", and the orientation of the interior of a face extends to the closed face (Figure 9.1). §IX.16 formalized the linear algebra with $T_y\partial M$ and $e$ as data. Here they are built from $M$ (gap G.103, closed; V9.36–V9.38 explicit).

**The boundary operation** (Chapter9/BoundaryOperation). *Step 1, presented charts.* $T^R_yX$ and $C_yX$ are read in a chart with a presentation of its partial quadrant Chapter9/BoundaryOperation · MPolyfold.PresentedChart; a chart change acts on chart vectors over $y$ by a continuous linear map carrying reduced space to reduced space and cone to cone Chapter9/BoundaryOperation · MPolyfold.PresentedChart.exists_clm, so "reduced", "inward" and the reduced codimension $\dim(L/L \cap T^R_yX)$ are chart independent Chapter9/BoundaryOperation · MPolyfold.isReducedVec_iff Chapter9/BoundaryOperation · MPolyfold.isInwardVec_iff Chapter9/BoundaryOperation · MPolyfold.hasReducedCodim_iff. *Step 2.* For tame $M$, $d_M(y)$ is the reduced codimension of $T_yM$, the content of Lemma 9.3.7's proof Chapter9/BoundaryOperation · MPolyfold.subDegIndex_eq_quotDim_span. *Step 3, the cone spans.* Every vector of $T_yM$ is a difference of two vectors of $C_yM = T_yM \cap$

$C_yX$ Chapter9/BoundaryOperation · MPolyfold.exists_sub_eq_vec_submanifoldTangent; so if $T_yM$ has reduced codimension 1, $C_yM$ is a half-space of $T_yM$ bounded by $T_y^RM$ Chapter9/BoundaryOperation · exists_coeff_halfSpace, outward vectors exist Chapter9/BoundaryOperation · MPolyfold.IsBoundaryAdmissible.exists_outward, and two of them satisfy $e' = ce + k$ with $c > 0$, $k \in T_y^RM$ Chapter9/BoundaryOperation · MPolyfold.IsBoundaryAdmissible.exists_pos. *Step 4, the operation.* $T_y(M, o) \mapsto (T_y^RM, o_\partial)$, with the frames $a$ such that $(e, a)$ is positive for every outward $e$ Chapter9/BoundaryOperation · MPolyfold.boundaryOp, a function of the oriented tangent set with no choice; by Step 3 and §IX.16 it is Definition 9.3.8 for any single outward $e$ Chapter9/BoundaryOperation · MPolyfold.boundaryOp_eq. *Step 5, identification.* The chart image of its carrier is the reduced tangent space of the induced chart of $M$, $T_y\partial M = T_y^RM$ Chapter9/BoundaryOperation · MPolyfold.chartVec_image_boundaryCarrier, and "outward" means "not in the partial cone of $M$" Chapter9/BoundaryOperation · MPolyfold.isOutward_iff_not_mem_partialCone. *Step 6.* With it, $\hat{T}_{\partial\Theta}$ is the oriented sum filtered by reduced codimension 1, which needs no quantifier over submanifolds Chapter9/BoundaryOperation · EpGroupoid.boundaryOrientation; it agrees with Definition 9.3.10 for tame orientations Chapter9/BoundaryOperation · EpGroupoid.boundaryLift_boundaryOp_eq_boundaryOrientation, and Proposition 9.3.9 holds Chapter9/BoundaryOperation · EpGroupoid.proposition_9_3_9_boundaryOrientation. The situation occurs at the corner of $[0, \infty)$ Chapter9/BoundaryOperation · MPolyfold.exists_isBoundaryAdmissible_halfLine.

**Continuity of $o_{\partial M}$** Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.SubmanifoldOrientation.exists_continuous_boundaryFrames. The text says the orientations fit together "by definition". *Step 1.* The boundary stratum is the set of points where $T_yA$ has reduced codimension $1$ Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.boundaryStratum, $\{d_A = 1\}$ for tame $A$ Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.boundaryStratum_eq_of_tame. *Step 2.* Let $r$ be the chart sc⁺-retraction at $y_0$, $w$ a positive frame of $o_{\partial A}$ at $y_0$, $s \in C_{y_0}A$ off $T_{y_0}\partial A$. For $z$ in the stratum near $y_0$, $Dr(z)$ maps the partial quadrant and the reduced space at $y_0$ into those at $z$ Chapter9/BoundaryOrientationContinuity2021 · ScRetraction.scDeriv_mem_partialQuadrantAtWith Chapter9/BoundaryOrientationContinuity2021 · ScRetraction.scDeriv_mem_minimalSubspaceWith, so $Dr(z)w$ is a basis of $T_z\partial A$ by a dimension count, and $-Dr(z)s$ is outward. *Step 3.* The sign of $(-Dr(z)s, Dr(z)w)$ against a continuous positive frame of $o$ is a frame determinant, locally constant (§IX.15). So $o_{\partial M}$ is an orientation along the stratum Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.OrientationAlong Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.SubmanifoldOrientation.boundaryOrientation, for every oriented submanifold: tameness is not used, only the reduced codimension 1. Every oriented structure gives one on each sheet Chapter9/BoundaryOrientationContinuity2021 · EpGroupoid.OrientedBranchingStructure.exists_boundaryOrientation.

**The stratum.** *Step 1, open and dense.* $\{d_Q = 1\} = \partial Q \cap \{d_Q \le 1\}$ is open in $\partial Q$ for every M-polyfold Chapter9/GapP2_DBoundaryStratum · MPolyfold.setOf_degIndex_eq_one Chapter9/GapP2_DBoundaryStratum · MPolyfold.isOpen_degIndex_le_one, and dense in $\partial Q$ for a corner structure, since in corner coordinates a point with vanishing coordinates is a limit of points with exactly one of them vanishing Chapter9/GapP2_DBoundaryStratum · MPolyfold.HasCornerStructureOfDim.boundary_subset_closure_degIndex_eq_one; so for a tame sheet the stratum is open and dense in $\partial M$ Chapter9/GapP2_DBoundaryStratum · MPolyfold.exists_isOpen_boundaryStratum_eq Chapter9/GapP2_DBoundaryStratum · MPolyfold.submanifoldBoundary_subset_closure_boundaryStratum. *Step 2, a submanifold.* In corner coordinates $\kappa$, on the open piece where all coordinates but $y_i$ are positive Chapter9/GapP2_E6_StratumRetraction · MPolyfold.CornerCoords.facePiece, the map $\kappa^{-1} \circ (y_i \mapsto 0) \circ \kappa$ is an sc-smooth retraction onto the points with $d = 1$ Chapter9/GapP2_E6_StratumRetraction · MPolyfold.CornerCoords.faceRet. With $s$ the sc⁺-retraction of $M$, $\iota \circ \kappa^{-1} \circ (y_i \mapsto 0) \circ \kappa \circ s$ is an sc⁺-retraction of an open subset of $X$ onto the stratum, the sc⁺ property coming from $r = s \circ r$ Chapter9/GapP2_E6_StratumRetraction · MPolyfold.exists_stratum_coords Chapter9/GapP2_E6_StratumRetraction · MPolyfold.

exists_isScPlusRetractionOn_stratumOne Chapter9/GapP2_E6_StratumRetraction · MPolyfold.isSmoothSubmanifold_stratumOne Chapter9/GapP2_E6_StratumRetraction · MPolyfold.stratumOne_eq_boundaryStratum. *Step 3, dimension and tangent spaces.* In a chart the image of $T_y\partial_1 M$ is the image of the hyperplane $\{v_i = 0\}$ under the derivative of the inclusion Chapter9/GapP2_E6_StratumTangent · MPolyfold.chartTangent_stratumOne_eq, so $\partial_1 M$ is $(n-1)$-dimensional Chapter9/GapP2_E6_StratumTangent · MPolyfold.isSubmanifoldOfDim_stratumOne with $T_y\partial_1 M = T_y^R M$ Chapter9/GapP2_E6_StratumTangent · MPolyfold.submanifoldTangent_stratumOne. *Step 4, oriented.* $o_{\partial M}$ is an orientation of the submanifold $\partial_1 M$ in the sense of Definition 9.3.2's sheets Chapter9/GapP2_E6_BoundaryOrientation · MPolyfold.SubmanifoldOrientation.stratumOrientation Chapter9/GapP2_E6_BoundaryOrientation · MPolyfold.SubmanifoldOrientation.stratumOrientation_tangent. In corner coordinates where only $y_i$ vanishes, $T\iota(v) \in C_y M$ iff $v_i \geq 0$ Chapter9/GapP2_E6_BoundaryOrientation · MPolyfold.isInwardVec_incFrame_iff, so $T\iota(-e_i)$ is outward, and a frame $T\iota(v_1), \dots, T\iota(v_m)$ of $\{v_i = 0\}$ is positive for $o_{\partial M}$ iff $(T\iota(-e_i), T\iota(v_1), \dots)$ is positive for $o$ Chapter9/GapP2_E6_BoundaryOrientation · MPolyfold.incFrame_mem_boundaryFramesAt_iff: "outward normal first", Part I's rule for the faces of a manifold with corners.

**Figure 9.1** Chapter9/GapP2_D3FaceOrientation · MPolyfold.figure_9_1_existsUnique_faceOrientation. For a face-structured tame sheet, Part I reads the closed faces as manifolds with corners Chapter9/GapP1_Stokes2021 · MPolyfold.faceChartedSpace with the induced orientation, outward normal first Chapter9/GapP1_Stokes2021 · MPolyfold.faceOrientation. *Step 1, linear algebra.* $\det(-e_i, \iota_i v_1, \dots, \iota_i v_m) = (-1)^{i+1} \det v$ for the face embedding $\iota_i$ Chapter9/GapP2_D3FaceOrientation · det_cons_neg_stdFrame_faceEmbedding, and the basis $(-e_i, \iota_i b)$ Chapter9/GapP2_D3FaceOrientation · consFaceBasis has a given orientation iff $b$ has the one Part I's rule assigns Chapter9/GapP2_D3FaceOrientation · orientation_consFaceBasis_eq_iff. *Step 2, agreement.* At every point of a face with $d_M = 1$, a basis $b$ of a face chart is positive for Part I's face orientation iff the frame $T\iota(\iota_i b_1), \dots, T\iota(\iota_i b_m)$ is positive for $o_{\partial M}$ Chapter9/GapP2_D3FaceOrientation · MPolyfold.faceOrientation_eq_iff_mem_stratumOrientation Chapter9/GapP2_D3FaceOrientation · MPolyfold.faceOrientation_eq_iff_mem_stratumOrientation_pair. *Step 3, extension.* The points with $d_M = 1$ are dense in each closed face Chapter9/GapP2_D3FaceOrientation · MPolyfold.dense_faceStratum, and two continuous tangent orientations agreeing on a dense set are equal Chapter9/GapP2_D3FaceOrientation · TangentOrientation.eq_of_eqOn_dense. So the orientation of the open face extends to exactly one continuous orientation of the closed face, Part I's face orientation. The text's "interiors of the faces" are read as the points with $d_M = 1$, and "extends" as the unique continuous extension. The orientations of the faces of the faces (the caption's last sentence) are not formalized.

## IX.31 Branched integration for an abstract sheet integration: (9.20), Lemma 9.5.7 (2017: 9.17), (9.21), Lemmas 9.5.9–9.5.11 (2017: 9.18, 9.19), Theorems 9.5.4, 9.5.5 (2017: 9.2, 9.3), Definition 9.5.6 and Proposition 9.5.13 (1)

**The text.** For an oriented local branching structure $(M_i, o_i)$, $(\sigma_i)$ on $U(x)$ with the natural $G_x$-action and a measurable $K \subseteq |M_U|$ with preimages $K_i \subseteq M_i$, the local measures are $\mu_\omega^U(K) = \frac{1}{\#G_x} \sum_i \sigma_i \int_{K_i} \omega | (M_i, o_i)$ and $\mu_\omega^{\mathrm{eff},U}$ with $\#G_x^{\mathrm{eff}}$ (9.21). They do not depend on the structure (Lemma 9.5.7, via (9.20)), on shrinking $U$ (Lemmas 9.5.9, 9.5.10) or on moving along a morphism (Lemma 9.5.11); covering $S$ by finitely many $|M_{U_p}|$ and summing gives a unique linear $\omega \mapsto \mu_\omega$ into the signed measures on $(S, L(S))$ with the local formula (Theorems 9.5.4, 9.5.5), and $\int_{\hat\Theta} \omega = \mu_\omega(S)$, $\oint_{\hat\Theta} \omega = \mu_\omega^{\mathrm{eff}}(S)$ (Definition 9.5.6). Rows V9.54–V9.57 and V9.66 are explicit or restated (gap G.107, closed).

**The integration as a structure.** *Step 1.* The measurable subsets of a sheet $N$ are the sets whose trace differs from a Borel trace by a null set of $N$ `Chapter9/GapP2_E1_SheetIntegration · SheetNullSets.IsMeasIn`; two sheets induce the same measurable subsets of their intersection `Chapter9/GapP2_E1_SheetIntegration · SheetNullSets.IsMeasIn.transfer`. *Step 2.* A `SheetIntegration` gives numbers $\int_A \omega|(N,o)$ for the forms $\omega$ of a real vector space $\Omega$, the oriented sheets $N$ of a class, and measurable $A \subseteq N$ `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration`, with the properties the text uses: linearity in $\omega$, countable additivity, vanishing on null sets, equal (opposite) integrals over a common subset of two sheets where their oriented tangent spaces agree (are opposite), invariance under the sc-diffeomorphisms whose tangent maps are those of morphisms `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.IsMorTangentOn`, Lindelöf sheets, and Lemma 9.4.6 with the note of Definition 9.4.7. It is built for sc-forms in §IX.32.

**(9.20) and Lemma 9.5.7** `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration.sum_integ_eq_zero` `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration.sum_integ_eq_sum_integ`. *A finite family of oriented sheets with weights whose weighted sum of oriented tangent spaces vanishes at every point of $K$ has $\sum_k w_k \int_{K\cap N_k} \omega|(N_k,o_k) = 0$.* **The text's proof** removes a null set and compares on the pieces of Proposition 9.4.10. **The formalized proof** redoes that decomposition for measurable pieces. *Step 1.* The points where two sheets through a point have different tangent spaces form a null set in every sheet `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration.isNull_tangentBad`. *Step 2.* The rest of $K$ is cut into the finitely many measurable pieces on which the set of sheets through $y$ and the pairs with equal oriented tangent space are constant `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration.piece` `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration.isMeasIn_piece` `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration.pairwise_disjoint_piece`. *Step 3.* On a piece all the integrals are $\pm$ the integral over one sheet, and the signed sum of the weights vanishes because a tangent space has two orientations `Chapter9/GapP2_E1_SheetIntegration · EpGroupoid.SheetIntegration.sum_sign_weight_eq_zero`. The families need not pass through one point. For two oriented tame structures describing $\hat{T}_\Theta$ this gives Lemma 9.5.7 `Chapter9/GapP2_E1_LocalMeasure · lemma_9_5_7`.

**(9.21) and Lemmas 9.5.9–9.5.11.** *The local measures.* An oriented tame structure is a tame structure (natural representation, properness, tame sheets) with $n$-dimensional oriented sheets `Chapter9/GapP2_E1_LocalMeasure · OrientedTameStructure`; it describes $\hat{T}_\Theta$ if its oriented sum is $\hat{T}_\Theta$ near $x$ `Chapter9/GapP2_E1_LocalMeasure · Represents`. The weighted sum $\sum_i \sigma_i \int_{A\cap M_i} \omega|(M_i,o_i)$ `Chapter9/GapP2_E1_LocalMeasure · sheetSum` divided by $\#G_x$ or $\#G_x^{\mathrm{eff}}$ is $\mu_\omega^U$, $\mu_\omega^{\mathrm{eff},U}$ `Chapter9/GapP2_E1_LocalMeasure · localMeasure` `Chapter9/GapP2_E1_LocalMeasure · localMeasureEff`, linear in $\omega$ and additive in the set `Chapter9/GapP2_E1_LocalMeasure · sheetSum_hasSum`. *Lemma 9.5.9* `Chapter9/GapP2_E1_LocalMeasure · lemma_9_5_9`: with the slice $V$ of §IX.20 and the decomposition $M_U \cap \pi^{-1}(K) = \bigsqcup_{r\in R} \varphi_r(M_U \cap V \cap \pi^{-1}(K))$, each $\varphi_r$ preserves the integrals, its tangent maps being those of morphisms `Chapter9/GapP2_E1_LocalMeasure · NaturalRep.isMorTangentOn_Φ` `Chapter9/GapP2_E1_LocalMeasure · sheetSum_image`, so the sum over $U$ is $\#R$ times the sum over $V$ `Chapter9/GapP2_E1_LocalMeasure · sheetSum_preimage_eq_card_mul_slice`; with $\#R \cdot \#G_y = \#G_x$ the normalized measures agree. The text takes the restricted structure on $V$; by Lemma 9.5.7 any structure at $y$ describing $\hat{T}_\Theta$ serves, and the text's proof uses Lemma 9.5.7 and the naturality of $\hat{T}_\Theta$ here. *Lemma 9.5.10* `Chapter9/GapP2_E1_LocalMeasure · lemma_9_5_10`: the same with (9.23), $\#R \cdot \#G_y^{\mathrm{eff}} = \#G_x^{\mathrm{eff}}$ under the local unique continuation property. *Lemma 9.5.11* `Chapter9/GapP2_E1_LocalMeasure · lemma_9_5_11`: the local sc-diffeomorphism $\hat{\varphi} : U \to U'$ of a morphism has, at each point, the tangent map of a morphism, so it carries the structure on $U$ to one on $U'$ with the same sums `Chapter9/GapP2_E1_LocalMeasure · sheetSum_preimage_eq_of_morDiffeo`, and $\#G_x = \#G_{x'}$ (§IX.20). The effective clause holds without the local unique continuation property, only $\#G_x^{\mathrm{eff}} = \#G_{x'}^{\mathrm{eff}}$ being used. Remark 9.5.8 has no content beyond Lemma 9.5.11.

**Theorems 9.5.4 and 9.5.5** `Chapter9/GapP2_E1_BranchedIntegral · theorem_9_5_4` `Chapter9/GapP2_E1_BranchedIntegral · theorem_9_5_5`. *Step 1, measurability.* The local sc-diffeomorphism $t \circ (s|U(\varphi))^{-1}$ of a morphism has the tangent maps of the morphisms of its sheet `Chapter9/GapP2_E1_BranchedIntegral · isMorTangentOn_hat`, and the preimage of $K \in L(S)$ has measurable traces on the sheets, by Lemma 9.5.2 `Chapter9/GapP2_E1_BranchedIntegral · OrientedTameStructure.isMeas_preimage`. *Step 2, overlaps.* Near two points $z \in U$, $z' \in U'$ over one point of $|X|$ the two local measures agree, combining Lemmas 9.5.9 and 9.5.11 `Chapter9/GapP2_E1_BranchedIntegral · exists_slice_pair` `Chapter9/GapP2_E1_BranchedIntegral · exists_local_agreement`; since the sheets are Lindelöf, countably many such neighbourhoods cover a measurable $K \subseteq |M_U| \cap |M_{U'}|$, and the two structures give the same measure of $K$ `Chapter9/GapP2_E1_BranchedIntegral · overlap`. The text does not mention that countably many pieces suffice. *Step 3, gluing.* Each structure gives a signed measure on $(S, L(S))$, linear in $\omega$ `Chapter9/GapP2_E1_BranchedIntegral · localSignedMeasure`. On a compact space with a σ-algebra containing the Borel sets, linear local expressions on the members of an open cover that agree on overlaps glue to exactly one linear map into signed measures: cut $K$ into the pieces $W_k \setminus \bigcup_{j<k} W_j$ of a finite subcover and add `Chapter9/GapP2_E2MeasureGlue · existsUnique_linearMap_signedMeasure_of_local` `Chapter9/GapP2_E2MeasureGlue · existsUnique_linearMap_signedMeasure_of_openCover`. This is the text's "defines our desired signed measure". *Step 4.* Hence exactly one linear $\omega \mapsto \mu_\omega$ with the local formula for every oriented tame structure with sheets in the class describing $\hat{T}_\Theta$ and every measurable $K \subseteq |U(x)|$ `Chapter9/GapP2_E1_BranchedIntegral · existsUnique_branchedMeasure` `Chapter9/GapP2_E1_BranchedIntegral · branchedMeasure` `Chapter9/GapP2_E1_BranchedIntegral · branchedMeasureEff`, given such a structure at every point of the support (`hcov` in this abstract form, the class condition of the integration). *Definition 9.5.6* `Chapter9/GapP2_E1_BranchedIntegral · branchedIntegral` `Chapter9/GapP2_E1_BranchedIntegral · branchedIntegralEff`, and the identity after Theorem 9.5.5, $\mu^{\mathrm{eff},U} = (\#G_x/\#G_x^{\mathrm{eff}})\,\mu^U$ `Chapter9/GapP2_E1_BranchedIntegral · OrientedTameStructure.localMeasureEff_eq`. *Proposition 9.5.13 (1)*: for effective $X$ the two integrals agree `Chapter9/GapP2_E1_BranchedIntegral · proposition_9_5_13_1`. A witness in dimension 0: points as sheets, the integral the signed value of an invariant function `Chapter9/GapP2_E1_PointIntegration · pointIntegration`.

## IX.32 The integral of sc-differential forms over oriented sheets, and Theorems 9.5.4 and 9.5.5 as printed

**The text** integrates "$\omega|(M_i, o_i)$" over measurable subsets of the sheets without constructing the integral; the sheets are manifolds with boundary and corners and $\omega$ an sc-differential form on the ep-groupoid (Definition 8.2.1). Here the integral is built for $n \geq 1$ and shown to be a `SheetIntegration` of §IX.31, first on sheets in one chart, then on sheets covered by countably many charts (gap G.107, closed).

**Evaluating forms.** *Step 1.* An evaluation of forms is a real vector space $\Omega$ with values $\omega(\alpha_1, \dots, \alpha_n)$ on tangent vectors, linear in $\omega$, alternating and multilinear on the tangent space of a submanifold (stated through determinants), invariant under the tangent maps of morphisms, and continuous along continuous frame fields `Chapter9/GapP2_E3_GraphChart · FormEval`. *Step 2.* The forms $\Omega^n_{ep}(X)$ of Definition 8.2.1 are one: the chart expression of $\omega$ is sc-smooth, hence continuous along level-1 lifts, which are continuous along a submanifold since the chart retraction is $\mathrm{sc}^+$ `Chapter9/GapP2_E3_FormEvalEp · MPolyfold.continuousWithinAt_levelLift` `Chapter9/GapP2_E3_FormEvalEp · epFormEval`. *Step 3.* The direct limit $\Omega^n_{ep,\infty}(X)$ of Definition 8.2.2 is one too: a submanifold of $X$ in $X_\infty$ is one of every level $X^i$ with the same tangent spaces and continuous frame fields, so the forms of $X^i$ are evaluated on lifted tangent vectors, compatibly with the steps of the limit `Chapter9/GapP2_E3_FormEvalInfty · IsSmoothSubmanifold.levelOne` `Chapter9/GapP2_E3_FormEvalInfty · epFormsInftyEval`.

**Graph charts.** *Step 1.* A graph chart of an $n$-dimensional submanifold $N$ is a chart $\varphi$ of the M-polyfold containing $N$ and a continuous linear $p : E_0 \to \mathbb{R}^n$, injective on $\varphi(N)$, whose inverse is differentiable within $p(\varphi(N))$ with derivative a linear isomorphism onto the chart image of $T_yN$, continuous in $y$ Chapter9/GapP2_E3_GraphChart · MPolyfold.GraphChart. No corner structure of $N$ is used, and derivatives are taken within a set that need not be open. *Step 2.* Near every point of an $n$-dimensional submanifold there is one: the tangent projection of §IX.28 is bi-Lipschitz on $N$ near the point, and its image is closed Chapter9/GapP2_E3_GraphChartExists · ScRetraction.exists_graphData Chapter9/GapP2_E3_GraphChartExists · exists_graphChart. *Step 3, the integral in a chart.* The density is $\pm\omega(\partial_1, \dots, \partial_n)$ for the coordinate frame, with sign $+$ where the frame is positive for the orientation Chapter9/GapP2_E3_ChartIntegral · density, and $\int_A$ is its Lebesgue integral over the coordinate image of $A \cap N$ Chapter9/GapP2_E3_ChartIntegral · chartInteg; the sign is locally constant on $N$ Chapter9/GapP2_E3_FrameSign · GraphChart.exists_isOpen_isPos. *Step 4, the comparison of two charts.* At a common point of two sheets with $T_yN = T_yN'$ the coordinate change is differentiable within the coordinate image of $N \cap N'$, with derivative expressing one frame in the other Chapter9/GapP2_E3_Transition · GraphChart.exists_transition; the change of variables for maps differentiable on a measurable set Chapter9/GapP2_E3_GraphChart · integral_image_eq_of_det Chapter9/GapP2_E3_Transition · GraphChart.setIntegral_coord_image_eq gives equal integrals where the oriented tangent spaces agree and opposite ones where they are opposite Chapter9/GapP2_E3_ChartIntegral · FormEval.chartInteg_congr Chapter9/GapP2_E3_ChartIntegral · FormEval.chartInteg_neg. With $N = N'$ this is the independence of the graph chart; no smoothness of $N \cap N'$ is needed. This is how the proof of Lemma 9.5.7 compares sheets. *Step 5.* Invariance under the sc-diffeomorphisms whose tangent maps are those of morphisms Chapter9/GapP2_E3_ChartIntegral · FormEval.chartInteg_mapDiffeo.

**The one-chart integration** Chapter9/GapP2_E3_SheetIntegration · FormEval.sheetIntegration. The class consists of the $n$-dimensional submanifolds in one graph chart on which every form is absolutely integrable Chapter9/GapP2_E3_SheetIntegration · FormEval.sheets, the null sets are the $n$-dimensional Hausdorff null sets Chapter9/GapP2_E3_SheetIntegration · FormEval.sheetNull, and the integral is that of Step 3 Chapter9/GapP2_E3_SheetIntegration · FormEval.integ; all fields of a `SheetIntegration` hold, among them the Lindelöf property, through the coordinate homeomorphism Chapter9/GapP2_E3_SheetIntegration · FormEval.hereditarilyLindelofSpace, and the integral vanishes where $\omega$ vanishes on the sheet Chapter9/GapP2_E3_SheetIntegration · FormEval.integ_eq_zero_of_ev_eq_zero. With it, §IX.31 gives Theorems 9.5.4, 9.5.5 for structures whose sheets lie in one graph chart Chapter9/GapP2_E3_TameOpen · theorem_9_5_4_tame Chapter9/GapP2_E3_TameOpen · theorem_9_5_5_tame Chapter9/GapP2_E3_TameOpen · tameBranchedIntegral; such structures exist near every point of the support once some oriented tame structure does, by restriction to a smaller domain Chapter9/GapP2_E3_Covering · exists_restrict_mem_sheets Chapter9/GapP2_E3_TameOpen · HasTameInducedStructure.inter_isOpen Chapter9/GapP2_E3_TameOpen · hcov_of_loc.

**Several charts.** *Step 1, $\int |\omega|$.* In a graph chart, $\int_A |\omega|$ bounds $|\int_A \omega|$, does not depend on the chart, is invariant under morphisms and countably additive Chapter9/GapP2_E4_ChartAbs · abs_chartInteg_le Chapter9/GapP2_E4_ChartAbs · chartAbs_congr Chapter9/GapP2_E4_ChartAbs · chartAbs_mapDiffeo. *Step 2.* A sheet covered by countably many open $V_k$ with $N \cap V_k$ in the one-chart class has the disjoint Borel pieces $P_k = N \cap V_k \setminus \bigcup_{j<k} V_j$ Chapter9/GapP2_E4_MultiChart · ChartCover; the cover is finite in the sense $\sum_k \int_{P_k} |\omega| < \infty$ Chapter9/GapP2_E4_MultiChart · ChartCover.IsFinite, and then $\int_A \omega = \sum_k \int_{A \cap P_k} \omega$ converges absolutely Chapter9/GapP2_E4_MultiChart · ChartCover.gInteg. *Step 3, independence.* Two covers, or two sheets over a common set where their oriented tangent spaces agree or are opposite, give equal or opposite sums, by a double series over the pieces of the two covers Chapter9/GapP2_E4_MultiChart · hasSum_of_double Chapter9/GapP2_E4_MultiChart · ChartCover.gInteg_eq_of. *Step 4.* The class of sheets with a finite cover Chapter9/GapP2_E4_MultiChart · FormEval.msheets carries a `SheetIntegration` Chapter9/

GapP2_E4_MultiChartInteg · FormEval.msInteg Chapter9/GapP2_E4_MultiChartInteg · FormEval.msheetIntegration that contains the one-chart class and agrees with it there Chapter9/GapP2_E4_MultiChartInteg · FormEval.sheets_subset_msheets Chapter9/GapP2_E4_MultiChartInteg · FormEval.msInteg_eq_integ; Theorems 9.5.4, 9.5.5 follow for structures with sheets in this class Chapter9/GapP2_E4_MultiChartTheorem · theorem_9_5_4_multi Chapter9/GapP2_E4_MultiChartTheorem · theorem_9_5_5_multi Chapter9/GapP2_E4_MultiChartTheorem · multiBranchedIntegral Chapter9/GapP2_E4_MultiChartTheorem · multiBranchedIntegralEff. Compact sheets are in it Chapter9/GapP2_E4_MultiChartCompact · msheets_of_isCompact.

**The finiteness the text uses tacitly** Chapter9/GapP2_E4_FiniteTheorem · msheets_of_orientedTameStructure. The text integrates over measurable subsets of the sheets, which presupposes $\int |\omega| < \infty$. For the sheets of an oriented tame structure of a compact $n$-dimensional $\Theta$ it holds. *Step 1.* The support is locally compact, being locally a finite union of finite-dimensional submanifolds; a compact $K' \subseteq \operatorname{supp}(\Theta)$ meets every orbit of the support since $|\operatorname{supp}(\Theta)|$ is compact; and $\operatorname{supp}(\Theta) \cap U \subseteq s(\{\varphi \mid s(\varphi) \in \operatorname{cl} U,\ t(\varphi) \in K'\})$, compact by the properness of $\operatorname{cl} U$ (Definition 7.5.8) Chapter9/GapP2_E4_FiniteTheorem · exists_isCompact_supp_inter. *Step 2.* An $n$-dimensional submanifold inside a compact part of the support is covered by finitely many one-chart sheets of structures at points of it Chapter9/GapP2_E4_FiniteTheorem · msheets_of_subset_isCompact, and $\int |\omega|$ over it is dominated by the finitely many one-chart integrals Chapter9/GapP2_E4_FiniteDominated · chartAbs_eq_of_borel Chapter9/GapP2_E4_FiniteDominated · isFinite_of_dominated. *Step 3, Theorems 9.5.4 and 9.5.5 as printed* Chapter9/GapP2_E4_FiniteTheorem · theorem_9_5_4_full Chapter9/GapP2_E4_FiniteTheorem · theorem_9_5_5_full: there is exactly one linear $\omega \mapsto \mu_\omega$ into the signed measures on $(S, L(S))$ with $\mu_\omega(K) = \mu_\omega^U(K)$ for every oriented tame structure describing $\hat{T}_\Theta$ and every measurable $K \subseteq |U(x)|$; the $n$-dimensionality of $\Theta$ follows from the hypothesis Chapter9/GapP2_E4_FiniteTheorem · isNDimensional_of_loc.

**The reading `hloc` of "oriented, tame".** The theorems take as hypothesis that every point of the support has an oriented tame structure describing $\hat{T}_\Theta$. Without boundary it follows from the text's hypotheses Chapter9/GapP2_E4_Hloc · hloc_of_degIndex_eq_zero: the oriented structure of Definition 9.3.2 has $n$-dimensional sheets, and restricted to the domain of a natural representation inside an open set with the properness property it is tame, every submanifold having a tame induced structure when $d_X = 0$ Chapter9/GapP2_E4_Hloc · exists_orientedTameStructure_of_degIndex_eq_zero Chapter9/GapP2_E4_FiniteCircle · theorem_9_5_4_full_of_degIndex_eq_zero. With boundary, Definition 9.3.6 gives tame sheets of *some* structure and Definition 9.3.2 oriented sheets of *another*; that the oriented structure can be taken tame is the reading of "tame".

**Part I's integral, and a witness.** Over a tame sheet the integral equals Part I's manifold integral of the restricted form for the corresponding tangent orientation Chapter9/GapP2_E3_ManifoldBridge · bridgeOrientation Chapter9/GapP2_E3_ManifoldBridge · isContinuous_bridgeOrientation Chapter9/GapP2_E3_ManifoldBridge2 · FormEval.integ_eq_manifoldIntegral, chart by chart of a partition of unity by the change of variables between graph and corner coordinates Chapter9/GapP2_E3_ManifoldBridge · FormEval.chartInteg_eq_setIntegral_corner; for classes of $\Omega^n_{ep,\infty}(X)$ represented on any level see Chapter9/GapP2_E5_LevelRestrict · epFormsInftyEval_integ_eq_integral. The unit circle in $\mathbb{R}^2$, the image of the radial retraction $r \mapsto r/|r|$ of $\mathbb{R}^2 \setminus \{0\}$, oriented counter-clockwise Chapter9/GapP2_E4_Circle · rnPolyfold_isSubmanifoldOfDim_unitCircle2 Chapter9/GapP2_E4_Circle · circleOrientation, satisfies all hypotheses with compact nonempty support Chapter9/GapP2_E4_Circle · circle_witness Chapter9/GapP2_E4_FiniteCircle · circle_theorem_9_5_4_full, and with $U(x) = \mathbb{R}^2$ and the whole circle as one sheet $\int_{\hat{\Theta}} \omega = \int_{S^1} \omega$ Chapter9/GapP2_E4_CircleMulti · circle_multiBranchedIntegral.

## IX.33 (9.22) and Proposition 9.5.13 (2)

**(9.22)** (V9.69, explicit). *If $\tau = f \cdot \omega$, then $d\mu_\tau = \bar{f}\, d\mu_\omega$, with $\bar{f} \circ \pi = f$.* **Reading.** $f \cdot \omega$ is read through the evaluation, $\tau(\alpha) = f(y)\,\omega(\alpha)$ on the frames of the tangent spaces $T_yN$ of the sheets `Chapter9/GapP2_E3_Density · IsMul`, witnessed for constant $f$ `Chapter9/GapP2_E3_Density · isMul_smul`; the form $f \cdot \omega$ for a non-constant sc-smooth $f$ is not built. *Step 1, local, exact.* In a graph chart, $\int_A \tau$ is the integral of $f$ times the density of $\omega$ `Chapter9/GapP2_E3_Density · integ_eq_setIntegral_mul`, so $\mu_\tau^U(K)$ is (9.21) for $\omega$ with $f$ inserted in the integrands `Chapter9/GapP2_E3_Density · OrientedTameStructure.localMeasure_eq_sum_setIntegral_mul`. *Step 2, the estimate.* If $|\bar{f} - a| \le \varepsilon$ on $K$, then $|\mu_\tau(K) - a\,\mu_\omega(K)| \le \varepsilon\, M_\omega(K)$ with $M_\omega(K) = \frac{1}{\#G_x} \sum \sigma_i \int_{K_i} |\omega|$, additive in $K$ `Chapter9/GapP2_E3_Density · OrientedTameStructure.abs_localMeasure_sub_le` `Chapter9/GapP2_E3_Density · abs_branchedMeasure_sub_le` `Chapter9/GapP2_E4_Density · OrientedTameStructure.absSheetSum_union`. *Step 3, integrals against a signed measure.* Mathlib's integral against a vector measure is, for a signed measure $s$, $\int_K g\, ds^+ - \int_K g\, ds^-$ for the Jordan decomposition `Chapter9/GapP2_E4_DensitySigned · SignedDensity.setIntegral_eq_jordan`. If a second signed measure $t$ satisfies the estimate of Step 2 against $s$ on the measurable subsets of a set, then $t(K) = \int_K g\, ds$, by Riemann–Stieltjes sums over the partition of $K$ by the values of $g$ and $\varepsilon \to 0$ `Chapter9/GapP2_E4_DensitySigned · SignedDensity.eq_setIntegral_of_estimate`; on a compact space, estimates near every point suffice `Chapter9/GapP2_E4_DensitySigned · SignedDensity.eq_setIntegral_of_local_estimate`. *Step 4.* So for $\bar{f}$ bounded and $L(S)$-measurable, continuous for instance, $\mu_\tau(K) = \int_K \bar{f}\, d\mu_\omega$ for every $K \in L(S)$, and the same for the effective measures `Chapter9/GapP2_E4_Density · eq_setIntegral_of_sheetSum` `Chapter9/GapP2_E4_Density · branchedMeasure_eq_setIntegral_mul` `Chapter9/GapP2_E4_Density · branchedMeasureEff_eq_setIntegral_mul` `Chapter9/GapP2_E4_Density · branchedMeasure_eq_jordan`.

**Proposition 9.5.13 (2)** (V9.57, restated). *Under the local unique continuation property, $\oint_{\hat{\Theta}} \omega = \int_{\hat{\Theta}'} \omega'$ on the reduced ep-groupoid $X_R$.* **The text's proof**: $X$ and $X_R$ have the same object M-polyfold, $\Theta' = \Theta$, "we can take the same local branching structure", and $\#G_x^{\mathrm{eff}} = \#G'^{\mathrm{eff}}_x$. **The formalized proof.** *Step 1, the natural action of $G_x^R$,* which "the same local branching structure" needs and the text does not construct. For a natural representation of $G_x$ on $U(x)$ with the property of Definition 7.2.18, $\Phi^R([g]) = \Phi(g)$ and $\Gamma^R([g], y) = [\Gamma(g, y)]$ define one of $G_x^R = G_x/T_x$ on the same $U(x)$ `Chapter9/GapP2_E4_ReducedRep · NaturalRep.toReduced`: $\Phi$ descends because $T_x = \ker \Phi$ `Chapter9/GapP2_E4_ReducedRep · Φ_eq_of_reducedIsotropyHom_eq`, $\Gamma$ because $\Gamma(gk, y) = \Gamma(g, y) \circ \Gamma(k, y)$ with $\Gamma(k, y)$ trivial in $X_R$ `Chapter9/GapP2_E4_ReducedRep · reducedπ_Γ_eq_of_reducedIsotropyHom_eq`, and the uniqueness in Theorem 7.1.13 (3) holds because $[\Gamma(a, y)] = [\Gamma(b, y)]$ forces $a = bc$ with $\Phi(c) = \mathrm{id}$ near $y$, and the local unique continuation property gives $c \in T_x$ `Chapter9/GapP2_E4_ReducedRep · reducedIsotropyHom_eq_of_reducedπ_Γ_eq`. *Step 2, the same structures.* The properness property, tame and oriented tame structures carry over with the same $U$, sheets, weights and orientations `Chapter9/GapP2_E4_ReducedRep · HasPropernessProperty.toReduced` `Chapter9/GapP2_E4_ReducedRep · OrientedTameStructure.toReduced`, and $\#G_x^R = \#G_x^{\mathrm{eff}}$ `Chapter9/GapP2_E4_ReducedRep · card_isotropy_reduced_eq_card_effIsotropy`. *Step 3, the same integrals.* The tangent map of $[\varphi]$ is that of $\varphi$ `Chapter9/GapP2_E4_ReducedRep · tangentMor_reducedπ`, so a form of $X$ is one of $X_R$ and a sheet integration on $X$ is one on $X_R$ with the same integrals `Chapter9/GapP2_E4_ReducedRep · SheetIntegration.toReduced`; the orientation carries over `Chapter9/GapP2_E4_ReducedRep · IsOrientation.toReduced`. *Step 4, the same σ-algebra.* $|X| \cong |X_R|$ and $|\mathrm{supp}(\Theta)|$ in both are identified `Chapter9/GapP2_E4_ReducedIntegral · orbitHomeomorphReduced` `Chapter9/GapP2_E4_ReducedIntegral · orbitSuppHomeomorphReduced`, carrying $L(S)$ to $L(S)$ `Chapter9/GapP2_E4_ReducedIntegral · measurableSet_image_orbitSuppHomeomorphReduced`. *Step 5.* By the uniqueness in Theorem 9.5.4 on $X_R$, $\mu^{\mathrm{eff}}_{\hat{\Theta},\omega}(K) = \mu_{\hat{\Theta}',\omega'}(K)$ for $K \in L(S)$ `Chapter9/GapP2_E4_ReducedIntegral · proposition_9_5_13_2_measure`, and with $K = S$ the proposition `Chapter9/GapP2_E4_ReducedIntegral · proposition_9_5_13_2`. The local unique continuation property also makes the morphism space of $X_R$ Hausdorff, so that $X_R$ is

an ep-groupoid. No hypothesis is added. Witness: the circle Chapter9/GapP2_E4_ReducedCircle · circle_proposition_9_5_13_2.

## IX.34 Definition 9.5.14 (2017: 9.17) with $(n-1)$-dimensional null sets, and Theorems 9.5.15, 9.5.16 (2017: 9.4, 9.5): the boundary measures

**The text.** For $K \subseteq |\partial M_U|$ the preimages lie in the $\partial M_i$, manifolds with boundary and corners, and measure zero is $(n-1)$-dimensional Lebesgue measure zero in every boundary face (p. 448). Then (Theorem 9.5.15) there is a linear map $\omega \mapsto \mu^{\partial}_{\omega}$ into the signed measures on $(\partial S, L(\partial S))$, "uniquely characterized" by $\mu^{\partial}_{\omega}(K) = \frac{1}{\#G_x}\sum_i \sigma_i \int_{K_i} \omega|(\partial M_i, o_{\partial M_i})$, and Theorem 9.5.16 the same with $\#G_x^{\mathrm{eff}}$ under the local unique continuation property; the proof is "using the same techniques" as Theorem 9.5.4.

**The null sets of the boundaries** (gap G.108, closed; V9.52, V9.53). *Step 1.* As for the sheets (§IX.28), $(n-1)$-dimensional measure zero is read in the M-polyfold charts with the $(n-1)$-dimensional Hausdorff measure. *Step 2, positivity.* In corner coordinates of dimension $m+1$, near a point whose only vanishing coordinate is $y_i$, the chart image of $\{y_i = 0\}$ is an open piece of a coordinate hyperplane, not $m$-null Chapter9/GapP2_E5_BoundaryNull · MPolyfold.CornerCoords.not_hNull_chart_coord_eq_zero; so a nonempty $\partial M$ of a tame $n$-dimensional sheet is not $(n-1)$-null Chapter9/GapP2_E5_BoundaryNull · MPolyfold.HasCornerStructureOfDim.not_isHNull_boundary Chapter9/GapP2_E5_BoundaryNull · MPolyfold.not_isHNull_submanifoldBoundary. *Step 3.* These null sets form a null-set system on the class of the boundaries of tame sheets with no hypothesis Chapter9/GapP2_E5_BoundaryNullSigma · MPolyfold.tameBoundaries Chapter9/GapP2_E5_BoundaryNullSigma · MPolyfold.boundaryNullSets; for $n = 1$ they are the counting null sets Chapter9/GapP2_E5_BoundaryNullSigma · MPolyfold.boundaryNullSets_one_isNull_iff Chapter9/GapP2_E2BoundaryNull · SheetNullSets.point. *Step 4, well defined.* For $n$-dimensional $\Theta$, $K$ has measure zero in $\partial S$ iff near each point of $\mathrm{supp}(\partial\Theta)$ a tame structure has $\partial M_i \cap \pi^{-1}(K)$ null for all $i$, and then every tame structure has Chapter9/GapP2_E5_BoundaryNullSigma · EpGroupoid.isMeasureZeroBoundary_boundaryNullSets_iff Chapter9/GapP2_E5_BoundaryNullSigma · EpGroupoid.isHNull_boundary_of_isMeasureZeroBoundary; Definition 9.5.14 is Chapter9/GapP2_E5_BoundaryNullSigma · EpGroupoid.lebesgueSigmaBoundaryH. *Step 5, the corners.* A codimension-2 coordinate subspace of $\mathbb{R}^n$ is $(n-1)$-null, so the points with $d_M \geq 2$ are $(n-1)$-null Chapter9/GapP2_E5_BoundaryNullSigma · MPolyfold.isHNull_two_le_subDegIndex and $\partial M$ minus the stratum $\{d_M = 1\}$ is null Chapter9/GapP2_E5_BoundaryNullSigma · MPolyfold.isHNull_submanifoldBoundary_diff_boundaryStratum: the boundary integral lives on the stratum. *Face by face.* The text's notion for a face-structured sheet is formalized as a proper σ-ideal Chapter9/GapP2_E2BoundaryNull · MPolyfold.IsFaceNull Chapter9/GapP2_E2BoundaryNull · MPolyfold.not_isFaceNull_face; its comparison with the ambient reading is not formalized, and no statement needs it. *Witness with $\partial S \neq \emptyset$:* $\Theta = 1$ on $[0,\infty)$, tame and 1-dimensional Chapter9/GapP2_E2BoundaryNull · halfLineTame, with $|0| \in \partial S$ Chapter9/GapP2_E2BoundaryNull · halfLine_orbitSupp_boundary_nonempty and $L(\partial S)$ the Borel sets Chapter9/GapP2_E2BoundaryNull · halfLine_lebesgueSigmaBoundary.

**The boundary sheets.** *Step 1.* For an oriented tame structure of dimension $m+1$, $\partial_1 M_i = \{d_{M_i} = 1\}$ is an $m$-dimensional submanifold (§IX.30) with the orientation $o_{\partial M_i}$ Chapter9/GapP2_E6_BoundarySheets · OrientedTameStructure.boundarySheet Chapter9/GapP2_E6_BoundarySheets · OrientedTameStructure.boundaryOrient. *Step 2, (9.12) with the stratum.* If the structure describes $\hat{T}_\Theta$, then $\hat{T}_{\partial\Theta}(y) = \sum_{y\in\partial_1 M_i} \sigma_i T_y(\partial_1 M_i, o_{\partial M_i})$ Chapter9/GapP2_E6_BoundarySheets · OrientedTameStructure.boundaryOrientation_eq_sum; the left side depends on $\hat{T}_\Theta$ only. *Step 3.* Near each of its points a boundary sheet lies in the one-chart class for $m$-forms Chapter9/GapP2_E6_BoundarySheets · OrientedTameStructure.exists_boundarySheet_mem_

sheets. *Step 4, the traces.* For $K \in L(\partial S)$ the trace of $\pi^{-1}(K)$ on $\partial_1 M_i$ is measurable, by induction over the generators of $L(\partial S)$: Borel generators have Borel traces, null generators null traces by Step 4 above and $\partial_1 M_i \subseteq \partial M_i$ Chapter9/GapP2_E7_BoundaryMeasure · OrientedTameStructure.isMeasIn_boundarySheet.

**The machinery of §IX.31 for sheet families.** The boundary sheets are not the sheets of a branching structure, so (9.20), (9.21) and Lemmas 9.5.7–9.5.11 are proved again for an oriented sheet family: a natural representation on $U(x)$ and finitely many $m$-dimensional oriented submanifolds with rational weights whose union is $D \cap U(x)$ for a fixed set $D$ closed under morphisms Chapter9/GapP2_E7_SheetFamily · IsMorSaturated Chapter9/GapP2_E7_SheetFamily · OrientedSheetFamily, described by a field $T'$ carried along by morphisms Chapter9/GapP2_E7_SheetFamily · IsNaturalOn Chapter9/GapP2_E7_SheetFamily · OrientedSheetFamily.lemma_9_5_7 Chapter9/GapP2_E7_SheetFamily · OrientedSheetFamily.lemma_9_5_9 Chapter9/GapP2_E7_SheetFamily · OrientedSheetFamily.lemma_9_5_11. An oriented tame structure is the instance $D = \operatorname{supp}(\Theta)$ Chapter9/GapP2_E7_SheetFamily · OrientedTameStructure.toFamily Chapter9/GapP2_E7_SheetFamily · isNaturalOn_of_isOrientation. The overlaps and the gluing go as in §IX.31 Chapter9/GapP2_E7_FamilyMeasure · OrientedSheetFamily.overlap Chapter9/GapP2_E7_FamilyMeasure · existsUnique_familyMeasure. *The boundary family.* $D$ is the set of points of the support where $\hat{T}_\Theta$ has a term of reduced codimension $1$ Chapter9/GapP2_E7_BoundaryFamily · boundaryStratumSet, closed under morphisms Chapter9/GapP2_E7_BoundaryFamily · isMorSaturated_boundaryStratumSet since tangent maps of morphisms preserve the reduced codimension; for a structure describing $\hat{T}_\Theta$ it is the union of the $\partial_1 M_i$ in $U(x)$, the weights being positive. $\hat{T}_{\partial\Theta}$ is natural Chapter9/GapP2_E7_BoundaryFamily · isNaturalOn_boundaryOrientation, and the $(\partial_1 M_i, o_{\partial M_i})$, $(\sigma_i)$ form a family describing it Chapter9/GapP2_E7_BoundaryFamily · OrientedTameStructure.boundaryFamily Chapter9/GapP2_E7_BoundaryFamily · OrientedTameStructure.represents_boundaryFamily.

**Theorems 9.5.15 and 9.5.16, first form** (V9.67; the form with the text's hypotheses is in §§IX.36–IX.38) Chapter9/GapP2_E7_BoundaryMeasure · theorem_9_5_15_hyp Chapter9/GapP2_E7_BoundaryMeasure · theorem_9_5_16_hyp. *For every sheet integration whose null sets are the $m$-dimensional ones, $n = m + 1$, $m \geq 1$, there is exactly one linear $\omega \mapsto \mu^\partial_\omega$ into the signed measures on $(\partial S, L(\partial S))$ with the printed local formula,* and for the integration over countably many graph charts Chapter9/GapP2_E7_BoundaryMeasure · theorem_9_5_15_msheets Chapter9/GapP2_E7_BoundaryMeasure · theorem_9_5_16_msheets. $\partial S$ is compact, being closed in $S$ Chapter9/GapP2_E2MeasureGlue · EpGroupoid.isCompact_orbitSupp_boundaryTheta, and the gluing is Chapter9/GapP2_E2MeasureGlue · EpGroupoid.theorem_9_5_15_of_overlap. The sum runs over the nonempty $\partial_1 M_i$, the empty set need not belong to the class Chapter9/GapP2_E7_BoundaryMeasure · OrientedSheetFamily.nonemptyPart, and $n$-dimensionality is stated explicitly. **Added hypothesis `hcov`**, not derived from the text: at every point of $\operatorname{supp}(\partial\Theta)$ an oriented tame structure describing $\hat{T}_\Theta$ whose nonempty boundary sheets lie in the class of the integration. For the multi-chart class this is the finiteness of $\int |\omega|$ over $\partial_1 M_i$ near the corners of $M_i$, which the text uses tacitly; the difficulty lies where $\partial_1 M_i$ approaches the corners, and compact boundary sheets satisfy it Chapter9/GapP2_E7_BoundaryMeasure · OrientedTameStructure.boundarySheet_mem_msheets_of_isCompact Chapter9/GapP2_E7_BoundaryMeasure · hcov_msheets_of_isCompact. Witnesses: $\Theta = 0$ Chapter9/GapP2_E7_BoundaryMeasure · hcov_zero, and with $\operatorname{supp}(\partial\Theta) \neq \emptyset$ the closed upper hemisphere of §IX.35 Chapter9/GapP2_E8W_Witness · e8w_hcov. Since 2026-11-13 `hcov` is derived from the text's hypotheses (§§IX.36–IX.37), and the case $m = 0$ is §IX.38 (gap G.109, closed).

# IX.35 Theorems 9.5.17, 9.5.18 (2017: 9.6, 9.7): Stokes' theorem, and the hemisphere

**The text, paraphrased (pp. 449–451).** For $X$ with sc-smooth partitions of unity, $\Theta$ tame, oriented and compact of dimension $n$, and $\omega$ an sc-differential form of degree $n-1$, $\int_{\hat{\Theta}} d\omega = \mu^{\partial}_{\omega}(\partial S)$; the effective version asks in addition for the local unique continuation property. The proof covers $S$ by finitely many $|U(x_k)|$ carrying oriented branching structures $(M_i, o_i)_{i\in I_k}$, takes an sc-smooth partition of unity $\beta_0, \dots, \beta_\ell$ constant on the orbits and subordinate to the saturations $U_k^*$ and a further saturated $U_0^*$, puts $\omega_k = \beta_k\omega$, applies the classical Stokes theorem on each sheet, and sums: $\mu_{d\omega}(S) = \sum_k c_k \sum_{i\in I_k} \sigma_i \int_{(M_i,o_i)} d\omega_k = \sum_k c_k \sum_{i\in I_k} \sigma_i \int_{\partial(M_i,o_i)} \omega_k = \sum_k \mu^{\partial}_{\omega_k}(\partial S) = \mu^{\partial}_{\omega}(\partial S)$, with $c_k = 1/\sharp G_{x_k}$ or $1/\sharp G^{\mathrm{eff}}_{x_k}$.

**The Lean proof, step by step.**

*Step 1 (the cover and the partition of unity).* From the sc-smooth partitions of unity of Definition 7.5.5 and the compactness of $S$ one gets finitely many structures and $\beta_0, \dots, \beta_\ell$ as in the text, with $U_0^* = \pi^{-1}(|X| \setminus S)$ `Chapter9/GapP2_E5_Stokes · EP.exists_stokes_partition`. The product $\beta\omega$ is a form of $\Omega^m_{ep,\infty}(X)$ `Chapter9/GapP2_E5_FunForm · EP.epFormsInftySmul`, and the restriction to a level commutes with the evaluation used by the integral `Chapter9/GapP2_E5_LevelRestrict · formsInftyRestrict_epFormsInftyToObj Chapter9/GapP2_E5_LevelRestrict · epFormsInftyEval_integ_eq_integral`.

*Step 2 (Stokes on one sheet).* The sheets $M_i$ need not be compact. What makes the classical theorem applicable is that $M_i \cap \operatorname{supp}\beta_k$ is compact, which the text does not state; it follows from the properness of $U(x) \to |U(x)|$ and the compactness of $\Theta$ `Chapter9/GapP2_E2Stokes · EpGroupoid.NaturalRep.exists_ultrafilter_le_nhds Chapter9/GapP2_E2Stokes · EpGroupoid.TameBranchingStructure.isCompact_inter`. Then $\int_M (d\omega)|_M = \int_{\partial M} \omega|_M$ on a σ-compact oriented sheet with corners when the restriction has compact support `Chapter9/GapP2_E2Stokes · MPolyfold.stokes_formsInftyRestrict_of_isCompact Chapter9/GapP2_E2Stokes · MPolyfold.OrientedSheet.stokes`. A sheet of an oriented tame structure is such a sheet, with the tangent orientation of $o_i$ `Chapter9/GapP2_E5_StokesSheet · OrientedTameStructure.orientedSheet`, so $\int_{(M_i,o_i)} d(\beta\omega) = \int_{\partial(M_i,o_i)} \beta\omega$ `Chapter9/GapP2_E5_StokesSheet · EP.integ_d_smul_eq_boundaryIntegral`, and on one structure $\Phi(d(\beta\omega))(S) = c(x)\sum_i \sigma_i \int_{\partial(M_i,o_i)} \beta\omega$ for every $\Phi$ with the local formula of Theorem 9.5.4 or 9.5.5 `Chapter9/GapP2_E5_StokesSheet · EP.stokes_one_structure` (the second and third equalities of the display). The term $k = 0$ vanishes `Chapter9/GapP2_E5_StokesSheet · EP.stokes_off_support`.

*Step 3 (the sum).* Summing over $k$ gives the first three equalities of the display `Chapter9/GapP2_E5_Stokes · EP.stokes_of_partition`, hence Theorems 9.5.17 and 9.5.18 with the right-hand side written as the sum of boundary integrals `Chapter9/GapP2_E5_Stokes · EP.theorem_9_5_17_boundary_sum Chapter9/GapP2_E5_Stokes · EP.theorem_9_5_18_boundary_sum`. The measure $\mu_{d\omega}$ is local: it vanishes on sets above which the integrals vanish `Chapter9/GapP2_E5_MeasureLocal · measure_eq_zero_of_integ_eq_zero Chapter9/GapP2_E5_MeasureLocal · measure_univ_eq_of_integ_eq_zero`. The last two equalities of the display are two properties of $\omega \mapsto \mu^{\partial}_{\omega}(\partial S)$: the local formula for $\beta\omega$ with $\operatorname{supp}\beta \subseteq U(x)^*$, and vanishing when $\operatorname{supp}\beta$ misses $\operatorname{supp}\Theta$. Every linear functional with these two properties satisfies Stokes' theorem `Chapter9/GapP2_E5_Stokes · EP.theorem_9_5_17_of_boundary Chapter9/GapP2_E5_Stokes · EP.theorem_9_5_18_of_boundary`, and such a functional exists `Chapter9/GapP2_E5_Stokes · EP.exists_boundary_functional`.

*Step 4 (every dimension $n \geq 1$).* The statements of Step 3 were first proved for $n \geq 2$, because in degree 0 the evaluation on frames is cut off. The product rule $(\beta\omega)|_N = \beta\,(\omega|_N)$ is proved in every degree `Chapter9/GapP2_E6_StokesDim1 · EP.formsInftyRestrict_epFormsInftySmul_all`, and the

statements hold for $n = 1$ as well Chapter9/GapP2_E6_StokesDim1 · EP.stokes_one_structure_all Chapter9/GapP2_E6_StokesDim1b · EP.stokes_of_partition_all Chapter9/GapP2_E6_StokesDim1b · EP.theorem_9_5_17_boundary_sum_all Chapter9/GapP2_E6_StokesDim1b · EP.theorem_9_5_18_boundary_sum_all Chapter9/GapP2_E6_StokesDim1b · EP.exists_boundary_functional_all.

*Step 5 (the classical boundary integral is the integral over the stratum).* The text's $\int_{\partial(M_i,o_i)}$ is Part I's integral over the faces $\{y_l = 0\}$ of corner charts with signs; $\mu^\partial$ of §IX.34 integrates over the stratum $\partial_1 M_i = \{d_{M_i} = 1\}$ with the boundary orientation $o_{\partial M_i}$ of Definition 9.3.8. The text identifies the two without comment. The face frame $T\iota(\iota_l e_1), \dots, T\iota(\iota_l e_m)$ is positive for $o_{\partial M}$ exactly when the sign of the face matches the reading of $o$ in the chart Chapter9/GapP2_E8F_FaceFrame · MPolyfold.faceFrame_mem_stratumOrientation_iff; the change of variables on a face is Chapter9/GapP2_E8F_FaceChange · FormEval.face_change; the open face piece and the face domain differ by a null set, the corners Chapter9/GapP2_E8F_FaceSum · CornerCoords.faceSet_ae_eq; one term of Part I's integral is an integral over a face piece Chapter9/GapP2_E8F_FaceSum · FormEval.face_term_eq, and a weighted integral over the stratum splits over the faces Chapter9/GapP2_E8F_FaceSum · FormEval.integral_density_eq_sum_face. Hence $\int_{\partial(M_i,o_i)} a = \int_{\partial_1 M_i} a|(\partial_1 M_i, o_{\partial M_i})$ when the nonempty boundary sheet lies in the multi-chart class Chapter9/GapP2_E8F_BoundaryIntegral · EP.boundaryIntegral_eq_msInteg.

*Step 6 (the boundary measures are local).* $\mu^\partial_\omega(D) = 0$ when the integrals of $\omega$ over the boundary sheets above $D$ vanish Chapter9/GapP2_E8_BoundaryLocal · boundaryMeasure_eq_zero_of_integ_eq_zero, and $\mu^\partial_\omega(\partial S) = c(x)\sum_i \sigma_i \int_{\partial_1 M_i} \omega$ for one structure when the integrals vanish off a closed saturated subset of $U(x)^*$ Chapter9/GapP2_E8_BoundaryLocal · boundaryMeasure_univ_eq. For structures at points off $\mathrm{supp}(\partial\Theta)$ the structure is first restricted into a neighbourhood that avoids $\mathrm{supp}(\partial\Theta)$ Chapter9/GapP2_E8_BoundaryLocal · OrientedTameStructure.exists_restrict_subset, and both sides vanish.

*Step 7 (the theorems).* Steps 5 and 6 give the two properties of Step 3 for $\omega \mapsto \mu^\partial_\omega(\partial S)$. The generic statement is Chapter9/GapP2_E8_Stokes · EP.stokes_of_boundaryMeasure; for every functional with the local formula of Theorem 9.5.15 or 9.5.16 it is Chapter9/GapP2_E8_Stokes · EP.theorem_9_5_17_hyp Chapter9/GapP2_E8_Stokes · EP.theorem_9_5_18_hyp; with the boundary measures themselves Chapter9/GapP2_E8_Stokes · EP.boundaryBranchedMeasure Chapter9/GapP2_E8_Stokes · EP.boundaryBranchedMeasureEff it reads $\int_{\hat\Theta} d\omega = \mu^\partial_\omega(\partial S)$ and $\oint_\Theta d\omega = \oint_{\partial S} \omega\, d\mu^\partial$ Chapter9/GapP2_E8_Stokes · EP.theorem_9_5_17_msheets Chapter9/GapP2_E8_Stokes · EP.theorem_9_5_18_msheets, for $\omega \in \Omega^m_{ep,\infty}(X)$ with $m \geq 1$. **Added hypothesis `hcovS`**, not derived from the text: at every point of $\mathrm{supp}\,\Theta$ an oriented tame structure describing $\hat T_\Theta$ whose sheets lie in one graph chart and whose nonempty boundary sheets lie in the multi-chart class; it gives the `hcov` of §IX.34 Chapter9/GapP2_E8_Stokes · EP.hcovB_of_hcovS. It is derived from the text's hypotheses in §IX.36. For Theorem 9.5.18 the hypotheses of Theorem 9.5.5 are kept as in §IX.32.

**The witness.** $X$ is the trivial ep-groupoid of the half-space $[0,\infty)\times\mathbb{R}^2$, and $\Theta = 1$ on the closed upper hemisphere $\{|p| = 1\}$, a compact 2-dimensional tame sheet whose boundary stratum is the equator Chapter9/GapP2_E8W_Hemisphere · hemi Chapter9/GapP2_E8W_Hemisphere · isSubmanifoldOfDim_hemi Chapter9/GapP2_E8W_Hemisphere · boundaryStratum_hemi, with its orientation and $\hat T_\Theta$ Chapter9/GapP2_E8W_Witness · thetaH Chapter9/GapP2_E8W_Witness · hemiOrientation Chapter9/GapP2_E8W_Witness · TH. $X$ has sc-smooth partitions of unity Chapter9/GapP2_E8W_Witness · e8w_hasScPartitionsOfUnity, $\hat T_\Theta$ is an orientation Chapter9/GapP2_E8W_Witness · e8w_isOrientation, `hcov` and `hcovS` hold Chapter9/GapP2_E8W_Witness · e8w_hcov, and $\partial S \neq \emptyset$ Chapter9/GapP2_E8W_Witness · e8w_boundary_nonempty; together Chapter9/GapP2_E8W_Witness · e8w_stokes_hypotheses. So Theorem 9.5.17 applies with a nonempty boundary Chapter9/GapP2_E8_Hemisphere · E8W.e8_hemisphere_stokes. The witness has $m = 1$ and no corners, branching or isotropy. As a trivial ep-groupoid it has the local unique continua-

tion property, so it is a witness for Theorem 9.5.18 as well (§IX.38).

**Since 2026-11-13.** `hcovS` is derived (§IX.36), the case $m = 0$ holds with $\mu^{\partial}$ (§IX.38), and the effective version has the witness of §IX.38; row V9.68 is restated. What remains of gap G.110 is Remark 9.5.20 (§IX.39).

## IX.36 The class conditions derived: the corner estimate, and Theorems 9.5.15–9.5.18 for $n \geq 2$

The first proofs of §§IX.34–IX.35 assume `hcov` and `hcovS`: the nonempty boundary sheets $\partial_1 M_i$ of the structures lie in the multi-chart class, so that $\int |\omega|$ over them is finite. Both follow from the text's hypotheses. The earlier estimate for this (800–1,200 lines) assumed that the closed faces of a sheet would be built as submanifolds; the route taken needs none.

*Step 1, forms are continuous up to the corners.* On a tame sheet $M$, Part I's chart form of $\omega \in \Omega^m_{ep,\infty}(X)$ is a $C^\infty$ form on the Mathlib manifold with corners $M$, read in its corner charts, which lie in the maximal atlas (§IX.30).

*Step 2, the face change of variables with absolute values.* The change of variables of §IX.35, Step 5, is redone with absolute values and lower integrals: on a face set of a corner chart $\kappa$, $|\omega|$ on the graph-chart frame is $|\det|$ times the absolute value of the face pull-back `Chapter9/GapP3_K1_CornerFinite · EpGroupoid.FormEval.abs_ev_faceFrame`. On the preimage of a compact subset of the chart domain the face pull-back is a continuous function on a compact set, so it is integrable `Chapter9/GapP3_K1_CornerFinite · EpGroupoid.integrableOn_facePullback_of_isCompact`.

*Step 3, the corner estimate* `Chapter9/GapP3_K1_CornerFinite · EpGroupoid.FormEval.boundaryStratum_inter_mem_msheets`. If $\partial_1 M \cap V \subseteq K \subseteq M$ with $K$ compact, then $\partial_1 M \cap V$ is in the multi-chart class. Finitely many corner charts, each with a compact piece of its domain, cover $K$. $\partial_1 M \cap V$ is a countable union of face images, so it is Lindelöf and has a cover by countably many graph charts `Chapter9/GapP3_K1_CornerFinite · EpGroupoid.FormEval.exists_chartCover_of_isLindelof`. Each piece of that cover is split by corner chart and pulled back by Step 2. For a fixed corner chart the pieces are disjoint and lie in one compact set of face parameters, so the sum of the integrals of $|\omega|$ is at most a finite sum of integrals over compact sets.

*Step 4, shrinking a structure.* Every oriented tame structure at $x$ restricts to one with the same natural representation, describing the same $\widehat{T}_\Theta$, whose $U(x)$ lies in the interior of a closed neighbourhood $t$ `Chapter9/GapP3_K1_BoundaryCover · EpGroupoid.OrientedTameStructure.exists_restrict_eq`. Then $\partial_1 M_i \subseteq M_i \cap t \cap K$, where $K$ is the compact set containing $\operatorname{supp}\Theta \cap U(x)$ given by the properness property, and $M_i$ meets a compact subset of $U(x)$ in a compact set because $M_i \to U(x)$ is proper `Chapter9/GapP3_K1_BoundaryCover · EpGroupoid.LocalBranchingStructure.isCompact_M_inter`. Step 3 applies `Chapter9/GapP3_K1_BoundaryCover · EP.boundaryStratum_inter_mem_msheets` `Chapter9/GapP3_K1_BoundaryCover · EP.exists_restrict_boundary_mem_msheets`. The natural representation is kept, and with it the local unique continuation property.

*Step 5, the theorems.* So `hcov` and `hcovS` follow from `hloc`, the reading (d) of "oriented, tame" (§IX.23) `Chapter9/GapP3_K1_BoundaryCover · EP.hcov_of_loc` `Chapter9/GapP3_K1_BoundaryCover · EP.hcov_of_loc_isLUC` `Chapter9/GapP3_K1_BoundaryCover · EP.hcovS_of_loc` `Chapter9/GapP3_K1_BoundaryCover · EP.hcovS_of_loc_isLUC`. Theorems 9.5.15 and 9.5.16 hold with the local formula for the structures whose nonempty boundary sheets

are in the class Chapter9/GapP3_K1_Theorems · EP.theorem_9_5_15_class Chapter9/GapP3_K1_Theorems · EP.theorem_9_5_16_class; such structures exist at every point and every structure restricts to one, so the uniqueness is the text's. The canonical measures are Chapter9/GapP3_K1_Theorems · EP.boundaryMeasure Chapter9/GapP3_K1_Theorems · EP.boundaryMeasureEff Chapter9/GapP3_K1_Theorems · EP.boundaryMeasure_spec. **Theorems 9.5.17 and 9.5.18 as printed,** $m \geq 1$ (V9.68) Chapter9/GapP3_K1_Theorems · EP.theorem_9_5_17 Chapter9/GapP3_K1_Theorems · EP.theorem_9_5_18. *For $X$ with sc-smooth partitions of unity, $\Theta$ compact, branched, with an orientation $\hat{T}_\Theta$ and* `hloc`*, and $\omega \in \Omega^m_{ep,\infty}(X)$: $\int_{\hat{\Theta}} d\omega = \mu^\partial_\omega(\partial S)$, with the measure of Theorem 9.5.15; with the local unique continuation property of $X$, the same for $\oint$ and the measure of Theorem 9.5.16* Chapter9/GapP3_K1_Theorems · EP.hloc_hasLUC. The hemisphere of §IX.35 satisfies these hypotheses with $\partial S \neq \emptyset$ Chapter9/GapP3_K1_Hemisphere · E8W.k1_hemisphere_stokes.

## IX.37 The local formula for every oriented tame structure

The text states the formula of Theorem 9.5.15 for every structure. Step 4 of §IX.36 handles a structure after shrinking it; toward the frontier of $U(x)$ the corner charts of $M_i$ give no uniform bound. The formula holds for every structure all the same, by comparison with other structures.

*Step 1, a finite cover.* $\partial_1 M_i$ lies in a compact subset of $\operatorname{supp}\Theta$, by the properness property. Cover it by finitely many $U(p)'$ of structures as in §IX.36, whose nonempty boundary sheets are in the class.

*Step 2, a common tangent space* Chapter9/GapP3_K1_AllStructures · MPolyfold.k1_exists_tangent_eq. At $z \in \partial_1 M_i \cap U(p)'$ some sheet $M'_k$ of the other structure passes through $z$ with $T_z M'_k = T_z M_i$. Otherwise there is an open $O \ni z$ on which, at every common point, the tangent spaces of $M_i$ and of each $M'_k$ differ. Since $M_i \subseteq \operatorname{supp}\Theta$, every point of $M_i \cap O$ lies on some $M'_k$, and there the relative tangent space of $M_i \cap M'_k$ is not full. Such points form an $(m+1)$-null set, by the argument of Lemma 9.4.6 (§IX.28), while an open piece of an $(m+1)$-dimensional submanifold is not null.

*Step 3, the same stratum.* $\{d = 1\}$ is defined through the tangent space alone, so $z \in \partial_1 M'_k$ Chapter9/GapP3_K1_AllStructures · MPolyfold.k1_mem_boundaryStratum_of_tangent_eq. Whether $d_{M_i} = d_X$, which the tame induced structure does not give, does not matter.

*Step 4, domination.* $\partial_1 M_i$ lies in the union of finitely many sheets of the class, and a submanifold covered by finitely many sheets of the class is in the class Chapter9/GapP3_K1_AllStructures · EpGroupoid.FormEval.k1_msheets_of_subset_list. So every nonempty boundary sheet of every oriented tame structure is in the class Chapter9/GapP3_K1_AllStructures · EP.boundarySheet_mem_msheets_of_loc.

**Theorems 9.5.15 and 9.5.16 as printed,** $m \geq 1$ (V9.67, restated) Chapter9/GapP3_K1_AllStructures · EP.theorem_9_5_15 Chapter9/GapP3_K1_AllStructures · EP.theorem_9_5_16. *For $n = m+1$ with $m \geq 1$, $\Theta$ compact, branched, with an orientation $\hat{T}_\Theta$ and* `hloc`*, there is exactly one linear $\omega \mapsto \mu^\partial_\omega$ from $\Omega^m_{ep,\infty}(X)$ to the signed measures on $(\partial S, L(\partial S))$ with $\mu^\partial_\omega(K) = \frac{1}{\sharp G_x} \sum_i \sigma_i \int_{K_i} \omega|(\partial_1 M_i, o_{\partial M_i})$ for every $x \in \operatorname{supp}(\partial\Theta)$, every oriented tame structure at $x$ describing $\hat{T}_\Theta$ and every measurable $K$ with $|K| \subseteq |U(x)|$*; with $\sharp G_x^{\text{eff}}$ when $X$ has the local unique continuation property. The canonical measure has the formula for every structure Chapter9/GapP3_K1_AllStructures · EP.boundaryMeasure_apply_all. The sum runs over the nonempty $\partial_1 M_i$, which contribute nothing in the text when empty, and $n$-dimensionality follows from `hloc`.

# IX.38 Dimension one: 0-forms, finite sets of boundary points, and the half-circle

For $n = 1$ the boundary sheets are 0-dimensional and $\omega \in \Omega^0_{ep,\infty}(X)$; the multi-chart class of §IX.34 needs $m > 0$.

*Step 1, 0-forms as functions.* A 0-form is read as a function on objects, its level-0 value at a smooth point, invariant under morphisms `Chapter9/GapP3_K2_ZeroForms · EP.zeroFormEval`; its restriction to a sheet is this value `Chapter9/GapP3_K2_ZeroForms · EP.formsInftyRestrict_zero_eq`. The integral over an oriented point $p$ is $\pm\omega(p)$, the sign from the orientation: the point integration of §IX.31 through this evaluation `Chapter9/GapP3_K2_ZeroForms · EP.zeroFormIntegration`.

*Step 2, finite sets of oriented points* `Chapter9/GapP3_K2_FinPoints · EpGroupoid.finPointIntegration`. For a finite set $N$ of smooth points, $\int_A \omega = \sum_{p \in N \cap A} \pm\omega(p)$. This is a sheet integration in the sense of §IX.31 for the class of finite sets `Chapter9/GapP3_K2_FinPoints · EpGroupoid.finPointSheets`, with the null sets of dimension 0: the tangent space of a finite set at one of its points is that of the point `Chapter9/GapP3_K2_FinPoints · EpGroupoid.submanifoldTangent_finite_eq`, and every subset of a finite set is measurable. On sets with at most one point it is the point integration `Chapter9/GapP3_K2_FinPoints · EpGroupoid.finPointInteg_eq_pointInteg`. For 0-forms: `Chapter9/GapP3_K2_FinPoints · EP.finZeroFormIntegration`.

*Step 3, the boundary sheets are finite.* In a corner chart with values in $[0, \infty)$ the face is one point, so near every point a 1-dimensional tame sheet has at most one point of its boundary stratum `Chapter9/GapP3_K2_BoundaryDim1 · MPolyfold.exists_isOpen_boundaryStratum_subsingleton`. Every oriented tame structure restricts to one whose boundary sheets have at most one point `Chapter9/GapP3_K2_BoundaryDim1 · EpGroupoid.OrientedTameStructure.exists_boundarySheet_subsingleton`, and the class conditions follow from `hloc` `Chapter9/GapP3_K2_BoundaryDim1 · EpGroupoid.hcov_dim1_of_loc` `Chapter9/GapP3_K2_BoundaryDim1 · EpGroupoid.hcovEff_dim1_of_loc`. The covering argument of §IX.37 then shows that every $\partial_1 M_i$ is finite `Chapter9/GapP3_K2_AllStructuresDim1 · EP.boundarySheet_finite_of_loc` `Chapter9/GapP3_K2_AllStructuresDim1 · EP.boundarySheet_mem_finPointSheets_of_loc`.

*Step 4, Part I's boundary integral.* For a 1-dimensional sheet with one boundary point $p$, Part I's $\int_{\partial M} \omega$ is $\pm\omega(p)$: the Lebesgue measure on $\mathbb{R}^0$ is a Dirac measure, and the face sign is that of §IX.35, Step 5, at $m = 0$ `Chapter9/GapP3_K2_FaceDim1 · EpGroupoid.boundaryIntegral_eq_pointInteg` `Chapter9/GapP3_K2_FaceDim1 · EP.boundaryIntegral_eq_zeroFormIntegration`.

*Step 5, Stokes.* The locality argument of §IX.35, Step 6, is redone for any sheet integration `Chapter9/GapP3_K2_StokesDim1 · EpGroupoid.boundaryMeasure_univ_eqI` and combined with the boundary-sum form of §IX.35, Step 4 `Chapter9/GapP3_K2_StokesDim1 · EP.stokes_of_boundaryMeasure_dim1`.

**The theorems for $n = 1$ (V9.67, V9.68).** Theorems 9.5.15 and 9.5.16 with the formula for every oriented tame structure, $\int_{K_i} \omega|(\partial_1 M_i, o) = \sum_{p \in K_i} \pm\omega(p)$ `Chapter9/GapP3_K2_AllStructuresDim1 · EP.theorem_9_5_15_dim1_all` `Chapter9/GapP3_K2_AllStructuresDim1 · EP.theorem_9_5_16_dim1_all`, first proved for the structures whose boundary sheets are points `Chapter9/GapP3_K2_BoundaryDim1 · EP.theorem_9_5_15_dim1` `Chapter9/GapP3_K2_BoundaryDim1 · EP.theorem_9_5_16_dim1` `Chapter9/GapP3_K2_LUCDim1 · EP.theorem_9_5_16_dim1_isLUC`. Theorems 9.5.17 and 9.5.18: $\int_{\hat\Theta} d\omega = \mu^{\partial}_{\omega}(\partial S)$ `Chapter9/GapP3_K2_StokesDim1 · EP.theorem_9_5_17_dim1` `Chapter9/GapP3_K2_StokesDim1 · EP.theorem_9_5_18_dim1` `Chapter9/GapP3_K2_LUCDim1 · EP.theorem_9_5_18_dim1_isLUC`, with the measures `Chapter9/GapP3_K2_StokesDim1 · EP.boundaryBranchedMeasureDim1` `Chapter9/GapP3_K2_StokesDim1 · EP.boundaryBranchedMeasureEffDim1`, which are those of the $n = 1$ Theorems 9.5.15 and 9.5.16 `Chapter9/GapP3_K2_AllStructuresDim1 · EP.boundaryBranchedMeasureDim1_apply_all` `Chapter9/GapP3_K2_AllStructuresDim1 · EP.`

boundaryBranchedMeasureEffDim1_apply_all. The hypotheses are the text's.

**Witnesses with nonempty boundary.** $n = 1$. $X = [0,\infty) \times \mathbb{R}$ as a trivial ep-groupoid, and $\Theta = 1$ on the closed half-circle $\{|p| = 1\}$, the image of the sc⁺-retraction $p \mapsto p/|p|$ of $X \setminus \{0\}$: a compact, tame, 1-dimensional submanifold Chapter9/GapP3_K2_Arc · K2Arc.isSubmanifoldOfDim_arc Chapter9/GapP3_K2_Arc · K2Arc.isTameSubMPolyfold_arc whose boundary stratum is its two end points on $\{0\} \times \mathbb{R}$ Chapter9/GapP3_K2_Arc · K2Arc.boundaryStratum_arc, oriented by a field that is a unit tangent on the circle Chapter9/GapP3_K2_ArcWitness · K2Arc.arcOrientation. All hypotheses of Theorems 9.5.15–9.5.18 for $n = 1$ hold, with $\operatorname{supp}(\partial\Theta) \neq \emptyset$ Chapter9/GapP3_K2_ArcWitness · K2Arc.arc_hloc Chapter9/GapP3_K2_ArcWitness · K2Arc.arc_isLUC Chapter9/GapP3_K2_ArcWitness · K2Arc.k2_arc_hypotheses, and Stokes' theorem applies Chapter9/GapP3_K2_ArcWitness · K2Arc.k2_arc_stokes Chapter9/GapP3_K2_ArcWitness · K2Arc.k2_arc_stokes_eff. $n = 2$, *the effective version.* Every natural representation of a trivial ep-groupoid has the local unique continuation property Chapter9/GapP3_K2_HemisphereEff · E8W.hasLUC_trivial, so the hemisphere of §IX.35 has it Chapter9/GapP3_K2_HemisphereLUC · E8W.e8w_isLUC, and Theorem 9.5.18 applies with $\partial S \neq \emptyset$ Chapter9/GapP3_K2_HemisphereLUC · E8W.k2_hemisphere_stokes_eff Chapter9/GapP3_K2_HemisphereEff · E8W.e8_hemisphere_stokes_eff. Neither witness has corners, branching or nontrivial isotropy; Remark 9.5.20 is about the last (§IX.39).

## IX.39 Remark 9.5.20 (2021 pp. 451–452): the example, in part

**The text.** $X = \{(-1,0)\} \cup ((-1,1) \times B) \cup \{(1,0)\} \subseteq \mathbb{R} \times \mathbb{C}$ with an M-polyfold structure, $\mathbb{Z}_3$ rotating the $\mathbb{C}$-factor, $M = [-1,1] \times \{0\}$, $\Theta(t,z) = \delta_0(z)$ oriented by $[-1,1]$, and $f(t,z) = \phi(t)$. The remark computes $\oint_\Theta df = \frac{1}{3}(\phi(1) - \phi(-1))$ and observes that $\partial X = \{(\pm 1, 0)\}$ is a sub-ep-groupoid whose points have trivial effective isotropy, while in $X$ it is $\mathbb{Z}_3$: the boundary integral of $f$ is $\frac{1}{3}(\phi(1) - \phi(-1))$ read in $X$, in accordance with Stokes' theorem, and $\phi(1) - \phi(-1)$ read in $\partial X$.

**The M-polyfold** Chapter9/GapP3_K3_MPolyfold · K3.P. The text says only that $X$ can be given an M-polyfold structure. It is given one with one chart, onto an sc-smooth retract of the partial quadrant $[0,\infty)^2 \times (c_0 \oplus c_0)$ in $E = \mathbb{R}^2 \oplus c_0 \oplus c_0$, with the weighted scale of Remark 4.1.3's example (§IV.27). The retraction is $R(p,a,b) = (2p/|p|, \pi_{\tau(p)}a, \pi_{\tau(p)}b)$ Chapter9/GapP3_K3_Retract · K3.Rt Chapter9/GapP3_K3_Retract · K3.mem_0_iff, with Remark 4.1.3's family $\pi_\tau$ (rank one for $\tau > 0$, zero for $\tau \leq 0$) composed with $\tau(p) = p_0 p_1$ Chapter9/GapP3_K3_Retract · IsLossFamily.comp_contDiff Chapter9/GapP3_K3_Retract · IsLossFamily.diag. Its retract is the quarter circle of radius 2 carrying a plane over each interior point and a point over each end. The chart is $\psi(t,z) = (2(\cos\theta, \sin\theta), \operatorname{Re} w\,\beta_\tau, \operatorname{Im} w\,\beta_\tau)$ with $\theta = \pi(1+t)/4$ and $w = z/\sqrt{1-|z|^2}$ Chapter9/GapP3_K3_MPolyfold · K3.chart. The degeneracy index is 1 at $(\pm 1, 0)$ and 0 elsewhere Chapter9/GapP3_K3_MPolyfold · K3.degIndex_eq; the lower bound at the ends is Theorem 2.4.12, from an inward vector in the tangent space of the retract Chapter9/GapP3_K3_MPolyfold · K3.scDeriv_end. So $\partial X = \{(-1,0),(1,0)\}$ Chapter9/GapP3_K3_MPolyfold · K3.boundary_eq.

**The groupoid** Chapter9/GapP3_K3_Groupoid · K3.ε. $\mathbb{Z}_3$ acts by $k \cdot (t,z) = (t, \omega^k z)$, sc-smoothly, since in the chart it is the restriction of the rotation of $(a,b)$, an sc-operator commuting with $R$ Chapter9/GapP3_K3_Groupoid · K3.isScSmoothMap_smul. Its non-effective part is $\{1\} \times X$ Chapter9/GapP3_K3_Groupoid · K3.nonEffectivePart_ε, which is closed, so $\varepsilon = \mathbb{Z}_3 \ltimes X$ has the local unique continuation property Chapter9/GapP3_K3_Groupoid · K3.isLUC_ε. The effective isotropy has three elements on $M$ Chapter9/GapP3_K3_Groupoid · K3.card_effIsotropy_of_snd_eq_zero and one off $M$ Chapter9/GapP3_K3_Groupoid · K3.card_effIsotropy_of_snd_ne_zero. $\partial X$ is a saturated sub-M-polyfold Chapter9/GapP3_K3_Groupoid · K3.isEpSubgroupoid_boundary, and $\varepsilon_B$ is the sub-ep-groupoid of Proposition 7.1.22 on it Chapter9/GapP3_K3_Groupoid · K3.εB. At a point of $\partial X$ the isotropy group has three elements in $\varepsilon$ and in $\varepsilon_B$ Chapter9/GapP3_K3_Groupoid · K3.card_isotropy_εB, and the effective

isotropy has three in $\varepsilon$ and one in $\varepsilon_B$ `Chapter9/GapP3_K3_Groupoid · K3.card_effIsotropy_εB`: every morphism of $\varepsilon_B$ acts trivially near its points. Collected: `Chapter9/GapP3_K3_Groupoid · K3.remark_9_5_20_effIsotropy`.

**Not formalized** (gap G.110, in part). $\Theta$ with its orientation and `hloc` at the infinite-dimensional ends; sc-smooth partitions of unity of $\varepsilon$ (Corollary 7.6.6 asks for a Hilbert space on level 0, and here $E_0$ contains $c_0$); $f$ as a 0-form; $\oint_{\partial\Theta} df = \frac{1}{3}(\phi(1) - \phi(-1))$, by the $n = 1$ theorems of §IX.38 and the local formula at the two ends; the integral of $f$ over $\partial\Theta$ in $\varepsilon_B$; and that $\partial X$ is a smooth submanifold, shown at each of its two points only. Estimated at 2,340–3,790 lines (Findings, Chapter 9, item 120).

# Part X. Equivalences and localization (HWZ Chapter 10, §§10.1–10.5 and Appendix §§10.6.1–10.6.3 (2017 §§10.1–10.5))

Chapter 10 continues Part II; its numbers and pages are those of the book (Springer 2021, pp. 461–520), with the 2017 text's in parentheses (2017 pp. 333–374). It studies the sc-smooth functors between ep-groupoids that should become invertible once ep-groupoids are taken as models of polyfolds. §10.1 defines equivalences (Definition 10.1.1 (2017: 10.1)), characterizes them as the equivalences of categories that are local sc-diffeomorphisms on objects (Lemma 10.1.2 (2017: 10.1)), builds the refinement along an open covering (Proposition 10.1.3 (2017: 10.1)), shows that natural equivalence, the two-out-of-three property and the tangent functor respect equivalences (Definition 10.1.4, Propositions 10.1.5, 10.1.6, Theorem 10.1.7 (2017: 10.2, 10.2, 10.3, 10.1)) and, mostly new in 2021, that the properties of Chapter 7 are invariant (Remarks 10.1.8, 10.1.11, 10.1.12, Propositions 10.1.9, 10.1.10, 10.1.13, 10.1.14, 10.1.16–10.1.19, Example 10.1.15). §10.2 constructs the weak fibred product (Definition 10.2.1 (2017: 10.3), Theorem 10.2.2 (2017: 10.2), Remark 10.2.3 (2017: 10.2)). §10.4 carries all this to strong bundles: strong bundle equivalences (Definition 10.4.1 (2017: 10.10)), the pull-back and push-forward of section functors (Propositions 10.4.2–10.4.4, Theorem 10.4.5 (2017: 10.7–10.9, 10.8)), natural transformations of strong bundle functors (Definitions 10.4.8, 10.4.10, Remark 10.4.9 (2017: 10.11, 10.12, 10.4)) and the category $\mathcal{SEP}$ (Definition 10.4.11 (2017: 10.13)). §10.3, treated here after §10.4 in the order of the verified edition, inverts the equivalences: diagrams $X \xleftarrow{F} A \xrightarrow{\Phi} Y$ with $F$ an equivalence, their refinement and equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4, 10.3.6, Definition 10.3.7 (2017: 10.4–10.7, 10.3–10.5)), the composition of classes (Theorem 10.3.8 (2017: 10.3), Appendix §10.6.1), generalized isomorphisms (Definition 10.3.9, Theorem 10.3.10 (2017: 10.8, 10.4), Appendix §10.6.2), the category $\mathcal{EP}(E^{-1})$ with the functors $i$ and $|\cdot|$ (Proposition 10.3.11, Definition 10.3.12, Theorems 10.3.14, 10.3.15 (2017: 10.6, 10.9, 10.5, 10.6)) and the invariance of the properties of Chapter 7 under generalized isomorphisms (Propositions 10.3.16, 10.3.17, Remark 10.3.18, Theorem 10.3.19, Definition 10.3.20, Theorem 10.3.21). §10.5, treated after §10.3, repeats the localization for strong bundles: diagrams $W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W'$ with $\Phi$ a strong bundle equivalence, their classes, the maps $[d]$, $|[D]|$ and the homomorphisms $\gamma$ (pp. 504–506), the category $\mathcal{SEP}(F^{-1})$ and generalized strong bundle isomorphisms (Definitions 10.5.1, 10.5.2 (2017: 10.14, 10.15), Remark 10.5.3), the push-forward of section functors (Theorem 10.5.4 (2017: 10.9), Remark 10.5.5) and the local regularity property (Proposition 10.5.6, Theorem 10.5.7 (2017: 10.10)). Appendix §10.6.3, new

in 2021, gives the example behind Remark 10.1.8 on a weighted Sobolev scale; §10.1 also introduces the full subcategory $\mathcal{EP}_{lreg}$ (p. 461) and §10.4 the strong bundle versions of Propositions 10.1.9 and 10.1.10 (Propositions 10.4.6, 10.4.7).

**What is proved, and what is not.** Since 2026-11-12 every gap of Chapter 10 is closed (G.111–G.127): each statement of the chapter is proved, and two, Theorems 10.3.19 and 10.5.7, do not hold as we read them and are proved for an effective target. Nothing in the chapter is open or formalized only in part. *§10.1.* All definitions are formalized. Lemma 10.1.2 (with its converse), Propositions 10.1.3, 10.1.5, 10.1.6, 10.1.13, 10.1.14, 10.1.16–10.1.19, Theorem 10.1.7, Remarks 10.1.11 and 10.1.12 and Example 10.1.15 are proved, several in a stronger form. Proposition 10.1.9 is proved, and Proposition 10.1.10 with (LR) on one side only (§§X.5, X.7; G.113–G.115). The subcategory $\mathcal{EP}_{lreg}$ is §X.27 (G.111). Remark 10.1.8 with the example of Appendix §10.6.3, on the weighted Sobolev scale as printed, is §X.28 (G.112). Two results the text quotes or leaves to the reader are proved: Proposition 10.1.5, and the fullness and faithfulness of $TF$ in Theorem 10.1.7. *§10.2.* All claims of Theorem 10.2.2 are proved, including that the structures are those induced from the triple products (§X.29; G.116), and so is Remark 10.2.3. *§10.4.* All definitions are formalized, and Propositions 10.4.2–10.4.4, Remark 10.4.9 and the equivalence of Definitions 10.4.8 and 10.4.10 are proved. Theorem 10.4.5 is proved with its Fred part (§X.31; G.121); the sc-smoothness of the push-forward, "easily verified" in the text, is proved too. Propositions 10.4.6 and 10.4.7 are §X.32 (G.122), with Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$; Proposition 10.4.7 is proved with the property for the target only. *§10.3.* All definitions are formalized, with the proofs of the appendix. Proved: (E1)–(E3), Lemmas 10.3.2, 10.3.4, 10.3.6, Theorems 10.3.8, 10.3.10, 10.3.14, 10.3.15, Propositions 10.3.11, 10.3.16, 10.3.17, Remark 10.3.18 and Theorem 10.3.21 (§§X.16–X.20; G.118 closed by §X.20, G.119 by §X.23). The universal property of the localization (G.117) is §X.30. Theorem 10.3.19 does not hold as we read it, and holds for an effective target (§X.20; G.120, settled). *§10.5.* Formalized: the diagrams, their refinements and classes, $[d]$, $|[D]|$, the square $|\mathcal{D}|$, $\gamma$, Definitions 10.5.1 and 10.5.2, and $\mathcal{SEP}(F^{-1})$ as a category through the weak fibred product of strong bundles (§§X.21, X.22). (10.24) with generalized vertical maps and the local expressions $q$, $Q$ are §X.33 (G.124). Proved: Remark 10.5.3, with both directions of the strong bundle Theorem 10.3.10 (§§X.22, X.34; G.123); Theorem 10.5.4 with its Fred part and as a linear isomorphism (§§X.23, X.31; G.125); Remark 10.5.5, with an example; and Proposition 10.5.6 (§§X.23, X.32; G.126). Theorem 10.5.7 does not hold as we read it, and holds over an effective base of the target (§X.23; G.127, settled). Remark 10.3.13 and the sentences on "smaller models" in Remark 10.2.3 have no formal content. *The narrower row.* There is one: V10.66, Theorem 10.3.19 for an effective target. The effectiveness of $Y$ is an added hypothesis, not derived from the text; without it the theorem does not hold as we read it (`not_theorem_10_3_19`). Its satisfiability witness is the point (§X.20). The strong bundle analogue is part of the restated row V10.67, with the same reason: Theorem 10.5.7 over an effective base of the target, with (LR) of a base or the local regularity property of the target bundle (§X.23).

**Six structural choices.** (1) *Equivalences as a structure.* `EpFunctor.IsEquivalence` has the four conditions of Definition 10.1.1 as fields, with full and faithful stated on the sets of morphisms; Mathlib's `Functor.IsEquivalence` of the underlying functor is derived (§X.1). (2) *Indexed coverings.* The refinement $X_{\mathcal{U}}$ takes a family $(U_i)_{i\in\iota}$ of open sets; objects and morphisms are open subsets of products with the discrete $\iota$ (§X.2). (3) *Iterated fibred products.* The weak fibred product is built from Part I's fibred products, twice, so that its M-polyfold structures come with their universal properties; the admissibility, $F$ or $G$ a local sc-diffeomorphism on objects, is all that the construction uses (§X.9). (4) *Total pull-backs and push-forwards.* $\Phi^*$ and $\Phi_*$ are defined on all maps, by the zero of the fibre where needed, and the push-forward with one choice of $x$ and $\sigma$ made for all; all statements are about section functors, where the choice does not matter

(§X.12). (5) *Diagrams one universe up.* A diagram contains its middle ep-groupoid as an object of $\mathcal{EP}$, so the classes of diagrams form a type in the universe of the objects of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$ is a large category like $\mathcal{EP}$; compositions are written in Mathlib's order, $d$.comp $e$ for the text's $[e] \circ [d]$ (§X.16). (6) *The weak fibred product of strong bundles as a pull-back.* The text leaves it to the reader; it is the pull-back of $W''$ along $\pi_Z$, so that one projection is a strong bundle equivalence by construction and the squares in the proofs of the composition commute strictly, by the fibrewise injectivity of strong bundle equivalences; the generalized strong bundle maps, defined as the classes of the equivalence relation generated by having a common refinement, are the text's classes by the transitivity (§§X.21, X.22).

**Map.**

- `Chapter10/Equivalence`: local sc-diffeomorphisms at a point, Definition 10.1.1, Lemma 10.1.2, Definition 10.1.4, Propositions 10.1.5, 10.1.6, Theorem 10.1.7, the pair groupoid of $\mathbb{Z}$ over the point.
- `Chapter10/EquivalenceRefinement`: open coverings and Proposition 10.1.3, the covering of $\mathbb{Z}/2 \ltimes \mathbb{R}$.
- `Chapter10/EquivalenceReduced`: Remark 10.1.12, Propositions 10.1.13, 10.1.14, 10.1.16, 10.1.18.
- `Chapter10/EquivalenceExamples`: Remark 10.1.11, Example 10.1.15.
- `Chapter10/EquivalenceZhou`: Proposition 10.1.17 and the reduction of Proposition 10.1.19.
- `Chapter10/EquivalenceLR`: transport charts $Q = \widehat{\psi} \circ F$, Propositions 10.1.9, 10.1.10 and 10.1.19.
- `Chapter10/EquivalenceLREffective`: local families over an effective target and the natural equivalence they give.
- `Chapter10/WeakFibredProduct`: Definition 10.2.1 and the first claim of Theorem 10.2.2.
- `Chapter10/WeakFibredProductTheorem`: the projections, Theorem 10.2.2, Remark 10.2.3, witnesses.
- `Chapter10/StrongBundleEquivalence`: Definition 10.4.1, the pull-back and the push-forward, Propositions 10.4.2–10.4.4, Theorem 10.4.5.
- `Chapter10/StrongBundleEquivalenceNat`: Definitions 10.4.8, 10.4.10, 10.4.11, Remark 10.4.9.
- `Chapter10/StrongBundleEquivalenceExamples`: the strong bundle equivalence over $-\mathrm{id}$ and its uses.
- `Chapter10/Localization`: diagrams, Definitions 10.3.1, 10.3.3, 10.3.5, 10.3.7, Lemmas 10.3.2, 10.3.4, 10.3.6, Theorem 10.3.8 with Appendix §10.6.1, the diagonal of Appendix §10.6.2.
- `Chapter10/LocalizationCategory`: (E1), Definitions 10.3.9, 10.3.12, Theorem 10.3.10 with Appendix §10.6.2, Proposition 10.3.11, Theorems 10.3.14, 10.3.15, witnesses.
- `Chapter10/LocalizationProperties`: Proposition 10.3.17, Remark 10.3.18, Definition 10.3.20, Theorem 10.3.21.
- `Chapter10/LocalizationLR`: Proposition 10.3.16, Theorem 10.3.21 for all five properties.
- `Chapter10/LocalizationLRCounterexample`: Theorems 10.3.19 and 10.5.7 as statements, their negations over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$, Theorem 10.3.19 for an effective target.
- `Chapter10/StrongBundleLocalization`: diagrams of strong bundles, their refinements and classes, $[d]$, $|[D]|$, the square $|\mathcal{D}|$, $\gamma$, Definitions 10.5.1, 10.5.2, Remark 10.5.3.
- `Chapter10/StrongBundleLocalizationSections`: the push-forward and pull-back, Theorem 10.5.4, the reduction of Proposition 10.5.6.
- `Chapter10/StrongBundleLocalizationExamples`: the relation of the homomorphisms $\gamma$, the witnesses over $-\mathrm{id}$, Remark 10.5.5 over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$.
- `Chapter10/StrongBundleWeakFibredProductPullback`: pull-backs of strong bundles along functors, transports along natural transformations, fibrewise inverse lifts.
- `Chapter10/StrongBundleWeakFibredProduct`: the weak fibred product of strong bundles and the

strong bundle Theorem 10.2.2.

- `Chapter10/StrongBundleLocalizationCategory`: the strong bundle Lemma 10.3.4 and Theorem 10.3.8, $\mathcal{SEP}(F^{-1})$ as a category with its localization functor, one direction of the strong bundle Theorem 10.3.10.
- `Chapter10/LocalizationLREffectiveBundle`: Theorem 10.5.7 over an effective base of the target, with (LR) of a base.
- `Chapter10/LocalizationLRBundleBase`: Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$, Proposition 10.4.7 on the bases, Theorem 10.5.7 over an effective base from that property.
- `Chapter10/GapP2_F_LReg`: the subcategory $\mathcal{EP}_{lreg}$ of $\mathcal{EP}$ and of $\mathcal{EP}(E^{-1})$.
- `Chapter10/GapP2_A2TenSix`: Remark 10.1.8 and Appendix §10.6.3 as printed, on the scale $\mathbb{R} \oplus H^{m,\delta_m}(\mathbb{R})$.
- `Chapter10/GapP2_F_ZhouExample`: the same example with the projections of Remark 4.1.3.
- `Chapter10/GapP2_F_TripleProduct`: the structures of Theorem 10.2.2 induced from the triple products.
- `Chapter10/GapP2_F_LocUniversal`: the universal property of $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$.
- `Chapter10/FredholmEquivalence`: the Fred parts of Theorems 10.4.5 and 10.5.4.
- `Chapter10/SectionFunctorLinear2021`: the push-forwards of Theorems 10.4.5 and 10.5.4 as linear isomorphisms.
- `Chapter10/GapP2_GLocalIso`: local strong bundle isomorphisms between two strong bundles, transports along families of morphisms.
- `Chapter10/GapP2_GBundleLR`: Propositions 10.4.6 and 10.5.6.
- `Chapter10/GapP2_GBundleFamily`: Proposition 10.4.7.
- `Chapter10/GapP2_GDiagram`: the square (10.24) in $\mathcal{EP}(E^{-1})$ and the germ $q$.
- `Chapter10/GapP2_GDiagramLift`: the lift $Q$ and (10.25).
- `Chapter10/GapP2_F2_SBConverse`: the converse half of the strong bundle Theorem 10.3.10.

Throughout, $X = (X, \mathbf{X})$, $Y$, $Z$ are ep-groupoids as in Part VII, $s, t$ the source and target, $|X|$ the orbit space and $|F| : |X| \to |Y|$ the induced map, $G_x$ the isotropy group, $T_x$ its subgroup acting trivially near $x$ and $\mathbf{X}^T$ the non-effective part (§VII.3). Functors are sc-smooth functors (`EpFunctor`), and natural transformations those of Chapter 8 (`EpNatTrans`).

## X.1 Definition 10.1.1 (2017: 10.1) and Lemma 10.1.2 (2017: 10.1)

**Definition 10.1.1 (2017: 10.1)** (V10.1) `Chapter10/Equivalence · EpFunctor.IsEquivalence` `Chapter10/Equivalence · EP.equivalences`. The four fields are fullness and faithfulness on the sets of morphisms, $|F|$ a homeomorphism, and $F$ a local sc-diffeomorphism on objects. Fullness and faithfulness are those of Mathlib for the underlying functor `Chapter10/Equivalence · GroupoidFunctor.full_iff` `Chapter10/Equivalence · GroupoidFunctor.faithful_iff`, and so is essential surjectivity `Chapter10/Equivalence · GroupoidFunctor.essSurj_iff`. *Step 1, essential surjectivity.* For $y \in Y$, $|F|$ is onto, so $[y] = [F(x)]$ for some $x$, that is a morphism $F(x) \to y$ exists `Chapter10/Equivalence · EpFunctor.IsEquivalence.essSurj` `Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso`. *Step 2, the morphism sets.* Every $\psi : F(x) \to F(x')$ is $F(\varphi)$ `Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_mor_eq` and $F$ is injective on morphisms with the same ends `Chapter10/Equivalence · EpFunctor.IsEquivalence.mor_injective`. *Step 3, the class $E$.* The identity is an equivalence `Chapter10/Equivalence · EpFunctor.IsEquivalence.id`, composites are `Chapter10/Equivalence · EpFunctor.IsEquivalence.comp` (the composite of homeomorphisms and of local sc-diffeomorphisms, with $|G \circ F| = |G| \circ |F|$ `Chapter10/Equivalence · GroupoidFunctor.toFunctor_`

comp), so `EP.equivalences` is a multiplicative class. Witness: the functor from the pair groupoid of $\mathbb{Z}$ (one morphism between any two integers) to the point Chapter10/Equivalence · EpGroupoid.Examples.pairToPoint Chapter10/Equivalence · EpGroupoid.Examples.isEquivalence_pairToPoint, a local sc-diffeomorphism because $\mathbb{Z}$ is discrete Chapter10/Equivalence · EpGroupoid.Examples.isLocalScDiffeo_discrete.

**Local sc-diffeomorphisms at a point.** The local property of Definition 10.1.1 (3) is localized Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt Chapter10/Equivalence · MPolyfold.isLocalScDiffeo_iff_forall Chapter10/Equivalence · MPolyfold.IsLocalScDiffeo.at, with the rules used below: composition Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.comp, agreement near the point Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.congr, and the two cancellation rules: if $g \circ f$ and $g$ are local sc-diffeomorphisms at $x$ and $f(x)$, so is $f$ at $x$ Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.of_comp_left; if $g \circ f$ and $f$ are, so is $g$ at $f(x)$ Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.of_comp_right.

**Lemma 10.1.2 (2017: 10.1)** (V10.2) Chapter10/Equivalence · EpFunctor.lemma_10_1_2 Chapter10/Equivalence · EpFunctor.isEquivalence_iff. *Step 1, $|F|$ is bijective.* For a functor of groupoids, $|F|$ is injective when $F$ is full ($F(x) \cong F(x')$ lifts to $x \cong x'$) Chapter10/Equivalence · GroupoidFunctor.injective_orbitMap_of_full, surjective exactly when $F$ is essentially surjective Chapter10/Equivalence · GroupoidFunctor.surjective_orbitMap_iff, hence bijective for an equivalence of categories Chapter10/Equivalence · GroupoidFunctor.bijective_orbitMap_of_isEquivalence. *Step 2, $|F|$ is a homeomorphism.* $|F|$ is continuous by the universal property of the quotient Chapter10/Equivalence · EpFunctor.continuous_orbitMap; it is open because $F$ is a local sc-diffeomorphism, hence open, on objects and the orbit maps are open (Proposition 7.1.8 (2017: 7.1)) Chapter7/Basic · EpGroupoid.proposition_7_1_8 Chapter10/Equivalence · EpFunctor.isOpenMap_orbitMap. *Step 3, the converse.* An equivalence is full, faithful and essentially surjective (§X.1), so an equivalence of categories Chapter10/Equivalence · EpFunctor.IsEquivalence.isEquivalence_toFunctor. *Not invertible.* The functor $\mathbb{Z} \to \mathrm{pt}$ above has no left inverse on objects Chapter10/Equivalence · EpGroupoid.Examples.not_exists_leftInverse_pairToPoint.

## X.2 Proposition 10.1.3 (2017: 10.1): the refinement $X_{\mathcal{U}}$

**The construction** Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.objSet Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.morSet Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.groupoidStr. A covering is a family of open sets $U_i$ with every object in some $U_i$; the trivial covering by $X$ is one Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.univ. The text indexes the covering by itself and writes $(U, x)$; the index type is the formal reading. The objects $(i, x)$, $x \in U_i$, form an open subset of $\iota \times X$ Chapter10/EquivalenceRefinement · isOpen_of_discrete_left, the morphisms $(i, \varphi, j)$ one of $(\iota \times \iota) \times \mathbf{X}$, and both get the open-subset M-polyfold structures Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.objPolyfold Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.morPolyfold, which need a nonempty set: the object set of $X$ is assumed nonempty (for an empty $X$ there is nothing to prove). Products with a discrete space keep the paracompactness and metrizability Chapter10/EquivalenceRefinement · paracompactSpace_discrete_prod.

**$X_{\mathcal{U}}$ is an ep-groupoid** Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.epGroupoid. *Step 1, local sc-diffeomorphisms.* On a product with a discrete factor a map that is a local sc-diffeomorphism on each sheet is one Chapter10/EquivalenceRefinement · isLocalScDiffeo_sheet Chapter10/EquivalenceRefinement · isLocalScDiffeo_snd_discrete; $s(i, \varphi, j) = (i, s(\varphi))$ and $t(i, \varphi, j) = (j, t(\varphi))$ are local sc-diffeomorphisms Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isLocalScDiffeo_src Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isLocalScDiffeo_tgt, through the restriction

rules for open subsets Chapter10/EquivalenceRefinement · MPolyfold.isScSmoothOn_restrictOpen_iff Chapter10/EquivalenceRefinement · MPolyfold.isScSmoothMap_restrict Chapter10/EquivalenceRefinement · MPolyfold.isLocalScDiffeoAt_restrict. *Step 2, unit, inverse, multiplication.* They are those of $X$ on each sheet, sc-smooth by the continuity on the discrete factor Chapter10/EquivalenceRefinement · isScSmoothOn_discrete_of_continuousOn Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothMap_unit Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothMap_inv Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothCompSheet. *Step 3, properness.* At $(i, x)$ take a neighbourhood $V_0$ of $x$ with the properness property of $X$ and an open $V(x)$ with $\mathrm{cl}\, V(x) \subseteq U_i \cap V_0$; then $t$ is proper on $s^{-1}(\mathrm{cl}(\{i\} \times V(x)))$ Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.exists_isProperMap. The proof uses ultrafilters in place of the text's sequences; the index $j$ of a limit is determined by the sheet, which is clopen.

**Proposition 10.1.3 (2017: 10.1)** (V10.3) Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.proposition_10_1_3. $F(i, x) = x$, $F(i, \varphi, j) = \varphi$ Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.functor is a local sc-diffeomorphism on objects, faithful because a morphism of $X_{\mathcal{U}}$ is determined by its ends and its $\varphi$, full because $\psi : x \to x'$ lifts to $(i, \psi, j)$, and essentially surjective because every $x$ lies in some $U_i$; Lemma 10.1.2 applies. The text writes the morphism as $(U, \varphi, v)$ in the formula for $F$; we read $(U, \varphi, V)$ (possible misprint). *Witness.* The covering $\{x > -1\}$, $\{x < 1\}$ of $\mathbb{Z}/2 \ltimes \mathbb{R}$ Chapter10/EquivalenceRefinement · EpGroupoid.Examples.reflectionCover gives an equivalence Chapter10/EquivalenceRefinement · EpGroupoid.Examples.isEquivalence_reflectionCover that is not injective on objects Chapter10/EquivalenceRefinement · EpGroupoid.Examples.not_injective_reflectionCover_obj.

## X.3 Definition 10.1.4 (2017: 10.2), Propositions 10.1.5 and 10.1.6 (2017: 10.2 and 10.3)

**Definition 10.1.4 (2017: 10.2)** (V10.4) Chapter10/Equivalence · definition_10_1_4. The definition is Chapter 8's Definition 8.1.13 Chapter8/Basic · EpNatTrans Chapter8/Basic · EpFunctor.NaturallyEquivalent; the claims after it, that $\simeq$ is an equivalence relation and compatible with composition, are Chapter 8's Chapter8/NatTrans · EpFunctor.naturallyEquivalent_equivalence Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_left Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right. A natural transformation of ep-groupoids is a natural isomorphism of the underlying functors, every morphism being invertible Chapter10/Equivalence · EpNatTrans.toNatIso.

**Proposition 10.1.5 (2017: 10.2)** (V10.5) Chapter10/Equivalence · EpFunctor.proposition_10_1_5 Chapter10/Equivalence · EpFunctor.IsEquivalence.of_natTrans. The text quotes it from the literature. *Step 1.* $\tau : X \to \mathbf{X}'$ is continuous Chapter10/Equivalence · EpNatTrans.continuous_app and, since $s \circ \tau = F$ with $F$ and $s$ local sc-diffeomorphisms, a local sc-diffeomorphism Chapter10/Equivalence · EpNatTrans.isLocalScDiffeo_app. *Step 2.* $G = t \circ \tau$ is a composite of local sc-diffeomorphisms Chapter10/Equivalence · EpNatTrans.isLocalScDiffeo_obj. *Step 3.* The underlying functors are naturally isomorphic, so $G$ is an equivalence of categories with $F$, and Lemma 10.1.2 applies. The symmetric statement follows from the symmetry of $\simeq$.

**Proposition 10.1.6 (2017: 10.3)** (V10.6) Chapter10/Equivalence · EpFunctor.proposition_10_1_6. *Step 1, categories.* $\Phi$ is an equivalence of categories since $F$ and $\Phi \circ F$ are (Mathlib's two-out-of-three). *Step 2, objects in the image.* At $F(x)$, $\Phi \circ F$ and $F$ are local sc-diffeomorphisms at $x$, so $\Phi$ is one at $F(x)$ (the right cancellation rule). *Step 3, transport.* The text's argument stops here; an arbitrary $y$ receives a morphism $\sigma : F(x) \to y$ Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso, and the local sc-diffeomorphism property of a functor passes from $s(\sigma)$ to $t(\sigma)$ Chapter10/Equivalence · EpFunctor.isLocalScDiffeoAt_obj_tgt: near $\sigma$, $t \circ \tau = \Phi \circ t \circ \dots$ is written with a local inverse $\tau$ of $s$, and on morphisms $\Phi$ is a local sc-

diffeomorphism where it is one on objects `Chapter10/Equivalence · EpFunctor.isLocalScDiffeoAt_mor` `Chapter10/Equivalence · EpFunctor.isLocalScDiffeo_mor`. *Step 4.* Lemma 10.1.2. The text's $|\Phi| : |X| \to |Y|$ in the proof is read $|\Phi| : |Y| \to |Z|$ (possible misprint).

## X.4 Theorem 10.1.7 (2017: 10.1): the tangent of an equivalence

**Theorem 10.1.7 (2017: 10.1)** (V10.7) `Chapter10/Equivalence · EpFunctor.theorem_10_1_7`. $TF$ is Chapter 8's tangent functor `Chapter8/TangentEp · EpFunctor.tangent`; the text proves the homeomorphism and the local sc-diffeomorphism property and leaves full and faithful to the reader. Here all goes through Lemma 10.1.2. *Step 1, local sc-diffeomorphism.* The tangent of a local sc-diffeomorphism is one (Part I). *Step 2, faithful.* A tangent morphism $(\varphi, v)$ is determined by $\varphi$ and its source vector `Chapter8/TangentEp · EpGroupoid.tangent_ext`, and $F$ is faithful. *Step 3, full.* For $(\psi, w) : TF(x,a) \to TF(x',a')$, $\psi = F(\varphi)$ by fullness of $F$; $\varphi$ lifts to a tangent morphism with source $(x,a)$ `Chapter8/TangentEp · EpGroupoid.exists_tSrc_eq'`, whose image under $TF$ has the same source as $(\psi, w)$, and the targets agree since $TF$ is injective on the tangent fibre at $x'$ `Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.injOn_tangentFiber`. *Step 4, essentially surjective.* For $(y,b)$ take $\psi : F(x) \to y$, lift $\psi^{-1}$ with source $(y,b)$ and use that $TF$ maps $T_xX$ onto $T_{F(x)}Y$ `Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.tangentFiber_subset_image`, then invert `Chapter8/Basic · EpGroupoid.tInv`.

## X.5 Remarks 10.1.8 and 10.1.11 (2017: 10.1), Propositions 10.1.9 and 10.1.10 (2017: 10.4)

**Formalized since 2026-11-12.** The full subcategory $\mathcal{EP}_{lreg}$ of p. 461 (G.111) is §X.27. Remark 10.1.8 (new in the 2021 edition), with the example of Appendix §10.6.3 on its weighted Sobolev sc-retract (G.112), is §X.28. Propositions 10.1.9 and 10.1.10 (G.113, G.114) are the next two paragraphs.

**Proposition 10.1.9** (V10.62) `Chapter10/EquivalenceLR · EpFunctor.proposition_10_1_9`, for (LR) as in Definition 7.4.19 `Chapter7/Zhou · EpGroupoid.IsLR`. *The transport chart* `Chapter10/EquivalenceLR · EpFunctor.TransportChart` `Chapter10/EquivalenceLR · EpFunctor.exists_transportChart`: for $\psi : F(x) \to y$, $Q = \widehat{\psi} \circ F$ with $\widehat{\psi} = t \circ (s|\mathbf{U}(\psi))^{-1}$ is an sc-diffeomorphism of a neighbourhood of $x$ onto one of $y$, with morphisms $\psi_u : F(u) \to Q(u)$. *The isomorphism* $g \mapsto \psi \circ F(g) \circ \psi^{-1}$, $G_x \to G_y$ `Chapter10/EquivalenceLR · EpFunctor.TransportChart.conj` `Chapter10/EquivalenceLR · EpFunctor.TransportChart.bijective_conj`; on p. 469 its second factor $\sigma \mapsto \psi \circ \sigma \circ \psi^{-1}$ is printed with target $U(y)$, read $G_y$ (possible misprint). The equivariance (10.2)–(10.3), $Q(g * z) = (\psi \circ F(g) \circ \psi^{-1}) * Q(z)$ `Chapter10/EquivalenceLR · EpFunctor.TransportChart.Q_Φ`, comes from the uniqueness of $\Gamma$ on connected sets `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.eqOn_Γ`. *The step the text takes for granted,* $U(y) = Q(U(x))$ "with the natural action": a natural representation at $y$ (Theorem 7.1.13) restricted to $Q(U(x))$ is one, invariant under $G_y$ by the equivariance and the surjectivity of the isomorphism `Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_push`; the converse direction restricts one at $x$ to $Q^{-1}(U(y))$ `Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_pull` `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.eqOn_Φ`. *The conjugation.* The condition of (LR) on one $U(x)$ `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.HasLR` `Chapter10/EquivalenceLR · EpGroupoid.isLR_iff_hasLR` passes both ways along an equivariant sc-diffeomorphism over a surjection of the groups: $\beta = Q^{-1} \circ \gamma \circ Q = \hat{g}$ gives $\gamma = (\psi \circ F(g) \circ \psi^{-1})^\wedge$ `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.hasLR_iff_of_equivariant`. *Witness* `Chapter10/EquivalenceLR · EquivalenceLRExamples.nonempty_transportChart_pointEP`.

**Proposition 10.1.10 (2017: 10.4)** (V10.63) `Chapter10/EquivalenceLR · EpFunctor.proposition_10_1_10_one` `Chapter10/EquivalenceLR · EpFunctor.proposition_10_1_10_one_of_isLR_right` `Chapter10/EquivalenceLR · EpFunctor.`

proposition_10_1_10_two. (1) As in the text: $\beta = G^{-1} \circ \hat{\psi} \circ F$ of (10.4) is an sc-diffeomorphism near $x$ with $|\beta(z)| = |z|$, (LR) for $X$ gives $\beta = \hat{g}$, and $\alpha_z = G(\Gamma(g,z))^{-1} \circ \psi_z$. Only (LR) for $X$ is used; with (LR) for $Y$, Proposition 10.1.9 supplies it. (2) Differentiating $\alpha$, $T\alpha(v)$ is a morphism $TF(v) \to TG(v)$, so $|TF| = |TG|$. *Witness* Chapter10/EquivalenceLR · EquivalenceLRExamples.proposition_10_1_10_hypotheses. The families are in general not natural (§X.20).

**Remark 10.1.11 (2017: 10.1)** (V10.8) Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11. The ep-groupoid is $\mathbb{Z}/2$ acting on $\mathbb{R}^2$ by $z \mapsto -z$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.pointReflectionEpGroupoid, sc-smooth by Chapter10/EquivalenceExamples · EpGroupoid.Examples.isScSmoothMap_units_smul_rn; $G = -\mathrm{id}$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.negFunctor is an equivalence Chapter10/EquivalenceExamples · EpGroupoid.Examples.isEquivalence_negFunctor with $|\mathrm{id}| = |G|$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.orbitMap_id_eq_negFunctor. *Step 1.* $\alpha_z = (-1, z)$ is an sc-smooth family $F(z) \to G(z)$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11_family: the conclusion of Proposition 10.1.10 (1) holds in the example. *Step 2.* A family with $\alpha_0 = 1_0$ and $\alpha_z : z \to -z$ near 0 must take the value $(-1, z)$ at every $z \neq 0$, because the only morphism $z \to -z$ is the group element $-1$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.rn_eq_zero_of_eq_neg Chapter10/EquivalenceExamples · EpGroupoid.Examples.not_eventually_eq_neg; continuity at 0 fails. Hence no sc-smooth family exists either Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11_scSmooth.

## X.6 Remark 10.1.12, Propositions 10.1.13 and 10.1.14 (2017: 10.5), Example 10.1.15 (2017: 10.1)

**Remark 10.1.12** (V10.9) Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.remark_10_1_12. *Step 1.* $g \mapsto F(g)$ is a homomorphism $G_x \to G_{F(x)}$ Chapter10/EquivalenceReduced · EpFunctor.isotropyHom Chapter10/EquivalenceReduced · EpFunctor.isotropyHom_val, bijective for an equivalence (fullness and faithfulness at $(x,x)$) Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.bijective_isotropyHom. *Step 2.* $g \in T_x$ iff $F(g) \in T_{F(x)}$ Chapter10/EquivalenceReduced · EpFunctor.isotropyHom_mem_trivialSet_iff, since $F$ is a local sc-diffeomorphism on objects and on morphisms, and the local actions correspond. *Step 3.* Conjugation by $\psi : F(x) \to y$ Chapter7/Groupoid · GroupoidStr.isotropyConj maps $T_{F(x)}$ onto $T_y$ Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet_iff. The text's $G_x \to Y_y$ is read $G_x \to G_y$ (possible misprint).

**Proposition 10.1.13** (V10.10) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_13. *Step 1,* $\mathbf{X}^T$ *along* $F$. For a local sc-diffeomorphism on objects, $\varphi \in \mathbf{X}^T$ iff $F(\varphi) \in \mathbf{Y}^T$ Chapter10/EquivalenceReduced · EpFunctor.mor_mem_nonEffectivePart Chapter10/EquivalenceReduced · EpFunctor.mem_nonEffectivePart_of_mor, that is $\mathbf{X}^T = F^{-1}(\mathbf{Y}^T) \cap \{s = t\}$ Chapter10/EquivalenceReduced · EpFunctor.nonEffectivePart_eq. *Step 2, (1).* So $F_R$ is well defined Chapter10/EquivalenceReduced · EpFunctor.reducedMap Chapter10/EquivalenceReduced · EpFunctor.reducedMap_π, continuous Chapter10/EquivalenceReduced · EpFunctor.continuous_reducedMap and open, $F$ being open on morphisms and the quotient maps open Chapter10/EquivalenceReduced · EpFunctor.isOpenMap_reducedMap; only the local sc-diffeomorphism property of $F$ is used. *Step 3, closedness.* For an equivalence, $\mathbf{X}^T$ is closed iff $\mathbf{Y}^T$ is Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.isClosed_nonEffectivePart_iff. From $Y$ to $X$: $\mathbf{X}^T$ is the preimage of a closed set intersected with a closed set. From $X$ to $Y$: a $\psi \in \mathrm{cl}(\mathbf{Y}^T)$ at $y$ is conjugated, by a local inverse of $t$ at a morphism $\sigma : F(x) \to y$, into a morphism at $F(x)$ in $\mathrm{cl}(F(\mathbf{X}^T))$, with the continuity of the composition at a point Chapter10/EquivalenceReduced · EpGroupoid.continuousAt_comp; closures replace the text's sequences. *Step 4, (2).* Theorem 7.2.22 says $\mathbf{X}^R$ is Hausdorff iff $\mathbf{X}^T$ is closed Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_22; with Step 3 this gives (2), the route of the text.

**Proposition 10.1.14 (2017: 10.5)** (V10.11) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_14. With

both reduced morphism spaces Hausdorff the reduced ep-groupoids of Chapter 7 exist Chapter7/Reduced · EpGroupoid.reducedEpGroupoid. $F_R$ is $F$ on objects and $F_R$ of Step 2 on morphisms Chapter10/EquivalenceReduced · EpFunctor.reducedGroupoidFunctor Chapter10/EquivalenceReduced · EpFunctor.reduced; it is a functor since $[\cdot]$ is Chapter10/EquivalenceReduced · EpFunctor.reducedπ_eq_of_mor, sc-smooth since the M-polyfold structure of $\mathbf{X}^R$ is the one for which the quotient map is a local sc-diffeomorphism, full and faithful by Remark 10.1.12, and $|F_R| = |F|$. By Proposition 10.1.13 (2) either Hausdorff hypothesis implies the other; both are kept, as in the text.

**Example 10.1.15 (2017: 10.1)** (V10.12) Chapter10/EquivalenceExamples · EpGroupoid.Examples.example_10_1_15. $X = \mathbb{Z}/4 \ltimes \mathbb{R}$ with $1, 3$ acting by $-1$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.z4LineAction Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X, $Y = \mathbb{Z}/4 \ltimes \mathbb{R}^2$ by quarter turns Chapter10/EquivalenceExamples · EpGroupoid.Examples.z4PlaneAction Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y, $\Phi$ to the origin Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Functor. *Step 1.* 2 acts trivially on $\mathbb{R}$, not on $\mathbb{R}^2$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X_two_mem_nonEffectivePart Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y_two_not_mem_nonEffectivePart. *Step 2.* Hence $(1, x)$ and $(3, x) = (2, x) \circ (1, x)$ have the same class Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X_reducedπ_one_eq_three, their images not Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y_reducedπ_one_ne_three, so no map of sets $f$ with $f([\varphi]) = [\Phi(\varphi)]$ exists. The text speaks of an induced map of the reduced ep-groupoids; the statement here needs no topology.

## X.7 Propositions 10.1.16–10.1.19: (LUC), (ZHOU), (SLUC), (SLR)

**Proposition 10.1.16** (V10.13) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_16. (LUC) Chapter7/ReducedCriterion · EpGroupoid.IsLUC is the closedness of $\mathbf{X}^T$ (Proposition 7.2.21, part of Theorem 7.2.22) Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_21, and the closedness passes along equivalences (§X.6, Step 3). The text cites Theorem 7.2.22 here and in Proposition 10.1.13; the citation is the theorem on p. 323.

**Proposition 10.1.17** (V10.14) Chapter10/EquivalenceZhou · EpFunctor.proposition_10_1_17. *Step 1, smooth points.* $x$ is smooth iff $F(x)$ is, and iff a $y \cong F(x)$ is Chapter10/EquivalenceZhou · EpGroupoid.mem_smoothSet_src_iff Chapter10/EquivalenceZhou · EpGroupoid.mem_smoothSet_tgt_iff. *Step 2, $Y$ to $X$.* For $g \in G_x$ with $Tg = \mathrm{id}$ on $T_xX$: $T(F(g)) \circ TF = TF \circ Tg$ Chapter10/EquivalenceZhou · EpFunctor.tangentMor_mor and $TF$ is onto $T_{F(x)}Y$, so $T(F(g)) = \mathrm{id}$, $F(g) \in T_{F(x)}$ by (ZHOU) for $Y$, and $g \in T_x$ by Remark 10.1.12 Chapter10/EquivalenceZhou · EpFunctor.IsEquivalence.isZhou_of_isZhou_right. *Step 3, $X$ to $Y$.* For $k \in G'_y$ with $Tk = \mathrm{id}$ and $\psi : F(x) \to y$, $k' = \psi^{-1}k\psi \in G'_{F(x)}$ has $Tk' = \mathrm{id}$ by the composition rule for $T$ Chapter8/TangentEp · EpGroupoid.tangentMor_comp; $k' = F(g)$ with $g \in G_x$ (Remark 10.1.12), $Tg = \mathrm{id}$ by the injectivity of $TF$ on $T_xX$, so $g \in T_x$, $k' \in T_{F(x)}$ and $k \in T_y$ Chapter10/EquivalenceZhou · EpFunctor.IsEquivalence.isZhou_of_isZhou. The text argues with natural representations instead; it writes $U(F(y))$ and "$x \in U(x)$" in (b), read $U(F(x))$ and $a \in U(x)$ (possible misprints).

**Proposition 10.1.18** (V10.15) Chapter10/EquivalenceReduced · EP.proposition_10_1_18. *Step 1.* The restriction of $F$ to level 1 is an sc-smooth functor $X^1 \to Y^1$ Chapter10/EquivalenceReduced · EpFunctor.liftOne, an equivalence when $F$ is Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.liftOne (orbit spaces of $X^1$ carry the level-1 topology, and $|F^1|$ is a homeomorphism by Lemma 10.1.2 again). *Step 2.* By induction, every $X^i \to Y^i$ Chapter10/EquivalenceReduced · EP.exists_isEquivalence_lift over Chapter 7's lifts Chapter7/Zhou · EP.lift. *Step 3.* Proposition 10.1.16 at each level. The text cites Proposition 10.1.9 at Step 3; we read Propo-

sition 10.1.16 (possible misprint).

**Proposition 10.1.19** (V10.16) `Chapter10/EquivalenceLR · EP.proposition_10_1_19` `Chapter10/EquivalenceZhou · EP.isSLR_iff_of_isLR_iff`. Steps 1–2 as above, then Proposition 10.1.9 at each level (§X.5). The reduction alone holds when the two ep-groupoids agree `Chapter10/EquivalenceZhou · EP.isLR_iff_lift_self`; G.115 is closed.

## X.8 Definition 10.2.1 (2017: 10.3): the weak fibred product

**Definition 10.2.1 (2017: 10.3)** (V10.17) `Chapter10/WeakFibredProduct · WeakFibredProduct.groupoid`. *Objects.* $\{(\phi, z) \mid t(\phi) = G(z)\} \subseteq \mathbf{Y} \times Z$ `Chapter10/WeakFibredProduct · WeakFibredProduct.objRight`, then $\{(x, (\phi, z)) \mid F(x) = s(\phi)\}$ `Chapter10/WeakFibredProduct · WeakFibredProduct.objSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.Obj`, with projections and extensionality `Chapter10/WeakFibredProduct · WeakFibredProduct.Obj.ext`. *Morphisms.* The same with $\mathbf{X}$, $\mathbf{Z}$ and $F \circ s$, $G \circ s$ `Chapter10/WeakFibredProduct · WeakFibredProduct.morRight` `Chapter10/WeakFibredProduct · WeakFibredProduct.morSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.Mor` `Chapter10/WeakFibredProduct · WeakFibredProduct.Mor.ext`. *Structure maps.* The conjugate $G(k) \circ \phi \circ F(h)^{-1}$ `Chapter10/WeakFibredProduct · WeakFibredProduct.conj` has the ends $F(t(h))$, $G(t(k))$ `Chapter10/WeakFibredProduct · WeakFibredProduct.src_conj` `Chapter10/WeakFibredProduct · WeakFibredProduct.tgt_conj`; it is the target's middle entry `Chapter10/WeakFibredProduct · WeakFibredProduct.tgt`, $s$ keeps $\phi$ `Chapter10/WeakFibredProduct · WeakFibredProduct.src`, the unit is $(1_x, \phi, 1_z)$ `Chapter10/WeakFibredProduct · WeakFibredProduct.unit` and the inverse $(h^{-1}, \mathrm{conj}, k^{-1})$ `Chapter10/WeakFibredProduct · WeakFibredProduct.inv`. The text prints $\iota(h, \psi, k) = (h^{-1}, G(k) \circ \phi \circ F(h)^{-1}, k^{-1})$; we read $\psi = \phi$ (possible misprint). The product of a composable pair is $(h \circ h', \phi', k \circ k')$ `Chapter10/WeakFibredProduct · WeakFibredProduct.compOf`, extended to a total map that returns its first argument off composable pairs `Chapter10/WeakFibredProduct · WeakFibredProduct.comp`. *Groupoid laws.* By the rules $\mathrm{conj}(h \circ h', \phi', k \circ k') = \mathrm{conj}(h, \mathrm{conj}(h', \phi', k'), k)$ `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_comp`, conj of units `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_unit` and of inverses `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_inv`, with $\phi$ recovered from the target `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_recover` `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_comp_mor`; a morphism is a loop iff its ends agree `Chapter10/WeakFibredProduct · WeakFibredProduct.src_eq_tgt_iff`.

## X.9 Theorem 10.2.2 (2017: 10.2): $X \times_Y Z$ is an ep-groupoid

**The admissibility** `Chapter10/WeakFibredProduct · WeakFibredProduct.Admissible`. The construction uses of $F$ (or of $G$) only that it is a local sc-diffeomorphism on objects, which every equivalence is `Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Admissible.of_isEquivalence_left` `Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Admissible.of_isEquivalence_right`. *Step 1, sub-M-polyfolds.* Part I's fibred product of an sc-smooth map with a local sc-diffeomorphism is a sub-M-polyfold (Proposition 2.5.20); $t : \mathbf{Y} \to Y$ is one, so $\mathbf{Y}_t \times_G Z$ is a sub-M-polyfold of $\mathbf{Y} \times Z$ `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_objRight` `Chapter10/WeakFibredProduct · WeakFibredProduct.PA`, and its projection to $Z$ is a local sc-diffeomorphism `Chapter10/WeakFibredProduct · MPolyfold.isLocalScDiffeo_snd_fibredProduct`; then the second fibred product with $F$ or with that projection `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_objSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.PObj`, and the same for morphisms `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_morRight` `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_morSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.PB` `Chapter10/WeakFibredProduct · WeakFibredProduct.PMor`. The text says the sets are sub-M-polyfolds of

the triple products. They are, and the iterated structures are the induced ones (§X.29; G.116); the construction here does not use it. *Step 2, the projections.* $x$, $\phi$, $z$, $h$, $\phi$, $k$ are sc-smooth Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_x Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_z Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_k, and a map into $X \times_Y Z$ is sc-smooth when its entries are Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothOn_obj_of Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothOn_mor_of. For $F$ admissible, $(x, \phi, z) \mapsto z$ and $(h, \phi, k) \mapsto k$ are local sc-diffeomorphisms ((10.9), (10.10)) Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_z Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_k; for $G$ admissible the projections to $X$ and $\mathbf{X}$ are Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_x Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_h Chapter10/WeakFibredProduct · MPolyfold.isLocalScDiffeo_fst_fibredProduct. *Step 3, $s$ and $t$.* $\pi_Z \circ s = s_Z \circ \pi_{\mathbf{Z}}$ and $\pi_Z \circ t = t_Z \circ \pi_{\mathbf{Z}}$ with three local sc-diffeomorphisms make $s$, $t$ ones Chapter10/WeakFibredProduct · MPolyfold.IsLocalScDiffeo.of_comm Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_src Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_tgt; their sc-smoothness uses that of the conjugate Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_conj Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_src Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_tgt. *Step 4, unit, inverse, multiplication* Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_unit Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_inv Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothComp: entries are multiplications of $X$ and $Z$ and projections, sc-smooth on the composable pairs Chapter10/WeakFibredProduct · EpGroupoid.isScSmoothOn_comp_of. *Step 5, properness.* For open $U \ni x$, $V \ni z$ with the properness property in $X$ and $Z$, $W = \{x' \in U, z' \in V\}$ has $t$ proper on $s^{-1}(\mathrm{cl}\, W)$ Chapter10/WeakFibredProduct · WeakFibredProduct.exists_isProperMap: subsequences of $h_n$, then of $k_n$, converge, and $\phi_n = G(k_n)^{-1} \circ (G(k_n) \circ \phi_n \circ F(h_n)^{-1}) \circ F(h_n)$ converges by continuity of the multiplication Chapter10/WeakFibredProduct · EpGroupoid.tendsto_comp Chapter10/WeakFibredProduct · WeakFibredProduct.continuous_conj. The text writes $t : s^{-1}(U) \to X$; Definition 7.1.3 (3) asks for the closure, which is the reading. No admissibility is needed here. *Conclusion* (V10.18) Chapter10/WeakFibredProduct · WeakFibredProduct.epGroupoid.

## X.10 Theorem 10.2.2 (2017: 10.2): the projections, and Remark 10.2.3 (2017: 10.2)

**The functors** Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projLeft Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projRight Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projLeft_obj. $\sigma(x, \phi, z) = \phi$ is a natural transformation $F \circ \pi_X \to G \circ \pi_Z$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.natTrans Chapter10/WeakFibredProductTheorem · WeakFibredProduct.natTrans_app: its naturality is the definition of the target, and it is sc-smooth as a projection. Hence $F \circ \pi_X \simeq G \circ \pi_Z$ whenever $X \times_Y Z$ is defined Chapter10/WeakFibredProductTheorem · WeakFibredProduct.naturallyEquivalent.

**$F$ an equivalence** (V10.19) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projRight. *Step 1, faithful.* A morphism over $k$ is $(h, \phi, k)$ with $F(h)$ determined by $k$ and the ends, and $F$ is faithful. *Step 2, full.* For $k : z \to z'$ between $(x, \phi, z)$ and $(x', \phi', z')$, $\phi'^{-1} \circ G(k) \circ \phi : F(x) \to F(x')$ is $F(h_0)$ by fullness of $F$, and $(h_0, \phi, k)$ lies over $k$. The text writes the composite as $\phi' \circ G(k) \circ \phi^{-1}$, which does not compose; we read $\phi'^{-1} \circ G(k) \circ \phi$ (possible misprint). *Step 3, essentially surjective.* $G(z)$ receives $\psi : F(x) \to G(z)$ Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso, and $(x, \psi, z)$ lies over $z$. *Step 4.* Lemma 10.1.2 with Step 2 of §X.9. *The rest of the theorem* Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2: an equivalence preserves the degeneracy

index Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.degIndex_obj, so $d(x,\phi,z) = d_Z(z)$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.degIndex_eq_z, and pulls back tameness Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.isTame_of_isTame and the paracompactness of the orbit space, through $|\pi_Z|$ Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.hasParacompactOrbitSpace. *Witness.* $\mathbb{Z} \to$ pt and $\mathrm{id}_{\mathrm{pt}}$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Examples.theorem_10_2_2_pairToPoint.

**$G$ an equivalence, and both** (V10.20) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projLeft Chapter10/WeakFibredProductTheorem · WeakFibredProduct.degIndex_eq_x Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2_right Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2_both. The symmetric argument with $\phi^{-1}$. The admissibility is a proposition, so the ep-groupoids built from $F$ and from $G$ coincide and the two cases combine.

**Remark 10.2.3 (2017: 10.2)** (V10.21) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.remark_10_2_3. Both projections are equivalences and $\tau = \sigma$ is natural. *Witness.* id and $-\mathrm{id}$ on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Examples.remark_10_2_3_negFunctor. The sentences on "smaller models" have no formal content.

## X.11 Definition 10.4.1 (2017: 10.10): strong bundle equivalences

**The orbit space at level $[1]$** Chapter10/StrongBundleEquivalence · EpStrongBundle.oneOrbitSetoid Chapter10/StrongBundleEquivalence · EpStrongBundle.OneOrbitSpace Chapter10/StrongBundleEquivalence · EpStrongBundle.oneOrbitMap. $W[1]$ is not built as an ep-groupoid; $|W[1]|$ is the quotient of $W_{0,1}$ with its level-1 topology by the orbit relation of $W$, and a strong bundle functor induces a continuous map of these quotients Chapter10/StrongBundleEquivalence · EpStrongBundle.continuous_oneOrbitMap.

**Definition 10.4.1 (2017: 10.10)** (V10.22) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence. A strong bundle functor Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor with $\phi$ an equivalence, $\Phi$ an equivalence at level $[0]$, $|\Phi|$ a homeomorphism of $|W[1]|$, and local strong bundle isomorphisms on objects and morphisms Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn. Clause (1) at level $[1]$ is read in this form: fullness and faithfulness at $[1]$ are those at $[0]$ restricted, and the local sc-diffeomorphism property at $[1]$ follows from (2) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_one Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_obj. *Consequences.* $P' \circ \Phi = \phi \circ P$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.covers; $\Phi_x : W_x \to W'_{\phi(x)}$ is a linear isomorphism, the local strong bundle isomorphism at $x$ restricted Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.P_inv Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.fiberEquiv, so $\Phi$ is injective on fibres Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.eq_of_obj_eq. *Class.* The identity Chapter10/StrongBundleEquivalence · EpStrongBundle.isStrongBundleEquivalence_id; composites Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.comp, with strong bundle functors Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleFunctor.comp and local strong bundle isomorphisms composing on open sets Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.restrict Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.comp Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.exists_comp; isomorphisms with an inverse strong bundle functor Chapter10/StrongBundleEquivalenceNat · StrongBundle.isStrongBundleIsoOn_

univ_of_inverse Chapter10/StrongBundleEquivalenceNat · EpFunctor.isEquivalence_of_inverse Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.isStrongBundleEquivalence_of_inverse. *Witness.* Over $-\mathrm{id}$ of $\mathbb{R}^n$ Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negMap, the projection of the pull-back of the trivial bundle Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negBundle Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negEpBundle, with inverse Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negInv Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.isStrongBundleMap_negInv, is a strong bundle equivalence Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.isStrongBundleEquivalence_pullbackProj' not over the identity Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExa trivialNeg_obj_ne_id.

## X.12 The pull-back and the push-forward, (10.15)–(10.17) and (10.22)

**Pull-back** (V10.23) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback. $\Phi^* f'(x)$ is the unique $w \in W_x$ with $\Phi(w) = f'(\phi(x))$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.obj_pullback Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_eq, a section Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.P_pullback Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isSection_pullback, functorial when $f'$ is Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pullback ((10.17), by the injectivity of $\Phi$ on fibres and of $\mu'(\sigma, \cdot)$ Chapter10/StrongBundleEquivalence · EpStrongBundle.μ_injective_snd).

**Push-forward** Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.preObj Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.preMor Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_def. For $x'$ choose $x$ and $\sigma : \phi(x) \to x'$ (essential surjectivity of $\phi$) and set $\Phi_* f(x') = \mu'(\sigma, \Phi(f(x)))$, a section Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.P_pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isSection_pushforward. *(10.22).* For a section functor $f$ and any $y$, $\sigma' : \phi(y) \to x'$, the value is $\mu'(\sigma', \Phi(f(y)))$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq: $\sigma'^{-1} \circ \sigma : \phi(x) \to \phi(y)$ is $\phi(\tau)$ by fullness, $f(y) = \mu(\tau, f(x))$ by functoriality, and $\Phi$ intertwines $\mu$ and $\mu'$ Chapter10/StrongBundleEquivalence · EpStrongBundle.μ_mk_congr. Hence $\Phi_* f$ is a section functor Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pushforward, and $\Phi^* \Phi_* f = f$, $\Phi_* \Phi^* f' = f'$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_pullback.

**(10.15)** Chapter10/StrongBundleEquivalence · EpStrongBundle.exists_μ_eq_of_orbitMap_eq Chapter10/StrongBundleEquivalence · EpStrongBundle.orbitMap_eq_of_forall_exists. $|\Phi| = |\Psi|$ iff every $\Psi(w)$ is $\mu'(\sigma, \Phi(w))$ for some $\sigma$.

## X.13 Propositions 10.4.2–10.4.4 (2017: 10.7–10.9)

**Proposition 10.4.2 (2017: 10.7)** (V10.24) `Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_2` `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_eq_of_orbitMap_eq`. With $g = \Psi^* f'$, (10.15) gives $\sigma : \phi(x) \to \psi(x)$ with $\mu'(\sigma, \Phi(g(x))) = \Psi(g(x)) = f'(\psi(x))$; functoriality of $f'$ gives $f'(\psi(x)) = \mu'(\sigma, f'(\phi(x))) = \mu'(\sigma, \Phi(\Phi^* f'(x)))$; the injectivity of $\mu'(\sigma, \cdot)$ and of $\Phi_x$ gives $g(x) = \Phi^* f'(x)$. sc-smoothness is not used. The text's "$f' : W' \to X'$" is read $f' : X' \to W'$ (possible misprint).

**Proposition 10.4.3 (2017: 10.8)** (V10.25) `Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_3` `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq_of_orbitMap_eq`. With $\sigma : \phi(x) \to \psi(x)$ from (10.15) at $f(x)$, and (10.22) for $\Psi$ applied to the composite of $\sigma^{-1}$ with the choice for $\Phi$.

**Proposition 10.4.4 (2017: 10.9)** (V10.26) `Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_4`. The pull-back identity pointwise from the uniqueness in the definition; the push-forward identity from (10.22) for $\Psi \circ \Phi$.

## X.14 Theorem 10.4.5 (2017: 10.8)

**Theorem 10.4.5 (2017: 10.8)** (V10.27) `Chapter10/StrongBundleEquivalence · EpStrongBundle.theorem_10_4_5`. *Step 1, pull-back.* Near $x_0$, $\Phi^* f'$ agrees with $\Phi_U^{-1} \circ f' \circ \phi$ for the local strong bundle isomorphism $\Phi_U$ `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.eqOn_pullback`, so it is sc-smooth `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pullback` and $\mathrm{sc}^+$ when $f'$ is `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pullback`, smoothness being a local property in charts `Chapter10/StrongBundleEquivalence · MPolyfold.isScSmoothOn_of_forall_charts`. *Step 2, push-forward.* The text says the sc-smoothness is easily verified. Near $x_0'$, with a local inverse $\tau$ of $t'$ at the chosen $\sigma_0 : \phi(y_0) \to x_0'$ and a local inverse $g_1$ of $\phi$ at $y_0$, (10.22) gives $\Phi_* f = \mu' \circ (\tau, \Phi \circ f \circ g_1 \circ s' \circ \tau)$ `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.exists_pushforward_local`, sc-smooth `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pushforward`. At level $[1]$ the pair is sc-smooth into the pull-back's level $[1]$ by its universal property on an open set `Chapter10/StrongBundleEquivalence · StrongBundle.isScSmoothOn_to_pullback_one`, so $\mathrm{sc}^+$ is preserved `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pushforward`. *Step 3.* With §X.12 and §X.13, bijections between Chapter 8's sets of section functors `Chapter8/EpStrongBundle · EpStrongBundle.sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors`, independent of $\Phi$ within $|\Phi|$. The text's second display, $\Phi^* : \Gamma(P', \mu') \to \Gamma(P, \mu)$ in the $\Gamma^+$ line, is read with $\Gamma^+$ (a reading point). *The sc-Fredholm part* (G.121) is §X.31 `Chapter10/FredholmEquivalence · EpStrongBundle.theorem_10_4_5_fred`, and the bijections are linear isomorphisms there too. *Witness* `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.bijOn_pushforward_example`.

## X.15 Definitions 10.4.8, 10.4.10 and 10.4.11, Remark 10.4.9 (2017: 10.11–10.13, 10.4)

**Definition 10.4.8 (2017: 10.11)** (V10.28) `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans` `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatEquiv`. $T$ is data with $T(w) : \Phi(w) \to \Psi(w)$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.mem` and naturality, over an `EpNatTrans` $\tau$. Reflexivity `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.refl`; a witness with $\Phi \neq \Psi$ `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.strongBundleNatEquiv_example`.

**Remark 10.4.9 (2017: 10.4)** (V10.29). *Step 1.* A morphism of $\mathbf{W}'$ is $(\sigma, w')$ with $\sigma \in \mathbf{X}'$; $T(w)$ lies over $\tau(P(w))$ and starts at $\Phi(w)$, so $T(w) = (\tau(P(w)), \Phi(w))$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.T_val` `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.T_eq` and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.obj_eq`. The text prints the second line of (10.23) with $\mu$; we read $\mu'$. *Step 2.* $T$ and $\Psi$ on objects are determined by $\Phi$ and $\tau$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.unique`, and $\Psi$ on morphisms too, if $\Psi$ is a strong bundle functor `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.unique_functor`. *Step 3.* $T$ is a strong bundle map covering $\tau$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.isStrongBundleMap_T`.

**Definition 10.4.10 (2017: 10.12)** (V10.30) `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.IsStrongBundleNatTransMap`. Definition 10.4.8 gives it by Remark 10.4.9 `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.isStrongBundleNatTransMap`; conversely the naturality of $T$ follows from (10.23) and the functoriality of $\Phi$, $\Psi$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.IsStrongBundleNatTransMap.toNatTrans`. $\Phi \simeq \Psi$ implies $|\Phi| = |\Psi|$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.orbitMap_eq` `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatEquiv.orbitMap_eq`.

**Definition 10.4.11 (2017: 10.13)** (V10.31) `Chapter10/StrongBundleEquivalenceNat · SEP` `Chapter10/StrongBundleEquivalenceNat · SEP.Hom` `Chapter10/StrongBundleEquivalenceNat · SEP.Hom.ext'`. A morphism is a pair $(\phi, \Phi)$ with $\Phi$ a strong bundle functor covering $\phi$; $\Phi$ alone determines $\phi$ only on the image of $P$. The strong bundle equivalences form a multiplicative class `Chapter10/StrongBundleEquivalenceNat · SEP.equivalences`, mapped into $E$ by the forgetful functor `Chapter10/StrongBundleEquivalenceNat · SEP.toEP` `Chapter10/StrongBundleEquivalenceNat · SEP.toEP_map_mem_equivalences`. *Witness* `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEP` `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.epSEP` `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEPHom` `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEPHom_mem_equivalences`.

## X.16 §10.3: (E1)–(E3), diagrams and their equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4 (2017: Definitions 10.4–10.6, Lemmas 10.3, 10.4))

**(E1)–(E3)** (V10.32). (E1): an isomorphism of $\mathcal{EP}$ has an inverse sc-smooth functor, so it is full, faithful, a homeomorphism on orbit spaces and an sc-diffeomorphism on objects `Chapter10/`

LocalizationCategory · EPLocalization.isEquivalence_of_isIso. (E2) and (E3) are §X.1 and Proposition 10.1.5 Chapter10/Equivalence · EpFunctor.IsEquivalence.comp Chapter10/Equivalence · EpFunctor.proposition_10_1_5.

**Diagrams** (V10.33) Chapter10/Localization · EPLocalization.Diagram. The middle ep-groupoid is an object of $\mathcal{EP}$, so that a diagram, and later its class, lives one universe above the object and morphism types; no class-sized collection occurs. The weak fibred product of §X.9 is made an object of $\mathcal{EP}$ when admissible Chapter10/Localization · EPLocalization.wfp.

**Definition 10.3.1 (2017: 10.4)** Chapter10/Localization · EPLocalization.Diagram.IsRefinement Chapter10/Localization · EPLocalization.Diagram.IsRefinement.refl. Both examples after the definition: precomposition with an equivalence $H$ Chapter10/Localization · EPLocalization.Diagram.precomp Chapter10/Localization · EPLocalization.Diagram.isRefinement_precomp, with $H$ itself refining and the natural equivalences equalities Chapter10/Localization · EpFunctor.NaturallyEquivalent.of_eq; and naturally equivalent legs, with $H = 1_A$ Chapter10/Localization · EPLocalization.Diagram.isRefinement_of_naturallyEquivalent.

**Lemma 10.3.2 (2017: 10.3)** (V10.34) Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_2 Chapter10/Localization · EPLocalization.Diagram.IsRefinement.trans. For $H : d' \to d$ and $H' : d'' \to d'$, $H \circ H'$ is an equivalence (§X.1) and $F \circ H \circ H' \simeq F' \circ H' \simeq F''$, by the compatibility of $\simeq$ with composition Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right and its transitivity; the same for $\Phi$.

**Definition 10.3.3 (2017: 10.5) and Lemma 10.3.4 (2017: 10.4)** (V10.35, V10.36) Chapter10/Localization · EPLocalization.Diagram.IsCommonRefinement Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_4. *Step 1.* For $W$ refining $d$ by $H$ and $d'$ by $H'$, and $W'$ refining $d'$ by $K$ and $d''$ by $K'$, $H'$ and $K$ are equivalences into $A'$, so the weak fibred product $W \times_{A'} W'$ is defined and its projections are equivalences (Theorem 10.2.2). *Step 2.* The diagram (10.11) over $W \times_{A'} W'$ refines $d$ by $H \circ \pi_W$; it refines $d''$ by $K' \circ \pi_{W'}$, because $F' \circ H' \circ \pi_W \simeq F' \circ K \circ \pi_{W'}$ by the natural transformation $\sigma$ of the weak fibred product (§X.10) and Lemma 10.3.2. A refinement is equivalent to the refined diagram Chapter10/Localization · EPLocalization.Diagram.IsRefinement.equiv.

**Definition 10.3.5 (2017: 10.6)** (V10.37) Chapter10/Localization · EPLocalization.Diagram.Equiv Chapter10/Localization · EPLocalization.Diagram.setoid Chapter10/Localization · EPLocalization.Diagram.equiv_iff. Reflexivity and symmetry are immediate Chapter10/Localization · EPLocalization.Diagram.Equiv.refl Chapter10/Localization · EPLocalization.Diagram.Equiv.symm, transitivity is Lemma 10.3.4 Chapter10/Localization · EPLocalization.Diagram.Equiv.trans. *Witness.* pt $\leftarrow$ $\mathbb{Z}$-pair $\to$ pt, both legs the equivalence of §X.1, is equivalent to $1_{\text{pt}}$ through the refinement by that equivalence, which is not an isomorphism Chapter10/LocalizationCategory · EPLocalization.Examples.equiv_pairToPoint_id.

## X.17 Lemma 10.3.6 (2017: 10.5), Definition 10.3.7 (2017: 10.7) and Theorem 10.3.8 (2017: 10.3) with Appendix §10.6.1

**Lemma 10.3.6 (2017: 10.5)** (V10.38) Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_6. $|d| = |\Phi| \circ |F|^{-1}$, with $|F|^{-1}$ the inverse of the homeomorphism of Definition 10.1.1 (2) Chapter10/Localization · EPLocalization.Diagram.orbitMap Chapter10/Localization · EPLocalization.Diagram.continuous_orbitMap Chapter10/Localization · EPLocalization.Diagram.orbitMap_orbitMap_F Chapter10/Localization · EPLocalization.Diagram.orbitMap_eq_of. For $H : d' \to d$, $|F'| = |F \circ H| = |F| \circ |H|$ Chapter10/Localization · EpFunctor.orbitMap_comp Chapter10/Localization · EpFunctor.orbitMap_id, since naturally equivalent functors have the same orbit map Chapter10/Localization · EpFunctor.NaturallyEquivalent.

`orbitMap_eq`, and likewise $|\Phi'| = |\Phi| \circ |H|$; the factors $|H|$ cancel. Equivalent diagrams have a common refinement, hence the same map `Chapter10/Localization · EPLocalization.Diagram.Equiv.orbitMap_eq`.

**Definition 10.3.7 (2017: 10.7)** (V10.39) `Chapter10/Localization · EPLocalization.Diagram.orbitMapQ` `Chapter10/Localization · EPLocalization.Diagram.orbitMapQ_mk`. When $\Phi$ is an equivalence, $|\Phi| \circ |F|^{-1}$ is a composite of homeomorphisms `Chapter10/Localization · EPLocalization.Diagram.isHomeomorph_orbitMap`.

**The composite** `Chapter10/Localization · EPLocalization.Diagram.comp` `Chapter10/Localization · EPLocalization.Diagram.id`. For $d$ and $e : Y \xleftarrow{G} B \xrightarrow{\Psi} Z$, the weak fibred product of $\Phi$ and $G$ is admissible since $G$ is an equivalence `Chapter10/Localization · EPLocalization.Diagram.admissible`, and $\pi_A$ is an equivalence (Theorem 10.2.2), so $F \circ \pi_A$ is one. `d.comp e` is the text's $[e] \circ [d]$.

**Theorem 10.3.8 (2017: 10.3)** (V10.40) `Chapter10/Localization · EPLocalization.Diagram.theorem_10_3_8`, the proof of Appendix §10.6.1. *Step 1, the functor $L$.* For refinements $H : d' \to d$, $K : e' \to e$ with natural transformations $\tau_{A'} : \Phi' \to \Phi \circ H$ and $\tau_{B'} : G' \to G \circ K$, $L(a', \phi, b') = (H(a'), \tau_{B'}(b') \circ \phi \circ \tau_{A'}(a')^{-1}, K(b'))$ ((10.28), (10.30)) `Chapter10/Localization · EPLocalization.mapMid` `Chapter10/Localization · EPLocalization.mapObj` `Chapter10/Localization · EPLocalization.mapMor` `Chapter10/Localization · EPLocalization.wfpMap` is an sc-smooth functor, entry by entry, with the bookkeeping identity `Chapter10/Localization · GroupoidStr.comp_comp_inv_comp_inv`. In the text's naturality of $\tau_{B'}$ the letter $\tau_A$ appears; we read $\tau_{B'}$ (possible misprint). *Step 2, $L$ is an equivalence* `Chapter10/Localization · EPLocalization.isEquivalence_wfpMap`. $\pi_A \circ L = H \circ \pi_{A'}$ holds on the nose `Chapter10/Localization · EPLocalization.projLeft_comp_wfpMap` `Chapter10/Localization · EPLocalization.projRight_comp_wfpMap`; $\pi_A$, $\pi_{A'}$ and $H$ are equivalences, and the two-out-of-three rule on the other side from Proposition 10.1.6 `Chapter10/Localization · EpFunctor.IsEquivalence.of_comp_left` applies. The text checks directly that $L$ is full, faithful and a local sc-diffeomorphism. *Step 3, the refinement.* $(F \circ \pi_A) \circ L = F \circ H \circ \pi_{A'} \simeq F' \circ \pi_{A'}$, and $(\Psi \circ \pi_B) \circ L = \Psi \circ K \circ \pi_{B'} \simeq \Psi' \circ \pi_{B'}$ `Chapter10/Localization · EPLocalization.Diagram.IsRefinement.comp`; the text writes the second as "$(\Phi \circ \pi_B) \circ L \simeq \Psi' \circ \pi_{B'}$", read with $\Psi$ (possible misprint). With common refinements on both sides, the class of the composite depends only on the classes `Chapter10/Localization · EPLocalization.Diagram.Equiv.comp`. *Step 4, associativity.* The text identifies $(A \times_Y A') \times_Z A''$ and $A \times_Y (A' \times_Z A'')$; here the reassociation $((a, \phi, a'), \psi, a'') \mapsto (a, \phi, (a', \psi, a''))$ is a functor `Chapter10/Localization · EPLocalization.assocObj` `Chapter10/Localization · EPLocalization.assocMor` `Chapter10/Localization · EPLocalization.assocFunctor`, an equivalence by two out of three `Chapter10/Localization · EPLocalization.projLeft_comp_assocFunctor` `Chapter10/Localization · EPLocalization.isEquivalence_assocFunctor`, and a refinement with identity natural transformations `Chapter10/Localization · EPLocalization.Diagram.isRefinement_assoc` `Chapter10/Localization · EPLocalization.Diagram.equiv_assoc`. *Step 5, units.* $X \xleftarrow{\pi_X} X \times_X A \xrightarrow{\Phi \circ \pi_A} Y$ refines $d$ by $\pi_A$, with $F \circ \pi_A \simeq \pi_X$ by $\tau(x, \phi, a) = \phi$ `Chapter10/Localization · EPLocalization.Diagram.isRefinement_id_comp`, and symmetrically on the other side `Chapter10/Localization · EPLocalization.Diagram.isRefinement_comp_id`. The text's proof announces $[1_X] \circ [d] = [d]$ for $d : X \to Y$, which in its order of composition is $[d] \circ [1_X] = [d]$, writes $\tau(a, \phi, a)$ for $\tau(x, \phi, a)$ and "$\psi = \varphi$" for $\psi = \phi$ (possible misprints).

## X.18 Definitions 10.3.9 and 10.3.12, Theorem 10.3.10 (2017: Definitions 10.8, 10.9, Theorem 10.4) with Appendix §10.6.2

**Definition 10.3.12 (2017: 10.9)** (V10.44) `Chapter10/LocalizationCategory · EPLocalization.EPLoc`. The objects are a copy of those of $\mathcal{EP}$ `Chapter10/LocalizationCategory · EPLocalization.EPLoc.toEP_of` `Chapter10/LocalizationCategory · EPLocalization.EPLoc.of_toEP`, the morphisms the quotients of dia-

grams Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk, with $[d] \gg [e] = [d.\mathrm{comp}\, e]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_comp and $\mathrm{id} = [1_X]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.id_eq; the category laws are Theorem 10.3.8. $[d] = [d']$ iff $d$ and $d'$ are equivalent Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_eq_homMk_iff, and every morphism is a $[d]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_surjective.

**Definition 10.3.9 (2017: 10.8)** (V10.41) Chapter10/LocalizationCategory · EPLocalization.EPLoc.IsGeneralizedIso, the isomorphisms of the category Chapter10/LocalizationCategory · EPLocalization.EPLoc.isGeneralizedIso_iff_isIso.

**Theorem 10.3.10 (2017: 10.4)** (V10.42) Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_10, the proof of Appendix §10.6.2. *Step 1, $\Phi$ an equivalence.* The reversed diagram $Y \xleftarrow{\Phi} A \xrightarrow{F} X$ Chapter10/LocalizationCategory · EPLocalization.Diagram.symm. The composite $X \xleftarrow{F\circ\pi_1} A \times_Y A \xrightarrow{F\circ\pi_2} X$ is refined by $X \xleftarrow{F} A \xrightarrow{F} X$ through the diagonal $a \mapsto (a, 1_{\Phi(a)}, a)$ Chapter10/Localization · EPLocalization.diagObj Chapter10/Localization · EPLocalization.diagMor Chapter10/Localization · EPLocalization.diagFunctor, an equivalence since $\pi_1 \circ \Delta = 1_A$ Chapter10/Localization · EPLocalization.isEquivalence_diagFunctor Chapter10/Localization · EPLocalization.Diagram.isRefinement_comp_symm, and that diagram refines $1_X$ by $F$ Chapter10/Localization · EPLocalization.Diagram.isRefinement_id; so both composites are units Chapter10/LocalizationCategory · EPLocalization.Diagram.equiv_comp_symm Chapter10/LocalizationCategory · EPLocalization.Diagram.equiv_symm_comp. *Step 2, the converse.* Let $[b] = [Y \xleftarrow{G} B \xrightarrow{\Psi} X]$ be an inverse. A diagram $X \leftarrow C \xrightarrow{\Theta} X$ equivalent to $1_X$ has $\Theta$ an equivalence (Propositions 10.1.5 and 10.1.6 applied to a common refinement) Chapter10/LocalizationCategory · EPLocalization.Diagram.isEquivalence_Φ_of_equiv_id; so $\Psi \circ \pi_B$ and $\Phi \circ \pi_{A'}$ are equivalences. *Step 3, (C1).* $\Phi$ is an equivalence of categories Chapter10/LocalizationCategory · EPLocalization.isEquivalence_toFunctor_of_comp. Faithfulness is checked on morphisms $(h, \phi, k)$ with $k = 1_b$; in the text's injectivity argument $A \times_X B$ is read $A \times_Y B$ and $\mathrm{mor}_Y(\Psi(a), \Psi(a'))$ as $\mathrm{mor}_X(\Psi(b), \Psi(b'))$, and in the essential surjectivity "between $\Phi(a)$ and $x$" as "and $y$" (possible misprints). *Step 4, (C2).* The text defines $\Gamma(b') = (b', (s|U)^{-1}(b'), \ldots)$, read with $(s|U)^{-1}(\Psi(b'))$ (possible misprint), and then uses that $\Gamma' : U' \to A$ is a local sc-diffeomorphism; $\Gamma'$ is a composite with $\Psi$, so as we read it this needs $\Psi$ to be a local sc-diffeomorphism, the other half of (C2), and the step does not hold as we read it. The formal argument: a local inverse $\kappa$ of the equivalence $\Phi \circ \pi_{A'}$ at $(b, \phi, a)$ gives the local sc-smooth right inverse $\pi_{A'} \circ \kappa$ of $\Phi$ with value $a$ at $\Phi(a)$; $\Phi$ is injective near $a$ by a local section $a' \mapsto (a', \phi(a'), b(a'))$ of the local sc-diffeomorphism $\pi_A : A \times_Y B \to A$ and the local injectivity of $\Psi \circ \pi_B$, $s$ and $G$ Chapter10/LocalizationCategory · MPolyfold.IsLocalScDiffeoAt.exists_injOn; an injective map with such a right inverse is a local sc-diffeomorphism Chapter10/LocalizationCategory · MPolyfold.isLocalScDiffeoAt_of_injOn Chapter10/LocalizationCategory · EPLocalization.isLocalScDiffeo_obj_of_comp. *Step 5.* Lemma 10.1.2 Chapter10/LocalizationCategory · EPLocalization.Diagram.isEquivalence_Φ_of_equiv. *(2).* An inverse is unique, so every inverse is the class of the reversed diagram Chapter10/LocalizationCategory · EPLocalization.EPLoc.isIso_homMk_iff Chapter10/LocalizationCategory · EPLocalization.EPLoc.isIso_homMk Chapter10/LocalizationCategory · EPLocalization.EPLoc.inv_homMk.

## X.19 Proposition 10.3.11, Theorems 10.3.14 and 10.3.15 (2017: Proposition 10.6, Theorems 10.5, 10.6)

**Proposition 10.3.11 (2017: 10.6)** (V10.43) Chapter10/LocalizationCategory · EPLocalization.EPLoc.proposition_10_3_11. $||[d]||$ on the quotient Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitMap Chapter10/

LocalizationCategory · EPLocalization.EPLoc.orbitMap_homMk Chapter10/LocalizationCategory · EPLocalization.EPLoc.continuous_orbitMap Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitMap_id. The computation before the proposition: an object $(a, \phi, b)$ of $A \times_Y B$ has $\phi : \Phi(a) \to G(b)$, so $|\Phi(a)| = |G(b)|$, and $|d.\mathrm{comp}\, e| = |e| \circ |d|$ Chapter10/Localization · EPLocalization.Diagram.orbitMap_comp.

**Theorem 10.3.14 (2017: 10.5)** (V10.45) Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_14. *Step 1.* $d_\Phi = X \xleftarrow{1} X \xrightarrow{\Phi} Y$ Chapter10/LocalizationCategory · EPLocalization.Diagram.ofFunctor and $i$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor_obj Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor_map. *Step 2, functoriality.* The composite (10.13) $X \xleftarrow{\pi_X} X \times_Y Y \xrightarrow{\Psi \circ \pi_Y} Z$ refines $d_{\Psi\circ\Phi}$ by $\pi_X$, since $\Phi \circ \pi_X \simeq \pi_Y$ (§X.10) Chapter10/LocalizationCategory · EPLocalization.Diagram.isRefinement_ofFunctor_comp. The text states the refinement in the other direction; with Definition 10.3.1 (the refining equivalence goes out of the refinement) it is this one, and the equality of classes is the same. *Step 3, inverses.* For $F \in E$, Theorem 10.3.10 with the legs 1 and $F$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.isInvertedBy_localizationFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.inv_localizationFunctor_map, and every generalized map is $i(\Phi) \circ i(F)^{-1}$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_eq. *Witness.* The pair groupoid of $\mathbb{Z}$ and the point as objects of $\mathcal{EP}$ Chapter10/LocalizationCategory · EPLocalization.Examples.pairEP Chapter10/LocalizationCategory · EPLocalization.Examples.pointEP Chapter10/LocalizationCategory · EPLocalization.Examples.pairToPointEP Chapter10/LocalizationCategory · EPLocalization.Examples.isEquivalence_pairToPointEP: the functor is not an isomorphism of $\mathcal{EP}$ Chapter10/LocalizationCategory · EPLocalization.Examples.not_isIso_pairToPointEP, its image is invertible Chapter10/LocalizationCategory · EPLocalization.Examples.isGeneralizedIso_pairToPoint, with the inverse of Theorem 10.3.14 Chapter10/LocalizationCategory · EPLocalization.Examples.inv_localizationFunctor_map_pairToPoint. *The universal property* mentioned on p. 483 (G.117) is §X.30: $i$ is a localization of $\mathcal{EP}$ at $E$ Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.isLocalization.

**Theorem 10.3.15 (2017: 10.6)** (V10.46) Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_15. A functor into Mathlib's `TopCat` by Proposition 10.3.11.

## X.20 Propositions 10.3.16, 10.3.17, Remark 10.3.18 (2017: 10.3), Theorem 10.3.19 (2017: 10.7), Definition 10.3.20, Theorem 10.3.21

**The reduction** Chapter10/LocalizationProperties · EPLocalization.EPLoc.exists_diagram_of_isGeneralizedIso Chapter10/LocalizationProperties · EPLocalization.EPLoc.iff_of_isGeneralizedIso. A generalized isomorphism is some $[d]$, and by Theorem 10.3.10 its $\Phi$ is an equivalence; a property invariant under equivalences passes from $X$ to $A$ to $Y$, as on p. 492.

**Proposition 10.3.17** (V10.47) Chapter10/LocalizationProperties · EPLocalization.EPLoc.proposition_10_3_17, by Proposition 10.1.16. **Proposition 10.3.16** (V10.64) Chapter10/LocalizationLR · EPLocalization.EPLoc.proposition_10_3_16, by Proposition 10.1.9. **Theorem 10.3.21** (V10.50) Chapter10/LocalizationProperties · EPLocalization.EPLoc.theorem_10_3_21 Chapter10/LocalizationLR · EPLocalization.EPLoc.theorem_10_3_21_all, by Propositions 10.1.9, 10.1.16, 10.1.19, 10.1.18 and 10.1.17; the text leaves the proof to the reader (G.118, closed).

**Theorem 10.3.19 (2017: 10.7) does not hold as we read it** (V10.65) Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.Theorem_10_3_19 Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexa
not_theorem_10_3_19. *The example.* $X = \mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action has (LR): its orbits are points,

so an orbit-preserving $\beta$ is the identity, the action of $1$ Chapter10/LocalizationLRCounterexample · EpGroupoid.Examples.isLR_trivialZ3. $[X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X]$ and $[1_X]$ are generalized isomorphisms with the same orbit map that differ Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.theorem_10_3_19_counterexample, the base part of Remark 10.5.5 (§X.23). *The step on p. 494.* The proof arranges $|H| = |H'|$ for equivalences $H, H' : A \to Y$ and uses the families $\alpha_a$ of Proposition 10.1.10 (1) as a natural transformation $H_{\mathcal{U}} \to H'_{\mathcal{U}}$. Here $z \mapsto 1_z : z \to \mathrm{Inv}(z)$ is such a family, but there is no natural transformation $1 \to \mathrm{Inv}$: naturality at the generator $g$ of $G_z$ would give $g = g^{-1}$ Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_naturallyEquivalent_id_inv. $X$ is not effective Chapter7/NonVacuity · EpGroupoid.Examples.not_isEffective_trivialZ3. The 2017 Theorem 10.7, without (LR), does not hold as we read it either (G.120, settled).

**Theorem 10.3.19 for an effective target** (V10.66) Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.theorem_10_3_19_of_isEffective, the strongest correct version we prove. *Step 1.* In an effective ep-groupoid a continuous family $v \mapsto k(v) \in G_v$ on an open set is the family of units Chapter10/EquivalenceLREffective · EpGroupoid.eqOn_unit_of_isEffective, so two continuous families $\alpha_z, \alpha'_z : H(z) \to H'(z)$ on an open set agree when $H$ is a local sc-diffeomorphism on objects Chapter10/EquivalenceLREffective · EpFunctor.eqOn_of_isEffective. *Step 2.* The families of Proposition 10.1.10 (1) therefore glue on all of $X$, and they are natural: at a morphism $h$, $z \mapsto H'(\sigma(z))^{-1} \circ \alpha(t(\sigma(z))) \circ H(\sigma(z))$, with $\sigma$ the local inverse of $s$ through $h$, is another such family Chapter10/EquivalenceLREffective · EpFunctor.naturallyEquivalent_of_isEffective. *Step 3.* For $e = f' \circ f^{-1}$, represented by $Y \xleftarrow{F} A \xrightarrow{\Phi} Y$ with $|\Phi| = |F|$, Step 2 gives $F \simeq \Phi$, so $e = 1_Y$ and $f = f'$, without the weak fibred product and the cover $A_{\mathcal{U}}$ of the text. *Witnesses*, the point Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.theorem_10_3_19_of_isEffective_hypotheses Chapter10/EquivalenceLREffective · EquivalenceLREffectiveExamples.naturallyEquivalent_of_isEffective_hypotheses.

**Remark 10.3.18 (2017: 10.3)** (V10.48) Chapter10/LocalizationProperties · EPLocalization.OrbitChart Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel. Reflexive by $1_X$ Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.refl, symmetric by the inverse Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.symm and transitive by the composite Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.trans, each with Proposition 10.3.11 Chapter10/LocalizationProperties · EPLocalization.OrbitChart.setoid. *Witness* Chapter10/LocalizationProperties · EPLocalization.OrbitChart.rel_pairEP_pointEP. The metrizability of $Z$ is not used. The non-uniqueness of $f$ is shown by the base classes of the example of Remark 10.5.5 (§X.23; G.119, closed).

**Definition 10.3.20** (V10.49) Chapter10/LocalizationProperties · EPLocalization.EPreg Chapter10/LocalizationProperties · EPLocalization.EPLocReg, Mathlib full subcategories, with the point as an object of both Chapter10/LocalizationProperties · EPLocalization.Examples.pointReg Chapter10/LocalizationProperties · EPLocalization.Examples.pointLocReg. Remark 10.3.13 has no formal content.

## X.21 §10.5: diagrams of strong bundles, $[d]$, $|[D]|$ and $\gamma$

**Diagrams and refinements** (V10.51) Chapter10/StrongBundleLocalization · SBLocalization.SDiagram Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement. A diagram is a middle object of $\mathcal{SEP}$ with a strong bundle equivalence $\Phi$ and a morphism $\Psi$ of $\mathcal{SEP}$. A refinement is a strong bundle equivalence $H$ with $\Phi \circ H \simeq \Phi'$ and $\Psi \circ H \simeq \Psi'$, natural equivalence of morphisms of $\mathcal{SEP}$ being Definition 10.4.8 Chapter10/StrongBundleLocalization · SEP.NatEquiv, an equivalence re-

lation compatible with composition on the left Chapter10/StrongBundleLocalization · SEP.NatEquiv.comp_left. The two examples Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.isRefinement_precomp Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.isRefinement_of_natEquiv and the strong bundle Lemma 10.3.2 Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.trans follow: $H \circ H'$ is a strong bundle equivalence and $\Phi \circ H \circ H' \simeq \Phi' \circ H' \simeq \Phi''$. *The classes.* Having a common refinement Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv is reflexive and symmetric Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.refl Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.symm; its transitivity, the strong bundle Lemma 10.3.4, is proved in §X.22 through the weak fibred product of strong bundles. $[D]$ is the class for the generated equivalence relation Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.cls, which is having a common refinement (§X.22); every invariant below is proved invariant under `Equiv`, hence defined on the classes, and a refinement has the class of the refined diagram Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.equiv Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.cls_eq_cls_of_equiv.

**$[d]$ and $|[D]|$** (V10.52) Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.total Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.base Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.orbitMap. Forgetting the bundles gives the diagram $d$ of the base ep-groupoids, and the total spaces as ep-groupoids $(W, \mathbf{W})$ Chapter10/StrongBundleLocalization · SEP.totalEP give a diagram of total spaces; refinements and equivalences pass to both Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.total, since a natural equivalence of strong bundle functors covers one of the base functors and is one of the total-space functors. $|[D]|$ is §10.3's orbit map $|\Psi| \circ |\Phi|^{-1}$ of the diagram of total spaces Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap, continuous Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.continuous_orbitMap and the same for equivalent diagrams by Lemma 10.3.6 Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.orbitMap_eq.

**The square $|\mathcal{D}|$** (V10.53) Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap_square Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.orbitMap_square. Every orbit of $|W|$ is $|\Phi(w)|$ for some $w \in W''$, $|\Phi|$ being onto; then $|[D]|(|\Phi(w)|) = |\Psi(w)|$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap_orbitMap_Φ, $|P'(\Psi(w))| = |\psi(P''(w))|$ since $\Psi$ covers $\psi$, and $|[d]|(|\phi(P''(w))|) = |\psi(P''(w))|$ by Lemma 10.3.6. The printed $|X| \to |X|$ is read with $X'$.

$\gamma$ (V10.54) Chapter10/StrongBundleLocalization · EPLocalization.Diagram.exists_isotropyData Chapter10/StrongBundleLocalization · EPLocalization.Diagram.isotropyHom. The data: $x''$ and $a$ from the essential surjectivity of $\phi$, $b$ from $|[d]|(|x|) = |x'|$. $\gamma$ is the composite of the conjugation by $a^{-1}$, the inverse of the isomorphism $G_{x''} \to G_{\phi(x'')}$ of the equivalence $\phi$ (Remark 10.1.12), $\psi$ on isotropy groups and the conjugation by $b$, with its values Chapter10/StrongBundleLocalization · EPLocalization.Diagram.isotropyHom_spec; for $\psi$ an equivalence each factor is bijective Chapter10/StrongBundleLocalization · EPLocalization.Diagram.bijective_isotropyHom. The germ $q$ and its lift $Q$ of pp. 505–506 are §X.33 (G.124) Chapter10/GapP2_GDiagram · EPLocalization.Diagram.exists_germ_q Chapter10/GapP2_GDiagramLift · SBLocalization.SDiagram.exists_lift_Q.

## X.22 Definitions 10.5.1 and 10.5.2 (2017: 10.14 and 10.15), Remark 10.5.3

**The weak fibred product of strong bundles: Theorem 10.2.2 for strong bundles** (V10.68) `Chapter10/StrongBundleWeakFibredProduct · SBWeakFibredProduct.wfpSEP` `Chapter10/StrongBundleWeakFibredProduct · SBWeakFibredProduct.theorem_10_2_2`. The text introduces refinements and classes "following our earlier construction" (p. 504) and gives no weak fibred product of strong bundles. *The tools.* For a strong bundle $W$ over $X$ and a functor $\pi : A \to X$, the pull-back $\pi^*W$ over $A$, with $\mu_\pi(\gamma,(a,w)) = (t(\gamma),\mu(\pi(\gamma),w))$ `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.pullbackAlong`, whose projection is a strong bundle equivalence when $\pi$ is an equivalence `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.isStrongBundleEquivalence_pullbackProjFunctor`; the transport $w \mapsto \mu'(\tau(Pw),\Psi(w))$ of a strong bundle functor along a natural transformation $\tau$ `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.transportFunctor`; and, for a strong bundle equivalence $\Phi$ covering $\phi$ and a strong bundle functor $\Theta$ covering $\phi\circ\kappa$, the unique lift $\Lambda$ covering $\kappa$ with $\Phi\circ\Lambda = \Theta$, fibrewise $\Lambda(c) = \Phi^{-1}_{\kappa(Pc)}(\Theta(c))$, well defined because $\Phi$ is a linear isomorphism on fibres (Proposition 10.4.4) and sc-smooth because it is locally $\Phi^{-1}\circ\Theta$ `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.invLiftFunctor`. *The construction.* $W\times_{W'}W'' = \pi_Z^*W''$ over $X\times_{X'}X''$; $\Pi_Z$ is the projection of the pull-back, an equivalence since $\pi_Z$ is one (Theorem 10.2.2), and $\Pi_X$ the lift of the transport of $\Psi\circ\Pi_Z$ along the natural transformation $\psi\circ\pi_Z \to \phi\circ\pi_X$ `Chapter10/StrongBundleWeakFibredProduct · SEP.coneHom`, so that $\Phi\circ\Pi_X \simeq \Psi\circ\Pi_Z$ `Chapter10/StrongBundleWeakFibredProduct · SEP.natEquiv_cone`. When $\Psi$ is an equivalence, $\Pi_X$ is an isomorphism onto $\pi_X^*W$ followed by its projection, an equivalence `Chapter10/StrongBundleWeakFibredProduct · SBWeakFibredProduct.theorem_10_2_2_right`. *Witness* `Chapter10/StrongBundleWeakFibredProduct · StrongBundleWeakFibredProductExamples.theorem_10_2_2_negSEPHom`: $\Phi$ over $-\mathrm{id}$ and $\Psi = 1$, with both projections strong bundle equivalences.

**Transitivity: Lemma 10.3.4 for strong bundles** (V10.69) `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.lemma_10_3_4` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.Equiv.trans`. As in §X.16: if $w$ refines $D$, $D'$ by $H$, $H'$ and $w'$ refines $D'$, $D''$ by $K$, $K'$, the diagram on the weak fibred product of $H'$ and $K$ refines $D$ by $H\circ\Pi_X$ and $D''$ by $K'\circ\Pi_Z$, both strong bundle equivalences since $K$ is one. Hence $[D] = [D']$ iff $D$ and $D'$ have a common refinement `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.cls_eq_cls_iff`, and the classes of §X.21 are the text's.

**The composition: Theorem 10.3.8 for strong bundles** (V10.70) `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.comp` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.Equiv.comp`. For $D$ and $E : W' \xleftarrow{\Phi'} V'' \xrightarrow{\Psi'} W'''$ the middle of the composite is the pull-back of $W''$ along $\pi_X$ over the weak fibred product of $\psi$ and $\phi'$ `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.compMid`, so that its projection $\Pi_{W''}$ is a strong bundle equivalence `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.compLeft`, and $\Pi_{V''}(p,w) = \Phi'^{-1}_z(\mu'(\varphi,\Psi(w)))$ is a lift as above `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.compRight`. If $D'$ refines $D$ by $H$ and $E'$ refines $E$ by $K$, the lift of $H\circ\Pi_{W''}$ over the functor $L$ of (10.28), (10.30) refines the composite `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.IsRefinement.comp`; associativity is the lift over the reassociation functor of §X.17 `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.isRefinement_assoc` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_assoc`, and the units come from sections into the middle `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.unitSection` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_comp_`

id Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_id_comp. In each proof the square with the projections to the middle of $E$, or of the last diagram, commutes strictly, because a strong bundle equivalence is injective on fibres. *Witness* Chapter10/StrongBundleLocalizationCategory · StrongBundleLocalizationCategoryExamples.homMk_foldDiagram_comp Chapter10/StrongBundleLocalizationCategory · StrongBundleLocalizationCategoryExamples.nonempty_comp_foldDiagram: for $F = [W' \xleftarrow{P'} (-\mathrm{id})^*W' \xrightarrow{P'} W']$, $F \circ F = 1$, with a nonempty middle.

**Definition 10.5.1 (2017: 10.14)** (V10.55) Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.idHom Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.localizationPrefunctor. The objects are a one-field structure around $\mathcal{SEP}$, the morphisms the classes, the identities $[1_W]$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.id; with the composition $[E] \circ [D] = [D \circ E]$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.GenSBMap.comp the laws of the previous paragraph make it a category Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.category (G.123, closed). $\Psi \mapsto [1, \Psi]$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.ofHom, first a prefunctor mapping $1_W$ to $[1_W]$ Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.localizationPrefunctor_map_id, is a functor by $[1, \Psi'] \circ [1, \Psi] = [1, \Psi' \circ \Psi]$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_ofHom_comp Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.localizationFunctor, and it inverts the strong bundle equivalences Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.isInvertedBy_localizationFunctor.

**Definition 10.5.2 (2017: 10.15)** (V10.56) Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.IsGeneralizedIso: some representative has $\Psi$ in $F$ ($\Phi$ is in $F$ for every diagram). The reversed diagram Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.symm of such a representative, and $[1, \Psi]$ for $\Psi$ in $F$, give examples Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.isGeneralizedIso_symm Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.isGeneralizedIso_ofHom. *Witness* Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.negDiagram Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isGeneralizedIso_negDiagram: $[1, P']$ for the projection $P' : (-\mathrm{id})^*W' \to W'$ of §X.11, between two different strong bundles.

**Remark 10.5.3** (V10.57) Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.remark_10_5_3. For a representative with $\Phi$, $\Psi$ in $F$, $\phi$ and $\psi$ are equivalences and $\Phi$, $\Psi$ are equivalences of the total spaces (Definition 10.4.1); Theorem 10.3.10 makes $[d]$ and the class of the diagram of total spaces generalized isomorphisms, and Lemma 10.3.6 makes $|[D]|$ a homeomorphism. The strong bundle version of Theorem 10.3.10, which the remark builds into Definition 10.5.2, is read as: the generalized strong bundle isomorphisms are exactly the isomorphisms of $\mathcal{SEP}(F^{-1})$ (next paragraph and §X.34).

**Theorem 10.3.10 for strong bundles, the first direction** (V10.71; the converse is §X.34) Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.theorem_10_3_10 Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.isIso_of_isGeneralizedIso. If $\Psi$ is a strong bundle equivalence, $D \circ D^{-1}$ and $D^{-1} \circ D$, for the reversed diagram $D^{-1}$, are refined through sections into their middles Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_comp_symm Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_symm_comp, so $[D]$ is an isomorphism with inverse $[D^{-1}]$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.inv_homMk. *Towards the converse.* $[D] \mapsto [d]$ is a functor to $\mathcal{EP}(E^{-1})$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.baseFunctor, so an isomorphism $[D]$ gives an isomorphism $[d]$, and Theorem 10.3.10 for the underlying diagram makes $\psi$ an equivalence Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.isEquivalence_base_of_isIso. That $\Psi$ itself is a strong bundle equivalence is §X.34 (G.123,

closed for this half too), by a route that replaces the claims (C1), (C2) of Appendix §10.6.2 Chapter10/GapP2_F2_SBConverse · SBLocalization.SEPLoc.isIso_iff_isGeneralizedIso. *Witness* Chapter10/StrongBundleLocalizationCategory · StrongBundleLocalizationCategoryExamples.isIso_homMk_negDiagram: the class over $-\mathrm{id}$ of Definition 10.5.2.

## X.23 Theorem 10.5.4 (2017: 10.9), Remark 10.5.5, Proposition 10.5.6

**The push-forward along a diagram** (V10.58) Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pullback. For $\Phi$, $\Psi$ in $F$, $D_*f = \Psi_*\Phi^*f$ with §10.4's pull-back and push-forward, and $D^*g$ is the push-forward along the reversed diagram. That they map section functors to section functors, are mutually inverse and do the same on $\mathrm{sc}^+$-section functors Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.mapsTo_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.invOn_pullback_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.bijOn_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.bijOn_pushforward_plus comes from Theorem 10.4.5. *The text's step* Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward_eq_of_isRefinement: for a refinement by $H$, $\Psi_*\Phi^*f = \Psi_*H_*H^*\Phi^*f = (\Psi\circ H)_*(\Phi\circ H)^*f = \Psi'_*\Phi'^*f$, by Theorem 10.4.5 for $H$, Proposition 10.4.4, and Propositions 10.4.2, 10.4.3 for the naturally equivalent pairs, which have the same orbit maps. *Through the orbit spaces.* Step 1: $|\mu(\sigma,w)| = |w|$ Chapter10/StrongBundleLocalizationSections · EpStrongBundle.π_μ, so a section functor takes isomorphic objects to one orbit Chapter10/StrongBundleLocalizationSections · EpStrongBundle.π_eq_π_of_isFunctorialSection, and two section functors with $|f(x)| = |g(x)|$ for all $x$ agree: a morphism $(\sigma, f(x)) : f(x) \to g(x)$ has $\sigma \in G_x$, and $f(x) = \mu(\sigma, f(x))$ since $f$ is functorial Chapter10/StrongBundleLocalizationSections · EpStrongBundle.eq_of_forall_π_eq. Step 2: $|(D_*f)(x')| = |[D]|(|f(x)|)$ whenever $|[d]|(|x|) = |x'|$ Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.π_pushforward. Step 3: $|[d]|$ is determined by $|[D]|$ through the square, $|P|$ being onto Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.base_orbitMap_eq_of_orbitMap_eq. Hence diagrams with the same $|D|$ have the same push-forward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward_eq_of_orbitMap_eq and, reversed, the same pull-back Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pullback_eq_of_orbitMap_eq.

**Theorem 10.5.4 (2017: 10.9)** (V10.59) Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.theorem_10_5_4. $[D]_*$ and $[D]^*$ use a representative with $\Psi$ in $F$ chosen once Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.rep Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pullback; every other such representative has the same $|D|$, an invariant of the class, hence the same push-forward and pull-back Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pushforward_eq Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pullback_eq. The rest is the previous paragraph for the chosen representative, the last assertion with Step 3. The Fred part (G.125) Chapter10/FredholmEquivalence · SBLocalization.GenSBMap.theorem_10_5_4_fred and "isomorphism" as a linear isomorphism Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforwardEquiv are §X.31. *Witnesses* Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.bijOn_pushforward_negDiagram Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.pushforward_foldDiagram: $W' \xleftarrow{P'} (-\mathrm{id})^*W' \xrightarrow{P'} W'$ refines $1_{W'}$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isRefinement_foldDiagram Chapter10/

StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.cls_foldDiagram, so its push-forward is that of $[1_{W'}]$, the identity Chapter10/StrongBundleLocalizationExamples · SBLocalization.SDiagram.pushforward_id.

**Remark 10.5.5** (V10.60) Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.remark_10_5_5. The positive half is the last assertion of Theorem 10.5.4 Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.pushforward_inv_eq_pushforward_id; for the negative half the text gives no example. *The invariant* Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.IsotropyRel: $g \in G_x$ and $g' \in G_{x'}$ are related by $d$ if $g = a \circ \phi(k) \circ a^{-1}$ and $g' = b \circ \psi(k) \circ b^{-1}$ for some $z$, $k \in G_z$, $a$, $b$; the relation contains the graph of every $\gamma$ Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.isotropyRel_isotropyHom, and it does not change under refinement, through the conjugations by the natural transformations Chapter10/StrongBundleLocalizationExamples · EpNatTrans.isotropyHom_eq Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.IsRefinement.isotropyRel_iff, nor under equivalence Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.Equiv.isotropyRel_iff. *The example.* $W = \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with $\mu((g,x),w) = w$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3EpBundle, and Inv over $(g,x) \mapsto (g^{-1},x)$, a functor since $\mathbb{Z}/3$ is abelian and a strong bundle equivalence as its own inverse Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3InvHom Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3InvHom_mem_equivalences. Inv is the identity on objects, so $|[\mathrm{Inv}]| = |[1_W]|$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.orbitMap_ofHom_z3InvHom and the base orbit maps agree Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.base_orbitMap_ofHom_z3InvHom. For $[1_W]$ the generator of $G_0 = \mathbb{Z}/3$ is related to itself Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isotropyRel_id; for $[\mathrm{Inv}]$ a related pair is $(k,k^{-1})$ up to conjugation, which is trivial in $\mathbb{Z}/3$, and the generator is not its own inverse Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.not_isotropyRel_inv. So the base classes differ, and with them the classes. By Remark 10.5.3 the base classes are two distinct generalized isomorphisms with the same orbit map: the non-uniqueness claimed in Remark 10.3.18 (G.119, closed).

**Proposition 10.5.6** (V10.61) Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.iff_of_isGeneralizedIso. A representative with $\Phi$, $\Psi$ in $F$ carries a property invariant under strong bundle equivalences from $W$ to $W''$ to $W'$. For the local regularity property this is Proposition 10.4.6 Chapter10/GapP2_GBundleLR · EpStrongBundle.proposition_10_4_6, with Definition 8.4.2 read as `IsLRLocal`, and Proposition 10.5.6 follows Chapter10/GapP2_GBundleLR · SBLocalization.proposition_10_5_6 (G.122, G.126, closed since 2026-11-12); the text's "see Proposition 10.4.7" is read as Proposition 10.4.6.

**Theorem 10.5.7 (2017: 10.10) does not hold as we read it** (V10.67) Chapter10/LocalizationLRCounterexample · SBLocalization.Theorem_10_5_7 Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_theorem_10_5_7. The strong bundle of Remark 10.5.5 has the local regularity property of Definition 8.4.2, its orbits being points Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.isLR_z3EpBundle, and $[\mathrm{Inv}] \neq [1_W]$ with the same orbit map. The text's proof, "immediately from Proposition 10.4.7", meets the step of Theorem 10.3.19 (§X.20); the 2017 Theorem 10.10, without local regularity, does not hold as we read it either (G.127, settled).

**Theorem 10.5.7 over an effective base of the target** (V10.67) Chapter10/LocalizationLREffectiveBundle · SBLocalization.theorem_10_5_7_of_isEffective Chapter10/LocalizationLRBundleBase · SBLocalization.theorem_10_5_7_of_isEffective_of_isLRLocal, the strong bundle analogue of the effective Theorem 10.3.19. *Step 1, the bases.* $|\Phi| = |\Psi|$ on total spaces gives $|\phi| = |\psi|$ on the bases Chapter10/LocalizationLREffectiveBundle · SEP.orbitMap_

base_eq, since the projection of a strong bundle is open Chapter10/LocalizationLREffectiveBundle · StrongBundle.isOpenMap_proj. *Step 2, the natural equivalence.* For strong bundle equivalences $F, G : W'' \to W'$ with $|F| = |G|$ and an effective base of $W'$, the base transformation $\tau : \phi_F \to \phi_G$ is that of §X.20, Step 2; then $G(w) = \mu'(\tau(P(w)), F(w))$: both sides are continuous in $w$ and agree over the regular points, where only one morphism $\phi_F(P(w)) \to \phi_G(P(w))$ exists, and these are dense Chapter10/LocalizationLREffectiveBundle · SEP.natEquiv_of_isEffective. The families of Step 2 come from (LR) of a base (Proposition 10.1.10) in the first form, and in the second from Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$ Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal through Proposition 10.4.7 on the bases Chapter10/LocalizationLRBundleBase · SEP.exists_family_of_isLRLocal (§X.32). *Step 3.* For $D$, $D'$ with $|[D]| = |[D']|$, the weak fibred product of $\Phi$ and $\Phi'$ refines both: $\Phi \circ \Pi_X \simeq \Phi' \circ \Pi_Z$ by the strong bundle Theorem 10.2.2 and $\Psi \circ \Pi_X \simeq \Psi' \circ \Pi_Z$ by Step 2. *Hypotheses.* The effectiveness of the base of the target is added in both forms and is not derived from the text; (LR) of a base is added in the first form and is not derived from the text either. The second form needs only the local regularity property of the target bundle, and it implies the formal (LR) of strong bundles Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal.isLR. Effectiveness cannot be dropped under this reading: the bundle over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ has the property Chapter10/LocalizationLRBundleBase · LocalizationLRBundleBaseExamples.isLRLocal_z3EpBundle, and the conclusion fails there Chapter10/LocalizationLRBundleBase · LocalizationLRBundleBaseExamples.not_theorem_10_5_7_of_isLRLocal. *Witnesses,* $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ with $[1_W]$ Chapter10/LocalizationLREffectiveBundle · LocalizationLREffectiveBundleExamples.theorem_10_5_7_of_isEffective_hypotheses Chapter10/LocalizationLRBundleBase · LocalizationLRBundleBaseExamples.theorem_10_5_7_of_isEffective_of_isLRLocal_hypotheses, and the local regularity property of a trivial bundle Chapter10/LocalizationLRBundleBase · EpStrongBundle.isLRLocal_trivial.

## X.24 Witnesses and the hypotheses

**Hypotheses taken, with their witnesses.** The nonempty object set in Proposition 10.1.3: the covering of $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§X.2). `Admissible`: every equivalence (§X.9), and the pair groupoid of $\mathbb{Z}$ over the point, and $\pm$id on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ (§X.10). Proposition 10.1.9 for the lifts, once a hypothesis of Proposition 10.1.19, is proved (§X.5). The two Hausdorff hypotheses of Proposition 10.1.14 are the text's. The effectiveness of the target in Theorem 10.3.19: the point (§X.20). The effectiveness of the base of the target in Theorem 10.5.7, with (LR) of a base or the local regularity property of the target bundle: $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ (§X.23). Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$ is a reading, not an added hypothesis; it holds for trivial bundles (§§X.23, X.32). Otherwise §§10.3–10.5 and the new work of §§X.27–X.34 take no hypothesis.

**Every definition has a non-identity witness.** Equivalences: $\mathbb{Z} \to$ pt, the refinement of $\mathbb{Z}/2 \ltimes \mathbb{R}$, $-$id of $\mathbb{Z}/2 \ltimes \mathbb{R}^2$; the weak fibred product: §X.10; strong bundle equivalences, push-forwards, natural equivalences of strong bundle functors, morphisms of $\mathcal{SEP}$: the bundle over $-$id (§§X.11–X.15); equivalent diagrams, generalized isomorphisms that do not come from isomorphisms, equivalent pairs $(X, \beta)$ and objects of $\mathcal{EP}_{reg}$: the pair groupoid of $\mathbb{Z}$ and the point (§§X.16–X.20); refinements of diagrams of strong bundles, weak fibred products of strong bundles, composites in $\mathcal{SEP}(F^{-1})$, generalized strong bundle isomorphisms with their inverses and push-forwards: the bundle over $-$id, and distinct classes with the same orbit map: the bundle over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§§X.22, X.23). The subcategory $\mathcal{EP}_{lreg}$: the point (§X.27); an orbit-preserving sc-diffeomorphism that is no $\hat{g}$, and an ep-groupoid without (LR): $\eta$ on $\mathbb{Z}/2 \ltimes Y$ of Appendix §10.6.3 (§X.28); the hypotheses of Proposition 10.4.7: $\Phi = \Psi = 1$ on $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ (§X.32). The other results of §§X.29–X.34 add no hypothesis. The hypotheses of Proposition 10.1.10 and of the

effective version of Theorem 10.3.19 are witnessed on the point (§§X.5, X.20).

## X.25 What is not formalized

Nothing. Since 2026-11-12 every gap of Chapter 10 is closed: no item is open, none is formalized only in part, and none is set aside as having nothing to formalize. The list below says where each former gap is now proved. The only narrower row is V10.66, Theorem 10.3.19 for an effective target (§X.20).

- §10.1: $\mathcal{EP}_{lreg}$ (G.111, §X.27) `Chapter10/GapP2_F_LReg · EP.lreg` `Chapter10/GapP2_F_LReg · EP.lreg_iff_of_isEquivalence`; Remark 10.1.8 and Appendix §10.6.3 with the appendix's own projections (G.112, §X.28) `Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.remark_10_1_8`. Propositions 10.1.9, 10.1.10 and 10.1.19 (G.113–G.115) are closed by §§X.5, X.7.
- §10.2: the structures induced from the triple products (G.116, §X.29) `Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.theorem_10_2_2_triple`.
- §10.3: the universal property of the localization (G.117, §X.30) `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.lift` `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.liftMap_comp`. Proposition 10.3.16 and Theorem 10.3.21 (G.118) are closed by §X.20, and the non-uniqueness in Remark 10.3.18 (G.119) by §X.23. Theorem 10.3.19 (G.120) is settled by §X.20: it does not hold as we read it, and holds for an effective target. Remark 10.3.13 has no formal content.
- §10.4: the Fred part of Theorem 10.4.5 (G.121, §X.31) `Chapter10/FredholmEquivalence · EpStrongBundle.theorem_10_4_5_fred`; Propositions 10.4.6 and 10.4.7 (G.122, §X.32), with Definition 8.4.2 read as `IsLRLocal` `Chapter10/GapP2_GBundleLR · EpStrongBundle.proposition_10_4_6` `Chapter10/GapP2_GBundleFamily · EpStrongBundle.proposition_10_4_7`.
- §10.5: the converse half of the strong bundle Theorem 10.3.10 (G.123, §X.34; the other parts are closed by §X.22) `Chapter10/GapP2_F2_SBConverse · SBLocalization.SEPLoc.isIso_iff_isGeneralizedIso`; (10.24) with generalized vertical maps, and the local expressions $q$, $Q$ (G.124, §X.33) `Chapter10/GapP2_GDiagram · SBLocalization.GenSBMap.square_10_24` `Chapter10/GapP2_GDiagram · EPLocalization.Diagram.exists_germ_q` `Chapter10/GapP2_GDiagramLift · SBLocalization.SDiagram.exists_lift_Q`; the Fred part of Theorem 10.5.4 (G.125, §X.31) `Chapter10/FredholmEquivalence · SBLocalization.GenSBMap.theorem_10_5_4_fred`; Proposition 10.5.6 (G.126, §X.32) `Chapter10/GapP2_GBundleLR · SBLocalization.proposition_10_5_6`. Theorem 10.5.7 (G.127) is settled by §X.23: it does not hold as we read it, and holds over an effective base of the target.

## X.26 What the formalization taught us

The equivalences of §10.1 are the sc-smooth versions of equivalences of categories, and most proofs of the section are categorical arguments plus one local fact: a functor which is a local sc-diffeomorphism on objects at $s(\varphi)$ is one at $t(\varphi)$. The text uses this fact silently in Proposition 10.1.6, where the argument covers only the image of $F$; with it the proof is two lines. Proposition 10.1.5, quoted without proof, and the fullness and faithfulness of $TF$, left to the reader, are short once tangent morphisms are known to be determined by their base and source vector. The invariance results new in 2021 split into two kinds: those about $\mathbf{X}^T$ ((LUC), the reduced groupoids, (SLUC)) go through Theorem 7.2.22 and one closure argument; those about natural representations ((LR), (SLR), Proposition 10.1.10) need natural representations carried along $Q = \hat{\psi} \circ F$, which is done by restricting a natural representation at the target rather than by transporting $\Gamma$. The local families of Proposition 10.1.10 are not natural in general, and that is where Theorem 10.3.19 does not hold as we read it: on $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action,

which has (LR) because its orbits are points, the identity and the inversion have the same orbit map and are not naturally equivalent; over an effective target the families are unique and glue, and the theorem holds without the weak fibred product of its proof. In §10.2 the weak fibred product needs much less than an equivalence: one local sc-diffeomorphism on objects for the structure, nothing for properness. In §10.3 the two places where the text argues with spaces, "the same space" for the associativity and the direct check that $L$ is an equivalence, become functors and two-out-of-three arguments, and the local half of the hard direction of Theorem 10.3.10 needs a different route, through a local right inverse and local injectivity. In §10.4 the push-forward is well defined on section functors because of fullness (10.22), and once (10.22) is in hand its sc-smoothness is a local formula with two local inverses; Propositions 10.4.2–10.4.4 do not need sc-smoothness at all. In §10.5 the push-forward along a generalized strong bundle isomorphism is best seen through the orbit spaces: a section functor is determined by the orbits of its values, so the push-forward depends only on $|[D]|$, which gives the independence of the representative and the last assertion of Theorem 10.5.4 at once; and the example the text does not give for Remark 10.5.5 needs an invariant finer than the orbit maps, the relation of the isotropy homomorphisms $\gamma$, which also settles the non-uniqueness left informal in Remark 10.3.18. The weak fibred product of strong bundles, which the text leaves to the reader, is best taken as a pull-back: one projection is then a strong bundle equivalence by construction, and since strong bundle equivalences are injective on fibres, the squares in the proofs of transitivity, composition and associativity commute strictly, so that the proofs of §10.3 carry over.

The work of 2026-11-12 adds the following. First, the example of Appendix §10.6.3 needs no more than the scale and the loss families of Chapter 1: $\pi_{|s|}$ and the map $\eta$ are differences and sums of loss families, and the failure of (LR) comes down to the observation that in a natural representation of a translation groupoid each $\Phi(g)$ is the action of $g$ itself, by connectedness. Second, the universal property of the localization reduces to one fact: every functor that inverts the equivalences identifies naturally equivalent functors, through the cylinder $X \times_Y Y$, whose projection to $X$ is an equivalence with sections $s_\tau$. Third, Definition 8.4.2 must be read for strong bundle isomorphisms of $W|U(x)$ alone. Asked only of maps defined on all of $W$, it cannot be applied to the conjugates $H^{-1} \circ B \circ H$ of the proof of Proposition 10.4.6, which live over $U(x)$; read locally, Propositions 10.4.6, 10.4.7 and 10.5.6 follow the proofs of Propositions 10.1.9 and 10.1.10 with transports along families of morphisms in place of $\hat{\psi}$. Fourth, the converse half of the strong bundle Theorem 10.3.10 does not need strong bundle versions of the claims (C1), (C2) of the appendix: the two composites with the inverse class give fibrewise injectivity and local right inverses separately. The Fred parts of Theorems 10.4.5 and 10.5.4, called clear in the text, are the one place where a strong bundle equivalence is not a pull-back along a global isomorphism; the push-forward is handled with the action of the morphisms $\sigma : \phi(y) \to x'$, which is itself a local strong bundle isomorphism.

## X.27 The subcategory $\mathcal{EP}_{lreg}$ (2021 p. 461)

**The subcategory** (V10.74) `Chapter10/GapP2_F_LReg · EP.lreg Chapter10/GapP2_F_LReg · EP.Lreg`. The text introduces, new in 2021, the full subcategory of $\mathcal{EP}$ generated by the ep-groupoids with the local regularity property (Definition 7.4.19). It is an object property of $\mathcal{EP}$ with Mathlib's full subcategory, whose inclusion is fully faithful `Chapter10/GapP2_F_LReg · EP.fullyFaithful_ι_lreg`. *Step 1, along equivalences.* For an equivalence $F : X \to Y$, $X \in \mathcal{EP}_{lreg}$ iff $Y \in \mathcal{EP}_{lreg}$, by Proposition 10.1.9 (§X.5) `Chapter10/GapP2_F_LReg · EP.lreg_iff_of_isEquivalence`; this is the closing sentence of the proof of Proposition 10.1.9 on p. 469. *Step 2, isomorphisms of $\mathcal{EP}$.* An isomorphism of $\mathcal{EP}$ is an equivalence by (E1), so the property is closed under isomorphisms `Chapter10/GapP2_F_LReg · EP.isClosedUnderIsomorphisms_lreg`. *Step 3, in $\mathcal{EP}(E^{-1})$.* The same

object property on the localization Chapter10/GapP2_F_LReg · EPLocalization.EPLoc.lreg is closed under the isomorphisms there, since an isomorphism is $[X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ with $F$, $\Phi$ equivalences (Theorem 10.3.10) and Step 1 applies twice Chapter10/GapP2_F_LReg · EPLocalization.EPLoc.isClosedUnderIsomorphisms_lreg; the middle of a diagram out of an object of $\mathcal{EP}_{lreg}$ is in $\mathcal{EP}_{lreg}$ Chapter10/GapP2_F_LReg · EPLocalization.EPLoc.lreg_A_of_lreg. *Witness*, the point Chapter10/GapP2_F_LReg · EP.lreg_pointEP Chapter10/GapP2_F_LReg · EP.Lreg.point.

## X.28 Remark 10.1.8 and Appendix §10.6.3 (2021 pp. 467, 519)

**The example as printed** (V10.75) Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.remark_10_1_8. Remark 10.1.8, new in 2021, says that an sc-diffeomorphism $\beta$ of $U(x)$ with $|\beta(y)| = |y|$ need not be of the form $y \mapsto g * y$, and Appendix §10.6.3 gives the example. *Step 1, the scale and the retraction.* $F = \mathbb{R} \oplus L^2(\mathbb{R})$ with $F_m = \mathbb{R} \oplus H^{m,\delta_m}(\mathbb{R})$ Chapter7/GapP2_A2Line · ZhouLine.lineSc, and the family $\pi_t$ of Subsection 1.4.4 Chapter7/GapP2_A2Family · ZhouLine.zhouFam Chapter7/GapP2_A2Family · ZhouLine.isLossFamily_zhouFam. $\pi_{|s|} = \pi_s + \pi_{-s}$ is a loss family Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.isLossFamily_piAbs, by sums and reflections of loss families Chapter10/GapP2_F_ZhouExample · IsLossFamily.add Chapter10/GapP2_F_ZhouExample · IsLossFamily.comp_neg, so $r(s,f) = (s, \pi_{|s|}f)$ is an sc-smooth retraction Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.Rt₂ with image $Y$ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.O₂; $\mathbb{Z}/2$ acts by $(-1) * (s,f) = (s,-f)$, sc-smoothly Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.isScSmoothMap_smul₂, and $X = \mathbb{Z}/2 \ltimes Y$ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.X₂. *Step 2, $\eta$.* $\eta(s,f) = (s,f)$ for $s \le 0$ and $(s,-f)$ for $s > 0$ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.ηO_of_nonpos Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.ηO_of_pos is, on $Y$, the map $(s,f) \mapsto (s, (\pi_{-s} - \pi_s)f)$, with $\pi_{-s} - \pi_s$ a loss family Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.isLossFamily_famη Chapter10/GapP2_F_ZhouExample · IsLossFamily.sub; it is its own inverse Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.ηO_ηO, hence an sc-diffeomorphism Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.isScDiffeoOn_ηO fixing $0$ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.ηO_o₂, and $\eta(y)$ is $y$ or $(-1) * y$, so $|\eta(y)| = |y|$ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.π_ηO. *Step 3, no $g$.* The points $(\pm\frac{1}{n+1}, \frac{1}{n+1}\beta(\cdot + e^{n+1}))$ lie in $Y$ and tend to $0$ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.ptσ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.tendsto_ptσ; on those with $s > 0$, $\eta(y) = -y \neq y$, and on those with $s < 0$, $\eta(y) = y \neq -y$, so on no neighbourhood of $0$ is $\eta$ the action of one $g$ Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.exists_ηO_ne_smul. *Step 4, (LR) fails.* In a natural representation at $0$, $\Gamma(g,\cdot)$ has a constant group component on the connected $U$, so $\Phi(g)$ is $y \mapsto g * y$ there Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.naturalRep_Φ_eq. $\eta$ maps $U$ into itself, being the identity or the action of $-1$ pointwise, so it is an orbit-preserving sc-diffeomorphism of $U$; (LR) would make it $\Phi(g)$ on $U$, against Step 3 Chapter10/GapP2_A2TenSix · EpGroupoid.Examples.ZhouTenSix.not_isLR. *The variant* with the moving projections of Remark 4.1.3 and $\eta$ of the opposite sign convention Chapter10/GapP2_F_ZhouExample · ZhouSigma.remark_10_1_8 Chapter10/GapP2_F_ZhouExample · ZhouSigma.not_isLR is kept.

## X.29 Theorem 10.2.2 (2017: 10.2): the structures induced from the triple products

**The last sentence of the statement** Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.theorem_10_2_2_triple. The text says that the object set $L \subseteq X \times \mathbf{Y} \times Z$ and the morphism set $\mathbf{L} \subseteq \mathbf{X} \times \mathbf{Y} \times \mathbf{Z}$ are sub-M-polyfolds

and carry the induced structures; §X.9 builds the structures iteratively, $L = X \,{}_F\!\times_s (\mathbf{Y}_t \times_G Z)$ inside $X \times (\mathbf{Y}_t \times_G Z)$. *Step 1, products with a sub-M-polyfold.* For a sub-M-polyfold $A \subseteq N$, $X \times A$ is a sub-M-polyfold of $X \times N$ `Chapter10/GapP2_F_TripleProduct · MPolyfold.isSubMPolyfold_univ_prod`, and the subtype $X \times A$ with the induced structure is sc-diffeomorphic to $X \times A$ with the product of the structure of $A$ `Chapter10/GapP2_F_TripleProduct · MPolyfold.prodSubtypeMk Chapter10/GapP2_F_TripleProduct · MPolyfold.isScDiffeoOn_prodSubtype`. *Step 2, transitivity.* A sub-M-polyfold of an M-polyfold sc-diffeomorphic to a sub-M-polyfold $S \subseteq M$ is a sub-M-polyfold of $M$ `Chapter10/GapP2_F_TripleProduct · MPolyfold.isSubMPolyfold_image_of_isScDiffeoOn`. *Step 3, the triple sets.* The triple-product sets `Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.objSetTriple Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.morSetTriple` are the images of the iterated ones under the reassociation $(x,(\phi,z)) \mapsto (x,\phi,z)$ `Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.Obj.toTriple Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.objSetTriple_eq`, hence sub-M-polyfolds by Steps 1, 2 `Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.isSubMPolyfold_objSetTriple Chapter10/GapP2_F_TripleProduct · WeakFibredProduct.isSubMPolyfold_morSetTriple`. *Step 4.* The reassociations are sc-diffeomorphisms of the iterated structures onto the induced ones, so the two structures agree. Only the admissibility of §X.9 is used (G.116).

## X.30 The universal property of the localization (2021 p. 483 and Theorem 10.3.14)

$i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ **is a localization** `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.isLocalization Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.universal_property`. The text introduces $\mathcal{EP}(E^{-1})$ as the category in which the equivalences become invertible universally. *Step 1, the cylinder.* For a natural transformation $\tau : F \to F'$ of functors $X \to Y$, the weak fibred product $X \times_Y Y$ of $F$ and $1_Y$ receives the sc-smooth section $s_\tau(x) = (x, \tau(x), F'(x))$ `Chapter10/GapP2_F_LocUniversal · EPLocalization.secObj Chapter10/GapP2_F_LocUniversal · EPLocalization.secFunctor`, with $\pi_X \circ s_\tau = 1_X$ and $\pi_Y \circ s_\tau = F'$ `Chapter10/GapP2_F_LocUniversal · EPLocalization.secHom_cylLeft Chapter10/GapP2_F_LocUniversal · EPLocalization.secHom_cylRight`. *Step 2, natural equivalence is inverted.* $\pi_X$ is an equivalence (Theorem 10.2.2), so for every functor $L$ inverting $E$, $L(s_\sigma) = L(\pi_X)^{-1}$ for every $\sigma$, and $L(F') = L(s_\tau) \circ L(\pi_Y) = L(s_{1_F}) \circ L(\pi_Y) = L(F)$ `Chapter10/GapP2_F_LocUniversal · EPLocalization.map_eq_of_naturallyEquivalent`. *Step 3, the lift.* $L'[X \xleftarrow{F} A \xrightarrow{\Phi} Y] = L(F)^{-1} \circ L(\Phi)$ `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.liftMap`. For a refinement by $H$, $L(F') = L(H) \circ L(F)$ and $L(\Phi') = L(H) \circ L(\Phi)$ by Step 2, and the factors $L(H)$ cancel `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.liftMap_eq_of_isRefinement Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.liftMap_eq_of_equiv`. *Step 4, functoriality.* The identity `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.liftMap_id`, and composites through the square $L(\pi_A) \circ L(\Phi) = L(\pi_B) \circ L(G)$, which is Step 2 for the natural equivalence of the weak fibred product (§X.10) `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.liftMap_comp Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.lift`. *Step 5, factorization and uniqueness.* $i \circ L' = L$ `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.fac`, and $L'$ is unique because every generalized map is $i(\Phi) \circ i(F)^{-1}$ (Theorem 10.3.14) `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.uniq`; this is Mathlib's strict universal property for every target category `Chapter10/GapP2_F_LocUniversal · EPLocalization.EPLoc.strictUniversalPropertyFixedTarget` (G.117).

## X.31 Theorems 10.4.5 and 10.5.4 (2017: 10.8 and 10.9): the Fred parts, and linearity

**Part I's half.** An sc-Fredholm germ stays one along a local strong bundle isomorphism: such an isomorphism `Chapter3/FredholmPullbackIso · StrongBundle.LocalIso` transports charts `Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.transportChart` `Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.isChart_transportChart` and germs `Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.isScFredholmGerm`, and the sc-Fredholm property of a section is local `Chapter3/FredholmPullbackIso · StrongBundle.isScFredholmSection_of_forall`.

**Theorem 10.4.5, the Fred part** (V10.27) `Chapter10/FredholmEquivalence · EpStrongBundle.theorem_10_4_5_fred`. The text calls it clear. *Step 1.* A local strong bundle isomorphism of Definition 10.4.1 (2) is one of Part I `Chapter10/FredholmEquivalence · StrongBundle.IsStrongBundleIsoOn.toLocalIso`. *Step 2, $\Phi^*$.* Near $x_0$, $\Phi^* f'$ is the pull-back of $f'$ along that isomorphism (§X.14, Step 1) `Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.exists_localIso_pullback`, so it is sc-Fredholm when $f'$ is, in the germ reading of Definition 3.1.18 and with tame germ data `Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSection_pullback` `Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSectionTame_pullback`. *Step 3, the action of a morphism.* For a local section $\alpha$ of $s$, $w \mapsto \mu(\alpha(P(w)), w)$ `Chapter10/FredholmEquivalence · EpStrongBundle.sectionAct` is a local strong bundle isomorphism $W|t(T) \to W|s(T)$ along which every section functor is its own pull-back, by functoriality (10.16) `Chapter10/FredholmEquivalence · EpStrongBundle.exists_localIso_mor`. *Step 4, $\Phi_*$.* The push-forward is not a pull-back along a global isomorphism, $\phi$ being only essentially surjective. Near $x'$ with $\sigma : \phi(y) \to x'$, $\Phi_* f$ is $\Phi \circ f \circ \phi^{-1}$ near $\phi(y)$, sc-Fredholm by Step 2 for the inverse isomorphism, and near $x'$ it is its own pull-back along the action of $\sigma$ (Step 3) `Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.exists_localIso_pushforward` `Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSection_pushforward` `Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSectionTame_pushforward`. *Step 5.* With §X.12, $\Phi_*$ and $\Phi^*$ restrict to inverse bijections of the sets of sc-Fredholm section functors (G.121).

**Theorem 10.5.4, the Fred part** (V10.59) `Chapter10/FredholmEquivalence · SBLocalization.GenSBMap.theorem_10_5_4_fred`. $D_* = \Psi_* \Phi^*$ preserves the sc-Fredholm property by the above `Chapter10/FredholmEquivalence · SBLocalization.SDiagram.mapsTo_pushforward_fred`, and so does $D^*$ `Chapter10/FredholmEquivalence · SBLocalization.SDiagram.mapsTo_pullback_fred`; with §X.23 they are inverse bijections `Chapter10/FredholmEquivalence · SBLocalization.SDiagram.bijOn_pushforward_fred`, on the chosen representative of the class (G.125).

**"Isomorphism" as a linear isomorphism** `Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforwardEquiv` `Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.plusPushforwardEquiv`. *Step 1.* $\Gamma(P, \mu)$ and $\Gamma^+(P, \mu)$ of Definition 8.3.4 are realized as submodules of $\prod_x W_x$ `Chapter10/SectionFunctorLinear2021 · EpStrongBundle.sectionSpace` `Chapter10/SectionFunctorLinear2021 · EpStrongBundle.plusSectionSpace`. *Step 2.* Pull-back and push-forward are additive and homogeneous pointwise, for all sections, without functoriality `Chapter10/SectionFunctorLinear2021 · EpStrongBundle.IsStrongBundleEquivalence.pullback_sectionAdd` `Chapter10/SectionFunctorLinear2021 · EpStrongBundle.IsStrongBundleEquivalence.pushforward_sectionAdd` `Chapter10/SectionFunctorLinear2021 · EpStrongBundle.IsStrongBundleEquivalence.pushforward_sectionSmul`, hence along diagrams and classes `Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforward_sectionAdd`. *Step 3.* With the bijections of Theorem 10.5.4, $[D]_*$ is a linear isomorphism with inverse $[D]^*$ `Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforwardEquiv_symm_apply`,

and the same on $\Gamma^+$.

## X.32 Propositions 10.4.6, 10.4.7 and 10.5.6 (new in 2021)

**The reading of Definition 8.4.2** `Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal`. The text's $B$ is a strong bundle isomorphism of $W|U(x)$ with $|B(w)| = |w|$, and the conclusion is $B(w) = g * w$. We read it for maps defined on $W|U(x)$ only, with no condition off $W|U(x)$ `Chapter10/LocalizationLRBundleBase · StrongBundle.IsLocalIsoOn`. Read for strong bundle maps defined on all of $W$, the conjugate $H^{-1} \circ B \circ H$ of the proof of Proposition 10.4.6 would not be in its scope. The local reading implies the global one `Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal.isLR`; we do not know whether the converse holds. The condition on one $U(x)$ is recorded separately `Chapter10/GapP2_GBundleLR · EpStrongBundle.HasLRLocal` `Chapter10/GapP2_GBundleLR · EpStrongBundle.isLRLocal_iff_hasLRLocal`.

**Local isomorphisms between two bundles** `Chapter10/GapP2_GLocalIso · StrongBundle.IsLocalIso₂`. The proofs compose local isomorphisms $Y|U \to Y'|U'$ between different bundles and invert them: inverse, composite and restriction `Chapter10/GapP2_GLocalIso · StrongBundle.IsLocalIso₂.symm` `Chapter10/GapP2_GLocalIso · StrongBundle.IsLocalIso₂.comp` `Chapter10/GapP2_GLocalIso · StrongBundle.IsLocalIso₂.restrict`, with the inverse linear on the fibres `Chapter10/GapP2_GLocalIso · StrongBundle.IsLocalIsoOn.exists_isLocalIso₂`; a strong bundle equivalence restricts to one `Chapter10/GapP2_GLocalIso · StrongBundle.IsStrongBundleIsoOn.exists_isLocalIso₂`. For an sc-smooth family $m(y) : y \to \tau(y)$ over $U$ with $\tau$ an sc-diffeomorphism, $w \mapsto \mu(m(Pw), w)$ is one covering $\tau$, with inverse $w \mapsto \mu(m(\tau^{-1}(Pw))^{-1}, w)$ `Chapter10/GapP2_GLocalIso · EpStrongBundle.isLocalIso₂_lrAct` `Chapter10/GapP2_GLocalIso · EpStrongBundle.lrAct_comp`; the natural lift of $g$ is such a map `Chapter10/GapP2_GLocalIso · EpStrongBundle.natLift_eq_lrAct`.

**Proposition 10.4.6** (V10.72) `Chapter10/GapP2_GBundleLR · EpStrongBundle.proposition_10_4_6`. *Step 1, the transport.* For $\psi : F(x) \to y$ and the transport chart $Q = \hat{\psi} \circ F$ of §X.5, $v \mapsto \nu(\psi_{P(v)}, \Phi(v))$, the text's $H_{\hat{\psi}} \circ \Phi$, is a local strong bundle isomorphism $V|S \to W|Q(S)$ covering $Q$ `Chapter10/GapP2_GBundleLR · EpStrongBundle.IsStrongBundleEquivalence.exists_transport`, equivariant along $g \mapsto \psi \circ F(g) \circ \psi^{-1}$ `Chapter10/GapP2_GBundleLR · EpStrongBundle.IsStrongBundleEquivalence.transport_Γ` and preserving and reflecting orbit equality `Chapter10/GapP2_GBundleLR · EpStrongBundle.IsStrongBundleEquivalence.π_transport_iff`. *Step 2, the natural representations.* $U(y) = Q(U(x))$ with the natural action, as in Proposition 10.1.9 `Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_push` `Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_pull`. *Step 3, the conjugation.* For any such $H$ along a surjection $G_x \to G_y$, $B \mapsto H^{-1} \circ B \circ H$ carries the orbit-preserving strong bundle isomorphisms of $W|U(y)$ to those of $V|U(x)$ and the natural lifts to the natural lifts, both ways `Chapter10/GapP2_GBundleLR · EpStrongBundle.hasLRLocal_iff_of_equivariant`. "$\Phi : W \to W$" in the statement is read $\Phi : V \to W$.

**Proposition 10.4.7** (V10.73) `Chapter10/GapP2_GBundleFamily · EpStrongBundle.proposition_10_4_7` `Chapter10/GapP2_GBundleFamily · EpStrongBundle.proposition_10_4_7'`, with the property for the target only and $U(x)$ inside any given neighbourhood: a stronger form. *Step 1.* $|\Phi| = |\Psi|$ gives $\psi_0 : \psi(x) \to \phi(x)$; $Q = \hat{\psi}_0 \circ \psi$ and the transport $H(v) = \nu(\psi_{P(v)}, \Psi(v))$ of Proposition 10.4.6, Step 1. *Step 2.* $B = H \circ \Phi^{-1}$ is a strong bundle isomorphism of $W|U(\phi(x))$ covering $\gamma = Q \circ \phi^{-1}$, with $\gamma(U) = U$ by Lemma 7.4.20 and $|B(w)| = |w|$ by $|\Phi| = |\Psi|$. *Step 3.* Definition 8.4.2 gives $B = k*\cdot$, so $\nu(\psi_z, \Psi(v)) = \nu(\Gamma(k, \phi(z)), \Phi(v))$ and $\alpha_z = \psi_z^{-1} \circ \Gamma(k, \phi(z))$. On the bases this is `Chapter10/LocalizationLRBundleBase · SEP.exists_family_of_isLRLocal`; the proof shows that orbit equality on the total spaces determines it on the bases `Chapter10/GapP2_GBundleFamily · EpStrongBundle.π_base_eq_of_orbitMap_eq`. *Witness,* $\Phi = \Psi = 1$ on $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ `Chapter10/GapP2_GBundleFamily · GapP2_GBundleFamilyExamples.proposition_10_4_7_hypotheses`.

**Proposition 10.5.6** (V10.61) `Chapter10/GapP2_GBundleLR · SBLocalization.proposition_10_5_6`. A representative with $\Phi$, $\Psi$ strong bundle equivalences carries the property from $W$ to $W''$ to $W'$, by Proposition 10.4.6 twice (§X.23) (G.122, G.126).

## X.33 (10.24) and the local expressions $q$, $Q$ (2021 pp. 504–506)

**The square (10.24)** `Chapter10/GapP2_GDiagram · SBLocalization.SDiagram.square_10_24` `Chapter10/GapP2_GDiagram · SBLocalization.GenSBMap.square_10_24`. The text says the square with vertical arrows $P$, $P'$ "does not make too much sense", and that with the vertical arrows replaced by "a suitable generalized map version" it commutes, leaving this to the reader. We take $i(P) = [W \xleftarrow{1} W \xrightarrow{P} X]$, with $P$ a morphism of $\mathcal{EP}$ `Chapter10/GapP2_GDiagram · SEP.projEP`. *Step.* $P' \circ \Psi = \psi \circ P''$ on the representative, and $i$ is a functor (Theorem 10.3.14), so $[D]_{\text{total}} \circ i(P') = i(P) \circ [d]$ in $\mathcal{EP}(E^{-1})$.

**The germ** $q$ `Chapter10/GapP2_GDiagram · EPLocalization.Diagram.exists_germ_q`. For the data $(x'', a, b)$ of $\gamma$ (§X.21), $q$ is the composite $U(x) \leftarrow U(\phi(x'')) \leftarrow U(x'') \rightarrow U(\psi(x'')) \rightarrow U(x')$ of $t \circ (s|U(a))^{-1}$, of $\phi^{-1}$, of $\psi$ and of $t \circ (s|U(b))^{-1}$. *Step 1.* A functor intertwines natural representations at $u$ and $F(u)$ along $F : G_u \to G_{F(u)}$: both sides are sections of $s$ agreeing at $u$ `Chapter10/GapP2_GDiagram · EpFunctor.mor_Γ` `Chapter10/GapP2_GDiagram · EpFunctor.obj_Φ`. *Step 2.* The conjugations by $a$ and $b$ are equivariant for the conjugation homomorphisms `Chapter10/GapP2_GDiagram · EPLocalization.Diagram.isotropyHom_conj`. *Step 3.* Composing, $q(x) = x'$ and $q(g * z) = \gamma(g) * q(z)$ on natural representations $U(x)$, $U(x')$, the actions in which the equivariance is read.

**The lift $Q$ and (10.25)** `Chapter10/GapP2_GDiagramLift · SBLocalization.SDiagram.exists_lift_Q`. "A strong bundle map $W|O(X, x) \rightarrow W'|O(X', x')$" is read as a local strong bundle map on $W|U(x)$, with no condition off it `Chapter10/GapP2_GDiagramLift · StrongBundle.IsLocalMap₂` `Chapter10/GapP2_GDiagramLift · StrongBundle.IsLocalMap₂.comp` `Chapter10/GapP2_GDiagramLift · StrongBundle.IsLocalIso₂.isLocalMap₂`. *Step.* $Q = H_b \circ \Psi \circ H_a^{-1}$ with the transports $H_a(v) = \mu(a_{P(v)}, \Phi(v))$ and $H_b(u) = \mu'(b_{P'(u)}, u)$ of §X.32, Step 1. It covers $q = Q_b \circ \psi \circ Q_a^{-1}$, so $P' \circ Q = q \circ P$, (10.25), and $Q(g * w) = \gamma(g) * Q(w)$ for the natural lifts (G.124). As the text says, $q$ and $Q$ depend on the representative.

## X.34 Theorem 10.3.10 (2017: 10.4) for strong bundles: the converse

**Isomorphisms of $\mathcal{SEP}(F^{-1})$ are generalized strong bundle isomorphisms** (V10.71) `Chapter10/GapP2_F2_SBConverse · SBLocalization.SEPLoc.equivalences_Ψ_of_isIso` `Chapter10/GapP2_F2_SBConverse · SBLocalization.SEPLoc.isIso_homMk_iff` `Chapter10/GapP2_F2_SBConverse · SBLocalization.SEPLoc.isIso_iff_isGeneralizedIso`. Remark 10.5.3 says "a similar result can be proved" and builds it into Definition 10.5.2. Let $[E] = [W' \xleftarrow{\Phi'} V'' \xrightarrow{\Psi'} W]$ be the inverse of $[D]$. *Step 1, the base.* $\psi$ is an equivalence (§X.22). *Step 2, fibrewise injectivity from* $[D] \circ [E] = 1$. A strong bundle equivalence $H$ into the middle $W'' \times_{W'} V''$ makes $\Psi' \circ \Pi_{V''} \circ H$ naturally equivalent to a strong bundle equivalence `Chapter10/GapP2_F2_SBConverse · SBLocalization.SDiagram.exists_natEquiv_of_equiv`. Since $\Pi_{V''}(m)$ is determined by $\Psi(\Pi_{W''}(m))$, $\Psi$ is injective on the fibres over the points $\Pi_{W''}(H(c))$, and on all fibres by transport along morphisms `Chapter10/GapP2_F2_SBConverse · SEP.FibInj` `Chapter10/GapP2_F2_SBConverse · SBLocalization.SDiagram.fibInj_Ψ_of_equiv`. *Step 3, a natural equivalence from* $[E] \circ [D] = 1$. Likewise $\Psi \circ G$ is naturally equivalent to a strong bundle equivalence $K$ for $G = \Pi_{W''} \circ H$. *Step 4, local inverses.* For $x$ choose $\gamma : g(c) \to x$. The transport along $\gamma$ gives $G' = H_{\hat{\gamma}} \circ G$ (§X.32), and $\Psi \circ G'$ is the transport of $K$, a local strong

bundle isomorphism near $c$. So $G' \circ (\Psi \circ G')^{-1}$ is a local right inverse of $\Psi$ near $\psi(x)$, at levels $[0]$ and $[1]$. With the fibrewise injectivity and the local injectivity of $\psi$, $\Psi$ is a strong bundle isomorphism near $x$ `Chapter10/GapP2_F2_SBConverse · SEP.isStrongBundleIsoOn_of_rightInv` `Chapter10/GapP2_F2_SBConverse · SEP.exists_isStrongBundleIsoOn`, and on morphisms through the pull-back projections `Chapter10/GapP2_F2_SBConverse · SEP.isLocalStrongBundleIso_mor_of_obj`. *Step 5, the total space and* $|W[1]|$. The total functor is full `Chapter10/GapP2_F2_SBConverse · SEP.full_total` and an equivalence by Lemma 10.1.2 `Chapter10/GapP2_F2_SBConverse · SEP.isEquivalence_total`. $|\Psi|$ on $|W''[1]|$ is a continuous bijection with the continuous inverse $|G| \circ |K|^{-1}$ `Chapter10/GapP2_F2_SBConverse · SEP.NatEquiv.oneOrbitMap_eq` `Chapter10/GapP2_F2_SBConverse · SEP.isHomeomorph_oneOrbitMap_of`. Together, the saturation step `Chapter10/GapP2_F2_SBConverse · SEP.equivalences_of_natEquiv_comp`: a strong bundle functor covering an equivalence, injective on fibres, with $\Psi \circ G \simeq K$ for a strong bundle equivalence $K$, is a strong bundle equivalence. This replaces the claims (C1), (C2) of Appendix §10.6.2, which the text does not restate for strong bundles (G.123).

# Part XI. Geometry up to equivalences (HWZ Chapter 11, §§11.1–11.6 (2017 §§11.1–11.6))

Chapter 11 closes the ep-groupoid half of Part II; its numbers and pages are those of the book (Springer 2021, pp. 521–566), with the 2017 text's in parentheses (2017 pp. 375–416). It asks which of the notions built on ep-groupoids in Chapters 8 and 9 survive the passage to $\mathcal{EP}(E^{-1})$ of Chapter 10, where generalized isomorphisms are the isomorphisms. §11.1 extends the tangent functor (Theorem 11.1.1 (2017: 11.1)), shows that the degeneracy index is invariant (Proposition 11.1.2 (2017: 11.1)) and treats faces (Definition 11.1.3, Example 11.1.4, Remark 11.1.5, Proposition 11.1.6, Remark 11.1.7, Proposition 11.1.8 (2017: Definition 11.1, Example 11.1, Remark 11.1, Proposition 11.2, Remark 11.2, Proposition 11.3)). §11.2 does the same for sc-differential forms (Theorem 11.2.1 to Proposition 11.2.9 (2017: Theorem 11.2 to Proposition 11.5)). §11.3 pushes forward and pulls back functors $\Theta : X \to \mathbb{Q}^+$ (Definition 11.3.1, Lemma 11.3.2, Theorem 11.3.3 (2017: Definition 11.4, Lemma 11.5, Theorem 11.3)) and continues with the bundles of tangent sets and orientations (Lemma 11.3.4 to Theorem 11.3.9 (2017: Lemma 11.6 to Theorem 11.7)). §11.4 transforms the canonical measures (Theorems 11.4.1, 11.4.2, 11.4.4, 11.4.5, Remark 11.4.3 (2017: Theorems 11.8, 11.9, Remark 11.3)). §11.5 carries auxiliary norms and sc$^+$-multisections along generalized strong bundle isomorphisms (Proposition 11.5.1, Lemma 11.5.2, Proposition 11.5.3 (2017: Proposition 11.6, Lemma 11.7, Proposition 11.7)). §11.6 studies proper covering functors up to equivalence (Lemmas 11.6.1–11.6.3, Definitions 11.6.4, 11.6.5, 11.6.7, Theorem 11.6.6, Proposition 11.6.8, Remark 11.6.9 (2017: Lemmas 11.8, 11.9, Definitions 11.6–11.8, Theorem 11.10, Proposition 11.8, Remark 11.4); Lemma 11.6.1 is new in the 2021 edition).

**What is proved, and what is not.** Since 2026-11-13 all eight gaps of Chapter 11 are closed (G.128–G.135). None is set aside as having nothing to formalize. *§11.1.* Theorem 11.1.1, Proposition 11.1.2, Definition 11.1.3, Example 11.1.4 (for $[0,\infty)^2 \oplus W$ with any $W$), Remark 11.1.5 (with the result it quotes), Remark 11.1.7 and Proposition 11.1.8 are proved (§§XI.1–XI.3). Proposition 11.1.6 is proved except for its intersection clause, which does not hold as we read it and holds in a union form (§§XI.3, XI.19, XI.20; G.128): the faces of a face structured tame ep-groupoid are tame ep-groupoids, and a generalized isomorphism induces generalized isomorphisms $[d]_F$ of faces. *§11.2.* All of it is proved: Theorem 11.2.1, Lemmas 11.2.2, 11.2.4–11.2.6, Definitions 11.2.3, 11.2.7, (11.5) and Propositions 11.2.8, 11.2.9 (§XI.4; G.129). *§11.3.* Proved: Definition 11.3.1, Lemma 11.3.2 in a stronger form, the push-forward and pull-back along generalized maps, and all six parts of Theorem 11.3.3, (6) for every dimensional decomposition (§§XI.5–XI.7). Also proved: Lemma 11.3.4 with (11.8) and (11.9), Theorem 11.3.5, Definition 11.3.6 with (11.10), and Theorems 11.3.7 and 11.3.8, whose proofs the text leaves to the reader (§XI.8). They were first proved under

Chapter 9's hypothesis that the tangent sums of Theorem 9.2.1 are independent, which Chapter 9 now proves for every $\Theta$. Theorem 11.3.9 is proved for the intrinsic boundary orientation of Definition 9.3.10, independently of the representative and with no hypothesis (§XI.8; G.130). *§11.4.* Formalized (§XI.9; G.131). Theorems 11.4.1 and 11.4.2 and Remark 11.4.3 are proved, with no added hypothesis, for the canonical measures of Chapter 9 built from its concrete sheet integrals. Theorems 11.4.4 and 11.4.5 are proved with the text's hypotheses, for every $n \geq 1$, for the boundary measures of Theorems 9.5.15 and 9.5.16. *§11.5.* Proposition 11.5.1 is proved with conditions (1) and (2) of Definition 8.3.5 (§XI.10; G.132), and Lemma 11.5.2 and Proposition 11.5.3 for sc$^+$-multisections with their local section structures (§XI.11; G.133). *§11.6.* Proved: Lemmas 11.6.1–11.6.3, Definitions 11.6.4, 11.6.5, 11.6.7, all claims of Theorem 11.6.6 and Proposition 11.6.8 (§§XI.12–XI.15). Also proved: the localization $\mathcal{PC}(E^{-1})$ with $I$ (§XI.21; G.134), and the category of proper strong bundle coverings of Remark 11.6.9 with its localization (§XI.22; G.135), both sketched in the text with the details left to the reader. *The narrower row.* V11.6, Proposition 11.1.6: the clause that two faces meet in a face of each does not hold as we read it. The Lean counterexample is `FaceMeetExample.not_facesMeetInFaces`, two faces of $([0,\infty)^3 \oplus W) \setminus K$ meeting in two pieces of an edge. What is proved is that they meet in a union of faces of each (§XI.20). V11.29, Theorems 11.4.4 and 11.4.5, is restated since 2026-11-13: the class hypothesis `hcov` is derived in Chapter 9 and $n = 1$ is covered (§XI.9).

**Six structural choices.** (1) *Faces by their object sets.* A face $X_\theta$ of $X$ is the saturated set $\pi^{-1}(\theta)$ with the morphisms whose source lies in it; the faces of $|X|$ are Chapter 2's faces for the index $d_{|X|}$ on the open set $|X|$, so Chapter 2's lemmas on faces apply to them (§XI.2). (2) *Total push-forwards.* $F_*\Theta$ is defined for every sc-smooth functor and every map on objects, with one choice of $x$ and of a morphism $F(x) \to y$, and $0$ where there is none; the statements say when the choice does not matter (§XI.5). (3) *Classes through representatives.* $f_*$ and $f^*$ on $\mathcal{EP}(E^{-1})$ are computed on a chosen representative with an equivalence as left leg (`rep`), and shown independent of it (§XI.6). (4) *Multisections as functors.* A multisection functor on a strong bundle is a map on the total object space constant along the morphisms of the total ep-groupoid; the local section structures of Definition 13.2.1 are a predicate on top of it (§XI.11). (5) *Functors of proper coverings as pairs.* A functor $F \to F'$ is a structure with the two functors and the two commutation equations, on objects and on morphisms, and $\mathcal{PC}$ is a category with composition associative on the nose (§§XI.13, XI.15). (6) *Sections and boundary operations.* Section functors of $\widehat{Gr}$ are functions on all objects, $0$ off $X_\infty$; Chapter 9's boundary lift depends on a boundary operation given as data, so Theorem 11.3.9 is first proved for a representative under a hypothesis relating the boundary operations on the two sides, and then, for the intrinsic boundary orientation of Definition 9.3.10, with no hypothesis (§XI.8).

**Map.**

- `Chapter11/EpEquivalenceGeometry`: Theorem 11.1.1 (diagrams, refinements, the comparison functor `wfpLift`, the functor $T$ on $\mathcal{EP}(E^{-1})$) and Proposition 11.1.2.
- `Chapter11/EpEquivalenceGeometryFaces`: Definition 11.1.3, Example 11.1.4, Remark 11.1.5, Propositions 11.1.6, 11.1.8, Remark 11.1.7, faces of open sets for Chapter 2's index.
- `Chapter11/BranchedEquivalence`: Definition 11.3.1, Lemma 11.3.2, local branching structures moved by sc-diffeomorphisms, Theorem 11.3.3 for an equivalence.
- `Chapter11/BranchedEquivalenceLocalization`: push-forward and pull-back along diagrams and classes, base change in the weak fibred product, Theorem 11.3.3, the witnesses on $\mathbb{Z}/2 \ltimes \mathbb{R}$.
- `Chapter11/BranchedEquivalenceTangentSection`: section functors of $\widehat{Gr}$, (11.8), (11.9), Lemma 11.3.4, the witness of Example 9.3.5.
- `Chapter11/BranchedEquivalenceTangent`: Chapter 9's tangents, orientations and boundaries under sc-diffeomorphisms between two M-polyfolds, the independence of the tangent sums carried

along, Theorems 11.3.7 and 11.3.8.

- `Chapter11/BranchedEquivalenceTangentOrientation`: Theorem 11.3.5, Definition 11.3.6, (11.10), the boundary-transfer hypothesis, Theorem 11.3.9 and its witnesses.
- `Chapter11/StrongBundleUpToEquivalence`: $|D|$ on $|W[1]|$, Proposition 11.5.1, multisection functors, Lemma 11.5.2, Proposition 11.5.3.
- `Chapter11/CoveringEquivalence`: Lemmas 11.6.1–11.6.3, Definitions 11.6.4 and 11.6.5, the identity coverings of the pair groupoid of $\mathbb{Z}$ and the point.
- `Chapter11/CoveringEquivalenceTheorem`: the functor $H$ and Theorem 11.6.6.
- `Chapter11/CoveringEquivalenceDiagram`: the category $\mathcal{PC}$, Definition 11.6.7, Proposition 11.6.8.
- `Chapter11/GapP2_HFaces`: Proposition 11.1.6, faces as tame ep-groupoids.
- `Chapter11/GapP2_HFaceIso`: the generalized isomorphisms $[d]_F$ of (11.1), Proposition 11.1.8 for them.
- `Chapter11/GapP2_H2FaceMeet`: the intersection clause of Proposition 11.1.6 and its counterexample.
- `Chapter11/GapP2_H2FaceMeetUnion`: the intersection as a union of faces.
- `Chapter11/GapP2_B2_FormDescent`: forms along surjective local sc-diffeomorphisms, (11.4).
- `Chapter11/GapP2_B2_EquivForms`: Lemma 11.2.2, Definition 11.2.3, Lemma 11.2.4 (3) on one level.
- `Chapter11/GapP2_B2_EquivFormsInfty`: Definition 11.2.3 on $\Omega^*_{ep,\infty}$, (11.5), Lemmas 11.2.4–11.2.6.
- `Chapter11/GapP2_B2_GeneralizedForms`: Definition 11.2.7, Theorem 11.2.1 (1)–(3).
- `Chapter11/GapP2_B2_OrbitForms`: Propositions 11.2.8, 11.2.9, Theorem 11.2.1 (4).
- `Chapter11/BoundaryOrientationEquiv`: Theorem 11.3.9 for the intrinsic boundary orientation.
- `Chapter11/GapP2_H4Compat`, `GapP2_H4CompatInst`: evaluations of forms compatible with the pullback.
- `Chapter11/GapP2_H4CrossChart`, `GapP2_H4Move`, `GapP2_H4Pairs`, `GapP2_H4MSheets`: structures and sheet integrals moved between two ep-groupoids.
- `Chapter11/GapP2_H4Measure`, `GapP2_H4Diagram`: Theorem 11.4.1, Remark 11.4.3.
- `Chapter11/GapP2_H4Eff`, `GapP2_H4EffDiagram`: Theorem 11.4.2.
- `Chapter11/GapP2_H4BdGeom`, `GapP2_H4Bd`, `GapP2_H4BdDiagram`: Theorems 11.4.4 and 11.4.5.
- `Chapter11/AuxiliaryNormTransport2021`: conditions (1) and (2) in Proposition 11.5.1.
- `Chapter11/MultisectionLocalStructure2021`: local section structures, Lemma 11.5.2 and Proposition 11.5.3 for sc$^+$-multisections.
- `Chapter11/GapP2_HPCLocalization`: $\mathcal{PC}(E^{-1})$ and $I$.
- `Chapter11/GapP2_H2SBPC`, `GapP2_H3SBPullback`, `GapP2_H3SBFibred`, `GapP2_H3SBLocalization`: the category of Remark 11.6.9, Theorem 11.6.6 for it, its localization.
- `Chapter11/GapP3_K1_BoundaryLoc`: Theorems 11.4.4 and 11.4.5 with the text's hypotheses, $n \geq 2$; `Chapter11/GapP3_K2_H4BdDim1`, `GapP3_K2_H4BdDiagramDim1`: the same for $n = 1$ (§XI.9).

## XI.1 Theorem 11.1.1 (2017: 11.1): the tangent functor on $\mathcal{EP}(E^{-1})$

**The tangent of a diagram** (V11.1) `Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.tangent`. For $d = [X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ the diagram $Td = [TX \xleftarrow{TF} TA \xrightarrow{T\Phi} TY]$; $TF$ is an equivalence by Theorem 10.1.7. Chapter 8's functoriality of $T$ is used in the form $T(G \circ F) = TG \circ TF$ `Chapter11/EpEquivalenceGeometry · EpFunctor.tangent_comp` and $T1 = 1$ `Chapter11/EpEquivalenceGeometry · EpFunctor.tangent_id`. *Step 1, refinements.* A refinement $H : d' \to d$ with natural transformations $\sigma : F' \to F \circ H$, $\tau : \Phi' \to \Phi \circ H$ gives $TH$ with $T\sigma$, $T\tau$ (Proposition 8.1.14), so $TH : Td' \to Td$ is a refinement

Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.IsRefinement.tangent, and equivalent diagrams have equivalent tangents Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.Equiv.tangent; this is the argument of the text. *Step 2, identities.* $T$ of the identity diagram refines the identity diagram of $TX$ Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.isRefinement_tangent_id. *Step 3, composition, left to the reader.* For functors $P : C \to X, Q : C \to Z$ and a natural transformation $\Phi \circ P \to G \circ Q$, $c \mapsto (P(c), \tau(c), Q(c))$ is a functor into the weak fibred product Chapter11/EpEquivalenceGeometry · EPLocalization.wfpLift with the right projections Chapter11/EpEquivalenceGeometry · EPLocalization.projLeft_comp_wfpLift Chapter11/EpEquivalenceGeometry · EPLocalization.projRight_comp_wfpLift. Applied to $T\pi_A$, $T\pi_B$ and $T$ of the transformation of $A \times_Y B$, it gives $T(A \times_Y B) \to TA \times_{TY} TB$; its composite with the projection to $TA$ is the equivalence $T\pi_A$, so it is an equivalence by two out of three, and it refines $Td \circ Te$ by $T(d \circ e)$ Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.isRefinement_tangent_comp. *Step 4, the functor.* `tangentFunctor` on classes Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor_obj Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor_map_homMk, compatible with the localization functor Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor_map_localizationFunctor; it maps generalized isomorphisms to generalized isomorphisms Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.isGeneralizedIso_tangent. **Theorem 11.1.1** Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.theorem_11_1_1 collects these.

## XI.2 Proposition 11.1.2 (2017: 11.1), Definition 11.1.3 (2017: 11.1) and Example 11.1.4 (2017: 11.1)

**Proposition 11.1.2 (2017: 11.1)** (V11.2) Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.proposition_11_1_2. *Step 1, equivalences.* An equivalence is a local sc-diffeomorphism on objects, and a local sc-diffeomorphism preserves the degeneracy index, so $d_Y(F(x)) = d_X(x)$, and $d$ is constant on orbits Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.degIndexOrbit_orbitMap Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit. *Step 2, diagrams.* For a diagram with both legs equivalences, $|d| = |\Phi| \circ |F|^{-1}$ preserves $d_{|\cdot|}$ Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.degIndexOrbit_orbitMap. *Step 3, classes.* A generalized isomorphism has such a representative (Theorem 10.3.10; the proof says "Proposition 10.3.10", a possible misprint) Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.degIndexOrbit_orbitMap.

**Definition 11.1.3 (2017: 11.1)** (V11.3) Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsOrbitFace Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.faceObj Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.faceMor Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsEpFace Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsEpFaceStructured Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsWeaklyFaceStructured. The object set of $X_\theta$ is saturated Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.isSaturated_faceObj and its morphisms are those between its objects Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.faceMor_eq. A face of $|X|$ is exactly a face, in Chapter 2's sense, of the open set $|X|$ for the index $d_{|X|}$ Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.isOrbitFace_iff_isFaceIn, so the faces through $z$ Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.orbitFacesThrough are Chapter 2's Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.orbitFacesThrough_eq and face-structuredness is Chapter 2's condition for $|X|$ Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.isEpFaceStructured_iff. The number of faces through $z$ is counted in $\mathbb{N} \cup \{\infty\}$, so that infinitely many faces do not count as $0$.

**Example 11.1.4 (2017: 11.1)** (V11.4) Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.

example_11_1_4. *The trivial groupoid.* With only identities, $\pi$ is a homeomorphism Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.isHomeomorph_π_trivial and the face structure of $C = [0,\infty)^2 \oplus W$ is that of the quadrant Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.isFaceStructured_quadrant. *The swap.* $\mathbb{Z}/2 \ltimes C$ with $-1$ exchanging the two coordinates Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.swapEpGroupoid; its object M-polyfold is $C$, face structured. In $|X|$ the set $\{d_{|X|} = 1\}$ is the image of one connected stratum, since the swap identifies the two open half-axes Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.setOf_degIndexOrbit_eq_one_swap; $|X|$ has exactly one face Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.isOrbitFace_swap_iff, and $|0|$, of index 2, lies in one face only Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.not_isEpFaceStructured_swap. The text gives no reason; this is it.

## XI.3 Remark 11.1.5 (2017: 11.1), Proposition 11.1.6 (2017: 11.2), Remark 11.1.7 (2017: 11.2), Proposition 11.1.8 (2017: 11.3)

**The quoted result** (V11.5). The text quotes [56, Proposition 3.15] for the fact that faces correspond under a homeomorphism preserving the index. For the faces of Chapter 2 on open sets: such a homeomorphism maps faces to faces Chapter11/EpEquivalenceGeometryFaces · TameBoundary.isFaceIn_univ_image, every face is the image of one Chapter11/EpEquivalenceGeometryFaces · TameBoundary.exists_isFaceIn_univ_preimage, the orbit map of an equivalence is such a homeomorphism Chapter11/EpEquivalenceGeometryFaces · TameBoundary.isDHomeoOn_univ, and face-structuredness transfers Chapter11/EpEquivalenceGeometryFaces · TameBoundary.isFaceStructuredOpen_univ_iff. *Equivalences.* $|\Phi|$ induces the bijection of faces Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.bijOn_faces, with $\Phi_*(X_\theta) = \text{saturation}(\Phi(X_\theta)) = Y_{|\Phi|(\theta)}$ Chapter11/EpEquivalenceGeometryFaces · EpFunctor.saturation_image_faceObj and $\Phi^{-1}(Y_{\theta'}) = X_{|\Phi|^{-1}(\theta')}$ Chapter11/EpEquivalenceGeometryFaces · EpFunctor.preimage_faceObj. *Generalized isomorphisms.* $||[d]||$ is a homeomorphism Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.isHomeomorph_orbitMap; `faceMap` and `faceComap` are the image and preimage of faces Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceMap Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceComap, computed on a representative as $\Psi_* \circ \Phi^*$ and $\Phi_* \circ \Psi^*$ Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceMap_homMk Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceComap_homMk, and mutually inverse bijections Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.bijOn_faceMap. **Remark 11.1.5** Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.remark_11_1_5.

**Proposition 11.1.6 (2017: 11.2), the part on $[d]_*$** (V11.6) Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.proposition_11_1_6. $[d]_*$ and $[d]^*$ map faces to faces Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.isOrbitFace_faceMap Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.isOrbitFace_faceComap and are inverse Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceComap_faceMap Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceMap_faceComap; $||[d]||$ restricts to homeomorphisms $\theta \to [d]_*\theta$. No face structure is used. The faces as tame ep-groupoids and the generalized isomorphisms $[d]_F$ are §XI.19, and the intersection clause §XI.20 (G.128, closed).

**Remark 11.1.7 (2017: 11.2)** (V11.7) Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.remark_11_1_7. Chapter 10 transfers tameness from $Y$ to $X$ along an equivalence Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.isTame_of_isTame. The converse goes through $X \times_Y Y$ for $F$ and $1_Y$: its projection to $X$ is an equivalence, so $X \times_Y Y$ is tame by Chapter 10's direction, and its projection to $Y$ is a surjective local sc-diffeomorphism on objects, by the essential surjectivity of $F$, which carries tameness down Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.isTame_

of_isTame_left Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.isTame_iff. With §XI.2 and the transfer of face-structuredness above, a generalized isomorphism preserves "tame and face structured" both ways.

**Proposition 11.1.8 (2017: 11.3)** (V11.8) Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.proposition_11_1_8. $f_*$ and $f^*$ depend only on $|f|$, by their definition as image and preimage; on a face, $|f|$ and $|g|$ agree. The text's maps $|F| \to |f_*F|$ come from the generalized isomorphisms of (11.1). They are the restrictions of $|f|$, and the second claim, for the generalized isomorphisms $[f]_F$ themselves, is §XI.19 Chapter11/GapP2_HFaceIso · EPLocalization.EPLoc.proposition_11_1_8_faceIso.

## XI.4 §11.2: sc-differential forms (Theorem 11.2.1 to Proposition 11.2.9 (2017: Theorem 11.2 to Proposition 11.5))

Since 2026-11-12 §11.2 is formalized (G.129), for Chapter 8's forms $\Omega^k_{ep}$ of an ep-groupoid and their direct limit $\Omega^*_{ep,\infty}$ over the index lifts $X^i$. The pattern is that of §11.3: a push-forward along equivalences by a choice of $x$ and $\psi : F(x) \to y$ that does not matter, composition rules, invariance under refinements, and generalized maps through $F_* \circ G^*$. The formal route differs in one point: $F_*$ is defined as the inverse of $F^*$, and the text's defining formula becomes a theorem.

**Descent along a surjective local sc-diffeomorphism.** For M-polyfolds and a surjective local sc-diffeomorphism $q : X \to Y$, the tangent spaces of $Y$ are images Chapter11/GapP2_B2_FormDescent · MPolyfold.IsLocalScDiffeo.exists_tangentFiber_lift, so $q^*$ is injective on families of alternating forms Chapter11/GapP2_B2_FormDescent · MPolyfold.formPullback_injective_of_isLocalScDiffeo. A family $\tau$ on $Y$ with $q^*\tau$ an sc-form is an sc-form: on the image $V$ of a sheet of $q$, with inverse $g$, $\tau|V = g^*(q^*\tau)$, the text's (11.4), and sc-smoothness of a form is local Chapter11/GapP2_B2_FormDescent · MPolyfold.isScForm_of_formPullback. A family $\eta$ on $X$ taking the same values on vectors with the same image under $Tq$ descends, $\tau_y = \eta_x \circ (Tq(x))^{-1}$, with $q^*\tau = \eta$ Chapter11/GapP2_B2_FormDescent · MPolyfold.formDescend Chapter11/GapP2_B2_FormDescent · MPolyfold.formPullback_formDescend.

**Lemma 11.2.2 and Definition 11.2.3 (2017: Lemma 11.1, Definition 11.2)** (V11.10). *Step 1, the full case.* For $F$ full and a local sc-diffeomorphism on objects, a form of the ep-groupoid $X$ takes the same values on vectors over $x$, $x'$ with $F(x) = F(x')$ and the same images under $TF$, since a morphism $x \to x'$ exists and forms of ep-groupoids are invariant under morphisms; if $F$ is also surjective on objects, every form on $X$ descends to a form $\tau$ on $Y$ with $F^*\tau$ the given one Chapter11/GapP2_B2_EquivForms · EpFunctor.exists_epForms_formPullback_eq_of_surjective. *Step 2, equivalences, through the weak fibred product.* For an equivalence $F$, $A = X \times_Y Y$ of $F$ and $1_Y$ has the objects $(x, \psi, y)$, $\psi : F(x) \to y$, of the text's definition; $\pi_Y : A \to Y$ is an equivalence surjective on objects and $F \circ \pi_X \simeq \pi_Y$. Pull-backs compose Chapter11/GapP2_B2_EquivForms · EpFunctor.formPullback_comp_functor, and $F^*$ is injective Chapter11/GapP2_B2_EquivForms · EpFunctor.IsEquivalence.injOn_formPullback and onto, $F_*\omega$ being the descent of $\pi_X^*\omega$ along $\pi_Y$ Chapter11/GapP2_B2_EquivForms · EpFunctor.IsEquivalence.exists_formPullback_eq. So $F^* : \Omega^k_{ep}(Y) \to \Omega^k_{ep}(X)$ is bijective Chapter11/GapP2_B2_EquivForms · EpFunctor.IsEquivalence.bijective_formPullback, and $F_*$ is its inverse Chapter11/GapP2_B2_EquivForms · EpFunctor.IsEquivalence.formPush. *Step 3, the text's formula.* $(F_*\omega)_y(T\psi(TF(x)l_1), \dots, T\psi(TF(x)l_k)) = \omega_x(l_1, \dots, l_k)$ for every $x$, $\psi : F(x) \to y$ and $l_j \in T_xX$ Chapter11/GapP2_B2_EquivForms · EpFunctor.IsEquivalence.definition_11_2_3, and every $h_1, \dots, h_k \in T_yY$ is of this form Chapter11/GapP2_B2_EquivForms · EpFunctor.IsEquivalence.exists_lift_tangent, so the formula determines $F_*\omega$. *Step 4, Lemma 11.2.2* Chapter11/GapP2_B2_EquivForms · EpFunctor.IsEquivalence.lemma_11_2_2: both sides equal $(F_*\omega)_y(h_1, \dots, h_k)$. Before Definition 11.2.3 the text's $\Omega(X)$ for the target is read $\Omega(Y)$.

**Lemmas 11.2.4–11.2.6 (2017: 11.2–11.4) and (11.5)** (V11.11). *Step 1, levels.* The index raises $F^i : X^i \to Y^i$ of an equivalence are equivalences (Chapter 10) Chapter11/GapP2_B2_EquivFormsInfty · EP.isEquivalence_liftFunctor, so $F_*$ exists on each level Chapter11/GapP2_B2_EquivFormsInfty · EP.epFormsPush and commutes with the maps of the directed system; this gives $F_*$ on $\Omega^k_{ep,\infty}$ Chapter11/GapP2_B2_EquivFormsInfty · EP.epFormsInftyPush, inverse to $F^*$ Chapter11/GapP2_B2_EquivFormsInfty · EP.epFormsInftyPullback_comp_epFormsInftyPush Chapter11/GapP2_B2_EquivFormsInfty · EP.epFormsInftyPush_comp_epFormsInftyPullback, Lemma 11.2.4 (3). *Step 2, (11.5).* $d_Y \circ F_* = F_* \circ d_X$, from $d$ commuting with $F^*$ and $F_* = (F^*)^{-1}$ Chapter11/GapP2_B2_EquivFormsInfty · EP.epFormsInftyD_comp_epFormsInftyPush. *Step 3, Lemma 11.2.4 (1), (2).* $(G \circ F)^* = F^* \circ G^*$ is Chapter 8's Chapter8/GapP2_B_EpFormsFunctor · EP.epFormsInftyPullback_comp, and inverting it gives $(G \circ F)_* = G_* \circ F_*$ Chapter11/GapP2_B2_EquivFormsInfty · EP.lemma_11_2_4_two. *Step 4, Lemma 11.2.5.* Naturally equivalent functors have the same pull-back (Chapter 8) Chapter8/GapP2_B_EpFormsFunctor · EP.epFormsInftyPullback_eq_of_natTrans; $F_*$ is the only right inverse of $F^*$ Chapter11/GapP2_B2_EquivFormsInfty · EP.epFormsInftyPush_eq_of_comp_eq_id, so naturally equivalent equivalences have the same push-forward Chapter11/GapP2_B2_EquivFormsInfty · EP.lemma_11_2_5_push. *Step 5, Lemma 11.2.6.* For $H : A' \to A$ with $F \circ H \simeq F'$ and $G \circ H \simeq G'$: $F_*G^* = F_*H_*H^*G^* = (F \circ H)_*(G \circ H)^* = F'_*G'^*$ by Steps 3 and 4 Chapter11/GapP2_B2_EquivFormsInfty · EP.lemma_11_2_6, and the same for $G_*F^*$ Chapter11/GapP2_B2_EquivFormsInfty · EP.lemma_11_2_6_push. In the proof of Lemma 11.2.6 the text's "Lemma 11.2.2 and 11.2.3" is read as Lemmas 11.2.4 and 11.2.5.

**Definition 11.2.7 (2017: 11.3) and Theorem 11.2.1 (1)–(3) (2017: 11.2)** (V11.12, V11.9). *Step 1, diagrams.* $d^* = F_* \circ \Phi^*$ and, for $\Phi$ an equivalence, $d_* = \Phi_* \circ F^*$ Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.Diagram.formsPullback Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.Diagram.formsPush. They are unchanged under refinement by Lemma 11.2.6 Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.Diagram.formsPullback_eq_of_isRefinement, hence on equivalent diagrams Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.Diagram.formsPullback_eq_of_equiv Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.Diagram.formsPush_eq_of_equiv. *Step 2, composites.* A base change square in the weak fibred product Chapter11/GapP2_B2_GeneralizedForms · EP.epFormsInftyPush_epFormsInftyPullback_of_square gives the pull-back of the composite diagram Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.Diagram.formsPullback_comp. *Step 3, classes.* $f^*$ for every generalized map, equal to $F_* \circ \Phi^*$ on every representative Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.formsPullback Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.formsPullback_homMk, with $(g \circ f)^* = f^* \circ g^*$ Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.formsPullback_comp, so $\Omega^k_{ep,\infty}$ is a contravariant functor on $\mathcal{EP}(E^{-1})$, beyond the text. For a generalized isomorphism with inverse $g$, $h_* = g^*$ Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.formsPush, which by Theorem 10.3.10 is $\Phi_* \circ F^*$ on every representative Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.formsPush_homMk, the text's definition. *Step 4, Theorem 11.2.1.* (1) $d \circ d = 0$ Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.theorem_11_2_1_one; (2) $d_X \circ f^* = f^* \circ d_Y$ from (11.5) and Chapter 8 Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.theorem_11_2_1_two; (3) $h_* \circ d_X = d_Y \circ h_*$, $h_* \circ h^* = 1$, $h^* \circ h_* = 1$ Chapter11/GapP2_B2_GeneralizedForms · EPLocalization.EPLoc.theorem_11_2_1_three.

**Propositions 11.2.8 and 11.2.9 (2017: 11.4 and 11.5), Theorem 11.2.1 (4)** (V11.13). *Step 1, local families.* The conclusion of Proposition 10.1.10 (1), an sc-smooth $z \mapsto \alpha_z : F(z) \to G(z)$ near every object Chapter11/GapP2_B2_OrbitForms · EpFunctor.HasLocalNatIso, holds for equivalences with $|F| = |G|$ when $X$ or $Y$ has the local regularity property Chapter11/GapP2_B2_OrbitForms · EpFunctor.hasLocalNatIso_of_orbitMap_eq, and passes to level 1 and to the index raises Chapter11/GapP2_B2_OrbitForms · EpFunctor.HasLocalNatIso.liftOne. *Step 2, the pull-backs agree.* Locally $F = s \circ \alpha$ and $G = t \circ \alpha$, and $s^*\omega = t^*\omega$ for a form of the ep-groupoid $Y$, so $F^*\omega = G^*\omega$ Chapter11/GapP2_B2_OrbitForms · EpFunctor.HasLocalNatIso.formPullback_eq Chapter11/GapP2_B2_

OrbitForms · EP.epFormsInftyPullback_eq_of_orbitMap_eq. The text's second computation for $F_* = G_*$ is replaced by inverting. **Proposition 11.2.8** Chapter11/GapP2_B2_OrbitForms · EP.proposition_11_2_8. *Step 3, generalized isomorphisms.* As in the text, through the weak fibred product $A \times_X A'$: Proposition 11.2.8 applies to the equivalences $G \circ \pi_A$ and $G' \circ \pi_{A'}$, where the property of $Y$ suffices by Proposition 10.1.9 Chapter11/GapP2_B2_OrbitForms · EPLocalization.EPLoc.formsPullback_eq_of_orbitMap_eq. **Proposition 11.2.9** Chapter11/GapP2_B2_OrbitForms · EPLocalization.EPLoc.proposition_11_2_9 and **Theorem 11.2.1 (4)** Chapter11/GapP2_B2_OrbitForms · EPLocalization.EPLoc.theorem_11_2_1_four. The hypotheses are the text's; the local regularity property of the lifts $X^i$ is neither assumed nor needed, the families of Step 1 being restricted to the levels. The property is satisfiable, by the point and by $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (Chapter 10).

## XI.5 Definition 11.3.1 (2017: 11.4) and Lemma 11.3.2 (2017: 11.5)

**Definition 11.3.1 (2017: 11.4)** (V11.14) Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta. $F^*\Theta' = \Theta' \circ F$ Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_apply. $F_*\Theta(y)$ is $\Theta$ at a chosen $x$ with $F(x) \cong y$, and $0$ if there is none Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_of_not_exists; a nonzero value implies the existence Chapter11/BranchedEquivalence · EpFunctor.exists_isomorphic_of_pushforwardTheta_ne_zero, and nonnegativity is kept Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_nonneg. The text writes the push-forward as a map $X' \to \mathbb{Q}^+$; we read $Y \to \mathbb{Q}^+$ (possible misprint).

**Lemma 11.3.2 (2017: 11.5)** (V11.15). *Step 1, the choice does not matter.* Functors are constant on isomorphism classes Chapter11/BranchedEquivalence · EpGroupoid.IsFunctorial.eq_of_isomorphic. If $F$ is full and $F(x) \cong y \cong F(x')$, then $F(x) \cong F(x')$ Chapter11/BranchedEquivalence · EpFunctor.isomorphic_obj lifts to $x \cong x'$ Chapter11/BranchedEquivalence · EpFunctor.isomorphic_of_full, so $F_*\Theta(y) = \Theta(x)$ for every such $x$ Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_eq Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_obj Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_eq_of_isomorphic; this is the text's argument with $\sigma$ lifting $\psi^{-1} \circ \psi'$. *Step 2, functors.* $F_*\Theta$ is a functor for full $F$ Chapter11/BranchedEquivalence · EpFunctor.isFunctorial_pushforwardTheta, $F^*\Theta'$ for every $F$ Chapter11/BranchedEquivalence · EpFunctor.isFunctorial_pullbackTheta. *Step 3, (1) and (2).* $F^*F_*\Theta = \Theta$ for full $F$ Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_pushforwardTheta; $F_*F^*\Theta' = \Theta'$ for essentially surjective $F$ Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_pullbackTheta. *Step 4, (3).* Naturally equivalent full functors have equal push-forwards Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_eq_of_naturallyEquivalent, naturally equivalent functors equal pull-backs Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_eq_of_naturallyEquivalent. *Step 5, composition.* $(G \circ F)^* = F^* \circ G^*$ Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_comp, $(G \circ F)_* = G_* \circ F_*$ for full functors Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_comp, $1_* = 1$ Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_id.

## XI.6 The push-forward and pull-back along generalized maps, pp. 536–537

**Diagrams** (V11.16) Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pushTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pullTheta. For $d = [X \xleftarrow{F} A \xrightarrow{\Phi} Y]$, $d_* = \Phi_* F^*$ and $d^* = F_*\Phi^*$. *Step 1, refinements.* For a refinement $H : d' \to d$, $H$ is an equivalence and $F' \simeq F \circ H$, $\Phi' \simeq \Phi \circ H$; Lemma 11.3.2 (3) and the composition rules

give $d'^{*} = d^{*}$, and $d'_{*} = d_{*}$ when $\Phi$ is an equivalence, which then holds for $\Phi'$ as well Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.IsRefinement.isEquivalence_Φ_iff Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pullTheta_eq_of_isRefinement Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pushTheta_eq_of_isRefinement. *Step 2, equivalent diagrams* have a common refinement Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pullTheta_eq_of_equiv Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pushTheta_eq_of_equiv.

**Classes.** A chosen representative Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.rep Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.homMk_rep Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.equiv_rep_homMk, with an equivalence as right leg for a generalized isomorphism Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isEquivalence_rep_Φ, defines $f^{*}$ for every generalized map and $f_{*}$ for generalized isomorphisms Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta, computed on any representative Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta_homMk Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_homMk, and mapping functors to functors Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isFunctorial_pullTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isFunctorial_pushTheta. *Composition, "easily verified" in the text.* The composite of $[d]$ and $[e]$ is represented through the weak fibred product $A \times_Y B$ (Theorem 10.3.8). In it, $\Theta$ satisfies base change: $(\pi_B)_{*}(\pi_A)^{*}\Theta = G^{*}F_{*}\Theta$ when $F$ is an equivalence, and symmetrically Chapter11/BranchedEquivalenceLocalization · WeakFibredProduct.pushforwardTheta_projRight_pullbackTheta Chapter11/BranchedEquivalenceLocalization · WeakFibredProduct.pushforwardTheta_projLeft_pullbackTheta; the proof evaluates both sides at $(a, \psi, b)$ and uses that a functor $\Theta$ takes equal values on isomorphic objects. Hence $(f \gg g)^{*} = f^{*} \circ g^{*}$ for all generalized maps Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta_comp and $(f \gg g)_{*} = g_{*} \circ f_{*}$ for generalized isomorphisms Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_comp, with $(f^{-1})_{*} = f^{*}$ Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_inv and the inverse laws Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta_pushTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_pullTheta. Nothing here uses Theorem 10.3.19, which does not hold as we read it (Chapter 10).

## XI.7 Theorem 11.3.3 (2017: 11.3)

**Moving a local branching structure** (V11.17). Given an sc-diffeomorphism $f : W \to W'$ between open subsets of the object spaces of two ep-groupoids, with $\Theta' \circ f = \Theta$ on $W$, a local branching structure $(U, M_i, \sigma_i)$ of $\Theta$ at $x$ gives one of $\Theta'$ at $f(x)$: the open set $f(U \cap W)$ Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.mapU Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.mem_image_iff_of_mem_mapU, the sheets $f(M_i \cap W)$, first restricted to $W$, with the same weights Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.mapDiffeo. The sheets keep their dimension Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.isOfDim_mapDiffeo, because the dimension of a submanifold at a point is invariant under sc-diffeomorphisms Chapter11/BranchedEquivalence · MPolyfold.HasDimAt.of_image Chapter11/BranchedEquivalence · MPolyfold.hasDimAt_image_iff_of_isScDiffeoOn Chapter11/BranchedEquivalence · MPolyfold.IsSubmanifoldOfDim.image_isScDiffeoOn.

**The step for an equivalence.** *Step 1, supports.* $|\mathrm{supp}(F^{*}\Theta')| = |F|^{-1}(|\mathrm{supp}(\Theta')|)$ Chapter11/BranchedEquivalence · EpFunctor.orbitSupp_pullbackTheta and, for full $F$, $|\mathrm{supp}(F_{*}\Theta)| = |F|(|\mathrm{supp}(\Theta)|)$ Chapter11/BranchedEquivalence · EpFunctor.orbitSupp_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.

exists_mem_supp_of_mem_supp_pushforwardTheta; since $|F|$ is a homeomorphism, the paracompactness of 2021 Definition 9.1.2 (1) transfers, a point the proof does not mention. *Step 2, local branching structures.* For $F_*$: at $y$ with $\psi : F(x) \to y$, Chapter 10's transport chart $\widehat{\psi} \circ F$ Chapter10/EquivalenceLR · EpFunctor.TransportChart is an sc-diffeomorphism near $x$ with $F_*\Theta \circ \widehat{\psi} \circ F = \Theta$, and the structure is moved Chapter11/BranchedEquivalence · EpFunctor.exists_localBranchingStructure_pushforwardTheta; for $F^*$, $F$ itself near $x$. Hence **branched ep⁺-subgroupoids go to branched ep⁺-subgroupoids** Chapter11/BranchedEquivalence · EpFunctor.isBranchedEpSubgroupoid_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isBranchedEpSubgroupoid_pullbackTheta. *Step 3, properties.* Closedness and compactness from Step 1 Chapter11/BranchedEquivalence · EpFunctor.isClosedBranched_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isClosedBranched_pullbackTheta Chapter11/BranchedEquivalence · EpFunctor.isCompactBranched_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isCompactBranched_pullbackTheta; orbifold-type from Step 2 Chapter11/BranchedEquivalence · EpFunctor.isOrbifoldType_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isOrbifoldType_pullbackTheta; manifold-type by comparing two morphisms $y \to y'$ through the faithfulness of $F$ Chapter11/BranchedEquivalence · EpFunctor.isManifoldType_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isManifoldType_pullbackTheta. In this step the text writes $\psi \circ \varphi' \circ \psi^{-1} : F(x) \to F(x)$ for $\psi : F(x) \to y$ and $\varphi' : y \to y$; we read $\psi^{-1} \circ \varphi' \circ \psi$ (possible misprint). *Step 4, dimension.* $n$-dimensionality Chapter11/BranchedEquivalence · EpFunctor.isNDimensional_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isNDimensional_pullbackTheta and dimensional decompositions, the push-forward of a sum being the sum of the push-forwards Chapter11/BranchedEquivalence · EpFunctor.exists_pushforwardTheta_eq_apply Chapter11/BranchedEquivalence · EpFunctor.isDimDecomposition_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isDimDecomposition_pullbackTheta.

**Theorem 11.3.3 (2017: 11.3)** (V11.18). On a representative with both legs equivalences, $f_* = \Phi_* F^*$ and $f^* = F_* \Phi^*$, and each part follows from the step for an equivalence: (1) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_one, (2) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_two, (3) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_three, (4) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_four, (5) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_five, (6) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_six, with $n$-dimensionality Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isNDimensional_pushTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isNDimensional_pullTheta and the support Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.orbitSupp_pushTheta. (6) speaks of "the" dimensional decomposition of Proposition 9.1.11, whose uniqueness does not hold as we read it Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn; it is proved for every dimensional decomposition.

**Witnesses** (V11.19). On $\mathbb{Z}/2 \ltimes \mathbb{R}$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionEP, $\Theta_0 = 1_{\{0\}}$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionTheta has the local branching structure $U = \mathbb{R}$, $M = \{0\}$ at the point with isotropy $\mathbb{Z}/2$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.localBranchingStructure_reflectionTheta Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.isBranchedEpSubgroupoid_reflectionTheta. Along the refinement of the covering $\{x > -1\}$, $\{x < 1\}$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionCoverEP, a functor not injective on objects, $F^*\Theta_0$ is a branched ep⁺-subgroupoid Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.isBranchedEpSubgroupoid_pullbackTheta_reflectionCover, equal to $1$ at the two objects over $0$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.pullbackTheta_reflectionCover_two_points, and $F_* F^* \Theta_0 = \Theta_0$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.

pushforwardTheta_pullbackTheta_reflectionCover. The generalized isomorphism of the refinement Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionCoverDiagram Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.isGeneralizedIso_reflectionCoverDiagram pushes $\Theta_0$ to $F^*\Theta_0$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.pushTheta_reflectionCoverDiagram.

## XI.8 Lemma 11.3.4 to Theorem 11.3.9 (2017: Lemma 11.6 to Theorem 11.7): sections, orientations, tangents, boundaries

**Section functors of $\widehat{Gr}$** (V11.20) Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.IsGrHatSection Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.GrHatSections. $h(x)$ is a finite formal sum of oriented tangent sets at $x \in X_\infty$, natural along morphisms, with no continuity; it is recorded on all objects, $0$ off $X_\infty$. Every orientation of Chapter 9 is one Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.IsOrientation.isGrHatSection. *Step 1, (11.8).* $F_*h(y)$ moves $h(x)$ by $T\psi \circ TF(x)$ for a chosen $\psi : F(x) \to y$, the formal sums by the image of carriers and frames Chapter11/BranchedEquivalenceTangent · MPolyfold.OrTangent.map Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection. For an equivalence the value is the same for every $\psi$ Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_eq: two choices differ by $T$ of a morphism of $X$, which $h$ absorbs by naturality, as the text says. *Step 2, (11.9).* $F^*h'(x) = TF(x)^{-1}h'(F(x))$, with the inverse of $TF$ on $T_xX$ Chapter11/BranchedEquivalenceTangentSection · MPolyfold.fibreInv Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection. *Step 3, equivalences.* Section functors go to section functors Chapter11/BranchedEquivalenceTangentSection · EpFunctor.isGrHatSection_pushSection Chapter11/BranchedEquivalenceTangentSection · EpFunctor.isGrHatSection_pullSection; $F^*F_* = 1$, $F_*F^* = 1$ Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection_pushSection Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_pullSection; naturally equivalent equivalences agree Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_eq_of_natTrans Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection_eq_of_natTrans; composition Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_comp Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection_comp; base change in the weak fibred product of two equivalences Chapter11/BranchedEquivalenceTangentSection · WeakFibredProduct.pushSection_projRight_pullSection Chapter11/BranchedEquivalenceTangentSection · WeakFibredProduct.pushSection_projLeft_pullSection. *Step 4, Lemma 11.3.4.* On diagrams Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pushSection Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pullSection, unchanged on equivalent diagrams Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pushSection_eq_of_equiv Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pullSection_eq_of_equiv; on classes Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pullSection, computed on any representative Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_homMk Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pullSection_homMk, with the composition laws for generalized isomorphisms through the base change Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_comp Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pullSection_comp and $(f^{-1})_* = f^*$ Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_inv. **Lemma 11.3.4** Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.lemma_11_3_4.

**Chapter 9 across two M-polyfolds.** Chapter 9 moves submanifolds, tangent sets and orientations by sc-diffeomorphisms of one M-polyfold; equivalences need them between open

subsets $U \subseteq X$, $U' \subseteq Y$ of two. For such an $f$: $T_{f(a)}f(A) = Tf(T_aA)$ Chapter11/BranchedEquivalenceTangent · MPolyfold.submanifoldTangent_image_of_isScDiffeoOn, chart vectors in transported charts Chapter11/BranchedEquivalenceTangent · MPolyfold.chartVec_tangentMapOn_of_isScDiffeoOn, oriented tangent sets Chapter11/BranchedEquivalenceTangent · MPolyfold.IsOrientedAt.map_tangentMapOn, orientations of submanifolds, their frames moved by $Tf$ and their continuity read in the chart $\varphi \circ f^{-1}$ Chapter11/BranchedEquivalenceTangent · MPolyfold.SubmanifoldOrientation.mapDiffeo Chapter11/BranchedEquivalenceTangent · MPolyfold.SubmanifoldOrientation.mapDiffeo_tangent, the reduced tangent condition Chapter11/BranchedEquivalenceTangent · MPolyfold.IsTangentReducedAt.image_of_isScDiffeoOn Chapter11/BranchedEquivalenceTangent · MPolyfold.isTangentReducedAt_image_iff_of_isScDiffeoOn and the boundary of a submanifold Chapter11/BranchedEquivalenceTangent · MPolyfold.mem_submanifoldBoundary_image_iff_of_isScDiffeoOn. For local branching structures moved as in §XI.7: the tangent sums are moved by $Tf$ Chapter11/BranchedEquivalenceTangent · EpGroupoid.LocalBranchingStructure.tangentSum_mapDiffeo; the independence of the tangent sums of Theorem 9.2.1 is local and carried along, by moving structures back by $f^{-1}$, comparing and moving forward Chapter11/BranchedEquivalenceTangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isScDiffeoOn; $T\Theta'(Tf\,\alpha) = T\Theta(\alpha)$ Chapter11/BranchedEquivalenceTangent · EpGroupoid.tangentTheta_tangentMapOn and $\partial\Theta'(f(x)) = \partial\Theta(x)$ Chapter11/BranchedEquivalenceTangent · EpGroupoid.boundaryTheta_apply_of_isScDiffeoOn; oriented structures Chapter11/BranchedEquivalenceTangent · EpGroupoid.OrientedBranchingStructure.mapDiffeo Chapter11/BranchedEquivalenceTangent · EpGroupoid.OrientedBranchingStructure.orientedSum_mapDiffeo.

**Theorem 11.3.5 (2017: 11.4)** (V11.21). Along a transport chart $Q = \hat{\psi} \circ F$, $T\psi_u \circ TF = TQ$ on $T_uX$ Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.TransportChart.tangentMor_ψ_tangent_obj, so an oriented branching structure at $x$ representing $\hat{T}_\Theta$, moved by $Q$, represents $F_*\hat{T}_\Theta$ near $y$ Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.isOrientation_pushSection; for $F^*$, a structure at $F(x)$ is moved back by a local inverse of $F$ Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.isOrientation_pullSection. This is the text's proof. For generalized isomorphisms Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_5_push Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_5_pull. The independence of the tangent sums, which Chapter 9 now proves for every $\Theta$ Chapter9/GeneralCorollaries · EpGroupoid.hasIndependentTangentSums, was assumed for $\Theta$ and is carried to $F_*\Theta$, $F^*\Theta'$ Chapter11/BranchedEquivalenceTangent · EpFunctor.hasIndependentTangentSums_pushforwardTheta Chapter11/BranchedEquivalenceTangent · EpFunctor.hasIndependentTangentSums_pullbackTheta and to $f_*\Theta$, $f^*\Theta'$ Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.hasIndependentTangentSums_pushTheta Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.hasIndependentTangentSums_pullTheta.

**Definition 11.3.6 (2017: 11.5) and (11.10)** (V11.22) Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.push Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.pull. The structure records $\Theta$, $\hat{T}_\Theta$ and the independence; push-forward and pull-back are oriented branched ep⁺-subgroupoids by Theorems 11.3.3 and 11.3.5; (11.10) Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.push_comp Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.pull_comp from Lemma 11.3.4 and §XI.6. Of the two inverse laws of (11.10), $f^*f_*\hat{\Theta} = \hat{\Theta}$ is proved for $\hat{T}_\Theta$ zero off $X_\infty$ Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.pull_push, and $f_*f^*\hat{\Theta}' = \hat{\Theta}'$ for its two components Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_pullTheta Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_pullSection.

**Theorem 11.3.7 (2017: 11.5)** (V11.23). *Step 1, equivalences.* $T(F_*\Theta)(F(x)) = TF(T\Theta(x))$ Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentValue_pushforwardTheta_obj and, for $\psi : F(x) \to y$,

$T(F_*\Theta)(y)$ is $T\Theta(x)$ moved by $T\psi \circ TF(x)$ Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentValue_pushforwardTheta; for the pull-back, with a local inverse of $F$ Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentValue_pullbackTheta. Hence $(TF)_*T\Theta = T(F_*\Theta)$, $(TF)^*T\Theta' = T(F^*\Theta')$, where $TF$ is an equivalence (Theorem 10.1.7) and $(TF)_*$ the push-forward of Definition 11.3.1 on $TX$ Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentTheta_pushforwardTheta Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentTheta_pullbackTheta. *Step 2, generalized isomorphisms.* $Tf$ is $[Td]$ for the chosen representative Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.tangentFunctor_map_eq (Theorem 11.1.1), and Step 1 applies to both legs Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_7_push Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_7_pull. The text leaves the proof to the reader.

**Theorem 11.3.8 (2017: 11.6)** (V11.24). By the local step, $\partial\Theta$ of Definition 9.2.16 is moved pointwise by every chart $\hat{\psi} \circ F$: the text's "$d_M(x)$ is preserved" is the invariance of $\partial M$ above. For equivalences Chapter11/BranchedEquivalenceTangent · EpFunctor.boundaryTheta_pushforwardTheta Chapter11/BranchedEquivalenceTangent · EpFunctor.boundaryTheta_pullbackTheta and generalized isomorphisms, (11.11) Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_8_push Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_8_pull.

**Theorem 11.3.9 (2017: 11.7), for a representative** (V11.25). Chapter 9's boundary orientation $\hat{T}_{\partial\Theta}$ is the boundary lift of $\hat{T}_\Theta$ by a boundary operation $bd$ on oriented tangent sets, taken as data. A linear map $T_xX \to T_yY$ transfers $bd$ to $bd'$ if it keeps degree 1 and commutes with them there Chapter11/BranchedEquivalenceTangentOrientation · MPolyfold.IsBoundaryTransfer; the identity does Chapter11/BranchedEquivalenceTangentOrientation · MPolyfold.isBoundaryTransfer_id. A fibre map that transfers them commutes with the degree-1 part and the boundary operation Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.mapDomain_boundary_mapDomain_map, so along an equivalence whose maps $T\psi \circ TF(x)$ transfer $bd$ to $bd'$, $F_*\hat{T}_{\partial\Theta} = \hat{T}_{\partial(F_*\Theta)}$ Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.pushSection_boundaryLift, and dually Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.pullSection_boundaryLift. With Theorem 11.3.8 for $\partial\Theta$, for a representative $X \xleftarrow{F} A \xrightarrow{\Phi} Y$ and boundary operations on $X$, $A$, $Y$ corresponding under $TF(a)^{-1}$ and $T\psi \circ T\Phi(a)$ Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_9_push Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_9_pull. That Definition 9.3.8's boundary operation satisfies the hypothesis along equivalences is the next paragraph (G.130, closed).

**Theorem 11.3.9 for the intrinsic boundary orientation** (V11.25) Chapter11/BoundaryOrientationEquiv · EPLocalization.EPLoc.theorem_11_3_9_push_boundaryOrientation Chapter11/BoundaryOrientationEquiv · EPLocalization.EPLoc.theorem_11_3_9_pull_boundaryOrientation, independent of the representative, with no transfer hypothesis (G.130). *Step 1, sc-diffeomorphisms move the boundary operation.* A presented chart $\psi$ at $f(x)$ gives the presented chart $\psi \circ f$ at $x$ with the same model, so $Tf$ preserves $T^R_xX$, $C_xX$ and the reduced codimension, and with them Definition 9.3.8's boundary operation: $\partial(Tf_*(L,o)) = Tf_*(\partial(L,o))$ Chapter11/BoundaryOrientationEquiv · MPolyfold.boundaryOp_map_tangentMapOn. "$d_M(x)$ is preserved", in the proof of Theorem 11.3.8, is read for the reduced codimension of $T_xM$, which for tame $M$ is $d_M(x)$ and is preserved by every sc-diffeomorphism, without tameness. *Step 2, equivalences.* Through the transport chart $\hat{\psi} \circ F$ of Chapter 10, the maps $T\psi \circ TF(x)$ move it for every $\psi : F(x) \to y$ Chapter11/BoundaryOrientationEquiv · EpFunctor.isReducedBoundaryTransfer_tangentMor, and so do the fibre inverses $TF(x)^{-1}$ Chapter11/BoundaryOrientationEquiv · EpFunctor.isReducedBoundaryTransfer_fibreInv. *Step 3, the intrinsic orientation.* $\hat{T}_{\partial\Theta}$ of Definition 9.3.10 is a function of $\hat{T}_\Theta$, computed with the

boundary operation pointwise. By Steps 1 and 2 the push-forward along a representative commutes with it `Chapter11/BoundaryOrientationEquiv · EPLocalization.Diagram.pushSection_boundaryOrientation` `Chapter11/BoundaryOrientationEquiv · EPLocalization.Diagram.pullSection_boundaryOrientation`. With Theorem 11.3.8 for $\partial\Theta$, $f_*(\partial\hat{\Theta}) = \partial(f_*\hat{\Theta})$ and the same for $f^*$. The statement holds for every section of $\widehat{Gr}$, oriented or not. *The boundary-lift form.* For Chapter 9's boundary lift with Definition 9.3.8's operation, the boundary-transfer hypothesis above follows from Steps 1 and 2 `Chapter11/BoundaryOrientationEquiv · MPolyfold.IsReducedBoundaryTransfer.isBoundaryTransfer` `Chapter11/BoundaryOrientationEquiv · EpFunctor.isBoundaryTransfer_tangentMor`, given that $f$ carries tangent sets of tame submanifolds to tangent sets of tame submanifolds `Chapter11/BoundaryOrientationEquiv · MPolyfold.CarriesTameTangents`. This form, which also asks that $f_*\hat{\Theta}$ be tame `Chapter11/BoundaryOrientationEquiv · EPLocalization.EPLoc.theorem_11_3_9_push_boundaryOp`, has those two conditions as hypotheses; they hold along the identity `Chapter11/BoundaryOrientationEquiv · MPolyfold.carriesTameTangents_id` and for Example 9.3.5 `Chapter11/BoundaryOrientationEquiv · EpGroupoid.Examples.theorem_11_3_9_push_boundaryOp_example935`. That sc-diffeomorphisms preserve tame induced structures on submanifolds is not formalized, and the intrinsic form does not need it.

**Witnesses** (V11.26). The orientation $\hat{T}^1$ of Example 9.3.5 is a section functor on $\mathbb{R}^2$ `Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.Examples.exampleT1_mem_grHatSections`, fixed by the identity `Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.Examples.pushSection_id_exampleT1`; $(\Theta, \hat{T}^1)$ on the trivial ep-groupoid of $\mathbb{R}^2$ `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.rn2EP` `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.example935Oriented` is pushed forward along the identity generalized isomorphism `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.isGeneralizedIso_id_rn2EP` `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.example935Oriented_push`, and the boundary-transfer hypothesis holds there `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.isBoundaryTransfer_id_rn2`.

## XI.9 §11.4: integration (Theorems 11.4.1, 11.4.2, 11.4.4, 11.4.5, Remark 11.4.3 (2017: Theorems 11.8, 11.9, Remark 11.3))

Since 2026-11-13 §11.4 is formalized (G.131). Theorems 11.4.1, 11.4.2 and Remark 11.4.3 hold with no added hypothesis. Theorems 11.4.4 and 11.4.5 were first proved under the class hypothesis of Chapter 9's boundary measures and for $n \geq 2$; they now hold with the text's hypotheses for every $n \geq 1$. The measures are Chapter 9's: the canonical measures of Theorems 9.5.4 and 9.5.5 built from the concrete sheet integrals `Chapter9/GapP2_E1_BranchedIntegral · EpGroupoid.theorem_9_5_4` `Chapter9/GapP2_E1_BranchedIntegral · EpGroupoid.theorem_9_5_5`, and the boundary measures of Theorems 9.5.15 and 9.5.16 on sheets covered by countably many graph charts `Chapter9/GapP2_E7_BoundaryMeasure · EpGroupoid.theorem_9_5_15_msheets` `Chapter9/GapP2_E7_BoundaryMeasure · EpGroupoid.theorem_9_5_16_msheets`. "Oriented, tame" is read as in §XI.8, and the independence of the tangent sums holds for every $\Theta$.

**What the proof uses of the forms.** For an equivalence $\Phi$ the identity $\mu^{\Phi^*\hat{\Theta}'}_{\Phi^*\omega'} = |\Phi|^*\mu^{\hat{\Theta}'}_{\omega'}$ uses two facts about the pull-back of forms, recorded as a structure `Chapter11/GapP2_H4Compat · EpGroupoid.FormEval.IsPullback`: $(\Phi^*\omega')(\alpha_1, \dots, \alpha_n) = \omega'(T\Phi\,\alpha_1, \dots, T\Phi\,\alpha_n)$ on the tangent spaces of finite-dimensional submanifolds, and every form on $X$ is a pull-back (§XI.4). Both hold for $\Omega^n_{ep}$ `Chapter11/GapP2_H4CompatInst · EpFunctor.IsEquivalence.epFormEval_isPullback` and for the direct limit $\Omega^n_{ep,\infty}$ `Chapter11/GapP2_H4CompatInst · EP.epFormsInftyEval_isPullback`.

**Theorem 11.4.1 for an equivalence** (V11.27) `Chapter11/GapP2_H4Measure · EpFunctor.theorem_11_4_1_pull_equiv` `Chapter11/GapP2_H4Measure · EpFunctor.theorem_11_4_1_push_equiv`. The text takes $U(x)$, $U(F(x))$ on which $F$

is an sc-diffeomorphism, both with the properness property and the natural actions, and moves the local branching structure by $F$. *Step 1, between two M-polyfolds.* An sc-diffeomorphism $f : U \to U'$ between open subsets of two M-polyfolds carries graph charts to graph charts in the transported chart Chapter11/GapP2_H4CrossChart · MPolyfold.GraphChart.mapDiffeoCross, null sets to null sets Chapter11/GapP2_H4CrossChart · MPolyfold.IsHNull.image_of_isScDiffeoOn_cross, and tame induced structures to tame induced structures Chapter11/GapP2_H4CrossChart · MPolyfold.HasTameInducedStructure.image_of_isScDiffeoOn. *Step 2, moving structures.* An oriented tame structure is restricted to a smaller natural representation Chapter11/GapP2_H4Move · EpGroupoid.OrientedTameStructure.restrictRep and moved, $(f(M_i \cap W), f_* o_i)$, $(\sigma_i)$ Chapter11/GapP2_H4Move · EpGroupoid.OrientedTameStructure.move Chapter11/GapP2_H4Move · EpGroupoid.OrientedTameStructure.represents_move. *Step 3, (11.12) = (11.13).* When $\omega_T(Tf \circ \beta) = \omega_S(\beta)$ on the frames of a sheet Chapter11/GapP2_H4Move · EpGroupoid.FormEval.EvCompatOn, $\int_{f(A)} \omega_T|(f(N), f_* o) = \int_A \omega_S|(N, o)$ Chapter11/GapP2_H4Move · EpGroupoid.FormEval.integ_image_cross, and so the weighted sums of a structure and of its moved structure agree Chapter11/GapP2_H4Move · EpGroupoid.OrientedTameStructure.sheetSum_move; the same holds for sheets covered by several graph charts Chapter11/GapP2_H4MSheets · EpGroupoid.FormEval.image_mem_msheets_cross Chapter11/GapP2_H4MSheets · EpGroupoid.FormEval.msInteg_image_cross. *Step 4, the transport chart.* For $Q = \hat{\psi} \circ \Phi$ (Chapter 10), $Q$ preserves $X_\infty$ and $\Phi^*\Theta'$ Chapter11/GapP2_H4Pairs · EpFunctor.TransportChart.pullbackTheta_Q, moves the orientations Chapter11/GapP2_H4Pairs · EpFunctor.TransportChart.pullSection_Q and moves $\omega'$ to $\Phi^*\omega'$ Chapter11/GapP2_H4Pairs · EpFunctor.TransportChart.evCompatOn_Q. For small $U(x)$ there are natural representations on $U(x)$ and on $Q(U(x))$ Chapter11/GapP2_H4Pairs · EpFunctor.TransportChart.exists_rep_push. A structure at $y$ is restricted to $Q(U(x))$ and moved to $x$ by $Q^{-1}$ Chapter11/GapP2_H4Pairs · EpGroupoid.OrientedTameStructure.pullStructure Chapter11/GapP2_H4Pairs · EpGroupoid.OrientedTameStructure.represents_pullStructure, and one at $x$ is moved to $y$ by $Q$ Chapter11/GapP2_H4Pairs · EpGroupoid.OrientedTameStructure.exists_push_structure; this is the text's "$F^*\hat{\Theta}'$, $F_*\hat{\Theta}$ are oriented and tame" Chapter11/GapP2_H4Measure · EpFunctor.hcov_pull Chapter11/GapP2_H4Measure · EpFunctor.hcov_push. *Step 5, the isotropy factor.* $\sharp G_x = \sharp G_y$ by the isomorphism $g \mapsto \psi \circ \Phi(g) \circ \psi^{-1}$ Chapter10/EquivalenceLR · EpFunctor.TransportChart.bijective_conj. *Step 6, $\mathcal{L}$.* $|\Phi|$ and its inverse map sets of measure zero (Definition 9.5.1) to sets of measure zero Chapter11/GapP2_H4Measure · EpFunctor.isMeasureZeroSupp_image Chapter11/GapP2_H4Measure · EpFunctor.isMeasureZeroSupp_preimage, so $|\Phi| : |\mathrm{supp}\, \Phi^*\Theta'| \to |\mathrm{supp}\, \Theta'|$ is an isomorphism of the $\sigma$-algebras $\mathcal{L}$ Chapter11/GapP2_H4Measure · EpFunctor.measurable_suppMap Chapter11/GapP2_H4Measure · EpFunctor.measurable_suppInv. The text uses this without comment. *Step 7, gluing.* By Steps 3–5 the local formulas of Theorem 9.5.4 for $\mu^{\Phi^*\hat{\Theta}'}_{\Phi^*\omega'}$ and for $\mu^{\hat{\Theta}'}_{\omega'} \circ |\Phi|$ agree on the sets $|U(x)|$, and the measure with these local formulas is unique Chapter9/GapP2_E2MeasureGlue · existsUnique_linearMap_signedMeasure_of_openCover Chapter11/GapP2_H4Measure · EpFunctor.measure_pull_core. The push-forward identity is the same with $\Phi_*\omega$ any $\omega'$ with $\Phi^*\omega' = \omega$. The integrals follow with $K = |\mathrm{supp}|$ Chapter11/GapP2_H4Measure · EpFunctor.integral_pull_equiv Chapter11/GapP2_H4Measure · EpFunctor.integral_push_equiv.

**Theorem 11.4.1 (2017: 11.8) and Remark 11.4.3 (2017: 11.3)** (V11.27, V11.28) Chapter11/GapP2_H4Diagram · EPLocalization.EPLoc.theorem_11_4_1_pull Chapter11/GapP2_H4Diagram · EPLocalization.EPLoc.theorem_11_4_1_push. As in the text: for equivalences $F : A \to X$, $G : A \to Y$ and $\Theta_A = F^*\Theta_X = G^*\Theta_Y$, the equivalence case along $F$ and along $G$ gives $\mu^{\hat{\Theta}_X}_{\omega_X}(K) = \mu^{\hat{\Theta}_Y}_{\omega_Y}(|G|(|F|^{-1}(K)))$ when $F^*\omega_X = G^*\omega_Y$ Chapter11/GapP2_H4Diagram · EpFunctor.measure_span_core. With the forms of Definition 11.2.7 on the chosen representative this is the theorem for diagrams Chapter11/GapP2_H4Diagram · EPLocalization.Diagram.theorem_11_4_1_pull Chapter11/GapP2_H4Diagram · EPLocalization.Diagram.theorem_11_4_1_push and for generalized isomorphisms. Remark 11.4.3, in the form Chapter 11 uses it: two ep-groupoid models related by a generalized isomorphism give the same branched integrals Chapter11/GapP2_H4Diagram · EPLocalization.EPLoc.integral_pull Chapter11/GapP2_H4Diagram · EPLocalization.EPLoc.integral_push Chapter11/GapP2_H4Diagram · EPLocalization.

EPLoc.remark_11_4_3. The relation on pairs $(X, \beta)$ in Remark 11.4.3 is Chapter 10′s.

**Theorem 11.4.2 (new in 2021)** (V11.27) Chapter11/GapP2_H4EffDiagram · EPLocalization.EPLoc.theorem_11_4_2_pull Chapter11/GapP2_H4EffDiagram · EPLocalization.EPLoc.theorem_11_4_2_push. "The proof is similar to the previous theorem"; the one new step is $\sharp G_x^{eff} = \sharp G_y^{eff}$ along a transport chart Chapter11/GapP2_H4Eff · EpFunctor.TransportChart.card_effIsotropy_eq. The isomorphism of Step 5 carries $\ker \Phi_R = T_x$ onto $\ker \Phi_{R'} = T_y$ for natural representations on $U(x)$ and on $Q(U(x))$, by the equivariance of $Q$. Under the local unique continuation property every natural representation has the property used, and the property passes along equivalences (Proposition 10.1.16), so the covering hypotheses of Theorem 9.5.5 pass to $\Phi^*\widehat{\Theta}'$, $\Phi_*\widehat{\Theta}$ Chapter11/GapP2_H4Eff · EpFunctor.hcovL_pull Chapter11/GapP2_H4Eff · EpFunctor.hcovL_push. Then the proof of Theorem 11.4.1 Chapter11/GapP2_H4Eff · EpFunctor.measure_pull_core_eff Chapter11/GapP2_H4Eff · EpFunctor.theorem_11_4_2_pull_equiv Chapter11/GapP2_H4Eff · EpFunctor.theorem_11_4_2_push_equiv, and the diagram step, the middle $A$ having the property by Proposition 10.1.16 Chapter11/GapP2_H4EffDiagram · EpFunctor.measure_span_core_eff. The hypotheses are the text's.

**Theorems 11.4.4 and 11.4.5 (2017: 11.9)** (V11.29) Chapter11/GapP2_H4BdDiagram · EPLocalization.EPLoc.theorem_11_4_4_pull Chapter11/GapP2_H4BdDiagram · EPLocalization.EPLoc.theorem_11_4_4_push Chapter11/GapP2_H4BdDiagram · EPLocalization.EPLoc.theorem_11_4_5_pull Chapter11/GapP2_H4BdDiagram · EPLocalization.EPLoc.theorem_11_4_5_push. The text says that there are versions of Theorems 11.4.1 and 11.4.2 for the boundary case, relying on Theorems 9.5.15 and 9.5.16. *Step 1, boundary sheets move.* $\partial_1 f(M) = f(\partial_1 M)$, the reduced codimension being preserved Chapter11/GapP2_H4BdGeom · EpGroupoid.boundaryStratum_image_cross. The boundary orientation is moved by $Tf$ Chapter11/GapP2_H4BdGeom · EpGroupoid.OrientedTameStructure.boundarySheet_move Chapter11/GapP2_H4BdGeom · EpGroupoid.OrientedTameStructure.boundaryOrient_move_tangent, by the boundary operation of §XI.8, Step 1, and the boundary sums of Theorem 9.5.15 agree for a structure and its moved structure Chapter11/GapP2_H4BdGeom · EpGroupoid.OrientedTameStructure.boundarySheetSum_move, with the change of variables for the multi-chart integrals. *Step 2, the measures.* The boundary measures, the unique maps of Theorems 9.5.15 and 9.5.16 Chapter11/GapP2_H4Bd · EpGroupoid.boundaryMeasure Chapter11/GapP2_H4Bd · EpGroupoid.boundaryMeasureEff. $|\Phi|$ is measurable both ways on the boundary supports Chapter11/GapP2_H4Bd · EpFunctor.measurable_bsuppMap Chapter11/GapP2_H4Bd · EpFunctor.measurable_bsuppInv, and the gluing argument of Step 7 applies to any pair of measures with the local formulas of Theorem 9.5.15 Chapter11/GapP2_H4Bd · EpFunctor.measure_pull_core_bd Chapter11/GapP2_H4Bd · EpFunctor.theorem_11_4_4_pull_equiv Chapter11/GapP2_H4Bd · EpFunctor.theorem_11_4_5_push_equiv. *Step 3, diagrams* Chapter11/GapP2_H4BdDiagram · EpFunctor.measure_span_core_bd. *The added hypothesis.* The boundary measures of Chapter 9 carry the class hypothesis `hcov`: the nonempty boundary sheets of the structures lie in the class of integrated sheets, with finite $\int |\omega|$ over $\partial_1 M_i$. It is derived from the text's hypotheses since 2026-11-13 (§IX.36). It passes along equivalences and diagrams Chapter11/GapP2_H4Bd · EpFunctor.hcovB_pull Chapter11/GapP2_H4Bd · EpFunctor.hcovB_push Chapter11/GapP2_H4BdDiagram · EPLocalization.Diagram.hcovB_pullTheta Chapter11/GapP2_H4BdDiagram · EPLocalization.Diagram.hcovB_pushTheta. Its satisfiability witnesses: compact boundary sheets Chapter9/GapP2_E7_BoundaryMeasure · EpGroupoid.hcov_msheets_of_isCompact and, with nonempty boundary, the closed upper hemisphere in $[0,\infty) \times \mathbb{R}^2$ Chapter9/GapP2_E8W_Witness · E8W.e8w_hcov Chapter9/GapP2_E8_Stokes · EP.hcovB_of_hcovS. These first forms have $n = m + 1$ with $m > 0$.

**Theorems 11.4.4 and 11.4.5 with the text's hypotheses** (V11.29, restated; gap G.131, closed). $n \geq 2$ Chapter11/GapP3_K1_BoundaryLoc · EPLocalization.EPLoc.theorem_11_4_4_pull_loc Chapter11/GapP3_K1_BoundaryLoc · EPLocalization.EPLoc.theorem_11_4_4_push_loc Chapter11/GapP3_K1_BoundaryLoc · EPLocalization.EPLoc.theorem_11_4_5_pull_loc Chapter11/GapP3_K1_BoundaryLoc · EPLocalization.EPLoc.theorem_11_4_5_push_loc. For a generalized isomorphism $f$ and $\widehat{\Theta}'$ compact, branched, oriented and tame (`hloc`), $|f|(K)$ is measurable

and $\mu^{\partial f^*\hat\Theta'}_{f^*\omega'}(K) = \mu^{\partial\hat\Theta'}_{\omega'}(|f|(K))$; for $\hat\Theta$ with the same hypotheses $\mu^{\partial\hat\Theta}_{\omega}(K) = \mu^{\partial f_*\hat\Theta}_{f_*\omega}(|f|(K))$; the same for the effective measures when both ep-groupoids have the local unique continuation property. They are the versions above with `hloc` in place of `hcov`, the class condition being derived (§IX.36). The measures are those of Theorems 9.5.15 and 9.5.16 with the text's hypotheses, by definition Chapter11/GapP3_K1_BoundaryLoc · EP.boundaryMeasure_eq Chapter11/GapP3_K1_BoundaryLoc · EP.boundaryMeasureEff_eq; on the transported side the class conditions are the terms `hcovB_pull`, `hcovB_push` above. $n = 1$. The value of a 0-form moves with the point: $(F^*\omega)(p) = \omega(F(p))$ Chapter11/GapP3_K2_H4BdDim1 · EP.zeroFormEval_epFormsInftyPullback, and on a transport chart a boundary point moves to its image with the same frame sign Chapter11/GapP3_K2_H4BdDim1 · EpGroupoid.OrientedTameStructure.boundarySheetSum_move_dim1. The gluing of Step 2 is redone for the point sheets of §IX.38 Chapter11/GapP3_K2_H4BdDim1 · EpFunctor.measure_pull_core_bd_dim1 Chapter11/GapP3_K2_H4BdDim1 · EpFunctor.theorem_11_4_4_pull_equiv_dim1 Chapter11/GapP3_K2_H4BdDim1 · EpFunctor.theorem_11_4_4_push_equiv_dim1 Chapter11/GapP3_K2_H4BdDim1 · EpFunctor.theorem_11_4_5_pull_equiv_dim1 Chapter11/GapP3_K2_H4BdDim1 · EpFunctor.theorem_11_4_5_push_equiv_dim1, then for spans Chapter11/GapP3_K2_H4BdDiagramDim1 · EpFunctor.measure_span_core_bd_dim1 and for generalized isomorphisms Chapter11/GapP3_K2_H4BdDiagramDim1 · EPLocalization.EPLoc.theorem_11_4_4_pull_dim1 Chapter11/GapP3_K2_H4BdDiagramDim1 · EPLocalization.EPLoc.theorem_11_4_4_push_dim1 Chapter11/GapP3_K2_H4BdDiagramDim1 · EPLocalization.EPLoc.theorem_11_4_5_pull_dim1 Chapter11/GapP3_K2_H4BdDiagramDim1 · EPLocalization.EPLoc.theorem_11_4_5_push_dim1, with the measures of the $n = 1$ Theorems 9.5.15 and 9.5.16.

## XI.10 Proposition 11.5.1 (2017: 11.6): auxiliary norms

**The orbit map on $|W[1]|$.** $W_{0,1}$ with the morphisms of $W$ has orbit space $|W[1]|$; the map $|W[1]| \to |W|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneToOrbit is injective Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneToOrbit_injective. A strong bundle functor induces a continuous map on $|W[1]|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.continuous_oneMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneToOrbit_oneMap, and a diagram with both legs strong bundle equivalences the map $|D| = |\Psi| \circ |\Phi|^{-1}$ on $|W[1]|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneOrbitMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.continuous_oneOrbitMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneOrbitMap_oneMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneToOrbit_oneOrbitMap, determined by $|D|$ on $|W|$ because $|W[1]| \to |W|$ is injective Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneOrbitMap_eq_of_orbitMap_eq.

**Proposition 11.5.1** (V11.30) Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.proposition_11_5_1. A function on $W_{0,1}$ is invariant if it is constant along morphisms Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsOneInvariant Chapter11/StrongBundleUpToEquivalence · SBLocalization.isOneInvariant_iff Chapter11/StrongBundleUpToEquivalence · SBLocalization.isOneInvariant_const; auxiliary norms are, by condition (3) Chapter11/StrongBundleUpToEquivalence · EpStrongBundle.IsAuxNorm.isOneInvariant. *Step 1, the definition.* $[D]^*N'(e) = N'(e')$ for a representative $e'$ of $|D|(|e|)$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushNorm; for invariant $N'$ it does not depend on the representative Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq and equals the text's $N'(\Psi(e''))$ for every $e''$ with $\Phi(e'') \cong e$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq_Ψ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq_lift. *Step 2,*

*the properties.* It is invariant Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.isOneInvariant_pullNorm and continuous, as the map induced by $N'$ on $|W'[1]|$ composed with $|D|$ and the quotient map Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.continuous_pullNorm; the two are inverse Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushNorm_pullNorm Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_pushNorm and depend only on the orbit map Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq_of_orbitMap_eq Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushNorm_eq_of_orbitMap_eq. *Step 3, classes.* Hence well defined on generalized strong bundle isomorphisms Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pullNorm Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pushNorm. *Step 4, conditions (1) and (2) of Definition 8.3.5* (G.132) Chapter11/AuxiliaryNormTransport2021 · SBLocalization.GenSBMap.proposition_11_5_1_book, the statement as printed with no added hypothesis; the text calls it a consequence of the definitions. (1) On a fibre of $W_{0,1}$, $[D]^*N'$ is $N'$ composed with linear isomorphisms: the action $\mu(\sigma,\cdot)$ between fibres at level $1$ Chapter11/AuxiliaryNormTransport2021 · SBLocalization.oneFiberMor and a strong bundle equivalence on fibres Chapter11/AuxiliaryNormTransport2021 · SBLocalization.oneFiberObj. A complete norm stays complete along a linear isomorphism Chapter11/AuxiliaryNormTransport2021 · StrongBundle.IsCompleteFiberNorm.of_linearEquiv, so the norm is complete Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.isCompleteFiberNorm_pullNorm. (2) For $h_k$ with $P(h_k) \to x$ and $[D]^*N'(h_k) \to 0$: fix $\sigma : \phi(x'') \to x$, a local inverse $\tau$ of the target map at $\sigma$ and a local inverse of $\phi$ at $x''$, and lift $h_k$ to $e''_k$ over $g(s(\tau(P(h_k))))$. Condition (2) for $N'$ gives $\Psi(e''_k) \to 0$. A strong bundle equivalence is a local homeomorphism of $W[1]$ near $0$ (Definition 10.4.1 (2)) Chapter11/AuxiliaryNormTransport2021 · SBLocalization.tendsto_zeroOne_of_tendsto_obj, so $e''_k \to 0$, and $h_k = \mu(\tau(\cdot), \Phi(e''_k)) \to 0$ Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.tendsto_zeroOne_pullNorm. Hence $[D]^*N'$ is an auxiliary norm Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.isAuxNorm_pullNorm, and $[D]_*N$ is one as the pull-back along the reversed diagram Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.isAuxNorm_pushNorm. The sentence of p. 547 deriving $[D] = [D']$ from the local regularity property is Theorem 10.5.7, which does not hold as we read it (Chapter 10); Step 2 needs only $|[D]| = |[D']|$.

## XI.11 Lemma 11.5.2 and Proposition 11.5.3 (2017: 11.7 and 11.7): multi-section functors

**Functors on $W$** (V11.31) Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsWFunctor Chapter11/StrongBundleUpToEquivalence · SBLocalization.isWFunctor_const. A map $\Lambda$ on the total space constant along $\mu(\varphi, w)$ takes equal values over isomorphic objects Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsWFunctor.eq_of_π_eq. $\Psi^*\Lambda' = \Lambda' \circ \Psi$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.pullMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsWFunctor.pullMult. **Lemma 11.5.2 (2017: 11.7)** Chapter11/StrongBundleUpToEquivalence · SBLocalization.lemma_11_5_2: for a strong bundle equivalence, $\Phi_*\Lambda(w')$ is $\Lambda$ at a chosen $w$ with $\Phi(w) \cong w'$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.π_obj_choose, equal for every such $w$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.pushMult_eq, a functor; with the inverse laws Chapter11/StrongBundleUpToEquivalence · SBLocalization.pullMult_pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.pushMult_pullMult.

**Proposition 11.5.3 (2017: 11.7)** (V11.32) Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.proposition_11_5_3. On diagrams $[D]_* = \Psi_*\Phi^*$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult and $[D]^* = \Phi_*\Psi^*$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult;

$[D]_*\Lambda(w') = \Lambda(w)$ whenever $|D|(|w|) = |w'|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult_eq Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult_eq, so everything depends on $|D|$ only: functors go to functors Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.isWFunctor_pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.isWFunctor_pullMult, the two are inverse Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult_pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult_pullMult and agree for equal orbit maps Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult_eq_of_orbitMap_eq Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult_eq_of_orbitMap_eq; on classes Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pullMult. The text's "generalized strong bundle maps" in the last clause is read as generalized strong bundle isomorphisms, for which $[D]_*$ is defined.

**sc⁺-multisections: Lemma 11.5.2 and Proposition 11.5.3 as stated** (V11.31, V11.32) Chapter11/MultisectionLocalStructure2021 · SBLocalization.lemma_11_5_2_book Chapter11/MultisectionLocalStructure2021 · SBLocalization.GenSBMap.proposition_11_5_3_book, with no added hypothesis (G.133). *Definition 13.2.1.* An sc⁺-multisection is a functor $\Lambda : W \to \mathbb{Q}^+$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsScPlusMultisection with a local section structure around every $x_0$: on the domain $U(x_0)$ of a natural representation, sc⁺-sections $s_1, \dots, s_k$ with positive rational weights summing to 1 and $\Lambda(w) = \sum_{i:w=s_i(P(w))} \sigma_i$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.HasLocalSectionStructure. Such a representation on any open neighbourhood Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn gives a local section structure, by shrinking to a natural representation (Theorem 7.1.13). The weights and values are in $\mathbb{Q}_{\geq 0}$, which contains the value 0 off the sections. *Step 1, pull-back.* $\Lambda' \circ \Phi$ has the local sections $\Phi_U^{-1} \circ s_i \circ \phi$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.isScPlusMultisection_pullMult. *Step 2, push-forward, the text's construction.* Around $x'$, with $\sigma' : \phi(x) \to x'$, $s_i'(y') = \mu'(\sigma', \Phi(s_i(x)))$ with $\sigma'$ extended by the local inverse $\tau$ of $t'$ at $\sigma'$ and the local inverse $g$ of $\phi$. The sections are defined on $O' = \{y' : s(\tau(y')) \in \phi(U),\ g(s(\tau(y'))) \in O\}$ and take arbitrary values off $O'$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.exists_isLocalRepOn_pushMult. *Step 3, diagrams and classes.* $\Psi_*\Phi^*$ and $\Phi_*\Psi^*$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.SDiagram.isScPlusMultisection_pushMult Chapter11/MultisectionLocalStructure2021 · SBLocalization.SDiagram.isScPlusMultisection_pullMult, with the rest of Proposition 11.5.3 from the previous paragraph. *Witness*, the multisection of the zero section Chapter11/MultisectionLocalStructure2021 · SBLocalization.isScPlusMultisection_zeroMultisection.

## XI.12 Lemmas 11.6.1–11.6.3 (2017: Lemmas 11.8, 11.9): the fibres of a proper covering

**Lemma 11.6.1** (V11.33) Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lemma_11_6_1. For a proper covering functor $F : Y \to X$, $\alpha : \psi \mapsto (F(\psi), s(\psi))$ is an sc-diffeomorphism onto the fibred product (Definition 8.5.1), so every $(\varphi, y)$ with $F(y) = s(\varphi)$ has a unique lift Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lift Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.mor_lift Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.src_lift Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.eq_lift over $t(\varphi)$ Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.obj_tgt_lift; lifts with prescribed target are unique as well Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.existsUnique_lift_tgt. Surjectivity on morphisms comes from surjectivity on objects, and the local sc-diffeomorphism property from $\alpha$ and the source map.

**The action and Lemma 11.6.2 (2017: 11.8)** (V11.34) `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lemma_11_6_2`. The fibre $Y_x$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Fiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Fiber.val` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Fiber.mk` carries $g * y = t(\alpha^{-1}(g, y))$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.fiberSMul` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.fiberSMul_val_of`, an action ((11.15)) `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.instMulActionFiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.smul_val_of`: $\alpha^{-1}(g_2 g_1, y)$ and $\alpha^{-1}(g_2, g_1 * y) \circ \alpha^{-1}(g_1, y)$ have the same image under $\alpha$. In the text's display of $\alpha(\psi_2 \circ \psi_1)$ the second factor is printed $(F(\psi_2), s(\psi_1))$; we read $(F(\psi_1), s(\psi_1))$ (possible misprint). The full subcategory $Y(x)$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.FiberFull` and the translation groupoid `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Translation` are related by the functor that is the identity on objects and $\alpha$ on morphisms `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.toTranslation`, bijective on objects, full and faithful `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.bijective_toTranslation_obj` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.full_toTranslation` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.faithful_toTranslation`: the "natural identification".

**Lemma 11.6.3 (2017: 11.9)** (V11.35) `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lemma_11_6_3`. $B_\varphi(y) = t(\mathrm{lift}(\varphi, y))$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber_val_of`, with inverse $B_{\varphi^{-1}}$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber_inv_transportFiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.bijective_transportFiber`, equivariant along conjugation `Chapter7/Basic · EpGroupoid.conjIsotropy` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber_smul`, by uniqueness of lifts. $\tilde{\varphi}$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.translationMap` is bijective on objects, full and faithful.

## XI.13 Definitions 11.6.4 and 11.6.5 (2017: 11.6 and 11.7)

**Definition 11.6.4 (2017: 11.6)** (V11.36) `Chapter11/CoveringEquivalence · ProperCoveringHom` `Chapter11/CoveringEquivalence · ProperCoveringHom.ext`. $\Psi$ maps fibres into fibres `Chapter11/CoveringEquivalence · ProperCoveringHom.mapsTo_fiber`; identities and composites `Chapter11/CoveringEquivalence · ProperCoveringHom.id` `Chapter11/CoveringEquivalence · ProperCoveringHom.comp`. (2): the fibre condition `Chapter11/CoveringEquivalence · ProperCoveringHom.IsFiberBijective` and equivalences `Chapter11/CoveringEquivalence · ProperCoveringHom.IsEquivalence` `Chapter11/CoveringEquivalence · ProperCoveringHom.IsEquivalence.id` `Chapter11/CoveringEquivalence · ProperCoveringHom.IsEquivalence.comp`. *The fibre condition is redundant* `Chapter11/CoveringEquivalence · ProperCoveringHom.isFiberBijective_of_isEquivalence` `Chapter11/CoveringEquivalence · ProperCoveringHom.isEquivalence_iff`. *Injectivity.* If $\Psi(y_1) = \Psi(y_2)$ with $F(y_1) = F(y_2) = x$, fullness of $\Psi$ gives $\eta : y_1 \to y_2$ with $\Psi(\eta) = 1$; then $\Phi(F(\eta)) = F'(\Psi(\eta)) = 1 = \Phi(1_x)$ and faithfulness of $\Phi$ give $F(\eta) = 1_x$, so $\eta$ and $1_{y_1}$ lift the same $(1_x, y_1)$ and $y_2 = y_1$. *Surjectivity.* For $y' \in Y'_{\Phi(x)}$, essential surjectivity gives $y$ and $\eta' : \Psi(y) \to y'$; $F'(\eta') : \Phi(F(y)) \to \Phi(x)$ is $\Phi(\varphi)$ for some $\varphi : F(y) \to x$ by fullness; then $\Psi(\mathrm{lift}(\varphi, y))$ and $\eta'$ lift the same morphism with the same source, so $\Psi(B_\varphi(y)) = y'$. *Witness.* The functor from the pair groupoid of $\mathbb{Z}$ to the point, between the identity coverings `Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.pairToPointCovering`, is an equivalence `Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.isEquivalence_pairToPointCovering`.

**Definition 11.6.5 (2017: 11.7)** (V11.37) `Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans` `Chapter11/CoveringEquivalence · ProperCoveringHom.NaturallyEquivalent`. (11.20) objectwise; the square af-

ter the definition Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.naturality_F. Identity, inverse and composite Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.refl Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.symm Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.trans, whiskerings Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.whiskerLeft Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.whiskerRight; natural equivalence is an equivalence relation Chapter11/CoveringEquivalence · ProperCoveringHom.naturallyEquivalent_equivalence compatible with composition Chapter11/CoveringEquivalence · ProperCoveringHom.NaturallyEquivalent.comp_left Chapter11/CoveringEquivalence · ProperCoveringHom.NaturallyEquivalent.comp_right. Witness Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.naturallyEquivalent_pairToPointCovering.

## XI.14 Theorem 11.6.6 (2017: 11.10)

**The functor** $H$ (V11.38) Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hObj Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hMor Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hFunctor. On morphisms (11.19), with sources and targets checked through the commutation equations Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.mor_conj Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hMor_src Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hMor_tgt. Assume $\Psi$ and $\Phi$ are local sc-diffeomorphisms on objects and $\Psi$ maps each $Y_x$ onto $Y'_{\Phi(x)}$. *Step (A), objects.* Surjectivity: for $(x, \varphi', x'')$ lift $x''$ to $y''$, then $\varphi' : \Phi(x) \to \xi(x'')$ along $F'$ to $\psi'$ with target $\zeta(y'')$; the source of $\psi'$ lies over $\Phi(x)$, so it is $\Psi(y)$ for some $y \in Y_x$ by the hypothesis on fibres Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.surjective_hObj. The local sc-diffeomorphism property: the projections $P \to Y''$ and $Q \to X''$ are local sc-diffeomorphisms on objects because $\Psi$ and $\Phi$ are, $F''$ is one, and $\pi_{X''} \circ H = F'' \circ \pi_{Y''}$, so $H$ is one by two out of three Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isLocalScDiffeo_hObj. The text's heading of (A) has $P$ and $Q$ in the reverse order (possible misprint). *Step (B), $\alpha$.* The map $\psi \mapsto (H(\psi), s(\psi))$ for $H$ is an sc-diffeomorphism onto the fibred product, its inverse built from those of $F$ and $F''$ Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isScDiffeo_fibredMap_hFunctor. *Step (C), finite fibres and properness.* A fibre of $H$ sits in the product of fibres of $F$ and $F''$ Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.finite_fiber_hObj; a proper covering is proper on objects Chapter11/CoveringEquivalenceTheorem · EpFunctor.IsProperCovering.isProperMap, and so is $H$ Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isProperMap_hObj; a proper local sc-diffeomorphism is evenly covered near a point with finite fibre, by disjoint sheets around the points of the fibre and the closedness of proper maps Chapter11/CoveringEquivalenceTheorem · MPolyfold.IsLocalScDiffeo.evenlyCovered_of_isProperMap, the sheets cut down to the neighbourhood found Chapter11/CoveringEquivalenceTheorem · MPolyfold.IsScDiffeoOn.restrictTarget. The properness of $H$ is an ultrafilter argument from the properness of $F$ and $F''$ on objects and the continuity of the inverse of the map $\alpha$ of $F'$; it replaces the text's argument with sequences. **(1)** Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isProperCovering_hFunctor_of.

**Theorem 11.6.6** (V11.39) Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.theorem_11_6_6. An equivalence $(\Psi, \Phi)$ satisfies the hypotheses of (1), the fibre condition by §XI.13. *Step (D), (2).* $\pi_{Y''}$ and $\pi_{X''}$ are equivalences Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projRight, so the pair Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.projRightHom is an equivalence by §XI.13. *Step (E), (3).* $\tau_P(y, \psi', y'') = \psi'$ and $\tau_Q(x, \varphi', x'') = \varphi'$ satisfy (11.20) Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.natTransProj. *Step (F), tameness.* $P \to Y''$ and $Q \to X''$ are surjective local sc-diffeomorphisms on objects, and $Y'' \to X''$ is one; tameness pulls back along local sc-diffeomorphisms Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame. If $(\zeta, \xi)$ is an

equivalence, the other projection Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.projLeftHom is one by symmetry. Witness: the equivalence of §XI.13 and the identity of the point Chapter11/CoveringEquivalenceTheorem · CoveringEquivalence.Examples.theorem_11_6_6_pairToPoint, with nonempty fibres Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.nonempty_fiber_id.

## XI.15 The category $\mathcal{PC}$, Definition 11.6.7, Proposition 11.6.8 (2017: 11.8, 11.8) and Remark 11.6.9 (2017: 11.4)

$\mathcal{PC}$ (V11.40) Chapter11/CoveringEquivalenceDiagram · PC Chapter11/CoveringEquivalenceDiagram · ProperCoveringHom.comp_assoc Chapter11/CoveringEquivalenceDiagram · PC.equivalences. The weak fibred product as an object, with its projections Chapter11/CoveringEquivalenceDiagram · PC.fibred Chapter11/CoveringEquivalenceDiagram · PC.fibredFst Chapter11/CoveringEquivalenceDiagram · PC.fibredSnd. **Definition 11.6.7 (2017: 11.8)** Chapter11/CoveringEquivalenceDiagram · PC.Diagram Chapter11/CoveringEquivalenceDiagram · PC.Diagram.CommonRefinement Chapter11/CoveringEquivalenceDiagram · PC.Diagram.HasCommonRefinement. Witnesses: the pair and the point Chapter11/CoveringEquivalenceDiagram · PC.Examples.pair Chapter11/CoveringEquivalenceDiagram · PC.Examples.point Chapter11/CoveringEquivalenceDiagram · PC.Examples.equivalences_pairToPoint, the diagrams pair $\leftarrow$ pair $\rightarrow$ point Chapter11/CoveringEquivalenceDiagram · PC.Examples.diagram and pair $\leftarrow$ pair $\times_{\text{point}}$ pair $\rightarrow$ point Chapter11/CoveringEquivalenceDiagram · PC.Examples.fibredPair Chapter11/CoveringEquivalenceDiagram · PC.Examples.isEquivalence_fibredFst Chapter11/CoveringEquivalenceDiagram · PC.Examples.diagram'.

**Proposition 11.6.8 (2017: 11.8)** (V11.41) Chapter11/CoveringEquivalenceDiagram · PC.proposition_11_6_8. Given $(H, H')$ for $d, d'$ and $(K, K')$ for $d', d''$, take the weak fibred product $A'' \times_{A'} B''$ of $H'$ and $K$ (Theorem 11.6.6) with the equivalences $H \circ \pi_{A''}$ and $K' \circ \pi_{B''}$. For the $\Gamma$'s: $\Gamma \circ H \circ \pi_{A''} \simeq \Gamma' \circ H' \circ \pi_{A''} \simeq \Gamma' \circ K \circ \pi_{B''} \simeq \Gamma'' \circ K' \circ \pi_{B''}$, the middle step by Theorem 11.6.6 (3). For the $\varepsilon$'s the text's chain ends at $\varepsilon' \circ K \circ \pi_{B''}$; the last step to $\varepsilon'' \circ K' \circ \pi_{B''}$ uses $\varepsilon' \circ K \simeq \varepsilon'' \circ K'$ and is added. Reflexivity and symmetry are immediate Chapter11/CoveringEquivalenceDiagram · PC.Diagram.hasCommonRefinement_equivalence. Witnesses Chapter11/CoveringEquivalenceDiagram · PC.Examples.hasCommonRefinement_diagram Chapter11/CoveringEquivalenceDiagram · PC.Examples.hasCommonRefinement_diagram_diagram.

**The localization and Remark 11.6.9.** $\mathcal{PC}(E^{-1})$ and $I$, whose composition the text leaves to the reader, are §XI.21 (V11.43; G.134). Remark 11.6.9 (V11.42) recalls Definition 8.5.9 Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021 and sketches the category of proper strong bundle coverings and its localization, which is §XI.22 (G.135).

## XI.16 Witnesses and the hypotheses

**Hypotheses taken.** *§XI.8.* Chapter 9's independence of the tangent sums of Theorem 9.2.1 was assumed for the branched ep⁺-subgroupoid being moved and proved for its images. Chapter 9 now proves it for every $\Theta$ Chapter9/GeneralCorollaries · EpGroupoid.hasIndependentTangentSums, so it is no longer a restriction; it holds on $\mathbb{R}^n$ Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn. The boundary-transfer hypothesis of Theorem 11.3.9 for a representative holds along the identity, on every M-polyfold and on $\mathbb{R}^2$. Theorem 11.3.9 for the intrinsic boundary orientation needs no hypothesis. The boundary-lift form of Theorem 11.3.9 asks that $f$ carry tame tangent sets and that $f_*\widehat{\Theta}$ be tame; both hold along the identity and for Example 9.3.5. *§XI.9.* The first forms of Theorems 11.4.4 and 11.4.5 carry the class hypothesis `hcov` of Chapter 9; it is derived from the text's hypotheses (§IX.36), and the forms with `hloc` take none

beyond the text's. The local regularity property in Propositions 11.2.8, 11.2.9 and the local unique continuation property in Theorems 11.4.2, 11.4.5 are the text's. *Elsewhere.* The branched statements of §11.3 and the category $\mathcal{PC}$ take their ep-groupoids in one universe; the tangent functor on $\mathcal{EP}(E^{-1})$ lives where Chapter 8's $T$ does. In Remark 11.6.9 the class $E_{SBPC}$, which the text does not define, is read as "both components strong bundle equivalences" (§XI.22). Several statements need less than the text assumes: Lemma 11.3.2 (§XI.5), Proposition 11.1.6 (§XI.3), Theorem 11.6.6 (1) (§XI.14), and Proposition 11.2.8 with the property on one side (§XI.4).

**Every definition has a witness.** Faces and (weak) face structure: the quadrant with and without the swap (§XI.2). Push-forwards, pull-backs, branched ep⁺-subgroupoids along an equivalence that is not injective on objects and along a generalized isomorphism: $1_{\{0\}}$ on $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§XI.7). Section functors and oriented branched ep⁺-subgroupoids: Example 9.3.5 on $\mathbb{R}^2$ (§XI.8). Auxiliary norms and multisection functors: the constant functions (§§XI.10, XI.11). Functors, equivalences and natural transformations of proper coverings, Theorem 11.6.6, diagrams and common refinements: the pair groupoid of $\mathbb{Z}$ over the point (§§XI.13–XI.15). Two faces meeting in more than one face: $([0,\infty)^3 \oplus W) \setminus K$ (§XI.20). The local regularity property in Propositions 11.2.8 and 11.2.9: the point and $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§XI.4). The class hypothesis of Theorems 11.4.4 and 11.4.5: compact boundary sheets and the closed upper hemisphere (§XI.9). sc⁺-multisections: the zero section (§XI.11). $\mathcal{PC}(E^{-1})$: pair $\to$ point (§XI.21). Proper strong bundle coverings and their weak fibred products: $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ (§XI.22).

## XI.17 What is not formalized

Since 2026-11-13 all eight gaps of Chapter 11 are closed.

- Closed in wave K: §11.4 (G.131, §XI.9). Theorems 11.4.1, 11.4.2 and Remark 11.4.3 are proved with no added hypothesis, and Theorems 11.4.4 and 11.4.5 with the text's hypotheses for every $n \geq 1$.
- Closed: §11.1, Proposition 11.1.6 with the second claim of Proposition 11.1.8 (G.128, §§XI.3, XI.19, XI.20; the intersection clause does not hold as we read it, and holds in a union form); §11.2 (G.129, §XI.4); §11.3, Theorem 11.3.9 for the intrinsic boundary orientation (G.130, §XI.8); §11.5, conditions (1) and (2) in Proposition 11.5.1 (G.132, §XI.10) and the local section structures of sc⁺-multisections (G.133, §XI.11); §11.6, $\mathcal{PC}(E^{-1})$ and $I$ (G.134, §XI.21) and the category of Remark 11.6.9 with its localization (G.135, §XI.22).
- Within closed items, the boundary-lift form of Theorem 11.3.9 keeps two hypotheses that the intrinsic form does not need: that sc-diffeomorphisms carry tame induced structures on submanifolds to tame induced structures, and that $f_*\widehat{\Theta}$ is tame. The ordered face structures of Remark 11.1.5 have no formal content.

## XI.18 What the formalization taught us

Most of Chapter 11 is Chapter 10's machinery applied to one more kind of object, and the formal work went into the two places where an object is carried along an equivalence by a choice. For functors $\Theta$ the choice of $x$ and $F(x) \to y$ does not matter as soon as $F$ is full, which is all that the well-definedness in Lemma 11.3.2 uses; the statements for generalized maps then reduce, through one base change formula in the weak fibred product, to statements for equivalences, and the pull-back is defined for every generalized map. For branched ep⁺-subgroupoids the text's proof moves a local branching structure by $\widehat{\psi} \circ F$; written out, the sheets must first be cut down to the domain of that local sc-diffeomorphism, and

the paracompactness of the support, part of 2021 Definition 9.1.2, must be carried along too. Theorem 11.3.3 (6) refers to the uniqueness of dimensional decompositions, which does not hold as we read it (Chapter 9); the statement for every decomposition is what the proof gives. The rest of §11.3, whose proofs the text leaves to the reader, needed Chapter 9's notions between two M-polyfolds rather than on one, Chapter 9's independence hypothesis carried along, and, for Theorem 11.3.9, a condition on the boundary operations that Chapter 9 takes as data. Faces are a question about $|X|$ alone: once the faces of $|X|$ are Chapter 2's faces for $d_{|X|}$, Remark 11.1.5 and the bijections of Proposition 11.1.6 are statements about homeomorphisms preserving an index, and hold for every generalized isomorphism; only the ep-groupoid structure of a face needs more. In §11.6 the fibre condition of Definition 11.6.4 (2) comes for free for proper coverings, by unique lifting; and the chain of natural equivalences in Proposition 11.6.8 needs one more step than it displays.

The work of 2026-11-12 and 11-13 adds the following. Proposition 11.1.6 splits: its first and last claims hold, the faces being tame ep-groupoids because near each of its points $X_\theta$ is one face of the object M-polyfold, and the faces of a generalized isomorphism being related by the restricted diagram (11.1); but two faces can meet in several components, so the intersection is a union of faces of each, not one face. In §11.2 the forms behave best when $F_*$ is defined as the inverse of $F^*$: the bijectivity of $F^*$ is a descent along the projection $X \times_Y Y \to Y$, which is a surjective local sc-diffeomorphism, and the text's formula, composition rules, naturality and independence of the representative all follow by inverting statements about pull-backs. Theorem 11.3.9 needed no condition once the boundary orientation is taken intrinsically: Definition 9.3.8's operation is built from presented charts, which sc-diffeomorphisms transport, and "$d_M(x)$ is preserved" is read for the reduced codimension, which needs no tameness. §11.4 follows the text's proof, structures moved along transport charts and the sums (11.12), (11.13) compared, with two additions: $|f|$ preserves the $\sigma$-algebra $\mathcal{L}$, which the text uses silently, and in the effective case $\sharp G^{eff}$ is preserved because the transport isomorphism carries $T_x$ onto $T_y$. The boundary versions first inherited Chapter 9's class condition near corners; once Chapter 9 derived it, and covered $n = 1$, they hold with the text's hypotheses. In §11.5 conditions (1) and (2) need a local lift of a sequence through local inverses of $t$ and $\phi$, and the push-forward of local section structures needs the explicit domain on which the text's sections $s_i'$ are defined. In §11.6 the localizations of $\mathcal{PC}$ and of proper strong bundle coverings follow §10.3 once Theorem 11.6.6 is stated with the equivalence on the left; for strong bundles the weak fibred product is a pull-back, and conditions (2) and (3) of Definition 8.5.9 come for free over a proper covering.

## XI.19 Proposition 11.1.6 (2017: 11.2): faces as ep-groupoids, and Proposition 11.1.8 (2017: 11.3) for $[d]_F$

**Faces are tame ep-groupoids** (V11.6) `Chapter11/GapP2_HFaces · EpGroupoid.proposition_11_1_6_faces`. The text: for a face structured tame $X$, the object and morphism sets of a face $X_\theta$ are sub-M-polyfolds with tame induced structures, and $X_\theta$ is an ep-groupoid. The proof (pp. 524–525) passes through the faces of the object M-polyfold, and so does the formal one. *Step 1, local pieces.* $\pi$ maps the face of the object M-polyfold through $p$ into the face of $|X|$ through $|p|$ `Chapter11/GapP2_HFaces · EpGroupoid.image_faceAt_subset`. For the local pieces of $\{d_X = 1\}$ at $x$, the faces of $|X|$ through $|x|$ are those through the images of the pieces `Chapter11/GapP2_HFaces · EpGroupoid.orbitFacesThrough_eq_range`, and for face structured $X$ the map is injective `Chapter11/GapP2_HFaces · EpGroupoid.injective_orbitFaceAt_pt`. This is the text's count: if $x$ lay in fewer global faces of the object M-polyfold, fewer faces of $|X|$ would pass through $|x|$. So the object M-polyfold is weakly face structured `Chapter11/GapP2_HFaces · EpGroupoid.IsEpFaceStructured.isWeaklyFaceStructured`. *Step*

*2, one face near each point.* Every $x \in X_\theta$ lies in exactly one face of the object M-polyfold contained in $X_\theta$ `Chapter11/GapP2_HFaces · EpGroupoid.IsEpFaceStructured.existsUnique_face_subset`, and near $x$, $X_\theta$ is that face `Chapter11/GapP2_HFaces · EpGroupoid.IsEpFaceStructured.exists_nhds_eq_face`. The text's "disjoint union" is read in this local form, which is what is used. *Step 3, sub-M-polyfold.* A set that near each of its points agrees with a sub-M-polyfold is one `Chapter11/GapP2_HFaces · MPolyfold.isSubMPolyfold_of_forall_nhds`, and faces are sub-M-polyfolds with tame induced structure (Theorem 2.5.17); so $X_\theta$ is a closed `Chapter11/GapP2_HFaces · EpGroupoid.isClosed_faceObj` saturated sub-M-polyfold `Chapter11/GapP2_HFaces · EpGroupoid.isSubMPolyfold_faceObj`. *Step 4, the ep-groupoid.* With the morphisms between its points it is an ep-subgroupoid `Chapter11/GapP2_HFaces · EpGroupoid.isEpSubgroupoid_faceObj` (Proposition 7.1.22) `Chapter11/GapP2_HFaces · EpGroupoid.faceEpGroupoid`, tame on objects and on morphisms `Chapter11/GapP2_HFaces · EpGroupoid.isTame_faceEpGroupoid` `Chapter11/GapP2_HFaces · EpGroupoid.isTame_faceEpGroupoid_mor`, with $d_X(x) = d_{X_\theta}(x) + 1$ `Chapter11/GapP2_HFaces · EpGroupoid.degIndex_faceEpGroupoid`.

**The generalized isomorphisms $[d]_F$ of (11.1)** `Chapter11/GapP2_HFaceIso · EPLocalization.EPLoc.proposition_11_1_6_faceIso`. The text: for $X \xleftarrow{\Phi} A \xrightarrow{\Psi} Y$ and a face $F$ of $X$, $B = \Phi^{-1}(F)$ is a face of $A$, the saturation $C$ of $\Psi(B)$ is a face of $Y$, and $F \leftarrow B \rightarrow C$ is a generalized isomorphism by Theorem 10.3.10. *Step 1, restriction.* An sc-smooth functor restricts to ep-subgroupoids $A = G^{-1}(B)$ `Chapter11/GapP2_HFaceIso · EpFunctor.restrictSub`, and so do natural transformations `Chapter11/GapP2_HFaceIso · EpNatTrans.restrictSub`. For an equivalence and saturated $B$ the restriction is full, faithful, essentially surjective and a local sc-diffeomorphism, hence an equivalence by Lemma 10.1.2 `Chapter11/GapP2_HFaceIso · EpFunctor.IsEquivalence.restrictSub`; this is the text's argument. Face structure passes both ways along equivalences `Chapter11/GapP2_HFaceIso · EpFunctor.IsEquivalence.isEpFaceStructured_iff`. *Step 2, the diagram (11.1).* The face as an object of $\mathcal{EP}$ `Chapter11/GapP2_HFaceIso · EPLocalization.faceEP` and the restricted diagram `Chapter11/GapP2_HFaceIso · EPLocalization.Diagram.restrictFace`, with $C = Y_{|\Psi|(\theta_A)}$ (Remark 11.1.5). Refinements and equivalent diagrams restrict to refinements and equivalent diagrams `Chapter11/GapP2_HFaceIso · EPLocalization.Diagram.IsRefinement.restrictFace` `Chapter11/GapP2_HFaceIso · EPLocalization.Diagram.Equiv.restrictFace`, and $|d_F|$ is the restriction of $|d|$ `Chapter11/GapP2_HFaceIso · EPLocalization.Diagram.orbitMap_restrictFace`. *Step 3, the class.* $[d]_F$ `Chapter11/GapP2_HFaceIso · EPLocalization.EPLoc.faceIso` is the class of the restriction of every representative `Chapter11/GapP2_HFaceIso · EPLocalization.EPLoc.homMk_restrictFace_eq_faceIso`. It is a generalized isomorphism `Chapter11/GapP2_HFaceIso · EPLocalization.EPLoc.isGeneralizedIso_faceIso` whose orbit map is the restriction of $|[d]|$ `Chapter11/GapP2_HFaceIso · EPLocalization.EPLoc.orbitMap_faceIso`. The target face is passed as a set with a proof that it is $f_*\theta$, so that $[f]_F$ and $[g]_F$ have the same type when $f_* = g_*$.

**Proposition 11.1.8, the second claim** (V11.8) `Chapter11/GapP2_HFaceIso · EPLocalization.EPLoc.proposition_11_1_8_faceIso`. If $|f| = |g|$, then $f_* = g_*$ (§XI.3), and the generalized isomorphisms $F \to f_*F$ and $F \to g_*F$ induce the same map $|F| \to |f_*F|$, both being restrictions of $|f| = |g|$.

## XI.20 Proposition 11.1.6 (2017: 11.2): the intersection clause

**The clause as printed does not hold as we read it** (V11.6) `Chapter11/GapP2_H2FaceMeet · EpGroupoid.FacesMeetInFaces` `Chapter11/GapP2_H2FaceMeet · FaceMeetExample.not_facesMeetInFaces`. The text: if two faces $F$, $G$ meet, $F \cap G$ is a face in $F$ as well as in $G$. We read "face in $F$" in the ep-groupoid $X_\theta$ of §XI.19. *The example.* An open subset of a tame M-polyfold, with Part I's open-subset structure, has the same degeneracy index, is tame, and is face structured when the open set is `Chapter11/GapP2_H2FaceMeet · MPolyfold.isTame_restrictOpen` `Chapter11/GapP2_H2FaceMeet · MPolyfold.isFaceStructuredOpen_restrictOpen`; its trivial ep-groupoid

is then face structured Chapter11/GapP2_H2FaceMeet · EpGroupoid.isEpFaceStructured_trivial_restrictOpen. Take $V = ([0,\infty)^3 \oplus W) \setminus K$ with $K = \{p_0 + p_1 \le 1/4,\ 1 \le p_2 \le 2\}$, a closed block cutting the edge $p_0 = p_1 = 0$ at heights $[1,2]$ Chapter11/GapP2_H2FaceMeet · FaceMeetExample.V Chapter11/GapP2_H2FaceMeet · FaceMeetExample.isEpFaceStructured_XV. *Step 1.* $\theta_0 = \{p_0 = 0\}$ and $\theta_1 = \{p_1 = 0\}$ are distinct faces Chapter11/GapP2_H2FaceMeet · FaceMeetExample.isOrbitFace_θ₀ Chapter11/GapP2_H2FaceMeet · FaceMeetExample.θ₀_ne_θ₁, connected by paths in them Chapter11/GapP2_H2FaceMeet · FaceMeetExample.mem_face_of_path. *Step 2.* They meet in the two pieces $p_2 \in [0,1)$ and $p_2 \in (2,\infty)$ of the edge. A face of $X_{\theta_0}$ is the closure of one connected component of its index-one set, and the continuous function $p_2$ takes values below 1 or above 2 on the edge Chapter11/GapP2_H2FaceMeet · FaceMeetExample.cF_lt_or_gt, so the intersection is not one face Chapter11/GapP2_H2FaceMeet · FaceMeetExample.not_isEpFace_inter. With $W = 0$ this is a 3-manifold with boundary and corners.

**The union form** Chapter11/GapP2_H2FaceMeetUnion · EpGroupoid.IsEpFaceStructured.faceObj_inter_eq_sUnion Chapter11/GapP2_H2FaceMeetUnion · EpGroupoid.IsEpFaceStructured.facesMeetInUnions. For distinct faces $\theta \ne \theta'$ of $|X|$, $X_\theta \cap X_{\theta'}$ is the union of the faces of $X_\theta$ it contains, and likewise for $X_{\theta'}$ Chapter11/GapP2_H2FaceMeetUnion · EpGroupoid.FacesMeetInUnionsOfFaces. *Step 1, a ray in a tame chart.* A point $x$ of two distinct faces is a limit of points of $F_1 \cap F_2$ with $d_X = 2$: in a tame chart the faces are those of two distinct indices $i \ne j$ of Lemma 4.C, and the points $r(o + ty)$ with $\lambda_i(y) = \lambda_j(y) = 0$ and $\lambda_l(y) = 1$ otherwise have index 2 and lie in both closures Chapter11/GapP2_H2FaceMeetUnion · MPolyfold.mem_closure_inter_faces. *Step 2.* Near a point $y \in X_{\theta'}$ with $d_X(y) = 2$, every point with $d_X = 2$ lies in $X_{\theta'}$ Chapter11/GapP2_H2FaceMeetUnion · EpGroupoid.IsEpFaceStructured.exists_nhds_degIndex_two. *Step 3.* Those points lie in a local piece of $\{d_{X_\theta} = 1\}$ at $x$. The face of $X_\theta$ through that piece contains $x$, and it lies in $X_{\theta'}$ because $X_{\theta'} \cap \{d_{X_\theta} = 1\}$ is open and closed in $\{d_{X_\theta} = 1\}$.

## XI.21 The localization $\mathcal{PC}(E^{-1})$ and $I$ (2021 pp. 561–563)

$\mathcal{PC}(E^{-1})$ (V11.43) Chapter11/GapP2_HPCLocalization · PCLoc Chapter11/GapP2_HPCLocalization · PCLoc.I. The text sketches the localization of $\mathcal{PC}$ at $E_{PC}$ "as in the ep-groupoid case" and leaves the details to the reader. *Step 1, maps into weak fibred products.* For functors $L : P \to X$, $R : P \to Z$ and $\alpha : F \circ L \to G \circ R$, $p \mapsto (L(p), \alpha(p), R(p))$ is a functor into $X \times_Y Z$ with $\pi_X \circ \langle L, R\rangle = L$ and $\pi_Z \circ \langle L, R\rangle = R$ on the nose Chapter11/GapP2_HPCLocalization · WeakFibredProduct.pairFunctor. Its version for proper coverings, with the square (11.20) for $\alpha$ Chapter11/GapP2_HPCLocalization · PC.fibredLift, is an equivalence by two out of three Chapter11/GapP2_HPCLocalization · PC.IsEquivalence.of_comp_left Chapter11/GapP2_HPCLocalization · PC.isEquivalence_fibredLift, the projection to the side of the equivalence being one (Theorem 11.6.6 (2)). *Step 2, composition.* The composite of $d : F \leftarrow A \to F'$ and $d' : F' \leftarrow A' \to F''$ is taken through the weak fibred product $A' \times_{F'} A$ Chapter11/GapP2_HPCLocalization · PC.Diagram.comp. The text takes $A \times_{F'} A'$; Theorem 11.6.6 is stated here with the equivalence on the left, and the two products are isomorphic. Its class does not depend on the representatives Chapter11/GapP2_HPCLocalization · PC.Diagram.hasCommonRefinement_comp: a single equivalence $M$ with $\Gamma' \circ M \simeq \Gamma$ and $\varepsilon' \circ M \simeq \varepsilon$ gives a common refinement Chapter11/GapP2_HPCLocalization · PC.Diagram.hasCommonRefinement_of_map, and Step 1 gives such maps. Associativity and the units hold up to common refinement Chapter11/GapP2_HPCLocalization · PC.Diagram.hasCommonRefinement_assoc. *Step 3, $I$.* $I$ is the identity on objects with $I(\varepsilon) = [F \xleftarrow{1} F \xrightarrow{\varepsilon} F']$. Equivalences go to isomorphisms with inverse $[F' \xleftarrow{\Gamma} F \xrightarrow{1} F]$ Chapter11/GapP2_HPCLocalization · PCLoc.isIso_I_map Chapter11/GapP2_HPCLocalization · PCLoc.inv_I_map, $[d] = I(\Gamma)^{-1} \circ I(\varepsilon)$ Chapter11/GapP2_HPCLocalization · PCLoc.homMk_eq, and the inverse of $[F \xleftarrow{\Gamma} A \xrightarrow{\Gamma'} F']$ for an equivalence $\Gamma'$ is the reversed class Chapter11/GapP2_

HPCLocalization · PCLoc.isIso_homMk Chapter11/GapP2_HPCLocalization · PCLoc.inv_homMk (G.134). *Witness.* pair $\to$ point is not an isomorphism of $\mathcal{PC}$, and its image under $I$ is one Chapter11/GapP2_HPCLocalization · PCLoc.Examples.not_isIso_pairToPoint_isIso_I_map.

## XI.22 Remark 11.6.9 (2017: 11.4): proper strong bundle coverings up to equivalence

**The category $\mathcal{SBPC}$** (V11.42) Chapter11/GapP2_H2SBPC · SBPC Chapter11/GapP2_H2SBPC · SBPC.Hom Chapter11/GapP2_H2SBPC · instCategorySBPC. The remark views a strong bundle as a functor, recalls Definition 8.5.9 and describes the category of proper strong bundle coverings: its objects are pairs $(A, F)$ with $A$ a strong bundle functor covering a proper covering $F$, its morphisms commutative cubes, and its localization is left to the reader. A morphism is a pair of morphisms of $\mathcal{SEP}$ with $A' \circ a = b \circ A$, the top and bottom faces of the cube; the side faces commute because morphisms of $\mathcal{SEP}$ are strong bundle functors. There is a functor to $\mathcal{PC}$ Chapter11/GapP2_H2SBPC · SBPC.forget. *The class of equivalences.* The text does not define it. We take the morphisms whose two components are strong bundle equivalences Chapter11/GapP2_H2SBPC · SBPC.equivalences; the condition on the base face is then automatic Chapter11/GapP2_H3SBFibred · SBPC.equivalences_iff. Mathlib's localization gives $\mathcal{SBPC}[E^{-1}]$ with its universal property Chapter11/GapP2_H2SBPC · SBPC.Loc Chapter11/GapP2_H2SBPC · SBPC.isLocalization_Q Chapter11/GapP2_H2SBPC · SBPC.lift Chapter11/GapP2_H2SBPC · SBPC.uniq, with functors to $\mathcal{PC}(E^{-1})$ and to $\mathcal{SEP}(F^{-1})$ Chapter11/GapP2_H2SBPC · SBPC.toPCLoc Chapter11/GapP2_H2SBPC · SBPC.toSEPLocW Chapter11/GapP2_H2SBPC · SBPC.toSEPLocV. The identity covering of a strong bundle is an object Chapter11/GapP2_H2SBPC · SBPC.ofSEP.

**Theorem 11.6.6 for strong bundles** Chapter11/GapP2_H3SBFibred · SBPC.theorem_11_6_6. *Step 1, Definition 8.5.9 over a proper covering.* Condition (3) holds for every strong bundle and every proper covering: $(\psi, w) \mapsto ((F(\psi), s(\psi)), w)$ is a map of pull-backs along the sc-diffeomorphism of Lemma 11.6.1 Chapter11/GapP2_H3SBPullback · EpStrongBundle.isStrongBundleIso_fibredLift_of_isProperCovering. Condition (2) follows from the local strong bundle isomorphisms of (1) and the even covering Chapter11/GapP2_H3SBPullback · StrongBundle.evenlyCovered_of_isStrongBundleIsoOn. So over a proper covering Definition 8.5.9 is (0) and (1) Chapter11/GapP2_H3SBPullback · EpStrongBundle.isProperStrongBundleCovering2021_of. *Step 2, pull-backs.* The weak fibred product of strong bundles is a pull-back (Chapter 10). The pull-back $(p, w) \mapsto (H(p), A(w))$ of a proper strong bundle covering along an equivalence of proper coverings is one Chapter11/GapP2_H3SBPullback · SEP.pullbackCovering Chapter11/GapP2_H3SBPullback · SEP.isProperStrongBundleCovering2021_pullbackCovering. *Step 3, the theorem.* With natural transformations of morphisms of $\mathcal{SBPC}$, strong bundle natural transformations on both levels with (11.20) for the bases Chapter11/GapP2_H3SBFibred · SBPC.NatTrans Chapter11/GapP2_H3SBFibred · SBPC.NaturallyEquivalent: the weak fibred product Chapter11/GapP2_H3SBFibred · SBPC.fibred is a proper strong bundle covering covering $H$, its second projection is an equivalence, and $\Gamma \circ \mathrm{pr}_1 \simeq \Lambda \circ \mathrm{pr}_2$ Chapter11/GapP2_H3SBFibred · SBPC.fibredNatTrans.

**$\mathcal{SBPC}[E^{-1}]$ by diagrams** Chapter11/GapP2_H3SBLocalization · SBPCLoc. The procedure of §XI.21, with Step 3 in place of Theorem 11.6.6: maps into weak fibred products Chapter11/GapP2_H3SBLocalization · SEP.wfpLift Chapter11/GapP2_H3SBLocalization · SBPC.fibredLift, diagrams and common refinements Chapter11/GapP2_H3SBLocalization · SBPC.Diagram, Proposition 11.6.8 for them Chapter11/GapP2_H3SBLocalization · SBPC.proposition_11_6_8, composition Chapter11/GapP2_H3SBLocalization · SBPC.Diagram.comp Chapter11/GapP2_H3SBLocalization · SBPC.Diagram.hasCommonRefinement_comp, and $I$ with equivalences going to isomorphisms Chapter11/GapP2_H3SBLocalization · SBPCLoc.isIso_I_map Chapter11/GapP2_H3SBLocalization · SBPCLoc.homMk_eq Chapter11/GapP2_H3SBLocalization · SBPCLoc.inv_homMk. Mathlib's localization maps to it compatibly with

$I$ `Chapter11/GapP2_H3SBLocalization · SBPCLoc.fromLoc` (G.135). *Witness.* Theorem 11.6.6 for the identity covering of $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$, all conclusions, with a nonempty weak fibred product `Chapter11/GapP2_H3SBLocalization · SBPC.Examples.theorem_11_6_6_ep`.

# Part XII. Sc-Fredholm sections on ep-groupoids (HWZ Chapter 12, §§12.1–12.5)

Chapter 12 opens Part III of the book (Springer 2021, pp. 569–597); its numbers are the 2021 ones, with the 2017 text's in parentheses. It carries the Fredholm package of Part I (Chapters 5 and 6) from strong bundles over M-polyfolds to strong bundles $(P : W \to X, \mu)$ over ep-groupoids. §12.1 defines sc-Fredholm section functors (Definition 12.1.1 (2017: 12.1)). §12.2 builds auxiliary norms (Definition 12.2.1, Theorem 12.2.2, Definitions 12.2.3, 12.2.4, Proposition 12.2.5, Theorems 12.2.6, 12.2.7 (2017: Definition 12.2, Theorem 12.1, Definitions 12.3, 12.4, Proposition 12.1, Theorems 12.2, 12.3)) and their reflexive variant (Definitions 12.2.8, 12.2.10, Remarks 12.2.9, 12.2.12, Theorems 12.2.11, 12.2.13 (2017: Definitions 12.5, 12.6, Remarks 12.1, 12.2, Theorems 12.4, 12.5)). §12.3 supplies sc$^+$-section functors (Definition 12.3.1, Theorems 12.3.2, 12.3.3, Remarks 12.3.4, 12.3.5, Definition 12.3.6, Proposition 12.3.7, Corollary 12.3.8, Theorem 12.3.9 (2017: Definition 12.7, Theorems 12.6, 12.7, Remarks 12.3, 12.4, Definition 12.8, Proposition 12.2, Corollary 12.1, Theorem 12.8)). §12.4 treats compactness (Definition 12.4.1, Theorem 12.4.2, Definition 12.4.3, Lemma 12.4.4, Proposition 12.4.5, Theorem 12.4.6 (2017: Definition 12.9, Theorem 12.9, Definition 12.10, Lemma 12.1, Proposition 12.3, Theorem 12.10)), and §12.5 the orientation bundle $O_f$ (Lemma 12.5.1, Definition 12.5.2, Theorem 12.5.3, Definition 12.5.4, Remark 12.5.5, Proposition 12.5.6, Theorem 12.5.7 (2017: Lemma 12.2, Definition 12.11, Theorem 12.11, Definition 12.12, Remark 12.5, Proposition 12.4, Theorem 12.12)). This Part retells §12.1 (§XII.1), the auxiliary norms of §12.2 (§§XII.2–XII.5), §12.3 up to Corollary 12.3.8 (§§XII.6–XII.8), the first half of §12.4 (§XII.9) and §12.5 up to Proposition 12.5.6 (§§XII.10–XII.11); then the witnesses (§XII.12), what is not formalized (§XII.13) and what the formalization taught us (§XII.14).

**What is proved, and what is not.** Waves 1 and 2 of Part III (2026-11-13) formalize Chapter 12; what remains open is the topology of the orientation bundle under morphisms and equivalences in §12.5, and one local step of the reflexive extension theorem. *§12.1.* Definition 12.1.1 and the remark after it (regularizing on objects and on morphisms, sc-Fredholm germs) are formalized (§XII.1). *§12.2.* Definition 12.2.1, Theorem 12.2.2, Definitions 12.2.3, 12.2.4 with the remark between them, Proposition 12.2.5 and the good neighbourhood Theorem 12.2.6 are proved as printed (§§XII.2–XII.4). Theorem 12.2.7 is proved as printed: wave 1 proved its global part with the local step as a hypothesis (§XII.5), and wave 2 proves the local step, the extension of the norm near a boundary point by $N(rq(u), A(rq(u))w) + \|w - A(rq(u))w\|_{F_1}$, for every tame $X$ (§XII.15); gap G.136 is closed. The reflexive auxiliary norms: Definitions 12.2.8 and 12.2.10 are formalized, Remark 12.2.9 is proved in a stronger form (morphisms preserve mixed convergence), Theorem 12.2.11 is proved as printed with its

three clauses, and Remark 12.2.12 as printed, its clause (3) in a stronger form (§XII.16). Theorem 12.2.13 is proved with its local step taken as a hypothesis with a witness: the lower semicontinuity, under mixed convergence, of the local extension of §XII.15 (§XII.17; a new gap). *§12.3.* Definition 12.3.1 is formalized as printed and in a pointwise reading. Theorem 12.3.2 (1) is proved in each of the two consistent readings, and Theorem 12.3.2 (2) and Theorem 12.3.3 in the pointwise reading (§§XII.6–XII.7); the printed pairing of Definition 12.3.1 with Part I's pointwise bump functions would need Part I's open weak-to-pointwise question, which is recorded as a reading, not as formalization work. Remarks 12.3.4 and 12.3.5 are proved in explicit forms (§XII.7). Definition 12.3.6 is formalized in its local form, Proposition 12.3.7 in a stronger form and Corollary 12.3.8 explicitly (§XII.8); the ep-groupoid form of Definition 12.3.6 is formalized, and Theorem 12.3.9, the extension of sc$^+$-section functors from $\partial X$ with the bound $N(\widetilde{s}) \leq \delta$, is proved as printed (§XII.18); gap G.137 is closed. *§12.4.* Definition 12.4.1, Theorem 12.4.2 and Lemma 12.4.4 are proved (§XII.9); Definition 12.4.3 is formalized, and Proposition 12.4.5 and Theorem 12.4.6 are proved in stronger forms, several of the text's hypotheses being unused (§XII.19). *§12.5.* Lemma 12.5.1 and the orientation action $\varphi_*$ are proved with Part I's standing hypothesis that sc$^+$-sections pass through smooth fibre points (§XII.10). Definition 12.5.2 is $O_f$ of Part I, Definition 12.5.4 is formalized, Theorem 12.5.3 is proved under Part I's hypotheses except for its last sentence (gap G.138), and Proposition 12.5.6 is proved in the direction (□ ) and, in the direction (□ ), without the continuity of the orientation (gap G.139, which follows from G.138) (§XII.11). Remark 12.5.5 is prose. Theorem 12.5.7 is proved for one representative of the generalized strong bundle isomorphism, without its topology: a bijection $|O_f| \to |O_{f'}|$ over the bijection $|X_\infty| \to |X'_\infty|$, compatible with the diagram, and a unique invariant push-forward of an invariant orientation (§XII.20); that the bijection is a homeomorphism, that the push-forward is continuous, and that both are independent of the representative and functorial, is a new gap, which needs G.138. *Open:* G.138, G.139, G.140 (Part I's hypotheses carried by §12.5) and the two new gaps of §§XII.17 and XII.20. No statement of Chapter 12 was found not to hold as we read it.

**Four structural choices.** (1) *Part I on the object M-polyfold.* An sc-Fredholm section functor is an sc-smooth section functor that is a Part I sc-Fredholm section of $W \to X$; an auxiliary norm is a Part I auxiliary norm invariant under $\mu$ (Definition 8.3.5). So every Part I theorem applies to the objects, and Chapter 12's work is the passage through the morphisms. (2) *Norms are built from Part I norms.* The local norms that the text takes from a chart are Part I auxiliary norms (Proposition 5.1.3), and completeness of a glued fibre norm is not checked by hand: a new Part I lemma shows that a continuous fibrewise seminorm with condition (2) is an auxiliary norm (§XII.2). (3) *Bump functions in two readings.* Definition 12.3.1 asks for "not identically 0"; Part I's Definition 5.3.1 is read pointwise in this project (Part V). Both readings are kept for ep-groupoids (§XII.6). (4) *Orientations in chart points.* $\mathrm{Lin}(f, x)$ and $\mathrm{Or}_x^f$ are Part I's, read in a chosen chart point; a morphism $\varphi$ acts through a local strong bundle isomorphism built from the sheet of $\varphi$ (§XII.10).

**Map.**

- `Chapter12/P3S_CompactnessEp`: Definition 12.1.1 and its remark, Definition 12.4.1, Theorem 12.4.2, Lemma 12.4.4.
- `Chapter12/P3N_AuxNormEp`: the Part I lemma on automatic completeness, the lifted action on $W_{0,1}$, averaging, transport and gluing of norms, Definitions 12.2.3, 12.2.4, Proposition 12.2.5, Theorem 12.2.2.
- `Chapter12/P3N_GoodNeighborhood`: Theorem 12.2.6.
- `Chapter12/P3N_AuxNormExtension`: auxiliary norms over a subset, Theorem 12.2.7 with its witness.
- `Chapter12/P3S_BumpEp`: Definition 12.3.1 in two readings, Theorem 12.3.2.

- `Chapter12/P3S_ScPlusSectionFunctor`: the action of morphisms on sections, equivariant averages, the natural extension, Theorem 12.3.3, Remarks 12.3.4, 12.3.5.
- `Chapter12/P3B_LocalBoundaryExtension`: Definition 12.3.6 (local form), the inclusion–exclusion extension, Proposition 12.3.7, Corollary 12.3.8.
- `Chapter12/P3O_PullChart`, `Chapter12/P3O_MorphismIso`: chart points pulled back along a local strong bundle isomorphism; the isomorphism of a morphism.
- `Chapter12/P3O_OrientationBundleEp`: $T\varphi$, $\mu(\varphi,\cdot)$ in chart points, Lemma 12.5.1, $\varphi_*$ on orientations.
- `Chapter12/P3O_OrientationOrbit`: Definition 12.5.2, Theorem 12.5.3, Definition 12.5.4, Proposition 12.5.6.
- `Chapter12/P3E_AuxNormLocalExtension`: the local step of Theorem 12.2.7 near the boundary (§XII.15).
- `Chapter12/P3R2_ReflexiveAuxNormEp`: Definitions 12.2.8, 12.2.10, Remarks 12.2.9, 12.2.12, Theorem 12.2.11 (§XII.16).
- `Chapter12/P3R2_ReflexiveAuxNormExtension`: Theorem 12.2.13 with its local step as a hypothesis (§XII.17).
- `Chapter12/P3R2_ReflexiveCompactnessEp`: Definition 12.4.3, Proposition 12.4.5, Theorem 12.4.6 (§XII.19).
- `Chapter12/P3X_ScPlusExtension`: Definition 12.3.6 on ep-groupoids, Theorem 12.3.9 (§XII.18).
- `Chapter12/P3F_MixedPull`, `P3F_EquivalenceOr`, `P3F_Naturality`: orientation transport and Theorem 12.5.7 (§XII.20).

Throughout, $(P : W \to X, \mu)$ is a strong bundle over an ep-groupoid $X$, $B$ is the underlying Part I strong bundle on objects, $W_{0,1}$ is its level-$(0,1)$ part, $\pi : X \to |X|$ is the orbit map, $G_x$ the isotropy group, and a natural representation of $G_x$ on $U(x)$ is the one of Theorem 7.1.13 `Chapter7/NaturalRep · theorem_7_1_13`.

## XII.1 Definition 12.1.1 (2017: 12.1) and the remark after it

**Definition 12.1.1** `Chapter12/P3S_CompactnessEp · EpStrongBundle.definition_12_1_1` is the project's definition unfolded: an sc-smooth section functor is sc-Fredholm when it is an sc-Fredholm section of the object bundle. **The remark after it, p. 569.** *On objects:* $x \in X_m$ and $f(x) \in W_{m,m+1}$ give $x \in X_{m+1}$, which is Part I's regularizing property `Chapter12/P3S_CompactnessEp · EpStrongBundle.remark_12_1_1_regularizing_obj`. *On morphisms:* the text writes $f(\varphi)$ for a morphism; this is read as the section $\varphi \mapsto (\varphi, f(s(\varphi)))$ of the pull-back bundle $\mathbf{X}_s\times_P W \to \mathbf{X}$. If $\varphi \in \mathbf{X}_m$ and $f(\varphi)$ lies on level $(m, m+1)$, the projection to $W$, a strong bundle map, puts $f(s(\varphi))$ on level $(m, m+1)$, so $s(\varphi) \in X_{m+1}$, and $\varphi \in \mathbf{X}_{m+1}$ because levels of morphisms are those of their sources `Chapter12/P3S_CompactnessEp · EpStrongBundle.remark_12_1_1_regularizing_mor`. *Germs:* at a smooth $x$ the germ of $f$ is an sc-Fredholm germ `Chapter12/P3S_CompactnessEp · EpStrongBundle.remark_12_1_1_germ`.

## XII.2 Definition 12.2.1 (2017: 12.2) and Theorem 12.2.2 (2017: 12.1): auxiliary norms exist

**Definition 12.2.1** recalls Definition 8.3.5 and is not restated: an auxiliary norm is a Part I auxiliary norm $N : W_{0,1} \to [0,\infty)$ (continuous, a complete norm on each fibre, and $P(h_k) \to x$, $N(h_k) \to 0$ imply

$h_k \to 0_x$) with $N(\mu(\varphi, k)) = N(k)$ Chapter8/Book2021 · EpStrongBundle.IsAuxNorm.

**Completeness is automatic** (extra, Part I) Chapter12/P3N_AuxNormEp · StrongBundle.isAuxNorm_of_tendsto. The text says that the glued norms are complete on the fibres; the formal proof replaces this by a lemma about Part I's Definition 5.1.1. Let $N$ be continuous, a seminorm on each fibre Chapter12/P3N_AuxNormEp · StrongBundle.IsFiberSeminorm, with condition (2) Chapter12/P3N_AuxNormEp · StrongBundle.TendstoZeroOne. *Step 1, upper bound.* Near a point, in a strong bundle chart, continuity gives $N(w) \le c\,|w|_1$, where $|\cdot|_1$ is the chart norm of $\Phi[1]$. *Step 2, lower bound.* If no $c > 0$ had $|w|_1 \le c\,N(w)$ nearby, there would be $w_k$ over points converging to $x$ with $|w_k|_1 = 1$ and $N(w_k) \to 0$; condition (2) then gives $w_k \to 0_x$, against $|w_k|_1 = 1$ Chapter12/P3N_AuxNormEp · StrongBundle.exists_norm_liftOne_le_of_tendsto. *Step 3.* A seminorm squeezed between two multiples of a complete norm is a complete norm. A *local auxiliary norm* on $U$ asks for continuity, the seminorm property and condition (2) over $U$ only Chapter12/P3N_AuxNormEp · StrongBundle.IsLocalAuxNorm.

**Theorem 12.2.2** Chapter12/P3N_AuxNormEp · EpStrongBundle.theorem_12_2_2: every strong bundle over an ep-groupoid with paracompact orbit space admits an auxiliary norm; also for a bundled object of $\mathcal{SEP}$ Chapter12/P3N_AuxNormEp · SEP.theorem_12_2_2. The proof follows the text's three steps. *Step 0, the lifted action.* For a choice $\alpha$ of morphisms with $s(\alpha(y)) = y$, $w \mapsto \mu(\alpha(P(w)), w)$ is a map of $W_{0,1}$ Chapter12/P3N_AuxNormEp · EpStrongBundle.actOne, and along a natural representation it is a group action of $G_x$ Chapter12/P3N_AuxNormEp · EpStrongBundle.actOne_Γ_mul. *Step 1, average.* For a local norm $M$ on $U(x)$, $\bar{M}(w) = |G_x|^{-1} \sum_{g \in G_x} M(g{*}w)$ Chapter12/P3N_AuxNormEp · EpStrongBundle.avgNorm is $G_x$-invariant Chapter12/P3N_AuxNormEp · EpStrongBundle.avgNorm_actOne and again a local norm Chapter12/P3N_AuxNormEp · EpStrongBundle.isLocalAuxNorm_avgNorm. *Step 2, transport.* The saturation of $U(x)$ is the set of $y$ with a morphism into $U(x)$ Chapter12/P3N_AuxNormEp · EpGroupoid.satSet; choose one, $\varphi_y$ Chapter12/P3N_AuxNormEp · EpGroupoid.selMor, and put $T(w) = \bar{M}(\mu(\varphi_{P(w)}, w))$, and 0 off the saturation Chapter12/P3N_AuxNormEp · EpStrongBundle.transNorm. Two choices differ by an element of $G_x$ acting through the natural representation, because a morphism between points of $U(x)$ is $\Gamma(g, \cdot)$ for a unique $g$; so $T$ does not depend on the choice, by the invariance of $\bar{M}$ Chapter12/P3N_AuxNormEp · EpStrongBundle.transNorm_eq_of. The text uses Theorem 12.2.6 (5) at this point; the invariance makes it unnecessary. $T$ is a local norm on the saturation Chapter12/P3N_AuxNormEp · EpStrongBundle.isLocalAuxNorm_transNorm and invariant under every morphism Chapter12/P3N_AuxNormEp · EpStrongBundle.transNorm_μ. *Step 3, glue.* For a partition of unity $(\beta_i)$ of $|X|$ subordinate to the images $\pi$(saturation of $U(x_i)$), $N(w) = \sum_i \beta_i(\pi(P(w)))\,T_i(w)$ Chapter12/P3N_AuxNormEp · EpStrongBundle.glueNorm is continuous, a fibrewise seminorm, satisfies (2), and is invariant; so it is an auxiliary norm by the lemma above Chapter12/P3N_AuxNormEp · EpStrongBundle.isAuxNorm_glueNorm. The local norms are the restrictions of one Part I norm Chapter5/AuxNorm · proposition_5_1_3_exists, the natural representations come from Theorem 7.1.13 on all of $X$, and $|X|$ is metrizable, hence admits the partition of unity, by Theorem 7.3.1 Chapter7/OrbitTopology · theorem_7_3_1_metrizable.

## XII.3 Definitions 12.2.3, 12.2.4 (2017: 12.3, 12.4) and Proposition 12.2.5 (2017: 12.1)

**Definition 12.2.3** Chapter12/P3N_AuxNormEp · EpGroupoid.IsContPartitionOfUnity: continuous $\beta_\lambda : X \to [0,1]$, constant along morphisms, locally finite through saturated open neighbourhoods, with $\sum_\lambda \beta_\lambda = 1$. Supports are closures in $X$. The paracompactness of $|X|$ in the text's preamble is not a field, since the definition does not use it. **The remark after it.** A partition of unity of $|X|$ composed with $\pi$ is one of $X$ Chapter12/P3N_AuxNormEp · EpGroupoid.isContPartitionOfUnity_comp_π; conversely one of $X$ descends to $|X|$

Chapter12/P3N_AuxNormEp · EpGroupoid.IsContPartitionOfUnity.descend. In the second direction the functions descend by (1), and local finiteness on $|X|$ uses that $\pi$ is open (Proposition 7.1.8). **Definition 12.2.4** Chapter12/P3N_AuxNormEp · EpGroupoid.IsContPartitionOfUnity.IsSubordinate: supp $\beta_\lambda \subset U_\lambda$. **Proposition 12.2.5** Chapter12/P3N_AuxNormEp · EpGroupoid.proposition_12_2_5: for paracompact $|X|$, every cover of $X$ by saturated open sets has a subordinate continuous partition of unity. The proof is the text's: the sets $\pi(U_\lambda)$ are open and cover $|X|$, which is normal by Theorem 7.3.1; a partition of unity of $|X|$ subordinate to them, composed with $\pi$, is subordinate to $(U_\lambda)$ Chapter12/P3N_AuxNormEp · EpGroupoid.isSubordinate_comp_π.

## XII.4 Theorem 12.2.6 (2017: 12.2): good neighbourhoods

**Theorem 12.2.6** Chapter12/P3N_GoodNeighborhood · EpGroupoid.theorem_12_2_6. Every object $x$ has open neighbourhoods $U'(x) \subset U''(x)$ with (1) $\overline{U'} \subset U''$; (2) $U''$ is the domain of a natural representation of $G_x$, which leaves $U'$ invariant; (3) $U''$ lies in the base of one strong bundle chart, which restricted to $P^{-1}(U'')$ is the isomorphism with a local model; (4) a sequence of morphisms with sources in $\overline{U''}$ and convergent targets $t(\varphi_k) \to z$ has a subsequence converging to a $\varphi$ with $s(\varphi) \in \overline{U''}$, $t(\varphi) = z$; (5) $(\pi^{-1}(\pi(U')) \setminus U') \cap \overline{U''} = \emptyset$. *Step 1.* A neighbourhood $V_0$ of $x$ on which $t$ is proper on $s^{-1}(\overline{V_0})$, inside a chart domain. *Step 2.* A natural representation in $V_0$ (Theorem 7.1.13), shrunk twice Chapter7/NaturalRep · exists_naturalRep_shrink, to $U''$ and to $U'$ with $\overline{U'} \subset U''$; the image of an invariant set under the representation is computed once Chapter12/P3N_GoodNeighborhood · EpGroupoid.NaturalRep.image_Φ_eq. *Step 3, (4).* Properness of $t$ gives the subsequence Chapter12/P3N_GoodNeighborhood · EpGroupoid.exists_subseq_of_isProperMap. *Step 4, (5).* A point $y \in \overline{U''}$ with a morphism to a point of $U'$ is moved into $U'$ by a unique group element, by the uniqueness clause of Theorem 7.1.13, so $y \in U'$. No hypothesis is added.

## XII.5 Theorem 12.2.7 (2017: 12.3): extension of auxiliary norms from the boundary

**Auxiliary norms over a subset.** For $A \subset X$, Definition 12.2.1 over $A$ Chapter12/P3N_AuxNormExtension · EpStrongBundle.IsAuxNormOn: continuity on $W_{0,1}|A$, a complete norm on the fibres over $A$, condition (2) for sequences converging to points of $A$, and invariance where $P(k) \in A$. The text's $N$ is defined on $W_{0,1}|\partial X$; here it is a function on $W_{0,1}$ with conditions over $\partial X$ only, which is the same thing. A global norm restricts Chapter12/P3N_AuxNormExtension · EpStrongBundle.IsAuxNorm.isAuxNormOn.

**The added hypothesis** Chapter12/P3N_AuxNormExtension · EpStrongBundle.HasLocalExtensions: every $x \in \partial X$ has an open $V \ni x$ and a local auxiliary norm $M$ on $V$ equal to $N$ on $W_{0,1}|(V \cap \partial X)$. It is not derived from the text; it is the text's local step, the norm $N(rq(u), A(rq(u))w) + \|w - A(rq(u))w\|_{F_1}$ in a tame local model. Its satisfiability witness is `theorem_12_2_7_hyp_witness`: over every ep-groupoid with paracompact $|X|$ there is an $N$ with both properties, the restriction of a norm of Theorem 12.2.2 Chapter12/P3N_AuxNormExtension · EpStrongBundle.theorem_12_2_7_hyp_witness Chapter12/P3N_AuxNormExtension · EpStrongBundle.hasLocalExtensions_of_isAuxNorm.

**Theorem 12.2.7** (narrower) Chapter12/P3N_AuxNormExtension · EpStrongBundle.theorem_12_2_7_hyp: for paracompact $|X|$, an auxiliary norm $N$ over $\partial X$ with local extensions extends to an auxiliary norm $\tilde{N}$ of $W$ with $\tilde{N} = N$ over $\partial X$. *Step 1, data at each point.* At $x \in \partial X$: the local extension $M$, and a natural representation inside its $V$. At $x \notin \partial X$: a natural representation inside the open set $X \setminus \partial X$ Chapter12/P3N_AuxNormExtension · EpGroupoid.isOpen_compl_boundary carrying a Part I norm, restricted Chapter12/P3N_AuxNormExtension · StrongBundle.IsLocalAuxNorm.mono. *Step 2, average and transport,* as in §XII.2. Over $\partial X$ the

average of $M$ is still $N$, because $N$ is invariant and $\partial X$ is preserved by the morphisms Chapter12/P3N_AuxNormExtension · EpGroupoid.tgt_mem_boundary_iff. *Step 3, glue.* Over a boundary point every nonzero term of $\sum_i \beta_i T_i$ is $\beta_i N$: the saturation of an interior neighbourhood misses $\partial X$, and the terms from boundary points are $N$ there. So the sum is $N$ over $\partial X$. Tameness of $X$, which the text assumes, is not used in the global part; it would serve only to discharge the hypothesis.

## XII.6 Definition 12.3.1 (2017: 12.7) and Theorem 12.3.2 (2017: 12.6): bump functions

**Definition 12.3.1, as printed** Chapter12/P3S_BumpEp · EpGroupoid.AdmitsBumpFunctions: for every $x$ and saturated open $U \ni x$, an sc-smooth $f : X \to \mathbb{R}$, not identically 0, with supp $f \subset U$ and $f(s(\varphi)) = f(t(\varphi))$. **The pointwise reading** Chapter12/P3S_BumpEp · EpGroupoid.AdmitsBumpFunctionsPointwise asks $f(x) \neq 0$, as the project reads Part I's Definition 5.3.1; it implies the printed form Chapter12/P3S_BumpEp · EpGroupoid.AdmitsBumpFunctionsPointwise.weak.

**Theorem 12.3.2 (1)**, $X$ admits sc-smooth bump functions iff the object M-polyfold does, is proved in each consistent reading: pointwise on both sides Chapter12/P3S_BumpEp · EpGroupoid.theorem_12_3_2_1 and "not identically 0" on both sides Chapter12/P3S_BumpEp · EpGroupoid.theorem_12_3_2_1_weak. The printed form pairs Definition 12.3.1 with Part I's pointwise bump functions; its direction (□ ) would then be Part I's weak-to-pointwise question (Part V, §V.31), which is open. *(□ ).* Given an ep bump function $f$ near $x$, take an invariant $V(x)$ with the natural action inside $U$ and cut off: $\mathbf{1}_V f$ is sc-smooth on the objects Chapter12/P3S_BumpEp · EpGroupoid.exists_objBump_of_epBump. *(□ ).* Given a bump function $g$ of the object M-polyfold supported in an invariant $V$ with $\overline{V}$ in the domain of a natural representation, average over $G_x$ Chapter12/P3S_BumpEp · EpGroupoid.exists_natAverage and extend to the saturation by the morphisms Chapter12/P3S_BumpEp · EpGroupoid.exists_invariant_extension; the extension is sc-smooth because the saturation of the proper closed set $\overline{V}$ is closed Chapter7/ProperSubsets · isClosed_saturation_of_isProperSubset. The text averages $f$ itself, and that average can cancel to 0; the formal proof averages $g^2$, presumably what is intended, since the bump functions of Proposition 5.5.3 are nonnegative Chapter12/P3S_BumpEp · EpGroupoid.exists_epBump_of_objBump.

**Theorem 12.3.2 (2)** (narrower) Chapter12/P3S_BumpEp · EpGroupoid.theorem_12_3_2_2: from pointwise ep bump functions, for every $x$ and saturated open $U \ni x$ an invariant sc-smooth $f : X \to [0,1]$ with $f(x) = 1$ and supp $f \subset U$. The proof is the text's: a function $\equiv 1$ near $x$ from Proposition 5.5.3 Chapter5/BumpFunctions · proposition_5_5_3, averaged and extended; $f(x) = 1$ because every $\Phi(g)$ fixes $x$. The hypothesis is the pointwise reading of Definition 12.3.1. Witness: the trivial ep-groupoid on $\mathbb{R}^n$ Chapter12/P3S_BumpEp · EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn, and every ep-groupoid whose object M-polyfold admits bump functions Chapter12/P3S_BumpEp · EpGroupoid.admitsBumpFunctionsPointwise_of_objects.

## XII.7 Theorem 12.3.3 (2017: 12.7), Remarks 12.3.4 and 12.3.5 (2017: 12.3, 12.4)

**Morphisms acting on sections.** For a section $\sigma$ and a morphism $\varphi$, $\mu(\varphi, \sigma(s(\varphi)))$ Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.secAct, compatible with composition Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.secAct_comp. If $\tau$ is a local sc-smooth choice of morphisms, $y \mapsto \mu(\tau(y), \sigma(s(\tau(y))))$ is sc-smooth into $W[1]$ where $\tau$ is Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.

exists_isScSmoothOn_one_secAct, and sc⁺ is a local property Chapter12/P3S_ScPlusSectionFunctor · StrongBundle.isScPlusSection_of_locally.

**Remark 12.3.5** (explicit) Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.remark_12_3_5. Let $U(x)$ carry a natural representation, $\overline{V} \subset U(x)$ with $\overline{V}$ proper, and $\sigma$ an sc⁺-section supported in $\overline{V}$ and equivariant under the morphisms of $U(x)$. Then the natural extension Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.natExtension is an sc⁺-section functor Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.natExtension_isFunctorial, equal to $\sigma$ on $U(x)$, supported in the saturation of $\operatorname{supp}\sigma$, and every bound $N(\sigma) \le b$ for an auxiliary norm carries over Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.NormBound. *Proof.* At a point of the saturation, take a sheet $\tau$ of $t$ through a morphism into $U(x)$ and put $\mu(\tau(y), \sigma(s(\tau(y))))$; equivariance makes this independent of the sheet, and it is sc⁺ there. Off the saturation of $\operatorname{supp}\sigma$ it is $0$ Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.natExtension_eq_zero. The bound holds because $N$ is $\mu$-invariant Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.IsAuxNorm.secAct_eq. The text's $f \cdot s$ is $\sigma$, already cut off, and "controlled" is the bound $b$.

**Theorem 12.3.3** (narrower) Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.theorem_12_3_3. Hypotheses: pointwise ep bump functions; paracompact $|X|$; an auxiliary norm $N$; a smooth $x$; a smooth $e \in W_x$ with $\mu(g, e) = e$ for $g \in G_x$; an open $\hat{O} \ni |x|$; $\varepsilon > 0$. Conclusion: an sc⁺-section functor $\sigma$ with $\operatorname{supp}\sigma \subset \pi^{-1}(\hat{O})$, $\sigma(x) = e$ and $N(\sigma(y)) \le N(e) + \varepsilon$. *Step 1.* An invariant neighbourhood with the natural action, inside $\pi^{-1}(\hat{O})$ and with proper closure. *Step 2, a local section through $e$.* Part I's Lemma 5.3.3 Chapter5/BumpFunctions · lemma_5_3_3 gives an sc⁺-section through $e$, supported in that neighbourhood, with $N \le N(e) + \varepsilon$; it replaces the text's constant section in a chart times a cut-off $\beta$, so no separate cut-off is needed. *Step 3, average.* $\sigma'' = |G_x|^{-1} \sum_k h_k$ with $h_k(y) = \mu(\Gamma(k, \Phi(k^{-1})y), s(\Phi(k^{-1})y))$ Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.natTerm is equivariant, keeps the value $e$ at $x$ because $e$ is $G_x$-fixed, and keeps the bound Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.exists_equivariant_average. *Step 4.* Remark 12.3.5 extends it. Paracompactness is kept from the text and not used: the text needs it only for the existence of $N$, which is here a hypothesis.

**Remark 12.3.4** (explicit) Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.remark_12_3_4: for a functorial section $\sigma$ and $g \in G_x$, $\mu(g, \sigma(x)) = \sigma(x)$; so the invariance of $e$ in Theorem 12.3.3 cannot be dropped. The rest of the remark is commentary.

## XII.8 Definition 12.3.6 (2017: 12.8), Proposition 12.3.7 (2017: 12.2), Corollary 12.3.8 (2017: 12.1)

**Definition 12.3.6, local form.** In a tame local strong bundle model $K = R(U \lhd F) \to O$ with a tame presentation $T$, $\partial O$ is the set of $z \in O$ with $d_C(z) \neq 0$ Chapter12/P3B_LocalBoundaryExtension · retractBoundary, equivalently with nonempty index set $I(z)$ Chapter12/P3B_LocalBoundaryExtension · mem_retractBoundary_iff. A section over $D \subset \partial O$ is an sc⁺-section over the boundary when, for $x \in D$ and $i \in I(x)$, its restriction to the face germ $O \cap V \cap \{t_i = 0\}$ is sc⁺ for some open $V \ni x$ Chapter12/P3B_LocalBoundaryExtension · IsScPlusOnFaceNear Chapter12/P3B_LocalBoundaryExtension · IsBoundaryScPlusSectionOn. Witness of the quantifier over charts and presentations: the zero section Chapter12/P3B_LocalBoundaryExtension · isBoundaryScPlusSectionOn_zero. The M-polyfold form is Definition 5.4.1 (Part V, §V.54); the ep-groupoid wrapper is not stated.

**The extension by inclusion–exclusion.** For $z$ near $x_0$ and $\emptyset \neq I \subseteq \sigma = I(x_0)$, $z_I$ is $z$ with

the coordinates $t_i$, $i \in I$, set to 0 and retracted; then $\tilde{s}(z) = R(z, \sum_{\emptyset \neq I \subseteq \sigma} (-1)^{|I|+1} s(z_I))$ Chapter12/P3B_LocalBoundaryExtension · boundaryExtension, which is the text's sum Chapter12/P3B_LocalBoundaryExtension · boundaryExtension_toBase_snd and lies in $K[1]$ Chapter12/P3B_LocalBoundaryExtension · boundaryExtension_mem. Each term is sc-smooth through the corner retraction of the face Chapter12/P3B_LocalBoundaryExtension · exists_isScSmooth_faceTerm Chapter2/CornerGerms · corner_retraction. *The cancellation.* If $v(I \cup \{i_0\}) = v(I)$ for all $I$, then $\sum_{\emptyset \neq I \subseteq \sigma} (-1)^{|I|+1} v(I) = v(\emptyset)$, because $\sum_{I \subseteq \sigma} (-1)^{|I|} v(I) = 0$ by pairing $I$ with $I \cup \{i_0\}$ Chapter12/P3B_LocalBoundaryExtension · sum_powerset_erase_empty_alternating. On the face $t_{i_0} = 0$, killing $i_0$ changes nothing Chapter12/P3B_LocalBoundaryExtension · cornerKill_insert_of_coord_eq_zero, so $\tilde{s} = s$ there.

**Proposition 12.3.7** (stronger) Chapter12/P3B_LocalBoundaryExtension · proposition_12_3_7_of_faces_at: only the face germs at $x_0$ itself are needed. $V$ is the intersection of a neighbourhood where $I(z) \subseteq \sigma$, of the neighbourhoods where the $z_I$ stay in the domains, and of the face germs' domains. Sc-smoothness on $O \cap V$ goes through the restricted retraction; equality on $\partial O \cap V$ holds because at $z \in \partial O \cap V$ some $i_0 \in I(z) \subseteq \sigma$, and the cancellation applies. The text's form follows Chapter12/P3B_LocalBoundaryExtension · proposition_12_3_7, with an arbitrary tame presentation in place of $\mathbb{R}^n \oplus H$. Witness with $\partial O \neq \emptyset$: $C = [0, \infty) \oplus W$, $R = \mathrm{id}$, $x_0 = 0$ and the zero section Chapter12/P3B_LocalBoundaryExtension · proposition_12_3_7_nonvacuous.

**Corollary 12.3.8** (explicit) Chapter12/P3B_LocalBoundaryExtension · corollary_12_3_8. "$G_{x_0}$ acts on $O$ with a lift to $K$" is a structure: a finite group acting by sc-smooth maps of $O$, with an sc-smooth, fibre-linear lift to $K[1]$ covering the action, with unit and composition laws Chapter12/P3B_LocalBoundaryExtension · LocalBundleAction; witness, the trivial action Chapter12/P3B_LocalBoundaryExtension · LocalBundleAction.trivial. That the group fixes $x_0$, true of an isotropy group, is an explicit hypothesis; $\partial O \cap V_0$ is invariant and $s$ equivariant there. Then $\hat{s}(x) = |G|^{-1} \sum_g g^{-1} * \tilde{s}(g * x)$ Chapter12/P3B_LocalBoundaryExtension · LocalBundleAction.average, for the extension $\tilde{s}$ of Proposition 12.3.7 on an invariant neighbourhood, is sc$^+$ (sums of sc$^+$ terms stay in $K[1]$ Chapter12/P3B_LocalBoundaryExtension · mem_strongShiftOne_sum_smul), equivariant, and equal to $s$ on $\partial O \cap V$, where each term is $g^{-1} * s(g * x) = s(x)$. "Invariant" is read as equivariant.

## XII.9 Definition 12.4.1 (2017: 12.9), Theorem 12.4.2 (2017: 12.9), Lemma 12.4.4 (2017: 12.1)

**Lemma 12.4.4** Chapter12/P3S_CompactnessEp · EpGroupoid.lemma_12_4_4_1 Chapter12/P3S_CompactnessEp · EpGroupoid.lemma_12_4_4_2: $\mathrm{cl}_{|X|}\, \pi(\Sigma) = \pi(\mathrm{cl}_X\, \Sigma)$ for saturated $\Sigma$, and for $\Sigma$ with compact closure. For saturated $\Sigma$, $\pi$ open gives $\pi^{-1}(\mathrm{cl}\, \pi(\Sigma)) = \mathrm{cl}\, \pi^{-1}(\pi(\Sigma)) = \mathrm{cl}\, \Sigma$; for compact closure, $\pi(\mathrm{cl}\, \Sigma)$ is compact, hence closed. The image of a closed saturated set is closed Chapter12/P3S_CompactnessEp · EpGroupoid.isClosed_image_π_of_isSaturated.

**Definition 12.4.1** Chapter12/P3S_CompactnessEp · EpStrongBundle.HasCompactSolutionSetEp Chapter12/P3S_CompactnessEp · EpStrongBundle.IsProperFunctor Chapter12/P3S_CompactnessEp · EpStrongBundle.IsProperLtInftyFunctor: (1) $|S_f|$ compact; (2) proper: $|S_f|$ compact, and for every auxiliary norm $N$ a saturated open $U \supset S_f$ with $\mathrm{cl}_{|X|}\, |\{x \in U : N(f(x)) \leq 1\}|$ compact; (3) proper$_{<\infty}$: the same for every $0 < C < \infty$ with one $U$. $N(f(x)) \leq C$ is Part I's relation, and it passes along morphisms Chapter12/P3S_CompactnessEp · EpStrongBundle.auxLe_tgt_of_auxLe_src.

**Theorem 12.4.2** Chapter12/P3S_CompactnessEp · EpStrongBundle.theorem_12_4_2: for an sc-Fredholm section functor, (1), (2), (3) are equivalent. The text's proof labels its steps (3)□ (2), (2)□ (1), (1)□ (3); their arguments

are presumably those of (3)□ (1), (1)□ (2), (2)□ (3), and the formal proof follows the arguments. *(3)□ (1)* is $C = 1$ `Chapter12/P3S_CompactnessEp · EpStrongBundle.IsProperLtInftyFunctor.isProperFunctor`, and *(1)□ (2)* is the first clause `Chapter12/P3S_CompactnessEp · EpStrongBundle.IsProperFunctor.hasCompactSolutionSetEp`. *(2)□ (3)* `Chapter12/P3S_CompactnessEp · EpStrongBundle.isProperLtInftyFunctor_of_hasCompactSolutionSetEp`. *Step 1, local.* At each zero $x$, Part I's Theorem 5.2.2 `Chapter5/Book2021 · theorem_5_2_2` gives a neighbourhood $V(x)$ on which $\{N(f) \le C\}$ has compact closure, for each $C$ `Chapter12/P3S_CompactnessEp · exists_nhds_isCompact_closure_auxLe`. *Step 2.* The open sets $\pi(V(x))$ cover the compact $|S_f|$; take finitely many and $U = \pi^{-1}$ of their union. *Step 3.* $|\{x \in U : N(f(x)) \le C\}|$ lies in the finite union of the compact sets $\pi(\mathrm{cl}\{y \in V(x_i) : N(f(y)) \le C\})$, so its closure is compact. The text's shrinking to neighbourhoods with the natural action is not needed. Paracompactness of $|X|$ is kept from the text and not used, since (1)–(3) quantify over all auxiliary norms.

## XII.10 Lemma 12.5.1 (2017: 12.2): morphisms act on linearizations

**The isomorphism of a morphism.** For a morphism $\varphi$, let $\tau$ be the local inverse of $t$ on the sheet of $\varphi$. Then $y \mapsto s(\tau(y))$ on the base and $w \mapsto \mu(\tau(P(w))^{-1}, w)$ on the bundle form a local strong bundle isomorphism from a neighbourhood of $t(\varphi)$ to one of $s(\varphi)$ `Chapter12/P3O_MorphismIso · EpStrongBundle.morIso`, the inverse of the text's $\hat{\Phi}$; on the fibre over $t(\varphi)$ it is $\mu(\varphi^{-1}, \cdot)$ `Chapter12/P3O_MorphismIso · EpStrongBundle.morIso_Φ_eq`. A section functor is unchanged by it `Chapter12/P3O_MorphismIso · EpStrongBundle.eqOn_morIso`, and for composable morphisms the isomorphisms compose as germs `Chapter12/P3O_MorphismIso · EpStrongBundle.germAgree_morIso_comp`.

**Chart points pulled back.** Part I reads $\mathrm{Lin}(f, x)$ in a chart point. A local isomorphism pulls a chart point at $x$ back to one at $x'$ `Chapter12/P3O_PullChart · StrongBundle.LocalIso.pullPt`; if $f$ is unchanged by the isomorphism, the principal parts agree and so do the sets Lin `Chapter12/P3O_PullChart · StrongBundle.LocalIso.linSet_pullPt`. Pulling back commutes with Part I's identifications between chart points `Chapter12/P3O_PullChart · StrongBundle.LocalIso.transferT_pullPt`, also twice, for germs that agree `Chapter12/P3O_PullChart · StrongBundle.LocalIso.transferT_pullPt_pullPt` `Chapter12/P3O_PullChart · StrongBundle.LocalIso.GermAgree`.

$T\varphi$ **and** $\mu(\varphi, \cdot)$ `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.morTx` `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.morKx` are Part I's identifications from the chosen chart point at $x$ to a chart at $x$ inside the sheet, the identity to its pulled chart at $x'$, and from there to the chosen chart point at $x'$. They respect composition, units and inverses `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.morTx_comp` `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.morTx_unit` `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.morTx_inv`. Then $\varphi_* S = \mu(\varphi, \cdot) \circ S \circ (T\varphi)^{-1}$ `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.linMor`.

**Lemma 12.5.1** (explicit). The added hypothesis `hsec`, sc⁺-sections through every smooth fibre point, is not derived from the text; it is Part I's standing hypothesis for $\mathrm{Lin}(f, x) \neq \emptyset$, which §12.5 uses through Part I's Lin. Its witness: bump functions on the objects give it `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`. (1) $\varphi_* S \in \mathrm{Lin}(f, x')$ `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.lemma_12_5_1_one`: $S = (f - s_0)'(x)$ for an sc⁺-section $s_0$, and pushed by the isomorphism of $\varphi$ it is $(f - \hat{\Phi}_* s_0)'(x')$, by the agreement of Lin in a chart and its pulled chart. (2) $\varphi_*$ is a bijection $\mathrm{Lin}(f, x) \to \mathrm{Lin}(f, x')$ `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.lemma_12_5_1_two`, directly, as conjugation by isomorphisms; the text derives (2) from (1) and (3). (3) $(1_x)_* = \mathrm{Id}$ `Chapter12/P3O_OrientationBundleEp · EpStrongBundle.`

lemma_12_5_1_three_unit and $(\psi\varphi)_* = \psi_*\varphi_*$ Chapter12/P3O_OrientationBundleEp · EpStrongBundle.lemma_12_5_1_three_comp, from the germ identity of the composed isomorphisms. The text calls the lemma obvious. **On orientations** (p. 593) $\varphi_* o$ is conjugation of a line orientation by $T\varphi$ and $\mu(\varphi, \cdot)$ Chapter12/P3O_OrientationBundleEp · EpStrongBundle.orMor Chapter12/P3O_OrientationBundleEp · EpStrongBundle.orMor_apply, functorial Chapter12/P3O_OrientationBundleEp · EpStrongBundle.orMor_comp Chapter12/P3O_OrientationBundleEp · EpStrongBundle.orMor_unit Chapter12/P3O_OrientationBundleEp · EpStrongBundle.orMor_inv.

# XII.11 Definition 12.5.2 to Proposition 12.5.6 (2017: Definition 12.11 to Proposition 12.4): the orientation bundle

**Definition 12.5.2** (restated) Chapter12/P3O_OrientationOrbit · EpStrongBundle.definition_12_5_2: $O_f$ is Part I's orientation space of $f$ as a section of the object bundle, with projection $\sigma : O_f \to X_\infty$, surjective with exactly two points over each $x$ Chapter12/P3O_OrientationOrbit · orProj_two Chapter12/P3O_OrientationOrbit · exists_other_or (with hsec), and with Part I's topology. The morphisms act as a relation, an equivalence Chapter12/P3O_OrientationOrbit · EpStrongBundle.OrRel Chapter12/P3O_OrientationOrbit · EpStrongBundle.orRel_equivalence, with the projection $|\sigma| : |O_f| \to |X_\infty|$ Chapter12/P3O_OrientationOrbit · EpStrongBundle.orbitProj; $|X_\infty|$ is the quotient by the morphisms between smooth points Chapter12/P3O_OrientationOrbit · EpGroupoid.SmoothRel Chapter12/P3O_OrientationOrbit · EpGroupoid.smoothRel_equivalence. The text uses these orbit spaces in Proposition 12.5.6 without defining them; they carry the quotient topologies.

**Theorem 12.5.3** (narrower) Chapter12/P3O_OrientationOrbit · EpStrongBundle.theorem_12_5_3_hyp: there is exactly one topology on $O_f$ for which it is Hausdorff, $\sigma$ is a local homeomorphism, there are exactly two continuous sections near each point and they propagate; every fibre has two points. The proof is Part I's Theorem 6.6.17 Chapter6/OrientationMainClean · theorem_6_6_17 applied to the object bundle, with the two-point fibres. Its hypotheses are Part I's: $X$ tame, bump functions on the objects, and tame germ data for $f$. They are inherited from Theorem 6.6.17, not new, and the text assumes only a tame $X$. The last sentence, that the isomorphism of a morphism pushes continuous local sections of $\sigma$ to continuous local sections, is not proved (§XII.13); the fibre bijections and their functoriality are §XII.10.

**Definition 12.5.4** Chapter12/P3O_OrientationOrbit · EpStrongBundle.definition_12_5_4: an orientation is a section $x \mapsto o_x$ of $\sigma$, continuous for a given topology on $O_f$ (Part I's, by Theorem 12.5.3), with $\varphi_* o_{s(\varphi)} = o_{t(\varphi)}$ for every morphism between smooth points. **Remark 12.5.5** is prose pointing to Chapter 15, and is not formalized.

**Proposition 12.5.6.** *(□ )* Chapter12/P3O_OrientationOrbit · EpStrongBundle.proposition_12_5_6_mp: an orientation makes $|\sigma|$ two-to-one Chapter12/P3O_OrientationOrbit · EpStrongBundle.IsTwoToOne and gives the continuous section $|x| \mapsto |(x, o_x)|$ of $|\sigma|$, well defined by functoriality and continuous through the quotient. *(□ ), without continuity* (narrower) Chapter12/P3O_OrientationOrbit · EpStrongBundle.proposition_12_5_6_mpr_invariant: if $|\sigma|$ is two-to-one, every section of it lifts to a section $o$ of $\sigma$ with $\varphi_* o_x = o_{x'}$ for all morphisms and $|(x, o_x)| = s(|x|)$. Since $|\sigma|$ is two-to-one, exactly one of the two orientations over $x$ lies in the class $s(|x|)$; it is $o_x$, and $\varphi_* o_x$ lies in the same class, so it is $o_{x'}$. That $o$ is continuous when the section of $|\sigma|$ is continuous needs the last sentence of Theorem 12.5.3, and is a gap (§XII.13).

## XII.12 Witnesses and the hypotheses

**Hypotheses taken.** *§XII.5.* `HasLocalExtensions`, the local step of Theorem 12.2.7, not derived from the text; witness `theorem_12_2_7_hyp_witness`. Tameness of $X$ is dropped there. *§§XII.6–XII.7.* Theorem 12.3.2 (2) and Theorem 12.3.3 take pointwise ep bump functions, the project's reading of Part I's Definition 5.3.1; witness `admitsBumpFunctionsPointwise_trivial_rn`. *§§XII.10–XII.11.* `hsec`, sc$^+$-sections through smooth fibre points, not derived from the text, Part I's standing hypothesis for $\mathrm{Lin}(f,x)\neq\emptyset$; witness `MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`. Theorem 12.5.3 takes Part I's hypotheses of Theorem 6.6.17 (bump functions on the objects, tame germ data), inherited. *§XII.8.* Corollary 12.3.8 makes explicit that the group fixes $x_0$ and models "acts with a lift" as a structure. *Kept and unused.* Paracompactness of $|X|$ in Theorems 12.3.3 and 12.4.2. Several statements need less than the text assumes: Theorem 12.2.2 does not use Theorem 12.2.6 (5) (§XII.2), Proposition 12.3.7 uses the face germs at $x_0$ only (§XII.8).

**Every definition has a witness.** Auxiliary norms: Theorem 12.2.2 itself. Auxiliary norms over $\partial X$ with local extensions: restrictions of global ones (§XII.5). Ep bump functions, pointwise: the trivial ep-groupoid on $\mathbb{R}^n$ (§XII.6). sc$^+$-sections over the boundary, in every tame chart and presentation: the zero section, and with nonempty boundary $[0,\infty)\oplus W$ (§XII.8). Local bundle actions: the trivial action (§XII.8). `hsec`: bump functions on the objects (§XII.10).

## XII.13 What is not formalized

- *Closed in wave 2.* G.136, the local step of Theorem 12.2.7 (§XII.15); G.137, the ep-groupoid form of Definition 12.3.6 (§XII.18).
- Gap G.138: the last sentence of Theorem 12.5.3, that the isomorphism of a morphism keeps local orientation sections continuous (§XII.11). The pointwise data now exist also between different bundles (§XII.20); missing is the comparison of Part I's propagation along an sc-smooth path with the transport through a local strong bundle isomorphism along the image path, uniformly along the path; about 1–1.5k lines.
- Gap G.139: the continuity in Proposition 12.5.6 (□) (§XII.11); about 300–500 lines once G.138 is closed.
- Gap G.140: §12.5 with a tame $X$ only. Theorem 12.5.3 carries the hypotheses of Part I's Theorem 6.6.17 (bump functions on the objects, tame germ data), and Lemma 12.5.1, Proposition 12.5.6 and Theorem 12.5.7 the hypothesis that sc$^+$-sections pass through smooth fibre points. These are inherited from Part I; removing them needs Part I's orientation transport without bump functions and for untamed germ data. In the tame case, sc$^+$-sections through points would follow from Theorem 12.3.9, but only under the hypothesis that partitions of unity exist, which is no improvement.
- Gap G.147: the local step of Theorem 12.2.13, the lower semicontinuity under mixed convergence of the local extension of §XII.15 (§XII.17). The first term is lower semicontinuous once the transported points are shown to be mixed convergent in $W|\partial X$, the second once $w_k - A(rq(u_k))w_k \rightharpoonup w - A(rq(u))w$ in $F_1$; about 300–600 lines.
- Gap G.158: the topological clauses of Theorem 12.5.7 (§XII.20): $|\bar{\mathfrak{f}}|_*$ a homeomorphism, $\bar{\mathfrak{f}}_* o$ continuous, independence of the representative, functoriality. About 300 lines after G.138, and about 300 more for independence and functoriality.
- A reading, not formalization work: Theorem 12.3.2 (1) with Definition 12.3.1 as printed on one side and Part I's pointwise bump functions on the other; its (□) is Part I's open weak-to-pointwise

question (§XII.6).
- Remark 12.5.5 is prose.

## XII.14 What the formalization taught us

Chapter 12 is Part I seen through the morphisms, and most of its formal content is one pattern used again and again: build an object on a neighbourhood with the natural action of $G_x$, average it over $G_x$, transport it to the saturation by a choice of morphisms, and glue with a partition of unity of $|X|$. Norms (Theorems 12.2.2, 12.2.7), reflexive norms (Theorems 12.2.11, 12.2.13), bump functions (Theorem 12.3.2) and sc$^+$-sections (Theorems 12.3.3, 12.3.9, Remark 12.3.5) all follow it. What makes the transport well defined is invariance under $G_x$, not the good neighbourhood property (5) of Theorem 12.2.6 that the text invokes: two morphisms from $y$ into $U(x)$ differ by a unique element of $G_x$. Averaging has one trap. The average of a bump function over a finite group can vanish identically; the average of its square cannot, and the text's functions are nonnegative anyway. Completeness of glued fibre norms is not a separate check: a continuous fibrewise seminorm satisfying condition (2) of an auxiliary norm is squeezed, near every point, between two multiples of the chart norm, so completeness comes for free in Part I as well. For the reflexive variant one more fact carries the pattern: morphisms preserve mixed convergence, because they act boundedly on the fibres and continuously on level $0$, and a bounded, level-$0$ convergent sequence in a reflexive fibre is mixed convergent. With it, averaging, transport and gluing all keep lower semicontinuity, and the question whether lower semicontinuity survives the gluing (it does: near a point only finitely many terms matter, and the others are nonnegative) is settled.

Several places carried more, or less, than the text shows. The inclusion–exclusion formula of Proposition 12.3.7 restricts to $s$ on the boundary because an alternating sum over subsets collapses when one index is already on its face; written out, the face germs at $x_0$ are all that is used, so the proposition holds in a stronger form, and Part I's $\partial$-extension theorem is assembled from it (Part V, §V.55). The local step of Theorem 12.2.7 needs less than the text's tame local model: the retraction onto the boundary is built in a tame chart of the base, and the fibre formula can be evaluated in any strong bundle chart of the atlas, since only the continuity of the retraction and its two properties (values in $\partial X$, identity on $\partial X$) enter. Theorem 12.3.9 uses bump functions on the objects, which the text takes from Theorem 12.3.2 (2); they come from the assumed partitions of unity, so no hypothesis is added, and the second neighbourhood $V'$ of the text's bound is not needed. In §12.5 the text's "obvious" Lemma 12.5.1 took about 1.1k lines: a morphism acts on $\mathrm{Lin}(f,x)$ through a local strong bundle isomorphism built from its sheet, and Part I's linearizations, which live in chart points, have to be pulled back along it and compared; Theorem 12.5.7 repeats this between two different bundles. The orientation bundle itself is Part I's, and Part I's Theorem 6.6.17 supplies its topology; what remains open is the statement, used three times in §12.5, that compares Part I's propagation along paths with local strong bundle isomorphisms. Hypotheses of the text that turned out not to be used: paracompactness of $|X|$ in Theorems 12.3.3, 12.3.9 (where partitions of unity are assumed), 12.4.2 and 12.4.6; tameness in the global part of Theorem 12.2.7 and in Theorem 12.4.6; reflexivity, sc-smoothness and the regularizing property in Proposition 12.4.5. The step (3)$\square$ (1) of Proposition 12.4.5, "easily obtained" in the text, needs a lifting argument through the properness of $t$. The proof of Theorem 12.4.2 labels its three implications in an order that its arguments presumably permute.

## XII.15 Theorem 12.2.7 (2017: 12.3) as printed: the local step near the boundary

Wave 1 proved Theorem 12.2.7 with its local step as the hypothesis `HasLocalExtensions` (§XII.5). That hypothesis is now proved for every tame $X$, so the theorem holds as printed. The local step is a Part I statement about one strong bundle over a tame M-polyfold, the object bundle.

**The retraction onto the boundary** (extra, Part I) `Chapter12/P3E_AuxNormLocalExtension · MPolyfold.exists_retraction_boundary`. In a tame M-polyfold every boundary point $x_0$ has an open $V \ni x_0$ and a map $\rho$, continuous on $V$, with values in $\partial X$ and $\rho(y) = y$ for $y \in V \cap \partial X$. *The coordinates.* Let $C = [0,\infty)^n \oplus H$ be the partial quadrant of a tame chart $\psi$ at $x_0$, read through a presentation $T$. Put $\tau(t,h) = \min(t_1, \dots, t_n)$ `Chapter12/P3E_AuxNormLocalExtension · minCoord` and $q(t,h) = (t,h) - \tau(t,h)(1,\dots,1,0)$ `Chapter12/P3E_AuxNormLocalExtension · boundaryShift`. Then $q$ is continuous `Chapter12/P3E_AuxNormLocalExtension · continuous_boundaryShift` and takes values in $C$ `Chapter12/P3E_AuxNormLocalExtension · mem_quadrant_boundaryShift`, the coordinate where the minimum is attained becomes $0$, so the values lie in $\partial C$ `Chapter12/P3E_AuxNormLocalExtension · exists_coord_boundaryShift_eq_zero`, and on $\partial C$ the minimum is $0$, so $q$ is the identity there `Chapter12/P3E_AuxNormLocalExtension · boundaryShift_eq_self`. A point is on $\partial C$ iff some half-line coordinate vanishes `Chapter12/P3E_AuxNormLocalExtension · one_le_degIndexWith_iff`. *The retraction.* With $r$ the tame retraction of the chart, $\rho = \psi^{-1} \circ r \circ q \circ \psi$ on the open set where this is defined. Tameness of the chart gives $d_X = d_C$ on its domain, so $r$ maps $\partial C$ to boundary points and $\rho$ has values in $\partial X$; and $r$ fixes $O = r(U)$, so $\rho$ fixes $V \cap \partial X$. The text takes $x_0 = (0, h_0)$; this is not needed, since $q$ maps into $\partial C$ and fixes $\partial C$ whatever the coordinates of $x_0$.

**The norm $N'$** (extra, Part I) `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.retractNorm`. In a strong bundle chart $c$ of the atlas around $x_0$, $A(u)h$ is the fibre part of the retraction $R[1]$ `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.chartFibMap`, linear in $h$ `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.isLinearMap_chartFibMap` and the identity on the image $K[1]$ `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.chartFibMap_of_mem`. A vector $w$ over $y$ with chart coordinates $(u,h)$ is moved to the fibre over $\rho(y)$ by $h \mapsto A(\varphi(\rho(y)))h$, read back through $\Phi[1]^{-1}$ `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.retractTransport`, a linear fibre map. Then $N'(w) = N(\text{transported } w) + \|h - A(\varphi(\rho(y)))h\|_{F_1}$, the text's formula read in $\Phi[1]$ `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.retractNorm_toOne`.

**$N'$ is a local auxiliary norm** `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.isLocalAuxNorm_retractNorm`. The lemma is stated for any set $A$ of base points and any $N$ that is continuous on $W_{0,1}|A$, a seminorm on the fibres over $A$, and satisfies condition (2) for sequences in $A$, and any $\rho$ continuous on an open $V$ inside the chart domain with values in $A$. By the Part I lemma of §XII.2, a local auxiliary norm needs only three checks. *Continuity.* $(u,h) \mapsto A(u)h$ is continuous on $U \oplus F_1$, because it is level $0$ of the sc-smooth retraction $R : (U \lhd F)[1] \to (U \lhd F)[1]$ `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.continuousOn_shiftFiber_R`; so both terms are continuous on $W_{0,1}|V$. *Seminorm.* The transport is linear on each fibre, and the second term is a norm of a linear expression `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.isFiberSeminorm_retractNorm`. *Condition (2).* Let $P(w_k) \to y$ in $V$ and $N'(w_k) \to 0$. Both terms tend to $0$. The transported vectors lie over $\rho(P(w_k)) \to \rho(y) \in A$, so condition (2) for $N$, applied to a tail of the sequence, makes them converge to $0_{\rho(y)}$; hence $A(u_k)h_k \to 0$ in $F_1$, and with $\|h_k - A(u_k)h_k\| \to 0$ also $h_k \to 0$. Convergence in the chart gives $w_k \to 0_y$ `Chapter12/P3E_AuxNormLocalExtension · MChart.tendsto_of_tendsto_apply`. **$N' = N$ where $\rho$ is the identity** `Chapter12/P3E_AuxNormLocalExtension · StrongBundle.retractNorm_eq`: over $y \in V \cap A$ with $\rho(y) = y$, a vector of the fibre has $h \in K[1]$, so $Ah = h$, the transport is the identity and the second term vanishes.

**The local step** Chapter12/P3E_AuxNormLocalExtension · StrongBundle.exists_localExtension_of_isTame: for a tame M-polyfold and $N$ as above over $\partial X$, every $x_0 \in \partial X$ has an open $V \ni x_0$ and a local auxiliary norm $M$ on $V$ with $M = N$ over $V \cap \partial X$. Take $\rho$ from the retraction lemma on $V_1$, a chart $c$ at $x_0$, and $V = V_1 \cap \operatorname{dom} c \cap \rho^{-1}(\operatorname{dom} c)$, open by the continuity of $\rho$ and containing $x_0$ because $\rho(x_0) = x_0$; then $M = N'$. For an ep-groupoid this discharges the hypothesis of wave 1 Chapter12/P3E_AuxNormLocalExtension · EpStrongBundle.hasLocalExtensions_of_isTame.

**Theorem 12.2.7** Chapter12/P3E_AuxNormLocalExtension · EpStrongBundle.theorem_12_2_7: for a strong bundle over a tame ep-groupoid with paracompact $|X|$, every auxiliary norm $N$ over $\partial X$ extends to an auxiliary norm $\tilde{N}$ of $W$ with $\tilde{N} = N$ over $\partial X$. It is the theorem of §XII.5 with its hypothesis discharged; the hypotheses are the text's. Two readings. The text works in one local strong bundle model whose base retraction is tame; here the retraction onto the boundary lives in a tame chart of $X$ and the fibre formula in an arbitrary strong bundle chart, so no tame strong bundle chart is needed. The invariance of $N$ under morphisms is not used in the local step; it enters through the average over $G_{x_0}$ (§XII.5).

## XII.16 Definitions 12.2.8, 12.2.10, Remarks 12.2.9, 12.2.12 and Theorem 12.2.11 (2017: Definitions 12.5, 12.6, Remarks 12.1, 12.2, Theorem 12.4): reflexive auxiliary norms

**Reflexive 1-fibres and mixed convergence.** A strong bundle over an ep-groupoid has reflexive 1-fibres when the object bundle has them in the sense of Part I's Definition 5.1.4 (Part V, §V.49) Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.HasReflexiveFibers: the compatible charts whose $F_1$ is reflexive cover $X$. **Definition 12.2.8** Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.MixedTendsto Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.definition_12_2_8 is Part I's mixed convergence on the object bundle (§V.50): $P(w_k) \to P(w)$ in $X_0$ and, in a chart with reflexive $F_1$, the fibre parts converge weakly in $F_1$.

**Remark 12.2.9** (stronger) Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_9. For $w_k \to w$ mixed: (i) the fibre parts converge weakly in every compatible chart around $P(w)$ with reflexive $F_1$, which is the remark, from Part I's Lemma 5.1.5; (ii) $w_k \to w$ in $W$; (iii) for every sc-smooth local section $\alpha$ of the source map $s$ on an open $S \ni P(w)$, $\mu(\alpha(P(w_k)), w_k) \to \mu(\alpha(P(w)), w)$ mixed Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.mixedTendsto_actOne. So morphisms preserve mixed convergence; through a given morphism $\varphi$ with $s(\varphi) = P(w)$ such an $\alpha$ is the local inverse of $s$ on the sheet of $\varphi$, and then $\mu(\alpha(P(w_k)), w_k) \to \mu(\varphi, w)$ mixed Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.exists_mixedTendsto_μ. *Proof of (iii),* the remark's argument with the fibre maps of $\mu$ in place of a transition map. *Step 1, level 0.* $\mu$ is continuous, so the images converge on level 0. *Step 2, a bound.* For a Part I auxiliary norm $N_0$, near $P(w)$ one has $N_0(\mu(\alpha(\cdot), v)) \le C\, N_0(v)$ Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.exists_le_mul_auxNorm, and a mixed convergent sequence is eventually bounded in $N_0$ Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.MixedTendsto.exists_eventually_le; so the images are eventually bounded. *Step 3.* A bounded sequence that converges on level 0 is mixed convergent Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.mixedTendsto_of_bounded: by reflexivity every subsequence has a weakly convergent subsequence (Part V, §V.48), and its weak limit is identified by the level-0 limit, so the whole sequence converges weakly.

**Lower semicontinuity along mixed convergence** (extra) Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.MixedLscAt: $M$ is mixed lower semicontinuous at $y$ if for every $w_k \to y$ mixed and $\varepsilon > 0$, eventually

$M(y) - \varepsilon < M(w_k)$; this gives $M(y) \le \liminf M(w_k)$ Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.MixedLscAt.le_liminf. It is stable under sums Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.MixedLscAt.add Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.mixedLscAt_finset_sum, nonnegative constant multiples Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.MixedLscAt.const_mul and products with continuous positive functions of the base point Chapter12/P3R2_ReflexiveAuxNormEp · StrongBundle.MixedLscAt.mul_comp, and through morphisms Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_comp_actOne.

**Definition 12.2.10** Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.IsReflexiveAuxNorm: an auxiliary norm (Definition 12.2.1, with continuity, the functor property, complete fibre norms and condition (2)) with (3), $N(w) \le \liminf_k N(w_k)$ for $w_k \to w$ mixed, the liminf taken in the extended reals. The printed $\liminf_{k\to 0}$ is presumably $k \to \infty$, as in Part I (§V.50). A reflexive auxiliary norm is one of Part I's Definition 5.1.7 on the objects Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.IsReflexiveAuxNorm.toStrong, and (3) follows from mixed lower semicontinuity at every point Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.isReflexiveAuxNorm_of_mixedLscAt.

**Theorem 12.2.11** Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11, with exactly the text's hypotheses, paracompact $|X|$ and reflexive 1-fibres. *(1) Existence* Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11_exists. The proof is the construction of Theorem 12.2.2 (§XII.2), with lower semicontinuity carried through each step. The local norm on a neighbourhood with the natural action is a reflexive auxiliary norm of Part I's Proposition 5.1.8 (§V.51), where the text writes $\|\mathrm{pr}_2 \circ \Psi(g * h)\|_{F_1}$. *Average.* Each term $M(g * w)$ is mixed lower semicontinuous because $g*$ acts through morphisms, which preserve mixed convergence; so is the finite sum Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_avgNorm. *Transport.* Near a point of the saturation the chosen morphisms into $U(x)$ can be taken from a local sc-smooth section of $s$, and (iii) of Remark 12.2.9 applies Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_transNorm. *Glue.* Near $|P(w)|$ only finitely many $\beta_\lambda$ are nonzero; a term with $\beta_\lambda(|P(w)|) > 0$ is a continuous positive function times a lower semicontinuous one, and a term with $\beta_\lambda(|P(w)|) = 0$ contributes 0 at the limit and $\ge 0$ along the sequence Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_glueNorm. So the glued norm of §XII.2 is a reflexive auxiliary norm Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.isReflexiveAuxNorm_glueNorm. *(2)* $h \mapsto f(P(h))N(h)$ is reflexive for a continuous functorial $f > 0$ Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11_mul, which is Remark 12.2.12 (1). *(3) The sandwich* Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11_sandwich: for an auxiliary norm $N'$ and a reflexive $N$, Remark 12.2.12 (2) gives functorial $f, g > 0$ with $N \le (f \circ P)N'$ and $N' \le (g \circ P)N$; then $N_1 = (f \circ P)^{-1}N \le N' \le (g \circ P)N = N_2$, both reflexive by (2).

**Remark 12.2.12.** *(1)* Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_12_1 as printed: continuity, functor property and condition (2) pass to $(f \circ P)N$, and lower semicontinuity by the product rule above. *(2)* Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_12_2: for auxiliary norms $N_1, N_2$ and paracompact $|X|$, which the remark's context has, there is a continuous $f > 0$ constant along morphisms with $N_1 \le (f \circ P)N_2$. Near each point Part I's comparison of auxiliary norms (Lemma 5.1.2) gives a constant; invariance of both norms makes the bound hold on the saturation; a partition of unity of $|X|$ glues the constants into $f$. *(3)* (stronger) Chapter12/P3R2_ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_12_3: for $U$ open, $N \ge 0$ mixed lower semicontinuous over $U$ and $f \ge 0$ continuous, $\tilde{N} = f \cdot N$ on $U$ and 0 off $U$ satisfies $\tilde{N}(w) \le \liminf \tilde{N}(w_k)$ for $w_k \to w$ mixed. Off $U$, $\tilde{N}(w) = 0 \le \tilde{N}(w_k)$; on $U$ the sequence eventually lies over $U$. The saturation of $U$, the functoriality of $f$ and $\mathrm{supp}\, f \subset U$ serve the continuity and the functor property of $\tilde{N}$, not the estimate, and are not needed for it.

## XII.17 Theorem 12.2.13 (2017: 12.5): extension of reflexive auxiliary norms, with the local step as a hypothesis

**Over the boundary** `Chapter12/P3R2_ReflexiveAuxNormExtension · EpStrongBundle.IsReflexiveAuxNormOn`: an auxiliary norm over $A$ in the sense of §XII.5, with the liminf inequality for mixed convergent sequences over $A$; a global reflexive norm restricts `Chapter12/P3R2_ReflexiveAuxNormExtension · EpStrongBundle.IsReflexiveAuxNorm.isReflexiveAuxNormOn`.

**The added hypothesis** `Chapter12/P3R2_ReflexiveAuxNormExtension · EpStrongBundle.HasReflexiveLocalExtensions`: every $x \in \partial X$ has an open $V \ni x$ and a local auxiliary norm $M$ on $V$, mixed lower semicontinuous at every vector over $V$, with $M = N$ over $V \cap \partial X$. It is not derived from the text; it is the text's local step, the norm $N'$ of §XII.15, which the text calls well behaved under mixed limits "by inspection". It implies the hypothesis of Theorem 12.2.7 `Chapter12/P3R2_ReflexiveAuxNormExtension · EpStrongBundle.HasReflexiveLocalExtensions.hasLocalExtensions`. Its witnesses: the restriction of a global reflexive norm has reflexive local extensions `Chapter12/P3R2_ReflexiveAuxNormExtension · EpStrongBundle.hasReflexiveLocalExtensions_of_isReflexiveAuxNorm`, so every bundle with reflexive fibres and paracompact $|X|$ satisfies all hypotheses `Chapter12/P3R2_ReflexiveAuxNormExtension · EpStrongBundle.theorem_12_2_13_hyp_witness`; concretely the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid `Chapter12/P3R2_ReflexiveAuxNormExtension · TrivialBundle.epBundle_reflexive_witness`.

**Theorem 12.2.13** (narrower) `Chapter12/P3R2_ReflexiveAuxNormExtension · EpStrongBundle.theorem_12_2_13_hyp`: for paracompact $|X|$, reflexive 1-fibres, and a reflexive auxiliary norm $N$ over $\partial X$ with reflexive local extensions, there is a reflexive auxiliary norm $\tilde{N}$ with $\tilde{N} = N$ over $\partial X$. The proof is that of Theorem 12.2.7 (§XII.5) with the lower semicontinuity of §XII.16 carried along: at boundary points the local extensions of the hypothesis, at interior points the average of a reflexive norm of Part I's Proposition 5.1.8; average, transport and gluing keep mixed lower semicontinuity. Paracompactness of $|X|$, which the text uses through the partition of unity of Theorem 12.2.7, is a hypothesis; tameness enters only the local step and is dropped. **What is missing** (a new gap, §XII.13): that $N'$ of §XII.15 is mixed lower semicontinuous when $N$ is. For $w_k \to w$ mixed near $\partial X$, the transported vectors converge on level $0$ and are bounded, hence are mixed convergent in $W|\partial X$ by the argument of Remark 12.2.9, so the first term is lower semicontinuous; the second term is weakly lower semicontinuous once $h_k - A(u_k)h_k$ converges weakly in $F_1$.

## XII.18 Definition 12.3.6 (2017: 12.8) on ep-groupoids, and Theorem 12.3.9 (2017: 12.8): extension of sc⁺-section functors

**Definition 12.3.6, ep-groupoid form** `Chapter12/P3X_ScPlusExtension · EpStrongBundle.IsBoundaryScPlusSection`: a section $s$ over $\partial X$ is an sc⁺-section over the boundary when it is one for the object bundle in the sense of Part I's Definition 5.4.1 (Part V, §V.54): $P(s(x)) = x$, $s(x) \in W_{0,1}$, and in every tame strong bundle chart and tame presentation the local representative is sc⁺ on each face germ at the point. The local faces are read chart-wise, as in Definition 5.4.1, and the definition does not involve $\mu$. **An sc⁺-section functor of** $W|\partial X$, as Theorem 12.3.9 uses it `Chapter12/P3X_ScPlusExtension · EpStrongBundle.IsBoundaryScPlusSectionFunctor`, adds $s(t(\varphi)) = \mu(\varphi, s(s(\varphi)))$ for every morphism $\varphi$ with $s(\varphi) \in \partial X$; then $t(\varphi) \in \partial X$ as well, because morphisms preserve the degeneracy index `Chapter12/P3X_ScPlusExtension · EpGroupoid.tgt_mem_boundary_iff_p3x`. Witness: the zero section `Chapter12/P3X_ScPlusExtension · EpStrongBundle.isBoundaryScPlusSectionFunctor_fiberZero`.

**Bump functions from partitions of unity** (extra) Chapter12/P3X_ScPlusExtension · EpGroupoid.admitsBumpFunctionsPointwise_of_hasScPartitionsOfUnity: an ep-groupoid with sc-smooth partitions of unity (Definition 7.5.5) admits sc-smooth bump functions in the pointwise reading. For $x$ in a saturated open $U$, take an invariant $V \ni x$ with the natural action and proper closure inside $U$; the saturation $K$ of $\overline{V}$ is closed; the function attached to $U$ of a partition of unity for the cover $\{U, X \setminus K\}$ is 1 at $x$ and supported in $U$. The text uses this tacitly ("Using Theorem 12.3.2 (2)"); by Theorem 12.3.2 (1) the object M-polyfold then admits bump functions.

**Gluing sc⁺-sections** (extra). For a partition of unity $(\rho_i)$ and sections $\sigma_i$, $\sum_i \rho_i\sigma_i$ Chapter12/P3X_ScPlusExtension · glueSection is supported in $\bigcup_i \operatorname{supp}\rho_i \cap \operatorname{supp}\sigma_i$ Chapter12/P3X_ScPlusExtension · sectionSupport_glueSection_subset, is sc⁺ when the $\sigma_i$ are Chapter12/P3X_ScPlusExtension · isScPlusSection_glueSection (locally a finite sum), takes the value $v$ where every $\sigma_i$ with $\rho_i \neq 0$ does Chapter12/P3X_ScPlusExtension · glueSection_eq_of, and is an sc⁺-section functor when the $\sigma_i$ are and the $\rho_i$ are constant along morphisms Chapter12/P3X_ScPlusExtension · EpStrongBundle.isScPlusSectionFunctor_glueSection. An invariant function times a functorial section is functorial Chapter12/P3X_ScPlusExtension · EpStrongBundle.isFunctorialSection_sectionSmul_p3x.

**The local step** Chapter12/P3X_ScPlusExtension · EpStrongBundle.exists_local_functor_extension. For $x \in \partial X$ in a saturated open $U$ there are a saturated open $U' \ni x$ and an sc⁺-section functor $\sigma$ supported in $U$ with $\sigma = s$ on $U' \cap \partial X$. *Step 1.* An invariant $V \ni x$ with the natural action and proper closure inside $U$. *Step 2, Part I's local extension.* In a tame strong bundle chart at $x$, the local extension of Theorem 5.4.2 (Proposition 12.3.7 in the chart; §V.55) gives an sc⁺-section near $x$ equal to $s$ on the boundary Chapter5/P3B_BoundaryExtension2021 · StrongBundle.exists_local_extension; a bump function $\beta$, $\equiv 1$ near $x$ and supported in $V$ Chapter5/BumpFunctions · proposition_5_5_3, cuts it off. *Step 3, average.* The equivariant average over $G_x$ in the natural representation (§XII.7) Chapter12/P3X_ScPlusExtension · EpStrongBundle.exists_equivariant_average_p3x keeps the value $w$ at every point where all translates take the value $w$. At a boundary point $y$ near $x$ each translate is $\mu(\Gamma(k,\cdot), s(\cdot))$ applied at $\Phi(k^{-1})y$, which is $s(y)$ because $s$ is functorial on $\partial X$ and the morphisms preserve $\partial X$; so the average still extends $s$. This is Corollary 12.3.8 in its natural-representation form. *Step 4.* Remark 12.3.5 extends the average to the saturation Chapter12/P3S_ScPlusSectionFunctor · EpStrongBundle.remark_12_3_5.

**The extension** Chapter12/P3X_ScPlusExtension · EpStrongBundle.exists_extension_of_tameCharts. The sets $U'_x$ ($x \in \partial X$) and $X \setminus \partial X$, saturated and open Chapter12/P3X_ScPlusExtension · EpGroupoid.isSaturated_compl_boundary_p3x, cover $X$. Take an sc-smooth partition of unity $(\rho_i)$ subordinate to it, constant along morphisms (Definition 7.5.5), the local extensions $\sigma_x$ on the pieces $U'_x$, and 0 on the interior piece, and glue. On $\partial X$ the interior piece has $\rho = 0$ and every other $\sigma_x$ with $\rho \neq 0$ equals $s$, so $\tilde{s} = s$ there; the support lies in $U$; and $\tilde{s}$ is an sc⁺-section functor. **The bound** Chapter12/P3X_ScPlusExtension · EpStrongBundle.exists_extension_bounded_of_tameCharts. Given an auxiliary norm $N$ and $N(s(x)) < \delta$ on $\partial X$, $V = \{N(\tilde{s}) < \delta\}$ is open, saturated by the invariance of $N$, and contains $\partial X$. A partition of unity for the two-set cover $\{V, X \setminus \partial X\}$ gives $\beta$ with $\operatorname{supp}\beta \subset V$ and $\beta = 1$ on $\partial X$ Chapter12/P3X_ScPlusExtension · EpGroupoid.partitionOfUnity_true_eq_one_p3x, and $\beta\tilde{s}$ has $N \leq \delta$ everywhere. The text's second neighbourhood $V'$ is not needed.

**Theorem 12.3.9** Chapter12/P3X_ScPlusExtension · EpStrongBundle.theorem_12_3_9: for a strong bundle over a tame ep-groupoid (the object bundle tame, as in Theorem 5.4.2), $|X|$ paracompact, admitting sc-smooth partitions of unity, an sc⁺-section functor $s$ of $W|\partial X$ and a saturated open $U \supset \partial X$, there is an sc⁺-section functor $\tilde{s}$ with $\operatorname{supp}\tilde{s} \subset U$ and $\tilde{s} = s$ on $\partial X$; and given an auxiliary norm $N$ with $N(s(x)) < \delta$ on $\partial X$, also $N(\tilde{s}(x)) \leq \delta$ for all $x$. Bump functions on the objects come from the partitions of unity,

as above, with Theorem 12.3.2 (1) `Chapter12/P3S_BumpEp · EpGroupoid.theorem_12_3_2_1`; paracompactness is the text's hypothesis, kept and unused. No hypothesis is added. Witness for all hypotheses together `Chapter12/P3X_ScPlusExtension · EpStrongBundle.theorem_12_3_9_hypotheses_rn`: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid, the zero section, and $U = X$.

## XII.19 Definition 12.4.3, Proposition 12.4.5 and Theorem 12.4.6 (2017: Definition 12.10, Proposition 12.3, Theorem 12.10)

**Definition 12.4.3** `Chapter12/P3R2_ReflexiveCompactnessEp · EpStrongBundle.HasReflexiveLocalCompactnessEp`: a section functor $f$ has the ep-groupoid reflexive local compactness property if for every reflexive auxiliary norm $N$ and every object $x$ there is an open $U(x) \ni x$ with $\mathrm{cl}_X\{y \in U(x) : N(f(y)) \le 1\}$ compact. The text's standing assumptions (reflexive $(0,1)$-fibres, $f$ sc-smooth and regularizing) are hypotheses of the statements that use it, and $N(f(y)) \le 1$ is Part I's relation ($N = \infty$ off $W_{0,1}$). Part I's Definition 5.2.6 on the object bundle implies it `Chapter12/P3R2_ReflexiveCompactnessEp · EpStrongBundle.hasReflexiveLocalCompactnessEp_of_strong`.

**Two tools.** The set $\{y \in \tilde{U} : N(f(y)) \le 1\}$ is saturated when $\tilde{U}$ is, because $N$ is invariant and $f$ a functor `Chapter12/P3R2_ReflexiveCompactnessEp · EpStrongBundle.isSaturated_setOf_auxLe`. *Compact closures from orbit sets* `Chapter12/P3R2_ReflexiveCompactnessEp · EpGroupoid.isCompact_closure_of_isProperMap`: if $t : s^{-1}(\overline{V_0}) \to X$ is proper and $A \subset V_0$ has $\mathrm{cl}_{|X|}\,|A|$ compact, then $\mathrm{cl}_X A$ is compact. Given a sequence in $\mathrm{cl}\,A$, its orbits have a convergent subsequence in $|X|$; a convergent sequence of $|X|$ lifts to a convergent sequence of objects `Chapter12/P3R2_ReflexiveCompactnessEp · EpGroupoid.exists_lift_tendsto`; morphisms connect the given points to the lifts, with sources in $\overline{V_0}$ and convergent targets, and properness gives a convergent subsequence of these morphisms, whose sources converge.

**Proposition 12.4.5** (stronger) `Chapter12/P3R2_ReflexiveCompactnessEp · EpStrongBundle.proposition_12_4_5`: for an sc-smooth section functor the three conditions are equivalent: (1) Definition 12.4.3; (2) for every reflexive $N$ and $z \in |X|$ a saturated open $\tilde{U}$ with $z \in |\tilde{U}|$ and $|\,\mathrm{cl}_X\{y \in \tilde{U} : N(f(y)) \le 1\}|$ compact; (3) the same with $\mathrm{cl}_{|X|}\,|\{y \in \tilde{U} : N(f(y)) \le 1\}|$ compact. It holds for every section functor `Chapter12/P3R2_ReflexiveCompactnessEp · EpStrongBundle.proposition_12_4_5_of_functorial`; reflexive fibres, sc-smoothness and the regularizing property are not used. *(2)□ (3)* by Lemma 12.4.4 (1), the set being saturated (§XII.9). *(1)□ (3).* $\tilde{U}$ is the saturation of $U(x)$, which has the same orbit set, and Lemma 12.4.4 (2) applies to the compact closure. *(3)□ (1)*, "easily obtained" in the text, uses the tool above: with $V_0 \ni x$ on which $t$ is proper over $s^{-1}(\overline{V_0})$, put $U = \tilde{U} \cap V_0$, open `Chapter12/P3R2_ReflexiveCompactnessEp · EpGroupoid.isSaturated_inter`; its sublevel set lies in $V_0$ and has an orbit set with compact closure, so its closure in $X$ is compact.

**Theorem 12.4.6** (stronger) `Chapter12/P3R2_ReflexiveCompactnessEp · EpStrongBundle.theorem_12_4_6`. Hypotheses, as printed: $X$ tame, $|X|$ paracompact, reflexive $(0,1)$-fibres, $N$ reflexive, $f$ an sc-Fredholm section functor with compact $|S_f|$ and the reflexive local compactness property, and $U_\partial = W' \cap \partial X$ ($W'$ open, so $U_\partial$ is open in $\partial X$), saturated, containing $S_f \cap \partial X$, with $\mathrm{cl}_{|X|}\,|\{y \in U_\partial : N(f(y)) \le 1\}|$ compact. Conclusion: a saturated open $U \supset S_f$ with $U \cap \partial X = U_\partial$ and $\mathrm{cl}_{|X|}\,|\{x \in U : N(f(x)) \le 1\}|$ compact. It holds for every continuous section functor with local compactness at the given $N$ `Chapter12/P3R2_ReflexiveCompactnessEp · EpStrongBundle.theorem_12_4_6_of_continuous`; tameness, paracompactness and the Fredholm property beyond continuity are unused. *Proof*, the text's, in $|X|$. *Step 1.* Finitely many $|U(\bar{x}_i)|$, from local compactness at boundary points, cover the compact $\mathrm{cl}\,|U_\partial \cap \{N \circ f \le 1\}|$. *Step 2.* $\{N \circ f > 1\}$ is open (Part I's Proposition 5.2.7 (2)) and saturated. *Step 3.* Finitely many interior sets $|U(x_i) \setminus \partial X|$ cover the rest of the compact $|S_f|$. *Step 4.* $U$ is the preimage of the union of these orbit sets,

cut by $|W'| \cup |X \setminus \partial X|$; since $\partial X$ and $U_\partial$ are saturated, $U \cap \partial X = U_\partial$, and the sublevel set of $U$ lies in finitely many sets with compact closure.

## XII.20 Theorem 12.5.7 (2017: 12.12): naturality of the orientation bundle, without topology

**The setting.** A generalized strong bundle isomorphism is represented by a diagram $W \xleftarrow{h_1} V \xrightarrow{h_2} W'$ of strong bundle equivalences over $\phi_1$, $\phi_2$, with sc-Fredholm section functors $f$, $f'$, $f''$ such that $f'' = h_1^* f = h_2^* f'$. The text's $f'$ is the push-forward of $f$; any section with the same pull-back serves, and the push-forward is one `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward_witness`. Orientations are those of §XII.11; an orientation section is invariant if $\varphi_* o_{s(\varphi)} = o_{t(\varphi)}$ `Chapter12/P3F_Naturality · EpStrongBundle.IsOrInvariant`, the second clause of Definition 12.5.4 `Chapter12/P3F_Naturality · EpStrongBundle.definition_12_5_4_iff`.

**Linearizations between different bundles** (extra) `Chapter12/P3F_MixedPull · StrongBundle.LocalIso.mpullPt`. Wave 1 pulled chart points back along a local strong bundle isomorphism of a bundle to itself (§XII.10); the same construction works between two bundles over different bases. If $f = \Psi \circ f' \circ \phi$ near the point, the pulled chart point has the same set Lin `Chapter12/P3F_MixedPull · StrongBundle.LocalIso.linSet_mpullPt`, pulling back commutes with Part I's identifications between chart points, also for two composites with the same germ `Chapter12/P3F_MixedPull · StrongBundle.LocalIso.GermAgree₄` `Chapter12/P3F_MixedPull · StrongBundle.LocalIso.transferT_mpullPt₄`, and the resulting map of orientations `Chapter12/P3F_MixedPull · StrongBundle.LocalIso.morVia` does not depend on the auxiliary chart point `Chapter12/P3F_MixedPull · StrongBundle.LocalIso.morVia_eq`. For a bundle to itself it is wave 1's construction `Chapter12/P3F_MixedPull · StrongBundle.LocalIso.mpullPt_eq_pullPt`.

**One equivalence.** Let $h : V \to W$ be a strong bundle equivalence over $\phi$ with $f'' = h^* f$. At each object $a$, $h$ is a local strong bundle isomorphism near $a$ `Chapter12/P3F_EquivalenceOr · EpStrongBundle.IsStrongBundleEquivalence.locIso`, and conjugation by its linearizations is a bijection of orientation sets $\mathrm{orEqv}_a : \mathrm{Or}^f_{\phi(a)} \to \mathrm{Or}^{f''}_a$ `Chapter12/P3F_EquivalenceOr · EpStrongBundle.IsStrongBundleEquivalence.orEqv`. *Naturality* `Chapter12/P3F_EquivalenceOr · EpStrongBundle.IsStrongBundleEquivalence.orMor_orEqv`: for a morphism $\alpha : a \to b$, $\alpha_* \circ \mathrm{orEqv}_a = \mathrm{orEqv}_b \circ (\phi\alpha)_*$. The local isomorphism of $\alpha$ followed by that of $h$ at $b$ and the local isomorphism of $h$ at $a$ followed by that of $\phi(\alpha)$ have the same germ, because $h$ is a functor; so the two composite transfers agree, and so do the conjugations. The orbit maps: $\phi$ induces a bijection of the smooth orbit spaces $|V_\infty| \to |X_\infty|$ `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_bijective` (an equivalence is essentially surjective and fully faithful, and preserves smoothness), and $\mathrm{orEqv}^{-1}$ induces a bijection $|O_{f''}| \to |O_f|$ `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_bijective` over it `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.orbitProj_orbitOrMap`, well defined by naturality. *Orientations.* An invariant orientation of $f$ pulls back to an invariant one of $f''$ `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pullOr`, and an invariant orientation of $f''$ pushes forward `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.pushOr`: at $x$ choose $a$ with an isomorphism $\phi(a) \cong x$ and transport by it; invariance makes this independent of the choice, the result is invariant `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pushOr` and it is the only invariant orientation that corresponds to the given one at the points $\phi(a)$ `Chapter12/P3F_Naturality · EpStrongBundle.IsStrongBundleEquivalence.pushOr_unique`.

**Theorem 12.5.7** (narrower). *(1)* `Chapter12/P3F_Naturality · theorem_12_5_7_one_hyp`: for the diagram above, there are bijections $F : |O_f| \to |O_{f'}|$ and $G : |X_\infty| \to |X'_\infty|$ with $|\sigma'| \circ F = G \circ |\sigma|$, and $F$, $G$ are the maps of the diagram, composed from the bijections of $h_1$ and $h_2$. *(2)* `Chapter12/P3F_Naturality · theorem_12_5_7_two_invariant`: for an invariant orientation $o$ of $f$ there is exactly one invariant orientation $o'$ of $f'$ that corresponds to $o$ through the two equivalences at every smooth object of $V$: pull back along $h_1$, push forward along $h_2$. The added hypotheses are §12.5's standing hypothesis that sc⁺-sections pass through smooth fibre points, for $W$, $W'$ and $V$, not derived from the text, with witness that bump functions give it (§XII.10); and that $f''$ is an sc-Fredholm section functor, which holds for the pull-back of one (Theorem 10.4.5) `Chapter10/FredholmEquivalence · theorem_10_4_5_fred`. The identity diagram is a witness of the representative data. *Not proved* (a new gap, §XII.13): that $F$ is a homeomorphism and $o'$ continuous for Part I's topology, which needs G.138, and the independence of the representative and functoriality, which the text leaves to the reader.

## XII.21 Witnesses and the hypotheses of wave 2

**Hypotheses taken.** *§XII.17.* `HasReflexiveLocalExtensions`, the local step of Theorem 12.2.13, not derived from the text; witnesses `hasReflexiveLocalExtensions_of_isReflexiveAuxNorm` and `Trivial-Bundle.epBundle_reflexive_witness`. Tameness is dropped there. *§XII.20.* The hypothesis of §§XII.10–XII.11 that sc⁺-sections pass through smooth fibre points, for the three bundles of the diagram; witness from bump functions. *None added* to Theorems 12.2.7, 12.2.11, 12.3.9, Proposition 12.4.5 or Theorem 12.4.6. *Kept and unused.* Paracompactness of $|X|$ in Theorem 12.3.9 (partitions of unity being assumed) and in Theorem 12.4.6; tameness in Theorem 12.4.6; reflexive fibres, sc-smoothness and the regularizing property in Proposition 12.4.5.

**Every definition has a witness.** Reflexive auxiliary norms: Theorem 12.2.11 itself, for every bundle with reflexive fibres and paracompact $|X|$, for instance the trivial bundle of §XII.17. sc⁺-section functors of $W|\partial X$: the zero section (§XII.18). All hypotheses of Theorem 12.3.9 together: `theorem_12_3_9_hypotheses_rn`. The reflexive local compactness property: from Part I's Definition 5.2.6 (§XII.19). Invariant orientations, pushed forward: the identity diagram (§XII.20).

# Part XIII. Sc⁺-multisections (HWZ Chapter 13, §§13.1–13.3, 13.5)

Chapter 13 (Springer 2021, pp. 599–635; 2017 numbers in parentheses) builds the perturbations of the Fredholm theory on ep-groupoids. §13.1 constructs good systems of neighbourhoods, in which every neighbourhood carries the natural action of its isotropy group and the morphisms between two neighbourhoods fall into components on which source and target are sc-diffeomorphisms (Lemma 13.1.1, Definition 13.1.2, Proposition 13.1.3, Definitions 13.1.4, 13.1.5, Lemma 13.1.6, Theorems 13.1.7, 13.1.8, Proposition 13.1.9 (2017: Lemma 13.1, Definition 13.1, Proposition 13.1, Definitions 13.2, 13.3, Lemma 13.2, Theorems 13.1, 13.2, Proposition 13.2)). §13.2 defines sc⁺-multisections, their supports, symmetric section structures and their algebra (Definitions 13.2.1–13.2.5, Lemma 13.2.6, Definitions 13.2.7–13.2.10, Lemma 13.2.11, Definition 13.2.12, Lemma 13.2.13 (2017: Definitions 13.4–13.8, Lemma 13.3, Definitions 13.9–13.12, Lemma 13.4, Definition 13.13, Lemma 13.5)). §13.3 structures them (structurable sc⁺-multisections), §13.4 relates them along equivalences and coverings, and §13.5 constructs them. This Part retells §13.1 (§§XIII.1–XIII.5) and the functor-level part of §13.2 (§§XIII.6–XIII.9); then the witnesses (§XIII.10), what is not formalized (§XIII.11) and what the formalization taught us (§XIII.12).

**What is proved, and what is not.** Waves 1 and 2 of Part III (2026-11-13) formalize §§13.1–13.3 and the first half of §13.5; §13.4 is not yet formalized. *§13.1.* Everything is proved. Lemma 13.1.1 is proved in a stronger form, without the invariance of $V$ (§XIII.1). Definition 13.1.2 is formalized with "carries the natural $G_x$-action" read as an invariant open subset of the domain of a natural representation (§XIII.1). Proposition 13.1.3 holds as printed in that reading, and Definition 13.1.4 is formalized (§XIII.2). Definition 13.1.5, the natural bi-action, is formalized with its laws and its sc-smoothness, and Lemma 13.1.6 is proved in a stronger form (§XIII.3). Theorem 13.1.7 is proved as printed, with properness read in the full subcategory on the open set $Q$ (§XIII.13; gap G.141 is closed), and in a stronger form that reads properness in $X$ and needs neither $Q$ open nor $\mathrm{cl}_X U(q) \subset Q$ (§XIII.4). Theorem 13.1.8 is proved as printed (§XIII.4), and Proposition 13.1.9 in a stronger form, uniqueness holding among all maps (§XIII.5). *§13.2.* Definition 13.2.1 is Chapter 11's (Part XI, §XI.11). Definitions 13.2.2–13.2.4 and the claims after them are proved (§XIII.6). Definition 13.2.5 is formalized and Lemma 13.2.6 proved (§XIII.7). Definitions 13.2.9 and 13.2.10 and Lemmas 13.2.11 and 13.2.13 are proved at the level of functors with local section structures (§§XIII.8–XIII.9) and, in wave 2, at the level of decorated sc⁺-multisections: Definitions 13.2.7 and 13.2.8 are formalized, and the sum, the scalar multiples, the products with functions and the locally finite sums are decorated and compatible with the equivalence (§§XIII.14–XIII.15); gap G.142 is closed. The remark that there are in general no inverses is proved by an example (§XIII.15); gap G.143 is closed. In Definition 13.2.9 and Lemma 13.2.13 the intersection $U'(x) \cap U''(x)$ is read as a neighbourhood inside it carrying one natural action for both structures. *§13.3.* Definitions 13.3.1, 13.3.2, Example 13.3.3, Defi-

nition 13.3.4 and Remark 13.3.5 (correspondences) are formalized and proved, Definition 13.3.2 with its action laws (§XIII.16). Definitions 13.3.6, 13.3.8, 13.3.10, 13.3.11 and Remark 13.3.7 are formalized, Proposition 13.3.12 is proved, Lemma 13.3.14 as printed, and Remark 13.3.13 in its positive half (§XIII.17). The construction before Proposition 13.3.15 is proved with the compatibility the text does not check, Proposition 13.3.15 with bijections in place of equalities of index sets, Definition 13.3.17 and Corollary 13.3.18 as printed (§XIII.18). Remarks 13.3.9 and 13.3.16 are prose. Proposition 13.3.19, locally finite sums of structurable sc⁺-multisections, is not formalized (a new gap). *§13.5.* Definition 13.5.1 is formalized; Theorems 13.5.4 and 13.5.6 are proved, narrower only through the pointwise reading of bump functions and otherwise stronger; Theorem 13.5.5 is proved as printed for existence and uniqueness (with $I \neq \emptyset$, which the text presupposes); Theorem 13.5.3 and the last clause of Theorem 13.5.5, that atomic sc⁺-multisections are structurable, are proved with the correspondences as a hypothesis (§§XIII.19–XIII.20; a new gap). Remark 13.5.2 is prose. Theorem 13.5.7 is not yet formalized. No statement of these sections was found not to hold as we read it.

**Three structural choices.** (1) *Neighbourhoods with the natural action.* The project's natural representation of $G_x$ (Theorem 7.1.13) has a connected domain; the invariant subsets of §13.1 need not be connected. A neighbourhood with the natural action is therefore a natural representation together with an open $U \ni x$ inside its domain, invariant under every $\Phi(g)$ `Chapter13/P3G_GoodSystem · EpGroupoid.NatNbhd`. In this reading every claim of §13.1 that an invariant open subset carries the natural action holds as printed. (2) *Components named by a point.* A component of $\mathcal{U}(x, x')$ is the connected component in $\mathcal{U}(x, x')$ of one of its morphisms, and empty if the morphism is not in $\mathcal{U}(x, x')$; so no statement needs a separate index set of components. (3) *Multisections as functors.* §13.2 is formalized for functors $\Lambda : W \to \mathbb{Q}^+$ satisfying Chapter 11's predicate of sc⁺-multisection, with local section structures on open sets `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsScPlusMultisection` `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn`; the decorations $[\Lambda, \mathcal{U}, \mathcal{S}]$ wait for wave 2. The local section structures of a sum, a multiple and a locally finite sum are proved, which is the content of the decorations.

**Map.**

- `Chapter13/P3G_GoodSystem`: neighbourhoods with the natural action, Lemma 13.1.1, $\mathcal{U}(x, x')$ and its components, Definition 13.1.2, Proposition 13.1.3, Definition 13.1.4.
- `Chapter13/P3G_BiAction`: the bi-action of Definition 13.1.5, Lemma 13.1.6, stabilizers, $D_\Sigma$, Proposition 13.1.9.
- `Chapter13/P3G_GoodSystemExistence`: transports, Theorems 13.1.7 and 13.1.8, the witness.
- `Chapter13/P3M_ScPlusSectionOn`: sums, multiples and extensions by $0$ of local sc⁺-sections.
- `Chapter13/P3M_MultisectionSupport`: Definitions 13.2.2–13.2.4.
- `Chapter13/P3M_MultisectionSymmetric`: Definition 13.2.5, Lemma 13.2.6.
- `Chapter13/P3M_MultisectionConvolution`: the convolution of finitely many multisections.
- `Chapter13/P3M_MultisectionAlgebra`: Definitions 13.2.9, 13.2.10, Lemma 13.2.11.
- `Chapter13/P3M_MultisectionLocallyFinite`: Definition 13.2.12, Lemma 13.2.13.
- `Chapter13/P3E_GoodSystemRelative`: Theorem 13.1.7 as printed (§XIII.13).
- `Chapter13/P3D_SymmetricOn`, `P3D_Decorated`, `P3D_NoInverse`: decorated sc⁺-multisections, §13.2 decorated, no inverses (§§XIII.14–XIII.15).
- `Chapter13/P3D_Correspondence`, `P3D_Structurable`, `P3D_Tight`: §13.3 (§§XIII.16–XIII.18).
- `Chapter13/P3A_Atomic`, `P3A_AtomicConstruction`, `P3A_AtomicStructurable`: §13.5 (§§XIII.19–XIII.20).

Throughout, $X$ is an ep-groupoid with source and target maps $s$, $t$, $\pi : X \to |X|$ the orbit map, and $\mathcal{U}(x,x') = \{\varphi : s(\varphi) \in U(x),\ t(\varphi) \in U(x')\}$ `Chapter13/P3G_GoodSystem · EpGroupoid.morBetween`. In §§XIII.6–XIII.9, $(P : W \to X, \mu)$ is a strong bundle and $\Lambda : W \to \mathbb{Q}^+$.

## XIII.1 Lemma 13.1.1 (2017: 13.1) and Definition 13.1.2 (2017: 13.1)

**Lemma 13.1.1** (stronger) `Chapter13/P3G_GoodSystem · EpGroupoid.lemma_13_1_1`. Let $U(x)$ carry the natural action, with $t : s^{-1}(\mathrm{cl}\,U(x)) \to X$ proper, and $V \subset U(x)$. Then $\mathrm{cl}_X V \subset U(x)$ iff $\mathrm{cl}_{|X|}\,|V| \subset |U(x)|$. The text assumes $V$ invariant; that form follows `Chapter13/P3G_GoodSystem · EpGroupoid.lemma_13_1_1_book`. *(□ )* uses only properness: $\pi(\mathrm{cl}\,V) = \mathrm{cl}\,\pi(V)$ for a set with proper closure (Proposition 7.1.10). *(□ )* uses only the natural action. A point $y \in \mathrm{cl}\,V$ has $|y| \in |U(x)|$, so there is a morphism between $y$ and a point of $U(x)$; along it, the points of $V$ near $y$ are carried by morphisms between points of $U(x)$, each of the form $\Gamma(g,\cdot)$, and by pigeonhole over the finite $G_x$ one $g$ occurs along a sequence converging to $y$, which places $y$ in $U(x)$. Inside every open $W \ni x$ there is a neighbourhood with the natural action and proper closure `Chapter13/P3G_GoodSystem · EpGroupoid.exists_natNbhd_isProperSubset`.

**Definition 13.1.2** (restated) `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem`: a family of neighbourhoods $U(x)$ with the natural action (2), with $t : s^{-1}(\mathrm{cl}\,U(x)) \to X$ proper (1), and such that (3) for every component $\Sigma$ of $\mathcal{U}(x,x')$, $s|_\Sigma$ and $t|_\Sigma$ are sc-diffeomorphisms onto their images `Chapter13/P3G_GoodSystem · EpGroupoid.IsDiffeoComponent`. The images are open automatically. For an open component, (3) is the injectivity of $s$ and of $t$ on it, since both are local sc-diffeomorphisms `Chapter13/P3G_GoodSystem · EpGroupoid.isDiffeoComponent_iff_injOn` `Chapter13/P3G_GoodSystem · MPolyfold.IsLocalScDiffeo.isScDiffeoOn_invFunOn`. The inverses $(s|_\Sigma)^{-1}$ are functions on the objects `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.srcInv`.

## XIII.2 Proposition 13.1.3 (2017: 13.1) and Definition 13.1.4 (2017: 13.2)

**Proposition 13.1.3** `Chapter13/P3G_GoodSystem · EpGroupoid.proposition_13_1_3`: if $(U(x))$ is a good system and $\tilde{U}(x) \subset U(x)$ are invariant open neighbourhoods, then $(\tilde{U}(x))$ is a good system, with the same natural representations `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.restrict`. (2) holds in the reading of structural choice (1); under the reading "natural action on a connected domain" $\tilde{U}(x)$ could fail to be such a domain. (1) passes to the smaller set because a closed subset of a set with proper closure has proper closure. (3) passes because a component of $\mathcal{U}$ for smaller sets lies in a component for the larger ones, on which $s$ and $t$ are injective `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.isDiffeoComponent_of_subset`. *The step after the proposition.* Inside $U(x) \cap V(x)$ there is an invariant open neighbourhood, the largest invariant open subset `Chapter13/P3G_GoodSystem · EpGroupoid.NatNbhd.invariantCore`, so every good system has a refinement inside any family of open neighbourhoods `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.exists_refinement_subset`.

**Definition 13.1.4** `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.IsRefinement` `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.IsSubordinate`: $\mathcal{W}$ refines $\mathcal{U}$ if $W(x) \subset U(x)$ for all $x$; $\mathcal{U}$ is subordinate to an open cover if each $U(x)$ lies in a member. Restrictions are refinements `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.isRefinement_restrict`.

## XIII.3 Definition 13.1.5 (2017: 13.3) and Lemma 13.1.6 (2017: 13.2)

**Definition 13.1.5** (restated) `Chapter13/P3G_BiAction · EpGroupoid.biact`. The text names the map $G_{x'} \times \mathcal{U}(x,x') \times G_x \to \mathcal{U}(x,x')$; the formalization writes the formula that its proofs use: $g' * \varphi * g = \Gamma'(g', t(\varphi)) \circ \varphi \circ \Gamma(g, \Phi(g^{-1})(s(\varphi)))$, with source $\Phi(g^{-1})(s(\varphi))$ `Chapter13/P3G_BiAction · EpGroupoid.src_biact` and target $\Phi'(g')(t(\varphi))$ `Chapter13/P3G_BiAction · EpGroupoid.tgt_biact`, again in $\mathcal{U}(x,x')$ `Chapter13/P3G_BiAction · EpGroupoid.biact_mem`. It is an action, $g' * (h' * \varphi * h) * g = (g'h') * \varphi * (hg)$ `Chapter13/P3G_BiAction · EpGroupoid.biact_biact`, of $G_{x'} \times G_x^{\mathrm{op}}$ `Chapter13/P3G_BiAction · EpGroupoid.biactMulAction`, and each $\varphi \mapsto g' * \varphi * g$ is sc-smooth `Chapter13/P3G_BiAction · EpGroupoid.isScSmoothOn_biact`. On a good system `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.biact` it maps components onto components `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.image_biact_comp`, as the text says before Lemma 13.1.6; inversion maps components of $\mathcal{U}(x,x')$ onto components of $\mathcal{U}(x',x)$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.image_inv_comp`, and the components partition $\mathcal{U}(x,x')$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.components`.

**Two tools.** Two morphisms of $\mathcal{U}(x,x')$ with the same source differ by a left factor $g'$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.exists_biact_left_of_src`, because their composite $\psi \circ \varphi^{-1}$ is a morphism between points of $U(x')$, hence $\Gamma'(g', \cdot)$; similarly with the same target and a right factor `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.exists_biact_right_of_tgt`.

**Lemma 13.1.6** (stronger). (1) If $\psi \in \Sigma$ and $g' * \psi \in \Sigma$ then $g' = 1$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.eq_one_of_biact_left_mem`: both have the same source, and $s$ is injective on $\Sigma$, so $g' * \psi = \psi$, and $g' = 1$ by left cancellation. The text's (1), $g' * \Sigma = \Sigma \Rightarrow g' = 1$, follows `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_one`. (2) is the same on the right `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.eq_one_of_biact_right_mem` `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_two`. (3) For two components, $s(\Sigma_1)$ and $s(\Sigma_2)$ are equal or disjoint `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_three_src`, and so are $t(\Sigma_1)$, $t(\Sigma_2)$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_three_tgt`. If $s(\varphi_1) = s(\varphi_2)$ with $\varphi_i \in \Sigma_i$, then $\varphi_2 = g' * \varphi_1$, so $\Sigma_2 = g' * \Sigma_1$ and $s(\Sigma_2) = s(\Sigma_1)$. No membership hypotheses are needed, since the component of a morphism outside $\mathcal{U}(x,x')$ is empty.

## XIII.4 Theorems 13.1.7 and 13.1.8 (2017: 13.1 and 13.2): good systems exist

**Good on a set** `Chapter13/P3G_GoodSystem · EpGroupoid.IsGoodOn`. A family of neighbourhoods with the natural action is good on $Q$ if for $q, q' \in Q$: $U(q) \subseteq Q$, $t : s^{-1}(\mathrm{cl}_X U(q)) \to X$ is proper, and (3) holds for $\mathcal{U}(q,q')$. The text asks for a good system of the full sub-ep-groupoid on the open set $Q$, whose properness is read in $Q$; here it is read in $X$. The natural actions and the sets $\mathcal{U}(q,q')$ are the same in $Q$ and in $X$, because $Q$ is open and the subcategory full.

**Transports** `Chapter13/P3G_GoodSystemExistence · EpGroupoid.IsTransport`: a continuous injective choice $\sigma$ of morphisms on $U(y)$ with $t(\sigma(z)) = z$ and $s(\sigma(z)) \in U(q)$. Along transports $\sigma$, $\sigma'$ the map $\varphi \mapsto \sigma'(t\varphi)^{-1} \circ \varphi \circ \sigma(s\varphi)$ carries $\mathcal{U}(y,y')$ into $\mathcal{U}(q,q')$, so injectivity of $s$ and $t$ on components passes back `Chapter13/P3G_GoodSystemExistence · EpGroupoid.injOn_comp_of_transport`.

**Theorem 13.1.7** (narrower) `Chapter13/P3G_GoodSystemExistence · EpGroupoid.theorem_13_1_7`. Let $|Q| = |X|$, let the family $(U(q))_{q \in Q}$ be good on $Q$, and let $W(y) \ni y$ be open for $y \notin Q$. Then there is a good system of $X$ that keeps $U(q)$ for $q \in Q$ and has $U(y) \subset W(y)$ for $y \notin Q$. *Step 1, a transport at each $y \notin Q$* `Chapter13/P3G_GoodSystemExistence · EpGroupoid.exists_transport_of_isGoodOn`. Choose a morphism $\psi$ from some $q_y \in Q$

to $y$, and an open $O \ni \psi$ on which $s$ and $t$ are injective `Chapter13/P3G_GoodSystemExistence · EpGroupoid.exists_isOpen_injOn_src_tgt` and $s(O) \subset U(q_y)$. Then $(t|_O)^{-1}$ is a transport on any neighbourhood $U(y)$ with the natural action and proper closure inside $W(y) \cap t(O)$ (§XIII.1). *Step 2.* The family is good on all of $X$ by the transports, and a family good on $X$ is a good system. The text's further smallness condition on the closures of $U(y)$ is not used, and neither is the openness of $Q$. That $Q$-properness implies $X$-properness when $|Q| = |X|$ is presumably true and is the gap of this Part (§XIII.11).

**Theorem 13.1.8** `Chapter13/P3G_GoodSystemExistence · EpGroupoid.theorem_13_1_8`: if $|X|$ is paracompact, for every family of open $W(x) \ni x$ there is a good system with $U(x) \subset W(x)$. The text's "for all $\in X$" is read as "for all $x \in X$", presumably a dropped symbol. *Step 1.* Neighbourhoods $W'(x) \subset W(x)$ with the natural action and proper closure. *Step 2.* A locally finite open refinement $(\tilde{V}_a)$ of $(|W'(a)|)$, shrunk so that the closures are in it and still locally finite. *Step 3.* $V_a = \pi^{-1}(\tilde{V}_a) \cap W'(a)$ and $V = \bigcup_a V_a$, with $|V| = |X|$; by Lemma 13.1.1, $\operatorname{cl} V_a \subset W'(a)$, and the closures are locally finite. *Step 4, good on $V$.* Near each $x$ only finitely many $\operatorname{cl} V_a$ matter, and near a point there are finitely many morphisms out of a finite union of sets with proper closure `Chapter13/P3G_GoodSystemExistence · EpGroupoid.exists_nhds_injOn_of_isProperSubset`; a small neighbourhood $U(x)$, $x \in V$, makes $s$ and $t$ injective on the components. *Step 5.* Theorem 13.1.7 extends the system from $V$ to $X$. Metrizability of $|X|$ is not used.

## XIII.5 Proposition 13.1.9 (2017: 13.2)

**Stabilizers and $D_\Sigma$.** For a component $\Sigma$ with $O = s(\Sigma)$, $G_{x,\Sigma} = \{g : g * O = O\}$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.stabSrc` and $G_{x',\Sigma}$ with $t(\Sigma)$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.stabTgt`. $D_\Sigma = t \circ (s|_\Sigma)^{-1} : s(\Sigma) \to t(\Sigma)$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.D` is an sc-diffeomorphism with inverse $s \circ (t|_\Sigma)^{-1}$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.isScDiffeoOn_D`.

**Proposition 13.1.9** (stronger) `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.proposition_13_1_9`: there is a group isomorphism $g \mapsto g'(g)$, $G_{x,\Sigma} \to G_{x',\Sigma}$, with $(s|_\Sigma)^{-1}(g * z) = g'(g) * (s|_\Sigma)^{-1}(z) * g^{-1}$ for $z \in s(\Sigma)$, and every map with this property equals it. *Step 1, the partner.* For $g \in G_{x,\Sigma}$ and $\varphi \in \Sigma$, $\varphi * g^{-1}$ has source $g * s(\varphi) \in O$, so there is $\psi \in \Sigma$ with the same source, and $\psi = g' * \varphi * g^{-1}$ for some $g'$; by Lemma 13.1.6 (1) $g'$ is unique `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.partner`. *Step 2.* It does not depend on $\varphi$: the bi-action maps the component of $\varphi * g^{-1}$, which is $\Sigma * g^{-1}$, onto $\Sigma$, so the same $g'$ works for all of $\Sigma$. *Step 3.* $g \mapsto g'(g)$ is a homomorphism by the action law `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.partnerHom`, with inverse given by the same construction on $\Sigma^{-1}$; uniqueness holds among all maps by Lemma 13.1.6 (1). The text proves the identity through the smoothness of a map $\Theta_g$; the formal proof needs only Lemma 13.1.6 and the action laws. *Equivariance of $D_\Sigma$:* $\Phi'(g'(g)) \circ D_\Sigma = D_\Sigma \circ \Phi(g)$ on $s(\Sigma)$ `Chapter13/P3G_BiAction · EpGroupoid.GoodSystem.proposition_13_1_9_D`.

## XIII.6 Definitions 13.2.2–13.2.4 (2017: 13.5–13.7): supports, norms, triviality

$\Lambda : W \to \mathbb{Q}^+$ is an sc⁺-multisection in Chapter 11's sense: a functor with local section structures, finitely many local sc⁺-sections $s_i$ with weights $\sigma_i$ and $\Lambda(w) = \sum_{i : s_i(P(w)) = w} \sigma_i$ near each point (Definition 13.2.1, Part XI, §XI.11).

**Definition 13.2.2** `Chapter13/P3M_MultisectionSupport · SBLocalization.domSupp` `Chapter13/P3M_MultisectionSupport · SBLocalization.supp` `Chapter13/P3M_MultisectionSupport · SBLocalization.suppAt`:

dom-supp$(\Lambda) = \mathrm{cl}\{x : \exists w \in W_x,\ w \neq 0_x,\ \Lambda(w) > 0\}$, and supp$(\Lambda) = \{w : \Lambda(w) > 0\}$ with fibres supp$(\Lambda)(x)$. The claims after it are proved: supp$(\Lambda)(x)$ is $\{s_1(x), \dots, s_k(x)\}$ on the domain of a local structure `Chapter13/P3M_MultisectionSupport · SBLocalization.IsLocalRepOn.suppAt_eq`, hence finite `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.finite_suppAt` and nonempty `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.suppAt_nonempty`, with weights summing to 1 `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.sum_suppAt` `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.finsum_suppAt`; and dom-supp$(\Lambda) = \mathrm{cl}\, P(\mathrm{supp}(\Lambda) \setminus 0_W)$ `Chapter13/P3M_MultisectionSupport · SBLocalization.domSupp_eq_closure_image`.

**Definition 13.2.3** (restated) `Chapter13/P3M_MultisectionSupport · SBLocalization.pointwiseNorm`: $N(\Lambda)(x) = \max\{N(w) : w \in \mathrm{supp}(\Lambda)(x)\}$, written as a supremum `Chapter13/P3M_MultisectionSupport · SBLocalization.normW` and shown to be attained `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.isGreatest_pointwiseNorm`. The values of sc⁺-sections lie in $W_{0,1}$, where $N$ is defined `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.suppAt_subset_oneSet`. For an auxiliary norm, dom-supp$(\Lambda) = \mathrm{cl}\{N(\Lambda) > 0\}$ `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.domSupp_eq_closure_pointwiseNorm`.

**Definition 13.2.4** `Chapter13/P3M_MultisectionSupport · SBLocalization.IsTrivialOn`: $\Lambda$ is trivial on $V$ if $\Lambda(0_x) = 1$ for $x \in V$; "and hence" $\Lambda(w) = 0$ for $w \neq 0_x$, because the weights over $x$ sum to 1 `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.apply_fiberZero_eq_one_iff`. dom-supp$(\Lambda)$ is the smallest closed set off which $\Lambda$ is trivial `Chapter13/P3M_MultisectionSupport · SBLocalization.IsScPlusMultisection.isLeast_domSupp`. The zero multisection is trivial everywhere `Chapter13/P3M_MultisectionSupport · SBLocalization.isTrivialOn_zeroMultisection`.

## XIII.7 Definition 13.2.5 (2017: 13.8) and Lemma 13.2.6 (2017: 13.3): symmetric structures

**Definition 13.2.5** (restated) `Chapter13/P3M_MultisectionSymmetric · SBLocalization.SymmetricStructure`: at $x$, the domain $U(x)$ of a natural representation of $G_x$ with the properness property; a finite index set $I_x$ with an action of $G_x$; sc⁺-sections $s_i$ on $U(x)$ with (1) $s_{g(i)}(g*y) = \mu(\Gamma(g,y), s_i(y))$ and (2) $\Lambda(w) = |I_x|^{-1}\#\{i : s_i(P(w)) = w\}$ over $U(x)$. "$G_x$-invariant with the natural action" is read here as the domain of a natural representation, the reading of Chapter 11; the index set lives in the universe of the isotropy groups.

**Lemma 13.2.6** `Chapter13/P3M_MultisectionSymmetric · SBLocalization.lemma_13_2_6`: every sc⁺-multisection admits a symmetric structure at every point. The proof is the text's, before Definition 13.2.5. *Step 1.* A local structure $(s_i, \sigma_i)$ near $x$, on the domain of a natural representation with the properness property (Theorem 7.1.13). *Step 2, equal weights.* Rational weights $\sigma_i = n_i/d_i$ are made equal by repeating $s_i$ $n_i \prod_{j \neq i} d_j$ times, with common weight $1/\prod_j d_j$ `Chapter13/P3M_MultisectionSymmetric · SBLocalization.IsLocalRepOn.exists_equal_weights`; this is the text's $k_j q$. *Step 3, symmetrize.* $\hat{I} = I \times G_x$ with $s_{(i,g)}(z) = \mu(\Gamma(g, g^{-1} * z), s_i(g^{-1} * z))$ `Chapter13/P3M_MultisectionSymmetric · SBLocalization.transportSection` and $h \cdot (i,g) = (i, hg)$ `Chapter13/P3M_MultisectionSymmetric · SBLocalization.prodIsotropyAction`. The transported sections are sc⁺ `Chapter13/P3M_MultisectionSymmetric · SBLocalization.isScPlusSectionOn_transportSection`, (1) is the action law of $\Gamma$, and (2) holds because $\Lambda$ is a functor, so each $g$ contributes the same count. Before transport the local sections are extended by 0 off $U(x)$, so that they are functions on all objects `Chapter13/P3M_ScPlusSectionOn · SBLocalization.extendZero` `Chapter13/P3M_ScPlusSectionOn · SBLocalization.`

IsLocalRepOn.extendZero.

## XIII.8 Definitions 13.2.9, 13.2.10 (2017: 13.11, 13.12) and Lemma 13.2.11 (2017: 13.4): the algebra, at functor level

**Convolution.** For finitely many $\Lambda_i$, $(*_i\Lambda_i)(w) = \sum_{(w_i):\sum_i w_i=w} \prod_i \Lambda_i(w_i)$, the sum over tuples in the fibre of $w$ Chapter13/P3M_MultisectionConvolution · SBLocalization.convAt Chapter13/P3M_MultisectionConvolution · SBLocalization.conv; it is a functor Chapter13/P3M_MultisectionConvolution · SBLocalization.isWFunctor_conv. If each $\Lambda_i$ has a local structure $(s^i_j, \sigma^i_j)$ on $O$, the convolution is given there by the sections $\sum_i s^i_{j_i}$ with weights $\prod_i \sigma^i_{j_i}$ Chapter13/P3M_MultisectionConvolution · SBLocalization.conv_eq_sum_of_isLocalRepOn Chapter13/P3M_MultisectionConvolution · SBLocalization.isLocalRepOn_conv, which are sc⁺ Chapter13/P3M_ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.sectionAdd; so it is an sc⁺-multisection Chapter13/P3M_MultisectionConvolution · SBLocalization.isScPlusMultisection_conv.

**Definition 13.2.9** (narrower: no decoration). (1) is $\Lambda' \oplus \Lambda''(w) = \sum_{w'+w''=w} \Lambda'(w')\Lambda''(w'')$ verbatim Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionAdd, the convolution of two Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionAdd_eq_conv. It is an sc⁺-multisection Chapter13/P3M_MultisectionAlgebra · SBLocalization.definition_13_2_9, and (2)–(3) hold for local structures: sections $s'_i + s''_j$ with weights $\sigma'_i\sigma''_j$ on a common domain Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionAdd_eq_sum_of_isLocalRepOn. $\oplus$ is commutative for all functions Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionAdd_comm and associative on sc⁺-multisections Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionAdd_assoc, where the fibres are finitely supported.

**Definition 13.2.10** (narrower: no decoration) Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionSmul: $(0\Lambda)(0_x) = 1$, $(0\Lambda)(w) = 0$ for $w \neq 0$, so $0\Lambda$ is the zero multisection Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionSmul_zero; $(a\Lambda)(w) = \Lambda(w/a)$ for $a \neq 0$. It is an sc⁺-multisection Chapter13/P3M_MultisectionAlgebra · SBLocalization.definition_13_2_10, locally given by the sections $as_i$ with the same weights Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionSmul_eq_sum_of_isLocalRepOn Chapter13/P3M_ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.sectionSmul.

**Lemma 13.2.11** Chapter13/P3M_MultisectionAlgebra · SBLocalization.lemma_13_2_11: for an sc-smooth functor $\beta : X \to \mathbb{R}$ (sc-smooth on the objects and constant along morphisms, the text's hypotheses), $(\beta\Lambda)(w) = (\beta(P(w))\Lambda)(w)$ Chapter13/P3M_MultisectionAlgebra · SBLocalization.multisectionSmulFun is an sc⁺-multisection, given near a point by the sections $\beta s_i$ with the weights $\sigma_i$: where $\beta \neq 0$ this is Definition 13.2.10 pointwise, and where $\beta = 0$ all $\beta s_i$ are 0, with total weight 1.

## XIII.9 Definition 13.2.12 (2017: 13.13) and Lemma 13.2.13 (2017: 13.5): locally finite sums

**Definition 13.2.12** Chapter13/P3M_MultisectionLocallyFinite · SBLocalization.IsLocallyFiniteFamily: every $x$ has an open saturated $O(x)$ for which only finitely many $j$ have a $w \neq 0$ over $O(x)$ with $\Lambda_j(w) > 0$ Chapter13/P3M_MultisectionLocallyFinite · SBLocalization.nontrivIdx. The remark after it, that $O(x)$ can be chosen so that the index set does not change on smaller saturated neighbourhoods, is proved for open saturated $O' \ni x$, by taking an $O(x)$ whose index set has the fewest elements Chapter13/P3M_MultisectionLocallyFinite · SBLocalization.IsLocallyFiniteFamily.exists_stable.

**Lemma 13.2.13** `Chapter13/P3M_MultisectionLocallyFinite · SBLocalization.lemma_13_2_13`: the sum of a locally finite family of sc⁺-multisections is an sc⁺-multisection. The text does not write out $\bigoplus_j \Lambda_j$ for infinitely many $j$; it is read as the sum over finitely supported families $(w_j)$ in the fibre with $\sum_j w_j = w$ of $\prod_j \Lambda_j(w_j)$ `Chapter13/P3M_MultisectionLocallyFinite · SBLocalization.lfSum`, which for finite index sets is the convolution `Chapter13/P3M_MultisectionLocallyFinite · SBLocalization.lfSum_eq_conv`. *Proof.* Over $O(x)$ an index outside the finite set $J_x$ contributes only 0 with weight $\Lambda_j(0) = 1$, so there the sum equals the finite convolution over $J_x$ `Chapter13/P3M_MultisectionLocallyFinite · SBLocalization.lfSum_eq_conv_of_mem`, which has a local structure by §XIII.8.

## XIII.10 Witnesses and the hypotheses

**Hypotheses taken.** None is added to a statement of the text. Lemma 13.1.1 drops the invariance of $V$; the form with it is kept. Theorem 13.1.7 reads the properness of the partial system in $X$ (§XIII.4). Lemma 13.2.11 assumes $\beta$ sc-smooth and constant along morphisms, the text's "sc-smooth functor"; the identity for dom-supp in Definition 13.2.3 assumes $N$ an auxiliary norm, as the text does.

**Every definition has a witness.** Good systems: every M-polyfold seen as a trivial ep-groupoid has one `Chapter13/P3G_GoodSystemExistence · EpGroupoid.nonempty_goodSystem_trivial` (through Theorem 13.1.8). Neighbourhoods with the natural action and proper closure: inside every open set (§XIII.1). Families good on a set: every good system is good on $X$ `Chapter13/P3G_GoodSystem · EpGroupoid.GoodSystem.isGoodOn`. Symmetric structures: Lemma 13.2.6. Triviality: the zero multisection (§XIII.6).

## XIII.11 What is not formalized

- *Closed in wave 2.* G.141, Theorem 13.1.7 with properness in the full subcategory on $Q$ (§XIII.13); G.142, decorated sc⁺-multisections and their algebra (§§XIII.14–XIII.15); G.143, no inverses (§XIII.15).
- Gap G.148: Proposition 13.3.19, that a locally finite sum of structurable sc⁺-multisections is structurable. Present: the decorated locally finite sum, tight versions, products of two correspondences and of finitely many symmetric structures. Missing: a common good system with $U(x) \subset T_{\Lambda_l}$ for the indices not active at $x$, with tight versions over that system (about 120 lines); the finite product of correspondences over the indices active at $x$ or at $x'$, with the unit factors for indices active at only one of them, which the text leaves to the reader (about 300 lines); and the verification of Definitions 13.3.4 and 13.3.6 for it (about 150 lines).
- Gap G.149: Theorem 13.5.3 (and the last clause of Theorem 13.5.5), the correspondences of an atomic sc⁺-multisection. Present: the local alternatives at every point and the hypothesis form (§XIII.20). Missing: a good system refining the local transports, the actions of $G_y$ on $I_y$ through the partner homomorphism of Proposition 13.1.9, the correspondences $\tau_{y,y'}$ from the sheet index of $\varphi_{y'} \circ \varphi \circ \varphi_y^{-1}$ in $U(x_0)$ with unit diagrams at the trivial points, and the compatibility check; about 400–700 lines, now that the predicate "structurable" exists (§XIII.17).
- Remark 13.3.13, the negative half: that the structured sums $\Lambda_1 \oplus \Lambda_2$ and $\Lambda_2 \oplus \Lambda_1$ differ, because $I_x^1 \times I_x^2$ and $I_x^2 \times I_x^1$ differ, is a statement about inequality of types and has no formal counterpart.
- Not yet formalized: §13.4 (Theorems 13.4.1, 13.4.3, 13.4.4, Proposition 13.4.2, Definition 13.4.5: pullbacks and push-forwards along strong bundle equivalences and coverings); Theorem 13.5.7. Remarks 13.3.9, 13.3.16 and 13.5.2 are prose.

## XIII.12 What the formalization taught us

§13.1 is a study of the bi-action, and once it is written as a formula almost everything follows from two facts: morphisms between points of a neighbourhood with the natural action are $\Gamma(g,\cdot)$ for a unique $g$, and $s$, $t$ are injective on each component. Lemma 13.1.6 (1) then needs one morphism with $g' * \psi \in \Sigma$, not $g' * \Sigma = \Sigma$, and Proposition 13.1.9 follows from Lemma 13.1.6 with the action laws, without the smoothness argument of the text, and with uniqueness among all maps. The phrase "carries the natural $G_x$-action" needed a decision. Proposition 13.1.3 shrinks to invariant subsets that need not be connected, so a neighbourhood with the natural action cannot be the connected domain of a natural representation; it is an invariant open subset of one. With that reading the section holds as printed. In Lemma 13.1.1 the two directions use different halves of the hypotheses (properness one way, the natural action the other), and the invariance of $V$ is used by neither. Theorem 13.1.7 holds in two forms. With properness read in $X$ it needs neither the openness of $Q$ nor the smallness of the closures of $U(y)$. As printed, with properness read in the full subcategory on the open $Q$, the hypothesis turns out to be equivalent to properness in $X$ together with $\mathrm{cl}_X U(q) \subset Q$: a limit of points of $U(q)$ outside $Q$ is moved into $Q$ by a local section of $s$, and properness in $Q$ then places it in $Q$. No sub-ep-groupoid structure is needed; the full subcategory is its set of morphisms with the restricted source and target.

In §13.2 the multisection algebra is convolution: $\oplus$, scalar multiples and locally finite sums are all sums over decompositions of a vector in a fibre, and their local structures are sums of sections with products of weights. A locally finite sum over infinitely many indices, which the text does not write out, is a sum over finitely supported families, and over $O(x)$ it is a finite convolution. The decorations add one subtlety: the intersection $U'(x) \cap U''(x)$ of two neighbourhoods with natural actions need not carry one natural action for which both structures are symmetric, so the sum is decorated on a smaller neighbourhood inside it on which the two actions agree, which Definition 13.2.8 allows. The absence of inverses is a one-line computation once a neutral element is fixed: a two-valued multisection convolved with anything gives weight at most $\frac{1}{2}$ to the zero vector. §13.3 is mostly combinatorics of finite index sets, and two points needed care. The text writes equalities of index sets ($I'_x = \{*\}$, $I'_x = I_x$) where a formalization has bijections, and the correspondences are compared after transport along them. And the tight version of Proposition 13.3.15 needs a compatibility the text does not check; it holds because every section of a structure vanishes where $\Lambda$ is trivial. Definition 13.3.8 as printed puts no condition on the sections; presumably they are meant to agree as in Definition 13.2.8, and nothing proved depends on it. In §13.5 the atomic multisection is determined by its data, the count in Definition 13.5.1 (3) does not depend on the morphism used, and its local structure near any point is the given sections transported by a local inverse of $t$; what the text does not say is that the index set must be nonempty, since the weights of a multisection sum to 1.

## XIII.13 Theorem 13.1.7 (2017: 13.1) as printed: properness in the full subcategory on $Q$

**The full subcategory on $Q$.** For $Q \subset X$ the morphisms between points of $Q$, with source and target `Chapter13/P3E_GoodSystemRelative · EpGroupoid.subSrc` `Chapter13/P3E_GoodSystemRelative · EpGroupoid.subTgt`. A subset $A \subset Q$ is proper in $Q$ if $t : s^{-1}(A) \to Q$ is proper in the subcategory `Chapter13/P3E_GoodSystemRelative · EpGroupoid.IsProperSubsetIn`. A family $(U(q))_{q\in Q}$ is a good system for $Q$ `Chapter13/P3E_GoodSystemRelative · EpGroupoid.IsGoodIn` if $U(q) \subset Q$, $\mathrm{cl}_Q U(q)$ is proper in $Q$ (Definition 13.1.2 (1) read in $Q$), and (3) holds for $q, q' \in Q$; conditions (2) and (3) are the same in $Q$ and in $X$, since a natural rep-

resentation of the ep-groupoid $Q$ is one of $X$. Every good system of $X$ is one for $Q = X$ Chapter13/P3E_GoodSystemRelative · EpGroupoid.GoodSystem.isGoodIn_univ, which with §XIII.10 is the witness.

**Properness by ultrafilters** (extra). A map defined on a subspace is proper iff every ultrafilter on the ambient space that lives on the subspace and whose image converges has a limit in the subspace mapping to the image limit Chapter13/P3E_GoodSystemRelative · EpGroupoid.isProperMap_iff_ultrafilter_of_isInducing; so properness of $\mathrm{cl}\, U$ in $X$ Chapter13/P3E_GoodSystemRelative · EpGroupoid.isProperSubset_iff_ultrafilter and in $Q$ Chapter13/P3E_GoodSystemRelative · EpGroupoid.isProperSubsetIn_iff_ultrafilter become statements about ultrafilters of morphisms. Two tools: for $|Q| = |X|$ every object has a morphism to a point of $Q$ Chapter13/P3E_GoodSystemRelative · EpGroupoid.exists_mor_tgt_mem; and through a morphism $\chi$ with target in an open $O$ there is a local section $\sigma$ of $s$ with $\sigma(s(\chi)) = \chi$ and targets in $O$ Chapter13/P3E_GoodSystemRelative · EpGroupoid.exists_localSection, the local inverse of the étale map $s$ cut down by the continuity of $t$.

**Step 1: the closure stays in $Q$** Chapter13/P3E_GoodSystemRelative · EpGroupoid.closure_subset_of_isProperSubsetIn. Let $Q$ be open with $|Q| = |X|$, $U \subset Q$, and $\mathrm{cl}_Q U$ proper in $Q$; then $\mathrm{cl}_X U \subset Q$. A point $z \in \mathrm{cl}_X U$ has a morphism $\chi : z \to q''$ with $q'' \in Q$; take $\sigma$ through $\chi$ with targets in $Q$. For $y'$ in $U$ near $z$, $\sigma(y')$ is a morphism of the subcategory with source in $U$ and target near $q''$; its targets converge in $Q$, so properness in $Q$ gives a limit of the morphisms $\sigma(y')$ in the subcategory, which is $\chi$; its source $z$ is then in $Q$.

**Step 2: properness in $X$** Chapter13/P3E_GoodSystemRelative · EpGroupoid.isProperSubset_of_isProperSubsetIn. Let an ultrafilter of morphisms $\varphi$ with sources in $\mathrm{cl}_X U = \mathrm{cl}_Q U$ have targets converging to some $y \in X$. Choose $\chi : y \to q''$ into $Q$ and $\rho$ a local section of $s$ through $\chi$ with targets in $Q$; then $\rho(t(\varphi)) \circ \varphi$ is a morphism of the subcategory whose targets converge to $q''$. Properness in $Q$ gives a limit $\psi$; then $\chi^{-1} \circ \psi$ is the limit of the $\varphi$, with target $y$. The converse, that properness in $X$ with $\mathrm{cl}_X U \subset Q$ gives properness in $Q$, holds as well Chapter13/P3E_GoodSystemRelative · EpGroupoid.isProperSubsetIn_of_isProperSubset, so Chapter13/P3E_GoodSystemRelative · EpGroupoid.isGoodIn_iff: for open $Q$ with $|Q| = |X|$, a family is a good system for $Q$ iff it is good on $Q$ in the sense of §XIII.4 and $\mathrm{cl}_X U(q) \subset Q$ for all $q$.

**Theorem 13.1.7, as printed** Chapter13/P3E_GoodSystemRelative · EpGroupoid.theorem_13_1_7_book: let $Q$ be open with $|Q| = |X|$ and $(U(q))_{q \in Q}$ a good system for the ep-groupoid $Q$. Given open $W(y) \ni y$ for $y \notin Q$, there is a good system of $X$ which keeps $U(q)$ for $q \in Q$ and has $U(y) \subset W(y)$ for $y \notin Q$. *Proof.* A good system for $Q$ is good on $Q$ in the sense of §XIII.4 Chapter13/P3E_GoodSystemRelative · EpGroupoid.IsGoodIn.isGoodOn, and the stronger Theorem 13.1.7 of §XIII.4 applies Chapter13/P3G_GoodSystemExistence · EpGroupoid.theorem_13_1_7. No hypothesis is added; $|Q| = |X|$ is the text's $|Q| = |Y|$.

## XIII.14 Definitions 13.2.7 and 13.2.8 (2017: 13.9 and 13.10): decorated sc⁺-multisections

**Symmetric structures on a neighbourhood** Chapter13/P3D_SymmetricOn · SBLocalization.SymmetricStructureOn. Definition 13.2.5 on a neighbourhood $N$ of $x$ with the natural action of §XIII.1: a finite index set $I$ with an action of $G_x$, sc⁺-sections $s_i$ on $N$ with $s_{g(i)}(g * y) = \mu(\Gamma(g, y), s_i(y))$, and $\Lambda(w) = |I|^{-1}\#\{i : s_i(P(w)) = w\}$ over $N$. The structures of §XIII.7 restrict to such Chapter13/P3D_SymmetricOn · SBLocalization.SymmetricStructure.toOn, so they exist at every point Chapter13/P3D_SymmetricOn · SBLocalization.exists_symmetricStructureOn (Lemma 13.2.6), and $I \neq \emptyset$ Chapter13/P3D_SymmetricOn · SBLocalization.SymmetricStructureOn.card_pos. Where $\Lambda(0_y) = 1$ all sections vanish Chapter13/P3D_SymmetricOn · SBLocalization.SymmetricStructureOn.s_eq_fiberZero. A structure restricts to a smaller neigh-

bourhood with the same action Chapter13/P3D_SymmetricOn · SBLocalization.SymmetricStructureOn.restrict; "the same action" is a relation between neighbourhoods Chapter13/P3D_SymmetricOn · EpGroupoid.NatNbhd.Le, and inside finitely many neighbourhoods of $x$ with natural actions there is one whose action agrees with all of them Chapter13/P3D_SymmetricOn · EpGroupoid.NatNbhd.exists_le_of_finite, because two natural representations at $x$ agree on the connected component of $x$ in the intersection of their domains (the uniqueness of natural representations, Part VII, §VII.9), and an invariant neighbourhood can be taken inside it.

**Definition 13.2.7** Chapter13/P3D_Decorated · SBLocalization.Decoration Chapter13/P3D_Decorated · SBLocalization.DecoratedTuple: an sc⁺-multisection $\Lambda$ with a good system $\mathcal{U}$ and, for every $x$, a symmetric structure on $U(x)$. The good system carries a natural action at each point (§XIII.1), and the properness clause of Definition 13.2.5 is Definition 13.1.2 (1). Every sc⁺-multisection over an ep-groupoid with paracompact orbit space has a decoration Chapter13/P3D_Decorated · SBLocalization.exists_decoration: a good system from Theorem 13.1.8, a symmetric structure at each point from Lemma 13.2.6, a good system inside their domains from Theorem 13.1.8, shrunk at each point to a neighbourhood where the action of the good system agrees with that of the structure Chapter13/P3D_Decorated · EpGroupoid.GoodSystem.ofSubset, and the structures restricted to it. The paracompactness is the text's, through Theorem 13.1.8.

**Definition 13.2.8** Chapter13/P3D_Decorated · SBLocalization.Decoration.Rel: $(\Lambda, \mathcal{U}, S) \sim (\Lambda', \mathcal{U}', S')$ if $\Lambda = \Lambda'$ and there are a good system $\mathcal{U}''$ with $U''(x) \subset U(x) \cap U'(x)$ and $G_x$-equivariant bijections $b_x : I_x \to I'_x$ with $s'_{b_x(i)} = s_i$ on $U''(x)$. Reflexivity and symmetry are direct Chapter13/P3D_Decorated · SBLocalization.Decoration.Rel.refl Chapter13/P3D_Decorated · SBLocalization.Decoration.Rel.symm; transitivity composes the bijections on a common refinement of the two intermediate good systems Chapter13/P3D_Decorated · SBLocalization.Decoration.Rel.trans Chapter13/P3D_Decorated · EpGroupoid.GoodSystem.exists_subset_inter, which exists by the refinement step after Proposition 13.1.3. The classes are the decorated sc⁺-multisections Chapter13/P3D_Decorated · SBLocalization.DecoratedTuple.equivalence Chapter13/P3D_Decorated · SBLocalization.DecoratedMultisection, with the map to the underlying functor Chapter13/P3D_Decorated · SBLocalization.DecoratedMultisection.forget.

## XIII.15 Definitions 13.2.9, 13.2.10, Lemmas 13.2.11, 13.2.13 (2017: 13.11, 13.12, 13.4, 13.5) decorated, and the remark that there are no inverses

**Moving sections by morphisms.** $\mu(\psi, v)$ for $v$ over $s(\psi)$ Chapter13/P3D_SymmetricOn · SBLocalization.fibAct is additive and homogeneous in the section Chapter13/P3D_SymmetricOn · SBLocalization.fibAct_sectionAdd Chapter13/P3D_SymmetricOn · SBLocalization.fibAct_sectionSmul Chapter13/P3D_SymmetricOn · SBLocalization.fibAct_sectionSum, since $\mu(\psi, \cdot)$ is linear on fibres. This is what makes the new families equivariant.

**Definition 13.2.9, decorated** Chapter13/P3D_Decorated · SBLocalization.definition_13_2_9_decorated. Given decorations of $\Lambda'$ and $\Lambda''$, the decoration of $\Lambda' \oplus \Lambda''$ Chapter13/P3D_Decorated · SBLocalization.Decoration.add has index sets $I'_x \times I''_x$ Chapter13/P3D_Decorated · SBLocalization.Decoration.add_I with the diagonal action, sections $s_{(i,i')} = s'_i + s''_{i'}$ Chapter13/P3D_Decorated · SBLocalization.Decoration.add_s_apply, and neighbourhoods inside $U'(x) \cap U''(x)$ Chapter13/P3D_Decorated · SBLocalization.Decoration.add_U_subset. *Reading.* The text takes $U(x) = U'(x) \cap U''(x)$. The two natural actions on it need not agree, and a symmetric structure needs one action for which both families are equivariant; so $U(x)$ is an open neighbourhood of $x$ inside the intersection on which the two actions agree (§XIII.14), and the class does not depend on this choice, since Definition 13.2.8 allows shrinking. The symmetric structure of the sum Chapter13/P3D_SymmetricOn · SBLocalization.SymmetricStructureOn.add: equivariance of $s'_i + s''_{i'}$ is that of the summands with the additivity of $\mu(\psi, \cdot)$, and the count is the con-

volution formula of §XIII.8. *Well defined on classes* Chapter13/P3D_Decorated · SBLocalization.Decoration.add_rel: equivalent summands give equivalent sums, with the product bijections $b'_x \times b''_x$. So $\oplus$ is defined on decorated sc⁺-multisections Chapter13/P3D_Decorated · SBLocalization.DecoratedMultisection.add, compatibly with forgetting Chapter13/P3D_Decorated · SBLocalization.DecoratedMultisection.forget_add, and commutative Chapter13/P3D_Decorated · SBLocalization.DecoratedMultisection.add_comm and associative Chapter13/P3D_Decorated · SBLocalization.DecoratedMultisection.add_assoc there, through the swap and reassociation bijections of index sets Chapter13/P3D_Decorated · SBLocalization.Decoration.add_comm_rel Chapter13/P3D_Decorated · SBLocalization.Decoration.add_assoc_rel.

**Definition 13.2.10 and Lemma 13.2.11, decorated.** $a\Lambda$ is decorated by the sections $as_i$ on the same good system and index sets Chapter13/P3D_Decorated · SBLocalization.Decoration.smul Chapter13/P3D_Decorated · SBLocalization.definition_13_2_10_decorated; for $a = 0$ the sections are $0$ and the count gives the zero multisection. For an sc-smooth functor $\beta : X \to \mathbb{R}$, $\beta\Lambda$ is decorated by $\beta s_i$ Chapter13/P3D_Decorated · SBLocalization.Decoration.smulFun Chapter13/P3D_Decorated · SBLocalization.lemma_13_2_11_decorated; equivariance uses $\beta(t(\psi)) = \beta(s(\psi))$. Both are compatible with Definition 13.2.8 Chapter13/P3D_Decorated · SBLocalization.Decoration.smulFun_rel and hence defined on classes Chapter13/P3D_Decorated · SBLocalization.DecoratedMultisection.smul.

**Lemma 13.2.13, decorated** Chapter13/P3D_Decorated · SBLocalization.lemma_13_2_13_decorated. For a locally finite family $(\Lambda_j)$ with decorations, the decoration of $\oplus_j \Lambda_j$ Chapter13/P3D_Decorated · SBLocalization.Decoration.lfSum has, at $x$, a neighbourhood inside $O(x) \cap \bigcap_{j \in J_x} U_j(x)$ on which the natural actions agree, index set $\prod_{j \in J_x} I_x^j$ and sections $\sum_{j \in J_x} s_{i_j}^{j,x}$ Chapter13/P3D_Decorated · SBLocalization.lfSumStructure, the symmetric structure of a finite product Chapter13/P3D_SymmetricOn · SBLocalization.SymmetricStructureOn.pi. Over $O(x)$ the indices outside $J_x$ contribute only the zero vector (§XIII.9). The intersection is read as for Definition 13.2.9. When $J_x = \emptyset$ the text leaves the natural action on $O(x)$ open; it is taken from a given good system $\mathcal{U}_0$, an argument of the construction (any $\mathcal{U}_j$, or Theorem 13.1.8).

**The remark after Definition 13.2.9: no inverses.** The zero multisection is neutral Chapter13/P3D_NoInverse · SBLocalization.multisectionAdd_zeroMultisection. On the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid Chapter13/P3D_NoInverse · SBLocalization.NoInverse.A, for $c \neq 0$ let $\Lambda_c$ give weight $\frac{1}{2}$ to the zero section and $\frac{1}{2}$ to the constant section $c$ Chapter13/P3D_NoInverse · SBLocalization.NoInverse.twoValued, an sc⁺-multisection Chapter13/P3D_NoInverse · SBLocalization.NoInverse.isScPlusMultisection_twoValued. **Then $\Lambda_c \oplus \Lambda' \neq \Lambda_0$ for every sc⁺-multisection $\Lambda'$** Chapter13/P3D_NoInverse · SBLocalization.NoInverse.remark_13_2_9_no_inverse. Over $x = 0$, $(\Lambda_c \oplus \Lambda')(0_x) = \frac{1}{2}\Lambda'(0) + \frac{1}{2}\Lambda'(-c)$, and since $0 \neq -c$ and the weights of $\Lambda'$ over $x$ sum to 1, this is at most $\frac{1}{2}$ Chapter13/P3D_NoInverse · SBLocalization.NoInverse.add_le_one_of_ne, while $\Lambda_0(0_x) = 1$. Such a $\Lambda_c$ exists when $k \geq 1$ Chapter13/P3D_NoInverse · SBLocalization.NoInverse.exists_no_inverse. "Inverse" is read with respect to the neutral element $\Lambda_0$.

## XIII.16 Definitions 13.3.1, 13.3.2, Example 13.3.3, Definition 13.3.4, Remark 13.3.5 (2017: Definitions 13.14, 13.15, Example 13.1, Definition 13.16, Remark 13.1): correspondences

**Definition 13.3.1** Chapter13/P3D_Correspondence · SBLocalization.IndexDiagram: a diagram $I_x \xleftarrow{a} I \xrightarrow{b} I_{x'}$ of finite sets, $a$ and $b$ surjective, all fibres of $a$ of one size and all fibres of $b$ of one size. Two diagrams are equivalent if a bijection $c : I \to I'$ makes both triangles commute Chapter13/P3D_Correspondence · SBLocalization.

IndexDiagram.Equiv Chapter13/P3D_Correspondence · SBLocalization.IndexDiagram.equivalence; $J(x,x')$ is the set of classes, the correspondences Chapter13/P3D_Correspondence · SBLocalization.IndexCorr. A bijection $e : I_x \to I_x$ gives the class $[\mathrm{id}, e]$ Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.ofPerm, and relabelling the two index sets by bijections acts on classes Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.map.

**Definition 13.3.2** (stronger) Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.swap: the pseudo-inverse $\sharp$ reverses the diagram; it is an involution Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.swap_swap and so a bijection $J(x,x') \to J(x',x)$ Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.swapEquiv. The bi-action $g' * [a,b] * g = [g^{-1} \circ a, g' \circ b]$ Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.biact is well defined, and the text's maps satisfy the action laws, which it does not state: $1 * \tau * 1 = \tau$ Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.biact_one_one and $g' * (h' * \tau * h) * g = (g'h') * \tau * (hg)$ Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.biact_biact, an action of $G_{x'} \times G_x^{\mathrm{op}}$.

**Example 13.3.3** Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.example_13_3_3: if $I_x = \{*\}$, then $g' * \tau * g = \tau$ for all $g, g'$. A bijection $c : I \to I$ with $b \circ c = g' \circ b$ is built fibre by fibre from the equal sizes of the fibres of $b$; and $a$ is constant.

**Definition 13.3.4** Chapter13/P3D_Structurable · SBLocalization.Decoration.IsCorrespondence: for a decoration and points $x, x'$, a map $\tau$ from the morphisms to $J(x,x')$ is a correspondence if on $\mathcal{U}(x,x')$ it is (1) locally constant with finitely many values, (2) equivariant, $\tau(g' * \varphi * g) = g' * \tau(\varphi) * g$ with the bi-action of the good system (§XIII.3), and (3) compatible: for every representative $I_x \xleftarrow{a} I \xrightarrow{b} I_{x'}$ of $\tau(\varphi)$, $s^{x'}_{b(k)}(t(\psi)) = \mu(\psi, s^{x}_{a(k)}(s(\psi)))$ for all $k$ and all $\psi$ in the component of $\varphi$ Chapter13/P3D_Structurable · SBLocalization.Decoration.Compat. The action of $G_x$ on $I_x$ is the one of the symmetric structure. **Remark 13.3.5** Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.sat_iff: for any relation $R$ on $I_x \times I_{x'}$, "$R(a(k), b(k))$ for all $k$" is invariant under equivalence of diagrams Chapter13/P3D_Correspondence · SBLocalization.IndexDiagram.sat_iff_of_equiv (the bijection $c$ only renames $k$), so it is a property of the class Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.Sat, held by $\tau$ iff by every representative; condition (3) is of this form.

## XIII.17 Definitions 13.3.6–13.3.11, Proposition 13.3.12, Remark 13.3.13, Lemma 13.3.14 (2017: Definitions 13.17–13.20, Remarks 13.2–13.4, Proposition 13.3, Lemma 13.6): structurable sc⁺-multisections

**Definition 13.3.6** Chapter13/P3D_Structurable · SBLocalization.Decoration.Tau: a structure on a decoration is a family of correspondences $\tau_{x,x'}$ for all $x, x'$ with (1) $\tau_{x,x}(\Gamma(g,y)) = [\mathrm{id}, g]$ and (2) $\tau_{x',x}(\varphi) = \tau_{x,x'}(\varphi^{-1})^{\sharp}$ for $\varphi \in \mathcal{U}(x',x)$. A decorated sc⁺-multisection is structurable if a representative carries one Chapter13/P3D_Structurable · SBLocalization.DecoratedMultisection.IsStructurable, and a functor is structurable if it is an sc⁺-multisection with a decoration carrying one Chapter13/P3D_Structurable · SBLocalization.IsStructurable, equivalently if it underlies a structurable decorated class Chapter13/P3D_Structurable · SBLocalization.isStructurable_iff. $\tau_{x,x'}$ is a function on all morphisms; only its values on $\mathcal{U}(x,x')$ enter. Witness: the zero multisection, with all index sets $\{*\}$ and all $\tau$ the unit diagram, over a paracompact orbit space Chapter13/P3D_Structurable · SBLocalization.isStructurable_zeroMultisection.

**Remark 13.3.7, Definitions 13.3.8, 13.3.10, 13.3.11.** The tuples $(\Lambda, \mathcal{U}, S, \tau)$ Chapter13/P3D_Structurable · SBLocalization.StructuredTuple are equivalent Chapter13/P3D_Structurable · SBLocalization.

StructuredTuple.Equiv if $\Lambda = \Lambda'$, a good system $\mathcal{U}''$ lies in $U(x) \cap U'(x)$, $I_x = I'_x$, and $\tau = \tau'$ on $\mathcal{U}''(x, x')$. The equality of index sets is an equality of types, and $\tau'$ is compared with $\tau$ after transport along it; this is an equivalence relation Chapter13/P3D_Structurable · SBLocalization.StructuredTuple.equivalence. As printed, the definition puts no condition on the sections; presumably they are meant to agree on $U''(x)$ as in Definition 13.2.8. It is formalized as printed, and nothing below uses the sections of an equivalent tuple. The classes are the structured sc⁺-multisections Chapter13/P3D_Structurable · SBLocalization.StructuredMultisection. Definition 13.3.11: $\Gamma^+_m(P)$, $\Gamma^+_s(P)$, $\Gamma^+_{st}(P)$ Chapter13/P3D_Structurable · SBLocalization.Γm Chapter13/P3D_Structurable · SBLocalization.Γs Chapter13/P3D_Structurable · SBLocalization.Γst, with $\Gamma^+_s \subset \Gamma^+_m$ and forget mapping $\Gamma^+_{st}$ onto $\Gamma^+_s$ Chapter13/P3D_Structurable · SBLocalization.StructuredMultisection.range_forget, which is the text's statement that every structurable $\Lambda$ lifts.

**Proposition 13.3.12** Chapter13/P3D_Structurable · SBLocalization.proposition_13_3_12: the sum of two structurable sc⁺-multisections is structurable. *The product of correspondences* Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.prod: $[a_1, b_1] \times [a_2, b_2] = [a_1 \times a_2, b_1 \times b_2]$; the fibres of a product map are products of fibres Chapter13/P3D_Correspondence · SBLocalization.IndexDiagram.card_fiber_prodMap, so they have equal sizes; the product commutes with ♯ Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.prod_swap, with the bi-action Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.prod_biact and with $[\mathrm{id}, g]$ Chapter13/P3D_Correspondence · SBLocalization.IndexCorr.prod_ofPerm. *"We may assume $\mathcal{U}_1 = \mathcal{U}_2$."* Both structures are restricted to the neighbourhoods of the decorated sum (§XIII.15), with the same actions Chapter13/P3D_Structurable · SBLocalization.Decoration.Tau.restrict; the bi-action does not change on a smaller neighbourhood with the same action Chapter13/P3D_Structurable · EpGroupoid.GoodSystem.biact_congr_of_le. *The structure of the sum* Chapter13/P3D_Structurable · SBLocalization.Decoration.Tau.prod Chapter13/P3D_Structurable · SBLocalization.Decoration.Tau.add is $\tau = \tau^1 \times \tau^2$, with index sets $I^1_x \times I^2_x$ and sections $s^1_i + s^2_j$. Local constancy, finiteness, equivariance and (1)–(2) hold factorwise; compatibility (3) is the text's computation, $\mu(\psi, s^1_{a_1(k)} + s^2_{a_2(l)}) = s^1_{b_1(k)} + s^2_{b_2(l)}$ by the additivity of $\mu(\psi, \cdot)$ (§XIII.15). For structured tuples the sum is again structured Chapter13/P3D_Structurable · SBLocalization.StructuredTuple.add, over $\Lambda_1 \oplus \Lambda_2$ Chapter13/P3D_Structurable · SBLocalization.StructuredTuple.add_Λ.

**Remark 13.3.13** (narrower): the structured sums $\Lambda_1 \oplus \Lambda_2$ and $\Lambda_2 \oplus \Lambda_1$ have the same underlying functor (`StructuredTuple.add_Λ` with the commutativity of §XIII.8); that they differ as structured classes, the index sets being $I^1_x \times I^2_x$ and $I^2_x \times I^1_x$, is a statement that two types are unequal, which has no formal statement.

**Lemma 13.3.14** Chapter13/P3D_Structurable · SBLocalization.lemma_13_3_14 Chapter13/P3D_Structurable · SBLocalization.lemma_13_3_14_fun: $r\Lambda$, for every $r \in \mathbb{R}$ including 0, and $\beta\Lambda$ for an sc-smooth functor $\beta$, are structurable when $\Lambda$ is. The same $\tau$ serves Chapter13/P3D_Structurable · SBLocalization.Decoration.Tau.smul Chapter13/P3D_Structurable · SBLocalization.Decoration.Tau.smulFun: compatibility (3) is homogeneity of $\mu(\psi, \cdot)$, with $\beta(t(\psi)) = \beta(s(\psi))$ for $\beta$.

## XIII.18 The construction before Proposition 13.3.15, Proposition 13.3.15, Definition 13.3.17, Corollary 13.3.18 (2017: Proposition 13.4, Definition 13.21, Corollary 13.1): tight structures

$T_\Lambda$ Chapter13/P3D_Tight · SBLocalization.trivialSet: the set of points near which $\Lambda$ is trivial, $\Lambda(0_y) = 1$; it is open Chapter13/P3D_Tight · SBLocalization.isOpen_trivialSet and $\Lambda$ is trivial on it Chapter13/P3D_

Tight · SBLocalization.isTrivialOn_trivialSet. Triviality at $s(\psi)$ and at $t(\psi)$ are the same, $\Lambda$ being a functor Chapter13/P3D_Tight · SBLocalization.apply_fiberZero_tgt.

**The construction.** For $x \in T_\Lambda$, shrink $U(x)$ to a neighbourhood with the same action inside $U(x) \cap T_\Lambda$ Chapter13/P3D_Tight · SBLocalization.Decoration.exists_tightNbhd Chapter13/P3D_Tight · SBLocalization.Decoration.tightNbhd, and put the single zero section on $I'_x = \{*\}$; for $x \notin T_\Lambda$ keep $U(x)$, $I_x$ and the sections. The index set is one type, $\{*\}$ when $x \in T_\Lambda$ and $I_x$ otherwise Chapter13/P3D_Tight · SBLocalization.TightIdx, with bijections $\{*\} \simeq I'_x$ Chapter13/P3D_Tight · SBLocalization.TightIdx.eT and $I_x \simeq I'_x$ Chapter13/P3D_Tight · SBLocalization.TightIdx.eN; the text's equalities $I'_x = \{*\}$, $I'_x = I_x$ are these bijections. The new decoration Chapter13/P3D_Tight · SBLocalization.Decoration.tight replaces $\tau$ by the unit diagrams when $x$ or $x'$ lies in $T_\Lambda$, and keeps it, relabelled, otherwise Chapter13/P3D_Tight · SBLocalization.Decoration.tightτ Chapter13/P3D_Tight · SBLocalization.Decoration.tightτ_TN Chapter13/P3D_Tight · SBLocalization.Decoration.tightτ_NN. **It is again structured** (stronger) Chapter13/P3D_Tight · SBLocalization.Decoration.Tau.tight. The text does not check condition (3) of Definition 13.3.4 for the new $\tau'$. It holds: at a morphism $\psi$ with $t(\psi)$ or $s(\psi)$ in $T_\Lambda$, both are, since triviality passes along morphisms, and every section of a symmetric structure vanishes where $\Lambda(0_y) = 1$ (§XIII.14), so both sides of (3) are $0$. Constant correspondences are correspondences Chapter13/P3D_Tight · SBLocalization.Decoration.isCorrespondence_const, and the unit diagrams satisfy (1)–(2) of Definition 13.3.6 because relabelling commutes with $\sharp$ and the bi-action Chapter13/P3D_Tight · SBLocalization.IndexCorr.swap_map Chapter13/P3D_Tight · SBLocalization.IndexCorr.biact_map.

**Proposition 13.3.15** Chapter13/P3D_Tight · SBLocalization.proposition_13_3_15: from a structured representative $(\Lambda, \mathcal{U}, S, \tau)$ one gets a structured $(\Lambda, \mathcal{U}', S', \tau')$ with (1) for $x \in T_\Lambda$: $I'_x = \{*\}$, the zero section, and $U'(x) \subset T_\Lambda$ with the action of $U(x)$; (2) for $x \notin T_\Lambda$: $U'(x) = U(x)$ and equivariant bijections $e_x : I_x \to I'_x$ with $s'_{e_x(i)} = s_i$; (3) $\tau' = \tau$ relabelled by $e_x$, $e_{x'}$, for $x, x' \notin T_\Lambda$; (4)–(5) the unit diagrams otherwise. It is the construction above Chapter13/P3D_Tight · SBLocalization.Decoration.isTight_tight; (3) holds for all morphisms.

**Definition 13.3.17** Chapter13/P3D_Tight · SBLocalization.Decoration.IsTight Chapter13/P3D_Tight · SBLocalization.StructuredMultisection.IsTight: a representative is tight if (1) $I_x = \{*\}$ with the zero section for $x \in T_\Lambda$, (2) $\tau_{x,x'}$ is $[* \leftarrow I_{x'} \xrightarrow{\mathrm{id}} I_{x'}]$, resp. $[I_x \xleftarrow{\mathrm{id}} I_x \to *]$, when exactly one of $x, x'$ lies in $T_\Lambda$, and (3) $[\{*\} \leftarrow \{*\} \to \{*\}]$ when both do. $I_x = \{*\}$ is a bijection $\{*\} \simeq I_x$, and (2)–(3) are stated for every such bijection (all are equal). **Corollary 13.3.18** Chapter13/P3D_Tight · SBLocalization.corollary_13_3_18 Chapter13/P3D_Tight · SBLocalization.corollary_13_3_18_structured: every structurable $\Lambda$ has a tightly structured version, and every element of $\Gamma_s^+$ underlies a tightly structured class.

## XIII.19 Definition 13.5.1, Theorems 13.5.4, 13.5.5 and 13.5.6 (2017: Definition 13.23, Theorems 13.7, 13.8, 13.9): atomic sc⁺-multisections

**Definition 13.5.1** Chapter13/P3A_Atomic · SBLocalization.AtomicData Chapter13/P3A_Atomic · SBLocalization.IsAtomic. Atomic data are: a point $x$ and a natural representation of $G_x$ on $U(x)$ with the properness property; an open invariant $V \ni x$ with $\overline{V} \subset U(x)$; a finite index set $I$ with a $G_x$-action; and sc⁺-sections $s_i$ on $U(x)$, equivariant, $s_{g(i)}(g * y) = \mu(\Gamma(g, y), s_i(y))$, and vanishing on $U(x) \setminus V$. An sc⁺-multisection $\Lambda$ is related to the data Chapter13/P3A_Atomic · SBLocalization.AtomicData.IsRelated if, for $w$ over a point joined by a morphism $\varphi$ to $U(x)$, $\Lambda(w) = |I|^{-1}\#\{i : s_i(t(\varphi)) = \mu(\varphi, w)\}$ Chapter13/P3A_Atomic · SBLocalization.AtomicData.count, and $\Lambda$ is trivial over the other points; it is atomic if it is related to some data. The count is taken at $t(\varphi)$, with $w' = \mu(\varphi, w)$. *The count does not depend on* $\varphi$ Chapter13/

P3A_Atomic · SBLocalization.AtomicData.count_μ_eq: two morphisms into $U(x)$ from the same point differ by some $\Gamma(g,\cdot)$, and equivariance permutes the indices by $g$ Chapter13/P3A_Atomic · SBLocalization.AtomicData.count_μ_Γ. So $\Lambda$ is determined by the data Chapter13/P3A_Atomic · SBLocalization.AtomicData.IsRelated.eq_mult; the data form a symmetric structure of $\Lambda$ at $x$ in the sense of Definition 13.2.5 Chapter13/P3A_Atomic · SBLocalization.AtomicData.IsRelated.symmetricStructure, so (2) holds as printed; and $I \neq \emptyset$ Chapter13/P3A_Atomic · SBLocalization.AtomicData.IsRelated.nonempty. Witness: the zero multisection is atomic at every point, with one index and the zero section Chapter13/P3A_Atomic · SBLocalization.isAtomic_zeroMultisection.

**The multisection of the data** Chapter13/P3A_Atomic · SBLocalization.AtomicData.mult is the formula of (3). It is a functor Chapter13/P3A_Atomic · SBLocalization.AtomicData.isWFunctor_mult and an sc⁺-multisection when $I \neq \emptyset$ Chapter13/P3A_Atomic · SBLocalization.AtomicData.isScPlusMultisection_mult. *Near a point $y$ joined to $U(x)$* Chapter13/P3A_Atomic · SBLocalization.AtomicData.exists_isLocalRepOn_transport: a morphism from a point of $U(x)$ to $y$ has a sheet on which $t$ has an sc-smooth local inverse $\tau$, with $t(\tau(z)) = z$ and $s(\tau(z)) \in U(x)$, and $s_i'(z) = \mu(\tau(z), s_i(s(\tau(z))))$ is a local structure with weights $|I|^{-1}$. *Away from such points:* the saturation of $\overline{V}$ is closed, by properness Chapter13/P3A_Atomic · SBLocalization.AtomicData.isClosed_saturation_closure_V, and off it the multisection is the zero multisection Chapter13/P3A_Atomic · SBLocalization.AtomicData.mult_eq_zeroMultisection, because the sections vanish off $V$. So the domain support lies in the saturation of $\overline{V}$ Chapter13/P3A_Atomic · SBLocalization.AtomicData.domSupp_mult_subset, and the multisection of the data is atomic Chapter13/P3A_Atomic · SBLocalization.AtomicData.isAtomic_mult.

**Symmetric families** Chapter13/P3A_AtomicConstruction · SBLocalization.SymmetricFamily: global sc⁺-sections of $W \to X$ supported in $U(x)$, with a $G_x$-action on the index set and equivariance on $U(x)$. *From one seed* Chapter13/P3A_AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed: for an sc⁺-section $s_0$ supported in an invariant $V$, $I = G_x$ acting by left multiplication and $s_g$ the translate of $s_0$ by $g$ (the terms of the average of §XII.7); $s_1 = s_0$ Chapter13/P3A_AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_one, the family vanishes off $V$ Chapter13/P3A_AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_vanish, and for an invariant auxiliary norm the translates have the bound of the seed Chapter13/P3A_AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_normW_le.

**Theorem 13.5.4 (Construction I).** Setting: $X$ with bump functions (pointwise reading), $x$ smooth, $U(x)$ with the natural action and the properness property, $V$ open invariant with $\overline{V} \subset U(x)$. *(1)* Chapter13/P3A_AtomicConstruction · SBLocalization.theorem_13_5_4_1: for a smooth $w \in W_x$ there is a symmetric family vanishing off $V$ with $s_{i_0}(x) = w$. The seed is an sc⁺-section through $w$ supported in $V$, from Part I's Lemma 5.3.3 with a Part I auxiliary norm Chapter5/BumpFunctions · lemma_5_3_3. *(2)* Chapter13/P3A_AtomicConstruction · SBLocalization.theorem_13_5_4_2: for an sc⁺-section $s$ on an open $O \ni x$ there is such a family with $s_{i_0} = s$ near $x$. The seed is $\beta s$ with $\beta \equiv 1$ near $x$ and supported in $V \cap O$ Chapter13/P3A_AtomicConstruction · SBLocalization.exists_cutoff, from Proposition 5.5.3; "$s_{i_0} = s$ near $x$" needs $\beta \equiv 1$ near $x$, not only $\beta(x) = 1$. The conclusion is stronger than printed: the sections are global sc⁺-sections vanishing off $V$, not only on $U(x) \setminus V$. The properness property, and in (2) the smoothness of $x$, are kept and unused. Narrower only through the pointwise reading of bump functions, as Theorem 12.3.3 (§XII.7).

**Theorem 13.5.5 (Construction II)** Chapter13/P3A_AtomicConstruction · SBLocalization.theorem_13_5_5: a symmetric family on $U(x)$ with nonempty index set determines a unique atomic $\Lambda$ that is trivial off the saturation of $U(x)$ and equals $|I|^{-1}\#\{i : s_i(P(w)) = w\}$ over $U(x)$. *Step 1,* $V(x)$ Chapter13/P3A_AtomicConstruction · SBLocalization.SymmetricFamily.exists_vanishing_nbhd: by normality an open set around $\{x\} \cup \bigcup_i \operatorname{supp} s_i$ with closure in $U(x)$, intersected over the translates by $G_x$, is invariant, and the family vanishes off it. *Step 2.* The data so obtained Chapter13/P3A_AtomicConstruction · SBLocalization.AtomicData.ofFamily give the multisection of the data, which is atomic, trivial off the saturation, and has the formula

over $U(x)$ Chapter13/P3A_Atomic · SBLocalization.AtomicData.mult_of_mem. *Uniqueness* is that $\Lambda$ is determined by its data. The added $I \neq \emptyset$ is implicit in the text: with $I = \emptyset$ the printed formula gives $\Lambda = 0$ over $U(x)$, which is not a multisection; for atomic $\Lambda$ it is automatic, and the seed families have $I = G_x$. "Supported in the saturation of $U(x)$" is read as triviality over the points not isomorphic to a point of $U(x)$. Bump functions and the smoothness of $x$ are not used. The last clause of the theorem, that $\Lambda$ is structurable, rests on Theorem 13.5.3 (§XIII.20).

**Theorem 13.5.6.** Setting: $|X|$ paracompact, bump functions (pointwise reading), an auxiliary norm $N$. *Part 1* Chapter13/P3A_AtomicConstruction · SBLocalization.theorem_13_5_6_1: for a smooth $w \neq 0$ over $x$ and $U(x)$ with the natural action, an atomic $\Lambda$ with domain support in the saturation of $U(x)$ and $\Lambda(\mu(g, w)) > 0$ for all $g \in G_x$: Construction I (1) through $w$, then Construction II; each $\mu(g, w)$ is the value of the section $s_g$ at $x$. *Part 2* Chapter13/P3A_AtomicConstruction · SBLocalization.theorem_13_5_6_2: for an sc⁺-section $s$ on $U(x)$ and $\varepsilon > 0$, an atomic $\Lambda$ supported there with $N(\Lambda)(y) \leq N(s(x)) + \varepsilon$ for all $y$, whose local structure near $x$ contains $s$ with positive weight, read as: a local structure on an open $O \ni x$ one of whose sections equals $s$ on $O$ Chapter13/P3A_AtomicConstruction · SBLocalization.AtomicData.isLocalRepOn_mult. The seed $\beta s$ is chosen with $N(\beta s) \leq N(s(x)) + \varepsilon$ near $x$; the translates keep the bound, because $N$ is $\mu$-invariant Chapter13/P3A_AtomicConstruction · SBLocalization.normW_μ, and the pointwise norm of the multisection is a maximum over the transported sections Chapter13/P3A_AtomicConstruction · SBLocalization.AtomicData.pointwiseNorm_mult_le. Both parts together Chapter13/P3A_AtomicConstruction · SBLocalization.theorem_13_5_6. Paracompactness is kept and unused: the text uses it for the existence of $N$, which is a hypothesis here. **Remark 13.5.2** is prose; Theorems 13.5.4–13.5.6 use bump functions only, no partitions of unity, as the remark says.

## XIII.20 Theorem 13.5.3 (2017: 13.6): atomic sc⁺-multisections are structurable, with the correspondences as a hypothesis

**The local alternative** Chapter13/P3A_AtomicStructurable · SBLocalization.AtomicData.IsLocallyTransported Chapter13/P3A_AtomicStructurable · SBLocalization.theorem_13_5_3_local: for atomic data of $\Lambda$, at every point $y$ either $\Lambda$ is trivial on an open $O \ni y$, or on an open $O \ni y$ there is an sc-smooth choice $\tau$ of morphisms with $t(\tau(z)) = z$ and $s(\tau(z)) \in U(x_0)$ such that $\Lambda$ is represented there by $s'_i(z) = \mu(\tau(z), s_i(s(\tau(z))))$, $i \in I$, with weights $|I|^{-1}$. This is the case distinction of the text's proof, with index set $I$ near points joined to $U(x_0)$ and $\{*\}$ elsewhere; it comes from the local structure of §XIII.19 and from the closedness of the saturation of $\overline{V}$.

**Theorem 13.5.3** (narrower) Chapter13/P3A_AtomicStructurable · SBLocalization.theorem_13_5_3_hyp: for a predicate "structurable" and the hypothesis that every sc⁺-multisection related to atomic data and locally transported at every point is structurable, every atomic $\Lambda$ is structurable. The hypothesis is the remaining step of the text's proof: that the transported structures fit together through correspondences. It was stated before Definition 13.3.6 was available to this block; the predicate of §XIII.17 can be substituted. It is satisfiable Chapter13/P3A_AtomicStructurable · SBLocalization.theorem_13_5_3_hyp_witness (with "structurable" read as "is an sc⁺-multisection"), a witness of consistency only. The last clause of Theorem 13.5.5 follows in the same form Chapter13/P3A_AtomicStructurable · SBLocalization.theorem_13_5_5_structurable_hyp. What remains is a new gap (§XIII.11): a good system refining the sets $O$, the $G_y$-actions on $I_y = I$ through the partner homomorphism of Proposition 13.1.9, and the correspondences from the sheet indices in $U(x_0)$.

# Part XV. Transversality and invariants (HWZ Chapter 15, §§15.1–15.3)

Chapter 15 (Springer 2021, pp. 683–703; 2017 numbers in parentheses) is the perturbation theory of sc-Fredholm section functors by sc$^+$-multisections. §15.1 defines the linearization $T(f,\Lambda)$ of $\Lambda\circ f$, a section of the formal sums GrF of sc-Fredholm operators (Theorem 15.1.1, Remark 15.1.2, Proposition 15.1.3, Lemmas 15.1.4, 15.1.5, Remark 15.1.6, Definition 15.1.7 (2017: Theorem 15.1, Remark 15.1, Proposition 15.1, Lemmas 15.1, 15.2, Definition 15.1; Remark 15.1.6 is new in the 2021 edition)). §15.2 treats transversality and local solution sets, §15.3 perturbations (general position, Theorem 15.3.7, and the transversal extension, Theorem 15.3.9), §15.4 orientations and invariants. This Part retells §15.1: the separation lemmas (§§XV.1–XV.2), the core of the proof of Theorem 15.1.1 (§XV.3), Theorem 15.1.1 on a strong bundle model and in a chart (§XV.4) and with intrinsic operators, with Definition 15.1.7 (§XV.5); then the witnesses (§XV.6), what is not formalized (§XV.7) and what the formalization taught us (§XV.8), then the sections $f-s_i$ on $W|_O$ (§XV.9), the sc-Fredholm summands (§XV.10), §15.2 (§§XV.11–XV.12), products and $[0,1]^n\times X$ (§XV.13), the definitions of §15.3 (§XV.14), Corollary 15.3.10 (2) (§XV.15) and the wave-2 witnesses (§XV.16).

**What is proved, and what is not.** Waves 1 and 2 of Part III (2026-11-13) formalize §15.1, §15.2 and the definitions of §15.3; the existence theorems of §15.3 and §15.4 are not yet formalized. *Theorem 15.1.1* is proved in what is its substance, the independence of $T(f,\Lambda)(x)$ from the local sc$^+$-section structure and from the chart, in a stronger form: at all level-1 points, for any sc-smooth $f$, and on the model for any weights (§§XV.4–XV.5). The sc-Fredholm property of its summands is proved in wave 2, read in a chart, for $f$ with Part I's tame germ data (§XV.10); gap G.145 is closed. The functor property $T(f,\Lambda)(t(\varphi))=\varphi_*T(f,\Lambda)(s(\varphi))$, which makes $T(f,\Lambda)$ a section functor, is not formalized (G.144). *Remark 15.1.2* and *Definition 15.1.7* are formalized (§§XV.4–XV.5). *Lemmas 15.1.4 and 15.1.5* are proved in a stronger, topological form (§XV.1), and *Remark 15.1.6* in an explicit form (§XV.2). *Proposition 15.1.3* is proved with the added hypothesis that the base point is not isolated. At an isolated point, for instance when the local model is a single point, Proposition 15.1.3 does not hold as we read it: there is no sequence $x_k\neq 0$ with $x_k\to 0$. This is a degenerate case, and Theorem 15.1.1 is unaffected, because the formal proof does not need the sequence (§XV.2). The proof of Theorem 15.1.1 is read in two places: the separation and the limit run in the level-1 topology, with the "weak topology" read as pointwise convergence of operators; and the final sum is presumably over the $i$ with $f(0)=s_i(0)$ (§XV.4). *§15.2.* Definitions 15.2.1 and 15.2.4 are formalized, the first with "nonzero" and "sc-Fredholm" proved automatic (§§XV.11–XV.12). Theorems 15.2.2 and 15.2.5 and Remark 15.2.3 are proved with Part I's tame germ data for $f$, inherited from Theorem 3.6.8, on which they rest; their "tame" submanifolds and "tame" branched ep$^+$-subgroupoids (Definition 9.3.6) are not formalized, Theorem 15.2.5 is not formalized un-

der the general position requirement, and the global form of Remark 15.2.3 needs a paracompact support (§§XV.11–XV.12). Remark 15.2.6 is prose. *§15.3.* The paragraph before Definition 15.3.1, Definitions 15.3.1, 15.3.3, 15.3.5, the example after Definition 15.3.3 and the first sentence of Remark 15.3.6 are formalized and proved (§XV.14). Corollary 15.3.10 (2) is proved, with explicit $\widetilde{N}$, $\widetilde{U}$, over a tame ep-groupoid $[0,1]^n \times X$ built for it, whose section $\tilde{f}$ is sc-Fredholm under the tame germ data of $f$ (§§XV.13, XV.15). Product ep-groupoids exist (§XV.13); product strong bundles and $\Lambda \times \Lambda'$ (the products remark and Remark 15.3.4) do not. The rest of Remark 15.3.6, Theorems 15.3.7 and 15.3.9 and Corollary 15.3.10 (1), (3) are not yet formalized; Remarks 15.3.2, 15.3.8 and 15.3.11 are prose. *§15.4* is not yet formalized. *Open:* G.144 and the new gaps of §XV.7. No statement of §§15.2–15.3 was found not to hold as we read it.

**Three structural choices.** (1) *Formal sums are finitely supported functions.* $\mathrm{GrF}(x)$ is the set of finitely supported functions from operators $T_xX \to W_x$ to $\mathbb{Q}^+$ `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.GrF`, defined at every level-1 point; that the operators are sc-Fredholm is not part of the type. On a strong bundle model the operators are $T_zO \to F$ `Chapter15/P3T_LinearizationModel · Linearization15.GrFModel`. (2) *The separation argument is pure topology.* Propositions and lemmas about finitely many continuous maps $s_i$ are proved in topological or (pseudo-)metric spaces, with values in a Hausdorff space, and applied on the level-1 subspace $U_1 \subset E_1$ of the model, where the sections and their derivatives are continuous. (3) *Pairs, not operators.* The proof compares formal sums of pairs $(s_i(y), (f-s_i)'(y))$, over all $i$; the sum of Theorem 15.1.1 is then the part with first entry $f(y)$. A finite formal sum of points is $\mathrm{wsum}(a,\sigma)(q) = \sum_{i:a_i=q} \sigma_i$ `Chapter15/P3T_SeparationLemmas · Linearization15.wsum`, and the mass of a set is the sum over the $i$ with $a_i$ in it `Chapter15/P3T_SeparationLemmas · Linearization15.mass` `Chapter15/P3T_SeparationLemmas · Linearization15.mass_eq_of_wsum_eq`.

**Map.**

- `Chapter15/P3T_SeparationLemmas`: formal sums of points, Lemmas 15.1.4, 15.1.5, Proposition 15.1.3 with its witness and the isolated point, Remark 15.1.6, the core lemma.
- `Chapter15/P3T_LinearizationModel`: GrF on a strong bundle model, Theorem 15.1.1 and Remark 15.1.2 there.
- `Chapter15/P3T_Linearization`: Theorem 15.1.1 in a strong bundle chart, Definition 15.1.7 in a chart, Remark 15.1.2.
- `Chapter15/P3T_LinearizationIntrinsic`: $\mathrm{GrF}(x)$ with intrinsic operators, independence of the chart, Theorem 15.1.1, Definition 15.1.7.
- `Chapter15/P3L_LocalScPlus`, `P3L_LocalIFT`, `P3L_FredholmSummands`: the sections $f - s_i$ and the sc-Fredholm summands (§§XV.9–XV.10).
- `Chapter15/P3L_Transversality`, `P3L_BoundaryPerturbation`, `P3L_Witness`: §15.2 (§§XV.11–XV.12).
- `Chapter15/P3P_ProductEp`, `P3P_CubeEp`: products and $[0,1]^n \times X$ (§XV.13).
- `Chapter15/P3P_GeneralPosition`, `P3P_CubeAuxNorm`: Definitions 15.3.1–15.3.5, Corollary 15.3.10 (2) (§§XV.14–XV.15).

Throughout, $(P : W \to X, \mu)$ is a strong bundle over an ep-groupoid, $f$ an sc-Fredholm section functor and $\Lambda$ an sc$^+$-multisection with local sc$^+$-section structures $(s_i, \sigma_i)$ in Chapter 11's sense `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn` (Part XIII, §XIII.6).

## XV.1 Lemmas 15.1.4 and 15.1.5 (2017: 15.1 and 15.2): separating finitely many maps

**Lemma 15.1.4** (stronger) `Chapter15/P3T_SeparationLemmas · Linearization15.lemma_15_1_4`: if a ball $B_\varepsilon(y)$ of a pseudo-metric space is the union of finitely many sets closed in it, one of them has nonempty interior. The text works in $\mathcal{O}$ with the metric of $E_0$ and moves a point step by step; the formal proof is an induction over the index set: a finite union of closed sets with empty interiors has empty interior `Chapter15/P3T_SeparationLemmas · Linearization15.interior_biUnion_eq_empty`, so a nonempty open set covered by finitely many closed sets meets the interior of one `Chapter15/P3T_SeparationLemmas · Linearization15.exists_nonempty_interior_of_finite_closed_cover`. No Baire property is needed for a finite union.

**Lemma 15.1.5** (stronger) `Chapter15/P3T_SeparationLemmas · Linearization15.lemma_15_1_5`: for finitely many maps continuous on $B_\varepsilon(y)$ into a Hausdorff space there are $z$ with $d(z,y) < \varepsilon$, $\delta > 0$ and a partition $I = I_1 \sqcup \cdots \sqcup I_p$ with (1) $B_\delta(z) \subset B_\varepsilon(y)$, (2) $s_i = s_{i'}$ on $B_\delta(z)$ within a piece, and (3) $s_i(u) \neq s_j(u)$ for $u \in B_\delta(z)$ across pieces. The maps need only be continuous; they need not be sc⁺-sections. *Topological form* `Chapter15/P3T_SeparationLemmas · Linearization15.exists_isOpen_dichotomy`: inside any nonempty open set there is a nonempty open $W$ on which each pair of maps either coincides or differs at every point. The proof goes through the pairs one at a time. For a pair, the set where $s_i \neq s_j$ is open in $W$ (Hausdorff values); if it is empty the two maps coincide on $W$, and otherwise $W$ is shrunk to it. This route does not need Lemma 15.1.4. The pieces are the classes of "coincide on $W$" `Chapter15/P3T_SeparationLemmas · Linearization15.exists_fin_partition`. A point is *locally dichotomous* when every pair coincides near it or differs at it `Chapter15/P3T_SeparationLemmas · Linearization15.IsLocDich`, and such points lie in every nonempty open set `Chapter15/P3T_SeparationLemmas · Linearization15.exists_isLocDich_mem`.

## XV.2 Proposition 15.1.3 (2017: 15.1) and Remark 15.1.6

**The separation property** at a point $z$ for a partition $e : I \to \{1,\dots,p\}$ `Chapter15/P3T_SeparationLemmas · Linearization15.IsSeparatedAt`: (1) $s_i(z) = s_{i'}(z)$ iff $e(i) = e(i')$; (2) then $s_i = s_{i'}$ near $z$.

**Proposition 15.1.3** (narrower) `Chapter15/P3T_SeparationLemmas · Linearization15.proposition_15_1_3_hyp`. Let $Z$ be a first countable $T_1$ space, $s_i$ finitely many maps continuous on an open $V \ni x_0$ with values in a Hausdorff space, and $x_0$ not isolated. Then there are $x_k \in V$, $x_k \neq x_0$, $x_k \to x_0$, and one partition $e$ with the separation property at every $x_k$. The text states it on the local model $\mathcal{O}$ with $x_0 = 0$. *Step 1.* For a countable decreasing basis $(B_k)$ of neighbourhoods of $x_0$ inside $V$, each $B_k \setminus \{x_0\}$ is open and nonempty, because $x_0$ is not isolated and $Z$ is $T_1$; it contains a locally dichotomous point $z_k$ (§XV.1), and $z_k \to x_0$. *Step 2, the subsequence.* There are finitely many patterns of coincidence; along a subsequence the pattern is constant, and it defines $e$ `Chapter15/P3T_SeparationLemmas · Linearization15.exists_isSeparatedAt_subseq`.

**The added hypothesis**, that $x_0$ is not isolated, is not derived from the text; its witness is two maps on $\mathbb{R}$ at $0$ `Chapter15/P3T_SeparationLemmas · Linearization15.proposition_15_1_3_hyp_witness`. **At an isolated point, Proposition 15.1.3 does not hold as we read it.** If $\{x_0\}$ is open there is no sequence $x_k \neq x_0$ with $x_k \to x_0$ `Chapter15/P3T_SeparationLemmas · Linearization15.not_proposition_15_1_3_of_isolated`; this happens for the one-point local model $\mathcal{O} = \{0\}$ of a zero-dimensional $X$ `Chapter15/P3T_SeparationLemmas · Linearization15.not_proposition_15_1_3_point`. The case is degenerate: there the statement of Theorem 15.1.1 holds directly, and the formal proof of Theorem 15.1.1 does not use Proposition 15.1.3 (§XV.3).

**Remark 15.1.6** (explicit). *First part* Chapter15/P3T_SeparationLemmas · Linearization15.remark_15_1_6: in a metric space, from any $x'_k \to x_0$ in $V$ there are $x''_k \in V$ with $d(x'_k, x''_k) < 1/(k+1)$ and a subsequence of $(x''_k)$ with the properties of Proposition 15.1.3; "slightly" is the bound $1/(k+1)$. *Second part* Chapter15/P3T_SeparationLemmas · Linearization15.IsSeparatedAt.eventually: the separation property at $z$ holds at all $z'$ near $z$, for maps continuous at $z$; so small moves of the points keep it.

## XV.3 The core of Theorem 15.1.1: two families with the same counts

**The limit step** Chapter15/P3T_SeparationLemmas · Linearization15.wsum_eq_of_frequently: two finite formal sums of points $a_i(z)$, $b_j(z)$ of a Hausdorff space, continuous at $x_0$, which agree for $z$ frequently near $x_0$, agree at $x_0$. Near $x_0$ the points $a_i(z)$ lie close to the finitely many $a_i(x_0)$; separating these by disjoint neighbourhoods, the weight of each neighbourhood is the weight of the corresponding point at $x_0$.

**At a locally dichotomous point** Chapter15/P3T_SeparationLemmas · Linearization15.wsum_pair_eq_at_of_isLocDich. Let $c_\alpha$, $\alpha \in I \sqcup K$, be the branches of two families with weights $\sigma$, $\tau$ and equal weighted counts of every value at $u$, and let $D_\alpha(u)$ depend only on the germ of $c_\alpha$ at $u$. If $u$ is locally dichotomous for the combined family, all branches through a value $v$ coincide near $u$, so they have one $D$, say $L_v$. Hence the formal sum of pairs gives $(v, L)$ the weight of $v$ if $L = L_v$ and $0$ otherwise Chapter15/P3T_SeparationLemmas · Linearization15.wsum_pair_of_const, the same for both families.

**The core lemma** Chapter15/P3T_SeparationLemmas · Linearization15.wsum_pair_eq_of_isLocal. Let both families be continuous on an open $V$, with equal weighted counts of every value at every point of $V$, and let the "derivatives" $D_\alpha$ be continuous on $V$ and depend only on germs. Then at every $x_0 \in V$ the formal sums $\sum \sigma_i \cdot (c_i, D_i)$ and $\sum \tau_j \cdot (c_j, D_j)$ agree. *Proof.* Locally dichotomous points of the combined family lie in every neighbourhood of $x_0$ (§XV.1); the sums agree there, and the limit step carries the equality to $x_0$. The text separates the two structures by forming their union and halving the weights; here Lemma 15.1.5 is applied to the combined family $I \sqcup K$ at once. The points used may include $x_0$ itself, so no sequence $x_k \neq x_0$ is needed, and isolated points are covered.

## XV.4 Theorem 15.1.1 (2017: 15.1) and Remark 15.1.2 (2017: 15.1) on a strong bundle model and in a chart

**The model.** $U \lhd F$ is a strong bundle model with retraction $R$, $O = r(U)$, and principal parts $\hat{f}$, $\hat{s}_i$ with $\hat{f} \circ r$, $\hat{s}_i \circ r$ sc-smooth on $U$. At a level-1 point $z$, $(\hat{f} - \hat{s}_i)'(z)$ is $D((\hat{f} - \hat{s}_i) \circ r)(z)$ restricted to $T_z O$, Part I's linearization in a chart, and $\sum_{i:\hat{s}_i(z)=\hat{f}(z)} \sigma_i \cdot (\hat{f} - \hat{s}_i)'(z)$ is the formal sum Chapter15/P3T_LinearizationModel · Linearization15.linSumModel.

**Two readings of the proof.** (a) The text takes the points $x_k$ of Proposition 15.1.3 in the metric of $E_0$ and passes to the limit of $T(f - s_i)(x_k)$ "in the weak topology". The derivative of an sc¹ map is continuous only on level 1, and there pointwise as an operator $E_0 \to F_0$ (Definition 1.1.15 (3)). Presumably the limit is meant in this sense: the separation runs on the level-1 subspace, where the sections are continuous, and the derivatives are continuous into the topology of pointwise convergence Chapter15/P3T_LinearizationModel · Linearization15.continuousOn_scDeriv_coe. The derivative at a level-1 point depends only on the values near it in the level-1 topology Chapter15/P3T_LinearizationModel · Linearization15.scDeriv_eq_of_eventuallyEq_lvl. (b) The last display sums $\sigma_i T(f - s_i)(0)$ over all $i$ and uses $f(0) = s_i(0)$; this holds only for some $i$, and the sum is presumably over $\{i : f(0) = s_i(0)\}$. The formal proof compares the

formal sums of the pairs $(\hat{s}_i(z), (\hat{f}-\hat{s}_i)'(z))$ over all $i$ and then selects those with first entry $\hat{f}(z)$.

**On the model** `Chapter15/P3T_LinearizationModel · Linearization15.wsum_model_eq`: if two structures $(\hat{s}_i, \sigma_i)$, $(\hat{t}_j, \tau_j)$ give the same weighted count of every value at every point of $O \cap N$, $N$ open, then the formal sums of pairs agree at every level-1 point over $O \cap N$. This is the core lemma on $U_1$ with $c_\alpha = \hat{s}_\alpha \circ r$ and $D_\alpha$ the derivative of $(\hat{f}-\hat{s}_\alpha)\circ r$; the germ condition holds by locality of the derivative. **Theorem 15.1.1 on the model** `Chapter15/P3T_LinearizationModel · Linearization15.theorem_15_1_1_model`: the formal sums $\sum_{i:\hat{s}_i=\hat{f}} \sigma_i (\hat{f}-\hat{s}_i)'$ agree at every level-1 point of $O \cap N$ (selection of the pairs with first entry $\hat{f}(z)$), also after pushing forward by any map of operators `Chapter15/P3T_LinearizationModel · Linearization15.theorem_15_1_1_model_map`. Neither $\Lambda$ being a functor nor positivity of the weights is used.

**Remark 15.1.2** (restated) `Chapter15/P3T_LinearizationModel · Linearization15.remark_15_1_2_model`. In the formalization $(f-s_i)'$ is defined for every $i$, and the sum is filtered by $s_i(y) = f(y)$, which is the text's point that only defined terms are counted. With positive weights the formal sum is 0 where the weighted count of $\hat{f}(y)$ is 0, so (1) follows from (2); and the operators with nonzero coefficient are those $(\hat{f}-\hat{s}_i)'(z)$ with $\hat{s}_i = \hat{f}$ `Chapter15/P3T_LinearizationModel · Linearization15.mem_support_linSumModel_iff`.

**In a strong bundle chart.** A local sc⁺-section on $O$ has an sc-smooth principal part $\hat{s} \circ r$ in every chart with domain in $O$ `Chapter15/P3T_Linearization · SBLocalization.IsScPlusSectionOn.isScSmooth_principal_comp_r`, and in the chart the weighted count of $v$ at $y$ is $\Lambda(\Phi^{-1}(y,v))$ `Chapter15/P3T_Linearization · SBLocalization.IsLocalRepOn.count_eq`. So two structures of the same $\Lambda$ have the same counts, and the model applies: **Theorem 15.1.1 in a chart** `Chapter15/P3T_Linearization · SBLocalization.theorem_15_1_1` `Chapter15/P3T_Linearization · SBLocalization.linSumChart`: for two structures of $\Lambda$ on open $O$, $O'$, the formal sums agree at every level-1 point of a chart with domain in $O \cap O'$, for any sc-smooth $f$ `Chapter15/P3T_Linearization · SBLocalization.linSumChart_eq`. Every object lies in such a chart `Chapter15/P3T_Linearization · SBLocalization.exists_chart_isLocalRepOn`. *Definition 15.1.7 in a chart* `Chapter15/P3T_Linearization · SBLocalization.linearizationIn` `Chapter15/P3T_Linearization · SBLocalization.linearizationIn_eq`, with Remark 15.1.2 `Chapter15/P3T_Linearization · SBLocalization.remark_15_1_2`.

## XV.5 Theorem 15.1.1 with intrinsic operators, and Definition 15.1.7 (2017: 15.1)

**Intrinsic operators.** From an operator $L$ of a chart $c$, $\Phi_x^{-1} \circ L \circ T\varphi : T_xX \to W_x$ `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.toIntr`, read through the chart operators `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.chartOp`. The formal sum through $c$ is $\sum_{i:f(x)=s_i(x)} \sigma_i \cdot (f-s_i)'(x) \in \mathrm{GrF}(x)$ `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.linSumIntr`. *The reduction "without loss of generality $X = O$"* is a change of chart. A global section equal to $f-s$ on $O$ `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.sectionSubOn` has an sc-smooth principal part in every chart with domain in $O$, and at a zero of $f-s$ the intrinsic operator $(f-s)'(x)$ is the same through any two such charts `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.toIntr_chart_independent`, by Part I's chain rule for linearizations. So the formal sum does not depend on the chart `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.linSumIntr_chart_independent`.

**Theorem 15.1.1** (narrower) `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.theorem_15_1_1_intrinsic`: for an sc-Fredholm section functor $f$ and two local structures of $\Lambda$ on open $O$, $O'$, the formal sums in $\mathrm{GrF}(x)$ agree, through any two charts with domains in $O$ and $O'$, at every $x \in X_1$ in both domains. The two structures are compared through a common restricted chart (§XV.4), and each through the change of

chart. The theorem is stated for sc-Fredholm $f$, as in the text, and uses only that $f$ is sc-smooth.

**Definition 15.1.7** (restated) `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.linearization`. *Local data* at $x$ are a local structure of $\Lambda$ on an open $O$ and a chart around $x$ with domain in $O$ `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.LocalLinData`, with their formal sum `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.LocalLinData.sum`. $T(f,\Lambda)(x)$ is the sum of some local data, and $0$ if there are none. Data exist for an sc⁺-multisection `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.nonempty_localLinData`, and any data give the value `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.linearization_eq`; this is Theorem 15.1.1 (2). Theorem 15.1.1 (1) and Remark 15.1.2: $T(f,\Lambda)(x) = 0$ where $\Lambda(f(x)) = 0$, that is, off $\operatorname{supp}(\Lambda \circ f)$ `Chapter15/P3T_LinearizationIntrinsic · SBLocalization.linearization_eq_zero`. The definition is made at every level-1 point; it is not yet shown to be a section functor (§XV.7).

## XV.6 Witnesses and the hypotheses

**Hypotheses taken.** One: in Proposition 15.1.3, that $x_0$ is not isolated, not derived from the text, with witness `proposition_15_1_3_hyp_witness` (two maps on $\mathbb{R}$ at 0); without it the proposition does not hold as we read it (§XV.2). Theorem 15.1.1 takes no added hypothesis and holds at isolated points as well. It takes sc-Fredholm $f$ as the text does and uses only sc-smoothness.

**Every definition has a witness.** $\mathrm{GrF}(x)$ and $T(f,\Lambda)(x)$ exist at every level-1 point of a strong bundle with an sc⁺-multisection (`nonempty_localLinData`); sc⁺-multisections exist, for instance the zero multisection (Part XIII, §XIII.6). The separation property: the witness of Proposition 15.1.3.

## XV.7 What is not formalized

- *Closed in wave 2.* G.145, the sc-Fredholm property of the summands of $T(f,\Lambda)(x)$, in the chart reading and with tame germ data (§XV.10).
- Gap G.144: the functor property of $T(f,\Lambda)$. Present: $T(f,\Lambda)(x)$ at every $x \in X_1$, its summands, and wave 1's action of a morphism through the local strong bundle isomorphism of its sheet (Part XII, §XII.10). Missing: (i) $\varphi_*$ on GrF, $L \mapsto \mu(\varphi,\cdot) \circ L \circ (T\varphi)^{-1}$ on formal sums (about 40 lines); (ii) the transport of a local section structure along $t \circ s^{-1}$ near $\varphi$ (about 120 lines); (iii) the bridge from the chart operators of §XV.5 to the chart points of §XII.10 and the chain rule under the local strong bundle isomorphism (about 200–300 lines).
- Gap G.150: tame solution sets and tame branched ep⁺-subgroupoids in §15.2 (Remark 15.2.3, Theorem 15.2.5). The $M_i$ are submanifolds in the sense of Definition 4.2.1 and the local branching structure is given (§§XV.11–XV.12); the tameness of Definition 9.3.6, with the properness property, is not; about 150–300 lines.
- Gap G.151: Theorem 15.2.5 under general position (its closing sentence). Missing is Part I's step "onto on $T^R_x X$ □ good position", applied to $(f - s_i)|_O$; about 150–250 lines.
- Gap G.152: the restricted ep-groupoid $U(x)$ of Remark 15.2.3. The local branching structure is given on $X$ at the support points in $U(x)$; about 80 lines, not needed for §§15.3–15.5 as far as we can see.
- Gap G.160: product strong bundles $W \times W' \to X \times X'$, $f \times f'$, $N \times N' = \max$, $\Lambda \times \Lambda'$ and the admissibility of products (the remark after Definition 15.3.3, Remark 15.3.4). Product ep-groupoids exist (§XV.13); about 500–1,000 lines.

- Gap G.161: Remark 15.3.6 after its first sentence (general position gives good position at $\partial X$; $\Lambda \circ f$ is a tame branched ep$^+$-subgroupoid); about 150–300 lines on top of §§XV.11–XV.12.
- Gap G.162: Theorems 15.3.7, 15.3.9 and Corollary 15.3.10 (1), (3), the existence of admissible structurable perturbations in general position and the transversal extension. They need Theorem 13.5.6 (Part XIII, §XIII.19), §15.2, Theorem 14.1.9 (Chapter 14, not formalized) and partitions of unity; the data over $[0,1] \times X$ for (3) are ready (§§XV.13, XV.15).
- Gap G.159: Theorem 15.4.1 and Definition 15.4.2, the canonical oriented lift of $T(f,\Lambda)$. Missing: the category of formal sums of oriented sc-Fredholm operators, the orientation of $(f-s)'(y)$ induced by $o$, independence of the local structure, and continuity along $M_i$, which needs G.138 (Part XII, §XII.13); about 1,000 lines after G.138.
- Not yet formalized: Theorem 15.4.3, Corollary 15.4.4, Remark 15.4.5. Remarks 15.2.6, 15.3.2, 15.3.8 and 15.3.11 are prose.

## XV.8 What the formalization taught us

The proof of Theorem 15.1.1 is a statement about finitely many continuous maps and their germs, and it holds in that generality: two finite families of branches with the same weighted count of every value at every point have the same formal sums of (value, germ-invariant) pairs, at every point, as soon as the germ invariants vary continuously. Lemmas 15.1.4 and 15.1.5, Proposition 15.1.3 and Remark 15.1.6 are pure topology. Written this way two places of the text needed a reading. The derivative of an sc$^1$ map is continuous only on level 1, and there pointwise; so the argument runs in the level-1 topology, where it holds equally, and the text's "weak topology" is pointwise convergence. And the final sum of the text runs over all branches, as if every $s_i(0)$ equalled $f(0)$; comparing pairs (value, derivative) for all branches and then selecting the value $f(0)$ avoids this. Applying the separation to the two families together, rather than to their union with halved weights, gives the comparison in one step. Proposition 15.1.3 asks for a sequence $x_k \neq 0$ converging to 0, which a one-point local model does not have; at an isolated point the proposition does not hold as we read it. Nothing depends on it: the core lemma needs locally dichotomous points in every neighbourhood of $x_0$, and $x_0$ may be one of them. The step "without loss of generality $X = O$" is, formally, the independence of the intrinsic operator $(f-s)'(x)$ from the chart, which Part I's chain rule supplies at a zero of $f - s$.

§15.2 is Part I's implicit function theorem applied branch by branch, and its formal content is the passage from a local sc$^+$-section $s_i$ on $O$ to a global section problem: $s_i$ is a global sc$^+$-section of the restricted bundle $W|_O$, and $(f-s_i)|_O$ is an sc-Fredholm section there, so Part I's Theorems 3.1.18 and 3.6.8 apply. This also gives the sc-Fredholm property of the summands of $T(f,\Lambda)$, which wave 1 left open; the operators $T_xX \to W_x$ carry no sc-structure of their own, so the property is read in a chart. The price is the hypothesis of Part I's Theorem 3.6.8: tame germ data, without which that theorem does not hold as we read it (Part III, §III.65). It is inherited, not new, and it is the same hypothesis that §12.5 and the cylinder $[0,1] \times X$ carry. Definition 15.2.1 asks for less than it says: "nonzero" and "sc-Fredholm" are automatic, so transversality is the surjectivity of every summand; and the proof of Theorem 15.2.2, which writes $f(x) = s_i(x)$ for all $i$, presumably means the branches through $f(x)$, the others having empty solution sets near $x$. In §15.3 the cylinder needed more than the text says: the product with $[0,1]$ is built as the product with Part I's cube $[0,1]^n$, whose parametrized Fredholm theory is available, and the auxiliary norm of Corollary 15.3.10 (2), which the text does not give, is an explicit interpolation $\max(\alpha(t)N^0, \beta(t)N^1)$.

## XV.9 Local sc⁺-sections and the sections $f - s_i$ on $W|_O$ (prerequisite for §15.2)

The proofs of Theorems 15.2.2 and 15.2.5 apply the implicit function theorem to $f - s_i$ near $x$, where $s_i$ is a section of a local structure, defined only on an open $O$. The formalization makes $f - s_i$ a section problem of Part I.

**Restriction to $O$.** A local sc⁺-section on an open $O$ is a global sc⁺-section of the restricted strong bundle $W|_O$ `Chapter15/P3L_LocalScPlus · SBLocalization.IsScPlusSectionOn.restrictOpen`; in every chart with domain in $O$ its principal part is sc-smooth after the retraction `Chapter15/P3L_LocalScPlus · SBLocalization.IsScPlusSectionOn.isScSmoothRetractMap_localRepOne_of_isChart`. The section $f + (-1)s$, with $s$ extended by $0$ off $O$ `Chapter15/P3L_LocalScPlus · SBLocalization.subSec`, restricts to a section of $W|_O$ `Chapter15/P3L_LocalScPlus · SBLocalization.subRestrict`, and it is a tame sc-Fredholm section there when $f$ has Part I's tame germ data `Chapter15/P3L_LocalScPlus · SBLocalization.isScFredholmSectionTame_subRestrict`: restriction to an open set keeps tame germ data, and an sc⁺-perturbation of a tame sc-Fredholm section is one, by Part I's Theorem 3.1.18 `Chapter3/TameGermsAddG · theorem_3_1_18_tame`. Its zeros are the points with $f(y) = s(y)$ `Chapter15/P3L_LocalScPlus · SBLocalization.isZeroAt_subSec_iff`.

**The linearization in a chart.** At a level-1 point $z$ of a chart with domain in $O$, $D((\hat{f} - \hat{s}) \circ r)(z)$ `Chapter15/P3L_LocalIFT · SBLocalization.subDeriv`; "onto" means onto $K_z$ from $T_zO$ `Chapter15/P3L_LocalIFT · SBLocalization.ChartOnto`. The intrinsic operator $(f-s)'(x)$ of §XV.5 is onto $W_x$ iff the chart operator is onto in this sense `Chapter15/P3L_LocalIFT · SBLocalization.surjective_toIntr_subDeriv_iff`, and this is Part I's surjectivity of the linearization of $(f-s)|_O$ `Chapter15/P3L_LocalIFT · SBLocalization.isLinSurjAt_subRestrict` `Chapter15/P3L_LocalIFT · SBLocalization.chartOnto_of_isLinSurjAt_subRestrict`, since the restricted chart has the same principal parts `Chapter15/P3L_LocalIFT · SBLocalization.scDeriv_restrictTo_subRestrict`.

**Local solution sets.** At an interior point $x$ ($d_X(x) = 0$) with $f(x) = s(x)$ and $(f-s)'(x)$ onto `Chapter15/P3L_LocalIFT · SBLocalization.exists_solutionSet_interior`, and at a point where $(f - s)|_O$ is in good position (Part I's Definition 3.6.6) `Chapter15/P3L_LocalIFT · SBLocalization.exists_solutionSet_of_isGoodPosition`, there is an open $V \ni x$ such that $\{y \in V : f(y) = s(y)\}$ is a smooth submanifold (Definition 4.2.1) of smooth points, at whose points the linearization is onto, and in the second case in good position. Both come from Part I's Theorem 3.6.8 for $(f - s)|_O$ `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8'`, read back in $X$.

## XV.10 Theorem 15.1.1 (2017: 15.1): the summands are sc-Fredholm

**The chart operator is sc-Fredholm** `Chapter15/P3L_FredholmSummands · SBLocalization.subDeriv_isScFredholm`. Let $f$ have tame germ data, $s$ be sc⁺ on an open $O$ containing the domain of the chart, $f(x) = s(x)$, and $z$ the chart point over $x$. Then $D((\hat{f} - \hat{s}) \circ r)(z)$ is an sc-operator, $T_zO$ and $K_z$ are sc-subspaces, and its restriction $T_zO \to K_z$ is sc-Fredholm, Fredholm on every level with the index of level $0$. The proof is Part I's statement that the linearization of an sc-Fredholm section at a zero is sc-Fredholm `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm`, applied to $(f - s)|_O$ (§XV.9).

**Theorem 15.1.1 (2), the summands** (restated) `Chapter15/P3L_FredholmSummands · SBLocalization.linearization_summand_isScFredholm`: for $x \in X_1$ and any local data (a local structure and a chart, §XV.5), every operator in the support of $T(f, \Lambda)(x)$ is $(f - s_i)'(x)$ for some $i$ with $f(x) = s_i(x)$, and its chart operator is sc-Fredholm as above. *Reading.* $T_xX$ and $W_x$ carry no sc-structure in the project; "sc-Fredholm $T_xX \to W_x$" is read through the identifications $T_xX \cong T_zO$ and $W_x \cong K_z$ that define the intrinsic operators.

GrF is not cut down to a subtype; the lemma says what lies in the support. The tame germ data are Part I's (below, §XV.16).

## XV.11 Definition 15.2.1, Theorem 15.2.2 and Remark 15.2.3 (2017: Definition 15.2, Theorem 15.2, Remark 15.2): transversality and interior local perturbation

**Definition 15.2.1** (restated). $T(f,\Lambda)(x)$ is onto at $x \in X_1$ if every operator in its support is surjective `Chapter15/P3L_Transversality · SBLocalization.IsOntoAtLevel` `Chapter15/P3L_Transversality · SBLocalization.IsOntoAt`; $(f,\Lambda)$ is transversal at $x$ if $\Lambda(f(x)) > 0$ and $T(f,\Lambda)(x)$ is onto `Chapter15/P3L_Transversality · SBLocalization.IsTransversalAt`, and transversal if it is so at every point of $\operatorname{supp}(\Lambda \circ f)$ `Chapter15/P3L_Transversality · SBLocalization.IsTransversal`. In the text's terms `Chapter15/P3L_Transversality · SBLocalization.isOntoAtLevel_iff`: onto iff, for any local data, every $(f - s_i)'(x)$ with $s_i(x) = f(x)$ is surjective, because the support of the formal sum consists of exactly these operators `Chapter15/P3L_Transversality · SBLocalization.mem_support_linSumIntr_iff`. *Reading.* The text asks for a nonzero combination of surjective sc-Fredholm operators. "Nonzero" is automatic where $\Lambda(f(x)) > 0$ `Chapter15/P3L_Transversality · SBLocalization.linearization_ne_zero`, and "sc-Fredholm" by §XV.10; so the definition asks only for surjectivity. $T(f,\Lambda)$ lives on $X_1$, and under tame germ data the support of $\Lambda \circ f$ consists of smooth points `Chapter15/P3L_Transversality · SBLocalization.mem_smoothSet_of_pos`, since $f - s_i$ is regularizing.

**Theorem 15.2.2** (narrower) `Chapter15/P3L_Transversality · SBLocalization.theorem_15_2_2_hyp`. Let $f$ be an sc-Fredholm section functor with tame germ data, $\Lambda$ an sc$^+$-multisection, $d_X(x) = 0$ and $(f,\Lambda)$ transversal at $x$. Then there are a natural representation of $G_x$ on $U(x)$, a local structure $(s_i, \sigma_i)$ on an open $O$ and a chart with $U(x) \subset \mathrm{dom} \subset O$ such that, with $M_i = \{y \in U(x) : f(y) = s_i(y)\}$: (1) $d_X = 0$ on $U(x)$; (2) $T(f,\Lambda)$ is onto at every $y \in U(x)$ with $\Lambda(f(y)) > 0$; (a) each $M_i$ is a smooth submanifold; (b) properly embedded, the inclusion being a proper map; (c) $\{y \in U(x) : \Lambda(f(y)) > 0\} = \bigcup_i M_i$; (d) $\Lambda(f(y)) = \sum_{i:f(y)=s_i(y)} \sigma_i$ on $U(x)$; (e) $T(f,\Lambda)(y) = \sum_{f(y)=s_i(y)} \sigma_i (f - s_i)'(y)$ in the chart; and only the branches with $s_i(x) = f(x)$ meet $f$ in $U(x)$. *Proof. Step 1.* A local structure on an open $O \ni x$ and a chart inside it (§XV.5). *Step 2, separating the branches.* The branches with $s_i(x) \neq f(x)$ differ from $f$ on an open neighbourhood of $x$ `Chapter15/P3L_Transversality · SBLocalization.isOpen_sep_ne`, by continuity of local sc$^+$-sections `Chapter15/P3L_Transversality · SBLocalization.IsScPlusSectionOn.continuousOn`. *Step 3, each branch through $f(x)$.* By transversality $(f - s_i)'(x)$ is onto, and §XV.9 gives an open $V_i \ni x$ on which $\{f = s_i\}$ is a smooth submanifold with onto linearizations `Chapter15/P3L_Transversality · SBLocalization.exists_branch_interior`. *Step 4.* $\{d_X = 0\}$ is open around $x$; a natural representation inside all these sets and inside $O$, with the properness property (Theorem 7.1.13), gives $U(x)$. (c) and (d) are the local structure; (e) is Definition 15.1.7 with these data (§XV.5); properness of the inclusions is closedness of $\{f = s_i\}$ in $U(x)$. *Readings.* The index set is that of the local structure; branches with $s_i(x) \neq f(x)$ give $M_i = \emptyset$ (the last clause), where the text's sentence "by assumption, $f(x) = s_i(x)$" presumably restricts to the branches through $f(x)$. "Natural smooth manifold structure" in (a) is Definition 4.2.1 inside $\{d_X = 0\}$, so (a)'s "$d_{M_i} \equiv 0$" is (1) with Definition 4.2.1. (e) is read in one chart, which is no restriction (§XV.5). The added hypothesis is tame germ data (§XV.16); witness of all hypotheses together `Chapter15/P3L_Witness · SBLocalization.theorem_15_2_2_hyp_witness`.

**Remark 15.2.3** (narrower). *Locally* `Chapter15/P3L_Transversality · SBLocalization.remark_15_2_3_hyp`: at $y$ with $d_X(y) = 0$ and $(f,\Lambda)$ transversal at $y$, $z \mapsto \Lambda(f(z))$ has a local branching structure in the sense of Defi-

nition 9.1.2 (2)–(4) (Part IX, §IX.1) at $y$, with the $M_j$ of Theorem 15.2.2 for the branches through $f(y)$ and the weights $\sigma_j$. By (2) of the theorem every support point in $U(x)$ satisfies these hypotheses. *Globally* `Chapter15/P3L_Transversality · SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp`: if $(f, \Lambda)$ is transversal, $d_X = 0$ on $\mathrm{supp}(\Lambda \circ f)$ and the orbit space of the support is paracompact, then $\Lambda \circ f$ is a branched ep$^+$-subgroupoid. The paracompactness is Definition 9.1.2 (1), new in the 2021 edition, which the remark does not address. Witness `Chapter15/P3L_Witness · SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp_witness`, with the paracompactness from Part IX `Chapter9/Propositions · EpGroupoid.paracompactSpace_orbitSupp_trivial_rn`. Not formalized: "tame" (Definition 9.3.6), and the statement on the restricted ep-groupoid $U(x)$ rather than on $X$ (new gaps, §XV.7).

## XV.12 Definition 15.2.4, Theorem 15.2.5 and Remark 15.2.6 (2017: Definition 15.3, Theorem 15.3, Remark 15.3): good position and boundary local perturbation

**Definition 15.2.4** (restated). For one branch `Chapter15/P3L_BoundaryPerturbation · SBLocalization.IsBranchGoodPosition`: $(f - s)|_O$ is in good position at $x$ in the sense of Part I's Definition 3.6.6 on $W|_O$, which gives (1) tameness near $x$ in a tame chart and (3) a linearization onto with kernel in good position to the partial quadrant. $(f, \Lambda)$ is in good position at $x$ `Chapter15/P3L_BoundaryPerturbation · SBLocalization.IsGoodPositionAt` if $\Lambda(f(x)) > 0$, $T(f, \Lambda)(x)$ is onto, and for some local structure on an open $O \ni x$ every branch through $f(x)$ is in good position. The text calls this "a version of Definition 3.6.6"; it is read per branch, for one local structure (the text does not say whether it depends on the structure), and $x \in \partial X$ is not required.

**Theorem 15.2.5** (narrower) `Chapter15/P3L_BoundaryPerturbation · SBLocalization.theorem_15_2_5_hyp`: for $f$ with tame germ data and $(f, \Lambda)$ in good position at $x$, the data of Theorem 15.2.2 exist with: the $M_i$ smooth submanifolds, (1) properly embedded, $\{y \in U(x) : \Lambda(f(y)) > 0\} = \bigcup_i M_i$, (2) $(f, \Lambda)$ in good position, with the same local structure, at every $y \in U(x)$ with $\Lambda(f(y)) > 0$, (3) $\Lambda(f(y)) = \sum_{y \in M_i} \sigma_i$, the formula for $T(f, \Lambda)$, and only branches through $f(x)$ meet. The proof is that of Theorem 15.2.2 with the second local solution lemma of §XV.9 per branch `Chapter15/P3L_BoundaryPerturbation · SBLocalization.exists_branch_goodPosition`: good position is open along the solution set (Part I), so it persists at nearby solutions. Not formalized: "tame submanifolds" and "tame branched" (Definition 9.3.6; Definition 4.2.1 allows boundary and corners), and the closing variant with $T^R_y X$ (new gaps, §XV.7). **Remark 15.2.6** is prose pointing to §15.4; the data it needs, the $M_i$ and the surjective $(f - s_i)'(y)$, are those of Theorems 15.2.2 and 15.2.5.

## XV.13 Product ep-groupoids, and $[0, 1]^n \times X$ (prerequisites for §15.3)

**Products** (extra) `Chapter15/P3P_ProductEp · EpGroupoid.prod`. The product $X \times X'$ of ep-groupoids has componentwise structure maps `Chapter15/P3P_ProductEp · GroupoidStr.prod`; the étale property comes from products of local sc-diffeomorphisms `Chapter15/P3P_ProductEp · MPolyfold.IsLocalScDiffeo.prodMap`, properness from products of proper maps `Chapter15/P3P_ProductEp · isProperMap_domRestrict_prod`, and sc-smooth composition is checked in its local sheet form `Chapter15/P3P_ProductEp · EpGroupoid.prod_isScSmoothCompSheet`. The projections are functors `Chapter15/P3P_ProductEp · EpGroupoid.prodFst`, tameness passes to products `Chapter15/P3P_ProductEp · EpGroupoid.isTame_prod`, $d_{X \times X'} = d_X + d_{X'}$ `Chapter15/P3P_ProductEp · EpGroupoid.degIndex_prod`, and two objects are isomorphic iff their components are `Chapter15/P3P_ProductEp · EpGroupoid.prod_π_eq_π_iff`.

**The cylinder** Chapter15/P3P_CubeEp · EpGroupoid.cubeProd: $[0,1]^n \times X$, the product of the trivial ep-groupoid of Part I's cube with $X$; the text's $[0,1] \times X$ is $n = 1$. The cube is used because Part I's parametrized Fredholm theory is stated for it. The projection to $X$ Chapter15/P3P_CubeEp · EpGroupoid.cubeProj and the inclusions at $q \in [0,1]^n$ Chapter15/P3P_CubeEp · EpGroupoid.cubeIncl are functors; the orbit space is $[0,1]^n \times |X|$ in the sense that the orbit map is continuous Chapter15/P3P_CubeEp · EpGroupoid.continuous_cubeOrbitMap and compact sets of $|X|$ give compact sets Chapter15/P3P_CubeEp · EpGroupoid.isCompact_image_π_cubeProd. It is tame Chapter15/P3P_CubeEp · EpGroupoid.isTame_cubeProd; for $n = 1$ and $d_X \equiv 0$, $d(t,x) = 1$ iff $t \in \{0,1\}$, so $\partial([0,1] \times X) = \{0,1\} \times X$ Chapter15/P3P_CubeEp · EpGroupoid.degIndex_cubeProd_one_of_degIndex_eq_zero. Bump functions pass to it Chapter15/P3P_CubeEp · EpGroupoid.admitsBumpFunctionsPointwise_cubeProd. **The bundle and the section.** $\widetilde{W}$ is the pull-back of $W$ along the projection Chapter15/P3P_CubeEp · SEP.cubeSEP, and $\tilde{f}(q,x) = (q, f(x))$ Chapter15/P3P_CubeEp · SEP.cubeLiftEp is an sc-smooth section functor Chapter15/P3P_CubeEp · SEP.isScSmoothSectionFunctor_cubeLiftEp, sc-Fredholm when $f$ has tame germ data Chapter15/P3P_CubeEp · SEP.isScFredholmSectionFunctor_cubeLiftEp, by Part I's parametrized form for the cube (gap G.65 (a) of Part I concerns the untamed case), with $S_{\tilde{f}} = [0,1]^n \times S_f$ Chapter15/P3P_CubeEp · SEP.solutionSet_cubeLiftEp and $|S_{\tilde{f}}|$ compact when $|S_f|$ is Chapter15/P3P_CubeEp · SEP.hasCompactSolutionSetEp_cubeLiftEp.

## XV.14 The paragraph before Definition 15.3.1, Definitions 15.3.1, 15.3.3, 15.3.5 and Remark 15.3.6 (2017: Definitions 15.4, 15.5, 15.6, Remark 15.6)

**Definition 15.3.1** (restated) Chapter15/P3P_GeneralPosition · SBLocalization.ControlsCompactness: $(N,U)$ controls the compactness of $f$ if $N$ is an auxiliary norm, $U$ a saturated open neighbourhood of $S_f$, and $\mathrm{cl}_{|X|}\,|\{x \in U : N(f(x)) \le 1\}|$ is compact. $N$ lives on $W_{0,1}$, and "$N(f(x)) \le 1$" is Part I's relation, which is the text's extension of $N$ by $+\infty$. Paracompactness of $|X|$ is not part of the definition; the statements that need it carry it. A smaller saturated open $U' \supset S_f$ also controls Chapter15/P3P_GeneralPosition · SBLocalization.ControlsCompactness.mono. **The paragraph before the definition** Chapter15/P3P_GeneralPosition · SBLocalization.exists_controlsCompactness: for an sc-Fredholm section functor with compact $|S_f|$ and any auxiliary norm $N$, some $U$ controls the compactness, by Theorem 12.4.2 (Part XII, §XII.9) Chapter12/P3S_CompactnessEp · EpStrongBundle.isProperLtInftyFunctor_of_hasCompactSolutionSetEp.

**Definition 15.3.3** Chapter15/P3P_GeneralPosition · SBLocalization.IsAdmissible: an $(N,U)$-admissible perturbation is an sc$^+$-multisection $\Lambda$ with $N(\Lambda)(x) \le 1$ for all $x$ (Definition 13.2.3) and $\{x : \exists t \in [-1,1],\ (t\Lambda)(f(x)) > 0\} \subset U$ ($t\Lambda$ of Definition 13.2.10). The text's standing data are hypotheses of the statements that use it. **The example after it** Chapter15/P3P_GeneralPosition · SBLocalization.isAdmissible_of_domSupp_subset: if $S_f \subset U$, the domain support of $\Lambda$ lies in $U$ and $N(\Lambda) \le 1$, then $\Lambda$ is admissible. Off the domain support $\Lambda$ is trivial Chapter15/P3P_GeneralPosition · SBLocalization.eq_zero_of_not_mem_domSupp, so $(t\Lambda)(w) > 0$ forces $w = 0$ for every real $t$, $t = 0$ included Chapter15/P3P_GeneralPosition · SBLocalization.eq_fiberZero_of_multisectionSmul_pos; then $f(x) = 0$ and $x \in S_f \subset U$.

**Definition 15.3.5** (restated) Chapter15/P3P_GeneralPosition · SBLocalization.IsInGeneralPosition: $(f,\Lambda)$ is in general position if for every $x \in X_1$ with $\Lambda(f(x)) > 0$ and every operator $L$ in the support of $T(f,\Lambda)(x)$, the restriction of $L$ to the reduced tangent space $T^R_x X$ is surjective, $T^R_x X$ read in every chart around $x$ and every presentation of its partial quadrant (Part I's reduced tangent fibre). Charts and presentations exist at every point Chapter15/P3P_GeneralPosition · SBLocalization.exists_chart_presentation, and since $T^R_x X$ does not depend on them, one per point suffices Chapter15/P3P_GeneralPosition · SBLocalization.isInGeneralPosition_iff_exists. The reduced linearization $T^R_{(f,\Lambda)}(x)$ is the formal sum of the restrictions Chapter15/P3P_GeneralPosition · SBLocalization.reducedLinearization Chapter15/

P3P_GeneralPosition · SBLocalization.isInGeneralPosition_iff_reducedLinearization. The support of $\Lambda \circ f$ lies in $X_1$ Chapter15/P3P_GeneralPosition · SBLocalization.mem_levelSet_one_of_pos ($f(x)$ is then a value of an sc$^+$-section, in $W_{0,1}$, and $f$ is regularizing), so restricting to $X_1$ loses nothing. The all-charts form is satisfiable with nonempty support Chapter15/P3P_GeneralPosition · SBLocalization.isInGeneralPosition_witness: the zero section of $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ with the zero multisection. **Remark 15.3.6, first sentence** Chapter15/P3P_GeneralPosition · SBLocalization.IsInGeneralPosition.surjective: in general position every operator of $T(f,\Lambda)(x)$ is onto, since its restriction to a subspace is. The rest of the remark is a new gap (§XV.7).

## XV.15 Corollary 15.3.10 (2) (2017: Corollary 15.1 (2)): an auxiliary norm on $[0,1] \times X$

**The pulled-back norm** Chapter15/P3P_CubeAuxNorm · SEP.cubeAuxNorm: $\widetilde{N}(q,x,h) = N(h)$, Part I's pull-back of an auxiliary norm, is an auxiliary norm of $(\widetilde{W},\tilde{\mu})$ Chapter15/P3P_CubeAuxNorm · SEP.isAuxNorm_cubeAuxNorm, invariant under $\tilde{\mu}$ because $\tilde{\mu}$ acts on the $W$-component by $\mu$ Chapter15/P3P_CubeAuxNorm · SEP.cubeVal_μ. **The step of the proof** Chapter15/P3P_CubeAuxNorm · SEP.controlsCompactness_cube: if $(N,U)$ controls the compactness of $f$, then $(\widetilde{N},[0,1]^n \times U)$ controls that of $\tilde{f}$; the sublevel set is $[0,1]^n$ times that of $f$, and its orbit set has compact closure Chapter15/P3P_CubeAuxNorm · SEP.isCompact_closure_image_π_cube.

**Corollary 15.3.10 (2)** Chapter15/P3P_CubeAuxNorm · SEP.corollary_15_3_10_2: for $(N^0,U^0)$ and $(N^1,U^1)$ controlling the compactness of $f$, the norm $\widetilde{N} = \max(\alpha(t)N^0, \beta(t)N^1)$ with $\alpha(t) = \min(1, 3(1-t))$, $\beta(t) = \min(1,3t)$ Chapter15/P3P_CubeAuxNorm · SEP.interpAuxNorm and the set $\widetilde{U} = ([0,\frac{2}{3}) \times U^0) \cup ((\frac{1}{3},1] \times U^1)$ Chapter15/P3P_CubeAuxNorm · SEP.interpSet control the compactness of $\tilde{f}$ on $[0,1] \times X$. *Step 1.* $\max(aN_0, bN_1)$ is an auxiliary norm for continuous $a, b \geq 0$ never both 0 Chapter15/P3P_CubeAuxNorm · StrongBundle.isAuxNorm_max_mul: continuity and the seminorm property are clear, completeness passes to an equivalent fibre norm Chapter15/P3P_CubeAuxNorm · StrongBundle.IsCompleteFiberNorm.congr, and condition (2) holds for the larger of the two terms; so $\widetilde{N}$ is an auxiliary norm Chapter15/P3P_CubeAuxNorm · SEP.isAuxNorm_interpAuxNorm. *Step 2.* $\widetilde{U}$ is open, saturated and contains $S_{\tilde{f}} = [0,1] \times S_f$. *Step 3.* On $[0,\frac{2}{3}) \times U^0$, $\alpha = 1$ and $\widetilde{N} \geq N^0$, so the sublevel set lies in $[0,1]\times$ the sublevel set of $(N^0,U^0)$; similarly on the other piece; the orbit set lies in the union of two sets with compact closure. *Restrictions.* Over $t = 0$, $\widetilde{N} = N^0$ Chapter15/P3P_CubeAuxNorm · SEP.interpAuxNorm_of_cubeTime_eq_zero and $\widetilde{U}$ is $U^0$ Chapter15/P3P_CubeAuxNorm · SEP.mem_interpSet_zero; over $t = 1$ they are $N^1$ and $U^1$ Chapter15/P3P_CubeAuxNorm · SEP.interpAuxNorm_of_cubeTime_eq_one Chapter15/P3P_CubeAuxNorm · SEP.mem_interpSet_one. The text does not give $\widetilde{N}$; its proof uses a single $N$, which is the step above. The perturbations $\Lambda^i$ of the corollary play no role in (2). With §XV.13 ($\tilde{f}$ sc-Fredholm with compact solution set) all of (2) is covered, $\tilde{f}$ carrying the tame germ data of $f$.

## XV.16 Witnesses and the hypotheses of wave 2

**Hypotheses taken.** *Tame germ data for $f$* (Part I's `IsScFredholmSectionTame`), in Theorems 15.2.2, 15.2.5, Remark 15.2.3, the summands of §XV.10 and the sc-Fredholm property of $\tilde{f}$ in §XV.13. It is not new: it is inherited from Part I's Theorems 3.1.18 and 3.6.8, which are proved for tame germ data and on which these statements rest; with the 2021 untamed notion Theorem 3.6.8 does not hold as we read it Chapter3/GapP1_UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not (Part III, §III.65). It is the hypothesis of gap G.140 in §12.5 as well. No ep-level counterexample for the untamed §15.2 was built. Witness for all hypotheses of Theorem 15.2.2 together (`theorem_15_2_2_hyp_witness`): over $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, the graph section of 0, which has tame germ data by Part I Chapter3/NonVacuityG · isScFredholmSectionTame_graphSection,

with the zero multisection; it is transversal, $d_X \equiv 0$, and $\Lambda_0(f(x)) = 1$ everywhere. *The global form of Remark 15.2.3* has, besides, $d_X = 0$ on the support and a paracompact orbit space of the support, witnessed by the same example. *None added* in §§XV.13–XV.15 beyond the tame germ data of $\tilde{f}$.

**Every definition has a witness.** Transversality: the example above. Good position at a point: not witnessed separately (at interior points Part I's graph section would give one on $W|_O$). Controlling compactness: from any auxiliary norm when $|S_f|$ is compact (§XV.14). Admissible perturbations: the zero multisection, by the example after Definition 15.3.3. General position with nonempty support: `isInGeneralPosition_witness`. Product ep-groupoids and cylinders: of any ep-groupoid (§XV.13).

# Part XVI. Polyfolds (HWZ Chapter 16, §§16.1, 16.2, 16.4)

Chapter 16 (Springer 2021, pp. 705–740; 2017 numbers in parentheses) passes from ep-groupoids to polyfolds: a polyfold is a topological space with an equivalence class of ep-groupoid structures on it. §16.1 defines polyfold structures, their equivalence and polyfolds (Definitions 16.1.1–16.1.3 (2017: Definitions 16.1–16.3)), shows that bump functions and partitions of unity do not depend on the structure (Proposition 16.1.4, Definition 16.1.5 (2017: Proposition 16.2, Definition 16.4)), defines the morphisms of the polyfold category (Definitions 16.1.6, 16.1.8, 16.1.10, 16.1.11, Remarks 16.1.7, 16.1.9 (2017: Definitions 16.5–16.8, Remarks 16.1, 16.2)), and transports the degeneracy index, the boundary, tameness and faces (Lemma 16.1.13, Definition 16.1.14, Lemma 16.1.19, Definitions 16.1.17, 16.1.20 (2017: Lemma 16.1, Definitions 16.9, 16.11, Lemma 16.2, Definition 16.12), and Proposition 16.1.12, Definition 16.1.15, Remark 16.1.16, Example 16.1.18, Definition 16.1.21, Proposition 16.1.22). §16.2 builds the tangent and sc-differential forms, §16.3 strong polyfold bundles, §16.4 branched finite-dimensional orbifolds, §16.5 $\mathrm{sc}^+$-multisections and §16.6 the Fredholm theory on polyfolds. This Part retells the polyfold structures and polyfolds of §16.1 (§XVI.1), the uniqueness statement after Definition 16.1.2 and Remark 16.1.7 (§XVI.2), Proposition 16.1.4 and Definition 16.1.5 (§XVI.3), and the degeneracy index, boundary and tameness (§XVI.4); then the witnesses (§XVI.5), what is not formalized (§XVI.6) and what the formalization taught us (§XVI.7), then metrizability (§XVI.8), Lemma 16.2.1 (§XVI.9), the tangent of a polyfold (§XVI.10), sc-differential forms (§XVI.11), branched suborbifolds (§XVI.12), oriented ones and their integrals (§XVI.13), Stokes' theorem (§XVI.14) and the wave-2 witnesses (§XVI.15).

**What is proved, and what is not.** Waves 1 and 2 of Part III (2026-11-13) formalize part of §16.1, the core of §16.2 and §16.4; §§16.3, 16.5 and 16.6 are not yet formalized. *Formalized as printed:* Definitions 16.1.1, 16.1.2, 16.1.3, 16.1.5, 16.1.14, 16.1.17 and 16.1.20, and Lemma 16.1.13 (§§XVI.1, XVI.3, XVI.4); the construction of $T(Z, c)$ with the maps $T\alpha$ and (16.5), Lemma 16.2.2 and Definition 16.2.3 (§XVI.10); Definition 16.2.5 with the independence remark before it (§XVI.11); Definitions 16.4.1, 16.4.4 and 16.4.5, and Definition 16.4.2 (1)–(4) (§§XVI.12–XVI.13). Proposition 16.1.4 is proved, with the text's sketch for part (2) and its "easy exercise" written out (§XVI.3), and Lemma 16.1.19 as an equivalence (§XVI.4). The remark after Definition 16.1.1, that $Z$ is metrizable, is proved in wave 2 from Theorem 7.3.1, without the Nagata–Smirnov theorem; gap G.146 is closed (§XVI.8). *Narrower:* Theorem 16.2.6 for the de Rham complex, without the pull-back by sc-smooth maps (§XVI.11); Definition 16.4.2 (5), tameness, read for some representative (§XVI.12); Proposition 16.4.3 for every representative, without the comparison of two (§XVI.12); the integrals of p. 732 on tame compact representatives (§XVI.13). *With an added hypothesis:* Theorem 16.4.6, Stokes' theorem, is proved with sc-smooth partitions of unity, which the Chapter 9 Stokes theorem it reduces to assumes (§XVI.14). *Does not hold as we read it:* the sentence after Definition 16.1.2,

that the generalized isomorphism $f$ with $\alpha = \beta \circ |f|$ is unique as a consequence of (ZHOU), and Remark 16.1.7, that for a homeomorphism $h$ there is at most one generalized isomorphism $f$ with $h \circ \alpha = \beta \circ |f|$. A Lean counterexample satisfies every hypothesis of Definition 16.1.1: the ep-groupoid $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action, which satisfies (ZHOU) and has orbit space $\mathbb{R}$. The uniqueness holds when the target ep-groupoid is effective (§XVI.2). The same pair applies to the sentence on pp. 709–710 that an isomorphism $(f, \mathfrak{f})$ of polyfold structures is determined by $f$, and to Lemma 16.2.1, which does not hold as we read it either; Lemma 16.2.1 holds on orbit spaces, which is all that §16.2 uses, and for an effective target (§XVI.9). *Not formalized:* the second sentence of Remark 16.1.7, on continuous maps. *Not yet formalized:* Definitions 16.1.6, 16.1.8, 16.1.10, 16.1.11, Remark 16.1.9, Proposition 16.1.12, Definition 16.1.15, Remark 16.1.16, Example 16.1.18, Definition 16.1.21, Proposition 16.1.22; in §16.2 the raised polyfolds $(Z, c)^i$, the projection $\tau_Z$, Theorem 16.2.4, the second part of Theorem 16.2.6 and Remark 16.2.7; §16.3, §16.5 and §16.6.

**Two structural choices.** (1) *Polyfolds as quotients.* A polyfold structure is a structure with the ep-groupoid, its two properties and the homeomorphism; a polyfold is an element of the quotient of the polyfold structures on $Z$ by equivalence, a Lean quotient. Notions of a polyfold ("some structure in $c$ has the property") come with the theorem that then every structure in $c$ has it. (2) *Bump functions on the objects.* "Admits sc-smooth bump functions" in Proposition 16.1.4 (1) and Definition 16.1.5 (1) is read for the object M-polyfold (Definition 5.3.1, pointwise), as the proof uses it; by Theorem 12.3.2 (1) it is equivalent to Definition 12.3.1 in the pointwise reading (Part XII, §XII.6).

**Map.**

- `Chapter16/P3Z_PolyfoldStructure`: Definition 16.1.1, the equivalence of Definition 16.1.2, Definition 16.1.3, uniqueness for effective targets, Lemma 16.1.13, Definitions 16.1.14, 16.1.17, Lemma 16.1.19, Definition 16.1.20.
- `Chapter16/P3Z_ZhouUniquenessCheck`: the polyfold structure on $\mathbb{R}$ given by $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$, the uniqueness after Definition 16.1.2 and Remark 16.1.7, and their counterexample.
- `Chapter16/P3Z_PolyfoldBump`: bump points along equivalences, Proposition 16.1.4 (1), Definition 16.1.5 (1).
- `Chapter16/P3Z_PolyfoldPartitions`: partitions of unity read on $|X|$, along equivalences, Proposition 16.1.4 (2), Definition 16.1.5 (2).
- `Chapter16/P3Z_PolyfoldExamples`: the witnesses on the polyfold $\mathbb{R}$.
- `Chapter16/P3H_TangentLemma`: Lemma 16.2.1, tangents of generalized maps (§XVI.9).
- `Chapter16/P3H_TangentPolyfold`: metrizability, $T(Z, c)$, Lemma 16.2.2, Definition 16.2.3 (§§XVI.8, XVI.10).
- `Chapter16/P3H_FormsIndependence`, `P3H_PolyfoldForms`: the remark before Definition 16.2.5, Definition 16.2.5, Theorem 16.2.6 (§XVI.11).
- `Chapter16/P3H_BranchedSuborbifold`: §16.4 (§§XVI.12–XVI.14).

Throughout, a generalized isomorphism $f = [X \xleftarrow{F} A \xrightarrow{G} Y]$ (Chapter 10) induces $|f| = |G| \circ |F|^{-1} : |X| \to |Y|$, and $\mathcal{EP}$ is the category of ep-groupoids.

## XVI.1 Definitions 16.1.1–16.1.3 (2017: 16.1–16.3): polyfold structures and polyfolds

**Definition 16.1.1** `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure`: a polyfold structure on a space $Z$ is an ep-groupoid $X$ with (1) $|X|$ paracompact, (2) (ZHOU), and (3) a homeomorphism $\beta : |X| \to Z$. $Z$ lives in the universe of the object and morphism sets of $X$. The pair $(X, \beta)$ is an orbit chart in the sense of Remark 10.3.18 `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.toOrbitChart`, and (ZHOU) gives (LR) by Proposition 7.4.22 `Chapter7/SmoothRegularAssembly · proposition_7_4_22_of_isZhou Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.isLR`. The 2017 Definition 16.1 had neither (1) nor (2). The remark after the definition, that $Z$ is metrizable, is not formalized (§XVI.6).

**Definition 16.1.2** `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.Equiv`: $(X, \alpha) \sim (Y, \beta)$ if a generalized isomorphism $f : X \to Y$ has $\alpha = \beta \circ |f|$. It is an equivalence relation `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.Equiv.refl Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.Equiv.symm Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.Equiv.trans`: by $1_X$, by $f^{-1}$ with $|f^{-1}| = |f|^{-1}$, and by $g \circ f$ with $|g \circ f| = |g| \circ |f|$ (Proposition 10.3.11). It is the relation of Remark 10.3.18 on orbit charts `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.equiv_iff_rel`, and it forces $|f| = \beta^{-1} \circ \alpha$ `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.orbitMap_eq_of_equiv`.

**Definition 16.1.3** `Chapter16/P3Z_PolyfoldStructure · PolyfoldClass`: a polyfold is $(Z, c)$, $c$ an equivalence class of polyfold structures on $Z$, the quotient by the relation above `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.setoid`.

**The polyfold $\mathbb{R}$ with isotropy $\mathbb{Z}/3$.** The ep-groupoid $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action has $\pi$ injective `Chapter16/P3Z_ZhouUniquenessCheck · EpGroupoid.Examples.injective_π_trivialZ3`, hence a homeomorphism, being continuous and open (Proposition 7.1.8) `Chapter16/P3Z_ZhouUniquenessCheck · EpGroupoid.Examples.isHomeomorph_π_trivialZ3`, so $|X| \cong \mathbb{R}$ `Chapter16/P3Z_ZhouUniquenessCheck · EpGroupoid.Examples.trivialZ3OrbitHomeomorph` is paracompact `Chapter16/P3Z_ZhouUniquenessCheck · EpGroupoid.Examples.hasParacompactOrbitSpace_trivialZ3`. It satisfies (ZHOU), because every element of the isotropy group acts trivially `Chapter7/Zhou · EpGroupoid.Examples.isZhou_trivialZ3`. So it is a polyfold structure on $\mathbb{R}$ in the exact sense of Definition 16.1.1 `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.z3PolyfoldStructure`, and its class is a polyfold `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.z3Polyfold`. It is not effective `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.not_isEffective_z3PolyfoldStructure`.

## XVI.2 The uniqueness after Definition 16.1.2, and Remark 16.1.7 (2017: 16.1)

The text says, after Definition 16.1.2, that $f$ "is necessarily unique as a consequence of (ZHOU)", citing Theorem 10.3.19: for generalized isomorphisms $f, f' : X \to Y$, $|f| = |f'|$ implies $f = f'$ when one of the ep-groupoids satisfies (LR). The statement in Lean `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.Uniqueness_16_1_2`: for polyfold structures $(X, \alpha)$, $(Y, \beta)$ on $Z$, at most one generalized isomorphism $f$ has $\alpha = \beta \circ |f|$. Remark 16.1.7 `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.Remark_16_1_7`: for a homeomorphism $h : Z \to W$ and polyfold structures $(X, \alpha)$ on $Z$, $(Y, \beta)$ on $W$, at most one generalized isomorphism $f$ has $h \circ \alpha = \beta \circ |f|$.

**The uniqueness after Definition 16.1.2 does not hold as we read it** `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.not_uniqueness_16_1_2`. *The example.* On the polyfold structure of $\mathbb{R}$

given by $X = \mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§XVI.1), take the two generalized isomorphisms $[1_X]$ and $[X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X]$, where Inv inverts the morphisms `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.genId` `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.genInv`. Both are generalized isomorphisms `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.isGeneralizedIso_genInv`; both induce the identity of $|X|$, so both satisfy $\alpha = \alpha \circ |f|$ `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.β_eq_β_comp_orbitMap_genId` `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.β_eq_β_comp_orbitMap_genInv`; and they differ `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.genInv_ne_genId`, because there is no natural transformation from 1 to Inv. All hypotheses of Definition 16.1.1 hold for this $X$ `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.uniqueness_16_1_2_counterexample`. This is the pair of Part X, §X.20, where Theorem 10.3.19 does not hold as we read it `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_theorem_10_3_19`; (ZHOU) gives (LR), and the example shows that (LR) does not give the uniqueness.

**Remark 16.1.7 does not hold as we read it** `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.not_remark_16_1_7`: take $h = \mathrm{id}_{\mathbb{R}}$ in the same example. The same pair, with $h = f$, applies to the sentence on pp. 709–710 that an isomorphism $(f, \mathfrak{f})$ of polyfold structures is determined by $f$; it does not hold as we read it either.

**The form that holds: an effective target** `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.uniqueness_16_1_2_of_isEffective` `Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.remark_16_1_7_of_isEffective` `Chapter16/P3Z_ZhouUniquenessCheck · PolyfoldStructure.uniqueness_16_1_2_of_forall_isEffective`. When $Y$ is effective, Theorem 10.3.19 holds `Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.theorem_10_3_19_of_isEffective`, and two generalized isomorphisms with $\beta \circ |f| = h \circ \alpha = \beta \circ |f'|$ have $|f| = |f'|$, since $\beta$ is injective; so $f = f'$. Definitions 16.1.2, 16.1.3 and everything else in this Part are unaffected: none of them uses the uniqueness, and Definition 16.1.6 as printed carries its generalized isomorphisms as data. The second sentence of Remark 16.1.7, that for a continuous map there may be none or many, is an observation without formal content beyond an example and is not formalized.

## XVI.3 Proposition 16.1.4 (2017: 16.2) and Definition 16.1.5 (2017: 16.4)

**Proposition 16.1.4 (1)** `Chapter16/P3Z_PolyfoldBump · PolyfoldStructure.proposition_16_1_4_one`: for equivalent structures, the object M-polyfold of $X$ admits sc-smooth bump functions iff that of $Y$ does. Through a diagram $X \xleftarrow{F} A \xrightarrow{G} Y$ it is enough to show it along one equivalence `Chapter16/P3Z_PolyfoldBump · EpFunctor.IsEquivalence.admitsBumpFunctions_iff`. Bump functions are local: a point is a *bump point* if there are bump functions at it, and sc-smoothness of a real function is local `Chapter16/P3Z_PolyfoldBump · MPolyfold.isScSmoothMap_real_of_locally`. *Step 1.* Bump points are carried back along an sc-diffeomorphism of open sets: $g \circ \varphi$ on a small neighbourhood, extended by $0$ `Chapter16/P3Z_PolyfoldBump · MPolyfold.IsScDiffeoOn.hasBumpAt`. *Step 2.* An equivalence is a local sc-diffeomorphism on objects, so bump points pass from $F(a)$ to $a$ `Chapter16/P3Z_PolyfoldBump · EpFunctor.IsEquivalence.hasBumpAt_of_hasBumpAt_obj` and from $a$ to $F(a)$ `Chapter16/P3Z_PolyfoldBump · EpFunctor.IsEquivalence.hasBumpAt_obj_of_hasBumpAt`. *Step 3.* Every $y$ is isomorphic to some $F(a)$, and bump points are invariant under the local sc-diffeomorphism $\widehat{\psi} = t \circ (s|_{U(\psi)})^{-1}$ of a morphism `Chapter16/P3Z_PolyfoldBump · EpGroupoid.hasBumpAt_tgt`.

**Proposition 16.1.4 (2)** `Chapter16/P3Z_PolyfoldPartitions · PolyfoldStructure.proposition_16_1_4_two`: $X$ admits sc-smooth partitions of unity as an ep-groupoid (Definition 7.5.5) iff $Y$ does. *Step 1, read on* $|X|$. Definition 7.5.5 holds iff every open cover of $|X|$ has a subordinate partition of unity $(\sigma_\lambda)$ of $|X|$ with every $\sigma_\lambda \circ \pi$ sc-smooth `Chapter16/P3Z_PolyfoldPartitions · EpGroupoid.HasOrbitScPartitionsOfUnity` `Chapter16/P3Z_`

PolyfoldPartitions · EpGroupoid.hasScPartitionsOfUnity_iff_orbit: an invariant continuous function descends to $|X|$ Chapter16/P3Z_PolyfoldPartitions · EpGroupoid.descendP3Z, closures commute with $\pi^{-1}$ because $\pi$ is open Chapter16/P3Z_PolyfoldPartitions · EpGroupoid.preimage_closure_π, and partitions of unity pull back along continuous maps Chapter16/P3Z_PolyfoldPartitions · PartitionOfUnity.comapP3Z. *Step 2, the text's "easy exercise".* For an equivalence $F : A \to X$ and $g : |X| \to \mathbb{R}$, if $g \circ \pi_X \circ F$ is sc-smooth then so is $g \circ \pi_X$: near $F(a)$ through a local inverse of $F$, and near a $y$ isomorphic to $F(a)$ through $\hat{\psi}$ Chapter16/P3Z_PolyfoldPartitions · EpFunctor.IsEquivalence.isScSmoothMap_comp_π_of_comp_obj. *Step 3.* Partitions of unity on $|X|$ are transported by the homeomorphism $|F|$, so the property passes along an equivalence Chapter16/P3Z_PolyfoldPartitions · EpFunctor.IsEquivalence.hasScPartitionsOfUnity_iff, and along a diagram. Both parts together Chapter16/P3Z_PolyfoldPartitions · PolyfoldStructure.proposition_16_1_4. The text's proof of (1) goes through local inverses and $\hat{\psi}$ as here; its proof of (2) is a sketch, which the steps above fill.

**Definition 16.1.5** Chapter16/P3Z_PolyfoldBump · PolyfoldClass.AdmitsBumpFunctions Chapter16/P3Z_PolyfoldPartitions · PolyfoldClass.AdmitsScPartitionsOfUnity: a polyfold admits sc-smooth bump functions, or partitions of unity, if some structure in $c$ does; then every structure does, by Proposition 16.1.4 Chapter16/P3Z_PolyfoldBump · PolyfoldClass.definition_16_1_5_one Chapter16/P3Z_PolyfoldPartitions · PolyfoldClass.definition_16_1_5_two.

## XVI.4 Lemma 16.1.13 (2017: 16.1), Definitions 16.1.14, 16.1.17, 16.1.20 (2017: 16.9, 16.11, 16.12), Lemma 16.1.19 (2017: 16.2)

**The degeneracy index of a structure** Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.degIndex: $d_{(X,\beta)}(z) = d_{|X|}(\beta^{-1}(z))$, which is $d_X(x)$ for any $x$ with $\beta(|x|) = z$ Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.degIndex_eq. **Lemma 16.1.13** Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.lemma_16_1_13: equivalent structures have the same index, because the degeneracy index on orbit spaces is invariant under $|f|$ for a generalized isomorphism (Proposition 11.1.2, Part XI) and $|f| = \beta^{-1} \circ \alpha$.

**Definition 16.1.14** Chapter16/P3Z_PolyfoldStructure · PolyfoldClass.degIndex Chapter16/P3Z_PolyfoldStructure · PolyfoldClass.definition_16_1_14: $d_Z(z) = d_X(x)$ for every $(X, \beta) \in c$ and $\beta(|x|) = z$, well defined on the class by Lemma 16.1.13. **Definition 16.1.17** Chapter16/P3Z_PolyfoldStructure · PolyfoldClass.boundary: $\partial Z = \{z : d_Z(z) \geq 1\}$, computed in any representative Chapter16/P3Z_PolyfoldStructure · PolyfoldClass.mem_boundary_iff.

**Lemma 16.1.19** (stronger) Chapter16/P3Z_PolyfoldStructure · PolyfoldStructure.lemma_16_1_19: for equivalent structures, $X$ is tame iff $Y$ is, by Chapter 11's invariance of tameness under equivalences; the text states one direction, and the other follows by symmetry. **Definition 16.1.20** Chapter16/P3Z_PolyfoldStructure · PolyfoldClass.IsTame Chapter16/P3Z_PolyfoldStructure · PolyfoldClass.definition_16_1_20: a polyfold is tame if one, and then every, structure in $c$ has $X$ tame.

## XVI.5 Witnesses and the hypotheses

**Hypotheses taken.** None is added to a statement of the text. Proposition 16.1.4 (1) and Definition 16.1.5 (1) read bump functions on the object M-polyfold (structural choice (2)). The uniqueness after Definition 16.1.2 and Remark 16.1.7 are proved under the additional hypothesis that the target is effective; without it they do not hold as we read it (§XVI.2).

**Every definition has a witness**, on the polyfold $\mathbb{R}$ given by $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§XVI.1): polyfold structure and polyfold (`z3PolyfoldStructure`, `z3Polyfold`); tame `Chapter16/P3Z_PolyfoldExamples · PolyfoldStructure.z3Polyfold_isTame`; admits bump functions `Chapter16/P3Z_PolyfoldExamples · PolyfoldStructure.z3Polyfold_admitsBumpFunctions` and sc-smooth partitions of unity `Chapter16/P3Z_PolyfoldExamples · PolyfoldStructure.z3Polyfold_admitsScPartitionsOfUnity`; degeneracy index $\equiv 0$ `Chapter16/P3Z_PolyfoldExamples · PolyfoldStructure.z3Polyfold_degIndex` and empty boundary `Chapter16/P3Z_PolyfoldExamples · PolyfoldStructure.z3Polyfold_boundary`. The same polyfold is the counterexample of §XVI.2, so the hypotheses of Definition 16.1.1 are satisfiable there in a non-effective way.

## XVI.6 What is not formalized

- *Closed in wave 2.* G.146, the metrizability of polyfolds (§XVI.8).
- The second sentence of Remark 16.1.7, on continuous maps (no formal content beyond an example).
- Gap G.153: Stokes' theorem on polyfolds without partitions of unity (Theorem 16.4.6). Proved with sc-smooth partitions of unity, as Theorem 9.5.17 assumes (§XVI.14). Removing the hypothesis needs Stokes on ep-groupoids without partitions of unity, with a finite cut-off from the compactness of supp $\Theta$; about 300–600 lines.
- Gap G.154: tameness of branched suborbifolds is invariant (p. 729 and Definition 16.4.2 (5)). Present: `IsTame` for some representative; the integrals are defined on tame representatives, and their independence is proved among those (§XVI.13). Missing: the transport of the tame branching structures (Definition 9.3.6) and of the oriented tame structures along equivalences of ep-groupoids; about 250–500 lines.
- Gap G.155: naturality of the structure on supp $\theta$ (Proposition 16.4.3). Present for every representative: $\operatorname{supp}\theta \cong |\operatorname{supp}\Theta|$ and Proposition 9.1.9 (§XVI.12). Missing: that two representatives give sc-diffeomorphic structures; about 150–300 lines.
- Gap G.156: the raised polyfolds $(Z,c)^i = (Z_i, c_i)$ and the tangent projection $\tau_Z : T(Z,c) \to Z^1$ with $\tau_Z \circ T\alpha = \alpha \circ |\tau_X|$ (§16.2, pp. 715 and 717); about 100–200 lines.
- Gap G.157: Theorem 16.2.4 (the tangent functor on the polyfold category) and the second part of Theorem 16.2.6 (pull-back of $\Omega^*_{sc}$ by sc-smooth maps, commuting with $d$). Both need the morphisms of Definitions 16.1.6–16.1.11; the ep-level tools exist (§§XVI.9, XVI.11); about 300–500 lines with the category.
- Open question, not a gap: in the counterexample to Lemma 16.2.1, $|T\mathfrak{f}| = |T\mathfrak{g}|$ is proved; whether $T\mathfrak{f} \neq T\mathfrak{g}$ is not decided (§XVI.9).
- Not yet formalized: Definitions 16.1.6, 16.1.8, 16.1.10, 16.1.11, Remark 16.1.9 (the morphisms and the polyfold category), Proposition 16.1.12 (the lifts $Z_i$), Definition 16.1.15 and Remark 16.1.16 (subpolyfolds), Example 16.1.18, Definition 16.1.21 and Proposition 16.1.22 (faces); Remark 16.2.7; §16.3 (strong polyfold bundles), §16.5 (sc$^+$-multisections) and §16.6 (Fredholm theory on polyfolds).

## XVI.7 What the formalization taught us

A polyfold structure in the 2021 sense adds two conditions to the 2017 definition, paracompactness of $|X|$ and (ZHOU), and the text then derives from (ZHOU) that the generalized isomorphism realizing an equivalence of structures is unique. Formalized, (ZHOU) gives (LR), and (LR) is the hypothesis of Theorem 10.3.19; but Part X found that Theorem 10.3.19 does not hold as we read it, and its counterexample, $\mathbb{Z}/3$ acting trivially on $\mathbb{R}$, satisfies (ZHOU) and has a paracompact orbit space. It is therefore a

polyfold structure, and the two generalized isomorphisms $[1_X]$ and $[X \overset{1}{\leftarrow} X \overset{\mathrm{Inv}}{\longrightarrow} X]$ both realize the identity of $\mathbb{R}$. The uniqueness after Definition 16.1.2, Remark 16.1.7, the determination of isomorphisms on pp. 709–710 and Lemma 16.2.1 do not hold as we read them; they hold for an effective target. None of §16.1's definitions uses the uniqueness, since an equivalence of structures only asks that some generalized isomorphism exist, so the polyfold category can still be built if its morphisms carry their generalized isomorphisms, as Definition 16.1.6 already does.

§16.2 shows how little of the uniqueness is needed. The text builds $T(Z,c)$ by gluing along "the unique" $\mathfrak{f}$ with $\beta \circ |\mathfrak{f}| = \alpha$, and Lemma 16.2.1 is there to make the gluing maps canonical. What the gluing uses is only $|T\mathfrak{f}|$, and that is unique: (LR) gives, for $|F| = |G|$, local families from $F$ to $G$ (Proposition 10.1.10 (1)), and these give $|TF| = |TG|$ (Proposition 10.1.10 (2)). So $T(Z,c)$, $T\alpha$, (16.5), Lemma 16.2.2 and Definition 16.2.3 go through as printed, with one structure of $c$ as base point instead of the gluing over all; the same argument, Proposition 11.2.9, makes $h_*\omega$ independent of $h$, which is the remark before Definition 16.2.5. The invariance statements of §16.1 are statements about equivalences of ep-groupoids, transported to classes; bump functions are local, partitions of unity are best read on $|X|$, and the text's "easy exercise" is the descent of sc-smoothness along an equivalence. The metrizability of a polyfold, which wave 1 expected to need the Nagata–Smirnov theorem, follows from Theorem 7.3.1 and Part I's metrization theorem for locally metrizable paracompact Hausdorff spaces. §16.4 is Chapters 9 and 11 transported to classes in the same way, with two points the text leaves implicit. An orientation $\hat{T}_\Theta$ is read only on $X_\infty$, so a representative fixes it to be 0 elsewhere, without which $1^*\hat{T} = \hat{T}$ fails. And Theorem 16.4.6 reduces to Theorem 9.5.17, which is stated for ep-groupoids with sc-smooth partitions of unity; the polyfold version needs the same hypothesis, which by Definition 16.1.5 holds for every representative as soon as it holds for one.

## XVI.8 The remark after Definition 16.1.1 (2017: Proposition 16.1): polyfolds are metrizable

**The remark** `Chapter16/P3H_TangentPolyfold · PolyfoldStructure.metrizableSpace`: for a polyfold structure $(X,\beta)$ on $Z$, $Z$ is metrizable. *Proof.* $|X|$ is paracompact by Definition 16.1.1 (1); Theorem 7.3.1 gives that it is regular, Hausdorff and locally metrizable, hence metrizable `Chapter7/OrbitTopology · theorem_7_3_1_metrizable` (Part VII, §VII.18). The metrization step is Part I's theorem that a paracompact Hausdorff space which is locally metrizable is metrizable `Chapter2/Metrizable · metrizableSpace_of_locally`, through the pull-back of a metrizable topology along an embedding `Chapter2/Metrizable · metrizableSpace_of_pullback`; Mathlib's Nagata–Smirnov theorem is not needed. $Z$ is homeomorphic to $|X|$ through $\beta$. Gap G.146 is closed.

## XVI.9 Lemma 16.2.1 (2017: Lemma 16.3): uniqueness of isomorphisms of polyfold structures

The text: if two isomorphisms $(f,\mathfrak{f}),(g,\mathfrak{g}) : (X,\alpha) \to (Y,\beta)$ of polyfold structures have $f = g$, then $\mathfrak{f} = \mathfrak{g}$ and $T\mathfrak{f} = T\mathfrak{g}$. The proof: $|\mathfrak{f}| = |\mathfrak{g}|$, and Theorem 10.3.19. The statement in Lean `Chapter16/P3H_TangentLemma · PolyfoldStructure.Lemma_16_2_1`: $f, g$ homeomorphisms, $\mathfrak{f},\mathfrak{g}$ generalized isomorphisms with $f\circ\alpha = \beta\circ|\mathfrak{f}|$ and $g\circ\alpha = \beta\circ|\mathfrak{g}|$. The tangent $T\mathfrak{f}$ of $\mathfrak{f} = [X \overset{F}{\leftarrow} A \overset{\Phi}{\rightarrow} Y]$ is the class of $[TX \overset{TF}{\longleftarrow} TA \overset{T\Phi}{\longrightarrow} TY]$ `Chapter16/P3H_TangentLemma · EPLocalization.EPLoc.tangentHom`, a copy of Chapter 11's tangent functor for ep-groupoids in any universe `Chapter16/P3H_TangentLemma · EPLocalization.EPLoc.tangentHom_eq_tangentFunctor_`

map; it is compatible with identities and composition Chapter16/P3H_TangentLemma · EPLocalization.EPLoc.tangentHom_id Chapter16/P3H_TangentLemma · EPLocalization.EPLoc.tangentHom_comp and preserves generalized isomorphisms Chapter16/P3H_TangentLemma · EPLocalization.EPLoc.isGeneralizedIso_tangentHom.

**Lemma 16.2.1 does not hold as we read it** Chapter16/P3H_TangentLemma · PolyfoldStructure.not_lemma_16_2_1. *The example* Chapter16/P3H_TangentLemma · PolyfoldStructure.lemma_16_2_1_counterexample is the pair of §XVI.2: on the polyfold structure $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ of $\mathbb{R}$, $f = g = \mathrm{id}$, $\mathfrak{f} = [X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X]$ and $\mathfrak{g} = [1_X]$. Both squares commute, the space is paracompact and satisfies (ZHOU), and $\mathfrak{f} \neq \mathfrak{g}$. The last step of the text's proof is Theorem 10.3.19, which does not hold as we read it Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_theorem_10_3_19 (Part X, §X.20). Whether $T\mathfrak{f} \neq T\mathfrak{g}$ in this example is not decided; their orbit maps agree.

**The forms that hold.** *On orbit spaces* Chapter16/P3H_TangentLemma · PolyfoldStructure.lemma_16_2_1_orbit, under exactly the text's hypotheses: $|\mathfrak{f}| = |\mathfrak{g}|$ and $|T\mathfrak{f}| = |T\mathfrak{g}|$. *Step 1.* $|\mathfrak{f}| = \beta^{-1} \circ f \circ \alpha$ Chapter16/P3H_TangentLemma · PolyfoldStructure.orbitMap_eq_of_comp, so $f = g$ gives $|\mathfrak{f}| = |\mathfrak{g}|$. *Step 2* Chapter16/P3H_TangentLemma · EPLocalization.EPLoc.orbitMap_tangentHom_eq: if $X$ or $Y$ has (LR) and $|\mathfrak{f}| = |\mathfrak{g}|$, then $|T\mathfrak{f}| = |T\mathfrak{g}|$. With $\mathfrak{g}'$ an inverse of $\mathfrak{f}$, $e = \mathfrak{g}' \circ \mathfrak{g}$ is a generalized automorphism of $X$ with $|e| = 1$ (Proposition 10.3.11), and $\mathfrak{g} = \mathfrak{f} \circ e$; so it is enough that $|Te| = 1$ Chapter16/P3H_TangentLemma · EPLocalization.EPLoc.orbitMap_tangentHom_eq_id. On a representative $X \xleftarrow{F} A \xrightarrow{\Phi} X$ of $e$, $|F| = |\Phi|$, $A$ has (LR) by Proposition 10.1.9, and Proposition 10.1.10 (2) gives $|TF| = |T\Phi|$, that is $|Te| = 1$. (ZHOU) gives (LR) for $X$. *For an effective target* Chapter16/P3H_TangentLemma · PolyfoldStructure.lemma_16_2_1_of_isEffective: if $Y$ is effective, then $\mathfrak{f} = \mathfrak{g}$ by Remark 16.1.7 for an effective target (§XVI.2), and so $T\mathfrak{f} = T\mathfrak{g}$. Its hypotheses are satisfiable Chapter16/P3H_TangentLemma · PolyfoldStructure.lemma_16_2_1_of_isEffective_hypotheses: the point ep-groupoid is effective and satisfies (ZHOU). The orbit form is what the rest of §16.2 uses (§XVI.10).

## XVI.10 The tangent $T(Z, c)$ and (16.5), Lemma 16.2.2, Definition 16.2.3 (2017: Lemma 16.4, Definition 16.14)

**The construction (pp. 716–718).** The text glues $T(Z, c)$ from all structures in $c$ along canonical bijections. The formalization takes one structure $(X_0, \alpha_0)$ of $c$ as base point: $T(Z, c) = |TX_0|$ Chapter16/P3H_TangentPolyfold · PolyfoldClass.TangentSpace, with the orbit space topology Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangentSpaceTopology. For $(X, \alpha) \in c$, $T\alpha = |T\mathfrak{f}|$ for a generalized isomorphism $\mathfrak{f} : X \to X_0$ with $\alpha_0 \circ |\mathfrak{f}| = \alpha$ Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangentMap. *It does not depend on* $\mathfrak{f}$ Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangentMap_eq: two such $\mathfrak{f}$ have the same orbit map $\alpha_0^{-1} \circ \alpha$, hence the same $|T\mathfrak{f}|$ by §XVI.9, Step 2. The text says "the unique $\mathfrak{f}$"; $\mathfrak{f}$ need not be unique (§XVI.9), but $|T\mathfrak{f}|$ is, and that is what is used. **(16.5)** Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangentMap_comp: $T\beta \circ |Th| = T\alpha$ for every generalized isomorphism $h : X \to Y$ with $\beta \circ |h| = \alpha$, by composing with $\mathfrak{f}$ for $Y$ and $T(h \circ \mathfrak{f}) = T\mathfrak{f} \circ Th$. $T\alpha$ is a homeomorphism Chapter16/P3H_TangentPolyfold · PolyfoldClass.isHomeomorph_tangentMap Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangentHomeomorph, because the orbit map of a generalized isomorphism is Chapter16/P3H_TangentPolyfold · EPLocalization.EPLoc.isHomeomorph_orbitMap_of_isGeneralizedIso.

**Lemma 16.2.2** Chapter16/P3H_TangentPolyfold · PolyfoldClass.lemma_16_2_2: (1) every $T\alpha$ is a homeomorphism; (2) a topology on $T(Z, c)$ for which one $T\alpha$ is a homeomorphism is the given one, since both have as open sets those whose preimage under $T\alpha$ is open; (3) $T(Z, c)$ is metrizable, by Theorem 8.1.15 for $|TX_0|$ Chapter8/TangentOrbit · metrizableSpace_tangent_orbitSpace (Part VIII, §VIII.9).

**Definition 16.2.3** (restated) Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangent. $(TX, T\alpha)$ is a polyfold structure on $T(Z,c)$ Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangentStructure: $|TX|$ is paracompact by Theorem 8.1.15 Chapter16/P3H_TangentPolyfold · PolyfoldStructure.tangent_paracompact Chapter8/TangentOrbit · theorem_8_1_15, and $TX$ satisfies (ZHOU) by Proposition 8.1.16 Chapter16/P3H_TangentPolyfold · PolyfoldStructure.tangent_isZhou Chapter8/TangentZhouReflect2021 · proposition_8_1_16_2021 (Part VIII, §VIII.38). Two such structures are equivalent through $Th$ for any $h$ with $\beta \circ |h| = \alpha$, by (16.5) Chapter16/P3H_TangentPolyfold · PolyfoldClass.tangentStructure_equiv. $Tc$ is the class of $(TX_0, T\alpha_0)$, and every $(TX, T\alpha)$, $(X,\alpha) \in c$, lies in it Chapter16/P3H_TangentPolyfold · PolyfoldClass.definition_16_2_3. *Reading.* $T(Z,c)$ lives one universe up, as $TX$ does in Chapter 8. The tangent of the polyfold $\mathbb{R}$ of §XVI.1 is a witness. The raised polyfolds $(Z,c)^i$ and the projection $\tau_Z$ are not formalized (§XVI.6).

## XVI.11 The remark before Definition 16.2.5, Definition 16.2.5 and Theorem 16.2.6 (2017: Definition 16.15, Theorem 16.2): sc-differential forms on a polyfold

**The remark (p. 720)** Chapter16/P3H_FormsIndependence · EPLocalization.EPLoc.formsPush_eq_of_orbitMap_eq_or: for generalized isomorphisms $h, h' : X \to Y$ with $|h| = |h'|$, $h_* = h'_*$ on $\Omega^k_{ep,\infty}$ if $X$ or $Y$ has (LR); likewise for pull-backs Chapter16/P3H_FormsIndependence · EPLocalization.EPLoc.formsPullback_eq_of_orbitMap_eq_or. This is Proposition 11.2.9 Chapter11/GapP2_B2_OrbitForms · formsPush_eq_of_orbitMap_eq (Part XI, §XI.4), with (LR) on either side by Proposition 10.3.16; for polyfold structures (ZHOU) gives (LR), so $h_*\omega$ depends only on $\alpha$ and $\beta$.

**Definition 16.2.5** (restated). A representative is $(\omega, (X,\alpha))$ with $(X,\alpha) \in c$ and $\omega \in \Omega^k_{ep,\infty}(X)$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.FormRep, related to $(h_*\omega, (Y,\beta))$ for a generalized isomorphism $h$ with $\beta \circ |h| = \alpha$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.FormRep.Rel; this is an equivalence relation, by $1_* = 1$ and $(h' \circ h)_* = h'_* \circ h_*$ Chapter16/P3H_PolyfoldForms · EPLocalization.EPLoc.formsPush_id Chapter16/P3H_PolyfoldForms · EPLocalization.EPLoc.formsPush_comp. $\Omega^k_{sc}(Z,c)$ is the quotient Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm, with classes $[(\omega,(X,\alpha))]$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.mk. For every $(X,\alpha) \in c$ the representative on $(X,\alpha)$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.toStructure gives a bijection $\Omega^k_{sc}(Z,c) \cong \Omega^k_{ep,\infty}(X)$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.equivStructure, and representatives on related structures are related by $f^*$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.ScForm.toStructure_eq_formsPullback. *The vector space structure.* It is transported from one structure of $c$ and then satisfies the text's formulas for every representative: $[(\omega,q)] + [(\omega',q)] = [(\omega+\omega',q)]$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.mk_add and $r[(\omega,q)] = [(r\omega,q)]$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.mk_smul, so each bijection is linear Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.equivStructure□ .

**Theorem 16.2.6** (narrower) Chapter16/P3H_PolyfoldForms · PolyfoldClass.theorem_16_2_6_complex: $d : \Omega^k_{sc}(Z,c) \to \Omega^{k+1}_{sc}(Z,c)$ is linear with $d[(\omega,(X,\alpha))] = [(d\omega,(X,\alpha))]$ for every $(X,\alpha) \in c$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.d_mk, $d \circ d = 0$ Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.d_comp_d, and $\omega \mapsto [(\omega,(X,\alpha))]$ is a linear isomorphism commuting with $d$. *Proof.* $d$ is defined on representatives Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.d; it respects the relation because $h_*$ commutes with $d$ for a generalized isomorphism Chapter16/P3H_PolyfoldForms · PolyfoldClass.ScForm.formsPush_d (Part XI); $d \circ d = 0$ holds on $\Omega^*_{ep,\infty}(X)$. Not formalized: the second part, that sc-smooth maps of polyfolds pull the complex back; it needs the morphisms of Definitions 16.1.6–16.1.11 (§XVI.6).

## XVI.12 Definitions 16.4.1, 16.4.2 and Proposition 16.4.3 (2017: Definitions 16.29, 16.30, Proposition 16.6): branched suborbifolds

**Definition 16.4.1** (restated). A representative is $(\Theta,(X,\alpha))$ with $(X,\alpha) \in c$ and $\Theta$ a branched ep⁺-subgroupoid of $X$ `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedRep`; $(\Theta,(X,\alpha)) \sim (\Theta',(X',\alpha'))$ if a generalized isomorphism $f$ has $\alpha' \circ |f| = \alpha$ and $f^*\Theta' = \Theta$ `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedRep.Rel`, an equivalence relation `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedRep.Rel.trans`, reflexive because $1^*\Theta = \Theta$ for functorial $\Theta$ `Chapter16/P3H_BranchedSuborbifold · EPLocalization.EPLoc.pullTheta_id_of_isFunctorial`. A branched suborbifold is a class `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold`. $\theta = |\Theta| \circ \alpha^{-1}$ **is well defined** `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.toFun` `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_1_wellDefined`: $\theta(\alpha(|x|)) = \Theta(x)$ for every representative and every $x$. The key step is $(f^*\Theta')(x) = \Theta'(y)$ whenever $|f|(|x|) = |y|$ `Chapter16/P3H_BranchedSuborbifold · EPLocalization.EPLoc.pullTheta_eq_of_orbitMap`, the text's $|f^*\Theta'| = |\Theta'| \circ |f|$.

**Definition 16.4.2.** A property holds for $\theta$ if it holds for some representative `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.HasRepProperty`; for a property stable under pull-backs by generalized isomorphisms, some representative has it iff all have `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.hasRepProperty_iff`. (1) closed, (2) compact, (3) of manifold-type, (4) of orbifold-type are such properties by Theorem 11.3.3 (3)–(5) (Part XI, §XI.7), so they hold for every representative iff for one `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_closed` `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_compact` `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_manifold` `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_orbifold`, and so is the dimension, used in Theorem 16.4.6 `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_nDim`. (5) tame (narrower) `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.IsTame`: some representative is tame (Definition 9.3.6). The text says on p. 729 that tameness is preserved by $f_*$ and $f^*$; this is not formalized (§XVI.6), so only the "some representative" form is available.

**Proposition 16.4.3** (narrower). *The set* `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_supp`: $\operatorname{supp}\theta = \alpha(|\operatorname{supp}\Theta|)$ for every representative, and $\alpha$ restricts to a homeomorphism $|\operatorname{supp}\Theta| \cong \operatorname{supp}\theta$ `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_homeomorph`. *Manifold-type* `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_manifold`: with Proposition 9.1.9 (1), $|\operatorname{supp}\Theta|$ carries an M-polyfold structure for which $\operatorname{supp}\Theta \to |\operatorname{supp}\Theta|$ is a local sc-diffeomorphism. *Orbifold-type* `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_orbifold`: with Proposition 9.1.9 (2), $|\operatorname{supp}\Theta|$ is the orbit space of the ep-groupoid on $\operatorname{supp}\Theta$. This is done for every representative; that two representatives induce sc-diffeomorphic structures, the word "natural" of the text, is not formalized (§XVI.6).

## XVI.13 Definitions 16.4.4 and 16.4.5 (2017: Definitions 16.31, 16.34), and the integrals of p. 732

**Definition 16.4.4** (restated). A representative is $((\Theta,\hat{T}_\Theta),(X,\alpha))$ with $(\Theta,\hat{T}_\Theta)$ an oriented branched ep⁺-subgroupoid of $X$ (Chapter 11) and $\hat{T}_\Theta$ zero off $X_\infty$ `Chapter16/P3H_`

BranchedSuborbifold · PolyfoldClass.OrientedRep; related by $f^*(\Theta', \hat{T}_{\Theta'}) = (\Theta, \hat{T}_\Theta)$ and $\alpha' \circ |f| = \alpha$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.Rel. An oriented branched suborbifold is a class Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedSuborbifold, with an underlying branched suborbifold Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedSuborbifold.toBranched. *Reading.* The text's $\hat{T}_\Theta$ lives over $X_\infty$, and $f^*$ only reads it there; the formalization makes it $0$ elsewhere, which reflexivity needs: $1^*\hat{T} = \hat{T}$ holds for such $\hat{T}$ Chapter16/P3H_BranchedSuborbifold · EPLocalization.EPLoc.pullSection_id_of_mem Chapter16/P3H_BranchedSuborbifold · EPLocalization.EPLoc.pull_id_of_mem. Every pull-back of Chapter 11 has this property, so it is a normalization, not an added hypothesis.

**Definition 16.4.5** Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.definition_16_4_5_wellDefined: if two representatives are related through $f$, then $f^*\partial\Theta' = \partial\Theta$ and $f^*\hat{T}_{\partial\Theta'} = \hat{T}_{\partial\Theta}$, by Theorem 11.3.9 Chapter11/BoundaryOrientationEquiv · theorem_11_3_9_pull_boundaryOrientation (Part XI, §XI.8); so $\partial\hat{\theta} = [\partial(\Theta, \hat{T}_\Theta), (X, \alpha)]$ does not depend on the representative. The boundary function $\partial\theta = |\partial\Theta| \circ \alpha^{-1}$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryFun does not either Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryFun_eq_of_rel. *Readings.* The compatibility holds without tameness, which the text assumes. $\partial(\Theta, \hat{T}_\Theta)$ is formalized componentwise, as in Chapter 11.

**The integrals (p. 732).** On a representative that is compact and tame of dimension $n$ in the sense of Chapter 9 (oriented tame structures describing $\hat{T}_\Theta$ at every point of $\operatorname{supp}\Theta$) Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.IsTameOriented: $\int_{\hat{\theta}} \tau = \mu_\omega^{\hat{\Theta}}(|\operatorname{supp}\Theta|)$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.integral and $\oint_{\hat{\theta}} \tau$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.integralEff, $\omega$ the representative of $\tau$ on $(X, \alpha)$; for the effective integral (ZHOU) gives the local uniqueness condition (LUC) by Proposition 7.4.10. The boundary integrals Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegral Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralEff are the measures of Theorems 9.5.15 and 9.5.16 (Part IX, §IX.34), with separate forms for dimension $1$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralDim1 Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralEffDim1. **"The definition does not depend on the choice of the representatives"** (narrower), among tame compact representatives: $\int$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.integral_eq_of_rel and $\oint$ Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.integralEff_eq_of_rel. *Proof.* The representative of $\tau$ on the first structure is $f^*$ of that on the second, and the tame integrals are invariant under pull-back, Theorems 11.4.1 and 11.4.2 (Part XI, §XI.9) Chapter16/P3H_BranchedSuborbifold · EPLocalization.EPLoc.tameBranchedIntegral_pull Chapter16/P3H_BranchedSuborbifold · EPLocalization.EPLoc.tameBranchedIntegralEff_pull. For the boundary integrals the independence is obtained from Stokes' theorem, with sc-smooth partitions of unity Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegral_eq_of_rel Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralEff_eq_of_rel Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralDim1_eq_of_rel: both sides equal the integral of $d\tau$. The integrals are defined on tame representatives because tameness is not known to be invariant (§XVI.6).

## XVI.14 Theorem 16.4.6 (2017: Theorem 16.5): Stokes' theorem on polyfolds, with sc-smooth partitions of unity

**Theorem 16.4.6** (with sc-smooth partitions of unity) `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_hyp`: let $(Z, c)$ admit sc-smooth partitions of unity (Definition 16.1.5), $\hat{\theta} = [(\Theta, \hat{T}_\Theta), (X, \alpha)]$ oriented, compact and tame of dimension $n = m + 1 \geq 2$, and $\tau \in \Omega^m_{sc}(Z, c)$. Then $\int_{\hat{\theta}} d\tau = \int_{\partial\hat{\theta}} \tau$. The effective form `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_eff_hyp`: $\oint_{\hat{\theta}} d\tau = \oint_{\partial\hat{\theta}} \tau$; dimension 1 `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_dim1_hyp` `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_eff_dim1_hyp`; both identities together `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_both_hyp`.

*Proof. Step 1, to the representative.* The representative of $d\tau$ on $(X, \alpha)$ is $d$ of the representative $\omega$ of $\tau$ `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.toStructure_d` (§XVI.11), so both sides are integrals over $\Theta$ on $X$: $\mu^{\hat{\Theta}}_{d\omega}(|\operatorname{supp}\Theta|)$ and the boundary measure of $\omega$. *Step 2, the partitions of unity.* By Definition 16.1.5 (2) `Chapter16/P3Z_PolyfoldPartitions · PolyfoldClass.definition_16_1_5_two` (§XVI.3), $X$ admits sc-smooth partitions of unity. *Step 3, Stokes on $X$.* Theorem 9.5.17 `Chapter9/GapP3_K1_Theorems · theorem_9_5_17`, and for the effective form Theorem 9.5.18 `Chapter9/GapP3_K1_Theorems · theorem_9_5_18` with (LUC) from (ZHOU) (Part IX, §IX.35), give the identity on $X$. By §XVI.13 the result does not depend on the representative.

*The added hypothesis.* Theorem 16.4.6 does not state that $(Z, c)$ admits sc-smooth partitions of unity, but its proof reduces to Theorem 9.5.17, which assumes them for $X$; the polyfold theorem is proved with the same hypothesis, as Definition 16.1.5. Witness `Chapter16/P3H_BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_hyp_witness`: the polyfold $\mathbb{R}$ of §XVI.1 admits sc-smooth partitions of unity `Chapter16/P3Z_PolyfoldExamples · PolyfoldStructure.z3Polyfold_admitsScPartitionsOfUnity`. Without it the theorem is a new gap (§XVI.6). *Reading.* "Tame" is Chapter 9's, on the representative used (§XVI.13).

## XVI.15 Witnesses and the hypotheses of wave 2

**Hypotheses taken.** *Sc-smooth partitions of unity* in Theorem 16.4.6 and in the independence of the boundary integrals (§§XVI.13–XVI.14), witnessed by the polyfold $\mathbb{R}$ of §XVI.1 (`theorem_16_4_6_hyp_witness`). *An effective target* in `lemma_16_2_1_of_isEffective`, witnessed by the point ep-groupoid; without it Lemma 16.2.1 does not hold as we read it (§XVI.9), and the orbit form needs no added hypothesis. *A tame compact representative* for the integrals: this is the text's "tame, compact", read on a representative because tameness is not known to be invariant. The normalization $\hat{T}_\Theta = 0$ off $X_\infty$ is not a hypothesis (§XVI.13).

**Witnesses of the definitions.** The tangent polyfold: $T$ of the polyfold $\mathbb{R}$ of §XVI.1 (§XVI.10). Forms: $\Omega^k_{sc}(Z, c)$ is a vector space isomorphic to $\Omega^k_{ep,\infty}(X)$ for every structure in $c$ (§XVI.11). Branched and oriented suborbifolds: a class exists as soon as a representative does; no separate example was built in wave 2, and the hemisphere of Part IX (§IX.35) is the nearest ep-level example of a compact tame oriented one.

# Concordance

Every declaration cited in this book, by file, with its line in the source. 10730 declarations in 843 files. The web edition also shows each exact formal statement.

**Polyfold/Chapter1/Book2021.lean.** remark_1_1_2 (63); remark_1_1_2_openMapping (82); proposition_1_1_11 (107).

**Polyfold/Chapter1/ChainRule.lean.** IsSc1With.norm_sub_le_of_segment (68); IsSc1With.exists_delta (95); IsSc1With.hasFDerivWithinAt_comp (125); IsSc1With.tangentMapWith_comp (245); IsSc1With.comp (256); theorem_1_3_1' (269); theorem_1_3_1 (281); IsSc1.tangentMap_congr (289); IsSck.congr (298); IsSck.comp (306); IsScSmooth.comp (317); IsSck.comp_of_isRelOpenInQuadrant (323).

**Polyfold/Chapter1/Degeneracy.lean.** quadrantDeg (60); degIndexWith (65); coordMap (77); coordMap_surjective (82); finrank_quotient_ker_coordMap (91); mem_ker_coordMap (99); eq_zero_of_eventually_nonneg_mul (110); eventually_add_smul_mem_quadrant (122); map_ker_coordMap_le (141); quadrantDeg_eq_of_equiv (158); quadrantDeg_zero (177); lemma_1_3_4 (184); lemma_1_3_4_dim (204); degIndex (226); degIndex_eq (229); degIndex_quadrant (237).

**Polyfold/Chapter1/ExpositionLemmas.lean.** ScScale.finiteDimensional_of_surjective_incl (75); ScBanach.smooth_eq_top_of_finiteDimensional (96); ScBanach.toBase_bijective_of_finiteDimensional (102); ScBanach.isoConstOfFiniteDimensional (111); ScBanach.isoConstOfFiniteDimensional_apply (117); ScBanach.IsScSplitting.isTopCompl (128); ScBanach.IsScSplitting.scProj (136); ScBanach.IsScSplitting.scProj_apply (152); ScBanach.IsScSplitting.scProj_idem (156); ScBanach.IsScSplitting.range_scProj (161); ScBanach.IsScSplitting.ker_scProj (169); ScOperator.idem_lvl (177); ScOperator.levelSub_range_eq (182); ScBanach.isScSplitting_of_idem (198); ScBanach.isScSplitting_iff_exists_scProj (236); ScBanach.inclBound (256); ScBanach.one_le_inclBound (258); ScBanach.norm_incl_le (264); ScBanach.norm_toBase_le (267); ScBanach.norm_inclK_le (277); ScBanach.exists_smooth_mem_coset (295); ScBanach.mk_mem_quotient_smooth_iff (365); continuousOn_apply_iff_uniformOnCompact (392); IsSc1With.deriv_uniformOnCompact (442); ScBanach.lvlSet_const (458); isSc1_iff_contDiffOn_const (469); isSc1_iff_fderivWithin_const (483); isScSmooth_iff_contDiffOn_const (494); isSc1_iff_contDiffOn_of_isRelOpenInQuadrant (508); isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant (514).

**Polyfold/Chapter1/FredholmIndex.lean.** linIndex (93); HasFiniteIndex (97); finite_of_finite_quotient (103); finite_of_finite_ker_range (111); finiteDimensional_ker_of_comp (137); finiteDimensional_coker_of_comp (143); HasFiniteIndex.comp_and_linIndex (153); HasFiniteIndex.comp (215); linIndex_comp (218); HasFiniteIndex.equiv (231); linIndex_equiv (238); HasFiniteIndex.

of_finiteRange_sub_id_and_linIndex (250); linIndex_eq_zero_of_finiteRange_sub_id (306); HasFiniteIndex.of_bijective_add_and_linIndex (313); HasFiniteIndex.of_comp_of_comp (333); fredholmIndex (351); IsFredholm.hasFiniteIndex (355); finiteDimensional_range_sub_id_of_equiv (362); bijective_one_add_of_norm_lt (369); hasFiniteIndex_comp_add_of_norm_lt (384); fredholmIndex_add_of_norm_lt (401); fredholmIndex_add_isCompactOperator (439); ScBanach.finrank_levelSub_of_le_smooth (468); ScFredholmSplitting.C_le_smooth (480); ScFredholmSplitting.fredholmIndex_lvl (488); IsScFredholm.fredholmIndex_lvl (498); scIndex_eq_fredholmIndex (505); proposition_1_1_11_index (509); proposition_1_1_11_index_lvl (517).

**Polyfold/Chapter1/FredholmPerturbation.lean.** isStrictMap_of_isClosed_range (62); isFredholm_of_banach (73); isCompactOperator_of_finiteDimensional_range (84); finiteDimensional_ker_one_add (103); exists_bound_below_on_compl (130); isClosed_range_one_add (181); exists_unit_far (215); pick (243); pick_notMem (246); chain (255); chain_mono (260); chain_eq_sup (265); chain_ne_top (276); chain_isClosed (285); coFG_range_one_add (292); isFredholm_one_add (333); isCompactOperator_sub_one_of_equiv (344); isFredholm_add_isCompactOperator (354).

**Polyfold/Chapter1/GapP1_PartialCones.lean.** IsPartialCone (40); exists_toBase_mem_of_isOpen (47); interior_lvlSet_nonempty_of_interior (54); IsPartialCone.isScDomain_of_convex (68); IsPartialCone.isScDomain (81); IsPartialCone.tangentSet (87); IsPartialQuadrant.isPartialCone (106); remark_1_1_17_chainRule (128).

**Polyfold/Chapter1/GapP1_Sobolev1.lean.** GapP1Sobolev.cyl (39); GapP1Sobolev.IsTest (133); GapP1Sobolev.HasWeakDeriv (286).

**Polyfold/Chapter1/GapP1_Sobolev10.lean.** GapP1Sobolev.chi (119); GapP1Sobolev.tendsto_cut (232).

**Polyfold/Chapter1/GapP1_Sobolev11.lean.** GapP1Sobolev.trig (94); GapP1Sobolev.pD_trig (129); GapP1Sobolev.bc_eexp (165); GapP1Sobolev.tendsto_box_sq (228).

**Polyfold/Chapter1/GapP1_Sobolev12.lean.** GapP1Sobolev.approx_supp (80); GapP1Sobolev.dense_test (281).

**Polyfold/Chapter1/GapP1_Sobolev13.lean.** GapP1Sobolev.nonneg_of_strictMono (26); GapP1Sobolev.scaleH (30); GapP1Sobolev.toBase_toH (37); GapP1Sobolev.example_1_1_3 (49); GapP1Sobolev.example_1_1_3_lvl (58); GapP1Sobolev.level0_isometry (64); GapP1Sobolev.level0_surjective (82).

**Polyfold/Chapter1/GapP1_Sobolev2.lean.** GapP1Sobolev.bc (67); GapP1Sobolev.hasSum_bc (115); GapP1Sobolev.bc_ds (365); GapP1Sobolev.bc_dt (433); GapP1Sobolev.ae_eq_zero_of_forall_test (499); GapP1Sobolev.HasWeakDeriv.ae_eq (573).

**Polyfold/Chapter1/GapP1_Sobolev3.lean.** GapP1Sobolev.wt (28); GapP1Sobolev.MemH (108); GapP1Sobolev.Hsub (123); GapP1Sobolev.H (233); GapP1Sobolev.norm_sq_H (257); GapP1Sobolev.isometry_ΦH (264).

**Polyfold/Chapter1/GapP1_Sobolev4.lean.** GapP1Sobolev.isClosed_range_ΦH (42); GapP1Sobolev.completeSpace_H (95); GapP1Sobolev.inclH (154); GapP1Sobolev.valCLM_inclH (159); GapP1Sobolev.inclH_injective (163).

**Polyfold/Chapter1/GapP1_Sobolev5.lean.** GapP1Sobolev.totallyBounded_of_approx (30); GapP1Sobolev.totallyBounded_image_of_coords (45); GapP1Sobolev.exists_cutoff' (90);

GapP1Sobolev.summable_freq_bc (122); GapP1Sobolev.cutR (205); GapP1Sobolev.H1ball (208); GapP1Sobolev.norm_cutR_sq_le (213); GapP1Sobolev.tsum_sub_le (266); GapP1Sobolev.exists_N_eta (313); GapP1Sobolev.rellich (342).

**Polyfold/Chapter1/GapP1_Sobolev6.lean.** GapP1Sobolev.totallyBounded_of_isometry (23); GapP1Sobolev.dvCLM_mem_H1ball (64); GapP1Sobolev.cutWt (81); GapP1Sobolev.abs_cutWt_le (83); GapP1Sobolev.tail_le (94); GapP1Sobolev.isCompactOperator_inclH (106).

**Polyfold/Chapter1/GapP1_Sobolev7.lean.** GapP1Sobolev.SmPer.hasWeakDeriv (146); GapP1Sobolev.IsTest.mem_Hsub (177).

**Polyfold/Chapter1/GapP1_Sobolev8.lean.** GapP1Sobolev.leibniz (134).

**Polyfold/Chapter1/GapP1_Sobolev9.lean.** GapP1Sobolev.mulH (128); GapP1Sobolev.dv_mulH (131).

**Polyfold/Chapter1/Levels.lean.** exists_delta_of_continuousOn (117); continuousOn_clm_comp_of_isCompactOperator (144); contDiffOn_clm_comp_of_isCompactOperator (170); ScBanach.shift_lvlSet (310); ScBanach.shift_tangentSet (320); ScBanach.shift_inclK_mem_lvlSet (331); ScBanach.incl_shift_inclK (337); ScBanach.tangent_inclK (343); IsSc0.levelMap_shift (354); IsSc0.shift (364); IsScDomain.shift (377); levelDeriv (389); toBase_levelDeriv (394); levelDeriv_zero (400); IsSc1With.exists_levelDeriv (409); IsSc1With.toBase_levelDeriv (432); IsSc1With.levelMap_tangentMapWith (439); IsSc1With.continuousOn_levelDeriv (455); IsSc1With.continuousOn_levelDeriv_clm (468); IsSc1With.hasFDerivWithinAt_level (483); IsSc1With.contDiffOn_level (592); IsSc1With.levelDeriv_eq (604); IsSc1Levels (644); IsSc1Levels.continuousOn_clm (656); IsSc1Levels.contDiffOn (662); IsSc1Levels.toBase_apply (674); IsSc1Levels.isSc1With (704); IsSc1With.isSc1Levels (734); proposition_1_2_1 (744); IsSc1With.shift (752); IsSck.shift (769); IsScSmooth.shift (789); IsSck.contDiffOn_levels' (802); IsSck.contDiffOn_levels (856); IsSck.contDiffOn_levels_of_le (862); IsSck.contDiffOn_base (868); IsScSmooth.contDiffOn_levels (874); proposition_1_2_4 (884); corollary_1_2_5 (919); corollary_1_2_5_base (932); IsScSmooth.of_contDiffOn_levels (943); IsScSmooth.of_contDiffOn_finiteDimensional (954).

**Polyfold/Chapter1/LevelsWeak.lean.** ScBanach.castCLE (96); ScBanach.inclLE (111); ScBanach.toBase_inclLE (114); ScBanach.preimage_inclLE_lvlSet (127); ScBanach.shiftAdd (174); ScBanach.shiftAddZeroIso (185); ScBanach.shiftAddShiftIso (197); ScBanach.toBase_shiftAddShiftIso_symm (209); ScOperator.zeroOp (230); ScOperator.addOp (235); IsSck.zero (244); IsSck.add (248); IsSck.sum (254); cmm_eval_eq (272); hasFDerivAt_cmm_eval (279); IsDerivTower (322); levelDeriv-Tower (336); baseDerivTower (342); contDiffOn_levels_min (348); levelDerivTower_toBase (368); baseDerivTower_toBase (405); isDerivTower_of_contDiffOn_levels (419); IsDerivTower.eq_of_toBase (450); ElemData (467); ElemData.Good (483); ElemData.dom (489); ElemData.map (492); ElemData.mem_lvlSet_dom (496); ElemData.ρ₀ (502); ElemData.ρ (505); ElemData.ρ₀_incl (515); ElemData.map_toBase (532); ElemData.levelMap_eq (552); ElemData.levelMap_incl_eq (565); ElemData.isSc0 (584); ElemData.deriv (629); ElemData.deriv_apply (641); ElemData.deriv_toBase (651); ElemData.hasFDerivWithinAt_level (679); ElemData.continuousOn_deriv (719); ElemData.isSc1Levels (740); ElemData.isSc1With (746); IsScDomain.of_eq (753); IsSck.of_eq_set (763); ScOperator.tangentX (774); ScOperator.tangentH (779); ElemData.shiftData (809); ElemData.derivData (813); ElemData.slotData (818); ElemData.Good.shiftData (823); ElemData.Good.derivData (828); ElemData.Good.slotData (836); ElemData.tangentSet_dom (845); ElemData.tangentMap_eqOn (854); ElemData.isSck (900); ElemData.base (935); ElemData.base_dom (938); ElemData.base_good (942); proposition_1_2_4_text (949); IsScSmooth.of_contDiffOn_levels_text (973); corollary_1_2_5_text (984); IsScSmooth.

of_contDiffOn_levels_finiteDimensional_text (1005).

**Polyfold/Chapter1/Properties.lean.** IsSc1With.exists_delta_deriv (69); IsSc1With.continuousOn_deriv_clm (93); IsSc1With.contDiffOn_base (120); IsSc1With.scOperator_deriv (130); proposition_1_2_4_zero (165); corollary_1_2_5_zero (233).

**Polyfold/Chapter1/ScBanach.lean.** ScScale (84); ScScale.toBase (104); ScScale.toBase_injective (112); ScScale.inclK (117); ScScale.toBase_inclK (126); ScScale.smooth (135); ScScale.mem_smooth_iff (137); ScScale.smoothLvl (141); ScScale.exists_inclK_of_mem_smoothLvl (148); ScScale.smoothLvl_submodule_closed_sub (162); ScBanach (171); ScBanach.denseRange_inclK (181); ScBanach.denseRange_incl (184); ScBanach.constScale (191); ScBanach.constScale_toBase (200); ScBanach.constScale_smoothLvl (206); ScBanach.const (213); ScBanach.const_toBase (221); ScBanach.prodScale (227); ScBanach.prodScale_toBase (238); ScBanach.prodScale_smoothLvl (244); ScBanach.prod (267); ScBanach.prod_smoothLvl (279); ScBanach.shiftScale (285); ScBanach.shiftScale_toBase (291); ScBanach.shiftScale_smoothLvl (297); ScBanach.shift (315); ScBanach.shift_toBase (323); ScBanach.shift_smoothLvl (326); ScBanach.shiftK (330); ScOperator (341); ScOperator.toBase_comm (352); ScOperator.ext_base (359); ScOperator.id (370); ScOperator.comp (375); ScOperator.add (383); ScOperator.mapsTo_smoothLvl (393); IsScOperator (405); ScOperator.isScOperator (408); IsScOperator.toScOperator (412); IsScOperator.toScOperator_zero (419); ScIso (429); ScIso.toScOperator (441); ScIso.symm (449); ScIso.trans (459); ScIso.toBase_comm (464); ScIso.symm_toBase_comm (467); ScIso.refl (471); ScPlus (479); ScBanach.shiftIncl (484); ScPlus.toScOperator (495); ScPlus.toScOperator_apply (497); ScPlus.isCompactOperator (501); ScBanach.levelSub (513); ScBanach.mem_levelSub (516); ScBanach.levelSub_zero (519); ScBanach.incl_mem_levelSub (522); ScBanach.isClosed_levelSub (525); ScBanach.IsScSubspace (531); ScBanach.IsScSubspace.isClosed (535); ScBanach.ofSubspaceScale (539); ScBanach.ofSubspaceScale_coe_toBase (554); ScBanach.mem_ofSubspaceScale_smoothLvl (561); ScBanach.ofSubspaceScale_smoothLvl (578); ScBanach.ofSubspace (584); ScBanach.IsScSplitting (598); ScBanach.IsScSplitting.closedComplemented (601); ScBanach.IsScSplitting.symm (606); ScBanach.proposition_1_1_7_iff (613); ScBanach.levelRestrict (633); ScBanach.levelRestrict_bijective (636); ScBanach.levelEquiv (643); ScBanach.finProj (652); ScBanach.finProj_apply_of_mem (659); ScBanach.finProj_eq_zero_iff (672); ScBanach.proposition_1_1_7 (680); ScBanach.quotIncl (723); ScBanach.quotIncl_mk (728); ScBanach.quotIncl_injective (731); ScBanach.quotIncl_isCompactOperator (740); ScBanach.quotientScale (752); ScBanach.quotientScale_toBase_mk (759); ScBanach.quotient (768); ScBanach.quotientMap (781); ScBanach.proposition_1_1_8 (787).

**Polyfold/Chapter1/ScFredholm.lean.** ScOperator.ker_lvl (70); ScOperator.range_lvl_le (80); ScPlus.toScOperator_mem_range_toBase_succ (89); ScBanach.range_toBase_succ_le (96); ScFredholmSplitting (106); IsScFredholm (124); ScFredholmSplitting.index (127); scIndex (131); ScFredholmSplitting.index_eq (140); ScFredholmSplitting.K_le_smooth (148); ScFredholmSplitting.ker_lvl (152); ScFredholmSplitting.range_lvl (155); ScFredholmSplitting.isFredholm (161); proposition_1_1_10 (184); ScPlus.regularizing_add (208); proposition_1_1_11_preprint (226).

**Polyfold/Chapter1/ScFredholmCriterion.lean.** ScOperator.IsRegularizing (69); ScOperator.IsRegularizing.ker_le_smooth (73); ScOperator.IsRegularizing.surjective (82); IsScFredholm.isRegularizing (92); isScFredholm_of_isRegularizing (105); isScFredholm_of_isRegularizing_index (210); isScFredholm_iff (217); isScOperator_of_mapsTo (228); ScBanach.ofSubspace_mem_range_toBase_iff (253); ScOperator.restrict (272); ScOperator.IsRegularizing.restrict (284); ScOperator.IsRegularizing.isScFredholm_restrict (296); IsScFredholm.restrict (307).

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**Polyfold/Chapter10/GapP2_F_LReg.lean.** EP.lreg (40); EP.Lreg (44); EP.fullyFaithful_ι_lreg (47); EP.lreg_iff_of_isEquivalence (54); EP.isClosedUnderIsomorphisms_lreg (59); EP.lreg_pointEP (63); EP.Lreg.point (67); EPLocalization.EPLoc.lreg (74); EPLocalization.EPLoc.lreg_A_of_lreg (78); EPLocalization.EPLoc.isClosedUnderIsomorphisms_lreg (83).

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**Polyfold/Chapter10/GapP2_F_TripleProduct.lean.** MPolyfold.isSubMPolyfold_univ_prod (70); MPolyfold.prodSubtypeMk (91); MPolyfold.isScDiffeoOn_prodSubtype (101); MPolyfold.isSubMPolyfold_image_of_isScDiffeoOn (118); WeakFibredProduct.objSetTriple (199); WeakFibredProduct.morSetTriple (203); WeakFibredProduct.Obj.toTriple (209); WeakFibredProduct.objSetTriple_eq (222); WeakFibredProduct.isSubMPolyfold_objSetTriple (243); WeakFibredProduct.isSubMPolyfold_morSetTriple (258); WeakFibredProduct.theorem_10_2_2_triple (277).

**Polyfold/Chapter10/GapP2_F_ZhouExample.lean.** IsLossFamily.add (60); IsLossFamily.sub (71); IsLossFamily.comp_neg (82); ZhouSigma.remark_10_1_8 (351); ZhouSigma.not_isLR (374).

**Polyfold/Chapter10/GapP2_GBundleFamily.lean.** EpStrongBundle.π_base_eq_of_orbitMap_eq (67); EpStrongBundle.proposition_10_4_7 (90); EpStrongBundle.proposition_10_4_7' (177); GapP2_GBundleFamilyExamples.proposition_10_4_7_hypotheses (201).

**Polyfold/Chapter10/GapP2_GBundleLR.lean.** EpStrongBundle.HasLRLocal (121); EpStrongBundle.isLRLocal_iff_hasLRLocal (126); EpStrongBundle.hasLRLocal_iff_of_equivariant (136); EpStrongBundle.IsStrongBundleEquivalence.exists_transport (227); EpStrongBundle.IsStrongBundleEquivalence.transport_Γ (271); EpStrongBundle.IsStrongBundleEquivalence.π_transport_iff (289); EpStrongBundle.proposition_10_4_6 (308); SBLocalization.proposition_10_5_6 (375).

**Polyfold/Chapter10/GapP2_GDiagram.lean.** SEP.projEP (60); SBLocalization.SDiagram.square_10_24 (81); SBLocalization.GenSBMap.square_10_24 (112); EpFunctor.mor_Γ (138); EpFunctor.obj_Φ (151); EPLocalization.Diagram.isotropyHom_conj (165); EPLocalization.Diagram.exists_germ_q (197).

**Polyfold/Chapter10/GapP2_GDiagramLift.lean.** StrongBundle.IsLocalMap₂ (63); StrongBundle.IsLocalMap₂.comp (90); StrongBundle.IsLocalIso₂.isLocalMap₂ (111); SBLocalization.SDiagram.exists_lift_Q (144).

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**Polyfold/Chapter10/LocalizationCategory.lean.** MPolyfold.isLocalScDiffeoAt_of_injOn (109); MPolyfold.IsLocalScDiffeoAt.exists_injOn (129); EPLocalization.Diagram.isEquivalence_Φ_of_equiv_id (149); EPLocalization.isEquivalence_toFunctor_of_comp (165); EPLocalization.isLocalScDiffeo_obj_of_comp (197); EPLocalization.Diagram.isEquivalence_Φ_of_equiv (243); EPLocalization.Diagram.symm (253); EPLocalization.Diagram.equiv_comp_symm (257); EPLocalization.Diagram.equiv_symm_comp (263); EPLocalization.EPLoc (275); EPLocalization.EPLoc.toEP_of (283); EPLocalization.EPLoc.of_toEP (285); EPLocalization.EPLoc.homMk (296); EPLocalization.EPLoc.homMk_comp (298); EPLocalization.EPLoc.id_eq (301); EPLocalization.EPLoc.homMk_eq_homMk_iff (303); EPLocalization.EPLoc.homMk_surjective (306); EPLocalization.EPLoc.IsGeneralizedIso (314); EPLocalization.EPLoc.isGeneralizedIso_iff_isIso (318); EPLocalization.EPLoc.theorem_10_3_10 (325); EPLocalization.EPLoc.isIso_homMk_iff (340); EPLocalization.EPLoc.isIso_homMk (343); EPLocalization.EPLoc.inv_homMk (347); EPLocalization.EPLoc.orbitMap (356); EPLocalization.EPLoc.orbitMap_homMk (359); EPLocalization.EPLoc.continuous_orbitMap (361); EPLocalization.EPLoc.orbitMap_id (365); EPLocalization.EPLoc.proposition_10_3_11 (369); EPLocalization.EPLoc.orbitFunctor (375); EPLocalization.EPLoc.theorem_10_

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**Polyfold/Chapter11/BranchedEquivalenceTangent.lean.** MPolyfold.submanifoldTangent_image_of_isScDiffeoOn (144); MPolyfold.chartVec_tangentMapOn_of_isScDiffeoOn (168); MPolyfold.OrTangent.map (202); MPolyfold.IsOrientedAt.map_tangentMapOn (232); MPolyfold.SubmanifoldOrientation.mapDiffeo (273); MPolyfold.SubmanifoldOrientation.mapDiffeo_tangent (330); MPolyfold.IsTangentReducedAt.image_of_isScDiffeoOn (356); MPolyfold.isTangentReducedAt_image_iff_of_isScDiffeoOn (397); MPolyfold.mem_submanifoldBoundary_image_iff_of_isScDiffeoOn (412); EpGroupoid.LocalBranchingStructure.tangentSum_mapDiffeo (457); EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isScDiffeoOn (470); EpGroupoid.tangentTheta_tangentMapOn (492); EpGroupoid.boundaryTheta_apply_of_isScDiffeoOn (531); EpGroupoid.OrientedBranchingStructure.mapDiffeo (569); EpGroupoid.OrientedBranchingStructure.orientedSum_mapDiffeo (580); EpFunctor.hasIndependentTangentSums_pushforwardTheta (630); EpFunctor.hasIndependentTangentSums_pullbackTheta (641); EpFunctor.tangentValue_pushforwardTheta_obj (663); EpFunctor.tangentValue_pushforwardTheta (687); EpFunctor.tangentValue_pullbackTheta (699); EpFunctor.tangentTheta_pushforwardTheta (719); EpFunctor.tangentTheta_pullbackTheta (735); EpFunctor.boundaryTheta_pushforwardTheta (752); EpFunctor.boundaryTheta_pullbackTheta (764); EPLocalization.EPLoc.hasIndependentTangentSums_pushTheta (791); EPLocalization.EPLoc.hasIndependentTangentSums_pullTheta (799); EPLocalization.EPLoc.theorem_11_3_8_push (809); EPLocalization.EPLoc.theorem_11_3_8_pull (819); EPLocalization.EPLoc.tangentFunctor_map_eq (835); EPLocalization.EPLoc.theorem_11_3_7_push (844); EPLocalization.EPLoc.theorem_11_3_7_pull (860).

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**Polyfold/Chapter11/CoveringEquivalence.lean.** EpFunctor.IsProperCovering.lemma_11_6_1 (111); EpFunctor.IsProperCovering.lift (118); EpFunctor.IsProperCovering.mor_lift (123); EpFunctor.IsProperCovering.src_lift (125); EpFunctor.IsProperCovering.eq_lift (127); EpFunctor.IsProperCovering.obj_tgt_lift (130); EpFunctor.IsProperCovering.existsUnique_lift_tgt (134); EpFunctor.IsProperCovering.Fiber (152); EpFunctor.IsProperCovering.Fiber.val (159); EpFunctor.IsProperCovering.Fiber.mk (165); EpFunctor.IsProperCovering.fiberSMul (178); EpFunctor.IsProperCovering.fiberSMul_val_of (182); EpFunctor.IsProperCovering.instMulActionFiber (188); EpFunctor.IsProperCovering.smul_val_of (209); EpFunctor.IsProperCovering.FiberFull (217); EpFunctor.IsProperCovering.Translation (220); EpFunctor.IsProperCovering.toTranslation (252); EpFunctor.IsProperCovering.bijective_toTranslation_obj (271); EpFunctor.IsProperCovering.faithful_toTranslation (278); EpFunctor.IsProperCovering.full_toTranslation (286); EpFunctor.IsProperCovering.lemma_11_6_2 (304); EpFunctor.IsProperCovering.transportFiber (323); EpFunctor.IsProperCovering.transportFiber_val_of (327); EpFunctor.IsProperCovering.transportFiber_inv_transportFiber (332); EpFunctor.IsProperCovering.bijective_transportFiber (342); EpFunctor.IsProperCovering.transportFiber_smul (353); EpFunctor.IsProperCovering.translationMap (388); EpFunctor.IsProperCovering.lemma_11_6_3 (408); ProperCoveringHom (465); ProperCoveringHom.ext (477); ProperCoveringHom.mapsTo_fiber (481); ProperCoveringHom.id (488); ProperCoveringHom.comp (491); ProperCoveringHom.IsFiberBijective (509); ProperCoveringHom.IsEquivalence (515);

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**Polyfold/Chapter11/CoveringEquivalenceTheorem.lean.** MPolyfold.IsScDiffeoOn.restrictTarget (98); MPolyfold.IsLocalScDiffeo.evenlyCovered_of_isProperMap (106); EpFunctor.IsProperCovering.isProperMap (171); ProperCoveringHom.hObj (220); ProperCoveringHom.hMor (225); ProperCoveringHom.mor_conj (240); ProperCoveringHom.hMor_src (250); ProperCoveringHom.hMor_tgt (253); ProperCoveringHom.hFunctor (260); ProperCoveringHom.surjective_hObj (293); ProperCoveringHom.isLocalScDiffeo_hObj (305); ProperCoveringHom.finite_fiber_hObj (313); ProperCoveringHom.isProperMap_hObj (331); ProperCoveringHom.isScDiffeo_fibredMap_hFunctor (377); ProperCoveringHom.isProperCovering_hFunctor_of (454); ProperCoveringHom.projLeftHom (503); ProperCoveringHom.projRightHom (507); ProperCoveringHom.natTransProj (513); ProperCoveringHom.theorem_11_6_6 (527); CoveringEquivalence.Examples.theorem_11_6_6_pairToPoint (567).

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**Polyfold/Chapter11/GapP2_B2_EquivFormsInfty.lean.** EP.isEquivalence_liftFunctor (70); EP.epFormsPush (78); EP.epFormsInftyPush (103); EP.epFormsInftyPullback_comp_epFormsInftyPush (117); EP.epFormsInftyPush_comp_epFormsInftyPullback (126); EP.epFormsInftyPush_eq_of_comp_eq_id (149); EP.epFormsInftyD_comp_epFormsInftyPush (158); EP.lemma_11_2_4_two (170); EP.lemma_11_2_5_push (181); EP.lemma_11_2_6 (190); EP.lemma_11_2_6_push (204).

**Polyfold/Chapter11/GapP2_B2_FormDescent.lean.** MPolyfold.IsLocalScDiffeo.exists_tangentFiber_lift (57); MPolyfold.formPullback_injective_of_isLocalScDiffeo (71); MPolyfold.isScForm_of_formPullback (109); MPolyfold.formDescend (171); MPolyfold.formPullback_formDescend (179).

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EPLocalization.EPLoc.theorem_11_2_1_one (276); EPLocalization.EPLoc.theorem_11_2_1_two (283); EPLocalization.EPLoc.theorem_11_2_1_three (292).

**Polyfold/Chapter11/GapP2_B2_OrbitForms.lean.** EpFunctor.HasLocalNatIso (78); EpFunctor.hasLocalNatIso_of_orbitMap_eq (84); EpFunctor.HasLocalNatIso.liftOne (98); EpFunctor.HasLocalNatIso.formPullback_eq (114); EP.epFormsInftyPullback_eq_of_orbitMap_eq (172); EP.proposition_11_2_8 (192); EPLocalization.EPLoc.formsPullback_eq_of_orbitMap_eq (252); formsPush_eq_of_orbitMap_eq (262); EPLocalization.EPLoc.proposition_11_2_9 (279); EPLocalization.EPLoc.theorem_11_2_1_four (288).

**Polyfold/Chapter11/GapP2_B2_Poincare.lean.** EP.epFormsInftyToObj_epFormsInftyPullback (125); EP.epFormsInftyToObj_surjective_of_src_eq_tgt (161); EP.proposition_8_2_3_of_isOpen (229); EP.proposition_8_2_3 (261).

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**Polyfold/Chapter11/GapP2_H2FaceMeet.lean.** MPolyfold.isTame_restrictOpen (63); MPolyfold.isFaceStructuredOpen_restrictOpen (75); EpGroupoid.isEpFaceStructured_trivial_restrictOpen (88); EpGroupoid.FacesMeetInFaces (103); FaceMeetExample.V (125); FaceMeetExample.isEpFaceStructured_XV (144); FaceMeetExample.mem_face_of_path (227); FaceMeetExample.isOrbitFace_$\theta_0$ (252); FaceMeetExample.$\theta_0$_ne_$\theta_1$ (310); FaceMeetExample.cF_lt_or_gt (336); FaceMeetExample.not_isEpFace_inter (375); FaceMeetExample.not_facesMeetInFaces (413).

**Polyfold/Chapter11/GapP2_H2FaceMeetUnion.lean.** MPolyfold.mem_closure_inter_faces (57); EpGroupoid.IsEpFaceStructured.exists_nhds_degIndex_two (225); EpGroupoid.IsEpFaceStructured.faceObj_inter_eq_sUnion (266); EpGroupoid.FacesMeetInUnionsOfFaces (343); EpGroupoid.IsEpFaceStructured.facesMeetInUnions (354).

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**Polyfold/Chapter11/GapP2_H3SBLocalization.lean.** SEP.wfpLift (62); SBPC.fibredLift (104); SBPC.Diagram (135); SBPC.proposition_11_6_8 (164); SBPC.Diagram.comp (207); SBPC.Diagram.hasCommonRefinement_comp (299); SBPCLoc (406); SBPCLoc.isIso_I_map (464); SBPCLoc.homMk_eq (486); SBPCLoc.inv_homMk (505); SBPCLoc.fromLoc (513); SBPC.Examples.theorem_11_6_6_ep (528).

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**Polyfold/Chapter11/GapP2_H4Bd.lean.** EpGroupoid.boundaryMeasure (62); EpGroupoid.

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**Polyfold/Chapter11/GapP2_H4BdDiagram.lean.** EpFunctor.measure_span_core_bd (50); EPLocalization.Diagram.hcovB_pullTheta (167); EPLocalization.Diagram.hcovB_pushTheta (187); EPLocalization.EPLoc.theorem_11_4_4_pull (443); EPLocalization.EPLoc.theorem_11_4_4_push (476); EPLocalization.EPLoc.theorem_11_4_5_pull (508); EPLocalization.EPLoc.theorem_11_4_5_push (542).

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**Polyfold/Chapter11/GapP2_H4Compat.lean.** EpGroupoid.FormEval.IsPullback (51).

**Polyfold/Chapter11/GapP2_H4CompatInst.lean.** EpFunctor.IsEquivalence.epFormEval_isPullback (92); EP.epFormsInftyEval_isPullback (169).

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**Polyfold/Chapter11/GapP2_H4Diagram.lean.** EpFunctor.measure_span_core (77); EPLocalization.Diagram.theorem_11_4_1_pull (230); EPLocalization.Diagram.theorem_11_4_1_push (264); EPLocalization.EPLoc.theorem_11_4_1_pull (330); EPLocalization.EPLoc.theorem_11_4_1_push (361); EPLocalization.EPLoc.integral_pull (396); EPLocalization.EPLoc.integral_push (419); EPLocalization.EPLoc.remark_11_4_3 (447).

**Polyfold/Chapter11/GapP2_H4Eff.lean.** EpFunctor.TransportChart.card_effIsotropy_eq (63); EpFunctor.hcovL_pull (101); EpFunctor.hcovL_push (111); EpFunctor.measure_pull_core_eff (124); EpFunctor.theorem_11_4_2_pull_equiv (257); EpFunctor.theorem_11_4_2_push_equiv (282).

**Polyfold/Chapter11/GapP2_H4EffDiagram.lean.** EpFunctor.measure_span_core_eff (46); EPLocalization.EPLoc.theorem_11_4_2_pull (238); EPLocalization.EPLoc.theorem_11_4_2_push (272).

**Polyfold/Chapter11/GapP2_H4MSheets.lean.** EpGroupoid.FormEval.image_mem_msheets_cross (263); EpGroupoid.FormEval.msInteg_image_cross (359).

**Polyfold/Chapter11/GapP2_H4Measure.lean.** EpFunctor.hcov_pull (164); EpFunctor.hcov_push (175); EpFunctor.isMeasureZeroSupp_image (201); EpFunctor.isMeasureZeroSupp_preimage (233); EpFunctor.measurable_suppInv (320); EpFunctor.measurable_suppMap (351); EpFunctor.measure_pull_core (385); EpFunctor.theorem_11_4_1_pull_equiv (508); EpFunctor.theorem_11_4_1_push_equiv (531); EpFunctor.integral_pull_equiv (552); EpFunctor.integral_push_equiv (570).

**Polyfold/Chapter11/GapP2_H4Move.lean.** EpGroupoid.OrientedTameStructure.restrictRep (53); EpGroupoid.OrientedTameStructure.move (105); EpGroupoid.OrientedTameStructure.represents_move (143); EpGroupoid.FormEval.EvCompatOn (167); EpGroupoid.FormEval.integ_image_cross (224); EpGroupoid.OrientedTameStructure.sheetSum_move (265).

**Polyfold/Chapter11/GapP2_H4Pairs.lean.** EpFunctor.TransportChart.pullbackTheta_Q (85); EpFunctor.TransportChart.pullSection_Q (90); EpFunctor.TransportChart.evCompatOn_Q (127);

EpFunctor.TransportChart.exists_rep_push (175); EpGroupoid.OrientedTameStructure.exists_push_structure (211); EpGroupoid.OrientedTameStructure.pullStructure (260); EpGroupoid.OrientedTameStructure.represents_pullStructure (297).

**Polyfold/Chapter11/GapP2_HFaceIso.lean.** EpFunctor.restrictSub (130); EpFunctor.IsEquivalence.restrictSub (171); EpNatTrans.restrictSub (195); EpFunctor.IsEquivalence.isEpFaceStructured_iff (226); EPLocalization.faceEP (241); EPLocalization.Diagram.restrictFace (278); EPLocalization.Diagram.orbitMap_restrictFace (297); EPLocalization.Diagram.IsRefinement.restrictFace (317); EPLocalization.Diagram.Equiv.restrictFace (351); EPLocalization.EPLoc.faceIso (388); EPLocalization.EPLoc.homMk_restrictFace_eq_faceIso (395); EPLocalization.EPLoc.isGeneralizedIso_faceIso (403); EPLocalization.EPLoc.orbitMap_faceIso (411); EPLocalization.EPLoc.proposition_11_1_6_faceIso (439); EPLocalization.EPLoc.proposition_11_1_8_faceIso (460).

**Polyfold/Chapter11/GapP2_HFaces.lean.** MPolyfold.isSubMPolyfold_of_forall_nhds (74); EpGroupoid.image_faceAt_subset (188); EpGroupoid.orbitFacesThrough_eq_range (260); EpGroupoid.injective_orbitFaceAt_pt (273); EpGroupoid.IsEpFaceStructured.isWeaklyFaceStructured (295); EpGroupoid.IsEpFaceStructured.existsUnique_face_subset (310); EpGroupoid.IsEpFaceStructured.exists_nhds_eq_face (336); EpGroupoid.isClosed_faceObj (355); EpGroupoid.isSubMPolyfold_faceObj (360); EpGroupoid.isEpSubgroupoid_faceObj (368); EpGroupoid.faceEpGroupoid (378); EpGroupoid.isTame_faceEpGroupoid (407); EpGroupoid.isTame_faceEpGroupoid_mor (418); EpGroupoid.degIndex_faceEpGroupoid (424); EpGroupoid.proposition_11_1_6_faces (442).

**Polyfold/Chapter11/GapP2_HPCLocalization.lean.** WeakFibredProduct.pairFunctor (110); PC.IsEquivalence.of_comp_left (163); PC.fibredLift (198); PC.isEquivalence_fibredLift (219); PC.Diagram.comp (242); PC.Diagram.hasCommonRefinement_of_map (254); PC.Diagram.hasCommonRefinement_comp (314); PC.Diagram.hasCommonRefinement_assoc (323); PCLoc (391); PCLoc.I (429); PCLoc.isIso_I_map (456); PCLoc.inv_I_map (460); PCLoc.homMk_eq (482); PCLoc.isIso_homMk (497); PCLoc.inv_homMk (502); PCLoc.Examples.not_isIso_pairToPoint_isIso_I_map (518).

**Polyfold/Chapter11/GapP3_K1_BoundaryLoc.lean.** EP.boundaryMeasure_eq (45); EP.boundaryMeasureEff_eq (55); EPLocalization.EPLoc.theorem_11_4_4_pull_loc (78); EPLocalization.EPLoc.theorem_11_4_4_push_loc (109); EPLocalization.EPLoc.theorem_11_4_5_pull_loc (138); EPLocalization.EPLoc.theorem_11_4_5_push_loc (168).

**Polyfold/Chapter11/GapP3_K2_H4BdDiagramDim1.lean.** EpFunctor.measure_span_core_bd_dim1 (42); EPLocalization.EPLoc.theorem_11_4_4_pull_dim1 (390); EPLocalization.EPLoc.theorem_11_4_4_push_dim1 (416); EPLocalization.EPLoc.theorem_11_4_5_pull_dim1 (441); EPLocalization.EPLoc.theorem_11_4_5_push_dim1 (467).

**Polyfold/Chapter11/GapP3_K2_H4BdDim1.lean.** EP.zeroFormEval_epFormsInftyPullback (55); EpGroupoid.OrientedTameStructure.boundarySheetSum_move_dim1 (117); EpFunctor.measure_pull_core_bd_dim1 (287); EpFunctor.theorem_11_4_4_pull_equiv_dim1 (467); EpFunctor.theorem_11_4_4_push_equiv_dim1 (499); EpFunctor.theorem_11_4_5_pull_equiv_dim1 (538); EpFunctor.theorem_11_4_5_push_equiv_dim1 (571).

**Polyfold/Chapter11/MultisectionLocalStructure2021.lean.** SBLocalization.IsLocalRepOn (73); SBLocalization.HasLocalSectionStructure (80); SBLocalization.IsScPlusMultisection (86); SBLocalization.isScPlusMultisection_zeroMultisection (138); SBLocalization.isScPlusMultisection_

pullMult (199); SBLocalization.exists_isLocalRepOn_pushMult (214); SBLocalization.lemma_11_5_2_book (330); SBLocalization.SDiagram.isScPlusMultisection_pushMult (344); SBLocalization.SDiagram.isScPlusMultisection_pullMult (349); SBLocalization.GenSBMap.proposition_11_5_3_book (365).

**Polyfold/Chapter11/StrongBundleUpToEquivalence.lean.** SBLocalization.IsWFunctor (85); SBLocalization.isWFunctor_const (89); SBLocalization.IsWFunctor.eq_of_π_eq (92); SBLocalization.pullMult (100); SBLocalization.IsWFunctor.pullMult (102); SBLocalization.pushMult (112); SBLocalization.π_obj_choose (115); SBLocalization.pushMult_eq (122); SBLocalization.lemma_11_5_2 (130); SBLocalization.pullMult_pushMult (140); SBLocalization.pushMult_pullMult (145); SBLocalization.SDiagram.pushMult (154); SBLocalization.SDiagram.pullMult (158); SBLocalization.SDiagram.pushMult_eq (163); SBLocalization.SDiagram.pullMult_eq (174); SBLocalization.SDiagram.isWFunctor_pushMult (179); SBLocalization.SDiagram.isWFunctor_pullMult (183); SBLocalization.SDiagram.pullMult_pushMult (187); SBLocalization.SDiagram.pushMult_pullMult (192); SBLocalization.SDiagram.pushMult_eq_of_orbitMap_eq (200); SBLocalization.SDiagram.pullMult_eq_of_orbitMap_eq (210); SBLocalization.GenSBMap.pushMult (226); SBLocalization.GenSBMap.pullMult (229); SBLocalization.GenSBMap.proposition_11_5_3 (240); SBLocalization.IsOneInvariant (270); SBLocalization.isOneInvariant_const (275); SBLocalization.isOneInvariant_iff (279); EpStrongBundle.IsAuxNorm.isOneInvariant (296); SBLocalization.oneToOrbit (301); SBLocalization.oneToOrbit_injective (309); SBLocalization.oneMap (321); SBLocalization.continuous_oneMap (325); SBLocalization.oneToOrbit_oneMap (329); SBLocalization.SDiagram.oneOrbitMap (340); SBLocalization.SDiagram.continuous_oneOrbitMap (343); SBLocalization.SDiagram.oneOrbitMap_oneMap (346); SBLocalization.SDiagram.oneToOrbit_oneOrbitMap (353); SBLocalization.SDiagram.oneOrbitMap_eq_of_orbitMap_eq (360); SBLocalization.SDiagram.pullNorm (378); SBLocalization.SDiagram.pushNorm (381); SBLocalization.SDiagram.pullNorm_eq (386); SBLocalization.SDiagram.pullNorm_eq_Ψ (394); SBLocalization.SDiagram.isOneInvariant_pullNorm (400); SBLocalization.SDiagram.pullNorm_eq_lift (405); SBLocalization.SDiagram.continuous_pullNorm (412); SBLocalization.SDiagram.pushNorm_pullNorm (417); SBLocalization.SDiagram.pullNorm_pushNorm (425); SBLocalization.SDiagram.pullNorm_eq_of_orbitMap_eq (433); SBLocalization.SDiagram.pushNorm_eq_of_orbitMap_eq (438); SBLocalization.GenSBMap.pullNorm (451); SBLocalization.GenSBMap.pushNorm (454); SBLocalization.GenSBMap.proposition_11_5_1 (464).

**Polyfold/Chapter12/P3B_LocalBoundaryExtension.lean.** cornerKill_insert_of_coord_eq_zero (90); sum_powerset_erase_empty_alternating (110); retractBoundary (142); IsScPlusOnFaceNear (147); IsBoundaryScPlusSectionOn (156); mem_retractBoundary_iff (162); boundaryExtension (178); boundaryExtension_toBase_snd (186); boundaryExtension_mem (205); exists_isScSmooth_faceTerm (238); proposition_12_3_7_of_faces_at (277); proposition_12_3_7 (372); isBoundaryScPlusSectionOn_zero (415); proposition_12_3_7_nonvacuous (445); mem_strongShiftOne_sum_smul (481); LocalBundleAction (502); LocalBundleAction.trivial (518); LocalBundleAction.average (532); corollary_12_3_8 (545).

**Polyfold/Chapter12/P3E_AuxNormLocalExtension.lean.** one_le_degIndexWith_iff (67); minCoord (75); boundaryShift (78); continuous_boundaryShift (85); mem_quadrant_boundaryShift (101); exists_coord_boundaryShift_eq_zero (106); boundaryShift_eq_self (112); MPolyfold.exists_retraction_boundary (128); StrongBundle.chartFibMap (204); StrongBundle.chartFibMap_of_mem (239); StrongBundle.isLinearMap_chartFibMap (247); StrongBundle.continuousOn_shiftFiber_R (258); StrongBundle.retractNorm (270); StrongBundle.retractTransport (296); StrongBundle.

retractNorm_toOne (316); StrongBundle.isFiberSeminorm_retractNorm (333); MChart.tendsto_of_tendsto_apply (369); StrongBundle.isLocalAuxNorm_retractNorm (390); StrongBundle.retractNorm_eq (492); StrongBundle.exists_localExtension_of_isTame (504); EpStrongBundle.hasLocalExtensions_of_isTame (546); EpStrongBundle.theorem_12_2_7 (555).

**Polyfold/Chapter12/P3F_EquivalenceOr.lean.** EpStrongBundle.IsStrongBundleEquivalence.locIso (90); EpStrongBundle.IsStrongBundleEquivalence.orEqv (179); EpStrongBundle.IsStrongBundleEquivalence.orMor_orEqv (388).

**Polyfold/Chapter12/P3F_MixedPull.lean.** StrongBundle.LocalIso.mpullPt (68); StrongBundle.LocalIso.linSet_mpullPt (172); StrongBundle.LocalIso.GermAgree₄ (290); StrongBundle.LocalIso.transferT_mpullPt₄ (379); StrongBundle.LocalIso.morVia (505); StrongBundle.LocalIso.morVia_eq (559); StrongBundle.LocalIso.mpullPt_eq_pullPt (576).

**Polyfold/Chapter12/P3F_Naturality.lean.** EpStrongBundle.IsOrInvariant (75); EpStrongBundle.definition_12_5_4_iff (82); EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_bijective (113); EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward_witness (132); EpStrongBundle.IsStrongBundleEquivalence.orbitProj_orbitOrMap (163); EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_bijective (170); EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pullOr (210); EpStrongBundle.IsStrongBundleEquivalence.pushOr (236); EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pushOr (278); EpStrongBundle.IsStrongBundleEquivalence.pushOr_unique (290); theorem_12_5_7_one_hyp (351); theorem_12_5_7_two_invariant (382).

**Polyfold/Chapter12/P3N_AuxNormEp.lean.** StrongBundle.IsFiberSeminorm (79); StrongBundle.TendstoZeroOne (113); StrongBundle.exists_norm_liftOne_le_of_tendsto (121); StrongBundle.isAuxNorm_of_tendsto (191); StrongBundle.IsLocalAuxNorm (213); EpStrongBundle.actOne (268); EpGroupoid.satSet (324); EpGroupoid.selMor (359); EpStrongBundle.actOne_Γ_mul (411); EpStrongBundle.avgNorm (420); EpStrongBundle.avgNorm_actOne (429); EpStrongBundle.isLocalAuxNorm_avgNorm (440); EpStrongBundle.transNorm (476); EpStrongBundle.transNorm_eq_of (483); EpStrongBundle.isLocalAuxNorm_transNorm (629); EpStrongBundle.transNorm_μ (637); EpGroupoid.IsContPartitionOfUnity (693); EpGroupoid.IsContPartitionOfUnity.IsSubordinate (703); EpGroupoid.isContPartitionOfUnity_comp_π (711); EpGroupoid.isSubordinate_comp_π (726); EpGroupoid.IsContPartitionOfUnity.descend (735); EpGroupoid.proposition_12_2_5 (764); EpStrongBundle.glueNorm (795); EpStrongBundle.isAuxNorm_glueNorm (817); EpStrongBundle.theorem_12_2_2 (915); SEP.theorem_12_2_2 (942).

**Polyfold/Chapter12/P3N_AuxNormExtension.lean.** StrongBundle.IsLocalAuxNorm.mono (62); EpGroupoid.tgt_mem_boundary_iff (79); EpGroupoid.isOpen_compl_boundary (85); EpStrongBundle.IsAuxNormOn (111); EpStrongBundle.IsAuxNorm.isAuxNormOn (123); EpStrongBundle.HasLocalExtensions (134); EpStrongBundle.theorem_12_2_7_hyp (143); EpStrongBundle.hasLocalExtensions_of_isAuxNorm (215); EpStrongBundle.theorem_12_2_7_hyp_witness (222).

**Polyfold/Chapter12/P3N_GoodNeighborhood.lean.** EpGroupoid.NaturalRep.image_Φ_eq (56); EpGroupoid.exists_subseq_of_isProperMap (65); EpGroupoid.theorem_12_2_6 (90).

**Polyfold/Chapter12/P3O_MorphismIso.lean.** EpStrongBundle.morIso (146); EpStrongBundle.eqOn_morIso (209); EpStrongBundle.morIso_Φ_eq (226); EpStrongBundle.germAgree_morIso_comp (240).

**Polyfold/Chapter12/P3O_OrientationBundleEp.lean.** EpStrongBundle.morTx (308); EpStrongBundle.

morKx (316); EpStrongBundle.morTx_comp (382); EpStrongBundle.morTx_unit (436); EpStrongBundle.morTx_inv (452); EpStrongBundle.linMor (498); EpStrongBundle.lemma_12_5_1_one (515); EpStrongBundle.lemma_12_5_1_two (541); EpStrongBundle.lemma_12_5_1_three_unit (556); EpStrongBundle.lemma_12_5_1_three_comp (566); EpStrongBundle.orMor (584); EpStrongBundle.orMor_apply (592); EpStrongBundle.orMor_comp (605); EpStrongBundle.orMor_unit (622); EpStrongBundle.orMor_inv (632).

**Polyfold/Chapter12/P3O_OrientationOrbit.lean.** EpGroupoid.SmoothRel (55); EpGroupoid.smoothRel_equivalence (58); orProj_two (73); exists_other_or (88); EpStrongBundle.definition_12_5_2 (107); EpStrongBundle.OrRel (110); EpStrongBundle.orRel_equivalence (115); EpStrongBundle.orbitProj (129); EpStrongBundle.theorem_12_5_3_hyp (141); EpStrongBundle.definition_12_5_4 (157); EpStrongBundle.IsTwoToOne (171); EpStrongBundle.proposition_12_5_6_mp (179); EpStrongBundle.proposition_12_5_6_mpr_invariant (221).

**Polyfold/Chapter12/P3O_PullChart.lean.** StrongBundle.LocalIso.pullPt (79); StrongBundle.LocalIso.linSet_pullPt (184); StrongBundle.LocalIso.transferT_pullPt (256); StrongBundle.LocalIso.GermAgree (280); StrongBundle.LocalIso.transferT_pullPt_pullPt (363).

**Polyfold/Chapter12/P3R2_ReflexiveAuxNormEp.lean.** StrongBundle.mixedTendsto_of_bounded (82); StrongBundle.MixedTendsto.exists_eventually_le (98); StrongBundle.exists_le_mul_auxNorm (110); StrongBundle.MixedLscAt (136); StrongBundle.MixedLscAt.le_liminf (143); StrongBundle.MixedLscAt.add (157); StrongBundle.mixedLscAt_finset_sum (163); StrongBundle.MixedLscAt.const_mul (173); StrongBundle.MixedLscAt.mul_comp (186); EpStrongBundle.mixedTendsto_actOne (244); EpStrongBundle.mixedLscAt_comp_actOne (265); EpStrongBundle.exists_mixedTendsto_μ (274); EpStrongBundle.HasReflexiveFibers (304); EpStrongBundle.MixedTendsto (309); EpStrongBundle.definition_12_2_8 (313); EpStrongBundle.IsReflexiveAuxNorm (325); EpStrongBundle.IsReflexiveAuxNorm.toStrong (333); EpStrongBundle.isReflexiveAuxNorm_of_mixedLscAt (343); EpStrongBundle.remark_12_2_9 (368); EpStrongBundle.mixedLscAt_avgNorm (399); EpStrongBundle.mixedLscAt_transNorm (419); EpStrongBundle.mixedLscAt_glueNorm (446); EpStrongBundle.isReflexiveAuxNorm_glueNorm (480); EpStrongBundle.remark_12_2_12_1 (511); EpStrongBundle.remark_12_2_12_2 (523); EpStrongBundle.remark_12_2_12_3 (592); EpStrongBundle.theorem_12_2_11_exists (621); EpStrongBundle.theorem_12_2_11_mul (651); EpStrongBundle.theorem_12_2_11_sandwich (658); EpStrongBundle.theorem_12_2_11 (676).

**Polyfold/Chapter12/P3R2_ReflexiveAuxNormExtension.lean.** EpStrongBundle.IsReflexiveAuxNormOn (70); EpStrongBundle.IsReflexiveAuxNorm.isReflexiveAuxNormOn (79); EpStrongBundle.HasReflexiveLocalExtensions (88); EpStrongBundle.HasReflexiveLocalExtensions.hasLocalExtensions (94); EpStrongBundle.theorem_12_2_13_hyp (105); EpStrongBundle.hasReflexiveLocalExtensions_of_isReflexiveAuxNorm (187); EpStrongBundle.theorem_12_2_13_hyp_witness (195); TrivialBundle.epBundle_reflexive_witness (214).

**Polyfold/Chapter12/P3R2_ReflexiveCompactnessEp.lean.** EpGroupoid.exists_lift_tendsto (68); EpGroupoid.isCompact_closure_of_isProperMap (98); EpStrongBundle.HasReflexiveLocalCompactnessEp (140); EpStrongBundle.hasReflexiveLocalCompactnessEp_of_strong (149); EpStrongBundle.isSaturated_setOf_auxLe (154); EpGroupoid.isSaturated_inter (160); EpStrongBundle.proposition_12_4_5_of_functorial (170); EpStrongBundle.proposition_12_4_5 (223); EpStrongBundle.theorem_12_4_6_of_continuous (244); EpStrongBundle.theorem_12_4_6 (336).

**Polyfold/Chapter12/P3S_BumpEp.lean.** EpGroupoid.AdmitsBumpFunctions (76); EpGroupoid.AdmitsBumpFunctionsPointwise (83); EpGroupoid.AdmitsBumpFunctionsPointwise.weak (91); EpGroupoid.exists_natAverage (161); EpGroupoid.exists_invariant_extension (202); EpGroupoid.exists_objBump_of_epBump (236); EpGroupoid.exists_epBump_of_objBump (275); EpGroupoid.theorem_12_3_2_1 (306); EpGroupoid.theorem_12_3_2_1_weak (331); EpGroupoid.theorem_12_3_2_2 (361); EpGroupoid.admitsBumpFunctionsPointwise_of_objects (397); EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn (404).

**Polyfold/Chapter12/P3S_CompactnessEp.lean.** EpGroupoid.lemma_12_4_4_1 (77); EpGroupoid.lemma_12_4_4_2 (86); EpGroupoid.isClosed_image_π_of_isSaturated (93); exists_nhds_isCompact_closure_auxLe (115); EpStrongBundle.definition_12_1_1 (153); EpStrongBundle.remark_12_1_1_regularizing_obj (161); EpStrongBundle.remark_12_1_1_regularizing_mor (169); EpStrongBundle.remark_12_1_1_germ (184); EpStrongBundle.auxLe_tgt_of_auxLe_src (192); EpStrongBundle.HasCompactSolutionSetEp (211); EpStrongBundle.IsProperFunctor (217); EpStrongBundle.IsProperLtInftyFunctor (226); EpStrongBundle.IsProperLtInftyFunctor.isProperFunctor (234); EpStrongBundle.IsProperFunctor.hasCompactSolutionSetEp (241); EpStrongBundle.isProperLtInftyFunctor_of_hasCompactSolutionSetEp (249); EpStrongBundle.theorem_12_4_2 (284).

**Polyfold/Chapter12/P3S_ScPlusSectionFunctor.lean.** StrongBundle.isScPlusSection_of_locally (82); EpStrongBundle.secAct (135); EpStrongBundle.secAct_comp (149); EpStrongBundle.exists_isScSmoothOn_one_secAct (176); EpStrongBundle.NormBound (289); EpStrongBundle.IsAuxNorm.secAct_eq (293); EpStrongBundle.natTerm (301); EpStrongBundle.exists_equivariant_average (395); EpStrongBundle.natExtension (492); EpStrongBundle.natExtension_isFunctorial (544); EpStrongBundle.natExtension_eq_zero (572); EpStrongBundle.remark_12_3_5 (593); EpStrongBundle.theorem_12_3_3 (665); EpStrongBundle.remark_12_3_4 (693).

**Polyfold/Chapter12/P3X_ScPlusExtension.lean.** EpGroupoid.tgt_mem_boundary_iff_p3x (81); EpGroupoid.isSaturated_compl_boundary_p3x (87); EpGroupoid.partitionOfUnity_true_eq_one_p3x (102); EpGroupoid.admitsBumpFunctionsPointwise_of_hasScPartitionsOfUnity (114); glueSection (176); glueSection_eq_of (205); sectionSupport_glueSection_subset (221); isScPlusSection_glueSection (241); EpStrongBundle.IsBoundaryScPlusSection (272); EpStrongBundle.IsBoundaryScPlusSectionFunctor (278); EpStrongBundle.isBoundaryScPlusSectionFunctor_fiberZero (286); EpStrongBundle.isFunctorialSection_sectionSmul_p3x (299); EpStrongBundle.isScPlusSectionFunctor_glueSection (335); EpStrongBundle.exists_equivariant_average_p3x (369); EpStrongBundle.exists_local_functor_extension (418); EpStrongBundle.exists_extension_of_tameCharts (491); EpStrongBundle.exists_extension_bounded_of_tameCharts (525); EpStrongBundle.theorem_12_3_9 (622); EpStrongBundle.theorem_12_3_9_hypotheses_rn (640).

**Polyfold/Chapter13/P3A_Atomic.lean.** SBLocalization.AtomicData (77); SBLocalization.AtomicData.count (111); SBLocalization.AtomicData.IsRelated (120); SBLocalization.AtomicData.mult (126); SBLocalization.IsAtomic (135); SBLocalization.AtomicData.count_μ_Γ (158); SBLocalization.AtomicData.count_μ_eq (169); SBLocalization.AtomicData.mult_of_mem (221); SBLocalization.AtomicData.IsRelated.eq_mult (230); SBLocalization.AtomicData.isWFunctor_mult (238); SBLocalization.AtomicData.isClosed_saturation_closure_V (274); SBLocalization.AtomicData.mult_eq_zeroMultisection (290); SBLocalization.AtomicData.domSupp_mult_subset (303); SBLocalization.AtomicData.exists_isLocalRepOn_transport (321); SBLocalization.AtomicData.isScPlusMultisection_mult (407); SBLocalization.AtomicData.isAtomic_mult (415);

SBLocalization.AtomicData.IsRelated.symmetricStructure (420); SBLocalization.AtomicData.IsRelated.nonempty (434); SBLocalization.isAtomic_zeroMultisection (495).

**Polyfold/Chapter13/P3A_AtomicConstruction.lean.** SBLocalization.normW_μ (102); SBLocalization.SymmetricFamily (126); SBLocalization.SymmetricFamily.exists_vanishing_nbhd (166); SBLocalization.AtomicData.ofFamily (199); SBLocalization.theorem_13_5_5 (223); SBLocalization.SymmetricFamily.ofSeed (253); SBLocalization.SymmetricFamily.ofSeed_one (280); SBLocalization.SymmetricFamily.ofSeed_vanish (287); SBLocalization.SymmetricFamily.ofSeed_normW_le (292); SBLocalization.theorem_13_5_4_1 (312); SBLocalization.exists_cutoff (334); SBLocalization.theorem_13_5_4_2 (384); SBLocalization.AtomicData.pointwiseNorm_mult_le (405); SBLocalization.AtomicData.isLocalRepOn_mult (430); SBLocalization.theorem_13_5_6_1 (449); SBLocalization.theorem_13_5_6_2 (477); SBLocalization.theorem_13_5_6 (537).

**Polyfold/Chapter13/P3A_AtomicStructurable.lean.** SBLocalization.AtomicData.IsLocallyTransported (68); SBLocalization.theorem_13_5_3_local (83); SBLocalization.theorem_13_5_3_hyp (99); SBLocalization.theorem_13_5_3_hyp_witness (108); SBLocalization.theorem_13_5_5_structurable_hyp (116).

**Polyfold/Chapter13/P3D_Correspondence.lean.** SBLocalization.IndexDiagram (54); SBLocalization.IndexDiagram.Equiv (76); SBLocalization.IndexDiagram.equivalence (79); SBLocalization.IndexDiagram.card_fiber_prodMap (146); SBLocalization.IndexDiagram.sat_iff_of_equiv (188); SBLocalization.IndexCorr (205); SBLocalization.IndexCorr.map (221); SBLocalization.IndexCorr.swap (236); SBLocalization.IndexCorr.swap_swap (241); SBLocalization.IndexCorr.swapEquiv (244); SBLocalization.IndexCorr.biact (253); SBLocalization.IndexCorr.biact_one_one (256); SBLocalization.IndexCorr.biact_biact (263); SBLocalization.IndexCorr.example_13_3_3 (275); SBLocalization.IndexCorr.ofPerm (294); SBLocalization.IndexCorr.prod (309); SBLocalization.IndexCorr.prod_swap (315); SBLocalization.IndexCorr.prod_biact (319); SBLocalization.IndexCorr.prod_ofPerm (324); SBLocalization.IndexCorr.Sat (333); SBLocalization.IndexCorr.sat_iff (340).

**Polyfold/Chapter13/P3D_Decorated.lean.** EpGroupoid.GoodSystem.ofSubset (77); EpGroupoid.GoodSystem.exists_subset_inter (90); SBLocalization.Decoration (110); SBLocalization.DecoratedTuple (118); SBLocalization.Decoration.Rel (131); SBLocalization.Decoration.Rel.refl (140); SBLocalization.Decoration.Rel.symm (144); SBLocalization.Decoration.Rel.trans (154); SBLocalization.DecoratedTuple.equivalence (173); SBLocalization.DecoratedMultisection (182); SBLocalization.DecoratedMultisection.forget (186); SBLocalization.exists_decoration (198); SBLocalization.Decoration.add (224); SBLocalization.Decoration.add_U_subset (231); SBLocalization.Decoration.add_I (236); SBLocalization.Decoration.add_s_apply (239); SBLocalization.Decoration.add_rel (249); SBLocalization.Decoration.add_comm_rel (274); SBLocalization.Decoration.add_assoc_rel (284); SBLocalization.definition_13_2_9_decorated (312); SBLocalization.DecoratedMultisection.add (324); SBLocalization.DecoratedMultisection.forget_add (329); SBLocalization.DecoratedMultisection.add_comm (334); SBLocalization.DecoratedMultisection.add_assoc (340); SBLocalization.Decoration.smulFun (353); SBLocalization.Decoration.smul (359); SBLocalization.Decoration.smulFun_rel (363); SBLocalization.definition_13_2_10_decorated (384); SBLocalization.lemma_13_2_11_decorated (390); SBLocalization.DecoratedMultisection.smul (397); SBLocalization.lfSumStructure (434); SBLocalization.Decoration.lfSum (447); SBLocalization.lemma_13_2_13_decorated (460).

**Polyfold/Chapter13/P3D_NoInverse.lean.** SBLocalization.multisectionAdd_zeroMultisection

(54); SBLocalization.NoInverse.A (74); SBLocalization.NoInverse.twoValued (87); SBLocalization.NoInverse.isScPlusMultisection_twoValued (100); SBLocalization.NoInverse.add_le_one_of_ne (106); SBLocalization.NoInverse.remark_13_2_9_no_inverse (122); SBLocalization.NoInverse.exists_no_inverse (166).

**Polyfold/Chapter13/P3D_Structurable.lean.** EpGroupoid.GoodSystem.biact_congr_of_le (86); SBLocalization.Decoration.Compat (133); SBLocalization.Decoration.IsCorrespondence (142); SBLocalization.Decoration.Tau (155); SBLocalization.Decoration.Tau.restrict (172); SBLocalization.StructuredTuple (199); SBLocalization.IsStructurable (205); SBLocalization.DecoratedMultisection.IsStructurable (210); SBLocalization.isStructurable_iff (213); SBLocalization.StructuredTuple.Equiv (228); SBLocalization.StructuredTuple.equivalence (234); SBLocalization.StructuredMultisection (260); SBLocalization.Γm (271); SBLocalization.Γs (274); SBLocalization.Γst (277); SBLocalization.StructuredMultisection.range_forget (287); SBLocalization.Decoration.Tau.prod (300); SBLocalization.Decoration.Tau.add (354); SBLocalization.StructuredTuple.add (362); SBLocalization.StructuredTuple.add_Λ (365); SBLocalization.proposition_13_3_12 (370); SBLocalization.Decoration.Tau.smulFun (383); SBLocalization.Decoration.Tau.smul (405); SBLocalization.lemma_13_3_14 (413); SBLocalization.lemma_13_3_14_fun (420); SBLocalization.isStructurable_zeroMultisection (461).

**Polyfold/Chapter13/P3D_SymmetricOn.lean.** EpGroupoid.NatNbhd.Le (67); EpGroupoid.NatNbhd.exists_le_of_finite (88); SBLocalization.fibAct (116); SBLocalization.fibAct_sectionAdd (134); SBLocalization.fibAct_sectionSmul (148); SBLocalization.fibAct_sectionSum (160); SBLocalization.SymmetricStructureOn (198); SBLocalization.SymmetricStructureOn.card_pos (246); SBLocalization.SymmetricStructureOn.s_eq_fiberZero (261); SBLocalization.SymmetricStructureOn.restrict (278); SBLocalization.SymmetricStructure.toOn (307); SBLocalization.exists_symmetricStructureOn (317); SBLocalization.SymmetricStructureOn.add (347); SBLocalization.SymmetricStructureOn.pi (445).

**Polyfold/Chapter13/P3D_Tight.lean.** SBLocalization.trivialSet (59); SBLocalization.isOpen_trivialSet (61); SBLocalization.isTrivialOn_trivialSet (66); SBLocalization.apply_fiberZero_tgt (70); SBLocalization.TightIdx (78); SBLocalization.TightIdx.eT (101); SBLocalization.TightIdx.eN (111); SBLocalization.IndexCorr.swap_map (130); SBLocalization.IndexCorr.biact_map (135); SBLocalization.Decoration.exists_tightNbhd (201); SBLocalization.Decoration.tightNbhd (211); SBLocalization.Decoration.tight (225); SBLocalization.Decoration.tightτ (235); SBLocalization.Decoration.tightτ_TN (257); SBLocalization.Decoration.tightτ_NN (269); SBLocalization.Decoration.isCorrespondence_const (275); SBLocalization.Decoration.Tau.tight (287); SBLocalization.Decoration.IsTight (376); SBLocalization.Decoration.isTight_tight (389); SBLocalization.StructuredMultisection.IsTight (417); SBLocalization.proposition_13_3_15 (428); SBLocalization.corollary_13_3_18 (446); SBLocalization.corollary_13_3_18_structured (453).

**Polyfold/Chapter13/P3E_GoodSystemRelative.lean.** EpGroupoid.isProperMap_iff_ultrafilter_of_isInducing (66); EpGroupoid.isProperSubset_iff_ultrafilter (100); EpGroupoid.subSrc (116); EpGroupoid.subTgt (119); EpGroupoid.IsProperSubsetIn (126); EpGroupoid.IsGoodIn (133); EpGroupoid.isProperSubsetIn_iff_ultrafilter (151); EpGroupoid.exists_localSection (186); EpGroupoid.exists_mor_tgt_mem (203); EpGroupoid.closure_subset_of_isProperSubsetIn (213); EpGroupoid.isProperSubset_of_isProperSubsetIn (251); EpGroupoid.isProperSubsetIn_of_isProperSubset (322); EpGroupoid.IsGoodIn.isGoodOn (336); EpGroupoid.isGoodIn_iff (350);

EpGroupoid.theorem_13_1_7_book (361); EpGroupoid.GoodSystem.isGoodIn_univ (369).

**Polyfold/Chapter13/P3G_BiAction.lean.** EpGroupoid.biact (85); EpGroupoid.src_biact (118); EpGroupoid.tgt_biact (123); EpGroupoid.biact_mem (128); EpGroupoid.biact_biact (152); EpGroupoid.isScSmoothOn_biact (232); EpGroupoid.biactMulAction (263); EpGroupoid.GoodSystem.biact (281); EpGroupoid.GoodSystem.image_biact_comp (300); EpGroupoid.GoodSystem.exists_biact_left_of_src (341); EpGroupoid.GoodSystem.exists_biact_right_of_tgt (356); EpGroupoid.GoodSystem.components (378); EpGroupoid.GoodSystem.image_inv_comp (398); EpGroupoid.GoodSystem.eq_one_of_biact_left_mem (416); EpGroupoid.GoodSystem.eq_one_of_biact_right_mem (429); EpGroupoid.GoodSystem.lemma_13_1_6_one (442); EpGroupoid.GoodSystem.lemma_13_1_6_two (447); EpGroupoid.GoodSystem.lemma_13_1_6_three_src (453); EpGroupoid.GoodSystem.lemma_13_1_6_three_tgt (469); EpGroupoid.GoodSystem.stabSrc (487); EpGroupoid.GoodSystem.stabTgt (507); EpGroupoid.GoodSystem.D (527); EpGroupoid.GoodSystem.isScDiffeoOn_D (532); EpGroupoid.GoodSystem.partner (590); EpGroupoid.GoodSystem.partnerHom (621); EpGroupoid.GoodSystem.proposition_13_1_9 (676); EpGroupoid.GoodSystem.proposition_13_1_9_D (717).

**Polyfold/Chapter13/P3G_GoodSystem.lean.** MPolyfold.IsLocalScDiffeo.isScDiffeoOn_invFunOn (90); EpGroupoid.NatNbhd (154); EpGroupoid.NatNbhd.invariantCore (192); EpGroupoid.exists_natNbhd_isProperSubset (271); EpGroupoid.lemma_13_1_1 (282); EpGroupoid.lemma_13_1_1_book (345); EpGroupoid.morBetween (354); EpGroupoid.IsDiffeoComponent (368); EpGroupoid.isDiffeoComponent_iff_injOn (373); EpGroupoid.GoodSystem (416); EpGroupoid.IsGoodOn (429); EpGroupoid.GoodSystem.isGoodOn (440); EpGroupoid.GoodSystem.srcInv (499); EpGroupoid.GoodSystem.isDiffeoComponent_of_subset (558); EpGroupoid.GoodSystem.restrict (569); EpGroupoid.GoodSystem.IsRefinement (585); EpGroupoid.GoodSystem.IsSubordinate (589); EpGroupoid.GoodSystem.isRefinement_restrict (591); EpGroupoid.GoodSystem.exists_refinement_subset (598); EpGroupoid.proposition_13_1_3 (612).

**Polyfold/Chapter13/P3G_GoodSystemExistence.lean.** EpGroupoid.IsTransport (78); EpGroupoid.injOn_comp_of_transport (97); EpGroupoid.exists_isOpen_injOn_src_tgt (167); EpGroupoid.exists_nhds_injOn_of_isProperSubset (179); EpGroupoid.exists_transport_of_isGoodOn (235); EpGroupoid.theorem_13_1_7 (272); EpGroupoid.theorem_13_1_8 (290); EpGroupoid.nonempty_goodSystem_trivial (382).

**Polyfold/Chapter13/P3M_MultisectionAlgebra.lean.** SBLocalization.multisectionAdd (63); SBLocalization.multisectionAdd_eq_conv (78); SBLocalization.multisectionAdd_eq_sum_of_isLocalRepOn (126); SBLocalization.definition_13_2_9 (143); SBLocalization.multisectionAdd_comm (152); SBLocalization.multisectionAdd_assoc (182); SBLocalization.multisectionSmul (208); SBLocalization.multisectionSmul_zero (216); SBLocalization.multisectionSmul_eq_sum_of_isLocalRepOn (241); SBLocalization.definition_13_2_10 (277); SBLocalization.multisectionSmulFun (291); SBLocalization.lemma_13_2_11 (297).

**Polyfold/Chapter13/P3M_MultisectionConvolution.lean.** SBLocalization.convAt (76); SBLocalization.conv (81); SBLocalization.isWFunctor_conv (158); SBLocalization.conv_eq_sum_of_isLocalRepOn (182); SBLocalization.isLocalRepOn_conv (197); SBLocalization.isScPlusMultisection_conv (215).

**Polyfold/Chapter13/P3M_MultisectionLocallyFinite.lean.** SBLocalization.nontrivIdx (71); SBLocalization.IsLocallyFiniteFamily (80); SBLocalization.IsLocallyFiniteFamily.exists_stable (85); SBLocalization.lfSum (110); SBLocalization.lfSum_eq_conv (175); SBLocalization.lfSum_eq_

conv_of_mem (193); SBLocalization.lemma_13_2_13 (224).

**Polyfold/Chapter13/P3M_MultisectionSupport.lean.** SBLocalization.supp (77); SBLocalization.suppAt (80); SBLocalization.domSupp (84); SBLocalization.IsLocalRepOn.suppAt_eq (89); SBLocalization.IsScPlusMultisection.finite_suppAt (109); SBLocalization.IsScPlusMultisection.sum_suppAt (134); SBLocalization.IsScPlusMultisection.finsum_suppAt (146); SBLocalization.IsScPlusMultisection.suppAt_nonempty (151); SBLocalization.domSupp_eq_closure_image (161); SBLocalization.normW (176); SBLocalization.pointwiseNorm (186); SBLocalization.IsScPlusMultisection.suppAt_subset_oneSet (191); SBLocalization.IsScPlusMultisection.isGreatest_pointwiseNorm (199); SBLocalization.IsScPlusMultisection.domSupp_eq_closure_pointwiseNorm (230); SBLocalization.IsTrivialOn (256); SBLocalization.IsScPlusMultisection.apply_fiberZero_eq_one_iff (261); SBLocalization.IsScPlusMultisection.isLeast_domSupp (303); SBLocalization.isTrivialOn_zeroMultisection (315).

**Polyfold/Chapter13/P3M_MultisectionSymmetric.lean.** SBLocalization.transportSection (56); SBLocalization.isScPlusSectionOn_transportSection (66); SBLocalization.IsLocalRepOn.exists_equal_weights (108); SBLocalization.SymmetricStructure (155); SBLocalization.prodIsotropyAction (180); SBLocalization.lemma_13_2_6 (193).

**Polyfold/Chapter13/P3M_ScPlusSectionOn.lean.** SBLocalization.IsScPlusSectionOn.sectionAdd (153); SBLocalization.IsScPlusSectionOn.sectionSmul (212); SBLocalization.extendZero (296); SBLocalization.IsLocalRepOn.extendZero (317).

**Polyfold/Chapter15/P3L_BoundaryPerturbation.lean.** SBLocalization.IsBranchGoodPosition (68); SBLocalization.IsGoodPositionAt (82); SBLocalization.exists_branch_goodPosition (92); SBLocalization.theorem_15_2_5_hyp (131).

**Polyfold/Chapter15/P3L_FredholmSummands.lean.** SBLocalization.subDeriv_isScFredholm (58); SBLocalization.linearization_summand_isScFredholm (100).

**Polyfold/Chapter15/P3L_LocalIFT.lean.** SBLocalization.subDeriv (67); SBLocalization.ChartOnto (73); SBLocalization.surjective_toIntr_subDeriv_iff (91); SBLocalization.scDeriv_restrictTo_subRestrict (153); SBLocalization.isLinSurjAt_subRestrict (175); SBLocalization.chartOnto_of_isLinSurjAt_subRestrict (198); SBLocalization.exists_solutionSet_of_isGoodPosition (234); SBLocalization.exists_solutionSet_interior (283).

**Polyfold/Chapter15/P3L_LocalScPlus.lean.** SBLocalization.IsScPlusSectionOn.isScSmoothRetractMap_localRepOne_of_isChart (86); SBLocalization.IsScPlusSectionOn.restrictOpen (183); SBLocalization.subSec (201); SBLocalization.isZeroAt_subSec_iff (223); SBLocalization.subRestrict (253); SBLocalization.isScFredholmSectionTame_subRestrict (261).

**Polyfold/Chapter15/P3L_Transversality.lean.** SBLocalization.mem_support_linSumIntr_iff (68); SBLocalization.IsOntoAtLevel (103); SBLocalization.IsOntoAt (108); SBLocalization.IsTransversalAt (114); SBLocalization.IsTransversal (119); SBLocalization.isOntoAtLevel_iff (125); SBLocalization.linearization_ne_zero (141); SBLocalization.IsScPlusSectionOn.continuousOn (165); SBLocalization.isOpen_sep_ne (171); SBLocalization.mem_smoothSet_of_pos (196); SBLocalization.exists_branch_interior (216); SBLocalization.theorem_15_2_2_hyp (249); SBLocalization.remark_15_2_3_hyp (360); SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp (399).

**Polyfold/Chapter15/P3L_Witness.lean.** SBLocalization.theorem_15_2_2_hyp_witness (41); SBLocal-

ization.remark_15_2_3_isBranchedEpSubgroupoid_hyp_witness (76).

**Polyfold/Chapter15/P3P_CubeAuxNorm.lean.** StrongBundle.isAuxNorm_max_mul (59); StrongBundle.IsCompleteFiberNorm.congr (76); SEP.cubeAuxNorm (135); SEP.cubeVal_μ (171); SEP.isAuxNorm_cubeAuxNorm (189); SEP.isCompact_closure_image_π_cube (201); SEP.controlsCompactness_cube (219); SEP.interpAuxNorm (248); SEP.isAuxNorm_interpAuxNorm (262); SEP.interpAuxNorm_of_cubeTime_eq_zero (308); SEP.interpAuxNorm_of_cubeTime_eq_one (315); SEP.interpSet (323); SEP.mem_interpSet_zero (328); SEP.mem_interpSet_one (334); SEP.corollary_15_3_10_2 (344).

**Polyfold/Chapter15/P3P_CubeEp.lean.** EpGroupoid.cubeProd (93); EpGroupoid.cubeProj (104); EpGroupoid.cubeIncl (110); EpGroupoid.isTame_cubeProd (130); EpGroupoid.degIndex_cubeProd_one_of_degIndex_eq_zero (142); EpGroupoid.continuous_cubeOrbitMap (187); EpGroupoid.isCompact_image_π_cubeProd (206); EpGroupoid.admitsBumpFunctionsPointwise_cubeProd (215); SEP.cubeSEP (236); SEP.cubeLiftEp (254); SEP.isScSmoothSectionFunctor_cubeLiftEp (277); SEP.isScFredholmSectionFunctor_cubeLiftEp (288); SEP.solutionSet_cubeLiftEp (298); SEP.hasCompactSolutionSetEp_cubeLiftEp (309).

**Polyfold/Chapter15/P3P_GeneralPosition.lean.** SBLocalization.ControlsCompactness (83); SBLocalization.exists_controlsCompactness (90); SBLocalization.ControlsCompactness.mono (98); SBLocalization.IsAdmissible (110); SBLocalization.eq_zero_of_not_mem_domSupp (115); SBLocalization.eq_fiberZero_of_multisectionSmul_pos (121); SBLocalization.isAdmissible_of_domSupp_subset (143); SBLocalization.IsInGeneralPosition (162); SBLocalization.exists_chart_presentation (171); SBLocalization.isInGeneralPosition_iff_exists (181); SBLocalization.reducedLinearization (200); SBLocalization.isInGeneralPosition_iff_reducedLinearization (222); SBLocalization.IsInGeneralPosition.surjective (241); SBLocalization.mem_levelSet_one_of_pos (254); SBLocalization.isInGeneralPosition_witness (278).

**Polyfold/Chapter15/P3P_ProductEp.lean.** GroupoidStr.prod (53); MPolyfold.IsLocalScDiffeo.prodMap (121); isProperMap_domRestrict_prod (134); EpGroupoid.prod_isScSmoothCompSheet (169); EpGroupoid.prod (224); EpGroupoid.prodFst (256); EpGroupoid.isTame_prod (282); EpGroupoid.degIndex_prod (285); EpGroupoid.prod_π_eq_π_iff (290).

**Polyfold/Chapter15/P3T_Linearization.lean.** SBLocalization.IsScPlusSectionOn.isScSmooth_principal_comp_r (77); SBLocalization.IsLocalRepOn.count_eq (106); SBLocalization.linSumChart (141); SBLocalization.linSumChart_eq (148); SBLocalization.theorem_15_1_1 (170); SBLocalization.exists_chart_isLocalRepOn (184); SBLocalization.linearizationIn (202); SBLocalization.linearizationIn_eq (212); SBLocalization.remark_15_1_2 (227).

**Polyfold/Chapter15/P3T_LinearizationIntrinsic.lean.** SBLocalization.GrF (63); SBLocalization.chartOp (67); SBLocalization.toIntr (81); SBLocalization.linSumIntr (102); SBLocalization.sectionSubOn (135); SBLocalization.toIntr_chart_independent (248); SBLocalization.linSumIntr_chart_independent (323); SBLocalization.theorem_15_1_1_intrinsic (371); SBLocalization.LocalLinData (396); SBLocalization.LocalLinData.sum (414); SBLocalization.nonempty_localLinData (425); SBLocalization.linearization (436); SBLocalization.linearization_eq (441); SBLocalization.linearization_eq_zero (449).

**Polyfold/Chapter15/P3T_LinearizationModel.lean.** Linearization15.scDeriv_eq_of_eventuallyEq_lvl (74); Linearization15.continuousOn_scDeriv_coe (93); Linearization15.GrFModel (104); Linearization15.linSumModel (110); Linearization15.wsum_model_eq (136); Linearization15.theorem_15_1_1_model (193); Linearization15.theorem_15_1_1_model_map (252);

Linearization15.remark_15_1_2_model (290); Linearization15.mem_support_linSumModel_iff (305).

**Polyfold/Chapter15/P3T_SeparationLemmas.lean.** Linearization15.wsum (85); Linearization15.mass (88); Linearization15.mass_eq_of_wsum_eq (99); Linearization15.wsum_pair_of_const (119); Linearization15.wsum_eq_of_frequently (134); Linearization15.interior_biUnion_eq_empty (186); Linearization15.exists_nonempty_interior_of_finite_closed_cover (200); Linearization15.lemma_15_1_4 (218); Linearization15.exists_isOpen_dichotomy (250); Linearization15.exists_fin_partition (288); Linearization15.lemma_15_1_5 (307); Linearization15.IsLocDich (337); Linearization15.IsSeparatedAt (343); Linearization15.exists_isLocDich_mem (348); Linearization15.IsSeparatedAt.eventually (359); Linearization15.exists_isSeparatedAt_subseq (381); Linearization15.proposition_15_1_3_hyp (411); Linearization15.not_proposition_15_1_3_of_isolated (434); Linearization15.proposition_15_1_3_hyp_witness (442); Linearization15.not_proposition_15_1_3_point (450); Linearization15.remark_15_1_6 (458); Linearization15.wsum_pair_eq_at_of_isLocDich (492); Linearization15.wsum_pair_eq_of_isLocal (520).

**Polyfold/Chapter16/P3H_BranchedSuborbifold.lean.** EPLocalization.EPLoc.pullTheta_id_of_isFunctorial (85); EPLocalization.EPLoc.pullSection_id_of_mem (93); EPLocalization.EPLoc.pullTheta_eq_of_orbitMap (103); EPLocalization.EPLoc.pull_id_of_mem (130); EPLocalization.EPLoc.tameBranchedIntegral_pull (139); EPLocalization.EPLoc.tameBranchedIntegralEff_pull (156); PolyfoldClass.ScForm.toStructure_eq_formsPullback (189); PolyfoldClass.BranchedRep (203); PolyfoldClass.BranchedRep.Rel (217); PolyfoldClass.BranchedRep.Rel.trans (234); PolyfoldClass.BranchedSuborbifold (274); PolyfoldClass.BranchedSuborbifold.toFun (279); PolyfoldClass.BranchedSuborbifold.definition_16_4_1_wellDefined (284); PolyfoldClass.BranchedSuborbifold.HasRepProperty (298); PolyfoldClass.BranchedSuborbifold.hasRepProperty_iff (303); PolyfoldClass.BranchedSuborbifold.IsTame (337); PolyfoldClass.BranchedSuborbifold.definition_16_4_2_closed (341); PolyfoldClass.BranchedSuborbifold.definition_16_4_2_compact (347); PolyfoldClass.BranchedSuborbifold.definition_16_4_2_manifold (353); PolyfoldClass.BranchedSuborbifold.definition_16_4_2_orbifold (359); PolyfoldClass.BranchedSuborbifold.definition_16_4_2_nDim (365); PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_supp (373); PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_homeomorph (387); PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_manifold (398); PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_orbifold (410); PolyfoldClass.OrientedRep (426); PolyfoldClass.OrientedRep.Rel (441); PolyfoldClass.OrientedSuborbifold (483); PolyfoldClass.OrientedSuborbifold.toBranched (486); PolyfoldClass.definition_16_4_5_wellDefined (497); PolyfoldClass.OrientedRep.boundaryFun (512); PolyfoldClass.OrientedRep.boundaryFun_eq_of_rel (516); PolyfoldClass.OrientedRep.IsTameOriented (541); PolyfoldClass.OrientedRep.integral (546); PolyfoldClass.OrientedRep.integralEff (553); PolyfoldClass.OrientedRep.integral_eq_of_rel (561); PolyfoldClass.OrientedRep.integralEff_eq_of_rel (571); PolyfoldClass.OrientedRep.boundaryIntegral (583); PolyfoldClass.OrientedRep.boundaryIntegralEff (589); PolyfoldClass.OrientedRep.boundaryIntegralDim1 (596); PolyfoldClass.OrientedRep.boundaryIntegralEffDim1 (602); PolyfoldClass.OrientedRep.toStructure_d (608); PolyfoldClass.OrientedRep.theorem_16_4_6_hyp (625); PolyfoldClass.OrientedRep.theorem_16_4_6_eff_hyp (636); PolyfoldClass.OrientedRep.theorem_16_4_6_dim1_hyp (649); PolyfoldClass.OrientedRep.theorem_16_4_6_eff_dim1_hyp (660); PolyfoldClass.OrientedRep.boundaryIntegral_eq_of_rel (673); PolyfoldClass.OrientedRep.boundaryIntegralEff_eq_of_rel (684); PolyfoldClass.OrientedRep.boundaryIntegralDim1_eq_of_rel (695); PolyfoldClass.OrientedRep.theorem_16_4_

6_both_hyp (710); PolyfoldClass.OrientedRep.theorem_16_4_6_hyp_witness (720).

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**Polyfold/Chapter16/P3H_PolyfoldForms.lean.** EPLocalization.EPLoc.formsPush_id (54); EPLocalization.EPLoc.formsPush_comp (68); PolyfoldClass.FormRep (95); PolyfoldClass.FormRep.Rel (108); PolyfoldClass.ScForm (142); PolyfoldClass.ScForm.mk (147); PolyfoldClass.ScForm.toStructure (187); PolyfoldClass.ScForm.equivStructure (213); PolyfoldClass.ScForm.equivStructure□ (261); PolyfoldClass.ScForm.mk_add (270); PolyfoldClass.ScForm.mk_smul (275); PolyfoldClass.ScForm.formsPush_d (286); PolyfoldClass.ScForm.d (313); PolyfoldClass.ScForm.d_mk (322); PolyfoldClass.ScForm.d_comp_d (332); PolyfoldClass.theorem_16_2_6_complex (342).

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hasFDerivWithinAt_inverse_of_isUnit (63); contDiffOn_inverse_deriv_comp (88); lemma_3_8_21_contDiffOn (110); lemma_3_8_21 (140); lemma_3_8_21_smooth (155); lemma_3_8_21_printed (168).

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**Polyfold/Chapter3/GapP1_Remark338.lean.** ContractionGerm.fixLvl_zero_eq (42); ContractionGerm.exists_radius (50); exists_antitone_le (76); remark_3_3_8 (97).

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**Polyfold/Chapter3/GlobalIFT.lean.** ScBanach.qPartOp (102); ScBanach.wPartOp (112); scDeriv_eq_of_eqOn_nhds (131); finrank_ker_eq_fredholmIndex_of_surjective (162); solutionSet (716); theorem_3_6_11_of_tangentGerm (729); theorem_3_6_10_of_tangentGerm (742).

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**Polyfold/Chapter3/GlobalIFTNearbyGraphG.lean.** IsStrongBundleIso.pushforwardDeriv_at (98); IsStrongBundleIso.ker_deriv_eq_map_germ (112); theorem_3_6_8_chart_graph (158).

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**Polyfold/Chapter3/GlobalIFTNearbyOntoG.lean.** theorem_3_6_8_chart_onto (60); theorem_3_6_8_one' (508); theorem_3_6_8' (615); theorem_3_1_26' (631); theorem_3_1_22' (650).

**Polyfold/Chapter3/GlobalIFTNearbyOntoGerm.lean.** perturbEW (64); perturbProj (72); perturbMuProj (83); isQuadrantLike_perturb (92); ShiftedBasicData.levelMap_e_isScSmoothGerm (108); ShiftedBasicData.eW_eq_on (120); ShiftedBasicData.eW_isScSmoothGerm (136); ShiftedBasicData.levelB_eW (150); ShiftedBasicData.Radii (180); ShiftedBasicData.Radii.ν_anti_le (199); ShiftedBasicData.levelB_B_zero (207); ShiftedBasicData.exists_radii (214); ShiftedBasicData.perturbGerm (300); ShiftedBasicData.perturbB (322); ShiftedBasicData.perturbB_zero (328); ShiftedBasicData.perturbB_isScSmoothGerm (336); ShiftedBasicData.levelB_perturbB (387); ShiftedBasicData.perturbMu (407); ShiftedBasicData.perturbMu_pos (418); ShiftedBasicData.perturbMu_anti (430); ShiftedBasicData.perturbMu_mul_le (441); ShiftedBasicData.perturbContraction (474); ShiftedBasicData.mem_perturbContraction_N_iff (583); ShiftedBasicData.perturbContraction_isSc1GermWith (594); ShiftedBasicData.perturbContraction_sol_zero (604).

**Polyfold/Chapter3/GlobalIFTNearbyOntoLemmas.lean.** exists_pos_le_of_forall_pos_fin (58); exists_inward_of_isInGoodPosition (84); inward_add_smul (243); dist_curve_le (282); exists_curve_of_inward (305); hasDerivWithinAt_smul_right_of_continuousWithinAt (577); eq_of_hasDerivWithinAt_Ici (596); StrongRetr.mem_0_of_perturbed_zero (610).

**Polyfold/Chapter3/GlobalIFTNearbyOntoXi.lean.** isScSmooth_const_of_smooth (54); xiMap (81); xiDom (89); mem_xiDom (94); glue₀_preimage_relOpen (98); xiDom_relOpen (108); xiMap_isScSmooth (116); xiMap_smul (181); xiMap_zero (204); levelMap_xiMap_smul (210); ehat (232); ehat_isScSmooth (239); pushGerm₀_U_subset (255); ehat_isScSmoothGerm (263); levelMap_ehat (278); exists_xi_bounds (302); exists_xi_radii_bounds (412).

**Polyfold/Chapter3/GlobalIFTNearbyPerturb.lean.** prodCone (65); prodSub (68); mem_lineality_of_goodComplement_prod (76); sup_lineality_eq_top_of_goodComplement_prod (111); extProj (135); extParam (141); extendedKernel (152); extendedCompl (164); isInGoodPosition_extendedKernel (181).

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**Polyfold/Chapter3/GlobalIFTTheorems.lean.** MPolyfold.IsManifoldWithCornersLoc (125); MPolyfold.IsManifoldWithCorners.isManifoldWithCornersLoc (130); MPolyfold.theorem_3_2_3_loc_isSubMPolyfold (138); MPolyfold.tangentChartsLoc (147); MPolyfold.theorem_3_2_3_loc_isChart (155); MPolyfold.tangentChartsLoc_compatible (163); MPolyfold.germAtlasLoc (176); MPolyfold.theorem_3_2_3_loc_atlas_equiv (186); MPolyfold.theorem_3_2_3_loc_isTame (194); MPolyfold.

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**Polyfold/Chapter3/TameGermTransfer.lean.** StrongBundle.isScFredholmGermTame_of_chart_interior (354); StrongBundle.isScFredholmSectionTame_of_interior (455); CornerFilling.zero_not_mem_interior (508).

**Polyfold/Chapter3/TameGerms.lean.** ScRetraction.IsTame.congr (85); StrongBundle.IsTameChart (128); StrongBundle.isTameChart_of_eqOn_id (263).

**Polyfold/Chapter3/TameGermsAddG.lean.** StrongBundle.IsGoodPositionGerm.toTame (69); StrongBundle.IsScFredholmGermTame.sectionAdd (101); theorem_3_1_18_tame (115).

**Polyfold/Chapter3/TameGermsG.lean.** StrongBundle.IsScFredholmSection (54); StrongBundle.IsScFredholmSectionWholeChart.isScFredholmSection (60); IsScFredholmSection.mem_smoothSet_of_isZeroAt (66); StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt (66); StrongBundle.IsScFredholmGermTame (78); StrongBundle.IsScFredholmGermTame.isScFredholmGerm (87); StrongBundle.IsGoodPositionGermTame (104); StrongBundle.IsGoodPositionGermTame.isGoodPositionGerm (116); StrongBundle.IsScFredholmSectionTame (142).

**Polyfold/Chapter3/TameVacuity.lean.** ScRetraction.halfLineConst (70); ScRetraction.halfLineConst_retract (88); ScRetraction.halfLineConst_prod_not_isTame (102).

**Polyfold/Chapter3/TangentGerm.lean.** IsScInftyGermAt (118); IsScSmoothGerm.isScInftyGermAt (125); IsScSmooth.isScInftyGermAt (138); IsScInftyGermAt.contDiffOn_one (159); finSmoothLift (176); isScSmooth_subtype_val (195); MPolyfold.TangentGermChart (226); MPolyfold.HasTangentGermProperty (258); MPolyfold.TangentGermChart.isPartialQuadrant (268); MPolyfold.TangentGermChart.V_isRelOpenInQuadrant (272); MPolyfold.TangentGermChart.V_subset (276); MPolyfold.TangentGermChart.isTopCompl (280); MPolyfold.TangentGermChart.π (286); MPolyfold.TangentGermChart.Γ (297); MPolyfold.TangentGermChart.π_Γ (305); MPolyfold.TangentGermChart.symm_Γ_mem (308); MPolyfold.TangentGermChart.exists_eq_Γ (314); MPolyfold.TangentGermChart.IsRegular (324); MPolyfold.TangentGermChart.Sig (329); MPolyfold.TangentGermChart.t (332); MPolyfold.TangentGermChart.t_t (339); MPolyfold.TangentGermChart.t_eq_self_of_mem (342); MPolyfold.TangentGermChart.U₀ (346); MPolyfold.TangentGermChart.mem_U₀ (348); MPolyfold.TangentGermChart.Sig_eq (354); MPolyfold.TangentGermChart.isOpen_U₀ (363); MPolyfold.TangentGermChart.s (382); MPolyfold.TangentGermChart.s_s (394); MPolyfold.TangentGermChart.image_s (396); MPolyfold.TangentGermChart.Γ_isScSmooth (403); MPolyfold.TangentGermChart.π_isScSmooth (410); MPolyfold.TangentGermChart.t_isScSmoothRetractMap (415); MPolyfold.TangentGermChart.s_isScSmoothOn (428); theorem_3_2_3_isSubMPolyfold (450); MPolyfold.IsManifoldWithCorners (467); MPolyfold.TangentGermChart.model (477); MPolyfold.TangentGermChart.model_isTameRetract (483); MPolyfold.TangentGermChart.chartInv (488); MPolyfold.TangentGermChart.Γ_continuousOn (494); MPolyfold.TangentGermChart.chart (499); MPolyfold.TangentGermChart.chart_isCorner (546); MPolyfold.TangentGermChart.mem_chart_source (552); MPolyfold.TangentGermChart.Γ_isScSmoothRetractMap (558); MPolyfold.TangentGermChart.π_isScSmoothRetractMap (570); MPolyfold.TangentGermChart.chart_compatible_subChart (582); MPolyfold.theorem_3_2_3_isChart (649); MPolyfold.tangentCharts_compatible (657); MPolyfold.germAtlas (671); MPolyfold.theorem_3_2_3_atlas_equiv (681); MPolyfold.theorem_3_2_3_isTame (690); MPolyfold.theorem_3_2_3_isManifoldWithCorners (699); MPolyfold.theorem_3_2_3_degIndex (713); theorem_3_2_3_of_isRegular (730); MPolyfold.transportChart_model_0 (758); MPolyfold.TangentGermChart.transport (775); proposition_3_2_2_image (827); proposition_3_2_2 (845).

**Polyfold/Chapter3/TangentGermProjecting.lean.** interior_quadCone (123); IsGoodComplement.coordFun_eq_zero (155); MPolyfold.TangentGermChart.exists_projData (216); MPolyfold.TangentGermChart.regraphMap (222); MPolyfold.TangentGermChart.regraphMap_interior_iff (266); MPolyfold.TangentGermChart.hasFDerivWithinAt_regraphMap_zero (313); MPolyfold.TangentGermChart.exists_invData_ball (360); MPolyfold.TangentGermChart.regraphδ_regraphMap (434); MPolyfold.TangentGermChart.regraph (455); MPolyfold.TangentGermChart.π_eq_of_ker (558); MPolyfold.TangentGermChart.exists_projecting (579); MPolyfold.TangentGermChart.exists_isRegular (635); MPolyfold.HasTangentGermProperty.exists_isRegular (652); theorem_3_2_3 (676); theorem_3_2_3_2021def (689); theorem_3_2_3_loc (703).

**Polyfold/Chapter3/UnitIntervalPolyfold.lean.** unitIntervalPolyfold (242); MPolyfold.IsScSmoothMap.exists_rn_reparam (400).

**Polyfold/Chapter4/A3BoundaryForms2021.lean.** MPolyfold.formPullback_extD_eq_of_isAgreeingOver (83); BundledMPolyfold.IsTangentInjective (173); BundledMPolyfold.HasCornerCover (182); BundledMPolyfold.isAgreeingOver_extD (245).

**Polyfold/Chapter4/A3BoundaryForms2021b.lean.** MPolyfold.tangentChartMap_bdryMap (123);

MPolyfold.isTangentInjective_bdryMap (157); MPolyfold.CornerChart (240); MPolyfold.CornerChart.exists_tangent (547); MPolyfold.hasCornerCover_bdryMap (722); MPolyfold.isFaceCompatible_extD (749); MPolyfold.bForms_eq_agreeForms (756); MPolyfold.mem_bForms_iff (760).

**Polyfold/Chapter4/Book2021.lean.** MPolyfold.theorem_4_2_11 (84); theorem_4_3_1 (155).

**Polyfold/Chapter4/BookFormLieBracket2021.lean.** MPolyfold.proposition_4_4_5_book (52).

**Polyfold/Chapter4/BoundaryForms2021.lean.** MPolyfold.IsAgreeingOver (100); MPolyfold.IsFaceCompatible (463); MPolyfold.bForms (471); MPolyfold.le_bForms (480); MPolyfold.extD_mem_bForms (488); MPolyfold.bFormsInfty (499); MPolyfold.jStar (512); MPolyfold.jStar_comp_D (517); MPolyfold.formsPullback_mem_bForms (528); MPolyfold.RelForms (540); MPolyfold.relD (593); MPolyfold.relD_comp_relD (622); MPolyfold.relDeRham (645); MPolyfold.relDeRhamMk_eq_iff (653); MPolyfold.relDeRhamEquivOfNoBoundary (757).

**Polyfold/Chapter4/BoundaryFormsPullback2021.lean.** MPolyfold.CornerCoords.zero_coord_face_along (97); MPolyfold.faceAlongData (289); MPolyfold.faceAlongData_formsD (462); MPolyfold.isChartFaceForm_faceAlongData (495).

**Polyfold/Chapter4/BracketLevel.lean.** BracketLevel.wshift (73); BracketLevel.opS (105); BracketLevel.tw (113); BracketLevel.opT (138); retractTangent_univ (152); isRetractVectorField_scOperator (164); idRetraction (172); retractLieBracket_scOperator (178); BracketLevel.xPt (196); BracketLevel.bracket_not_mem_level_one (200); BracketLevel.isRetractVectorField_opS (221); BracketLevel.isRetractVectorField_opT (226); lieBracket_level_counterexample (235); not_forall_lieBracket_mem_level_one (247).

**Polyfold/Chapter4/Characterization.lean.** ScBanach.mem_smooth_of_finiteDimensional (109); tangentSpace_relOpen (121); isInGoodPosition_top (135); MChart.compRetract (162); MChart.compRetract_isChart (200); MChart.IsCornerChart (237); MChart.IsCornerChart.isRelOpenInQuadrant (248); MChart.IsCornerChart.finiteDimensional (254); MChart.IsCornerChart.isTameRetract (262); MChart.IsCornerChart.isScPlusRetract (267); MChart.IsCornerChart.mem_smooth (273); MPolyfold.HasCornerStructure (288); MPolyfold.hasCornerStructure_iff (296); MPolyfold.HasCornerStructure.isTame (314); MChart.isRelOpenInQuadrant_overlap (332); MChart.IsCornerChart.contDiffOn_transition (346); IsPartialQuadrant.toConst (373); IsRelOpenInQuadrant.toConst (384); IsRelOpenInQuadrant.isScDiffeoQuadrant_toConst (390); MChart.exists_isCornerChart_of_finiteDimensional (416); MPolyfold.hasCornerStructure_iff_finiteDimensional (445); MPolyfold.isManifoldWithCorners_iff (462); MPolyfold.IsManifoldWithCorners.hasCornerStructure (483); MPolyfold.isScSmoothMap_toLevelOne_of_cover (502); image_sub_const_eq_preimage_add' (536); MChart.IsCornerChart.exists_centred (550); MChart.IsCornerChart.exists_centred_tame (570); CornerChartData (600); CornerChartData.model (633); CornerChartData.chart (647); CornerChartData.chart_isCornerChart (664); CornerChartData.chart_isChart (669); CornerChartData.mem_chart_source (675); CornerChartData.chart_apply_eq_zero (678); exists_cornerChartData (690); MPolyfold.theorem_4_1_2_chart (723); MPolyfold.hasCornerStructure_of_centred (742); MPolyfold.theorem_4_1_2_general (769); MPolyfold.theorem_4_1_2 (796).

**Polyfold/Chapter4/EuclideanForms2021.lean.** stdFrame (76); topFormIntegral (92); alternatingMap_comp_eq_det_mul (131); formPullback (155); topFormIntegral_image (200); openQuadrant_ae_eq_quadrant (302); faceEmbedding (334); image_faceEmbedding_openQuadrant (396); facePullback (417);

faceSign (430); det_cons_neg_stdFrame_faceFrame (445).

**Polyfold/Chapter4/EuclideanStokes2021.lean.** integral_divergence_piQuadrant (84); stokes_quadrant (325); stokes_quadrant_inter_open (378).

**Polyfold/Chapter4/Families.lean.** ScBanach.prodInrOp (89); mem_smooth_rn_zero (94); isScSmooth_slice (103); scDeriv_slice (114); hasFDerivWithinAt_scDeriv_const' (132); exists_slice_retraction (169); isGeneralPositionRetractAt_slice (425); fibreSet (529); rnPres (533); rnPres_eq (537); IsStratRegularPoint (553); IsStratRegularValue (566); FibreLocal (573); exists_local_family (594); theorem_4_3_1_of_subset (889); theorem_4_3_1_preprint (941).

**Polyfold/Chapter4/FamiliesLocal.lean.** quadC (73); stratSub (77); mem_stratSub (84); zeroSetOf (88); mem_stratSub_zeroSetOf (93); finrank_stratSub (98); stratCritSet (129); measure_image_stratCritSet (140); ContractionData.exists_solution_smooth (209); theorem_3_8_22_smooth (252); quadAt (404); sliceEquiv (409); isPartialQuadrant_quadC (433); isPartialQuadrant_quadAt (441); convex_quadC (454); interior_quadC_nonempty (459); uniqueDiffWithinAt_quadC (470); exists_goodComplement_stratSub (481); exists_fibre_retraction (531).

**Polyfold/Chapter4/GapP1_Lemma4111.lean.** lemma_4_1_11 (64).

**Polyfold/Chapter4/GapP1_Lemma4111Split.lean.** isPartialQuadrant_preimage_quadrant (61); map_le_map_smooth (122); exists_splitData (140); splitW (173); isScSubspace_splitW (195); qOp (269); isConjugateToBasicGerm_qOp (316).

**Polyfold/Chapter4/GapP1_Lemma451.lean.** MPolyfold.restrictVectorField (57); MPolyfold.tangentMap_restrictVectorField (77); MPolyfold.IsScVectorField.restrictOpen (85); MPolyfold.tangentMap_lieBracket_restrictOpen (99); MPolyfold.lieBracket_restrictOpen (115); lemma_4_5_1_open (135); lemma_4_5_1_open_ambient (150).

**Polyfold/Chapter4/GapP1_Lemma451Bundle.lean.** tangentPullbackLocal (71); isScSmoothRetractMap_tangentPullbackLocal (128); MPolyfold.tangentPullback (183); MPolyfold.tangentPullbackChart (199); MPolyfold.mem_tangentPullback_iff (220); MPolyfold.isSubMPolyfold_tangentPullback (343); MPolyfold.tangentPullbackPolyfold (365); MPolyfold.isScSmoothMap_tangentPullbackProj (378); MPolyfold.tangentPullbackFiberEquiv (390); MPolyfold.isScSmoothMap_tangentPullback_mk (449); MPolyfold.lieBracketSection (465); proposition_4_4_5_section (471); MPolyfold.tangentPullbackMap (491); MPolyfold.isScSmoothMap_tangentPullbackMap (496); lemma_4_5_1_section (521).

**Polyfold/Chapter4/GapP1_LocalFaces.lean.** MPolyfold.exists_faces_chart (50); MPolyfold.theorem_4_2_11_faces (225).

**Polyfold/Chapter4/GapP1_LocalStrongBundle.lean.** isFilling2021_of_R_eq_self (56); LocalBundle.bundle (156); LocalBundle.sec (198); LocalBundle.isScSmoothSection_sec (203); LocalBundle.centred (242); LocalBundle.principal_centred (277).

**Polyfold/Chapter4/GapP1_OpenForms.lean.** MPolyfold.formRestrict (54); MPolyfold.extD_formRestrict (72); BundledMPolyfold.openSub (88); BundledMPolyfold.formsInftyResOpen (120); BundledMPolyfold.formsInftyResOpenOpen (125); BundledMPolyfold.FormsInftyEqOn (153); BundledMPolyfold.formsInftyEqOn_of_iff (158).

**Polyfold/Chapter4/GapP1_Question424.lean.** Question424.tSet (45); Question424.arc_side (51); Question424.arc_overlap (65); Question424.not_homeomorph_open (110); Question424.question_4_

2_4_scPlus (315).

**Polyfold/Chapter4/GapP1_Remark422.lean.** ScRetraction.interior_rangeS_scDeriv_zero_nonempty (73); ScRetraction.isInGoodPosition_rangeS_of_isTameWith (106); MPolyfold.IsSmoothSubmanifold.isSmoothSubmanifold_univ_subPolyfold (135); MPolyfold.remark_4_2_2_dA_zero (153); MPolyfold.remark_4_2_2_dA_zero_near (172); MPolyfold.hasCornerStructure_of_isTame (187); MPolyfold.remark_4_2_2_dA_one_of_isTame (227).

**Polyfold/Chapter4/GapP1_Remark422One.lean.** isInGoodPosition_halfSpace (54); MPolyfold.hasCornerStructure_of_degIndex_le_one (101); MPolyfold.IsSmoothSubmanifold.hasCornerStructure_of_degIndex_le_one (174); MPolyfold.isOpen_degIndex_le (190); MPolyfold.remark_4_2_2_dA_one (201); MPolyfold.remark_4_2_2_dA_one_near (220).

**Polyfold/Chapter4/GapP1_StokesFaceCharts.lean.** MPolyfold.CornerCoords.exists_faceData (202); MPolyfold.hasCornerStructureOfDim_face (353).

**Polyfold/Chapter4/GapP1_StokesFaceIntegral.lean.** restrictForm (103); FaceChartPair.chartForm_restrictForm (178); ManifoldForm.exists_integrationData_of_regions (290); GoodRegion.sum_integral_eq (486); sum_integral_restrictForm_eq_boundaryIntegral (531).

**Polyfold/Chapter4/GapP1_StokesFaceOrientation.lean.** FaceChartPair.chartChangeDeriv_face (193); FaceChartPair.inward (225); IsInducedOrientation (277); inducedOrientation (301); isContinuous_inducedOrientation (328); IsInducedOrientation.unique (345).

**Polyfold/Chapter4/GapP1_StokesFaces.lean.** MPolyfold.finite_setOf_isFace (55); MPolyfold.exists_face_eq_coordHyper (72).

**Polyfold/Chapter4/GapP1_Theorem4211.lean.** MPolyfold.theorem_4_2_11_one (54).

**Polyfold/Chapter4/GapP1_Theorem431.lean.** scDeriv_fst_eq_of_fst_eq (49); StratRegCond (86); StratRegCond.congr (92); StratRegCond.transport (128); transition_rnChart_prod_fst (185); stratRegCond_chart_change (200); IsStratRegularPoint.stratRegCond (304); isStratRegularPoint_iff_forall (330); isStratRegularValue_iff_forall (356).

**Polyfold/Chapter4/GapP3_K4_BoundaryWedge.lean.** MPolyfold.IsAgreeingOver.formWedge (47); BundledMPolyfold.agreeDFormsWedge (102); BundledMPolyfold.agreeDFormsInftyWedge (208); BundledMPolyfold.agreeDFormsInftyWedge_of_of (229); BundledMPolyfold.agreeDFormsInftyCast (262); BundledMPolyfold.agreeDFormsInftyD_agreeDFormsInftyWedge (328); BundledMPolyfold.agreeDFormsInftyWedge_assoc (383); BundledMPolyfold.agreeDFormsInftyOne (397); BundledMPolyfold.agreeDFormsInftyWedge_agreeDFormsInftyOne (426); BundledMPolyfold.agreeDFormsInftyCast_agreeDFormsInftyOne_wedge (435); BundledMPolyfold.agreeInftyPullback_formsInftyWedge (456); BundledMPolyfold.agreeDFormsInftyIncl_agreeDFormsInftyWedge (473); MPolyfold.jStar_formsInftyWedge (509).

**Polyfold/Chapter4/GapP3_K4_RelWedge.lean.** MPolyfold.relWedge (170); MPolyfold.relFst_relWedge (177); MPolyfold.relSnd_relWedge_succ (183); MPolyfold.relSnd_relWedge_zero (191); MPolyfold.relD_relWedge (286); MPolyfold.relWedge_assoc (341); MPolyfold.relOne (367); MPolyfold.relCast_relWedge_relOne (370); MPolyfold.relWedge_relOne (385); MPolyfold.relGMul (393); MPolyfold.relGNonUnitalNonAssocSemiring (398); MPolyfold.relGraded_of_mul_of (408); MPolyfold.relGraded_mul_assoc (415); MPolyfold.relGradedNonUnitalRing (427); MPolyfold.relGradedD (432); MPolyfold.relGradedD_comp_self (448); MPolyfold.relGradedD_of_mul_of (457).

**Polyfold/Chapter4/LocalStructure.lean.** exists_isClosed_isCompl_le (123); exists_proj_of_isCompl (153); mem_minimalSubspaceWith_zero (173); minimalSubspaceWith_zero_le (183); coord_add_of_mem_zero (190); degIndexWith_add_of_mem_zero (195); add_mem_iff_of_mem_zero (202); coordsOn (209); coordsOn_surjective (218); interior_nonempty_of_sup_eq_top (231); isGoodComplement_of_le_zero (254); exists_goodComplement_le_zero (260); degIndex_slice (272); exists_scComplement_le (327); isGeneralPositionRetractAt_of_sup_eq_top (421); sup_minimalSubspaceWith_eq_top (463); sup_minimalSubspaceWith_eq_top_graph (498); theorem_4_2_11_chart (542); quotDim_le_finrank_of_eq_sup (593); remark_4_3_2_chart (613); IsGeneralPositionRetractAt.isGoodPositionRetractAt (642); maskL (718); killOn (734); coordsOn_killOn (739); killOn_eq_self (748); killOn_idem (758); killOn_mem (764); opToConst (779); exists_scPlus_retraction_corner (797); exists_cornerModel_retraction (982); theorem_4_2_11_four_chart (1060); MPolyfold.isScPlusRetractionOn_of_chart (1227); MChart.cornerPreimage (1315); MPolyfold.theorem_4_2_11_four_of_chart (1332); MPolyfold.theorem_4_2_11_preprint (1541); MPolyfold.remark_4_3_2_codim (1676); MPolyfold.remark_4_3_2 (1700); MPolyfold.remark_4_3_2_not_mem_boundary (1723).

**Polyfold/Chapter4/ManifoldBoundary2021.lean.** exists_face_chartChange (199); orientation_face_chartChange (286).

**Polyfold/Chapter4/ManifoldFaceForms2021.lean.** ManifoldForm.extDeriv_extDeriv (190); IsChartFaceForm (771); ChartFaceForm (813); ChartFaceForm.ofForm (910); exists_extension (1102); ChartFaceForm.integral (1508); ChartFaceForm.integral_d_eq_zero (1656); ChartFaceForm.integral_d_eq_zero_of_compactSpace (1662).

**Polyfold/Chapter4/ManifoldForms2021.lean.** extDerivWithin_formPullback (68); ManifoldForm (102); chartForm (112); chartForm_eq_formPullback (203); ContMDiffForm (242); ManifoldForm.extDeriv (321); chartForm_extDeriv (387).

**Polyfold/Chapter4/ManifoldIntegral2021.lean.** signedOr (63); orSign (66); chartIntegral (163); chartIntegral_eq_integral_sign (236); IntegrationData (394); IntegrationData.integral_eq (481); exists_integrationData (527); ManifoldForm.integral (581); ManifoldForm.integral_eq_integrationData (590); ManifoldForm.integral_neg_orientation (630).

**Polyfold/Chapter4/ManifoldStokes2021.lean.** faceSign_mul_det_eq (126); chartBoundaryIntegral (203); chartIntegral_extDeriv_eq (222); IntegrationData.stokes (288); IntegrationData.boundaryIntegral_eq (315); ManifoldForm.boundaryIntegral (336); integral_extDeriv_eq_boundaryIntegral (370); stokes_of_compactSpace (383).

**Polyfold/Chapter4/Poincare2021.lean.** scDeriv_intervalIntegral (328); isSck_intervalIntegral (398); isScSmooth_intervalIntegral (412).

**Polyfold/Chapter4/Poincare2021b.lean.** homotopyForm (175); isScSmooth_homotopyForm (210); ambientD_homotopyForm (580); lemma_4_5_2 (666); retractD_eq_ambientD (723); hasPoincare_relOpen (796); hasPoincare_relOpen_of_zero_mem (844); localPoincare (853); lemma_4_4_8 (860).

**Polyfold/Chapter4/Poincare448.lean.** LocalPoincare (94); hasPoincare_lift (99); exists_smooth_mem_relOpen (108); isScSmoothMap_retractChart (152); exists_primitive_of_factor (240); exists_poincareChart (284); lemma_4_4_8_of_local (311).

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**Polyfold/Chapter4/Recentre.lean.** recentreNbhd (86); isOpen_recentreNbhd (88); mem_recentreNbhd (97); recentreNbhd_inter_eq (101); add_mem_partialQuadrantAtWith_iff (117); image_sub_partialQuadrantAtWith (129); zeroIdx (140); recentreDim (148); splitCoordsL (167); splitCoordsIso (189); recentreIso (200); partialQuadrantAtWith_eq_preimage (210); isPartialQuadrant_partialQuadrantAtWith (225); recentreIso_fst_sub (230); zero_of_mem_recentreNbhd (235); degIndexWith_recentreIso (242); degIndex_sub_recentre (258); minimalSubspaceWith_recentreIso (267); partialQuadrantAtWith_recentreIso (284); isScSmooth_add_const (310); isScSmooth_sub_const (320); relOpen_translate (328); isScDiffeoQuadrant_translate (350); ScRetraction.restrictIn (373); ScRetraction.restrictIn_retract (393); ScRetraction.scDeriv_restrictIn (404); ScRetraction.IsTameWith.restrictIn (410); ScRetraction.restrictIn_isTame (421); isScRetract_recentre (436); isTameRetract_recentre (450); MChart.recentreSource (473); MChart.recentre (494); MChart.recentre_apply_self (530); MChart.recentre_isChart (537); MChart.isTameRetract_recentre (544); MChart.chartDegIndex_recentre (552); subFst (563); addFst (566); mem_smooth_prod_zero (576); isScSmooth_addFst (583); isScSmooth_subFst (591); IsStrongBundleRetraction.restrictIn (621); IsStrongBundleRetraction.translate (651); StrongBundleChart.recentre (806); StrongBundleChart.recentre_overlap (887); StrongBundleChart.overlap_recentre_eq (912); StrongBundleChart.Compatible.recentre (934); StrongBundle.IsChart.recentre (1057); MPolyfold.exists_centredChart (1080); MPolyfold.IsTame.exists_centredChart (1096); StrongBundle.IsChart.exists_centred (1110).

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**Polyfold/Chapter4/ScFormsPullback2021.lean.** MPolyfold.tangentFiberMap (75); MPolyfold.formPullback (89); MPolyfold.chartForm_formPullback (162); MPolyfold.formPullback_mem_scForms (217); MPolyfold.formPullback_id (225); MPolyfold.formPullback_comp (234); MPolyfold.formLower_formPullback (246); MPolyfold.extD_formPullback (306); BundledMPolyfold.formsPullback (418); BundledMPolyfold.formsInftyPullback (444); BundledMPolyfold.formsInftyD_comp_formsInftyPullback (461); BundledMPolyfold.formsInftyPullback_comp (499); BundledMPolyfold.scDeRhamPullback (531).

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**Polyfold/Chapter4/VectorFieldsGlobal.lean.** IsRetractVectorField.diff_related (96); IsRetractVectorField.related_shift (192); retractLieBracket_natural (232); levelMap_levelMap_comp_eq (257); IsRetractVectorField.restrict (277); retractLieBracket_restrict (296); retractLieBracket_natural_inter (322); MPolyfold.tangentChartMap_change (376); MPolyfold.chartRetraction (385); MPolyfold.chartRetraction_retract (389); MPolyfold.principalPart (398); MPolyfold.tangentChartMap_eq_principalPart (402); MPolyfold.IsScVectorField.isRetractVectorField_principalPart (421); MPolyfold.IsScVectorField.principalPart_related (444); MPolyfold.chartLieBracket (476); MPolyfold.IsScVectorField.chartLieBracket_spec (480); MPolyfold.IsScVectorField.chartLieBracket_change (499); MPolyfold.levelLift_levelChart_spec (547); MPolyfold.lieBracketPre (561); MPolyfold.lieBracket (581); MPolyfold.tangentProj_lieBracket (586); MPolyfold.tangentChartMap_lieBracket (592); MPolyfold.tangentChartMap_lieBracket_eq (637); MPolyfold.continuous_lieBracket (648); proposition_4_4_5 (678); MPolyfold.liftLevelOneMap (713); MPolyfold.levelTwoMap (717); MPolyfold.levelMap_coord_levelLift (723); lemma_4_5_1 (736); MPolyfold.vectorFieldLevelOne (844); MPolyfold.raiseIndex (851); lemma_4_4_4 (858).

**Polyfold/Chapter5/AllowableSectionsMetric.lean.** StrongBundle.IsAuxNorm.neg (86); StrongBundle.IsAuxNorm.sub_le (92); StrongBundle.IsAuxNorm.sub_triangle (102); StrongBundle.allowableDist (110); StrongBundle.sectionDistAt_le_two (114); StrongBundle.allowableDist_le_two (131); StrongBundle.allowableDist_le_iff (142); StrongBundle.allowableDist_self (147); StrongBundle.allowableDist_comm (154); StrongBundle.allowableDist_triangle (162); StrongBundle.eq_of_allowableDist_eq_zero (172); StrongBundle.allowableMetric (184); StrongBundle.AllowableSpace (195); StrongBundle.subsingleton_allowableSections_empty (213); StrongBundle.completeSpace_allowableSections_empty (223); TrivialBundle.fibNorm (276); TrivialBundle.isAuxNorm_fibNorm (279); TrivialBundle.isScPlusSection_graphSection (349); TrivialBundle.differentiable_snd_of_isScPlusSection (372); TrivialBundle.abs_sqrt_sub_abs_le (425); TrivialBundle.approx (434); TrivialBundle.limitFn (437); TrivialBundle.abs_approx_sub_limitFn (453); TrivialBundle.graphSection_approx_mem (469); TrivialBundle.not_differentiableAt_limit (476); TrivialBundle.not_completeSpace_allowableSections (491); TrivialBundle.isScPlusSection_of_isSection_point (547); TrivialBundle.completeSpace_allowableSections_point (556).

**Polyfold/Chapter5/AuxNorm.lean.** StrongBundle.mem_oneSet_iff_of_isChart (120); StrongBundle.oneChartOf (182); StrongBundle.oneChart_eq_oneChartOf (290); StrongBundle.oneChartOf_transition (310); StrongBundle.oneChartOf_compatible (320); StrongBundle.isChart_oneChartOf (337); StrongBundle.fiberZero_mem_oneSet (351); StrongBundle.oneFiber (361); StrongBundle.toOne (385); StrongBundle.ofOne (388); StrongBundle.zeroOne (399); StrongBundle.liftOneFiber (437); StrongBundle.liftOneFiber_injective (458); StrongBundle.mem_range_liftOneFiber (480); StrongBundle.isClosed_slice (511); StrongBundle.complete_liftOneFiber (534); StrongBundle.IsCompleteFiberNorm (565); StrongBundle.IsAuxNorm (577); StrongBundle.IsAuxNorm.nonneg (600); StrongBundle.IsAuxNorm.smul (603); StrongBundle.exists_le_mul_norm_liftOne (616); StrongBundle.IsAuxNorm.exists_norm_liftOne_le (687); StrongBundle.IsAuxNorm.exists_bounds (759); lemma_5_1_2 (780); StrongBundle.exists_norm_liftOne_le_mul (828); StrongBundle.exists_uniform_comparison (844); StrongBundle.complete_of_bounds (877); StrongBundle.gluedNorm (902); StrongBundle.sum_partition_eq_one (906); StrongBundle.gluedNorm_eq_sum (910); StrongBundle.continuous_gluedNorm (921); StrongBundle.gluedNorm_bounds (940); StrongBundle.isCompleteFiberNorm_gluedNorm (972); proposition_5_1_3_exists (1023); proposition_5_1_3_

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**Polyfold/Chapter5/Book2021.lean.** StrongBundle.IsWeaklyProperOnU (110); StrongBundle.IsWeaklyProperLtInfty (119); StrongBundle.IsProperLtInfty (129); StrongBundle.IsAuxNorm.auxLe_fiberZero (134); StrongBundle.AuxLe.mono (143); StrongBundle.isZeroAt_of_not_mem_of_support_subset (149); StrongBundle.isWeaklyProperOnU_iff (156); StrongBundle.IsProperLtInfty.isProper (173); StrongBundle.IsWeaklyProperLtInfty.isWeaklyProperOnU (179); StrongBundle.IsProper.isWeaklyProperOnU (186); StrongBundle.IsProperLtInfty.isWeaklyProperLtInfty (192); StrongBundle.IsWeaklyProperOnU.isCompact_solutionSet (213); theorem_5_2_2 (224); theorem_5_2_2_tfae (255); isSubMPolyfold_solutionSet_of_manifold (277); theorem_5_3_5 (289); MPolyfold.HasSurjDerivAt (319); StrongBundle.IsCompactFredholmRestriction (331); StrongBundle.IsSubmersiveGenCompactHomotopy (346); StrongBundle.IsGenCompactHomotopy.isSubmersiveGenCompactHomotopy (359); StrongBundle.IsSubmersiveGenCompactHomotopy.isCompact_solutionSet_restrict (374); lemma_5_3_16 (400); isScSmoothRetractMap_of_domain_empty (439); StrongBundle.isCompactFredholmRestriction_of_eq_empty (445); GenCompactHomotopyWitness.arctanTime (470); GenCompactHomotopyWitness.contDiff_arctanTime (473); GenCompactHomotopyWitness.arctanTime_mem_Ioo (477); GenCompactHomotopyWitness.hasFDerivAt_arctanTime (492); GenCompactHomotopyWitness.hasSurjDerivAt_arctanTime (516); GenCompactHomotopyWitness.solutionSet_graphSection_id (543); GenCompactHomotopyWitness.arctanTime_preimage_eq_empty (549); GenCompactHomotopyWitness.isSubmersiveGenCompactHomotopy_witness (561); GenCompactHomotopyWitness.not_isGenCompactHomotopy_witness (575).

**Polyfold/Chapter5/BumpFunctions.lean.** realSc (128); realChart (134); realPolyfold (150); isChart_realChart (158); MPolyfold.isScSmoothMap_of_local (172); MPolyfold.IsScSmoothMap.real_chart (195); isScSmoothMap_real_of_local (205); isScSmoothMap_real_const (214); isScSmoothMap_real_of_contDiff (223); MPolyfold.IsScSmoothMap.comp_contDiff (232); MPolyfold.AdmitsBumpFunctions (248); MPolyfold.AdmitsBumpFunctionsWeak (254); MPolyfold.AdmitsBumpFunctions.weak (261); proposition_5_5_3 (271); proposition_5_3_2 (300); LocalModel.HasBumpProperty (316); closure_inter_ball_subset (325); theorem_5_5_7 (367); LocalModel.hasBumpProperty_of_quadratic (460); proposition_5_5_8 (502); MPolyfold.admitsBumpFunctions_of_hilbert (510); isScSmooth_const_of_mem_smooth (530); sectionSupport (540); sectionSmul (546); sectionSum (552); StrongBundle.scPlusSections (611); StrongBundle.allowableSections (616); StrongBundle.IsScPlusSection.mem_oneSet (624); StrongBundle.isScPlusSection_fiberZero (641); StrongBundle.IsScPlusSection.smul_left (656); StrongBundle.IsScPlusSection.add (691); StrongBundle.isScPlusSection_sectionSum (740); StrongBundle.cutoffSection (779); StrongBundle.isScPlusSection_cutoffSection (850); lemma_5_3_3 (907).

**Polyfold/Chapter5/BumpWeakPointwise.lean.** MPolyfold.HasBumpAt (72); MPolyfold.admitsBumpFunctions_iff_forall_hasBumpAt (78); MPolyfold.admitsBumpFunctionsWeak_of_dense (87); not_isSc0_add_const (105); scChart (129); scPolyfold (138); isScSmoothMap_scPolyfold_iff (157); MPolyfold.AdmitsBumpFunctionsWeak.hasBumpAt_of_mem_smooth (179).

**Polyfold/Chapter5/Compactness.lean.** theorem_5_2_3 (162); MPolyfold.continuous_smoothIncl (190); MPolyfold.continuous_smoothVal (196); MPolyfold.tendsto_smoothLevel_iff (202); theorem_5_2_4 (215); isCompact_solutionSet_level (242); ScNbhdGerm.shiftParam (266); IsSc0ContractionGerm.shift (284); IsBasicGerm.shift (306); IsConjugateToBasicGerm.shift (343); IsConjugateToBasicGerm.shift_germ (404); corollary_5_2_5_chart (482); IsSc0ContractionGerm.eq_delta_zero (527); IsSc0ContractionGerm.exists_isCompact_bdd (537); IsBasicGerm.exists_

isCompact_bdd (601); IsStrongBundleMap.shiftFiber_snd_zero (627); IsStrongBundleMap.shiftFiber_snd_smul (634); IsStrongBundleMap.exists_fiber_bound_one (645); exists_nbhd_isCompact_closure_bdd (695); StrongBundle.AuxLe (859); StrongBundle.IsWeaklyProper (866); StrongBundle.IsProper (875); StrongBundle.AuxLe.congr (880); StrongBundle.IsAuxNorm.auxLe_of_sectionAdd (885); theorem_5_2_2_preprint (920); StrongBundle.IsProper.isWeaklyProper (955); StrongBundle.IsWeaklyProper.isCompact_solutionSet (973); theorem_5_2_2_iff (978); remark_5_3_4 (993).

**Polyfold/Chapter5/Corollary58.lean.** IsScSmoothGerm.toBase_zero_mem_smooth (79); IsFillingGerm.mem_range_toBase_of_fillDefect (99); toBase_scDeriv_r_shift (205); toBase_ρCLM_shift (229); IsFillingGerm.germ_shift (245); StrongBundleChart.principal_sectionShift (359); StrongBundleChart.shift_φ_eq_zero (377); StrongBundleChart.shift_filling_eq_on (410); StrongBundleChart.shift_filling_mem_of_fixed (425); StrongBundleChart.shift_isFillingGerm (471); isScFredholmGerm_sectionShift (494); corollary_5_2_5_isScFredholmSection (536); corollary_5_2_5 (546); corollary_5_2_5_two (557); corollary_5_2_5_data (571).

**Polyfold/Chapter5/CylinderPerturbation2021.lean.** IsAuxNorm.max (79); allowableSections_max_subset (108); StrongBundle.IsAuxNorm.auxLe_sectionAdd (147); exists_adapted_margin (197); isGeneralPositionGerm_congr_nhds (241); isParamGP_elim0_of_isGeneralPositionGerm (265); isGeneralPositionGerm_paramSection_of_isGeneralPositionGerm (319); isGeneralPositionGerm_of_sup_eq (350); isGeneralPositionGerm_paramLift (509); exists_sections_isGeneralPositionGerm_near (652); exists_null_isGeneralPositionGerm_paramSlice (736); theorem_5_3_10_relative_hyp (946); StrongBundle.IsAuxNorm.pullback (1118); exists_admissible_subset (1236); exists_admissible (1254); exists_admissible_two (1266); cylCut (1305); cylInterp (1367); isScPlusSection_cylInterp (1509); isCompact_slab_paramLift_add (1524); isGeneralPositionGerm_paramLift_add_of_eq (1557); exists_cobordism (1599).

**Polyfold/Chapter5/GapP1_ApproxEveryChart.lean.** abs_sub_sum_mul_le (51); sum_mul_mem_Icc (66); exists_continuous_pos_iff (76); MPolyfold.AdmitsScPartitionsOfUnity.exists_urysohn (96); MPolyfold.AdmitsScPartitionsOfUnity.exists_approx (129); hasApproxProperty_of_isChart (187); theorem_5_5_10_every_chart (405).

**Polyfold/Chapter5/GapP1_BonicFrampton.lean.** BonicFrampton.kern_sub_le (196); BonicFrampton.taylor1_le (224); BonicFrampton.gK (258); BonicFrampton.hasDerivAt_gK (263); BonicFrampton.gK_taylor_le (277); BonicFrampton.holderTriple_qE (307); BonicFrampton.memLp_gK (382); BonicFrampton.Tj (415); BonicFrampton.continuous_Tj (430); BonicFrampton.hasFDerivAt_Tj (452); BonicFrampton.hasFTaylorSeriesUpTo_normRpow (540); BonicFrampton.contDiff_normRpow_of_lt (567); BonicFrampton.theorem_5_5_5 (582).

**Polyfold/Chapter5/GapP1_C0Base.lean.** IsRelC0Base (46); IsRelC0Base.comp (53); exists_isRelC0Base_of_admits (84); admitsContDiffPartitionsOfUnity_of_isRelC0Base (134); HasContDiffBump.exists_cutoff (199); isRelC0Base_glue (236).

**Polyfold/Chapter5/GapP1_DenseGoodPosition.lean.** paramSlice_apply_of_isZeroAt (49); sectionAdd_eq_of_not_mem_support (62); eventually_isZeroAt_nhdsSet_iff (72); theorem_5_3_12_param_rel (97); theorem_5_3_12_dense (409).

**Polyfold/Chapter5/GapP1_GTWZ.lean.** HasContDiffBump.submodule (42); IsWCG.of_proj (56); exists_dense_card_le (78); admitsContDiffPartitionsOfUnity_of_wcg_card (96); theorem_5_5_16 (199).

**Polyfold/Chapter5/GapP1_HilbertPOU.lean.** `l2_inverse_estimate` (43); `admitsContDiffPartitionsOfUnity_of_hilbertSpace` (144); `godefroyTroyanskiWhitfieldZizlerStatement_of_hilbertSpace` (237).

**Polyfold/Chapter5/GapP1_LindenstraussPair.lean.** `IsLPair` (44); `IsLPair.exists_proj` (144); `lclA` (247); `lclD` (250); `isLPair_lcl` (311); `lclA_finite` (393); `card_lcl` (454).

**Polyfold/Chapter5/GapP1_LpWeightedProduct.lean.** `BonicFrampton.eLpNorm_prod_le` (30); `BonicFrampton.wMul` (142); `BonicFrampton.norm_wMul_le` (166); `BonicFrampton.norm_rpow_eq_integral` (209); `BonicFrampton.theorem_5_5_5_even` (221).

**Polyfold/Chapter5/GapP1_Remark5314.lean.** `StrongBundle.IsProblemIso` (56); `StrongBundle.IsStrongBundleMap.map_fiberZero` (64); `StrongBundle.IsProblemIso.refl` (71); `StrongBundle.IsProblemIso.symm` (74); `StrongBundle.IsProblemIso.trans` (82); `StrongBundle.IsProblemIso.image_solutionSet` (91); `StrongBundle.IsProblemIso.isCompact_solutionSet_iff` (111); `StrongBundle.EndIso` (145); `StrongBundle.IsGenCompactHomotopyBetween` (161); `StrongBundle.EndIso.trans` (170); `StrongBundle.IsGenCompactHomotopyBetween.congr_left` (181); `StrongBundle.IsGenCompactHomotopyBetween.congr_right` (192); `StrongBundle.EndIso.isCompact_solutionSet` (204); `StrongBundle.IsGenCompactHomotopyBetween.isCompact_solutionSet_left` (221); `StrongBundle.IsGenCompactHomotopyBetween.isCompact_solutionSet_right` (229); `StrongBundle.IsSubmersiveGenCompactHomotopy.isGenCompactHomotopyBetween` (238).

**Polyfold/Chapter5/GapP1_Remark538.lean.** `exists_margin_of_isCompact` (65); `remark_5_3_8_relative` (106); `remark_5_3_8_arc` (172).

**Polyfold/Chapter5/GapP1_SeparablePOU.lean.** `IsCompactOperator.isSeparable_range` (53); `ScBanach.separableSpace_zero` (71); `HasContDiffBump.exists_one_on_ball` (87); `sum_range_mul_prod_one_sub` (137); `exists_contDiff_urysohn_of_separable` (153); `admitsContDiffPartitionsOfUnity_of_separable` (287); `godefroyTroyanskiWhitfieldZizlerStatement_of_separable` (311); `corollary_5_5_17_unconditional` (325); `MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional` (338).

**Polyfold/Chapter5/GapP1_SobolevCube1.lean.** `SobolevCube.abs_Vf_sub_le` (138); `SobolevCube.dist_V_le` (233); `SobolevCube.isCompactOperator_V` (275).

**Polyfold/Chapter5/GapP1_SobolevCube2.lean.** `SobolevCube.denseRange_poly` (185); `SobolevCube.eq_cst_of_weakDeriv_zero` (315); `SobolevCube.eq_cst_add_V` (341).

**Polyfold/Chapter5/GapP1_SobolevCube3.lean.** `SobolevCube.comp_eq_cst_add_V` (233).

**Polyfold/Chapter5/GapP1_SobolevCube4.lean.** `SobolevCube.abs_cst_le` (36); `SobolevCube.isCompactOperator_inclW_cube` (74).

**Polyfold/Chapter5/GapP1_SobolevCube5.lean.** `SobolevCube.exists_smooth_approx` (41); `SobolevCube.dense_smoothLvl_cube` (135); `SobolevCube.cubeSc` (156); `SobolevCube.remark_5_5_5_W14_cube` (165); `SobolevCube.remark_5_5_5_W14_instance` (175); `SobolevCube.remark_5_5_5_W1_3_2_cube` (183).

**Polyfold/Chapter5/GapP1_SobolevRemark1.lean.** `SobolevRemark.theorem_5_5_5_sum` (64); `SobolevRemark.bump_comp_rpow` (98); `SobolevRemark.hasContDiffBump_of_rpow` (115); `SobolevRemark.hasContDiffBump_Lp` (163); `LocalModel.hasBumpProperty_of_hasContDiffBump` (184); `MPolyfold.admitsBumpFunctions_of_hasContDiffBump` (230); `exists_isSck_bump` (240).

**Polyfold/Chapter5/GapP1_SobolevRemark2.lean.** SobolevRemark.wk (142); SobolevRemark.Wsub (157); SobolevRemark.isClosed_Wsub (164); SobolevRemark.norm_rpow_W (189); SobolevRemark.remark_5_5_5_sobolev (201); SobolevRemark.remark_5_5_5_sobolev_bump (210); SobolevRemark.Dop_zero_injective (233); SobolevRemark.inclW (279); SobolevRemark.inclW_injective (285).

**Polyfold/Chapter5/GapP1_SobolevRemark3.lean.** SobolevRemark.remark_5_5_5_W14 (137); SobolevRemark.remark_5_5_5_W1_3_2 (165); SobolevRemark.sobolevSc (191); SobolevRemark.sobolevSc_hyps_empty (241).

**Polyfold/Chapter5/GapP1_Torunczyk.lean.** abs_cz_sub_le (49); finite_cz_gt (69); contDiff_finsum_of_locallyFinite (81); admitsContDiffPartitionsOfUnity_of_urysohn (97); torunczyk_forward (129); thetaB (276); etaB (279); latticeBump (302); contDiff_latticeBump (365); locallyFinite_latticeBump (392); exists_latticeBump_eq_one (456); tsupport_latticeBump_subset_ball (480); exists_contDiff_urysohn_of_embedding (508); torunczyk_backward (670); theorem_5_5_13 (680).

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**Polyfold/Chapter5/GeneralPosition.lean.** isScPlus_of_isScSmooth_const (95); MPolyfold.TangentGermChart.Γ_isScPlus (141); MPolyfold.TangentGermChart.exists_scPlusRetraction (149); MPolyfold.TangentGermChart.exists_linearIndependent_tangentSpace (226); MPolyfold.TangentGermChart.exists_isScPlusRetractionOn (284); MPolyfold.isSmoothSubmanifold_of_isRegular (307); MPolyfold.IsGeneralPositionAt.congr_nhds (318); HasIndepTangent (371); HasIndepTangent.transport (378); StrongBundle.isSmoothSubmanifold_solutionSet_inter (422); StrongBundleChart.mem_linKer_iff (437); tangentSpace_subset_linKer (451); exists_tangentSpace_eq_linKer (508); StrongBundleChart.linKer_le (613); exists_compl_le_of_sup_eq (620); isGoodPositionGermTame_of_isGeneralPositionAt (653); exists_isGeneralPositionGerm_iff (715); exists_isOpen_isGeneralPositionGerm (787); StrongBundle.SolutionSetInGeneralPosition (827); theorem_5_3_10_of_family (864); StrongBundleChart.IsParamGP (1111); StrongBundleChart.IsParamGP.comp (1125); minimalSubspaceWith_rnPres (1145); pc0 (1166); ofProd_mem_rangeS (1173); isGeneralPositionGerm_paramSection_of_isParamGP (1197); isParamGP_of_isGeneralPositionGerm (1363); spanning_of_isLinSurjAt (1478); exists_sections_isParamGP (1528); theorem_5_3_10_param (1779); StrongBundle.IsAuxNorm.fiberZero_eq (1896); theorem_5_25_preprint (1914); theorem_5_3_10_dense (1968); sectionAdd_fiberZero (2088); theorem_5_3_12_param (2119).

**Polyfold/Chapter5/Homotopy.lean.** lemma_5_3_16_preprint (106); ne_zero_of_isCompl_ker (150); exists_isCompl_ker_of_ne_zero (162); restrictSection (205); isZeroAt_restrictSection_iff (212); StrongBundle.IsGenCompactHomotopy (234); StrongBundle.IsGenCompactHomotopy.isCompact_solutionSet_restrict (253); scDeriv_smul_apply (277); scDeriv_clm_sub_apply (303); chartLinCutoff (338); chartLinCutoff_of_mem (342); isScSmoothMap_chartLinCutoff (347); sectionSupport_sectionSmul_subset (390); principal_sectionSmul (399); exists_scPlusSection_linAt (416); lemma_5_3_17 (528).

**Polyfold/Chapter5/P3B_BoundaryExtension2021.lean.** StrongBundle.IsBoundaryScPlusSection (92); StrongBundle.boundarySupport (102); StrongBundle.isBoundaryScPlusSection_fiberZero (115); StrongBundle.degIndex_ne_zero_of_mem_boundary (132); StrongBundle.exists_local_extension (145); StrongBundle.aux_fiberZero (214); StrongBundle.isScPlusSection_of_locally_p3b (220); StrongBundle.exists_local_bounded (236); StrongBundle.exists_boundary_extension (278);

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**Polyfold/Chapter5/P3R_Reflexive2021.lean.** StrongBundleChart.liftOneSnd (108); StrongBundleChart.mem_K_of_tendsto (117); StrongBundle.HasReflexiveFibers (149); StrongBundle.hasReflexiveFibers_of_atlas (153); StrongBundle.exists_limit_of_weakTendsto (187); StrongBundle.weakTendsto_liftOneSnd_of_tendsto (233); StrongBundle.weakTendsto_transfer (255); lemma_5_1_5 (280); lemma_5_1_5_limit (297); StrongBundle.MixedTendsto (318); StrongBundle.MixedTendsto.weakTendsto_of_isChart (326); StrongBundle.MixedTendsto.tendsto_oneVal (337); StrongBundle.exists_mixedTendsto_subseq (354); StrongBundle.IsReflexiveAuxNorm (380); StrongBundle.IsReflexiveAuxNorm.le_of_frequently_le (386); StrongBundle.le_liminf_coe_of_eventually (395); StrongBundle.le_liminf_gluedNorm (429); StrongBundle.isReflexiveAuxNorm_gluedNorm (481); proposition_5_1_8_exists (506); proposition_5_1_8_compare (520); proposition_5_1_8 (528); StrongBundle.IsReflexiveAuxNorm.comp_mul (588); StrongBundle.exists_reflexiveAuxNorm_sandwich (611).

**Polyfold/Chapter5/P3R_ReflexiveCompactness2021.lean.** StrongBundle.HasReflexiveLocalCompactness (96); StrongBundle.exists_nhds_not_auxLe (105); StrongBundle.not_auxLe_of_not_mem_levelSet (141); StrongBundle.isClosed_setOf_auxLe (148); StrongBundle.closure_auxLe_subset (158); proposition_5_2_7 (174); theorem_5_2_8_of_continuous (190); theorem_5_2_8 (268); TrivialBundle.theorem_5_2_8_witness (307).

**Polyfold/Chapter5/P3R_WeakSeqCompact.lean.** WeakTendsto (55); WeakTendsto.exists_bound (87); WeakTendsto.eventually_norm_gt (103); exists_subseq_weakTendsto (129); exists_subseq_weakTendsto_of_eventually (246); WeakTendsto.tendsto_of_isCompactOperator (256); weakTendsto_of_bounded_of_tendsto (271); isReflexive_of_finiteDimensional (291).

**Polyfold/Chapter5/PartitionsOfUnity.lean.** IsScSmoothRetractMap.real_add (154); IsScSmoothRetractMap.real_mul (162); isScSmoothMap_real_of_eventuallyEq (192); MPolyfold.IsScSmoothMap.real_add (206); MPolyfold.IsScSmoothMap.real_mul (216); MPolyfold.IsScSmoothMap.real_sub (226); isScSmoothMap_real_finsetSum (236); isScSmoothMap_real_finsetProd (252); isScSmoothMap_real_finsum (268); isScSmoothMap_real_of_chart (277); PartitionOfUnity.abs_sub_finsum_mul_le (310); PartitionOfUnity.finsum_mul_mem_Icc (326); PartitionOfUnity.tsupport_finsum_mul_subset (339); MPolyfold.IsScSmoothPartitionOfUnity (363); MPolyfold.AdmitsScPartitionsOfUnity (370); LocalModel.HasApproxProperty (381); MChart.isClosed_inter_preimage (398); MChart.exists_small_nhds (417); LocalModel.HasApproxProperty.exists_approx (447); exists_isScSmoothMap_urysohn (518); admitsScPartitionsOfUnity_of_approx (602); hasApproxProperty_restrict (631); theorem_5_5_10 (747); AdmitsContDiffPartitionsOfUnity (777); HasContDiffBump (784); proposition_5_5_11 (798); cZero (879); mem_cZero_iff (892); norm_cZero (905); IsCoordinatewiseContDiffEmbedding (913); ToruńczykStatement (921); IsWCG (936); IsReflexive (940); span_closedBall_eq_top (944); proposition_5_5_15_reflexive (959); proposition_5_5_15_separable (986); hasContDiffBump_of_hilbert (1026); isReflexive_of_hilbert (1046); GodefroyTroyanskiWhitfieldZizlerStatement (1079); corollary_5_5_17 (1091); MPolyfold.admitsScPartitionsOfUnity_of_hilbert (1109); MPolyfold.AdmitsScPartitionsOfUnity.admitsBumpFunctions (1124).

**Polyfold/Chapter5/StrongBundleShift.lean.** IsStrongBundleRetraction.shift (99); StrongBundleChart.fiberK (172); StrongBundleChart.isClosed_fiberK (192); StrongBundleChart.fiberCLE (237); StrongBundleChart.isBanachable_fiber (277); MPolyfold.mem_shift_levelSet_iff (304); MPolyfold.mem_shift_smoothSet_iff (313); MPolyfold.levelChart_apply_levelOne (350); MPolyfold.IsScSmoothMap.levelOne (367); StrongBundle.ShiftSpace (482); StrongBun-

dle.shiftProj (485); StrongBundle.shiftFiberSubmodule (493); StrongBundleChart.shift (630); StrongBundleChart.shift_transition_one (723); StrongBundleChart.shift_compatible_range (800); StrongBundleChart.Compatible.shift (815); StrongBundle.shiftAtlas (829); StrongBundle.continuous_shiftProj (840); StrongBundle.surjective_shiftProj (861); StrongBundle.banachable_shift (865); StrongBundle.shift (877); StrongBundle.isChart_shift (887); StrongBundle.isChart_shift_toMPolyfold0_iff (905); StrongBundle.mem_shift_doubleLevel_iff (918); StrongBundle.sectionShift (940); StrongBundle.isSection_sectionShift (959); StrongBundle.IsScSmoothSection.shift (969); StrongBundle.solutionSet_sectionShift (994); StrongBundle.regularizing_sectionShift (1002).

**Polyfold/Chapter5/Theorem5315_Ends.lean.** scDeriv_apply_eq_self_of_retract_subset (72); subChartOf (122); subPullbackData (140); mem_minimalSubspaceWith_of_isWithoutBoundary (180); principal_restrictSection_subPullback (243); exists_scDeriv_eq_of_isLinSurjAt_restrict (283); isGeneralPositionGerm_of_restrict (323); StrongBundle.IsGeneralPositionAtEnds (401); StrongBundle.IsSubmersiveGenCompactHomotopy.isGeneralPositionGerm_of_mem_boundary (418); theorem_5_3_15_generalPosition (452); theorem_5_3_15_generalPosition_witness (475).

**Polyfold/Chapter5/Theorem5315_Final.lean.** theorem_5_3_15 (61).

**Polyfold/Chapter5/Theorem5315_GeneralPosition.lean.** exists_isOpen_isGeneralPositionGerm_of_forall (56); theorem_5_3_15_generalPosition_hyp (82).

**Polyfold/Chapter5/Theorem5315_LocalData.lean.** MPolyfold.IsSmoothParam.exists_isOpen_of_mem_nhds (65); StrongBundle.exists_isSmoothParam_solutionSet (90); StrongBundle.isRegularPointOn_of_extremum (141); IsScOperator.apply_mem_smooth (204); scDeriv_pc0_fst_apply (242); scDeriv_pc0_snd_apply (280); StrongBundle.exists_isSmoothParam_famZero (350); StrongBundle.hasFamilyParam (403); StrongBundle.exists_hasRegularFamilyParam (479); StrongBundle.hasMorseLocalData (611).

**Polyfold/Chapter5/Theorem5315_LocalInvariance.lean.** MPolyfold.IsSmoothParam.transition (42); MPolyfold.IsSmoothParam.hessian_injective_congr (58).

**Polyfold/Chapter5/Theorem5315_LocalInvarianceClause.lean.** MPolyfold.IsSmoothParam.hasFDerivWithinAt_transition (56); MPolyfold.IsSmoothParam.fderivWithin_transition_comp (81); MPolyfold.IsSmoothParam.fderivWithin_ne_zero_congr (97); MPolyfold.IsSmoothParam.mem_nhds_congr (125); MPolyfold.IsSmoothParam.morseClause_congr (157); MPolyfold.IsMorseOrRegularPoint.morseClause (182).

**Polyfold/Chapter5/Theorem5315_LocalParam.lean.** MPolyfold.TangentGermChart.exists_isSmoothParam (49).

**Polyfold/Chapter5/Theorem5315_LocalTangent.lean.** scDeriv_eq_of_hasFDerivWithinAt (56); MPolyfold.IsSmoothParam.exists_fderivWithin_eq_scDeriv (79); StrongBundle.exists_retract_range_eq_linKer (193); StrongBundle.exists_fderivWithin_eq_scDeriv_of_mem_linKer (227); mem_closure_levelOne_minimalSubspaceWith (251); scDeriv_eq_zero_of_extremum (284); MPolyfold.exists_scDeriv_eq_chart_change (370).

**Polyfold/Chapter5/Theorem5315_Morse.lean.** famZero (108); fiberComb_smul_eq (115); isZeroAt_paramSlice_smul_iff (136); MPolyfold.IsRegularPointOn (194); HasFamilyParam (221); HasRegularFamilyParam (230); not_isRegularPointOn_of_multiplier (258); abs_arctan_div_two_le_one (290); contDiff_arctan_div_two (297); exists_morse_perturbation_hyp (315); sectionAdd_sectionAdd_eq

(652); isZeroAt_sectionAdd_of_isZeroAt (666); StrongBundle.HasMorseLocalData (681); theorem_5_3_15_hyp (709).

**Polyfold/Chapter5/Theorem5315_MorseFamily.lean.** isOpen_setOf_surjective_clm (69); famConstraint (90); famGradient (130); fderiv_famGradient_mult (245); fderiv_famGradient_param (268); fderiv_famGradient_point (293); surjective_fderiv_famGradient (324); ae_isLagrangeNondegenerate_family (377).

**Polyfold/Chapter5/Theorem5315_MorseLagrange.lean.** fderiv_fderiv_comp_apply (58); exists_clm_comp_eq_of_ker_le (93); lagrangian (131); fderiv_lagrangian_eq_zero_iff (165); IsLagrangeNondegenerate (198); injective_hessian_comp_of_lagrange (256); exists_levelSet_param (347).

**Polyfold/Chapter5/Theorem5315_MorseParam.lean.** MPolyfold.IsSmoothParam (73); MPolyfold.IsSmoothParam.comp_equiv (110); MPolyfold.IsMorseOrRegularPoint (130); MPolyfold.HasOnlyMorseCriticalPoints (138); MPolyfold.isMorseOrRegularPoint_of_param (143); MPolyfold.IsSmoothParam.of_levelSet (187); isScSmoothMap_rn_of_components (248); famShift (273); isScSmoothMap_famShift (281); ae_isMorseOrRegularPoint_famSlice (318).

**Polyfold/Chapter5/Theorem5315_MorseTransfer.lean.** MPolyfold.IsSmoothParam.exists_isOpen_regular (92); MPolyfold.IsSmoothParam.isMorseOrRegularPoint_of_regular_of_eq (128); MPolyfold.IsSmoothParam.fderivWithin_eq_zero_of_multiplier (164); inclJ (225); exists_spanning_of_group (260).

**Polyfold/Chapter5/Theorem5315_ParametricMorse.lean.** exists_clm_prod_equiv (59); exists_isOpen_measure_image_fst_eq_zero (92); ae_forall_injective_of_surjective (194); HasOnlyMorseCriticalPointsOn (230); finrank_clm_real (234); fderiv_slice_eq (240); ae_hasOnlyMorseCriticalPointsOn (252); ae_hasOnlyMorseCriticalPointsOn_add_linear (295).

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**Polyfold/Chapter6/Book2021Continuity.lean.** isCompl_sup_inf_of_isCompl (76); exists_isClosed_isTopCompl_isTopCompl_of_finrank_eq (108); diagram_of_solveOp (137); projCod (258); solveOp (264); kerTransition (269); cokerTransition (274); solveOp_mem (277); kerTransition_apply (281); cokerTransition_apply (284); apply_solveOp_of_projRestrict_eq (288); disjoint_ker_comp_of_projRestrict_eq (299); isTopCompl_ker_comp_of_projRestrict_eq (312); continuousAt_solveOp (346); proposition_6_4_13_continuity (372); proposition_6_4_13_continuity_hyps (451).

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**Polyfold/Chapter6/CobordismOrientation2021.lean.** cobOr (60); isContinuous_cobOr (76); cobordismOrientation_of_ends (94).

**Polyfold/Chapter6/CobordismOrientationEnds2021.lean.** pullOr_wedgeExtLeft_eq_std_iff (66); MPolyfold.derivIn_prodRn (128).

**Polyfold/Chapter6/CobordismOrientationEnds2021b.lean.** surjRay_extConjDetFirst (108); pullOr_surjRay_extConjDetFirst_iff (164).

**Polyfold/Chapter6/CobordismOrientationEnds2021c.lean.** ChartPoint.linZero_congr_of_eventuallyEq (82); ChartPoint.linZero_prodPt_paramLift (131); ChartPoint.pullOr_tangentOrOf_prodOrFirst₀_iff (166); tangentOr_toChartOr_prodFirst (283); modelDeriv_prodRn (346).

**Polyfold/Chapter6/CobordismOrientationEnds2021d.lean.** IsInducedSectionOrientation (49); IsInducedSectionOrientation.sectionAdd (64); solReadOrOpen_prodFirst_iff (97); isEndProductOrientation_cobOr (232); cobordismOrientation_of_induced (340); cobordismOrientationHyp (372).

**Polyfold/Chapter6/CobordismRelativeForms2021.lean.** ProductEnd.isEndPullback_of_pullAt (165); ProductEnd.IsEndPullback.extDeriv (229); ProductEnd.IsEndProductOrientation.of_subEnd (281);

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**Polyfold/Chapter6/CobordismRelativeForms2021b.lean.** MPolyfold.pullOver (102); MPolyfold.faceRestriction_pullOver (183); exists_closed_form_of_ends (408); degIndex_cobPart (685); cobEndLeftIn (697); cobEndRightIn (710); cobordismRelativeFormsInner_zero (744); cobordismRelativeFormsInner_succ (792); cobordismRelativeFormsInner (853); hasCutEnd_leftIn (862); psiAt_eq_of_cobordism_of_orientation (913); psiAt_eq_of_admissible_of_orientation (961); psiAt_eq_of_homotopy_of_orientation (975); theorem_6_7_1_of_orientation (1021).

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**Polyfold/Chapter6/ContinuousOrientation.lean.** HasLocalPropagation.eq_trans (59); IsContinuousOrientation (126); exists_two_isContinuousOrientation (141); exists_orientableNbhd (201).

**Polyfold/Chapter6/DetBundle.lean.** DetFiber (71); LocalDetChart.coord (102); LocalDetChart.pretriv (141); exists_localDetChart (182); lemma_6_5_9_local (214); lemma_6_5_9 (259); detPrebundle (273); detBasis (309); proposition_6_5_7 (314); proposition_6_5_8 (337); fredholmIncl (345); DetOn (350); proposition_6_6_1 (356).

**Polyfold/Chapter6/DetLine.lean.** maxPow (72); maxPowDual (76); finrank_maxPow (79); dualVec (102); dualVec_self (110); eq_dualVec_of_apply_eq_one (140); wedge (162); wedge_eq_det_smul (205); wedge_append_basis_left (461); wedge_append_basis_right (472); wedge_append_add_right (483); wedge_append_add_left (499); maxPowMap (530); maxPowMap_wedge (534); maxPowMap_comp (544); maxPowMap_ne_zero_iff (562); maxPowDualMap (589); iotaDual (601); iotaDual_wedge_wedge (606); iotaDual_wedge_dualBasis (617); iotaDual_ne_zero (629); proposition_6_3_2 (691).

**Polyfold/Chapter6/DetVectorBundles2021.lean.** DetBundle.coord_wedgeAt_eq_det (178); DetBundle.lamPrebundle (246); DetBundle.lamDualPrebundle (369); DetBundle.tenMap_tenInv_eq (585); DetBundle.tenPrebundle (601); DetBundle.lemma_6_5_3 (643).

**Polyfold/Chapter6/DirectSumDet.lean.** kerInl (101); kerInr (107); cokerInl (113); cokerInr (120); kerProdEquiv (194); cokerProdEquiv (200); finrank_ker_prodMap (208); finrank_coker_prodMap (217); isFredholm_prodMap (228); directSumVec (258); directSumDet (274); directSumVec_eq_smul (293); definition_6_4_16 (338); directSumDet_bijective (376).

**Polyfold/Chapter6/ExactSequence.lean.** FourTermExact (101); FourTermExact.Choice (160); FourTermExact.Choice.wedgeC (215); FourTermExact.Choice.value (223); FourTermExact.Choice.exists_smul_source_value (284); FourTermExact.canonicalChoice (367); FourTermExact.phi (405); FourTermExact.lemma_6_3_3 (413); FourTermExact.phi_ne_zero (420); FourTermExact.phiEquiv (445); contractDual (567); FourTermExact.psi (585); FourTermExact.proposition_6_3_5 (591).

**Polyfold/Chapter6/FamilySections.lean.** reparamSection (37); pullbackSlice_reparamSection (46); rnProdMapOp (53); StrongBundle.isScPlusSection_of_local (61); StrongBundle.IsScPlusSection.reparam (82); StrongBundle.isGeneralPositionGerm_paramSection_of_comp (142).

**Polyfold/Chapter6/FredholmDet.lean.** linDet (50); linDetEquivOfEq (60); fredholmDet (133); finrank_fredholmDet (136); FredholmOrientation (144); nonempty_fredholmOrientation (147); exists_two_fredholmOrientation (154).

**Polyfold/Chapter6/GapP1_ClassicalDetBundle.lean.** derivFredholm (47); ClassicalDet (55); classicalDet_vectorBundle (66); classicalDet_two_orientations (74).

**Polyfold/Chapter6/GapP1_CubeBoundaryP.lean.** isGeneralPositionGerm_cubeSlice_iff (415); lemma_6_6_4_propertyP_cube (484).

**Polyfold/Chapter6/GapP1_CubeBoundarySlice.lean.** mem_minimalSubspaceWith_of_ hasFDerivWithinAt (345).

**Polyfold/Chapter6/GapP1_CubeBoundarySubmersion.lean.** lemma_6_6_5_submersion_cube (266).

**Polyfold/Chapter6/GapP1_CubeCharts.lean.** UnitCube.cChart (46).

**Polyfold/Chapter6/GapP1_CubeFredholm.lean.** StrongBundle.IsScFredholmSectionTame. paramCube (250).

**Polyfold/Chapter6/GapP1_CubeGerms.lean.** IsConjugateToBasicGerm.quadExtend (258).

**Polyfold/Chapter6/GapP1_CubeLemmas.lean.** lemma_6_6_4_cube (129); lemma_6_6_4_cube_family (171); lemma_6_6_5_isManifold_cube (214); lemma_6_6_5_isSubmanifoldOfDim_cube (258).

**Polyfold/Chapter6/GapP1_CubeSections.lean.** cubePolyfold (213); cubePolyfold_degIndex (285); proposition_6_6_3_three_cube (507); admitsBumpFunctions_prod_cube (569); proposition_6_6_3_ four_cube (605).

**Polyfold/Chapter6/GapP1_DetConventions.lean.** iotaDualEquiv (59); altLinDetEquiv (80); altLinDetEquiv_tmul_dualBasis (91); altFredholmDetEquiv (107); stdLine (128); stdLine_symm_wedge (140); stdLineDual_dualVec (155); tensorStdLine_tmul_stdWedge (170); tensorStdLineDual_tmul_dualVec (190).

**Polyfold/Chapter6/GapP1_DetNonOrientable.lean.** isEmpty_lineOrientation_of_loop (119); continuous_kerDetMk (268); isEmpty_lineOrientation_det_of_kernel_loop (306); isEmpty_ lineOrientation_det_of_complexStructure (392); exists_two_lineOrientation_det_of_ finiteDimensional (443); isEmpty_lineOrientation_det_prod (495).

**Polyfold/Chapter6/GapP1_DetNonOrientableEll2.lean.** isEmpty_lineOrientation_det_ellTwo (92).

**Polyfold/Chapter6/GapP1_DetNonOrientableTransport.lean.** nonempty_lineOrientation_det_iff_ of_equiv (119); isEmpty_lineOrientation_det_of_equiv_prod (142); isEmpty_lineOrientation_det_ hilbert (297); isEmpty_lineOrientation_det_of_equiv_hilbert (310).

**Polyfold/Chapter6/GapP1_OrientationBundle.lean.** exists_eq_orSec (115); continuous_orSec_iff (176); orNegHomeomorph (223); orMulAction (248); continuous_orUnitsAct (253); detCore (310); localTriv_detMk (342); detMk_eq_iff (360); isSectionOrientation_iff_continuous (422); isOrientableSection_iff_exists_continuous_section (441); detBundleCore (457).

**Polyfold/Chapter6/GapP1_PathCategoryInterval.lean.** ScHomotopicI (254); scHomotopicI_iff (384); betaConcat (431); ScPathI.concat (553); ScPathI.concat_toFun_of_le (578); ScPathI.concat_toFun_ of_ge (588); PathCatI (720); PathCatI.category (732); PathCatI.groupoid (759); PathCatI.toPathCat_ isEquivalence (800).

**Polyfold/Chapter6/GapP1_SlicePropertyP.lean.** fderiv_mem_minimalSubspaceWith_of_centre (71); isParamGP_slice_of_isParamGP (197); isGeneralPositionGerm_slice (330); lemma_6_6_4_propertyP (391); isScFredholmSectionTame_slice (426); lemma_6_6_5_fibre_isManifold (463).

**Polyfold/Chapter6/GapP1_SmoothL1.lean.** IsScSmooth.contDiffOn_levelMap_const (49); IsSc1With.apply_mem_smooth_of_finiteDimensional (158).

**Polyfold/Chapter6/GapP1_SmoothL2.lean.** SmoothSubspaceFamily.contMDiffVectorBundle (293); SmoothSubspaceFamily.infKer (367).

**Polyfold/Chapter6/GapP1_SmoothL3.lean.** exists_local_smooth_frame_linKer (85); lemma_6_6_5_smoothBundle (244).

**Polyfold/Chapter6/GapP1_SubmersionNearFamily.lean.** exists_nhds_surjective_linKer (151); lemma_6_6_5_submersion_nhds (231).

**Polyfold/Chapter6/GapP1_UnitIntervalSections.lean.** PullbackChartData.ofProdChart (120); exists_isScPlusSection_prodChart (146); proposition_6_6_3_three_interval (362); proposition_6_6_3_four_interval (509).

**Polyfold/Chapter6/GapP3_K4_Contractible.lean.** IsScContraction (78); exists_isScContractionRn (89); exists_isScContraction (223); proposition_6_6_3_one (259).

**Polyfold/Chapter6/GapP3_K4_IntervalHomotopy.lean.** chartSegment (52); isScSmoothOn_chartSegment (58); exists_isScHomotopyI (178); proposition_6_6_3_interval (267).

**Polyfold/Chapter6/GapP3_K4_SmoothLInterval.lean.** SmoothSubspaceFamily.comp (54); contMDiff_unitIntervalToOpens (79); lemma_6_6_5_smoothBundle_interval (110).

**Polyfold/Chapter6/GoodProjections.lean.** IsGoodLeftProj (67); goodLeftProjs (77); ProjLE (80); ProjLE.refl (84); ProjLE.antisymm (88); ProjLE.trans (93); IsGoodLeftProj.isFredholm (137); IsGoodLeftProj.fredholmIndex_eq_zero (142); IsGoodLeftProj.isFredholm_comp (150); IsGoodLeftProj.fredholmIndex_comp (158); IsGoodLeftProj.finiteDimensional_quotient_inf (167); exists_isGoodLeftProj (177); lemma_6_10_preprint (225); lemma_6_10_text_reading_preprint (259).

**Polyfold/Chapter6/InvariantCobordism2021.lean.** cutRise (66); integral_deriv_cutRise (81); tsupport_extDeriv_smulFun_subset (125); ProductEnd.IsCutEnd (156); HasCutEnd (165); integral_extDeriv_smulFun_cutRise (183); integral_extDeriv_smulFun_eq_zero_of_faces (218); integral_eq_of_cutCobordism (246); IsPsiIntegrand (326); exists_isPsiIntegrand (342); IsPsiIntegrand.psiAt_eq (382); cobPart (449); isSubmanifoldOfDim_cobPart (483); hasCornerStructureOfDim_cobPart (495); cobTime (515); isCompact_cobSlab (526); sigmaCompactSpace_cobPart (542); cobPart_end_left (578); cobPart_end_right (587); cobEndLeft (600); cobEndRight (612); CobordismRelativeForms (657); psiAt_eq_of_cobordism (715).

**Polyfold/Chapter6/InvariantProductEnds2021.lean.** ProductEnd (474); ProductEnd.IsEndProductOrientation (581); ProductEnd.integral_end_fubini (634); solutionSet_left_eq_prod (925); solutionSet_right_eq_prod (932).

**Polyfold/Chapter6/InvariantProductEnds2021b.lean.** isChart_prodEndChart (360); MPolyfold.CornerCoords.prodRn (399); endChart_eqOnSource (462); productEndOfSub (501); solProductEnd (565); isLinOfIndex_end_iff (637); MPolyfold.isEndPullback_formsInftyRestrict_snd (1338).

**Polyfold/Chapter6/InvariantSingle2021.lean.** ManifoldForm.integral_eq_sum_of_dim_zero (186); StrongBundle.IsAdmissible (262); StrongBundle.exists_isAdmissible_nbhd (285); StrongBundle.solPart (321); StrongBundle.isClosed_solPart (388); StrongBundle.isCompact_solPart (396); StrongBundle.solutionSet_eq_iUnion_solPart (403); StrongBundle.solPart_eq_empty_of_ne (420); StrongBundle.solPart_eq_solutionSet (428); isManifold_solPart (527); solPartOr (536);

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**Polyfold/Chapter6/KernelFamily.lean.** StrongBundle.IsSurjLinOfIndex.finrank_linKer (57); exists_local_frame_linKer (94); exists_subspaceFamily_linKer (166); hasFDerivAt_scDeriv_rn (197); fderiv_mem_linKer (217).

**Polyfold/Chapter6/Lemma629.lean.** isZeroAt_paramSection_zero (81); exists_sections_isGeneralPositionGerm_of_isCompact (102); lemma_6_6_4 (203).

**Polyfold/Chapter6/LinOrientation.lean.** ChartPoint.isFredholm_of_mem_linSet (52); ChartPoint.detLin (58); ChartPoint.Or (63); ChartPoint.exists_two_or (67); ChartPoint.orEquivRay (75).

**Polyfold/Chapter6/LinSet.lean.** ChartPoint (109); ChartPoint.T (129); ChartPoint.K (132); ChartPoint.IsScPlusOp (143); ChartPoint.linSet (153); ChartPoint.exists_isScPlusOp_linAt_eq (260); ChartPoint.mem_linSet_iff (309); ChartPoint.linSet_nonempty (323); ChartPoint.isScPlusOp_sub_of_mem_linSet (348); ChartPoint.isFredholm_isIndexAt_of_mem_linSet (362); ChartPoint.exists_index_linSet (388); ChartPoint.fredholmIndex_of_mem_linSet (397); ChartPoint.convex_linSet (404).

**Polyfold/Chapter6/LinSetFredholm.lean.** ChartPoint.exists_isScFredholm_of_mem_linSet (65); ChartPoint.definition_6_2_1_linSet_isScFredholm (98).

**Polyfold/Chapter6/LineBundleOrientation.lean.** IsLineOrientation (59); LineOrientation (66); LineOrientation.eq_of_apply_eq (171); OrCover (330); isCoveringMap_orCover (337); exists_lineOrientation_apply_eq (350); exists_two_lineOrientation (380).

**Polyfold/Chapter6/LocalDetBundle.lean.** FredholmOp (66); projRestrict (69); continuous_projRestrict (92); lemma_6_5_1 (129); LocalDetChart (180); LocalDetChart.ySol (216); LocalDetChart.kerMap (220); LocalDetChart.kerProj (223); LocalDetChart.lemma_6_5_2 (317); LocalDetChart.kerFrame (366); LocalDetChart.frame (379); LocalDetChart.frame_ne_zero (428); proposition_6_5_4 (524); LocalDetChart.gammaProj_frame_eq (760); lemma_6_5_5 (824).

**Polyfold/Chapter6/ManifoldOrientation.lean.** chartChangeDeriv (74); TangentOrientation.IsContinuous (359); IsOrientedAtlas (367); isOrientedAtlas_positiveCharts (506).

**Polyfold/Chapter6/OrientableNbhd.lean.** exists_contDiff_eq_id_near_Icc_mapsTo (140); exists_isScSmoothMap_eq_one_nhdsSet (179); exists_nhds_isCompact_zeros (249); pt1 (267); pt2 (270); IsScHomotopyRel (276); exists_isScPlusSection_pullback_of_mapsTo_of_bump (297); exists_isScPlusSection_pullback_of_family (479); hasScPlusSectionsThrough_pullback_rn_of_bump (511); proposition_6_6_3 (529).

**Polyfold/Chapter6/OrientationConvention2021.lean.** wedge_snoc_eq_smul_wedge_cons (72); kerSign_eq_indSign_mul_stabSign (105); extDetFirst (124); extDetFirst_wedge_tmul (138); extDetFirst_eq_indSign_smul_extDetN (165); eventually_fredholmIndex_eq (173); eventually_indSign_eq (183); isLineOrientation_extPullFirst (297); ChartPoint.existsUnique_prodOrFirst (455); ChartPoint.prodOrFirst (484); ChartPoint.exists_extConjDetFirst_eq_sign_smul (509).

**Polyfold/Chapter6/OrientationFunctor.lean.** nonempty_equiv_of_two (134); equiv_eq_of_apply_eq_of_two (153); LocalTransport.restrict (186); LocalTransport.transport_apply_eq_of_local (205);

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**Polyfold/Chapter6/OrientationMainClean.lean.** MPolyfold.hasConnectedSmoothNbhds (115); theorem_6_6_13 (131); theorem_6_6_17 (152); IsNaturalSolutionOrientation (187); theorem_6_6_18 (223).

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**Polyfold/Chapter9/GapP2_DSheetChartChange.lean.** ScRetraction.exists_lift_lipschitz (56); ScRetraction.hasFDerivWithinAt_of_isSc1 (92); hNull_image_of_pointwise_lipschitz (147); exists_hasFDerivWithinAt_transition (220); exists_clm_tangentCone_chart_change (252); isRelTangentFull_of_chart (337); isHNull_of_forall_exists_chart (348).

**Polyfold/Chapter9/GapP2_DSheetEssential.lean.** exists_mem_tangentConeAt_proj (63); exists_hNull_chart_not_isRelTangentFull (165); isHNull_setOf_not_isRelTangentFull (224); lemma_9_4_6_chart (243); sheetGeometry (256); sheetMeasureGeometry (275).

**Polyfold/Chapter9/GapP2_DSheetGeometryCor.lean.** lemma_9_4_8_of_isNDimensional (40); proposition_9_4_9_of_isNDimensional (51); proposition_9_4_10_of_isNDimensional (76).

**Polyfold/Chapter9/GapP2_DSheetMeasure.lean.** isHNull_mixed_dim (93); isOfDim_of_isNDimensional (123); lemma_9_4_4_of_isNDimensional (136); lemma_9_4_6_of_isNDimensional (145); IsMeasureZeroH (175); lemma_9_5_2_of_isNDimensional (187); hasTameSheetsIn_of_isNDimensional (203); lemma_9_5_2_tame (213); isSigmaIdeal_isMeasureZeroH (220); not_isMeasureZeroH_orbitSupp (234); remark_9_5_3_H (250); lebesgueSigmaH_eq (268).

**Polyfold/Chapter9/GapP2_DSheetNull.lean.** HNull (70); IsHNull (200); isHNull_iUnion (211); isHNull_of_local (225); exists_half_approx (299); not_isHNull_of_hasDimAt (361); sheetNullSets (436).

**Polyfold/Chapter9/GapP2_DSheetNullBridge.lean.** hNull_iff_volume_euclidean (54); isHNull_iff_isMeasureZero_cornerChartedSpace (102); isHNull_iff_isMeasureZero_sheet (197).

**Polyfold/Chapter9/GapP2_E1_BranchedIntegral.lean.** isMorTangentOn_hat (82); OrientedTameStructure.isMeas_preimage (111); exists_slice_pair (150); exists_local_agreement (208); overlap (277); localSignedMeasure (418); existsUnique_branchedMeasure (472); EpGroupoid.theorem_9_5_4 (535); theorem_9_5_4 (535); EpGroupoid.theorem_9_5_5 (564); theorem_9_5_5 (564); branchedMeasure (585); branchedMeasureEff (593); branchedIntegral (621); branchedIntegralEff (629); OrientedTameStructure.localMeasureEff_eq (637); proposition_9_5_13_1 (650).

**Polyfold/Chapter9/GapP2_E1_LocalMeasure.lean.** OrientedTameStructure (76); Represents (102); sheetSum (115); localMeasure (120); localMeasureEff (125); sheetSum_hasSum (157); lemma_9_5_7 (232); sheetSum_image (296); NaturalRep.isMorTangentOn_Φ (399); sheetSum_preimage_eq_card_mul_slice (420); lemma_9_5_9 (502); lemma_9_5_10 (523); sheetSum_preimage_eq_of_morDiffeo (554); lemma_9_5_11 (598).

**Polyfold/Chapter9/GapP2_E1_PointIntegration.lean.** pointIntegration (153).

**Polyfold/Chapter9/GapP2_E1_SheetIntegration.lean.** SheetNullSets.IsMeasIn (98); SheetNullSets.IsMeasIn.transfer (125); EpGroupoid.IsMorTangentOn (161); EpGroupoid.SheetIntegration (179); EpGroupoid.SheetIntegration.piece (362); EpGroupoid.SheetIntegration.isNull_tangentBad (368); EpGroupoid.SheetIntegration.isMeasIn_piece (396); EpGroupoid.SheetIntegration.pairwise_disjoint_piece (426); EpGroupoid.SheetIntegration.sum_sign_weight_eq_zero (457); EpGroupoid.SheetIntegration.sum_integ_eq_zero (508); EpGroupoid.SheetIntegration.sum_integ_eq_sum_integ (584).

**Polyfold/Chapter9/GapP2_E2BoundaryNull.lean.** MPolyfold.IsFaceNull (139); MPolyfold.not_isFaceNull_face (172); SheetNullSets.point (192); halfLineTame (294); halfLine_orbitSupp_boundary_nonempty (313); halfLine_lebesgueSigmaBoundary (337).

**Polyfold/Chapter9/GapP2_E2MeasureGlue.lean.** existsUnique_linearMap_signedMeasure_of_local (168); existsUnique_linearMap_signedMeasure_of_openCover (230); EpGroupoid.isCompact_orbitSupp_boundaryTheta (374); EpGroupoid.theorem_9_5_15_of_overlap (387).

**Polyfold/Chapter9/GapP2_E2Stokes.lean.** MPolyfold.stokes_formsInftyRestrict_of_isCompact (106); MPolyfold.OrientedSheet.stokes (188); EpGroupoid.NaturalRep.exists_ultrafilter_le_nhds (210); EpGroupoid.TameBranchingStructure.isCompact_inter (256).

**Polyfold/Chapter9/GapP2_E3_ChartIntegral.lean.** density (68); chartInteg (72); FormEval.chartInteg_congr (167); FormEval.chartInteg_neg (194); FormEval.chartInteg_mapDiffeo (235).

**Polyfold/Chapter9/GapP2_E3_Covering.lean.** exists_restrict_mem_sheets (200).

**Polyfold/Chapter9/GapP2_E3_Density.lean.** IsMul (67); isMul_smul (77); integ_eq_setIntegral_mul (88); OrientedTameStructure.localMeasure_eq_sum_setIntegral_mul (204); OrientedTameStructure.abs_localMeasure_sub_le (229); abs_branchedMeasure_sub_le (267).

**Polyfold/Chapter9/GapP2_E3_FormEvalEp.lean.** MPolyfold.continuousWithinAt_levelLift (153); epFormEval (256).

**Polyfold/Chapter9/GapP2_E3_FormEvalInfty.lean.** IsSmoothSubmanifold.levelOne (110); epFormsInftyEval (619).

**Polyfold/Chapter9/GapP2_E3_FrameSign.lean.** GraphChart.exists_isOpen_isPos (209).

**Polyfold/Chapter9/GapP2_E3_GraphChart.lean.** integral_image_eq_of_det (50); MPolyfold.GraphChart (75); FormEval (107).

**Polyfold/Chapter9/GapP2_E3_GraphChartExists.lean.** ScRetraction.exists_graphData (46); exists_graphChart (232).

**Polyfold/Chapter9/GapP2_E3_ManifoldBridge.lean.** bridgeOrientation (364); isContinuous_bridgeOrientation (402); FormEval.chartInteg_eq_setIntegral_corner (551).

**Polyfold/Chapter9/GapP2_E3_ManifoldBridge2.lean.** FormEval.integ_eq_manifoldIntegral (207).

**Polyfold/Chapter9/GapP2_E3_SheetIntegration.lean.** FormEval.sheets (55); FormEval.sheetNull (70); FormEval.integ (88); FormEval.hereditarilyLindelofSpace (142); FormEval.sheetIntegration (285); FormEval.integ_eq_zero_of_ev_eq_zero (384).

**Polyfold/Chapter9/GapP2_E3_TameOpen.lean.** HasTameInducedStructure.inter_isOpen (43); hcov_of_loc (94); theorem_9_5_4_tame (116); theorem_9_5_5_tame (130); tameBranchedIntegral (144); hcov_axis (207).

**Polyfold/Chapter9/GapP2_E3_Transition.lean.** GraphChart.exists_transition (50); GraphChart.setIntegral_coord_image_eq (87).

**Polyfold/Chapter9/GapP2_E4_ChartAbs.lean.** abs_chartInteg_le (57); chartAbs_congr (94); chartAbs_mapDiffeo (120).

**Polyfold/Chapter9/GapP2_E4_Circle.lean.** rnPolyfold_isSubmanifoldOfDim_unitCircle2 (219); circleOrientation (283); circle_witness (360).

**Polyfold/Chapter9/GapP2_E4_CircleMulti.lean.** circle_multiBranchedIntegral (130).

**Polyfold/Chapter9/GapP2_E4_Density.lean.** OrientedTameStructure.absSheetSum_union (84); eq_setIntegral_of_sheetSum (107); branchedMeasure_eq_setIntegral_mul (163); branchedMeasureEff_eq_setIntegral_mul (183); branchedMeasure_eq_jordan (204).

**Polyfold/Chapter9/GapP2_E4_DensitySigned.lean.** SignedDensity.setIntegral_eq_jordan (83); SignedDensity.eq_setIntegral_of_estimate (197); SignedDensity.eq_setIntegral_of_local_estimate (280).

**Polyfold/Chapter9/GapP2_E4_FiniteCircle.lean.** theorem_9_5_4_full_of_degIndex_eq_zero (45); circle_theorem_9_5_4_full (60).

**Polyfold/Chapter9/GapP2_E4_FiniteDominated.lean.** chartAbs_eq_of_borel (64); isFinite_of_dominated (126).

**Polyfold/Chapter9/GapP2_E4_FiniteTheorem.lean.** exists_isCompact_supp_inter (90); msheets_of_subset_isCompact (139); msheets_of_orientedTameStructure (155); isNDimensional_of_loc (167); theorem_9_5_4_full (181); theorem_9_5_5_full (197).

**Polyfold/Chapter9/GapP2_E4_Hloc.lean.** exists_orientedTameStructure_of_degIndex_eq_zero (64); hloc_of_degIndex_eq_zero (97).

**Polyfold/Chapter9/GapP2_E4_MultiChart.lean.** hasSum_of_double (39); ChartCover (96); ChartCover.IsFinite (147); FormEval.msheets (155); ChartCover.gInteg (185); ChartCover.gInteg_eq_of

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**Polyfold/Chapter9/GapP2_E4_MultiChartCompact.lean.** `msheets_of_isCompact` (38).

**Polyfold/Chapter9/GapP2_E4_MultiChartInteg.lean.** `FormEval.sheets_subset_msheets` (49); `FormEval.msInteg` (92); `FormEval.msheetIntegration` (302); `FormEval.msInteg_eq_integ` (368).

**Polyfold/Chapter9/GapP2_E4_MultiChartTheorem.lean.** `theorem_9_5_4_multi` (67); `theorem_9_5_5_multi` (81); `multiBranchedIntegral` (95); `multiBranchedIntegralEff` (102).

**Polyfold/Chapter9/GapP2_E4_ReducedCircle.lean.** `circle_proposition_9_5_13_2` (42).

**Polyfold/Chapter9/GapP2_E4_ReducedIntegral.lean.** `orbitHomeomorphReduced` (94); `orbitSuppHomeomorphReduced` (119); `measurableSet_image_orbitSuppHomeomorphReduced` (173); `proposition_9_5_13_2_measure` (215); `proposition_9_5_13_2` (282).

**Polyfold/Chapter9/GapP2_E4_ReducedRep.lean.** `Φ_eq_of_reducedIsotropyHom_eq` (78); `reducedπ_Γ_eq_of_reducedIsotropyHom_eq` (88); `reducedIsotropyHom_eq_of_reducedπ_Γ_eq` (104); `NaturalRep.toReduced` (159); `HasPropernessProperty.toReduced` (223); `OrientedTameStructure.toReduced` (277); `card_isotropy_reduced_eq_card_effIsotropy` (299); `tangentMor_reducedπ` (308); `SheetIntegration.toReduced` (336); `IsOrientation.toReduced` (391).

**Polyfold/Chapter9/GapP2_E5_BoundaryNull.lean.** `MPolyfold.CornerCoords.not_hNull_chart_coord_eq_zero` (95); `MPolyfold.HasCornerStructureOfDim.not_isHNull_boundary` (126); `MPolyfold.not_isHNull_submanifoldBoundary` (170).

**Polyfold/Chapter9/GapP2_E5_BoundaryNullSigma.lean.** `MPolyfold.isHNull_two_le_subDegIndex` (122); `MPolyfold.isHNull_submanifoldBoundary_diff_boundaryStratum` (135); `MPolyfold.tameBoundaries` (151); `MPolyfold.boundaryNullSets` (164); `MPolyfold.boundaryNullSets_one_isNull_iff` (187); `EpGroupoid.isMeasureZeroBoundary_boundaryNullSets_iff` (218); `EpGroupoid.isHNull_boundary_of_isMeasureZeroBoundary` (238); `EpGroupoid.lebesgueSigmaBoundaryH` (252).

**Polyfold/Chapter9/GapP2_E5_FunForm.lean.** `EP.epFormsInftySmul` (215).

**Polyfold/Chapter9/GapP2_E5_LevelRestrict.lean.** `formsInftyRestrict_epFormsInftyToObj` (160); `epFormsInftyEval_integ_eq_integral` (190).

**Polyfold/Chapter9/GapP2_E5_MeasureLocal.lean.** `measure_eq_zero_of_integ_eq_zero` (111); `measure_univ_eq_of_integ_eq_zero` (150).

**Polyfold/Chapter9/GapP2_E5_Stokes.lean.** `EP.exists_stokes_partition` (81); `EP.stokes_of_partition` (134); `EP.exists_boundary_functional` (228); `EP.theorem_9_5_17_boundary_sum` (267); `EP.theorem_9_5_18_boundary_sum` (306); `EP.theorem_9_5_17_of_boundary` (347); `EP.theorem_9_5_18_of_boundary` (382).

**Polyfold/Chapter9/GapP2_E5_StokesSheet.lean.** `OrientedTameStructure.orientedSheet` (69); `EP.integ_d_smul_eq_boundaryIntegral` (171); `EP.stokes_one_structure` (223); `EP.stokes_off_support` (254).

**Polyfold/Chapter9/GapP2_E6_BoundaryOrientation.lean.** `MPolyfold.isInwardVec_incFrame_iff` (217); `MPolyfold.incFrame_mem_boundaryFramesAt_iff` (284); `MPolyfold.SubmanifoldOrientation.stratumOrientation` (323); `MPolyfold.SubmanifoldOrientation.stratumOrientation_tangent` (337).

**Polyfold/Chapter9/GapP2_E6_BoundarySheets.lean.** OrientedTameStructure.boundarySheet (55); OrientedTameStructure.boundaryOrient (65); OrientedTameStructure.boundaryOrientation_eq_sum (77); OrientedTameStructure.exists_boundarySheet_mem_sheets (106).

**Polyfold/Chapter9/GapP2_E6_StokesDim1.lean.** EP.formsInftyRestrict_epFormsInftySmul_all (92); EP.stokes_one_structure_all (230).

**Polyfold/Chapter9/GapP2_E6_StokesDim1b.lean.** EP.stokes_of_partition_all (57); EP.exists_boundary_functional_all (153); EP.theorem_9_5_17_boundary_sum_all (193); EP.theorem_9_5_18_boundary_sum_all (233).

**Polyfold/Chapter9/GapP2_E6_StratumRetraction.lean.** MPolyfold.CornerCoords.facePiece (59); MPolyfold.CornerCoords.faceRet (63); MPolyfold.stratumOne_eq_boundaryStratum (203); MPolyfold.exists_stratum_coords (213); MPolyfold.exists_isScPlusRetractionOn_stratumOne (240); MPolyfold.isSmoothSubmanifold_stratumOne (306).

**Polyfold/Chapter9/GapP2_E6_StratumTangent.lean.** MPolyfold.chartTangent_stratumOne_eq (298); MPolyfold.submanifoldTangent_stratumOne (530); MPolyfold.isSubmanifoldOfDim_stratumOne (542).

**Polyfold/Chapter9/GapP2_E7_BoundaryFamily.lean.** isNaturalOn_boundaryOrientation (99); boundaryStratumSet (119); isMorSaturated_boundaryStratumSet (140); OrientedTameStructure.boundaryFamily (202); OrientedTameStructure.represents_boundaryFamily (216).

**Polyfold/Chapter9/GapP2_E7_BoundaryMeasure.lean.** OrientedSheetFamily.nonemptyPart (62); OrientedTameStructure.isMeasIn_boundarySheet (134); theorem_9_5_15_hyp (270); theorem_9_5_16_hyp (294); EpGroupoid.theorem_9_5_15_msheets (324); theorem_9_5_15_msheets (324); EpGroupoid.theorem_9_5_16_msheets (342); theorem_9_5_16_msheets (342); OrientedTameStructure.boundarySheet_mem_msheets_of_isCompact (364); EpGroupoid.hcov_msheets_of_isCompact (371); hcov_msheets_of_isCompact (371); hcov_zero (383).

**Polyfold/Chapter9/GapP2_E7_FamilyMeasure.lean.** OrientedSheetFamily.overlap (169); existsUnique_familyMeasure (394).

**Polyfold/Chapter9/GapP2_E7_SheetFamily.lean.** IsMorSaturated (50); IsNaturalOn (55); OrientedSheetFamily (65); OrientedSheetFamily.lemma_9_5_7 (216); OrientedSheetFamily.lemma_9_5_9 (479); OrientedSheetFamily.lemma_9_5_11 (558); isNaturalOn_of_isOrientation (589); OrientedTameStructure.toFamily (610).

**Polyfold/Chapter9/GapP2_E8F_BoundaryIntegral.lean.** EP.boundaryIntegral_eq_msInteg (220).

**Polyfold/Chapter9/GapP2_E8F_FaceChange.lean.** FormEval.face_change (260).

**Polyfold/Chapter9/GapP2_E8F_FaceFrame.lean.** MPolyfold.faceFrame_mem_stratumOrientation_iff (222).

**Polyfold/Chapter9/GapP2_E8F_FaceSum.lean.** CornerCoords.faceSet_ae_eq (54); FormEval.integral_density_eq_sum_face (92); FormEval.face_term_eq (187).

**Polyfold/Chapter9/GapP2_E8W_Hemisphere.lean.** hemi (160); isSubmanifoldOfDim_hemi (249); boundaryStratum_hemi (358).

**Polyfold/Chapter9/GapP2_E8W_Witness.lean.** hemiOrientation (107); thetaH (133); TH (137); E8W.e8w_hcov (274); e8w_hcov (274); e8w_boundary_nonempty (310); e8w_hasScPartitionsOfUnity (326);

e8w_isOrientation (334); e8w_stokes_hypotheses (339).

**Polyfold/Chapter9/GapP2_E8_BoundaryLocal.lean.** boundaryMeasure_eq_zero_of_integ_eq_zero (114); boundaryMeasure_univ_eq (156); OrientedTameStructure.exists_restrict_subset (266).

**Polyfold/Chapter9/GapP2_E8_Hemisphere.lean.** E8W.e8_hemisphere_stokes (34).

**Polyfold/Chapter9/GapP2_E8_Stokes.lean.** EP.stokes_of_boundaryMeasure (102); EP.theorem_9_5_17_hyp (188); EP.theorem_9_5_18_hyp (213); EP.hcovB_of_hcovS (241); EP.boundaryBranchedMeasure (252); EP.theorem_9_5_17_msheets (264); EP.boundaryBranchedMeasureEff (281); EP.theorem_9_5_18_msheets (296).

**Polyfold/Chapter9/GapP3_K1_AllStructures.lean.** MPolyfold.k1_exists_tangent_eq (51); MPolyfold.k1_mem_boundaryStratum_of_tangent_eq (80); EpGroupoid.FormEval.k1_msheets_of_subset_list (98); EP.boundarySheet_mem_msheets_of_loc (191); EP.theorem_9_5_15 (236); EP.theorem_9_5_16 (258); EP.boundaryMeasure_apply_all (279).

**Polyfold/Chapter9/GapP3_K1_BoundaryCover.lean.** EpGroupoid.LocalBranchingStructure.isCompact_M_inter (66); EpGroupoid.OrientedTameStructure.exists_restrict_eq (92); EP.boundaryStratum_inter_mem_msheets (130); EP.exists_restrict_boundary_mem_msheets (155); EP.hcovS_of_loc (191); EP.hcovS_of_loc_isLUC (223); EP.hcov_of_loc (237); EP.hcov_of_loc_isLUC (247).

**Polyfold/Chapter9/GapP3_K1_CornerFinite.lean.** EpGroupoid.FormEval.exists_chartCover_of_isLindelof (76); EpGroupoid.FormEval.abs_ev_faceFrame (104); EpGroupoid.integrableOn_facePullback_of_isCompact (175); EpGroupoid.FormEval.boundaryStratum_inter_mem_msheets (214).

**Polyfold/Chapter9/GapP3_K1_Hemisphere.lean.** E8W.k1_hemisphere_stokes (38).

**Polyfold/Chapter9/GapP3_K1_Theorems.lean.** EP.hloc_hasLUC (48); EP.theorem_9_5_15_class (59); EP.theorem_9_5_16_class (81); EP.boundaryMeasure (105); EP.boundaryMeasureEff (114); EP.boundaryMeasure_spec (122); EP.theorem_9_5_17 (143); theorem_9_5_17 (143); EP.theorem_9_5_18 (156); theorem_9_5_18 (156).

**Polyfold/Chapter9/GapP3_K2_AllStructuresDim1.lean.** EP.boundarySheet_finite_of_loc (50); EP.boundarySheet_mem_finPointSheets_of_loc (87); EP.theorem_9_5_15_dim1_all (102); EP.theorem_9_5_16_dim1_all (127); EP.boundaryBranchedMeasureDim1_apply_all (162); EP.boundaryBranchedMeasureEffDim1_apply_all (186).

**Polyfold/Chapter9/GapP3_K2_Arc.lean.** K2Arc.isSubmanifoldOfDim_arc (428); K2Arc.isTameSubMPolyfold_arc (502); K2Arc.boundaryStratum_arc (536).

**Polyfold/Chapter9/GapP3_K2_ArcWitness.lean.** K2Arc.arcOrientation (99); K2Arc.arc_hloc (243); K2Arc.arc_isLUC (296); K2Arc.k2_arc_stokes (304); K2Arc.k2_arc_stokes_eff (317); K2Arc.k2_arc_hypotheses (330).

**Polyfold/Chapter9/GapP3_K2_BoundaryDim1.lean.** MPolyfold.exists_isOpen_boundaryStratum_subsingleton (62); EpGroupoid.OrientedTameStructure.exists_boundarySheet_subsingleton (135); EpGroupoid.hcov_dim1_of_loc (172); EpGroupoid.hcovEff_dim1_of_loc (184); EP.theorem_9_5_15_dim1 (220); EP.theorem_9_5_16_dim1 (240).

**Polyfold/Chapter9/GapP3_K2_FaceDim1.lean.** EpGroupoid.boundaryIntegral_eq_pointInteg (163); EP.boundaryIntegral_eq_zeroFormIntegration (261).

**Polyfold/Chapter9/GapP3_K2_FinPoints.lean.** EpGroupoid.finPointSheets (44); EpGroupoid.finPointInteg_eq_pointInteg (90); EpGroupoid.submanifoldTangent_finite_eq (115); EpGroupoid.finPointIntegration (136); EP.finZeroFormIntegration (258).

**Polyfold/Chapter9/GapP3_K2_HemisphereEff.lean.** E8W.hasLUC_trivial (39); E8W.e8_hemisphere_stokes_eff (57).

**Polyfold/Chapter9/GapP3_K2_HemisphereLUC.lean.** E8W.e8w_isLUC (35); E8W.k2_hemisphere_stokes_eff (41).

**Polyfold/Chapter9/GapP3_K2_LUCDim1.lean.** EP.theorem_9_5_16_dim1_isLUC (46); EP.theorem_9_5_18_dim1_isLUC (70).

**Polyfold/Chapter9/GapP3_K2_StokesDim1.lean.** EpGroupoid.boundaryMeasure_univ_eqI (121); EP.stokes_of_boundaryMeasure_dim1 (283); EP.boundaryBranchedMeasureDim1 (357); EP.theorem_9_5_17_dim1 (369); EP.boundaryBranchedMeasureEffDim1 (391); EP.theorem_9_5_18_dim1 (403).

**Polyfold/Chapter9/GapP3_K2_ZeroForms.lean.** EP.zeroFormEval (235); EP.formsInftyRestrict_zero_eq (303); EP.zeroFormIntegration (365).

**Polyfold/Chapter9/GapP3_K3_Groupoid.lean.** K3.isScSmoothMap_smul (172); K3.ε (195); K3.nonEffectivePart_ε (258); K3.isLUC_ε (271); K3.card_effIsotropy_of_snd_eq_zero (306); K3.card_effIsotropy_of_snd_ne_zero (313); K3.isEpSubgroupoid_boundary (384); K3.εB (388); K3.card_isotropy_εB (401); K3.card_effIsotropy_εB (410); K3.remark_9_5_20_effIsotropy (418).

**Polyfold/Chapter9/GapP3_K3_MPolyfold.lean.** K3.chart (554); K3.P (584); K3.scDeriv_end (695); K3.degIndex_eq (781); K3.boundary_eq (789).

**Polyfold/Chapter9/GapP3_K3_Retract.lean.** IsLossFamily.comp_contDiff (48); IsLossFamily.diag (62); K3.Rt (191); K3.mem_0_iff (222).

**Polyfold/Chapter9/GeneralBoundaryPoints.lean.** exists_graph_unif_chart (49); exists_small_cone (77); hasOrientedGoodPointsAt (114); lemma_9_3_4_general (472).

**Polyfold/Chapter9/GeneralCorollaries.lean.** exists_slice_of_classGraphChart (88); exists_generic_slice_general (251); sum_class_eq (309); topMultiplicityAt (379); topMultiplicity (396); hasTangentSlicesAt (401); proposition_9_1_12 (425); LocalBranchingStructure.dimPartsAgreeNear (434); lemma_9_6_4_general (443); isTopBadPoint_iff_general (451); dimPart_topDim_eq (460); proposition_9_6_5_general (470); lemma_9_2_7_2_general (555); EpGroupoid.hasIndependentTangentSums (569); hasIndependentTangentSums (569); hasLabelledBranchingAt_general (585); isLocalDimDecomposition_dimComponent_general (597); existsUnique_localDimDecomposition_general (605); proposition_9_1_11_unique (613); dimComponent_locallyFinite_of_mem_general (620); proposition_9_1_11_local_of_le (629); lemma_9_3_4_of_hasGraphBallAt' (642); lemma_9_3_4_of_degIndex_eq_zero (654).

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**Mathlib/Analysis/Calculus/ContDiff/Basic.lean.** ContinuousLinearMap.iteratedFDerivWithin_comp_left (243); ContinuousLinearMap.iteratedFDerivWithin_comp_right (426).

**Mathlib/Analysis/Calculus/ContDiff/Comp.lean.** ContDiffWithinAt.iteratedFDerivWithin_right (694).

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**Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries.lean.** iteratedFDerivWithin (398); iteratedFDerivWithin_succ_apply_left (427); iteratedFDerivWithin_congr (571).

**Mathlib/Analysis/Calculus/FDeriv/Analytic.lean.** ContinuousMultilinearMap.hasFDerivAt (633).

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**Mathlib/Analysis/Calculus/LocalExtr/Basic.lean.** IsLocalMin.hasDerivAt_eq_zero (237).

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